3D2

process design review · 1211 steps · generated by CarbonEDA v0 on 2026-11-08 18:53 EST

853.4

kgCO2e per wafer start

620.6
EPW energy · 73%

65.8
MPW materials · 8%

167.0
GPW gases · 20%

wafer 300 mm · yield 1 · carbon intensity 0.43 kgCO2e/kWh · facility overhead 0.4 x tool energy
EPW 620.6 = tool 443.3 + facility overhead 177.3 kgCO2e (1,031 + 412 kWh)

flow technologies/dram/3D2/3D2_final.csv
layout technologies/dram/3D2/3D2_layout.gds
structure images from the spx run of 2026-11-08 13:21:08

CarbonEDA parameters

config/carbon_config.yaml and materials.yaml as used for every number in this report

EPW energy

(tool + 0.4 x tool facility overhead) x 0.43 kgCO2e/kWh

process area	kWh / step	steps
Dry_etch	1.33	68
RIE	1.33	109
Cryo_etch	0.835	104
Ion_implantation	1.33	25
Diffusion	1.33	47
Metallization	2	56
Wet_etch	0.28	29
Wafer_clean	0.28	228
Resist	0.28	104
CMP	0.28	37
Deposition	1.09	82
CVD	1.09	234
EUV	3.25	5
ArF-i	0.36	44
ArF	0.33	2
KrF	0.33	31
Hg	0.33	6

GPW process gases

kg x (1 - utilization) x (1 - abatement) x GWP, plus byproducts kg/kg x (1 - abatement) x GWP; defaults: utilization 1.0, abatement 0.9

area	gas	g / step	upstream*	utilization	abatement	byproducts (kg per kg gas)
CVD	NF3	2.54	12.5	0.8	0.95	CF4 0.046, C2F6 0.045, CHF3 0.025, CH2F2 0.00086,...
RIE	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
RIE	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05
Cryo_etch	CF4	0.0241	12.6	0.35	0.89	C2F6 0.079, C4F8 0.00063, CHF3 0.011, CH3F 0.008

* kgCO2e per kg of gas made, counted in MPW

GWP (kgCO2e per kg, 100-yr) and abatement of every gas emitted

gas	GWP	abatement
CF4	7390	0.89
C2F6	12200	0.95
C4F6	0	0.98
C4F8	10300	0.98
CHF3	14800	0.98
CH2F2	675	0.98
CH3F	116	0.98
NF3	17200	0.95
SF6	22800	0.95

MPW materials

blanket film over a 300 mm wafer: thickness x area x density x GWP; the wafer itself 26000 gCO2e (300 mm); yield 1

material	g/cm³	gCO2e/g
TiN	5.21	43.98
Resist	0.97	4.2
W	19.3	75.81
Ti	4.5	47.63
TaN	13.7	25.82
Cu	8.96	6.09
Co	8.9	44.5
Al	2.7	10.73

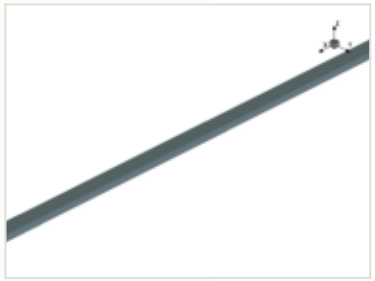
Step #20: Wafer

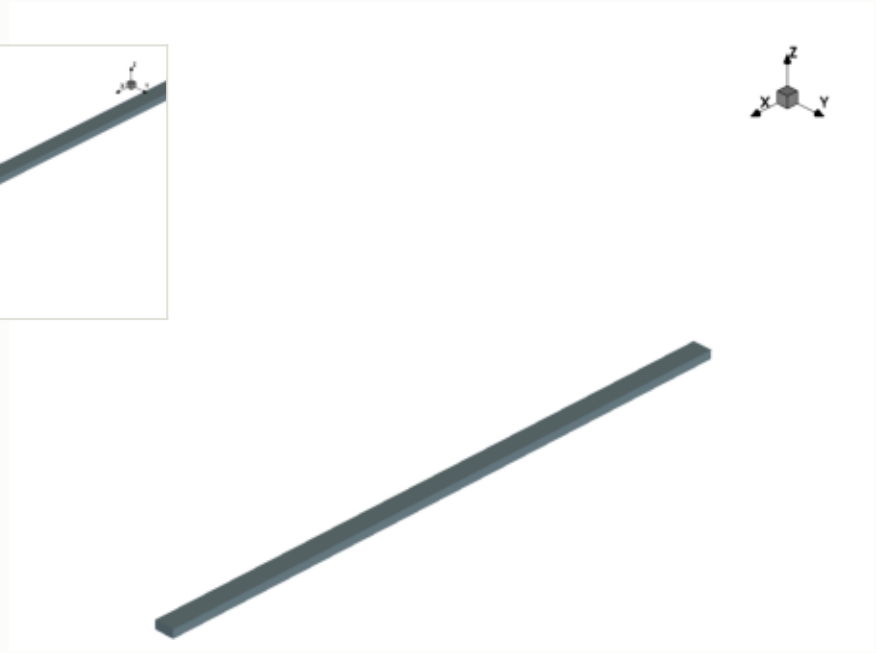
substrate · Wafer

26	26	3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



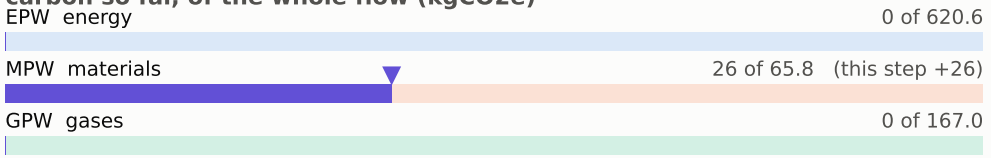


this step

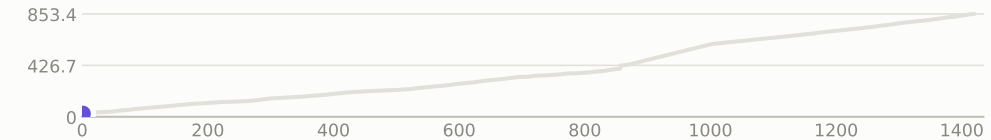
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	26 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**26 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

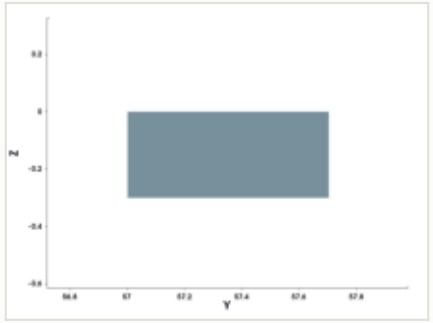
carbon so far, of the whole flow (kgCO2e)



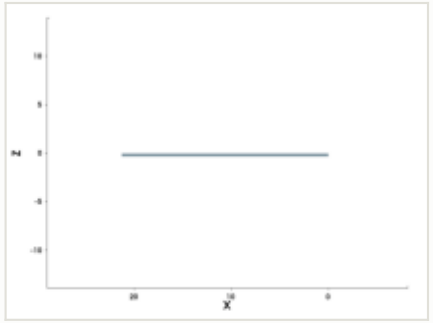
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



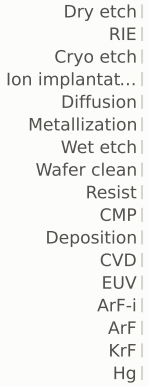
y cut at 57.351 µm



mask for this step

no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)

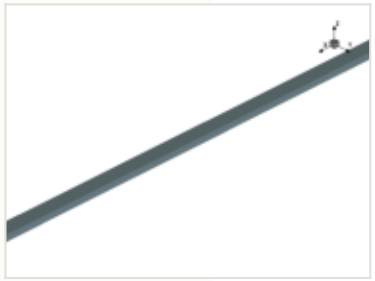
Step #40: Wafer_clean

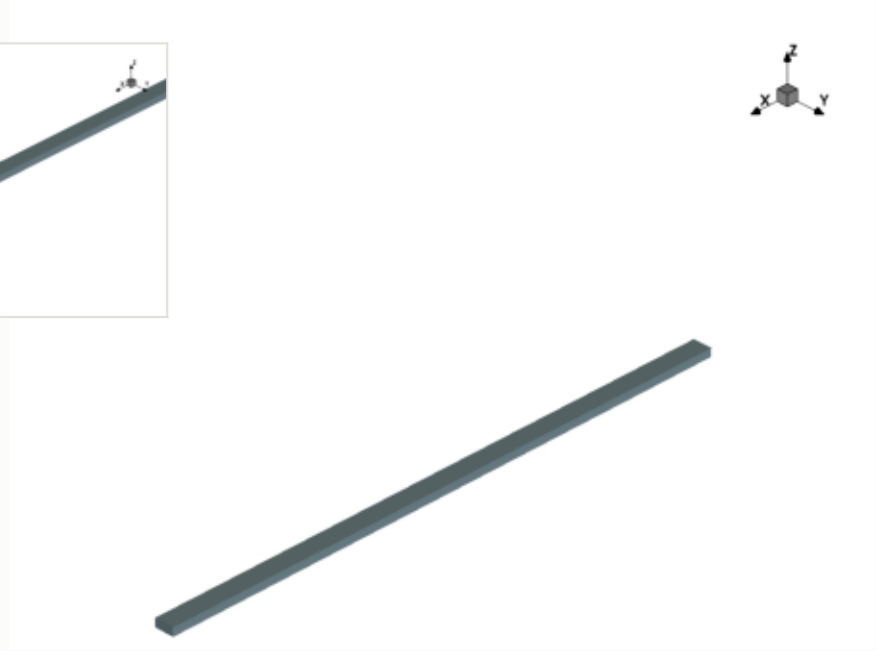
clean · Wafer_clean

0.169	26.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



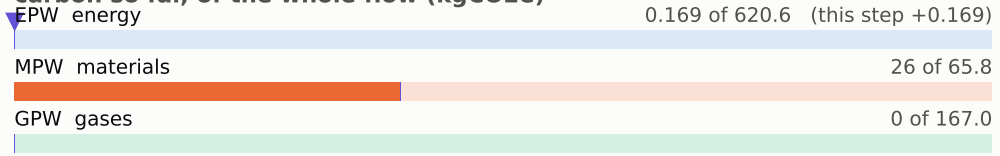


this step

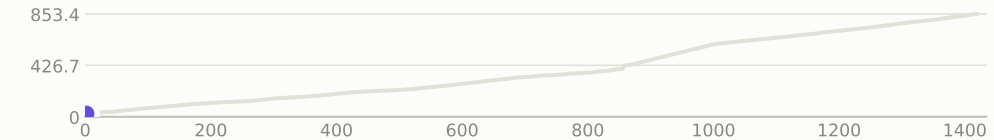
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

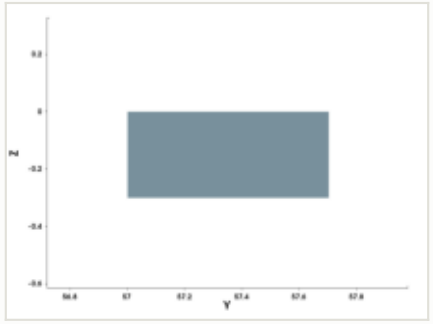
carbon so far, of the whole flow (kgCO2e)



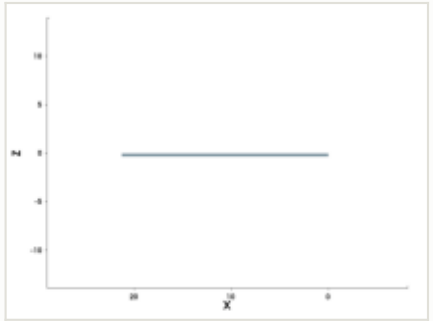
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)

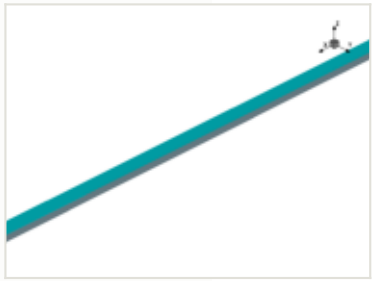
Step #50: Diffusion

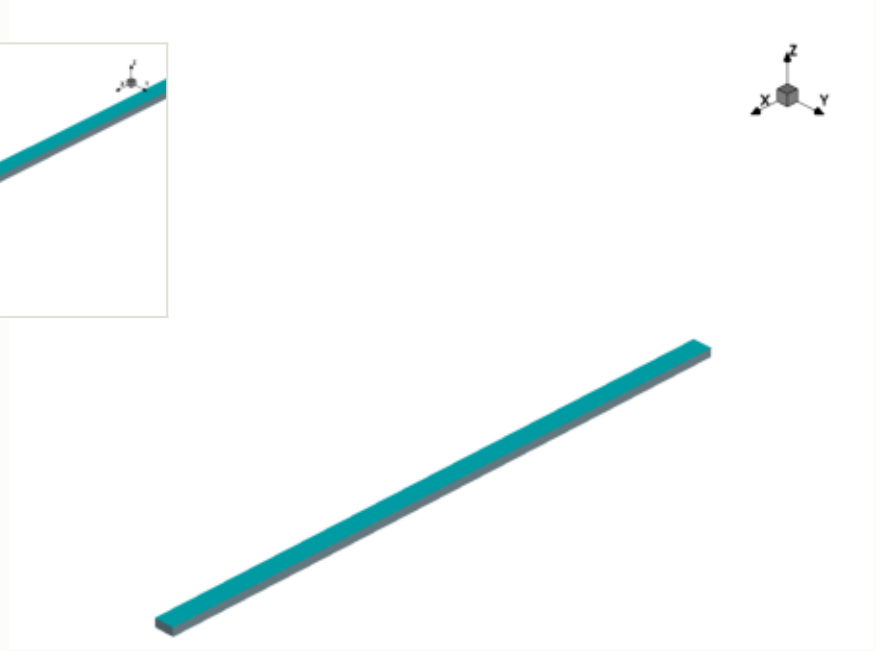
depo_isotropic · Diffusion

0.801	27	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

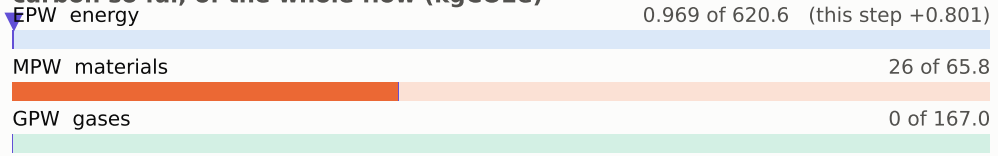




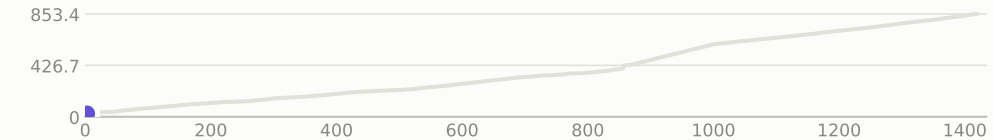
this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	0.0035 µm	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

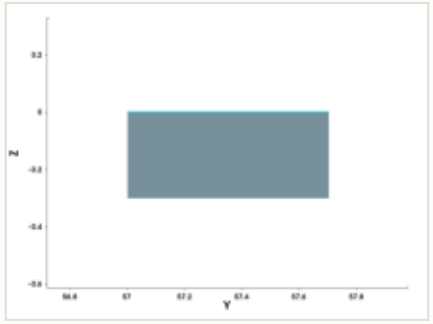
carbon so far, of the whole flow (kgCO2e)



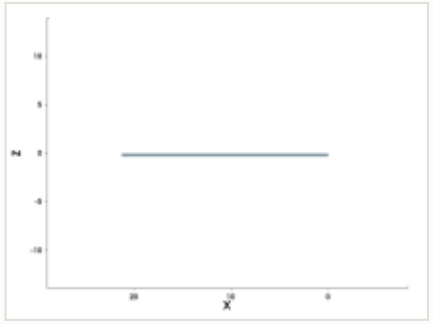
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



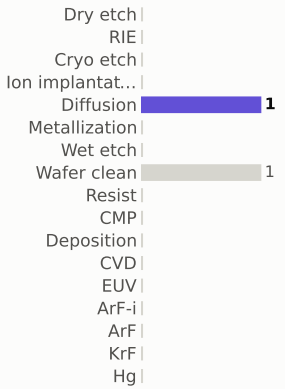
y cut at 57.351 µm



mask for this step

no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)

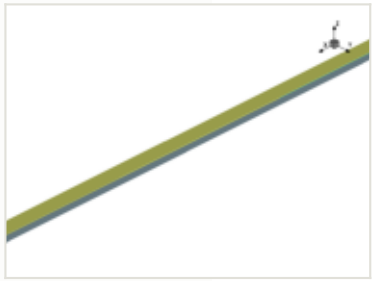
Step #60: CVD

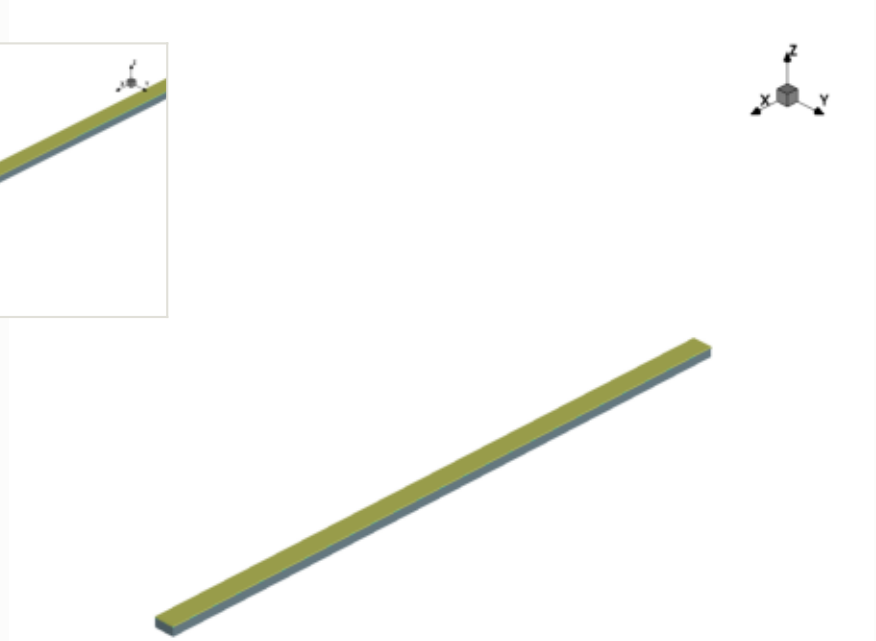
depo_isotropic · CVD

1.31	28.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



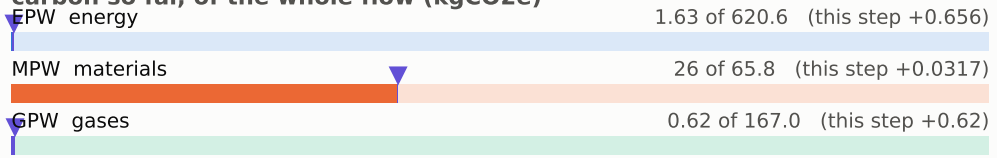


this step

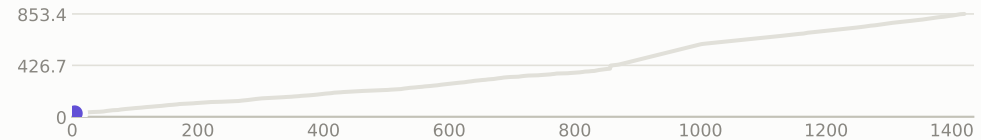
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

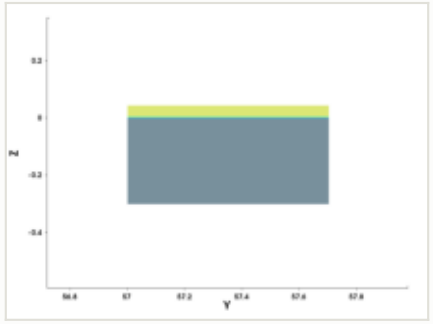
carbon so far, of the whole flow (kgCO2e)



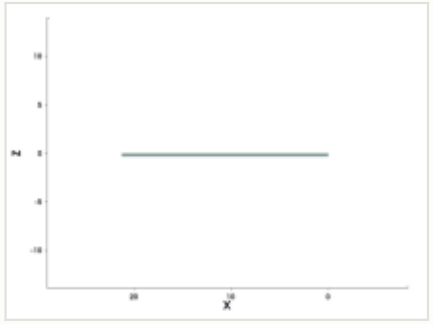
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



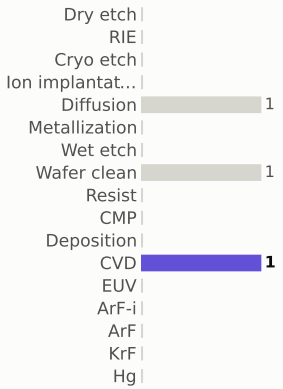
y cut at 57.351 µm



mask for this step

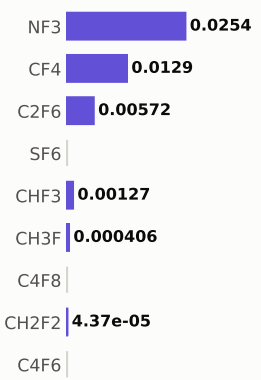
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



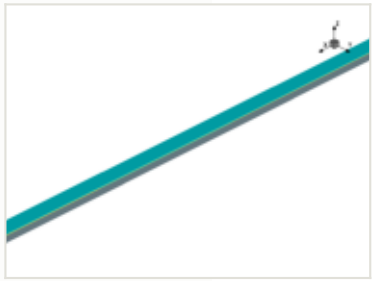
Step #70: CVD

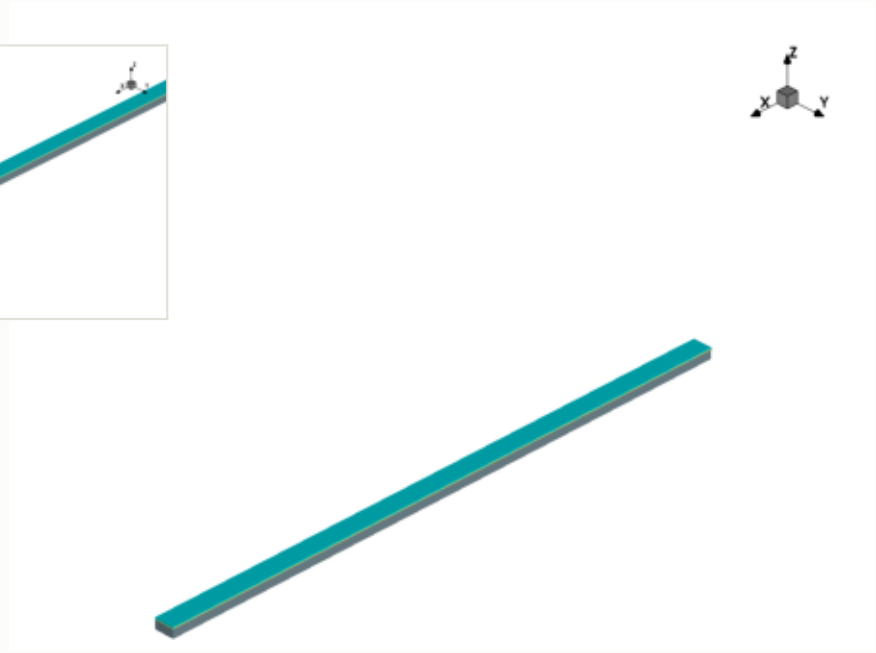
depo_isotropic · CVD

1.31	29.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



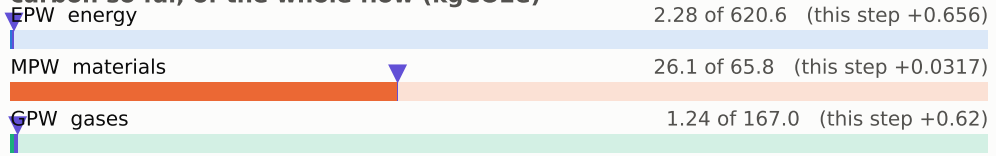


this step

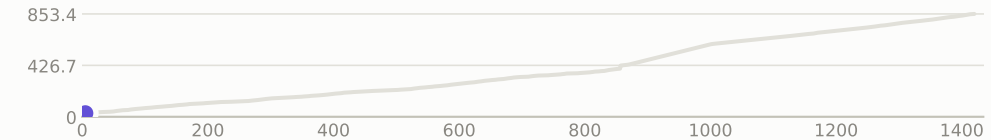
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

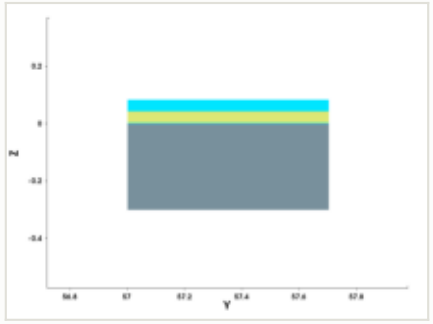
carbon so far, of the whole flow (kgCO2e)



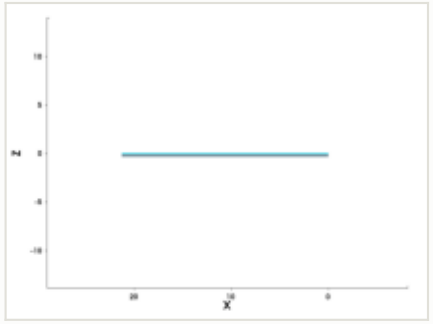
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



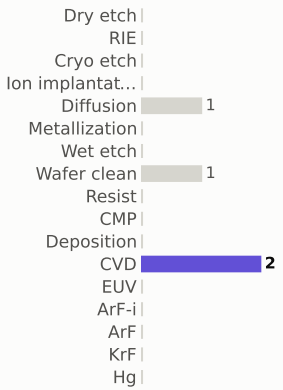
y cut at 57.351 µm



mask for this step

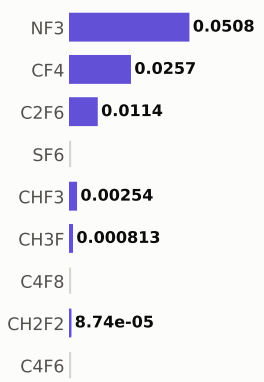
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



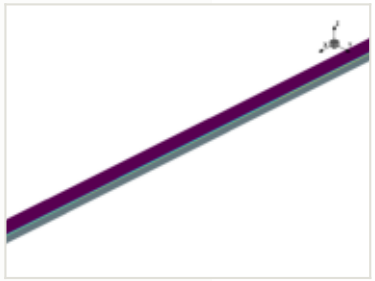
Step #80: CVD

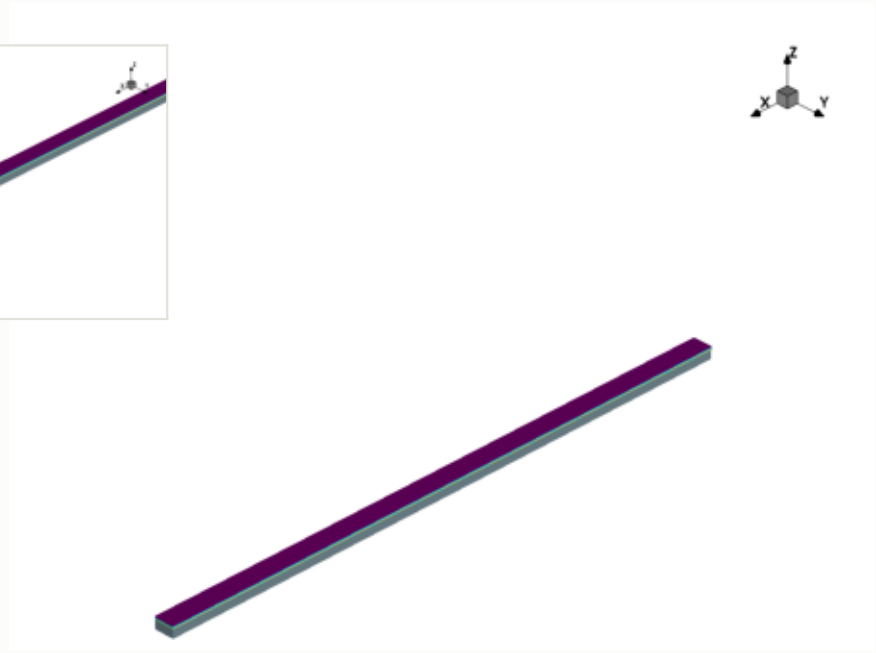
depo_isotropic · CVD

1.31	30.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



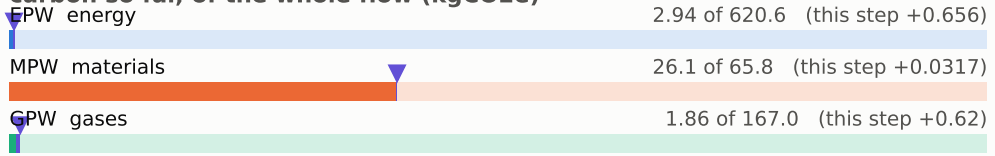


this step

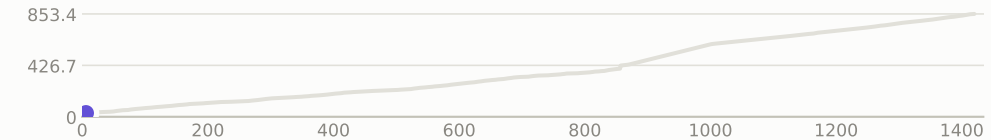
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

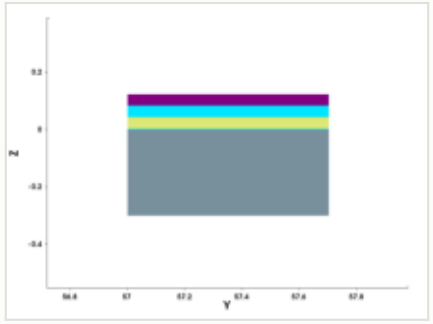
carbon so far, of the whole flow (kgCO2e)



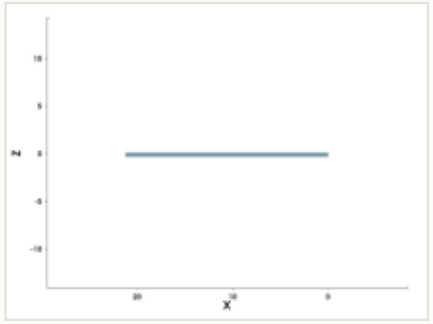
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



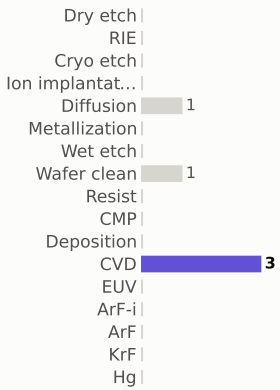
y cut at 57.351 µm



mask for this step

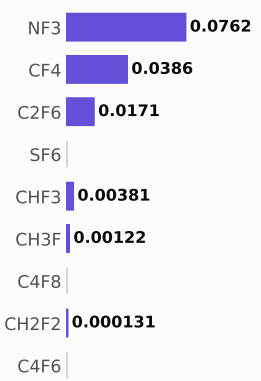
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



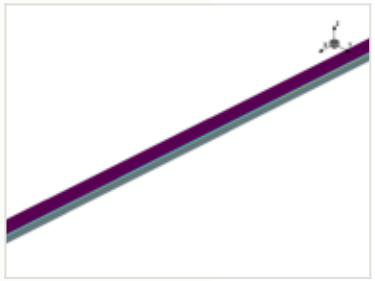
Step #90: Wet_etch

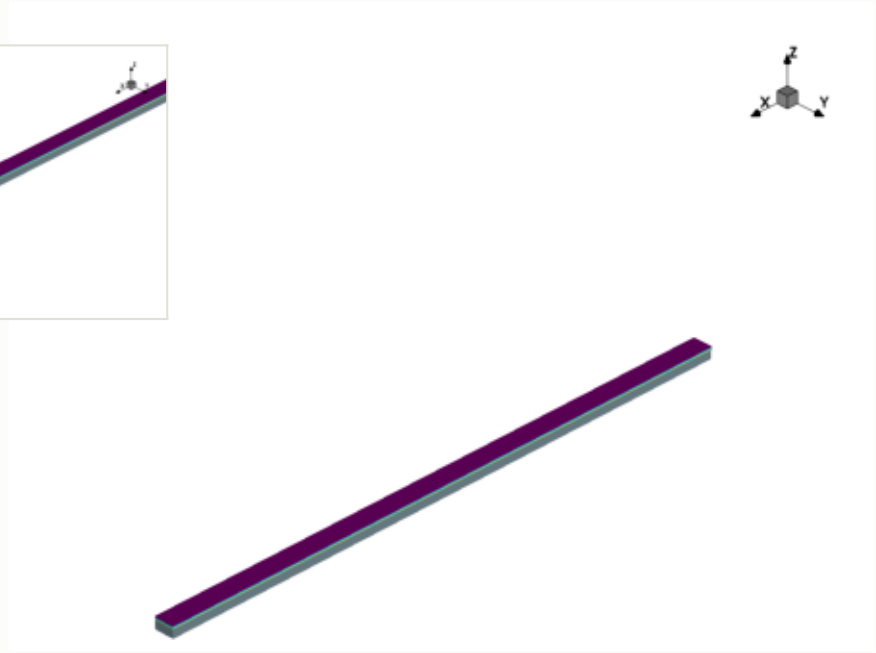
etch_isotropic · Wet_etch

0.169	31.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



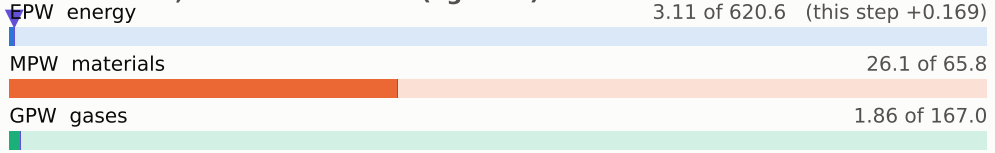


this step

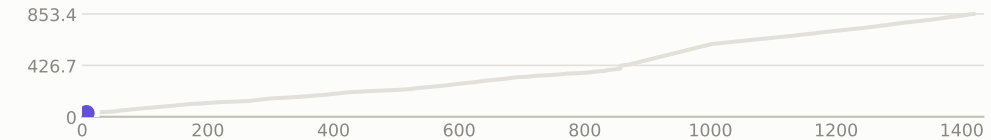
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

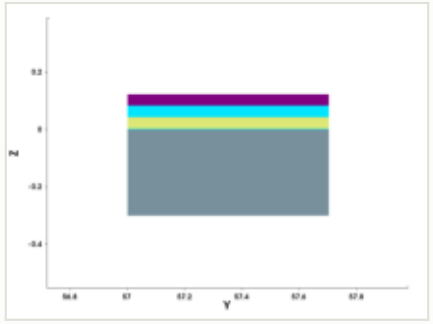
carbon so far, of the whole flow (kgCO2e)



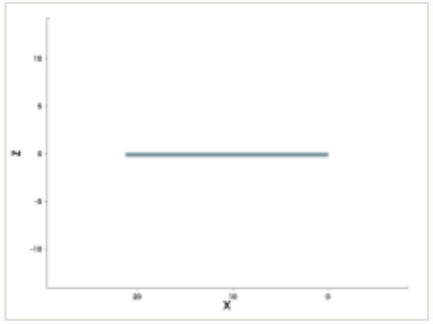
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



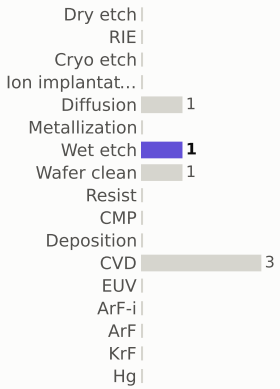
y cut at 57.351 µm



mask for this step

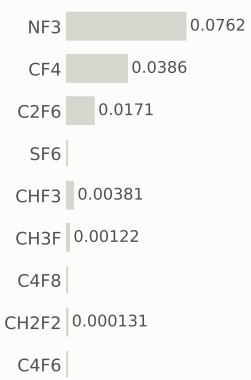
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



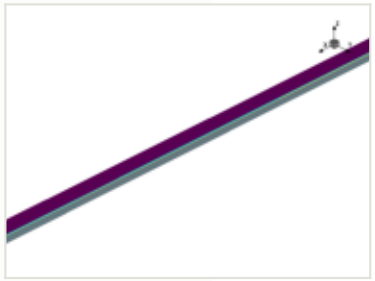
Step #100: Wet_etch

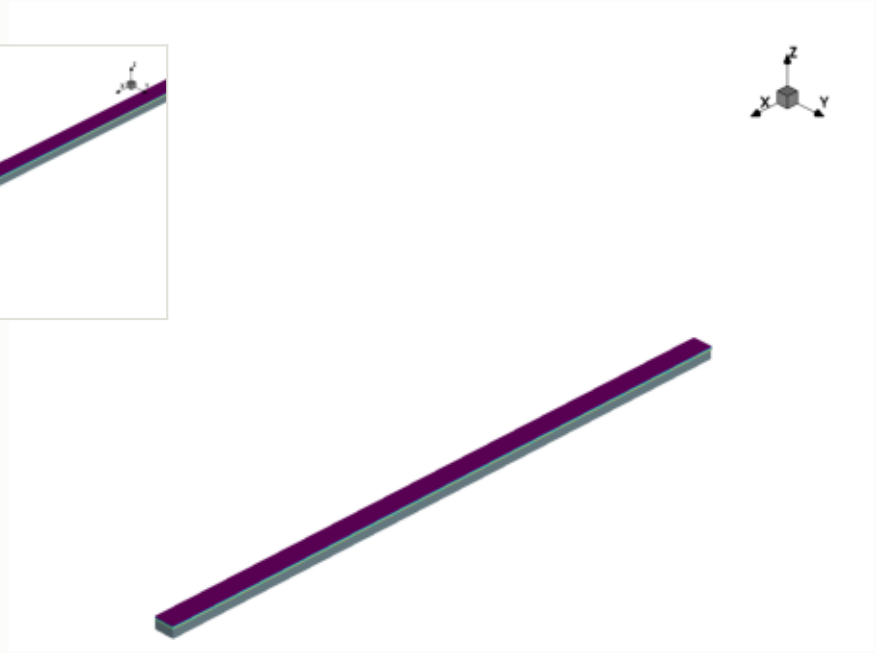
etch_isotropic · Wet_etch

0.169	31.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

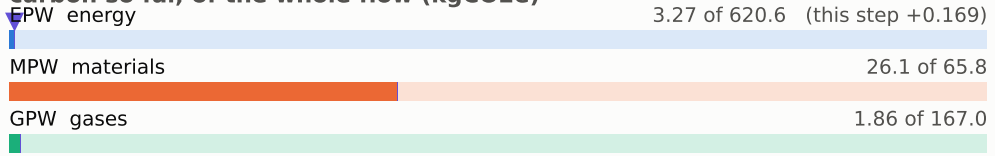




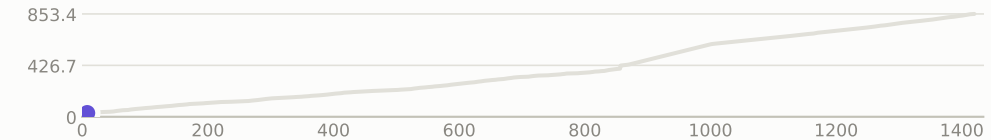
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GPW	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

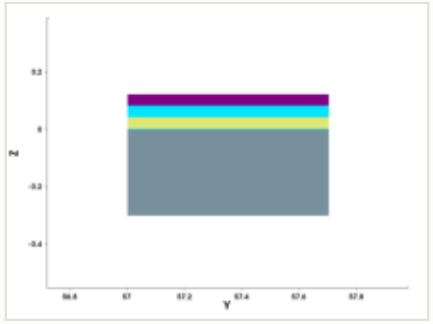
carbon so far, of the whole flow (kgCO2e)



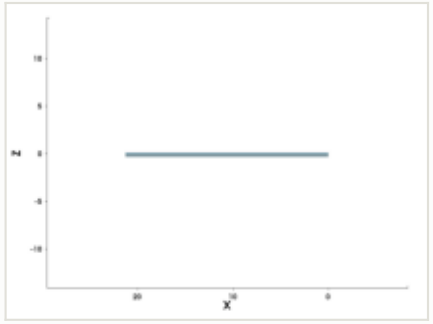
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



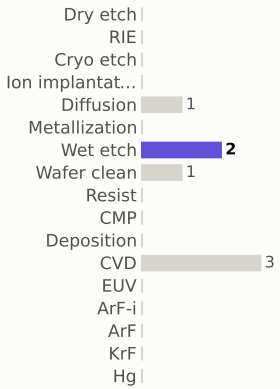
y cut at 57.351 µm



mask for this step

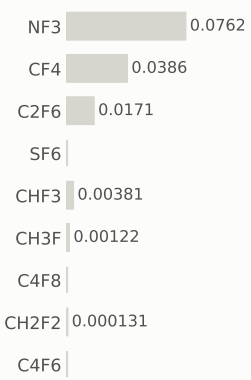
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



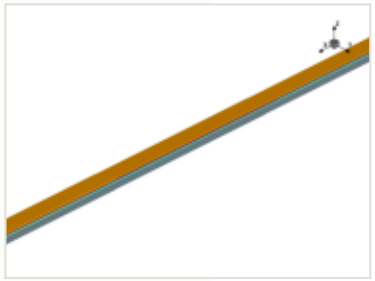
Step #110: Resist

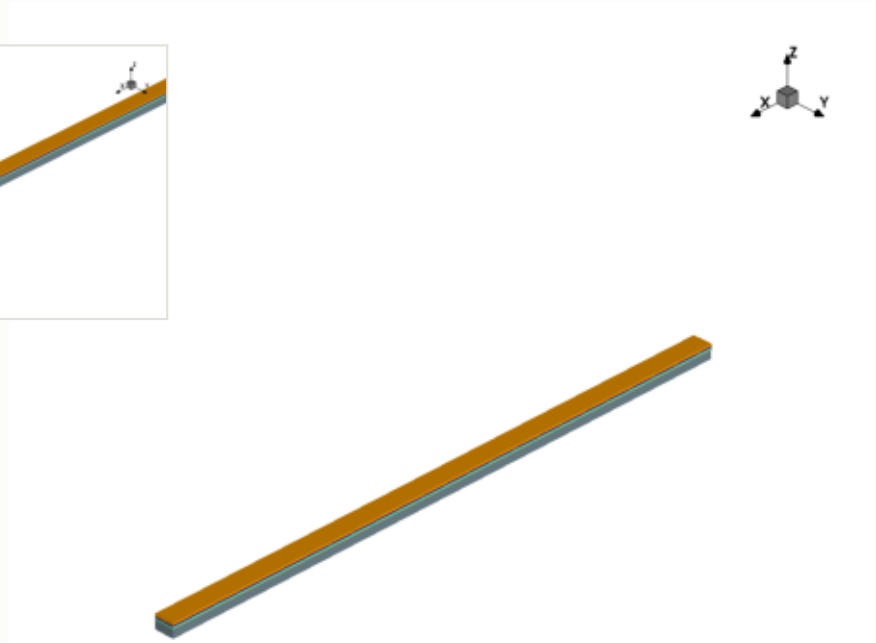
depo_fill · Resist

0.169	31.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



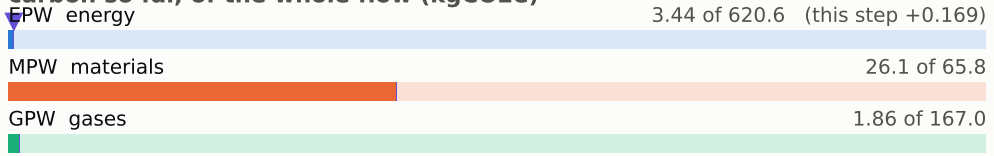


this step

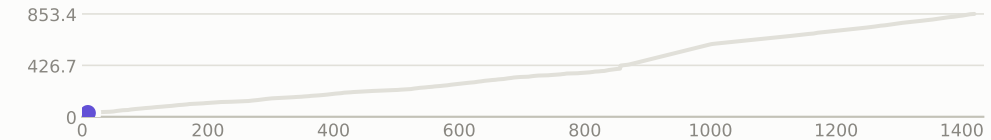
process area	Resist
material	—
thickness	0.08 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

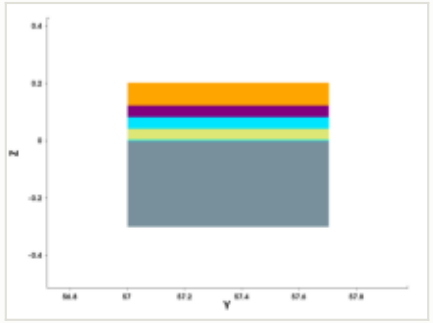
carbon so far, of the whole flow (kgCO2e)



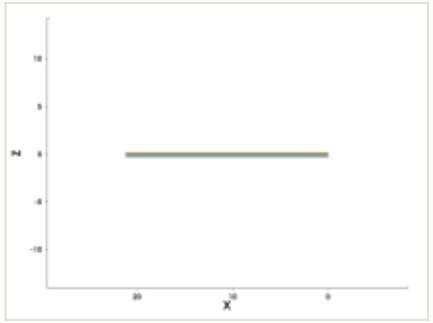
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



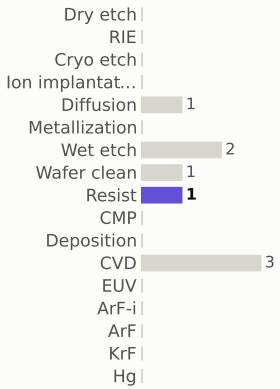
y cut at 57.351 µm



mask for this step

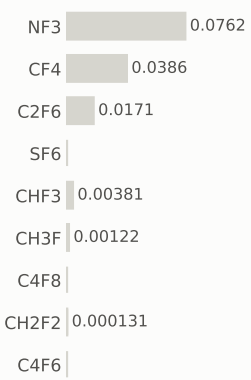
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



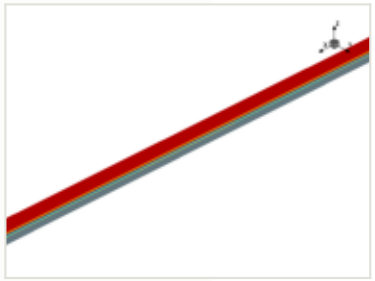
Step #120: CVD

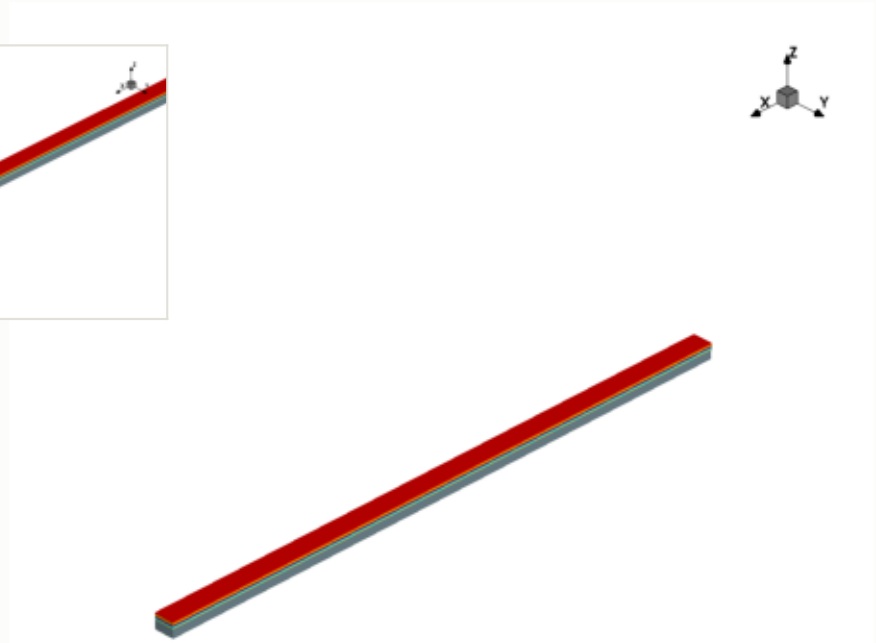
depo_isotropic · CVD

1.31	32.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



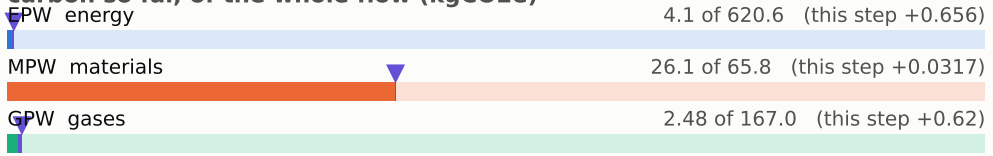


this step

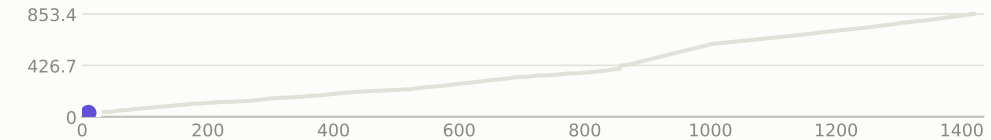
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

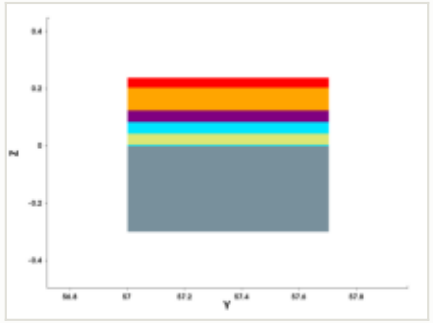
carbon so far, of the whole flow (kgCO2e)



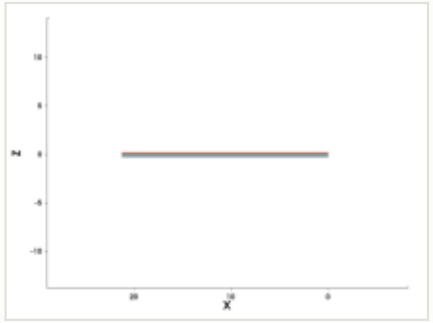
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



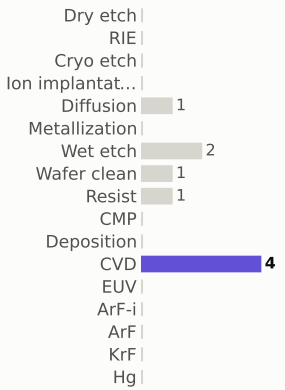
y cut at 57.351 µm



mask for this step

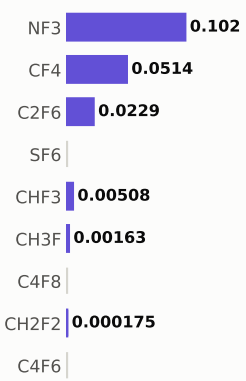
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



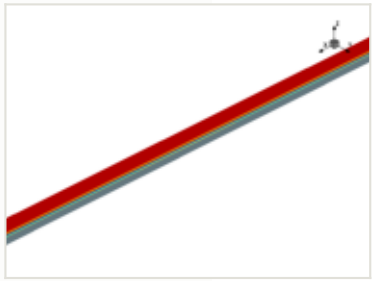
Step #130: Wafer_clean

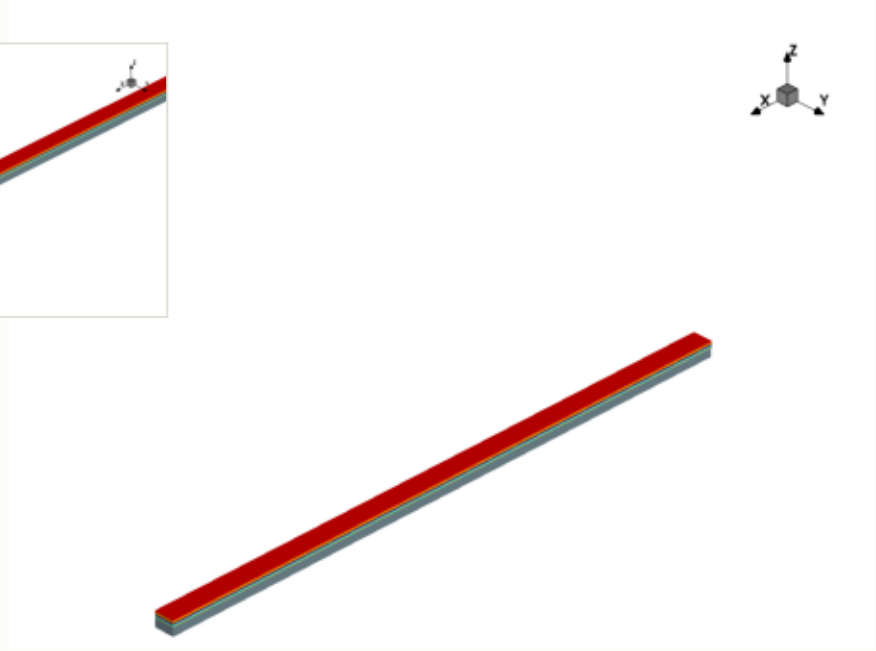
clean · Wafer_clean

0.169	32.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



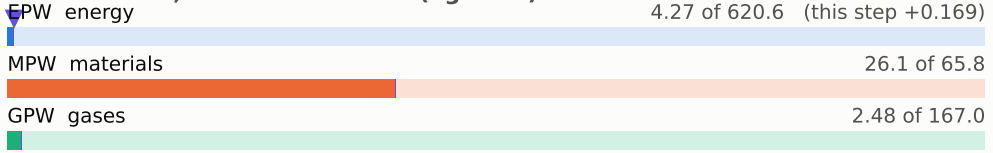


this step

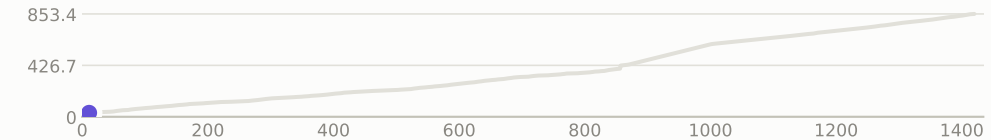
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GPW	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

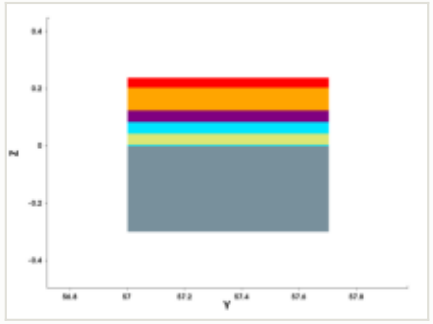
carbon so far, of the whole flow (kgCO2e)



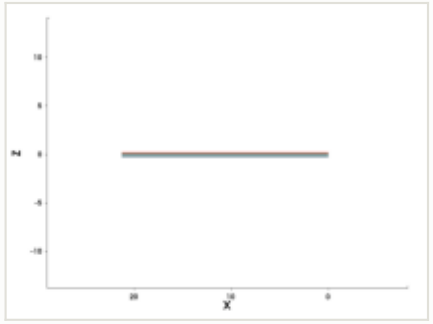
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



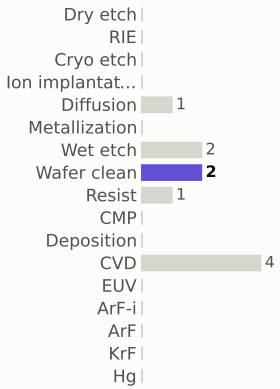
y cut at 57.351 µm



mask for this step

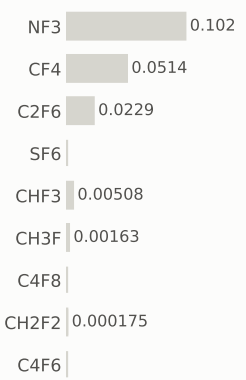
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



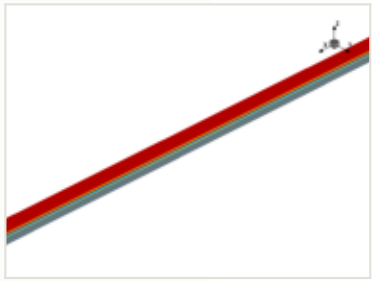
Step #140: Resist

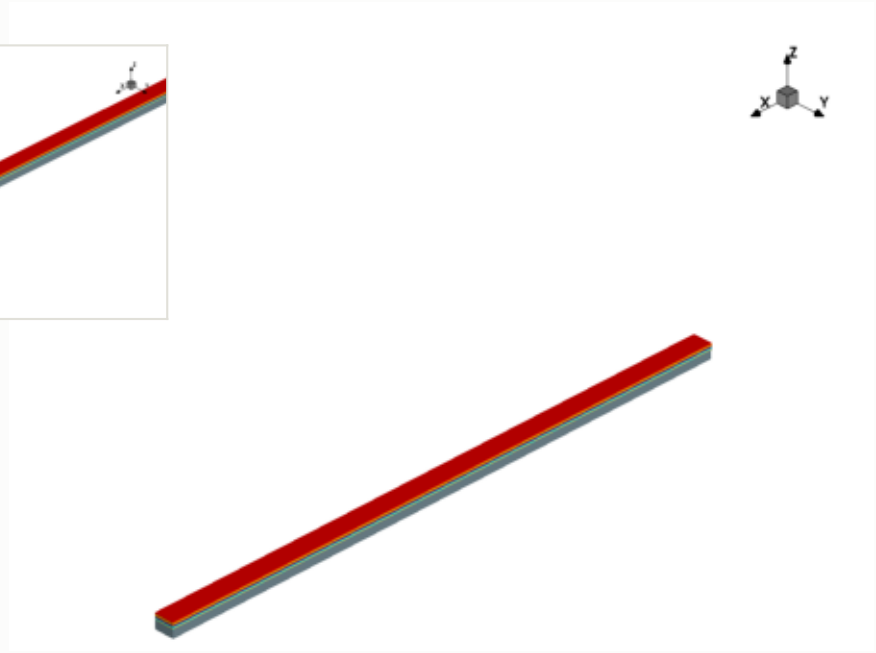
resist · Resist

0.169	33	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





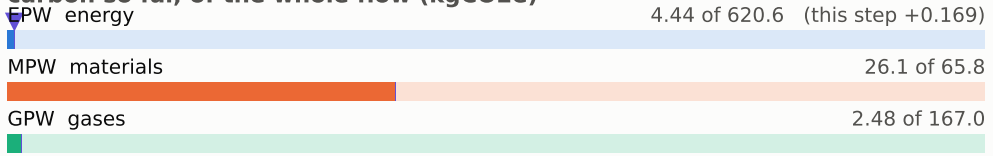
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

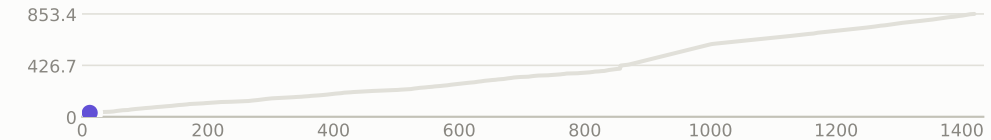
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

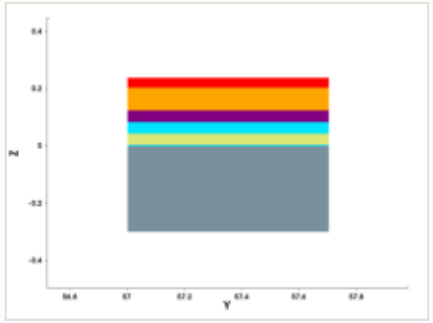
carbon so far, of the whole flow (kgCO2e)



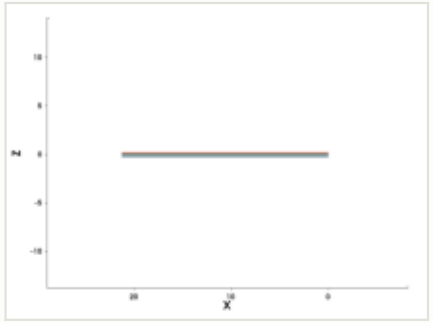
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



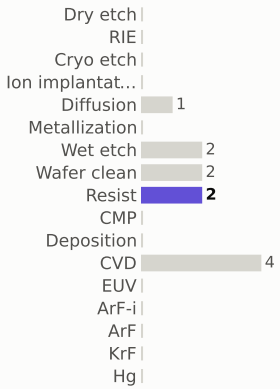
y cut at 57.351 µm



mask for this step

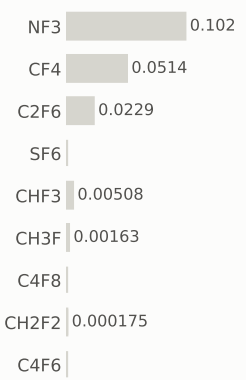
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



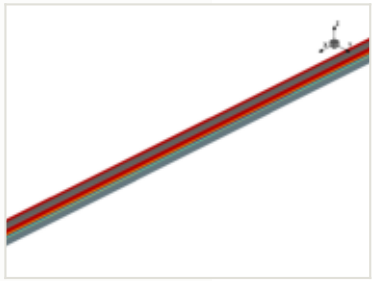
Step #150: ArF-i

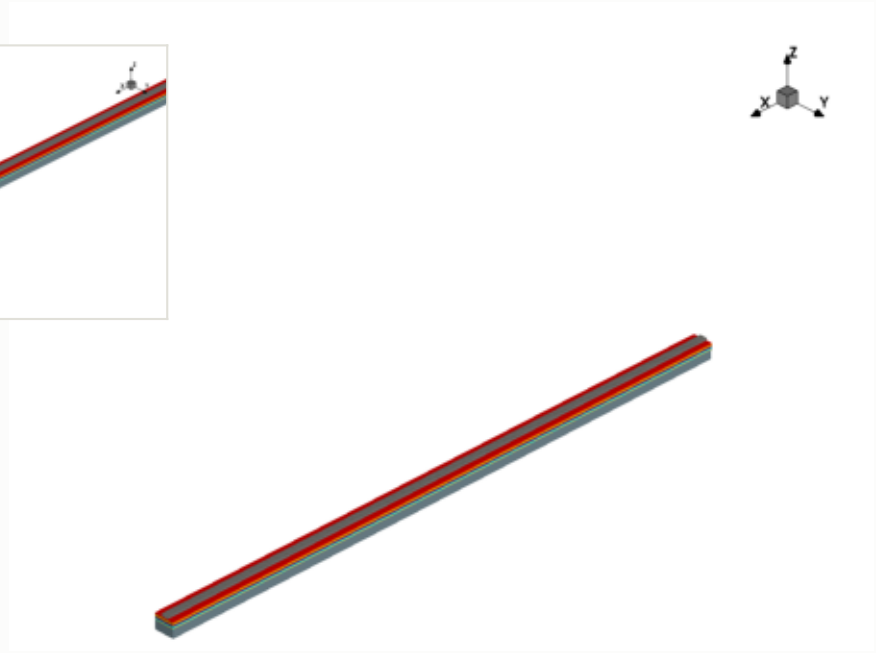
litho_patterning · ArF-i

0.217	33.3	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



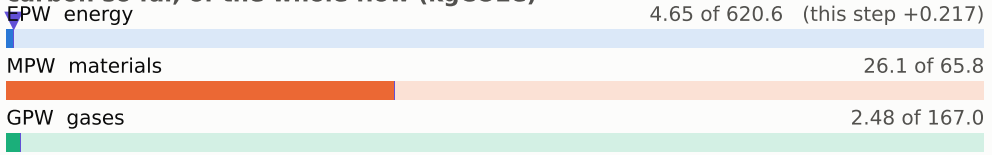


this step

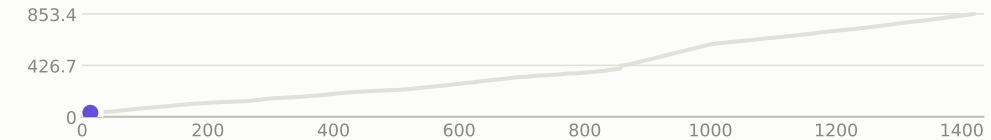
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

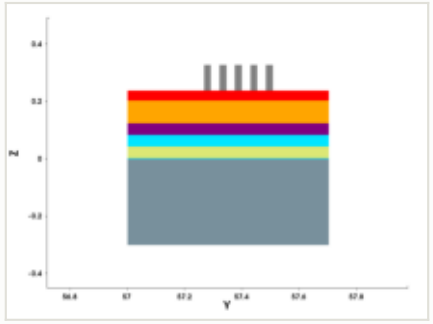
carbon so far, of the whole flow (kgCO2e)



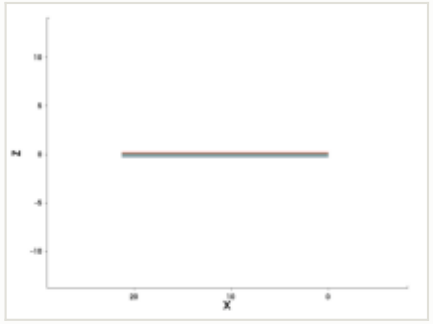
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

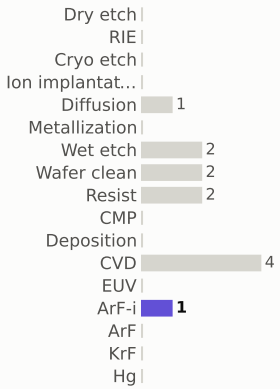


y cut at 57.351 µm



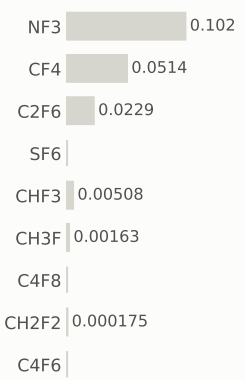
mask: layer 2401:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



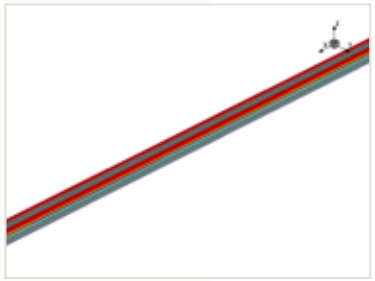
Step #160: Spx_only

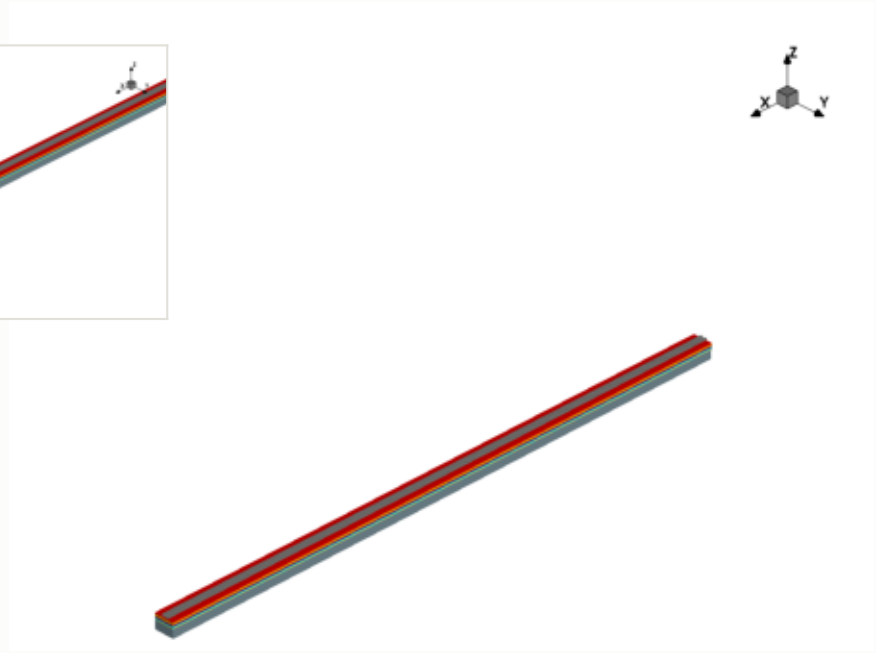
etch_isotropic · Spx_only

0	33.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





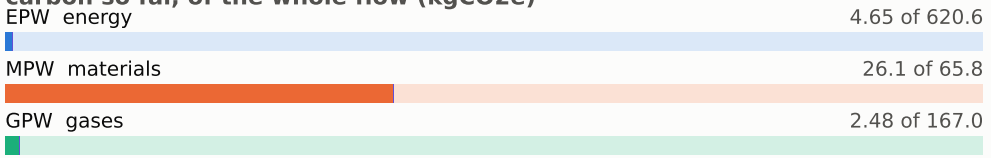
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

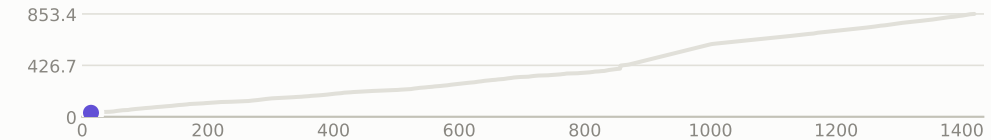
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

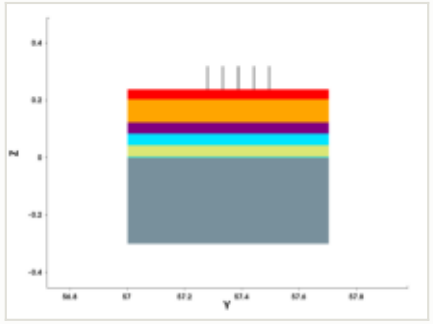
carbon so far, of the whole flow (kgCO2e)



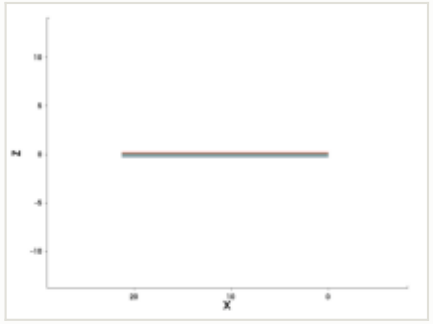
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



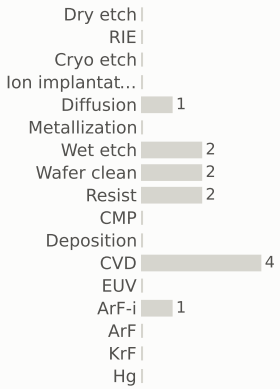
y cut at 57.351 µm



mask for this step

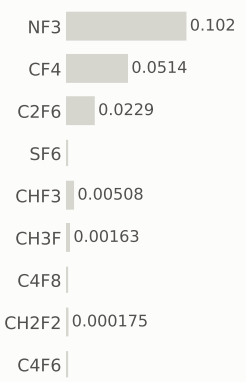
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



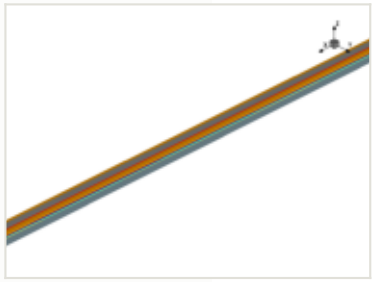
Step #170: Spx_only

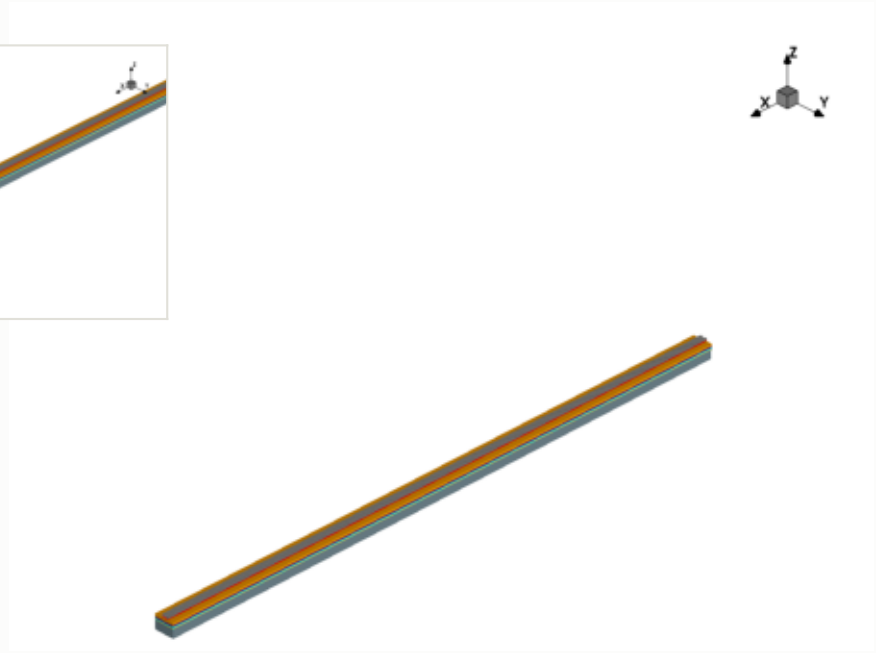
etch_anisotropic · Spx_only

0	33.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



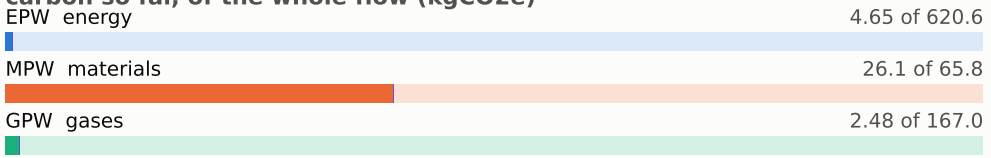


this step

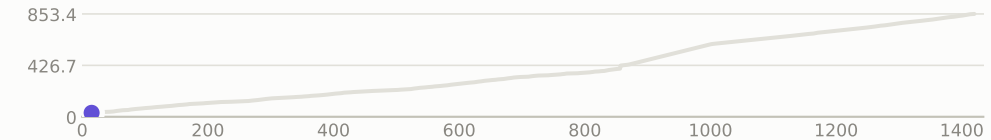
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

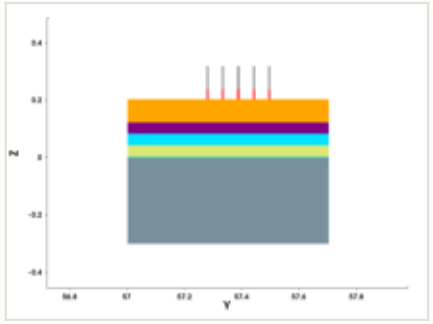
carbon so far, of the whole flow (kgCO2e)



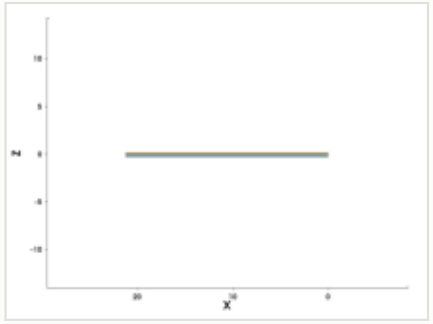
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



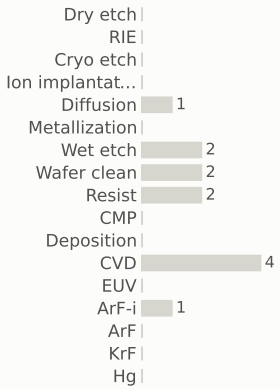
y cut at 57.351 µm



mask for this step

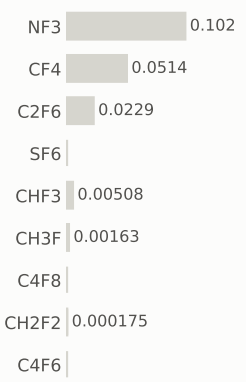
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



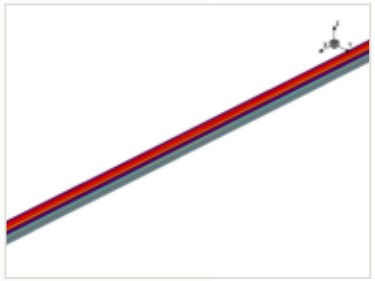
Step #180: RIE

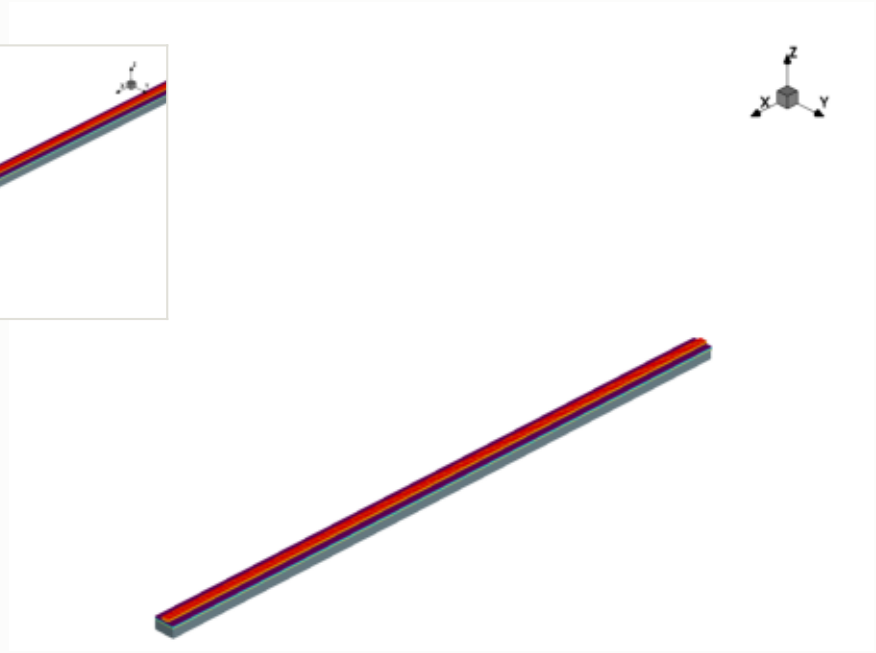
etch_anisotropic · RIE

1.04	34.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



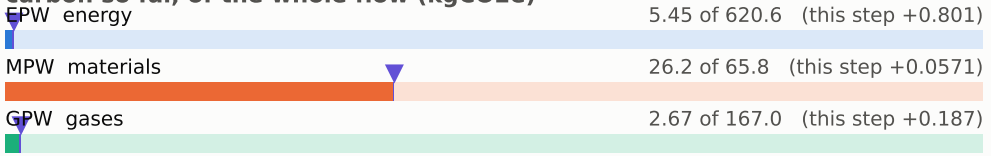


this step

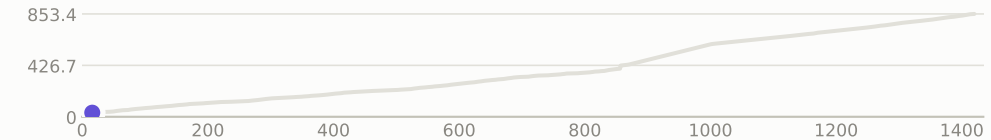
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

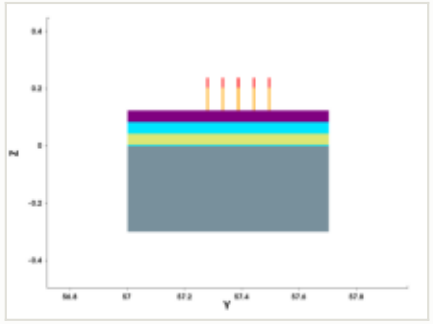
carbon so far, of the whole flow (kgCO2e)



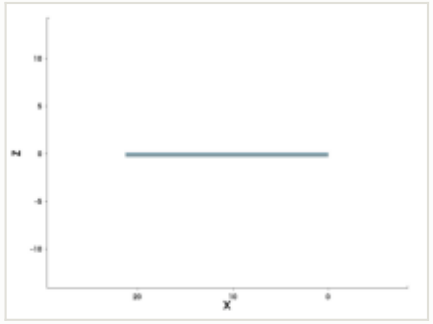
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



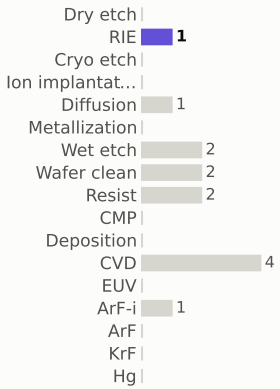
y cut at 57.351 µm



mask for this step

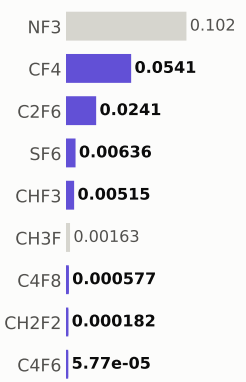
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



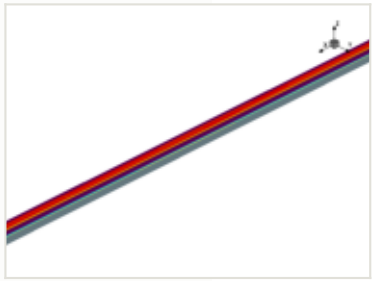
Step #190: Wafer_clean

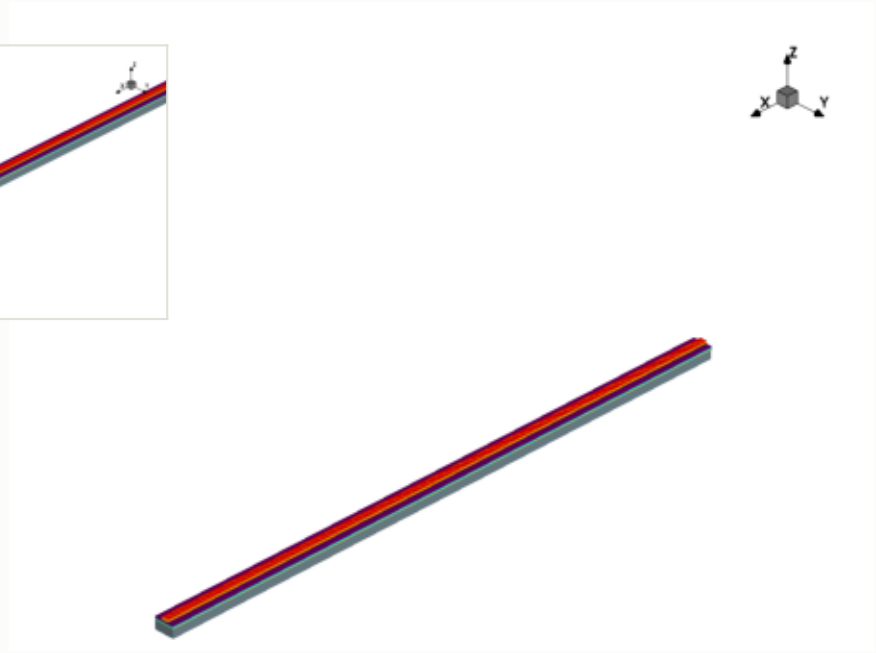
clean · Wafer_clean

0.169	34.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

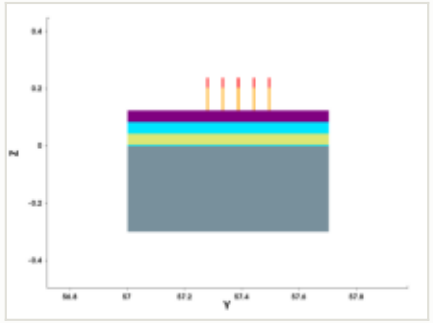
3D structure after this step

zoomed to fit

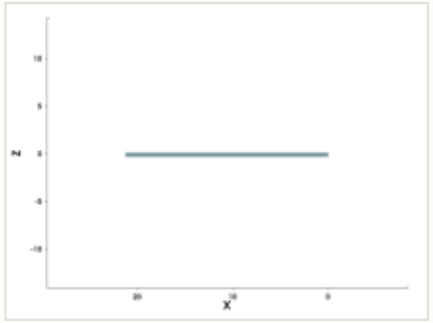




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

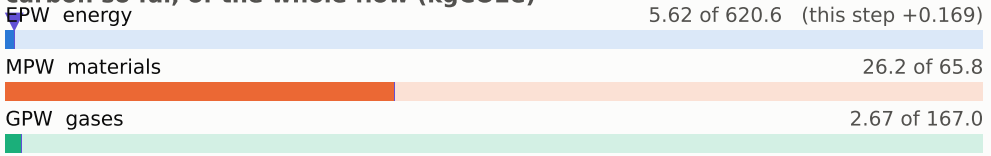
no mask for this step

this step

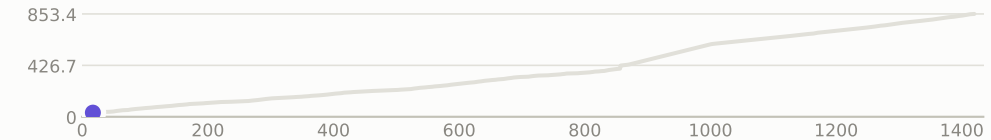
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GPW	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

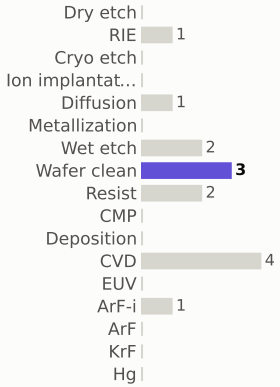
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

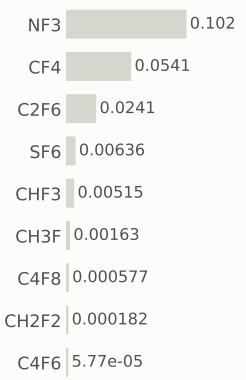


steps by process area



materials deposited (g)

gases emitted (g)



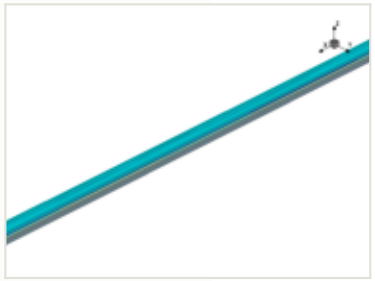
Step #200: Deposition

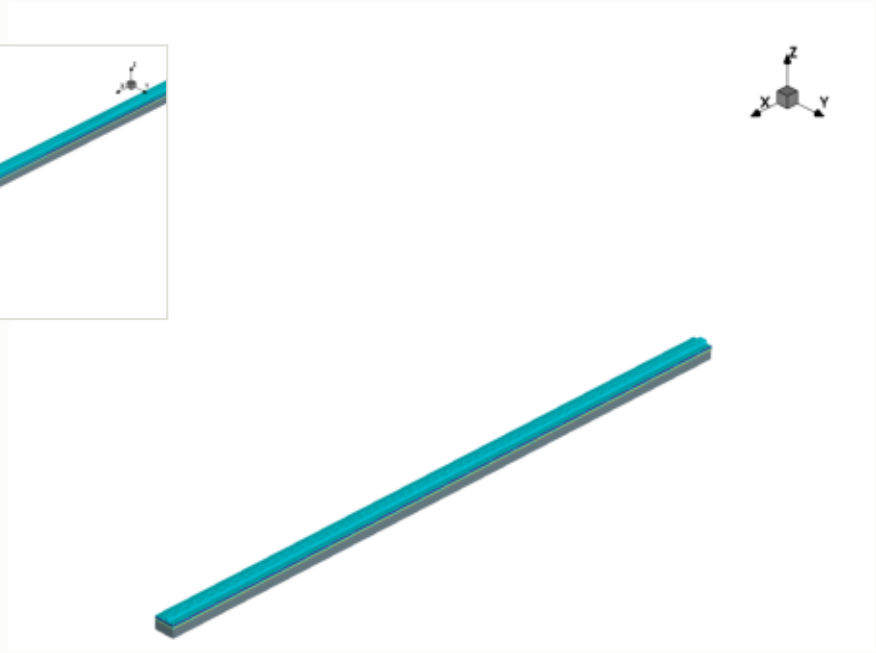
depo_isotropic · Deposition

0.656	35.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



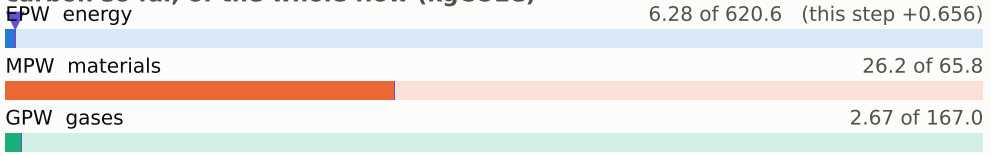


this step

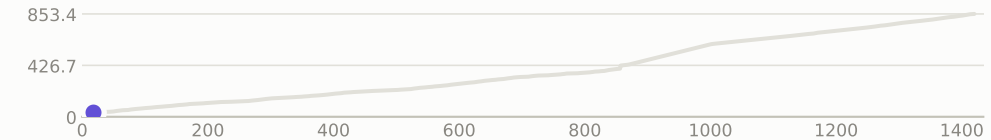
process area	Deposition
material	—
thickness	0.019 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

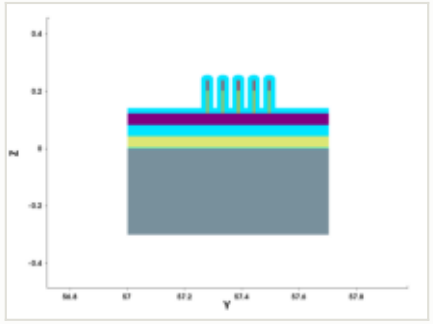
carbon so far, of the whole flow (kgCO2e)



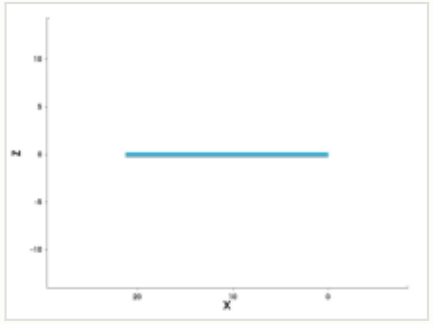
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



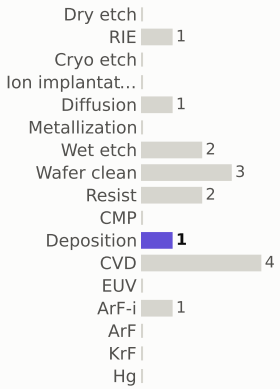
y cut at 57.351 µm



mask for this step

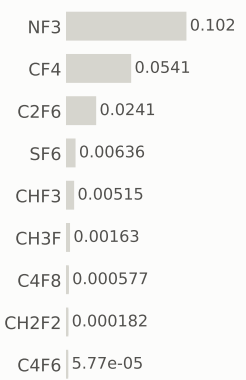
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



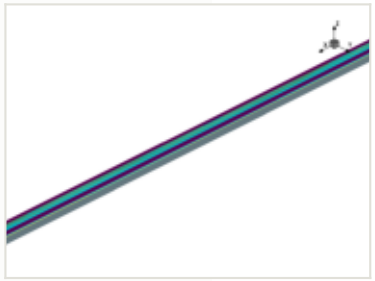
Step #210: RIE

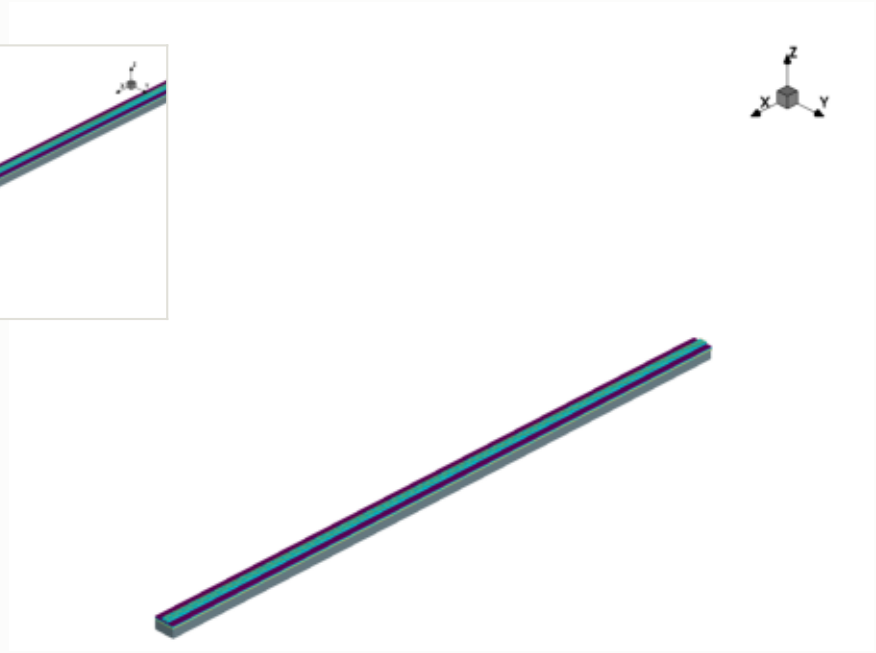
etch_ellipsoidal · RIE

1.04	36.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



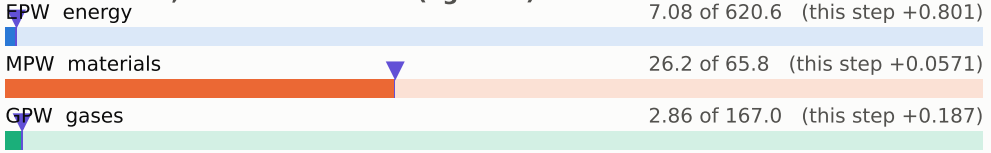


this step

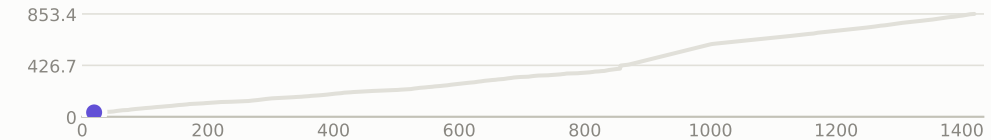
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

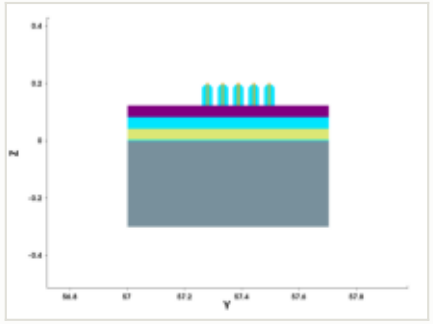
carbon so far, of the whole flow (kgCO2e)



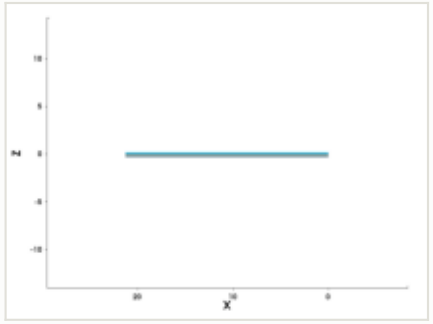
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



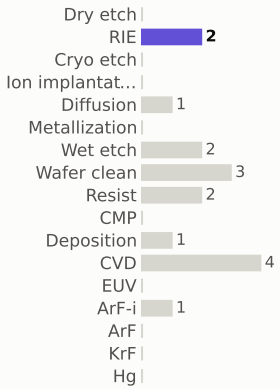
y cut at 57.351 µm



mask for this step

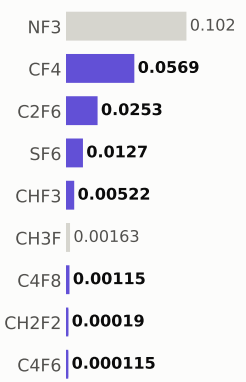
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



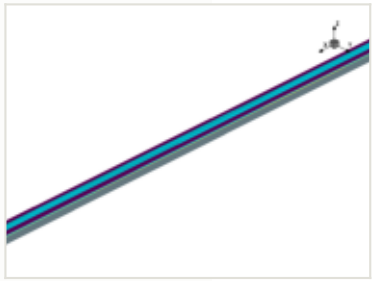
Step #220: Spx_only

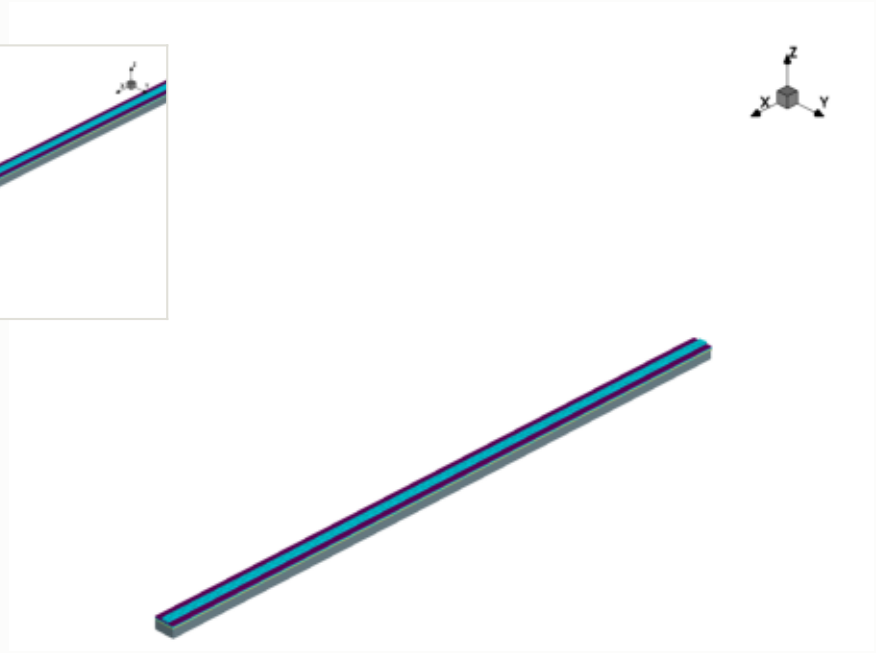
etch_isotropic · Spx_only

0	36.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





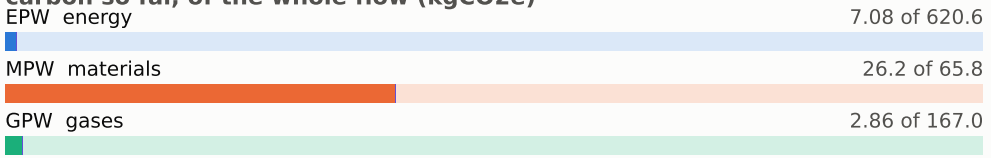
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

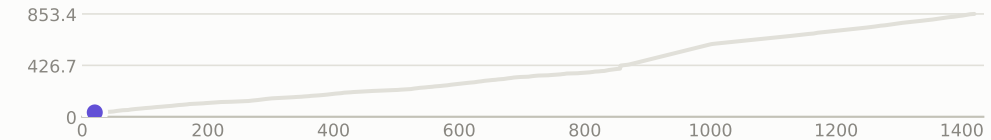
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

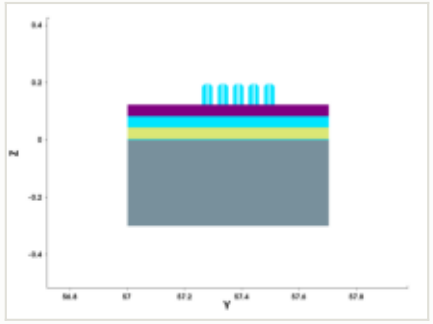
carbon so far, of the whole flow (kgCO2e)



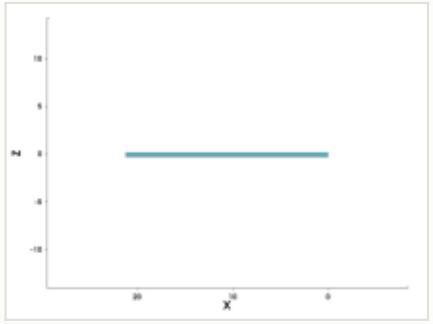
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



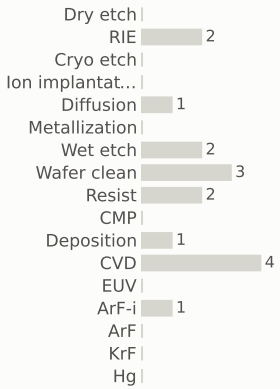
y cut at 57.351 µm



mask for this step

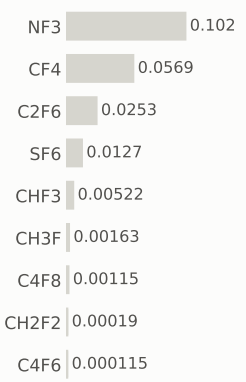
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



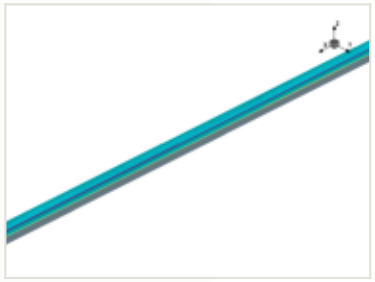
Step #230: Spx_only

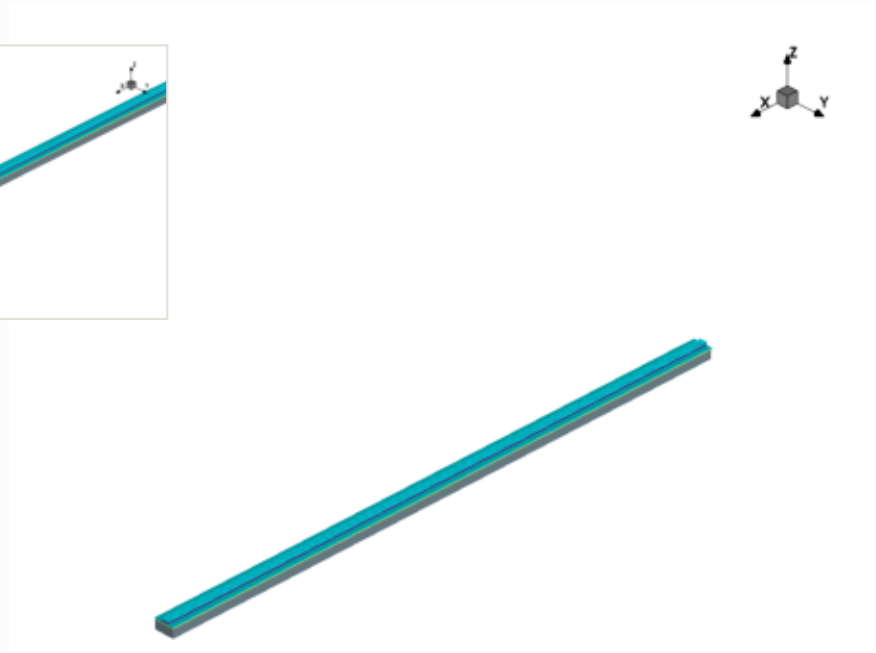
etch_anisotropic · Spx_only

0	36.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

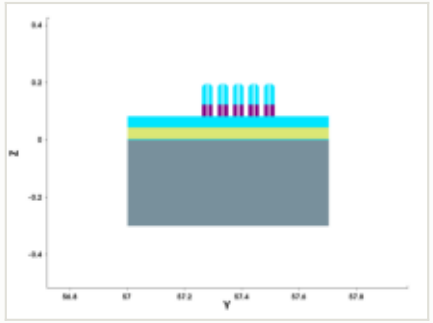
3D structure after this step

zoomed to fit

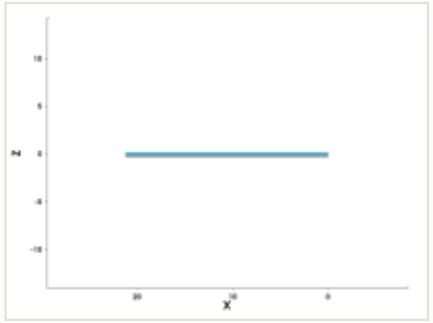




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

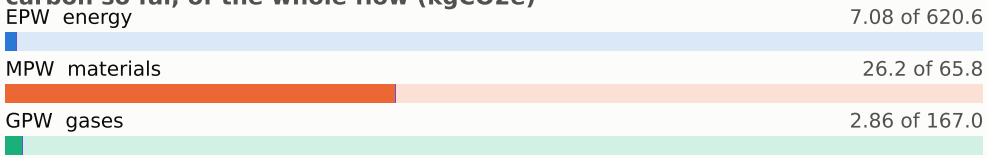
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

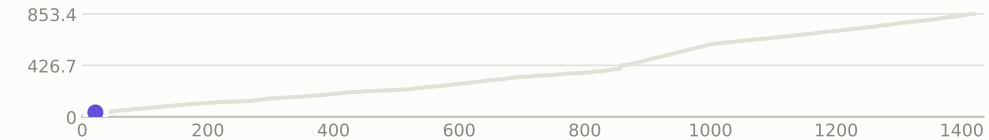
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

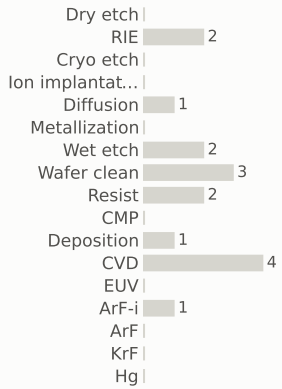
carbon so far, of the whole flow (kgCO2e)



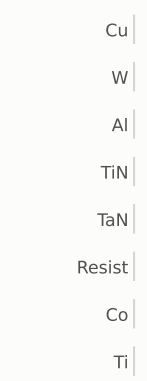
embodied carbon up to each step (kgCO2e per wafer)



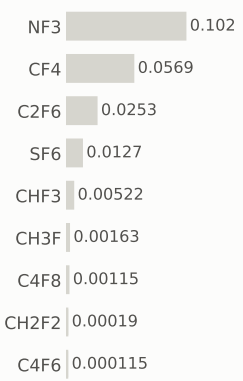
steps by process area



materials deposited (g)



gases emitted (g)



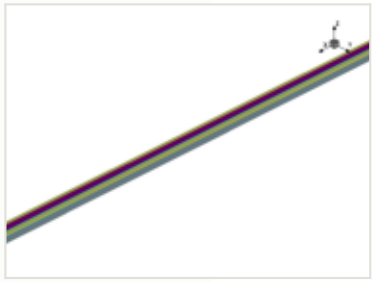
Step #240: Spx_only

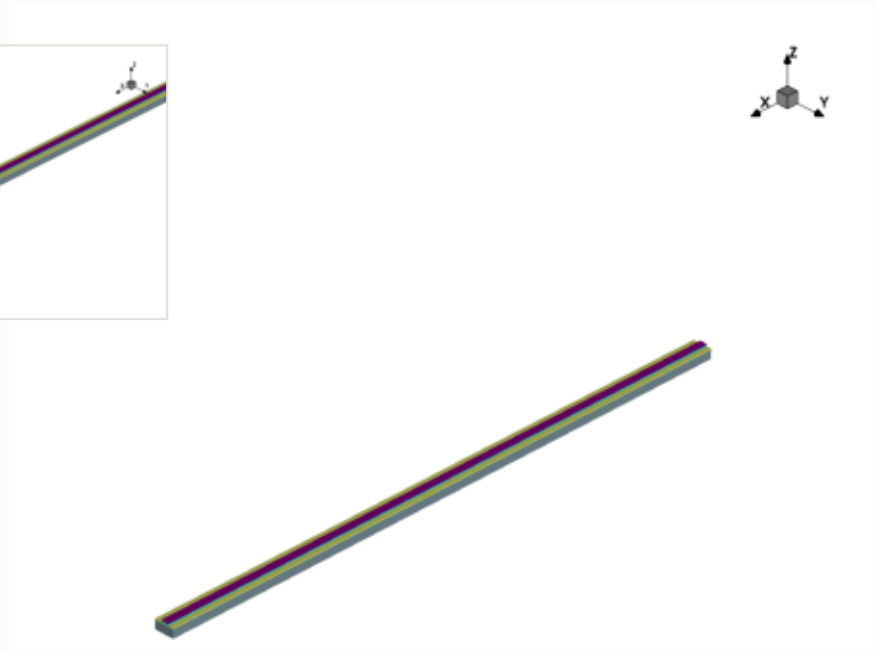
etch_anisotropic · Spx_only

0	36.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



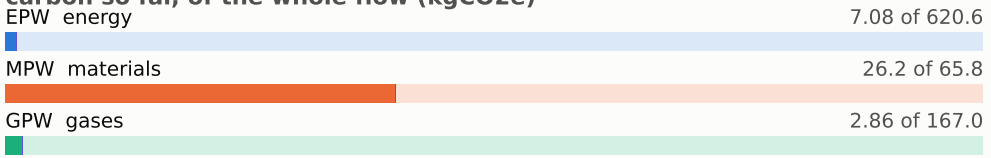


this step

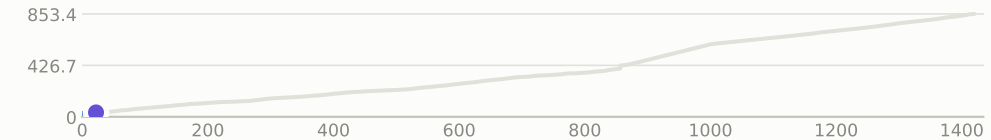
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

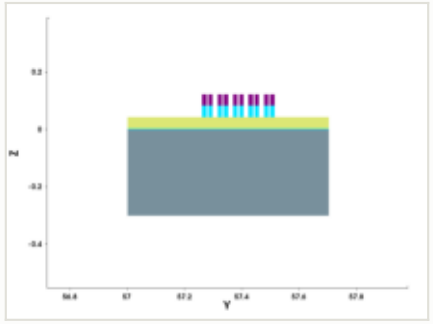
carbon so far, of the whole flow (kgCO2e)



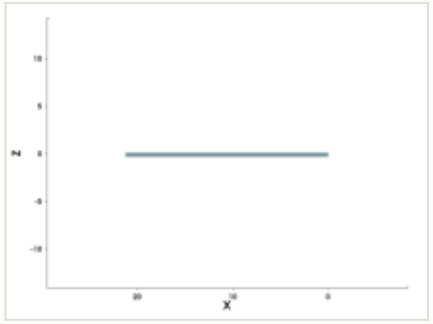
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



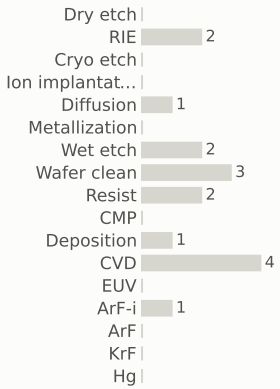
y cut at 57.351 µm



mask for this step

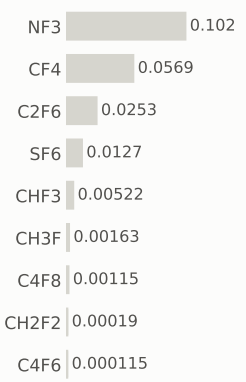
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



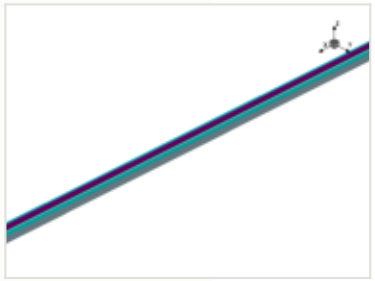
Step #250: Spx_only

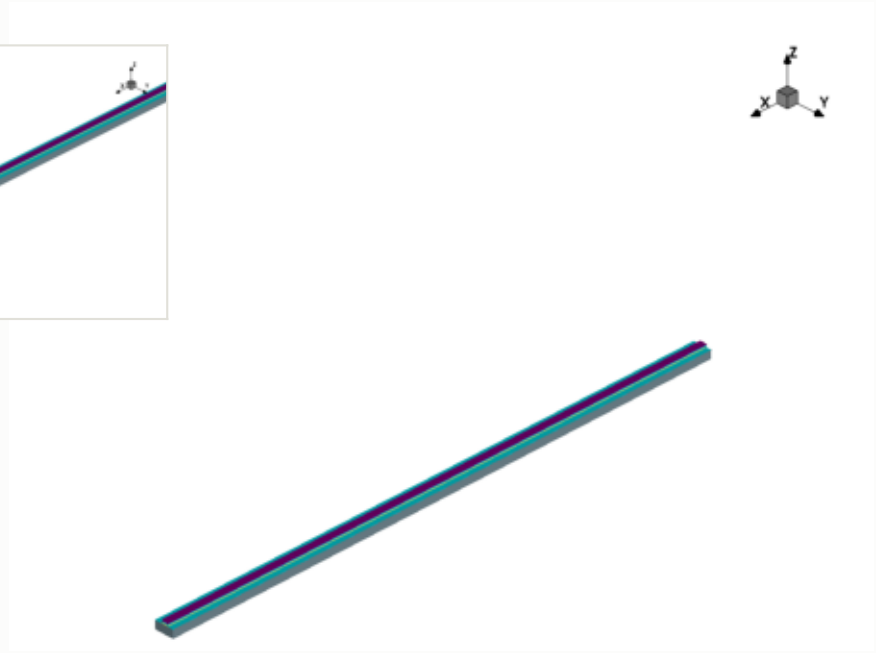
etch_anisotropic · Spx_only

0	36.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



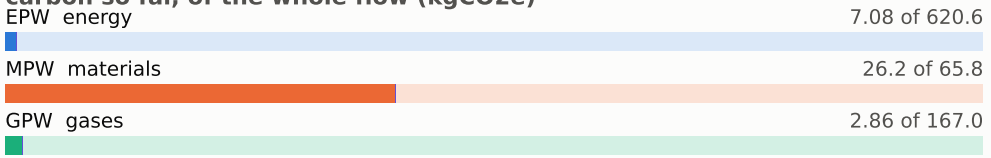


this step

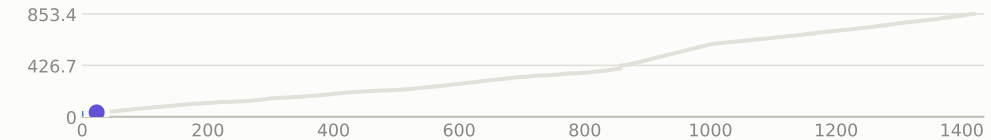
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

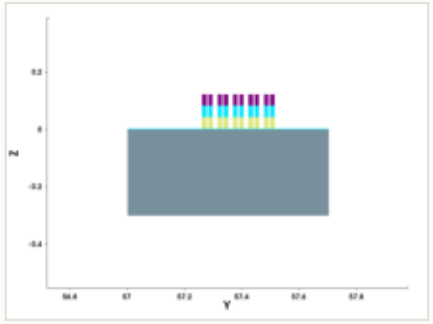
carbon so far, of the whole flow (kgCO2e)



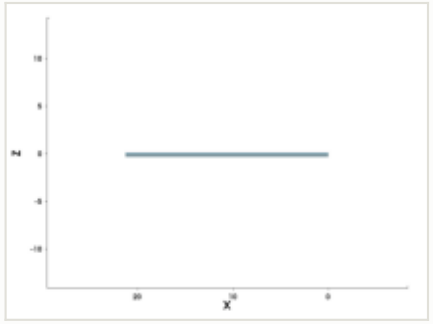
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



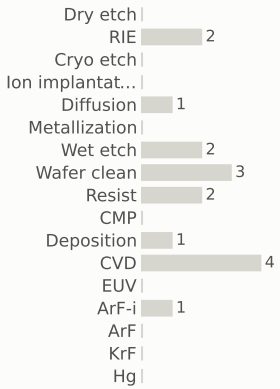
y cut at 57.351 µm



mask for this step

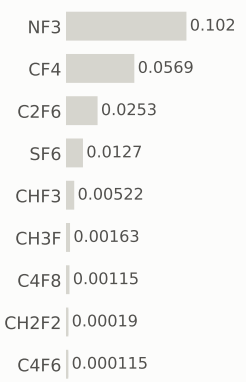
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



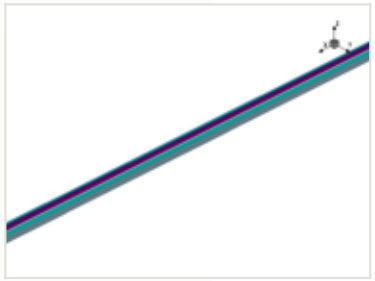
Step #260: Wafer_clean

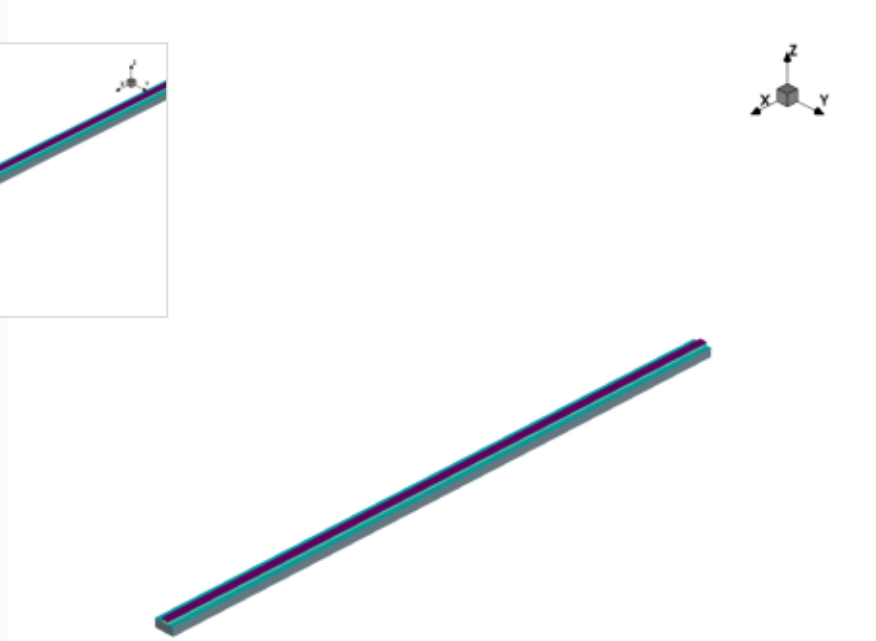
clean · Wafer_clean

0.169	36.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



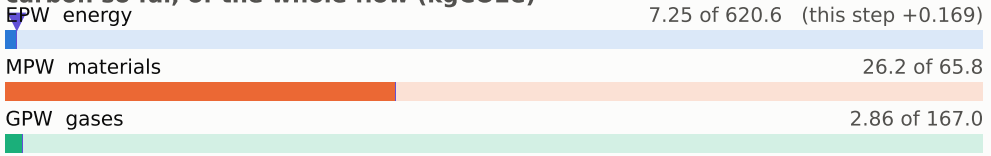


this step

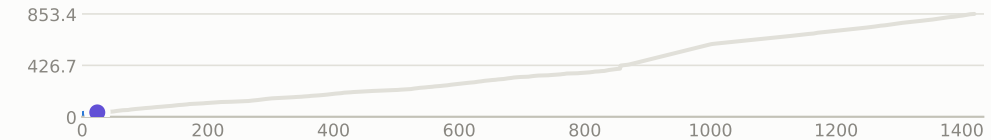
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

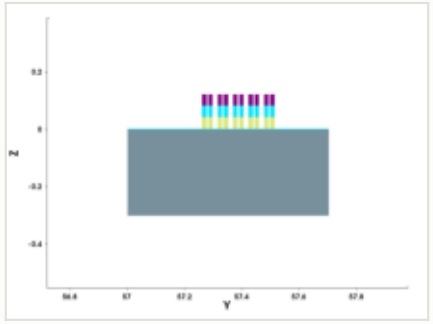
carbon so far, of the whole flow (kgCO2e)



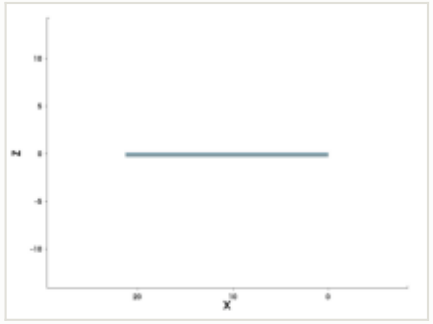
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



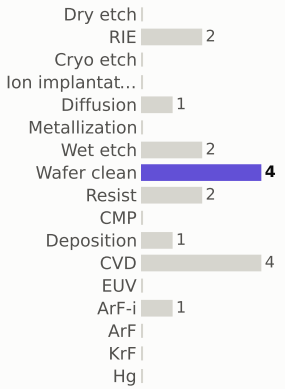
y cut at 57.351 µm



mask for this step

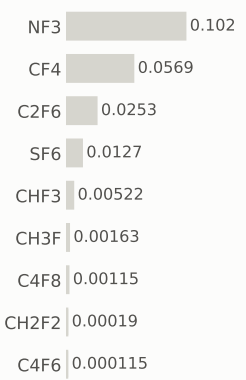
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



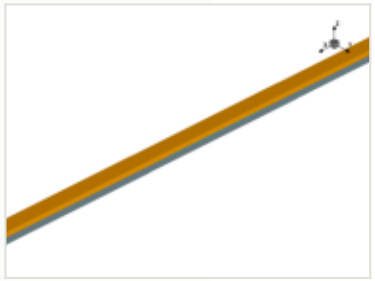
Step #270: Resist

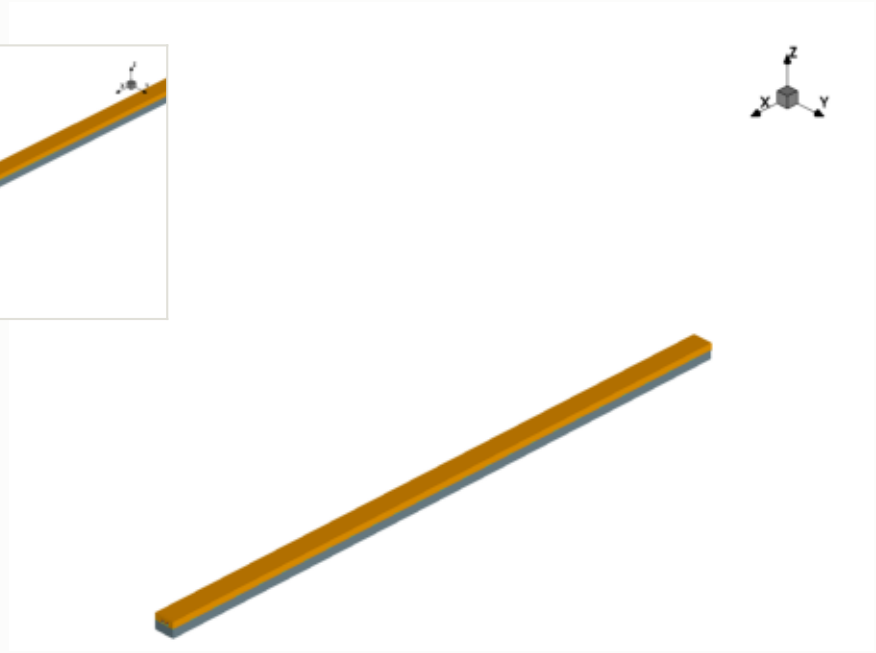
depo_fill · Resist

0.169	36.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



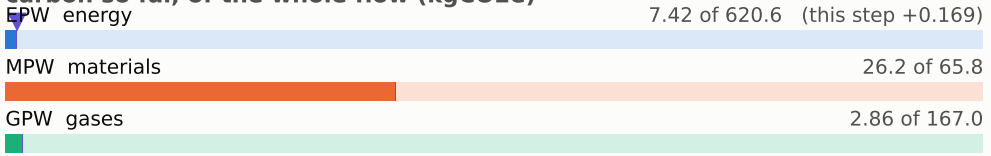


this step

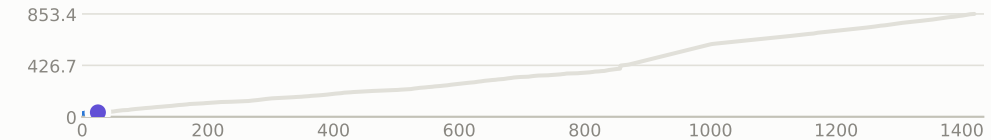
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

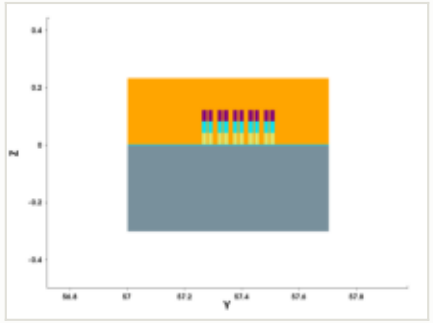
carbon so far, of the whole flow (kgCO2e)



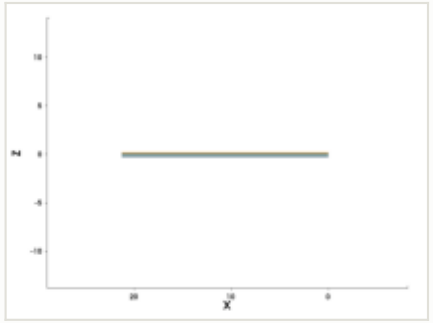
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



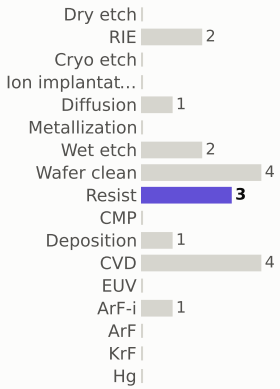
y cut at 57.351 µm



mask for this step

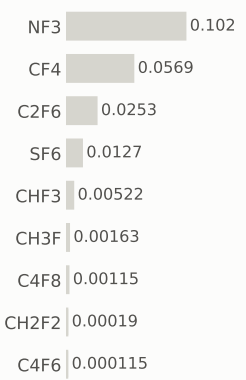
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



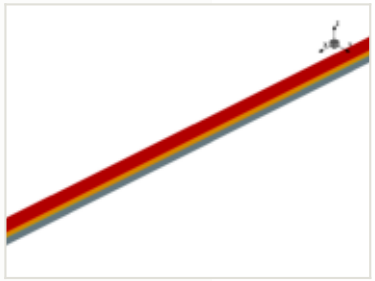
Step #280: CVD

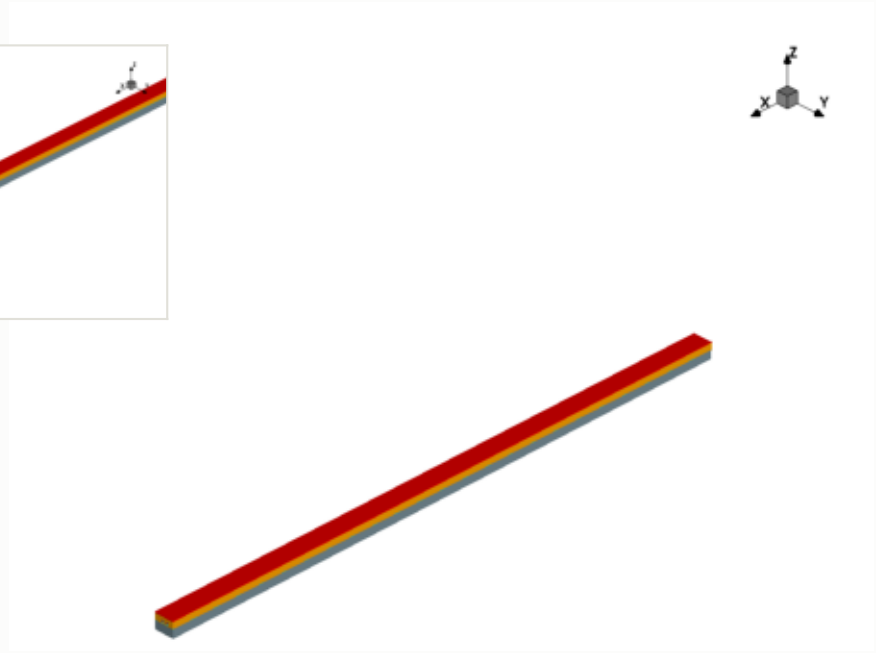
depo_isotropic · CVD

1.31	37.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



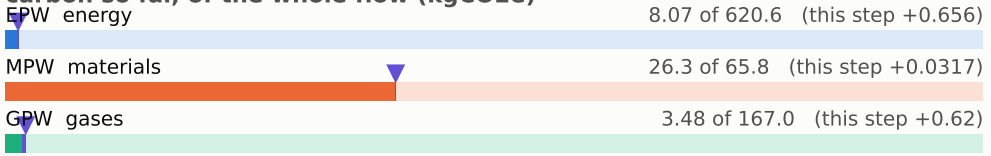


this step

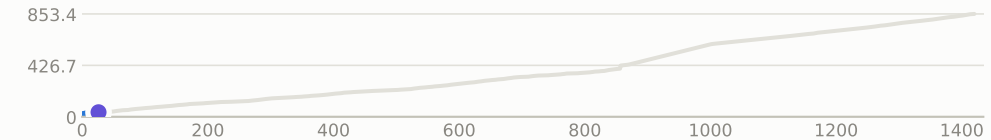
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

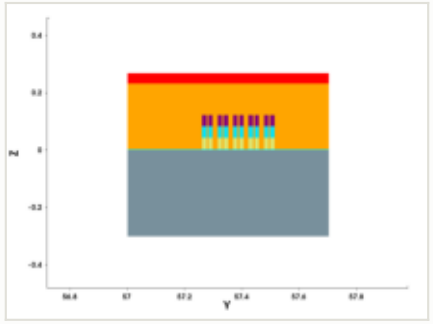
carbon so far, of the whole flow (kgCO2e)



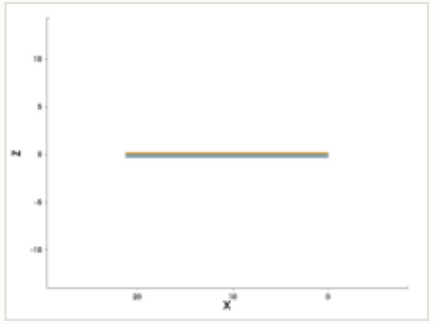
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



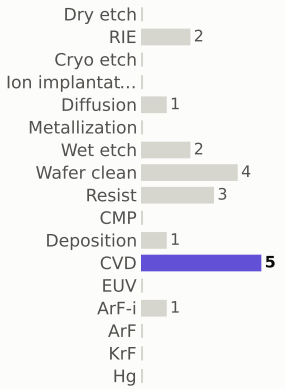
y cut at 57.351 µm



mask for this step

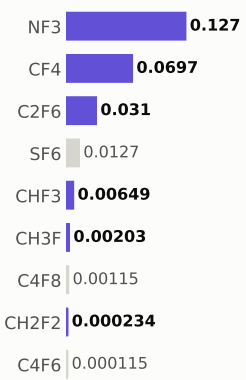
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



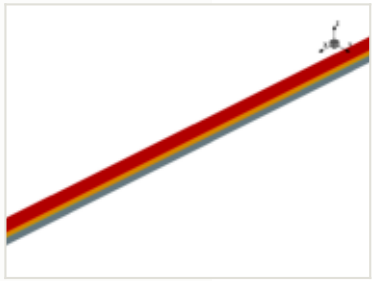
Step #290: Wafer_clean

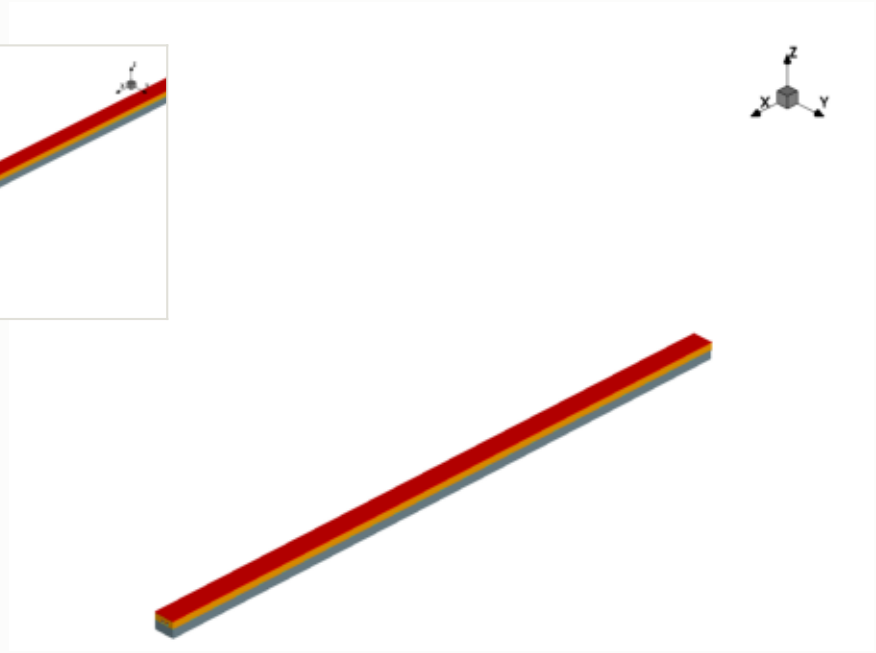
clean · Wafer_clean

0.169	38	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

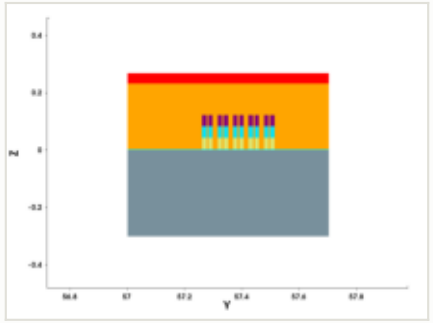
3D structure after this step

zoomed to fit

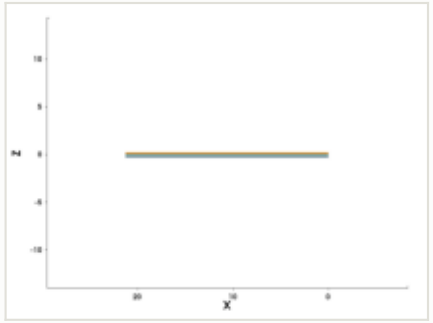




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

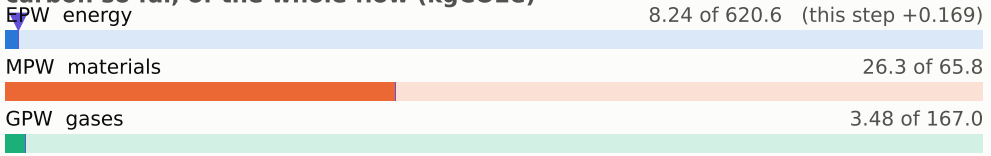
no mask for this step

this step

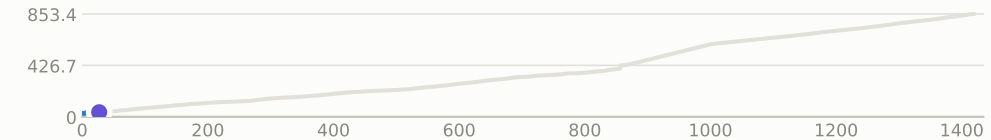
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

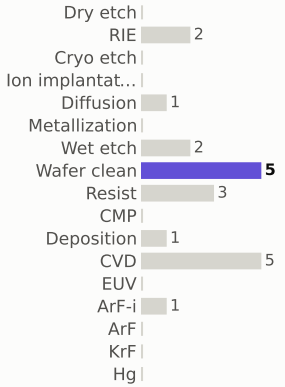
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

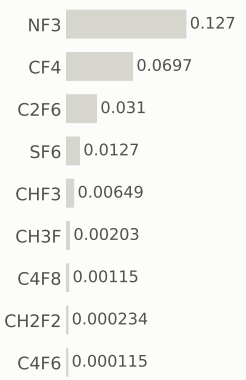


steps by process area



materials deposited (g)

gases emitted (g)



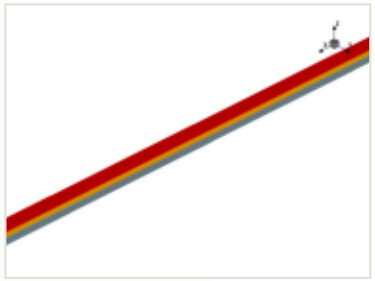
Step #300: Resist

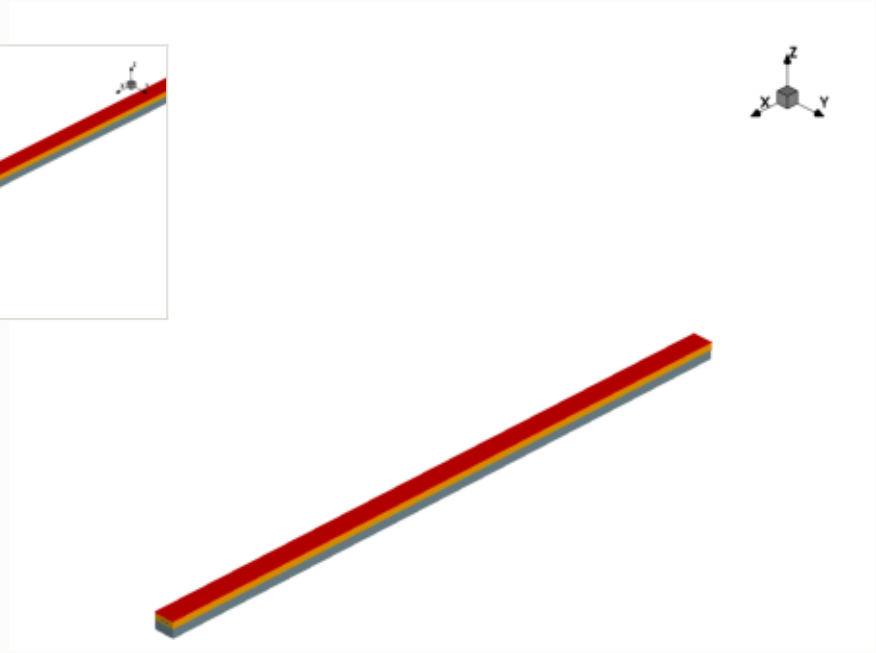
resist · Resist

0.169	38.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





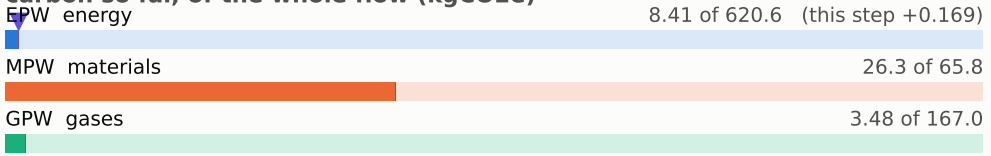
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

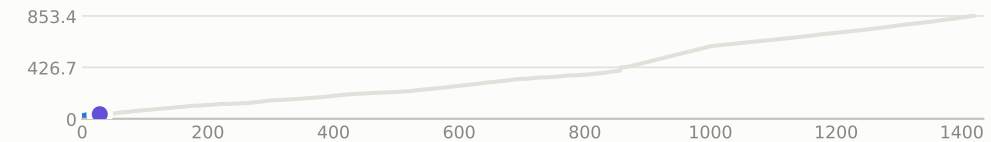
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

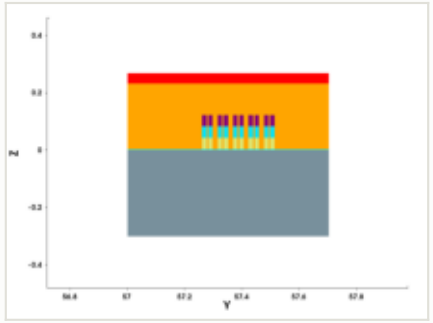
carbon so far, of the whole flow (kgCO2e)



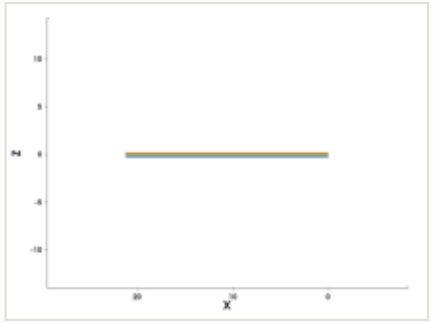
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



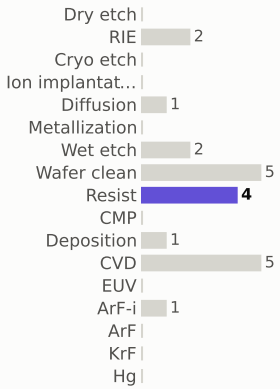
y cut at 57.351 µm



mask for this step

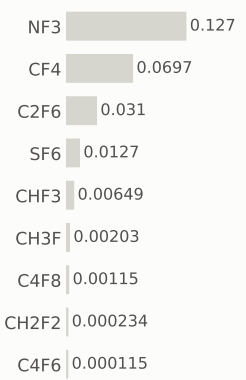
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



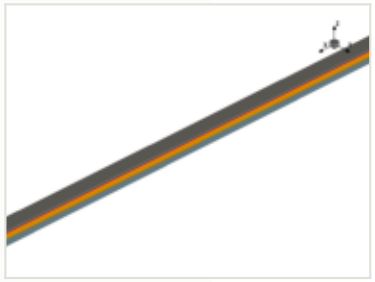
Step #310: ArF-i

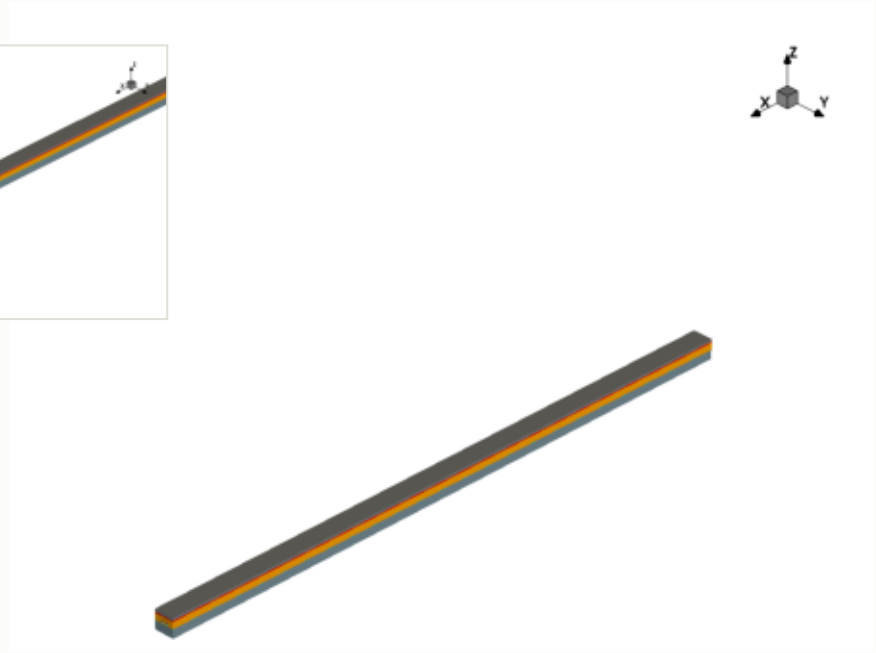
litho_patterning · ArF-i

0.217	38.4	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



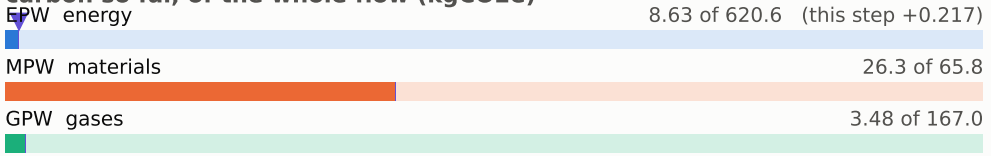


this step

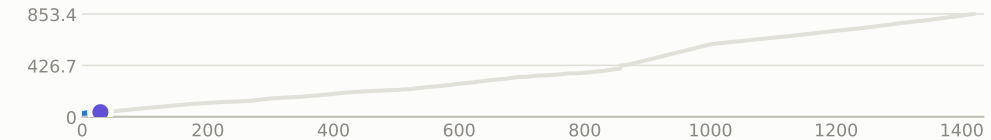
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

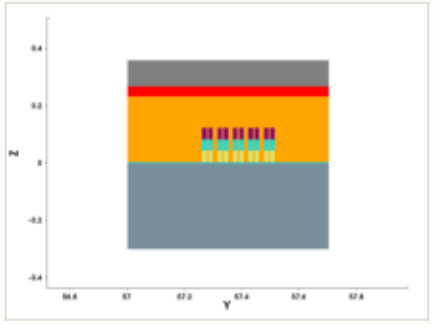
carbon so far, of the whole flow (kgCO2e)



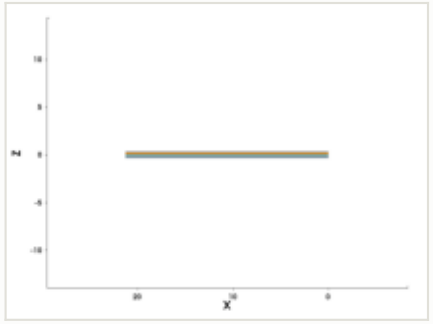
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



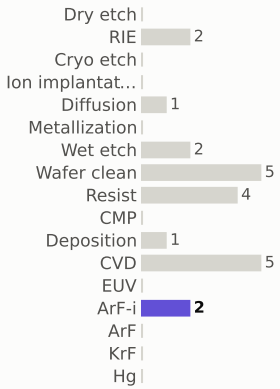
y cut at 57.351 µm



mask: layer 2039:0 (GDSII)

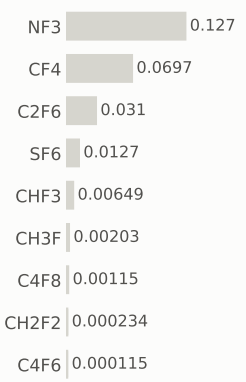
nothing in the clip window

steps by process area



materials deposited (g)

gases emitted (g)



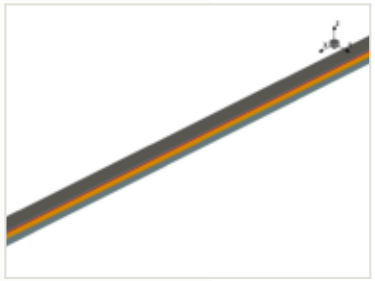
Step #320: RIE

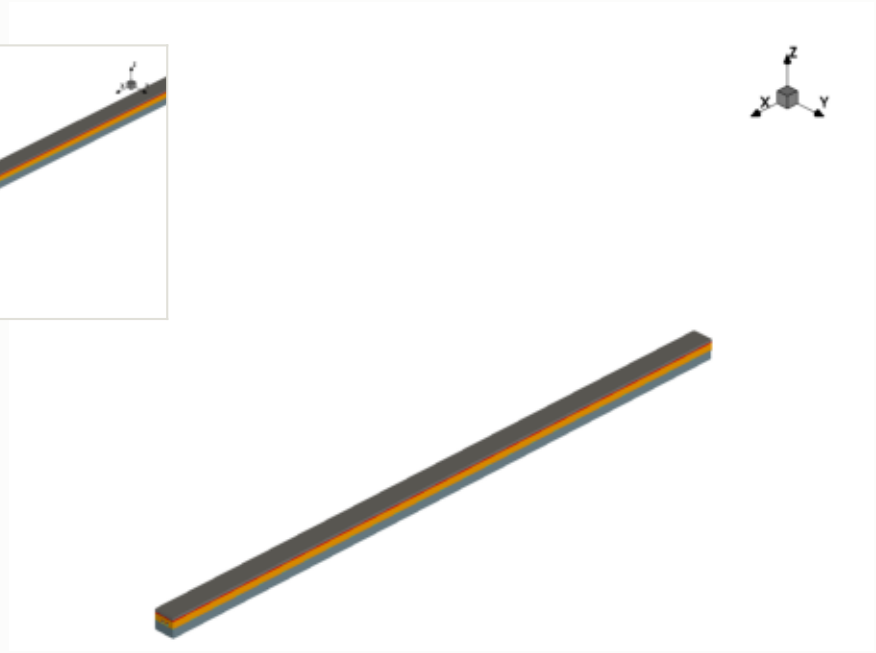
etch_anisotropic · RIE

1.04	39.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



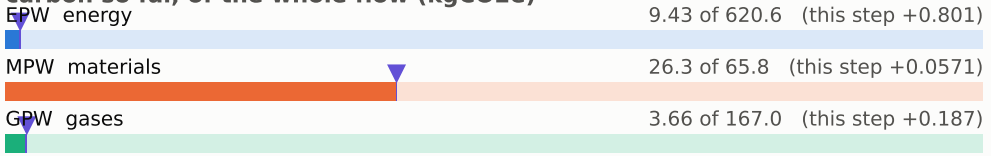


this step

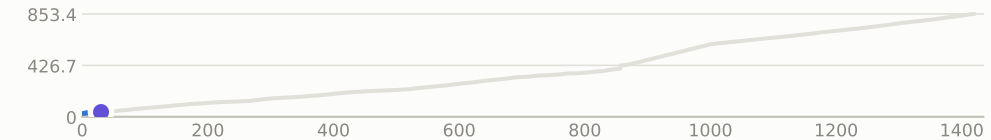
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

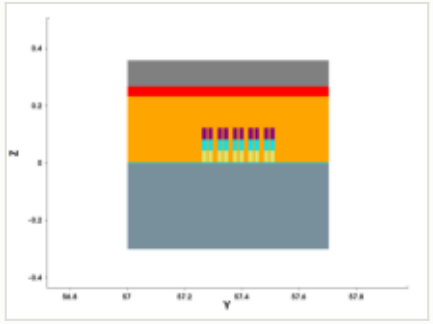
carbon so far, of the whole flow (kgCO2e)



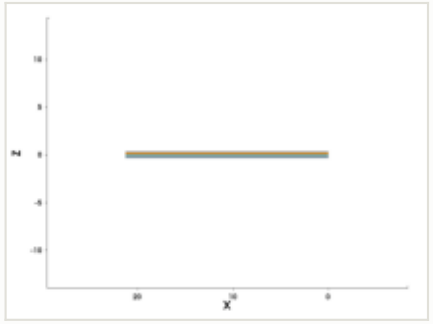
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



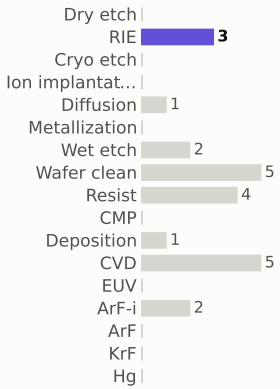
y cut at 57.351 µm



mask for this step

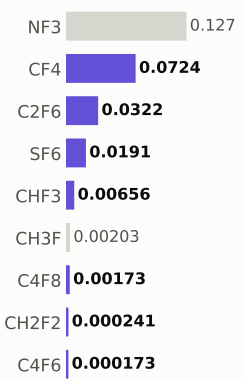
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



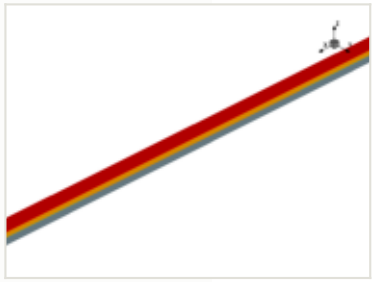
Step #330: Spx_only

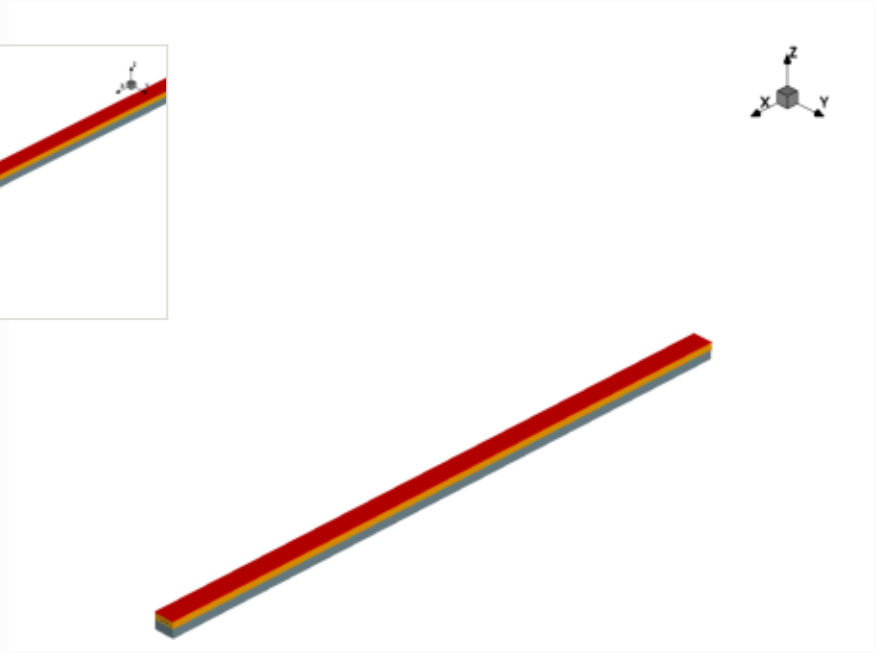
etch_anisotropic · Spx_only

0	39.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





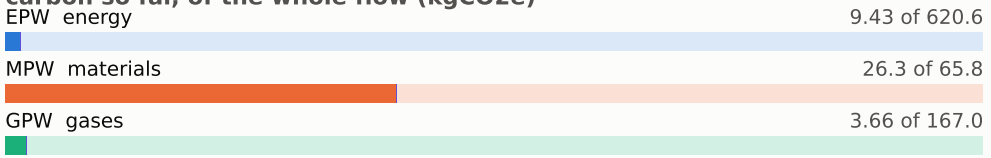
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

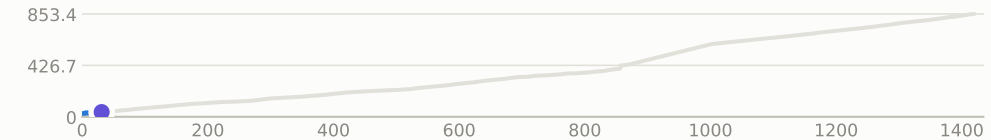
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

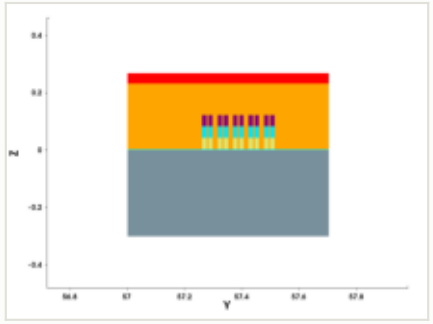
carbon so far, of the whole flow (kgCO2e)



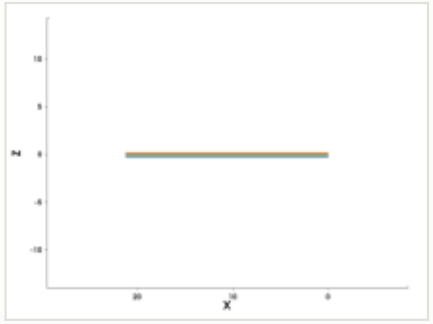
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



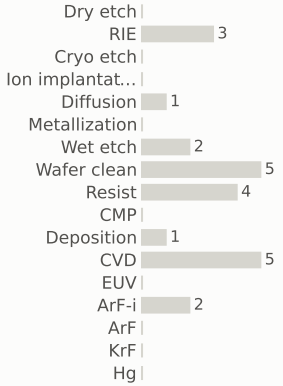
y cut at 57.351 µm



mask for this step

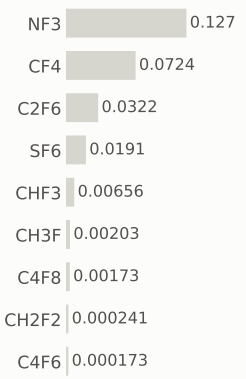
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



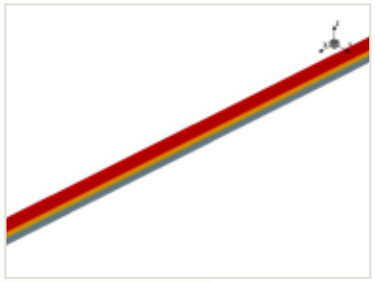
Step #340: Spx_only

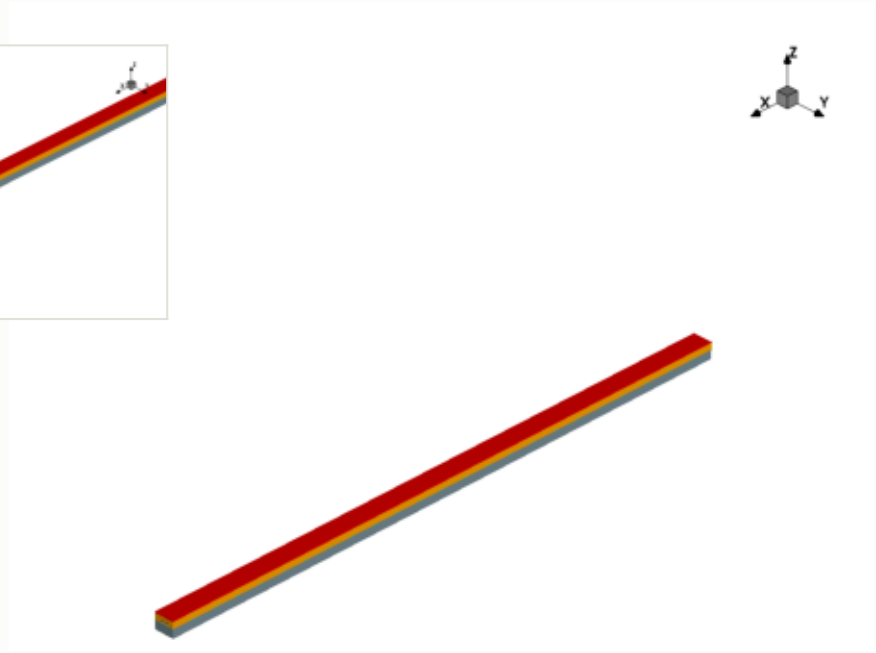
etch_anisotropic · Spx_only

0	39.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

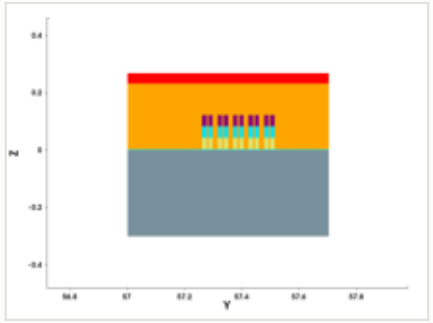
3D structure after this step

zoomed to fit

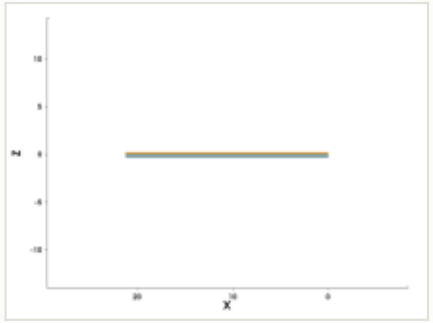




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

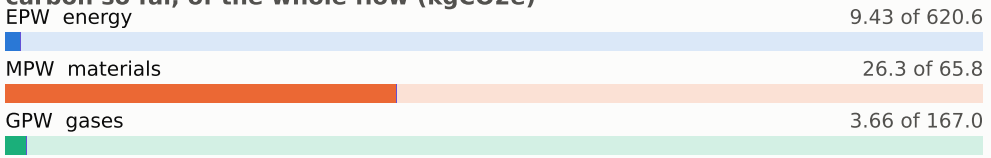
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

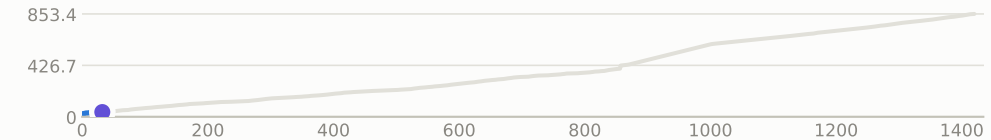
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

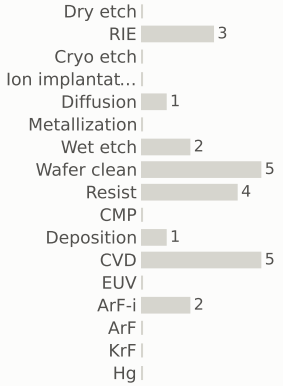
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

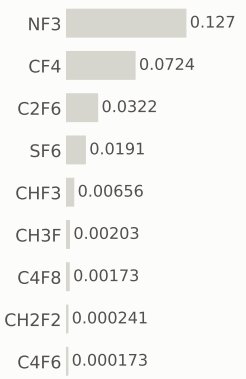


steps by process area



materials deposited (g)

gases emitted (g)



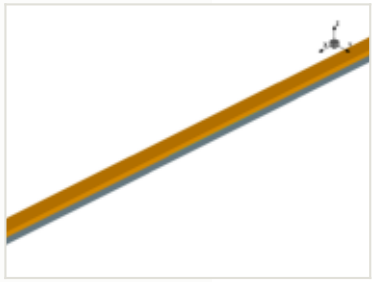
Step #350: Spx_only

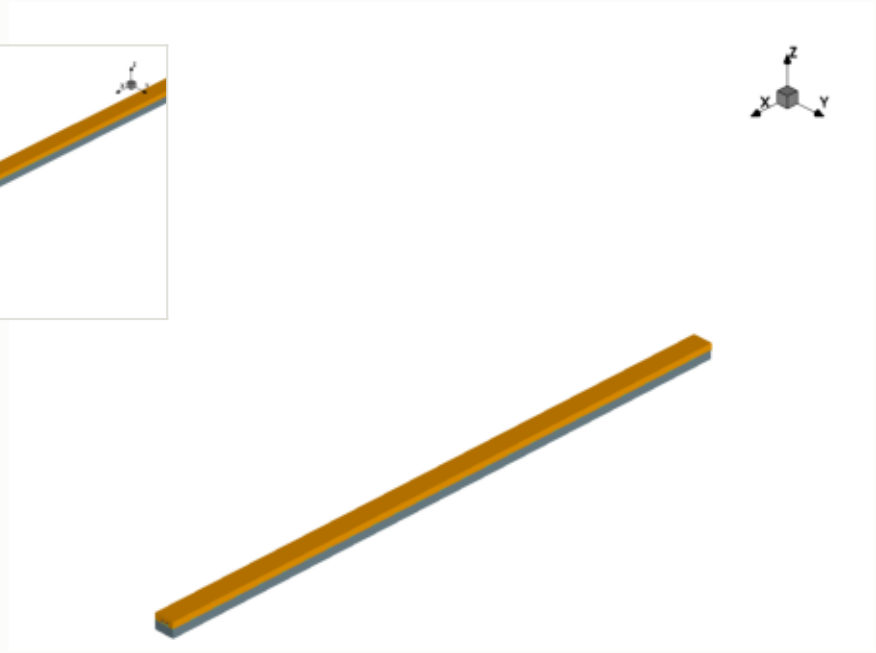
etch_anisotropic · Spx_only

0	39.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

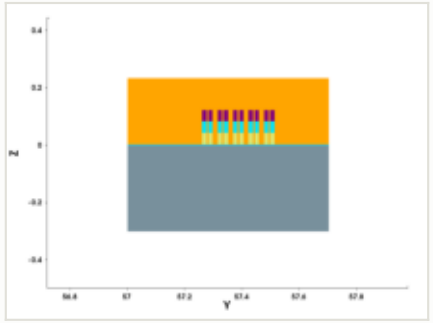
3D structure after this step

zoomed to fit

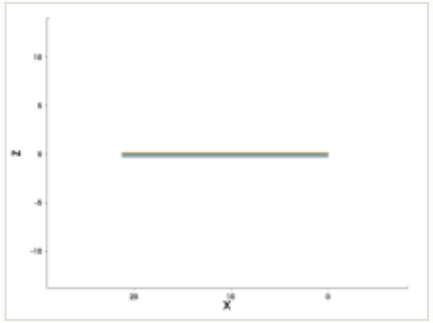




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

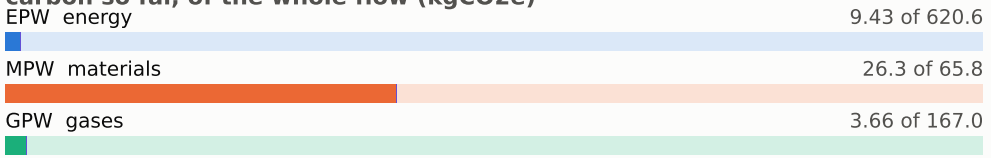
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

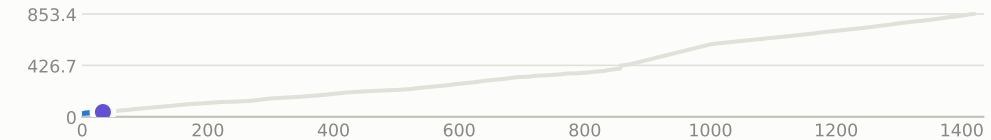
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

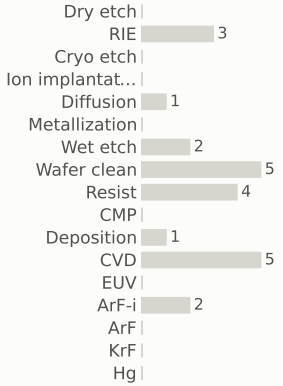
carbon so far, of the whole flow (kgCO2e)



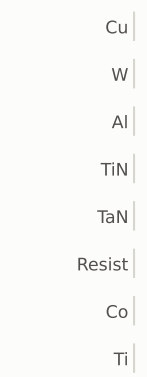
embodied carbon up to each step (kgCO2e per wafer)



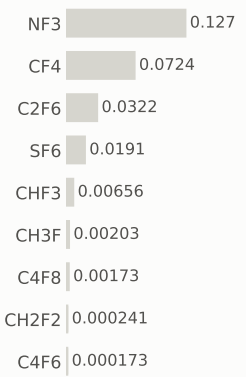
steps by process area



materials deposited (g)



gases emitted (g)



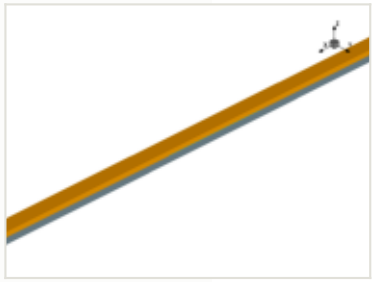
Step #360: Spx_only

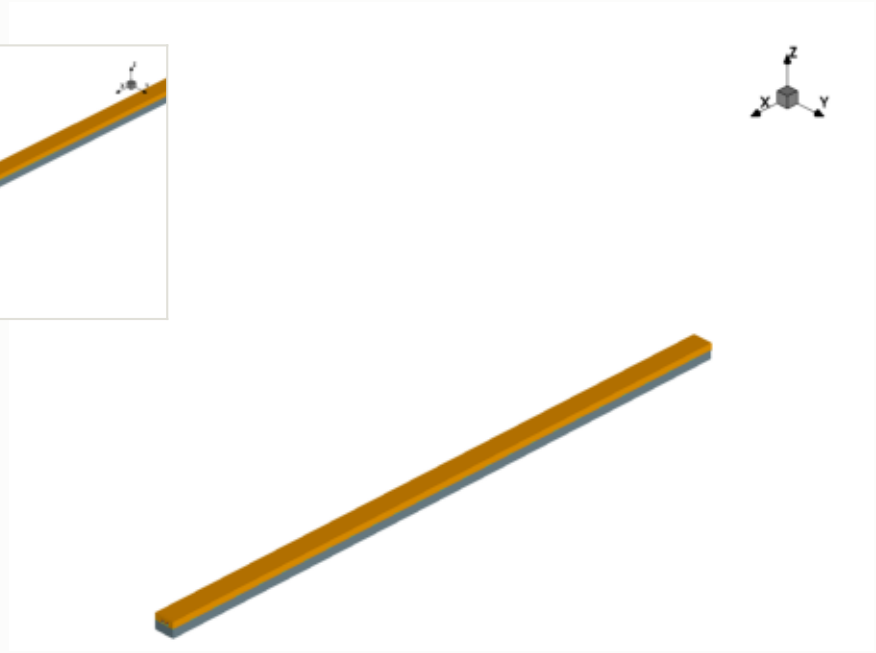
etch_anisotropic · Spx_only

0	39.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





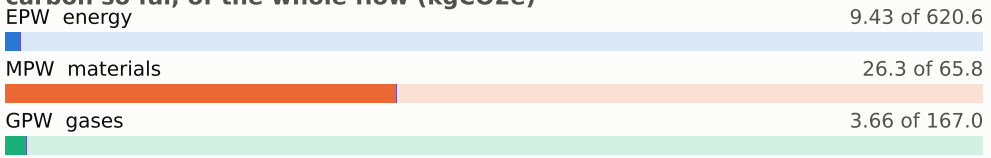
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

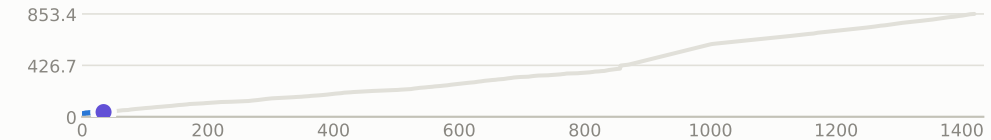
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

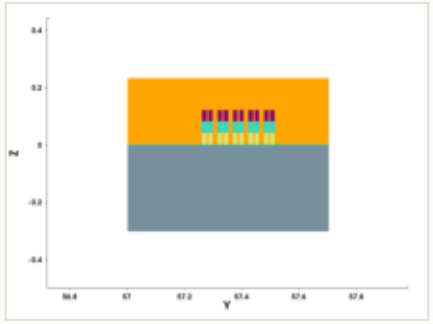
carbon so far, of the whole flow (kgCO2e)



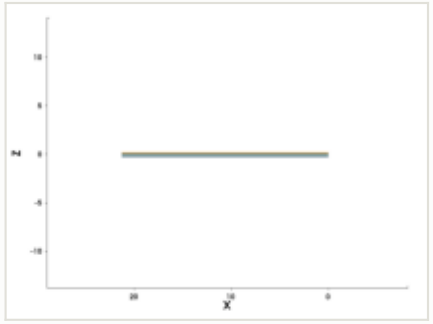
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



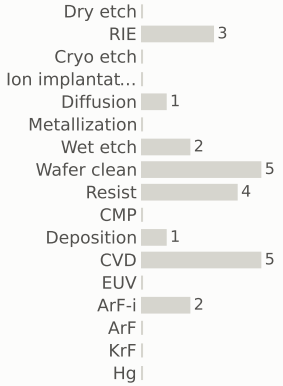
y cut at 57.351 µm



mask for this step

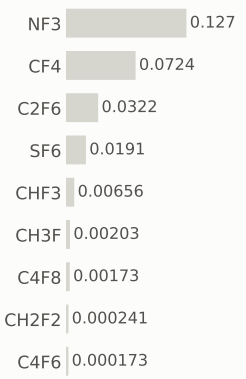
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



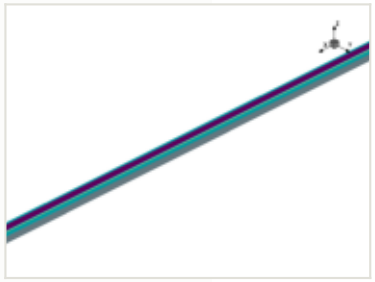
Step #370: Spx_only

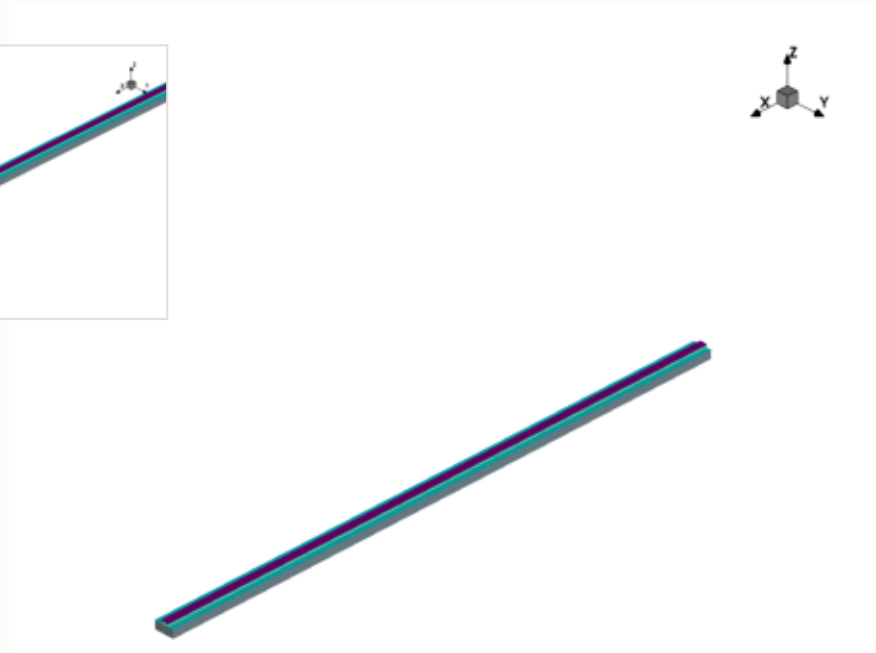
etch_strip · Spx_only

0	39.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

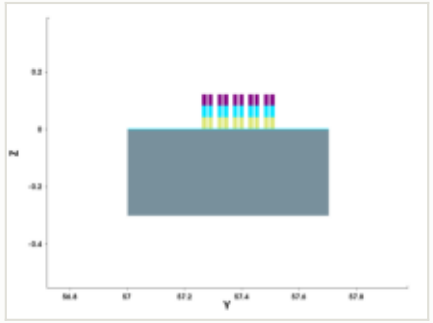
3D structure after this step

zoomed to fit

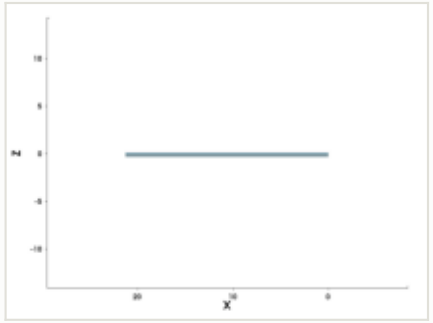




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

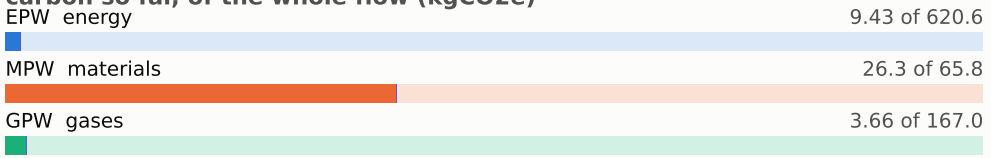
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

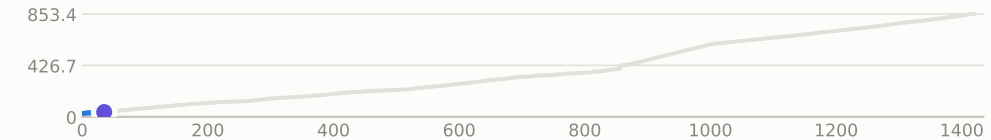
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

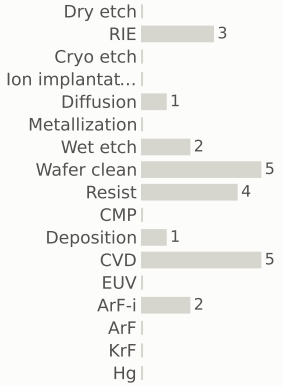
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

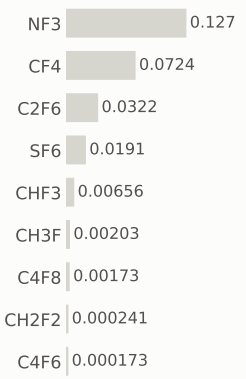


steps by process area



materials deposited (g)

gases emitted (g)



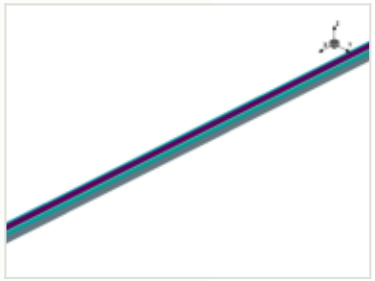
Step #380: Wafer_clean

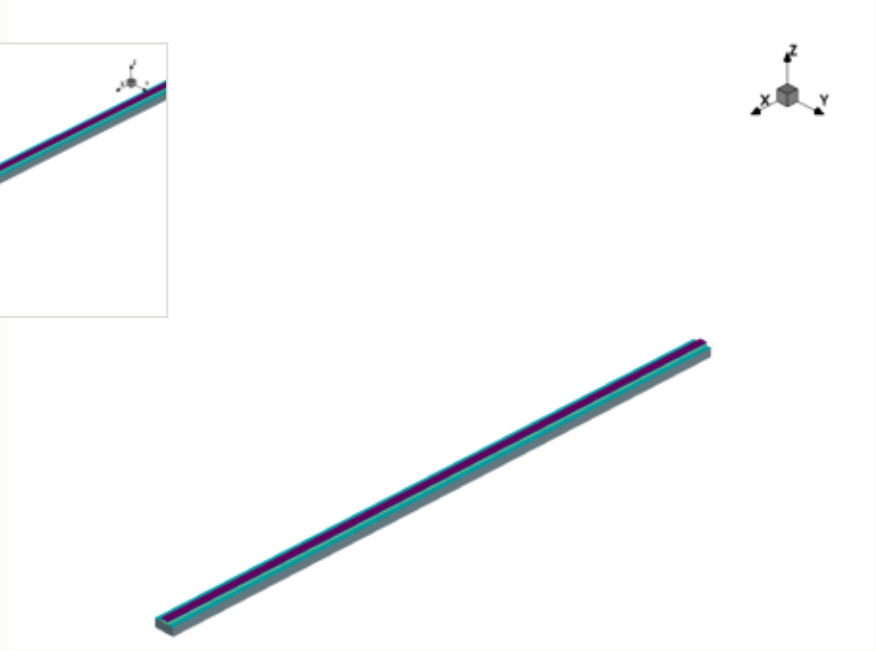
clean · Wafer_clean

0.169	39.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





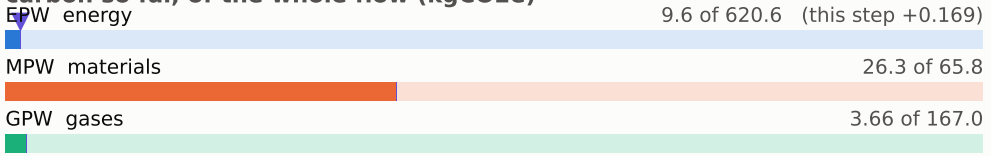
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

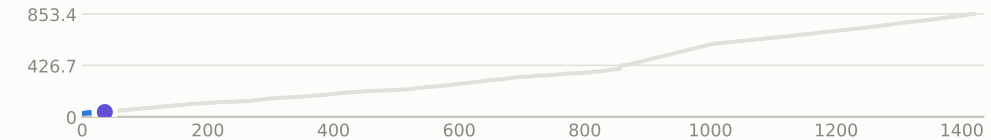
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

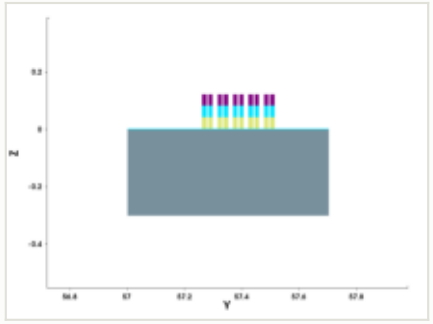
carbon so far, of the whole flow (kgCO2e)



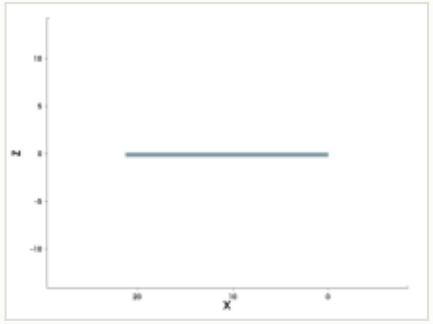
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



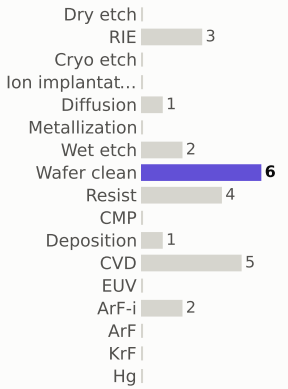
y cut at 57.351 µm



mask for this step

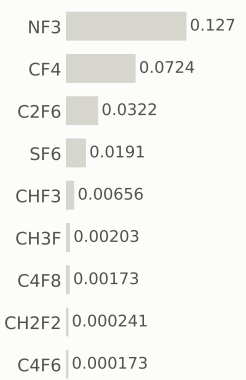
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



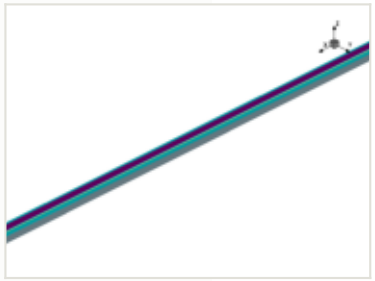
Step #390: Resist

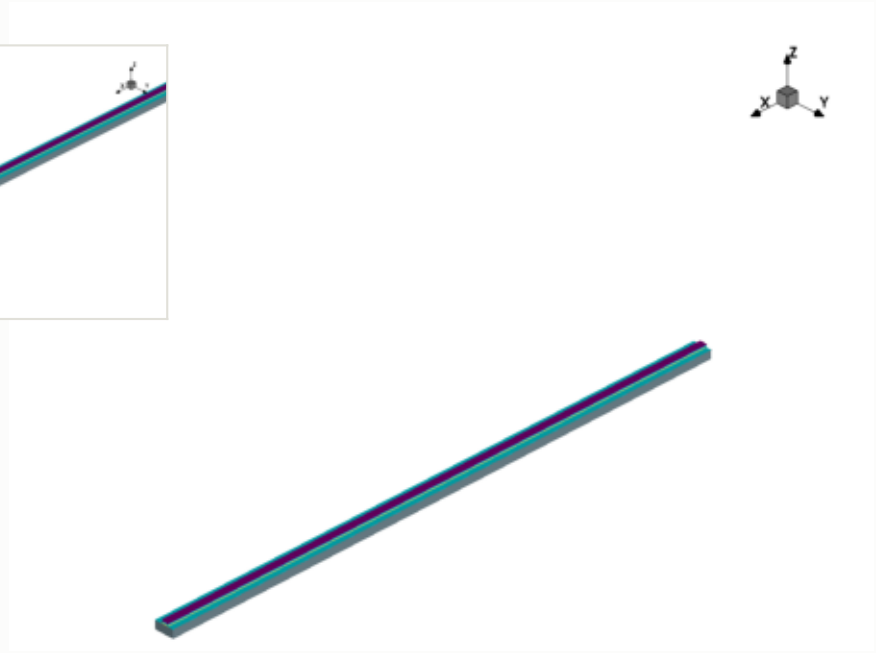
resist · Resist

0.169	39.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





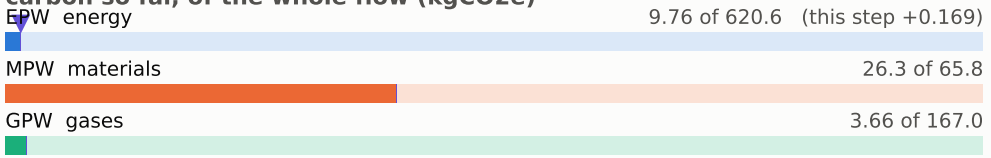
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

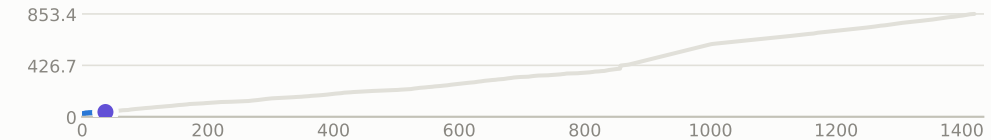
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

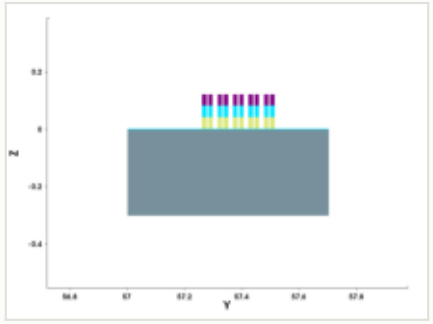
carbon so far, of the whole flow (kgCO2e)



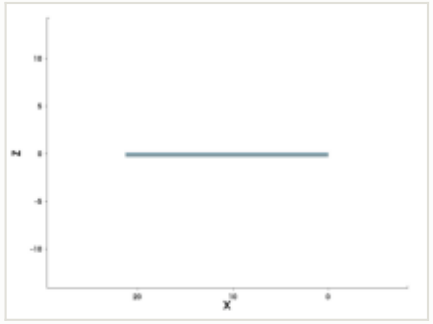
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



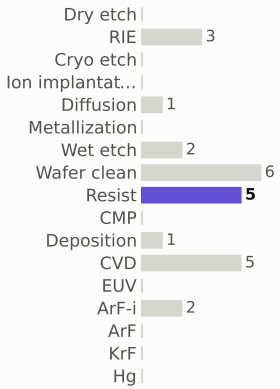
y cut at 57.351 µm



mask for this step

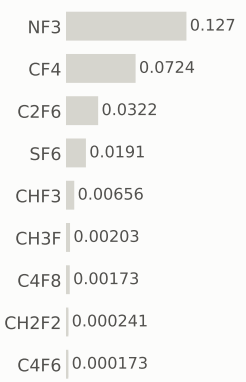
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



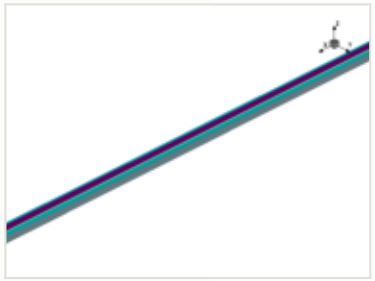
Step #400: Hg

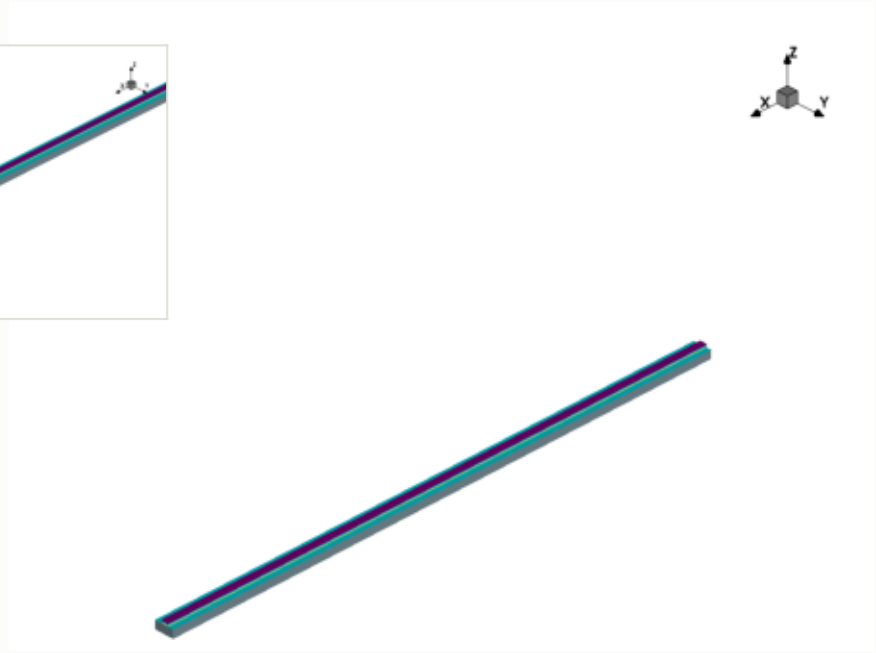
litho_patterning · Hg

0.199	40	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



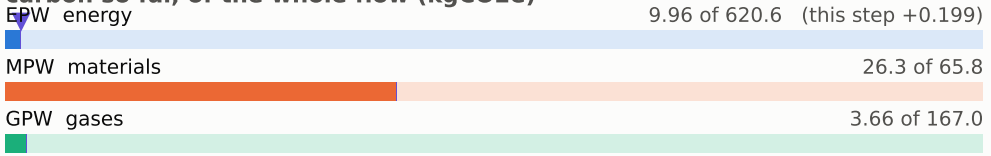


this step

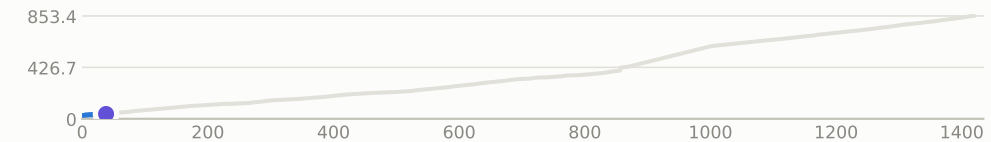
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

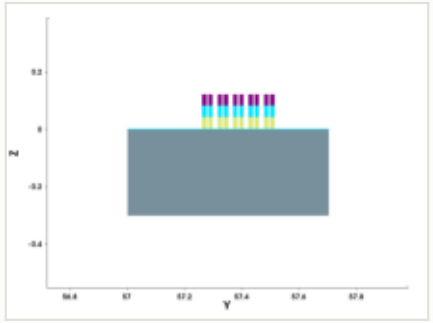
carbon so far, of the whole flow (kgCO2e)



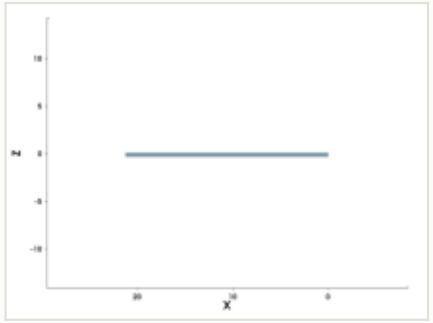
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



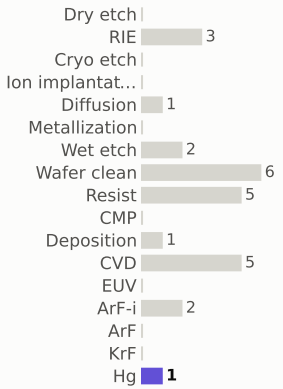
y cut at 57.351 µm



mask for this step

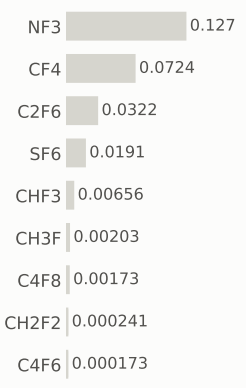
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



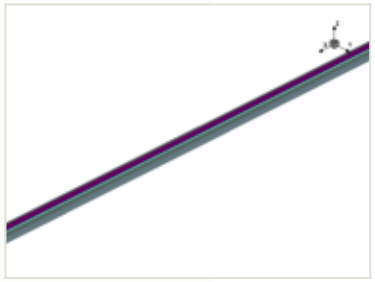
Step #410: Spx_only

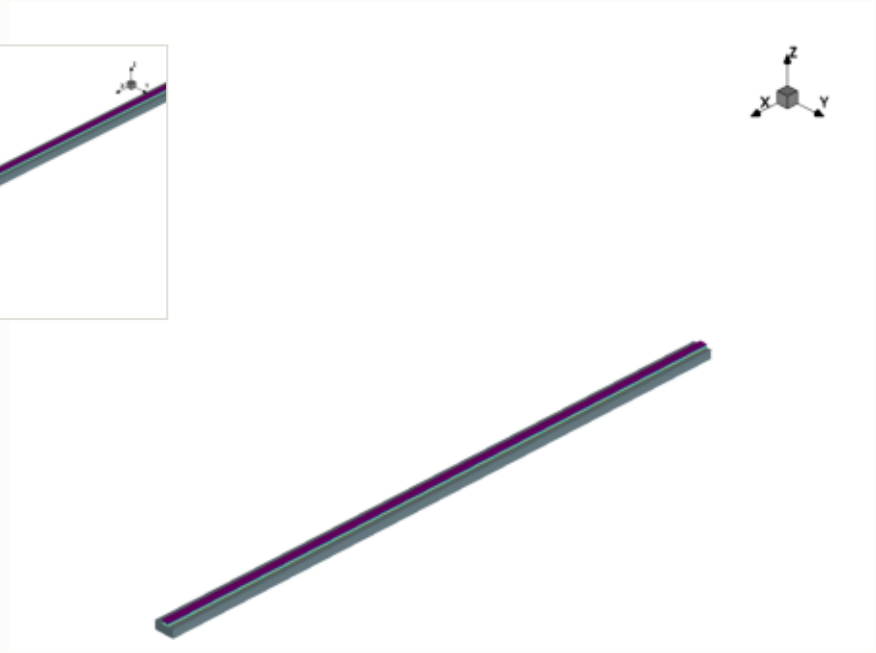
etch_anisotropic · Spx_only

0	40	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

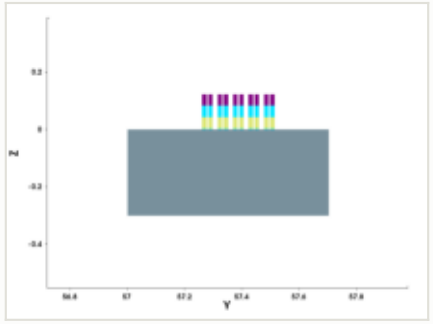
3D structure after this step

zoomed to fit

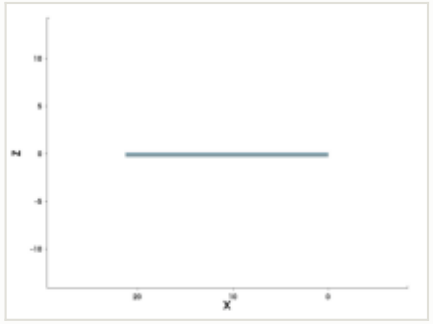




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

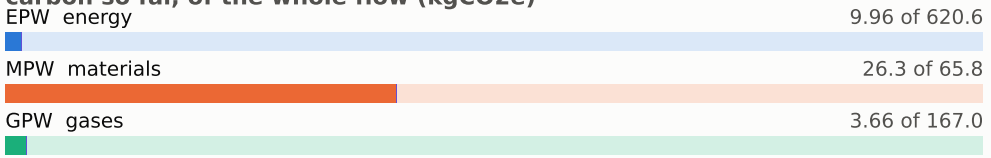
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

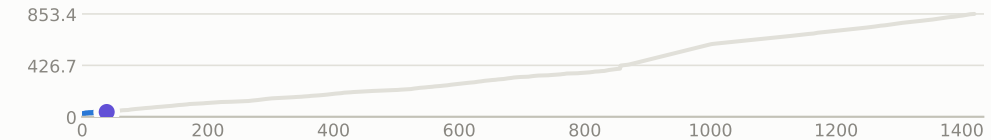
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

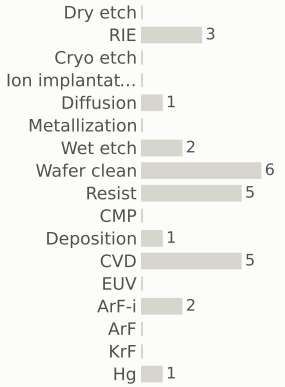
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

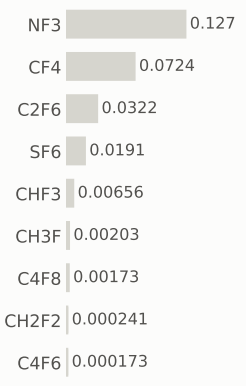


steps by process area



materials deposited (g)

gases emitted (g)



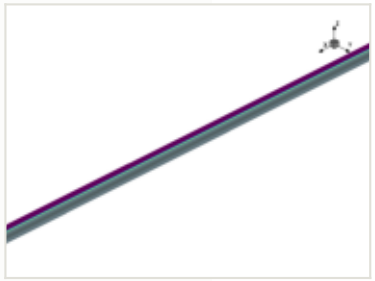
Step #420: RIE

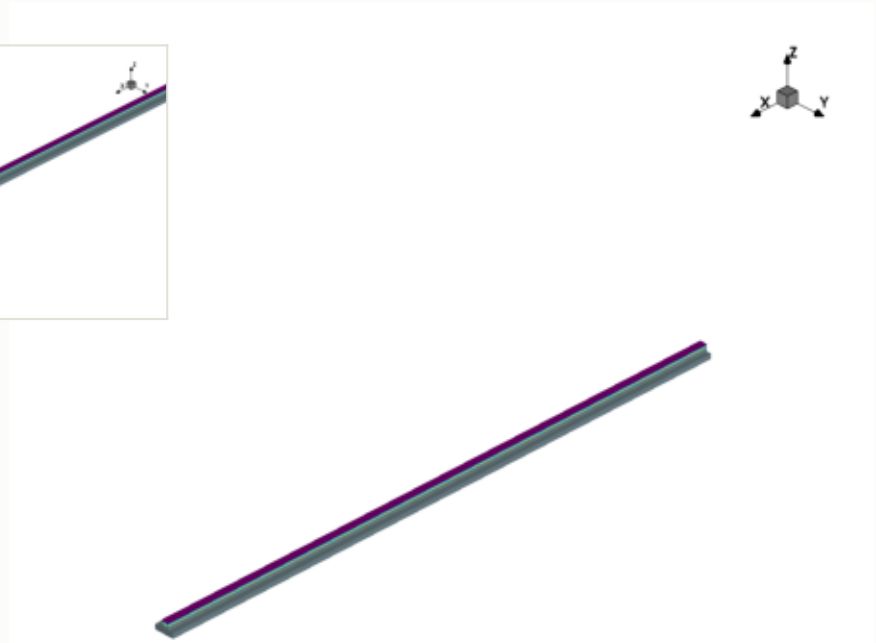
etch_anisotropic · RIE

1.04	41	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



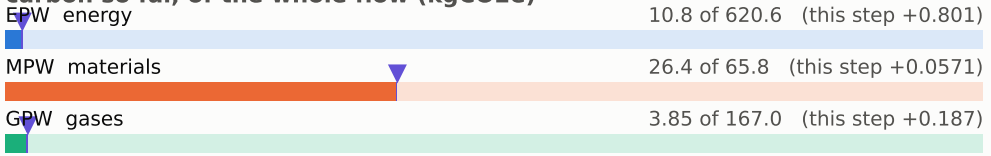


this step

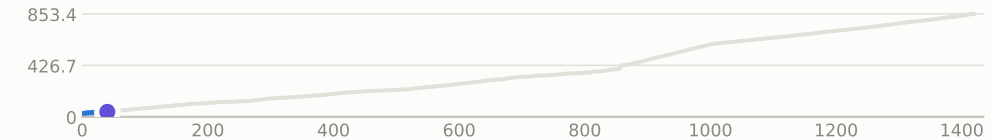
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

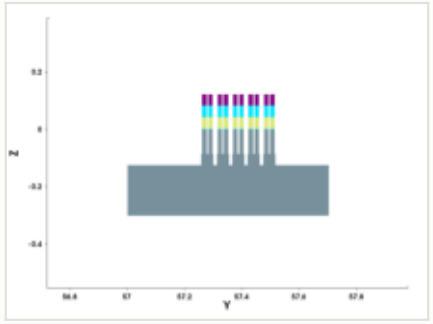
carbon so far, of the whole flow (kgCO2e)



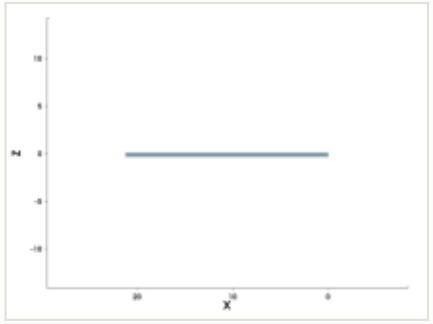
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



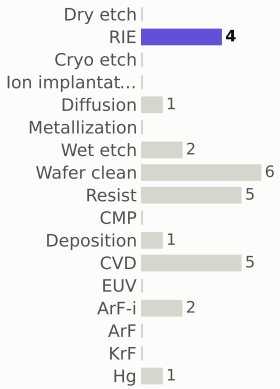
y cut at 57.351 µm



mask for this step

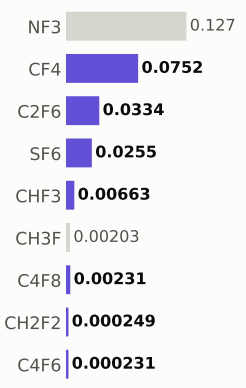
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



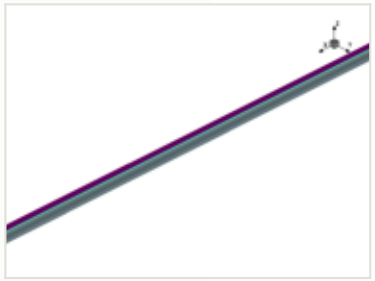
Step #430: Dry_etch

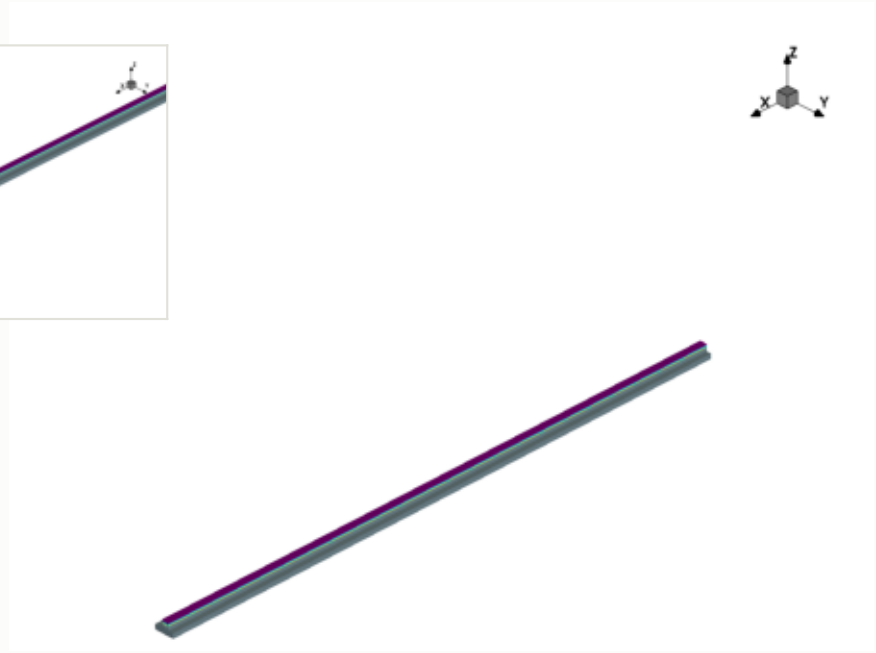
etch_isotropic · Dry_etch

0.801	41.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



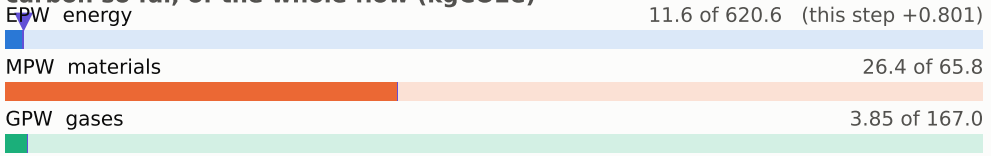


this step

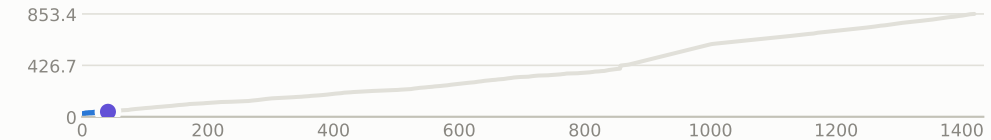
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

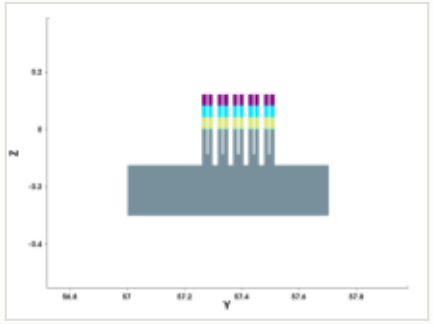
carbon so far, of the whole flow (kgCO2e)



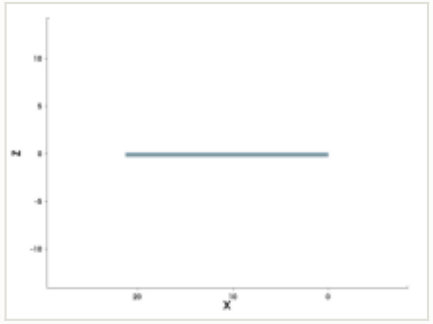
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



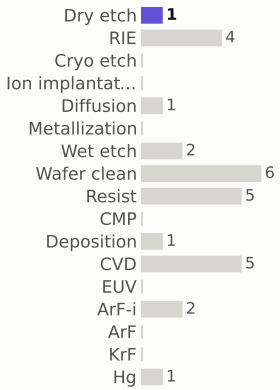
y cut at 57.351 µm



mask for this step

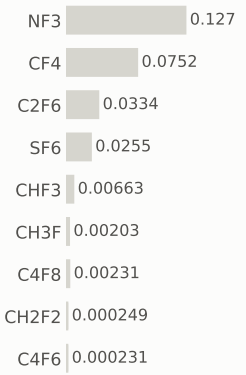
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



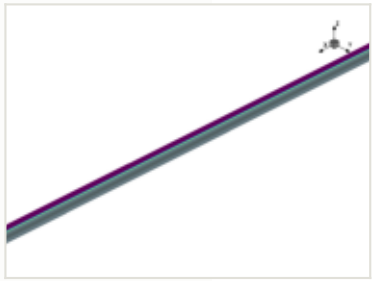
Step #440: Spx_only

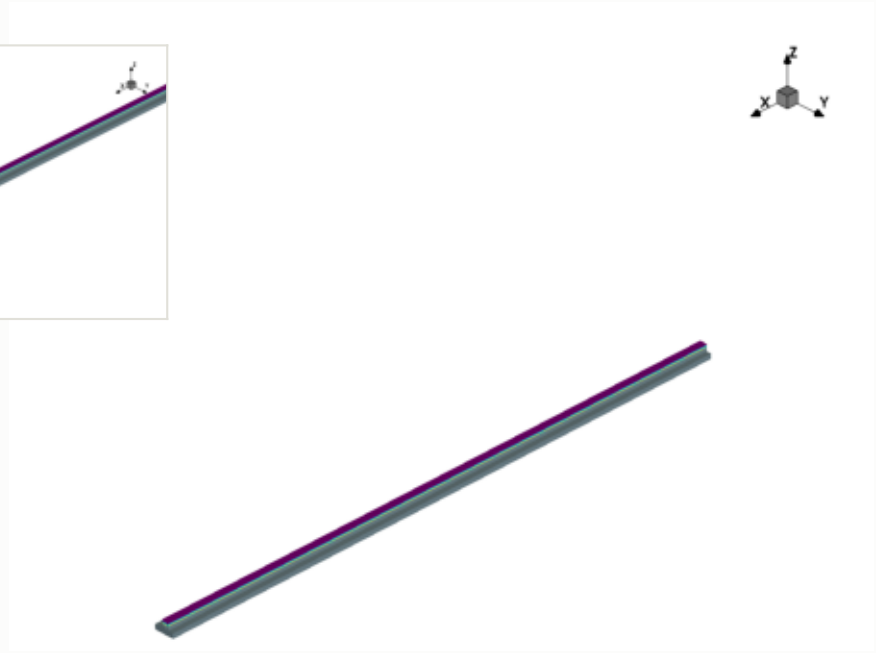
etch_microloading · Spx_only

0	41.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





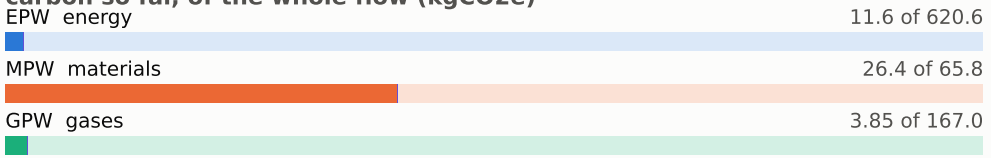
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

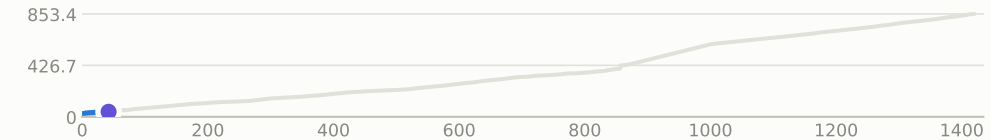
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

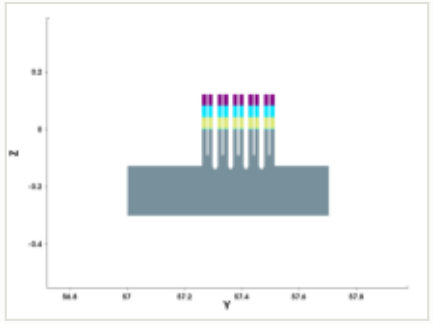
carbon so far, of the whole flow (kgCO2e)



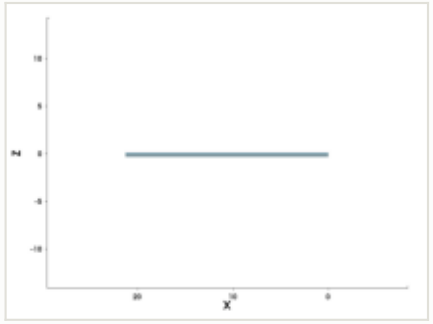
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



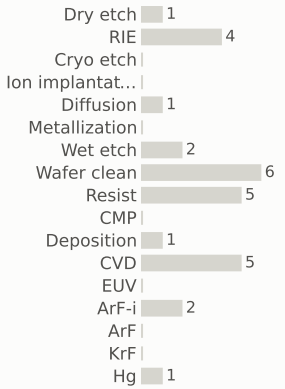
y cut at 57.351 µm



mask for this step

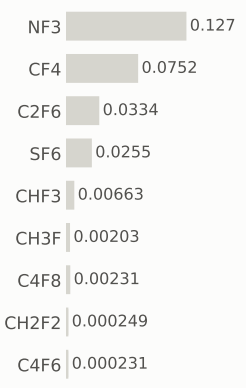
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



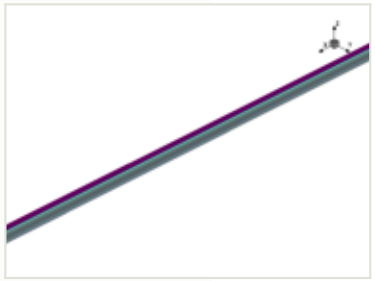
Step #450: Spx_only

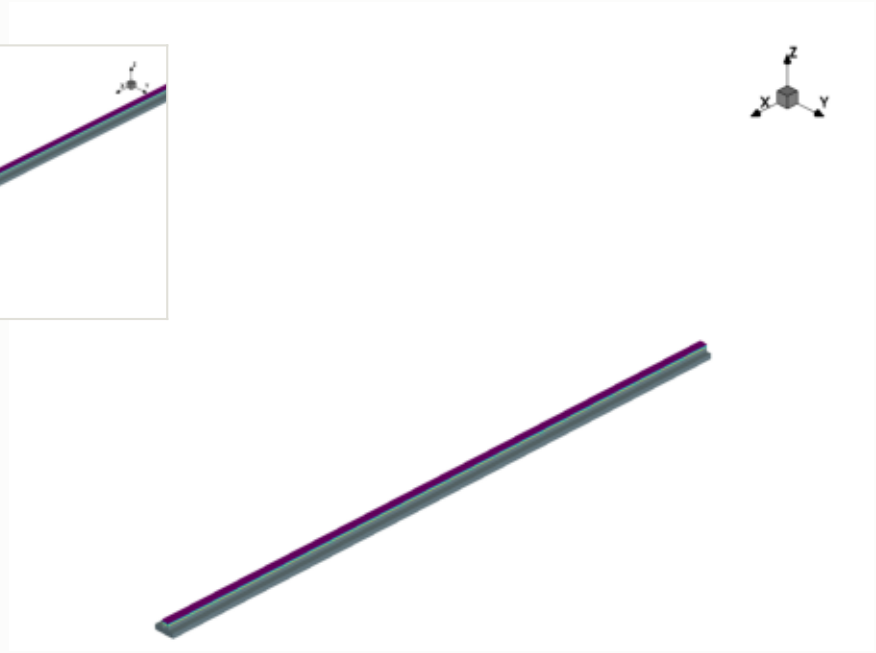
etch_isotropic · Spx_only

0	41.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



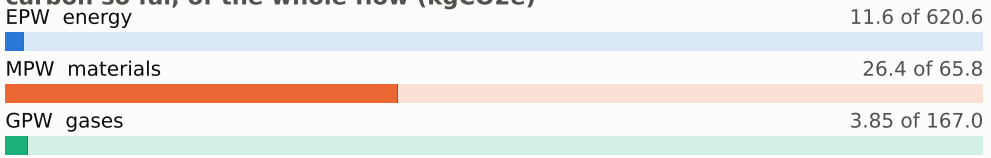


this step

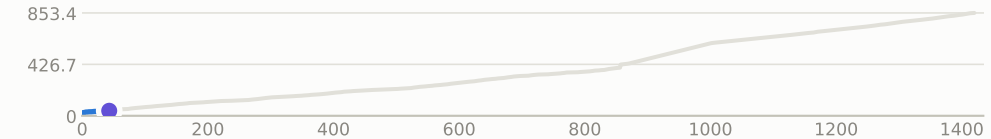
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

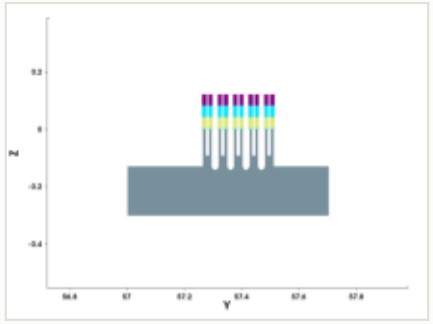
carbon so far, of the whole flow (kgCO2e)



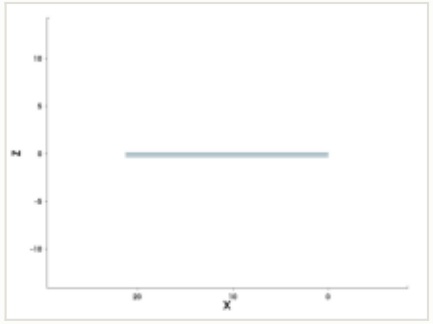
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



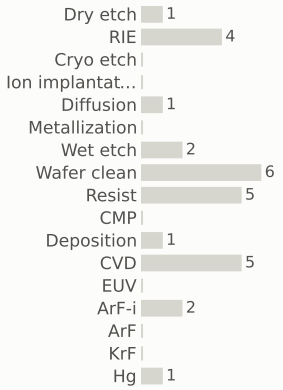
y cut at 57.351 µm



mask for this step

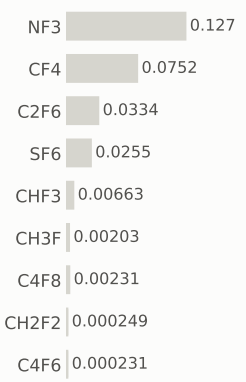
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



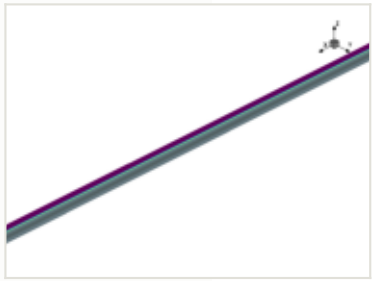
Step #460: Spx_only

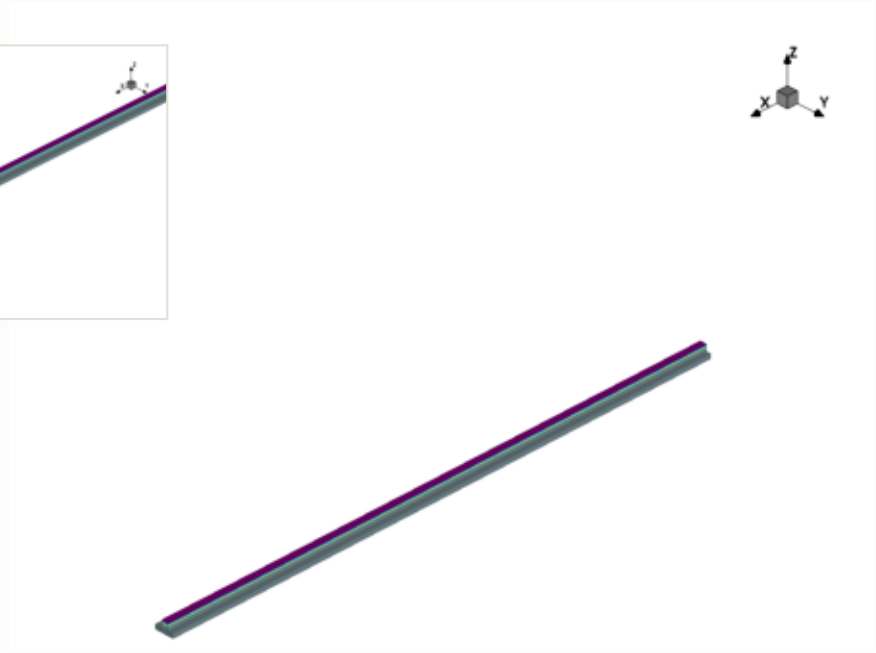
etch_strip · Spx_only

0	41.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

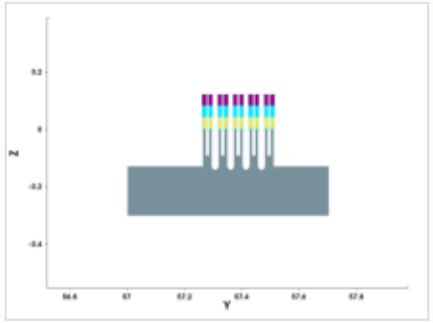
3D structure after this step

zoomed to fit

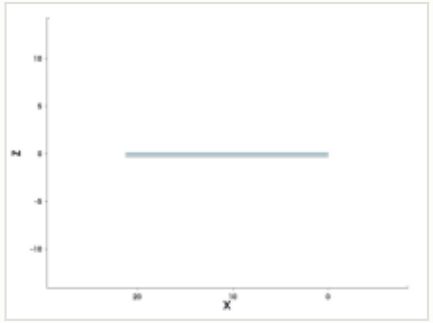




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

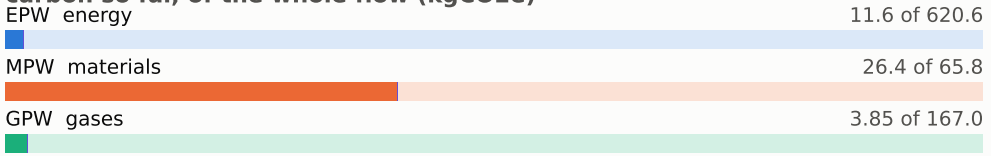
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

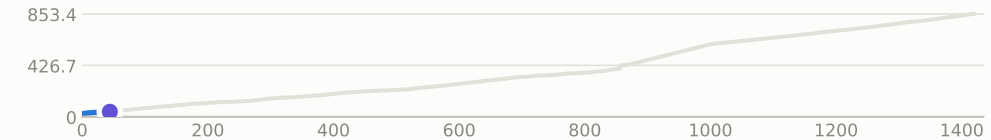
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

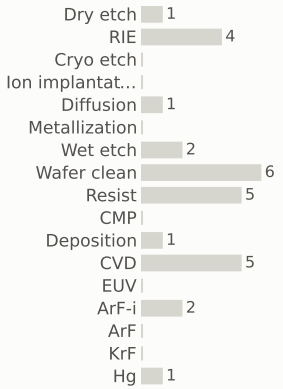
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

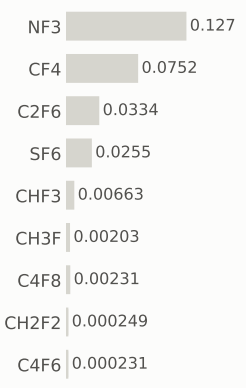


steps by process area



materials deposited (g)

gases emitted (g)



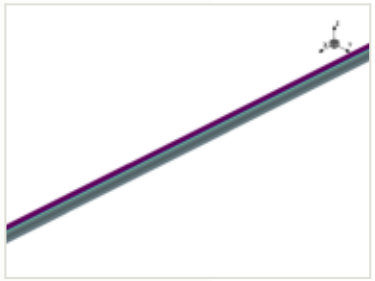
Step #470: Wafer_clean

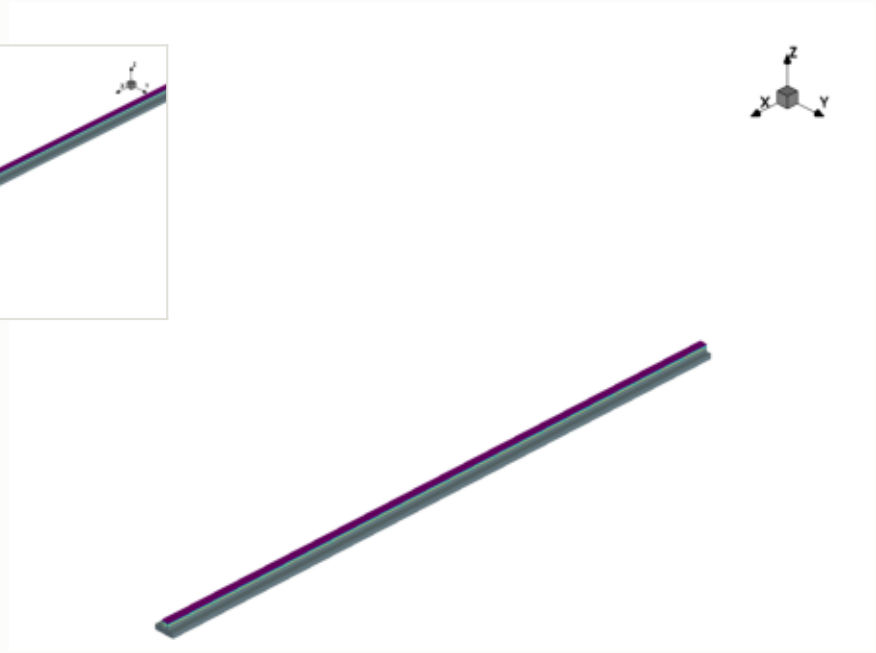
clean · Wafer_clean

0.169	42	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





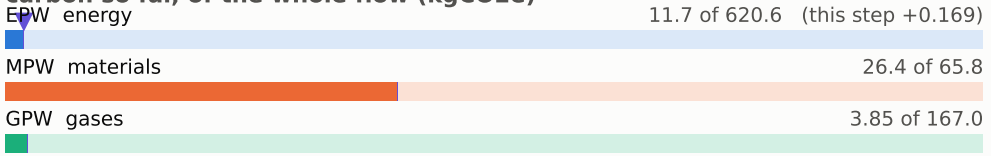
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

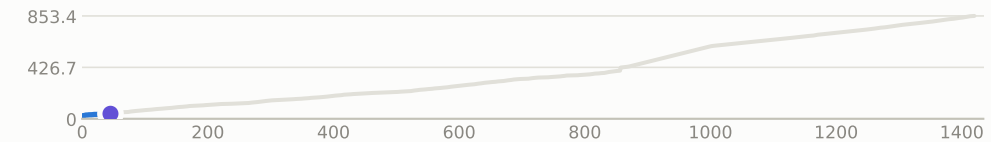
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

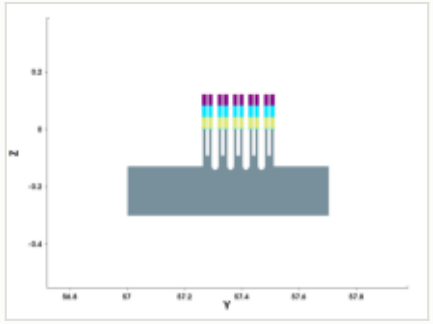
carbon so far, of the whole flow (kgCO2e)



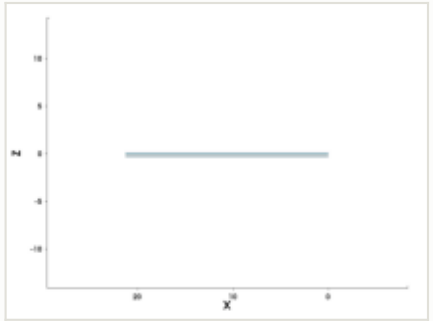
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



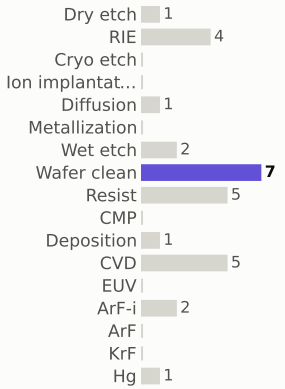
y cut at 57.351 µm



mask for this step

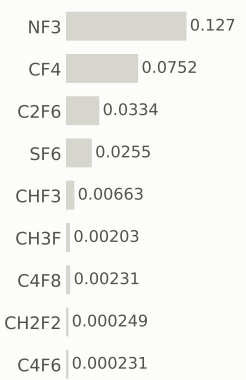
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



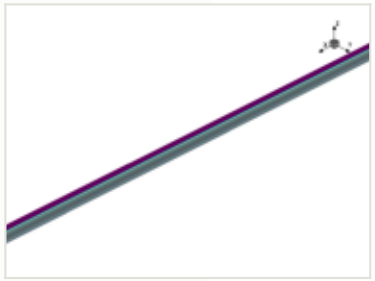
Step #480: Wafer_clean

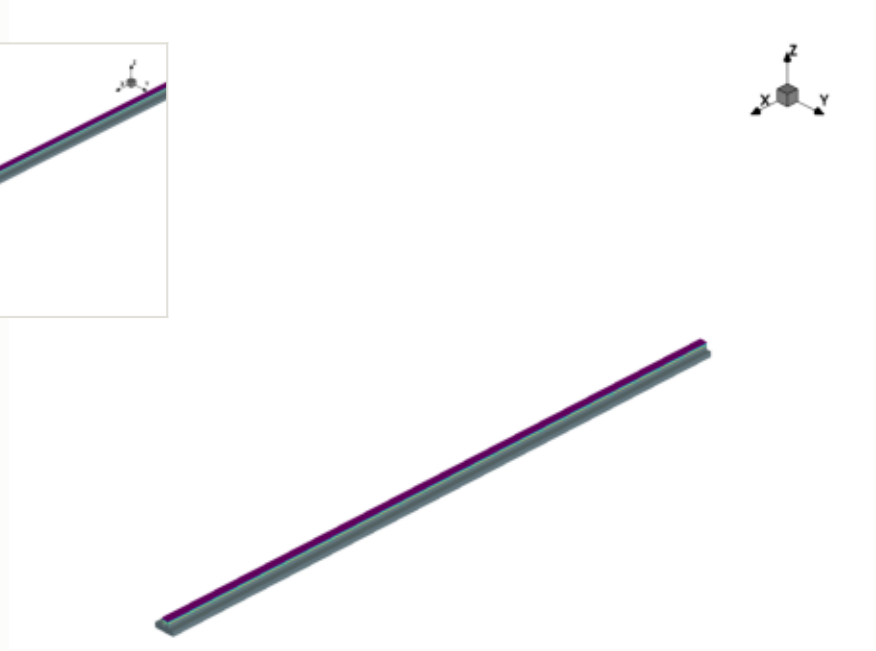
clean · Wafer_clean

0.169	42.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



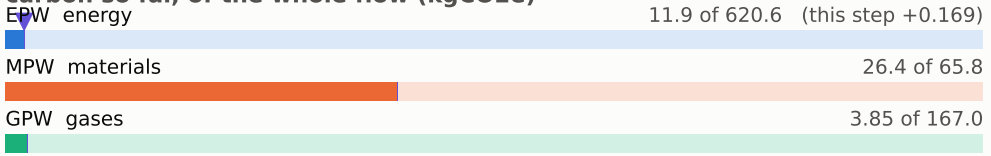


this step

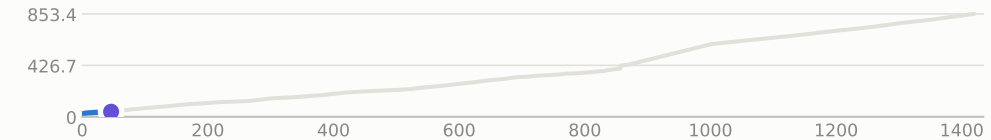
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

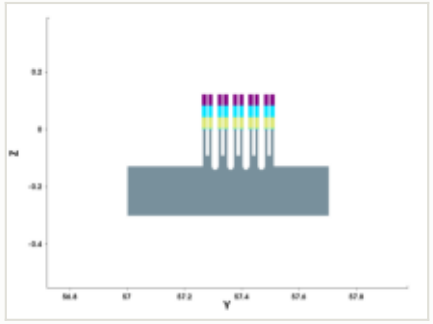
carbon so far, of the whole flow (kgCO2e)



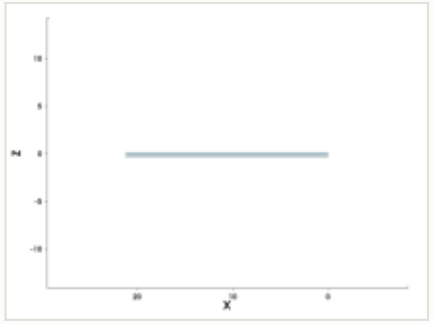
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



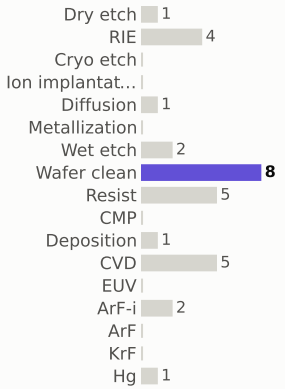
y cut at 57.351 µm



mask for this step

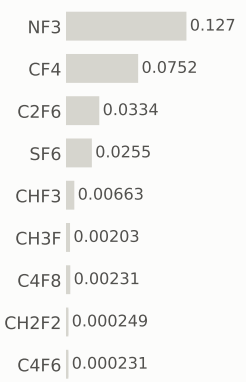
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



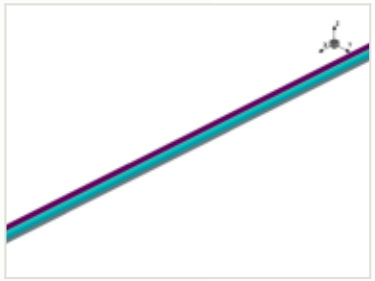
Step #490: Diffusion

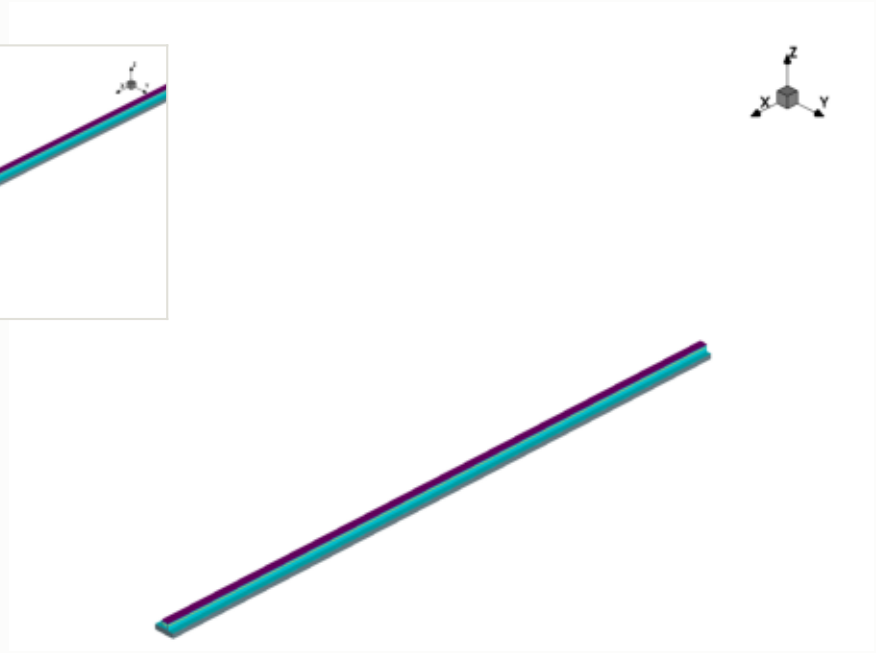
depo_isotropic · Diffusion

0.801	42.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



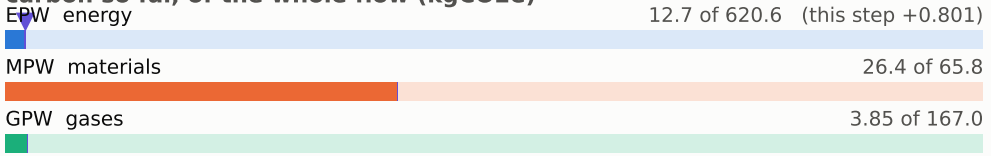


this step

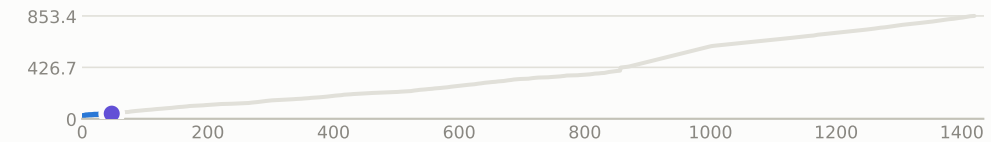
process area	Diffusion
material	—
thickness	0.0035 µm
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

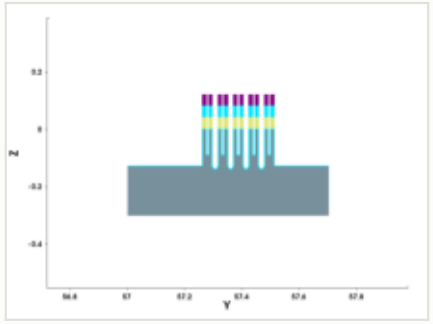
carbon so far, of the whole flow (kgCO2e)



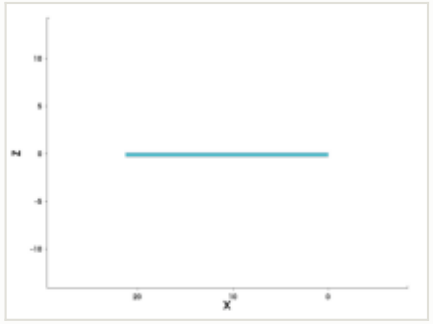
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



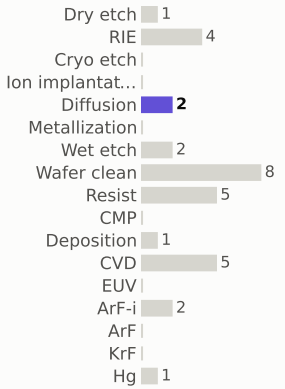
y cut at 57.351 µm



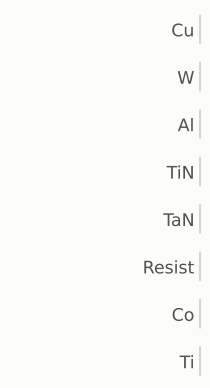
mask for this step

no mask for this step

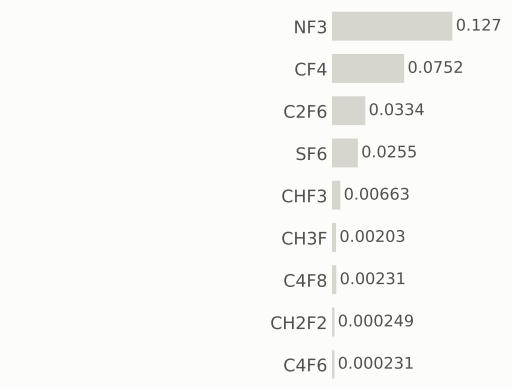
steps by process area



materials deposited (g)



gases emitted (g)



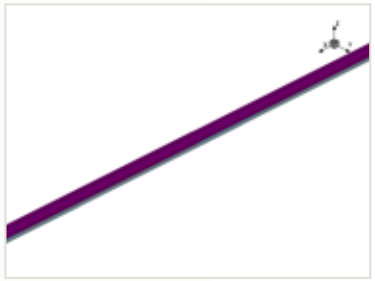
Step #500: CVD

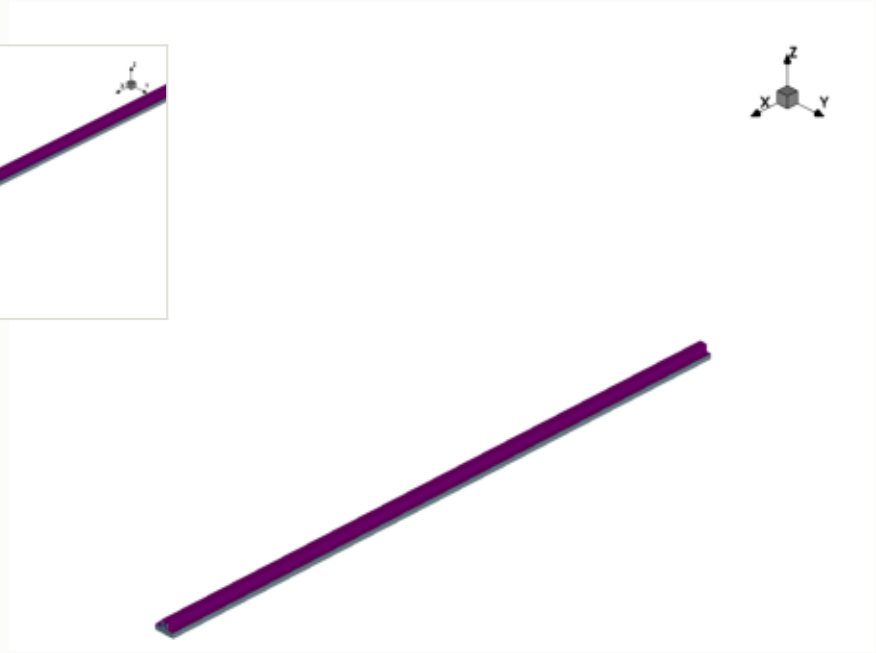
depo_isotropic · CVD

1.31	44.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



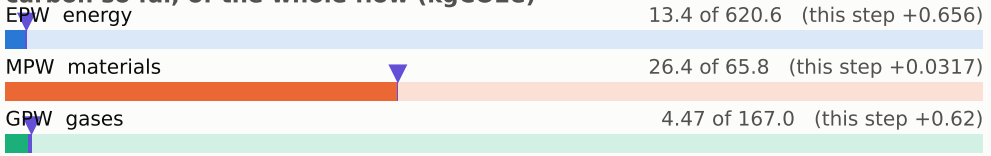


this step

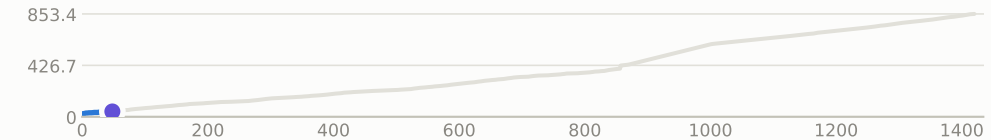
process area	CVD
material	—
thickness	0.002 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

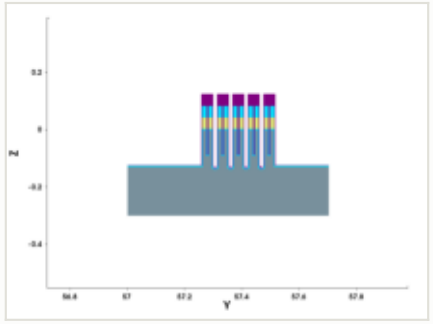
carbon so far, of the whole flow (kgCO2e)



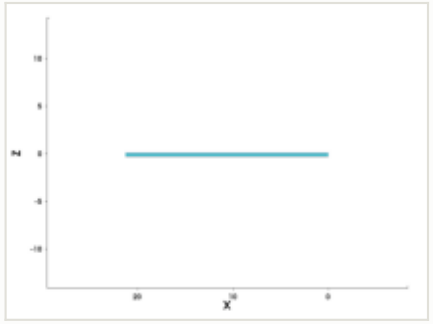
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



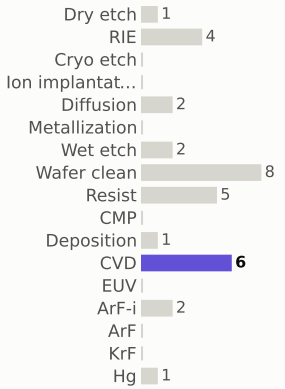
y cut at 57.351 µm



mask for this step

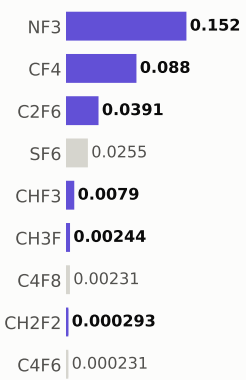
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



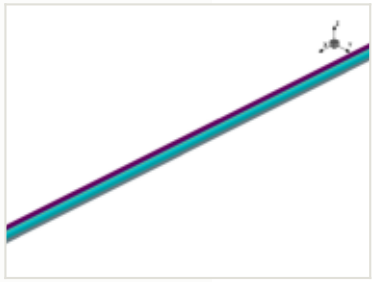
Step #510: Spx_only

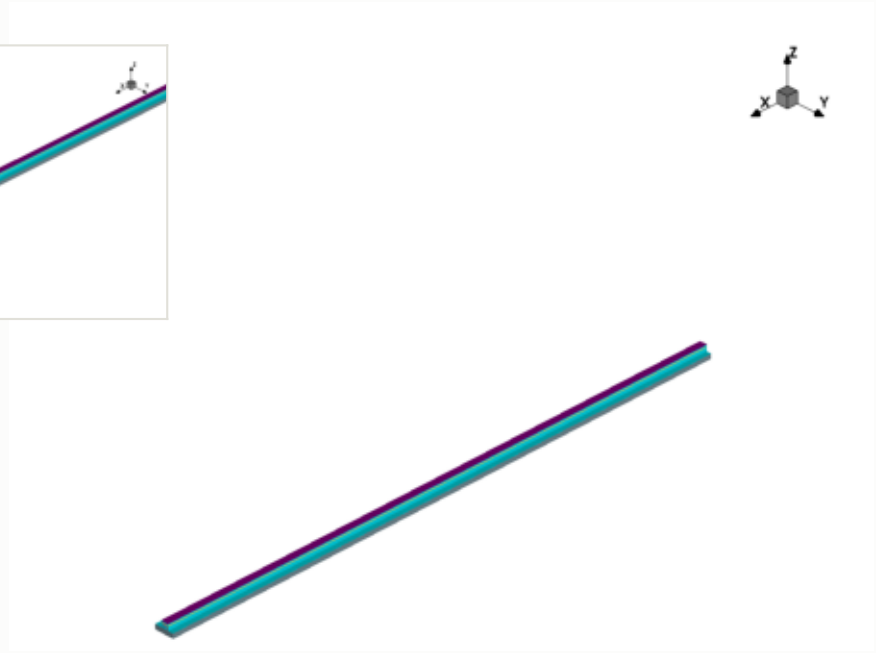
etch_isotropic · Spx_only

0	44.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





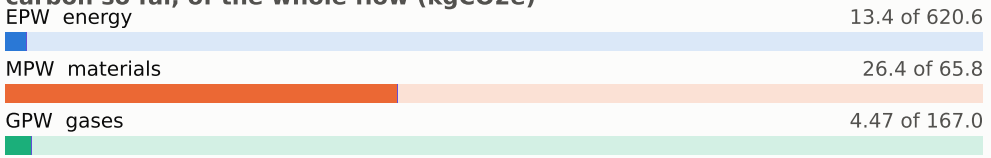
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

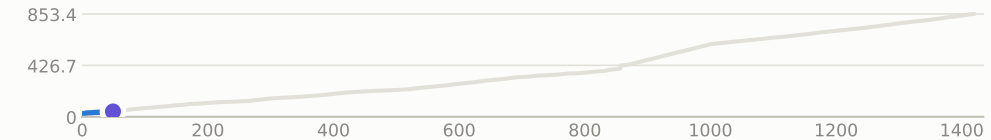
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

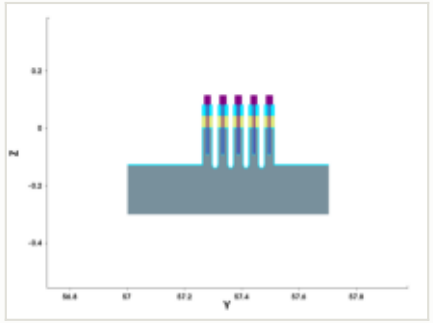
carbon so far, of the whole flow (kgCO2e)



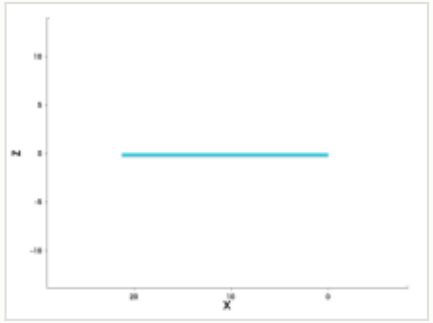
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



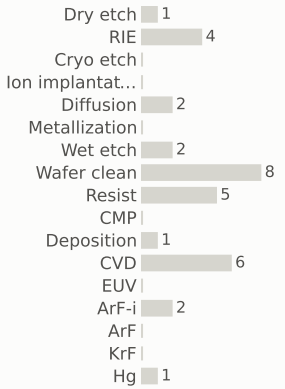
y cut at 57.351 µm



mask for this step

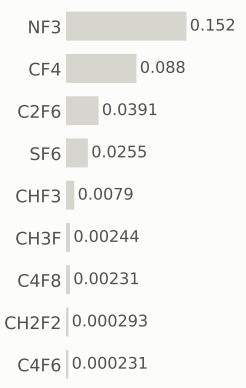
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



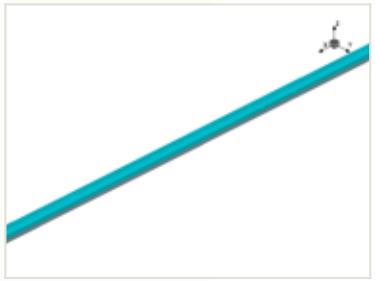
Step #520: Spx_only

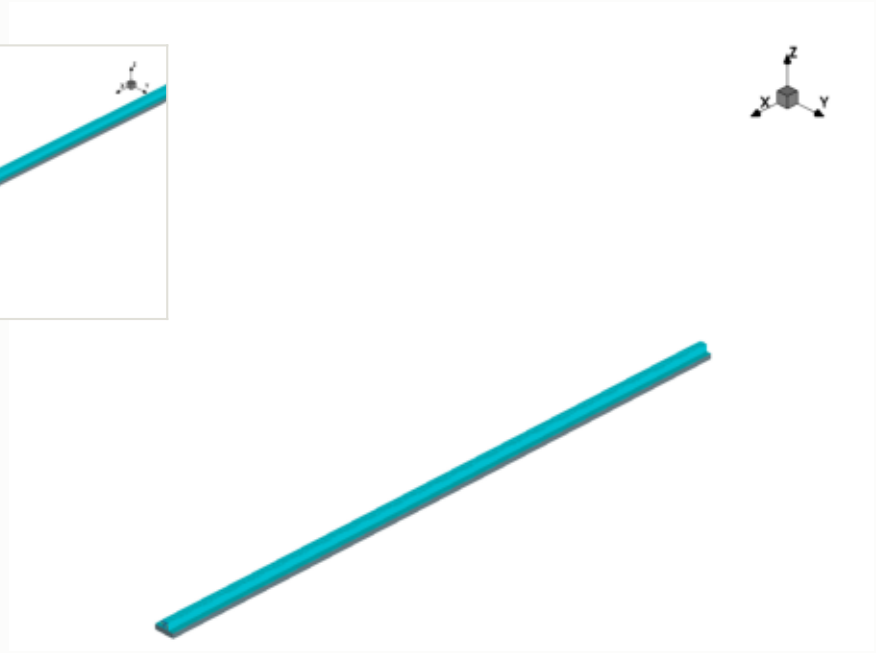
depo_isotropic · Spx_only

0	44.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

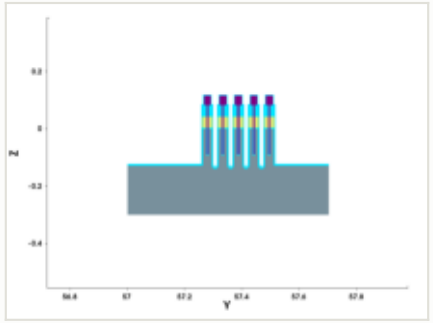
3D structure after this step

zoomed to fit

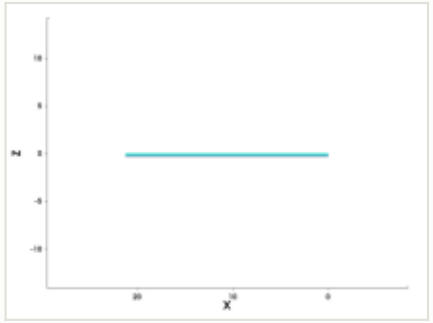




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

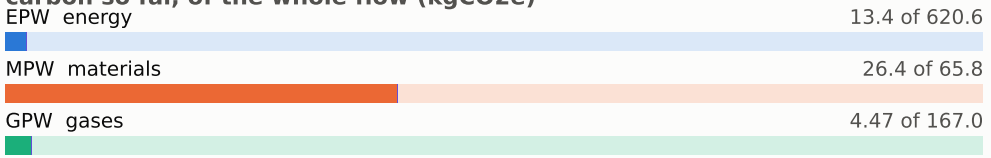
no mask for this step

this step

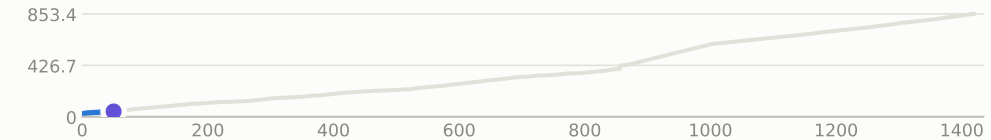
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	0.003 µm	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

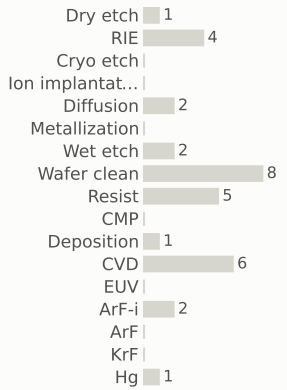
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

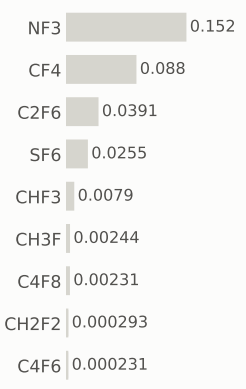


steps by process area



materials deposited (g)

gases emitted (g)



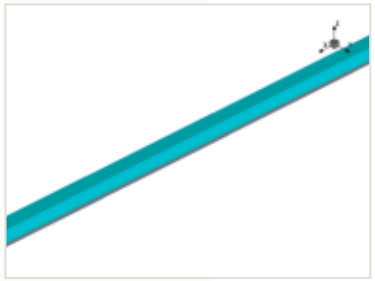
Step #530: CVD

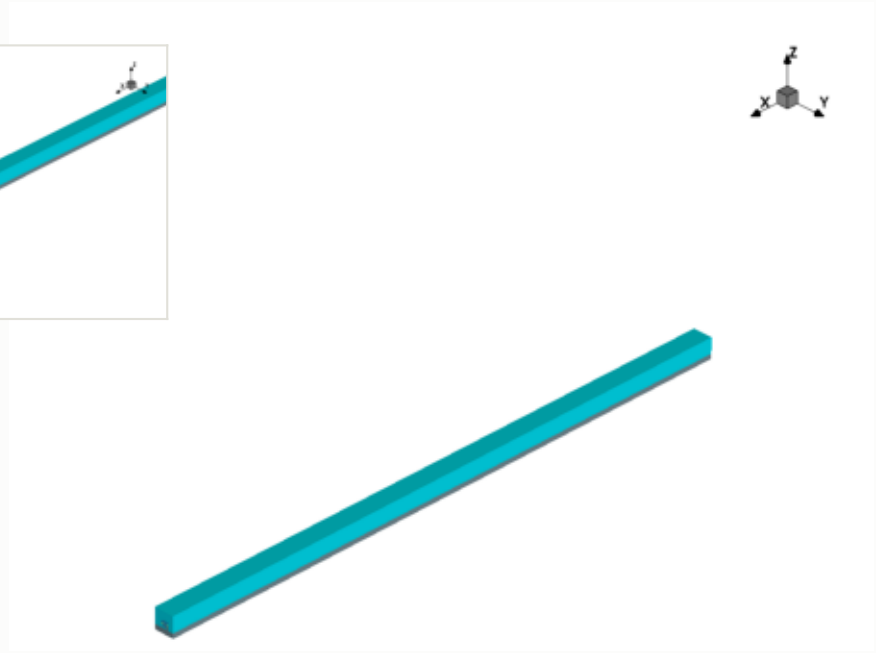
depo_fill · CVD

1.31	45.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



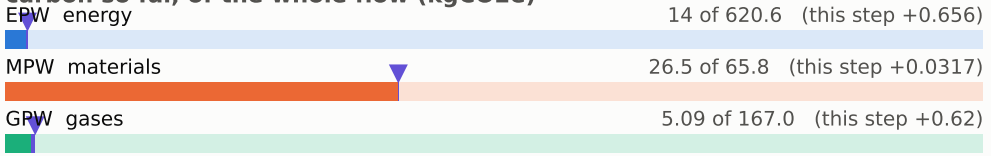


this step

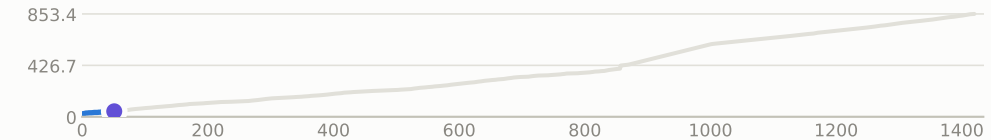
process area	CVD
material	—
thickness	0.3 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

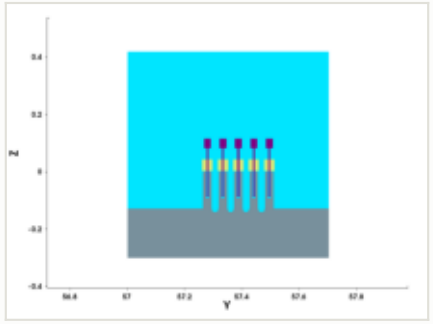
carbon so far, of the whole flow (kgCO2e)



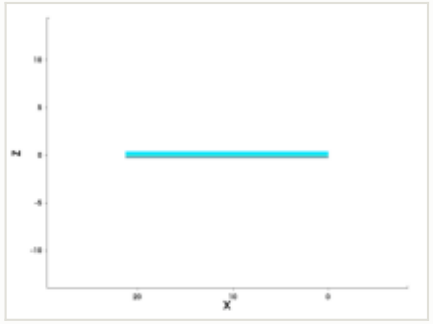
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



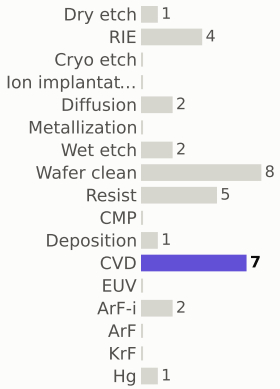
y cut at 57.351 µm



mask for this step

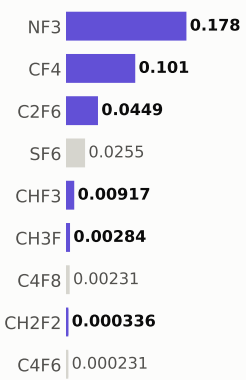
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



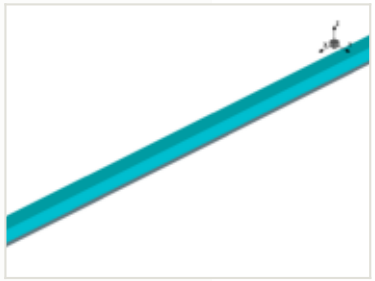
Step #540: Diffusion

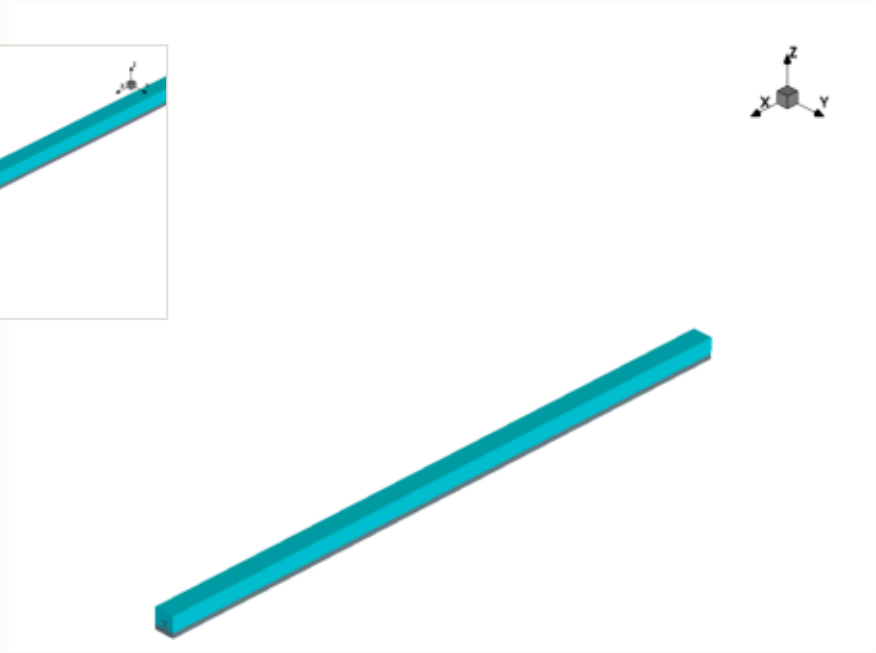
anneal · Diffusion

0.801	46.4	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





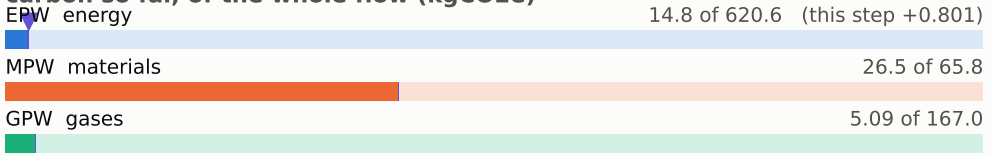
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

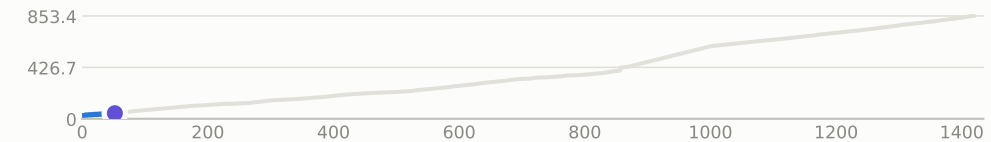
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

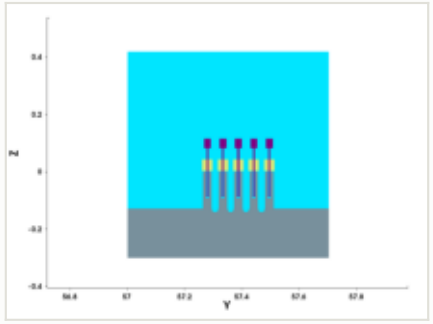
carbon so far, of the whole flow (kgCO2e)



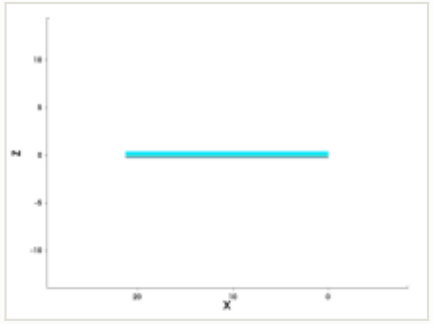
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



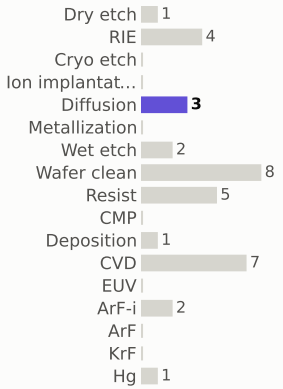
y cut at 57.351 µm



mask for this step

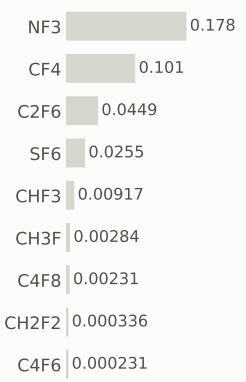
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



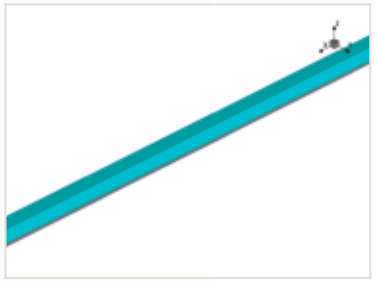
Step #550: Diffusion

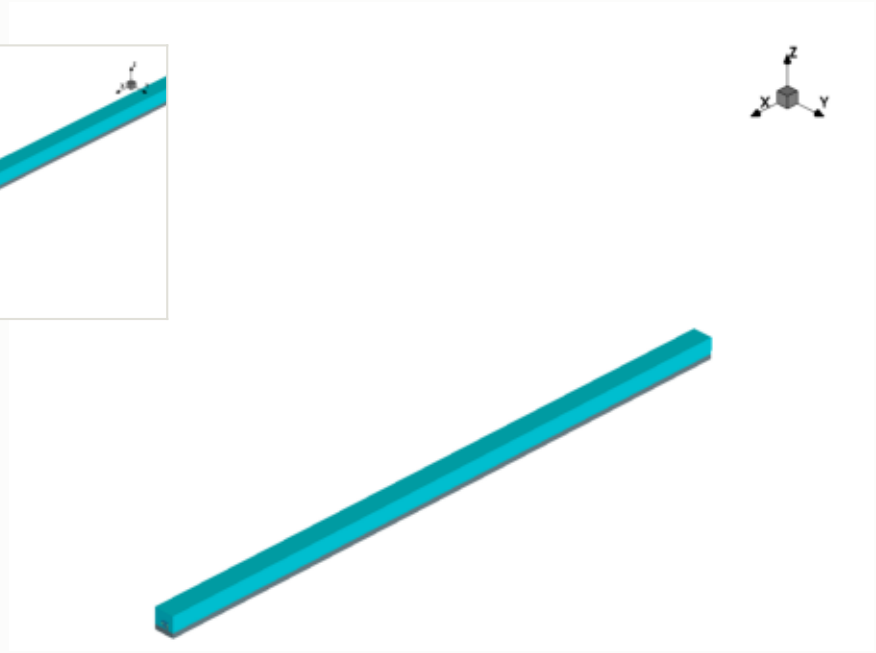
anneal · Diffusion

0.801	47.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





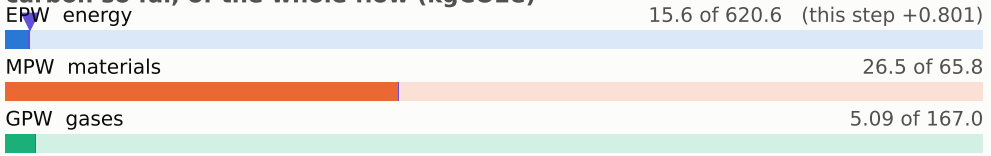
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

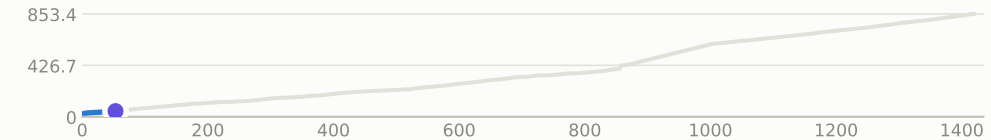
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

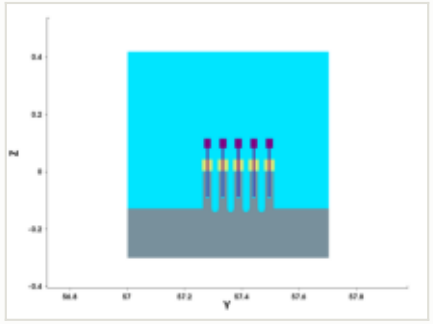
carbon so far, of the whole flow (kgCO2e)



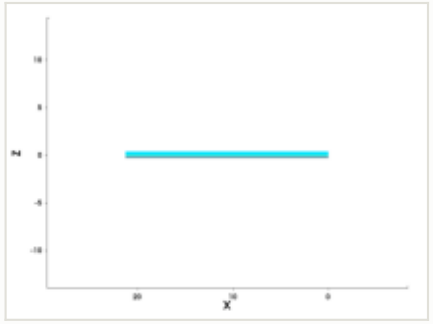
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



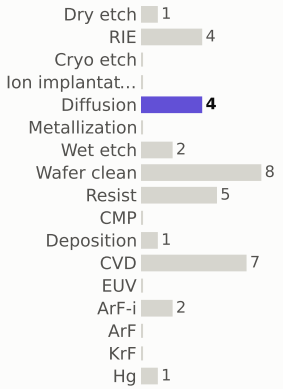
y cut at 57.351 µm



mask for this step

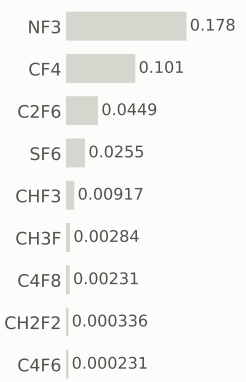
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



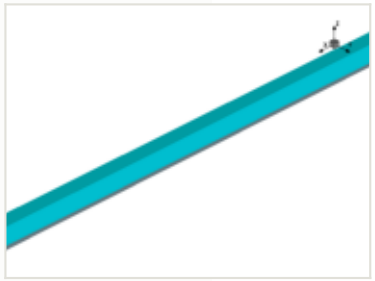
Step #560: CVD

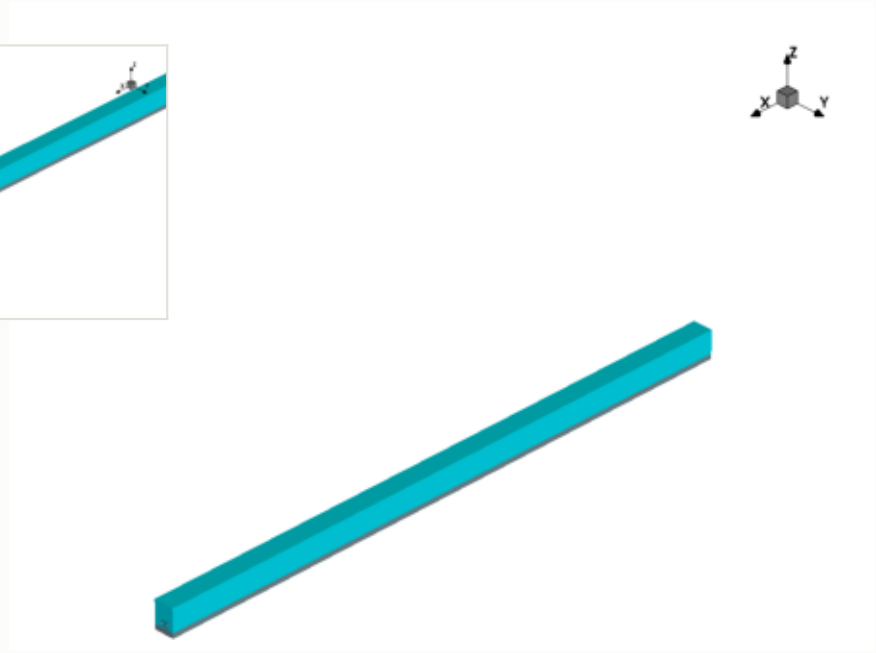
depo_isotropic · CVD

1.31	48.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



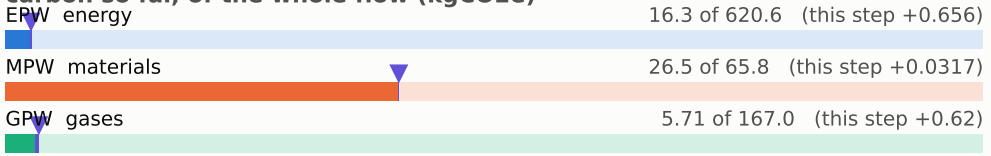


this step

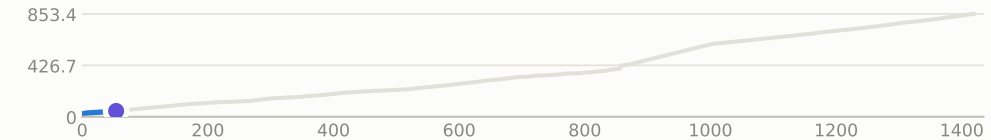
process area	CVD
material	—
thickness	0.25 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

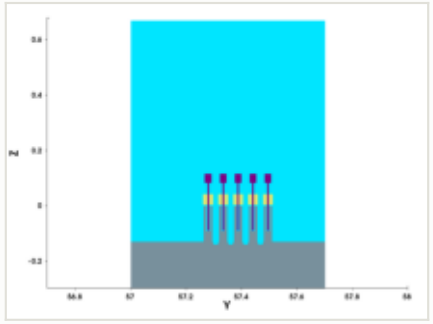
carbon so far, of the whole flow (kgCO2e)



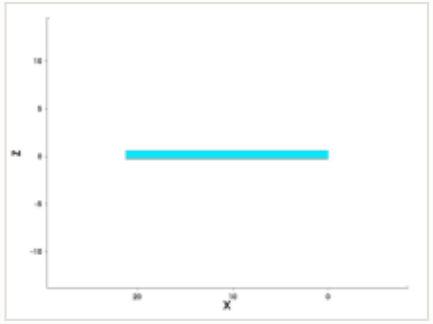
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



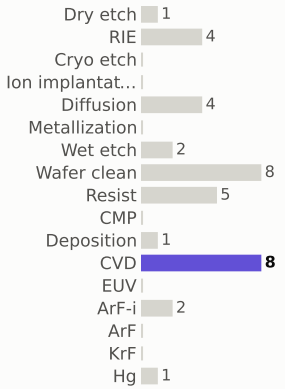
y cut at 57.351 µm



mask for this step

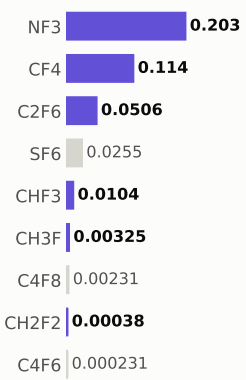
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



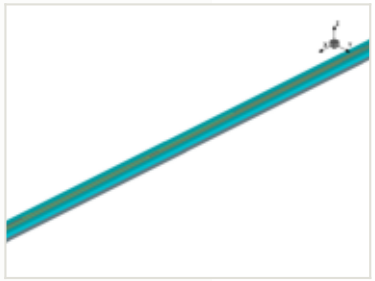
Step #570: CMP

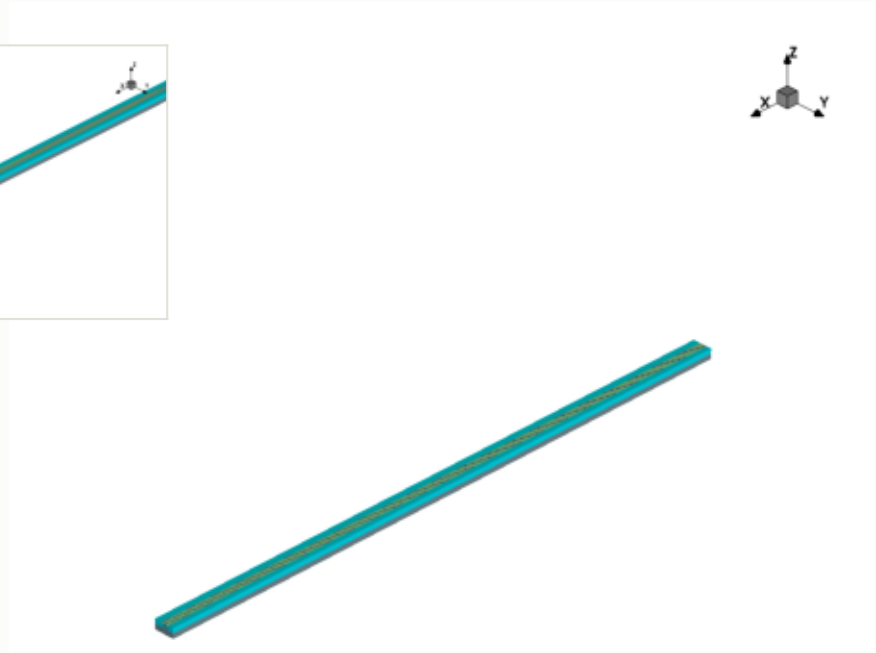
etch_cmp · CMP

0.169	48.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



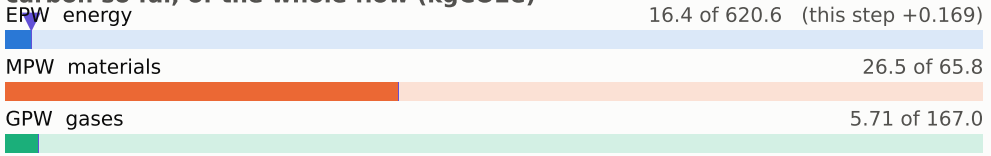


this step

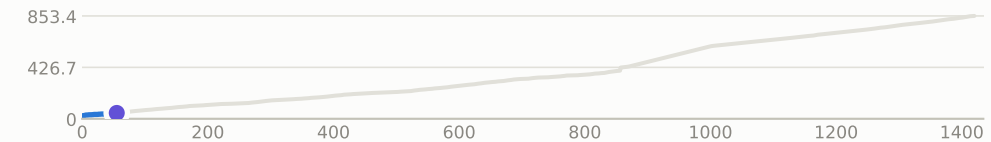
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

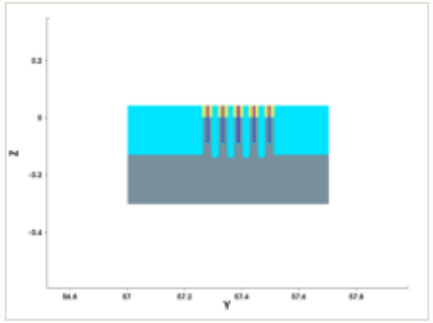
carbon so far, of the whole flow (kgCO2e)



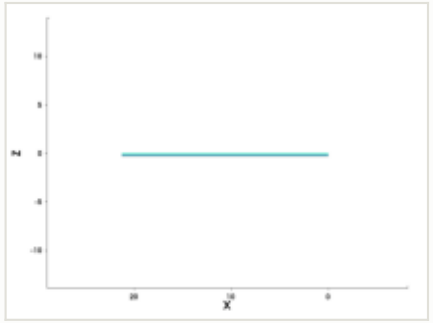
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



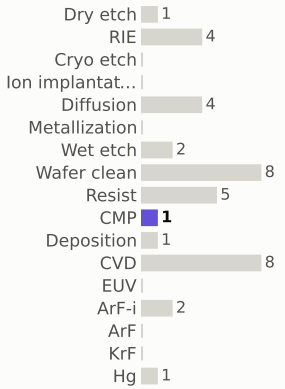
y cut at 57.351 µm



mask for this step

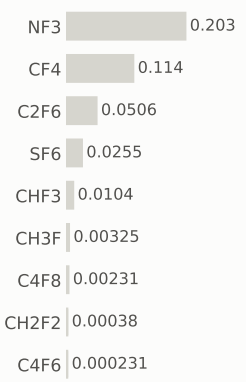
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



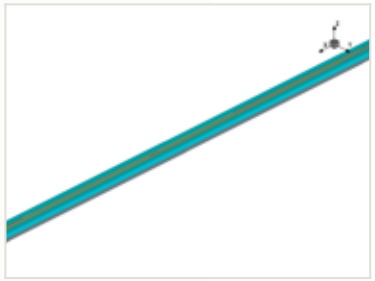
Step #580: Diffusion

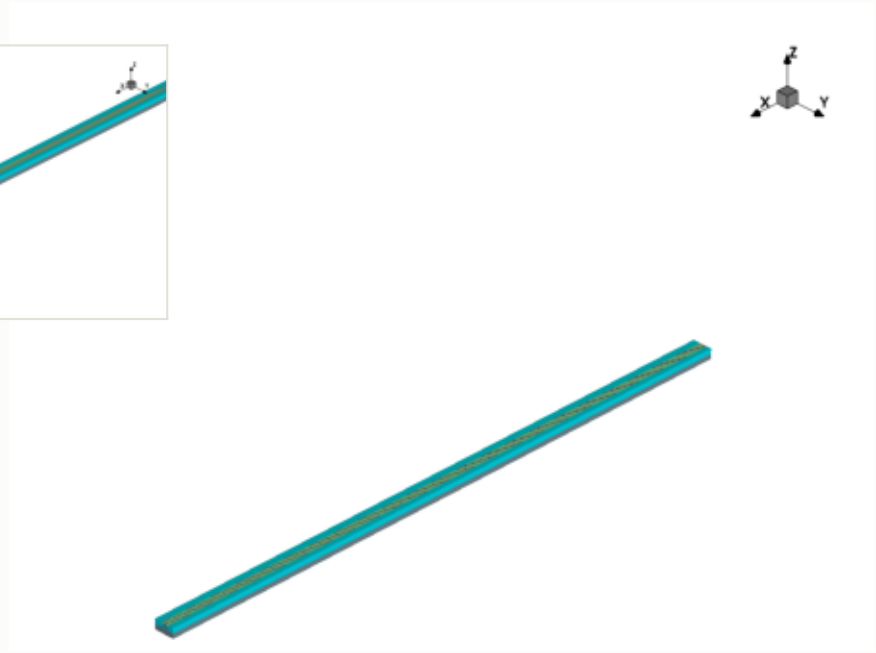
anneal · Diffusion

0.801	49.4	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





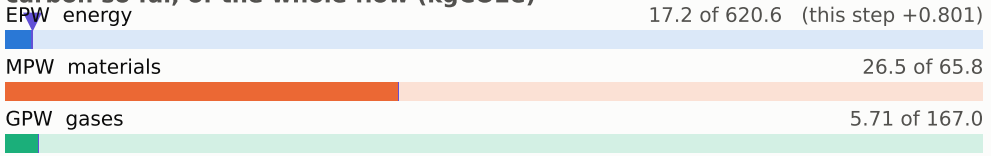
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

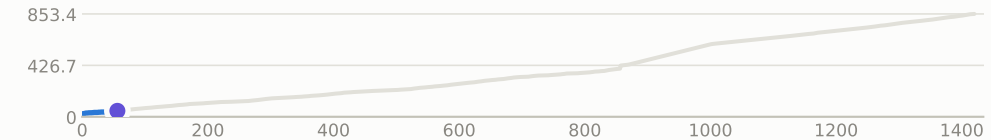
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

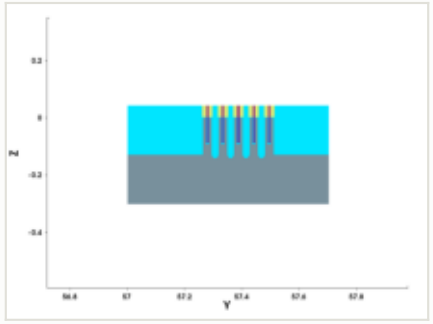
carbon so far, of the whole flow (kgCO2e)



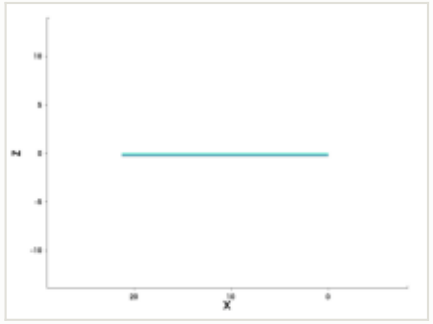
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



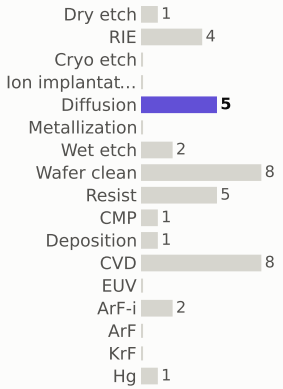
y cut at 57.351 µm



mask for this step

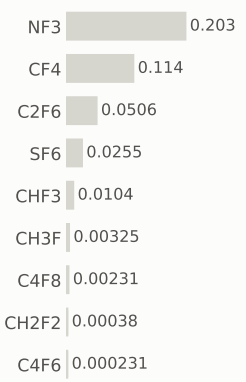
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



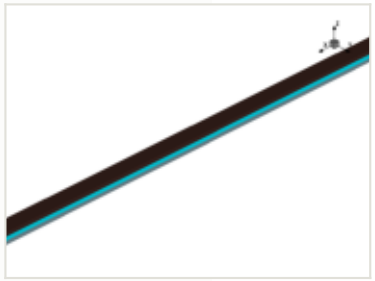
Step #590: CVD

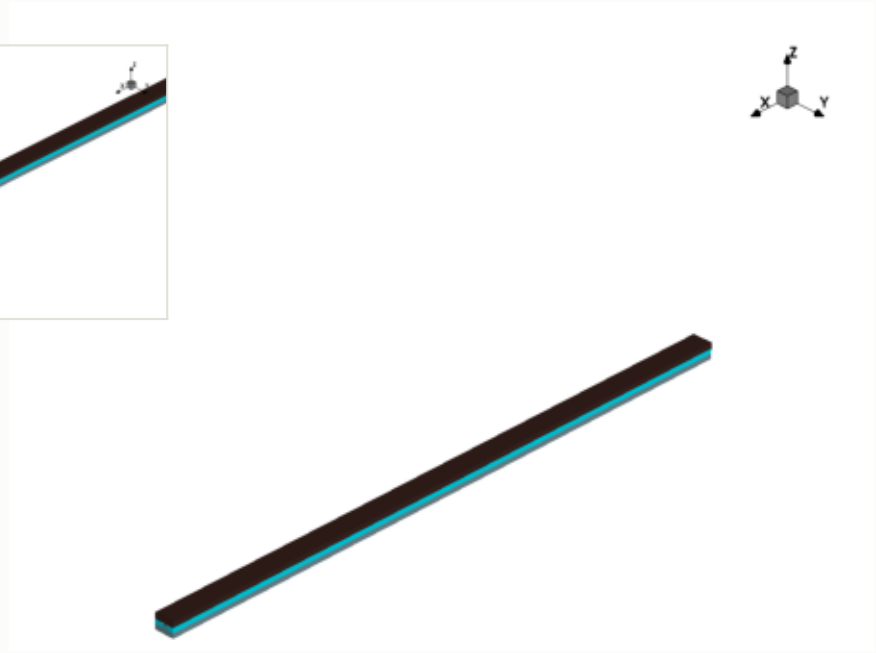
depo_isotropic · CVD

1.31	50.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



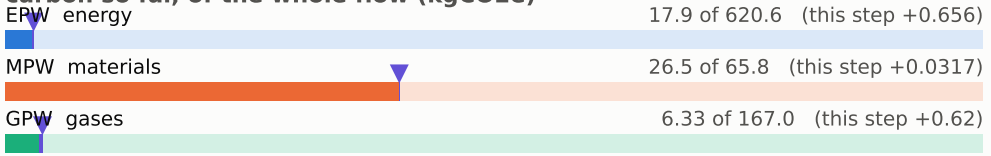


this step

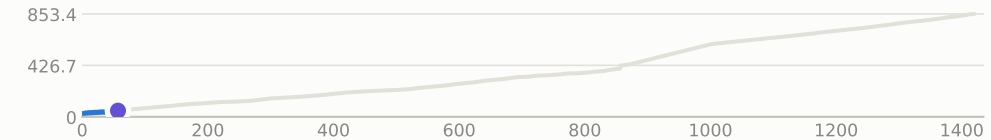
process area	CVD
material	—
thickness	0.2 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

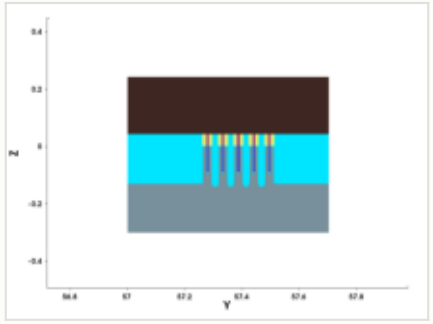
carbon so far, of the whole flow (kgCO2e)



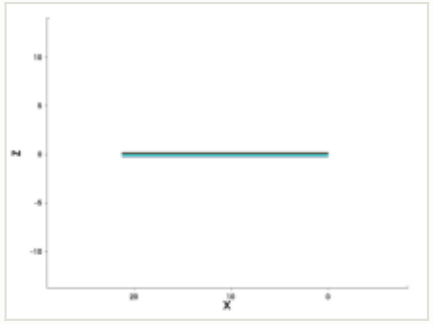
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



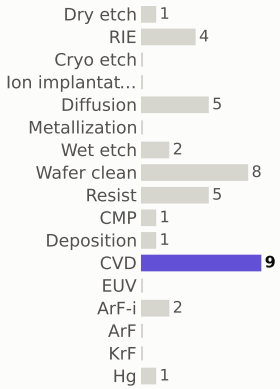
y cut at 57.351 µm



mask for this step

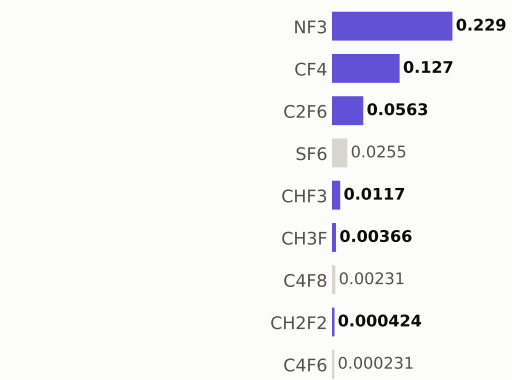
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



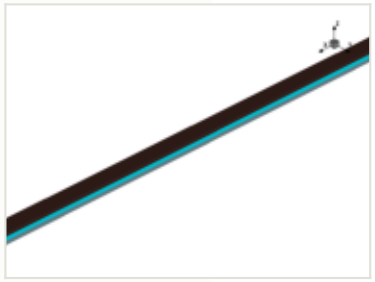
Step #600: RIE

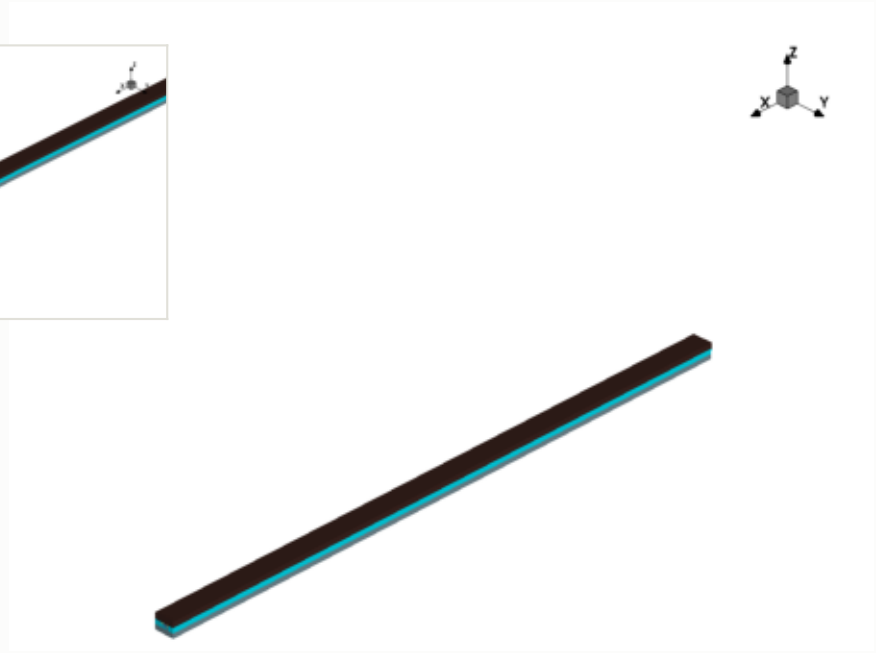
etch_anisotropic · RIE

1.04	51.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



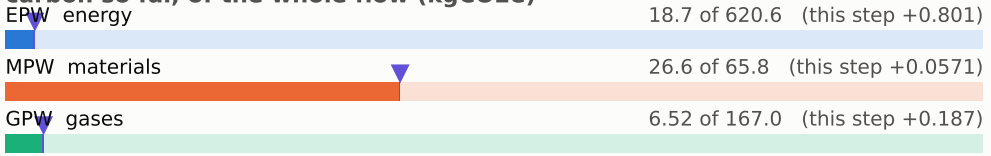


this step

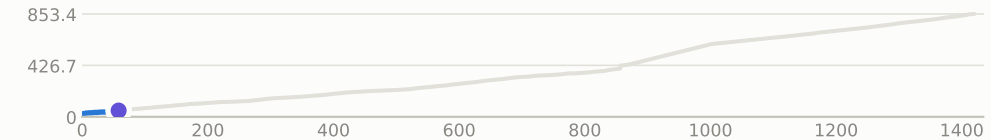
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

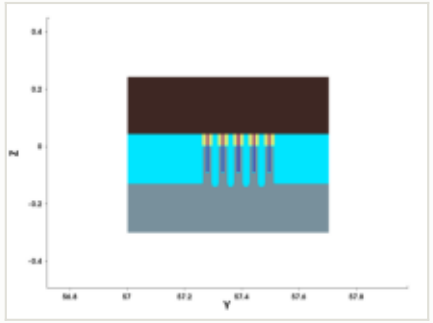
carbon so far, of the whole flow (kgCO2e)



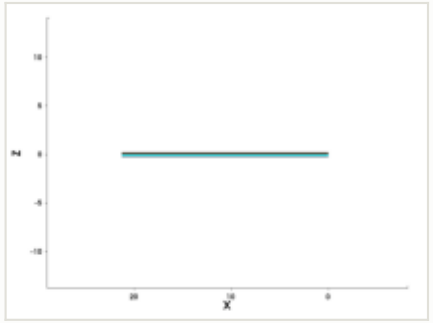
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



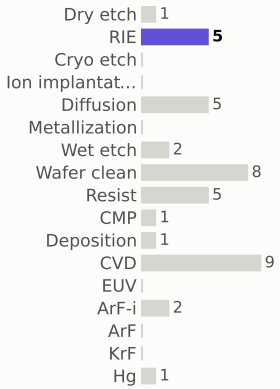
y cut at 57.351 µm



mask for this step

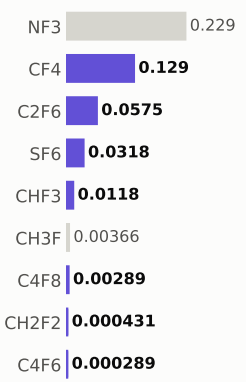
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



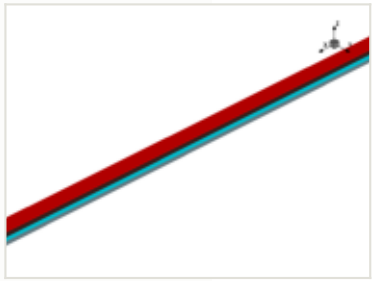
Step #610: CVD

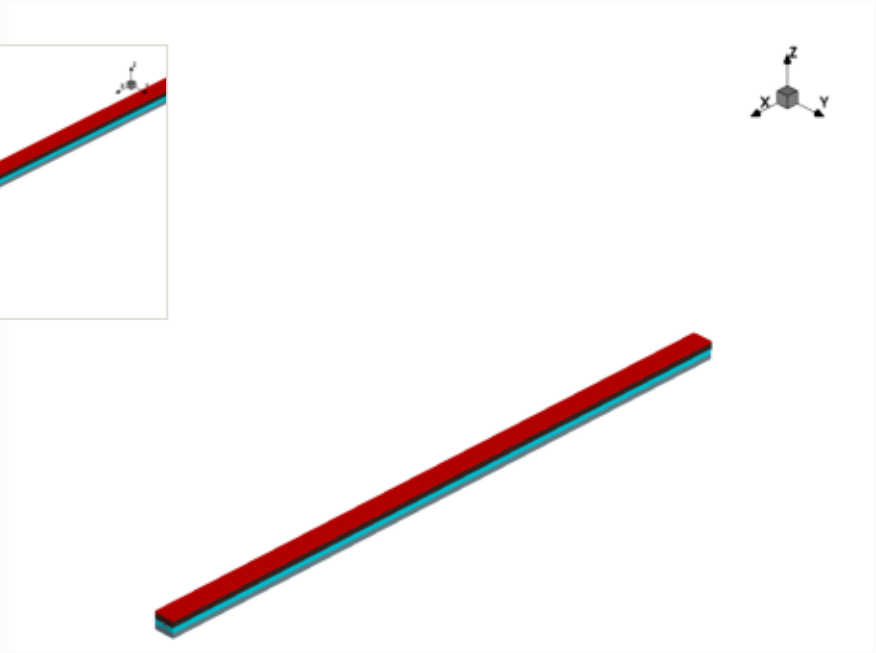
depo_isotropic · CVD

1.31	53.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



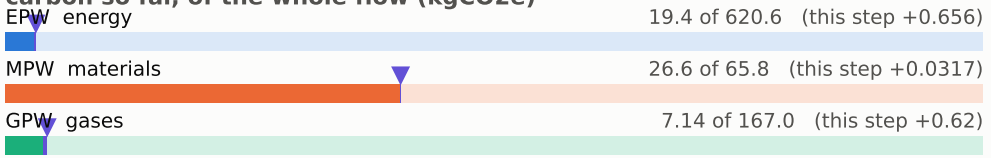


this step

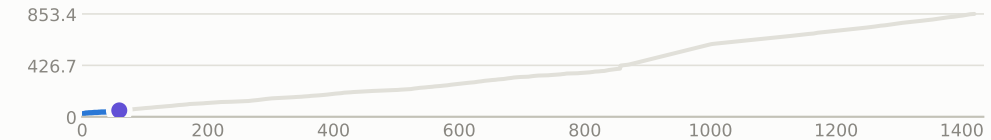
process area	CVD
material	—
thickness	0.045 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

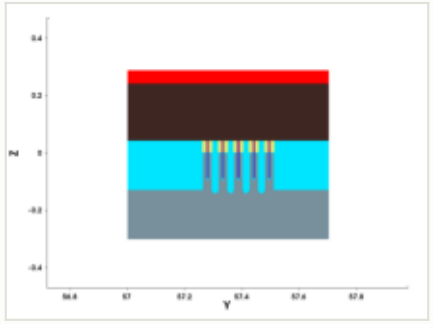
carbon so far, of the whole flow (kgCO2e)



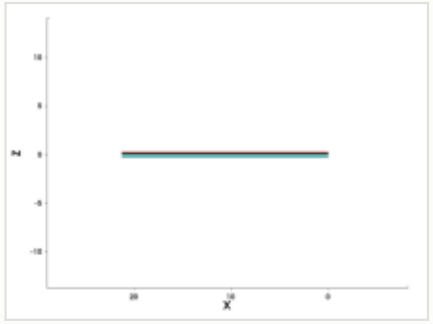
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



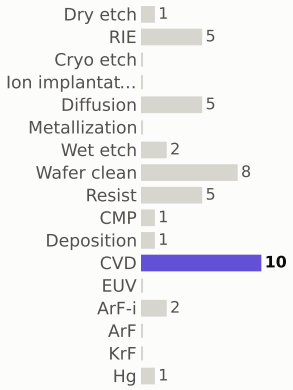
y cut at 57.351 µm



mask for this step

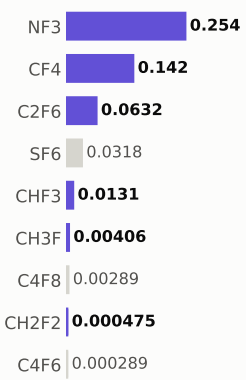
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



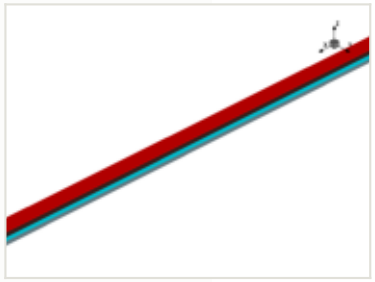
Step #620: Wafer_clean

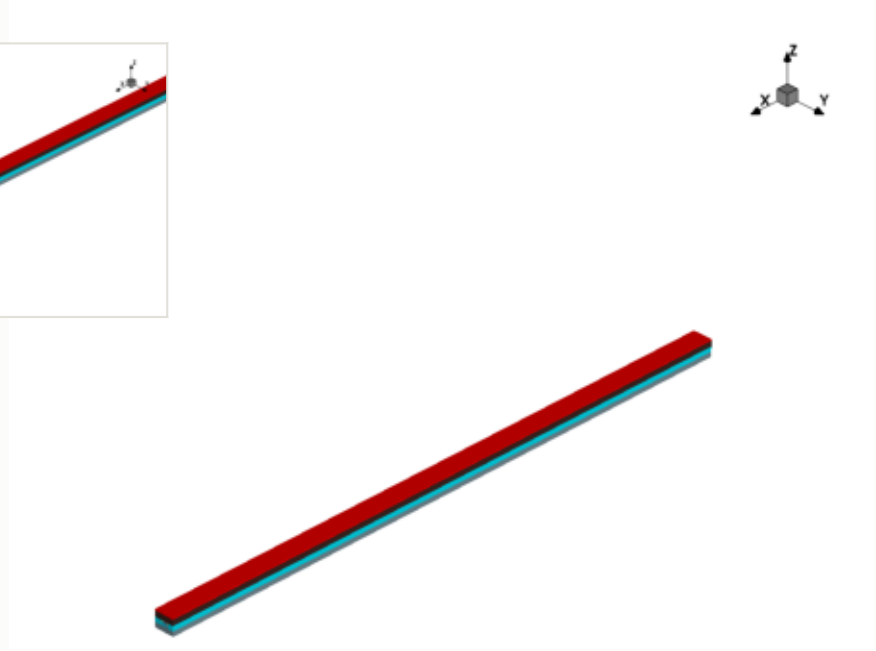
clean · Wafer_clean

0.169	53.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



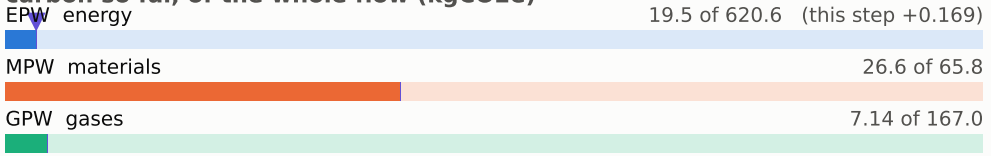


this step

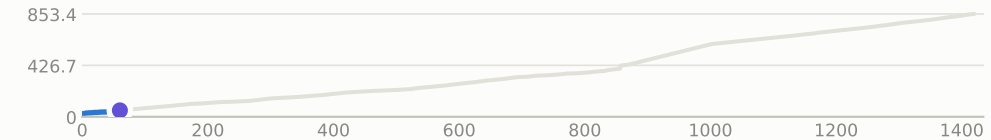
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GPW	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

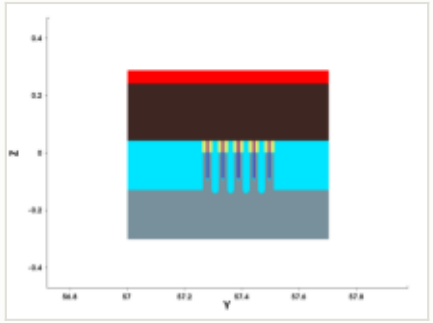
carbon so far, of the whole flow (kgCO2e)



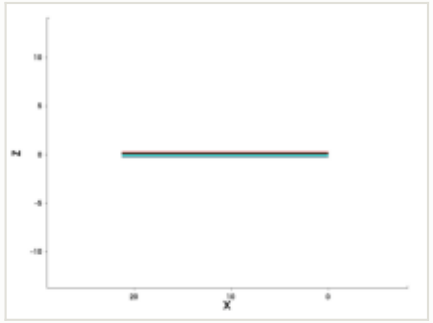
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



y cut at 57.351 µm



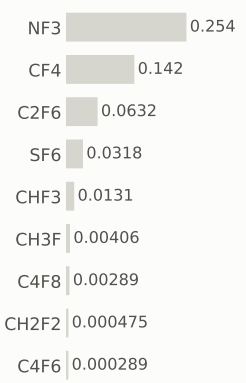
mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)



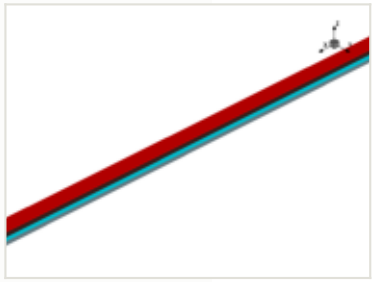
Step #630: Resist

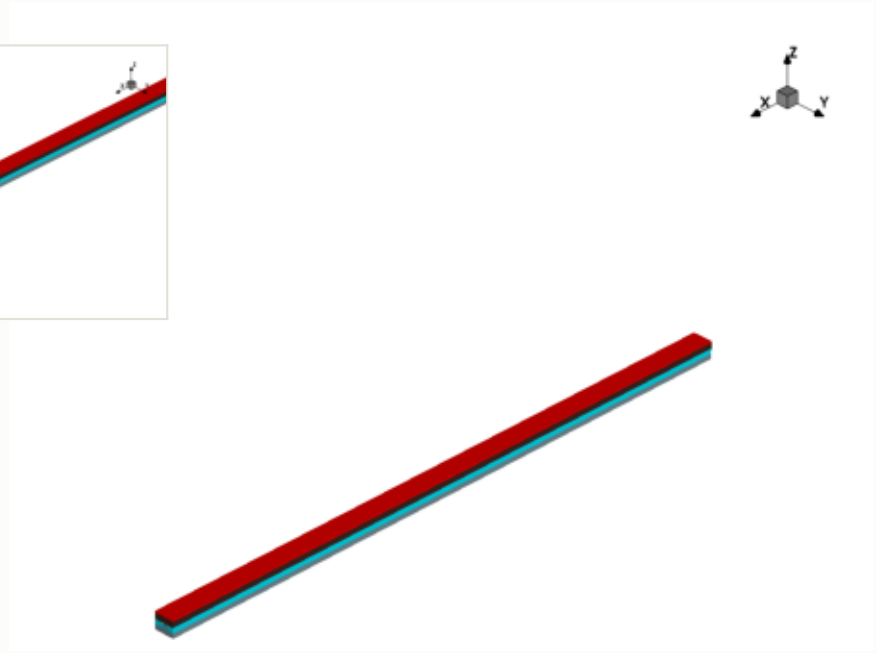
resist · Resist

0.169	53.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





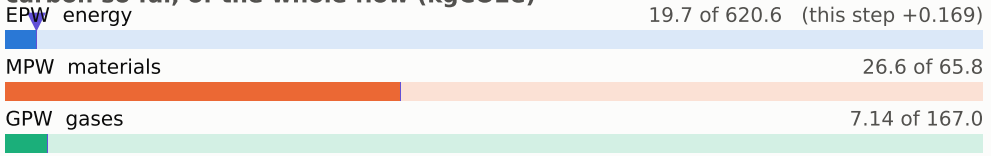
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

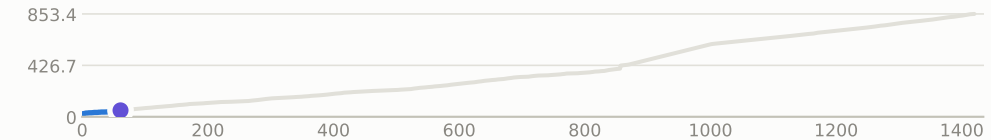
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

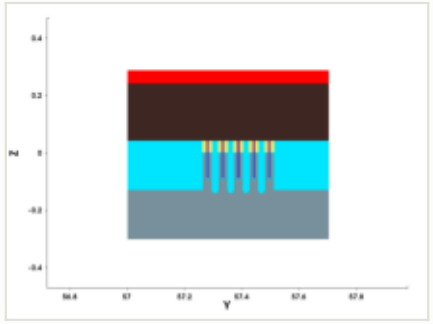
carbon so far, of the whole flow (kgCO2e)



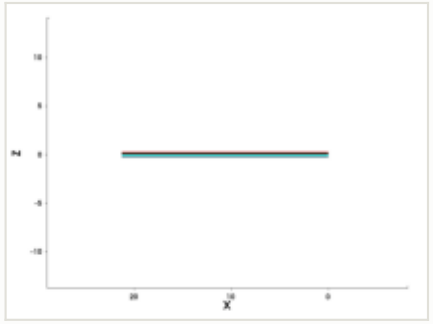
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



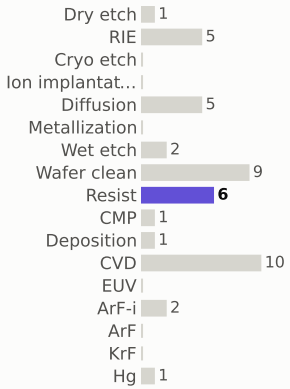
y cut at 57.351 µm



mask for this step

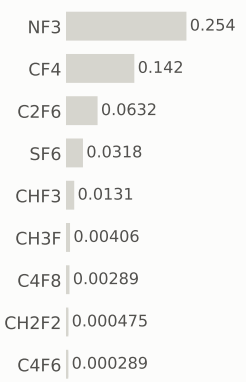
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



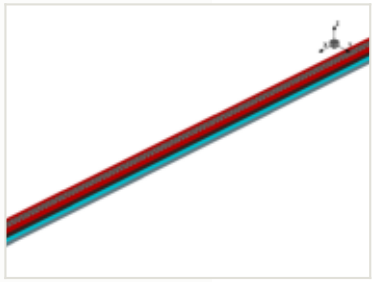
Step #640: ArF-i

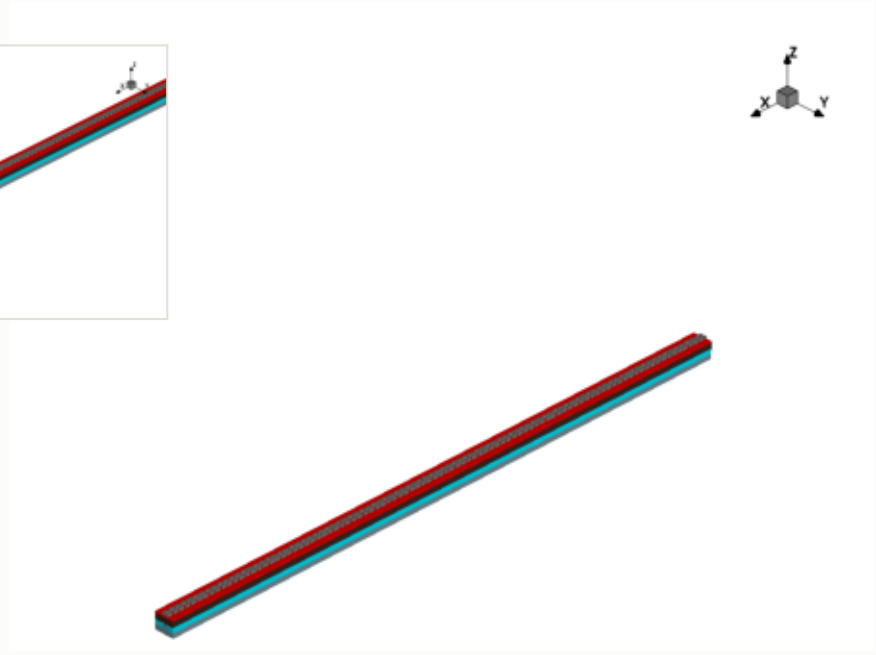
litho_patterning · ArF-i

0.217	53.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



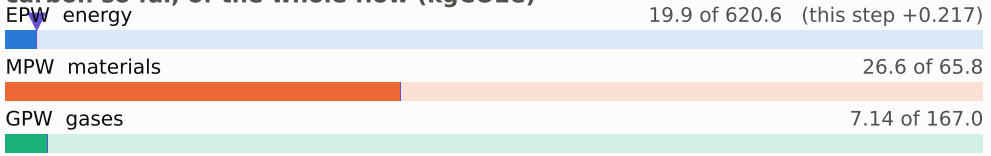


this step

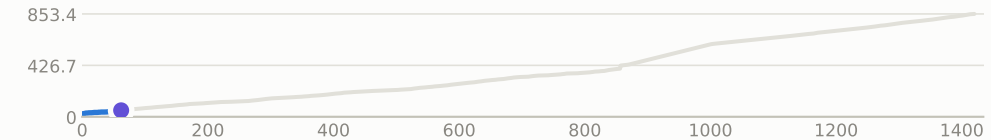
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

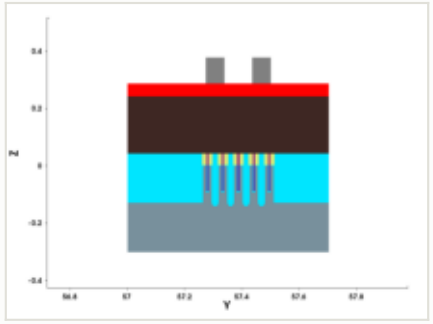
carbon so far, of the whole flow (kgCO2e)



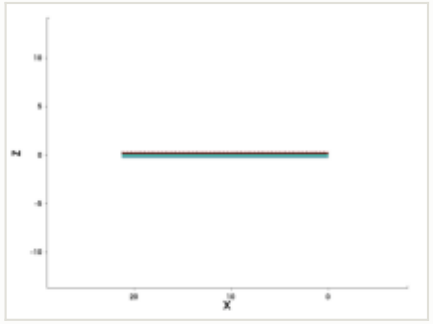
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

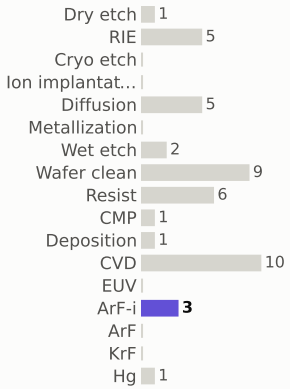


y cut at 57.351 µm



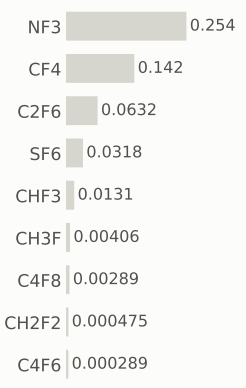
mask: layer 2011:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



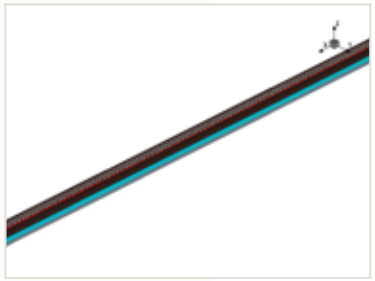
Step #650: RIE

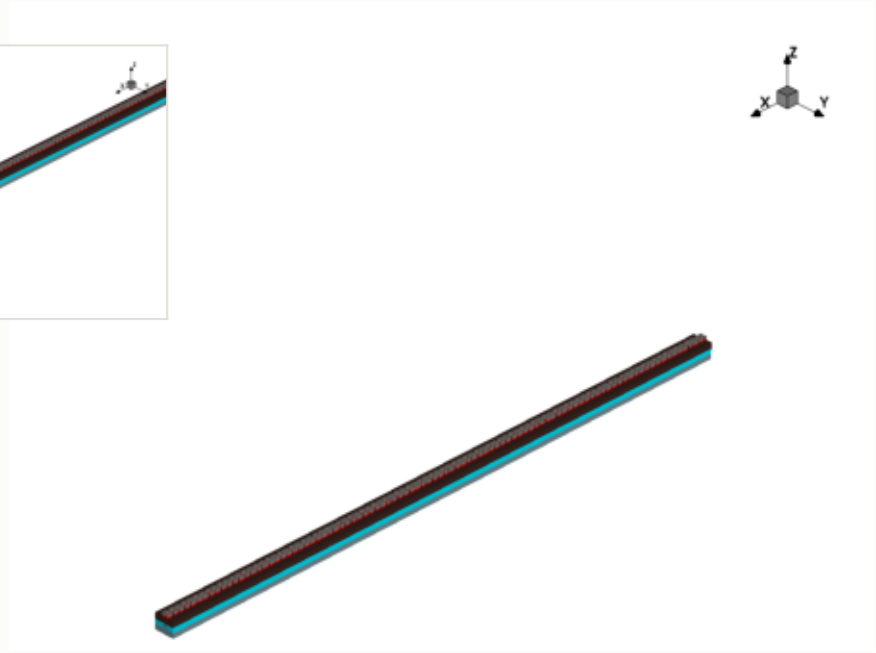
etch_anisotropic · RIE

1.04	54.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

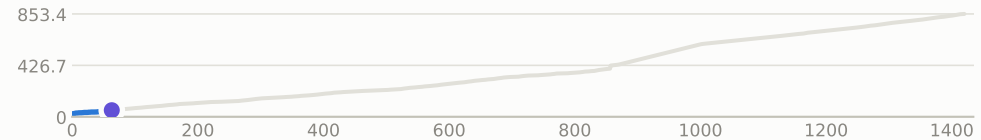
carbon so far, of the whole flow (kgCO2e)

EPW energy — 20.7 of 620.6 (this step +0.801)

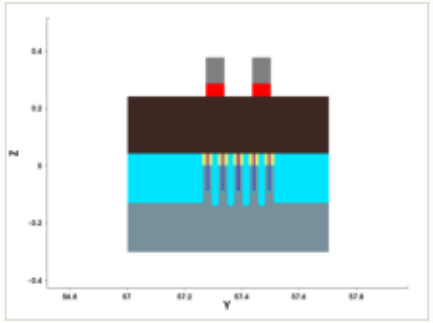
MPW materials — 26.7 of 65.8 (this step +0.0571)

GPW gases — 7.33 of 167.0 (this step +0.187)

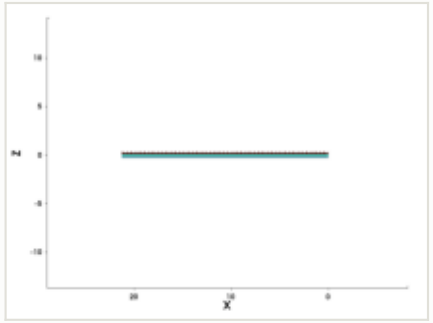
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



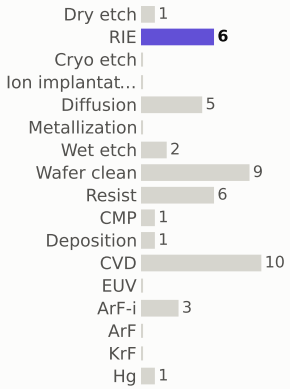
y cut at 57.351 µm



mask for this step

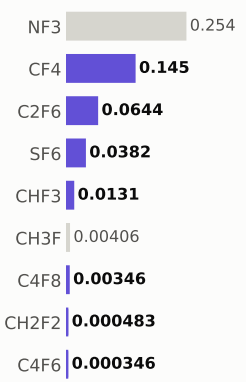
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



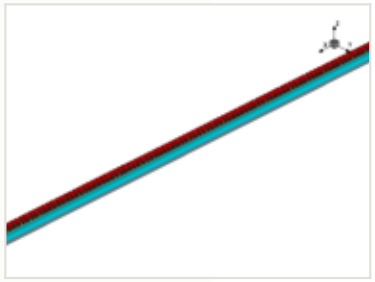
Step #660: Spx_only

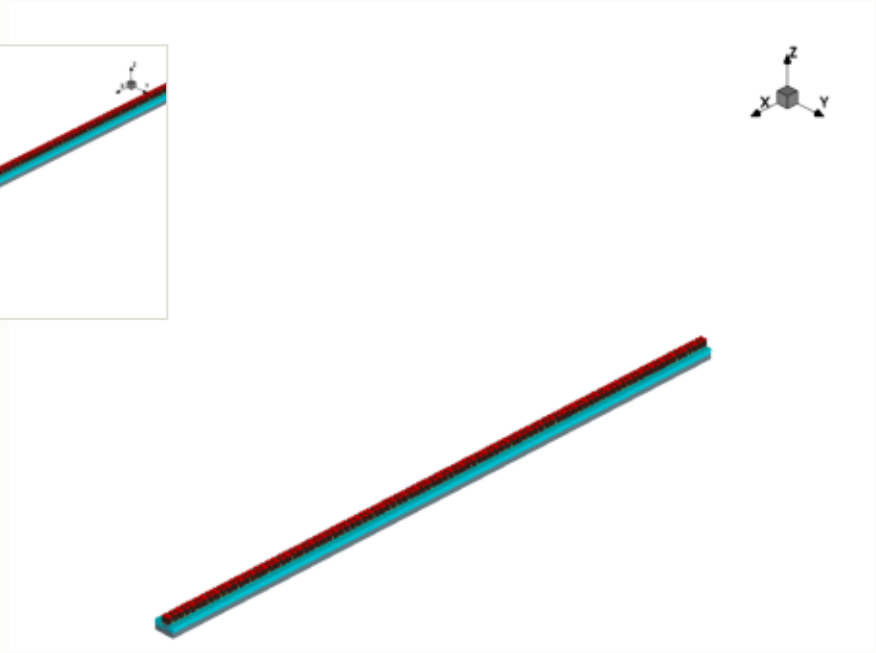
etch_anisotropic · Spx_only

0	54.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



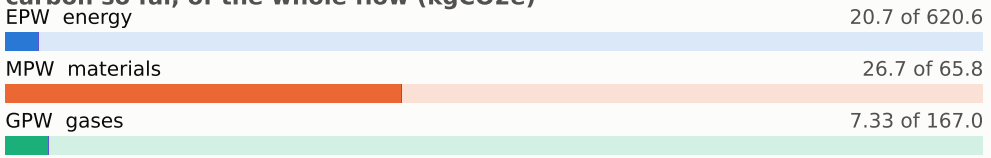


this step

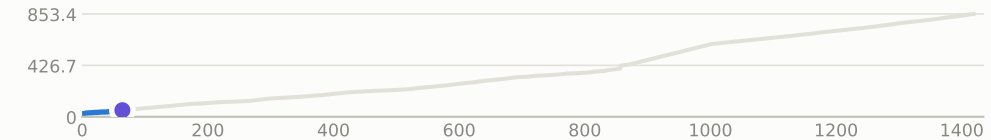
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

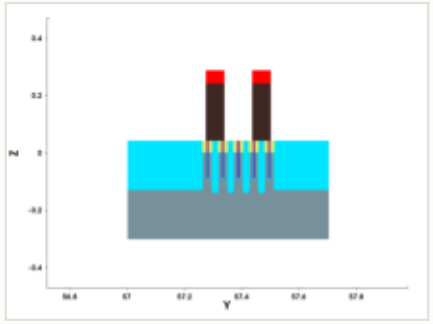
carbon so far, of the whole flow (kgCO2e)



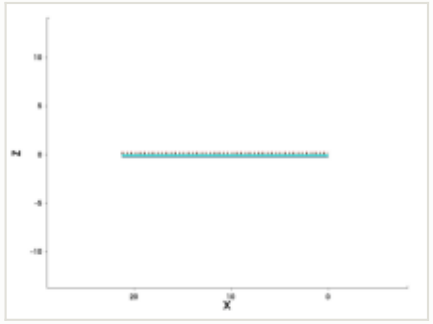
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



y cut at 57.351 µm



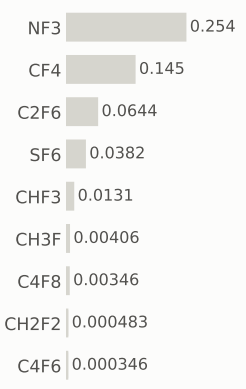
mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)



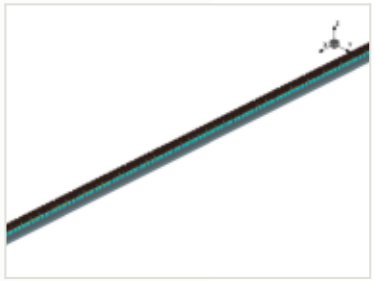
Step #670: Spx_only

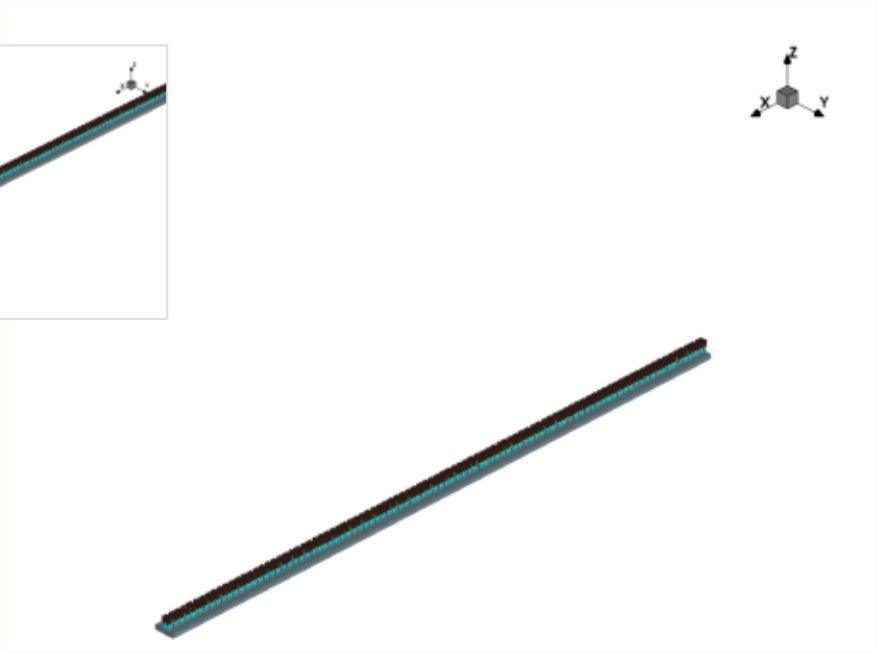
etch_anisotropic · Spx_only

0	54.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

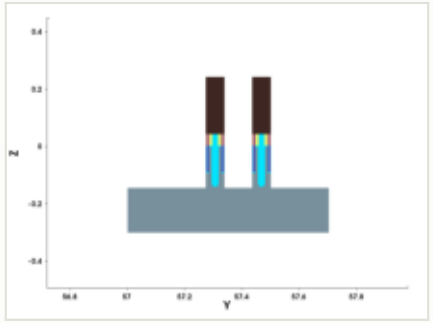
3D structure after this step

zoomed to fit

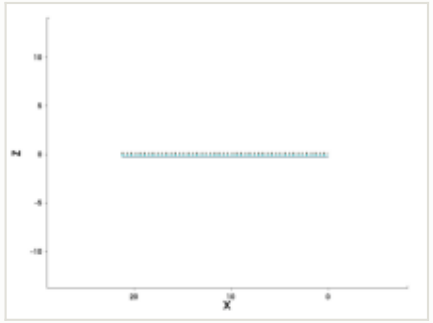




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

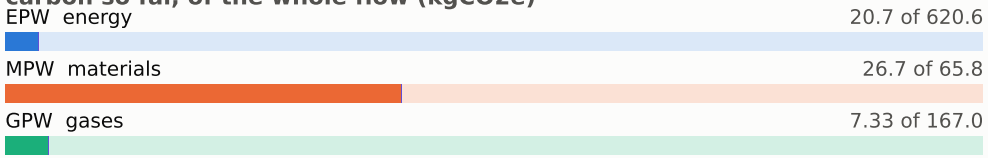
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

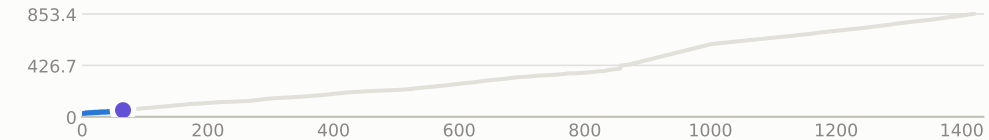
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

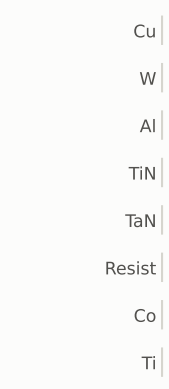


embodied carbon up to each step (kgCO2e per wafer)

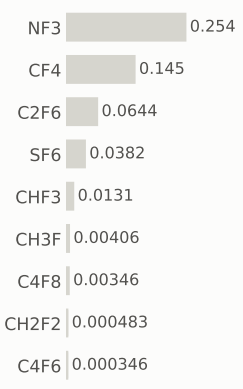


steps by process area

materials deposited (g)



gases emitted (g)



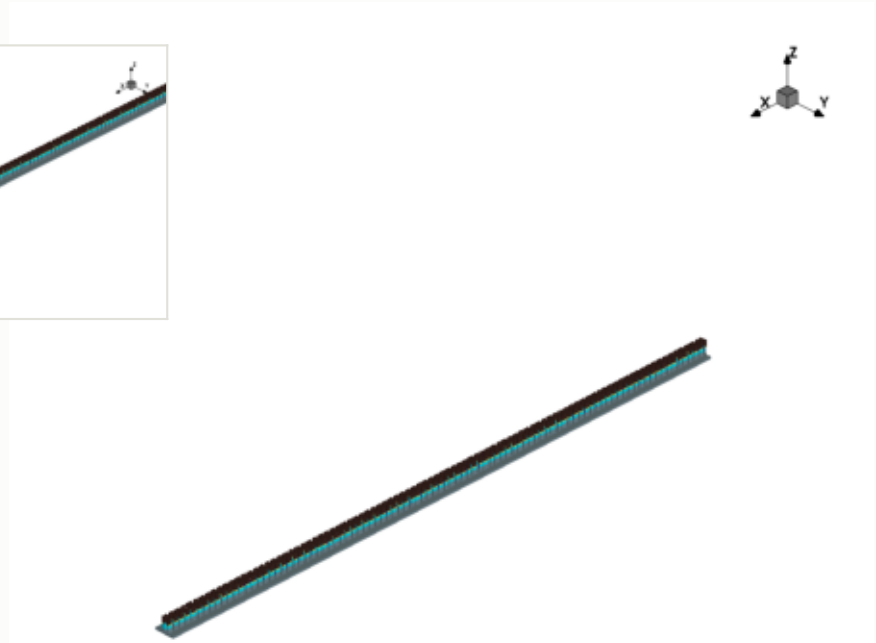
Step #680: Spx_only

etch_anisotropic · Spx_only

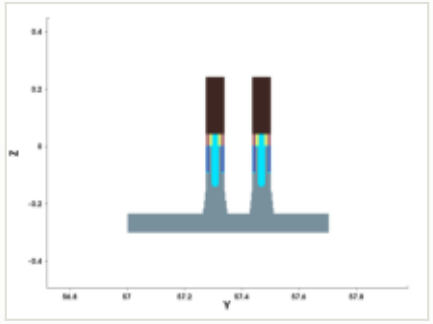
0	54.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

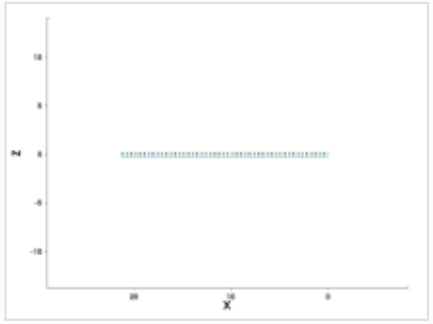
zoomed to fit



x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

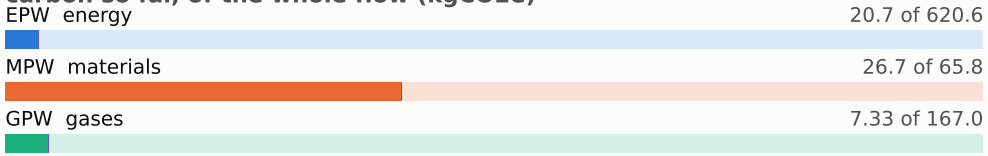
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

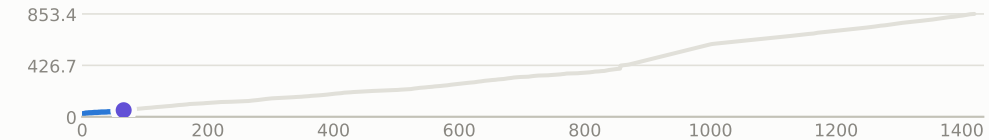
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

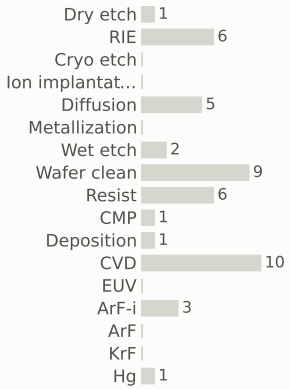
carbon so far, of the whole flow (kgCO2e)



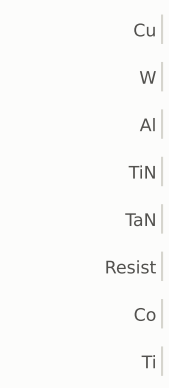
embodied carbon up to each step (kgCO2e per wafer)



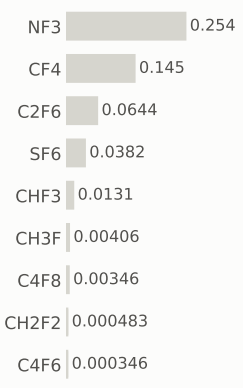
steps by process area



materials deposited (g)



gases emitted (g)



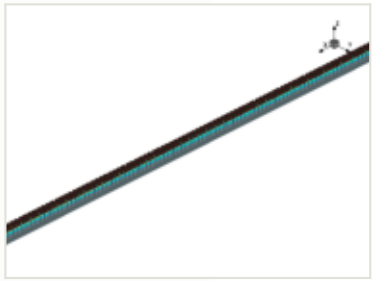
Step #690: Spx_only

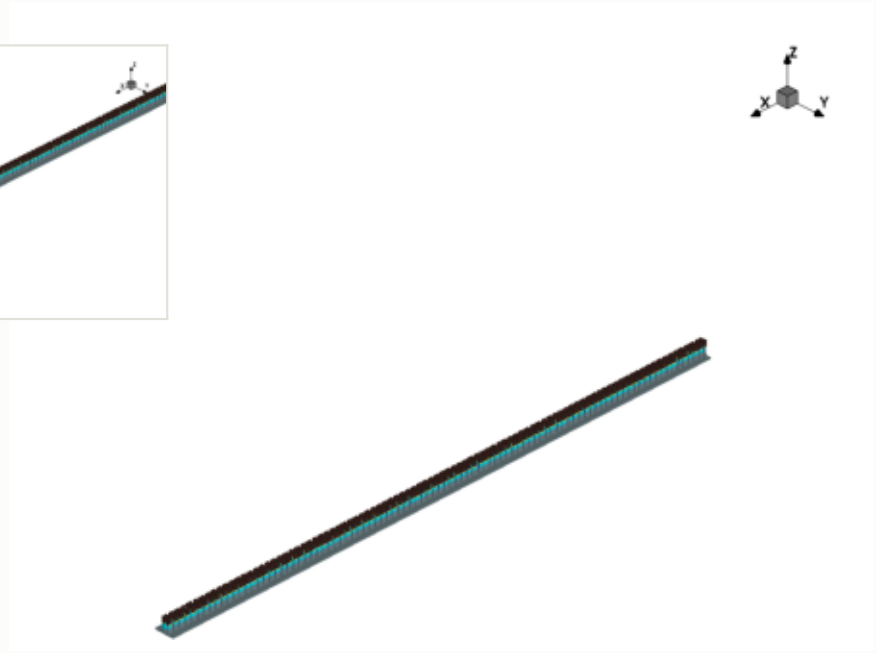
etch_microloading · Spx_only

0	54.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





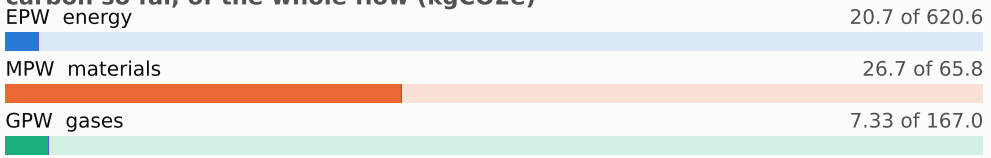
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

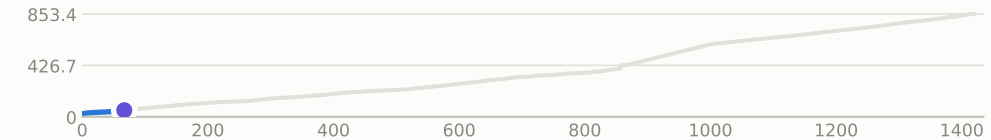
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

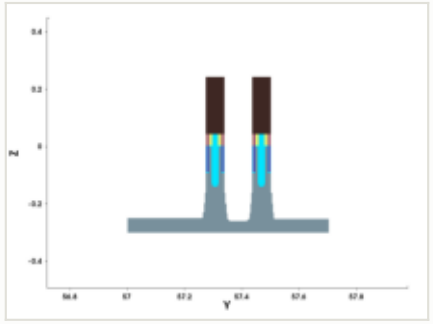
carbon so far, of the whole flow (kgCO2e)



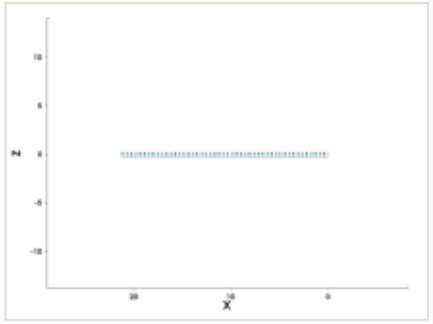
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



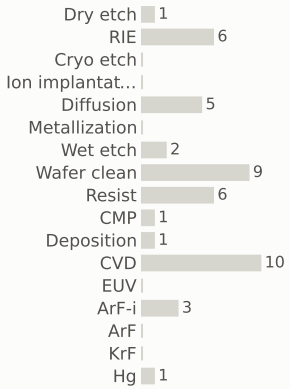
y cut at 57.351 µm



mask for this step

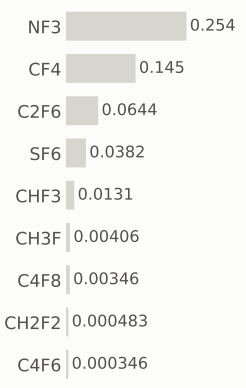
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



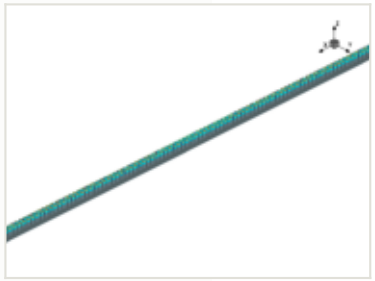
Step #700: Spx_only

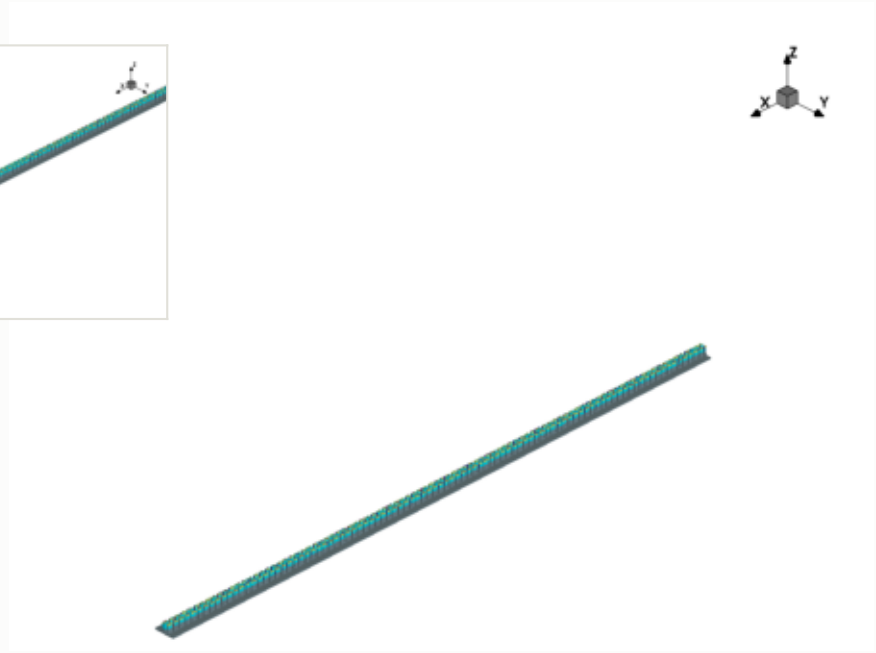
etch_strip · Spx_only

0	54.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

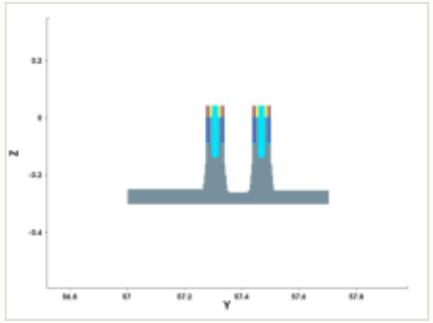
3D structure after this step

zoomed to fit

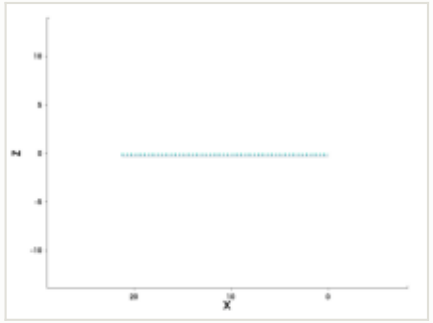




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

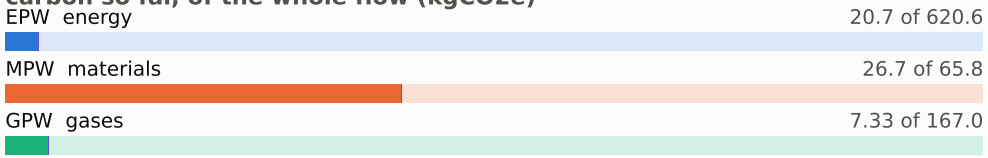
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

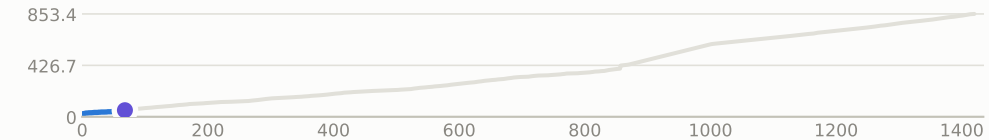
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)



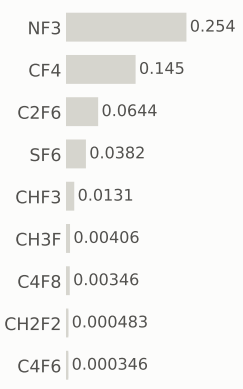
embodied carbon up to each step (kgCO2e per wafer)



steps by process area

materials deposited (g)

gases emitted (g)



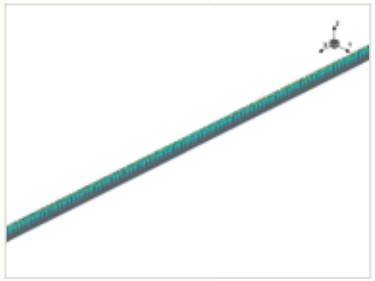
Step #710: Wafer_clean

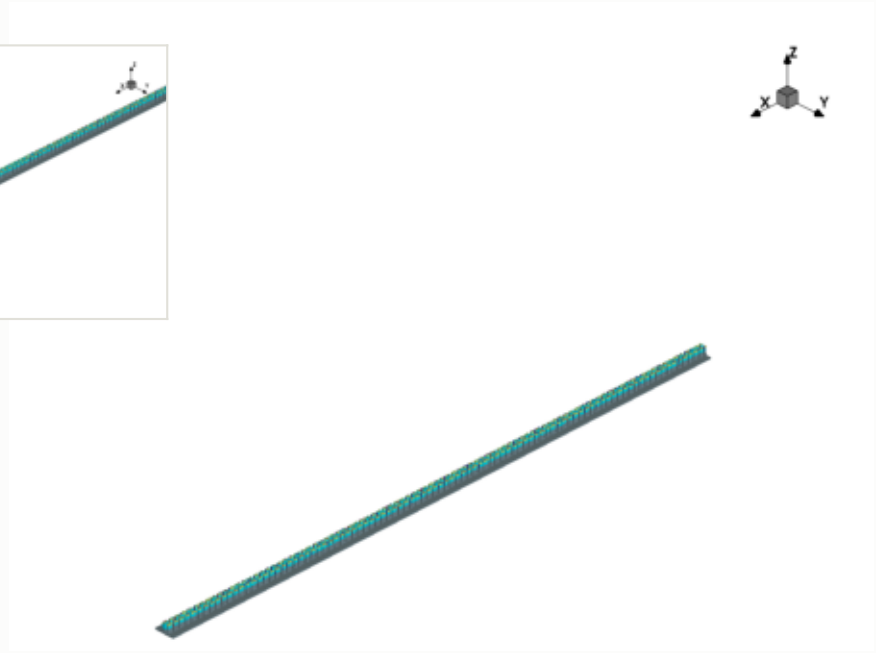
clean · Wafer_clean

0.169	54.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





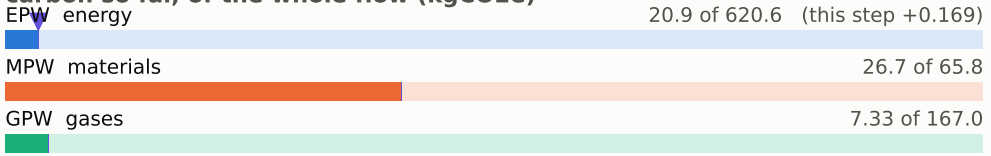
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

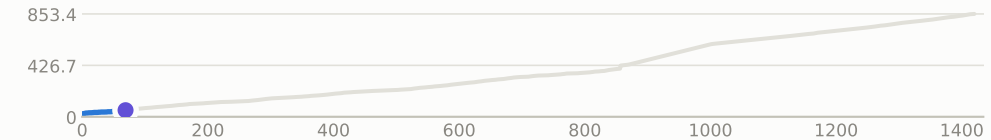
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

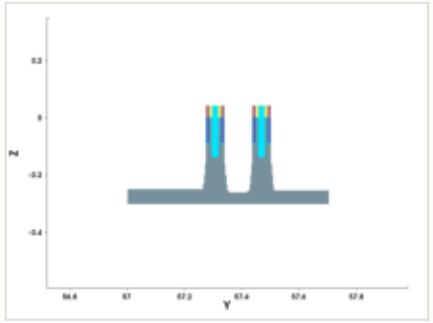
carbon so far, of the whole flow (kgCO2e)



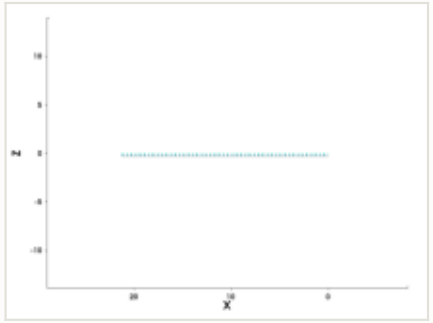
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



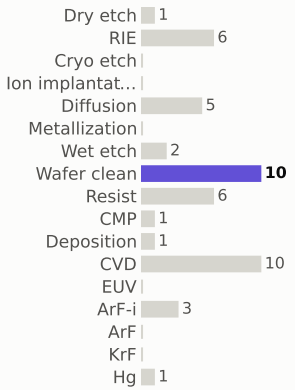
y cut at 57.351 µm



mask for this step

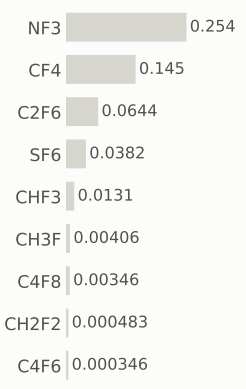
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



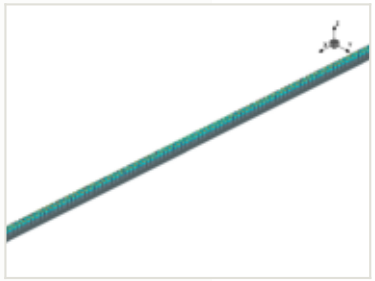
Step #720: Wafer_clean

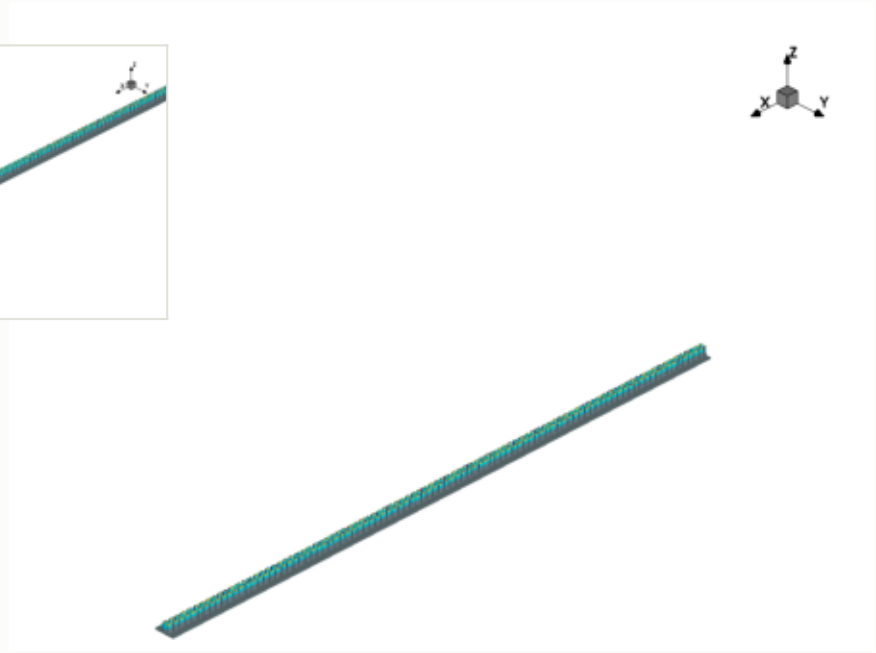
clean · Wafer_clean

0.169	55	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





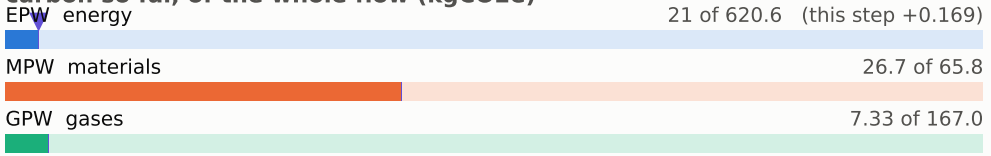
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

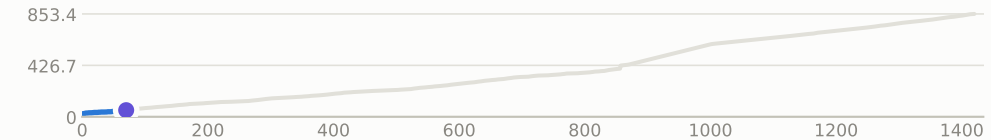
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

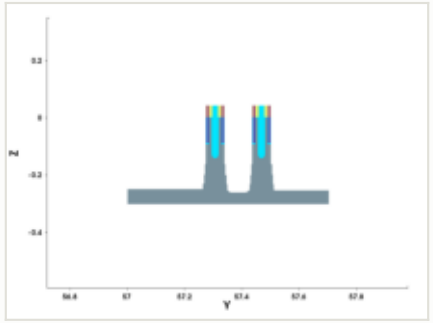
carbon so far, of the whole flow (kgCO2e)



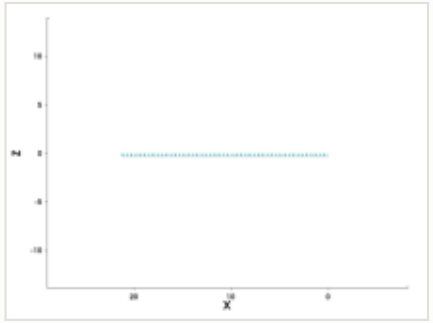
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



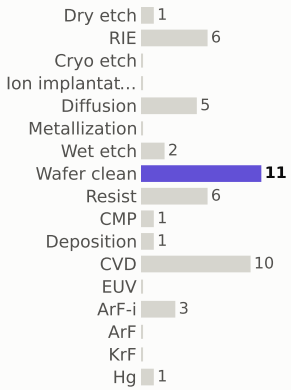
y cut at 57.351 µm



mask for this step

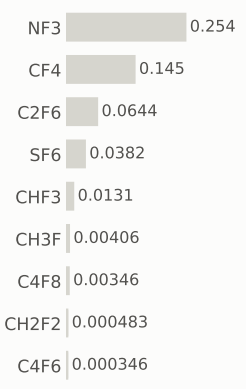
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



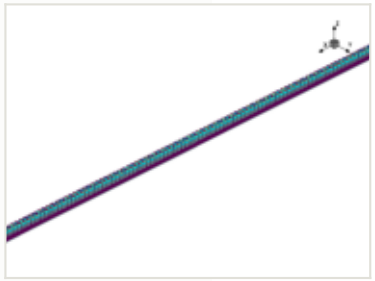
Step #730: CVD

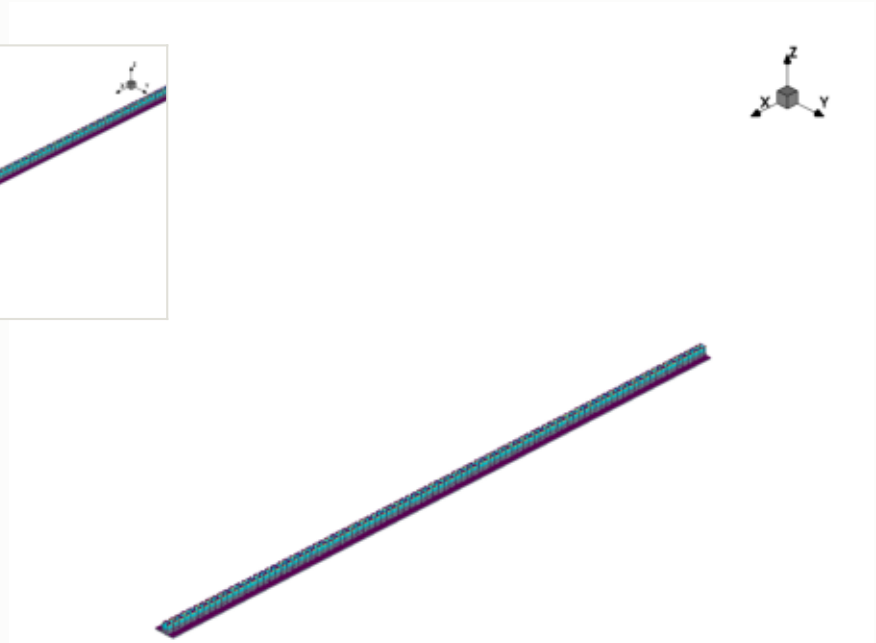
depo_isotropic · CVD

1.31	56.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



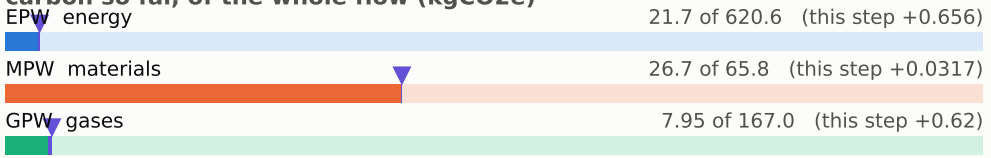


this step

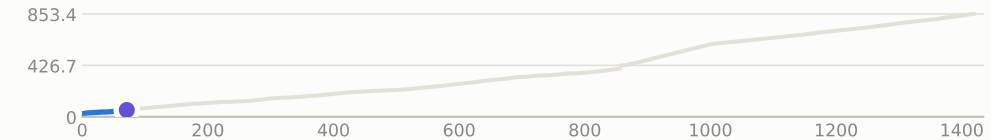
process area	CVD
material	—
thickness	0.0015 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

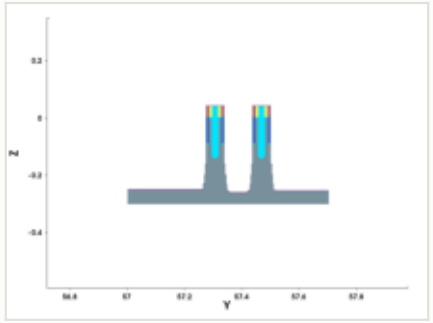
carbon so far, of the whole flow (kgCO2e)



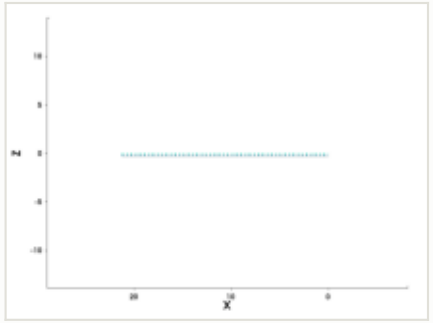
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



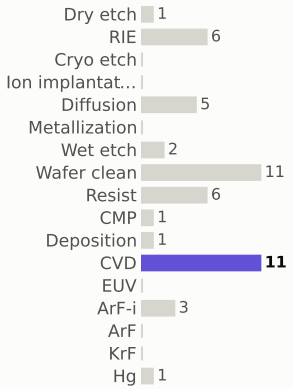
y cut at 57.351 µm



mask for this step

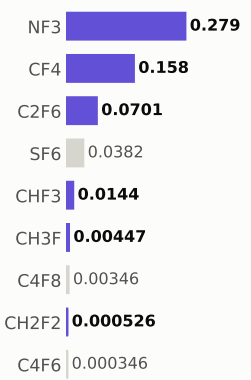
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



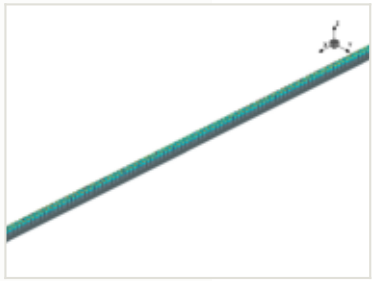
Step #740: Spx_only

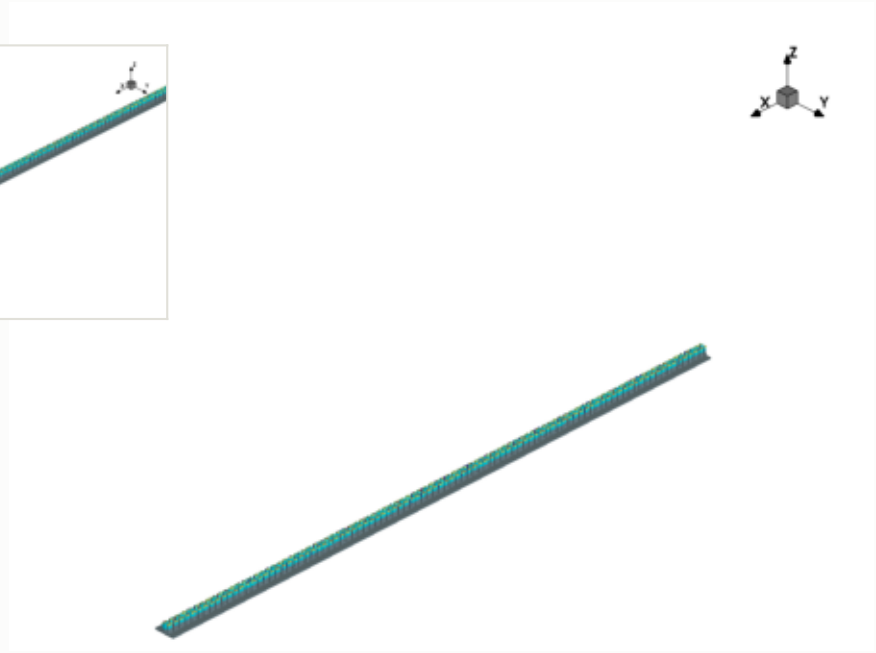
etch_isotropic · Spx_only

0	56.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

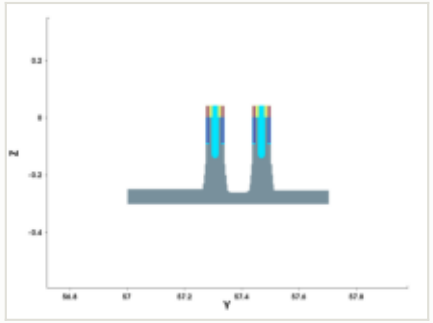
3D structure after this step

zoomed to fit

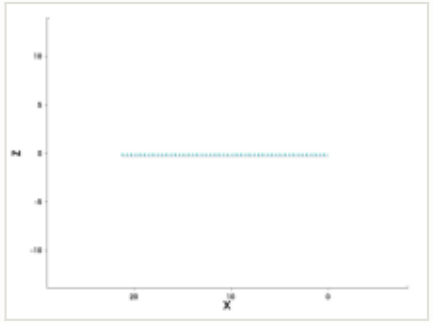




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

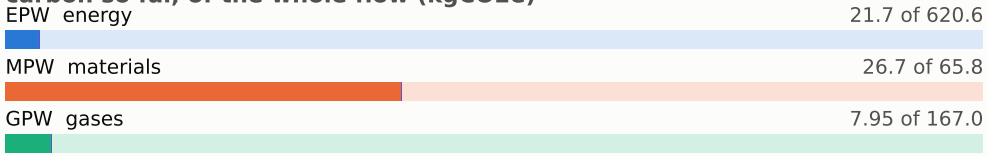
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

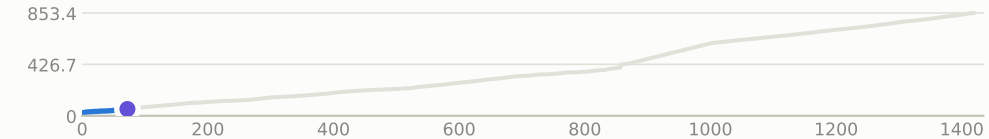
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

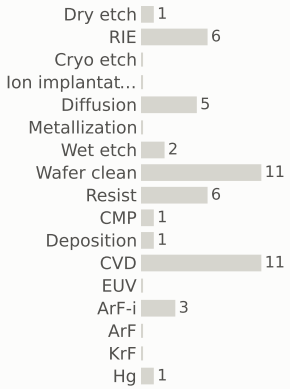
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

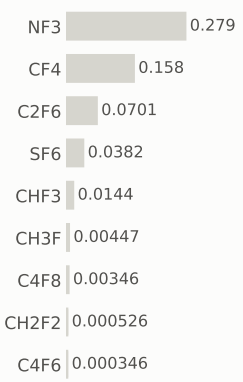


steps by process area



materials deposited (g)

gases emitted (g)



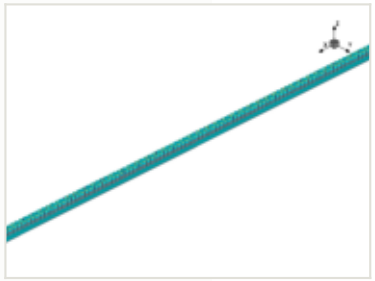
Step #750: Spx_only

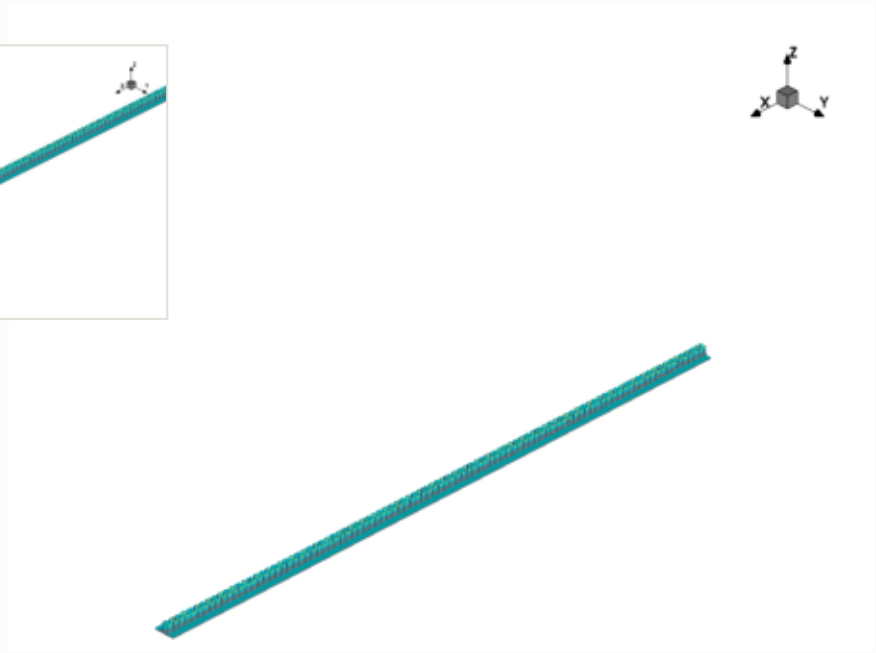
depo_isotropic · Spx_only

0	56.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

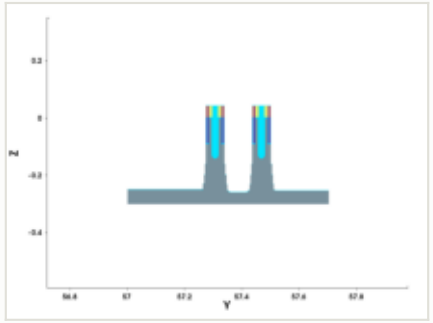
3D structure after this step

zoomed to fit

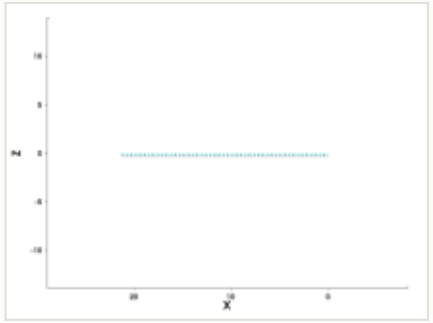




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

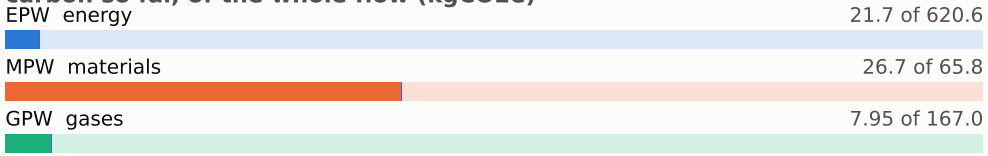
this step

process area	Spx_only
material	—
thickness	0.0015 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

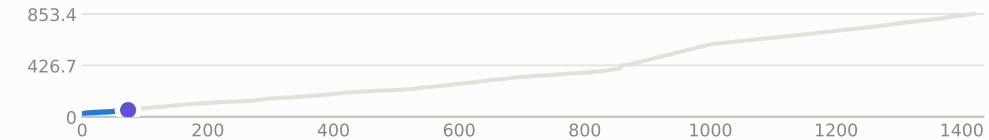
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

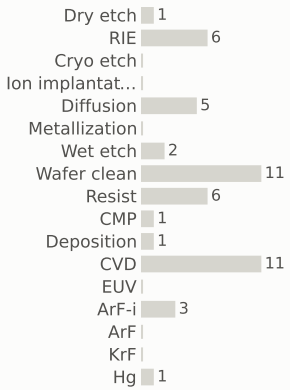
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

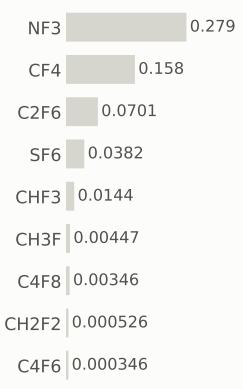


steps by process area



materials deposited (g)

gases emitted (g)



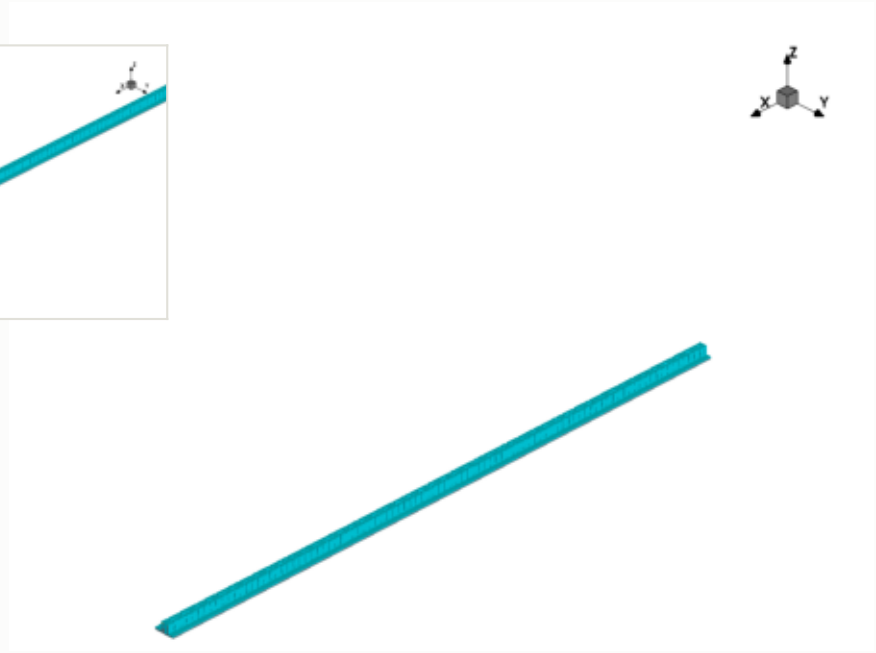
Step #760: CVD

depo_isotropic · CVD

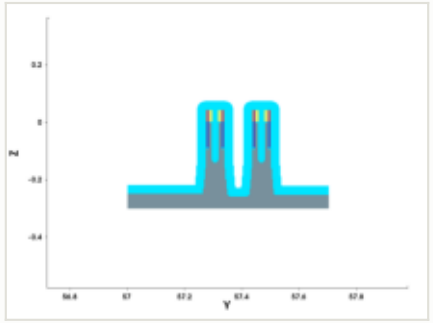
1.31	57.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

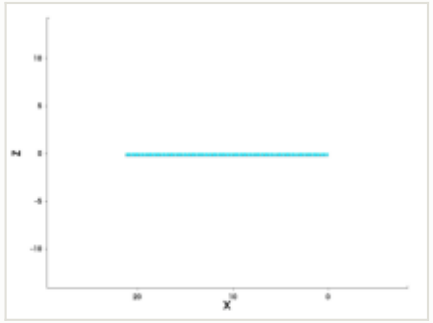
zoomed to fit



x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

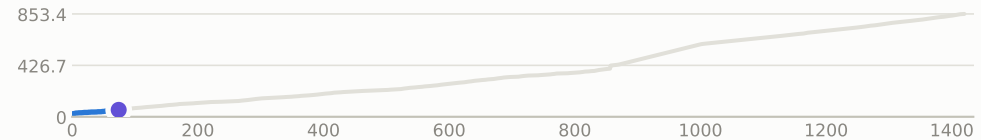
carbon so far, of the whole flow (kgCO2e)

EPW energy — 22.4 of 620.6 (this step +0.656)

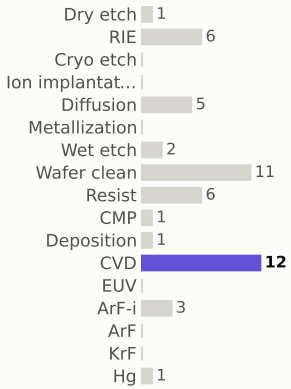
MPW materials — 26.7 of 65.8 (this step +0.0317)

GPW gases — 8.57 of 167.0 (this step +0.62)

embodied carbon up to each step (kgCO2e per wafer)

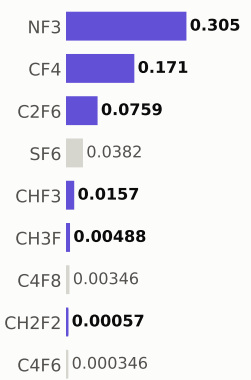


steps by process area



materials deposited (g)

gases emitted (g)



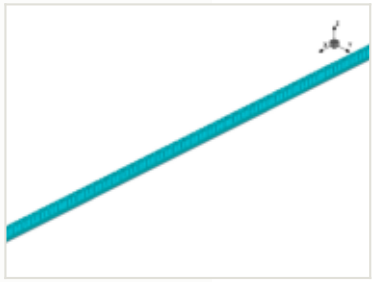
Step #770: RIE

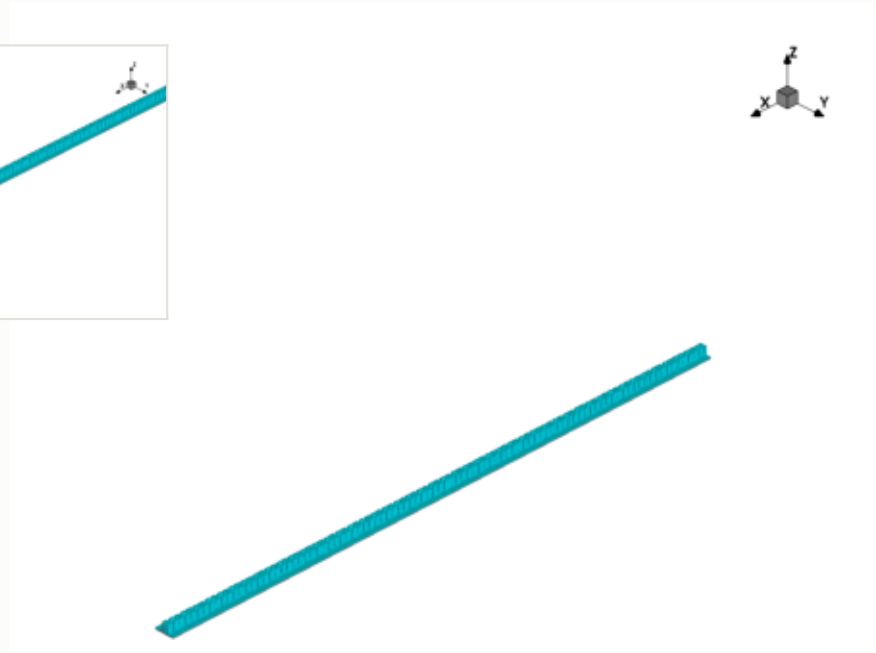
etch_isotropic · RIE

1.04	58.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



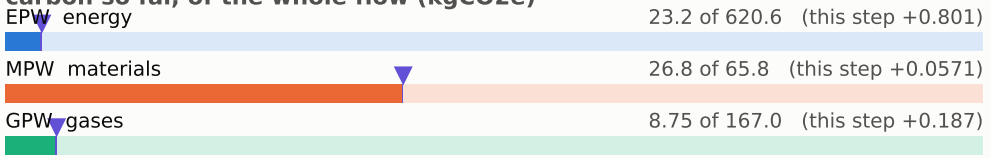


this step

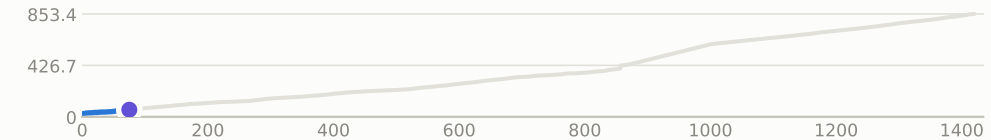
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

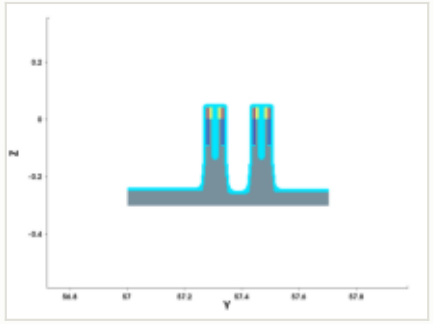
carbon so far, of the whole flow (kgCO2e)



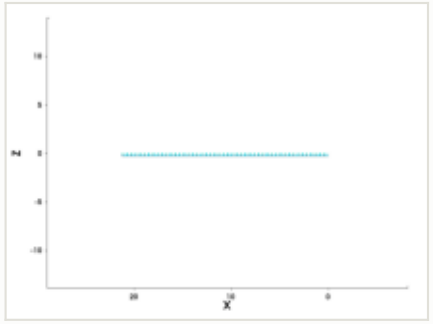
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



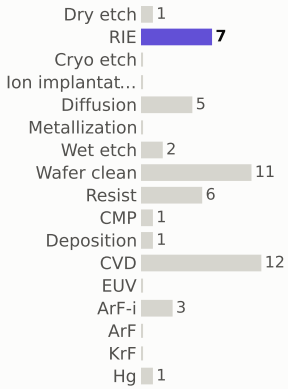
y cut at 57.351 µm



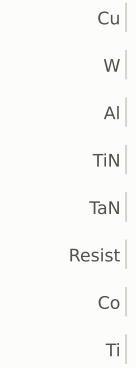
mask for this step

no mask for this step

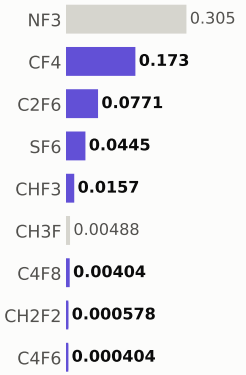
steps by process area



materials deposited (g)



gases emitted (g)



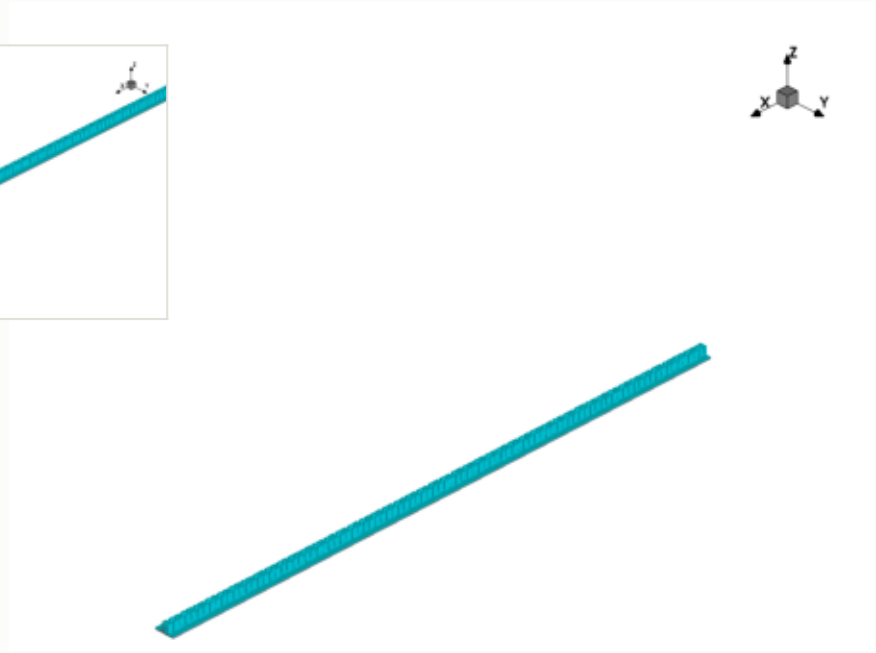
Step #780: Wafer_clean

clean · Wafer_clean

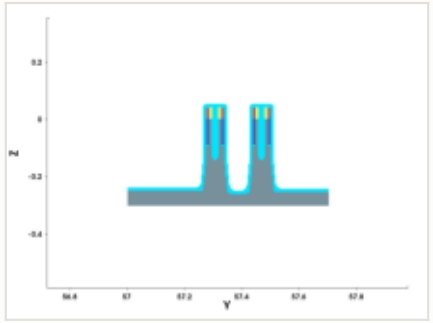
0.169	58.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

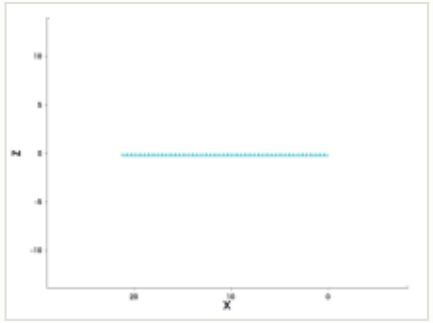
zoomed to fit



x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

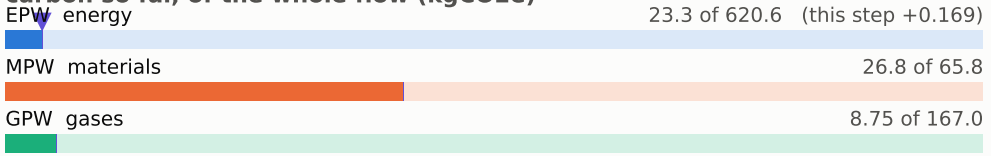
no mask for this step

this step

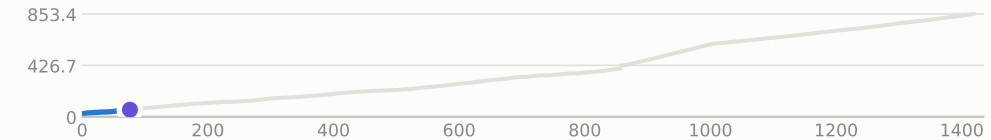
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

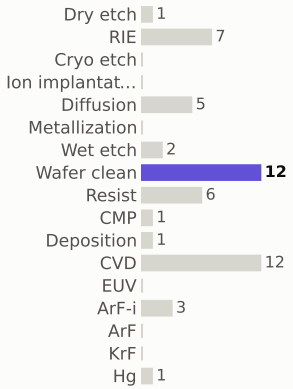
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

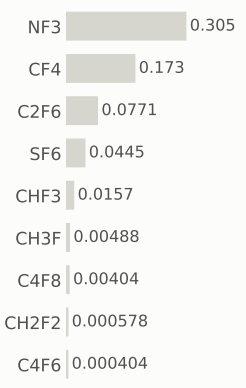


steps by process area



materials deposited (g)

gases emitted (g)



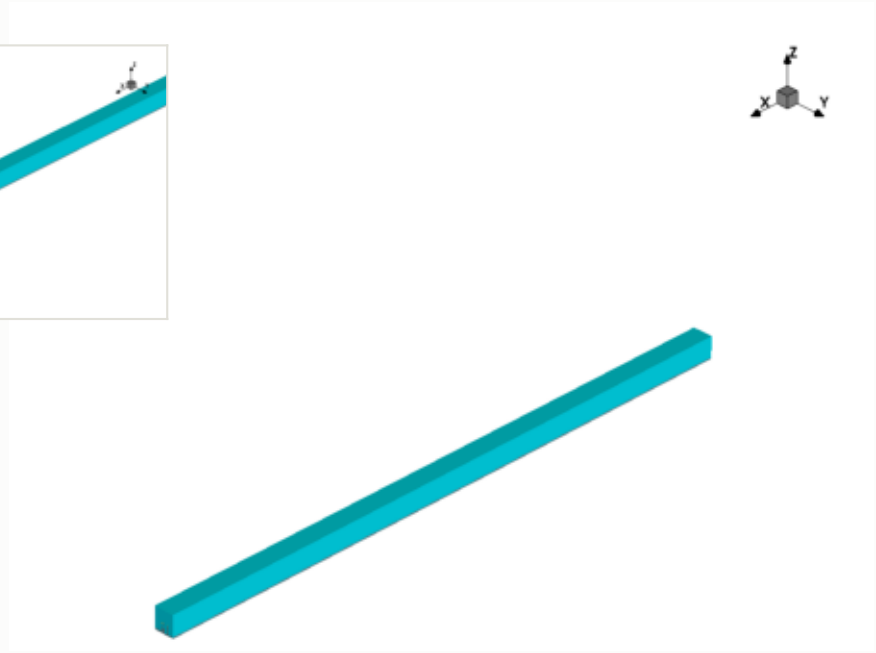
Step #790: CVD

depo_fill · CVD

1.31	60.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

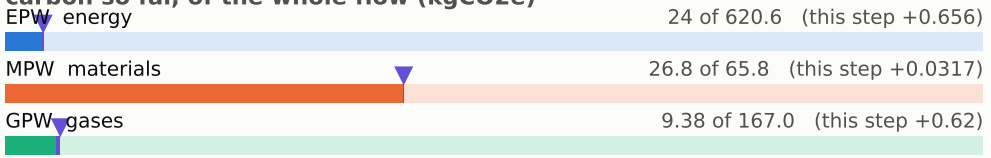


this step

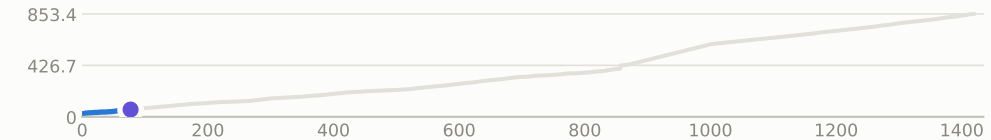
process area	CVD
material	—
thickness	0.4 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

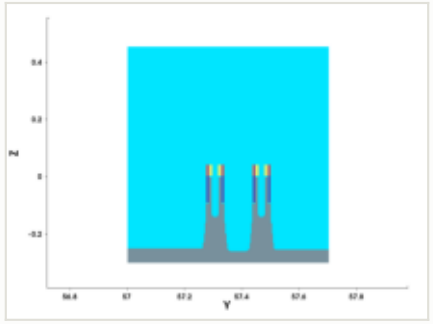
carbon so far, of the whole flow (kgCO2e)



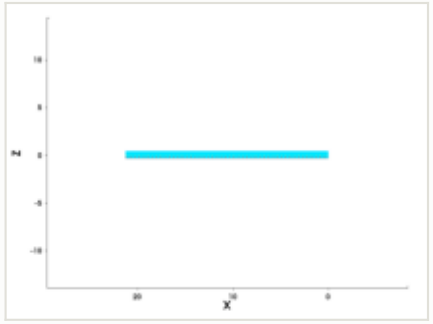
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



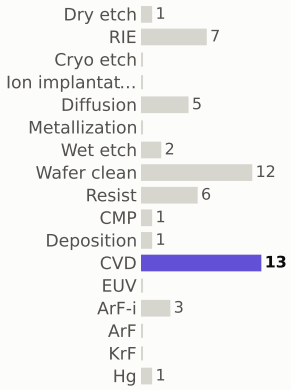
y cut at 57.351 µm



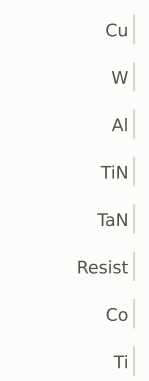
mask for this step

no mask for this step

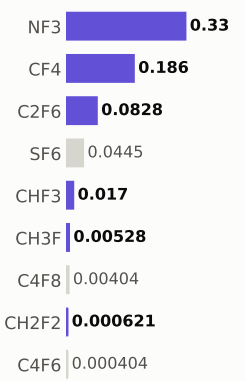
steps by process area



materials deposited (g)



gases emitted (g)



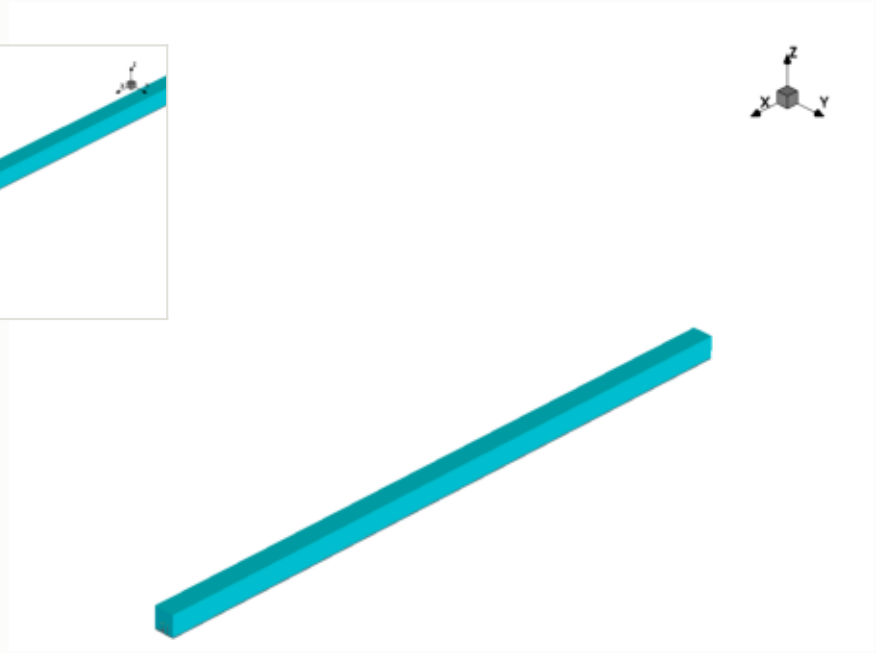
Step #800: Wafer_clean

clean · Wafer_clean

0.169	60.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



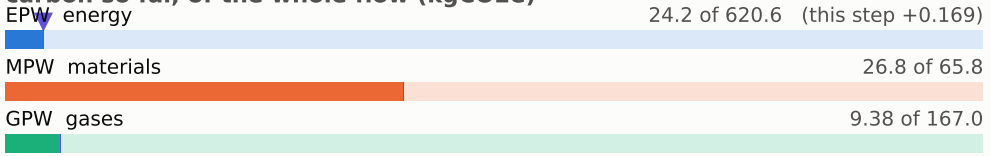
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

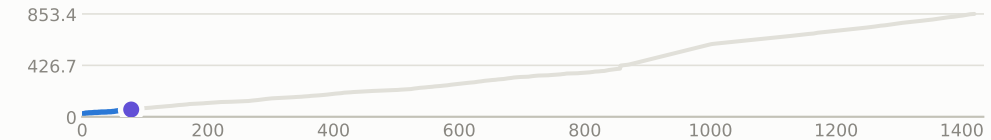
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

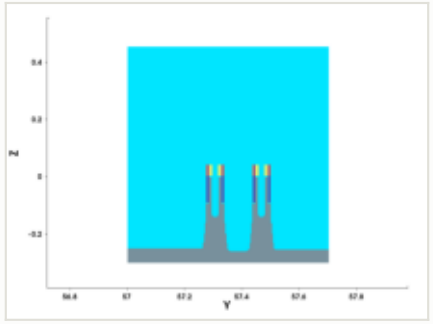
carbon so far, of the whole flow (kgCO2e)



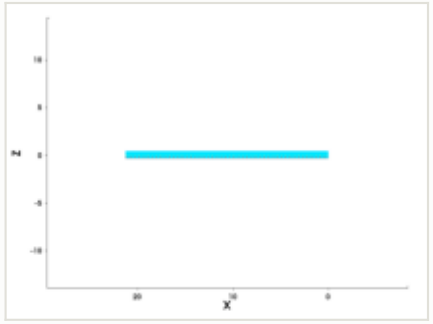
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



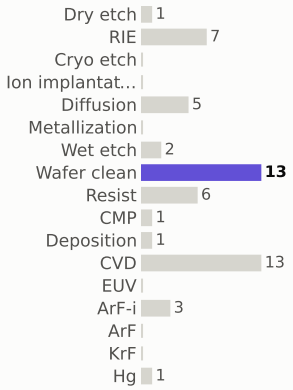
y cut at 57.351 µm



mask for this step

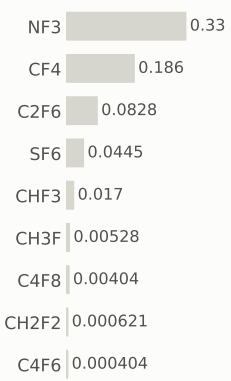
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



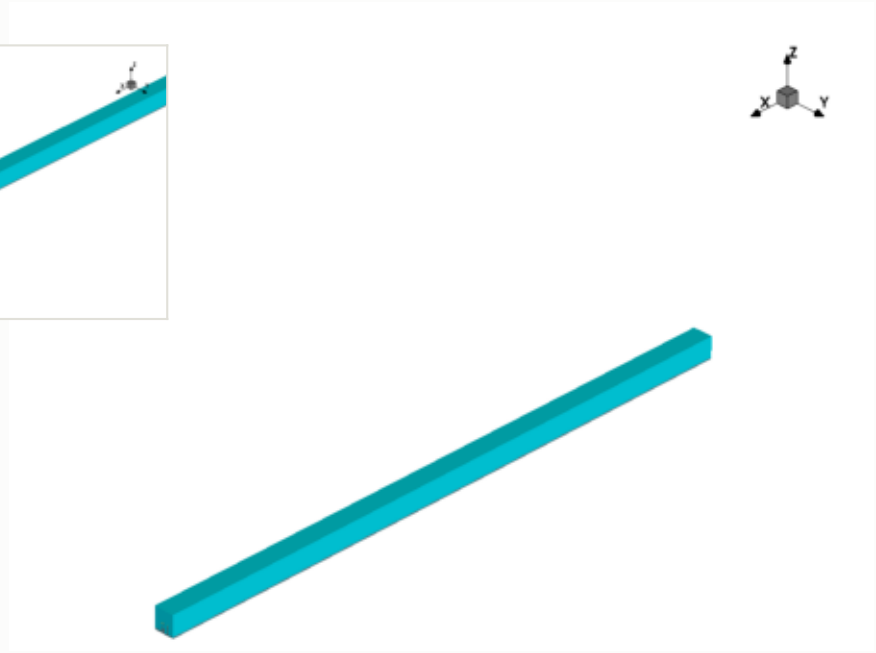
Step #810: Diffusion

anneal · Diffusion

0.801	61.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



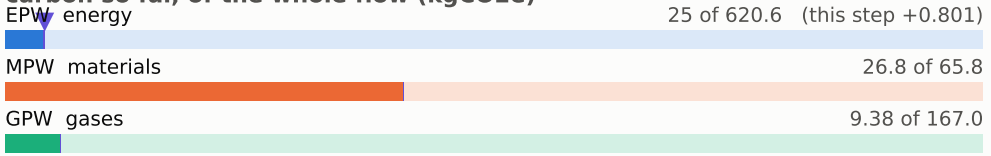
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

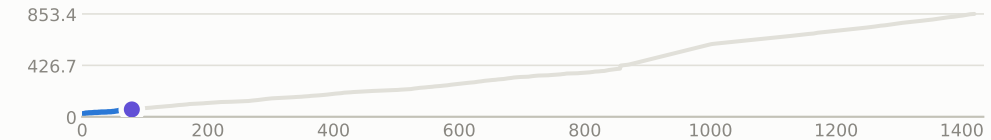
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

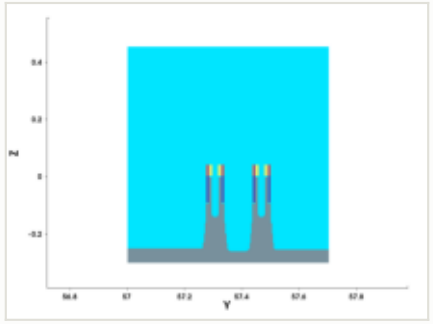
carbon so far, of the whole flow (kgCO2e)



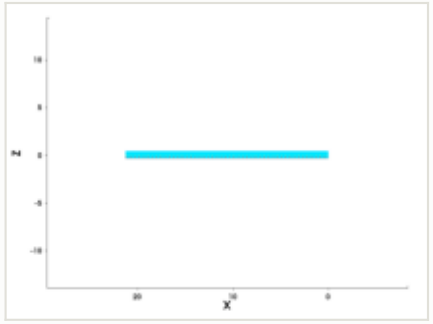
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



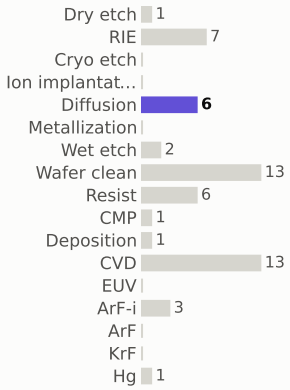
y cut at 57.351 µm



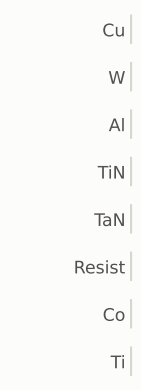
mask for this step

no mask for this step

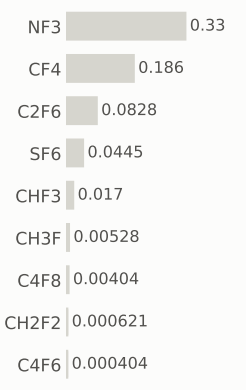
steps by process area



materials deposited (g)



gases emitted (g)



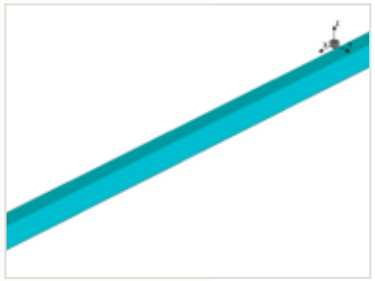
Step #820: CVD

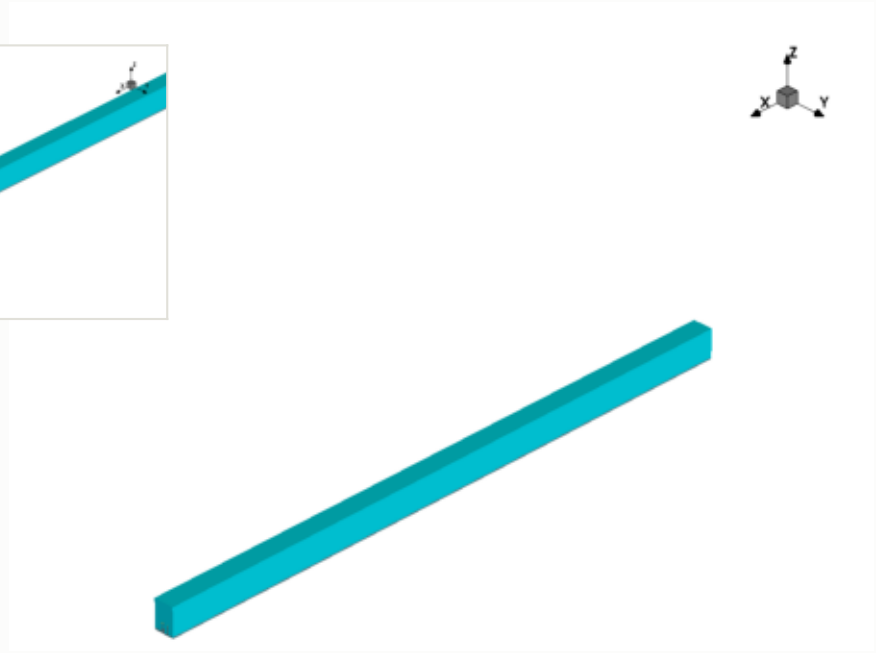
depo_isotropic · CVD

1.31	62.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



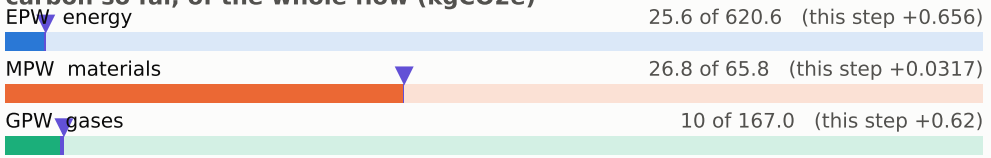


this step

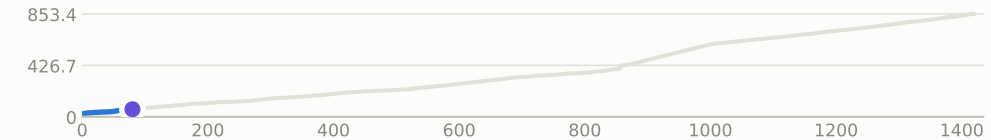
process area	CVD
material	—
thickness	0.25 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

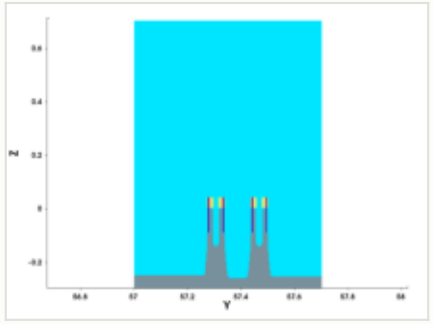
carbon so far, of the whole flow (kgCO2e)



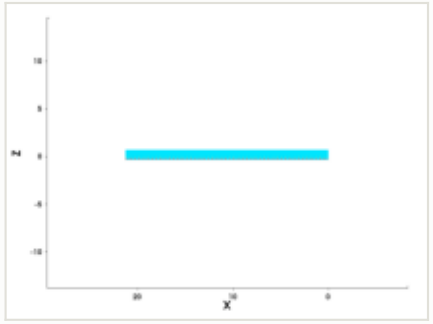
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



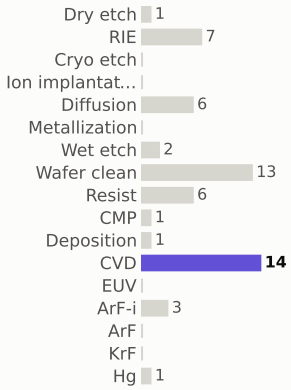
y cut at 57.351 µm



mask for this step

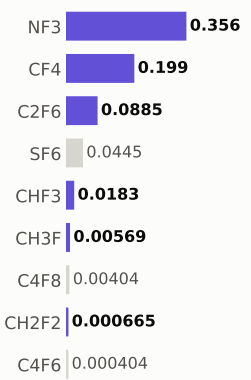
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



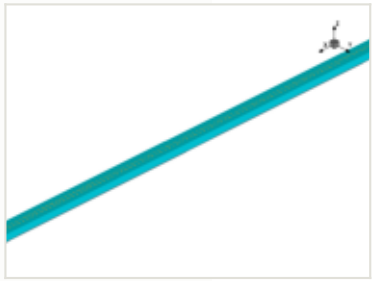
Step #830: CMP

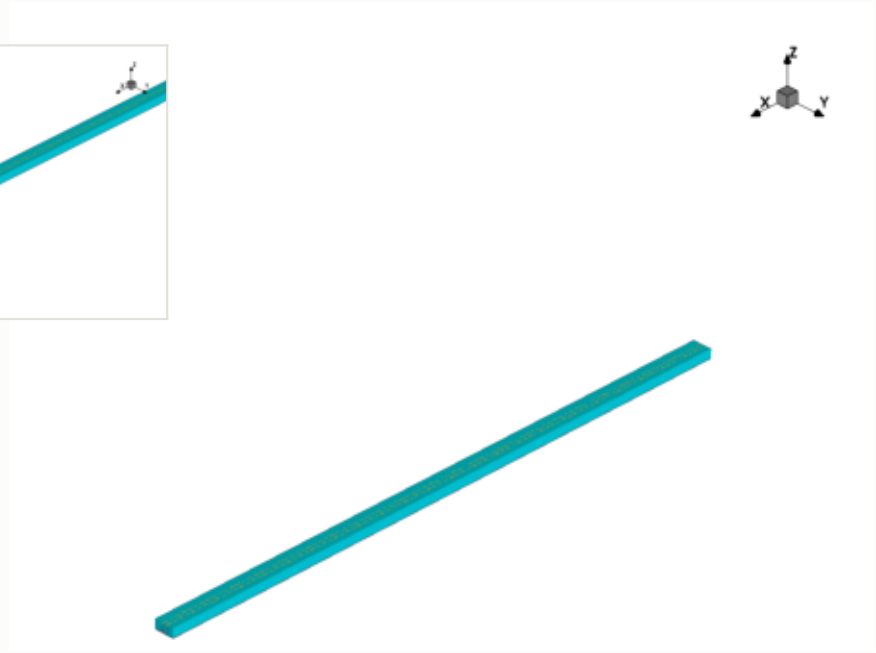
etch_cmp · CMP

0.169	62.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



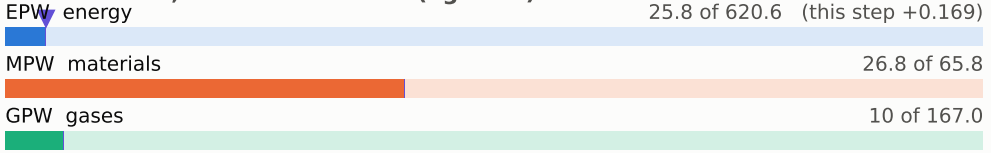


this step

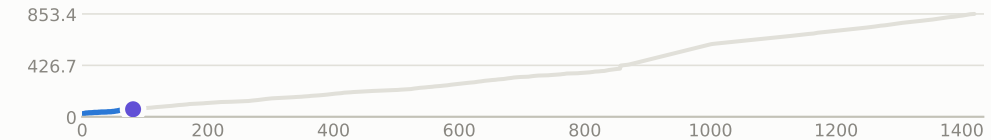
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

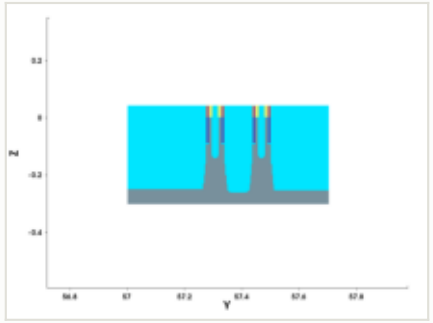
carbon so far, of the whole flow (kgCO2e)



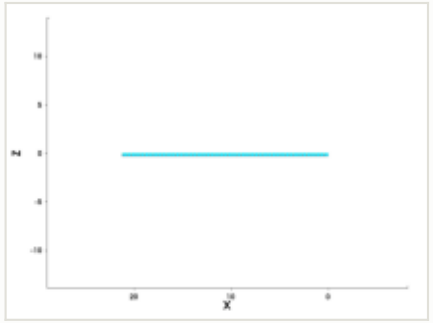
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



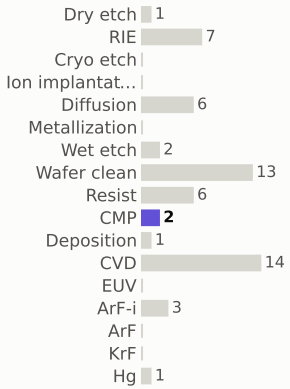
y cut at 57.351 µm



mask for this step

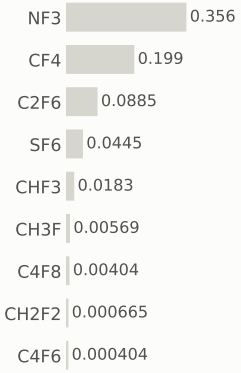
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



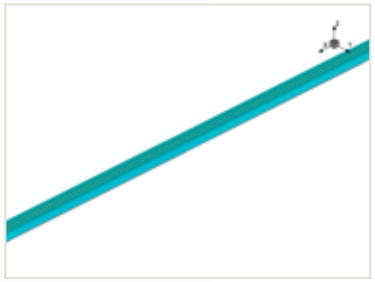
Step #840: Wafer_clean

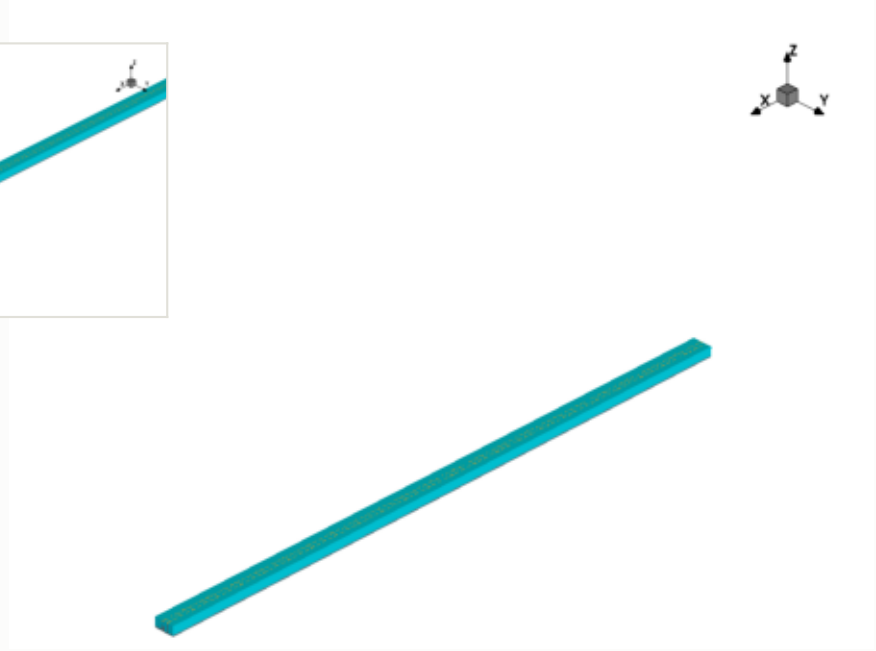
clean · Wafer_clean

0.169	62.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

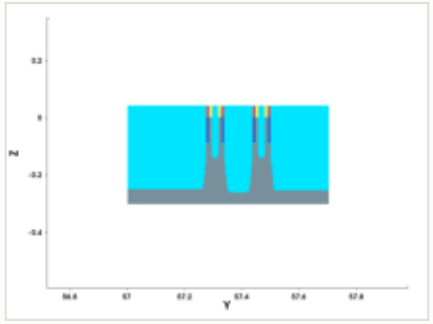
3D structure after this step

zoomed to fit

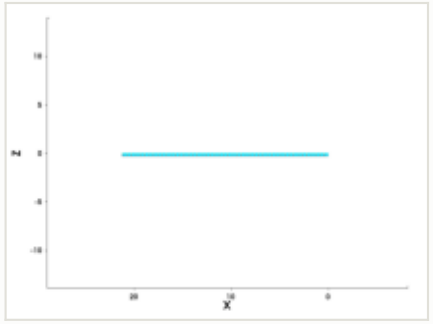




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

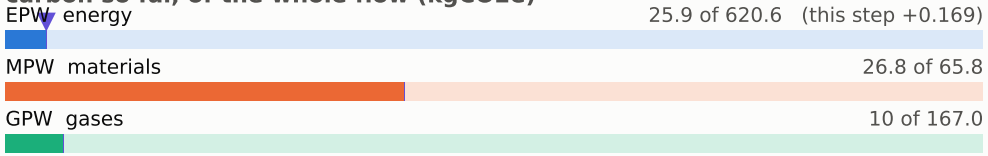
no mask for this step

this step

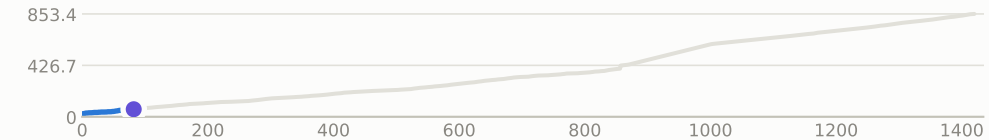
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

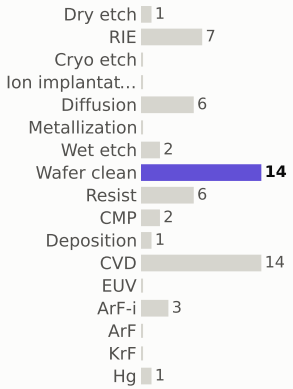
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

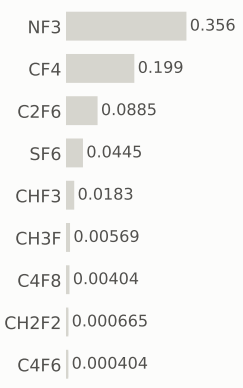


steps by process area



materials deposited (g)

gases emitted (g)



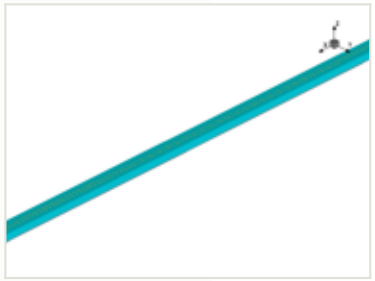
Step #850: Diffusion

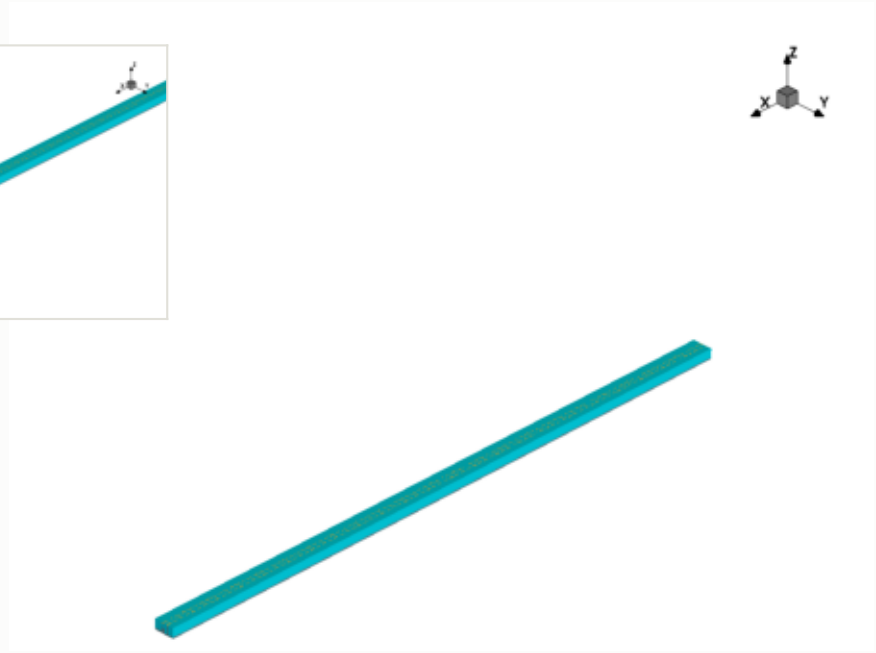
anneal · Diffusion

0.801	63.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





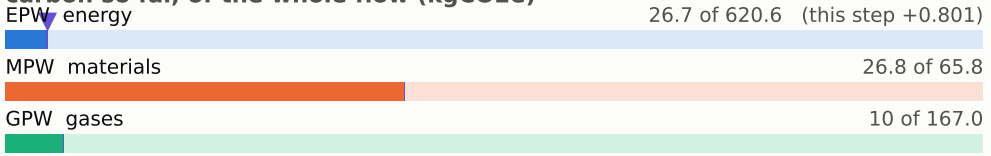
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

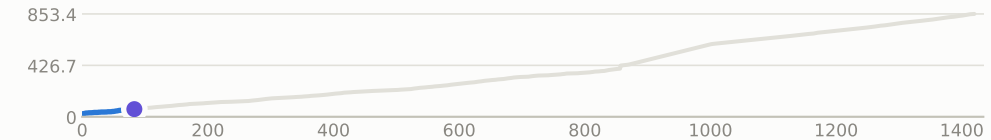
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

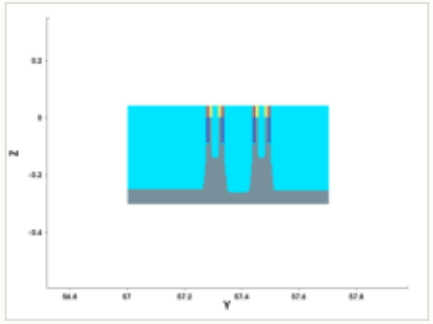
carbon so far, of the whole flow (kgCO2e)



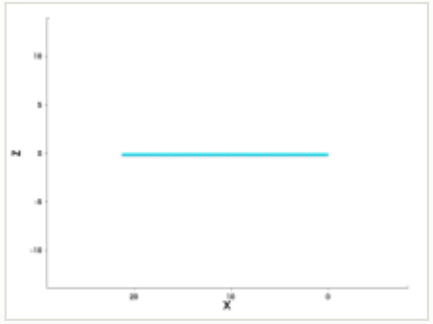
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



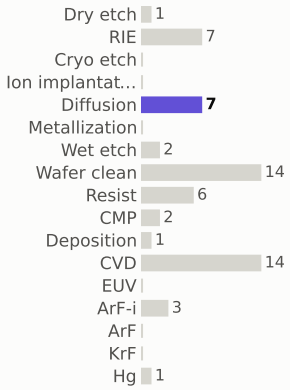
y cut at 57.351 µm



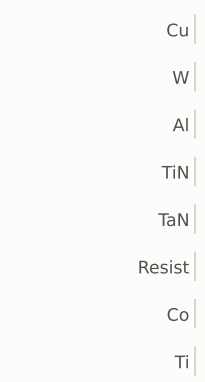
mask for this step

no mask for this step

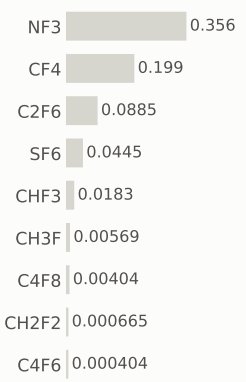
steps by process area



materials deposited (g)



gases emitted (g)



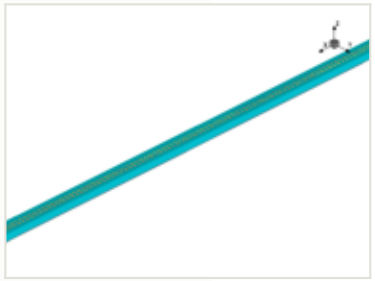
Step #860: RIE

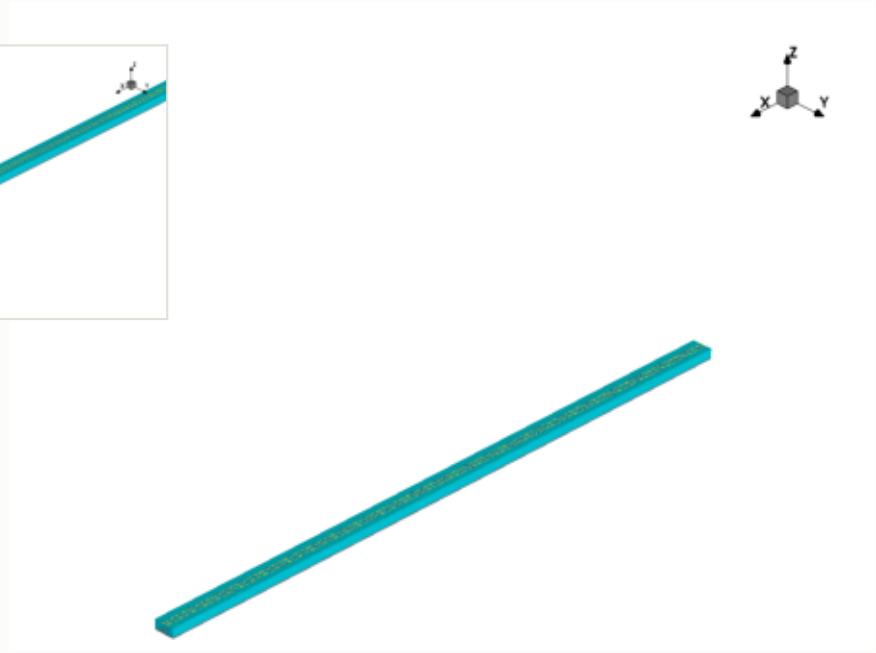
etch_anisotropic · RIE

1.04	64.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



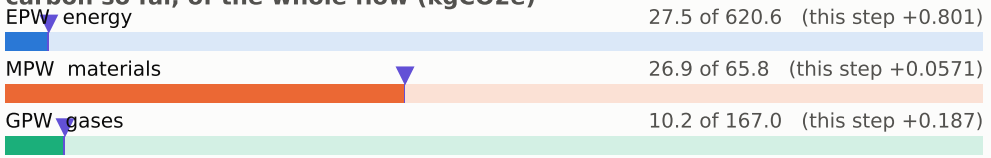


this step

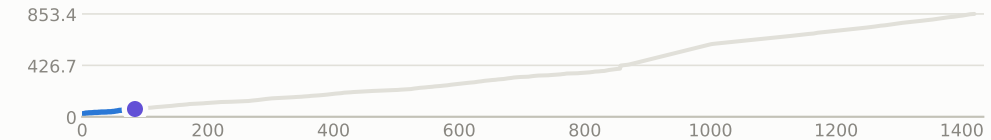
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

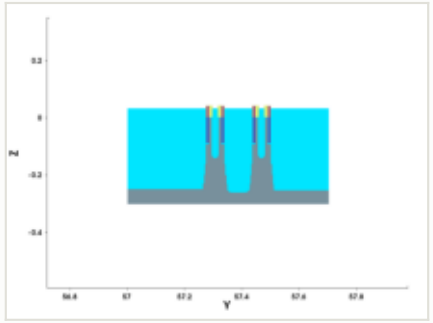
carbon so far, of the whole flow (kgCO2e)



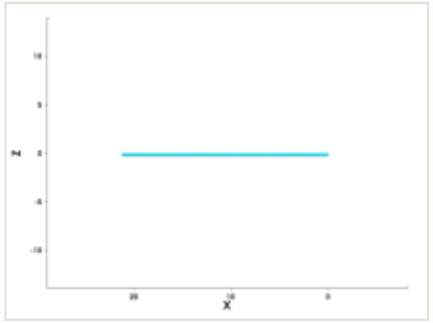
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



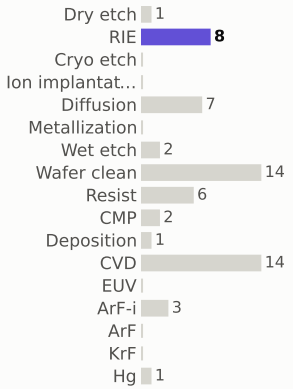
y cut at 57.351 µm



mask for this step

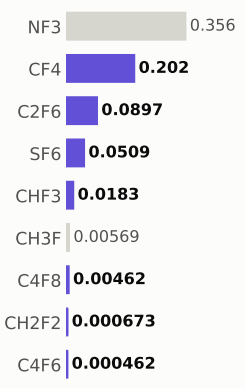
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



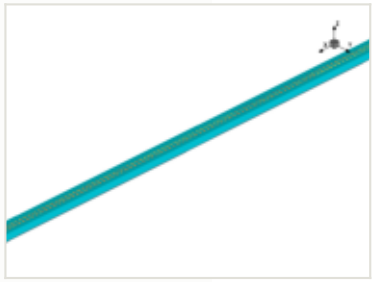
Step #870: Dry_etch

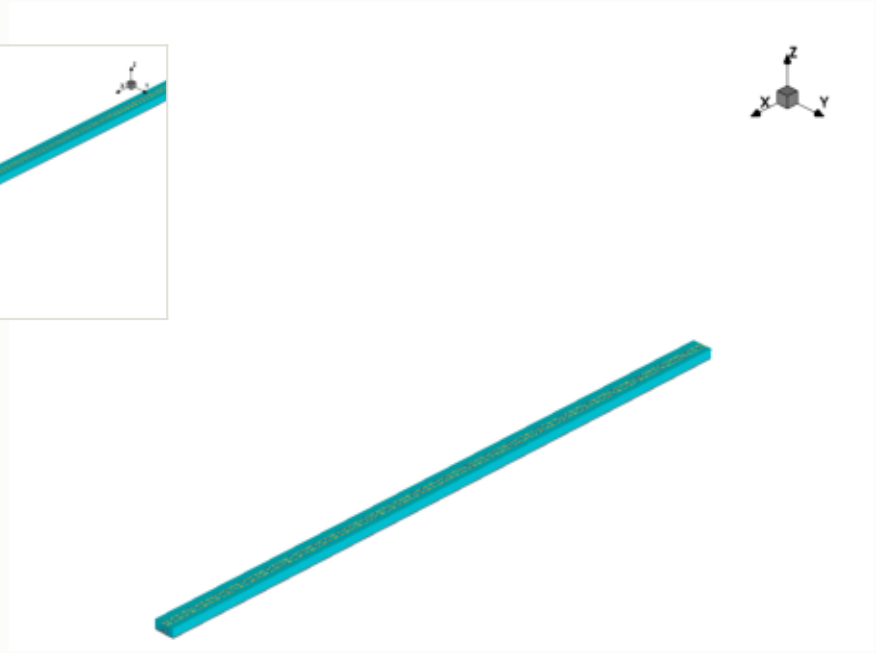
etch_strip · Dry_etch

0.801	65.4	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



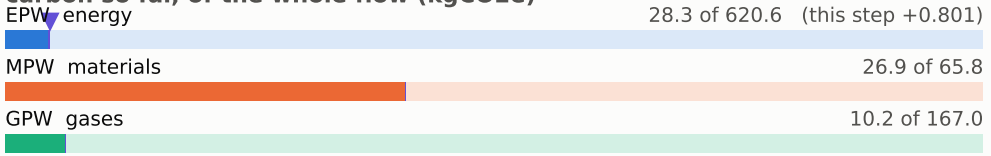


this step

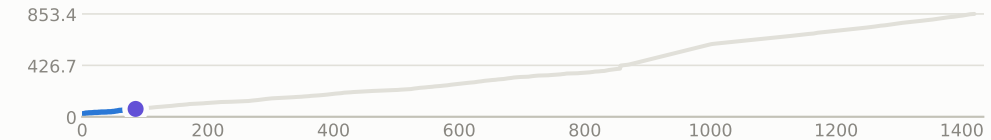
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

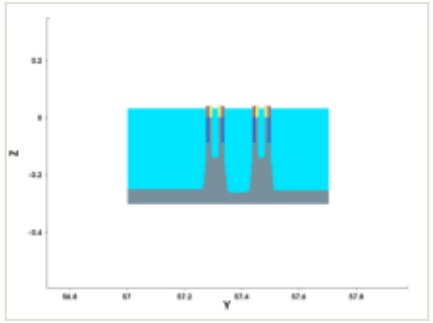
carbon so far, of the whole flow (kgCO2e)



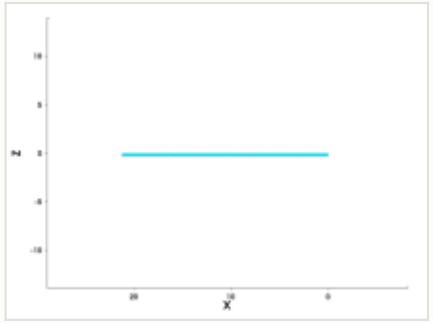
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



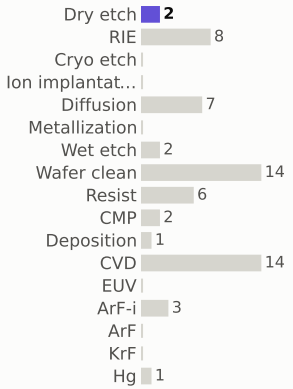
y cut at 57.351 µm



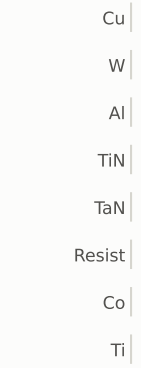
mask for this step

no mask for this step

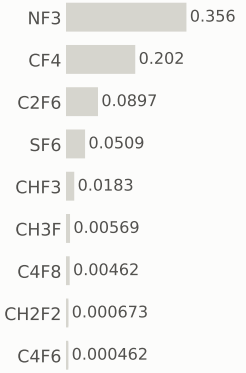
steps by process area



materials deposited (g)



gases emitted (g)



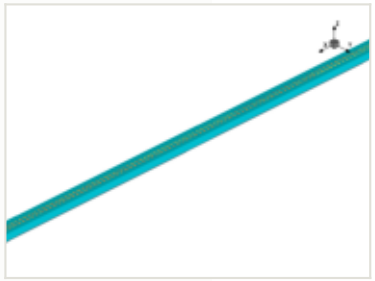
Step #880: Wafer_clean

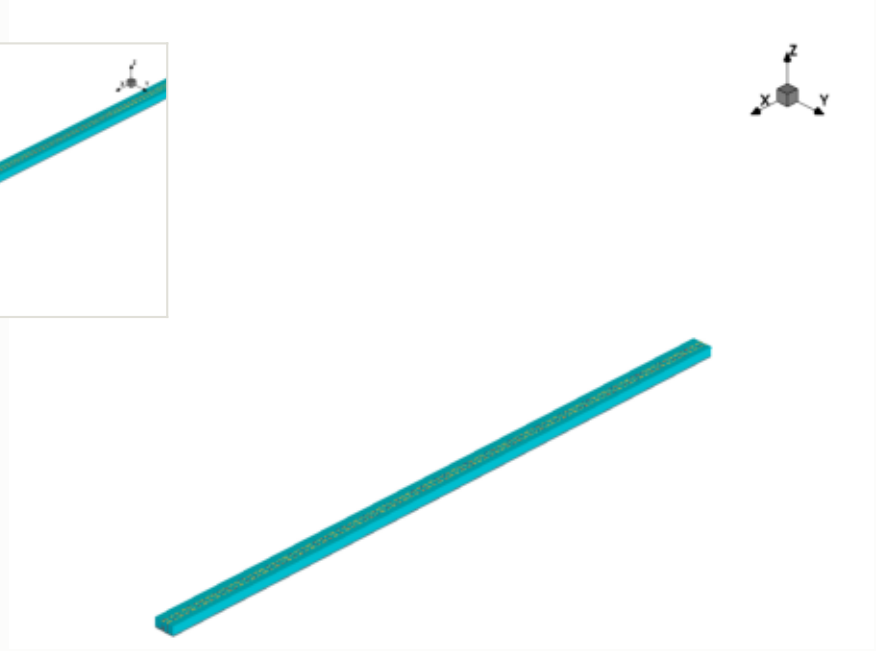
clean · Wafer_clean

0.169	65.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





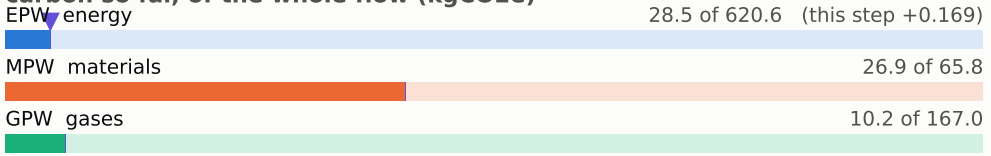
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

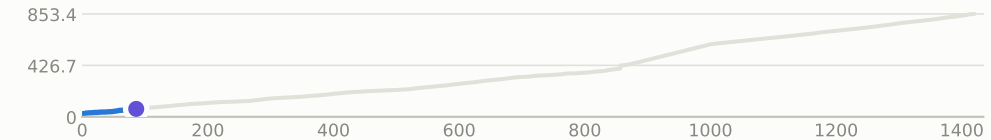
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

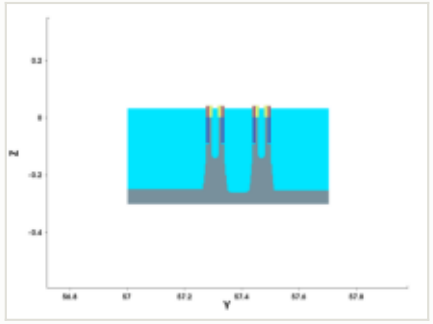
carbon so far, of the whole flow (kgCO2e)



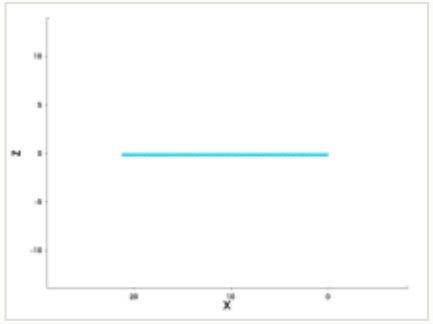
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



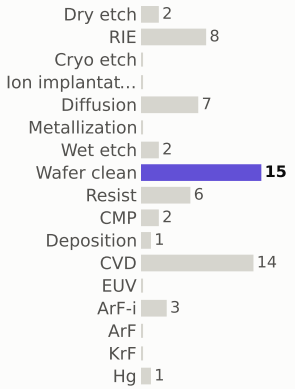
y cut at 57.351 µm



mask for this step

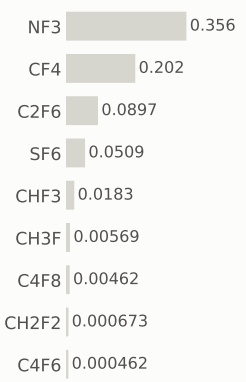
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



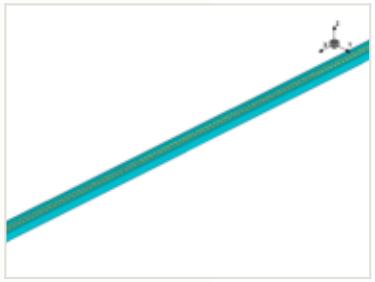
Step #890: Wet_etch

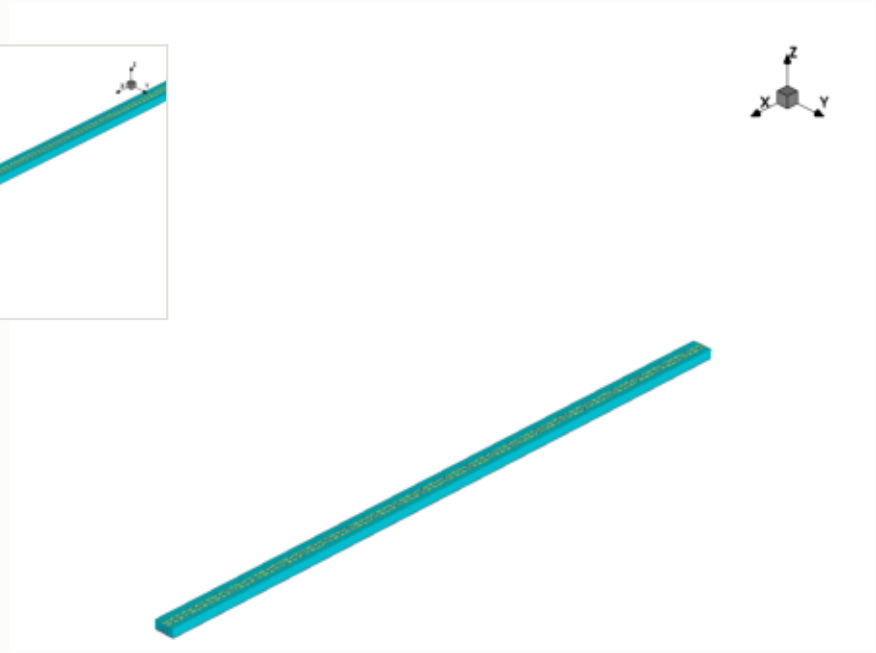
etch_isotropic · Wet_etch

0.169	65.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



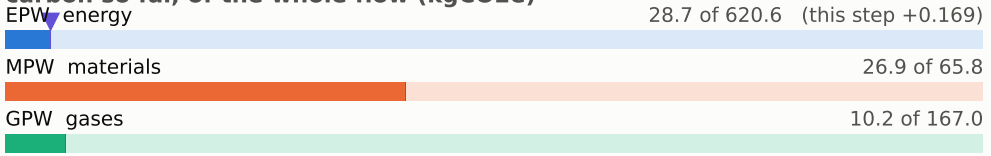


this step

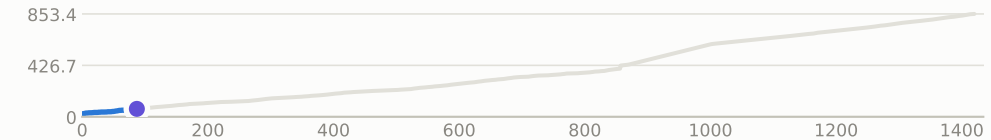
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

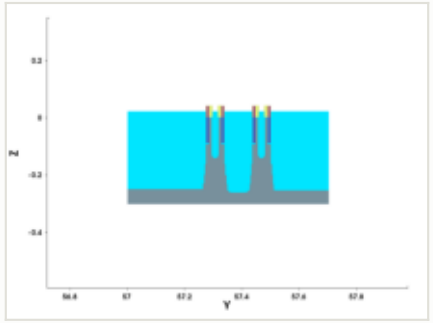
carbon so far, of the whole flow (kgCO2e)



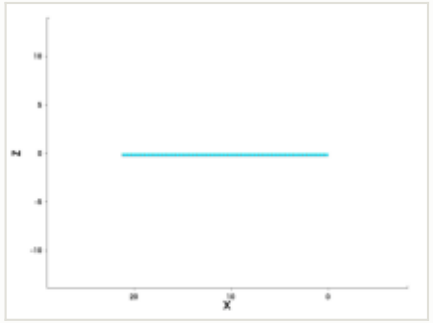
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



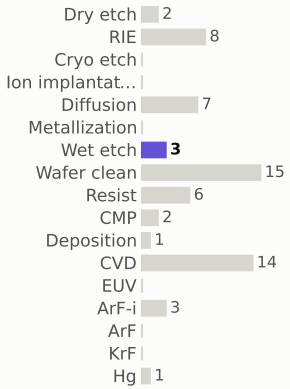
y cut at 57.351 µm



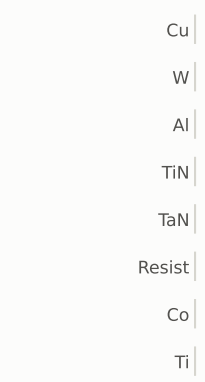
mask for this step

no mask for this step

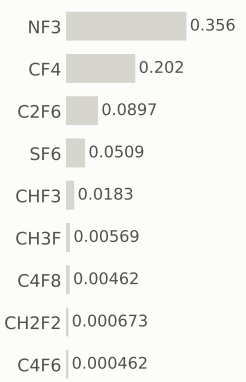
steps by process area



materials deposited (g)



gases emitted (g)



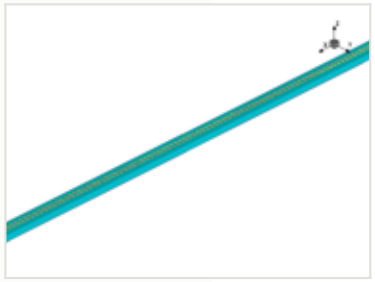
Step #900: Wet_etch

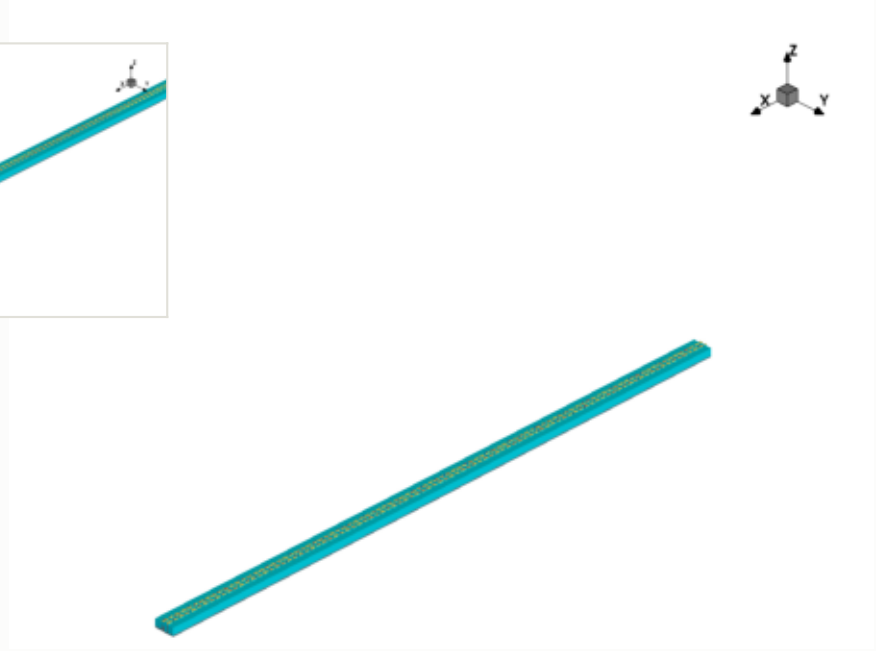
etch_microloading · Wet_etch

0.169	65.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



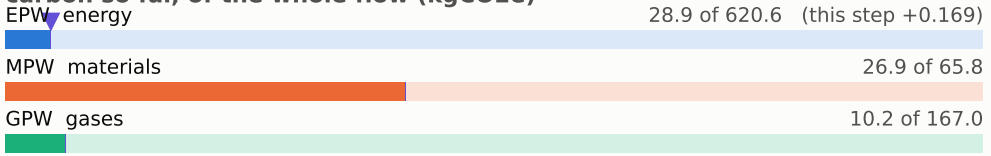


this step

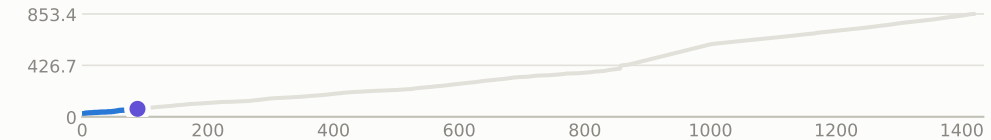
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

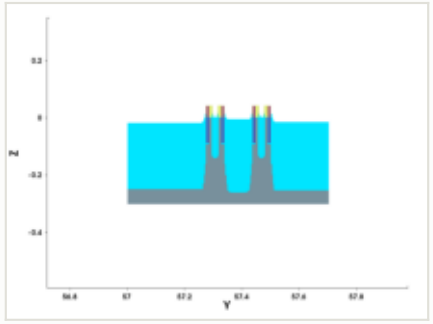
carbon so far, of the whole flow (kgCO2e)



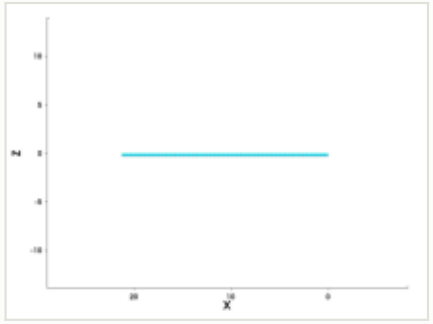
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



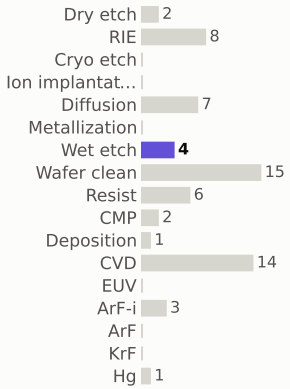
y cut at 57.351 µm



mask for this step

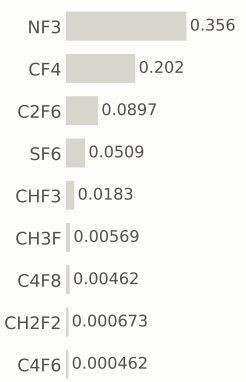
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



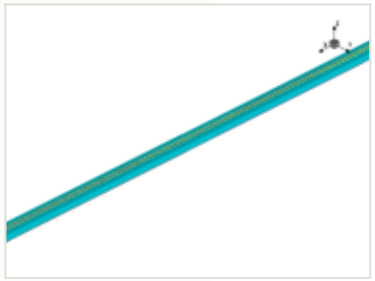
Step #910: Ion_implantation

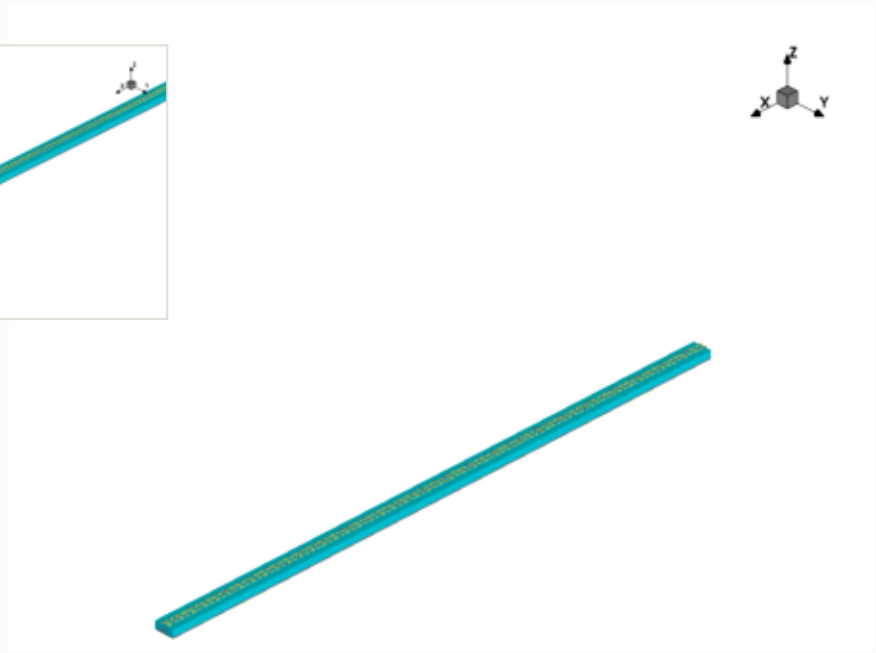
implant_ellipsoid · Ion_implantation

0.801	66.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

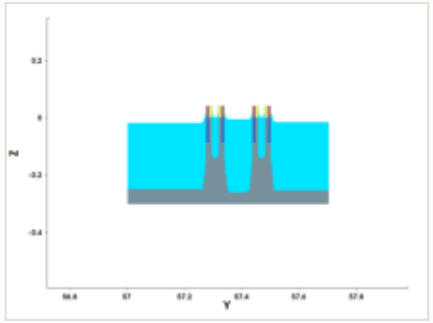
3D structure after this step

zoomed to fit

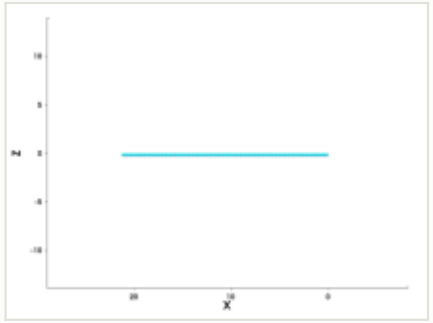




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

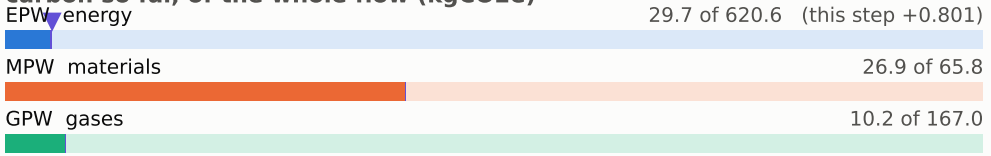
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

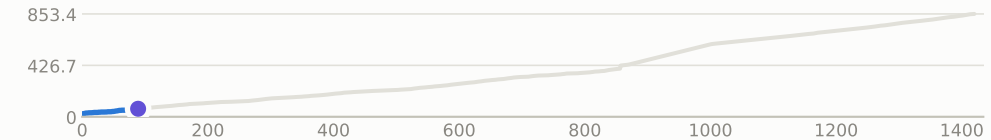
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

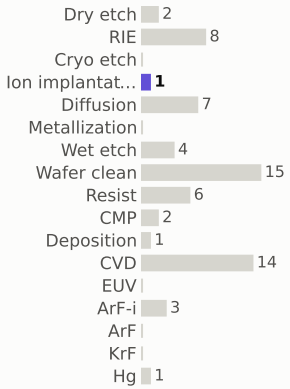
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

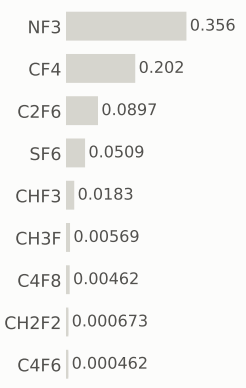


steps by process area



materials deposited (g)

gases emitted (g)



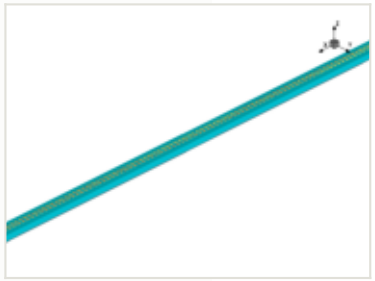
Step #920: Wafer_clean

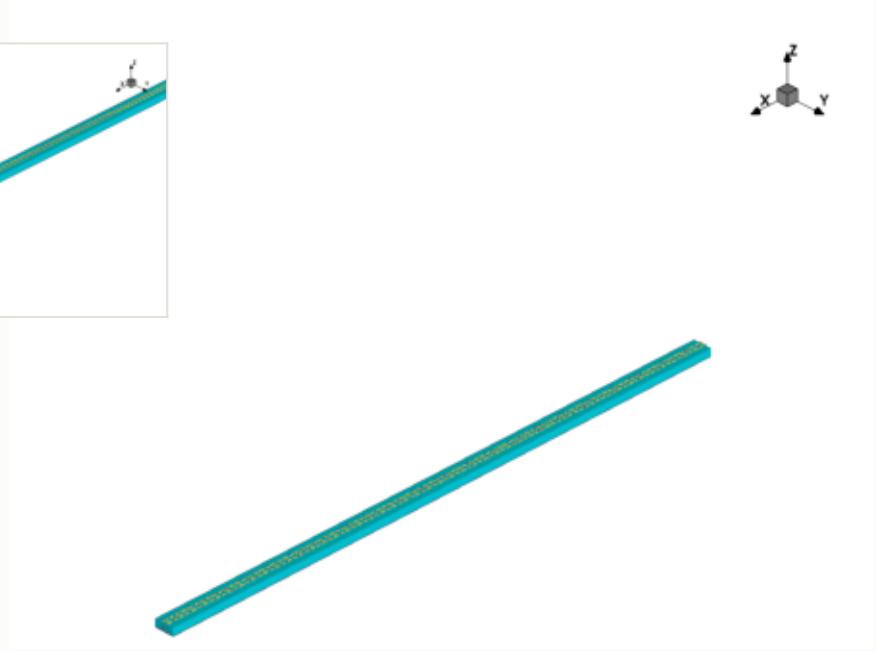
clean · Wafer_clean

0.169	66.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





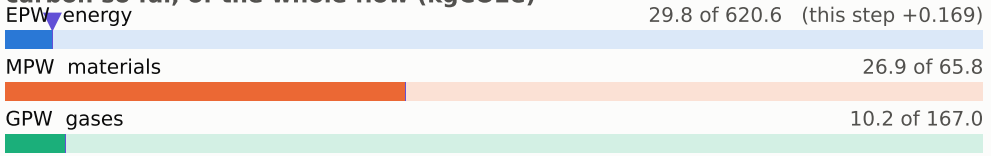
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

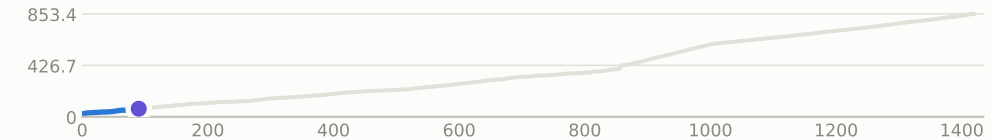
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

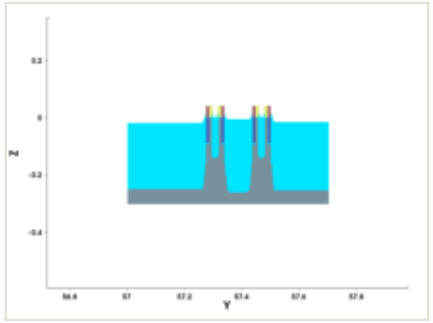
carbon so far, of the whole flow (kgCO2e)



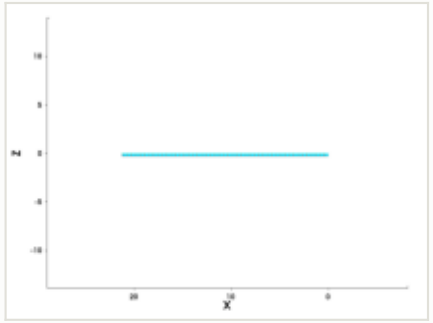
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



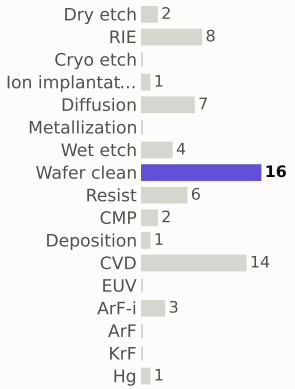
y cut at 57.351 µm



mask for this step

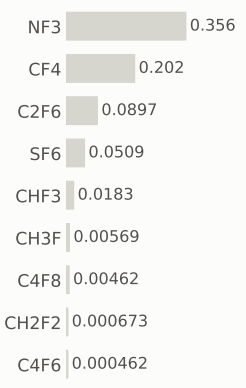
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



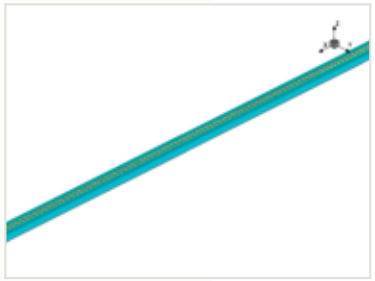
Step #930: Resist

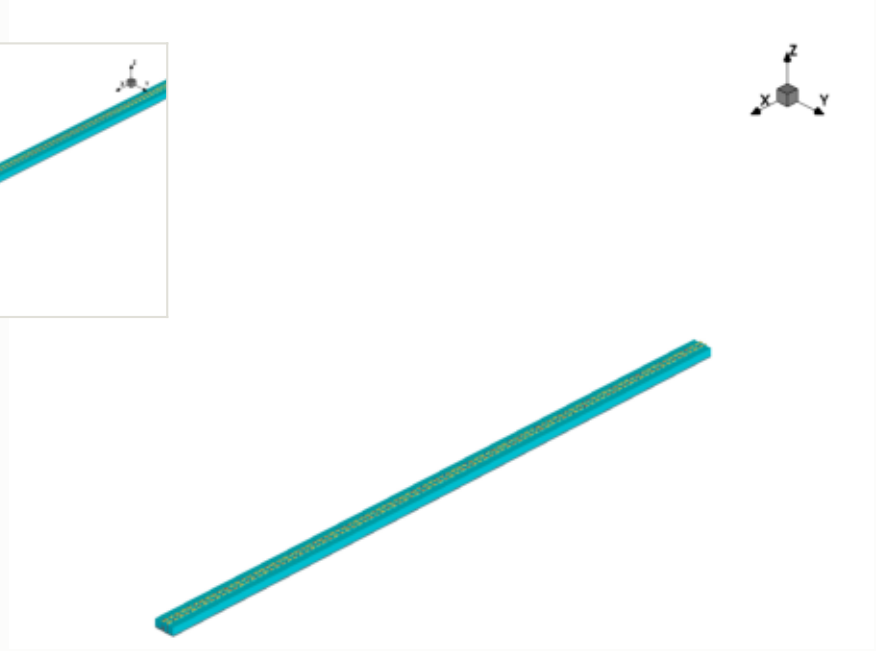
resist · Resist

0.169	67.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





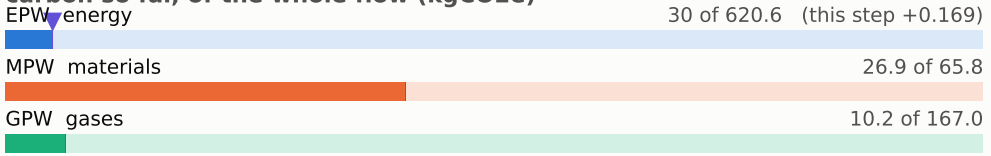
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

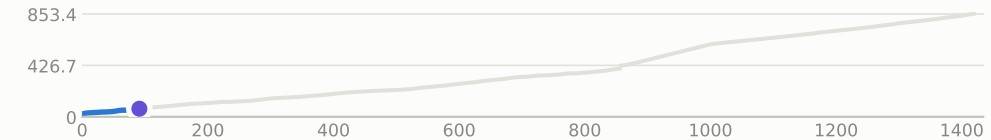
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

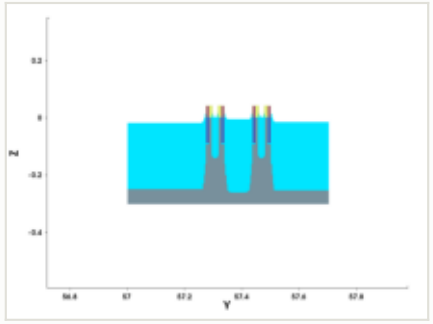
carbon so far, of the whole flow (kgCO2e)



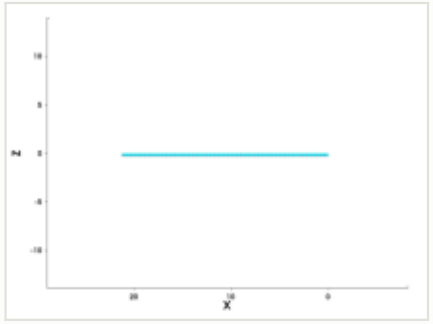
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



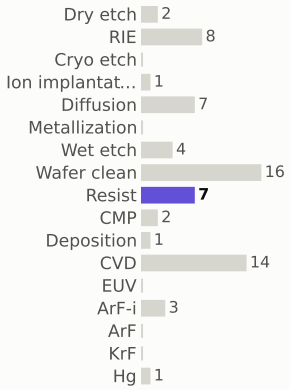
y cut at 57.351 µm



mask for this step

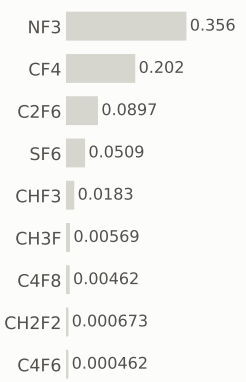
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



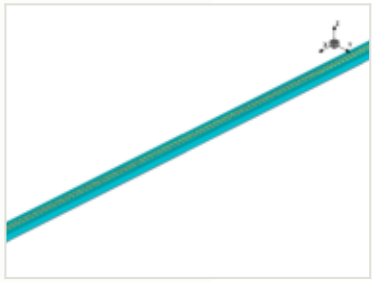
Step #940: KrF

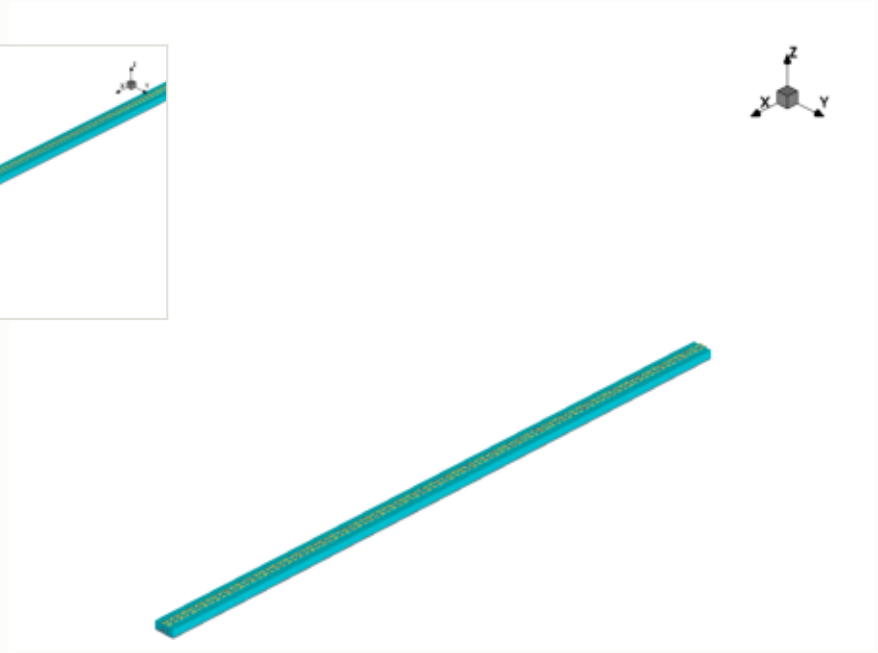
litho_patterning · KrF

0.199	67.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



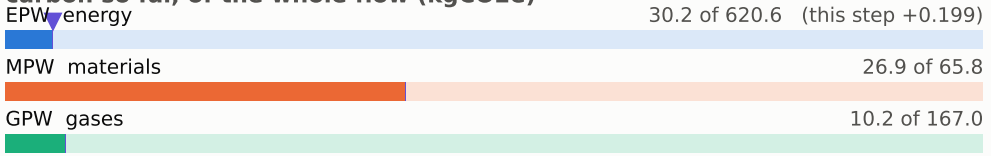


this step

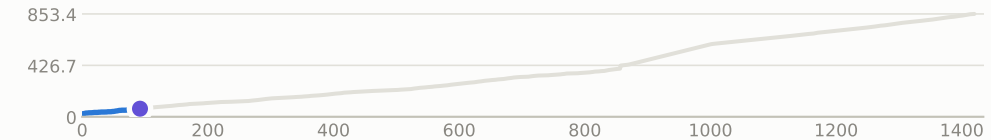
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

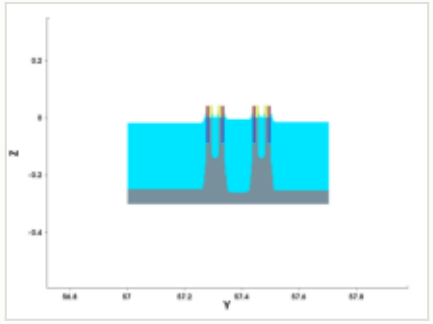
carbon so far, of the whole flow (kgCO2e)



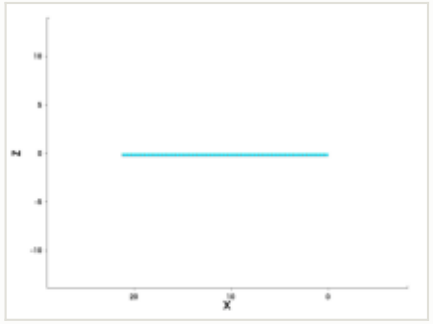
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

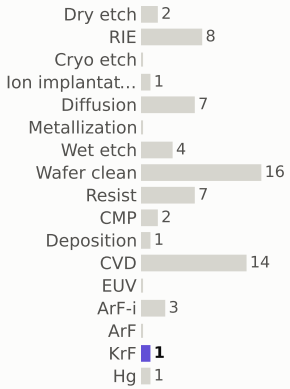


y cut at 57.351 µm



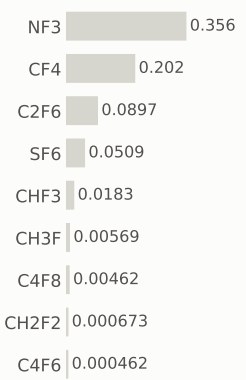
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



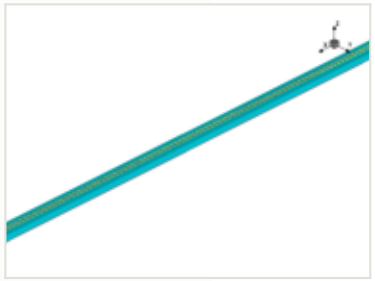
Step #950: Ion_implantation

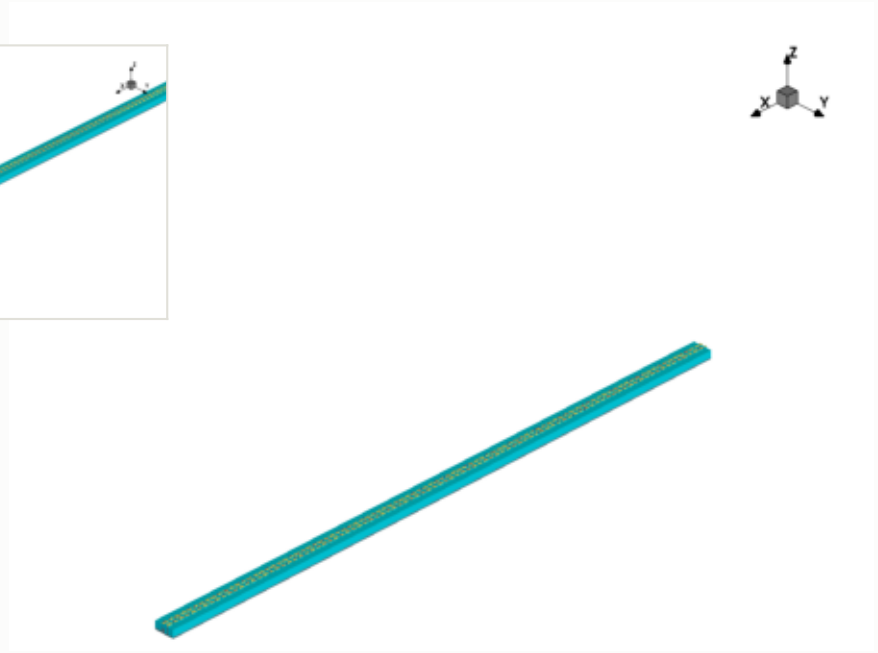
implant_ellipsoid · Ion_implantation

0.801	68.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

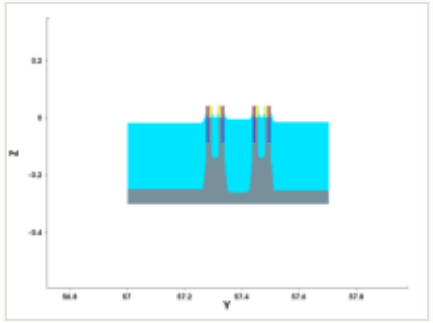
3D structure after this step

zoomed to fit

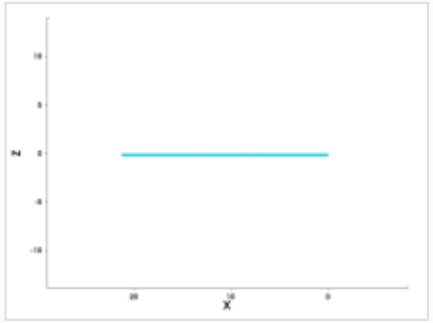




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

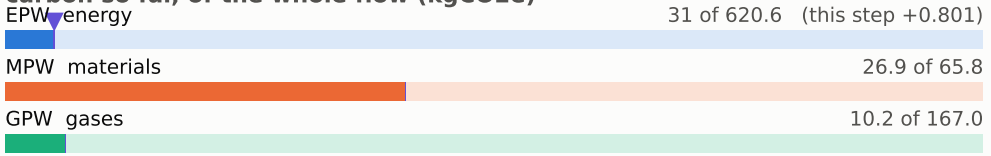
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

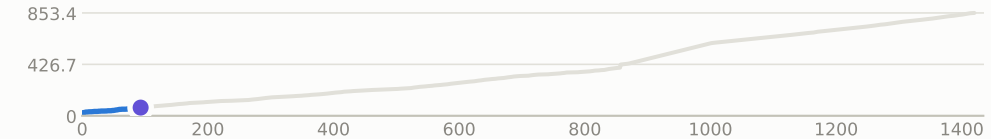
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

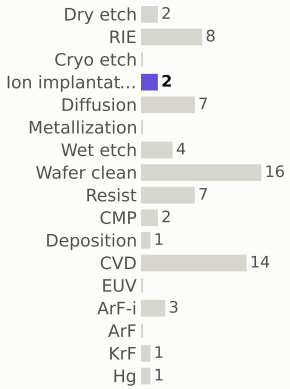
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

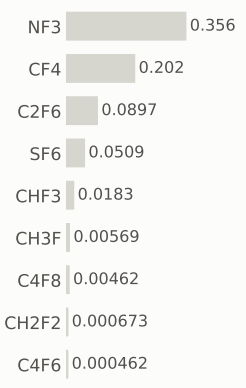


steps by process area



materials deposited (g)

gases emitted (g)



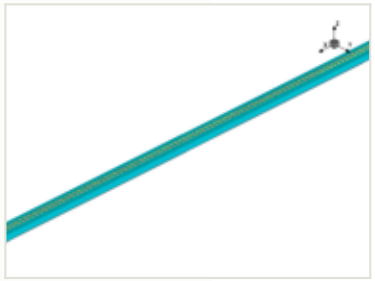
Step #960: Ion_implantation

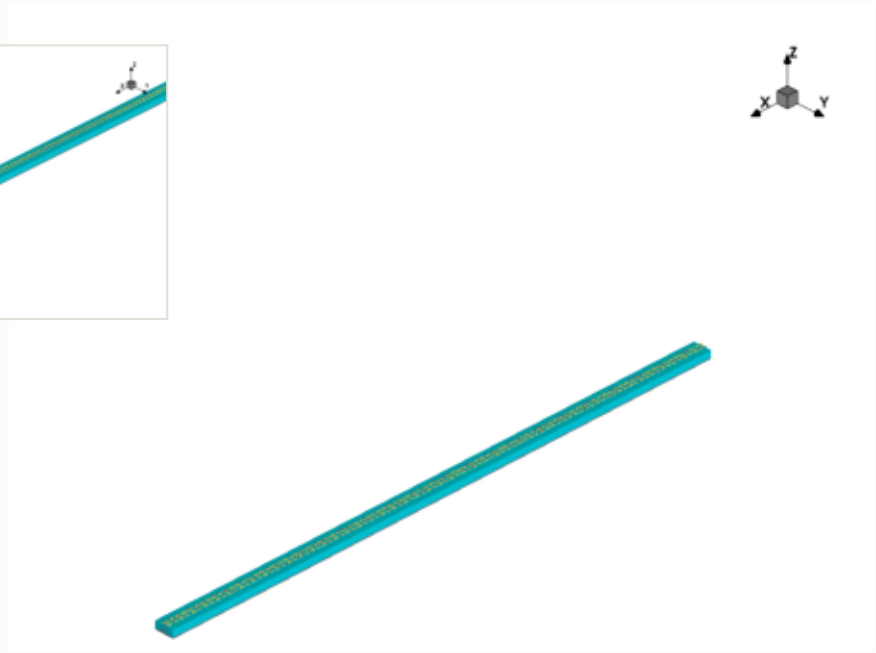
implant_ellipsoid · Ion_implantation

0.801	68.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

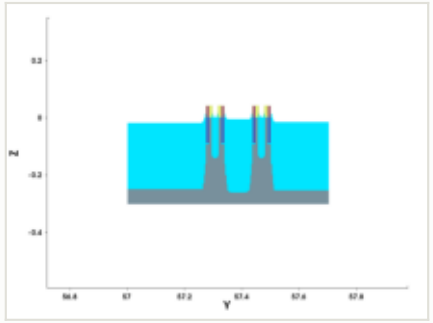
3D structure after this step

zoomed to fit

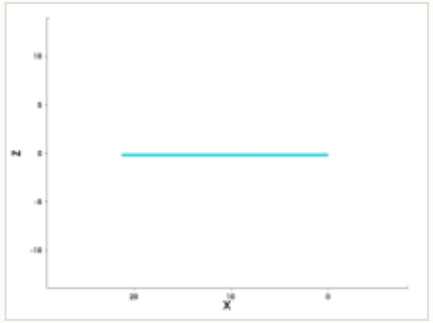




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

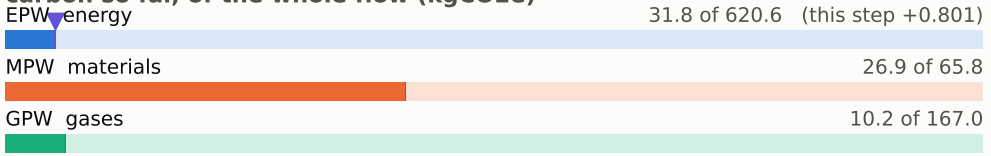
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

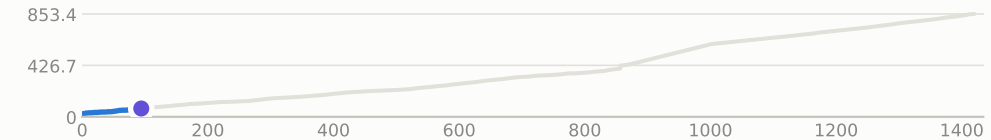
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

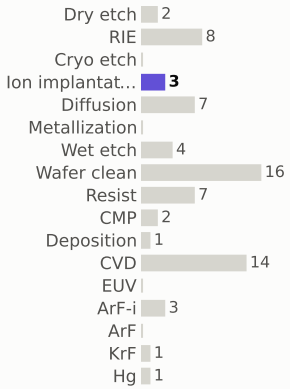
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

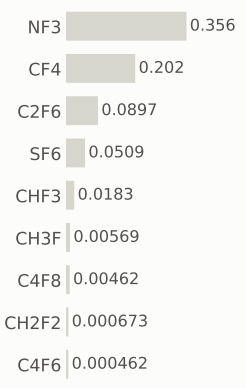


steps by process area



materials deposited (g)

gases emitted (g)



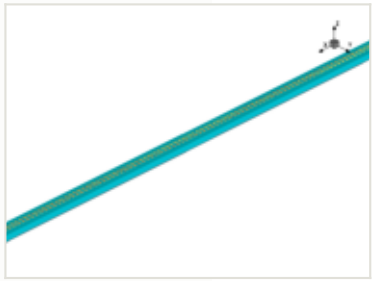
Step #970: Ion_implantation

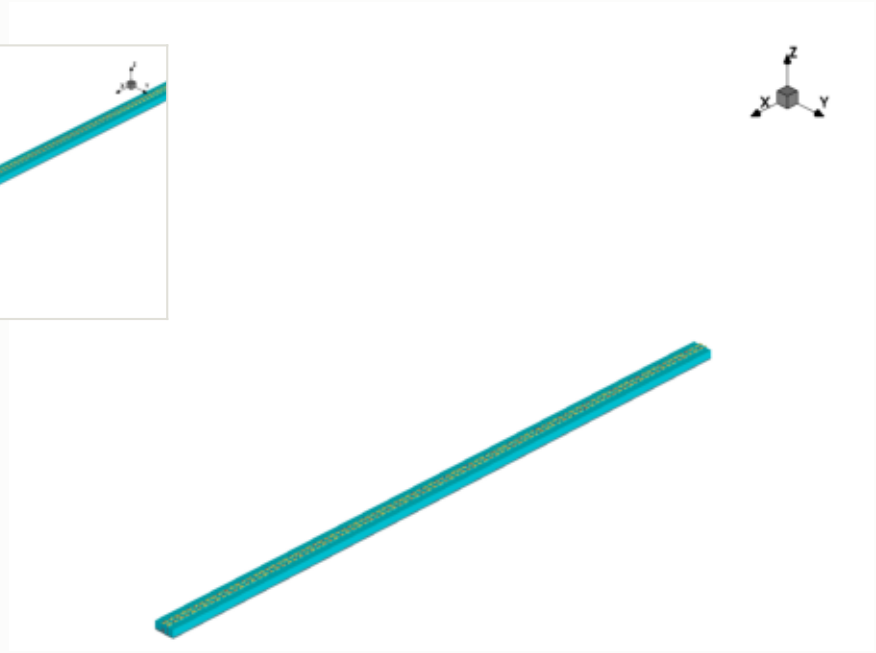
implant_ellipsoid · Ion_implantation

0.801	69.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

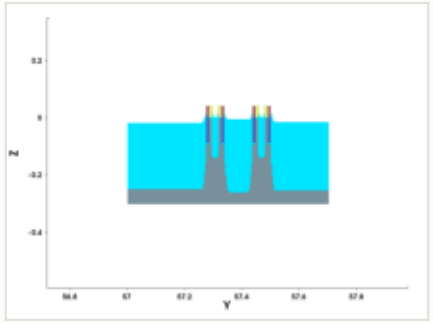
3D structure after this step

zoomed to fit

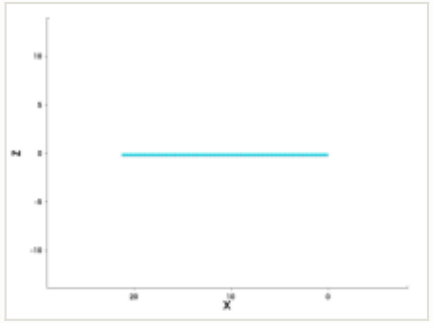




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

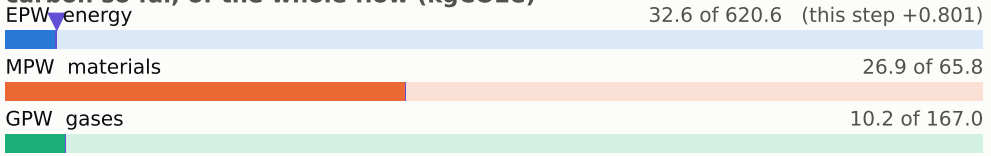
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

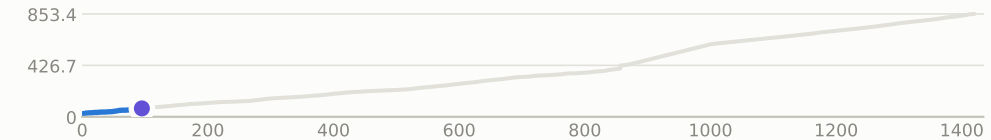
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

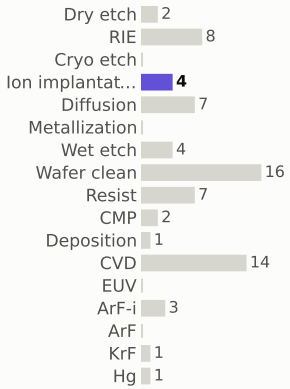
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

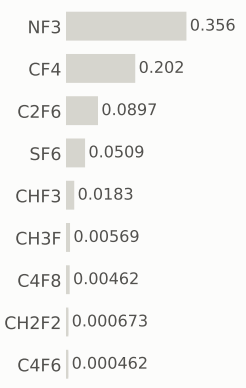


steps by process area



materials deposited (g)

gases emitted (g)



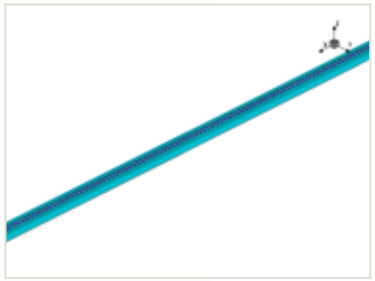
Step #980: Dry_etch

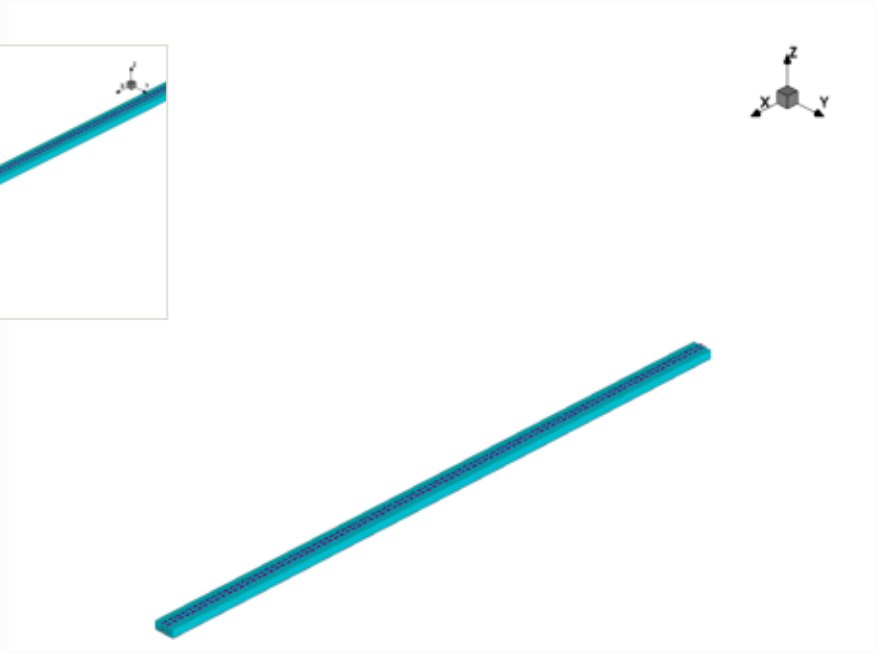
etch_strip · Dry_etch

0.801	70.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



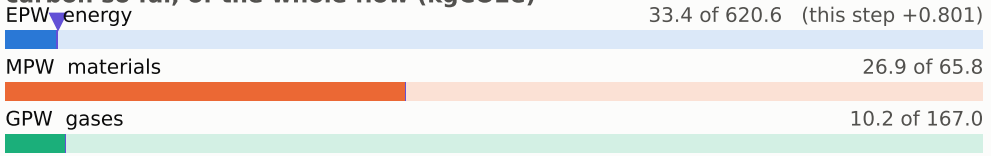


this step

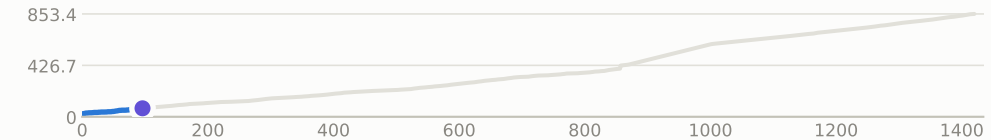
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

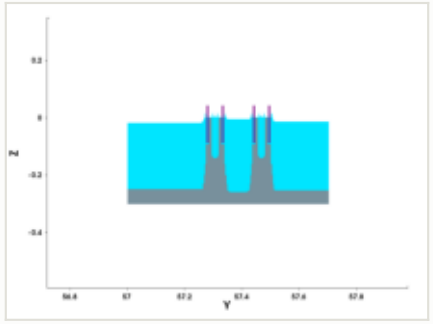
carbon so far, of the whole flow (kgCO2e)



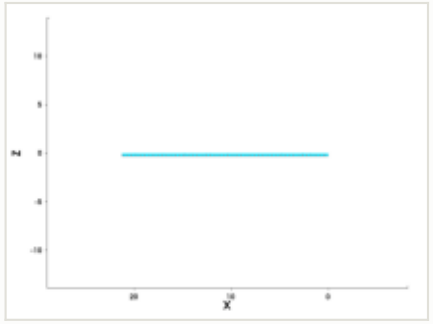
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



y cut at 57.351 µm



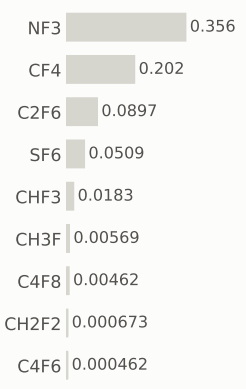
mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)



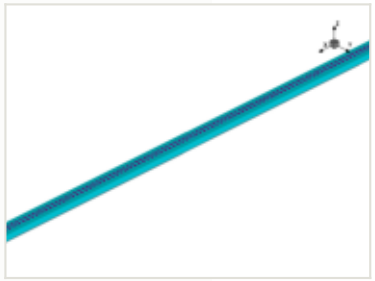
Step #990: Wafer_clean

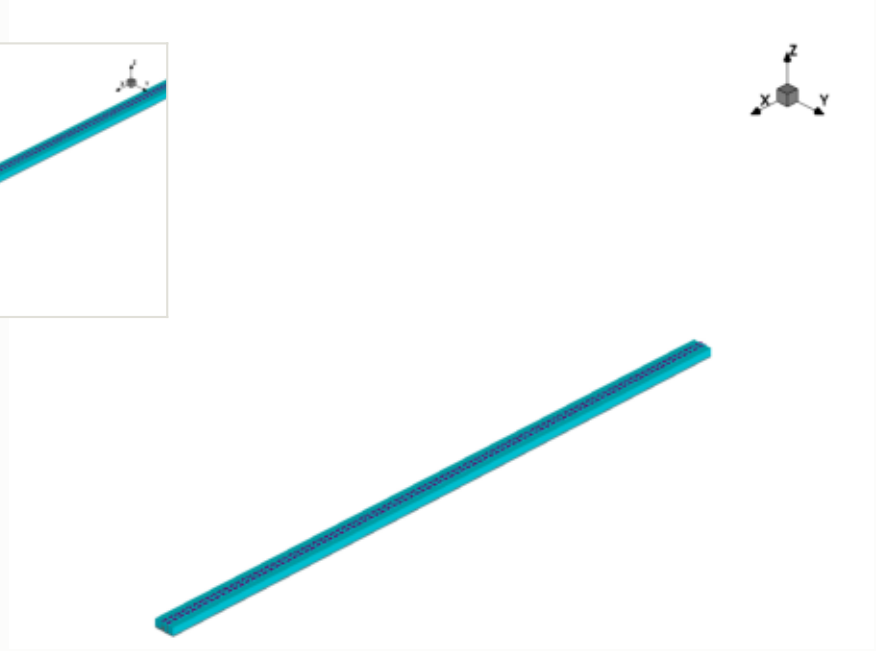
clean · Wafer_clean

0.169	70.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





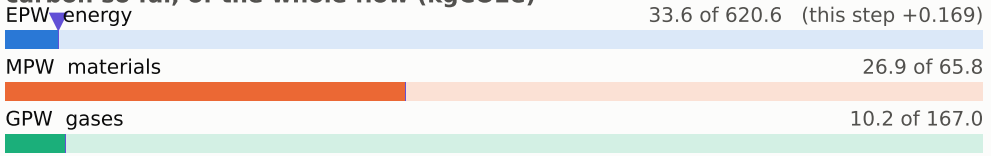
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

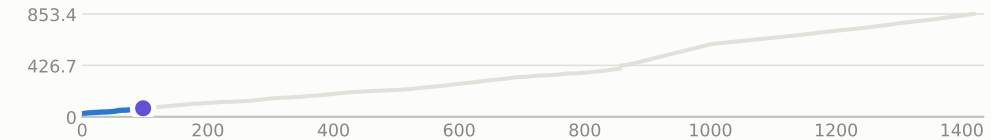
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

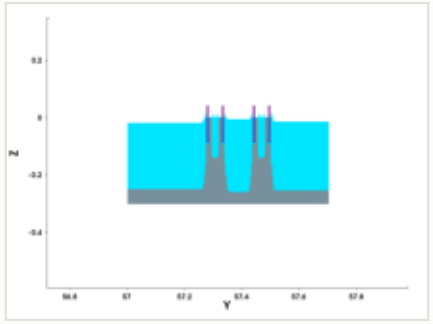
carbon so far, of the whole flow (kgCO2e)



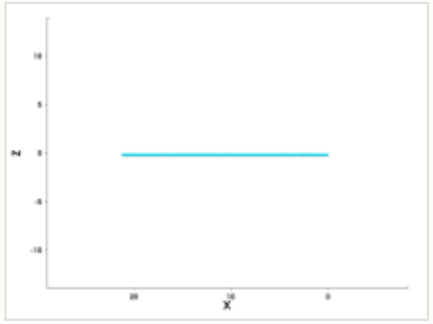
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



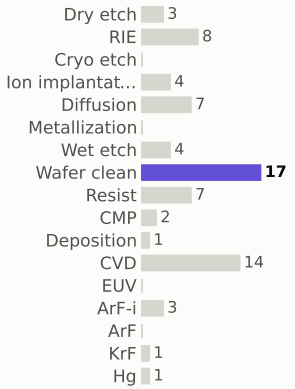
y cut at 57.351 µm



mask for this step

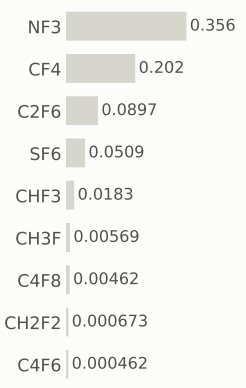
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



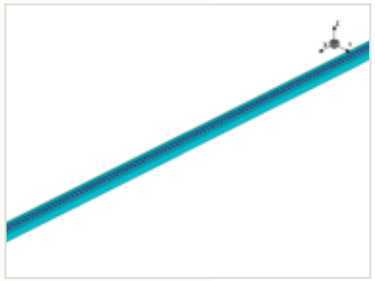
Step #1000: Resist

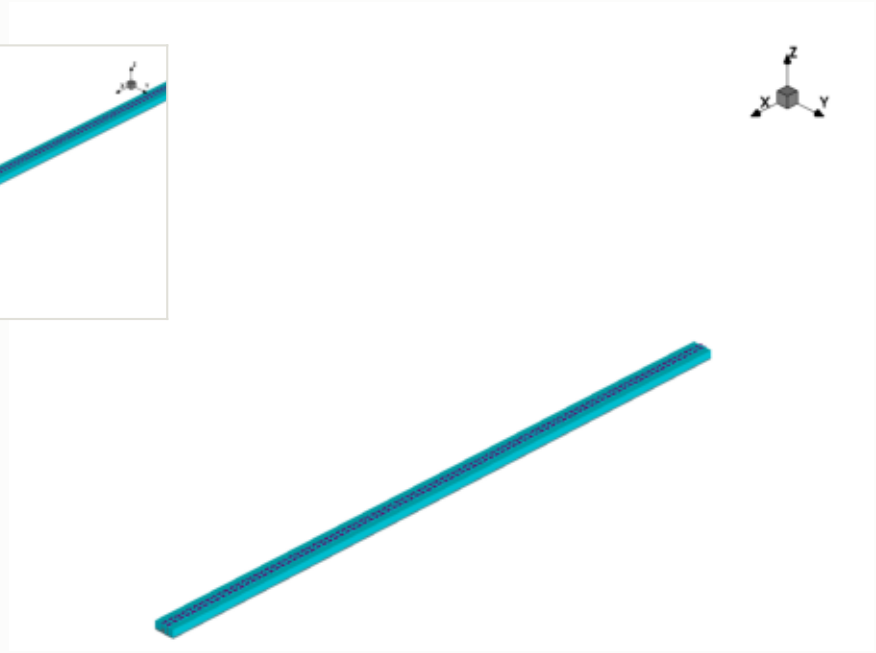
resist · Resist

0.169	70.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





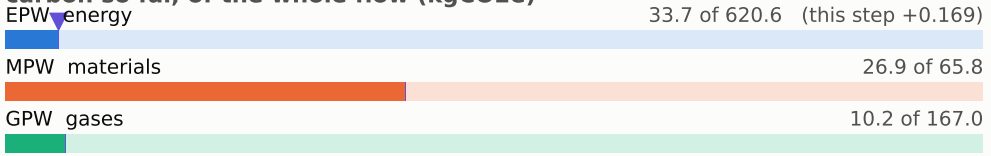
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

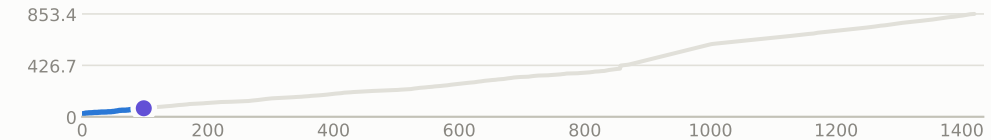
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

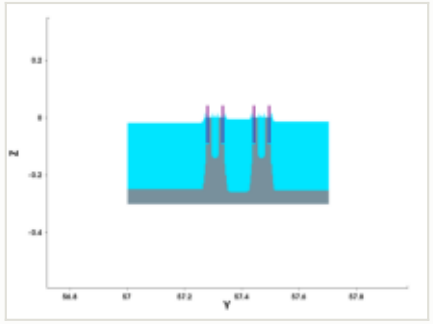
carbon so far, of the whole flow (kgCO2e)



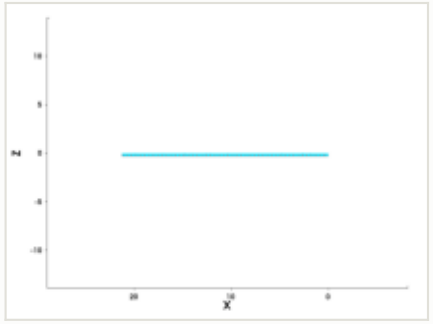
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



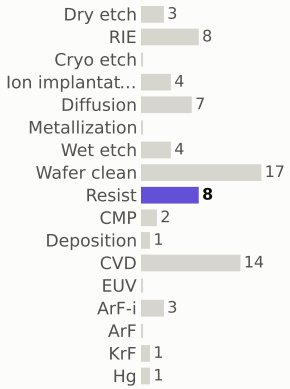
y cut at 57.351 µm



mask for this step

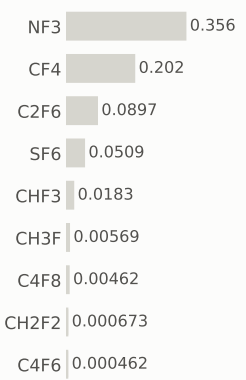
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



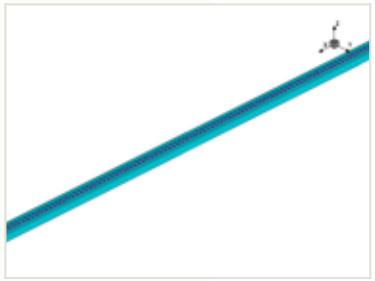
Step #1010: KrF

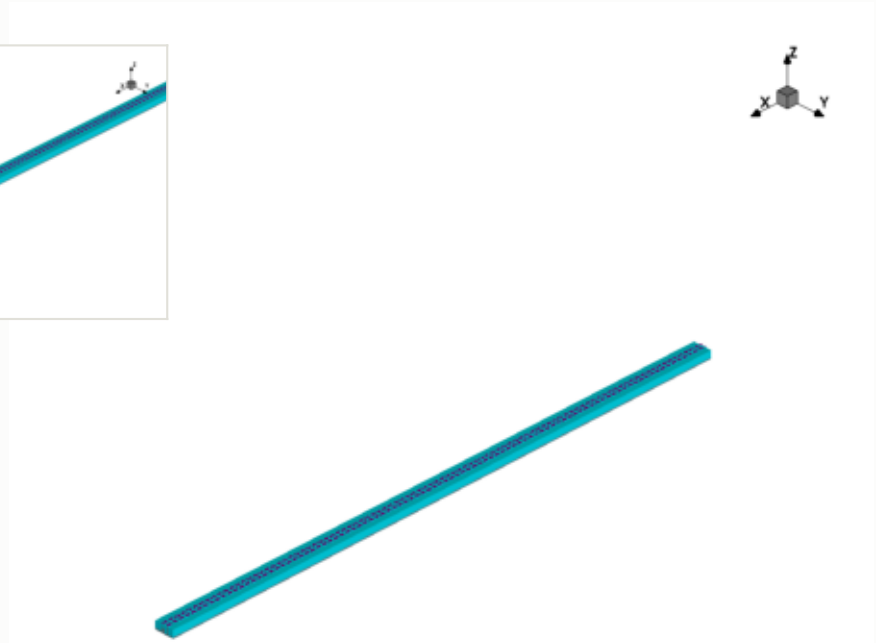
litho_patterning · KrF

0.199	71	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

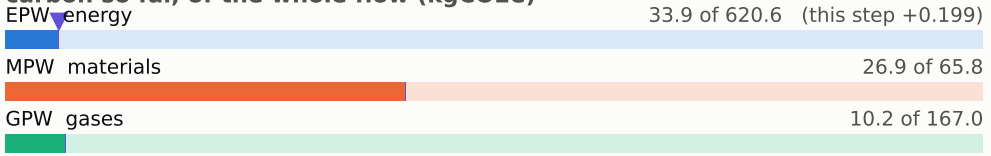




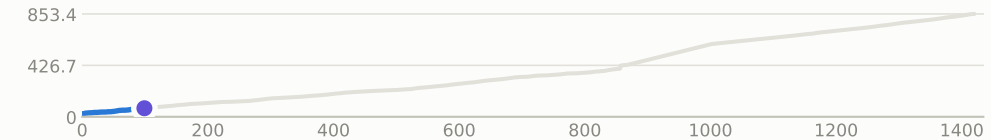
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

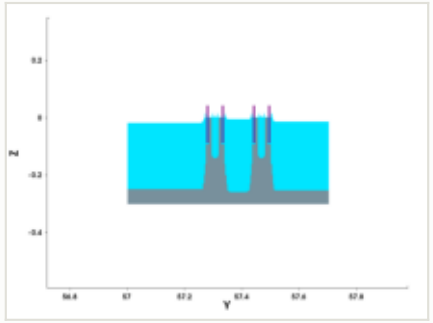
carbon so far, of the whole flow (kgCO2e)



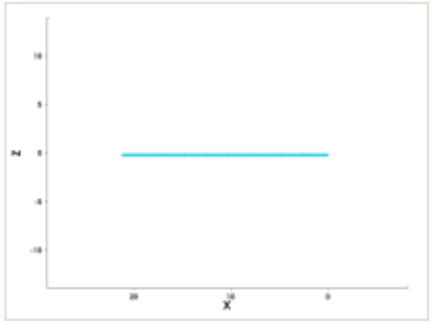
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

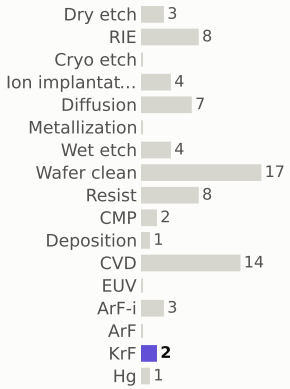


y cut at 57.351 µm

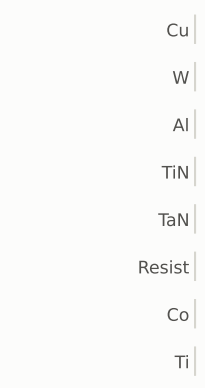


mask: layer 2100:0, positive (GDSII)

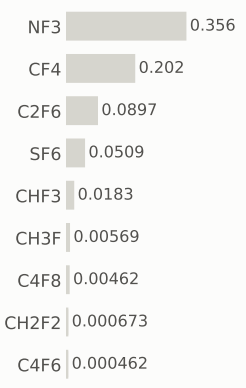
steps by process area



materials deposited (g)



gases emitted (g)



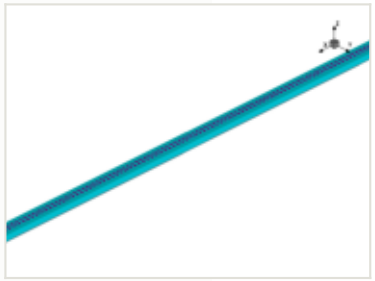
Step #1020: Ion_implantation

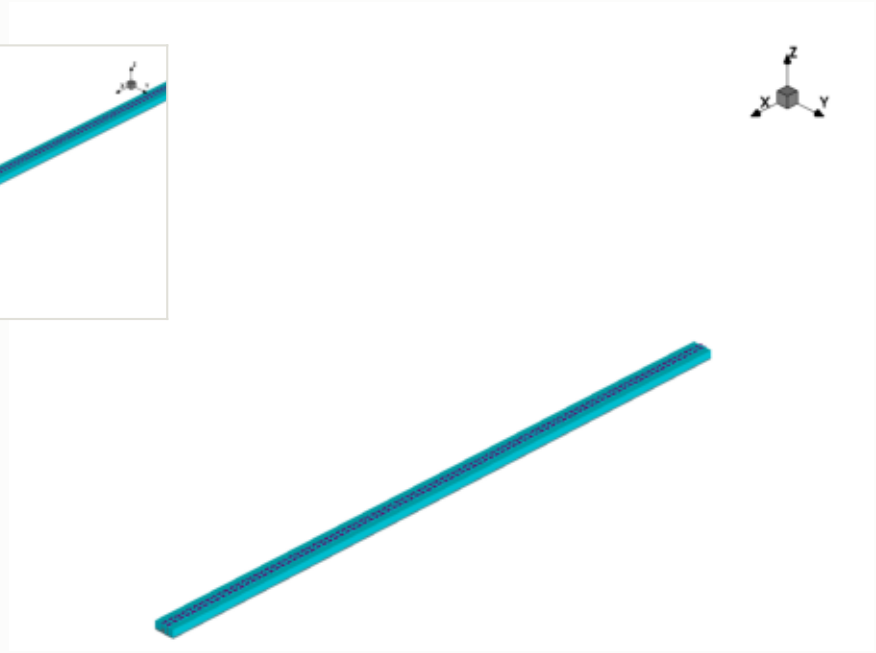
implant_ellipsoid · Ion_implantation

0.801	71.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

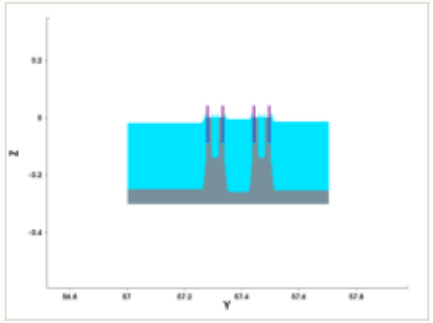
3D structure after this step

zoomed to fit

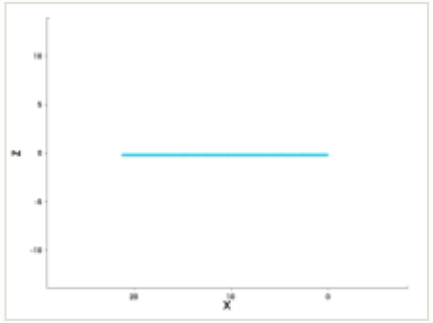




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

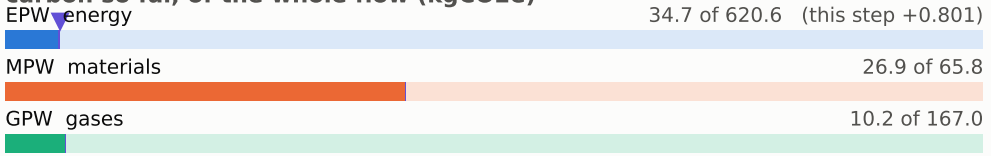
no mask for this step

this step

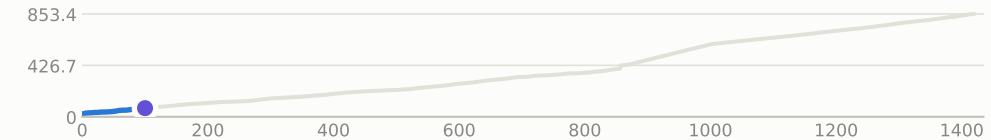
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

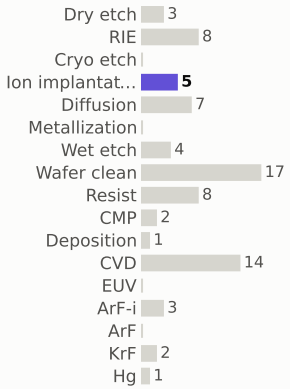
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

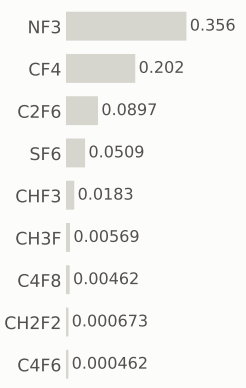


steps by process area



materials deposited (g)

gases emitted (g)



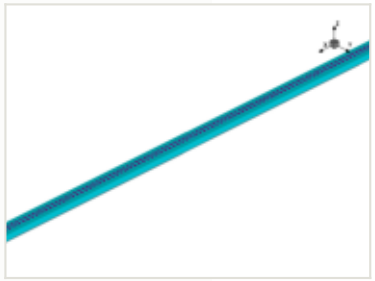
Step #1030: Dry_etch

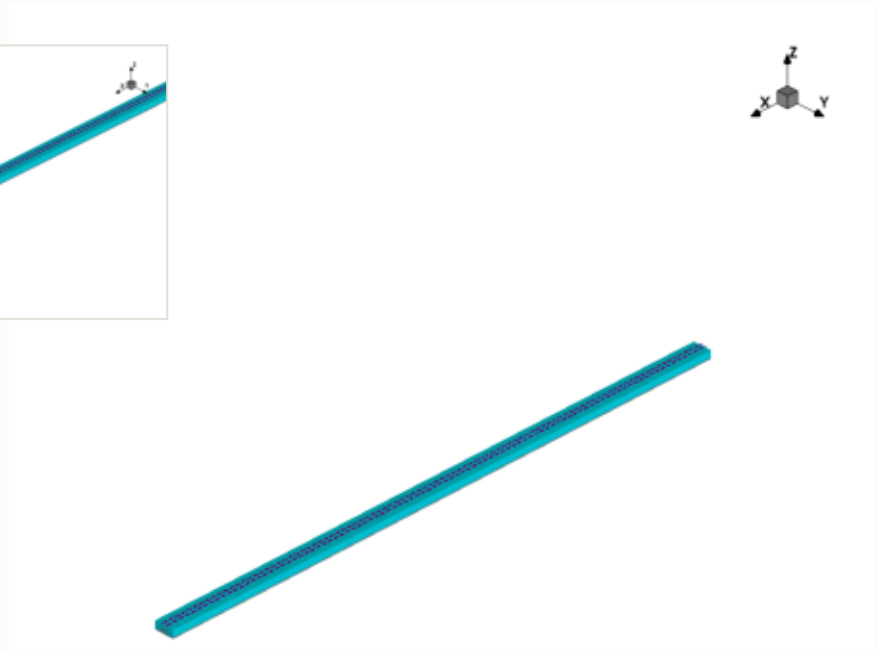
etch_strip · Dry_etch

0.801	72.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



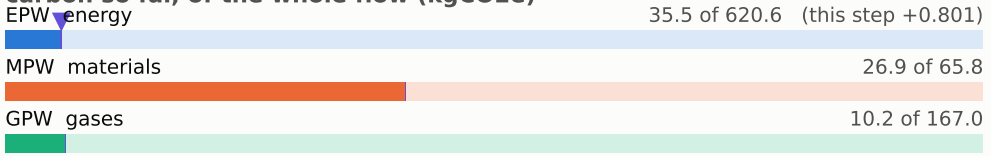


this step

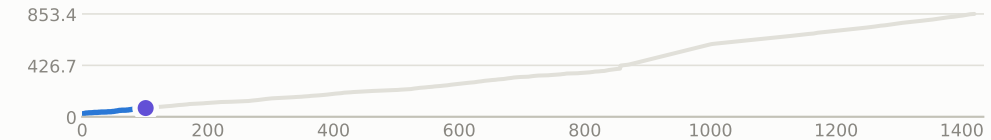
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

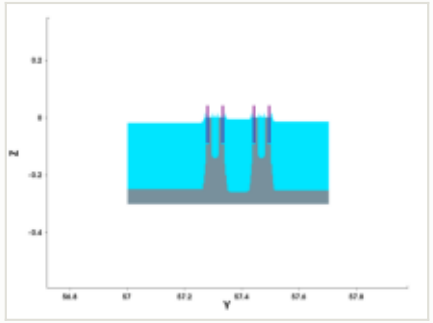
carbon so far, of the whole flow (kgCO2e)



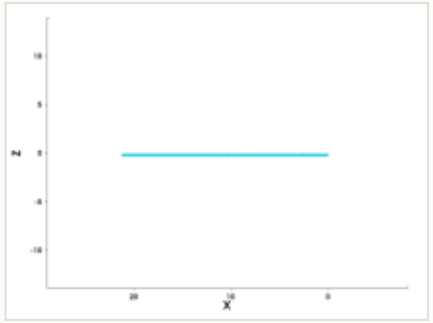
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



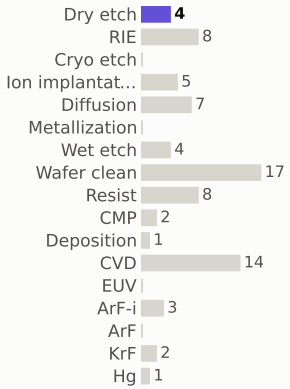
y cut at 57.351 µm



mask for this step

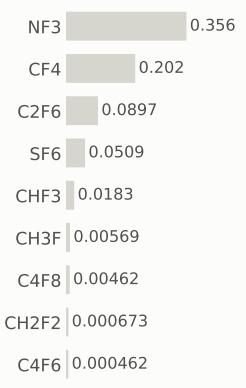
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



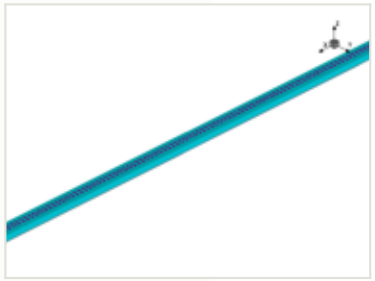
Step #1040: Wafer_clean

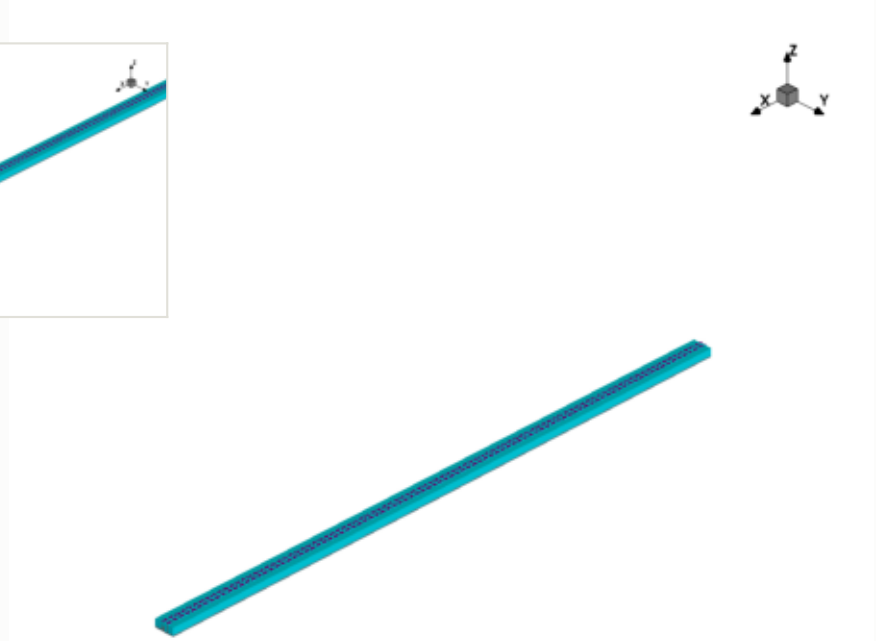
clean · Wafer_clean

0.169	72.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

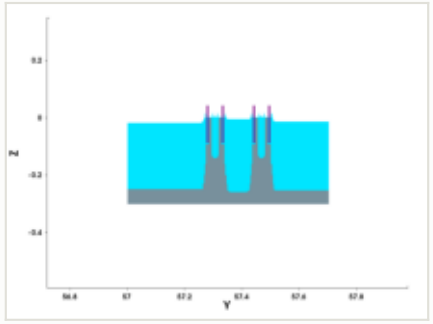
3D structure after this step

zoomed to fit

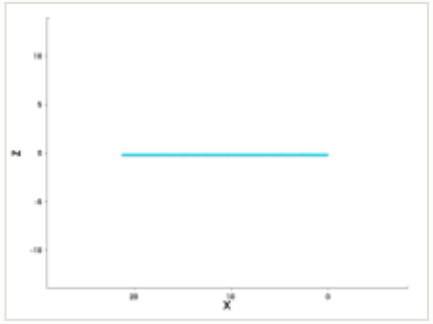




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

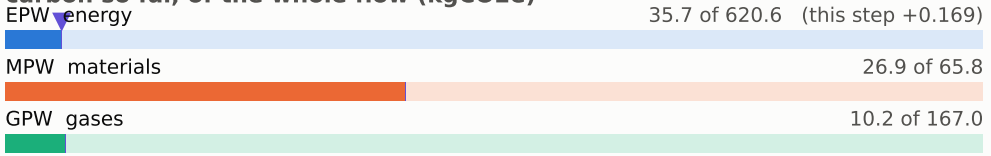
no mask for this step

this step

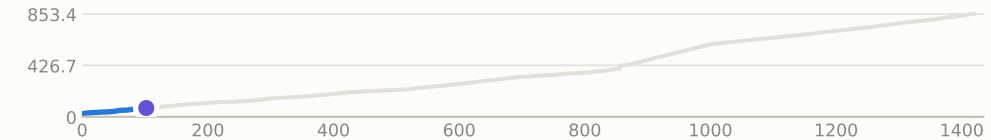
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

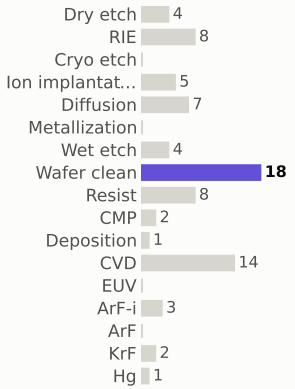
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

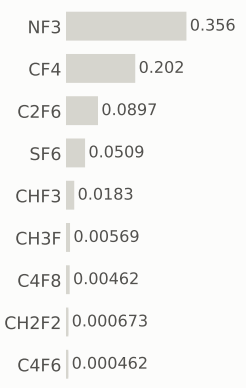


steps by process area



materials deposited (g)

gases emitted (g)



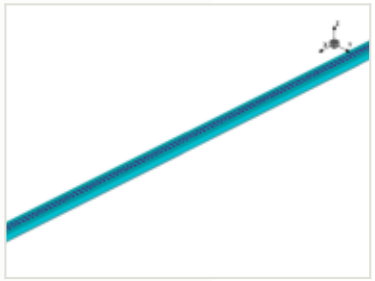
Step #1050: Resist

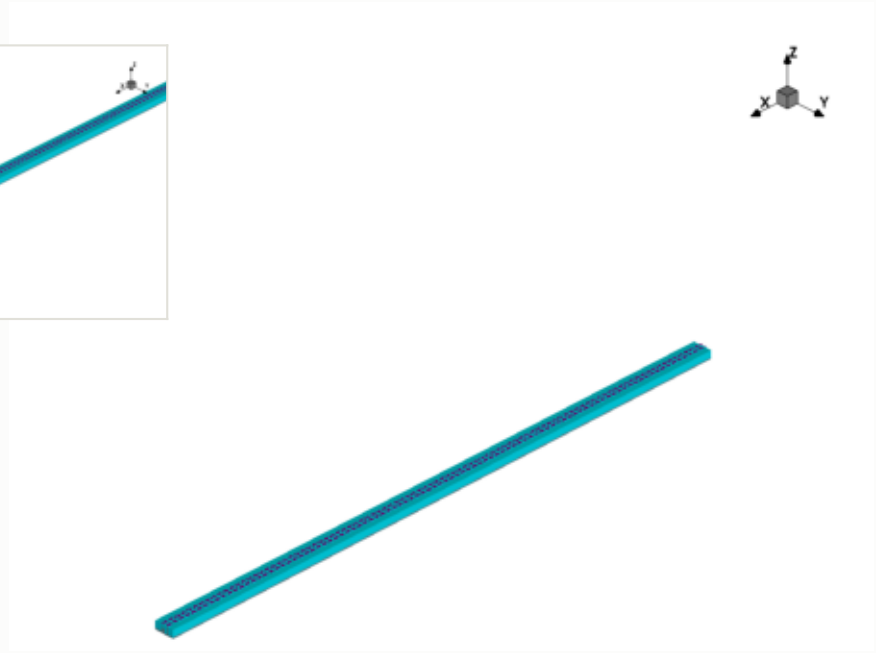
resist · Resist

0.169	73	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





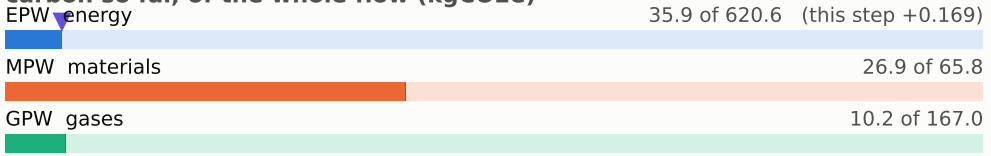
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

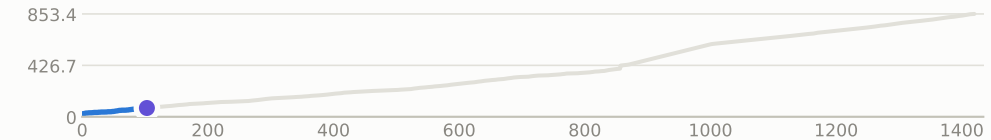
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

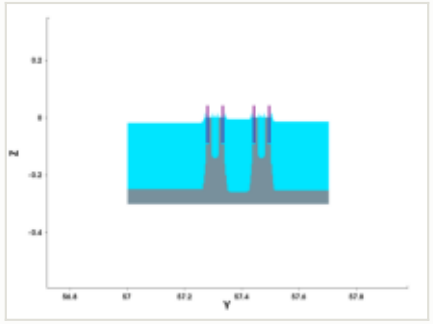
carbon so far, of the whole flow (kgCO2e)



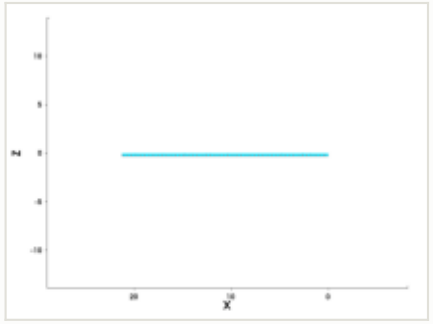
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



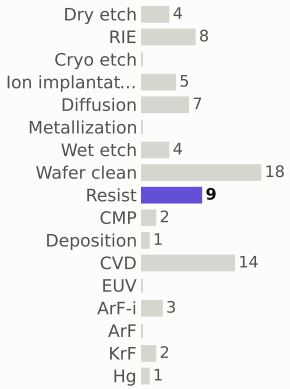
y cut at 57.351 µm



mask for this step

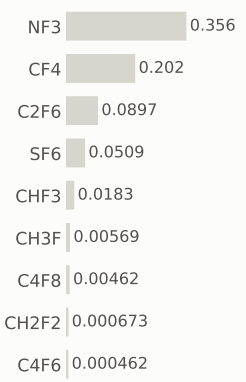
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



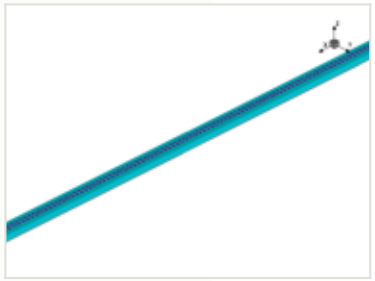
Step #1060: KrF

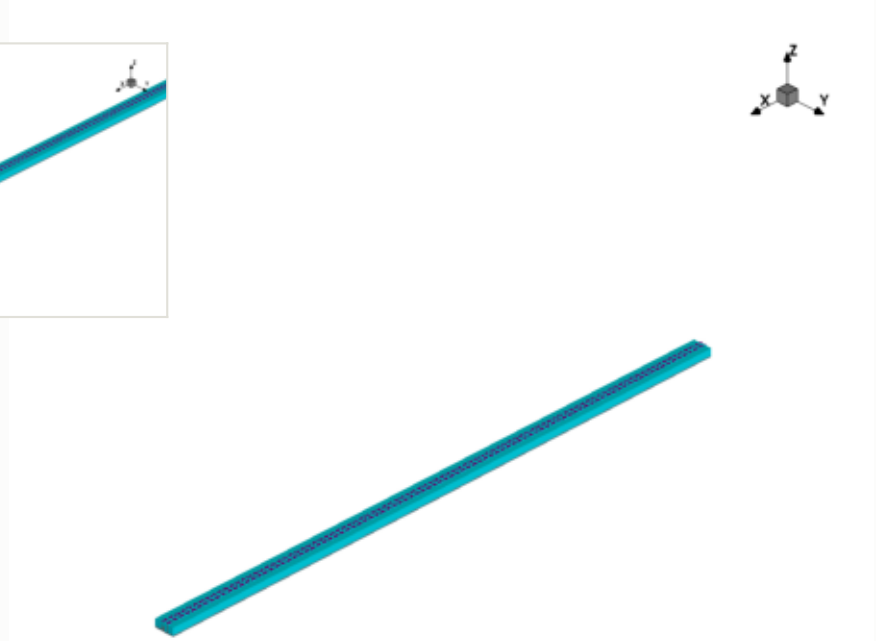
litho_patterning · KrF

0.199	73.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



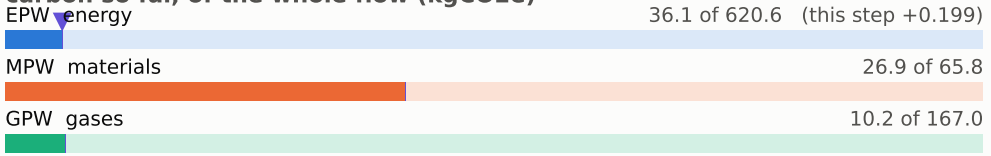


this step

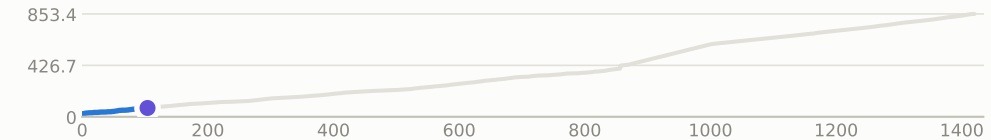
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

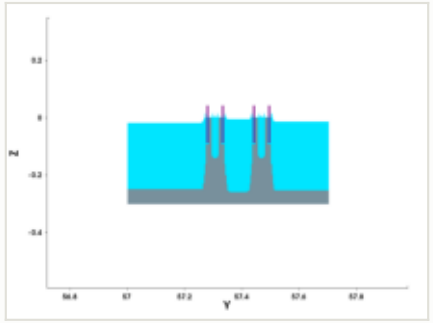
carbon so far, of the whole flow (kgCO2e)



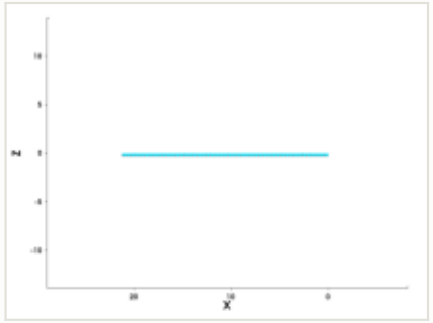
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

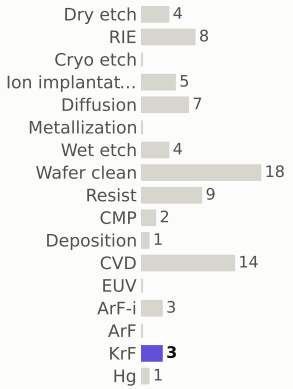


y cut at 57.351 µm

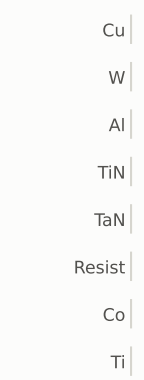


mask: layer 2100:0, positive (GDSII)

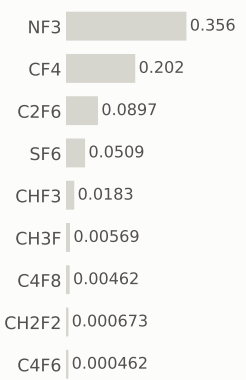
steps by process area



materials deposited (g)



gases emitted (g)



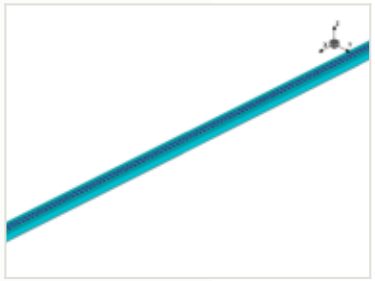
Step #1070: Ion_implantation

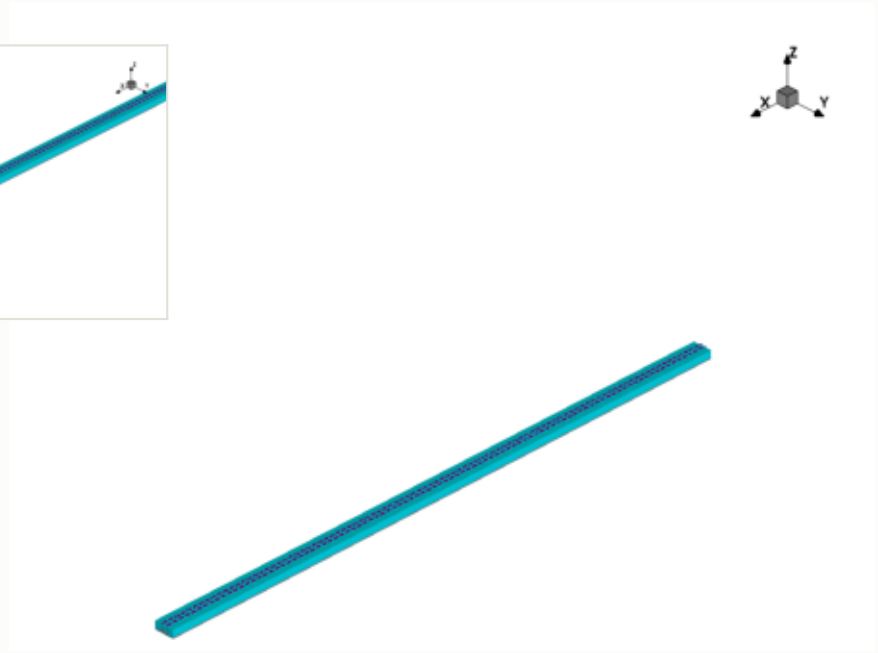
implant_ellipsoid · Ion_implantation

0.801	74	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

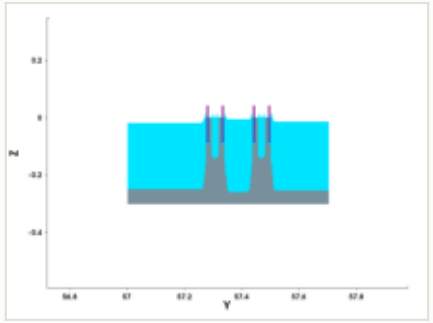
3D structure after this step

zoomed to fit

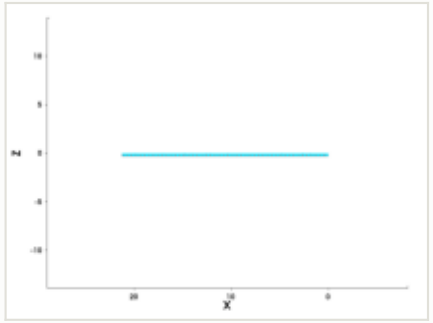




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

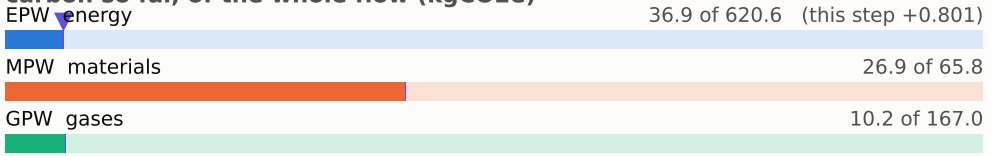
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

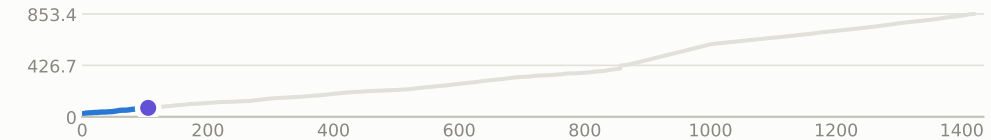
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

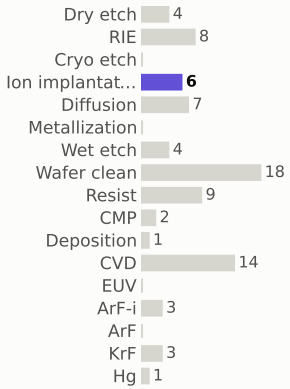
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

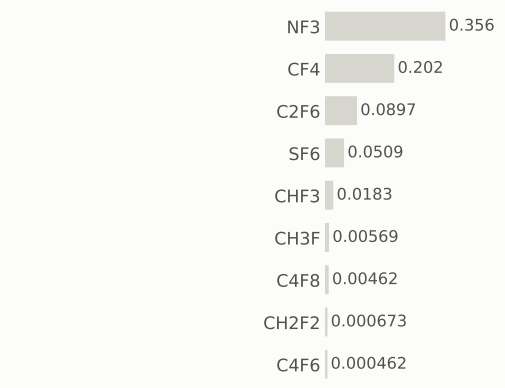


steps by process area



materials deposited (g)

gases emitted (g)



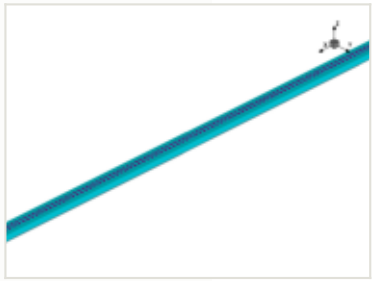
Step #1080: Dry_etch

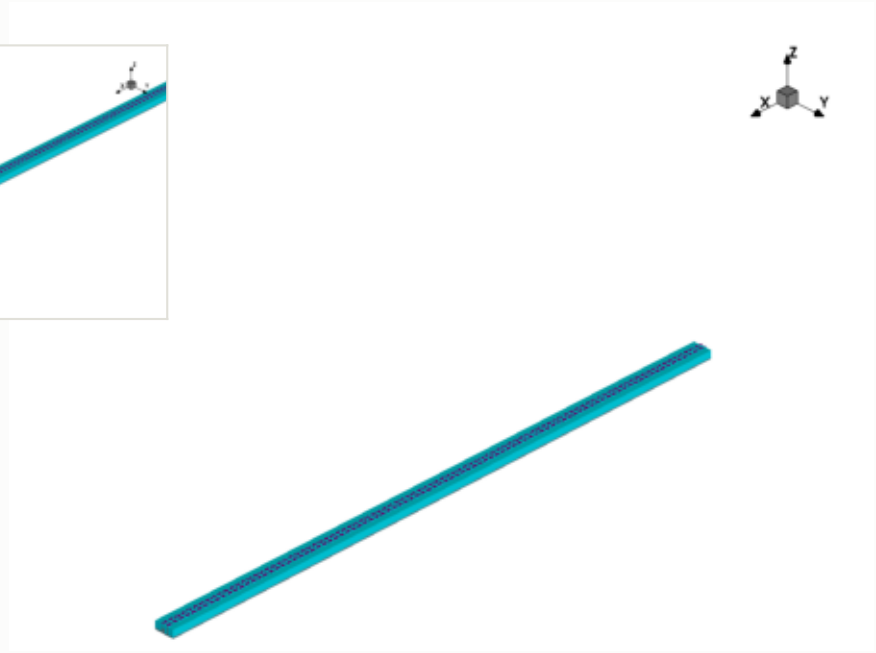
etch_strip · Dry_etch

0.801	74.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



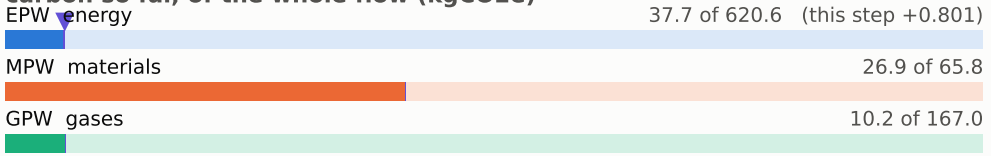


this step

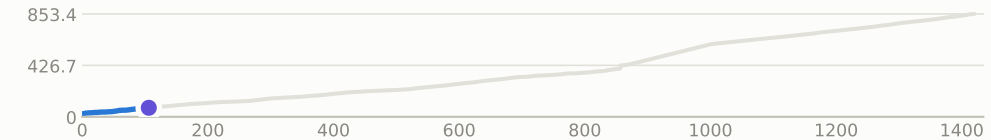
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

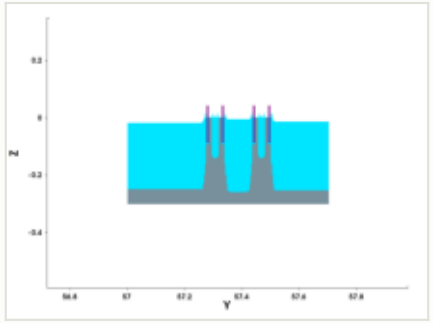
carbon so far, of the whole flow (kgCO2e)



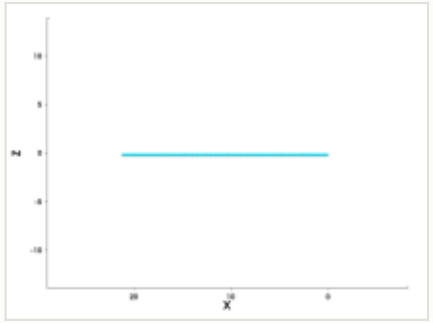
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



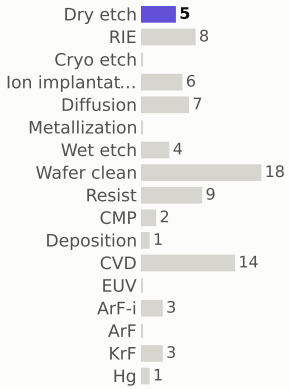
y cut at 57.351 µm



mask for this step

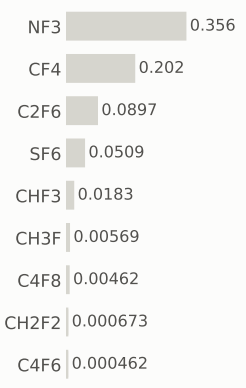
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



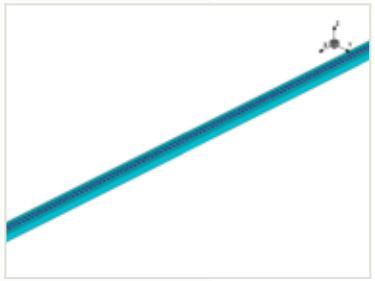
Step #1090: Wafer_clean

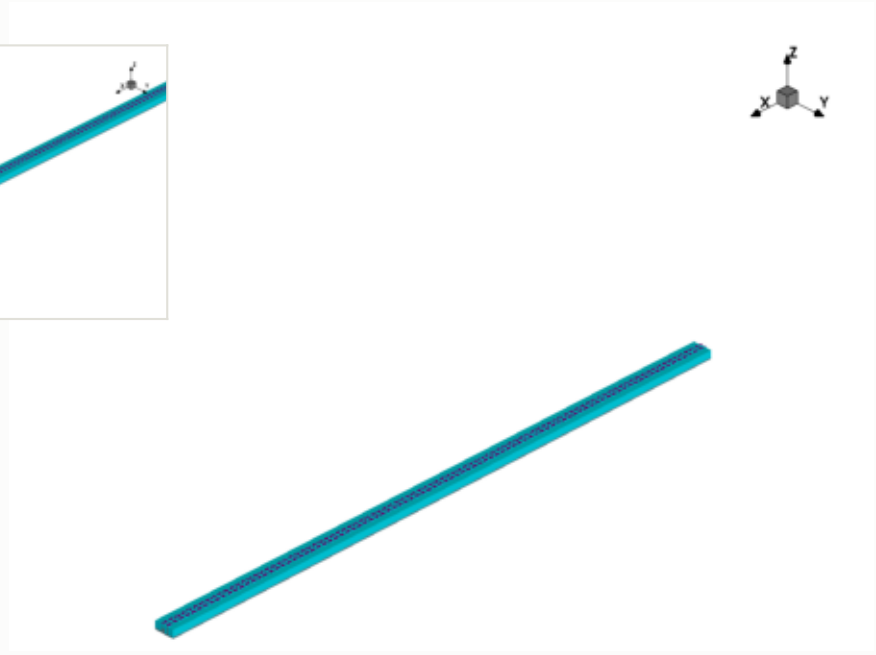
clean · Wafer_clean

0.169	74.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





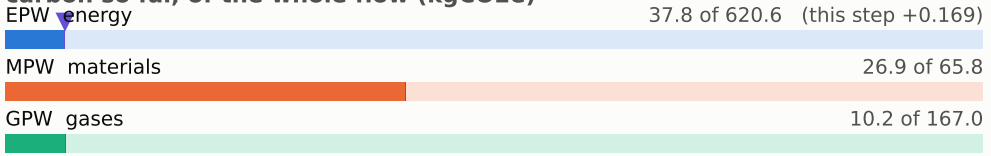
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

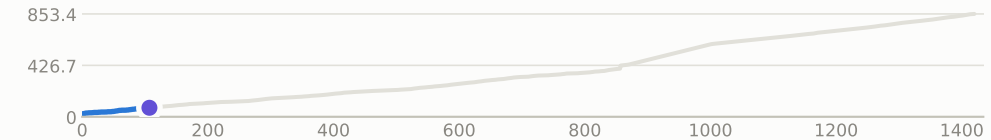
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

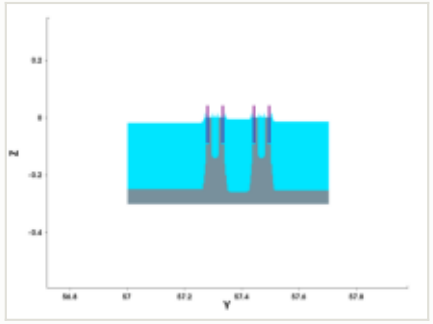
carbon so far, of the whole flow (kgCO2e)



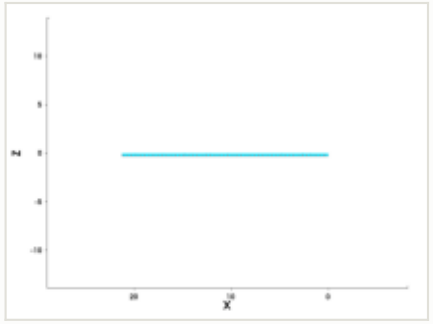
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



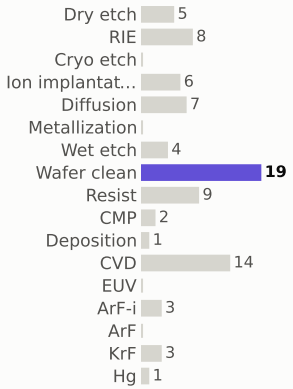
y cut at 57.351 µm



mask for this step

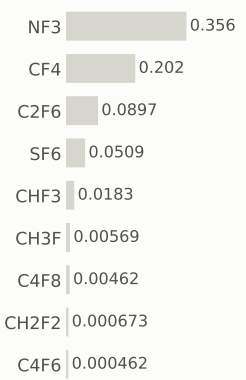
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



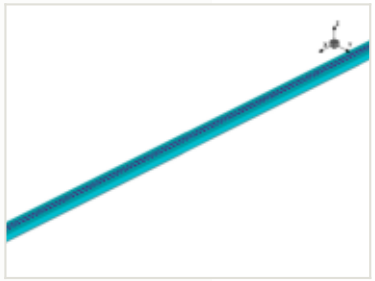
Step #1100: Resist

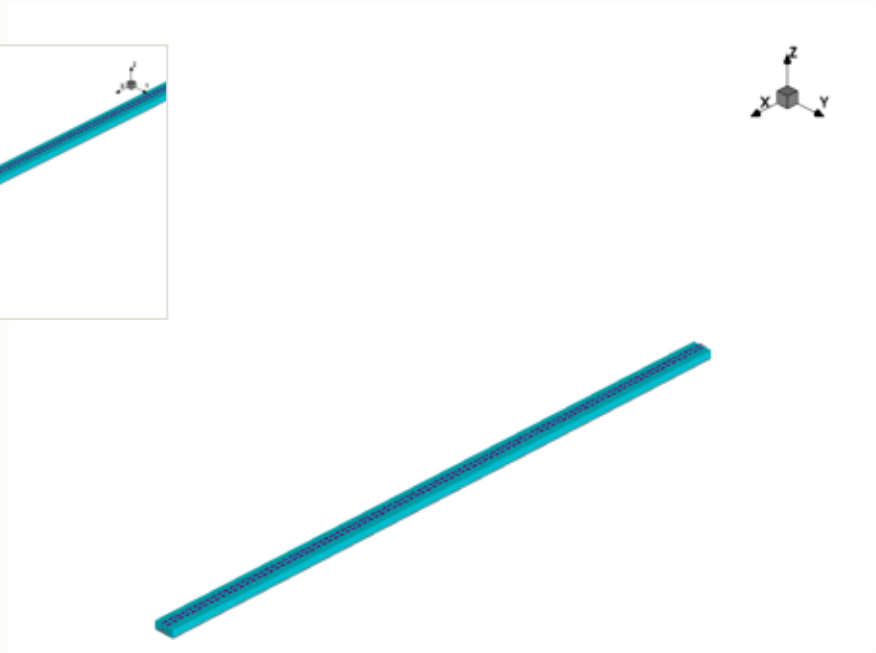
resist · Resist

0.169	75.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





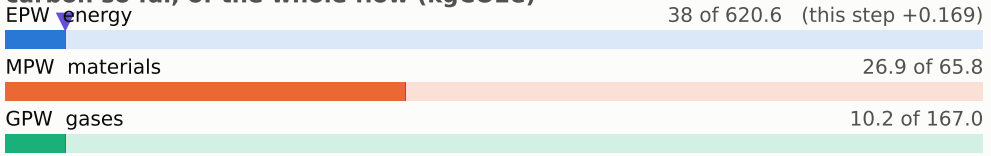
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

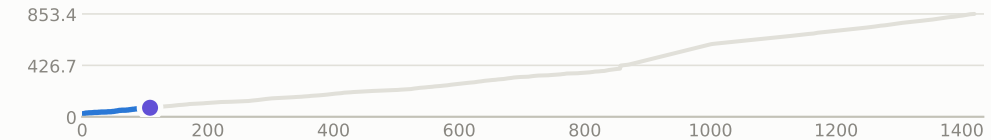
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

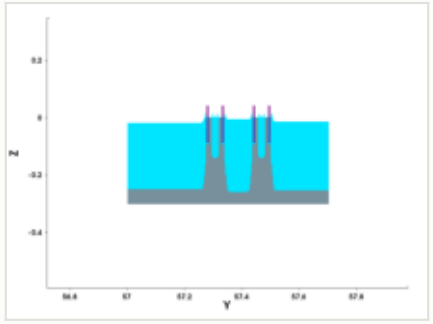
carbon so far, of the whole flow (kgCO2e)



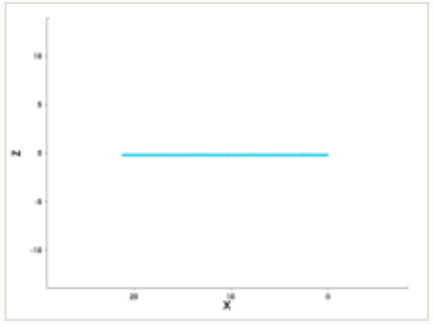
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



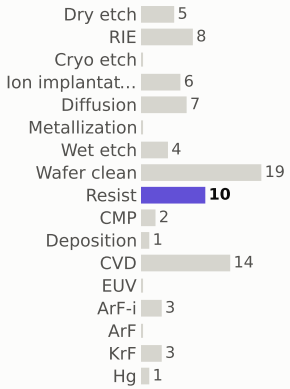
y cut at 57.351 µm



mask for this step

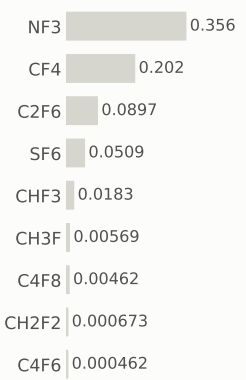
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



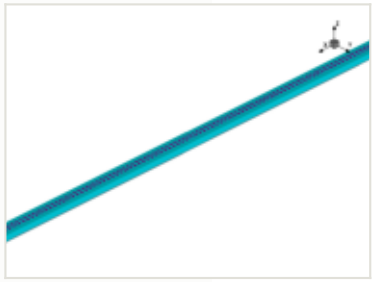
Step #1110: KrF

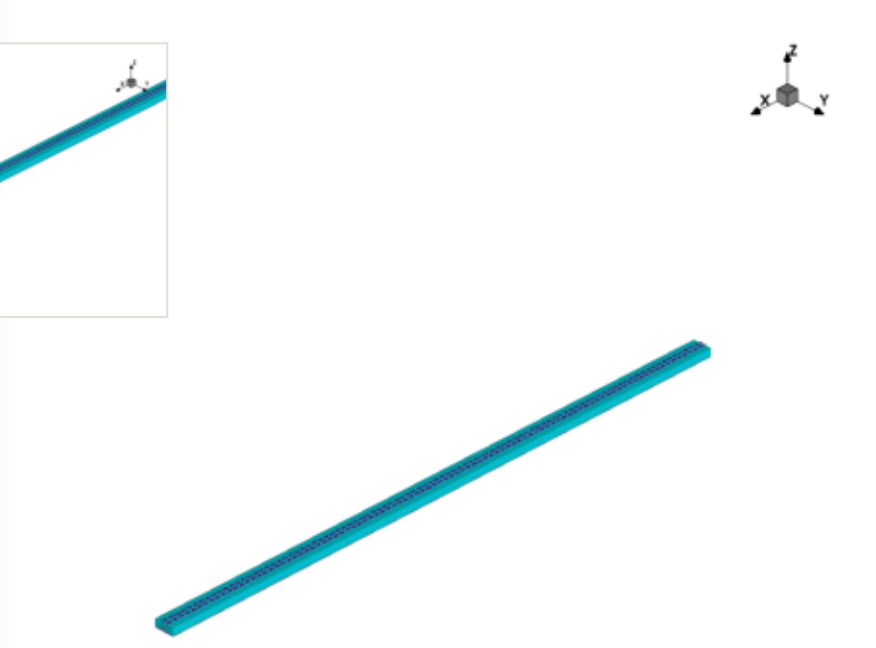
litho_patterning · KrF

0.199	75.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



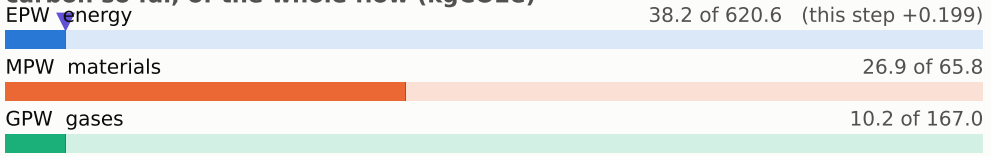


this step

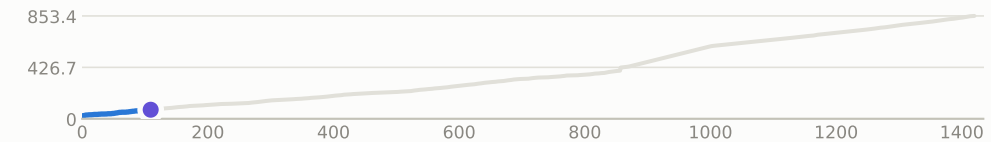
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

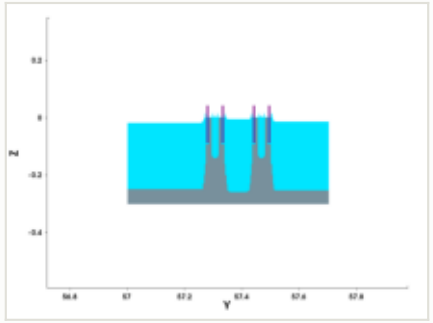
carbon so far, of the whole flow (kgCO2e)



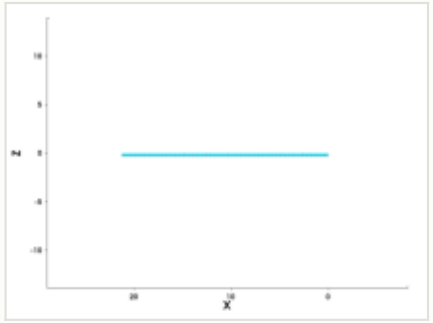
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

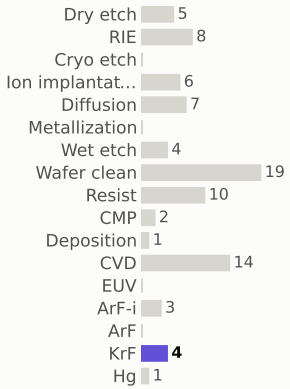


y cut at 57.351 µm



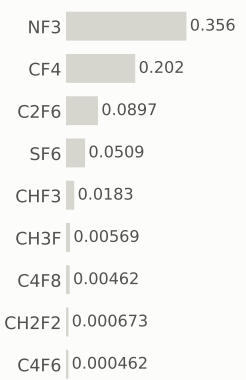
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



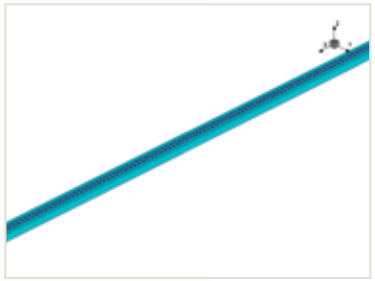
Step #1120: Ion_implantation

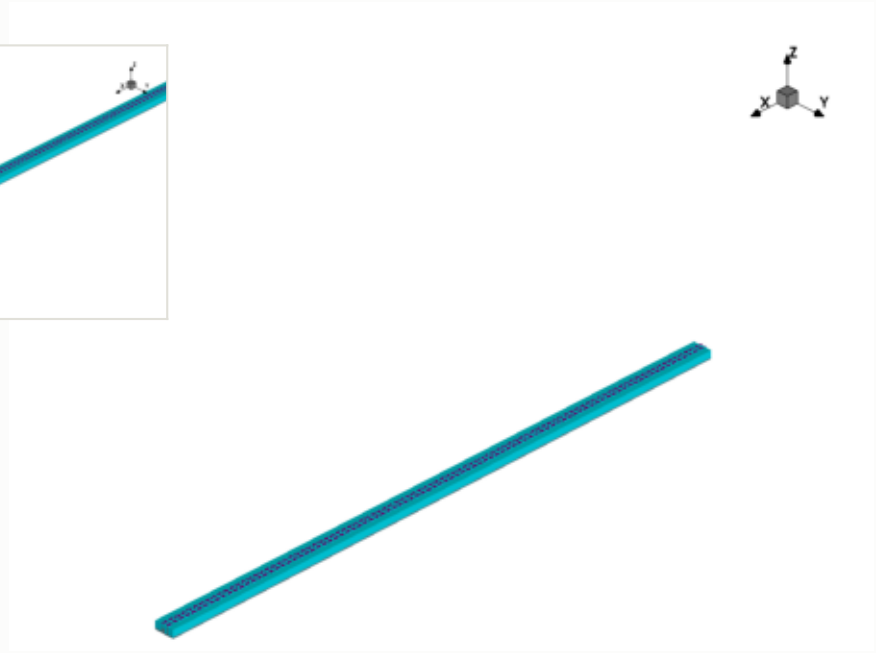
implant_ellipsoid · Ion_implantation

0.801	76.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

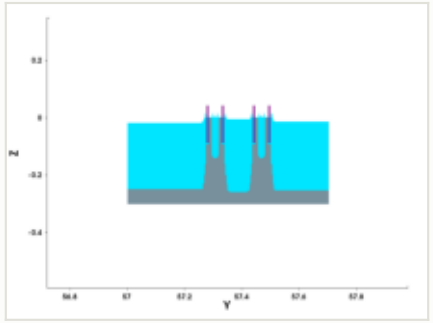
3D structure after this step

zoomed to fit

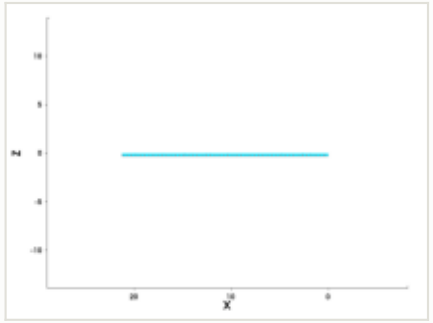




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

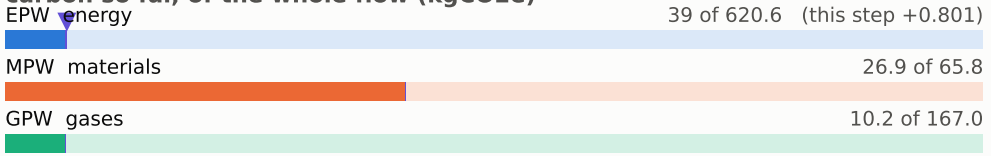
no mask for this step

this step

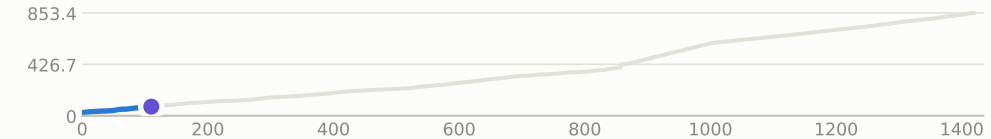
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

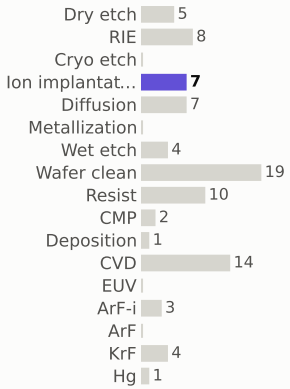
carbon so far, of the whole flow (kgCO2e)



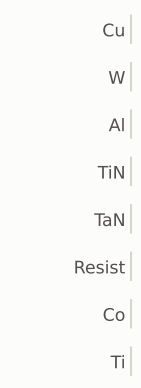
embodied carbon up to each step (kgCO2e per wafer)



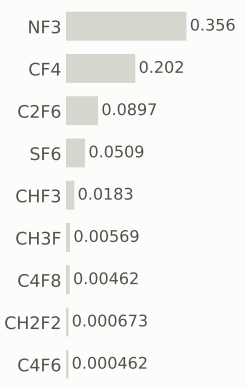
steps by process area



materials deposited (g)



gases emitted (g)



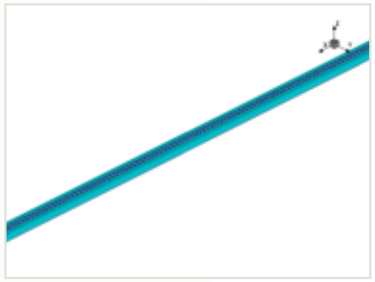
Step #1130: Dry_etch

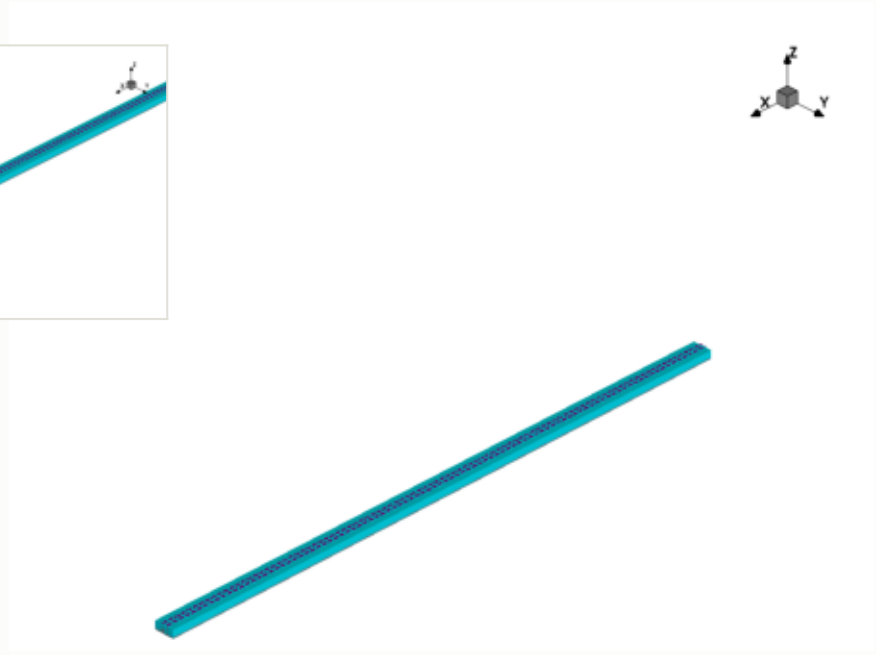
etch_strip · Dry_etch

0.801	76.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



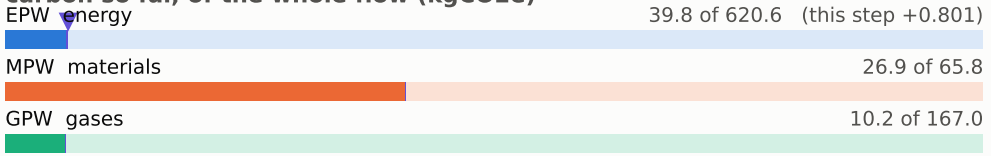


this step

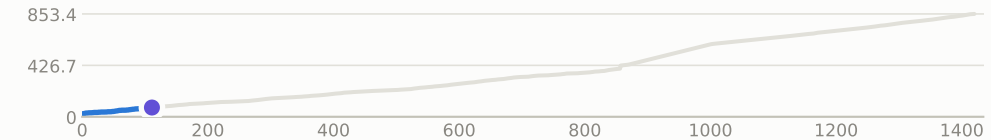
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

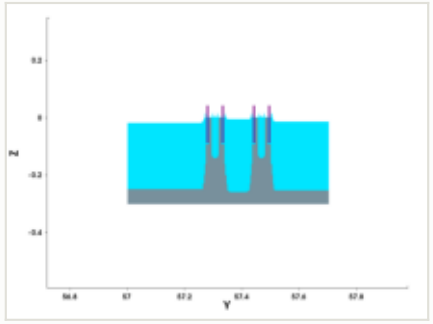
carbon so far, of the whole flow (kgCO2e)



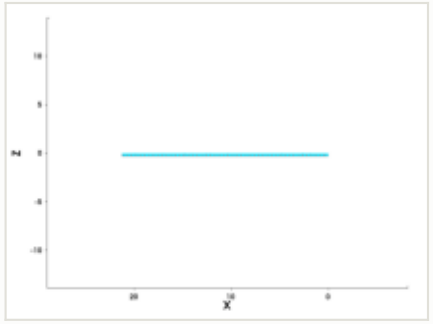
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



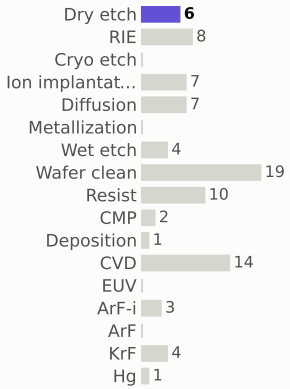
y cut at 57.351 µm



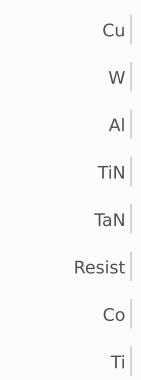
mask for this step

no mask for this step

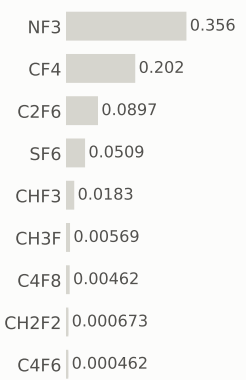
steps by process area



materials deposited (g)



gases emitted (g)



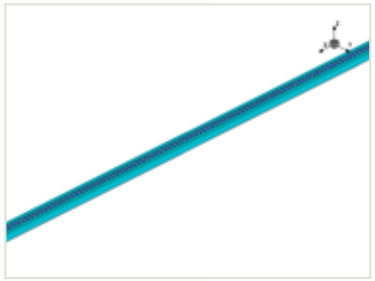
Step #1140: Wafer_clean

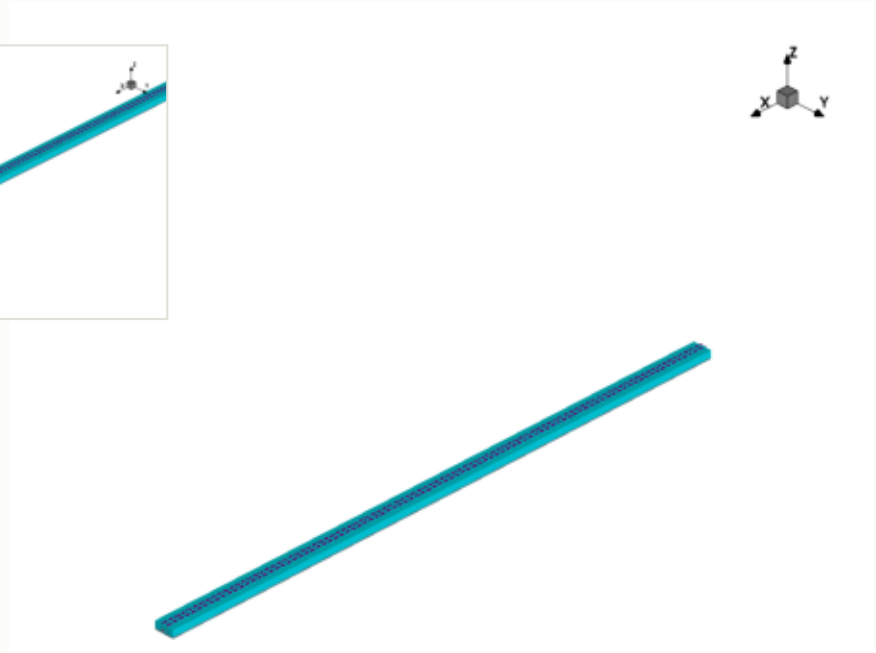
clean · Wafer_clean

0.169	77.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





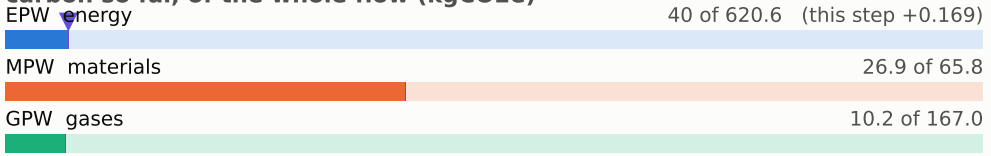
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

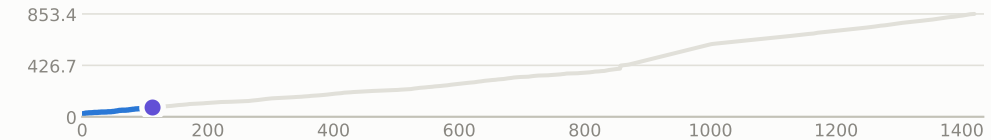
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

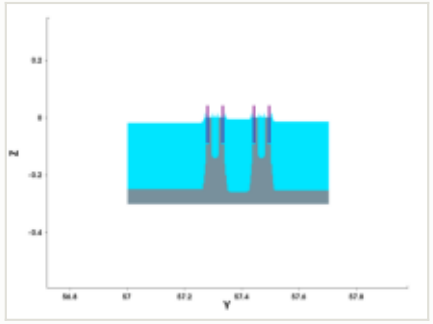
carbon so far, of the whole flow (kgCO2e)



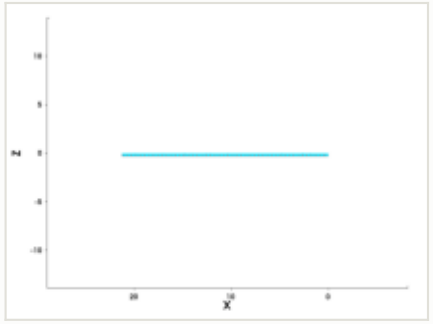
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



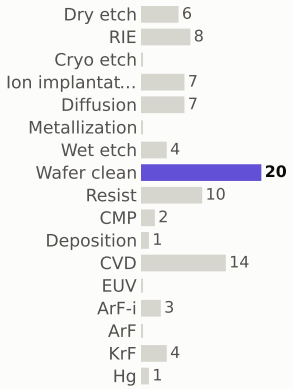
y cut at 57.351 µm



mask for this step

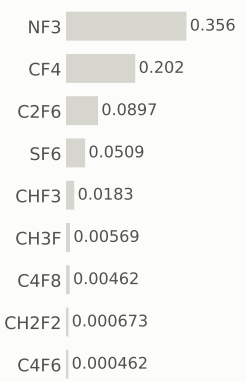
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



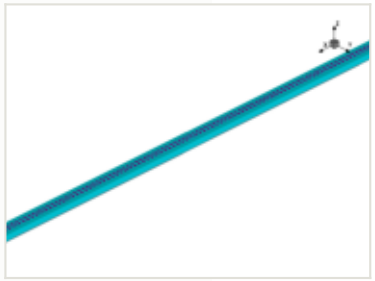
Step #1150: Resist

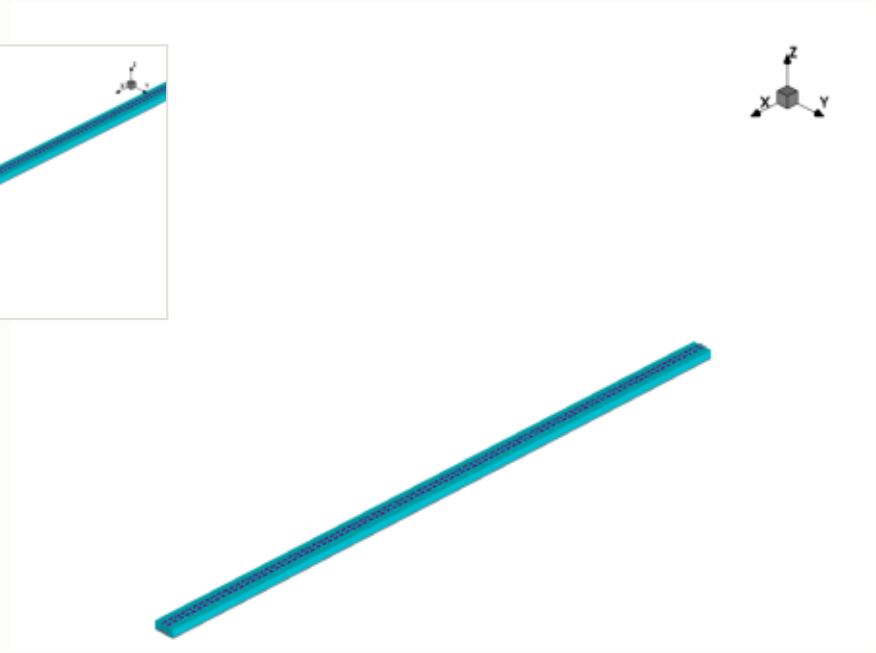
resist · Resist

0.169	77.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





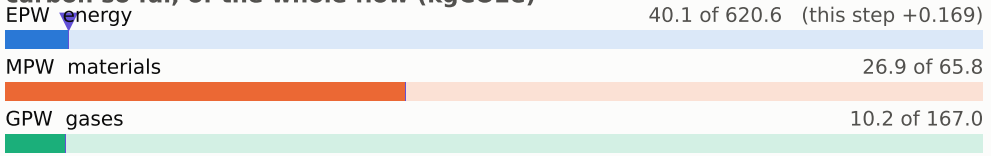
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

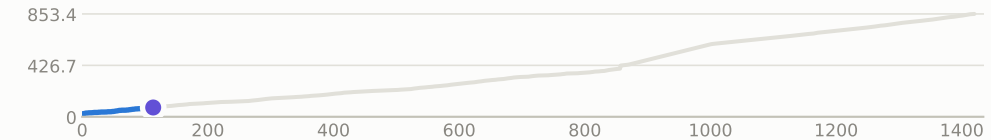
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

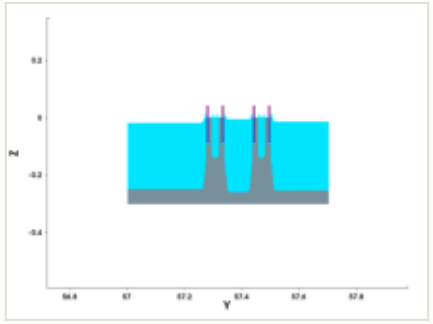
carbon so far, of the whole flow (kgCO2e)



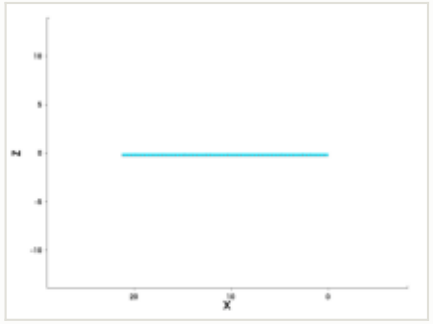
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



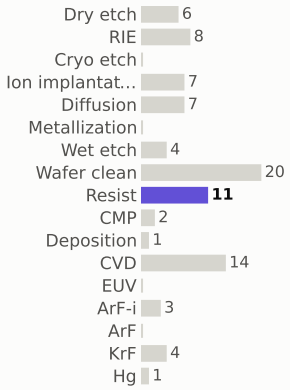
y cut at 57.351 µm



mask for this step

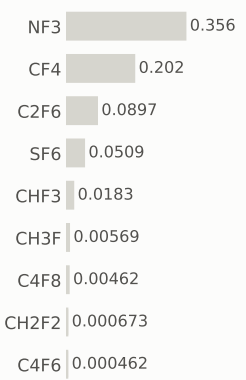
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



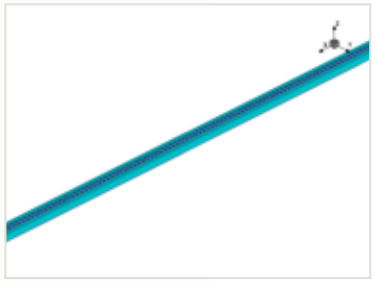
Step #1160: KrF

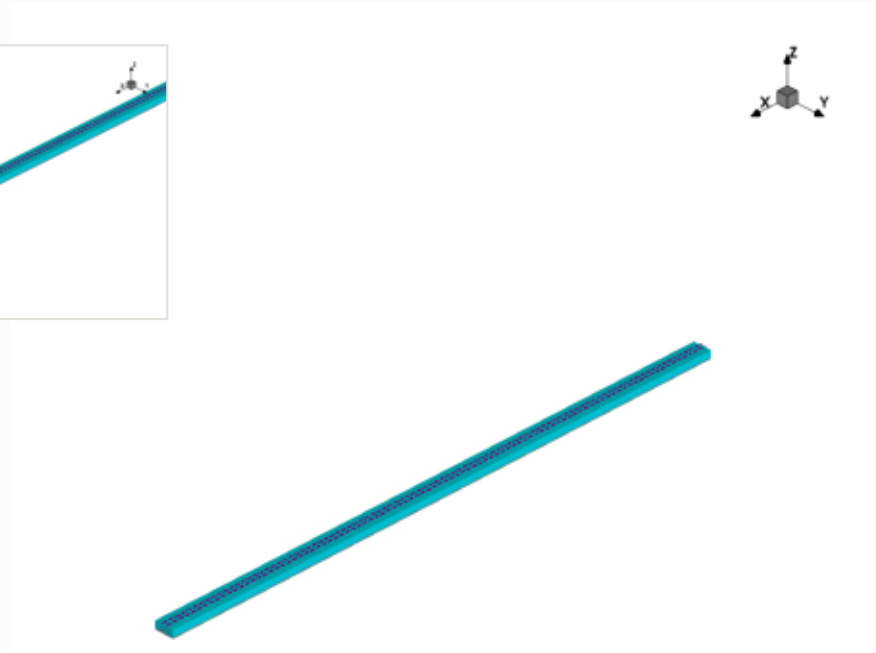
litho_patterning · KrF

0.199	77.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



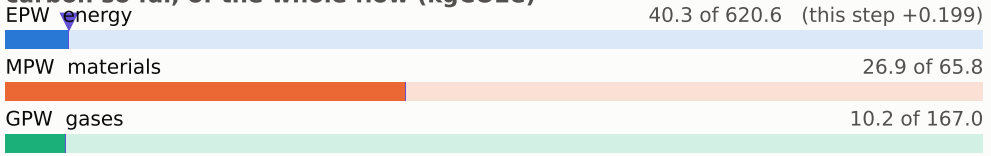


this step

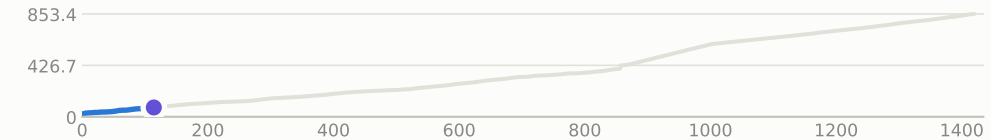
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

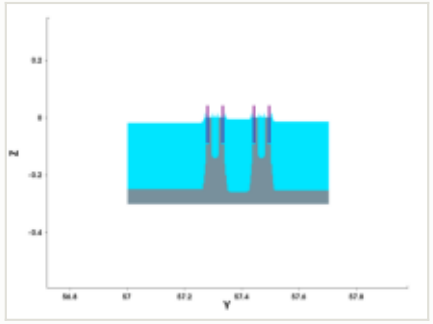
carbon so far, of the whole flow (kgCO2e)



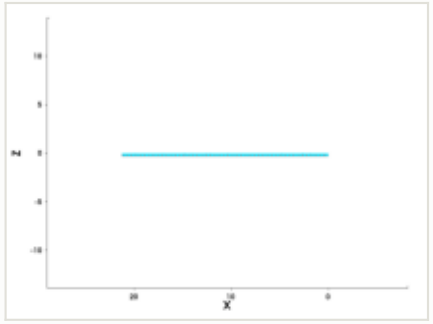
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

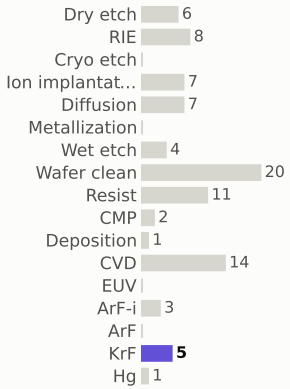


y cut at 57.351 µm



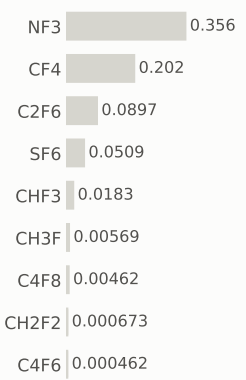
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



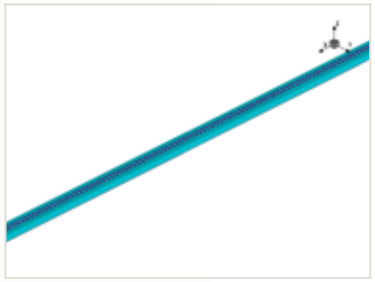
Step #1170: Ion_implantation

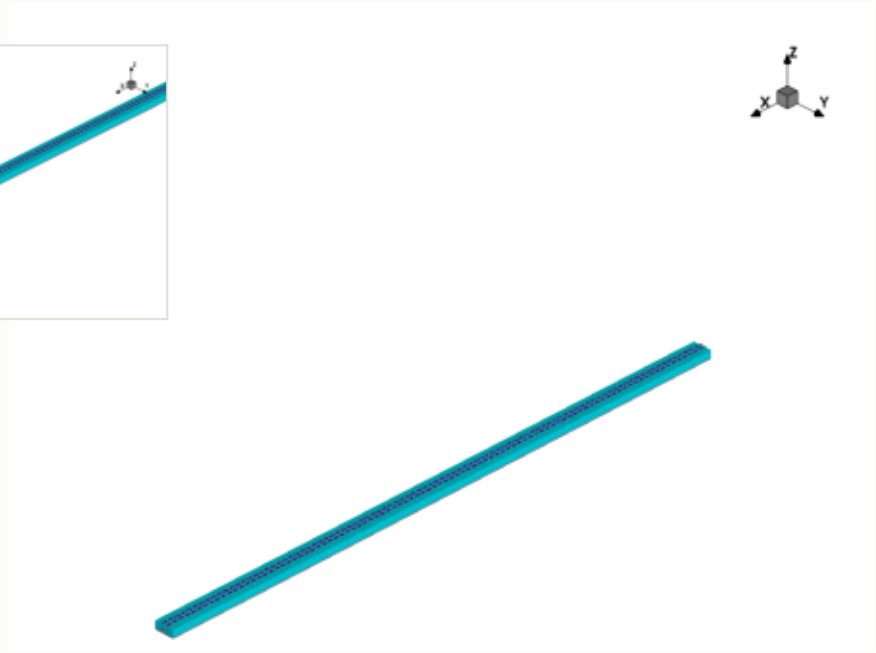
implant_ellipsoid · Ion_implantation

0.801	78.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

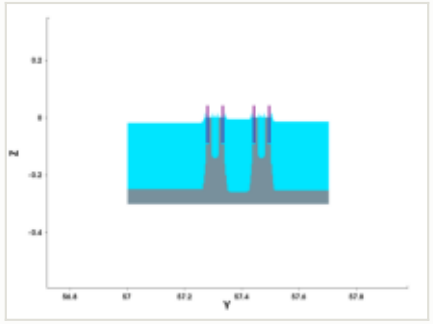
3D structure after this step

zoomed to fit

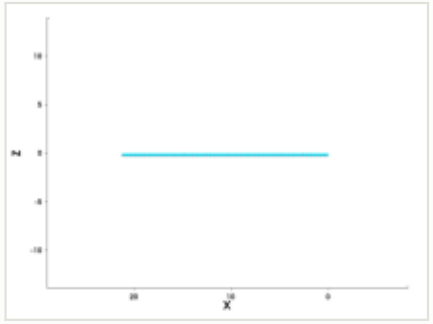




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

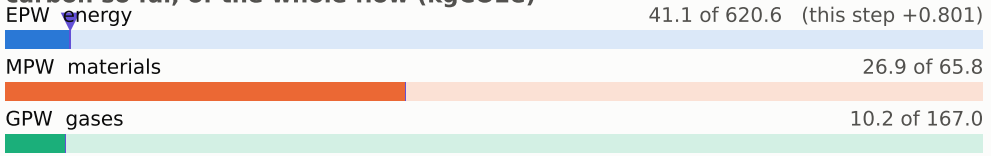
no mask for this step

this step

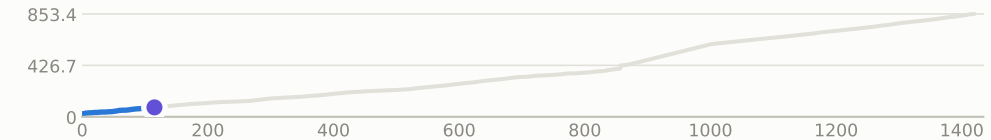
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

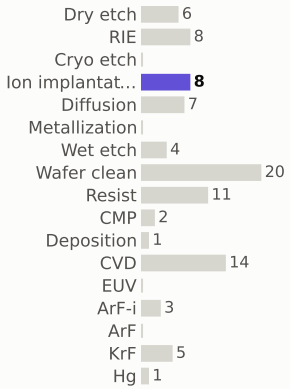
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

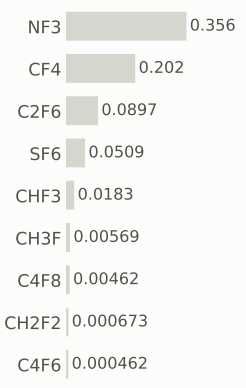


steps by process area



materials deposited (g)

gases emitted (g)



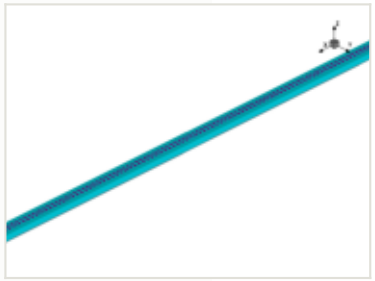
Step #1180: Ion_implantation

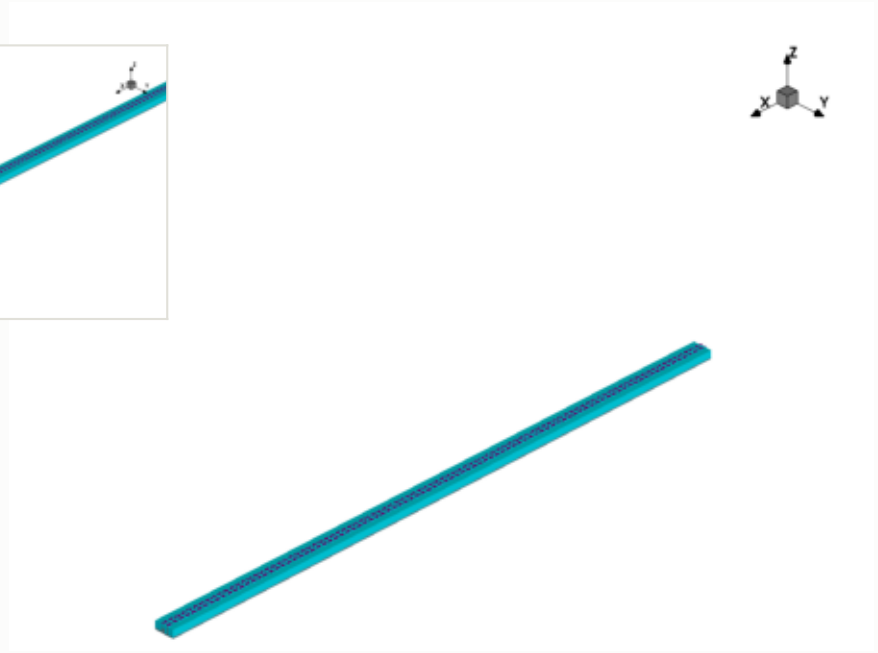
implant_ellipsoid · Ion_implantation

0.801	79	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

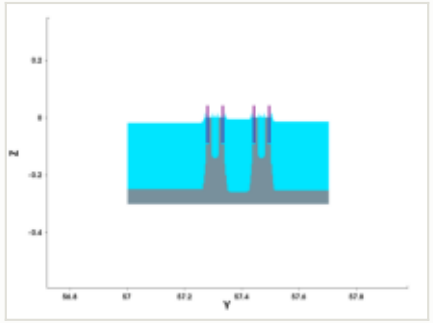
3D structure after this step

zoomed to fit

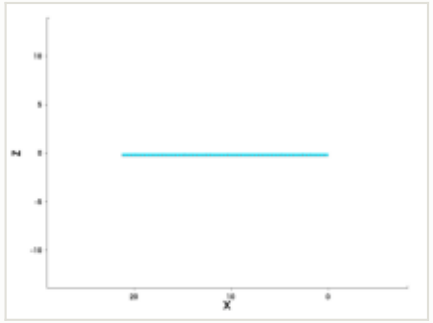




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

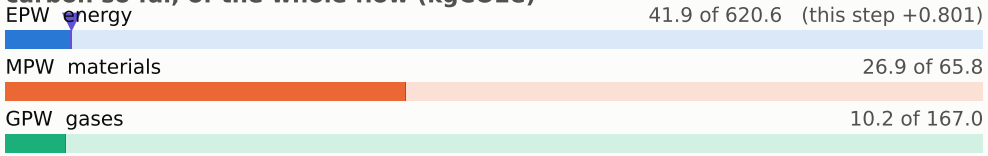
no mask for this step

this step

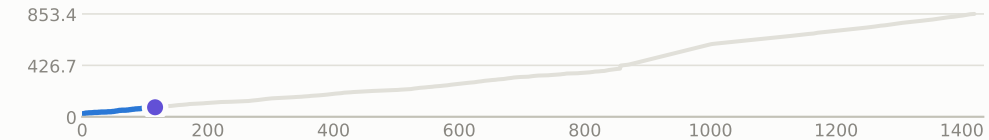
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

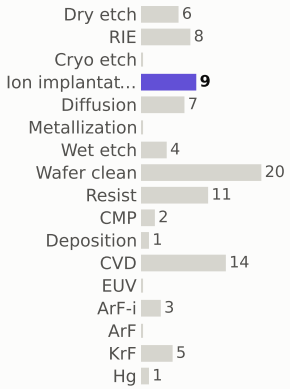
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

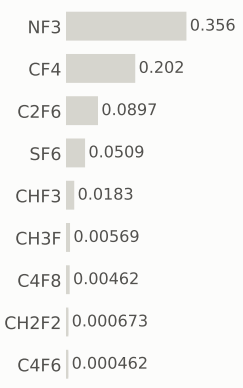


steps by process area



materials deposited (g)

gases emitted (g)



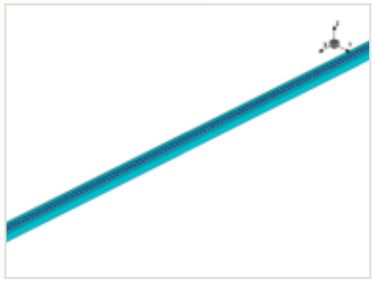
Step #1190: Ion_implantation

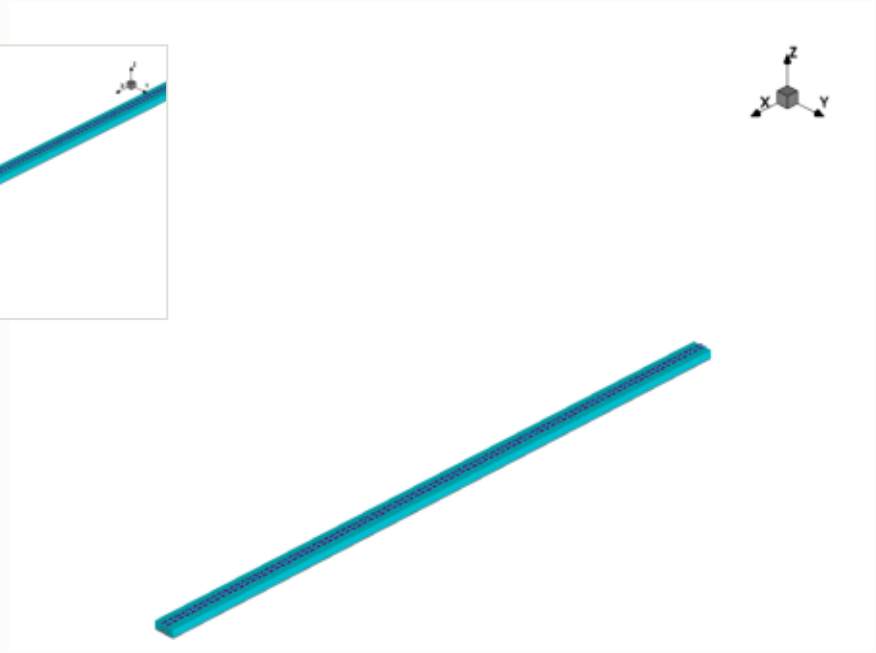
implant_ellipsoid · Ion_implantation

0.801	79.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

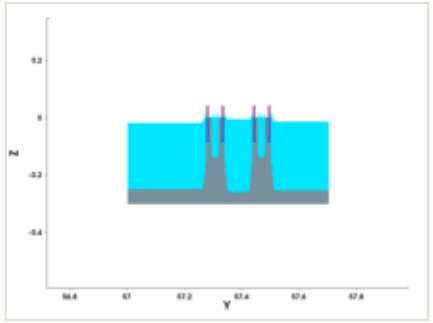
3D structure after this step

zoomed to fit

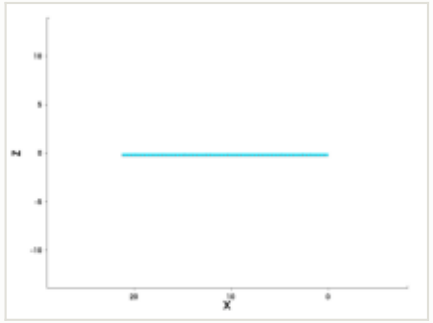




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

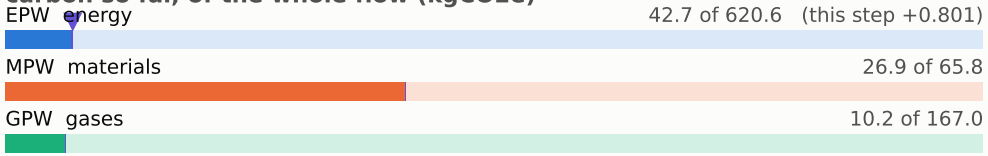
no mask for this step

this step

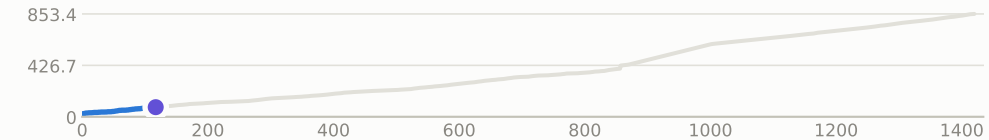
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

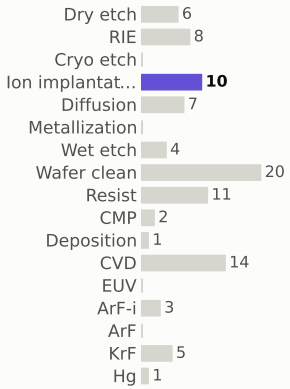
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

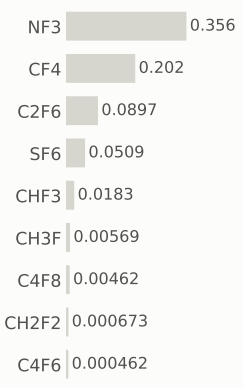


steps by process area



materials deposited (g)

gases emitted (g)



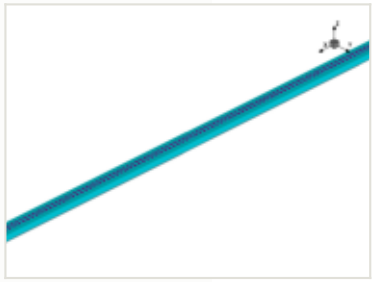
Step #1200: Dry_etch

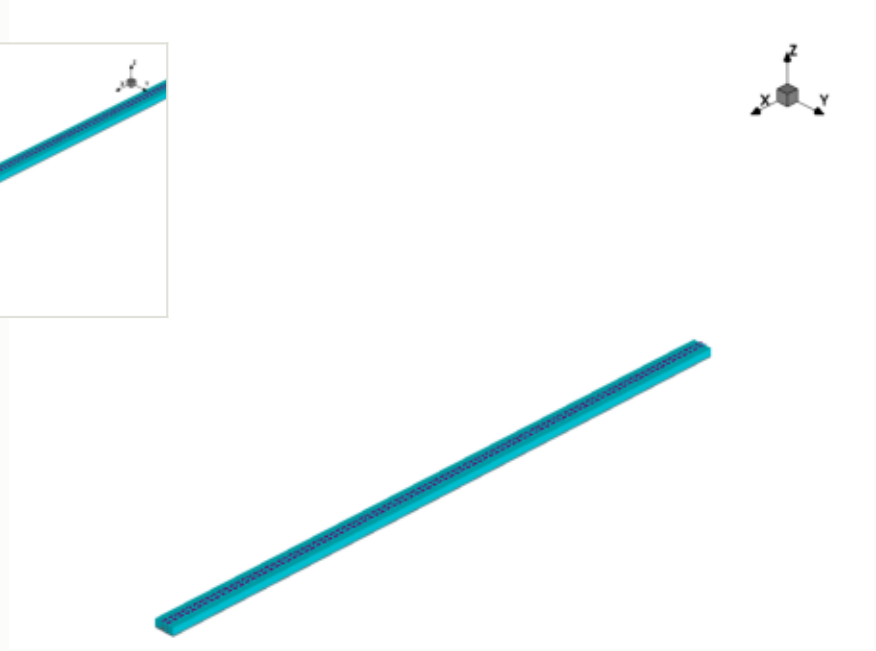
etch_strip · Dry_etch

0.801	80.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

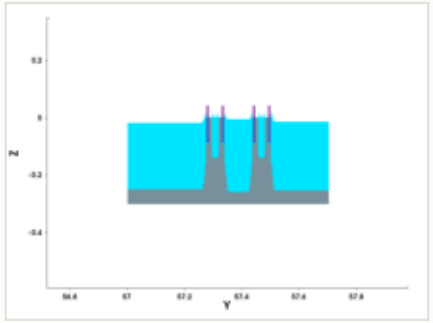
3D structure after this step

zoomed to fit

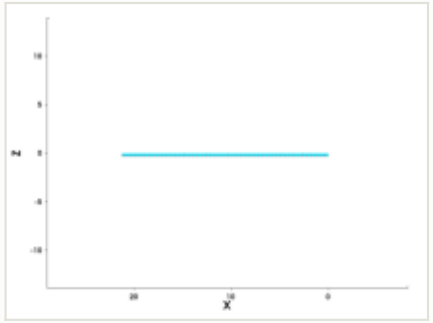




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

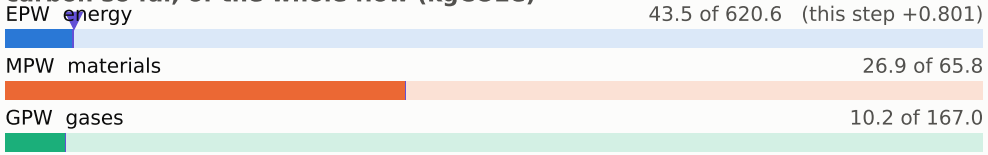
no mask for this step

this step

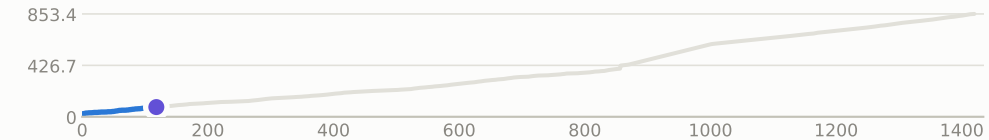
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

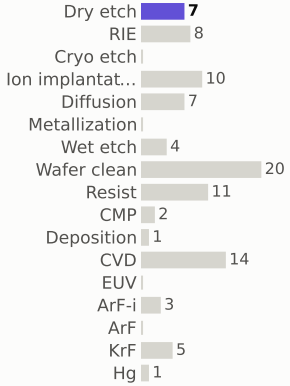
carbon so far, of the whole flow (kgCO2e)



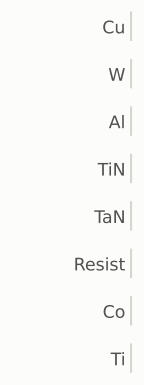
embodied carbon up to each step (kgCO2e per wafer)



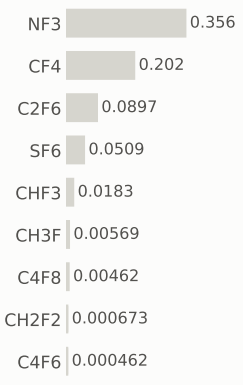
steps by process area



materials deposited (g)



gases emitted (g)



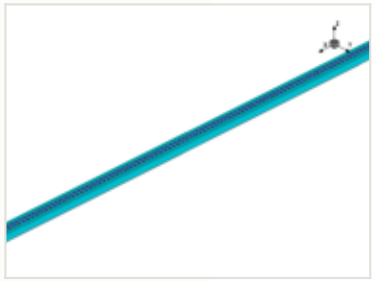
Step #1210: Wafer_clean

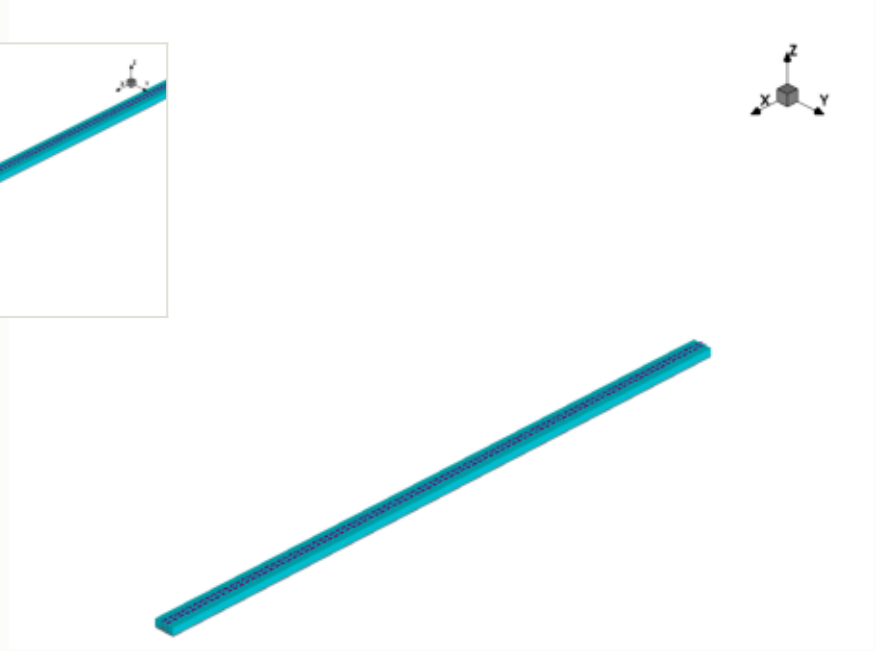
clean · Wafer_clean

0.169	80.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

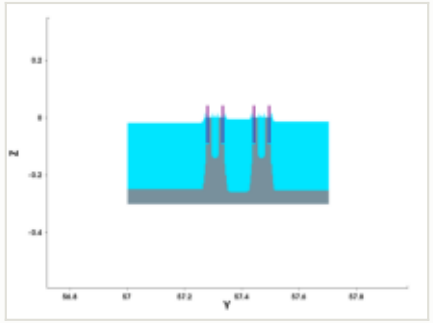
3D structure after this step

zoomed to fit

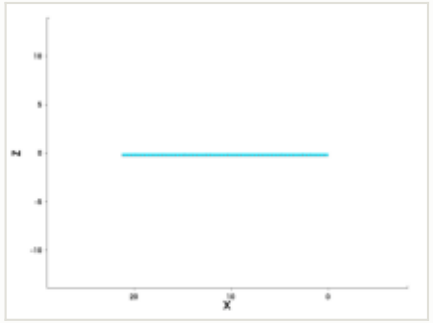




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

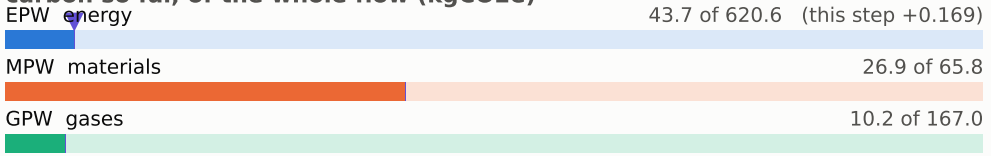
no mask for this step

this step

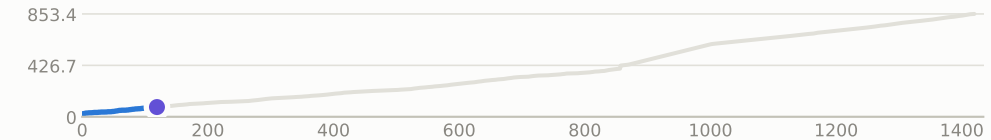
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

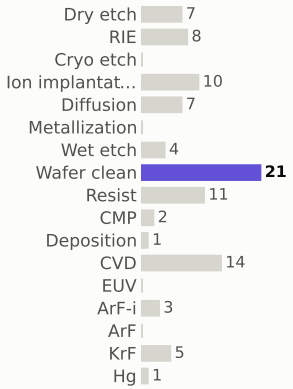
carbon so far, of the whole flow (kgCO2e)



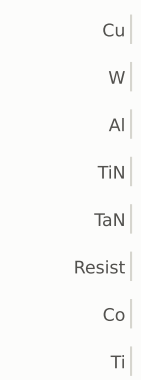
embodied carbon up to each step (kgCO2e per wafer)



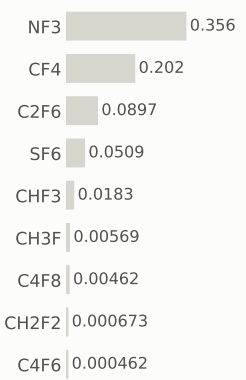
steps by process area



materials deposited (g)



gases emitted (g)



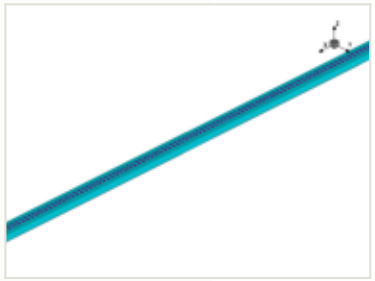
Step #1220: Resist

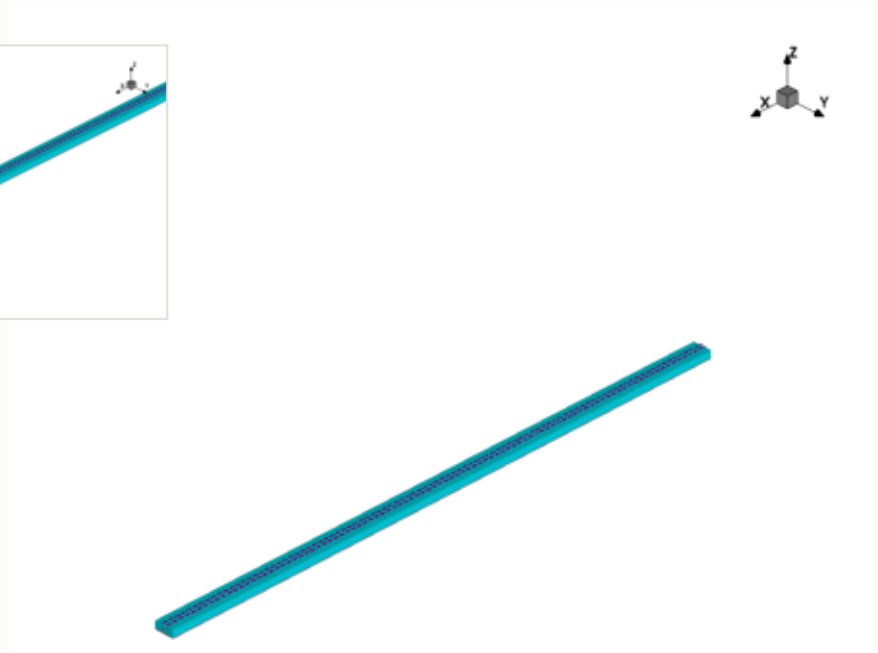
resist · Resist

0.169	81	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





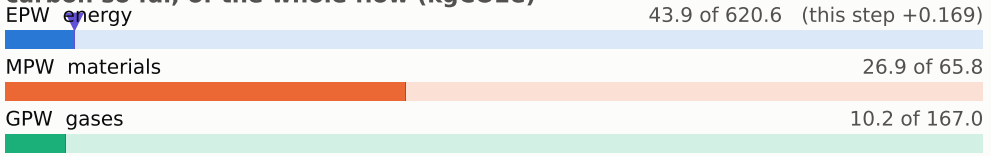
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

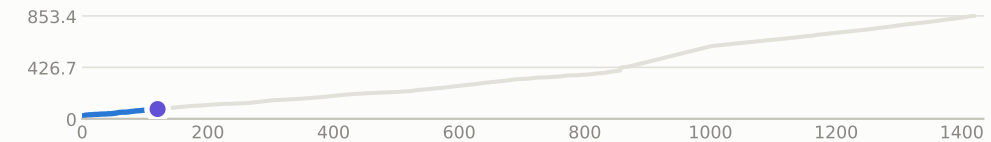
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

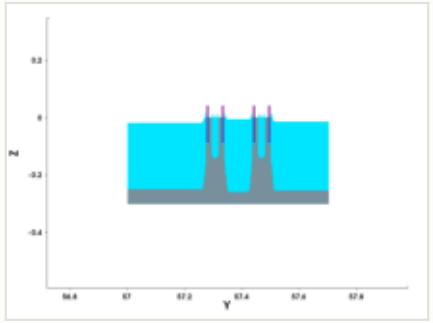
carbon so far, of the whole flow (kgCO2e)



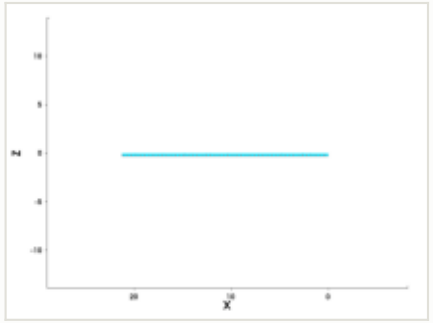
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



y cut at 57.351 µm



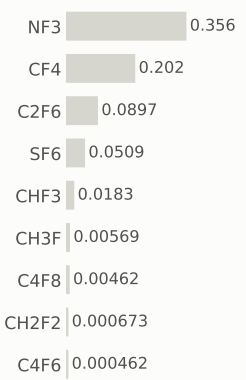
mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)



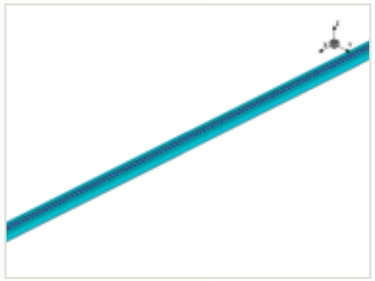
Step #1230: KrF

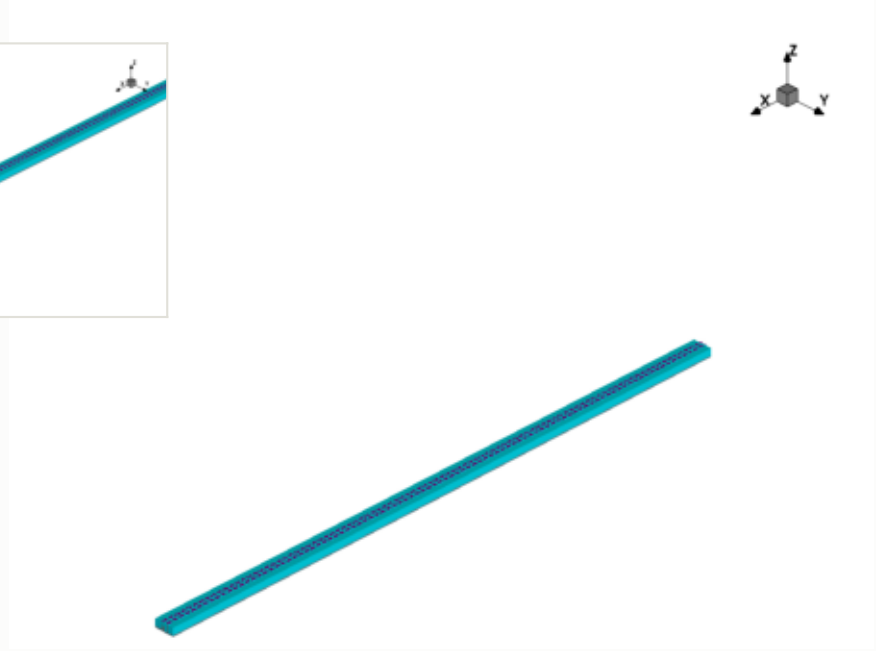
litho_patterning · KrF

0.199	81.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



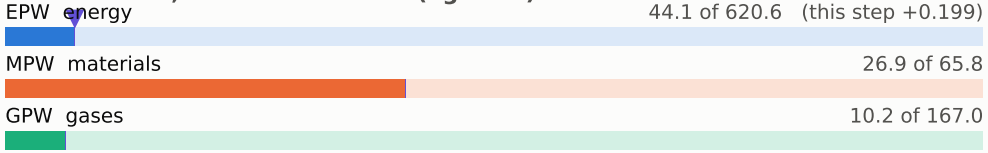


this step

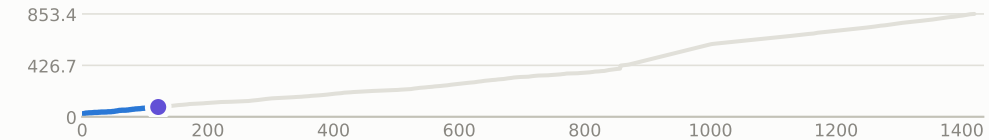
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

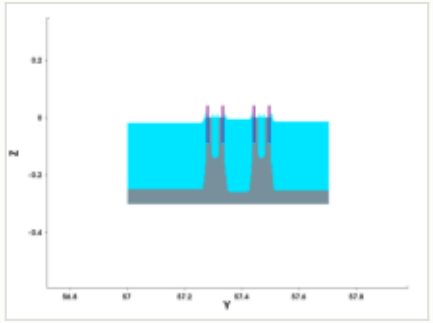
carbon so far, of the whole flow (kgCO2e)



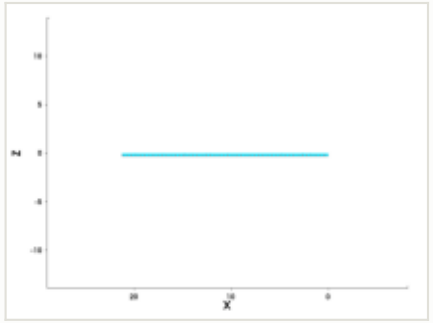
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

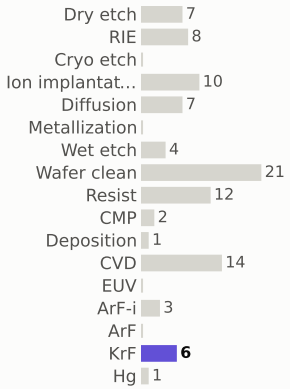


y cut at 57.351 µm

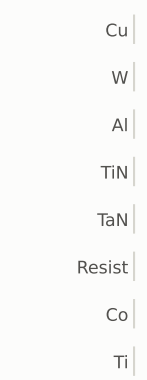


mask: layer 2100:0, positive (GDSII)

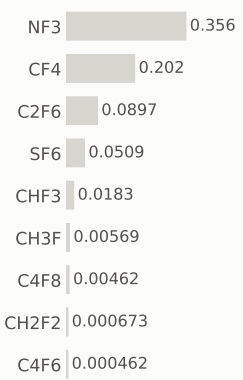
steps by process area



materials deposited (g)



gases emitted (g)



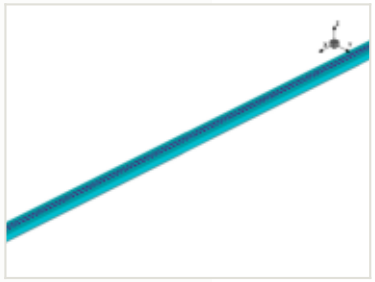
Step #1240: Ion_implantation

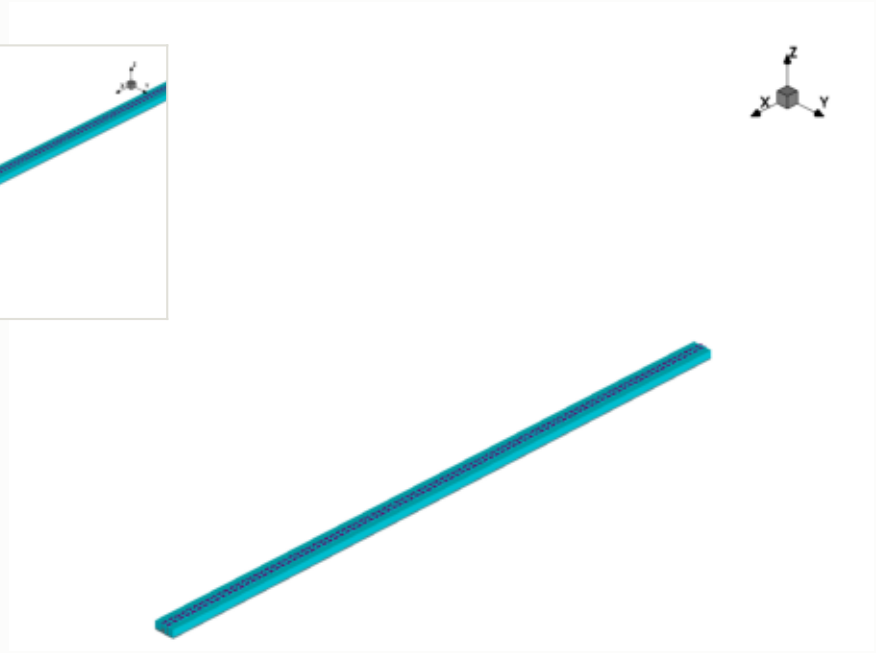
implant_ellipsoid · Ion_implantation

0.801	82	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

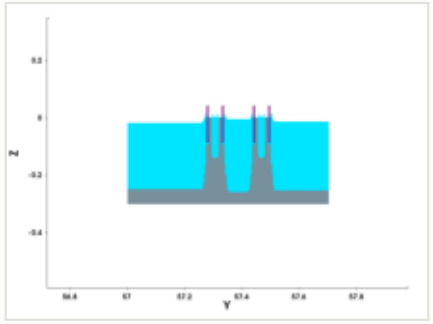
3D structure after this step

zoomed to fit

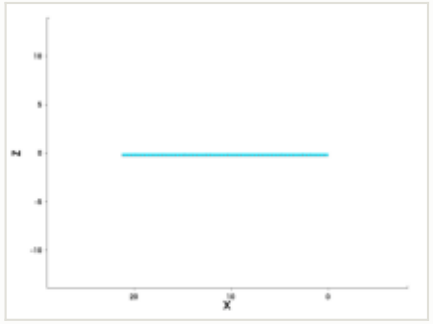




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

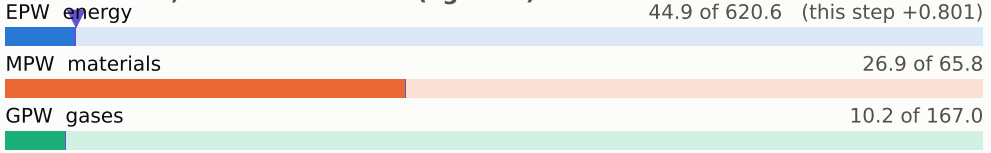
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

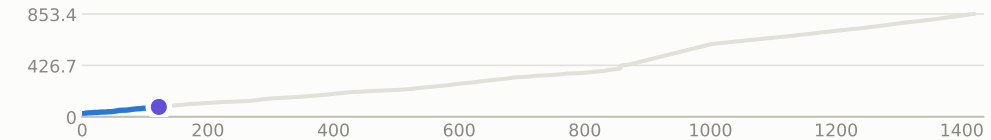
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

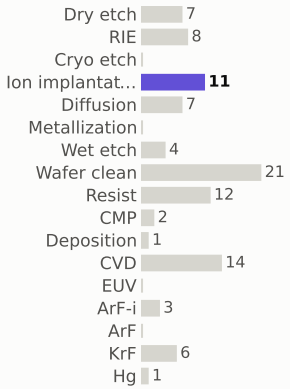
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

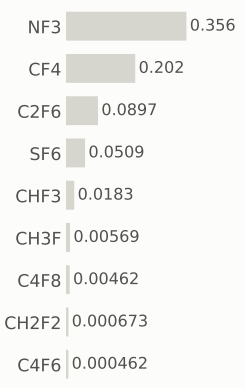


steps by process area



materials deposited (g)

gases emitted (g)



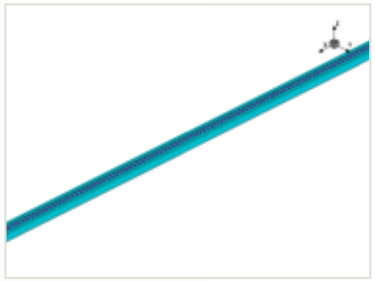
Step #1250: Dry_etch

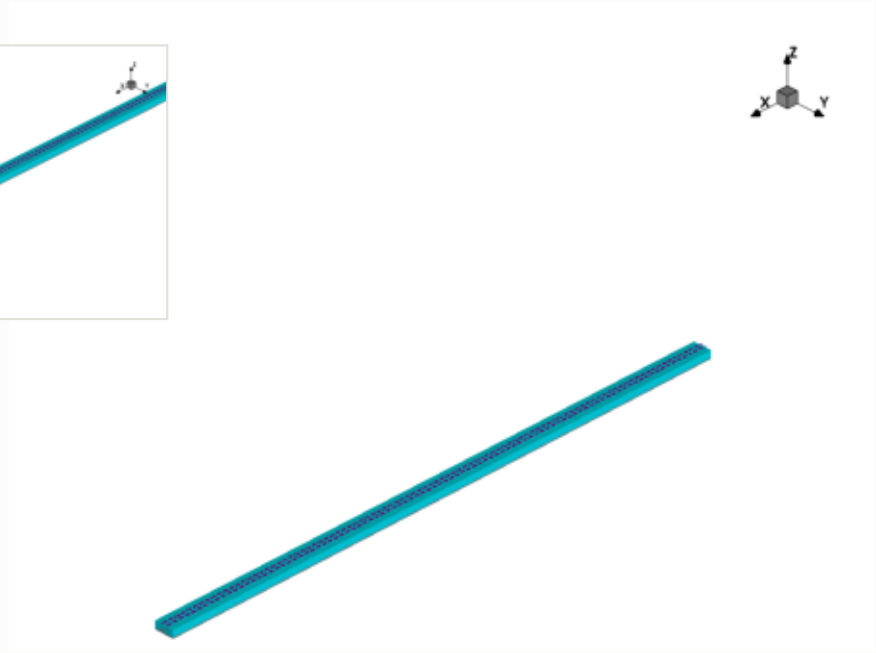
etch_strip · Dry_etch

0.801	82.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



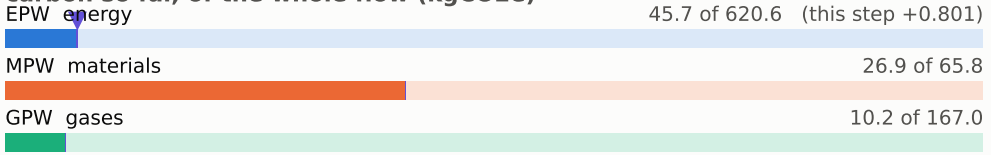


this step

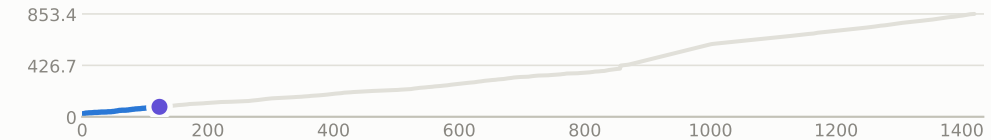
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

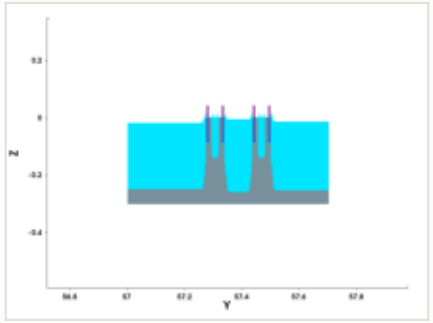
carbon so far, of the whole flow (kgCO2e)



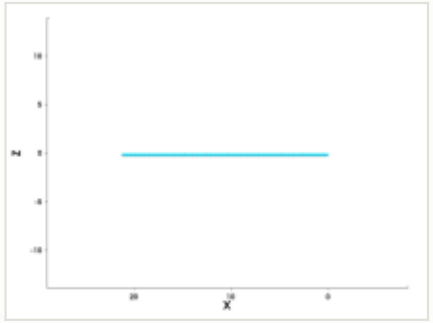
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



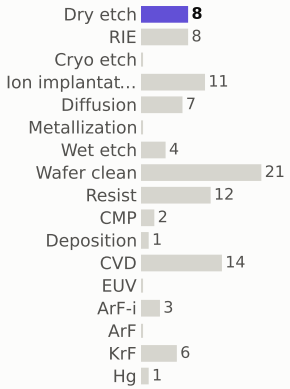
y cut at 57.351 µm



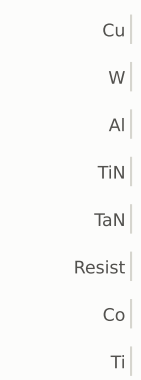
mask for this step

no mask for this step

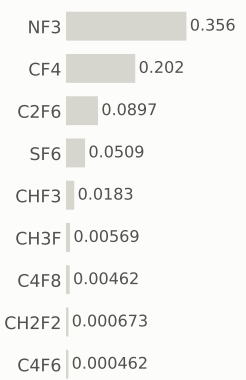
steps by process area



materials deposited (g)



gases emitted (g)



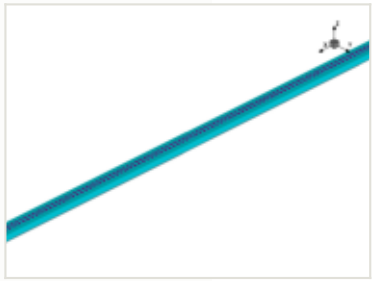
Step #1260: Wafer_clean

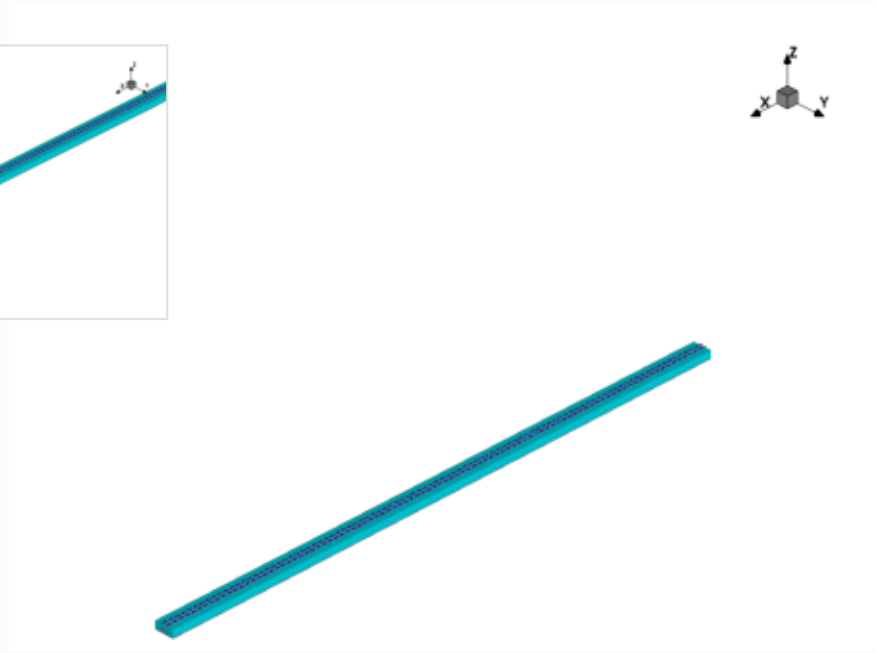
clean · Wafer_clean

0.169	82.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





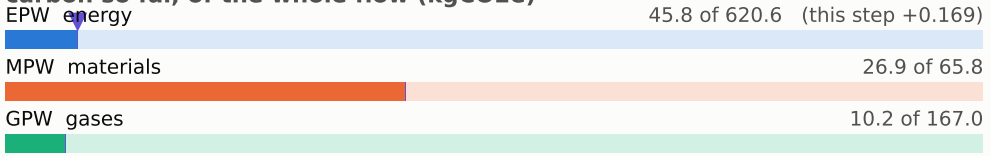
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

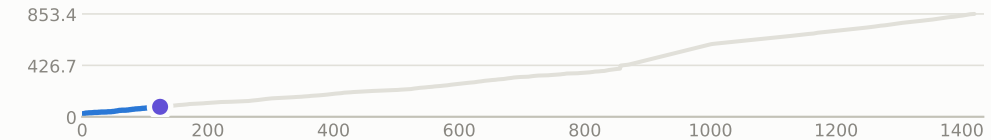
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

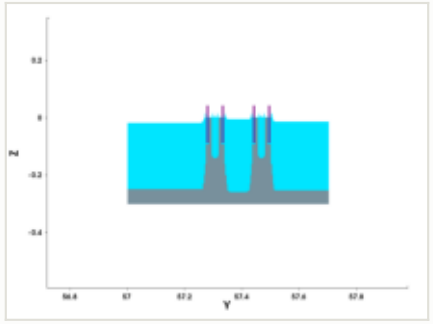
carbon so far, of the whole flow (kgCO2e)



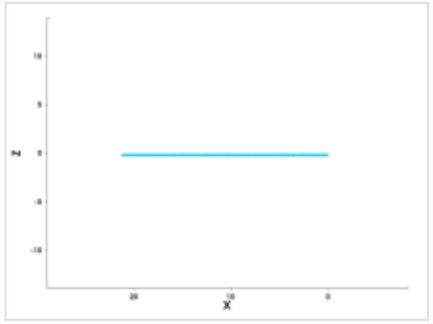
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



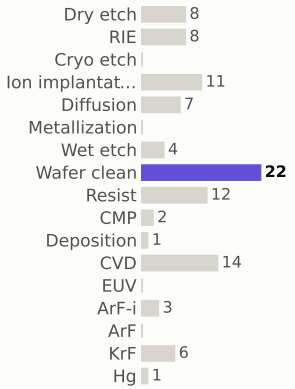
y cut at 57.351 µm



mask for this step

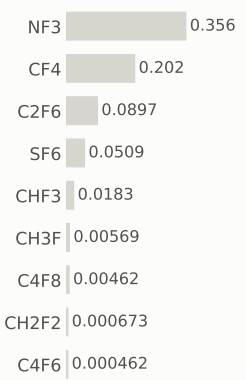
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



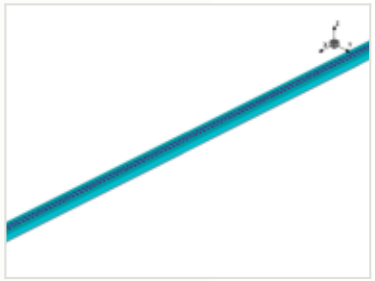
Step #1270: Resist

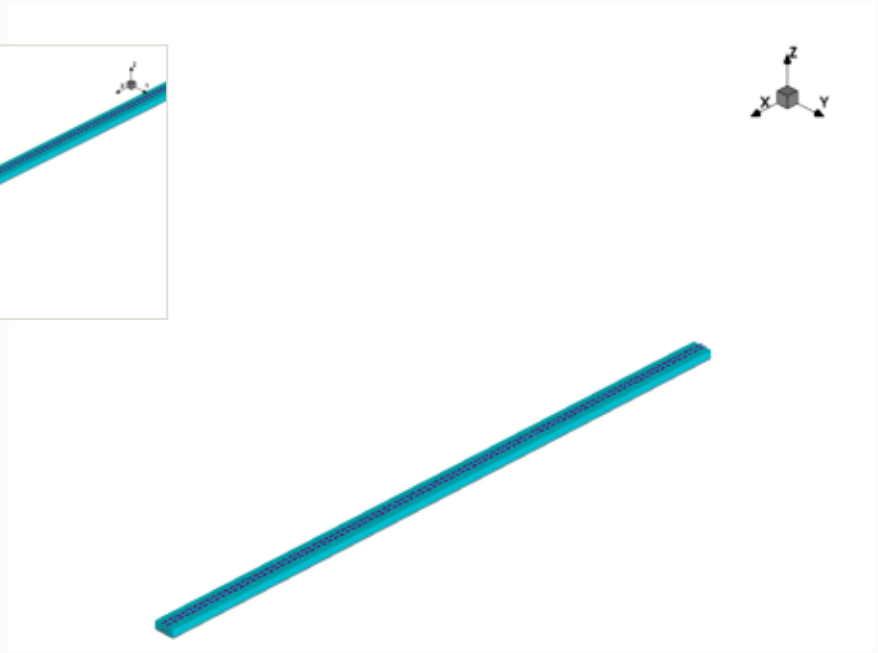
resist · Resist

0.169	83.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





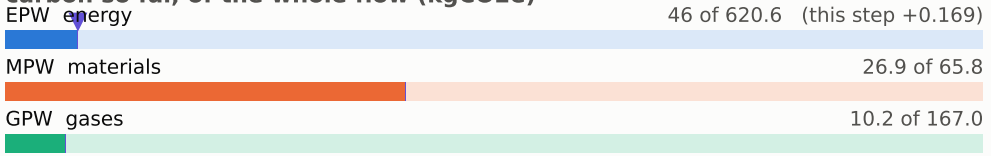
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

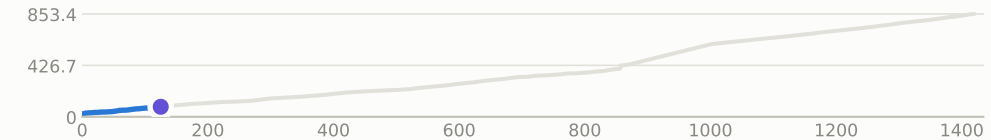
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

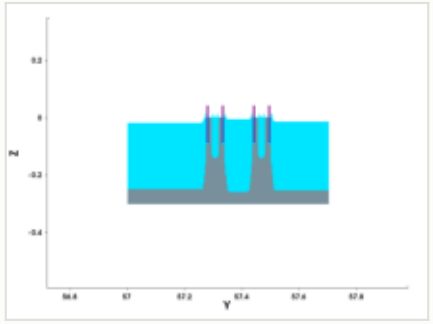
carbon so far, of the whole flow (kgCO2e)



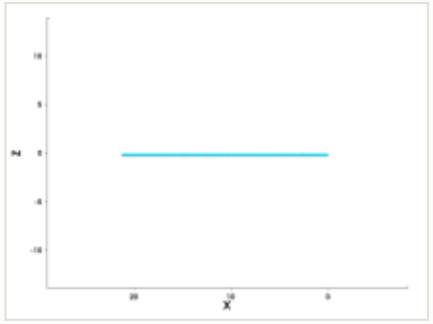
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



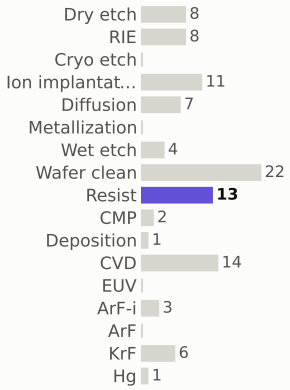
y cut at 57.351 µm



mask for this step

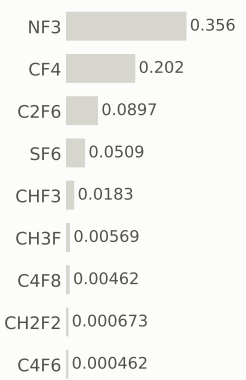
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



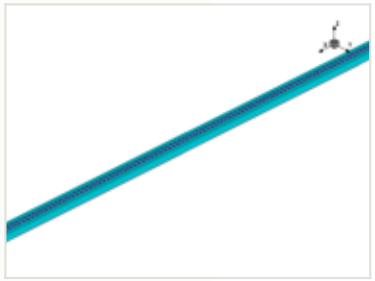
Step #1280: KrF

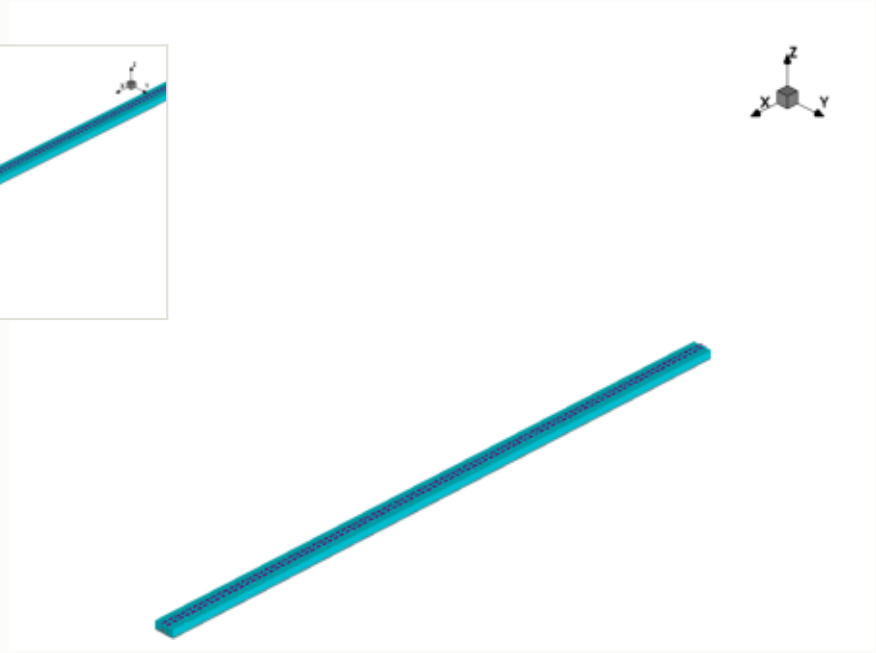
litho_patterning · KrF

0.199	83.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



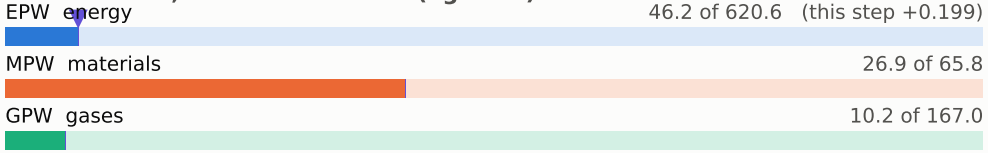


this step

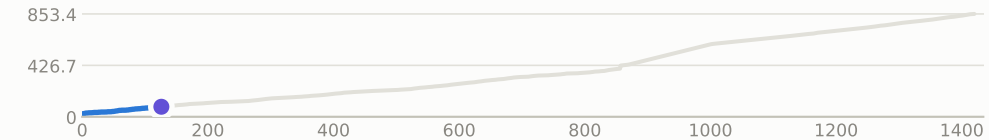
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

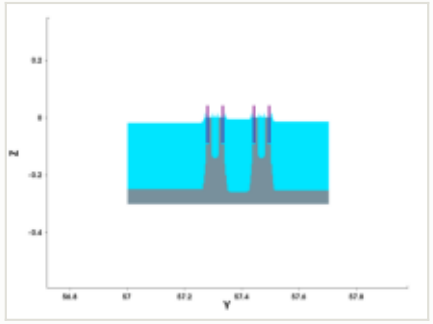
carbon so far, of the whole flow (kgCO2e)



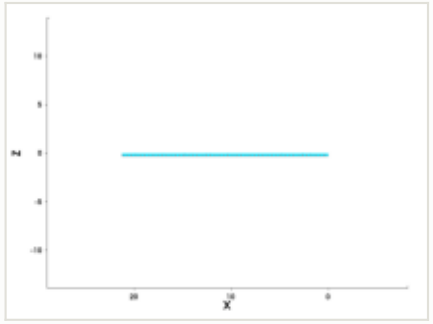
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

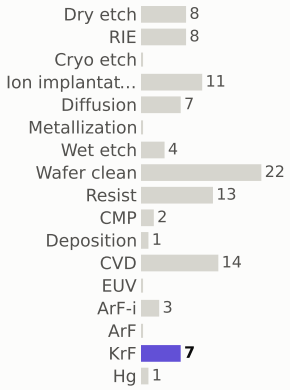


y cut at 57.351 µm



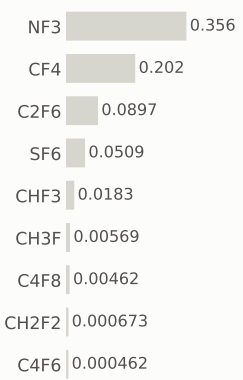
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



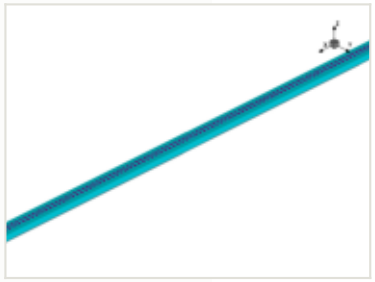
Step #1290: Ion_implantation

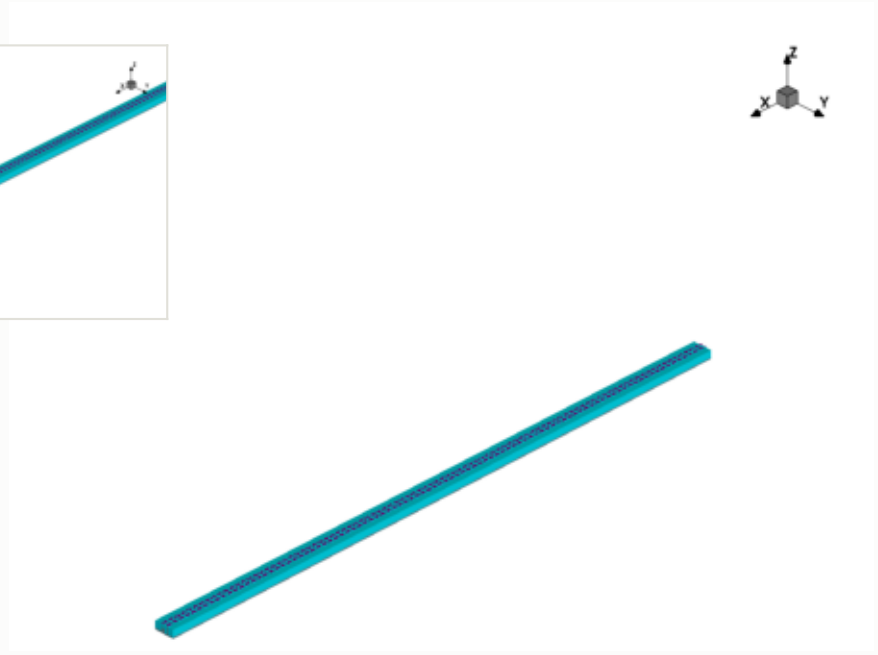
implant_ellipsoid · Ion_implantation

0.801	84.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

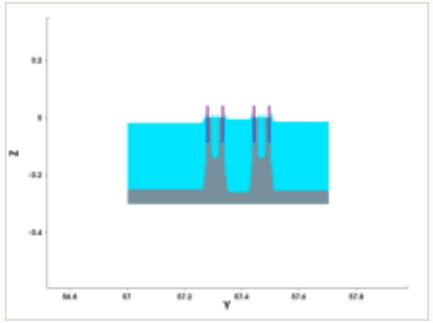
3D structure after this step

zoomed to fit

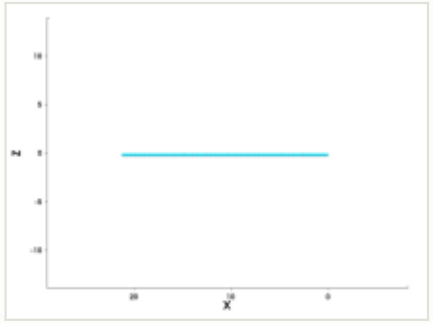




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

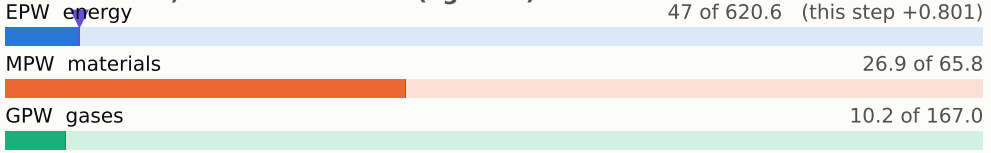
no mask for this step

this step

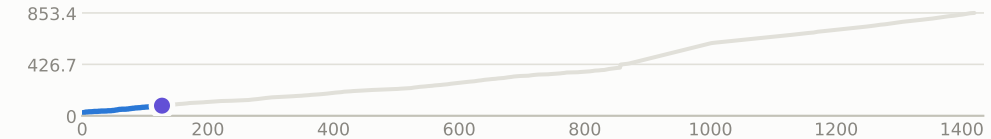
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

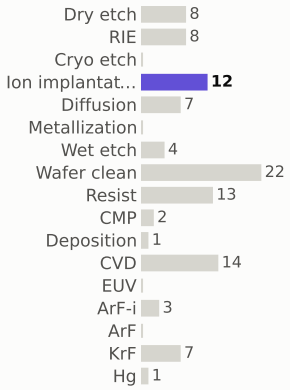
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

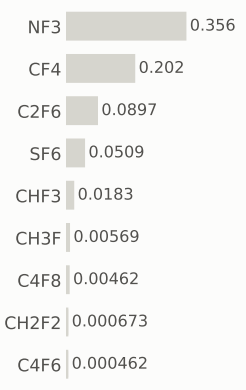


steps by process area



materials deposited (g)

gases emitted (g)



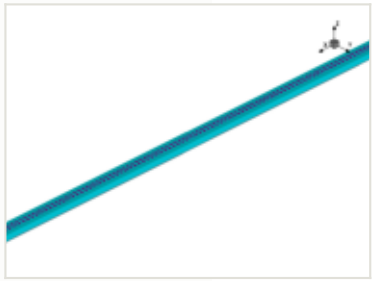
Step #1300: Dry_etch

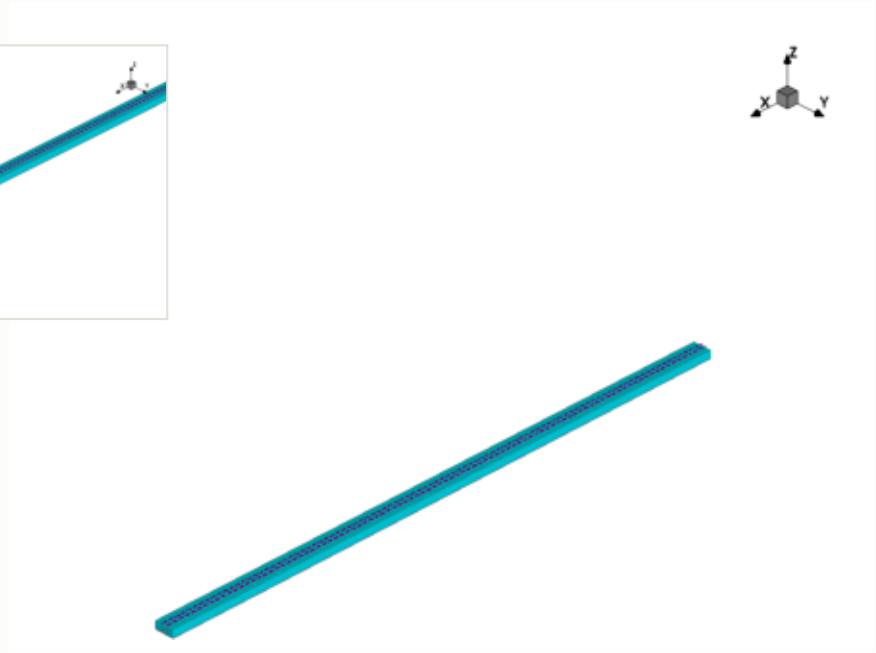
etch_strip · Dry_etch

0.801	84.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



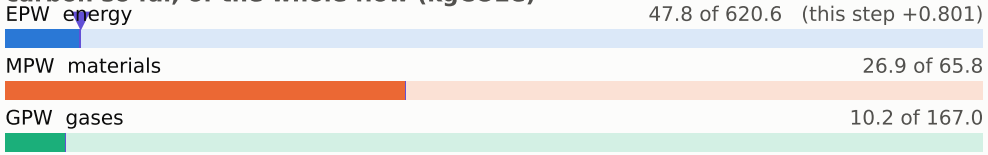


this step

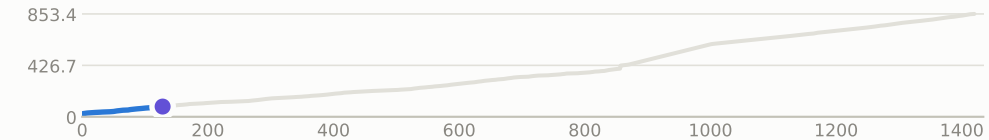
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

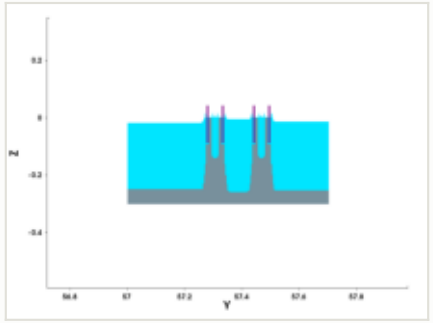
carbon so far, of the whole flow (kgCO2e)



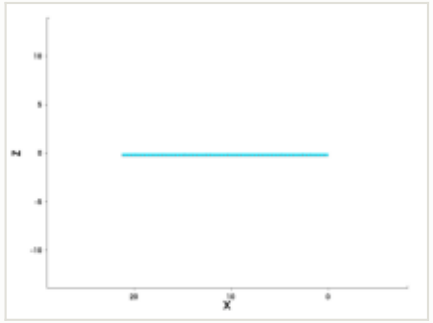
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



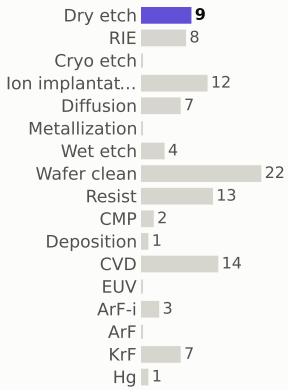
y cut at 57.351 µm



mask for this step

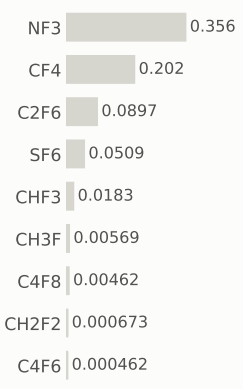
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



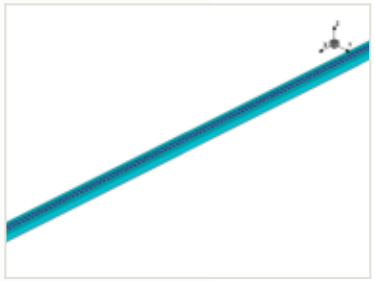
Step #1310: Wafer_clean

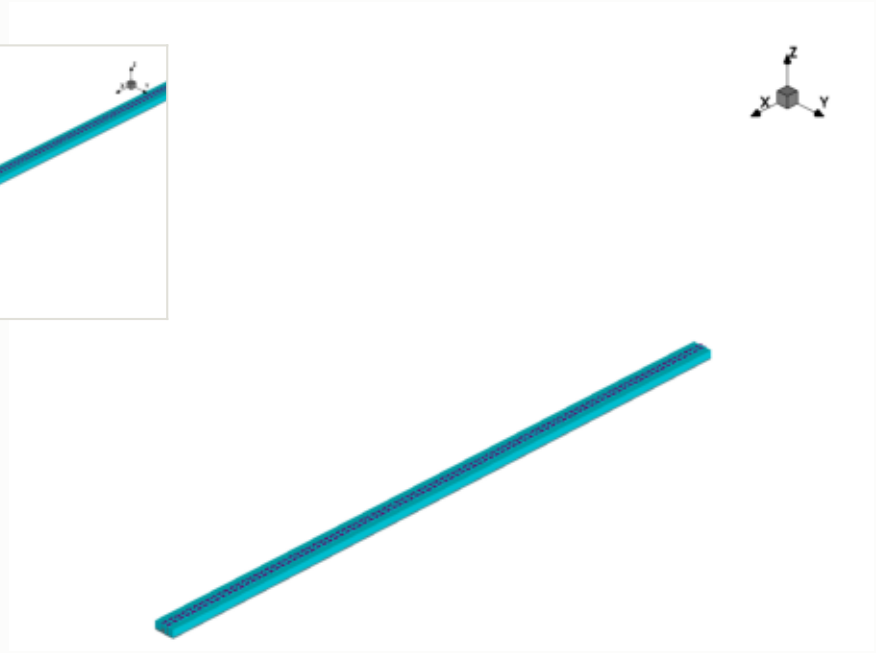
clean · Wafer_clean

0.169	85.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

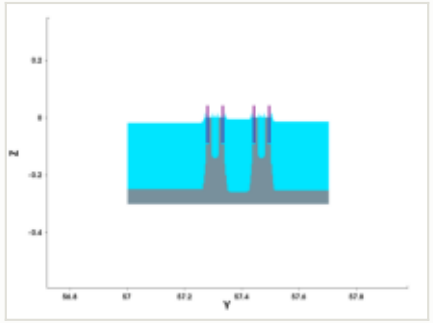
3D structure after this step

zoomed to fit

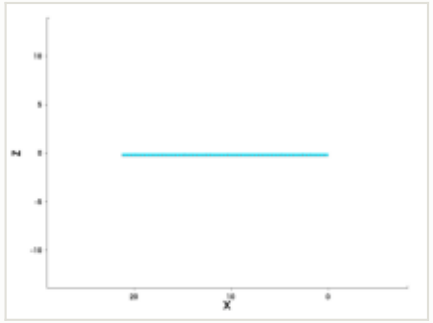




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

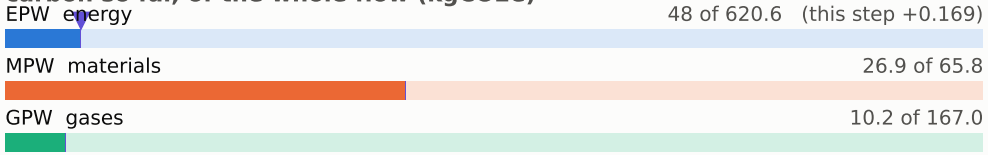
no mask for this step

this step

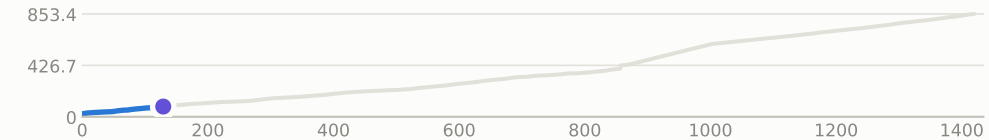
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

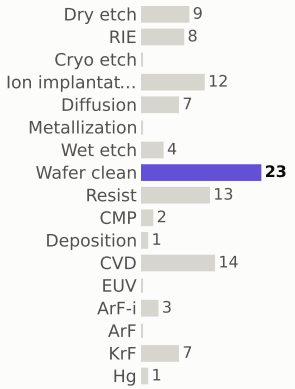
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

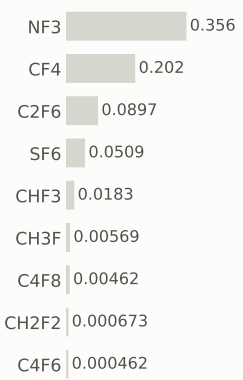


steps by process area



materials deposited (g)

gases emitted (g)



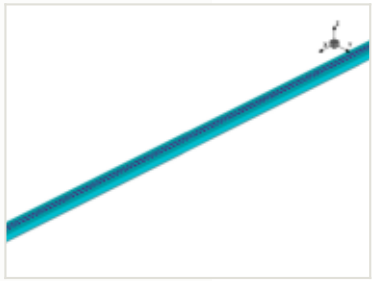
Step #1320: Resist

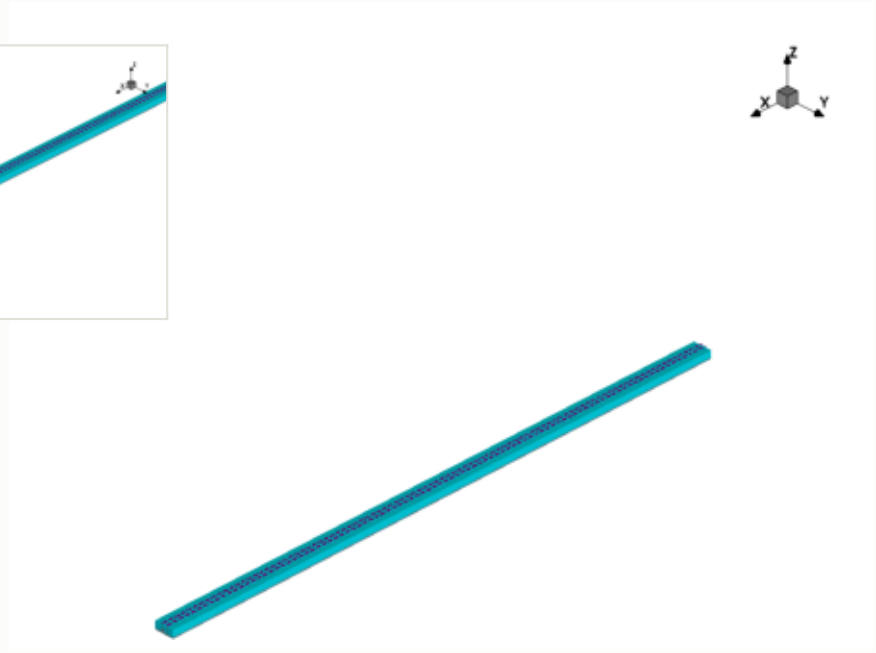
resist · Resist

0.169	85.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





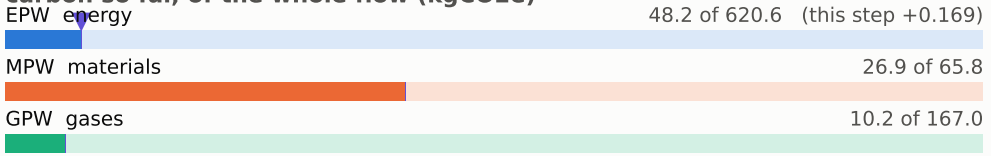
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

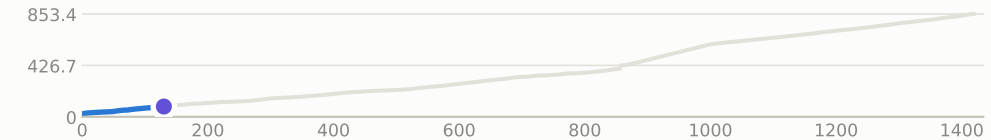
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

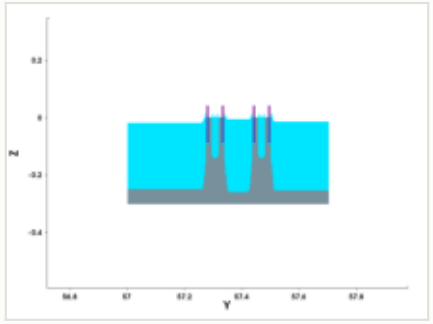
carbon so far, of the whole flow (kgCO2e)



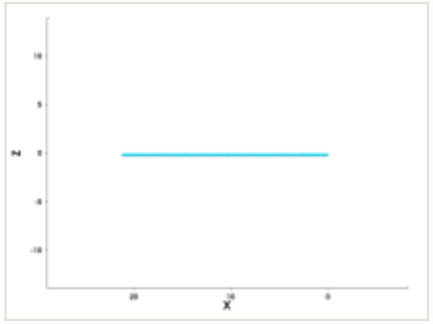
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



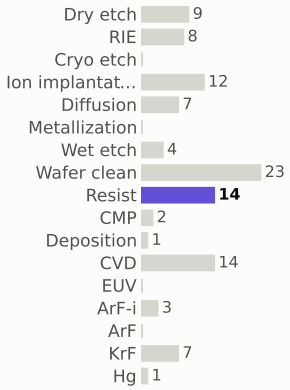
y cut at 57.351 µm



mask for this step

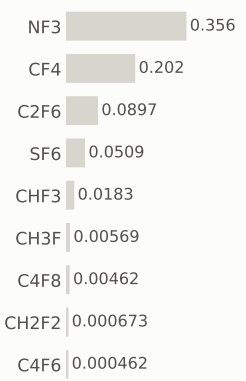
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



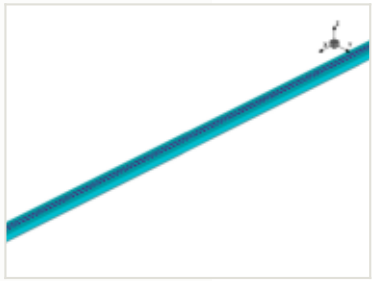
Step #1330: KrF

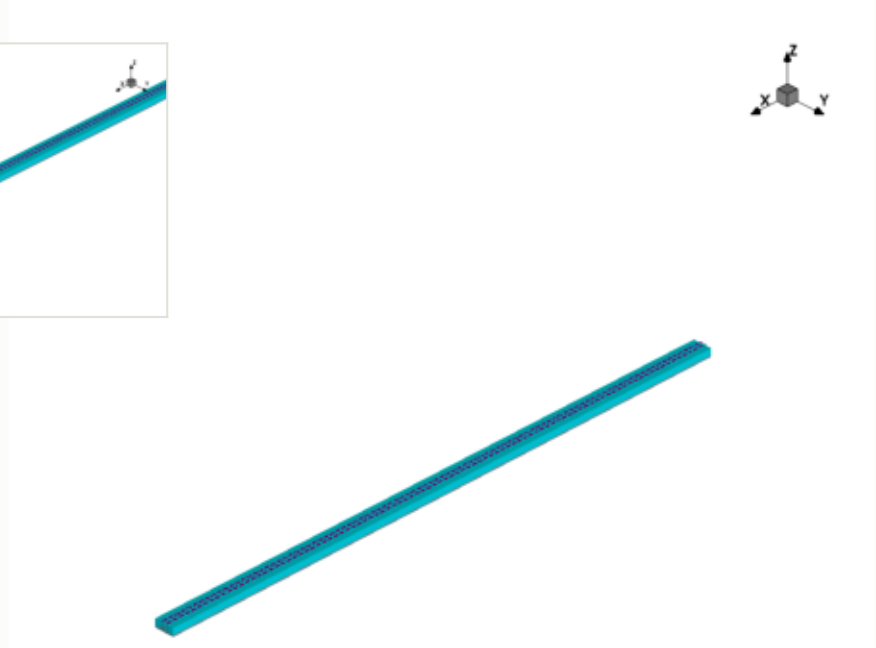
litho_patterning · KrF

0.199	85.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

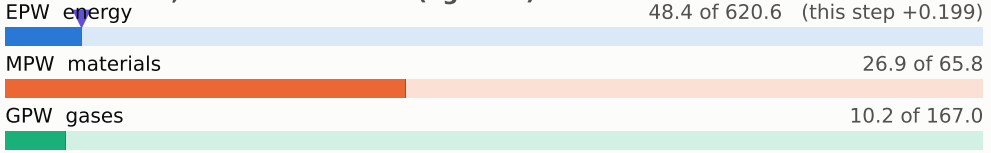




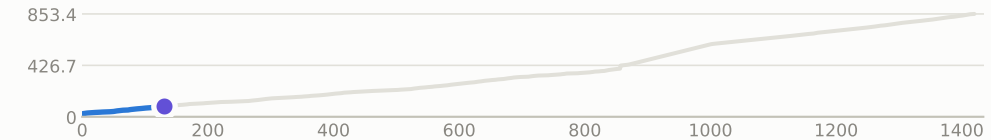
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

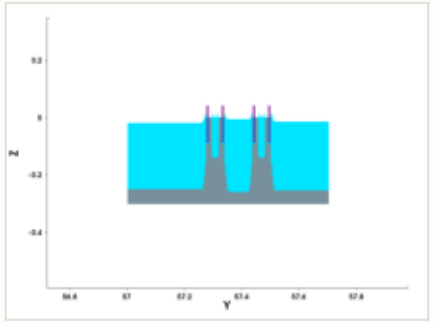
carbon so far, of the whole flow (kgCO2e)



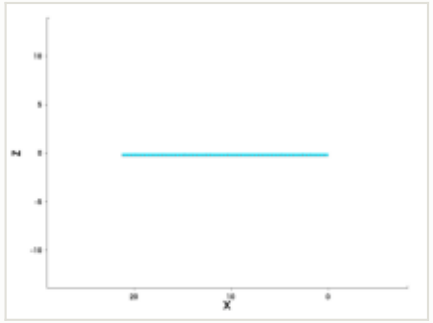
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

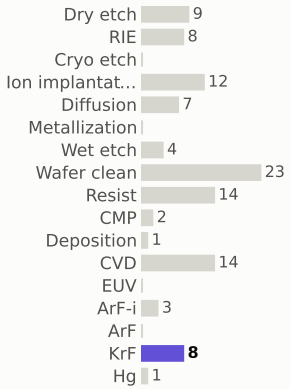


y cut at 57.351 µm



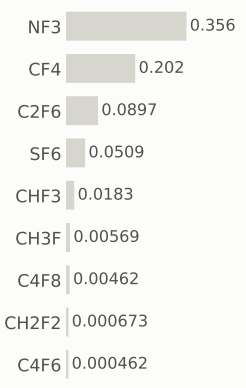
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



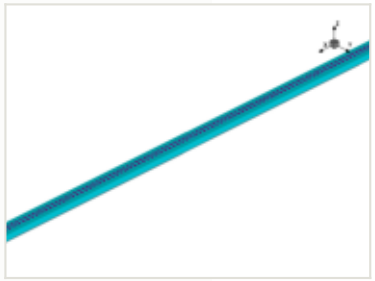
Step #1340: Ion_implantation

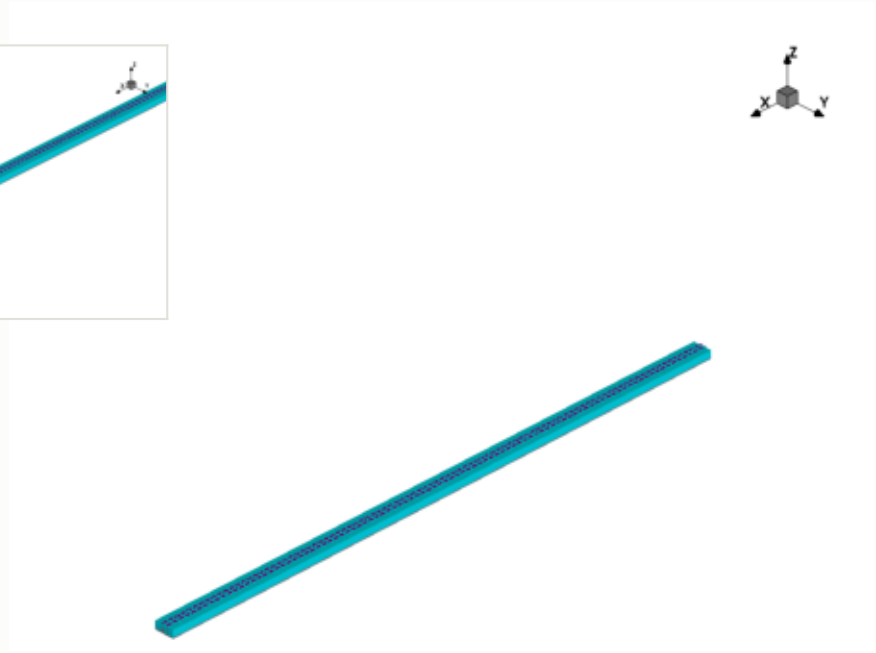
implant_ellipsoid · Ion_implantation

0.801	86.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

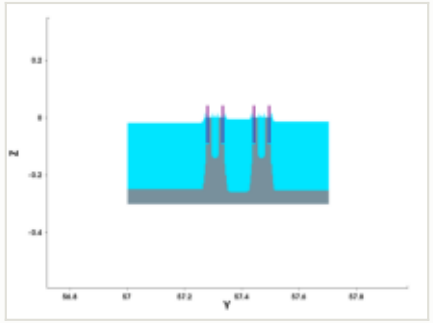
3D structure after this step

zoomed to fit

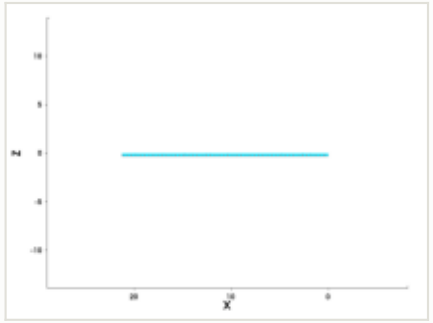




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

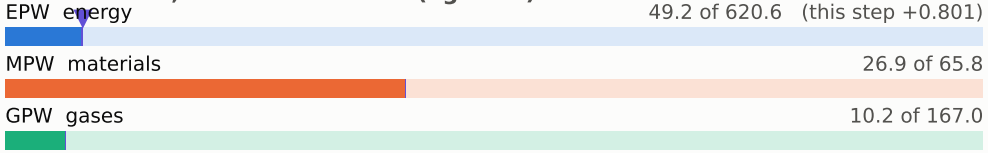
no mask for this step

this step

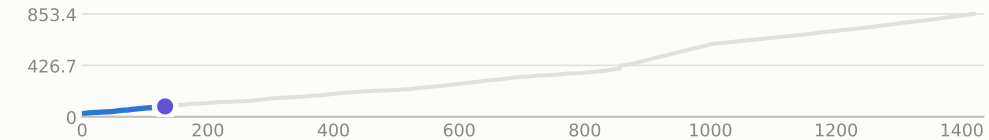
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

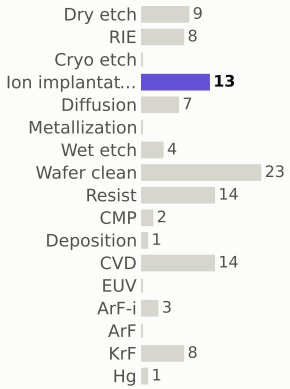
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

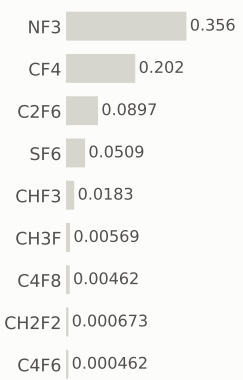


steps by process area



materials deposited (g)

gases emitted (g)



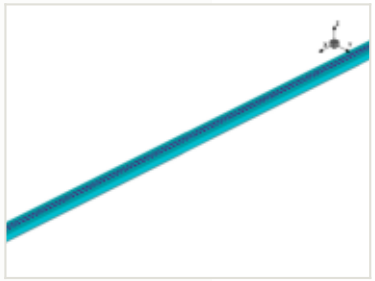
Step #1350: Dry_etch

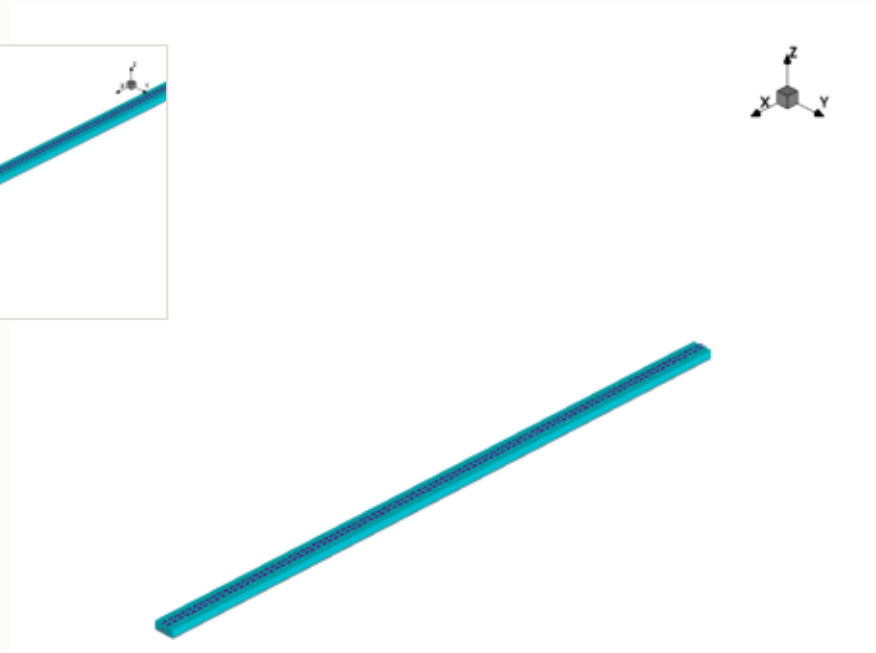
etch_strip · Dry_etch

0.801	87	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



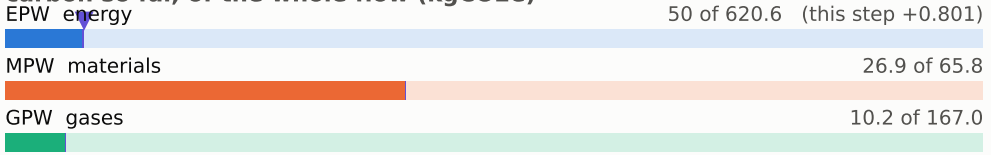


this step

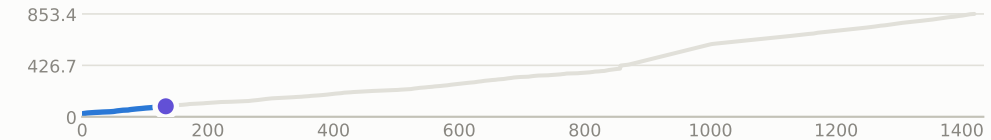
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

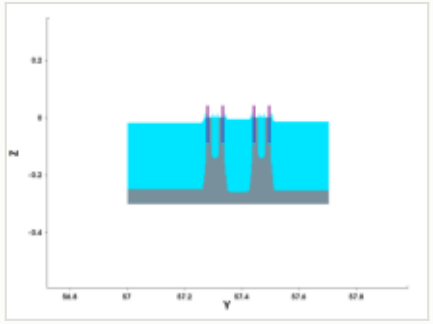
carbon so far, of the whole flow (kgCO2e)



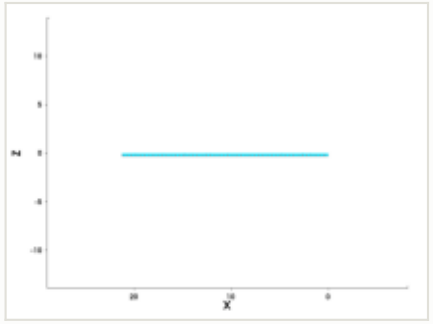
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



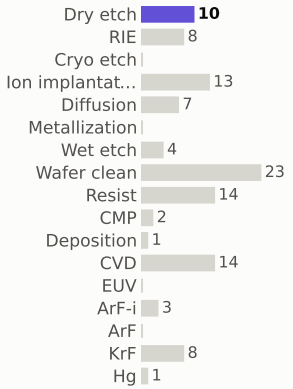
y cut at 57.351 µm



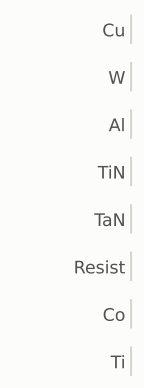
mask for this step

no mask for this step

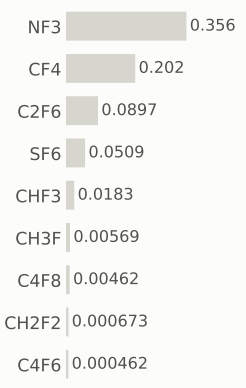
steps by process area



materials deposited (g)



gases emitted (g)



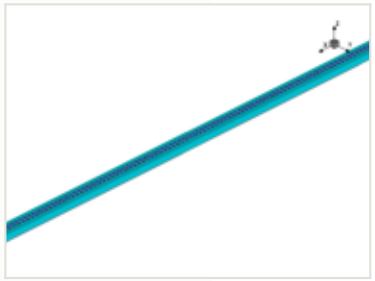
Step #1360: Wafer_clean

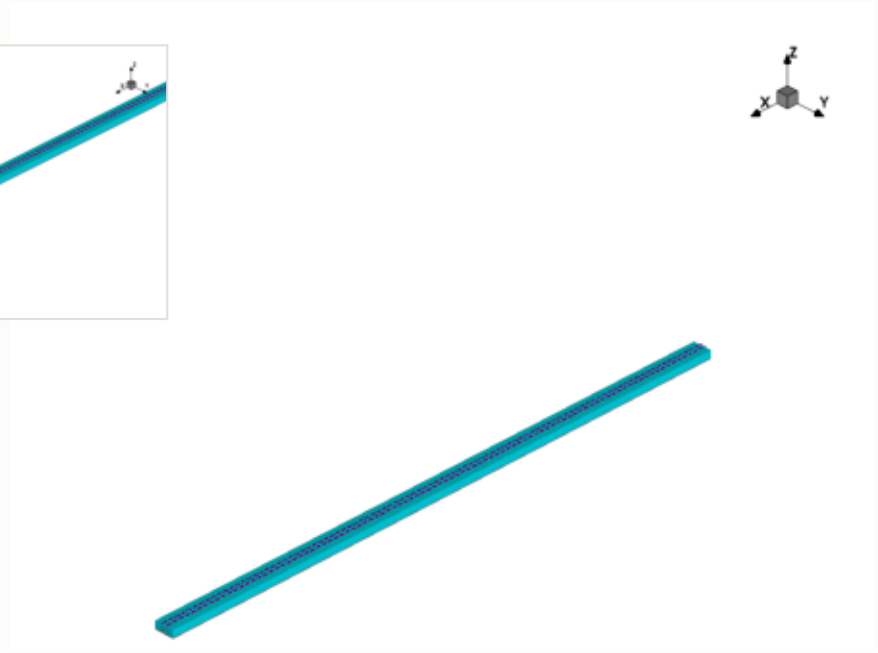
clean · Wafer_clean

0.169	87.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

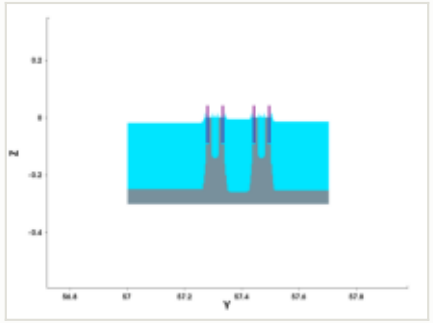
3D structure after this step

zoomed to fit

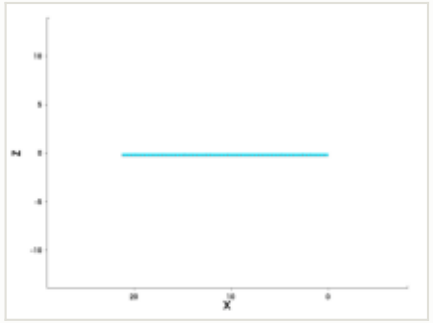




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

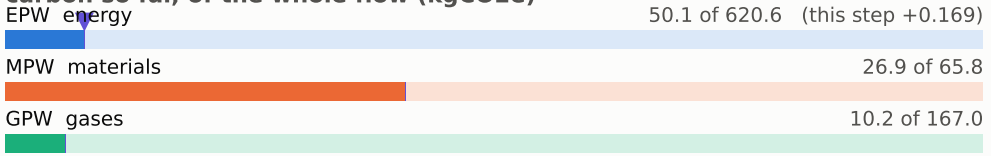
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

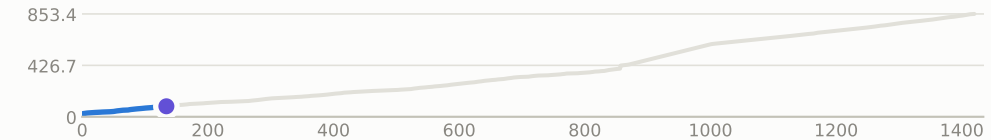
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

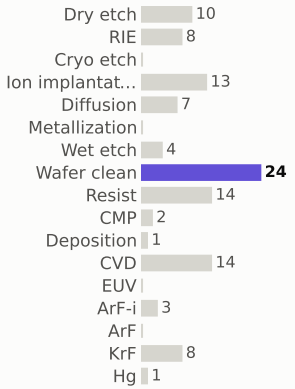
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

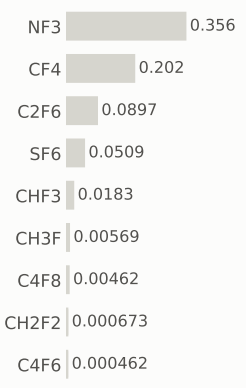


steps by process area



materials deposited (g)

gases emitted (g)



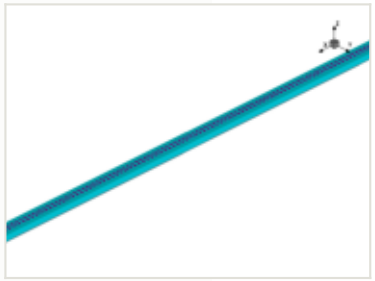
Step #1370: Resist

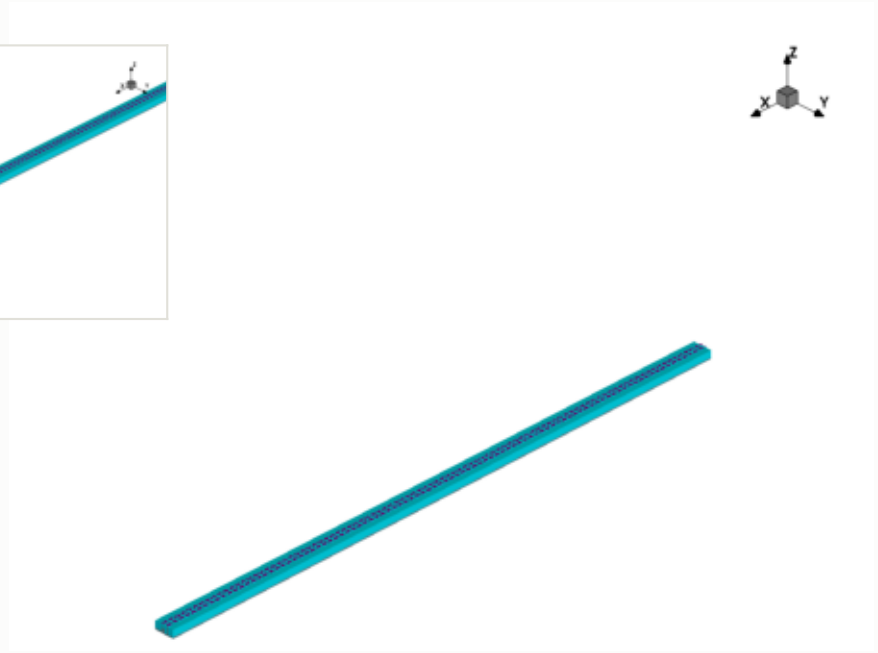
resist · Resist

0.169	87.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





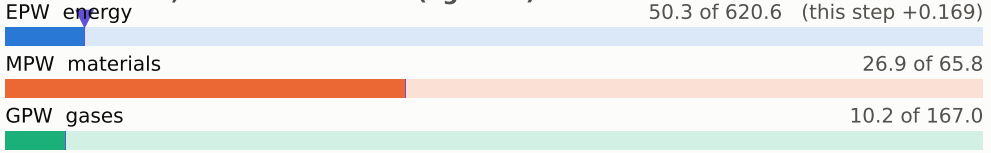
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

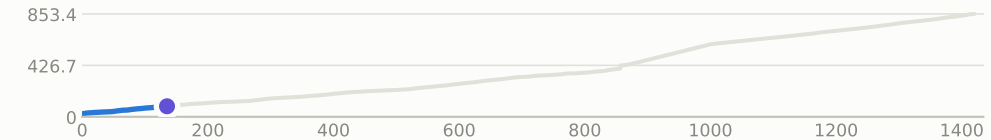
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

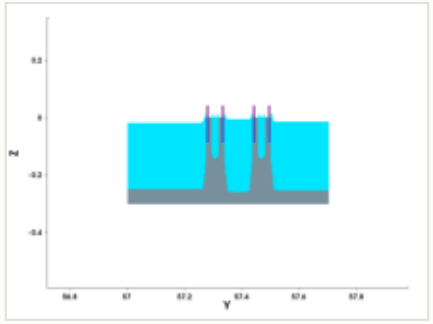
carbon so far, of the whole flow (kgCO2e)



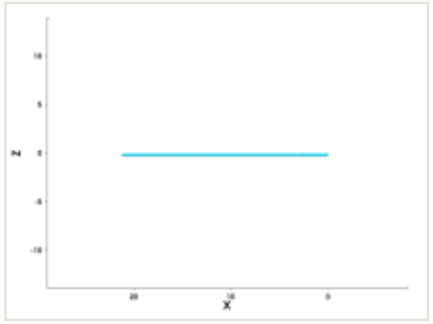
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



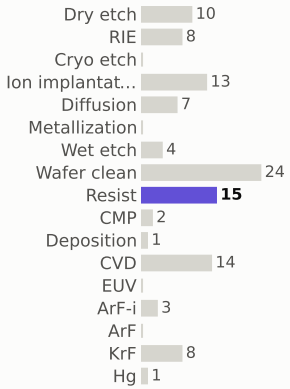
y cut at 57.351 µm



mask for this step

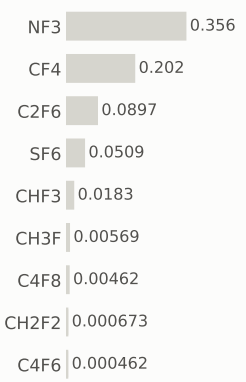
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



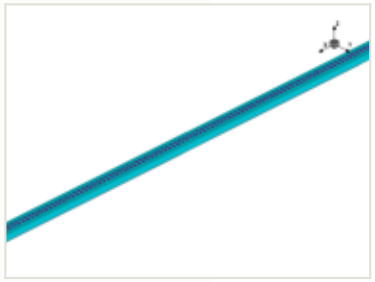
Step #1380: KrF

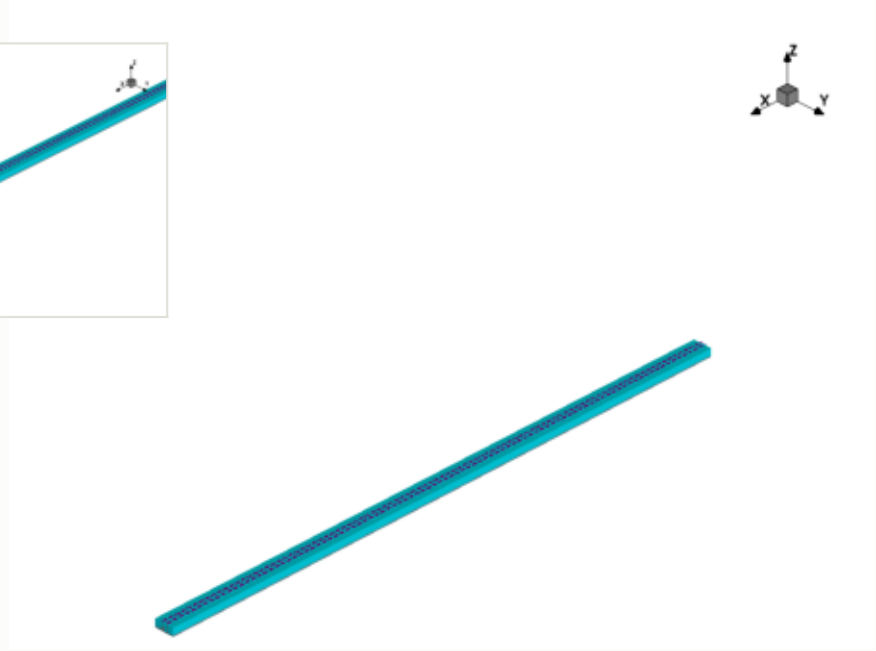
litho_patterning · KrF

0.199	87.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



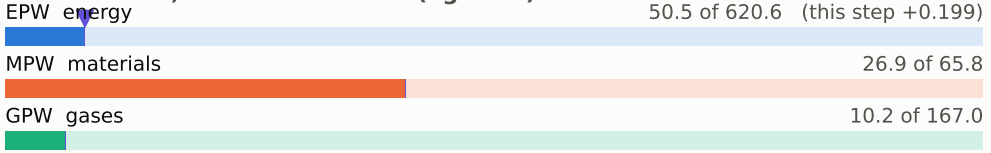


this step

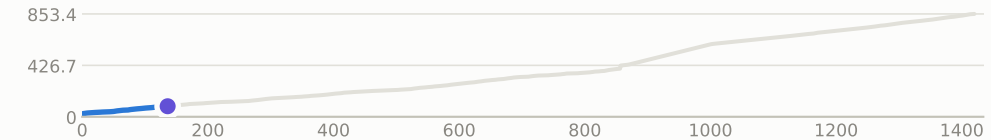
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

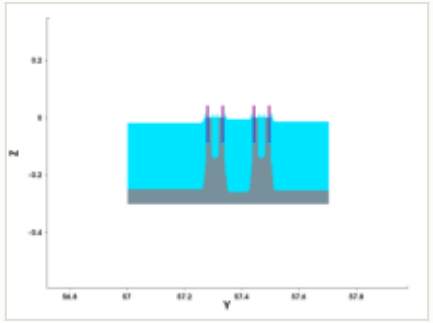
carbon so far, of the whole flow (kgCO2e)



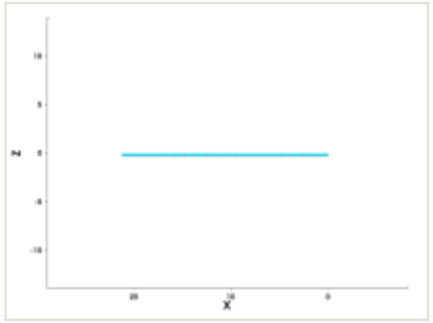
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

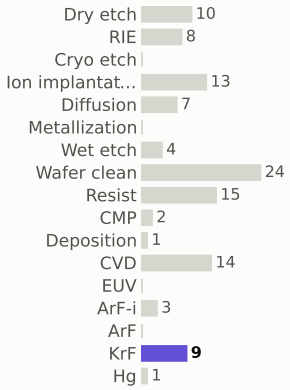


y cut at 57.351 µm



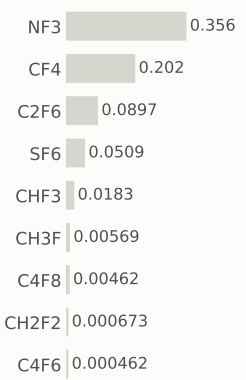
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



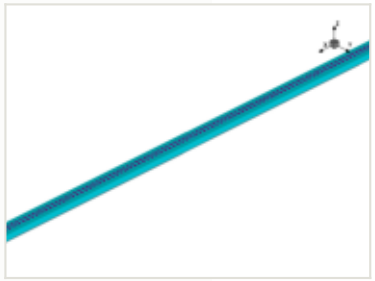
Step #1390: Ion_implantation

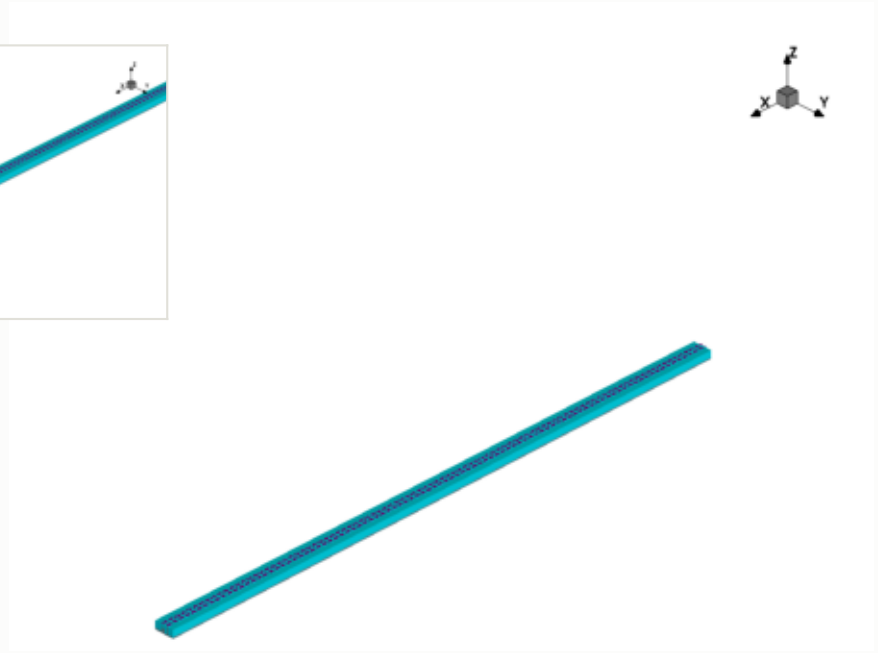
implant_ellipsoid · Ion_implantation

0.801	88.4	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

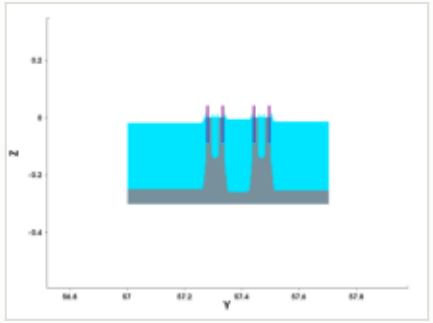
3D structure after this step

zoomed to fit

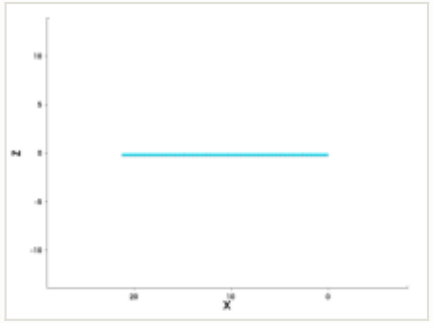




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

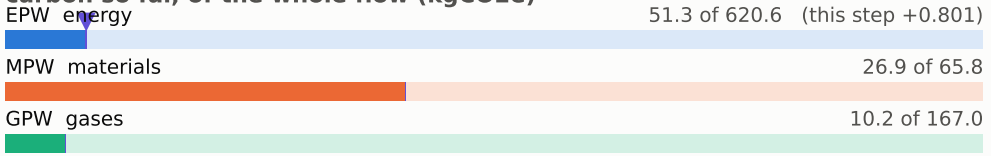
no mask for this step

this step

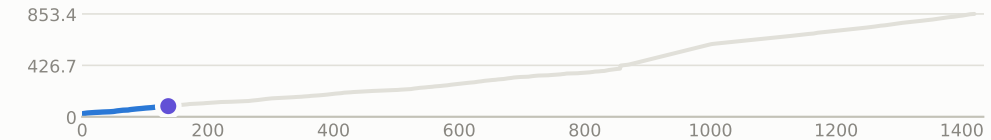
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

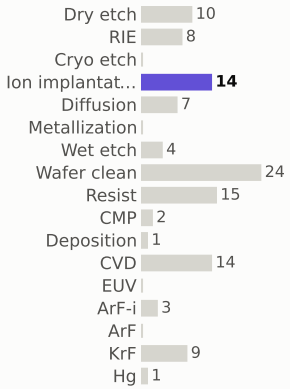
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

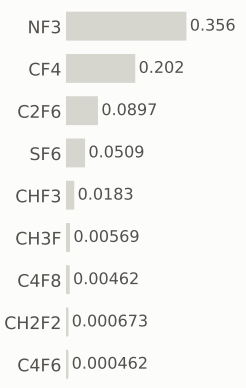


steps by process area



materials deposited (g)

gases emitted (g)



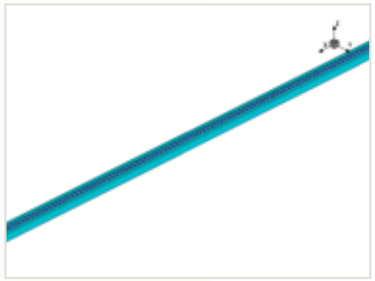
Step #1400: Ion_implantation

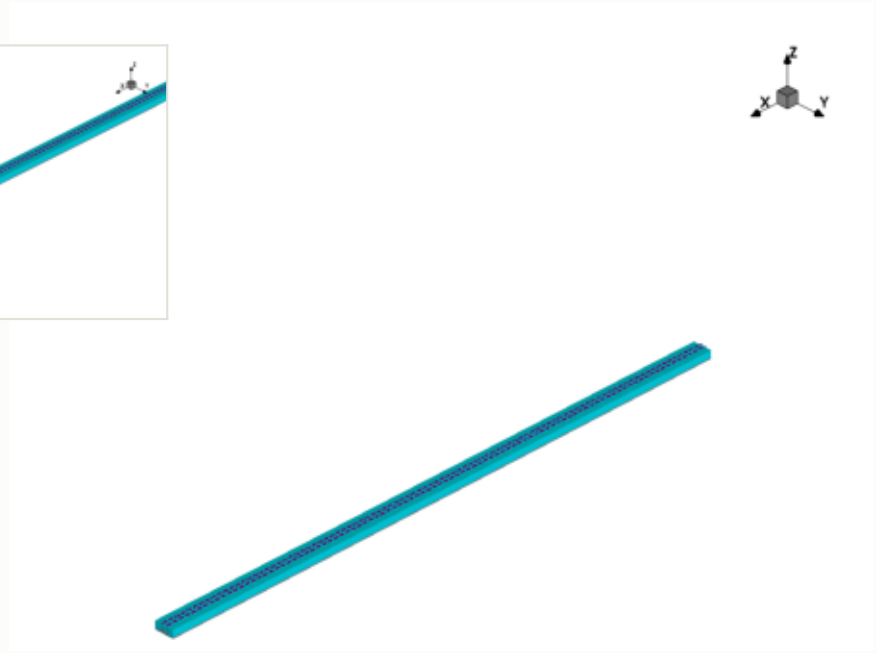
implant_ellipsoid · Ion_implantation

0.801	89.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

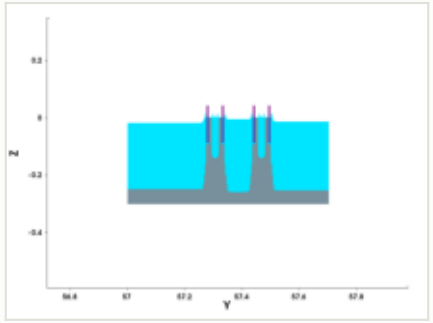
3D structure after this step

zoomed to fit

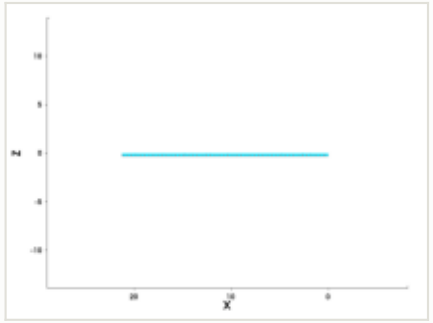




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

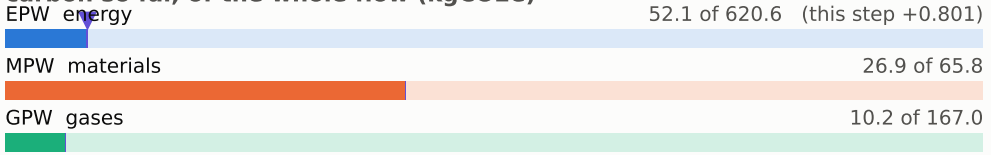
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

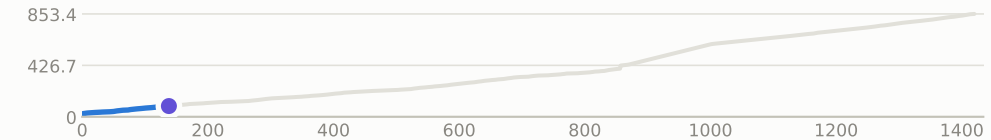
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

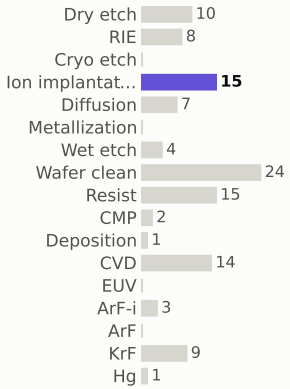
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

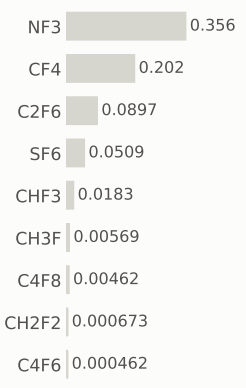


steps by process area



materials deposited (g)

gases emitted (g)



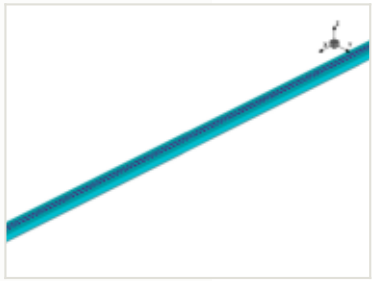
Step #1410: Dry_etch

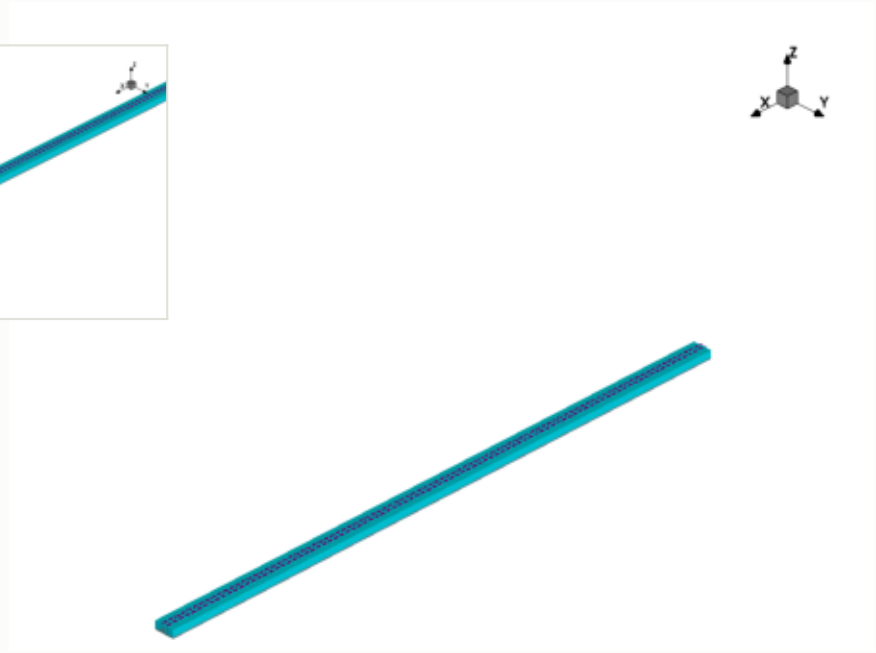
etch_strip · Dry_etch

0.801	90	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



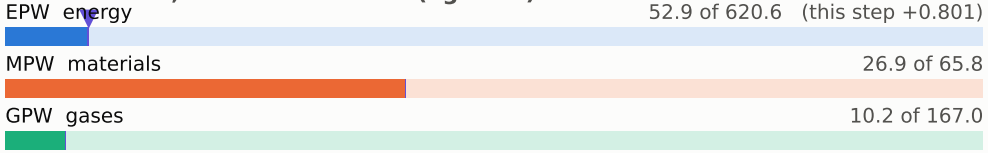


this step

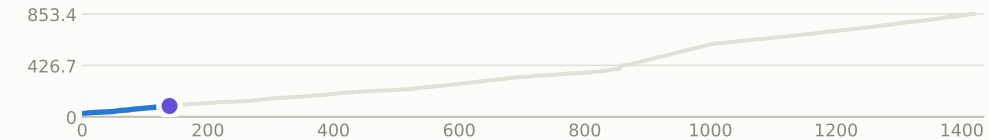
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

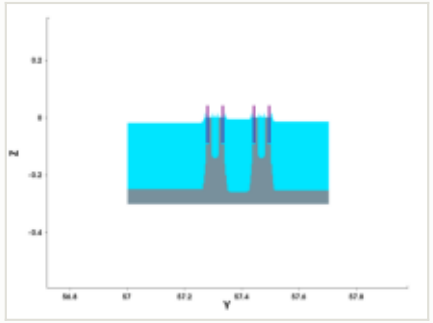
carbon so far, of the whole flow (kgCO2e)



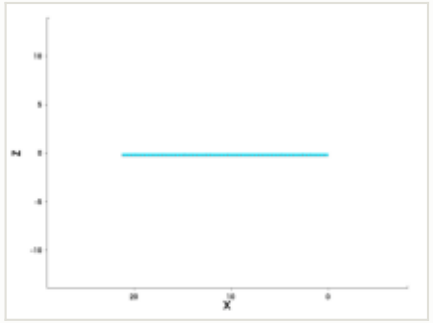
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



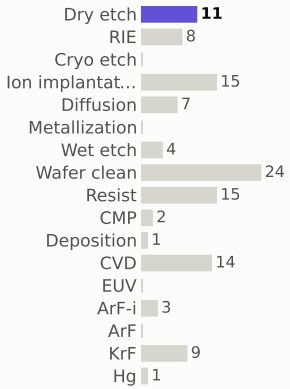
y cut at 57.351 µm



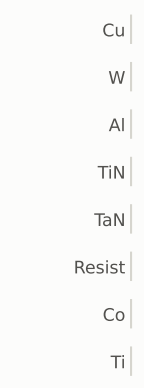
mask for this step

no mask for this step

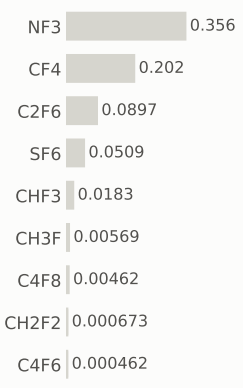
steps by process area



materials deposited (g)



gases emitted (g)



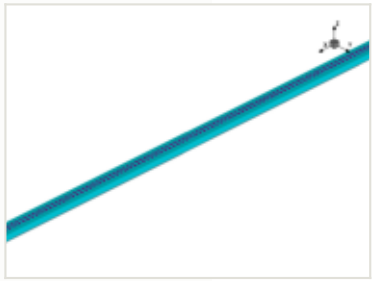
Step #1420: Wafer_clean

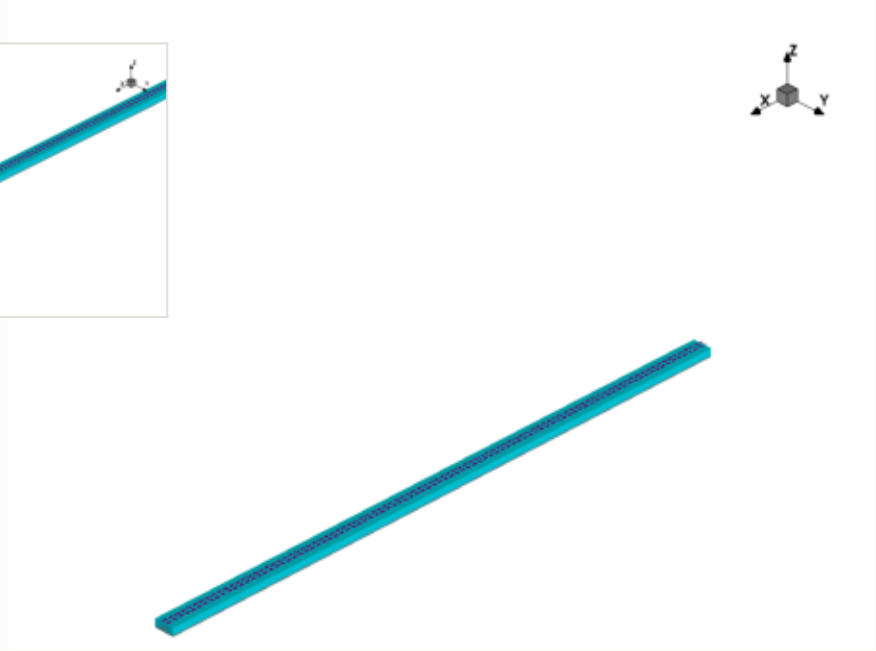
clean · Wafer_clean

0.169	90.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



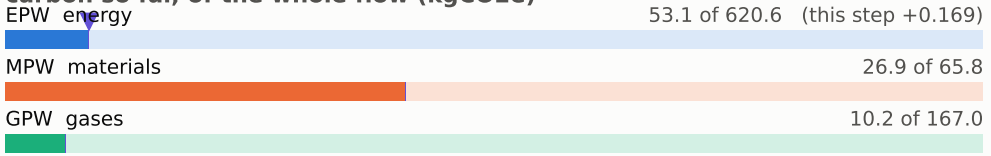


this step

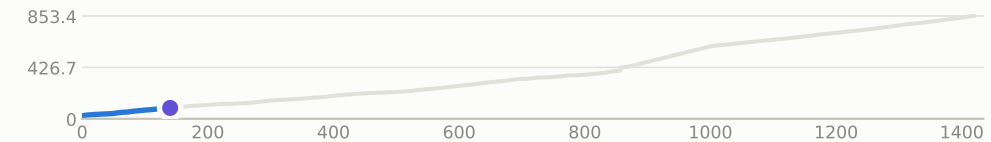
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

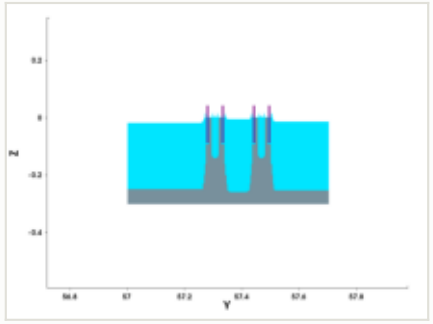
carbon so far, of the whole flow (kgCO2e)



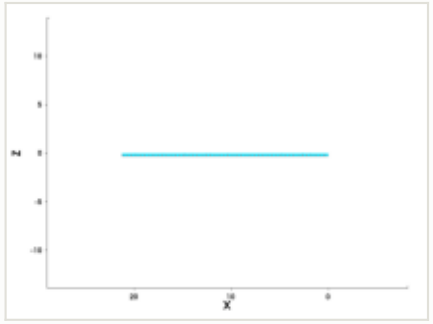
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



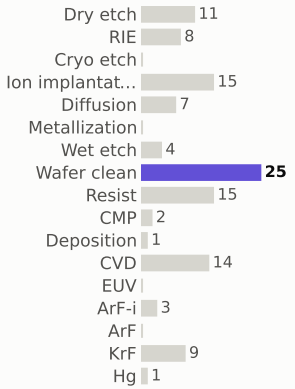
y cut at 57.351 µm



mask for this step

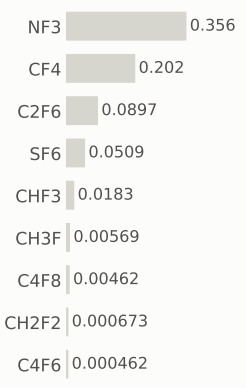
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



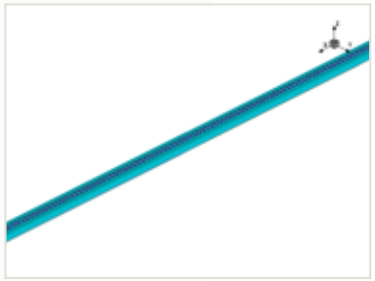
Step #1430: Resist

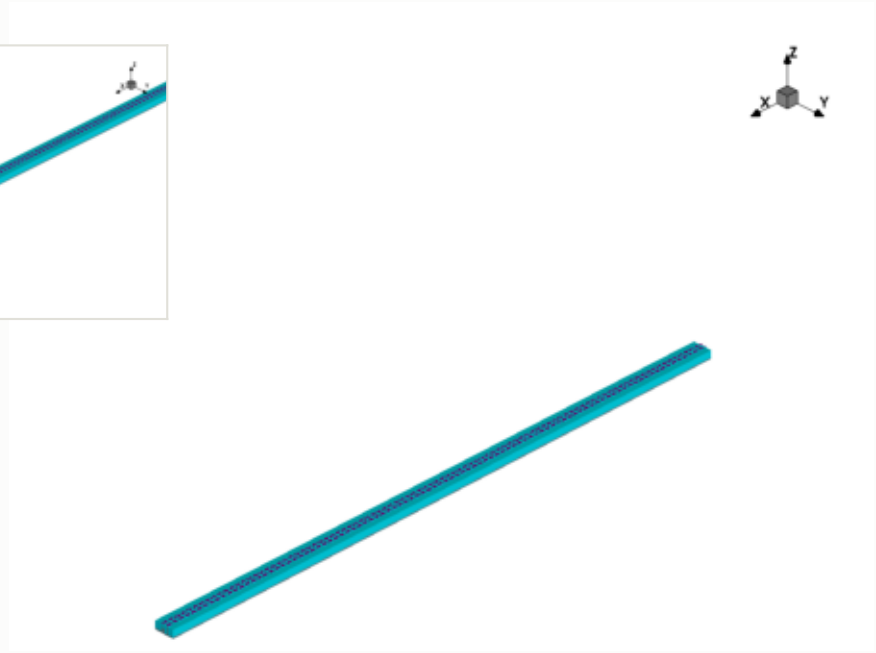
resist · Resist

0.169	90.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





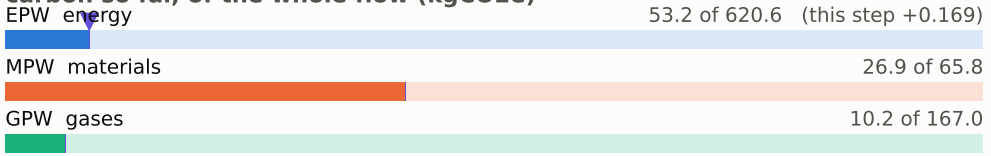
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

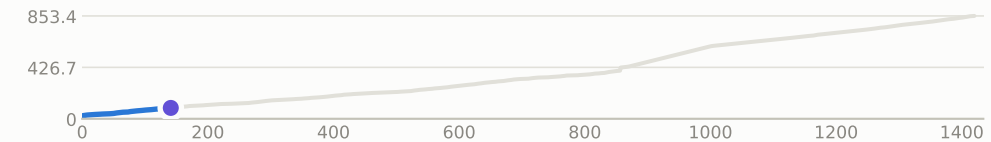
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

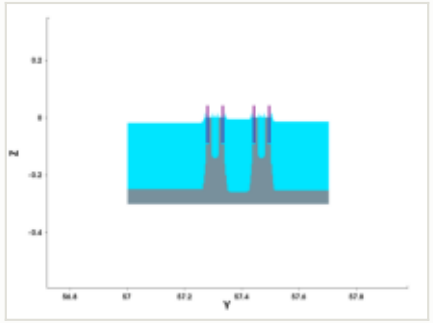
carbon so far, of the whole flow (kgCO2e)



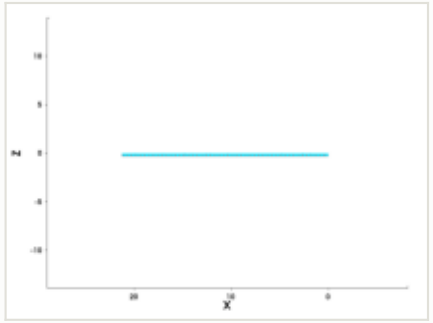
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



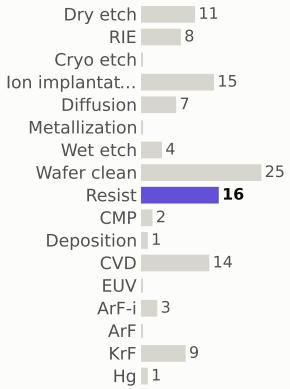
y cut at 57.351 µm



mask for this step

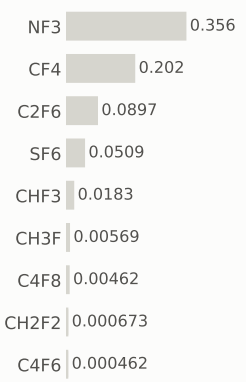
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



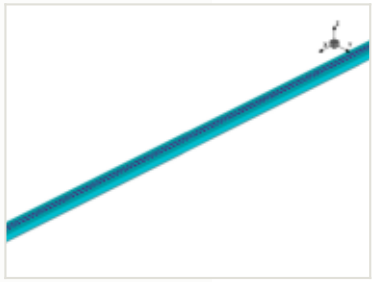
Step #1440: KrF

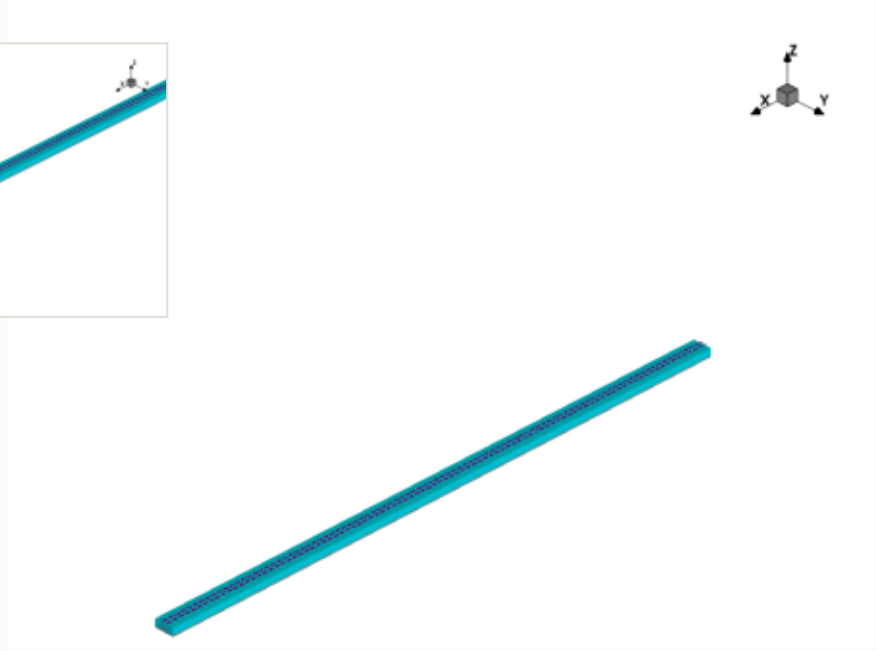
litho_patterning · KrF

0.199	90.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



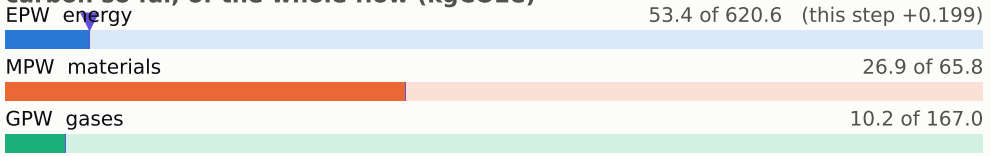


this step

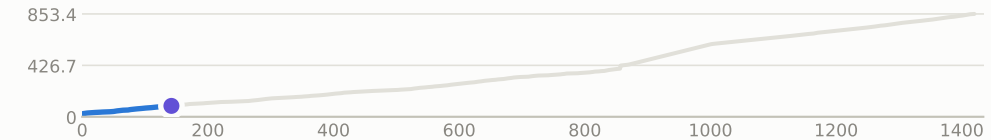
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

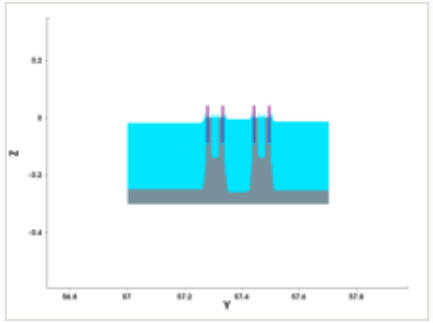
carbon so far, of the whole flow (kgCO2e)



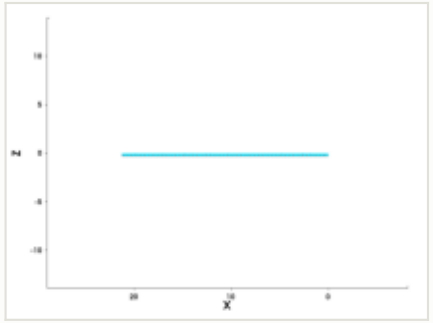
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

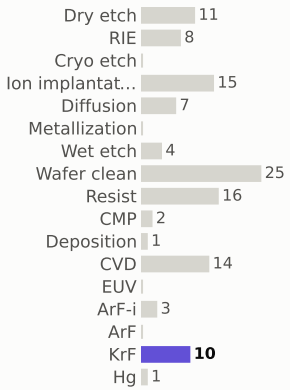


y cut at 57.351 µm



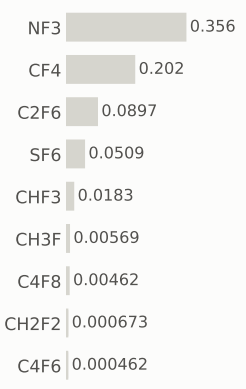
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



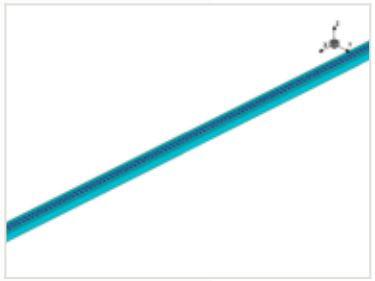
Step #1450: Ion_implantation

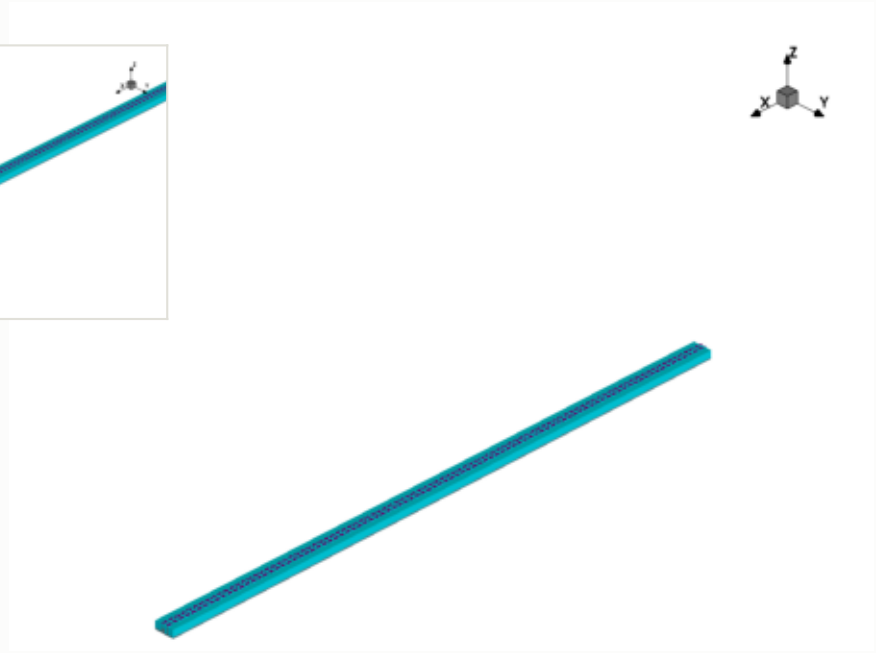
implant_ellipsoid · Ion_implantation

0.801	91.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

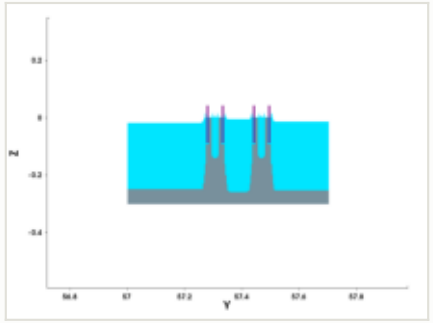
3D structure after this step

zoomed to fit

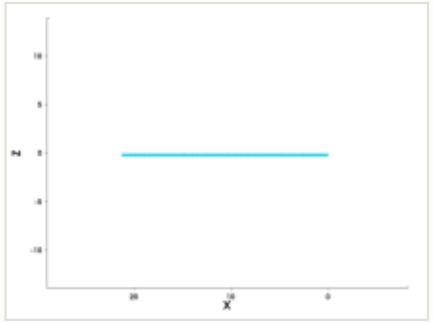




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

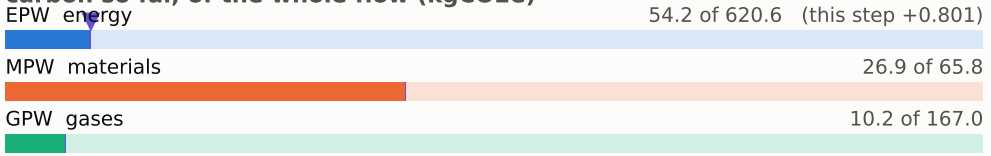
no mask for this step

this step

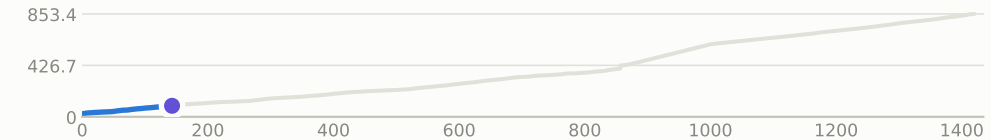
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

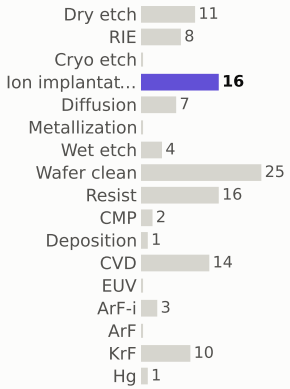
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

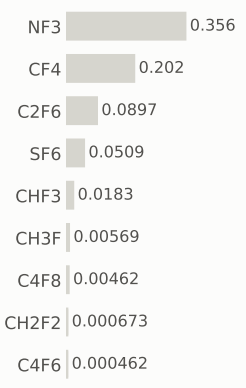


steps by process area



materials deposited (g)

gases emitted (g)



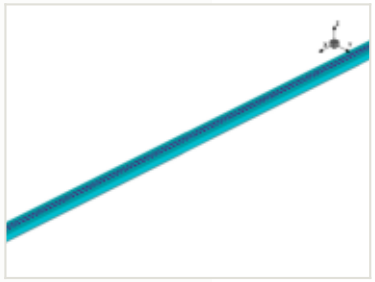
Step #1460: Ion_implantation

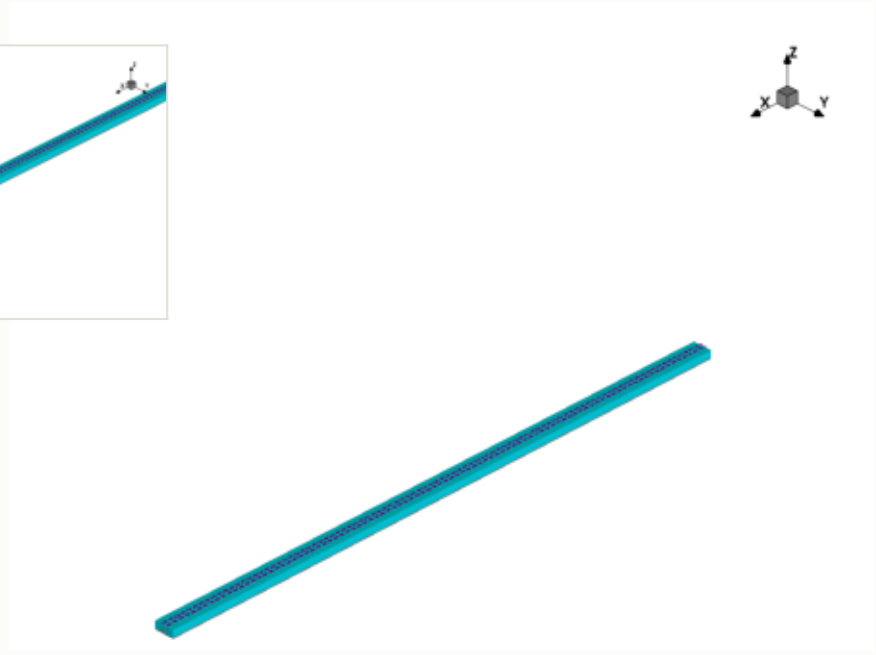
implant_ellipsoid · Ion_implantation

0.801	92.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

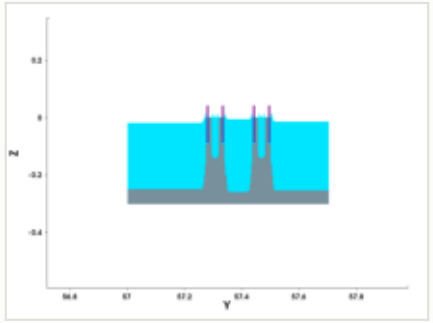
3D structure after this step

zoomed to fit

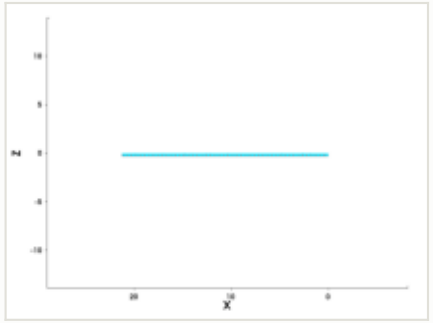




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

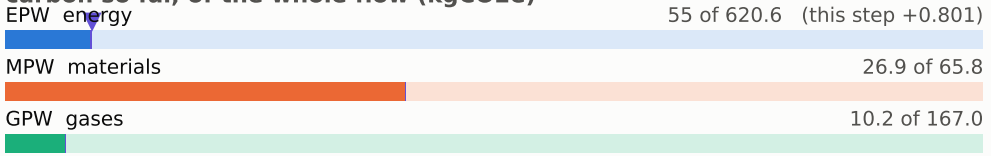
no mask for this step

this step

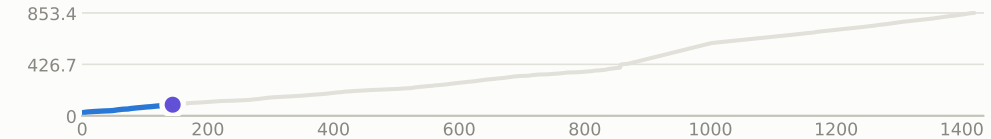
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

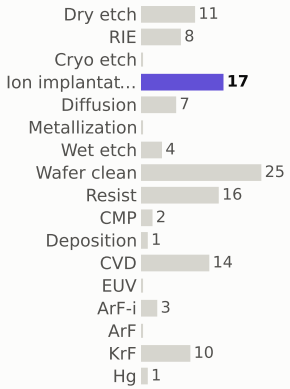
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

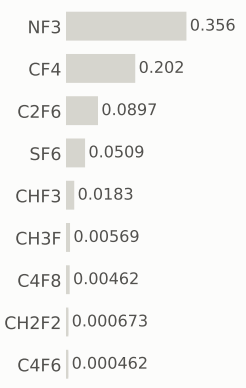


steps by process area



materials deposited (g)

gases emitted (g)



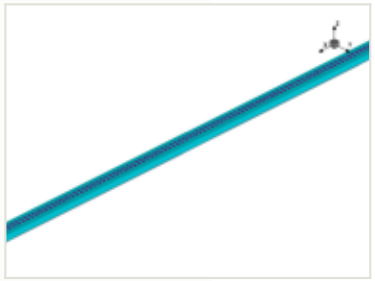
Step #1470: Dry_etch

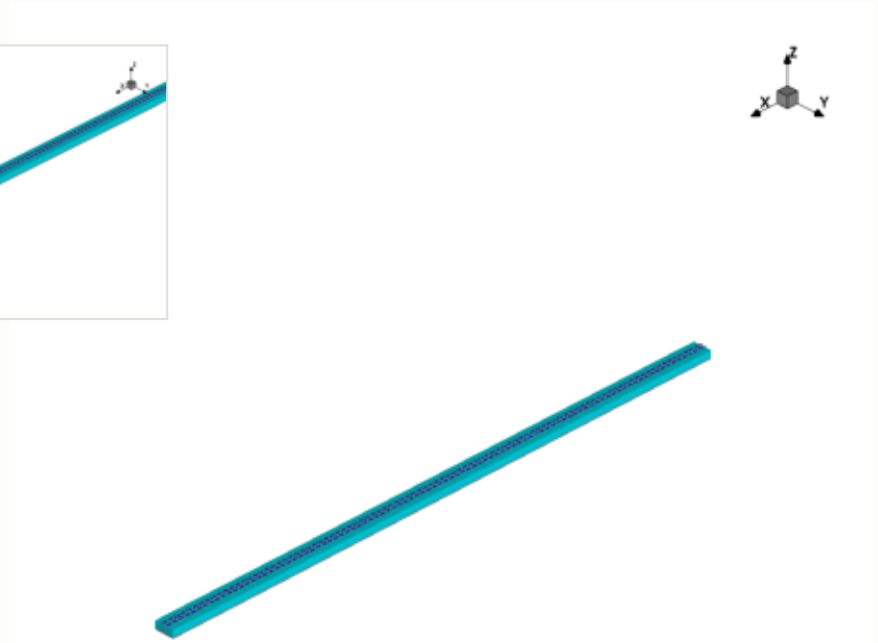
etch_strip · Dry_etch

0.801	92.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

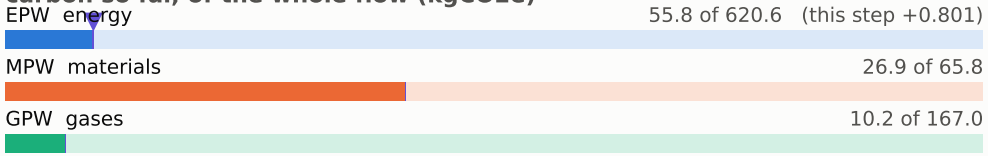




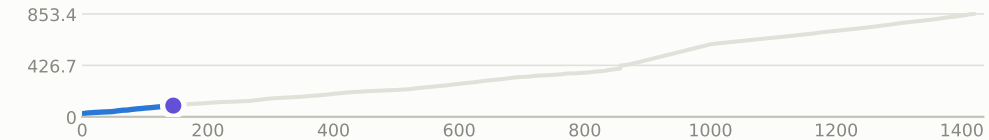
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

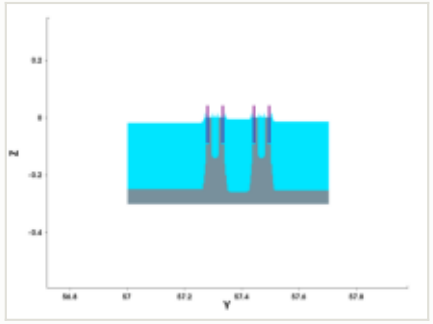
carbon so far, of the whole flow (kgCO2e)



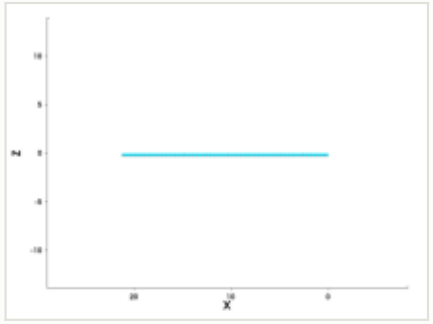
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



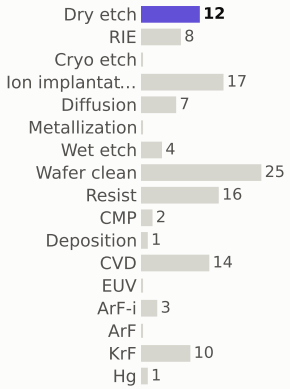
y cut at 57.351 µm



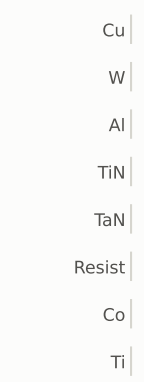
mask for this step

no mask for this step

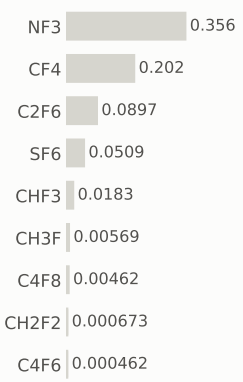
steps by process area



materials deposited (g)



gases emitted (g)



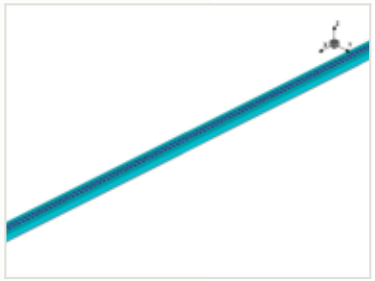
Step #1480: Wafer_clean

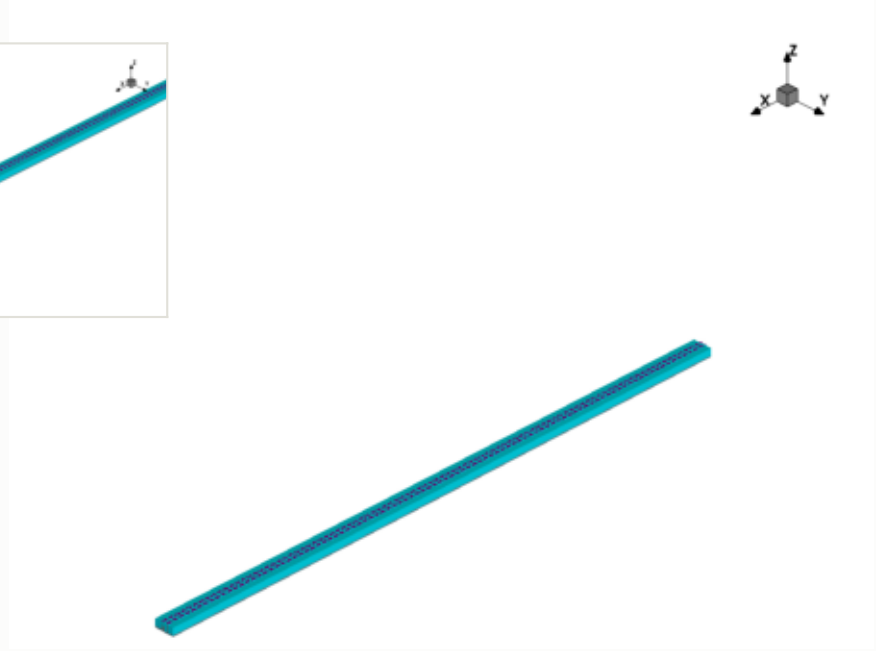
clean · Wafer_clean

0.169	93.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



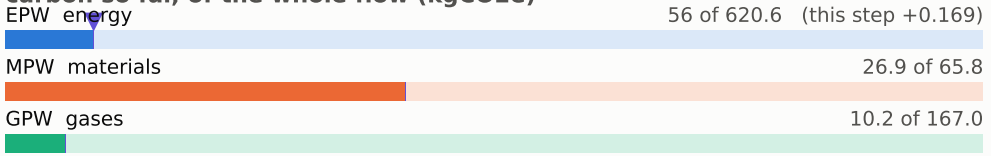


this step

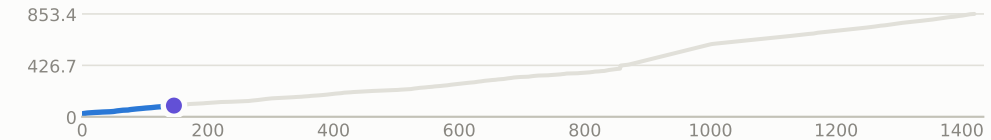
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

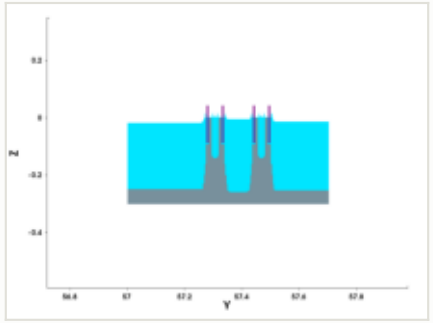
carbon so far, of the whole flow (kgCO2e)



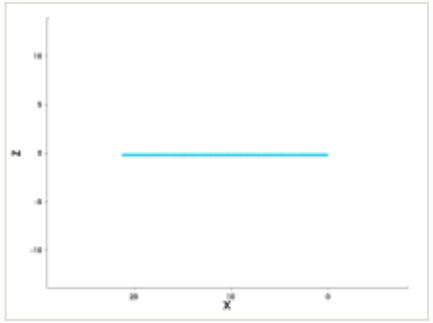
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



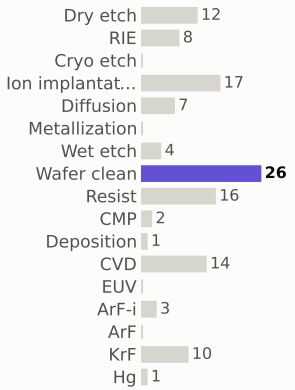
y cut at 57.351 µm



mask for this step

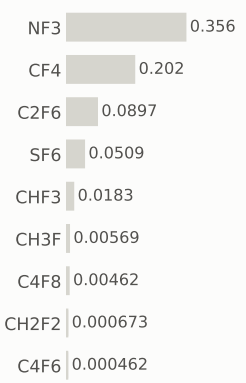
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



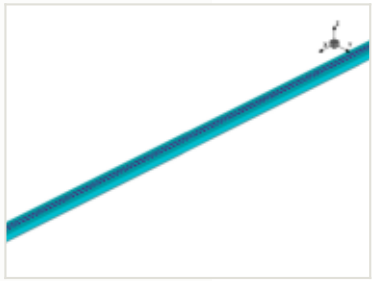
Step #1490: Diffusion

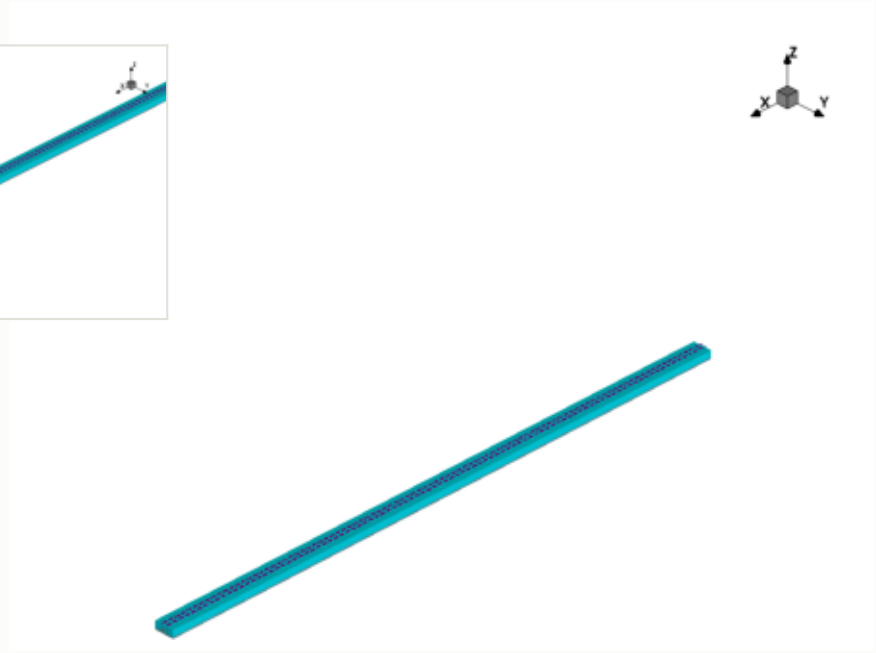
anneal · Diffusion

0.801	93.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

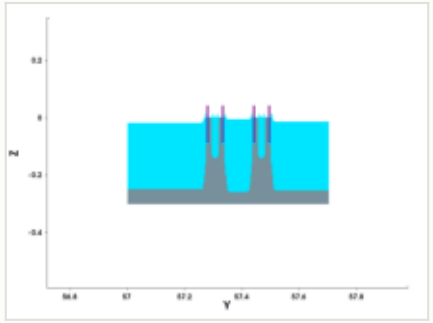
3D structure after this step

zoomed to fit

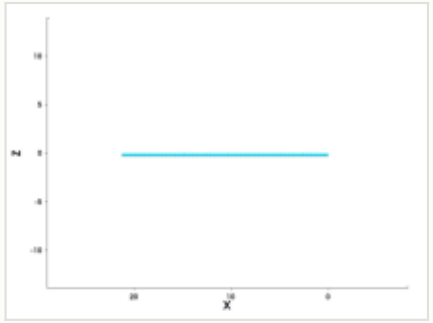




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

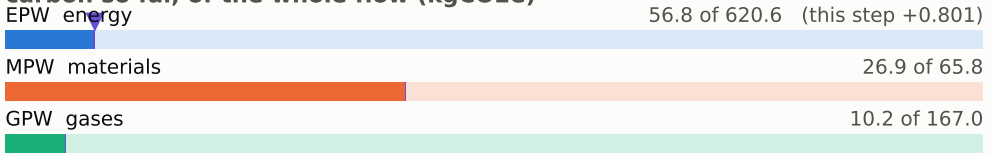
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

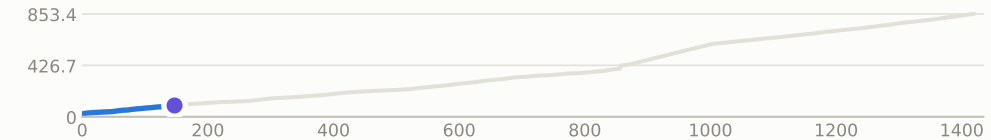
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

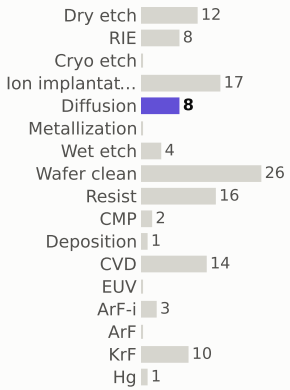
carbon so far, of the whole flow (kgCO2e)



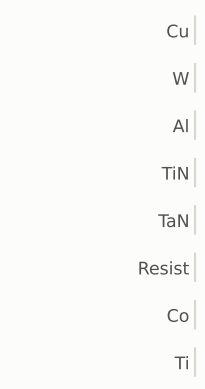
embodied carbon up to each step (kgCO2e per wafer)



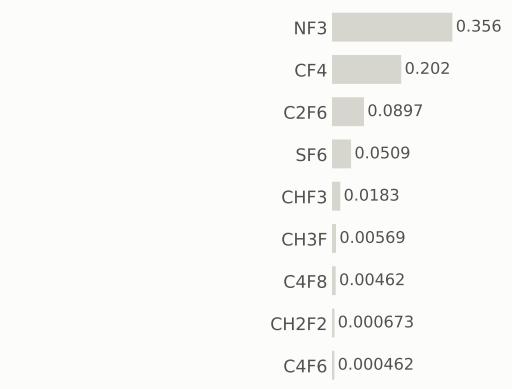
steps by process area



materials deposited (g)



gases emitted (g)



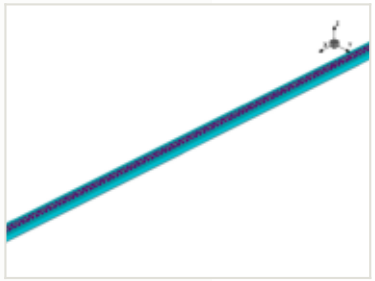
Step #1500: Dry_etch

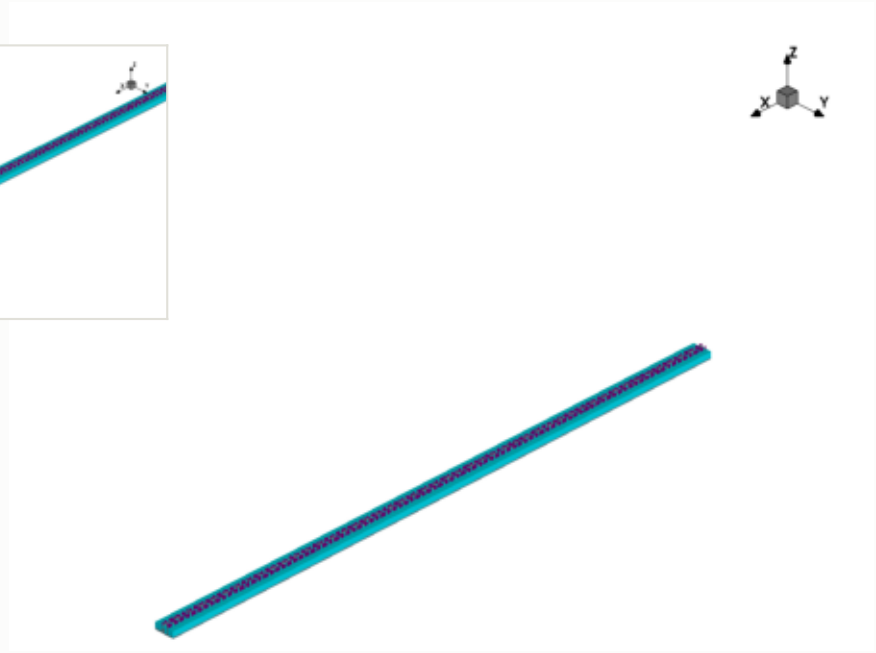
etch_isotropic · Dry_etch

0.801	94.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



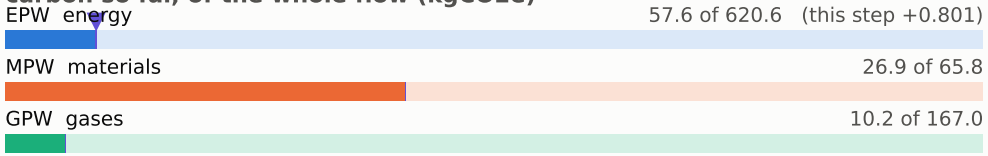


this step

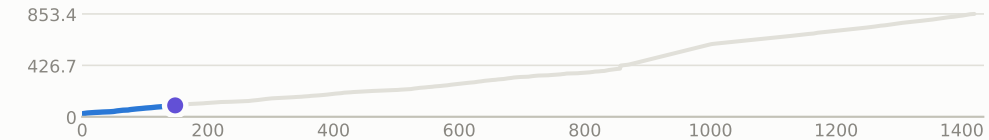
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

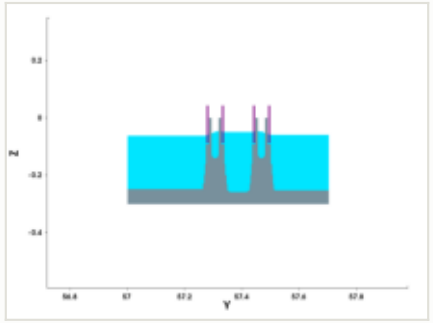
carbon so far, of the whole flow (kgCO2e)



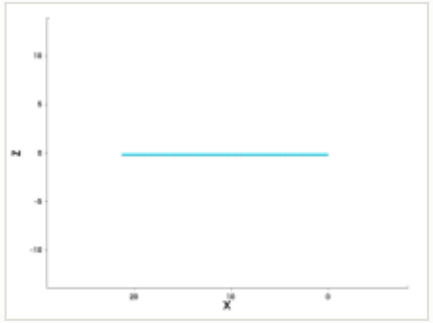
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



y cut at 57.351 µm



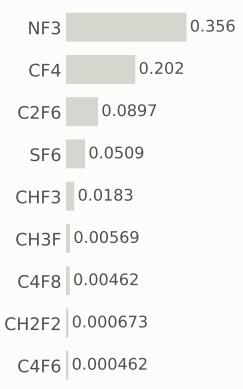
mask for this step

no mask for this step

steps by process area

materials deposited (g)

gases emitted (g)



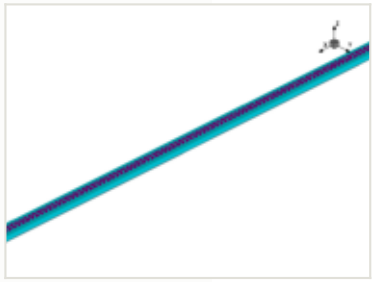
Step #1510: Spx_only

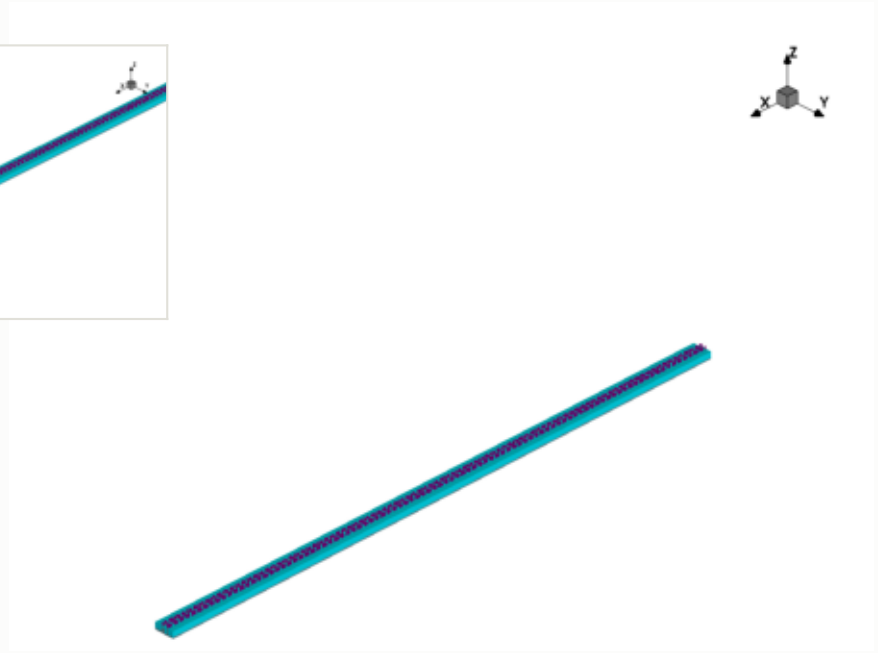
etch_isotropic · Spx_only

0	94.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

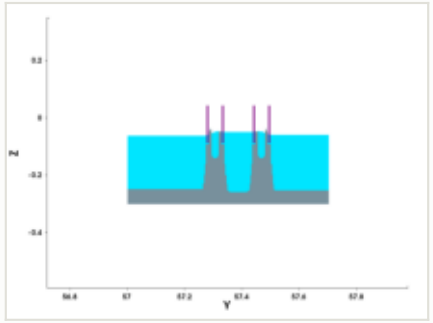
3D structure after this step

zoomed to fit

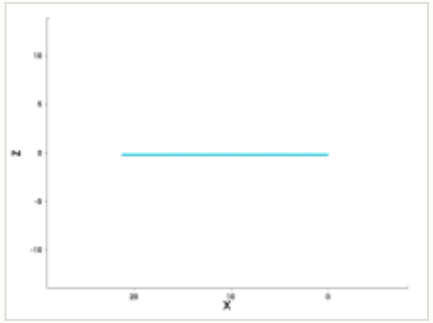




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

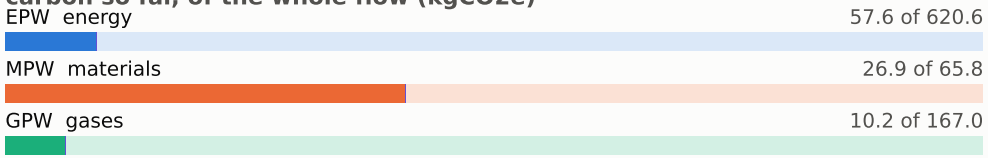
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

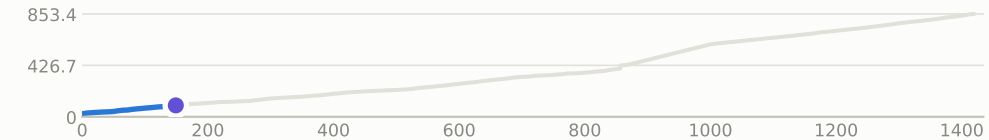
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

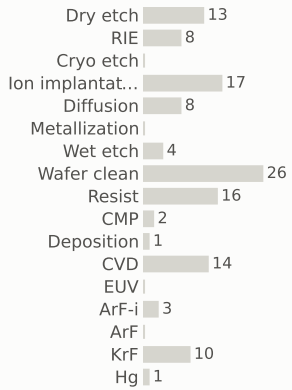
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

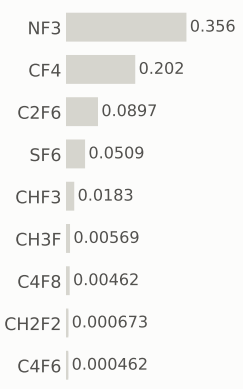


steps by process area



materials deposited (g)

gases emitted (g)



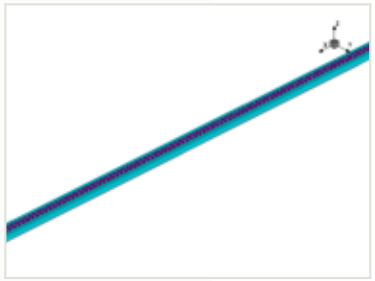
Step #1520: Wafer_clean

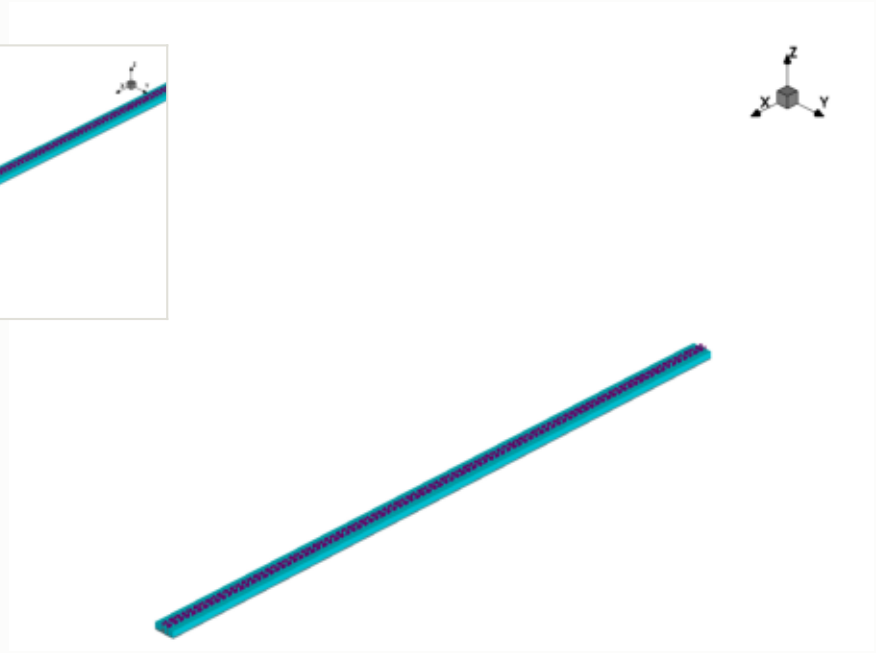
clean · Wafer_clean

0.169	94.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





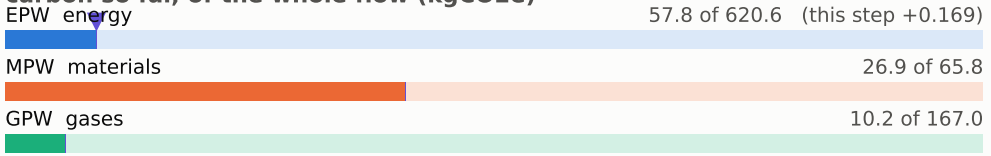
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

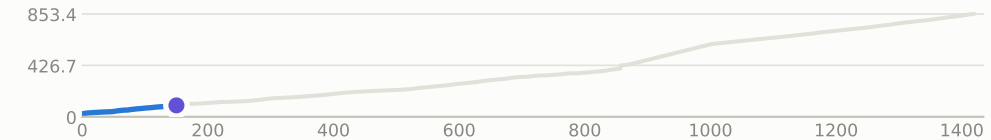
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

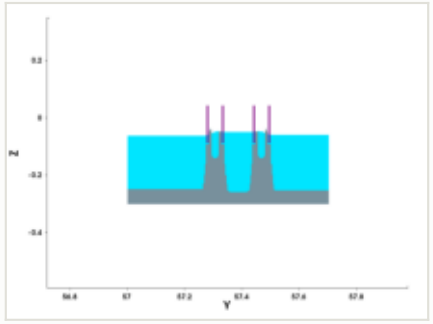
carbon so far, of the whole flow (kgCO2e)



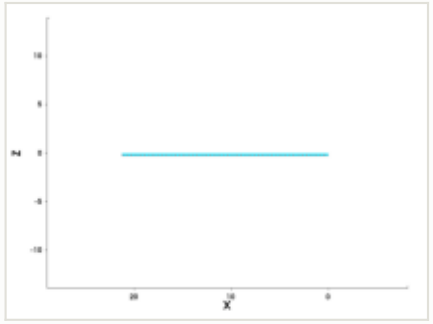
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



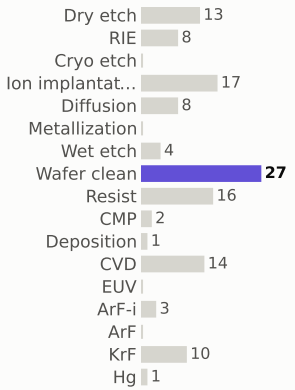
y cut at 57.351 µm



mask for this step

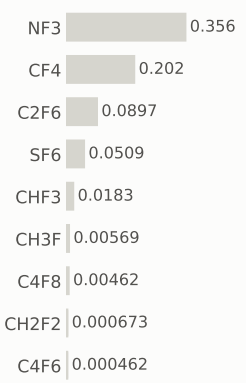
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



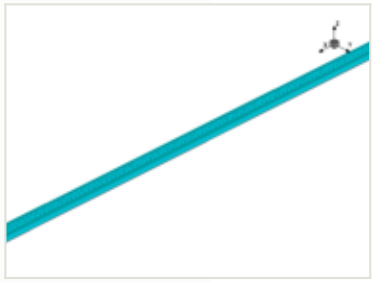
Step #1530: Diffusion

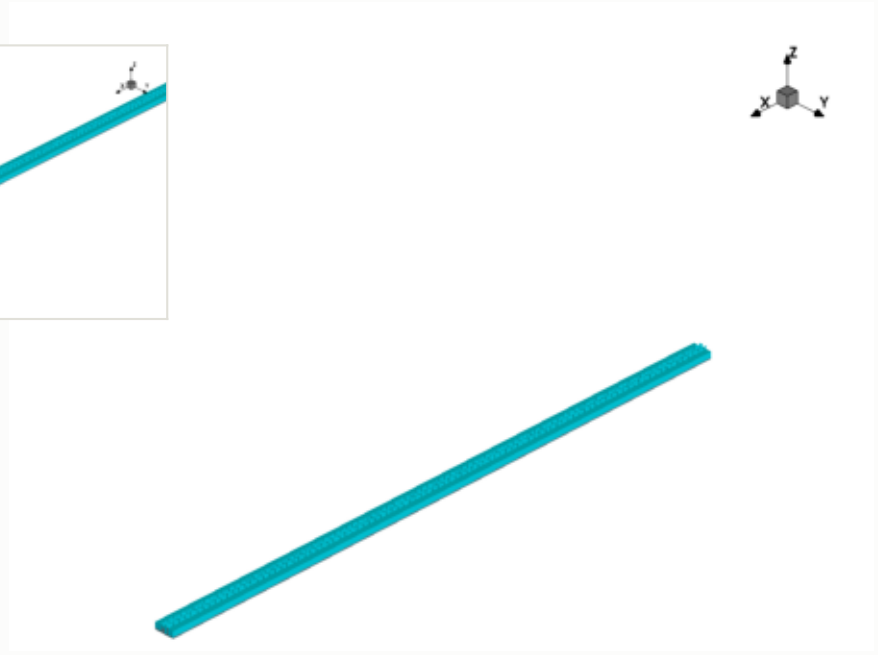
depo_isotropic · Diffusion

0.801	95.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

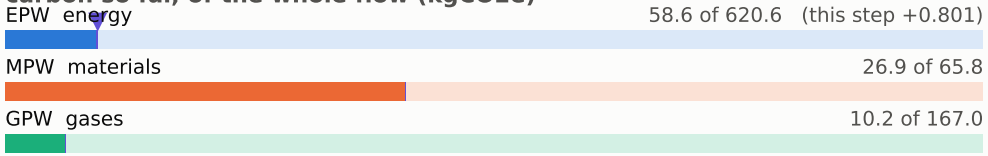




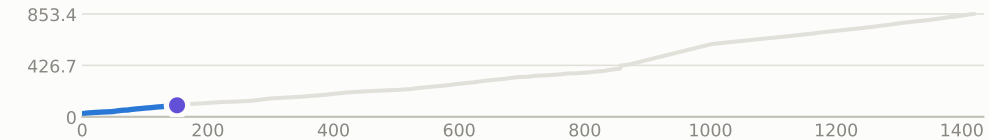
this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	0.0034 µm	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

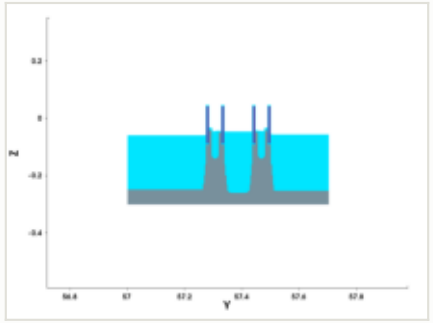
carbon so far, of the whole flow (kgCO2e)



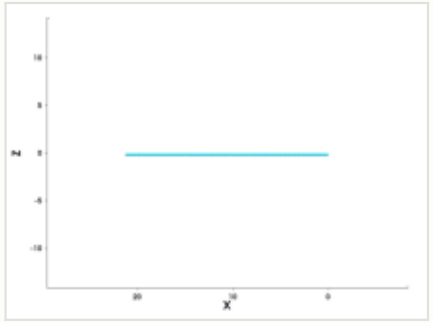
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



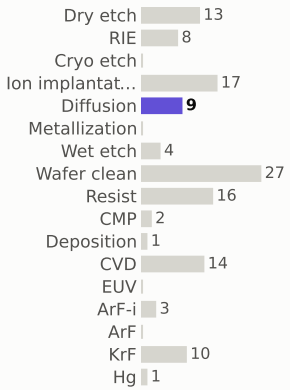
y cut at 57.351 µm



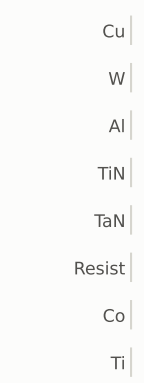
mask for this step

no mask for this step

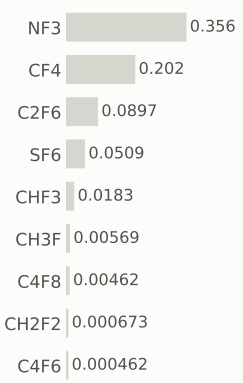
steps by process area



materials deposited (g)



gases emitted (g)



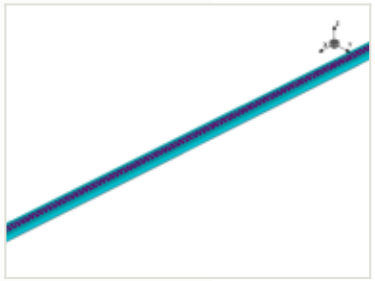
Step #1540: Dry_etch

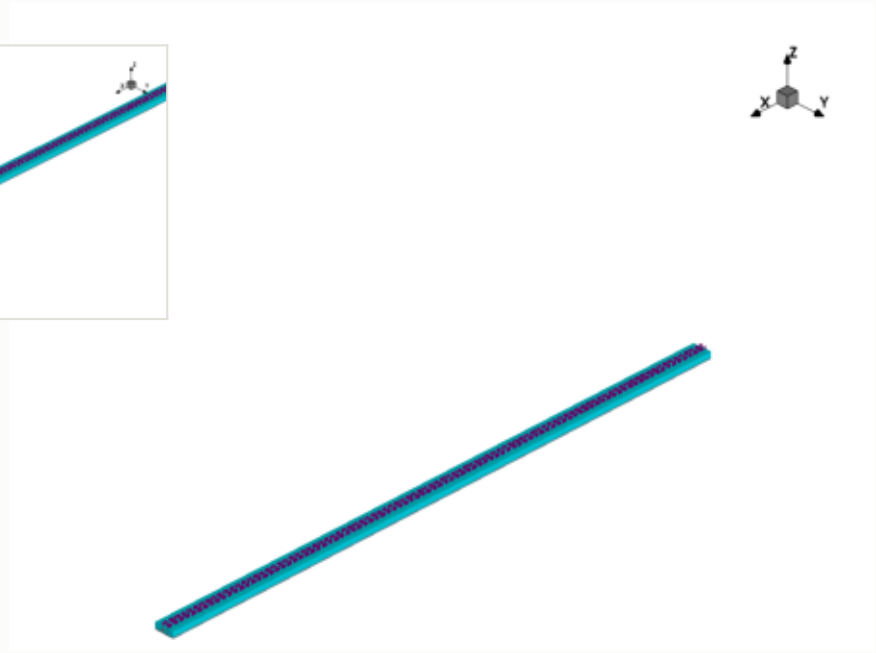
etch_isotropic · Dry_etch

0.801	96.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



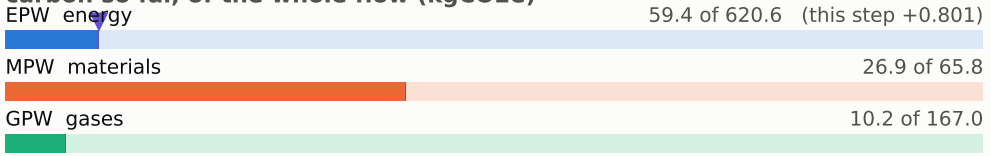


this step

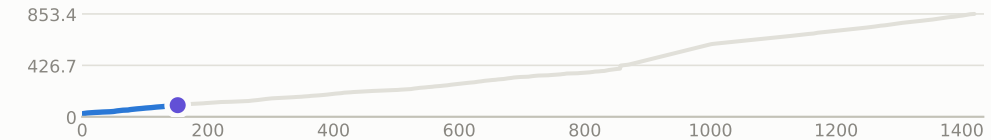
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

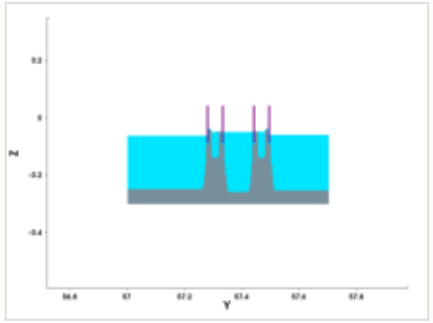
carbon so far, of the whole flow (kgCO2e)



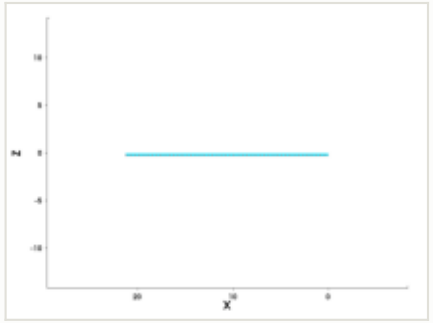
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



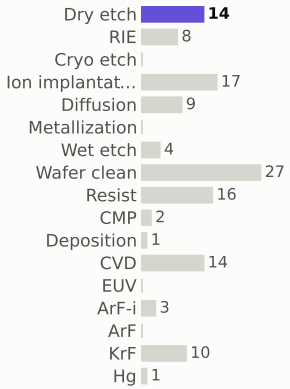
y cut at 57.351 µm



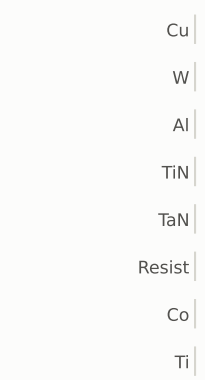
mask for this step

no mask for this step

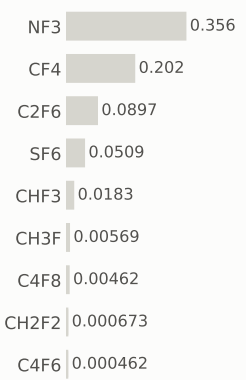
steps by process area



materials deposited (g)



gases emitted (g)



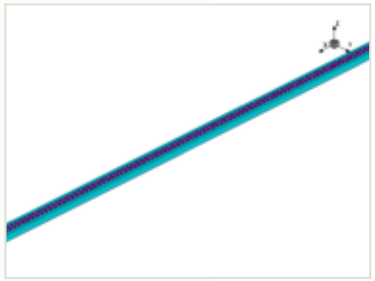
Step #1560: Diffusion

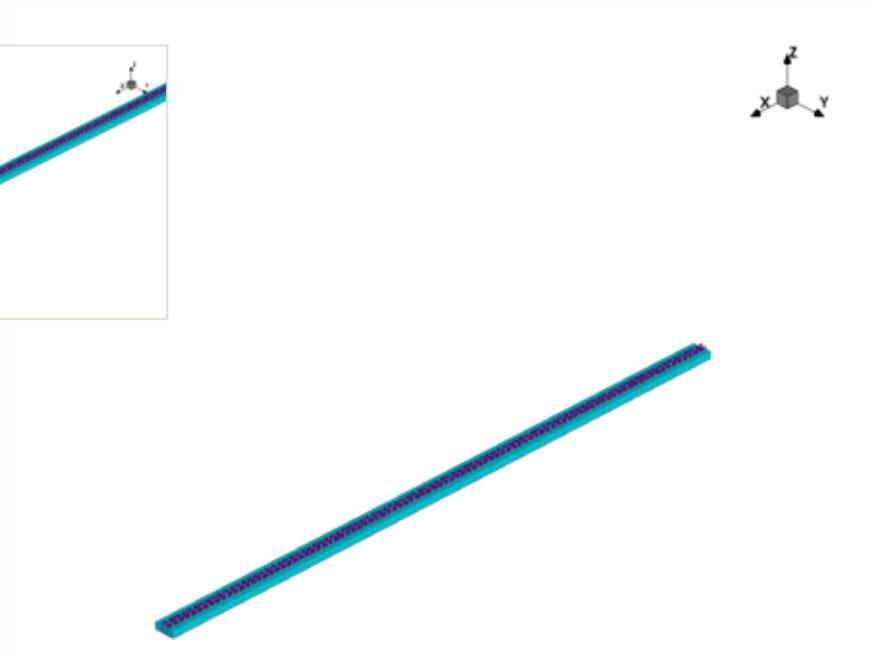
depo_isotropic · Diffusion

0.801	97.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



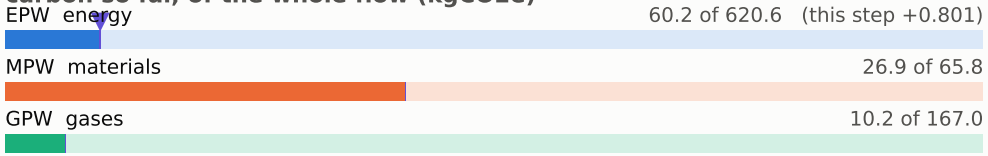


this step

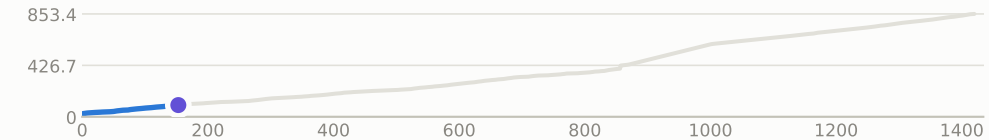
process area	Diffusion
material	—
thickness	0.0034 µm
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

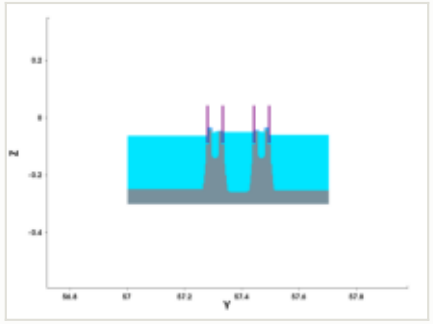
carbon so far, of the whole flow (kgCO2e)



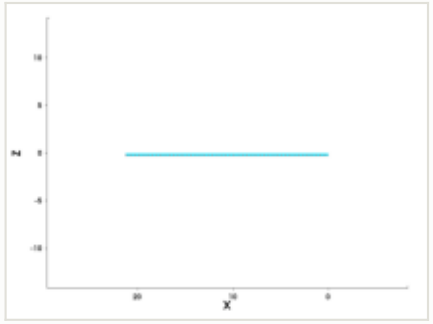
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



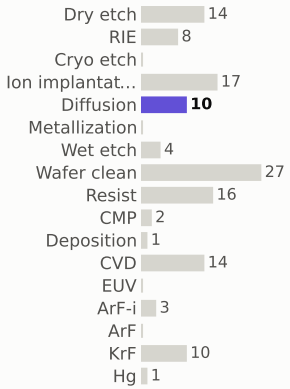
y cut at 57.351 µm



mask for this step

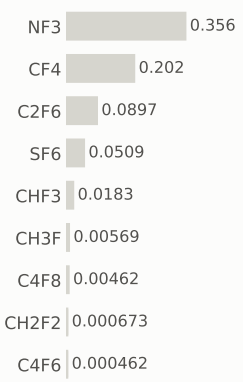
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



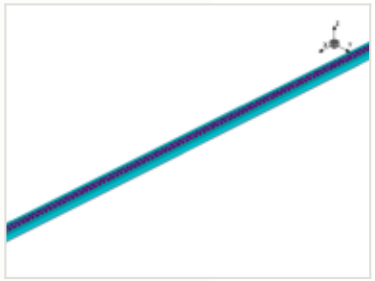
Step #1570: Spx_only

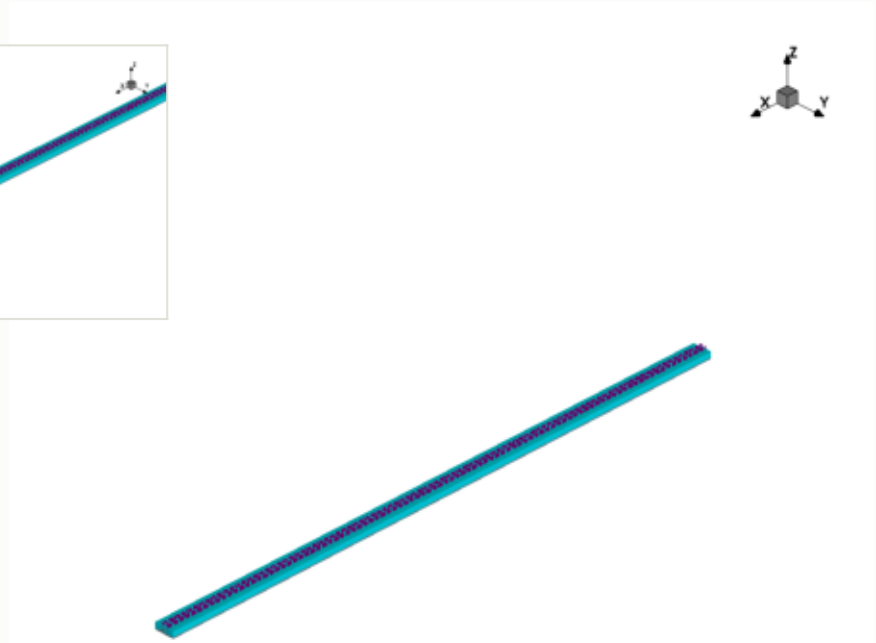
etch_isotropic · Spx_only

0	97.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





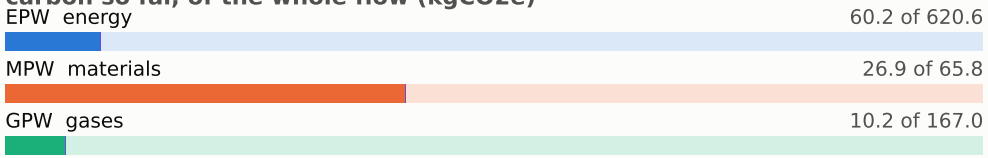
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

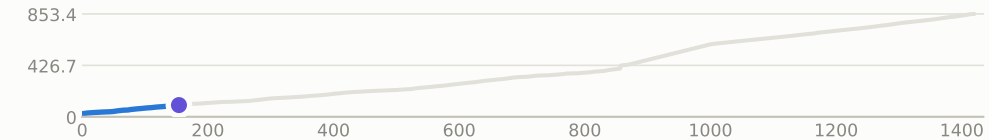
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

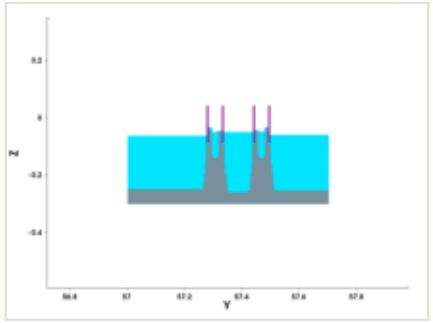
carbon so far, of the whole flow (kgCO2e)



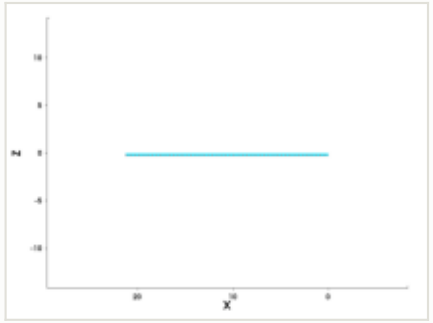
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



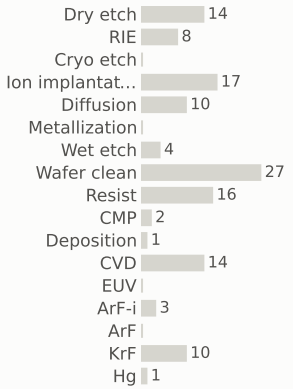
y cut at 57.351 µm



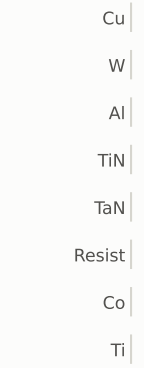
mask for this step

no mask for this step

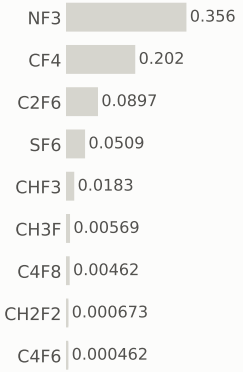
steps by process area



materials deposited (g)



gases emitted (g)



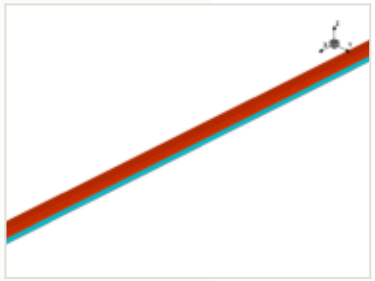
Step #1580: Spx_only

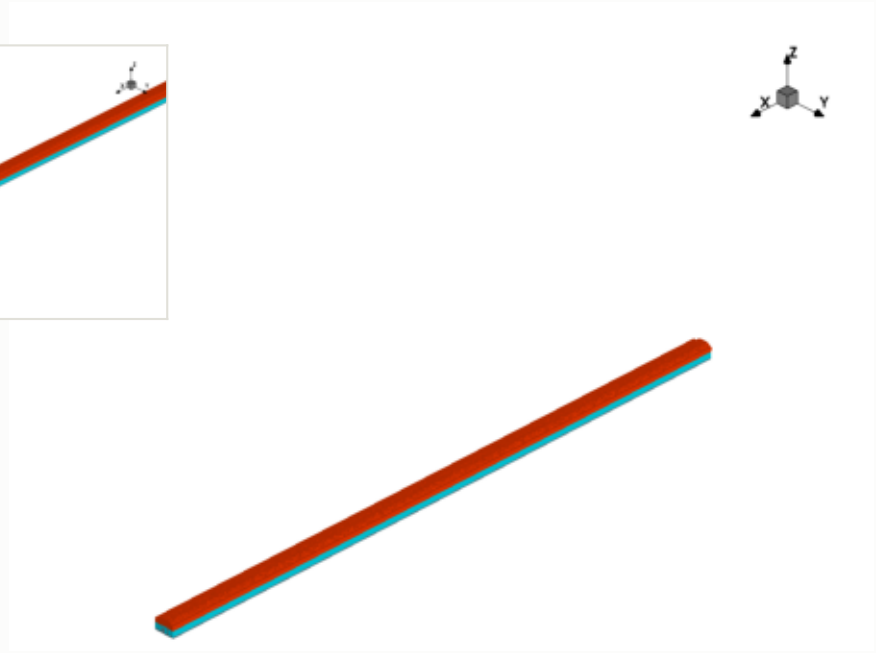
depo_isotropic · Spx_only

0	97.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



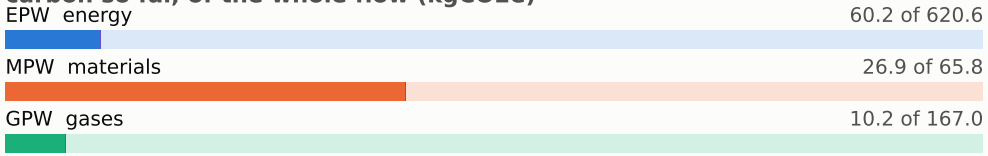


this step

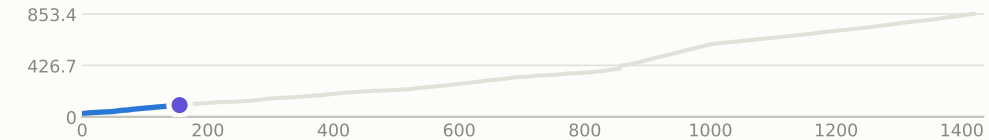
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	0.15 µm	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

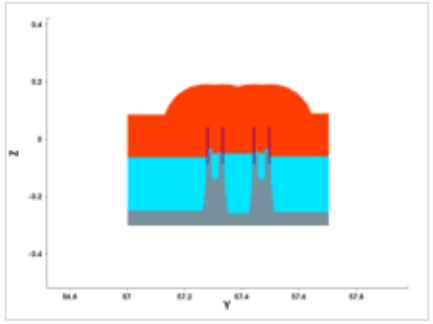
carbon so far, of the whole flow (kgCO2e)



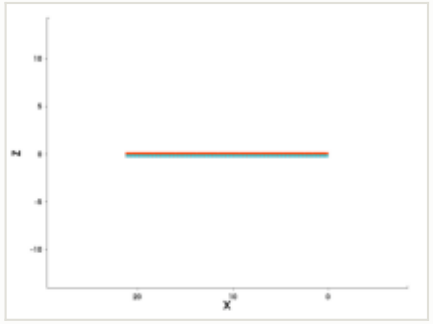
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



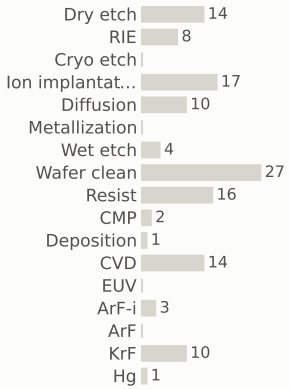
y cut at 57.351 µm



mask for this step

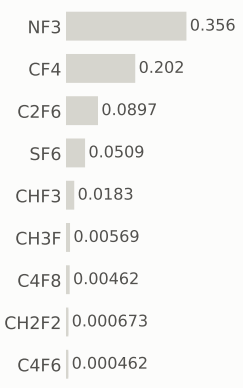
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



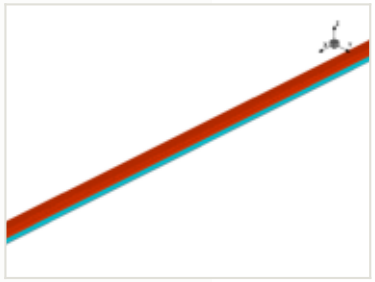
Step #1590: Diffusion

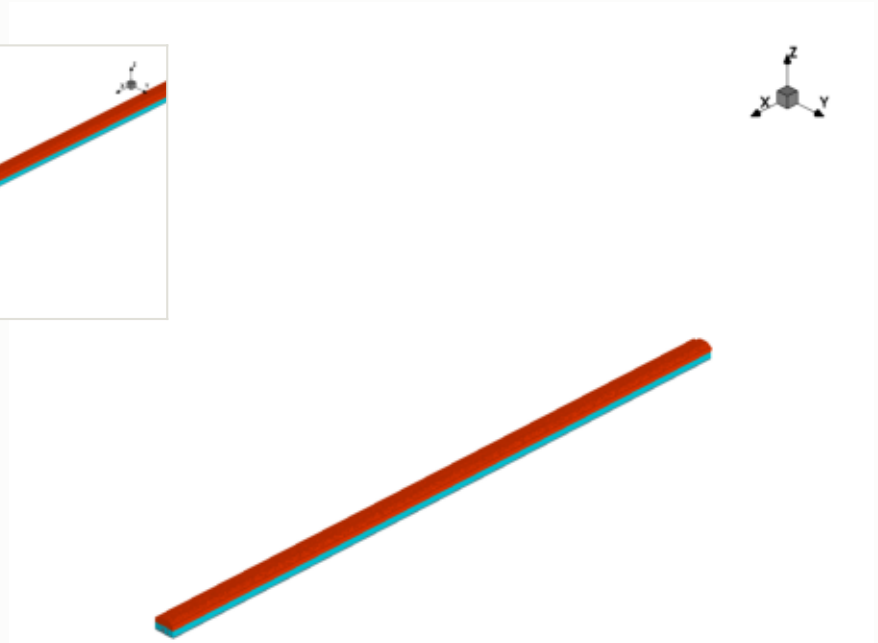
implant_ellipsoid · Diffusion

0.801	98.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

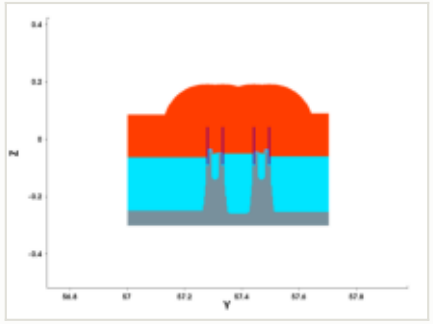
3D structure after this step

zoomed to fit

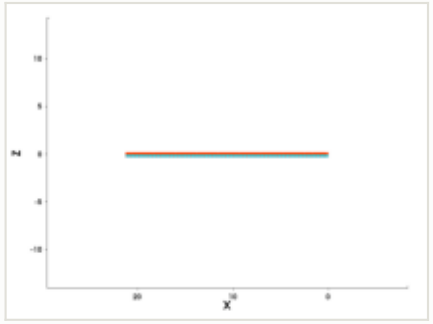




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

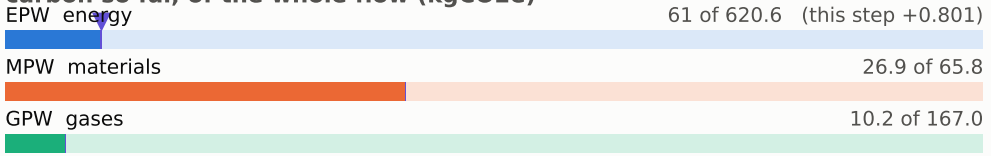
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

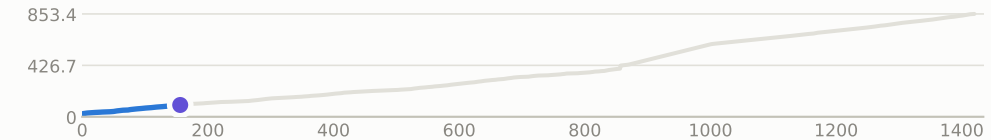
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

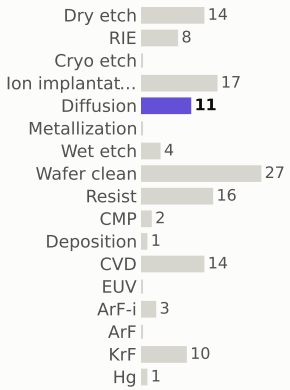
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

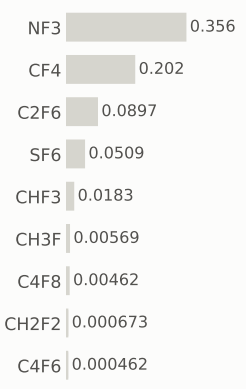


steps by process area



materials deposited (g)

gases emitted (g)



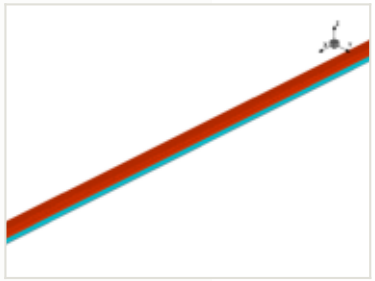
Step #1600: CVD

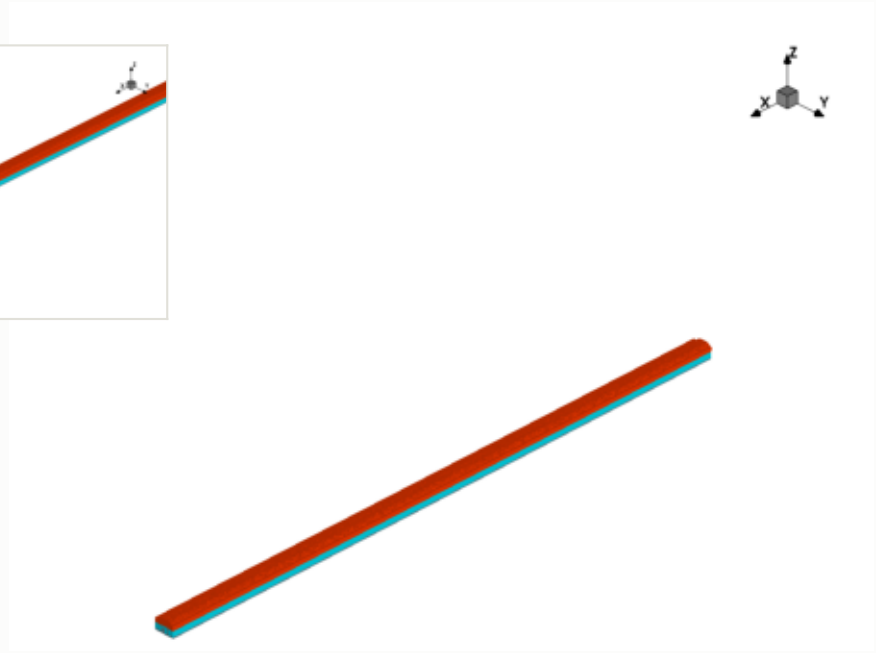
depo_isotropic · CVD

1.31	99.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

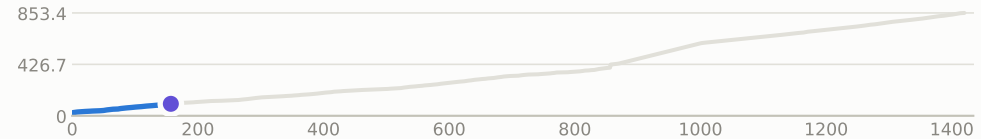
carbon so far, of the whole flow (kgCO2e)

EPW energy — 61.6 of 620.6 (this step +0.656)

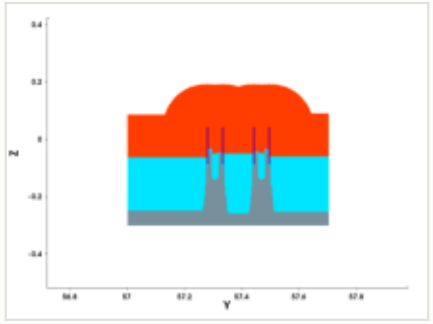
MPW materials — 26.9 of 65.8 (this step +0.0317)

GPW gases — 10.8 of 167.0 (this step +0.62)

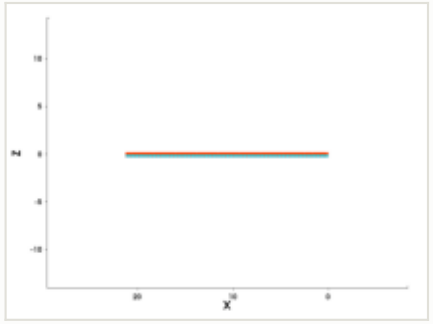
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



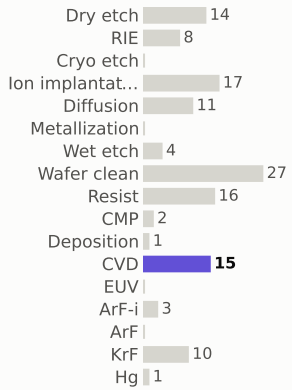
y cut at 57.351 µm



mask for this step

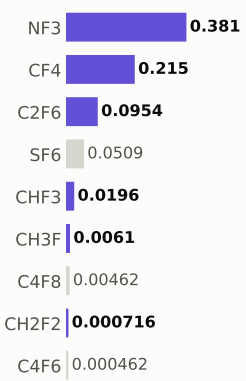
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



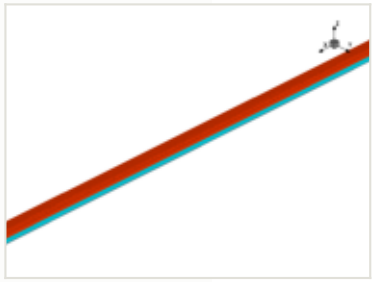
Step #1610: Wet_etch

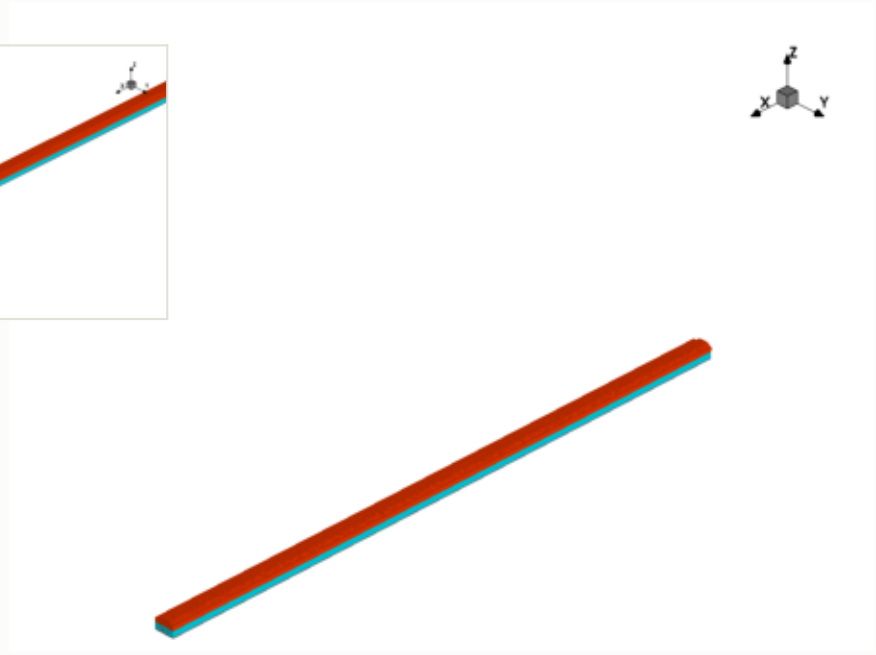
etch_isotropic · Wet_etch

0.169	99.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



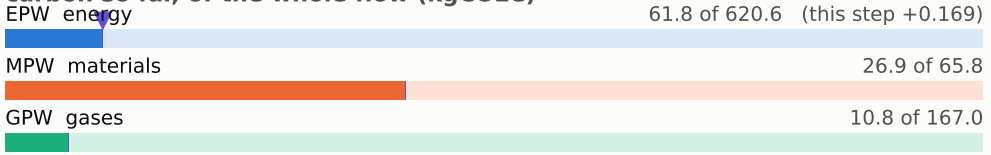


this step

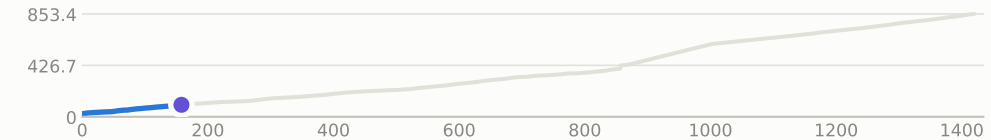
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

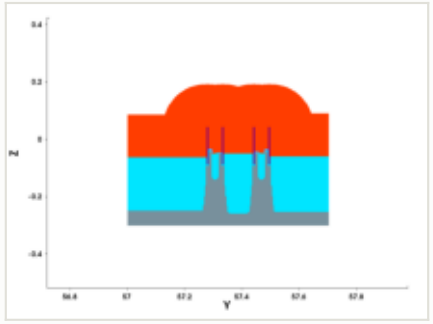
carbon so far, of the whole flow (kgCO2e)



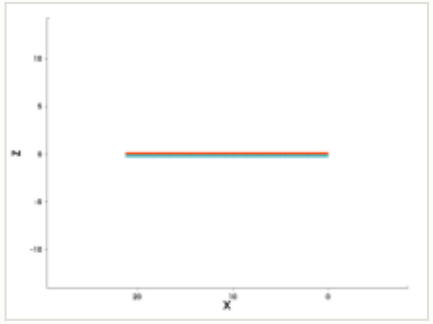
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



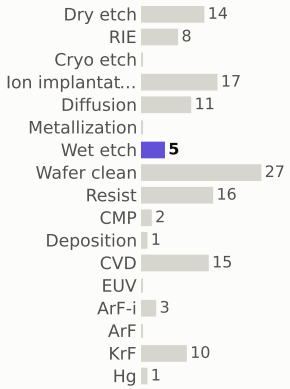
y cut at 57.351 µm



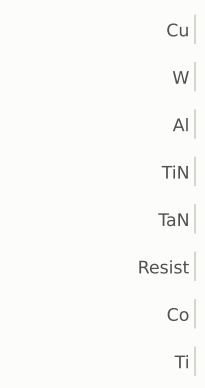
mask for this step

no mask for this step

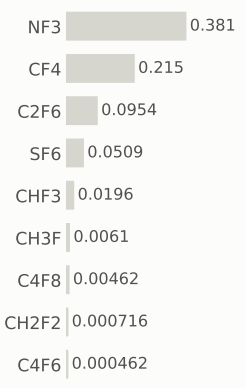
steps by process area



materials deposited (g)



gases emitted (g)



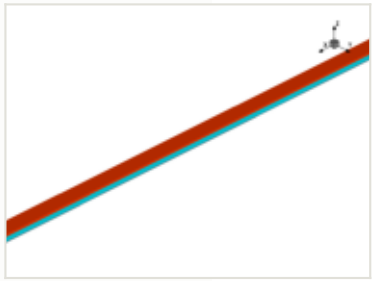
Step #1620: CMP

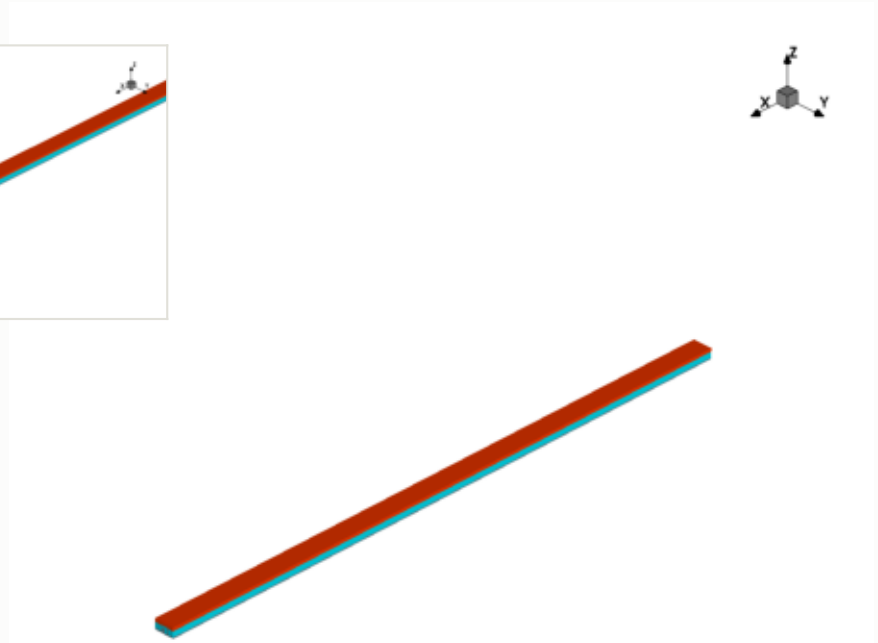
etch_cmp · CMP

0.169	99.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

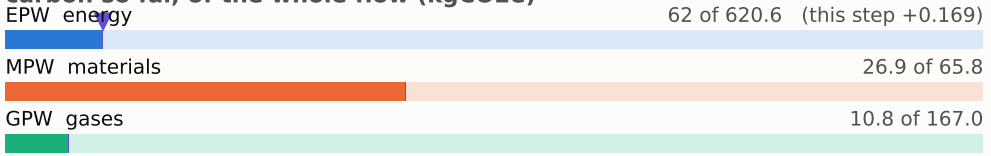




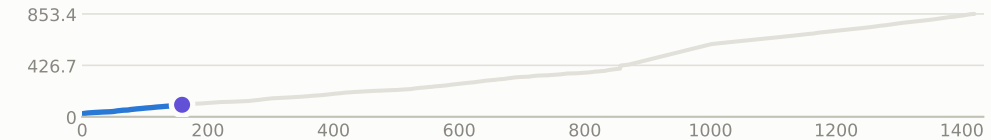
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

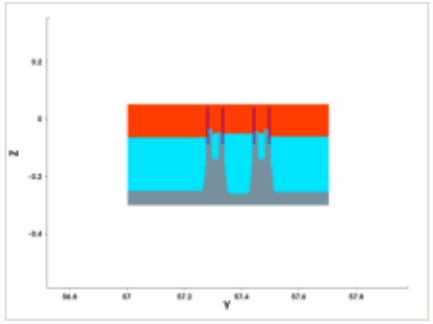
carbon so far, of the whole flow (kgCO2e)



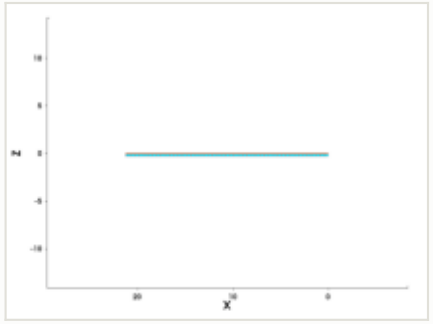
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



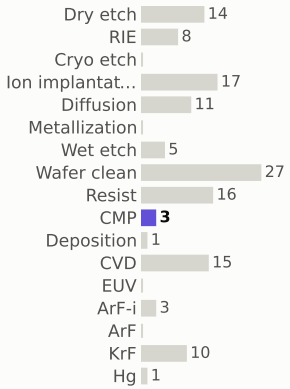
y cut at 57.351 µm



mask for this step

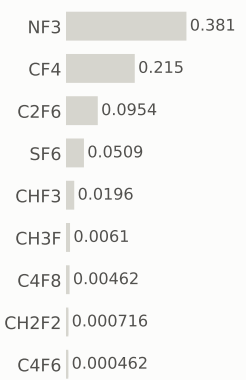
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



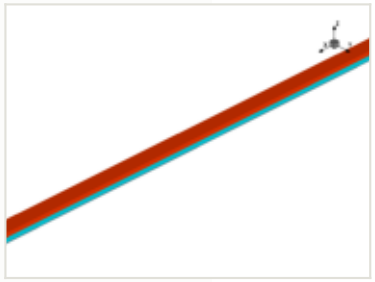
Step #1630: Spx_only

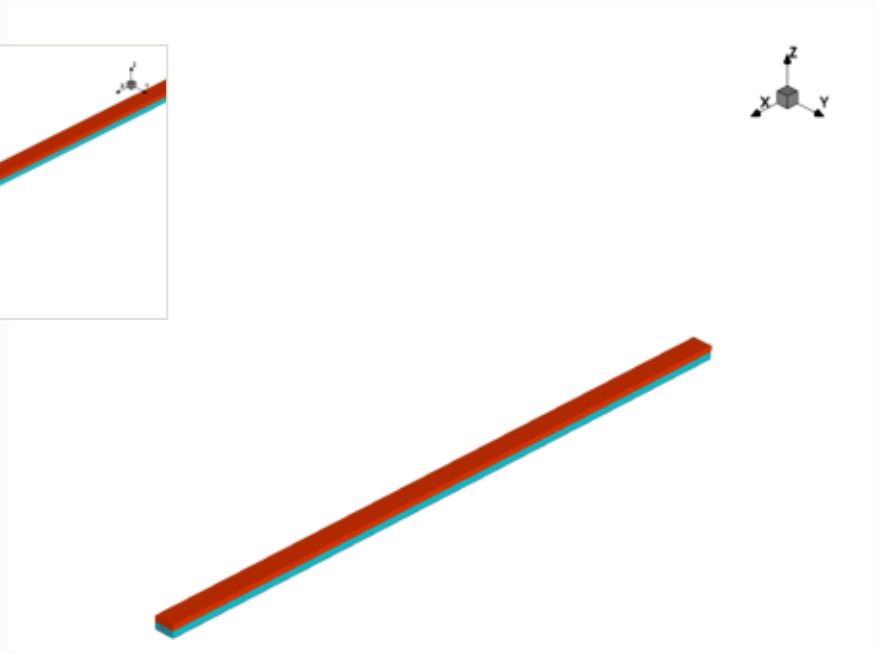
depo_isotropic · Spx_only

0	99.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





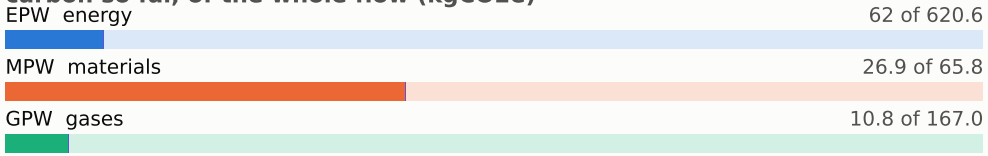
this step

process area	Spx_only
material	—
thickness	0.09 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

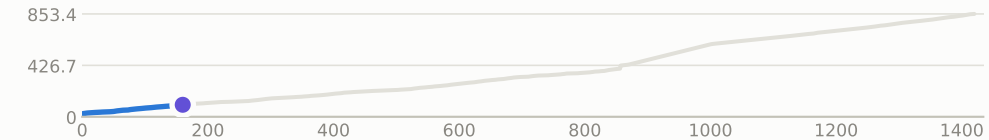
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

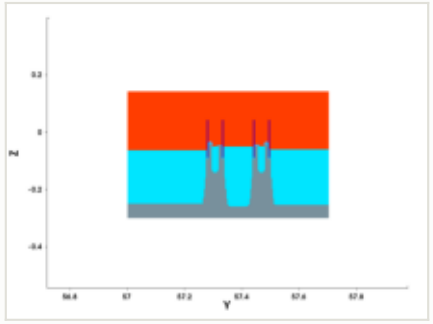
carbon so far, of the whole flow (kgCO2e)



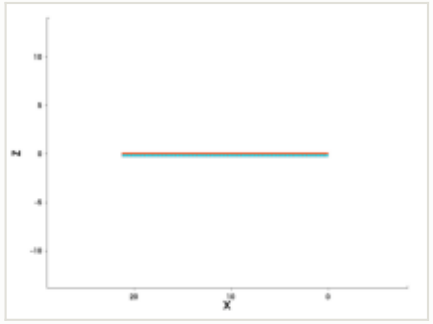
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



y cut at 57.351 µm

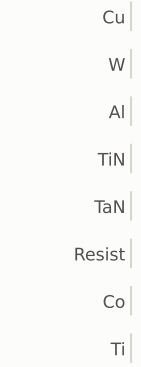


mask for this step

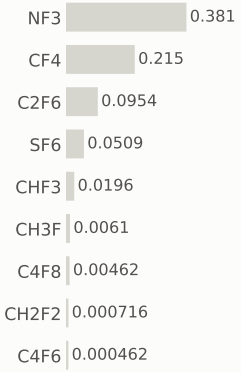
no mask for this step

steps by process area

materials deposited (g)



gases emitted (g)



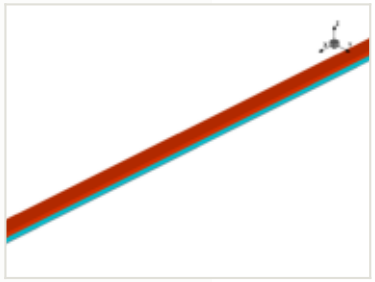
Step #1640: Wafer_clean

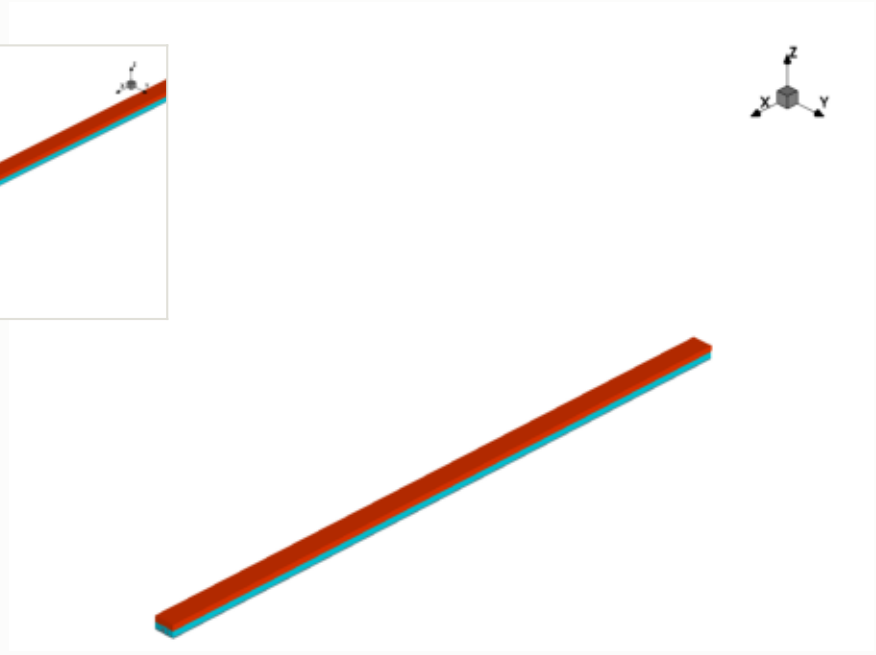
clean · Wafer_clean

0.169	99.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





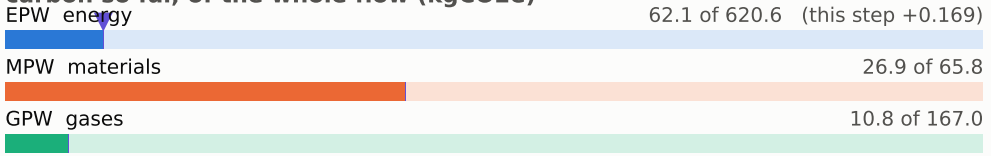
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

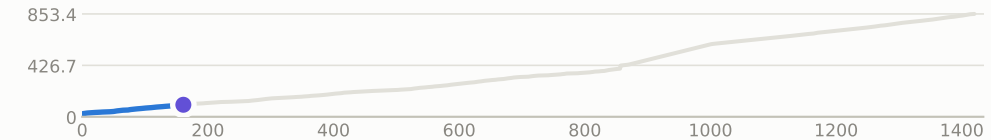
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

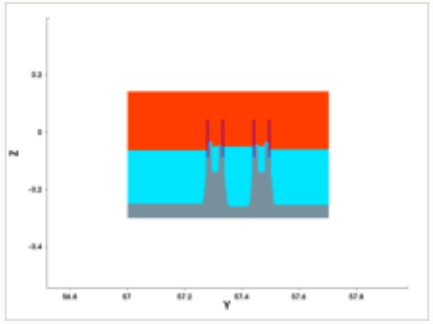
carbon so far, of the whole flow (kgCO2e)



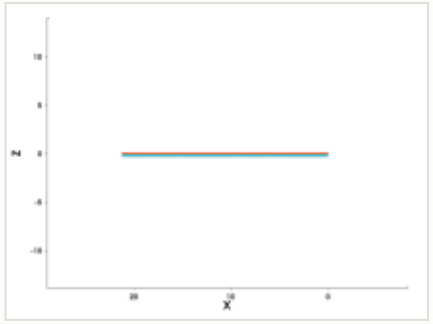
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



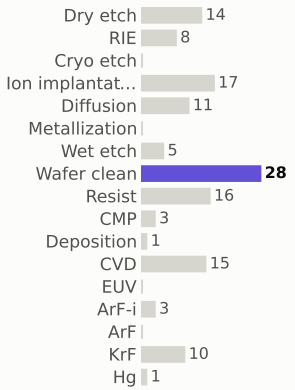
y cut at 57.351 µm



mask for this step

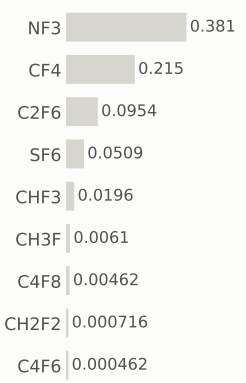
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



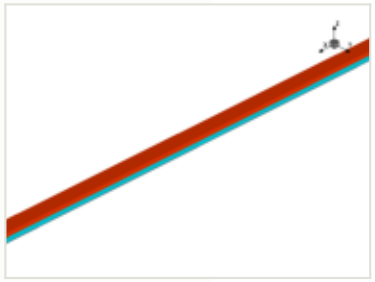
Step #1650: CVD

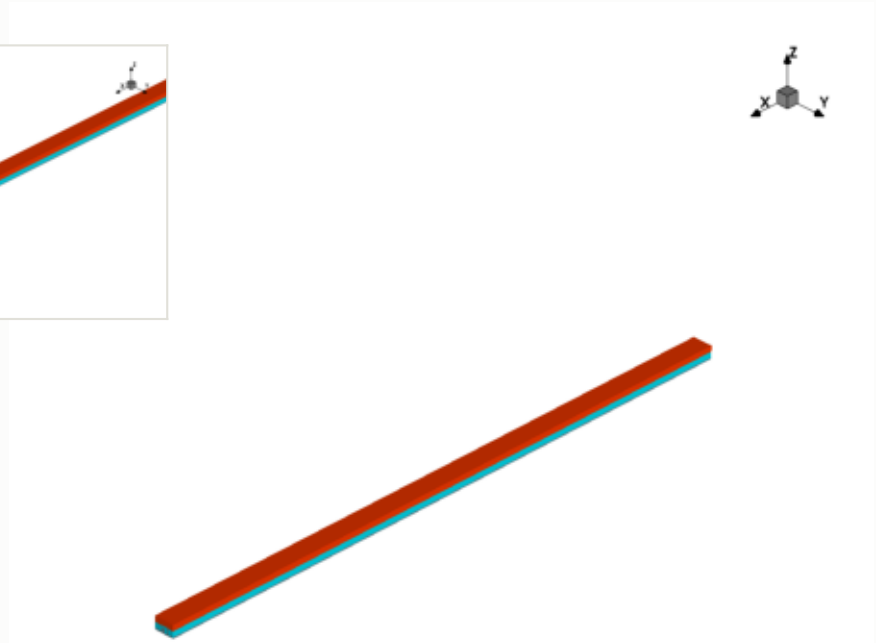
depo_isotropic · CVD

1.31	101.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



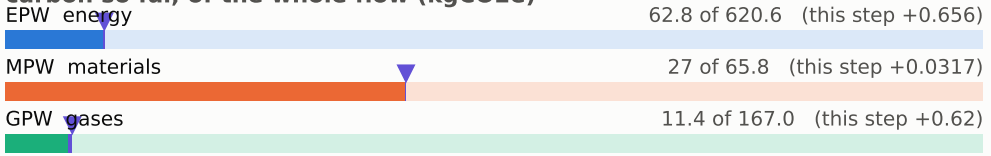


this step

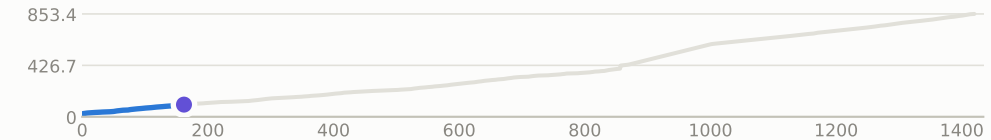
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

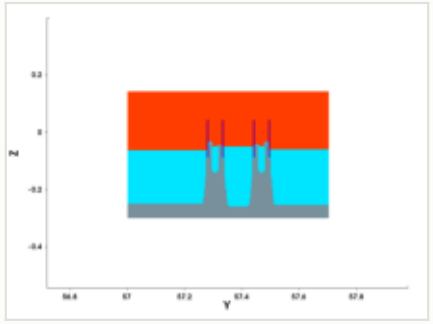
carbon so far, of the whole flow (kgCO2e)



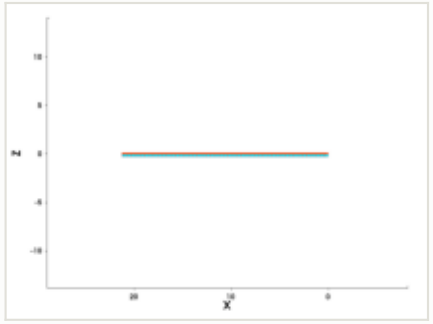
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



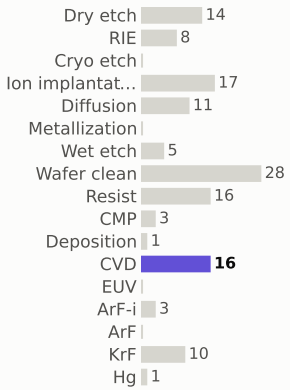
y cut at 57.351 µm



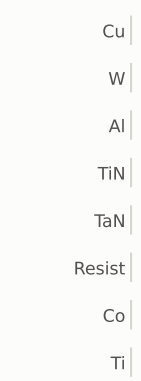
mask for this step

no mask for this step

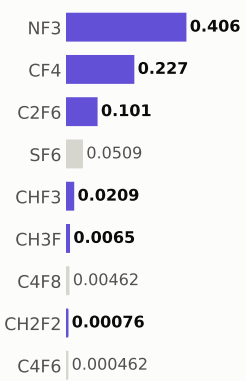
steps by process area



materials deposited (g)



gases emitted (g)



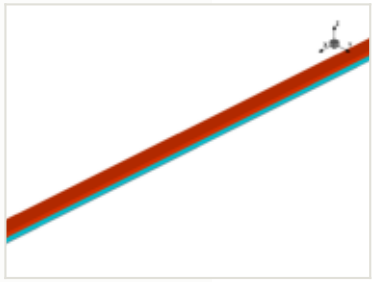
Step #1660: Wet_etch

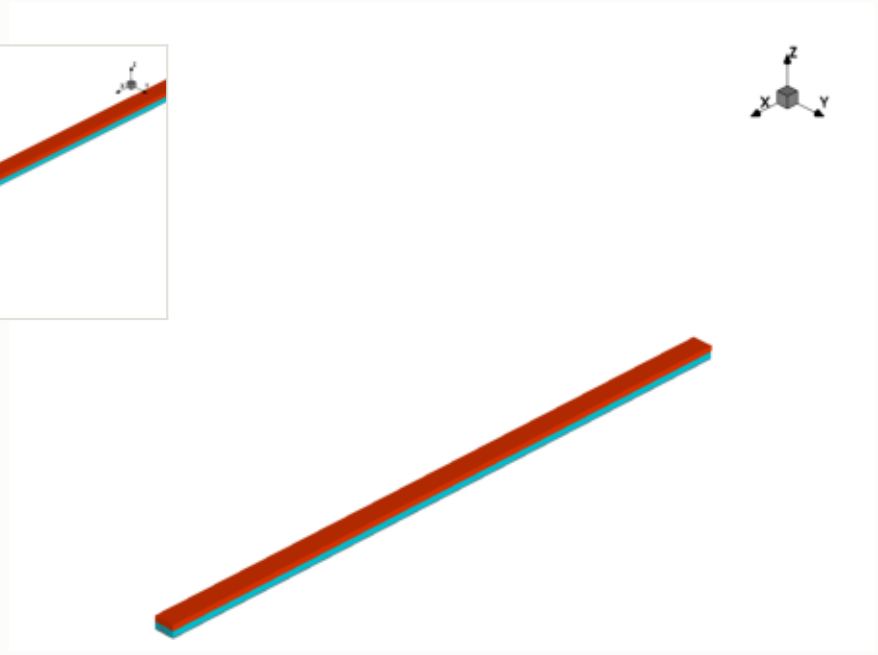
etch_isotropic · Wet_etch

0.169	101.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



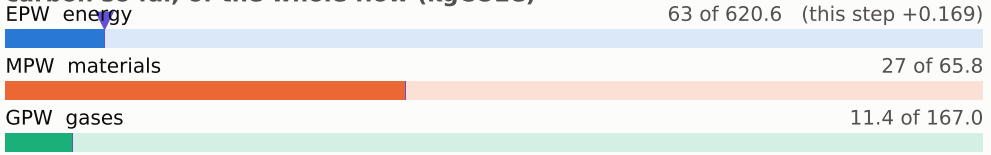


this step

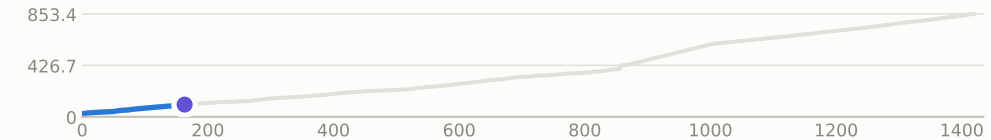
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

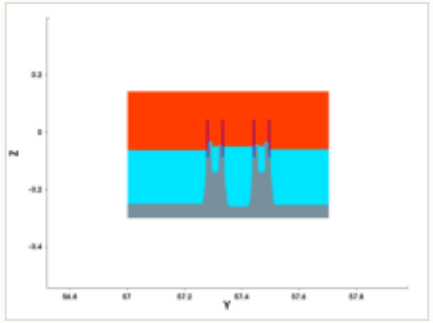
carbon so far, of the whole flow (kgCO2e)



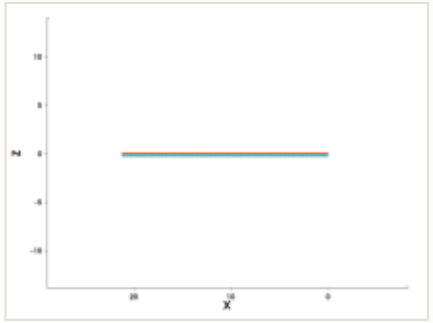
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



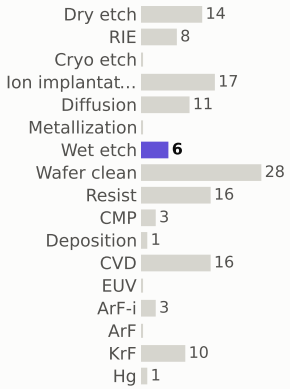
y cut at 57.351 µm



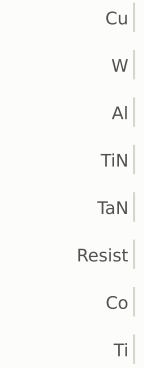
mask for this step

no mask for this step

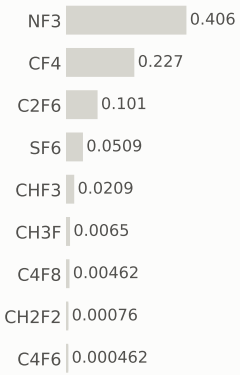
steps by process area



materials deposited (g)



gases emitted (g)



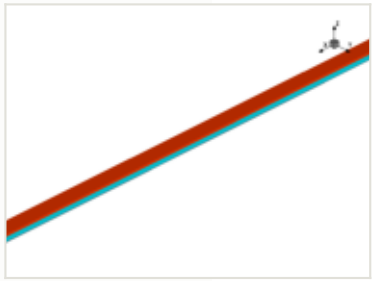
Step #1670: CMP

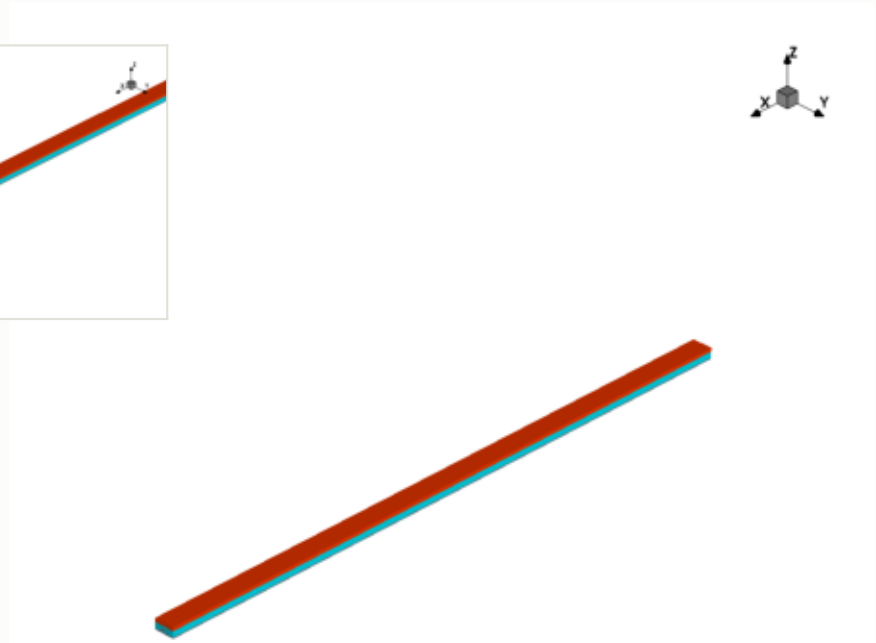
etch_cmp · CMP

0.169	101.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



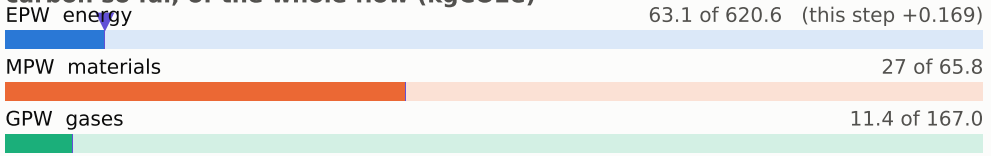


this step

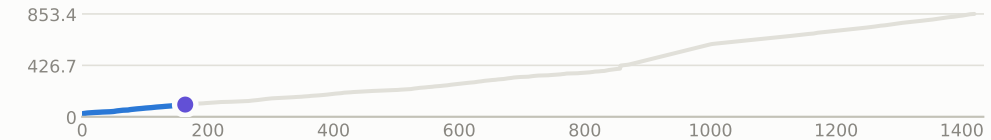
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

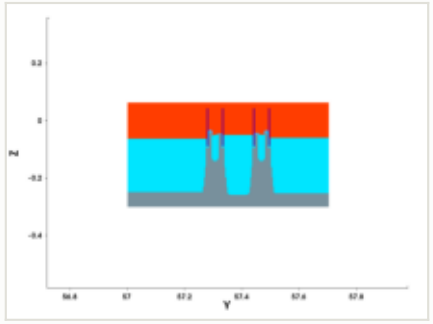
carbon so far, of the whole flow (kgCO2e)



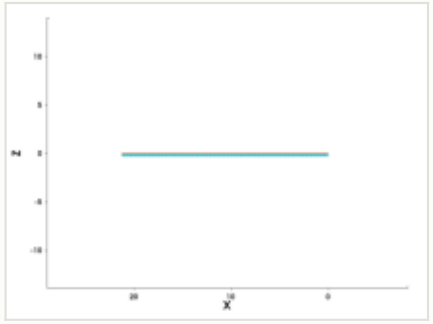
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



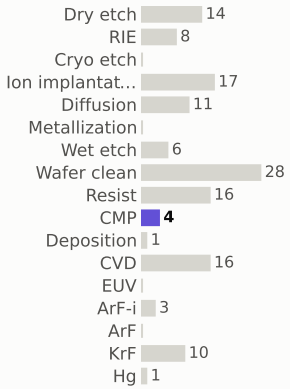
y cut at 57.351 µm



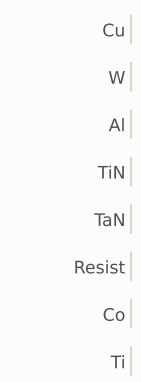
mask for this step

no mask for this step

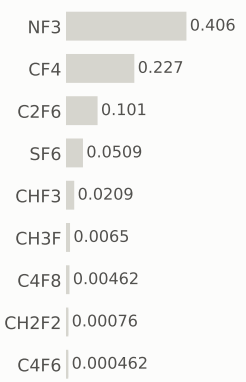
steps by process area



materials deposited (g)



gases emitted (g)



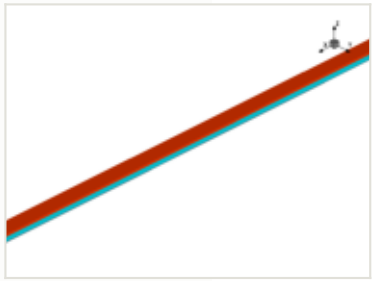
Step #1680: Wafer_clean

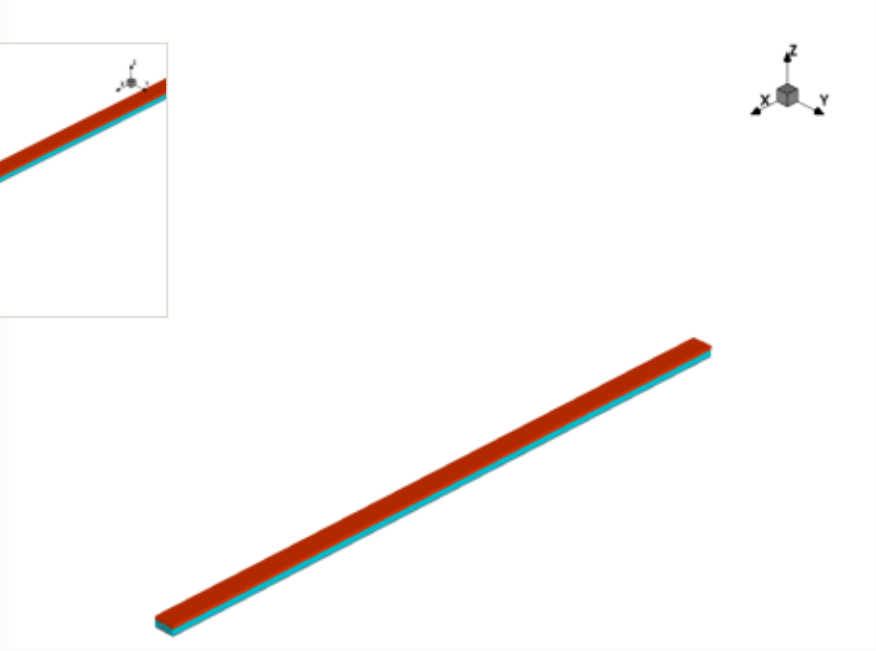
clean · Wafer_clean

0.169	101.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





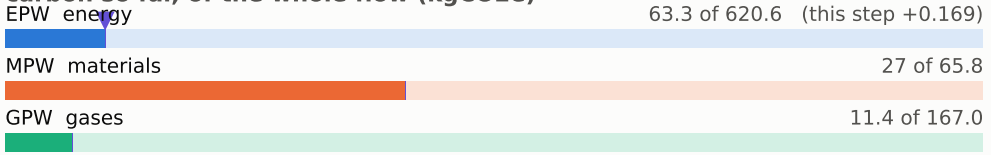
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

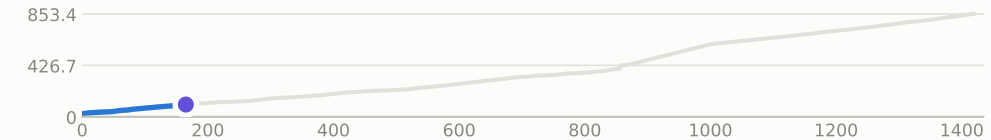
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

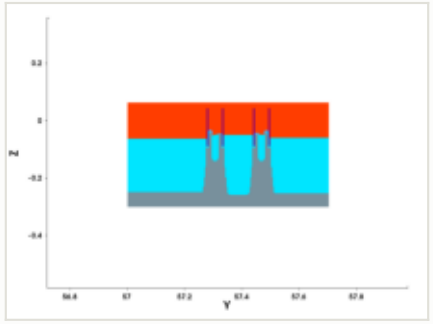
carbon so far, of the whole flow (kgCO2e)



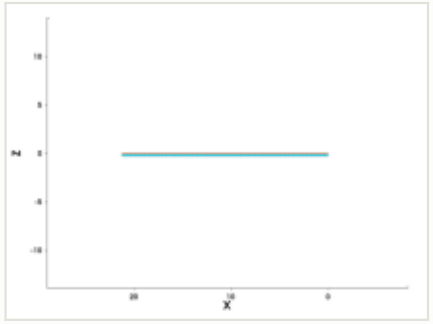
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



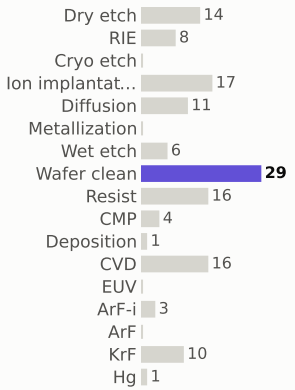
y cut at 57.351 µm



mask for this step

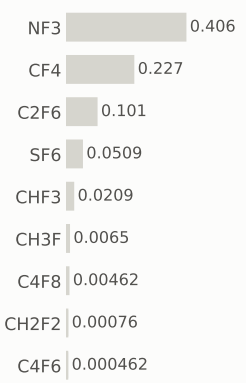
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



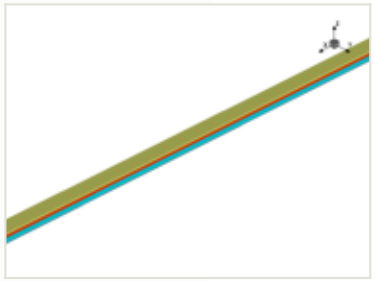
Step #1690: CVD

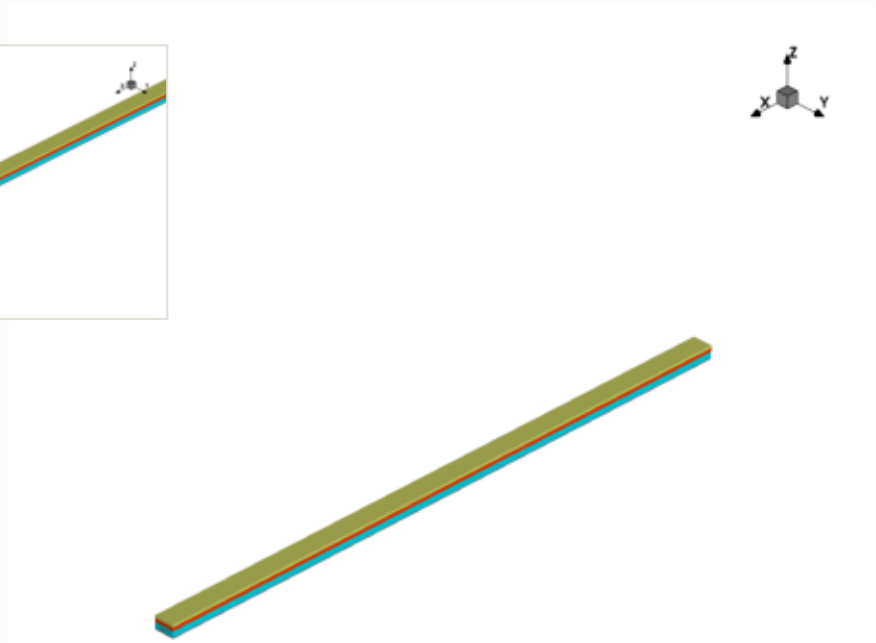
depo_isotropic · CVD

1.31	103.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



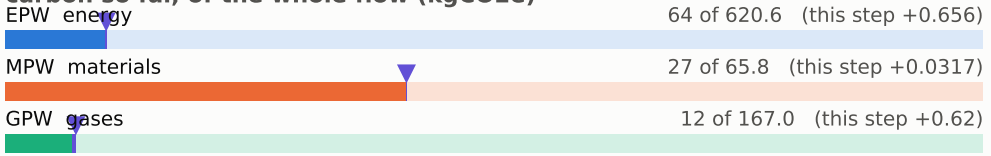


this step

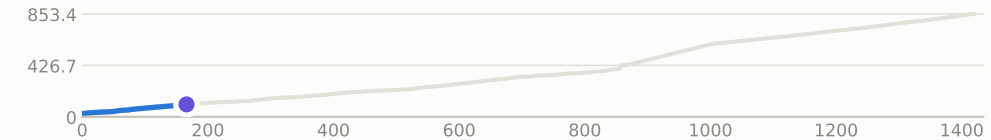
process area	CVD
material	—
thickness	0.087 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

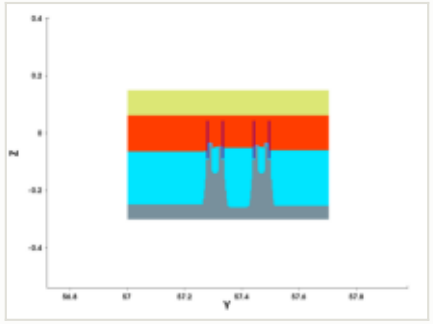
carbon so far, of the whole flow (kgCO2e)



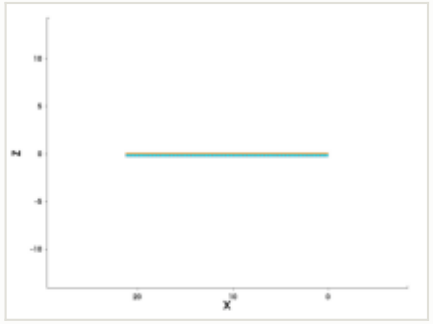
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



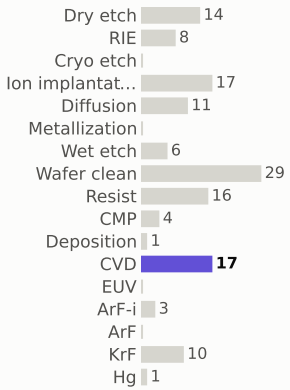
y cut at 57.351 µm



mask for this step

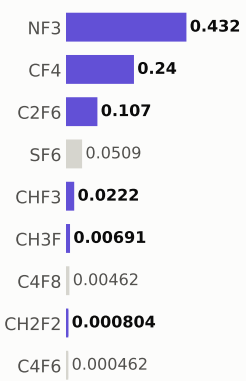
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



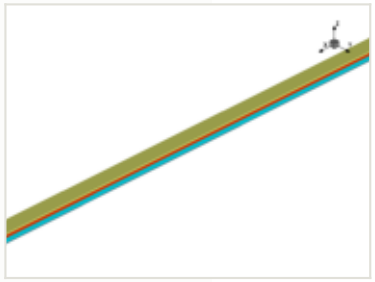
Step #1700: Wafer_clean

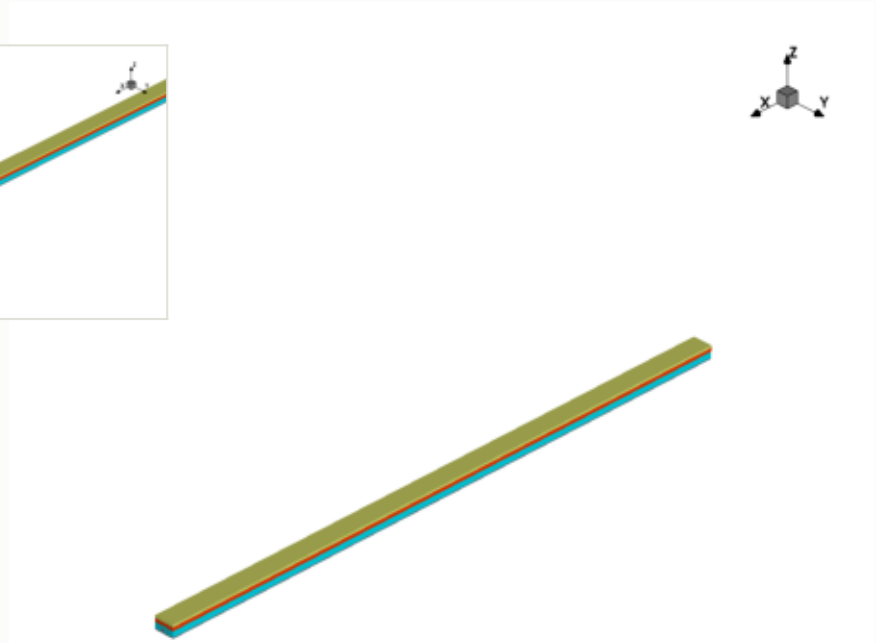
clean · Wafer_clean

0.169	103.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





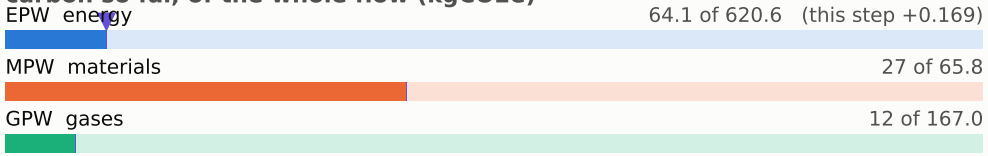
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

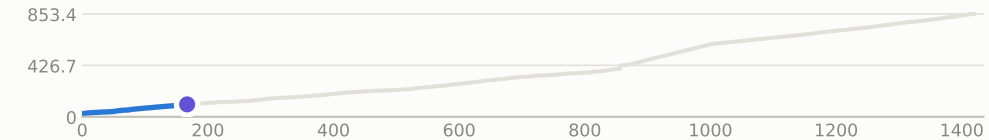
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

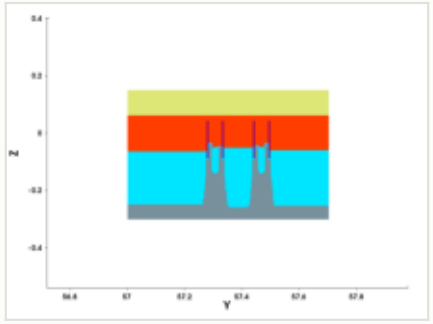
carbon so far, of the whole flow (kgCO2e)



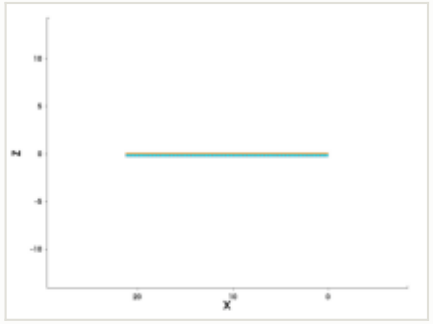
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



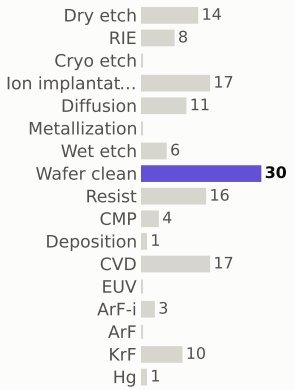
y cut at 57.351 µm



mask for this step

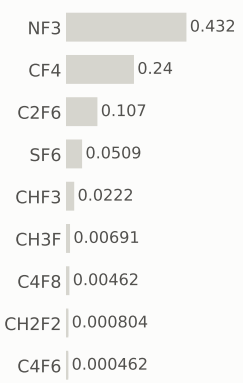
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



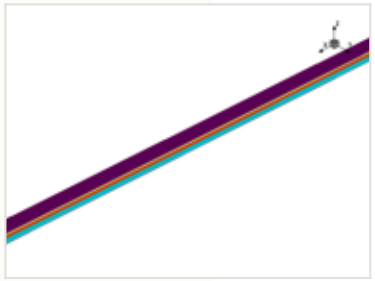
Step #1710: CVD

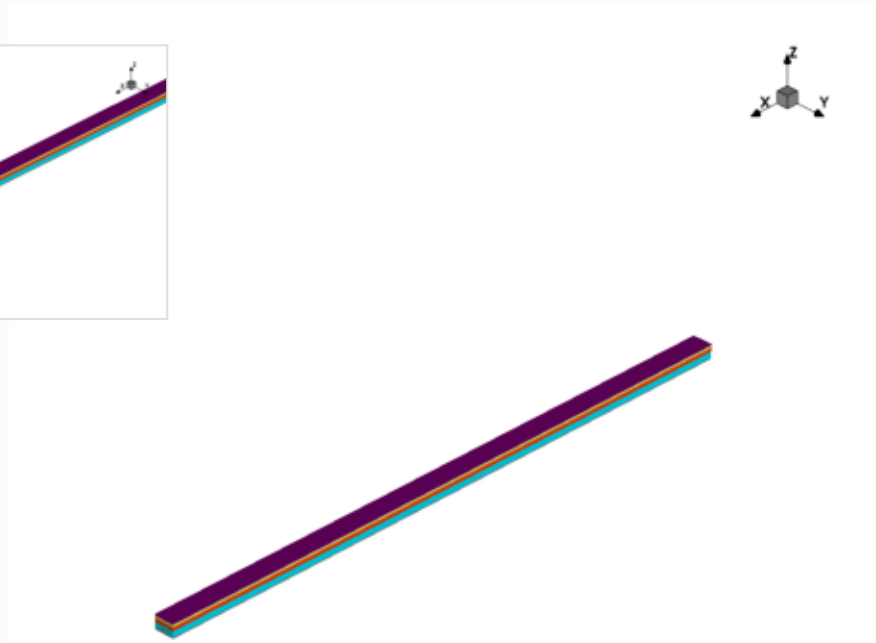
depo_isotropic · CVD

1.31	104.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



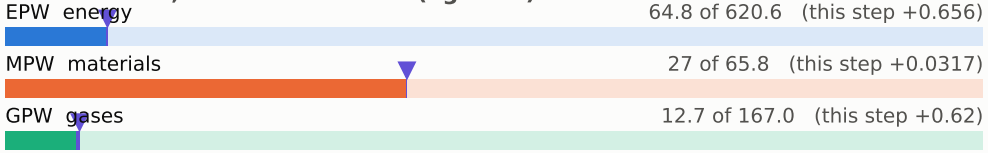


this step

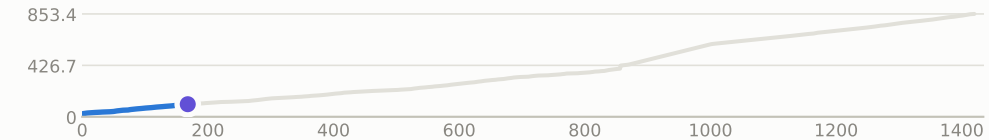
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

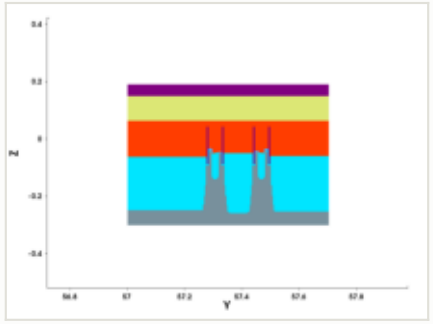
carbon so far, of the whole flow (kgCO2e)



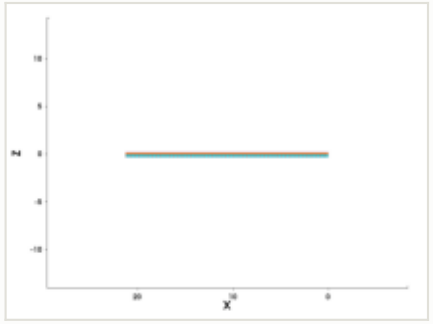
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



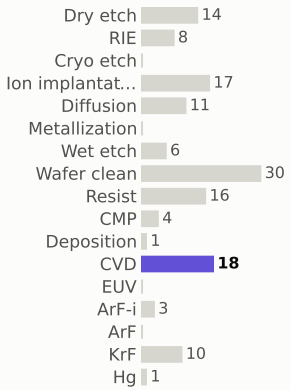
y cut at 57.351 µm



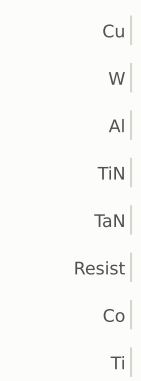
mask for this step

no mask for this step

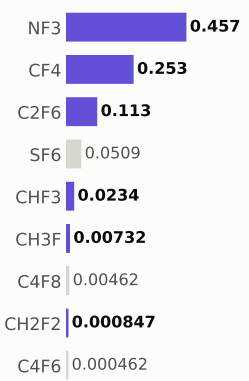
steps by process area



materials deposited (g)



gases emitted (g)



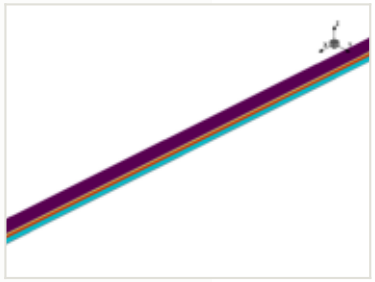
Step #1720: Resist

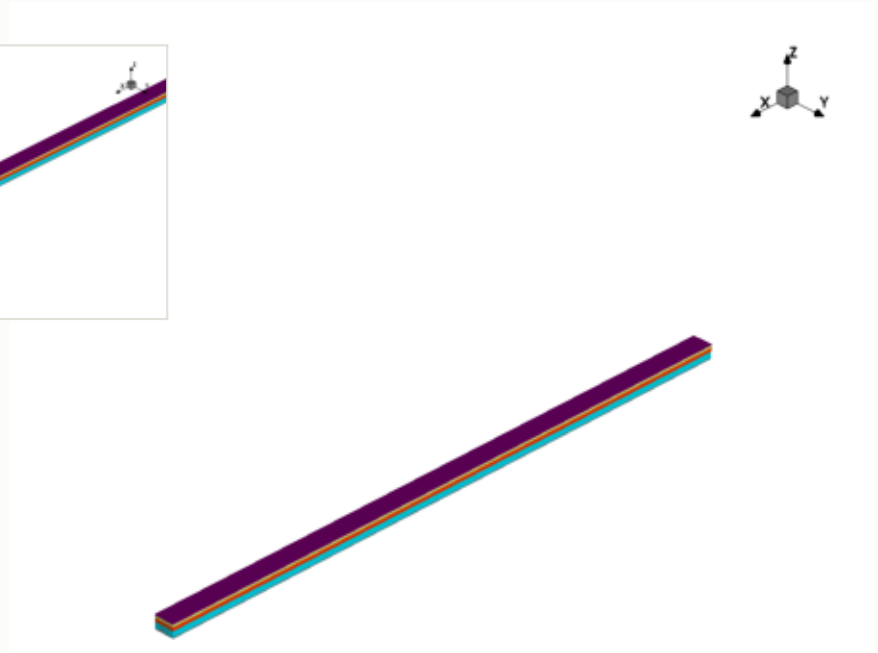
resist · Resist

0.169	104.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





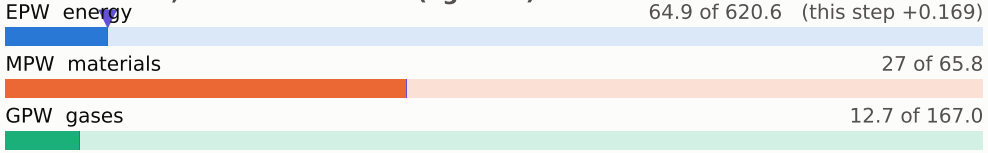
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

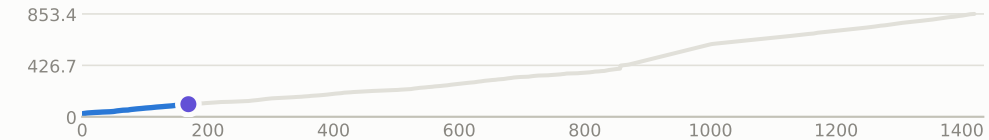
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

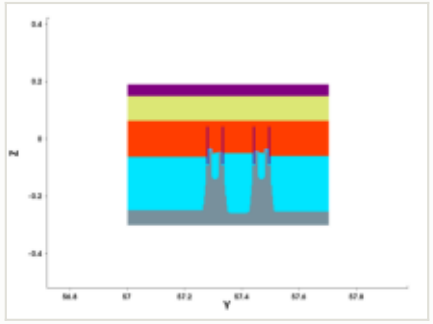
carbon so far, of the whole flow (kgCO2e)



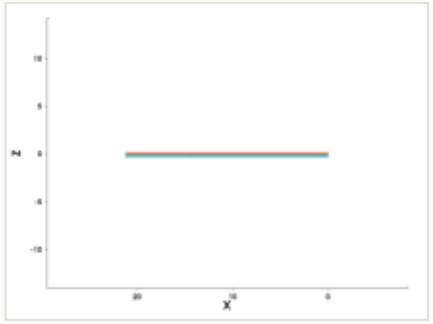
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



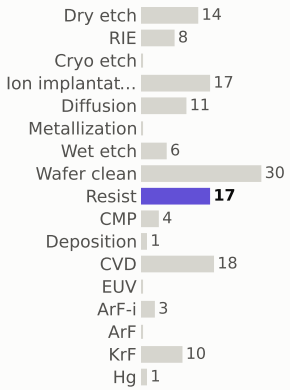
y cut at 57.351 µm



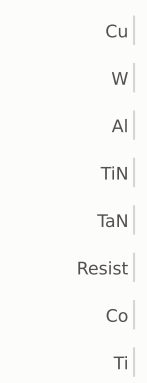
mask for this step

no mask for this step

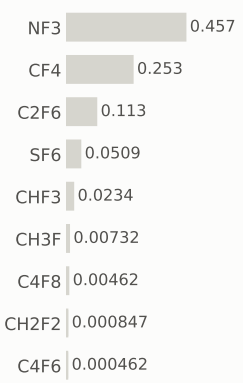
steps by process area



materials deposited (g)



gases emitted (g)



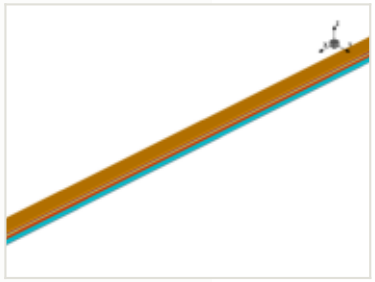
Step #1730: Resist

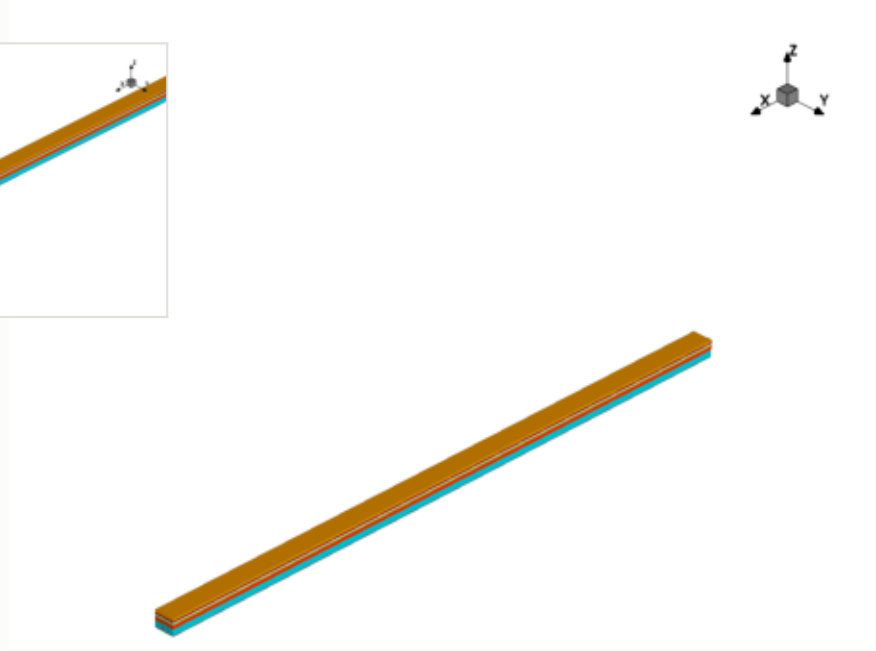
depo_fill · Resist

0.169	104.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

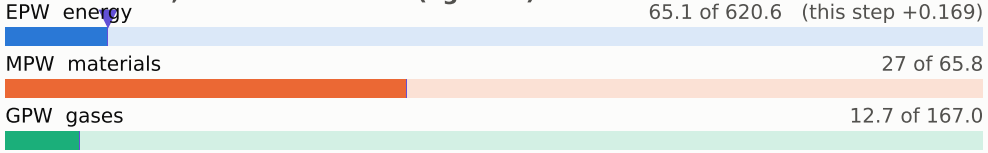




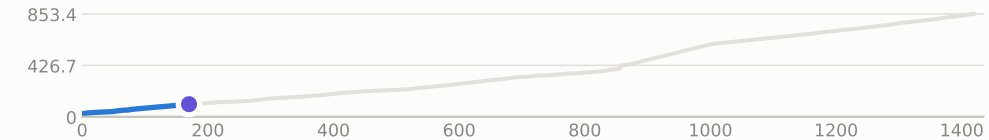
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.08 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

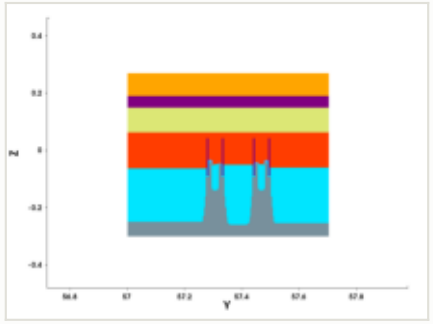
carbon so far, of the whole flow (kgCO2e)



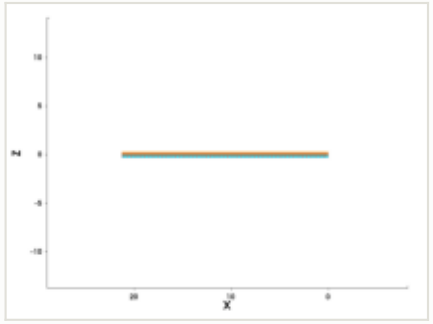
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



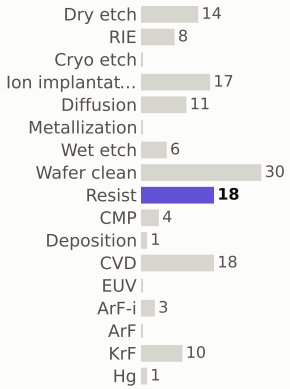
y cut at 57.351 µm



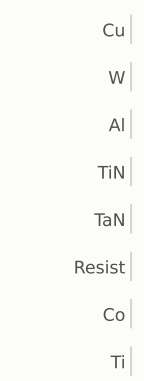
mask for this step

no mask for this step

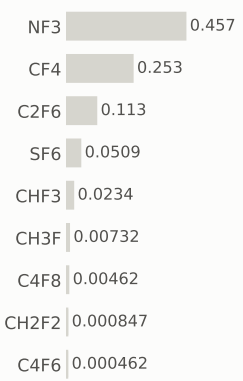
steps by process area



materials deposited (g)



gases emitted (g)



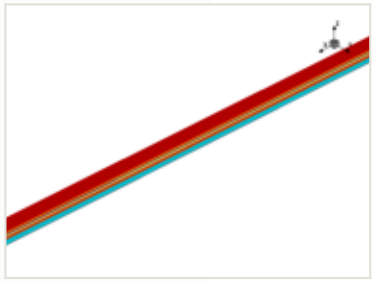
Step #1740: CVD

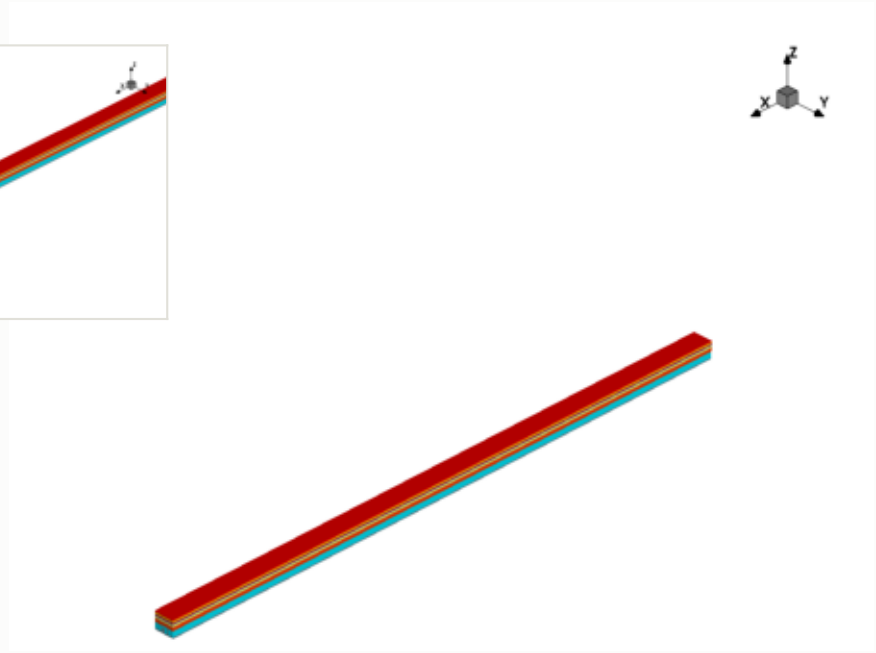
depo_isotropic · CVD

1.31	106.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



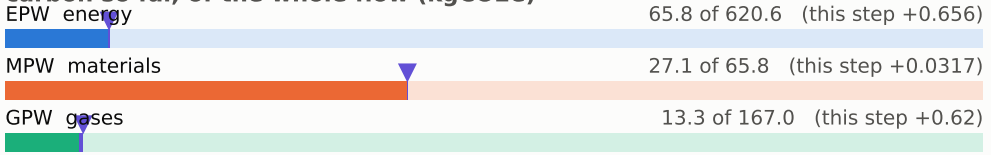


this step

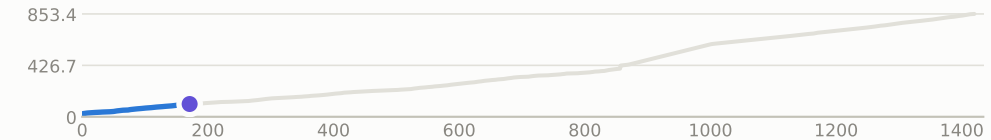
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

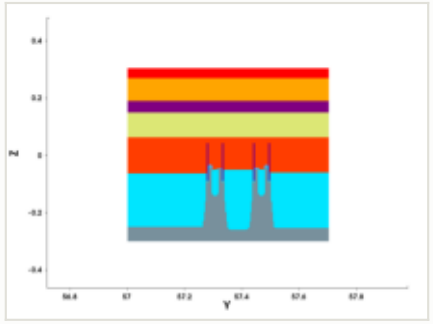
carbon so far, of the whole flow (kgCO2e)



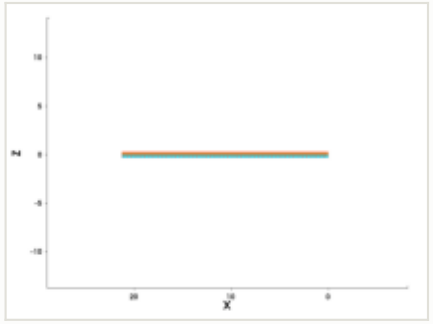
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



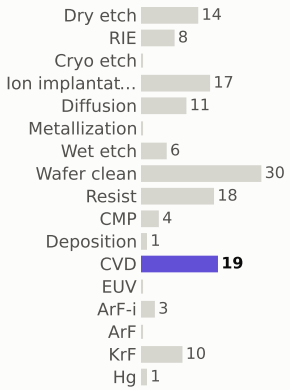
y cut at 57.351 µm



mask for this step

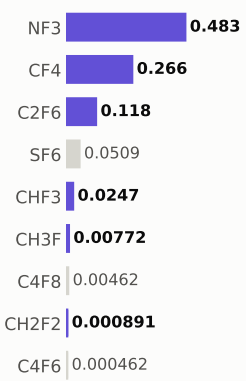
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



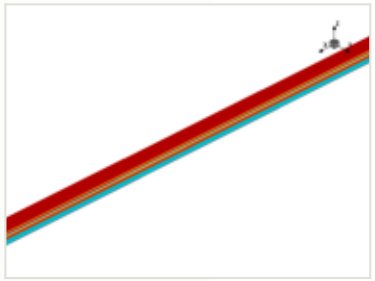
Step #1750: Wafer_clean

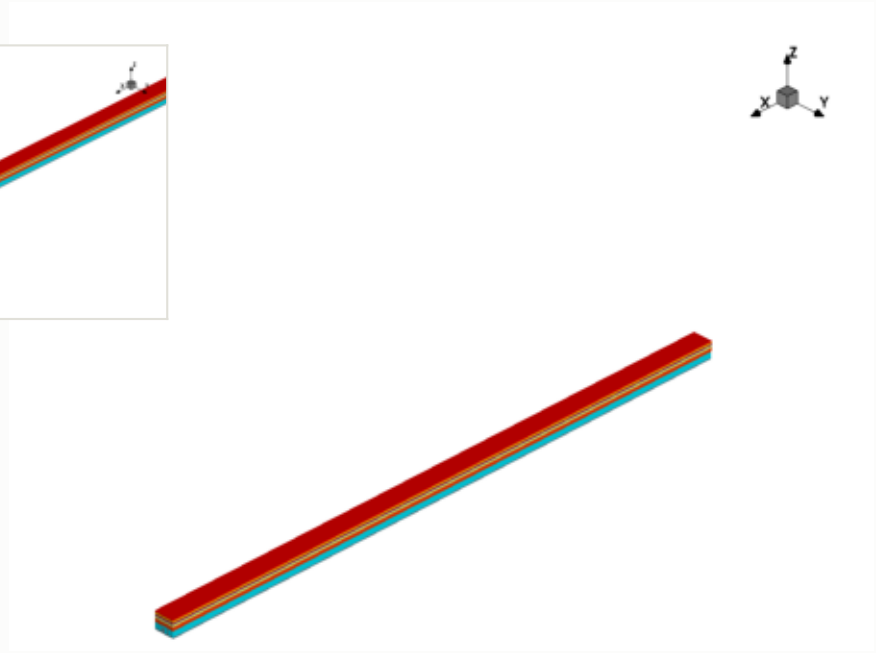
clean · Wafer_clean

0.169	106.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

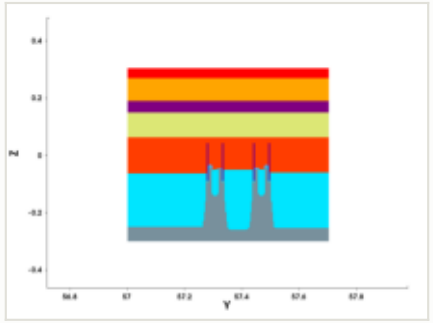
3D structure after this step

zoomed to fit

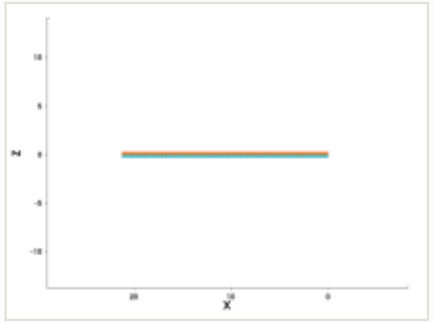




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

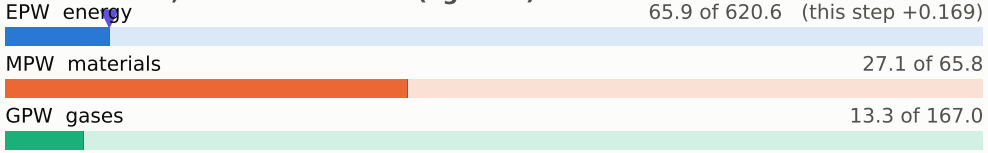
no mask for this step

this step

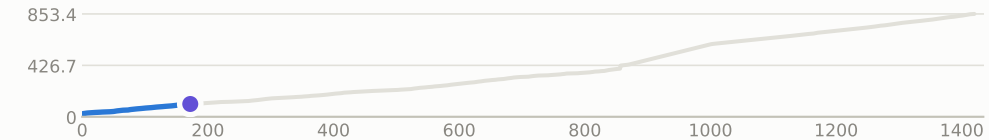
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

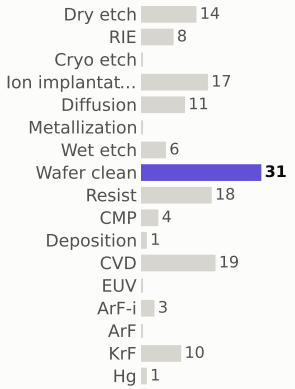
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

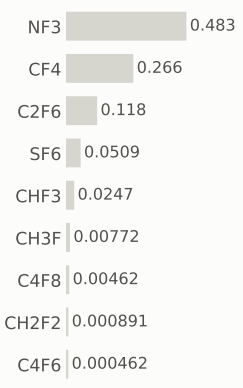


steps by process area



materials deposited (g)

gases emitted (g)



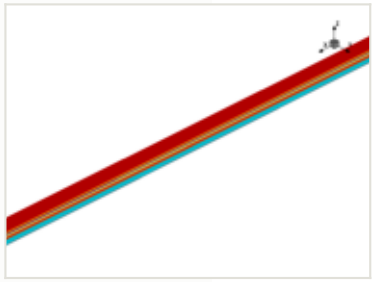
Step #1760: Resist

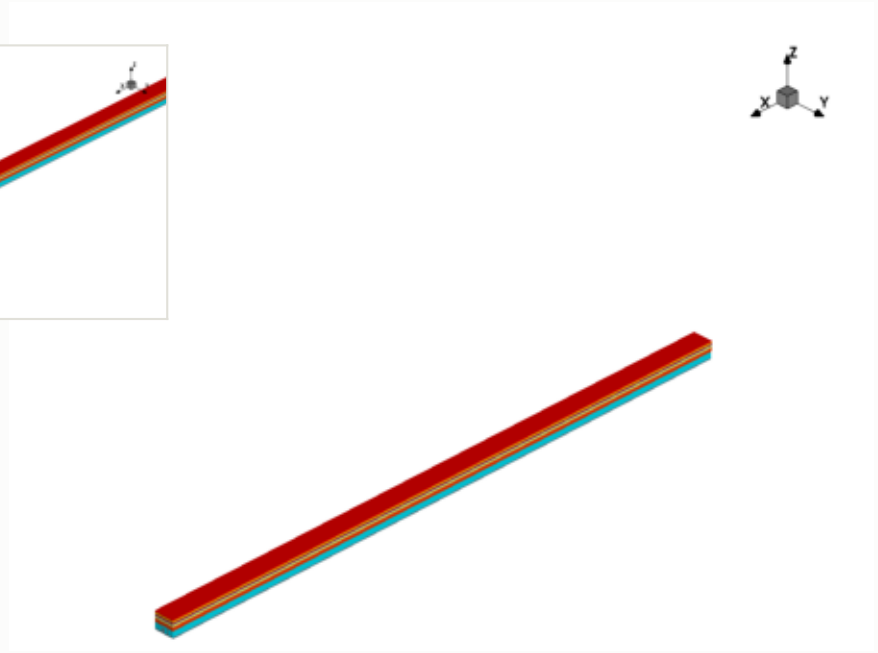
resist · Resist

0.169	106.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



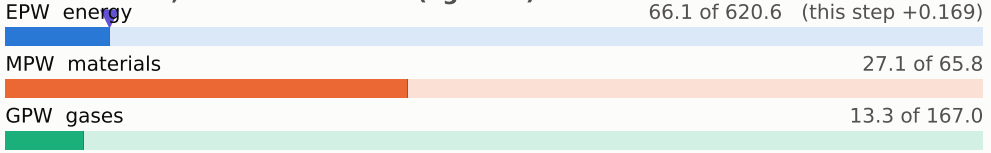


this step

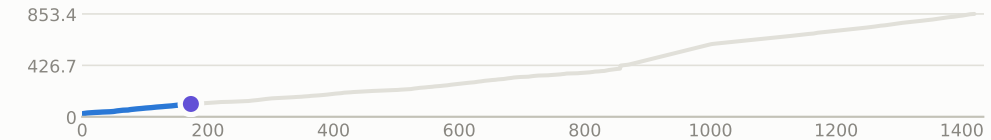
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

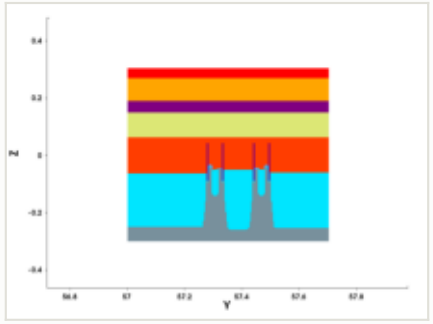
carbon so far, of the whole flow (kgCO2e)



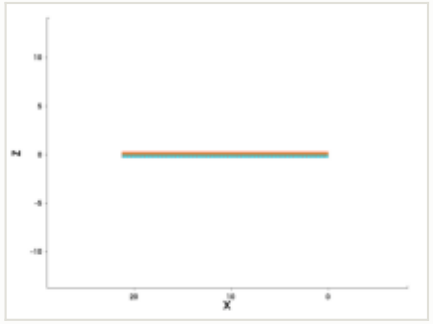
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



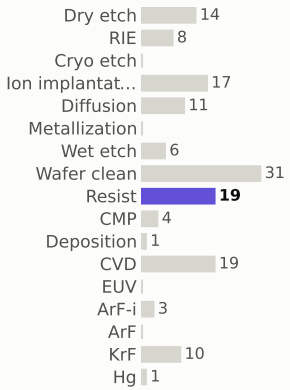
y cut at 57.351 µm



mask for this step

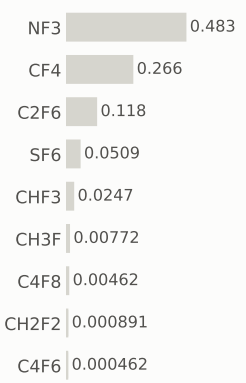
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



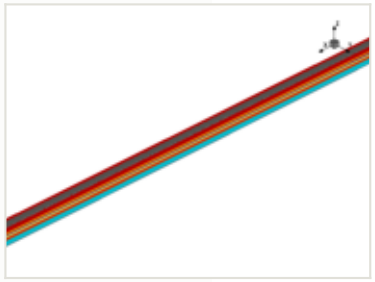
Step #1770: ArF-i

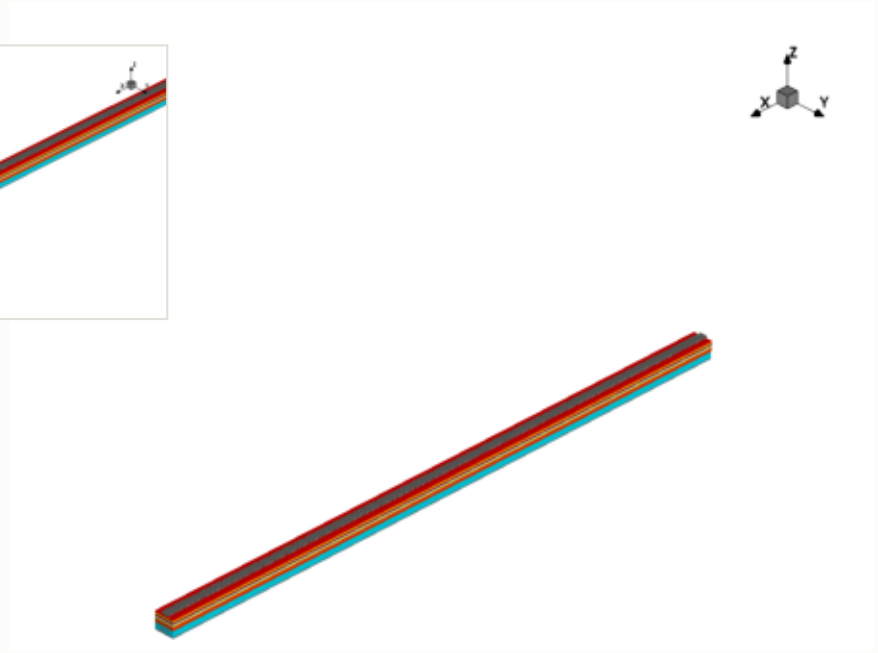
litho_patterning · ArF-i

0.217	106.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

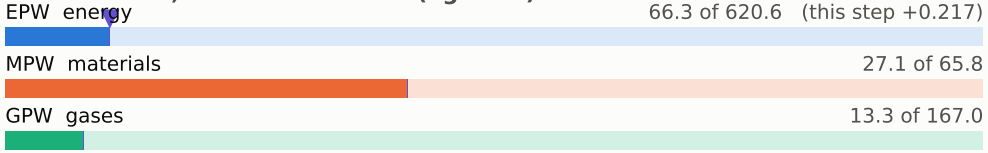




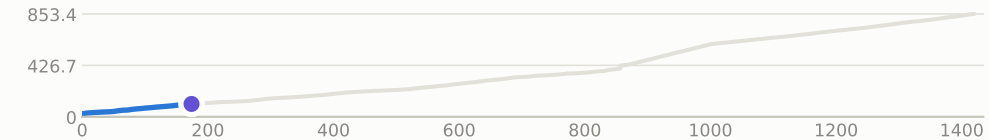
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

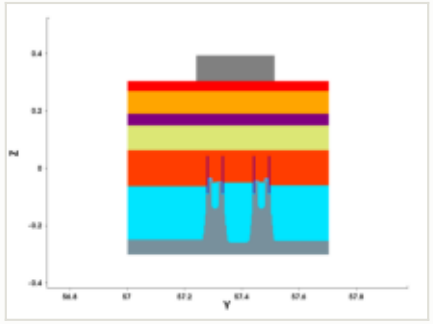
carbon so far, of the whole flow (kgCO2e)



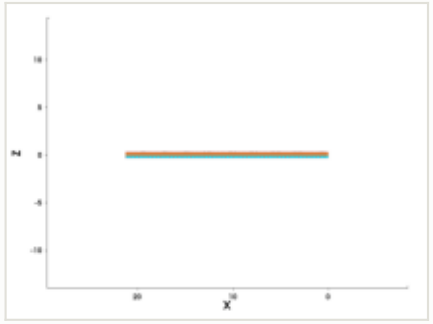
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

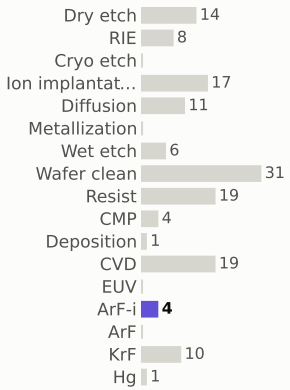


y cut at 57.351 µm



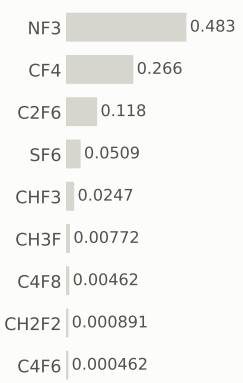
mask: layer 2400:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



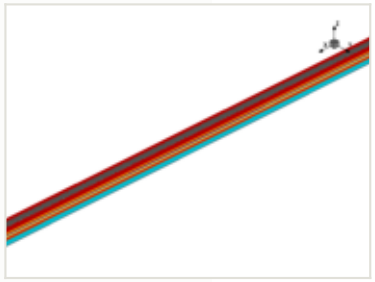
Step #1780: RIE

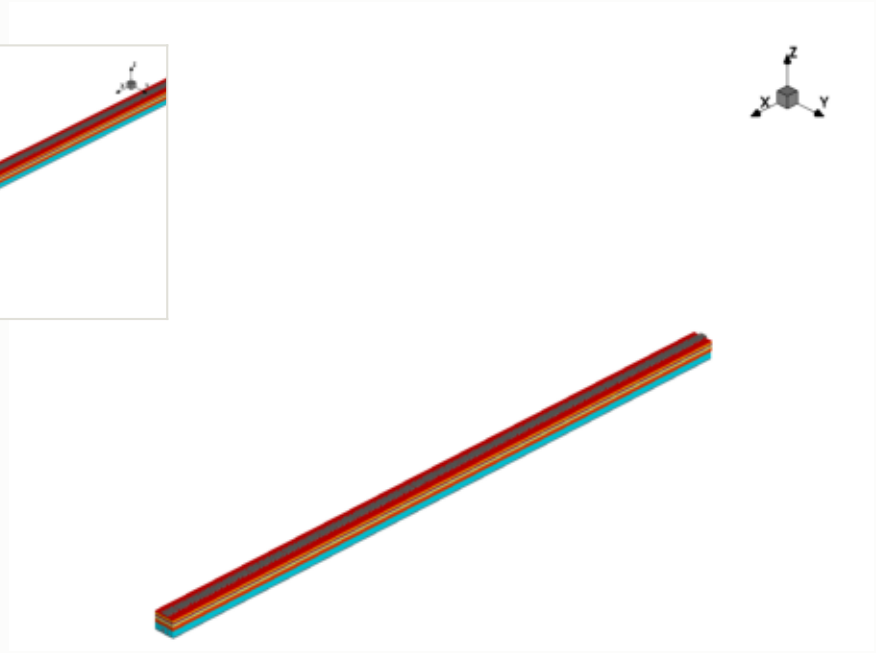
etch_isotropic · RIE

1.04	107.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



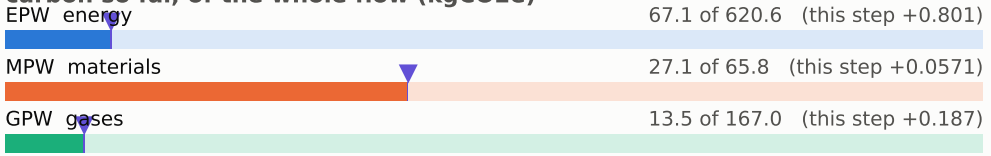


this step

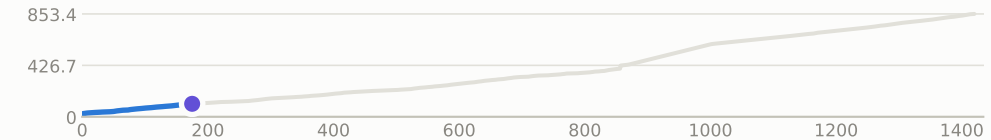
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

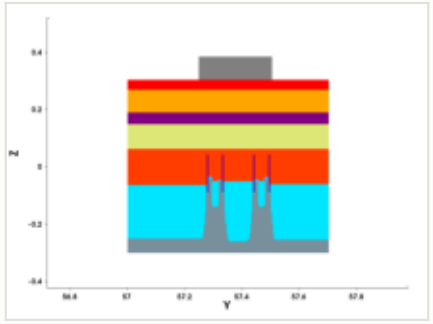
carbon so far, of the whole flow (kgCO2e)



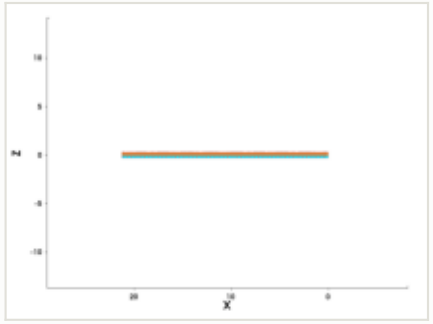
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



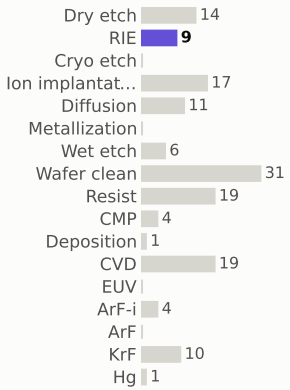
y cut at 57.351 µm



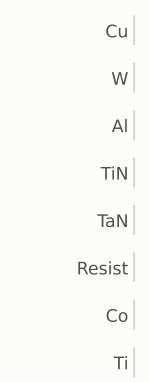
mask for this step

no mask for this step

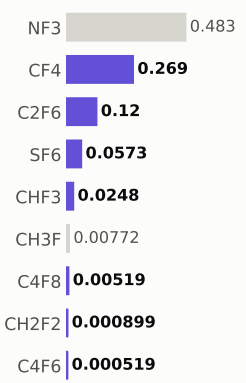
steps by process area



materials deposited (g)



gases emitted (g)



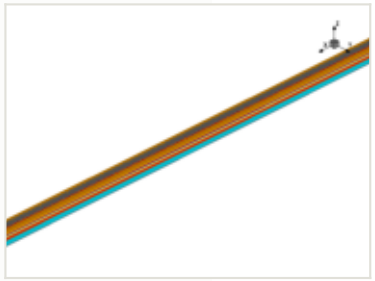
Step #1790: Spx_only

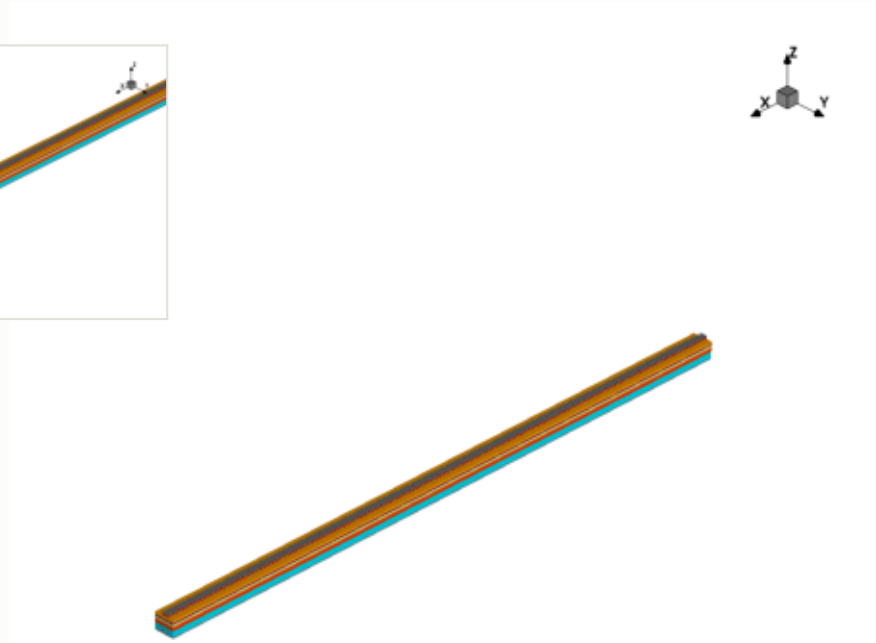
etch_anisotropic · Spx_only

0	107.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





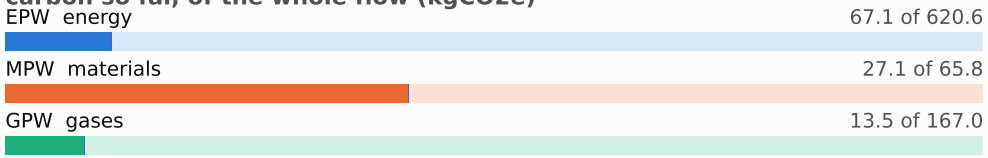
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

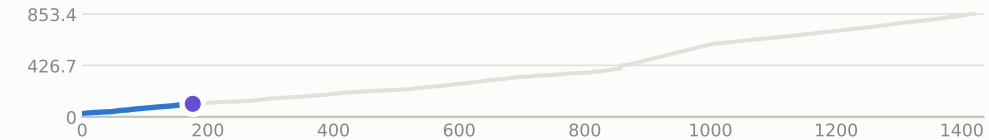
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

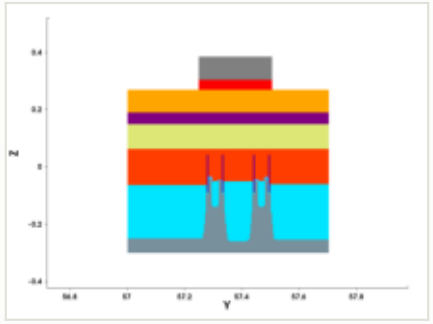
carbon so far, of the whole flow (kgCO2e)



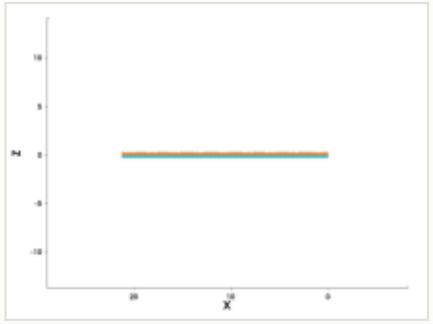
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



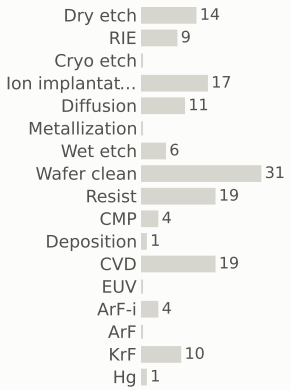
y cut at 57.351 µm



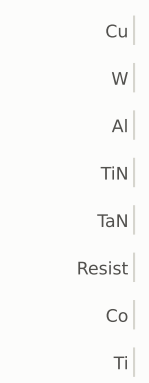
mask for this step

no mask for this step

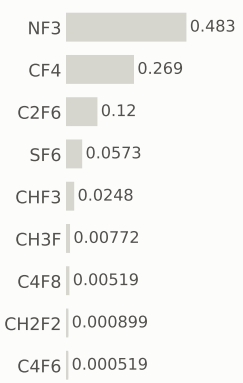
steps by process area



materials deposited (g)



gases emitted (g)



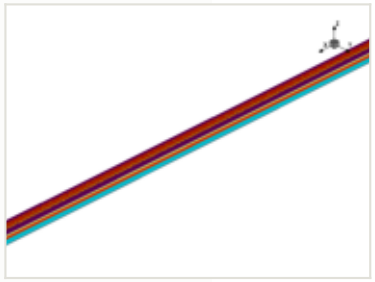
Step #1800: Spx_only

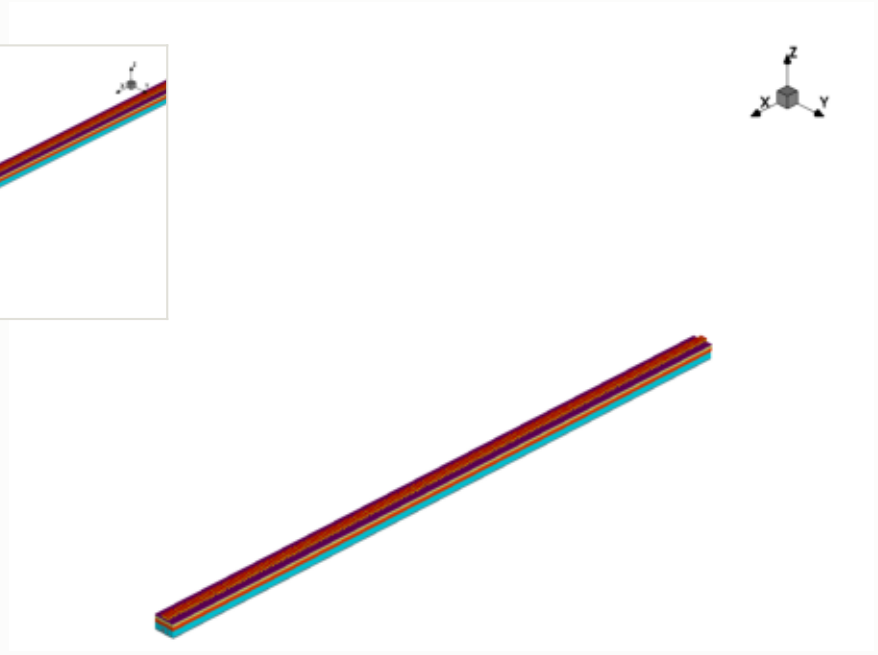
etch_anisotropic · Spx_only

0	107.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

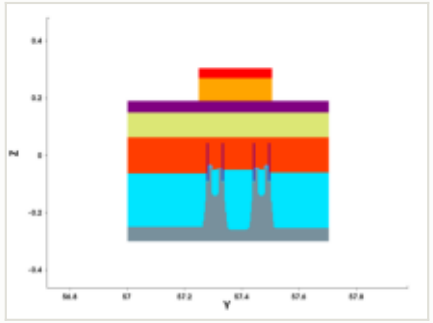
3D structure after this step

zoomed to fit

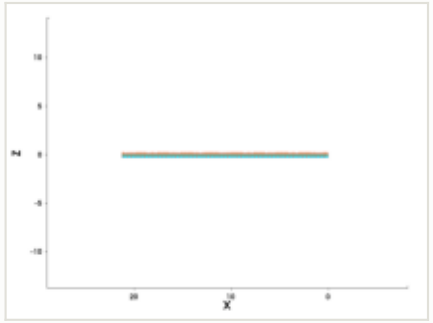




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

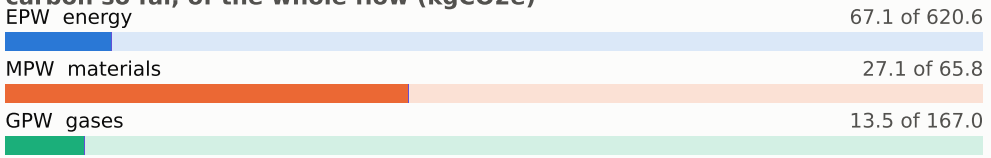
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

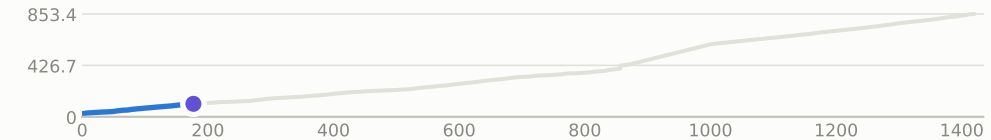
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

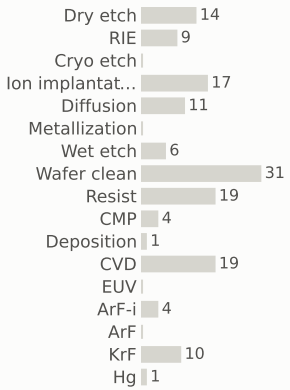
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

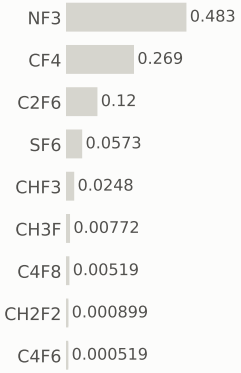


steps by process area



materials deposited (g)

gases emitted (g)



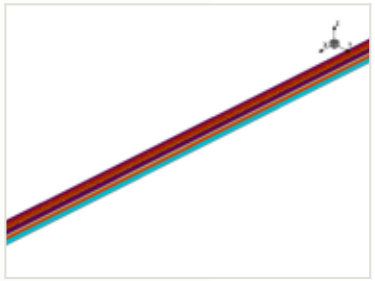
Step #1810: Wafer_clean

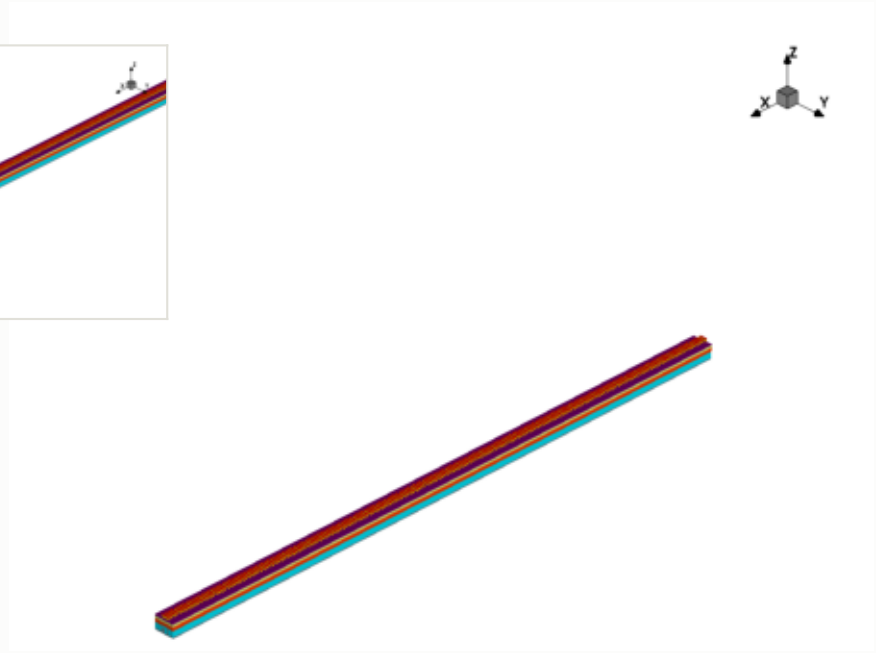
clean · Wafer_clean

0.169	107.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

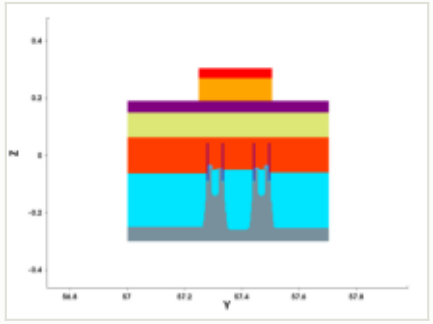
3D structure after this step

zoomed to fit

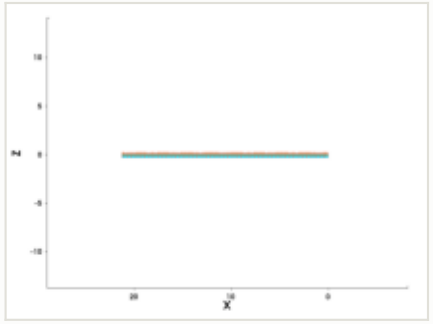




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

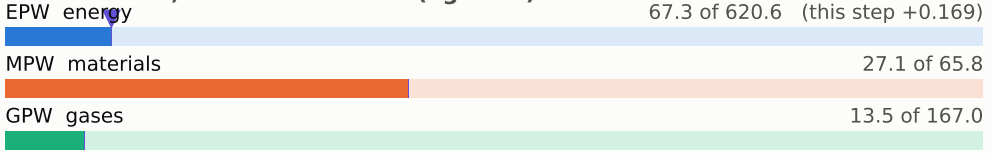
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

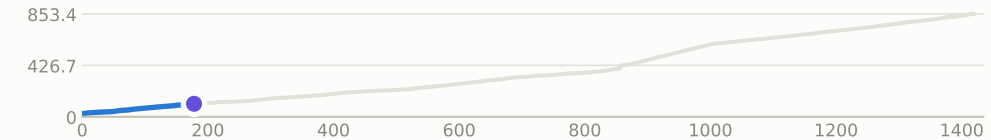
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

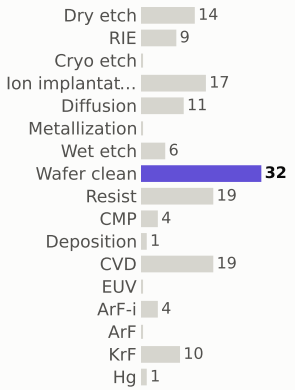
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

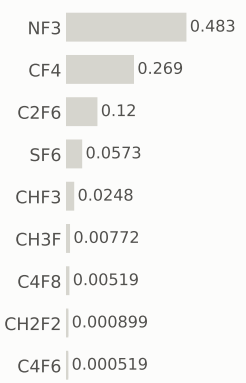


steps by process area



materials deposited (g)

gases emitted (g)



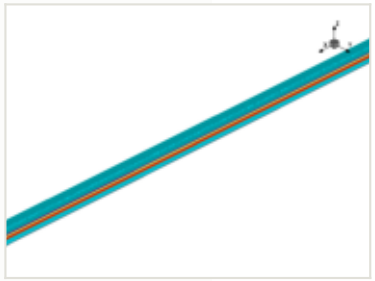
Step #1820: Deposition

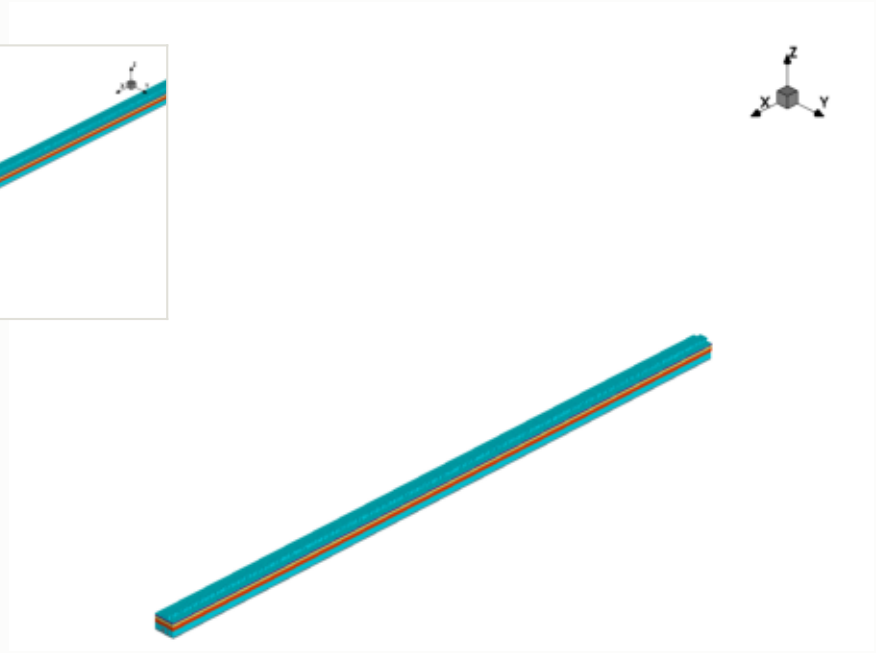
depo_isotropic · Deposition

0.656	108.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



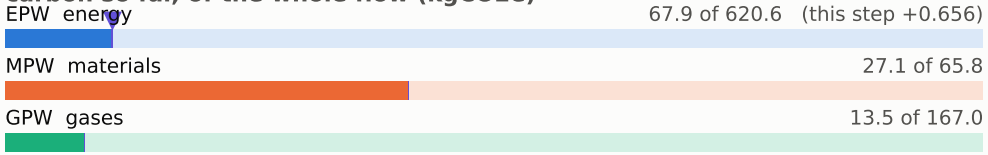


this step

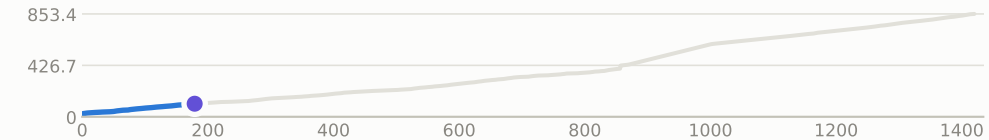
process area	Deposition
material	—
thickness	0.025 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

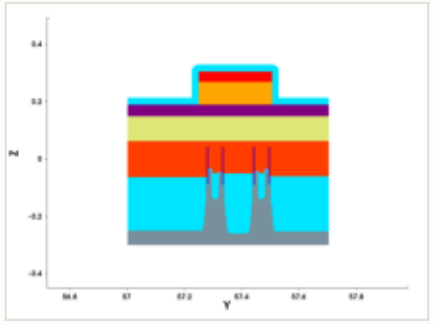
carbon so far, of the whole flow (kgCO2e)



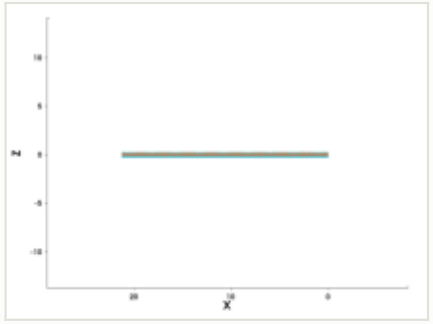
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



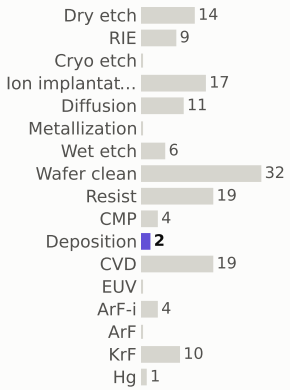
y cut at 57.351 µm



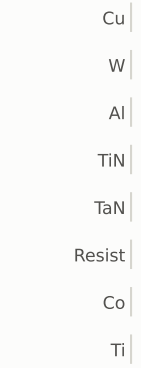
mask for this step

no mask for this step

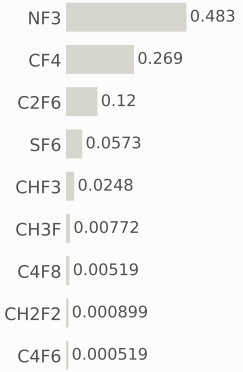
steps by process area



materials deposited (g)



gases emitted (g)



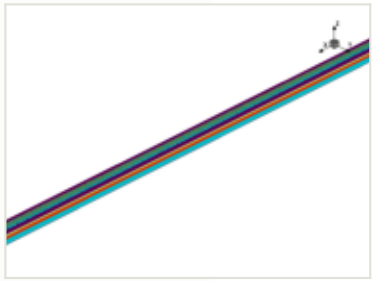
Step #1830: RIE

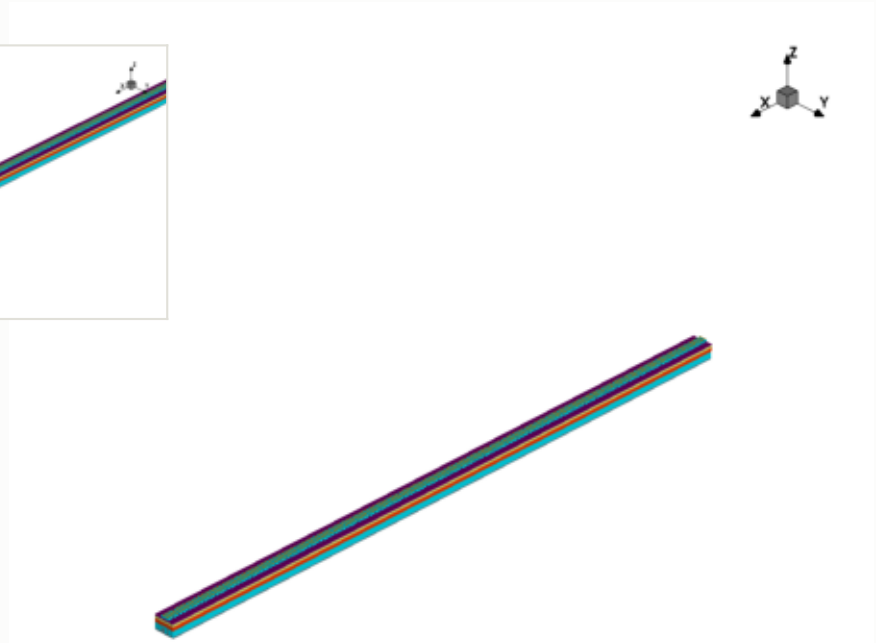
etch_ellipsoidal · RIE

1.04	109.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



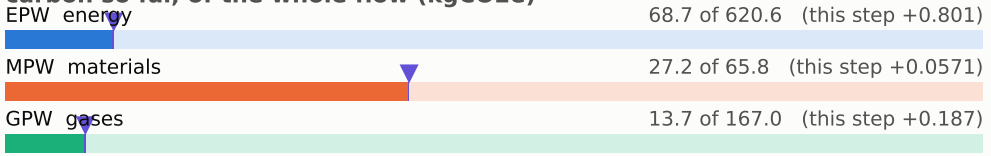


this step

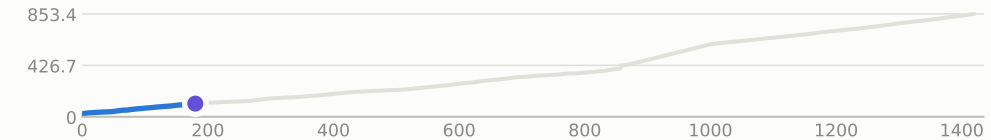
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

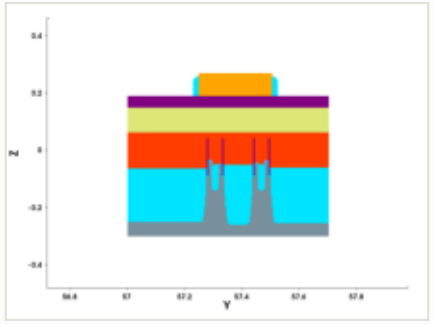
carbon so far, of the whole flow (kgCO2e)



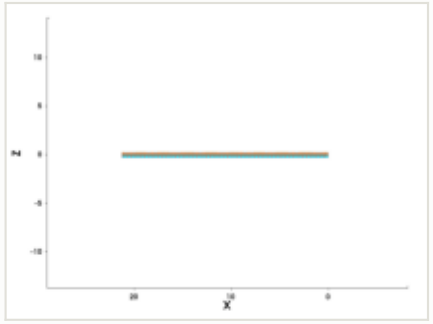
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



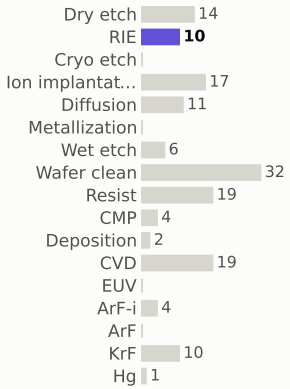
y cut at 57.351 µm



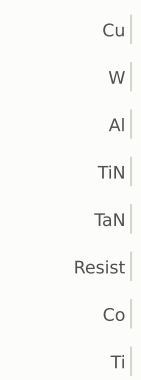
mask for this step

no mask for this step

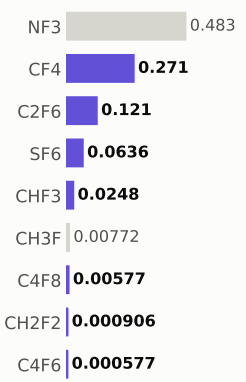
steps by process area



materials deposited (g)



gases emitted (g)



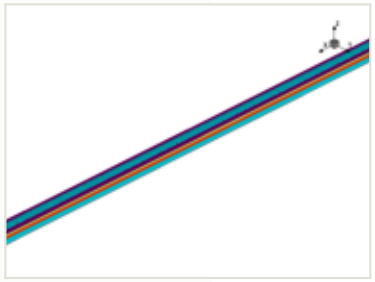
Step #1840: Spx_only

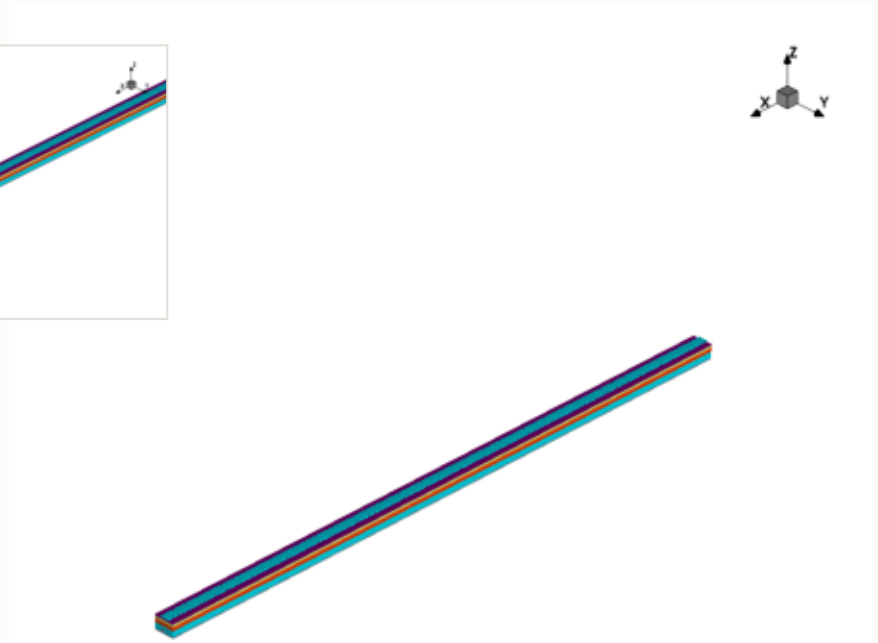
etch_isotropic · Spx_only

0	109.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

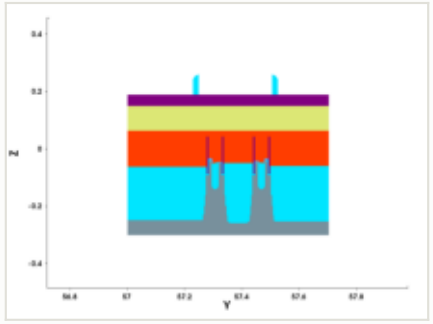
3D structure after this step

zoomed to fit

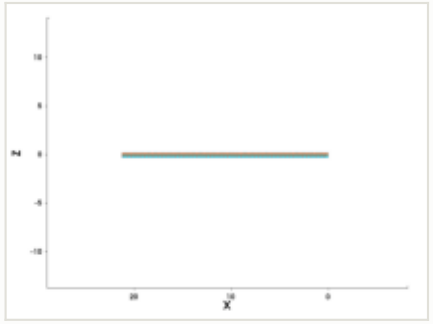




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

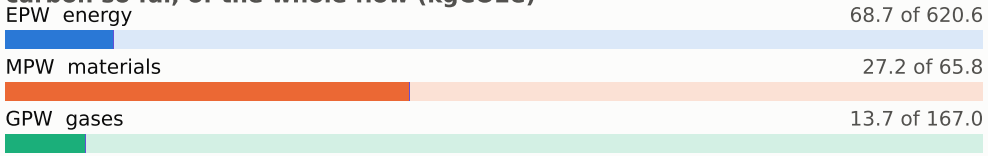
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

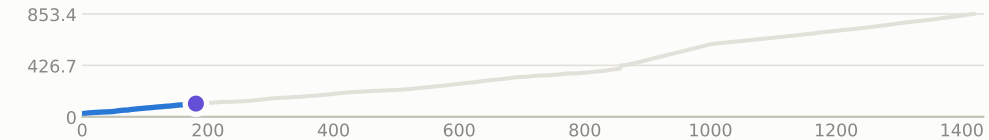
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

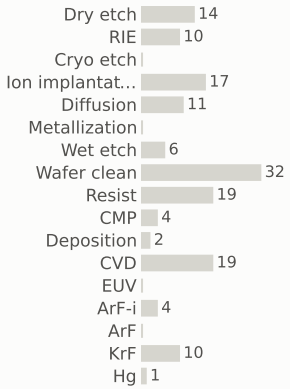
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

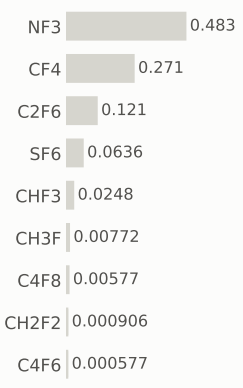


steps by process area



materials deposited (g)

gases emitted (g)



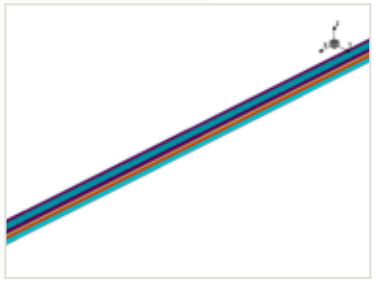
Step #1850: Wafer_clean

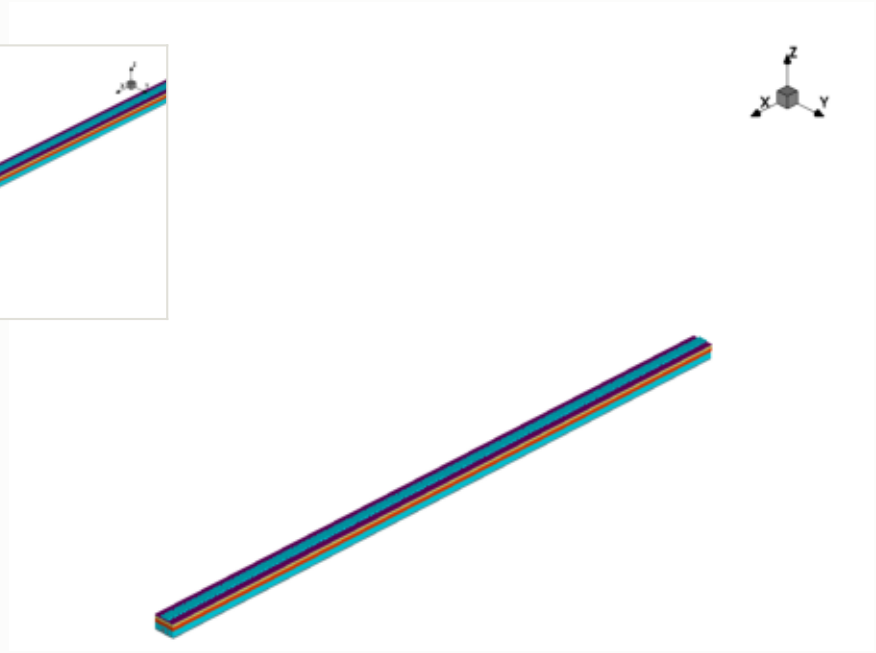
clean · Wafer_clean

0.169	109.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



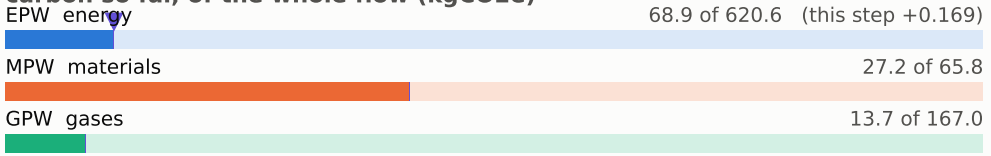


this step

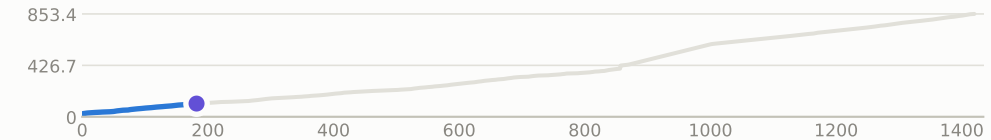
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

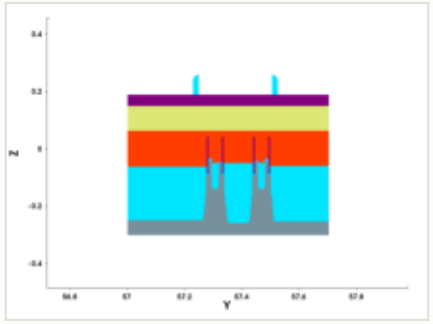
carbon so far, of the whole flow (kgCO2e)



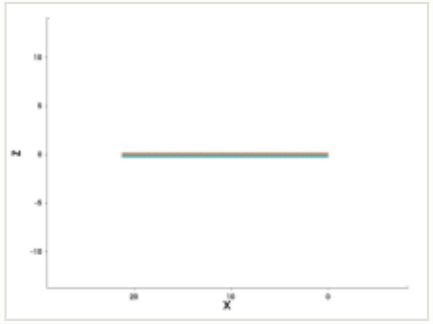
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



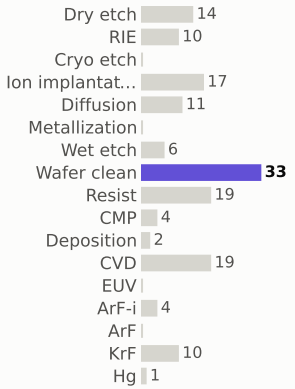
y cut at 57.351 µm



mask for this step

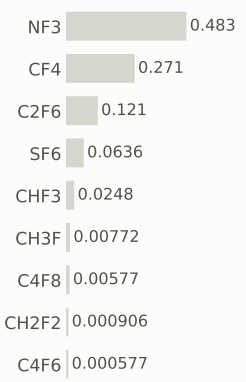
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



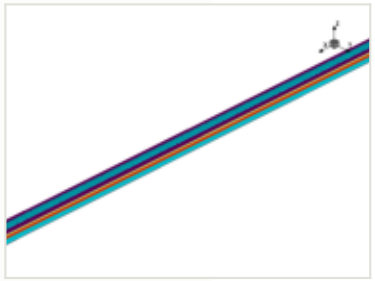
Step #1860: Wafer_clean

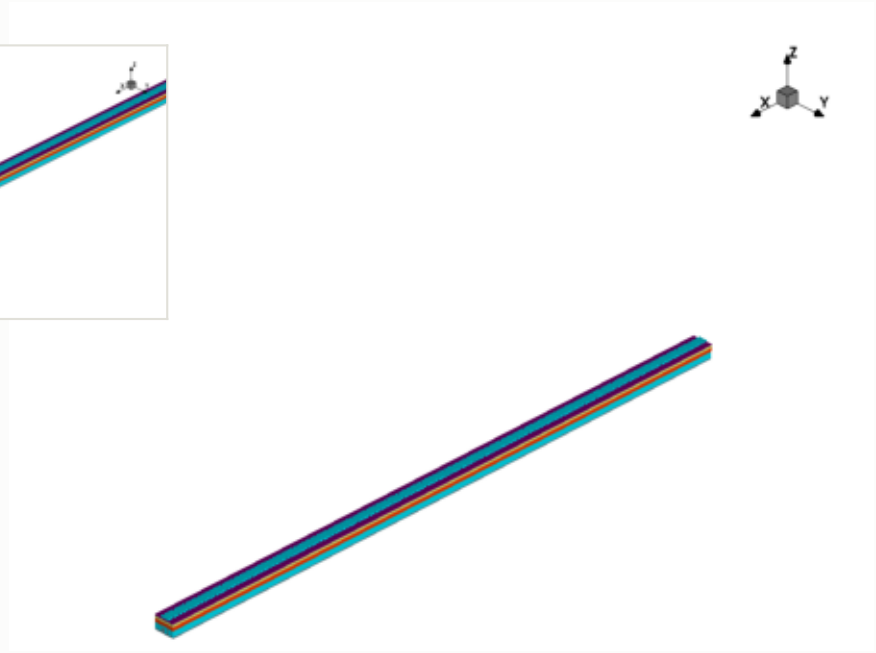
clean · Wafer_clean

0.169	109.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





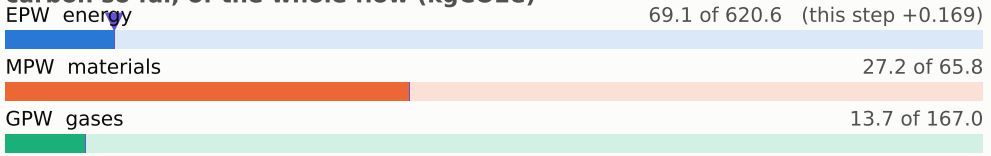
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

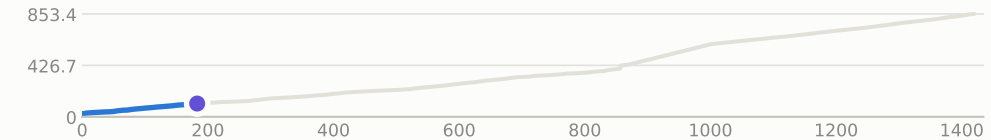
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

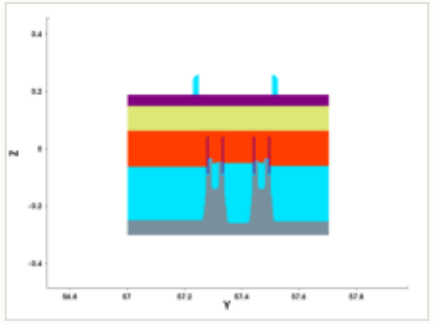
carbon so far, of the whole flow (kgCO2e)



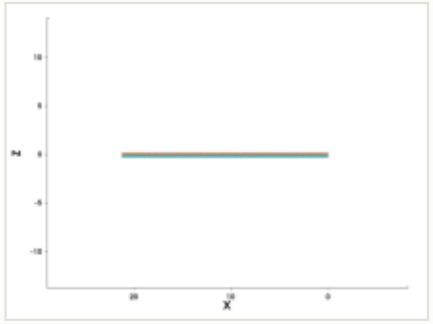
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



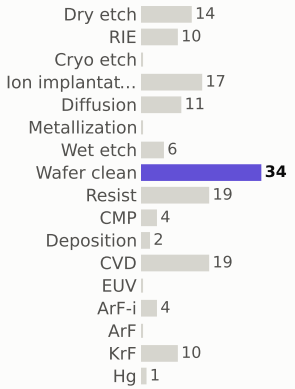
y cut at 57.351 µm



mask for this step

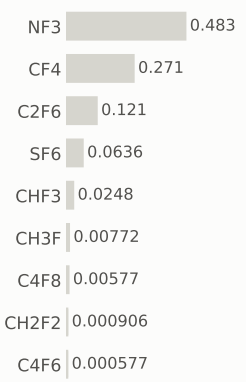
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



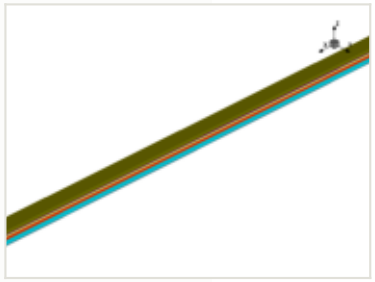
Step #1870: Resist

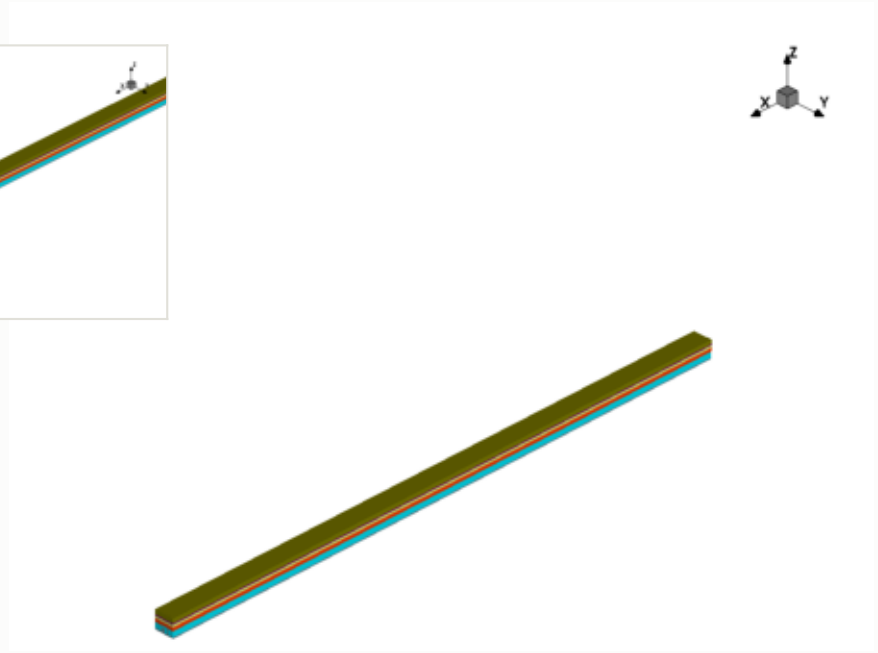
depo_fill · Resist

0.169	110.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



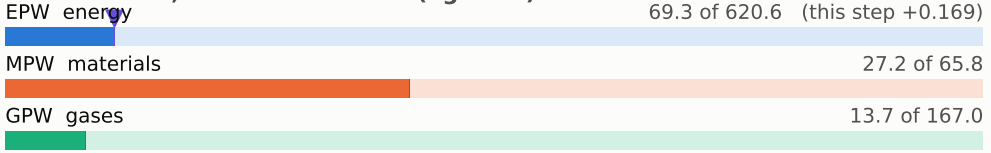


this step

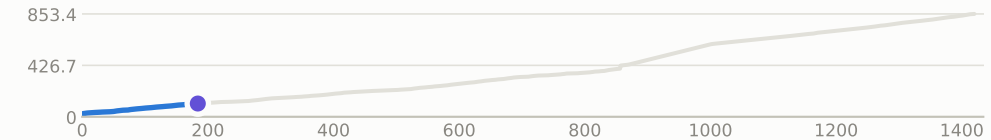
process area	Resist
material	—
thickness	0.08 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

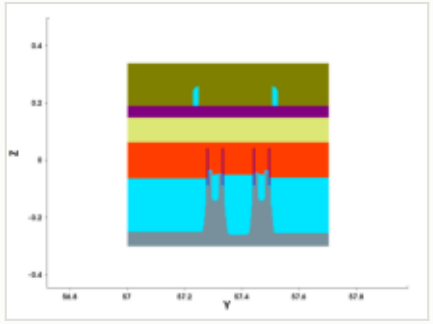
carbon so far, of the whole flow (kgCO2e)



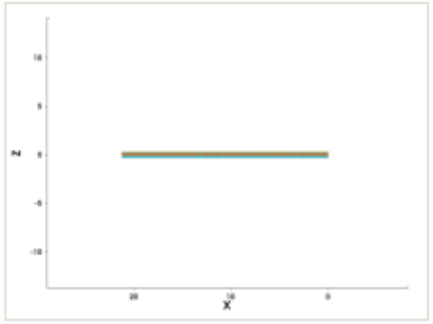
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



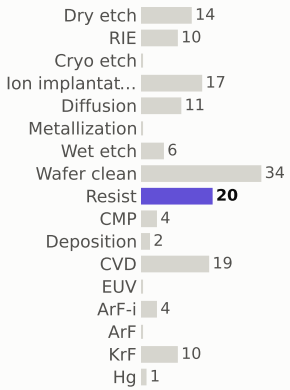
y cut at 57.351 µm



mask for this step

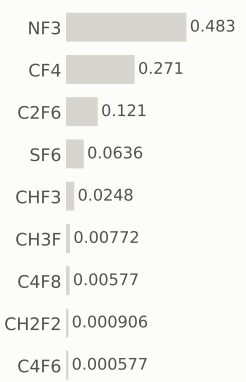
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



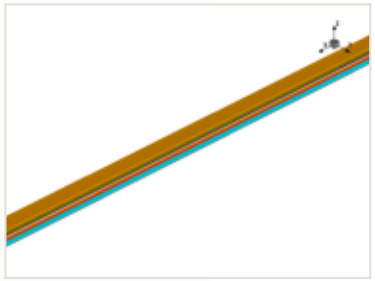
Step #1880: Spx_only

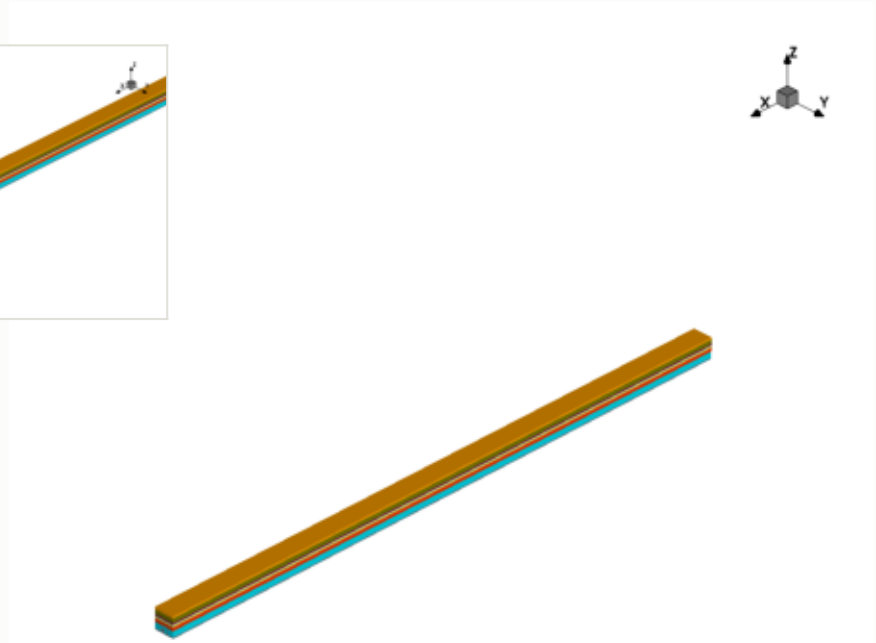
depo_fill · Spx_only

0	110.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





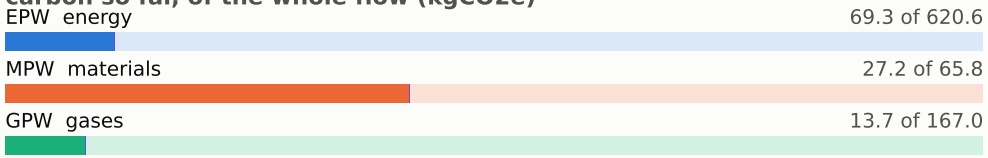
this step

process area	Spx_only
material	—
thickness	0.08 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

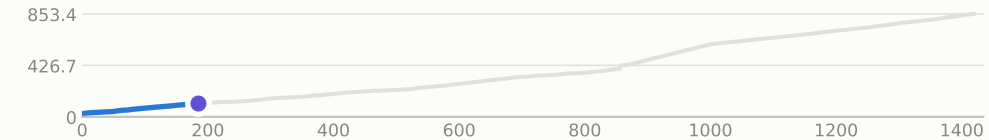
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

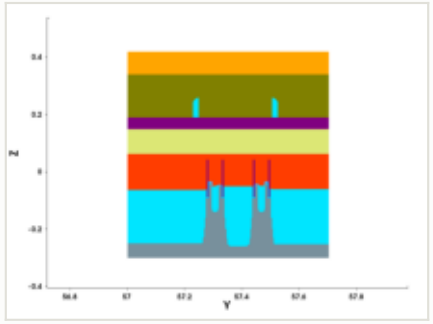
carbon so far, of the whole flow (kgCO2e)



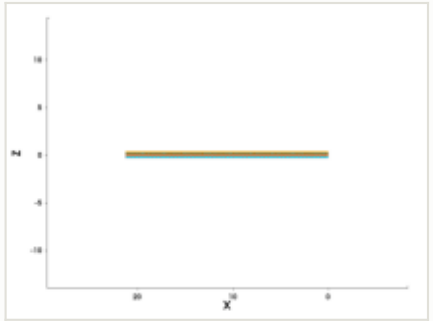
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



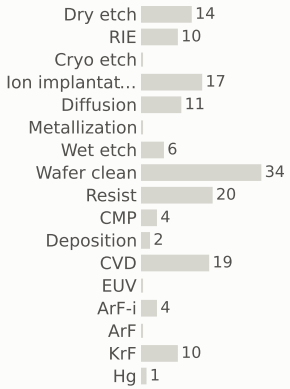
y cut at 57.351 µm



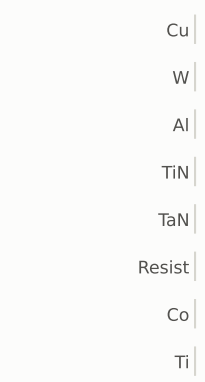
mask for this step

no mask for this step

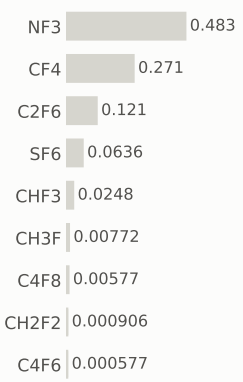
steps by process area



materials deposited (g)



gases emitted (g)



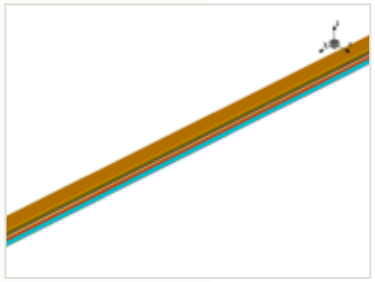
Step #1890: Resist

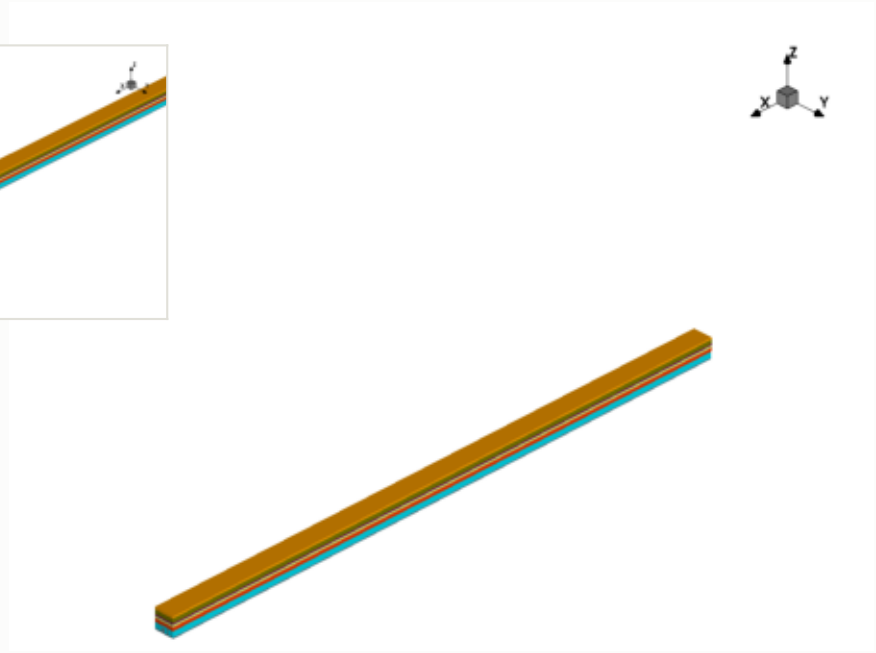
resist · Resist

0.169	110.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



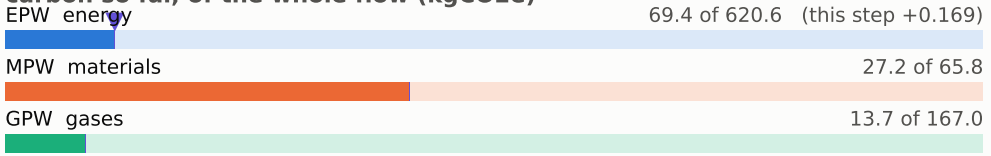


this step

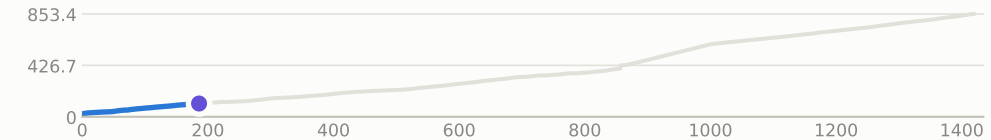
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

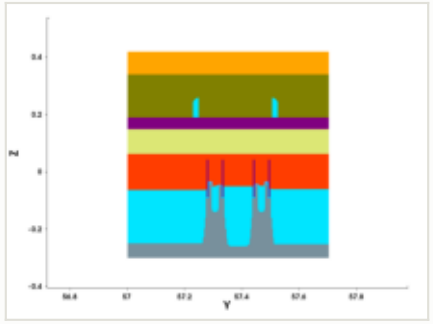
carbon so far, of the whole flow (kgCO2e)



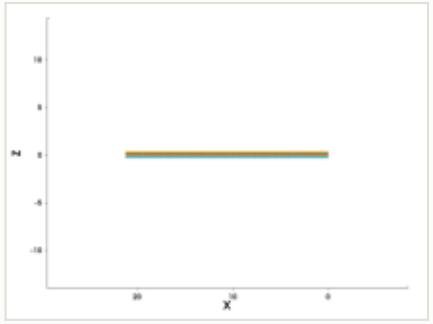
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



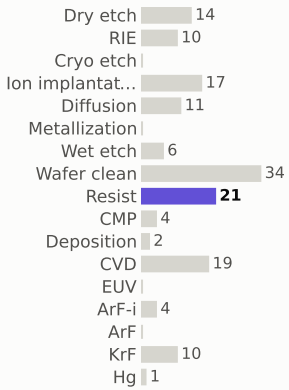
y cut at 57.351 µm



mask for this step

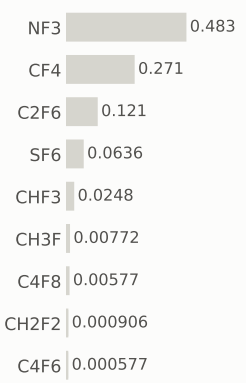
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



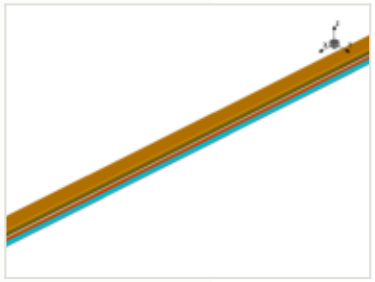
Step #1900: ArF

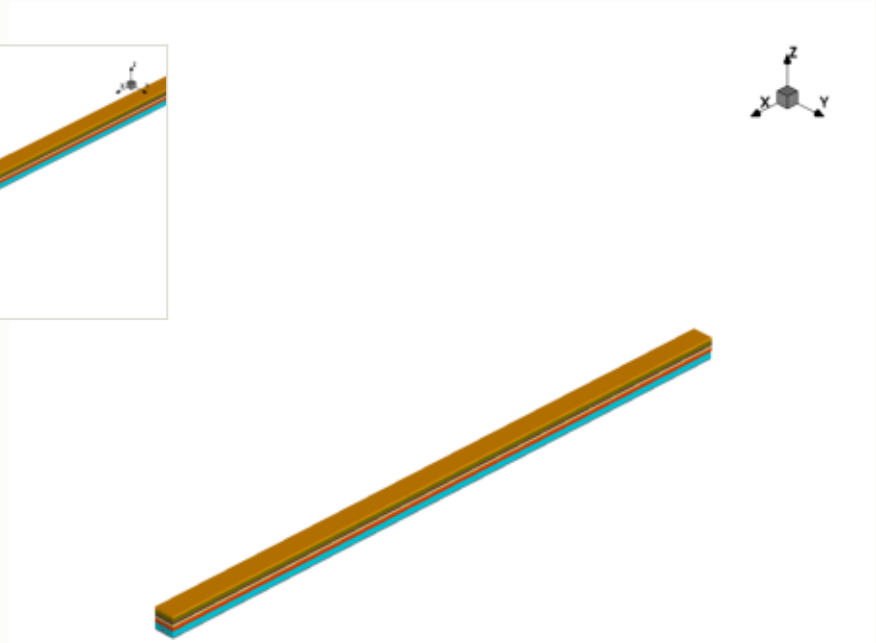
litho_patterning · ArF

0.199	110.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



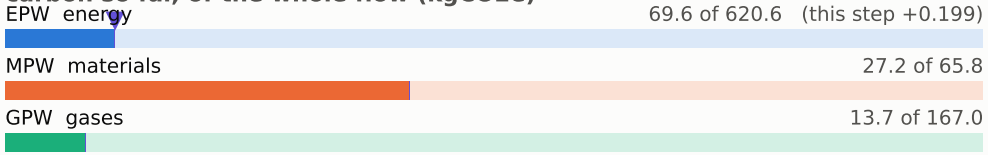


this step

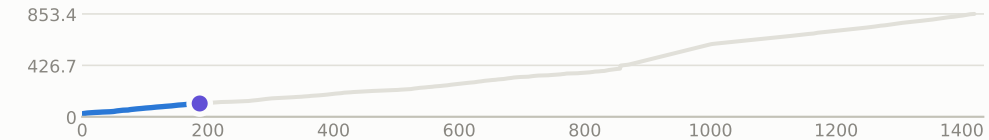
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

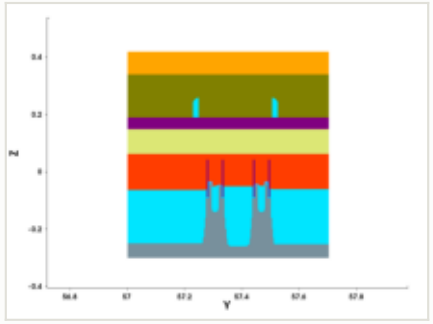
carbon so far, of the whole flow (kgCO2e)



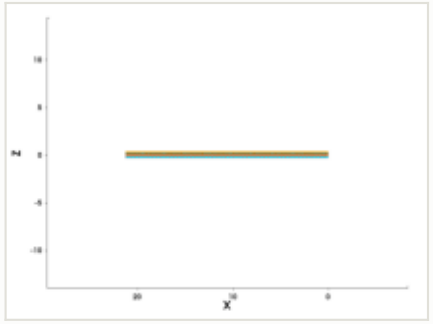
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



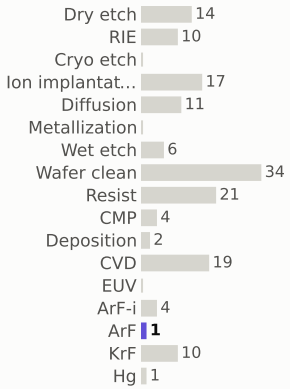
y cut at 57.351 µm



mask for this step

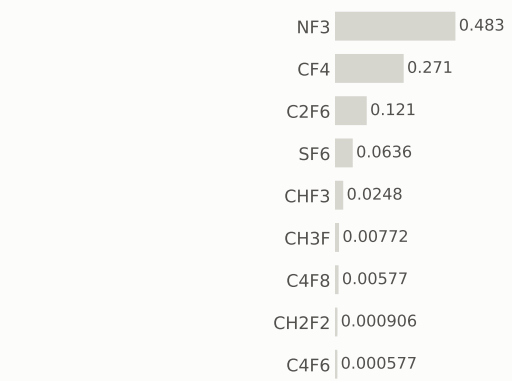
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



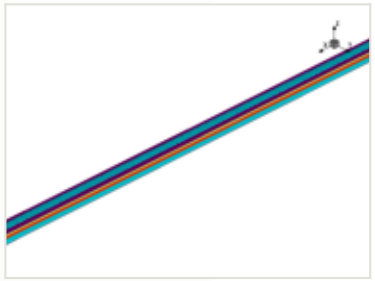
Step #1910: Spx_only

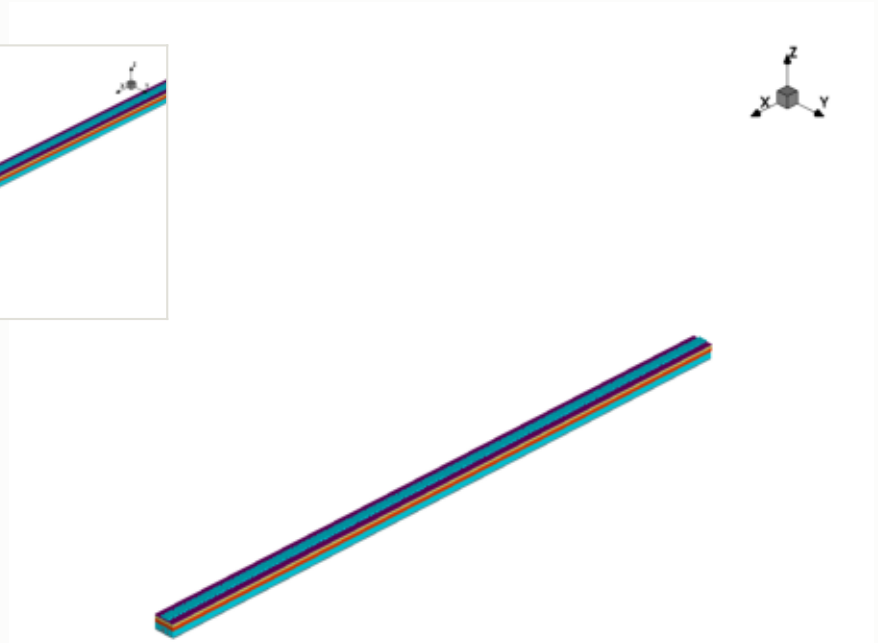
etch_anisotropic · Spx_only

0	110.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





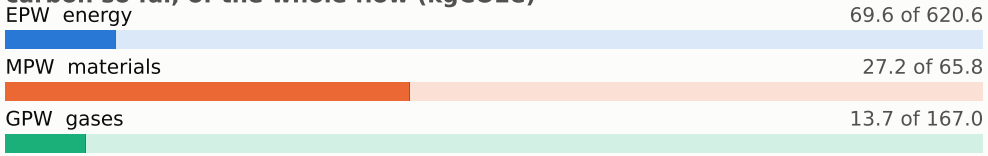
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

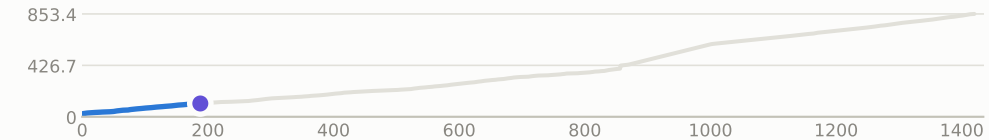
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

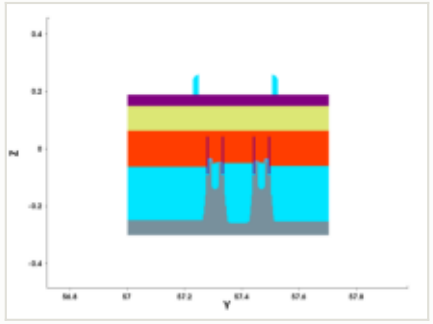
carbon so far, of the whole flow (kgCO2e)



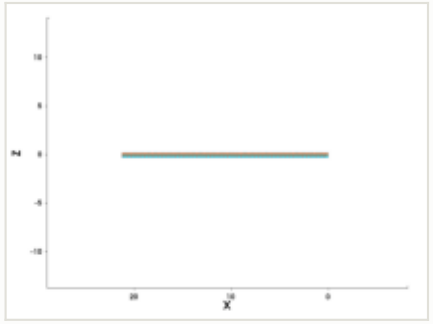
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



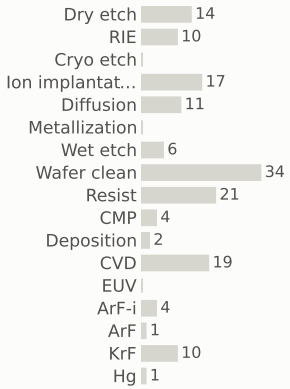
y cut at 57.351 µm



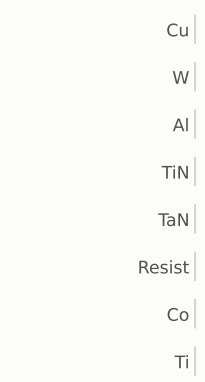
mask for this step

no mask for this step

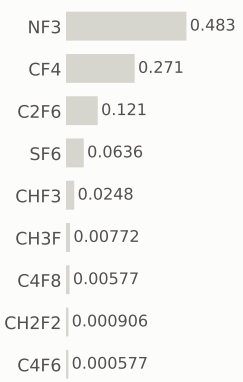
steps by process area



materials deposited (g)



gases emitted (g)



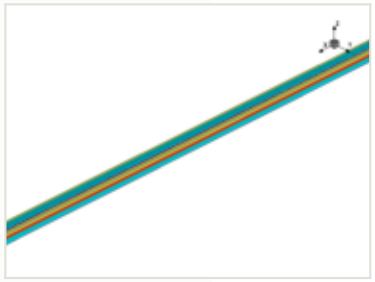
Step #1920: RIE

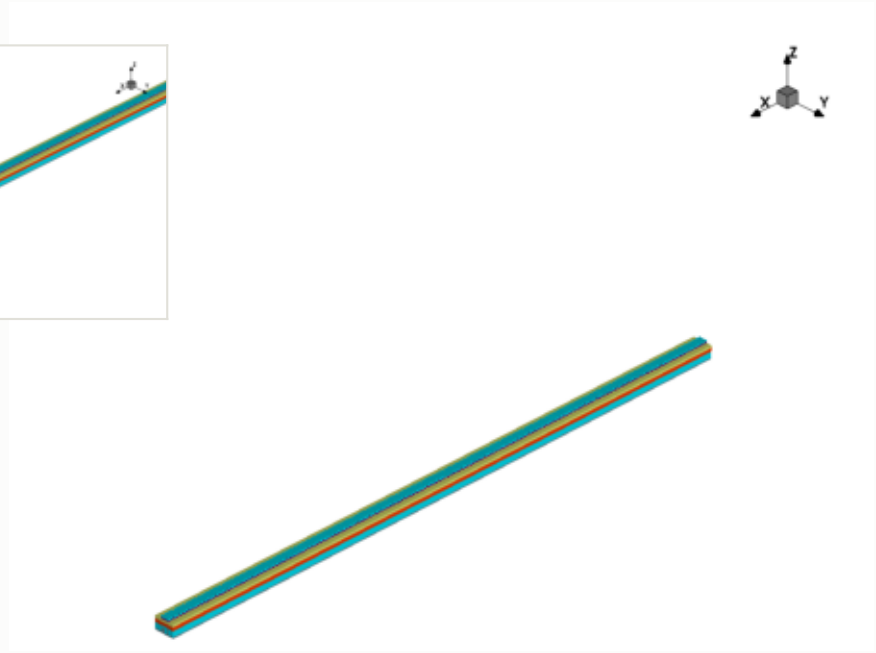
etch_anisotropic · RIE

1.04	111.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



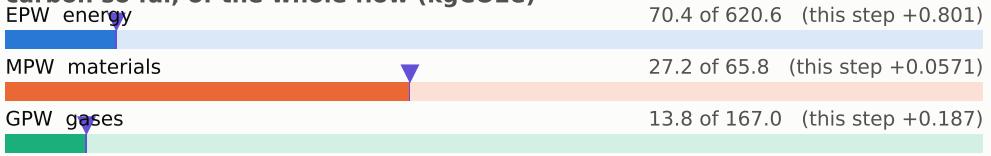


this step

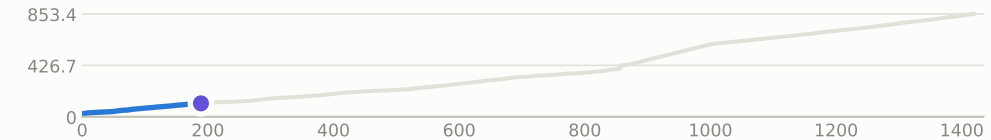
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

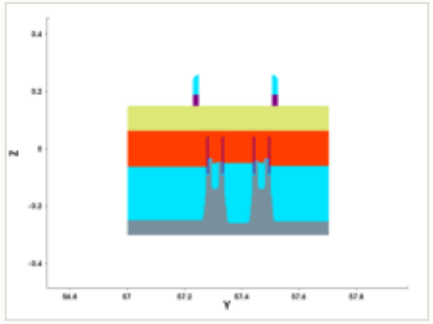
carbon so far, of the whole flow (kgCO2e)



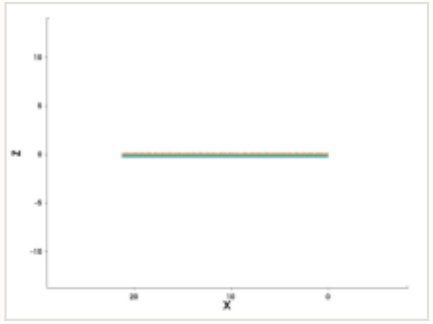
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



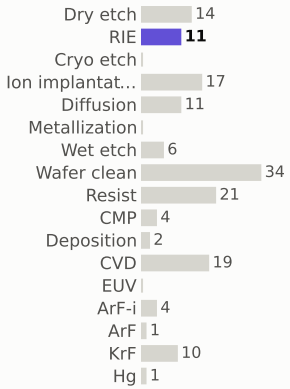
y cut at 57.351 µm



mask for this step

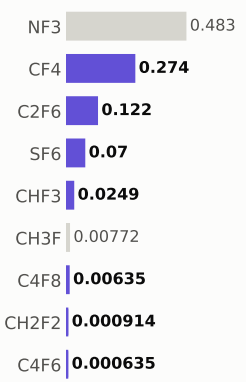
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



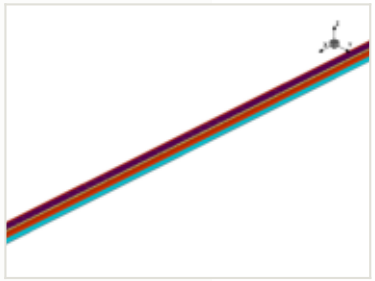
Step #1930: Spx_only

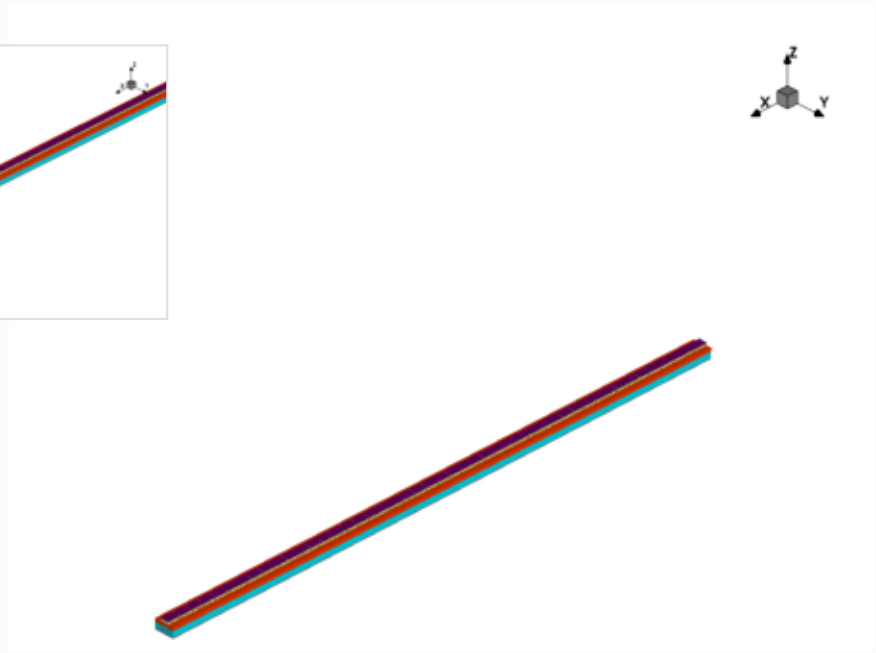
etch_anisotropic · Spx_only

0	111.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

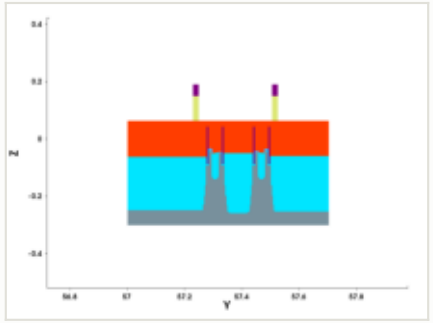
3D structure after this step

zoomed to fit

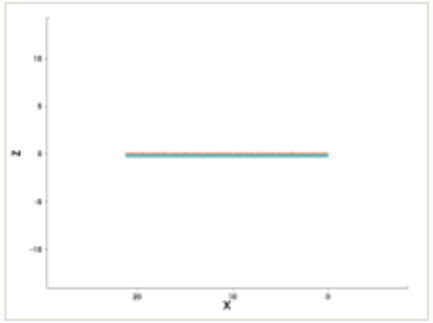




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

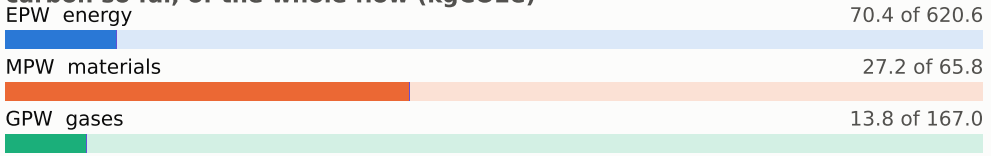
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

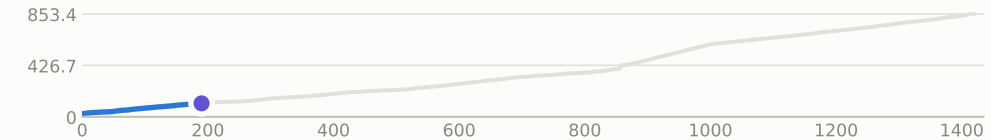
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

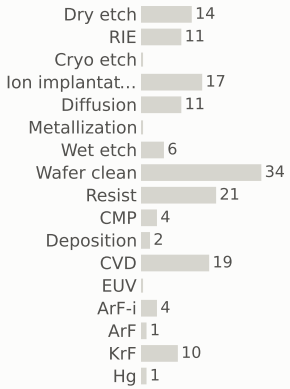
carbon so far, of the whole flow (kgCO2e)



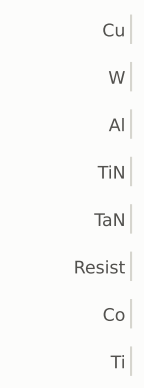
embodied carbon up to each step (kgCO2e per wafer)



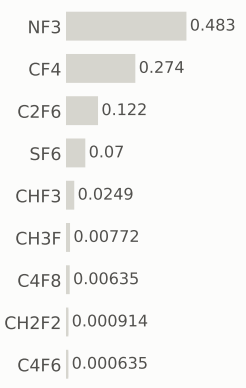
steps by process area



materials deposited (g)



gases emitted (g)



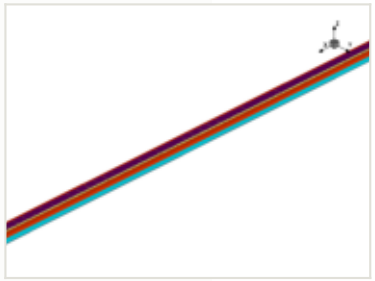
Step #1940: Spx_only

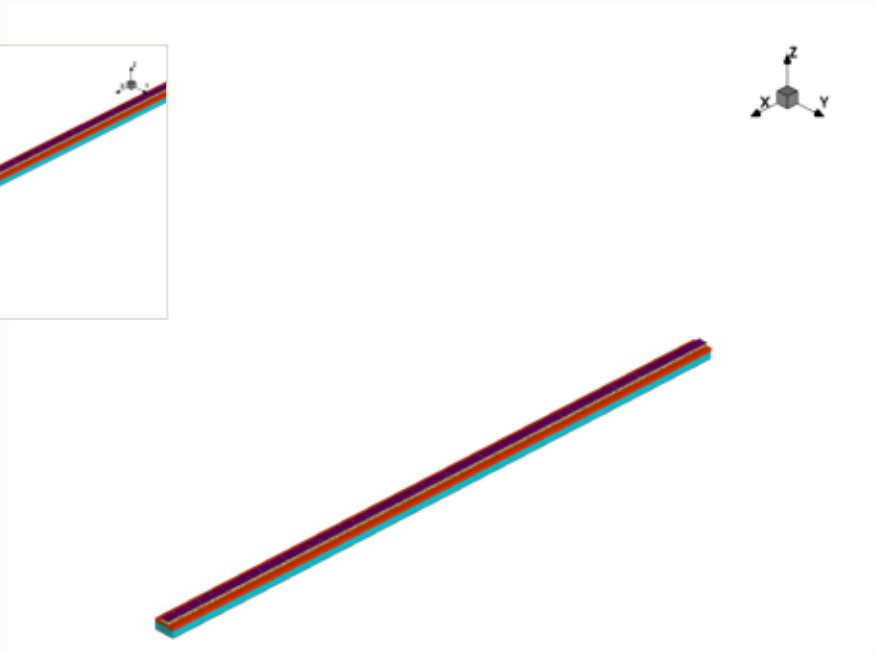
etch_strip · Spx_only

0	111.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





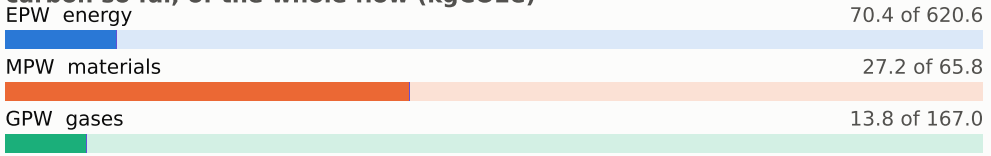
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

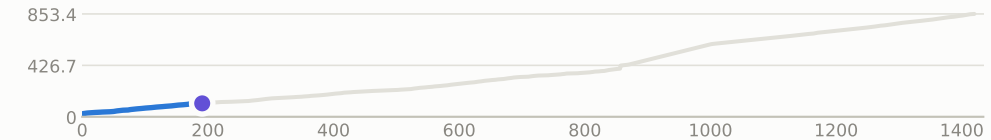
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

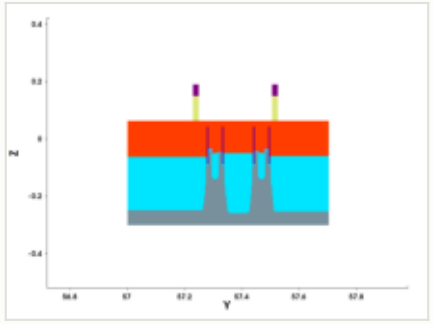
carbon so far, of the whole flow (kgCO2e)



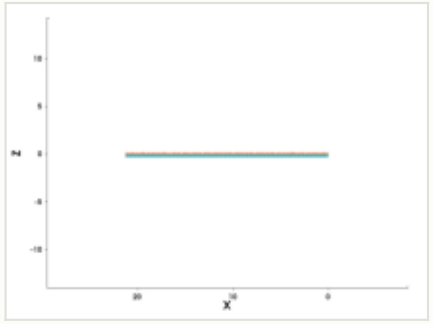
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



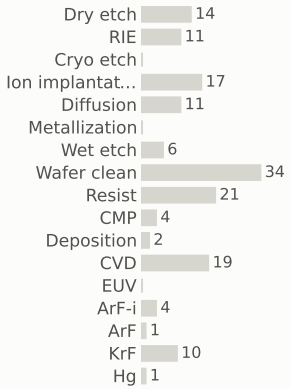
y cut at 57.351 µm



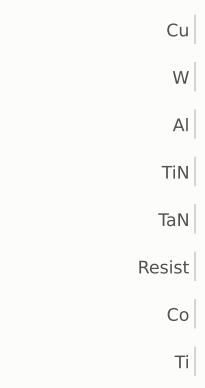
mask for this step

no mask for this step

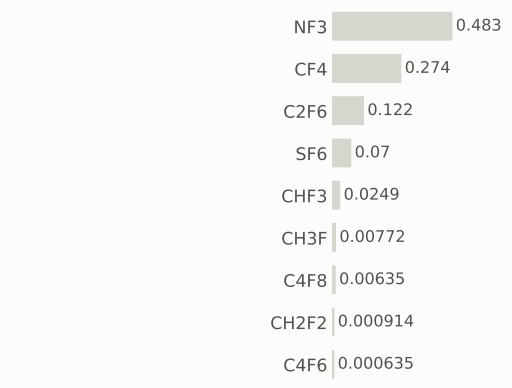
steps by process area



materials deposited (g)



gases emitted (g)



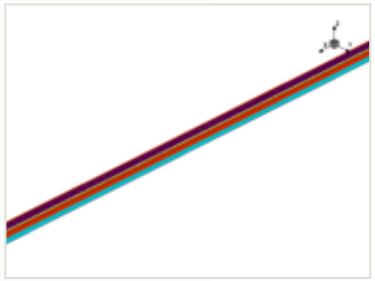
Step #1950: Wafer_clean

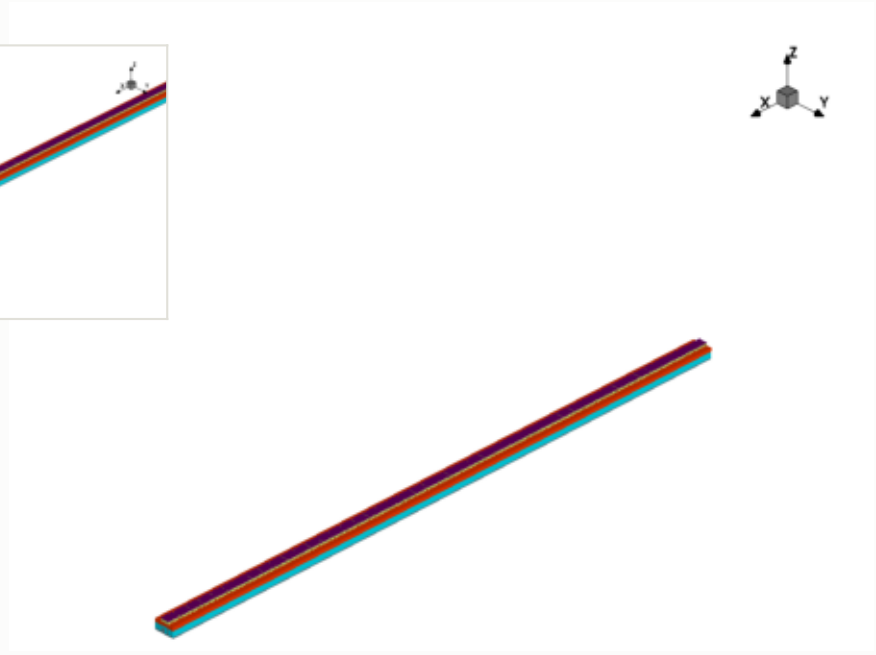
clean · Wafer_clean

0.169	111.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

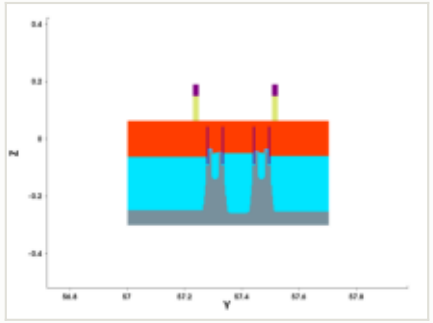
3D structure after this step

zoomed to fit

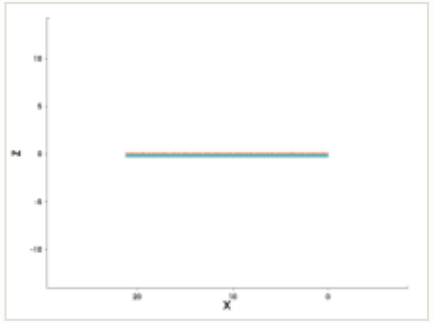




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

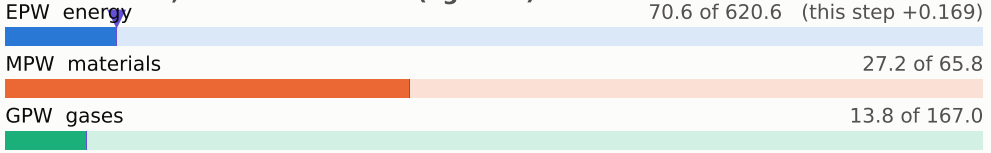
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

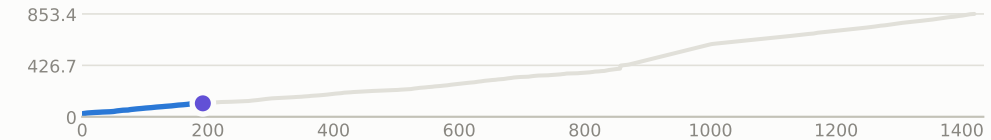
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

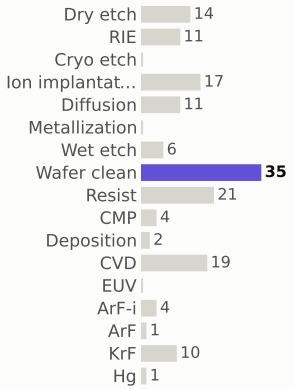
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

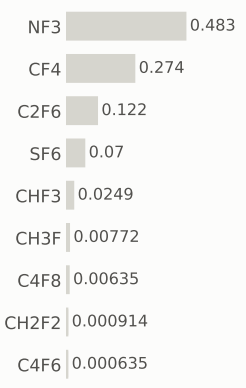


steps by process area



materials deposited (g)

gases emitted (g)



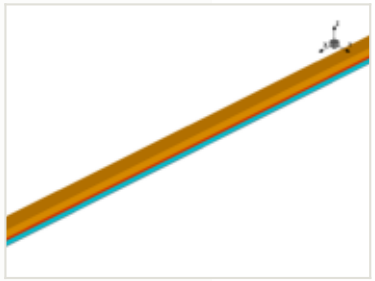
Step #1960: Resist

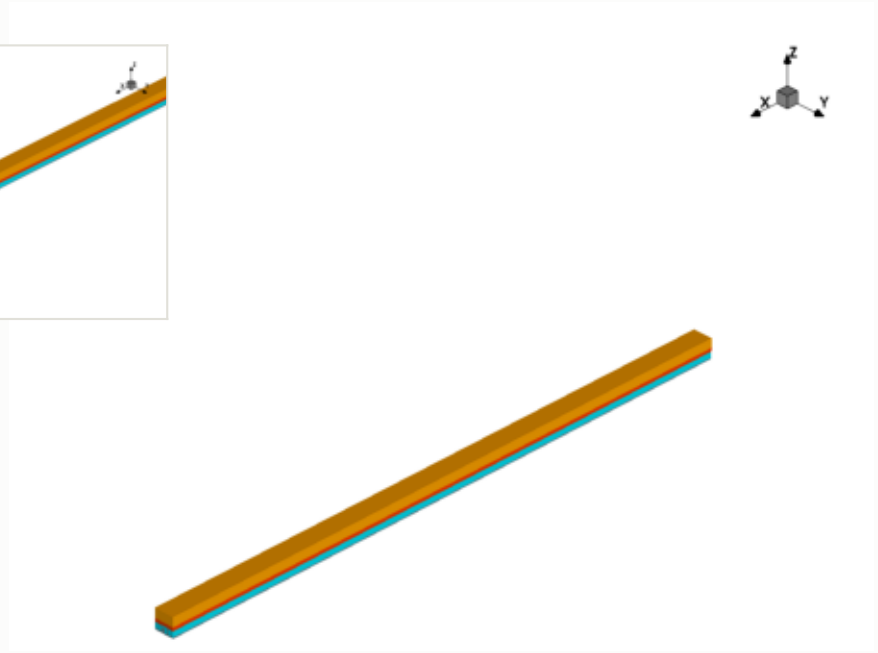
depo_fill · Resist

0.169	111.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



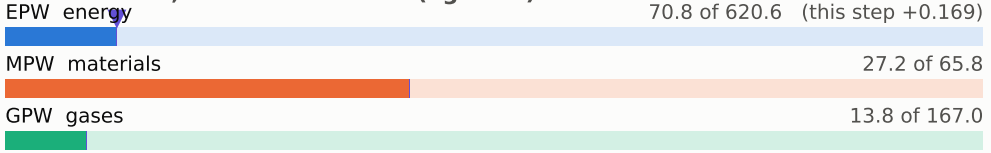


this step

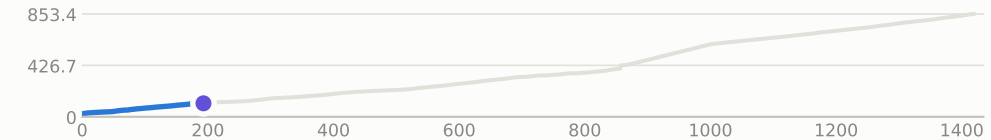
process area	Resist
material	—
thickness	0.2 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

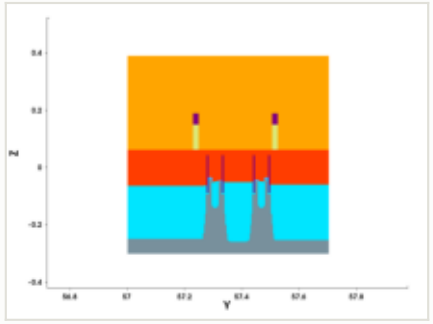
carbon so far, of the whole flow (kgCO2e)



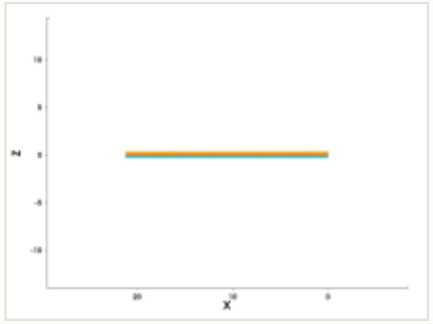
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



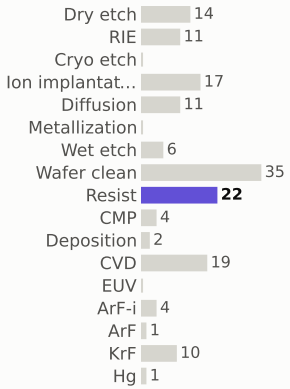
y cut at 57.351 µm



mask for this step

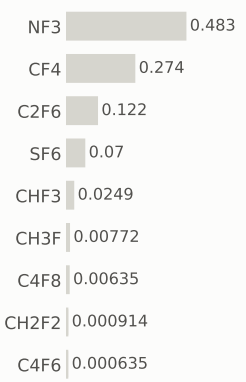
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



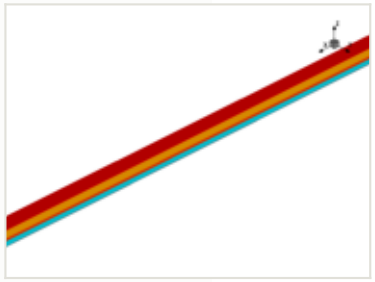
Step #1970: CVD

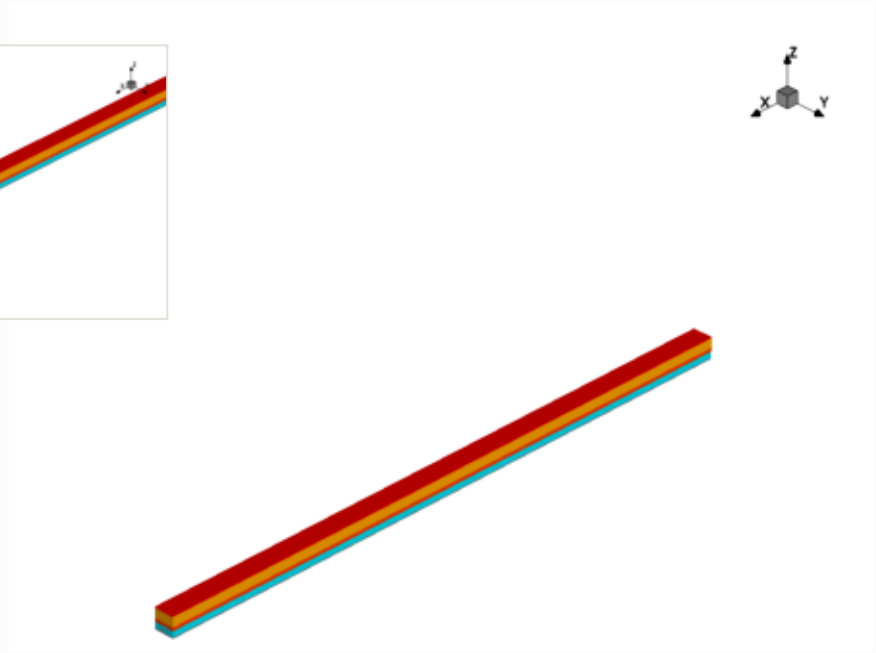
depo_isotropic · CVD

1.31	113.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



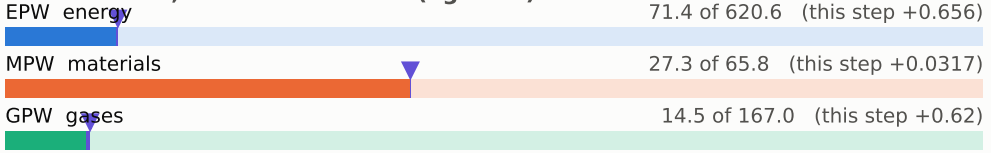


this step

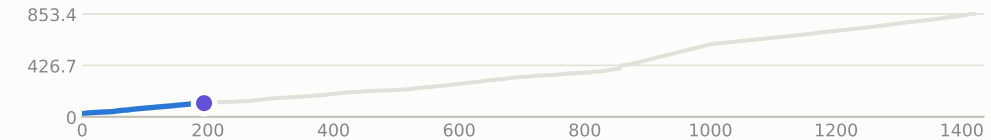
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

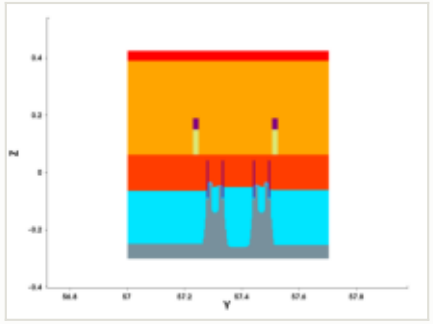
carbon so far, of the whole flow (kgCO2e)



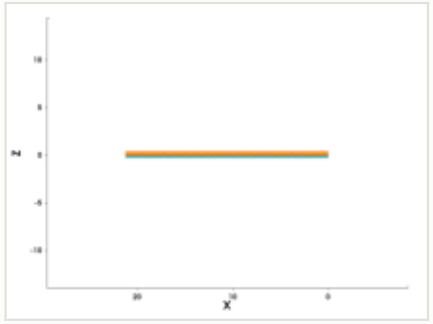
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



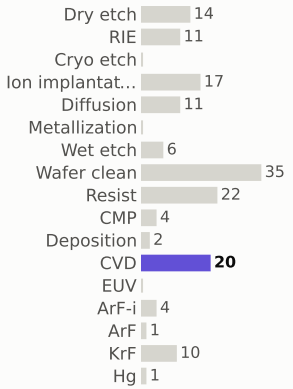
y cut at 57.351 µm



mask for this step

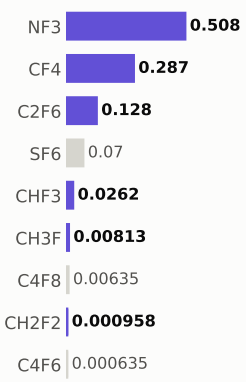
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



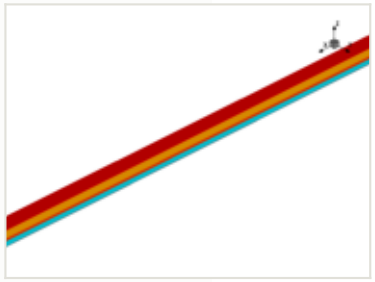
Step #1980: Wafer_clean

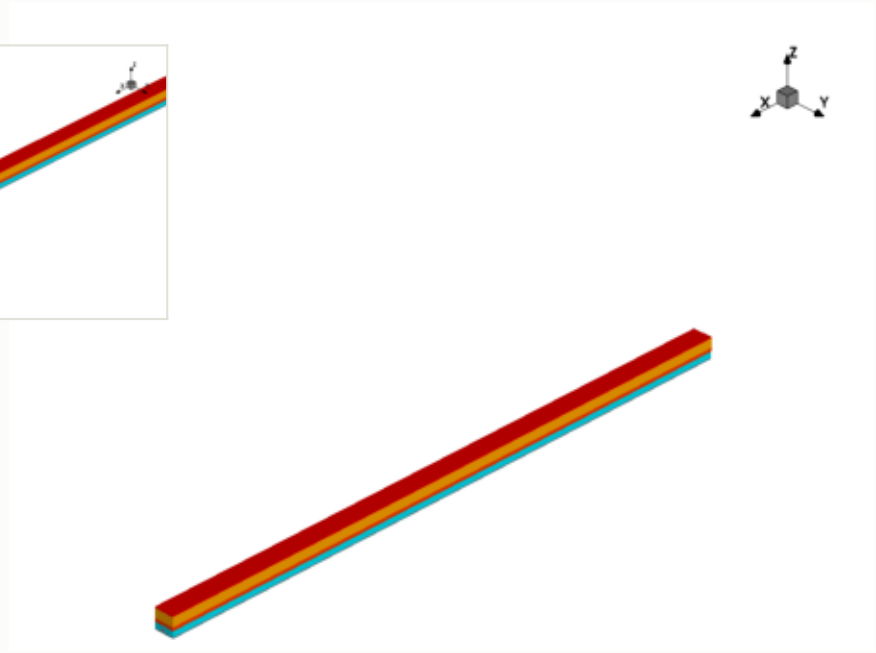
clean · Wafer_clean

0.169	113.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

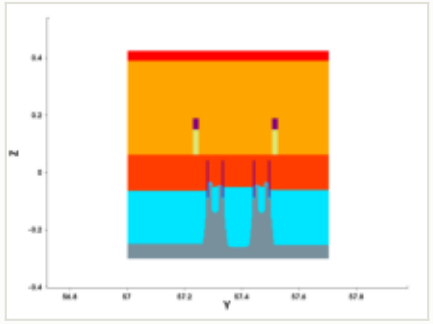
3D structure after this step

zoomed to fit

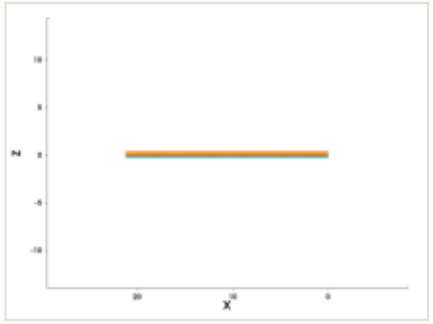




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

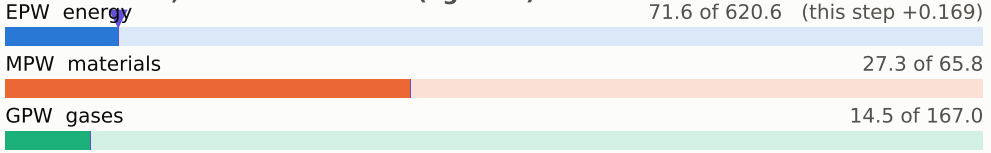
no mask for this step

this step

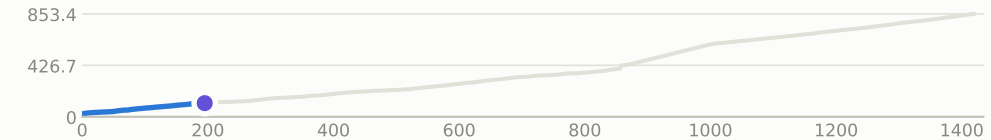
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

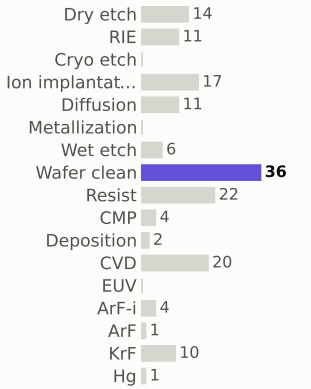
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

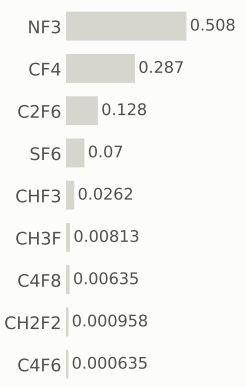


steps by process area



materials deposited (g)

gases emitted (g)



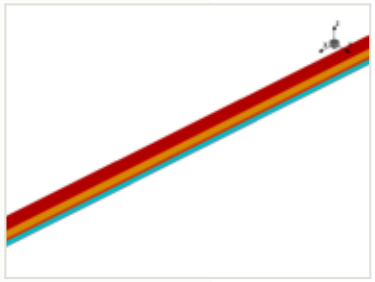
Step #1990: Resist

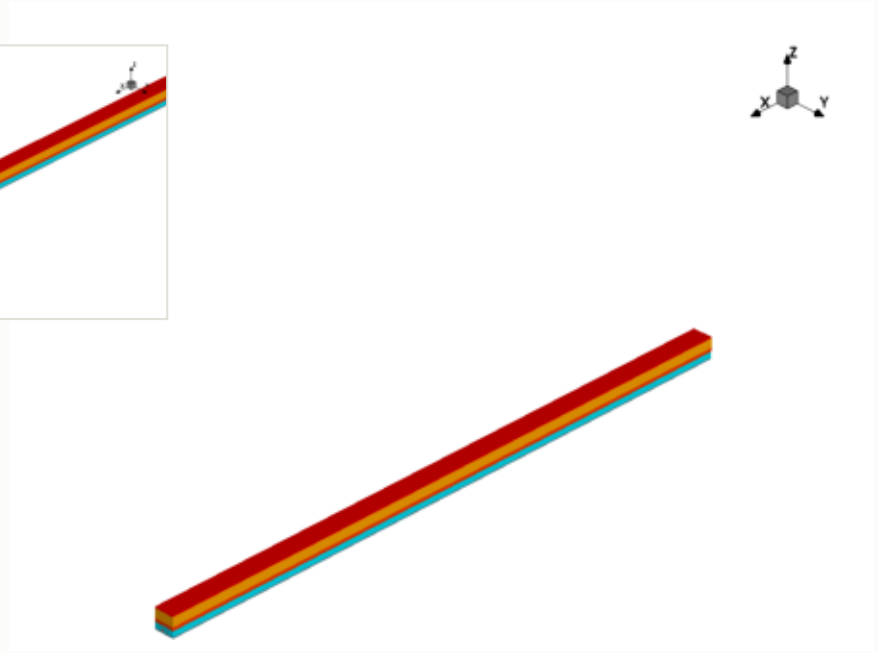
resist · Resist

0.169	113.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





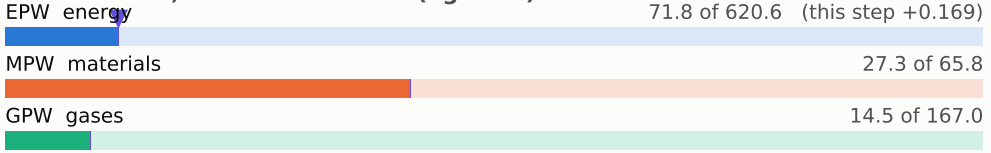
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

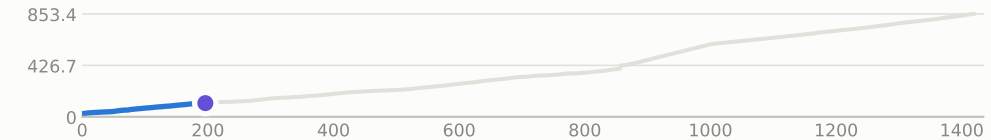
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

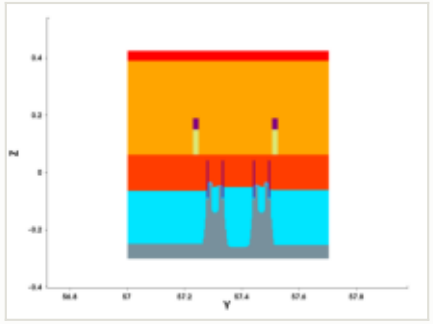
carbon so far, of the whole flow (kgCO2e)



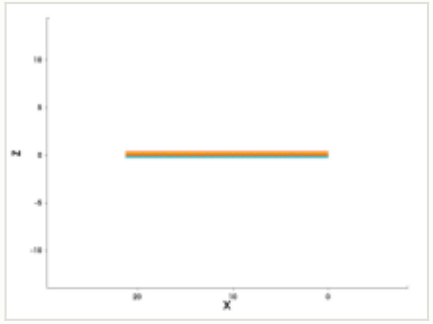
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



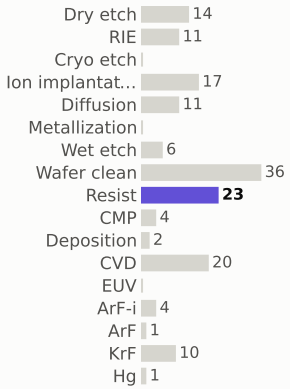
y cut at 57.351 µm



mask for this step

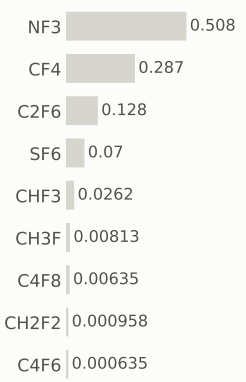
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



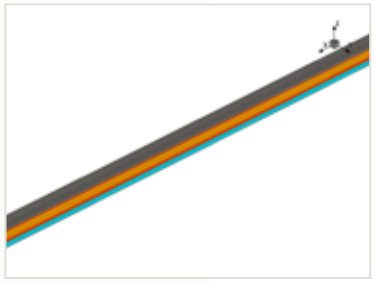
Step #2000: ArF-i

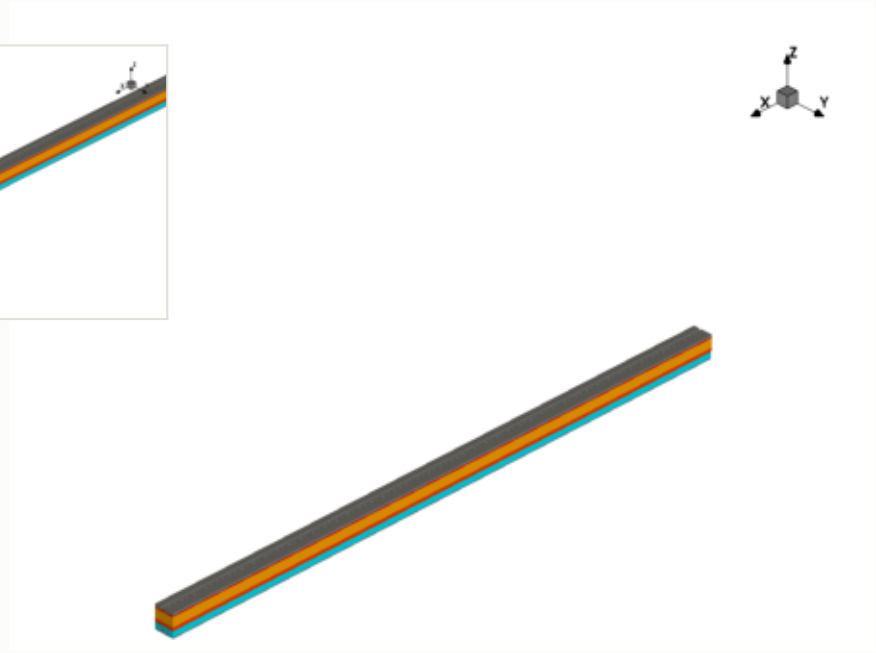
litho_patterning · ArF-i

0.217	113.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



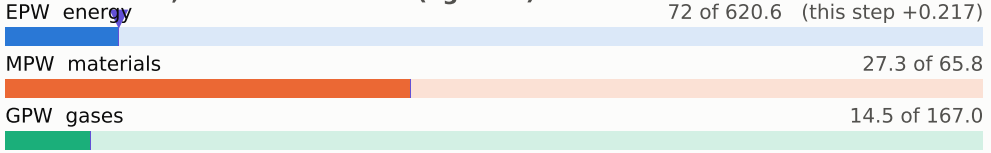


this step

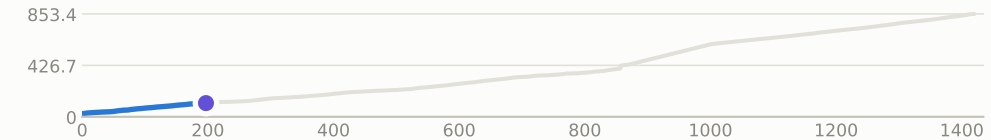
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

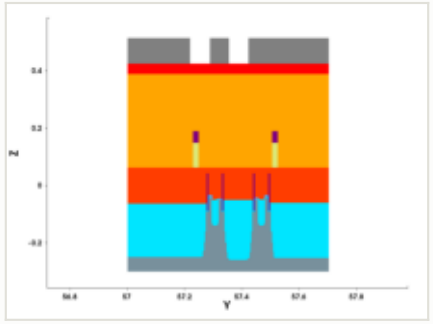
carbon so far, of the whole flow (kgCO2e)



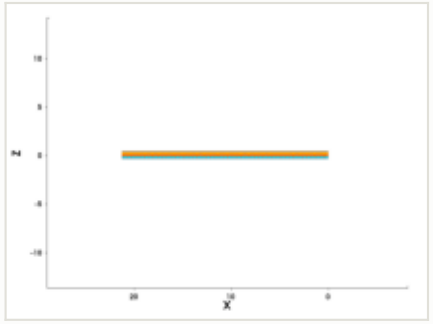
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

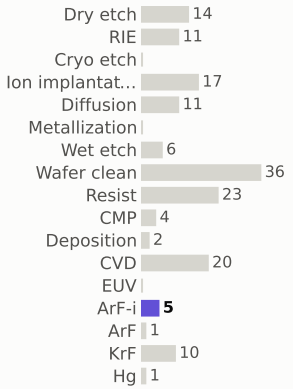


y cut at 57.351 µm



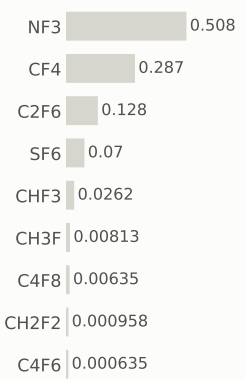
mask: layer 2010:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



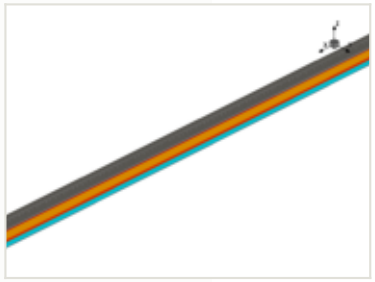
Step #2010: RIE

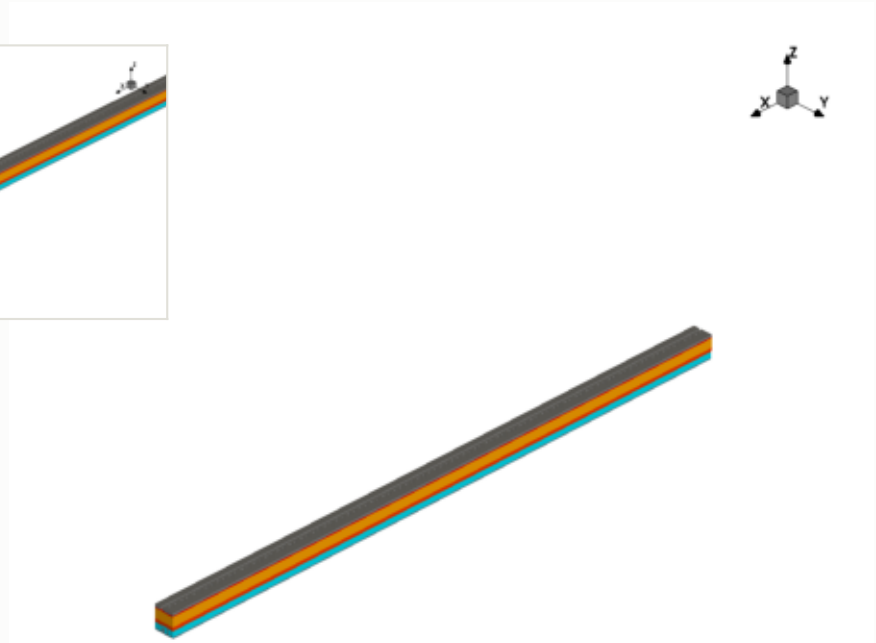
etch_anisotropic · RIE

1.04	114.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



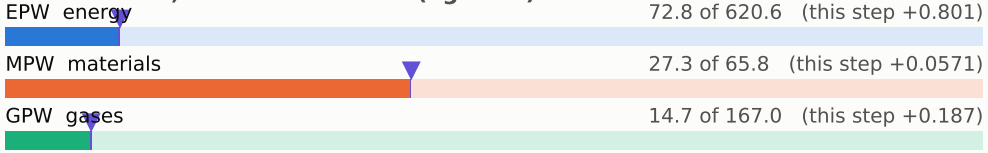


this step

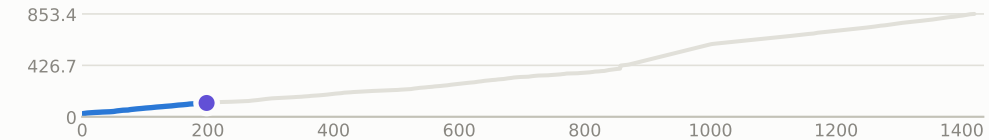
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

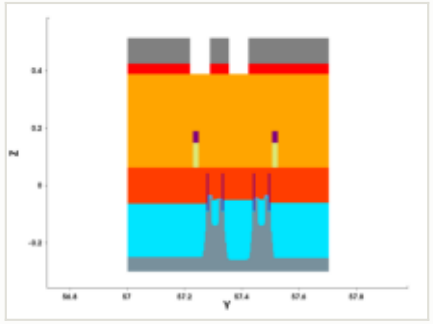
carbon so far, of the whole flow (kgCO2e)



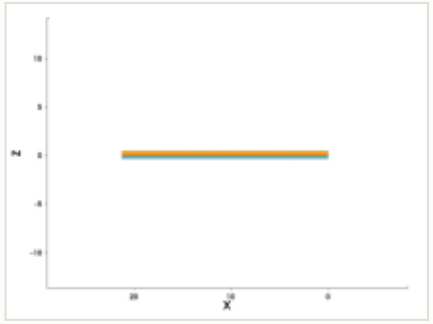
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



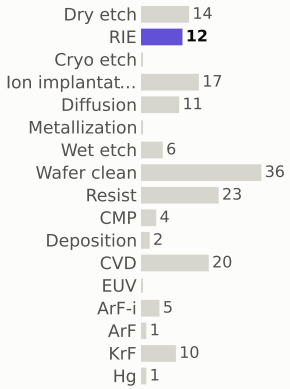
y cut at 57.351 µm



mask for this step

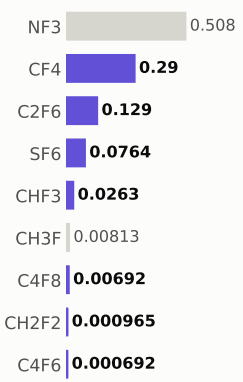
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



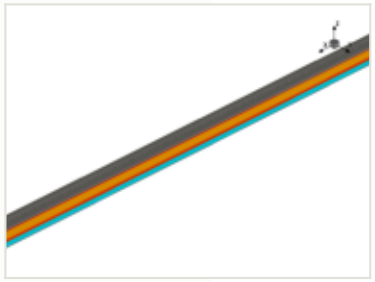
Step #2020: Wafer_clean

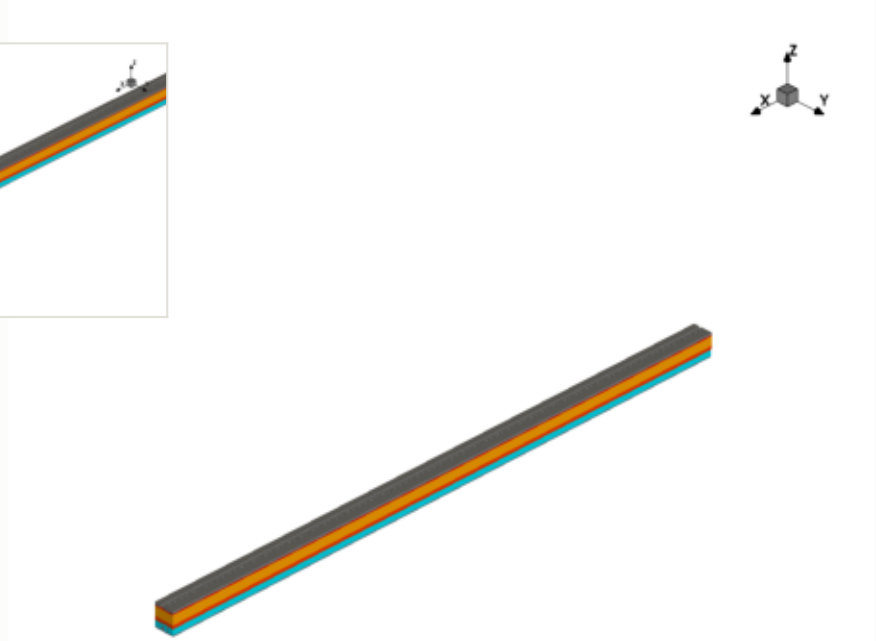
clean · Wafer_clean

0.169	114.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





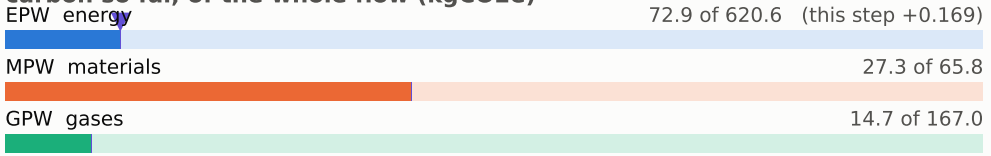
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

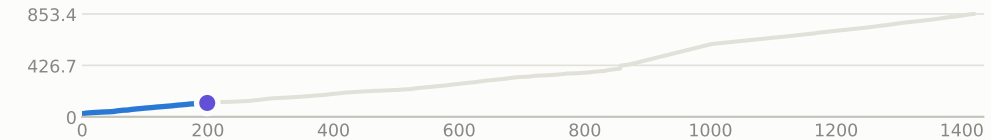
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

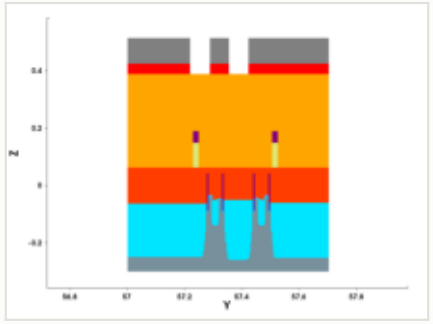
carbon so far, of the whole flow (kgCO2e)



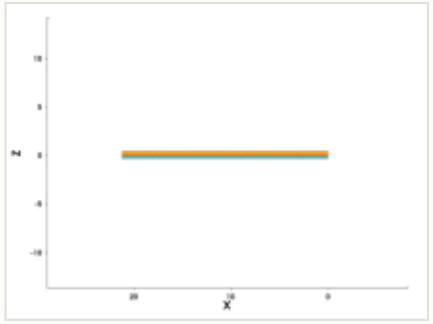
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



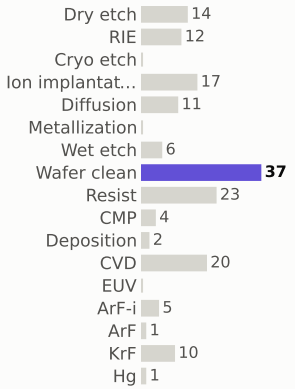
y cut at 57.351 µm



mask for this step

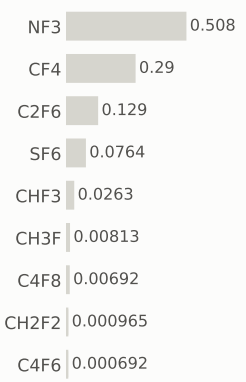
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



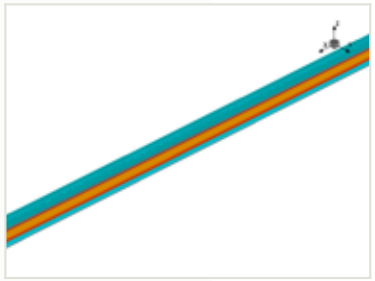
Step #2030: Deposition

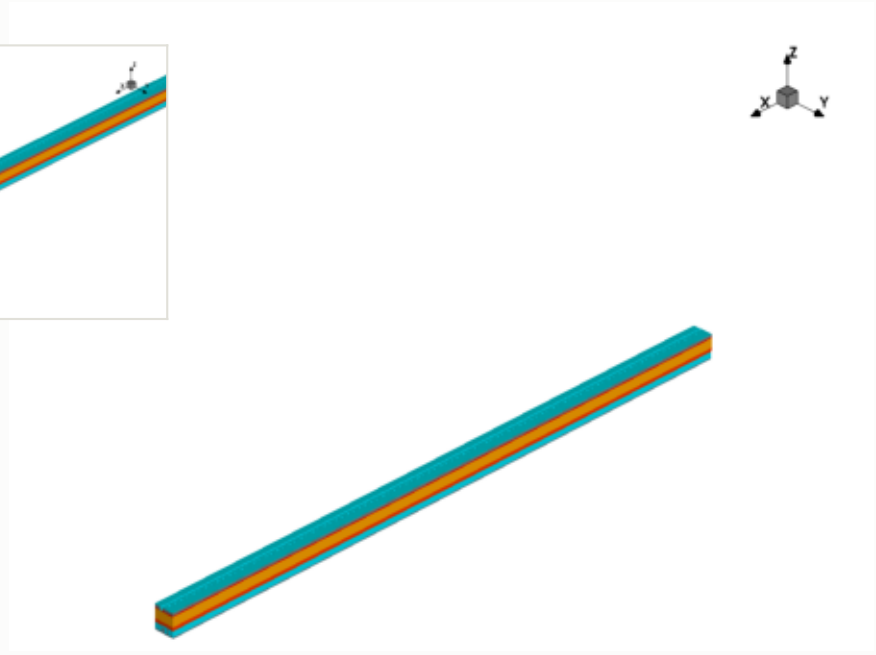
depo_isotropic · Deposition

0.656	115.6	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



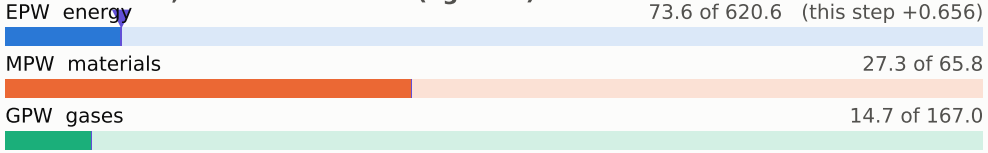


this step

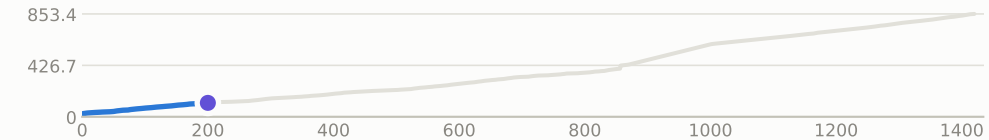
process area	Deposition
material	—
thickness	0.009 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

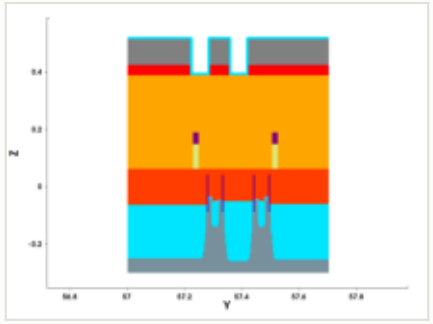
carbon so far, of the whole flow (kgCO2e)



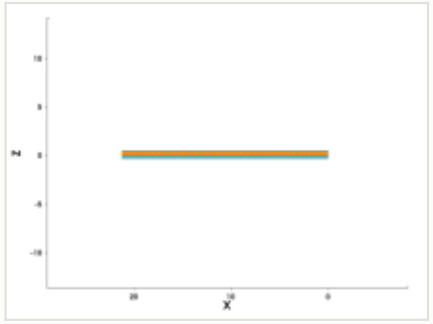
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



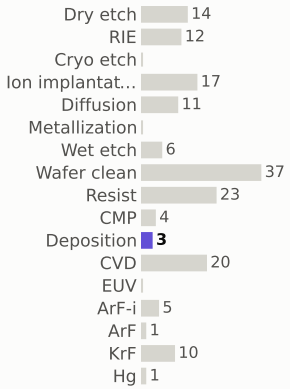
y cut at 57.351 µm



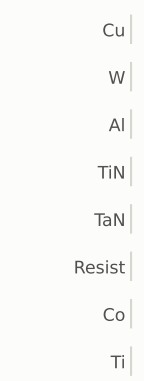
mask for this step

no mask for this step

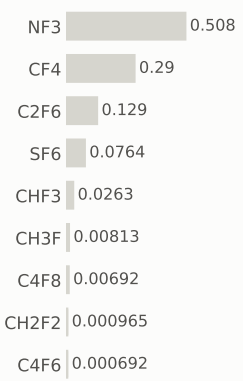
steps by process area



materials deposited (g)



gases emitted (g)



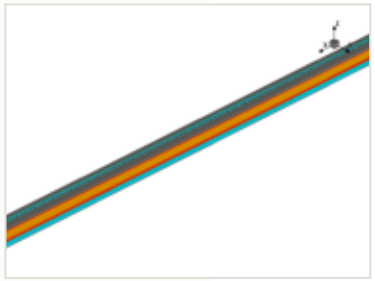
Step #2040: RIE

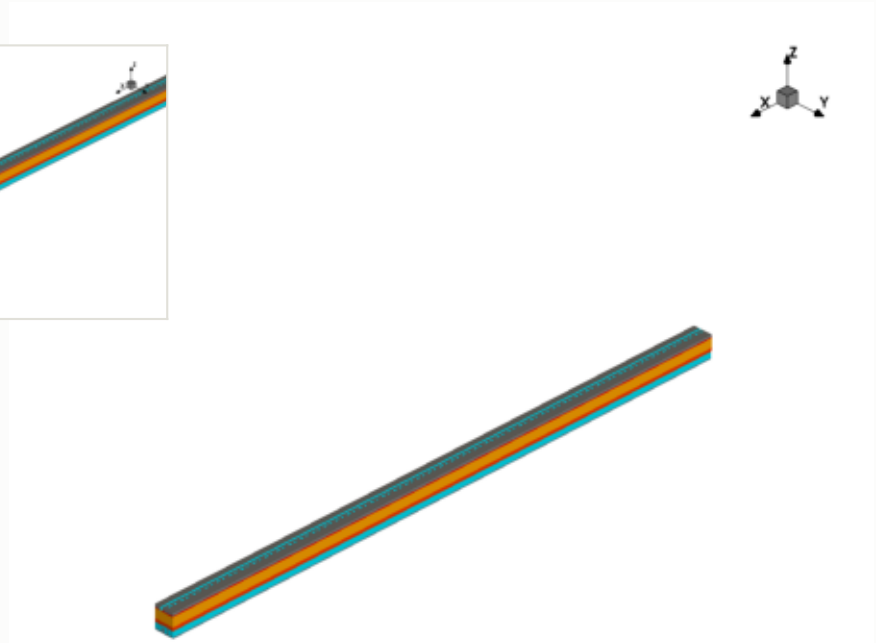
etch_anisotropic · RIE

1.04	116.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



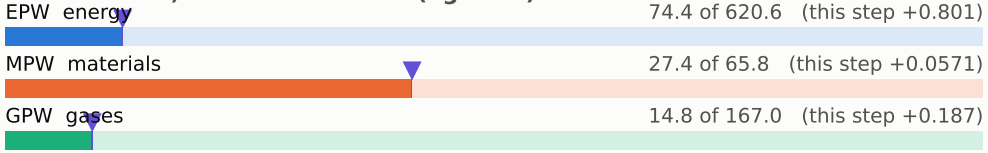


this step

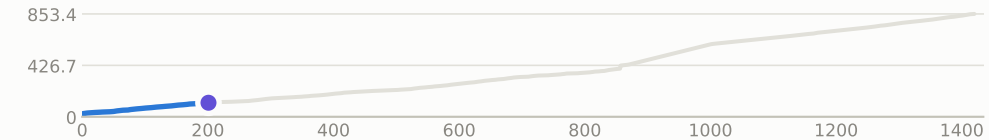
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

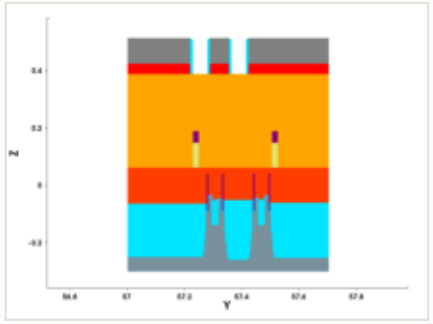
carbon so far, of the whole flow (kgCO2e)



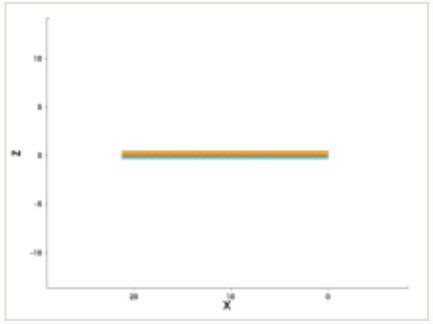
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



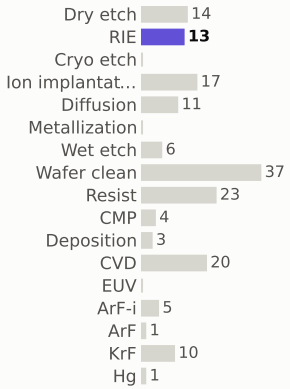
y cut at 57.351 µm



mask for this step

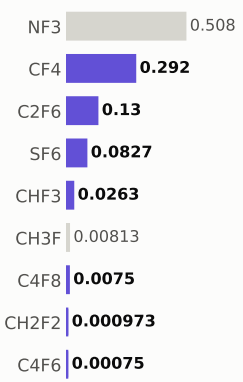
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



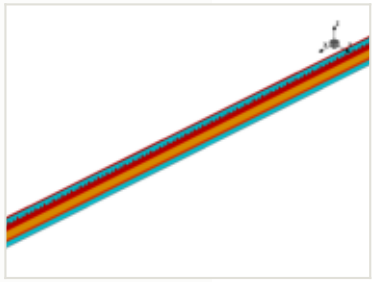
Step #2050: Spx_only

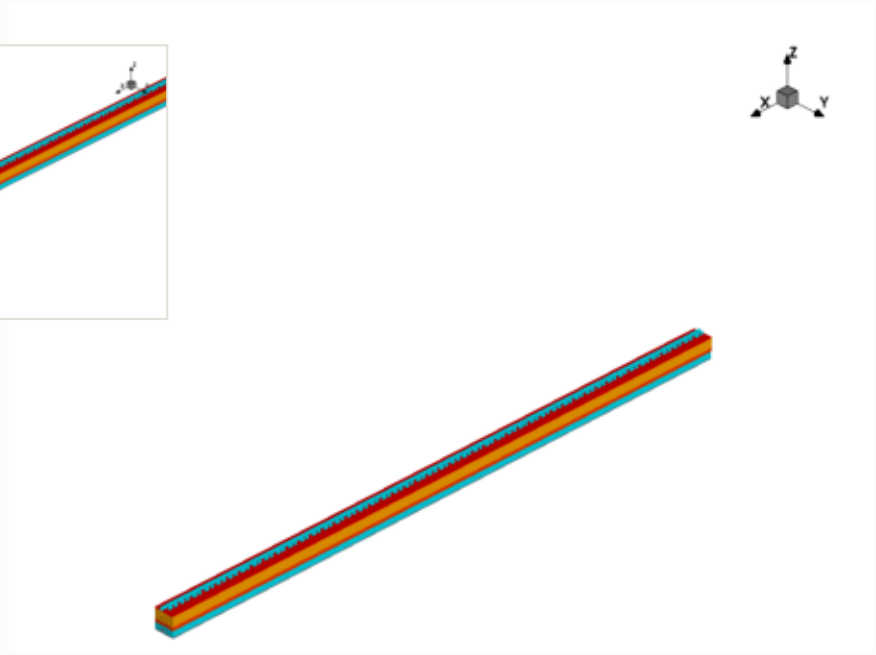
etch_anisotropic · Spx_only

0	116.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





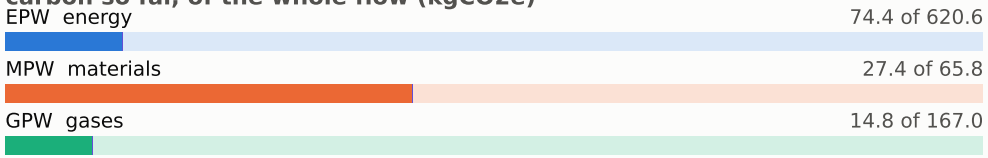
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

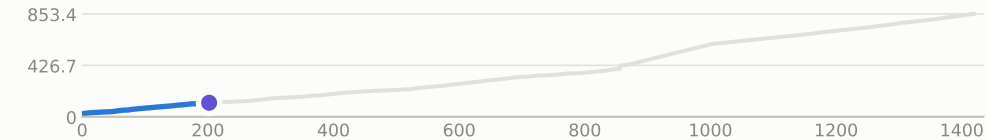
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

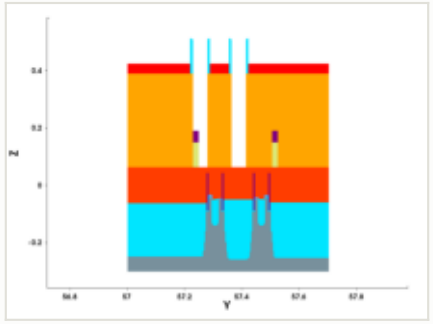
carbon so far, of the whole flow (kgCO2e)



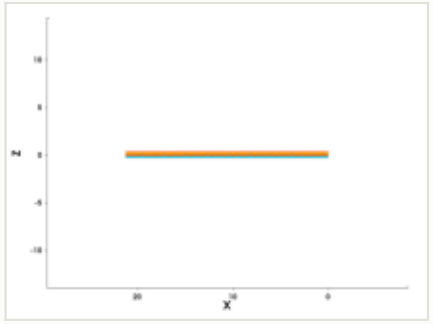
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



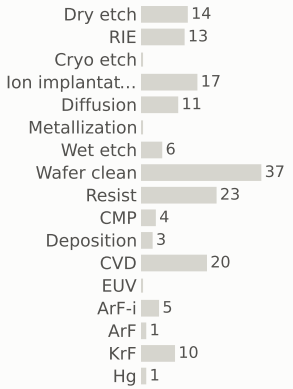
y cut at 57.351 µm



mask for this step

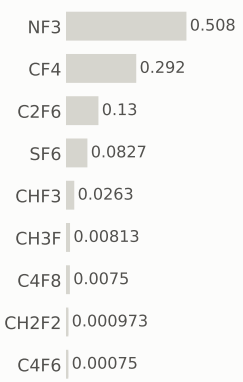
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



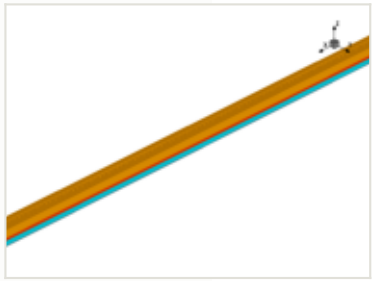
Step #2060: Spx_only

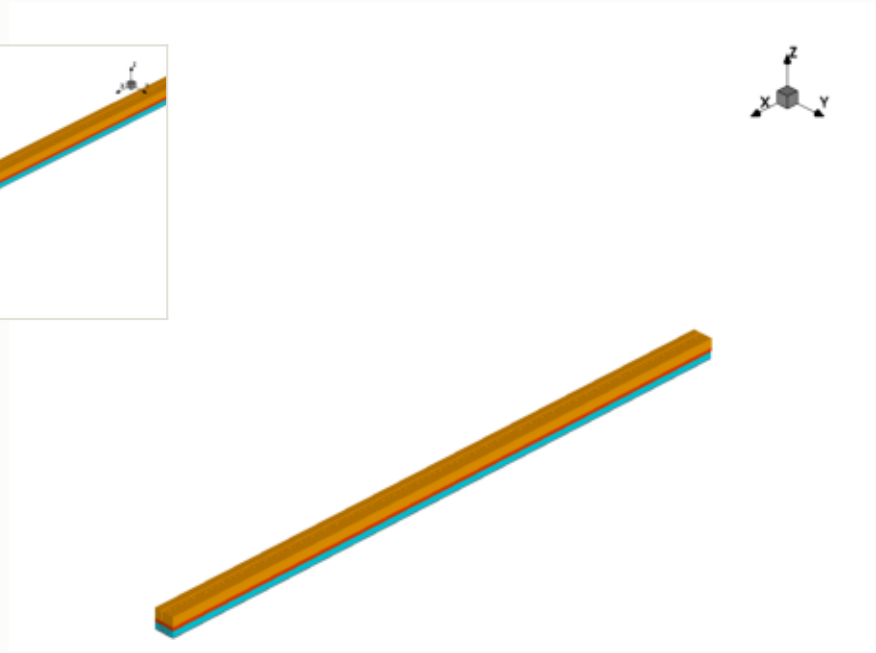
etch_anisotropic · Spx_only

0	116.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





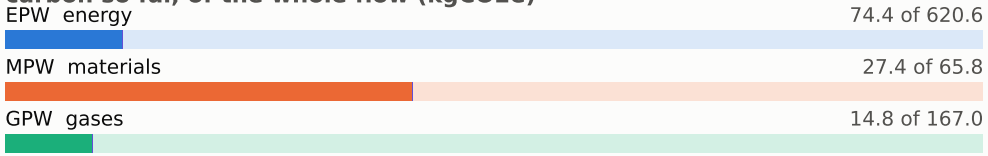
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

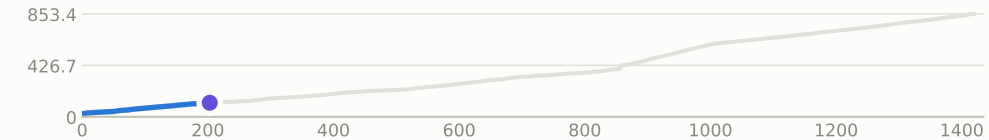
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

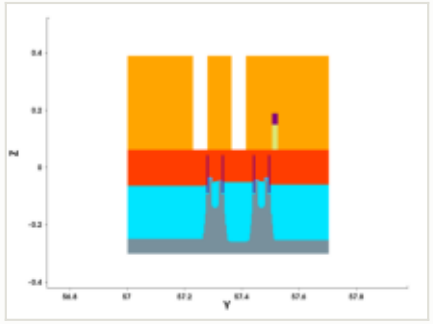
carbon so far, of the whole flow (kgCO2e)



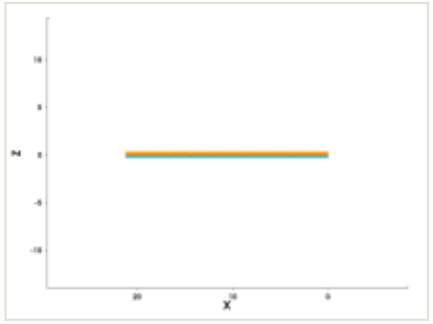
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



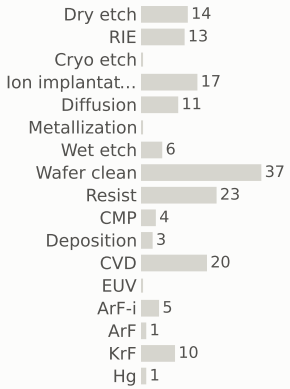
y cut at 57.351 µm



mask for this step

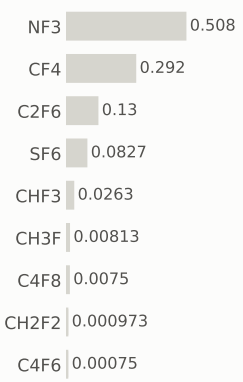
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



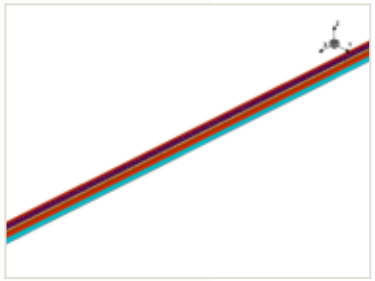
Step #2070: Spx_only

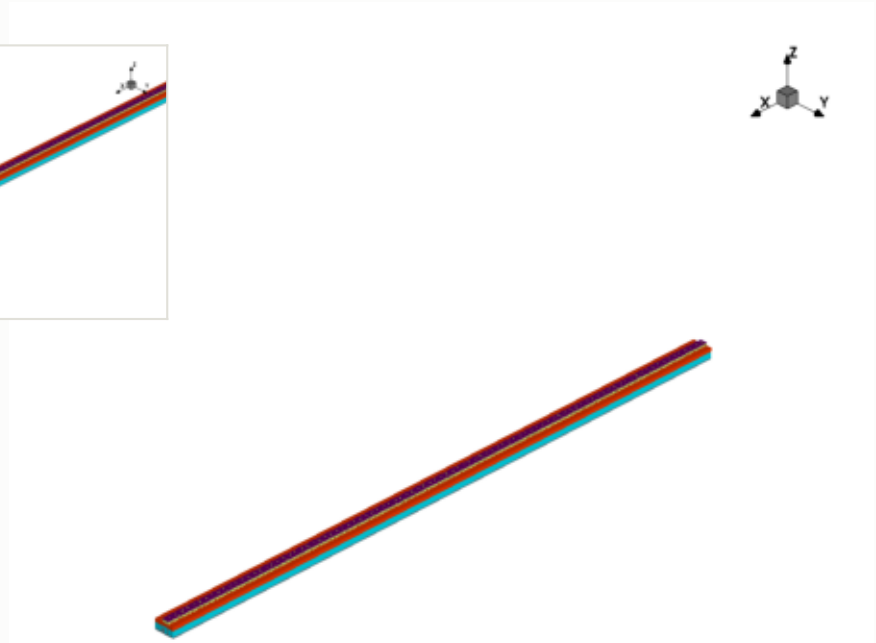
etch_strip · Spx_only

0	116.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



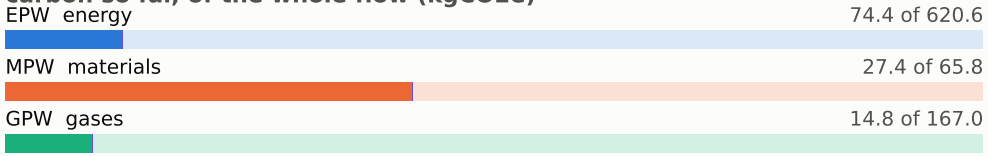


this step

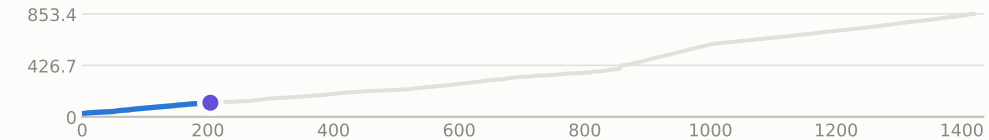
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

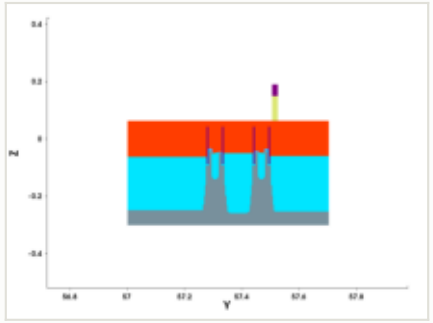
carbon so far, of the whole flow (kgCO2e)



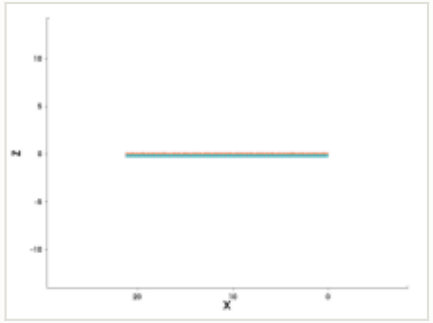
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



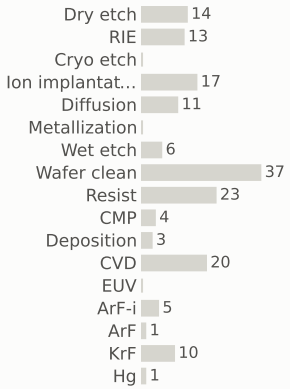
y cut at 57.351 µm



mask for this step

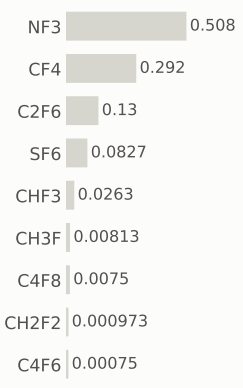
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



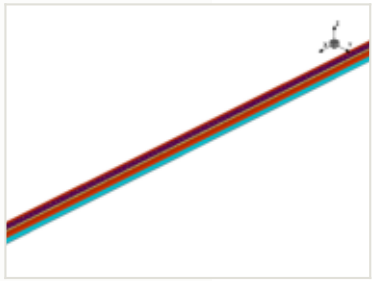
Step #2080: Wafer_clean

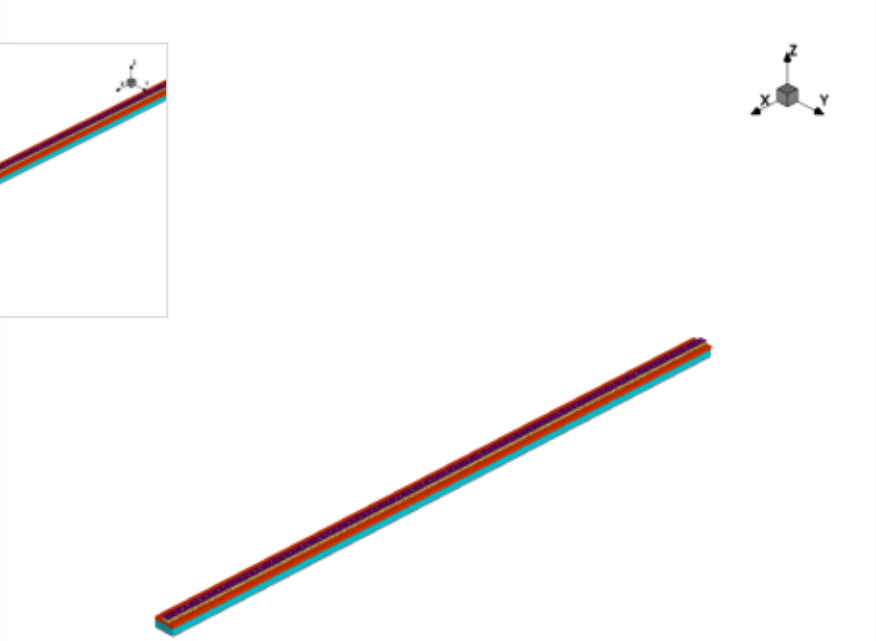
clean · Wafer_clean

0.169	116.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

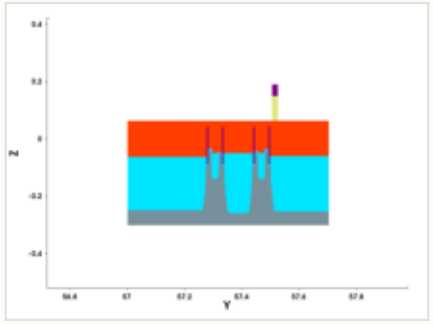
3D structure after this step

zoomed to fit

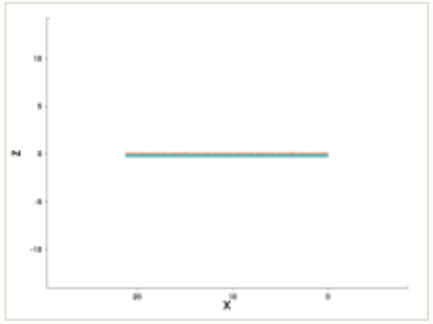




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

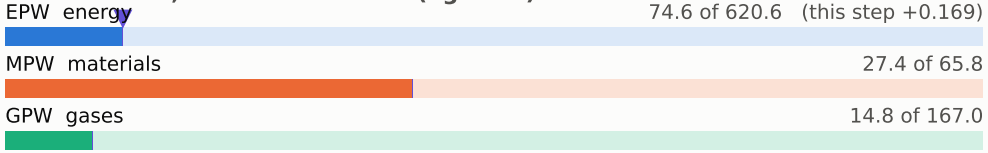
no mask for this step

this step

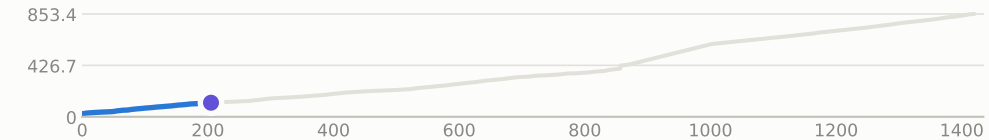
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

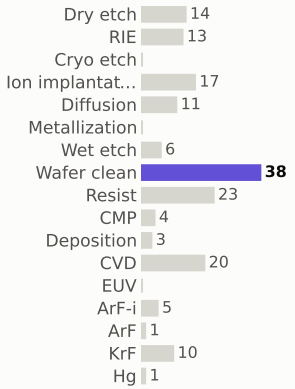
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

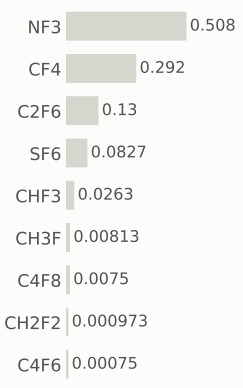


steps by process area



materials deposited (g)

gases emitted (g)



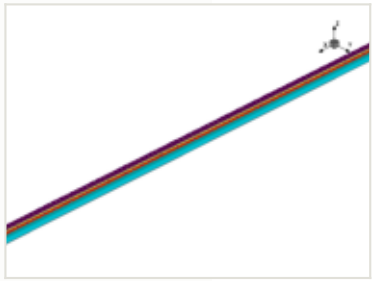
Step #2090: RIE

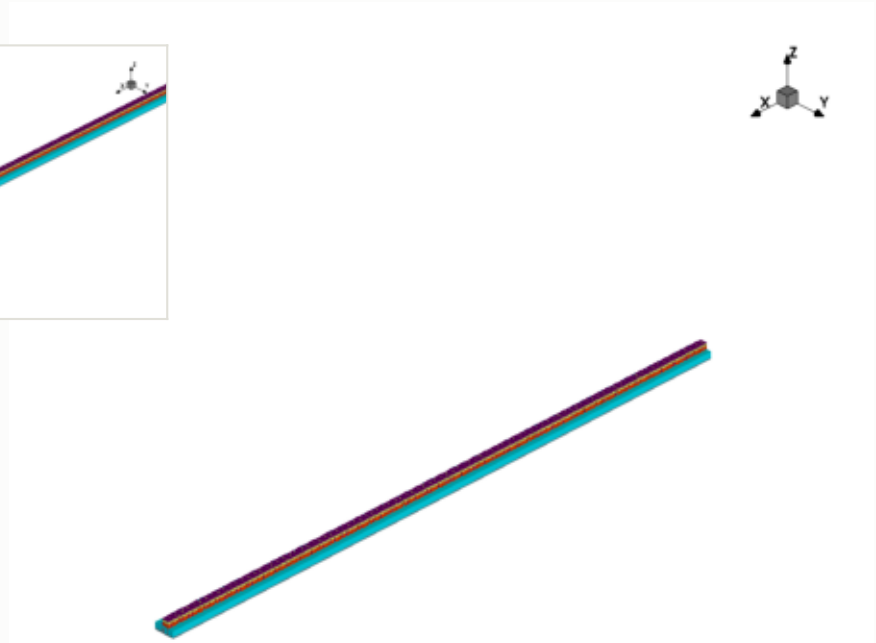
etch_anisotropic · RIE

1.04	117.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



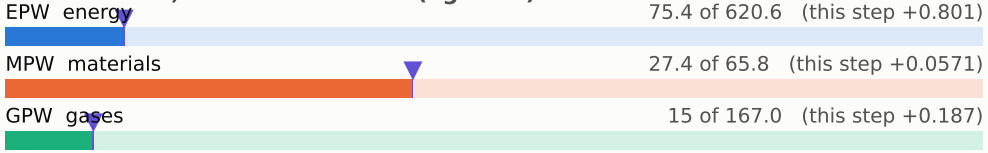


this step

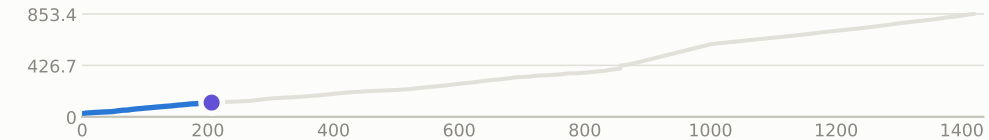
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

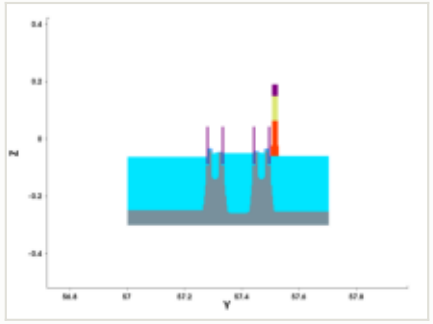
carbon so far, of the whole flow (kgCO2e)



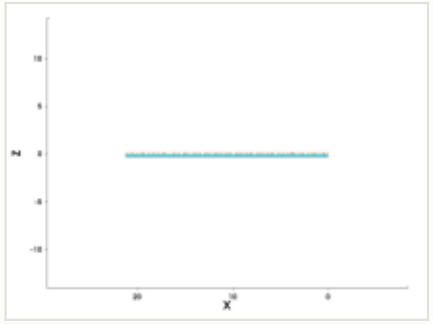
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



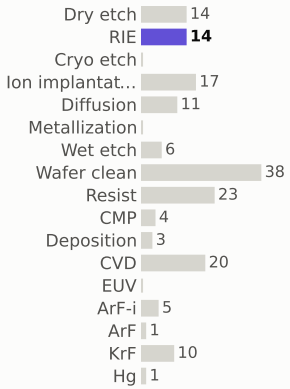
y cut at 57.351 µm



mask for this step

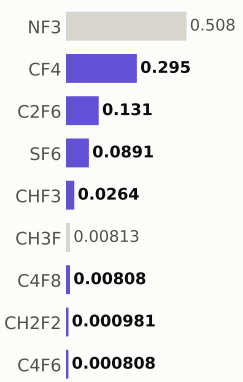
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



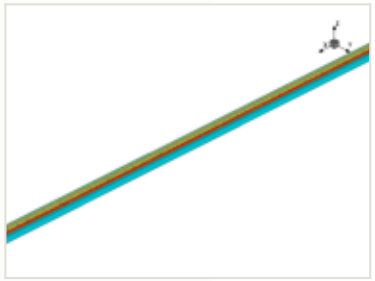
Step #2100: Spx_only

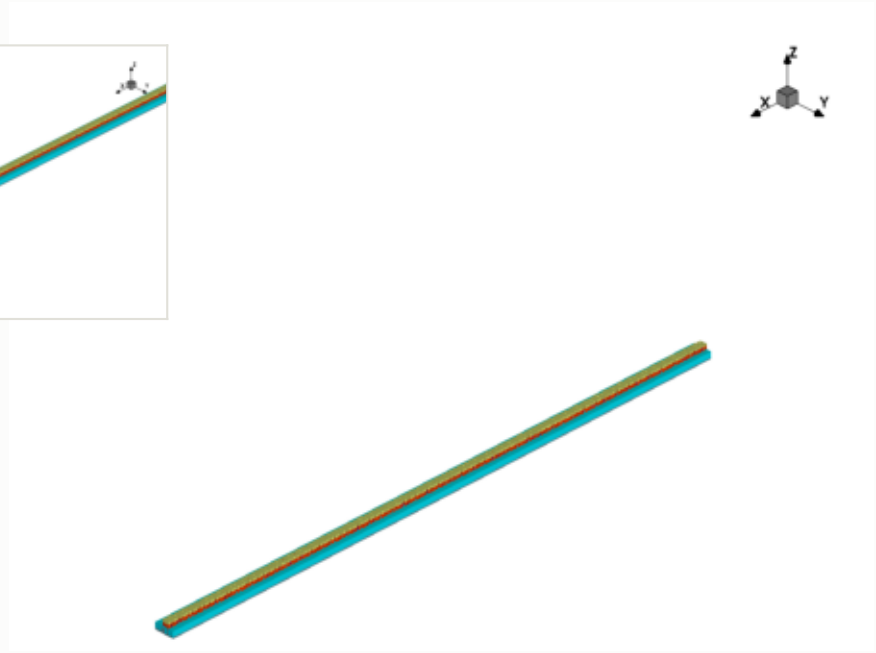
etch_isotropic · Spx_only

0	117.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



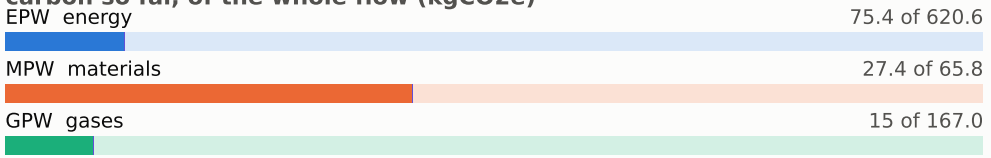


this step

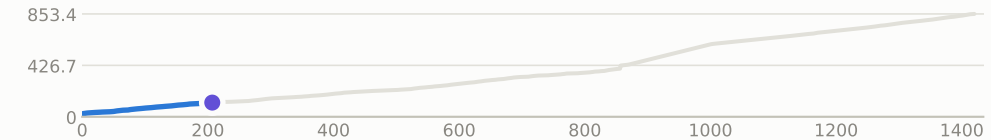
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

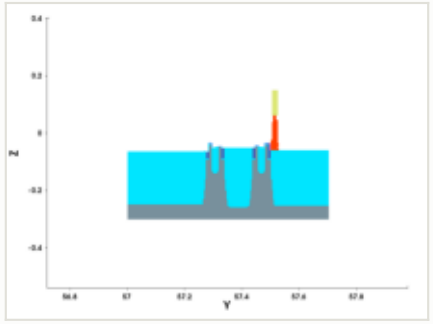
carbon so far, of the whole flow (kgCO2e)



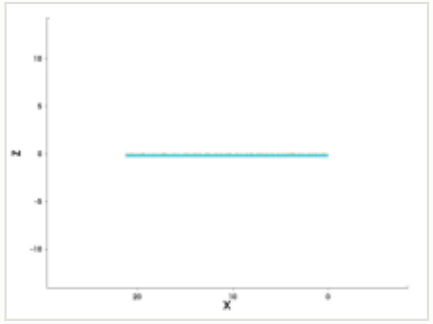
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



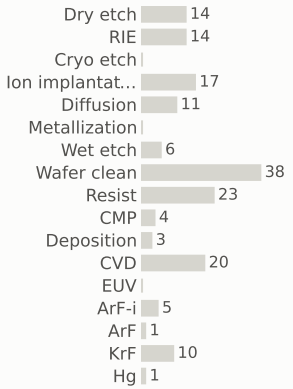
y cut at 57.351 µm



mask for this step

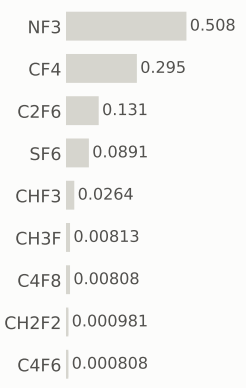
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



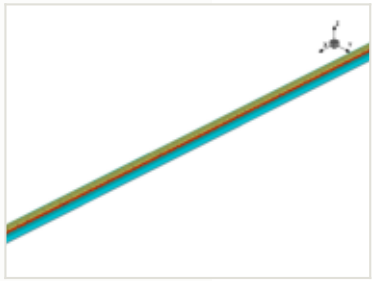
Step #2110: Spx_only

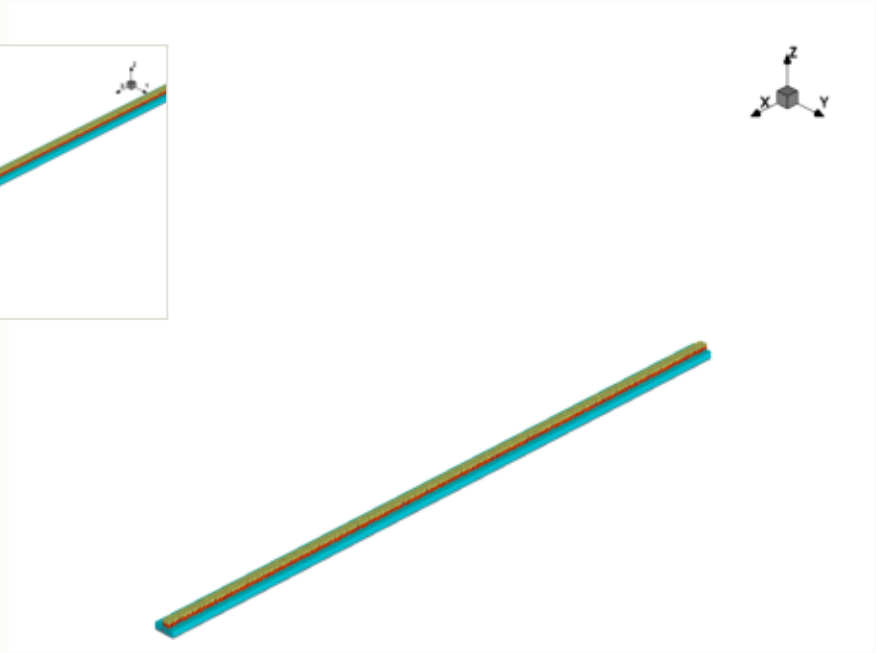
etch_cmp · Spx_only

0	117.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





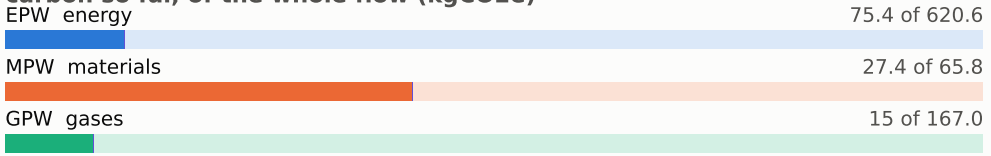
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

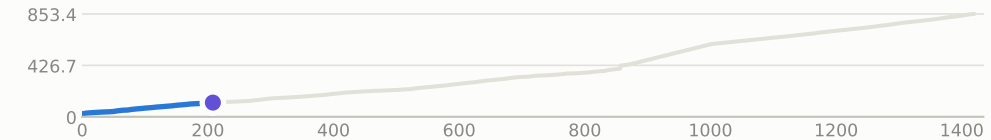
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

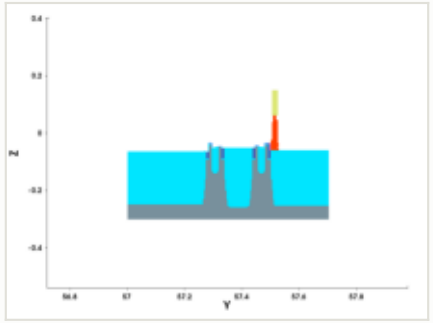
carbon so far, of the whole flow (kgCO2e)



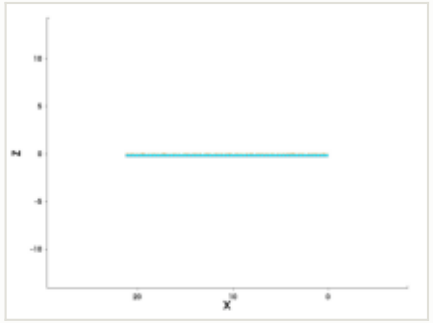
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



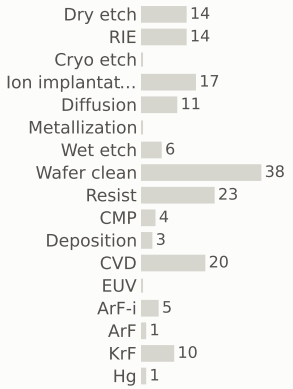
y cut at 57.351 µm



mask for this step

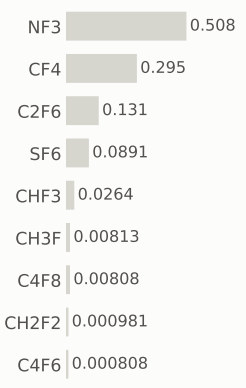
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



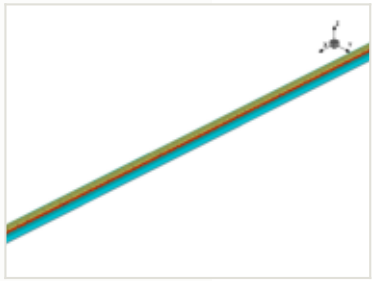
Step #2120: Spx_only

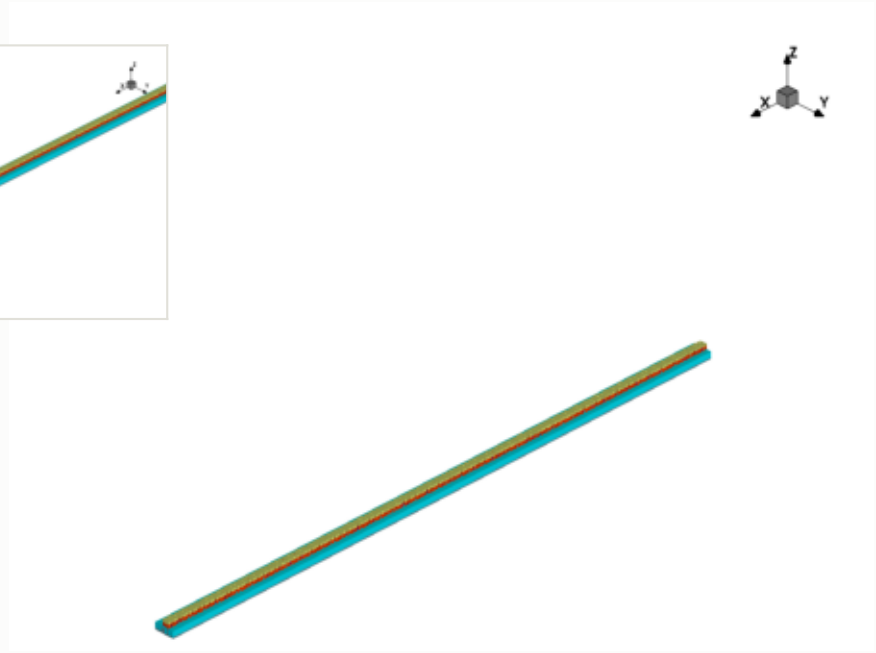
etch_isotropic · Spx_only

0	117.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





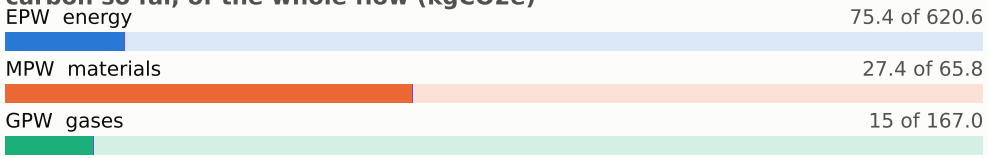
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

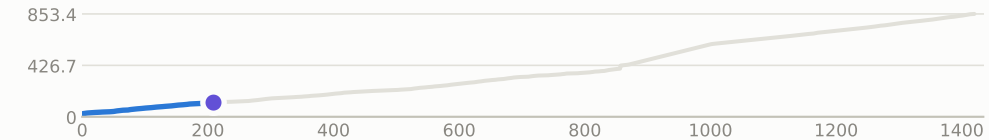
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

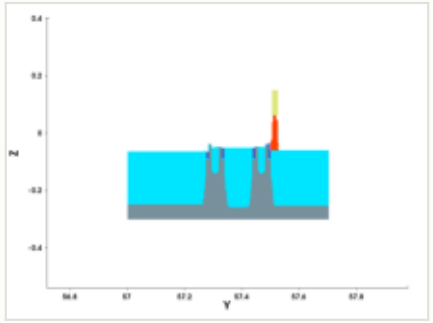
carbon so far, of the whole flow (kgCO2e)



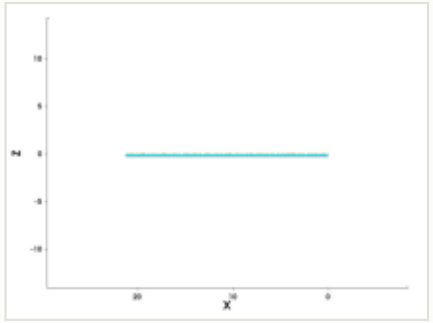
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



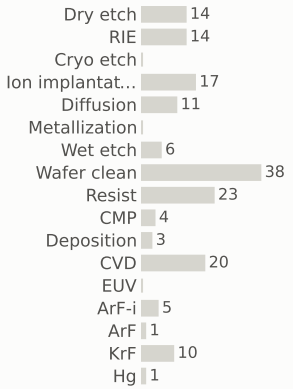
y cut at 57.351 µm



mask for this step

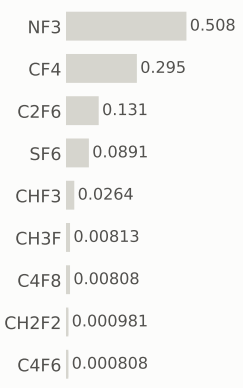
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



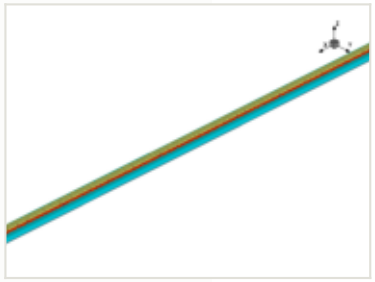
Step #2130: Wafer_clean

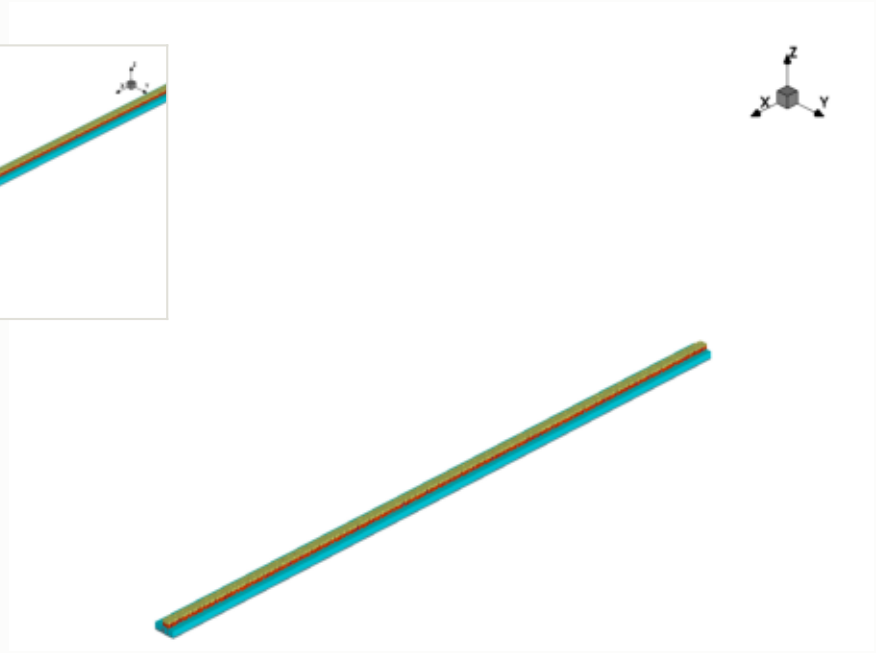
clean · Wafer_clean

0.169	118.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





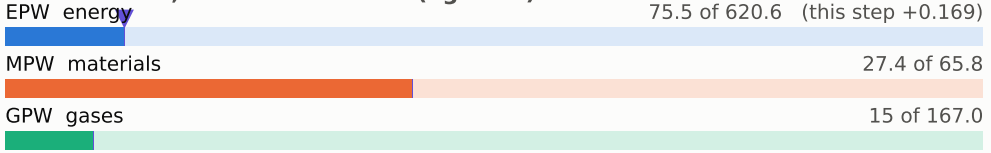
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

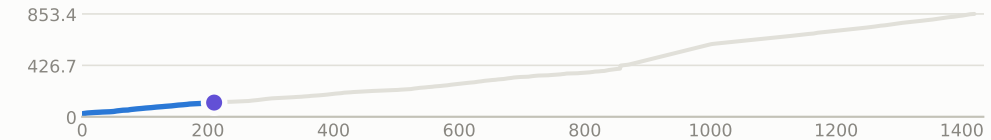
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

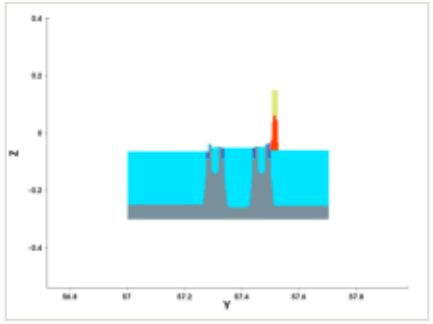
carbon so far, of the whole flow (kgCO2e)



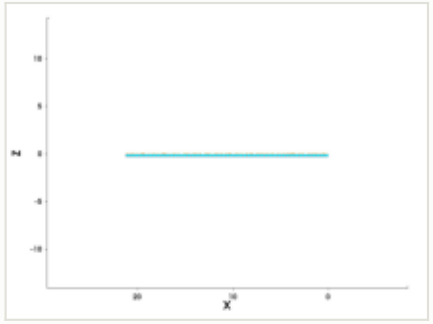
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



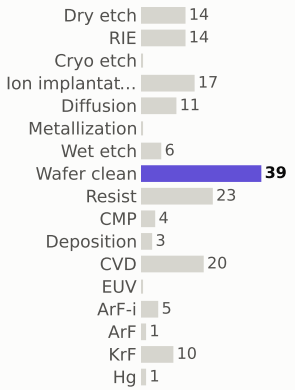
y cut at 57.351 µm



mask for this step

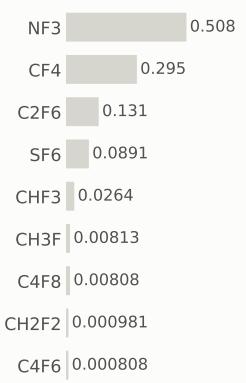
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



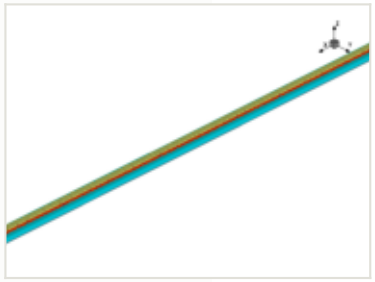
Step #2140: Wafer_clean

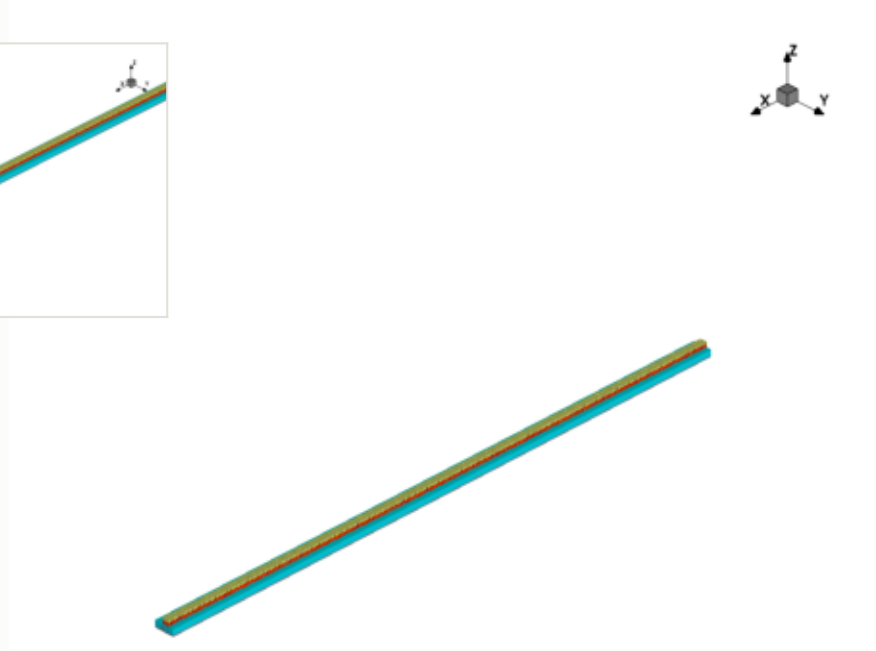
clean · Wafer_clean

0.169	118.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





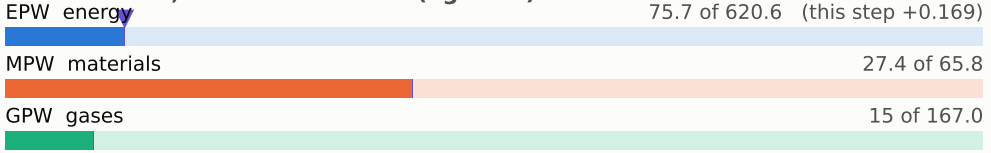
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

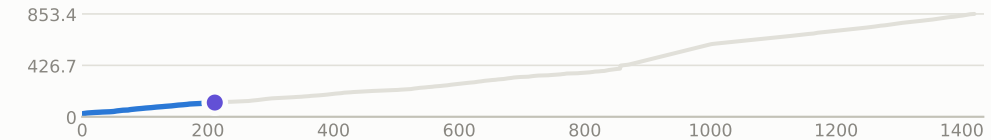
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

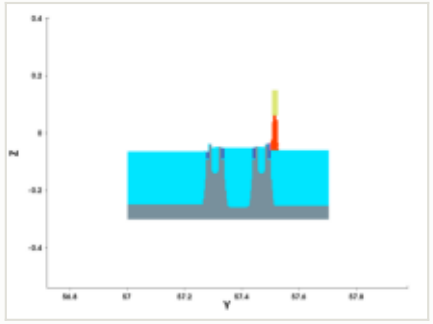
carbon so far, of the whole flow (kgCO2e)



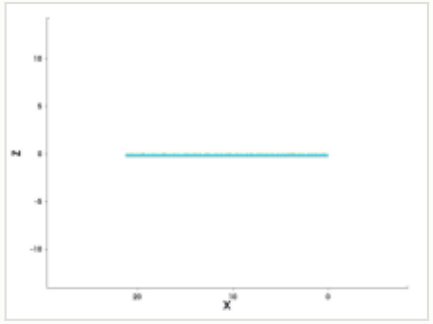
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



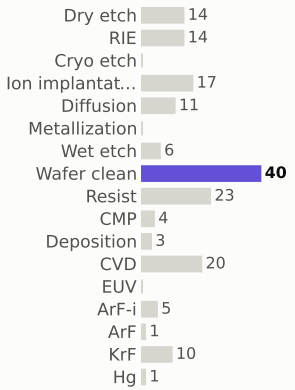
y cut at 57.351 µm



mask for this step

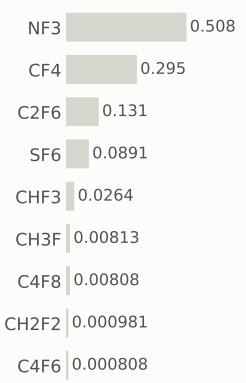
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



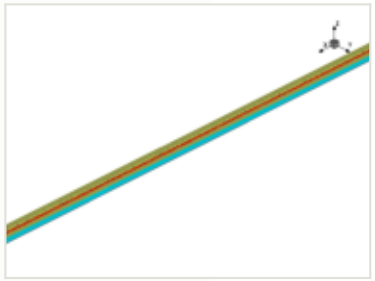
Step #2160: Deposition

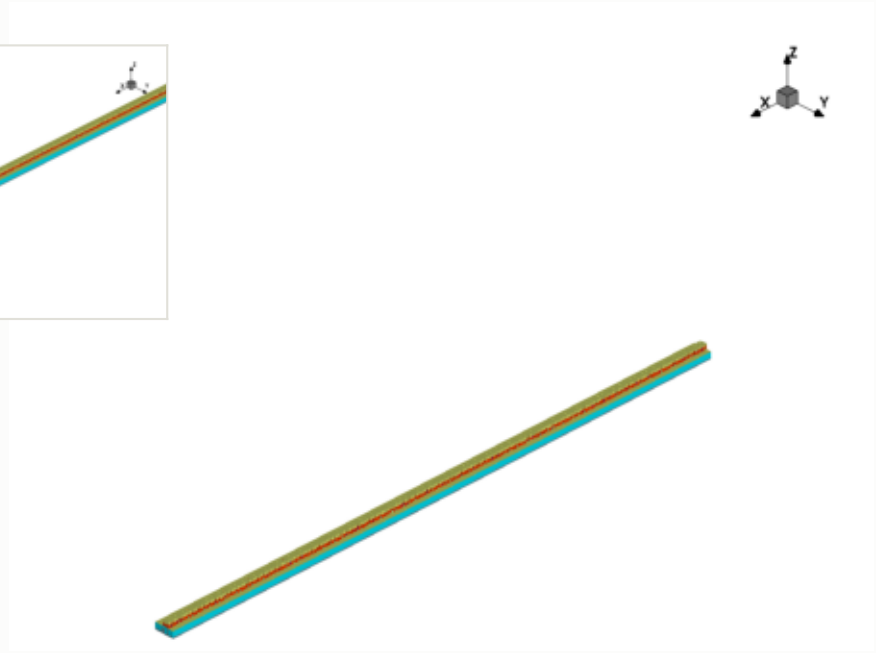
depo_isotropic · Deposition

0.656	118.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



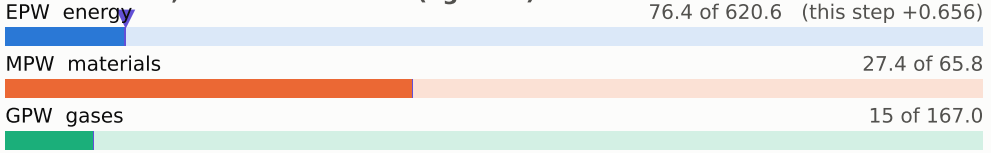


this step

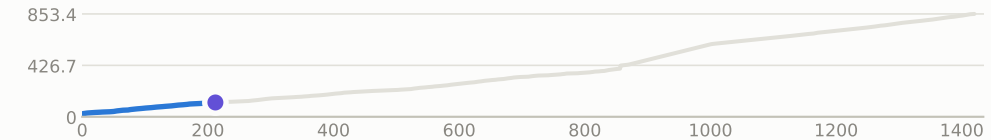
process area	Deposition
material	—
thickness	0.001 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

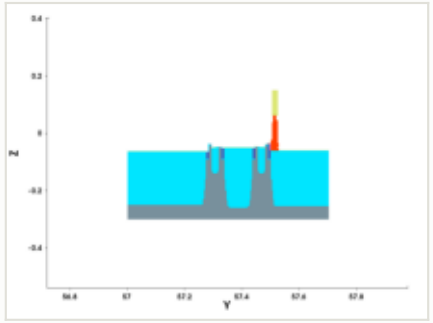
carbon so far, of the whole flow (kgCO2e)



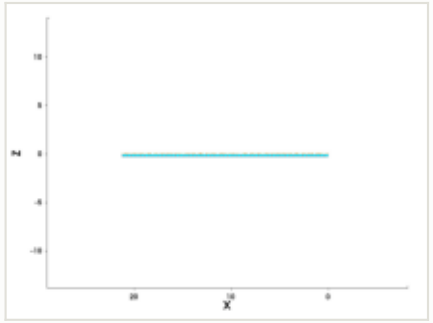
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



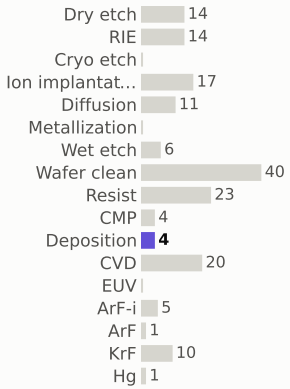
y cut at 57.351 µm



mask for this step

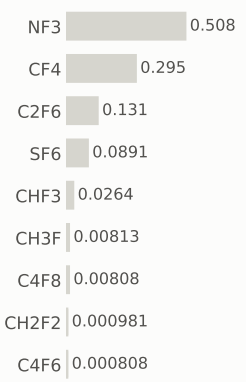
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



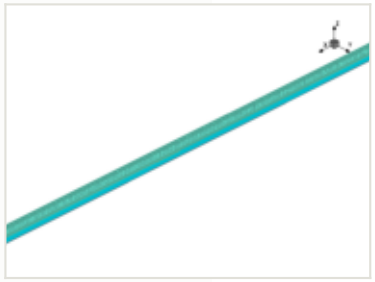
Step #2170: Deposition

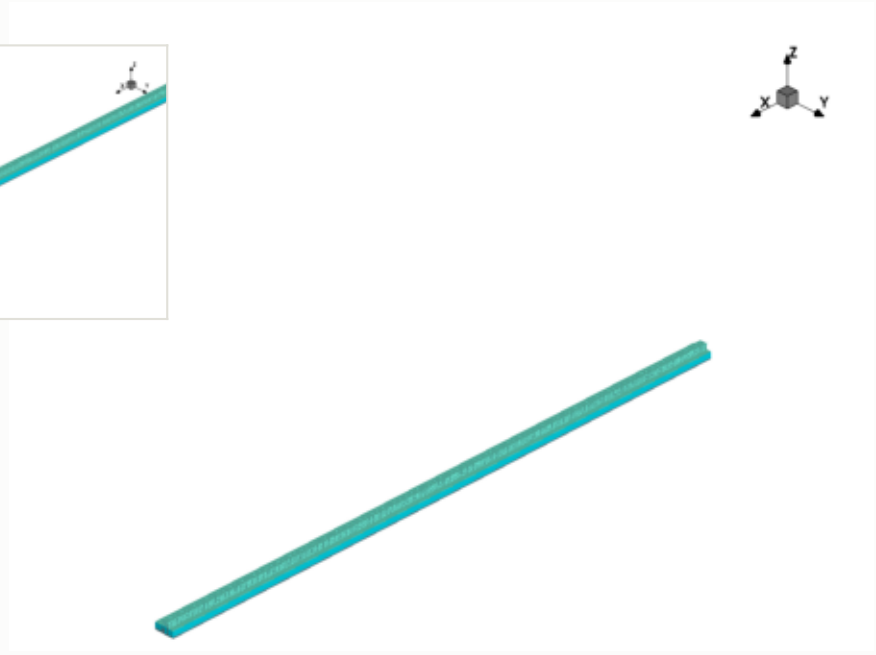
depo_isotropic · Deposition

0.656	119.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



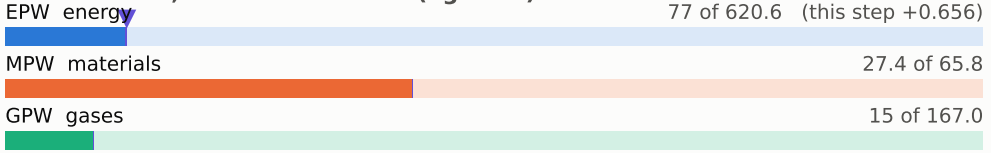


this step

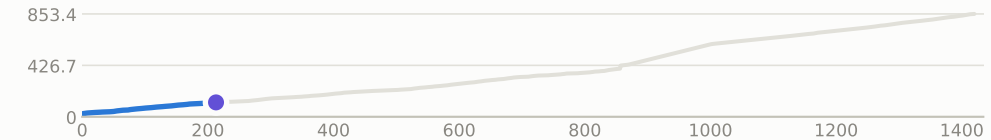
process area	Deposition
material	—
thickness	0.006 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

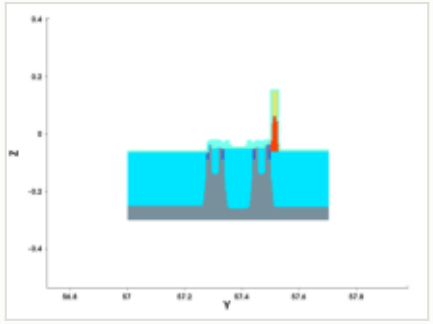
carbon so far, of the whole flow (kgCO2e)



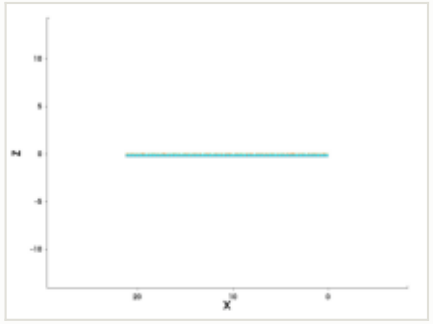
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



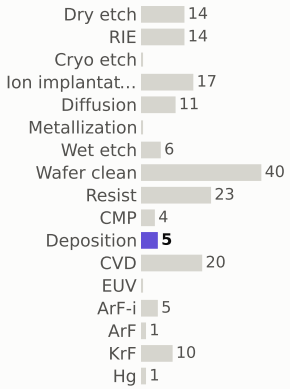
y cut at 57.351 µm



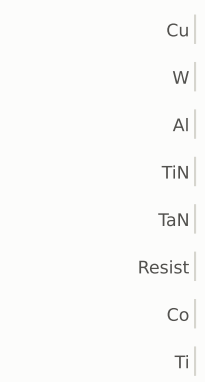
mask for this step

no mask for this step

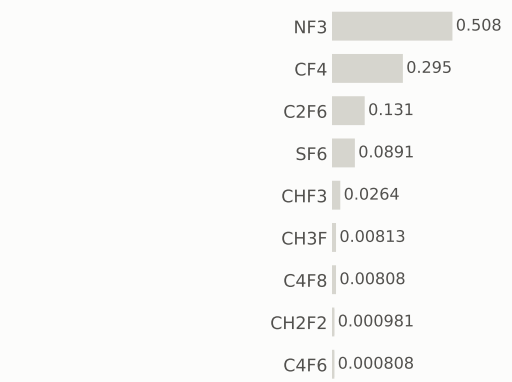
steps by process area



materials deposited (g)



gases emitted (g)



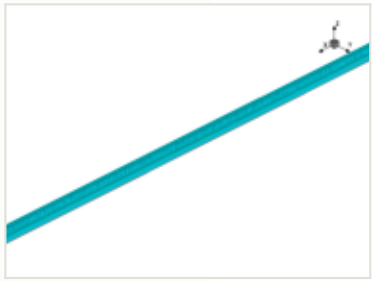
Step #2180: Deposition

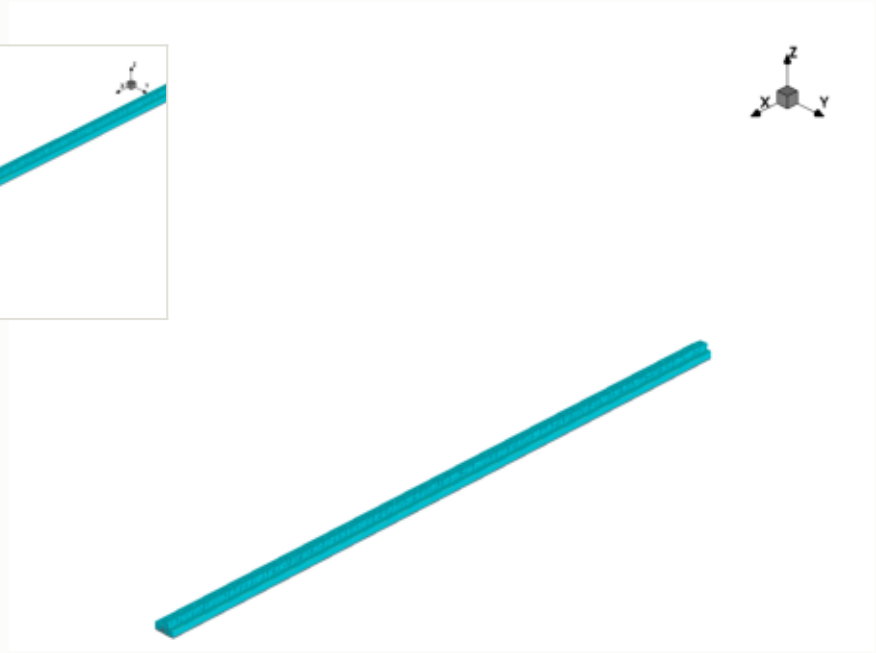
depo_isotropic · Deposition

0.656	120.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



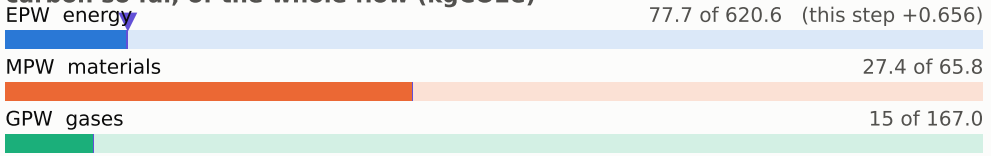


this step

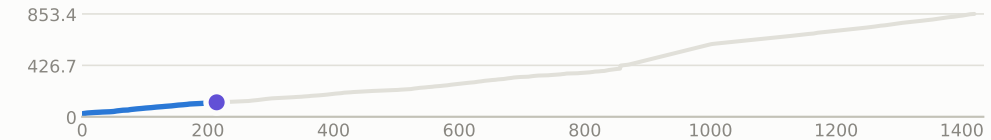
process area	Deposition
material	—
thickness	0.003 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

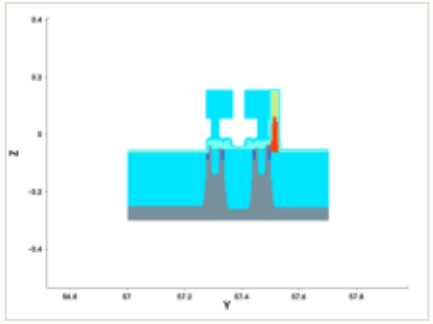
carbon so far, of the whole flow (kgCO2e)



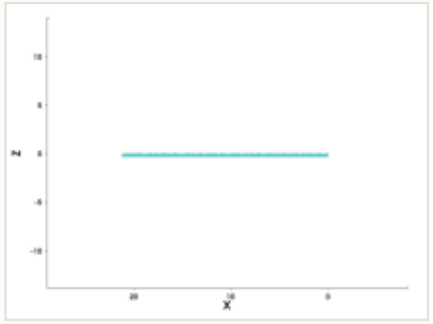
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



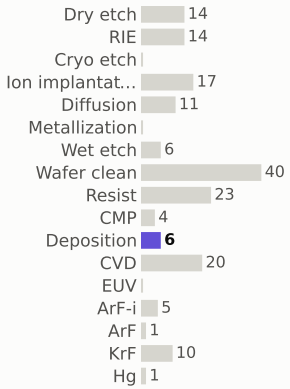
y cut at 57.351 µm



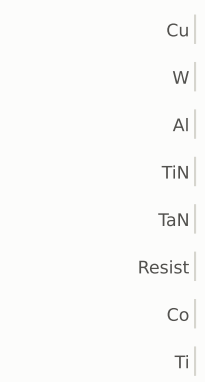
mask for this step

no mask for this step

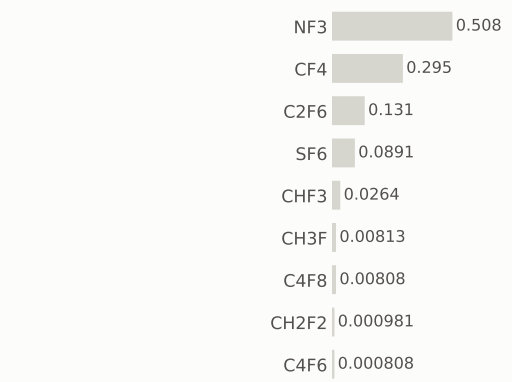
steps by process area



materials deposited (g)



gases emitted (g)



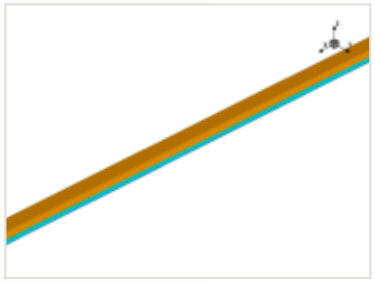
Step #2190: Resist

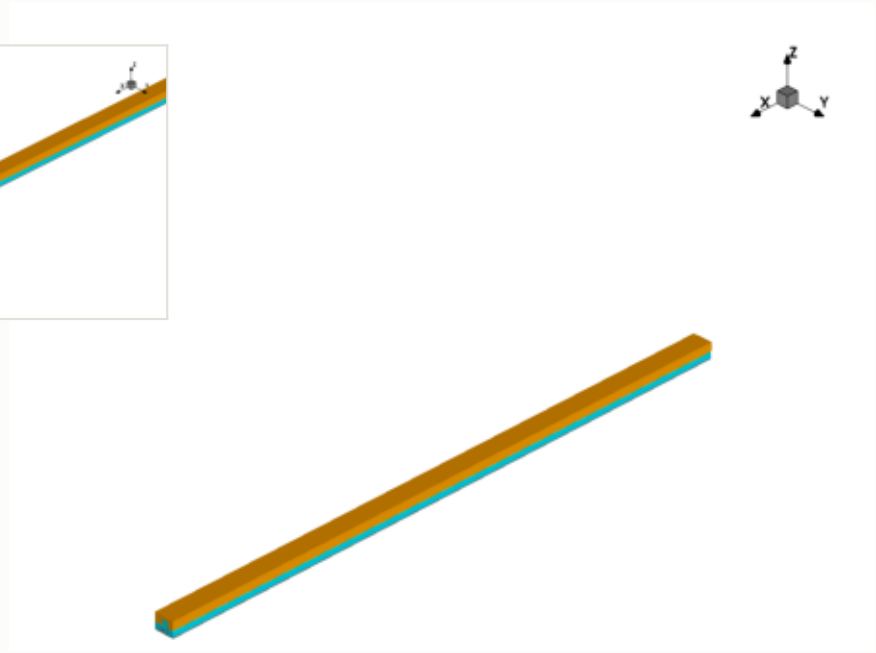
depo_fill · Resist

0.169	120.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



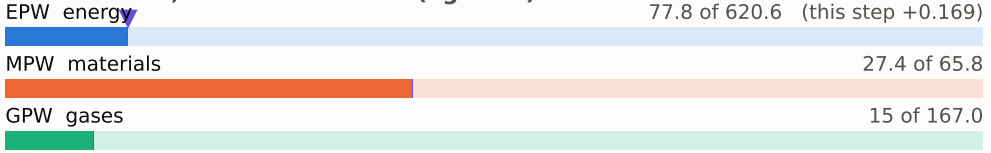


this step

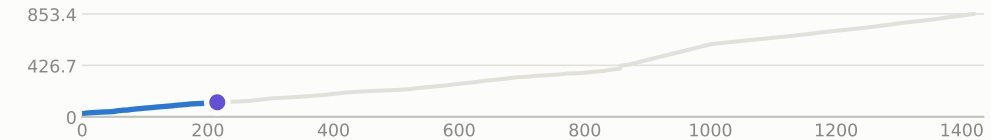
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

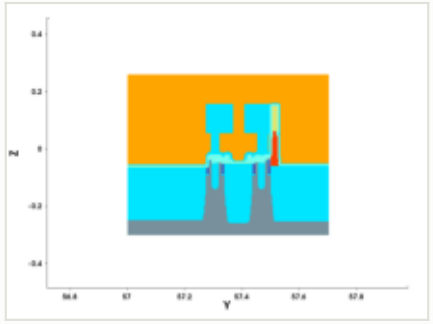
carbon so far, of the whole flow (kgCO2e)



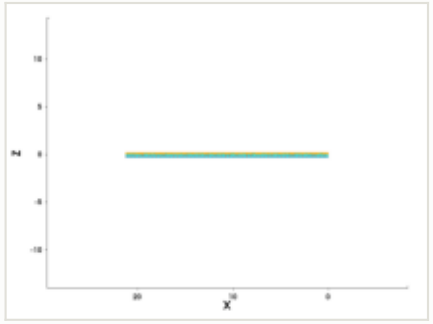
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



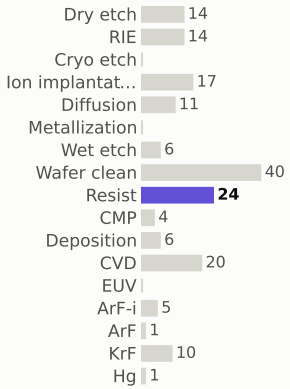
y cut at 57.351 µm



mask for this step

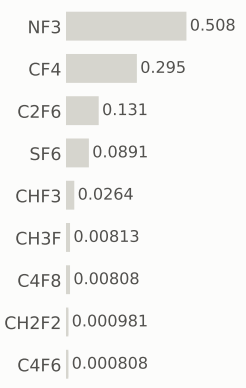
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



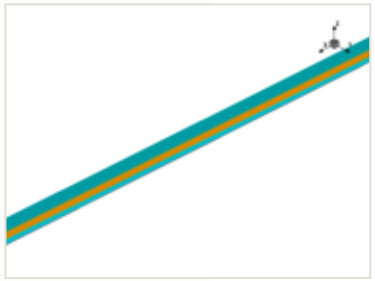
Step #2200: Deposition

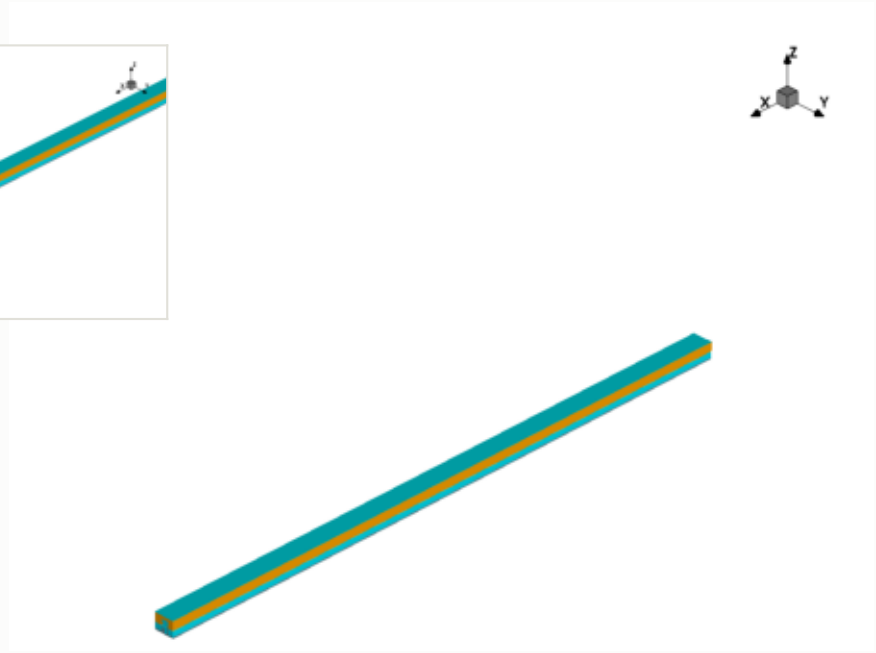
depo_isotropic · Deposition

0.656	121.0	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



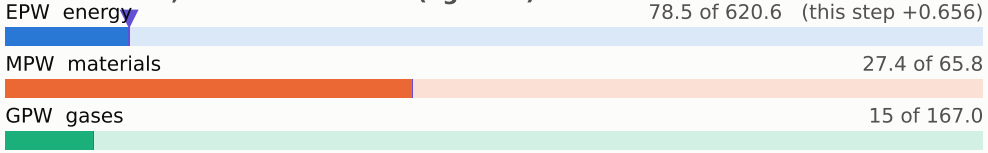


this step

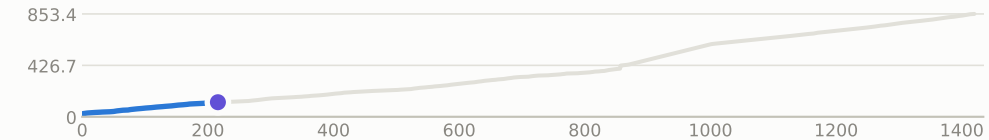
process area	Deposition
material	—
thickness	0.01 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

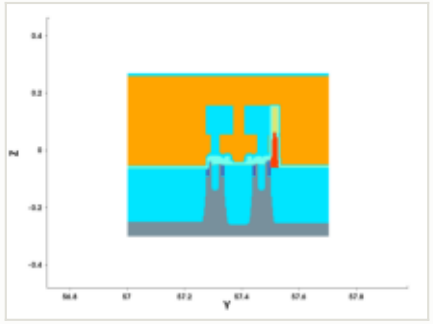
carbon so far, of the whole flow (kgCO2e)



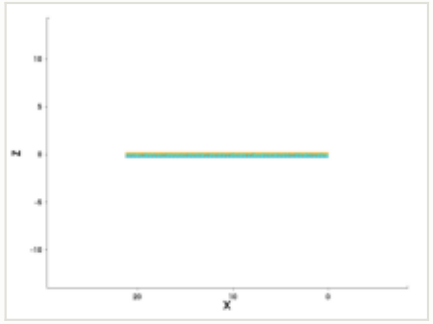
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



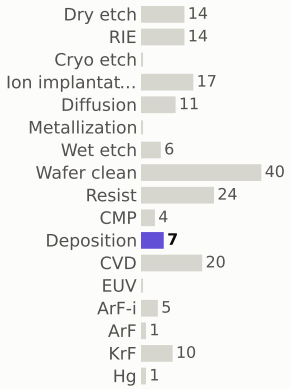
y cut at 57.351 µm



mask for this step

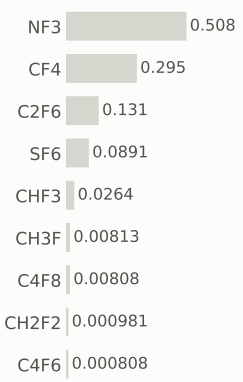
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



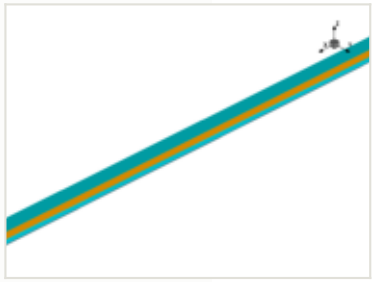
Step #2210: Wafer_clean

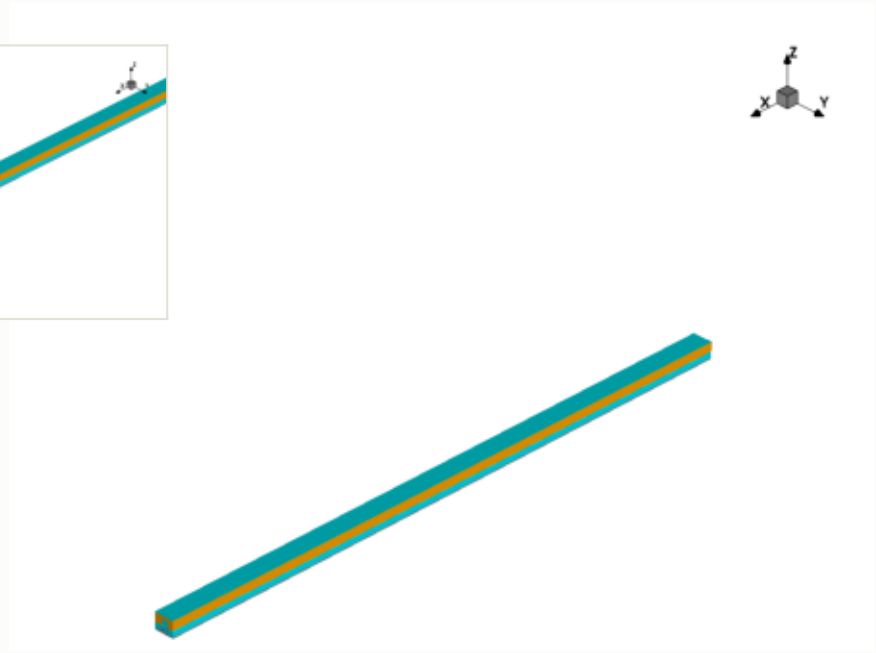
clean · Wafer_clean

0.169	121.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





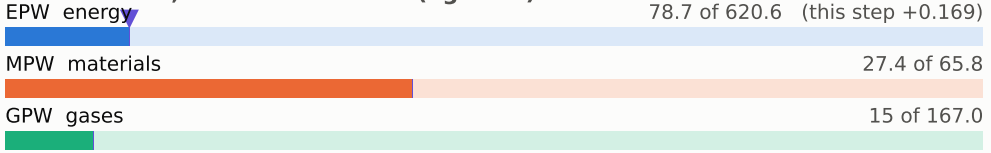
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

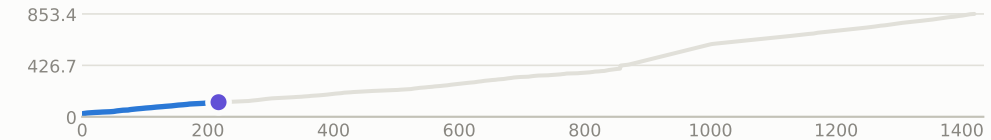
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

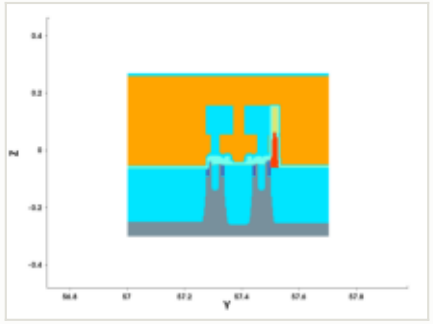
carbon so far, of the whole flow (kgCO2e)



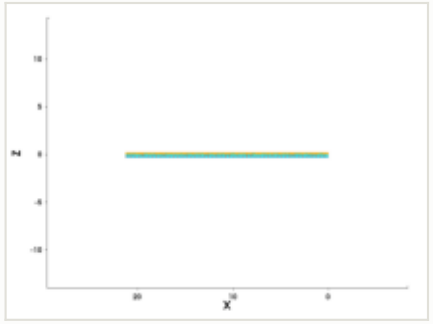
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



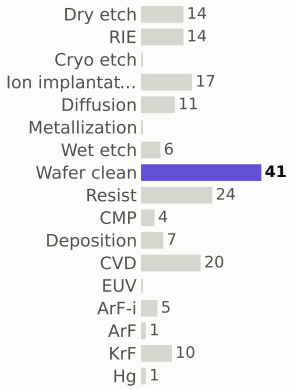
y cut at 57.351 µm



mask for this step

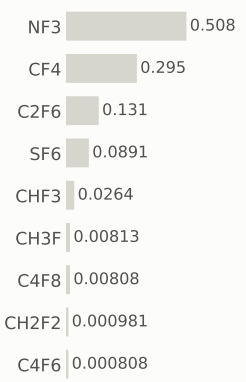
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



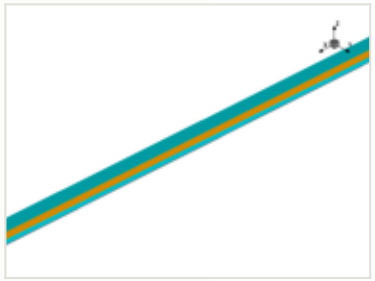
Step #2220: Resist

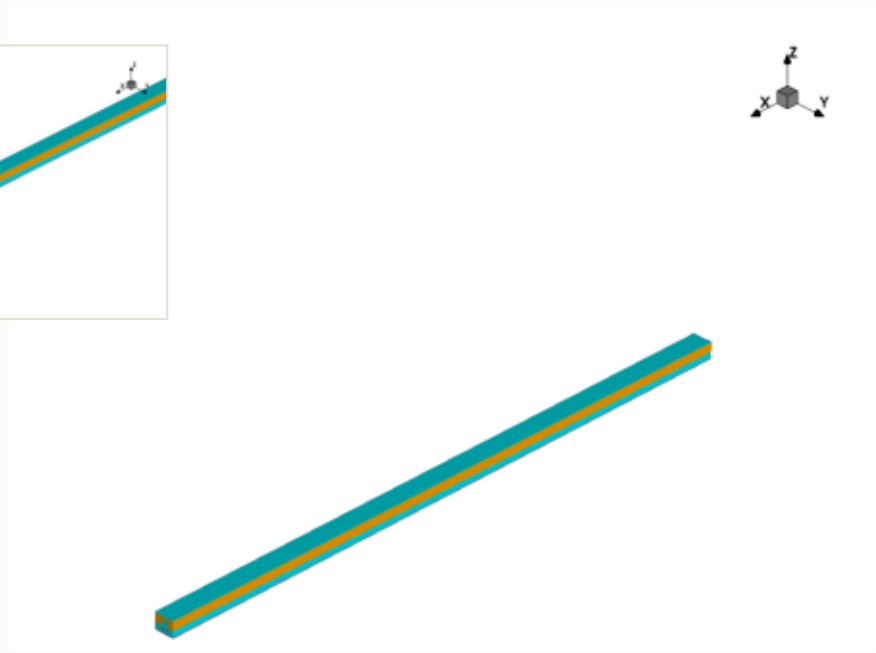
resist · Resist

0.169	121.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





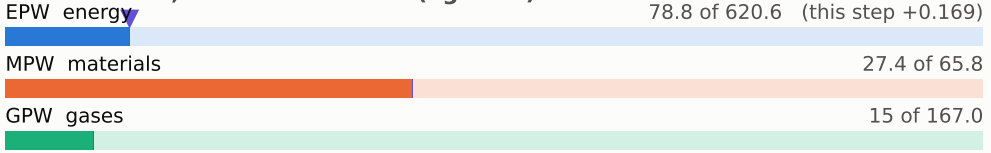
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

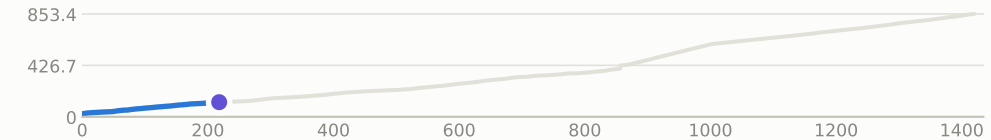
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

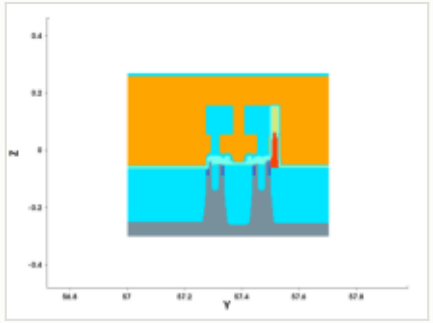
carbon so far, of the whole flow (kgCO2e)



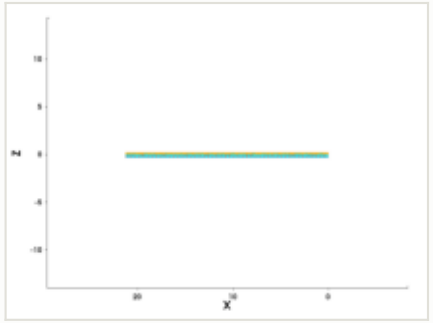
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



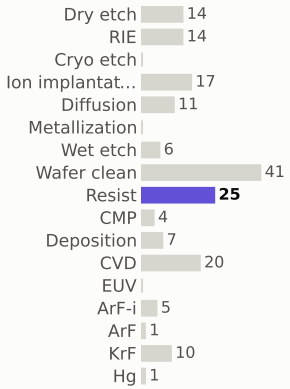
y cut at 57.351 µm



mask for this step

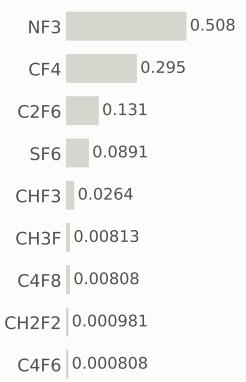
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



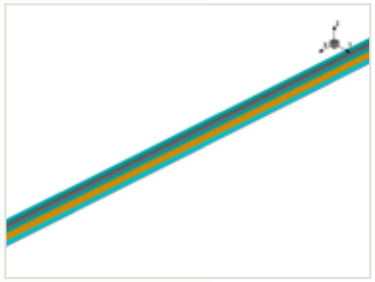
Step #2230: ArF-i

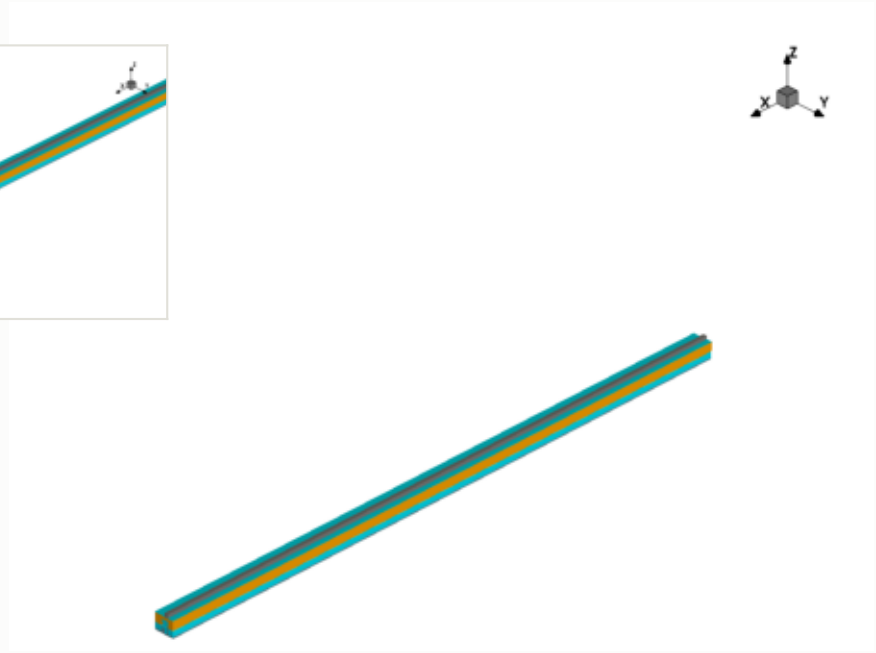
litho_patterning · ArF-i

0.217	121.5	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



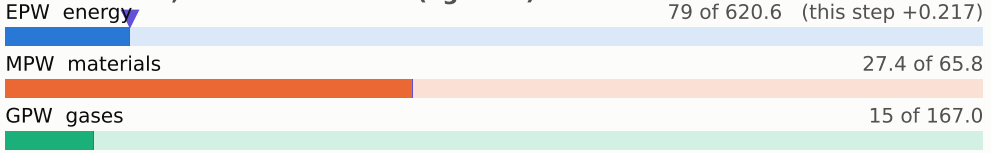


this step

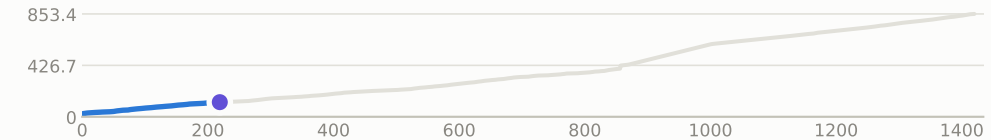
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

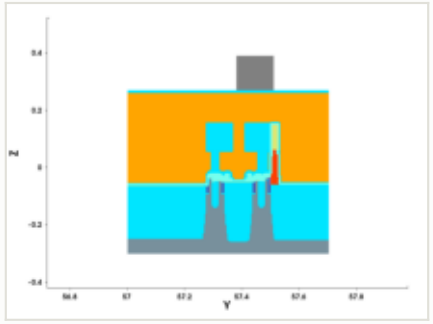
carbon so far, of the whole flow (kgCO2e)



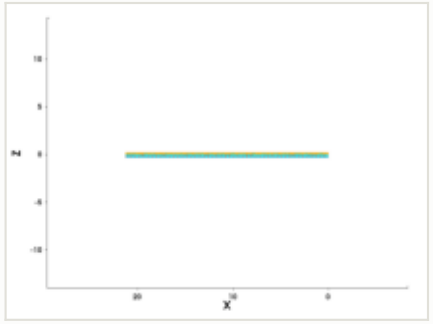
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

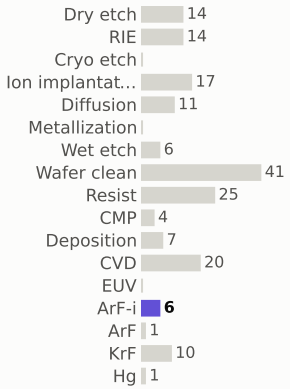


y cut at 57.351 µm

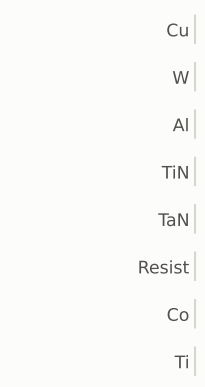


mask: layer 2013:0, positive (GDSII)

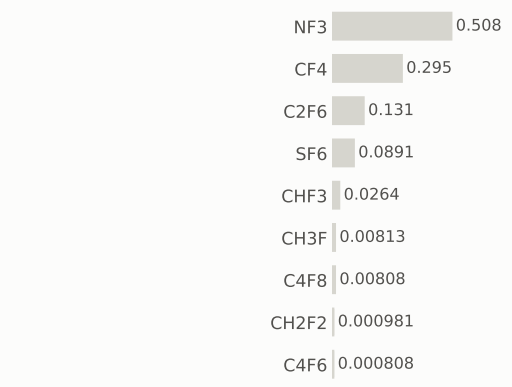
steps by process area



materials deposited (g)



gases emitted (g)



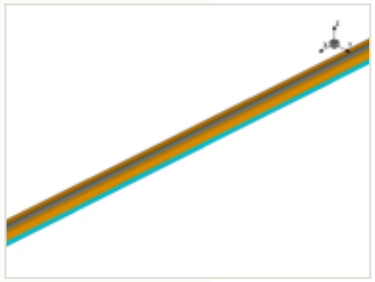
Step #2240: RIE

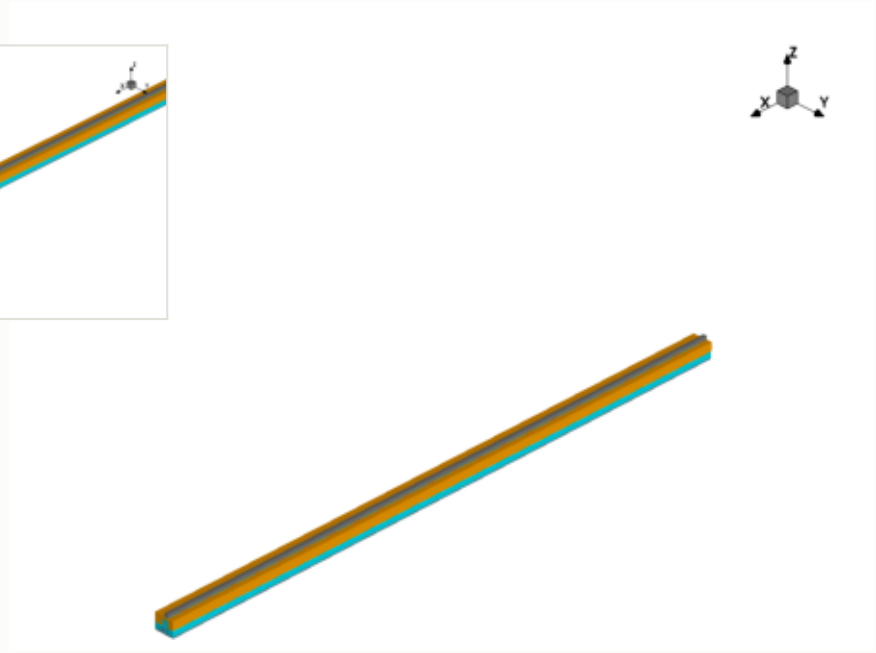
etch_anisotropic · RIE

1.04	122.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



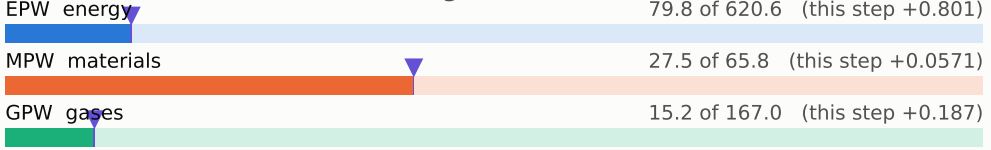


this step

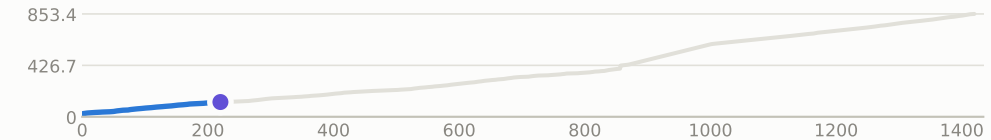
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

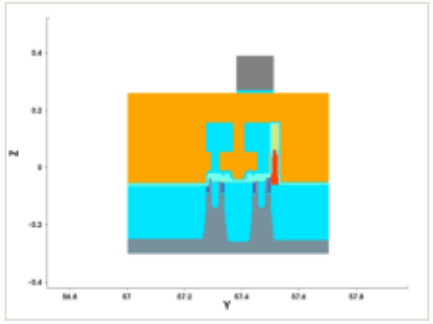
carbon so far, of the whole flow (kgCO2e)



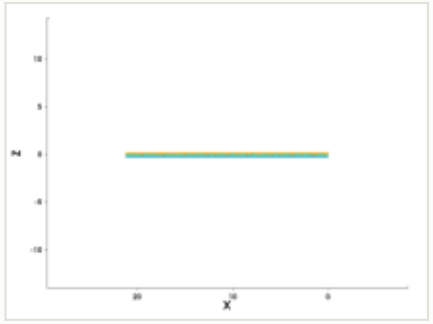
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



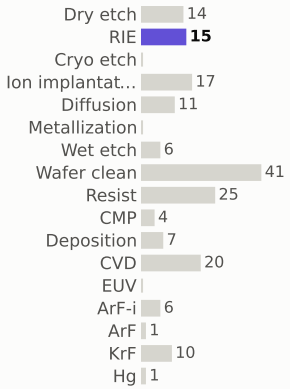
y cut at 57.351 µm



mask for this step

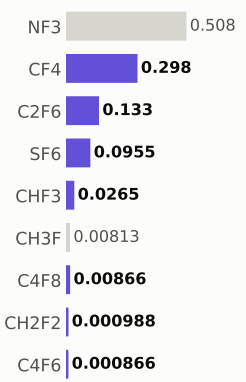
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



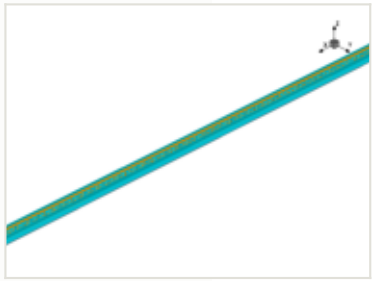
Step #2250: Spx_only

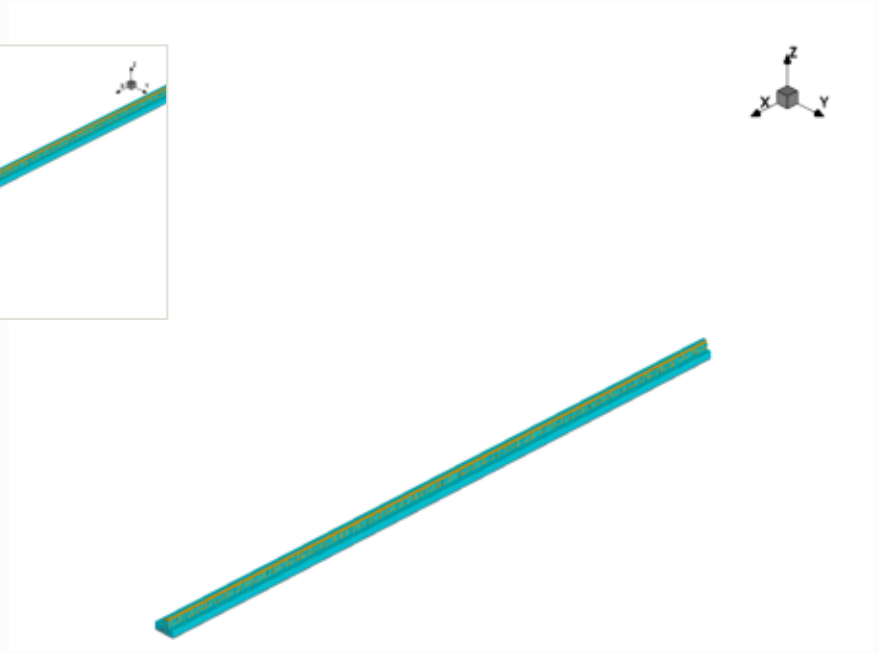
etch_anisotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





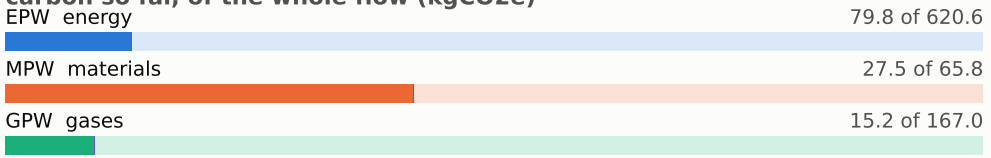
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

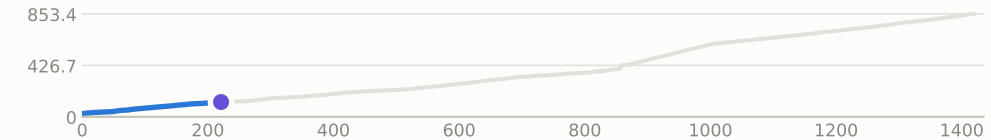
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

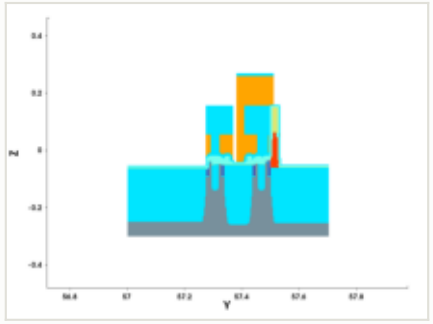
carbon so far, of the whole flow (kgCO2e)



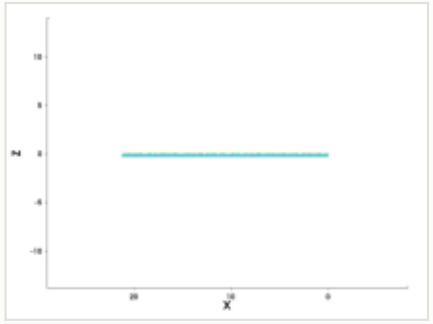
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



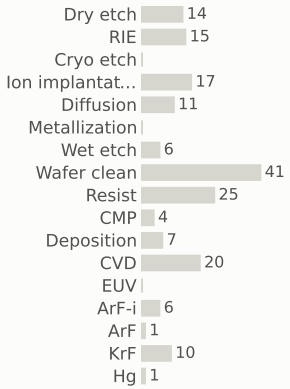
y cut at 57.351 µm



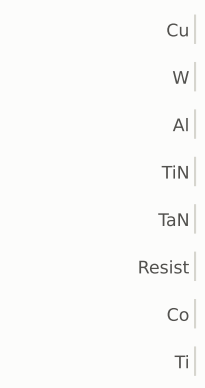
mask for this step

no mask for this step

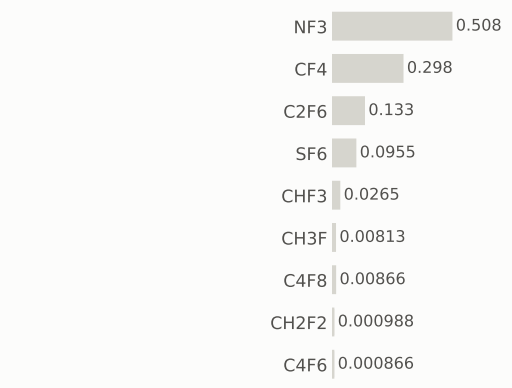
steps by process area



materials deposited (g)



gases emitted (g)



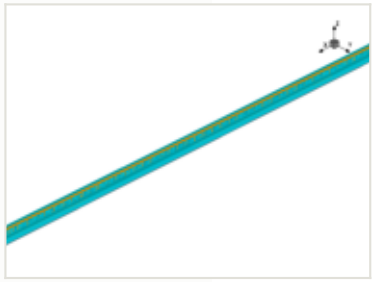
Step #2260: Spx_only

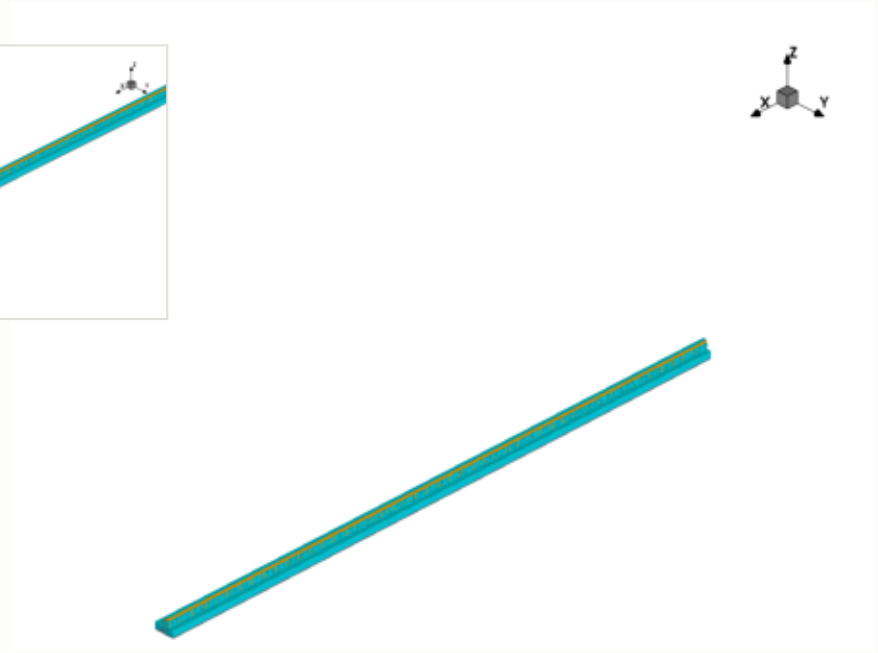
etch_isotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





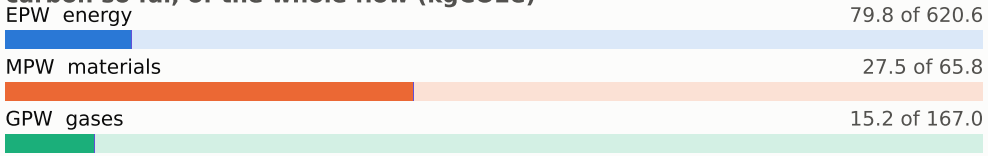
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

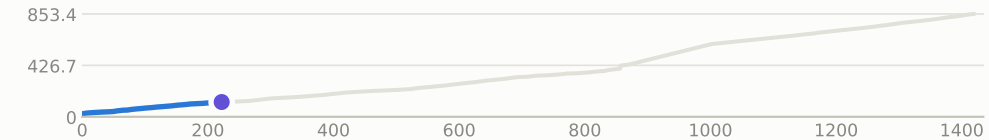
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

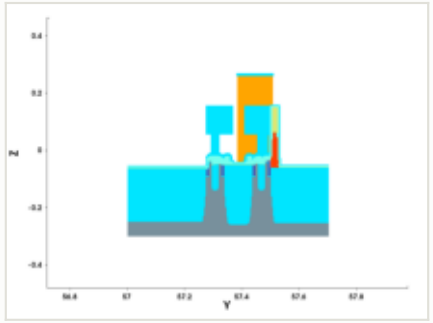
carbon so far, of the whole flow (kgCO2e)



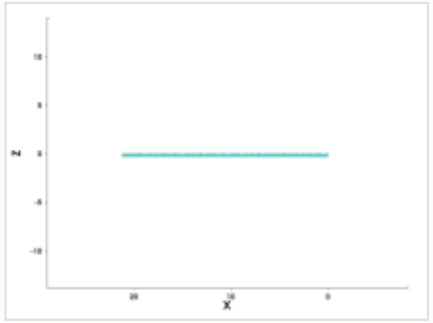
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



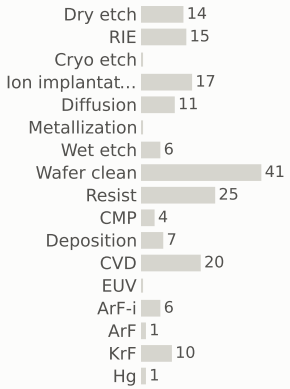
y cut at 57.351 µm



mask for this step

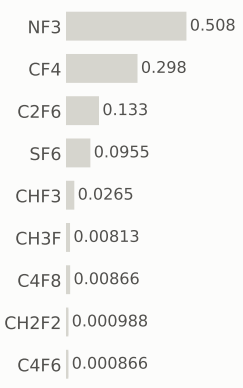
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



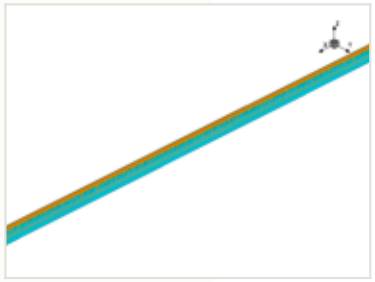
Step #2270: Spx_only

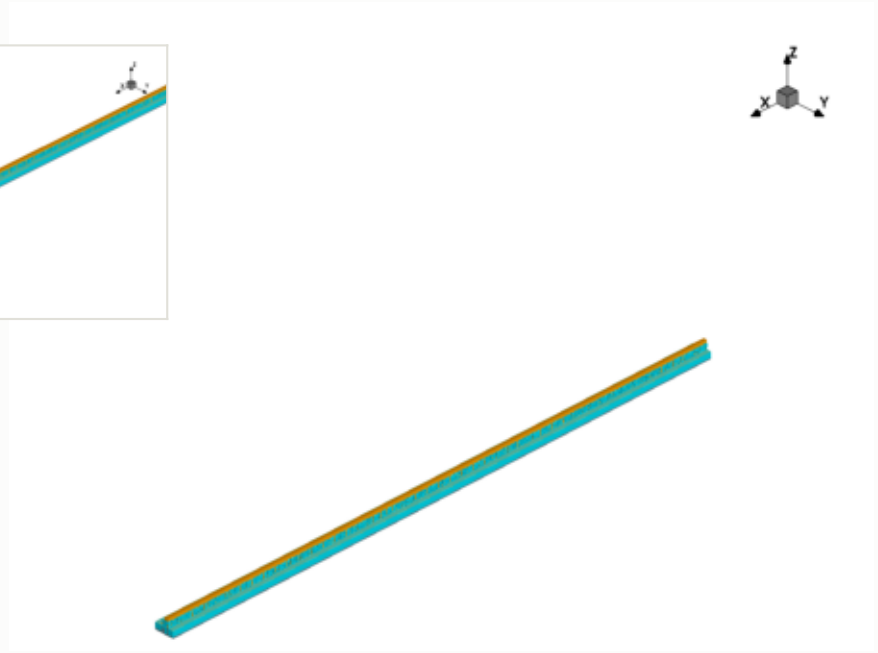
etch_anisotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





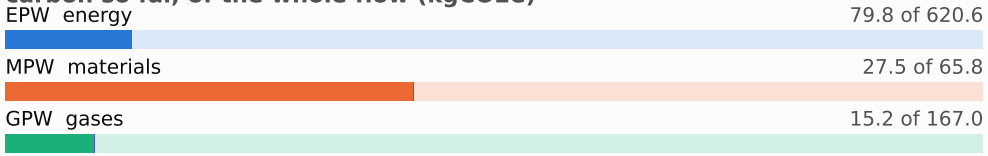
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

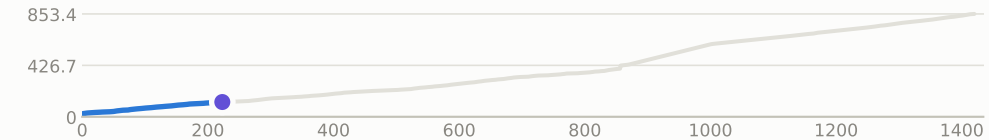
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

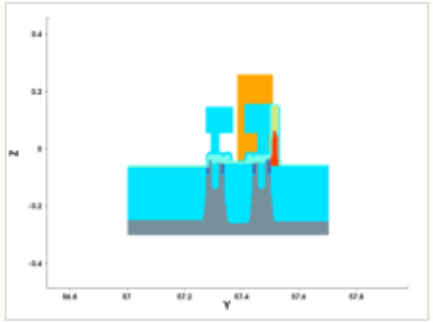
carbon so far, of the whole flow (kgCO2e)



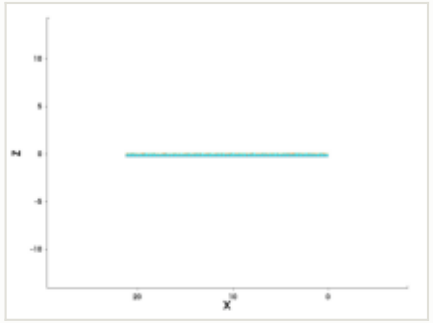
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



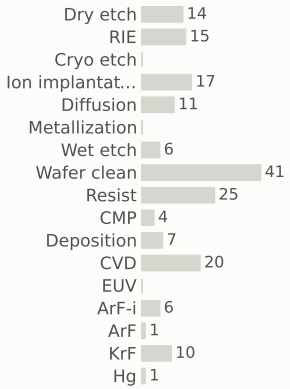
y cut at 57.351 µm



mask for this step

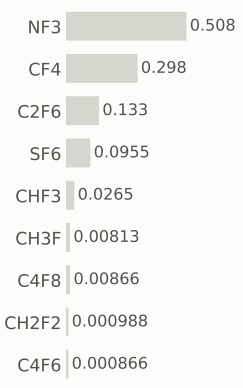
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



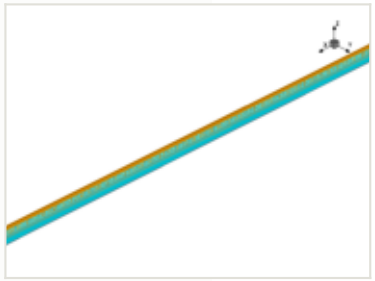
Step #2280: Spx_only

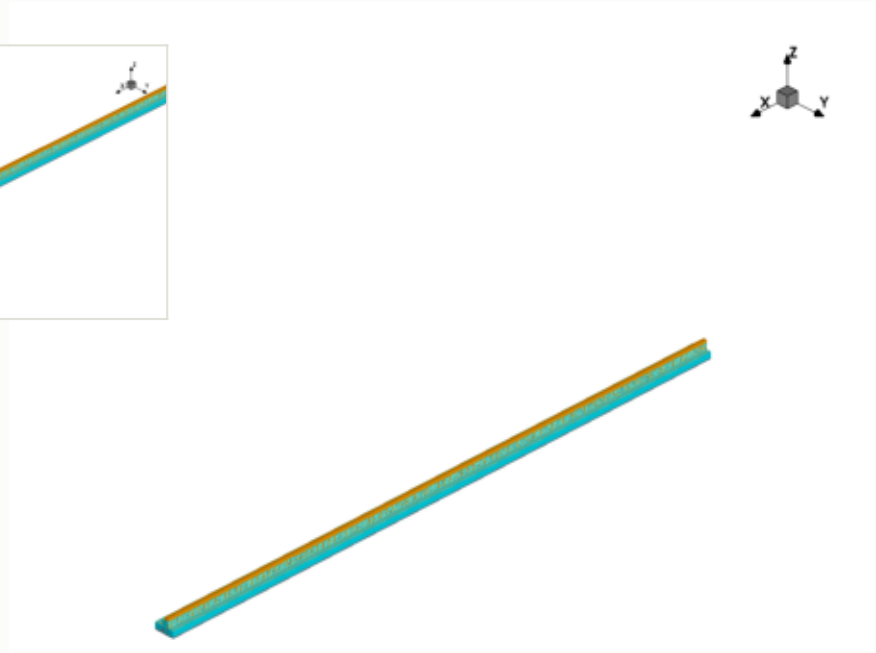
etch_isotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

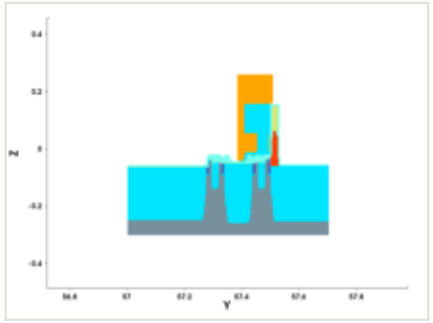
3D structure after this step

zoomed to fit

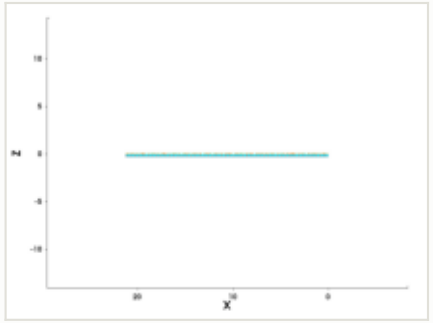




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

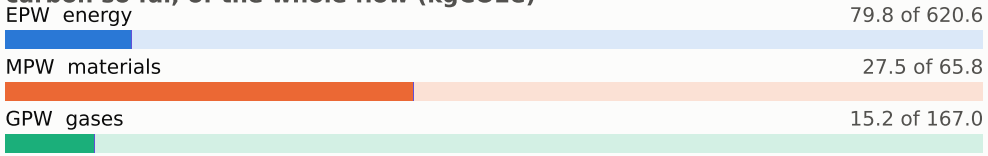
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

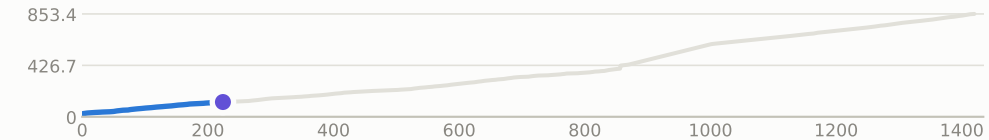
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

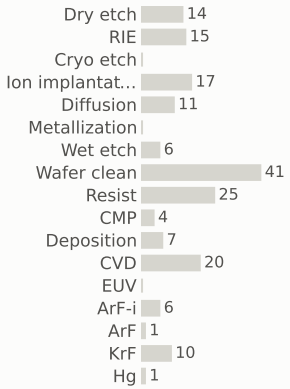
carbon so far, of the whole flow (kgCO2e)



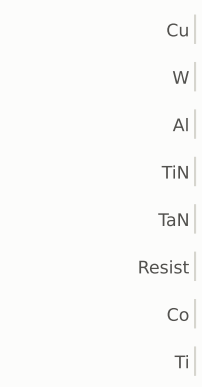
embodied carbon up to each step (kgCO2e per wafer)



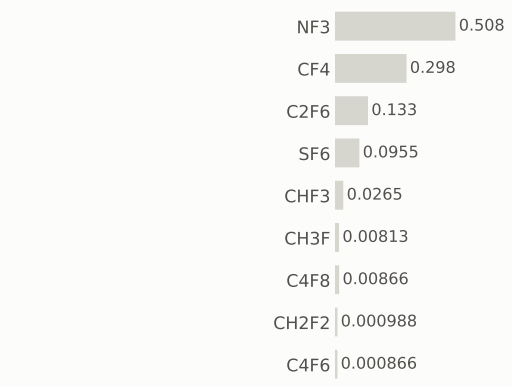
steps by process area



materials deposited (g)



gases emitted (g)



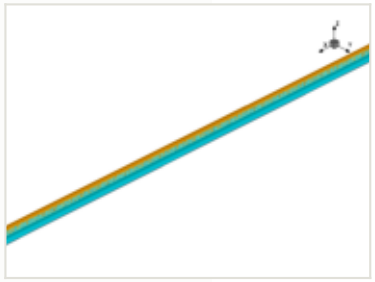
Step #2290: Spx_only

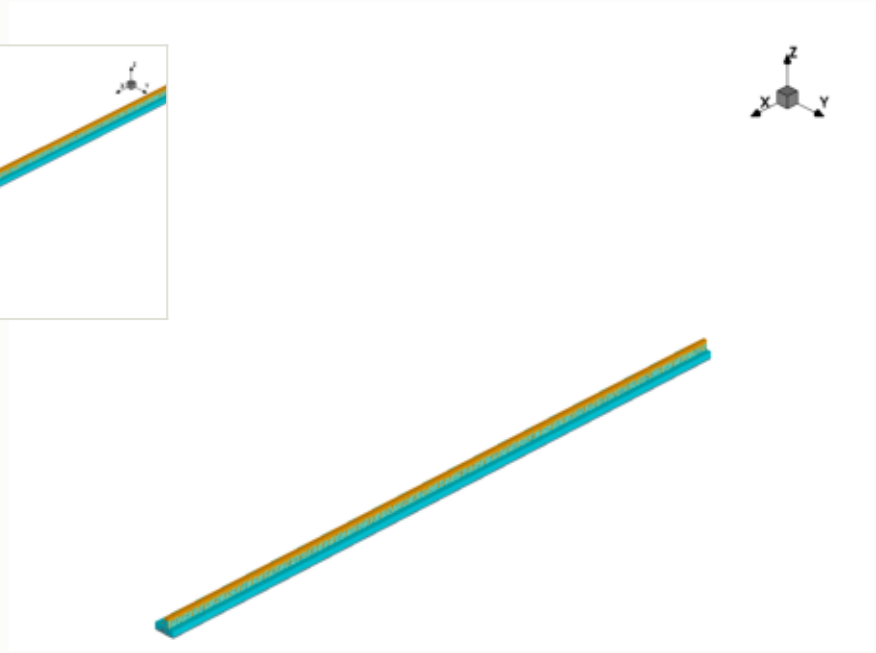
etch_anisotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

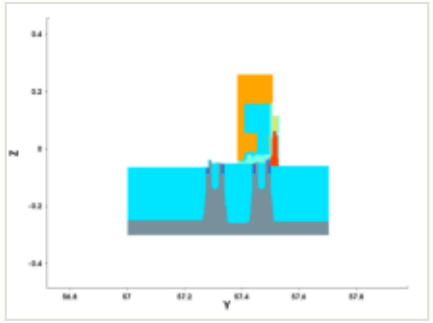
3D structure after this step

zoomed to fit

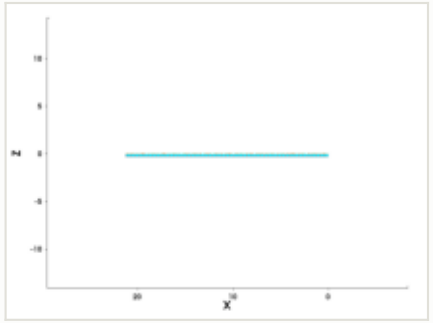




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

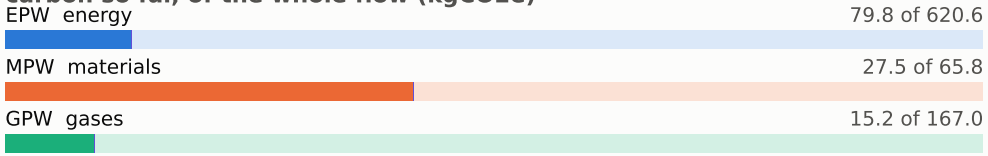
no mask for this step

this step

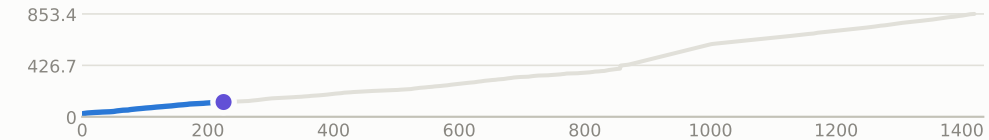
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

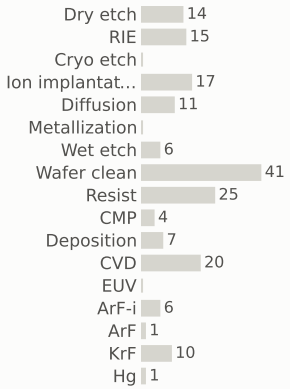
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

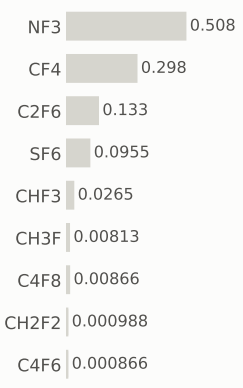


steps by process area



materials deposited (g)

gases emitted (g)



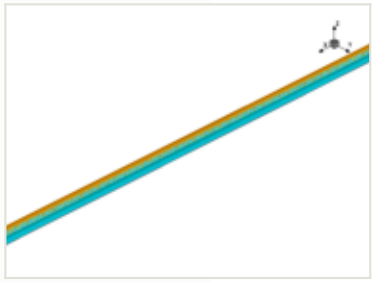
Step #2300: Spx_only

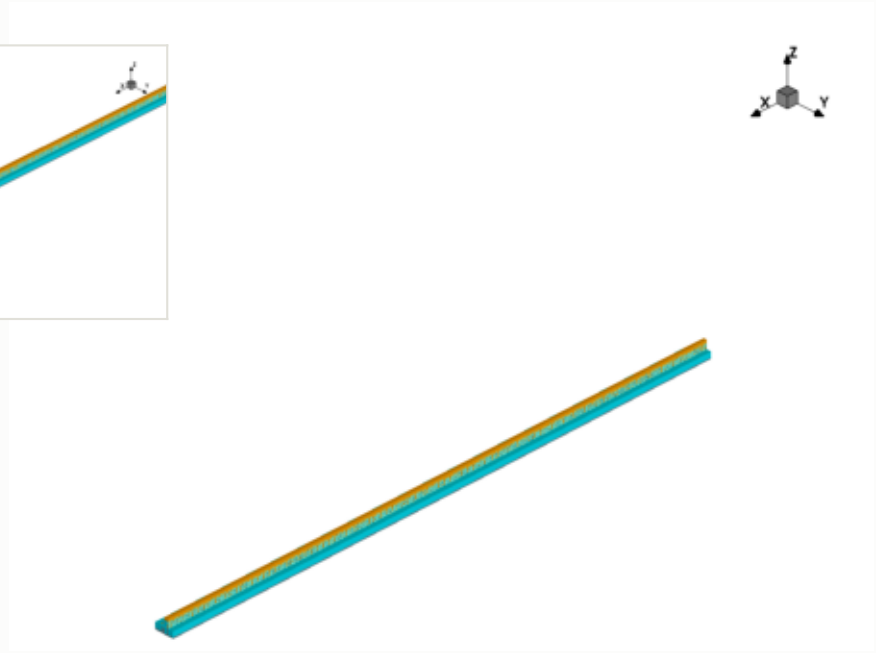
etch_anisotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





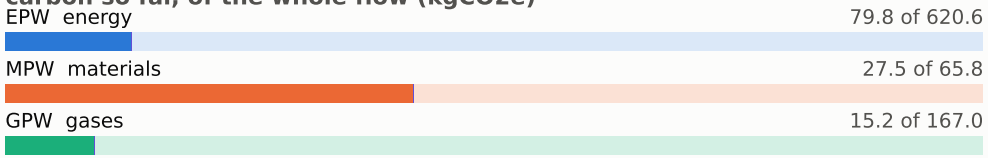
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

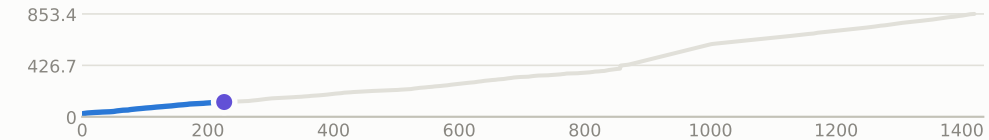
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

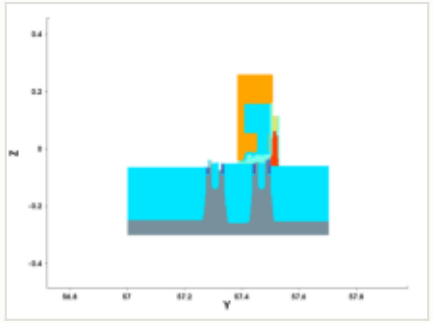
carbon so far, of the whole flow (kgCO2e)



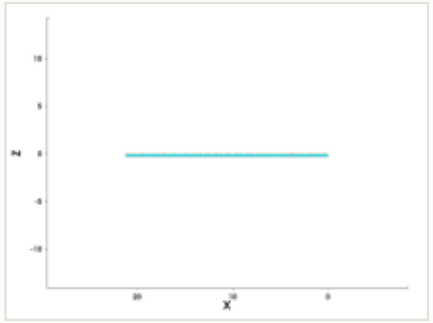
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



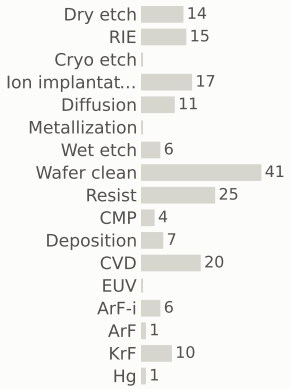
y cut at 57.351 µm



mask for this step

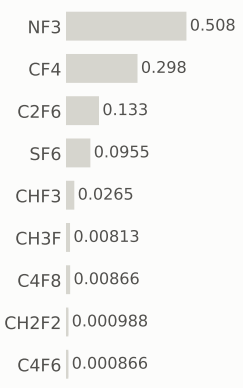
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



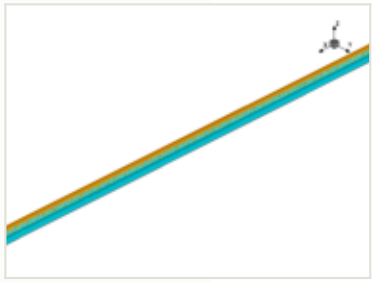
Step #2310: Spx_only

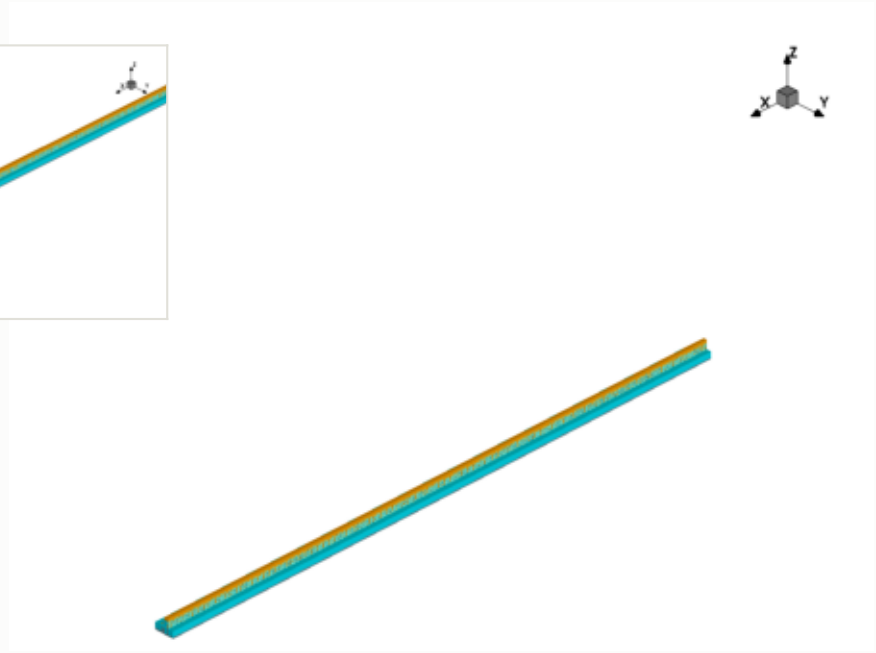
etch_anisotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





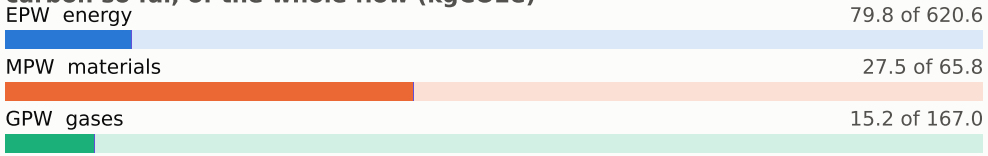
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

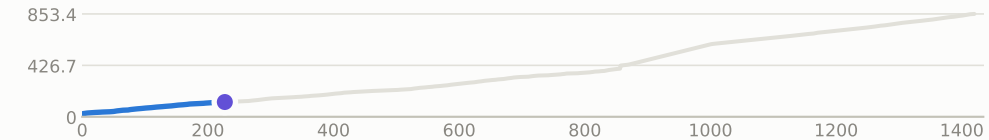
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

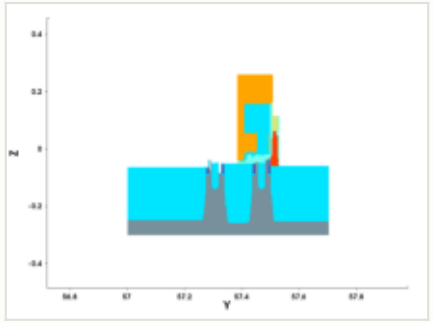
carbon so far, of the whole flow (kgCO2e)



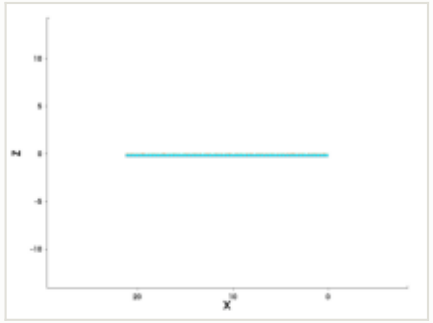
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



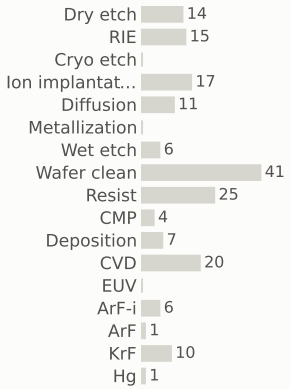
y cut at 57.351 µm



mask for this step

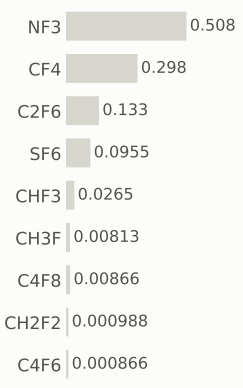
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



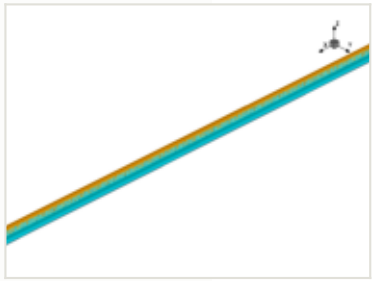
Step #2320: Spx_only

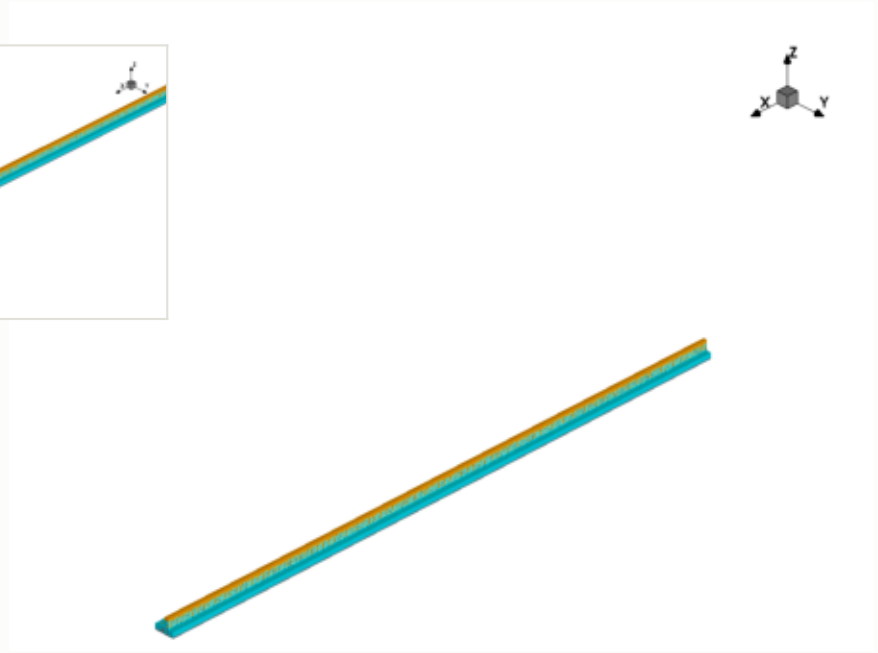
etch_microloading · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





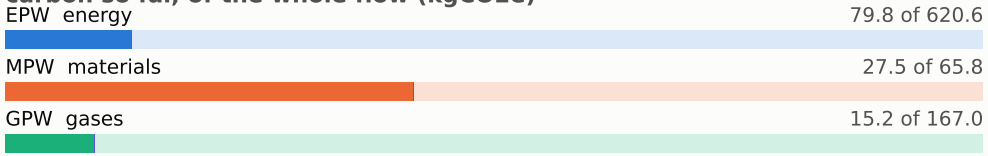
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

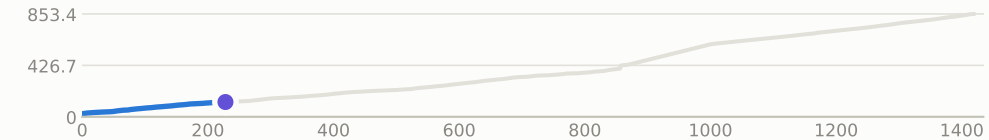
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

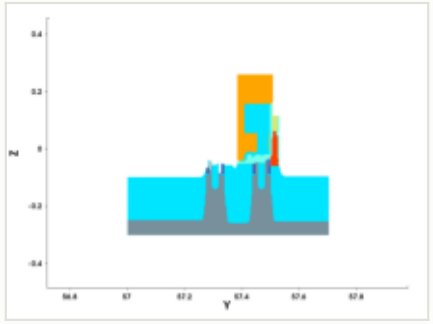
carbon so far, of the whole flow (kgCO2e)



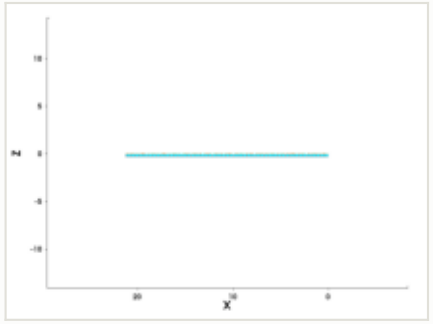
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



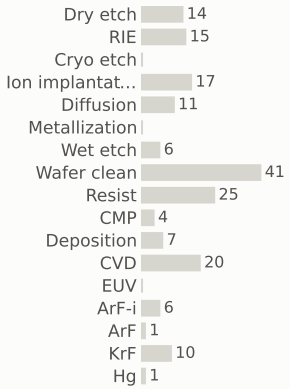
y cut at 57.351 µm



mask for this step

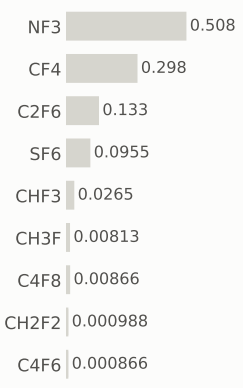
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



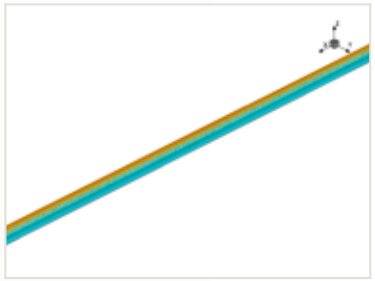
Step #2330: Spx_only

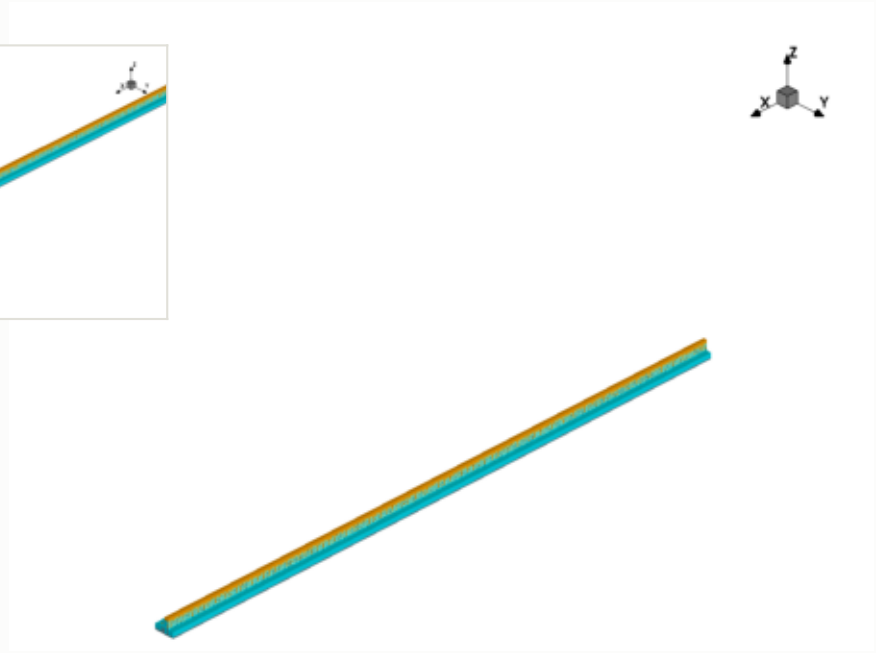
etch_anisotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





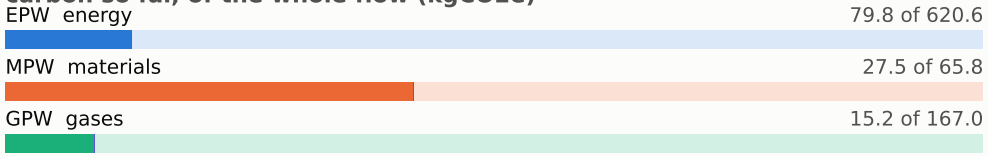
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

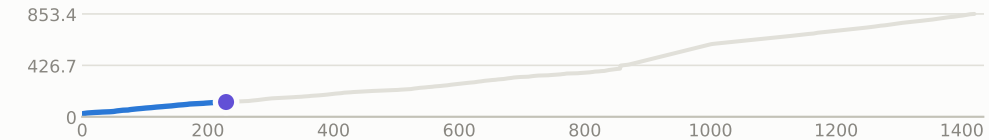
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

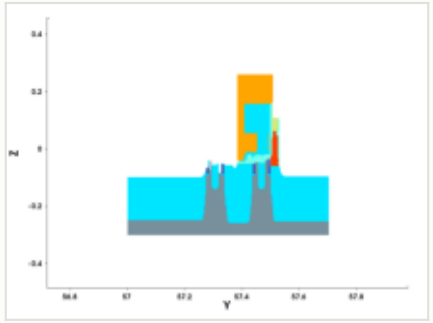
carbon so far, of the whole flow (kgCO2e)



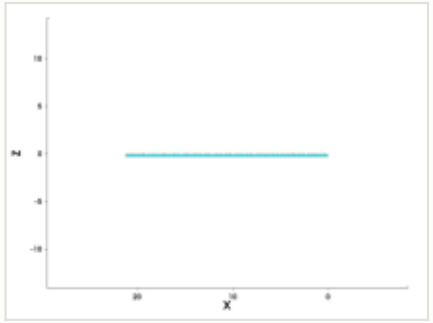
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



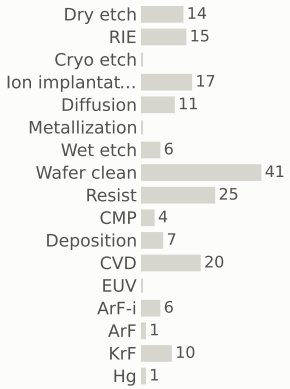
y cut at 57.351 µm



mask for this step

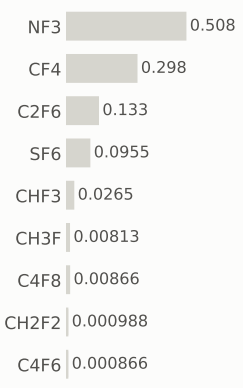
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



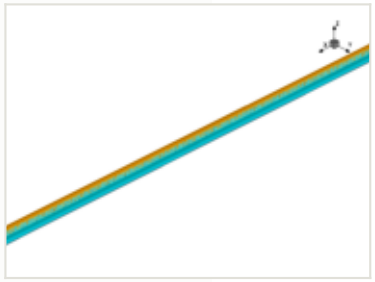
Step #2340: Spx_only

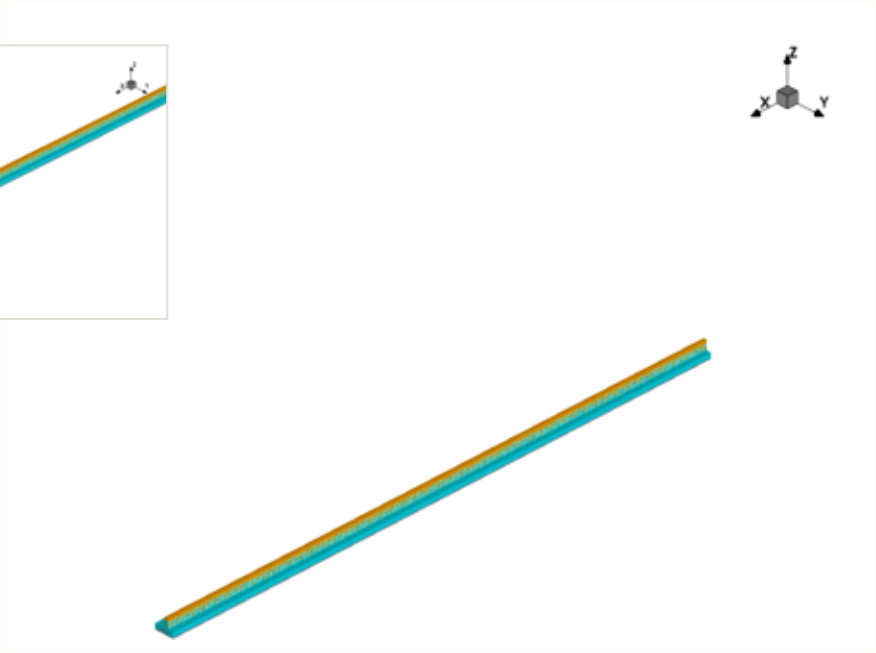
etch_isotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

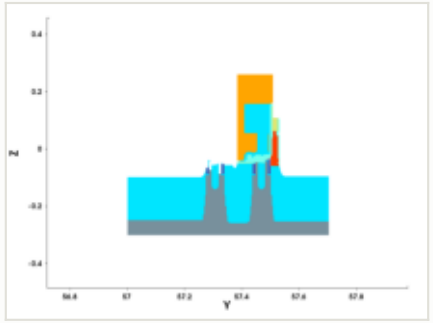
3D structure after this step

zoomed to fit

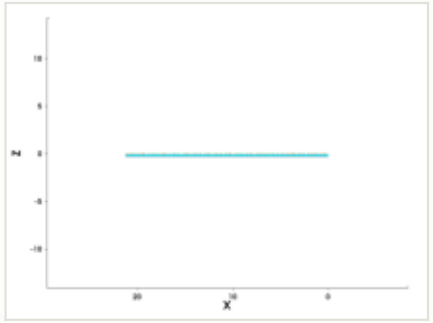




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

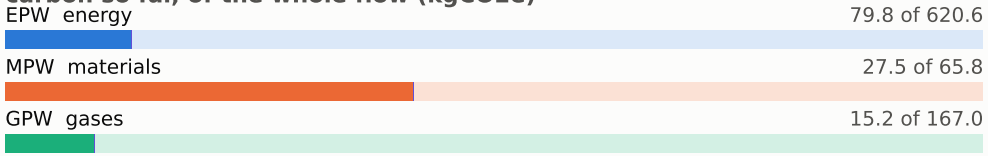
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

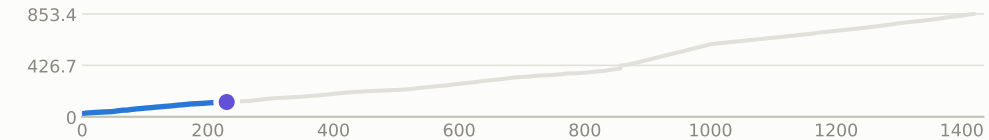
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

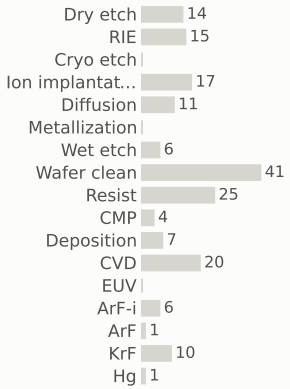
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

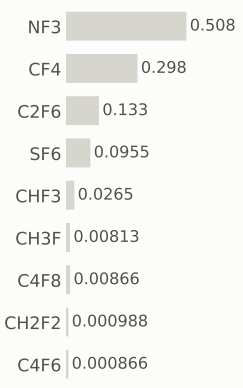


steps by process area



materials deposited (g)

gases emitted (g)



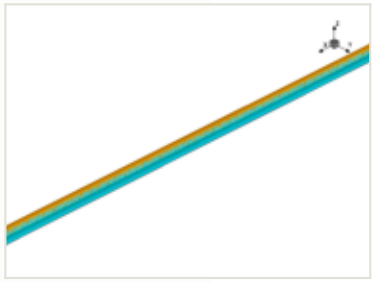
Step #2350: Spx_only

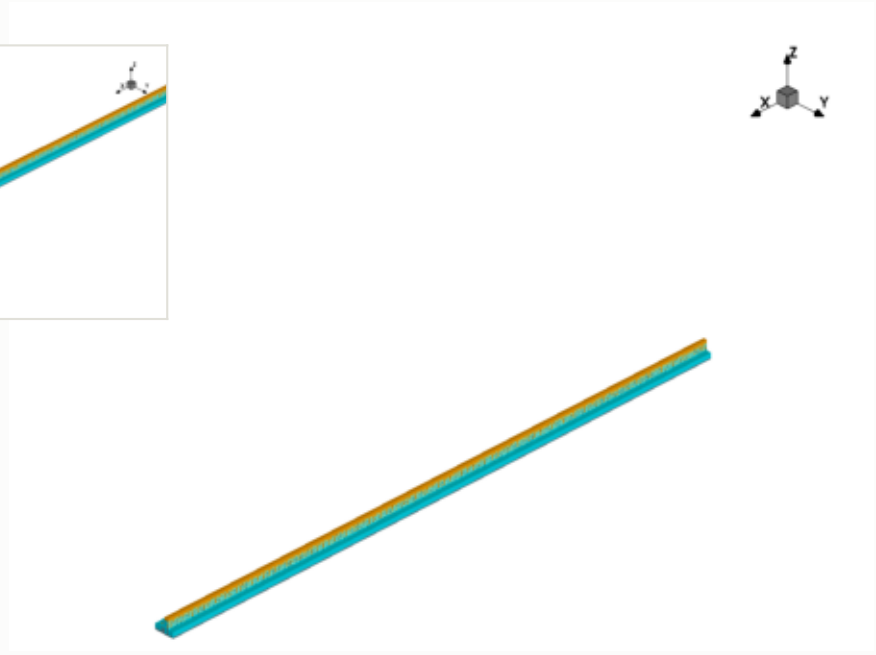
etch_anisotropic · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





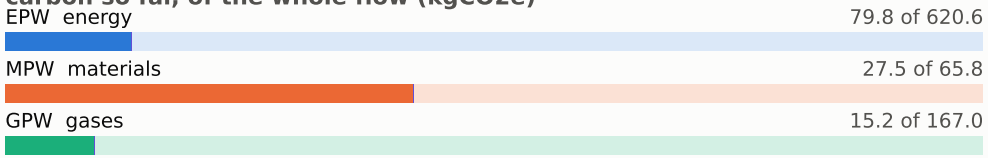
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

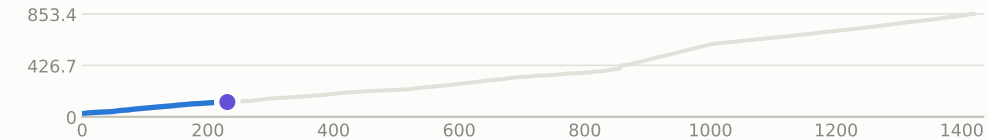
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

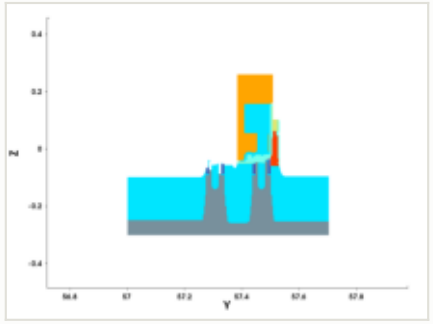
carbon so far, of the whole flow (kgCO2e)



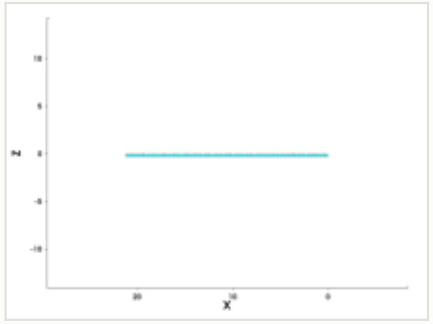
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



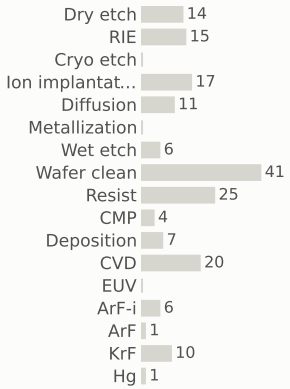
y cut at 57.351 µm



mask for this step

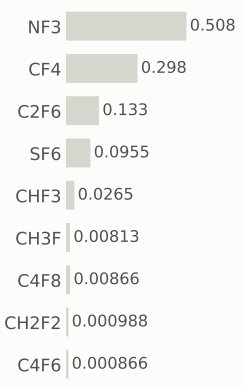
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



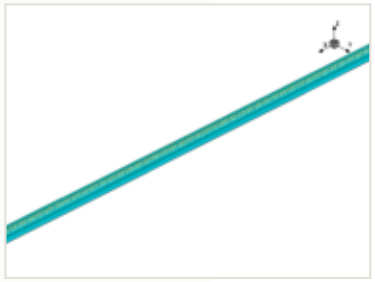
Step #2360: Spx_only

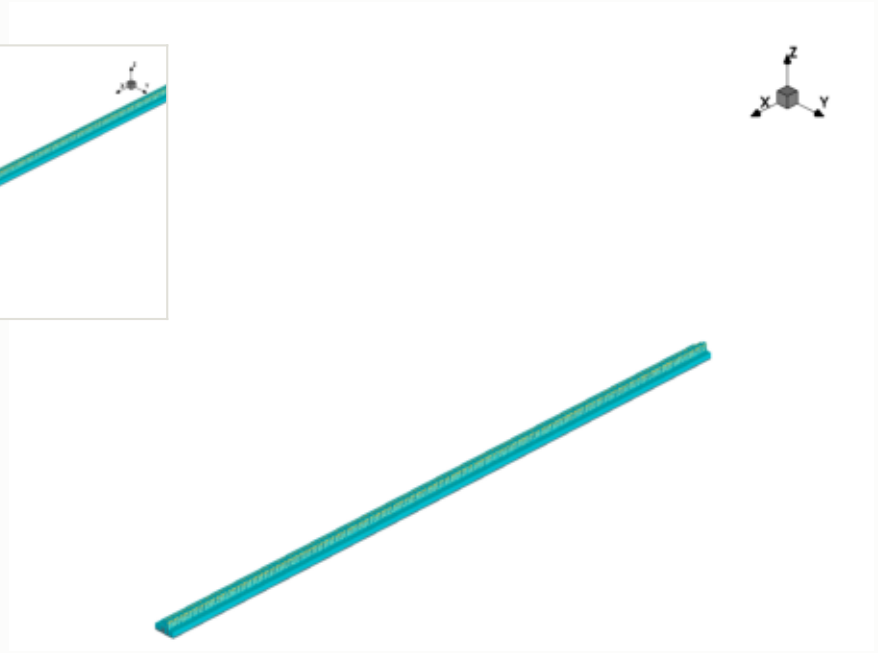
etch_strip · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





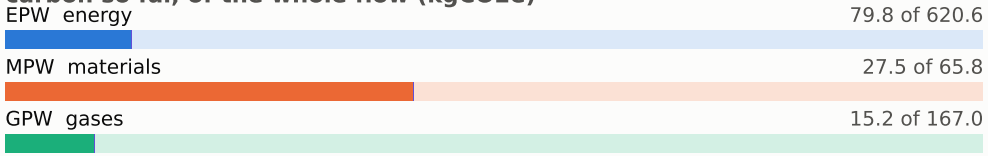
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

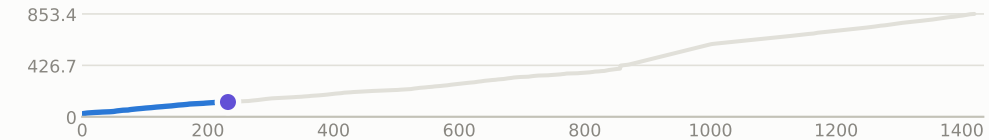
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

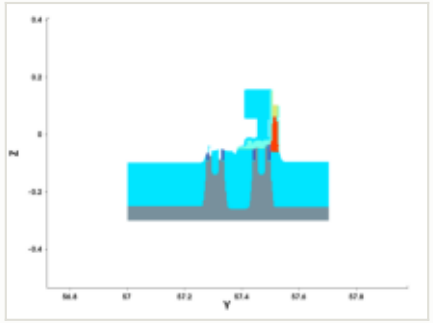
carbon so far, of the whole flow (kgCO2e)



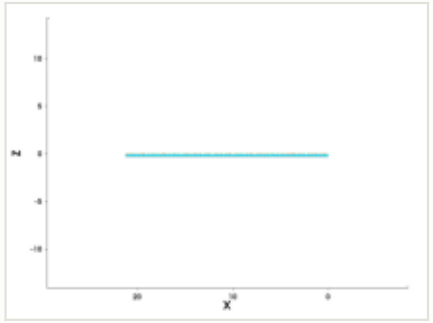
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



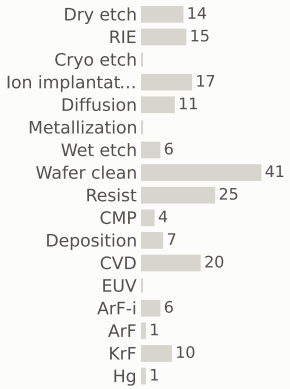
y cut at 57.351 µm



mask for this step

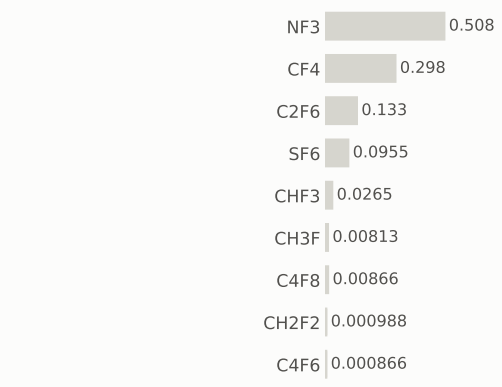
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



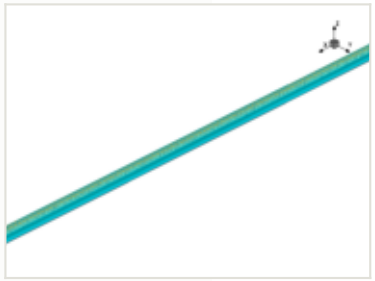
Step #2370: Spx_only

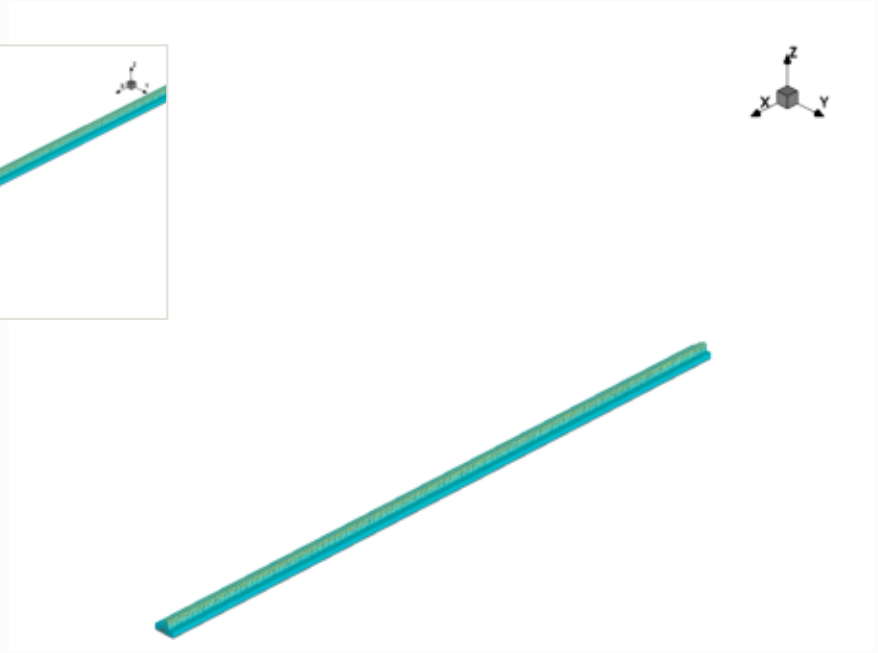
etch_ellipsoidal · Spx_only

0	122.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





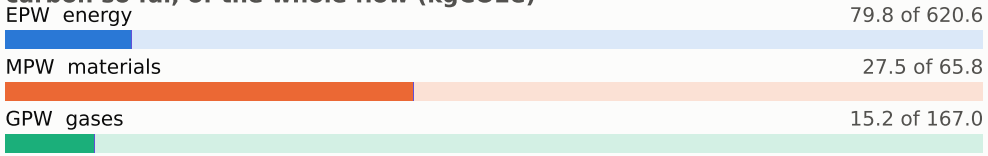
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

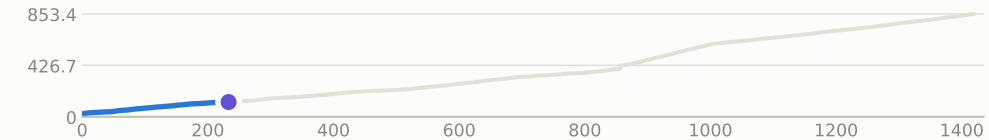
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

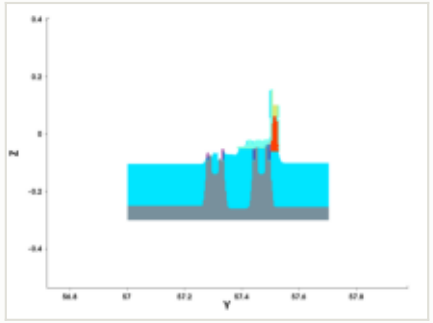
carbon so far, of the whole flow (kgCO2e)



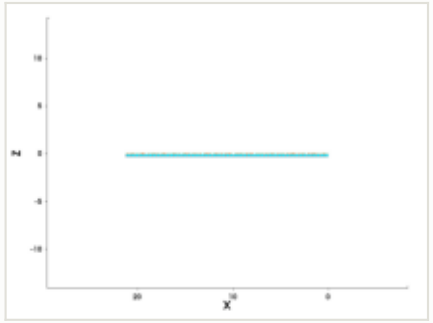
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



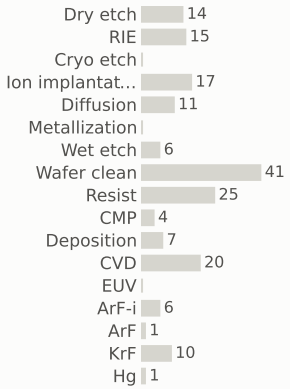
y cut at 57.351 µm



mask for this step

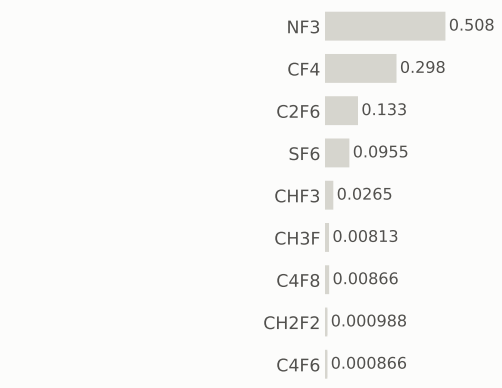
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



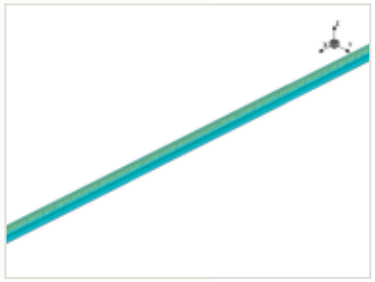
Step #2380: Wafer_clean

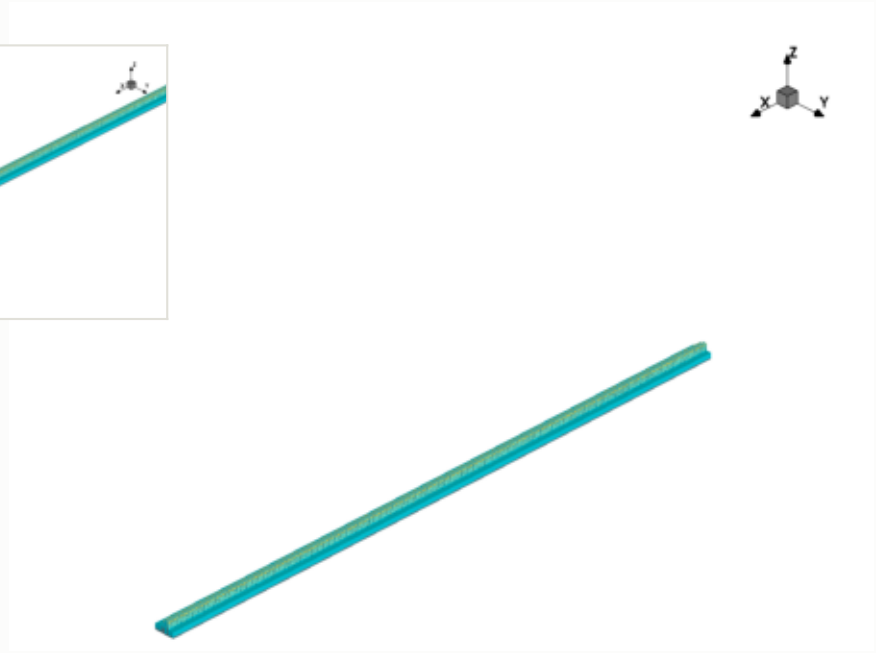
clean · Wafer_clean

0.169	122.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





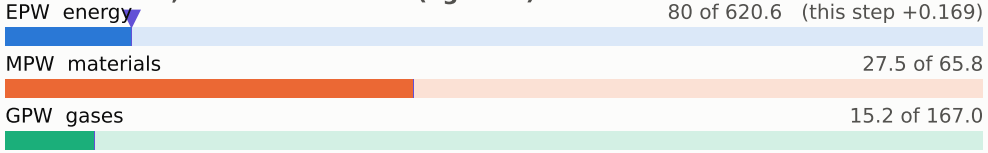
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

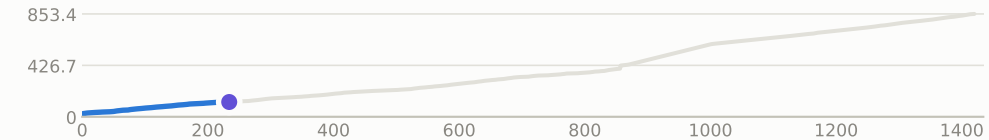
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

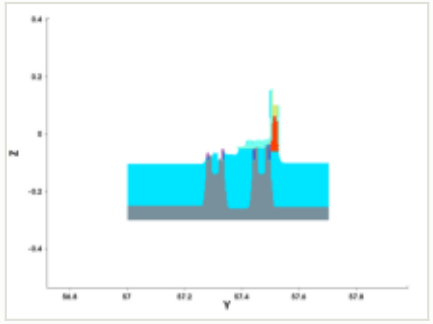
carbon so far, of the whole flow (kgCO2e)



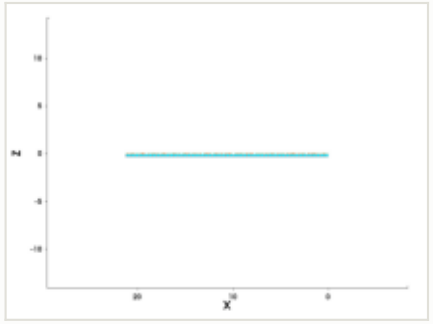
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



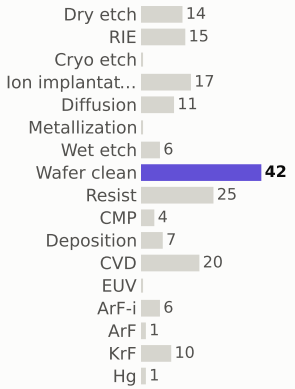
y cut at 57.351 µm



mask for this step

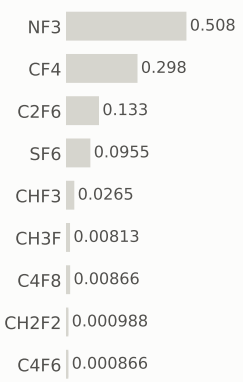
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



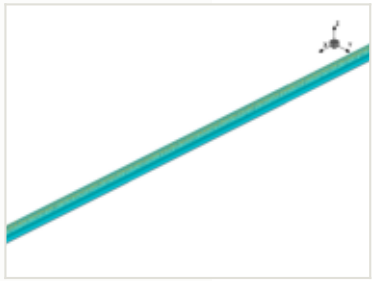
Step #2390: CVD

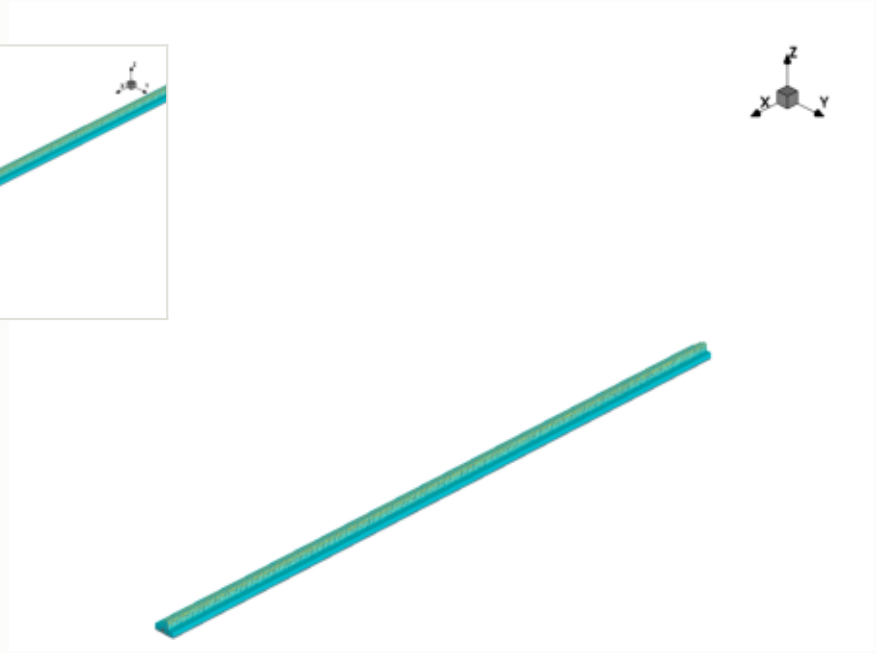
depo_epi · CVD

1.31	124.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



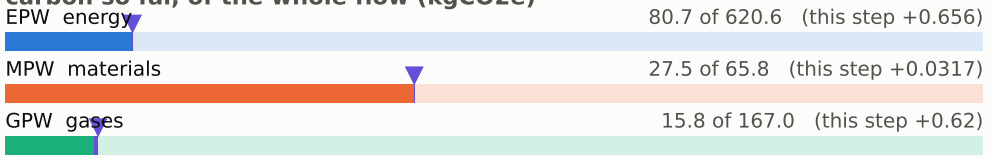


this step

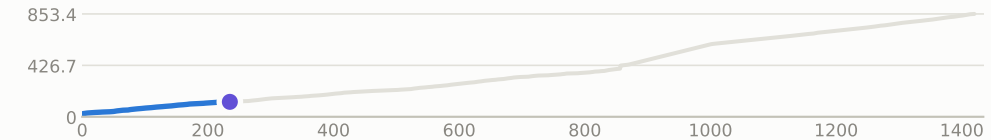
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

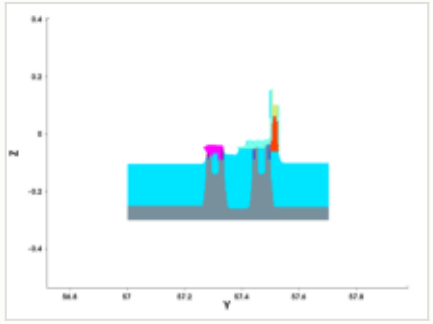
carbon so far, of the whole flow (kgCO2e)



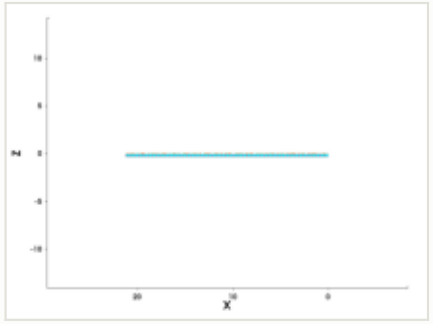
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



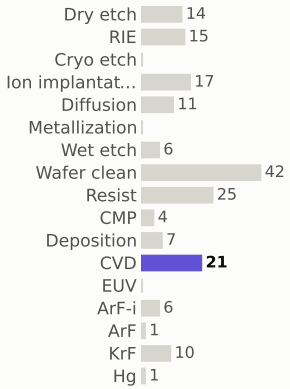
y cut at 57.351 µm



mask for this step

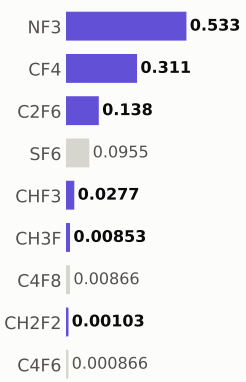
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



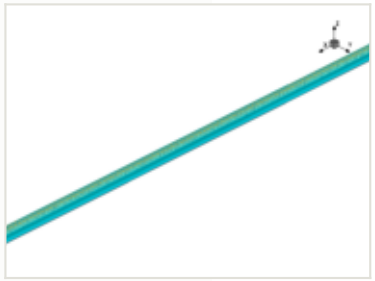
Step #2400: Spx_only

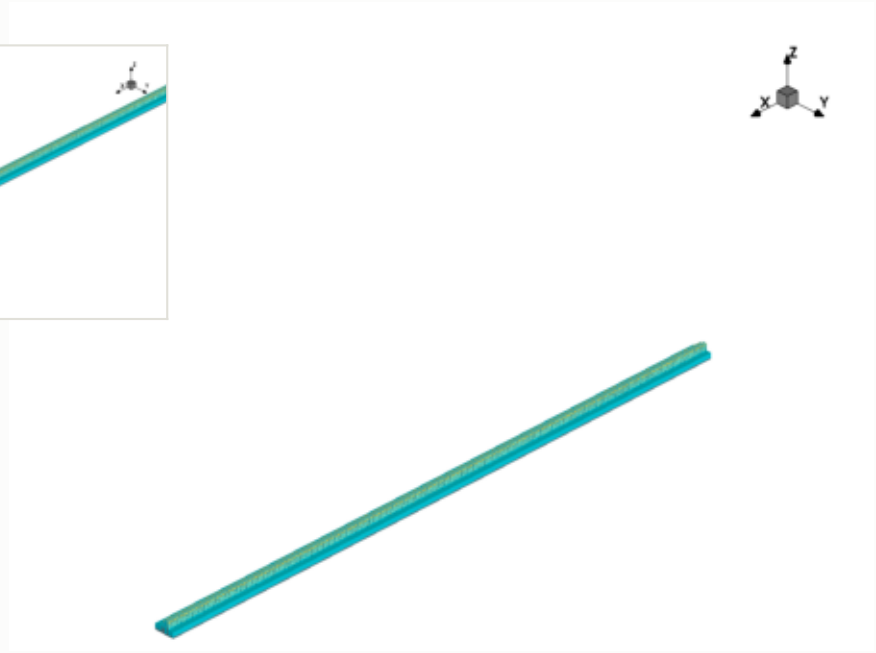
depo_epi · Spx_only

0	124.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





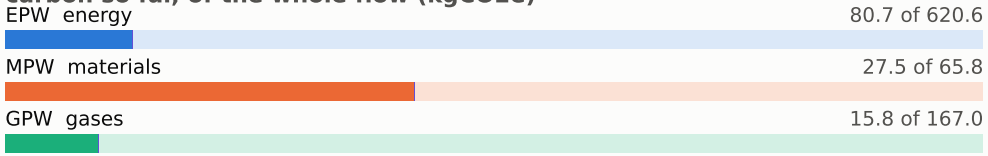
this step

process area	Spx_only
material	—
thickness	0.008 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

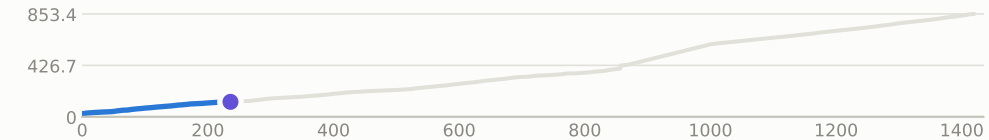
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

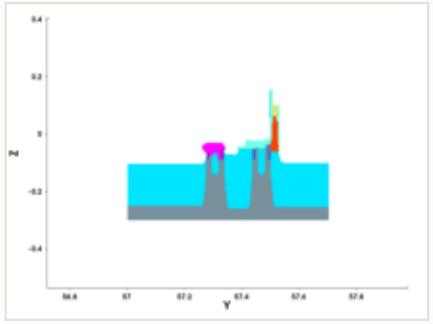
carbon so far, of the whole flow (kgCO2e)



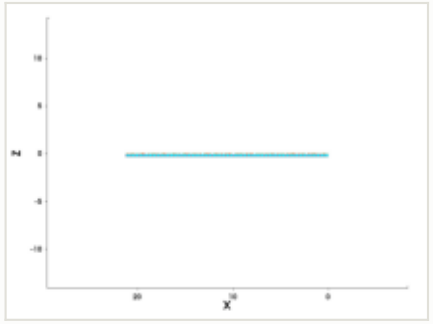
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



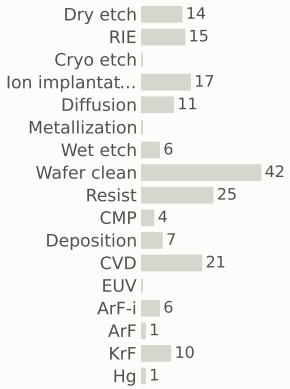
y cut at 57.351 µm



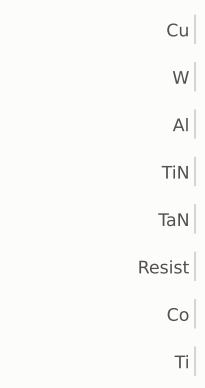
mask for this step

no mask for this step

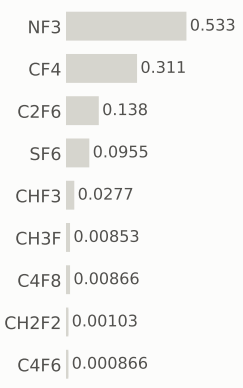
steps by process area



materials deposited (g)



gases emitted (g)



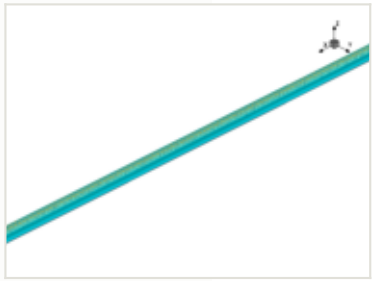
Step #2410: Wafer_clean

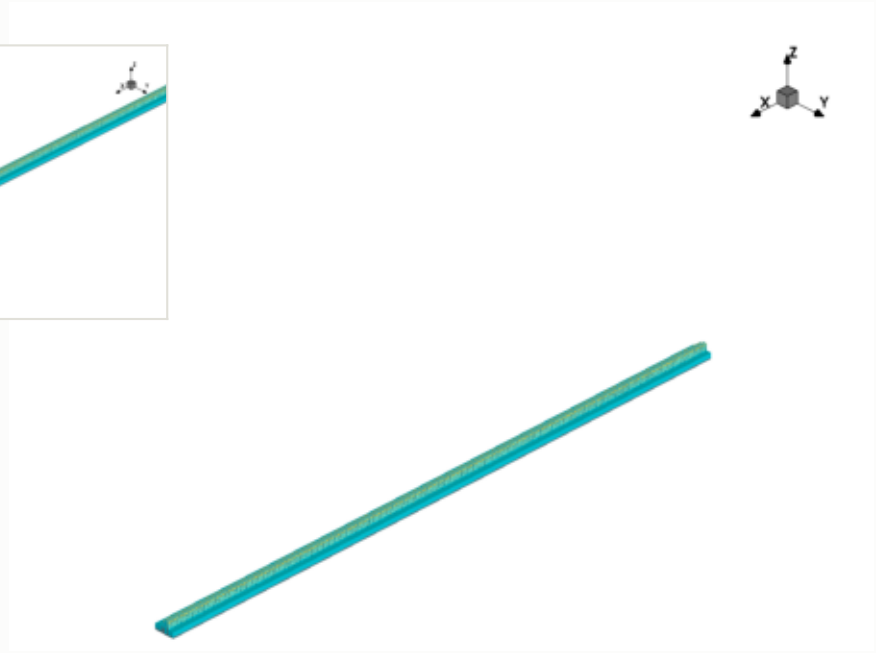
clean · Wafer_clean

0.169	124.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





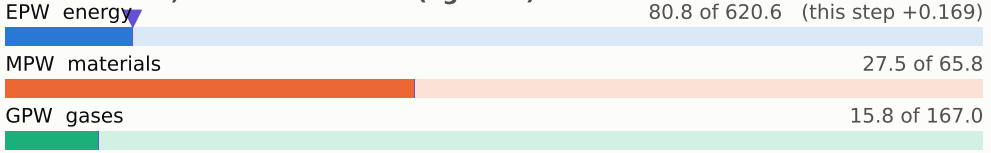
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

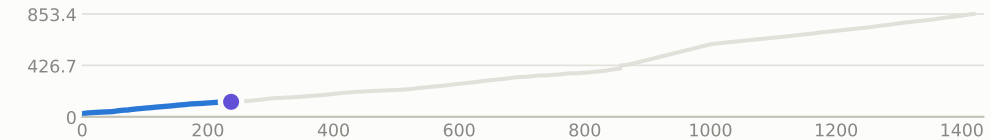
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

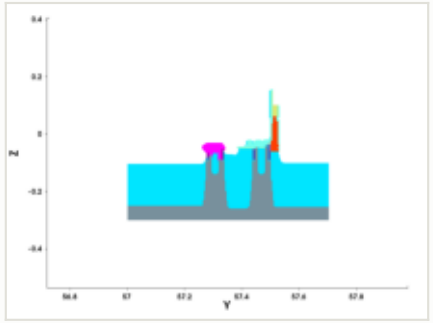
carbon so far, of the whole flow (kgCO2e)



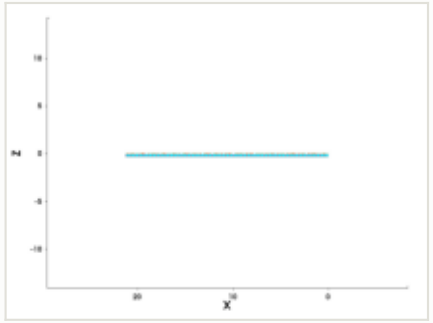
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



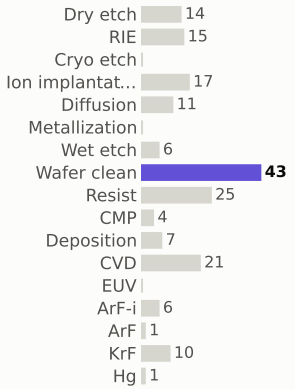
y cut at 57.351 µm



mask for this step

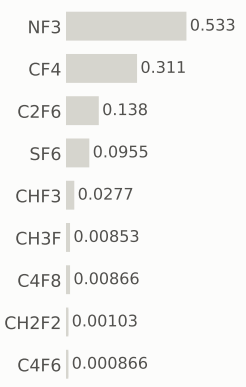
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



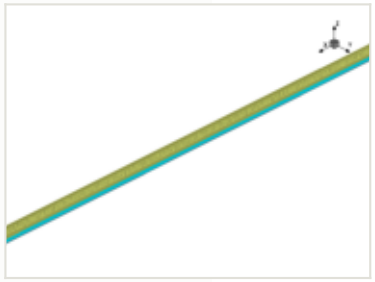
Step #2420: Deposition

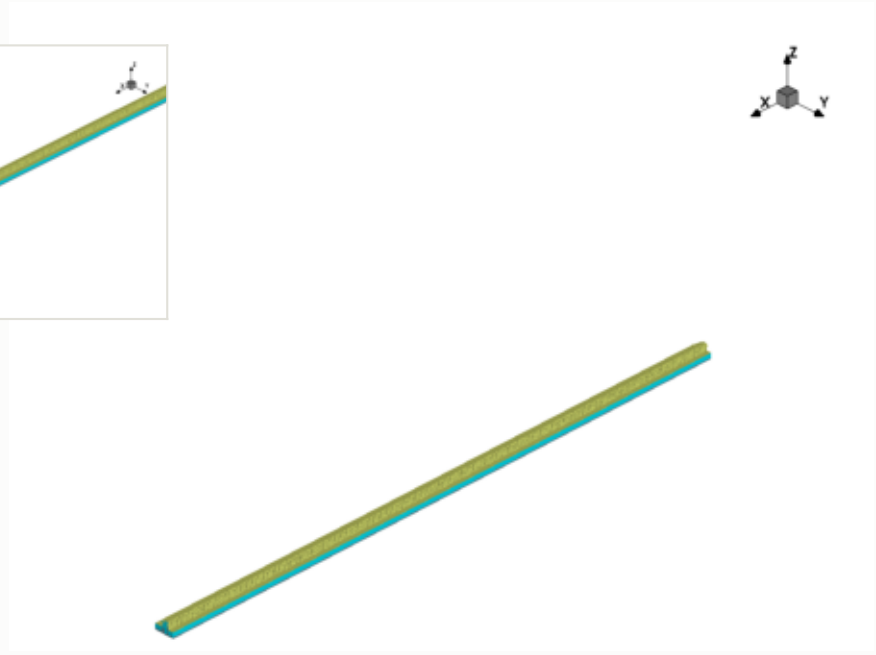
depo_isotropic · Deposition

0.656	124.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



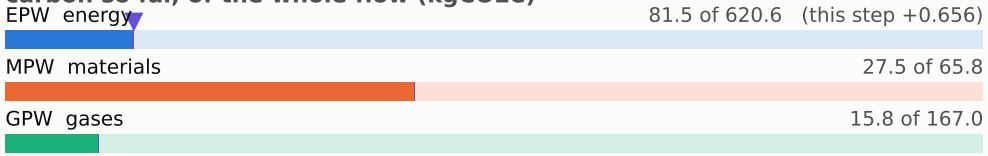


this step

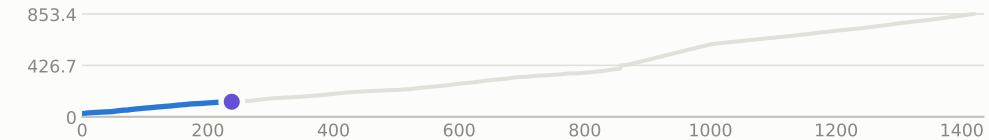
process area	Deposition
material	—
thickness	0.0025 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

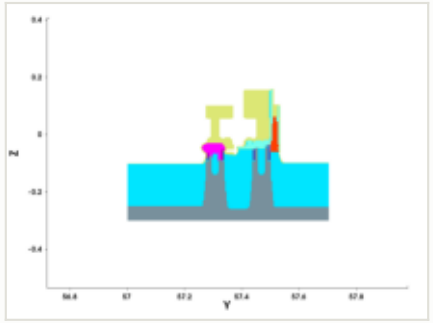
carbon so far, of the whole flow (kgCO2e)



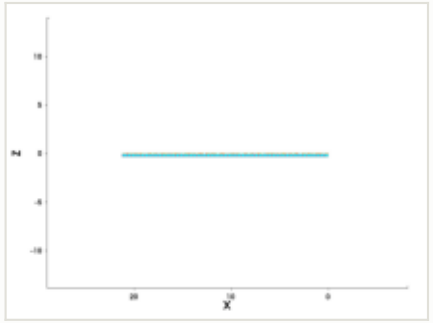
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



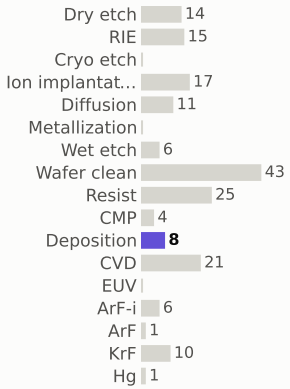
y cut at 57.351 µm



mask for this step

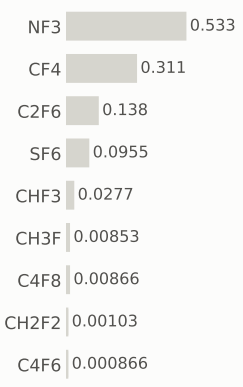
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



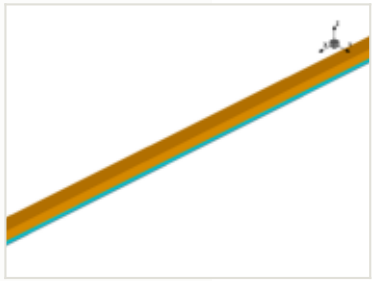
Step #2430: Resist

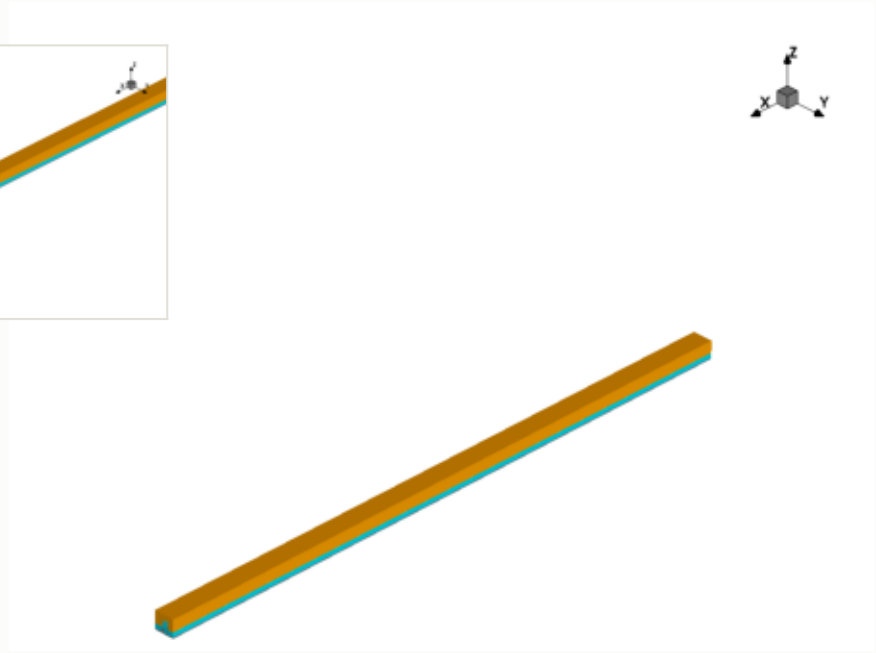
depo_fill · Resist

0.169	125.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



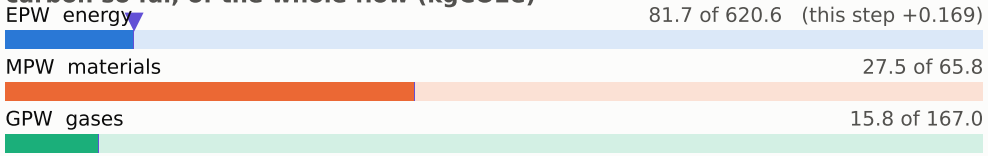


this step

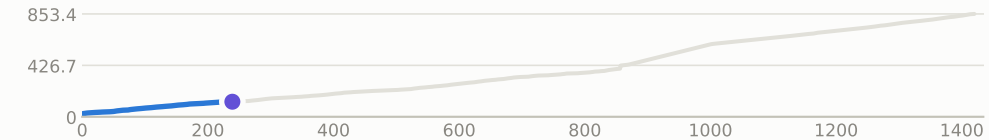
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

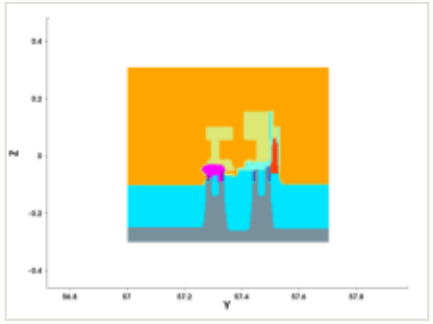
carbon so far, of the whole flow (kgCO2e)



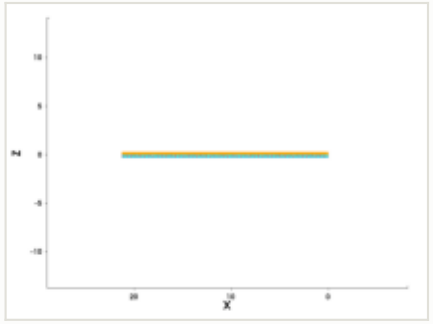
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



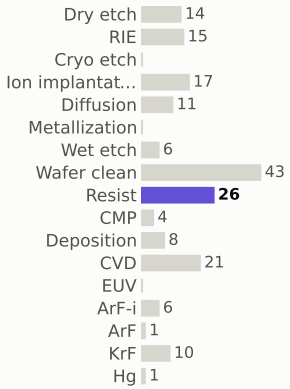
y cut at 57.351 µm



mask for this step

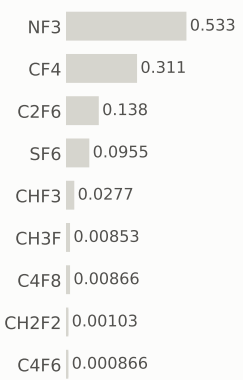
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



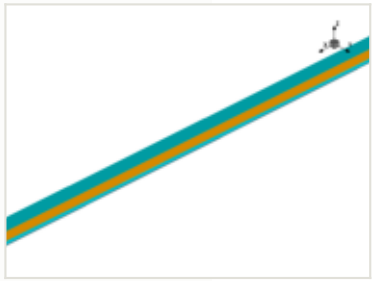
Step #2440: Deposition

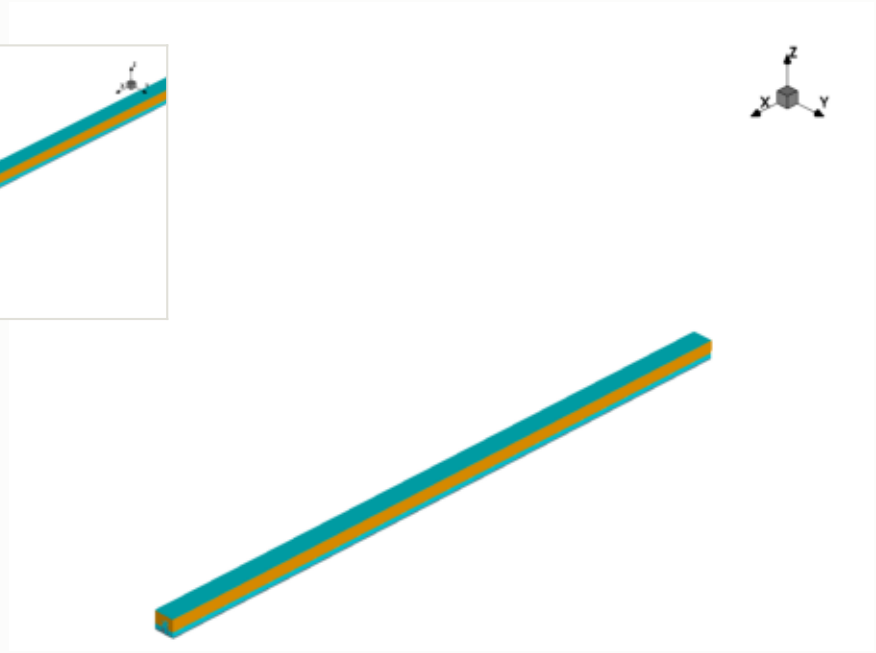
depo_isotropic · Deposition

0.656	125.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



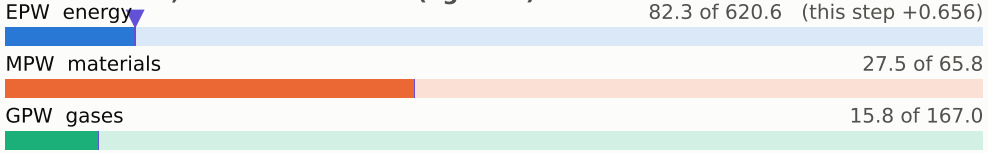


this step

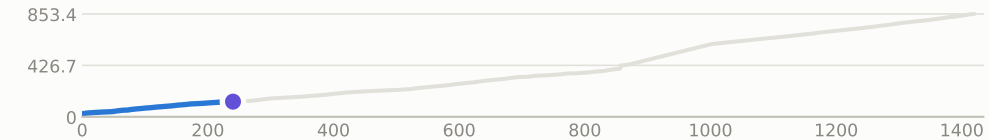
process area	Deposition
material	—
thickness	0.01 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

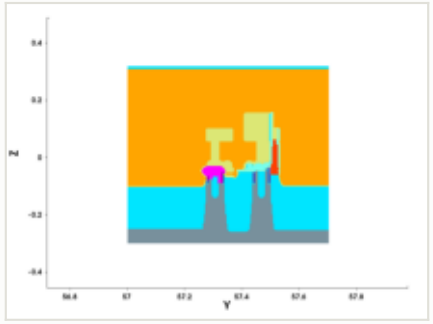
carbon so far, of the whole flow (kgCO2e)



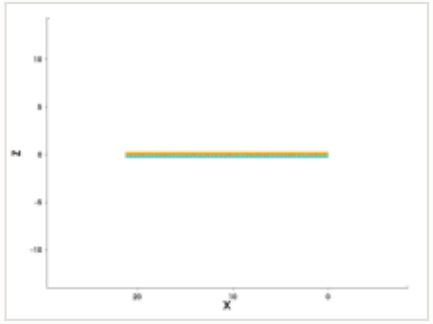
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



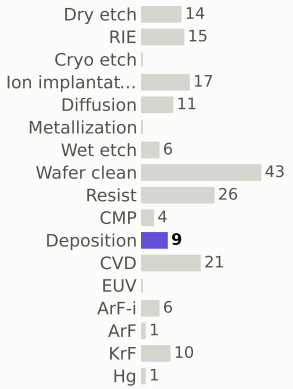
y cut at 57.351 µm



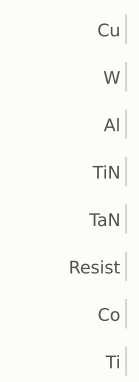
mask for this step

no mask for this step

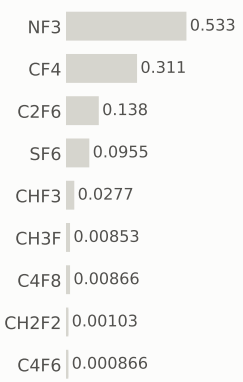
steps by process area



materials deposited (g)



gases emitted (g)



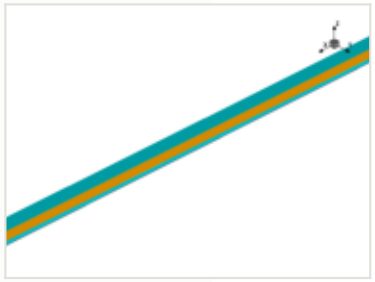
Step #2450: Wafer_clean

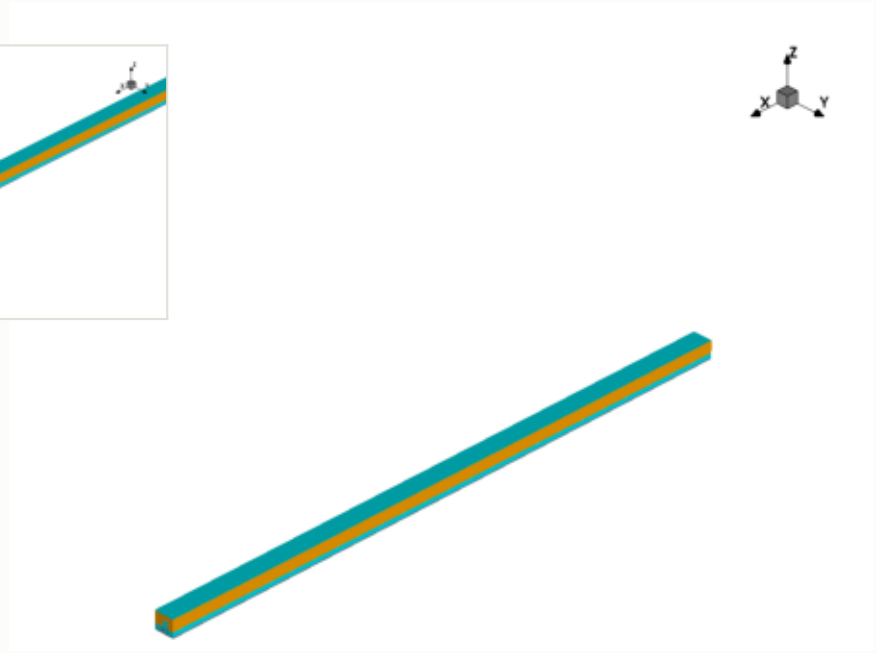
clean · Wafer_clean

0.169	125.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





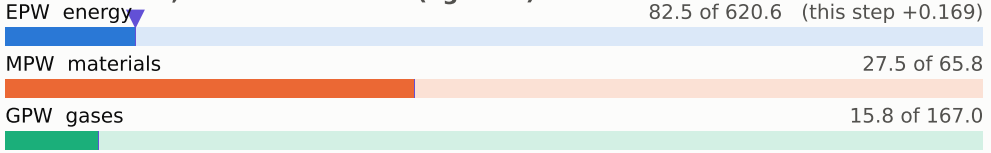
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

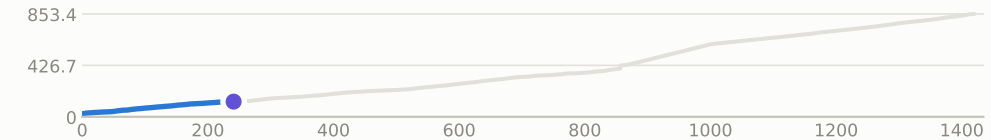
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

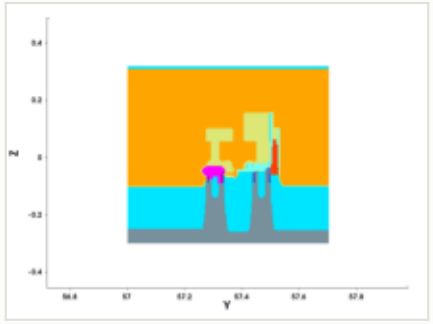
carbon so far, of the whole flow (kgCO2e)



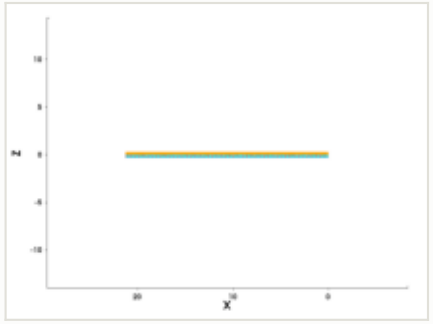
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



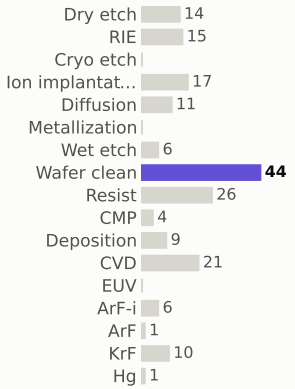
y cut at 57.351 µm



mask for this step

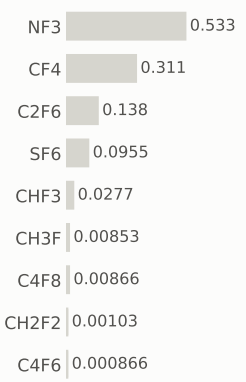
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



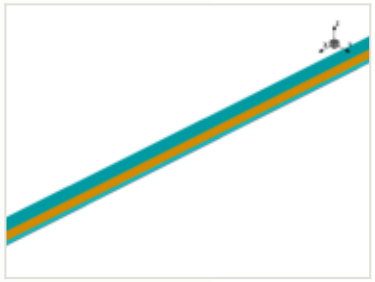
Step #2460: Resist

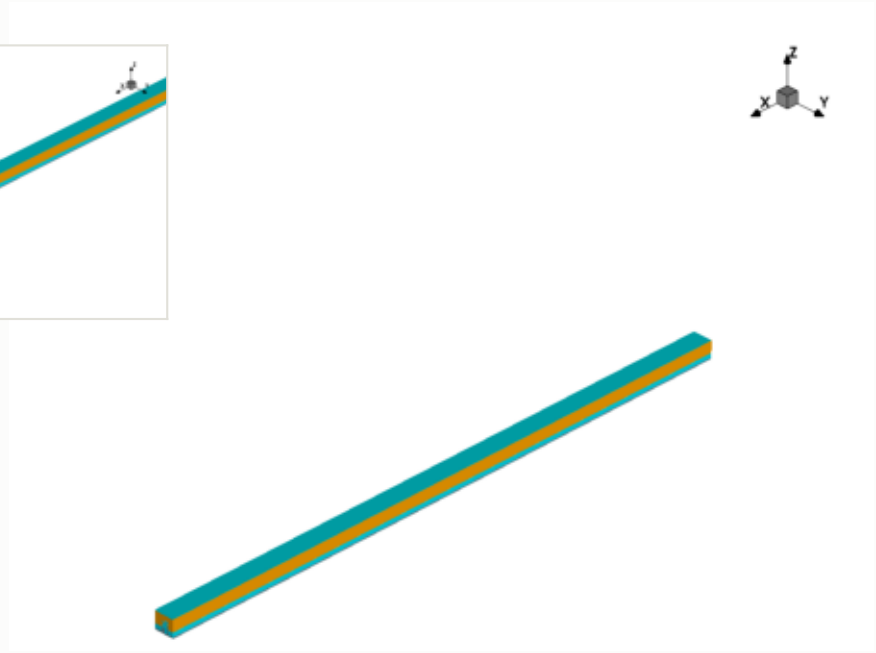
resist · Resist

0.169	126.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

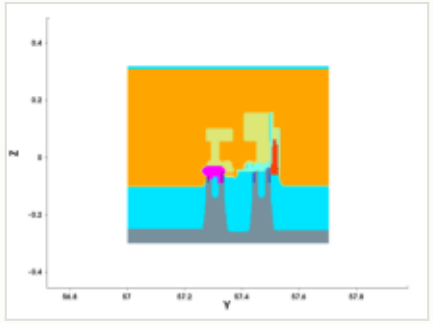
3D structure after this step

zoomed to fit

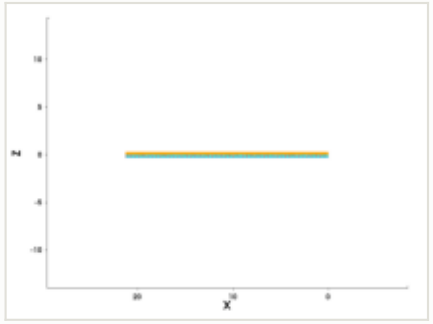




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

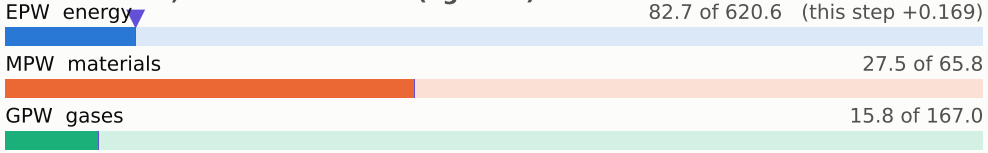
no mask for this step

this step

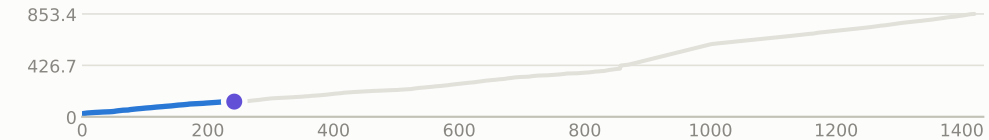
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

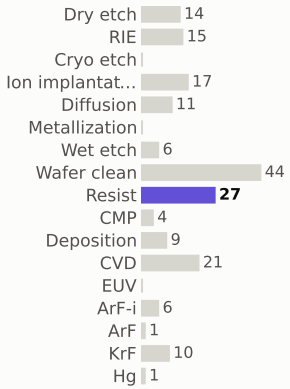
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

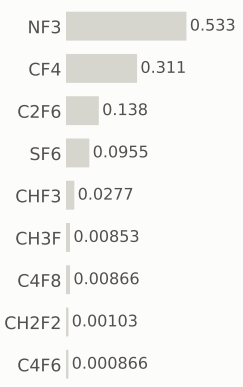


steps by process area



materials deposited (g)

gases emitted (g)



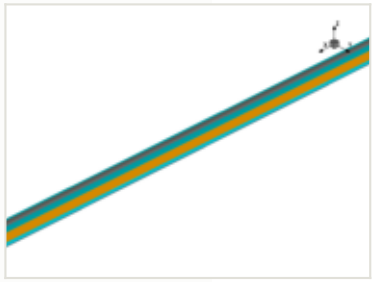
Step #2470: ArF-i

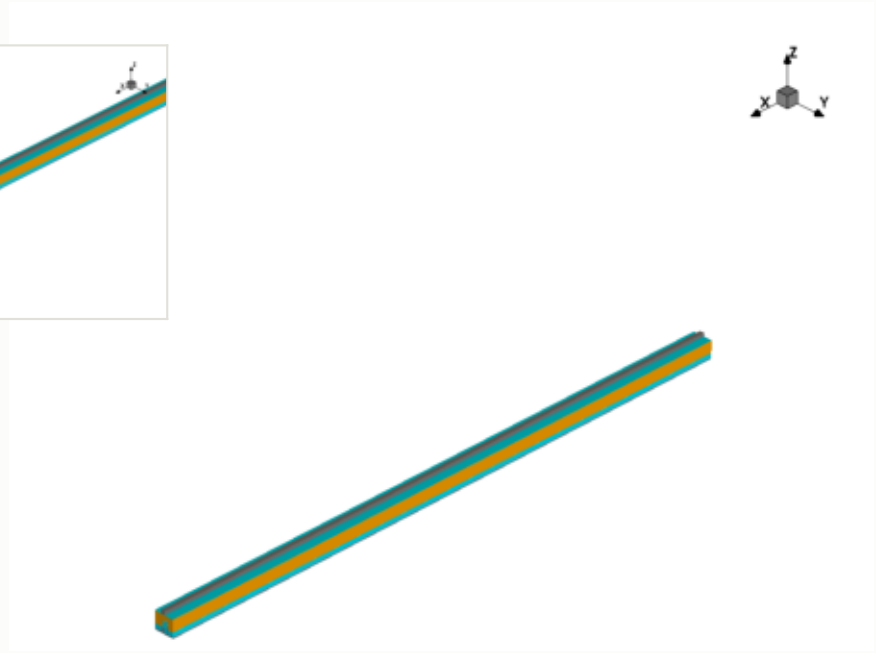
litho_patterning · ArF-i

0.217	126.2	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



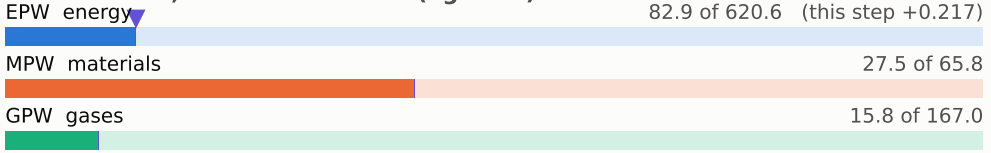


this step

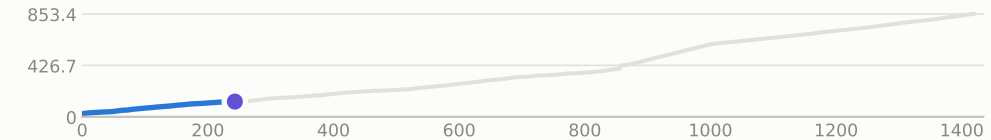
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

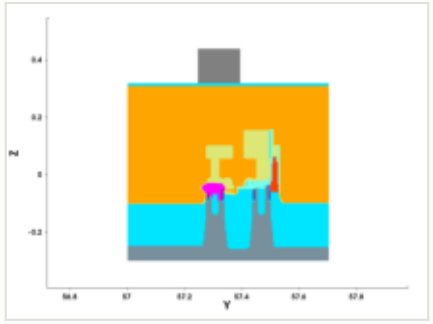
carbon so far, of the whole flow (kgCO2e)



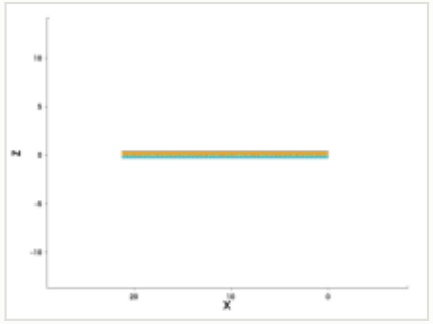
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

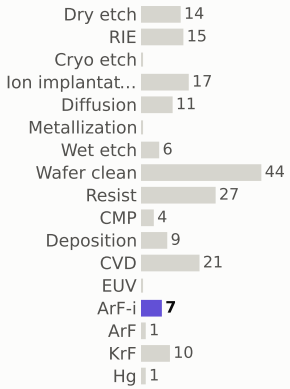


y cut at 57.351 µm

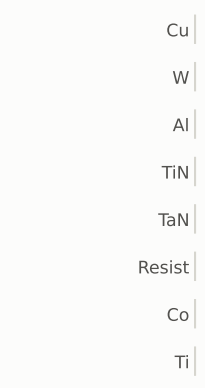


mask: layer 2012:0, positive (GDSII)

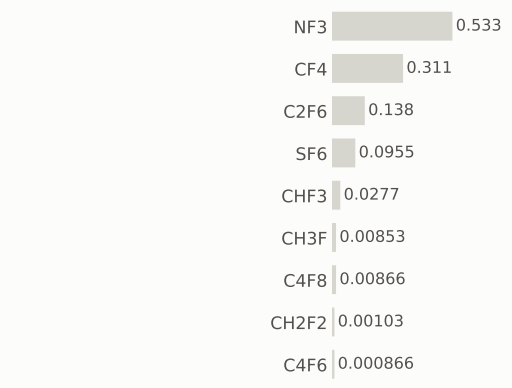
steps by process area



materials deposited (g)



gases emitted (g)



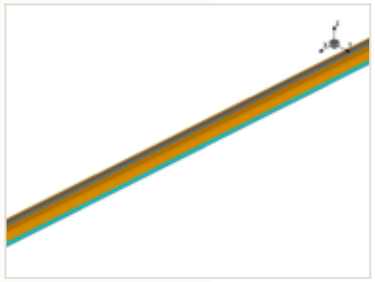
Step #2480: RIE

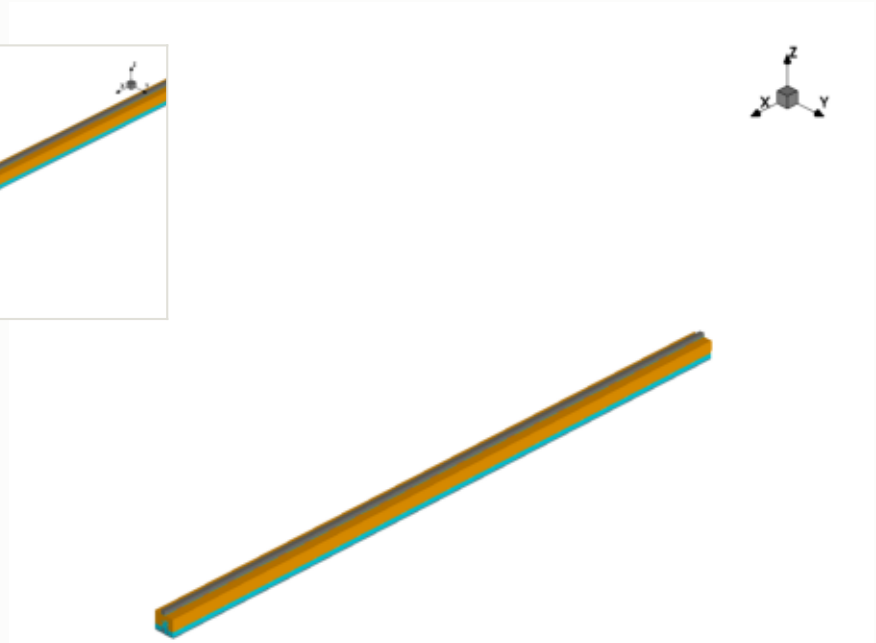
etch_anisotropic · RIE

1.04	127.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

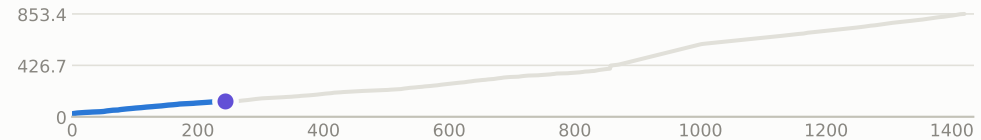
carbon so far, of the whole flow (kgCO2e)

EPW energy — 83.7 of 620.6 (this step +0.801)

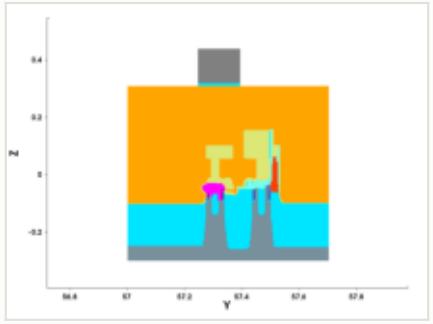
MPW materials — 27.6 of 65.8 (this step +0.0571)

GPW gases — 16 of 167.0 (this step +0.187)

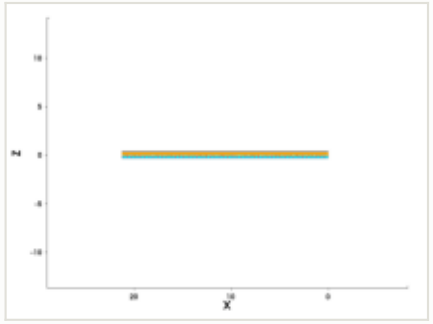
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



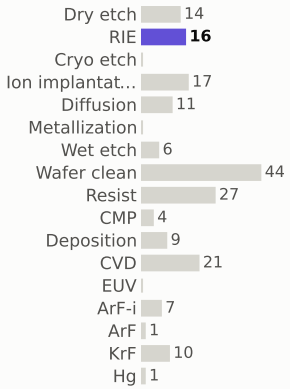
y cut at 57.351 µm



mask for this step

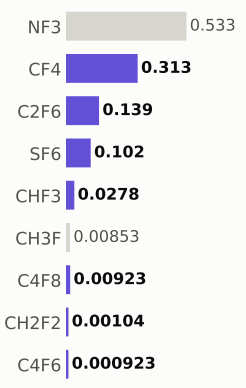
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



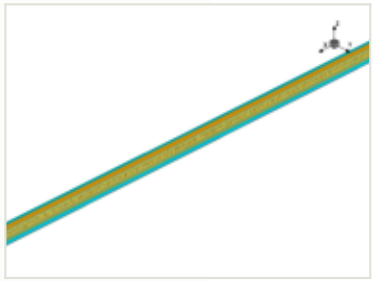
Step #2490: Spx_only

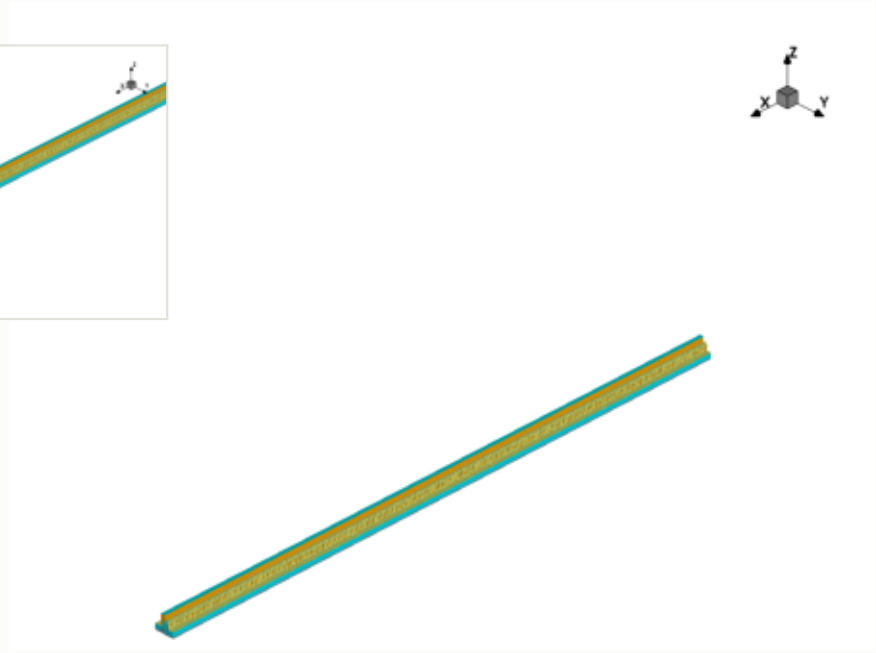
etch_anisotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

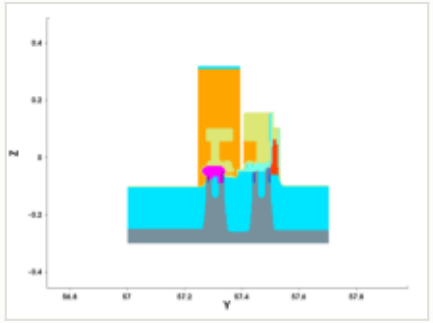
3D structure after this step

zoomed to fit

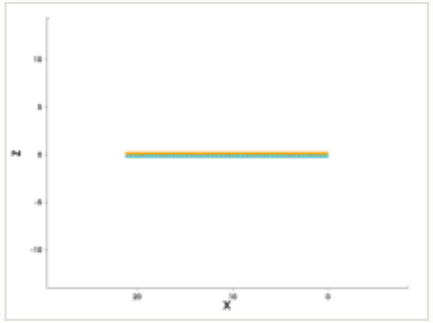




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

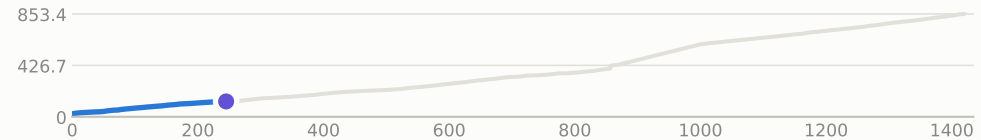
carbon so far, of the whole flow (kgCO2e)

EPW energy — 83.7 of 620.6

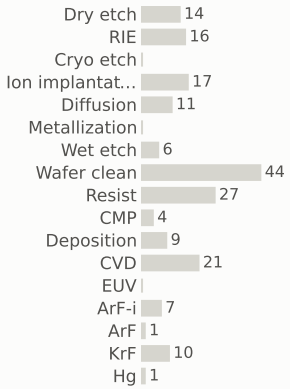
MPW materials — 27.6 of 65.8

GPW gases — 16 of 167.0

embodied carbon up to each step (kgCO2e per wafer)

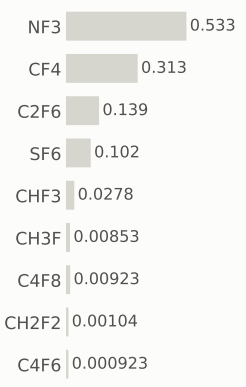


steps by process area



materials deposited (g)

gases emitted (g)



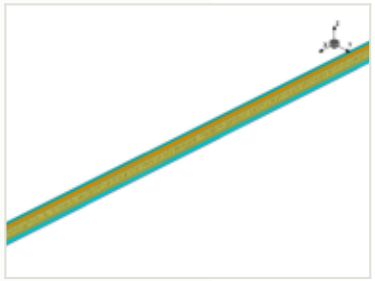
Step #2500: Spx_only

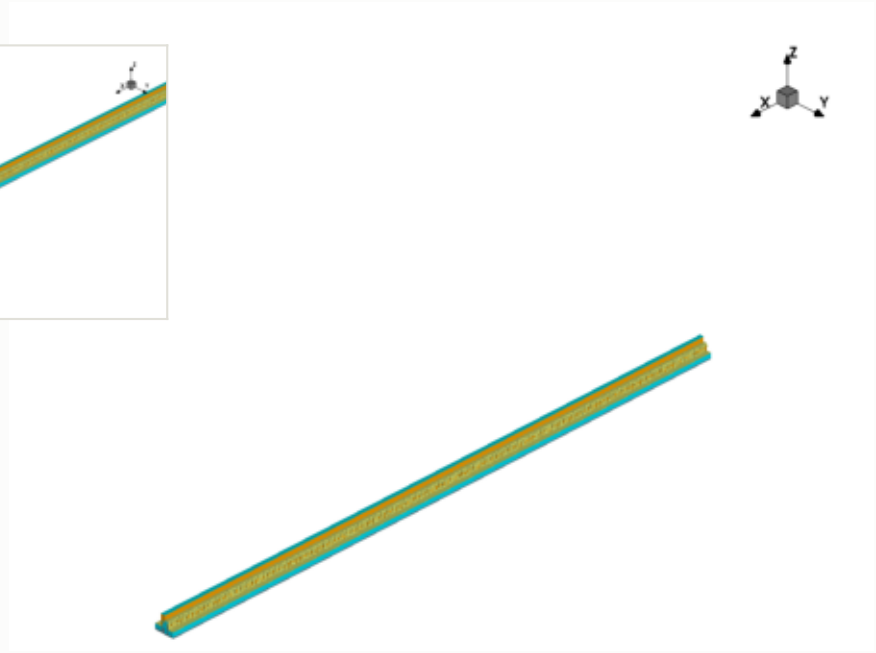
etch_isotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

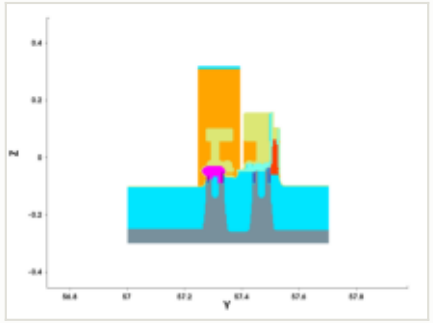
3D structure after this step

zoomed to fit

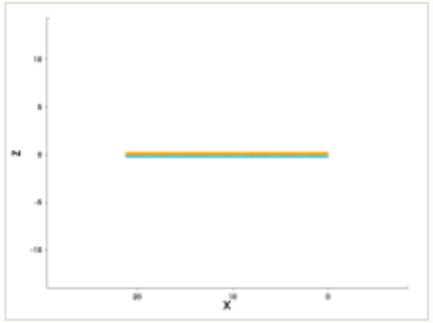




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

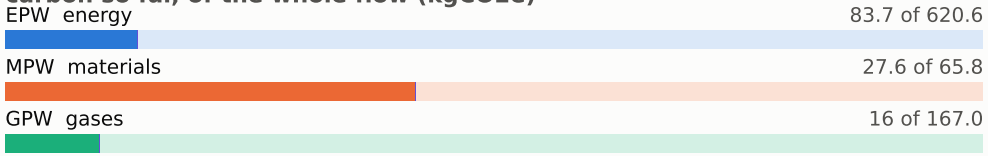
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

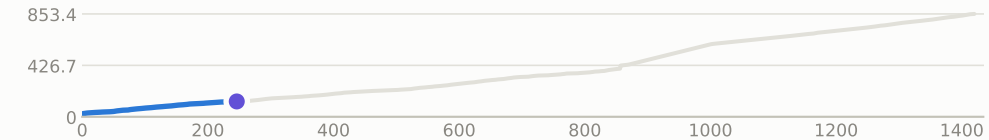
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

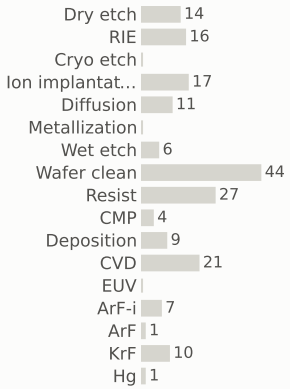
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

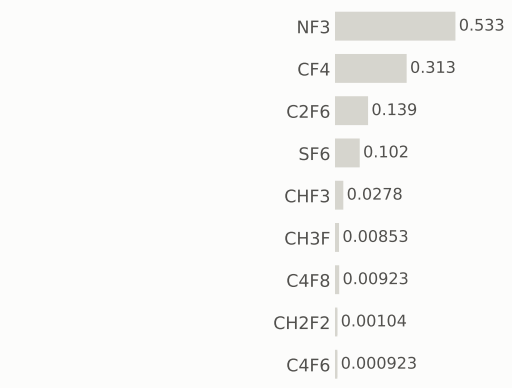


steps by process area



materials deposited (g)

gases emitted (g)



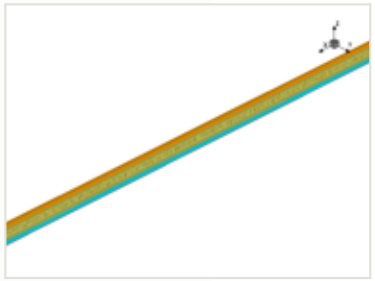
Step #2510: Spx_only

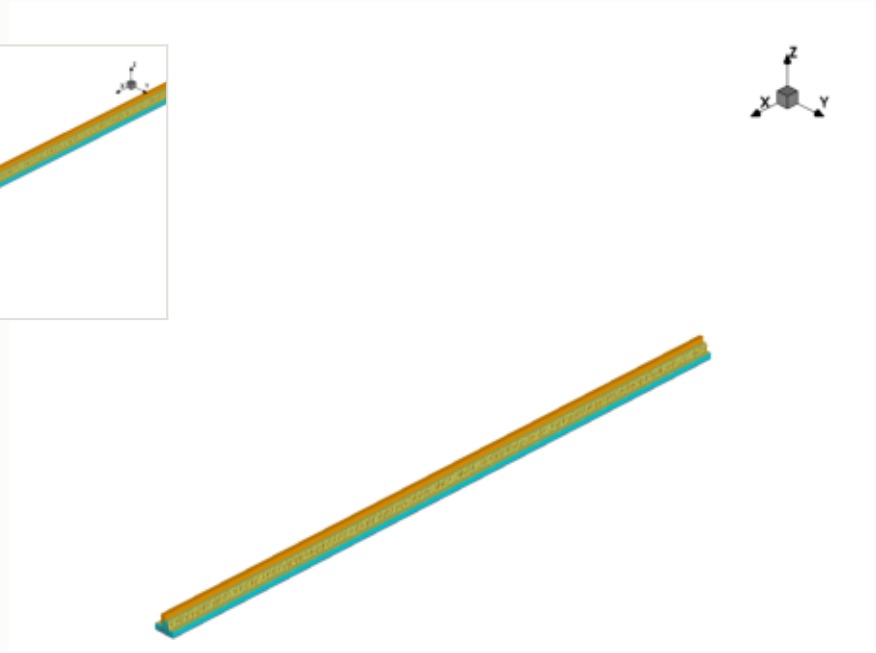
etch_anisotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





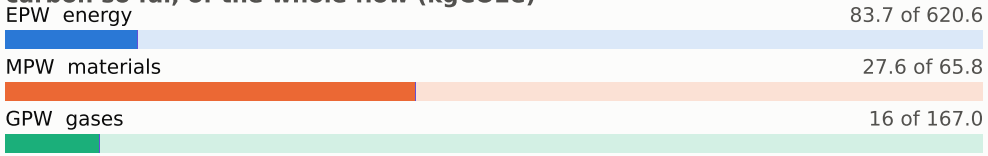
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

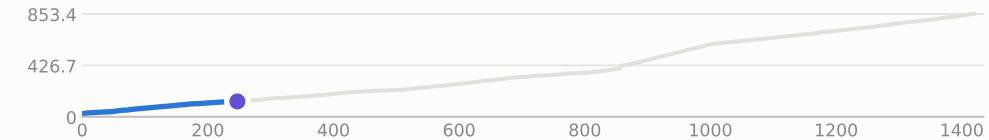
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

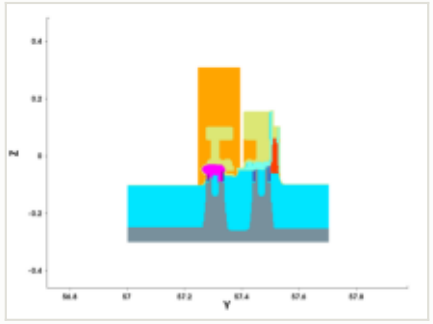
carbon so far, of the whole flow (kgCO2e)



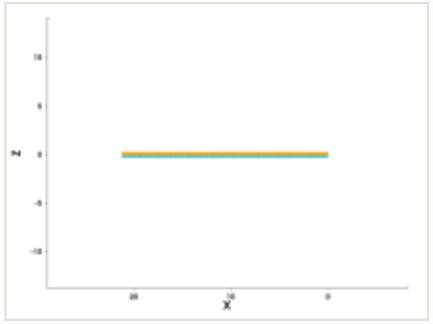
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



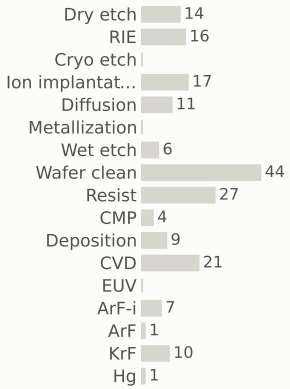
y cut at 57.351 µm



mask for this step

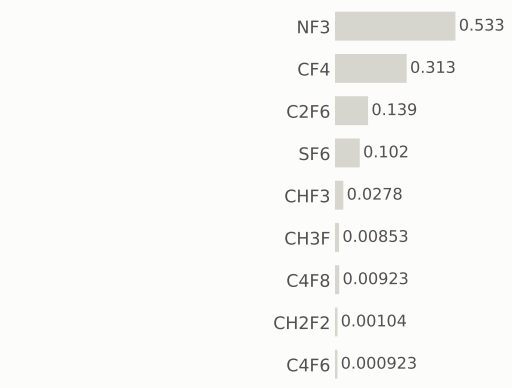
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



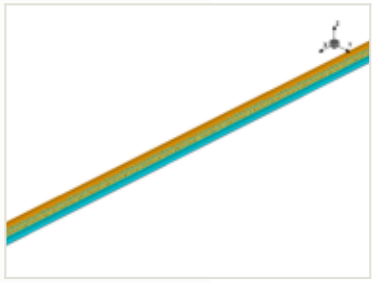
Step #2520: Spx_only

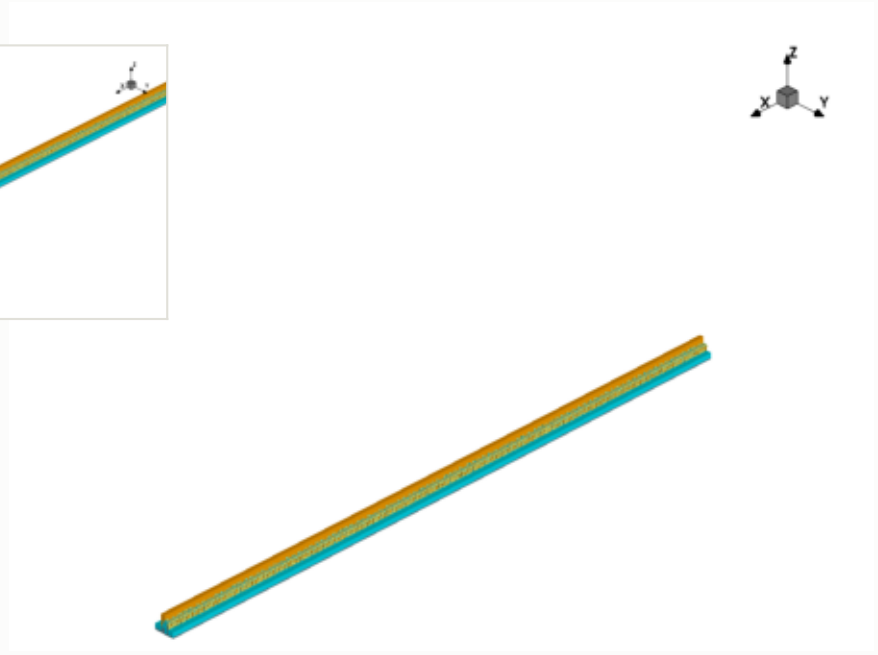
etch_anisotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





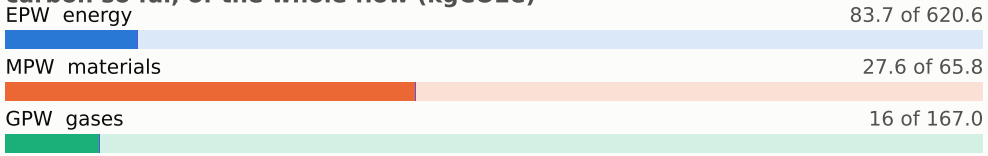
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

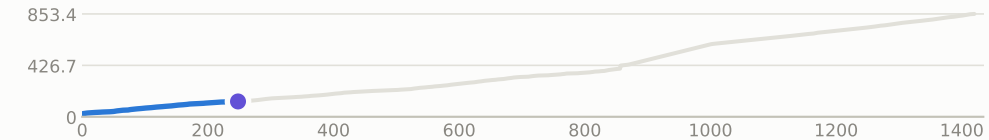
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

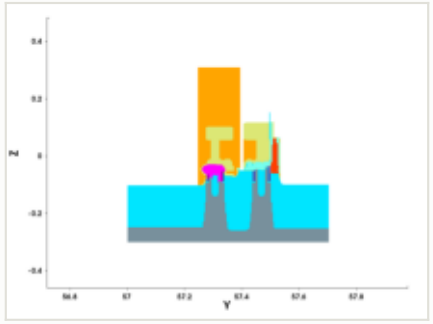
carbon so far, of the whole flow (kgCO2e)



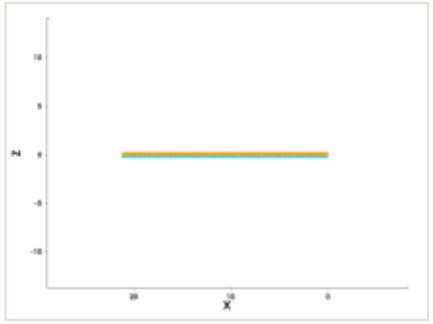
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



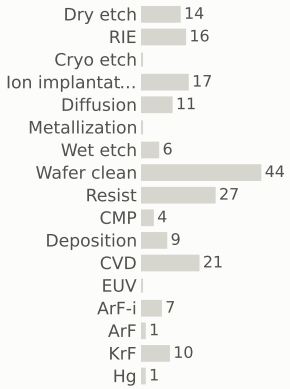
y cut at 57.351 µm



mask for this step

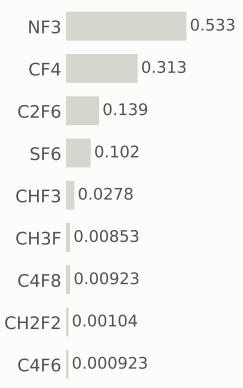
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



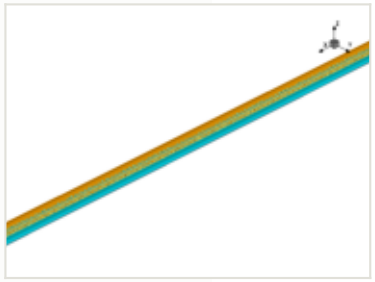
Step #2530: Spx_only

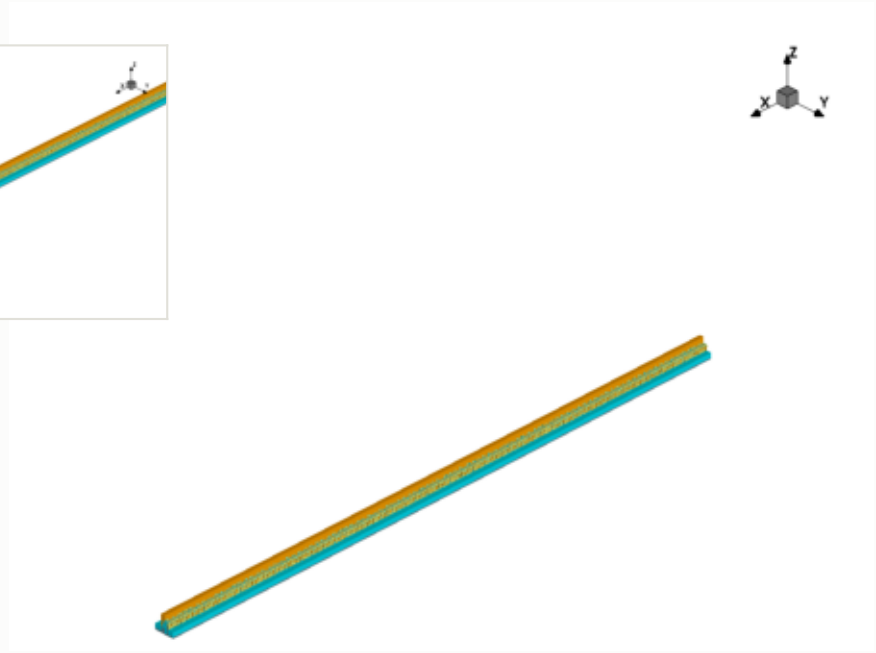
etch_anisotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





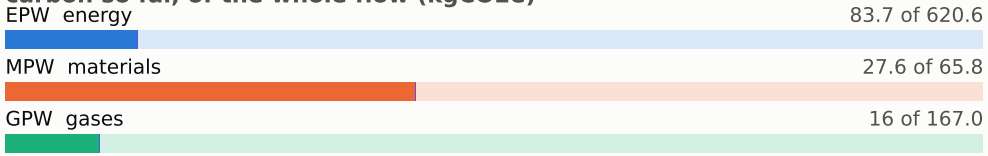
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

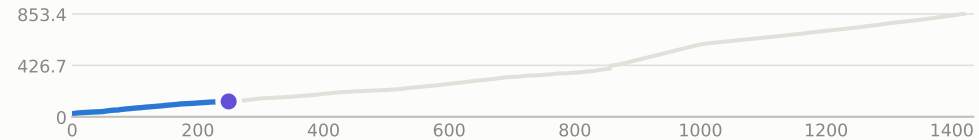
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

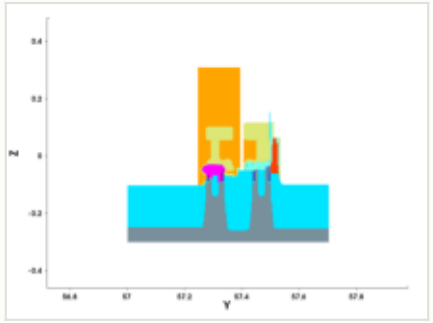
carbon so far, of the whole flow (kgCO2e)



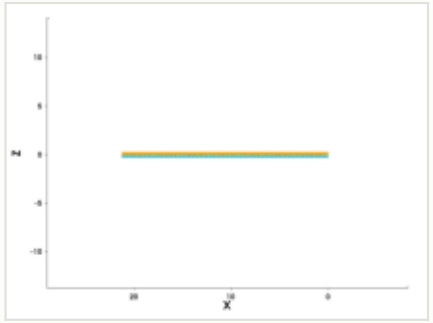
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



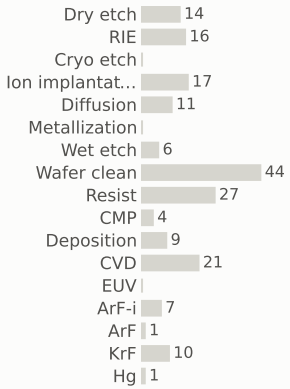
y cut at 57.351 µm



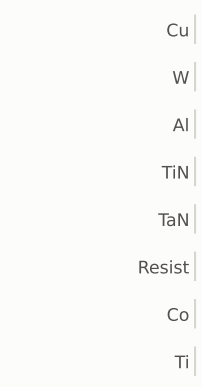
mask for this step

no mask for this step

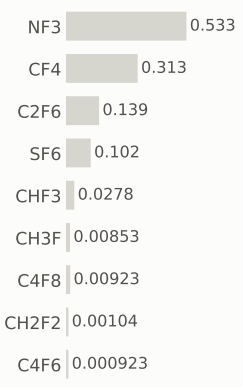
steps by process area



materials deposited (g)



gases emitted (g)



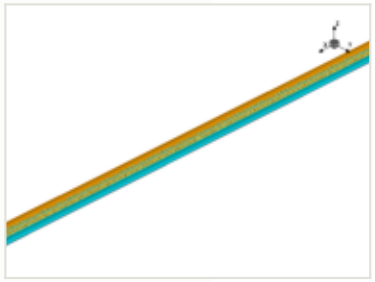
Step #2540: Spx_only

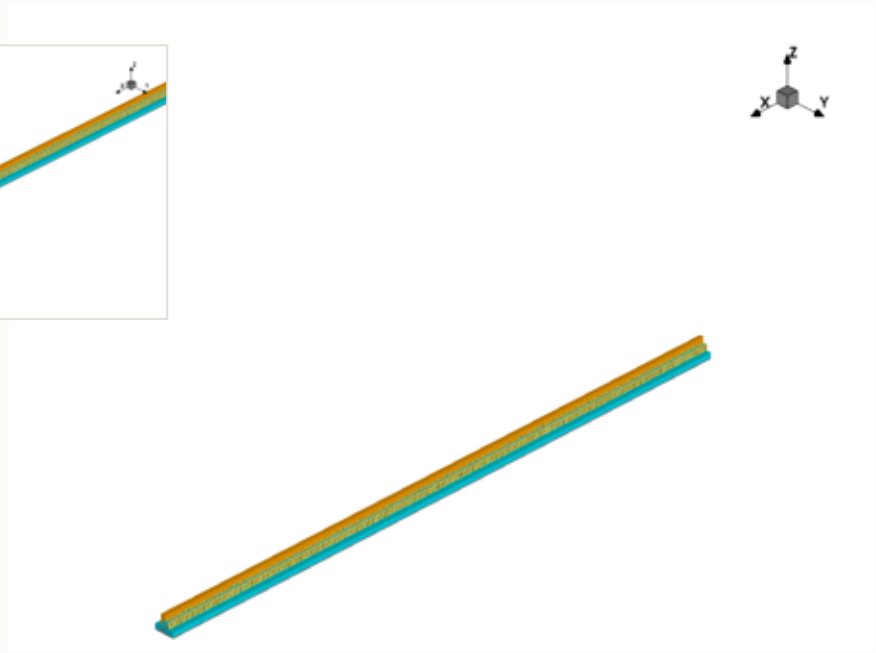
etch_anisotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





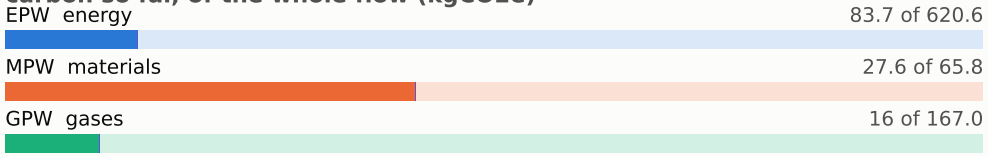
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

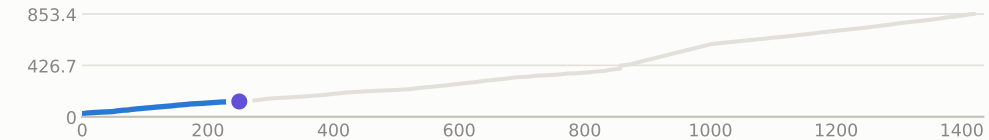
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

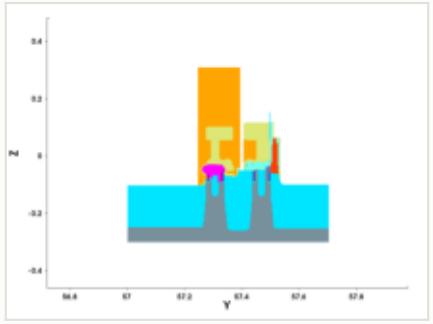
carbon so far, of the whole flow (kgCO2e)



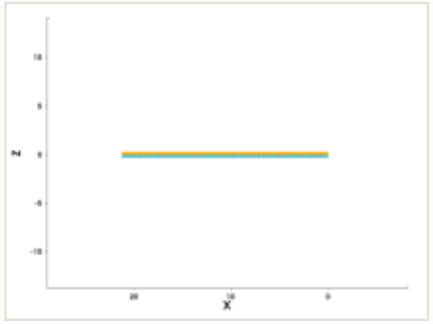
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



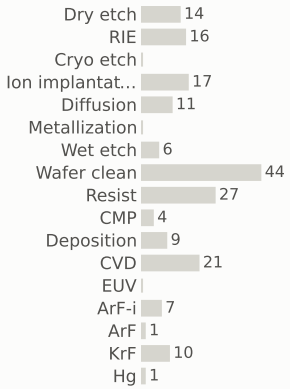
y cut at 57.351 µm



mask for this step

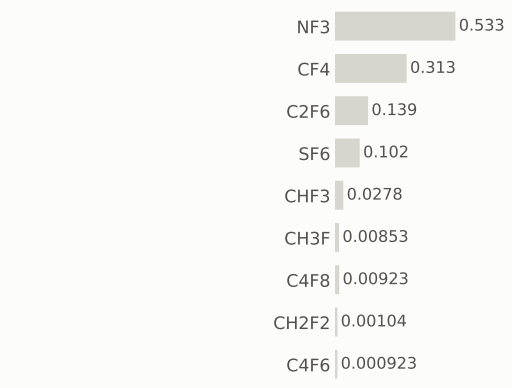
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



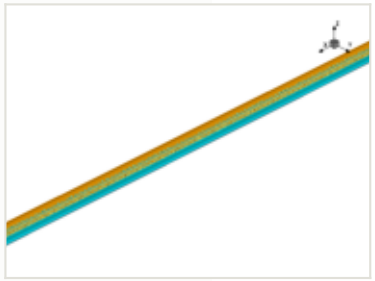
Step #2550: Spx_only

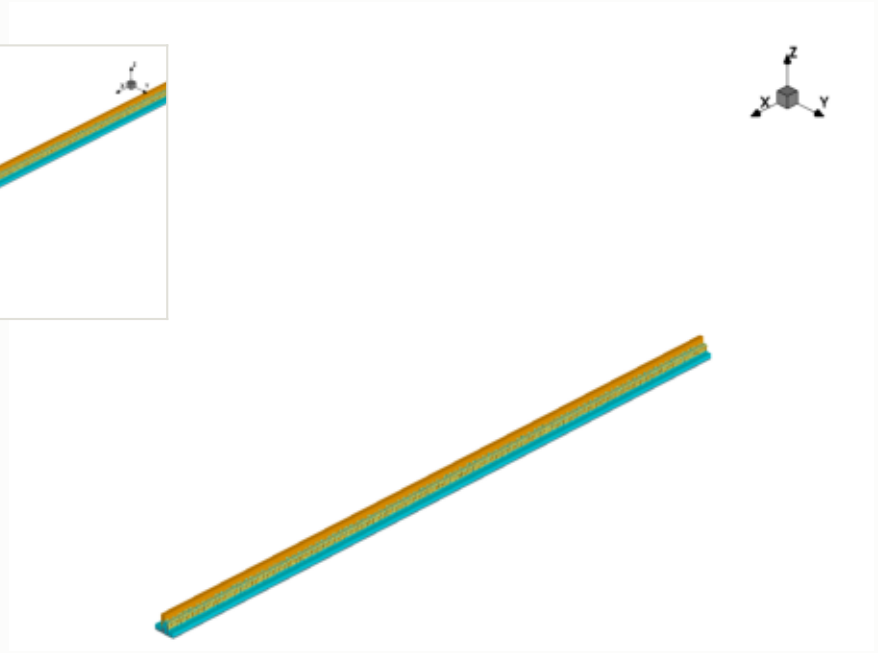
etch_microloading · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

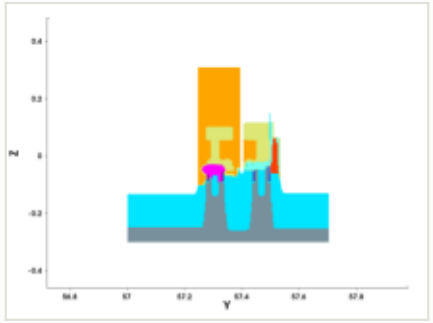
3D structure after this step

zoomed to fit

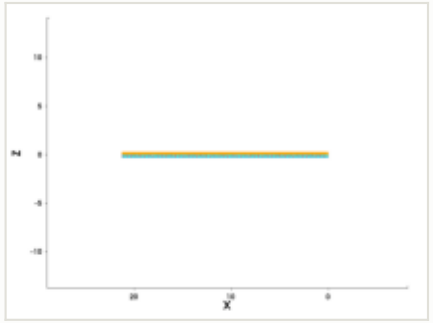




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

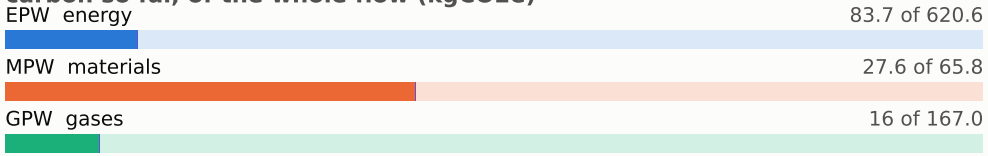
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

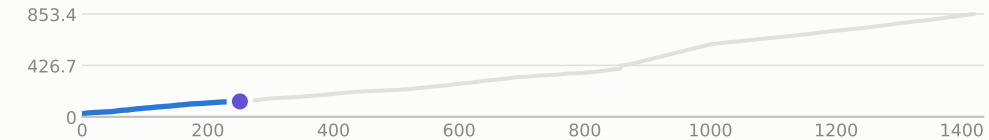
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

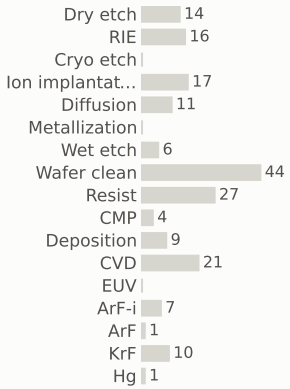
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

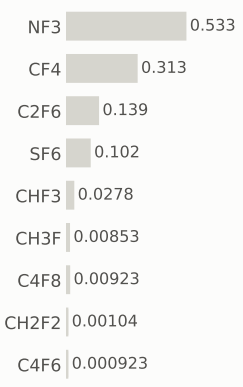


steps by process area



materials deposited (g)

gases emitted (g)



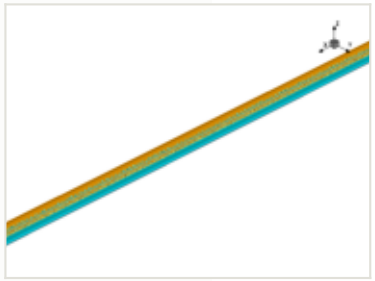
Step #2560: Spx_only

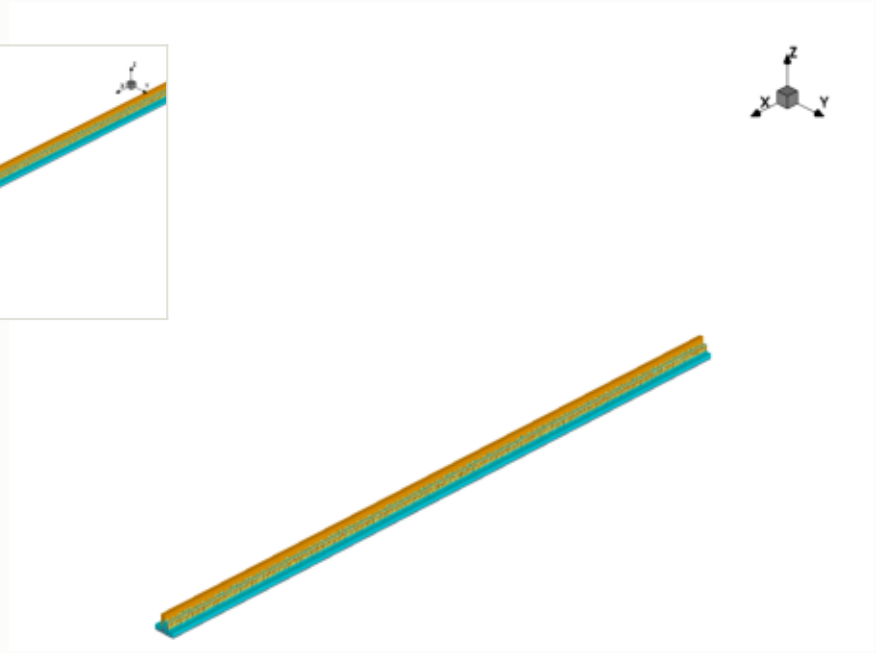
etch_anisotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





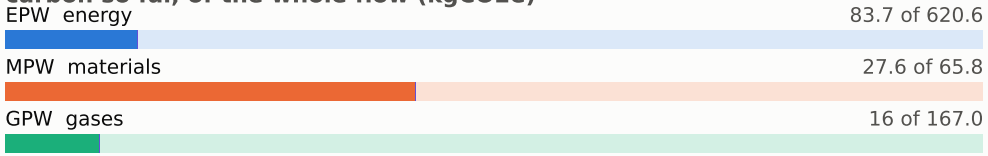
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

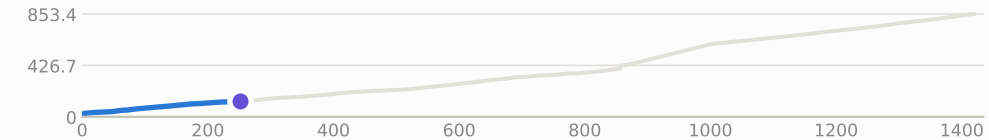
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

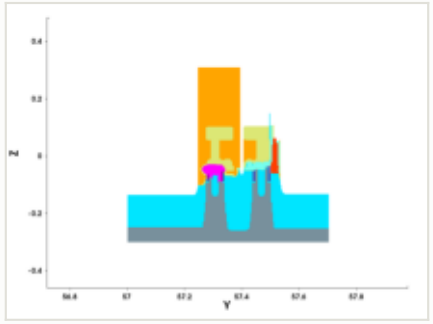
carbon so far, of the whole flow (kgCO2e)



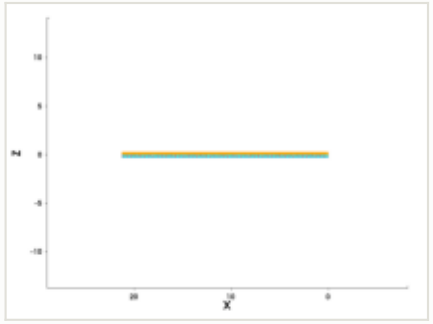
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



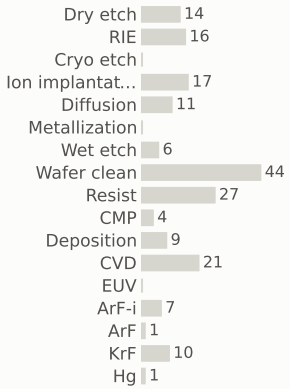
y cut at 57.351 µm



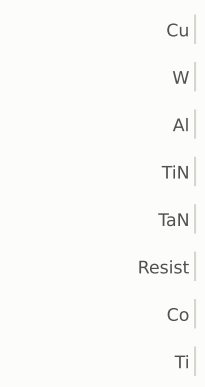
mask for this step

no mask for this step

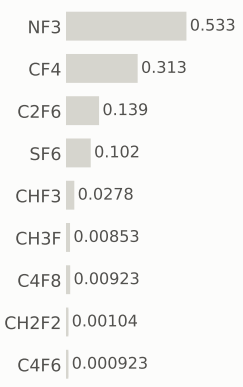
steps by process area



materials deposited (g)



gases emitted (g)



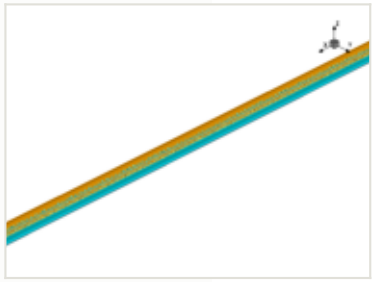
Step #2570: Spx_only

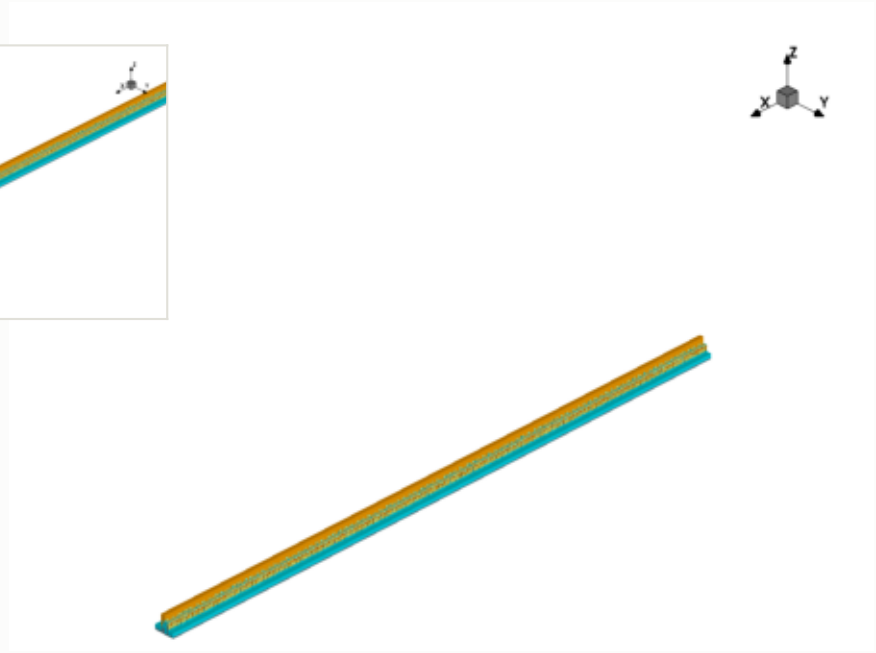
etch_isotropic · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



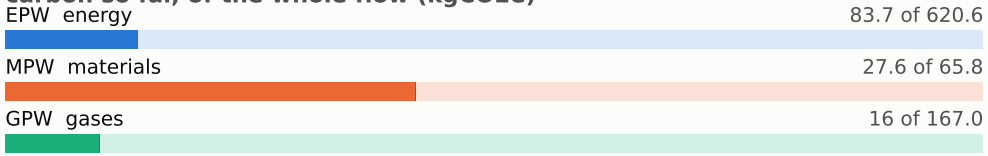


this step

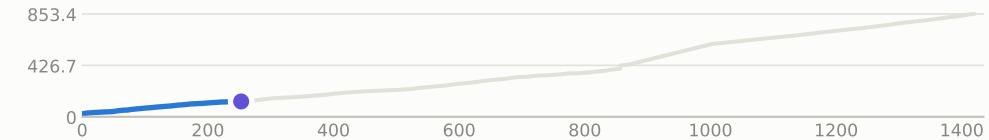
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

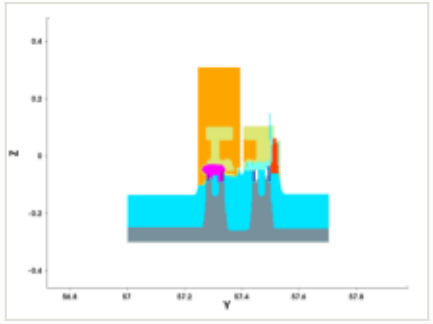
carbon so far, of the whole flow (kgCO2e)



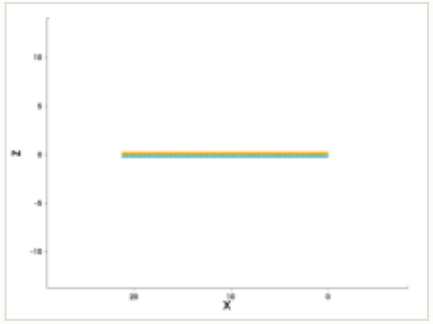
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



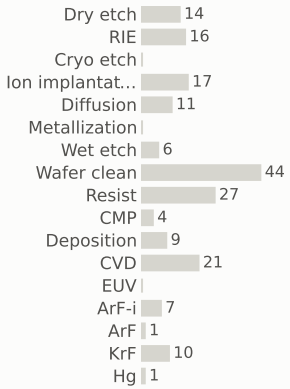
y cut at 57.351 µm



mask for this step

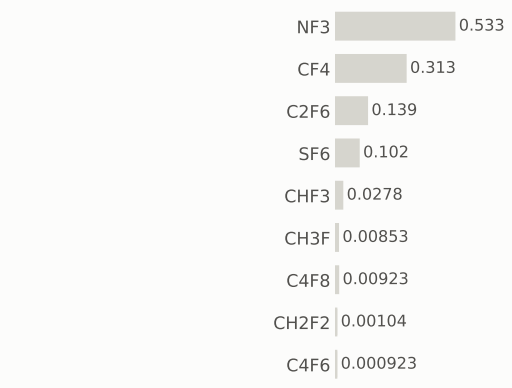
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



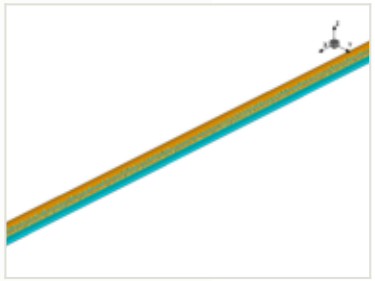
Step #2580: Spx_only

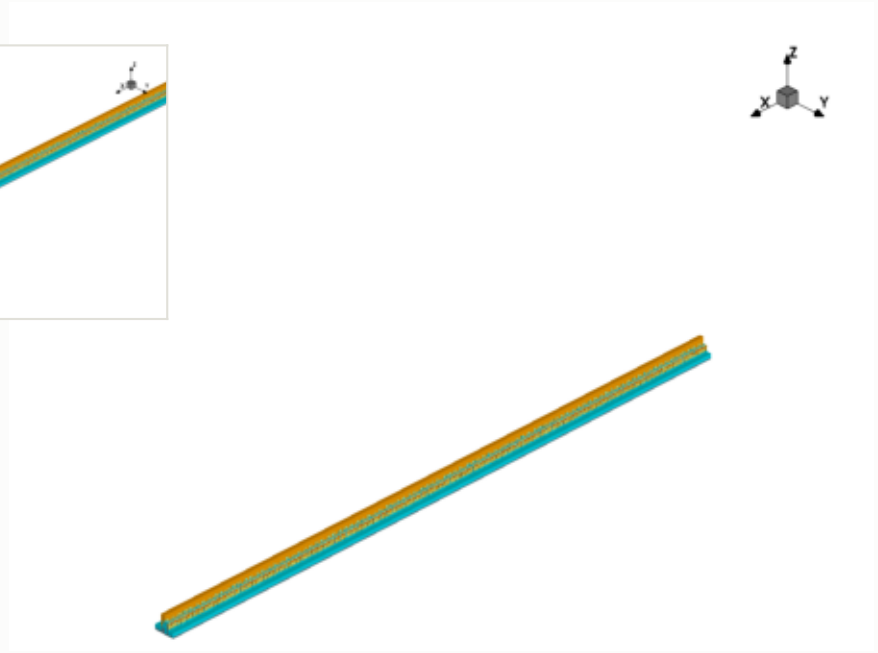
etch_ellipsoidal · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

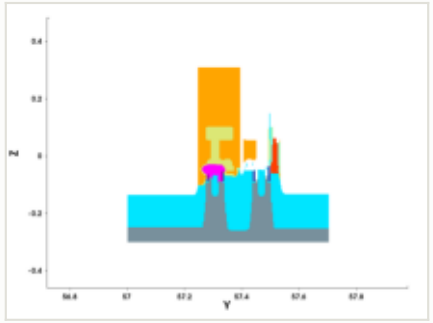
3D structure after this step

zoomed to fit

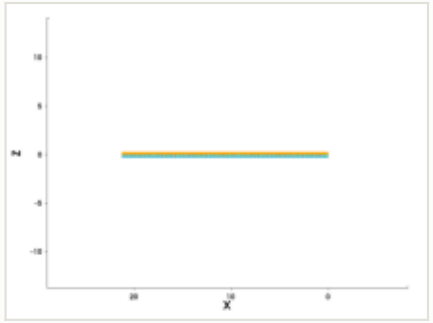




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

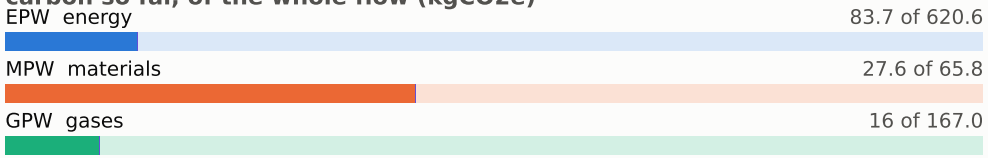
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

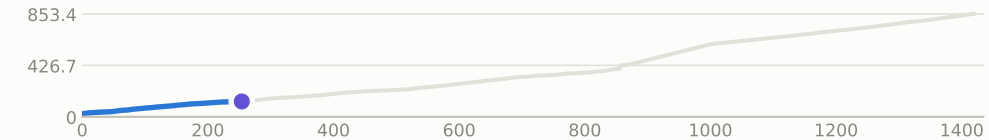
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

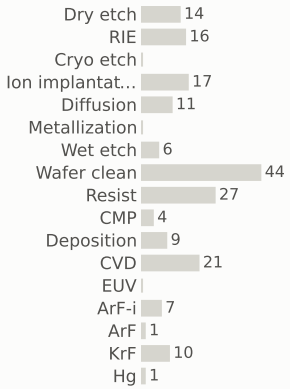
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

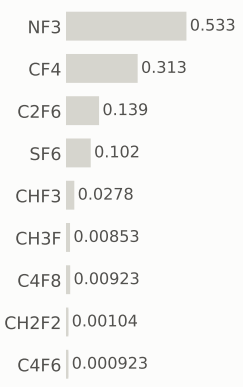


steps by process area



materials deposited (g)

gases emitted (g)



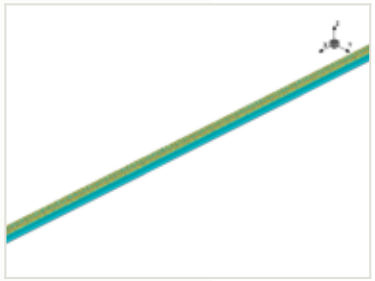
Step #2590: Spx_only

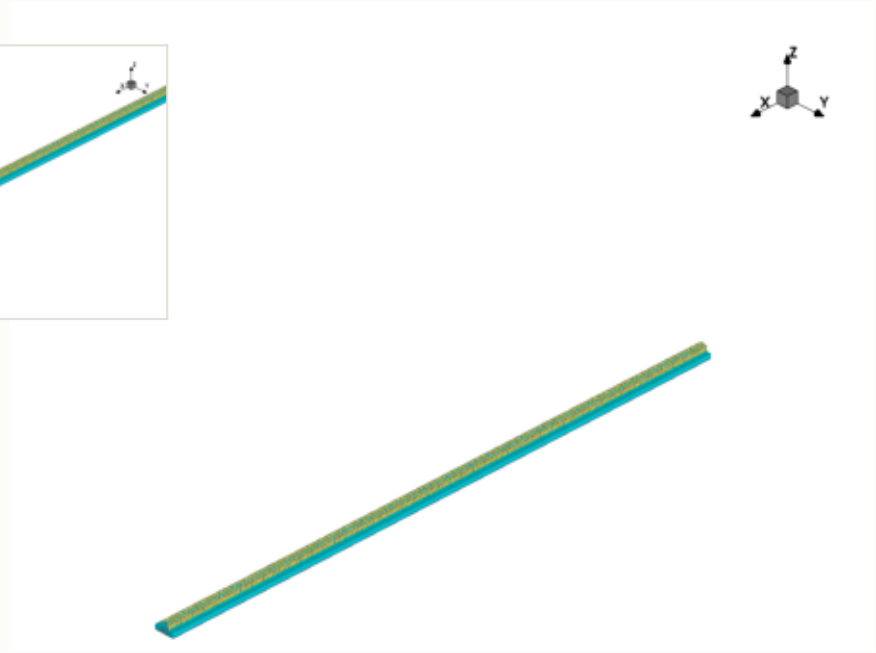
etch_strip · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





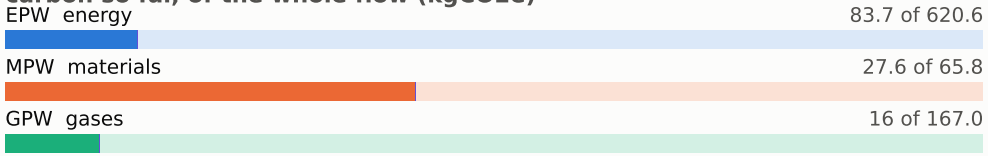
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

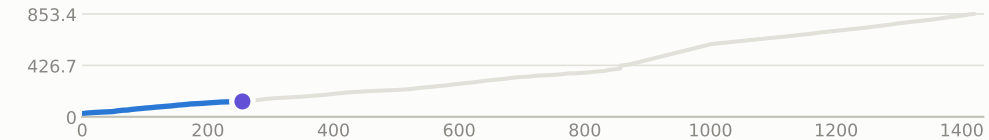
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

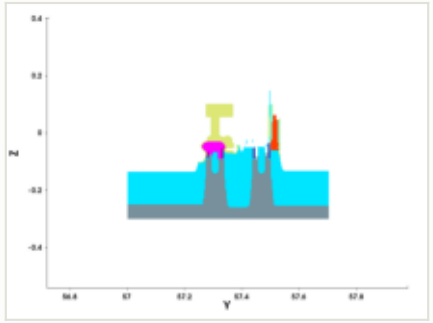
carbon so far, of the whole flow (kgCO2e)



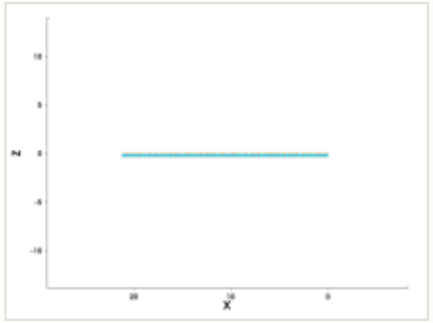
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



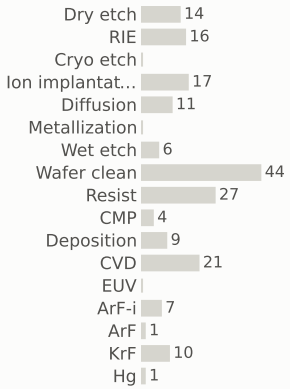
y cut at 57.351 µm



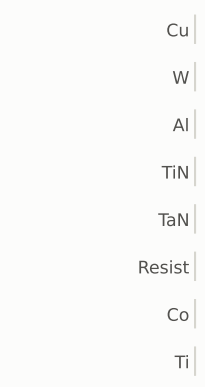
mask for this step

no mask for this step

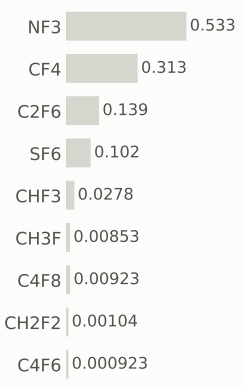
steps by process area



materials deposited (g)



gases emitted (g)



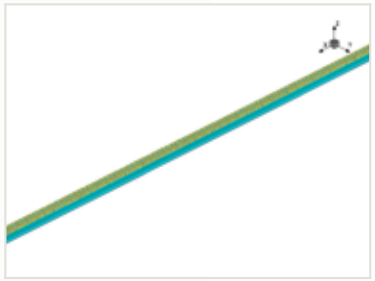
Step #2600: Spx_only

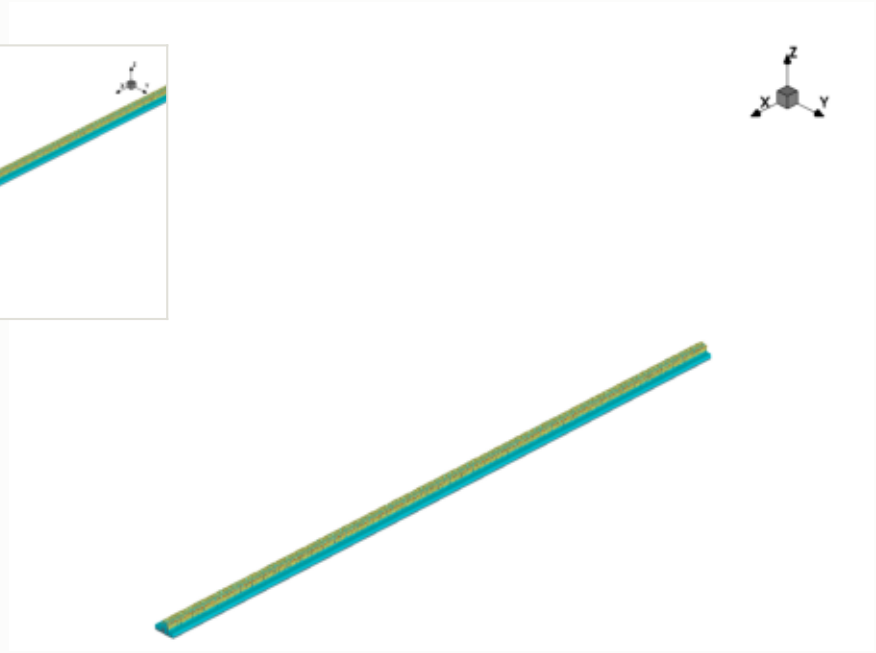
etch_ellipsoidal · Spx_only

0	127.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





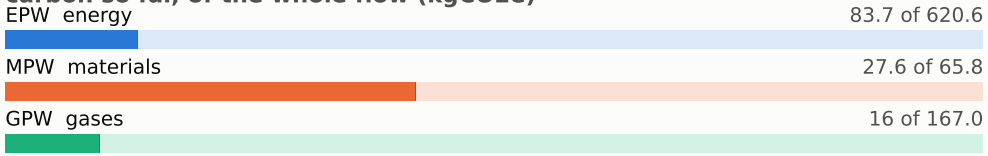
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

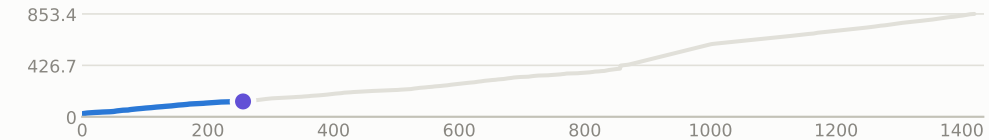
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

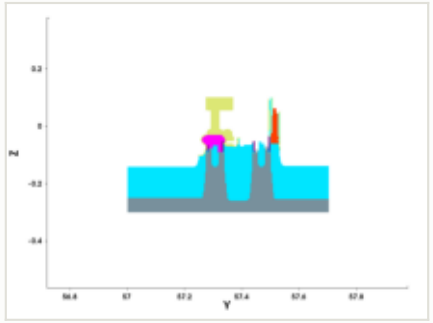
carbon so far, of the whole flow (kgCO2e)



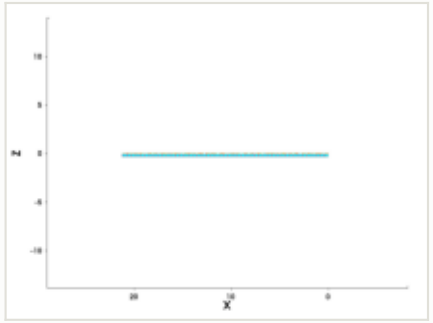
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



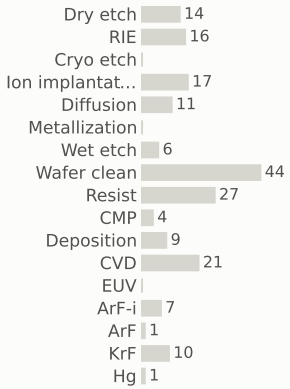
y cut at 57.351 µm



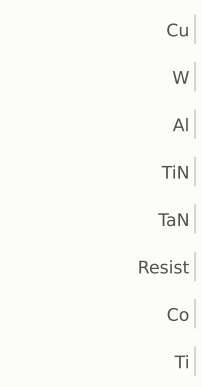
mask for this step

no mask for this step

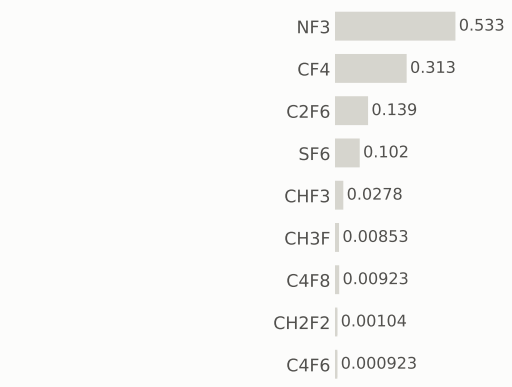
steps by process area



materials deposited (g)



gases emitted (g)



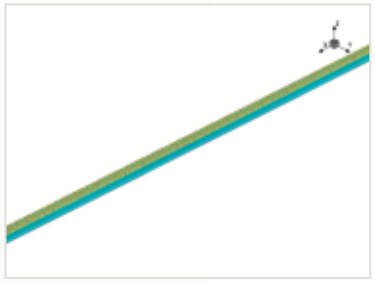
Step #2610: Wafer_clean

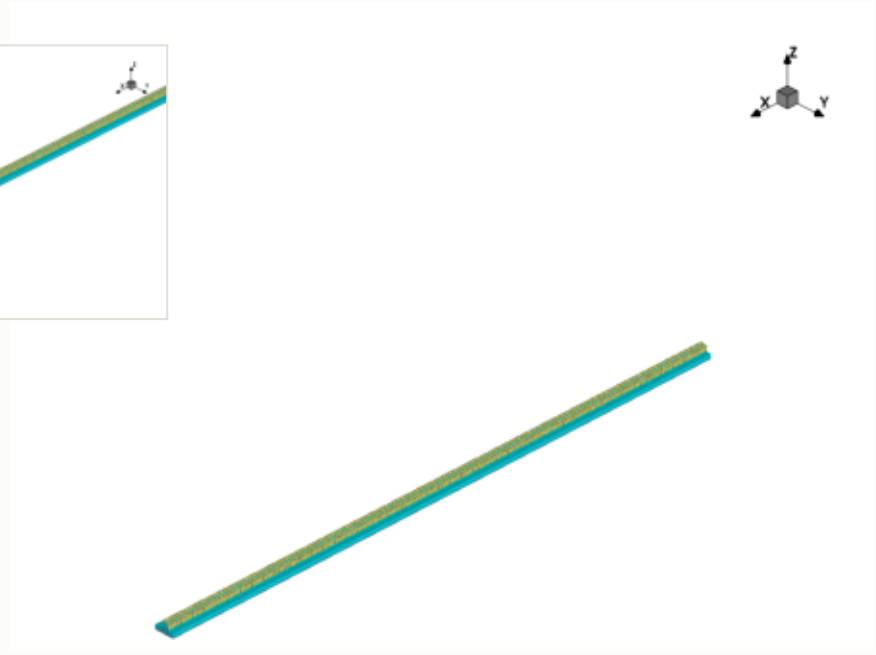
clean · Wafer_clean

0.169	127.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



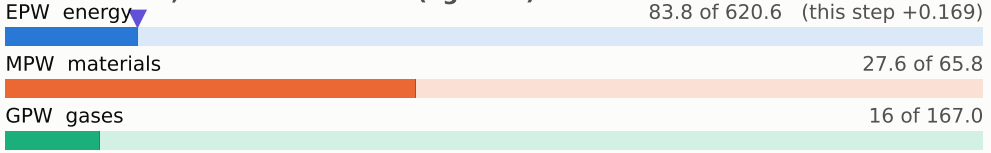


this step

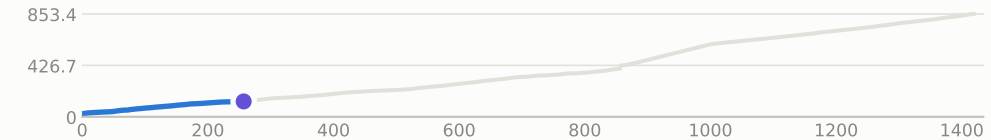
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

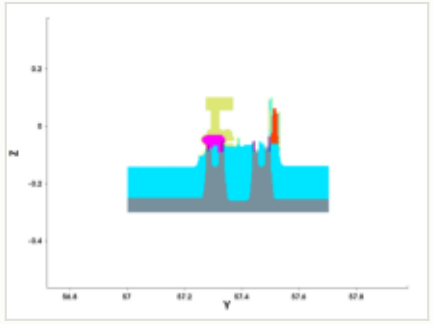
carbon so far, of the whole flow (kgCO2e)



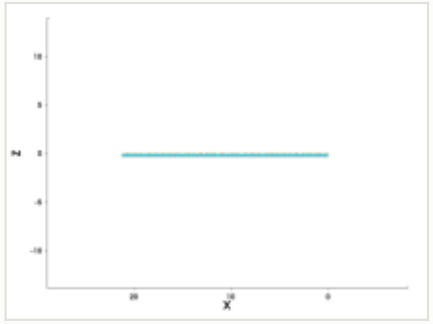
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



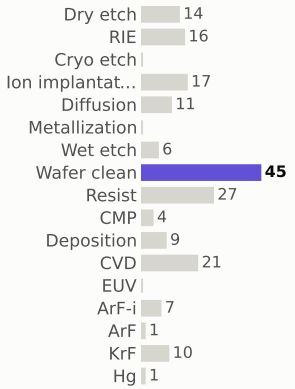
y cut at 57.351 µm



mask for this step

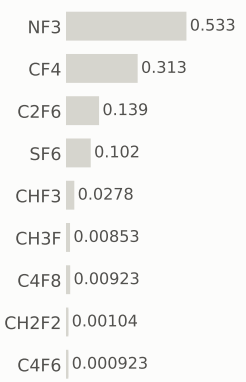
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



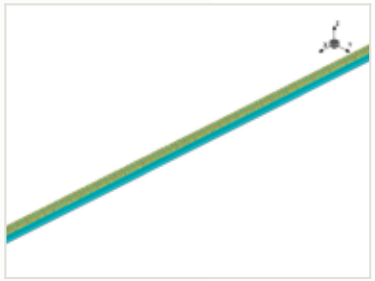
Step #2620: CVD

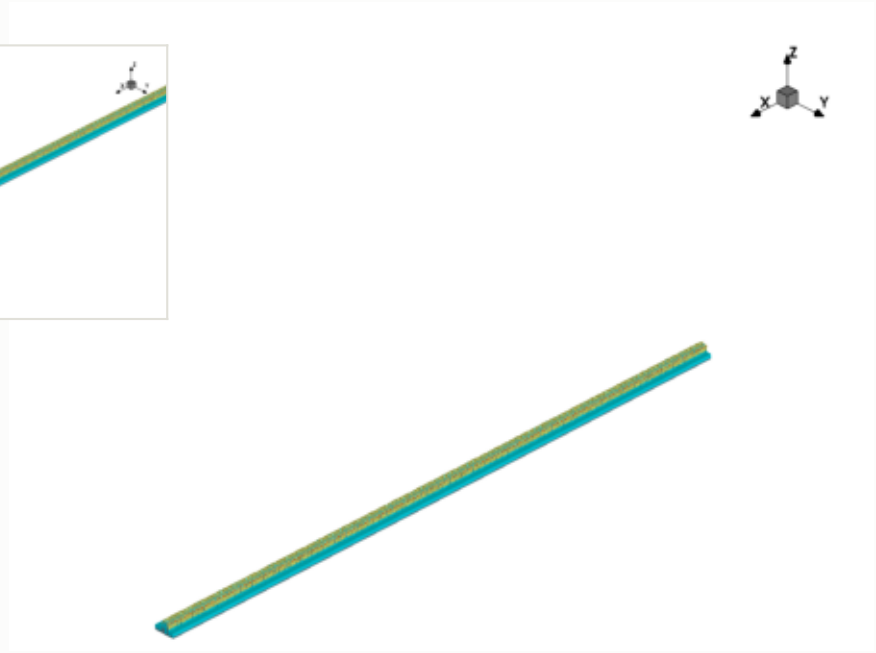
depo_epi · CVD

1.31	128.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



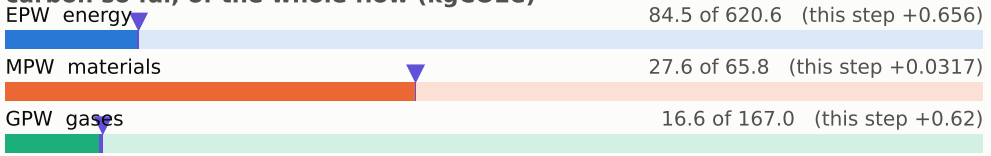


this step

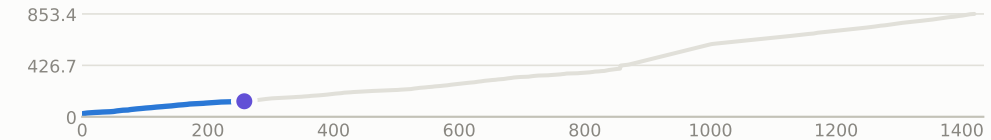
process area	CVD
material	—
thickness	0.0266 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

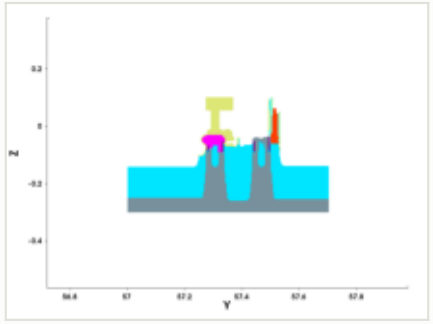
carbon so far, of the whole flow (kgCO2e)



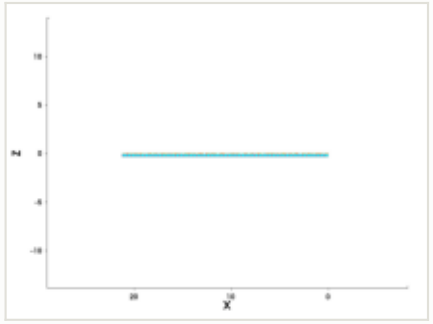
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



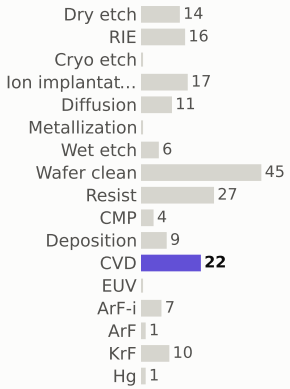
y cut at 57.351 µm



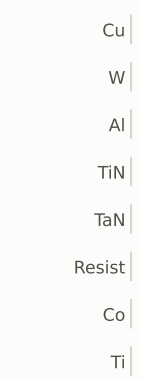
mask for this step

no mask for this step

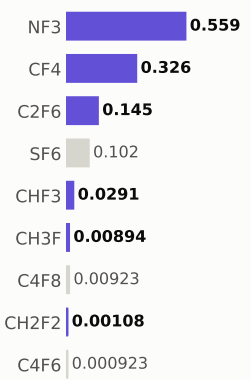
steps by process area



materials deposited (g)



gases emitted (g)



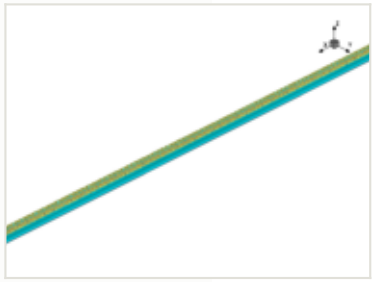
Step #2630: Spx_only

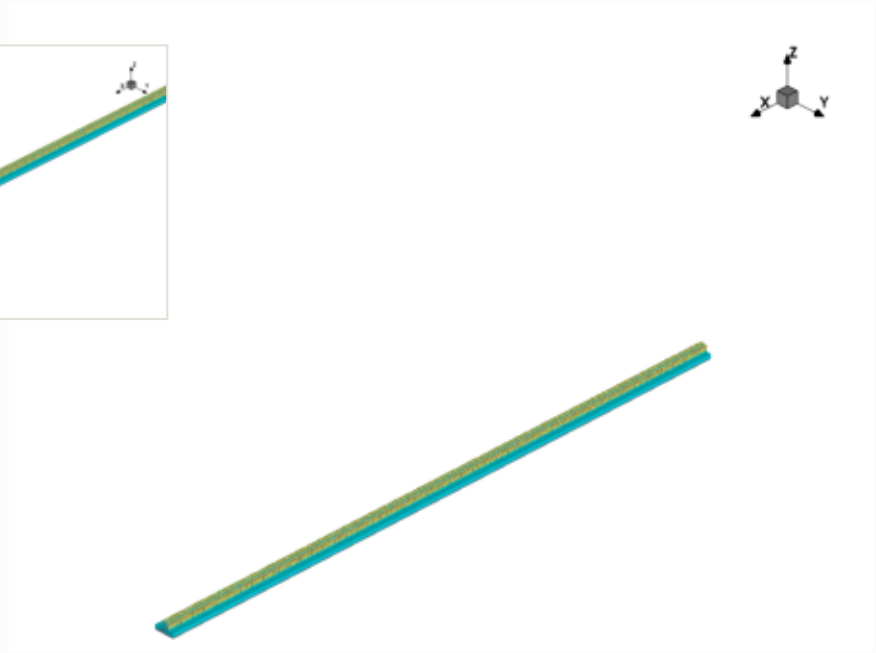
depo_epi · Spx_only

0	128.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





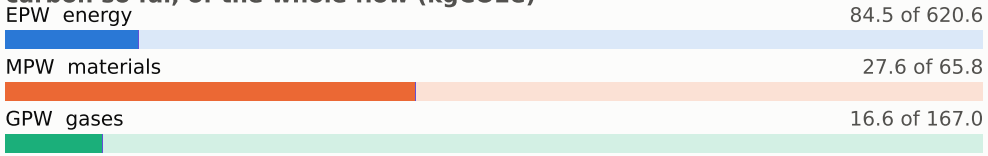
this step

process area	Spx_only
material	—
thickness	0.009 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

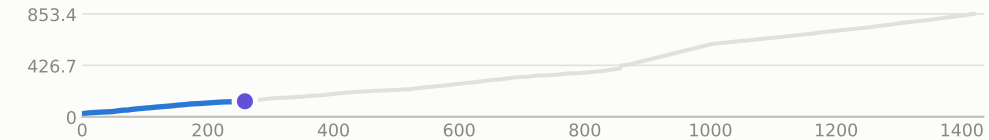
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

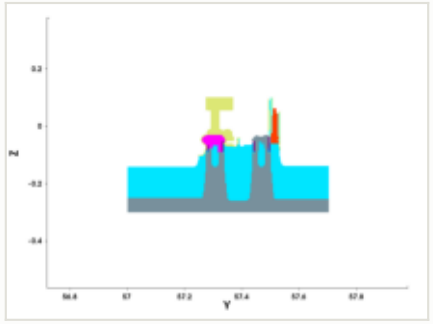
carbon so far, of the whole flow (kgCO2e)



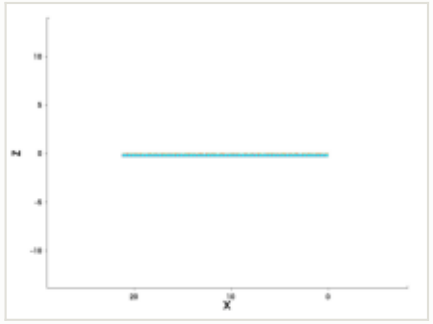
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



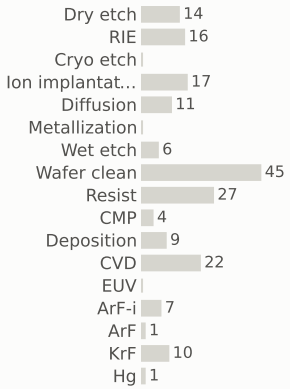
y cut at 57.351 µm



mask for this step

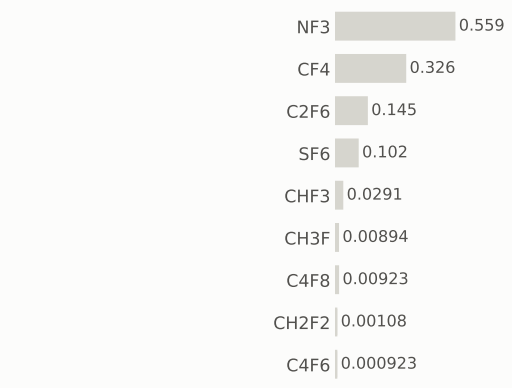
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



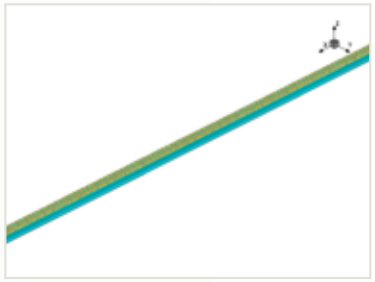
Step #2640: Wafer_clean

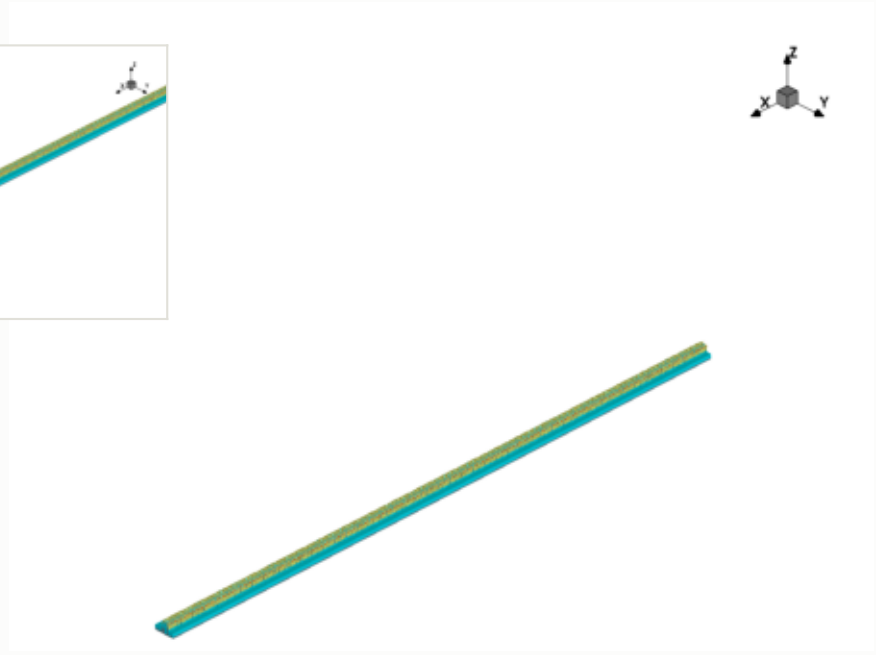
clean · Wafer_clean

0.169	128.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





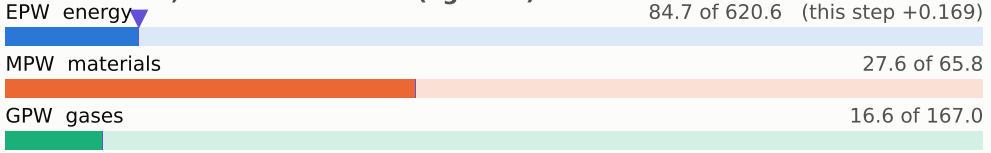
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

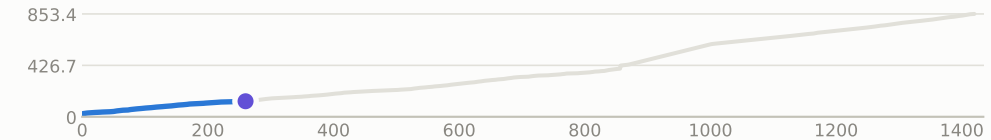
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

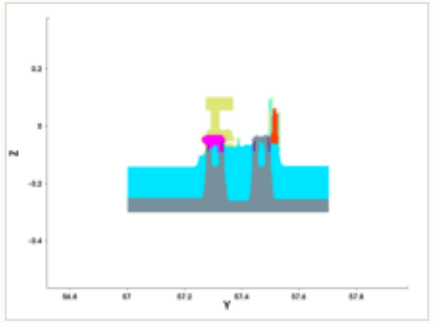
carbon so far, of the whole flow (kgCO2e)



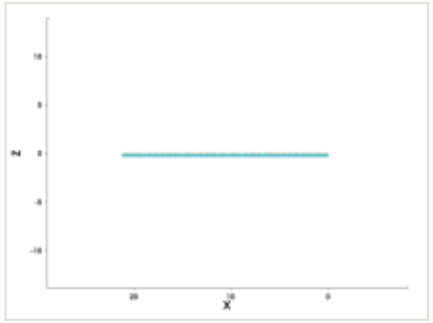
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



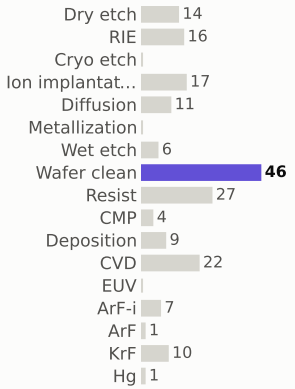
y cut at 57.351 µm



mask for this step

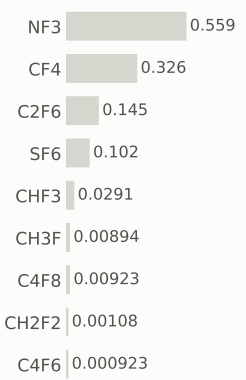
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



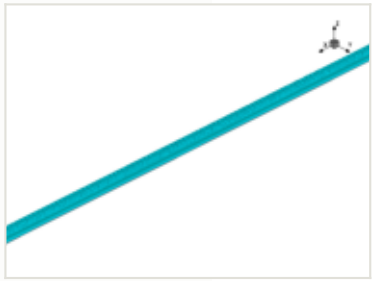
Step #2650: Deposition

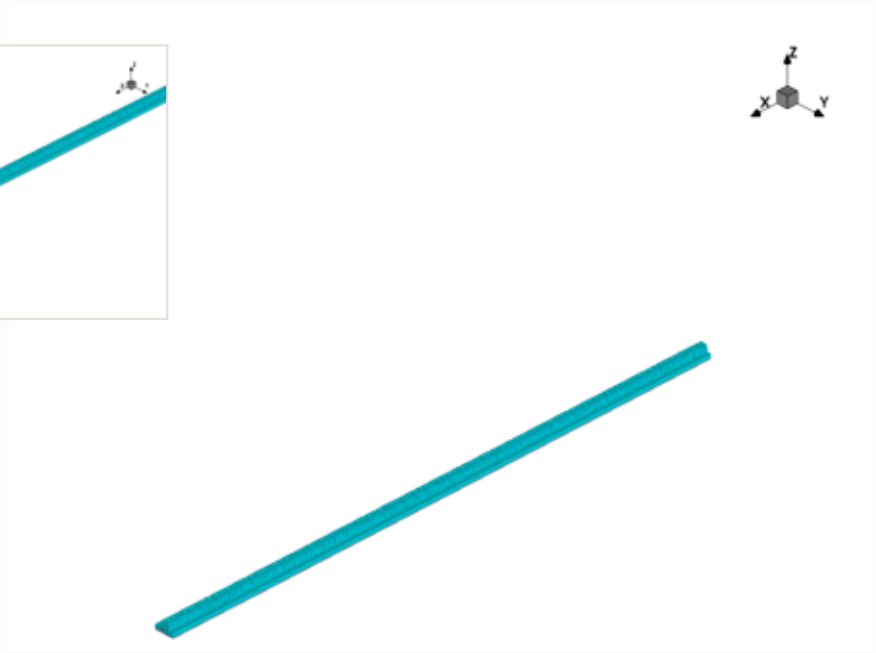
depo_isotropic · Deposition

0.656	129.6	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



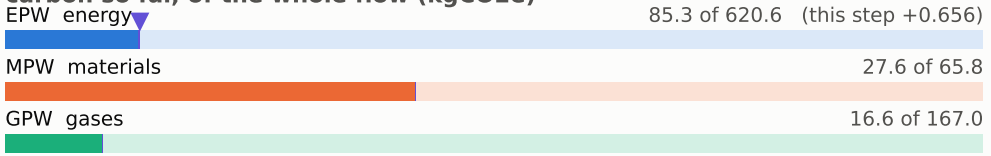


this step

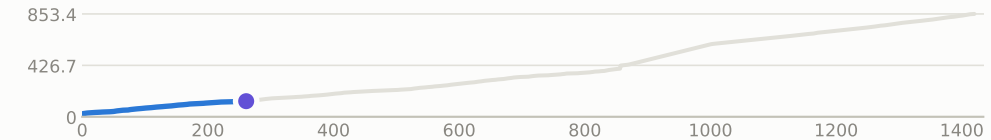
process area	Deposition
material	—
thickness	0.005 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

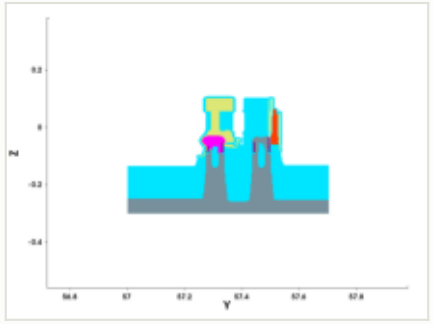
carbon so far, of the whole flow (kgCO2e)



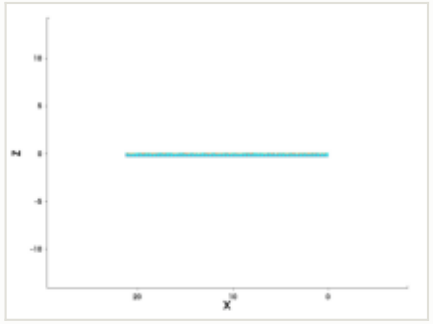
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



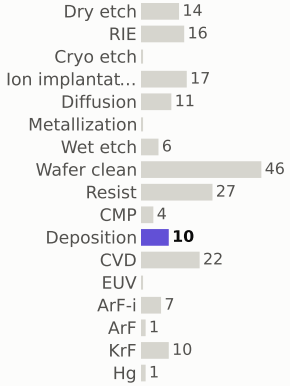
y cut at 57.351 µm



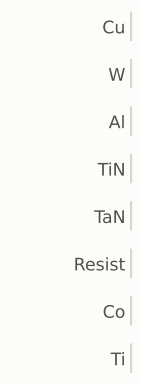
mask for this step

no mask for this step

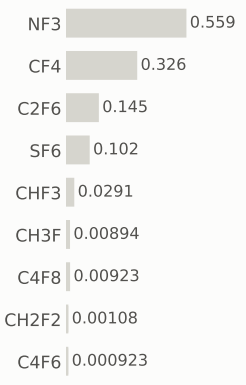
steps by process area



materials deposited (g)



gases emitted (g)



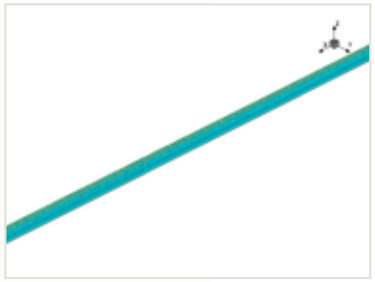
Step #2660: RIE

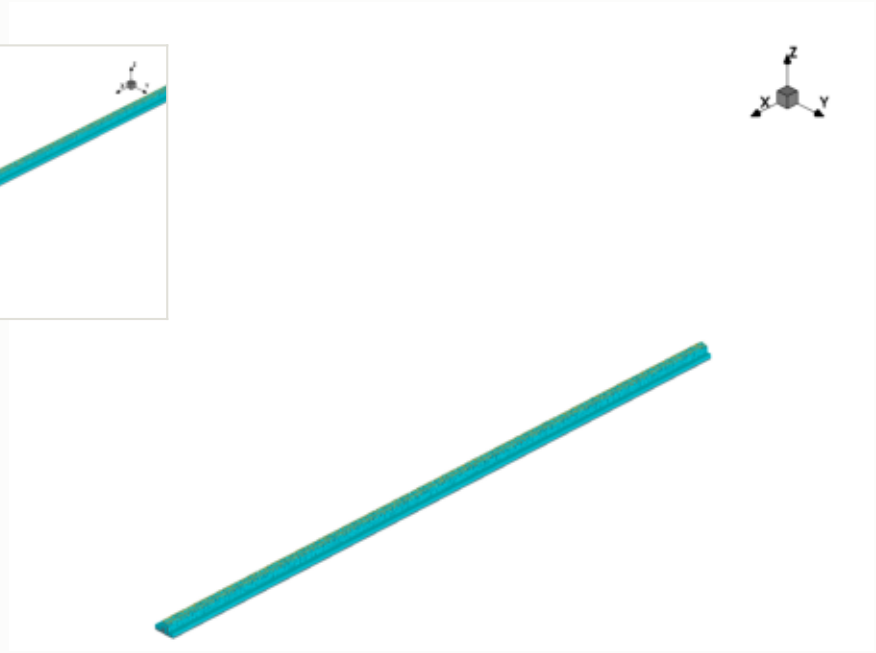
etch_anisotropic · RIE

1.04	130.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



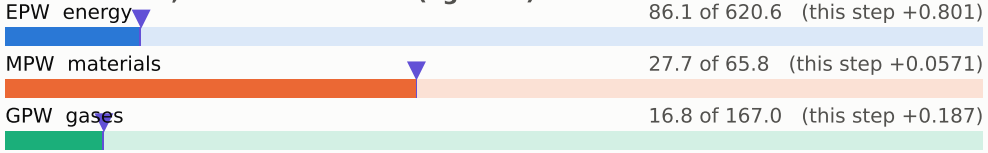


this step

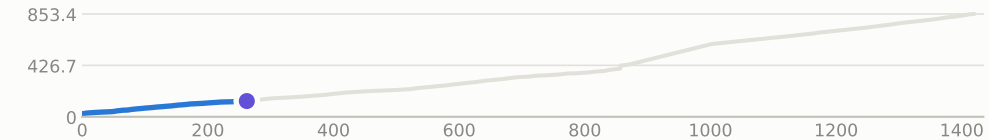
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

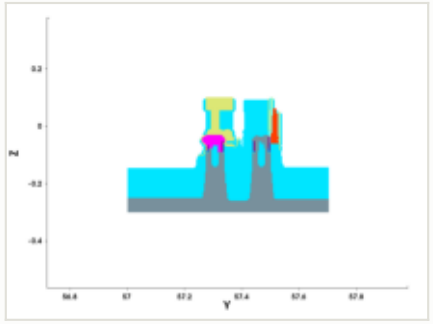
carbon so far, of the whole flow (kgCO2e)



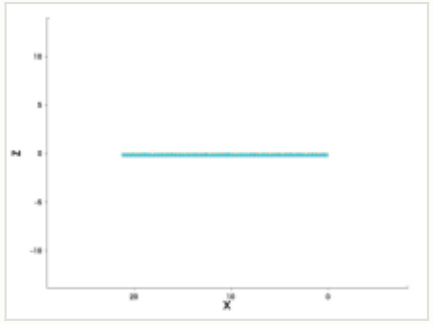
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



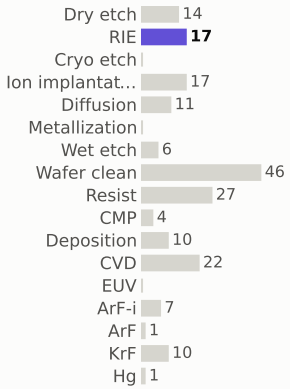
y cut at 57.351 µm



mask for this step

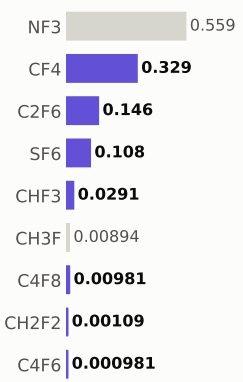
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



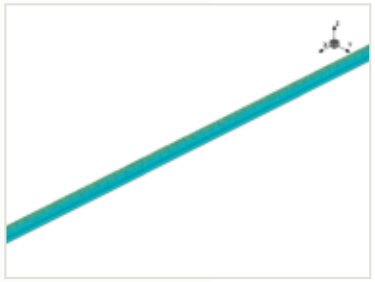
Step #2670: Wafer_clean

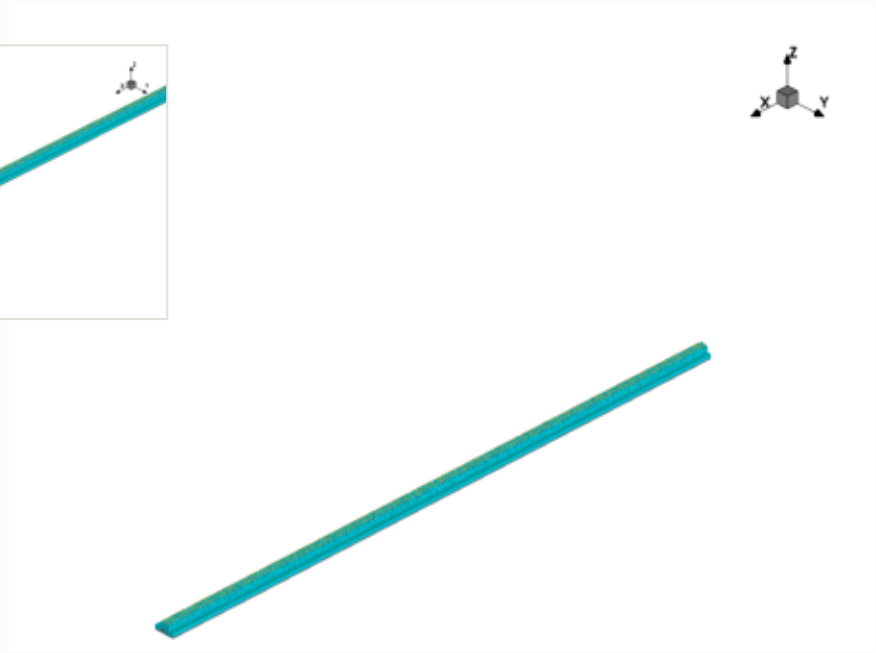
clean · Wafer_clean

0.169	130.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



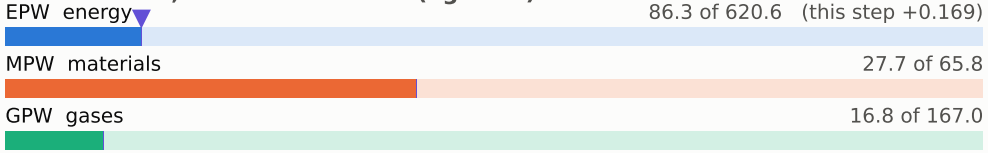


this step

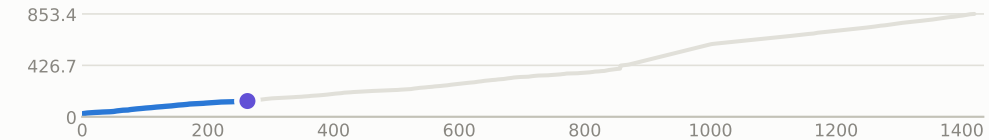
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

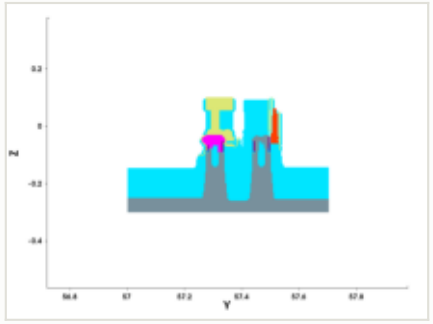
carbon so far, of the whole flow (kgCO2e)



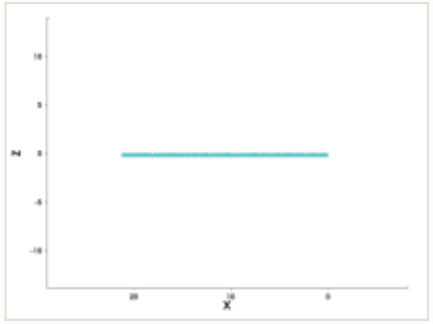
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



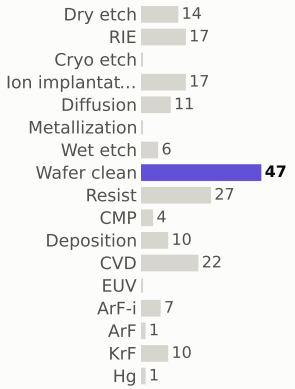
y cut at 57.351 µm



mask for this step

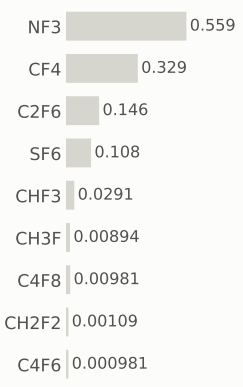
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



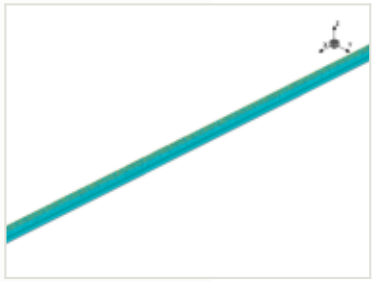
Step #2680: Resist

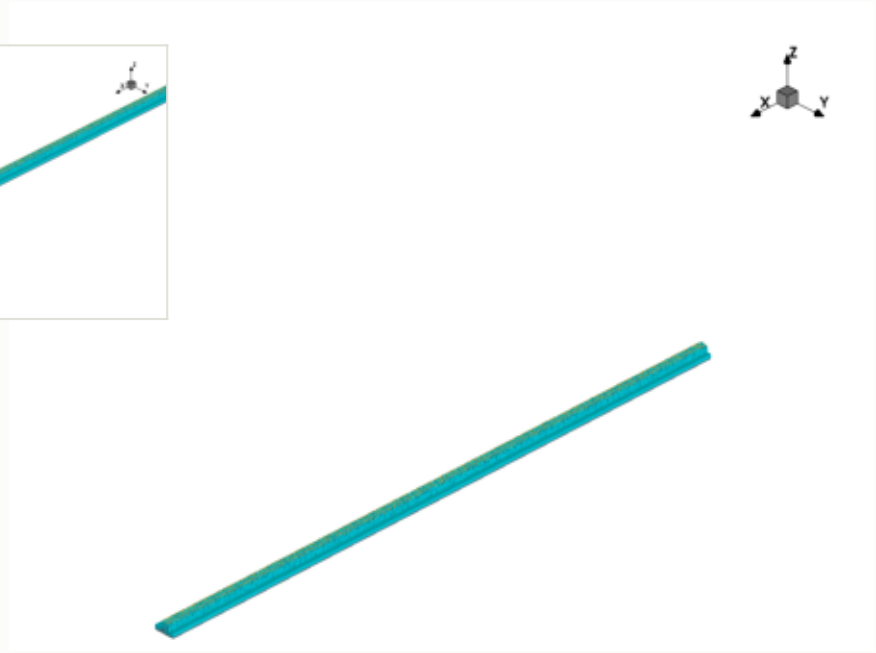
resist · Resist

0.169	131.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



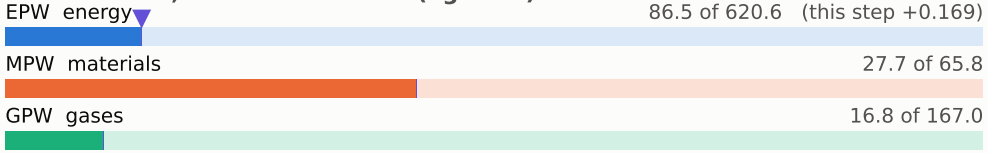


this step

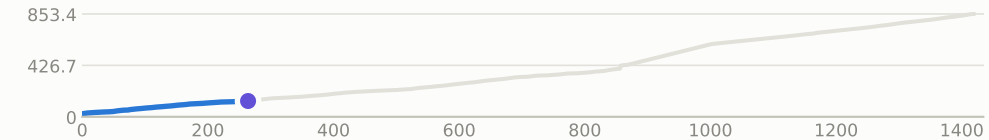
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

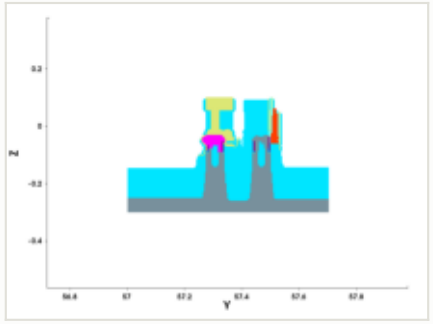
carbon so far, of the whole flow (kgCO2e)



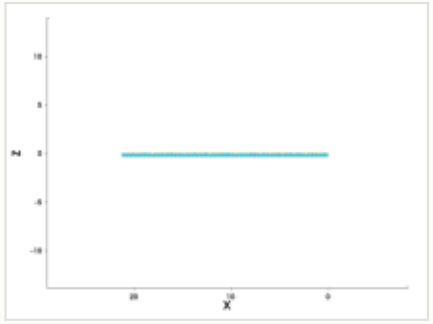
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



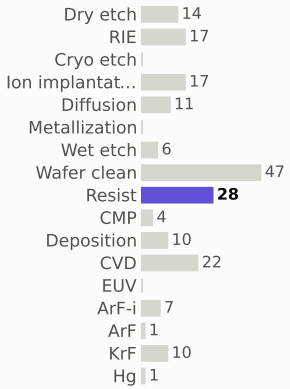
y cut at 57.351 µm



mask for this step

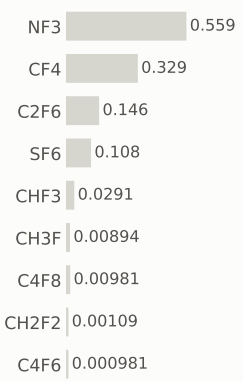
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



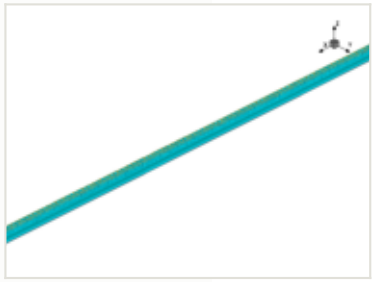
Step #2690: KrF

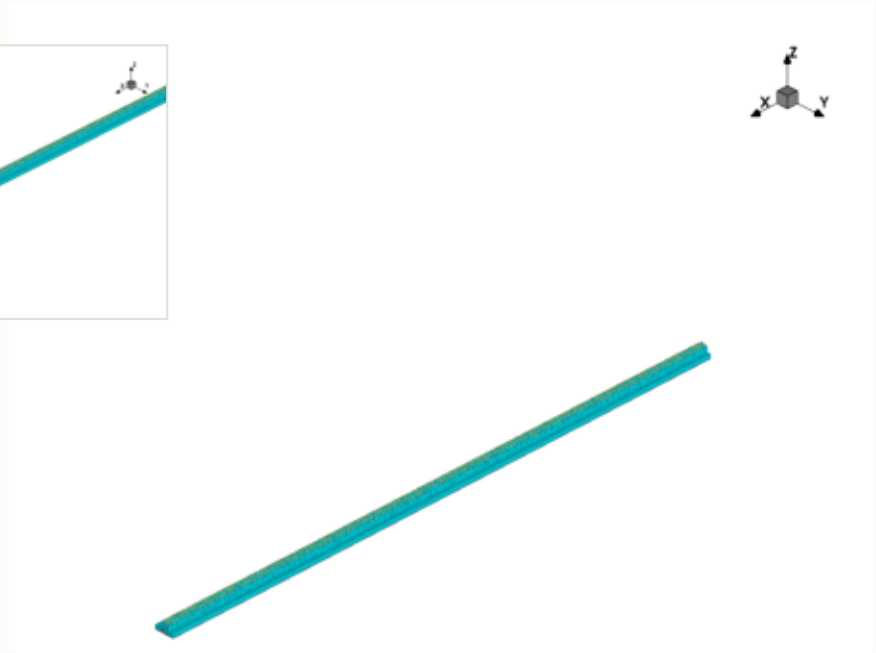
litho_patterning · KrF

0.199	131.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



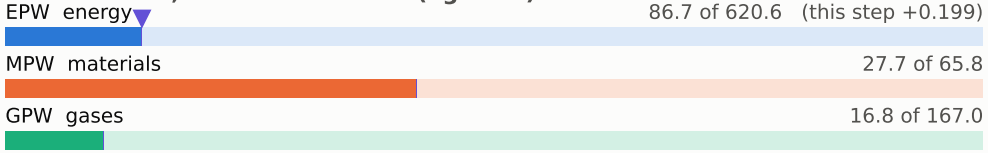


this step

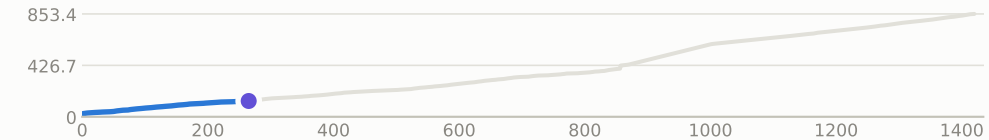
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

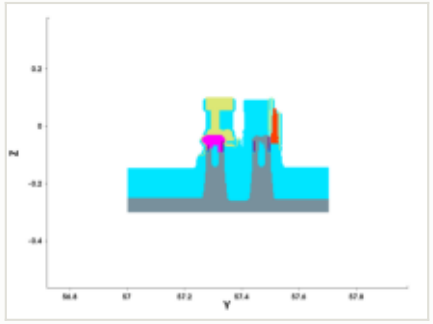
carbon so far, of the whole flow (kgCO2e)



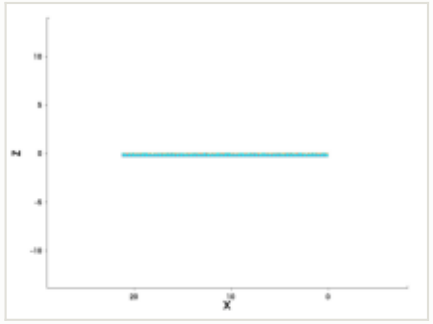
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

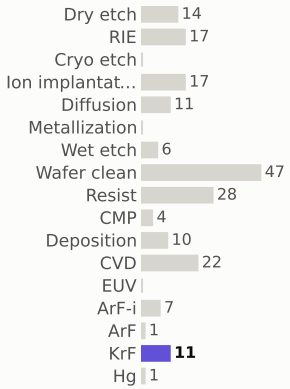


y cut at 57.351 µm



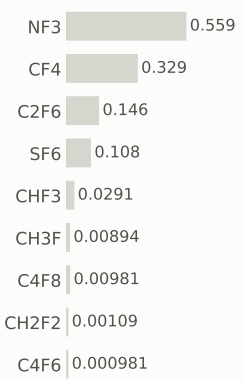
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



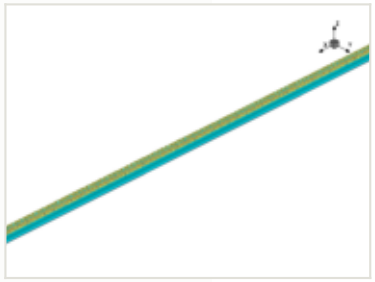
Step #2700: RIE

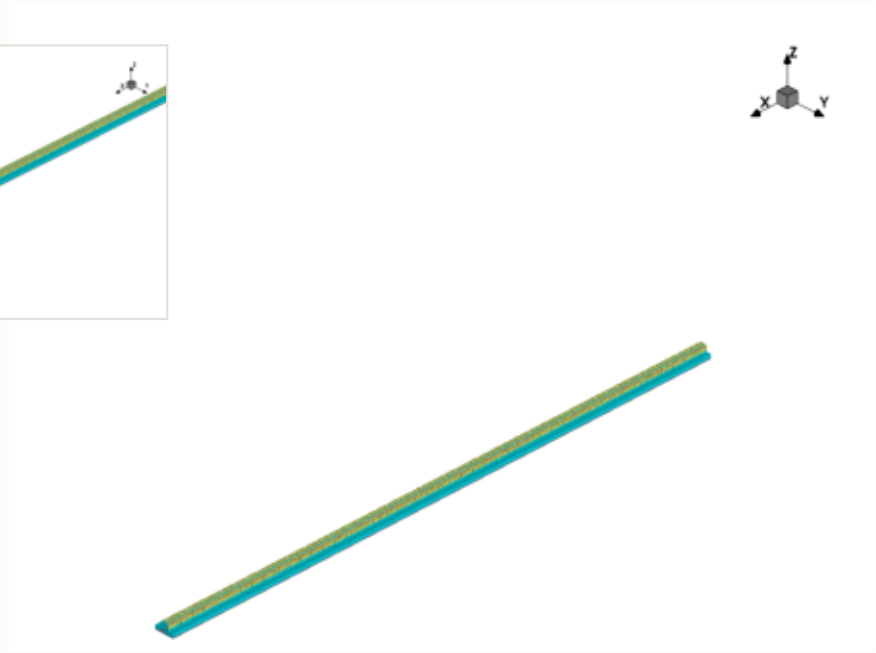
etch_isotropic · RIE

1.04	132.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



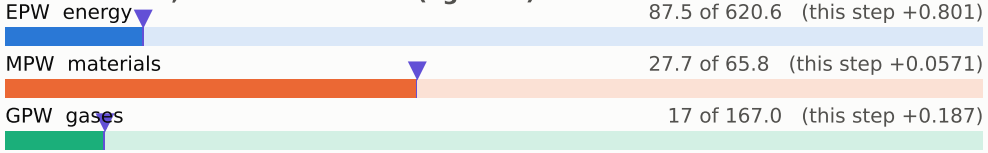


this step

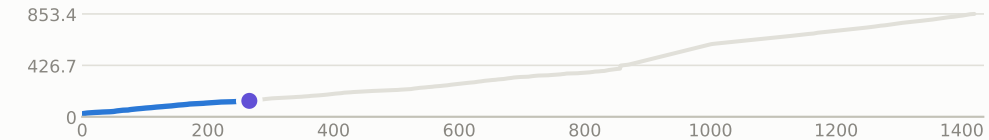
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

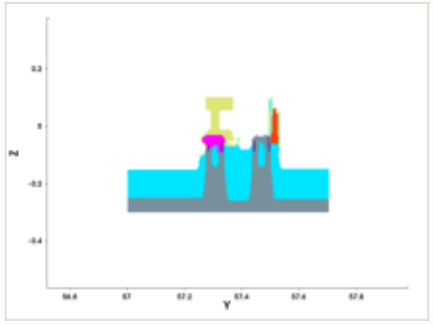
carbon so far, of the whole flow (kgCO2e)



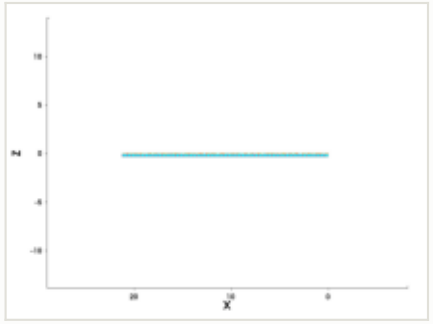
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



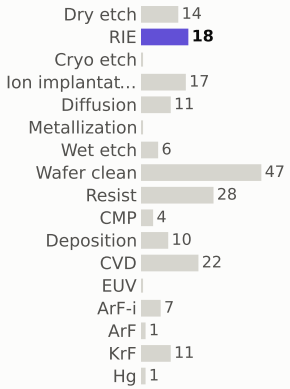
y cut at 57.351 µm



mask for this step

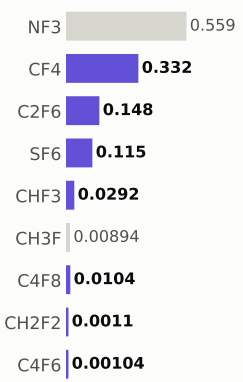
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



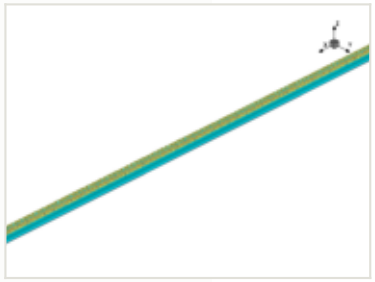
Step #2710: Ion_implantation

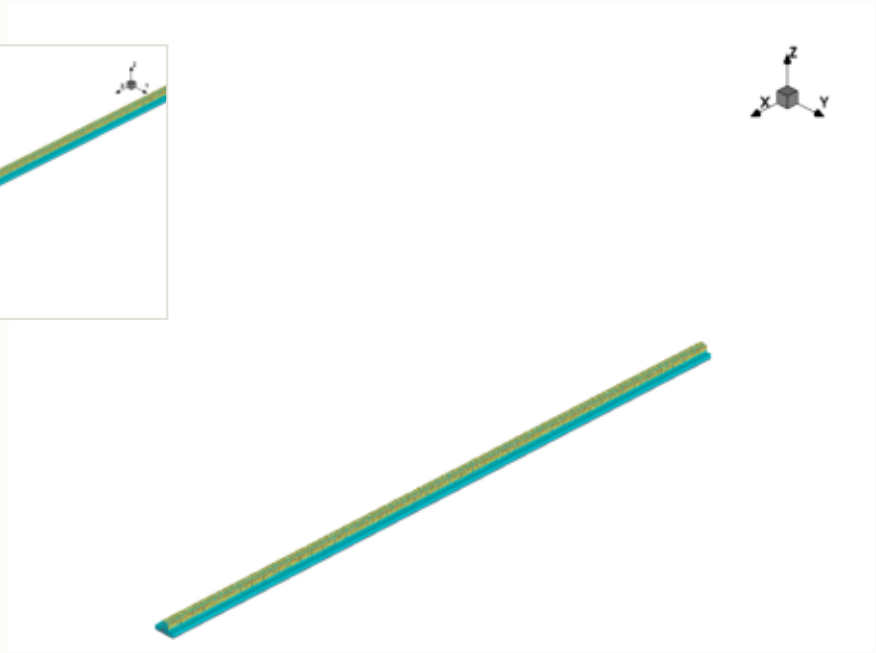
implant_ellipsoid · Ion_implantation

0.801	133.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

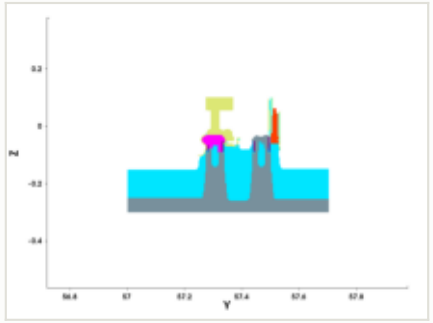
3D structure after this step

zoomed to fit

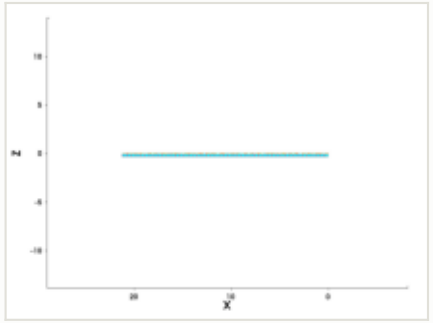




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

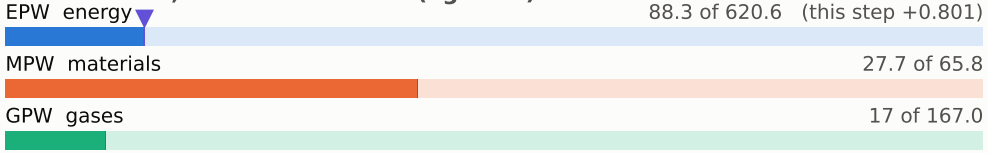
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

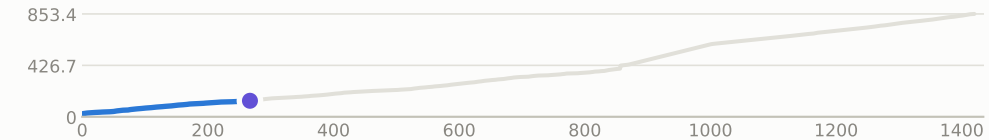
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

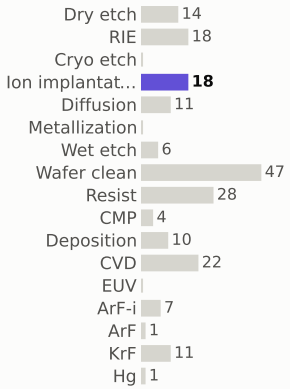
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

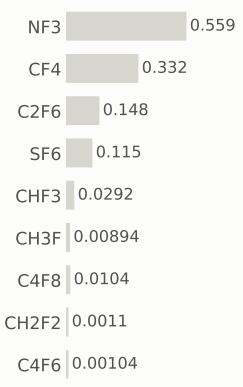


steps by process area



materials deposited (g)

gases emitted (g)



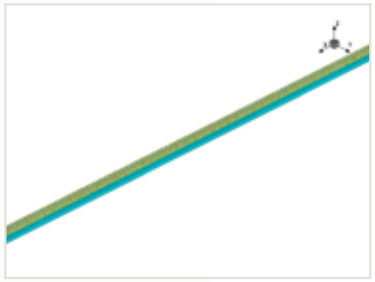
Step #2720: Ion_implantation

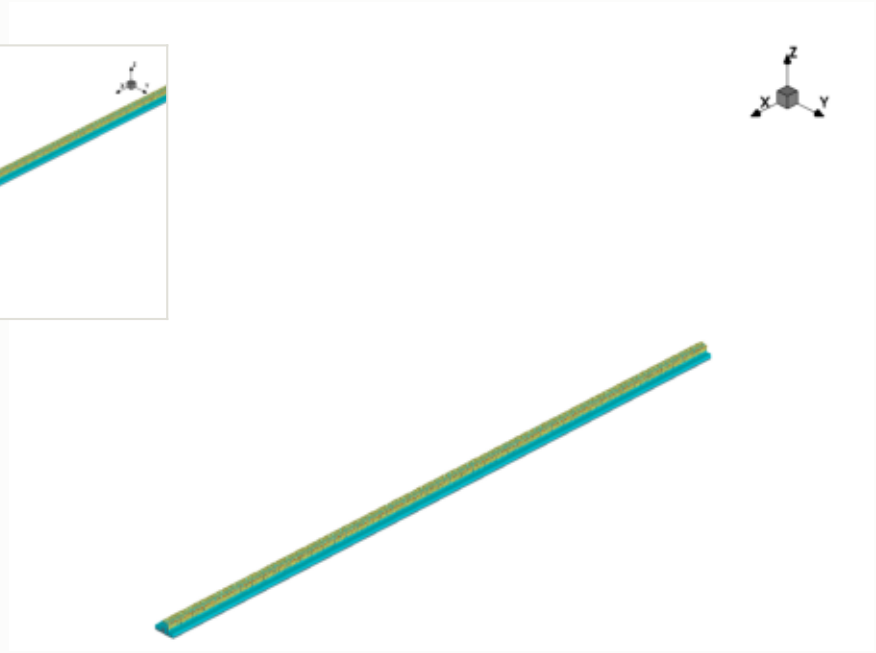
implant_ellipsoid · Ion_implantation

0.801	133.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

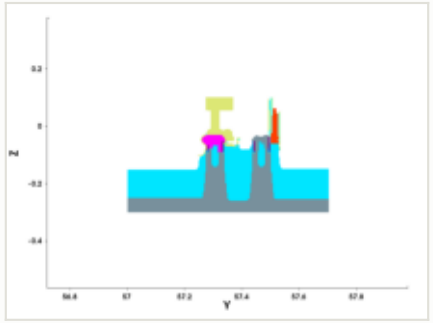
3D structure after this step

zoomed to fit

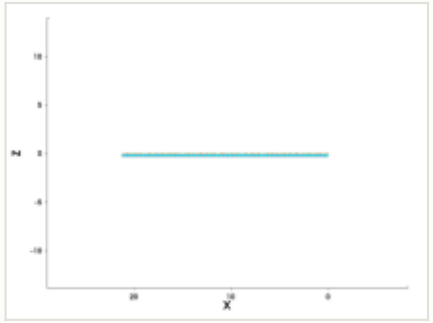




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

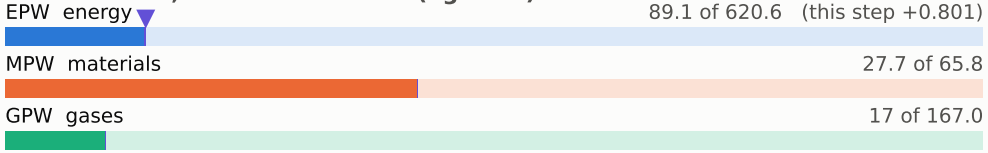
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

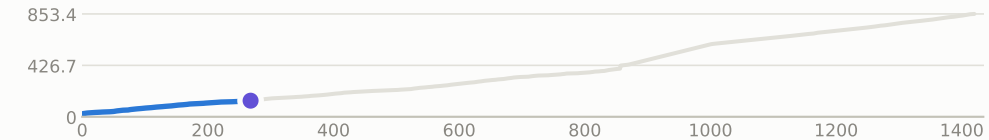
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

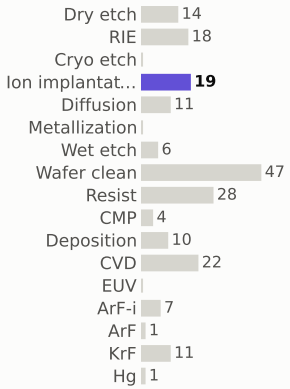
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

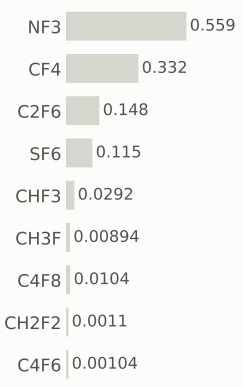


steps by process area



materials deposited (g)

gases emitted (g)



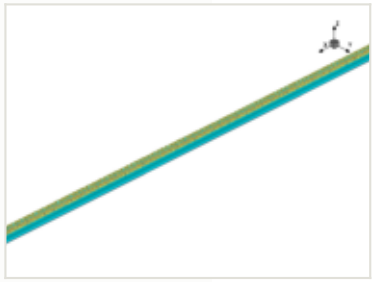
Step #2730: Dry_etch

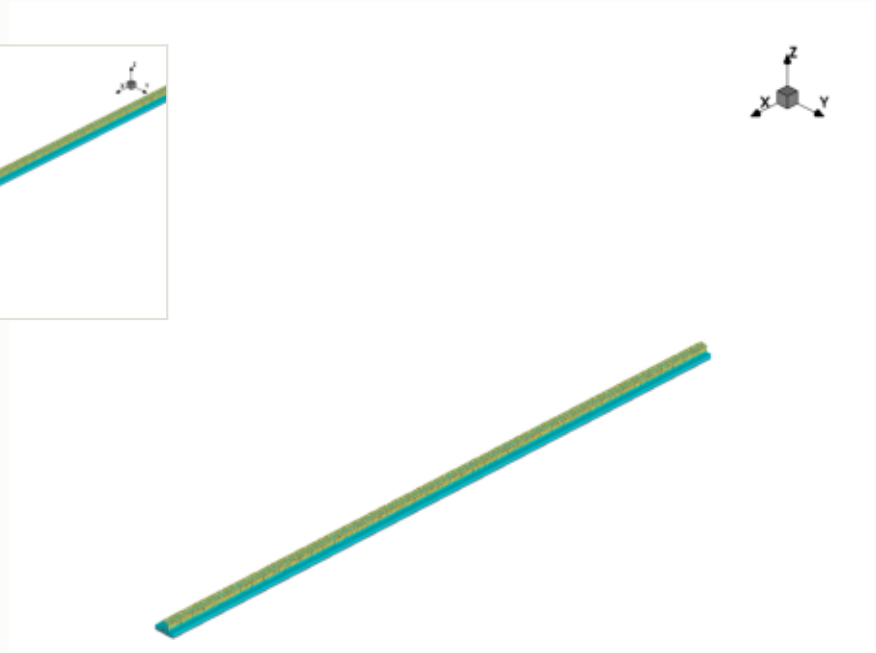
etch_strip · Dry_etch

0.801	134.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



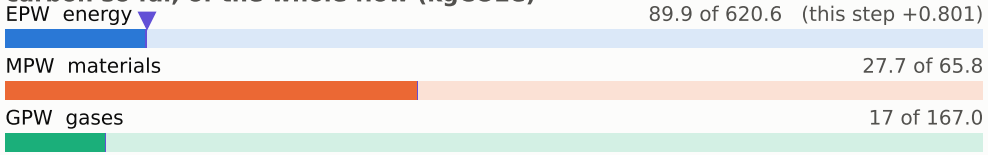


this step

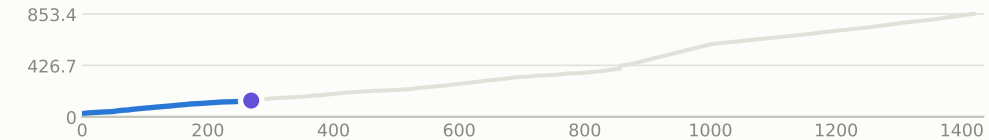
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

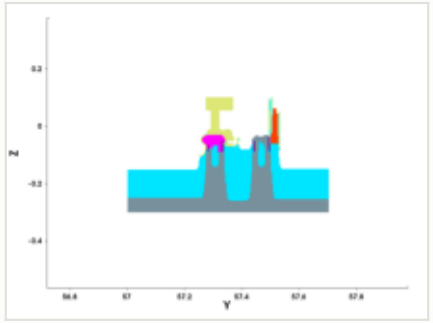
carbon so far, of the whole flow (kgCO2e)



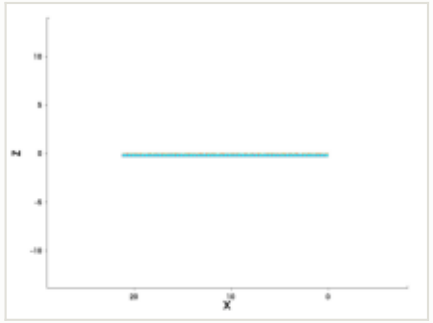
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



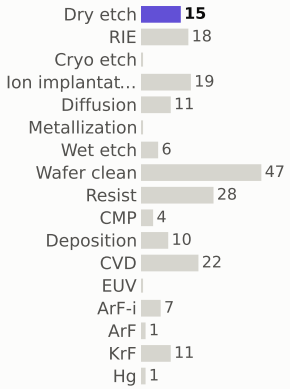
y cut at 57.351 µm



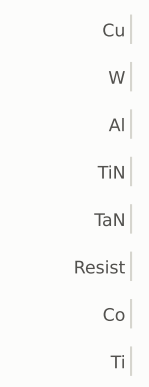
mask for this step

no mask for this step

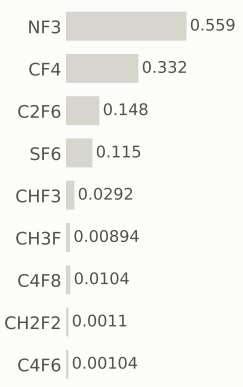
steps by process area



materials deposited (g)



gases emitted (g)



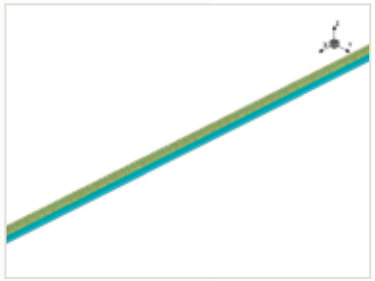
Step #2740: Wafer_clean

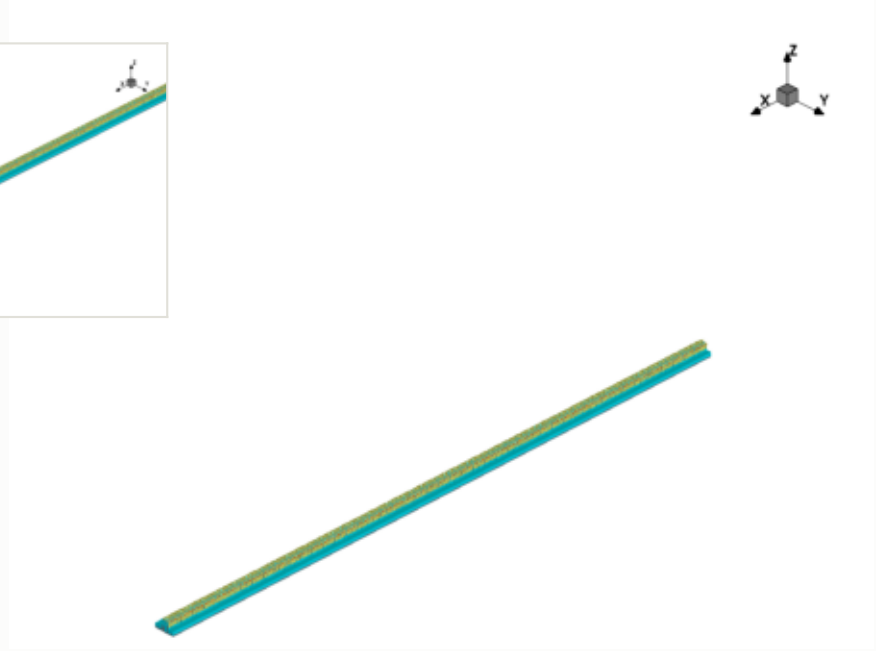
clean · Wafer_clean

0.169	134.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



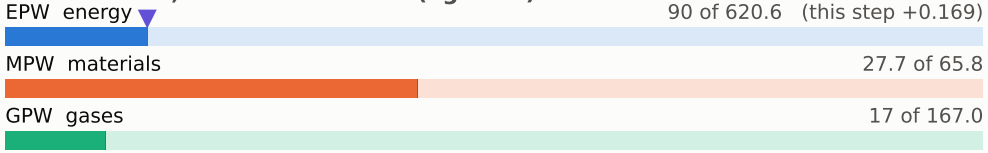


this step

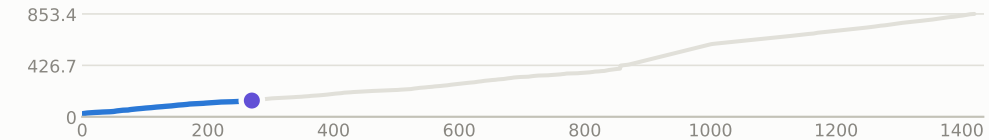
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

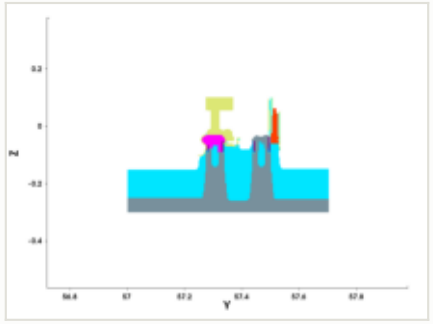
carbon so far, of the whole flow (kgCO2e)



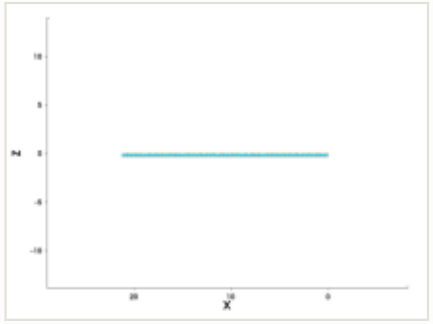
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



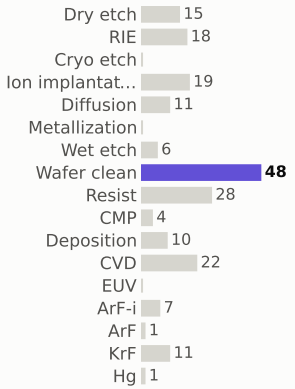
y cut at 57.351 µm



mask for this step

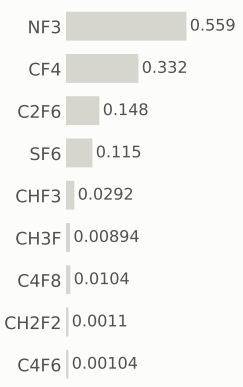
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



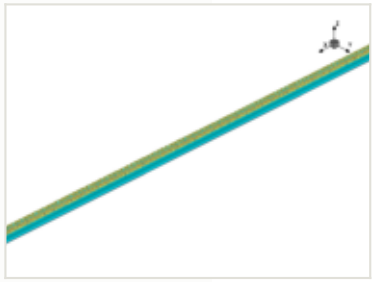
Step #2750: Resist

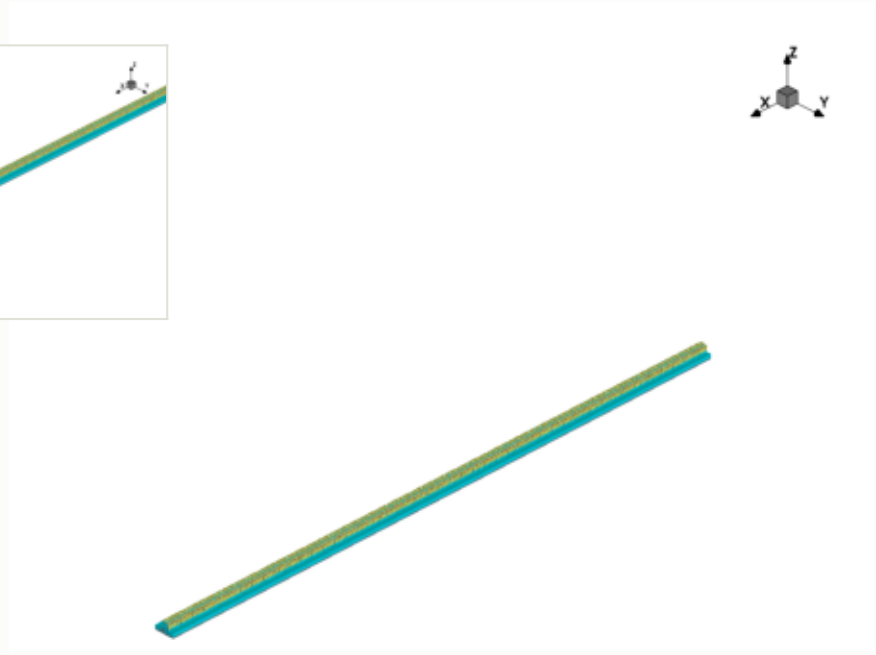
resist · Resist

0.169	134.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





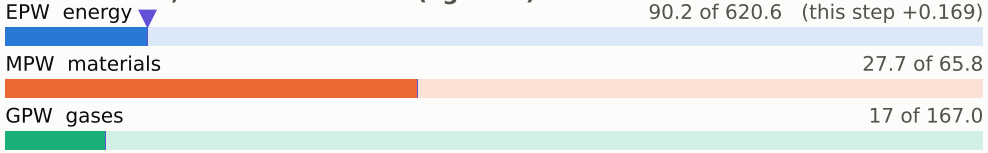
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

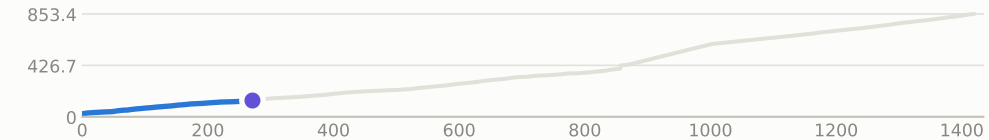
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

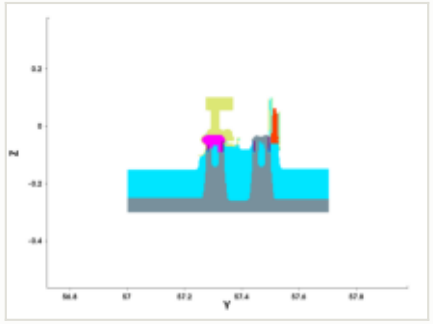
carbon so far, of the whole flow (kgCO2e)



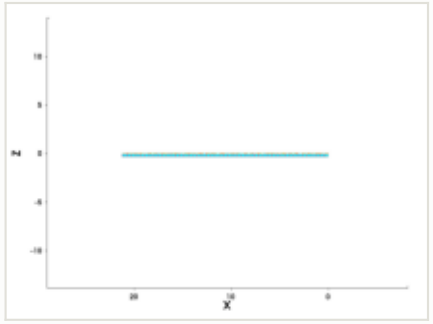
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



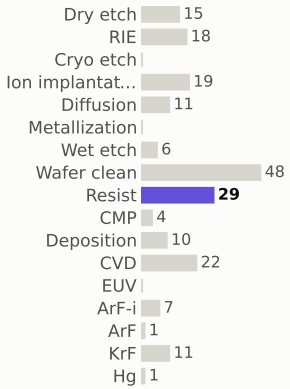
y cut at 57.351 µm



mask for this step

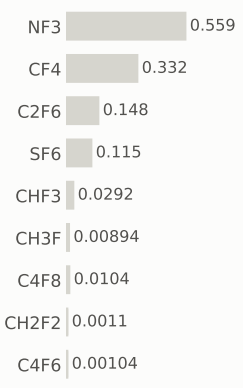
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



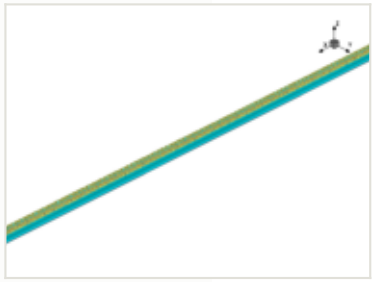
Step #2760: KrF

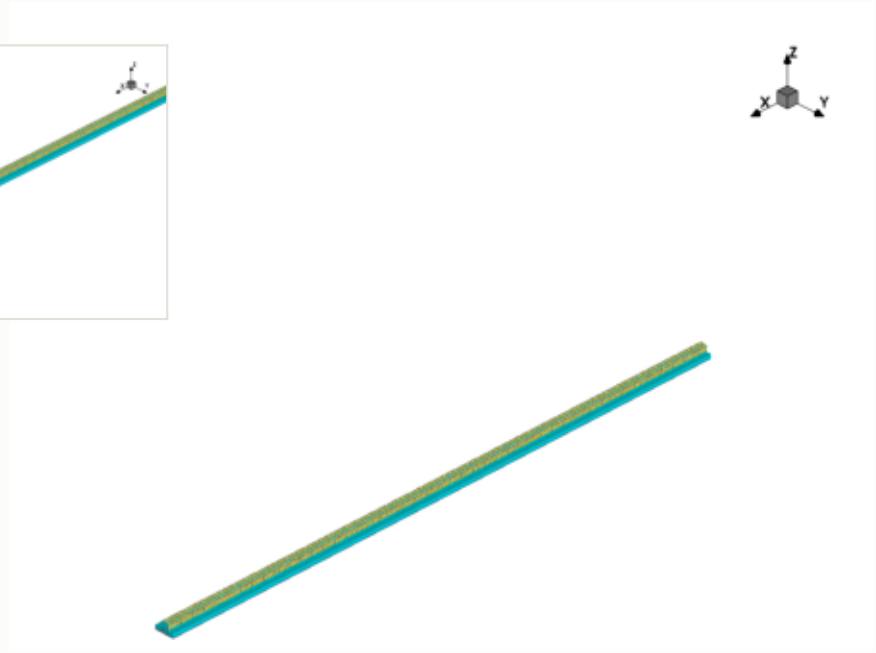
litho_patterning · KrF

0.199	135.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



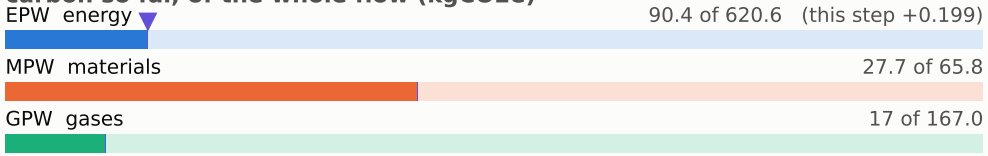


this step

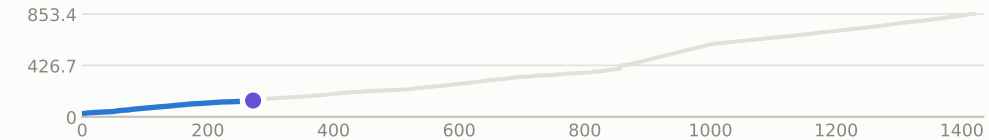
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

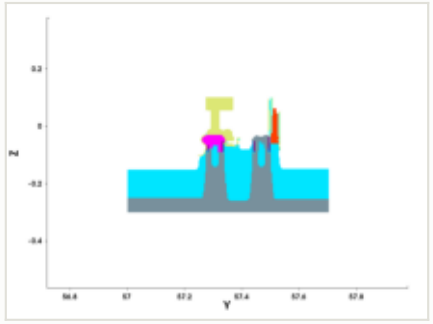
carbon so far, of the whole flow (kgCO2e)



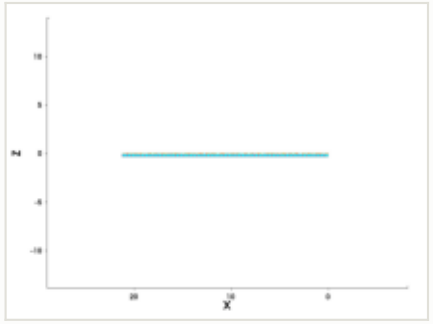
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

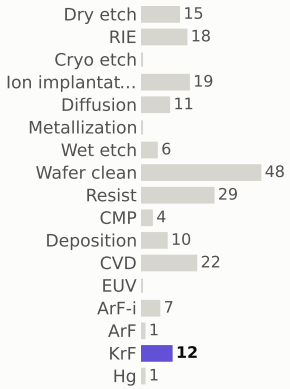


y cut at 57.351 µm

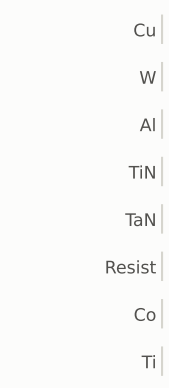


mask: layer 2100:0, positive (GDSII)

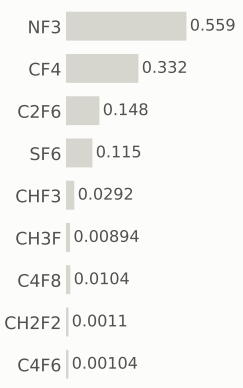
steps by process area



materials deposited (g)



gases emitted (g)



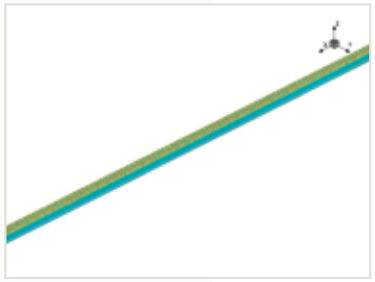
Step #2770: Ion_implantation

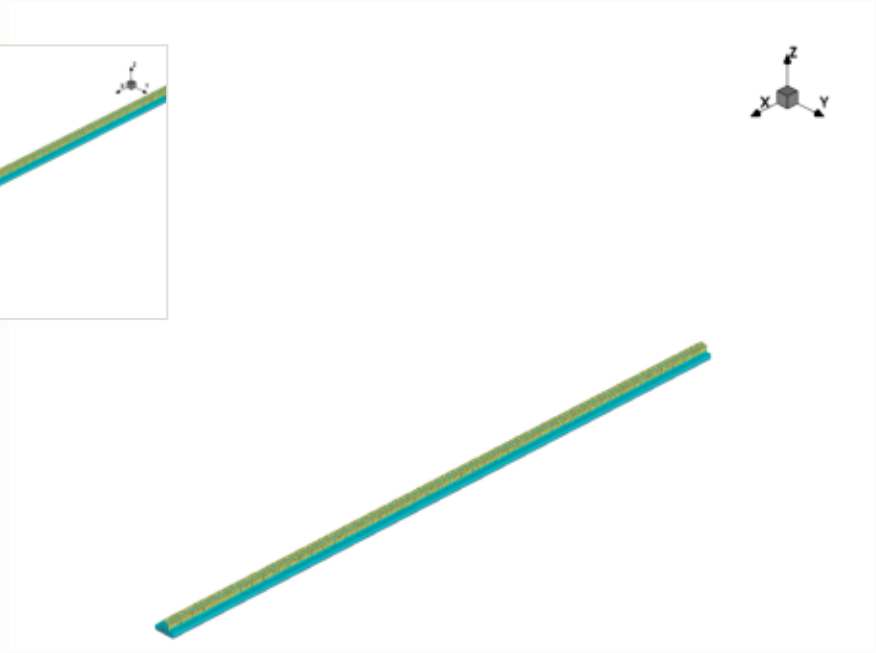
implant_ellipsoid · Ion_implantation

0.801	135.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

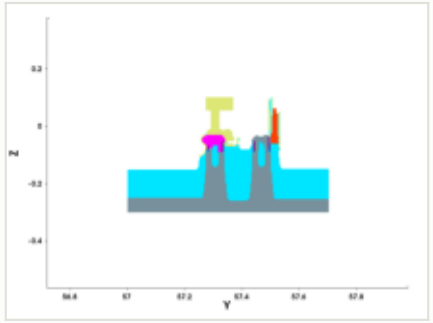
3D structure after this step

zoomed to fit

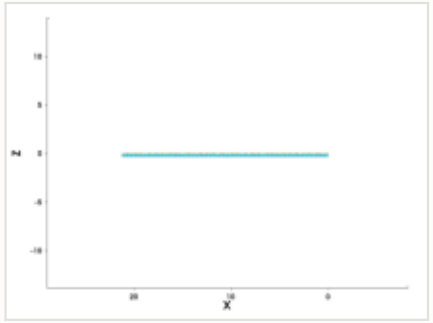




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

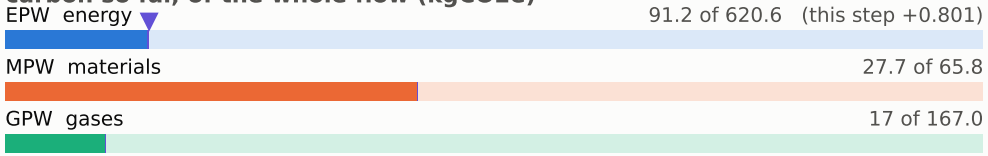
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

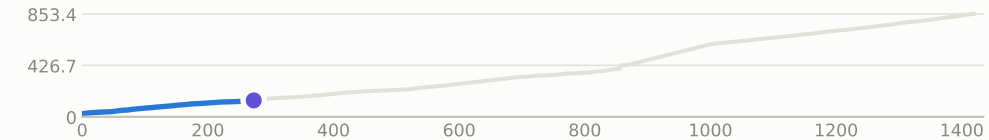
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

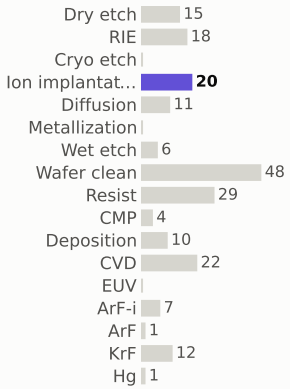
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

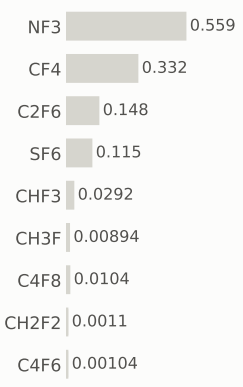


steps by process area



materials deposited (g)

gases emitted (g)



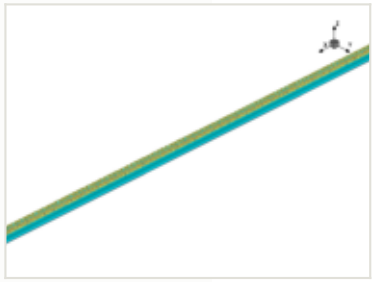
Step #2780: Ion_implantation

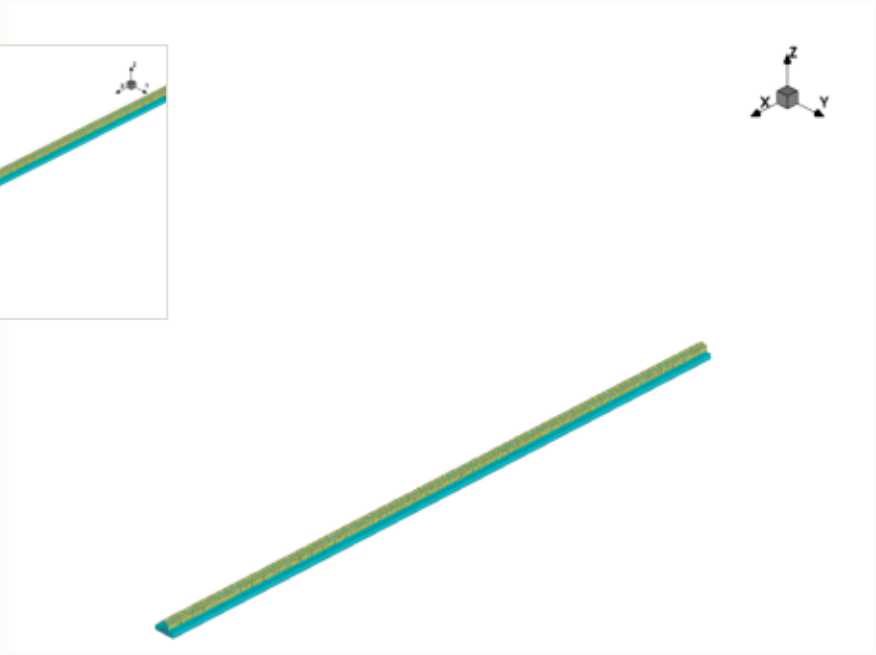
implant_ellipsoid · Ion_implantation

0.801	136.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

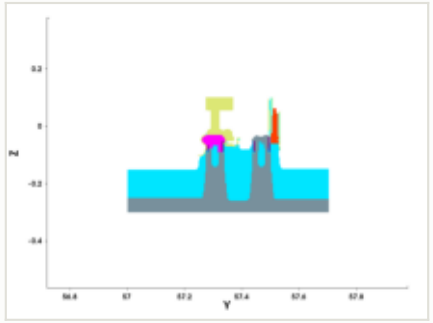
3D structure after this step

zoomed to fit

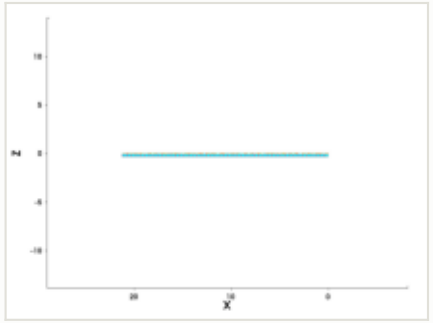




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

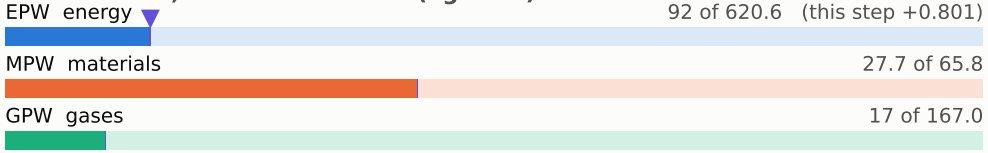
no mask for this step

this step

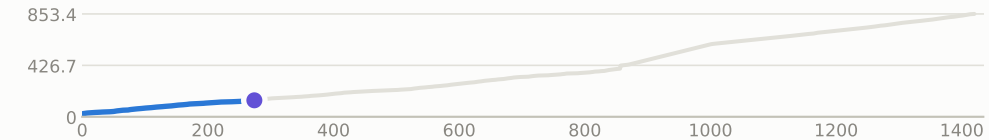
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

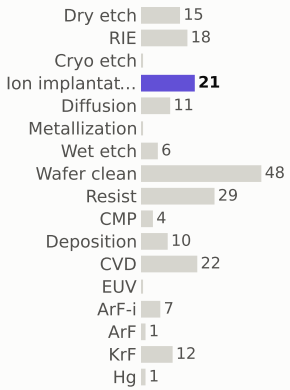
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

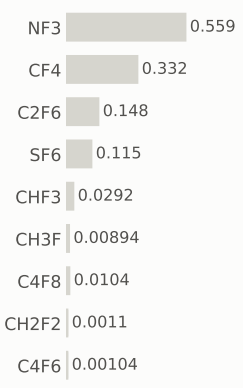


steps by process area



materials deposited (g)

gases emitted (g)



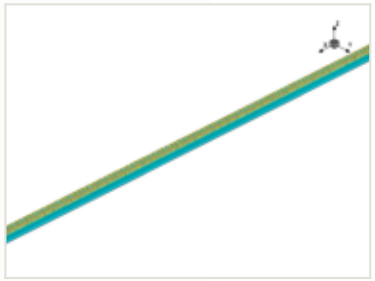
Step #2790: Dry_etch

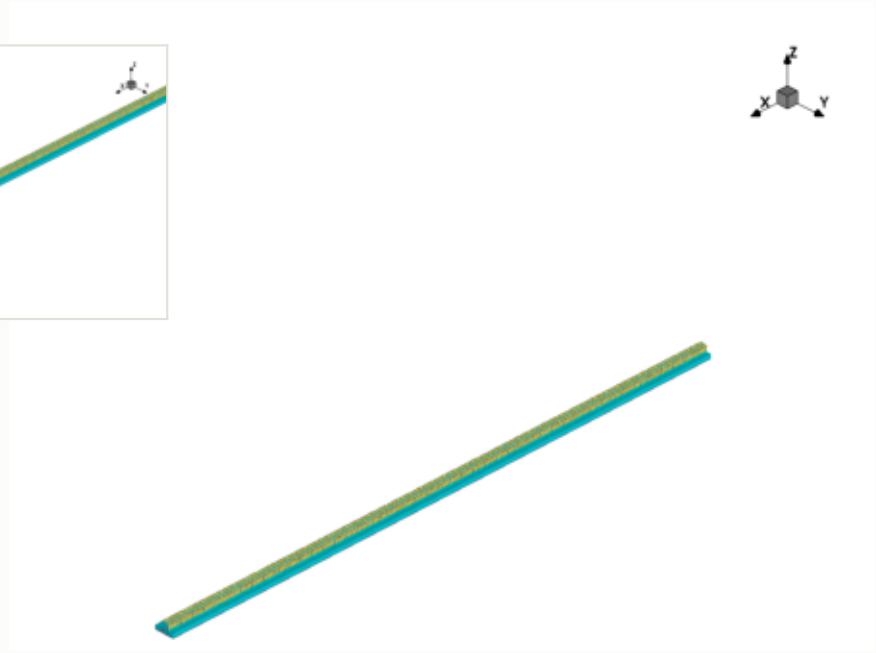
etch_strip · Dry_etch

0.801	137.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



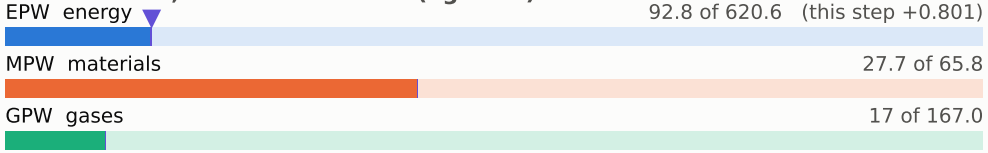


this step

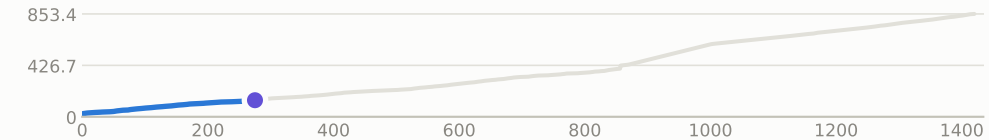
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

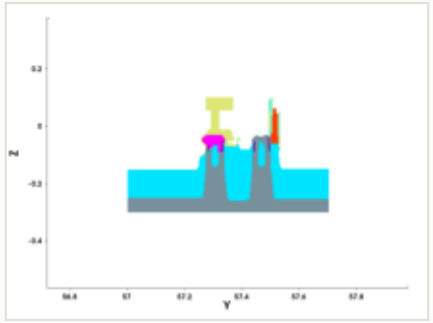
carbon so far, of the whole flow (kgCO2e)



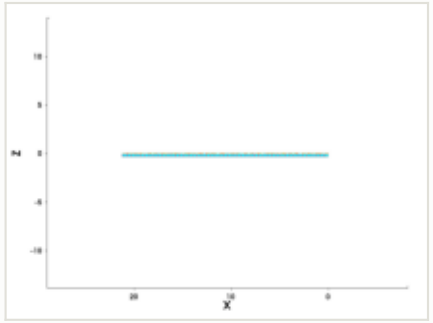
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



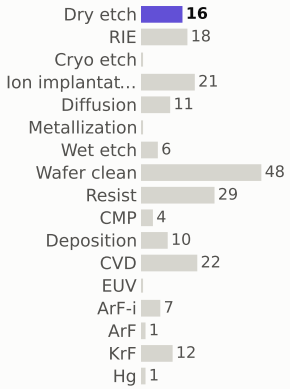
y cut at 57.351 µm



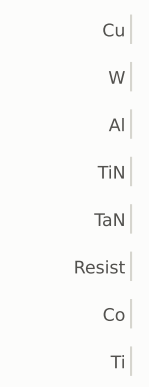
mask for this step

no mask for this step

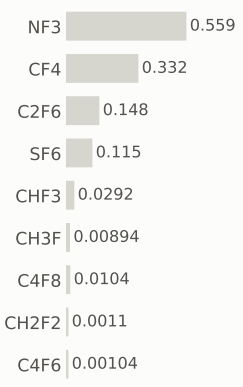
steps by process area



materials deposited (g)



gases emitted (g)



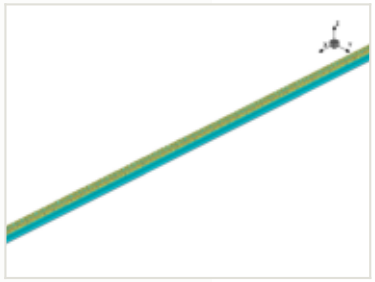
Step #2800: Wafer_clean

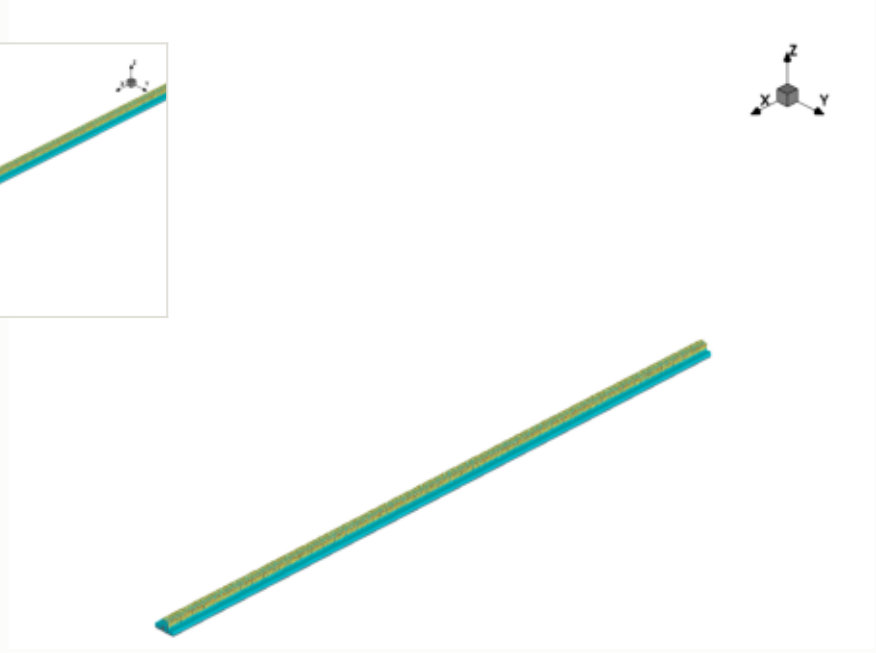
clean · Wafer_clean

0.169	137.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





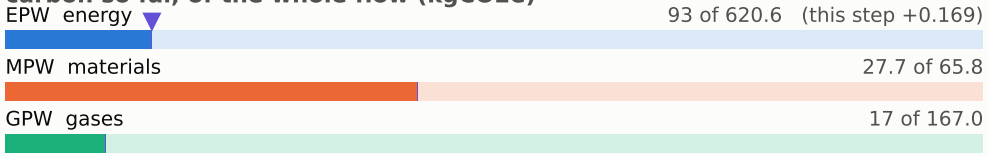
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

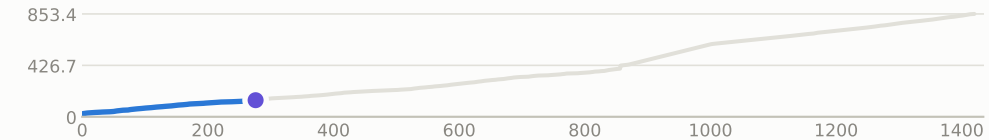
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

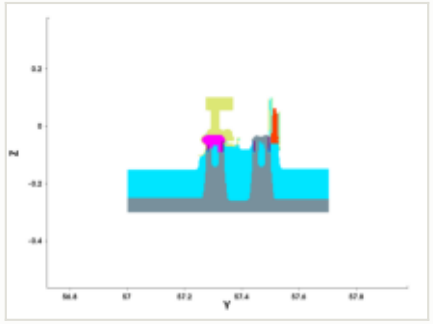
carbon so far, of the whole flow (kgCO2e)



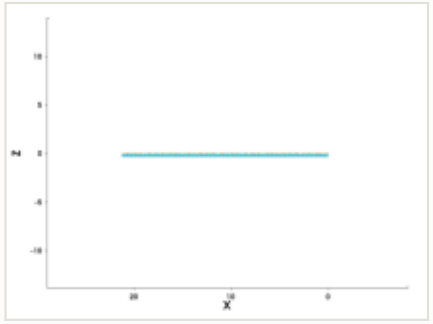
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



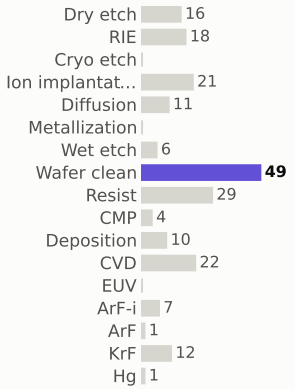
y cut at 57.351 µm



mask for this step

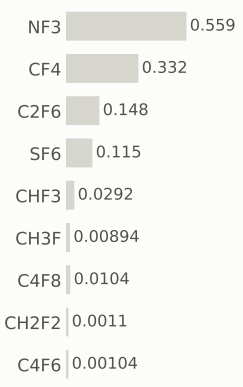
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



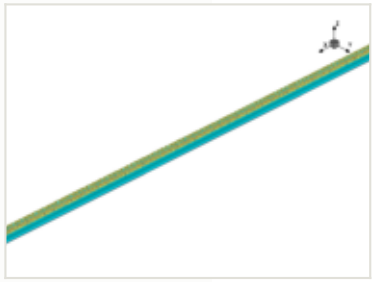
Step #2810: Resist

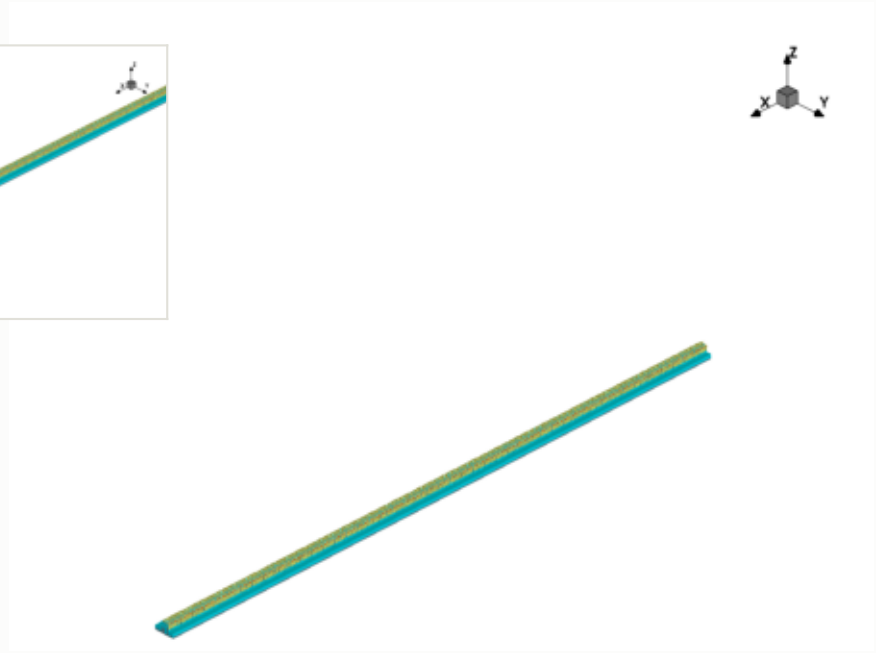
resist · Resist

0.169	137.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





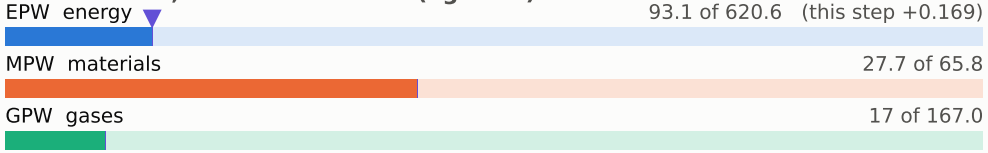
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

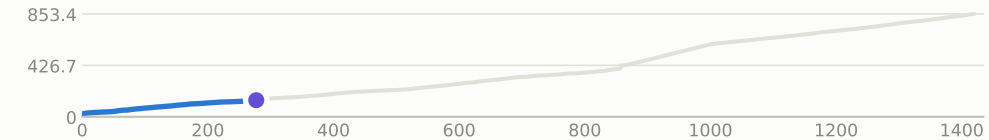
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

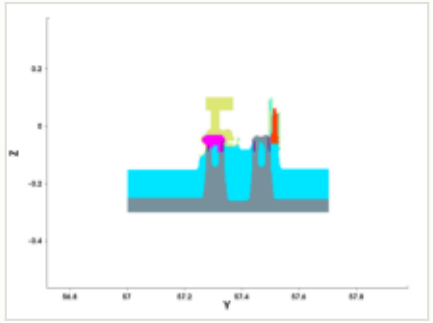
carbon so far, of the whole flow (kgCO2e)



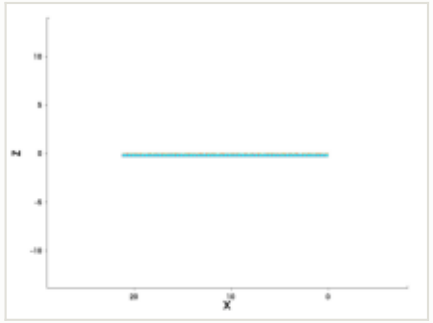
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



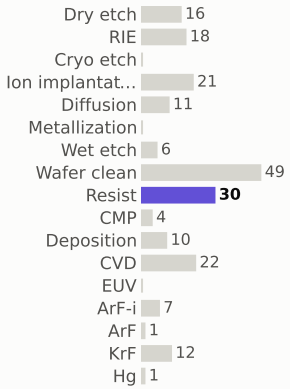
y cut at 57.351 µm



mask for this step

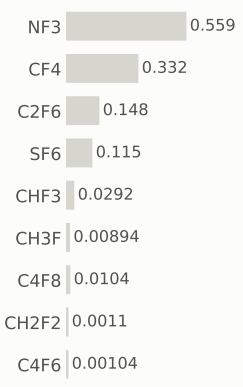
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



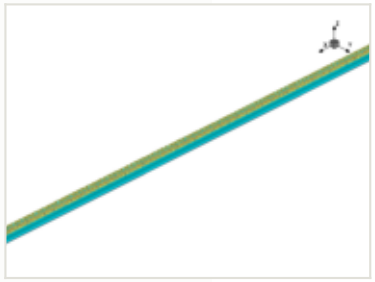
Step #2820: KrF

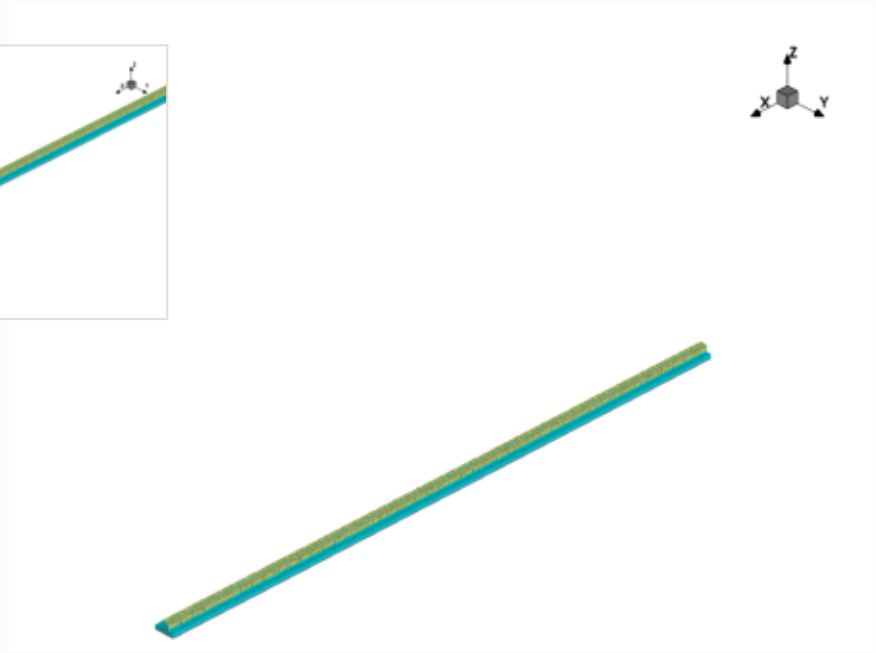
litho_patterning · KrF

0.199	138.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



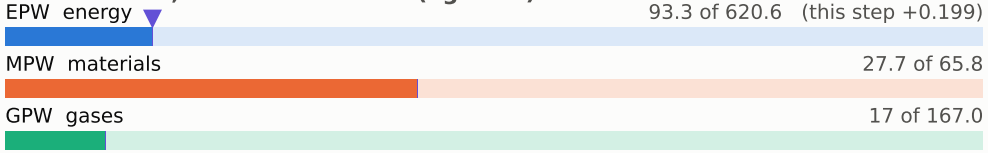


this step

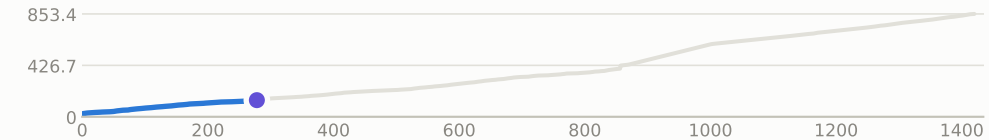
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

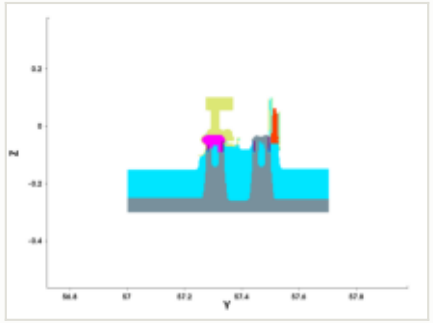
carbon so far, of the whole flow (kgCO2e)



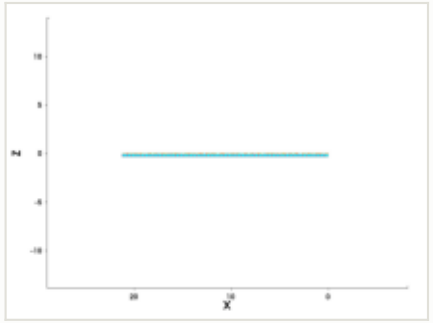
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

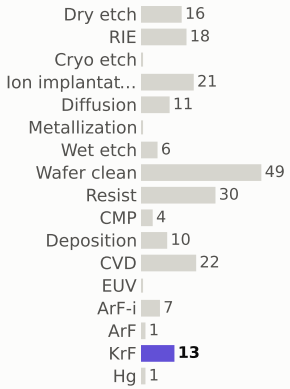


y cut at 57.351 µm



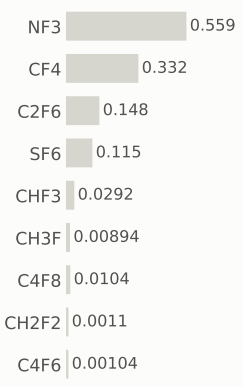
mask: layer 2100:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



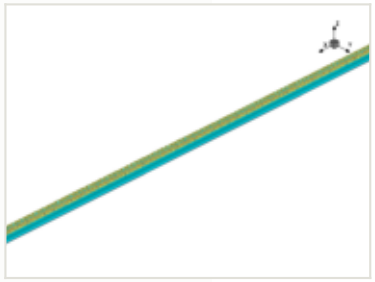
Step #2830: Ion_implantation

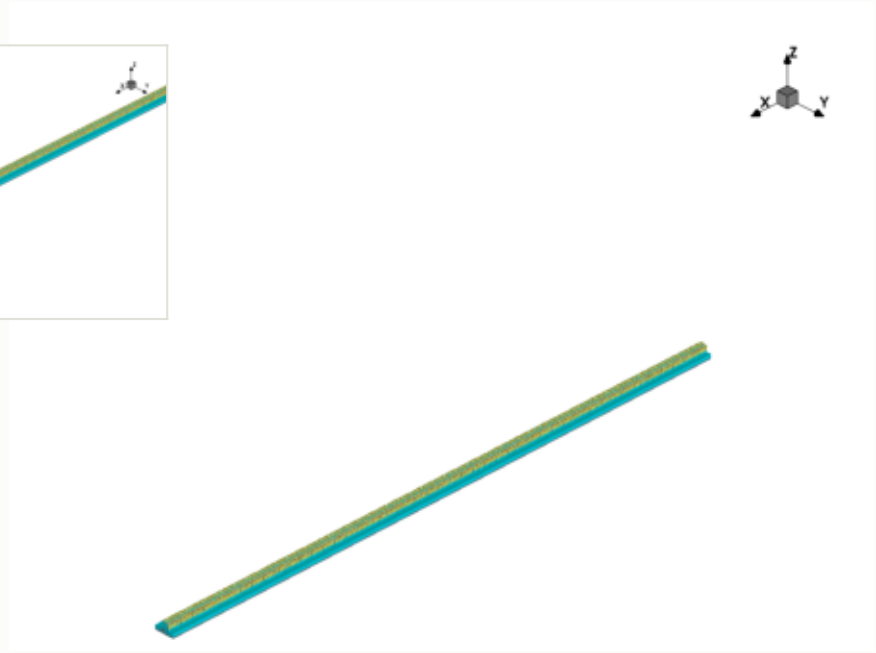
implant_ellipsoid · Ion_implantation

0.801	138.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

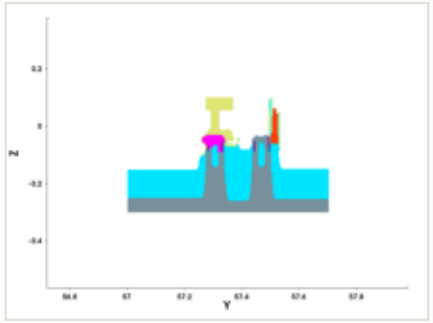
3D structure after this step

zoomed to fit

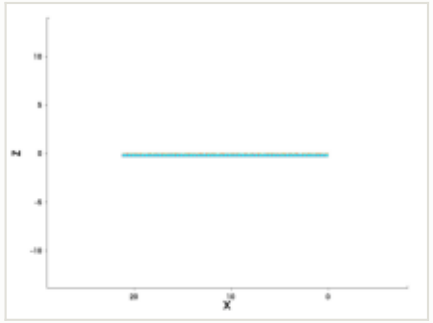




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

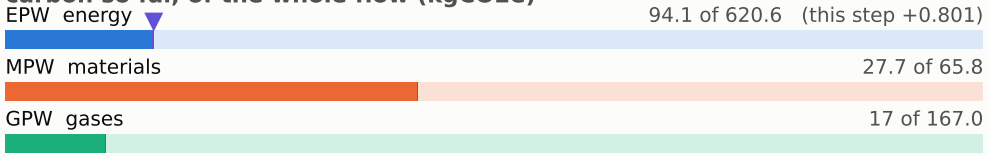
no mask for this step

this step

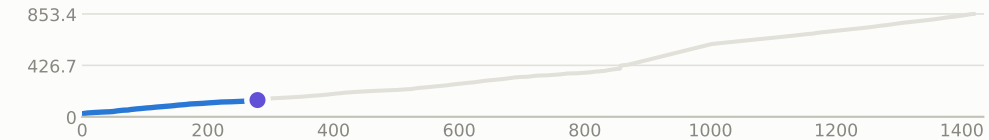
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

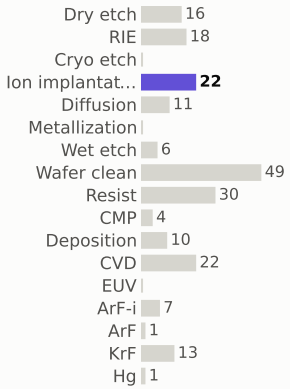
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

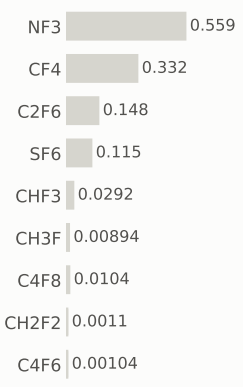


steps by process area



materials deposited (g)

gases emitted (g)



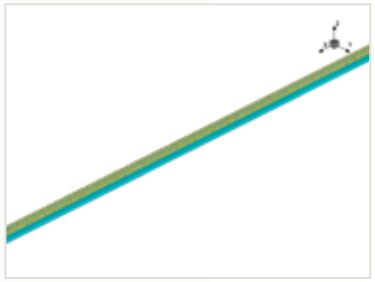
Step #2840: Ion_implantation

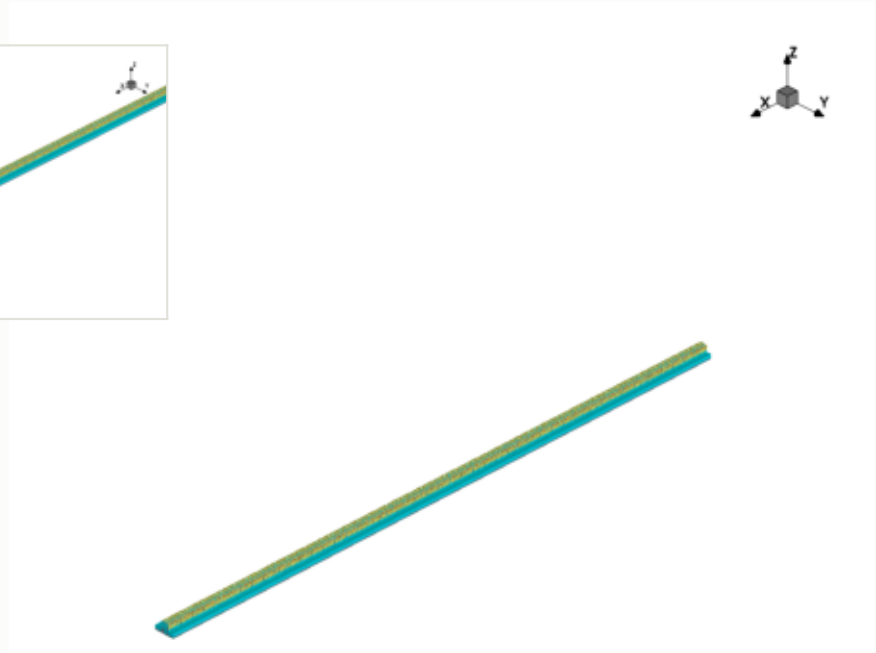
implant_ellipsoid · Ion_implantation

0.801	139.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

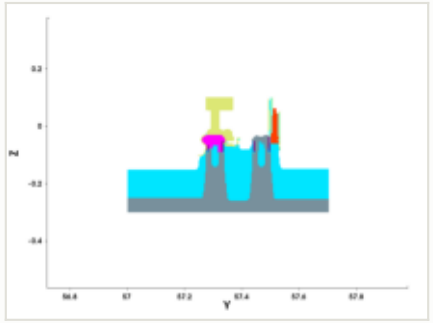
3D structure after this step

zoomed to fit

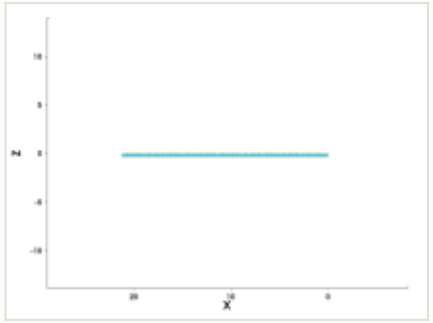




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

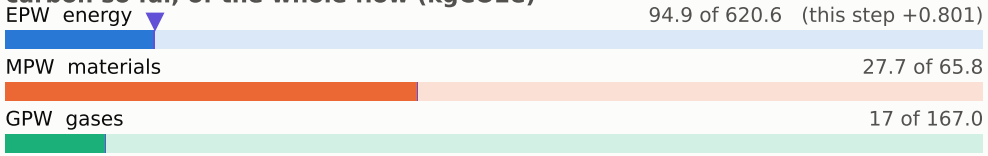
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

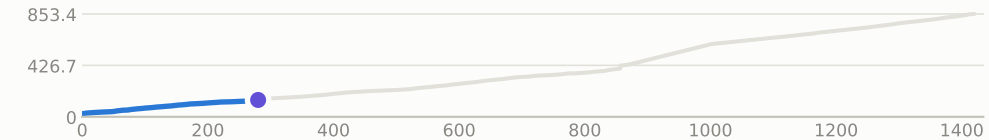
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

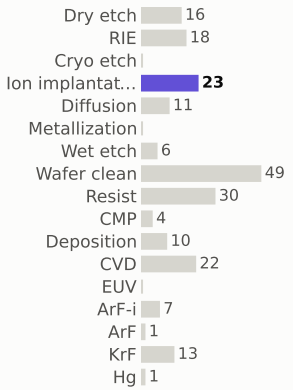
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

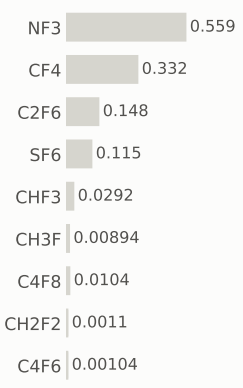


steps by process area



materials deposited (g)

gases emitted (g)



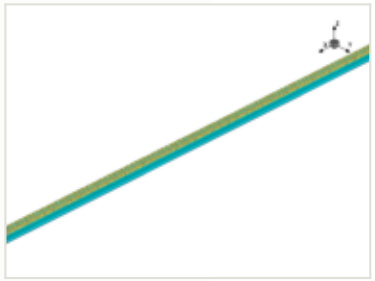
Step #2850: Dry_etch

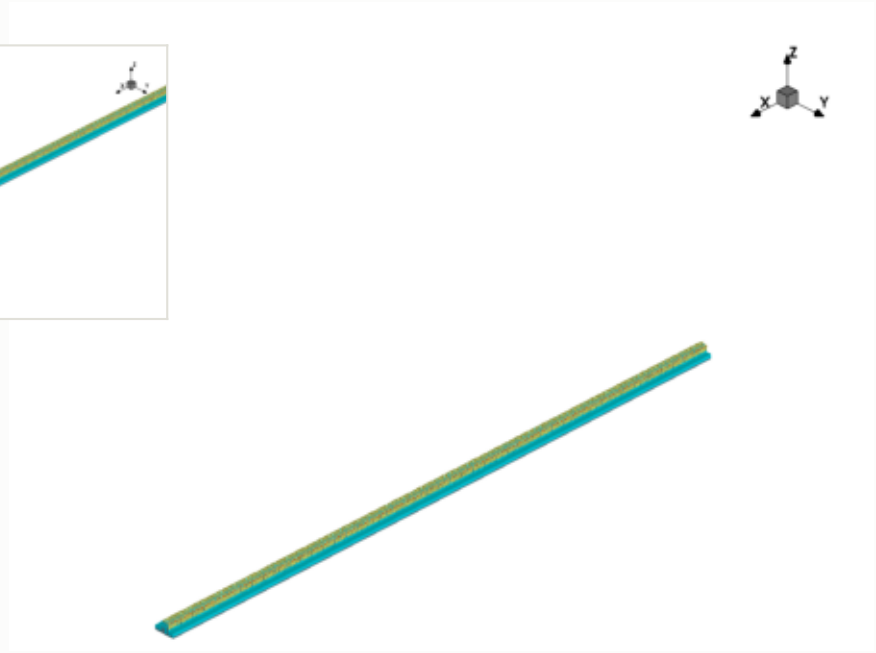
etch_strip · Dry_etch

0.801	140.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

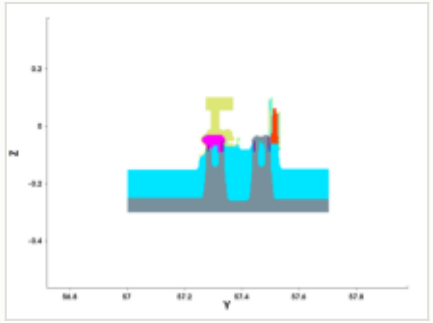
3D structure after this step

zoomed to fit

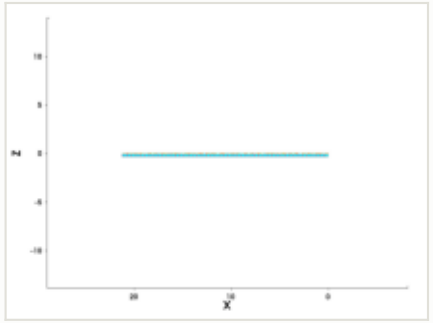




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

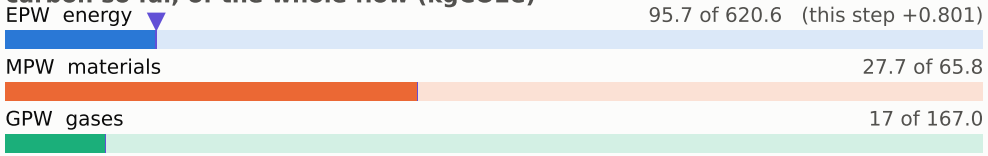
no mask for this step

this step

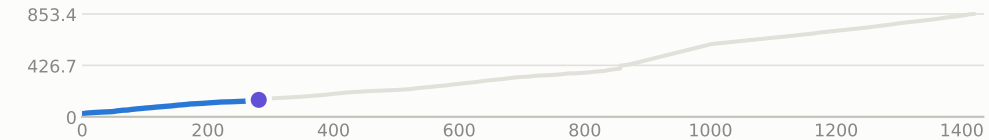
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

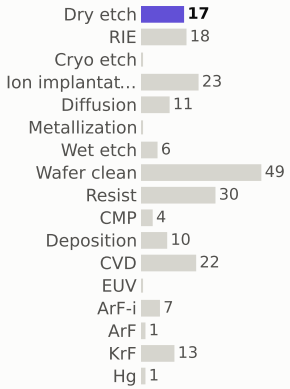
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

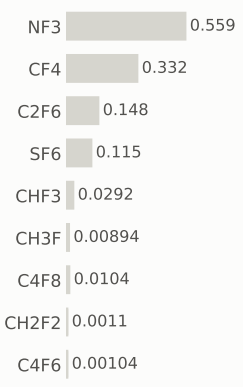


steps by process area



materials deposited (g)

gases emitted (g)



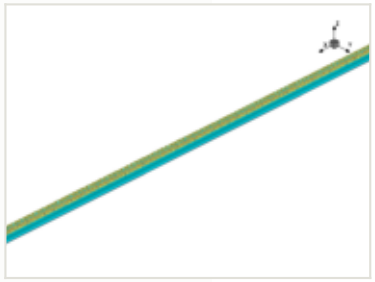
Step #2860: Wafer_clean

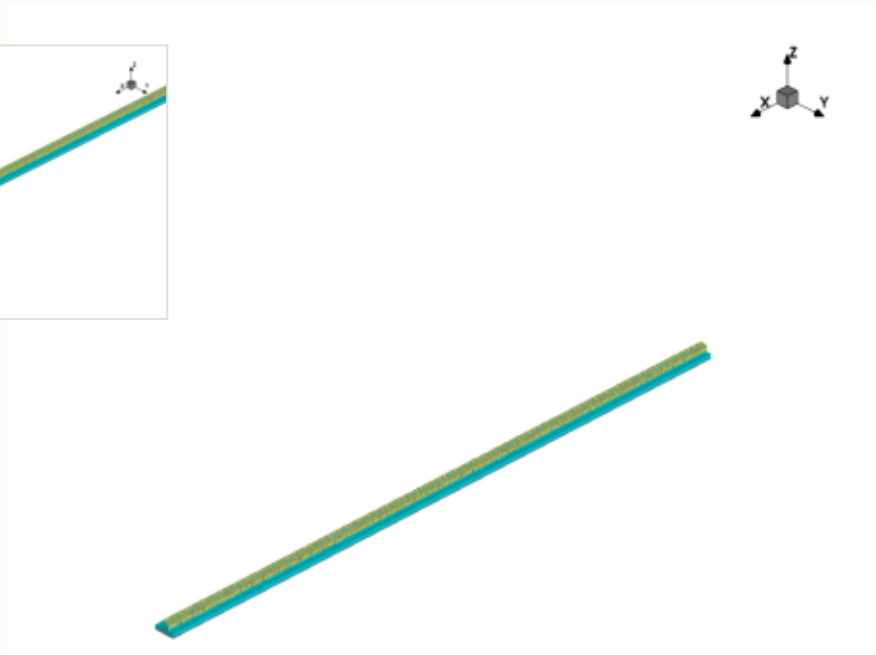
clean · Wafer_clean

0.169	140.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



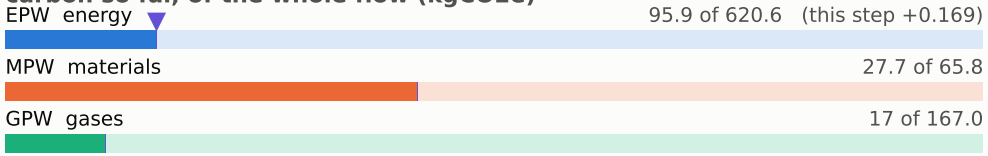


this step

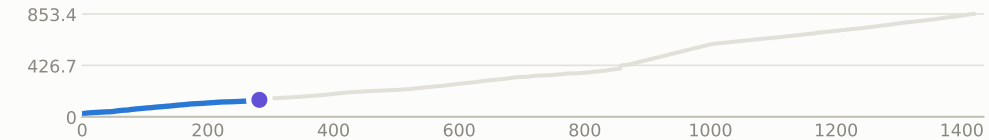
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

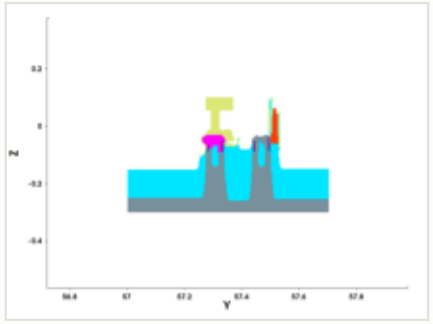
carbon so far, of the whole flow (kgCO2e)



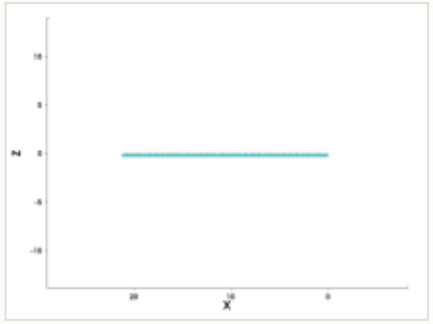
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



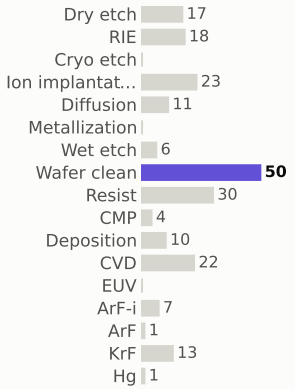
y cut at 57.351 µm



mask for this step

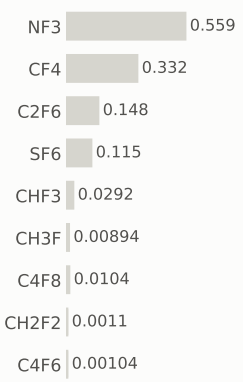
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



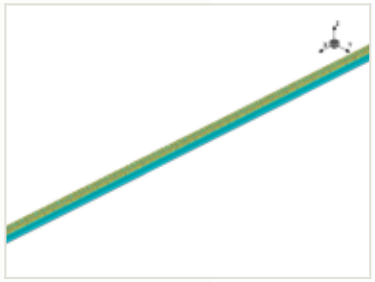
Step #2870: Diffusion

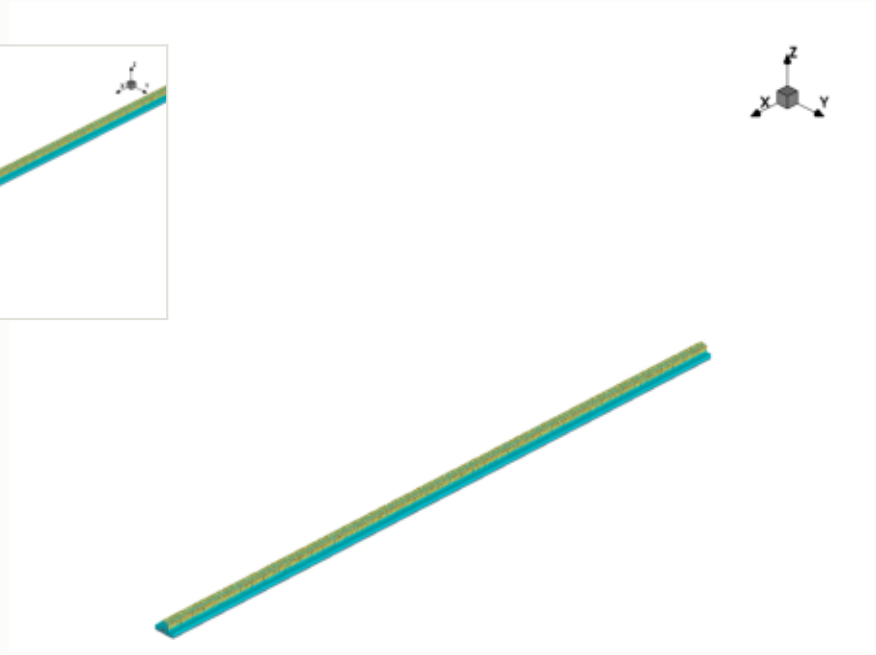
anneal · Diffusion

0.801	141.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





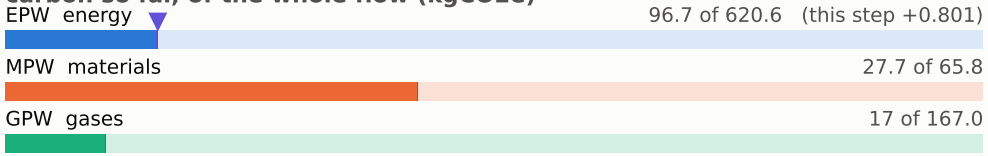
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

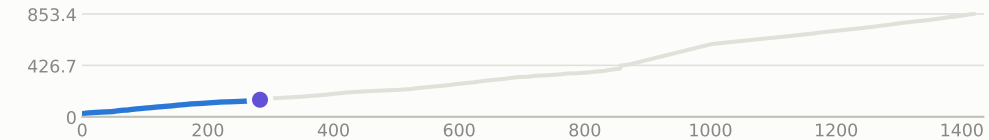
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

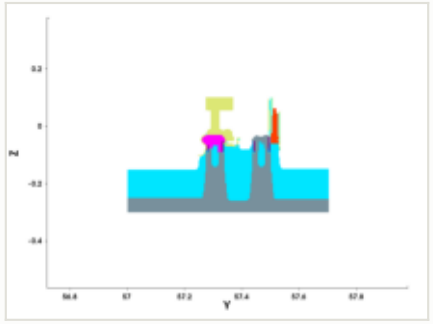
carbon so far, of the whole flow (kgCO2e)



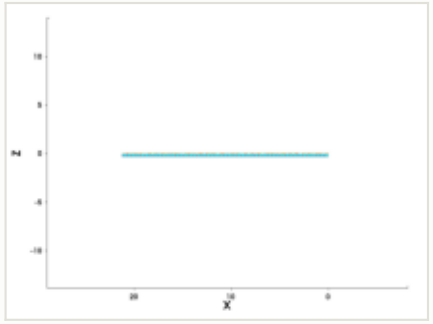
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



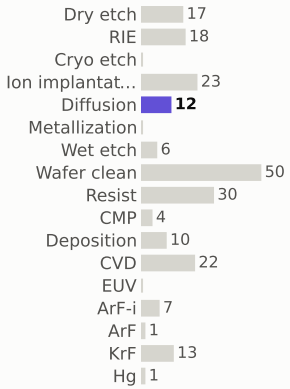
y cut at 57.351 µm



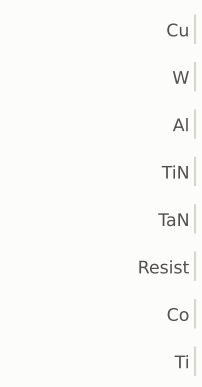
mask for this step

no mask for this step

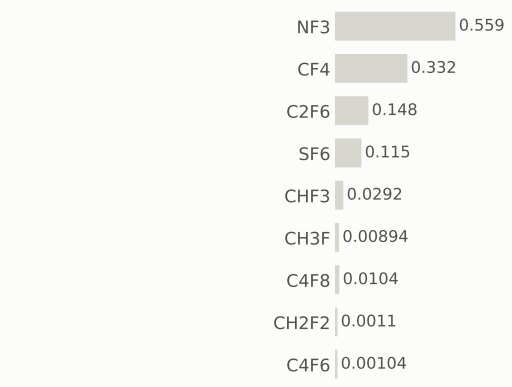
steps by process area



materials deposited (g)



gases emitted (g)



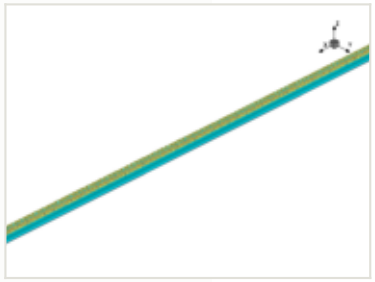
Step #2890: Wet_etch

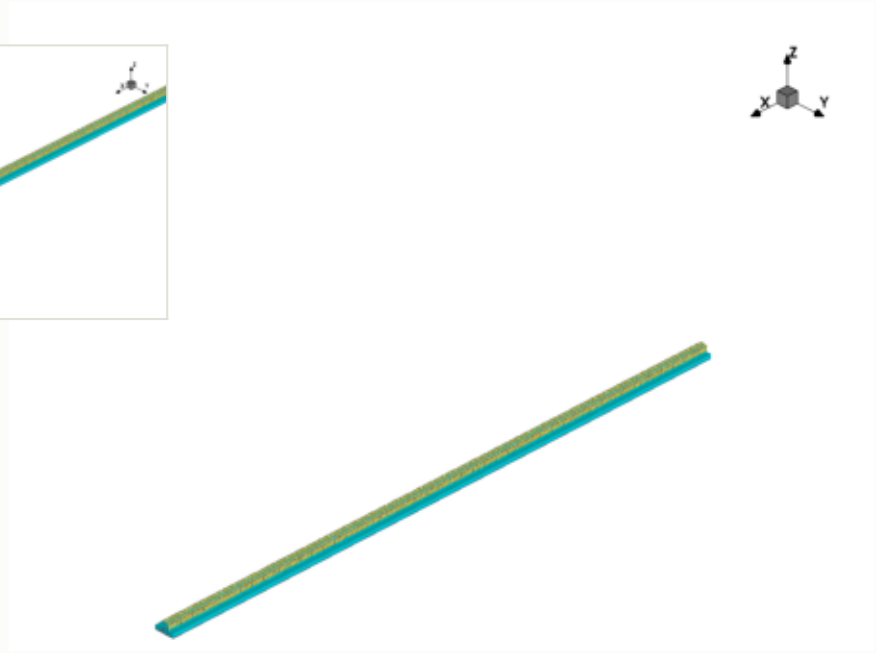
etch_isotropic · Wet_etch

0.169	141.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



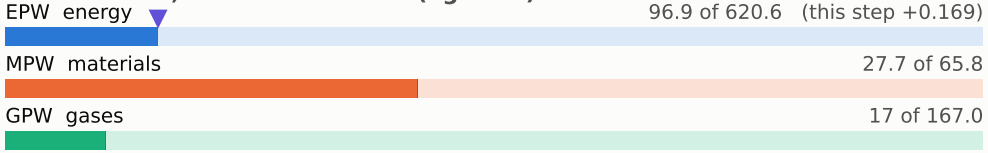


this step

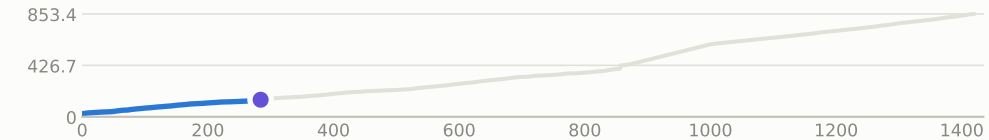
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

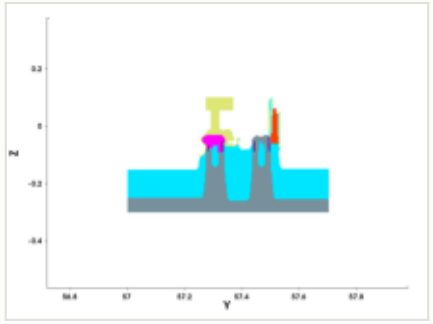
carbon so far, of the whole flow (kgCO2e)



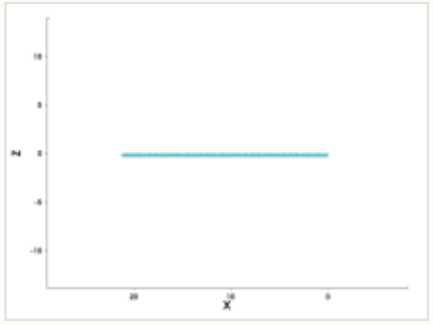
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



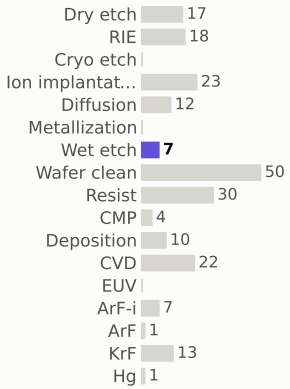
y cut at 57.351 µm



mask for this step

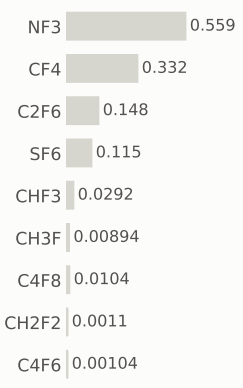
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



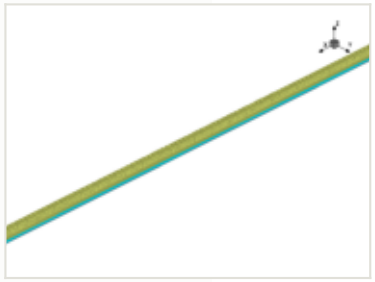
Step #2900: Deposition

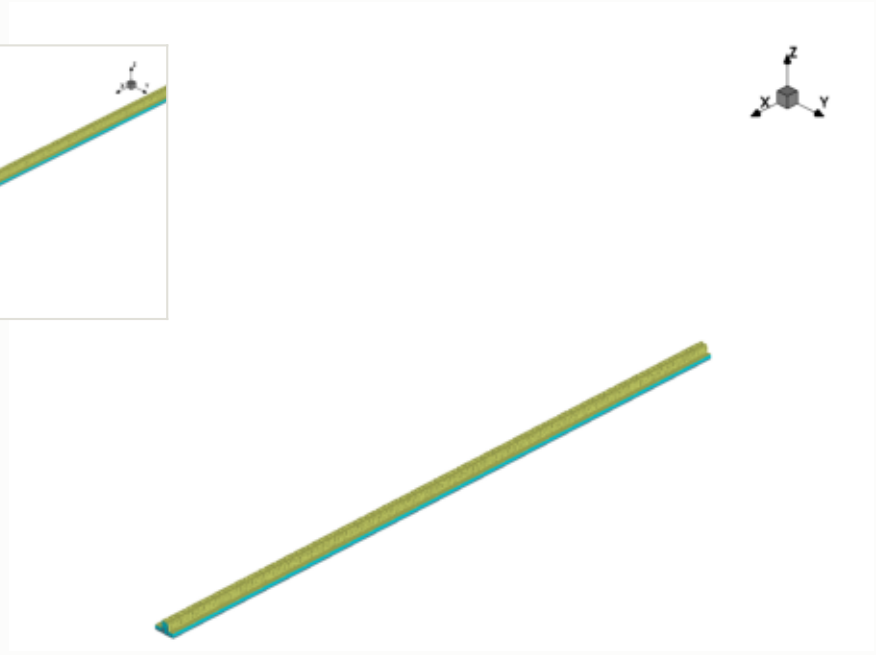
depo_isotropic · Deposition

0.656	142.3	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



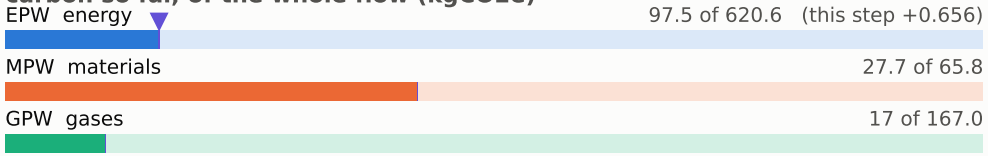


this step

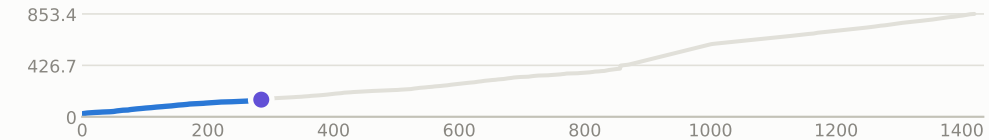
process area	Deposition
material	—
thickness	0.0025 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

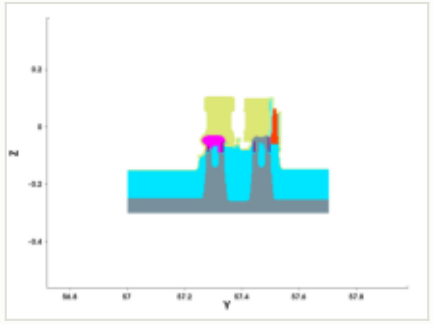
carbon so far, of the whole flow (kgCO2e)



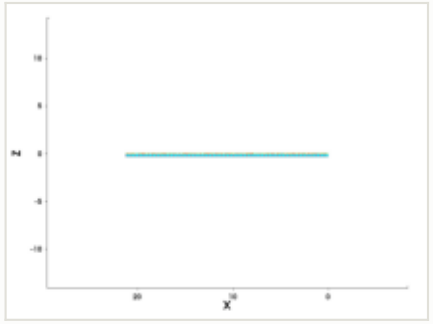
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



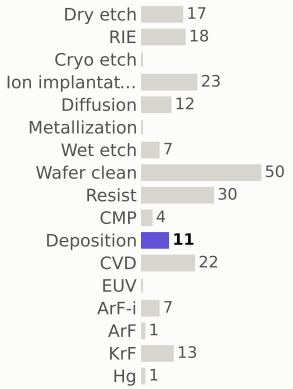
y cut at 57.351 µm



mask for this step

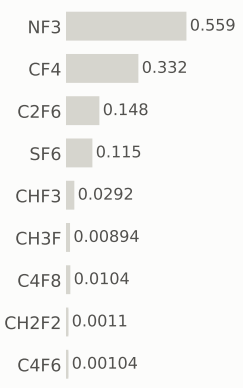
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



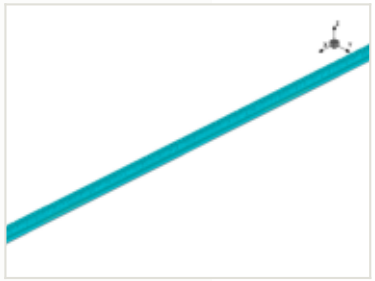
Step #2910: Deposition

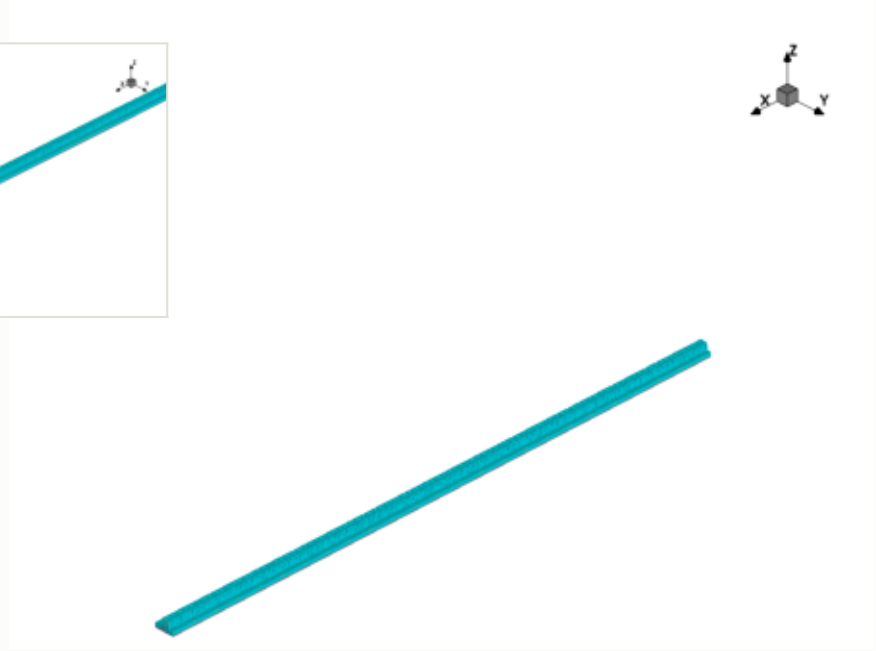
depo_isotropic · Deposition

0.656	142.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



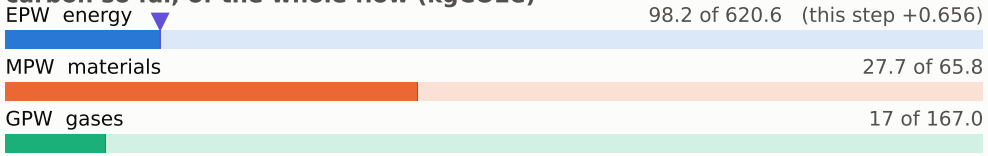


this step

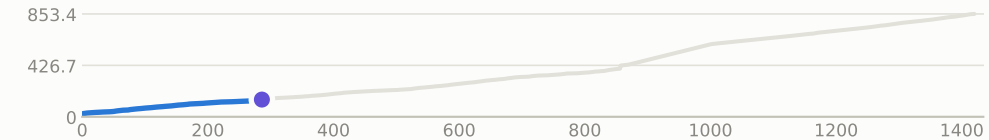
process area	Deposition
material	—
thickness	0.0035 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

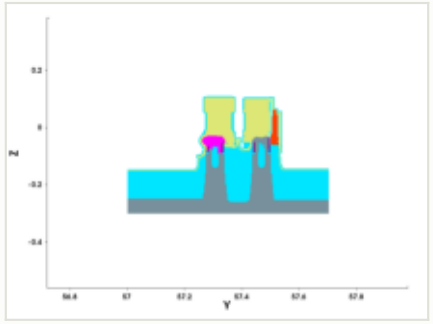
carbon so far, of the whole flow (kgCO2e)



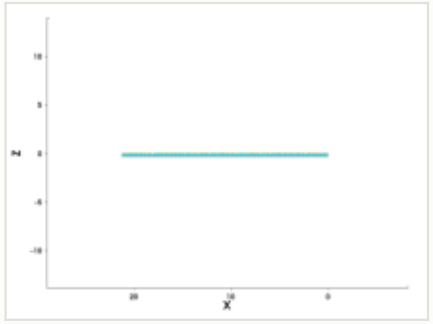
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



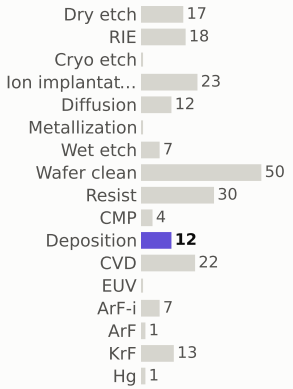
y cut at 57.351 µm



mask for this step

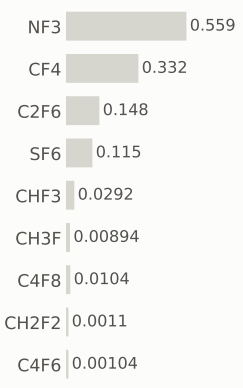
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



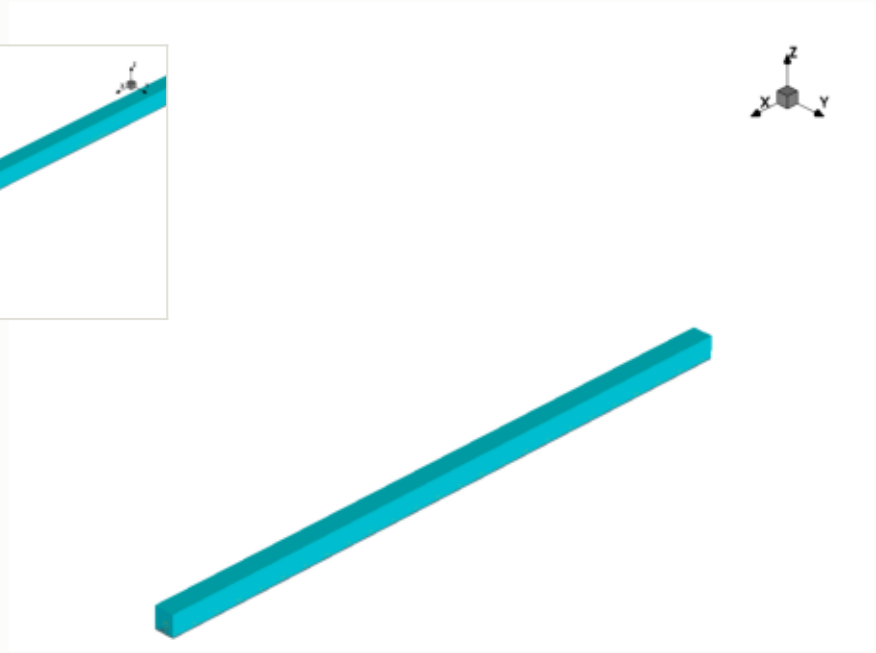
Step #2920: CVD

depo_fill · CVD

1.31	144.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

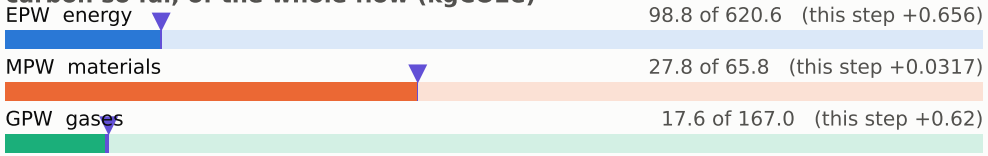


this step

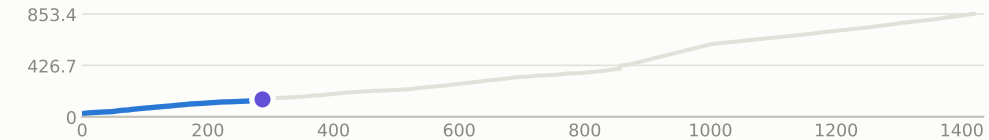
process area	CVD
material	—
thickness	0.3 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

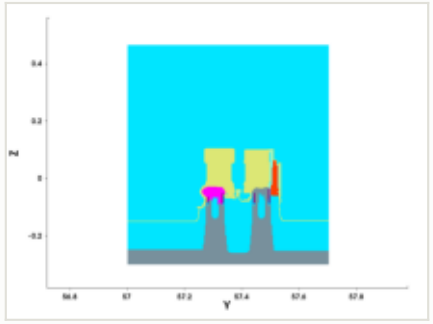
carbon so far, of the whole flow (kgCO2e)



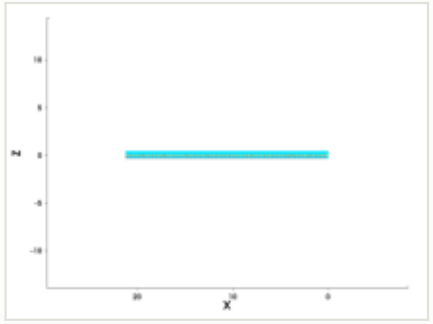
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



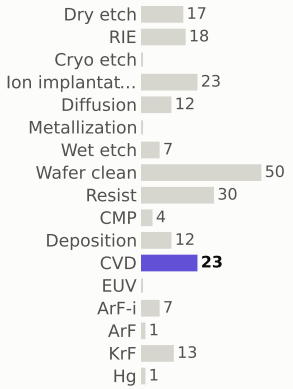
y cut at 57.351 µm



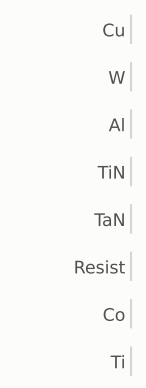
mask for this step

no mask for this step

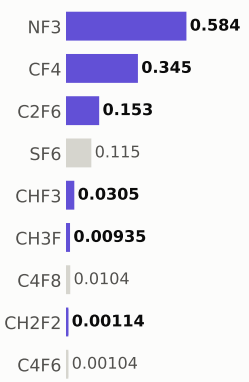
steps by process area



materials deposited (g)



gases emitted (g)



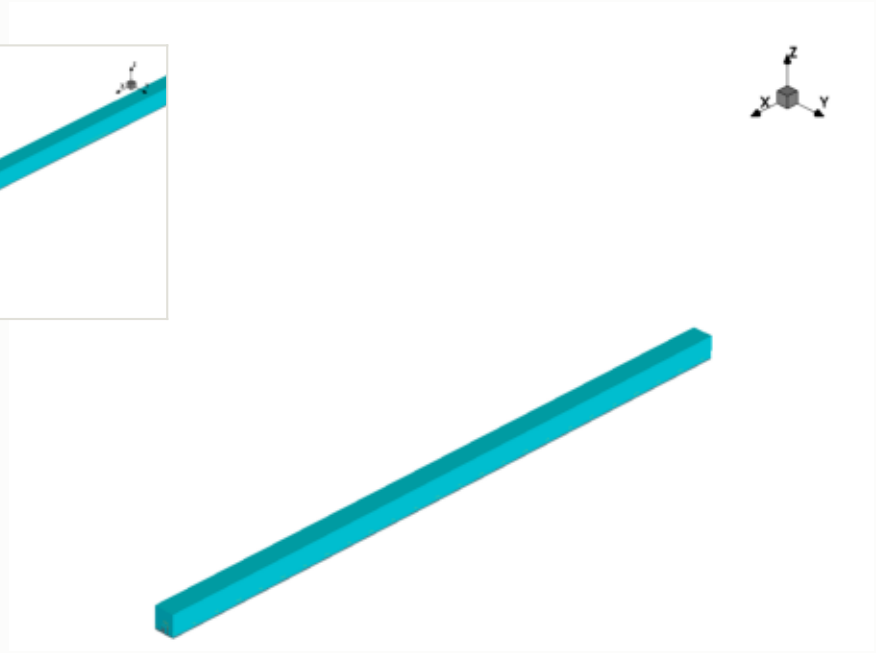
Step #2930: Diffusion

anneal · Diffusion

0.801	145.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



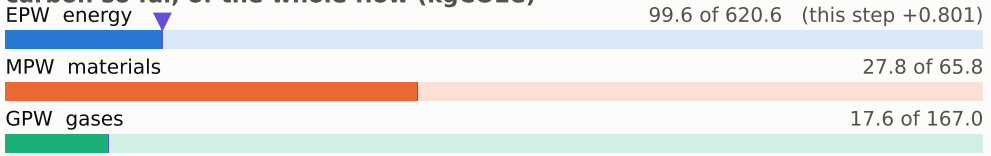
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

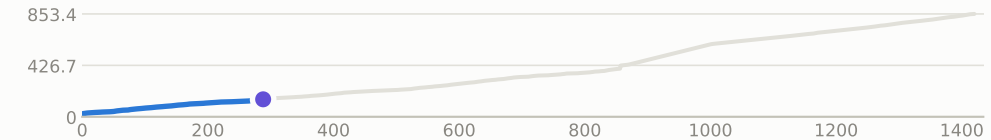
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

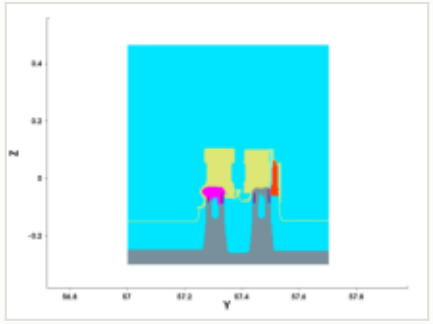
carbon so far, of the whole flow (kgCO2e)



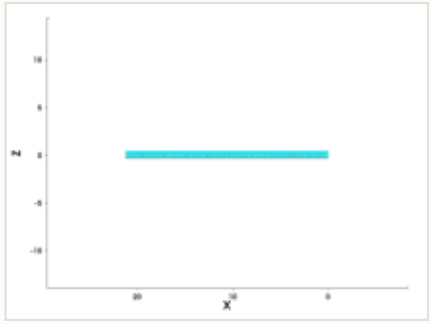
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



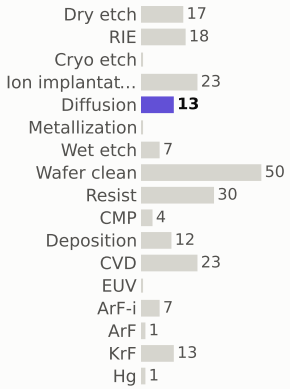
y cut at 57.351 µm



mask for this step

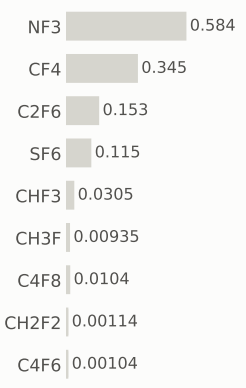
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



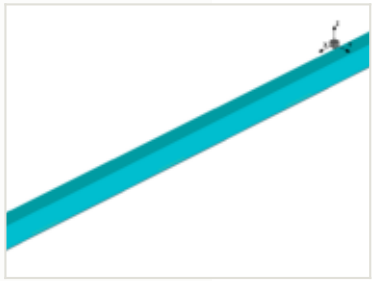
Step #2940: CVD

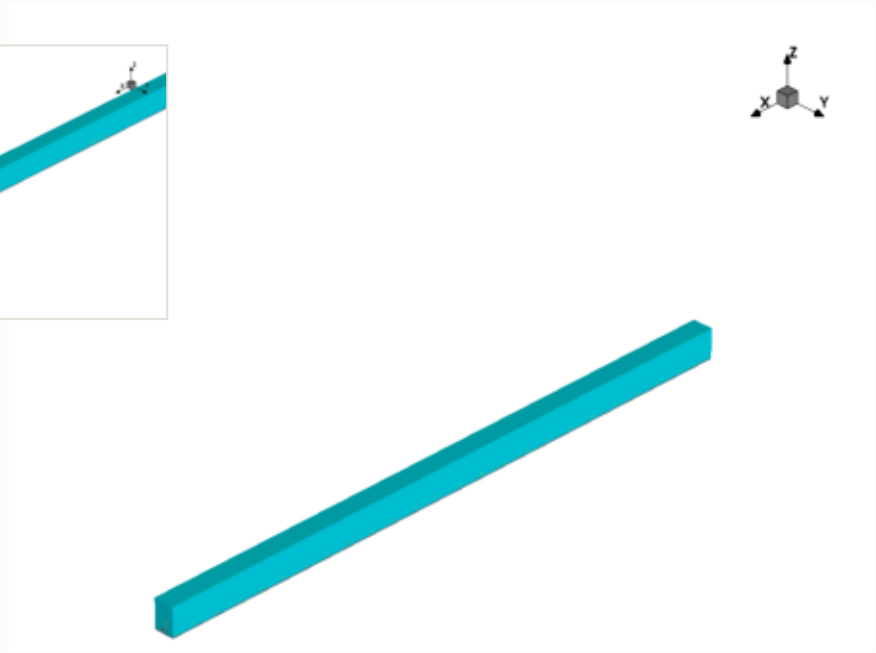
depo_fill · CVD

1.31	146.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



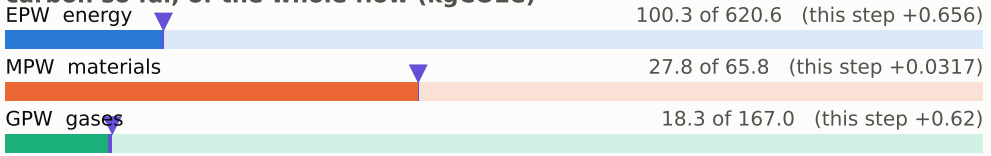


this step

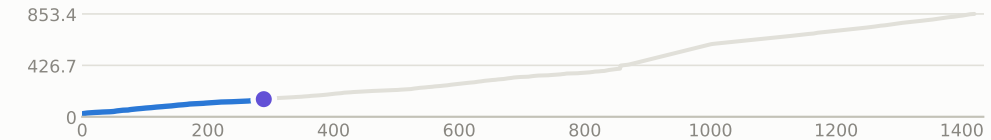
process area	CVD
material	—
thickness	0.25 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

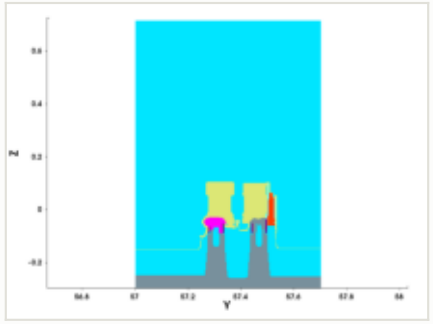
carbon so far, of the whole flow (kgCO2e)



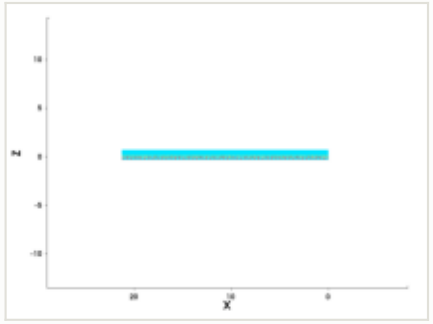
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



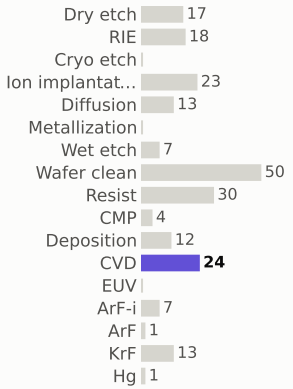
y cut at 57.351 µm



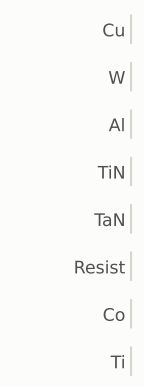
mask for this step

no mask for this step

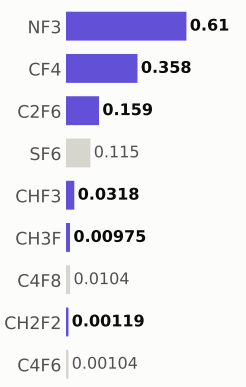
steps by process area



materials deposited (g)



gases emitted (g)



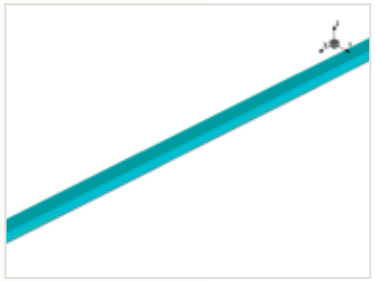
Step #2950: CMP

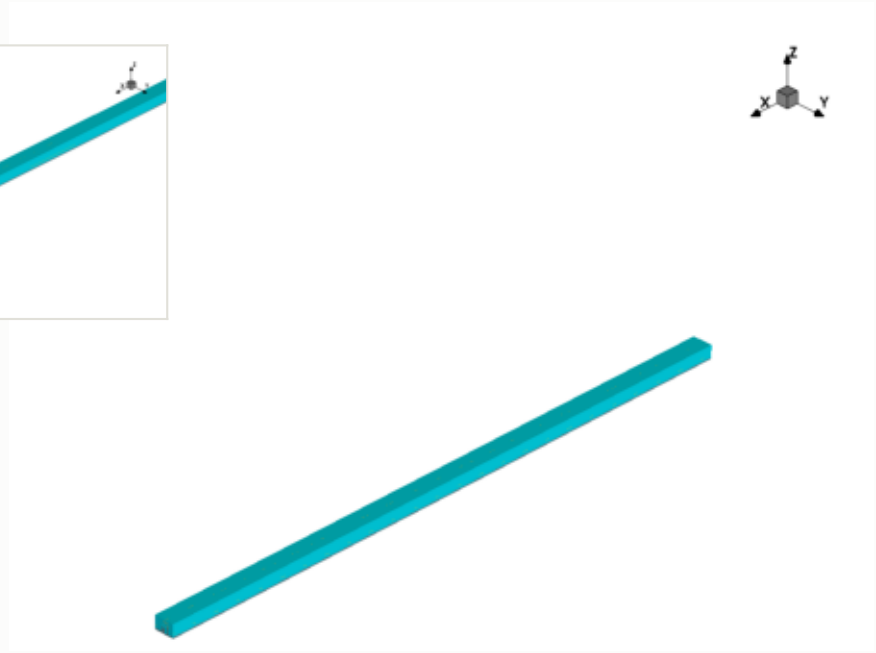
etch_cmp · CMP

0.169	146.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

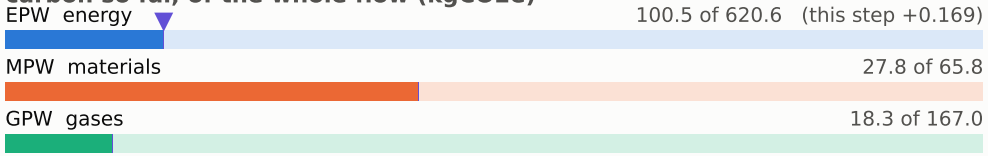




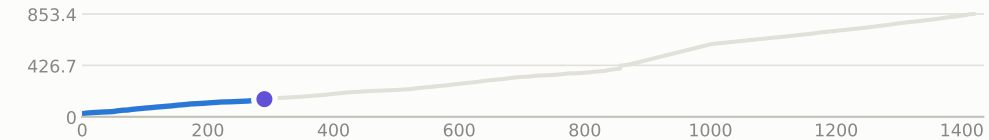
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

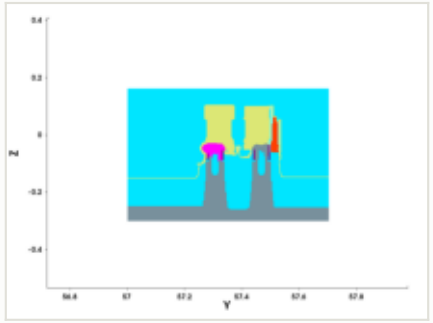
carbon so far, of the whole flow (kgCO2e)



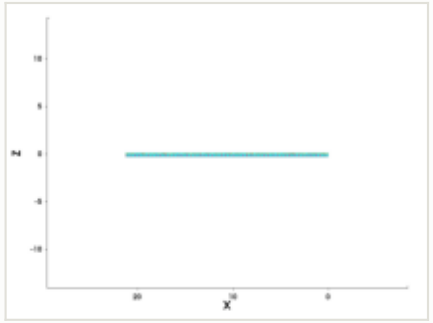
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



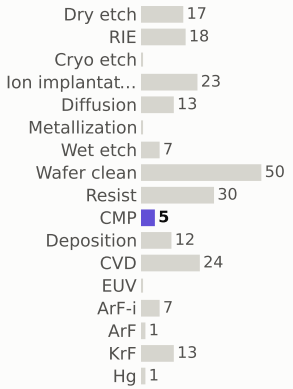
y cut at 57.351 µm



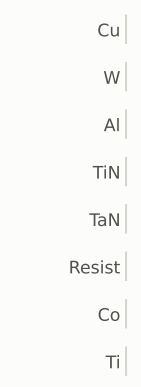
mask for this step

no mask for this step

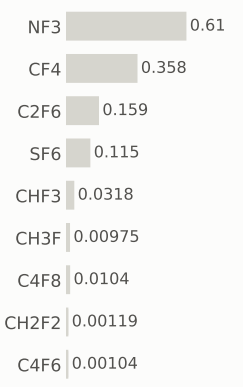
steps by process area



materials deposited (g)



gases emitted (g)



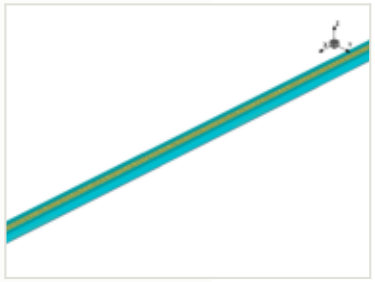
Step #2960: Dry_etch

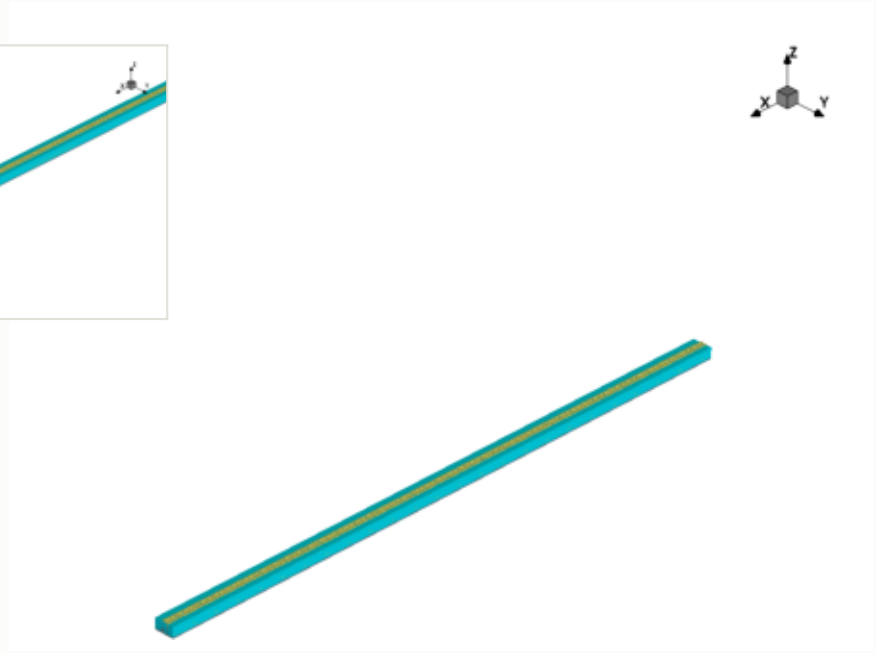
etch_isotropic · Dry_etch

0.801	147.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



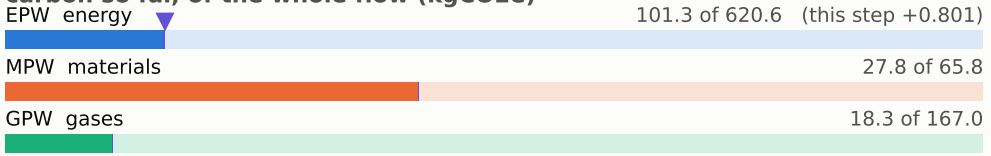


this step

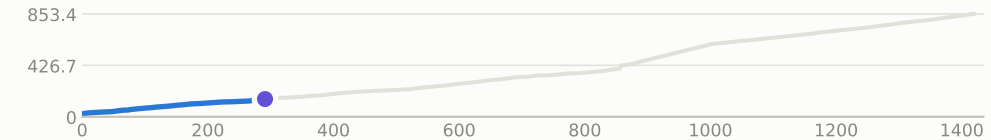
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

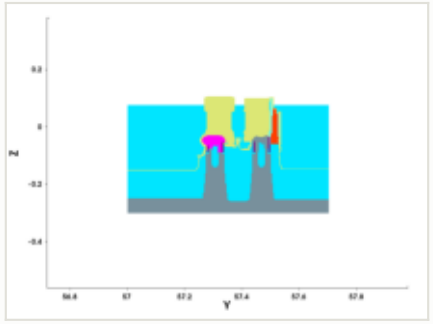
carbon so far, of the whole flow (kgCO2e)



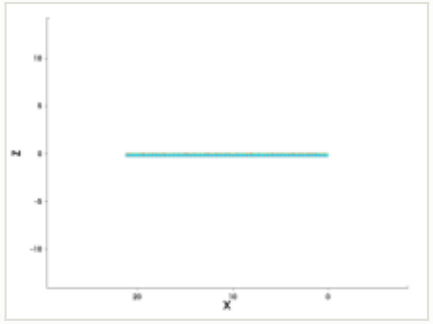
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



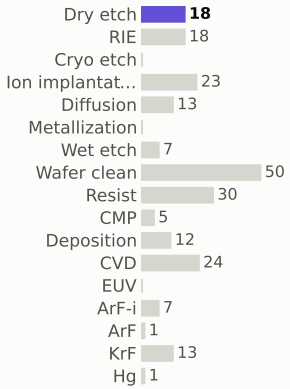
y cut at 57.351 µm



mask for this step

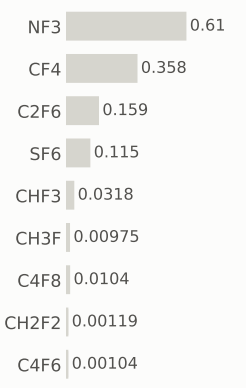
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



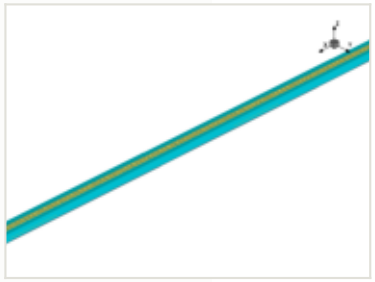
Step #2970: Wafer_clean

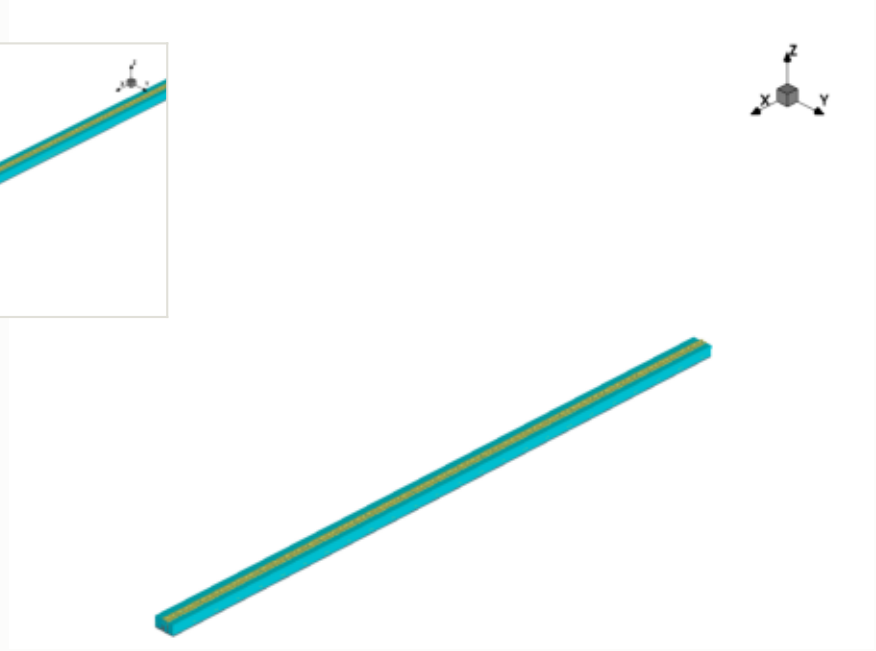
clean · Wafer_clean

0.169	147.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





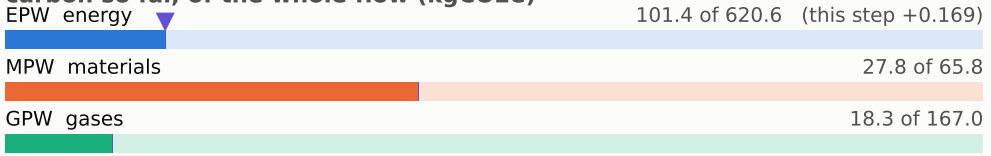
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

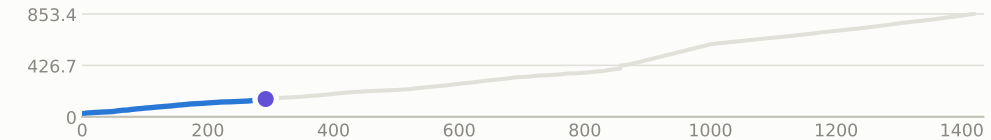
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

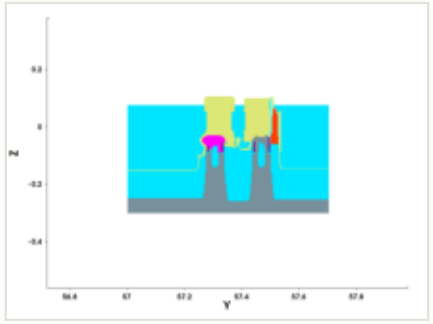
carbon so far, of the whole flow (kgCO2e)



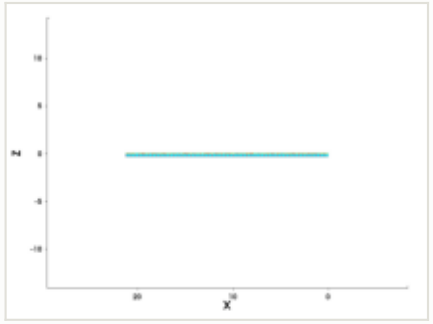
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



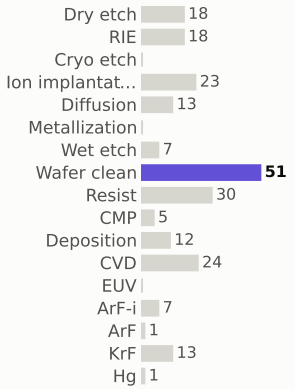
y cut at 57.351 µm



mask for this step

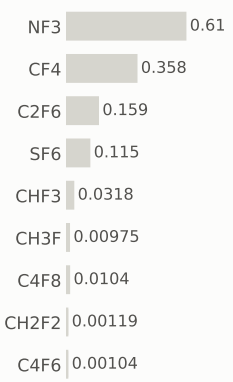
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



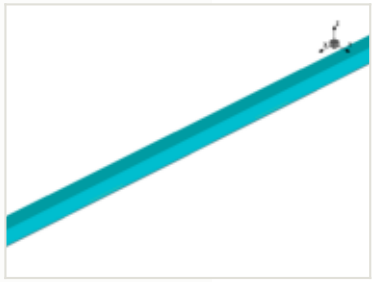
Step #2980: Deposition

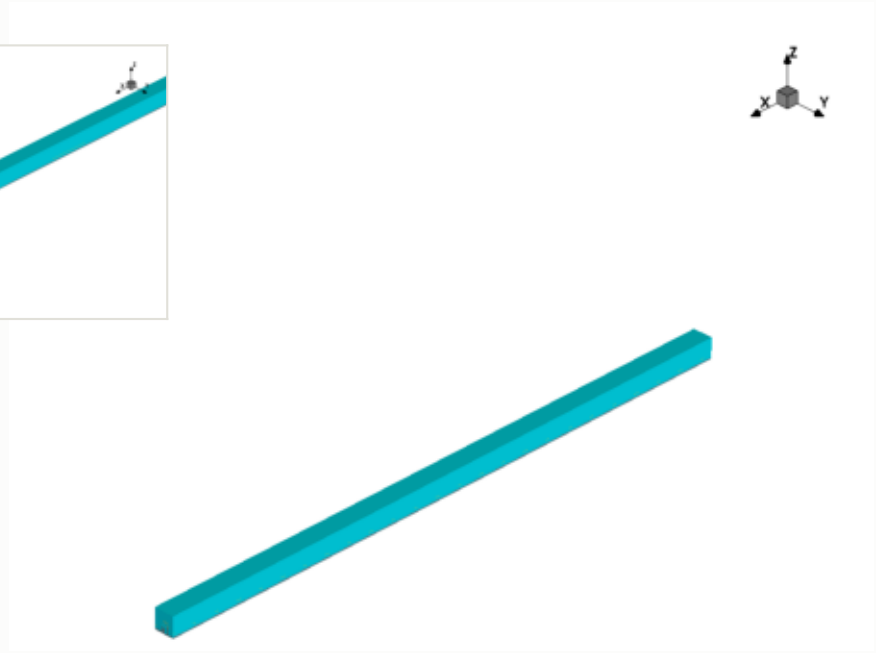
depo_fill · Deposition

0.656	148.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



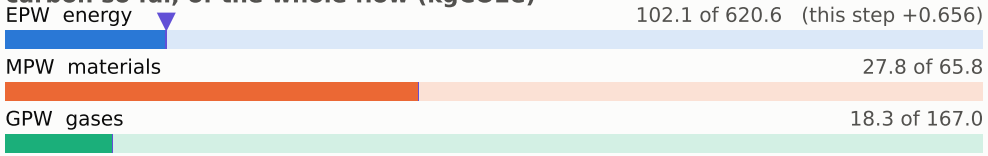


this step

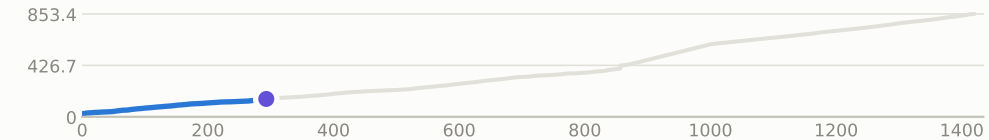
process area	Deposition
material	—
thickness	0.25 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

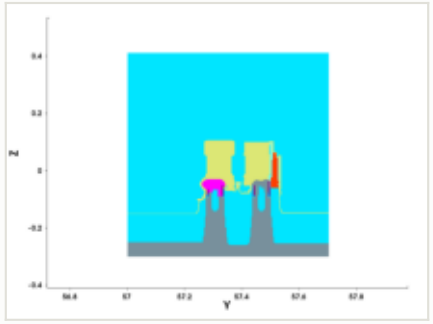
carbon so far, of the whole flow (kgCO2e)



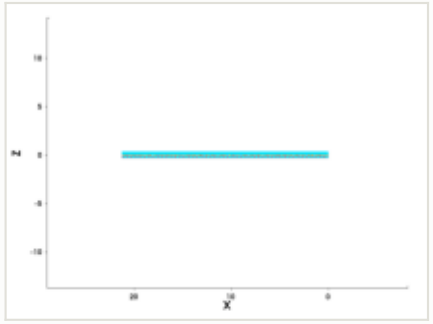
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



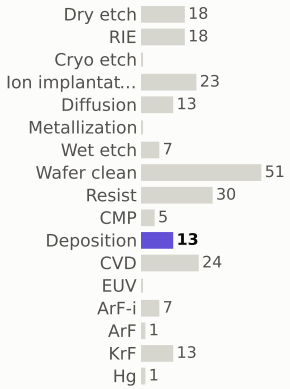
y cut at 57.351 µm



mask for this step

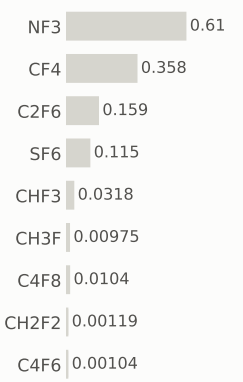
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



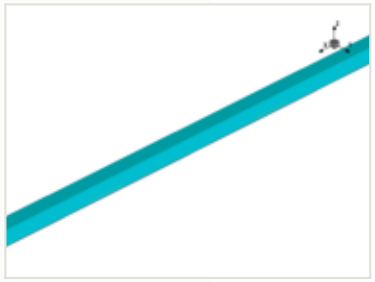
Step #2990: Wafer_clean

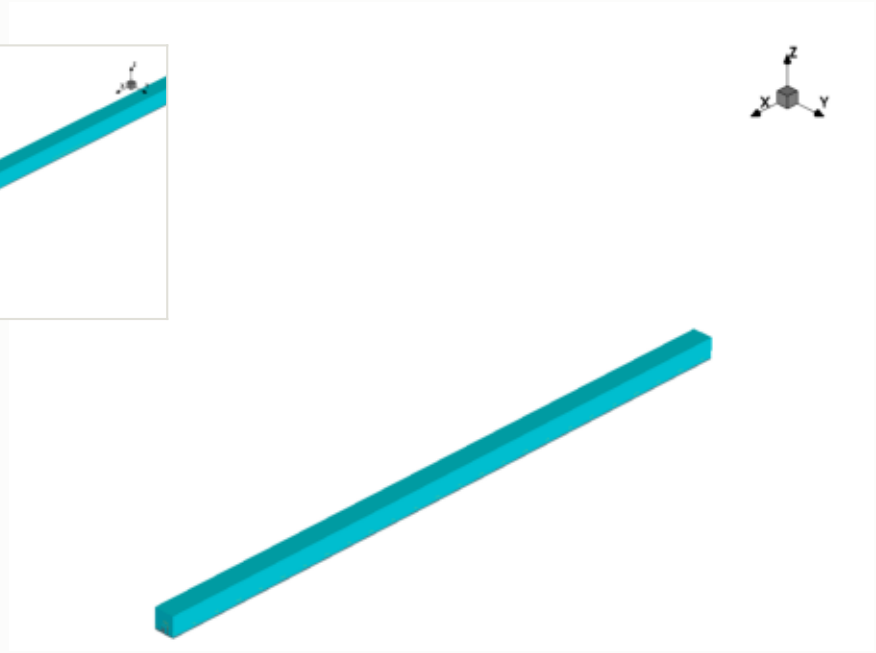
clean · Wafer_clean

0.169	148.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



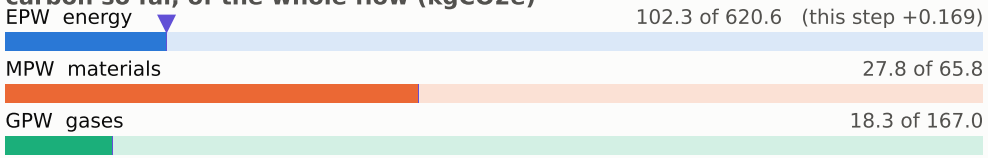


this step

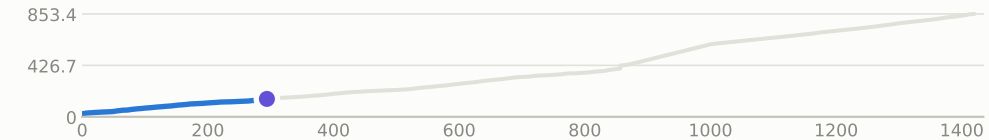
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

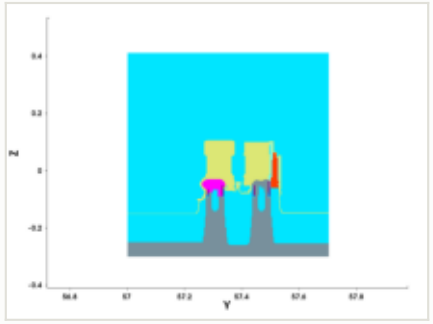
carbon so far, of the whole flow (kgCO2e)



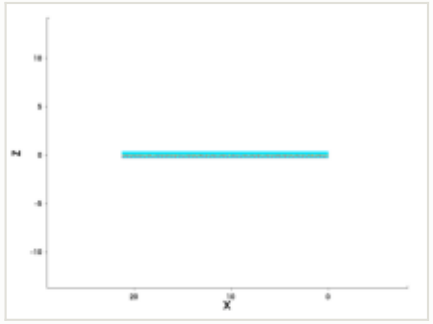
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



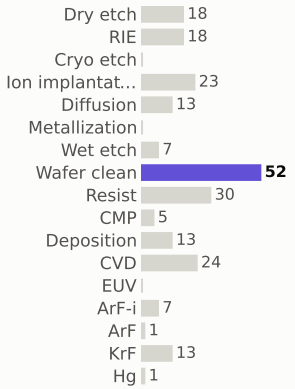
y cut at 57.351 µm



mask for this step

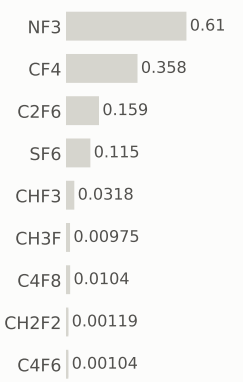
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



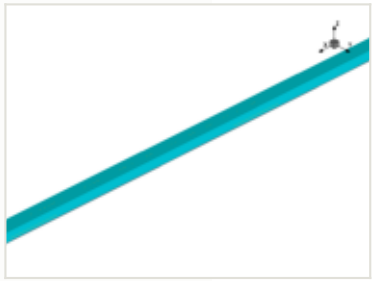
Step #3000: CMP

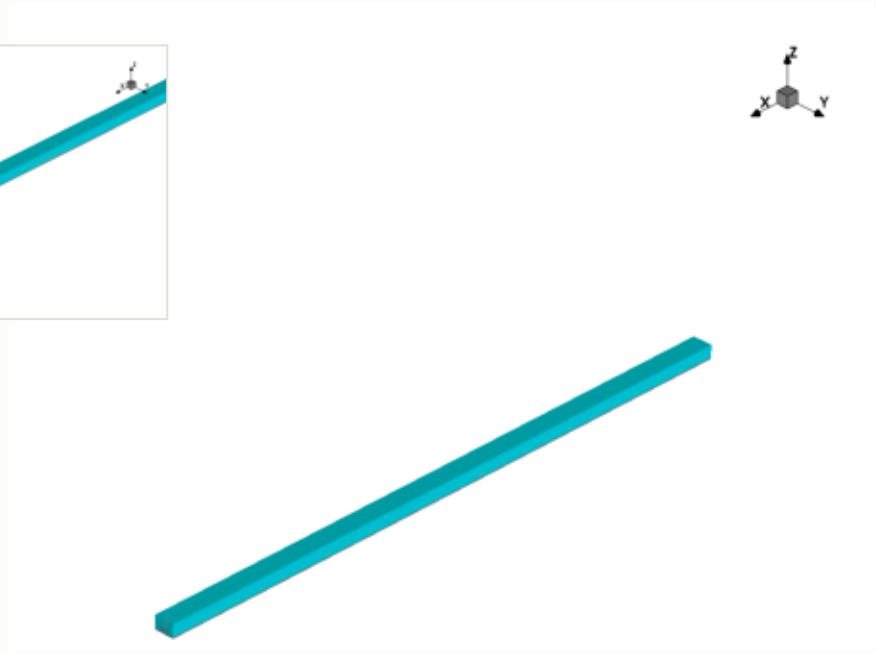
etch_cmp · CMP

0.169	148.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

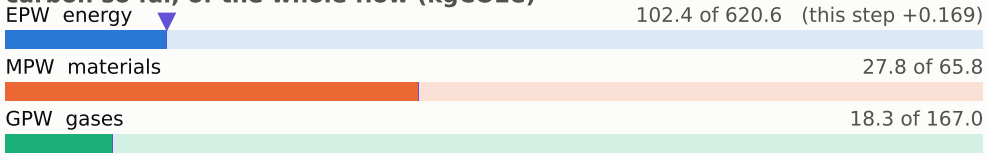




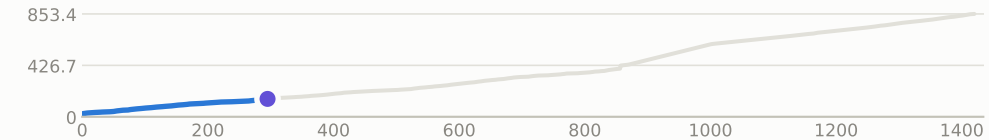
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

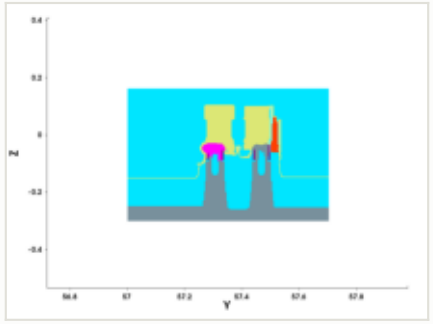
carbon so far, of the whole flow (kgCO2e)



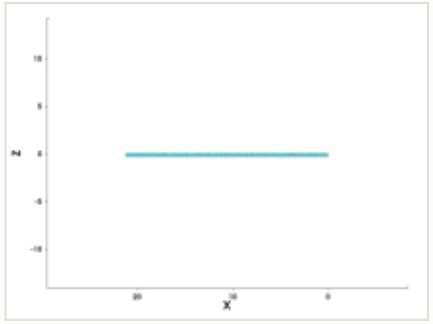
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



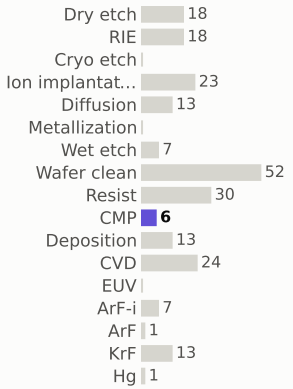
y cut at 57.351 µm



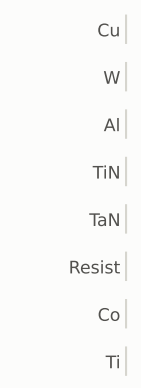
mask for this step

no mask for this step

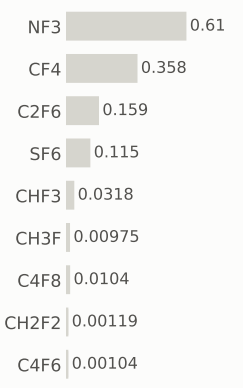
steps by process area



materials deposited (g)



gases emitted (g)



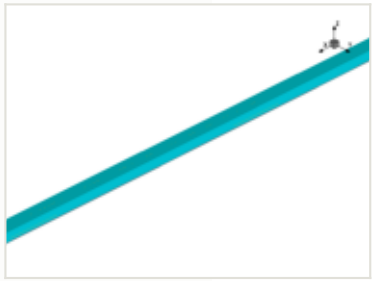
Step #3010: Resist

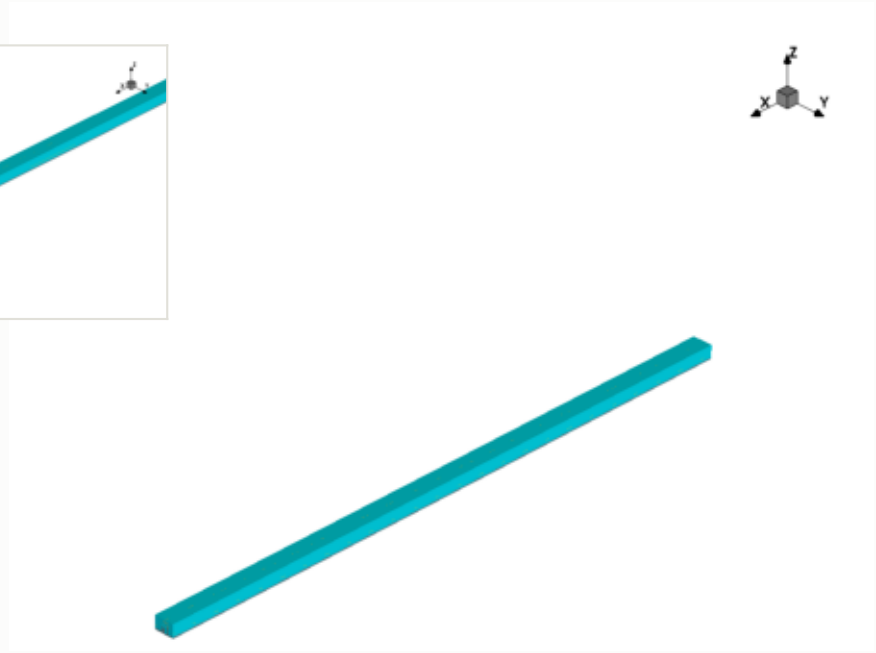
depo_fill · Resist

0.169	148.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



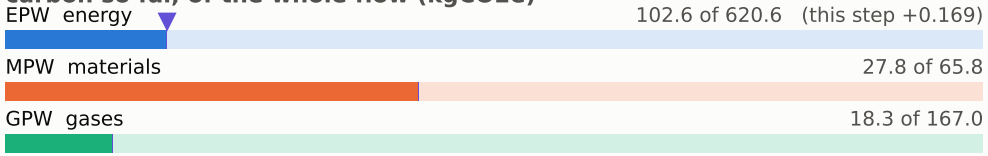


this step

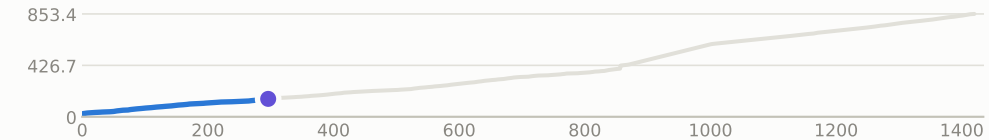
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

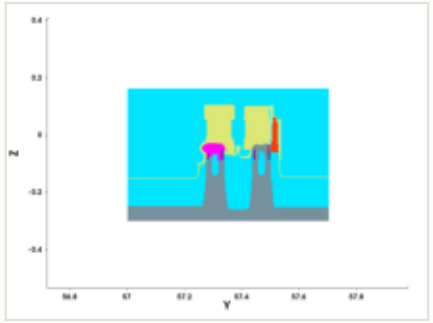
carbon so far, of the whole flow (kgCO2e)



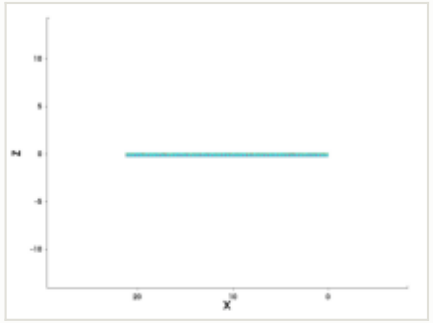
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



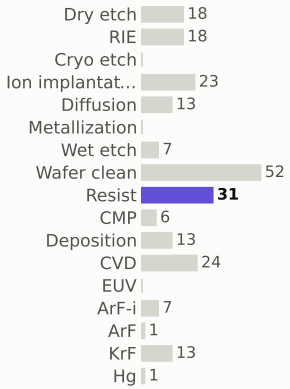
y cut at 57.351 µm



mask for this step

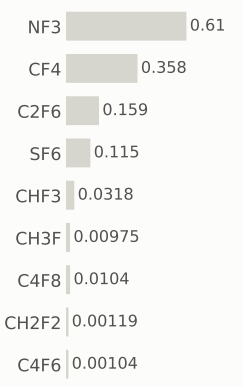
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



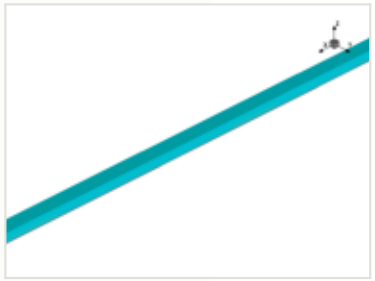
Step #3020: CVD

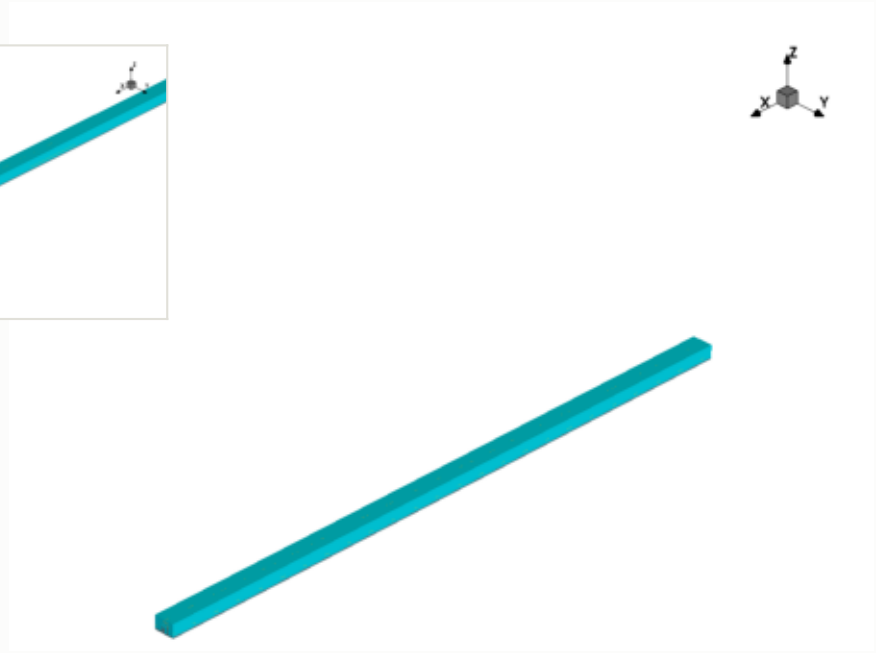
depo_fill · CVD

1.31	150.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



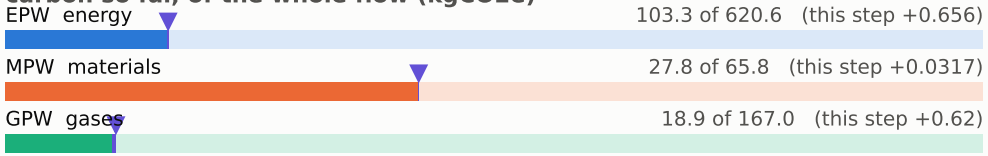


this step

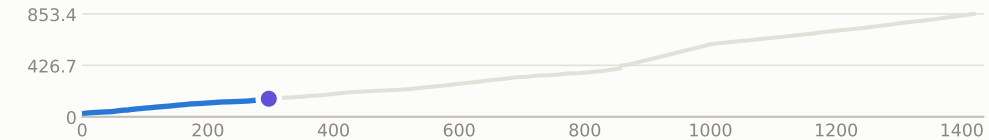
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

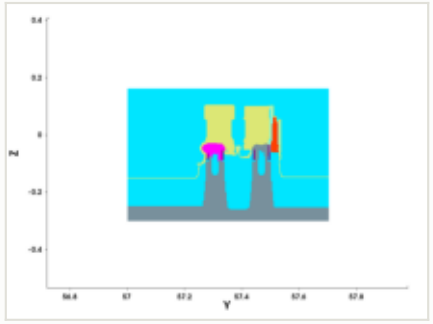
carbon so far, of the whole flow (kgCO2e)



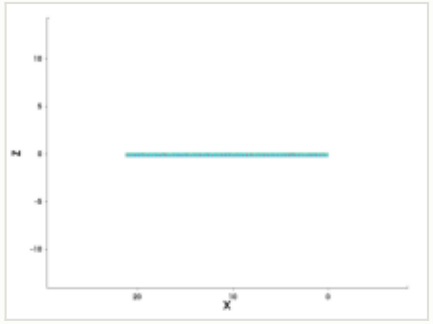
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



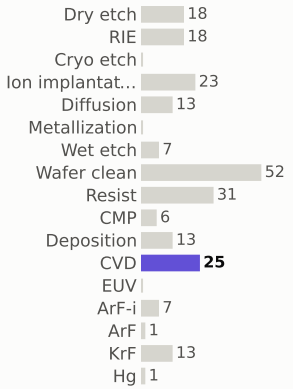
y cut at 57.351 µm



mask for this step

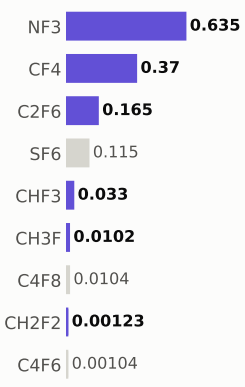
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



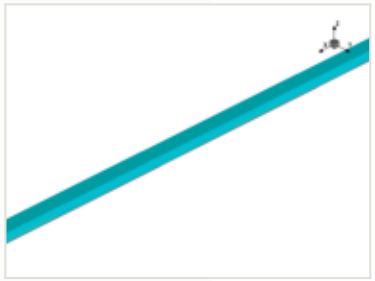
Step #3030: CVD

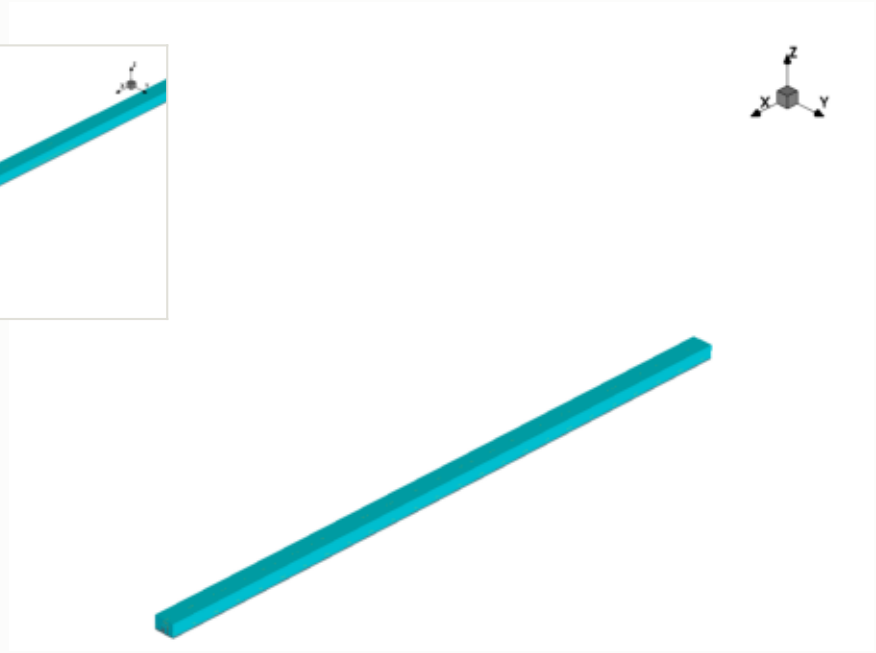
depo_fill · CVD

1.31	151.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



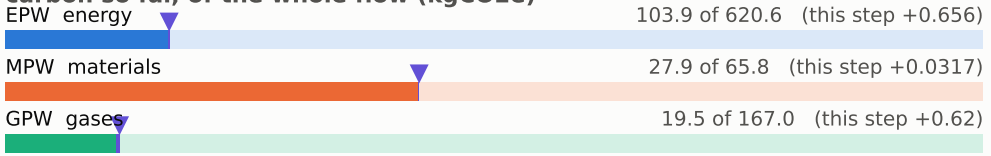


this step

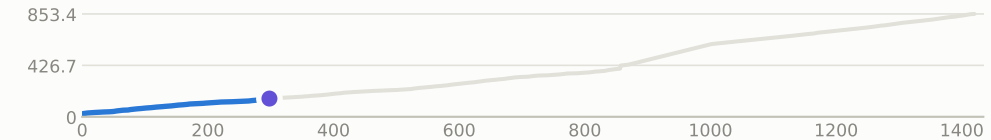
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

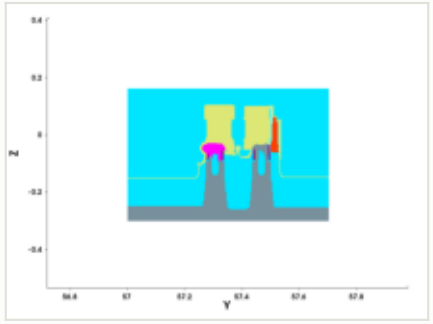
carbon so far, of the whole flow (kgCO2e)



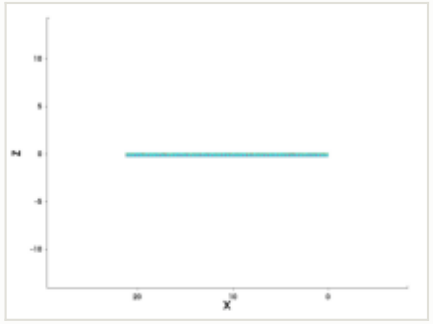
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



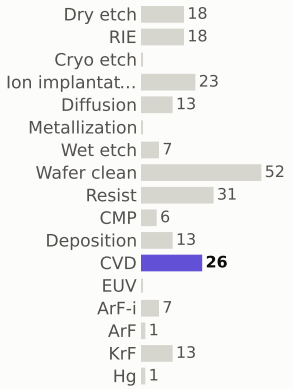
y cut at 57.351 µm



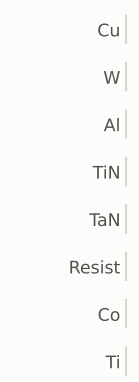
mask for this step

no mask for this step

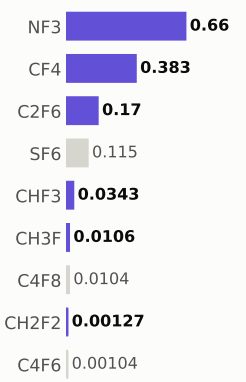
steps by process area



materials deposited (g)



gases emitted (g)



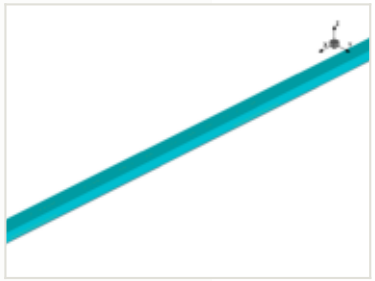
Step #3040: Resist

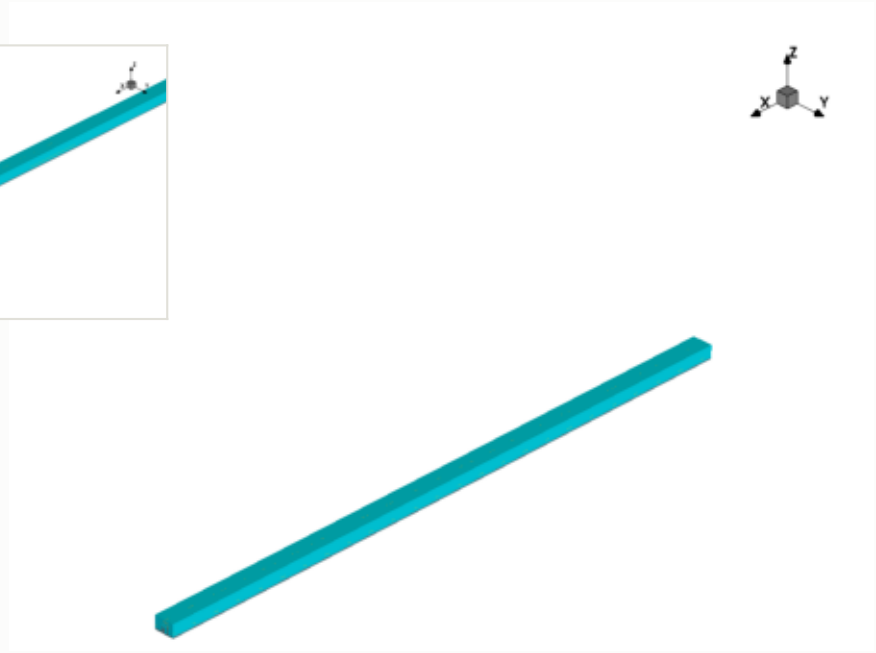
depo_fill · Resist

0.169	151.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



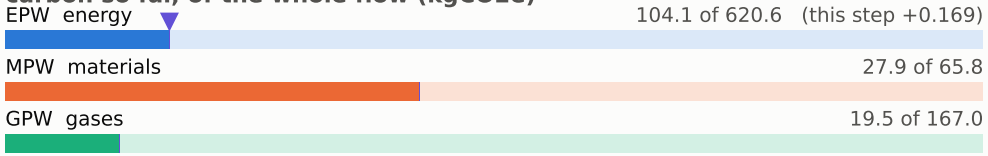


this step

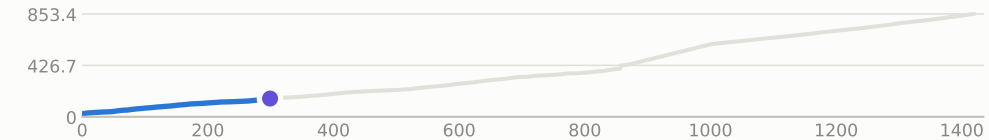
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

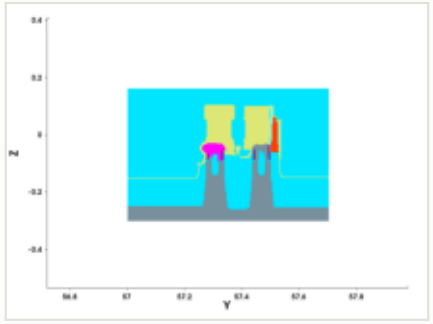
carbon so far, of the whole flow (kgCO2e)



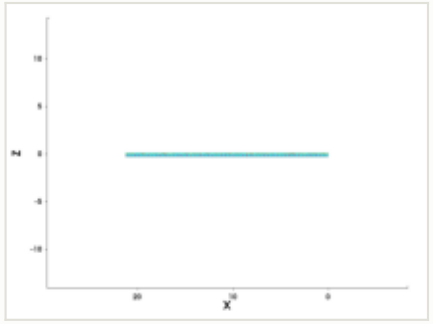
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



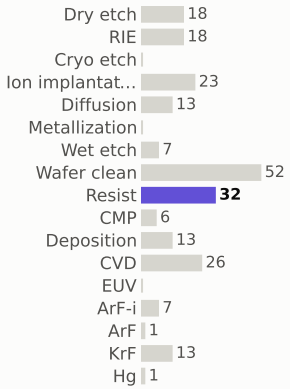
y cut at 57.351 µm



mask for this step

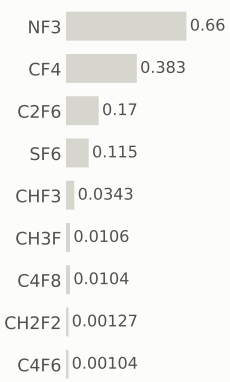
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



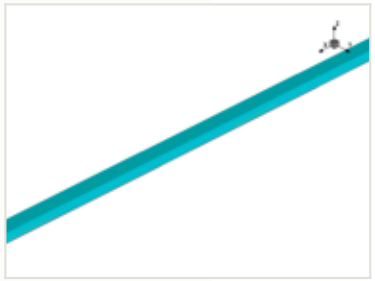
Step #3050: CVD

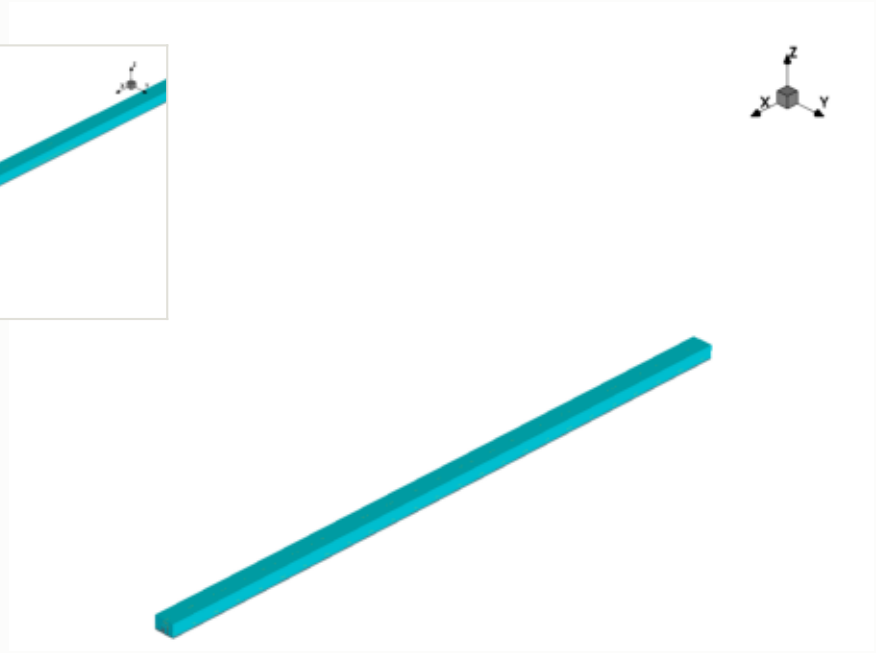
depo_fill · CVD

1.31	152.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



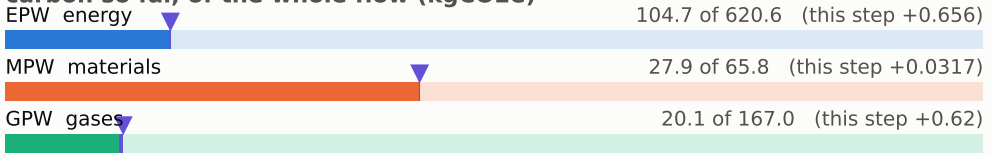


this step

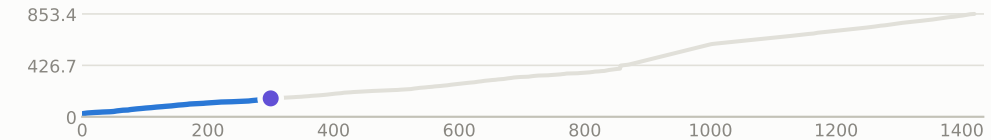
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

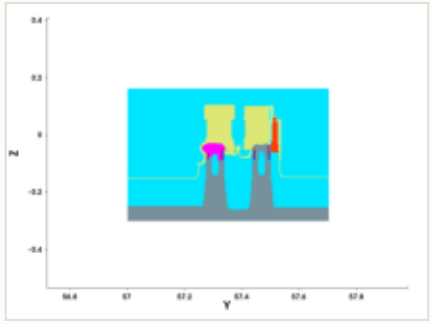
carbon so far, of the whole flow (kgCO2e)



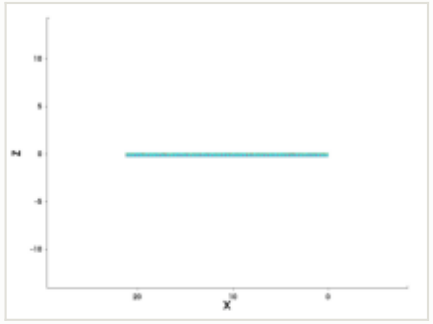
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



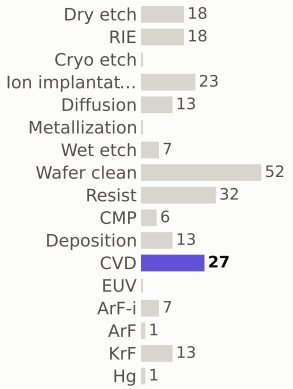
y cut at 57.351 µm



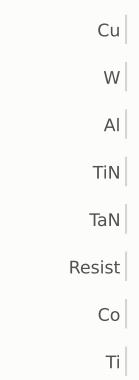
mask for this step

no mask for this step

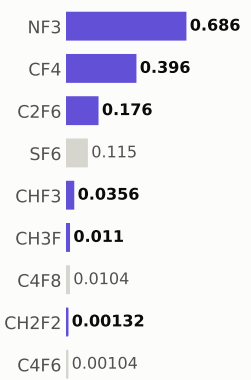
steps by process area



materials deposited (g)



gases emitted (g)



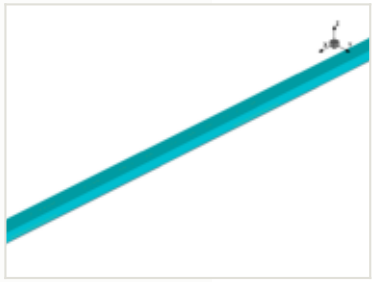
Step #3060: Wafer_clean

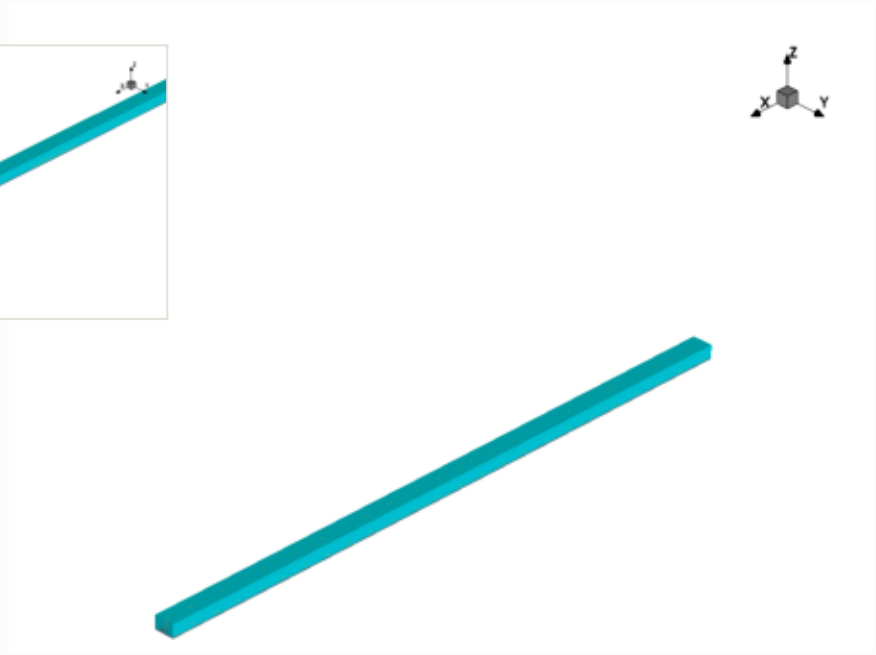
clean · Wafer_clean

0.169	152.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

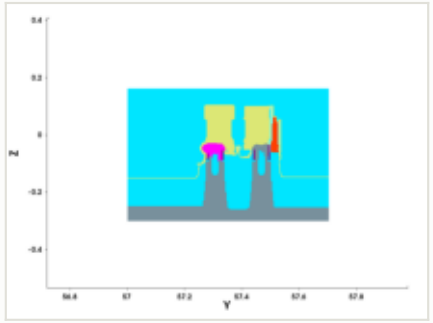
3D structure after this step

zoomed to fit

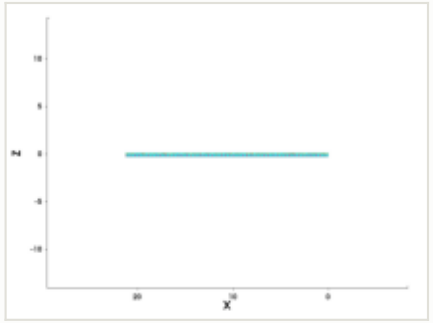




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

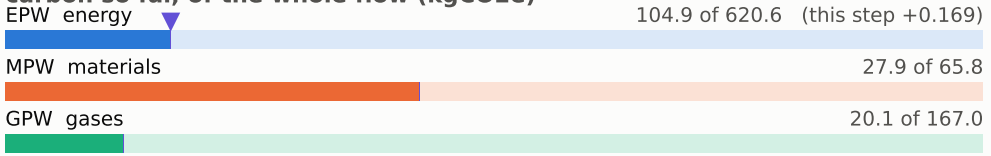
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

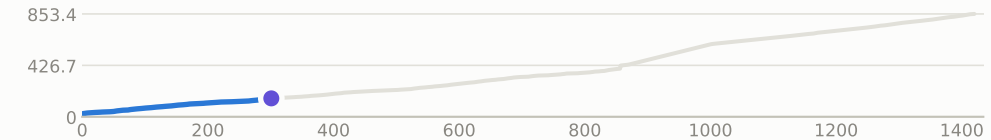
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

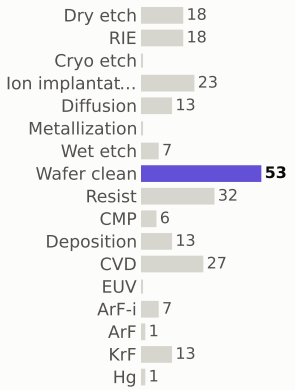
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

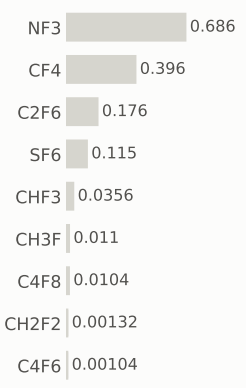


steps by process area



materials deposited (g)

gases emitted (g)



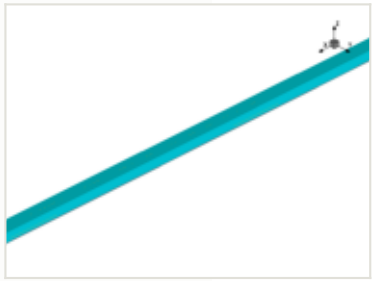
Step #3070: Resist

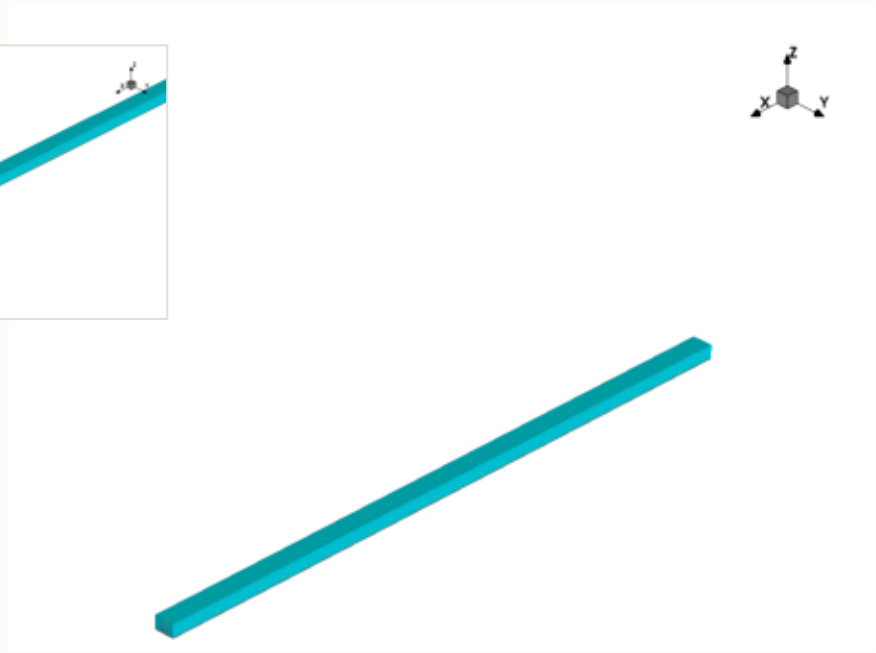
resist · Resist

0.169	153.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





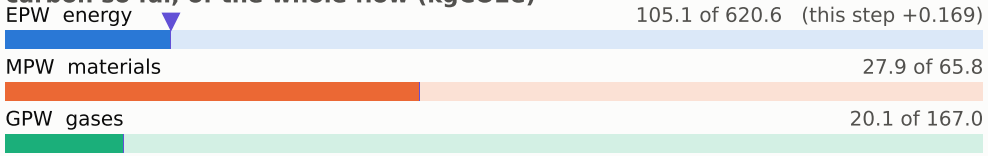
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

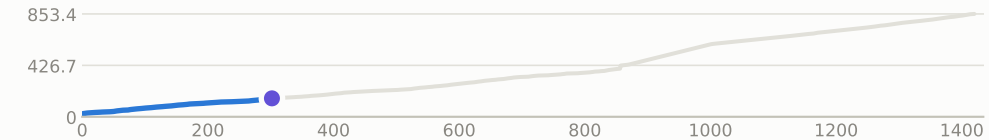
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

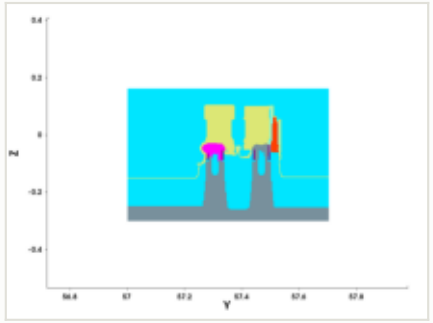
carbon so far, of the whole flow (kgCO2e)



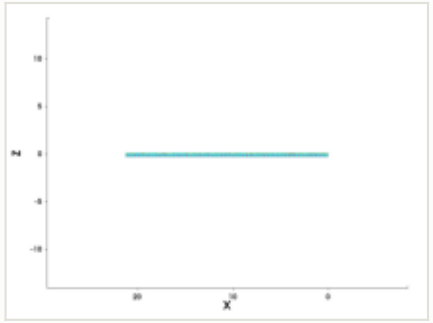
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



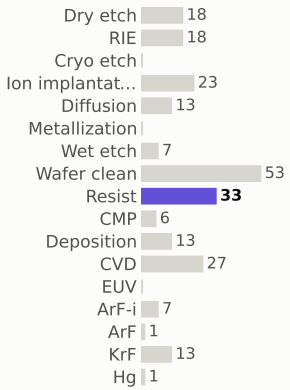
y cut at 57.351 µm



mask for this step

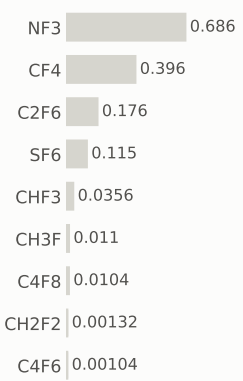
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



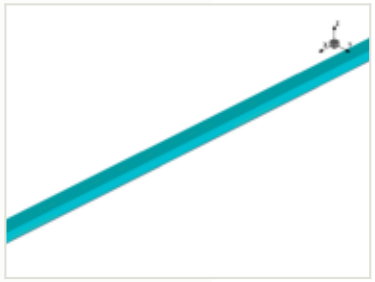
Step #3080: ArF-i

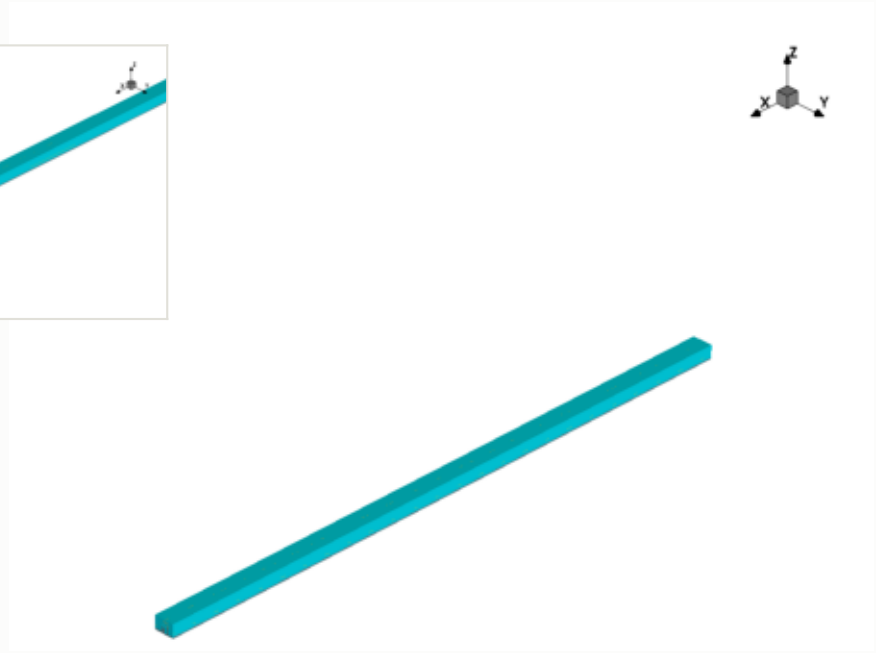
litho_patterning · ArF-i

0.217	153.3	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



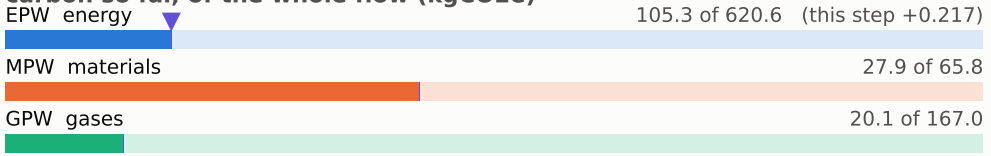


this step

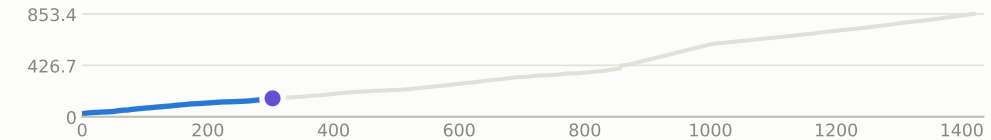
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

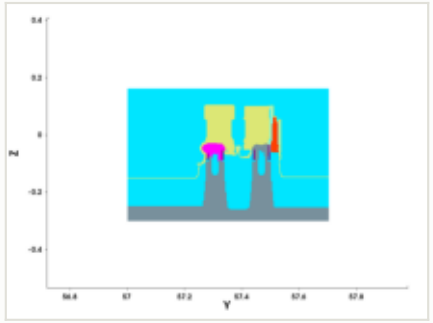
carbon so far, of the whole flow (kgCO2e)



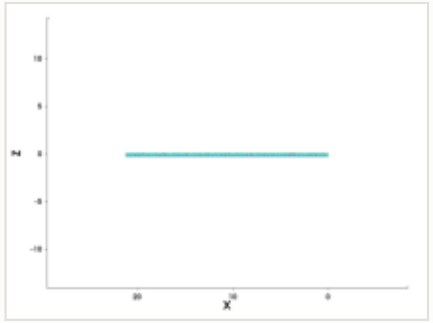
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

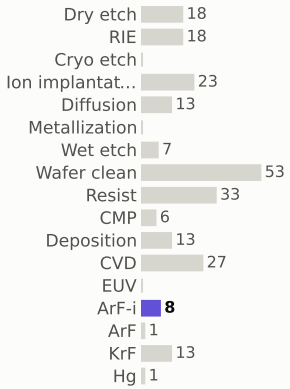


y cut at 57.351 µm



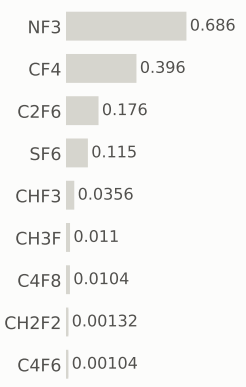
mask: layer 2088:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



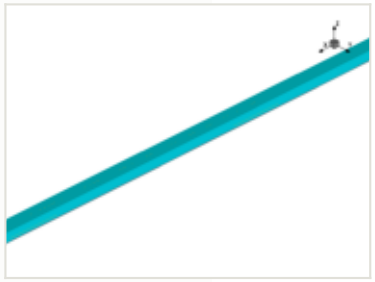
Step #3090: RIE

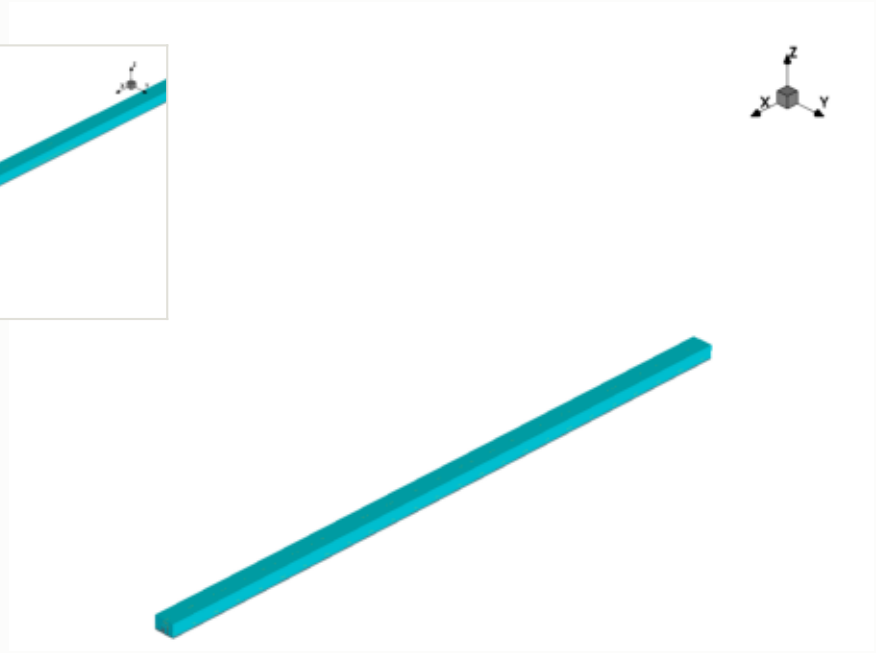
etch_anisotropic · RIE

1.04	154.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



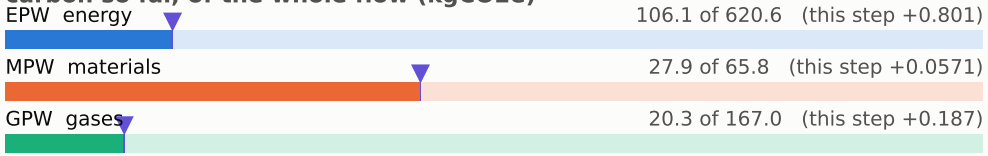


this step

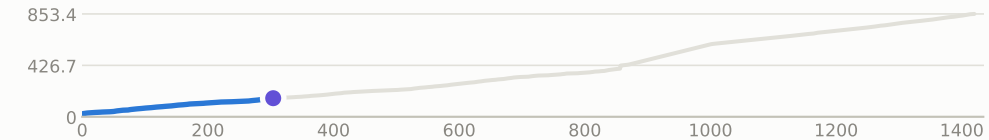
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

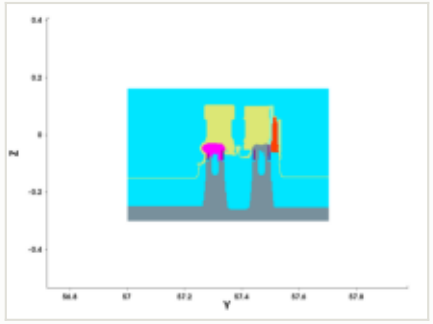
carbon so far, of the whole flow (kgCO2e)



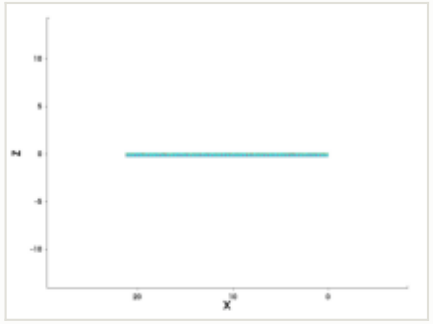
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



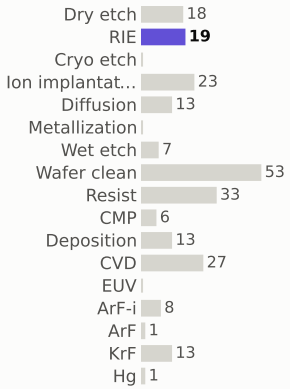
y cut at 57.351 µm



mask for this step

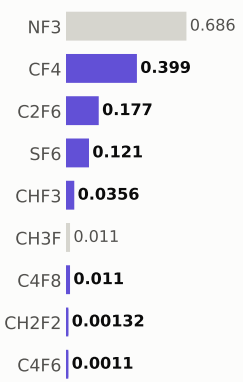
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



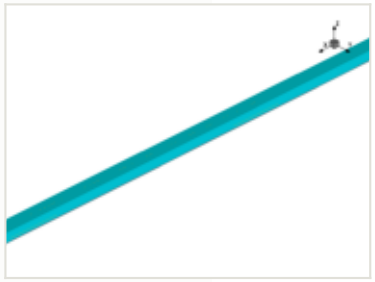
Step #3100: Spx_only

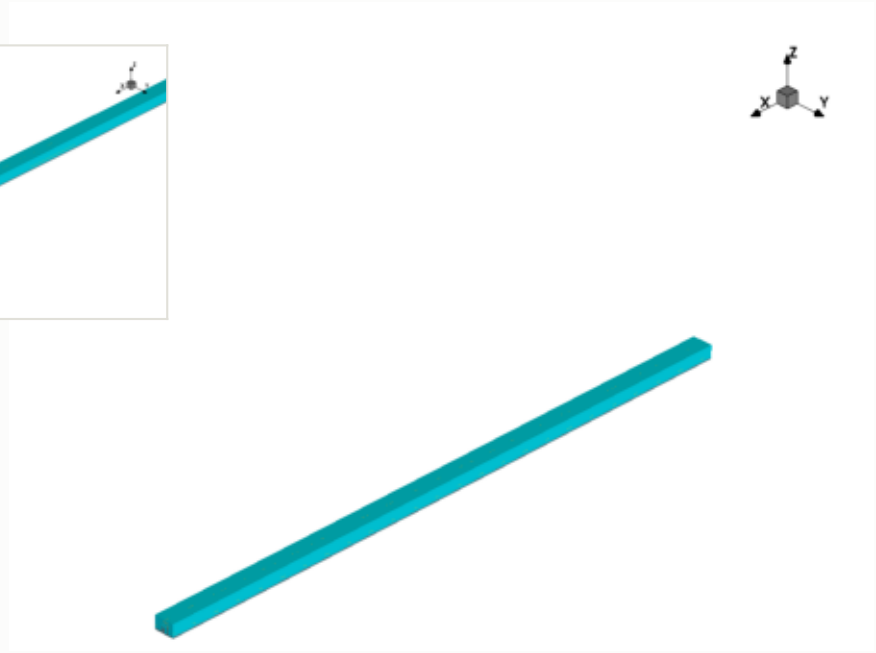
etch_anisotropic · Spx_only

0	154.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



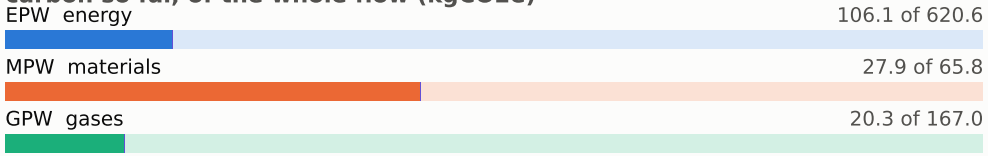


this step

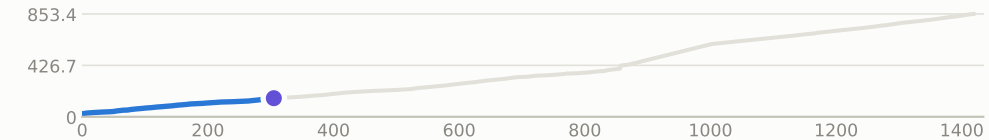
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

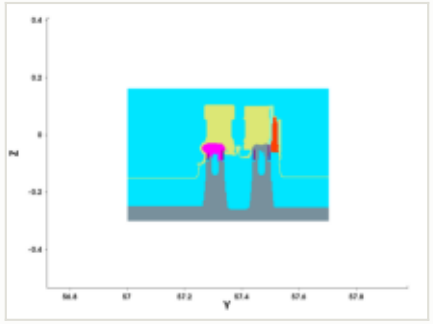
carbon so far, of the whole flow (kgCO2e)



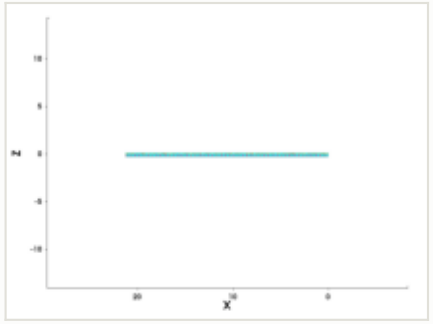
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



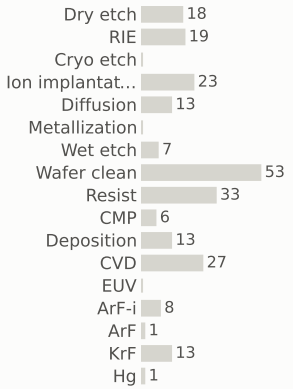
y cut at 57.351 µm



mask for this step

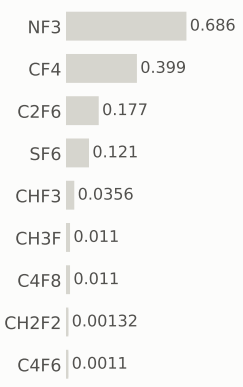
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



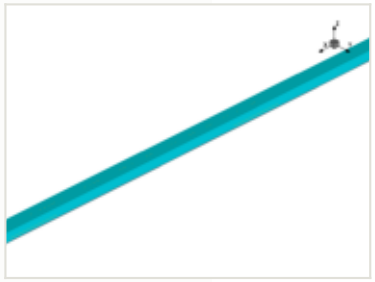
Step #3110: Spx_only

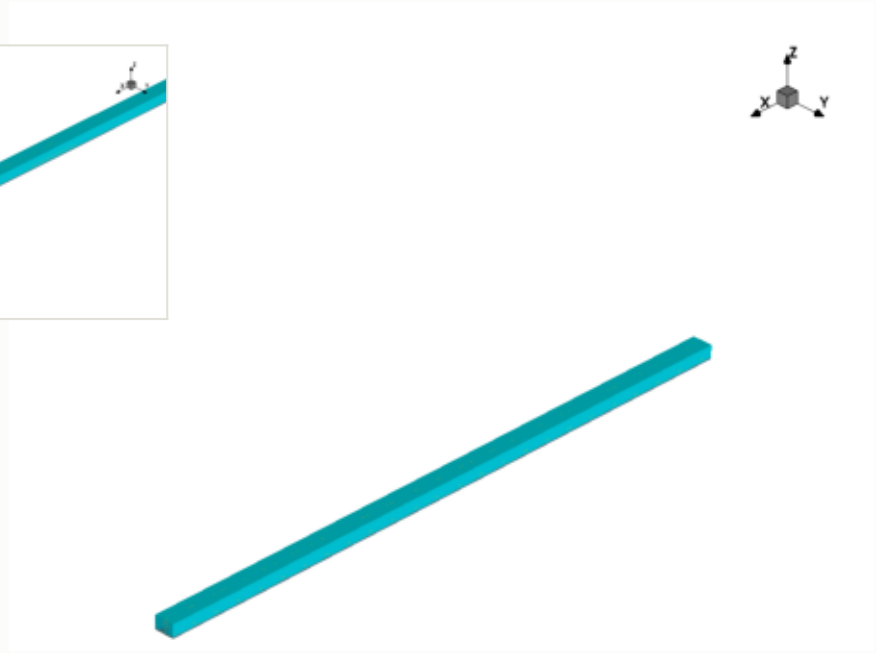
etch_anisotropic · Spx_only

0	154.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



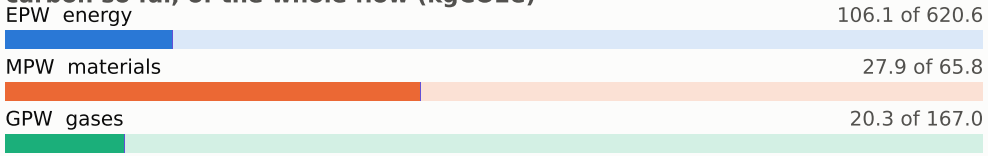


this step

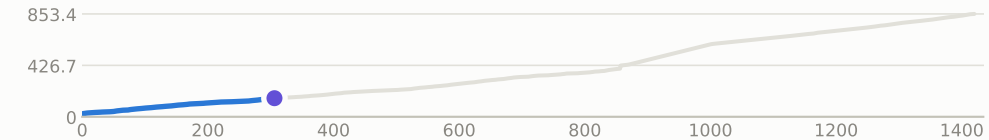
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

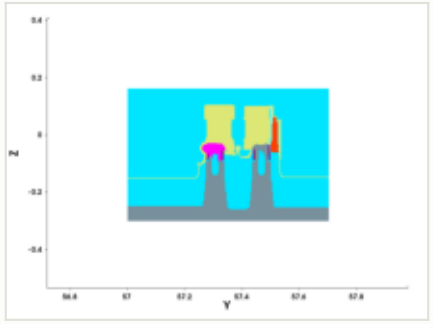
carbon so far, of the whole flow (kgCO2e)



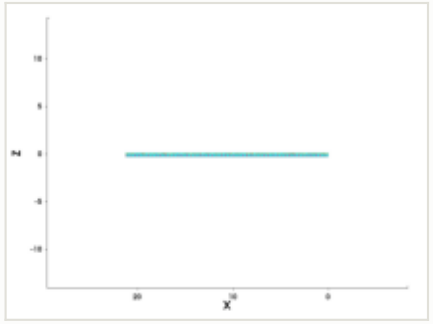
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



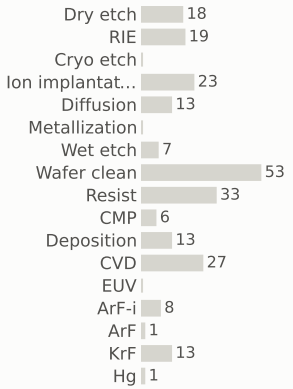
y cut at 57.351 µm



mask for this step

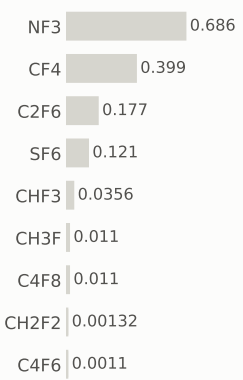
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



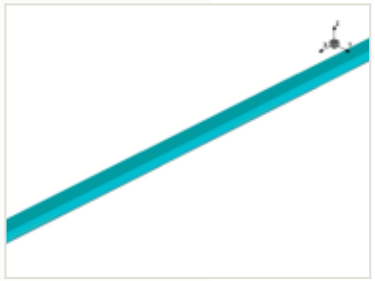
Step #3120: Spx_only

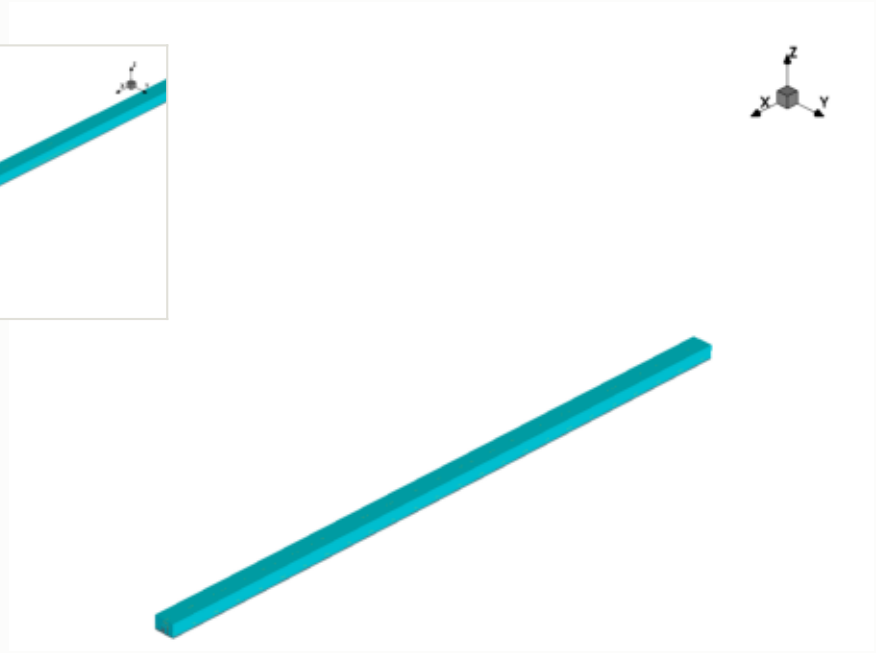
etch_strip · Spx_only

0	154.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



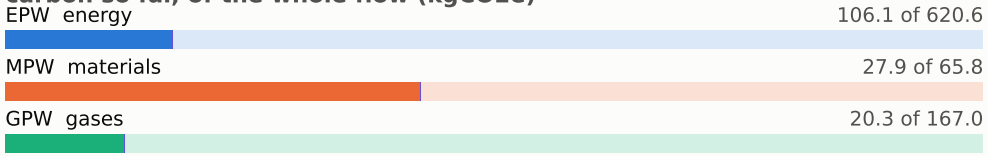


this step

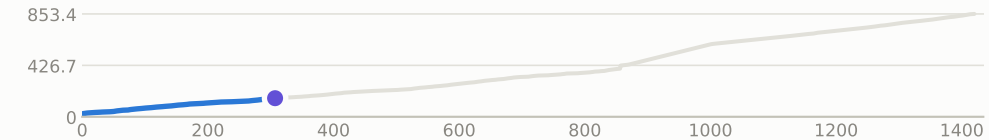
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

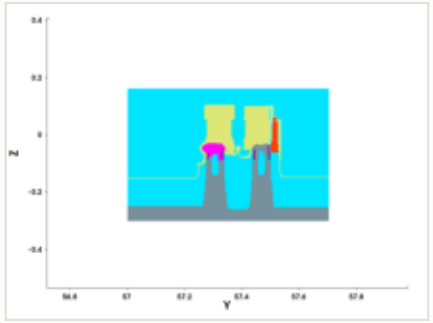
carbon so far, of the whole flow (kgCO2e)



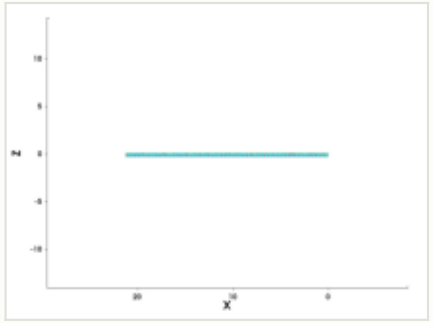
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



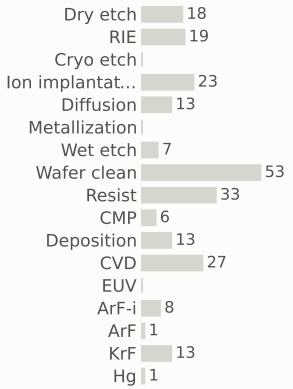
y cut at 57.351 µm



mask for this step

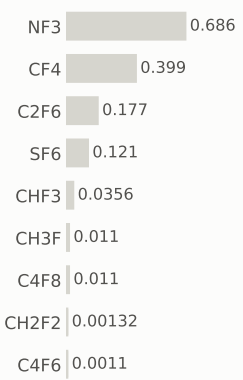
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



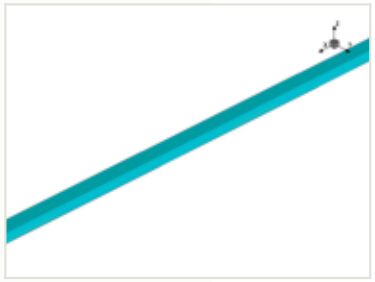
Step #3130: Wafer_clean

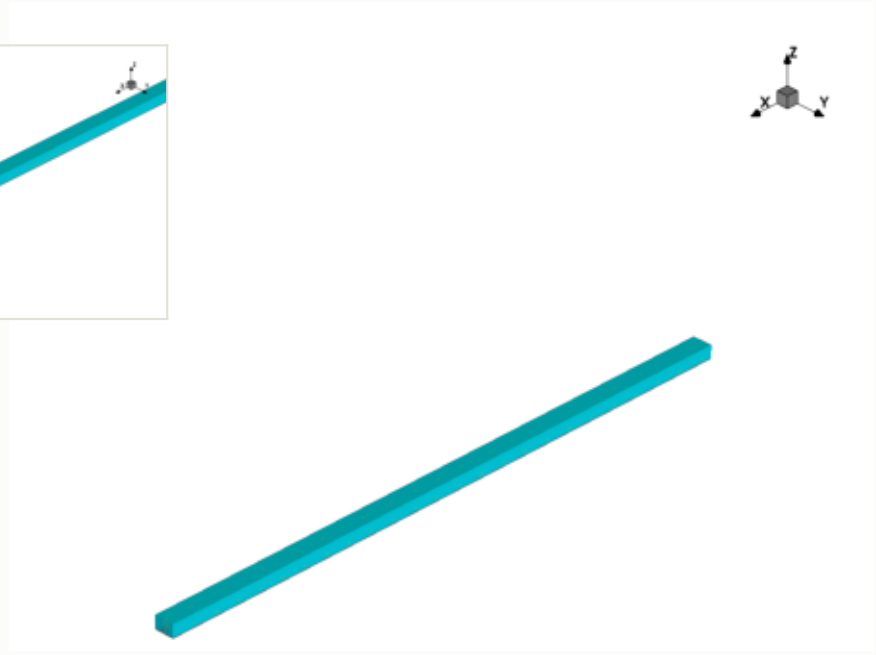
clean · Wafer_clean

0.169	154.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





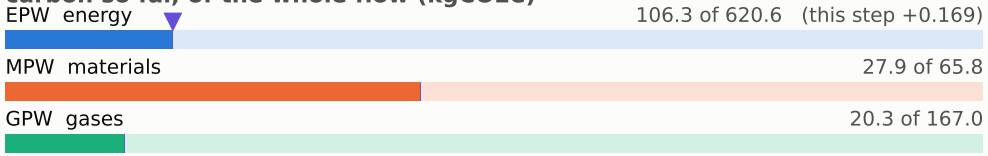
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

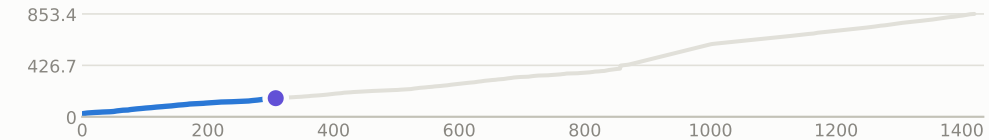
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

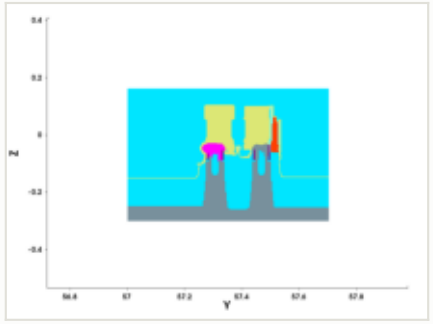
carbon so far, of the whole flow (kgCO2e)



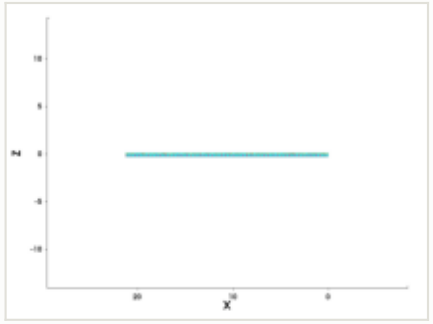
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



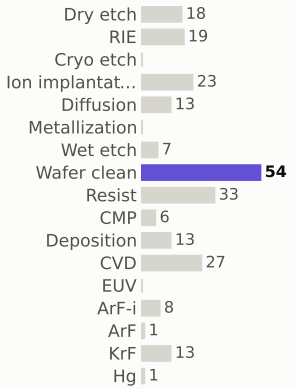
y cut at 57.351 µm



mask for this step

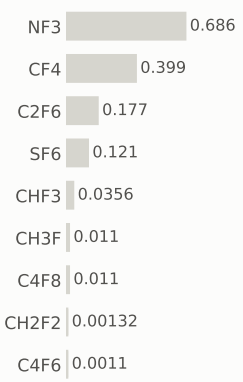
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



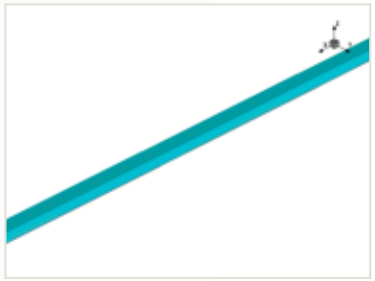
Step #3140: Resist

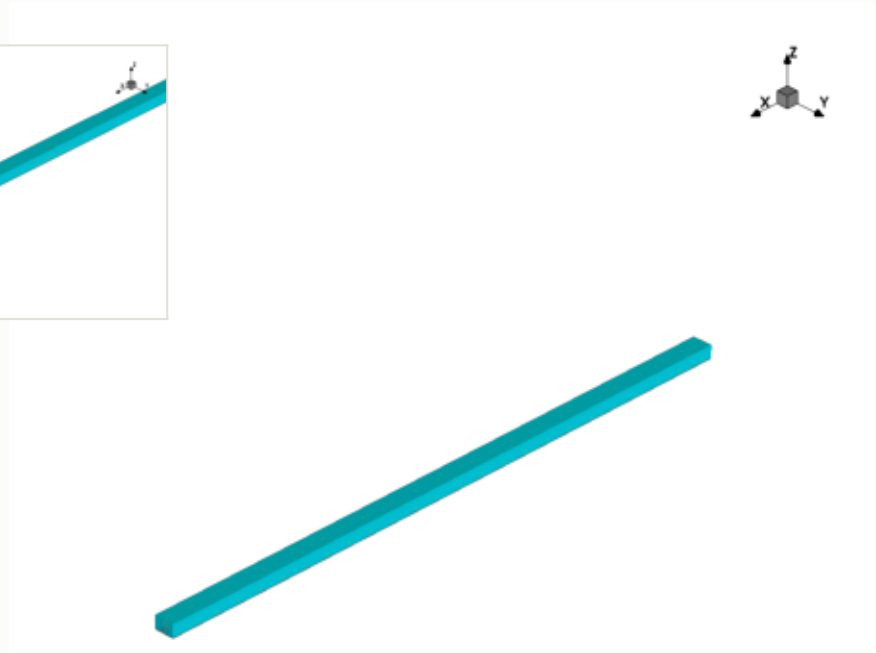
depo_fill · Resist

0.169	154.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



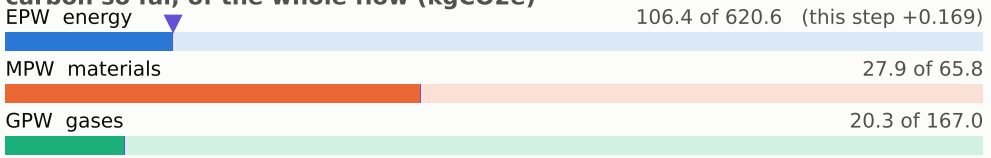


this step

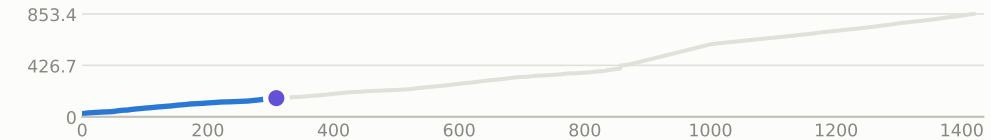
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

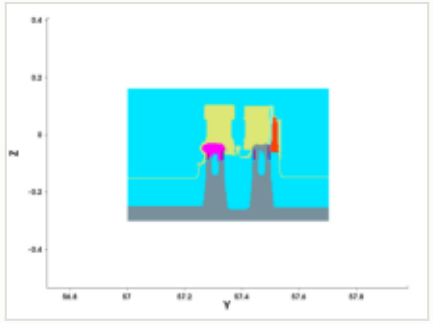
carbon so far, of the whole flow (kgCO2e)



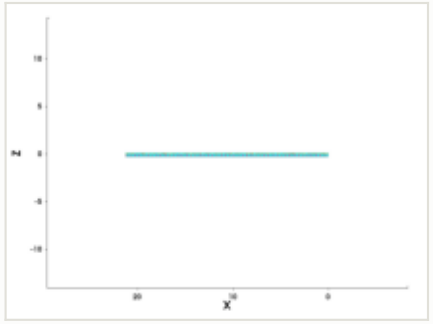
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



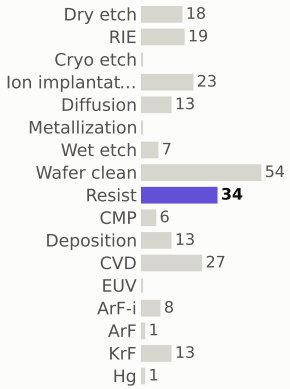
y cut at 57.351 µm



mask for this step

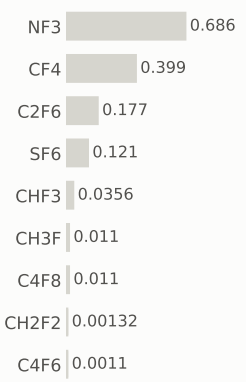
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



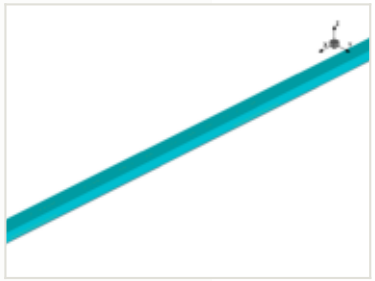
Step #3150: CVD

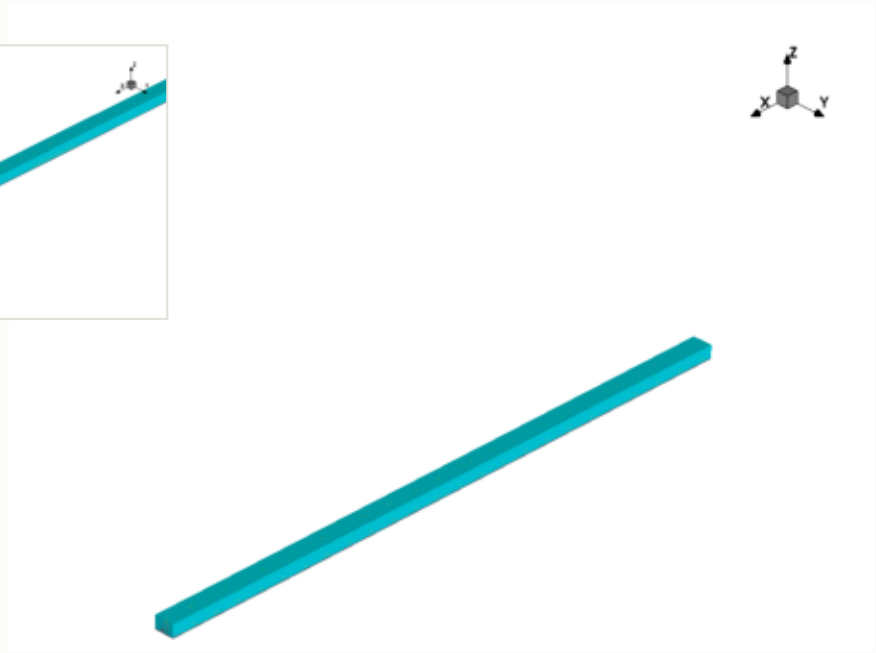
depo_fill · CVD

1.31	156.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



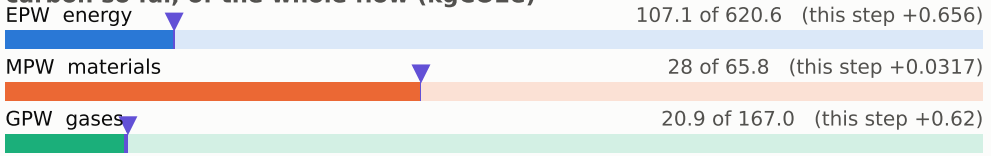


this step

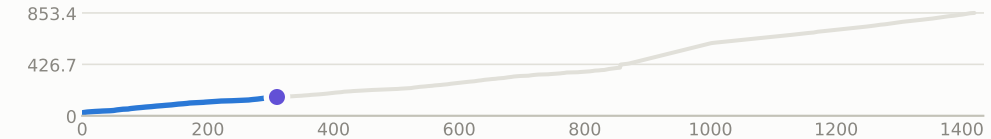
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

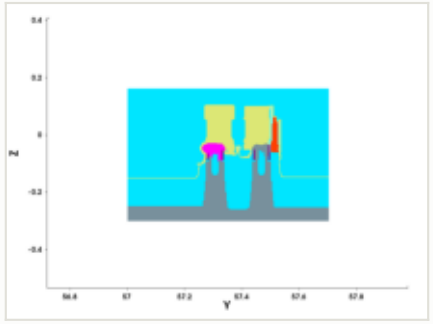
carbon so far, of the whole flow (kgCO2e)



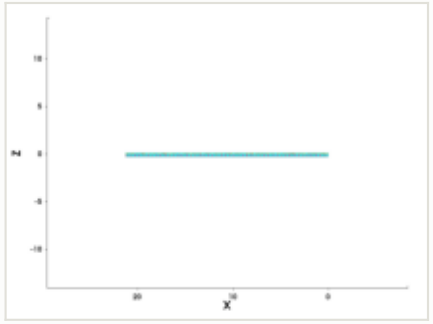
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



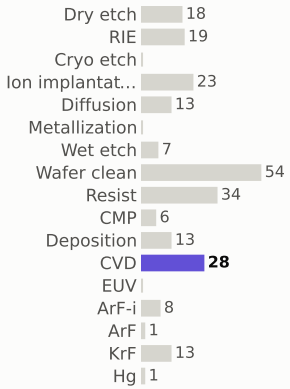
y cut at 57.351 µm



mask for this step

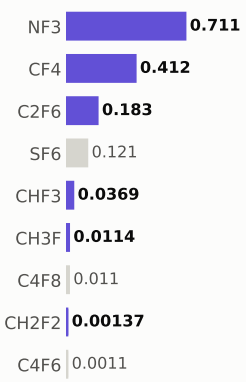
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



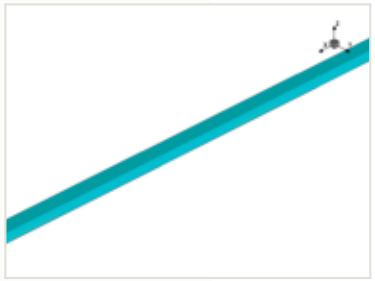
Step #3160: Wafer_clean

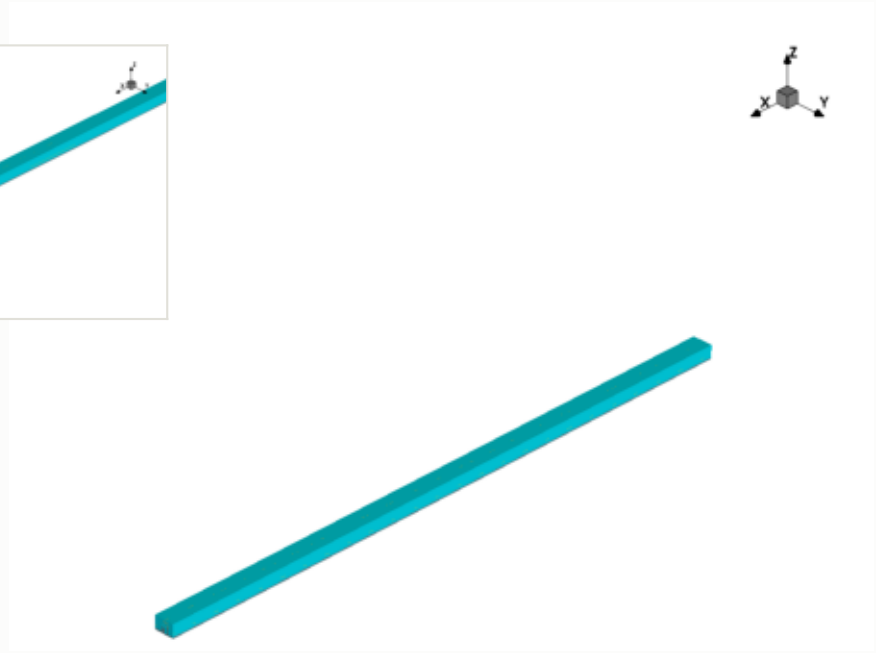
clean · Wafer_clean

0.169	156.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

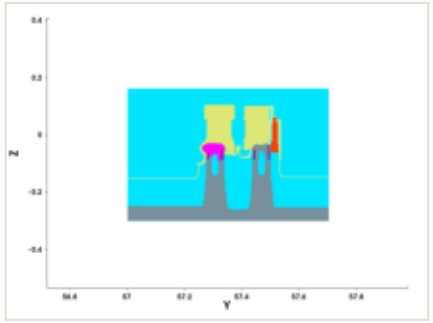
3D structure after this step

zoomed to fit

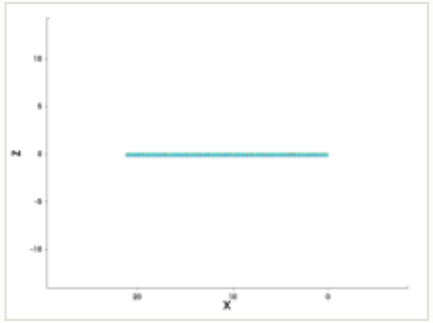




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

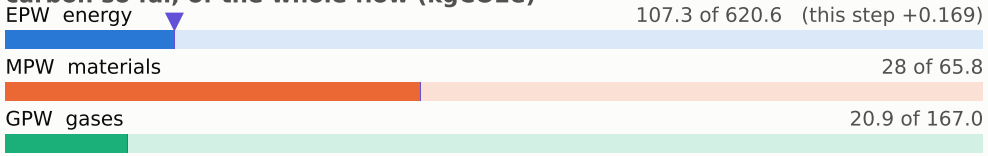
no mask for this step

this step

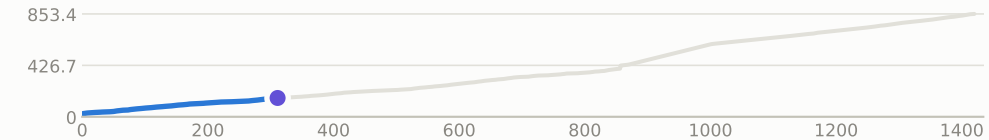
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

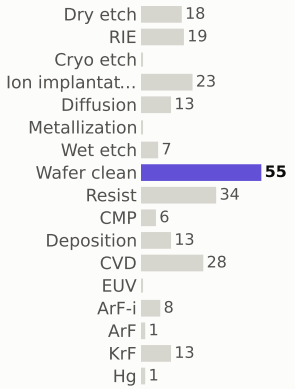
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

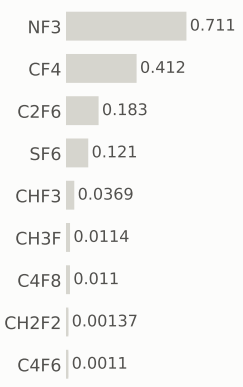


steps by process area



materials deposited (g)

gases emitted (g)



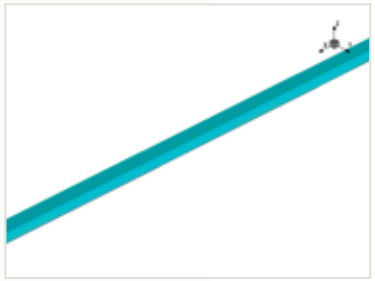
Step #3170: Resist

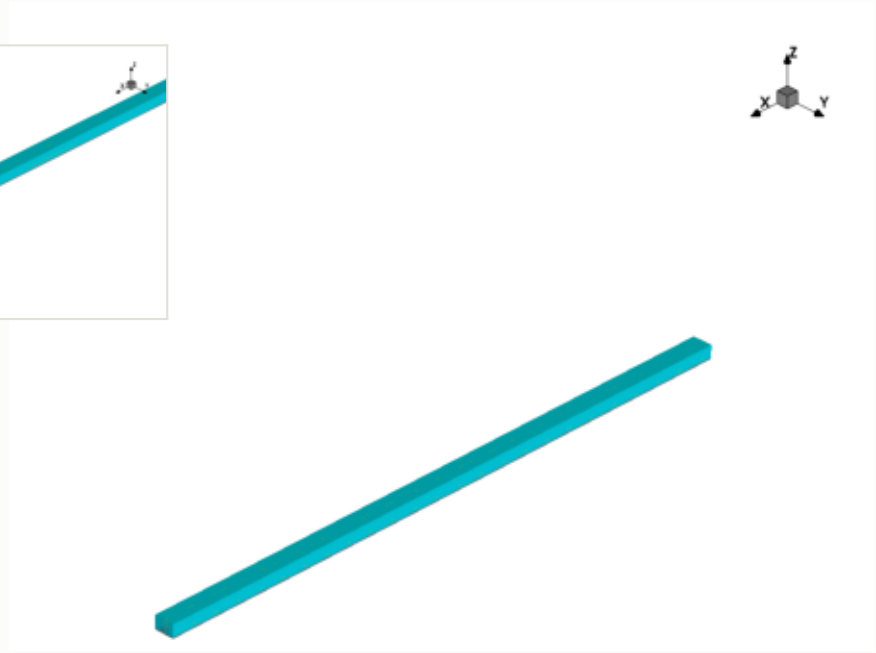
resist · Resist

0.169	156.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





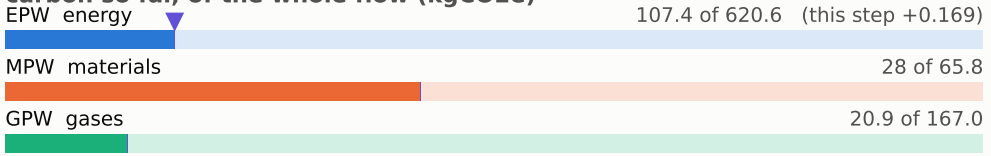
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

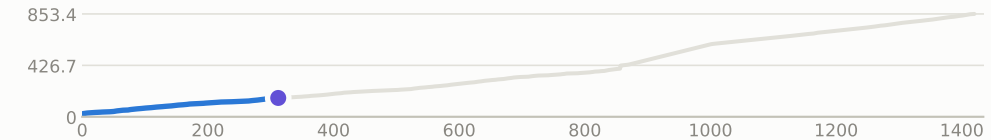
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

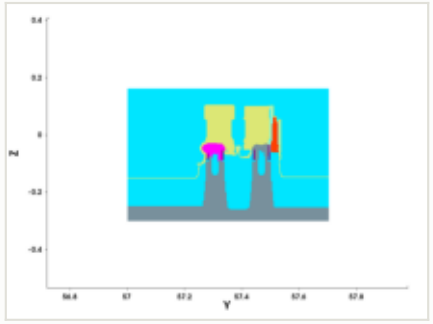
carbon so far, of the whole flow (kgCO2e)



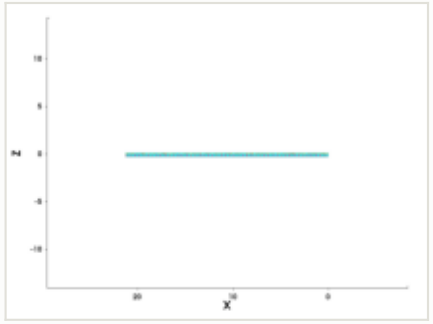
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



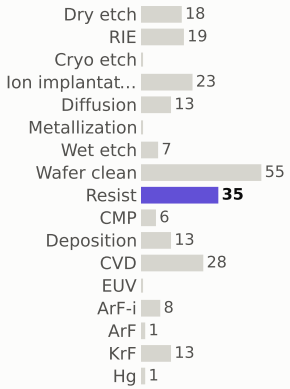
y cut at 57.351 µm



mask for this step

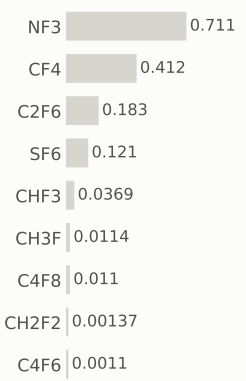
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



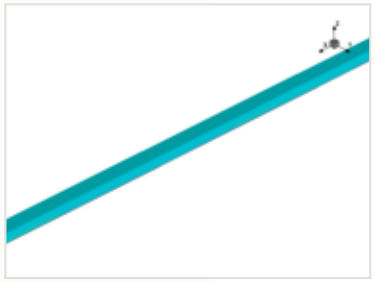
Step #3180: ArF-i

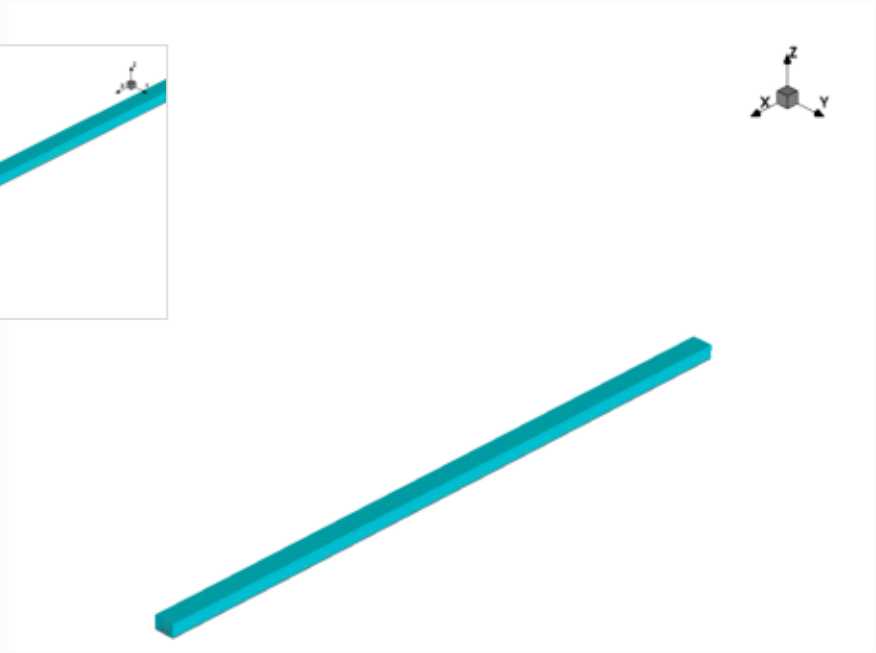
litho_patterning · ArF-i

0.217	156.5	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

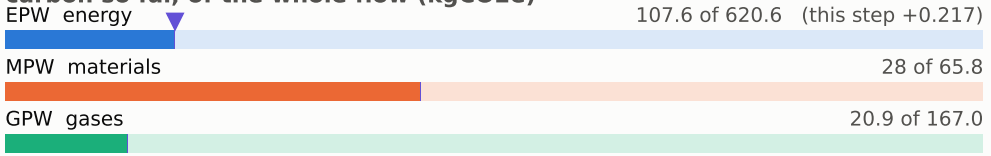




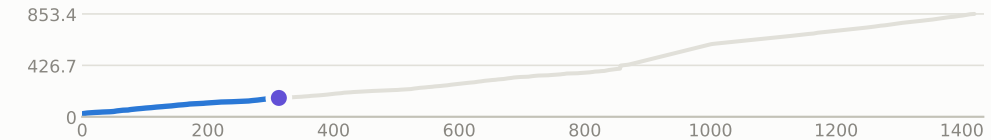
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

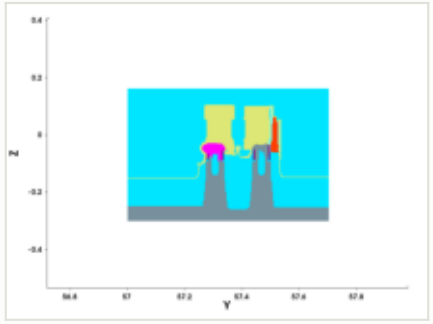
carbon so far, of the whole flow (kgCO2e)



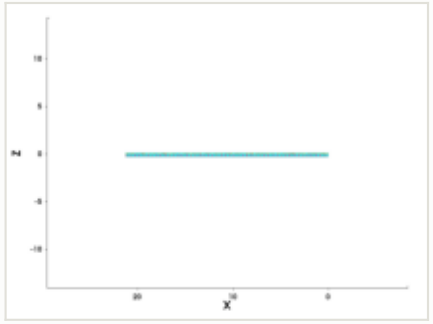
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

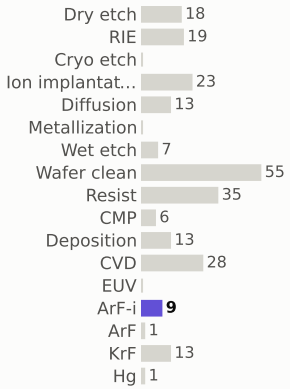


y cut at 57.351 µm



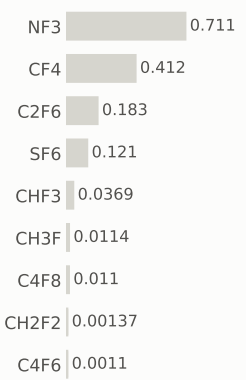
mask: layer 2088:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



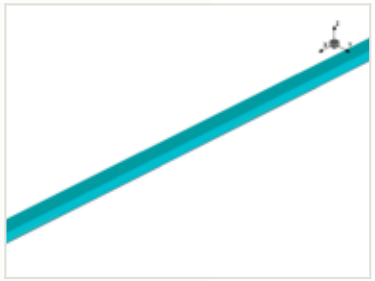
Step #3190: RIE

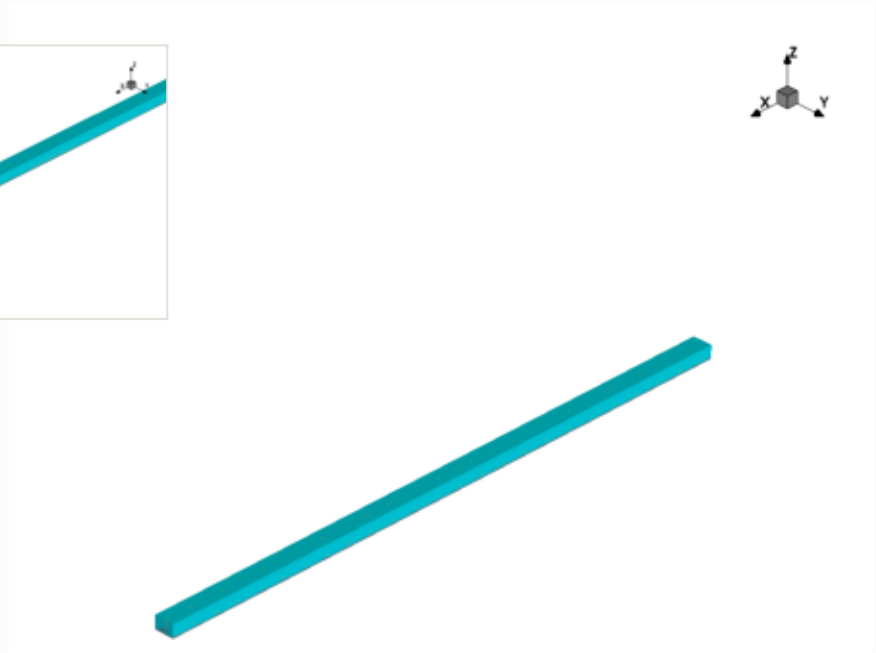
etch_anisotropic · RIE

1.04	157.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



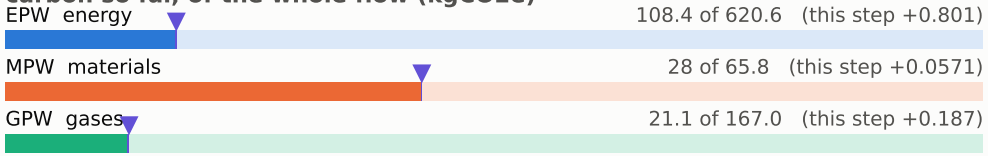


this step

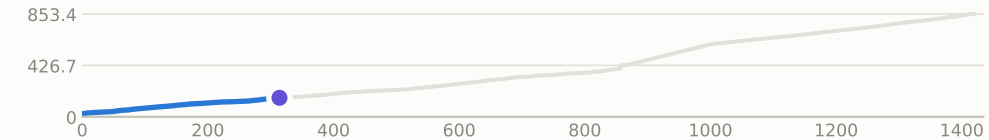
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

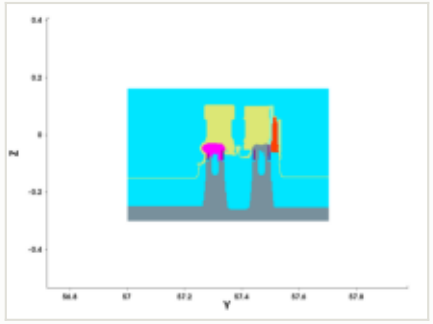
carbon so far, of the whole flow (kgCO2e)



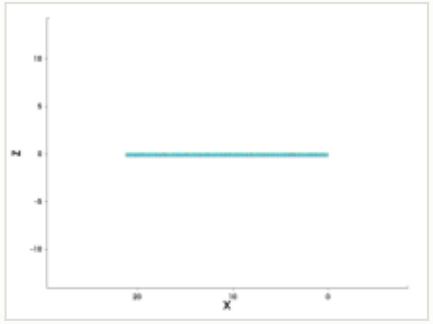
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



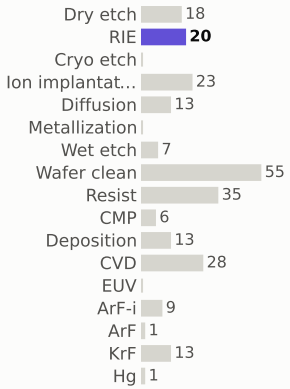
y cut at 57.351 µm



mask for this step

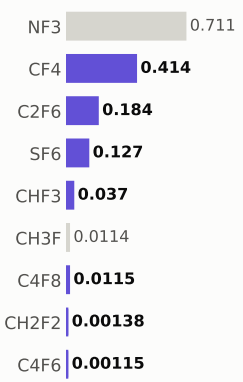
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



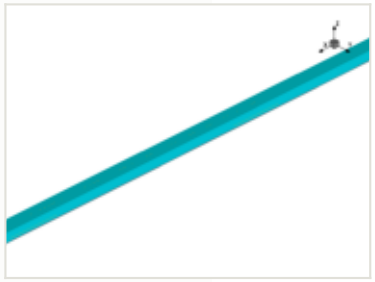
Step #3200: Spx_only

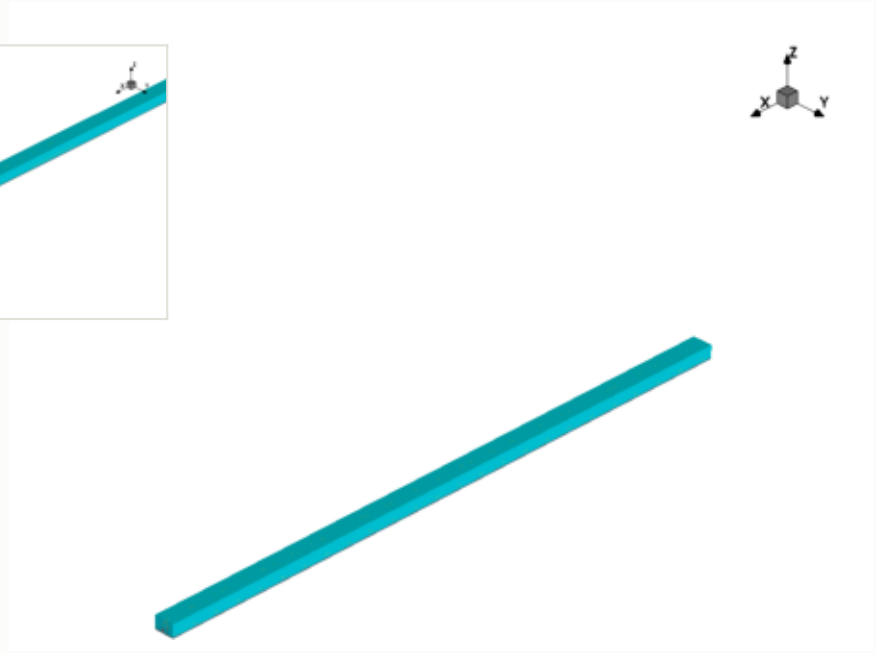
etch_anisotropic · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





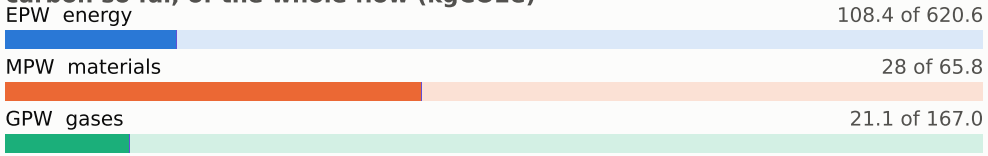
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

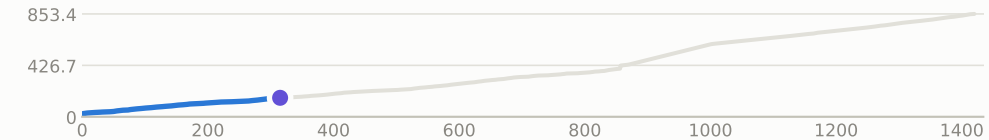
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

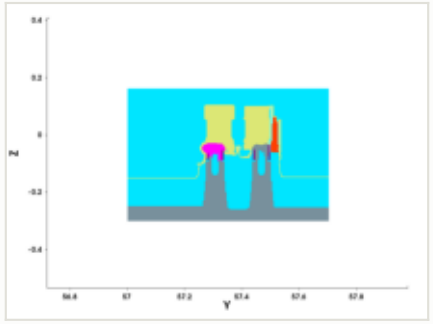
carbon so far, of the whole flow (kgCO2e)



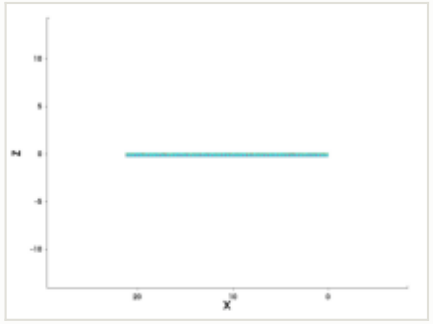
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



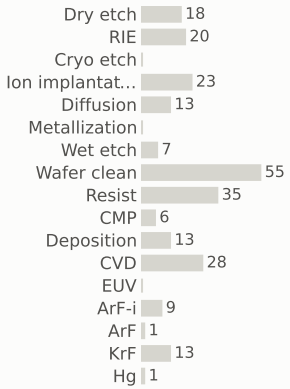
y cut at 57.351 µm



mask for this step

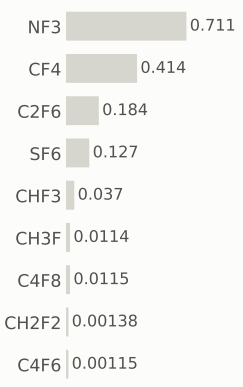
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



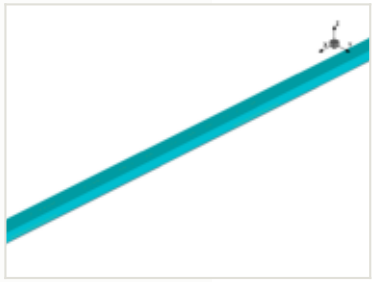
Step #3210: Spx_only

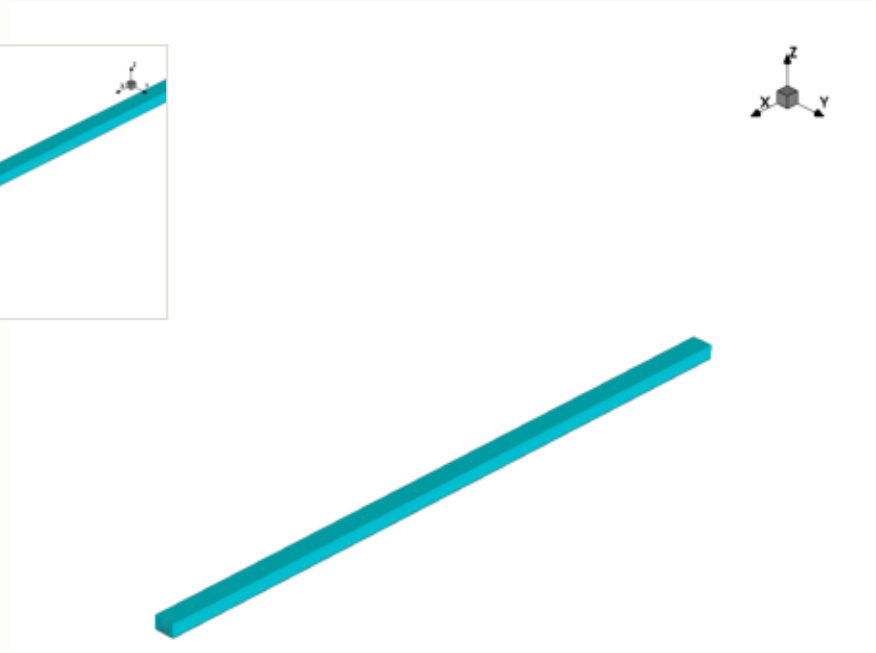
etch_anisotropic · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



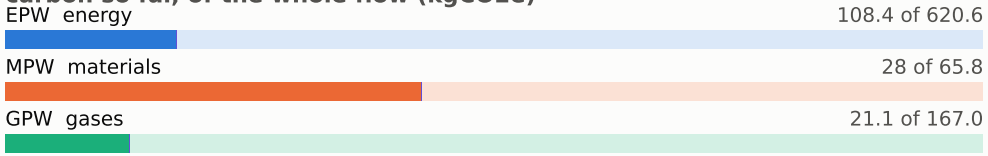


this step

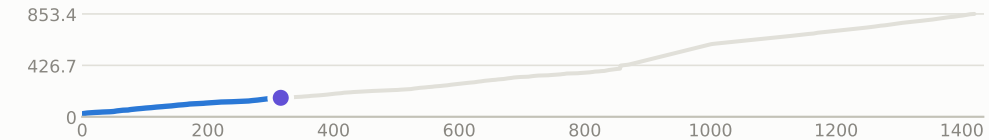
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

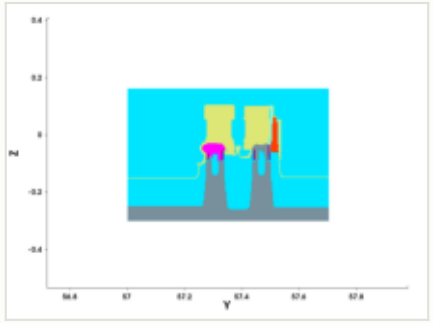
carbon so far, of the whole flow (kgCO2e)



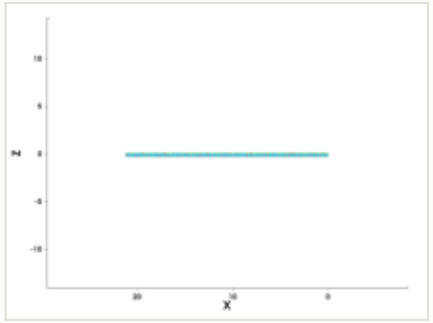
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



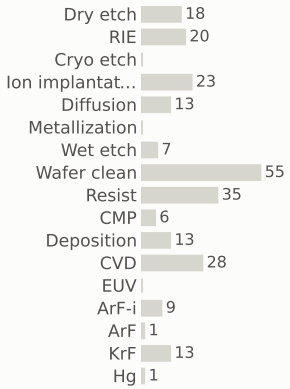
y cut at 57.351 µm



mask for this step

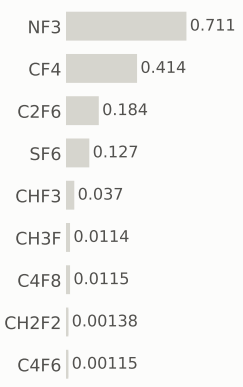
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



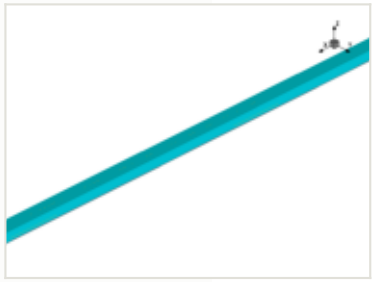
Step #3220: Spx_only

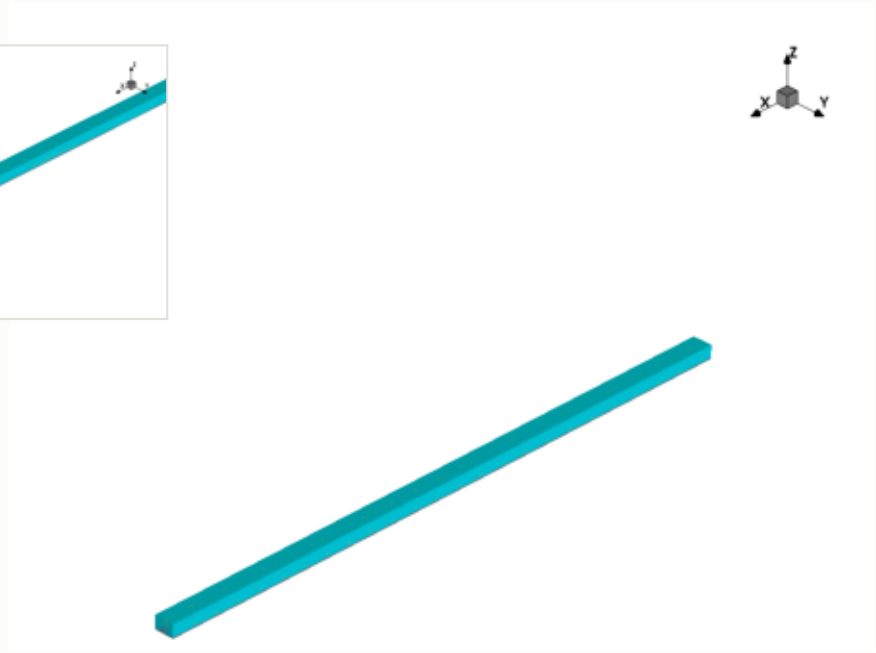
etch_strip · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



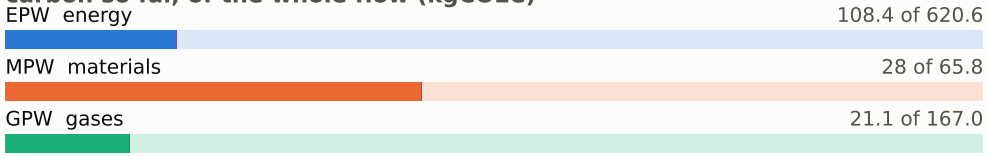


this step

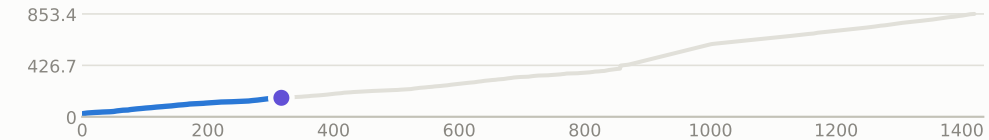
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

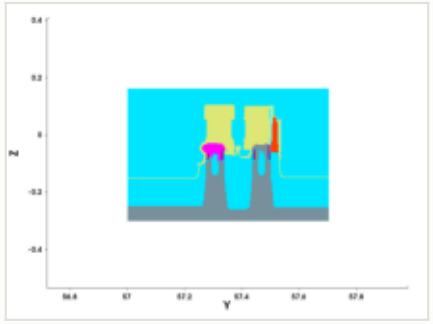
carbon so far, of the whole flow (kgCO2e)



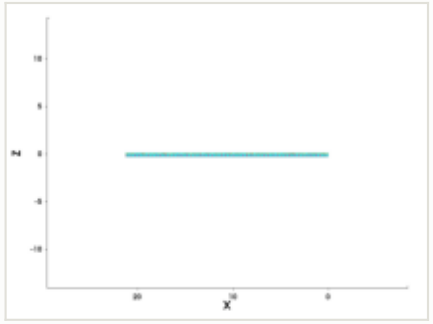
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



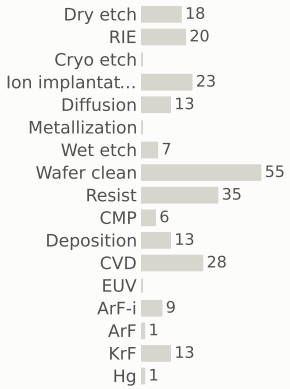
y cut at 57.351 µm



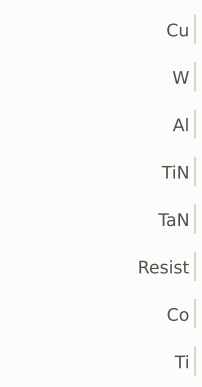
mask for this step

no mask for this step

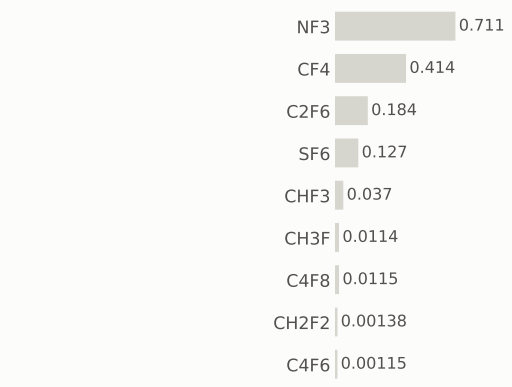
steps by process area



materials deposited (g)



gases emitted (g)



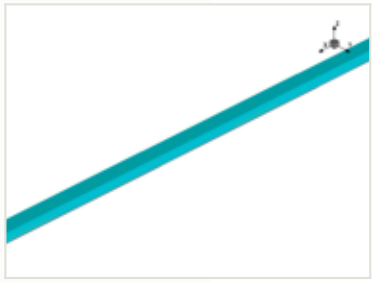
Step #3230: Spx_only

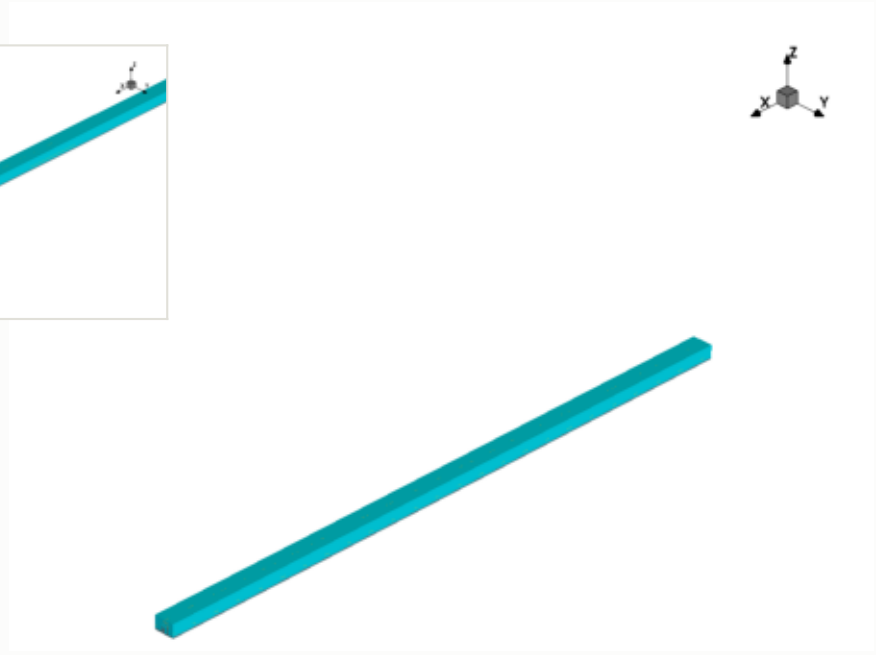
etch_anisotropic · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



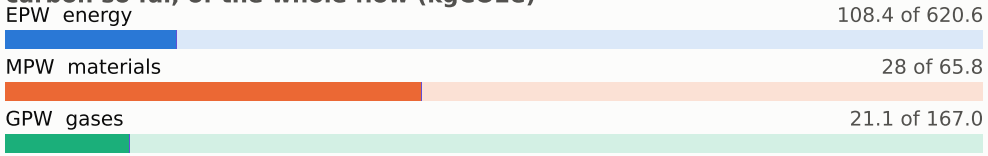


this step

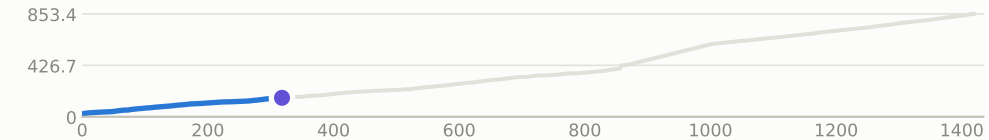
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

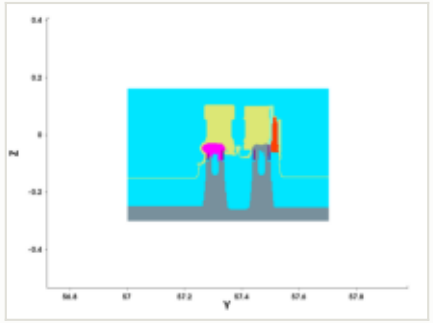
carbon so far, of the whole flow (kgCO2e)



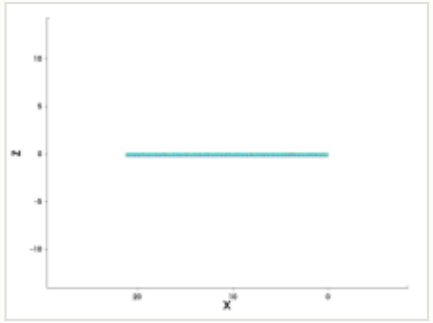
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



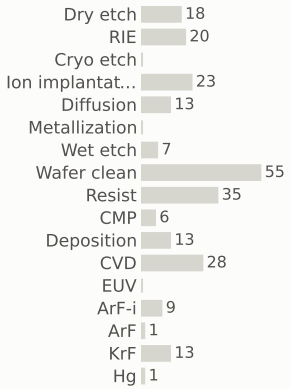
y cut at 57.351 µm



mask for this step

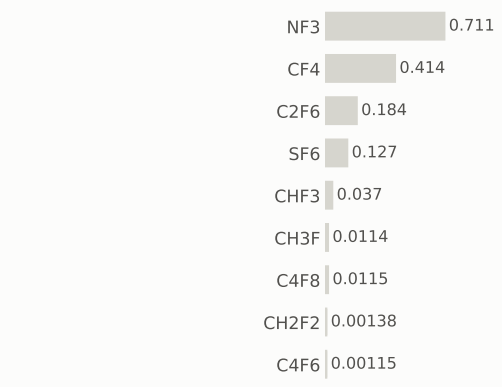
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



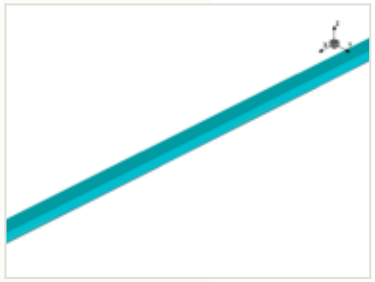
Step #3240: Spx_only

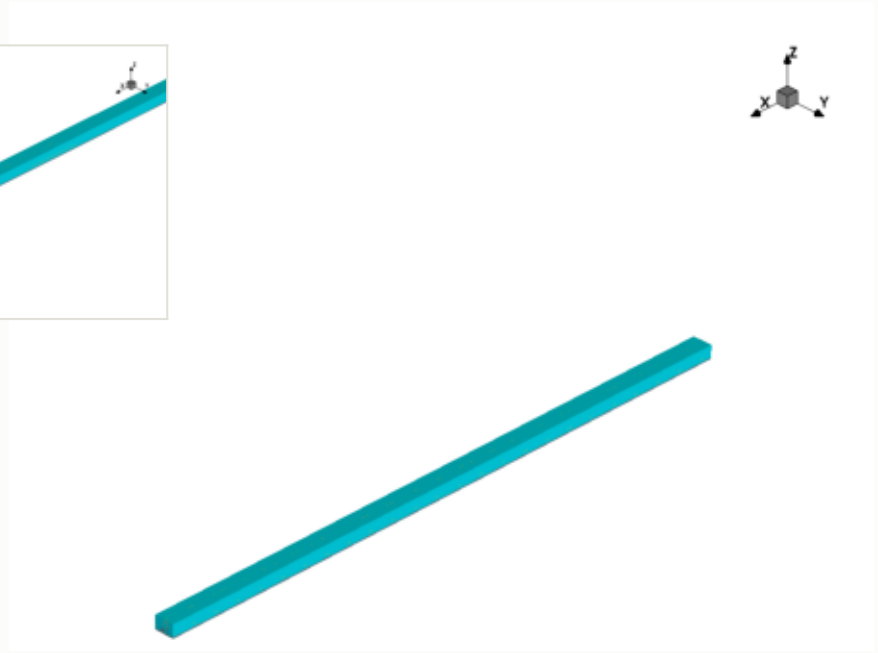
etch_anisotropic · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



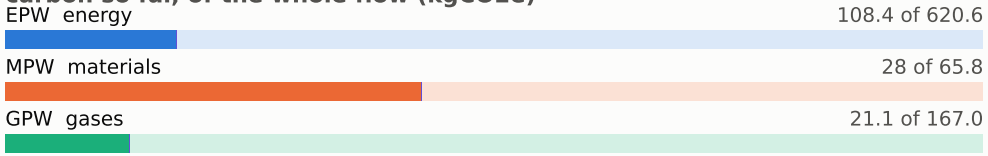


this step

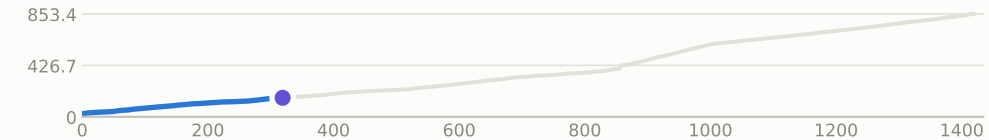
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

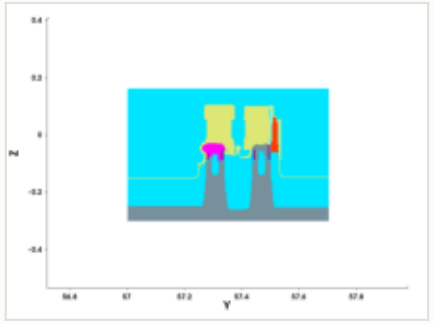
carbon so far, of the whole flow (kgCO2e)



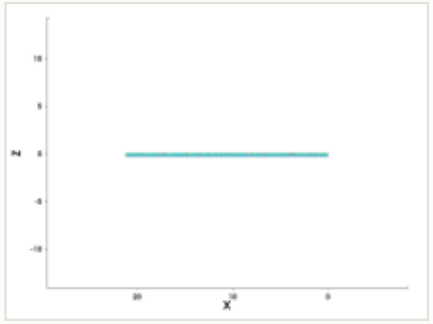
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



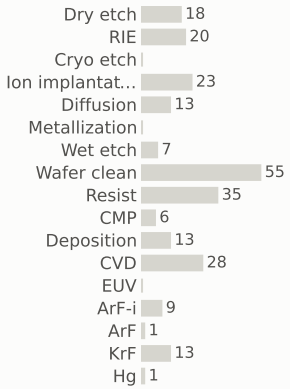
y cut at 57.351 µm



mask for this step

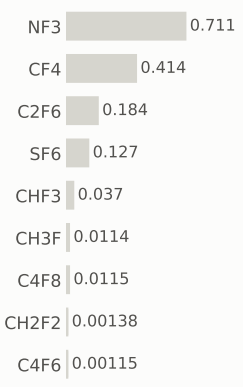
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



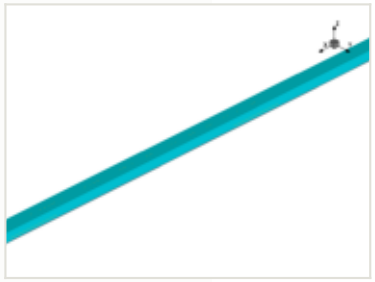
Step #3250: Spx_only

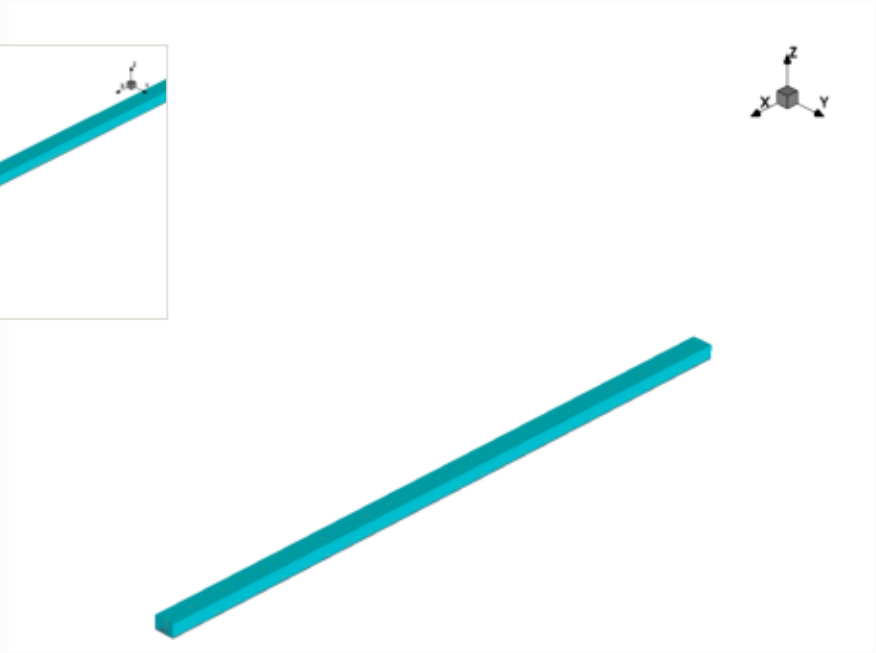
etch_anisotropic · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



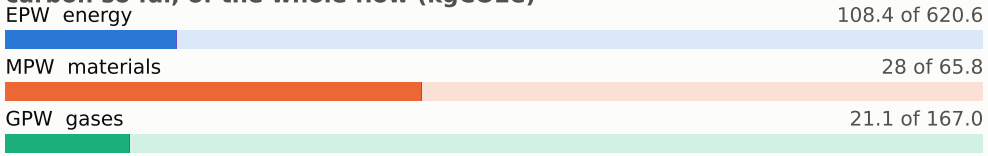


this step

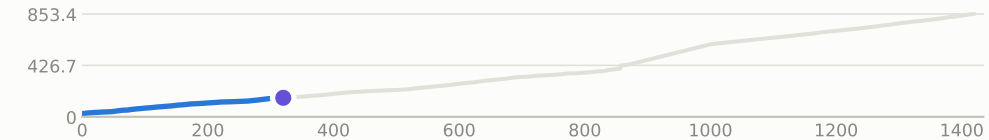
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

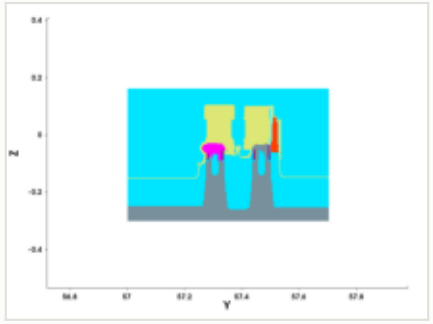
carbon so far, of the whole flow (kgCO2e)



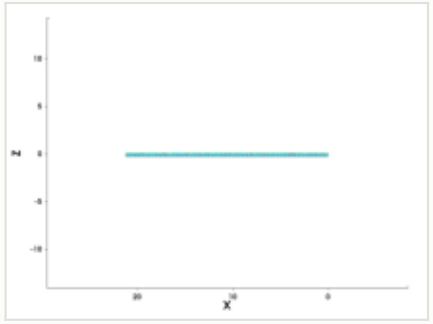
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



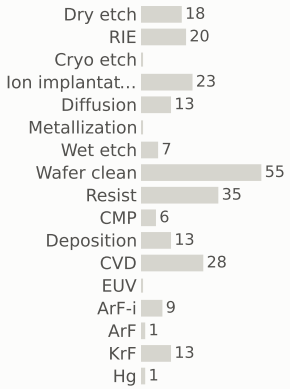
y cut at 57.351 µm



mask for this step

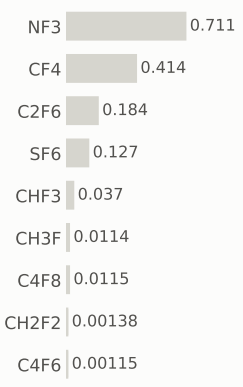
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



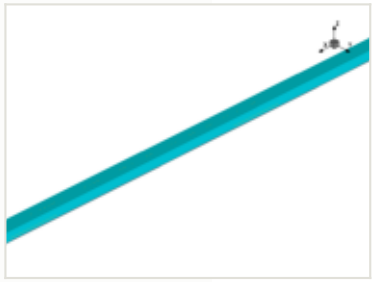
Step #3260: Spx_only

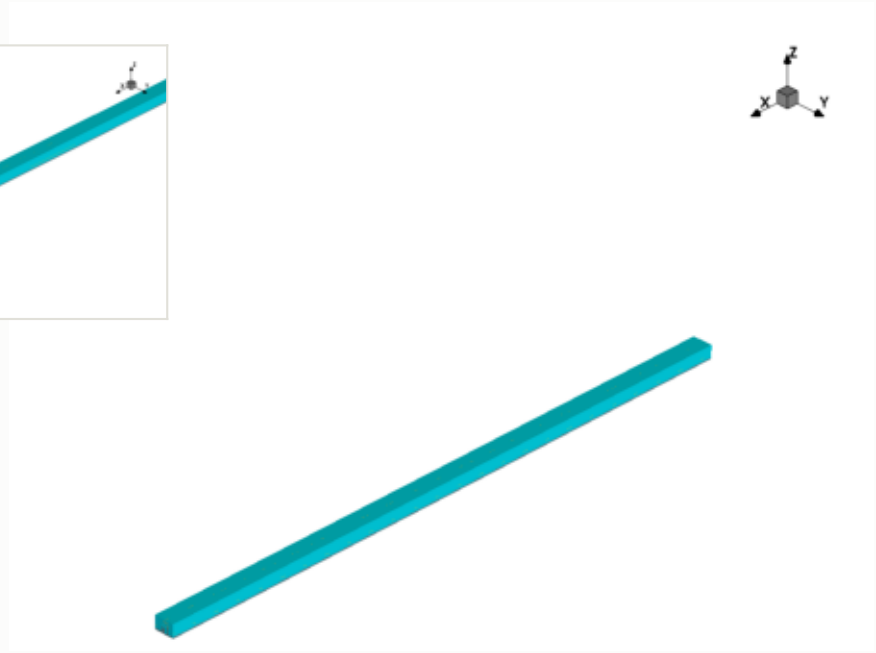
etch_microloading · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





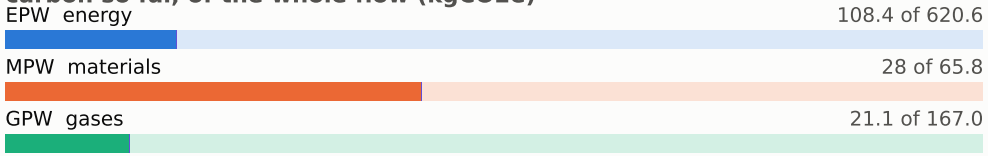
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

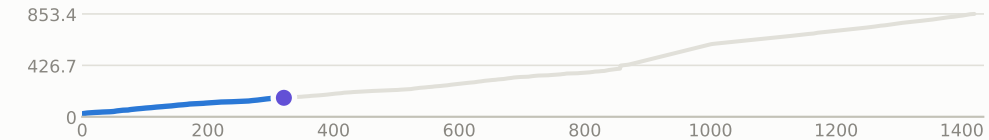
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

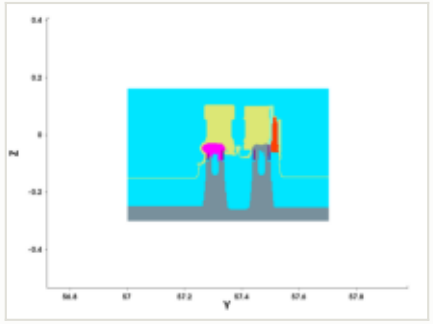
carbon so far, of the whole flow (kgCO2e)



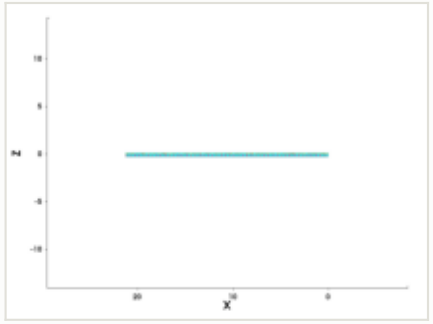
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



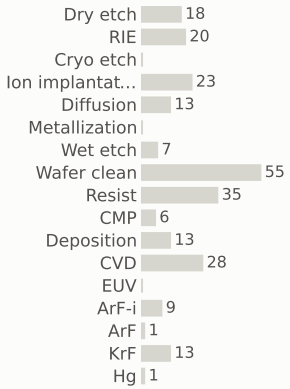
y cut at 57.351 µm



mask for this step

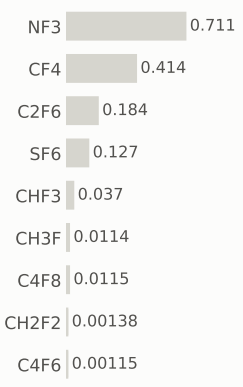
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



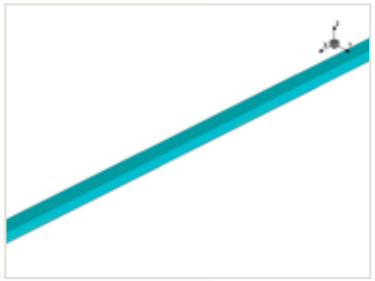
Step #3270: Spx_only

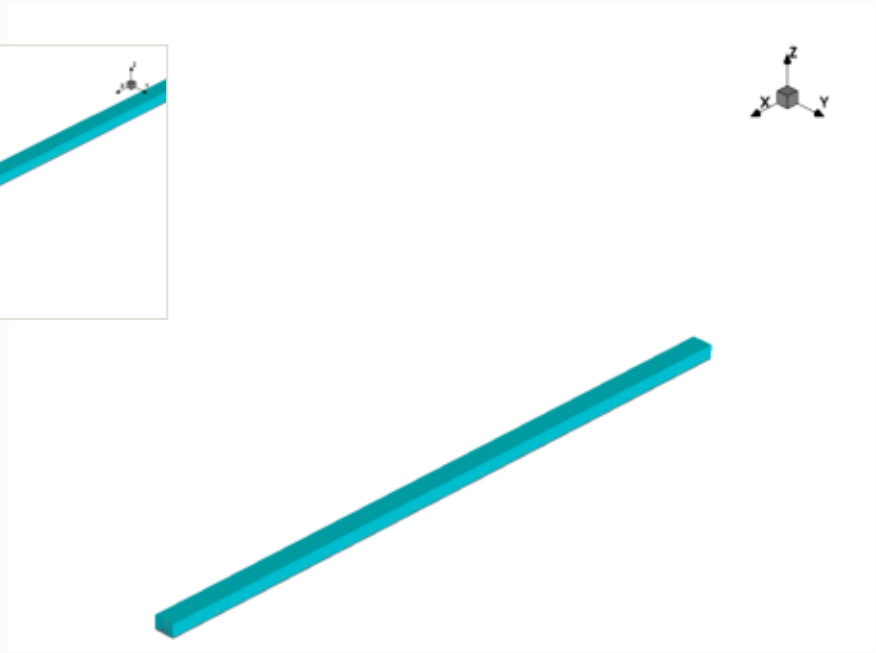
etch_ellipsoidal · Spx_only

0	157.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

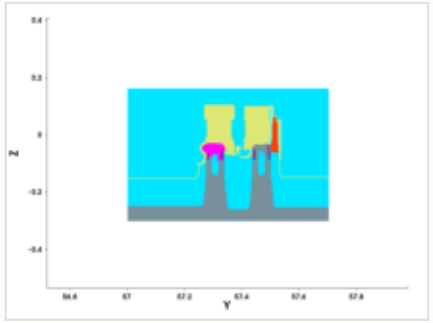
3D structure after this step

zoomed to fit

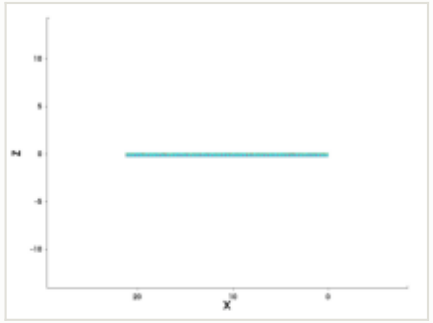




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

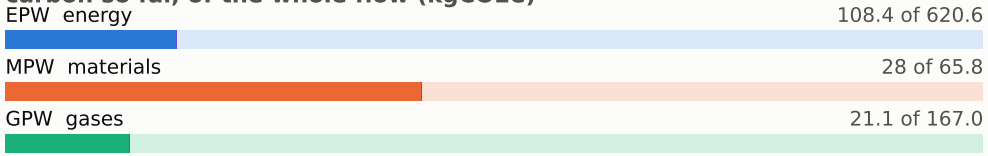
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

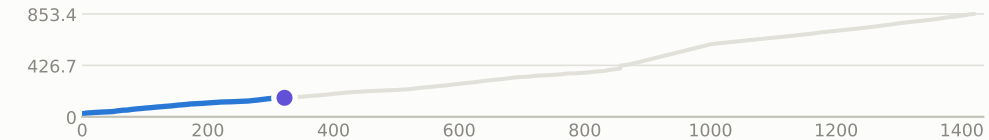
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

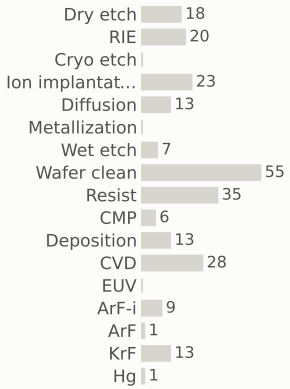
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

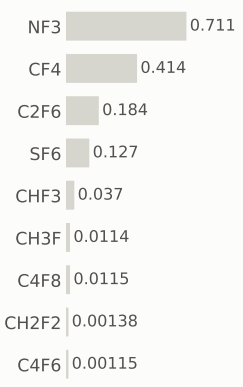


steps by process area



materials deposited (g)

gases emitted (g)



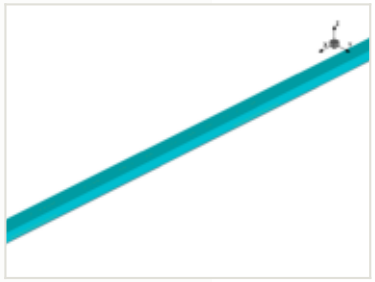
Step #3280: Wafer_clean

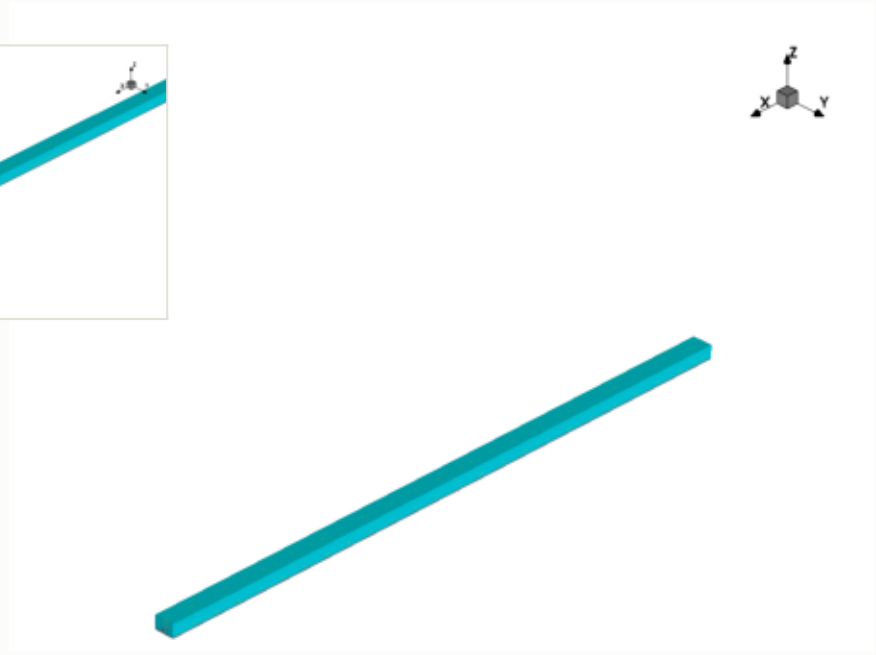
clean · Wafer_clean

0.169	157.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





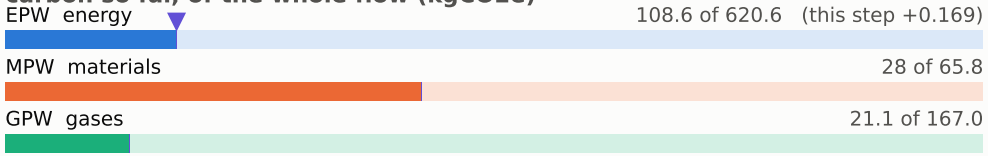
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

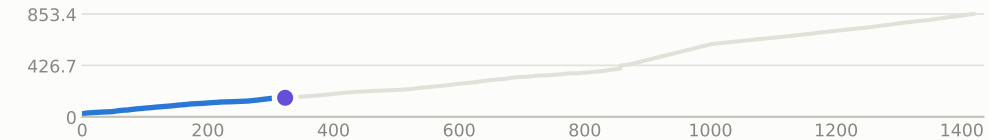
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

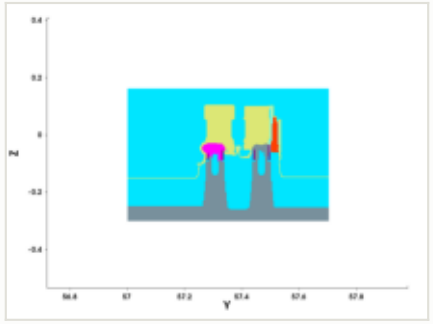
carbon so far, of the whole flow (kgCO2e)



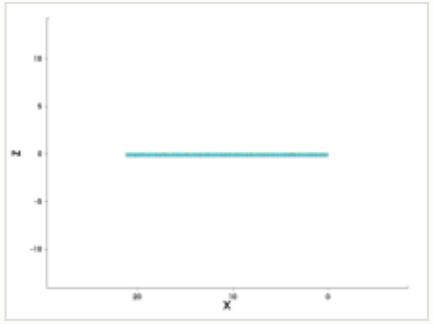
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



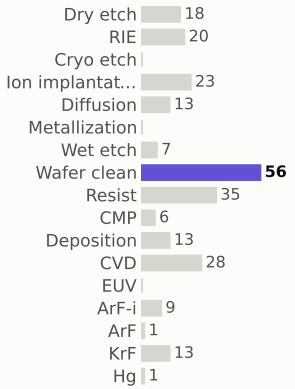
y cut at 57.351 µm



mask for this step

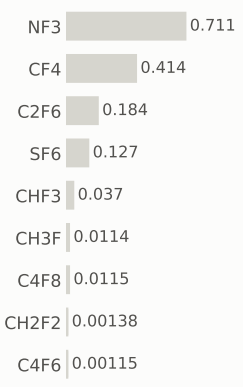
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



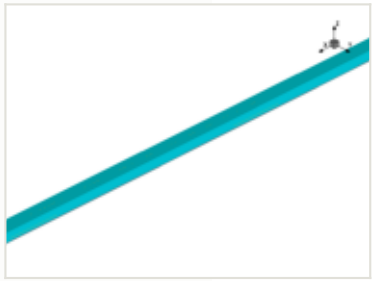
Step #3290: RIE

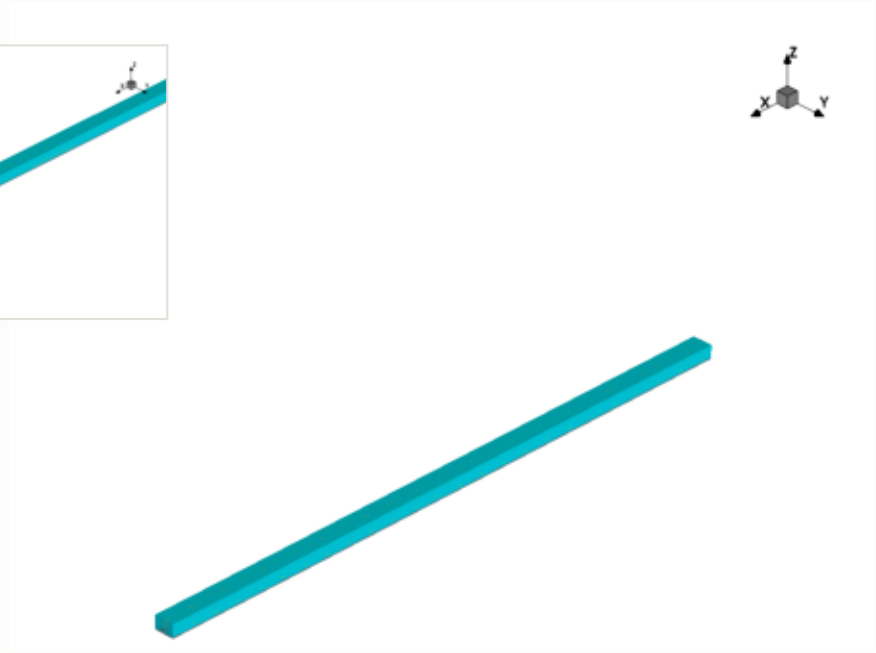
etch_ellipsoidal · RIE

1.04	158.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



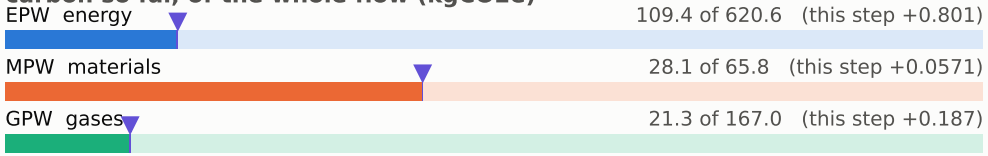


this step

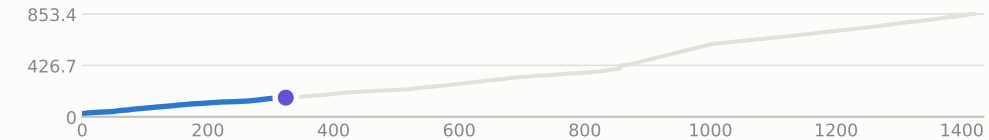
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

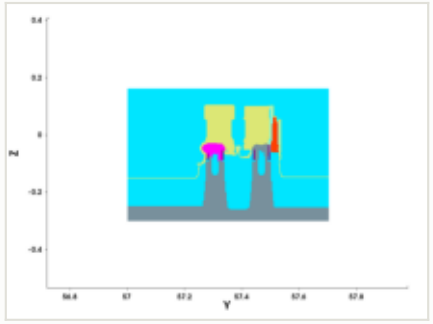
carbon so far, of the whole flow (kgCO2e)



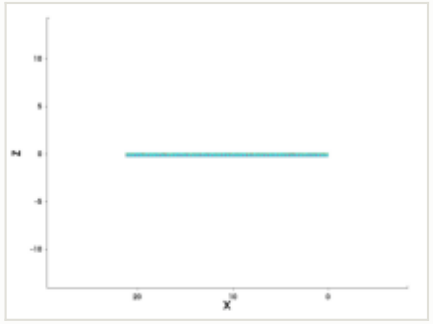
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



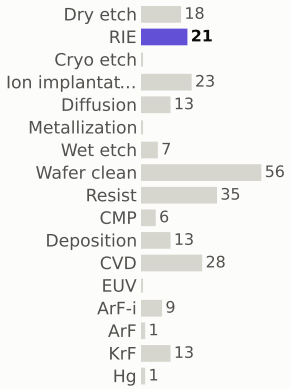
y cut at 57.351 µm



mask for this step

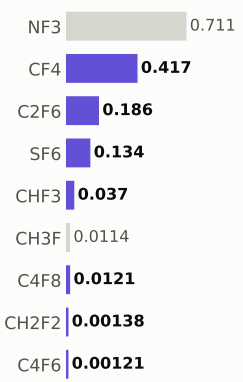
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



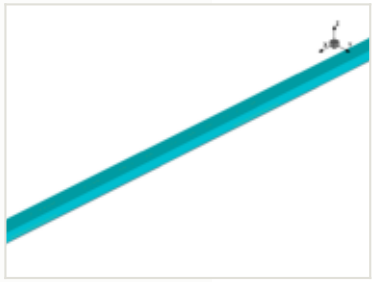
Step #3300: Spx_only

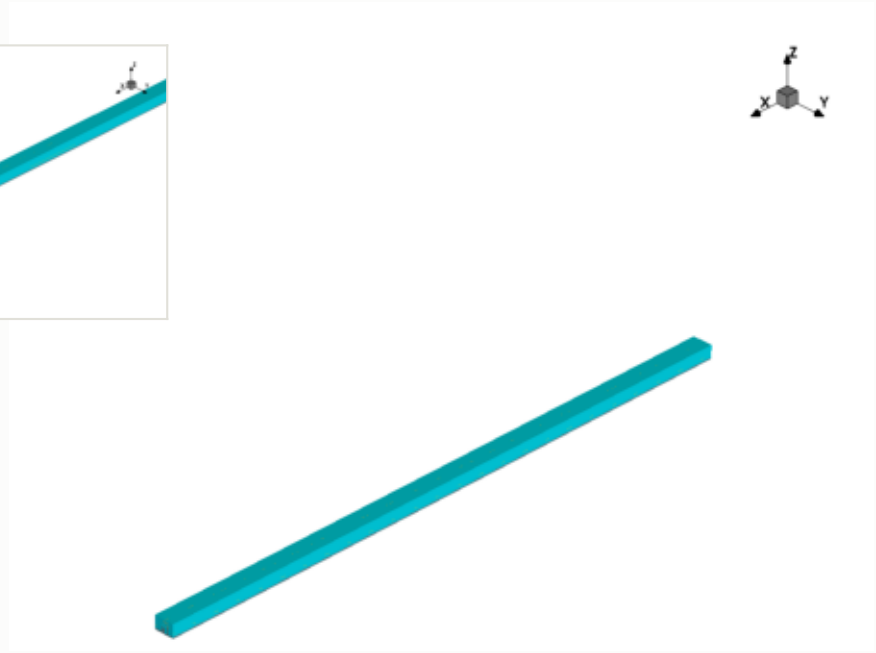
etch_isotropic · Spx_only

0	158.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

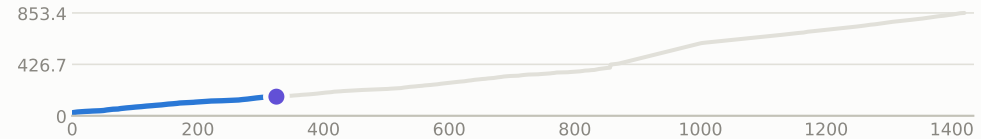
carbon so far, of the whole flow (kgCO2e)

EPW energy 109.4 of 620.6

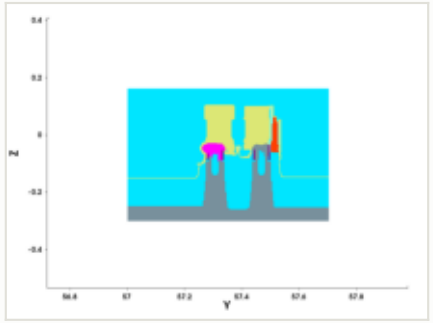
MPW materials 28.1 of 65.8

GPW gases 21.3 of 167.0

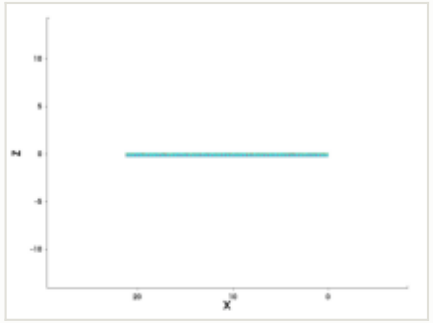
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



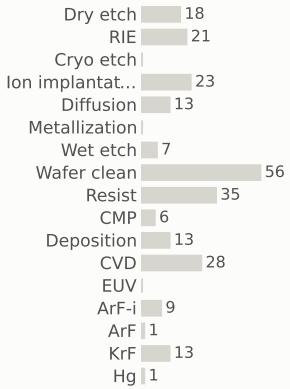
y cut at 57.351 µm



mask for this step

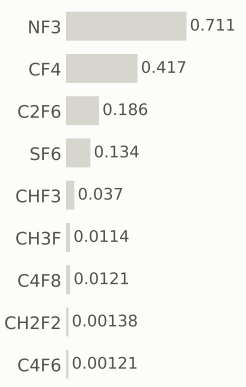
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



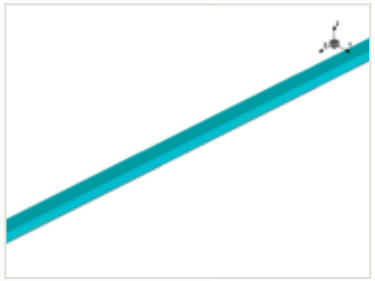
Step #3310: Spx_only

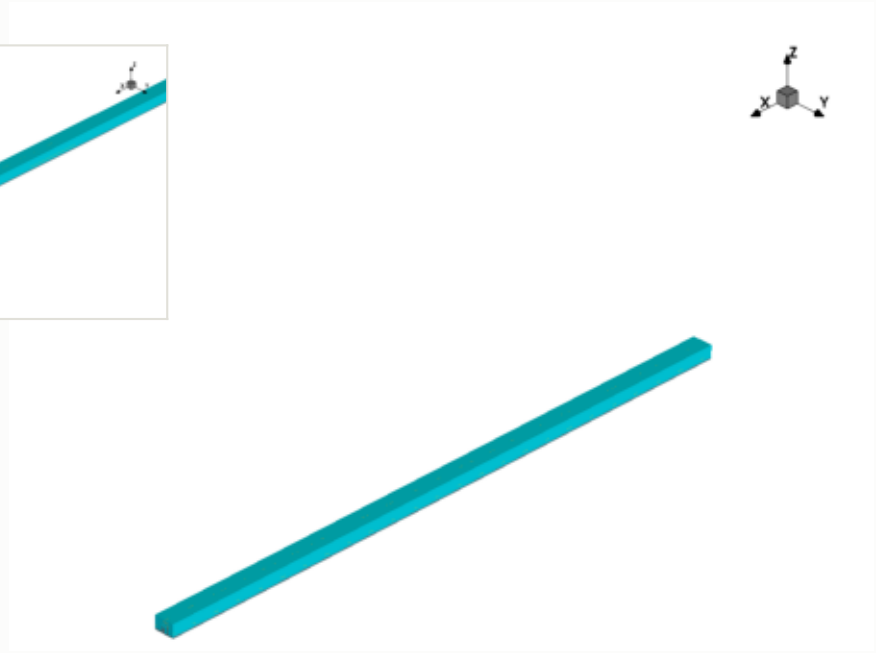
etch_strip · Spx_only

0	158.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

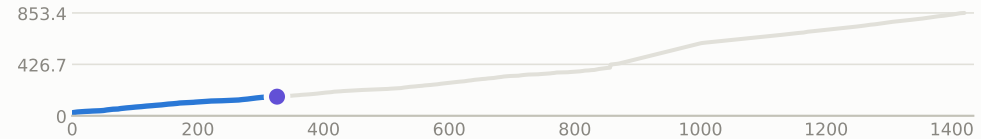
carbon so far, of the whole flow (kgCO2e)

EPW energy 109.4 of 620.6

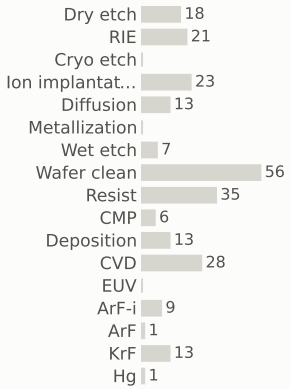
MPW materials 28.1 of 65.8

GPW gases 21.3 of 167.0

embodied carbon up to each step (kgCO2e per wafer)

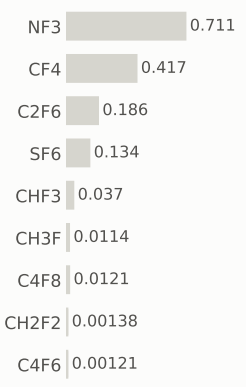


steps by process area

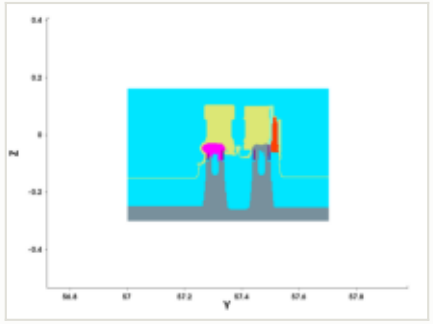


materials deposited (g)

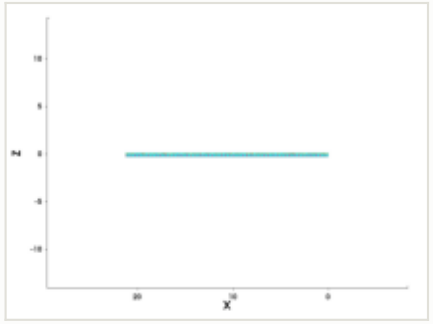
gases emitted (g)



x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

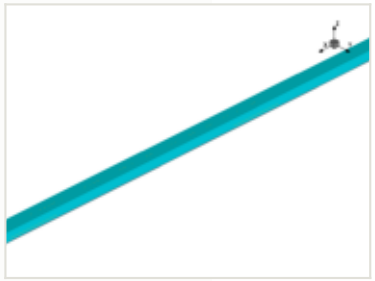
Step #3320: Wafer_clean

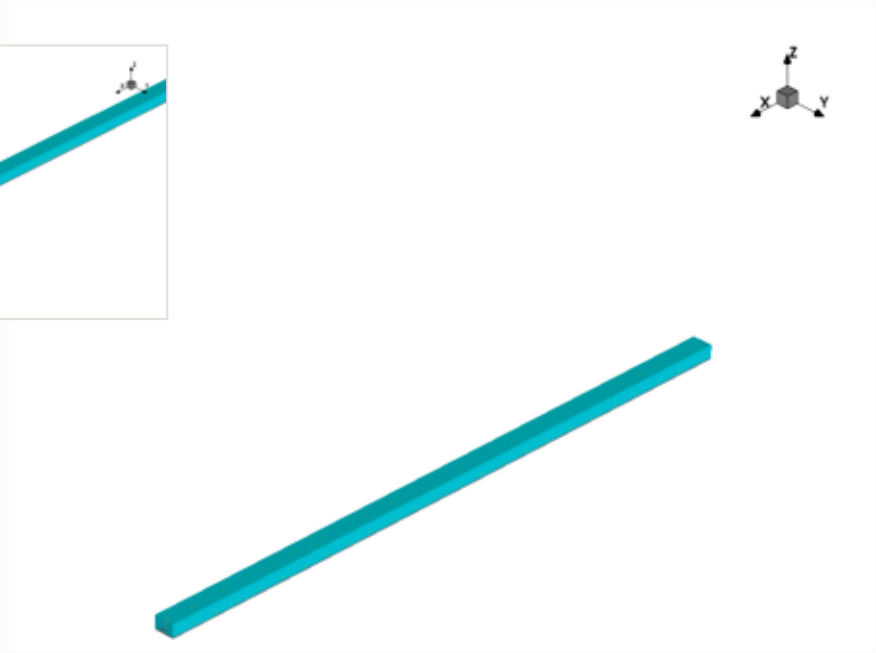
clean · Wafer_clean

0.169	159.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





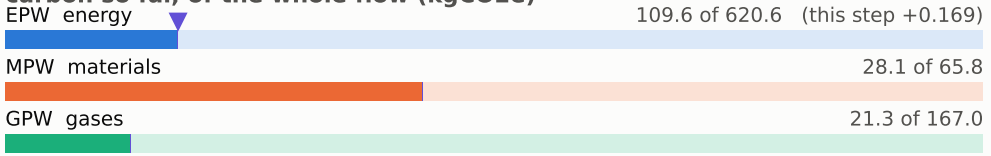
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

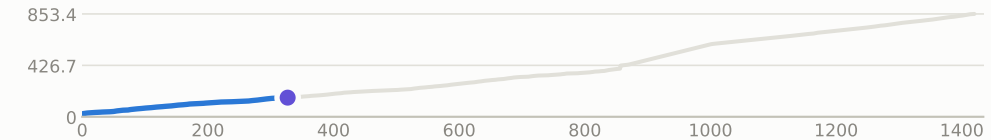
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

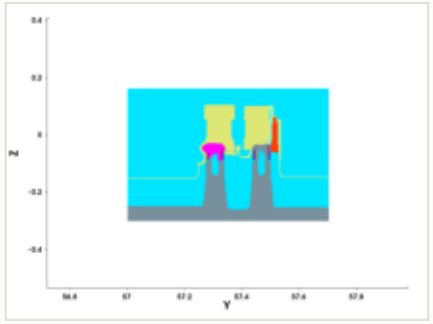
carbon so far, of the whole flow (kgCO2e)



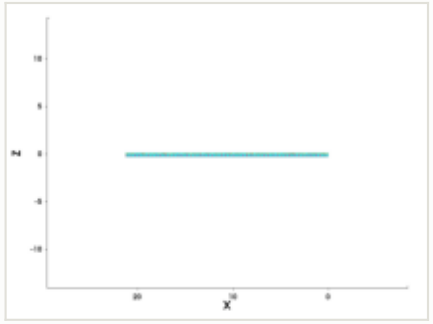
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



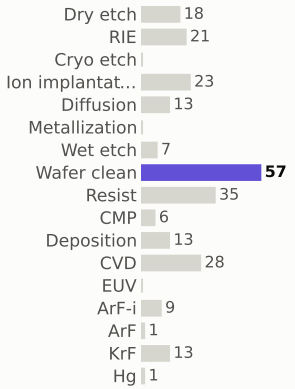
y cut at 57.351 µm



mask for this step

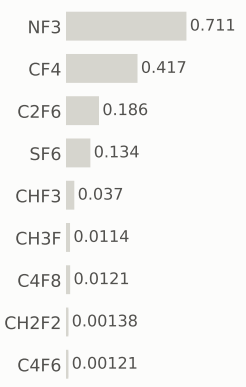
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



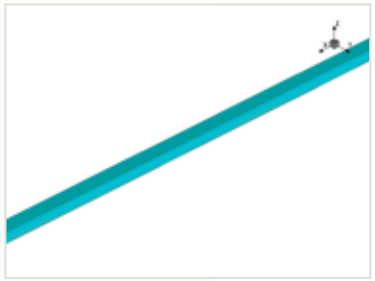
Step #3330: Dry_etch

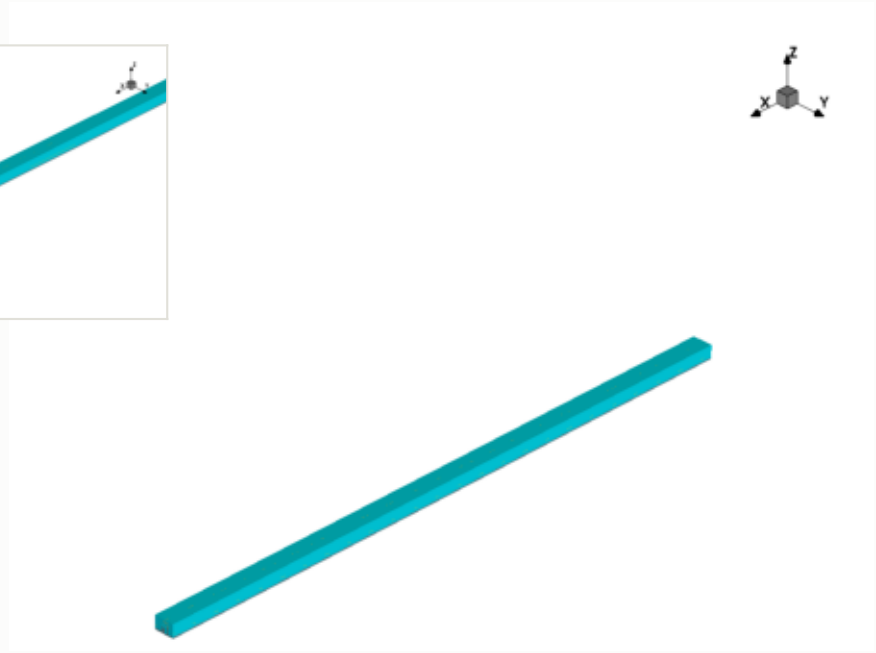
etch_isotropic · Dry_etch

0.801	159.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



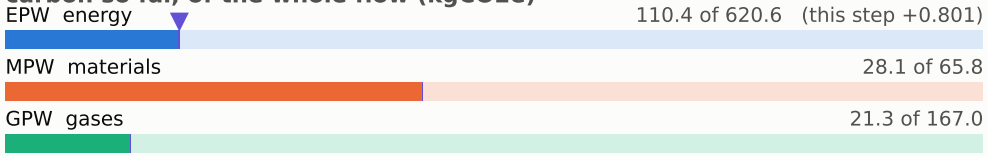


this step

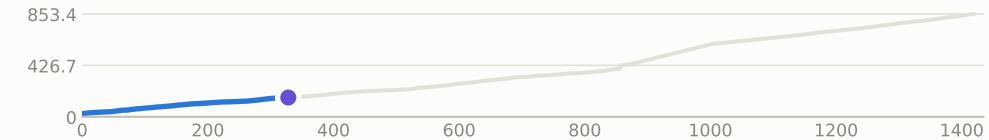
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

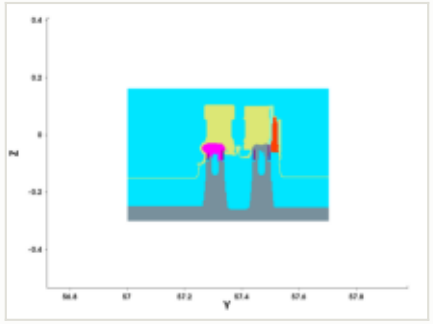
carbon so far, of the whole flow (kgCO2e)



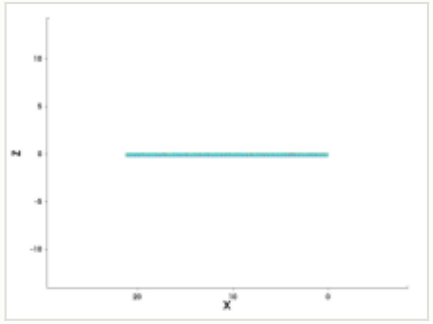
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



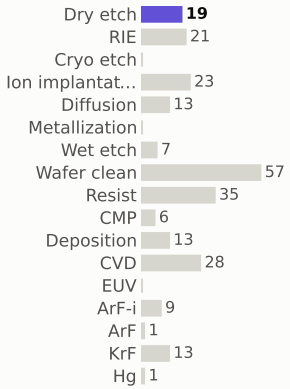
y cut at 57.351 µm



mask for this step

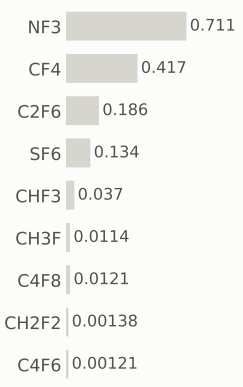
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



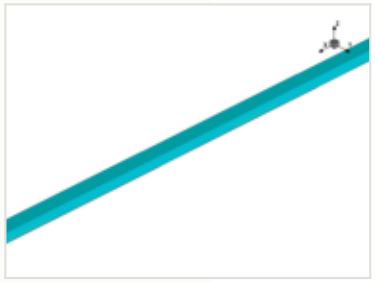
Step #3340: RIE

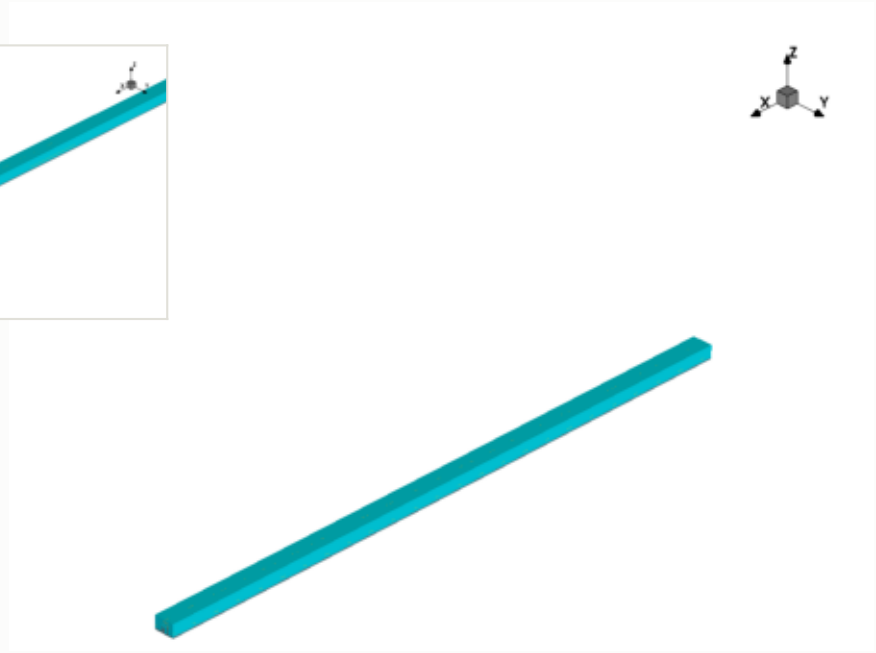
etch_anisotropic · RIE

1.04	160.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



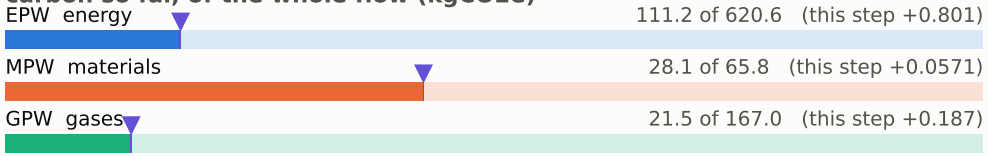


this step

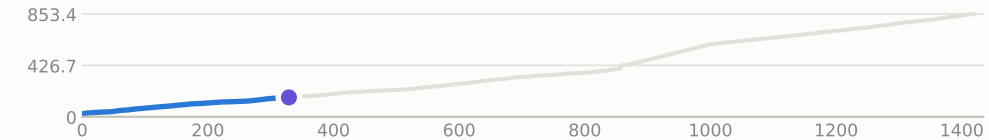
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

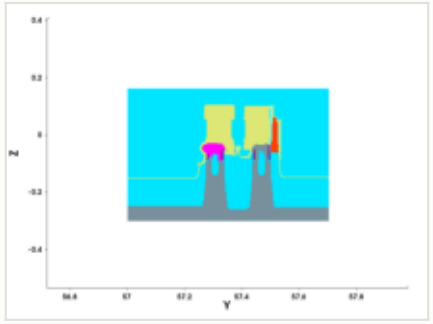
carbon so far, of the whole flow (kgCO2e)



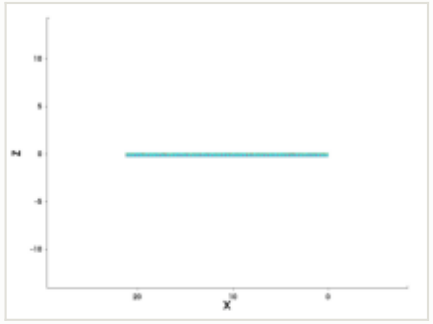
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



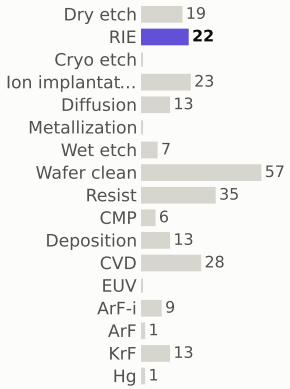
y cut at 57.351 µm



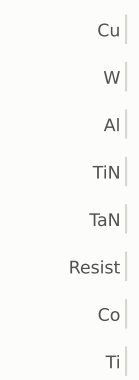
mask for this step

no mask for this step

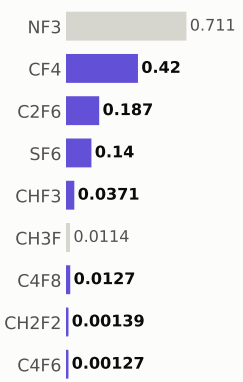
steps by process area



materials deposited (g)



gases emitted (g)



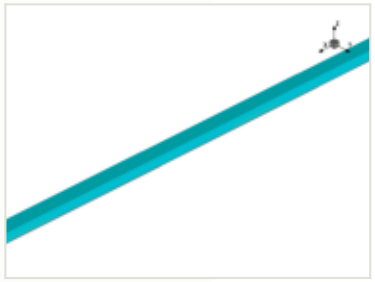
Step #3350: Spx_only

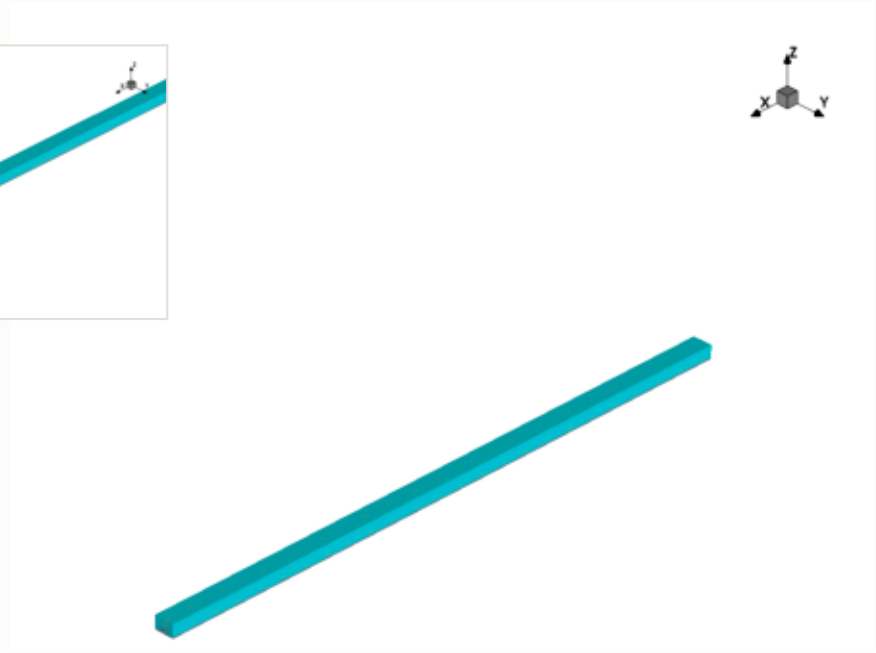
etch_microloading · Spx_only

0	160.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





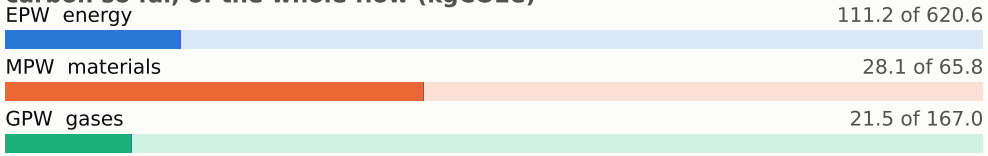
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

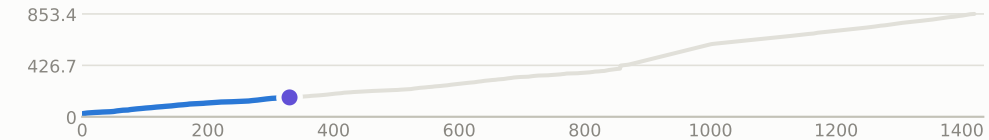
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

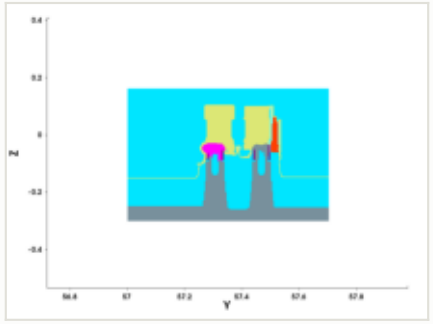
carbon so far, of the whole flow (kgCO2e)



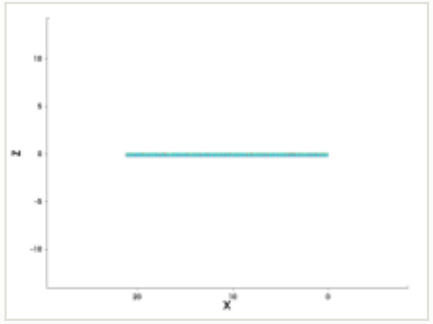
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



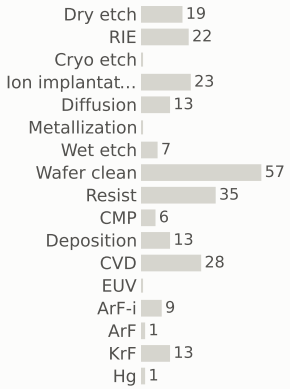
y cut at 57.351 µm



mask for this step

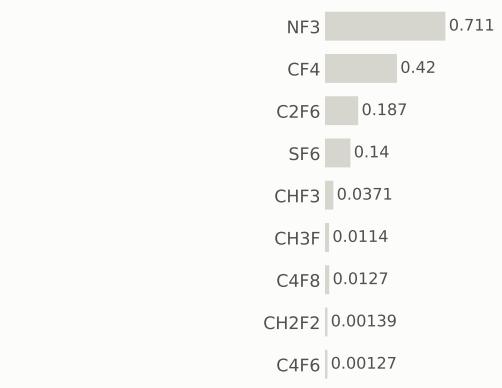
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



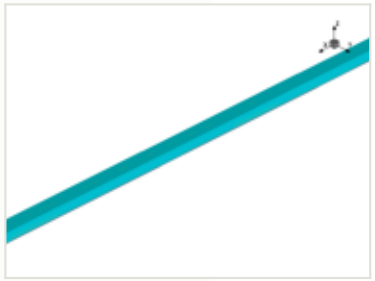
Step #3360: Spx_only

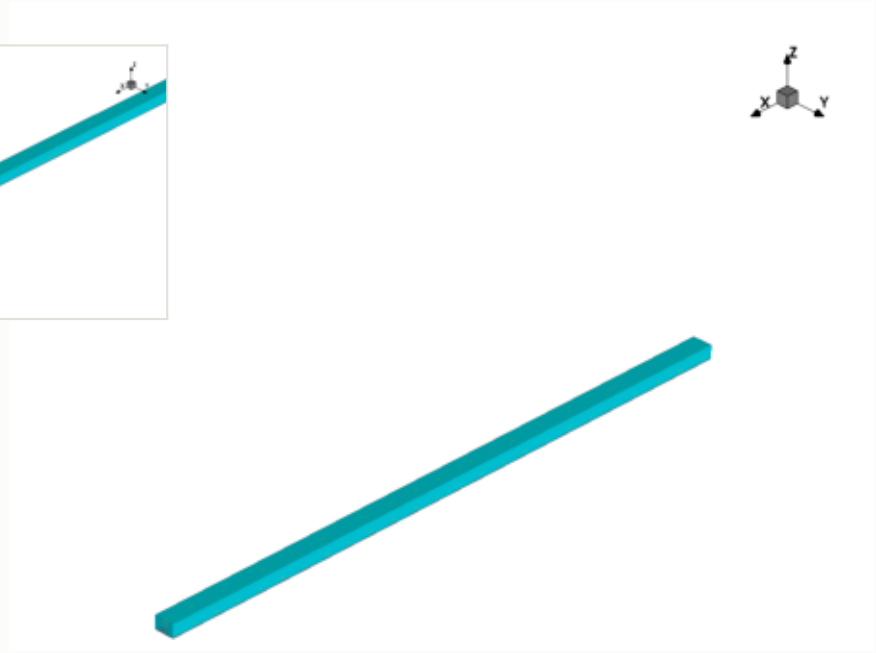
depo_isotropic · Spx_only

0	160.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





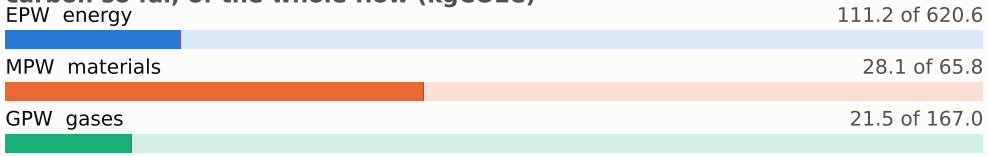
this step

process area	Spx_only
material	—
thickness	0.001 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

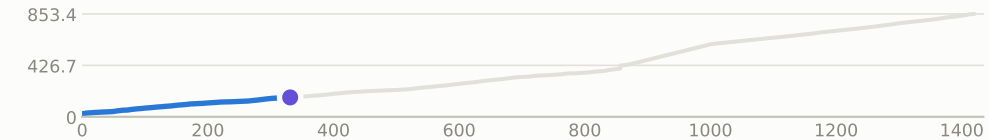
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

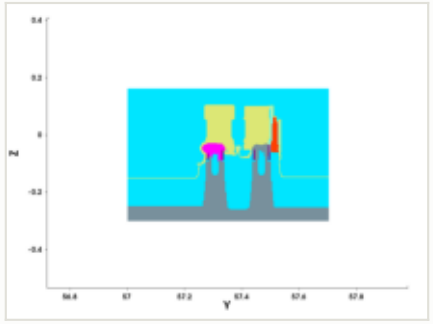
carbon so far, of the whole flow (kgCO2e)



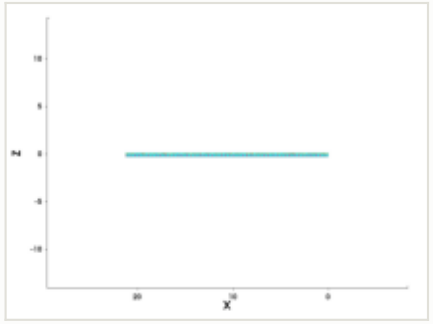
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



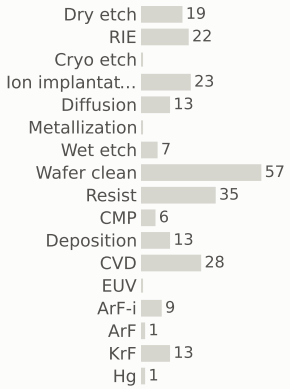
y cut at 57.351 µm



mask for this step

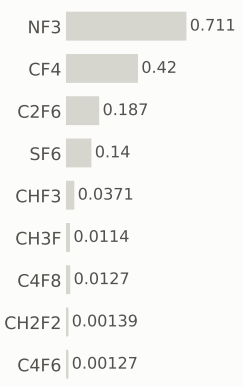
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



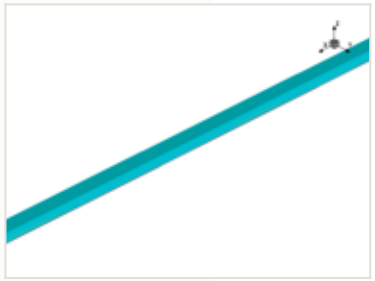
Step #3370: Wafer_clean

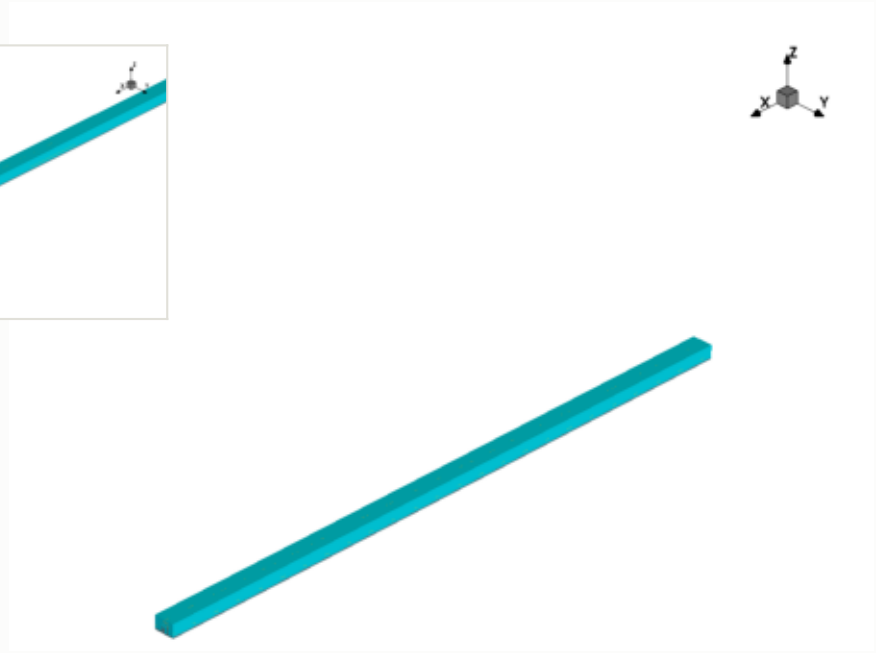
clean · Wafer_clean

0.169	161.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





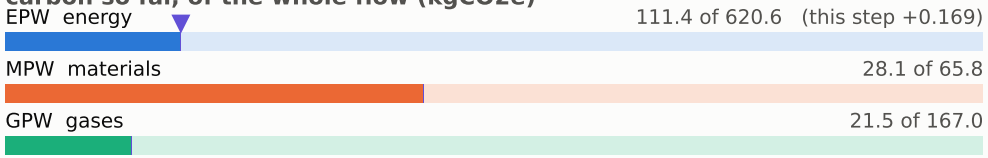
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

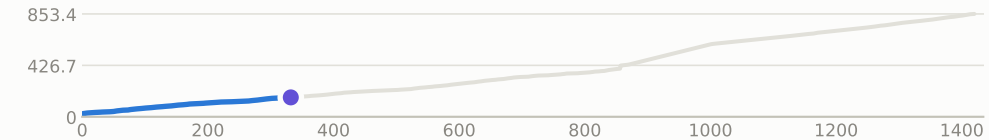
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

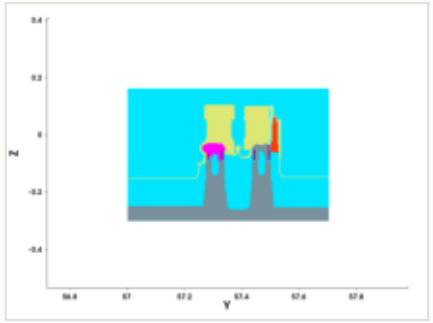
carbon so far, of the whole flow (kgCO2e)



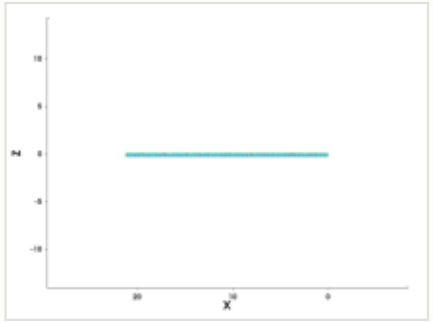
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



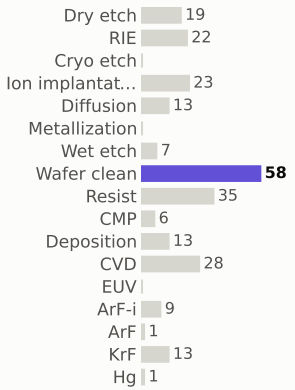
y cut at 57.351 µm



mask for this step

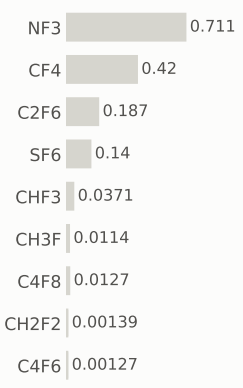
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



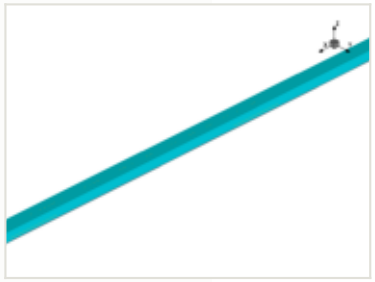
Step #3380: Wafer_clean

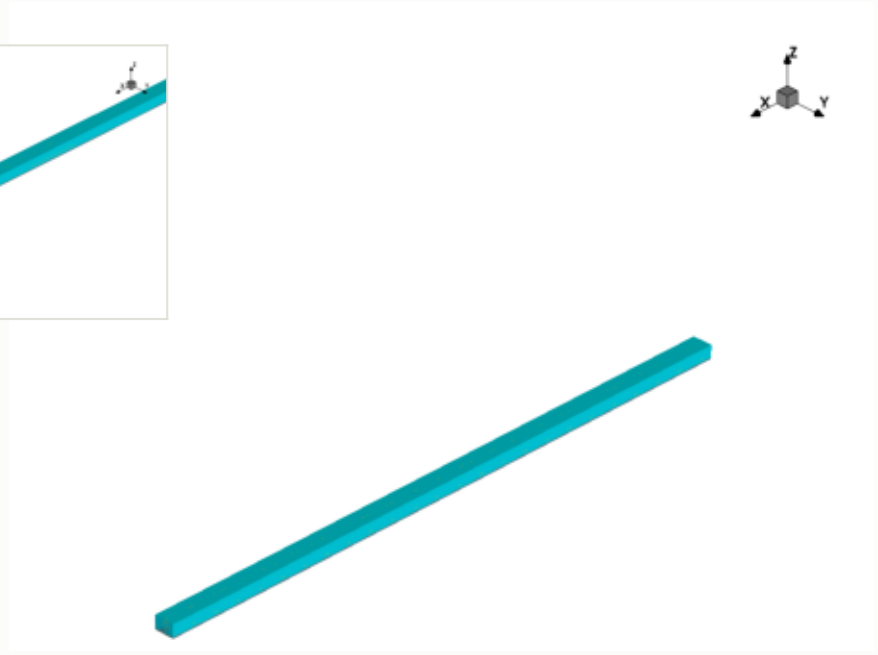
clean · Wafer_clean

0.169	161.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





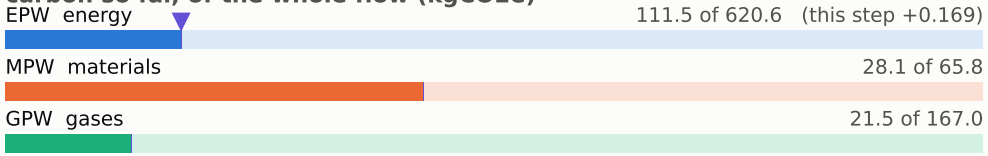
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

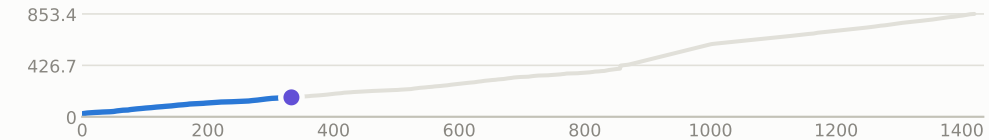
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

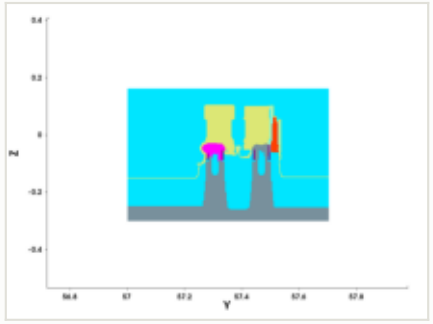
carbon so far, of the whole flow (kgCO2e)



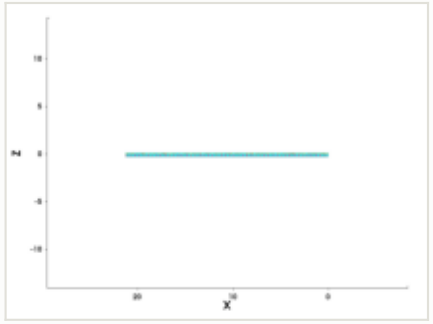
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



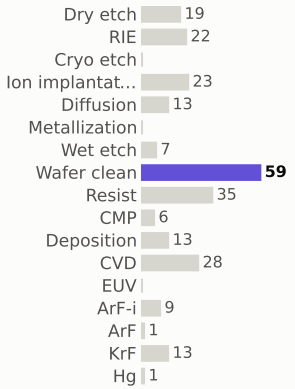
y cut at 57.351 µm



mask for this step

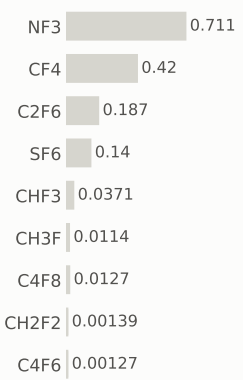
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



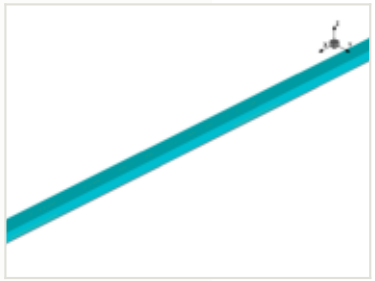
Step #3390: Deposition

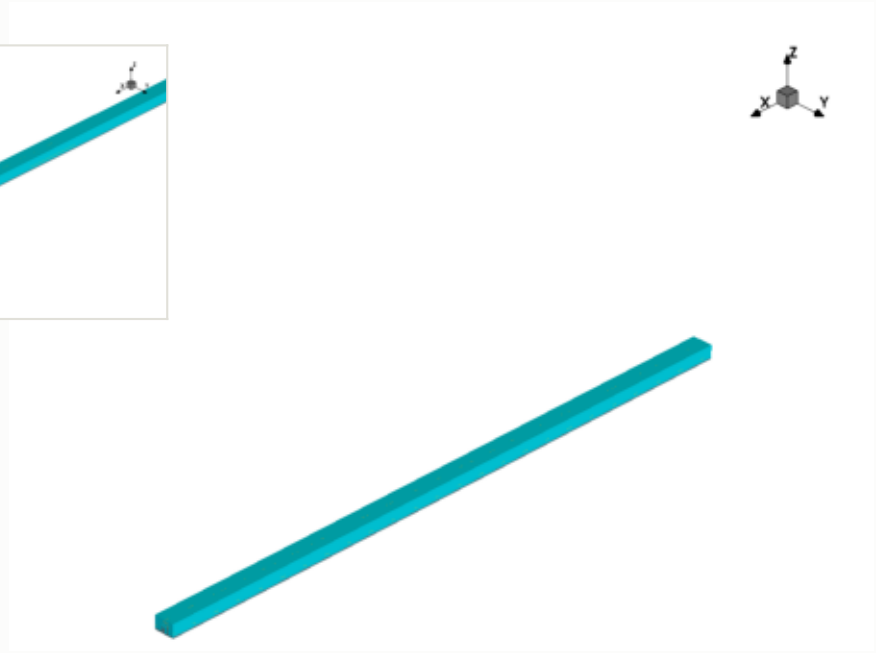
depo_isotropic · Deposition

0.656	161.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



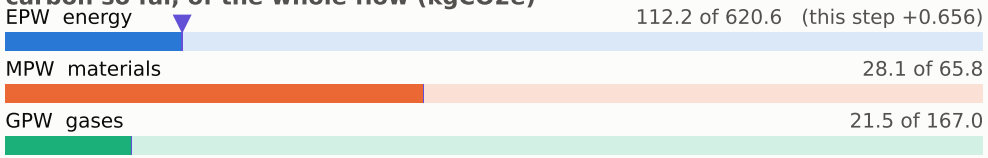


this step

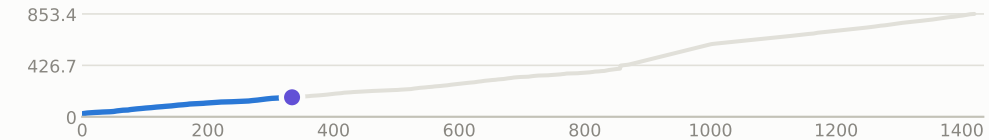
process area	Deposition
material	—
thickness	0.005 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

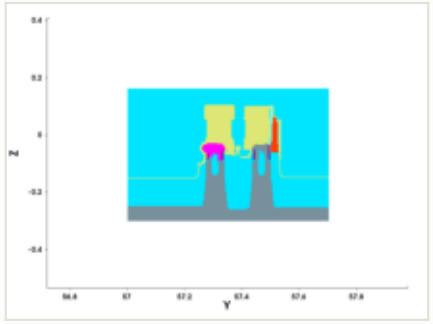
carbon so far, of the whole flow (kgCO2e)



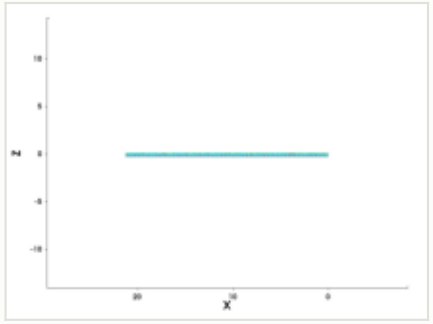
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



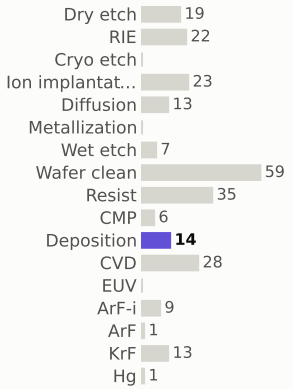
y cut at 57.351 µm



mask for this step

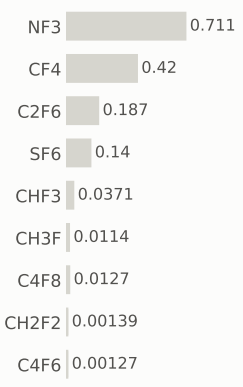
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



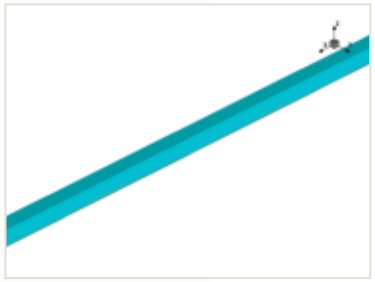
Step #3400: Deposition

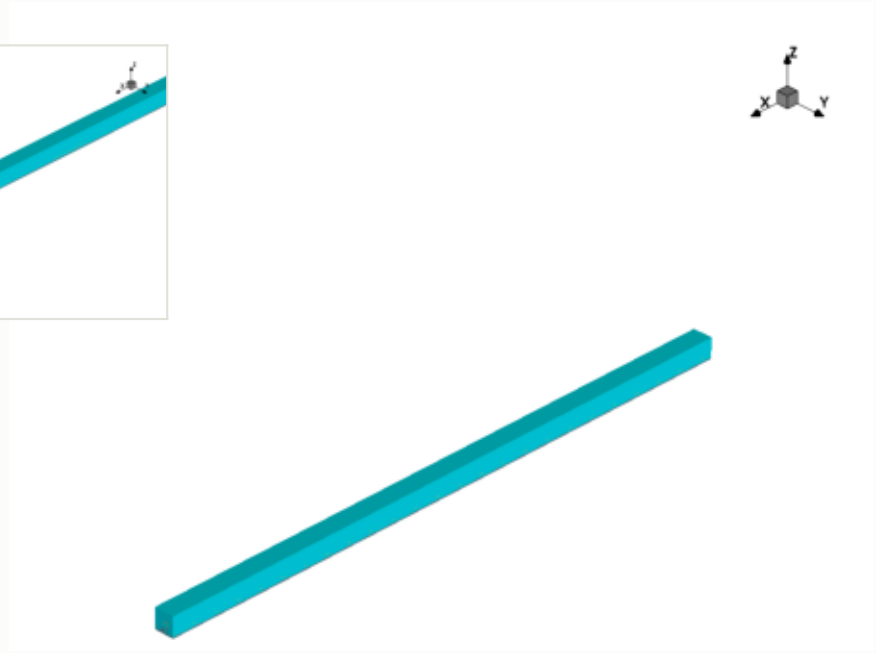
depo_fill · Deposition

0.656	162.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



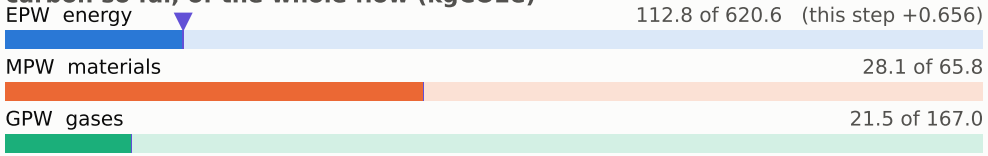


this step

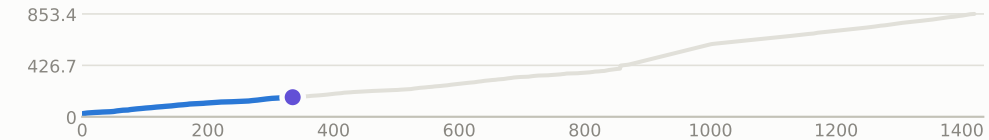
process area	Deposition
material	—
thickness	0.25 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

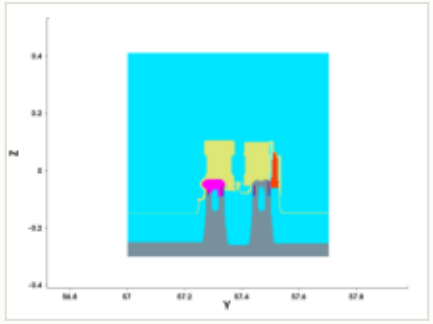
carbon so far, of the whole flow (kgCO2e)



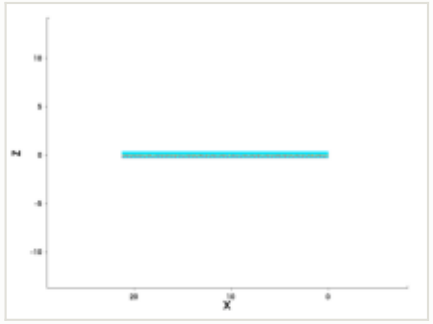
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



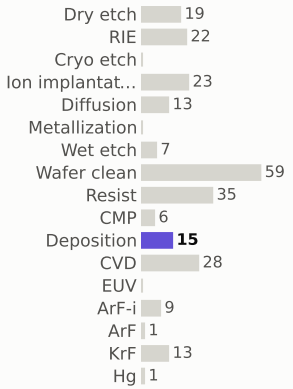
y cut at 57.351 µm



mask for this step

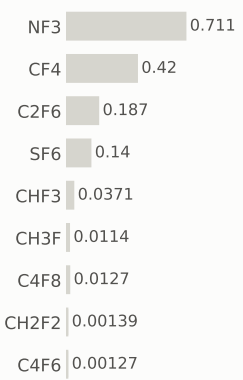
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



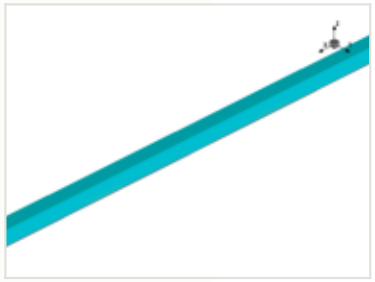
Step #3410: Wafer_clean

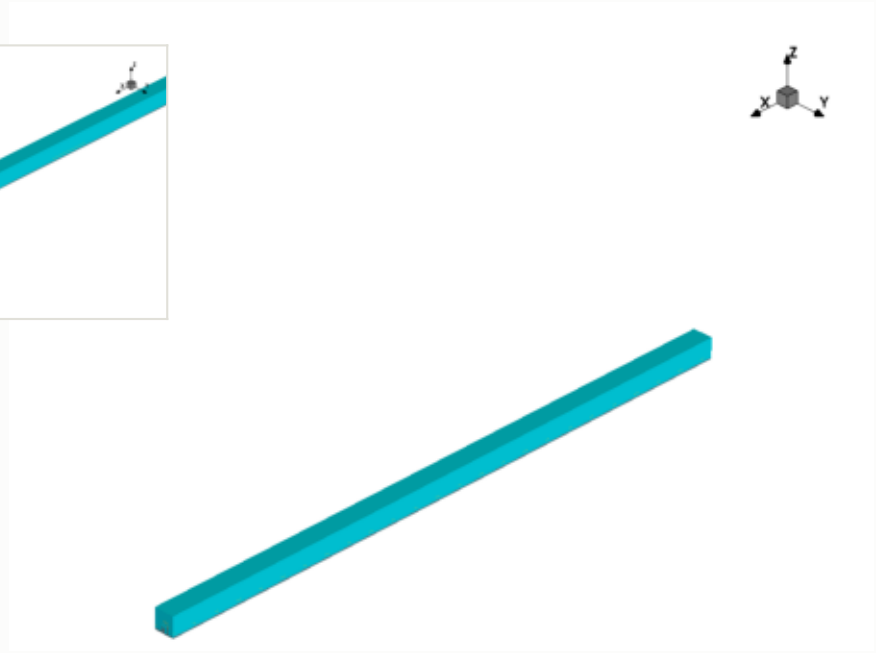
clean · Wafer_clean

0.169	162.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





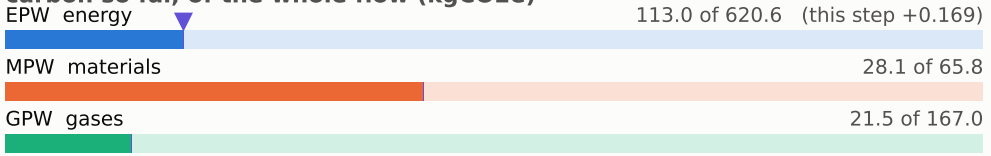
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

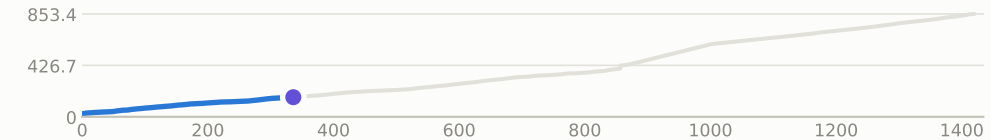
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

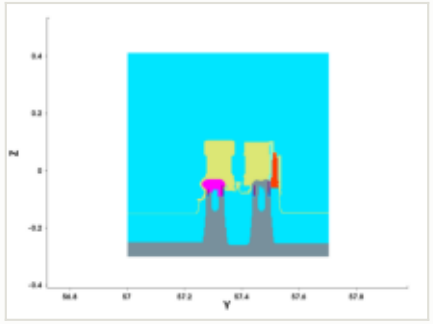
carbon so far, of the whole flow (kgCO2e)



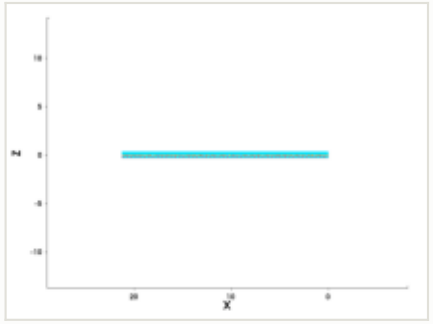
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



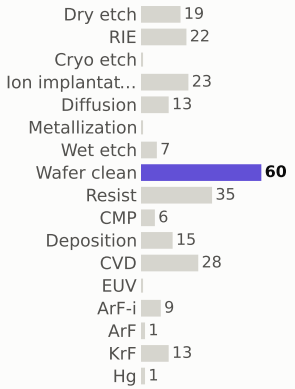
y cut at 57.351 µm



mask for this step

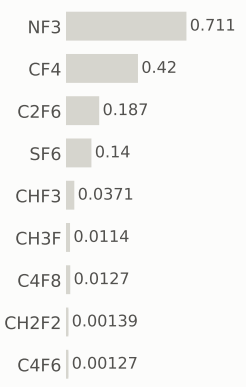
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



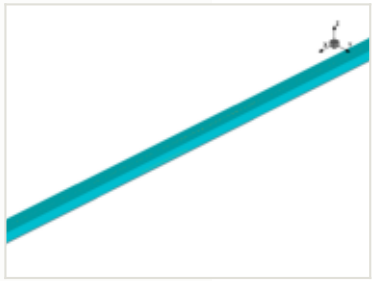
Step #3420: CMP

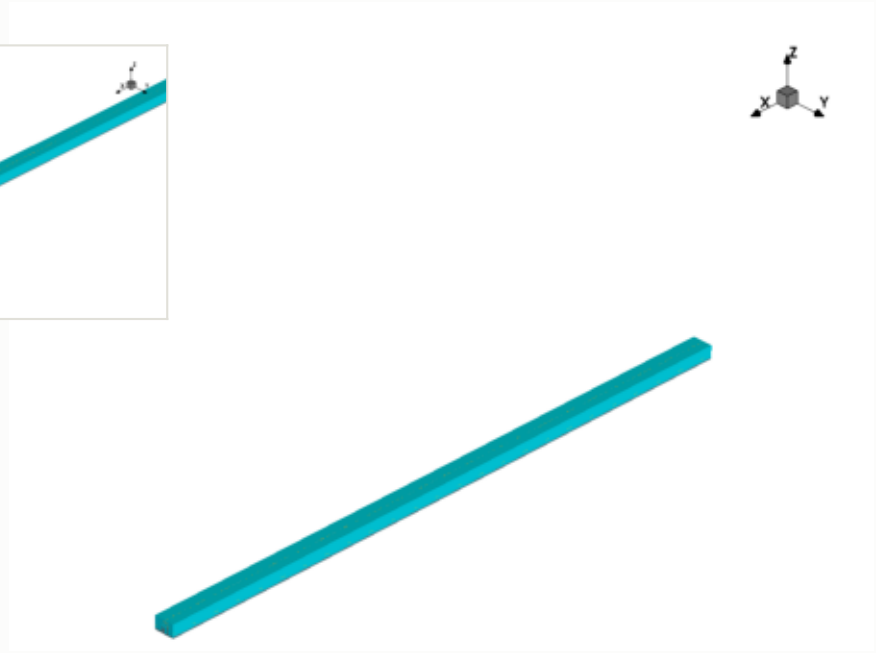
etch_cmp · CMP

0.169	162.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



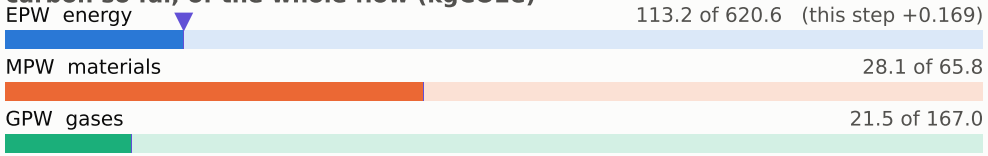


this step

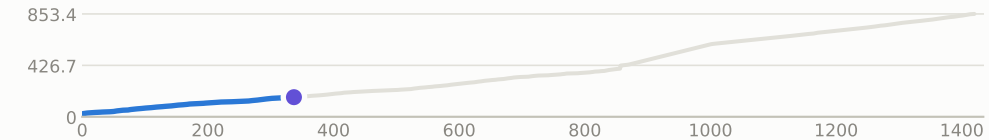
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

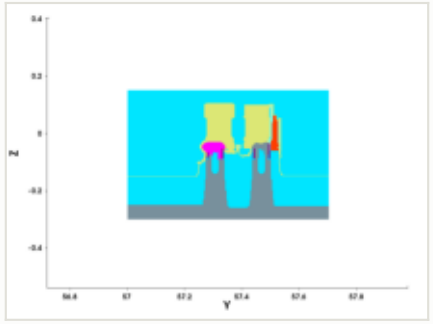
carbon so far, of the whole flow (kgCO2e)



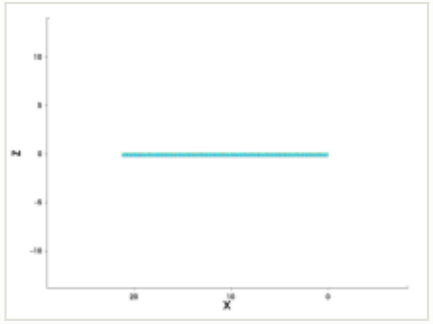
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



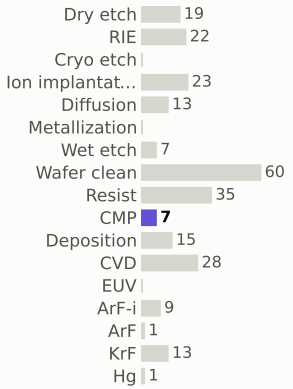
y cut at 57.351 µm



mask for this step

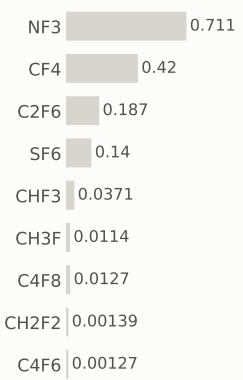
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



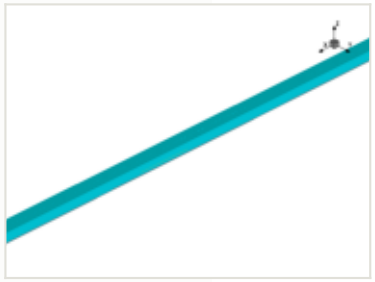
Step #3430: RIE

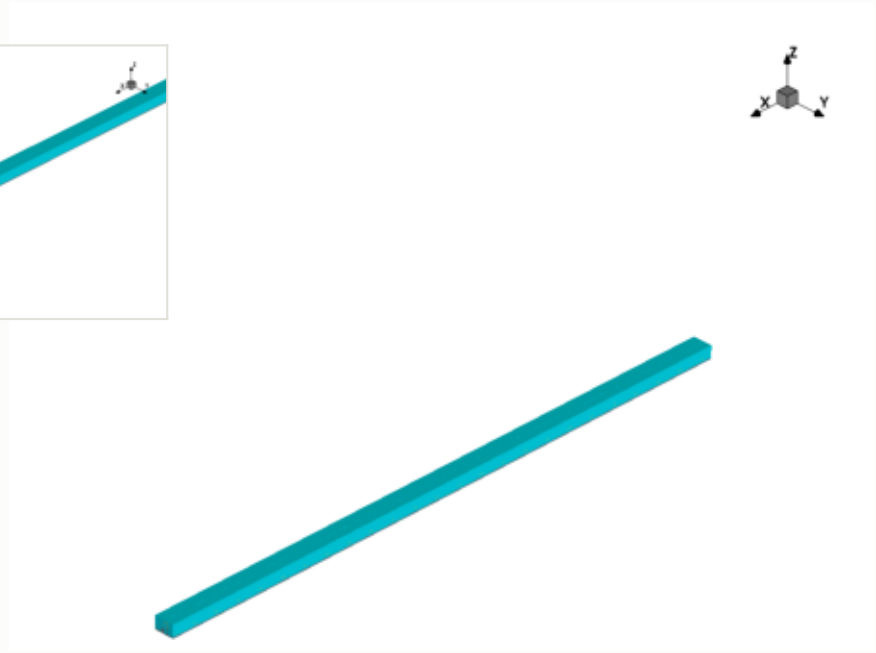
etch_ellipsoidal · RIE

1.04	163.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



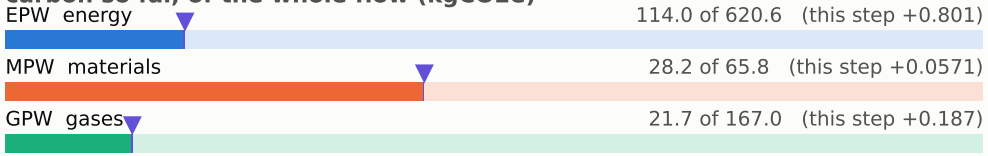


this step

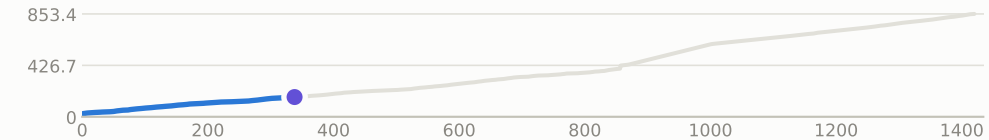
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

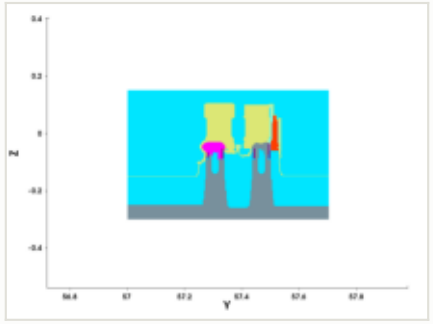
carbon so far, of the whole flow (kgCO2e)



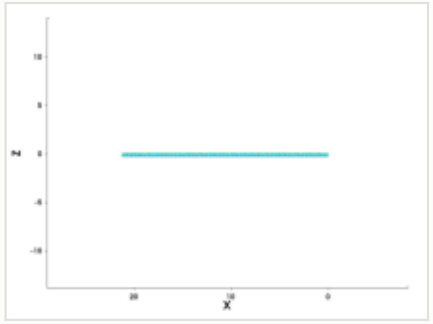
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



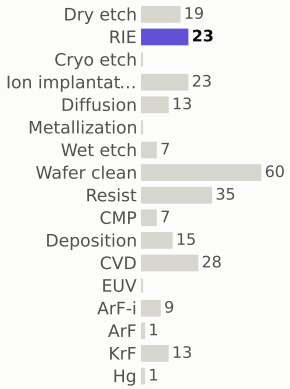
y cut at 57.351 µm



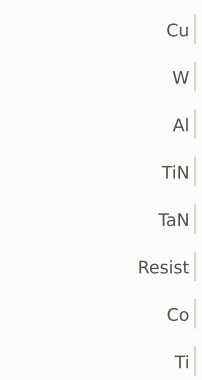
mask for this step

no mask for this step

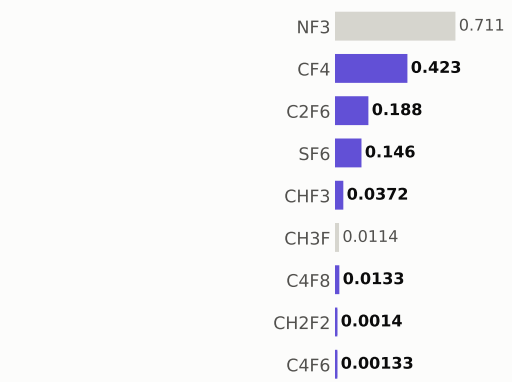
steps by process area



materials deposited (g)



gases emitted (g)



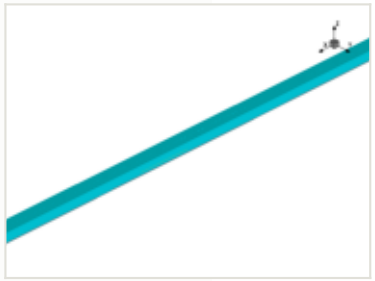
Step #3440: Wafer_clean

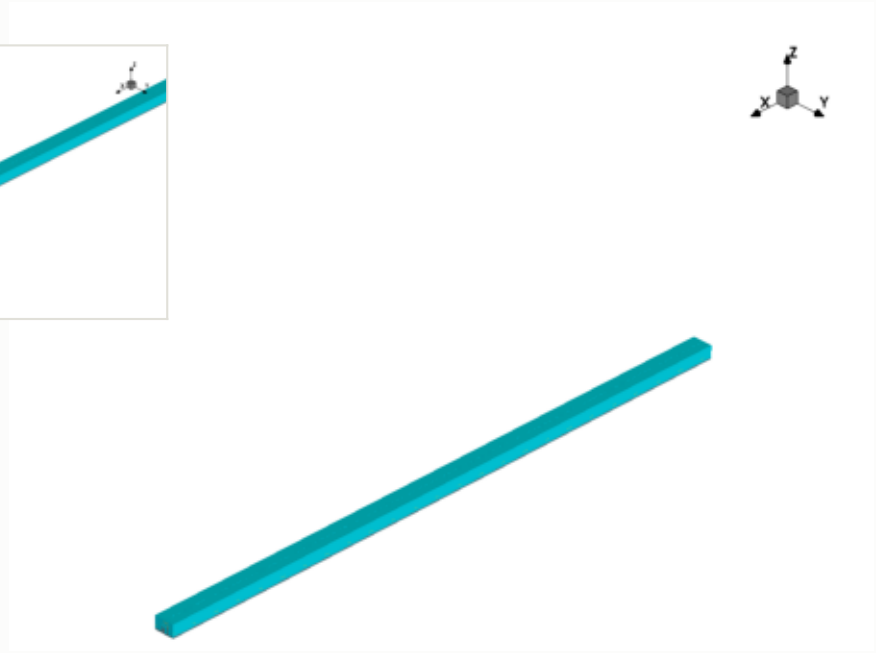
clean · Wafer_clean

0.169	164.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





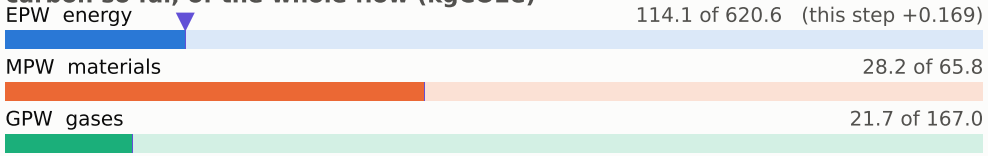
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

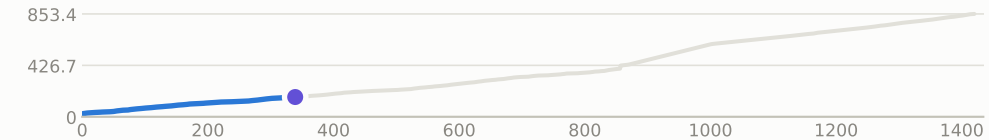
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

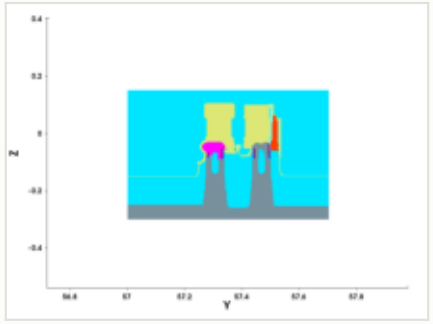
carbon so far, of the whole flow (kgCO2e)



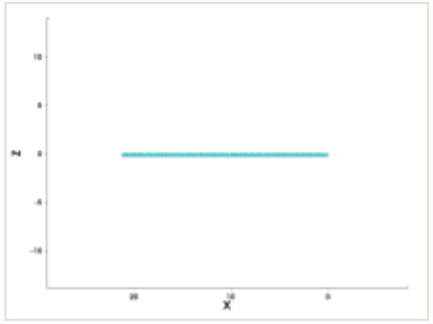
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



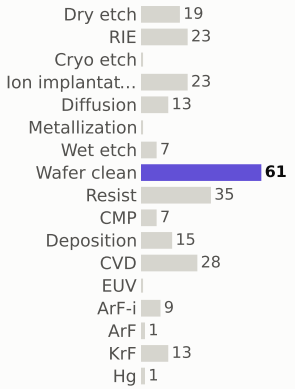
y cut at 57.351 µm



mask for this step

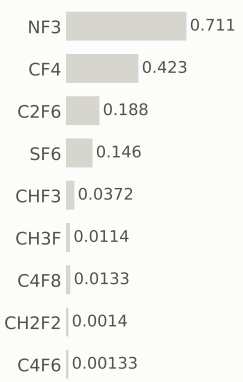
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



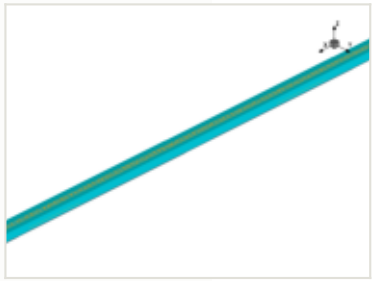
Step #3450: CMP

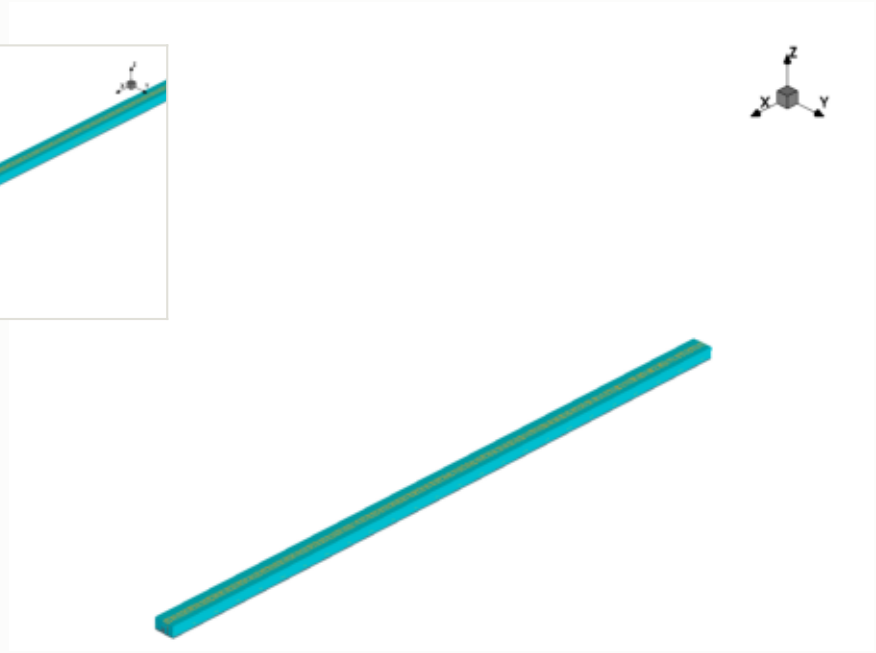
etch_cmp · CMP

0.169	164.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



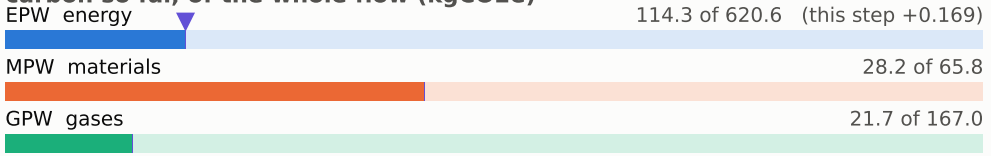


this step

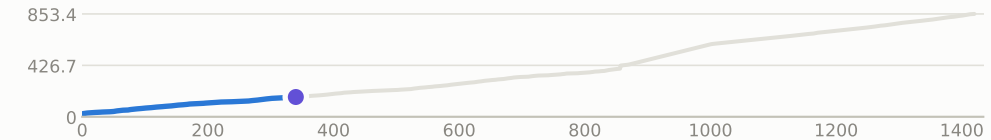
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

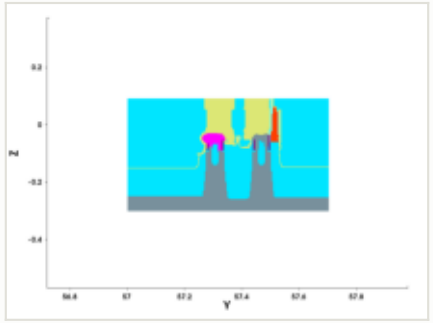
carbon so far, of the whole flow (kgCO2e)



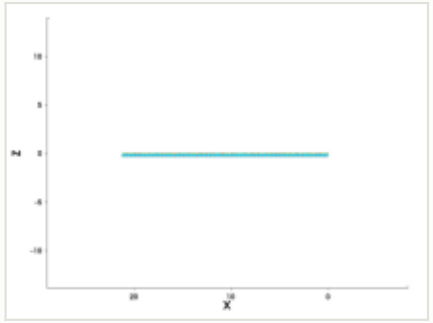
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



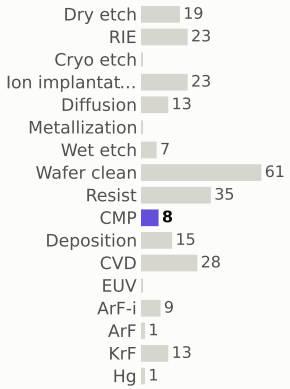
y cut at 57.351 µm



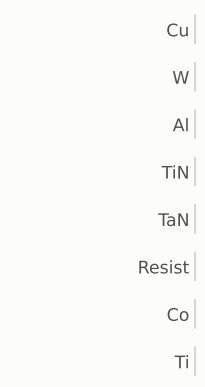
mask for this step

no mask for this step

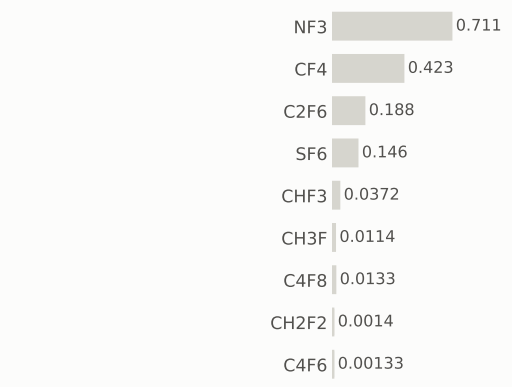
steps by process area



materials deposited (g)



gases emitted (g)



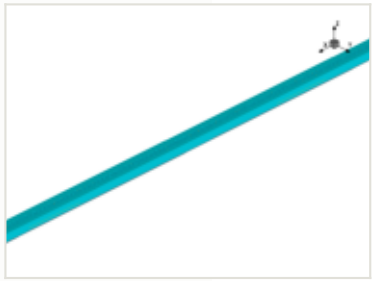
Step #3460: Spx_only

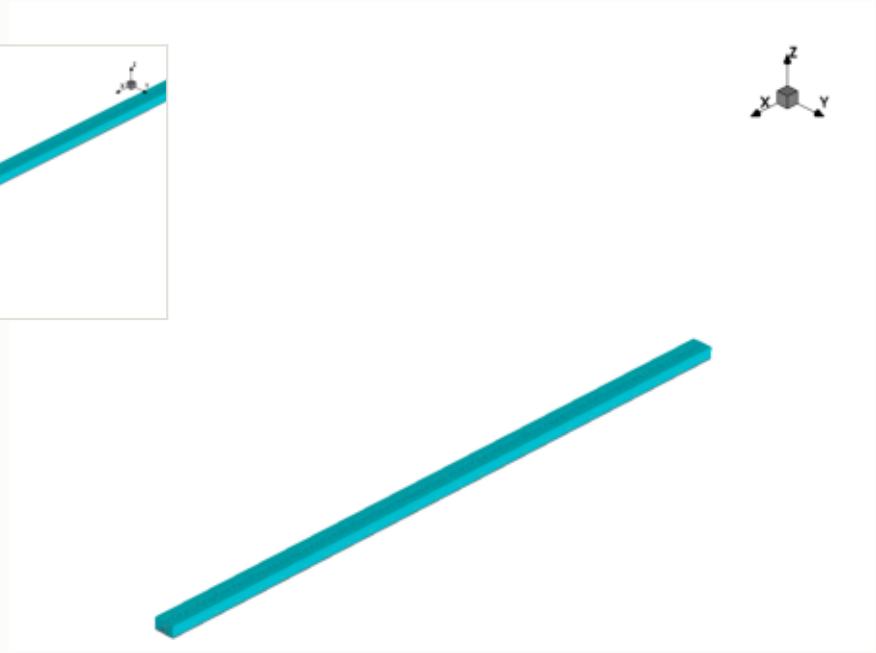
etch_anisotropic · Spx_only

0	164.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



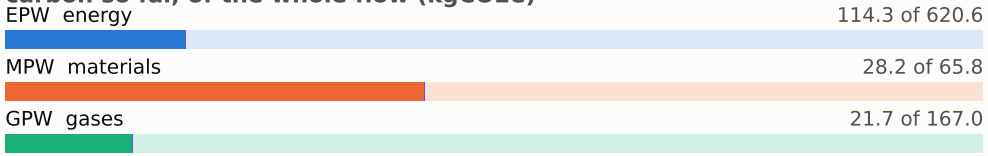


this step

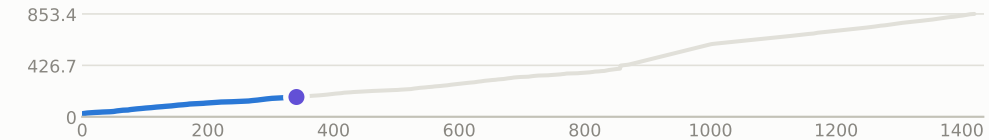
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

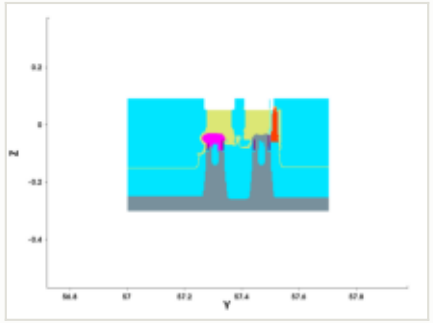
carbon so far, of the whole flow (kgCO2e)



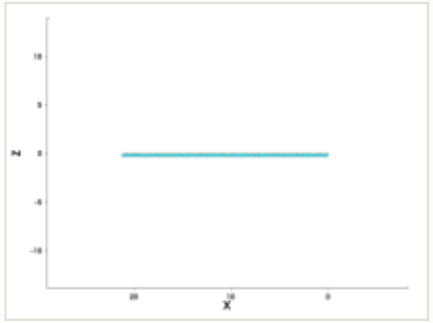
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



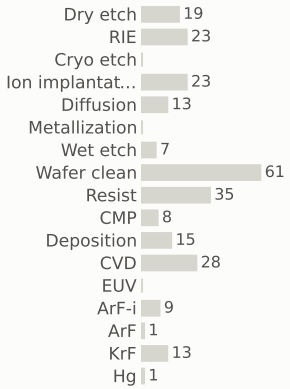
y cut at 57.351 µm



mask for this step

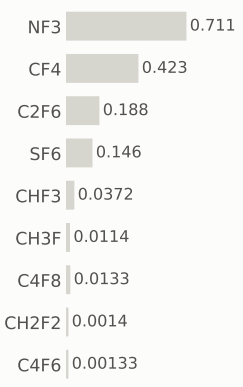
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



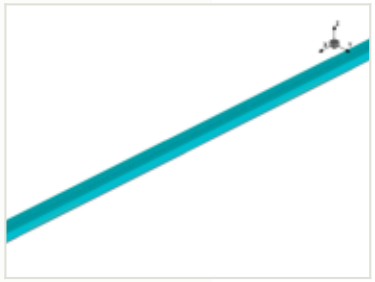
Step #3470: RIE

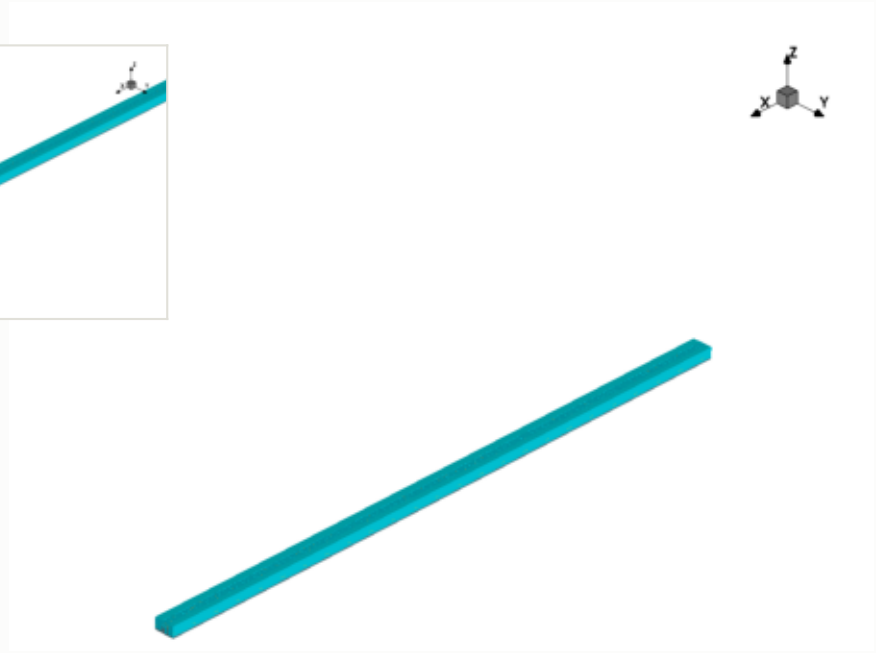
etch_anisotropic · RIE

1.04	165.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



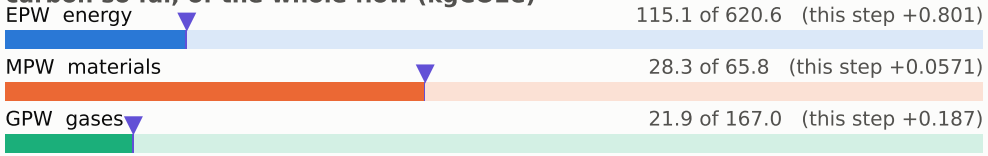


this step

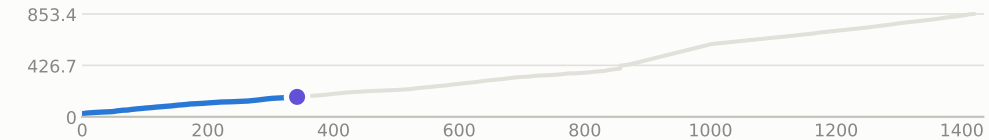
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

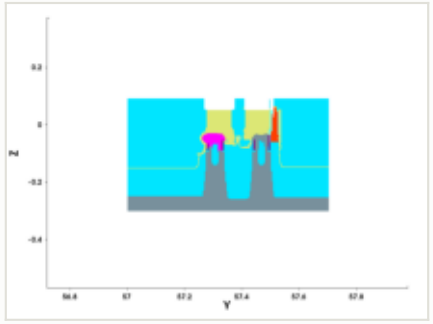
carbon so far, of the whole flow (kgCO2e)



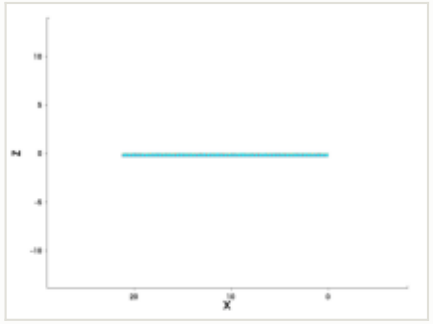
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



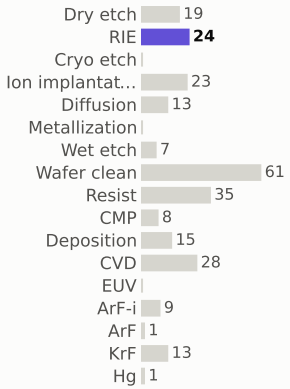
y cut at 57.351 µm



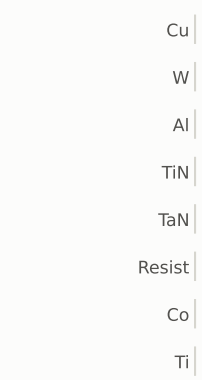
mask for this step

no mask for this step

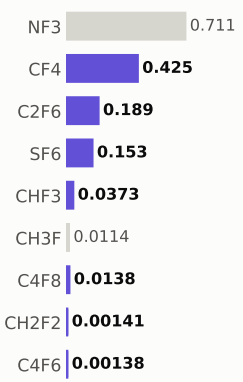
steps by process area



materials deposited (g)



gases emitted (g)



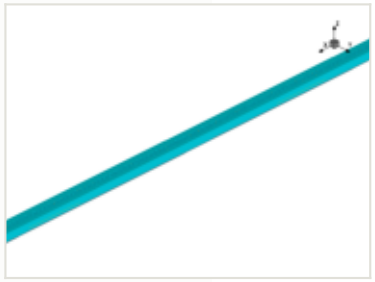
Step #3480: Wafer_clean

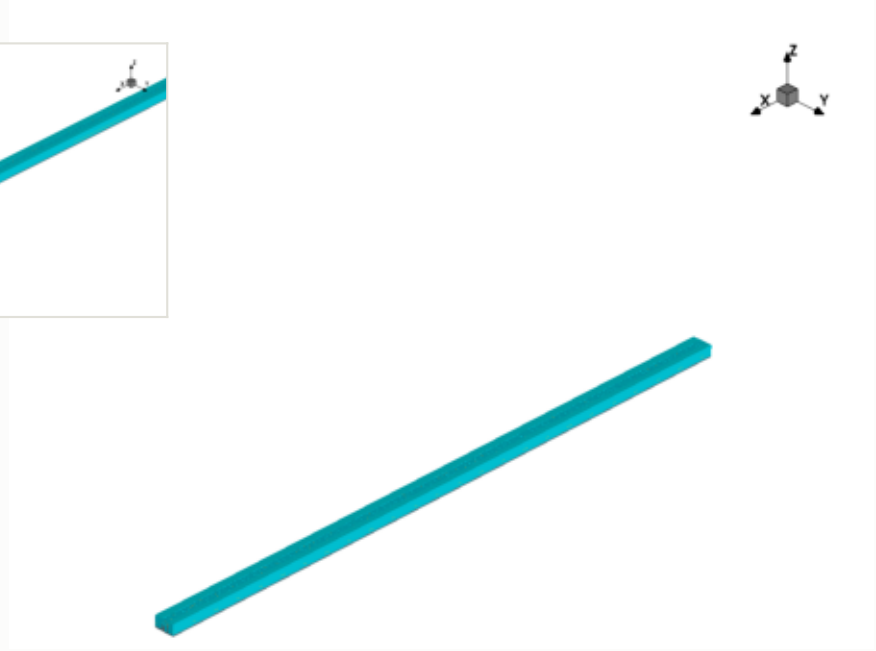
clean · Wafer_clean

0.169	165.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



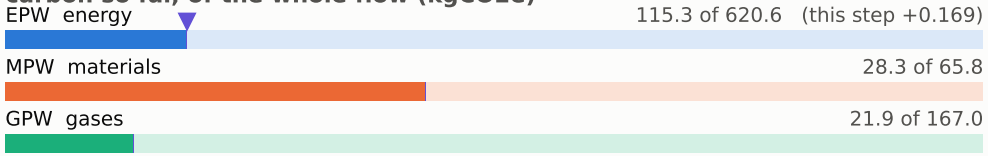


this step

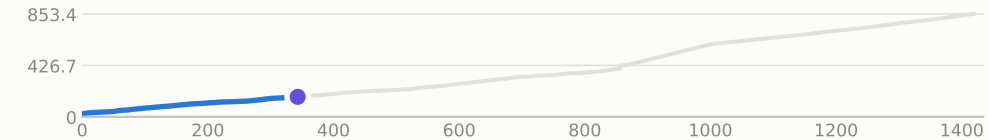
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

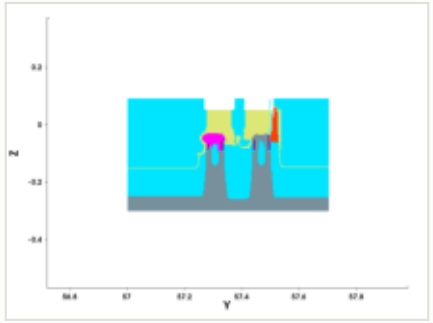
carbon so far, of the whole flow (kgCO2e)



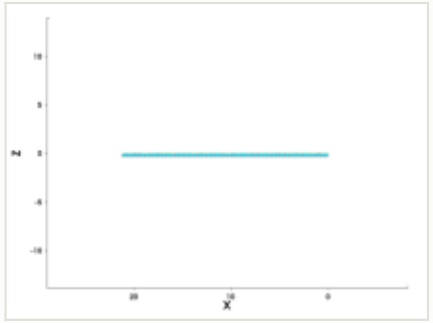
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



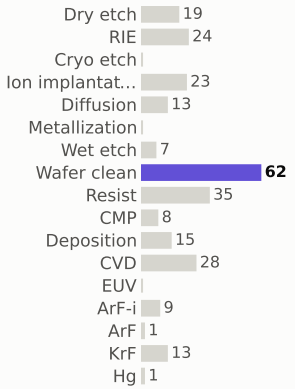
y cut at 57.351 µm



mask for this step

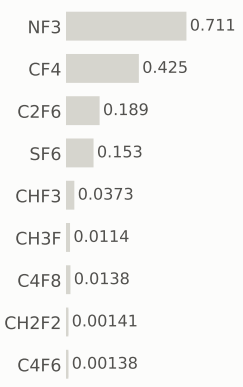
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



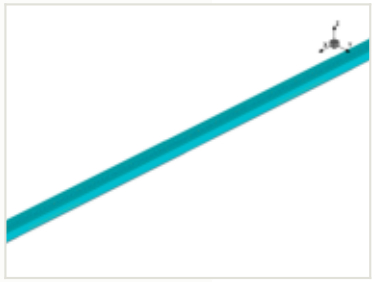
Step #3490: Wet_etch

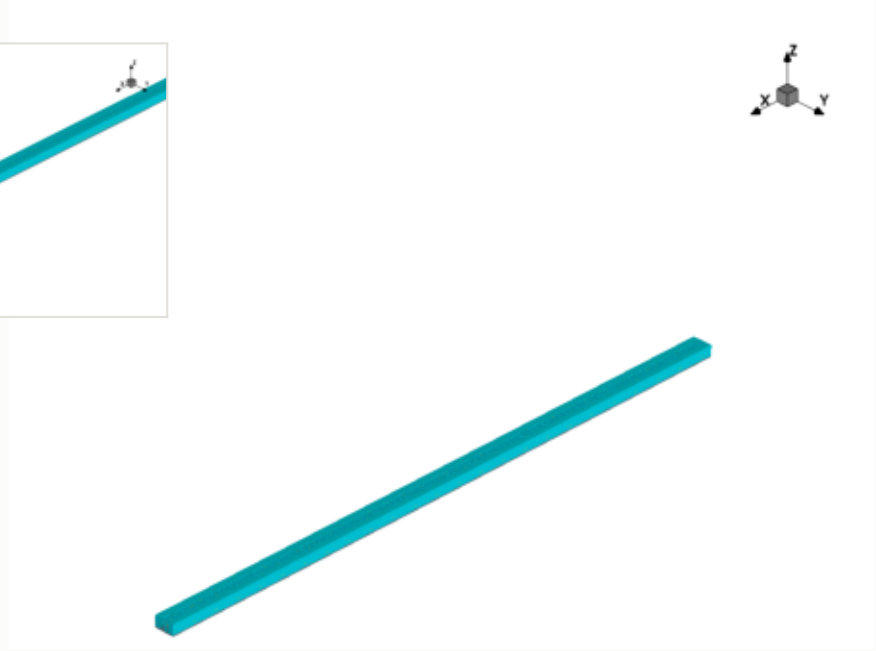
etch_isotropic · Wet_etch

0.169	165.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



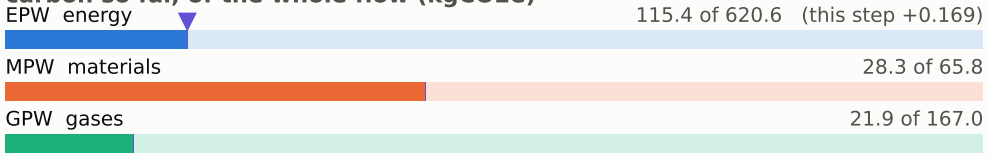


this step

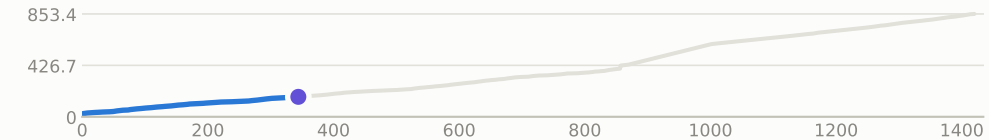
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

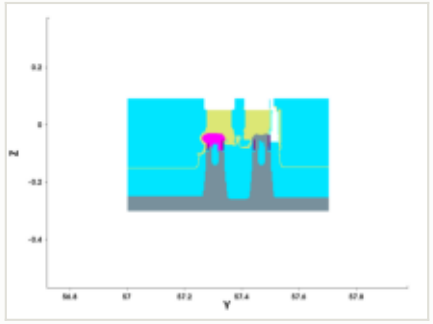
carbon so far, of the whole flow (kgCO2e)



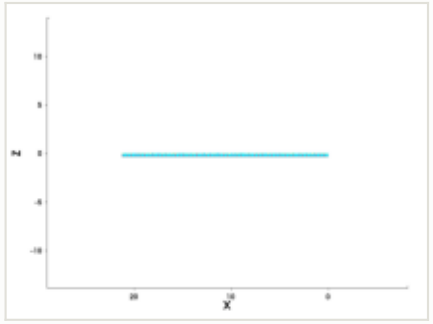
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



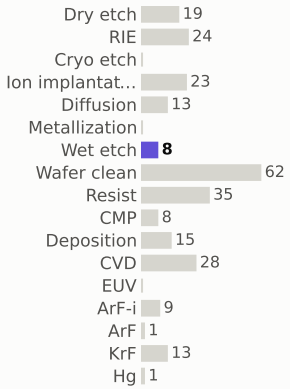
y cut at 57.351 µm



mask for this step

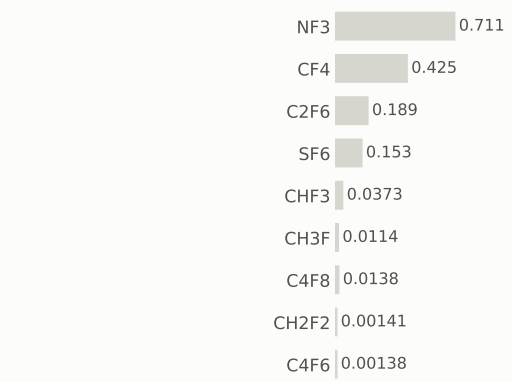
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



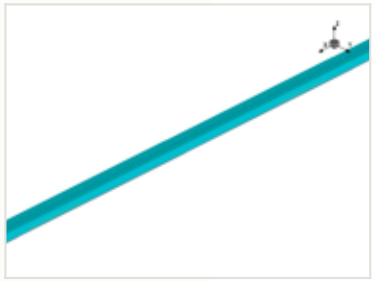
Step #3500: Spx_only

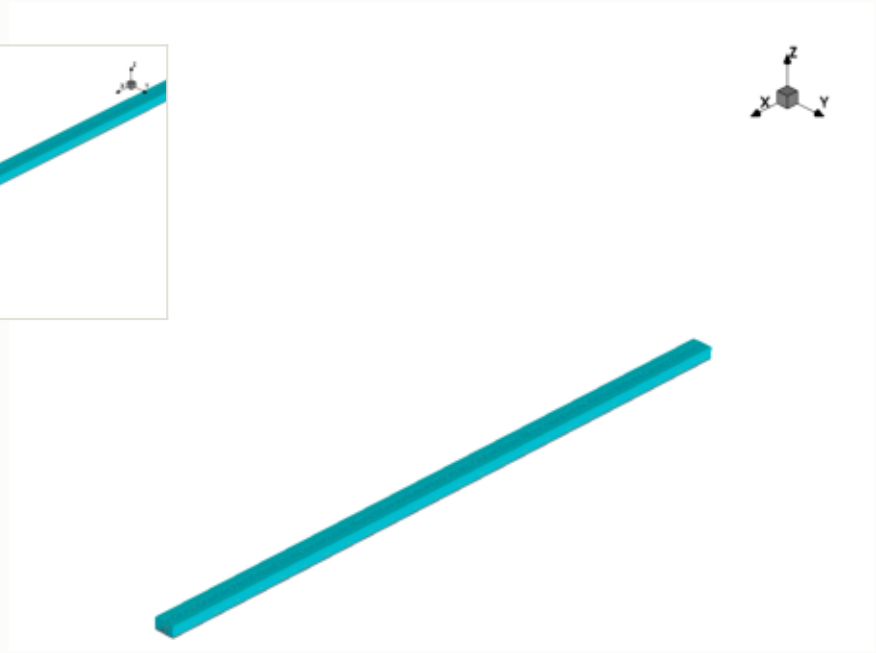
etch_isotropic · Spx_only

0	165.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



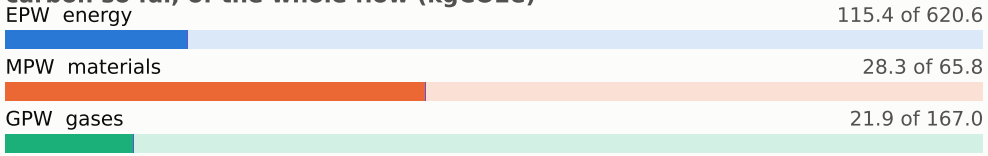


this step

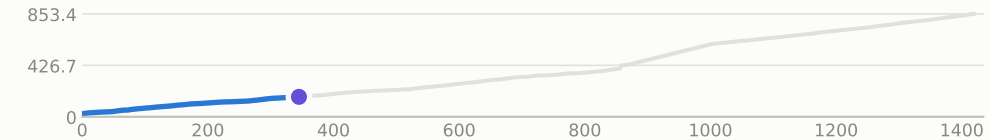
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

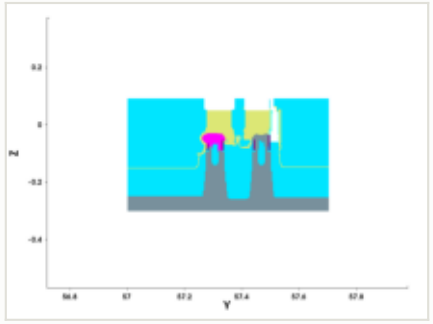
carbon so far, of the whole flow (kgCO2e)



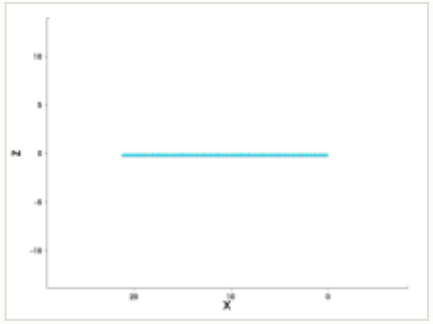
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



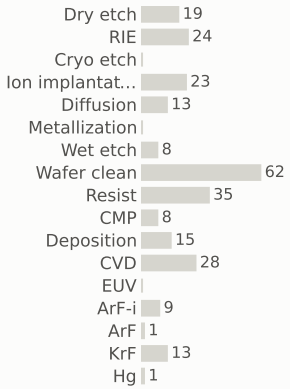
y cut at 57.351 µm



mask for this step

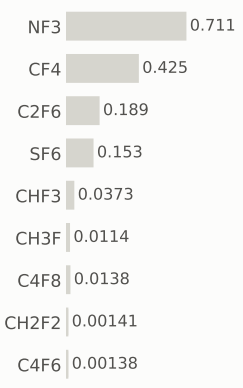
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



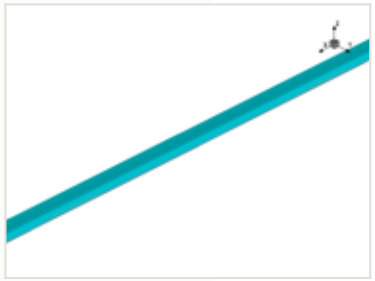
Step #3510: Spx_only

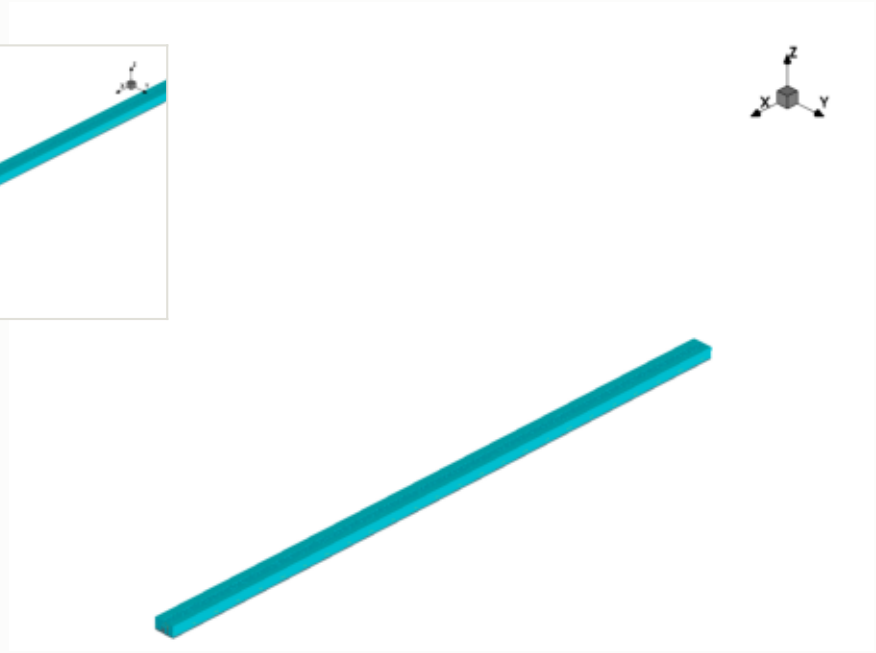
depo_isotropic · Spx_only

0	165.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





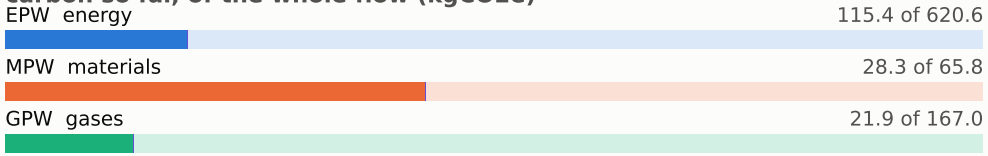
this step

process area	Spx_only
material	—
thickness	0.002 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

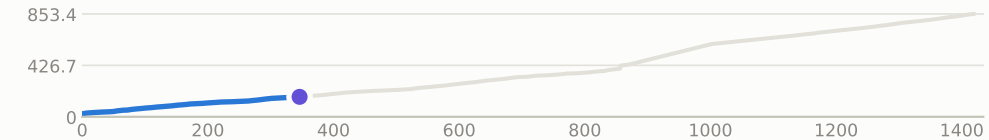
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

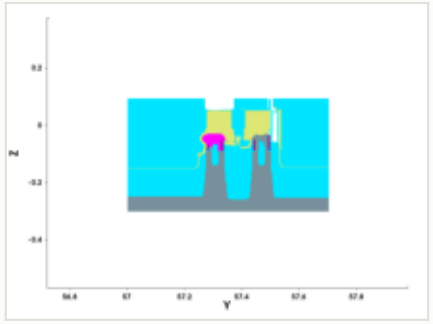
carbon so far, of the whole flow (kgCO2e)



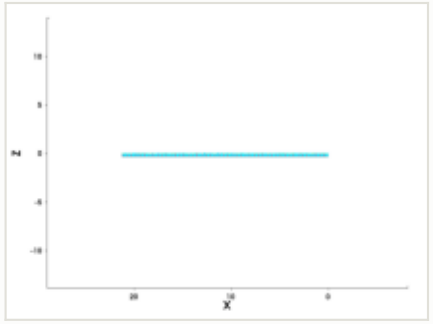
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



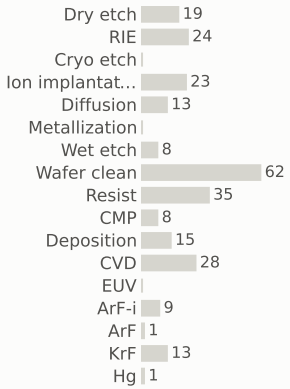
y cut at 57.351 µm



mask for this step

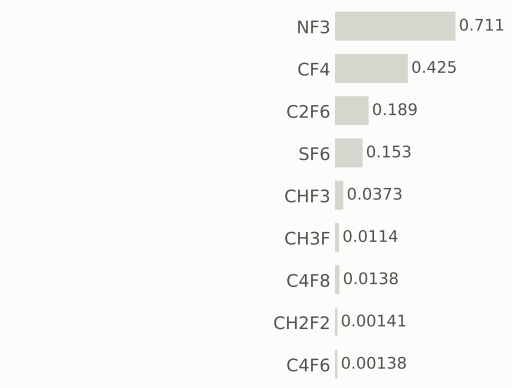
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



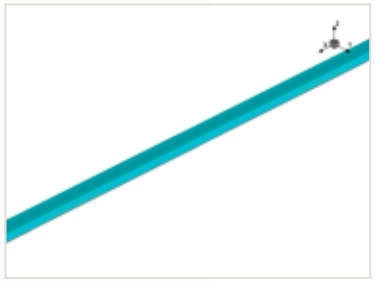
Step #3530: Deposition

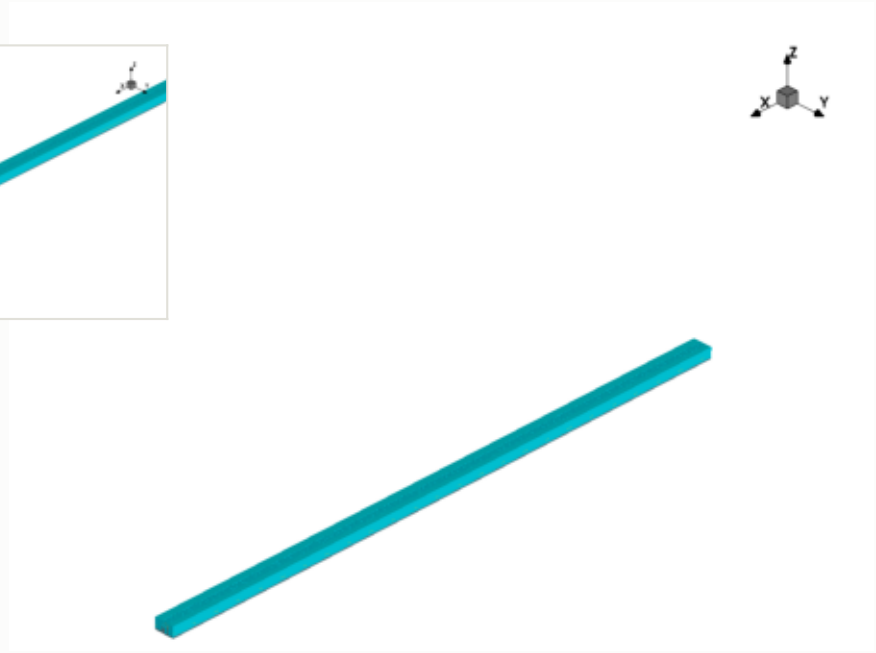
depo_isotropic · Deposition

0.656	166.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



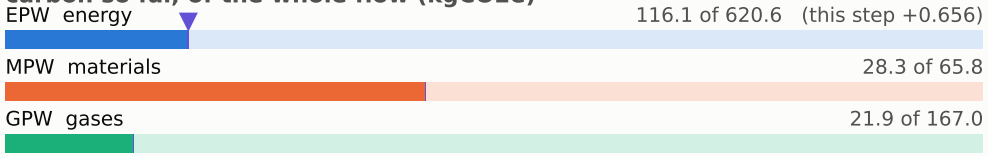


this step

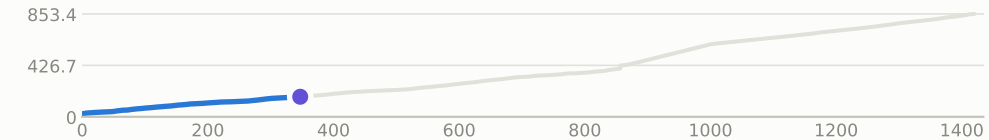
process area	Deposition
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

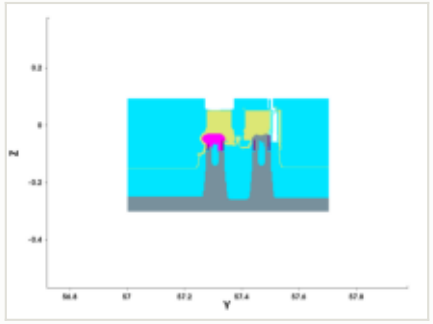
carbon so far, of the whole flow (kgCO2e)



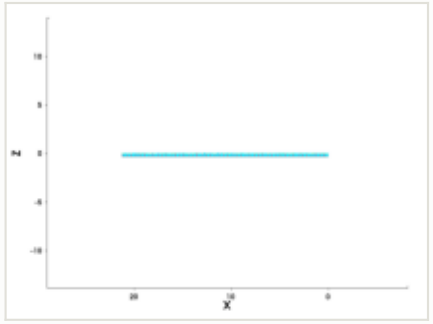
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



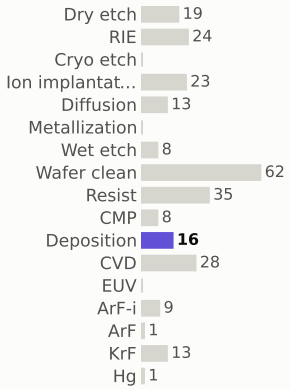
y cut at 57.351 µm



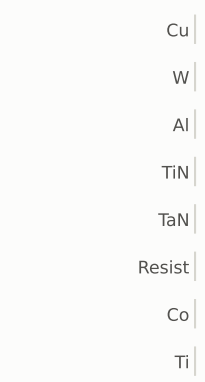
mask for this step

no mask for this step

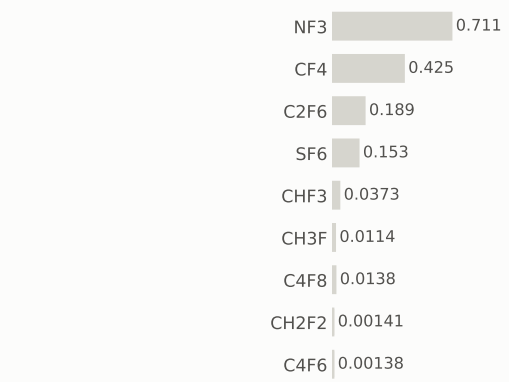
steps by process area



materials deposited (g)



gases emitted (g)



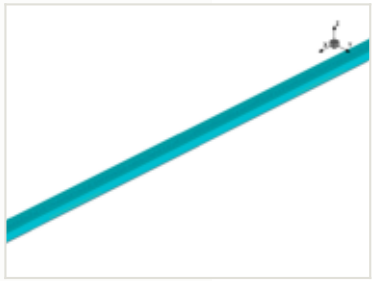
Step #3540: Resist

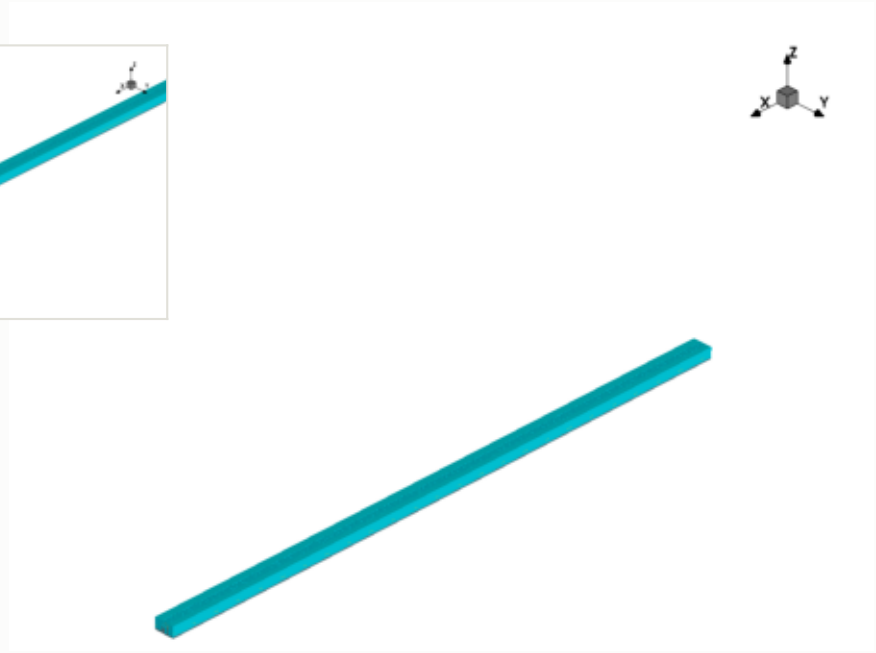
resist · Resist

0.169	166.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





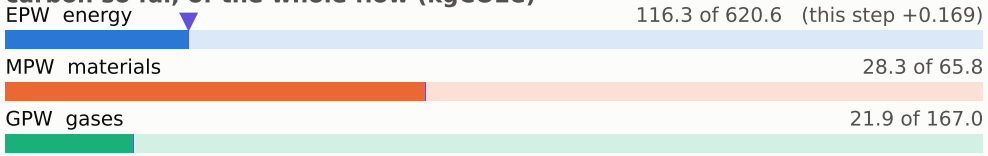
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

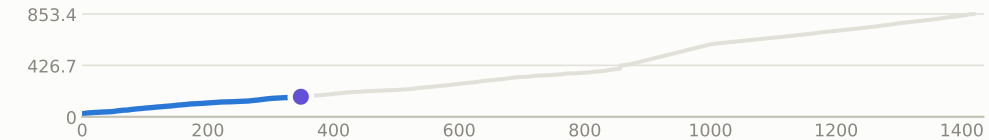
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

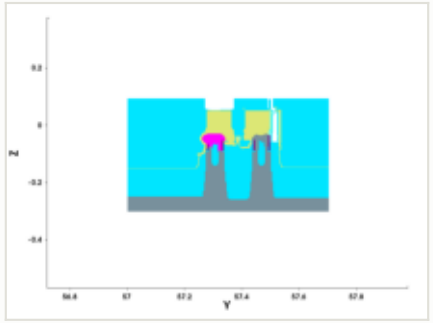
carbon so far, of the whole flow (kgCO2e)



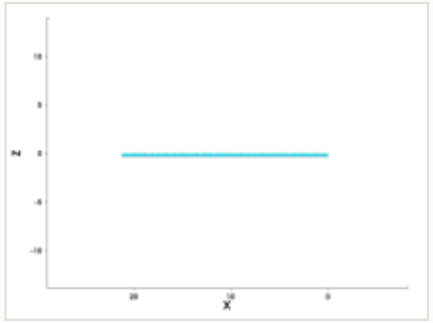
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



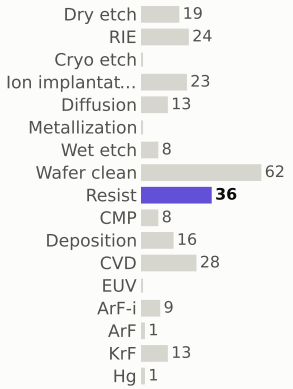
y cut at 57.351 µm



mask for this step

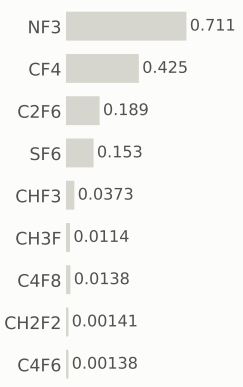
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



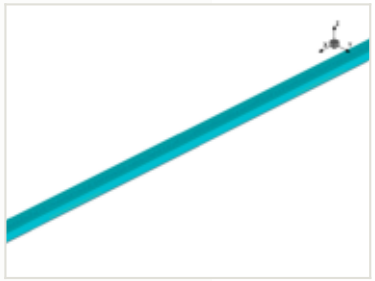
Step #3550: KrF

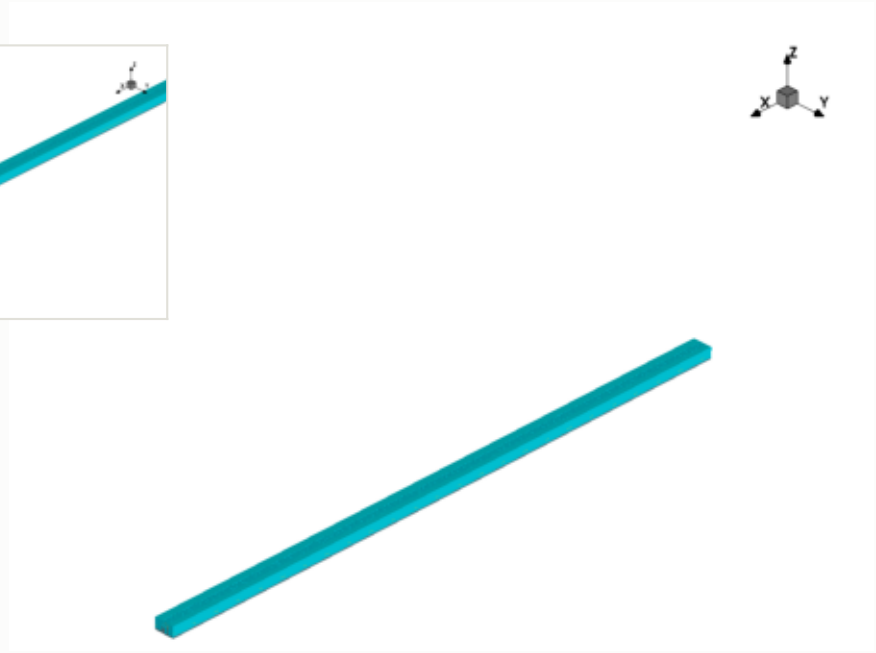
litho_patterning · KrF

0.199	166.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



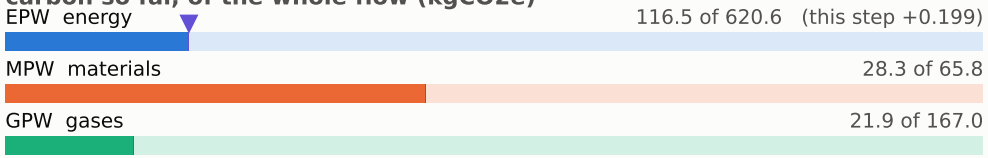


this step

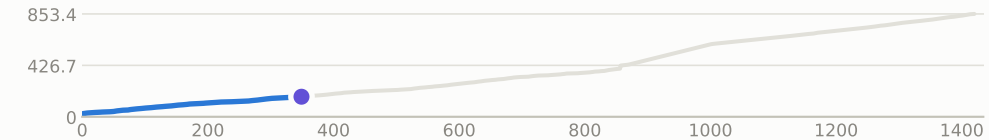
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

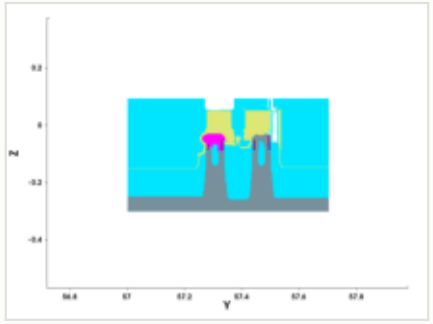
carbon so far, of the whole flow (kgCO2e)



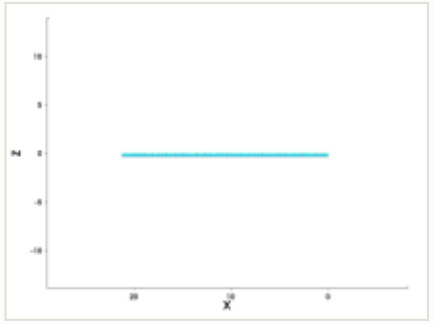
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



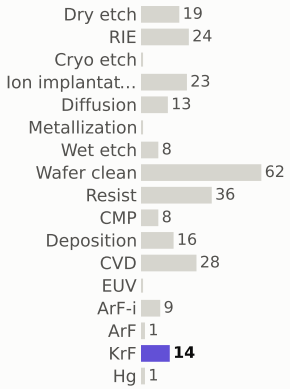
y cut at 57.351 µm



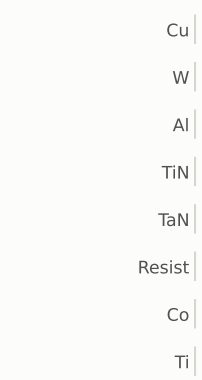
mask for this step

no mask for this step

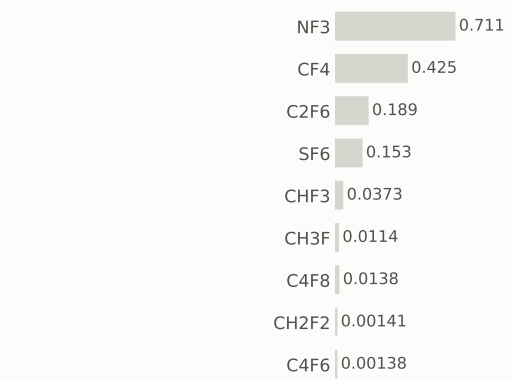
steps by process area



materials deposited (g)



gases emitted (g)



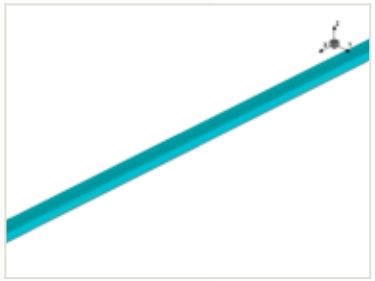
Step #3560: RIE

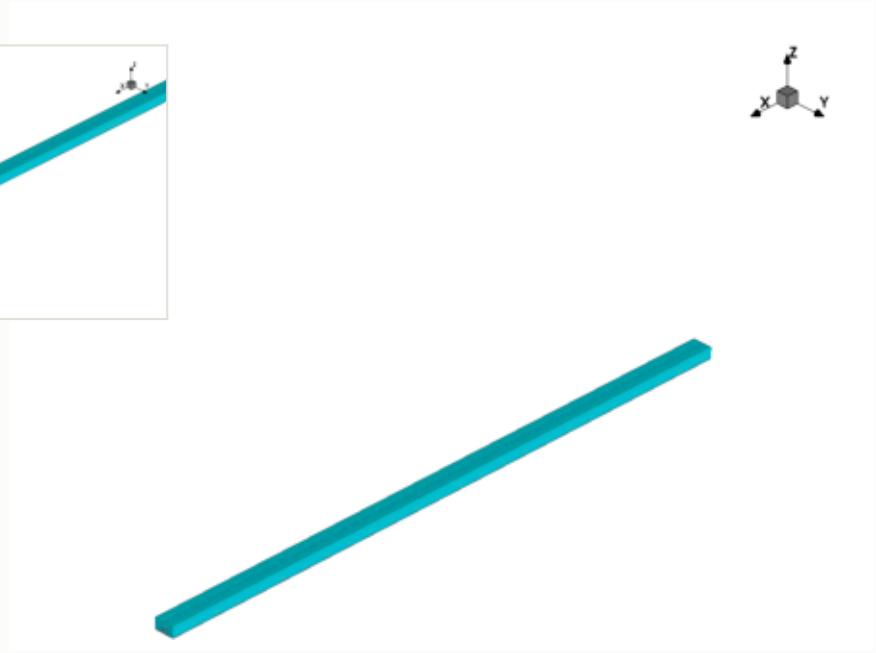
etch_anisotropic · RIE

1.04	167.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



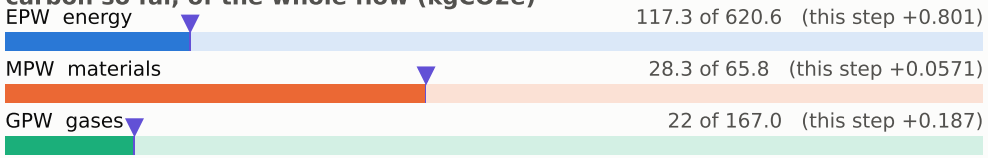


this step

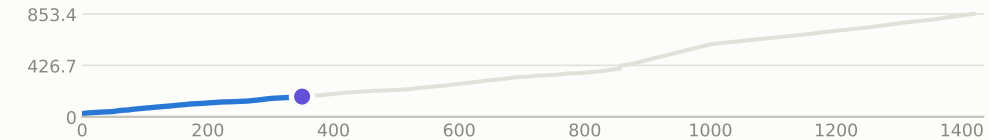
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

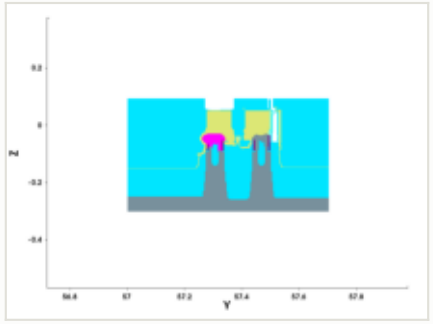
carbon so far, of the whole flow (kgCO2e)



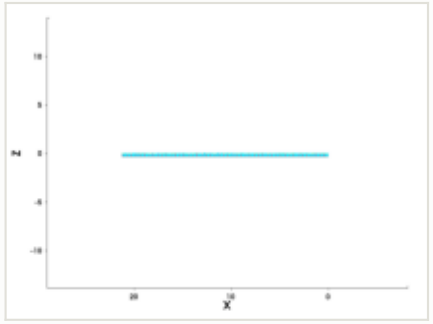
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



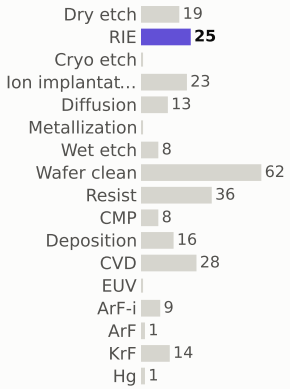
y cut at 57.351 µm



mask for this step

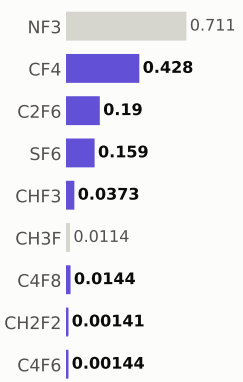
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



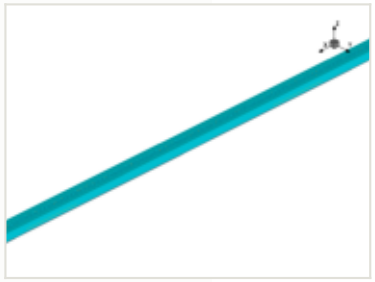
Step #3570: Wet_etch

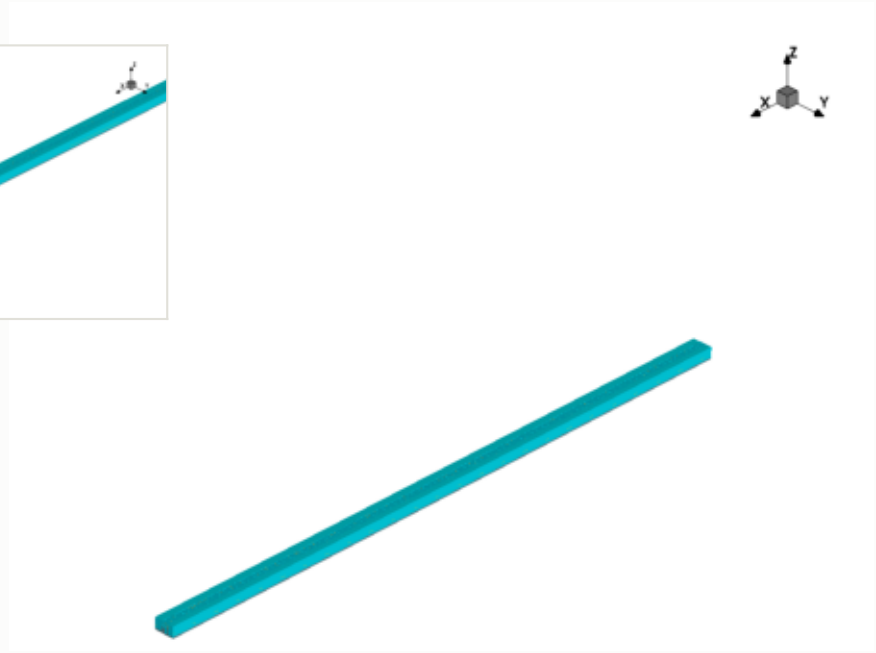
etch_isotropic · Wet_etch

0.169	167.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



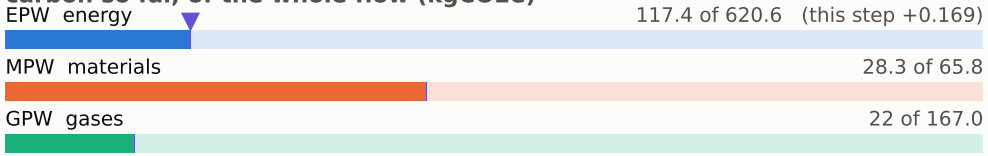


this step

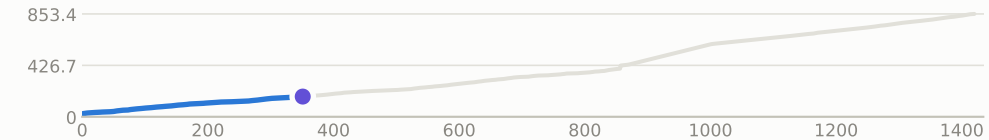
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

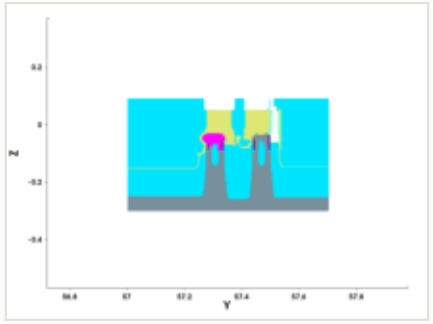
carbon so far, of the whole flow (kgCO2e)



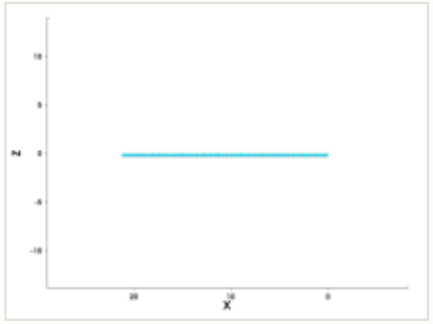
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



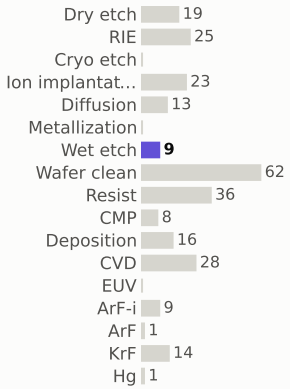
y cut at 57.351 µm



mask for this step

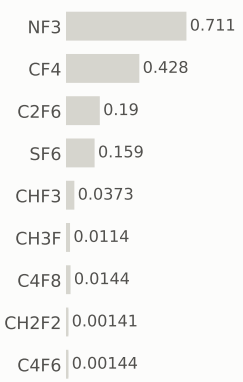
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



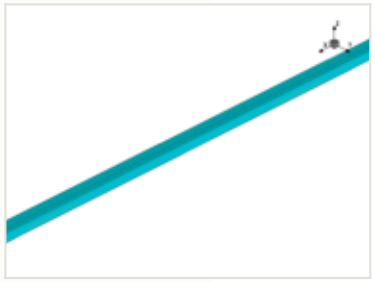
Step #3580: Dry_etch

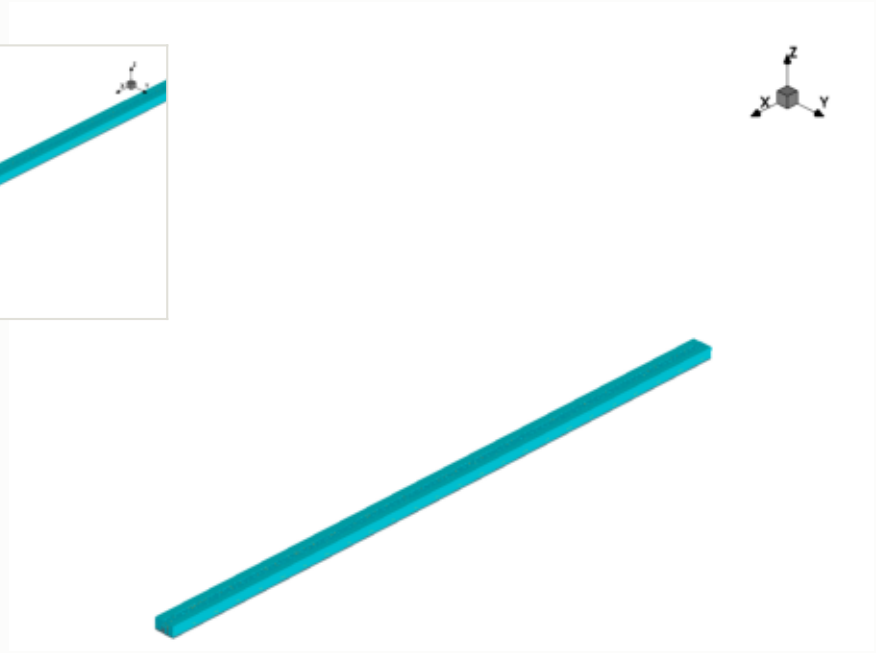
etch_isotropic · Dry_etch

0.801	168.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



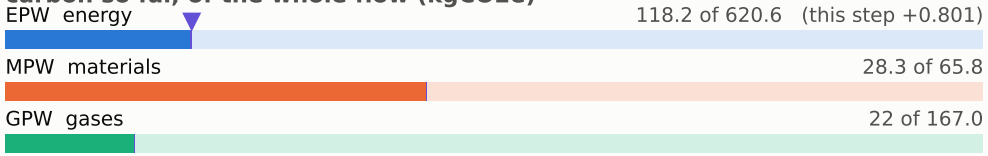


this step

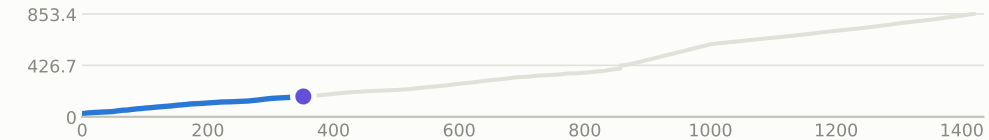
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

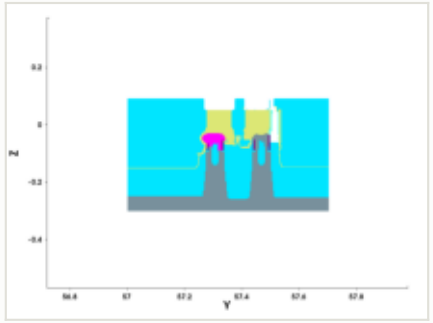
carbon so far, of the whole flow (kgCO2e)



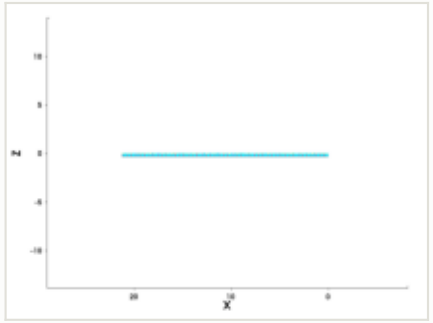
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



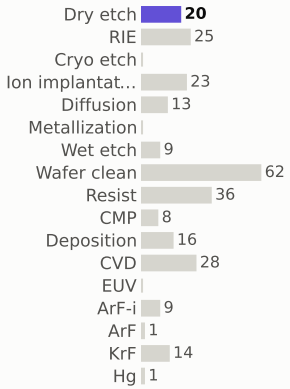
y cut at 57.351 µm



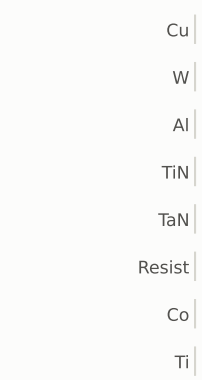
mask for this step

no mask for this step

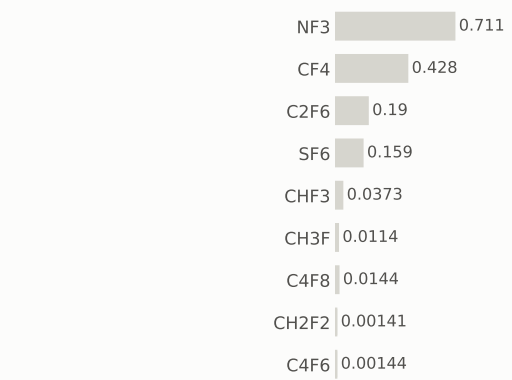
steps by process area



materials deposited (g)



gases emitted (g)



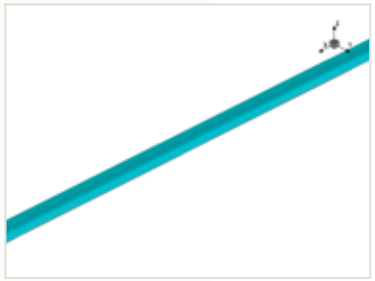
Step #3590: Spx_only

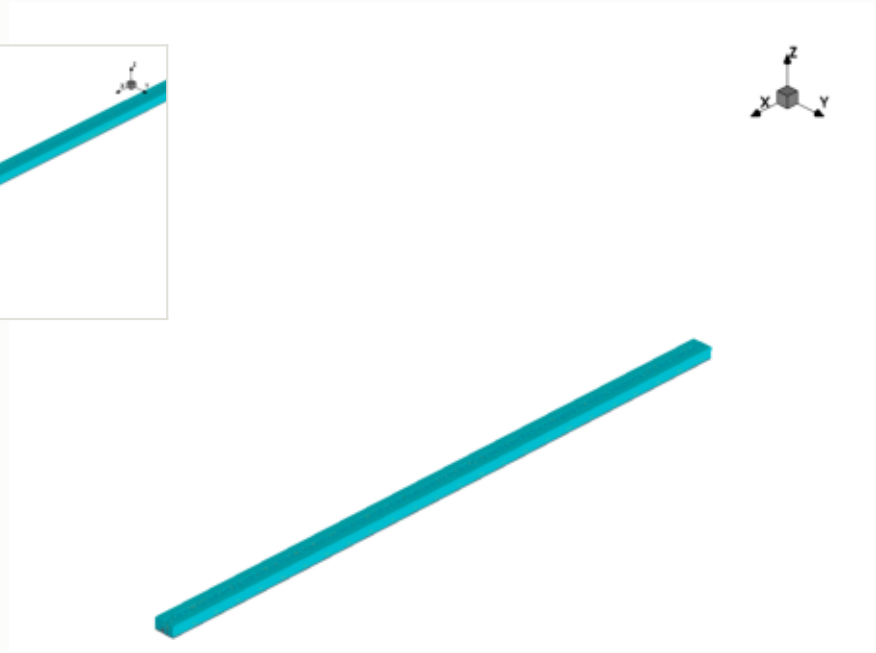
depo_isotropic · Spx_only

0	168.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





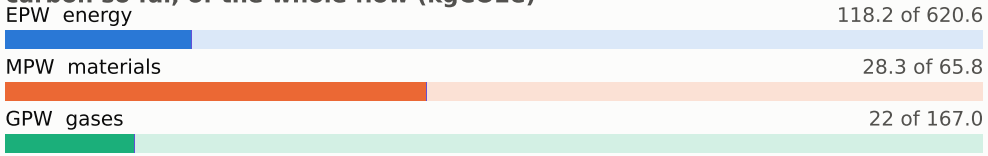
this step

process area	Spx_only
material	—
thickness	0.001 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

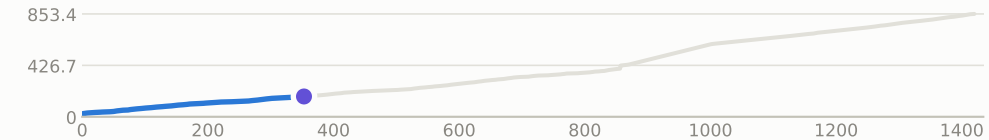
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

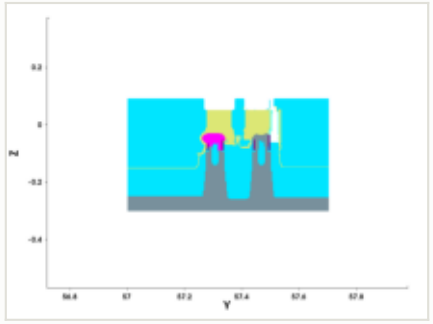
carbon so far, of the whole flow (kgCO2e)



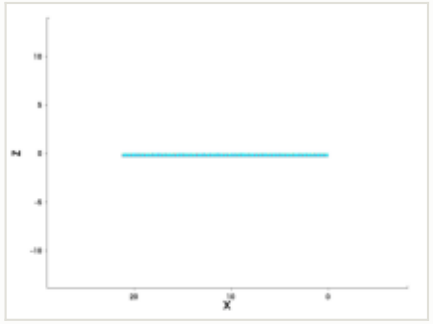
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



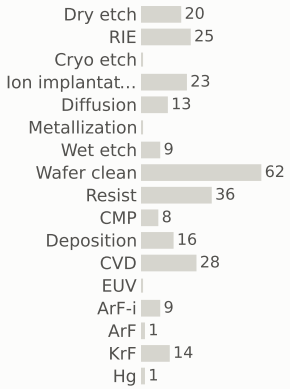
y cut at 57.351 µm



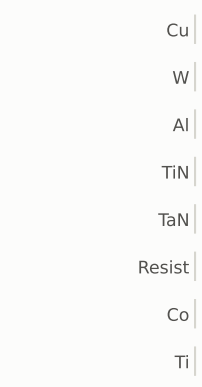
mask for this step

no mask for this step

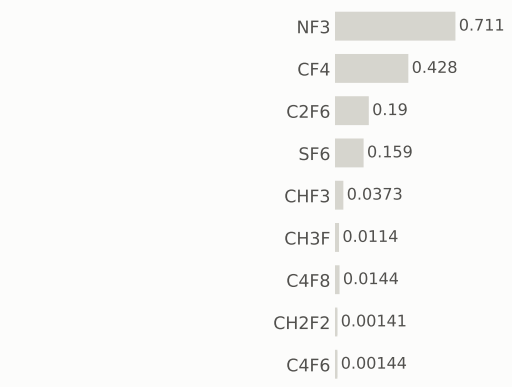
steps by process area



materials deposited (g)



gases emitted (g)



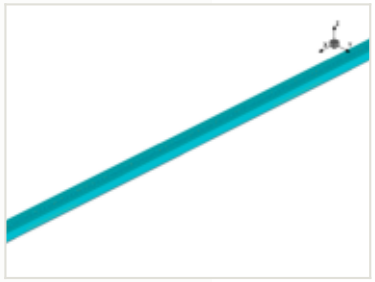
Step #3600: Wafer_clean

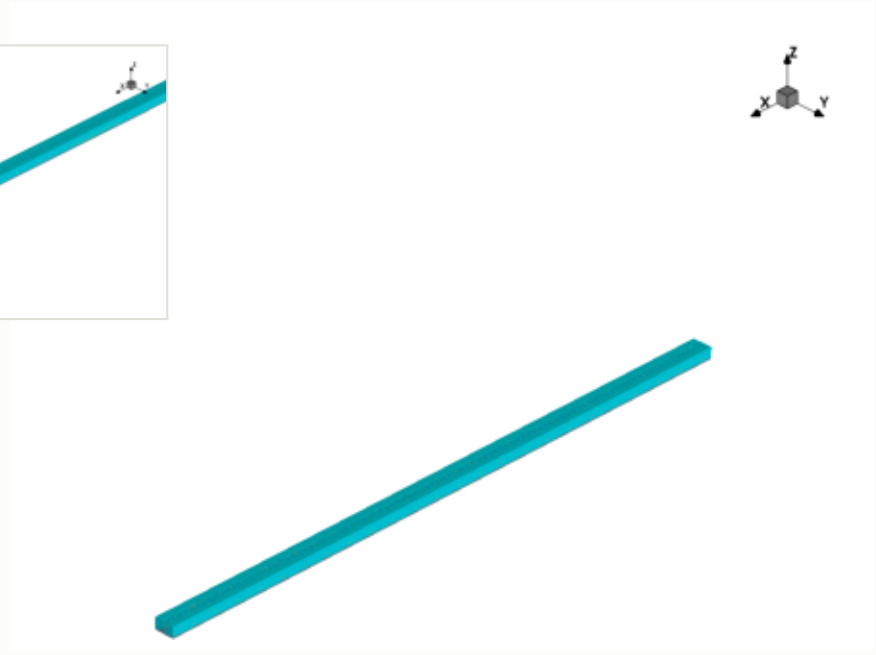
clean · Wafer_clean

0.169	168.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





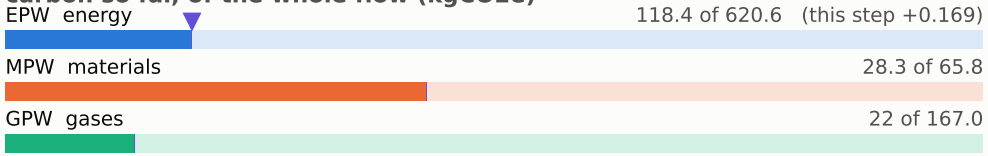
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

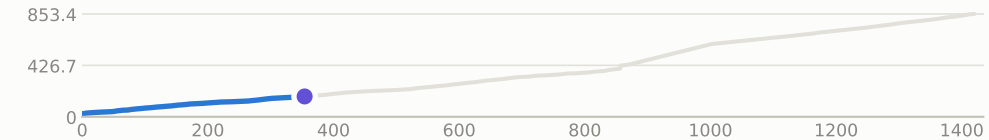
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

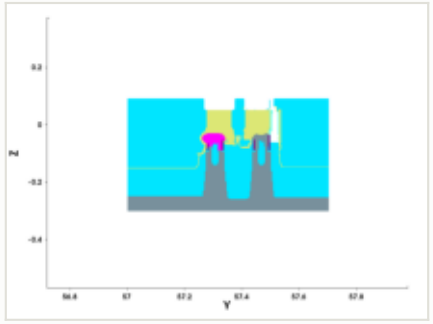
carbon so far, of the whole flow (kgCO2e)



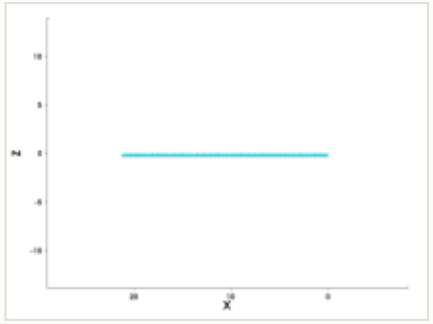
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



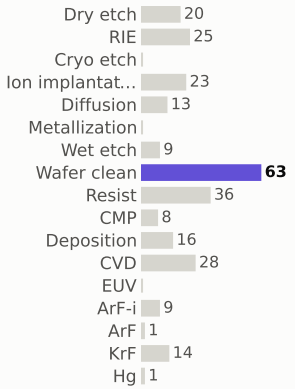
y cut at 57.351 µm



mask for this step

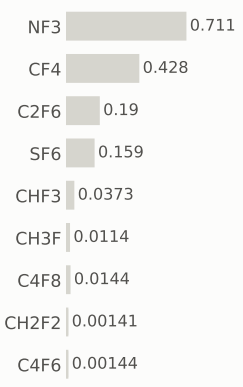
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



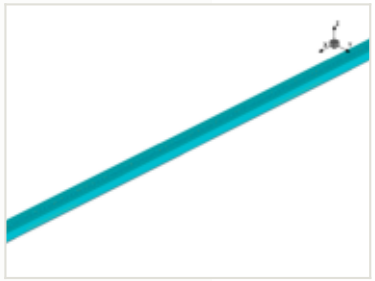
Step #3610: Wafer_clean

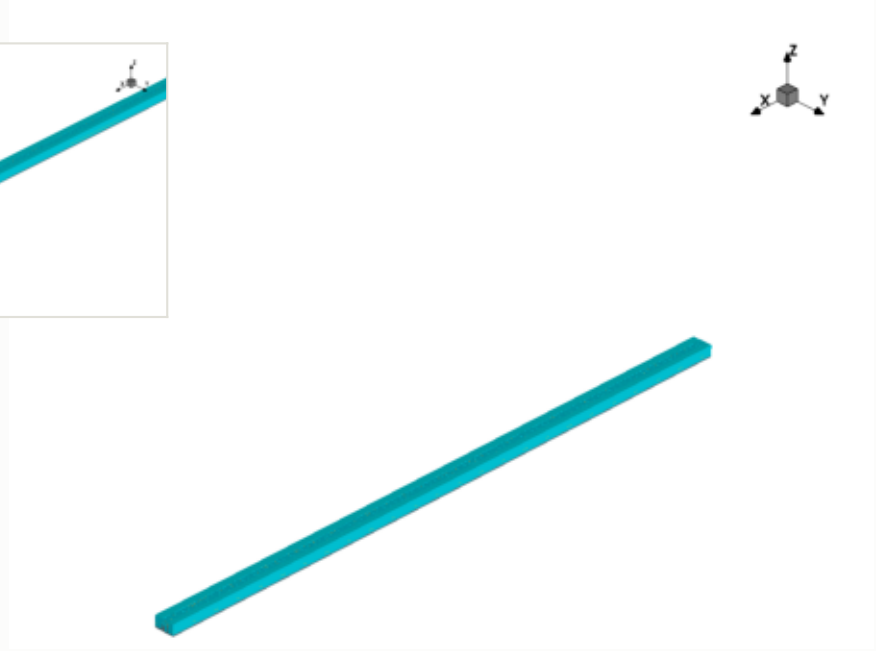
clean · Wafer_clean

0.169	168.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



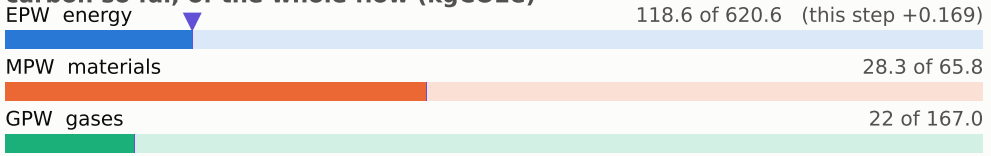


this step

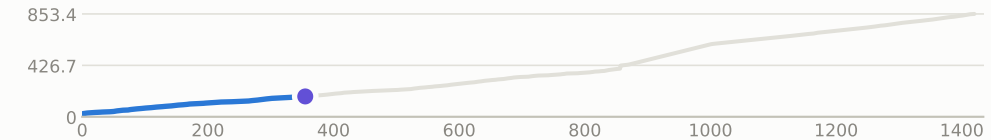
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

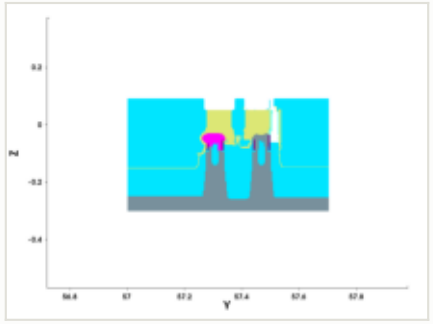
carbon so far, of the whole flow (kgCO2e)



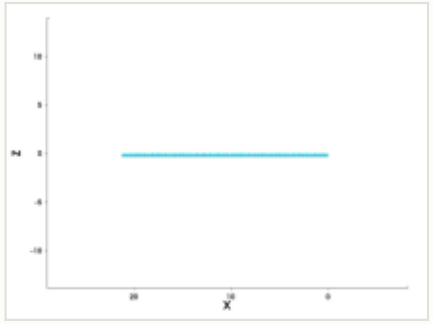
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



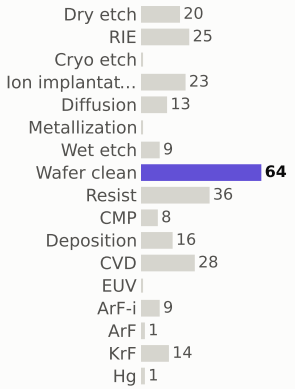
y cut at 57.351 µm



mask for this step

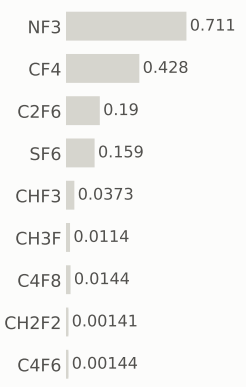
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



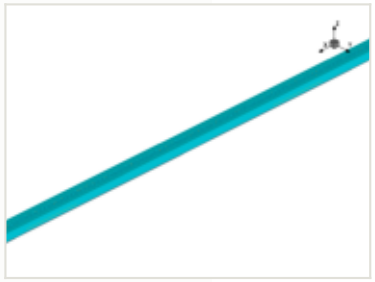
Step #3620: Wafer_clean

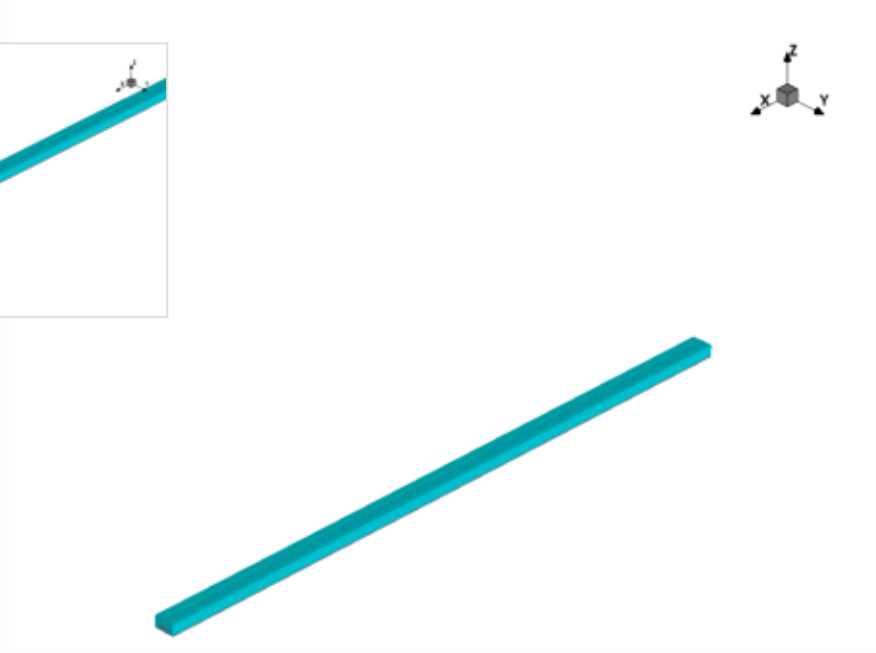
clean · Wafer_clean

0.169	169.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





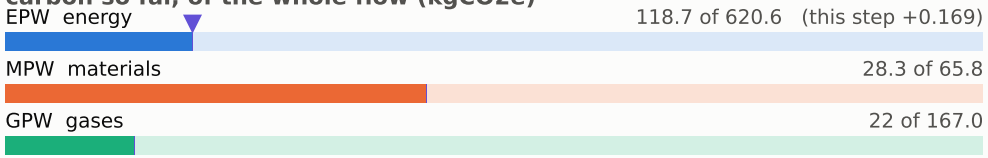
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

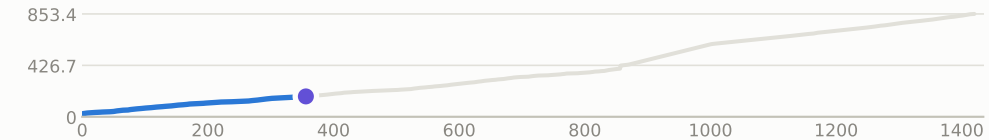
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

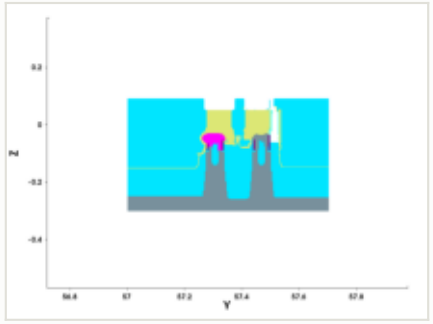
carbon so far, of the whole flow (kgCO2e)



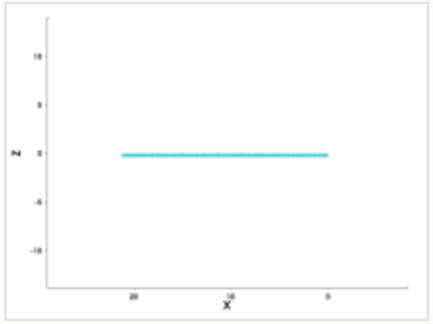
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



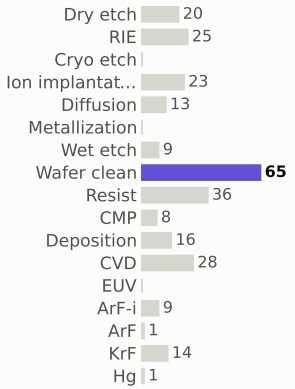
y cut at 57.351 µm



mask for this step

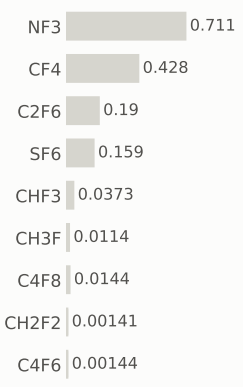
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



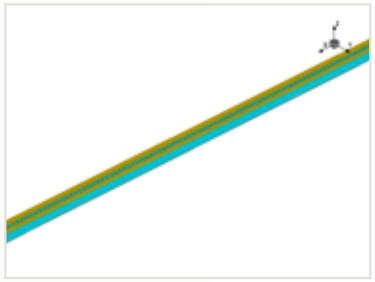
Step #3630: Deposition

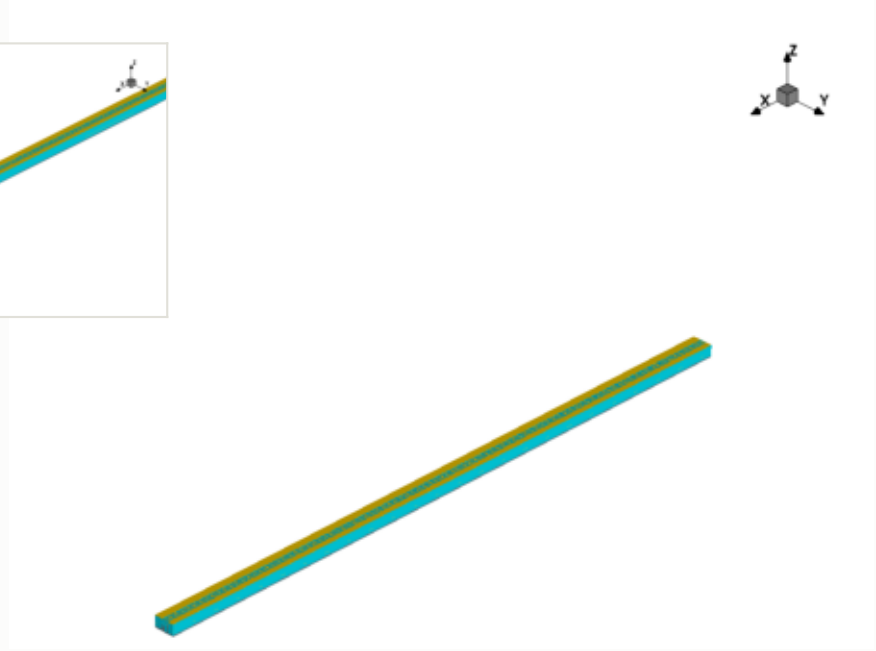
depo_isotropic · Deposition

0.656	169.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



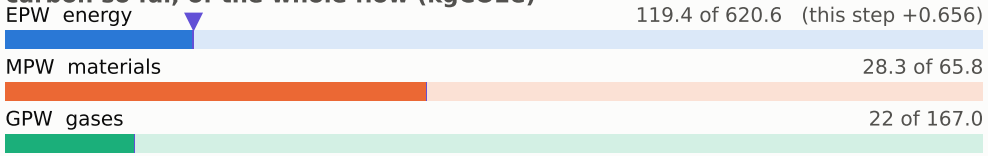


this step

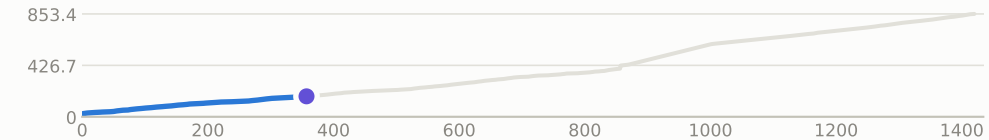
process area	Deposition
material	—
thickness	0.0015 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

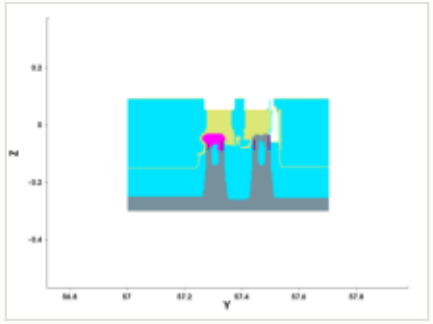
carbon so far, of the whole flow (kgCO2e)



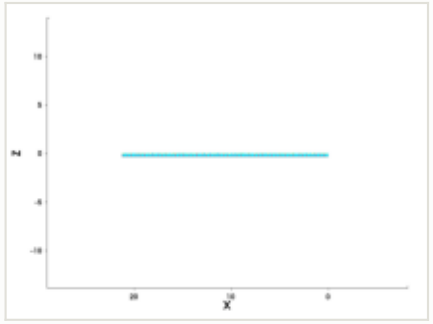
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



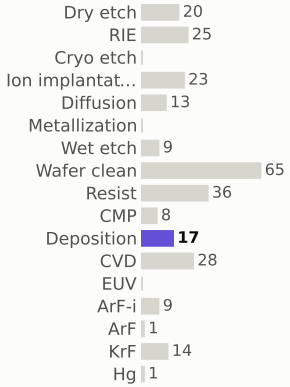
y cut at 57.351 µm



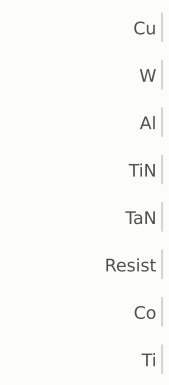
mask for this step

no mask for this step

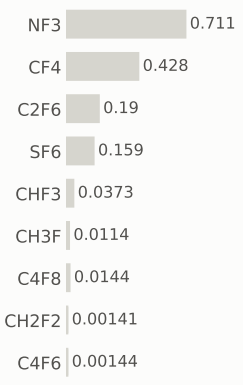
steps by process area



materials deposited (g)



gases emitted (g)



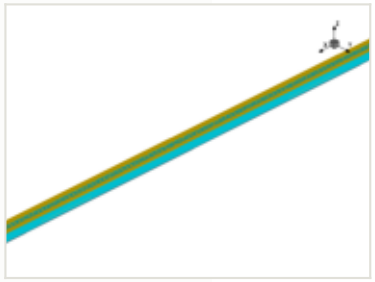
Step #3640: Diffusion

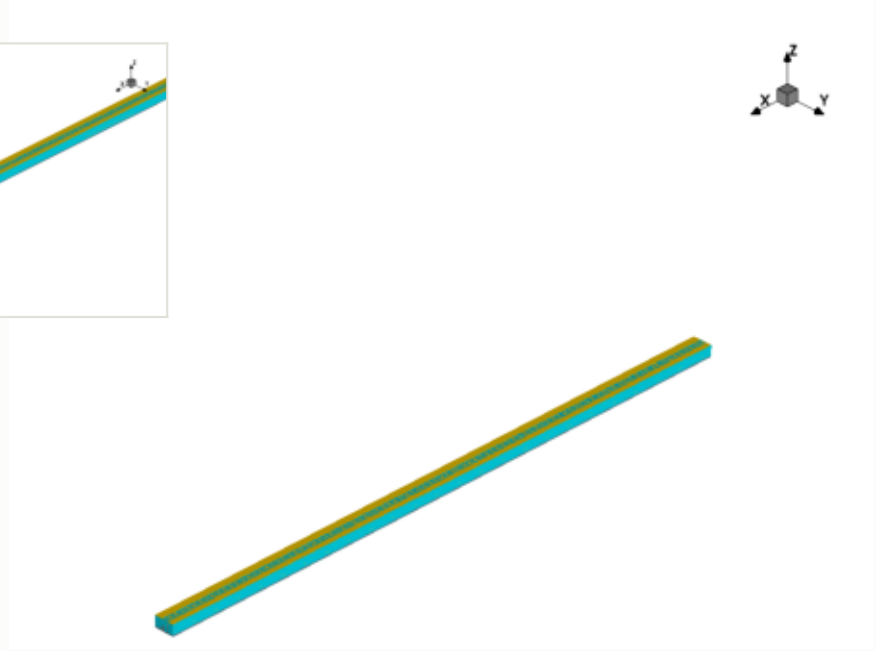
anneal · Diffusion

0.801	170.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





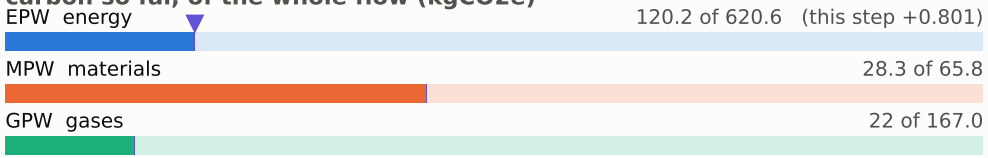
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

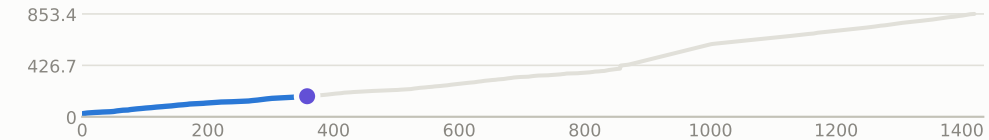
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

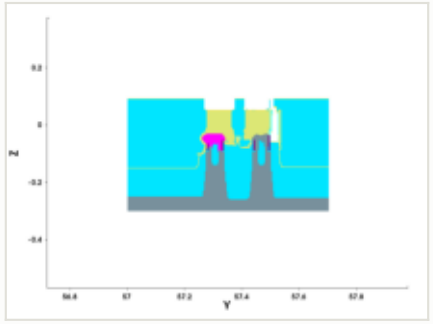
carbon so far, of the whole flow (kgCO2e)



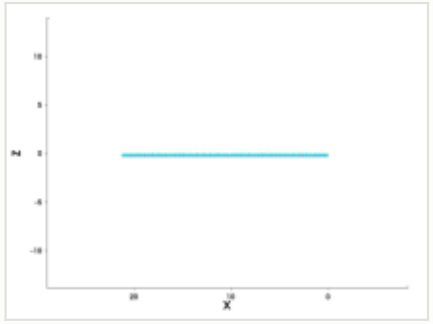
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



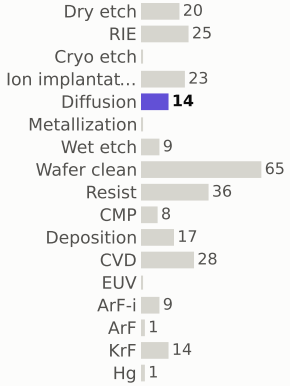
y cut at 57.351 µm



mask for this step

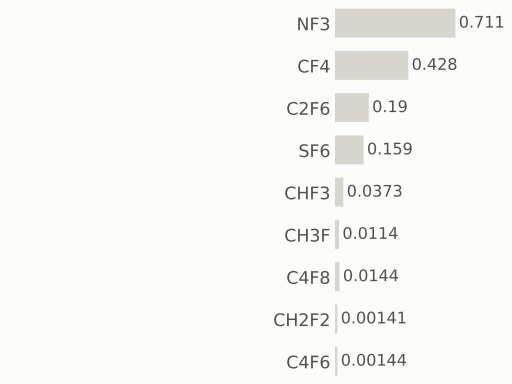
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



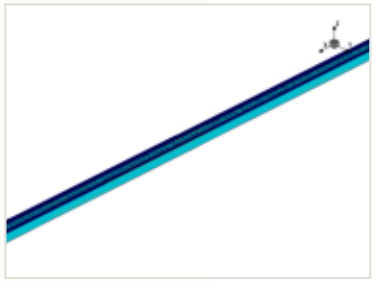
Step #3650: Deposition

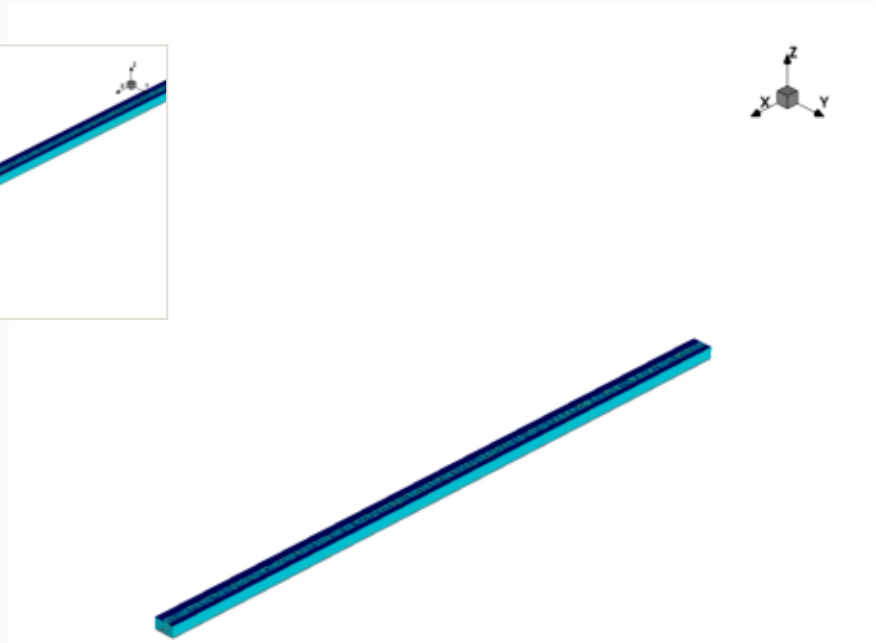
depo_isotropic · Deposition

0.656	171.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



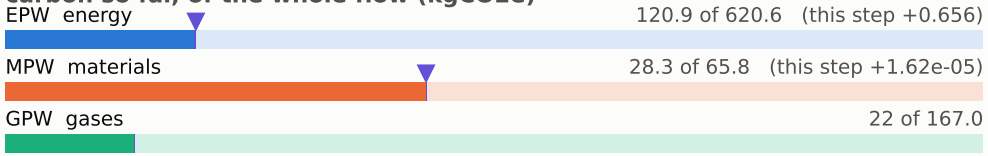


this step

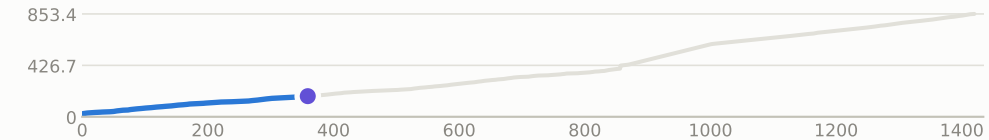
process area	Deposition
material	TiN
thickness	0.001 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	1.62e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

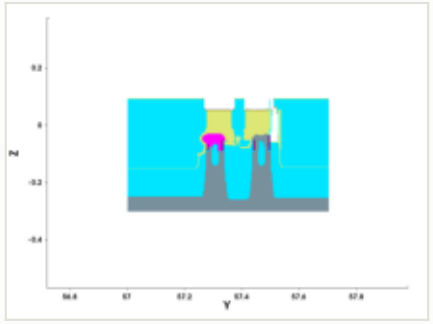
carbon so far, of the whole flow (kgCO2e)



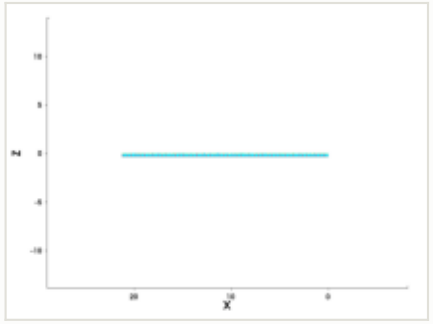
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



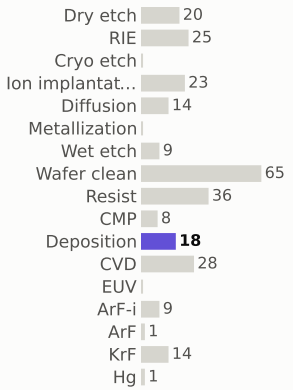
y cut at 57.351 µm



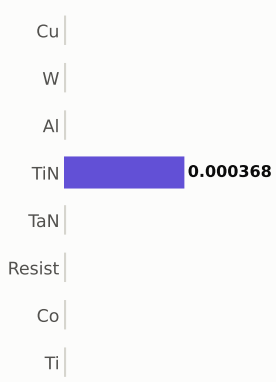
mask for this step

no mask for this step

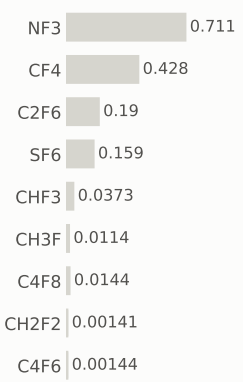
steps by process area



materials deposited (g)



gases emitted (g)



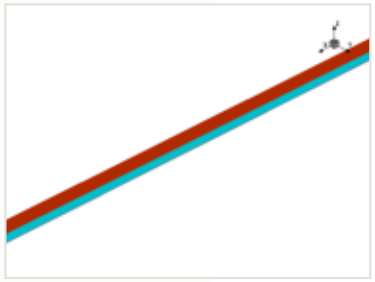
Step #3660: Spx_only

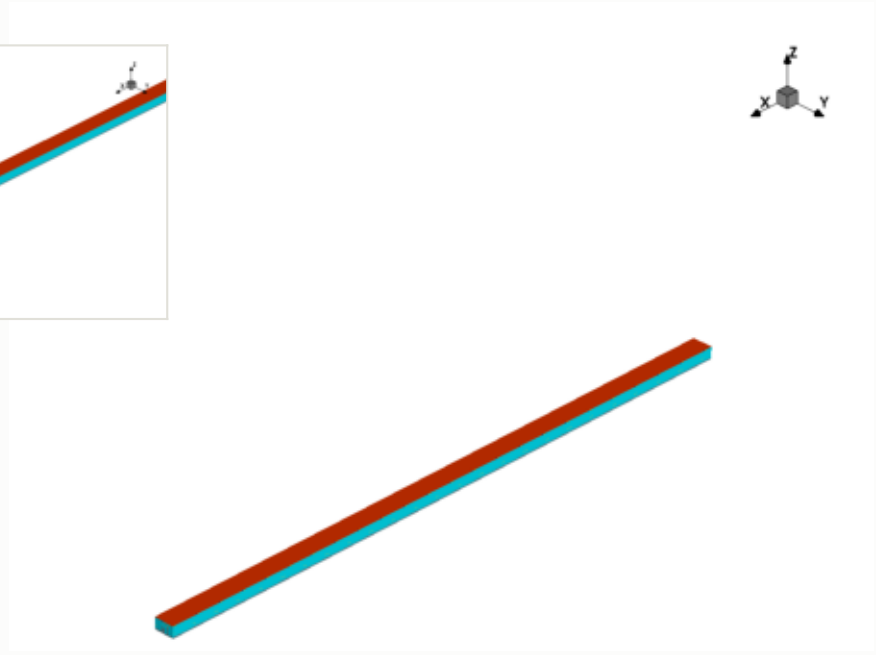
depo_fill · Spx_only

0	171.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





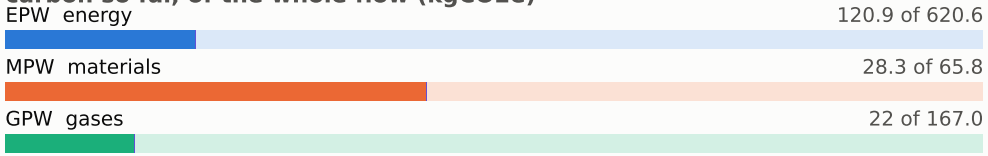
this step

process area	Spx_only
material	—
thickness	0.01 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

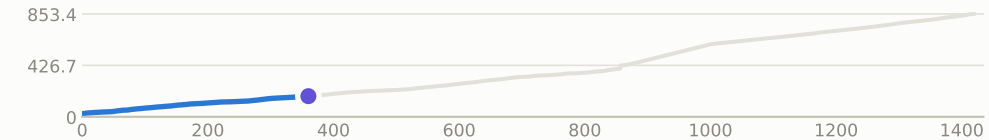
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

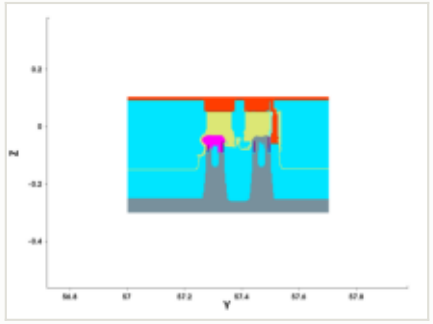
carbon so far, of the whole flow (kgCO2e)



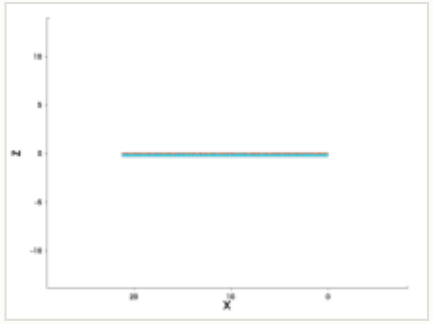
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



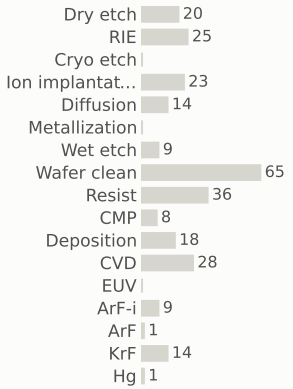
y cut at 57.351 µm



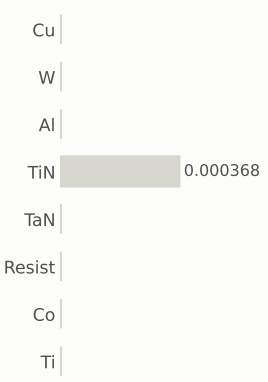
mask for this step

no mask for this step

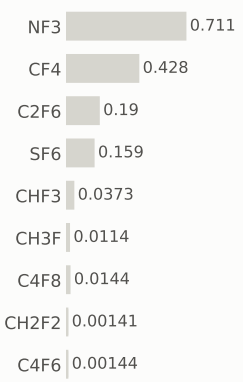
steps by process area



materials deposited (g)



gases emitted (g)



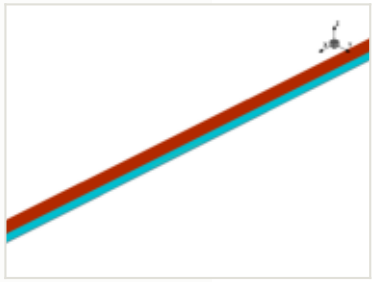
Step #3670: Wafer_clean

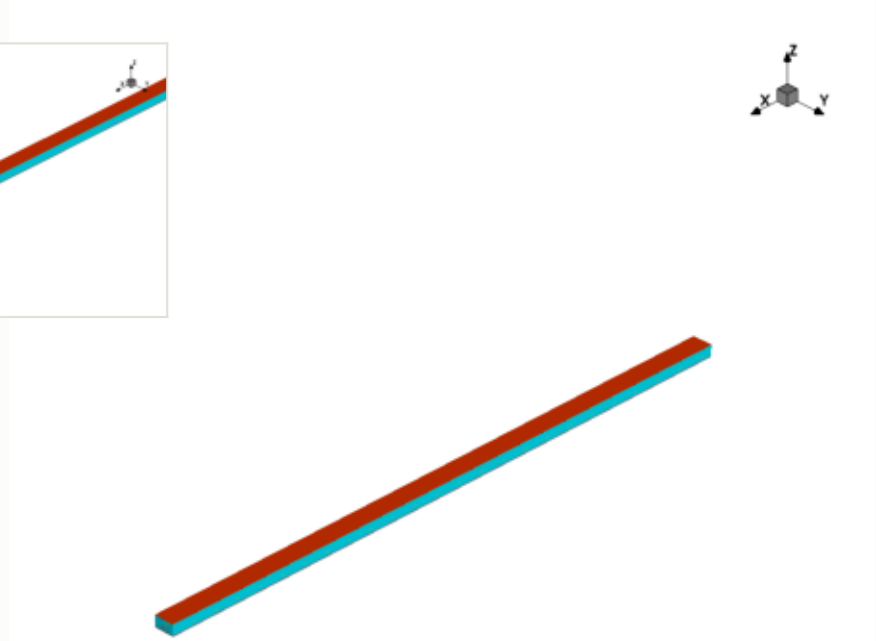
clean · Wafer_clean

0.169	171.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

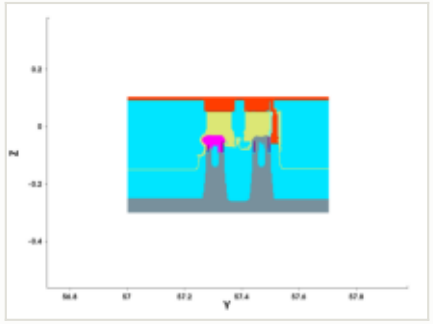
3D structure after this step

zoomed to fit

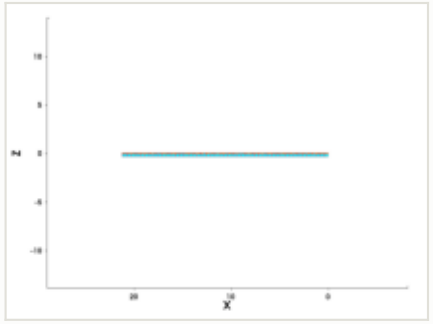




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

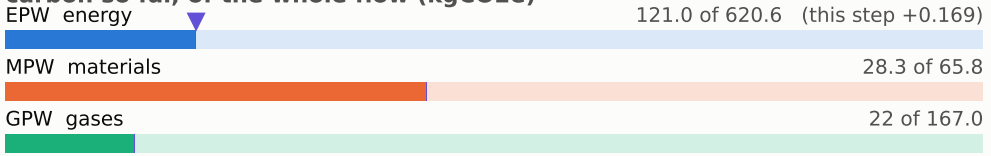
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

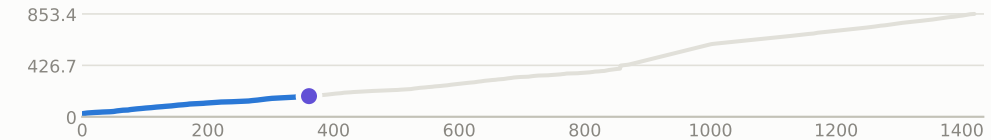
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

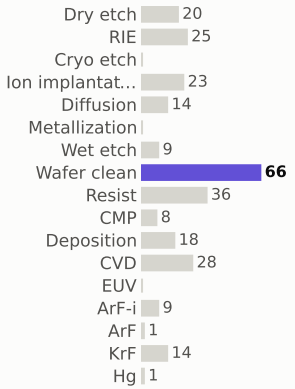
carbon so far, of the whole flow (kgCO2e)



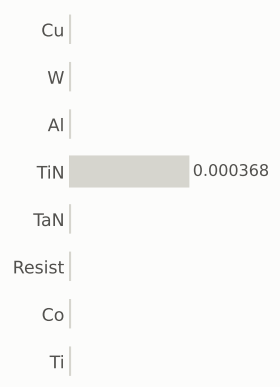
embodied carbon up to each step (kgCO2e per wafer)



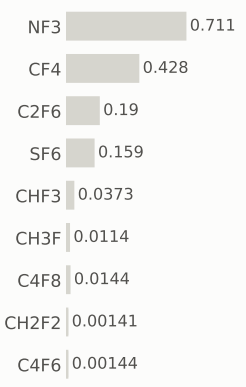
steps by process area



materials deposited (g)



gases emitted (g)



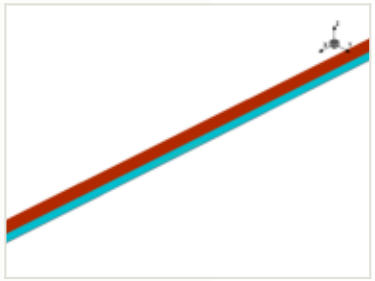
Step #3680: CVD

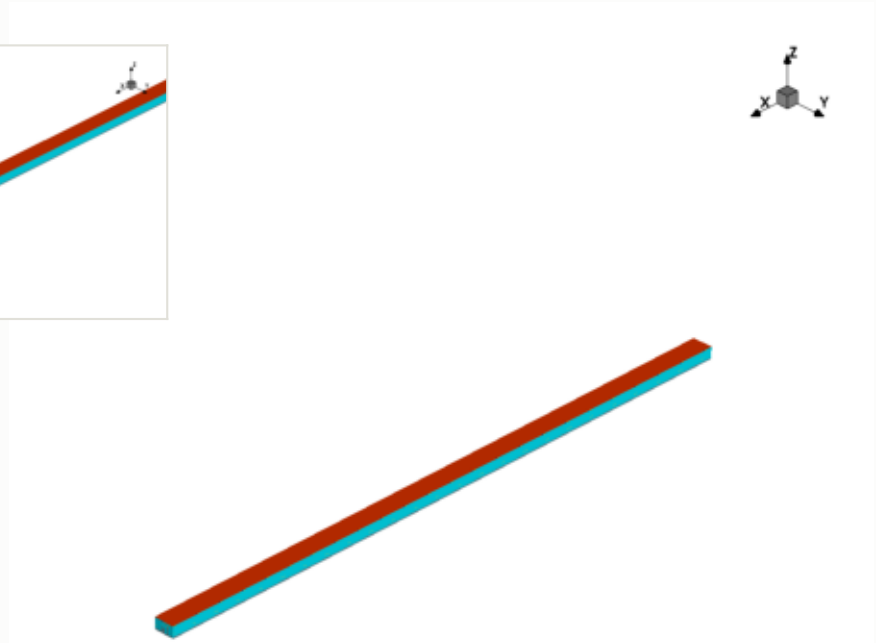
depo_isotropic · CVD

1.31	172.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



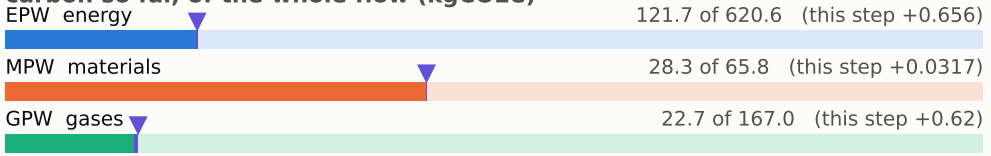


this step

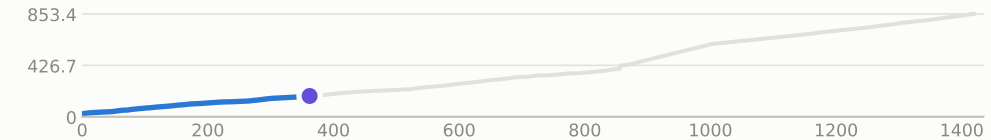
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

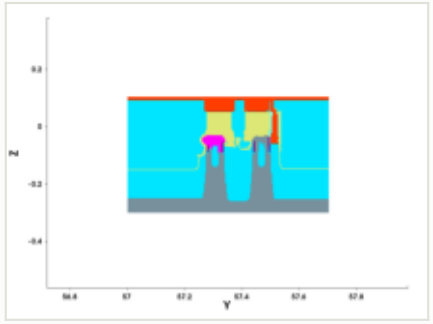
carbon so far, of the whole flow (kgCO2e)



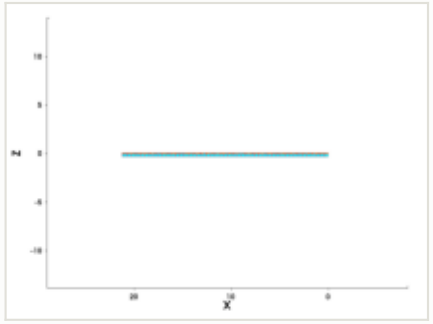
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



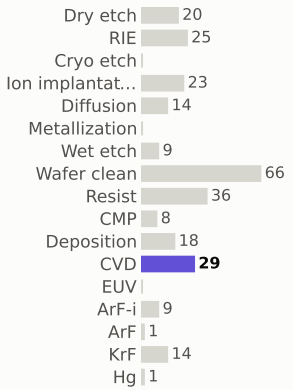
y cut at 57.351 µm



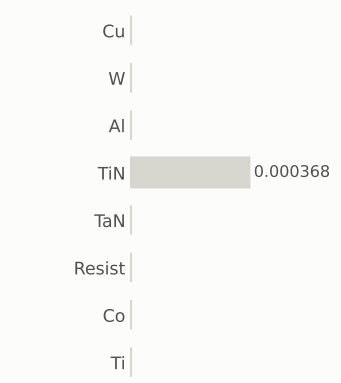
mask for this step

no mask for this step

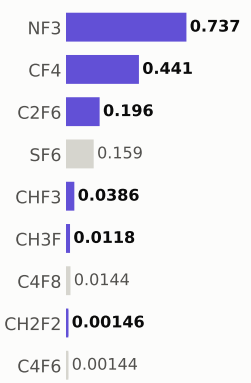
steps by process area



materials deposited (g)



gases emitted (g)



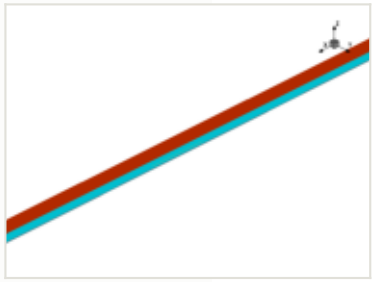
Step #3690: Wafer_clean

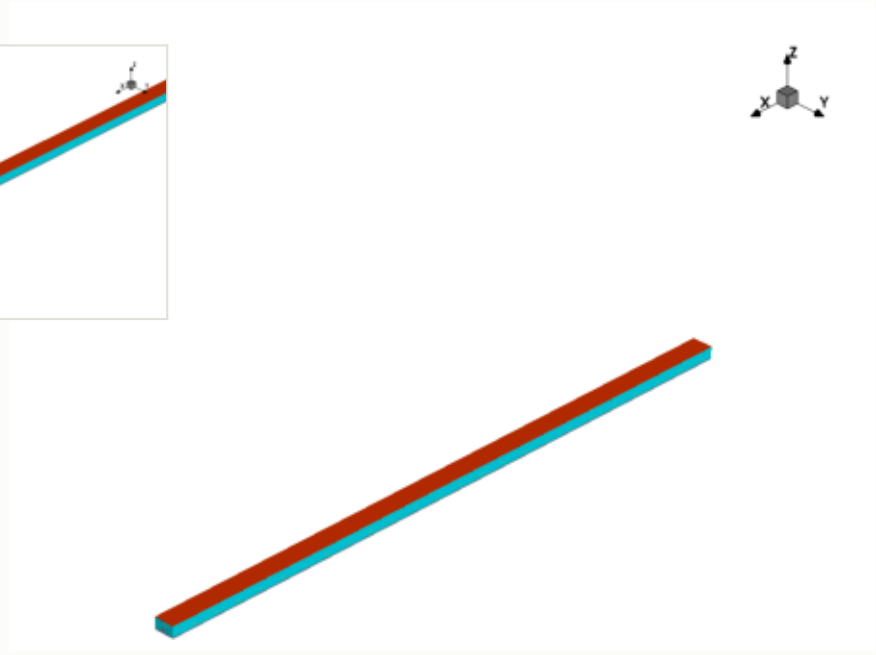
clean · Wafer_clean

0.169	172.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

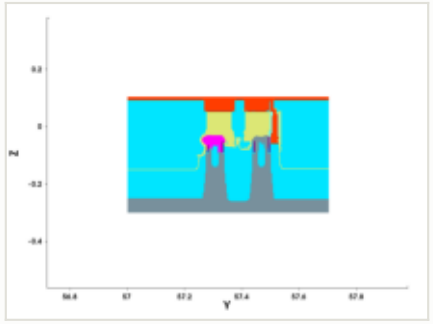
3D structure after this step

zoomed to fit

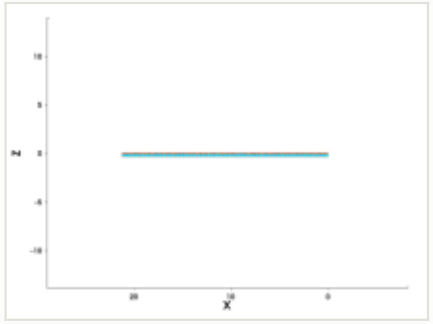




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

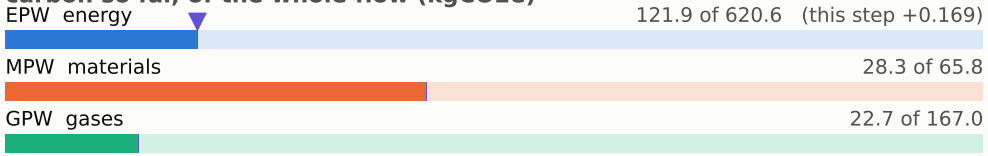
no mask for this step

this step

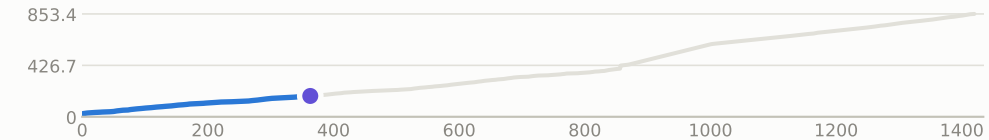
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

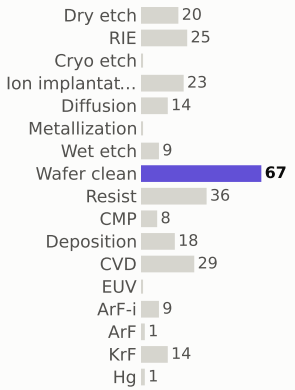
carbon so far, of the whole flow (kgCO2e)



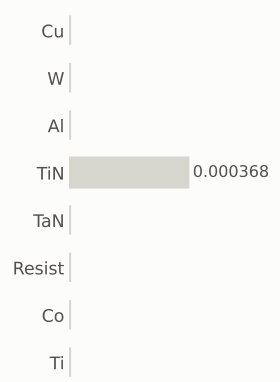
embodied carbon up to each step (kgCO2e per wafer)



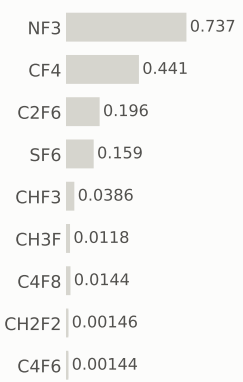
steps by process area



materials deposited (g)



gases emitted (g)



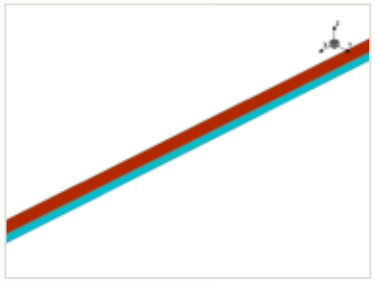
Step #3700: Diffusion

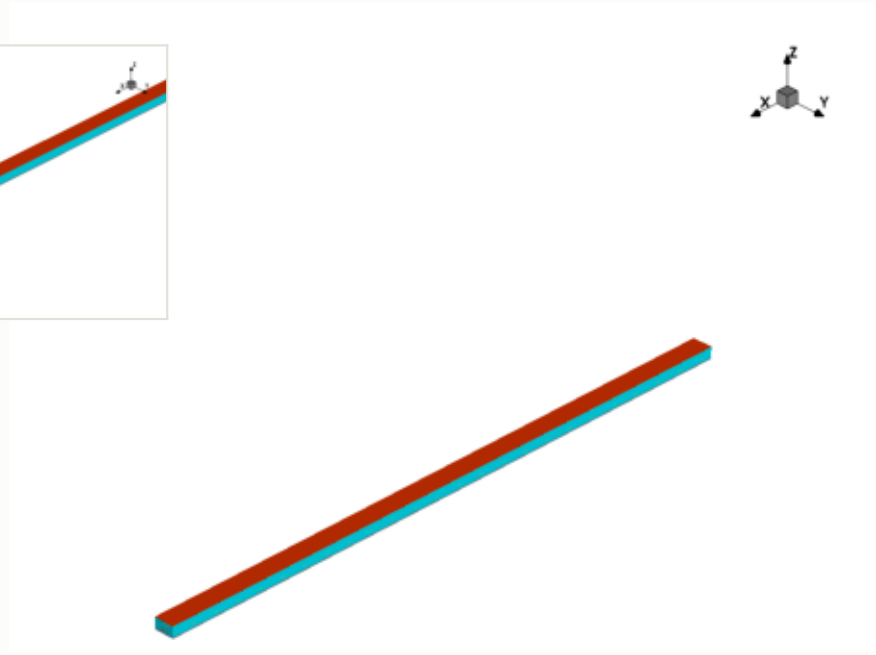
anneal · Diffusion

0.801	173.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





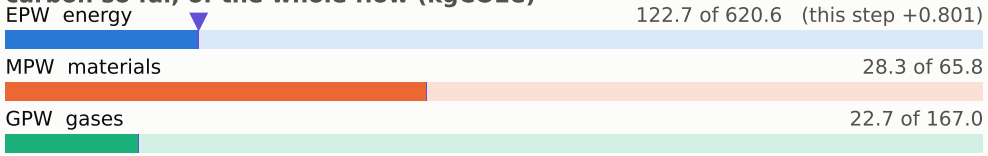
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

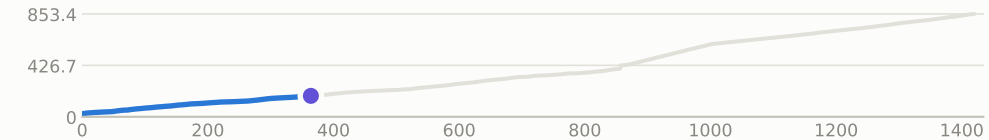
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

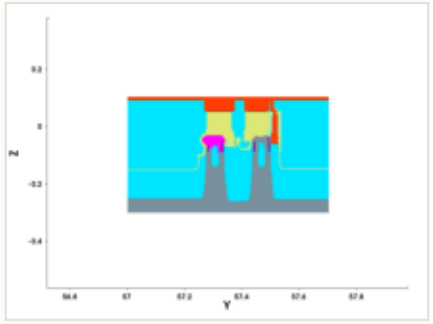
carbon so far, of the whole flow (kgCO2e)



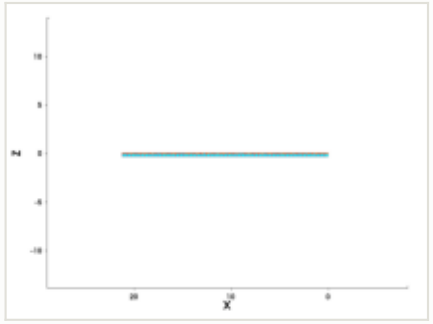
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



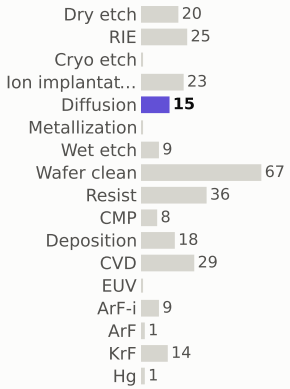
y cut at 57.351 µm



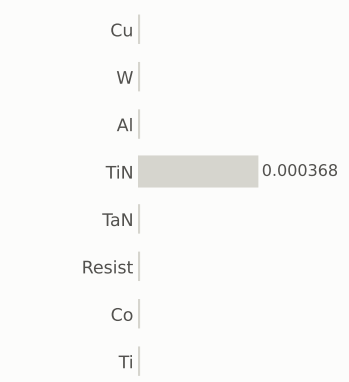
mask for this step

no mask for this step

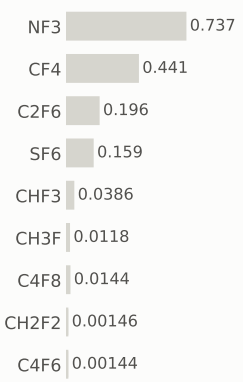
steps by process area



materials deposited (g)



gases emitted (g)



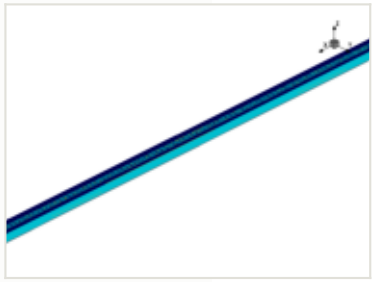
Step #3710: Wet_etch

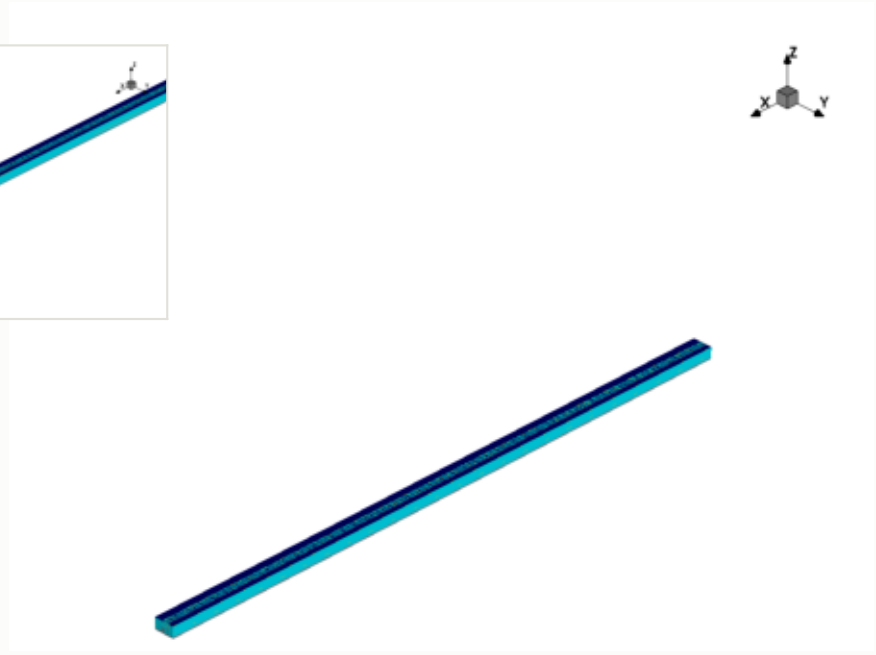
etch_isotropic · Wet_etch

0.169	173.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



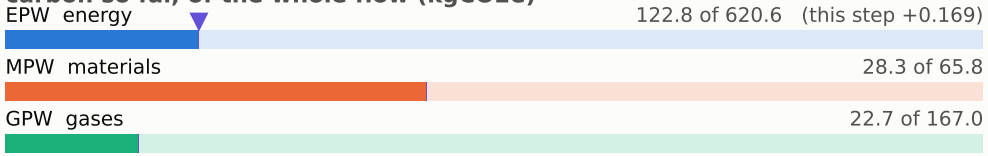


this step

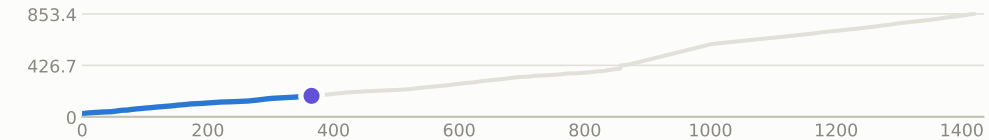
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

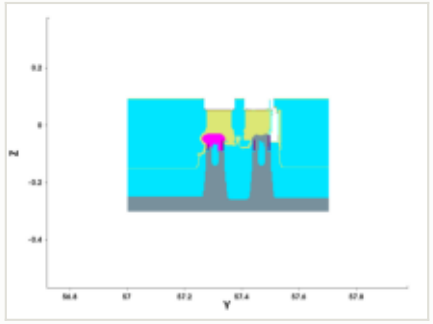
carbon so far, of the whole flow (kgCO2e)



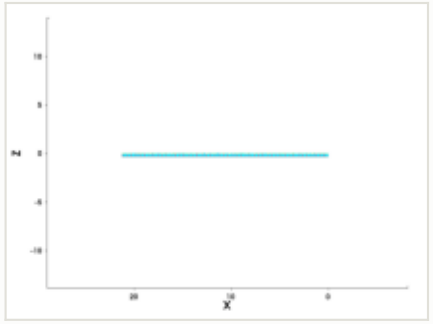
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



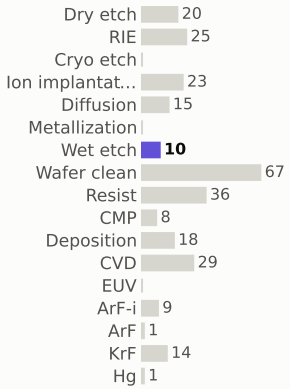
y cut at 57.351 µm



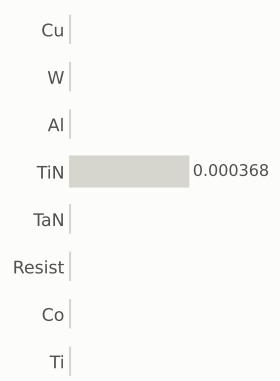
mask for this step

no mask for this step

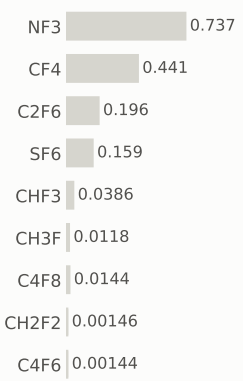
steps by process area



materials deposited (g)



gases emitted (g)



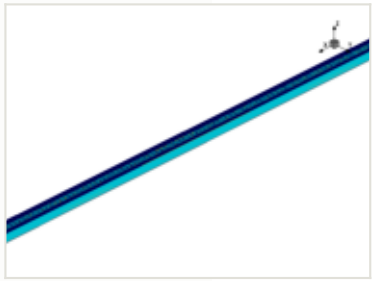
Step #3720: Spx_only

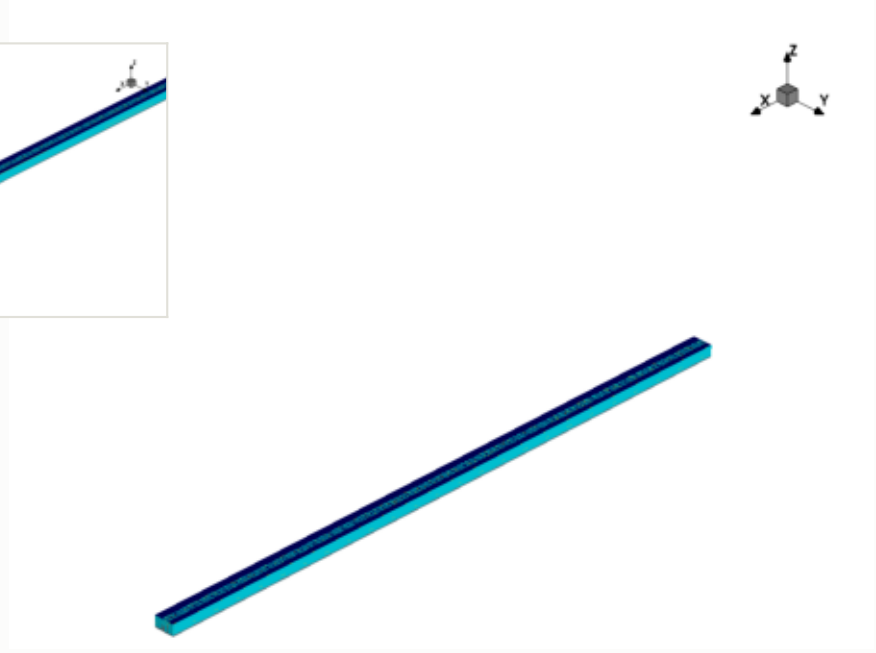
depo_isotropic · Spx_only

1.62e-05	173.8	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

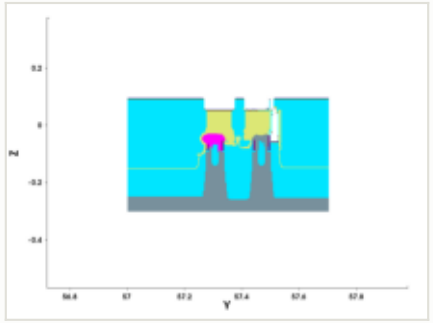
3D structure after this step

zoomed to fit

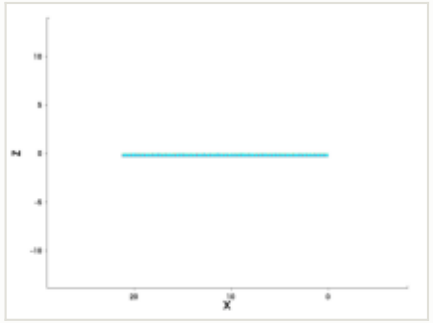




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

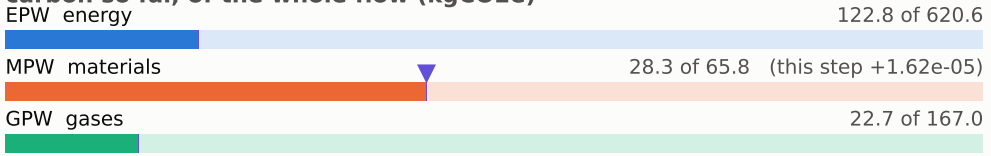
no mask for this step

this step

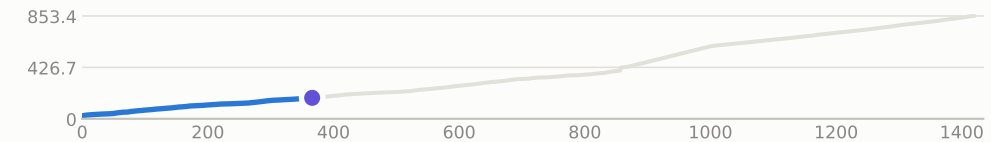
process area	Spx_only	EPW	0 kgCO2e
material	TiN	EPW tool	0 kgCO2e
thickness	0.001 µm	EPW overhead	0 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	1.62e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**1.62e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

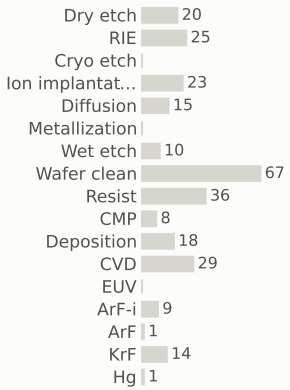
carbon so far, of the whole flow (kgCO2e)



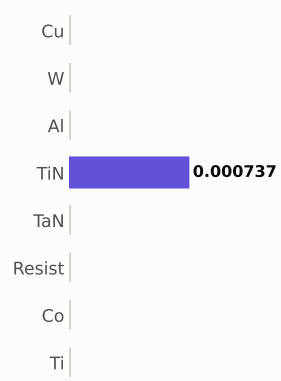
embodied carbon up to each step (kgCO2e per wafer)



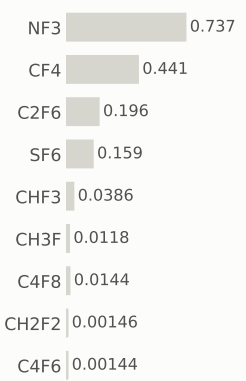
steps by process area



materials deposited (g)



gases emitted (g)



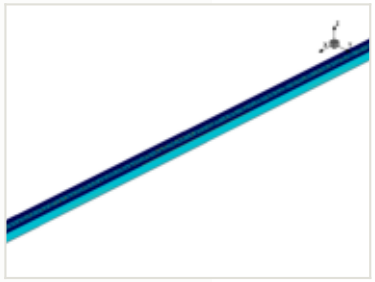
Step #3730: Wafer_clean

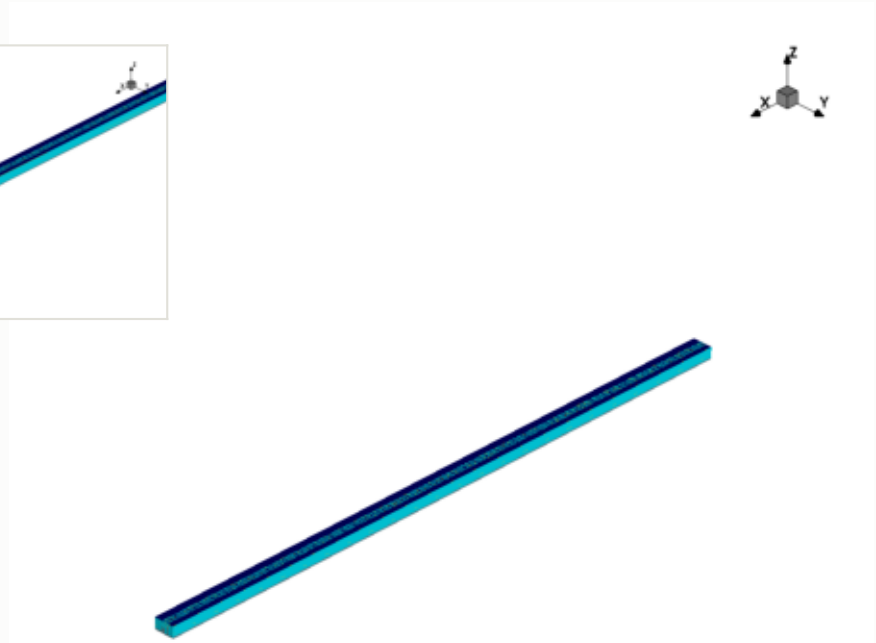
clean · Wafer_clean

0.169	174.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





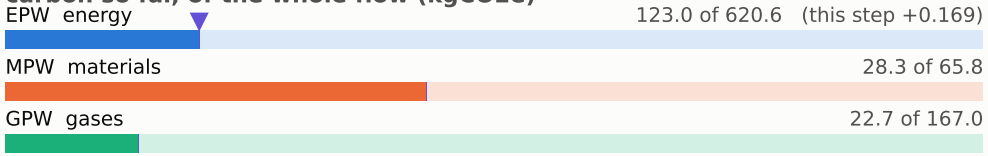
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

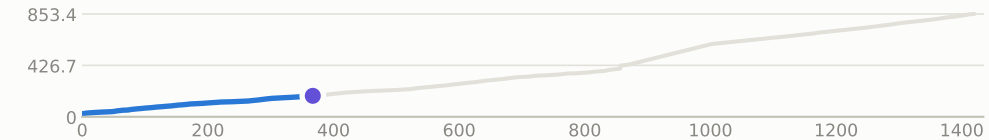
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

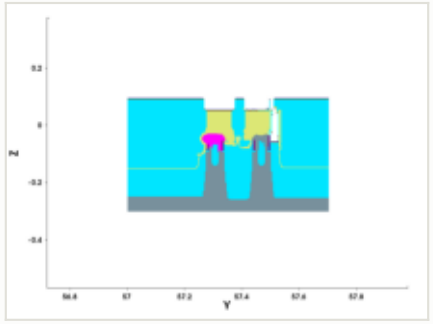
carbon so far, of the whole flow (kgCO2e)



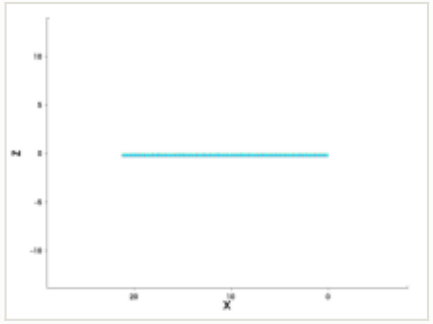
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



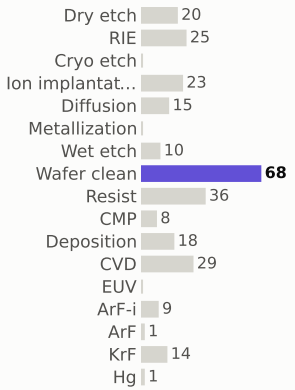
y cut at 57.351 µm



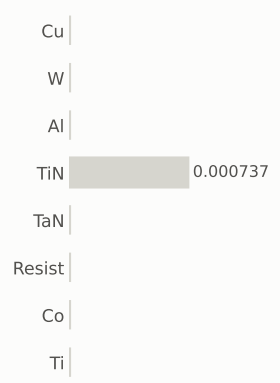
mask for this step

no mask for this step

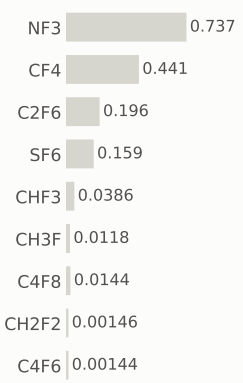
steps by process area



materials deposited (g)



gases emitted (g)



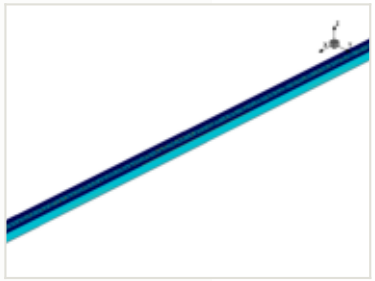
Step #3740: Deposition

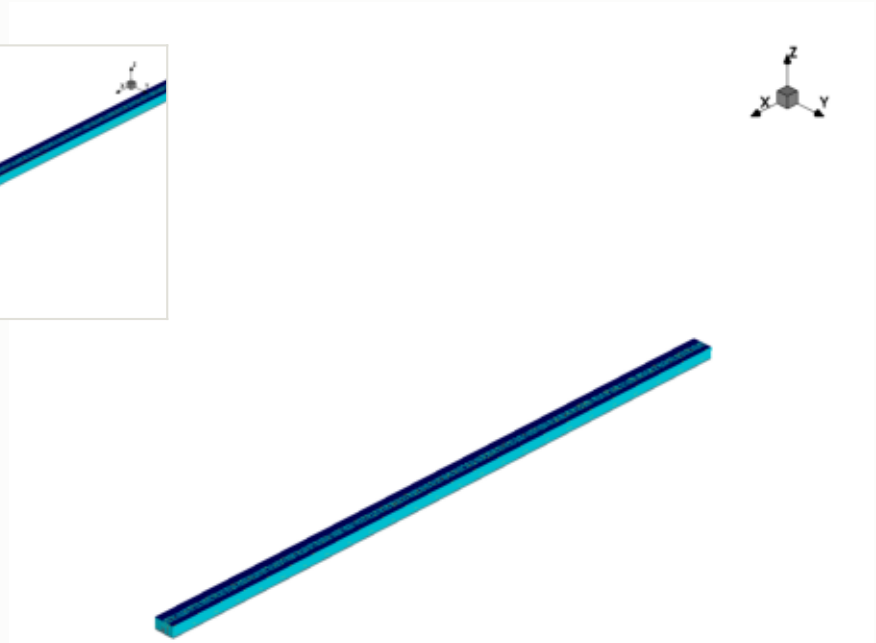
depo_isotropic · Deposition

0.656	174.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

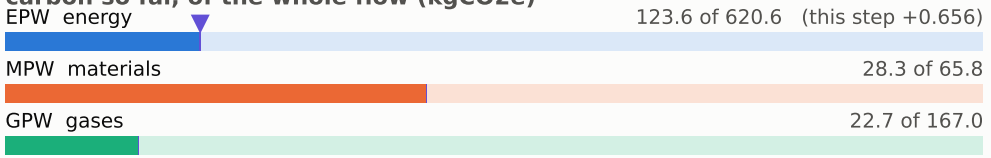




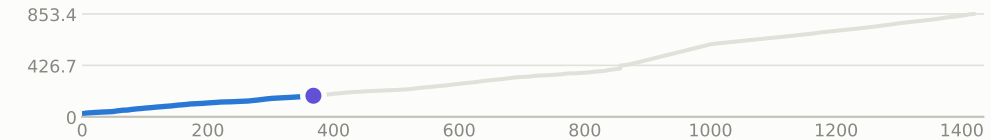
this step

process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

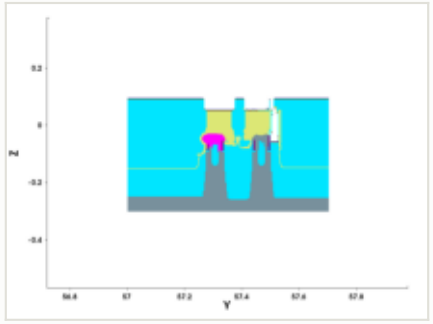
carbon so far, of the whole flow (kgCO2e)



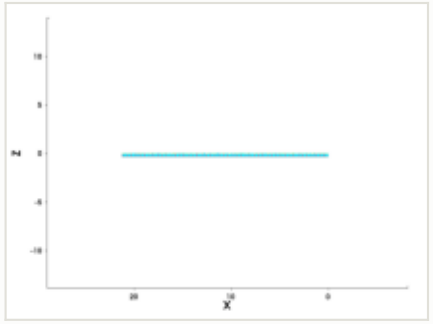
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



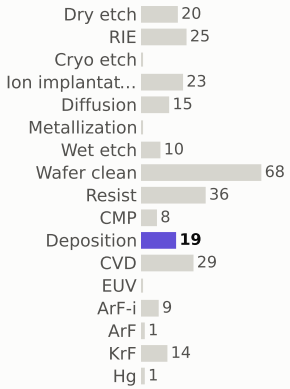
y cut at 57.351 µm



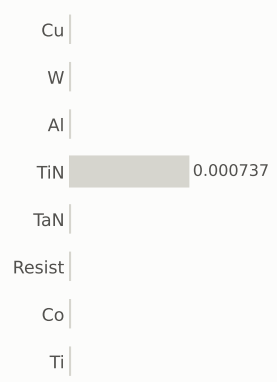
mask for this step

no mask for this step

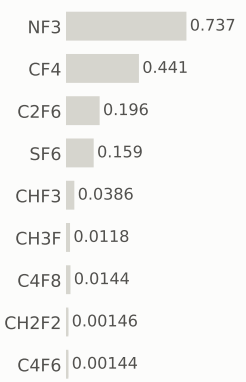
steps by process area



materials deposited (g)



gases emitted (g)



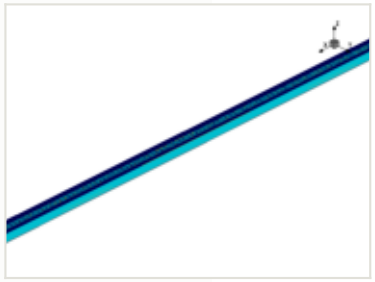
Step #3750: Wafer_clean

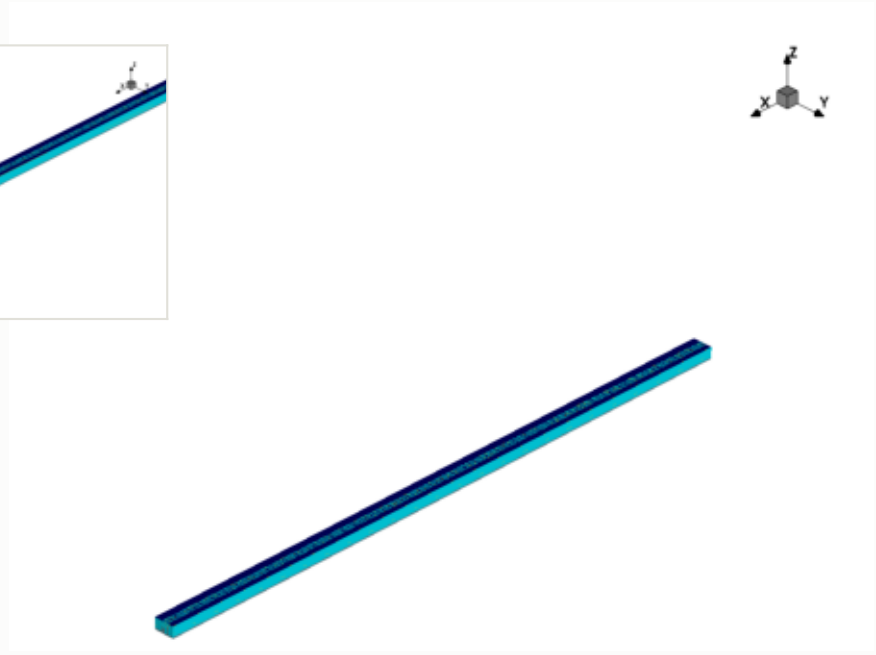
clean · Wafer_clean

0.169	174.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

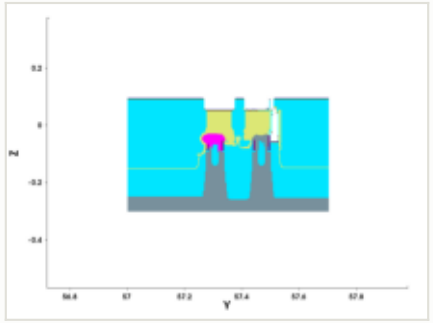
3D structure after this step

zoomed to fit

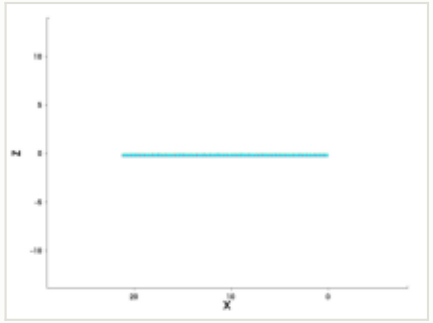




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

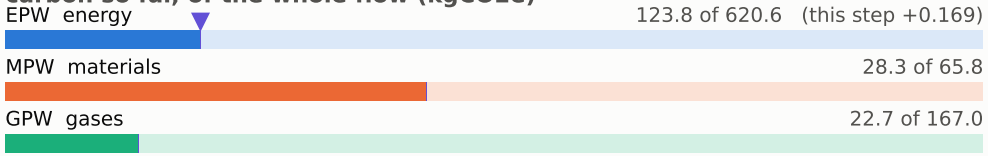
no mask for this step

this step

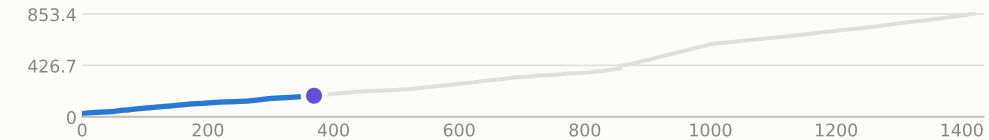
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

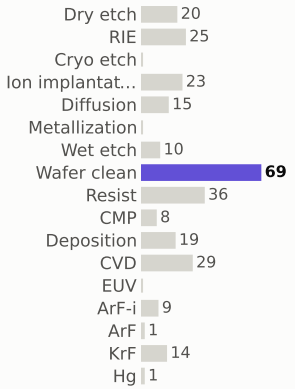
carbon so far, of the whole flow (kgCO2e)



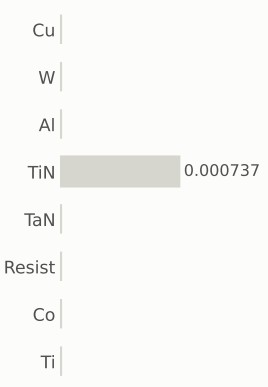
embodied carbon up to each step (kgCO2e per wafer)



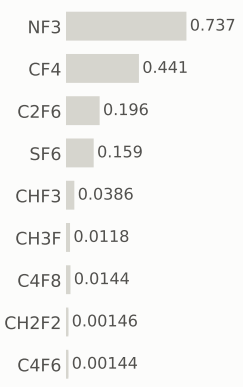
steps by process area



materials deposited (g)



gases emitted (g)



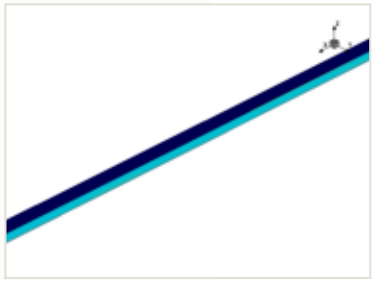
Step #3760: Deposition

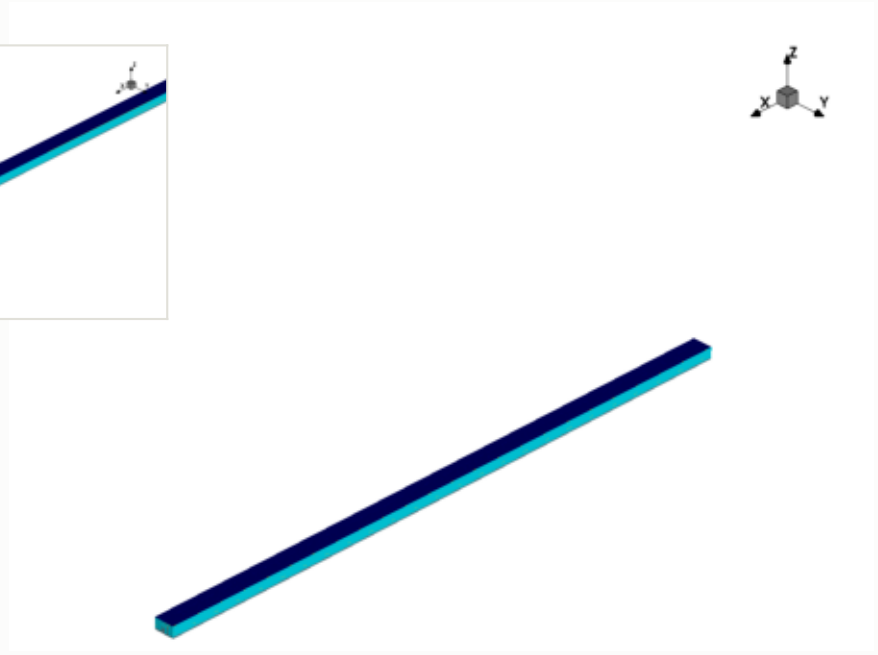
depo_isotropic · Deposition

0.656	175.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



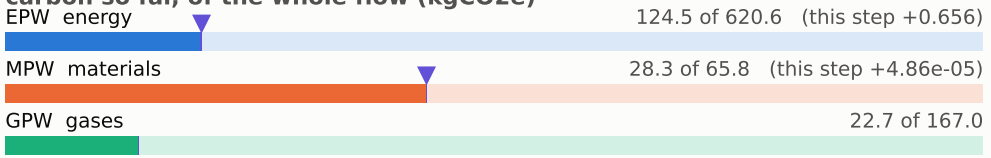


this step

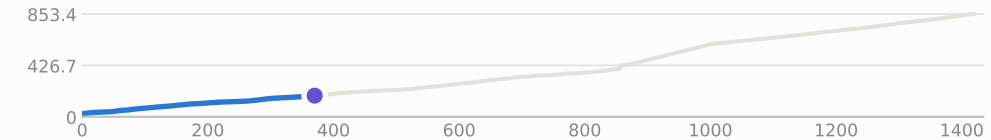
process area	Deposition
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

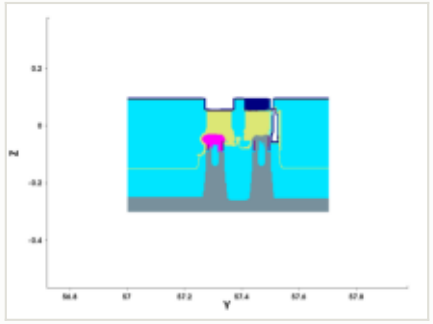
carbon so far, of the whole flow (kgCO2e)



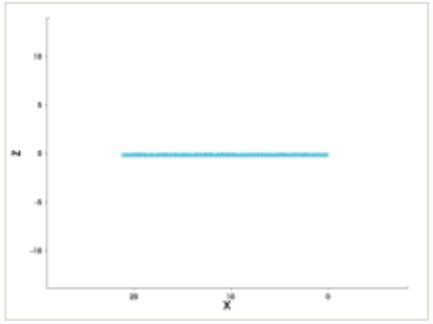
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



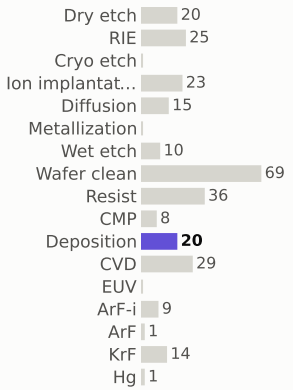
y cut at 57.351 µm



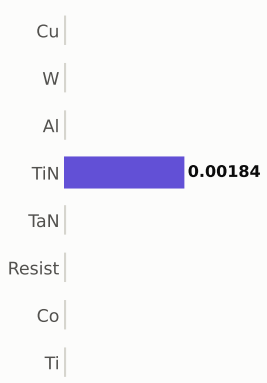
mask for this step

no mask for this step

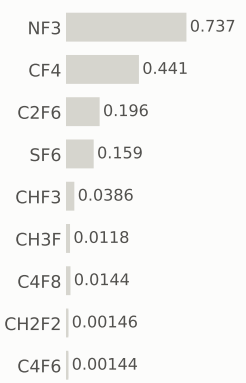
steps by process area



materials deposited (g)



gases emitted (g)



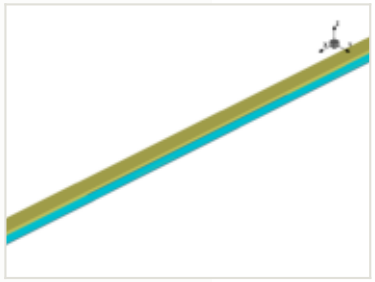
Step #3770: Spx_only

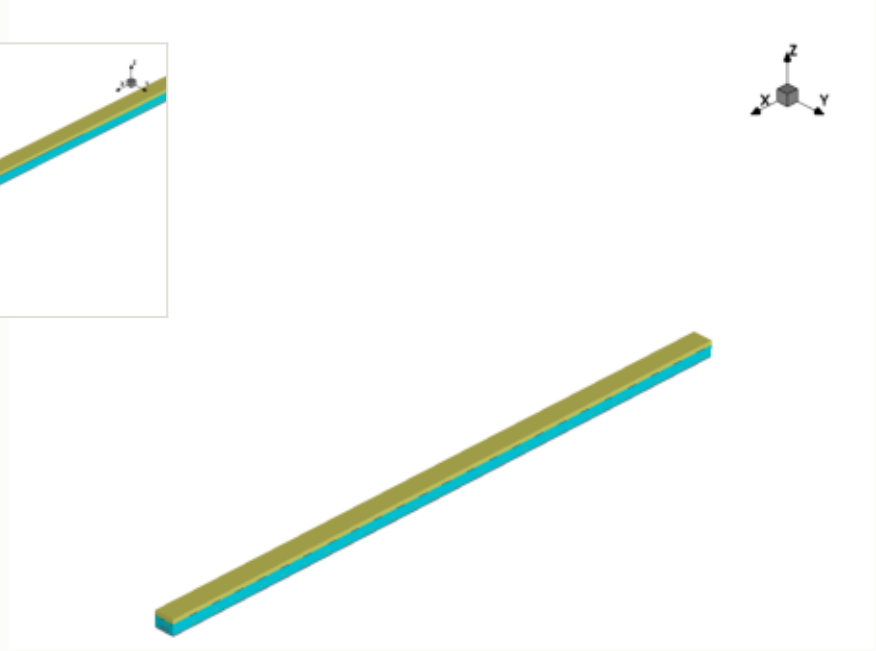
depo_fill · Spx_only

4.32e-05	175.5	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

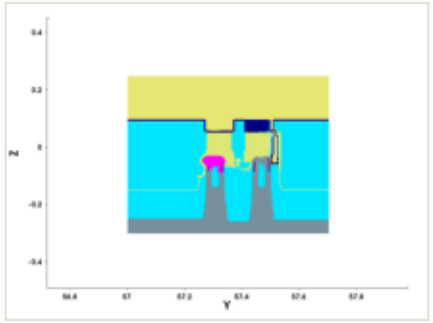
3D structure after this step

zoomed to fit

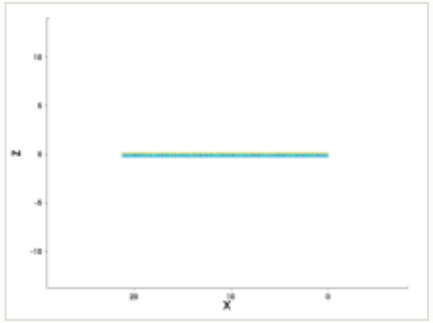




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

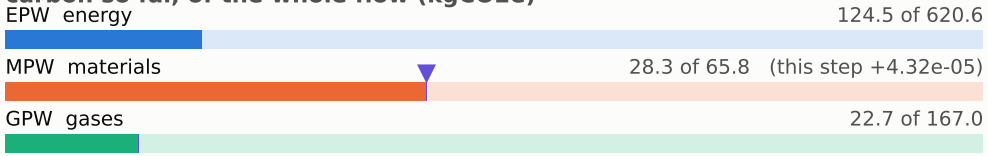
no mask for this step

this step

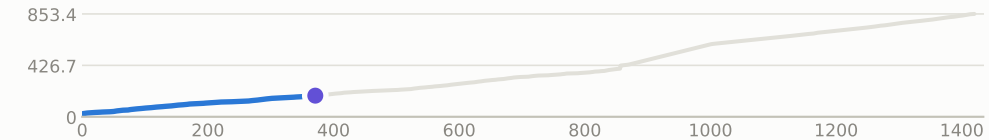
process area	Spx_only	EPW	0 kgCO2e
material	Resist	EPW tool	0 kgCO2e
thickness	0.15 µm	EPW overhead	0 kgCO2e
material GWP	4.2 gCO2e/g	MPW	4.32e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**4.32e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

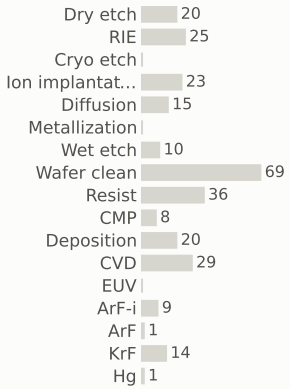
carbon so far, of the whole flow (kgCO2e)



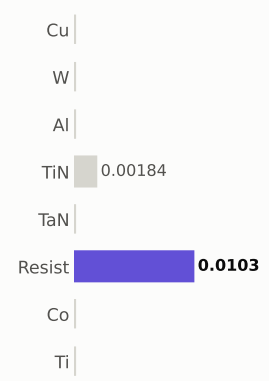
embodied carbon up to each step (kgCO2e per wafer)



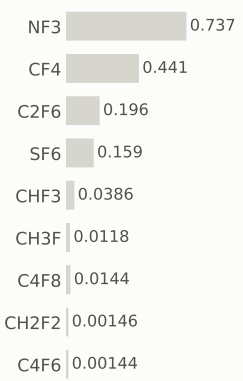
steps by process area



materials deposited (g)



gases emitted (g)



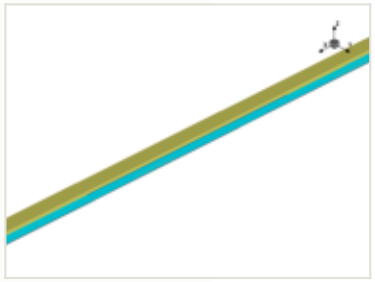
Step #3780: Wafer_clean

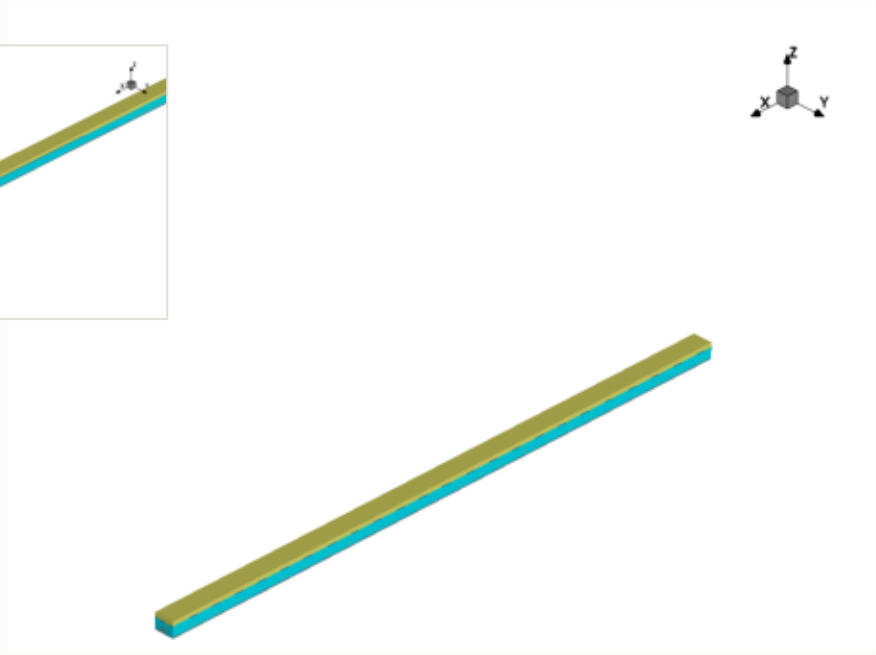
clean · Wafer_clean

0.169	175.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

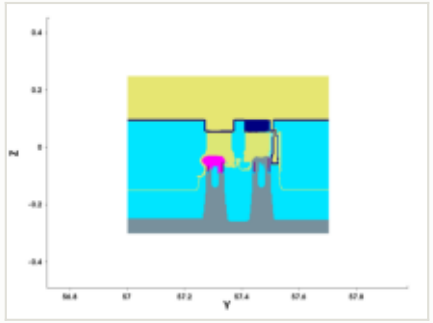
3D structure after this step

zoomed to fit

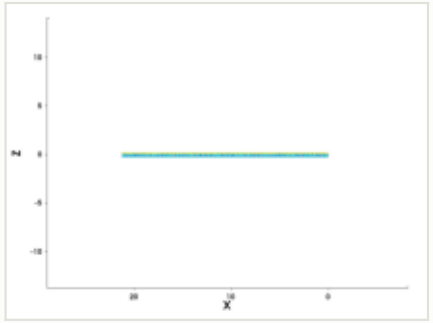




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

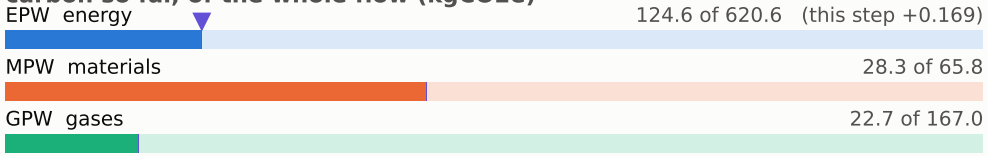
no mask for this step

this step

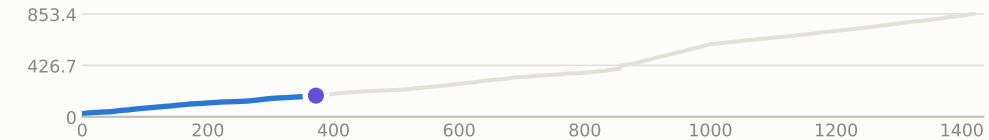
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

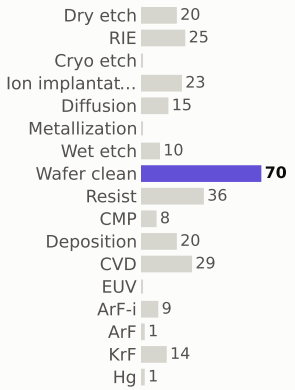
carbon so far, of the whole flow (kgCO2e)



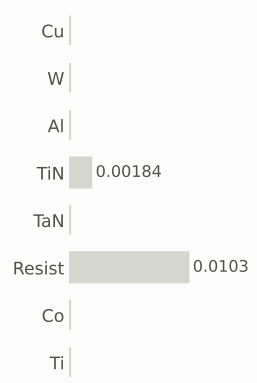
embodied carbon up to each step (kgCO2e per wafer)



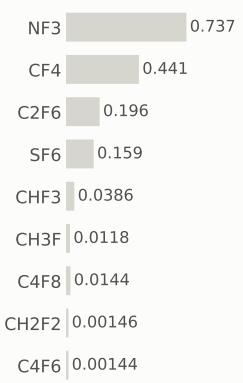
steps by process area



materials deposited (g)



gases emitted (g)



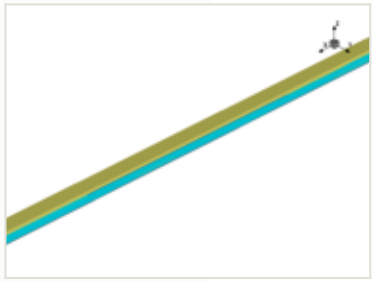
Step #3790: Resist

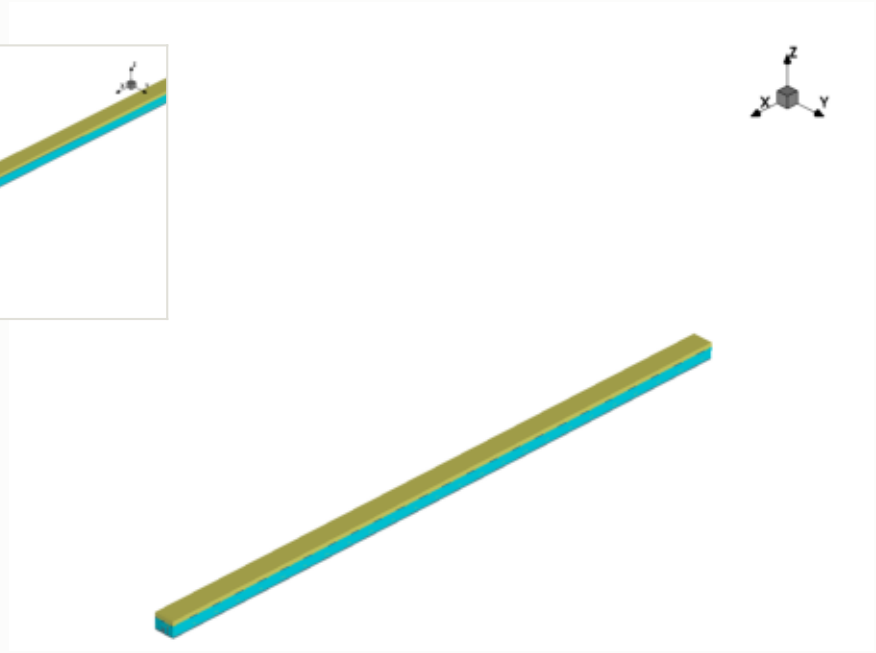
resist · Resist

0.169	175.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





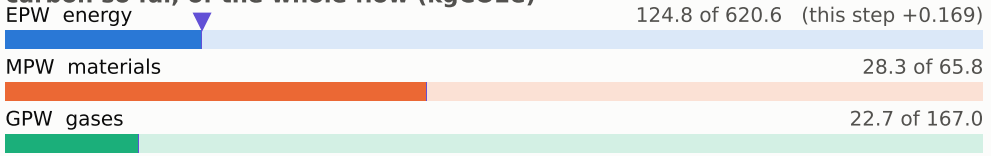
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

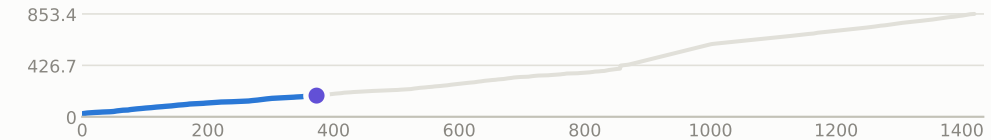
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

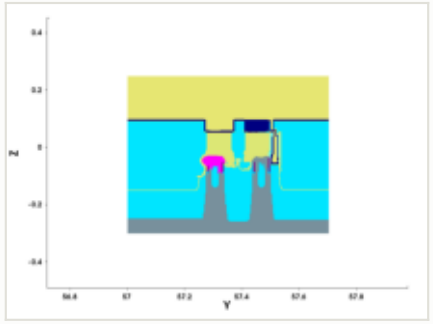
carbon so far, of the whole flow (kgCO2e)



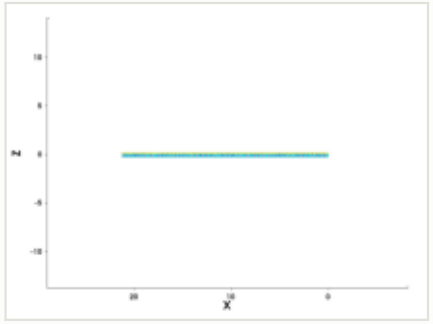
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



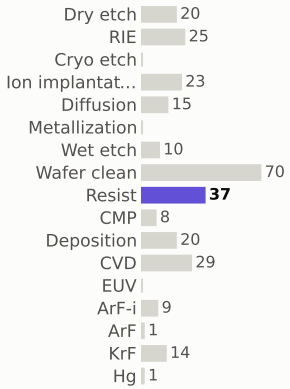
y cut at 57.351 µm



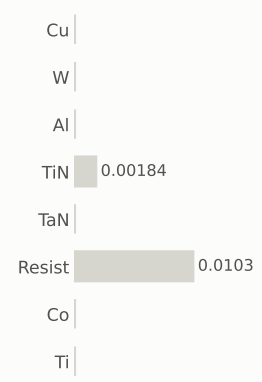
mask for this step

no mask for this step

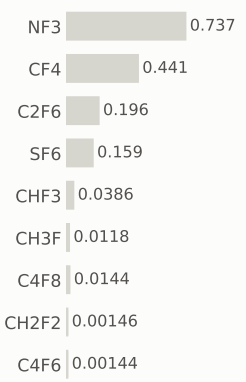
steps by process area



materials deposited (g)



gases emitted (g)



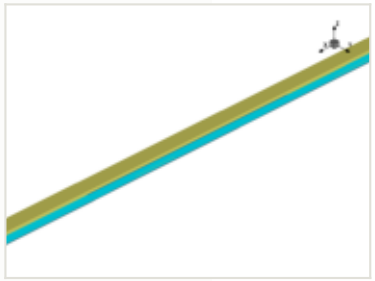
Step #3800: KrF

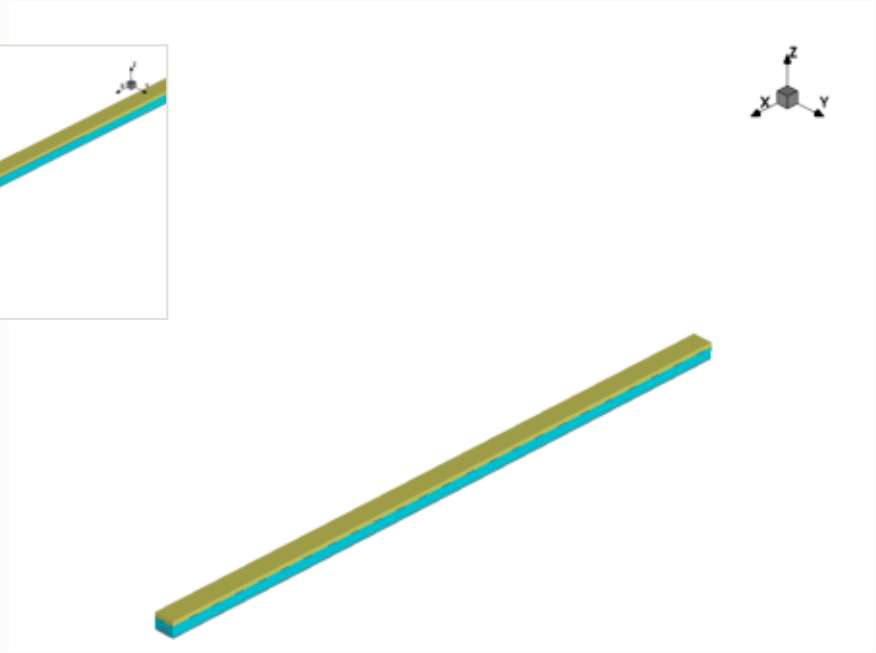
litho_patterning · KrF

0.199	176.0	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



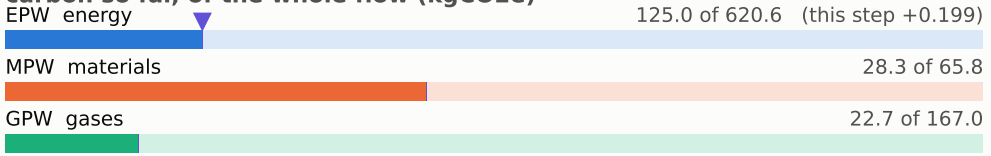


this step

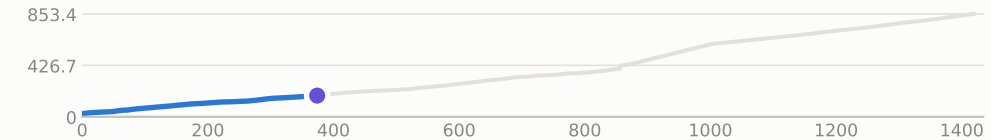
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

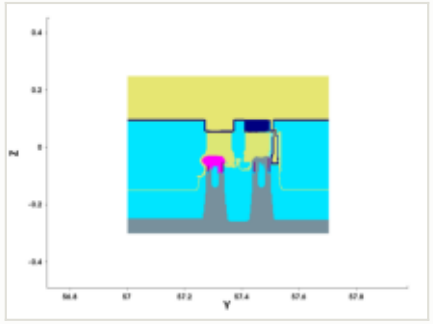
carbon so far, of the whole flow (kgCO2e)



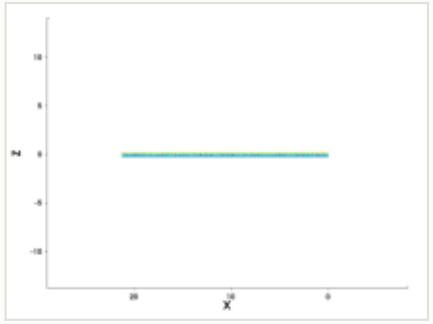
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



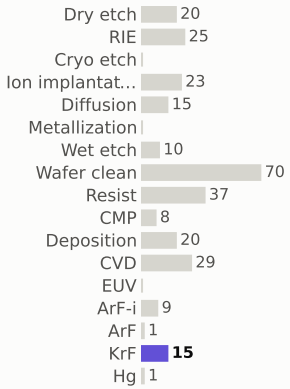
y cut at 57.351 µm



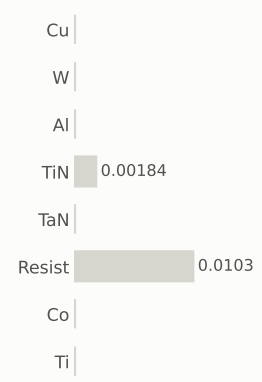
mask for this step

no mask for this step

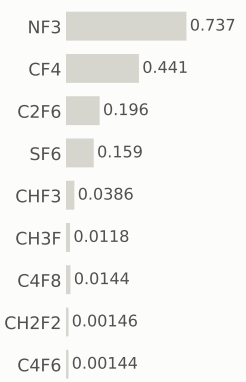
steps by process area



materials deposited (g)



gases emitted (g)



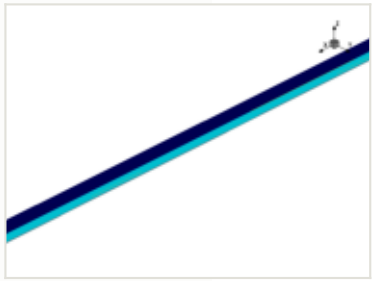
Step #3810: RIE

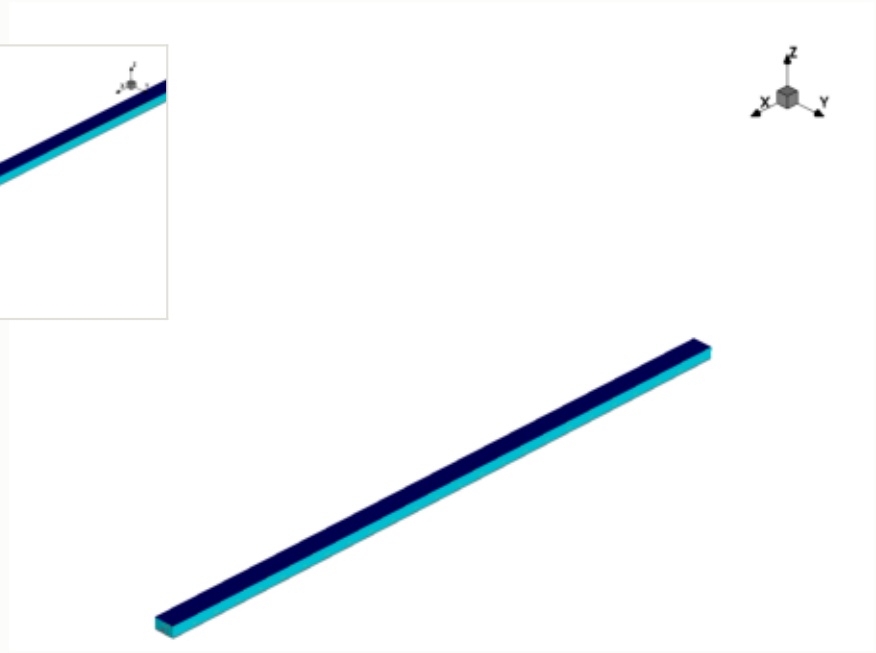
etch_ellipsoidal · RIE

1.04	177.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



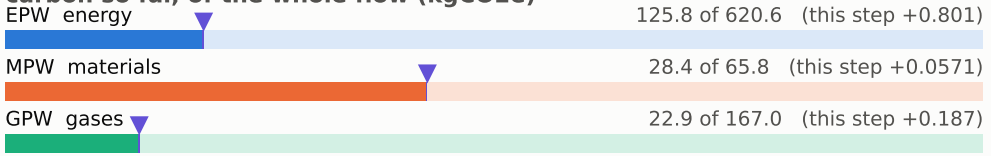


this step

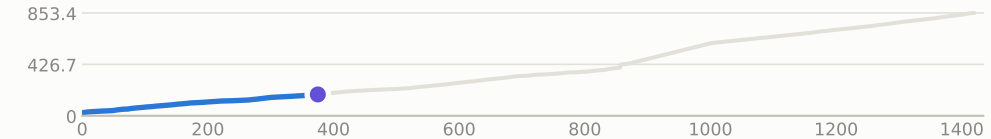
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

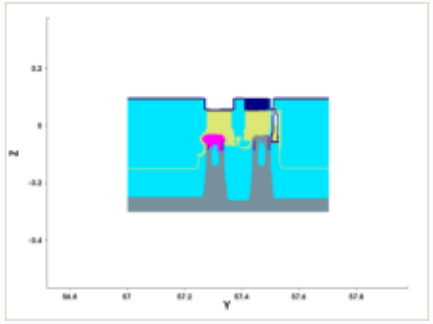
carbon so far, of the whole flow (kgCO2e)



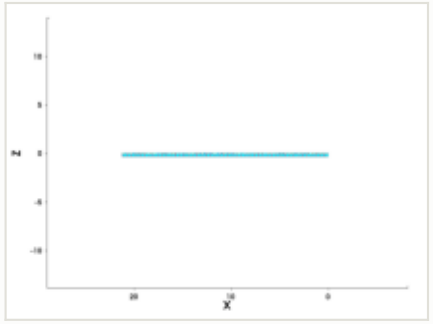
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



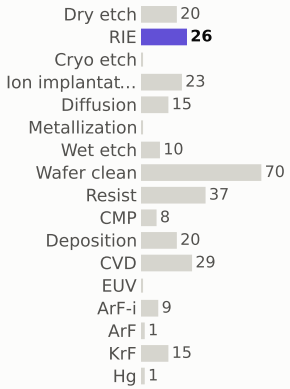
y cut at 57.351 µm



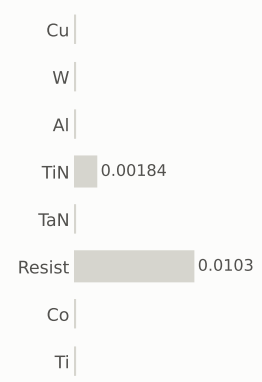
mask for this step

no mask for this step

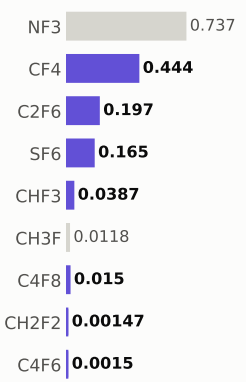
steps by process area



materials deposited (g)



gases emitted (g)



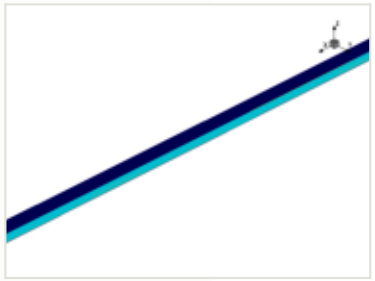
Step #3820: Dry_etch

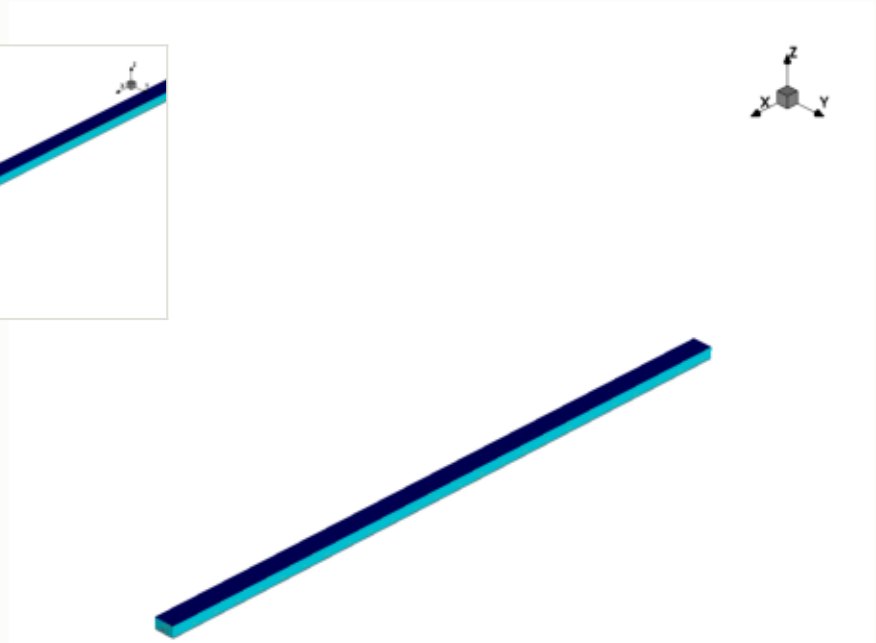
treatment · Dry_etch

0.801	177.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

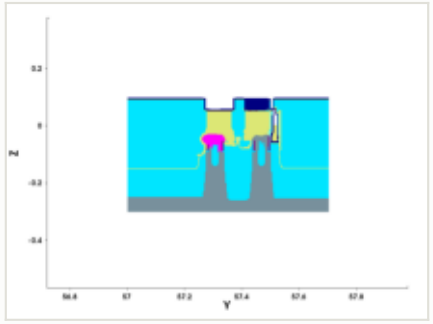
3D structure after this step

zoomed to fit

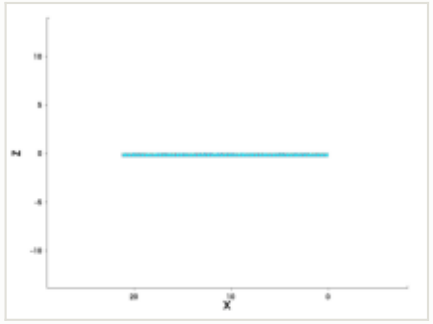




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

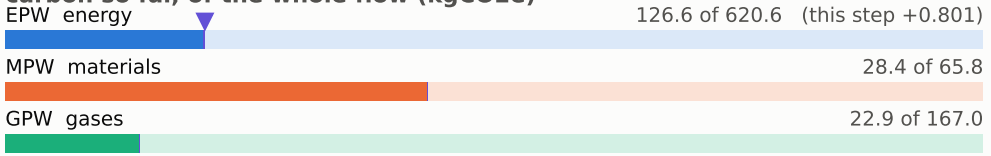
no mask for this step

this step

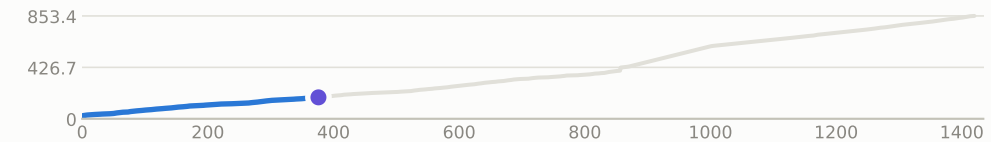
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

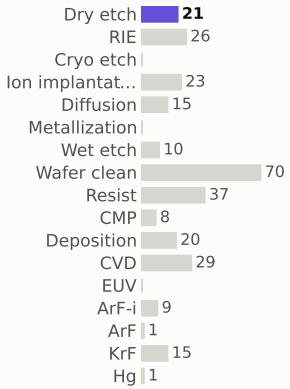
carbon so far, of the whole flow (kgCO2e)



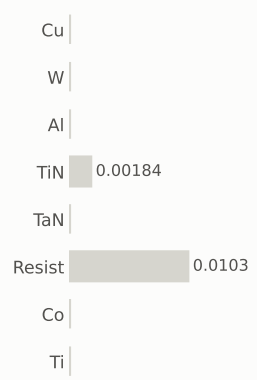
embodied carbon up to each step (kgCO2e per wafer)



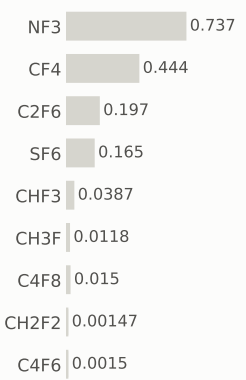
steps by process area



materials deposited (g)



gases emitted (g)



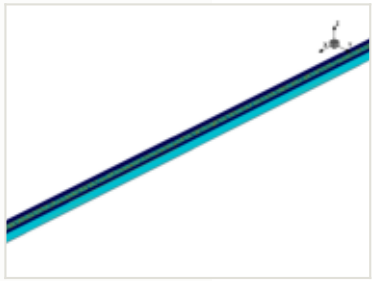
Step #3830: Wet_etch

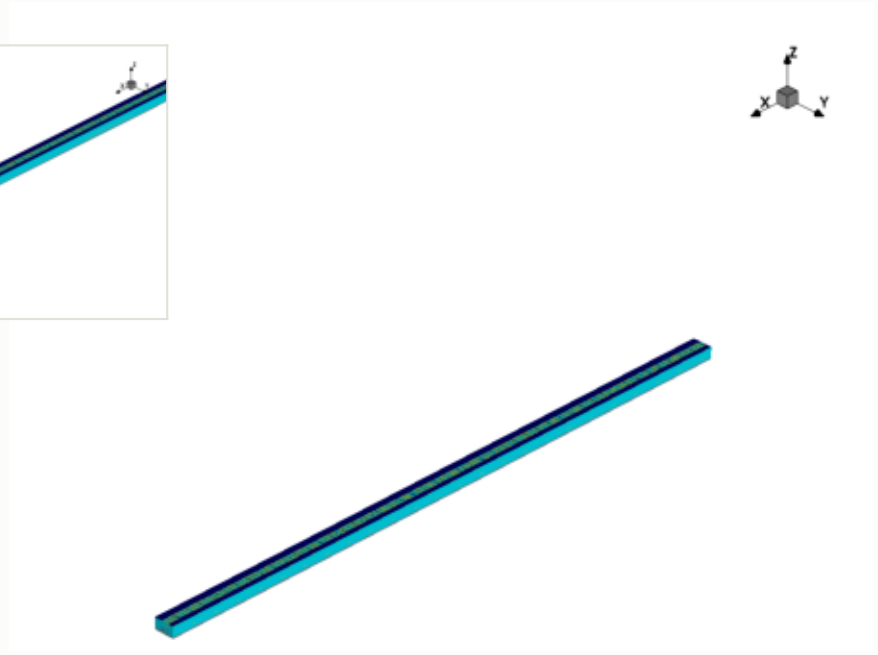
etch_isotropic · Wet_etch

0.169	178.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



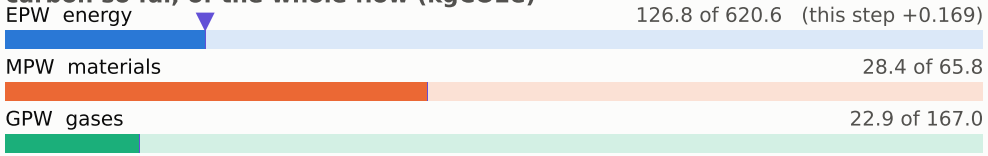


this step

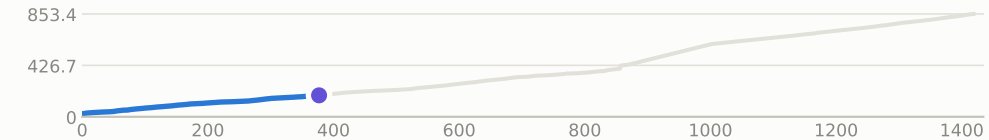
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

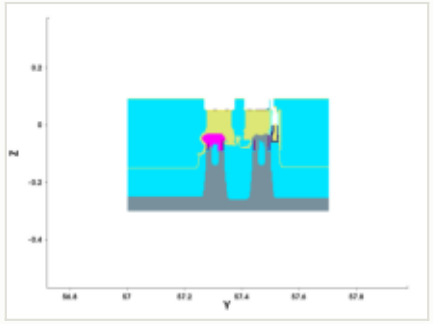
carbon so far, of the whole flow (kgCO2e)



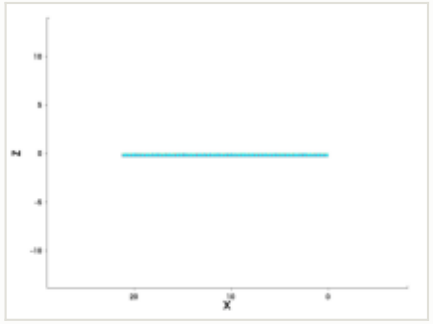
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



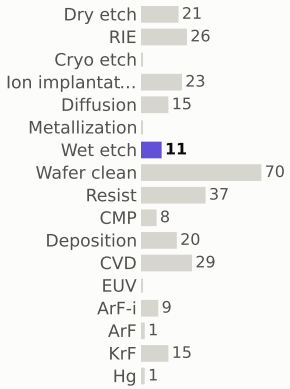
y cut at 57.351 µm



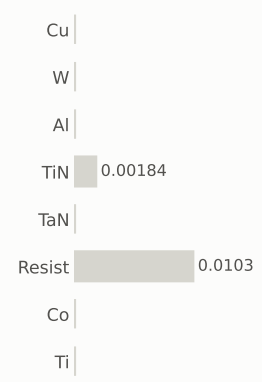
mask for this step

no mask for this step

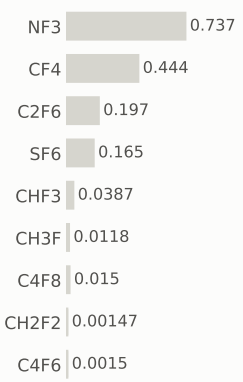
steps by process area



materials deposited (g)



gases emitted (g)



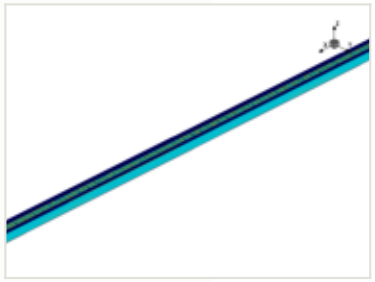
Step #3840: Dry_etch

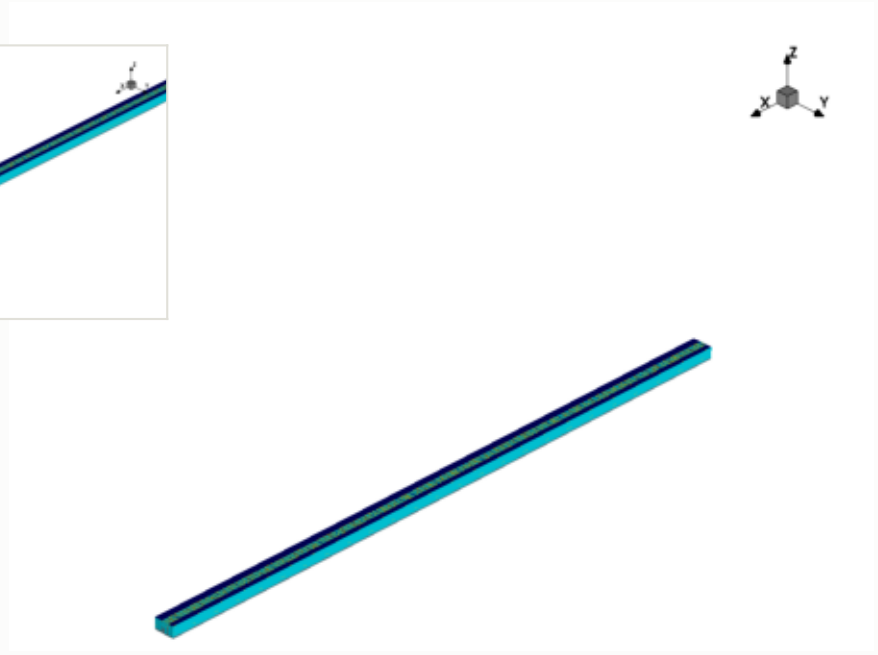
etch_strip · Dry_etch

0.801	178.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



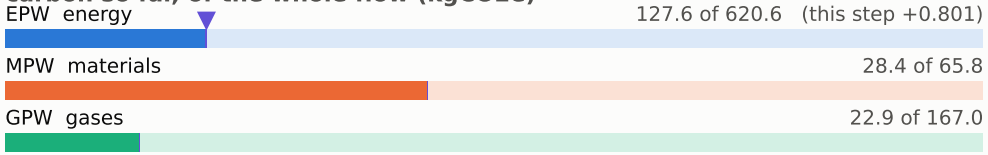


this step

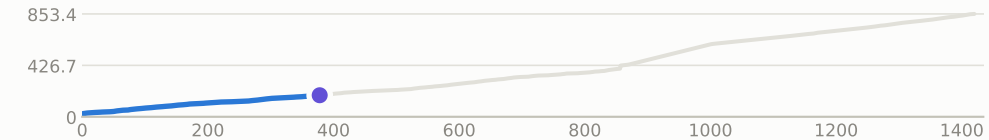
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

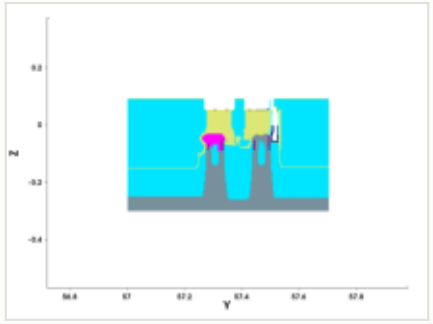
carbon so far, of the whole flow (kgCO2e)



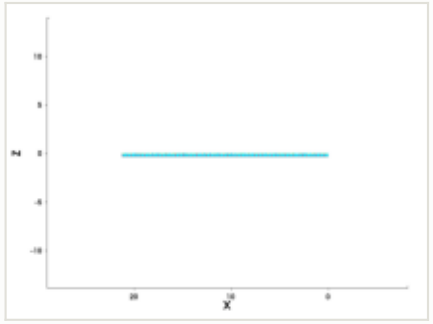
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



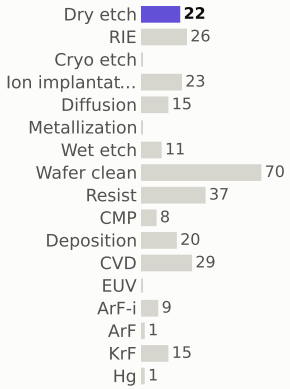
y cut at 57.351 µm



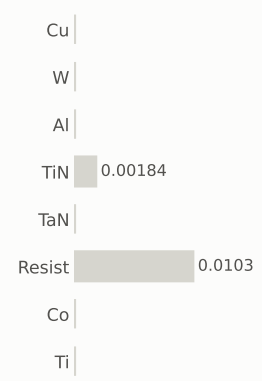
mask for this step

no mask for this step

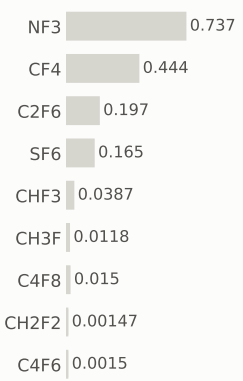
steps by process area



materials deposited (g)



gases emitted (g)



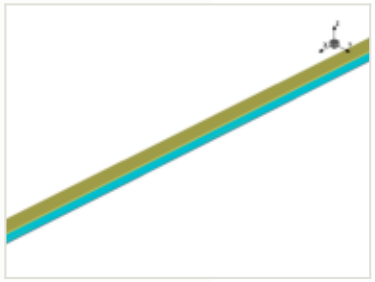
Step #3850: Spx_only

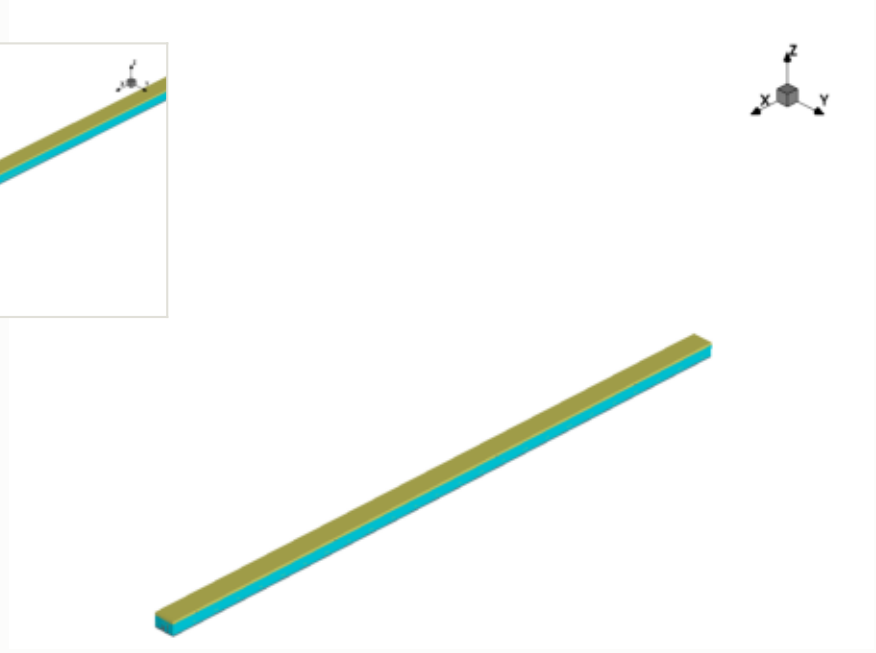
depo_fill · Spx_only

2.59e-05	178.8	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

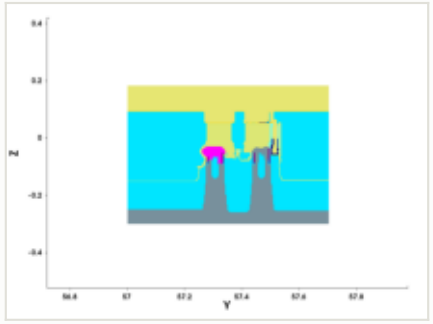
3D structure after this step

zoomed to fit

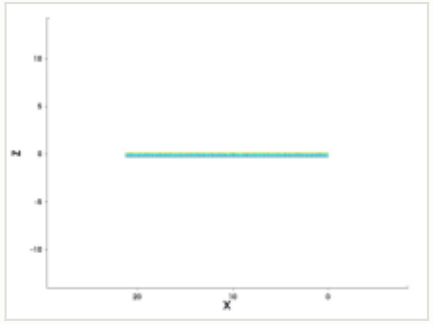




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

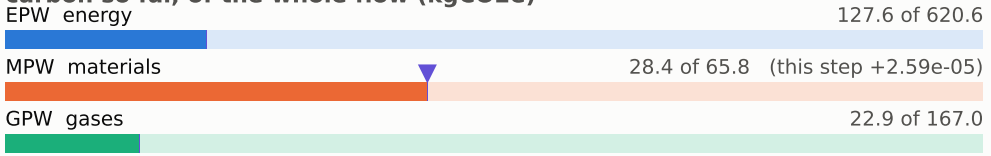
this step

process area	Spx_only
material	Resist
thickness	0.09 µm
material GWP	4.2 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

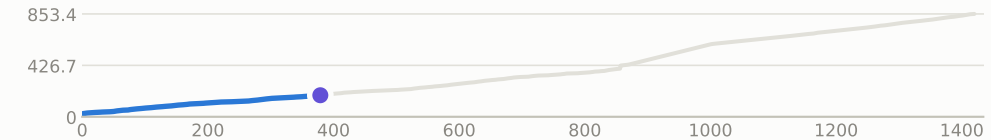
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	2.59e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**2.59e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

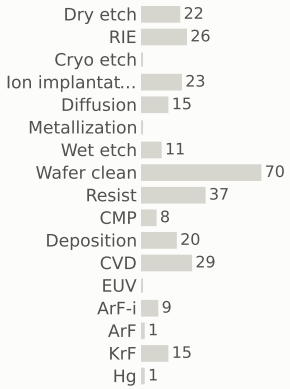
carbon so far, of the whole flow (kgCO2e)



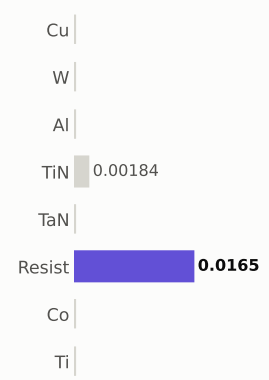
embodied carbon up to each step (kgCO2e per wafer)



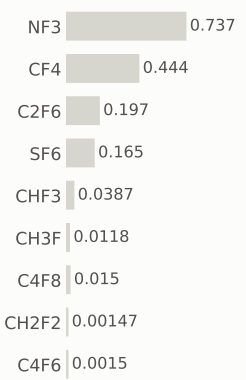
steps by process area



materials deposited (g)



gases emitted (g)



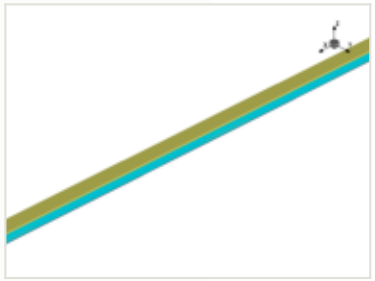
Step #3860: Resist

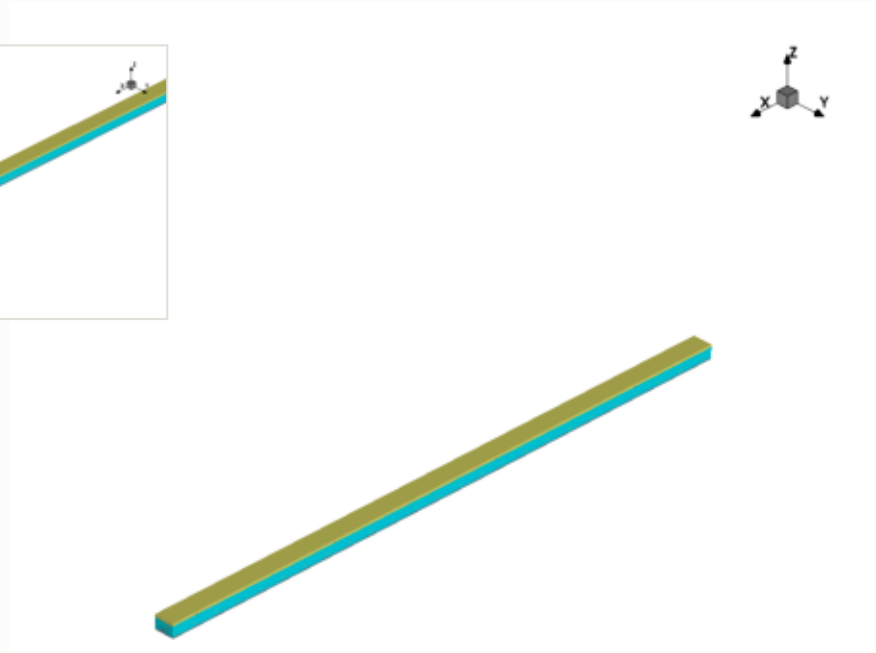
resist · Resist

0.169	179.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





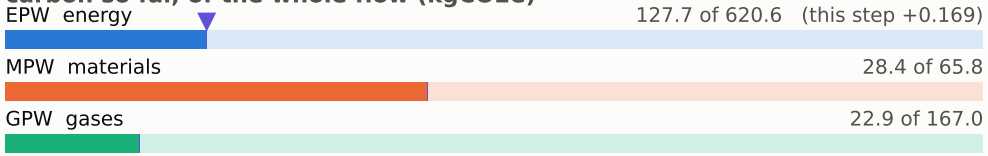
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

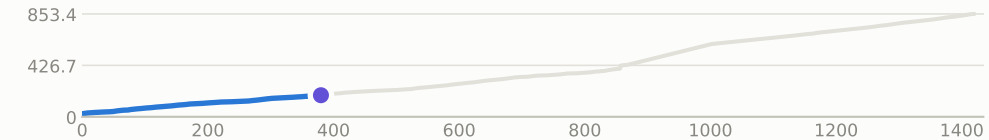
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

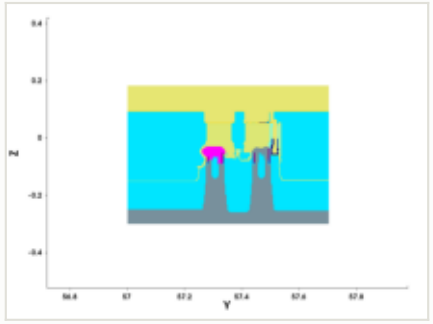
carbon so far, of the whole flow (kgCO2e)



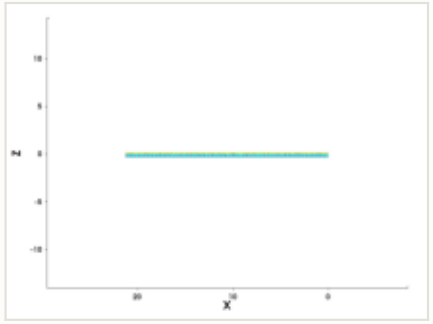
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



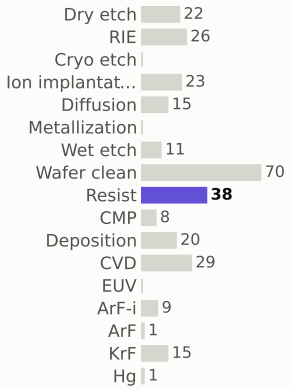
y cut at 57.351 µm



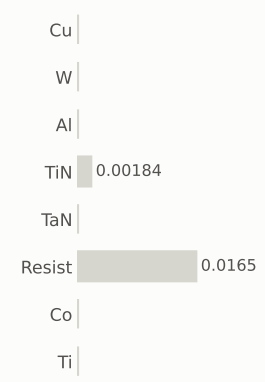
mask for this step

no mask for this step

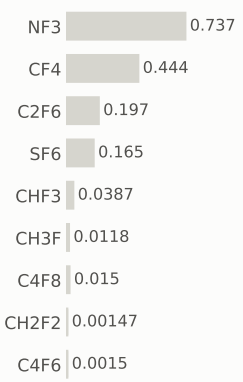
steps by process area



materials deposited (g)



gases emitted (g)



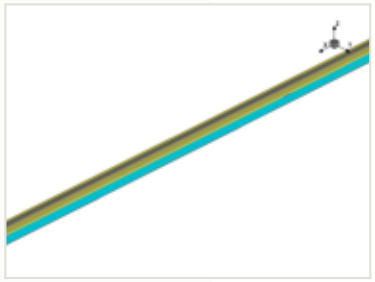
Step #3870: KrF

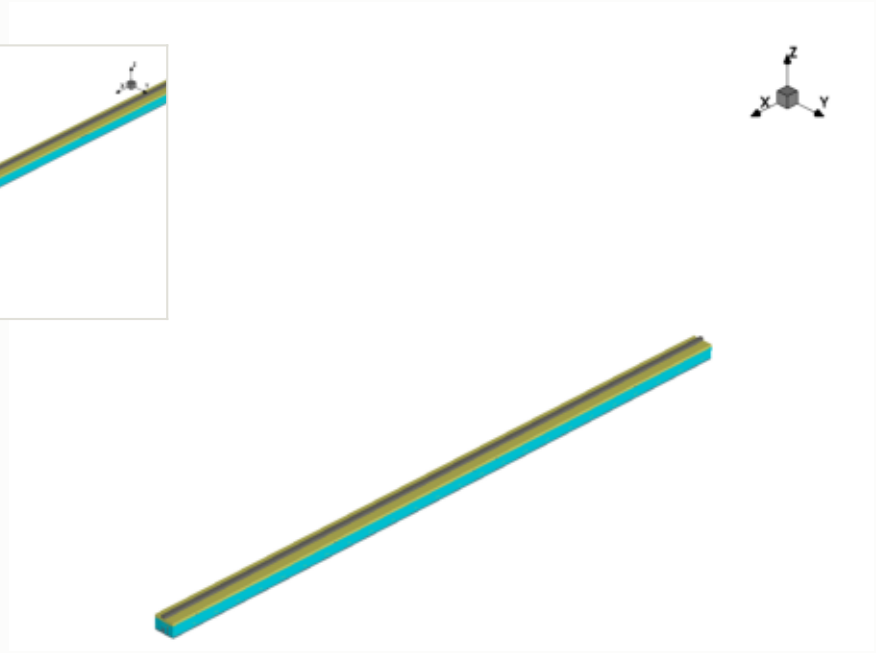
litho_patterning · KrF

0.199	179.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

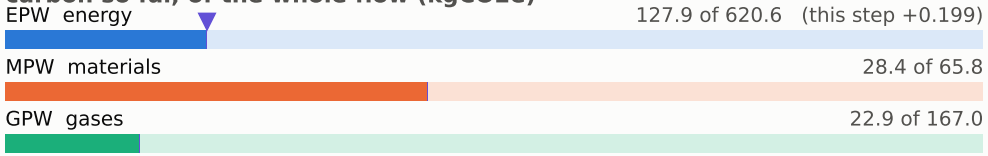




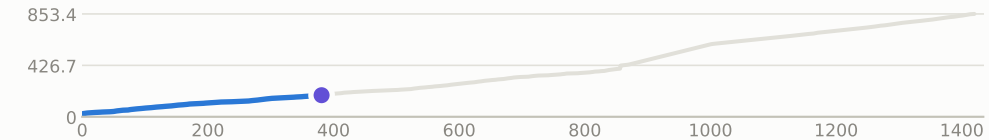
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

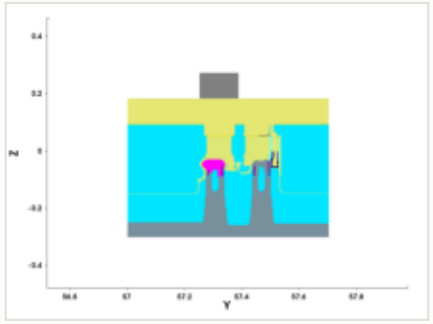
carbon so far, of the whole flow (kgCO2e)



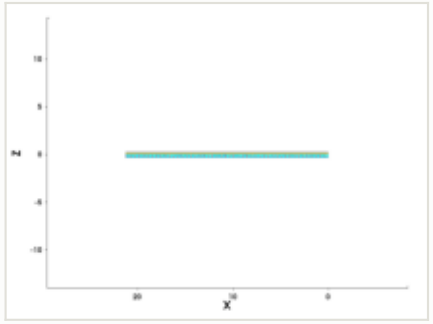
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

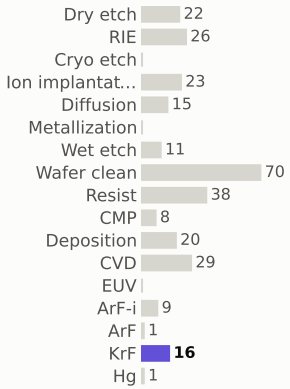


y cut at 57.351 µm

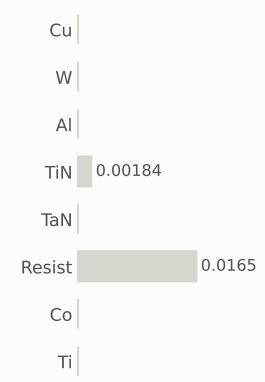


mask: layer 2012:0, positive (GDSII)

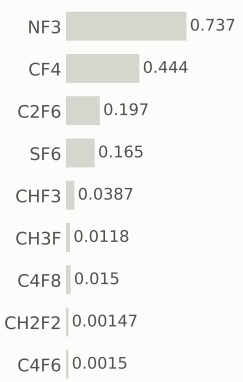
steps by process area



materials deposited (g)



gases emitted (g)



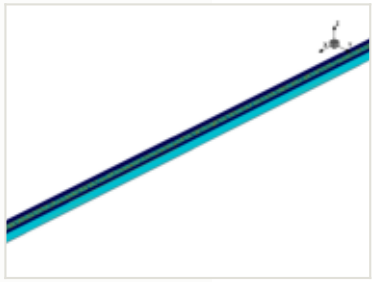
Step #3880: RIE

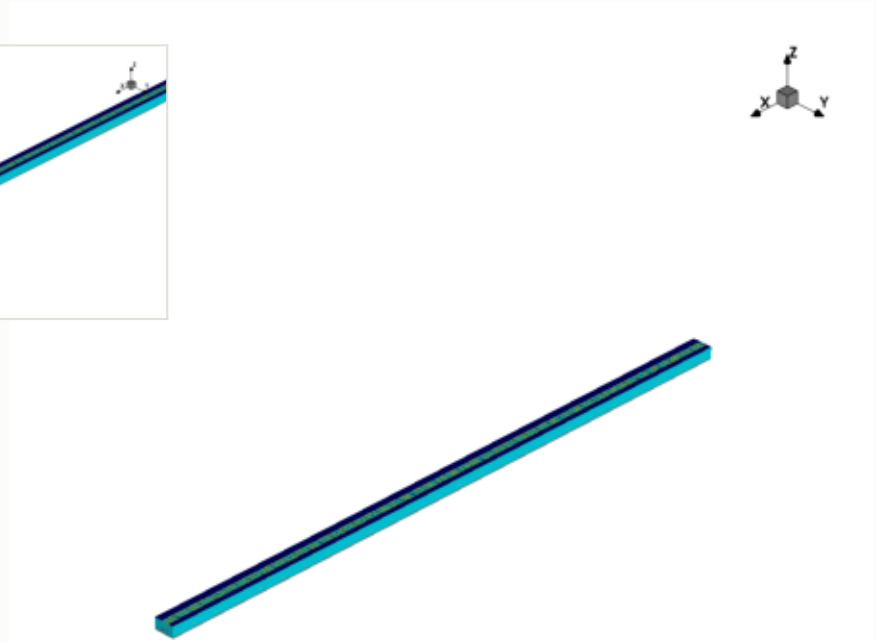
etch_ellipsoidal · RIE

1.04	180.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



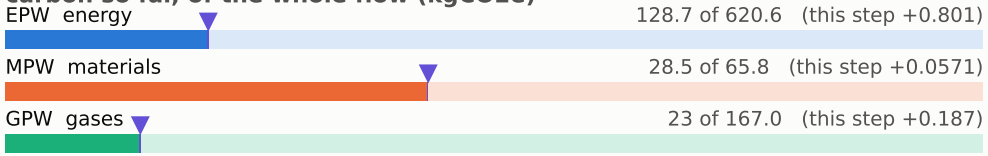


this step

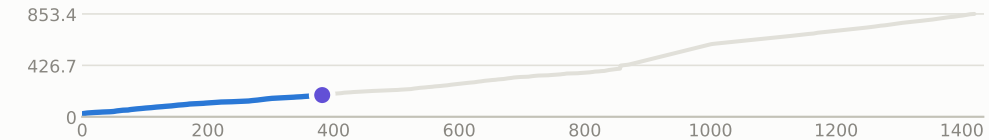
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

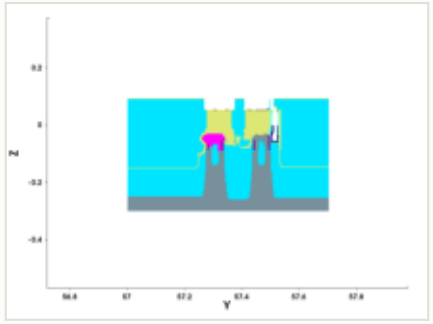
carbon so far, of the whole flow (kgCO2e)



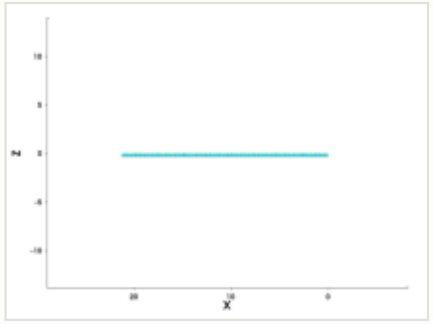
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



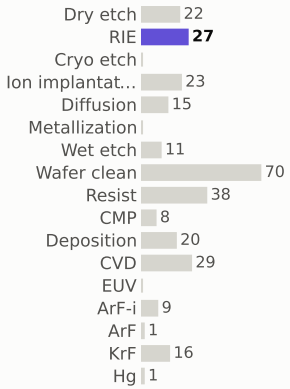
y cut at 57.351 µm



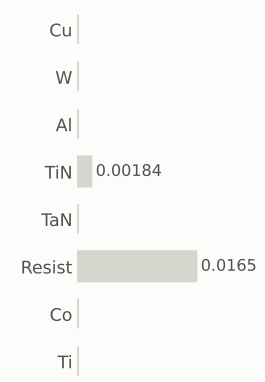
mask for this step

no mask for this step

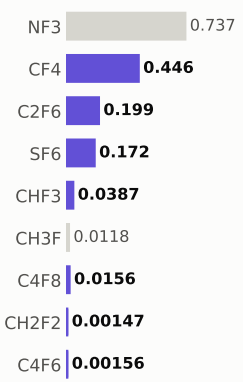
steps by process area



materials deposited (g)



gases emitted (g)



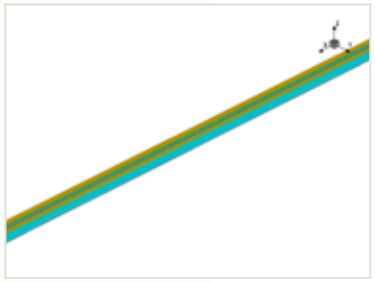
Step #3890: Wet_etch

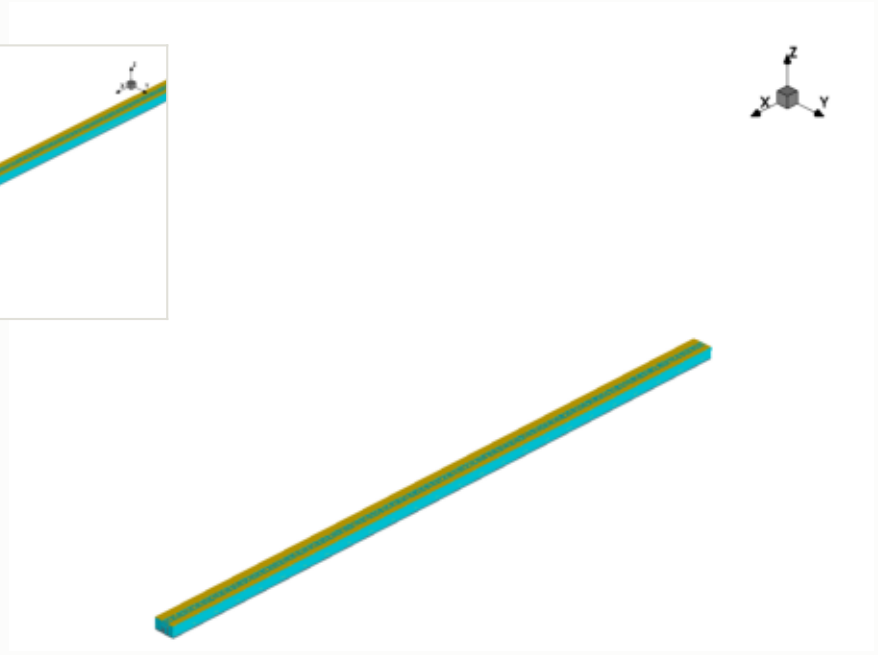
etch_ellipsoidal · Wet_etch

0.169	180.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



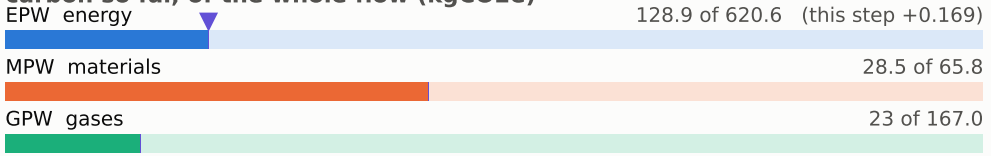


this step

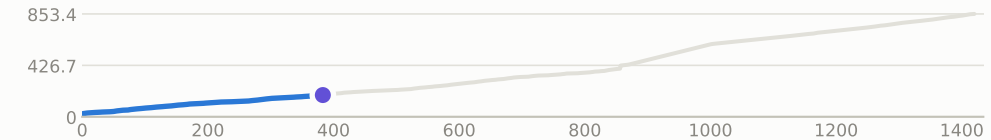
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

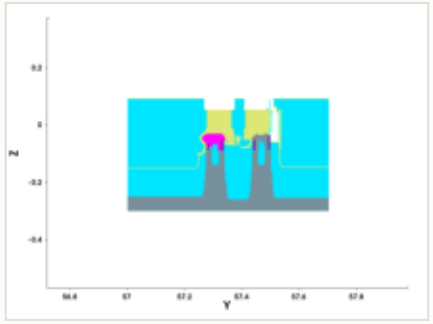
carbon so far, of the whole flow (kgCO2e)



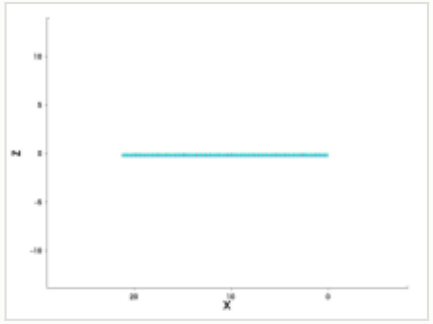
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



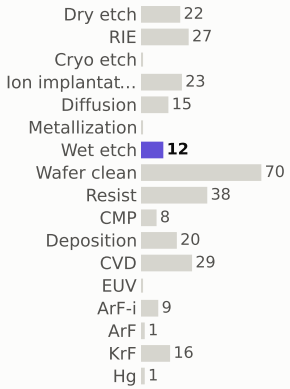
y cut at 57.351 µm



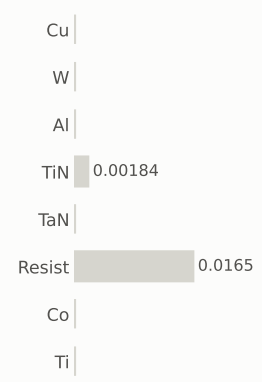
mask for this step

no mask for this step

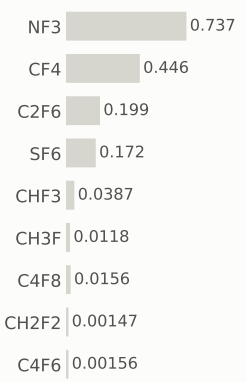
steps by process area



materials deposited (g)



gases emitted (g)



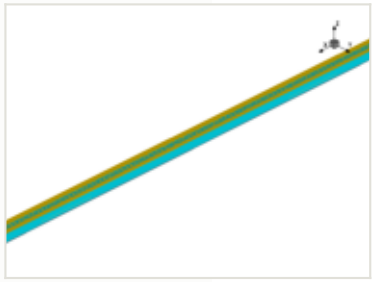
Step #3900: Dry_etch

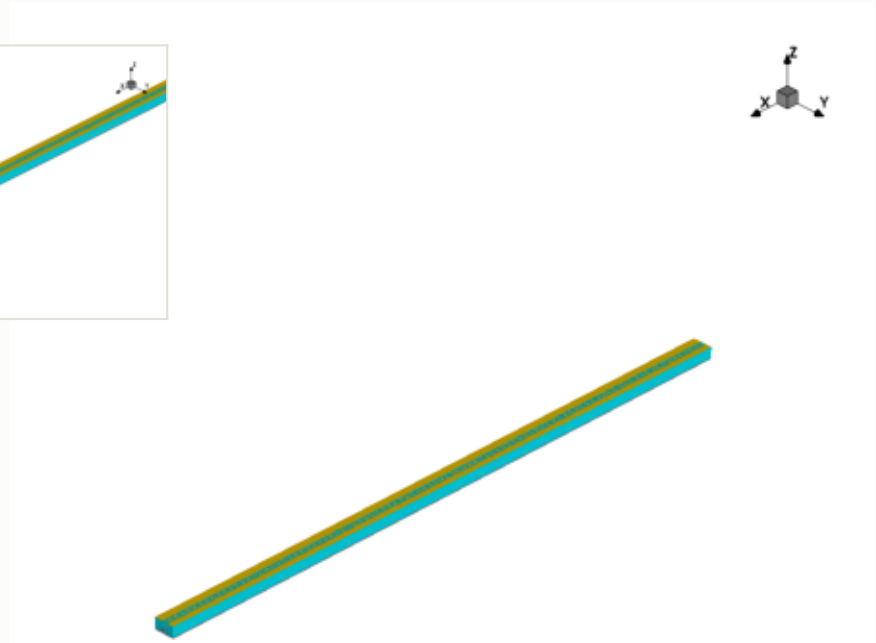
etch_strip · Dry_etch

0.801	181.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



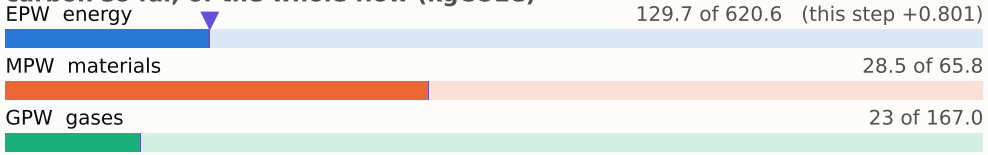


this step

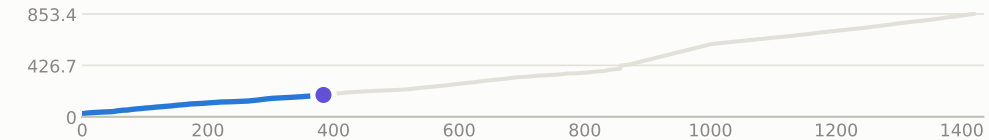
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

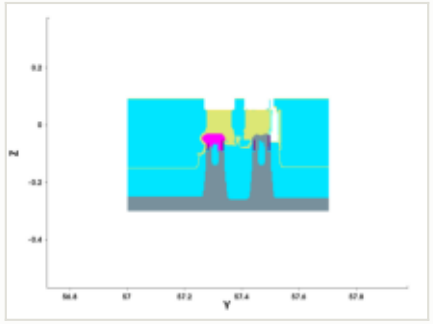
carbon so far, of the whole flow (kgCO2e)



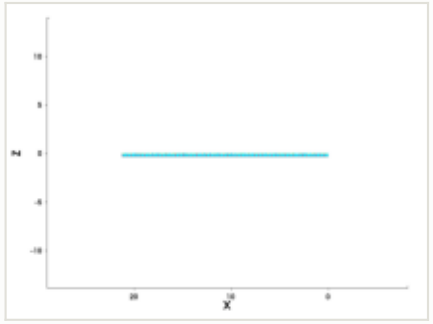
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



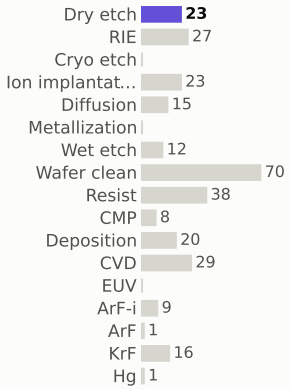
y cut at 57.351 µm



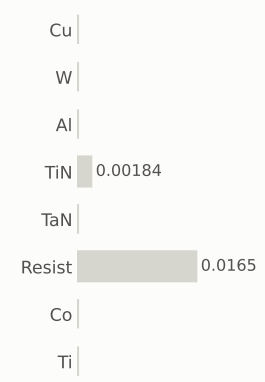
mask for this step

no mask for this step

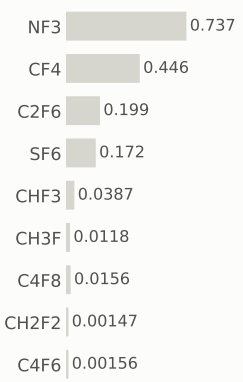
steps by process area



materials deposited (g)



gases emitted (g)



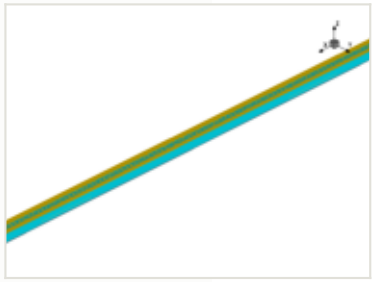
Step #3910: Wafer_clean

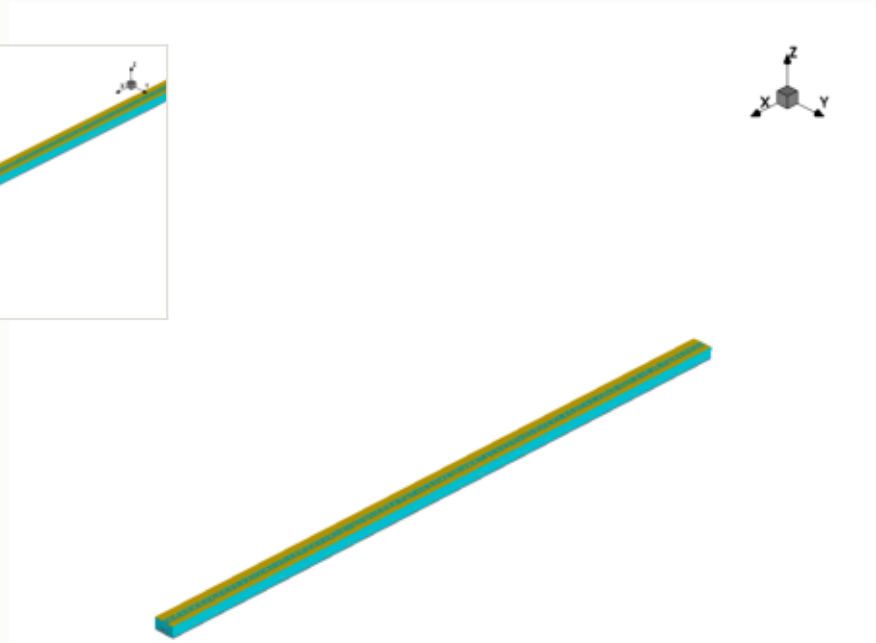
clean · Wafer_clean

0.169	181.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





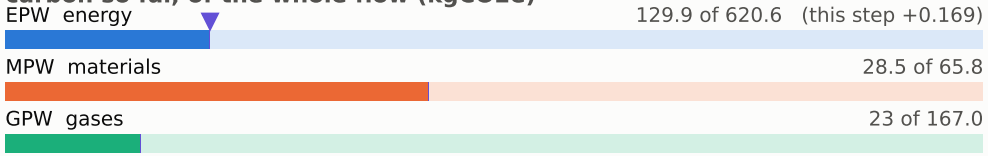
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

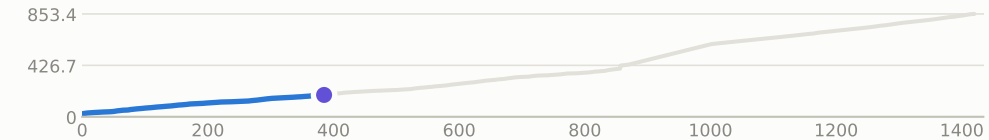
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

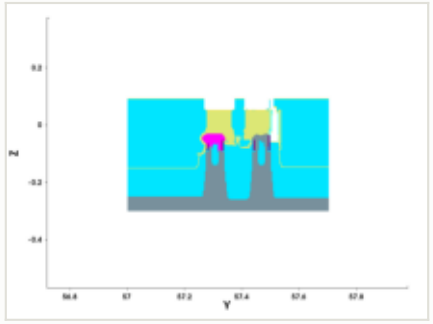
carbon so far, of the whole flow (kgCO2e)



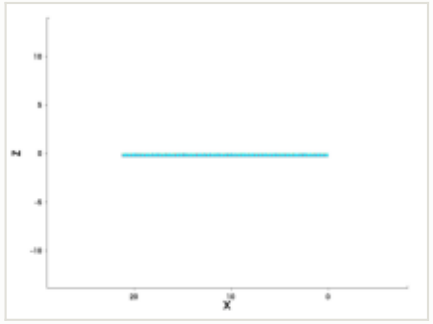
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



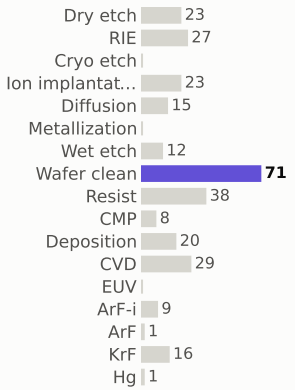
y cut at 57.351 µm



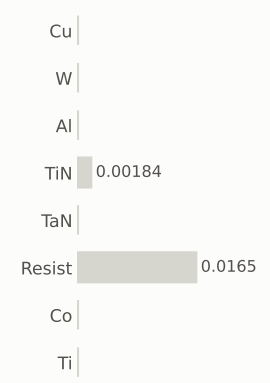
mask for this step

no mask for this step

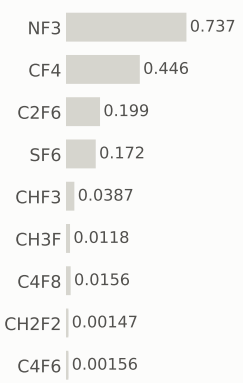
steps by process area



materials deposited (g)



gases emitted (g)



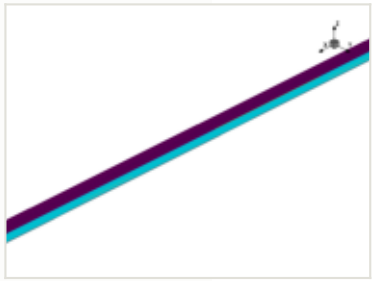
Step #3920: Deposition

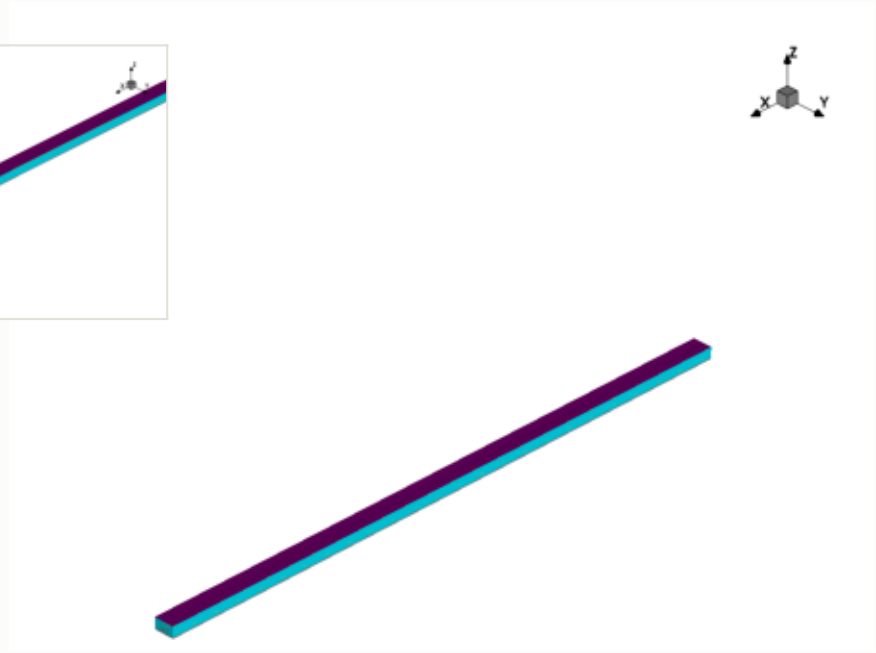
depo_isotropic · Deposition

0.656	182.0	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



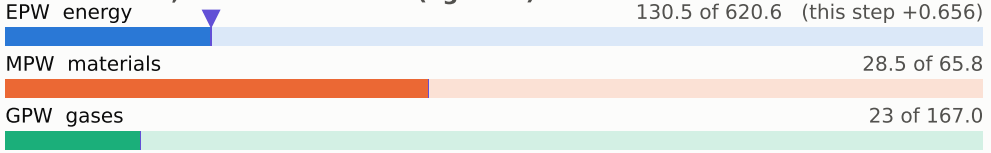


this step

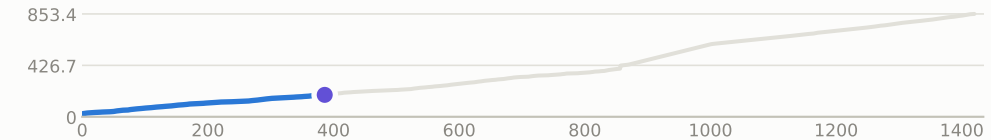
process area	Deposition
material	—
thickness	0.004 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

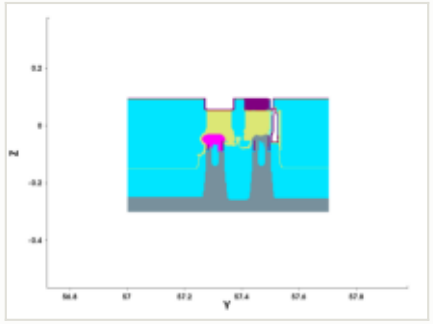
carbon so far, of the whole flow (kgCO2e)



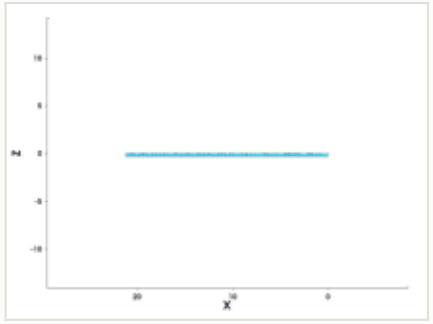
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



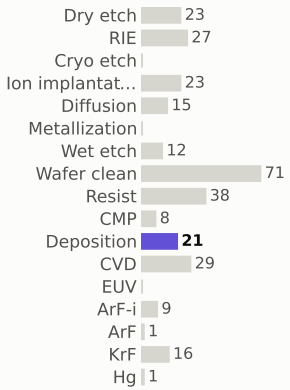
y cut at 57.351 µm



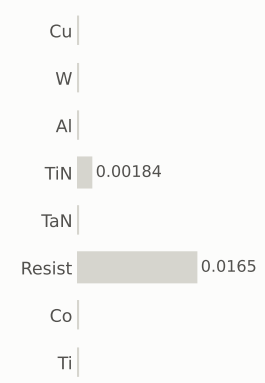
mask for this step

no mask for this step

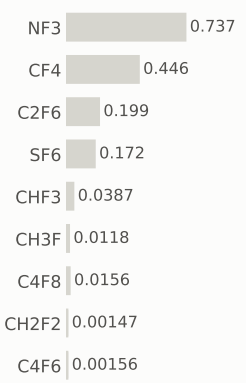
steps by process area



materials deposited (g)



gases emitted (g)



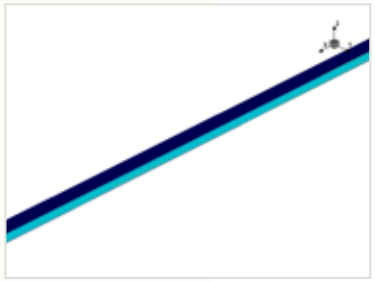
Step #3930: Deposition

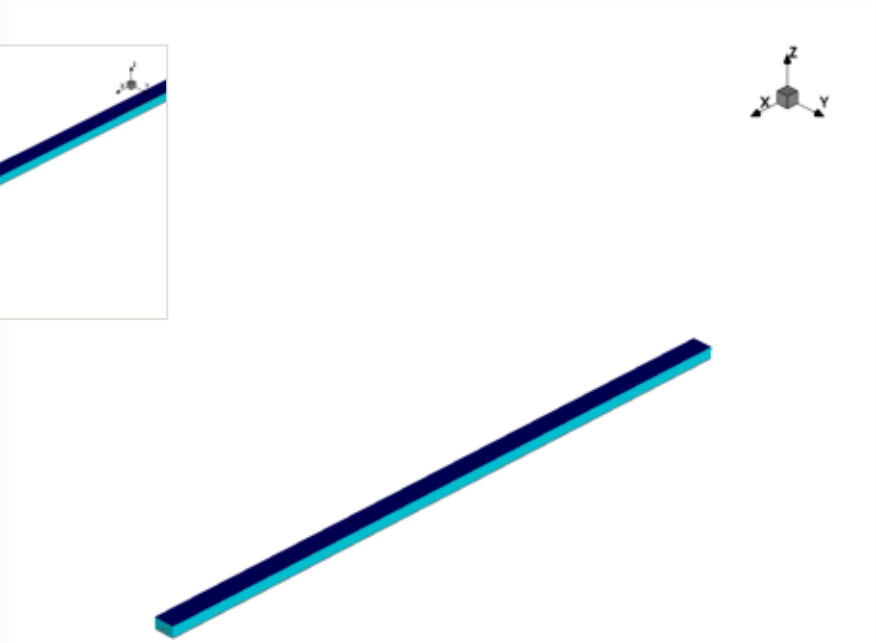
depo_isotropic · Deposition

0.656	182.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



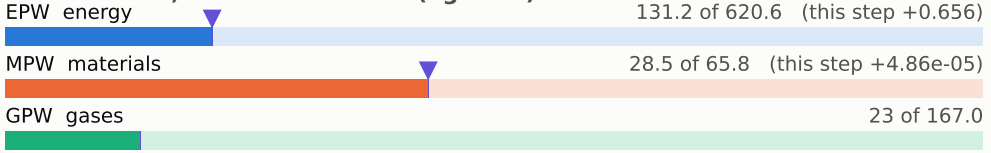


this step

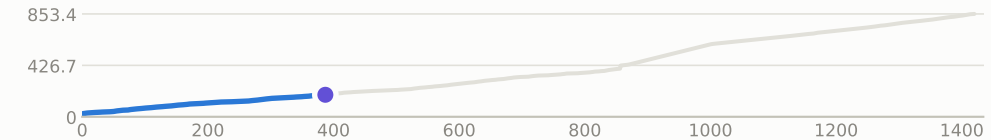
process area	Deposition
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

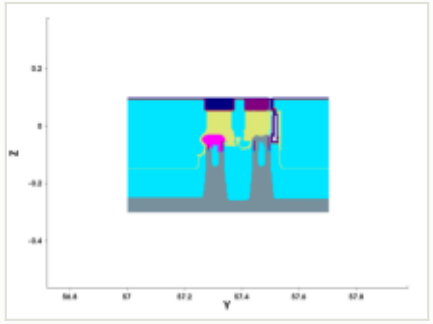
carbon so far, of the whole flow (kgCO2e)



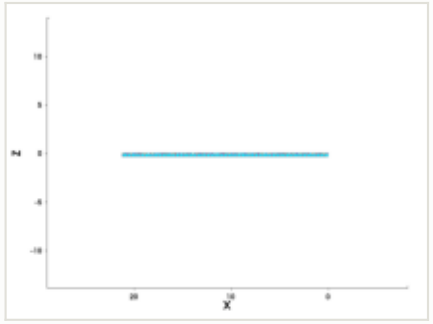
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



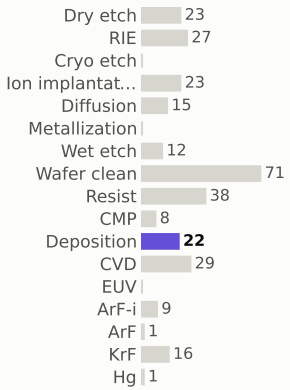
y cut at 57.351 µm



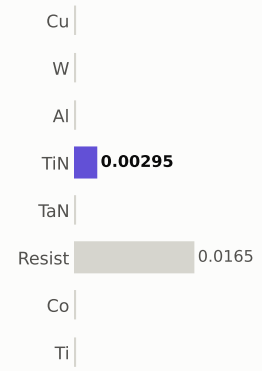
mask for this step

no mask for this step

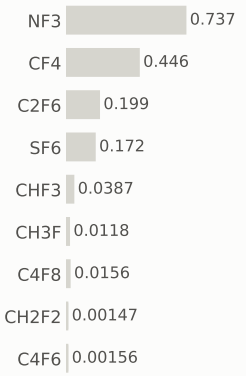
steps by process area



materials deposited (g)



gases emitted (g)



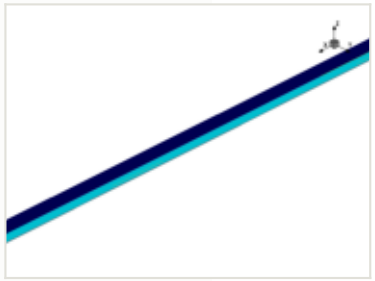
Step #3940: Wafer_clean

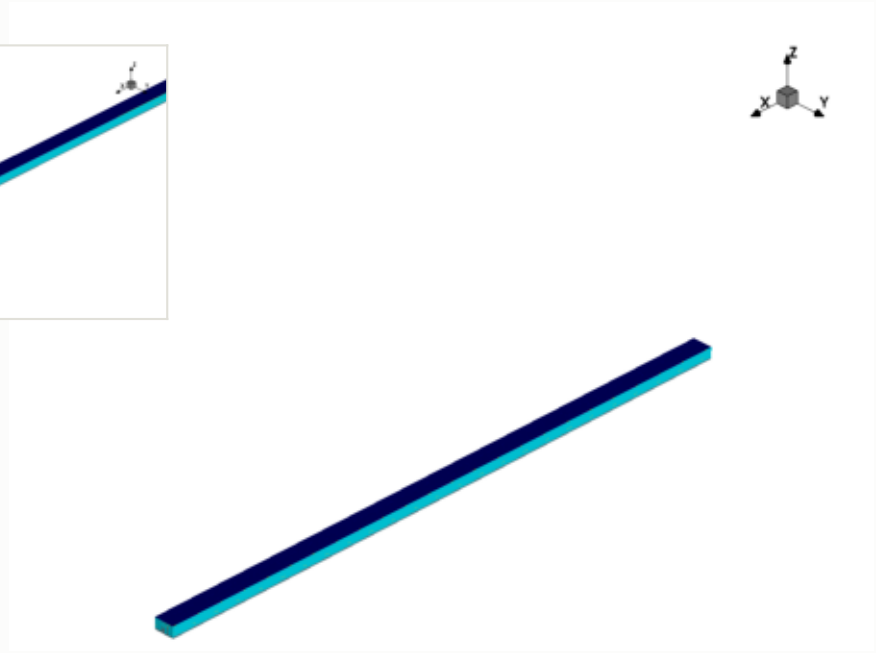
clean · Wafer_clean

0.169	182.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





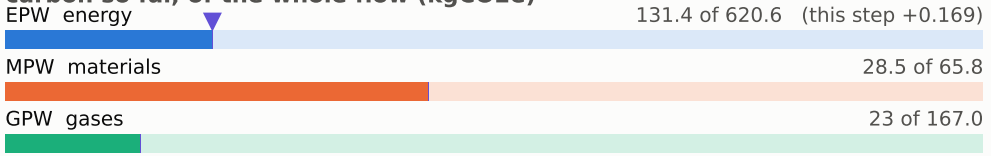
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

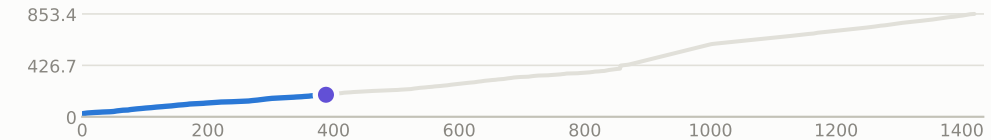
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

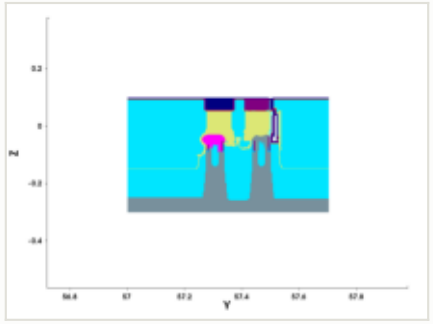
carbon so far, of the whole flow (kgCO2e)



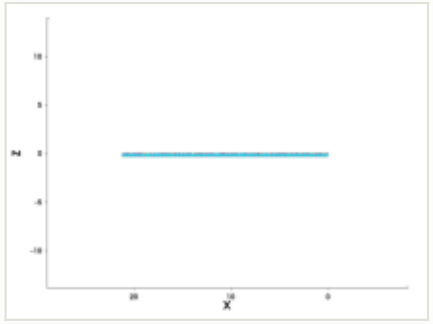
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



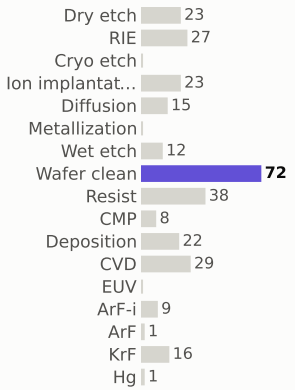
y cut at 57.351 µm



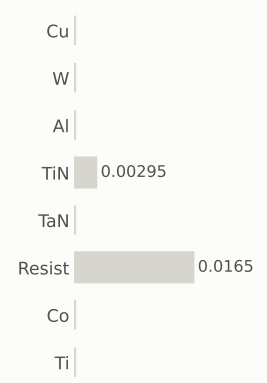
mask for this step

no mask for this step

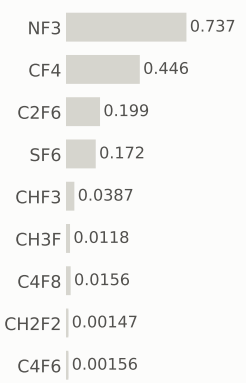
steps by process area



materials deposited (g)



gases emitted (g)



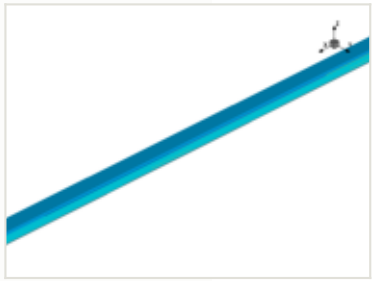
Step #3950: Metallization

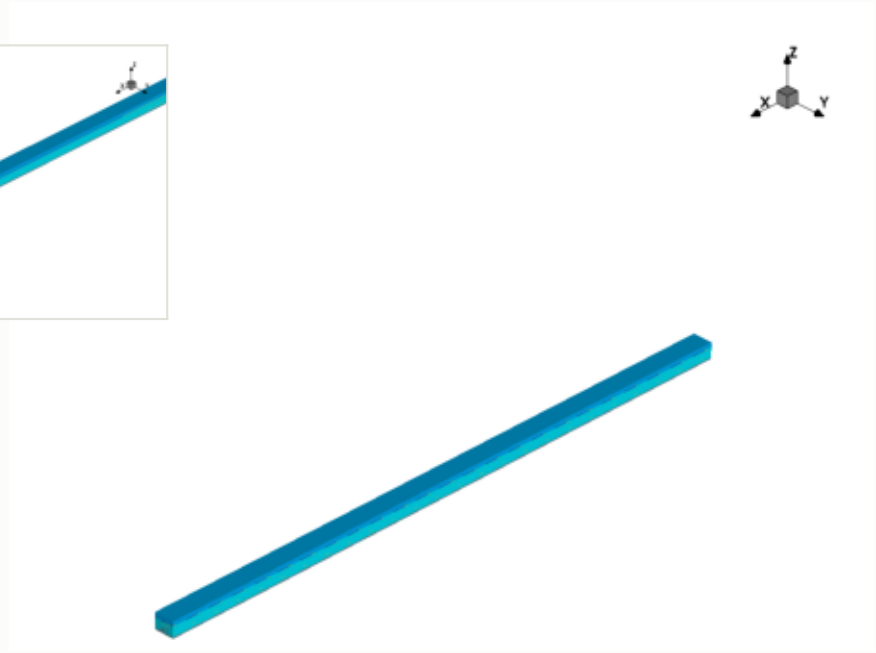
depo_fill · Metallization

1.22	184.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



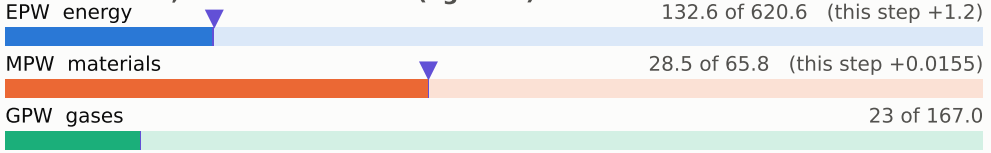


this step

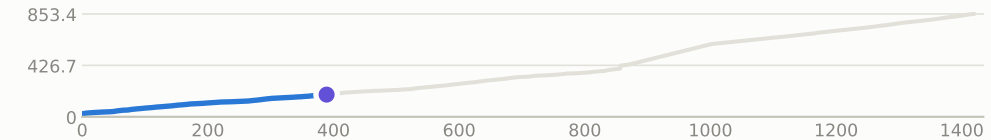
process area	Metallization
material	W
thickness	0.15 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0155 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.22 kgCO2e**

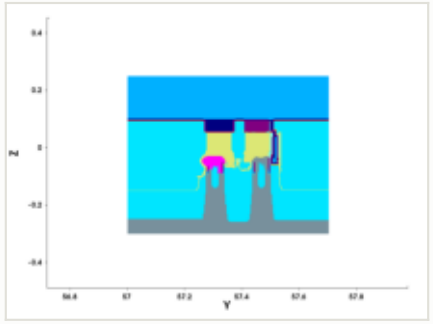
carbon so far, of the whole flow (kgCO2e)



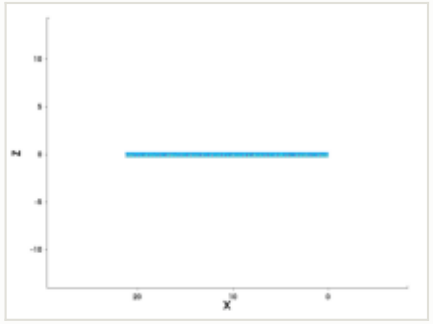
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



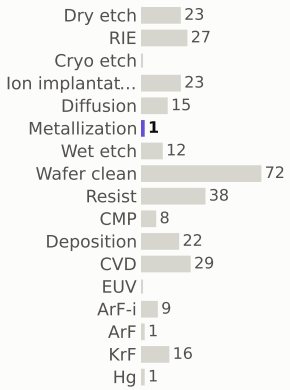
y cut at 57.351 µm



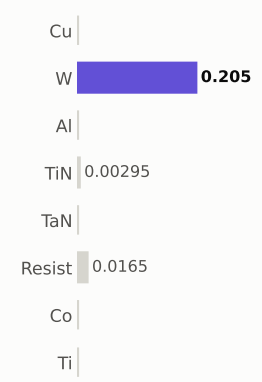
mask for this step

no mask for this step

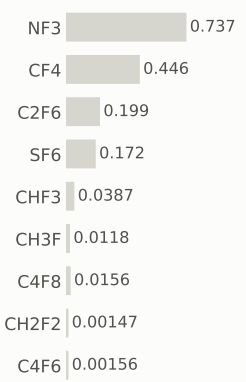
steps by process area



materials deposited (g)



gases emitted (g)



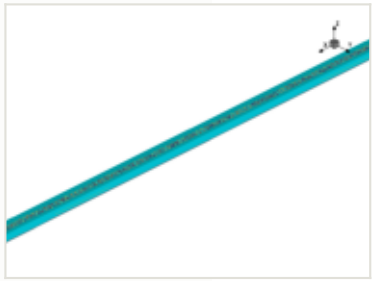
Step #3960: CMP

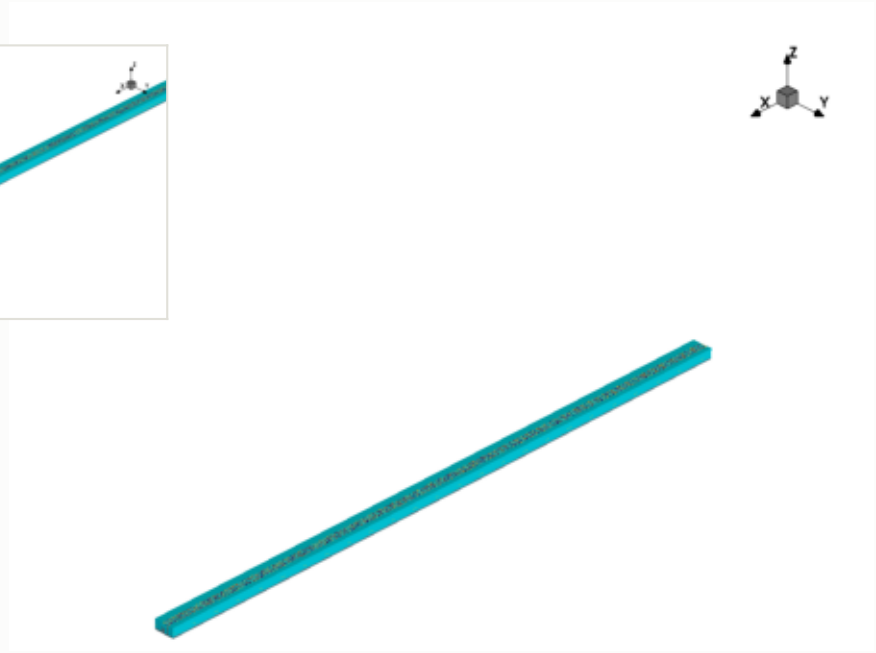
etch_cmp · CMP

0.169	184.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



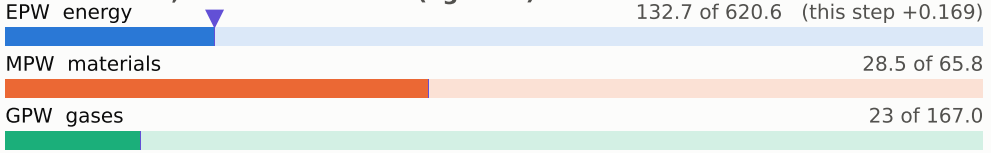


this step

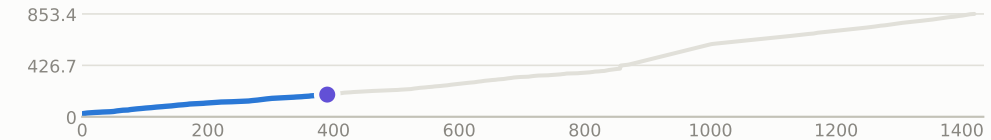
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

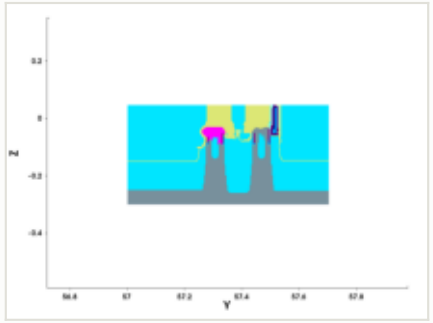
carbon so far, of the whole flow (kgCO2e)



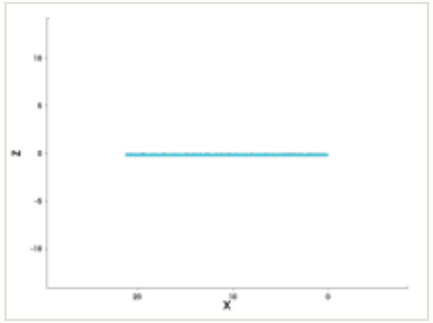
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



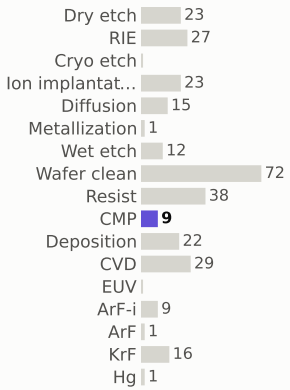
y cut at 57.351 µm



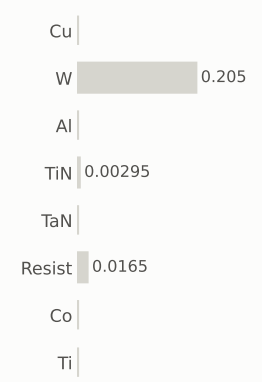
mask for this step

no mask for this step

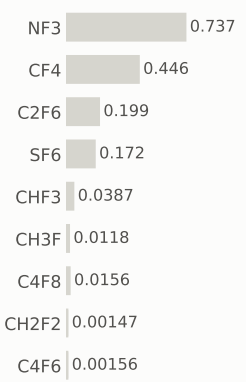
steps by process area



materials deposited (g)



gases emitted (g)



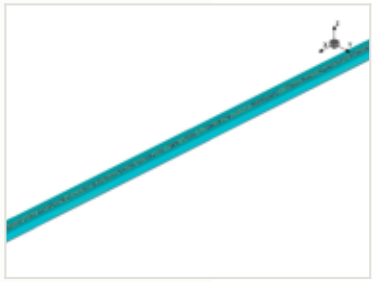
Step #3970: Dry_etch

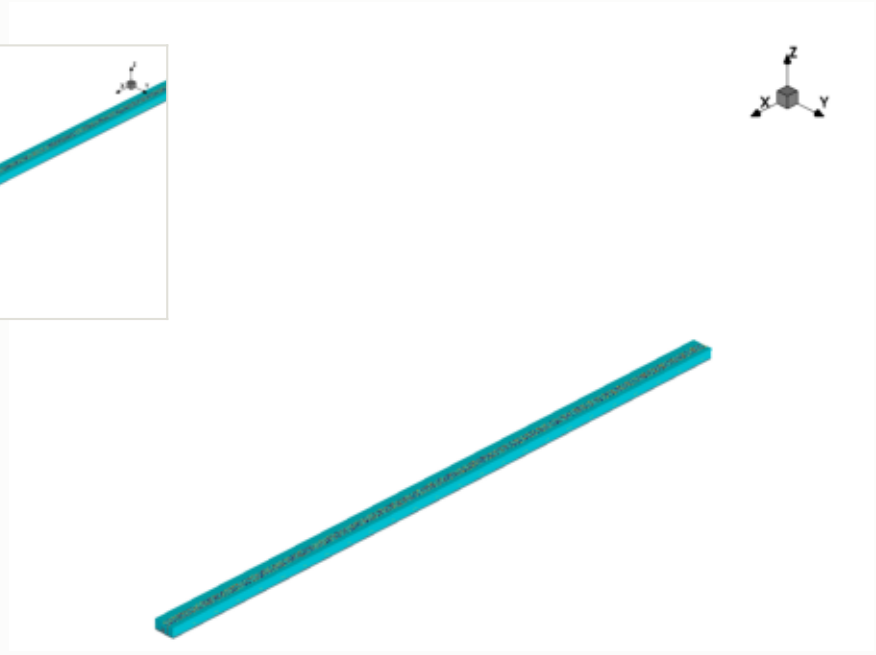
etch_strip · Dry_etch

0.801	185.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



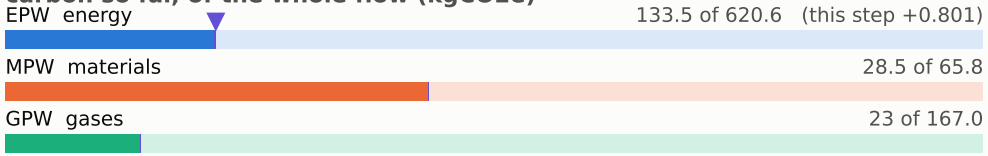


this step

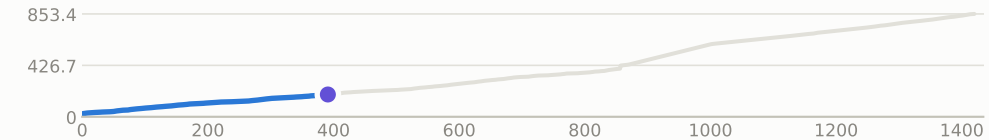
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

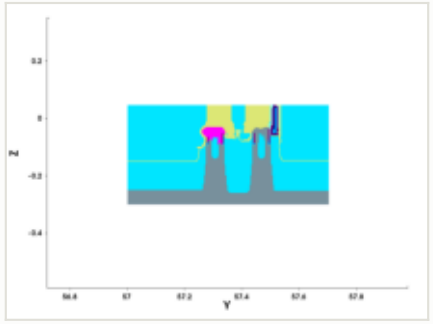
carbon so far, of the whole flow (kgCO2e)



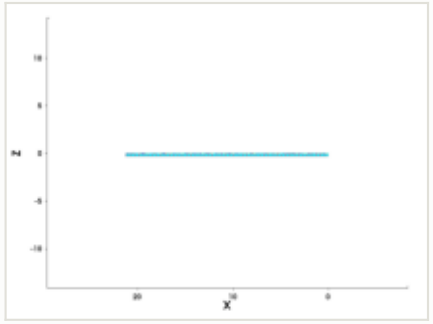
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



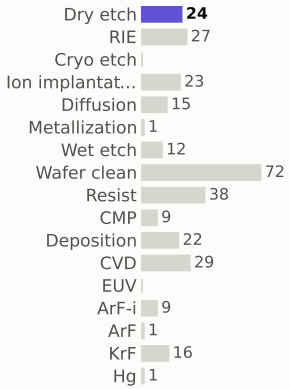
y cut at 57.351 µm



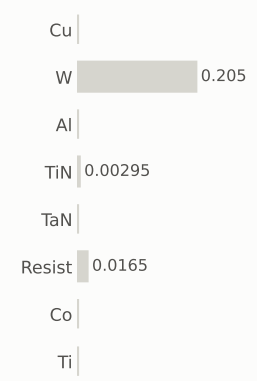
mask for this step

no mask for this step

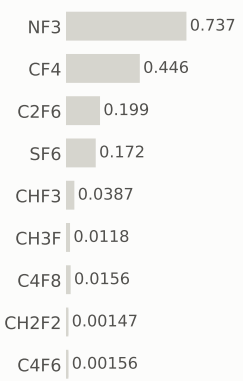
steps by process area



materials deposited (g)



gases emitted (g)



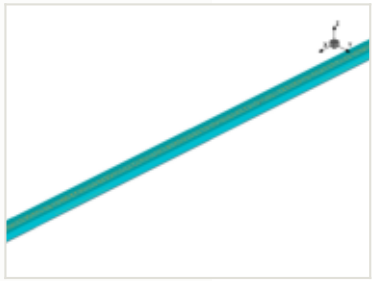
Step #3980: RIE

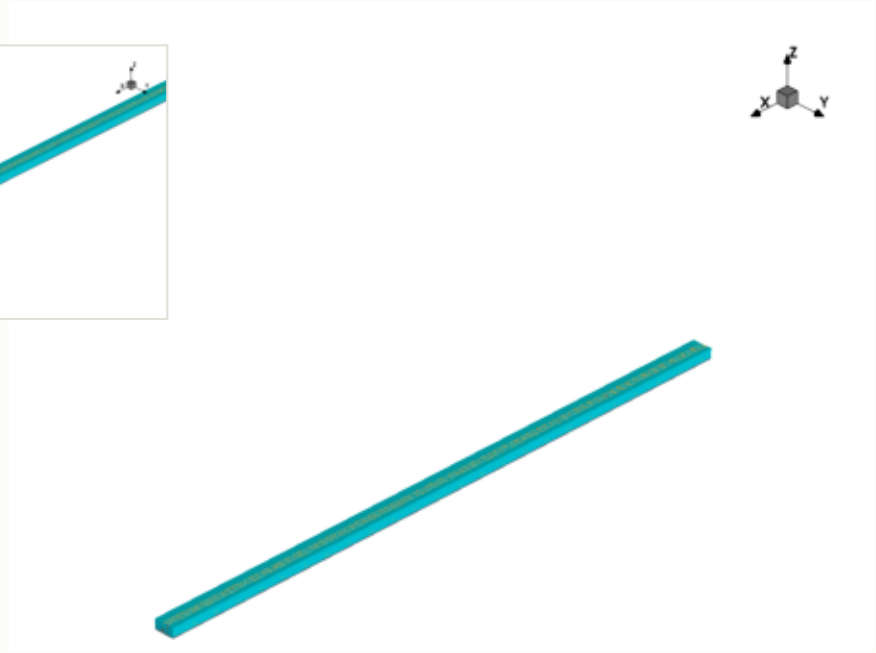
etch_isotropic · RIE

1.04	186.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



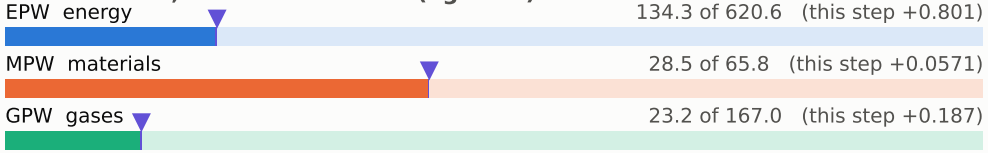


this step

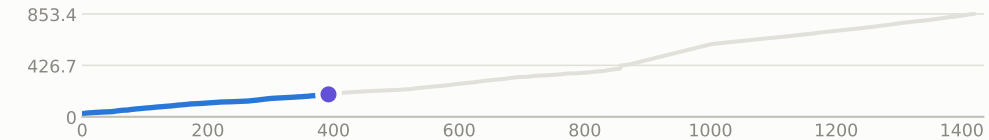
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

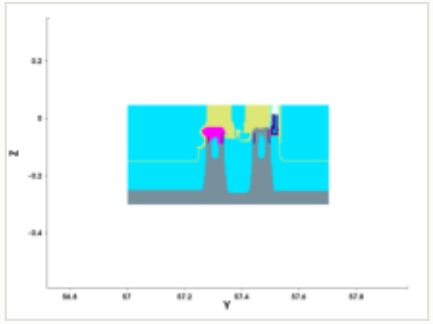
carbon so far, of the whole flow (kgCO2e)



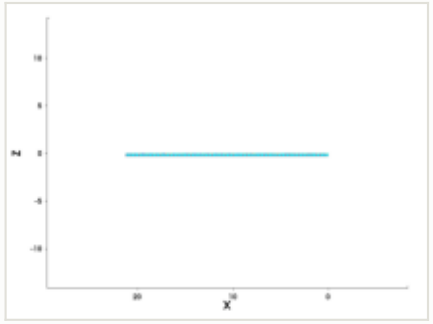
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



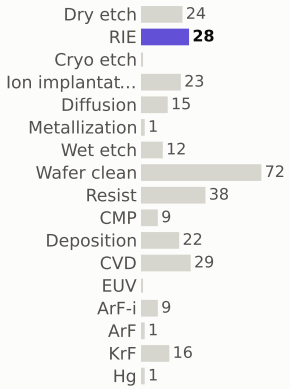
y cut at 57.351 µm



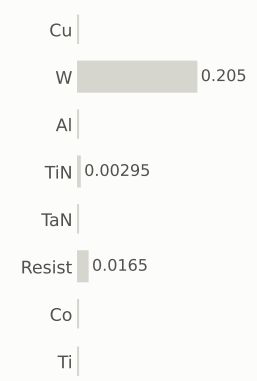
mask for this step

no mask for this step

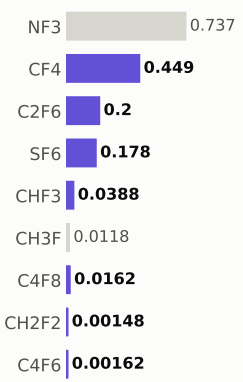
steps by process area



materials deposited (g)



gases emitted (g)



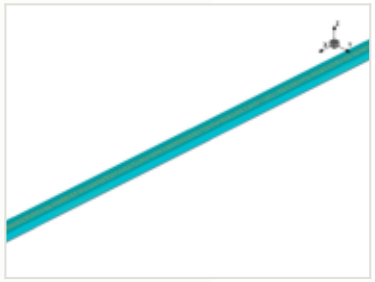
Step #3990: Dry_etch

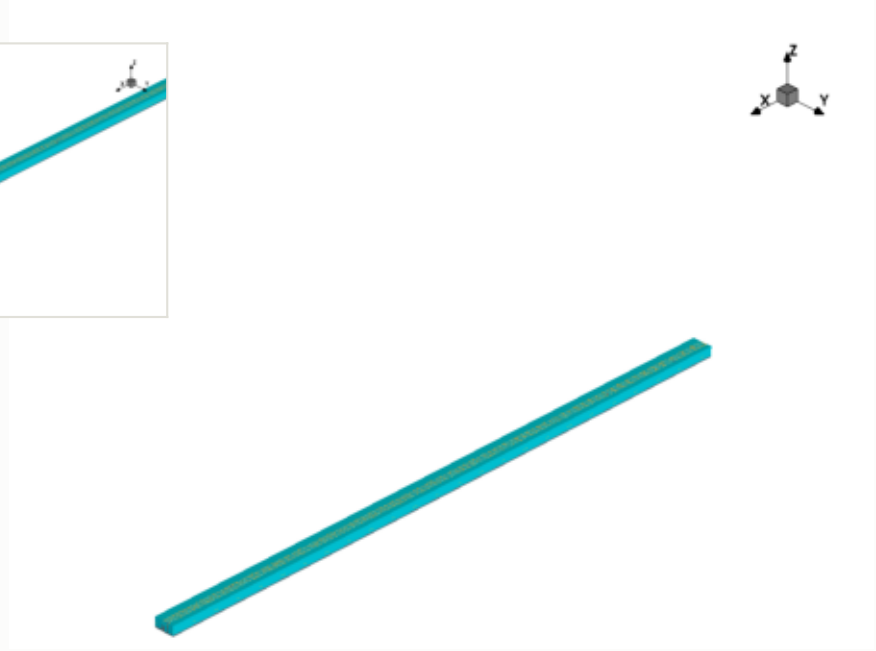
etch_strip · Dry_etch

0.801	186.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

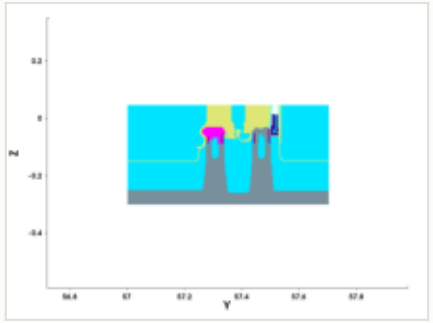
3D structure after this step

zoomed to fit

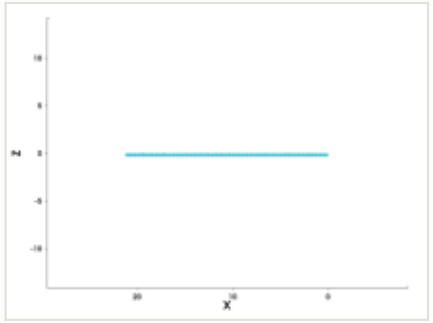




x cut at 10.650 µm



y cut at 57.351 µm



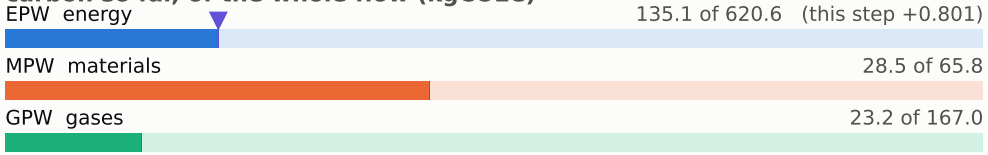
mask for this step

no mask for this step

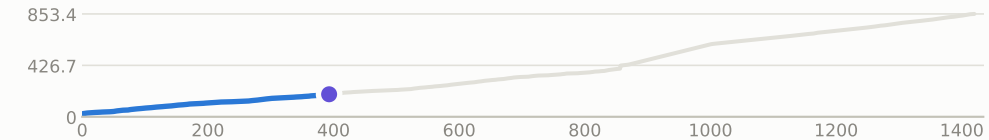
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

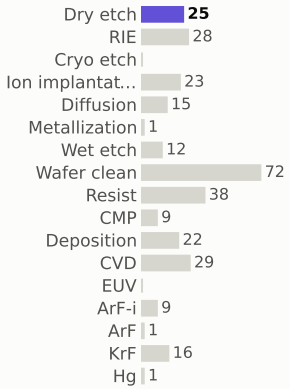
carbon so far, of the whole flow (kgCO2e)



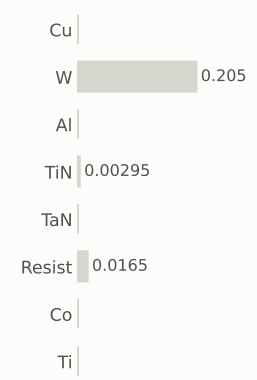
embodied carbon up to each step (kgCO2e per wafer)



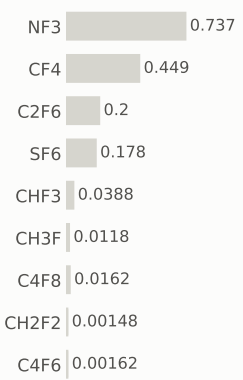
steps by process area



materials deposited (g)



gases emitted (g)



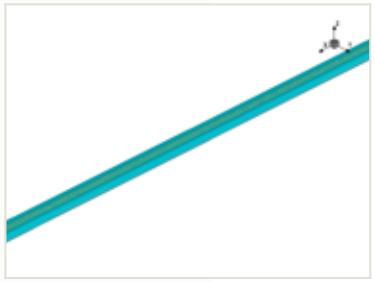
Step #4000: Wafer_clean

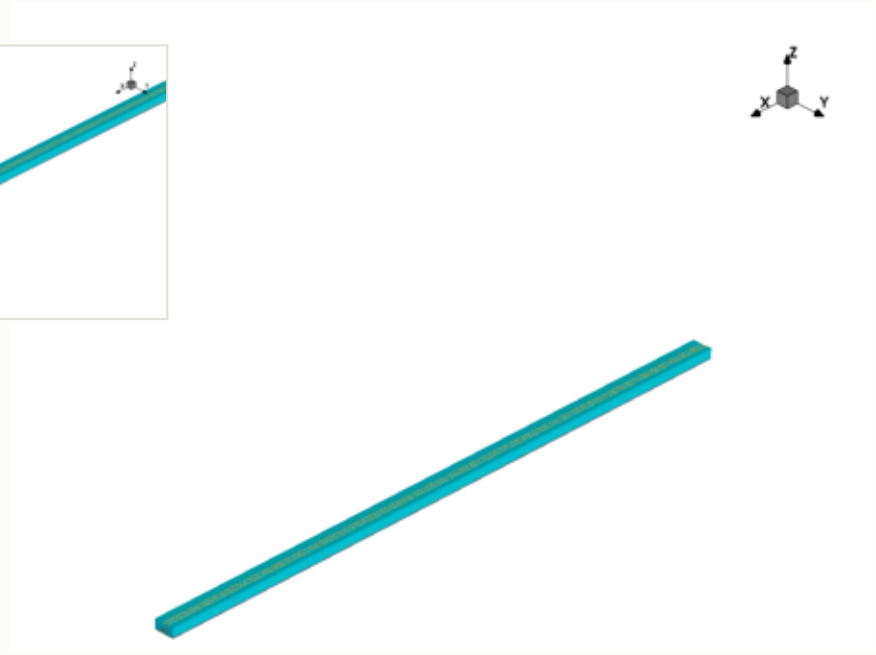
clean · Wafer_clean

0.169	187.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





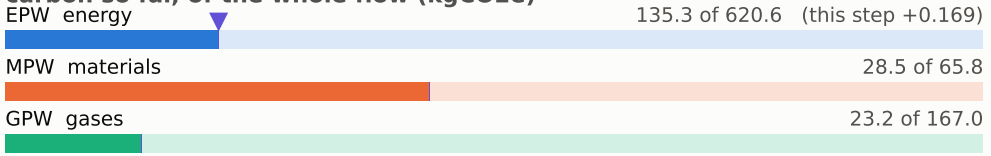
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

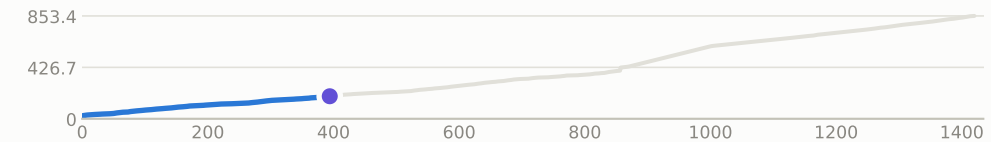
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

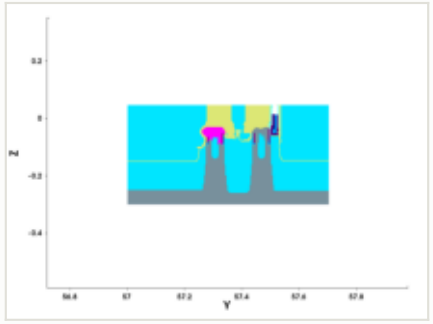
carbon so far, of the whole flow (kgCO2e)



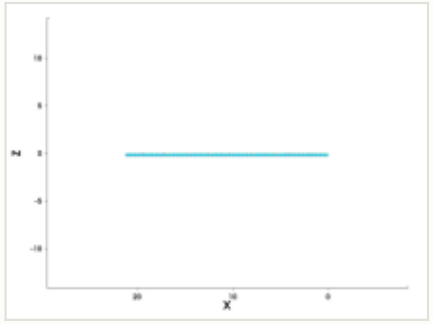
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



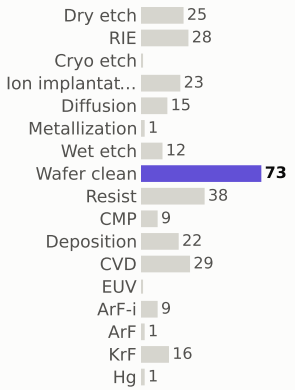
y cut at 57.351 µm



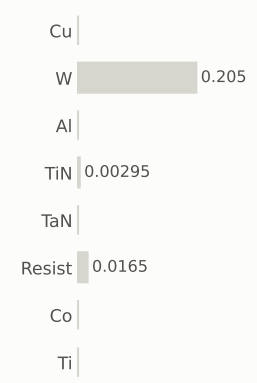
mask for this step

no mask for this step

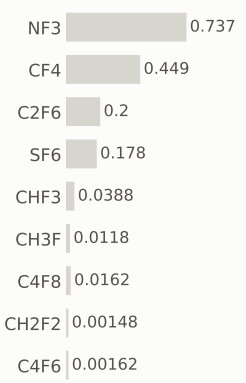
steps by process area



materials deposited (g)



gases emitted (g)



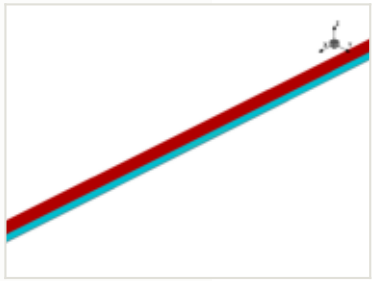
Step #4010: CVD

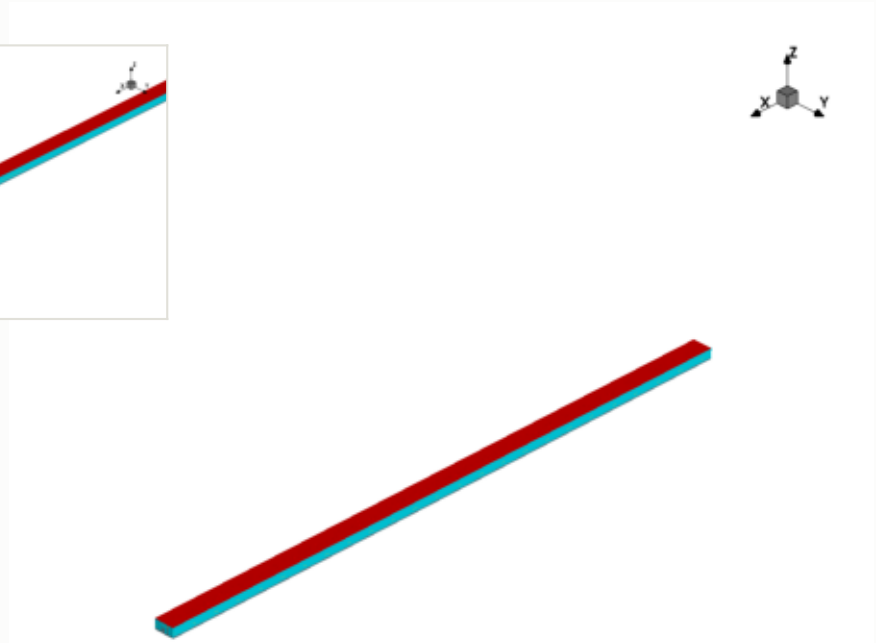
depo_isotropic · CVD

1.31	188.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



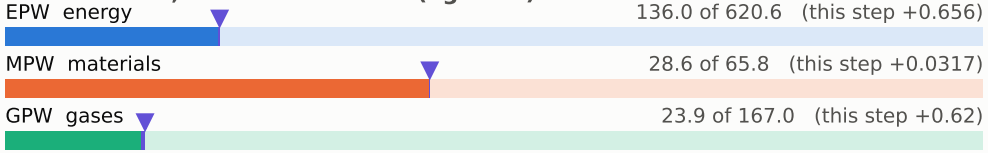


this step

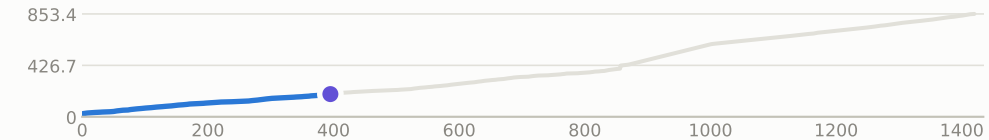
process area	CVD
material	—
thickness	0.011 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

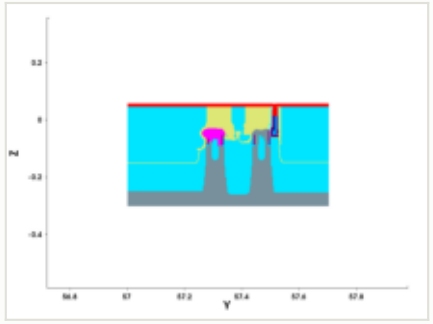
carbon so far, of the whole flow (kgCO2e)



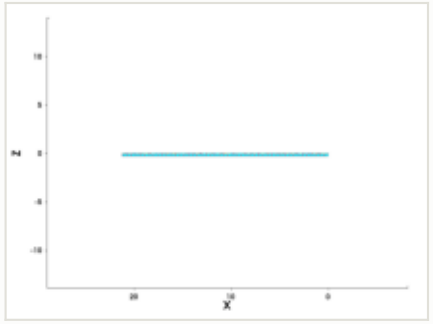
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



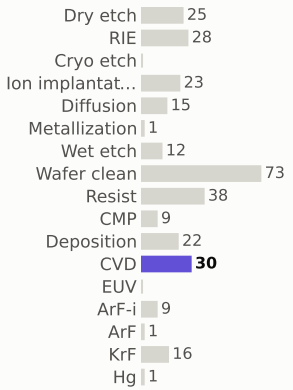
y cut at 57.351 µm



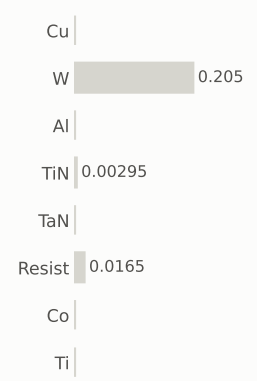
mask for this step

no mask for this step

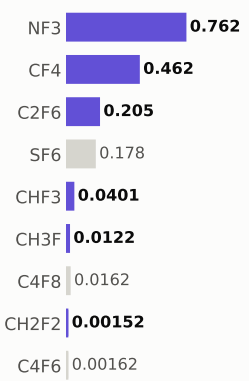
steps by process area



materials deposited (g)



gases emitted (g)



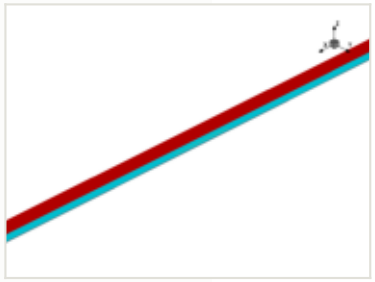
Step #4020: Wafer_clean

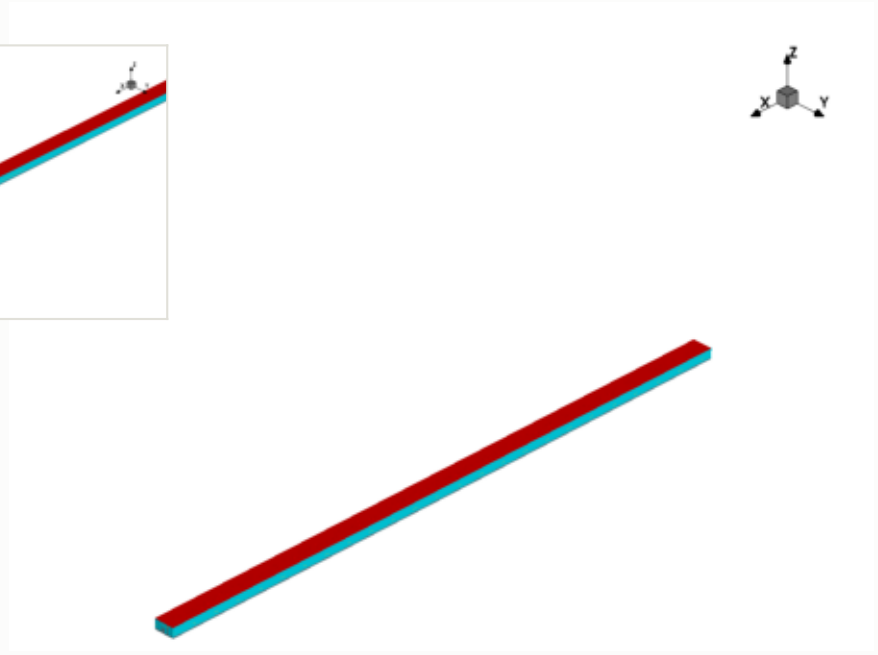
clean · Wafer_clean

0.169	188.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





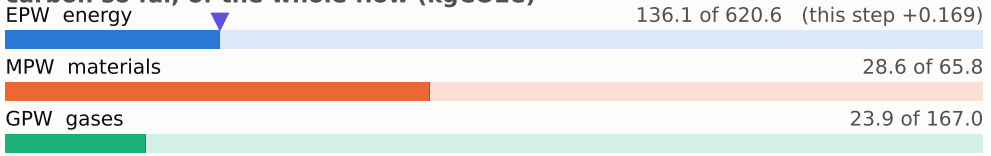
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

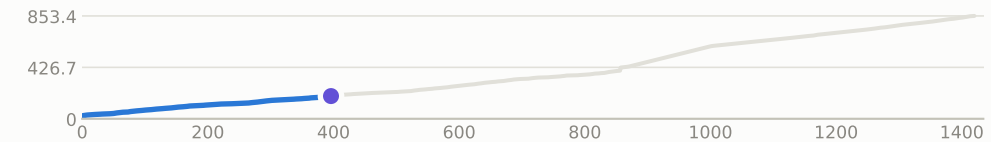
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

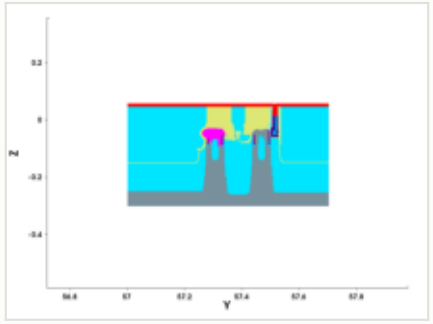
carbon so far, of the whole flow (kgCO2e)



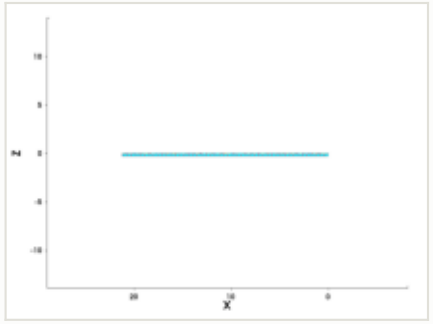
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



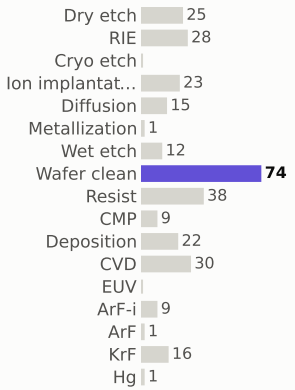
y cut at 57.351 µm



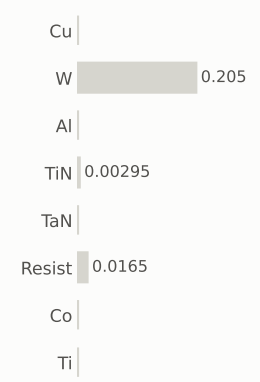
mask for this step

no mask for this step

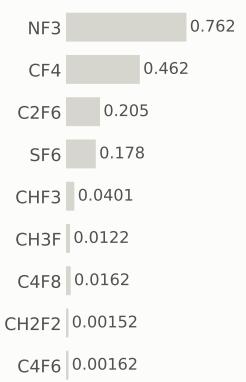
steps by process area



materials deposited (g)



gases emitted (g)



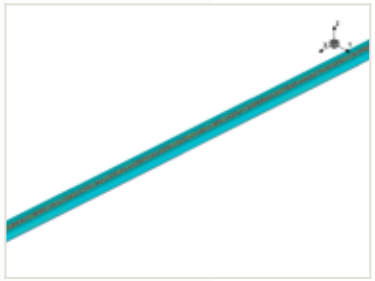
Step #4030: CMP

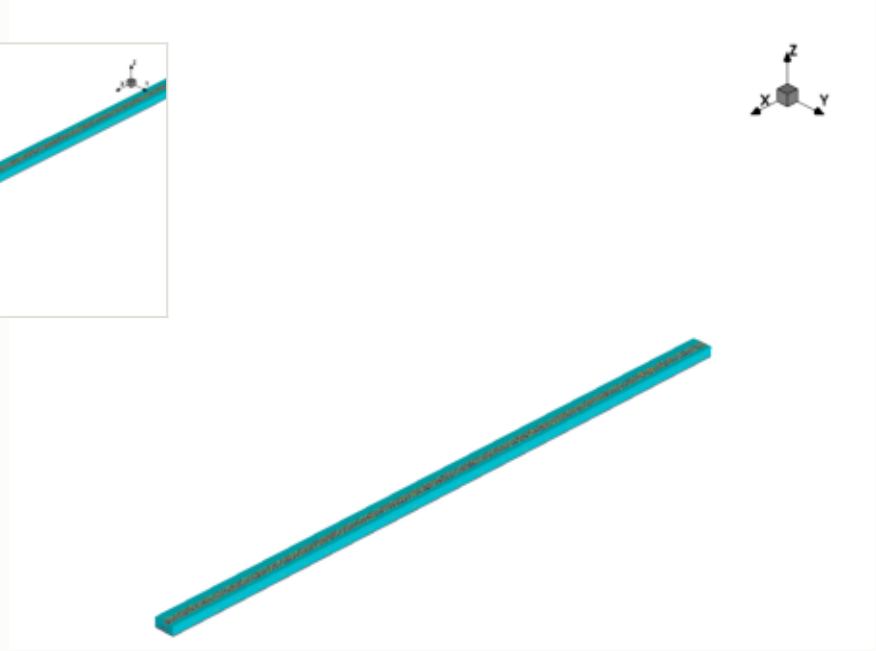
etch_cmp · CMP

0.169	188.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



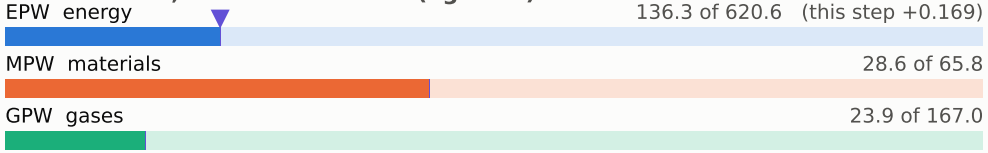


this step

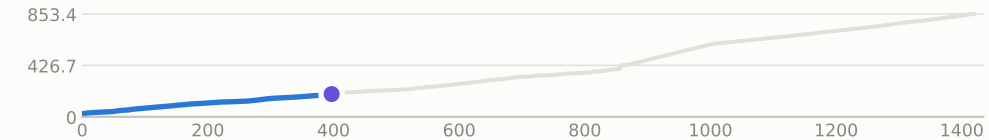
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

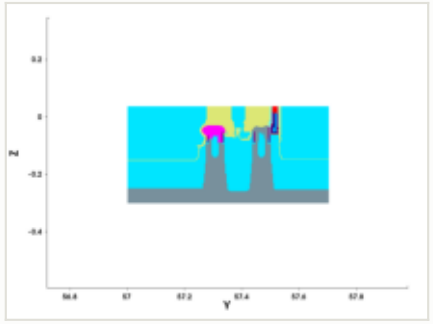
carbon so far, of the whole flow (kgCO2e)



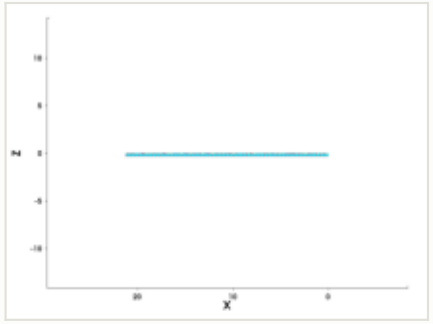
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



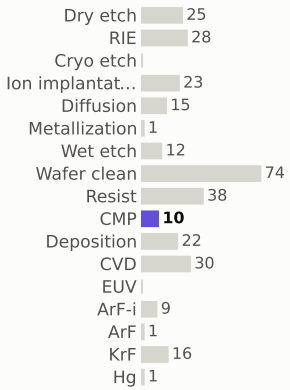
y cut at 57.351 µm



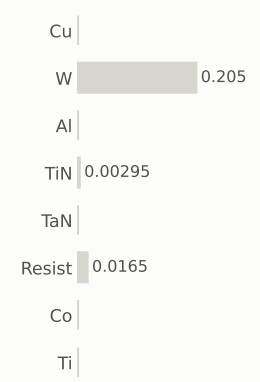
mask for this step

no mask for this step

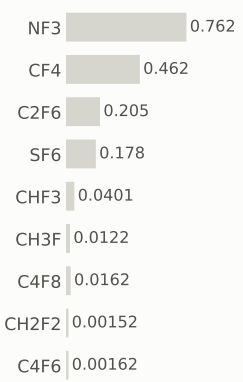
steps by process area



materials deposited (g)



gases emitted (g)



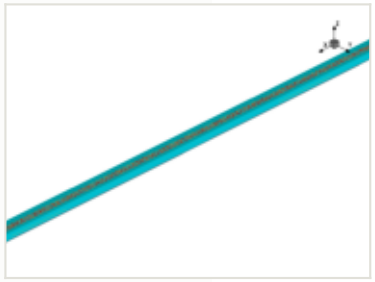
Step #4040: Wafer_clean

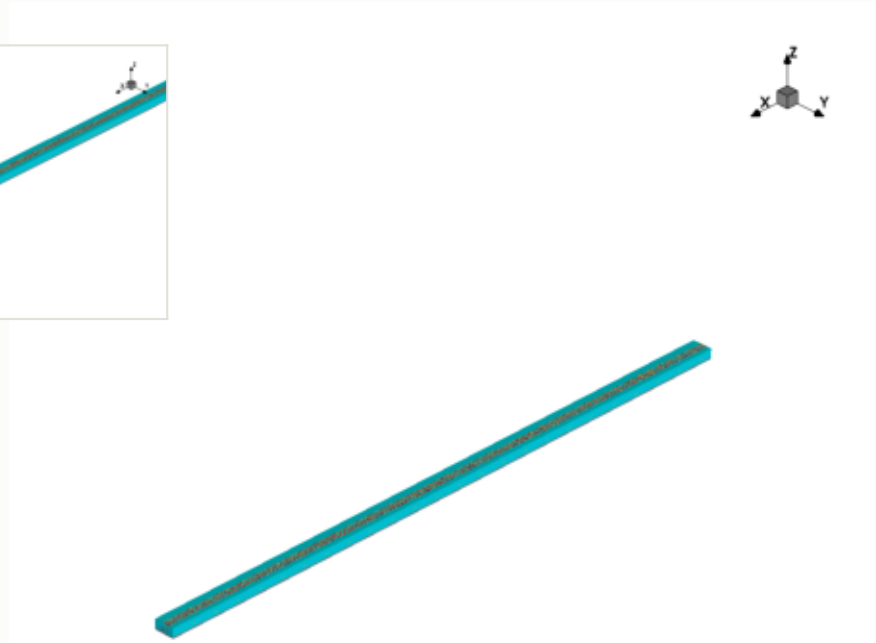
clean · Wafer_clean

0.169	188.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

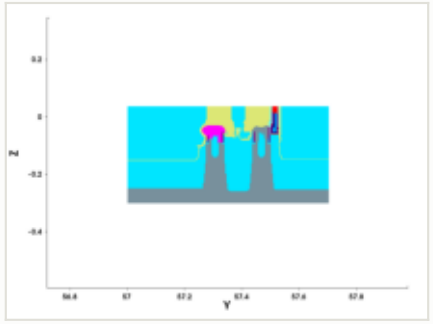
3D structure after this step

zoomed to fit

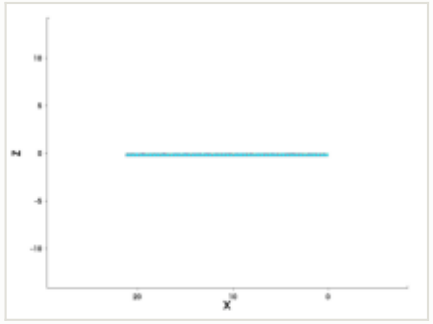




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

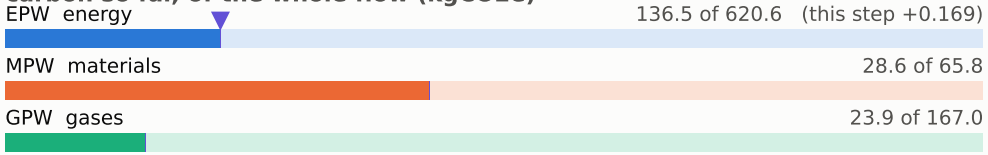
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

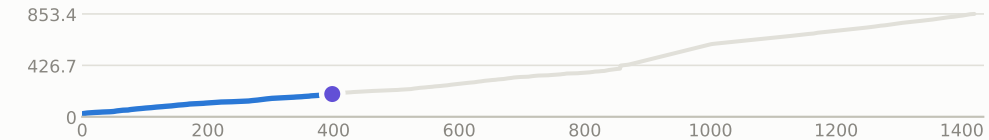
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

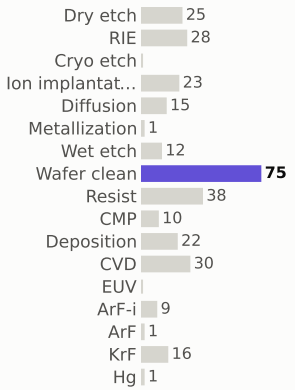
carbon so far, of the whole flow (kgCO2e)



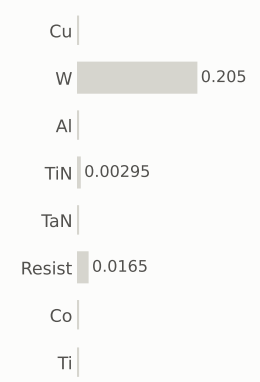
embodied carbon up to each step (kgCO2e per wafer)



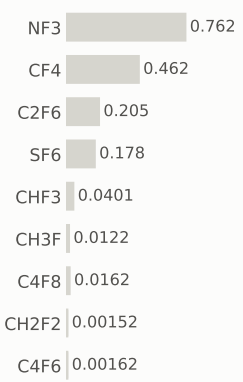
steps by process area



materials deposited (g)



gases emitted (g)



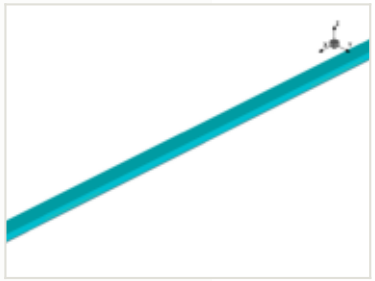
Step #4060: CVD

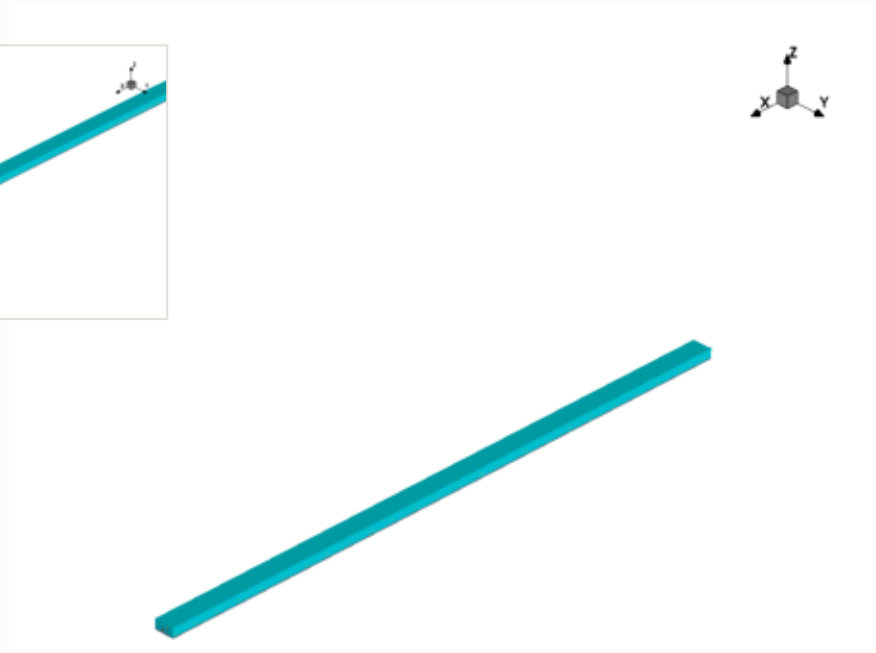
depo_fill · CVD

1.31	190.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



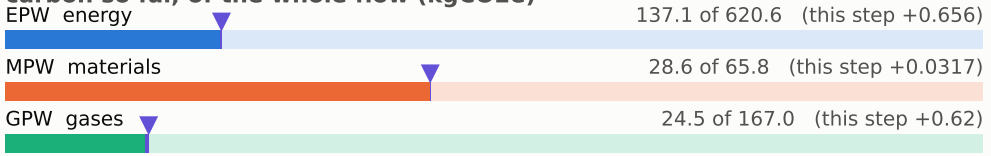


this step

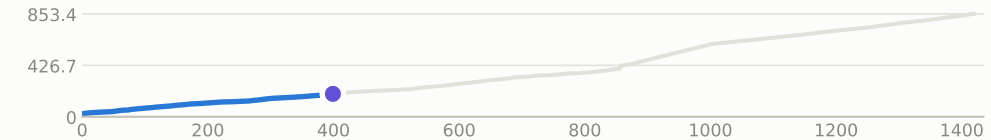
process area	CVD
material	—
thickness	0.003 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

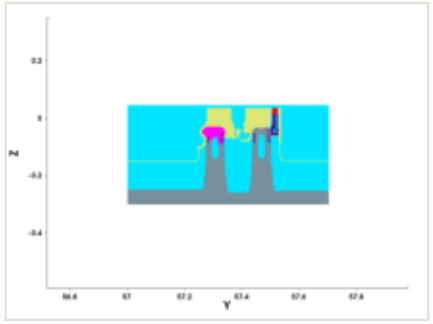
carbon so far, of the whole flow (kgCO2e)



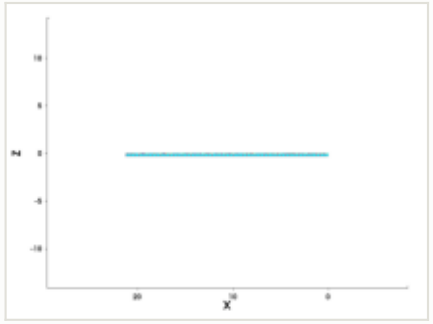
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



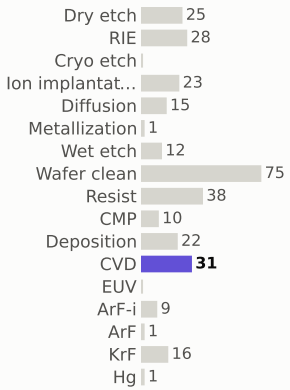
y cut at 57.351 µm



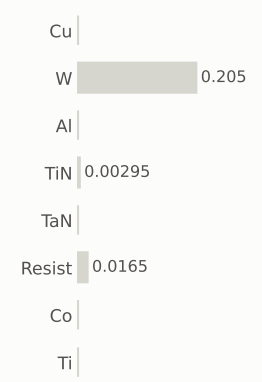
mask for this step

no mask for this step

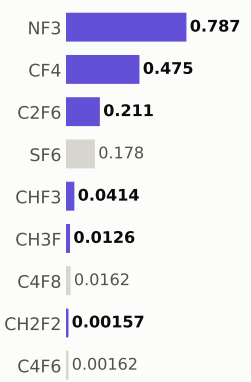
steps by process area



materials deposited (g)



gases emitted (g)



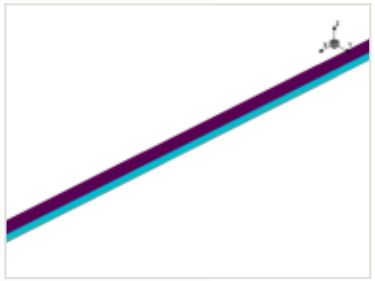
Step #4070: Deposition

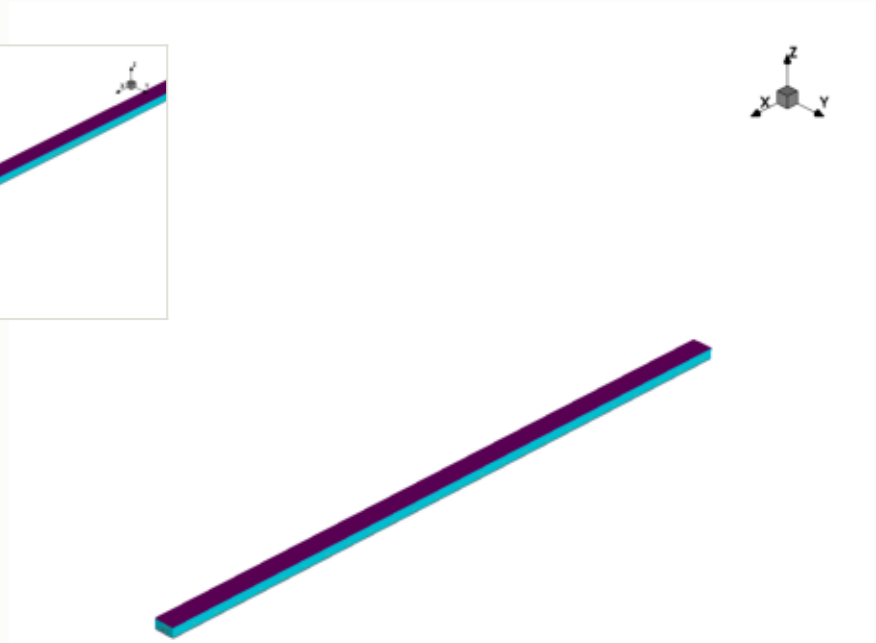
depo_isotropic · Deposition

0.656	190.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



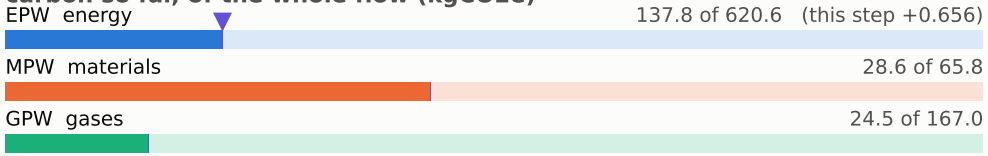


this step

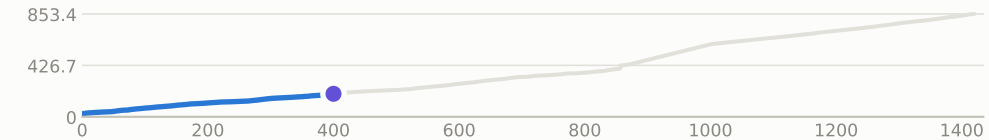
process area	Deposition
material	—
thickness	0.019 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

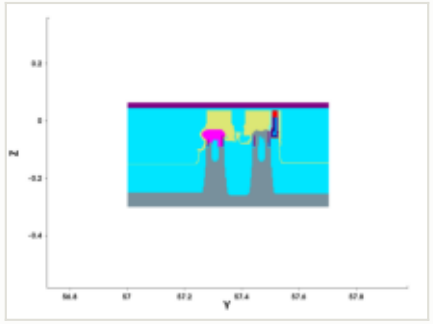
carbon so far, of the whole flow (kgCO2e)



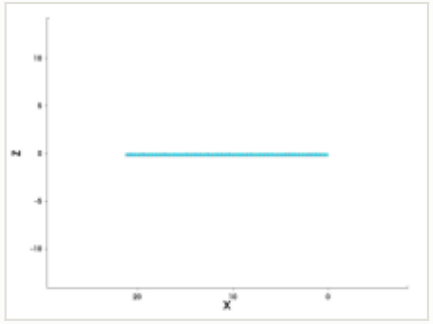
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



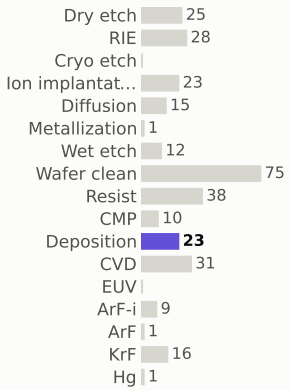
y cut at 57.351 µm



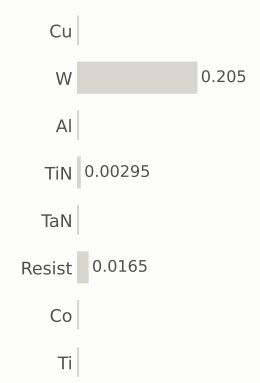
mask for this step

no mask for this step

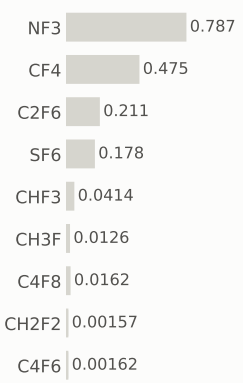
steps by process area



materials deposited (g)



gases emitted (g)



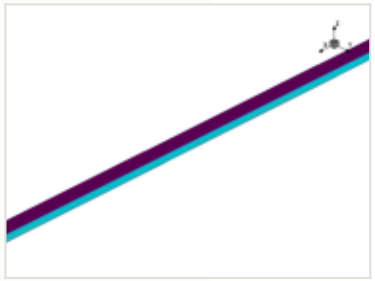
Step #4080: Wafer_clean

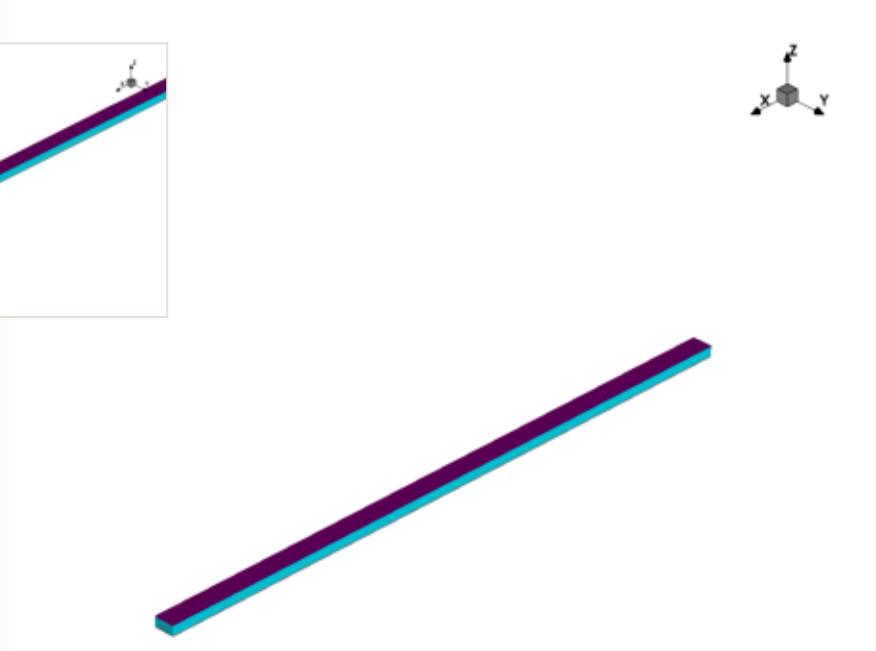
clean · Wafer_clean

0.169	191.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





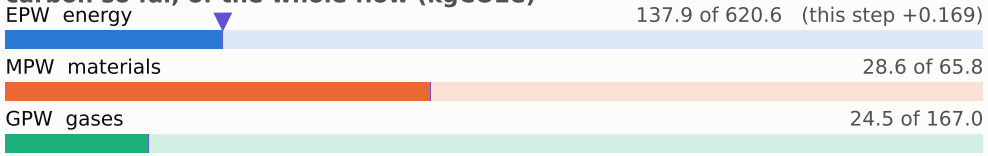
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

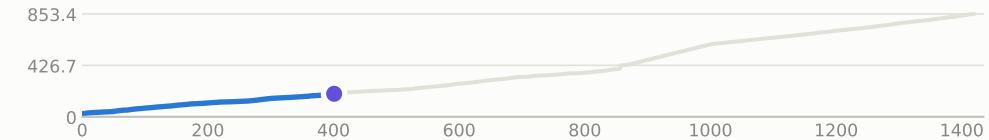
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

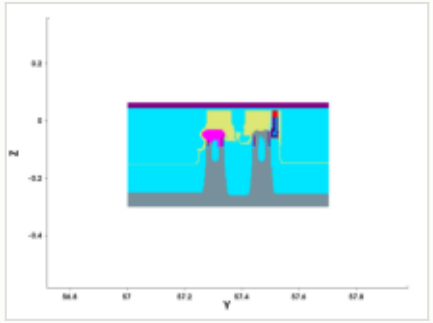
carbon so far, of the whole flow (kgCO2e)



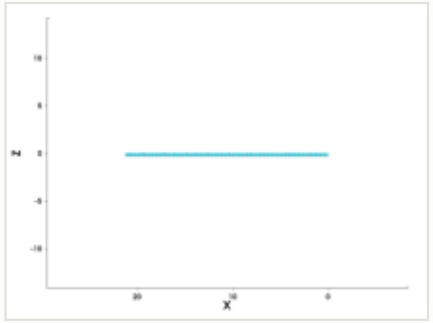
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



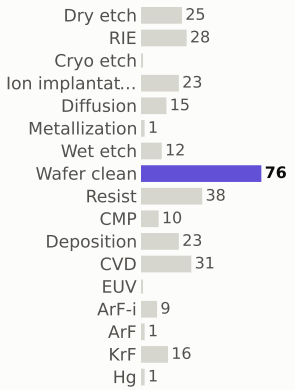
y cut at 57.351 µm



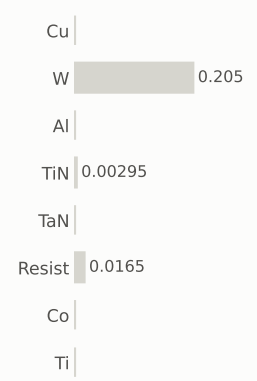
mask for this step

no mask for this step

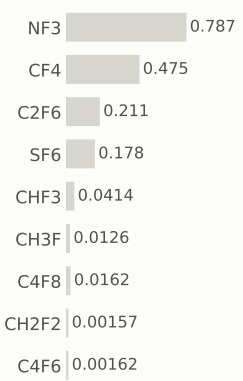
steps by process area



materials deposited (g)



gases emitted (g)



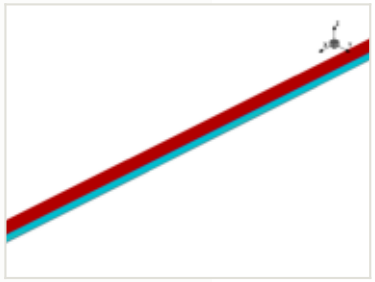
Step #4090: CVD

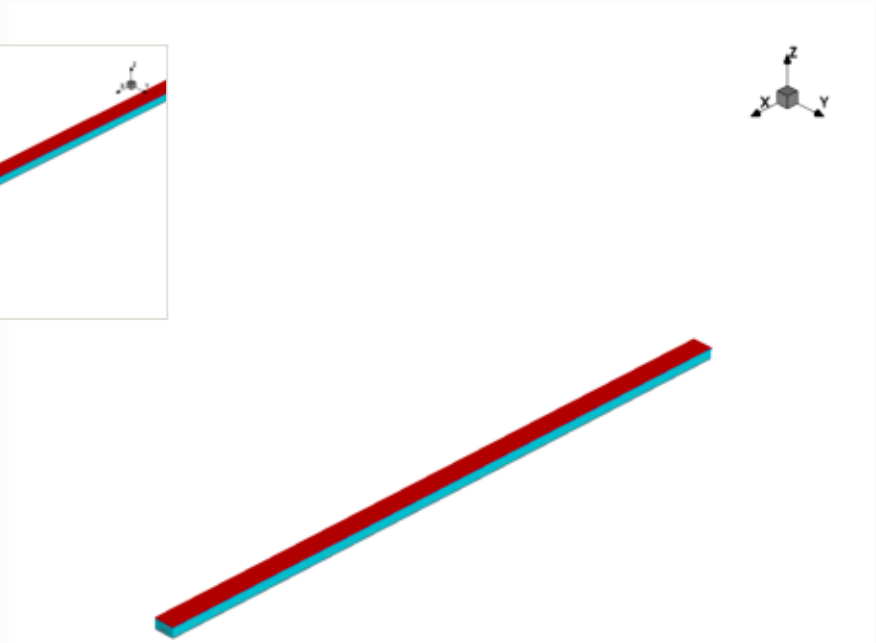
depo_isotropic · CVD

1.31	192.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



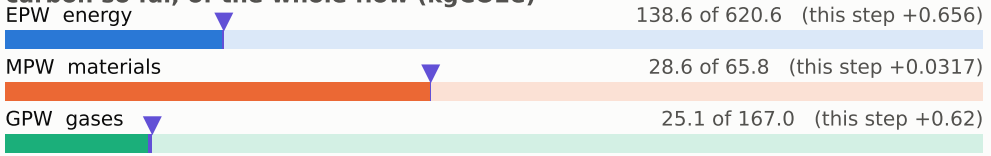


this step

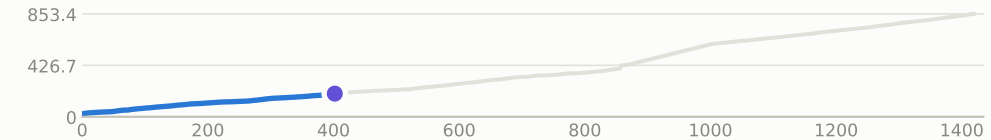
process area	CVD
material	—
thickness	0.022 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

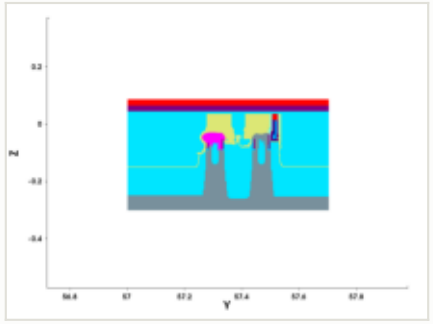
carbon so far, of the whole flow (kgCO2e)



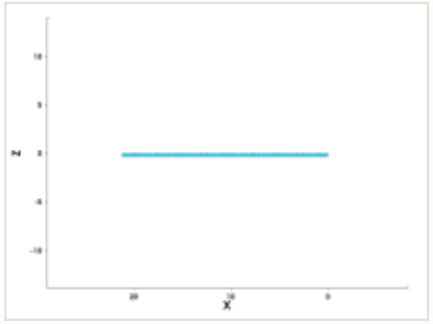
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



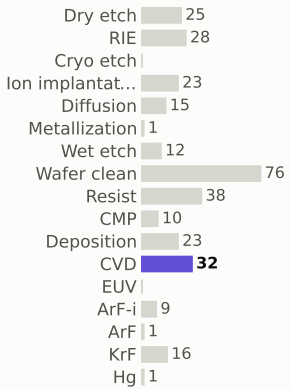
y cut at 57.351 µm



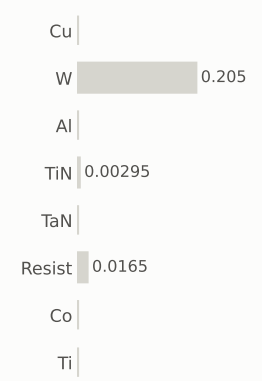
mask for this step

no mask for this step

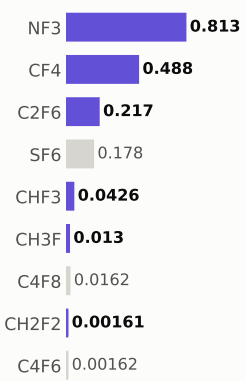
steps by process area



materials deposited (g)



gases emitted (g)



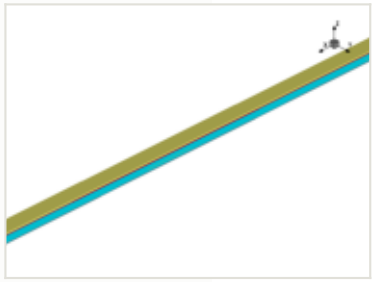
Step #4100: Spx_only

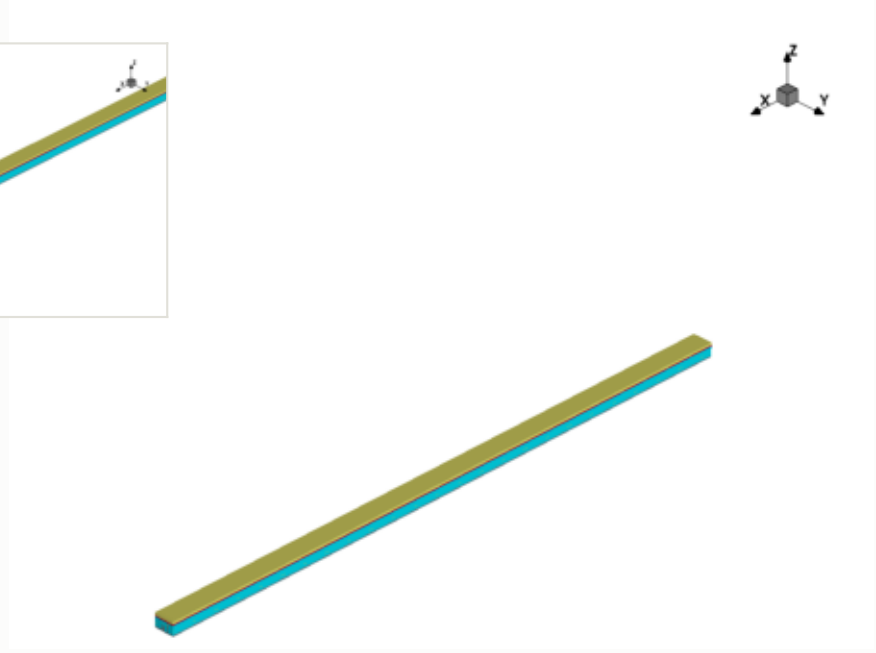
depo_fill · Spx_only

2.59e-05	192.3	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

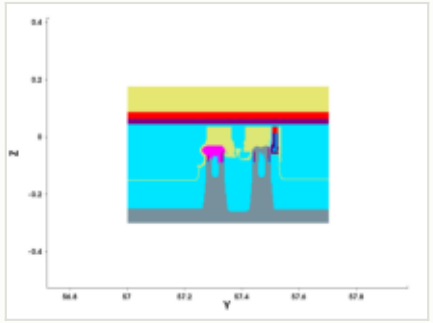
3D structure after this step

zoomed to fit

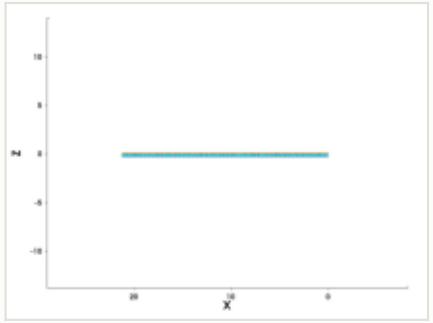




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

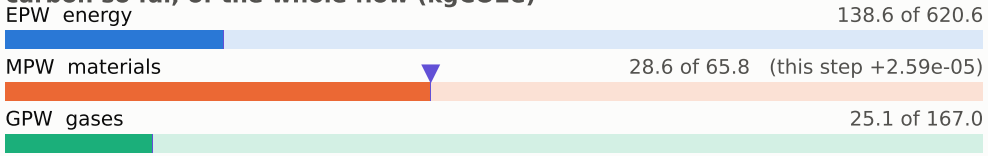
this step

process area	Spx_only
material	Resist
thickness	0.09 µm
material GWP	4.2 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

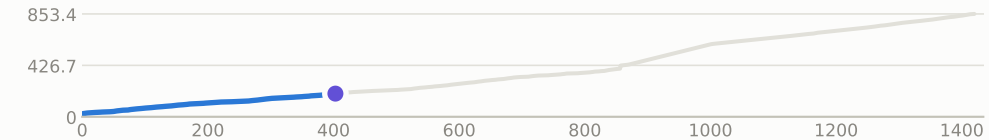
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	2.59e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**2.59e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

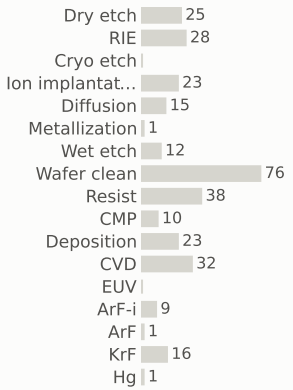
carbon so far, of the whole flow (kgCO2e)



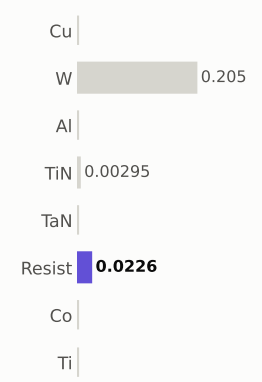
embodied carbon up to each step (kgCO2e per wafer)



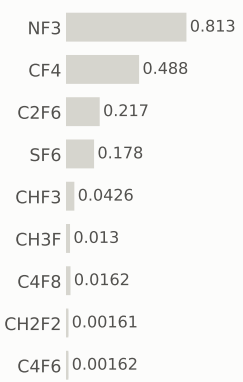
steps by process area



materials deposited (g)



gases emitted (g)



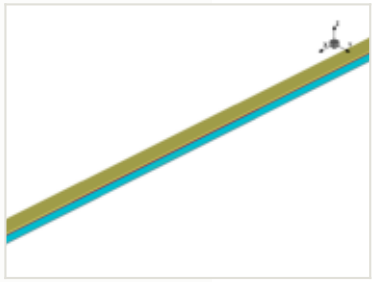
Step #4110: Resist

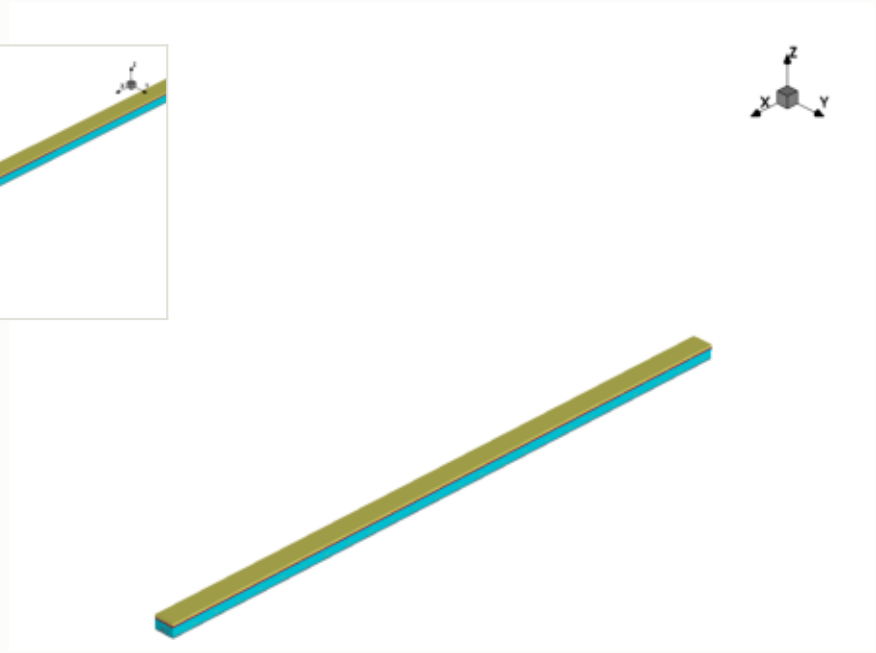
resist · Resist

0.169	192.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





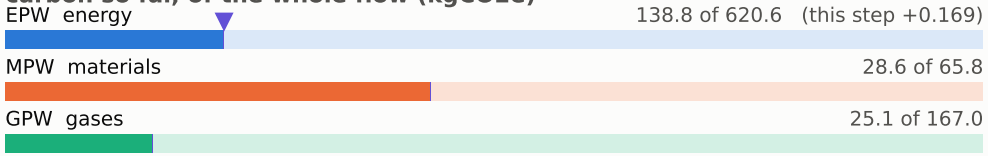
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

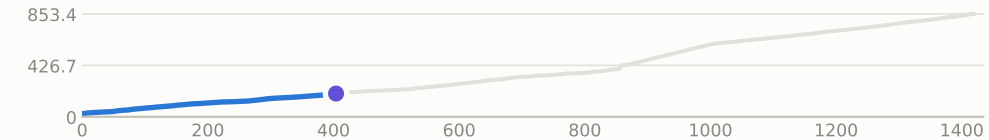
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

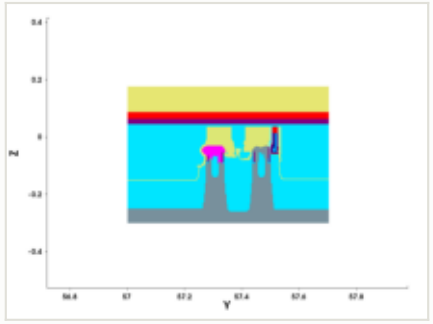
carbon so far, of the whole flow (kgCO2e)



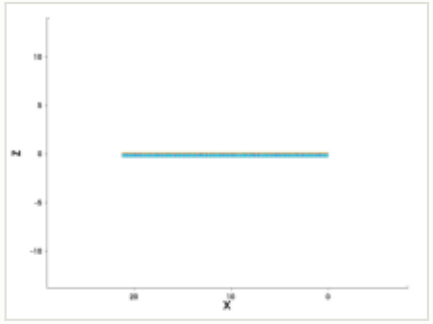
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



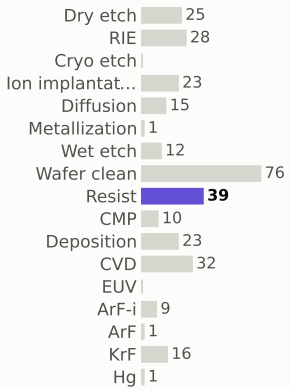
y cut at 57.351 µm



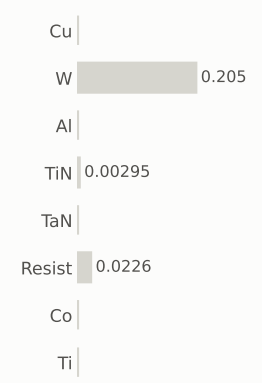
mask for this step

no mask for this step

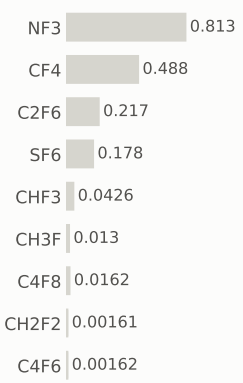
steps by process area



materials deposited (g)



gases emitted (g)



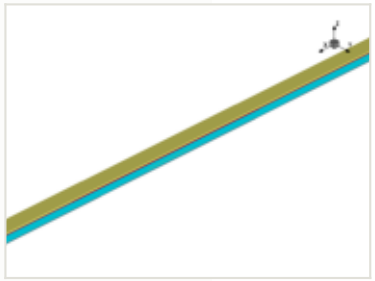
Step #4120: KrF

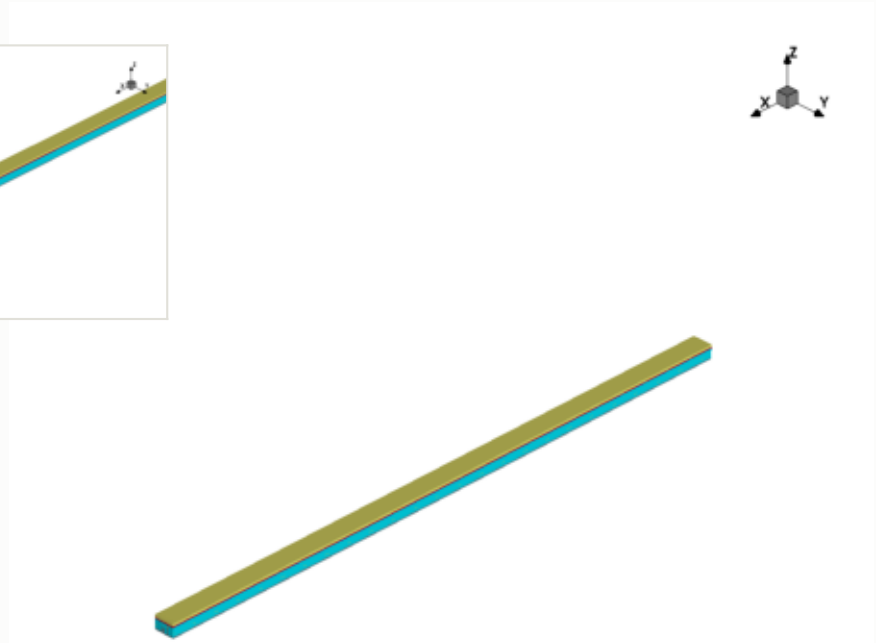
litho_patterning · KrF

0.199	192.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

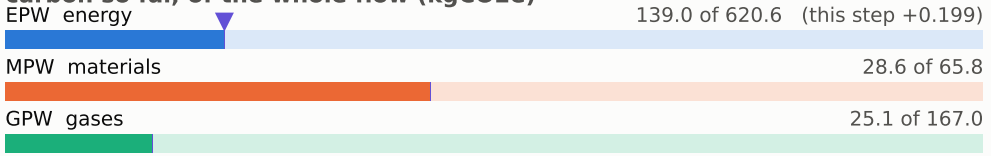




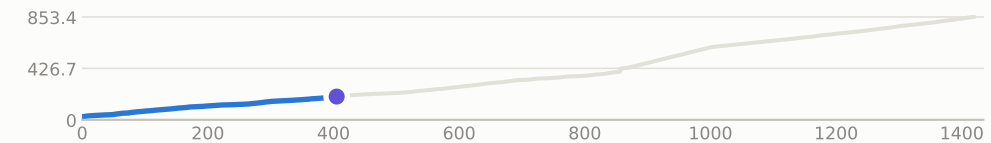
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

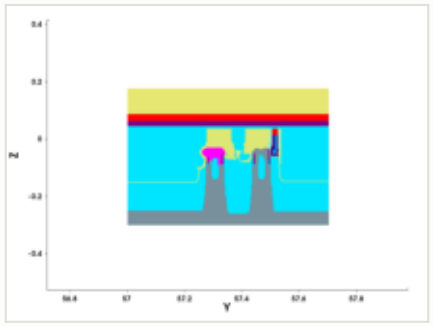
carbon so far, of the whole flow (kgCO2e)



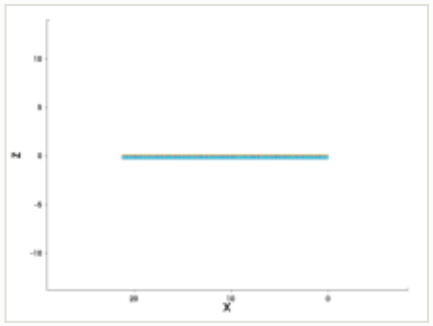
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



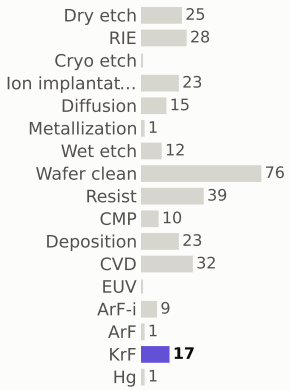
y cut at 57.351 µm



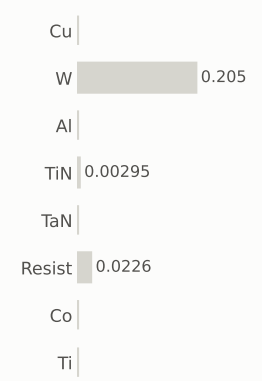
mask for this step

no mask for this step

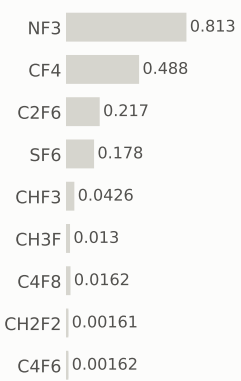
steps by process area



materials deposited (g)



gases emitted (g)



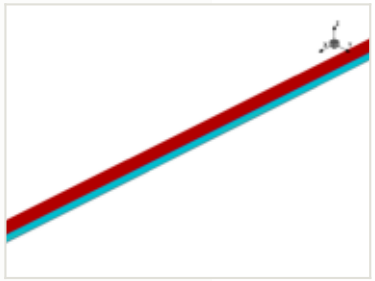
Step #4130: RIE

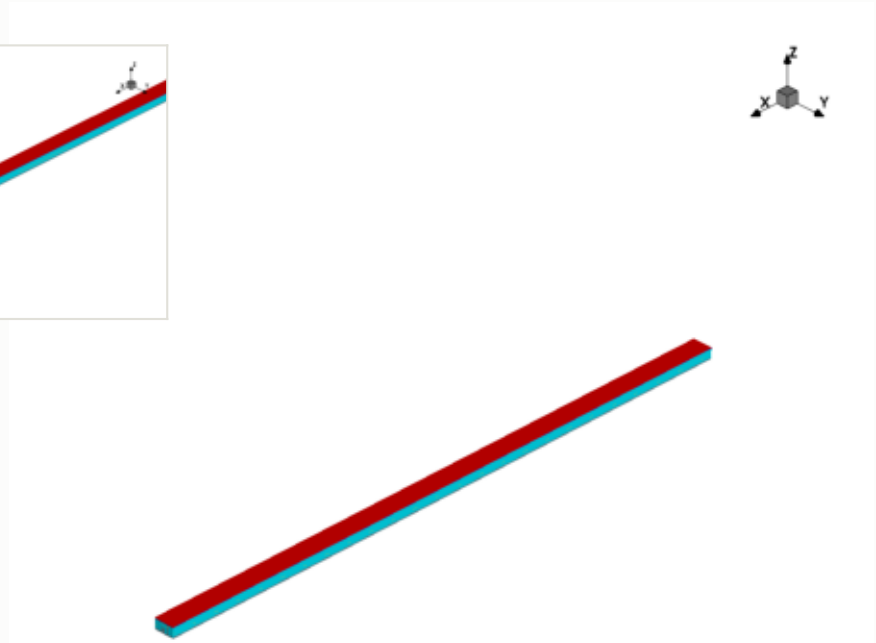
etch_anisotropic · RIE

1.04	193.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



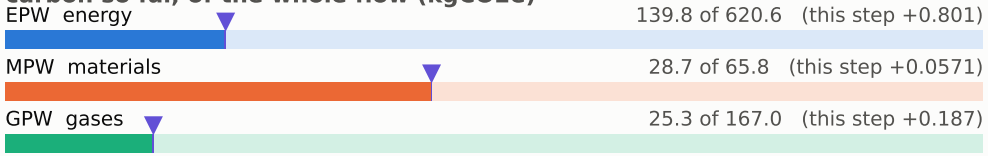


this step

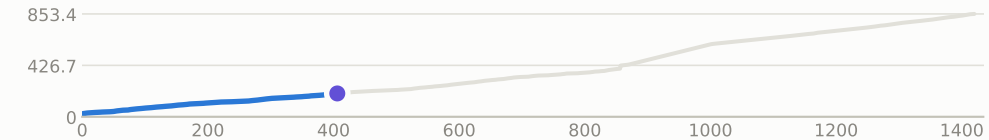
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

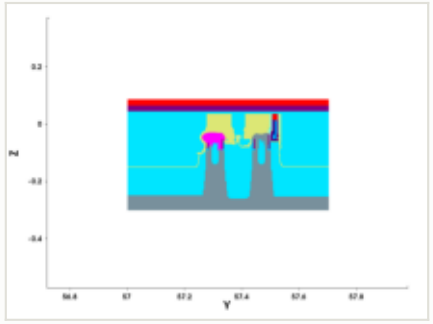
carbon so far, of the whole flow (kgCO2e)



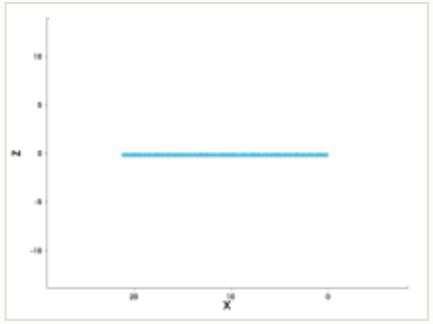
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



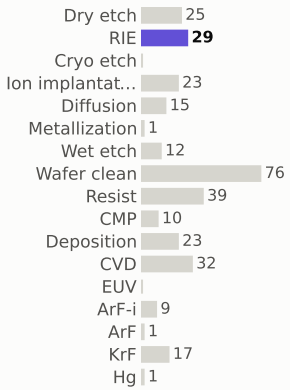
y cut at 57.351 µm



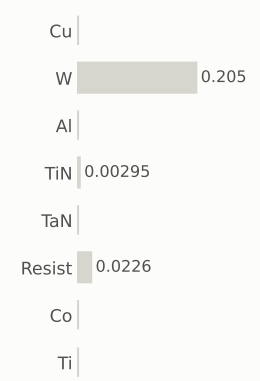
mask for this step

no mask for this step

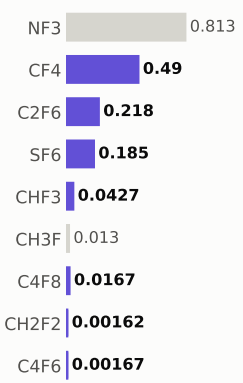
steps by process area



materials deposited (g)



gases emitted (g)



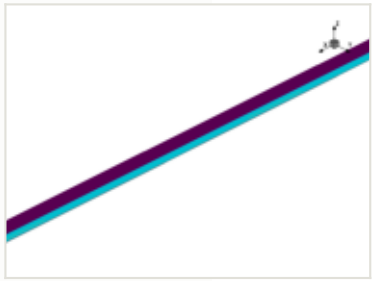
Step #4140: Spx_only

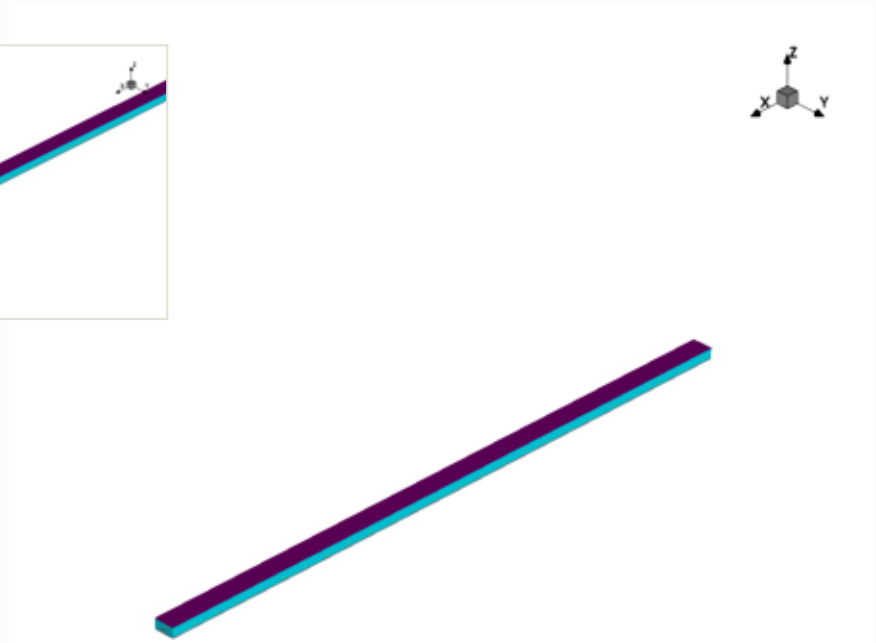
etch_anisotropic · Spx_only

0	193.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





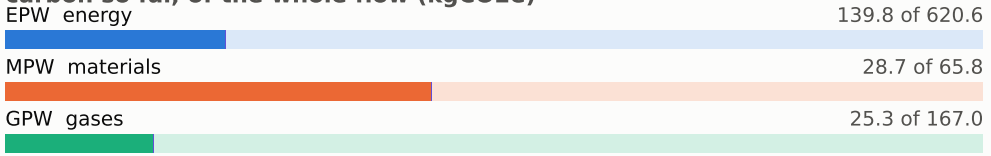
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

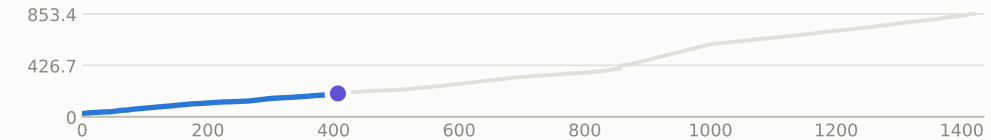
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

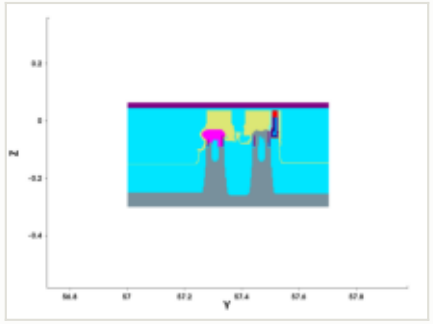
carbon so far, of the whole flow (kgCO2e)



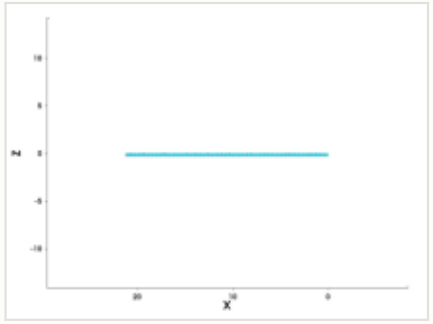
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



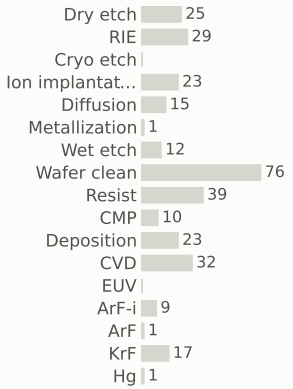
y cut at 57.351 µm



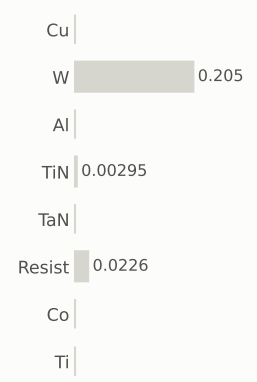
mask for this step

no mask for this step

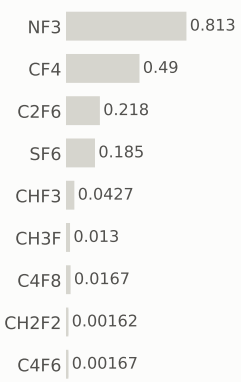
steps by process area



materials deposited (g)



gases emitted (g)



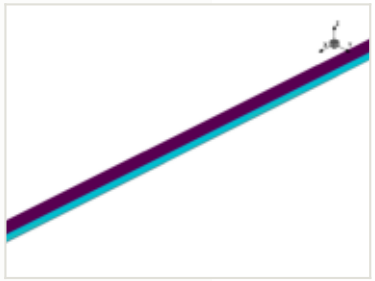
Step #4150: Dry_etch

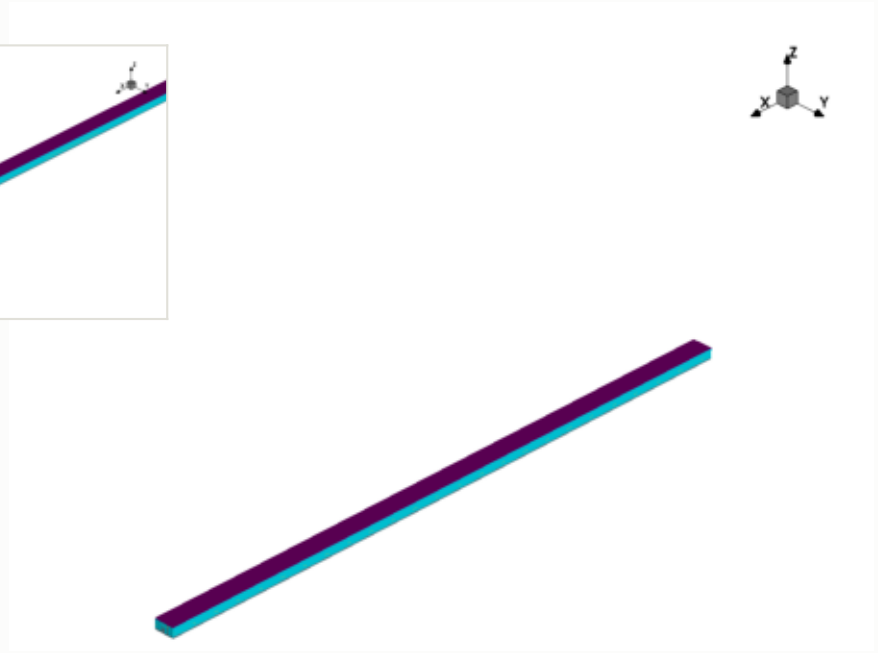
etch_strip · Dry_etch

0.801	194.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



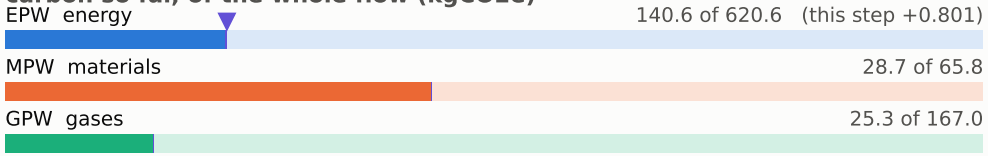


this step

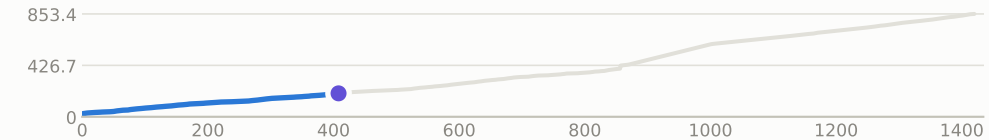
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

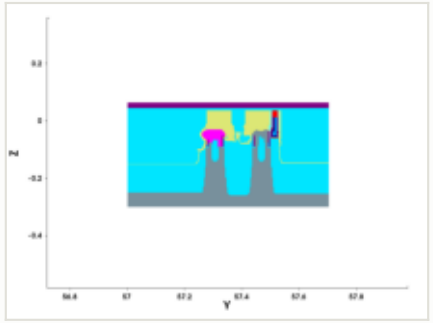
carbon so far, of the whole flow (kgCO2e)



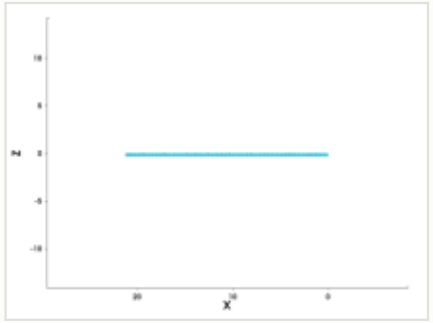
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



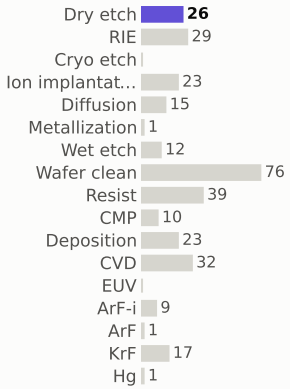
y cut at 57.351 µm



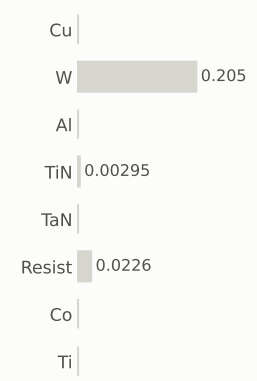
mask for this step

no mask for this step

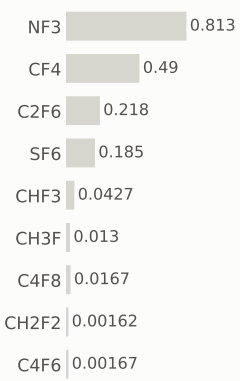
steps by process area



materials deposited (g)



gases emitted (g)



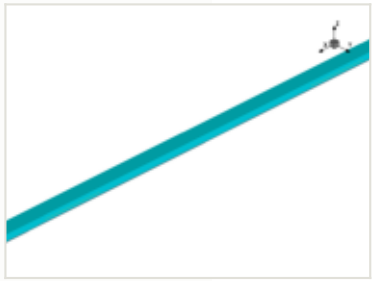
Step #4160: Spx_only

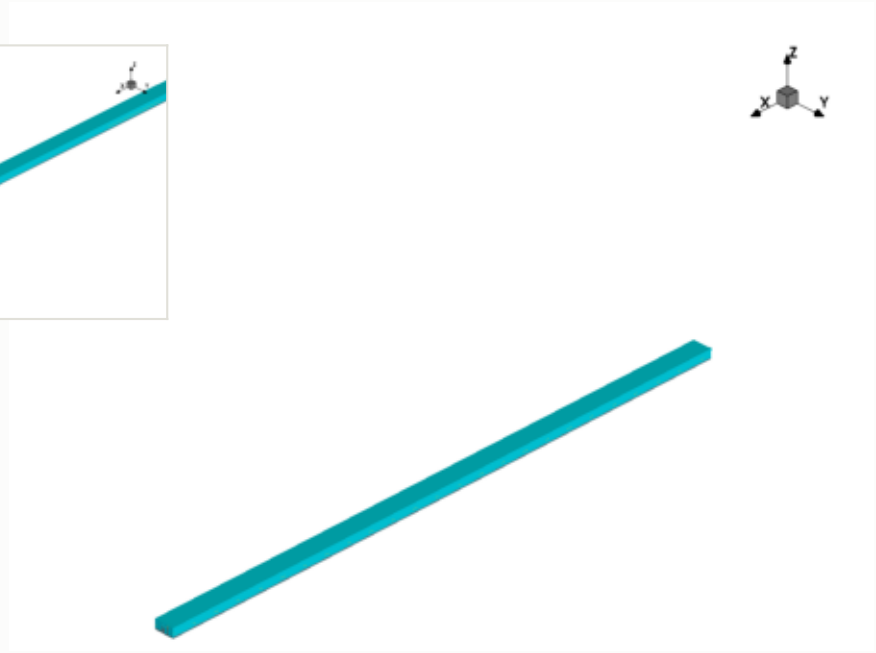
etch_anisotropic · Spx_only

0	194.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





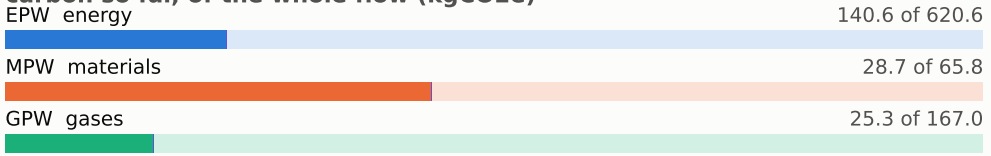
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

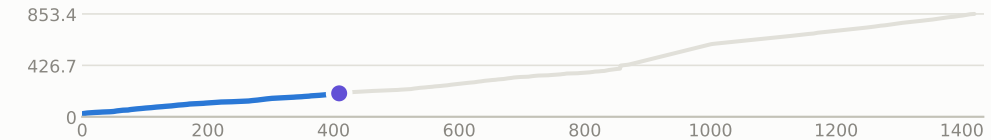
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

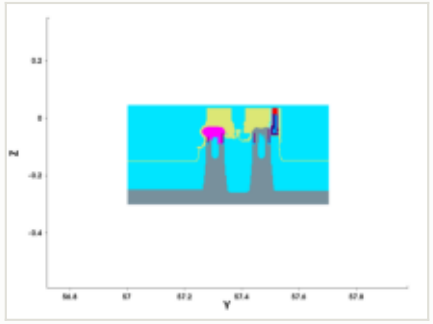
carbon so far, of the whole flow (kgCO2e)



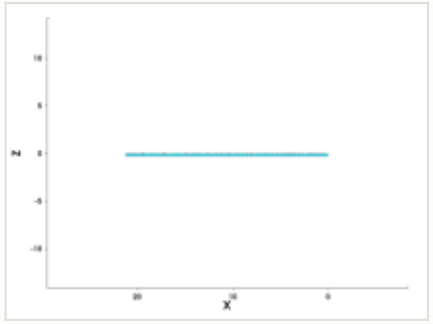
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



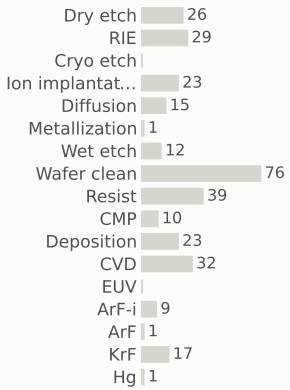
y cut at 57.351 µm



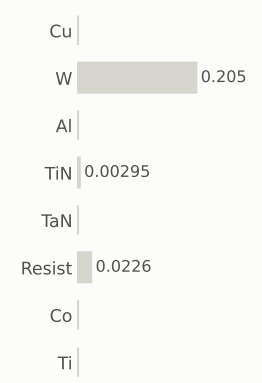
mask for this step

no mask for this step

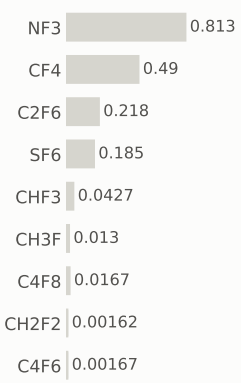
steps by process area



materials deposited (g)



gases emitted (g)



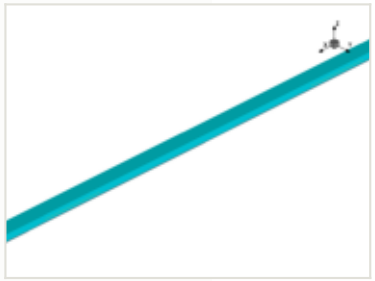
Step #4170: Wafer_clean

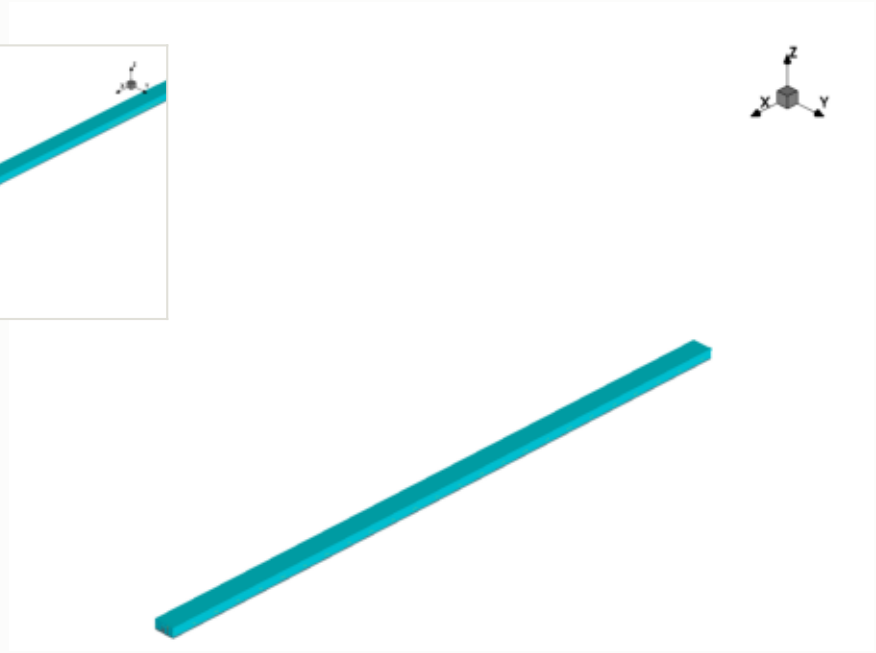
clean · Wafer_clean

0.169	194.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





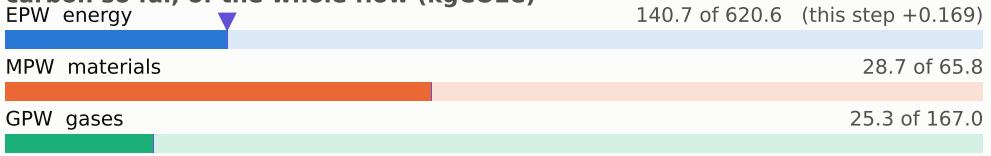
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

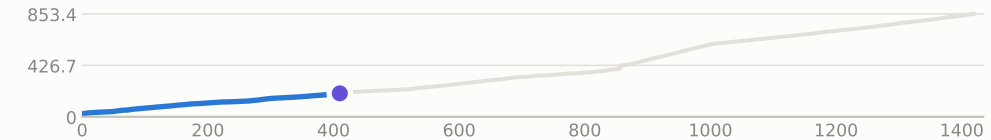
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

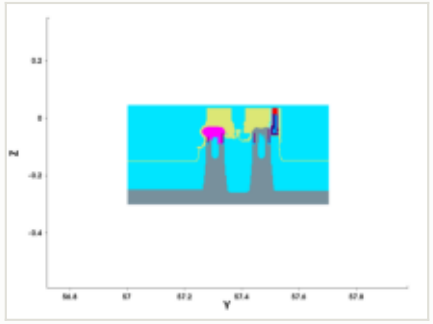
carbon so far, of the whole flow (kgCO2e)



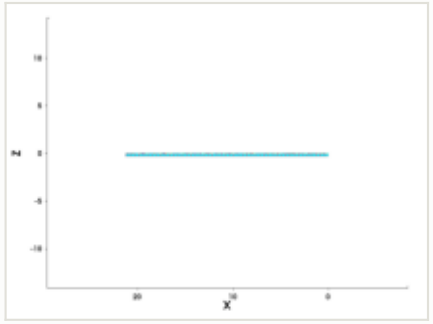
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



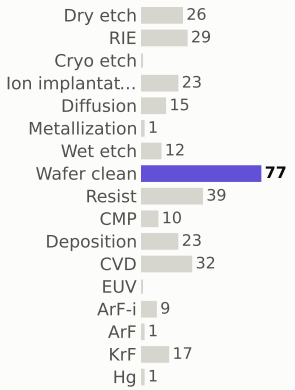
y cut at 57.351 µm



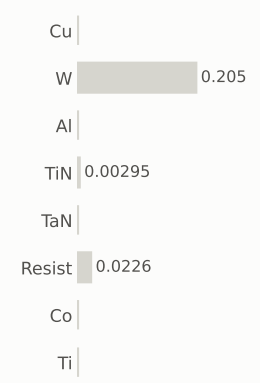
mask for this step

no mask for this step

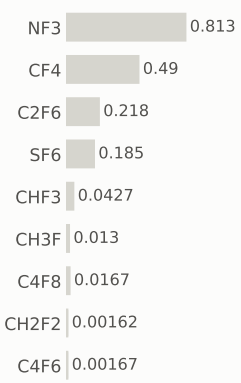
steps by process area



materials deposited (g)



gases emitted (g)



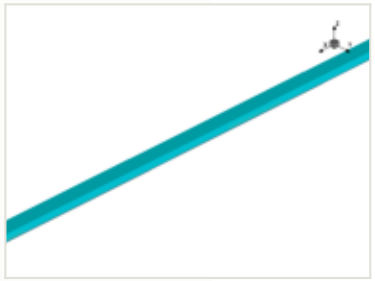
Step #4180: CVD

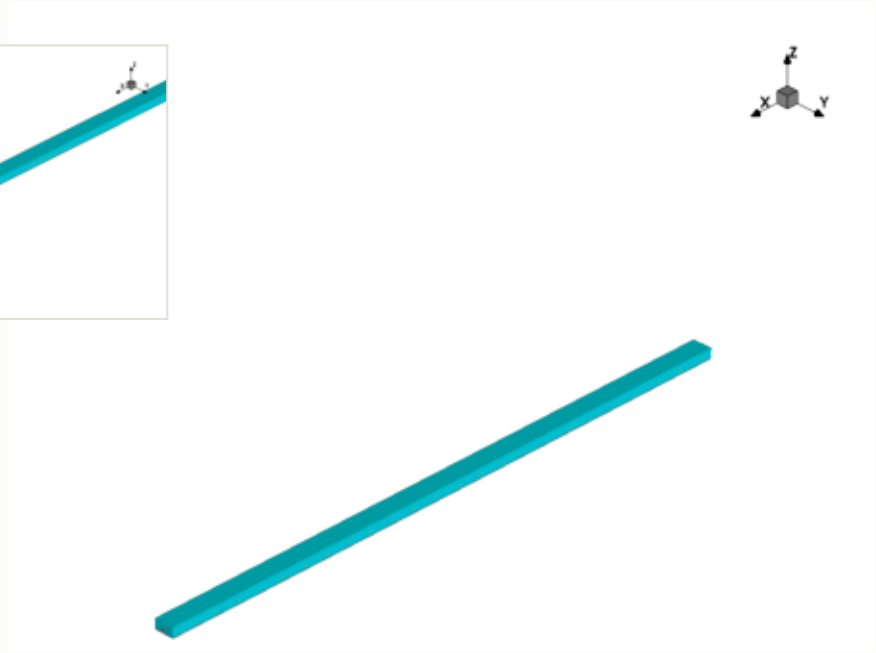
depo_fill · CVD

1.31	196.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



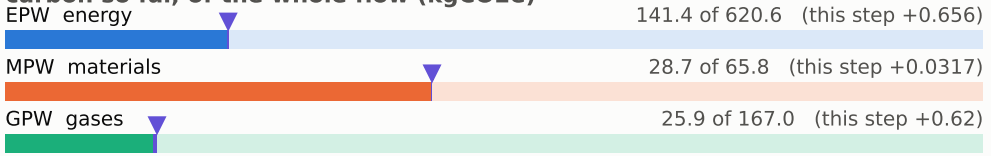


this step

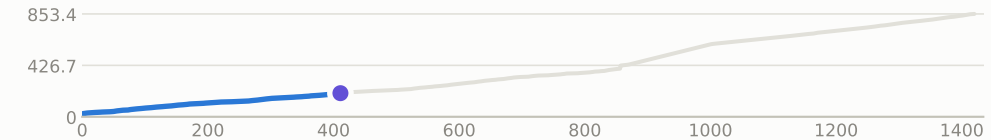
process area	CVD
material	—
thickness	0.017 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

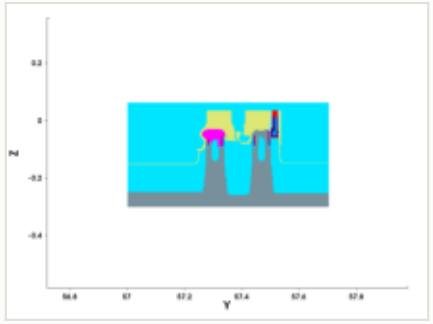
carbon so far, of the whole flow (kgCO2e)



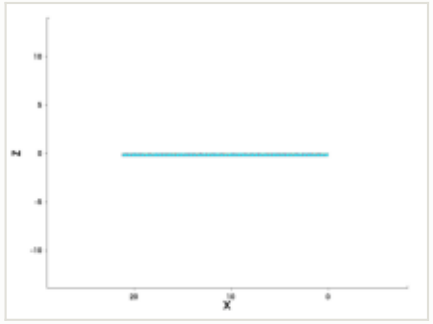
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



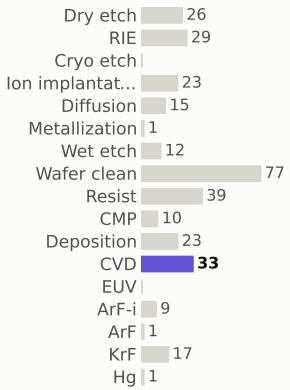
y cut at 57.351 µm



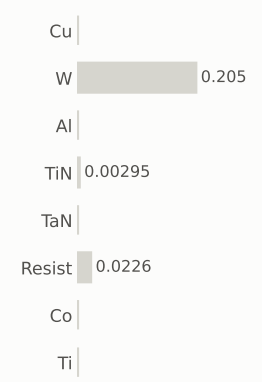
mask for this step

no mask for this step

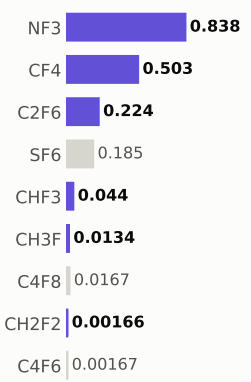
steps by process area



materials deposited (g)



gases emitted (g)



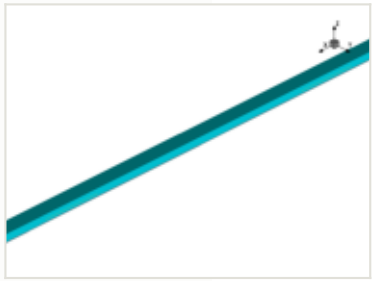
Step #4190: Spx_only

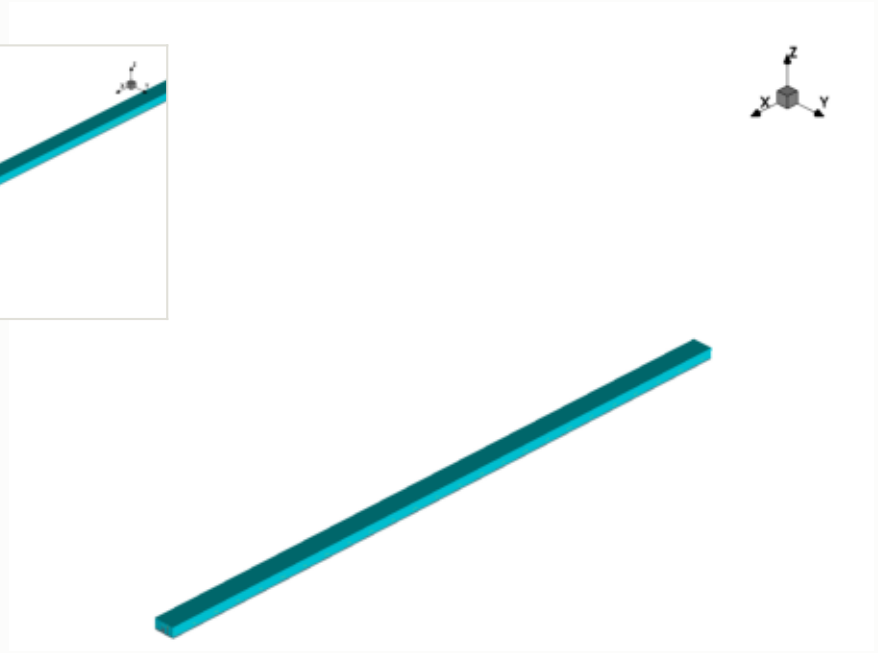
depo_fill · Spx_only

0	196.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





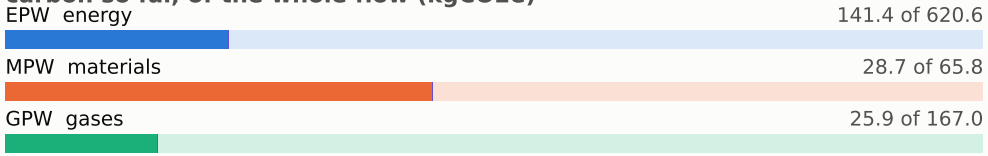
this step

process area	Spx_only
material	—
thickness	0.007 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

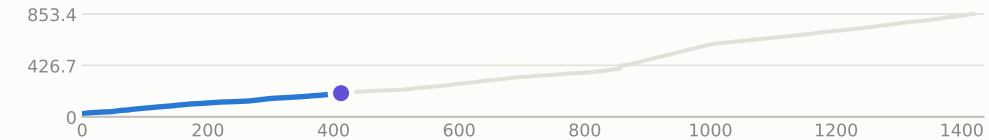
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

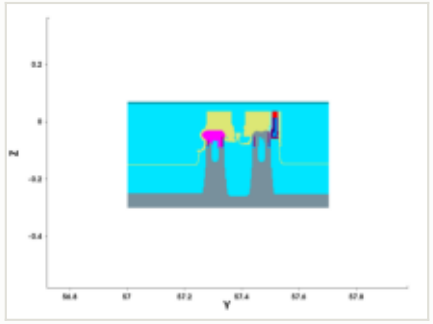
carbon so far, of the whole flow (kgCO2e)



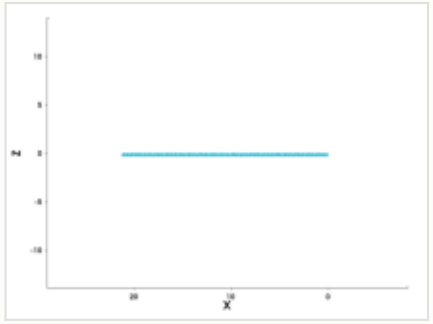
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



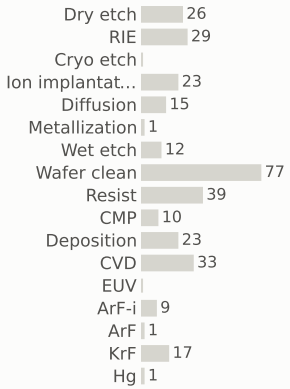
y cut at 57.351 µm



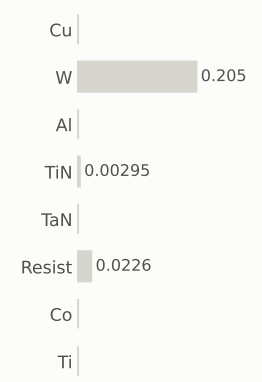
mask for this step

no mask for this step

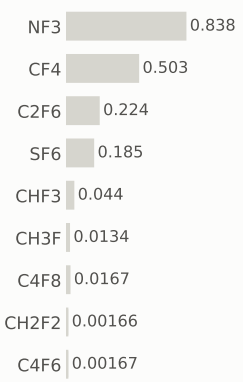
steps by process area



materials deposited (g)



gases emitted (g)



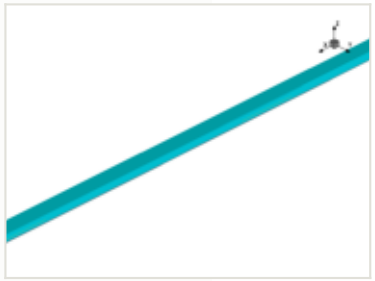
Step #4200: CVD

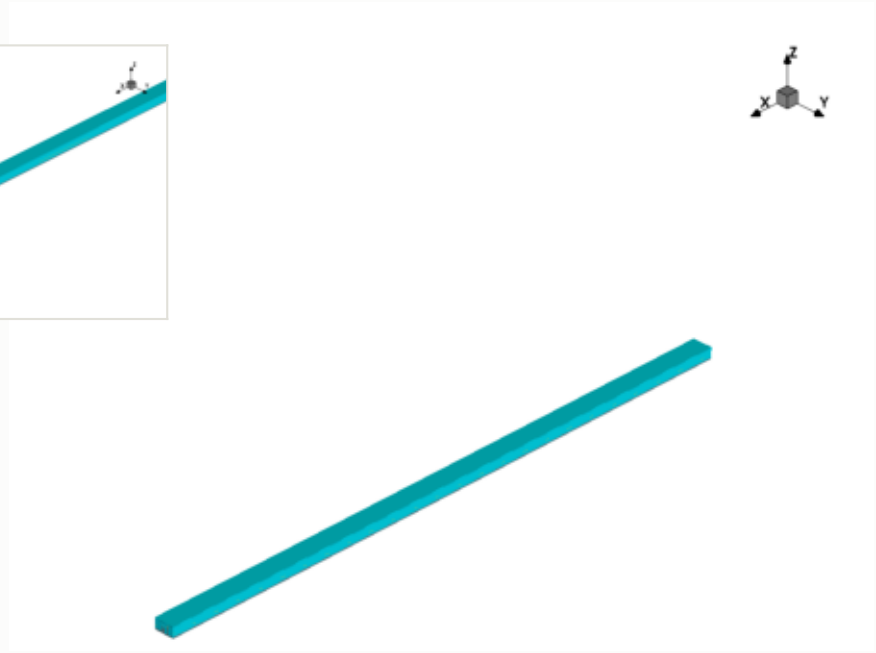
depo_fill · CVD

1.31	197.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



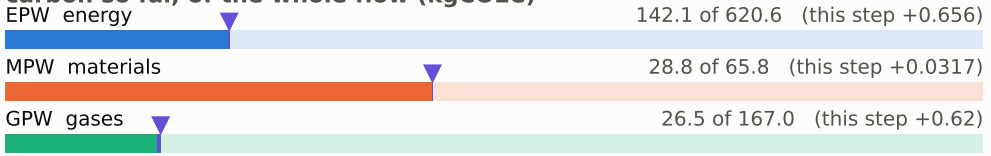


this step

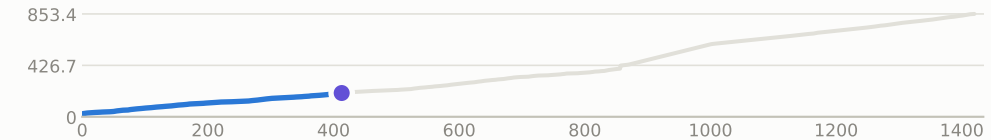
process area	CVD
material	—
thickness	0.019 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

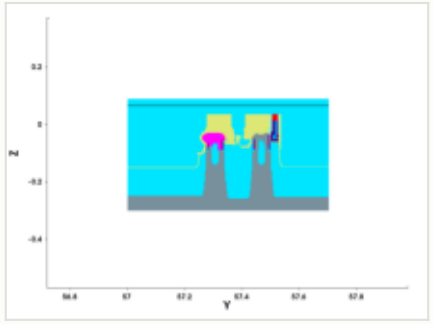
carbon so far, of the whole flow (kgCO2e)



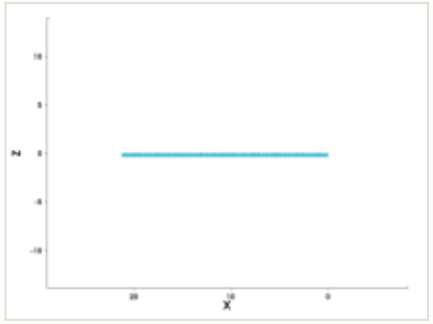
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



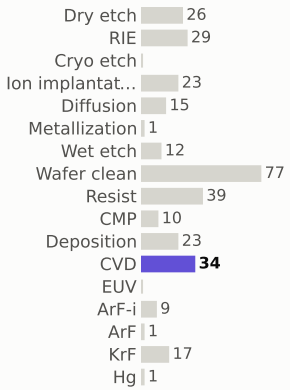
y cut at 57.351 µm



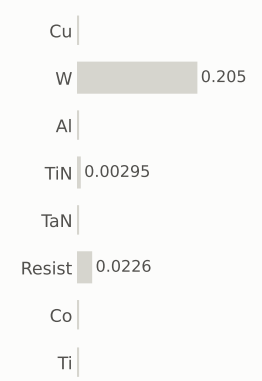
mask for this step

no mask for this step

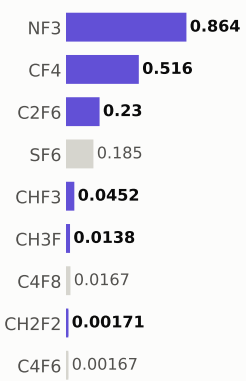
steps by process area



materials deposited (g)



gases emitted (g)



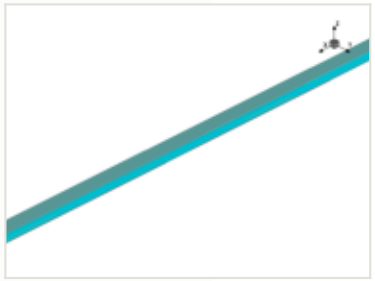
Step #4210: Spx_only

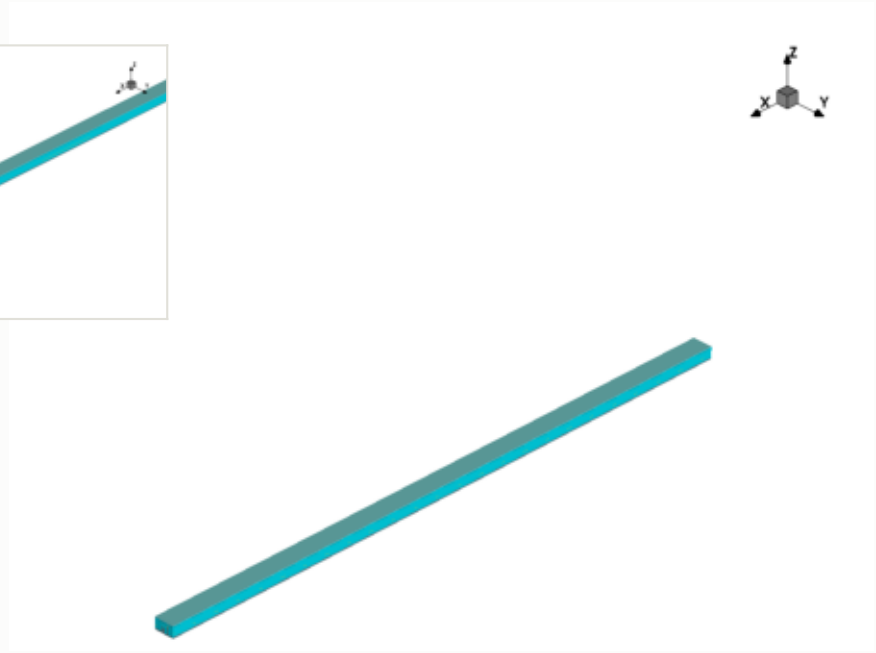
depo_fill · Spx_only

0	197.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

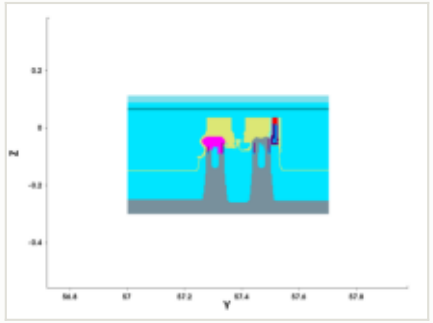
3D structure after this step

zoomed to fit

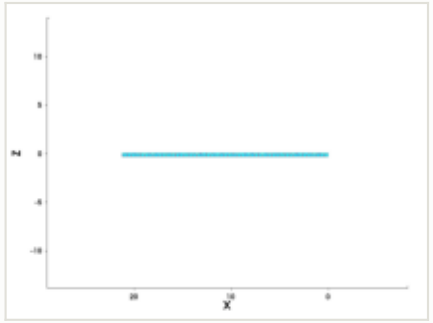




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

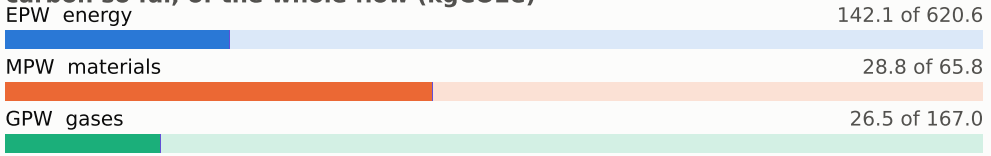
this step

process area	Spx_only
material	—
thickness	0.025 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

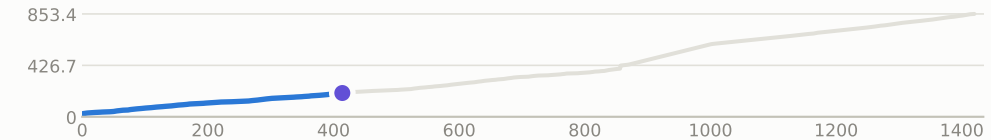
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

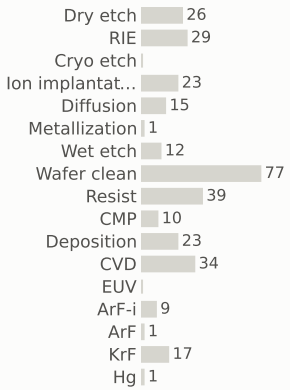
carbon so far, of the whole flow (kgCO2e)



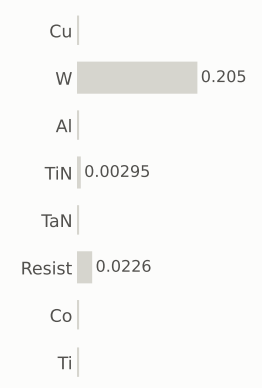
embodied carbon up to each step (kgCO2e per wafer)



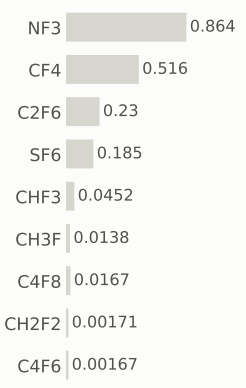
steps by process area



materials deposited (g)



gases emitted (g)



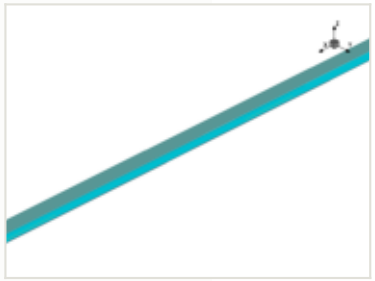
Step #4220: Resist

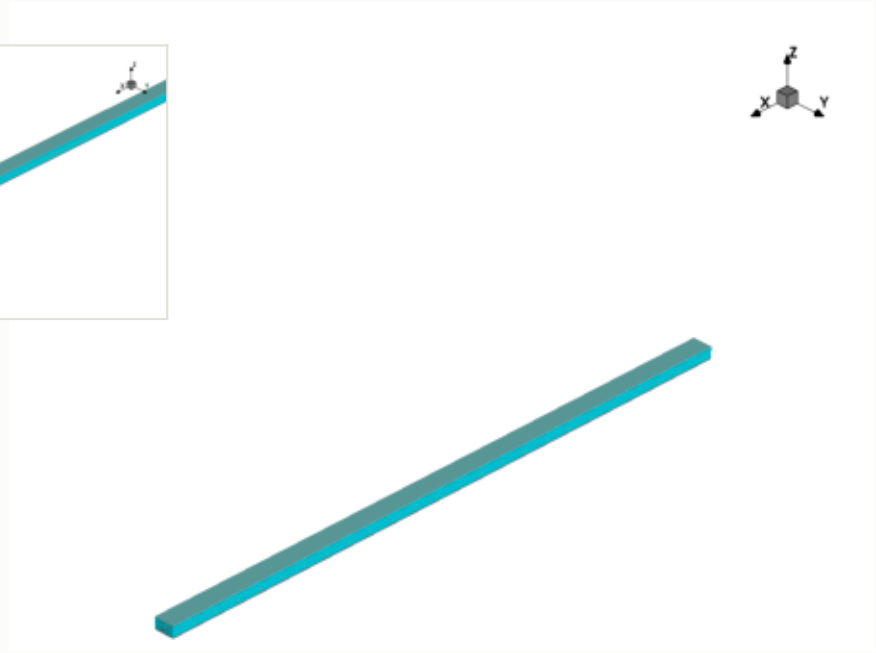
depo_fill · Resist

0.169	197.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



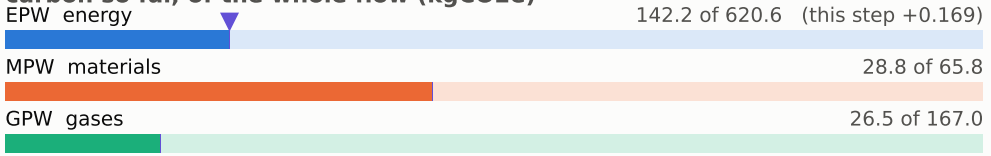


this step

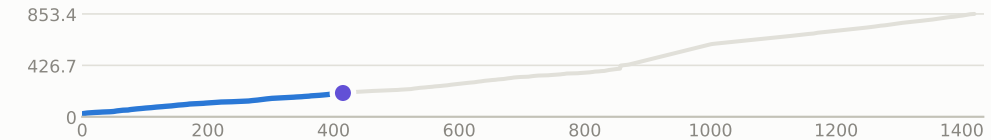
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

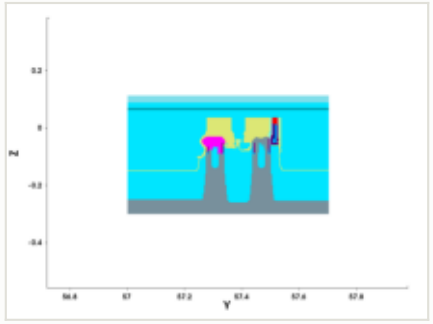
carbon so far, of the whole flow (kgCO2e)



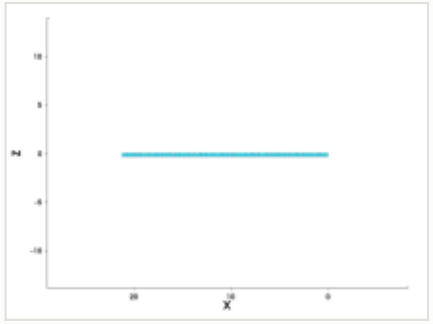
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



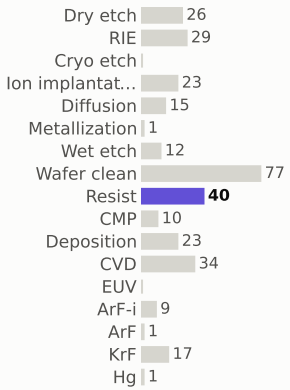
y cut at 57.351 µm



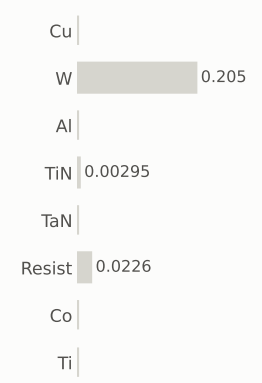
mask for this step

no mask for this step

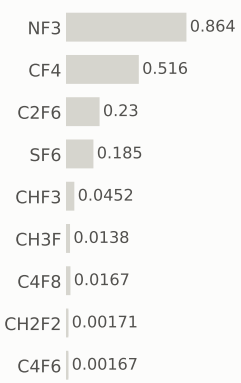
steps by process area



materials deposited (g)



gases emitted (g)



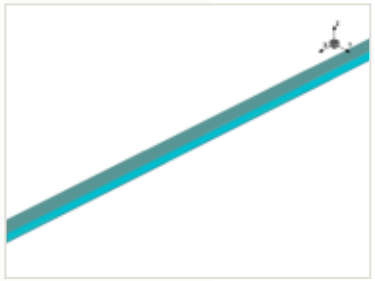
Step #4230: CVD

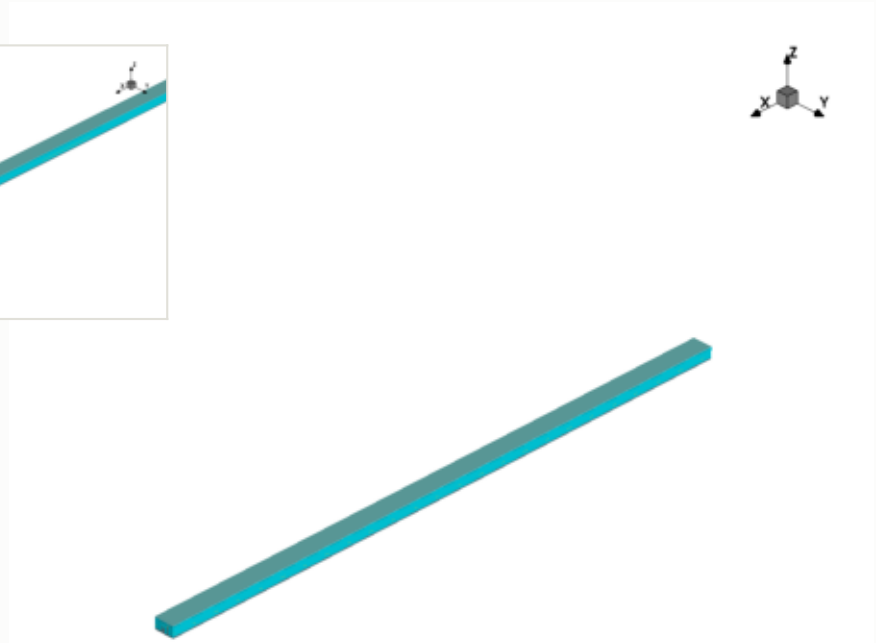
depo_fill · CVD

1.31	198.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



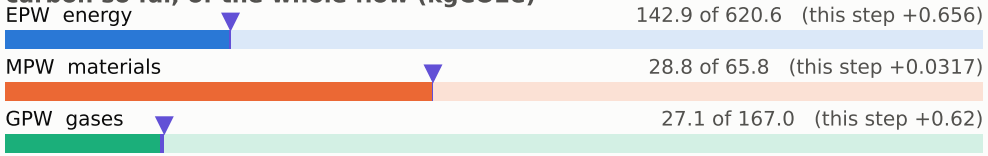


this step

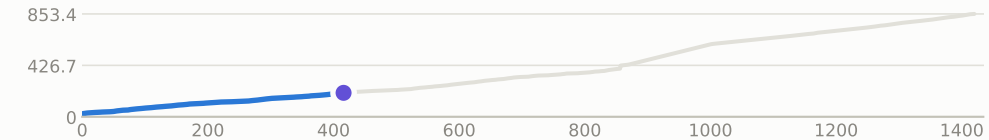
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

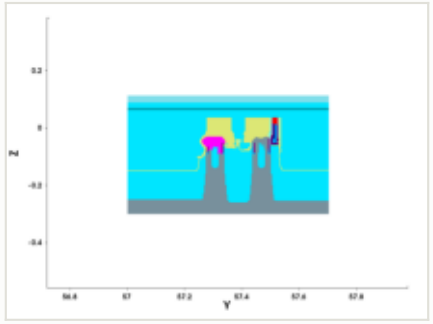
carbon so far, of the whole flow (kgCO2e)



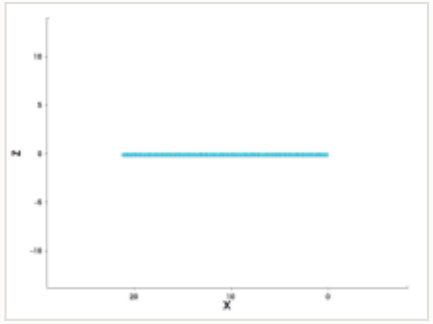
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



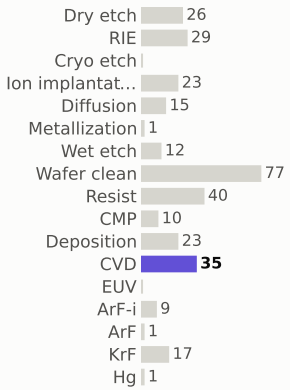
y cut at 57.351 µm



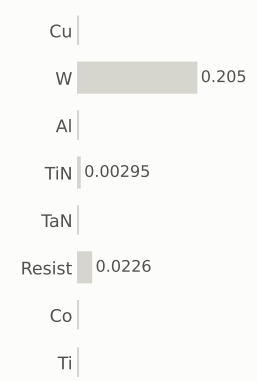
mask for this step

no mask for this step

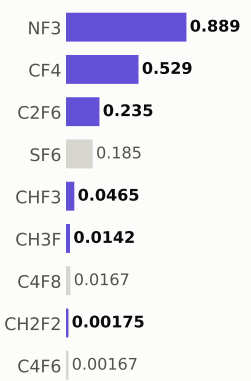
steps by process area



materials deposited (g)



gases emitted (g)



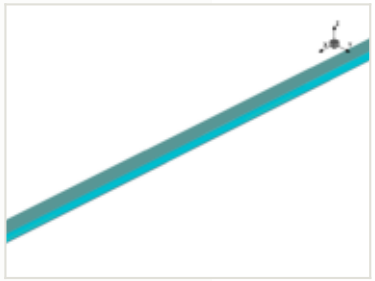
Step #4240: CVD

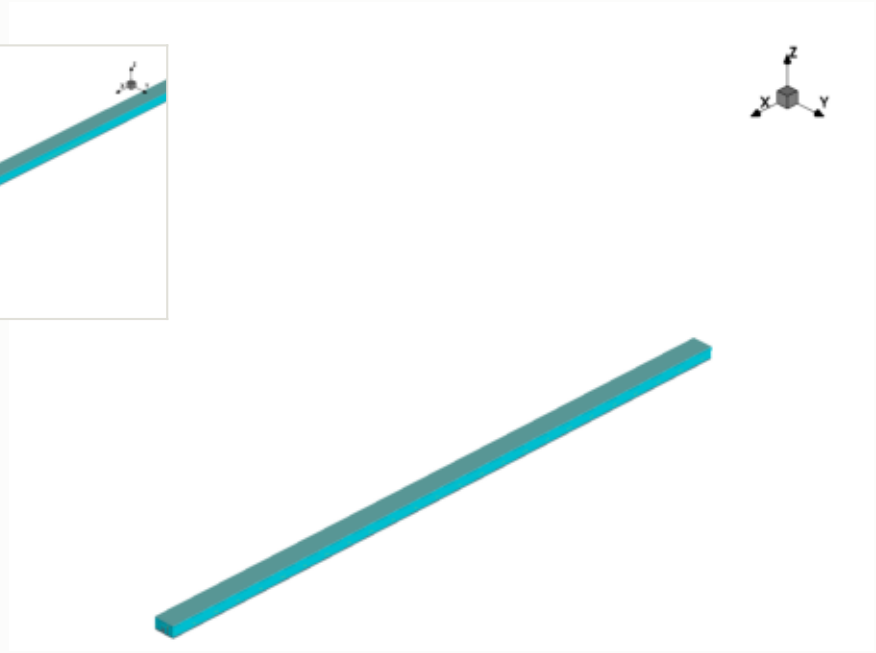
depo_fill · CVD

1.31	200.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



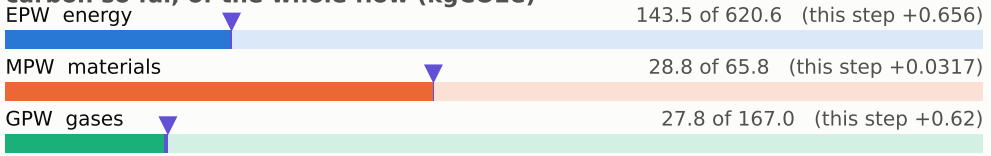


this step

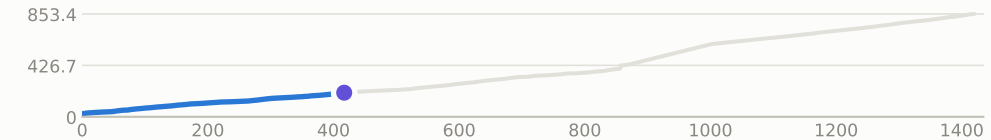
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

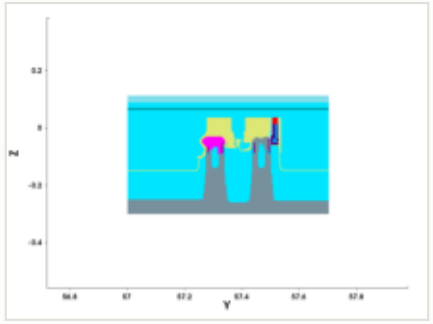
carbon so far, of the whole flow (kgCO2e)



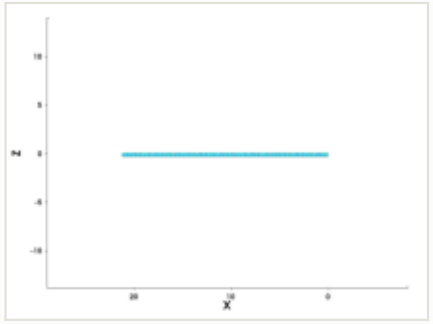
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



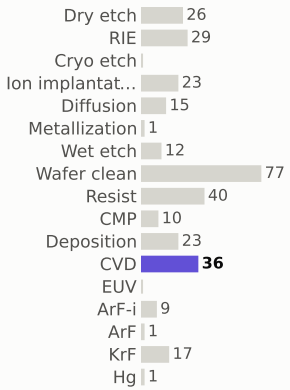
y cut at 57.351 µm



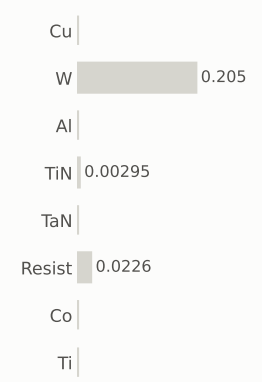
mask for this step

no mask for this step

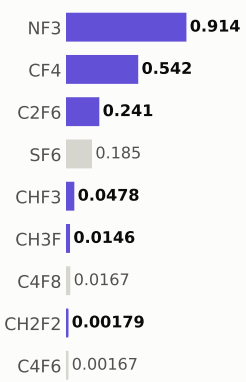
steps by process area



materials deposited (g)



gases emitted (g)



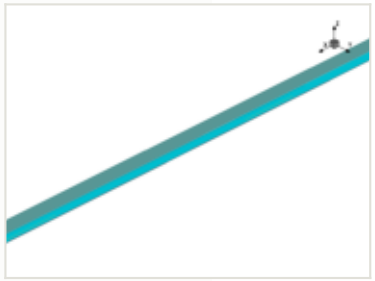
Step #4250: Resist

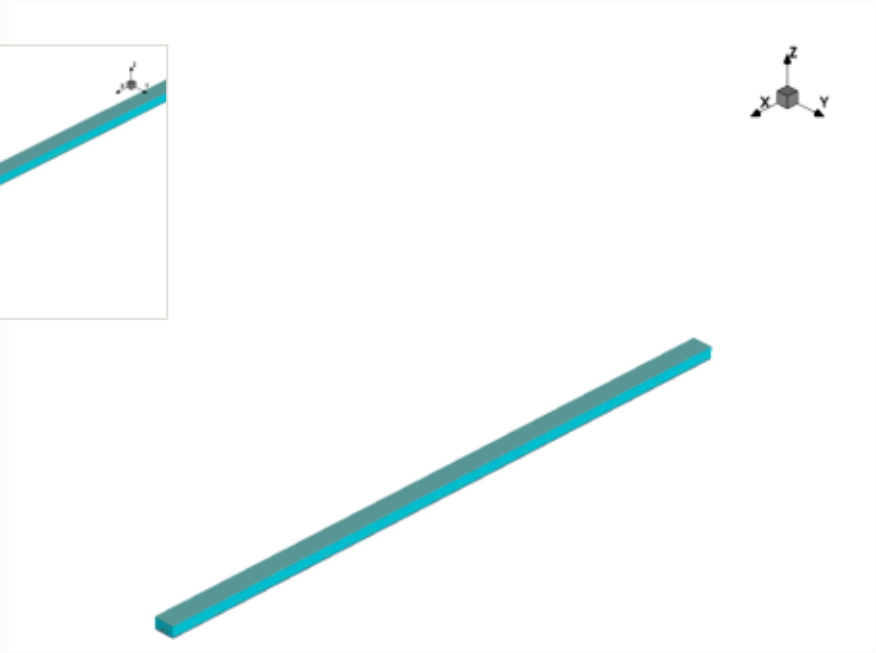
depo_fill · Resist

0.169	200.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



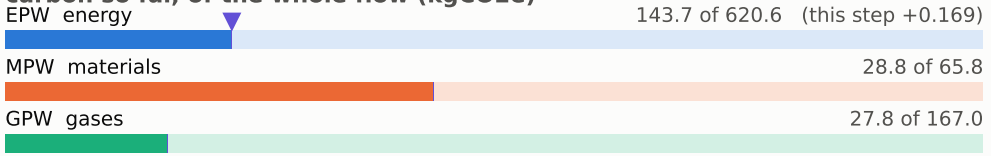


this step

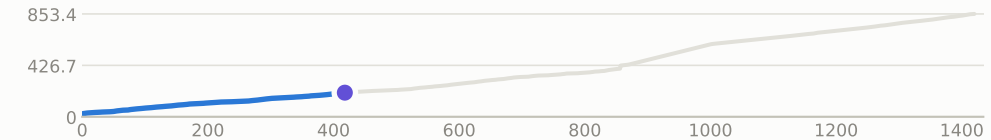
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

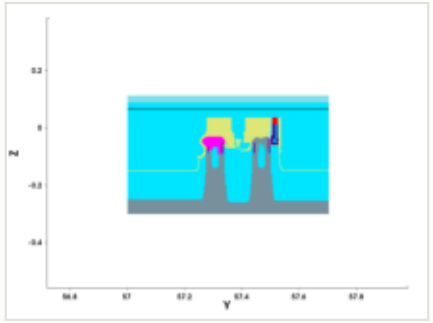
carbon so far, of the whole flow (kgCO2e)



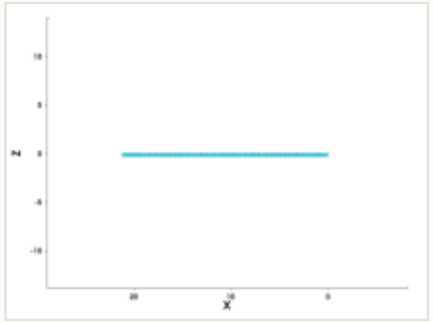
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



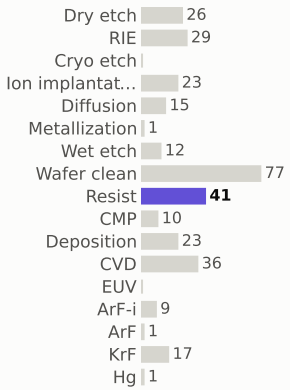
y cut at 57.351 µm



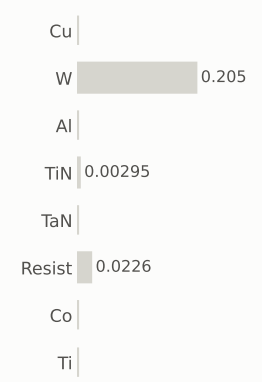
mask for this step

no mask for this step

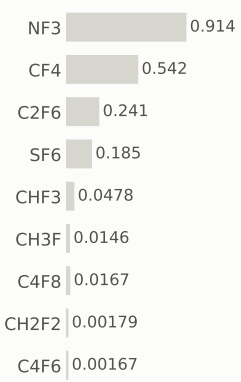
steps by process area



materials deposited (g)



gases emitted (g)



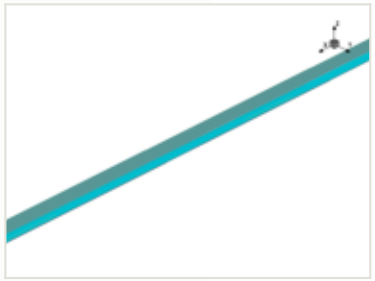
Step #4260: CVD

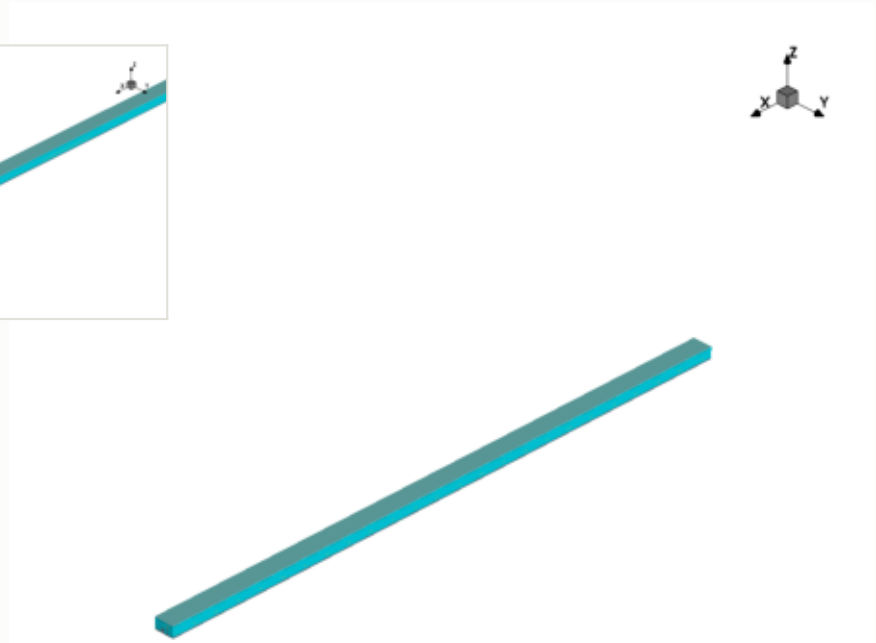
depo_fill · CVD

1.31	201.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



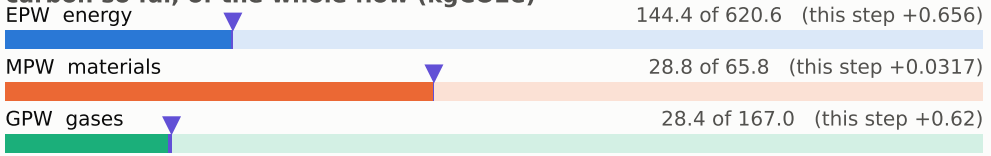


this step

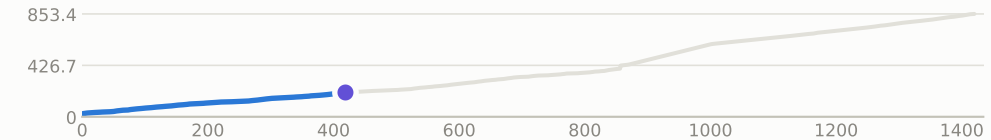
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

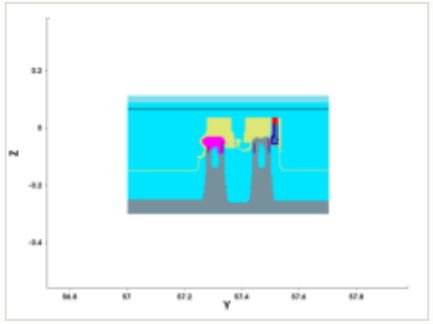
carbon so far, of the whole flow (kgCO2e)



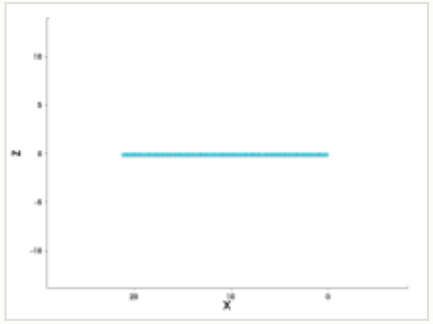
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



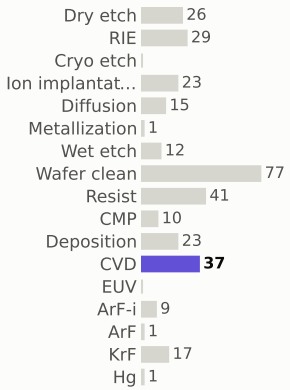
y cut at 57.351 µm



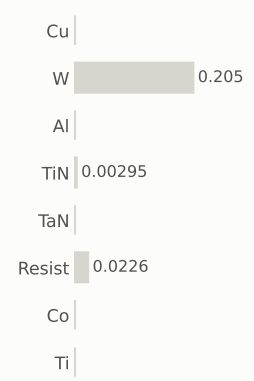
mask for this step

no mask for this step

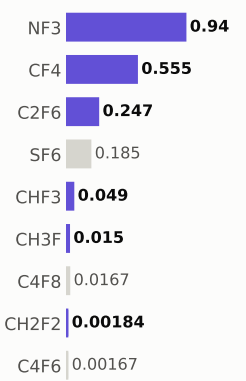
steps by process area



materials deposited (g)



gases emitted (g)



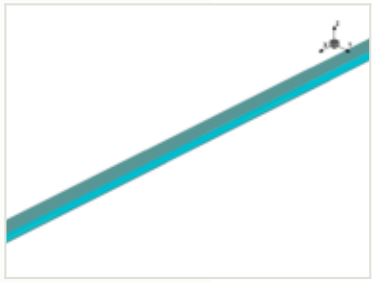
Step #4270: Wafer_clean

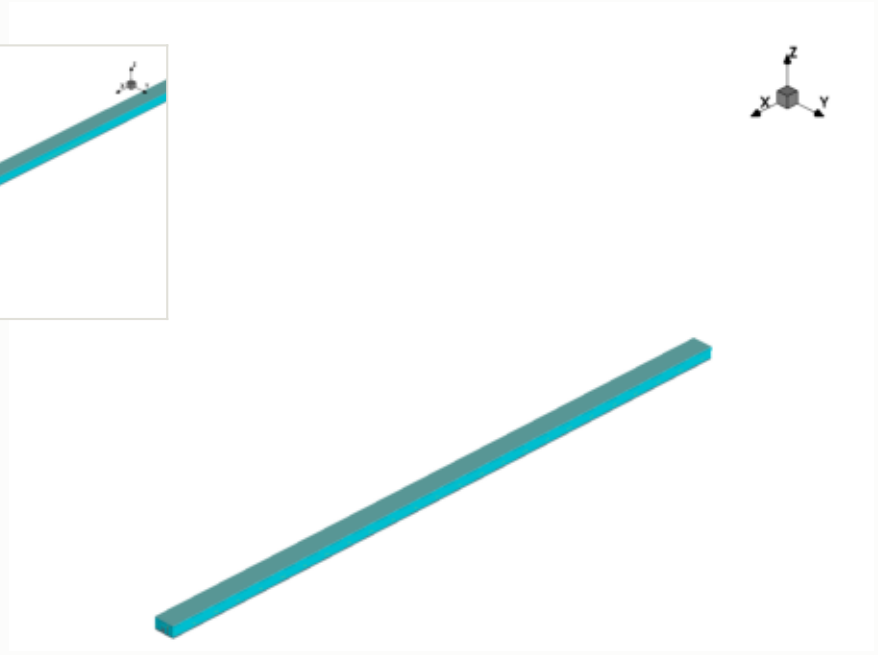
clean · Wafer_clean

0.169	201.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





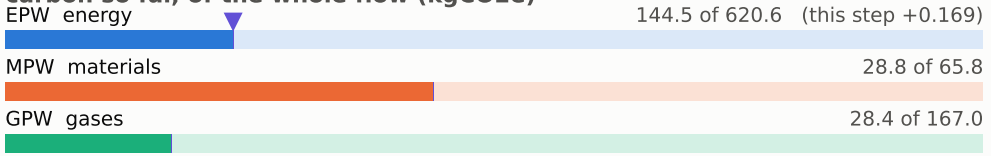
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

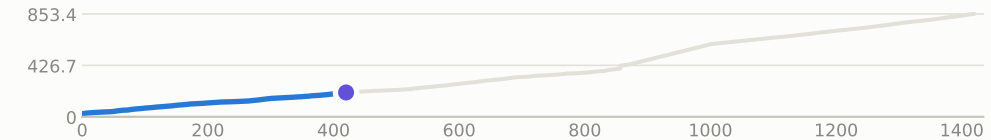
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

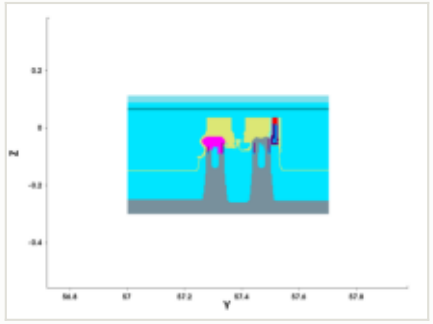
carbon so far, of the whole flow (kgCO2e)



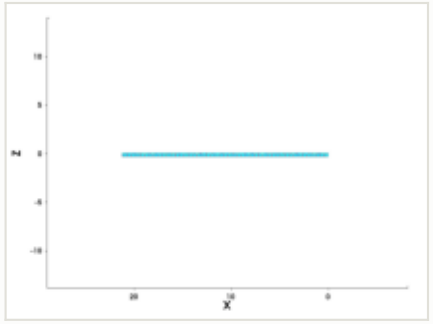
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



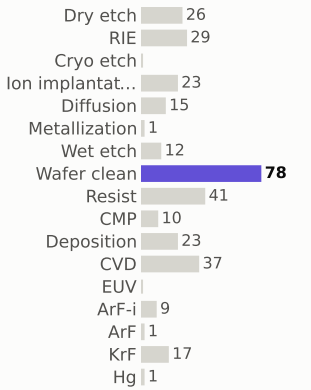
y cut at 57.351 µm



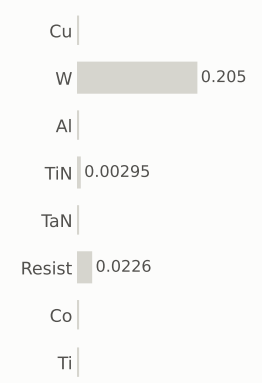
mask for this step

no mask for this step

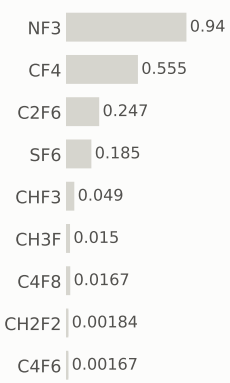
steps by process area



materials deposited (g)



gases emitted (g)



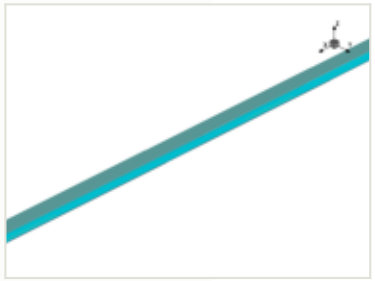
Step #4280: Resist

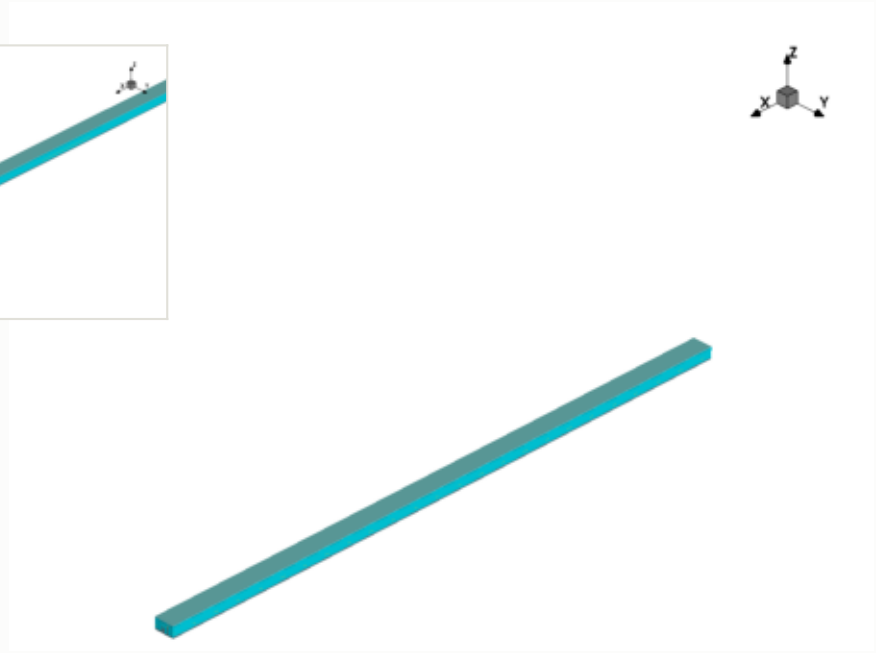
resist · Resist

0.169	201.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





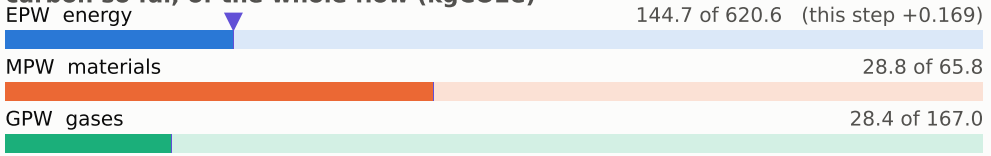
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

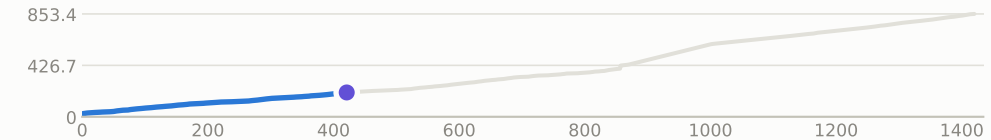
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

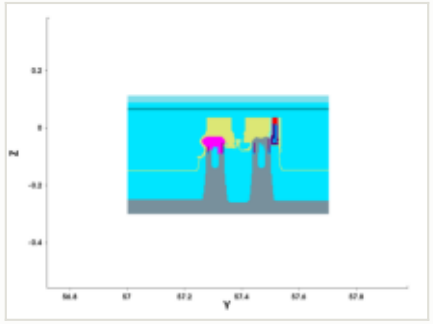
carbon so far, of the whole flow (kgCO2e)



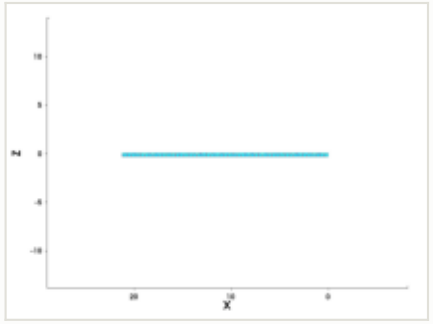
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



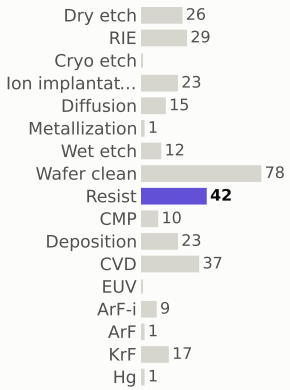
y cut at 57.351 µm



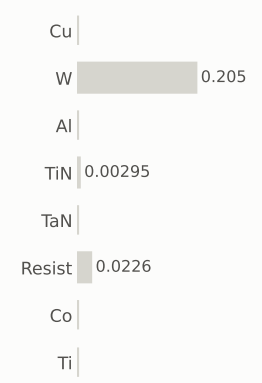
mask for this step

no mask for this step

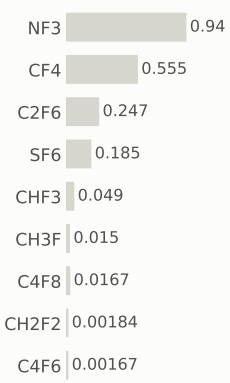
steps by process area



materials deposited (g)



gases emitted (g)



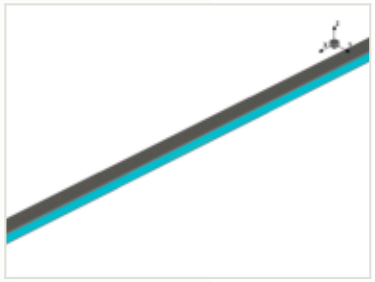
Step #4290: ArF-i

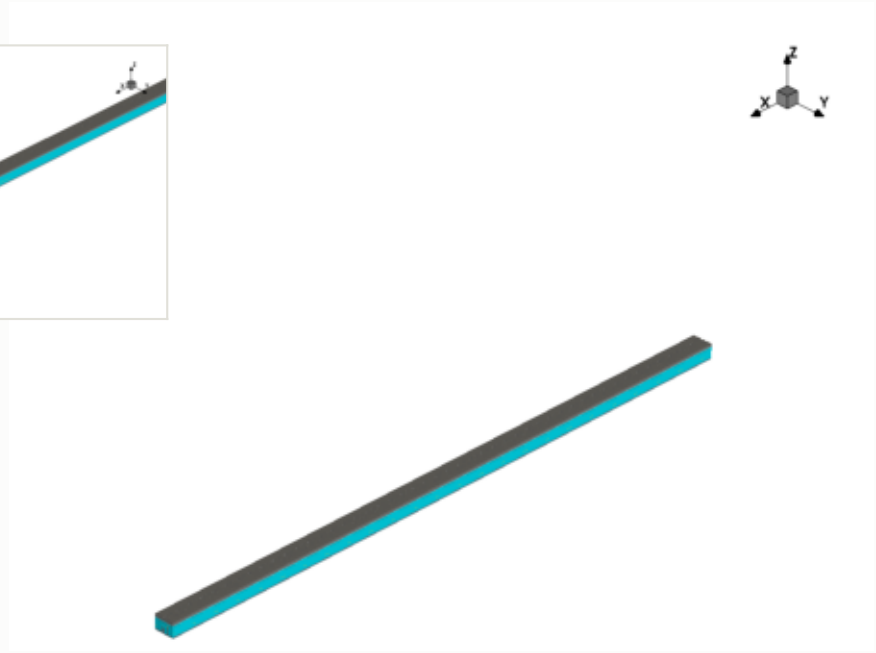
litho_patterning · ArF-i

0.217	202.1	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



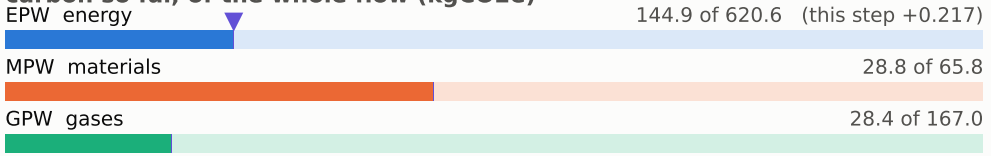


this step

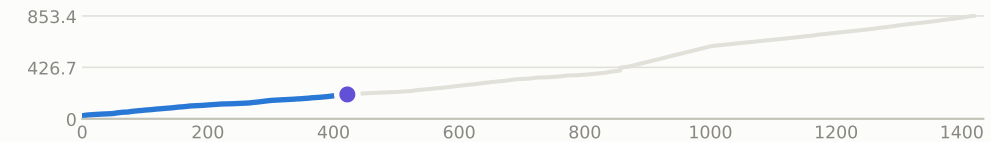
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

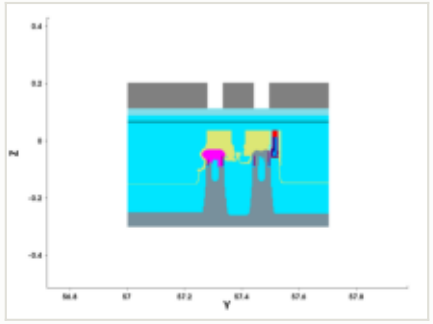
carbon so far, of the whole flow (kgCO2e)



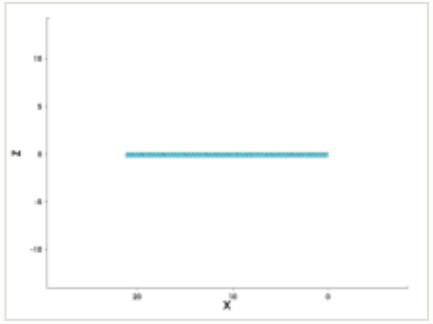
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

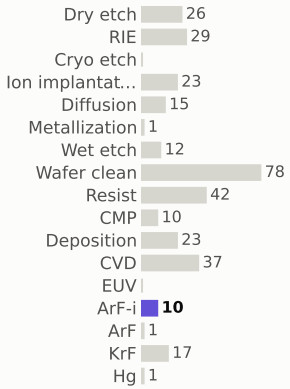


y cut at 57.351 µm

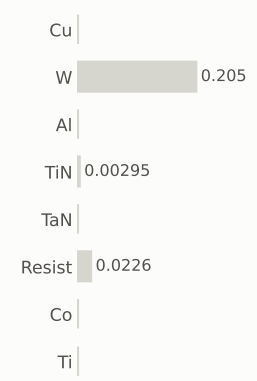


mask: layer 2017:0 (GDSII)

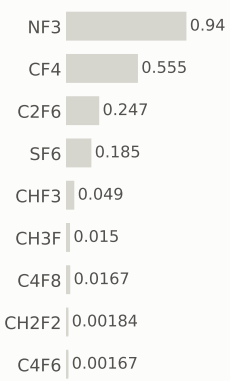
steps by process area



materials deposited (g)



gases emitted (g)



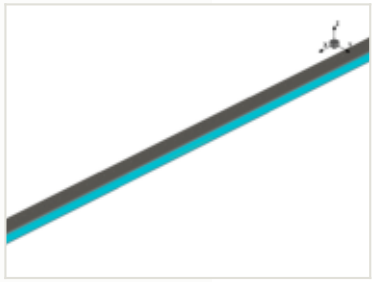
Step #4300: Spx_only

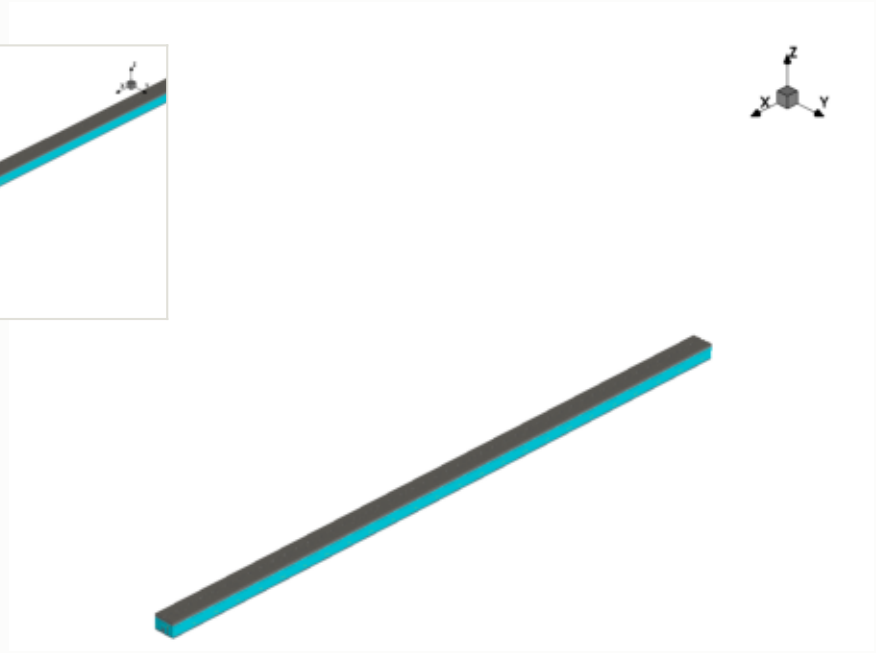
etch_anisotropic · Spx_only

0	202.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





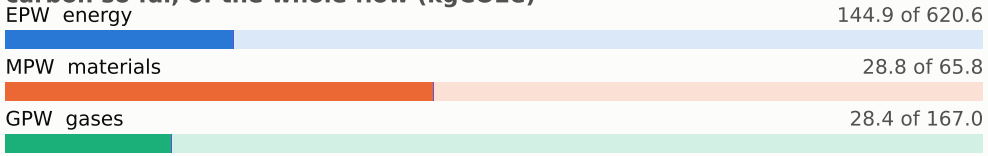
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

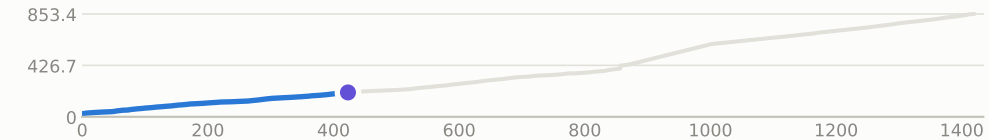
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

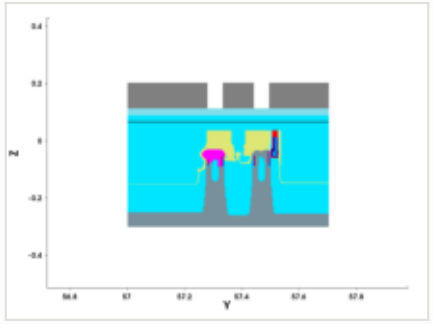
carbon so far, of the whole flow (kgCO2e)



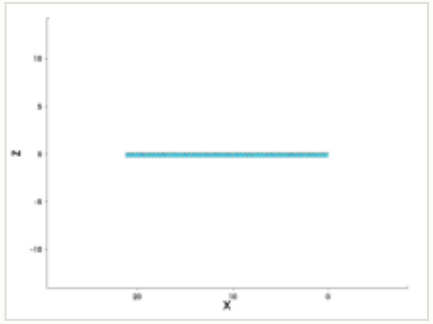
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



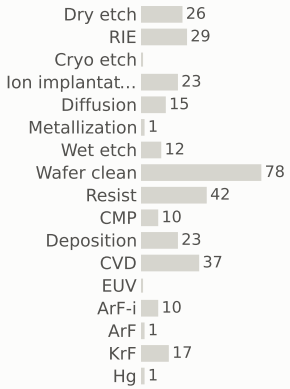
y cut at 57.351 µm



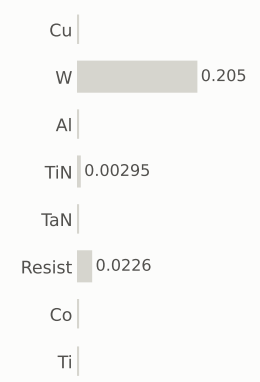
mask for this step

no mask for this step

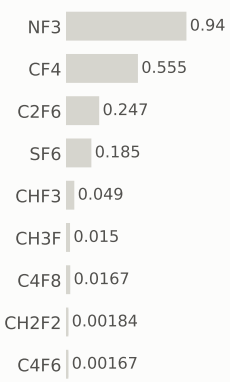
steps by process area



materials deposited (g)



gases emitted (g)



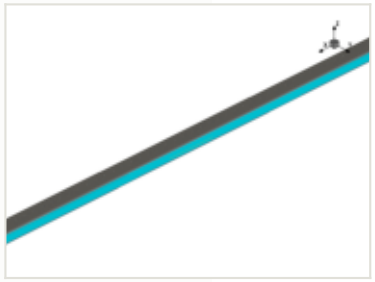
Step #4310: RIE

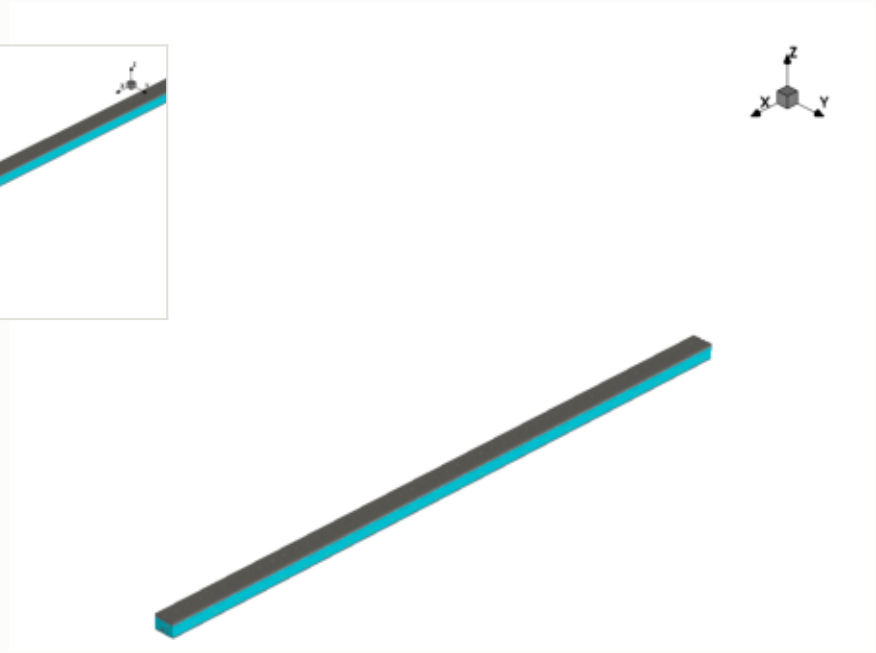
etch_anisotropic · RIE

1.04	203.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



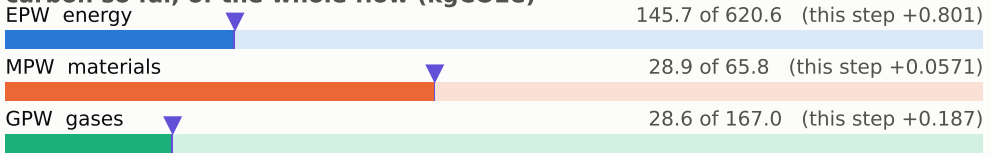


this step

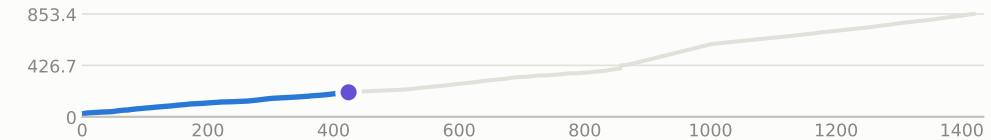
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

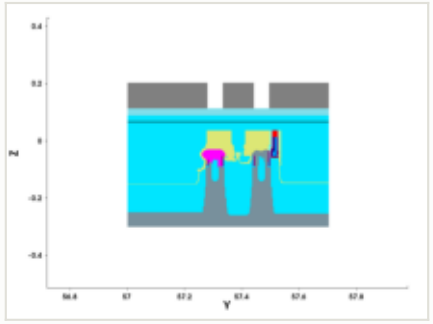
carbon so far, of the whole flow (kgCO2e)



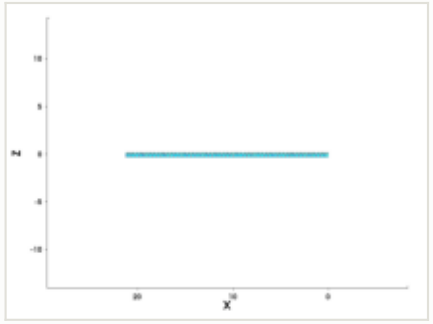
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



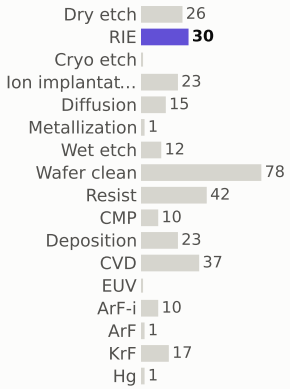
y cut at 57.351 µm



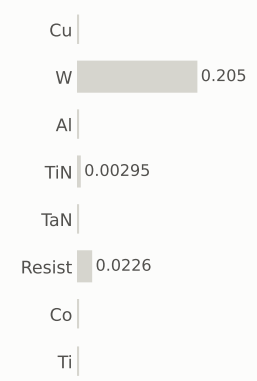
mask for this step

no mask for this step

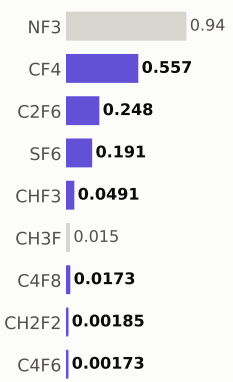
steps by process area



materials deposited (g)



gases emitted (g)



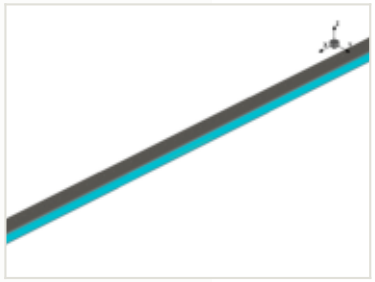
Step #4320: Spx_only

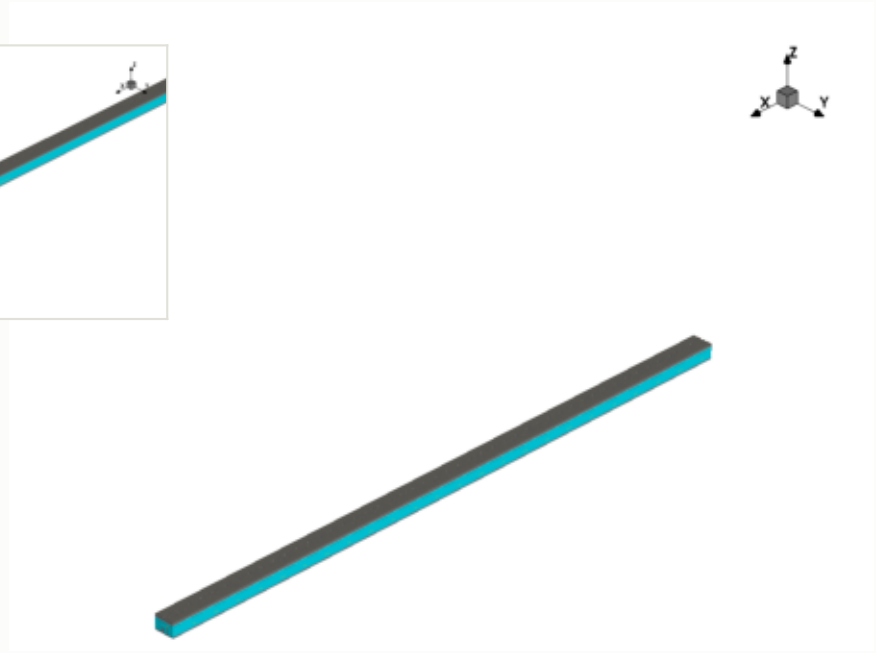
etch_anisotropic · Spx_only

0	203.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



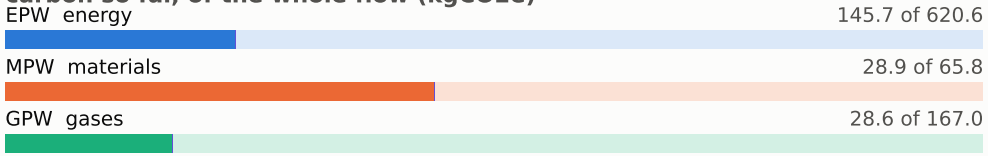


this step

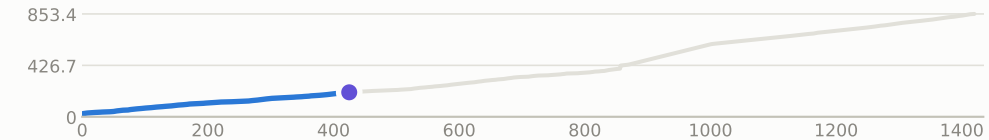
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

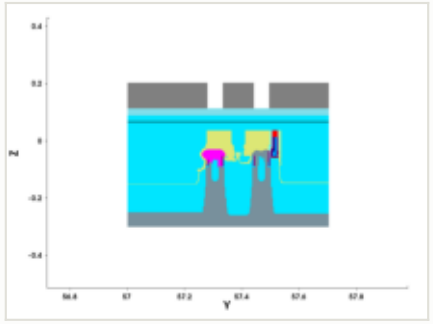
carbon so far, of the whole flow (kgCO2e)



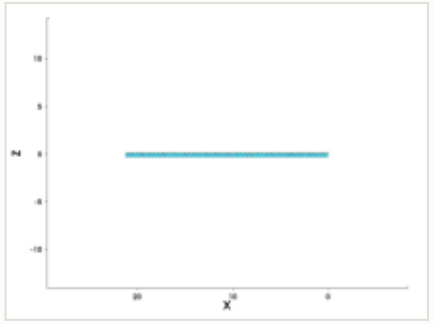
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



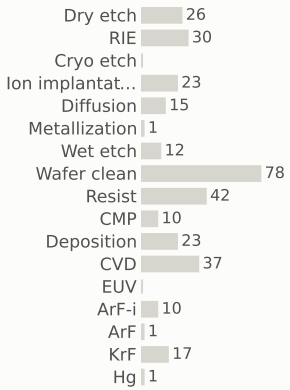
y cut at 57.351 µm



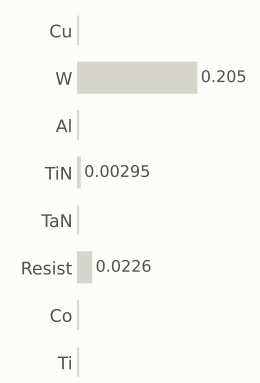
mask for this step

no mask for this step

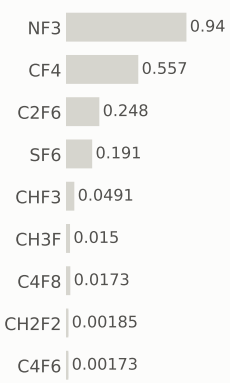
steps by process area



materials deposited (g)



gases emitted (g)



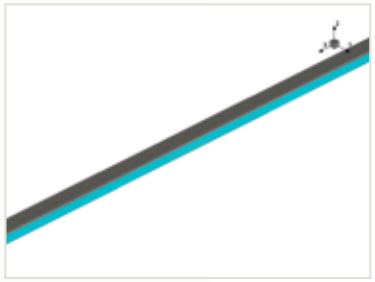
Step #4330: Spx_only

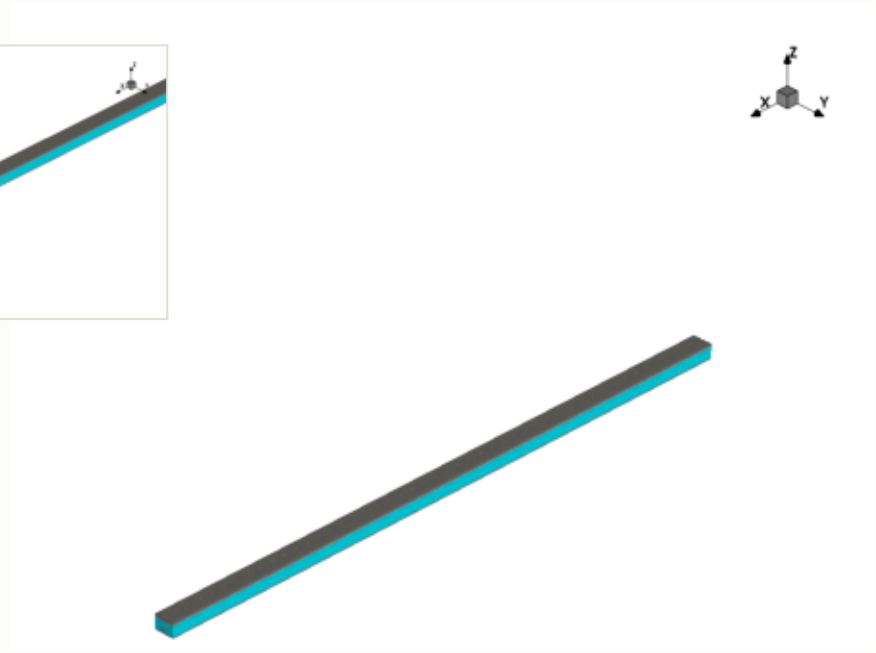
etch_strip · Spx_only

0	203.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

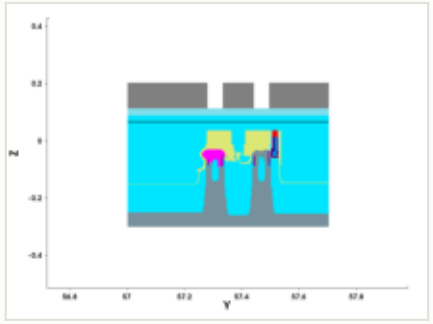
3D structure after this step

zoomed to fit

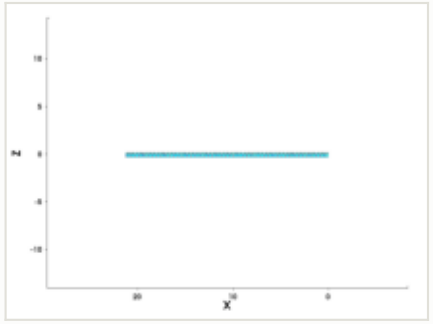




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

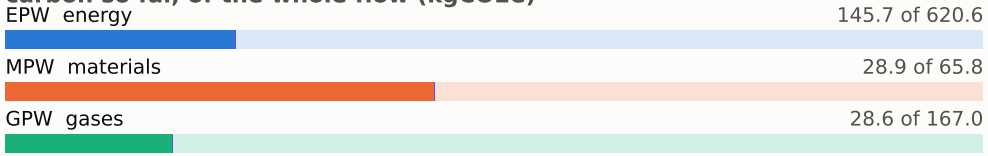
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

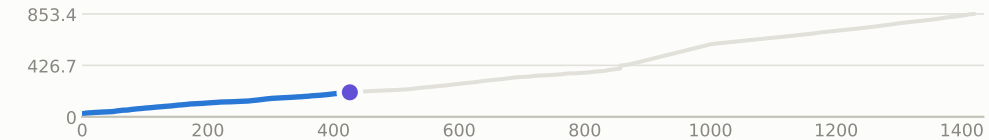
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

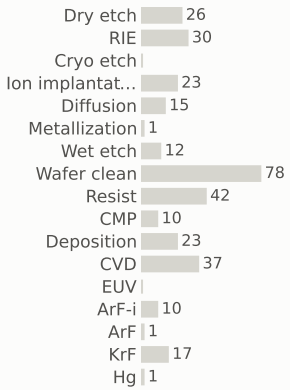
carbon so far, of the whole flow (kgCO2e)



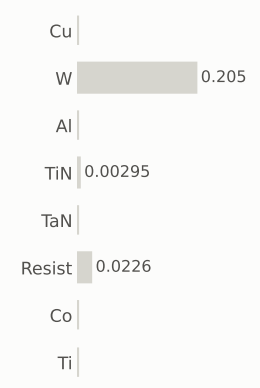
embodied carbon up to each step (kgCO2e per wafer)



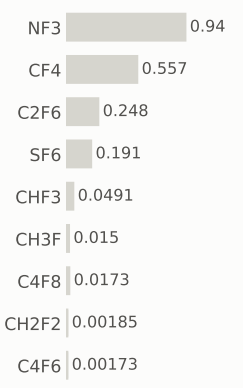
steps by process area



materials deposited (g)



gases emitted (g)



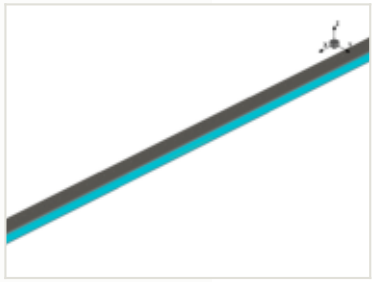
Step #4340: Wafer_clean

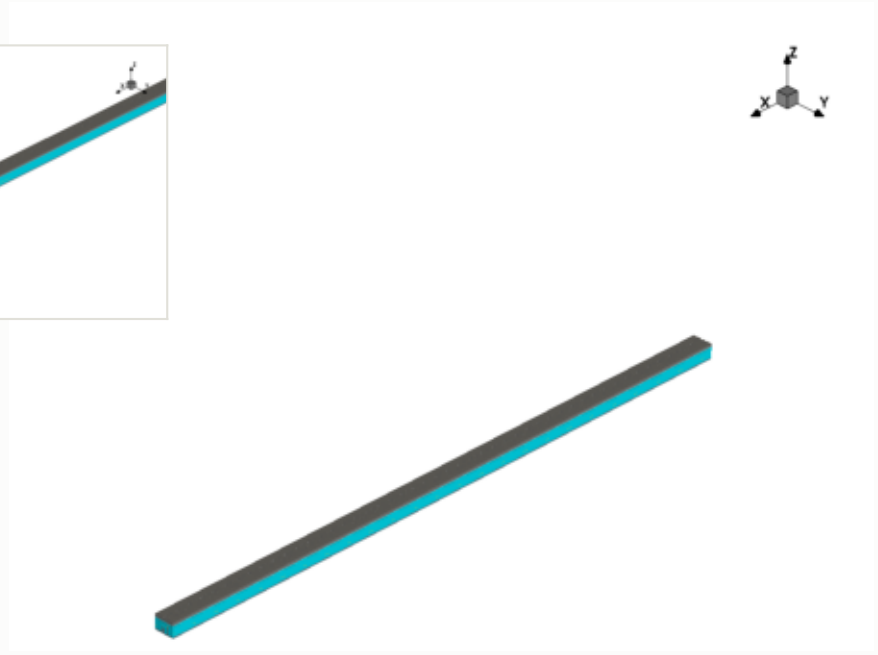
clean · Wafer_clean

0.169	203.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





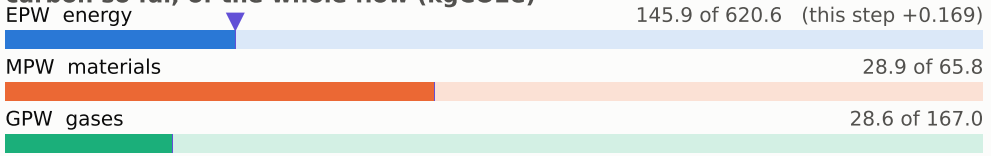
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

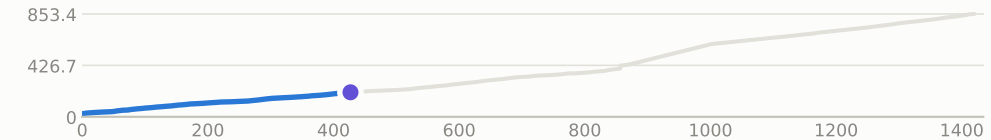
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

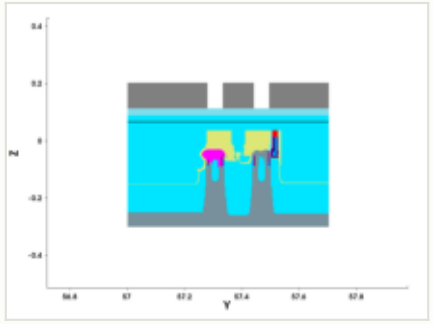
carbon so far, of the whole flow (kgCO2e)



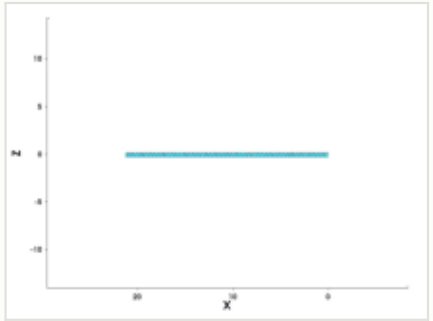
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



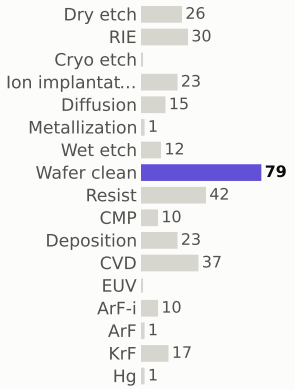
y cut at 57.351 µm



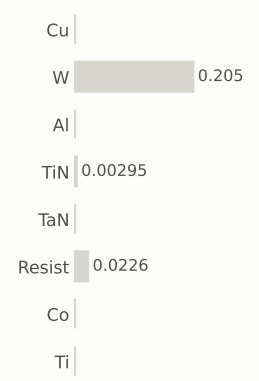
mask for this step

no mask for this step

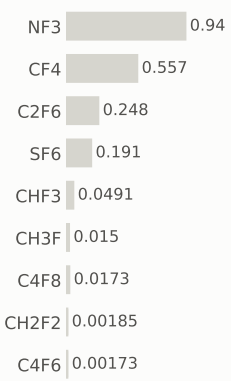
steps by process area



materials deposited (g)



gases emitted (g)



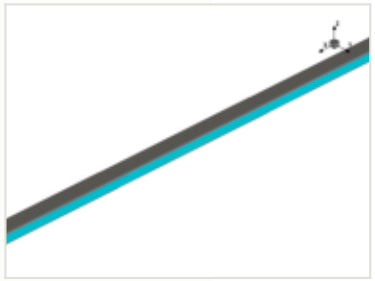
Step #4350: Resist

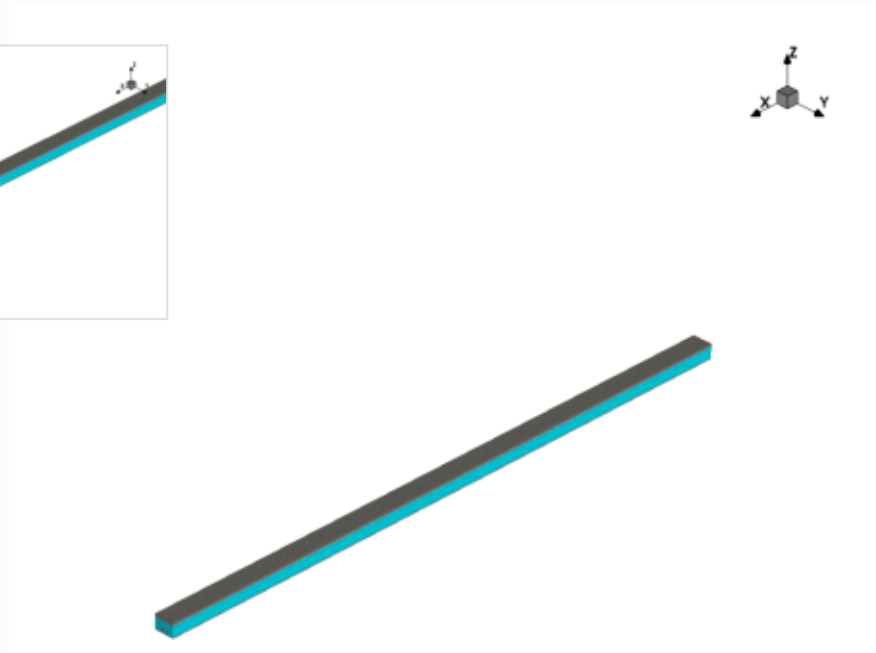
depo_fill · Resist

0.169	203.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



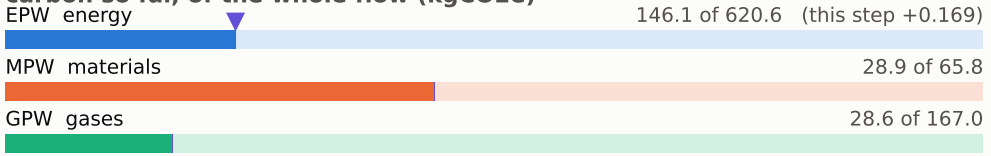


this step

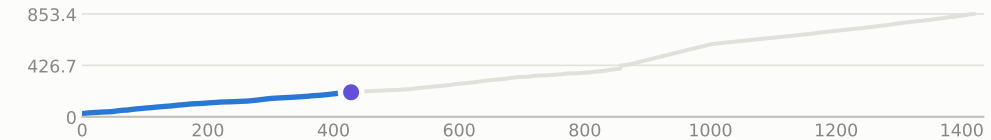
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

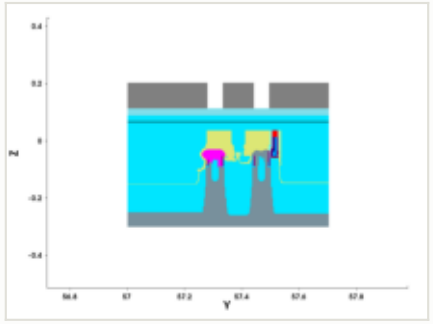
carbon so far, of the whole flow (kgCO2e)



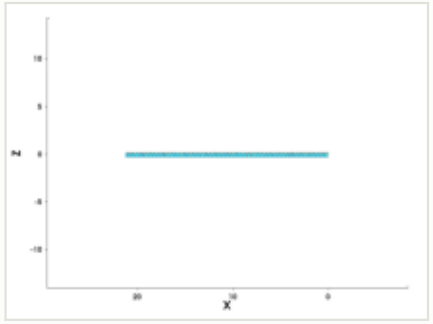
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



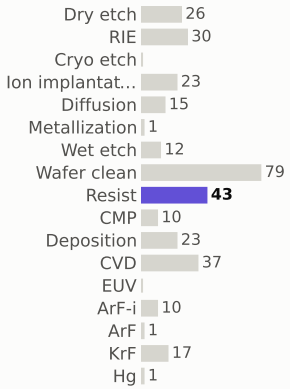
y cut at 57.351 µm



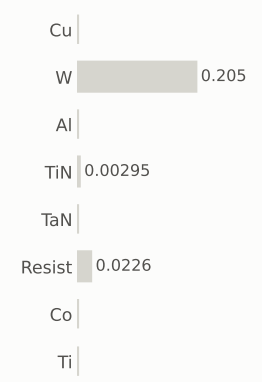
mask for this step

no mask for this step

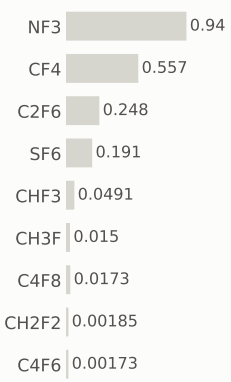
steps by process area



materials deposited (g)



gases emitted (g)



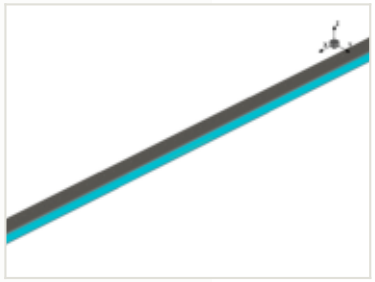
Step #4360: CVD

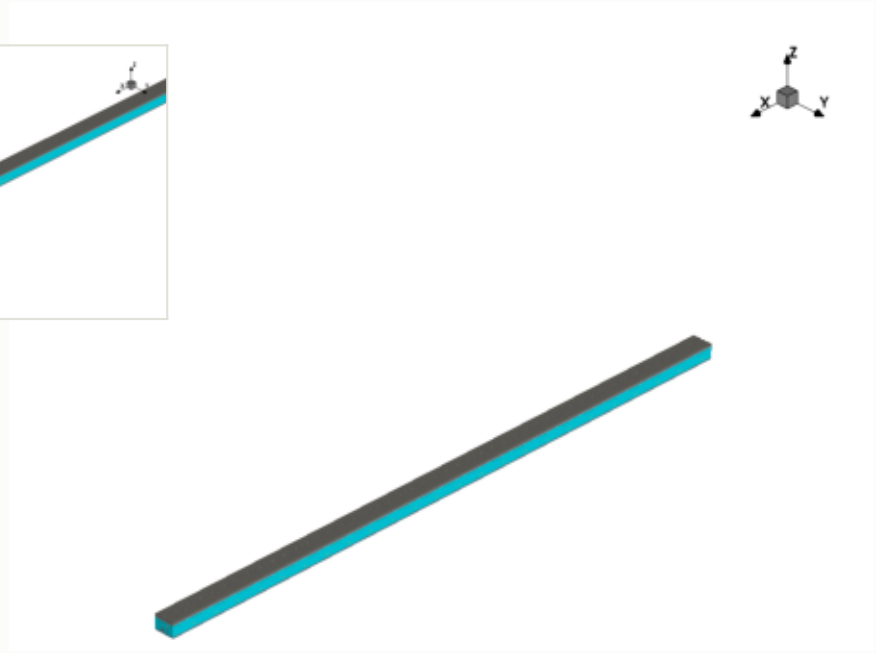
depo_fill · CVD

1.31	204.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



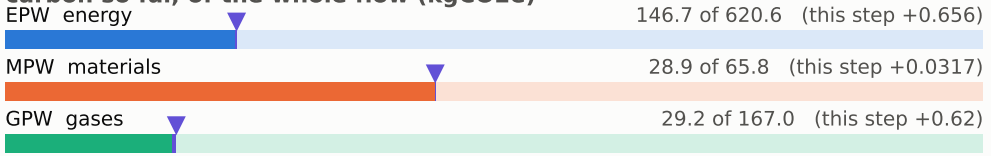


this step

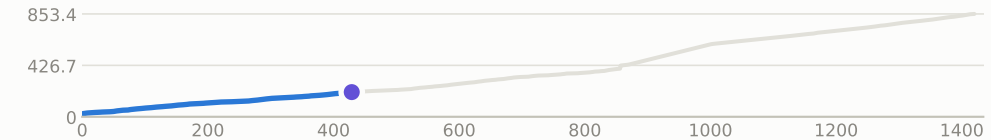
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

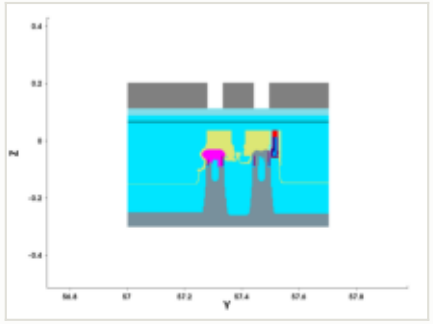
carbon so far, of the whole flow (kgCO2e)



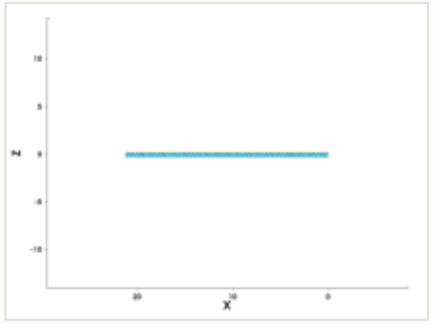
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



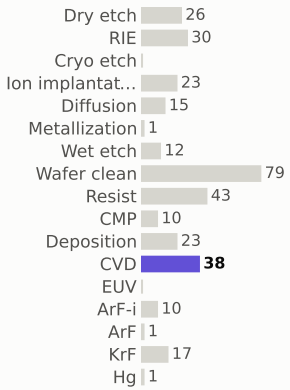
y cut at 57.351 µm



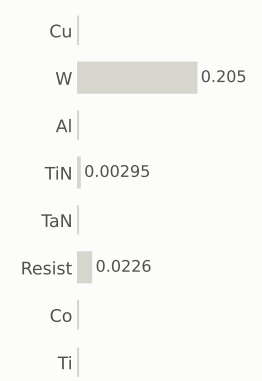
mask for this step

no mask for this step

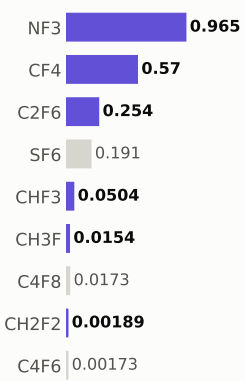
steps by process area



materials deposited (g)



gases emitted (g)



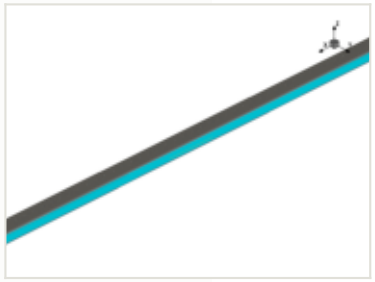
Step #4370: Wafer_clean

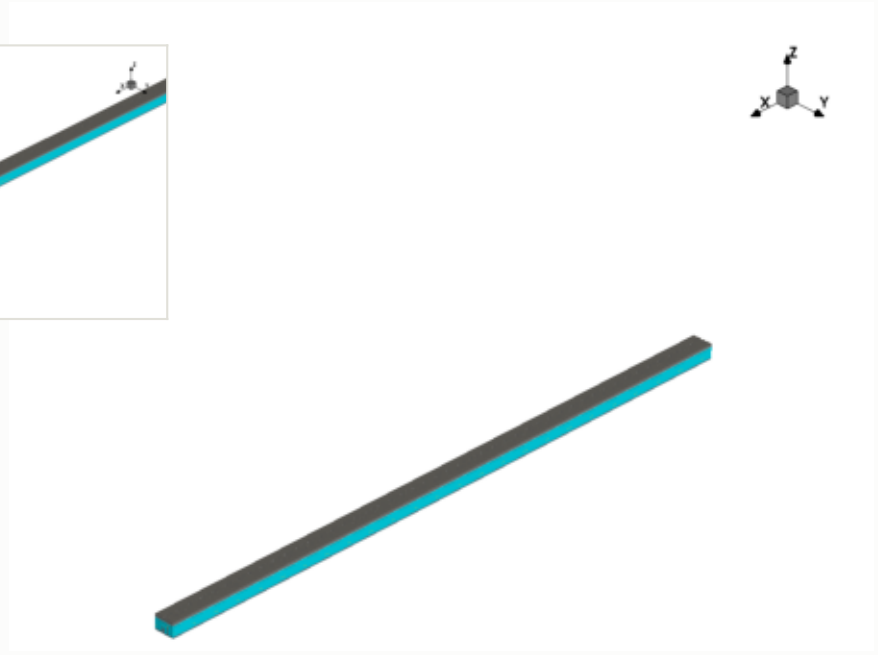
clean · Wafer_clean

0.169	205.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

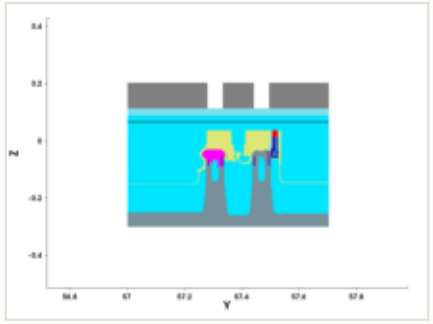
3D structure after this step

zoomed to fit

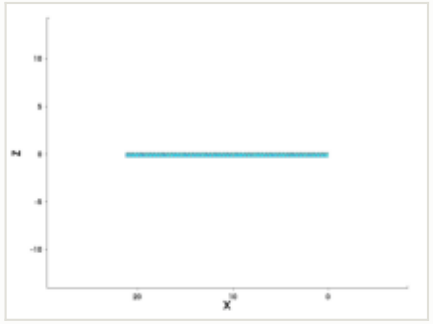




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

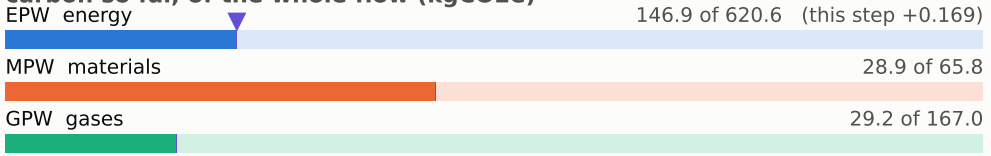
no mask for this step

this step

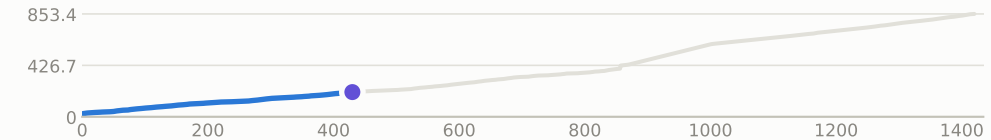
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

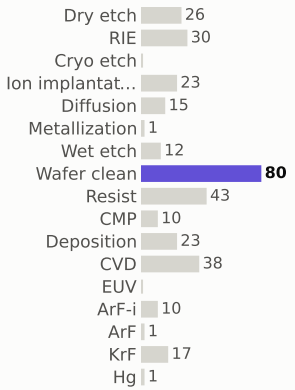
carbon so far, of the whole flow (kgCO2e)



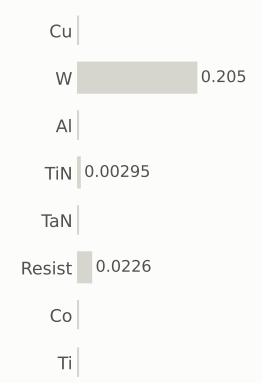
embodied carbon up to each step (kgCO2e per wafer)



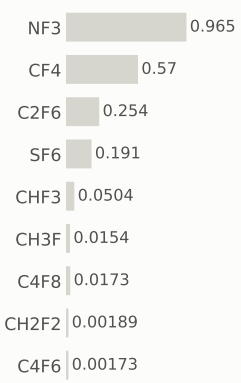
steps by process area



materials deposited (g)



gases emitted (g)



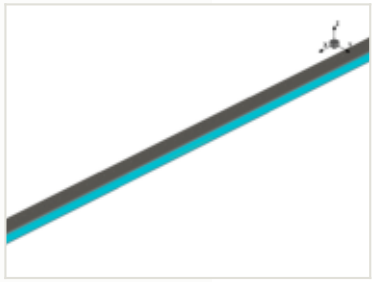
Step #4380: Resist

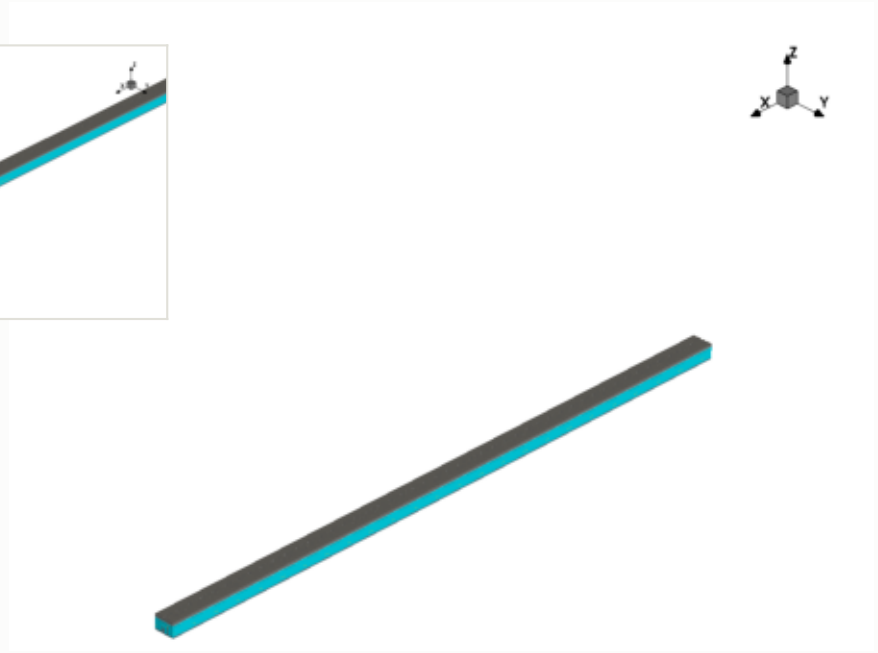
resist · Resist

0.169	205.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





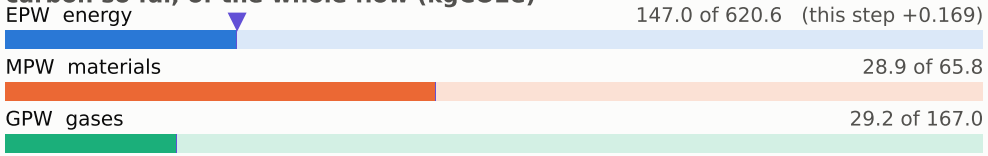
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

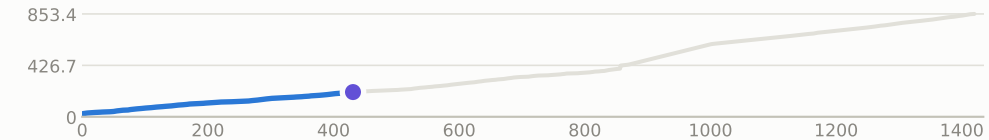
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

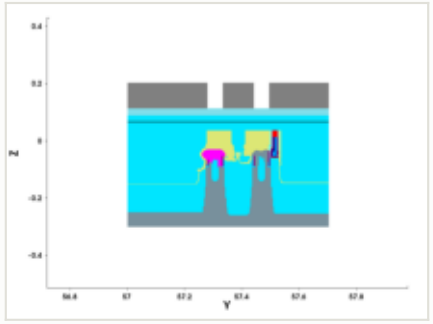
carbon so far, of the whole flow (kgCO2e)



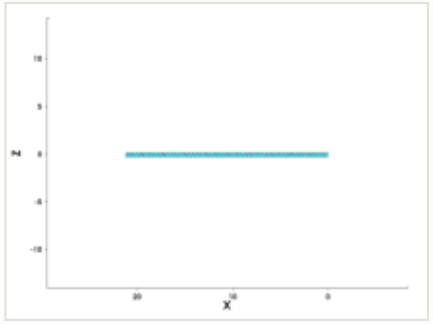
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



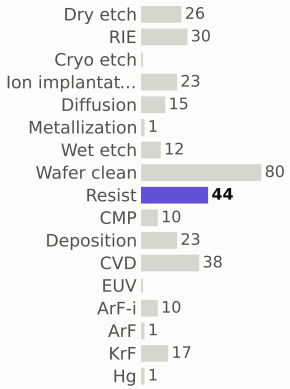
y cut at 57.351 µm



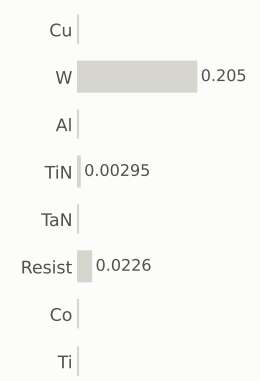
mask for this step

no mask for this step

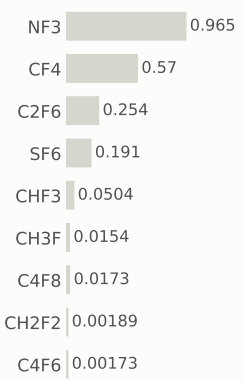
steps by process area



materials deposited (g)



gases emitted (g)



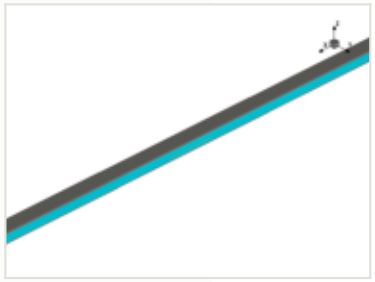
Step #4390: ArF-i

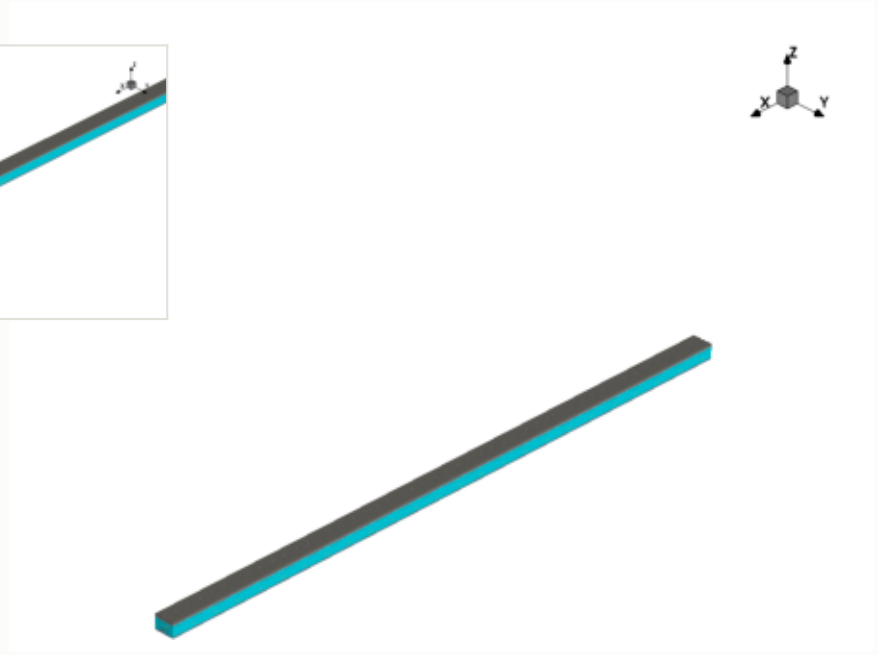
litho_patterning · ArF-i

0.217	205.4	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



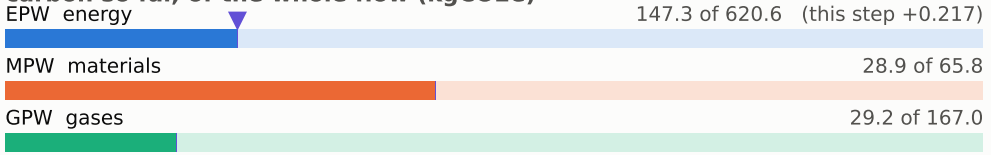


this step

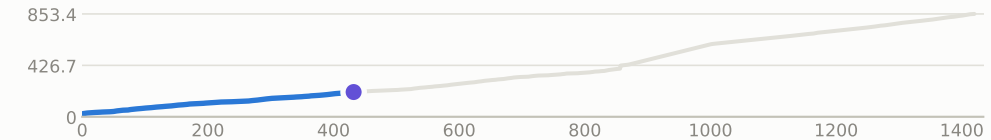
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

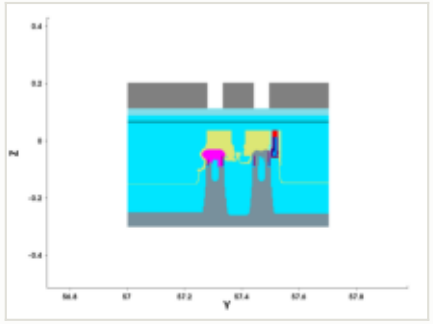
carbon so far, of the whole flow (kgCO2e)



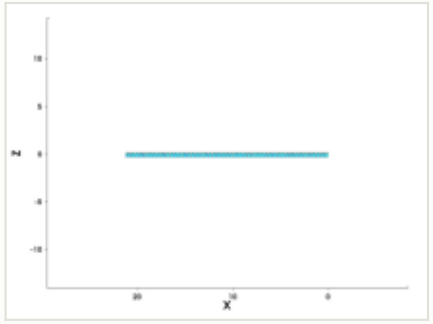
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

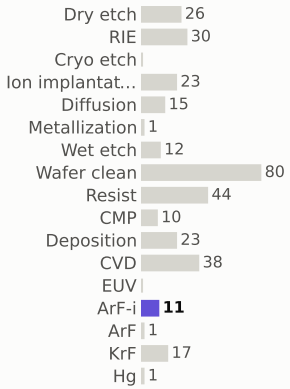


y cut at 57.351 µm

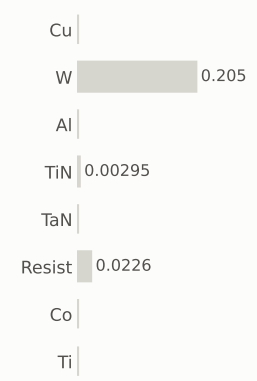


mask: layer 2017:0 (GDSII)

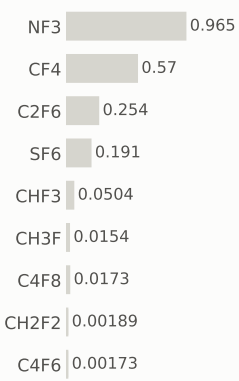
steps by process area



materials deposited (g)



gases emitted (g)



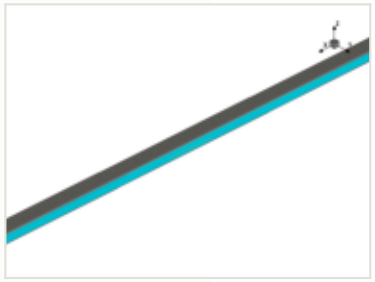
Step #4400: Spx_only

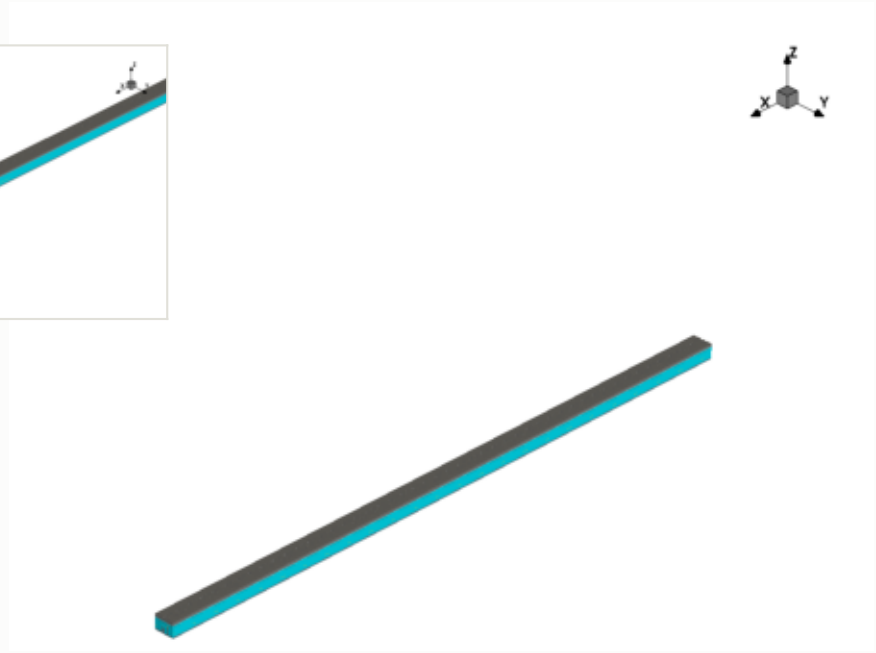
etch_anisotropic · Spx_only

0	205.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



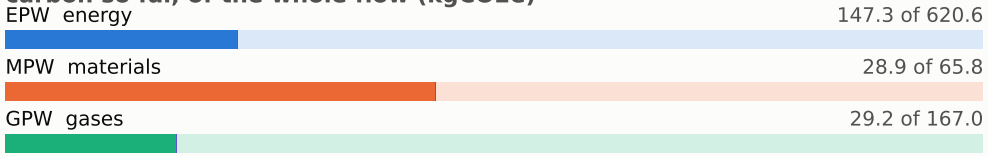


this step

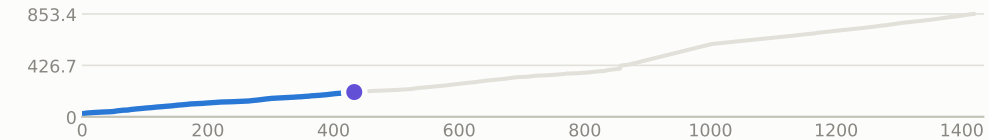
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

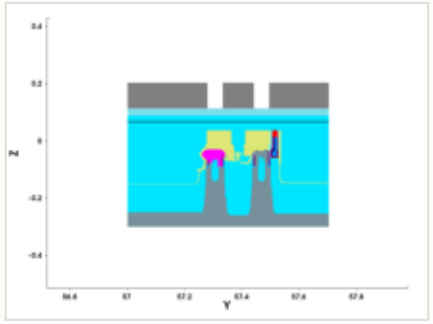
carbon so far, of the whole flow (kgCO2e)



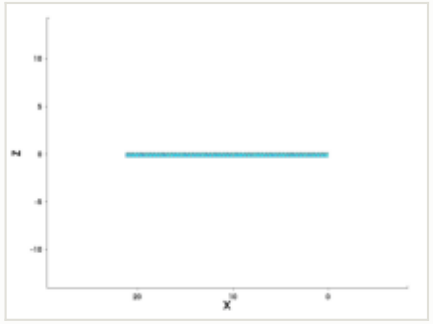
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



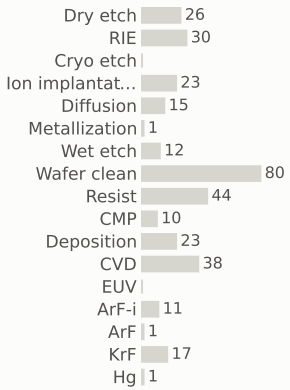
y cut at 57.351 µm



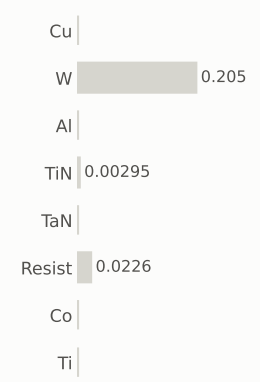
mask for this step

no mask for this step

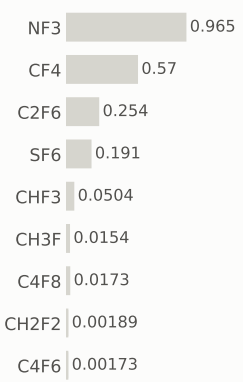
steps by process area



materials deposited (g)



gases emitted (g)



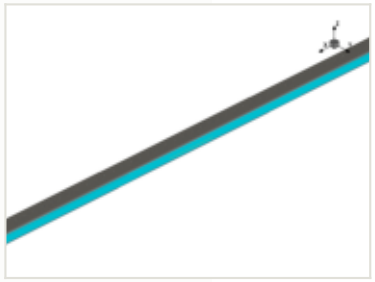
Step #4410: RIE

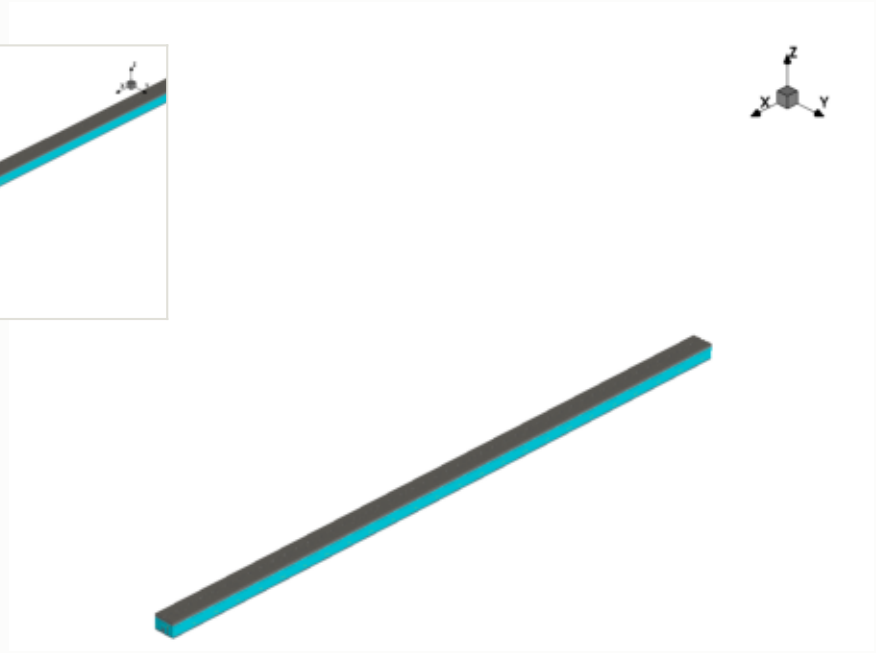
etch_anisotropic · RIE

1.04	206.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



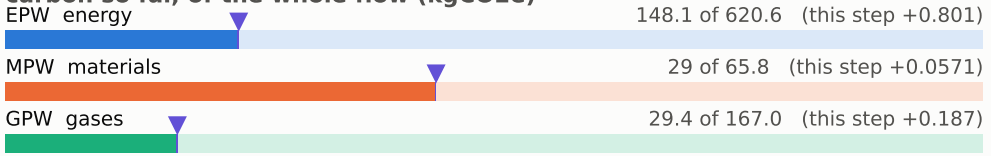


this step

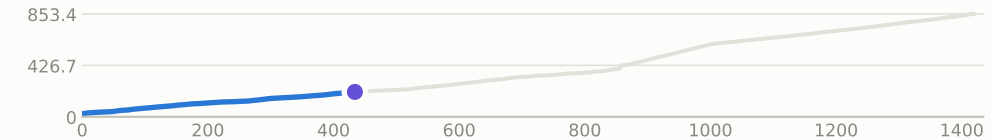
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

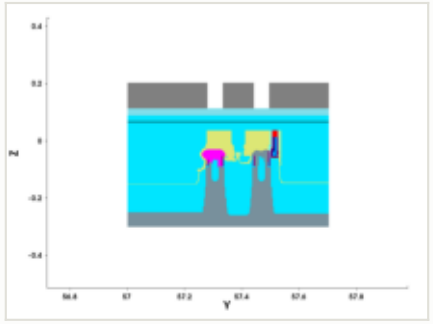
carbon so far, of the whole flow (kgCO2e)



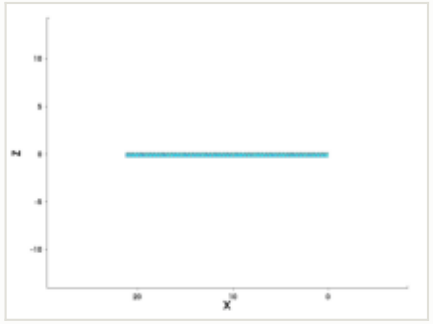
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



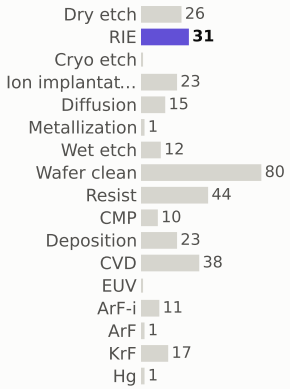
y cut at 57.351 µm



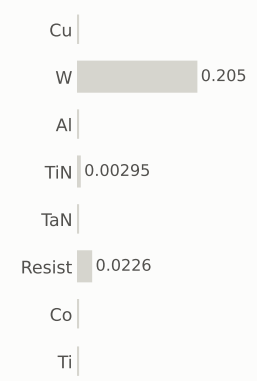
mask for this step

no mask for this step

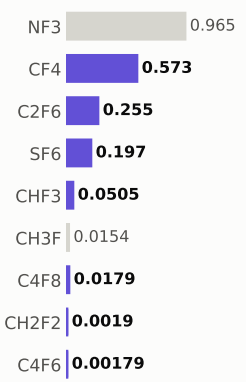
steps by process area



materials deposited (g)



gases emitted (g)



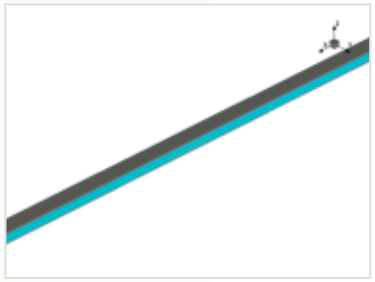
Step #4420: Spx_only

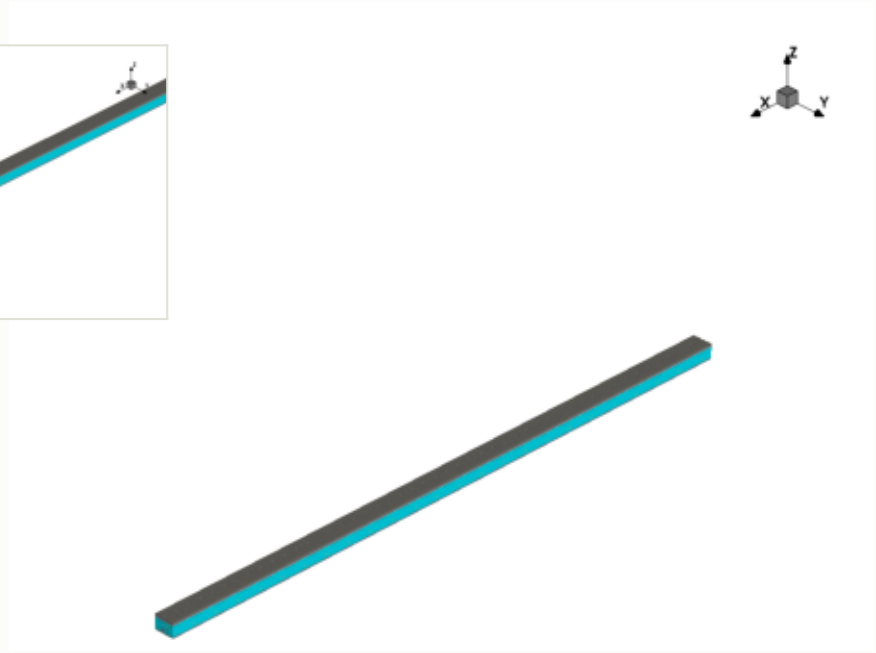
etch_anisotropic · Spx_only

0	206.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





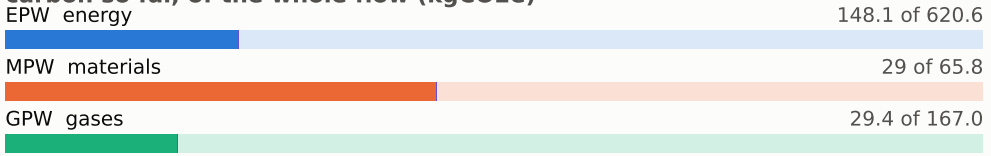
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

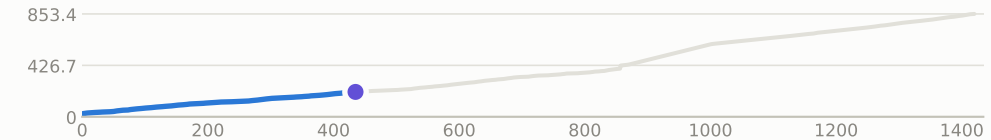
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

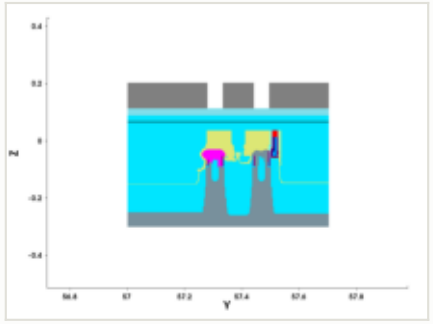
carbon so far, of the whole flow (kgCO2e)



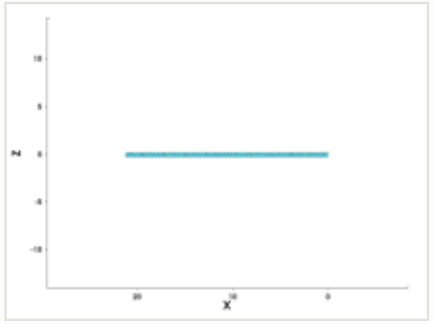
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



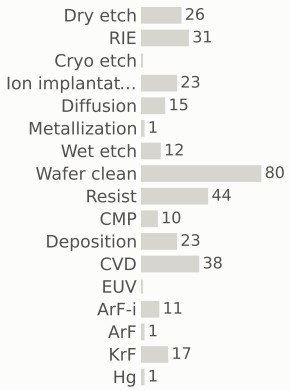
y cut at 57.351 µm



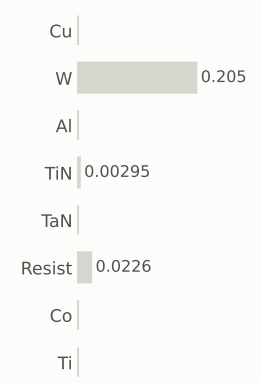
mask for this step

no mask for this step

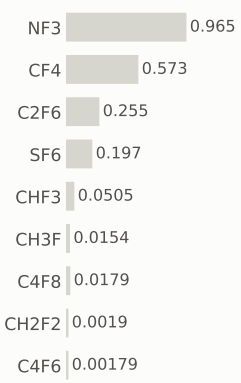
steps by process area



materials deposited (g)



gases emitted (g)



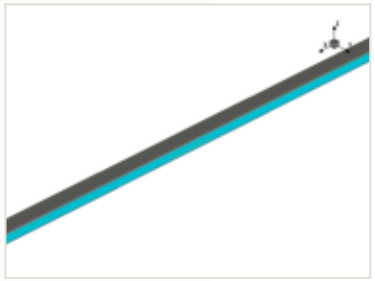
Step #4430: Spx_only

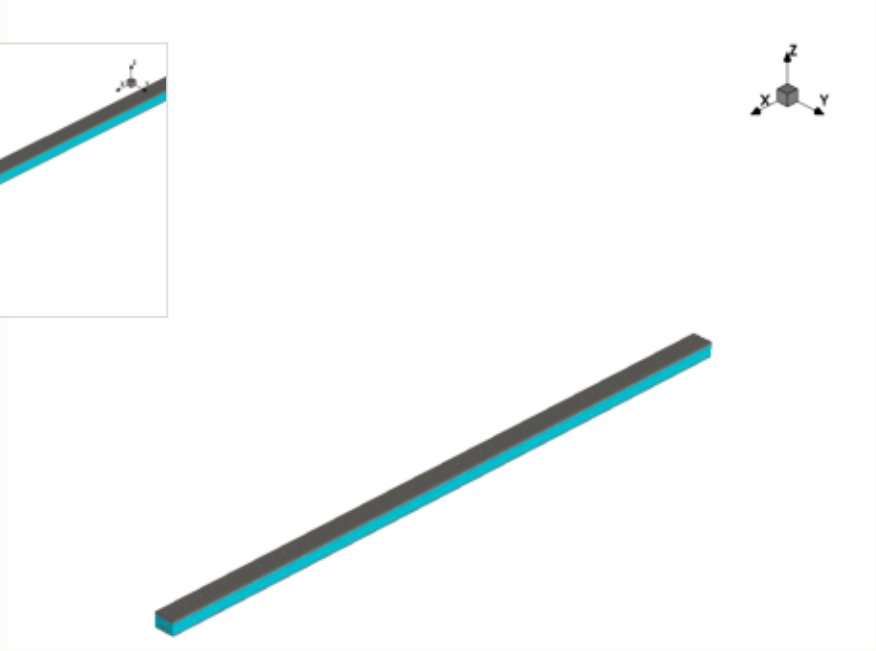
etch_strip · Spx_only

0	206.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





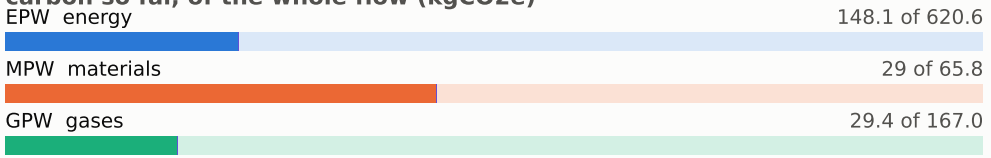
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

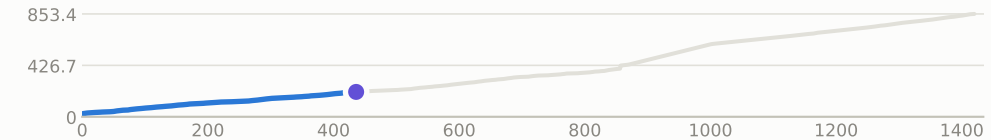
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

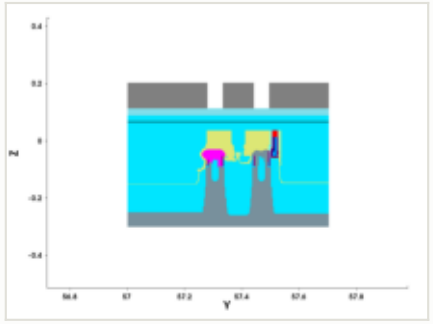
carbon so far, of the whole flow (kgCO2e)



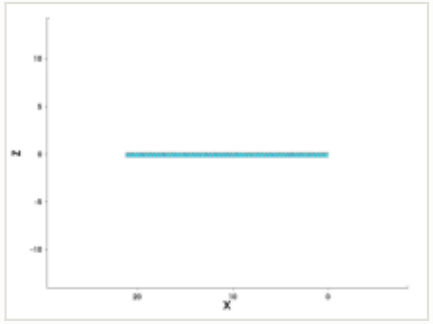
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



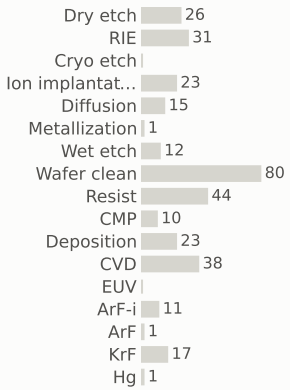
y cut at 57.351 µm



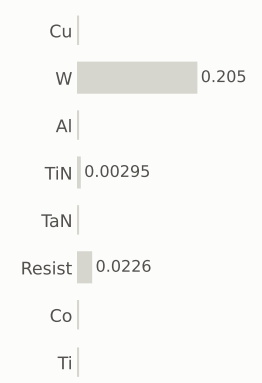
mask for this step

no mask for this step

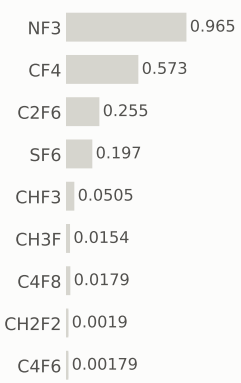
steps by process area



materials deposited (g)



gases emitted (g)



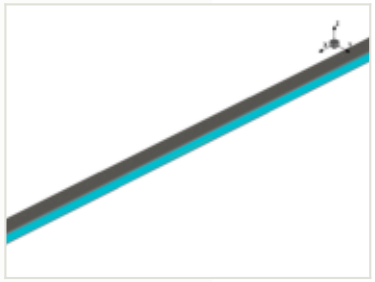
Step #4440: Wafer_clean

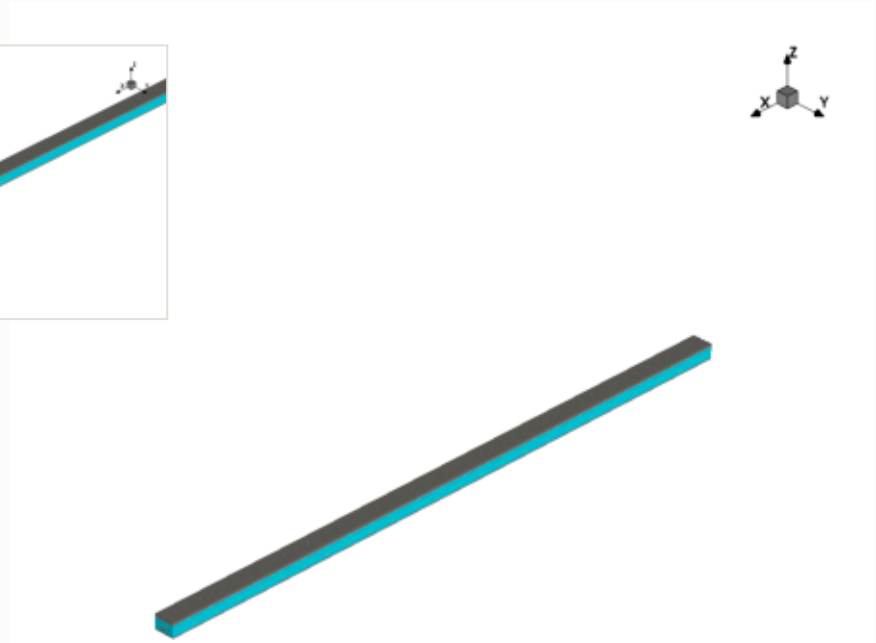
clean · Wafer_clean

0.169	206.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

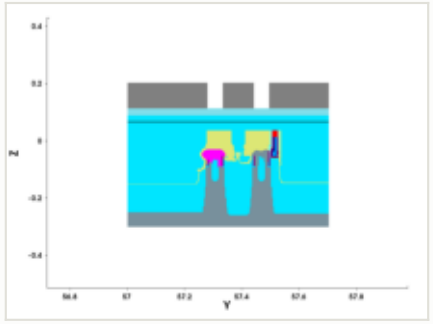
3D structure after this step

zoomed to fit

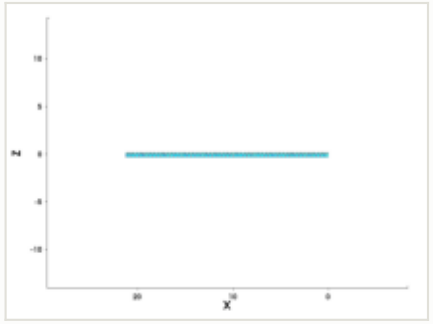




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

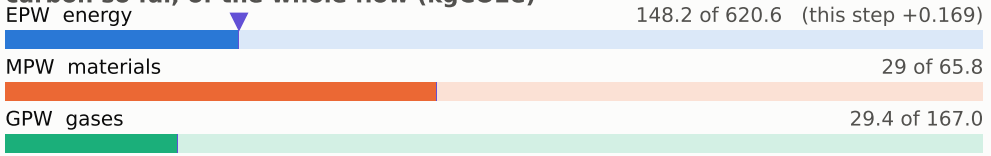
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

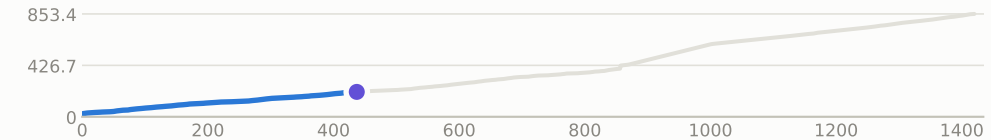
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

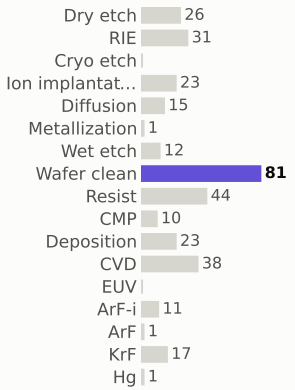
carbon so far, of the whole flow (kgCO2e)



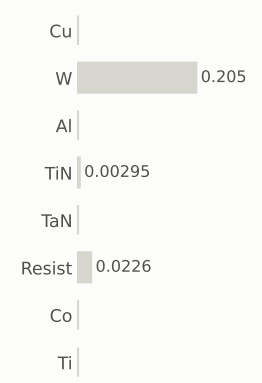
embodied carbon up to each step (kgCO2e per wafer)



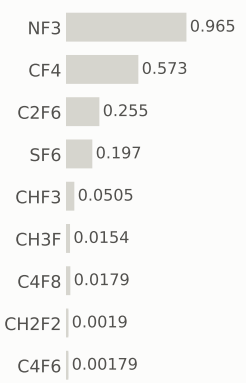
steps by process area



materials deposited (g)



gases emitted (g)



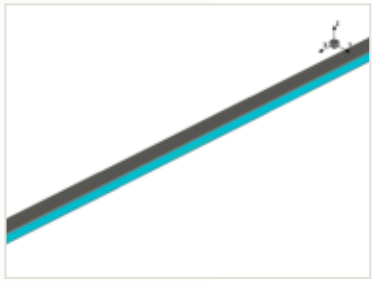
Step #4450: Resist

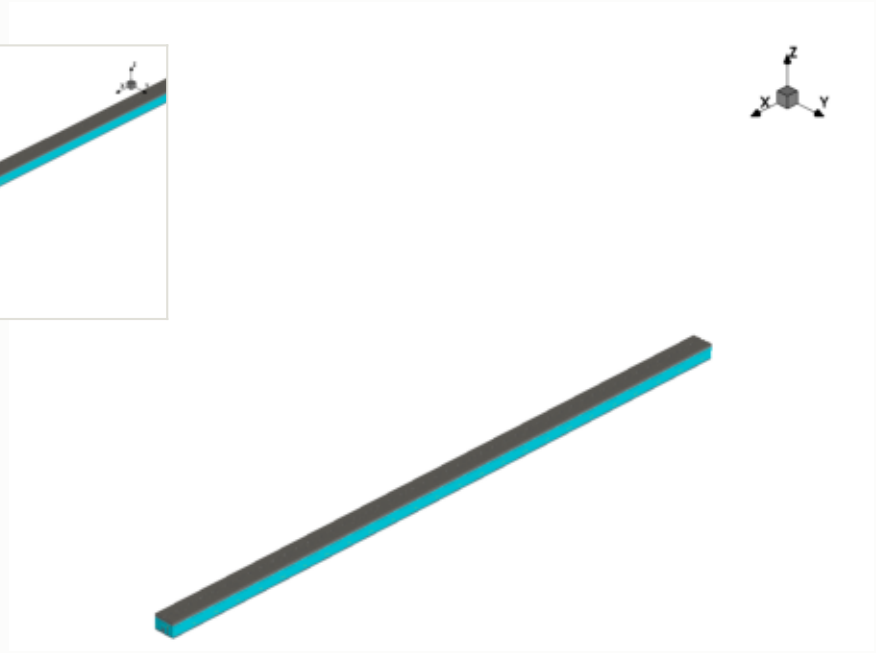
depo_fill · Resist

0.169	206.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



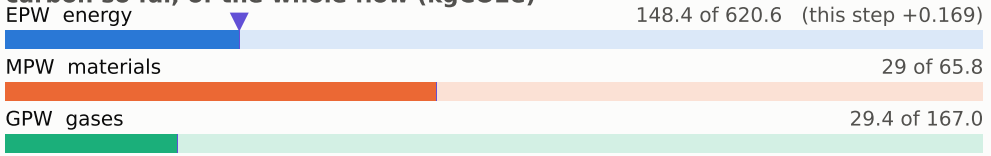


this step

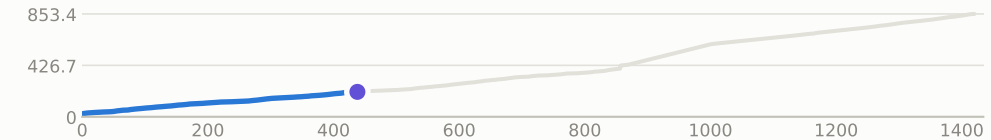
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

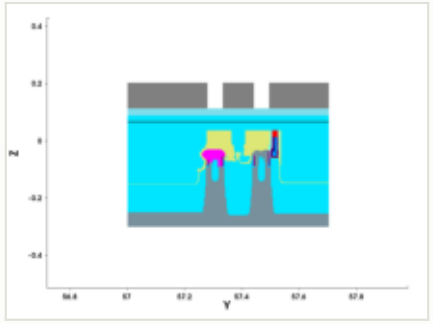
carbon so far, of the whole flow (kgCO2e)



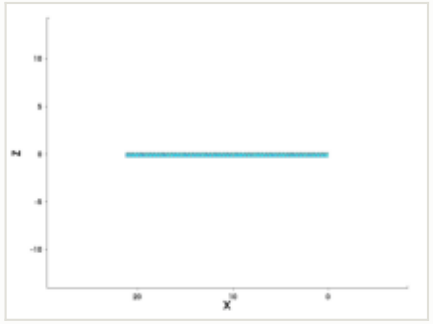
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



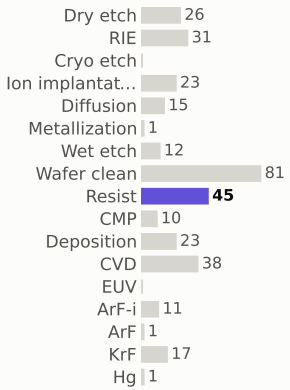
y cut at 57.351 µm



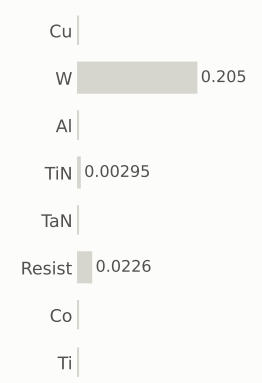
mask for this step

no mask for this step

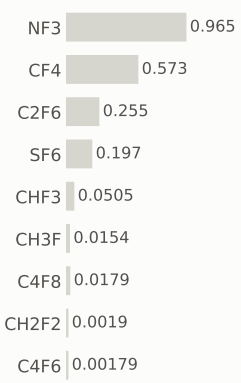
steps by process area



materials deposited (g)



gases emitted (g)



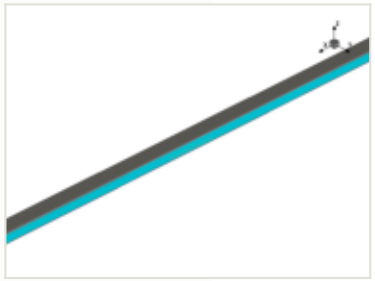
Step #4460: CVD

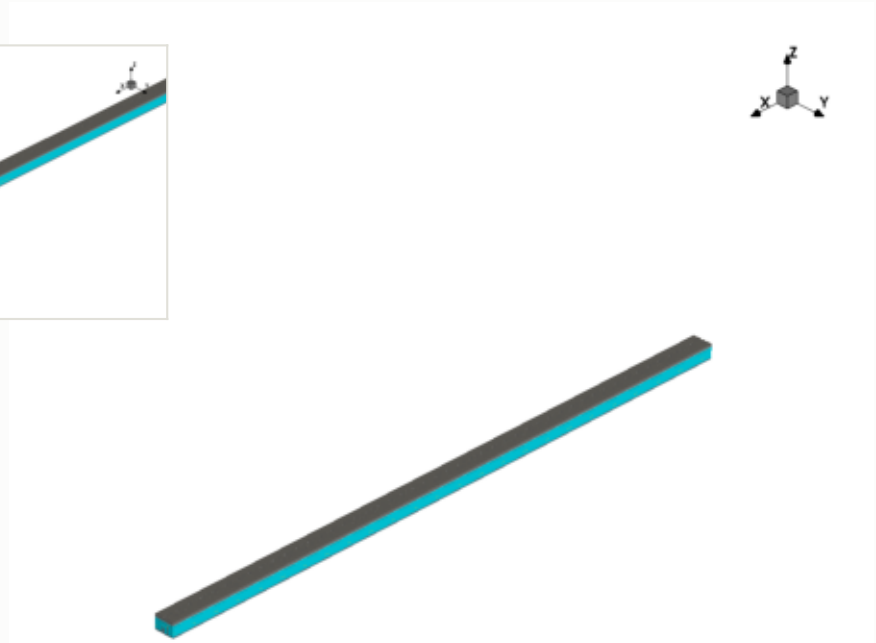
depo_fill · CVD

1.31	208.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



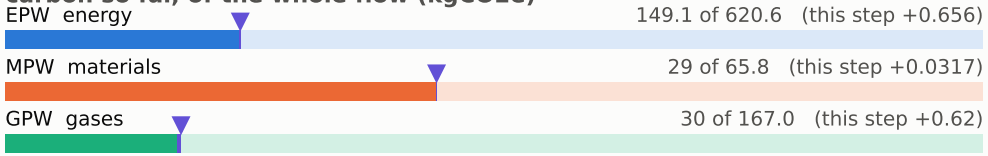


this step

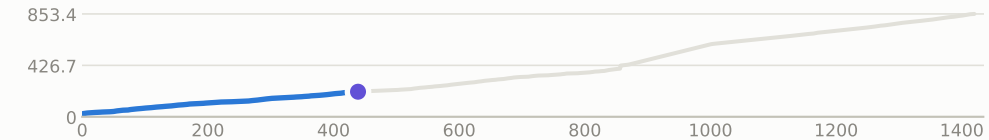
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

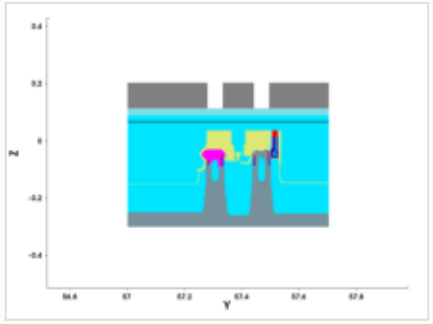
carbon so far, of the whole flow (kgCO2e)



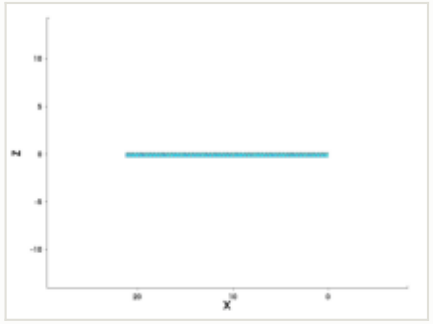
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



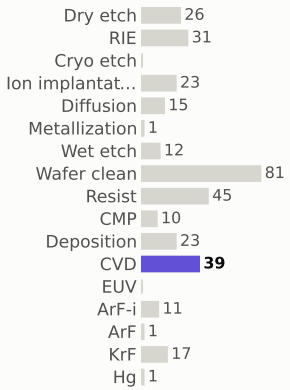
y cut at 57.351 µm



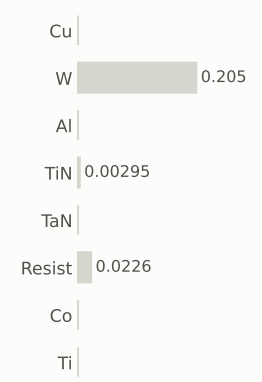
mask for this step

no mask for this step

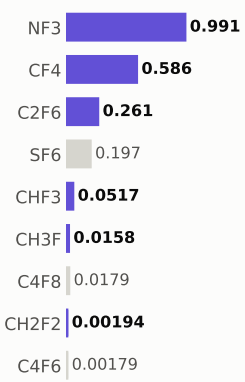
steps by process area



materials deposited (g)



gases emitted (g)



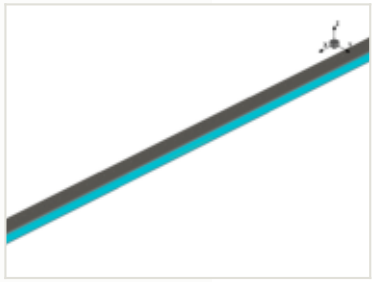
Step #4470: Wafer_clean

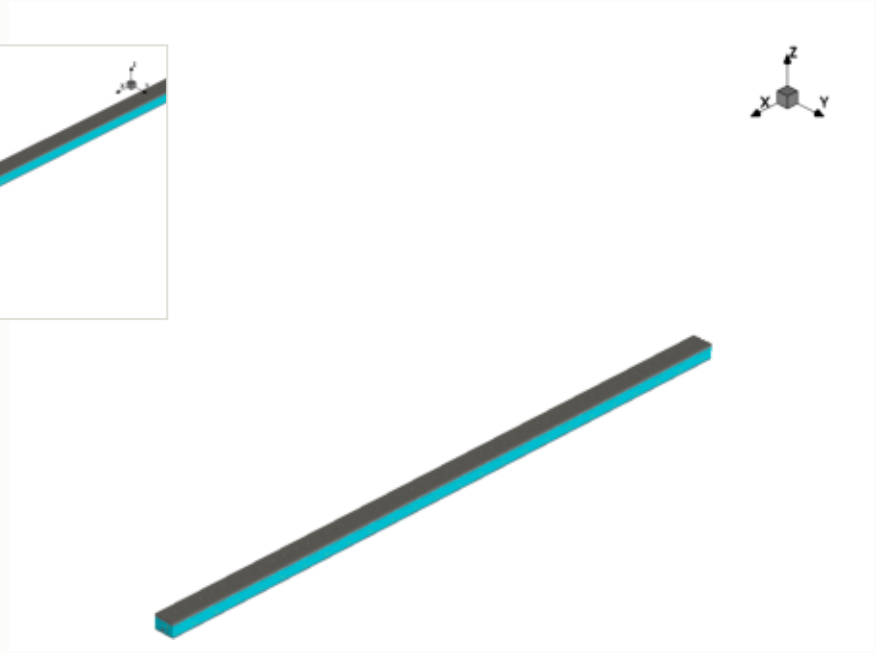
clean · Wafer_clean

0.169	208.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

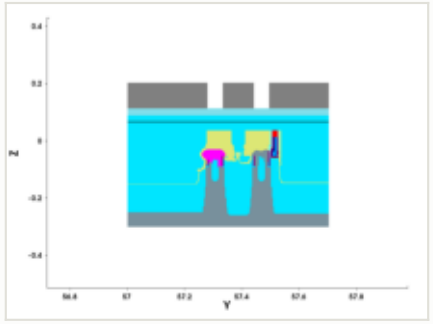
3D structure after this step

zoomed to fit

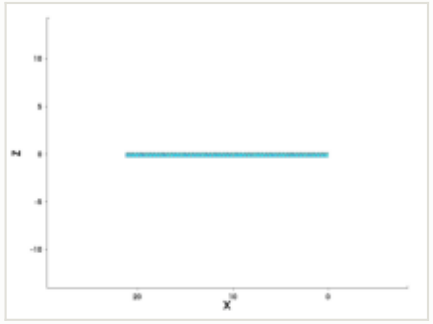




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

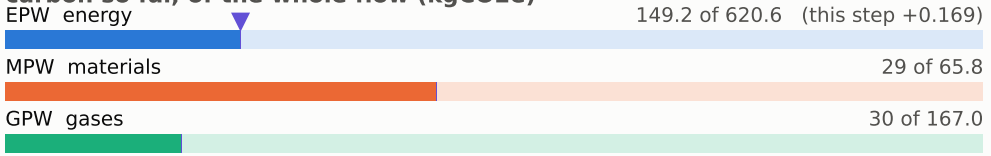
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

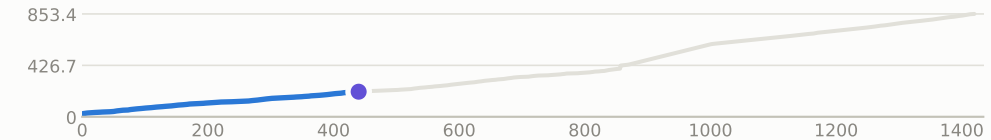
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

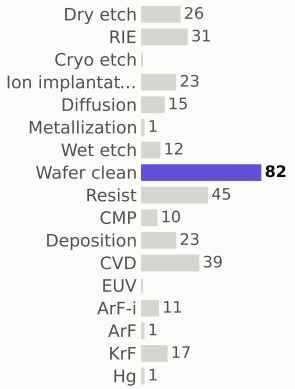
carbon so far, of the whole flow (kgCO2e)



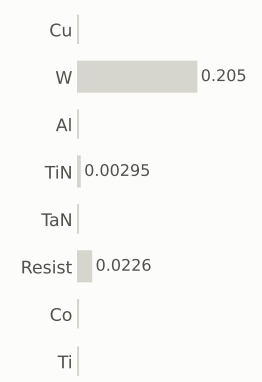
embodied carbon up to each step (kgCO2e per wafer)



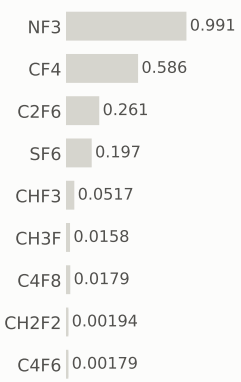
steps by process area



materials deposited (g)



gases emitted (g)



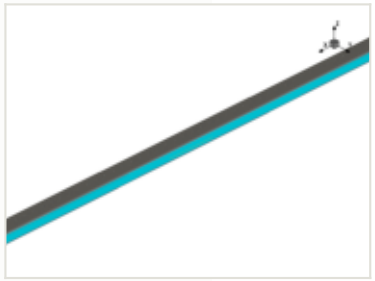
Step #4480: Resist

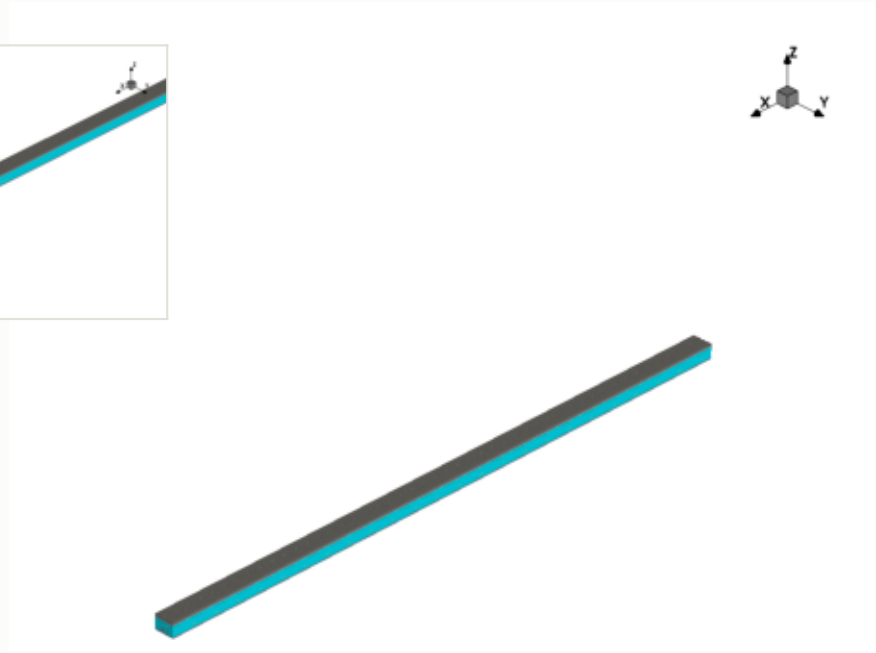
resist · Resist

0.169	208.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





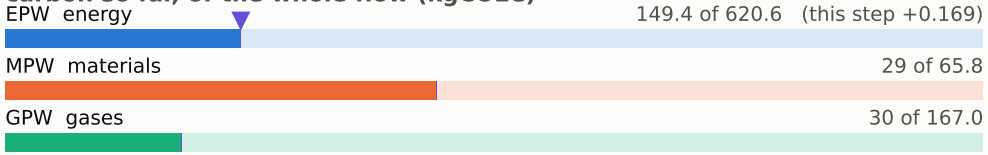
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

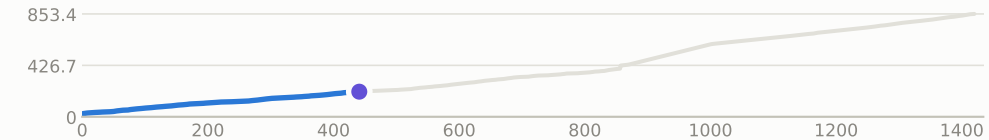
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

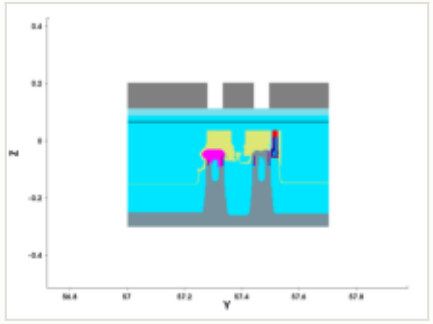
carbon so far, of the whole flow (kgCO2e)



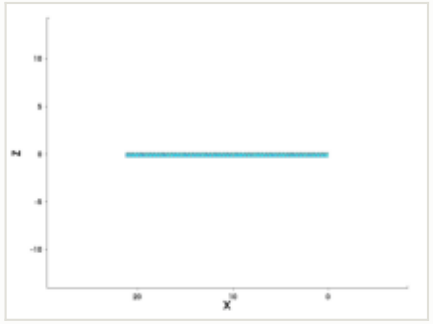
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



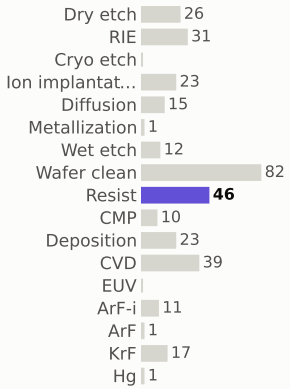
y cut at 57.351 µm



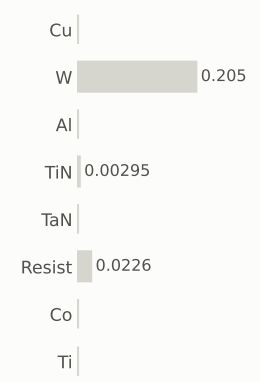
mask for this step

no mask for this step

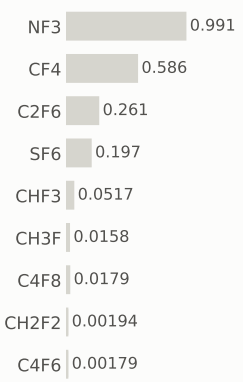
steps by process area



materials deposited (g)



gases emitted (g)



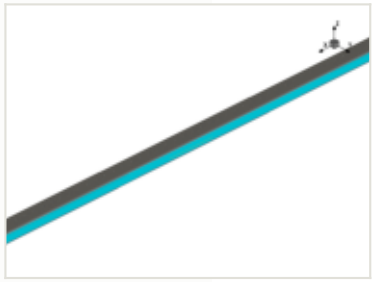
Step #4490: ArF-i

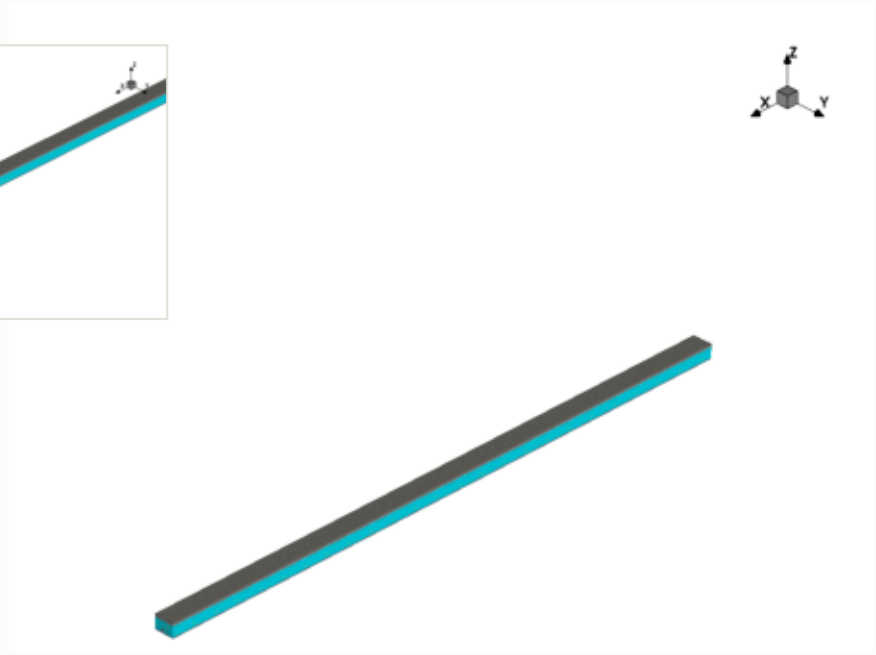
litho_patterning · ArF-i

0.217	208.6	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



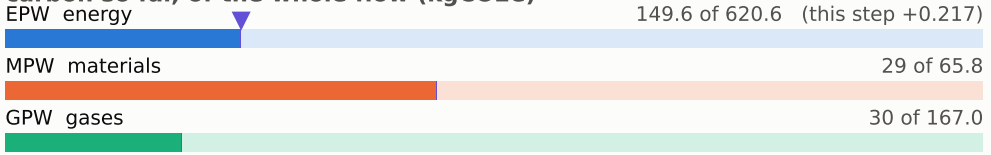


this step

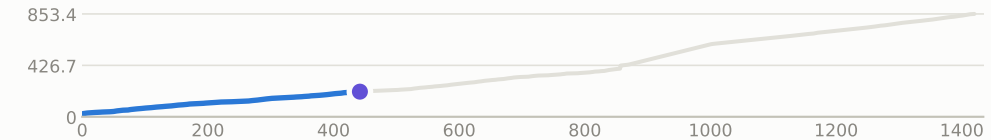
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

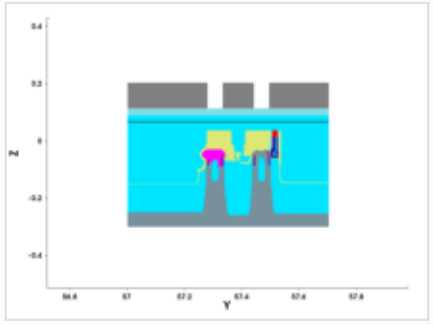
carbon so far, of the whole flow (kgCO2e)



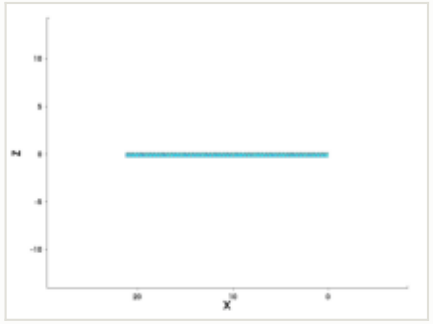
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

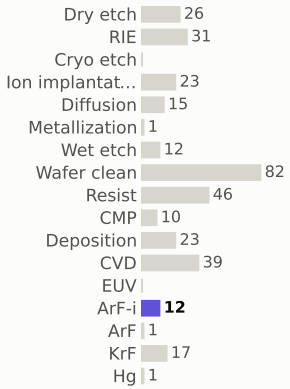


y cut at 57.351 µm

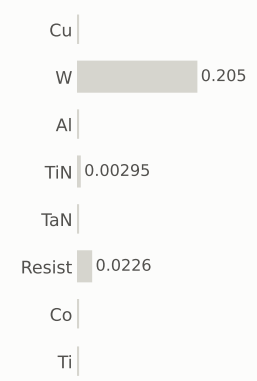


mask: layer 2017:0 (GDSII)

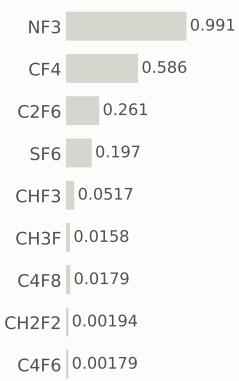
steps by process area



materials deposited (g)



gases emitted (g)



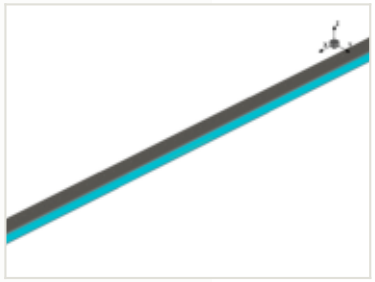
Step #4500: Spx_only

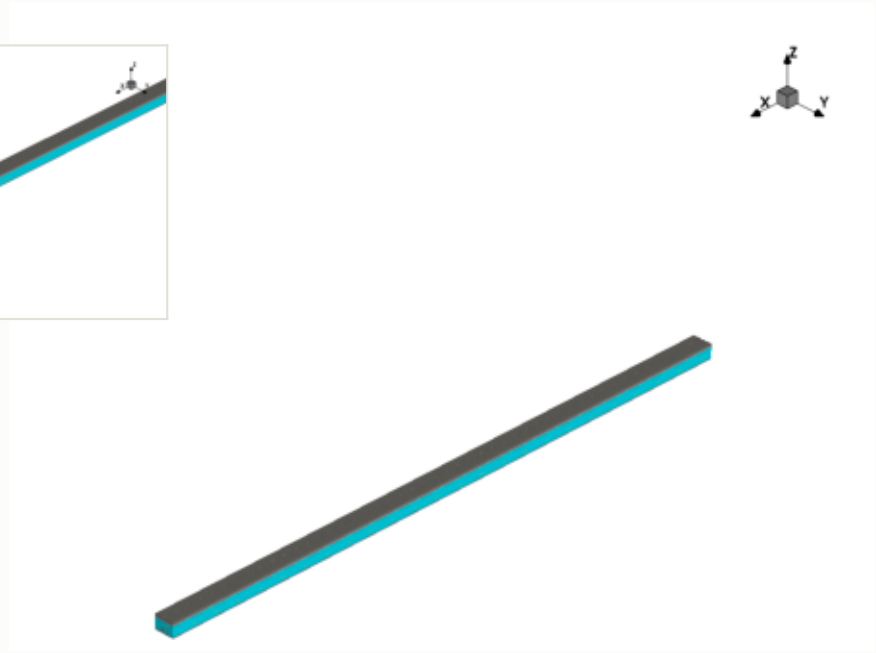
etch_anisotropic · Spx_only

0	208.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





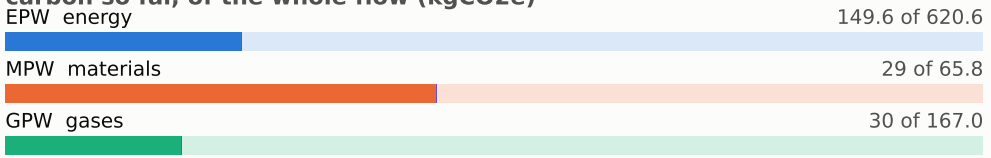
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

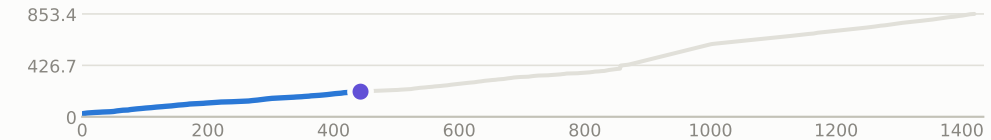
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

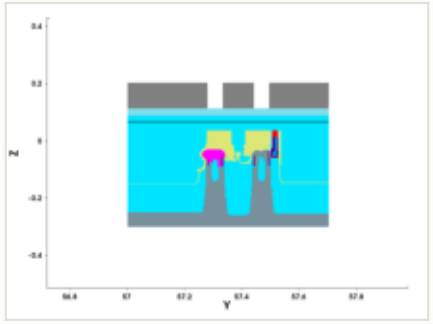
carbon so far, of the whole flow (kgCO2e)



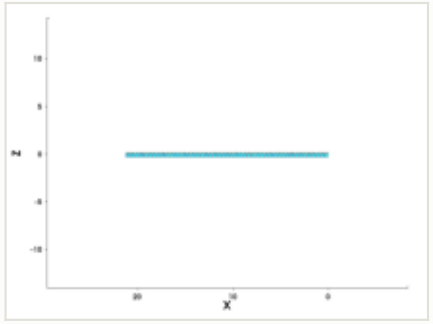
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



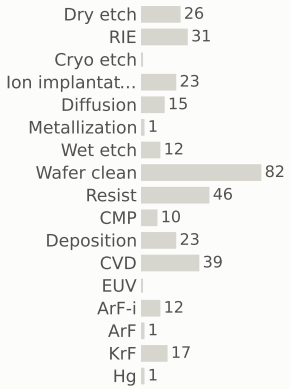
y cut at 57.351 µm



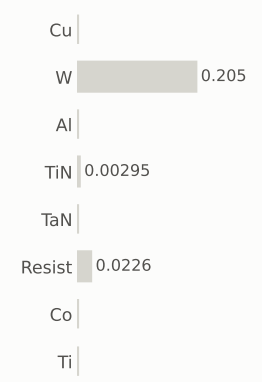
mask for this step

no mask for this step

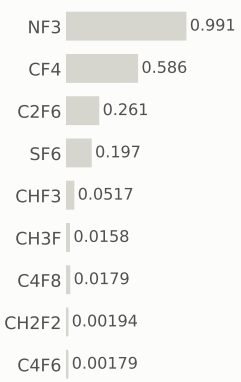
steps by process area



materials deposited (g)



gases emitted (g)



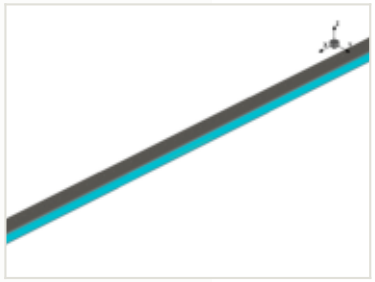
Step #4510: RIE

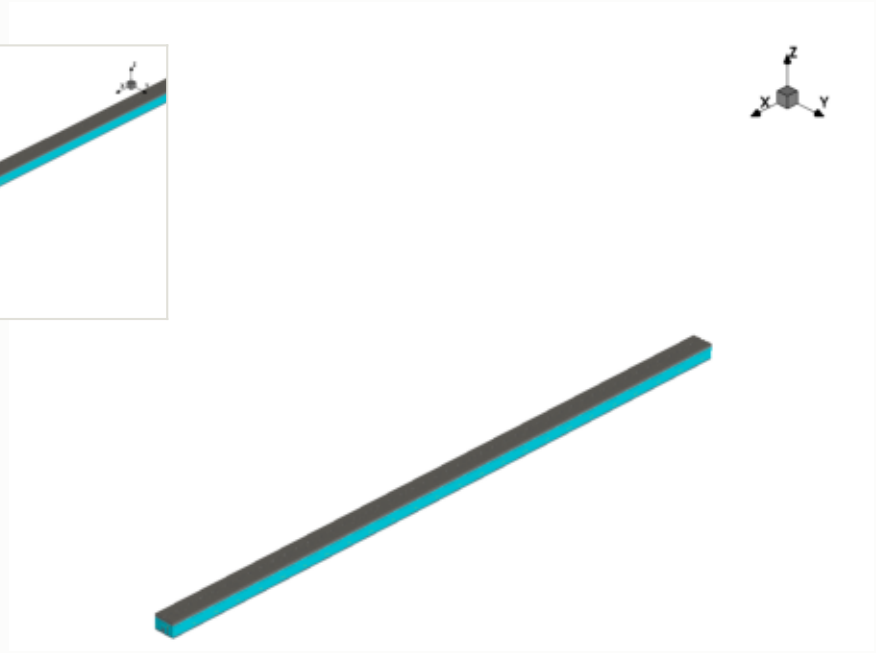
etch_anisotropic · RIE

1.04	209.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



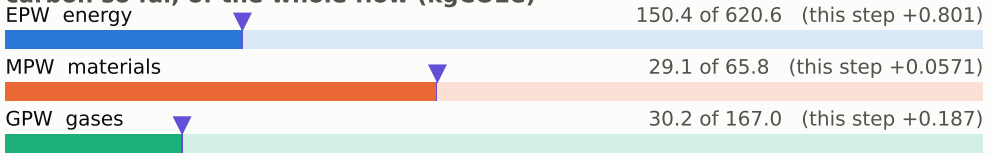


this step

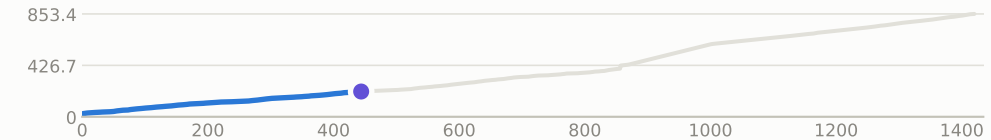
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

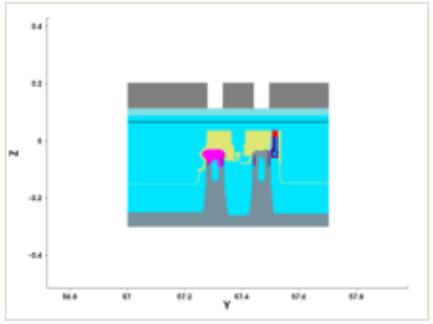
carbon so far, of the whole flow (kgCO2e)



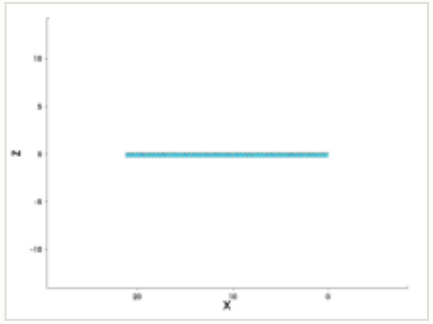
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



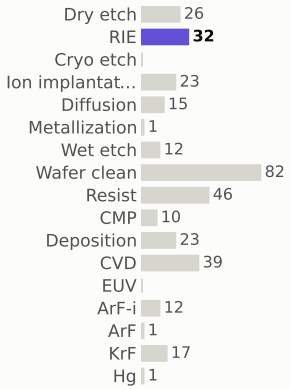
y cut at 57.351 µm



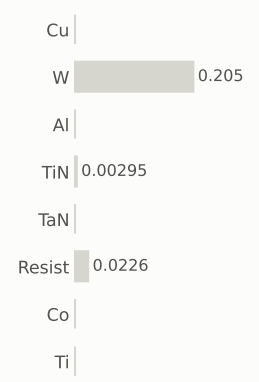
mask for this step

no mask for this step

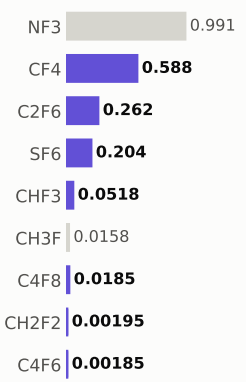
steps by process area



materials deposited (g)



gases emitted (g)



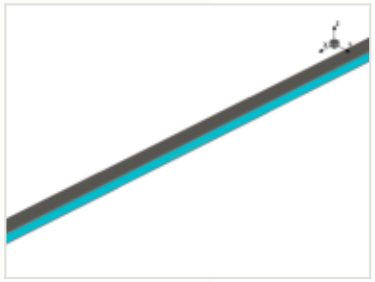
Step #4520: Spx_only

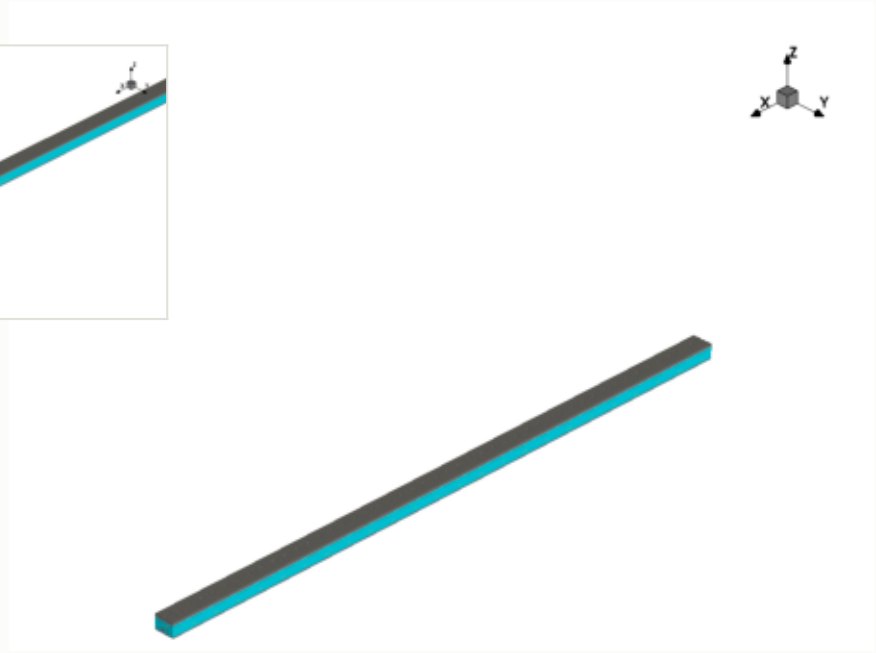
etch_anisotropic · Spx_only

0	209.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





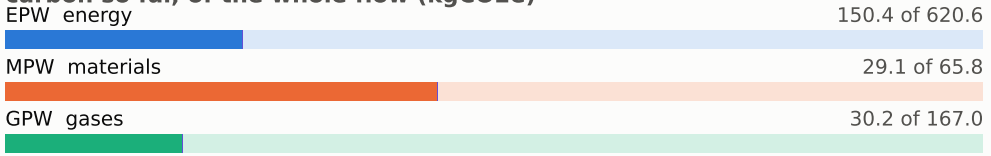
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

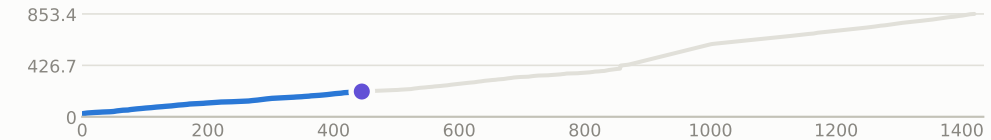
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

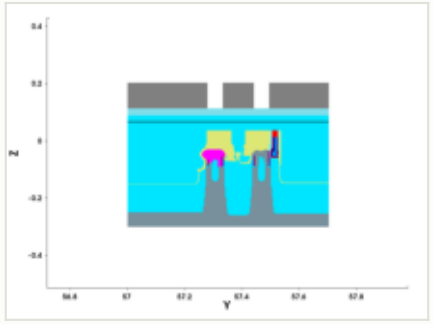
carbon so far, of the whole flow (kgCO2e)



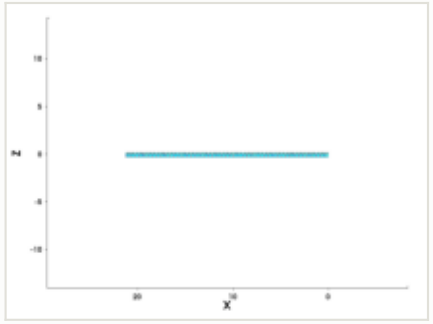
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



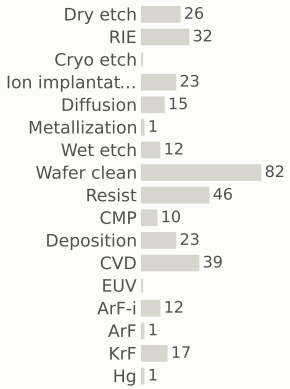
y cut at 57.351 µm



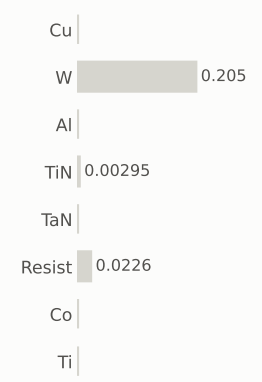
mask for this step

no mask for this step

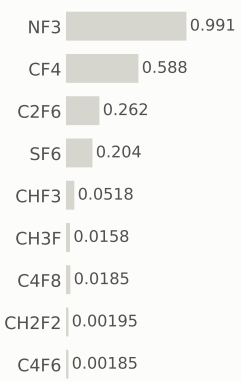
steps by process area



materials deposited (g)



gases emitted (g)



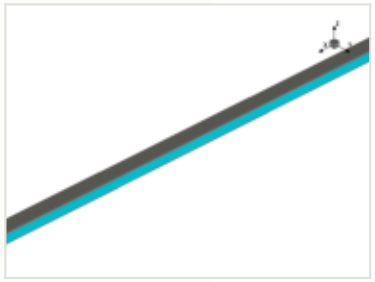
Step #4530: Spx_only

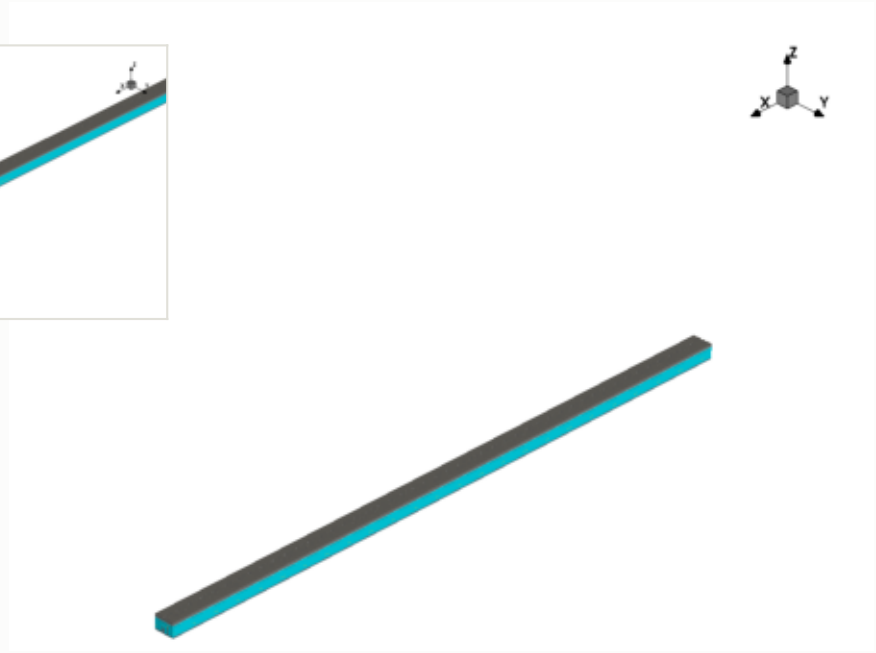
etch_strip · Spx_only

0	209.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





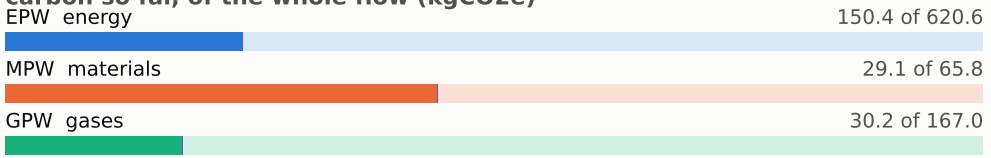
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

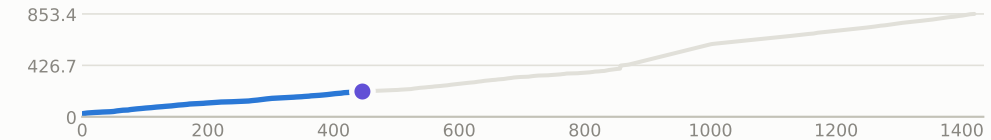
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

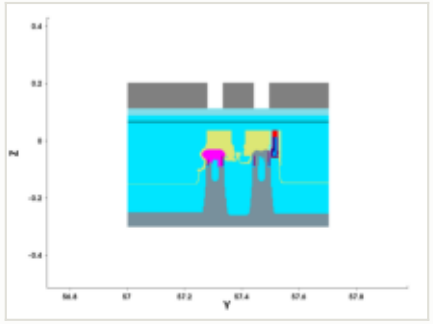
carbon so far, of the whole flow (kgCO2e)



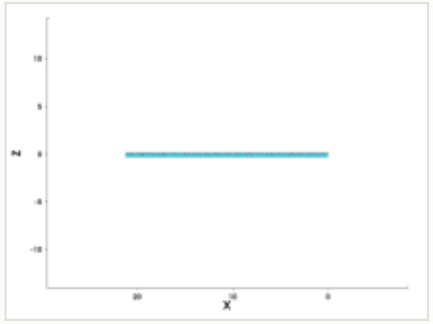
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



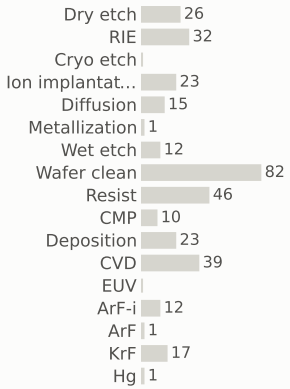
y cut at 57.351 µm



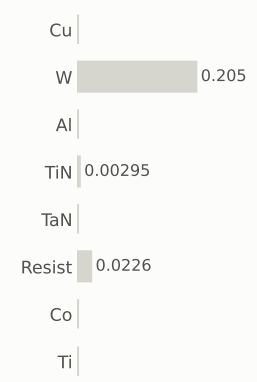
mask for this step

no mask for this step

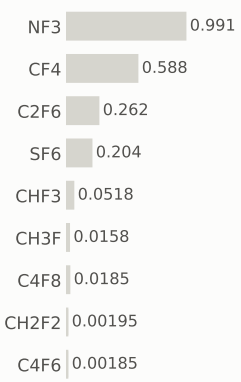
steps by process area



materials deposited (g)



gases emitted (g)



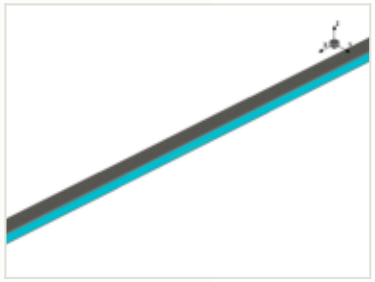
Step #4540: Wafer_clean

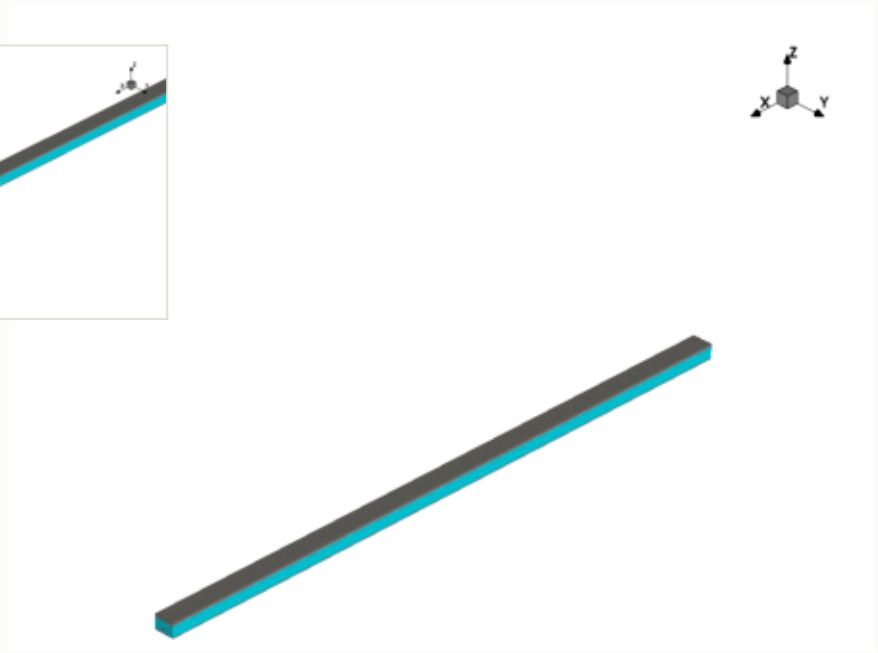
clean · Wafer_clean

0.169	209.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

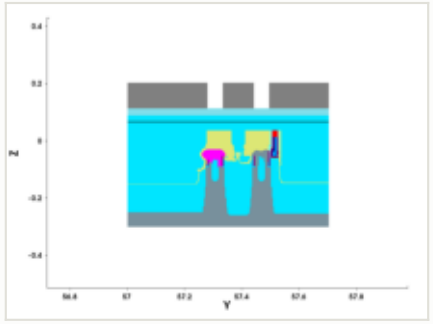
3D structure after this step

zoomed to fit

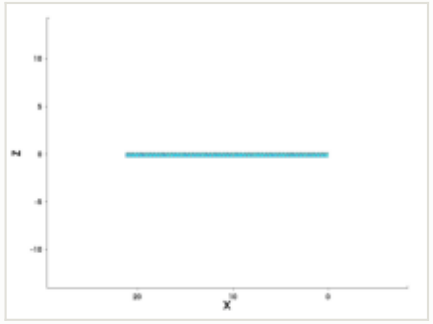




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

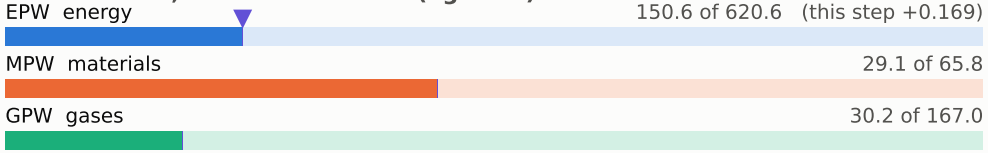
no mask for this step

this step

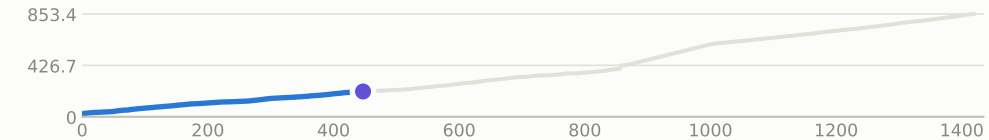
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

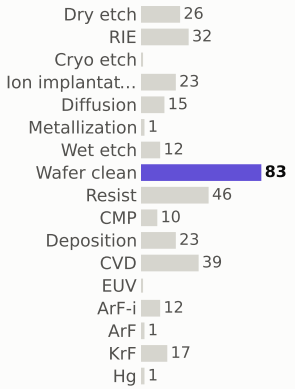
carbon so far, of the whole flow (kgCO2e)



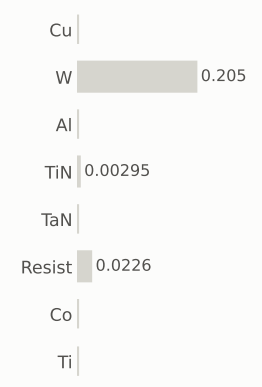
embodied carbon up to each step (kgCO2e per wafer)



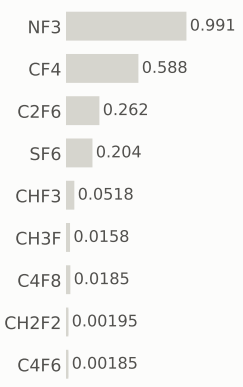
steps by process area



materials deposited (g)



gases emitted (g)



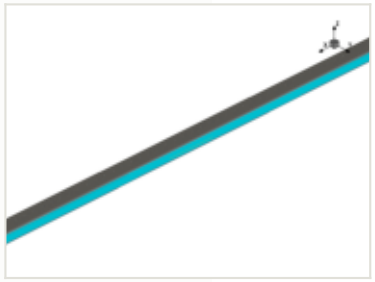
Step #4550: Resist

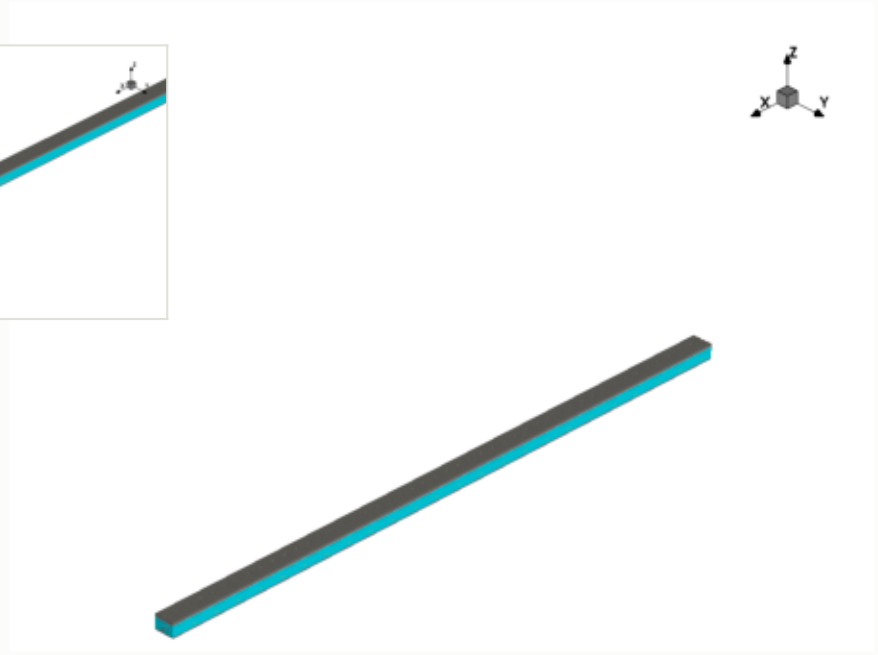
depo_fill · Resist

0.169	210.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



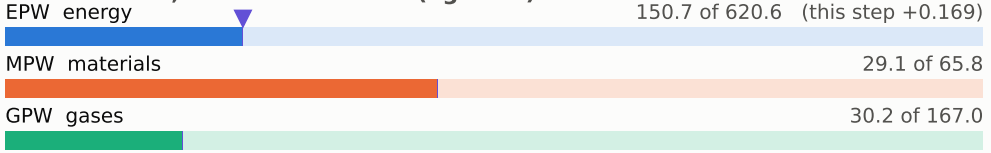


this step

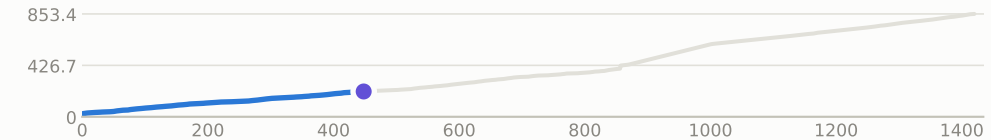
process area	Resist
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

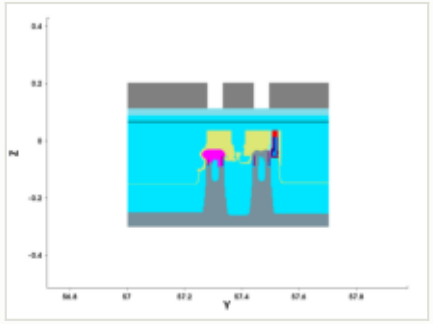
carbon so far, of the whole flow (kgCO2e)



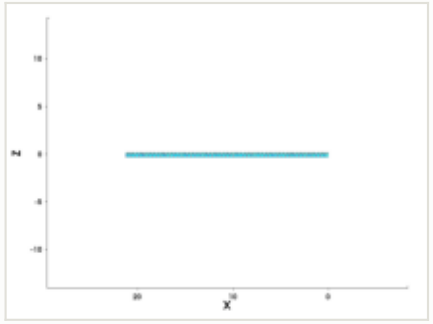
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



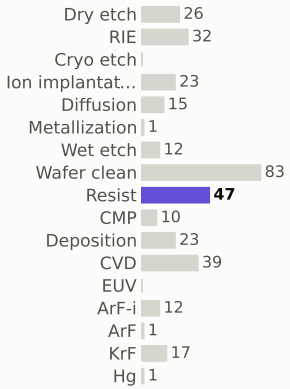
y cut at 57.351 µm



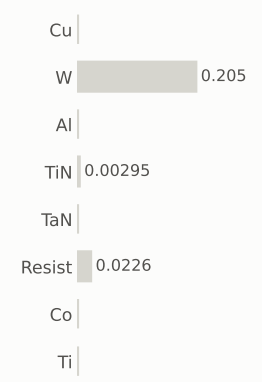
mask for this step

no mask for this step

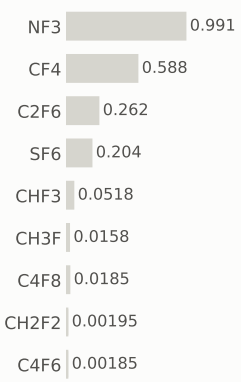
steps by process area



materials deposited (g)



gases emitted (g)



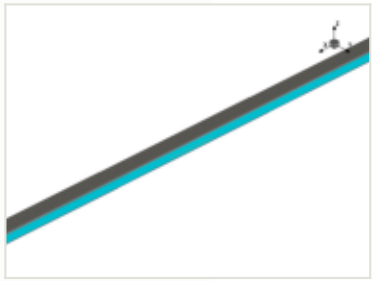
Step #4560: CVD

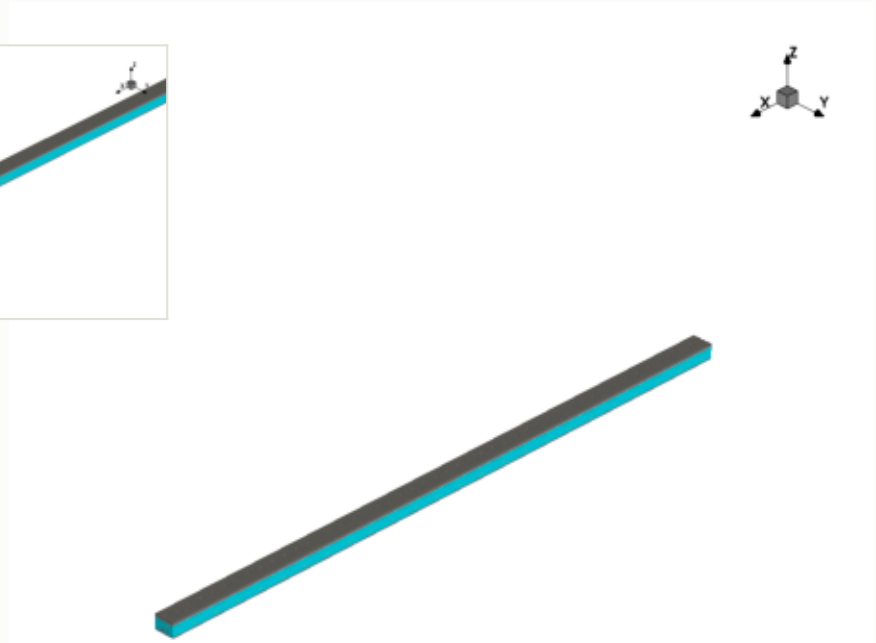
depo_fill · CVD

1.31	211.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



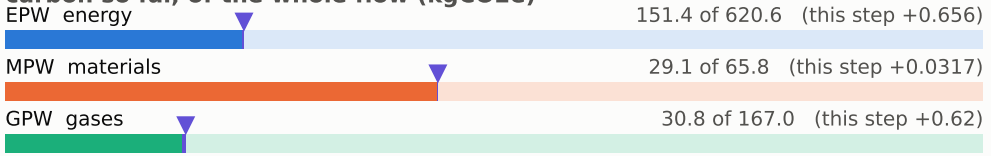


this step

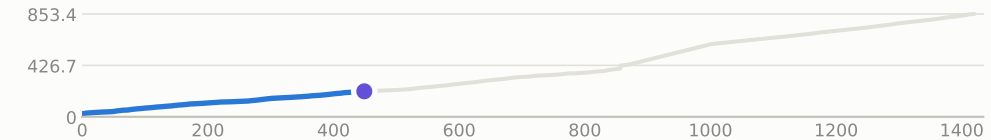
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

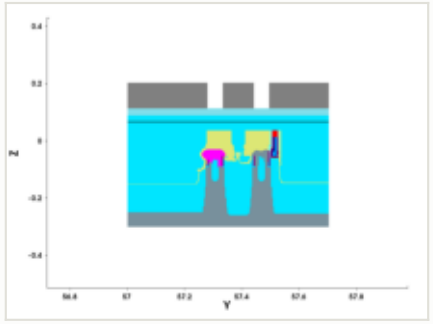
carbon so far, of the whole flow (kgCO2e)



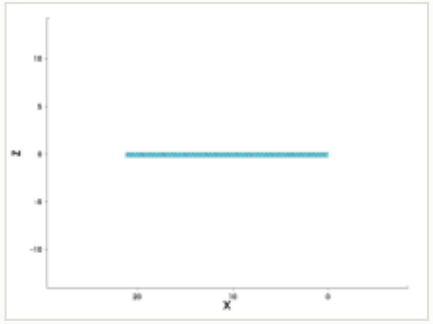
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



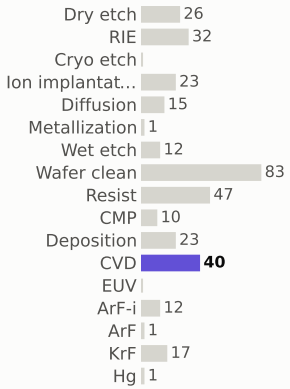
y cut at 57.351 µm



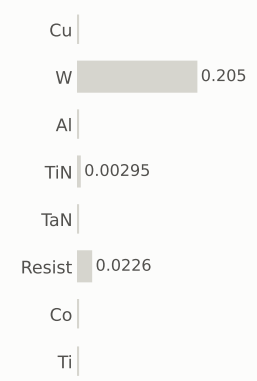
mask for this step

no mask for this step

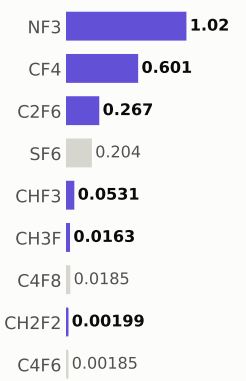
steps by process area



materials deposited (g)



gases emitted (g)



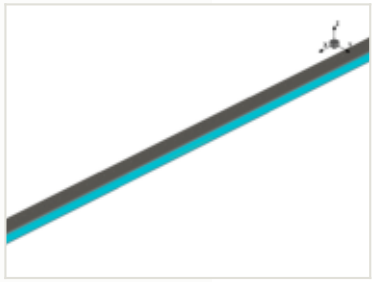
Step #4570: Wafer_clean

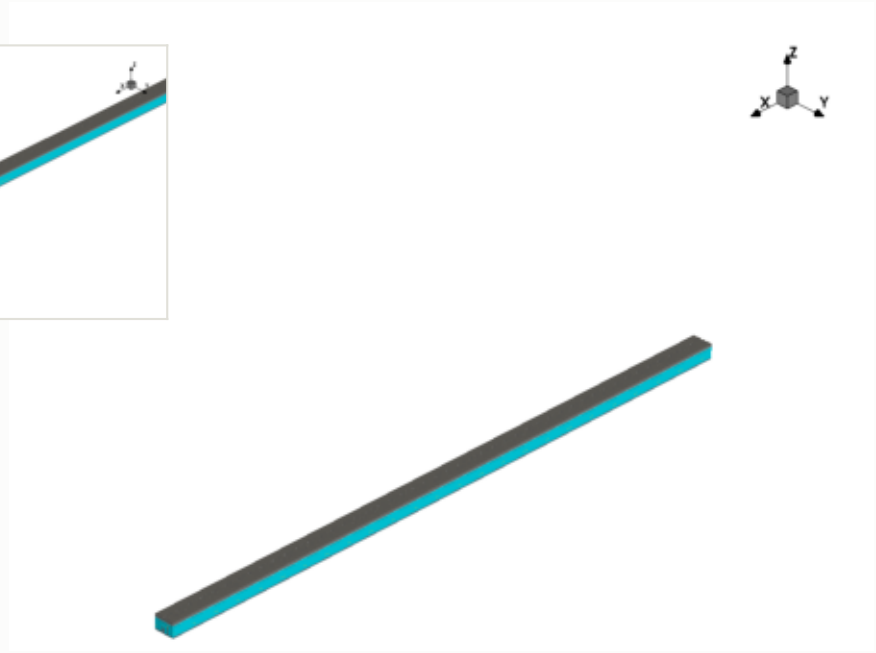
clean · Wafer_clean

0.169	211.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

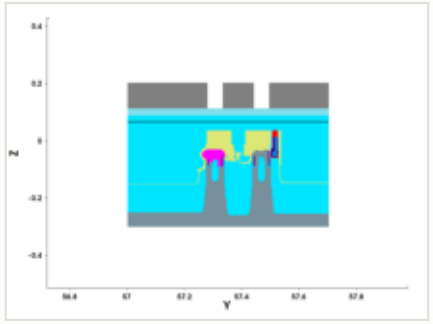
3D structure after this step

zoomed to fit

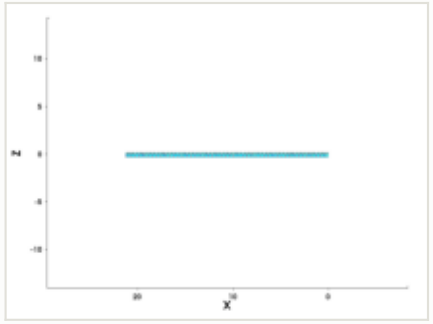




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

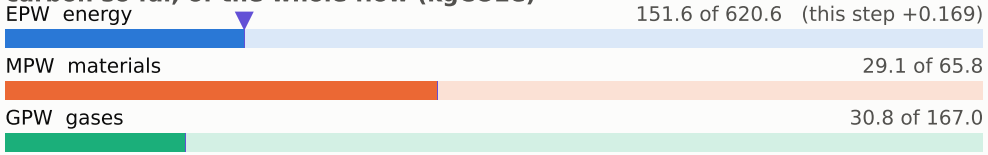
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

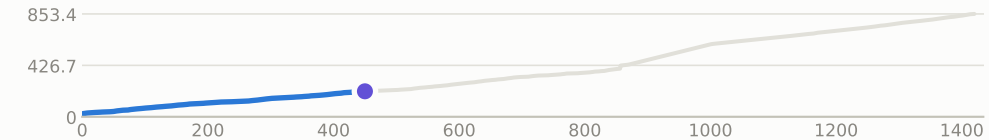
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

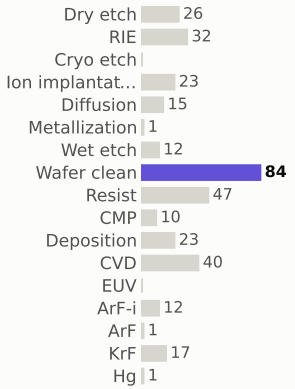
carbon so far, of the whole flow (kgCO2e)



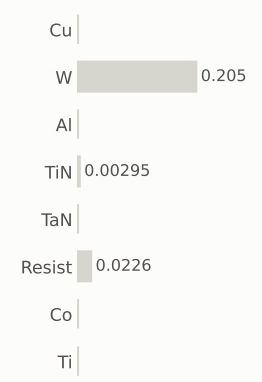
embodied carbon up to each step (kgCO2e per wafer)



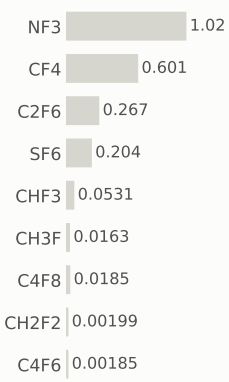
steps by process area



materials deposited (g)



gases emitted (g)



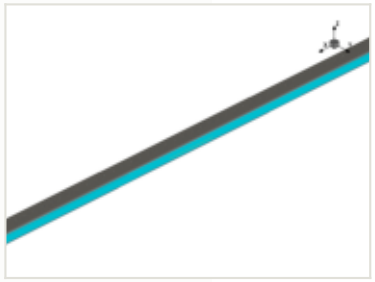
Step #4580: Resist

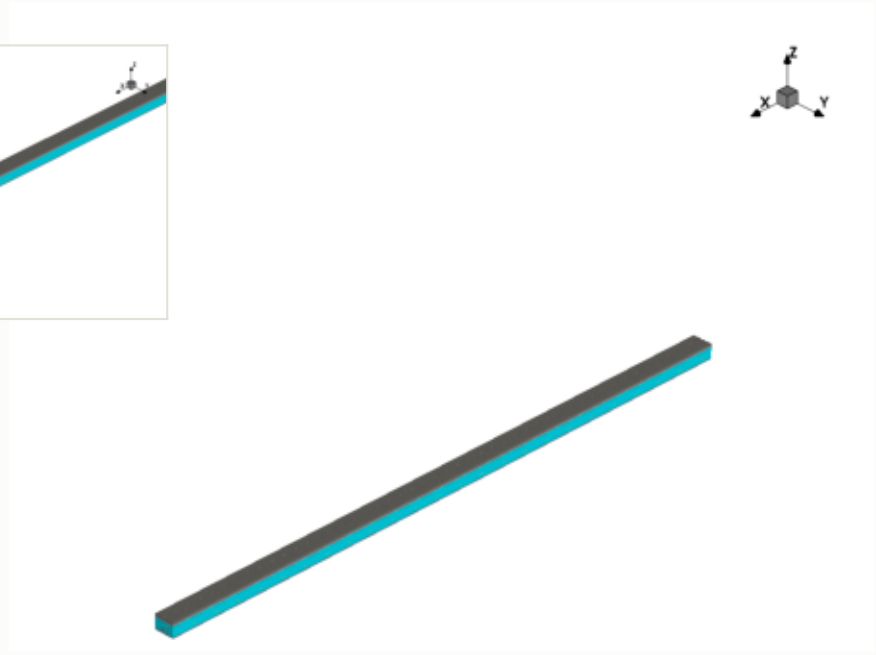
resist · Resist

0.169	211.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





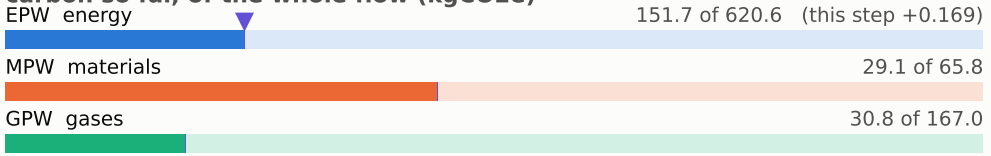
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

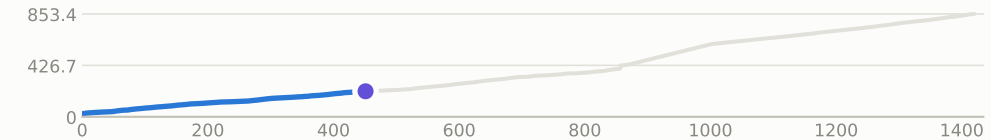
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

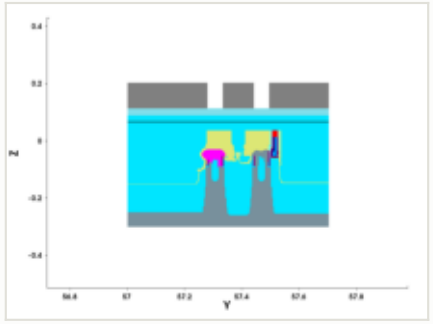
carbon so far, of the whole flow (kgCO2e)



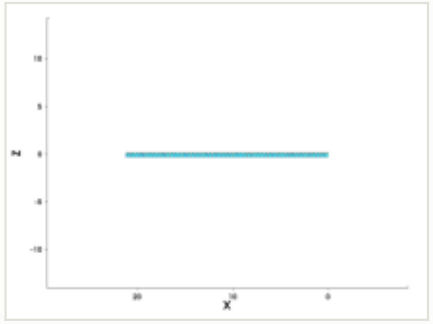
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



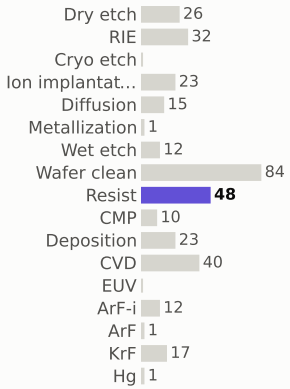
y cut at 57.351 µm



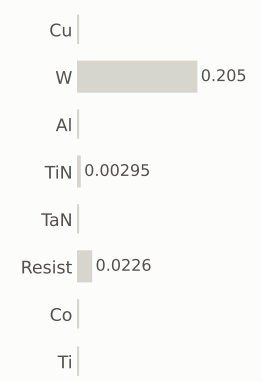
mask for this step

no mask for this step

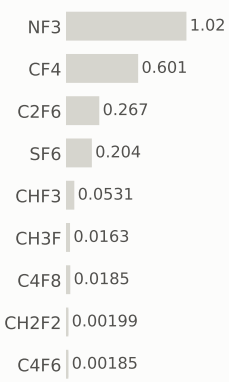
steps by process area



materials deposited (g)



gases emitted (g)



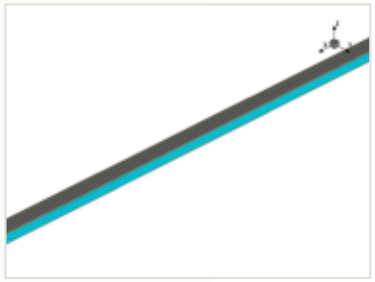
Step #4590: ArF-i

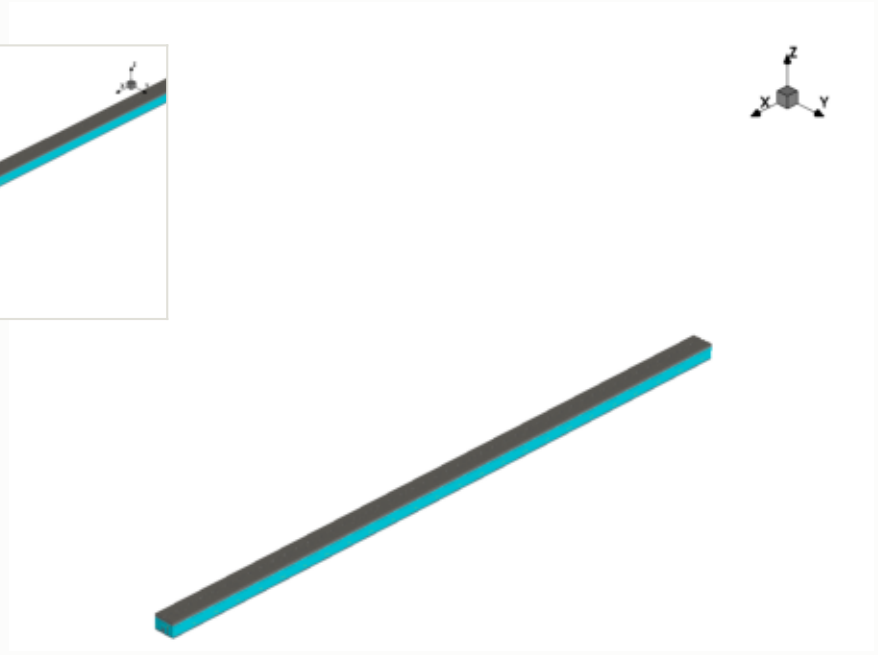
litho_patterning · ArF-i

0.217	211.9	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



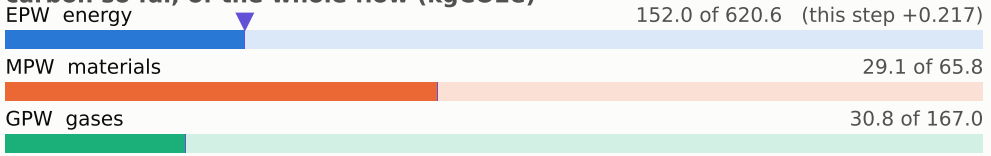


this step

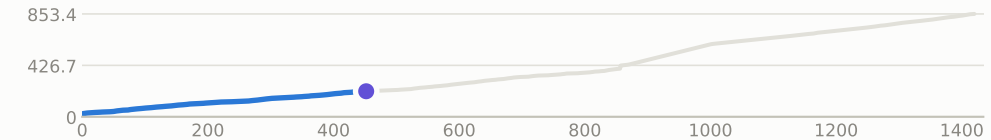
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

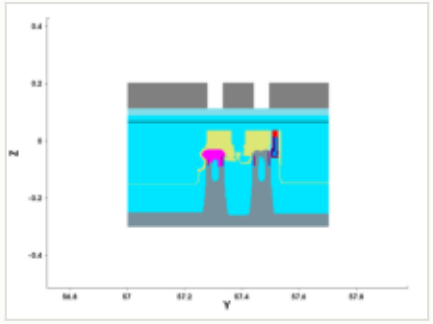
carbon so far, of the whole flow (kgCO2e)



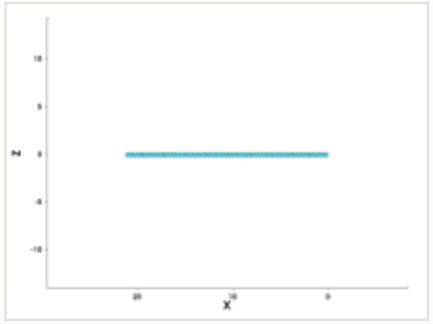
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

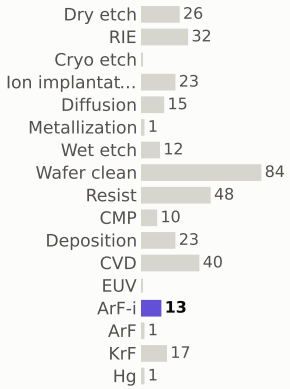


y cut at 57.351 µm

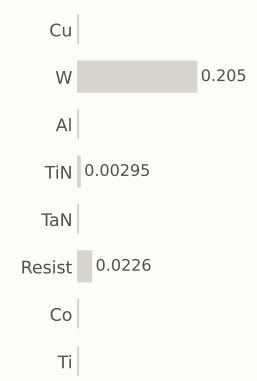


mask: layer 2017:0 (GDSII)

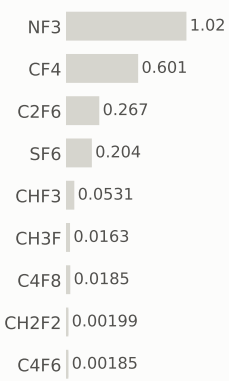
steps by process area



materials deposited (g)



gases emitted (g)



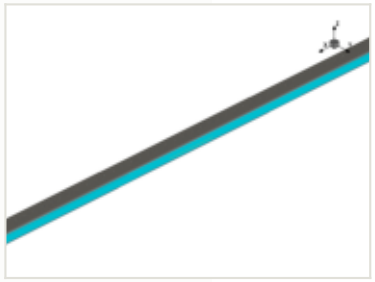
Step #4600: Spx_only

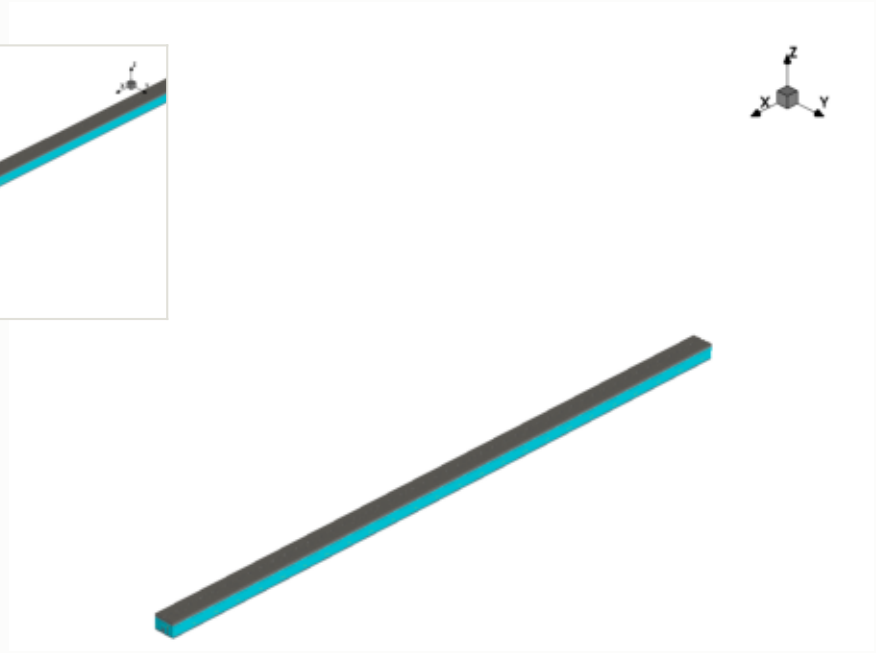
etch_anisotropic · Spx_only

0	211.9	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



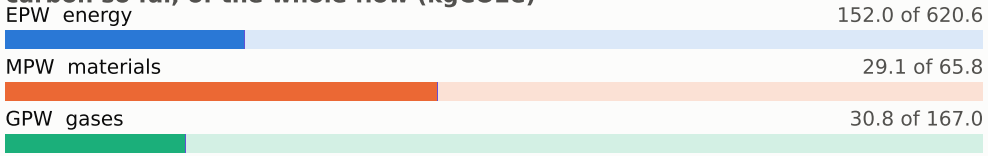


this step

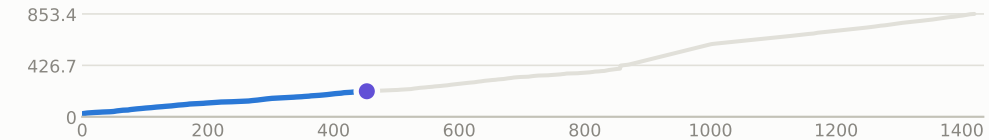
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

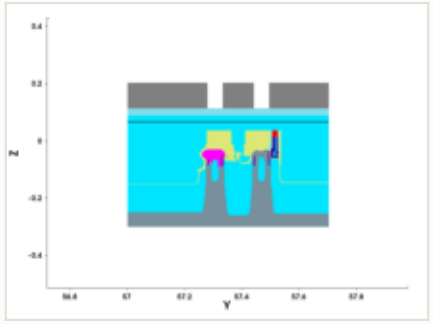
carbon so far, of the whole flow (kgCO2e)



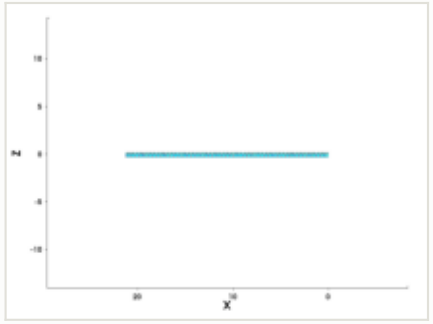
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



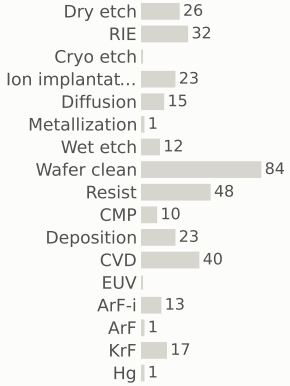
y cut at 57.351 µm



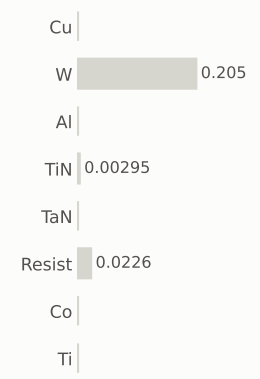
mask for this step

no mask for this step

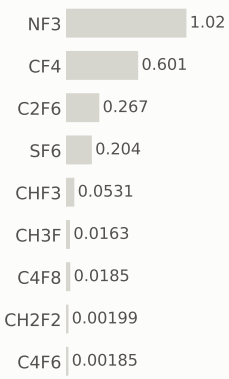
steps by process area



materials deposited (g)



gases emitted (g)



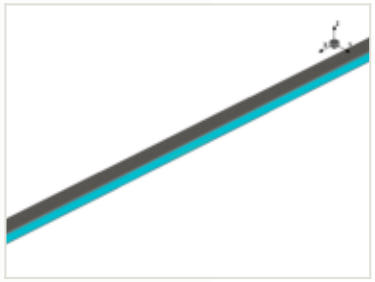
Step #4610: Spx_only

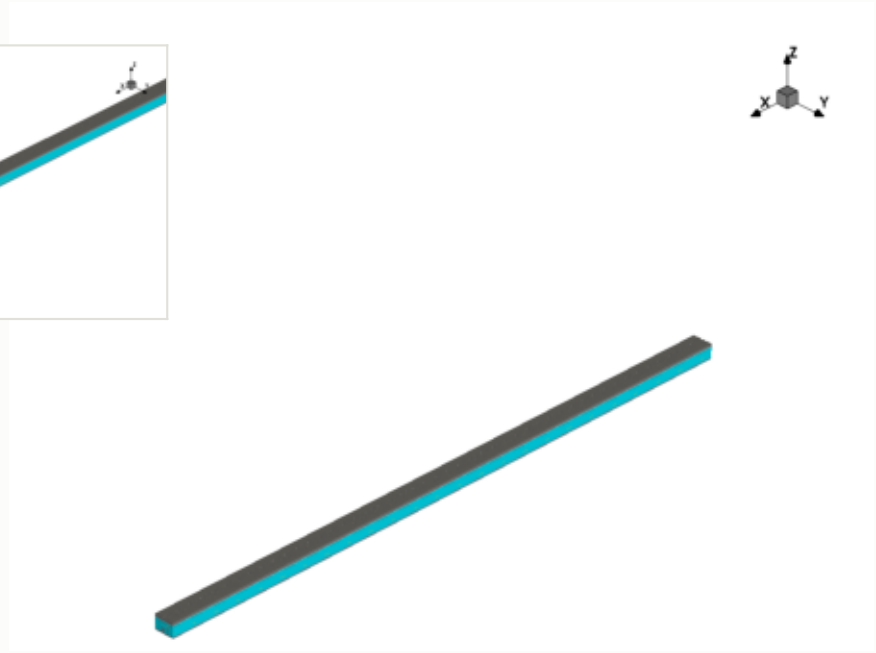
etch_anisotropic · Spx_only

0	211.9	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





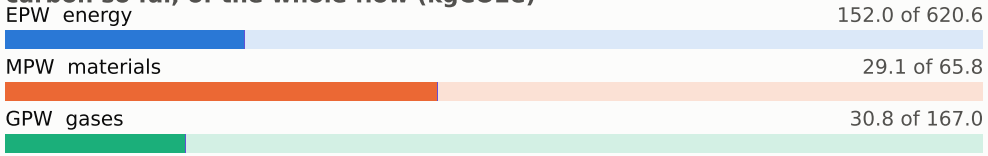
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

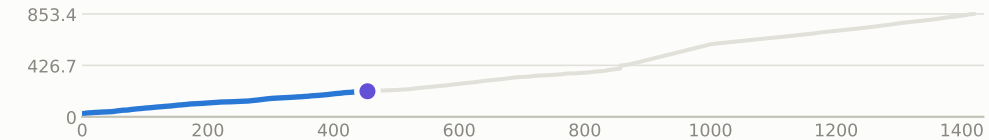
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

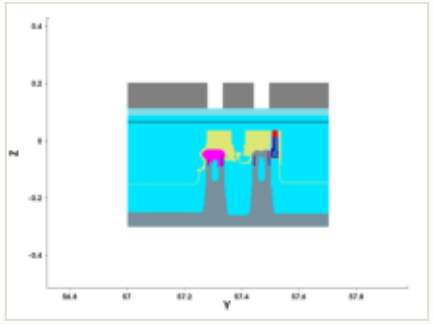
carbon so far, of the whole flow (kgCO2e)



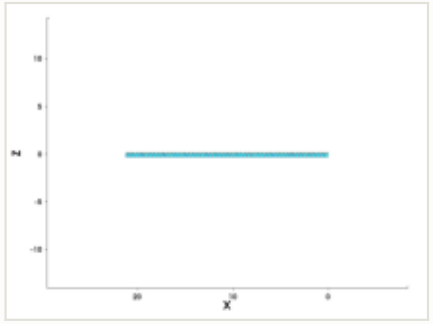
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



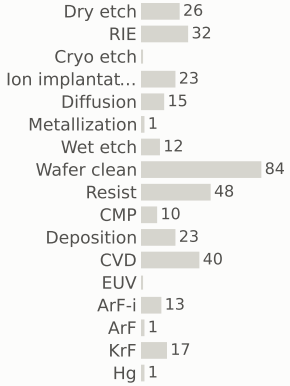
y cut at 57.351 µm



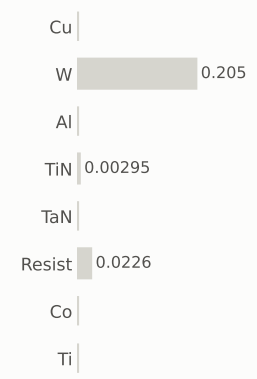
mask for this step

no mask for this step

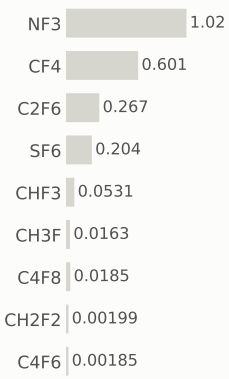
steps by process area



materials deposited (g)



gases emitted (g)



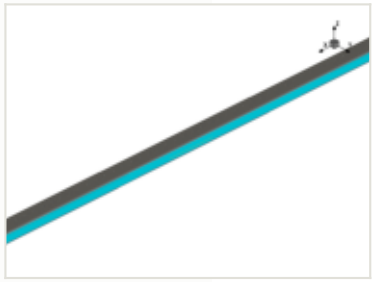
Step #4620: Spx_only

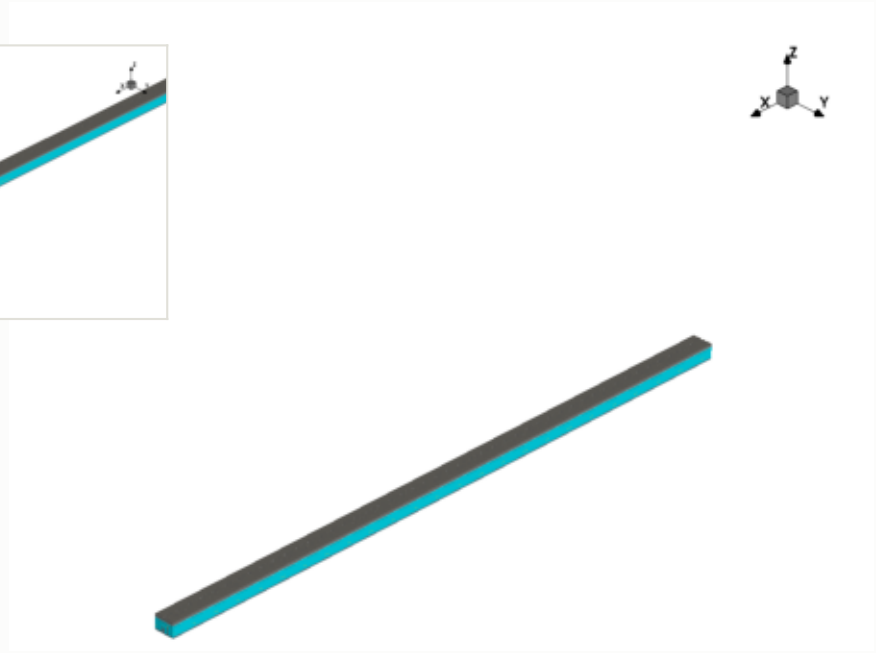
etch_anisotropic · Spx_only

0	211.9	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





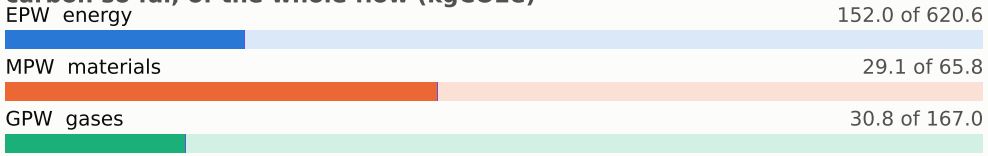
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

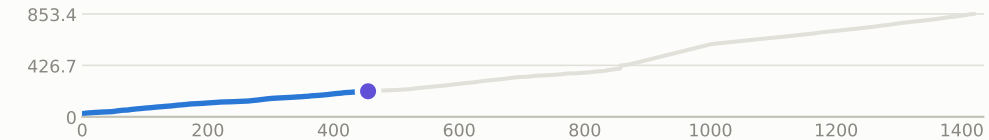
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

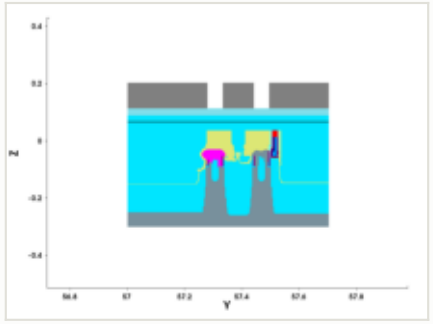
carbon so far, of the whole flow (kgCO2e)



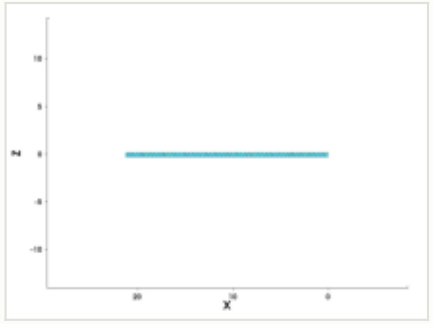
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



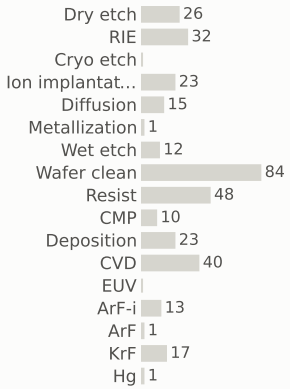
y cut at 57.351 µm



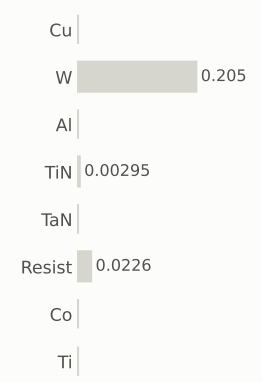
mask for this step

no mask for this step

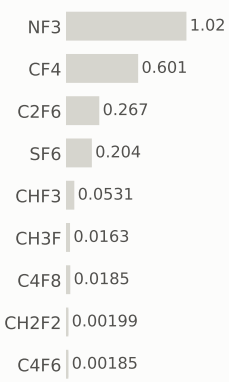
steps by process area



materials deposited (g)



gases emitted (g)



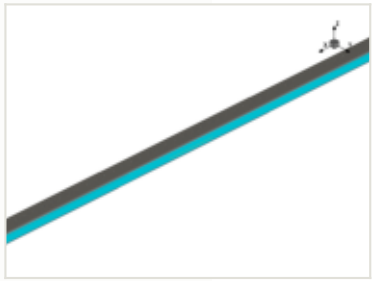
Step #4630: RIE

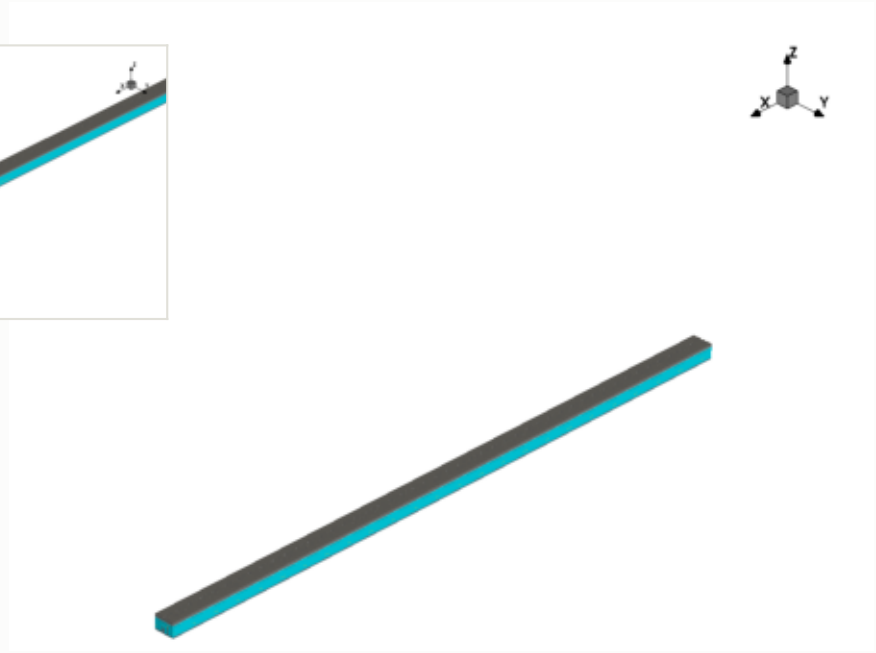
etch_strip · RIE

1.04	212.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



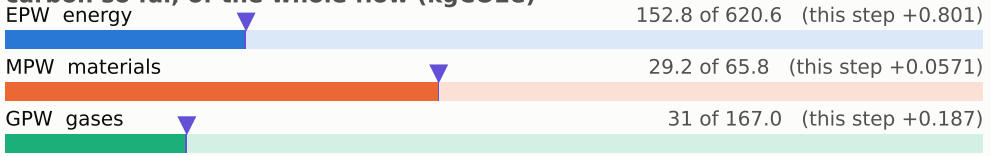


this step

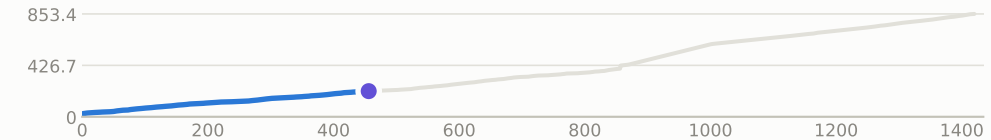
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

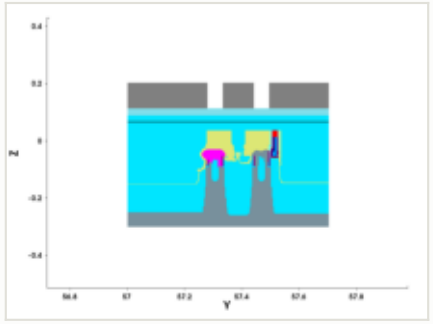
carbon so far, of the whole flow (kgCO2e)



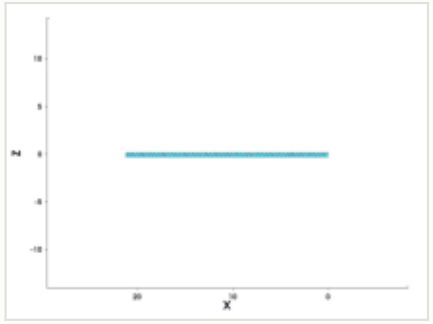
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



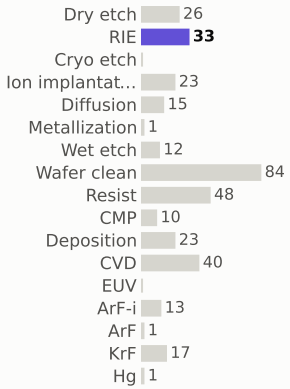
y cut at 57.351 µm



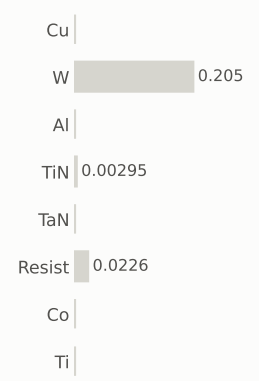
mask for this step

no mask for this step

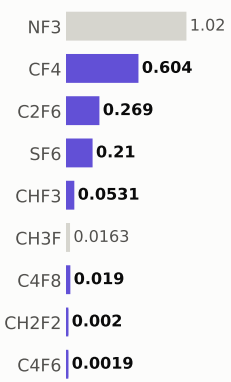
steps by process area



materials deposited (g)



gases emitted (g)



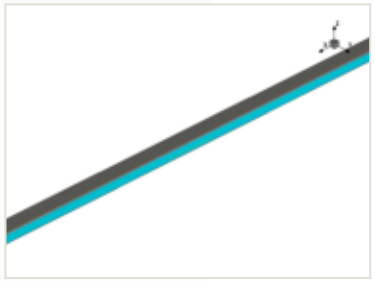
Step #4640: Spx_only

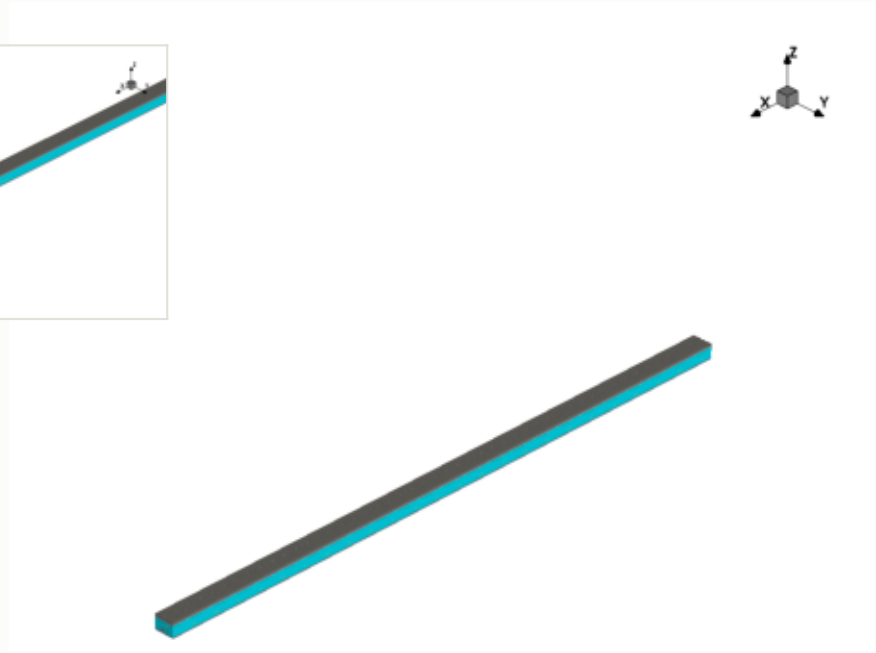
etch_anisotropic · Spx_only

0	212.9	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





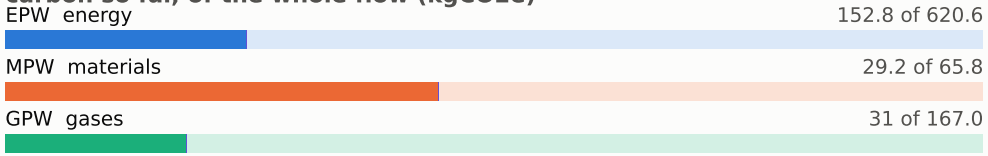
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

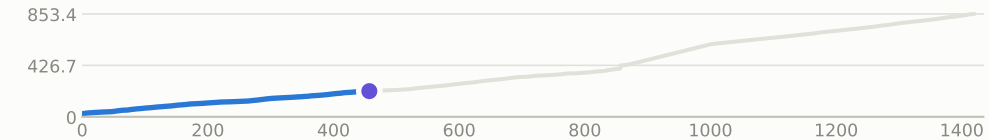
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

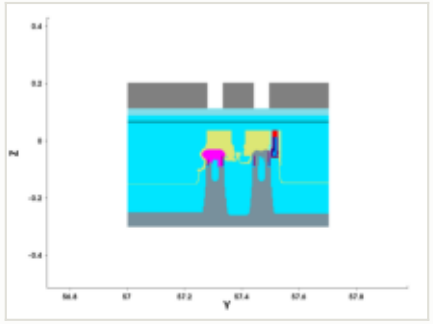
carbon so far, of the whole flow (kgCO2e)



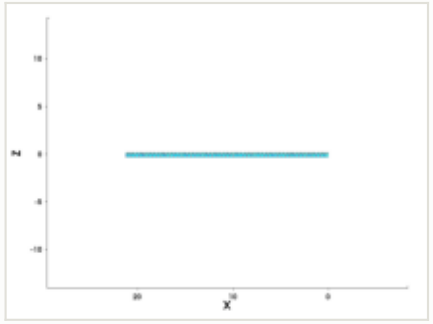
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



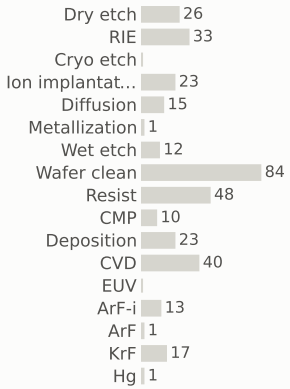
y cut at 57.351 µm



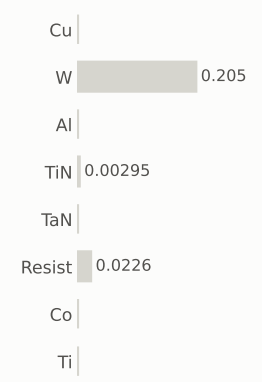
mask for this step

no mask for this step

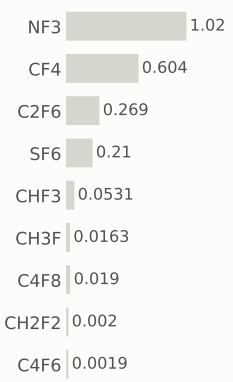
steps by process area



materials deposited (g)



gases emitted (g)



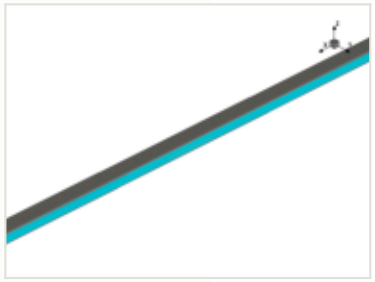
Step #4650: Wafer_clean

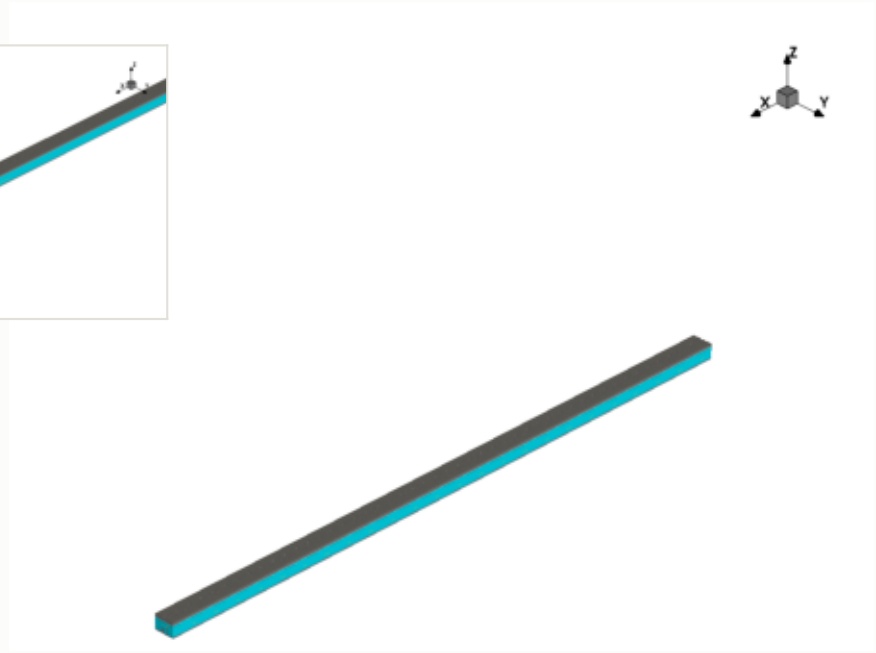
clean · Wafer_clean

0.169	213.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

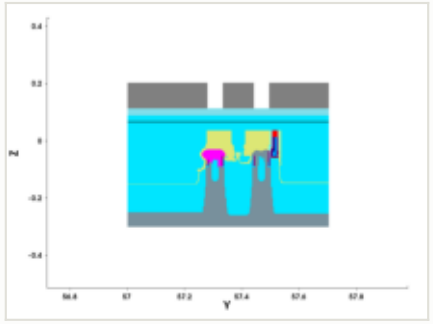
3D structure after this step

zoomed to fit

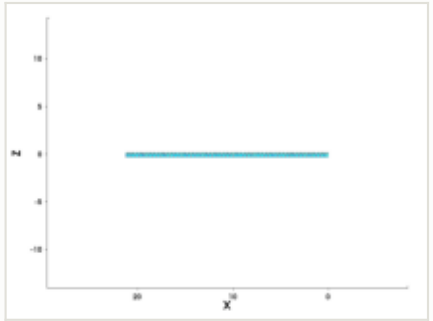




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

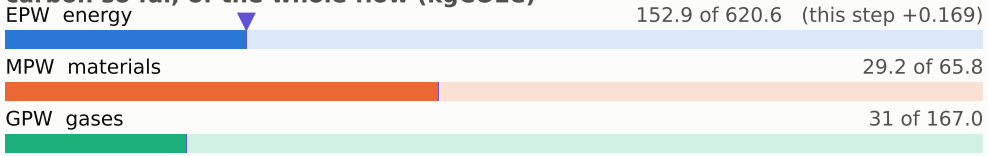
no mask for this step

this step

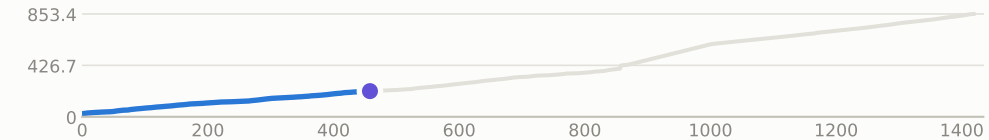
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

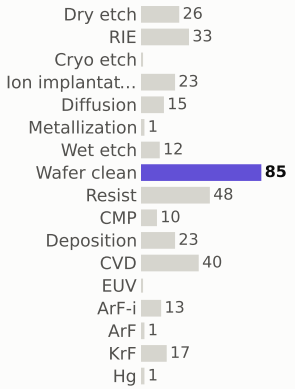
carbon so far, of the whole flow (kgCO2e)



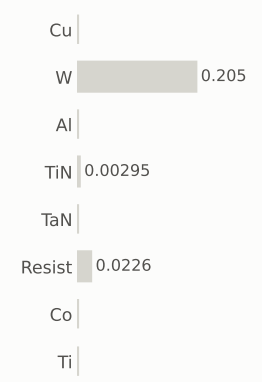
embodied carbon up to each step (kgCO2e per wafer)



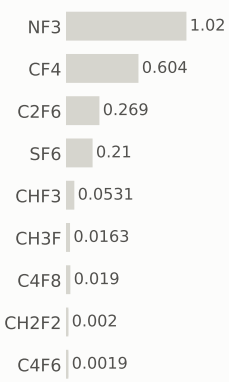
steps by process area



materials deposited (g)



gases emitted (g)



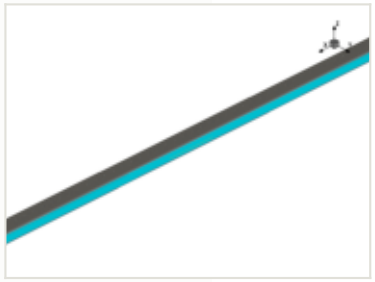
Step #4660: RIE

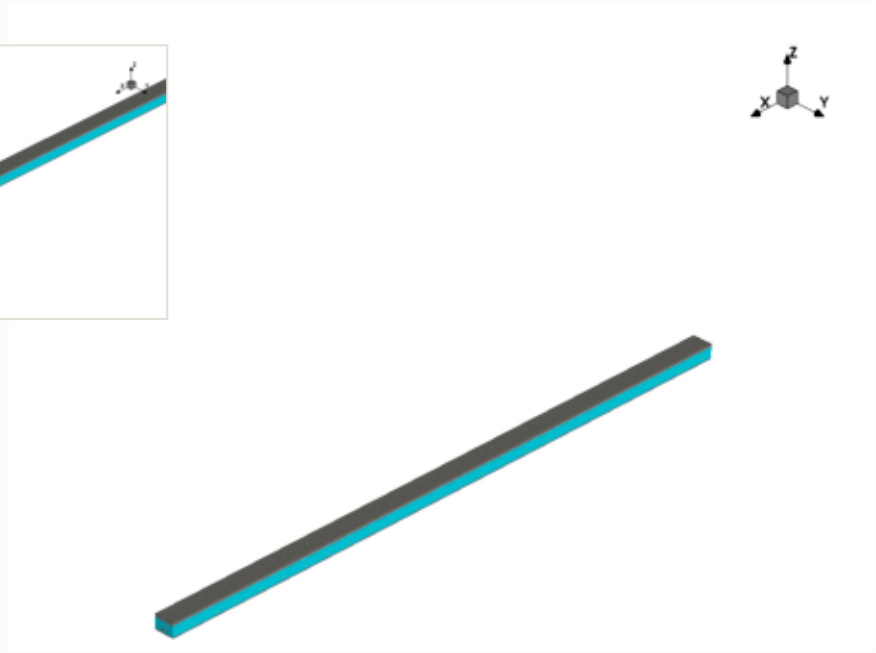
etch_anisotropic · RIE

1.04	214.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



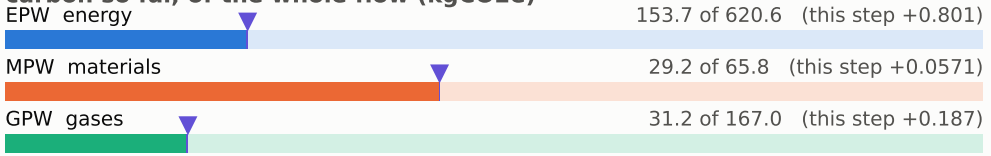


this step

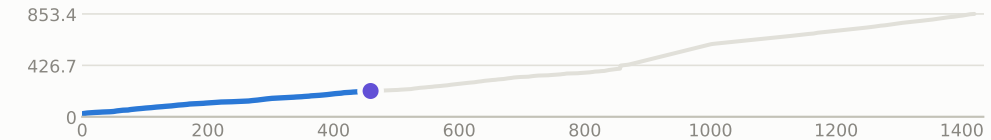
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

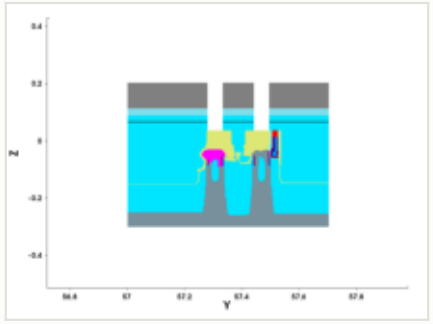
carbon so far, of the whole flow (kgCO2e)



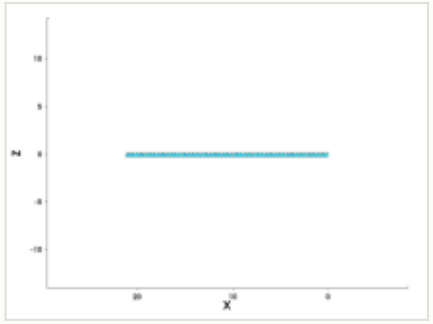
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



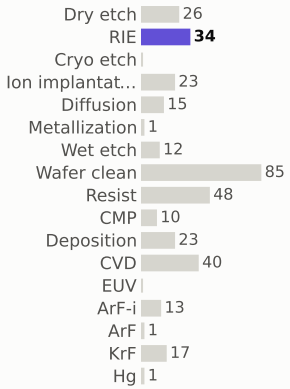
y cut at 57.351 µm



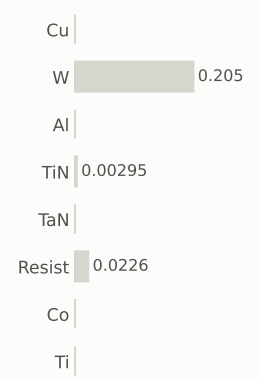
mask for this step

no mask for this step

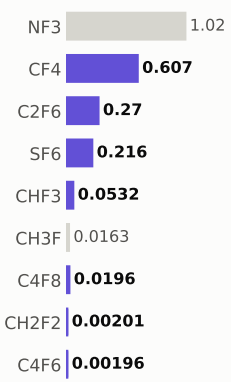
steps by process area



materials deposited (g)



gases emitted (g)



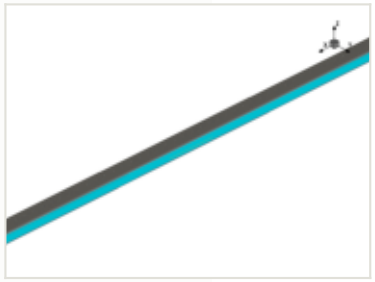
Step #4670: Spx_only

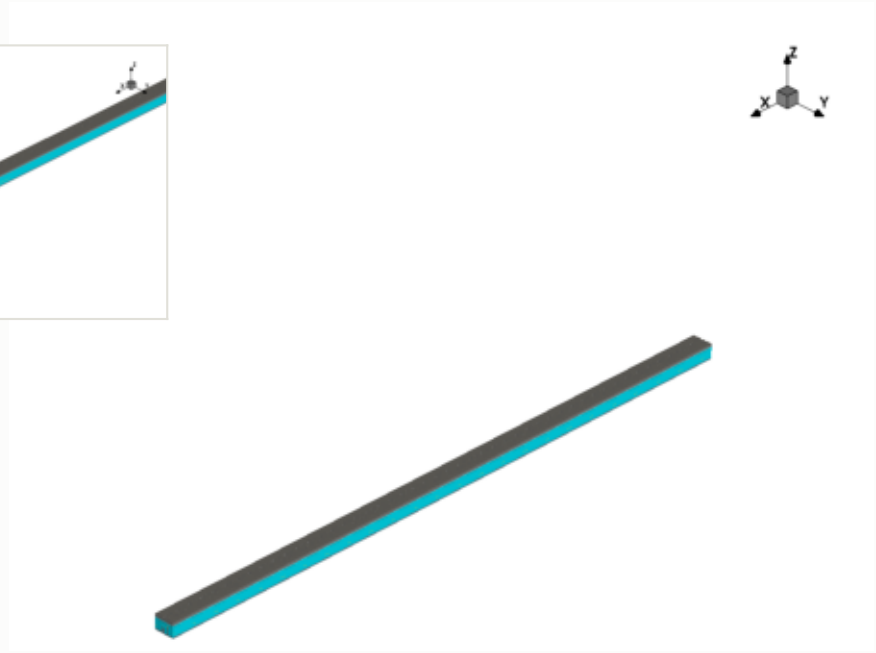
etch_anisotropic · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





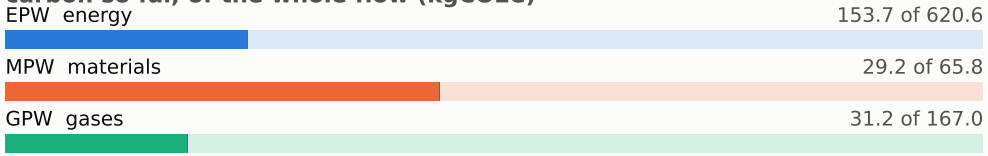
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

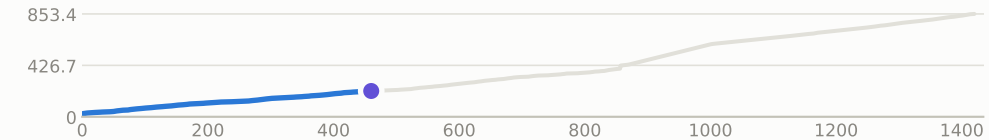
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

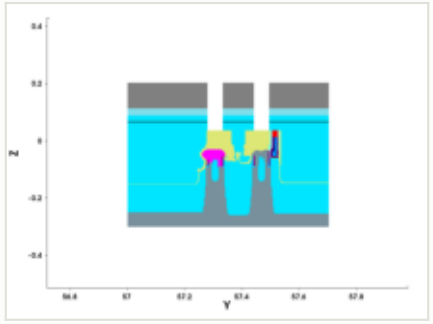
carbon so far, of the whole flow (kgCO2e)



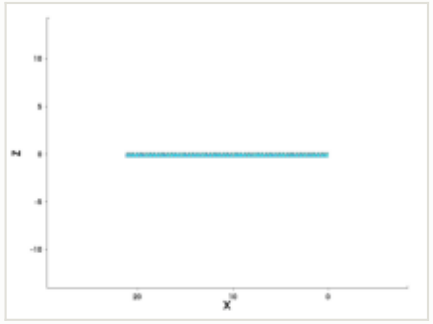
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



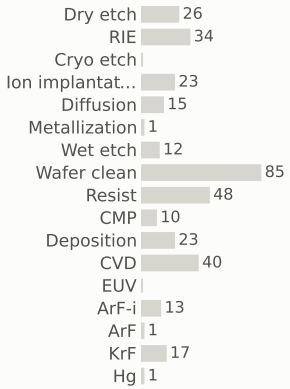
y cut at 57.351 µm



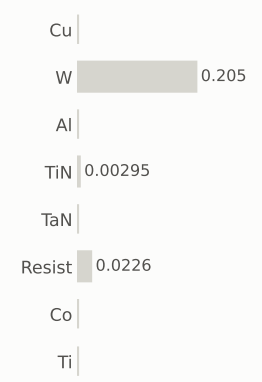
mask for this step

no mask for this step

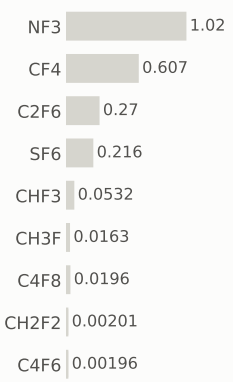
steps by process area



materials deposited (g)



gases emitted (g)



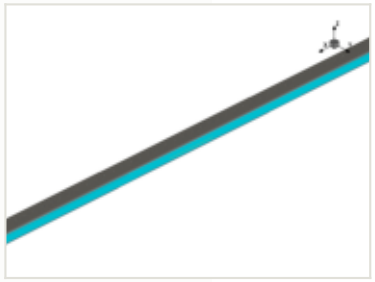
Step #4680: Spx_only

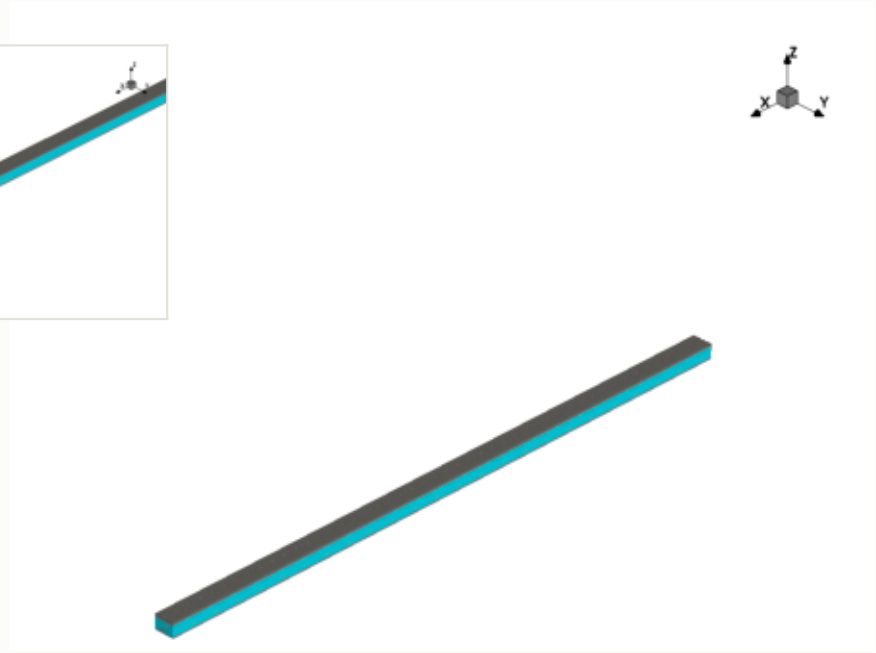
etch_anisotropic · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



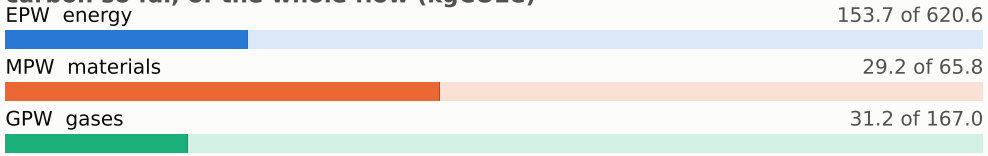


this step

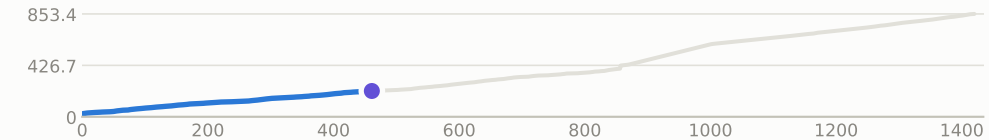
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

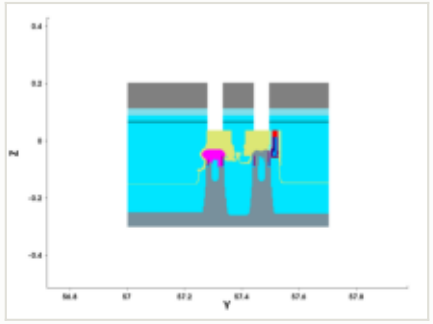
carbon so far, of the whole flow (kgCO2e)



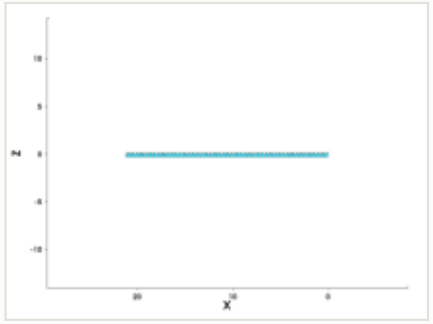
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



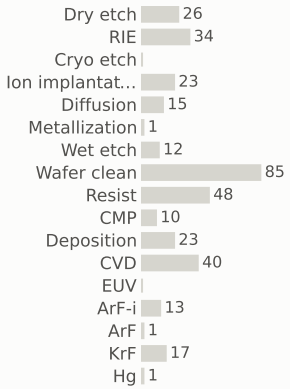
y cut at 57.351 µm



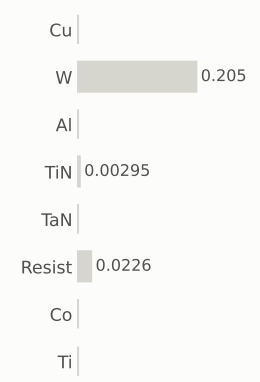
mask for this step

no mask for this step

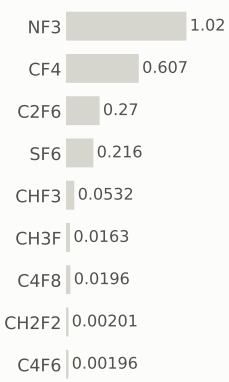
steps by process area



materials deposited (g)



gases emitted (g)



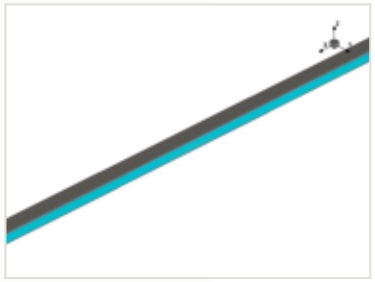
Step #4690: Spx_only

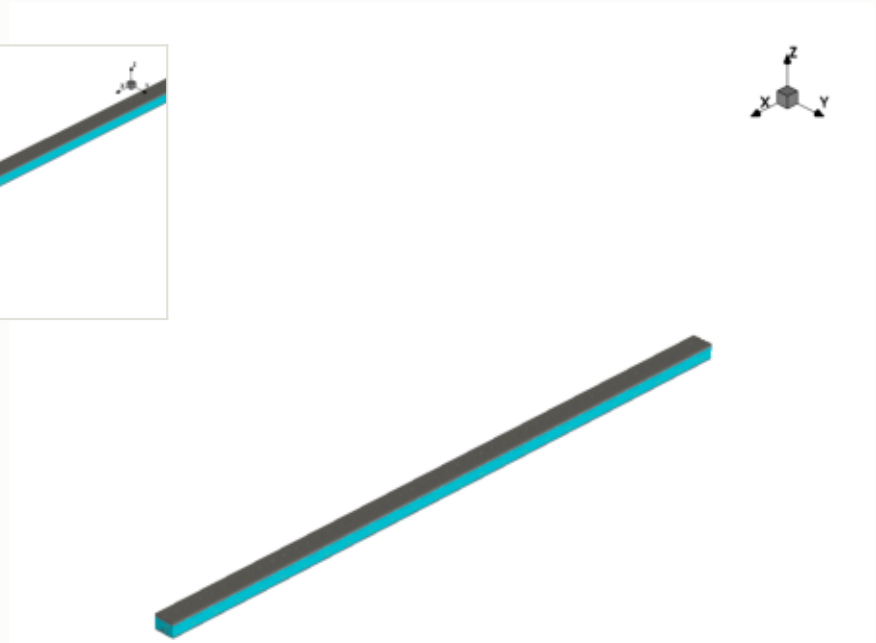
etch_anisotropic · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





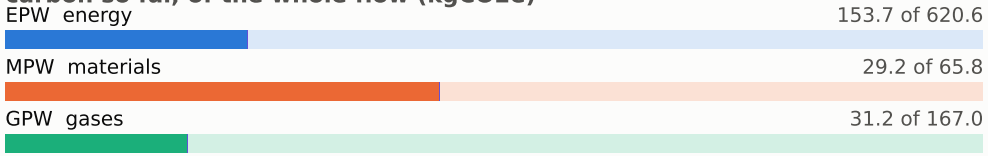
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

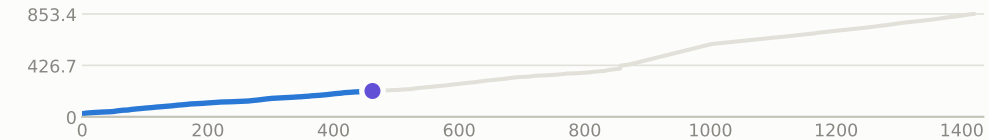
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

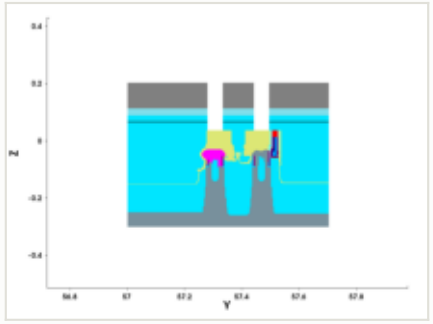
carbon so far, of the whole flow (kgCO2e)



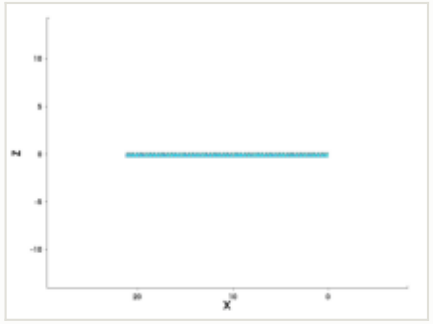
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



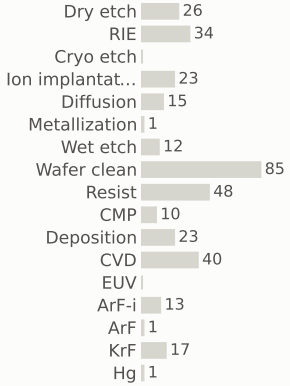
y cut at 57.351 µm



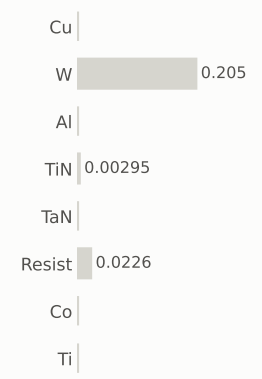
mask for this step

no mask for this step

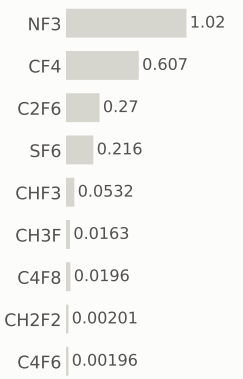
steps by process area



materials deposited (g)



gases emitted (g)



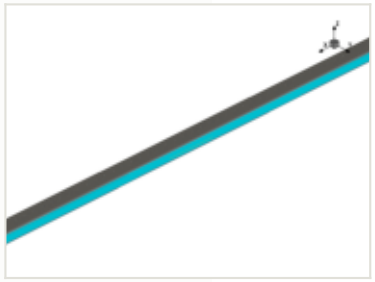
Step #4700: Spx_only

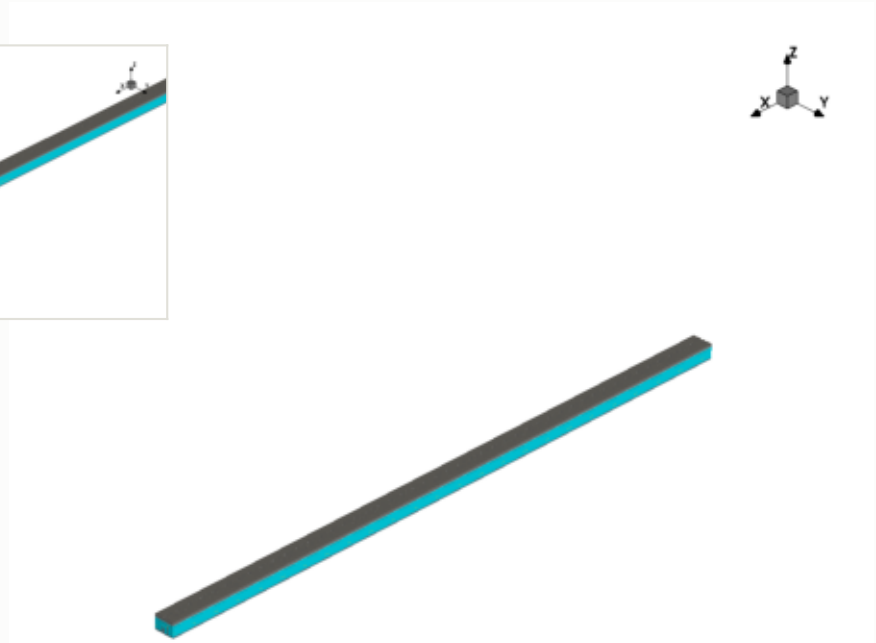
etch_anisotropic · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





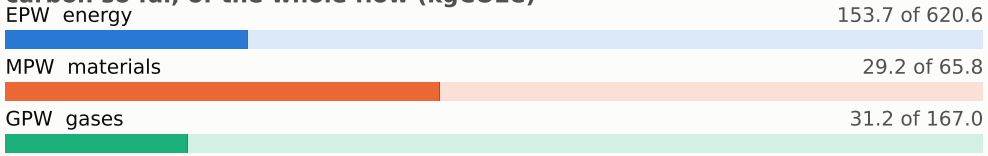
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

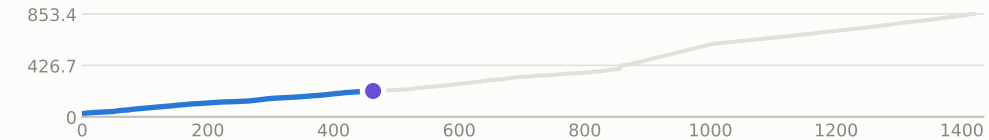
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

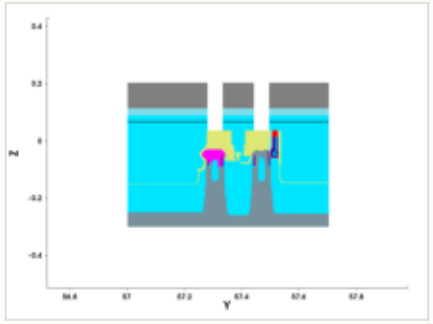
carbon so far, of the whole flow (kgCO2e)



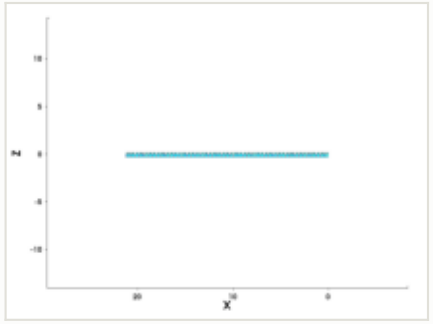
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



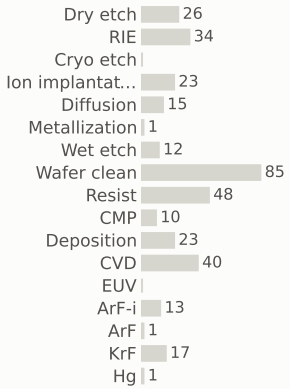
y cut at 57.351 µm



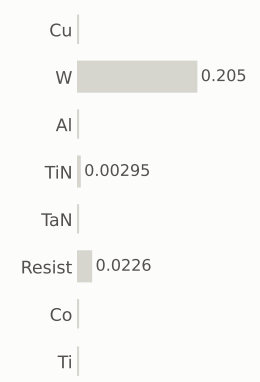
mask for this step

no mask for this step

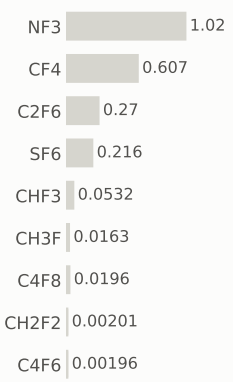
steps by process area



materials deposited (g)



gases emitted (g)



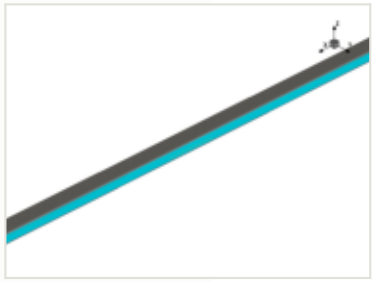
Step #4710: Spx_only

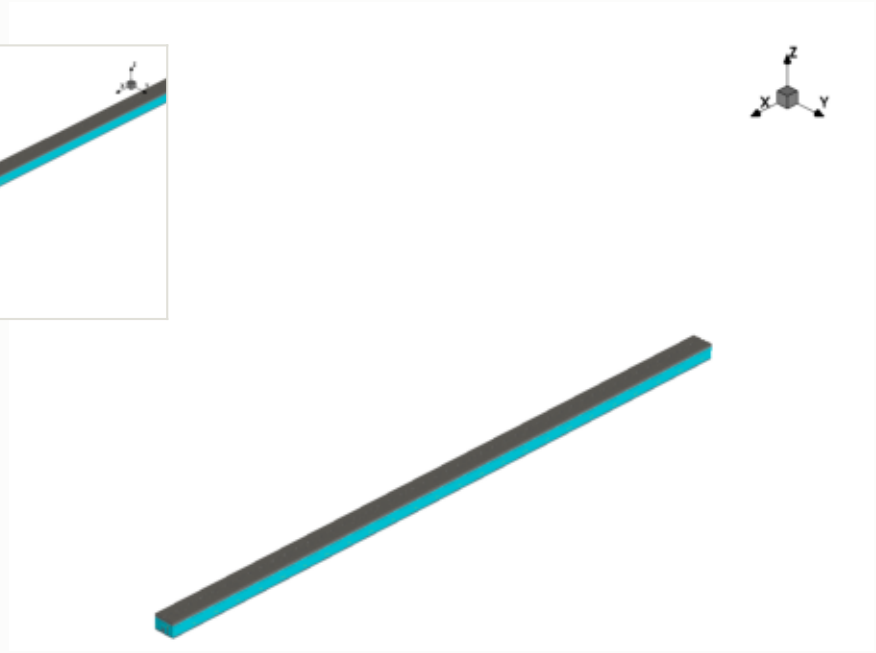
etch_anisotropic · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





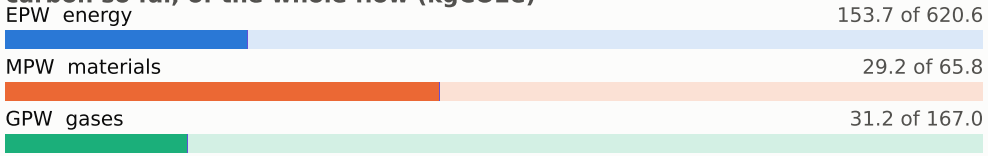
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

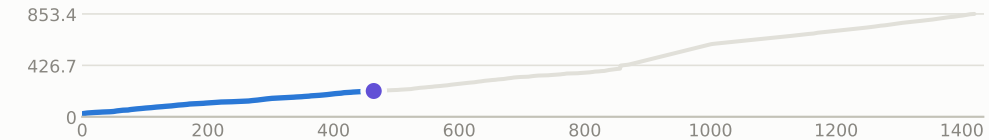
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

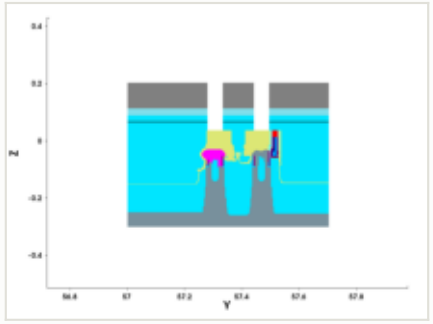
carbon so far, of the whole flow (kgCO2e)



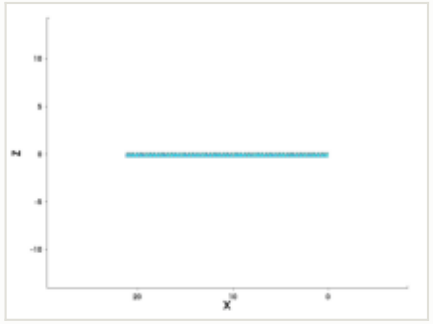
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



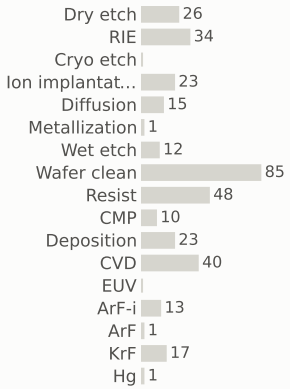
y cut at 57.351 µm



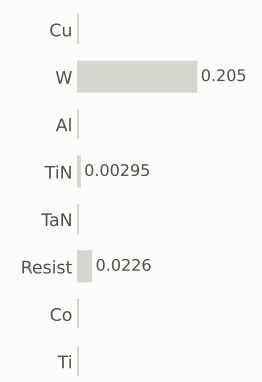
mask for this step

no mask for this step

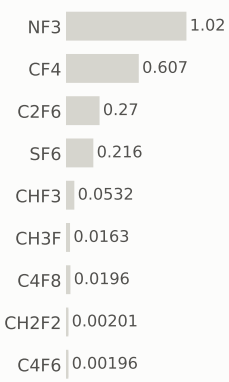
steps by process area



materials deposited (g)



gases emitted (g)



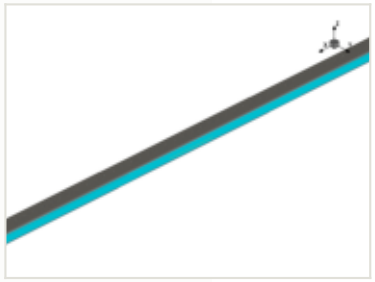
Step #4720: Spx_only

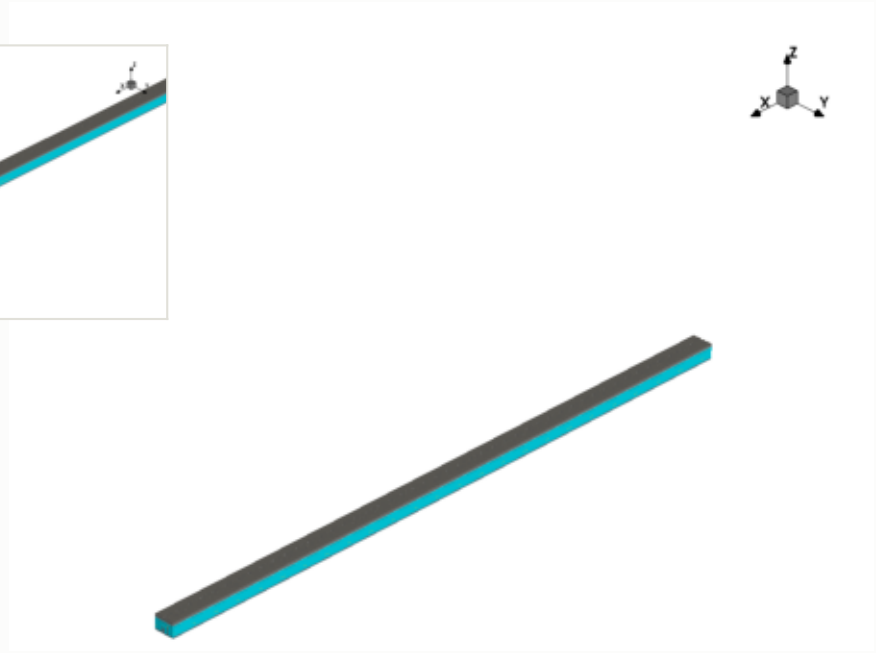
etch_anisotropic · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





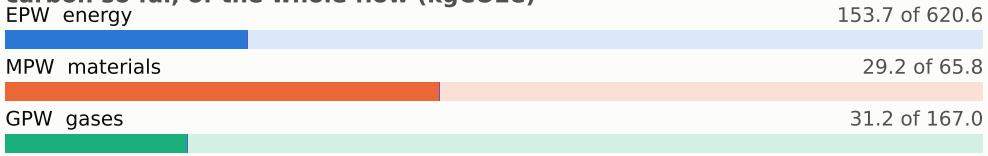
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

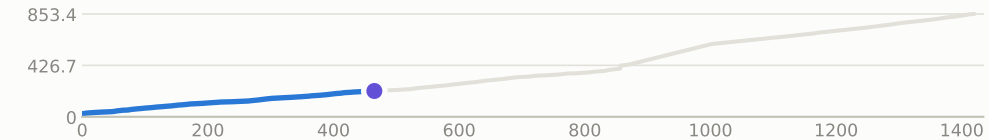
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

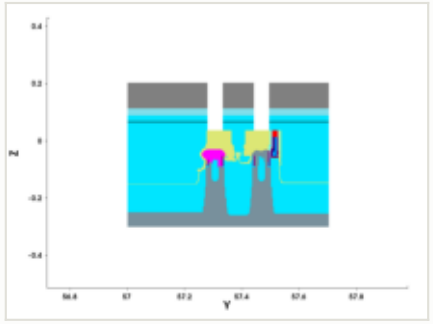
carbon so far, of the whole flow (kgCO2e)



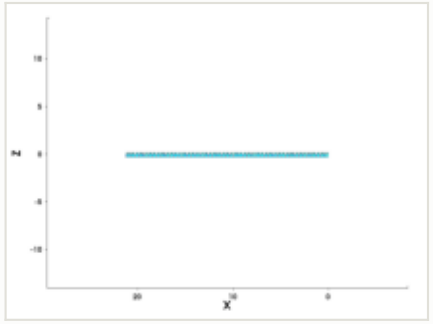
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



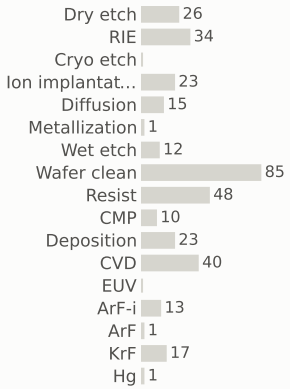
y cut at 57.351 µm



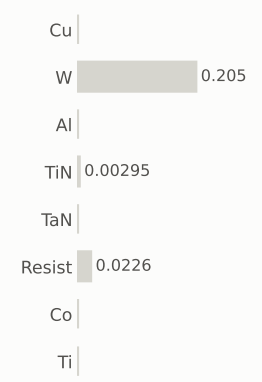
mask for this step

no mask for this step

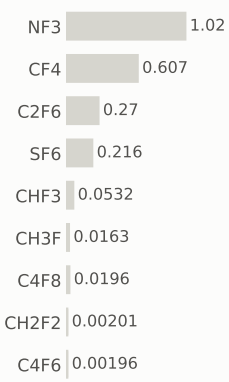
steps by process area



materials deposited (g)



gases emitted (g)



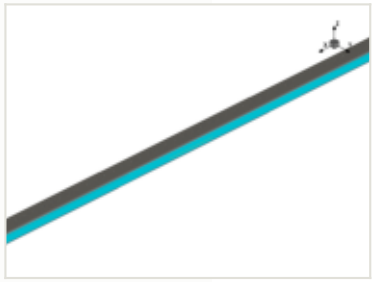
Step #4730: Spx_only

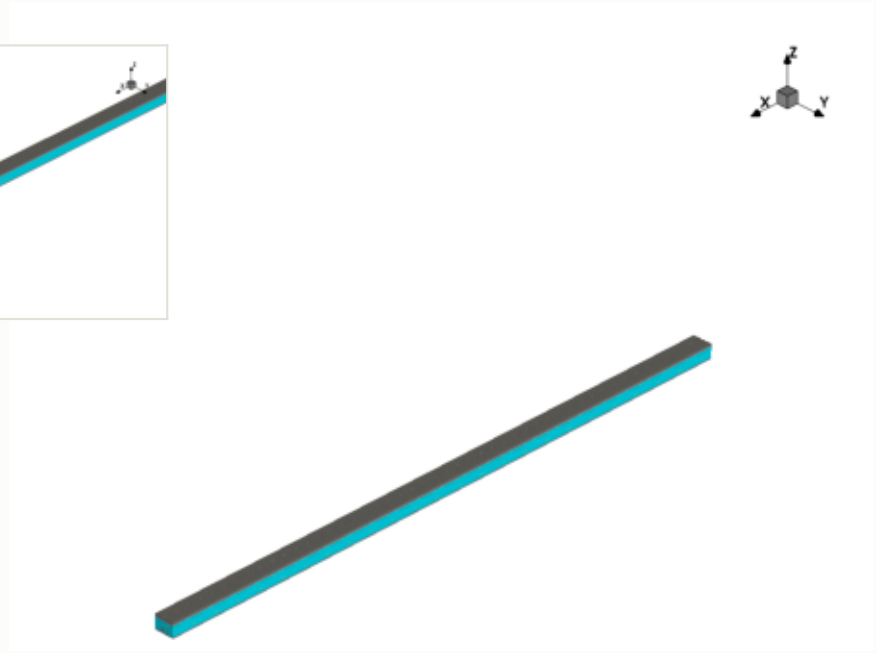
etch_anisotropic · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





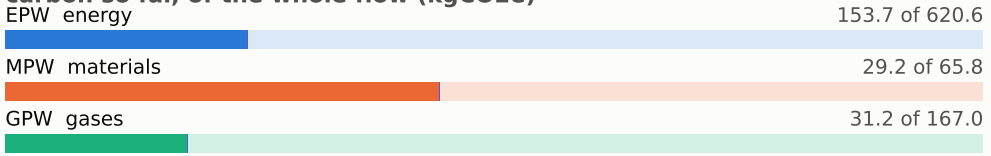
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

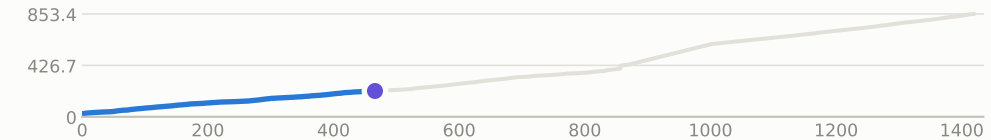
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

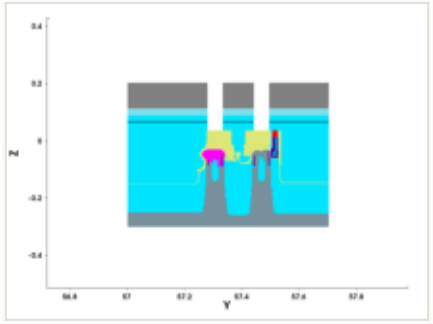
carbon so far, of the whole flow (kgCO2e)



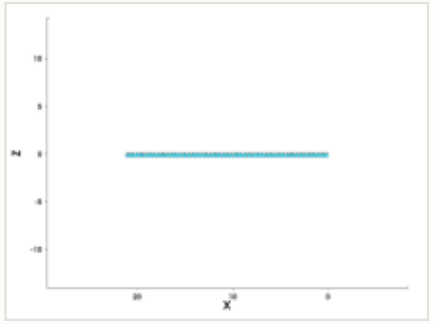
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



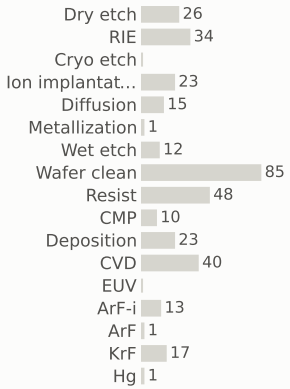
y cut at 57.351 µm



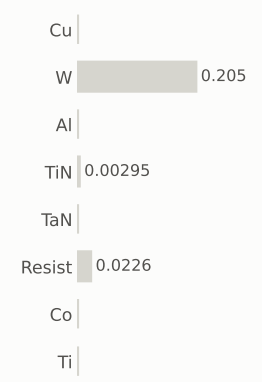
mask for this step

no mask for this step

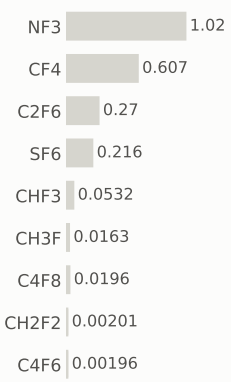
steps by process area



materials deposited (g)



gases emitted (g)



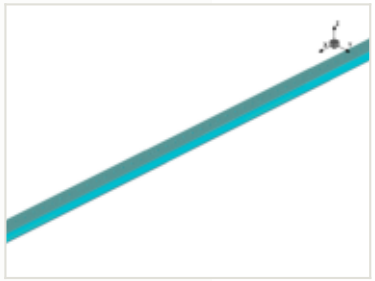
Step #4740: Spx_only

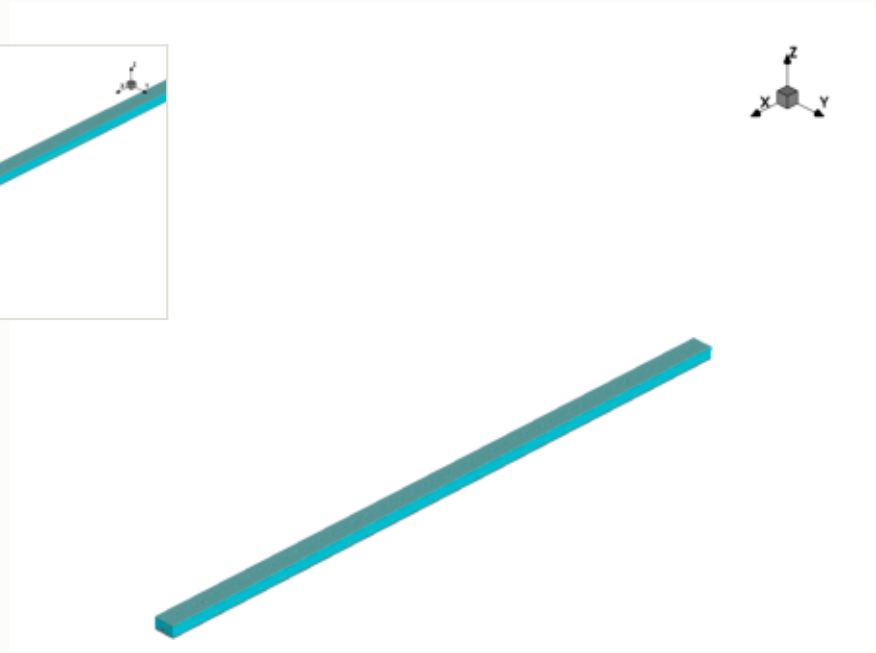
etch_strip · Spx_only

0	214.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





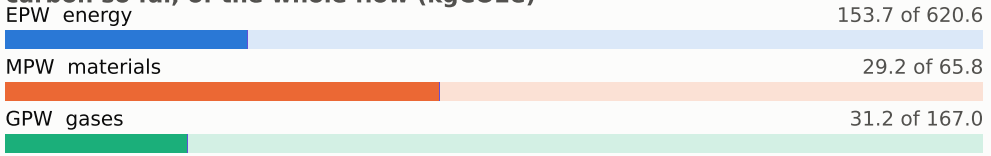
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

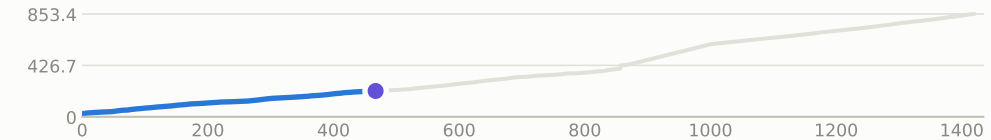
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

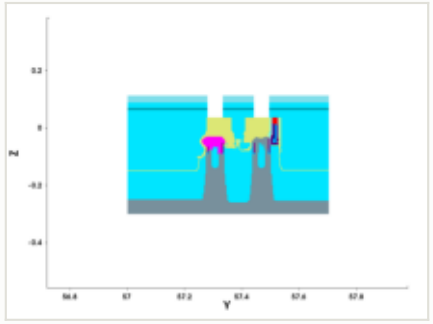
carbon so far, of the whole flow (kgCO2e)



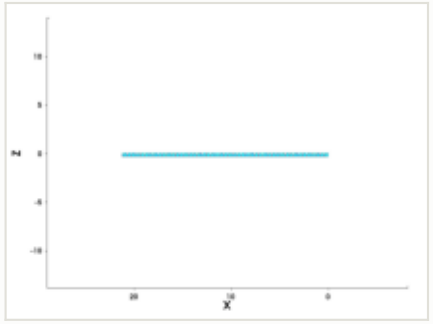
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



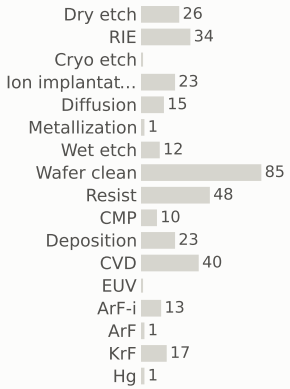
y cut at 57.351 µm



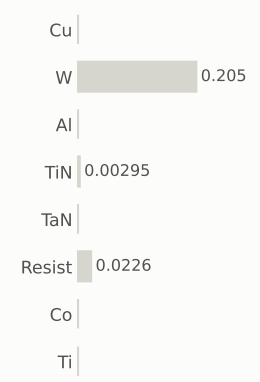
mask for this step

no mask for this step

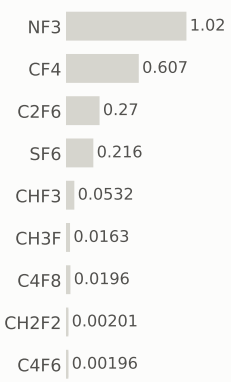
steps by process area



materials deposited (g)



gases emitted (g)



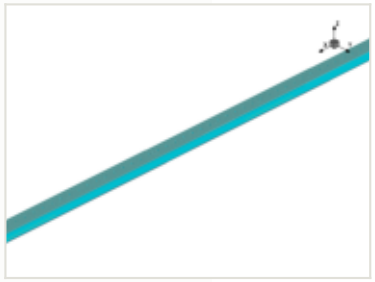
Step #4750: Wafer_clean

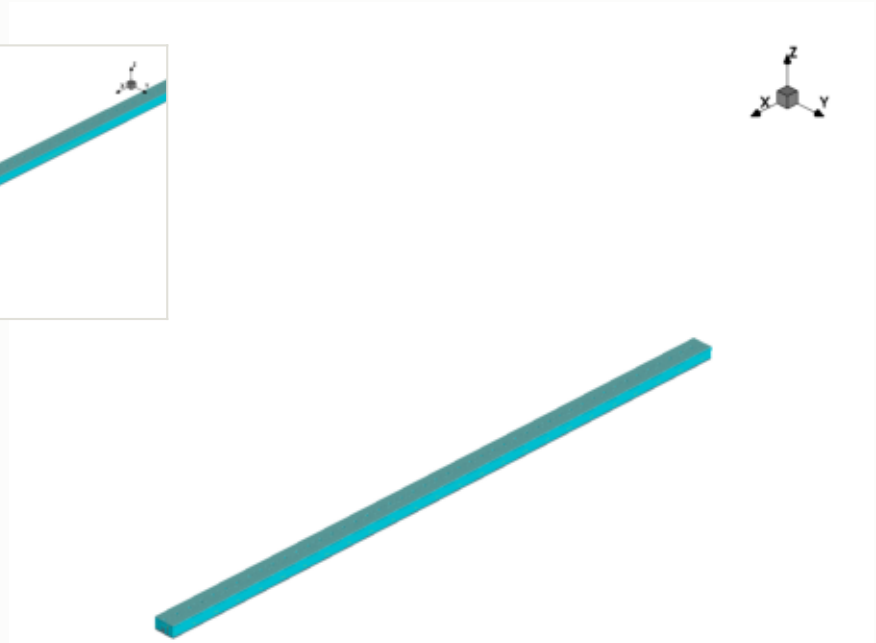
clean · Wafer_clean

0.169	214.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





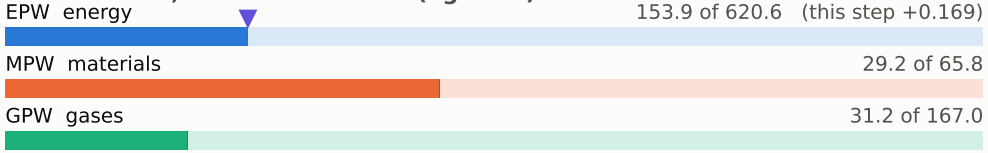
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

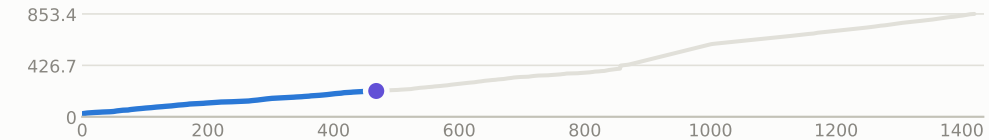
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

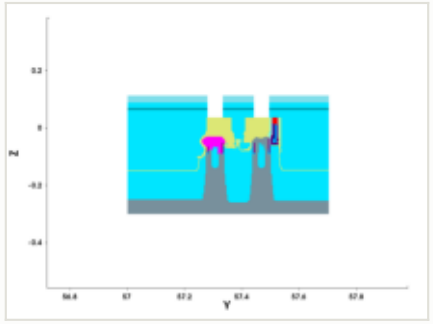
carbon so far, of the whole flow (kgCO2e)



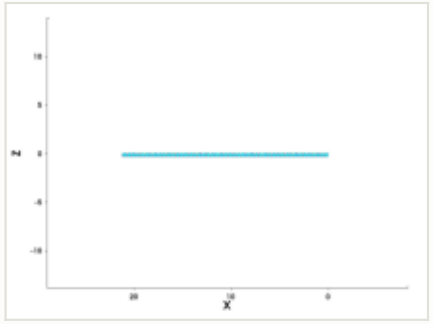
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



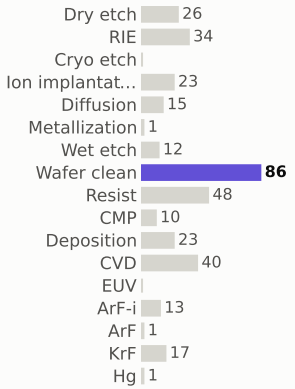
y cut at 57.351 µm



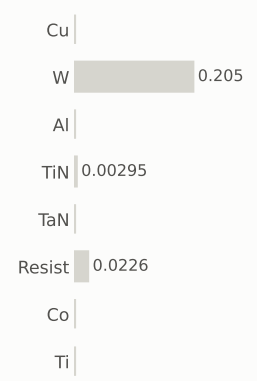
mask for this step

no mask for this step

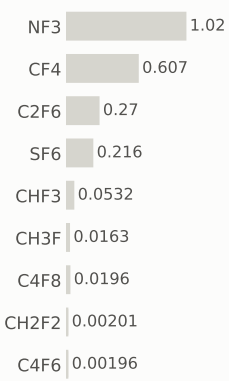
steps by process area



materials deposited (g)



gases emitted (g)



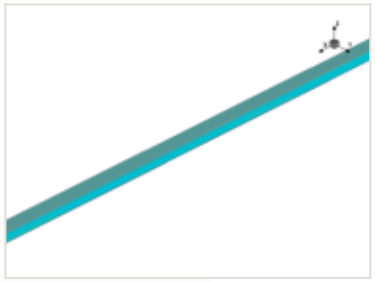
Step #4760: RIE

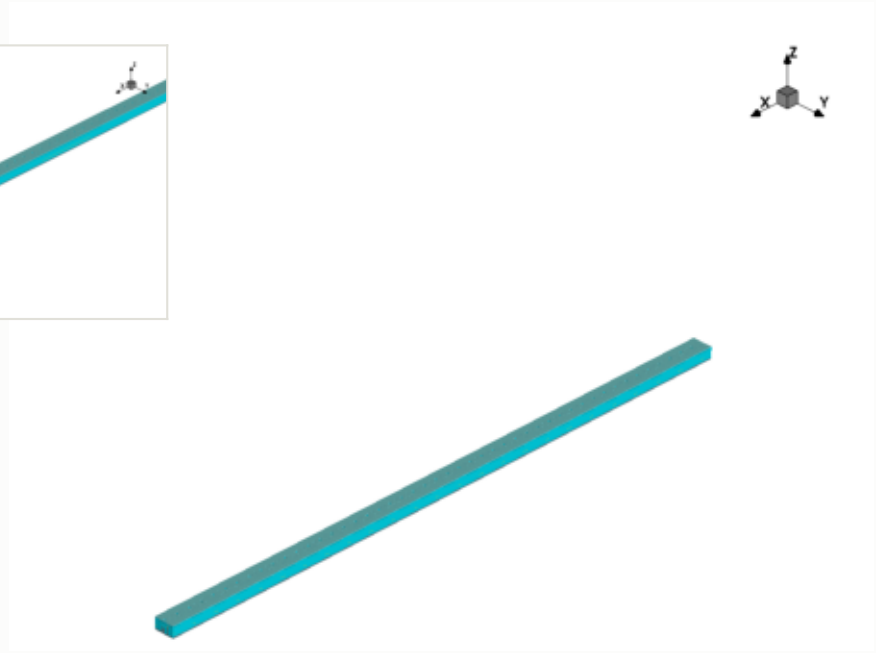
etch_anisotropic · RIE

1.04	215.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



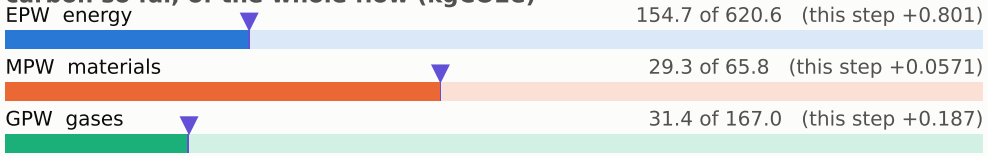


this step

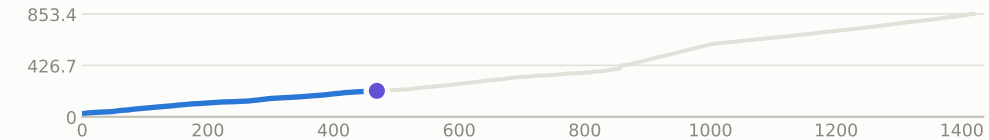
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

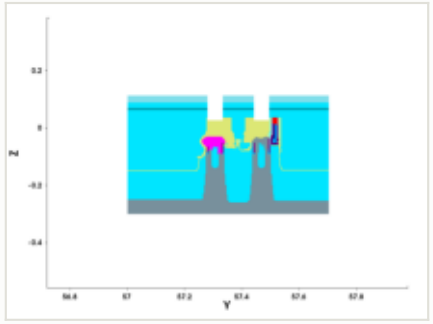
carbon so far, of the whole flow (kgCO2e)



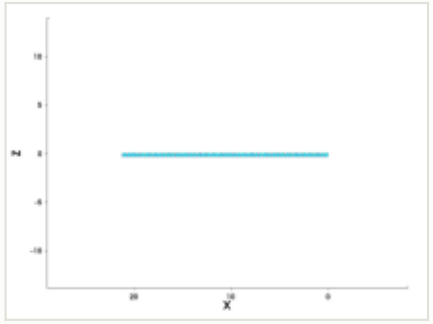
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



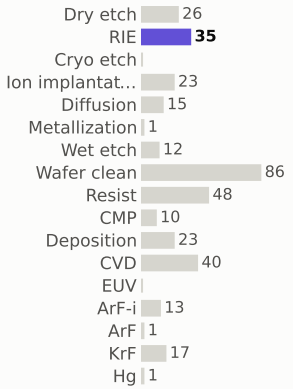
y cut at 57.351 µm



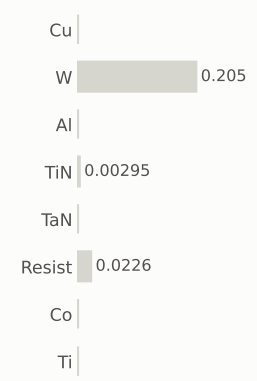
mask for this step

no mask for this step

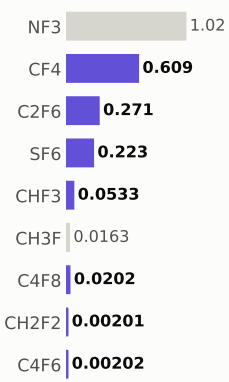
steps by process area



materials deposited (g)



gases emitted (g)



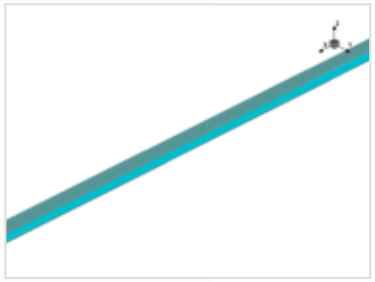
Step #4770: Wafer_clean

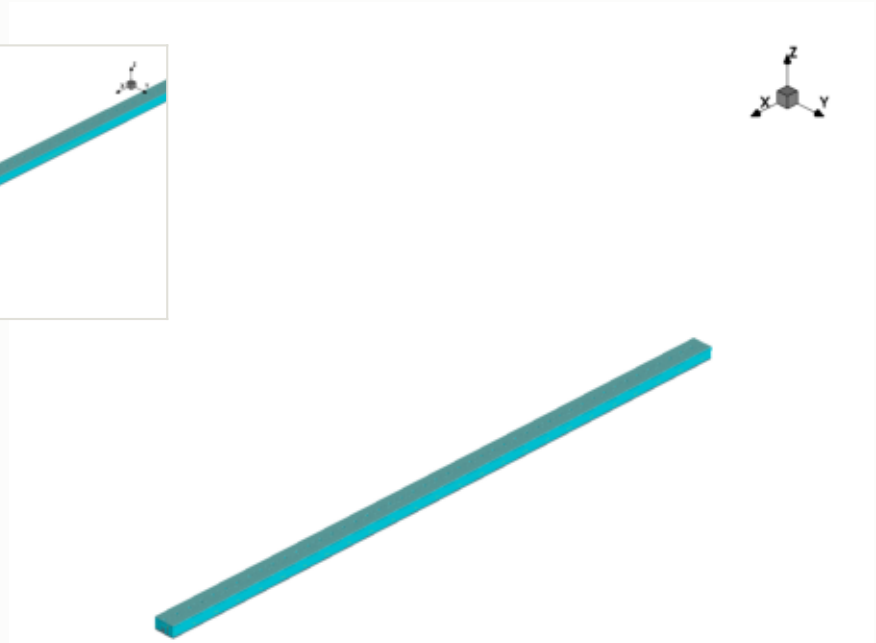
clean · Wafer_clean

0.169	215.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





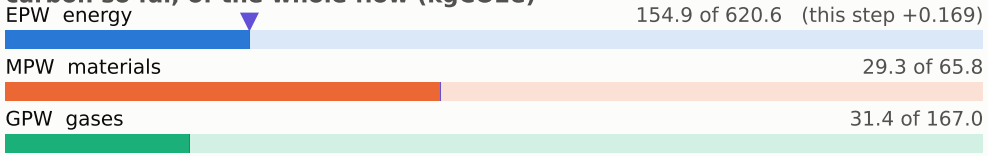
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

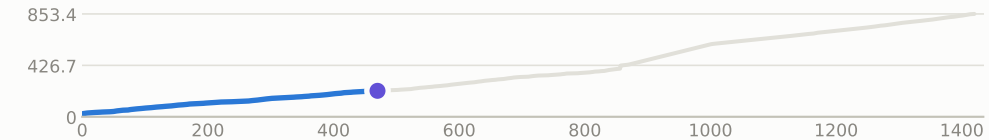
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

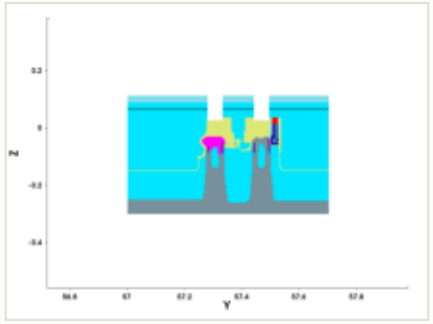
carbon so far, of the whole flow (kgCO2e)



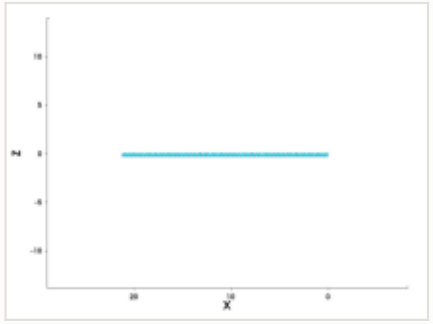
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



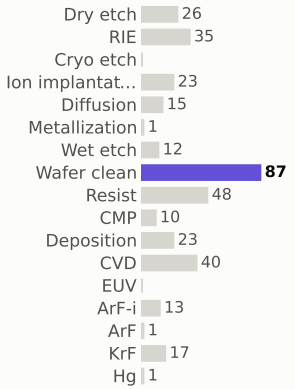
y cut at 57.351 µm



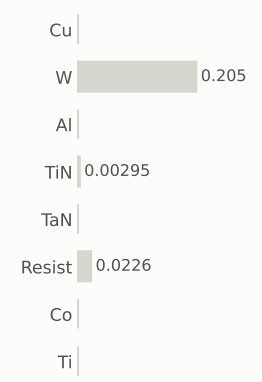
mask for this step

no mask for this step

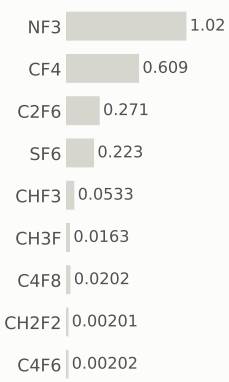
steps by process area



materials deposited (g)



gases emitted (g)



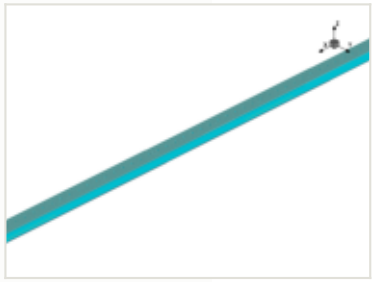
Step #4780: Ion_implantation

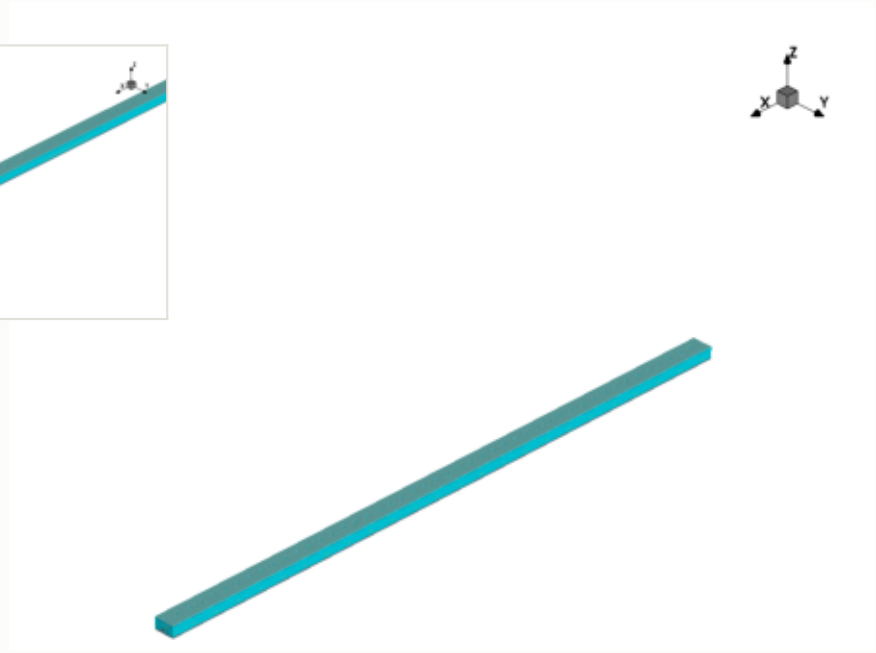
implant_ellipsoid · Ion_implantation

0.801	216.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

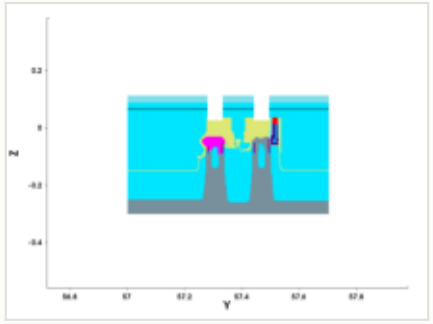
3D structure after this step

zoomed to fit

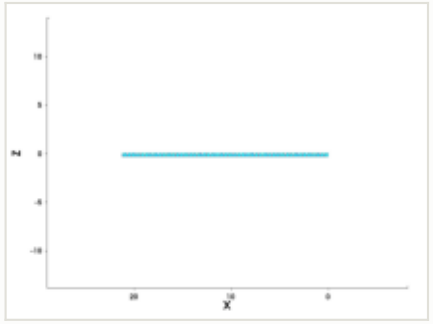




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

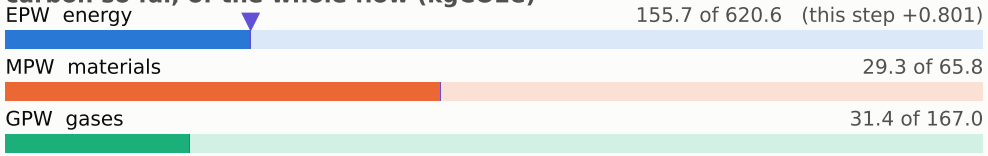
no mask for this step

this step

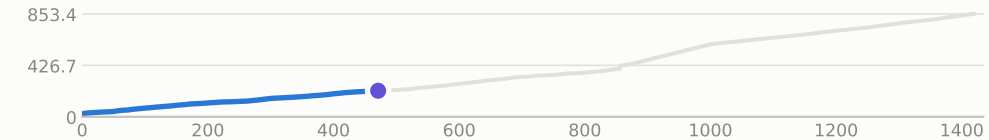
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

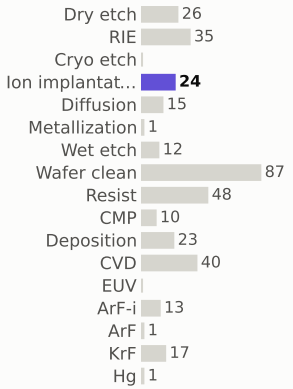
carbon so far, of the whole flow (kgCO2e)



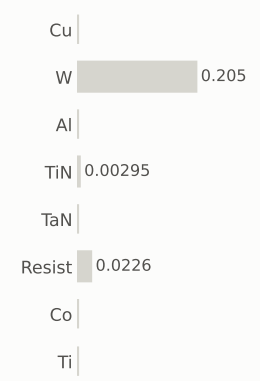
embodied carbon up to each step (kgCO2e per wafer)



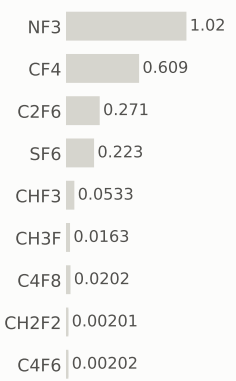
steps by process area



materials deposited (g)



gases emitted (g)



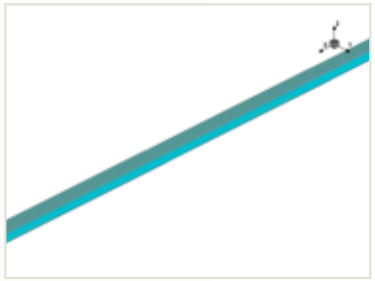
Step #4790: Wafer_clean

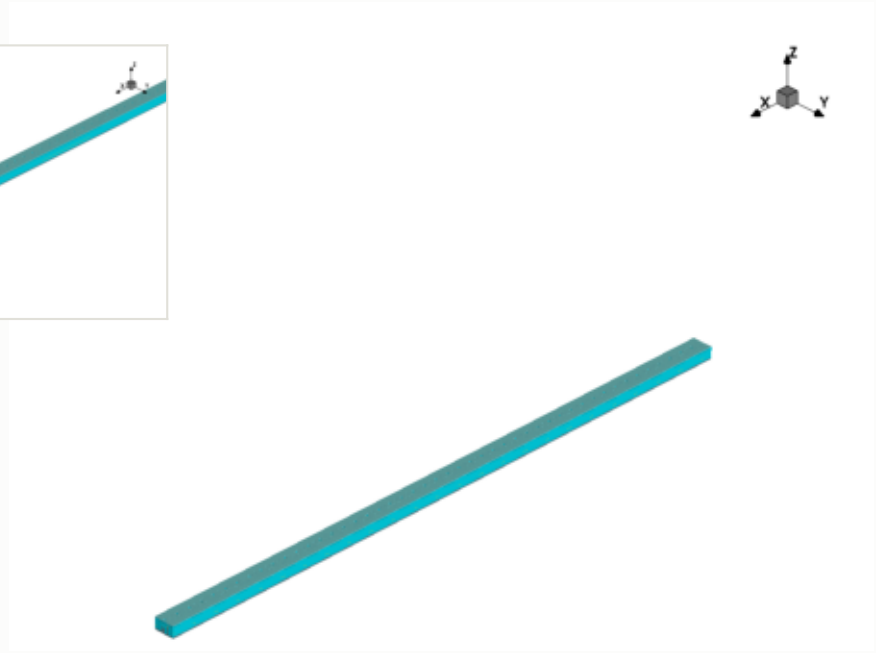
clean · Wafer_clean

0.169	216.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



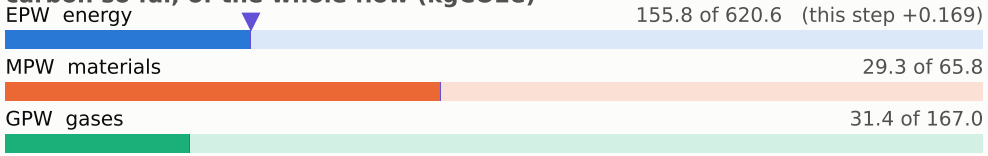


this step

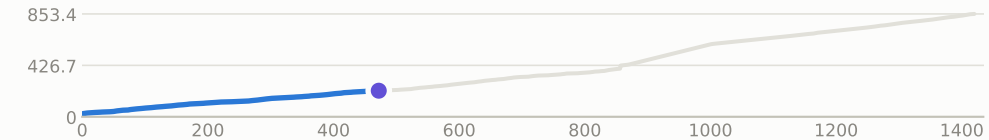
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

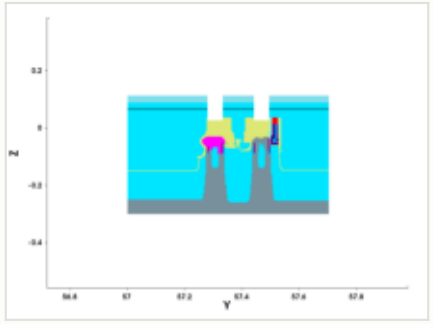
carbon so far, of the whole flow (kgCO2e)



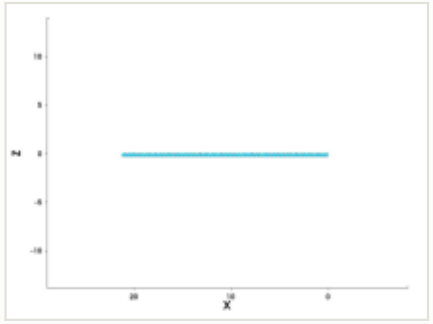
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



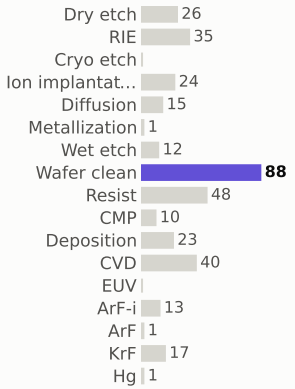
y cut at 57.351 µm



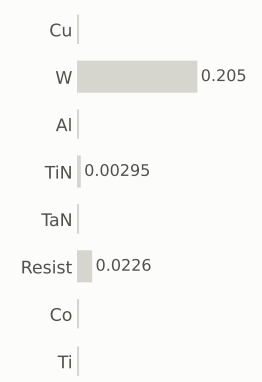
mask for this step

no mask for this step

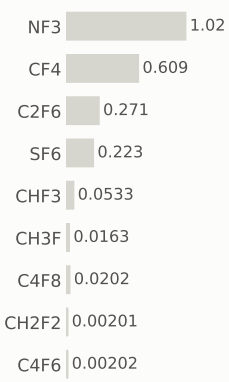
steps by process area



materials deposited (g)



gases emitted (g)



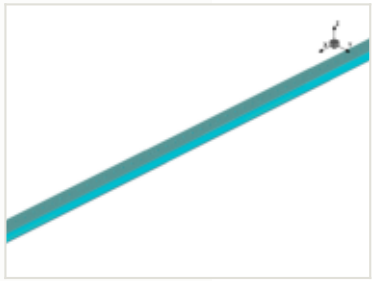
Step #4800: Resist

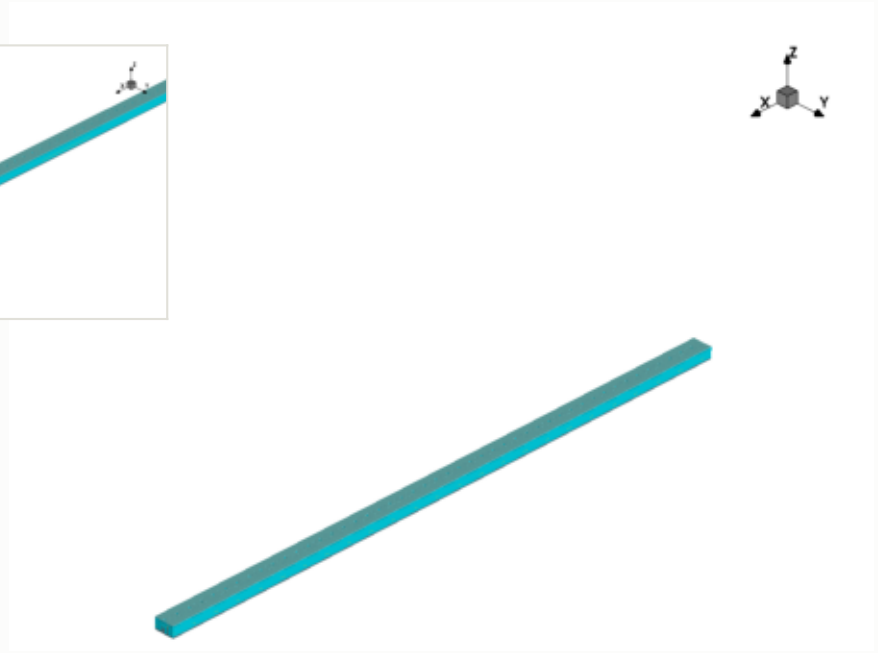
depo_fill · Resist

0.169	216.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



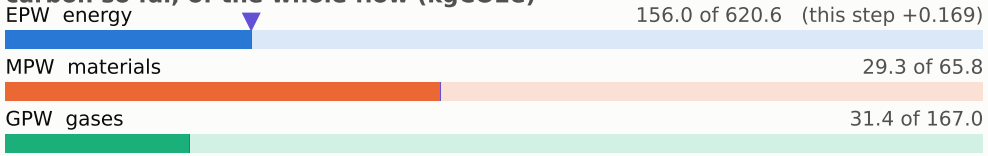


this step

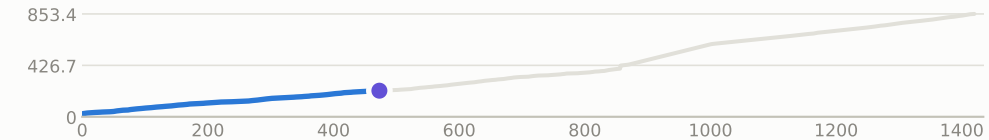
process area	Resist
material	—
thickness	0.12 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

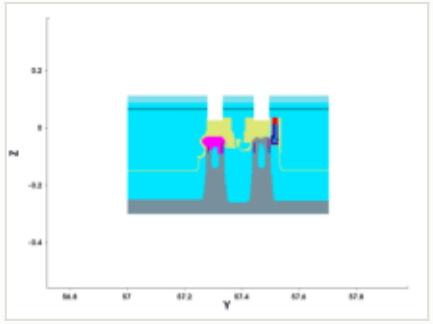
carbon so far, of the whole flow (kgCO2e)



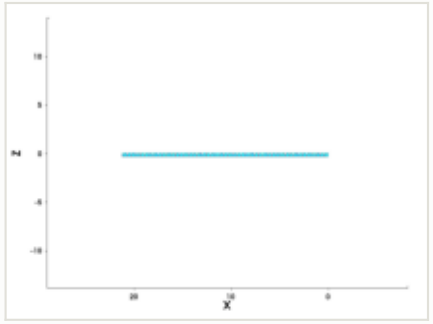
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



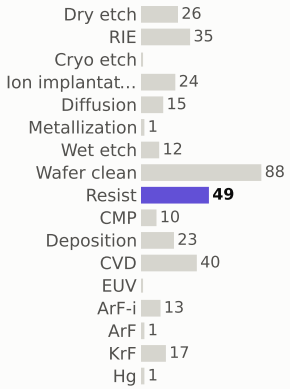
y cut at 57.351 µm



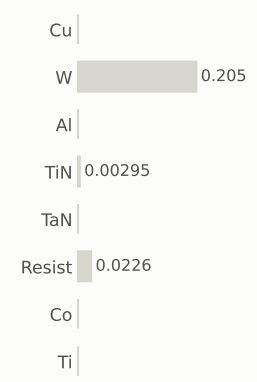
mask for this step

no mask for this step

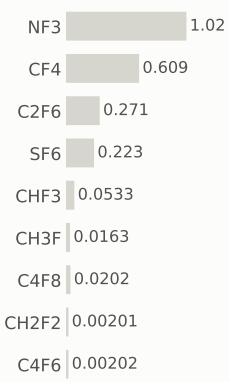
steps by process area



materials deposited (g)



gases emitted (g)



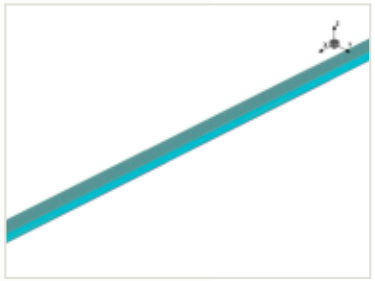
Step #4810: Spx_only

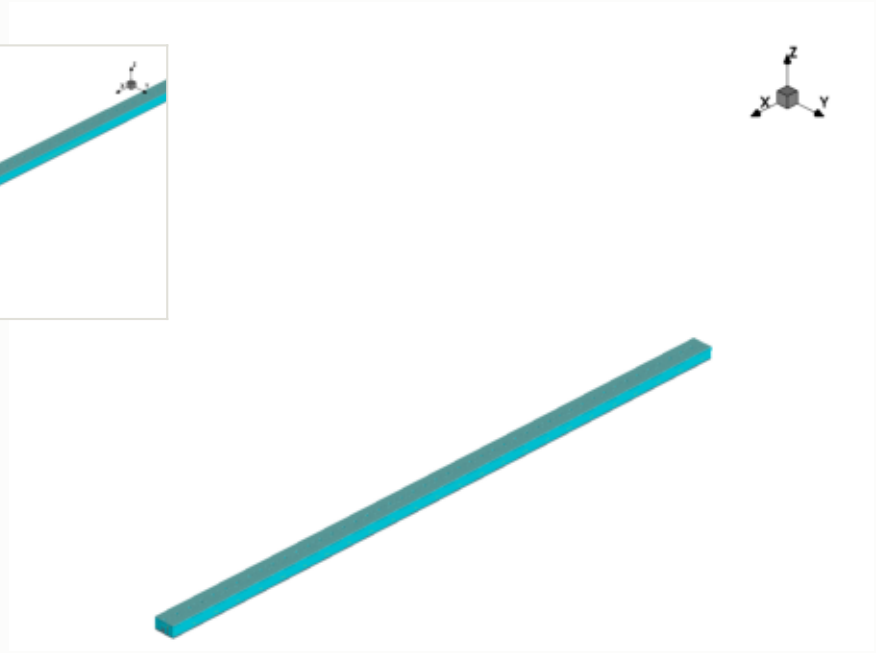
depo_fill · Spx_only

0	216.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





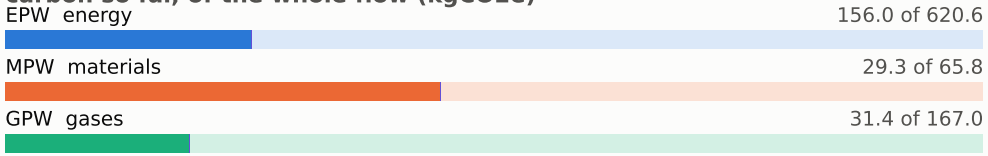
this step

process area	Spx_only
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

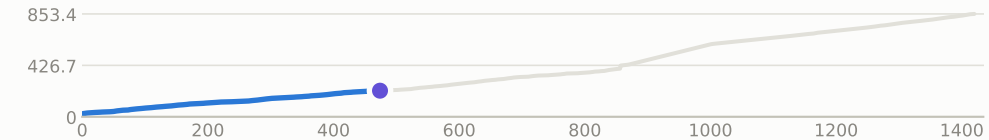
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

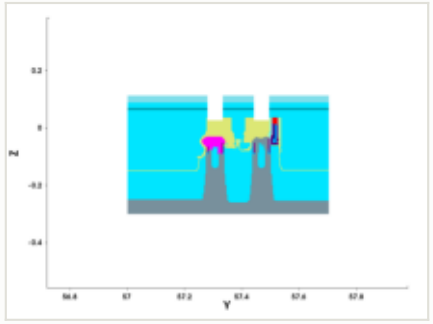
carbon so far, of the whole flow (kgCO2e)



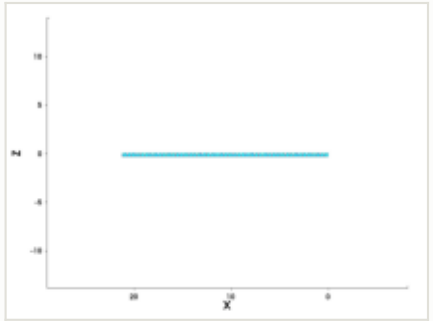
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



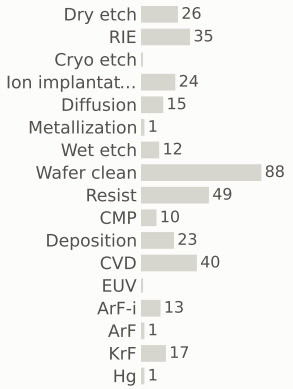
y cut at 57.351 µm



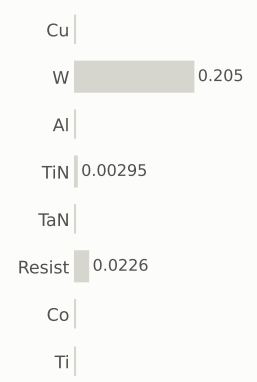
mask for this step

no mask for this step

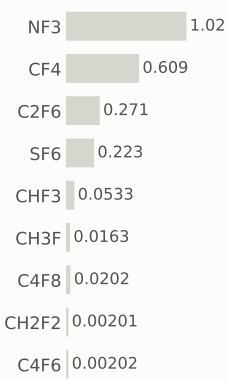
steps by process area



materials deposited (g)



gases emitted (g)



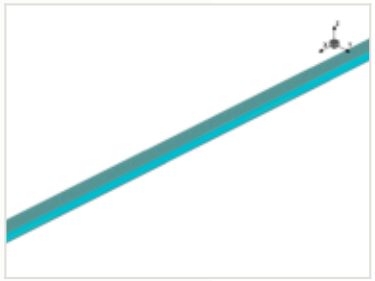
Step #4820: CVD

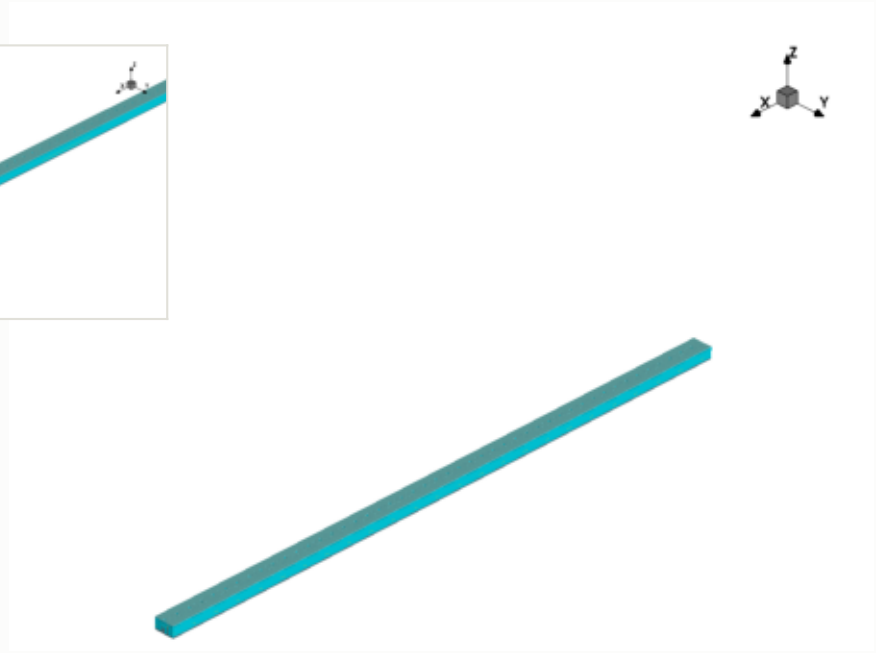
depo_fill · CVD

1.31	218.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



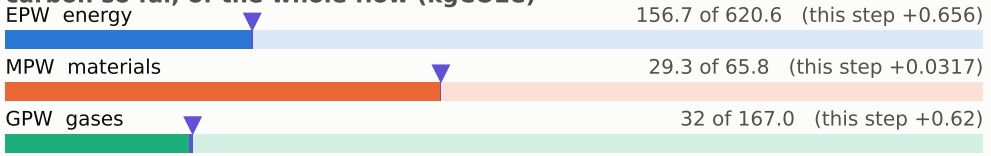


this step

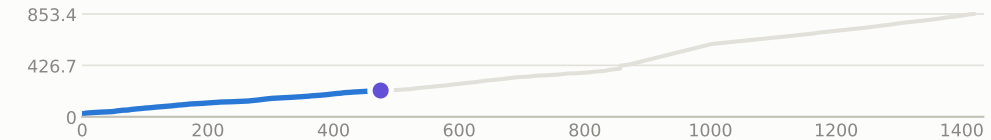
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

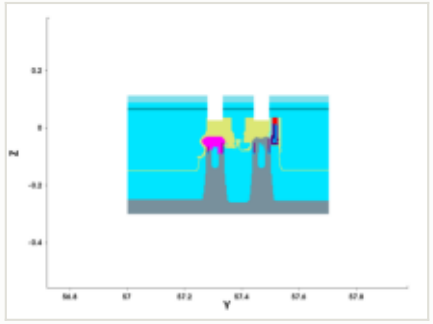
carbon so far, of the whole flow (kgCO2e)



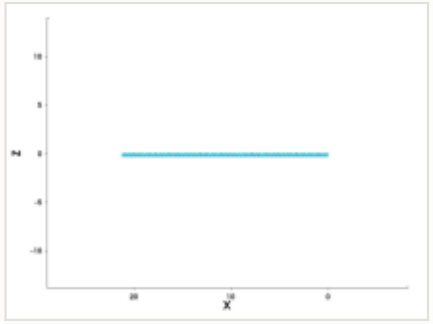
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



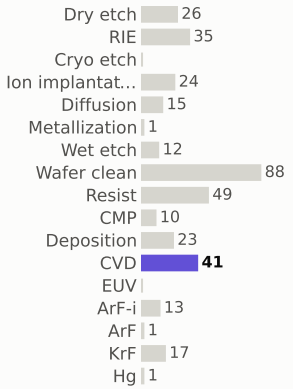
y cut at 57.351 µm



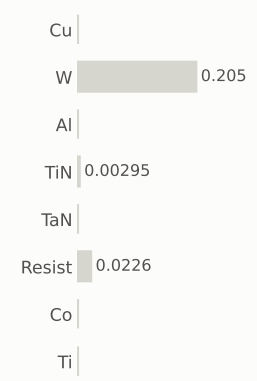
mask for this step

no mask for this step

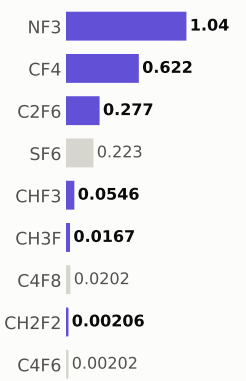
steps by process area



materials deposited (g)



gases emitted (g)



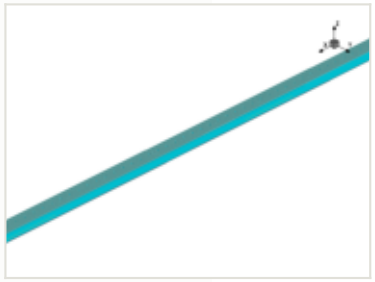
Step #4830: Wafer_clean

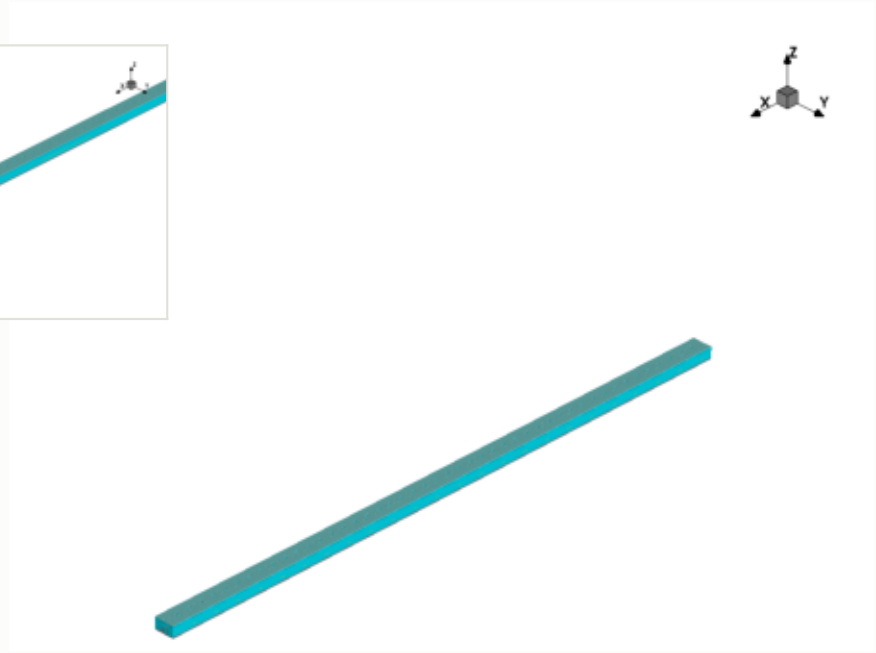
clean · Wafer_clean

0.169	218.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

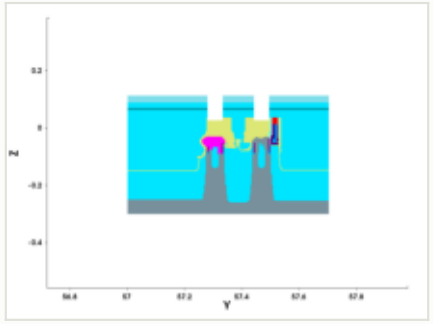
3D structure after this step

zoomed to fit

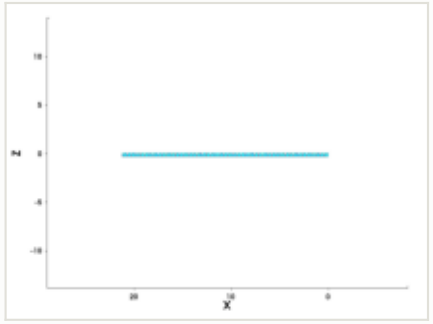




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

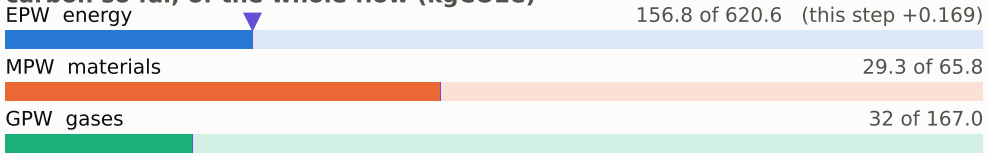
no mask for this step

this step

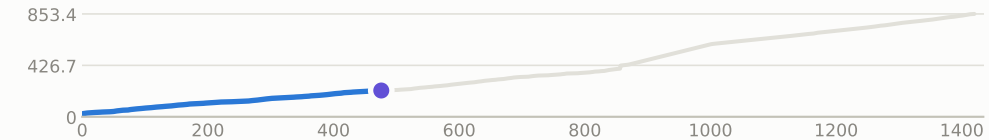
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

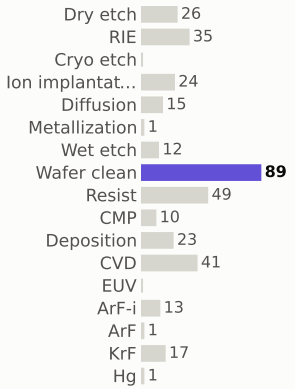
carbon so far, of the whole flow (kgCO2e)



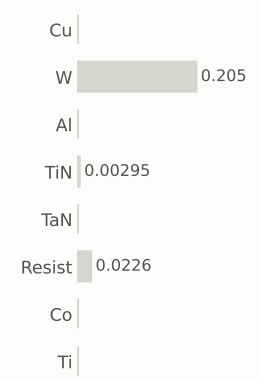
embodied carbon up to each step (kgCO2e per wafer)



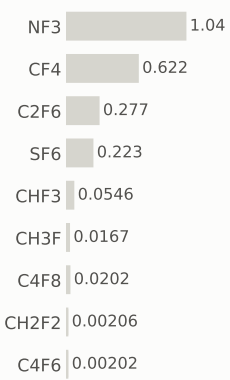
steps by process area



materials deposited (g)



gases emitted (g)



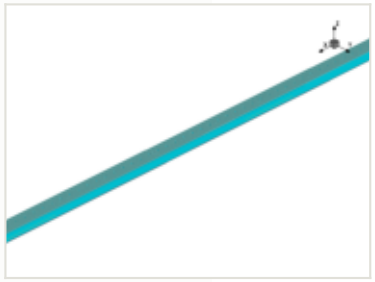
Step #4840: Resist

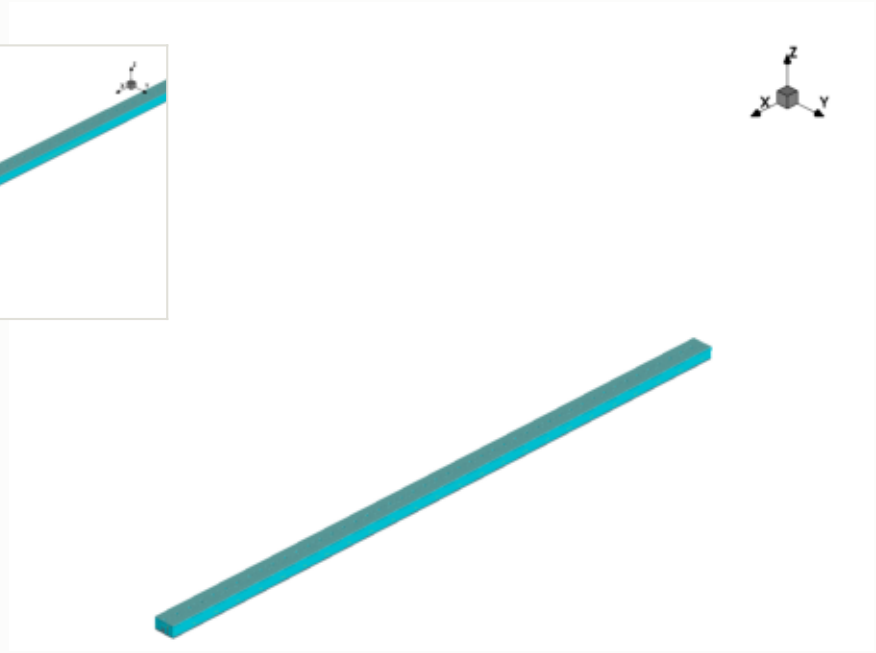
resist · Resist

0.169	218.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

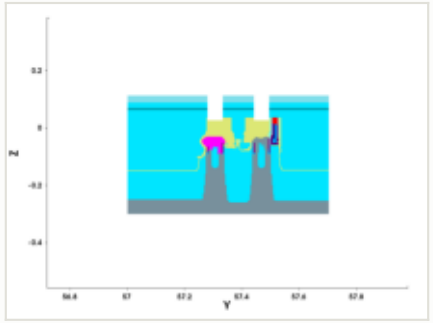
3D structure after this step

zoomed to fit

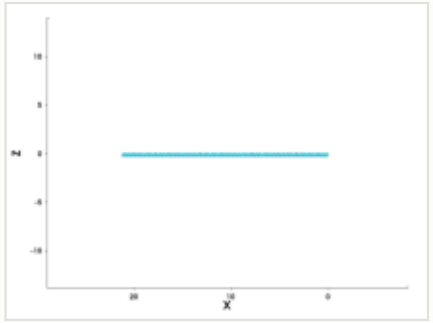




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

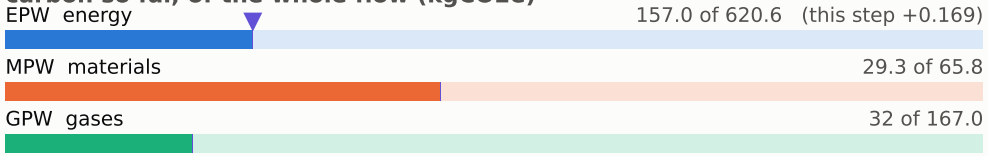
no mask for this step

this step

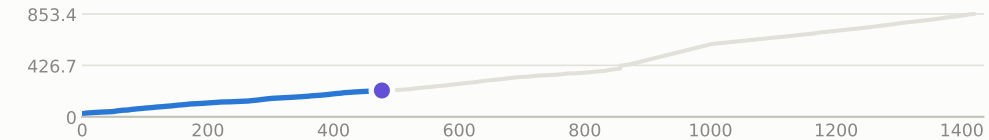
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

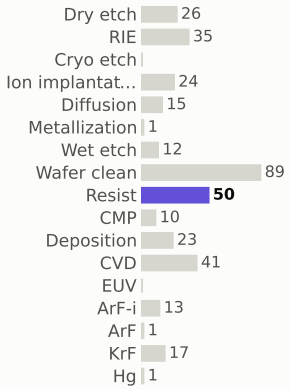
carbon so far, of the whole flow (kgCO2e)



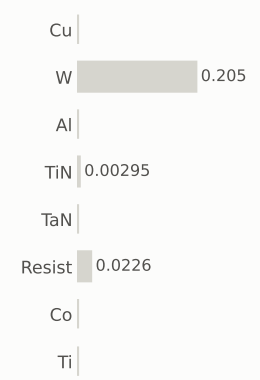
embodied carbon up to each step (kgCO2e per wafer)



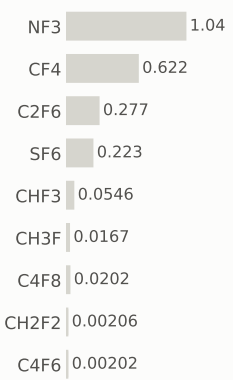
steps by process area



materials deposited (g)



gases emitted (g)



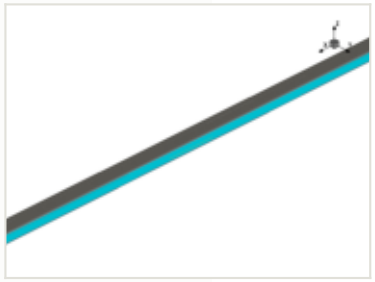
Step #4850: ArF-i

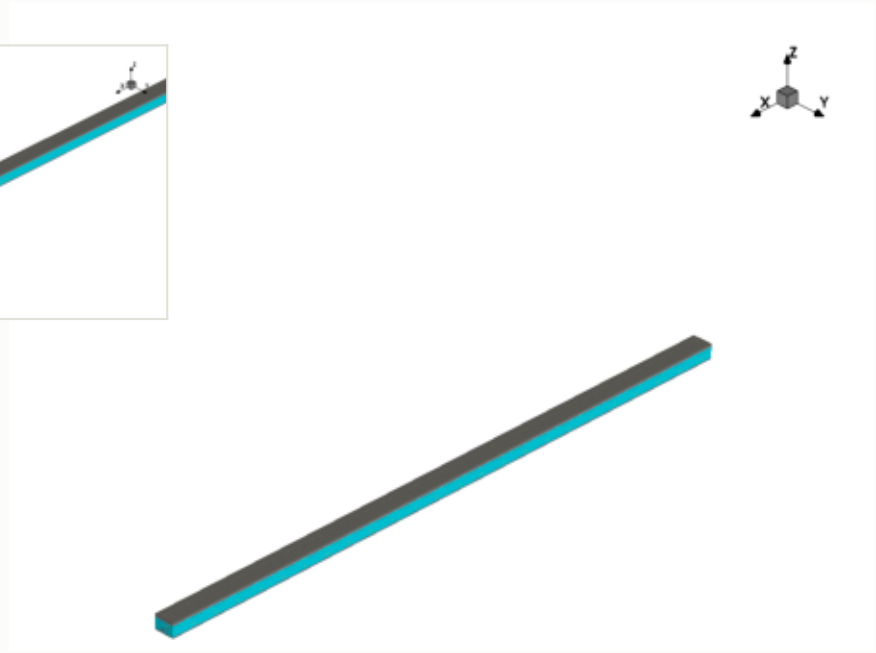
litho_patterning · ArF-i

0.217	218.5	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



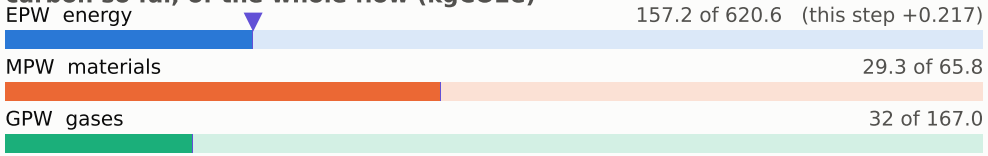


this step

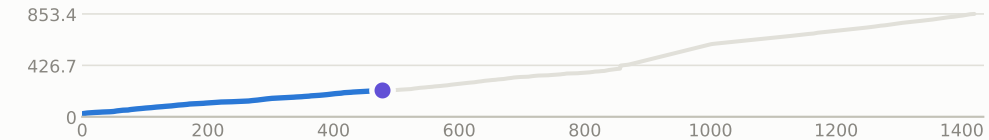
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

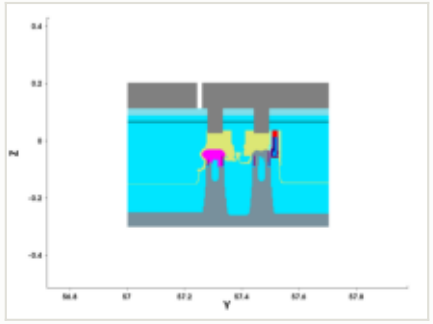
carbon so far, of the whole flow (kgCO2e)



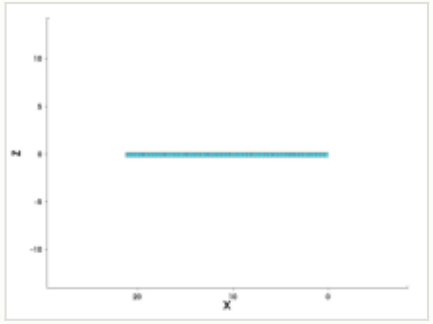
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

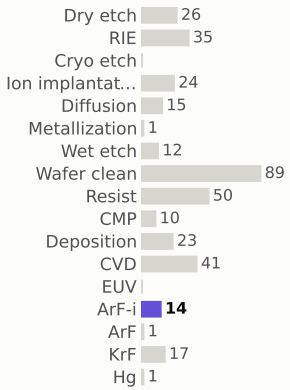


y cut at 57.351 µm

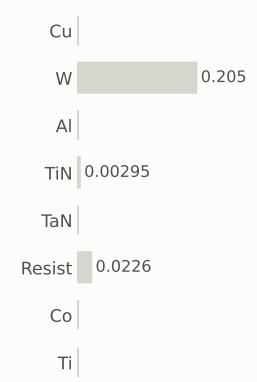


mask: layer 2016:0 (GDSII)

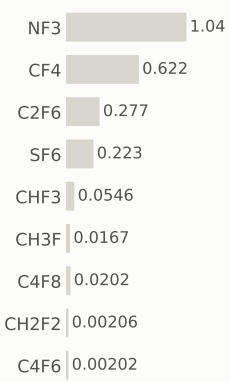
steps by process area



materials deposited (g)



gases emitted (g)



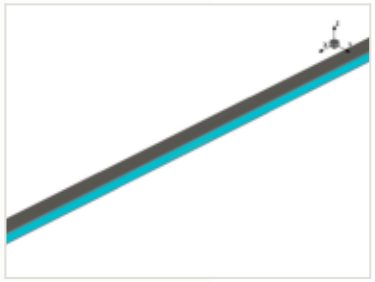
Step #4860: Spx_only

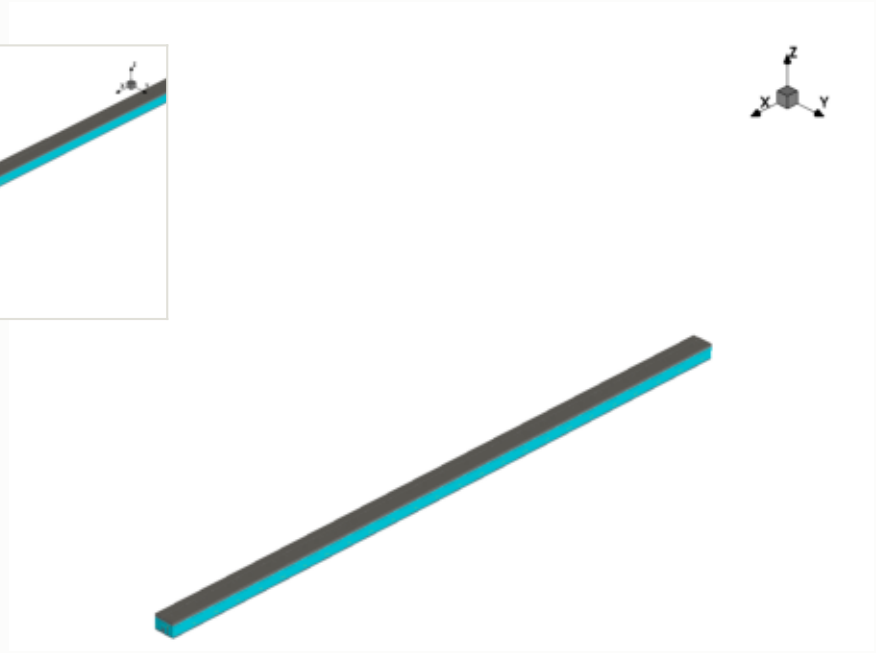
etch_anisotropic · Spx_only

0	218.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





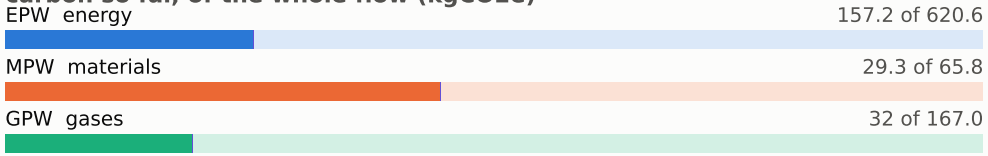
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

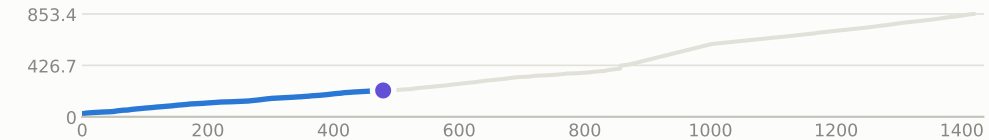
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

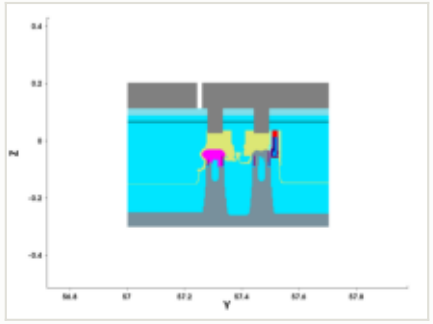
carbon so far, of the whole flow (kgCO2e)



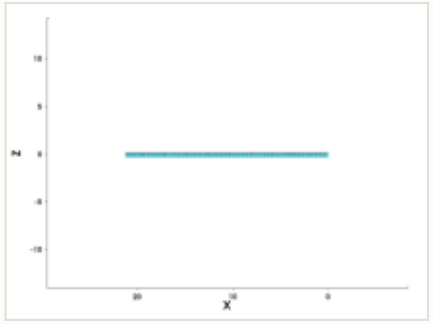
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



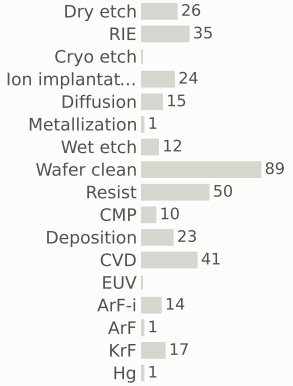
y cut at 57.351 µm



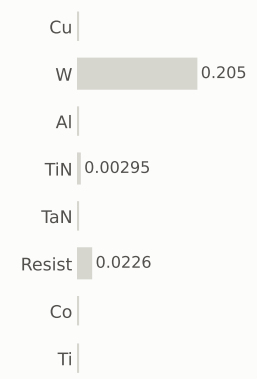
mask for this step

no mask for this step

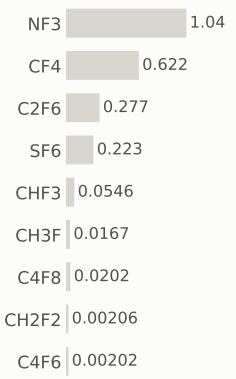
steps by process area



materials deposited (g)



gases emitted (g)



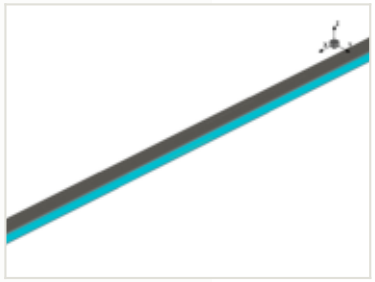
Step #4870: Spx_only

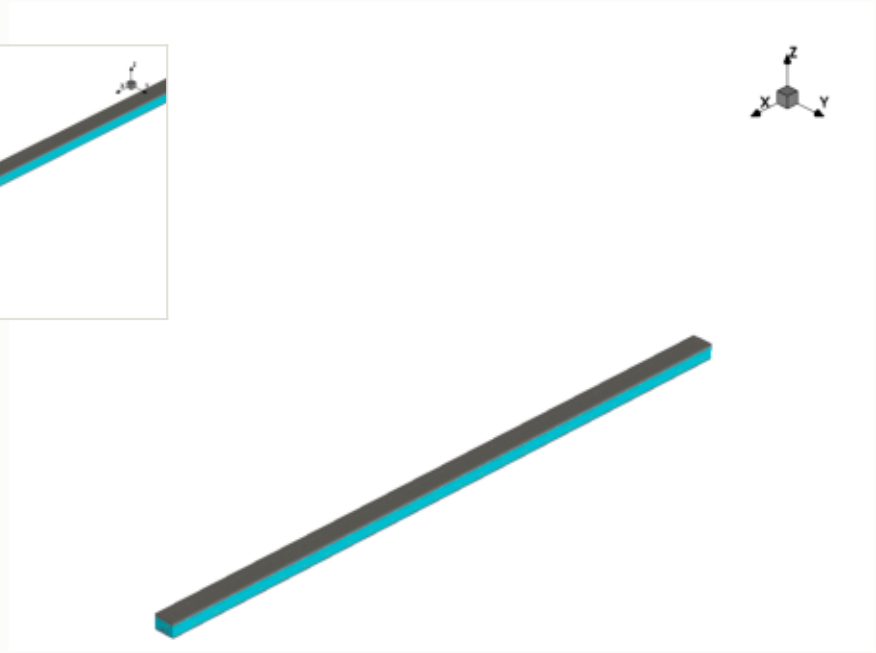
etch_anisotropic · Spx_only

0	218.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





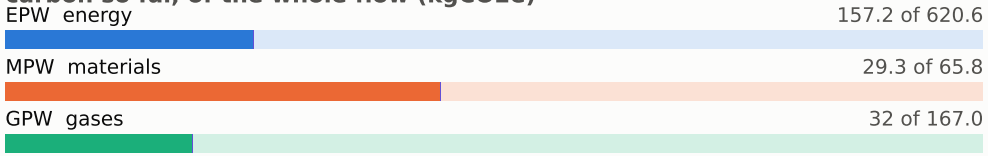
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

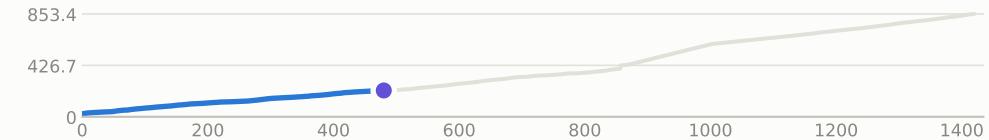
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

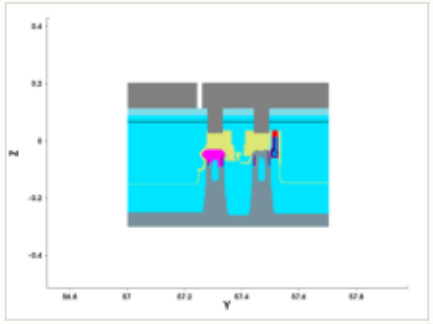
carbon so far, of the whole flow (kgCO2e)



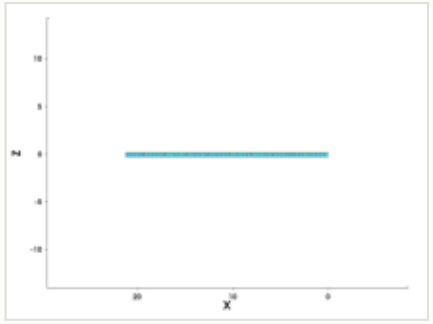
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



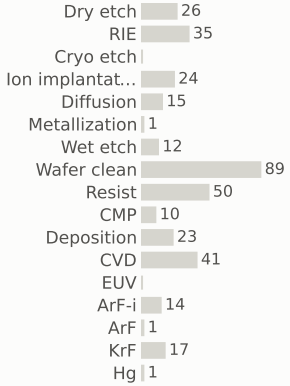
y cut at 57.351 µm



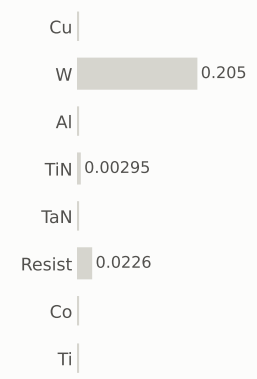
mask for this step

no mask for this step

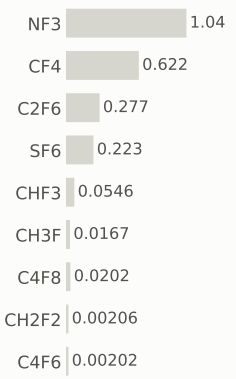
steps by process area



materials deposited (g)



gases emitted (g)



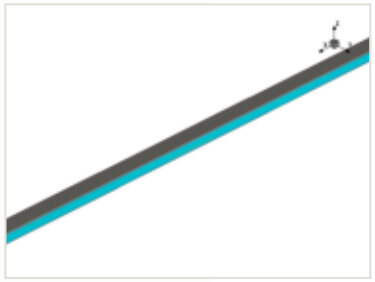
Step #4880: Spx_only

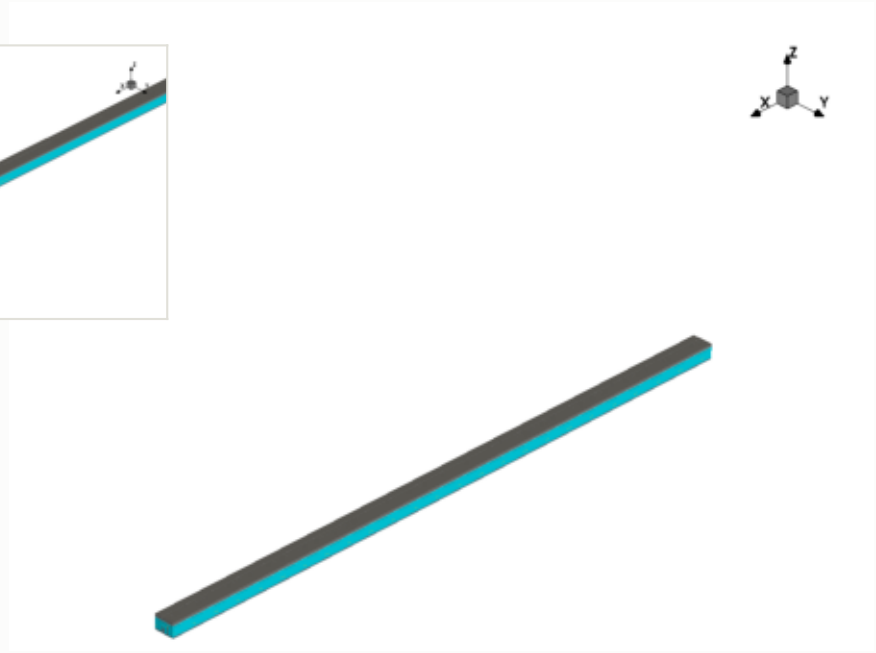
etch_anisotropic · Spx_only

0	218.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





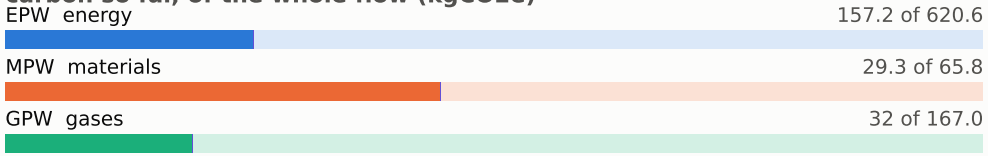
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

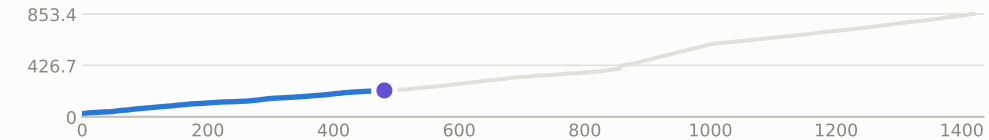
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

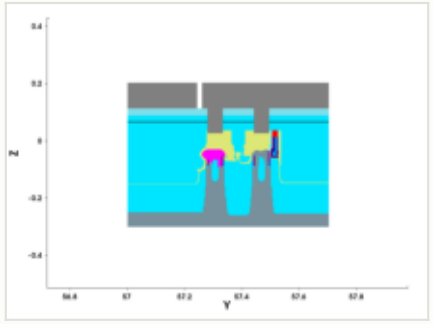
carbon so far, of the whole flow (kgCO2e)



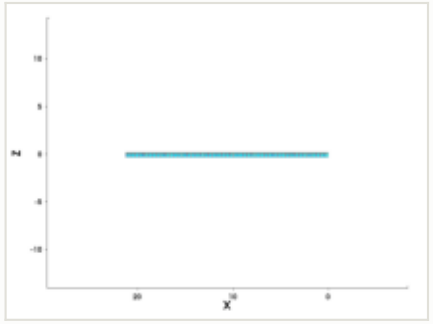
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



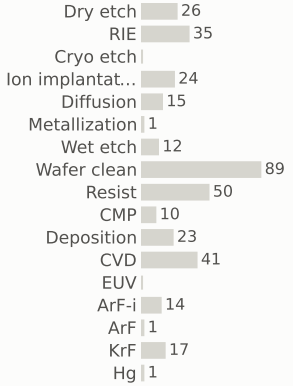
y cut at 57.351 µm



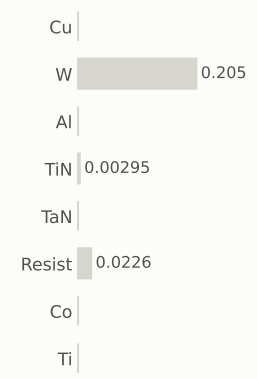
mask for this step

no mask for this step

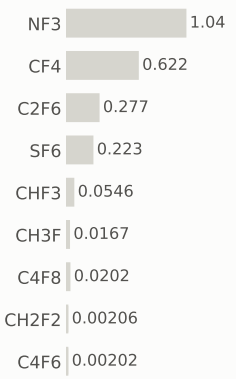
steps by process area



materials deposited (g)



gases emitted (g)



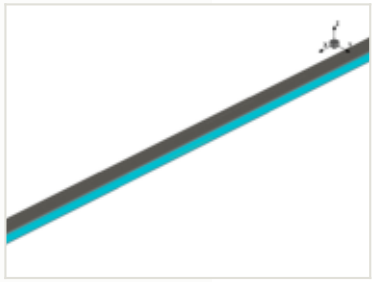
Step #4890: Spx_only

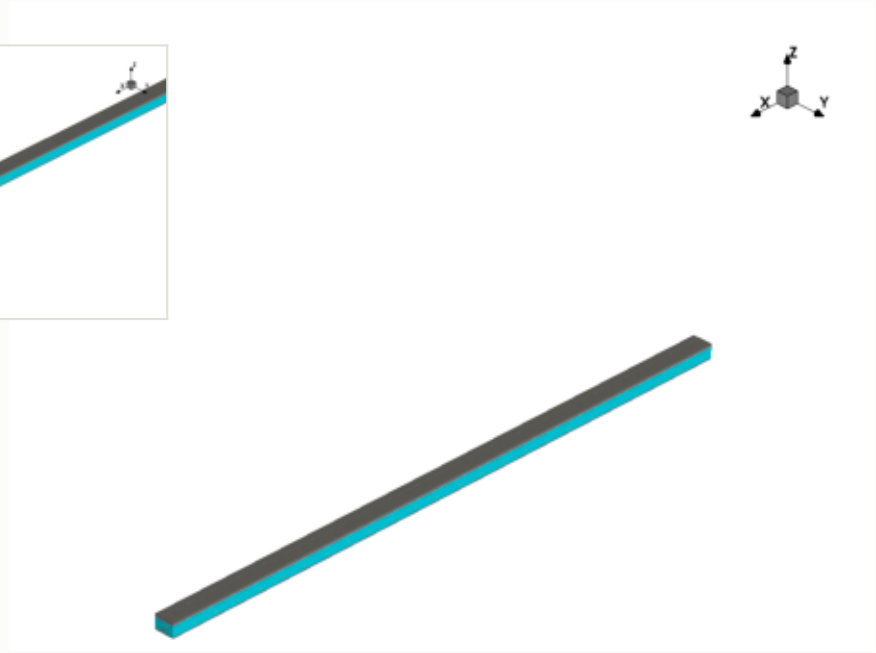
etch_anisotropic · Spx_only

0	218.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





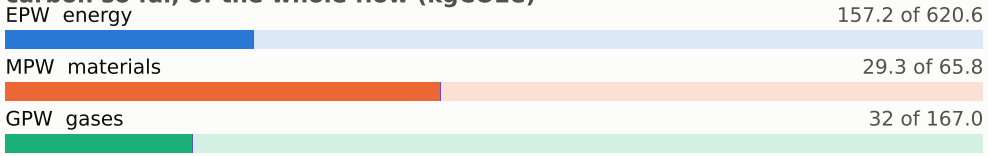
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

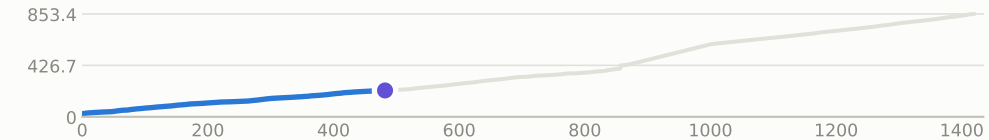
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

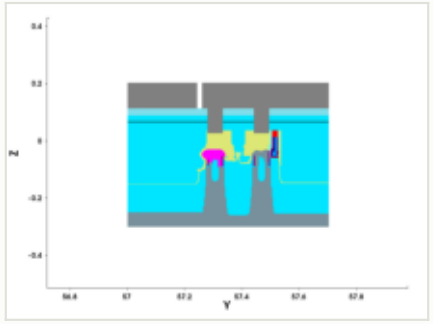
carbon so far, of the whole flow (kgCO2e)



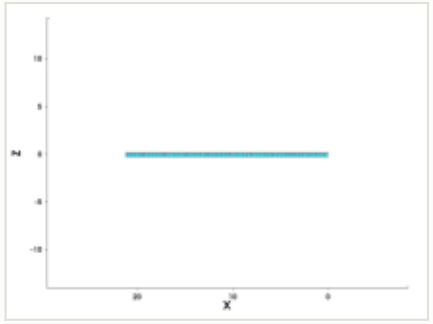
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



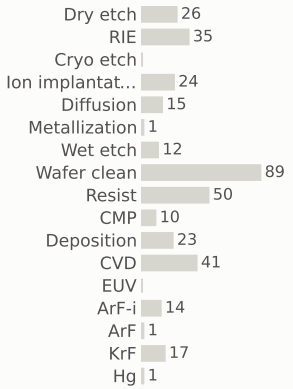
y cut at 57.351 µm



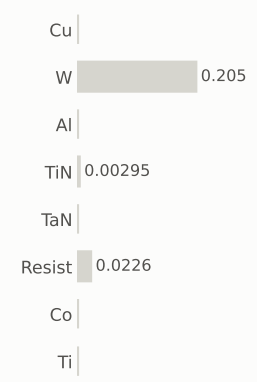
mask for this step

no mask for this step

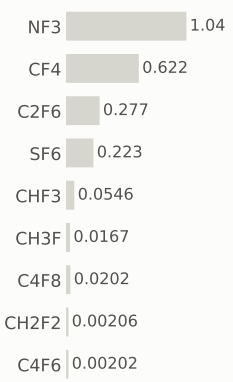
steps by process area



materials deposited (g)



gases emitted (g)



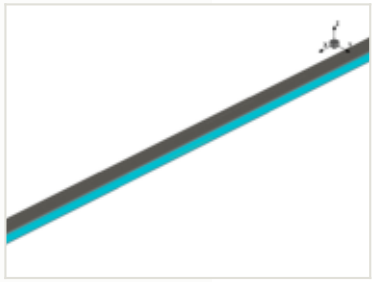
Step #4900: Spx_only

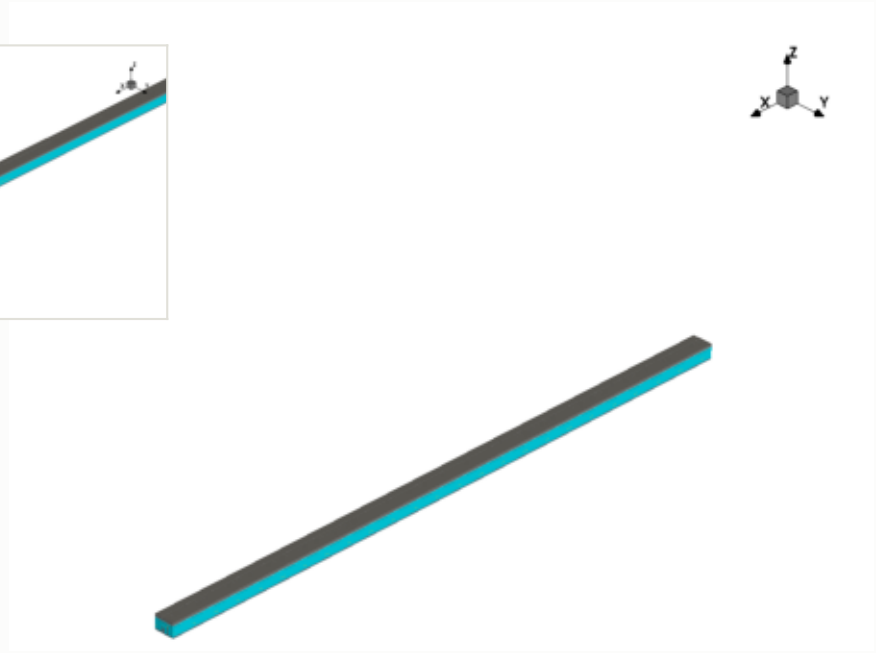
etch_anisotropic · Spx_only

0	218.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





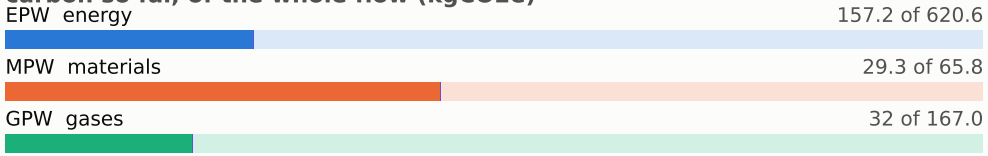
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

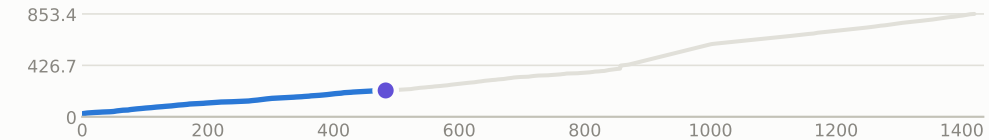
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

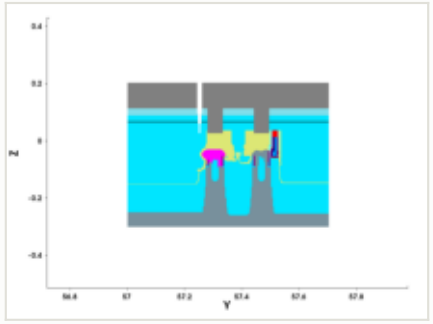
carbon so far, of the whole flow (kgCO2e)



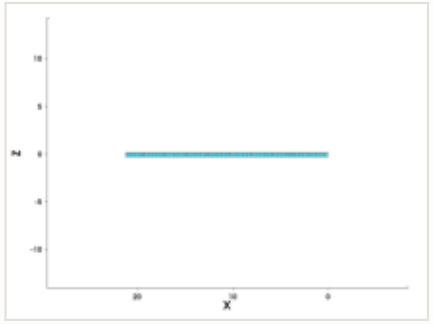
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



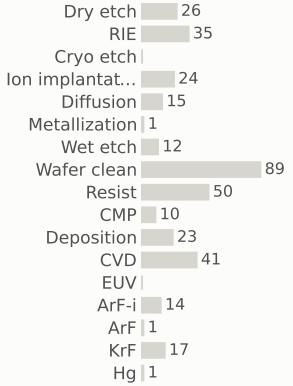
y cut at 57.351 µm



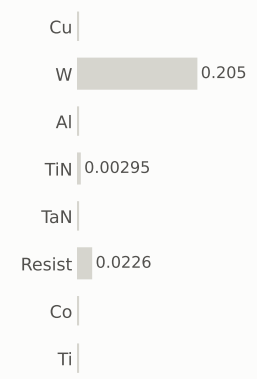
mask for this step

no mask for this step

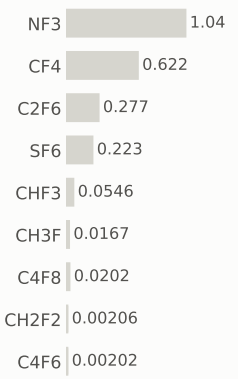
steps by process area



materials deposited (g)



gases emitted (g)



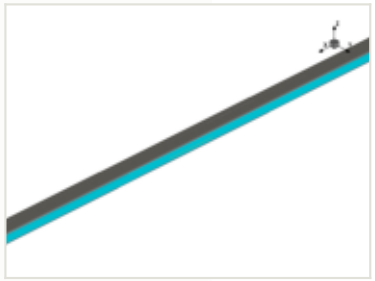
Step #4910: RIE

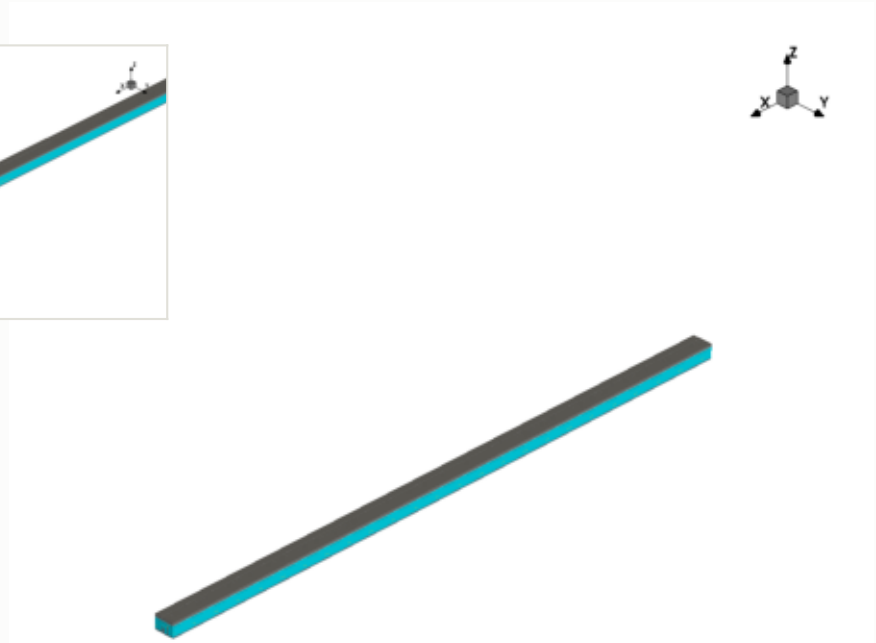
etch_anisotropic · RIE

1.04	219.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



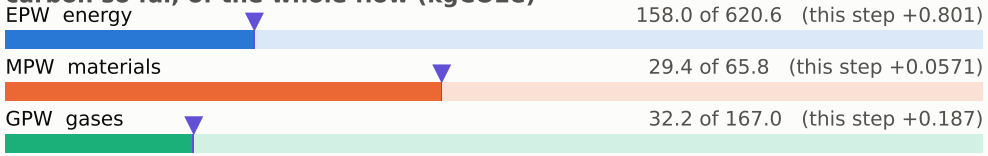


this step

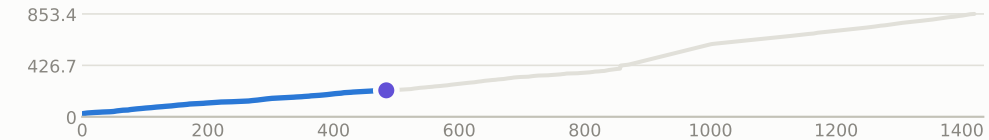
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

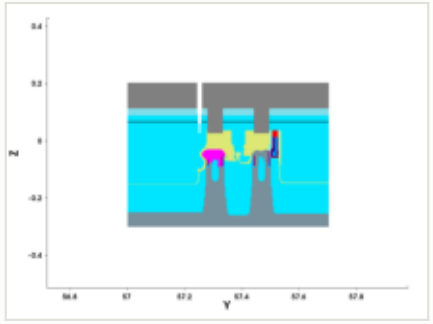
carbon so far, of the whole flow (kgCO2e)



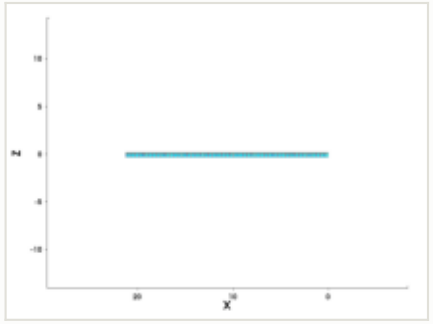
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



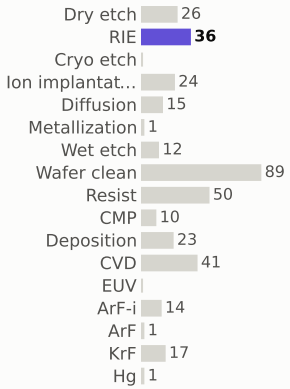
y cut at 57.351 µm



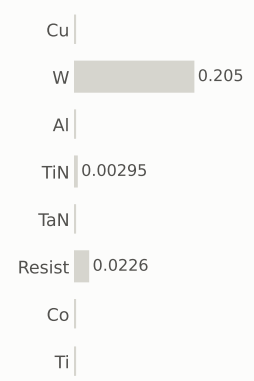
mask for this step

no mask for this step

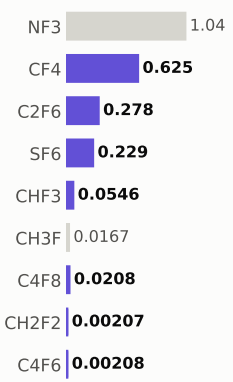
steps by process area



materials deposited (g)



gases emitted (g)



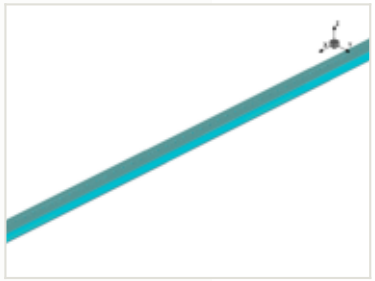
Step #4920: Spx_only

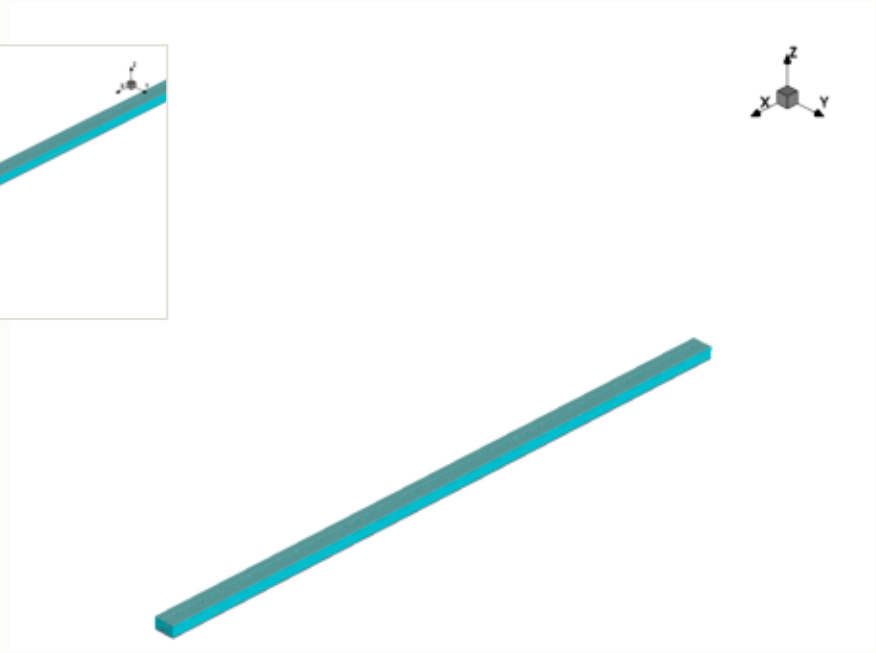
etch_strip · Spx_only

0	219.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



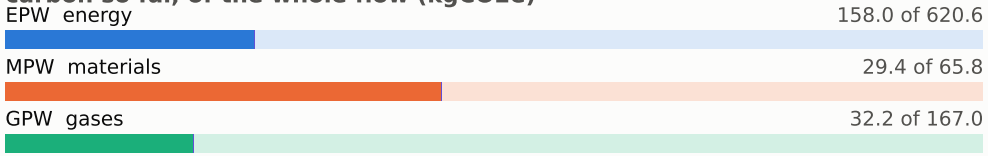


this step

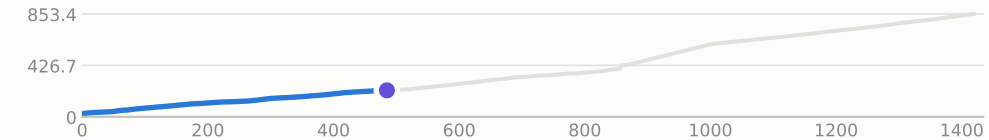
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

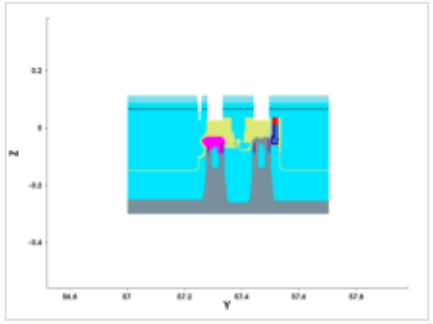
carbon so far, of the whole flow (kgCO2e)



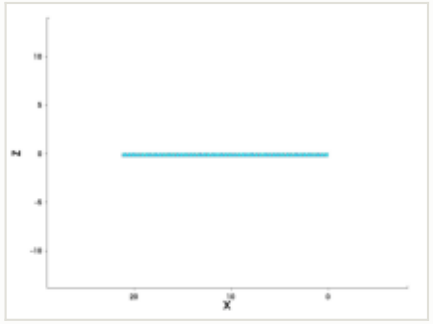
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



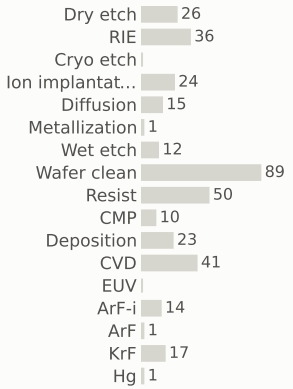
y cut at 57.351 µm



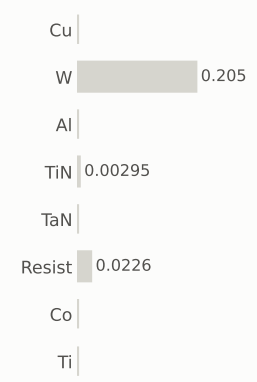
mask for this step

no mask for this step

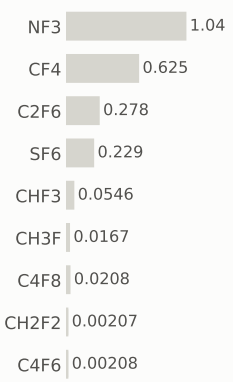
steps by process area



materials deposited (g)



gases emitted (g)



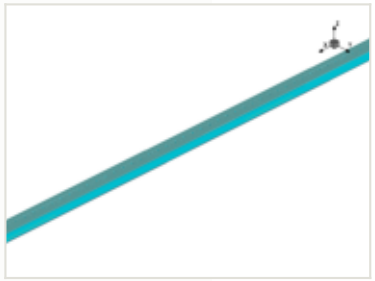
Step #4930: Wafer_clean

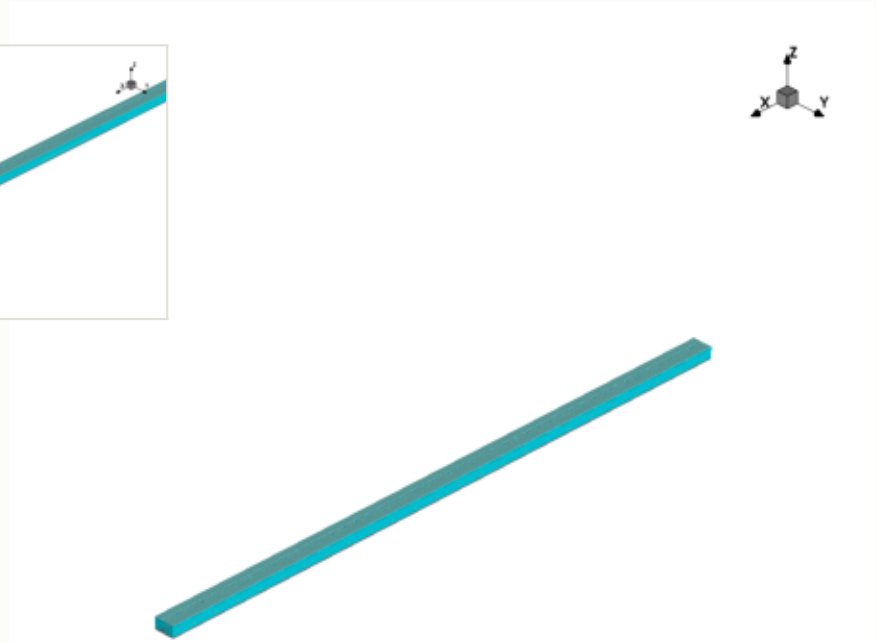
clean · Wafer_clean

0.169	219.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





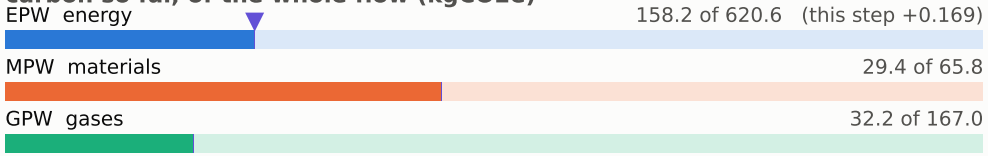
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

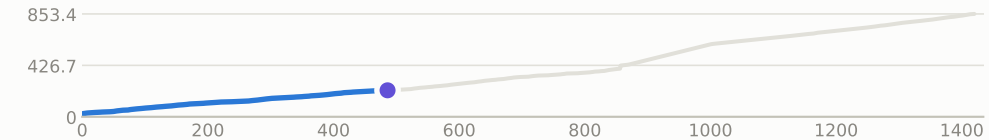
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

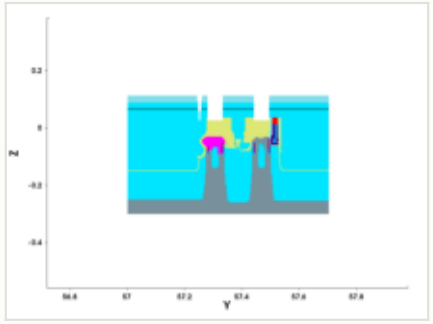
carbon so far, of the whole flow (kgCO2e)



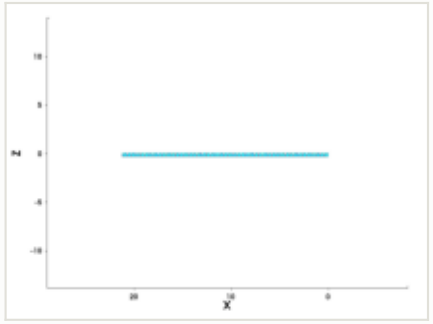
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



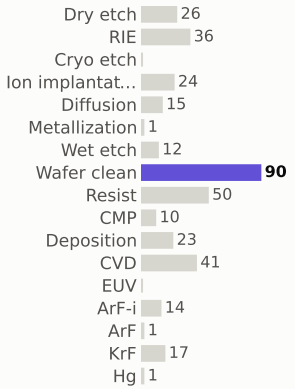
y cut at 57.351 µm



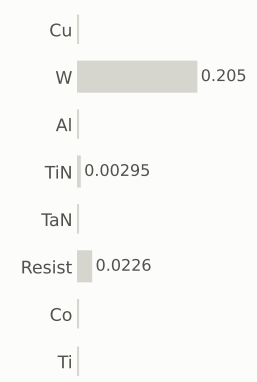
mask for this step

no mask for this step

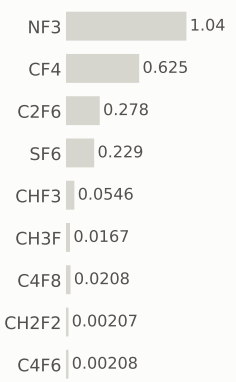
steps by process area



materials deposited (g)



gases emitted (g)



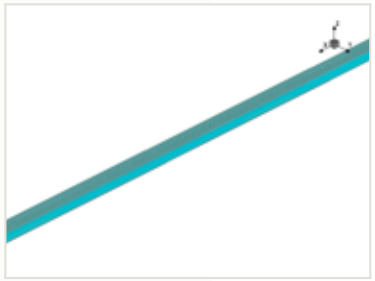
Step #4940: Resist

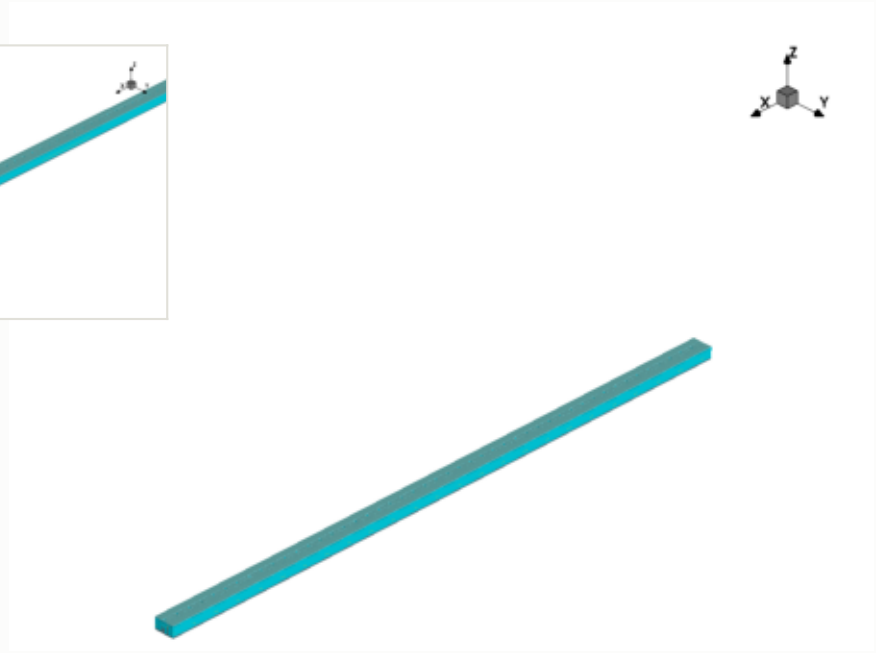
depo_fill · Resist

0.169	219.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



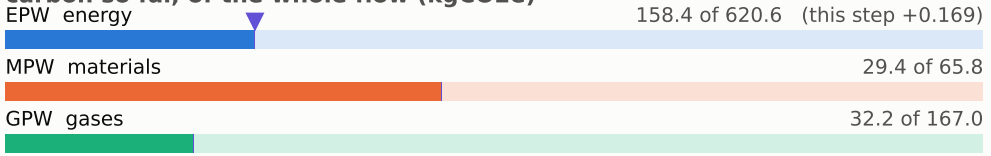


this step

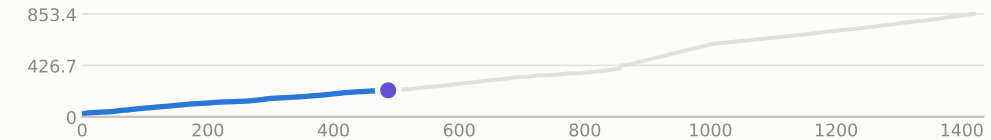
process area	Resist
material	—
thickness	0.12 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

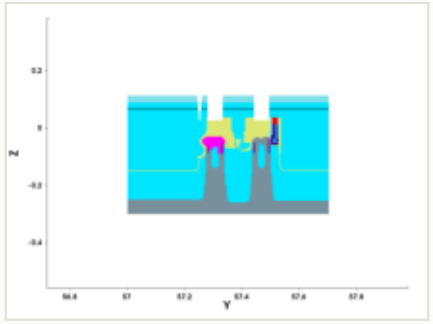
carbon so far, of the whole flow (kgCO2e)



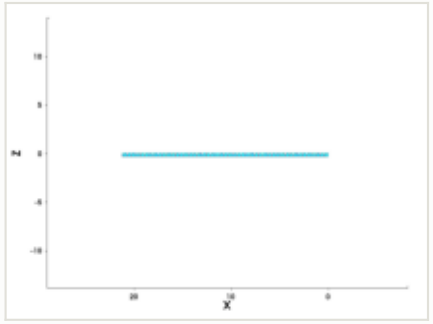
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



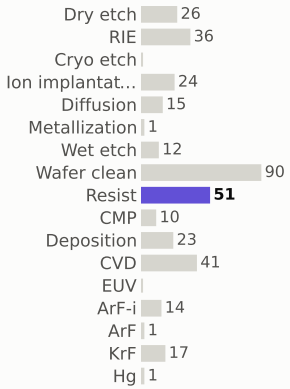
y cut at 57.351 µm



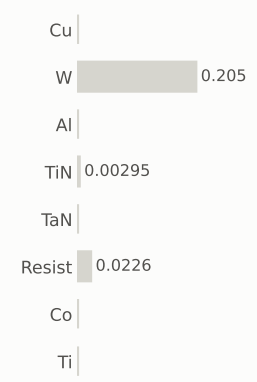
mask for this step

no mask for this step

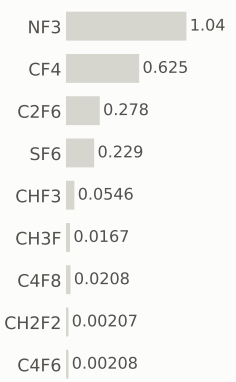
steps by process area



materials deposited (g)



gases emitted (g)



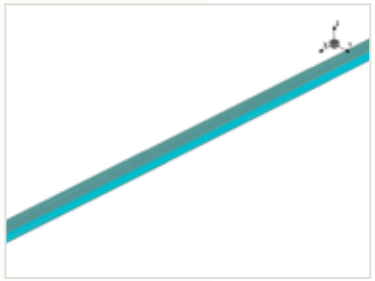
Step #4950: Spx_only

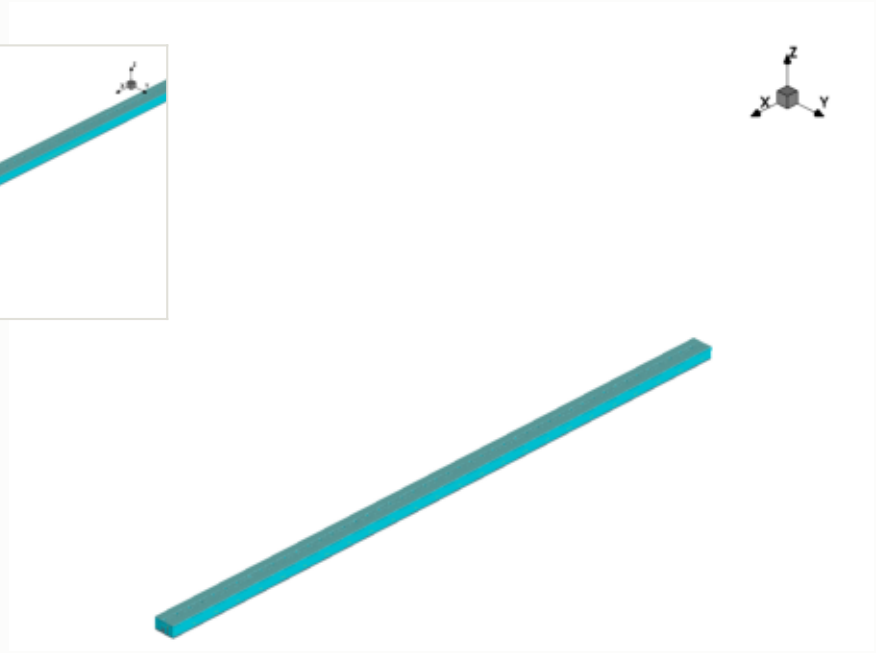
depo_fill · Spx_only

0	219.9	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





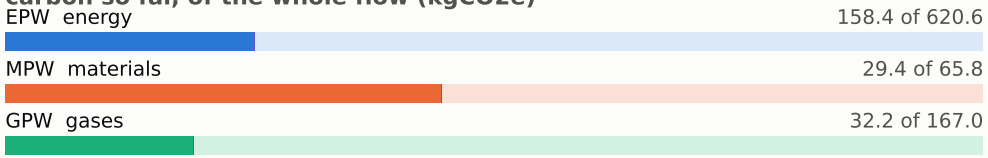
this step

process area	Spx_only
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

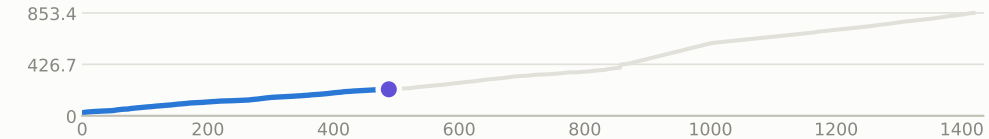
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

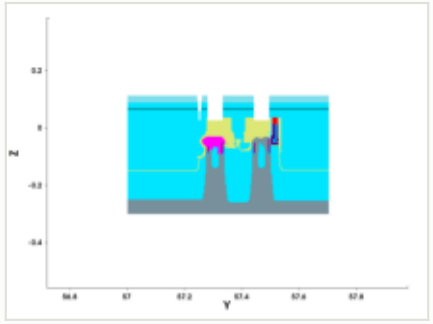
carbon so far, of the whole flow (kgCO2e)



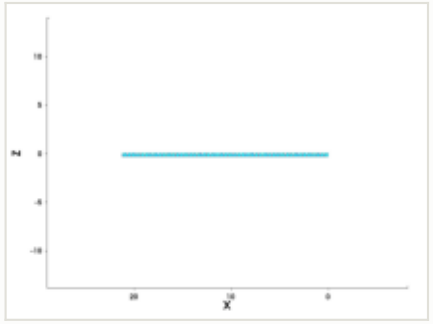
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



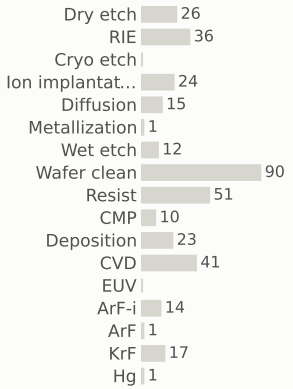
y cut at 57.351 µm



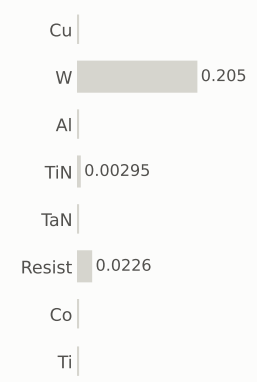
mask for this step

no mask for this step

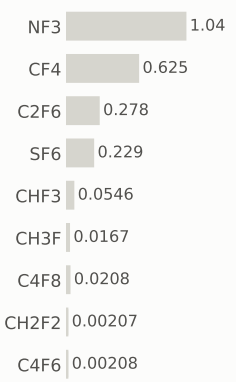
steps by process area



materials deposited (g)



gases emitted (g)



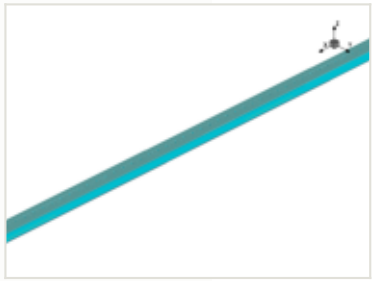
Step #4960: CVD

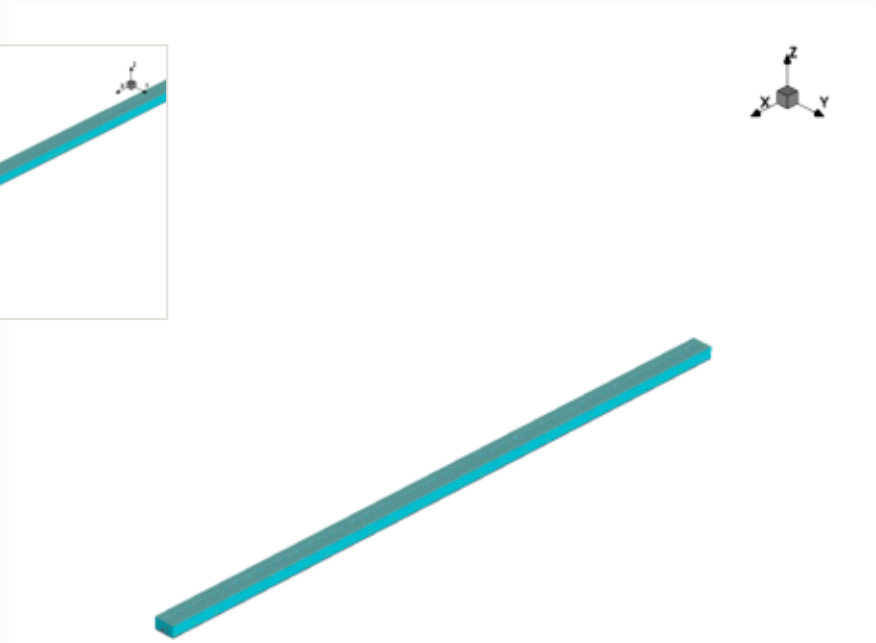
depo_fill · CVD

1.31	221.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



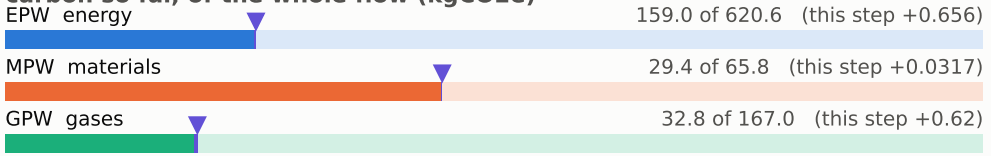


this step

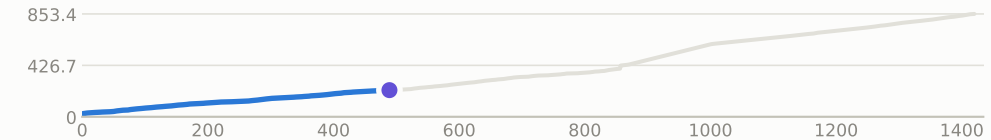
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

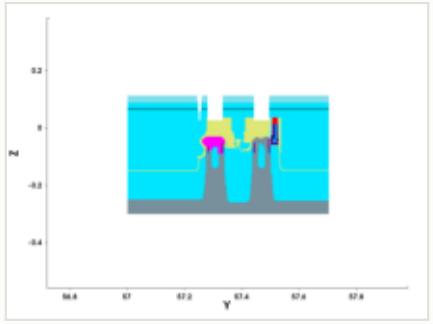
carbon so far, of the whole flow (kgCO2e)



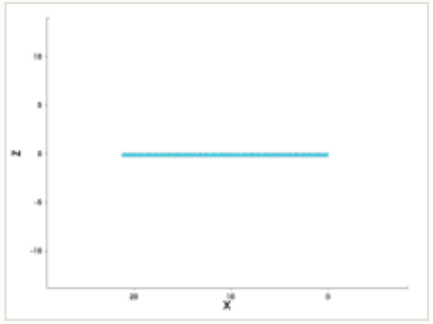
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



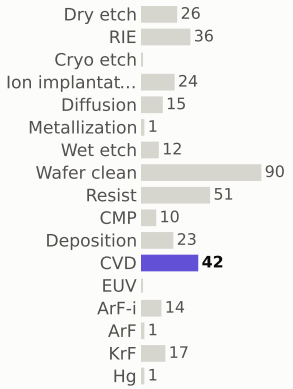
y cut at 57.351 µm



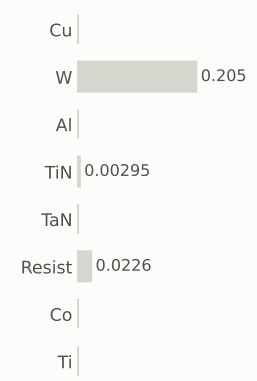
mask for this step

no mask for this step

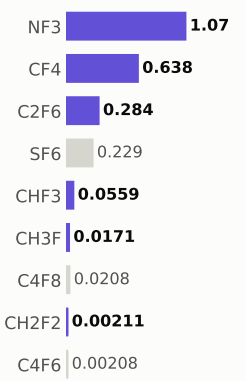
steps by process area



materials deposited (g)



gases emitted (g)



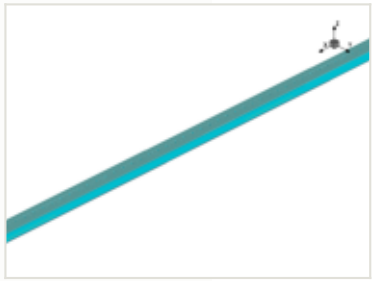
Step #4970: Wafer_clean

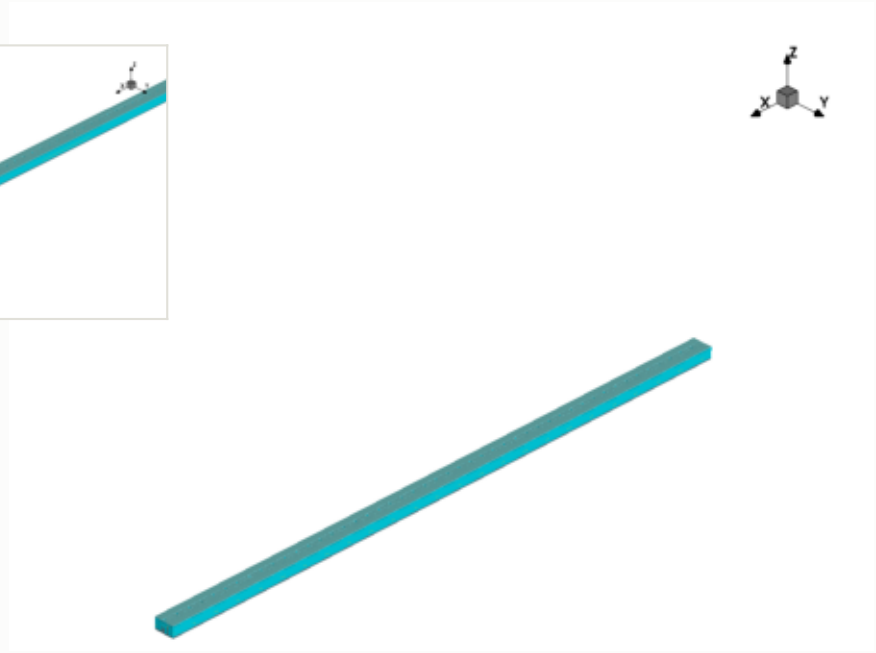
clean · Wafer_clean

0.169	221.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

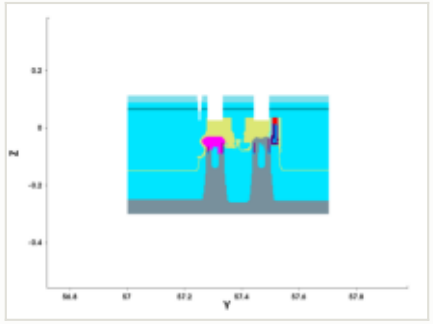
3D structure after this step

zoomed to fit

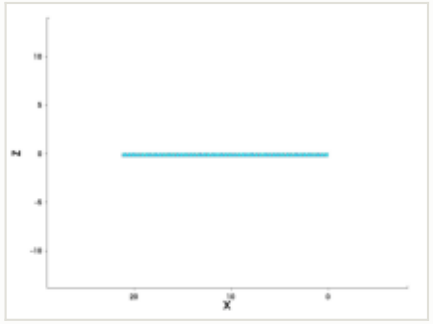




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

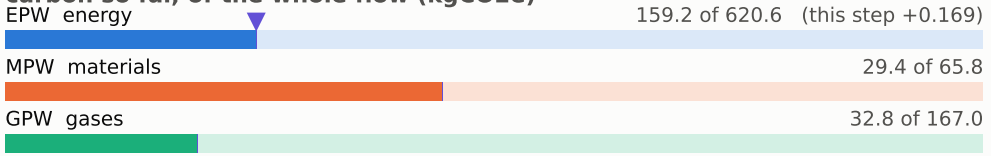
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

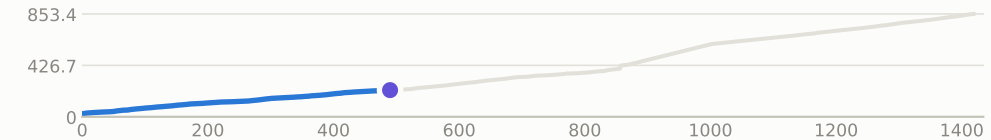
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

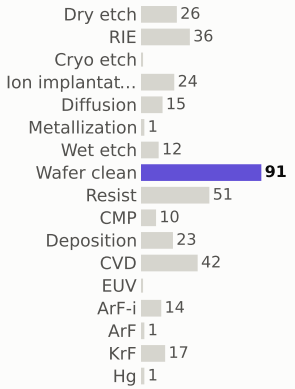
carbon so far, of the whole flow (kgCO2e)



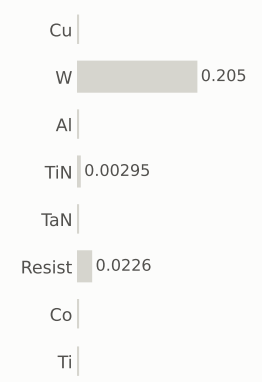
embodied carbon up to each step (kgCO2e per wafer)



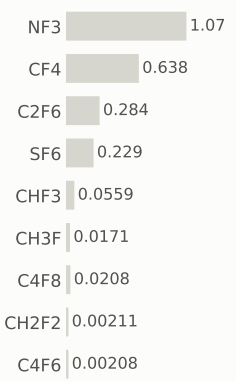
steps by process area



materials deposited (g)



gases emitted (g)



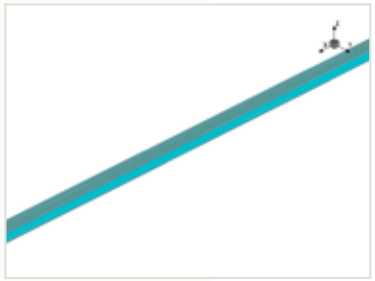
Step #4980: Resist

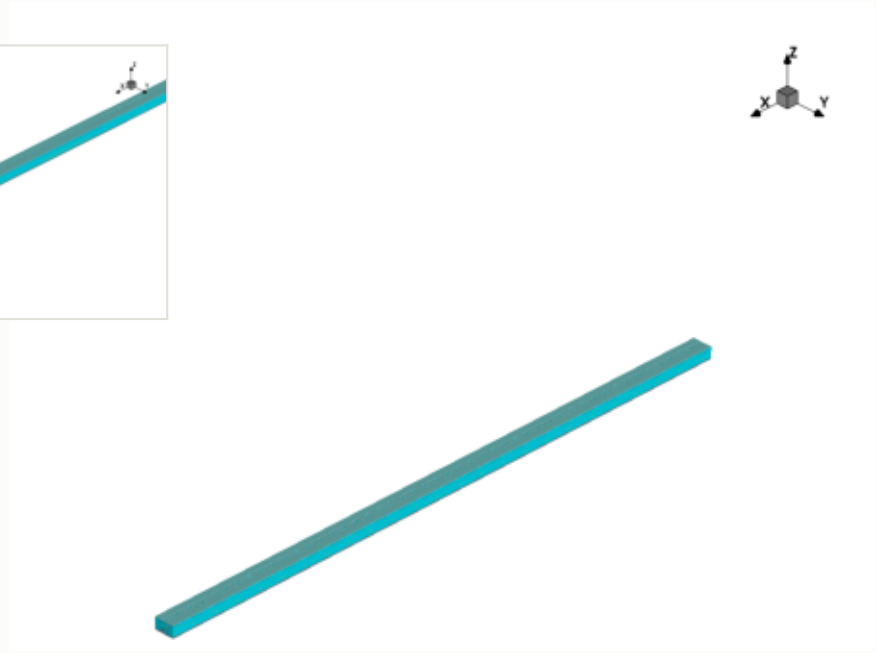
resist · Resist

0.169	221.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





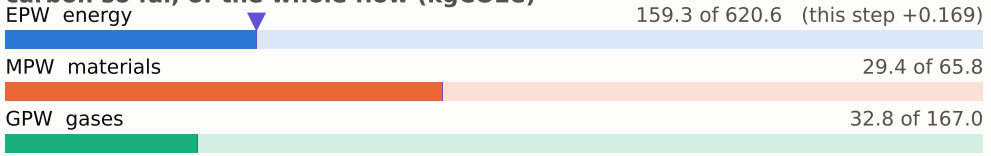
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

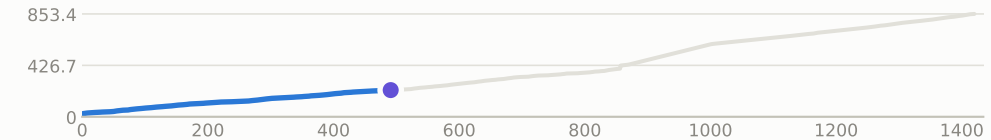
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

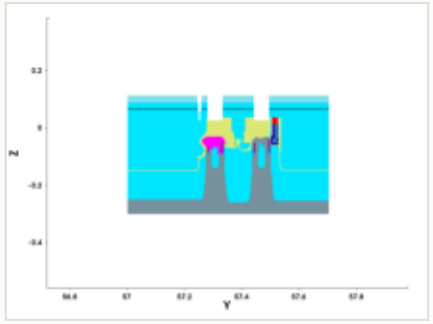
carbon so far, of the whole flow (kgCO2e)



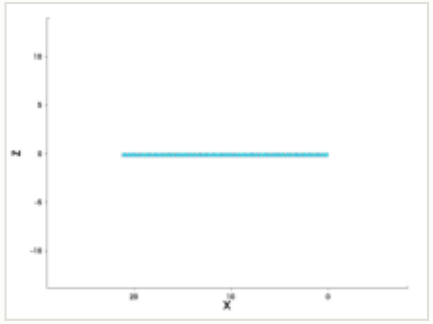
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



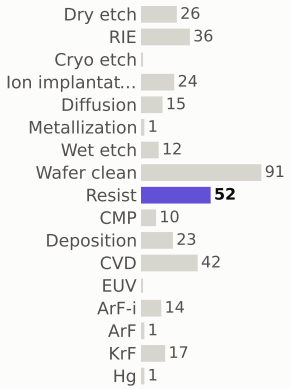
y cut at 57.351 µm



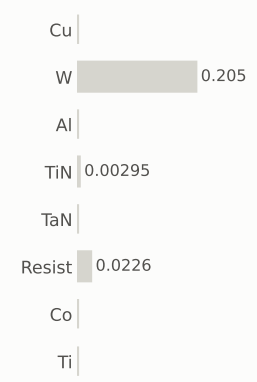
mask for this step

no mask for this step

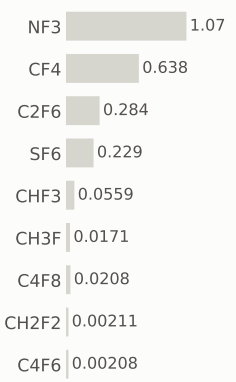
steps by process area



materials deposited (g)



gases emitted (g)



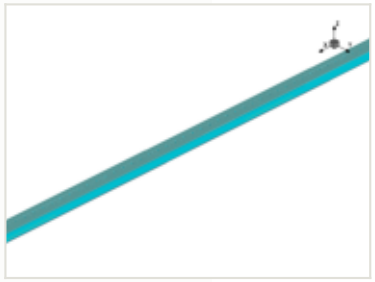
Step #4990: ArF-i

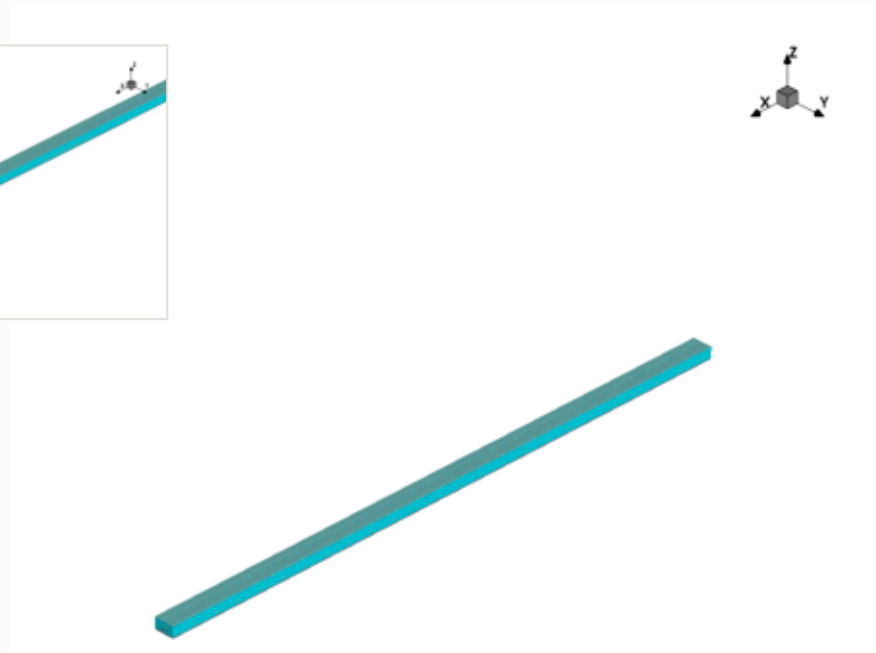
litho_patterning · ArF-i

0.217	221.8	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



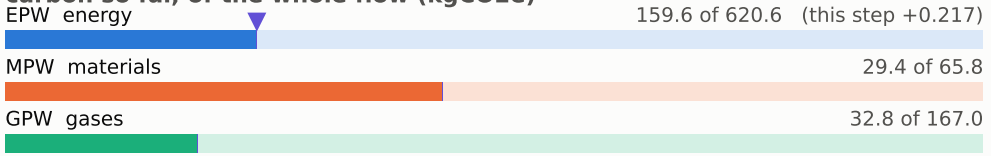


this step

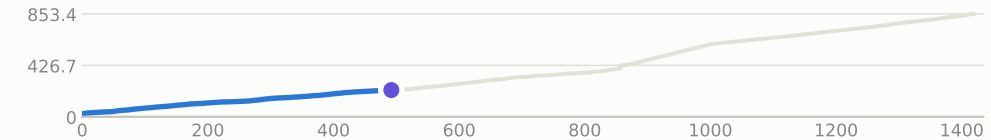
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

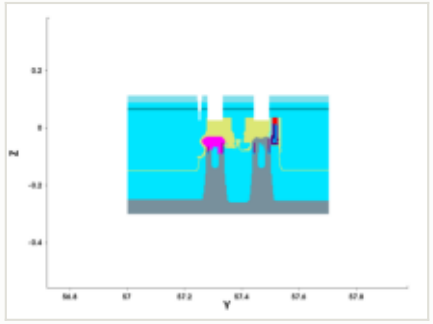
carbon so far, of the whole flow (kgCO2e)



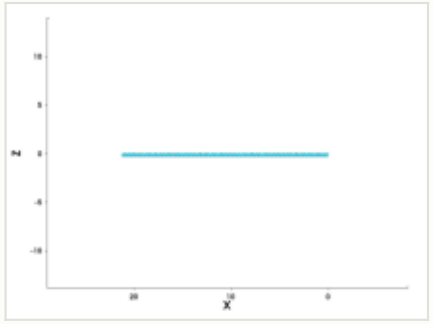
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

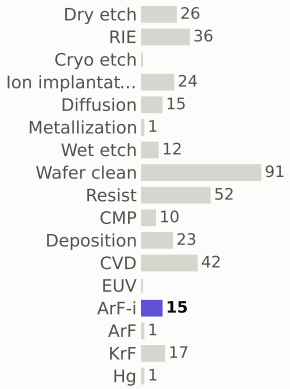


y cut at 57.351 µm

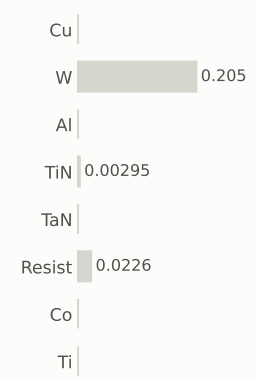


mask: layer 2016:0 (GDSII)

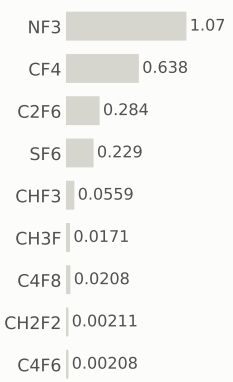
steps by process area



materials deposited (g)



gases emitted (g)



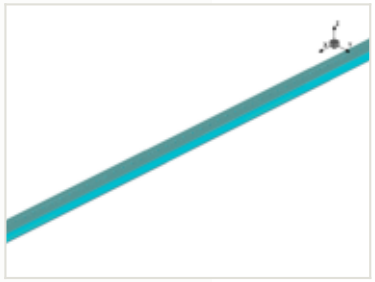
Step #5000: Spx_only

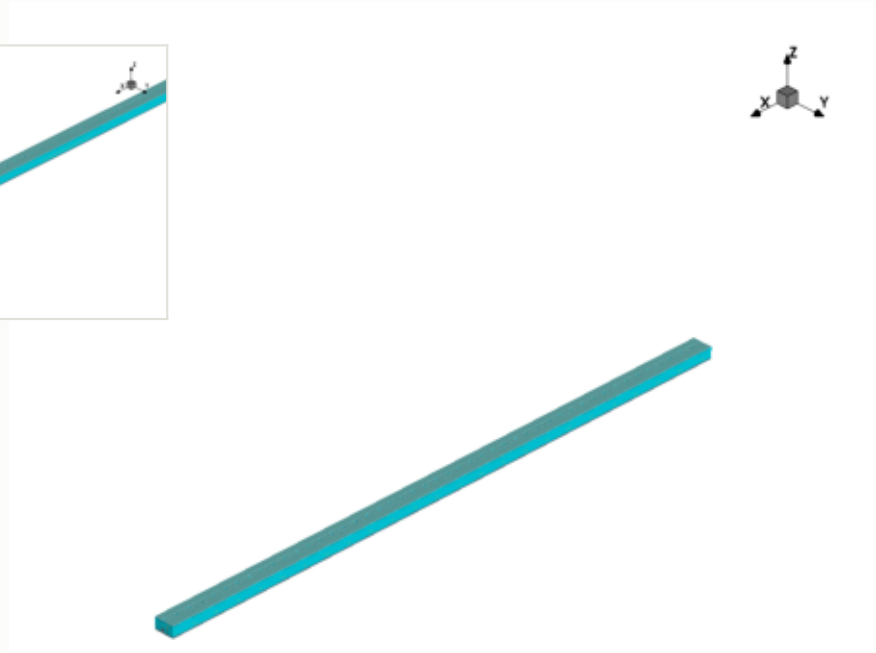
etch_anisotropic · Spx_only

0	221.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



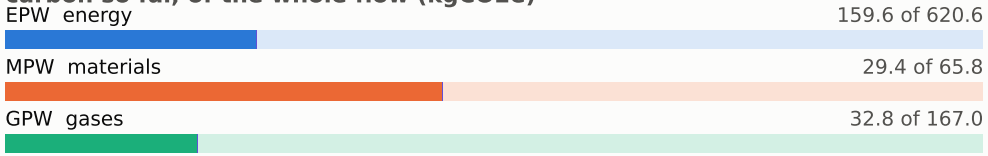


this step

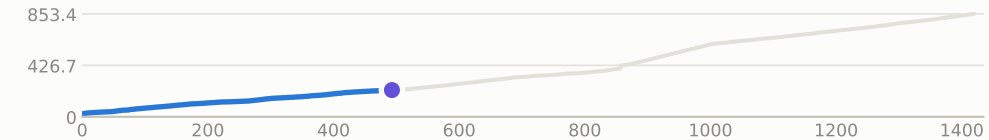
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

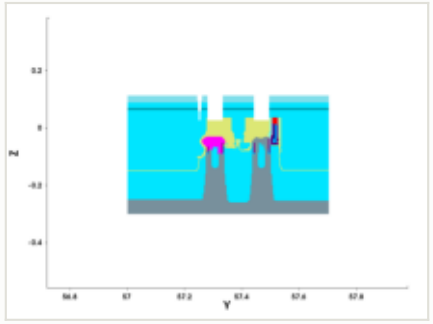
carbon so far, of the whole flow (kgCO2e)



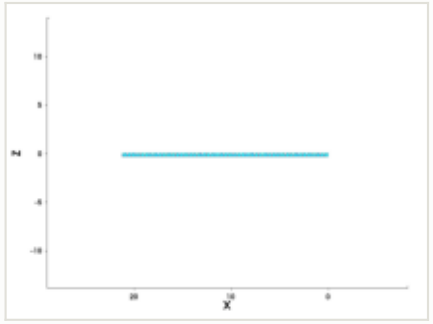
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



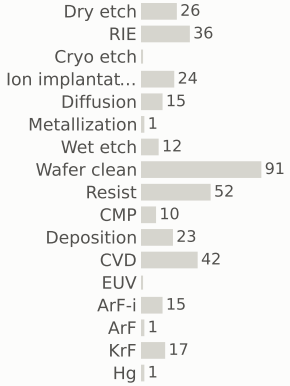
y cut at 57.351 µm



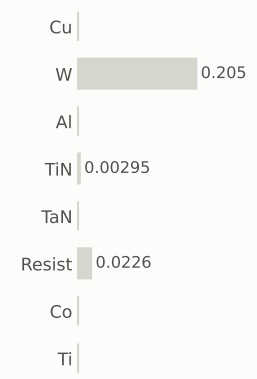
mask for this step

no mask for this step

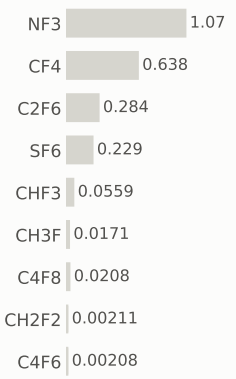
steps by process area



materials deposited (g)



gases emitted (g)



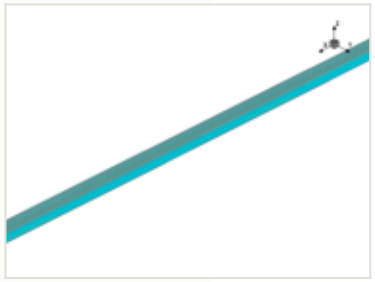
Step #5010: Spx_only

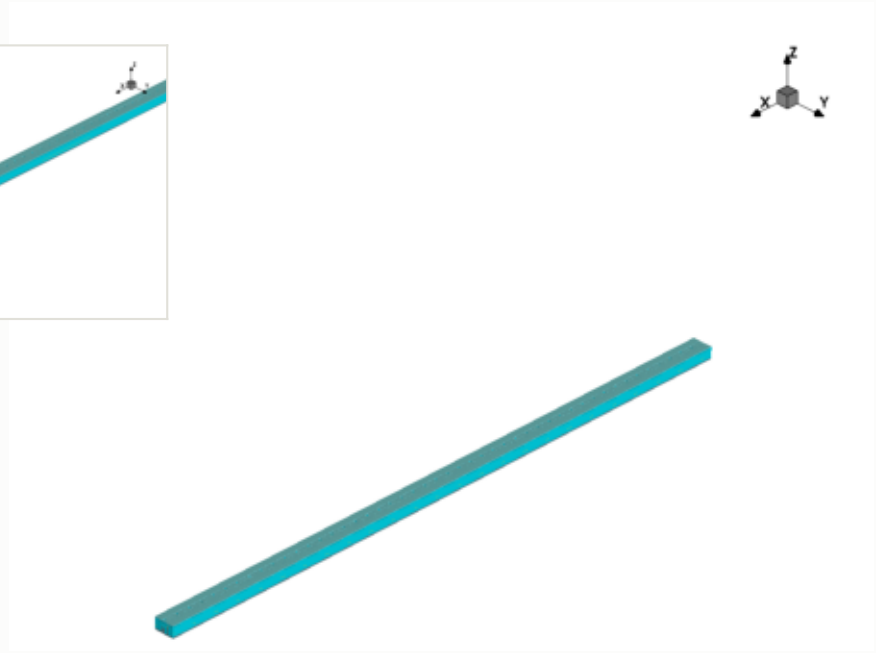
etch_anisotropic · Spx_only

0	221.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



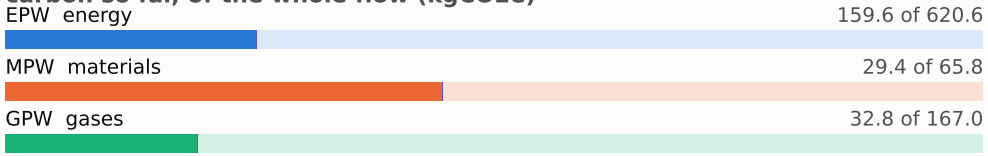


this step

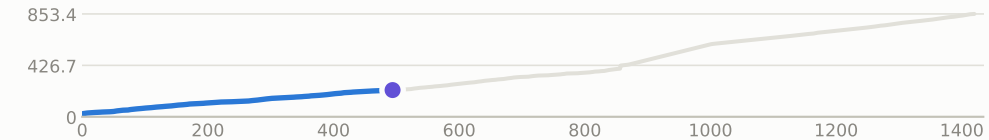
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

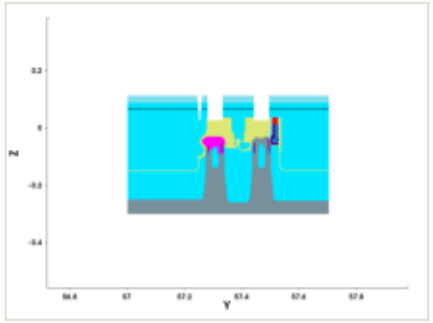
carbon so far, of the whole flow (kgCO2e)



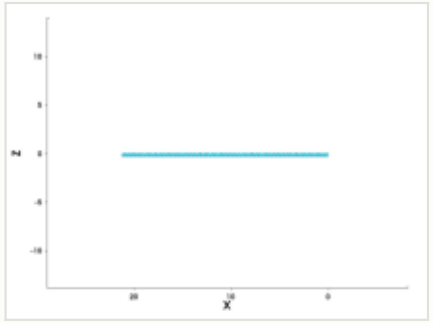
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



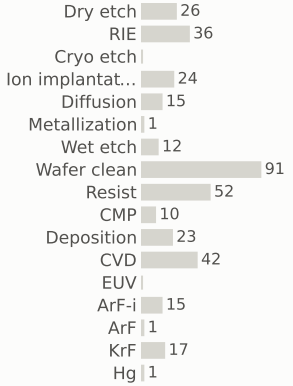
y cut at 57.351 µm



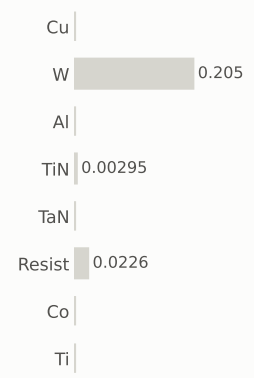
mask for this step

no mask for this step

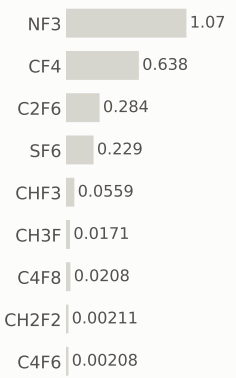
steps by process area



materials deposited (g)



gases emitted (g)



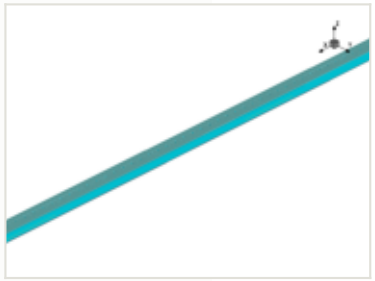
Step #5020: Spx_only

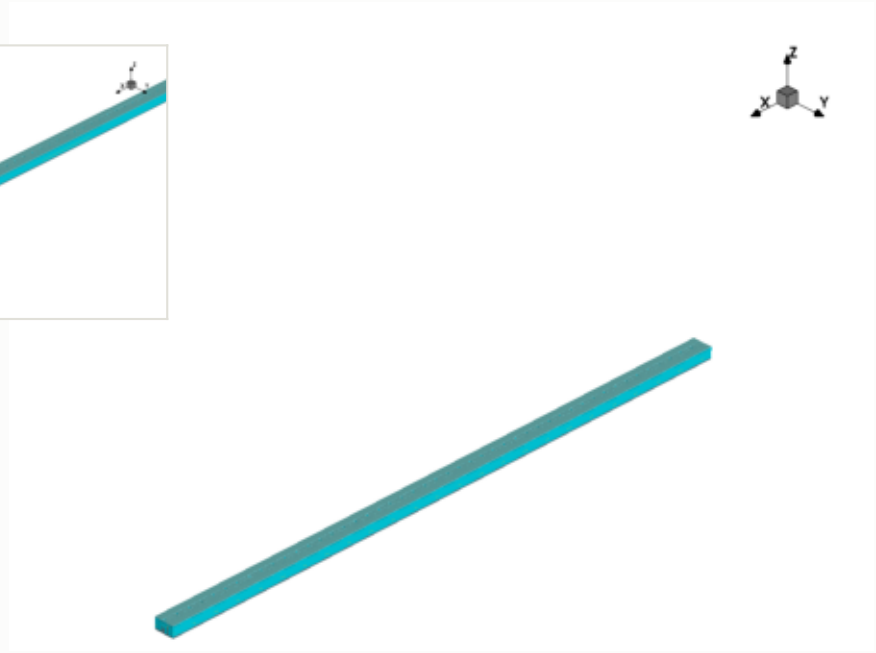
etch_anisotropic · Spx_only

0	221.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



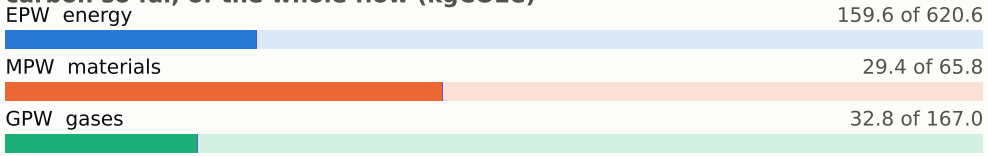


this step

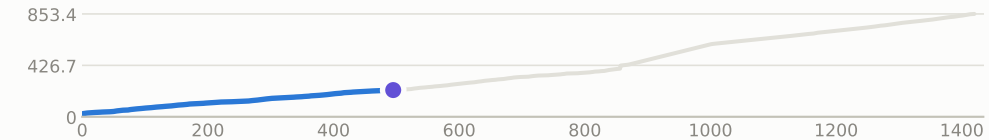
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

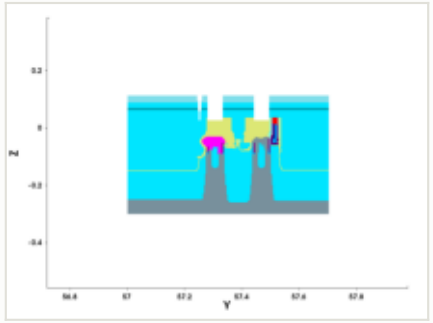
carbon so far, of the whole flow (kgCO2e)



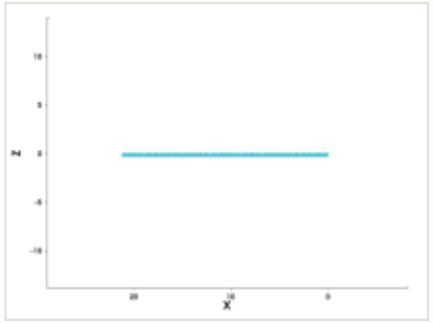
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



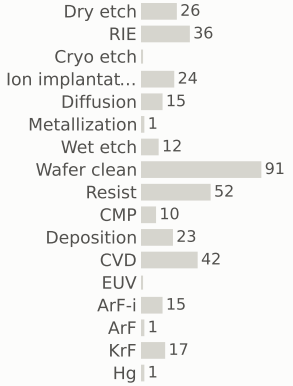
y cut at 57.351 µm



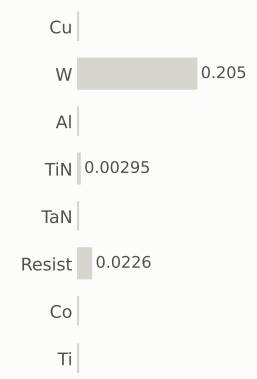
mask for this step

no mask for this step

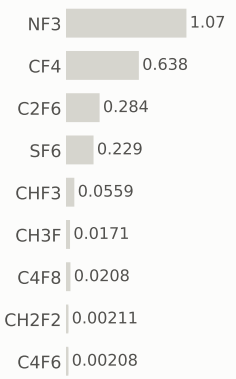
steps by process area



materials deposited (g)



gases emitted (g)



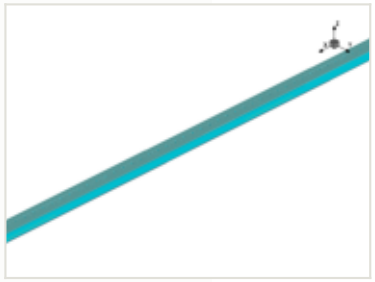
Step #5030: Spx_only

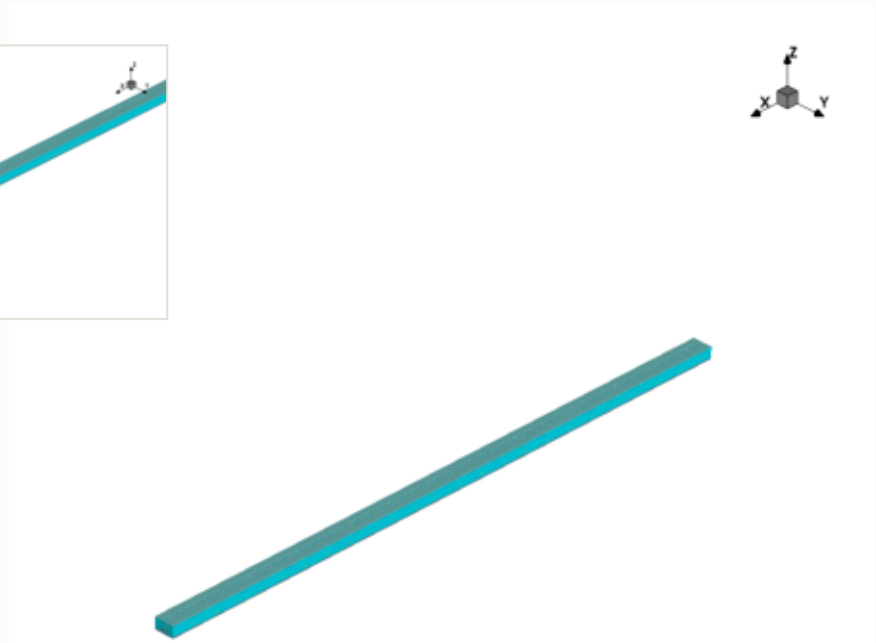
etch_anisotropic · Spx_only

0	221.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



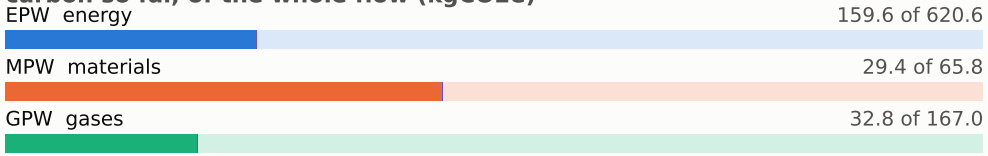


this step

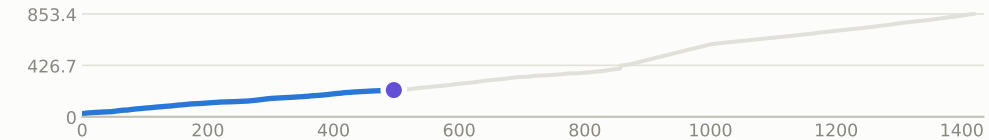
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

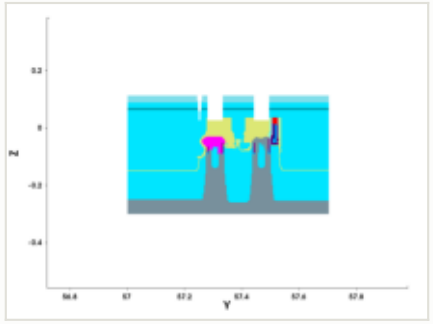
carbon so far, of the whole flow (kgCO2e)



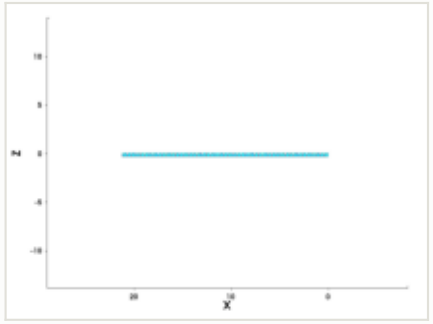
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



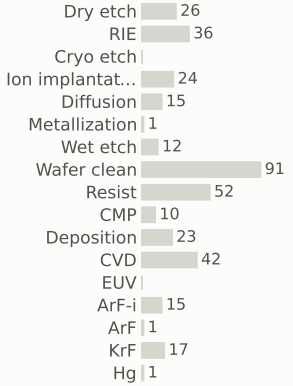
y cut at 57.351 µm



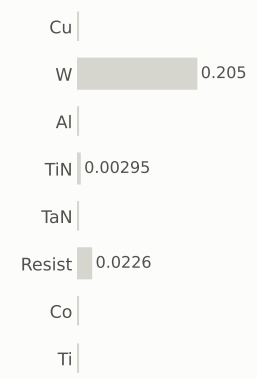
mask for this step

no mask for this step

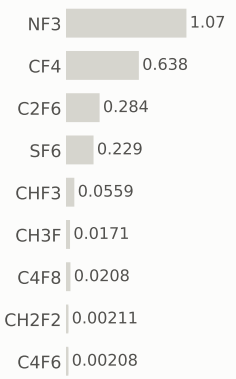
steps by process area



materials deposited (g)



gases emitted (g)



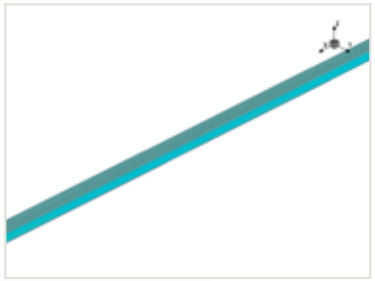
Step #5040: Spx_only

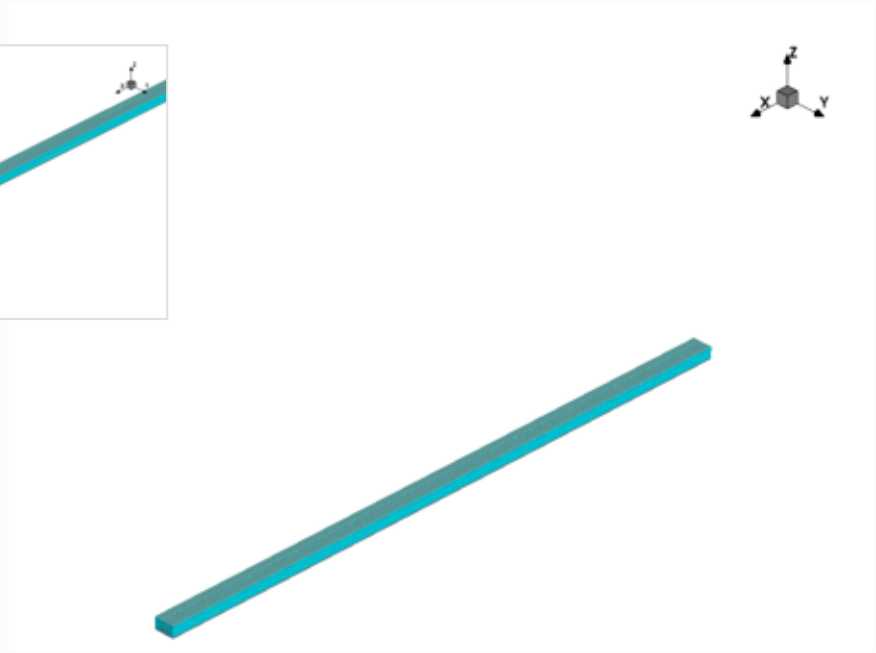
etch_anisotropic · Spx_only

0	221.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





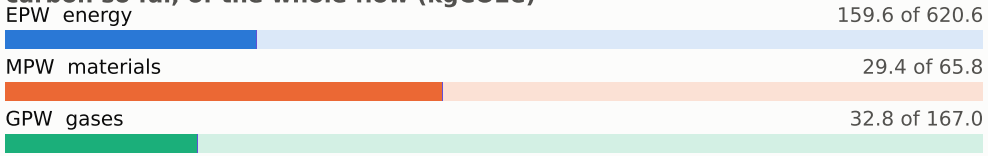
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

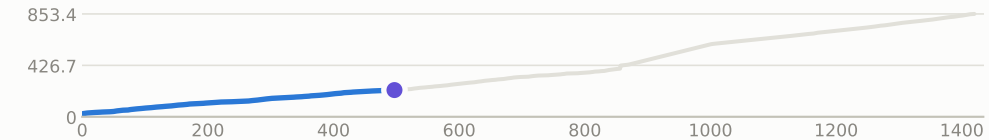
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

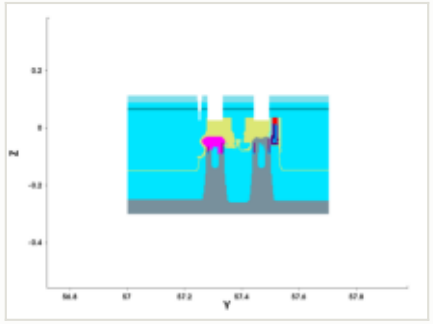
carbon so far, of the whole flow (kgCO2e)



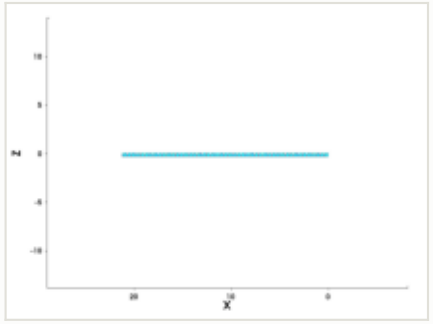
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



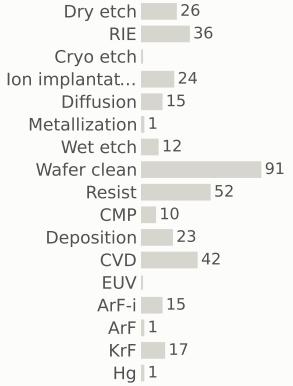
y cut at 57.351 µm



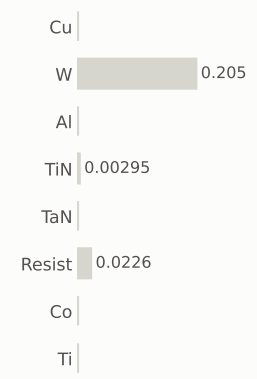
mask for this step

no mask for this step

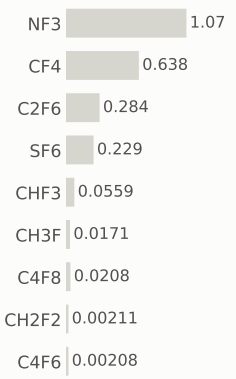
steps by process area



materials deposited (g)



gases emitted (g)



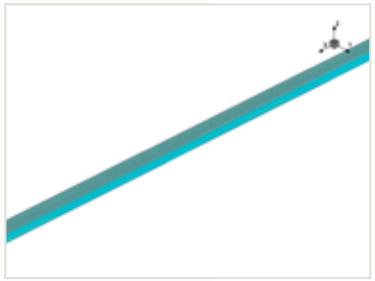
Step #5050: Spx_only

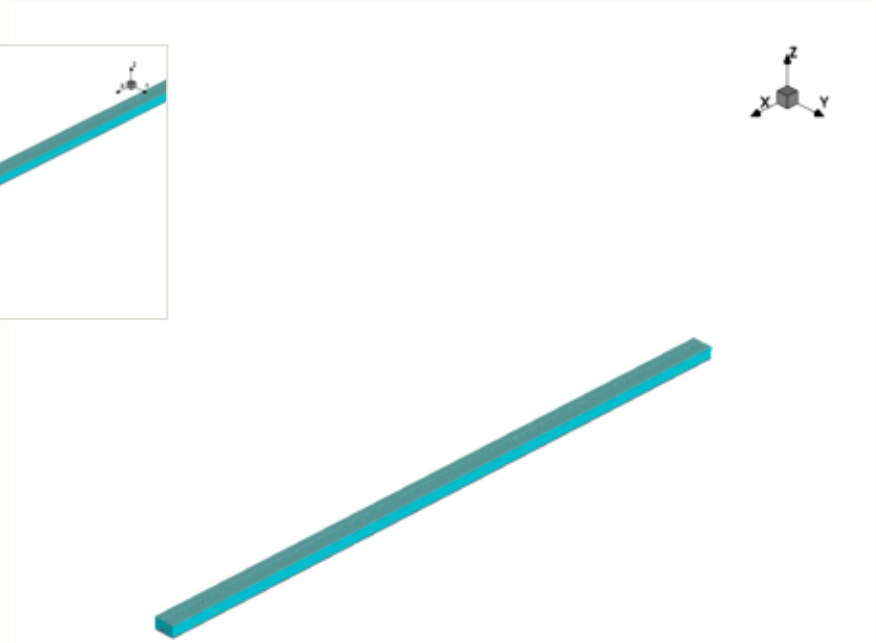
etch_ellipsoidal · Spx_only

0	221.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



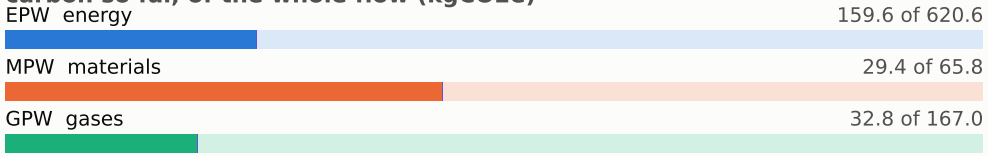


this step

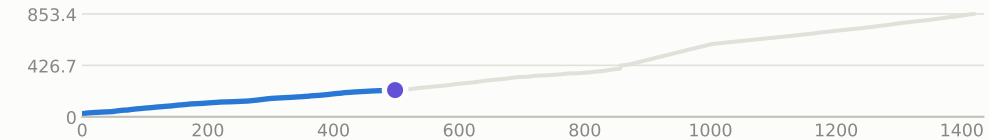
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

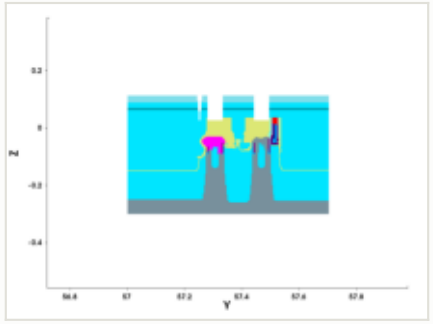
carbon so far, of the whole flow (kgCO2e)



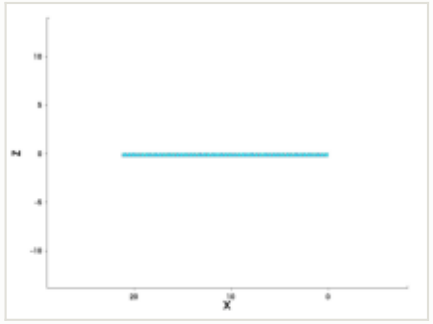
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



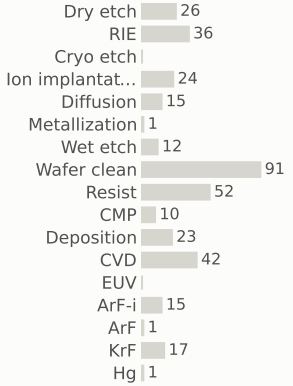
y cut at 57.351 µm



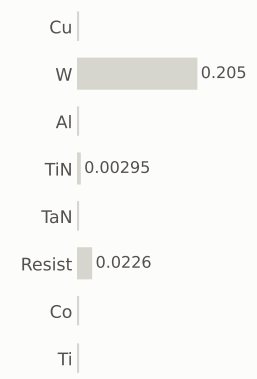
mask for this step

no mask for this step

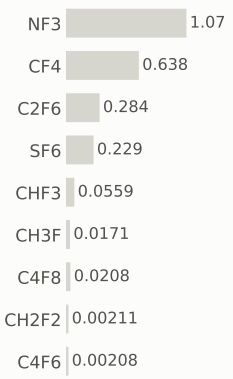
steps by process area



materials deposited (g)



gases emitted (g)



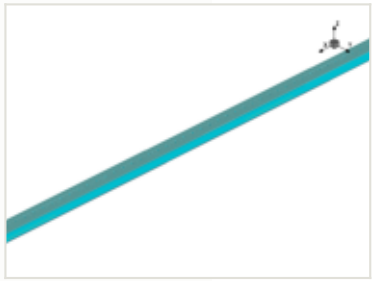
Step #5060: RIE

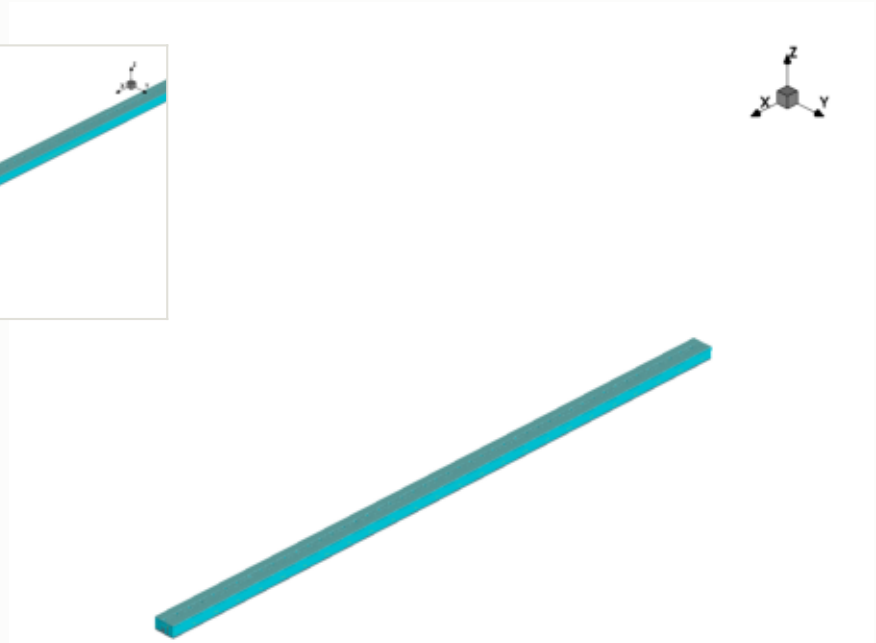
etch_anisotropic · RIE

1.04	222.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



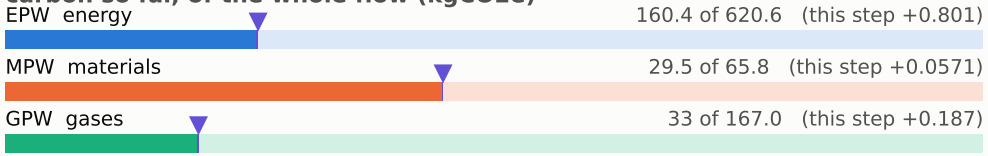


this step

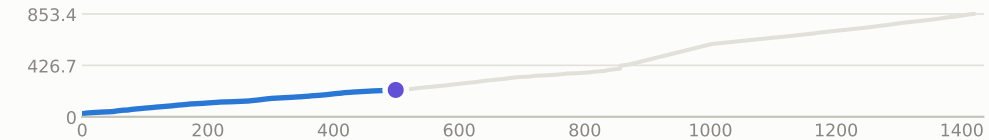
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

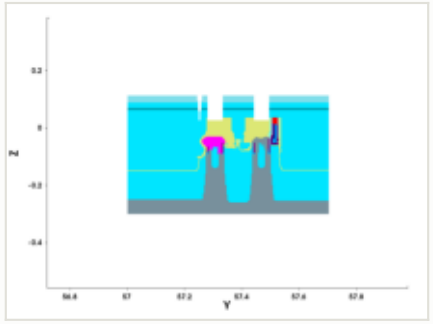
carbon so far, of the whole flow (kgCO2e)



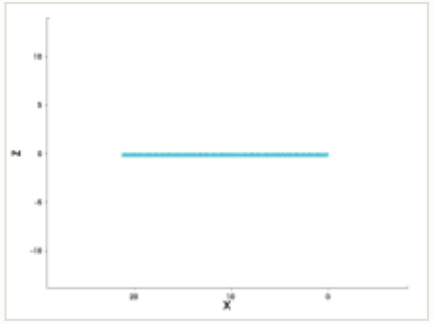
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



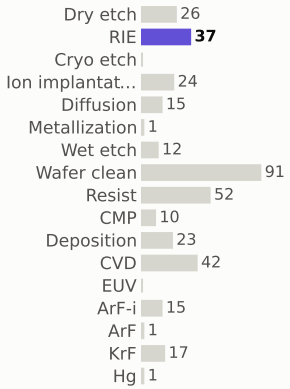
y cut at 57.351 µm



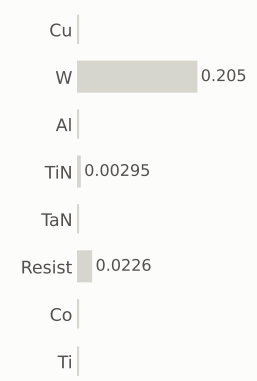
mask for this step

no mask for this step

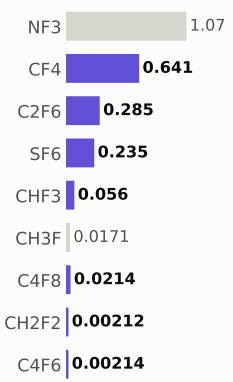
steps by process area



materials deposited (g)



gases emitted (g)



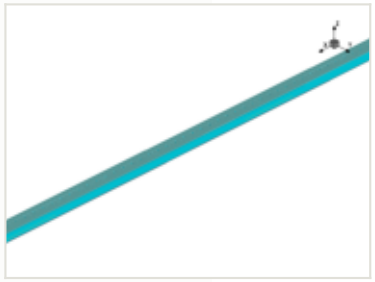
Step #5070: Spx_only

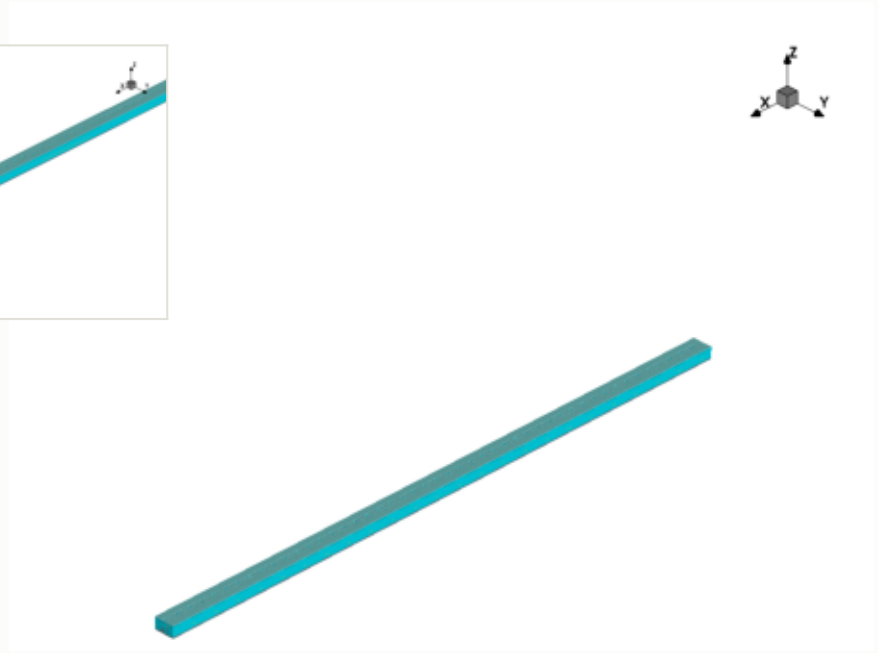
etch_strip · Spx_only

0	222.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





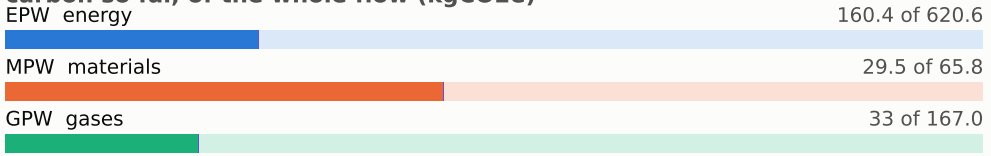
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

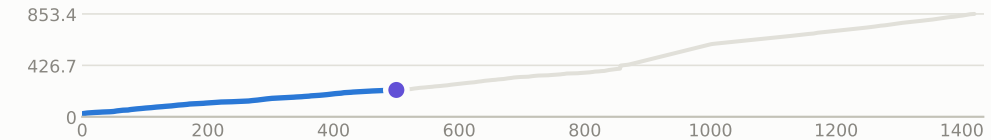
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

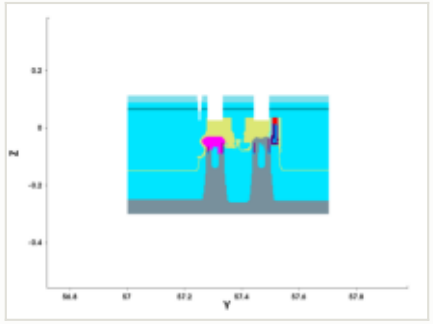
carbon so far, of the whole flow (kgCO2e)



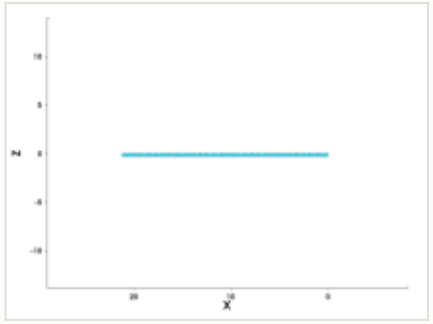
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



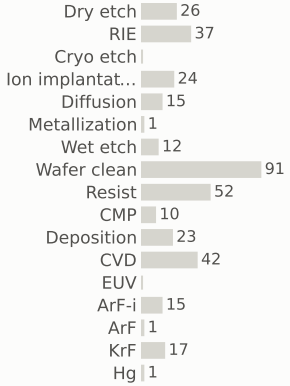
y cut at 57.351 µm



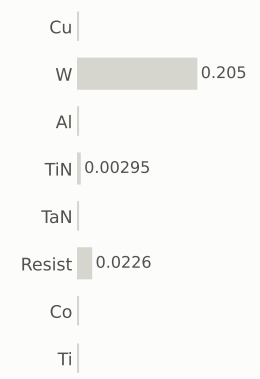
mask for this step

no mask for this step

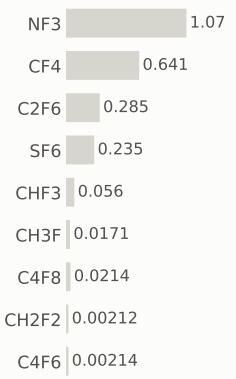
steps by process area



materials deposited (g)



gases emitted (g)



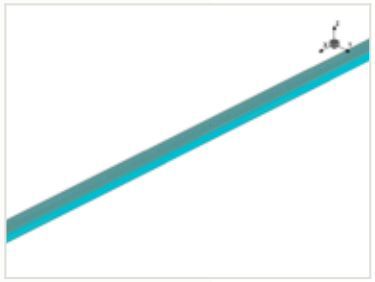
Step #5080: Wafer_clean

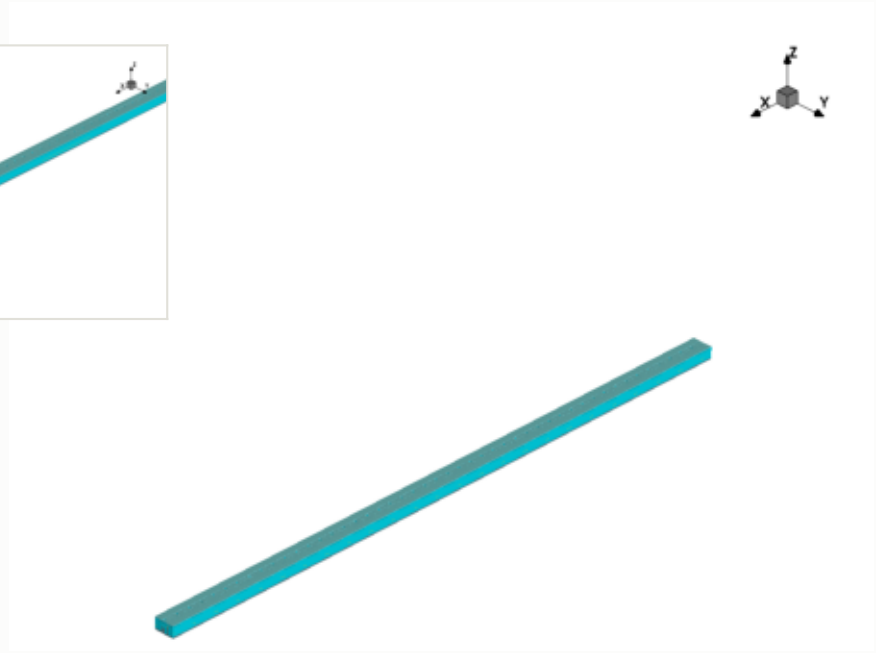
clean · Wafer_clean

0.169	223.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





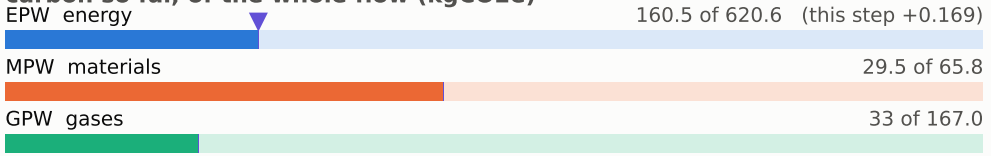
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

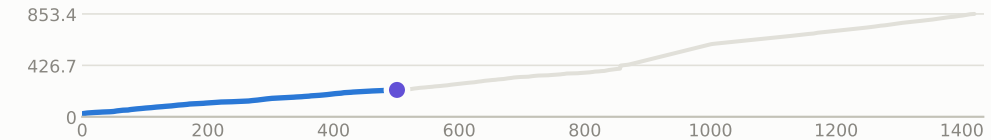
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

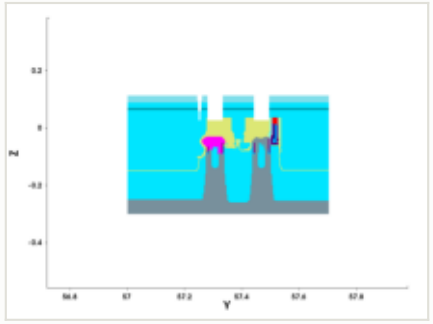
carbon so far, of the whole flow (kgCO2e)



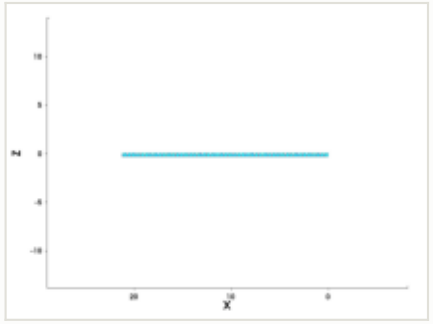
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



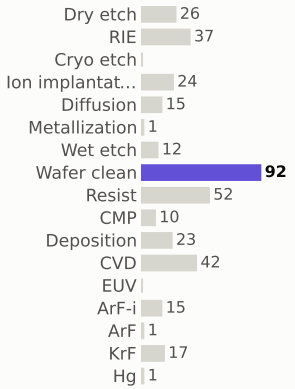
y cut at 57.351 µm



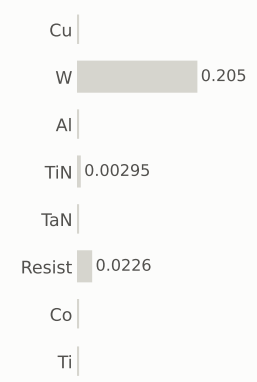
mask for this step

no mask for this step

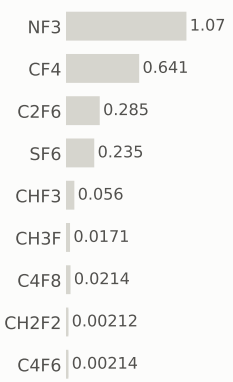
steps by process area



materials deposited (g)



gases emitted (g)



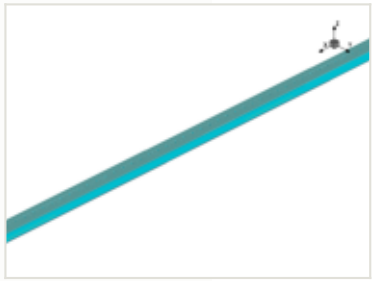
Step #5090: Resist

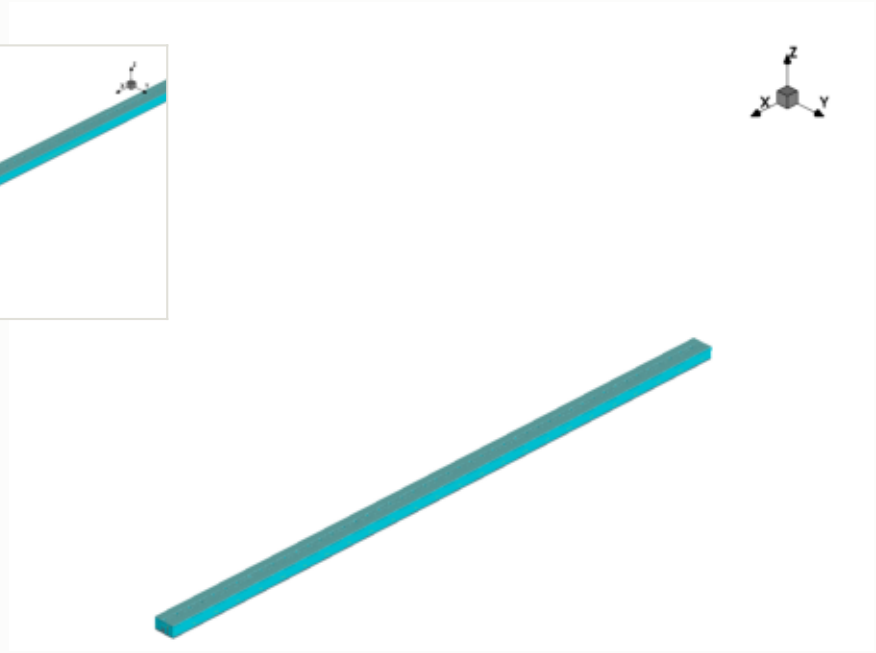
depo_fill · Resist

0.169	223.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



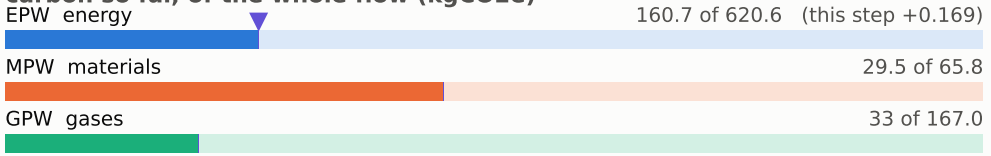


this step

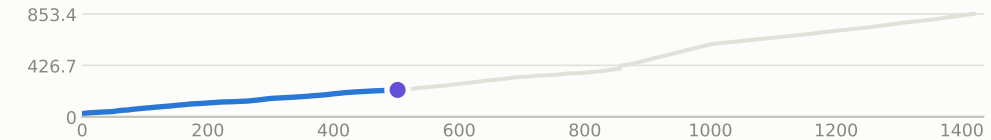
process area	Resist
material	—
thickness	0.12 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

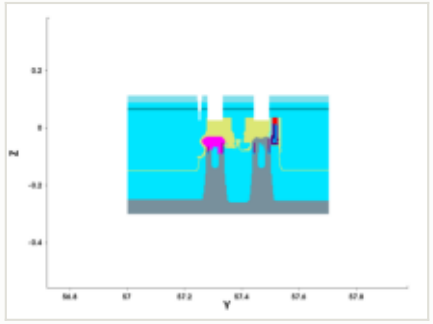
carbon so far, of the whole flow (kgCO2e)



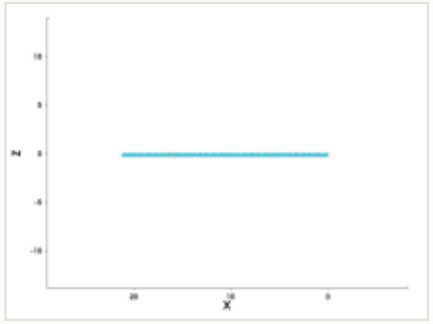
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



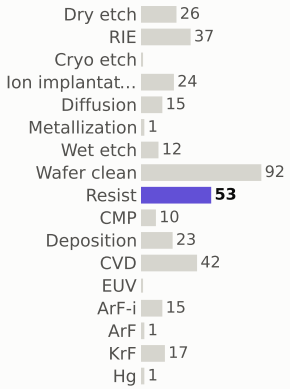
y cut at 57.351 µm



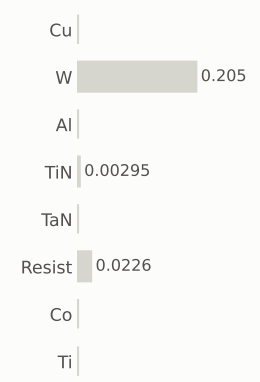
mask for this step

no mask for this step

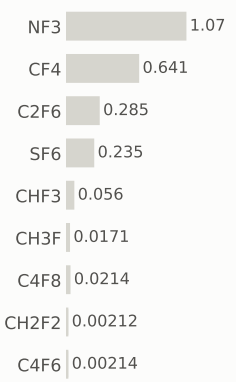
steps by process area



materials deposited (g)



gases emitted (g)



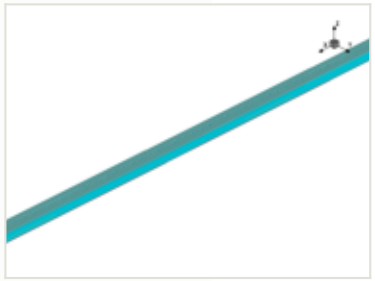
Step #5100: Spx_only

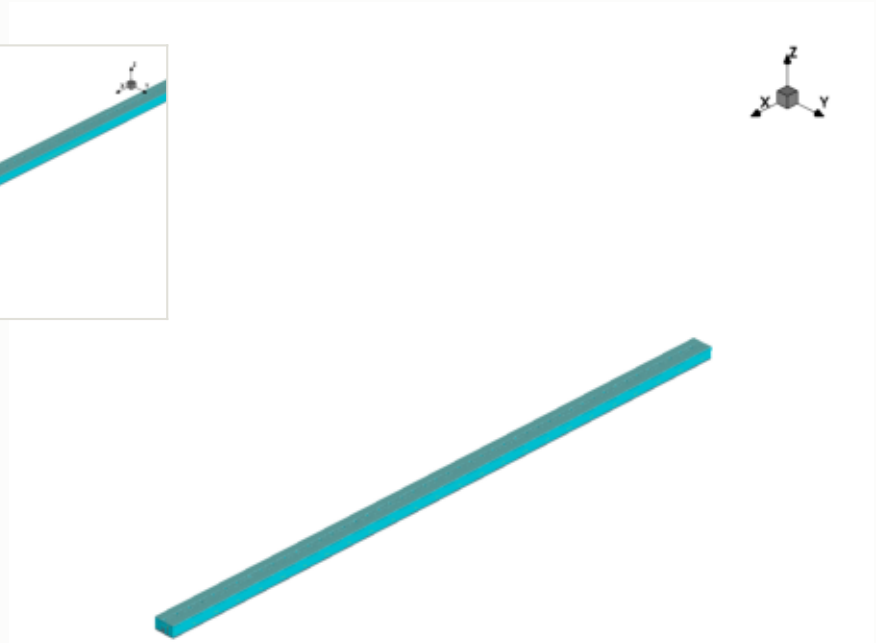
depo_fill · Spx_only

0	223.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





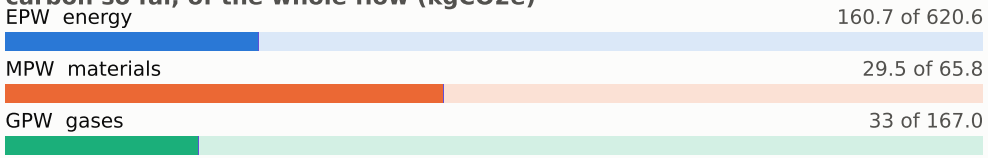
this step

process area	Spx_only
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

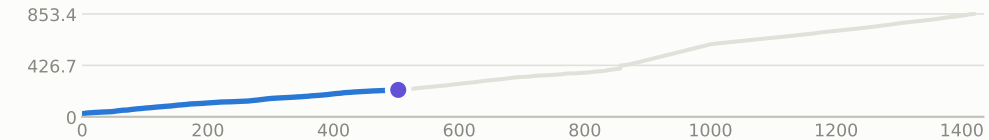
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

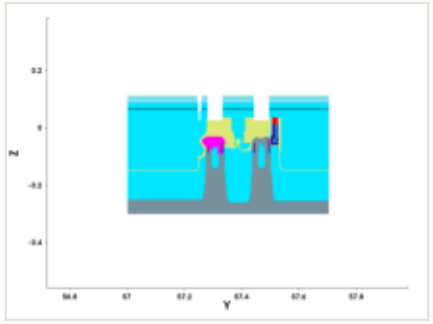
carbon so far, of the whole flow (kgCO2e)



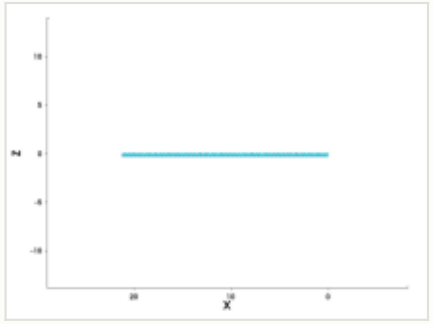
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



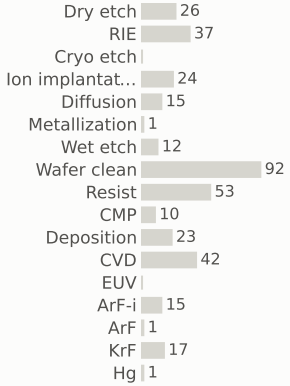
y cut at 57.351 µm



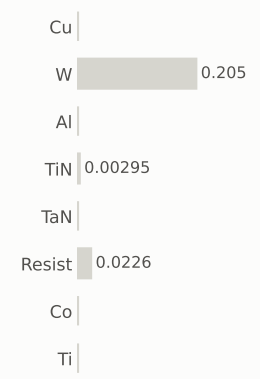
mask for this step

no mask for this step

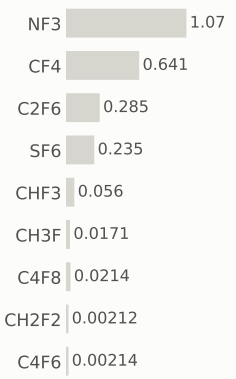
steps by process area



materials deposited (g)



gases emitted (g)



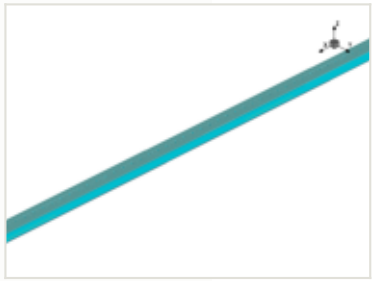
Step #5110: CVD

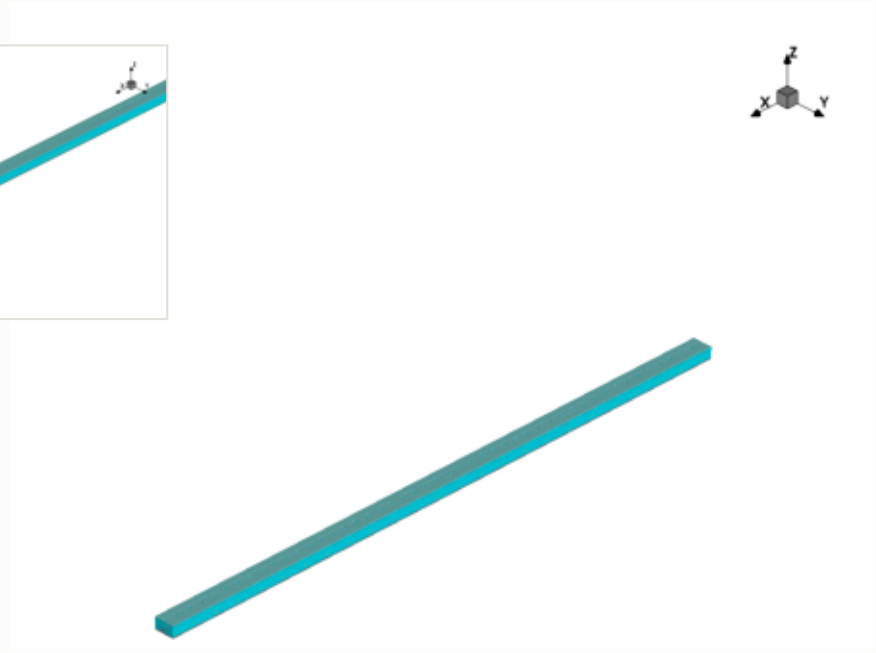
depo_fill · CVD

1.31	224.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



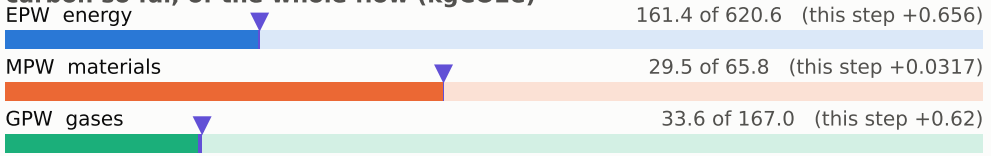


this step

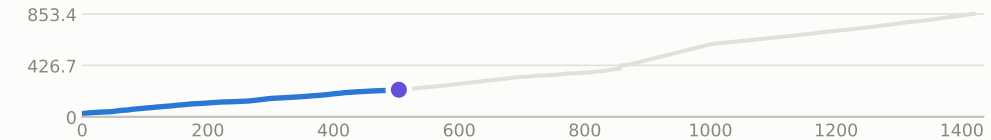
process area	CVD
material	—
thickness	0.035 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

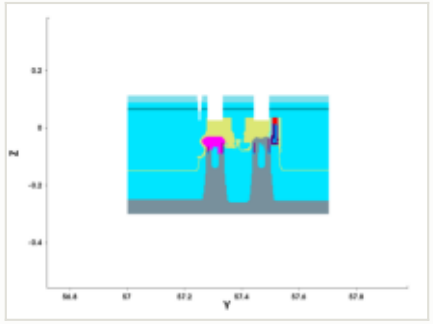
carbon so far, of the whole flow (kgCO2e)



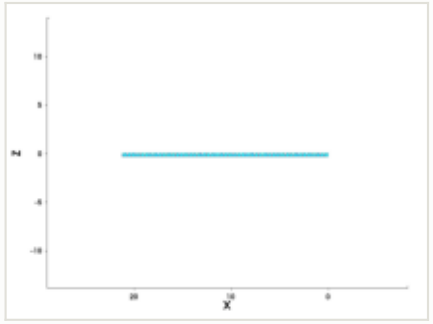
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



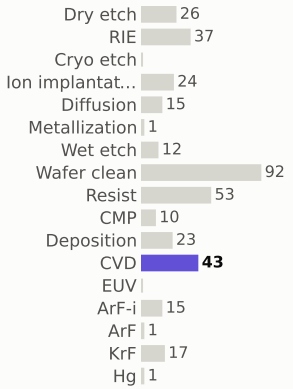
y cut at 57.351 µm



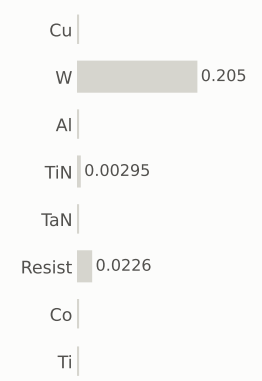
mask for this step

no mask for this step

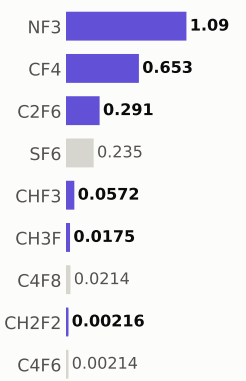
steps by process area



materials deposited (g)



gases emitted (g)



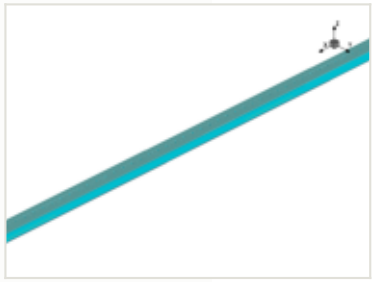
Step #5120: Wafer_clean

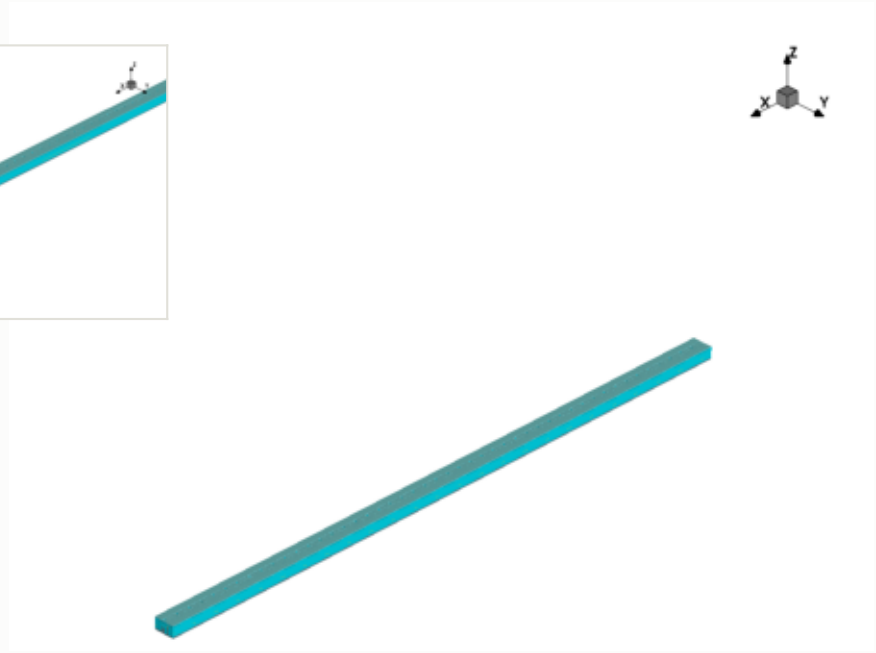
clean · Wafer_clean

0.169	224.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

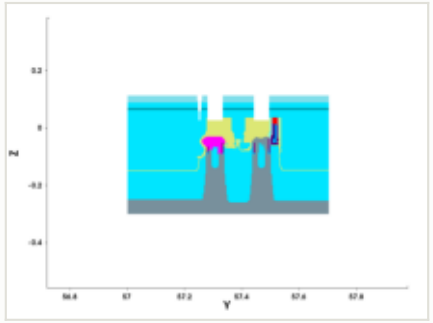
3D structure after this step

zoomed to fit

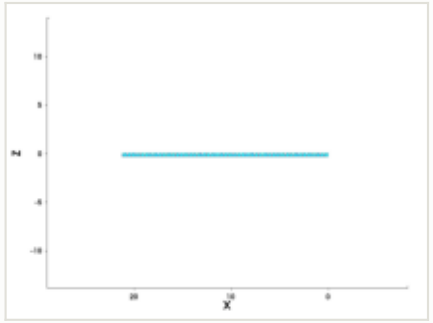




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

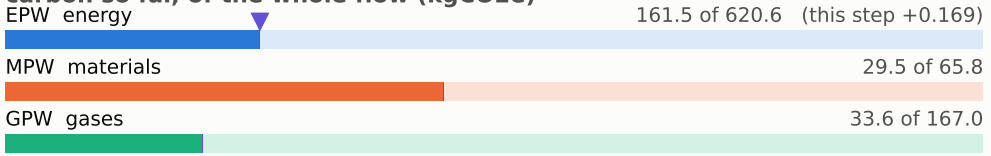
no mask for this step

this step

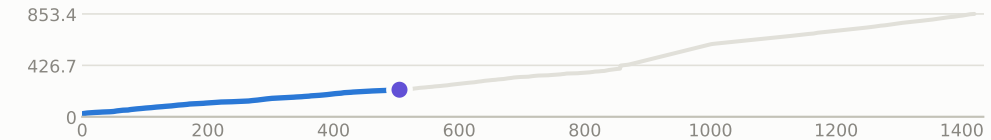
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

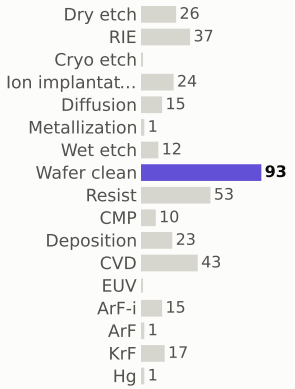
carbon so far, of the whole flow (kgCO2e)



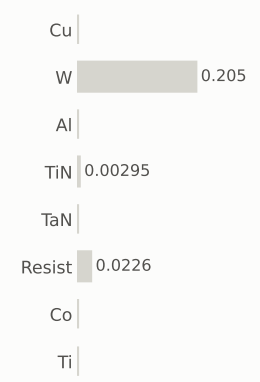
embodied carbon up to each step (kgCO2e per wafer)



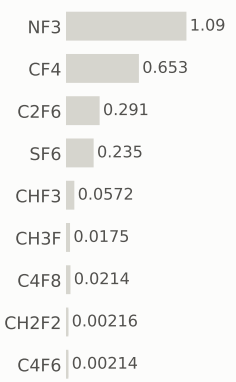
steps by process area



materials deposited (g)



gases emitted (g)



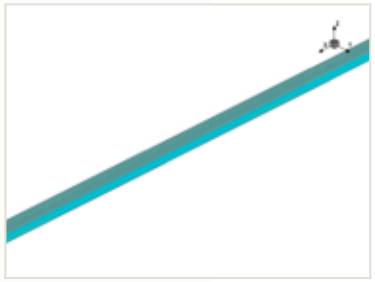
Step #5130: Resist

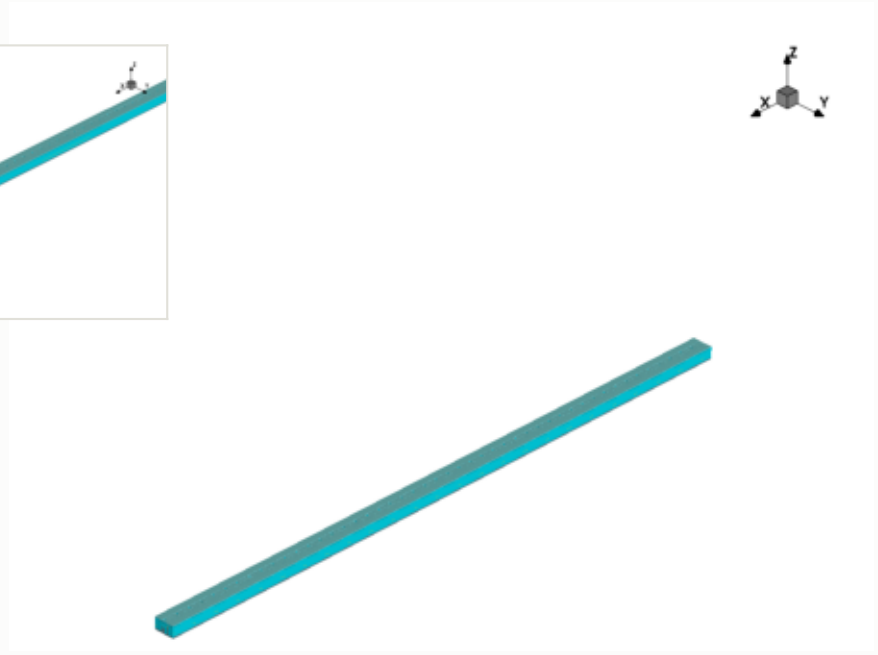
resist · Resist

0.169	224.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





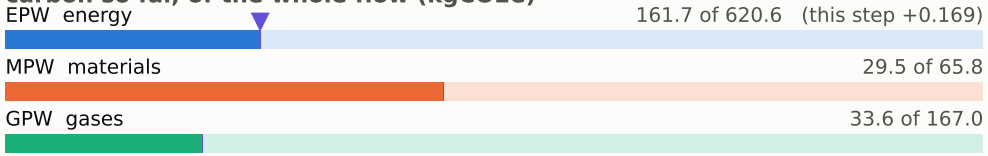
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

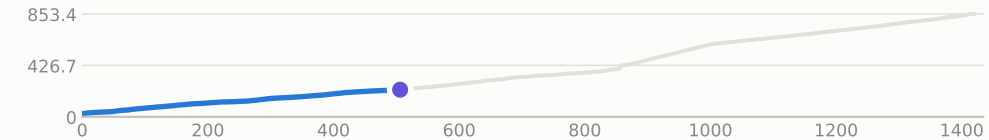
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

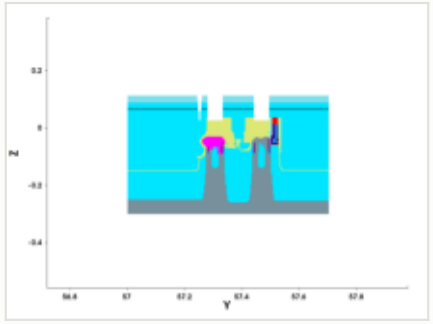
carbon so far, of the whole flow (kgCO2e)



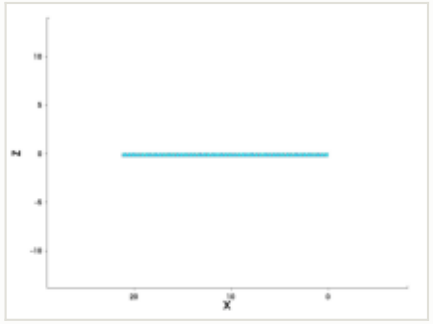
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



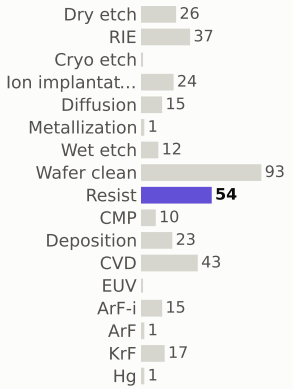
y cut at 57.351 µm



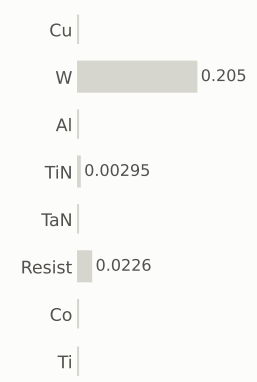
mask for this step

no mask for this step

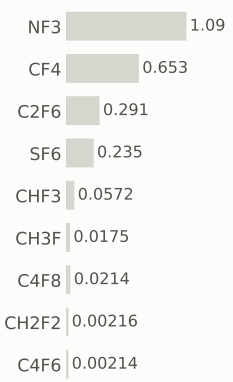
steps by process area



materials deposited (g)



gases emitted (g)



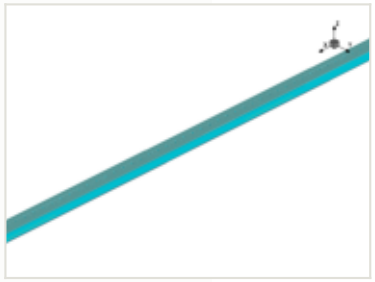
Step #5140: ArF-i

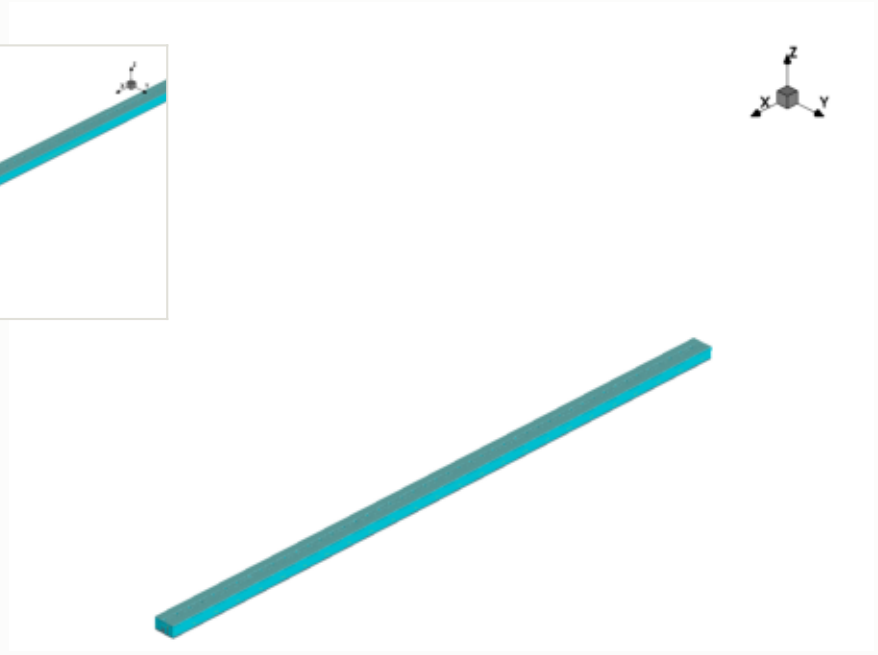
litho_patterning · ArF-i

0.217	225.0	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



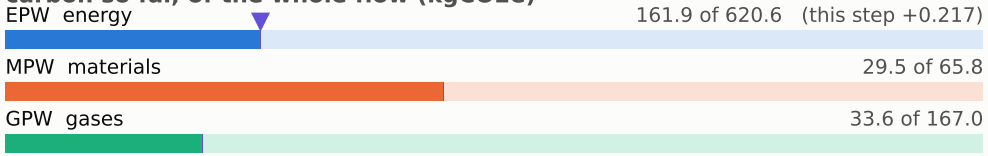


this step

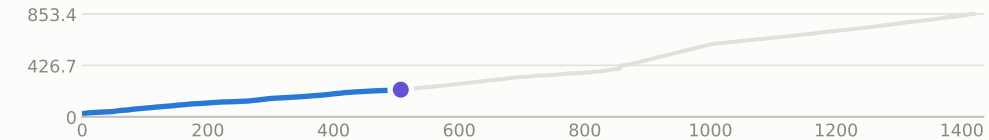
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

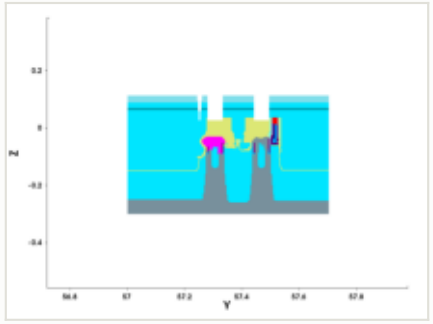
carbon so far, of the whole flow (kgCO2e)



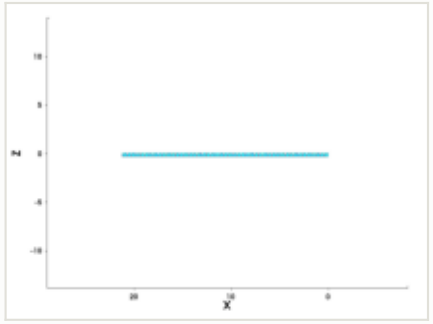
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

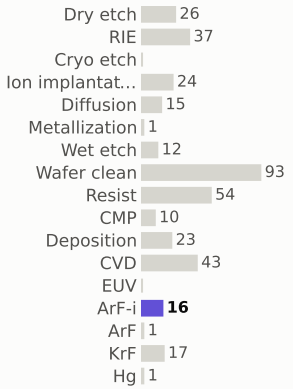


y cut at 57.351 µm

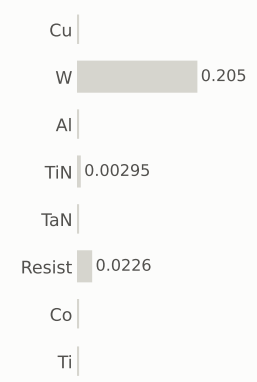


mask: layer 2016:0 (GDSII)

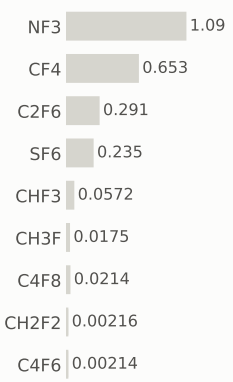
steps by process area



materials deposited (g)



gases emitted (g)



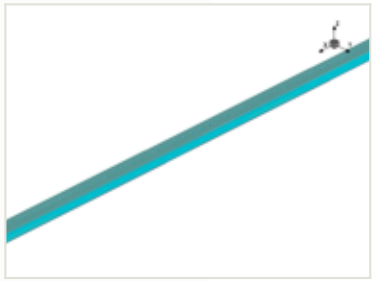
Step #5150: Spx_only

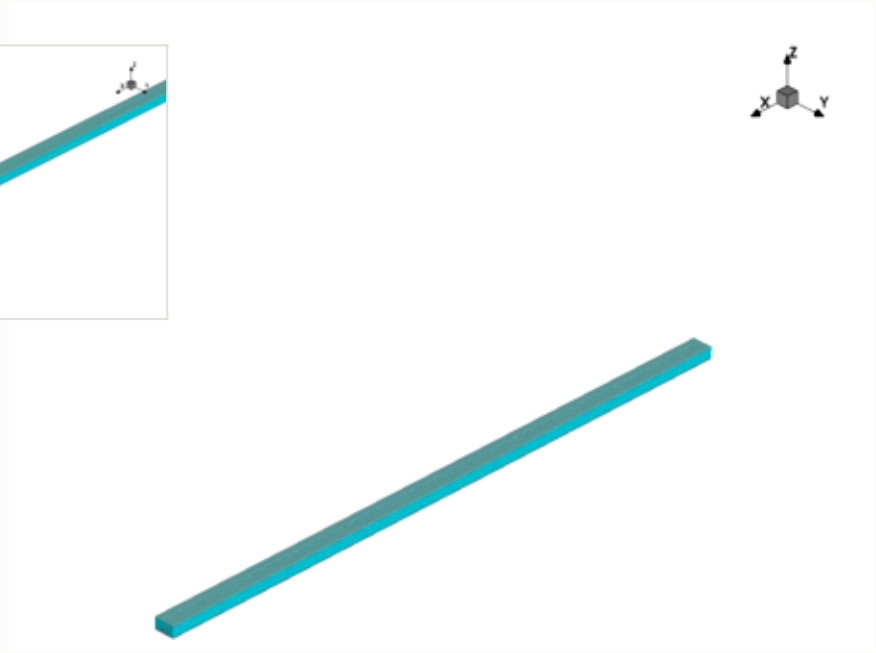
etch_anisotropic · Spx_only

0	225.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



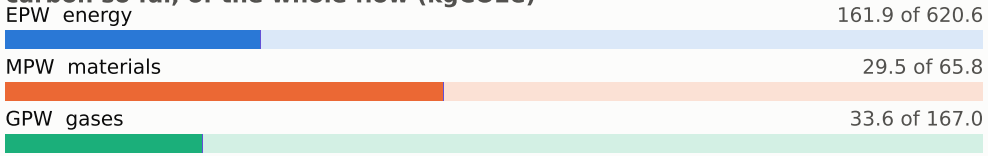


this step

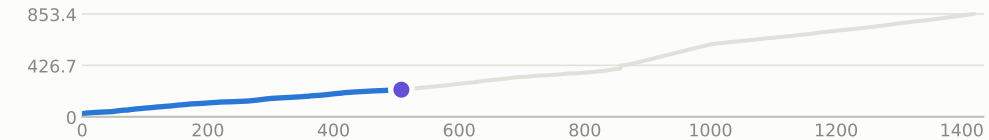
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

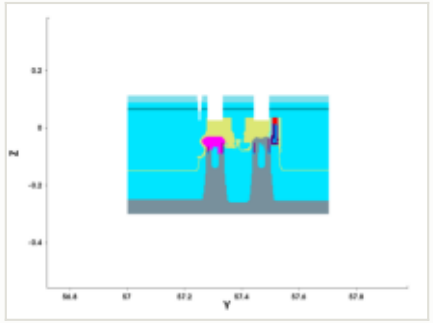
carbon so far, of the whole flow (kgCO2e)



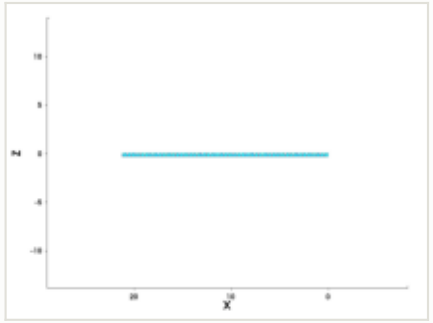
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



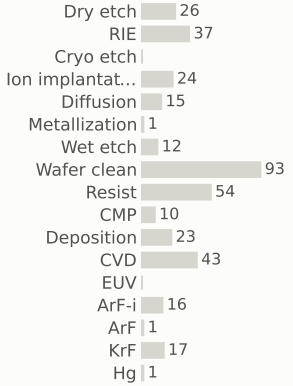
y cut at 57.351 µm



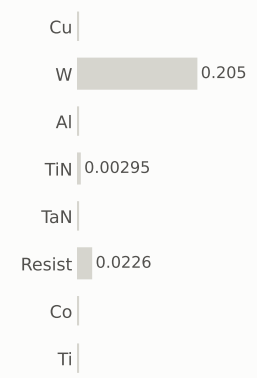
mask for this step

no mask for this step

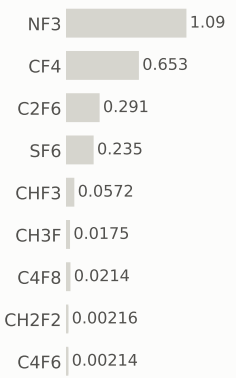
steps by process area



materials deposited (g)



gases emitted (g)



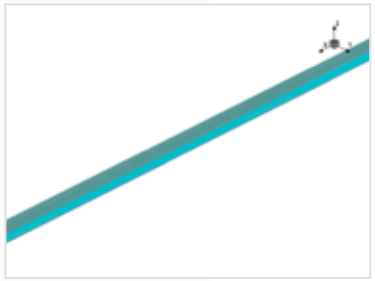
Step #5160: Spx_only

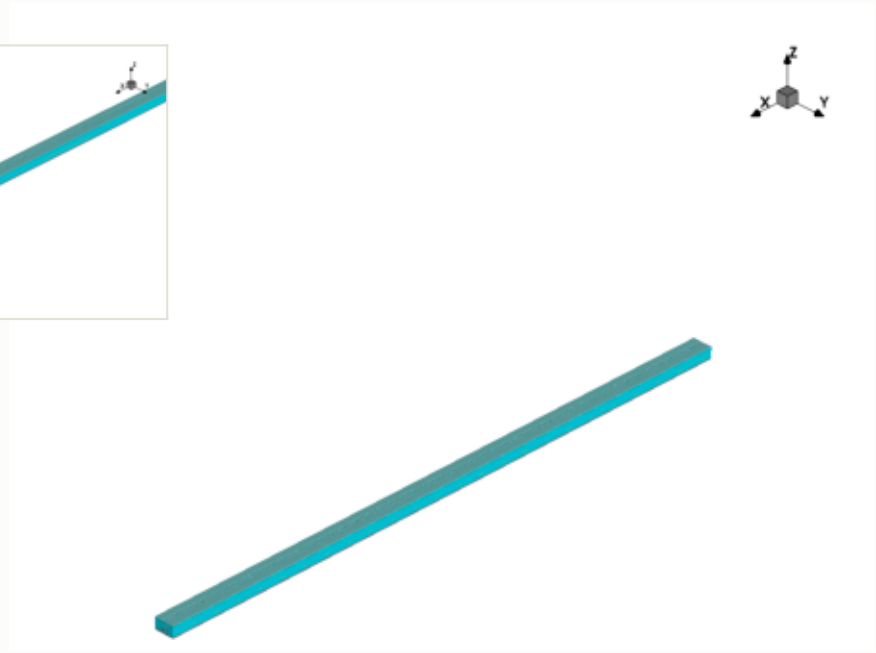
etch_anisotropic · Spx_only

0	225.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



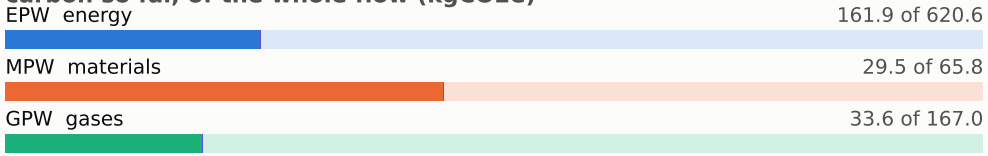


this step

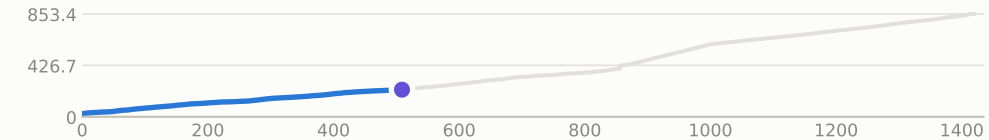
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

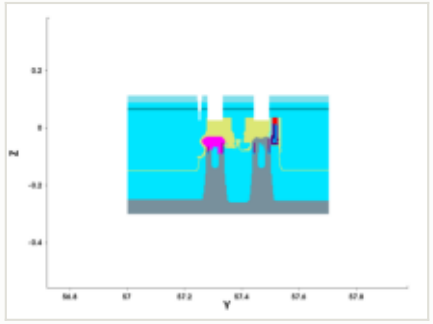
carbon so far, of the whole flow (kgCO2e)



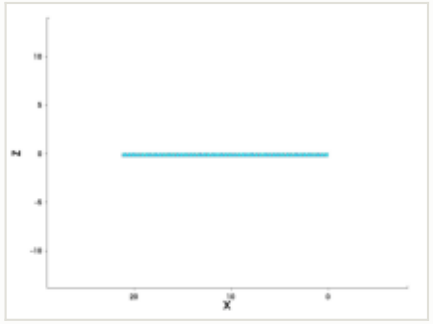
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



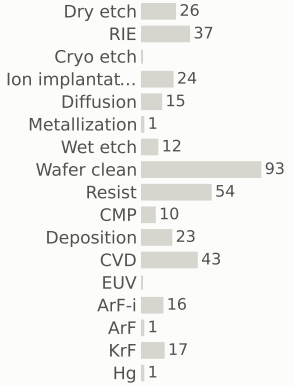
y cut at 57.351 µm



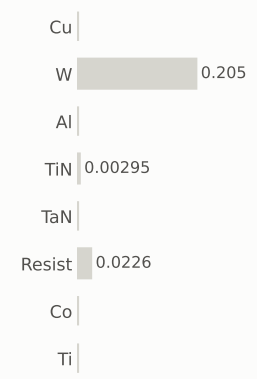
mask for this step

no mask for this step

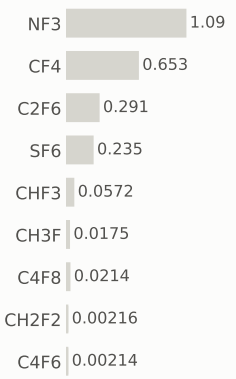
steps by process area



materials deposited (g)



gases emitted (g)



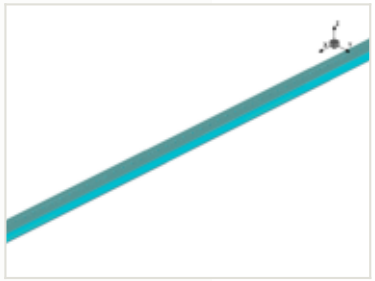
Step #5170: Spx_only

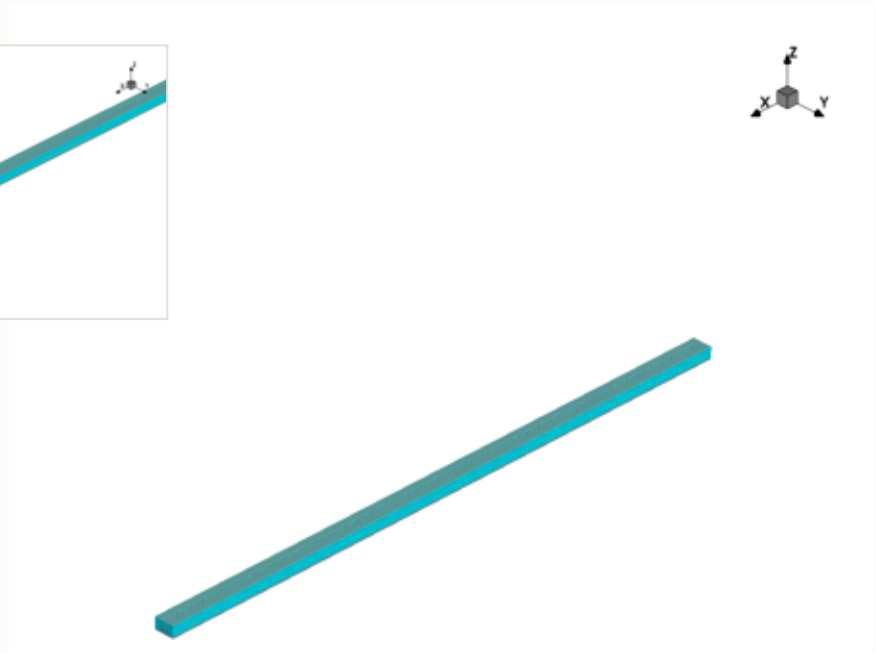
etch_anisotropic · Spx_only

0	225.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



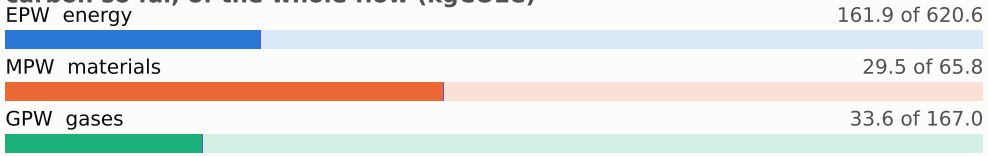


this step

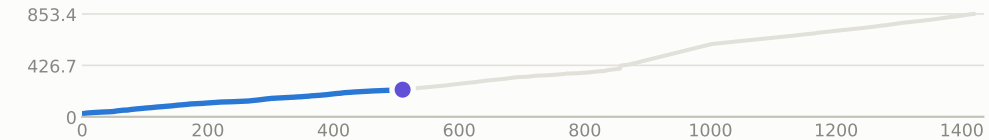
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

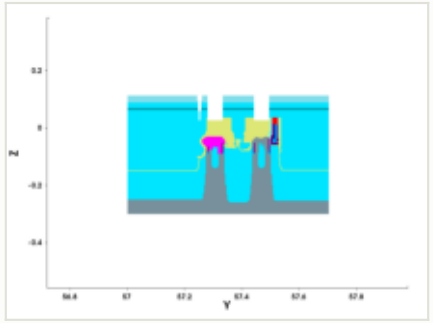
carbon so far, of the whole flow (kgCO2e)



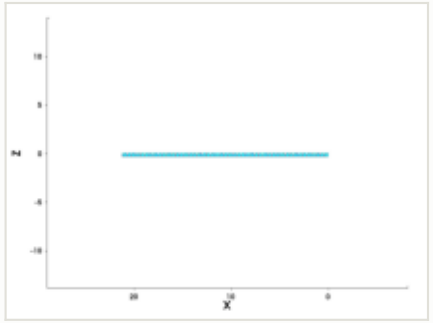
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



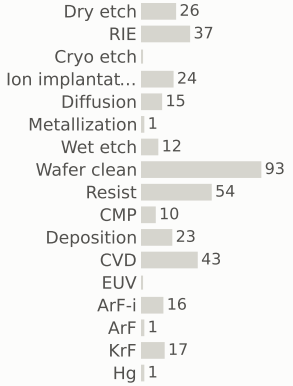
y cut at 57.351 µm



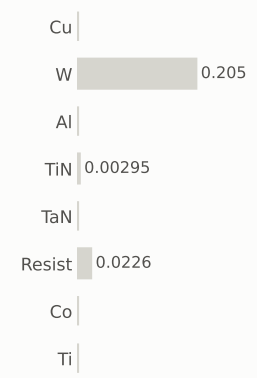
mask for this step

no mask for this step

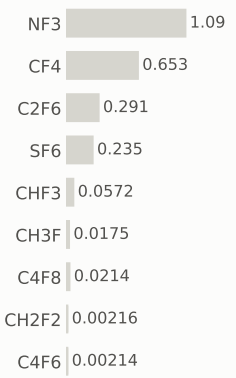
steps by process area



materials deposited (g)



gases emitted (g)



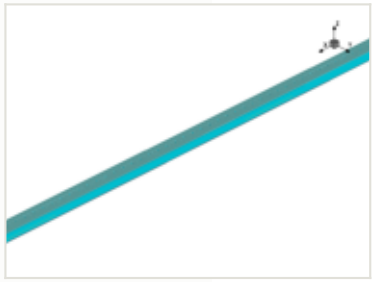
Step #5180: Spx_only

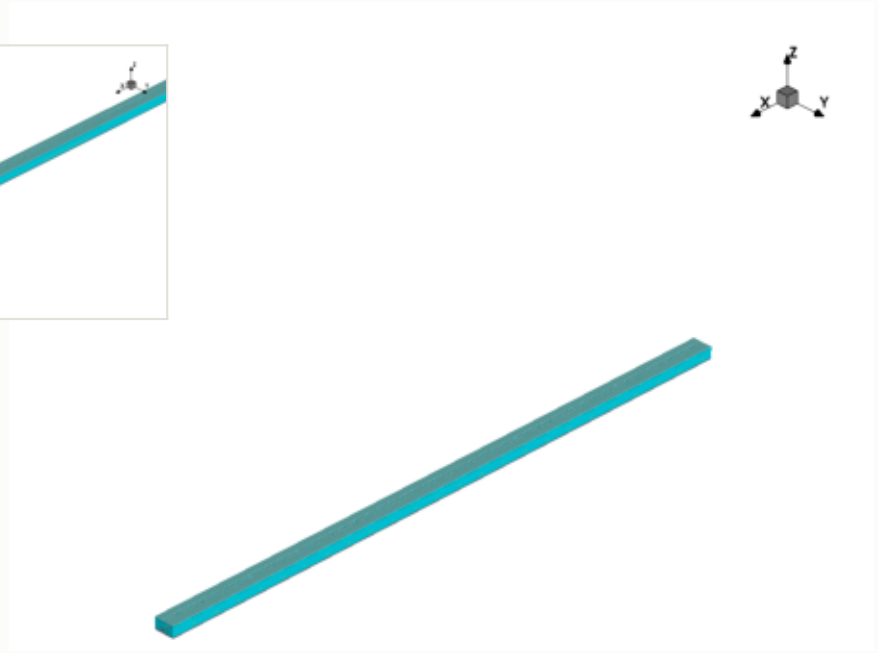
etch_anisotropic · Spx_only

0	225.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





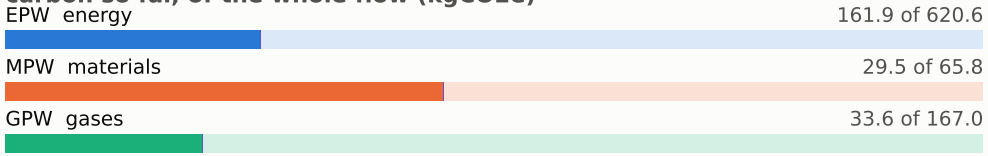
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

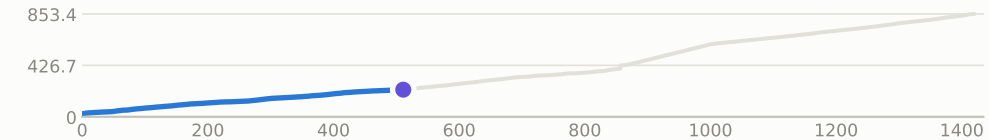
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

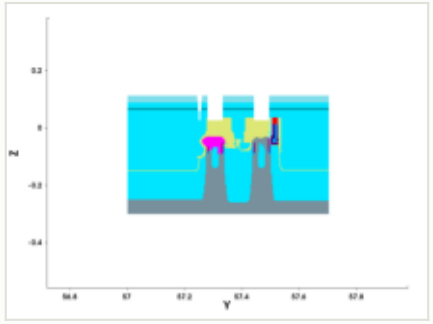
carbon so far, of the whole flow (kgCO2e)



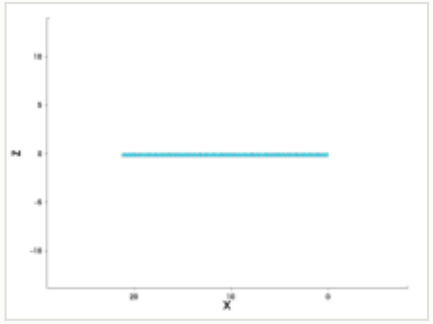
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



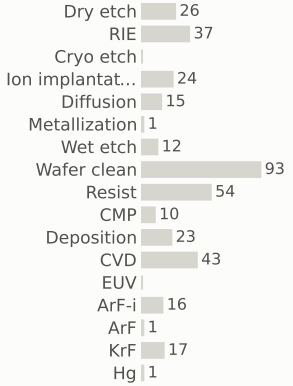
y cut at 57.351 µm



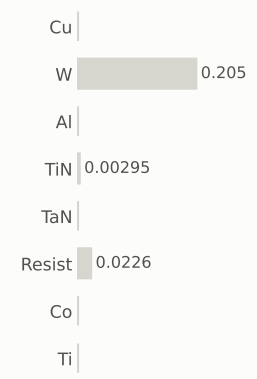
mask for this step

no mask for this step

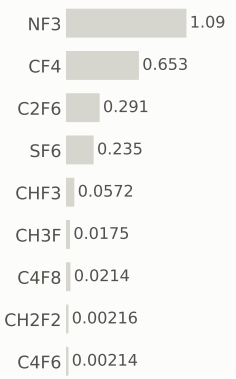
steps by process area



materials deposited (g)



gases emitted (g)



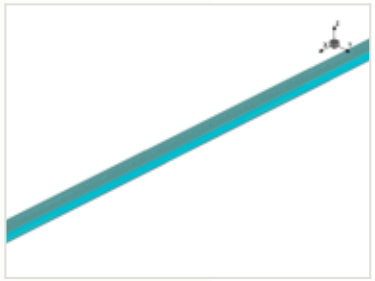
Step #5190: Spx_only

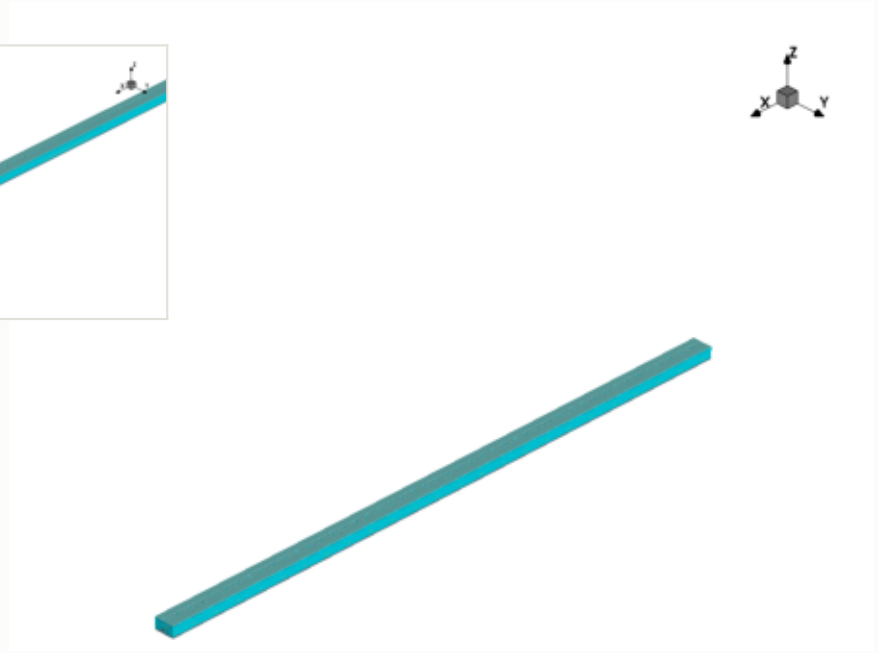
etch_anisotropic · Spx_only

0	225.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



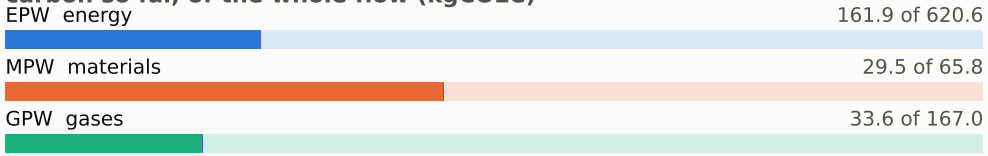


this step

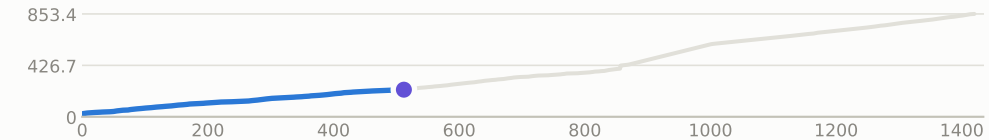
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

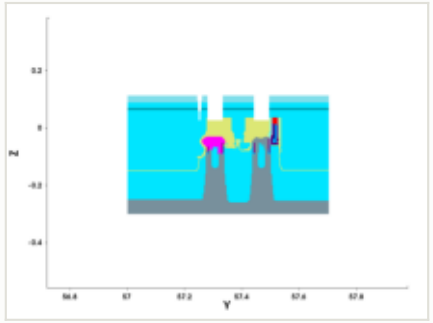
carbon so far, of the whole flow (kgCO2e)



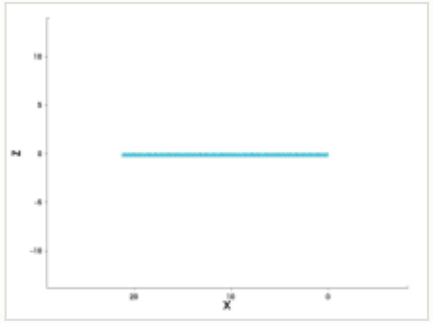
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



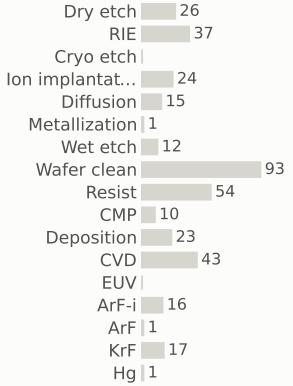
y cut at 57.351 µm



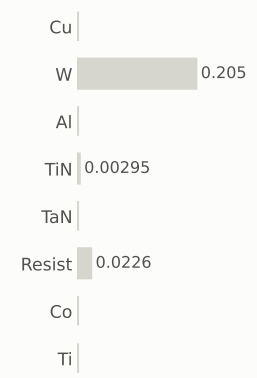
mask for this step

no mask for this step

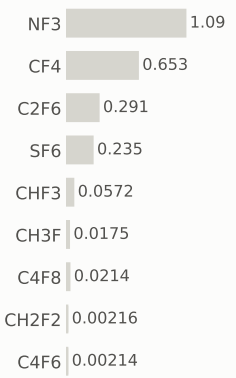
steps by process area



materials deposited (g)



gases emitted (g)



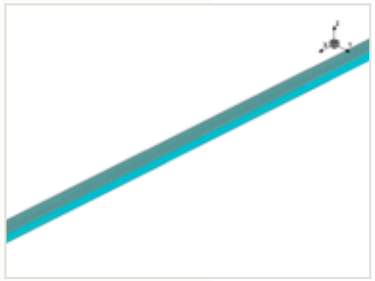
Step #5200: Spx_only

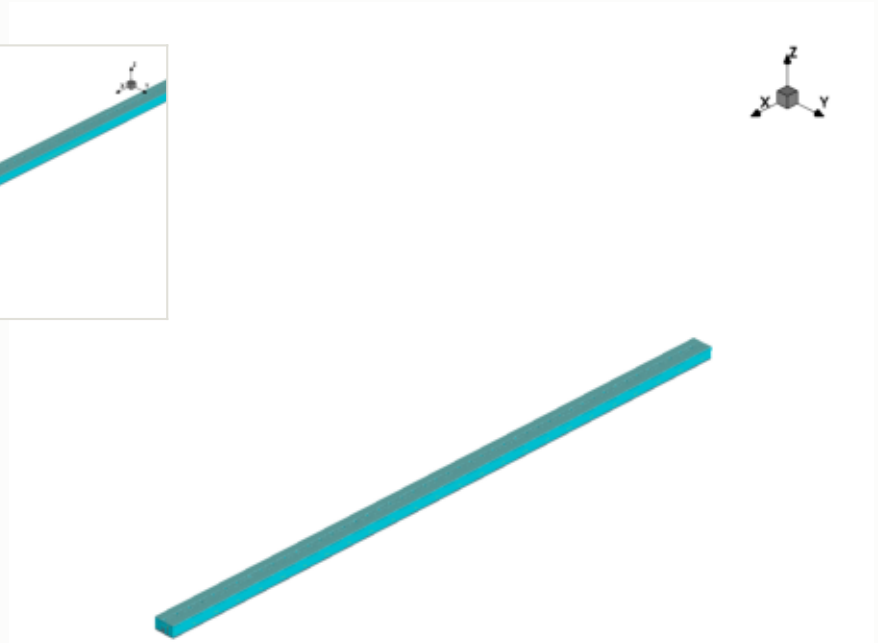
etch_ellipsoidal · Spx_only

0	225.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





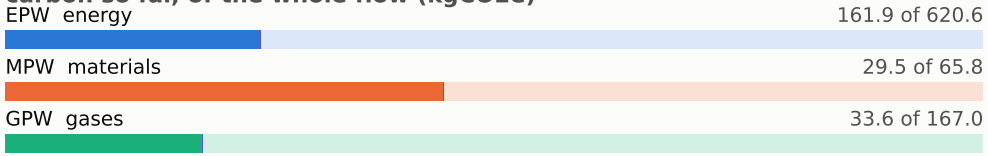
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

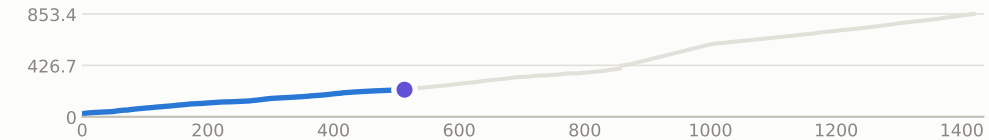
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

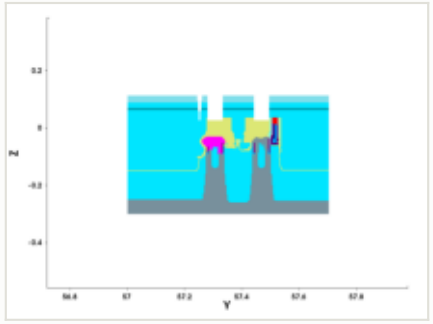
carbon so far, of the whole flow (kgCO2e)



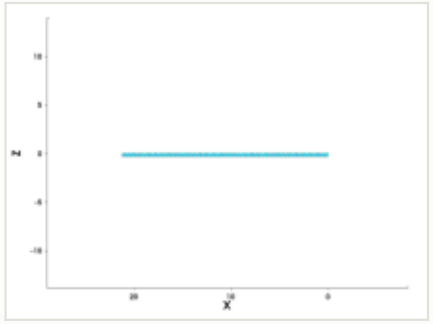
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



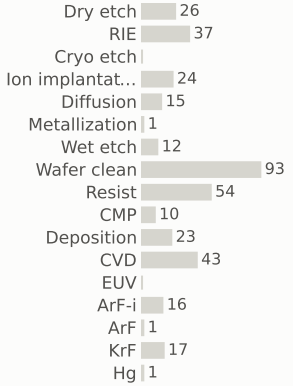
y cut at 57.351 µm



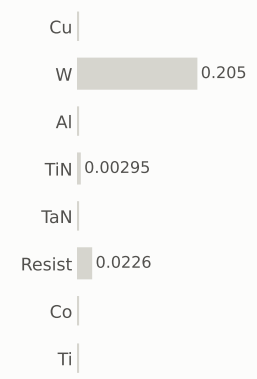
mask for this step

no mask for this step

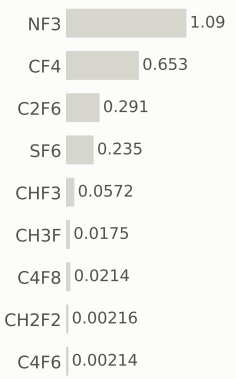
steps by process area



materials deposited (g)



gases emitted (g)



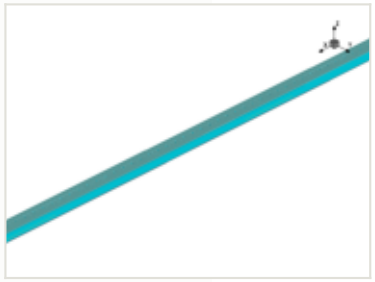
Step #5210: RIE

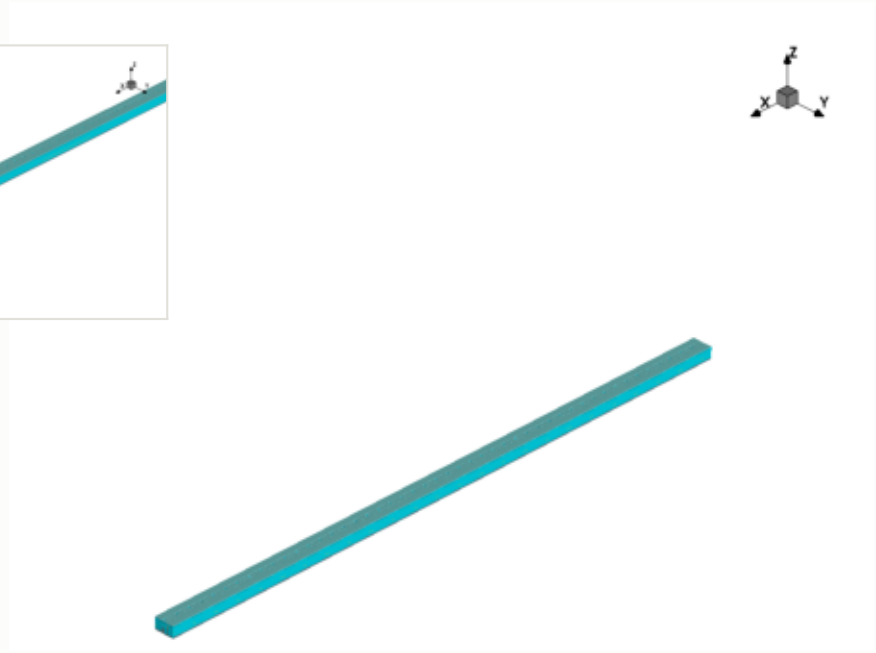
etch_anisotropic · RIE

1.04	226.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



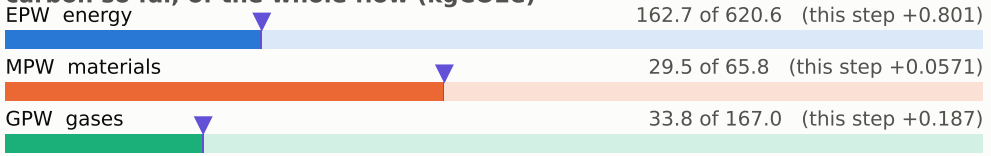


this step

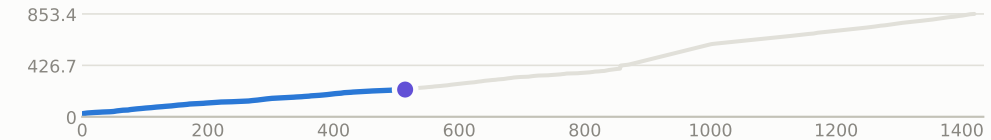
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

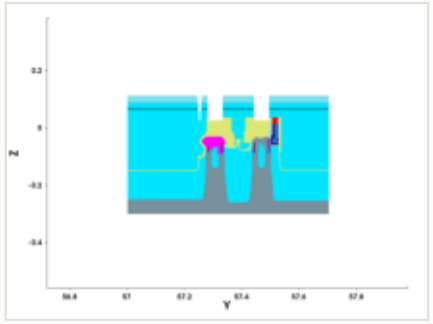
carbon so far, of the whole flow (kgCO2e)



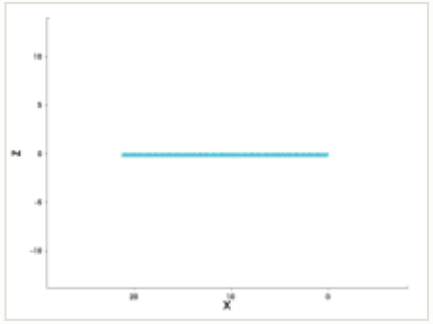
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



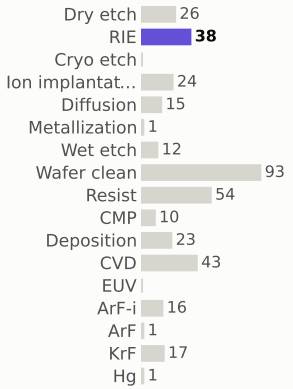
y cut at 57.351 µm



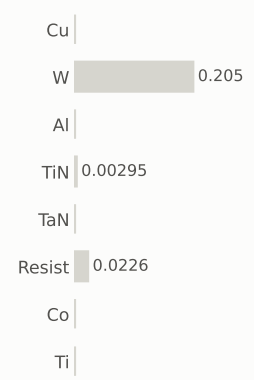
mask for this step

no mask for this step

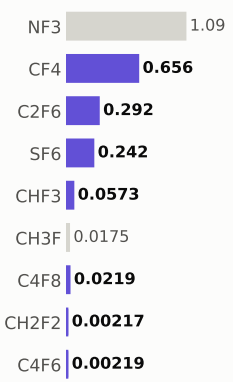
steps by process area



materials deposited (g)



gases emitted (g)



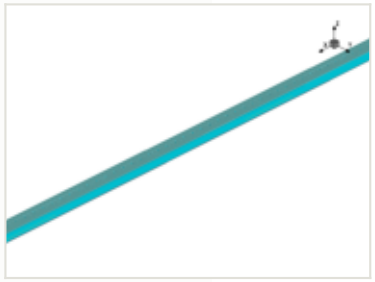
Step #5220: Dry_etch

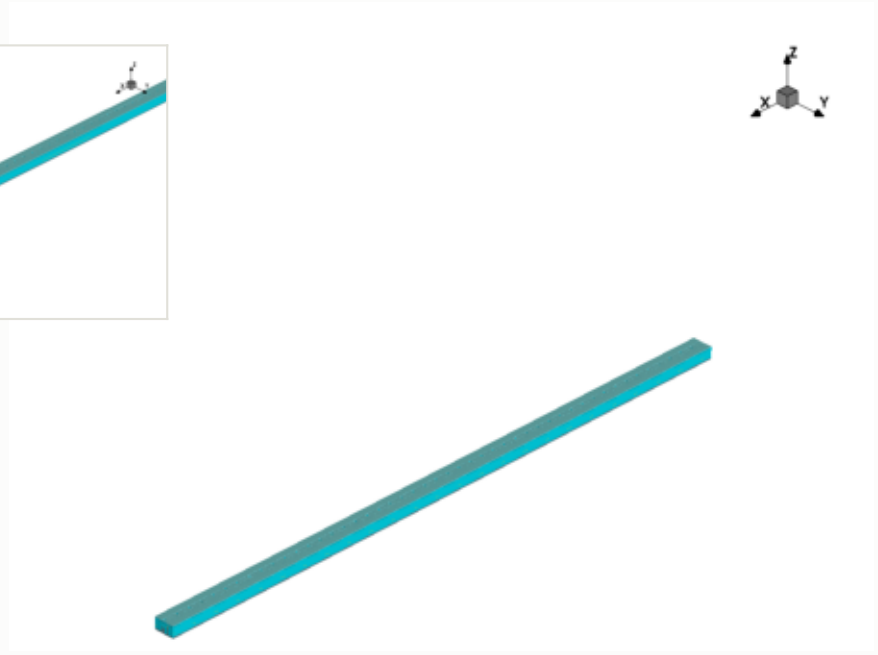
etch_strip · Dry_etch

0.801	226.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



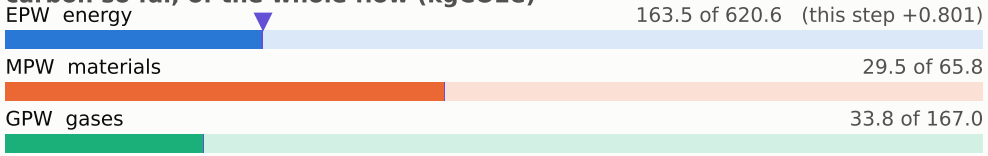


this step

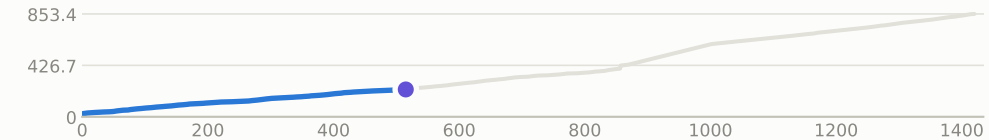
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

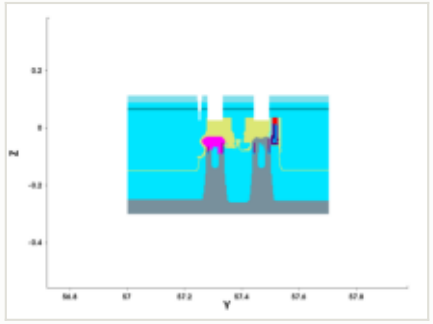
carbon so far, of the whole flow (kgCO2e)



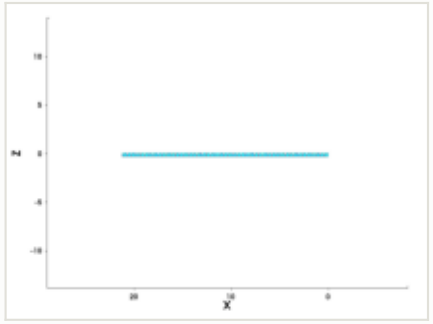
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



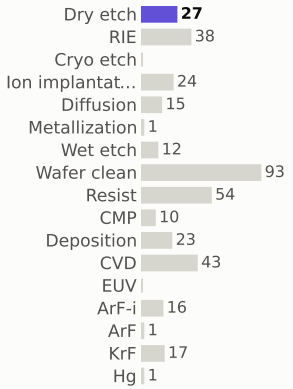
y cut at 57.351 µm



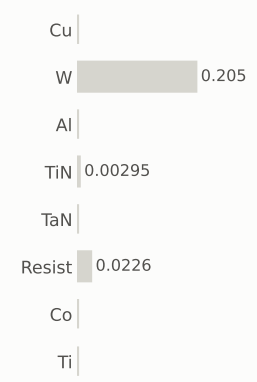
mask for this step

no mask for this step

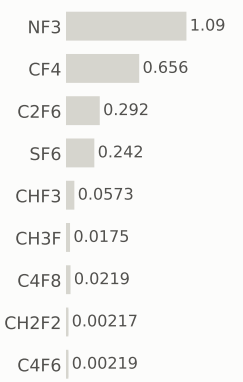
steps by process area



materials deposited (g)



gases emitted (g)



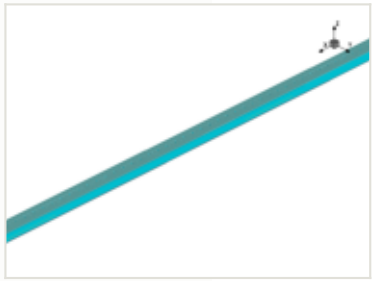
Step #5230: Spx_only

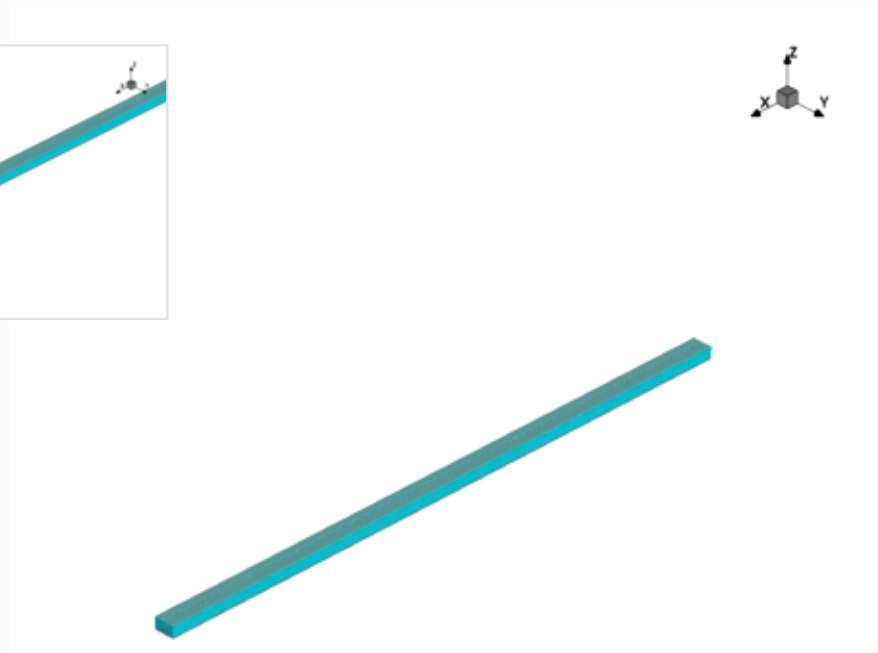
etch_anisotropic · Spx_only

0	226.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





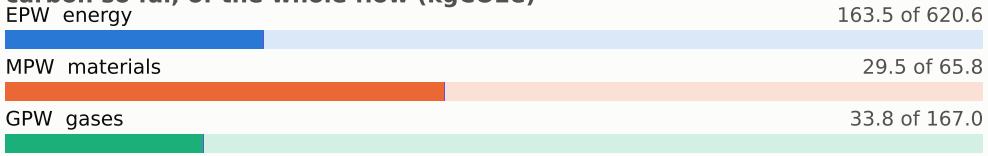
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

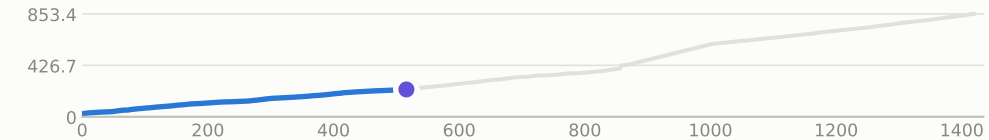
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

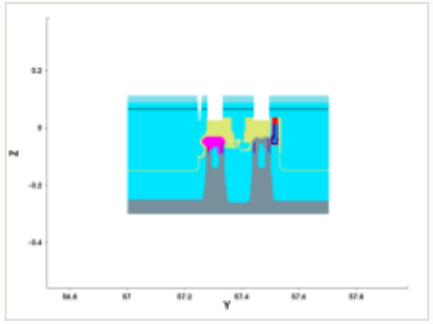
carbon so far, of the whole flow (kgCO2e)



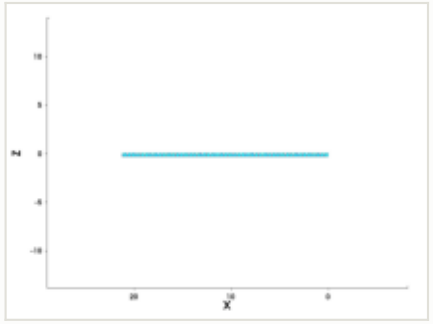
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



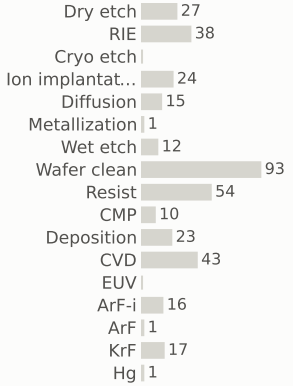
y cut at 57.351 µm



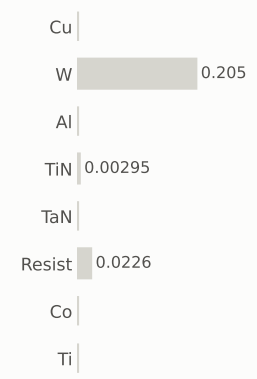
mask for this step

no mask for this step

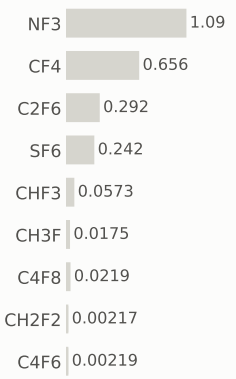
steps by process area



materials deposited (g)



gases emitted (g)



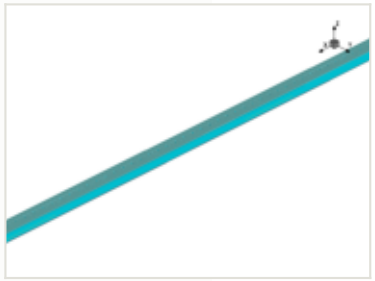
Step #5240: Spx_only

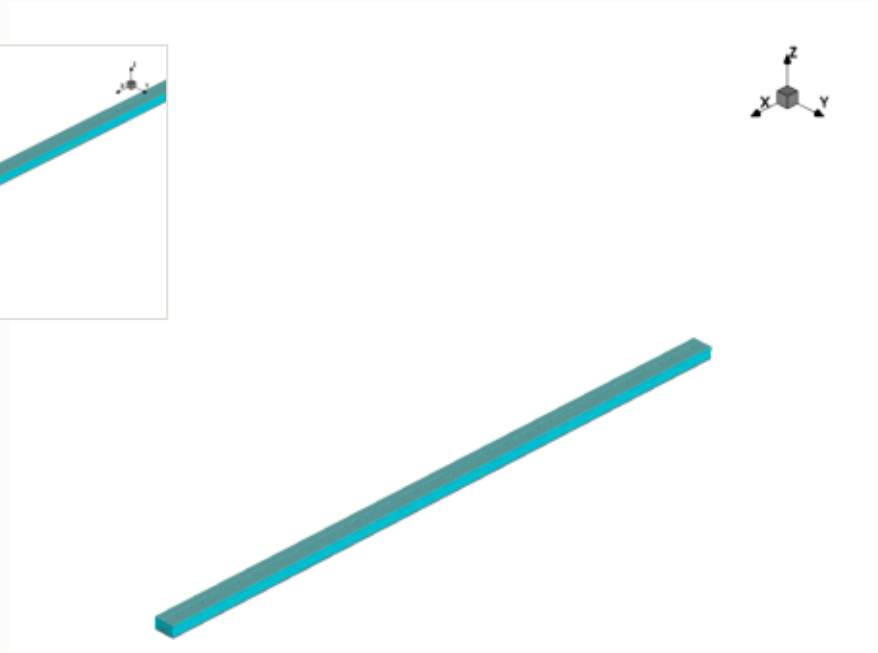
depo_isotropic · Spx_only

0	226.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

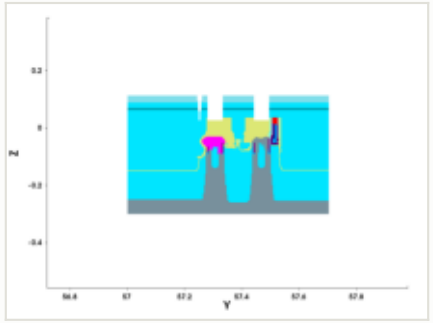
3D structure after this step

zoomed to fit

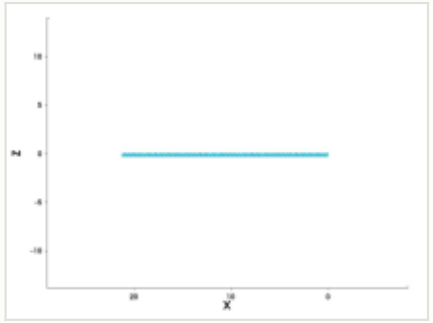




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

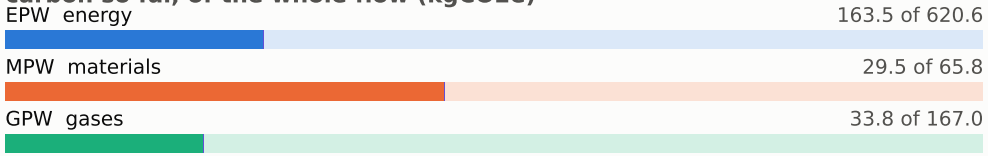
this step

process area	Spx_only
material	—
thickness	0.003 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

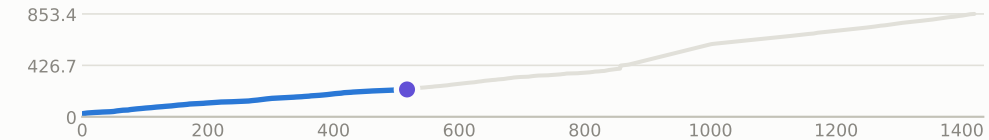
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

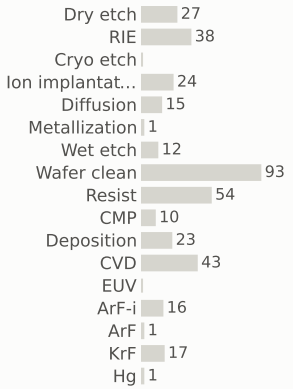
carbon so far, of the whole flow (kgCO2e)



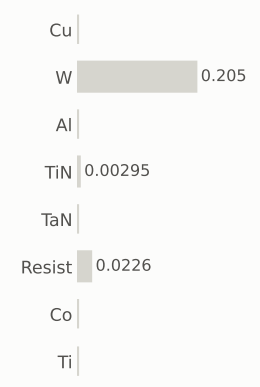
embodied carbon up to each step (kgCO2e per wafer)



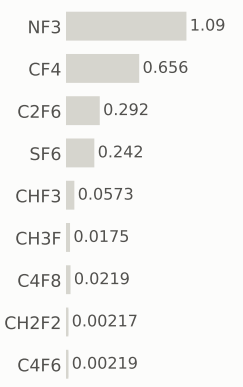
steps by process area



materials deposited (g)



gases emitted (g)



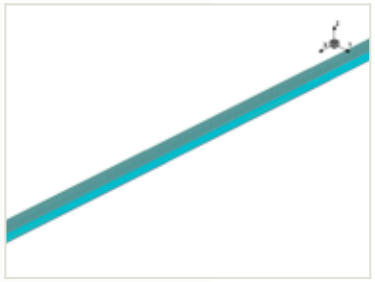
Step #5250: Wafer_clean

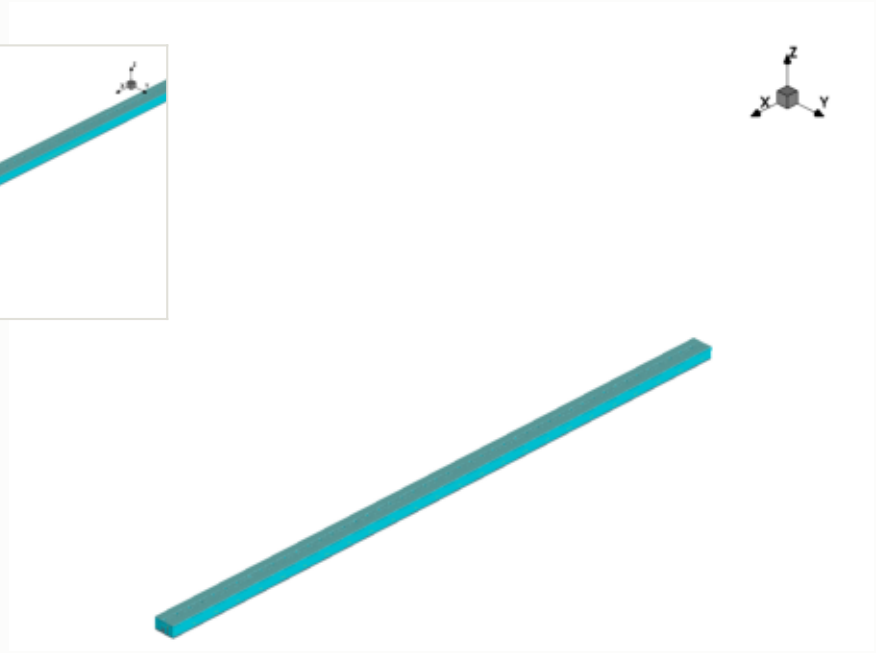
clean · Wafer_clean

0.169	227.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





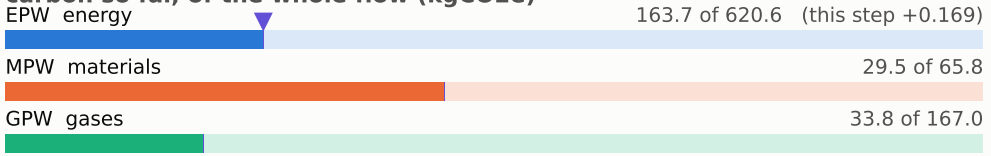
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

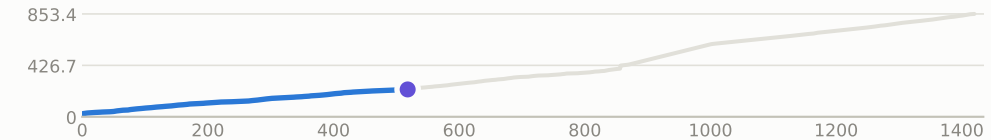
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

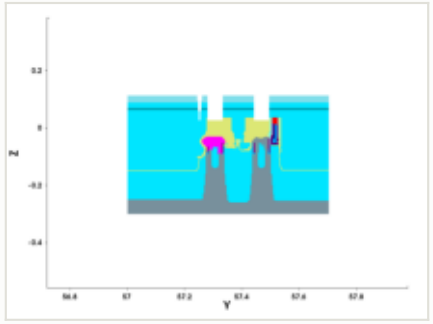
carbon so far, of the whole flow (kgCO2e)



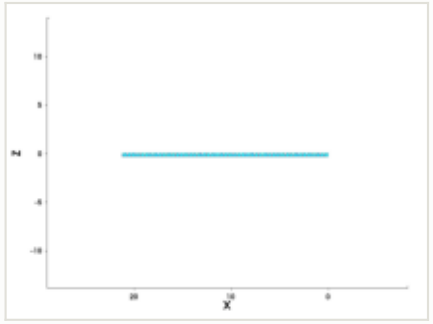
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



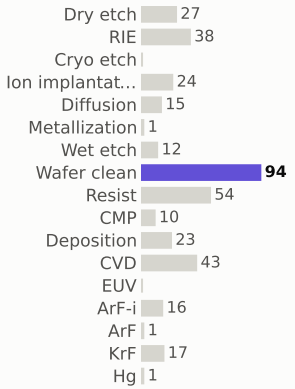
y cut at 57.351 µm



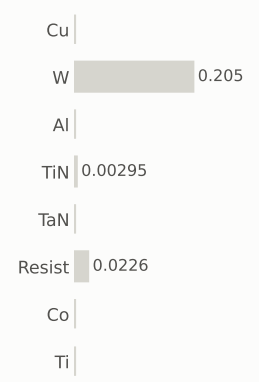
mask for this step

no mask for this step

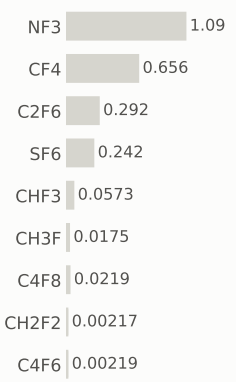
steps by process area



materials deposited (g)



gases emitted (g)



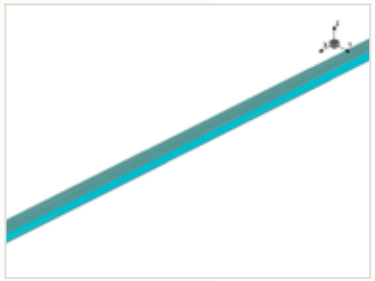
Step #5260: Dry_etch

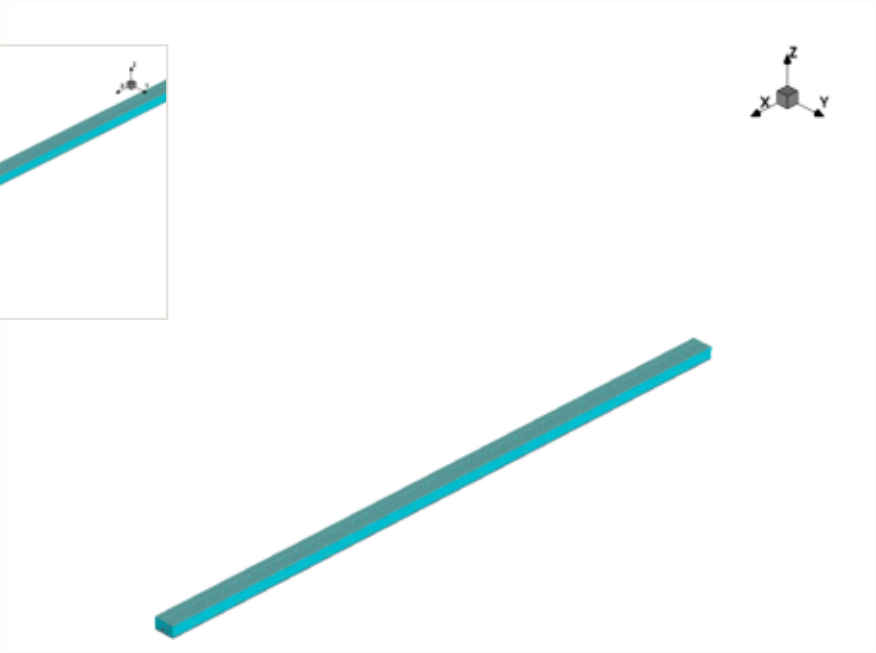
treatment · Dry_etch

0.801	227.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



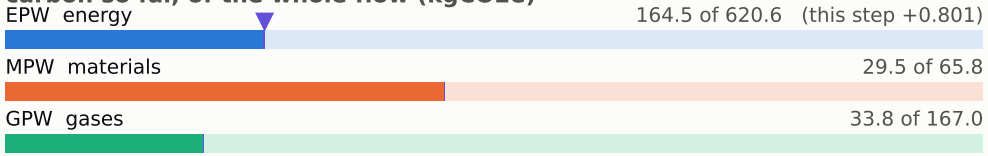


this step

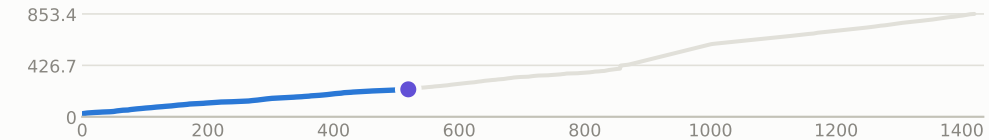
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

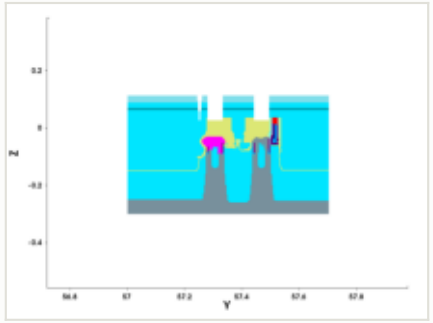
carbon so far, of the whole flow (kgCO2e)



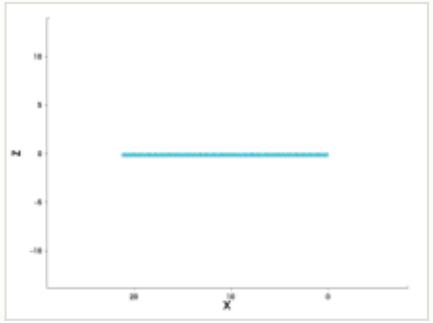
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



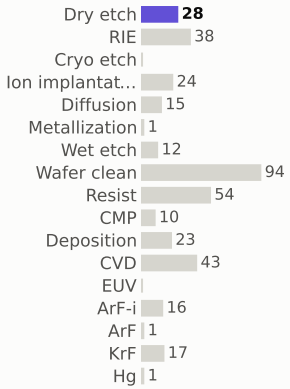
y cut at 57.351 µm



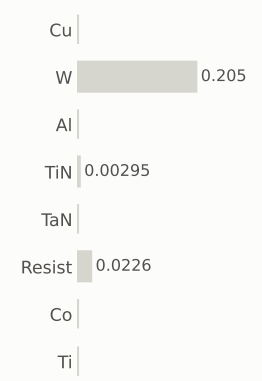
mask for this step

no mask for this step

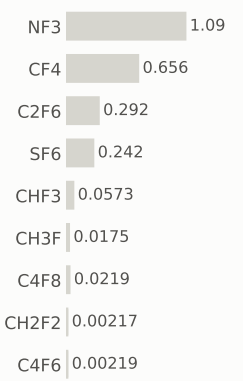
steps by process area



materials deposited (g)



gases emitted (g)



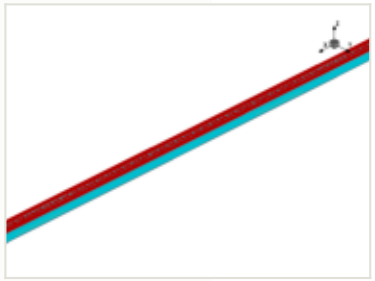
Step #5270: Metallization

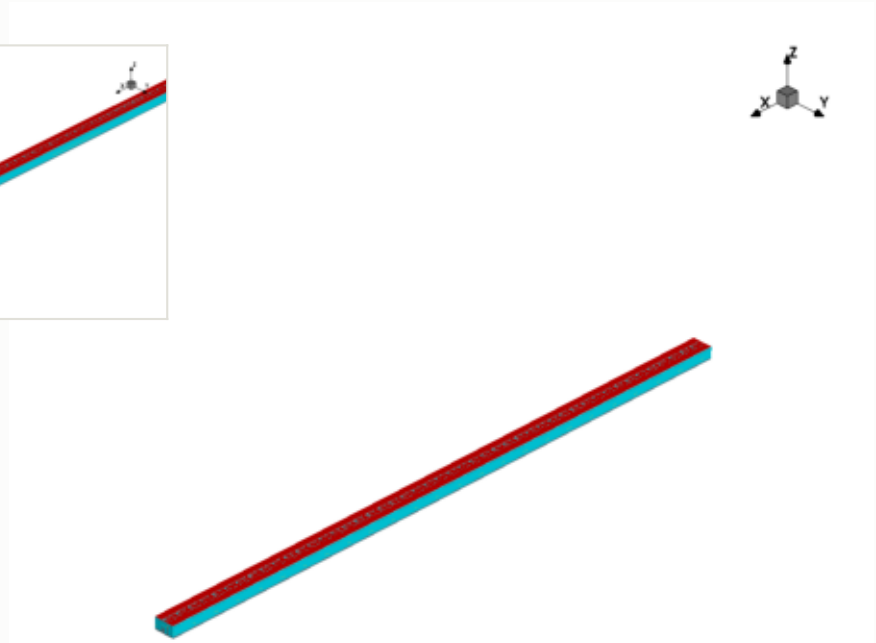
depo_isotropic · Metallization

1.2	229.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



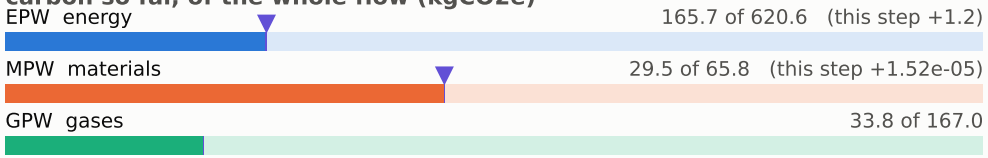


this step

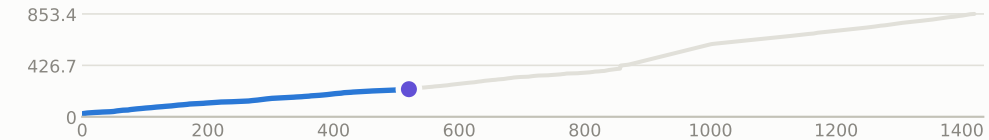
process area	Metallization
material	Ti
thickness	0.001 µm
material GWP	47.6334 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	1.52e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

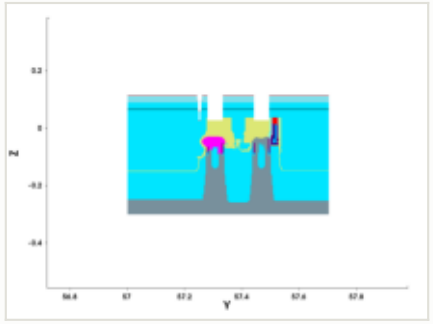
carbon so far, of the whole flow (kgCO2e)



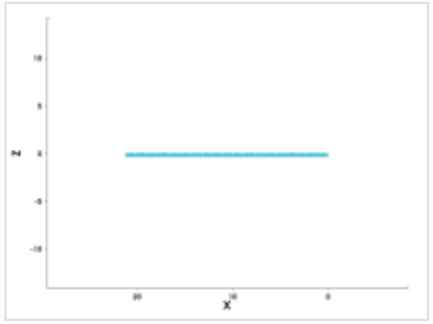
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



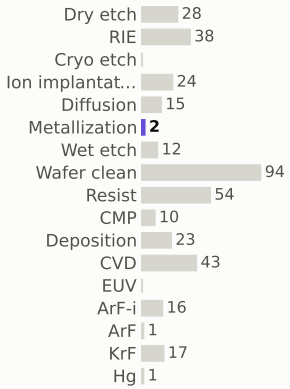
y cut at 57.351 µm



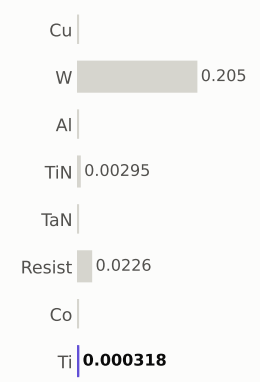
mask for this step

no mask for this step

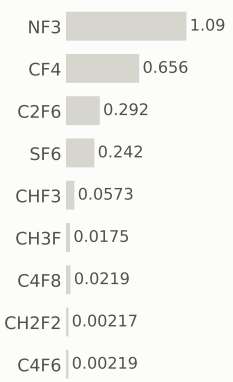
steps by process area



materials deposited (g)



gases emitted (g)



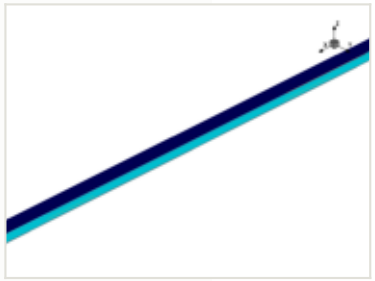
Step #5280: Spx_only

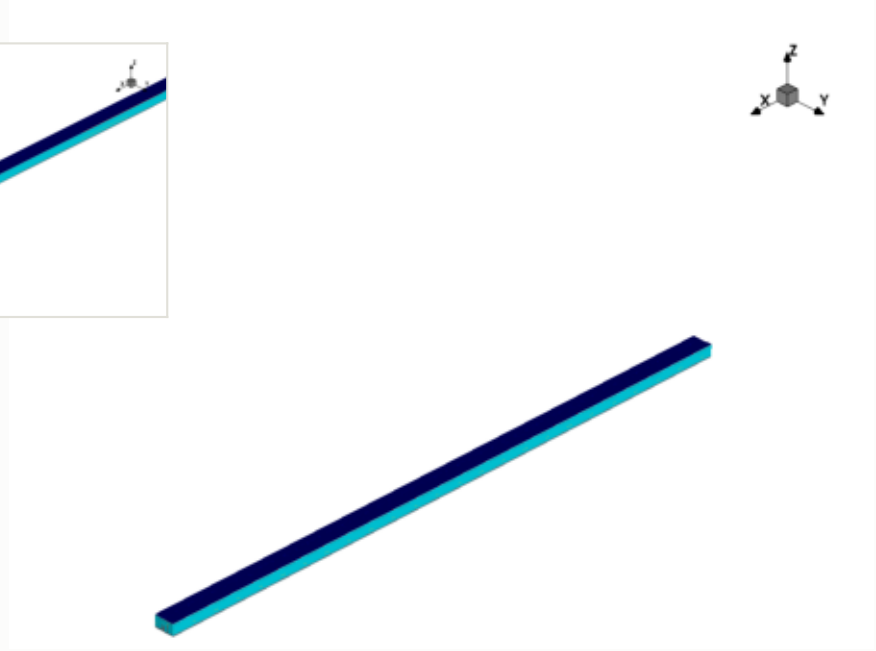
depo_isotropic · Spx_only

4.86e-05	229.0	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

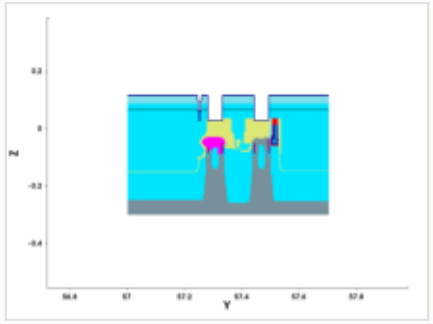
3D structure after this step

zoomed to fit

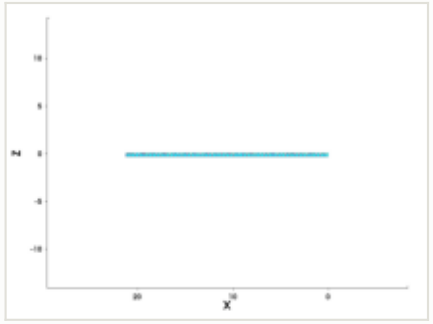




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

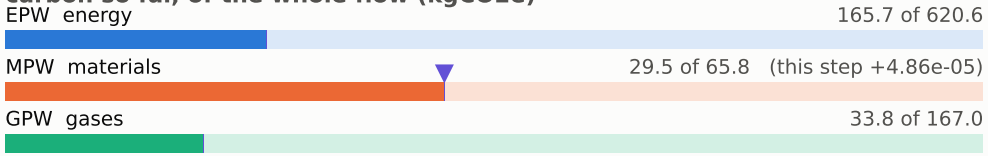
this step

process area	Spx_only
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

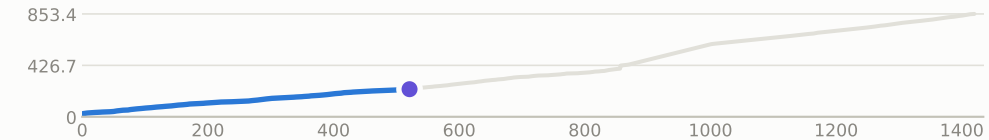
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**4.86e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

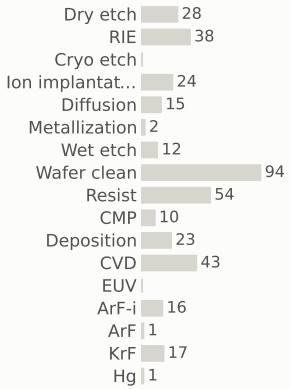
carbon so far, of the whole flow (kgCO2e)



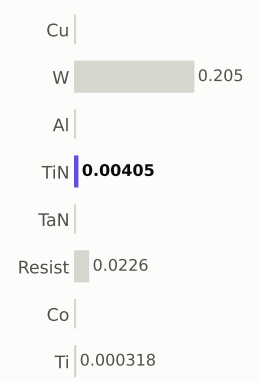
embodied carbon up to each step (kgCO2e per wafer)



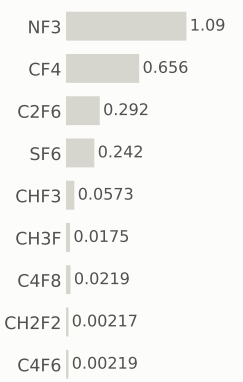
steps by process area



materials deposited (g)



gases emitted (g)



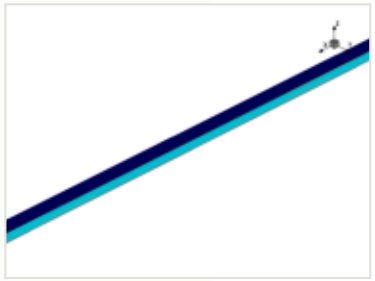
Step #5290: Wafer_clean

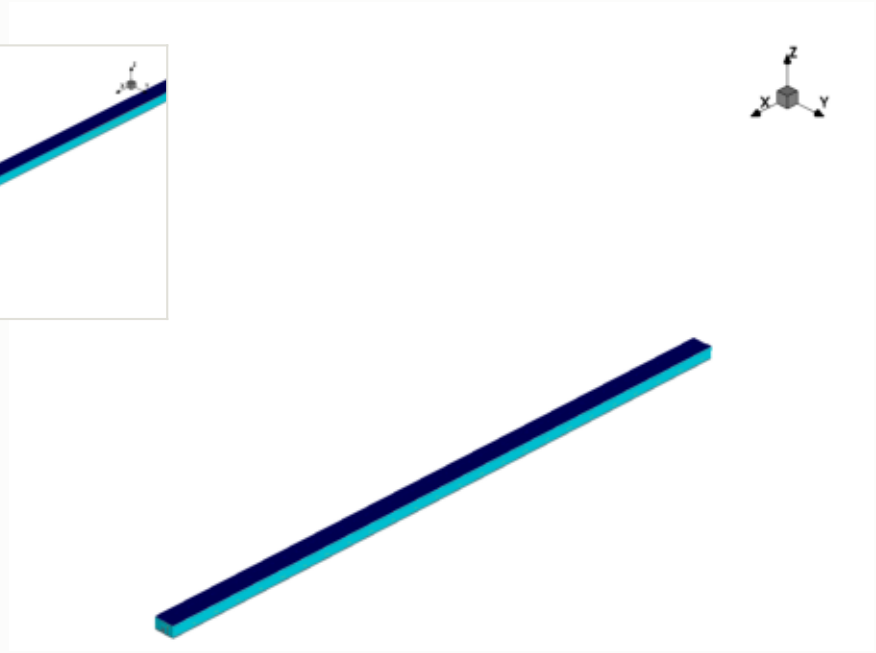
clean · Wafer_clean

0.169	229.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

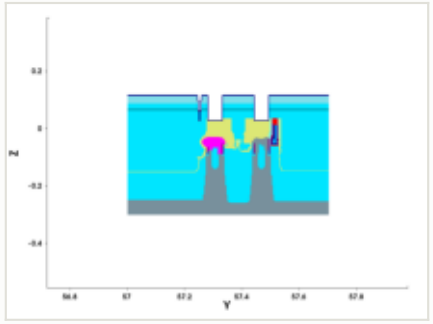
3D structure after this step

zoomed to fit

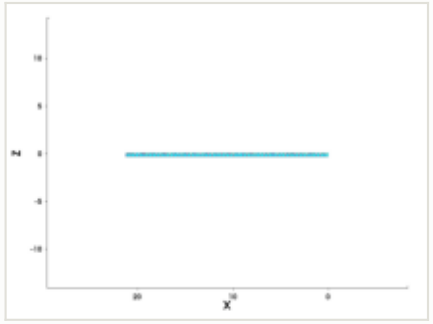




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

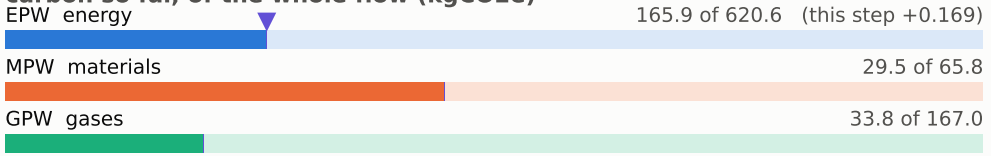
no mask for this step

this step

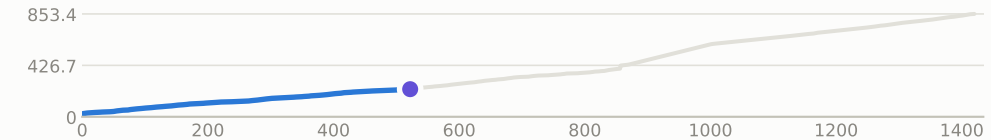
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

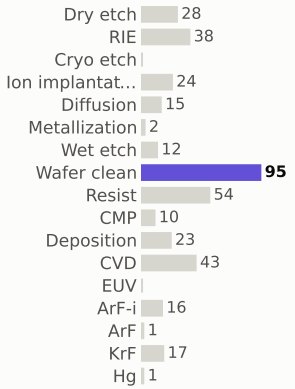
carbon so far, of the whole flow (kgCO2e)



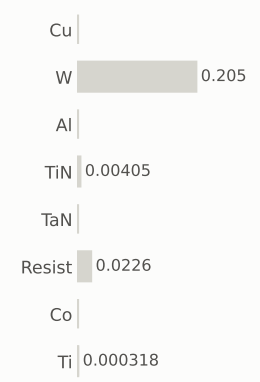
embodied carbon up to each step (kgCO2e per wafer)



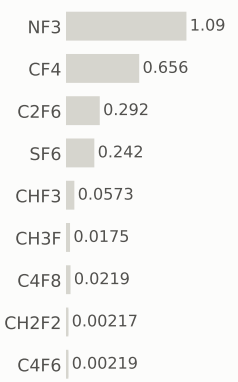
steps by process area



materials deposited (g)



gases emitted (g)



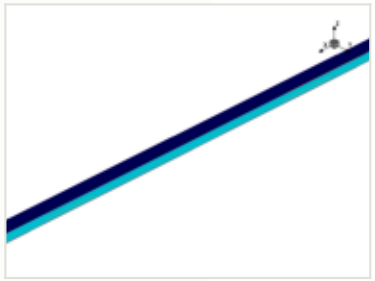
Step #5300: Diffusion

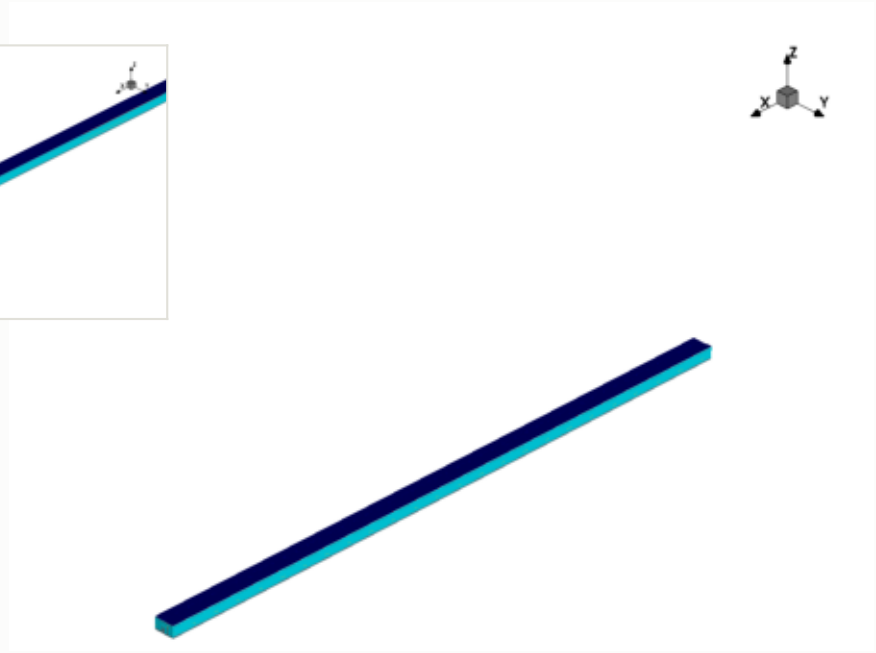
anneal · Diffusion

0.801	230.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





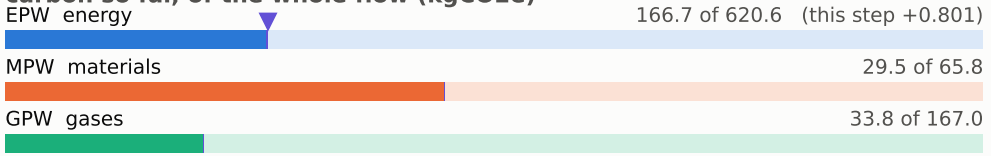
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

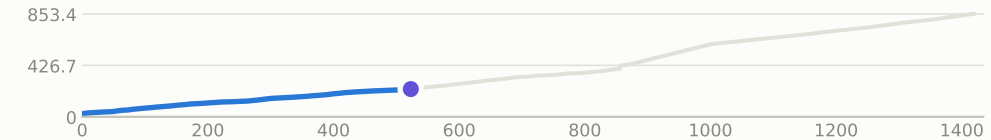
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

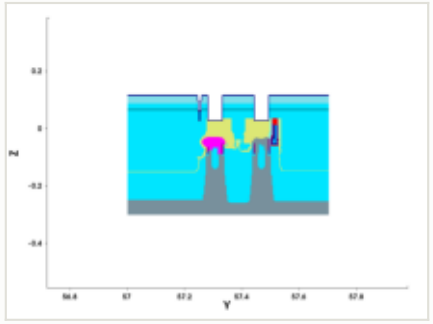
carbon so far, of the whole flow (kgCO2e)



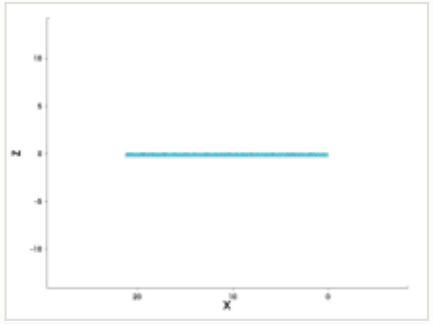
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



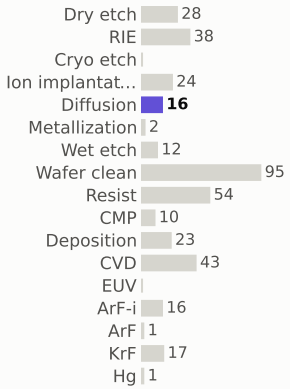
y cut at 57.351 µm



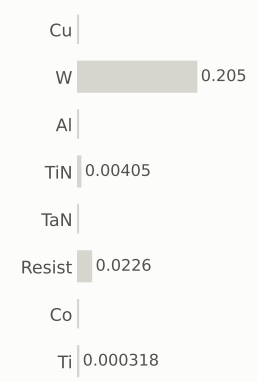
mask for this step

no mask for this step

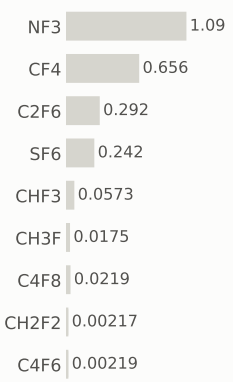
steps by process area



materials deposited (g)



gases emitted (g)



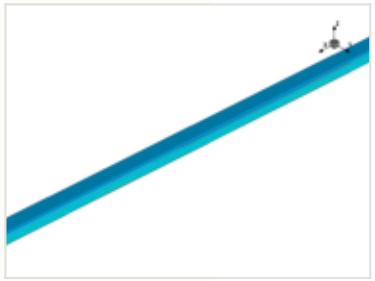
Step #5310: Metallization

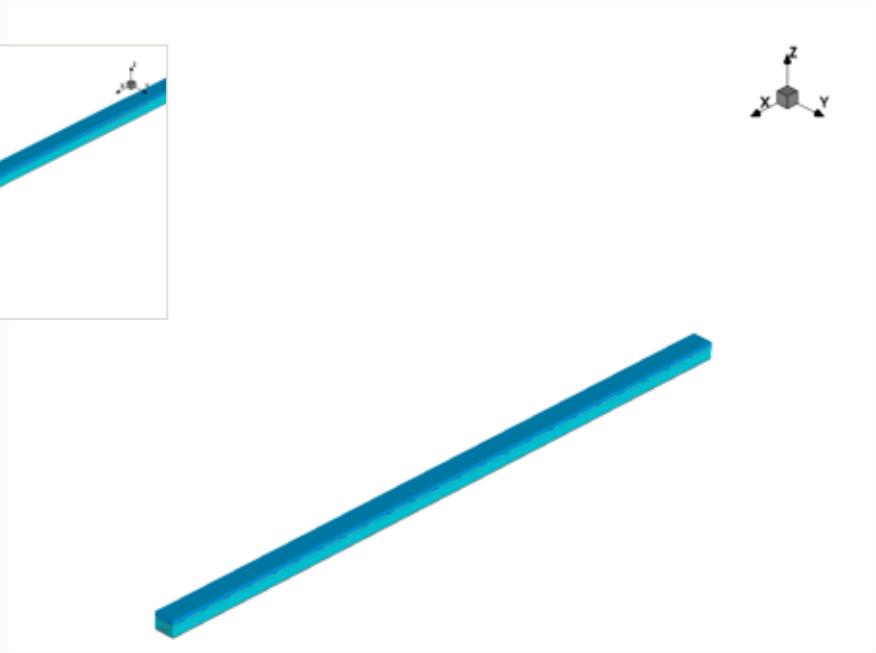
depo_fill · Metallization

1.22	231.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



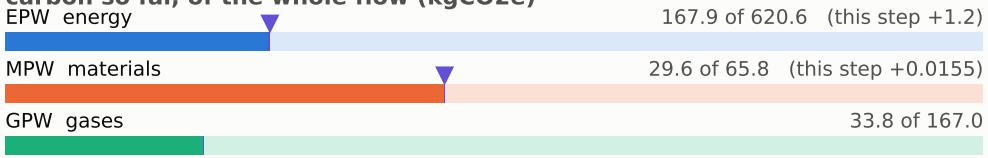


this step

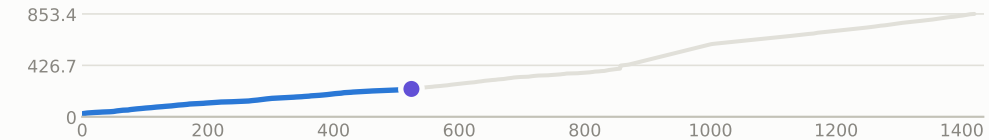
process area	Metallization
material	W
thickness	0.15 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0155 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.22 kgCO2e**

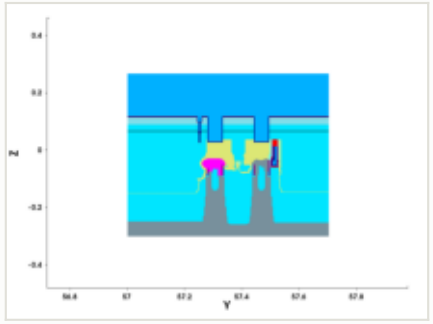
carbon so far, of the whole flow (kgCO2e)



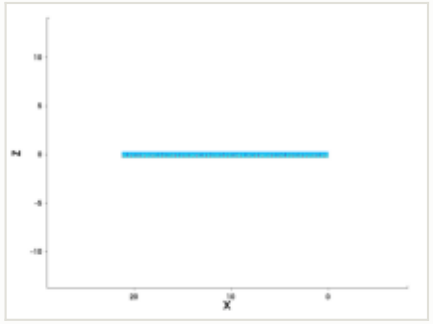
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



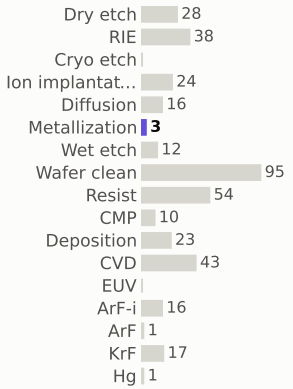
y cut at 57.351 µm



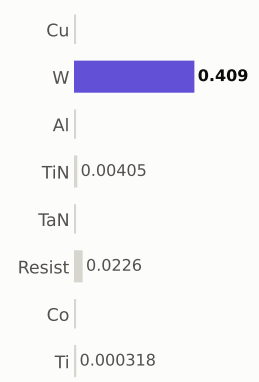
mask for this step

no mask for this step

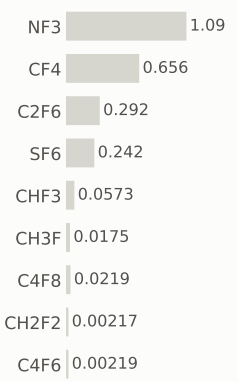
steps by process area



materials deposited (g)



gases emitted (g)



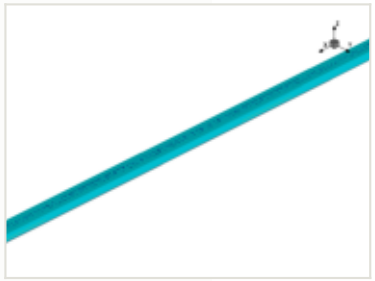
Step #5320: CMP

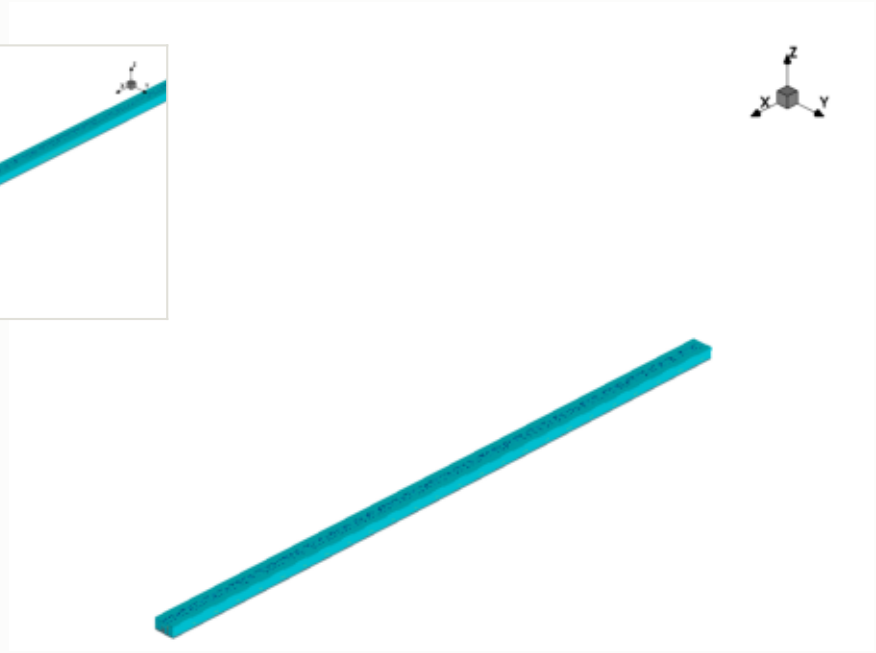
etch_cmp · CMP

0.169	231.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

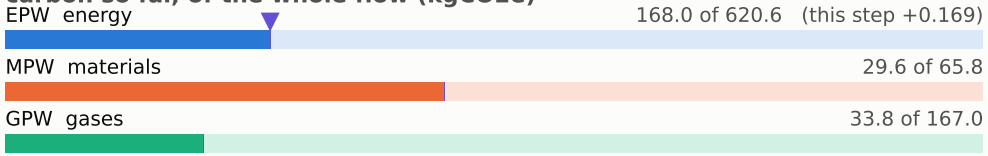




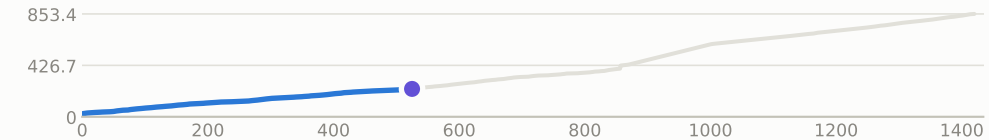
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

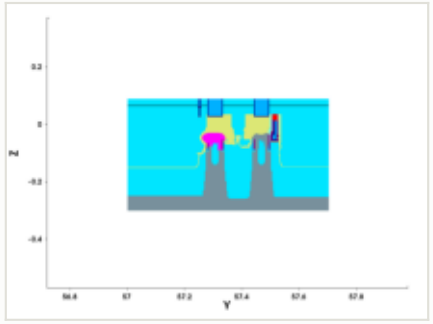
carbon so far, of the whole flow (kgCO2e)



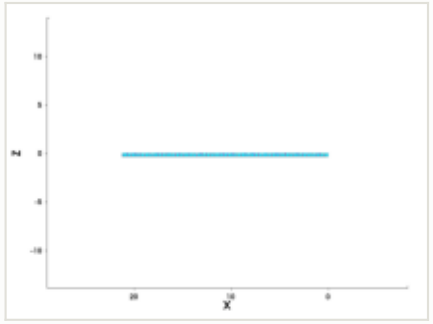
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



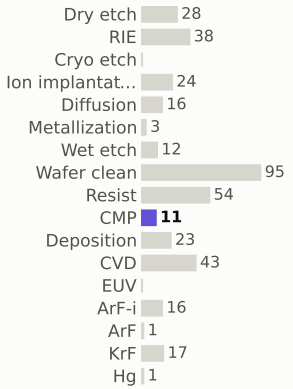
y cut at 57.351 µm



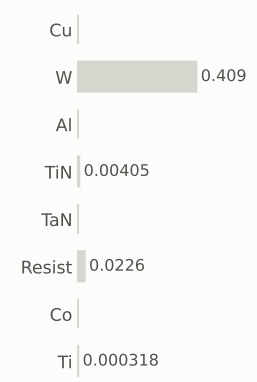
mask for this step

no mask for this step

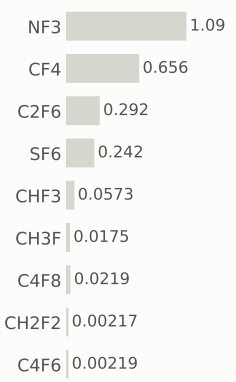
steps by process area



materials deposited (g)



gases emitted (g)



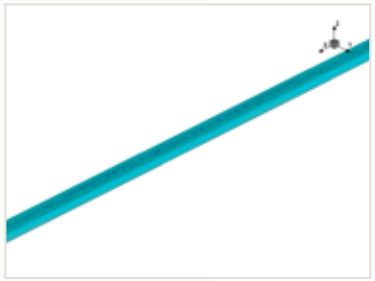
Step #5330: Diffusion

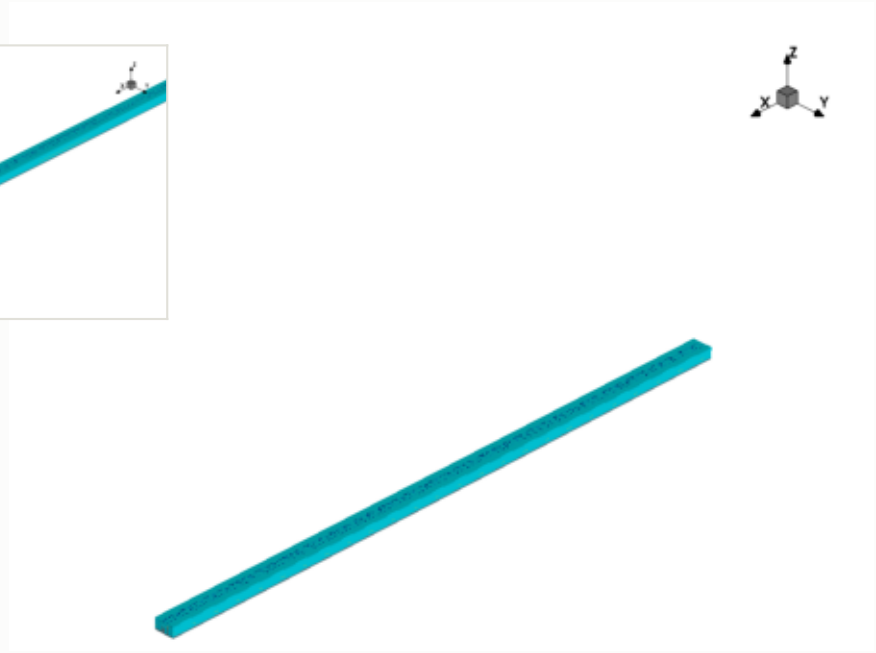
anneal · Diffusion

0.801	232.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





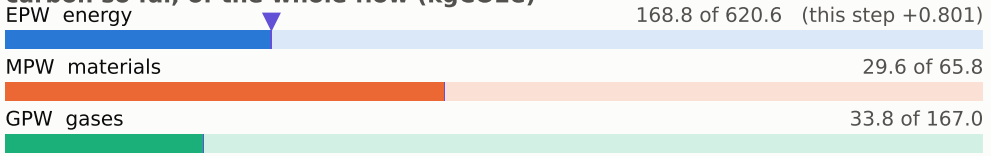
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

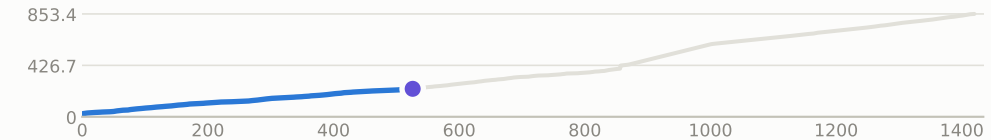
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

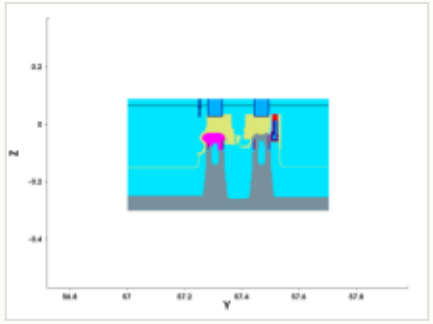
carbon so far, of the whole flow (kgCO2e)



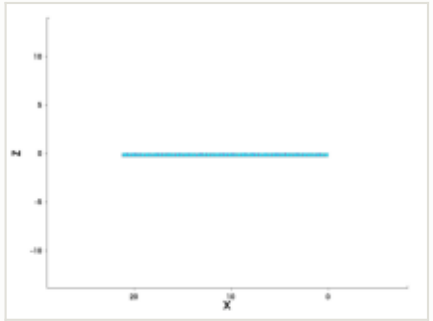
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



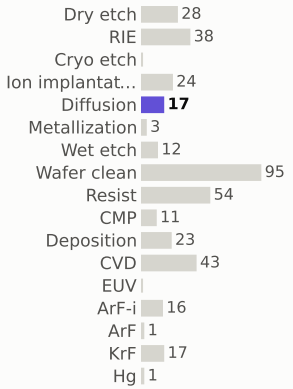
y cut at 57.351 µm



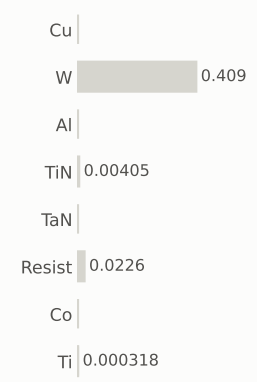
mask for this step

no mask for this step

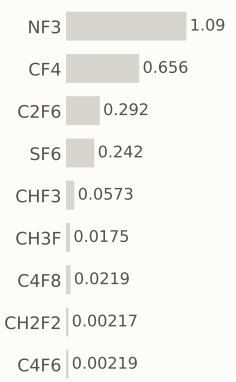
steps by process area



materials deposited (g)



gases emitted (g)



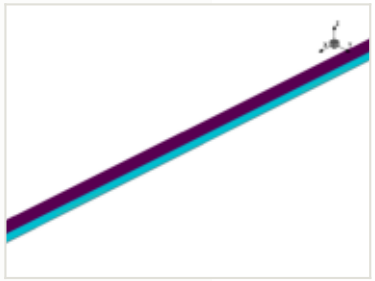
Step #5350: Deposition

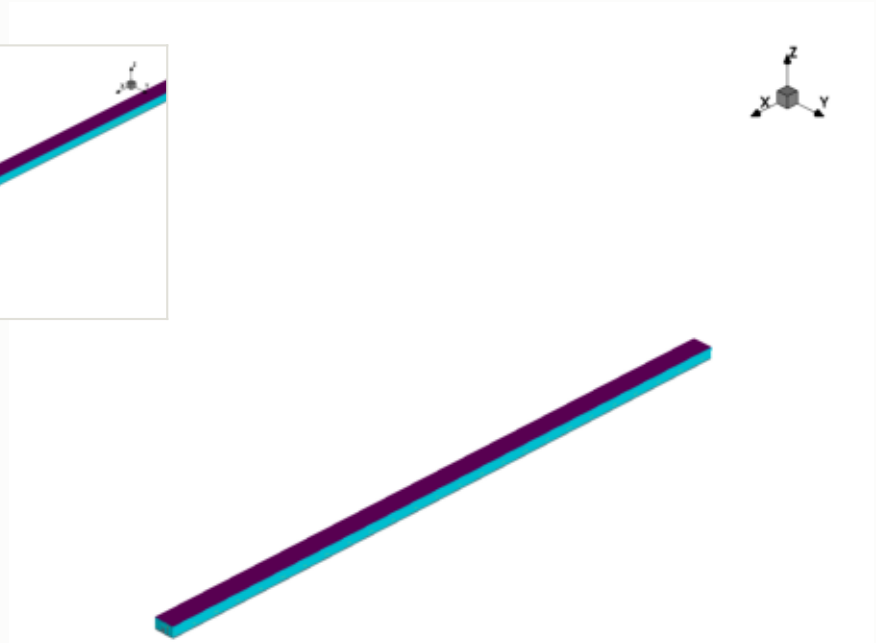
depo_fill · Deposition

0.656	232.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

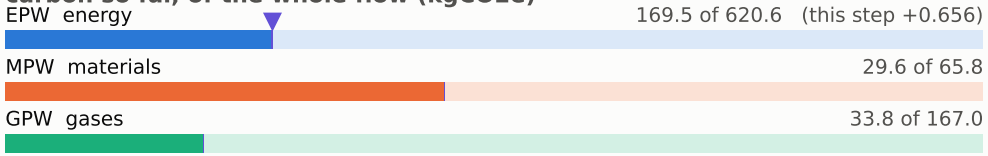




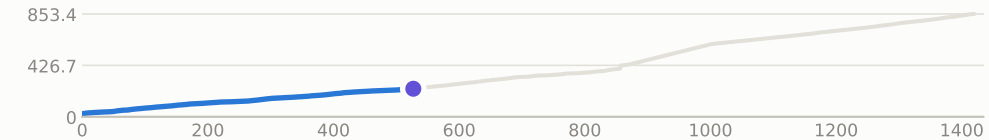
this step

process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.006 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

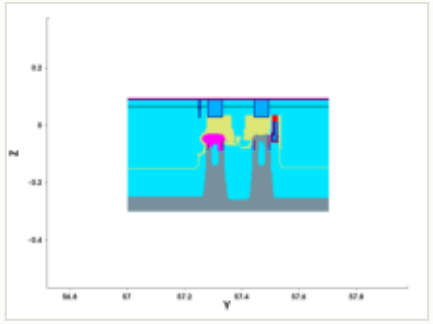
carbon so far, of the whole flow (kgCO2e)



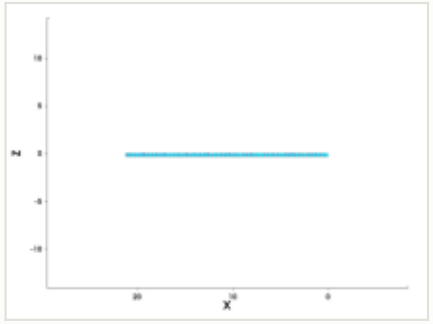
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



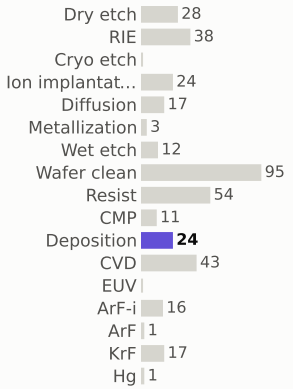
y cut at 57.351 µm



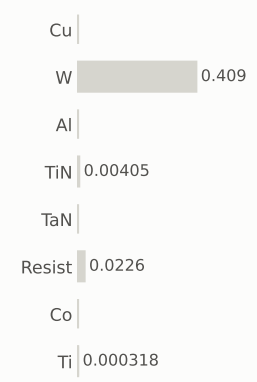
mask for this step

no mask for this step

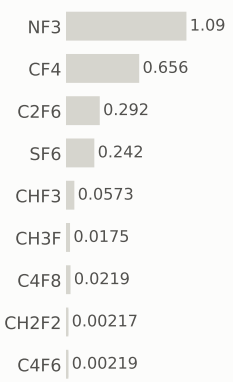
steps by process area



materials deposited (g)



gases emitted (g)



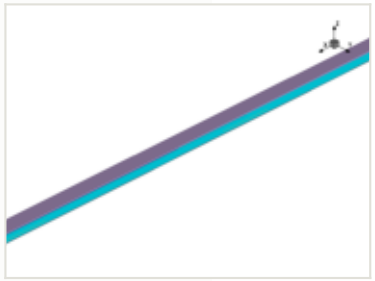
Step #5360: CVD

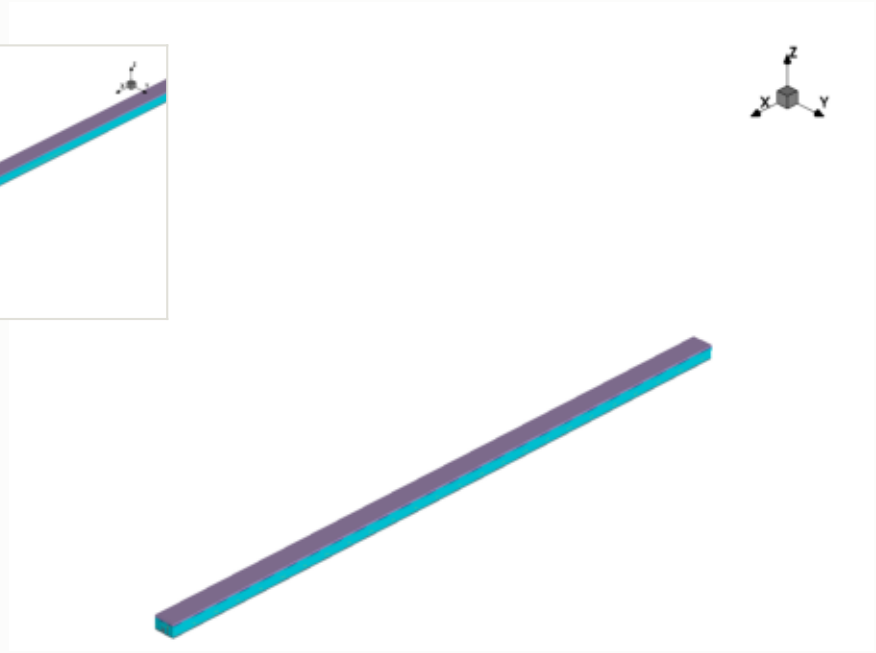
depo_fill · CVD

1.31	234.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



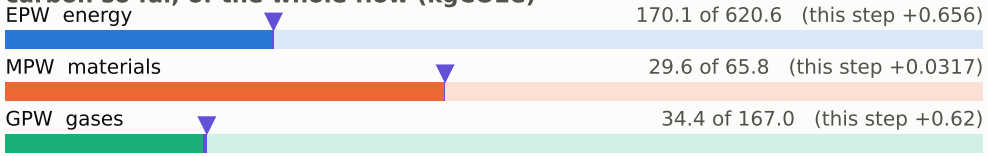


this step

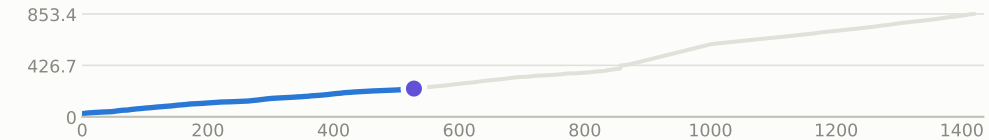
process area	CVD
material	—
thickness	0.066 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

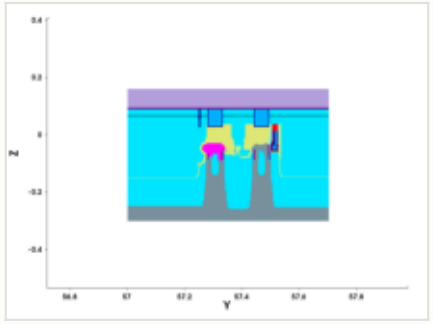
carbon so far, of the whole flow (kgCO2e)



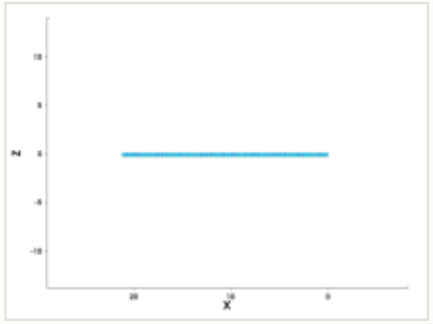
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



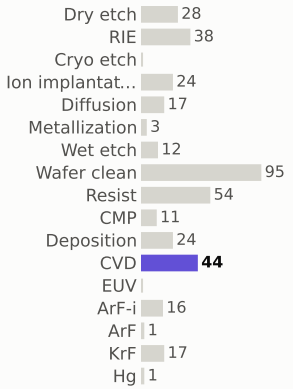
y cut at 57.351 µm



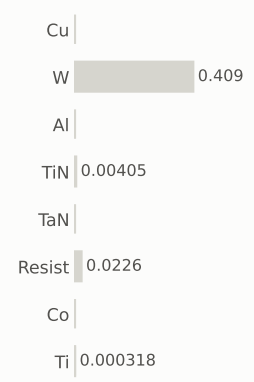
mask for this step

no mask for this step

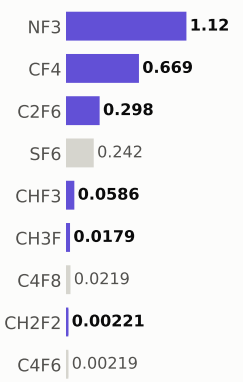
steps by process area



materials deposited (g)



gases emitted (g)



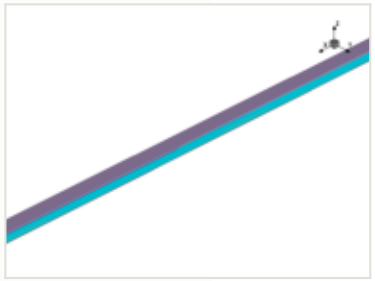
Step #5370: Diffusion

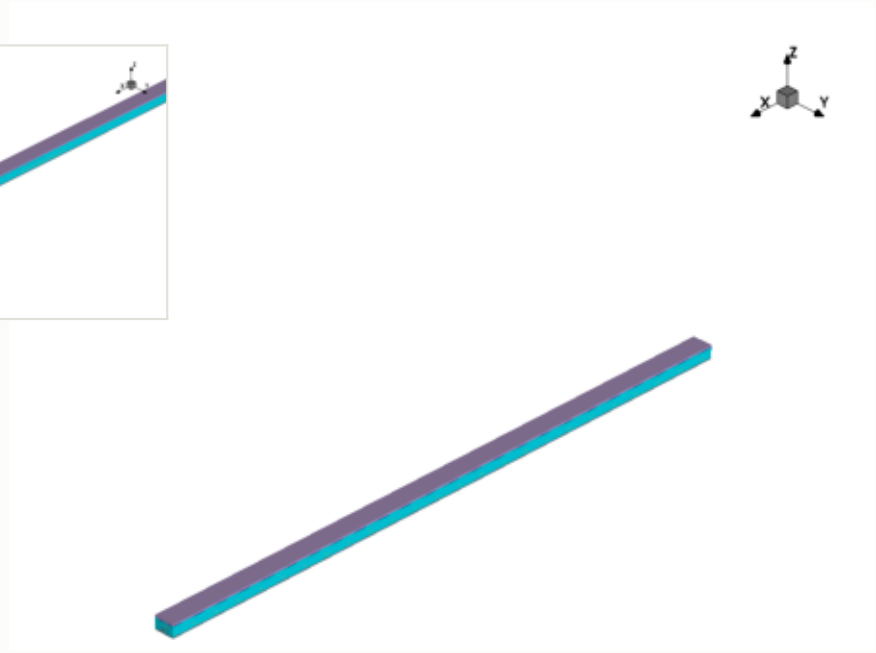
anneal · Diffusion

0.801	234.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





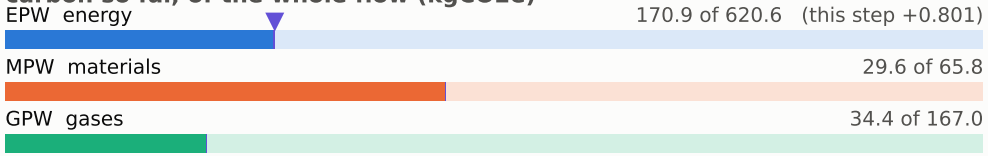
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

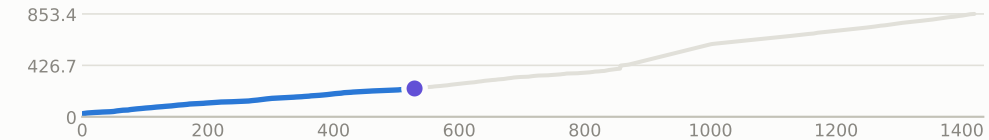
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

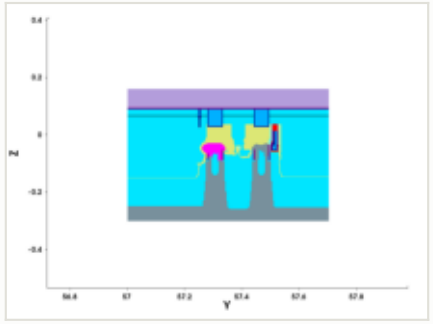
carbon so far, of the whole flow (kgCO2e)



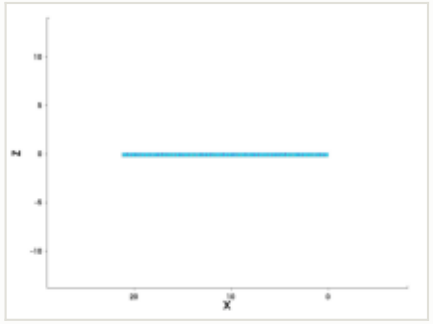
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



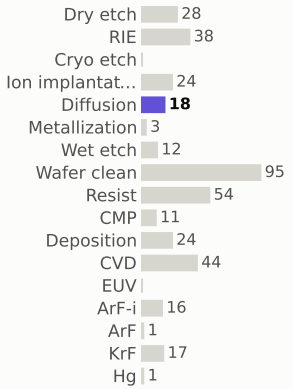
y cut at 57.351 µm



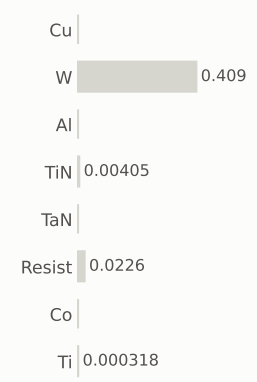
mask for this step

no mask for this step

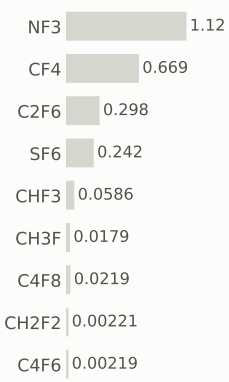
steps by process area



materials deposited (g)



gases emitted (g)



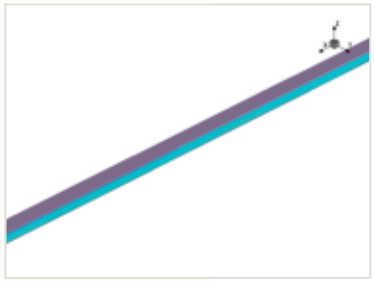
Step #5380: Wafer_clean

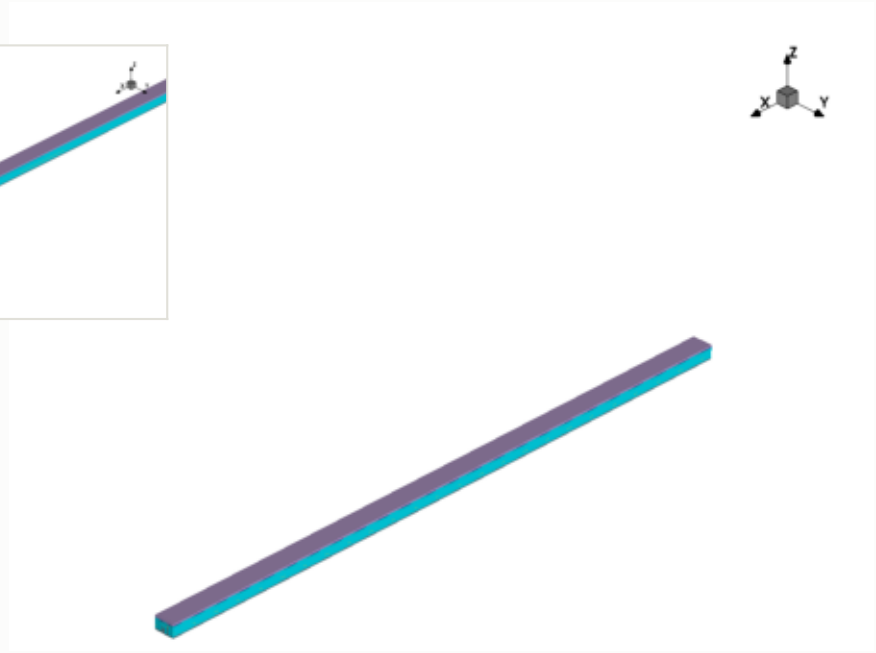
clean · Wafer_clean

0.169	235.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





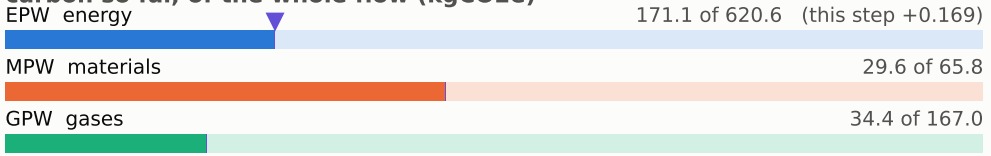
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

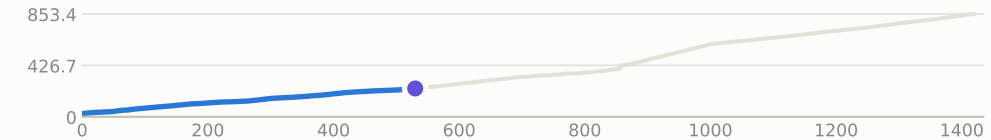
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

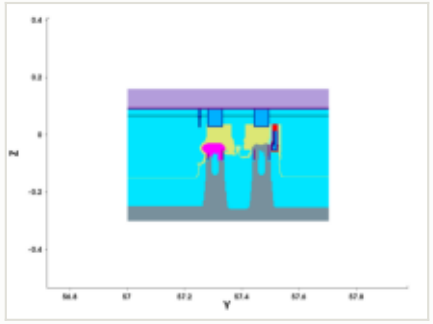
carbon so far, of the whole flow (kgCO2e)



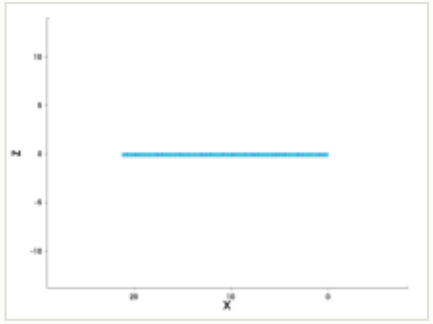
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



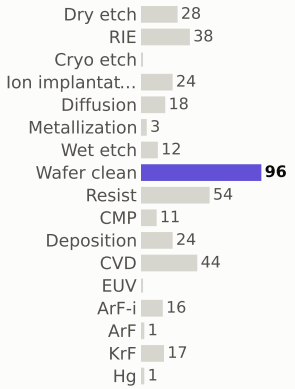
y cut at 57.351 µm



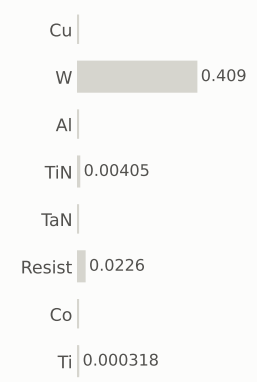
mask for this step

no mask for this step

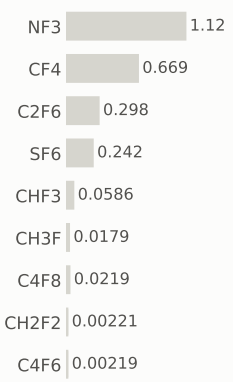
steps by process area



materials deposited (g)



gases emitted (g)



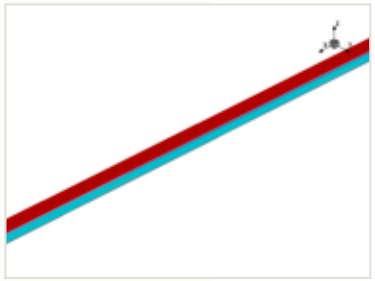
Step #5390: CVD

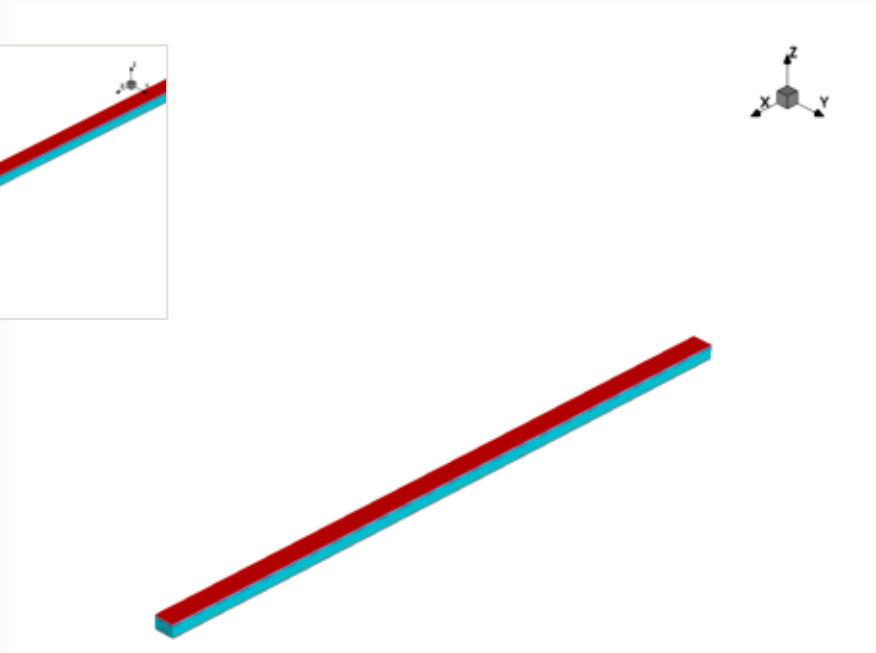
depo_fill · CVD

1.31	236.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



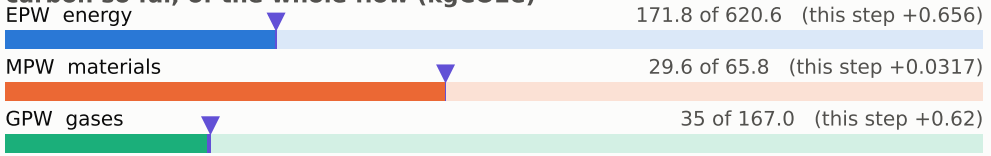


this step

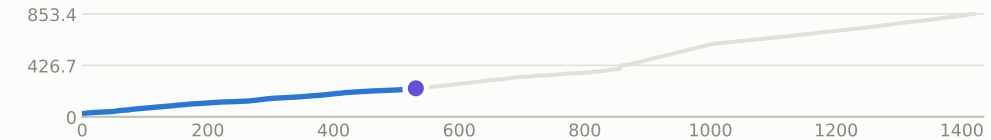
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

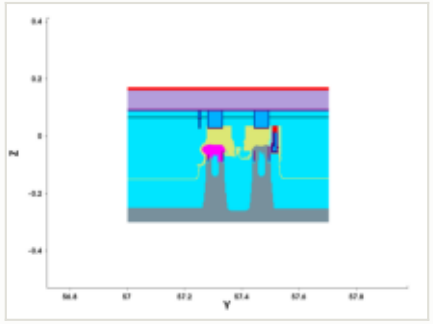
carbon so far, of the whole flow (kgCO2e)



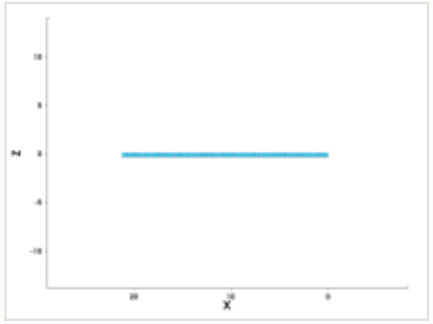
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



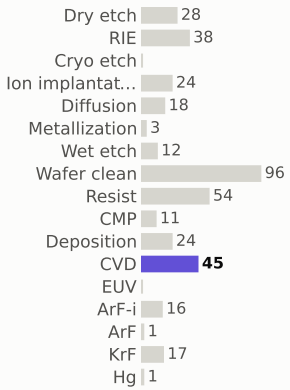
y cut at 57.351 µm



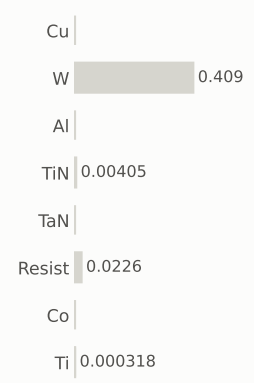
mask for this step

no mask for this step

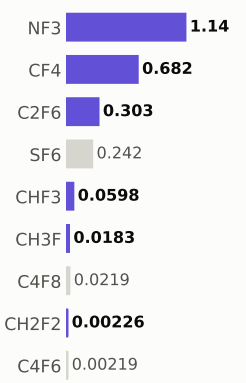
steps by process area



materials deposited (g)



gases emitted (g)



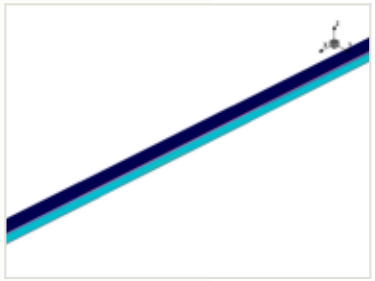
Step #5400: Deposition

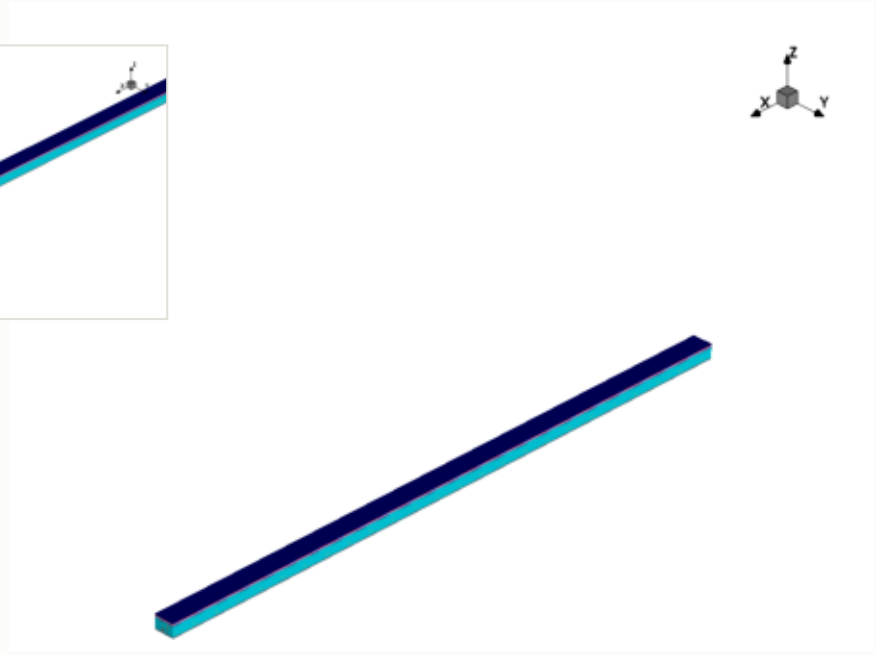
depo_fill · Deposition

0.657	237.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



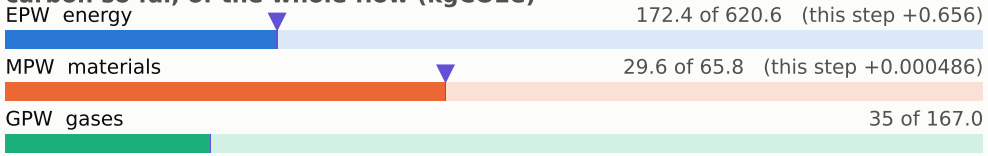


this step

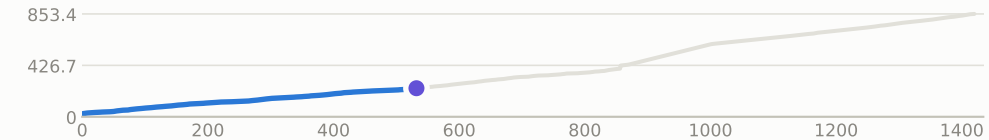
process area	Deposition
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000486 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

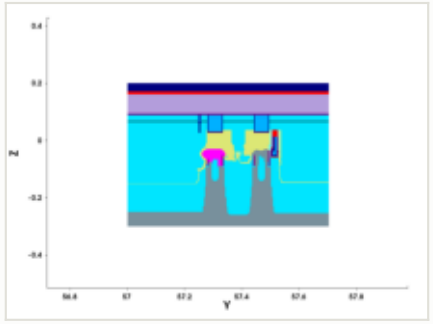
carbon so far, of the whole flow (kgCO2e)



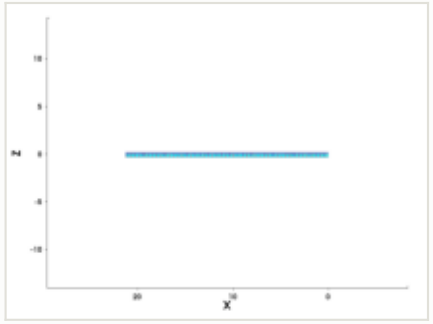
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



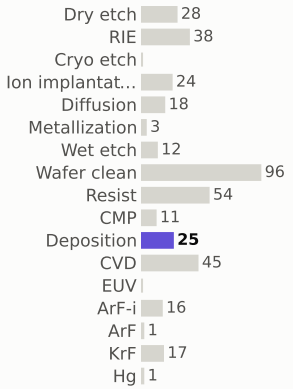
y cut at 57.351 µm



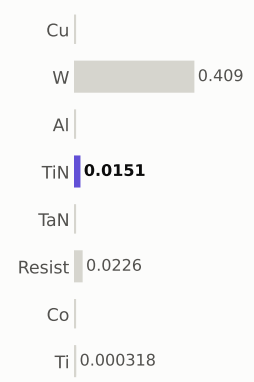
mask for this step

no mask for this step

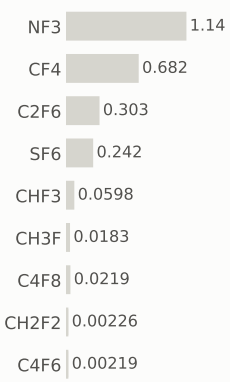
steps by process area



materials deposited (g)



gases emitted (g)



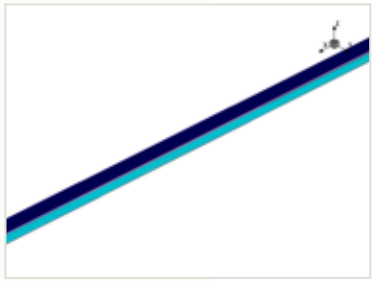
Step #5410: Wafer_clean

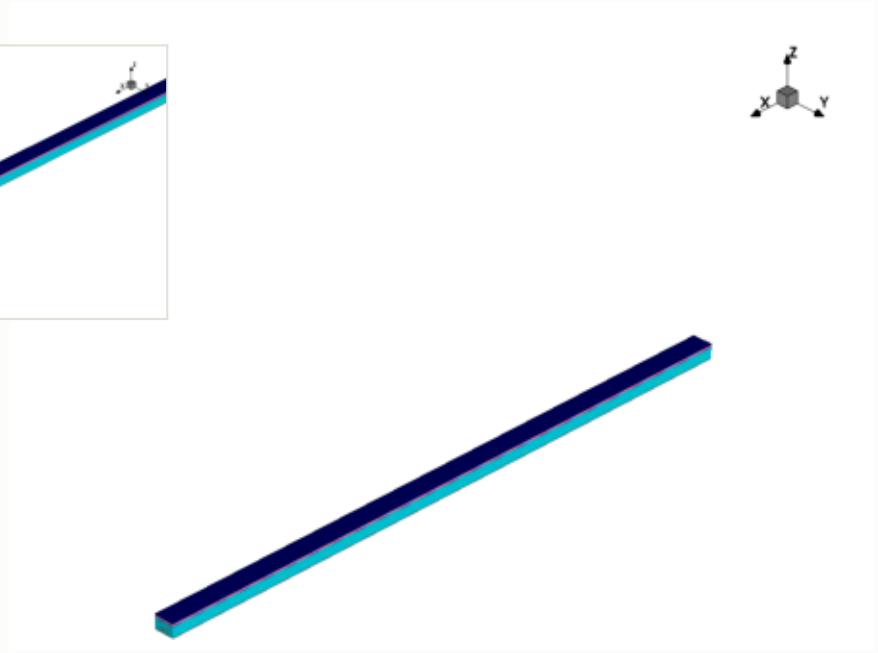
clean · Wafer_clean

0.169	237.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





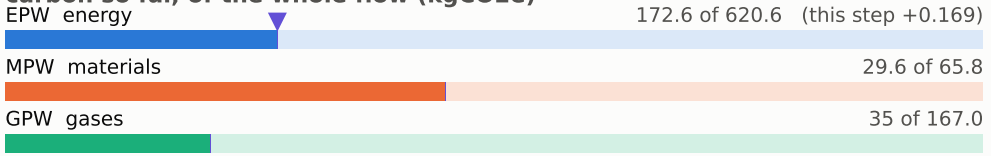
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

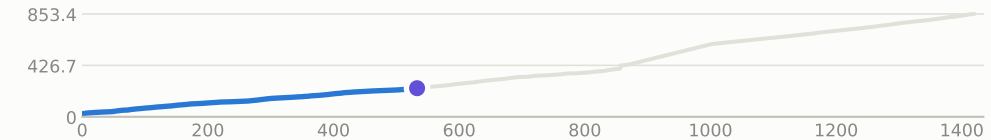
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

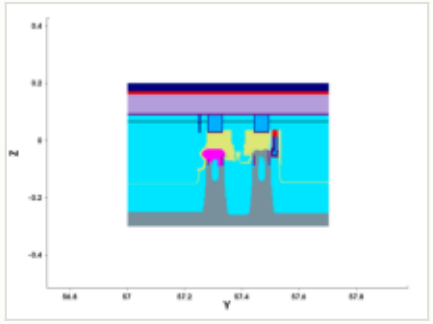
carbon so far, of the whole flow (kgCO2e)



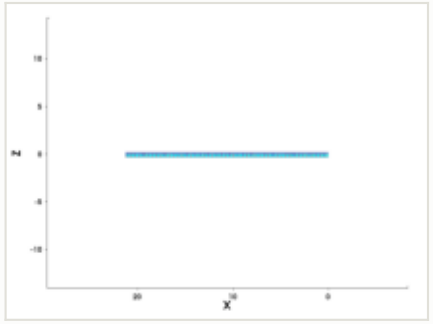
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



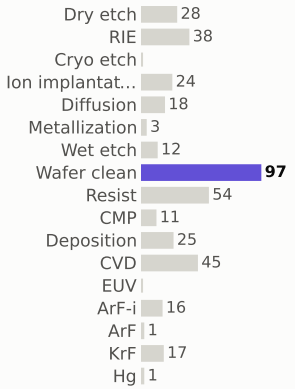
y cut at 57.351 µm



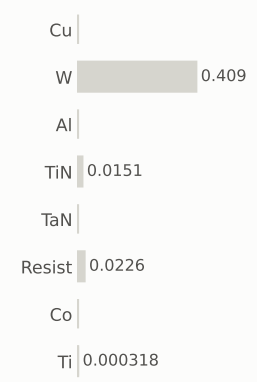
mask for this step

no mask for this step

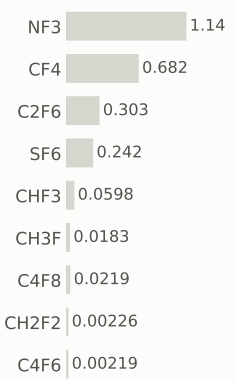
steps by process area



materials deposited (g)



gases emitted (g)



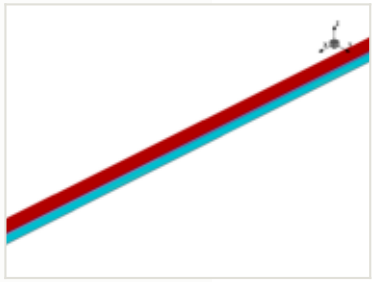
Step #5420: CVD

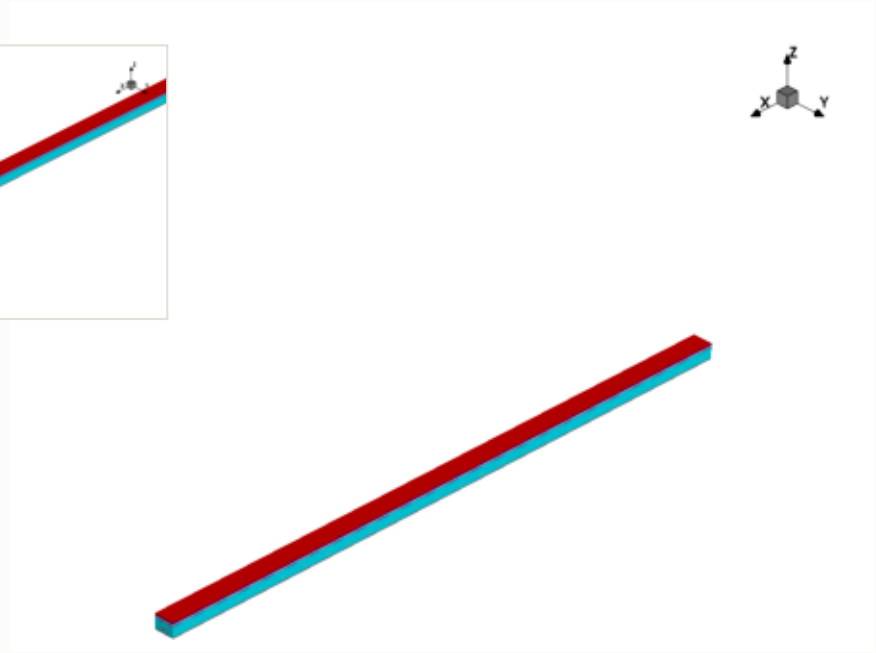
depo_fill · CVD

1.31	238.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



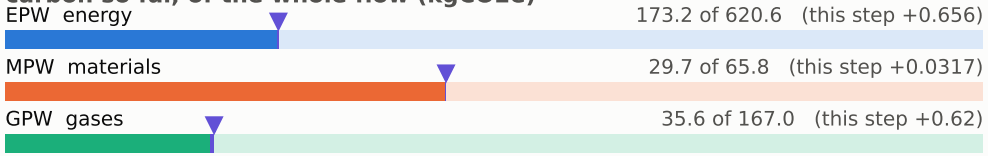


this step

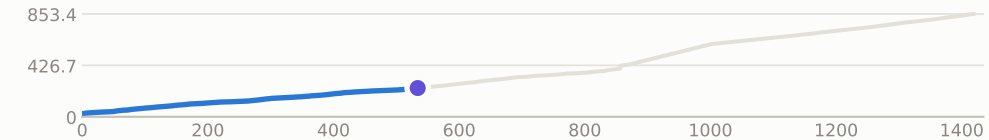
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

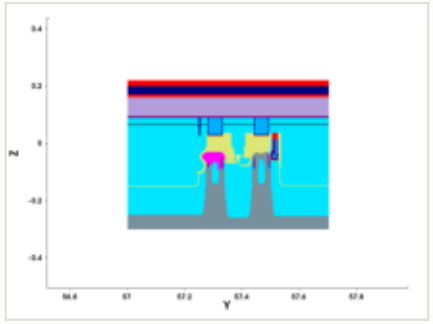
carbon so far, of the whole flow (kgCO2e)



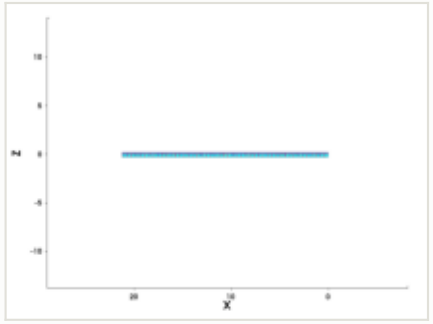
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



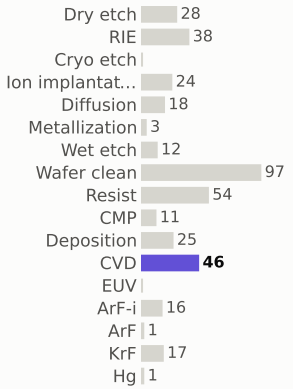
y cut at 57.351 µm



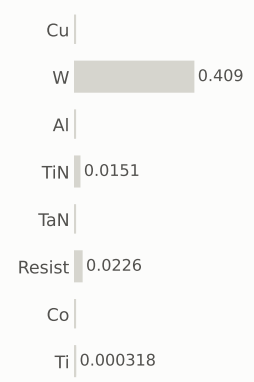
mask for this step

no mask for this step

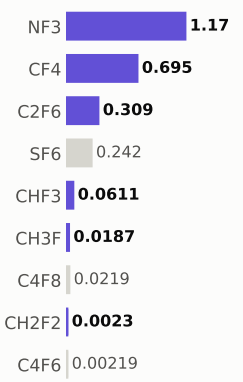
steps by process area



materials deposited (g)



gases emitted (g)



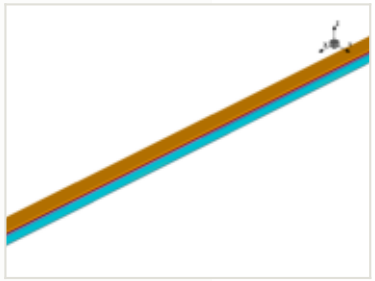
Step #5430: Resist

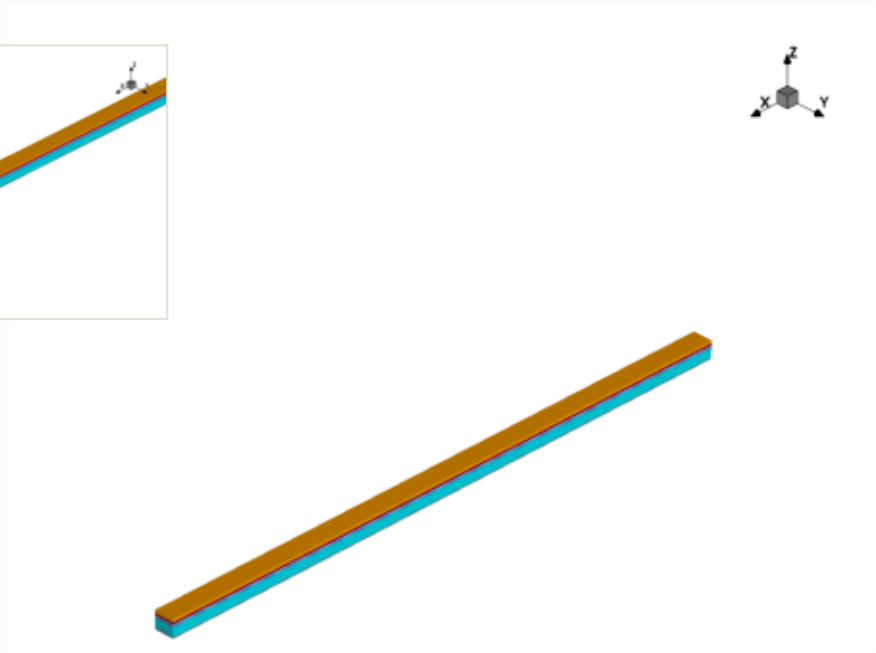
depo_fill · Resist

0.169	238.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



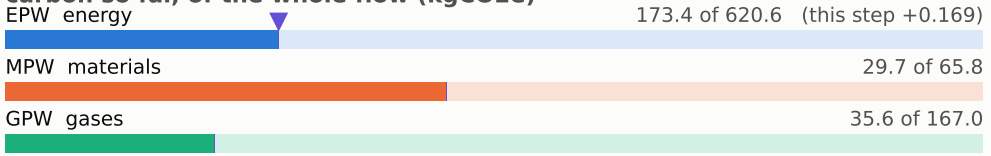


this step

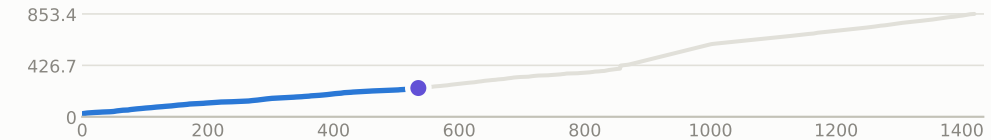
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

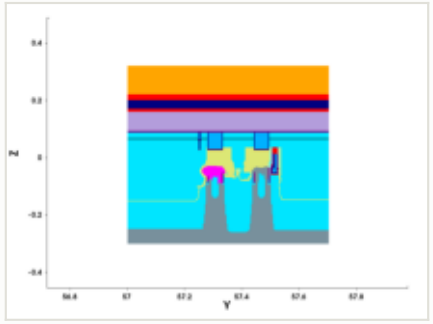
carbon so far, of the whole flow (kgCO2e)



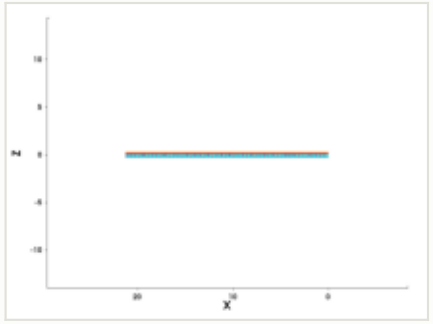
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



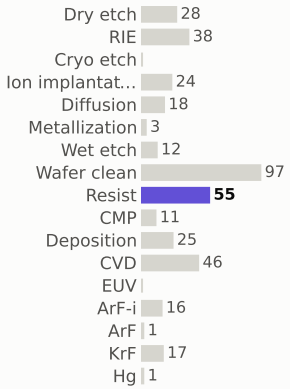
y cut at 57.351 µm



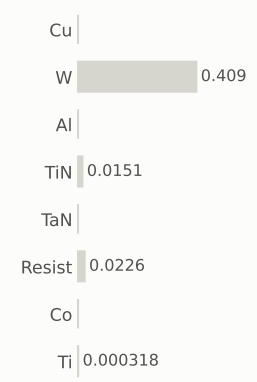
mask for this step

no mask for this step

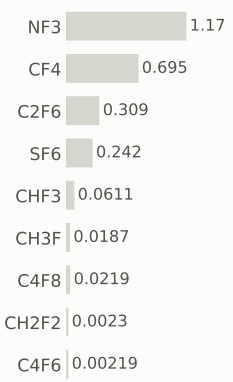
steps by process area



materials deposited (g)



gases emitted (g)



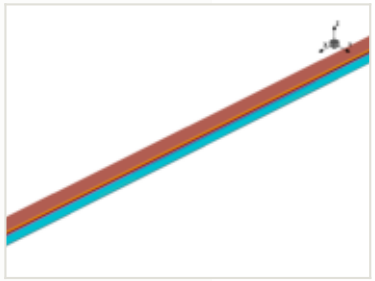
Step #5440: CVD

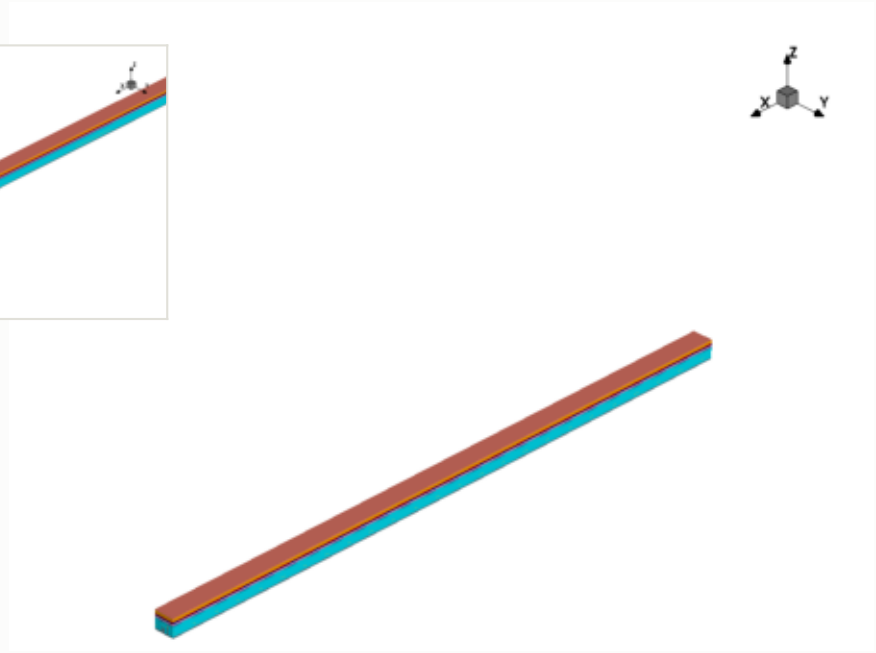
depo_fill · CVD

1.31	240.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



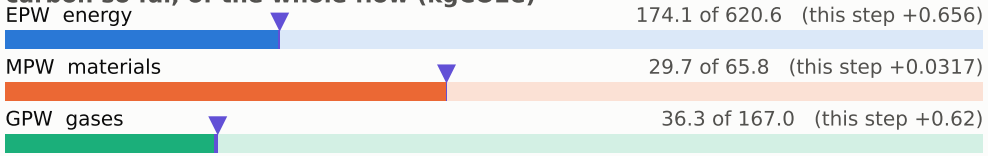


this step

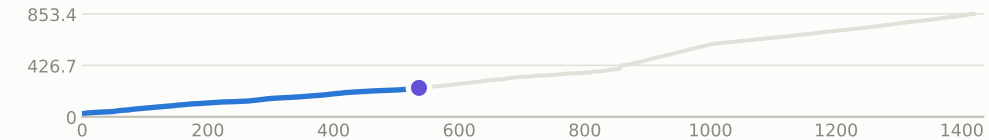
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

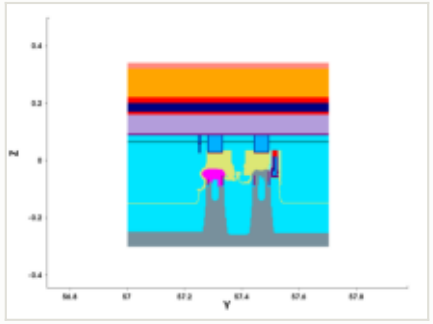
carbon so far, of the whole flow (kgCO2e)



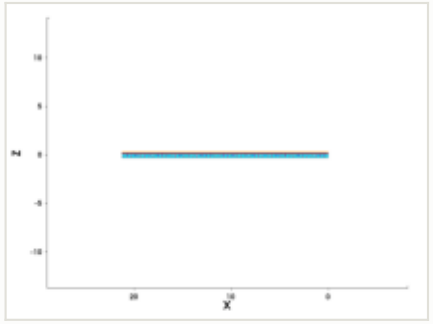
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



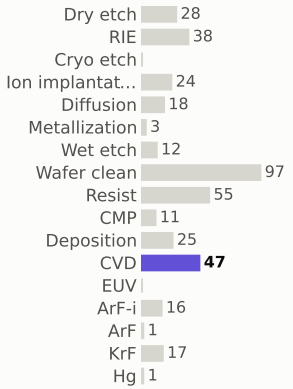
y cut at 57.351 µm



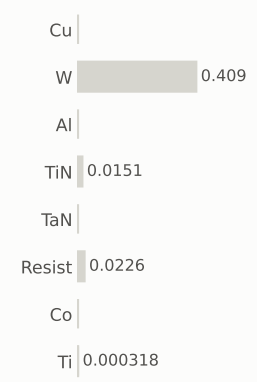
mask for this step

no mask for this step

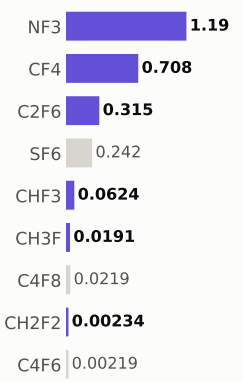
steps by process area



materials deposited (g)



gases emitted (g)



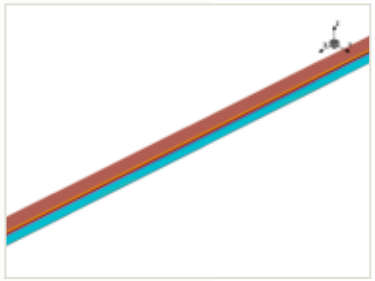
Step #5450: Wafer_clean

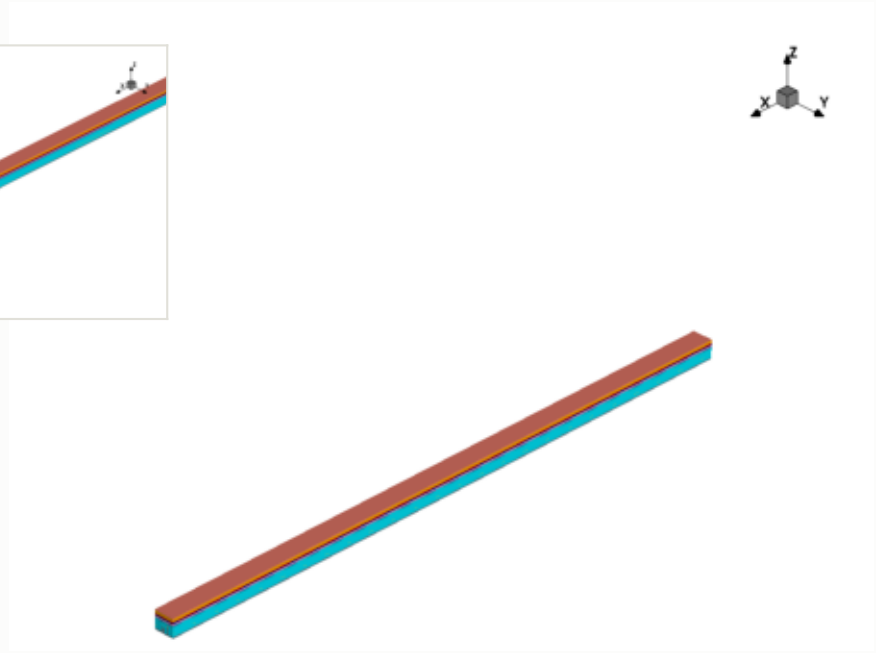
clean · Wafer_clean

0.169	240.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





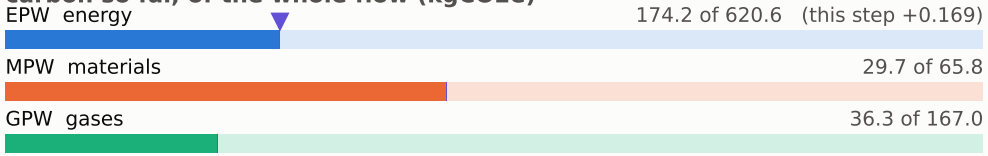
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

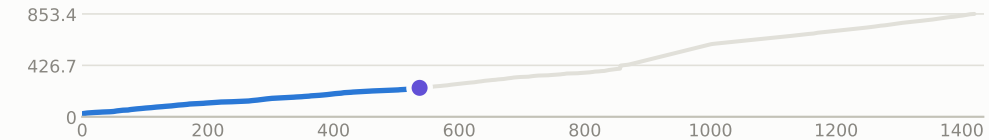
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

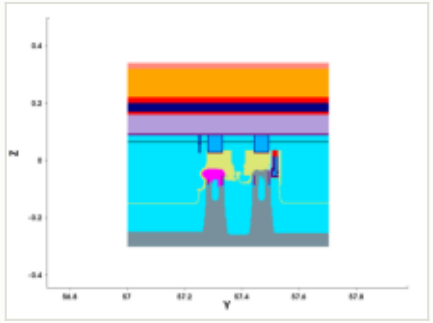
carbon so far, of the whole flow (kgCO2e)



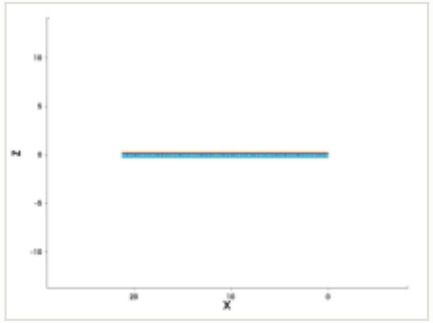
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



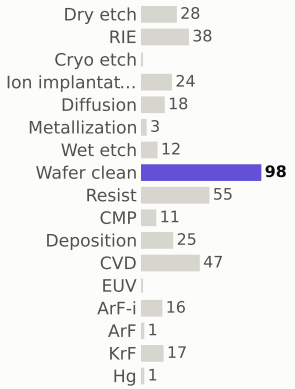
y cut at 57.351 µm



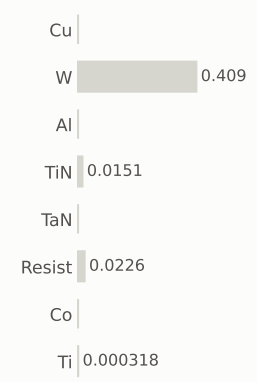
mask for this step

no mask for this step

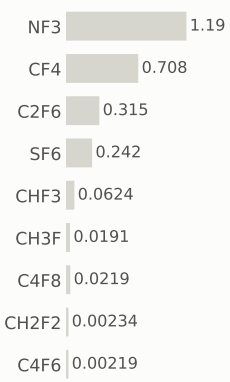
steps by process area



materials deposited (g)



gases emitted (g)



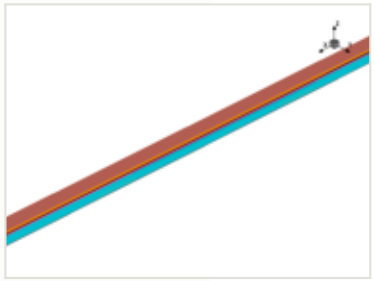
Step #5460: Resist

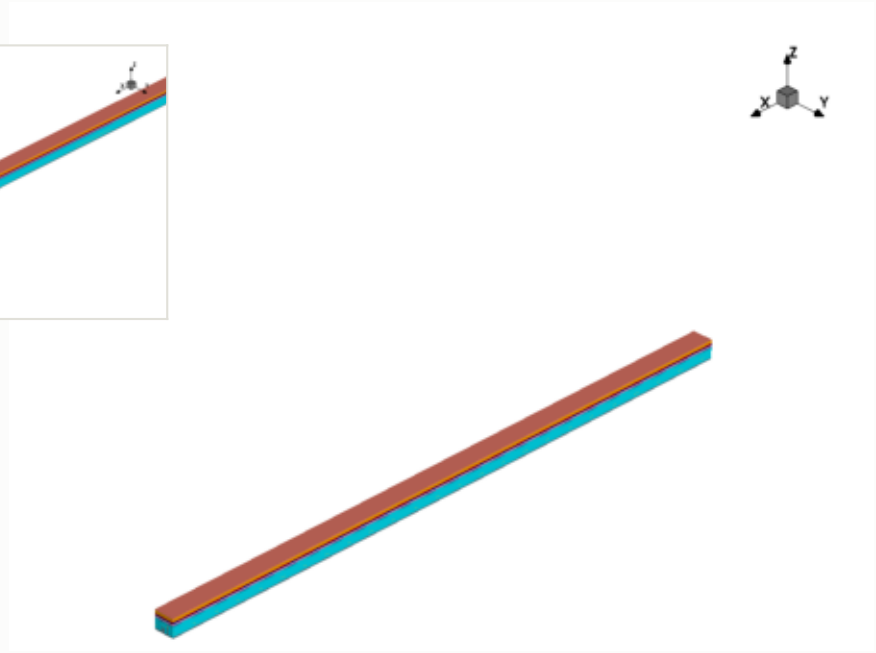
resist · Resist

0.169	240.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





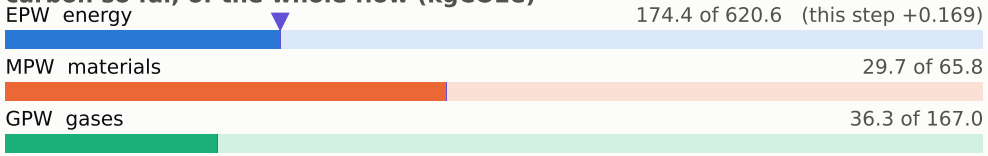
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

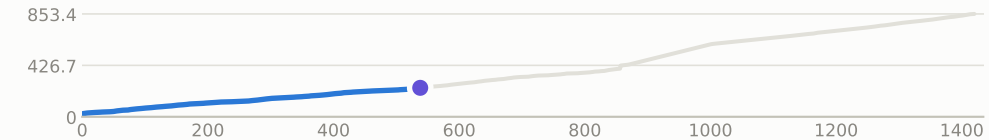
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

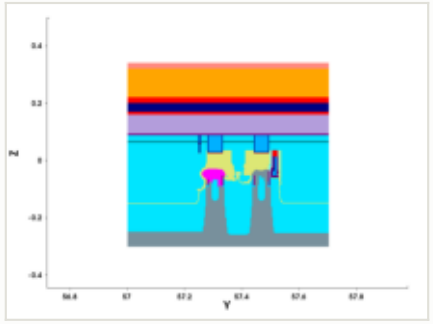
carbon so far, of the whole flow (kgCO2e)



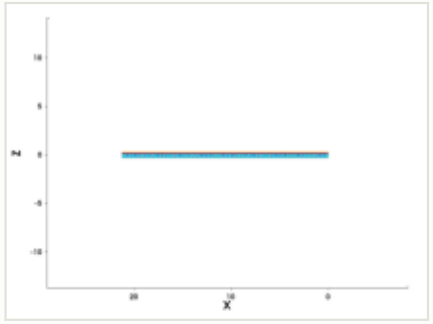
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



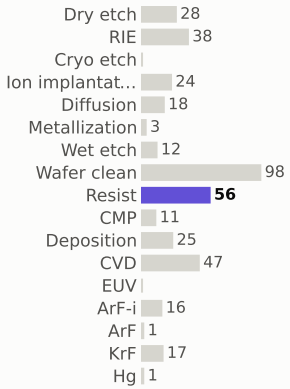
y cut at 57.351 µm



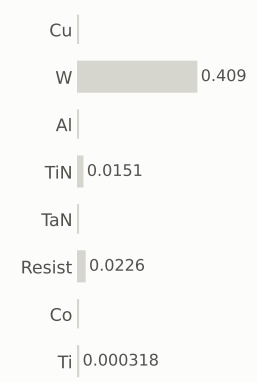
mask for this step

no mask for this step

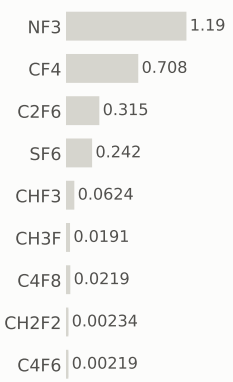
steps by process area



materials deposited (g)



gases emitted (g)



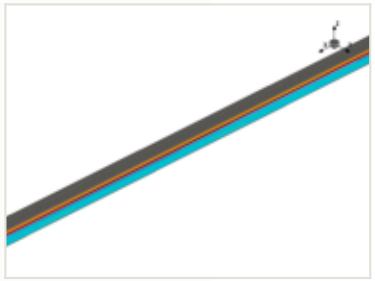
Step #5470: ArF-i

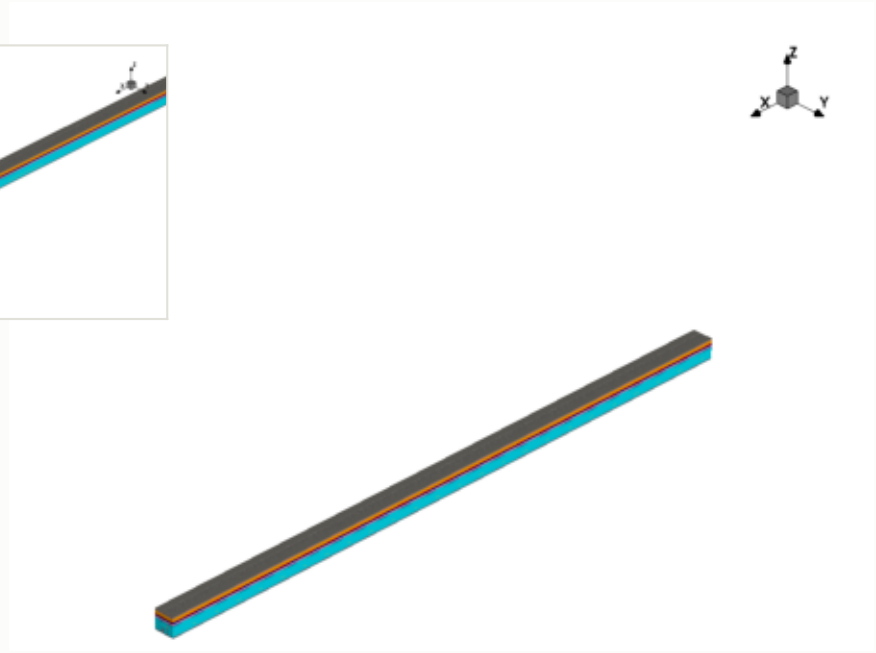
litho_patterning · ArF-i

0.217	240.6	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



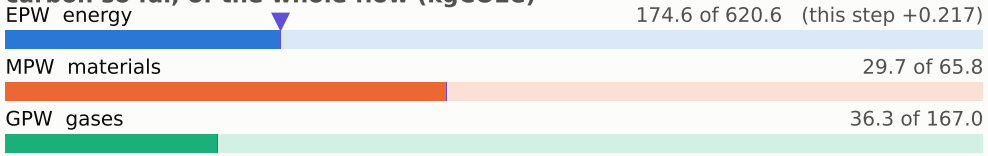


this step

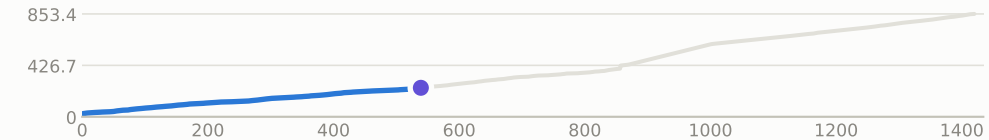
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

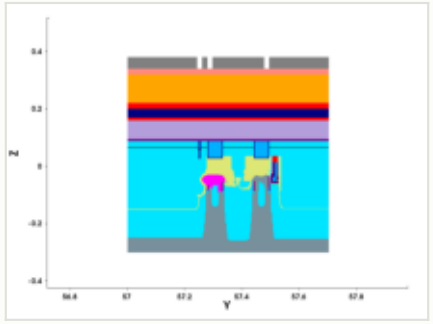
carbon so far, of the whole flow (kgCO2e)



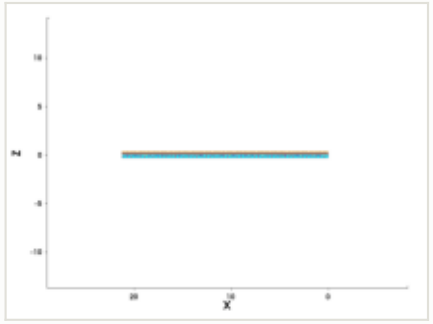
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

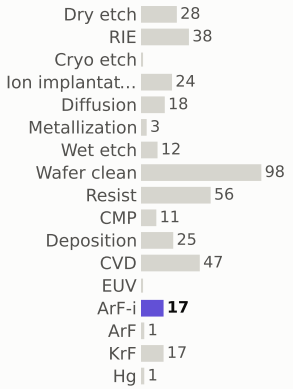


y cut at 57.351 µm

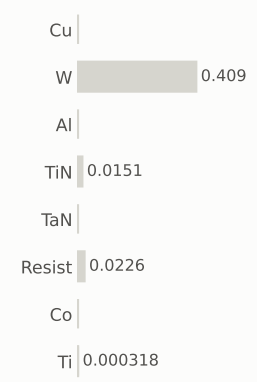


mask: layer 2019:0 (GDSII)

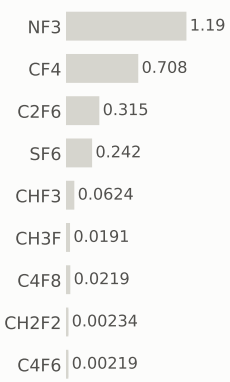
steps by process area



materials deposited (g)



gases emitted (g)



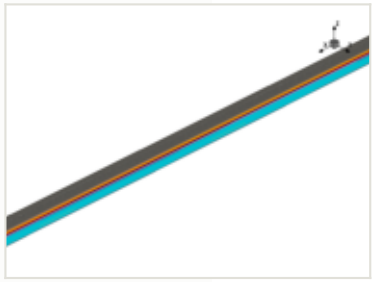
Step #5480: RIE

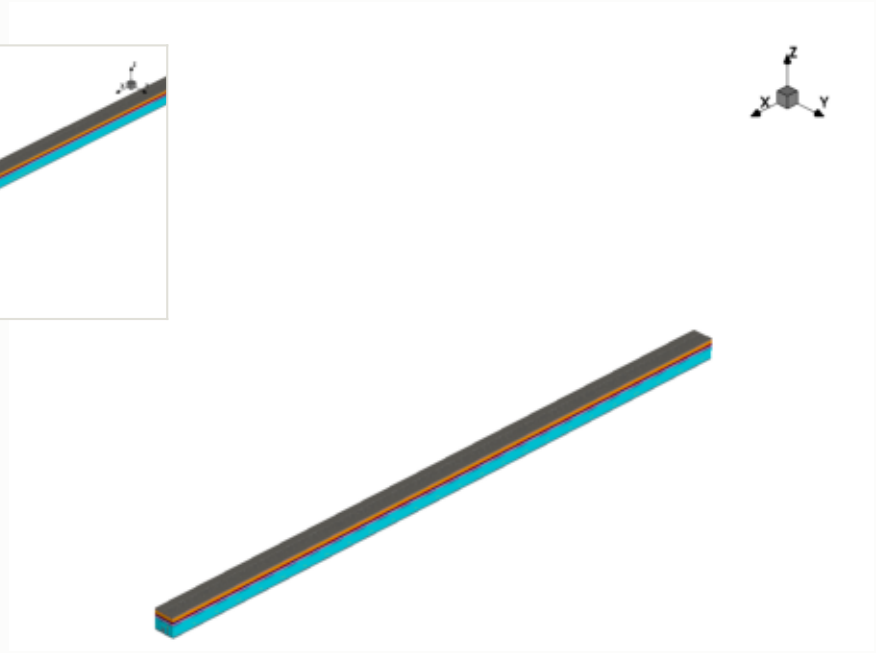
etch_anisotropic · RIE

1.04	241.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



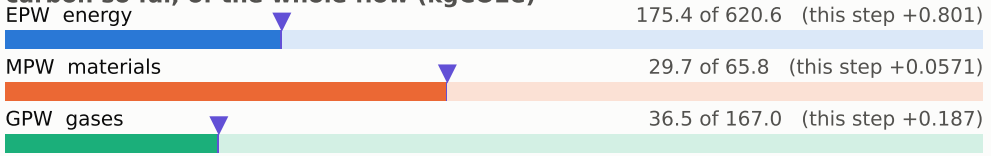


this step

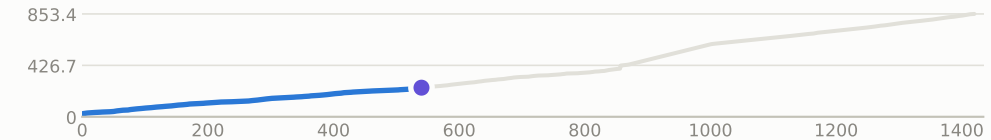
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

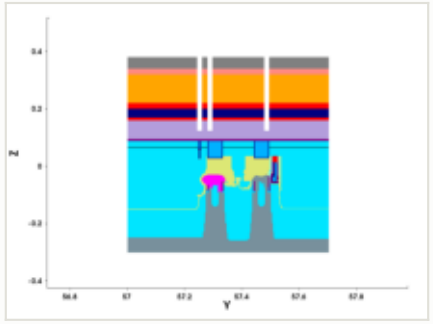
carbon so far, of the whole flow (kgCO2e)



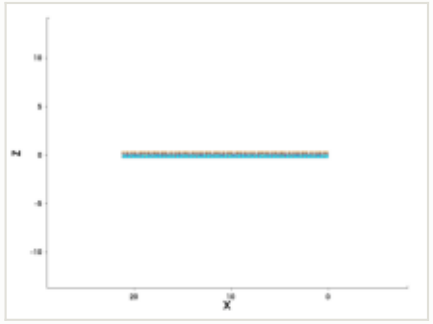
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



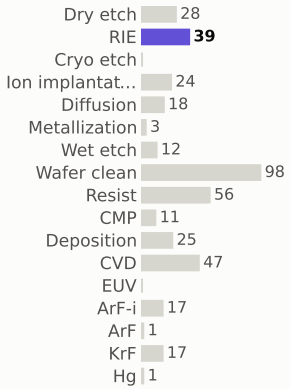
y cut at 57.351 µm



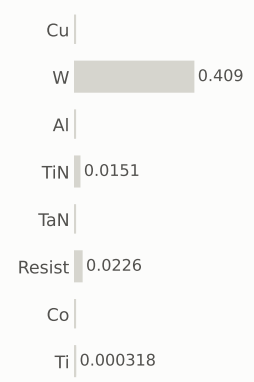
mask for this step

no mask for this step

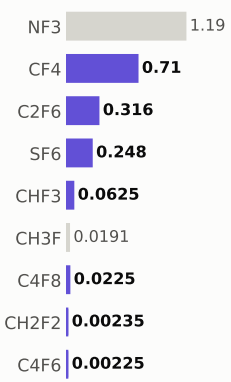
steps by process area



materials deposited (g)



gases emitted (g)



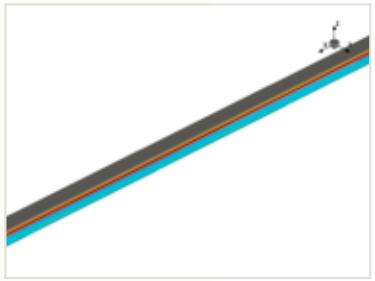
Step #5490: Wafer_clean

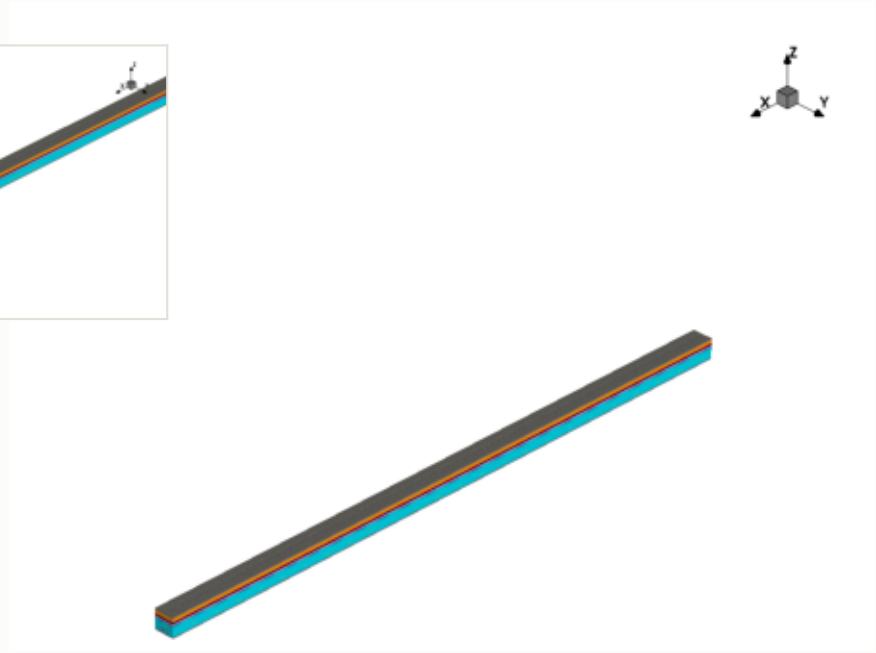
clean · Wafer_clean

0.169	241.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





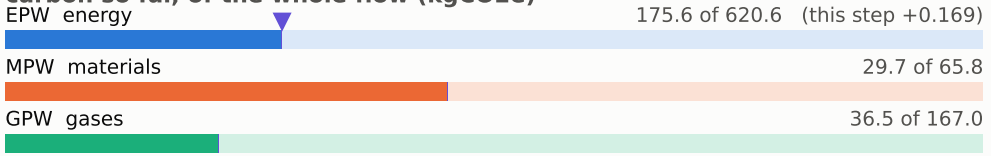
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

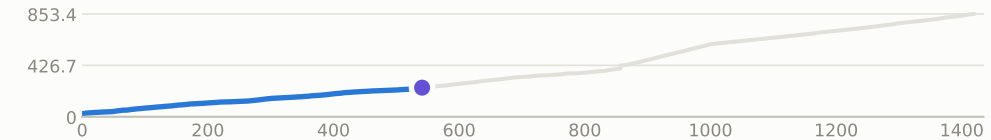
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

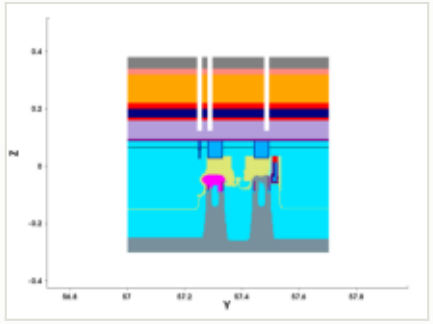
carbon so far, of the whole flow (kgCO2e)



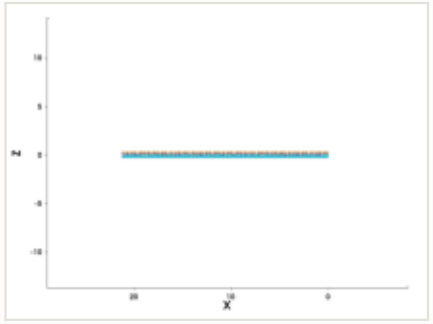
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



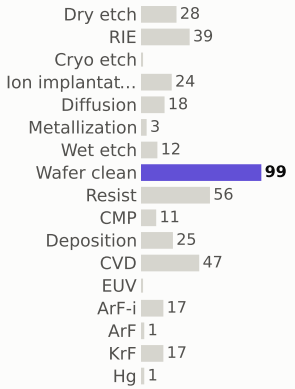
y cut at 57.351 µm



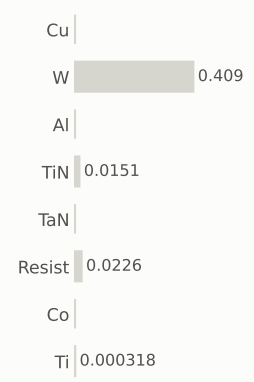
mask for this step

no mask for this step

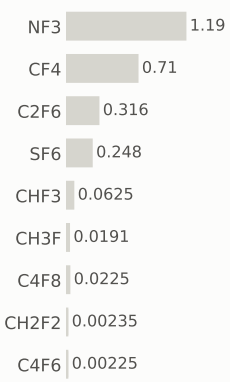
steps by process area



materials deposited (g)



gases emitted (g)



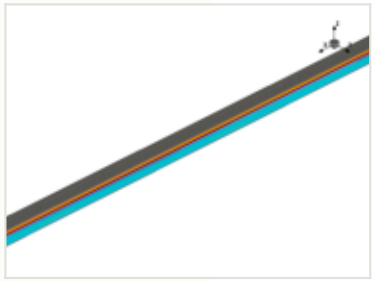
Step #5500: Resist

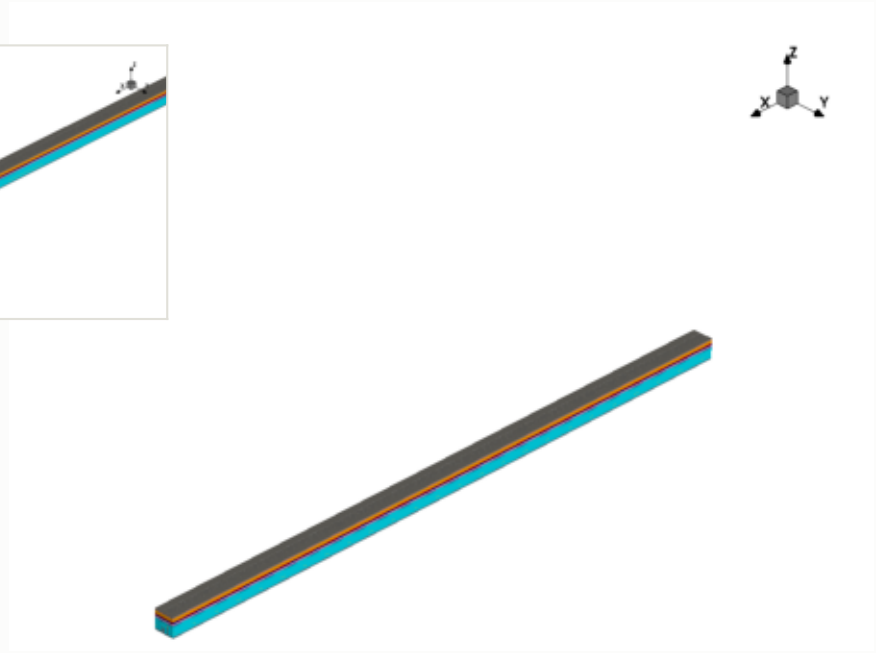
depo_fill · Resist

0.169	242.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



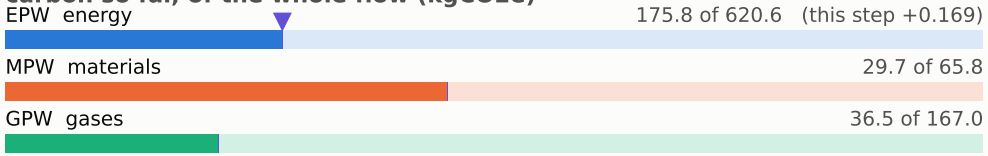


this step

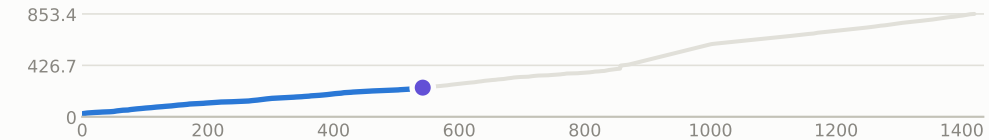
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

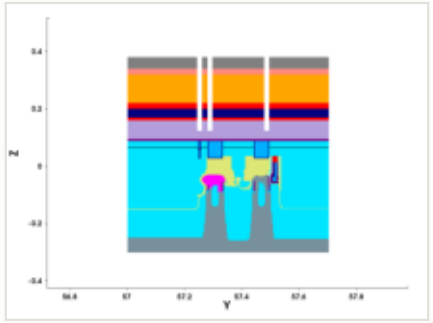
carbon so far, of the whole flow (kgCO2e)



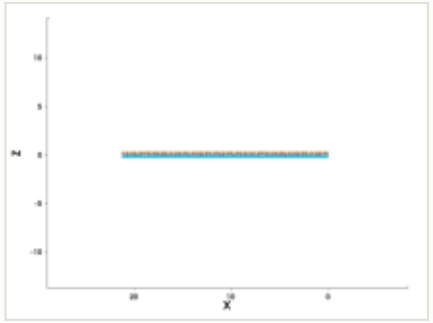
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



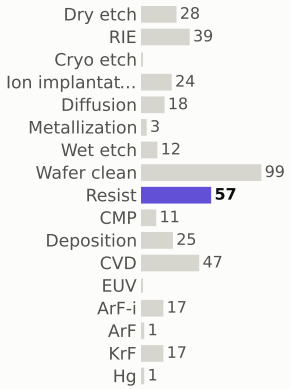
y cut at 57.351 µm



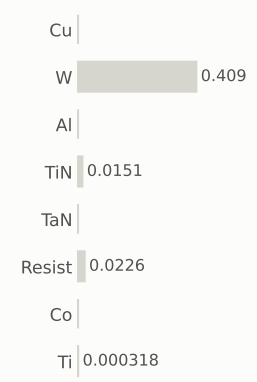
mask for this step

no mask for this step

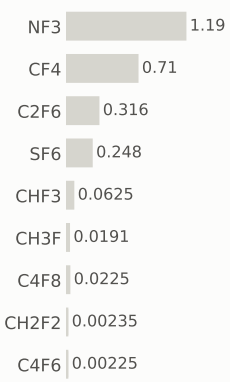
steps by process area



materials deposited (g)



gases emitted (g)



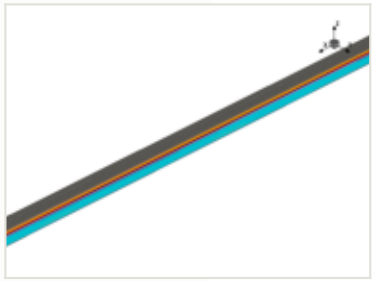
Step #5510: CVD

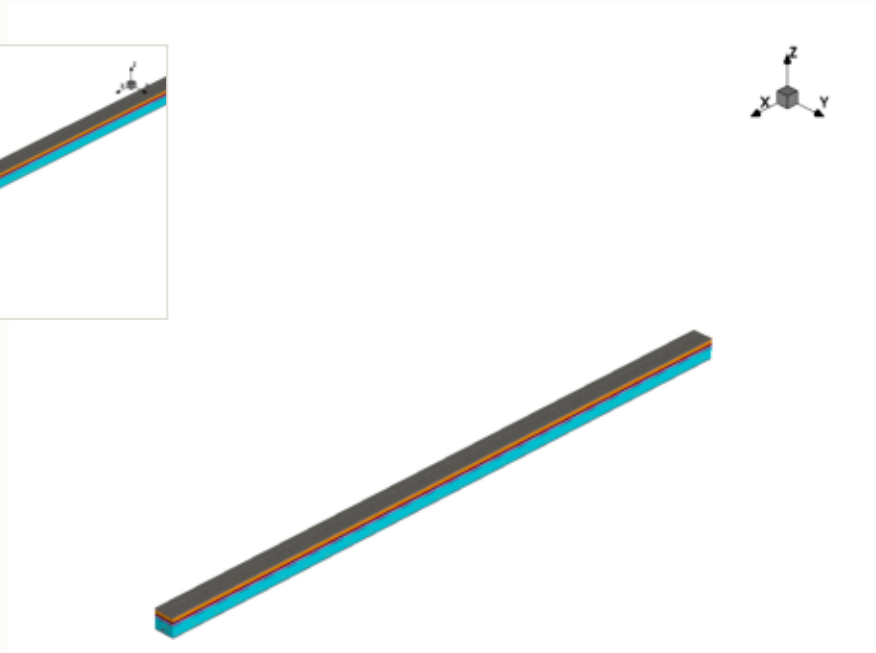
depo_isotropic · CVD

1.31	243.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



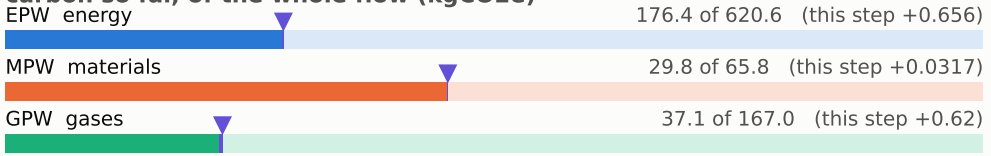


this step

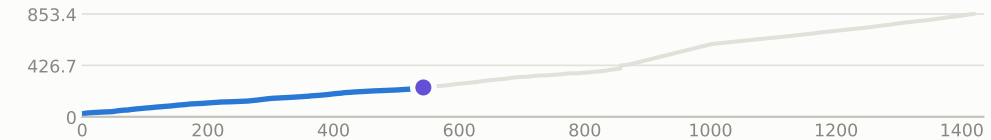
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

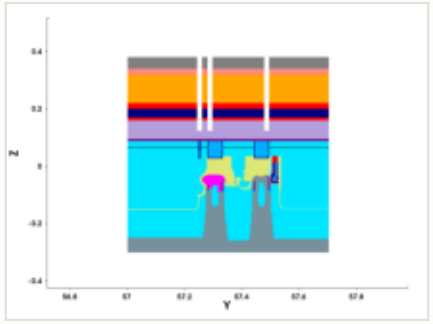
carbon so far, of the whole flow (kgCO2e)



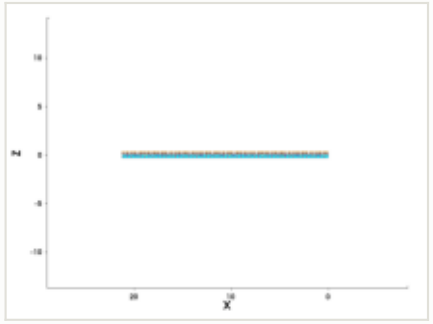
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



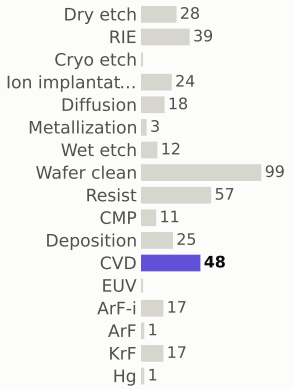
y cut at 57.351 µm



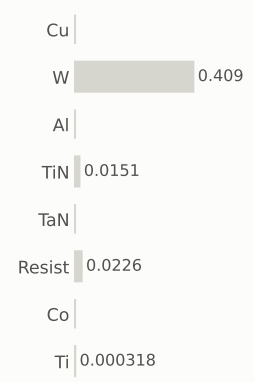
mask for this step

no mask for this step

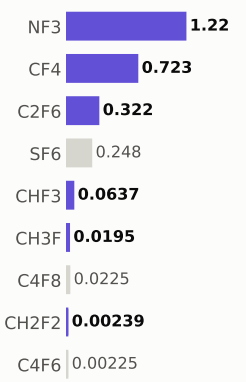
steps by process area



materials deposited (g)



gases emitted (g)



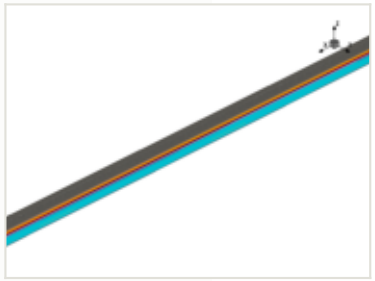
Step #5520: Wafer_clean

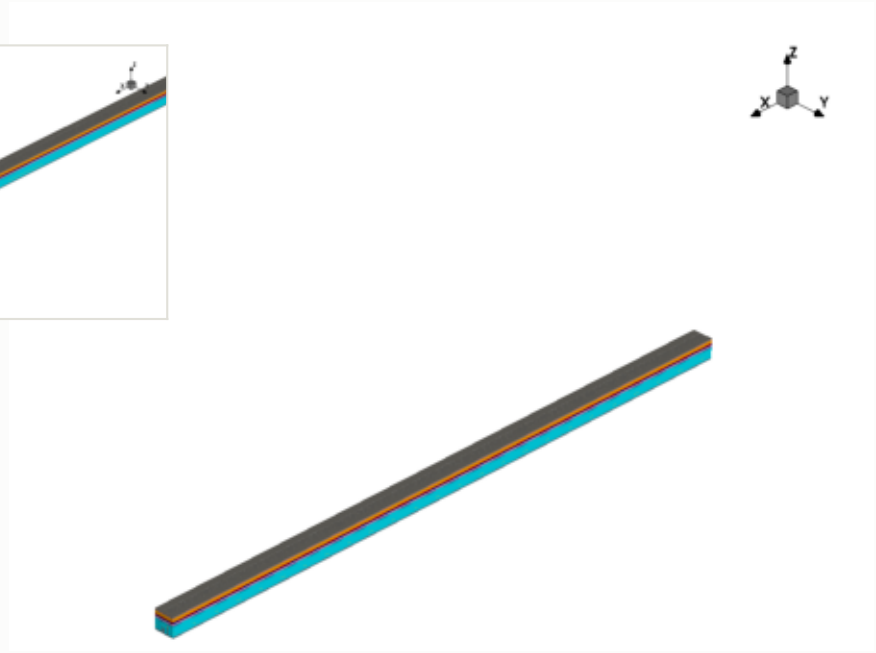
clean · Wafer_clean

0.169	243.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





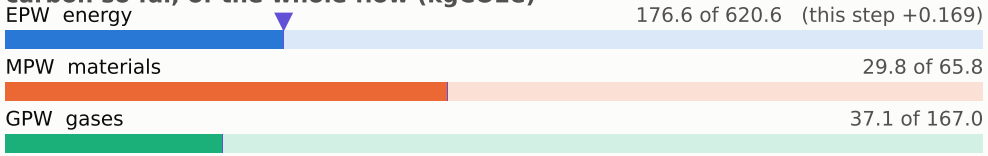
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

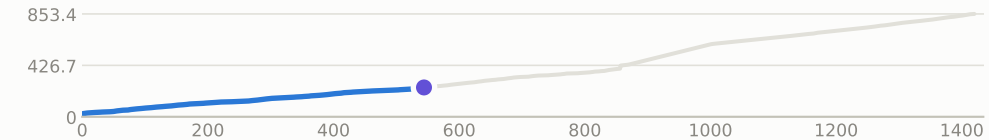
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

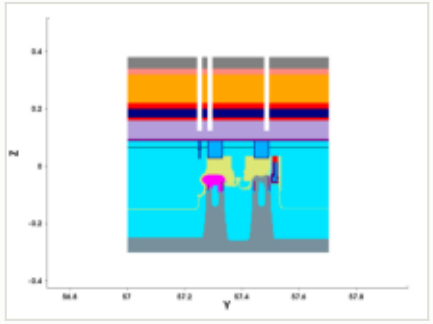
carbon so far, of the whole flow (kgCO2e)



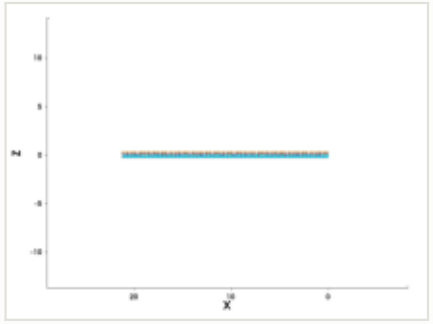
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



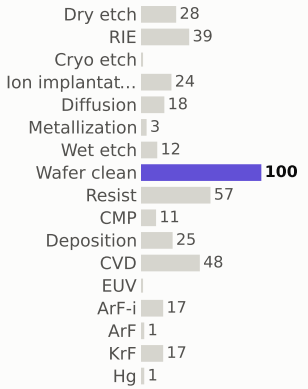
y cut at 57.351 µm



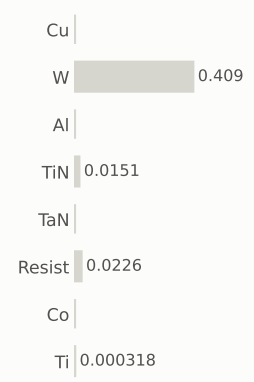
mask for this step

no mask for this step

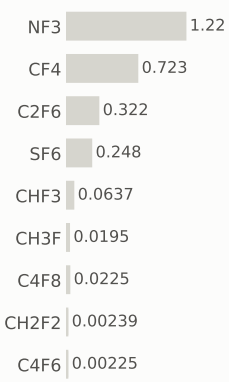
steps by process area



materials deposited (g)



gases emitted (g)



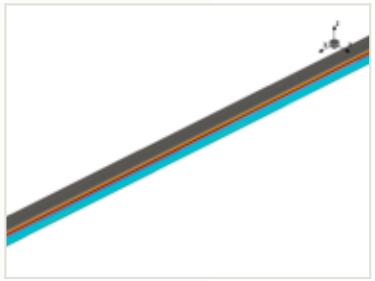
Step #5530: Resist

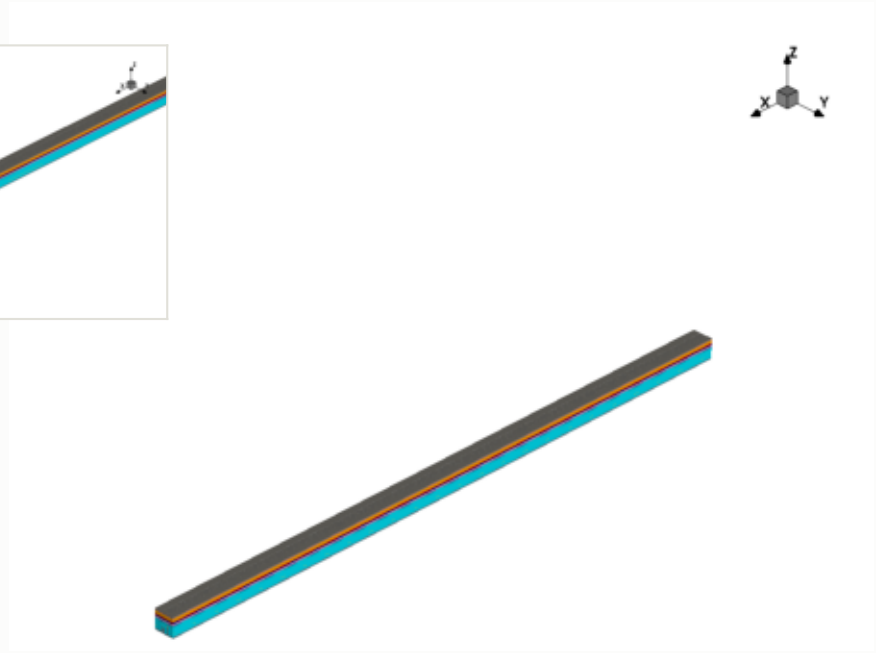
resist · Resist

0.169	243.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





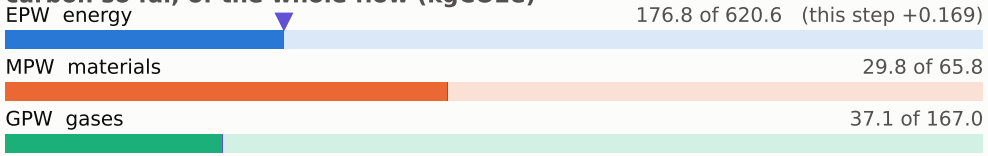
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

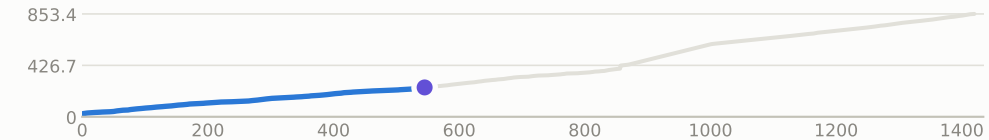
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

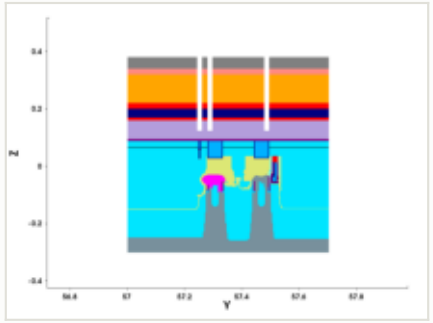
carbon so far, of the whole flow (kgCO2e)



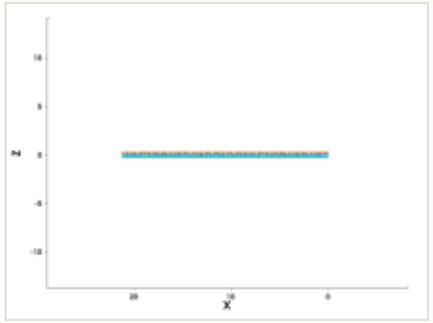
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



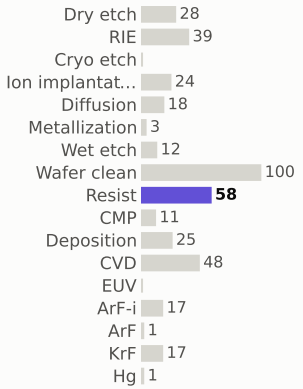
y cut at 57.351 µm



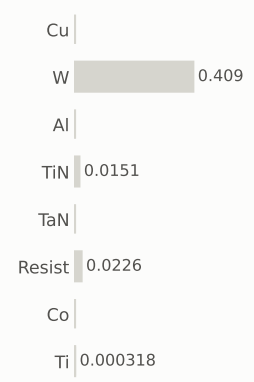
mask for this step

no mask for this step

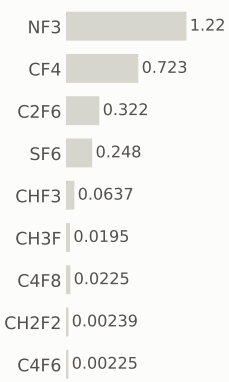
steps by process area



materials deposited (g)



gases emitted (g)



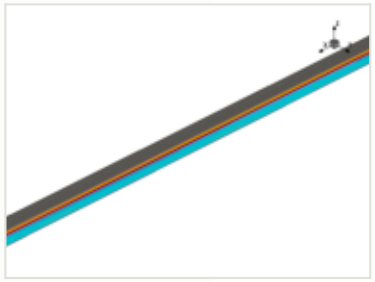
Step #5540: ArF-i

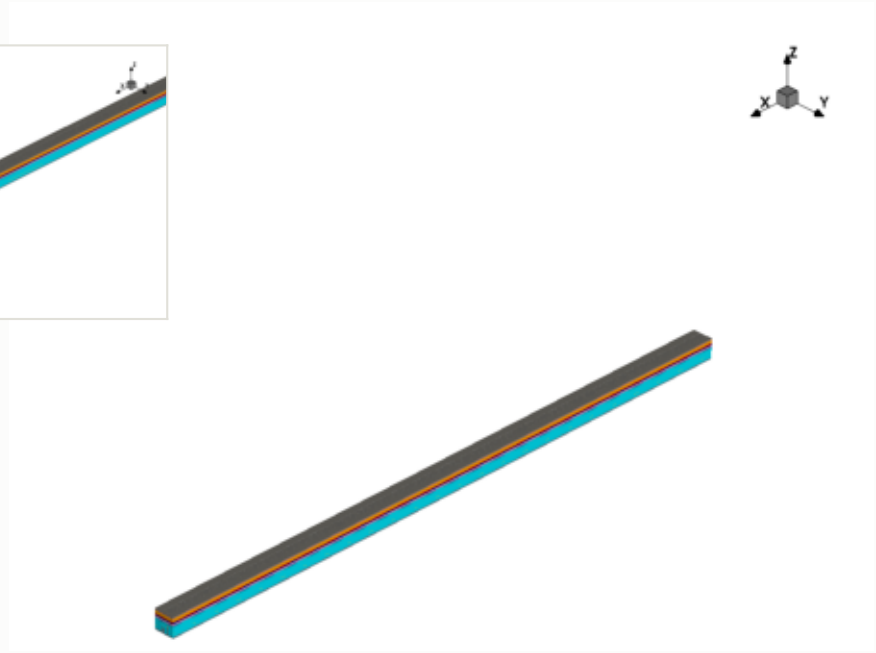
litho_patterning · ArF-i

0.217	243.8	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



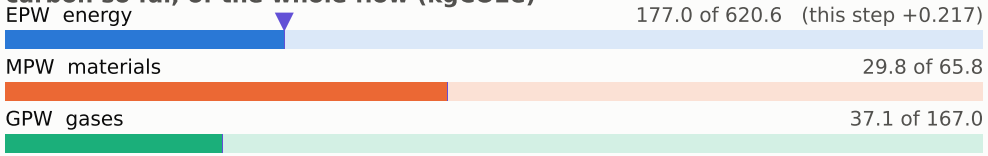


this step

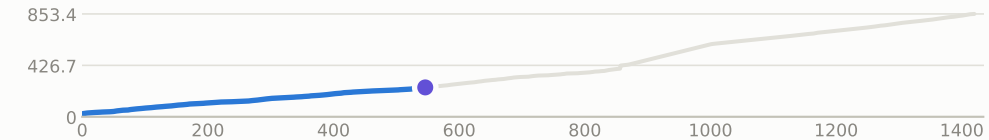
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

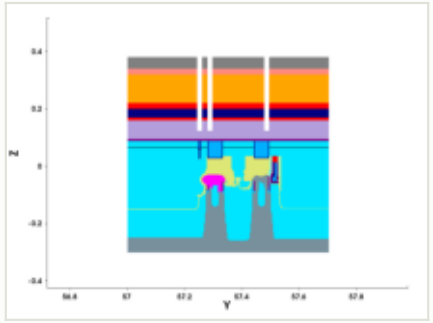
carbon so far, of the whole flow (kgCO2e)



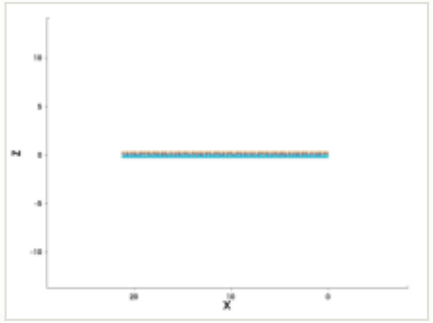
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

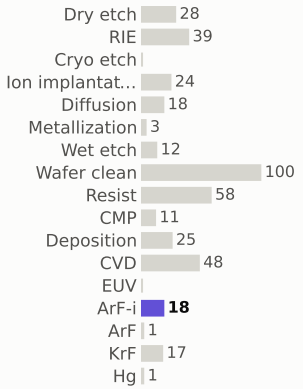


y cut at 57.351 µm

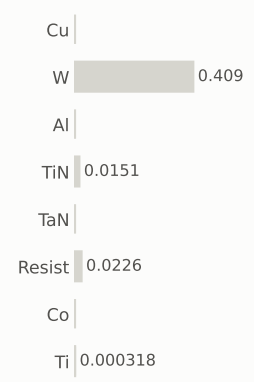


mask: layer 2100:0, positive (GDSII)

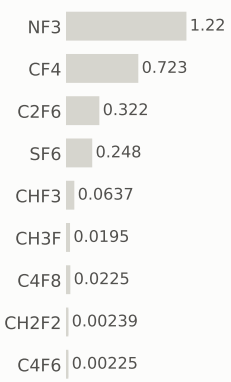
steps by process area



materials deposited (g)



gases emitted (g)



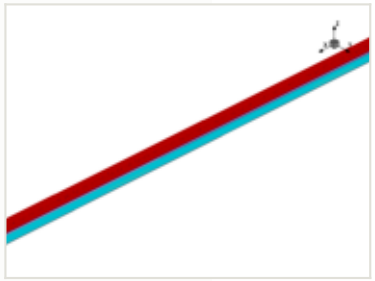
Step #5550: RIE

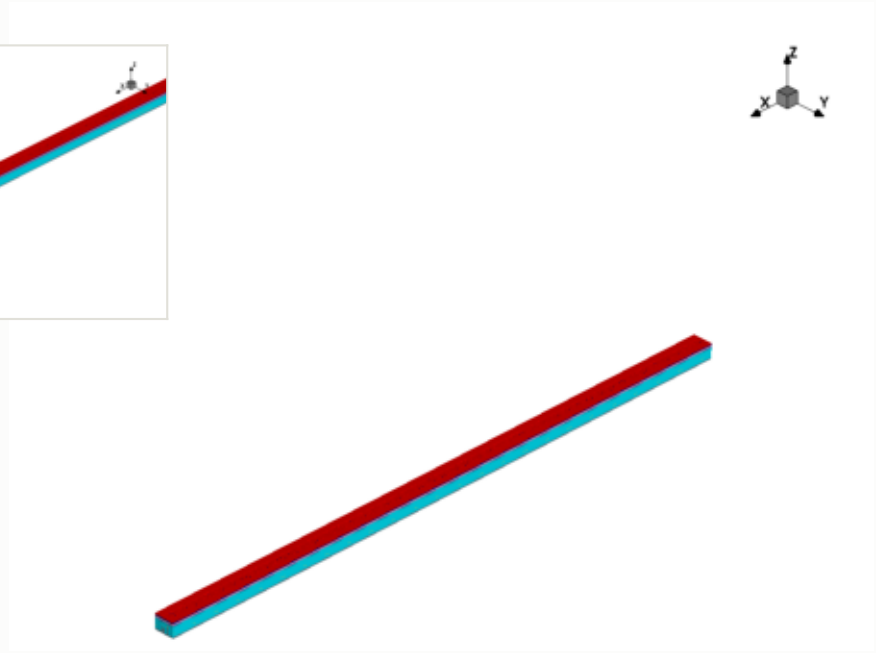
etch_strip · RIE

1.04	244.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



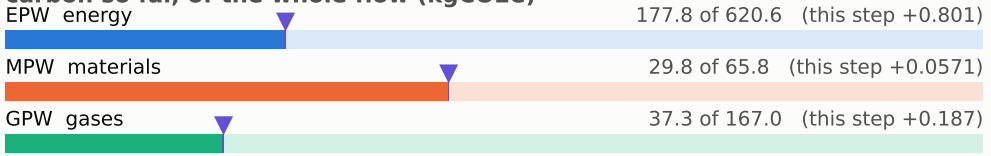


this step

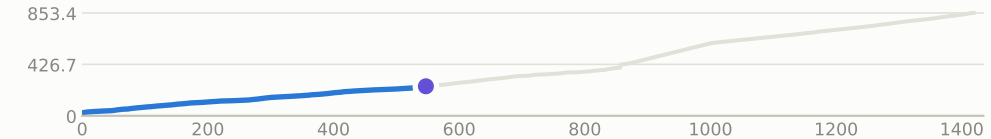
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

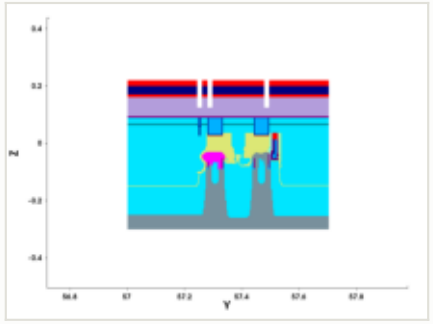
carbon so far, of the whole flow (kgCO2e)



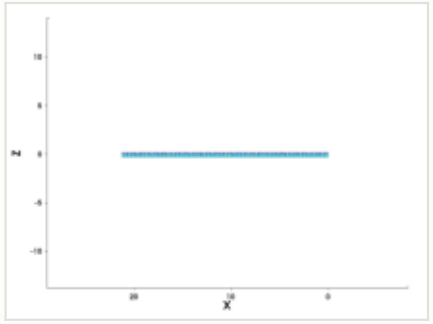
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



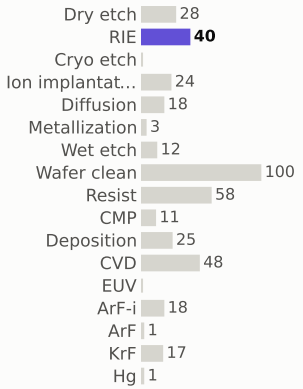
y cut at 57.351 µm



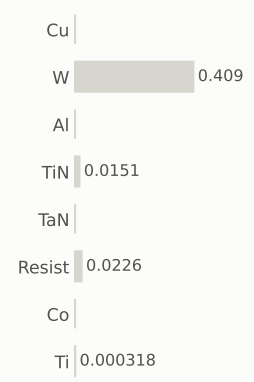
mask for this step

no mask for this step

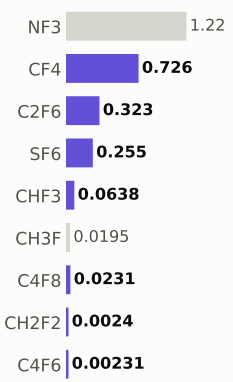
steps by process area



materials deposited (g)



gases emitted (g)



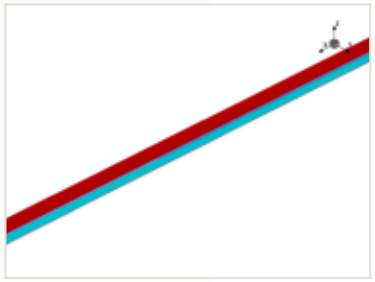
Step #5560: Wafer_clean

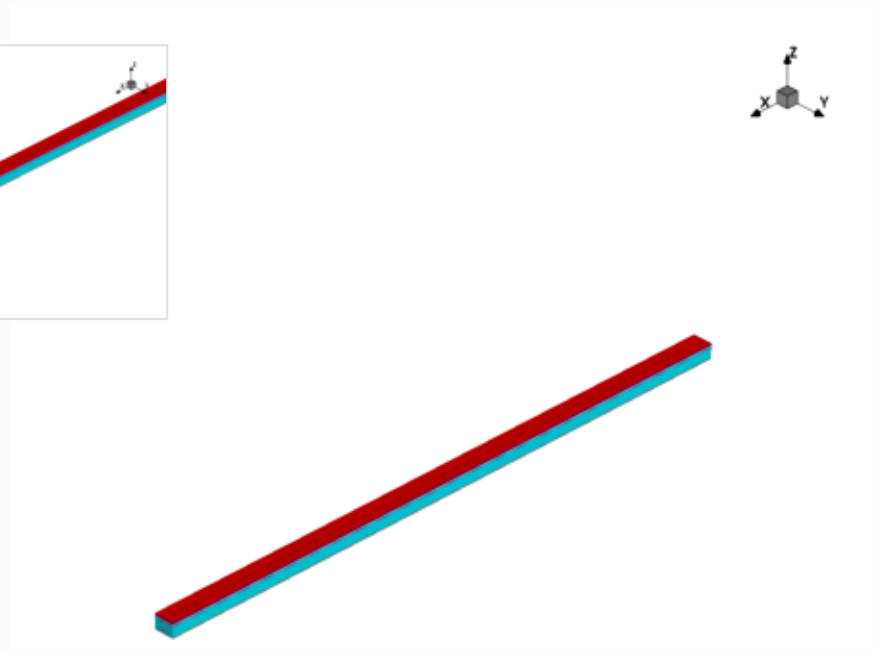
clean · Wafer_clean

0.169	245.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





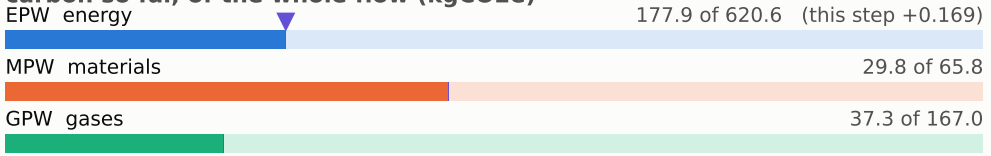
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

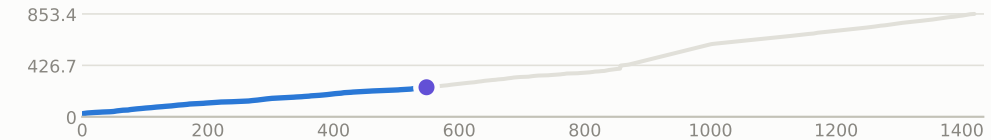
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

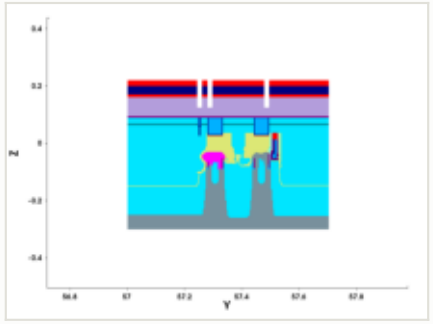
carbon so far, of the whole flow (kgCO2e)



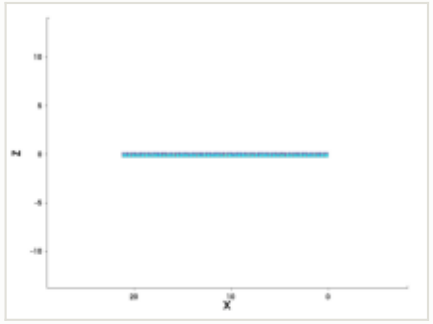
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



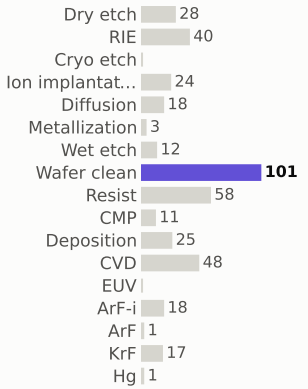
y cut at 57.351 µm



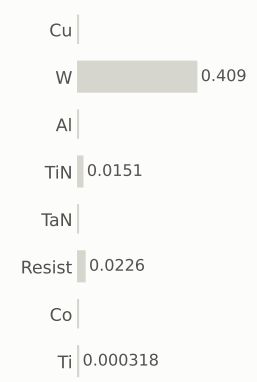
mask for this step

no mask for this step

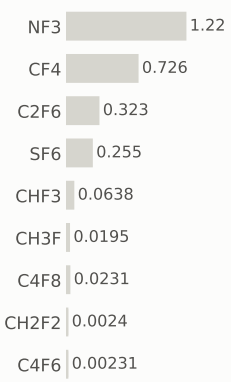
steps by process area



materials deposited (g)



gases emitted (g)



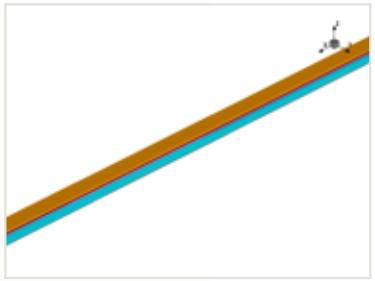
Step #5570: Resist

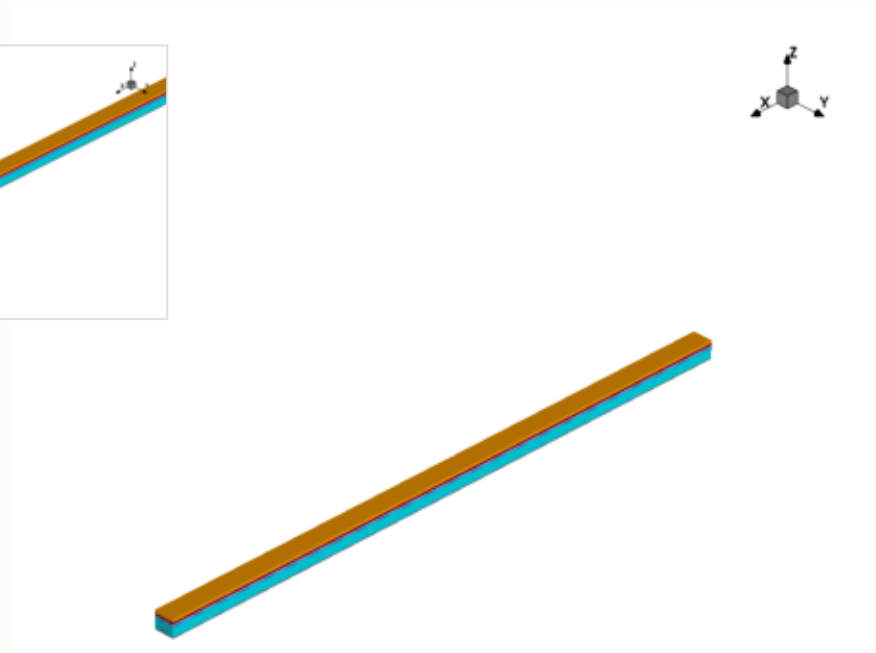
depo_fill · Resist

0.169	245.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



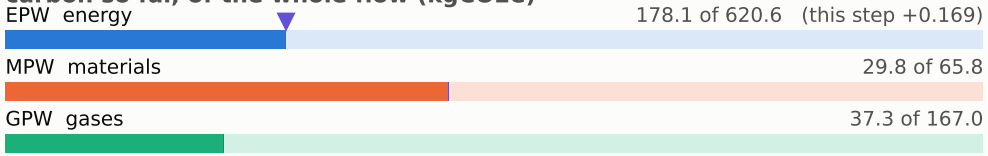


this step

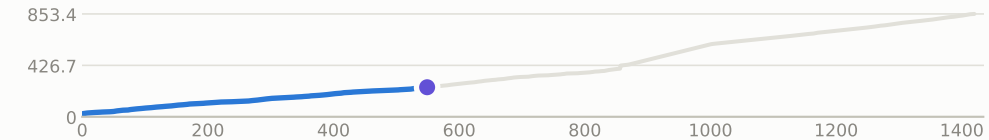
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

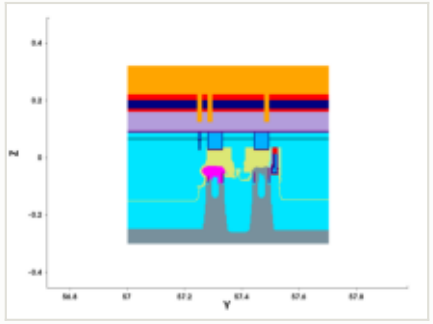
carbon so far, of the whole flow (kgCO2e)



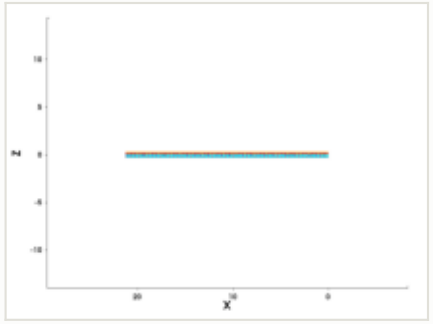
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



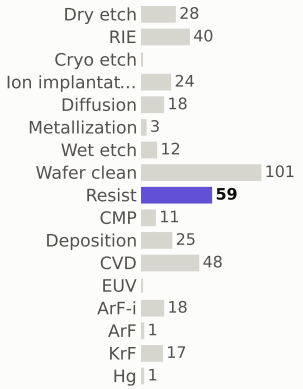
y cut at 57.351 µm



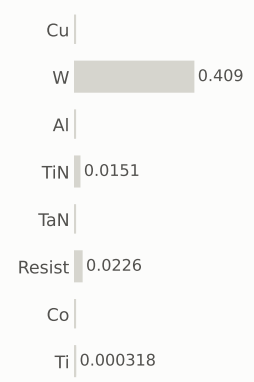
mask for this step

no mask for this step

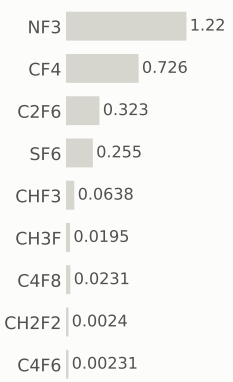
steps by process area



materials deposited (g)



gases emitted (g)



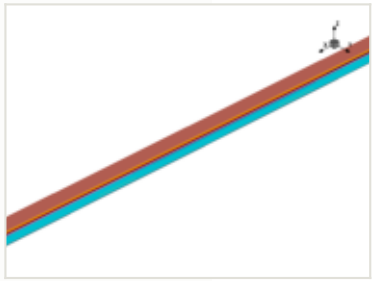
Step #5580: CVD

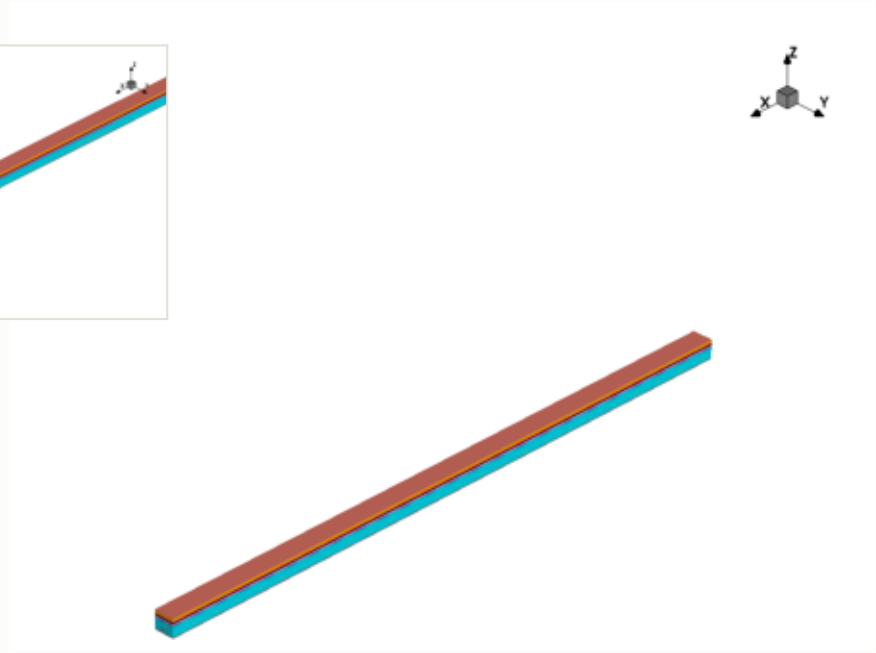
depo_fill · CVD

1.31	246.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



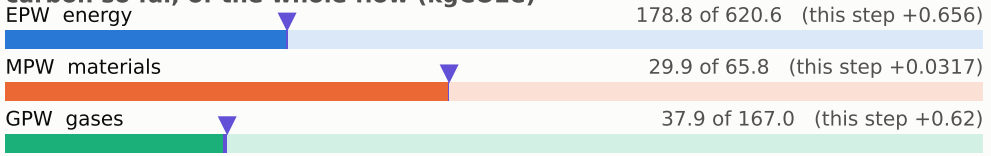


this step

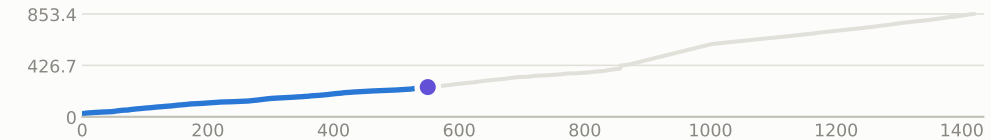
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

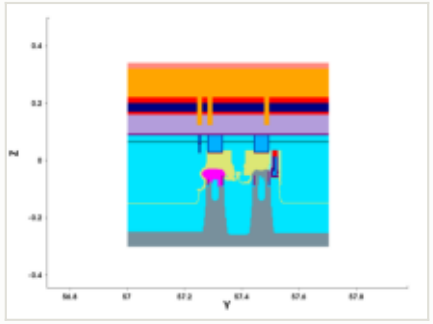
carbon so far, of the whole flow (kgCO2e)



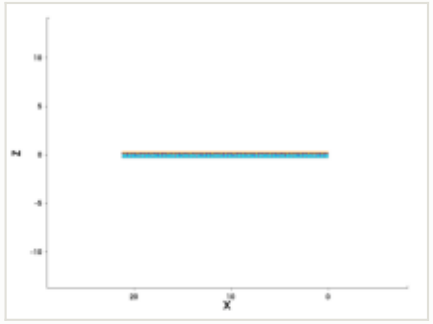
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



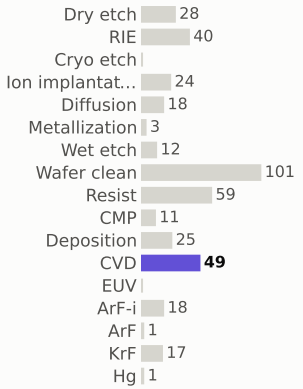
y cut at 57.351 µm



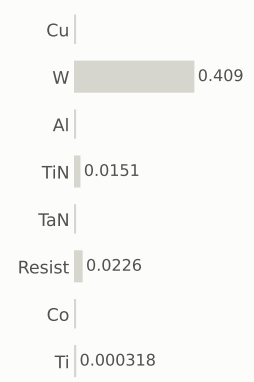
mask for this step

no mask for this step

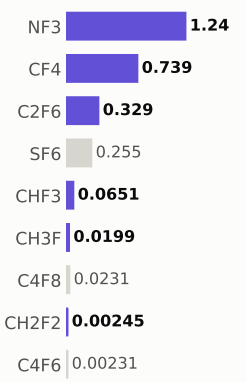
steps by process area



materials deposited (g)



gases emitted (g)



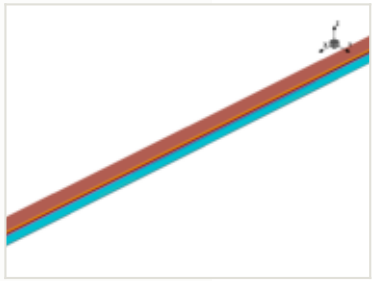
Step #5590: Wafer_clean

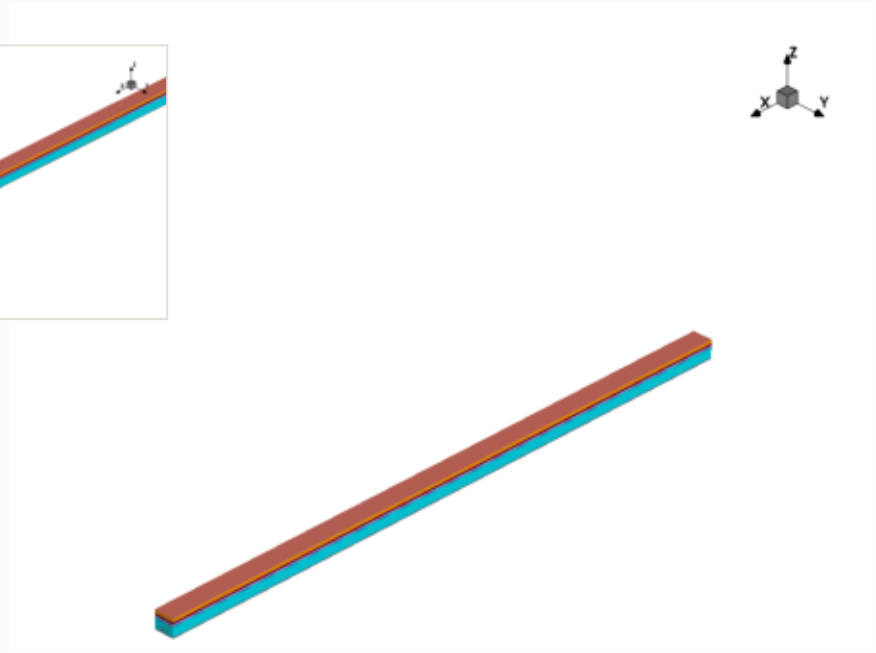
clean · Wafer_clean

0.169	246.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



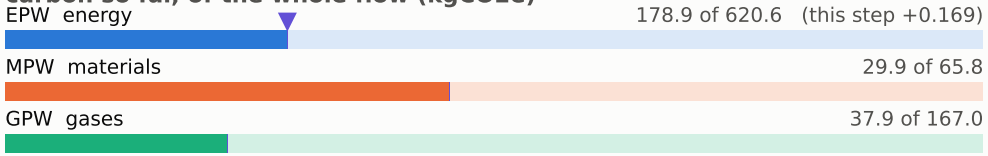


this step

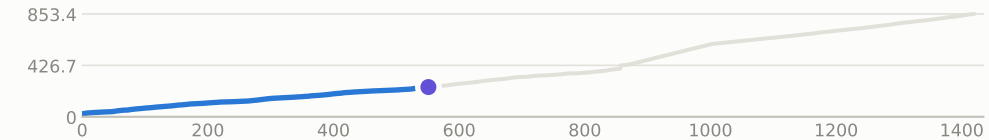
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

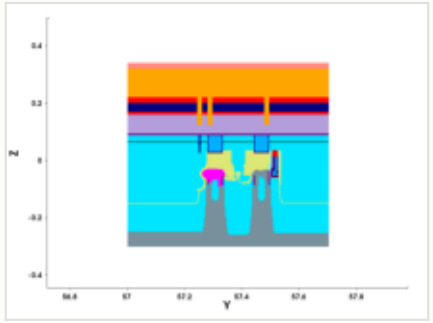
carbon so far, of the whole flow (kgCO2e)



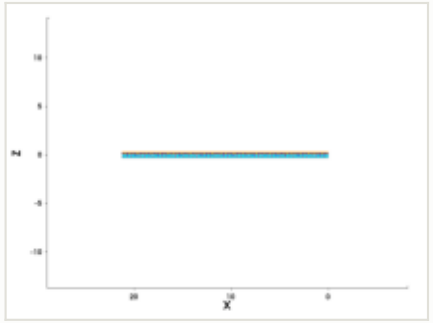
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



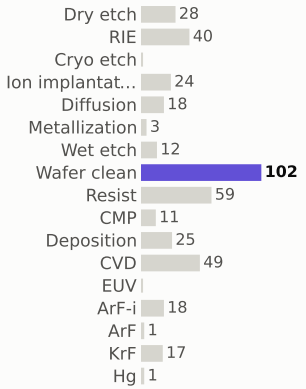
y cut at 57.351 µm



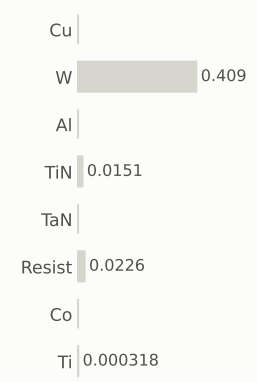
mask for this step

no mask for this step

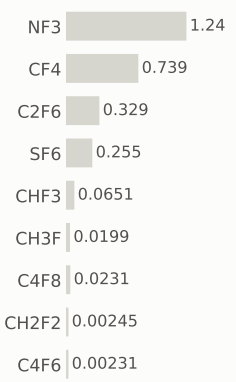
steps by process area



materials deposited (g)



gases emitted (g)



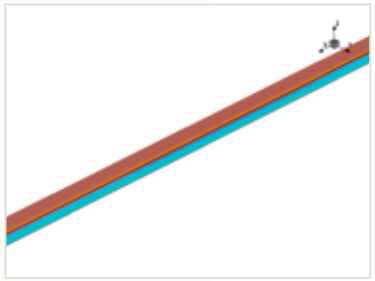
Step #5600: Resist

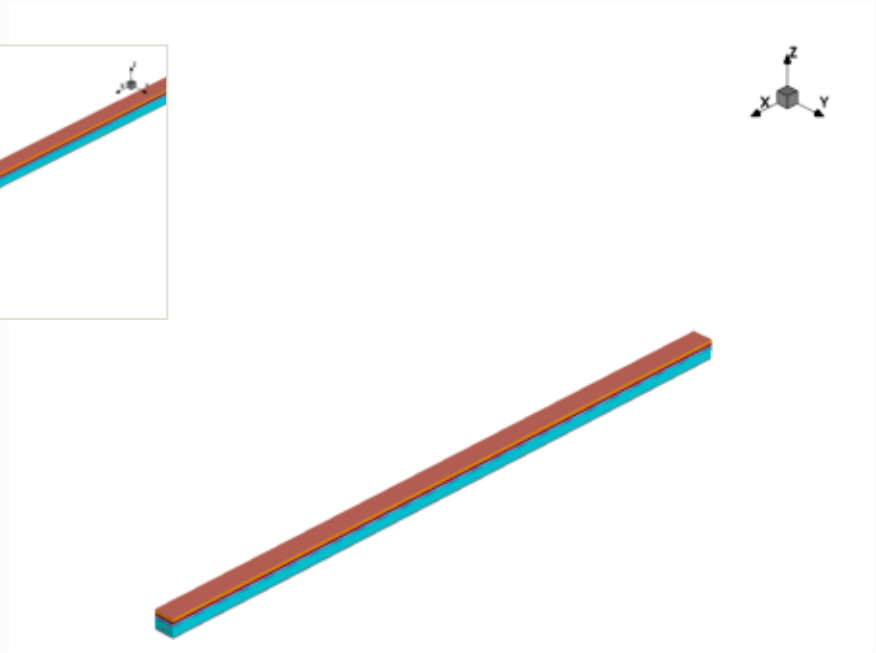
resist · Resist

0.169	246.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





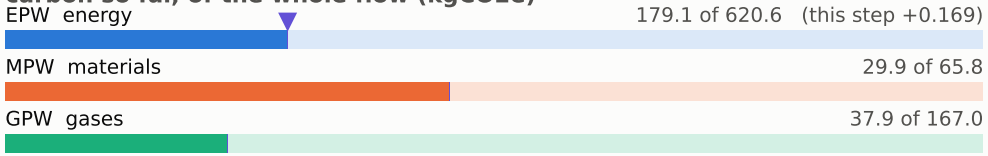
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

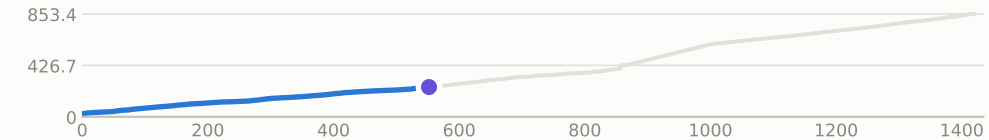
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

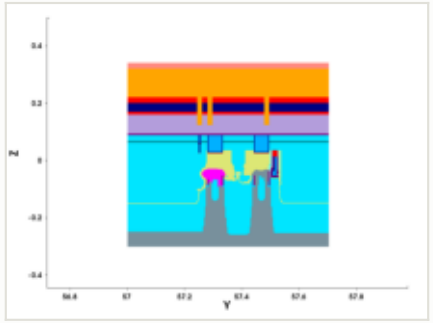
carbon so far, of the whole flow (kgCO2e)



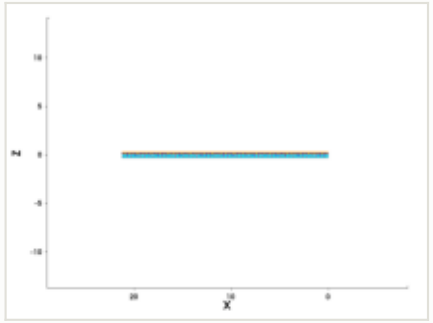
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



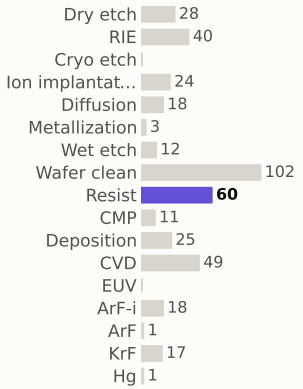
y cut at 57.351 µm



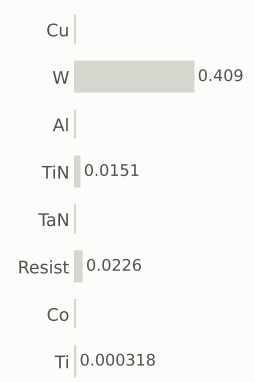
mask for this step

no mask for this step

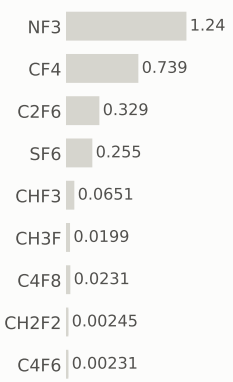
steps by process area



materials deposited (g)



gases emitted (g)



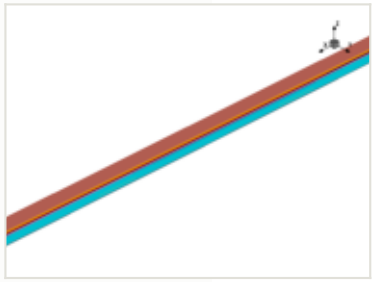
Step #5610: ArF-i

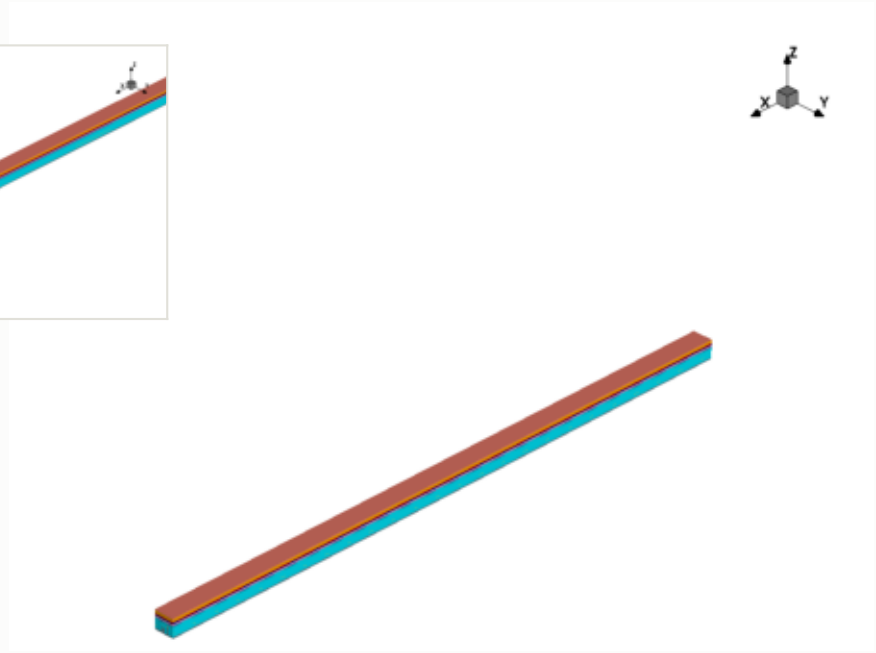
litho_patterning · ArF-i

0.217	247.1	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



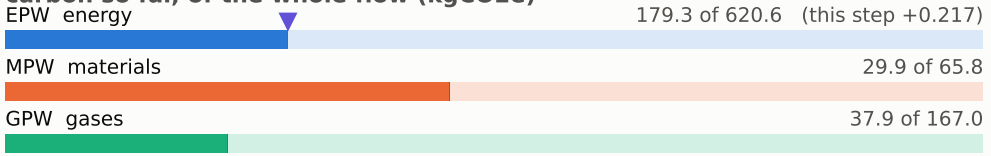


this step

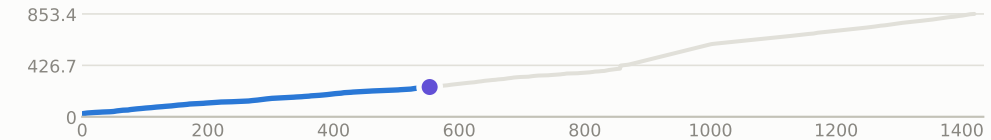
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

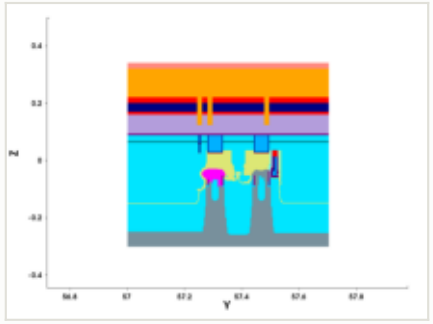
carbon so far, of the whole flow (kgCO2e)



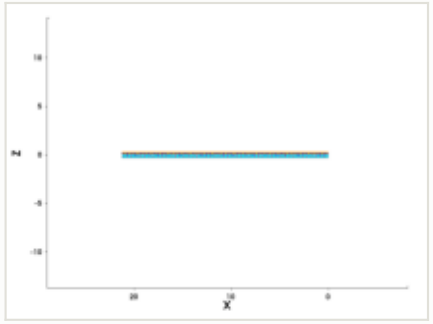
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

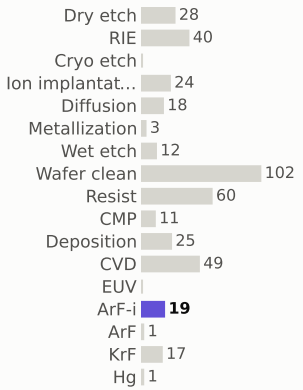


y cut at 57.351 µm

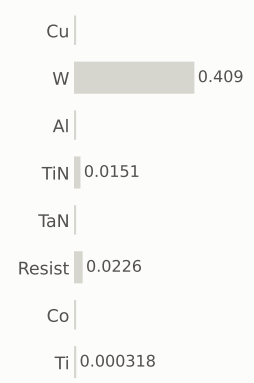


mask: layer 2100:0, positive (GDSII)

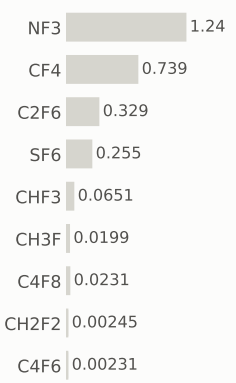
steps by process area



materials deposited (g)



gases emitted (g)



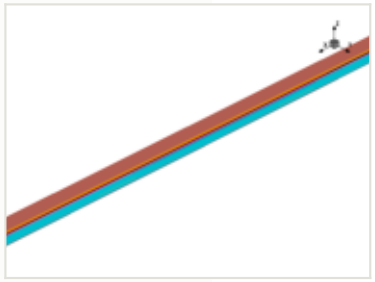
Step #5620: RIE

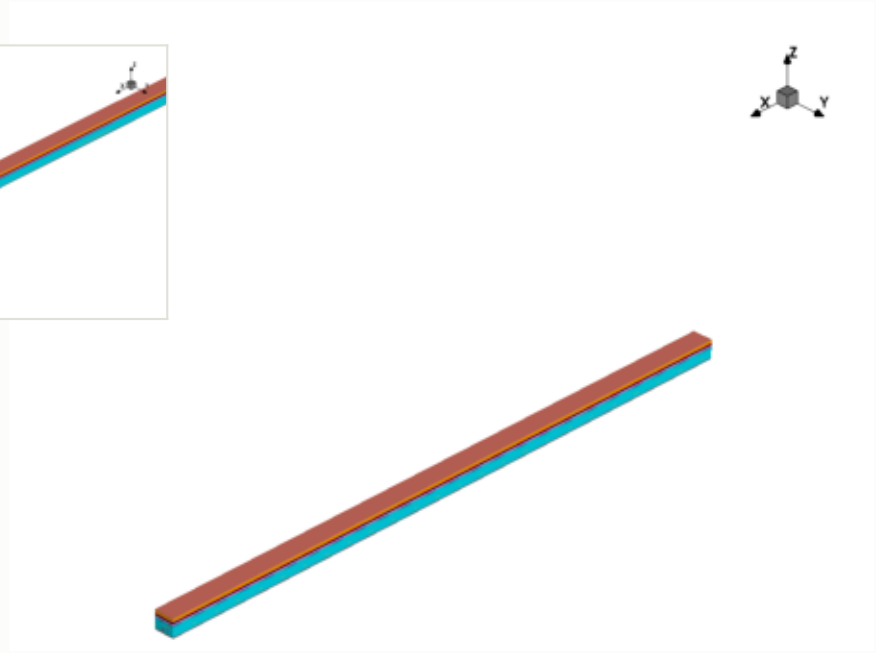
etch_anisotropic · RIE

1.04	248.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



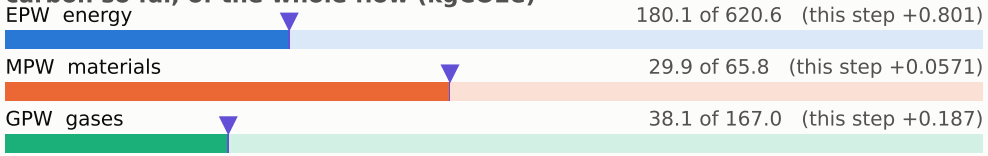


this step

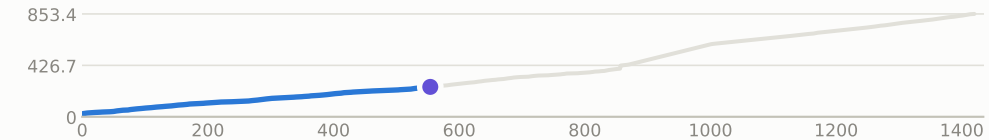
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

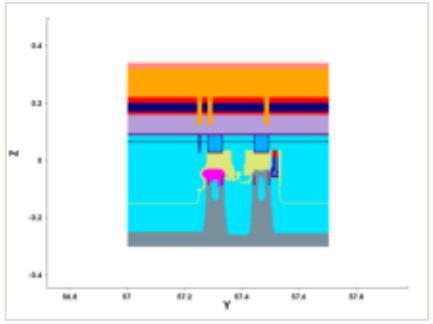
carbon so far, of the whole flow (kgCO2e)



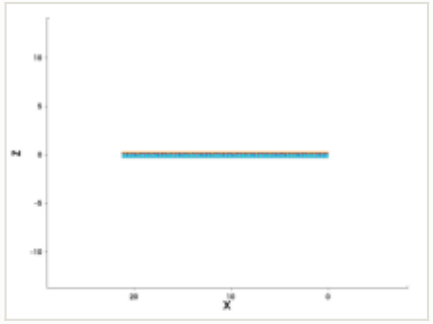
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



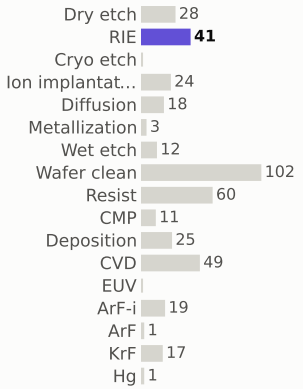
y cut at 57.351 µm



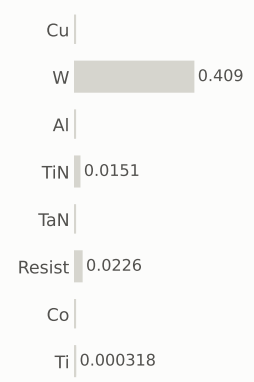
mask for this step

no mask for this step

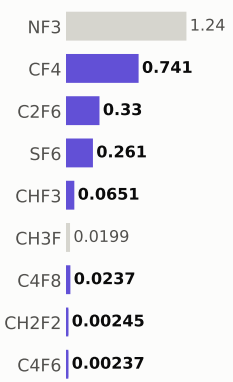
steps by process area



materials deposited (g)



gases emitted (g)



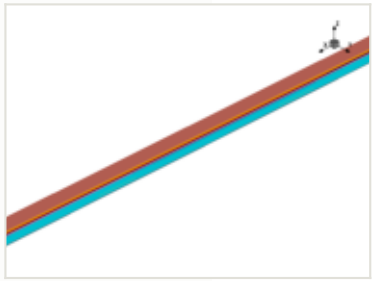
Step #5630: Wafer_clean

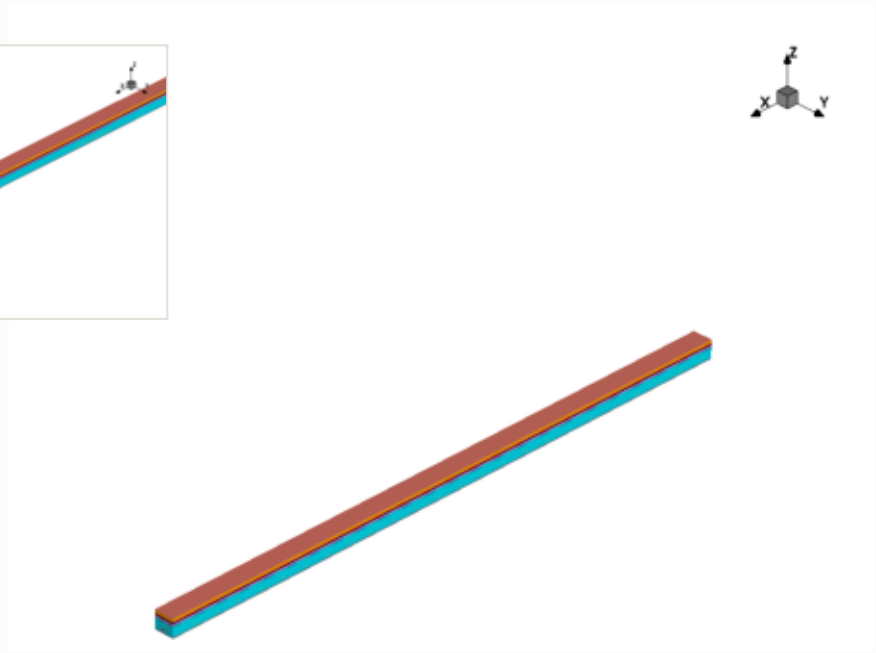
clean · Wafer_clean

0.169	248.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



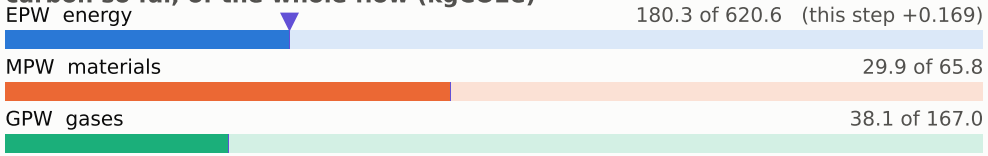


this step

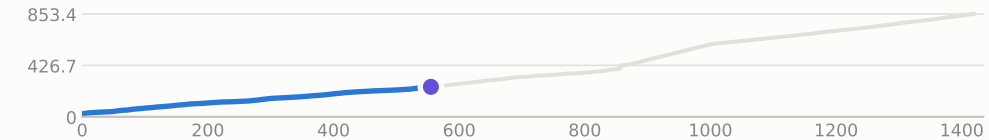
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

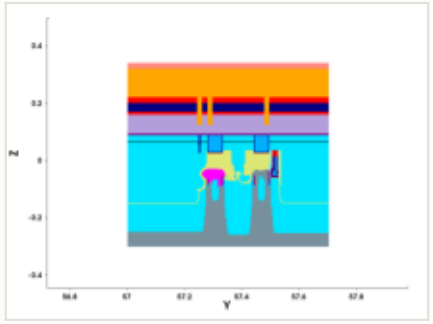
carbon so far, of the whole flow (kgCO2e)



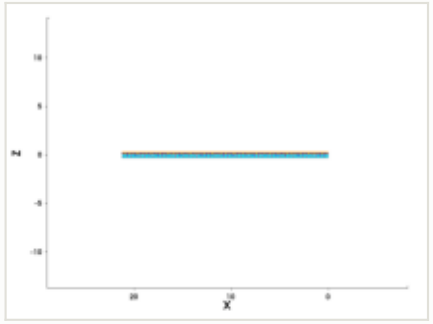
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



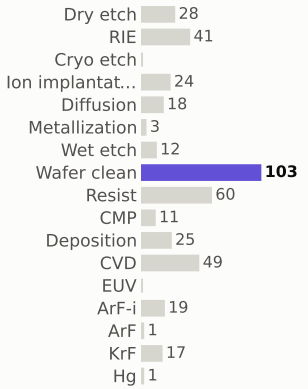
y cut at 57.351 µm



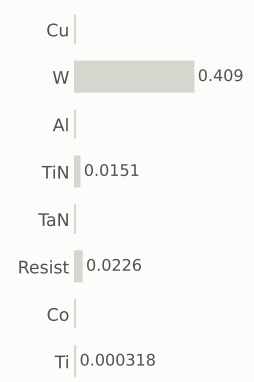
mask for this step

no mask for this step

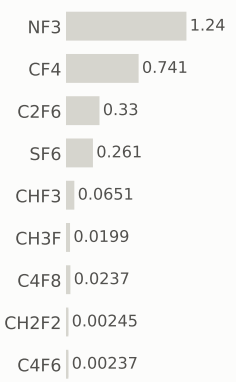
steps by process area



materials deposited (g)



gases emitted (g)



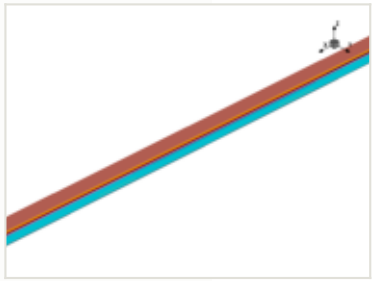
Step #5640: Resist

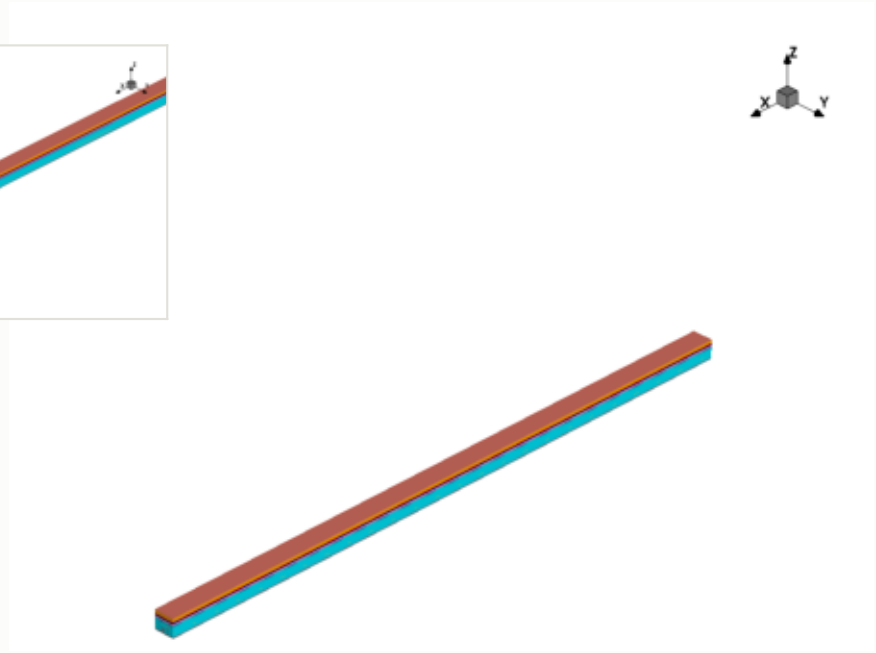
depo_fill · Resist

0.169	248.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



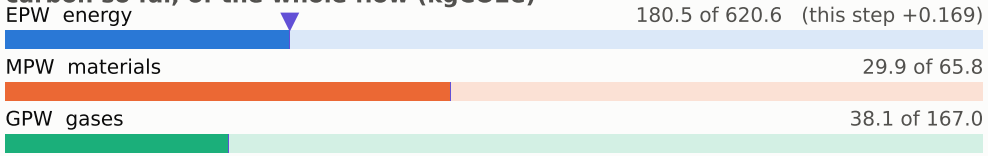


this step

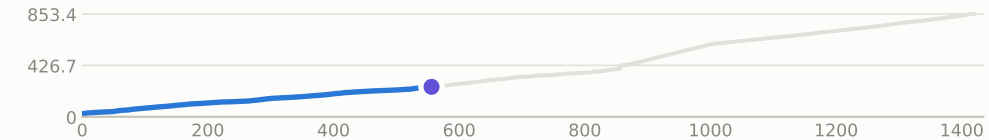
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

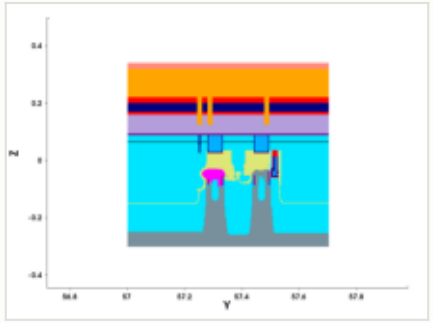
carbon so far, of the whole flow (kgCO2e)



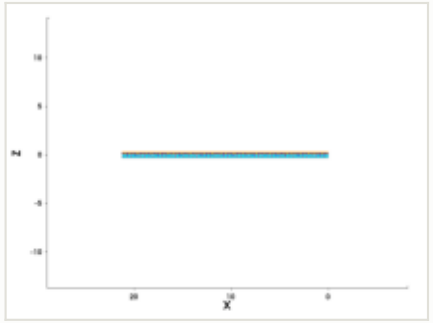
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



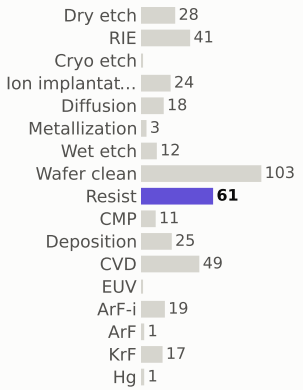
y cut at 57.351 µm



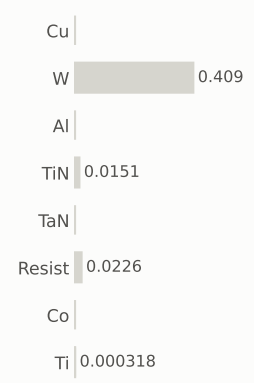
mask for this step

no mask for this step

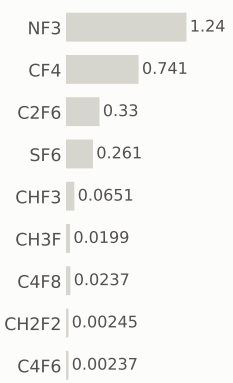
steps by process area



materials deposited (g)



gases emitted (g)



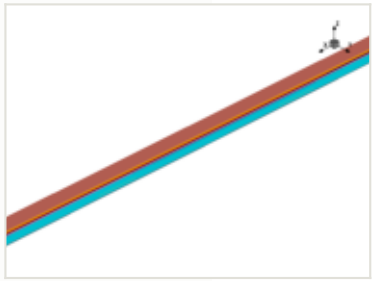
Step #5650: Deposition

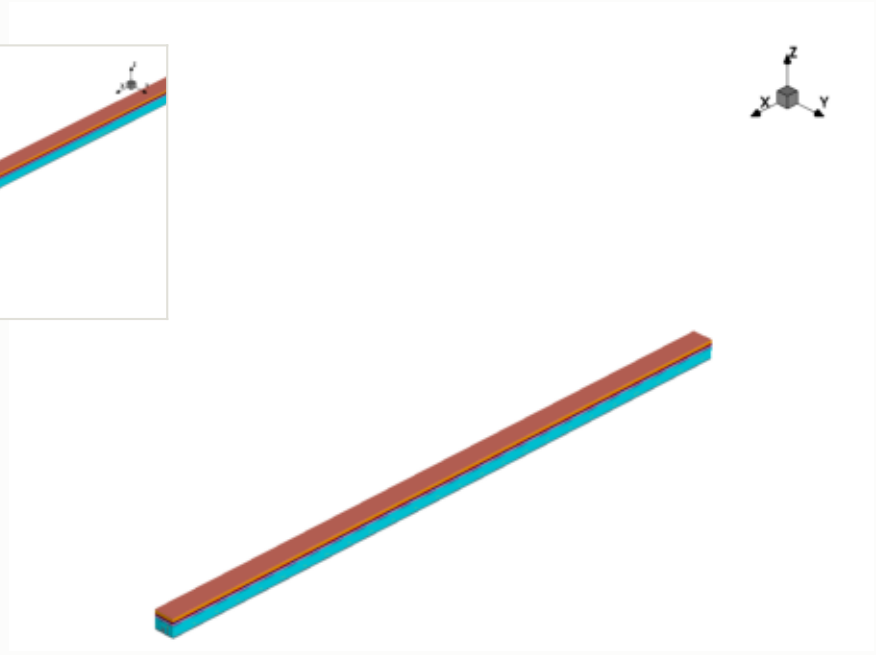
depo_isotropic · Deposition

0.656	249.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



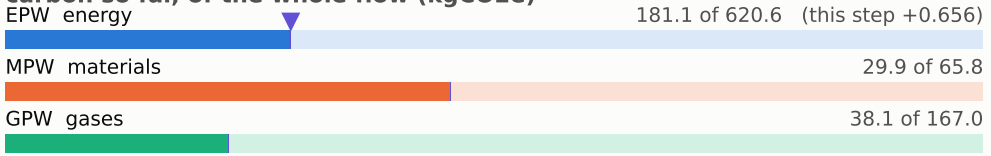


this step

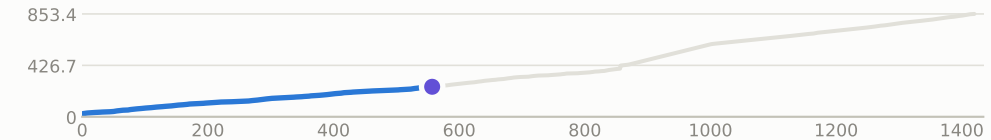
process area	Deposition
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

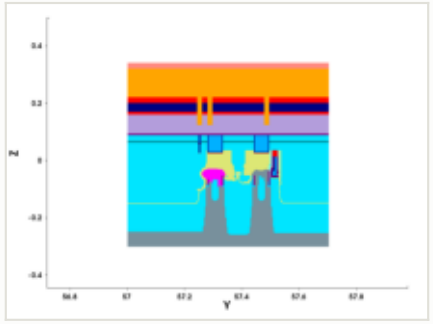
carbon so far, of the whole flow (kgCO2e)



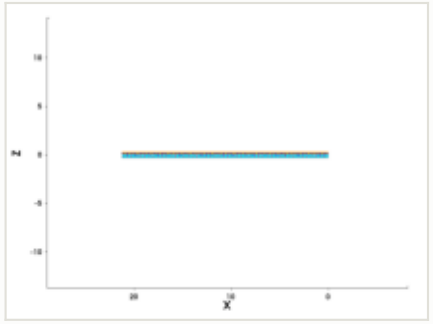
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



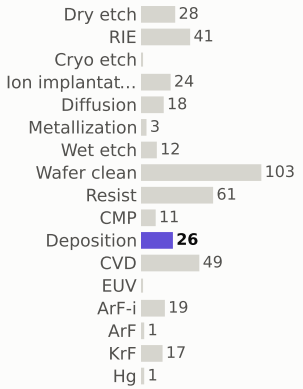
y cut at 57.351 µm



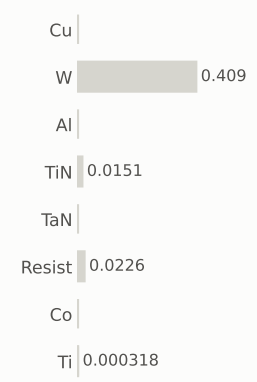
mask for this step

no mask for this step

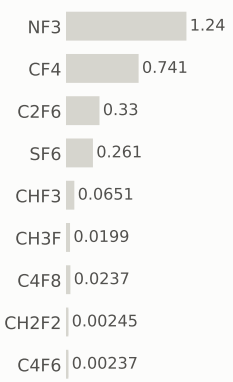
steps by process area



materials deposited (g)



gases emitted (g)



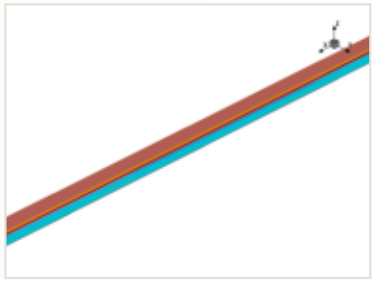
Step #5660: CVD

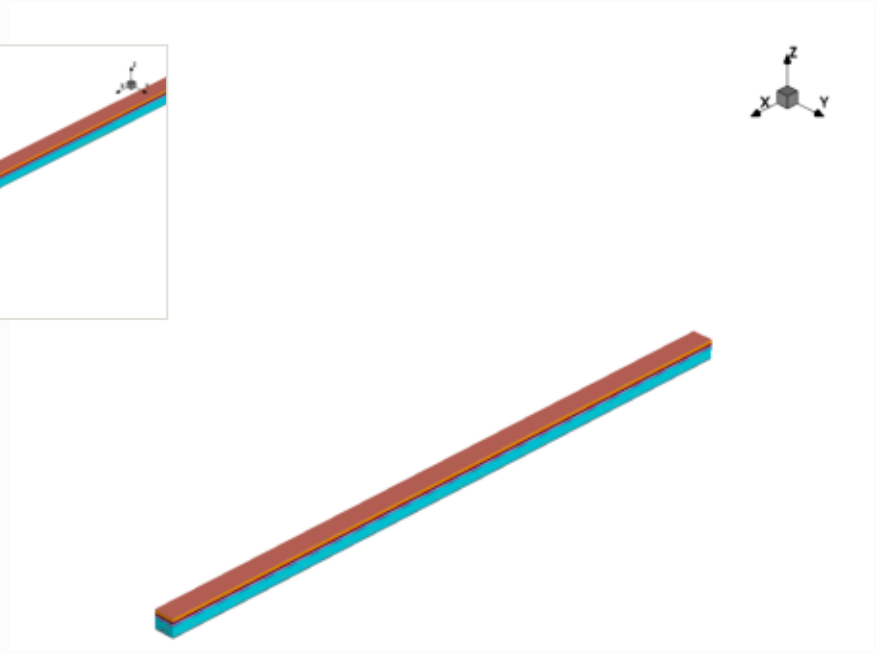
depo_isotropic · CVD

1.31	250.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



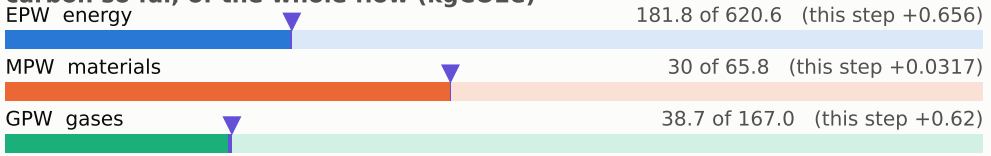


this step

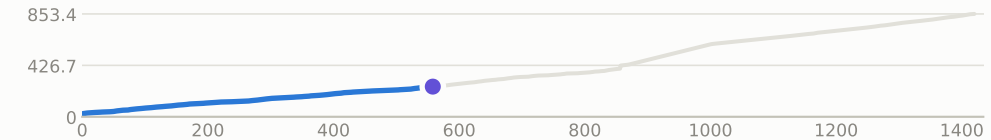
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

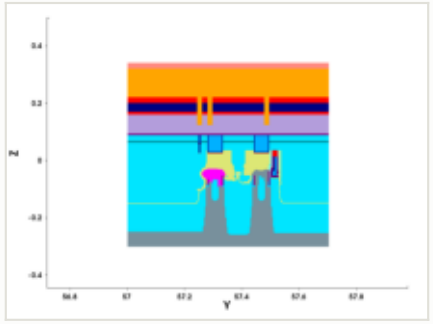
carbon so far, of the whole flow (kgCO2e)



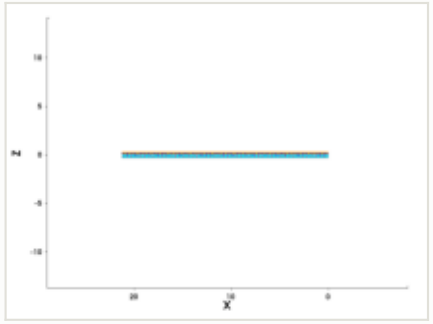
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



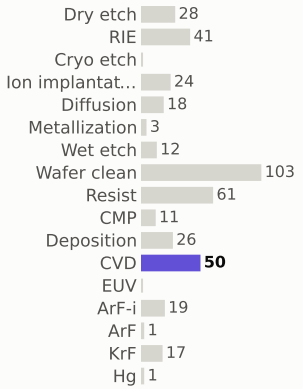
y cut at 57.351 µm



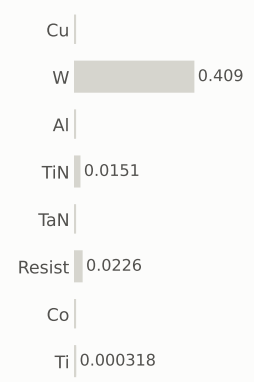
mask for this step

no mask for this step

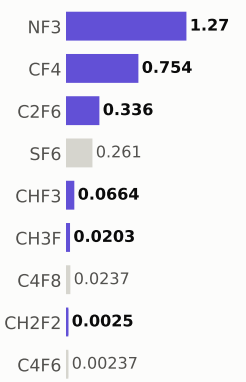
steps by process area



materials deposited (g)



gases emitted (g)



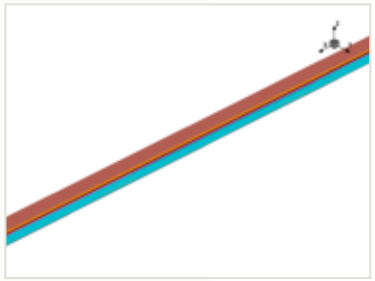
Step #5670: Resist

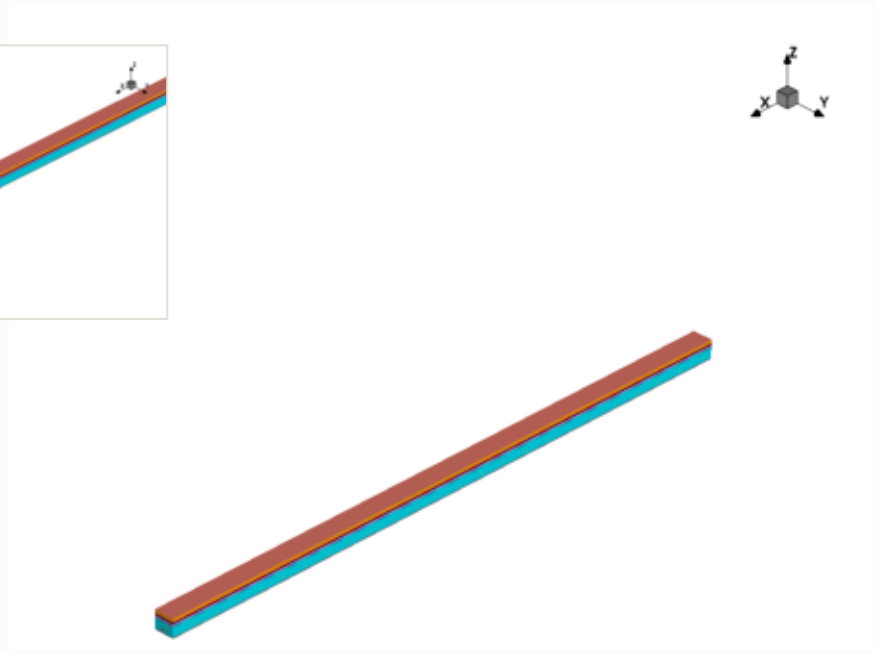
depo_fill · Resist

0.169	250.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



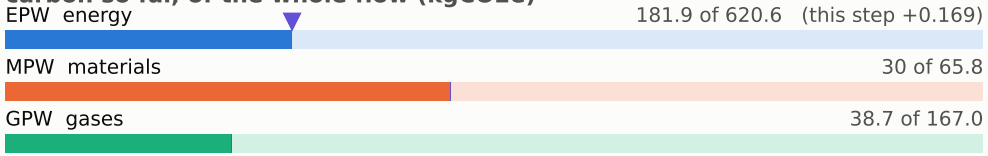


this step

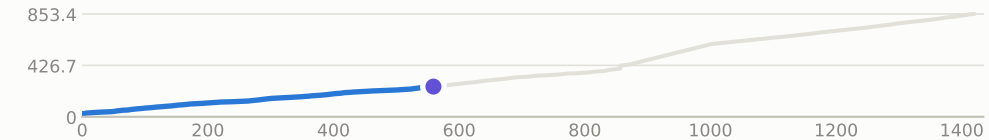
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

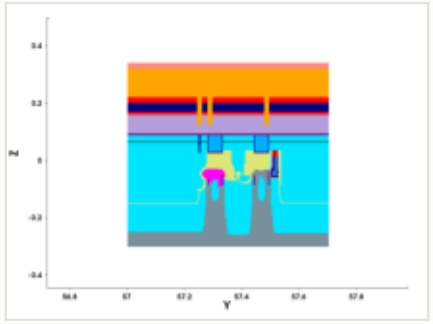
carbon so far, of the whole flow (kgCO2e)



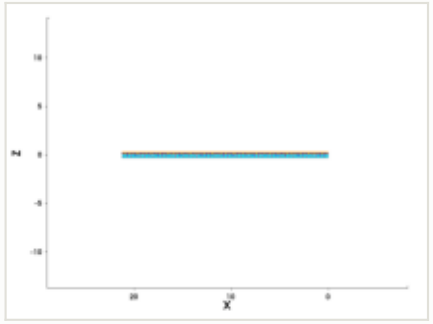
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



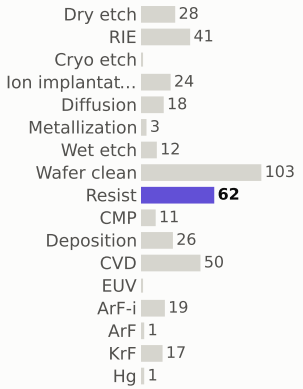
y cut at 57.351 µm



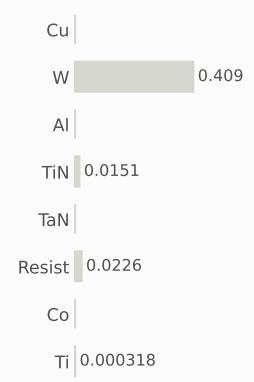
mask for this step

no mask for this step

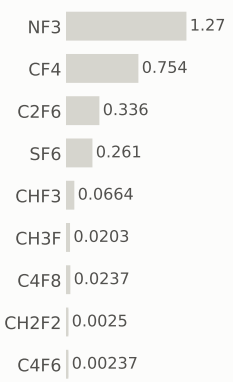
steps by process area



materials deposited (g)



gases emitted (g)



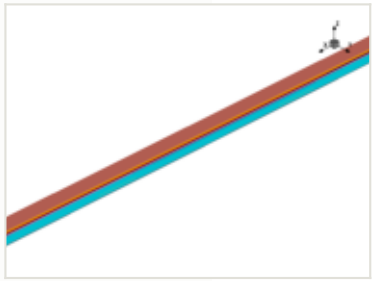
Step #5680: CVD

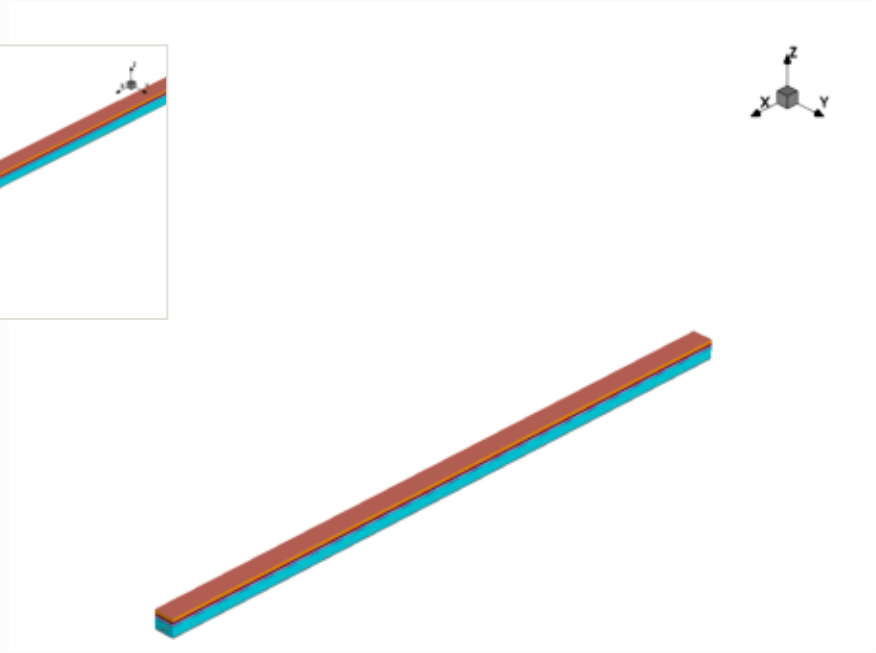
depo_isotropic · CVD

1.31	251.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



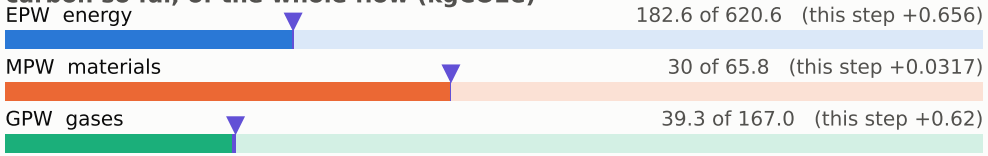


this step

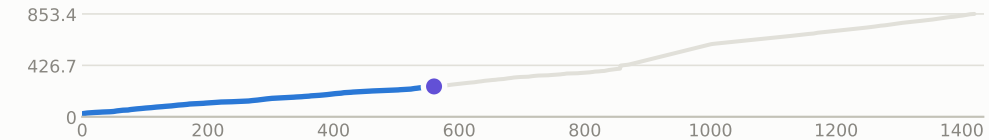
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

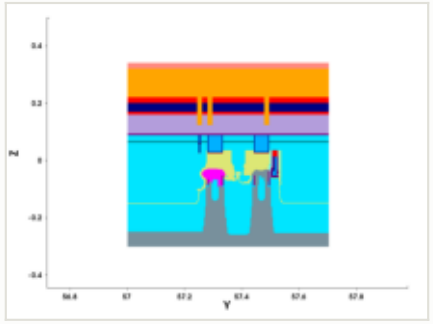
carbon so far, of the whole flow (kgCO2e)



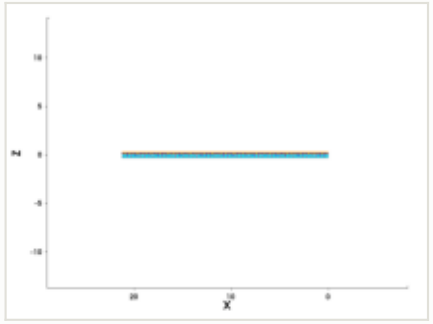
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



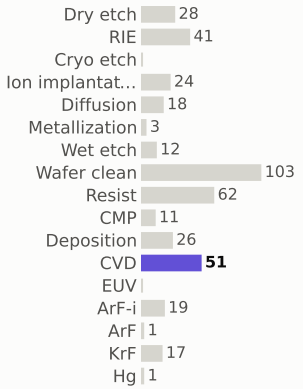
y cut at 57.351 µm



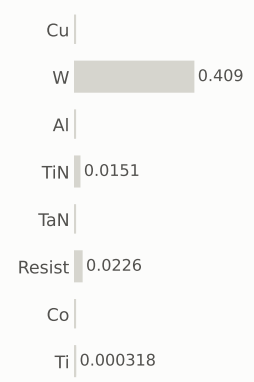
mask for this step

no mask for this step

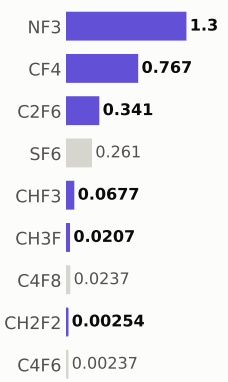
steps by process area



materials deposited (g)



gases emitted (g)



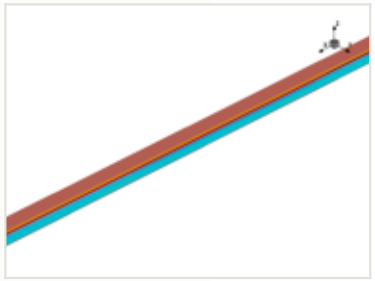
Step #5690: Wafer_clean

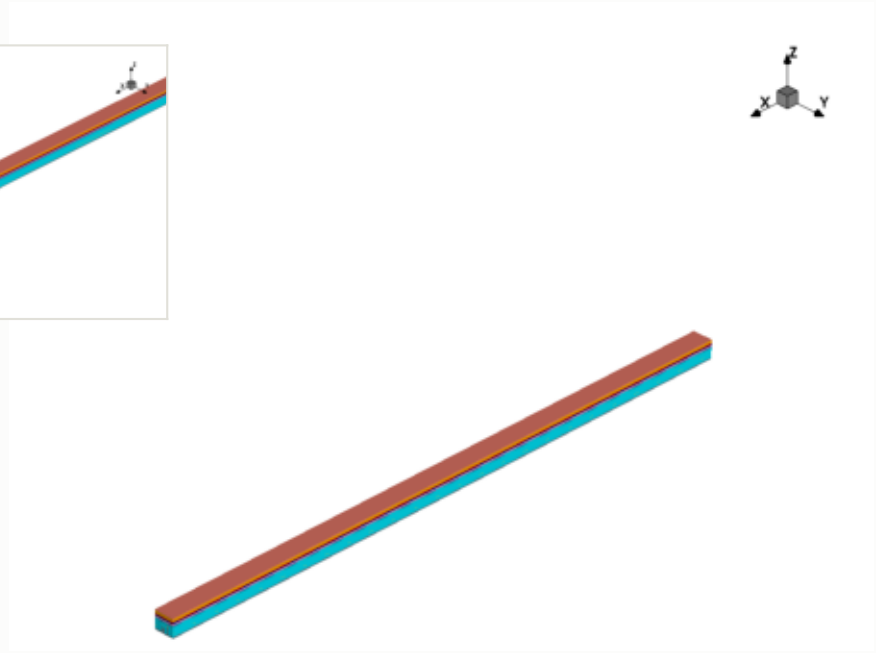
clean · Wafer_clean

0.169	252.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





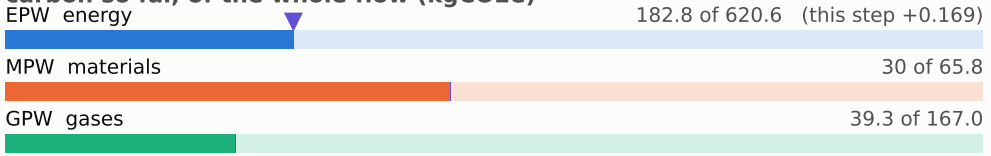
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

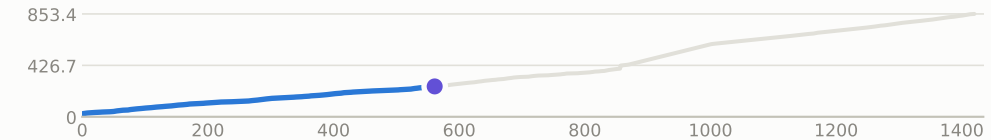
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

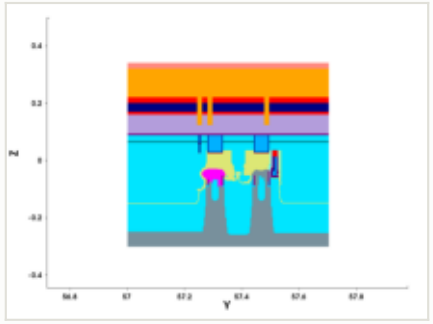
carbon so far, of the whole flow (kgCO2e)



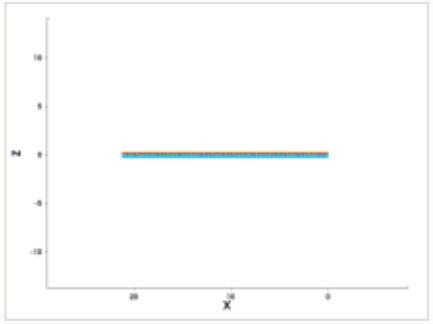
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



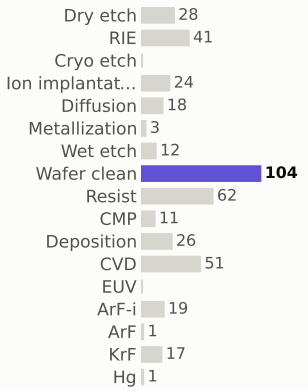
y cut at 57.351 µm



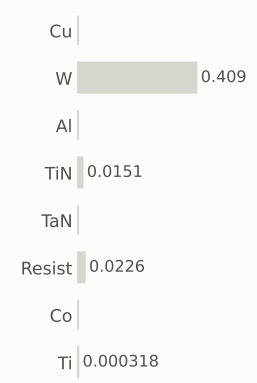
mask for this step

no mask for this step

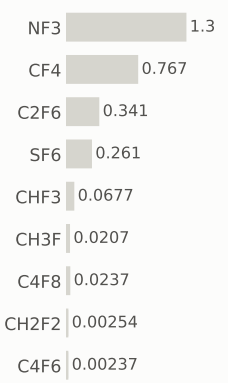
steps by process area



materials deposited (g)



gases emitted (g)



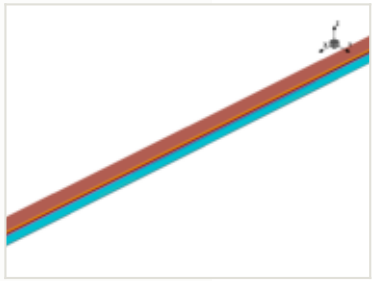
Step #5700: Resist

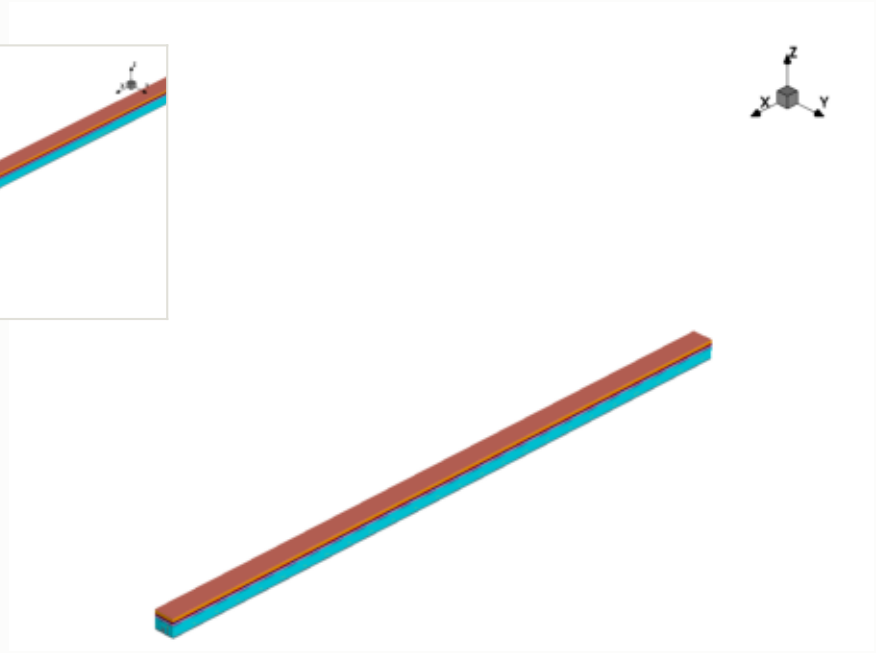
resist · Resist

0.169	252.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





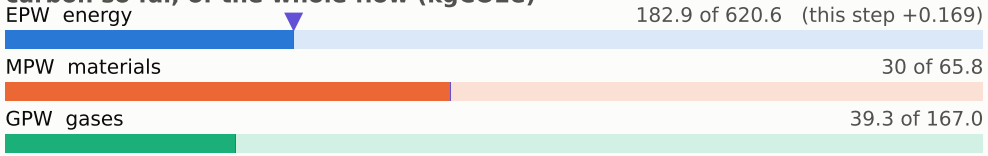
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

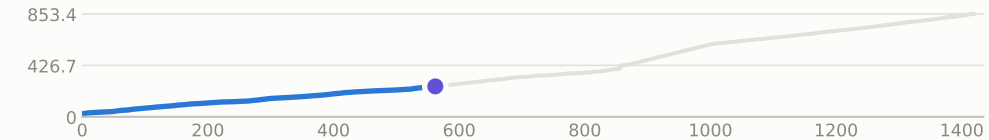
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

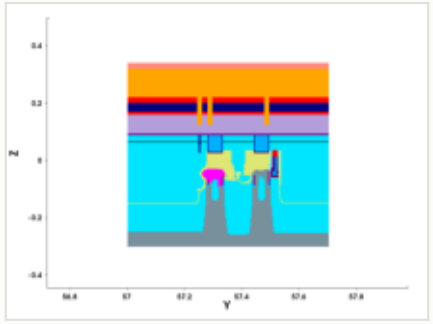
carbon so far, of the whole flow (kgCO2e)



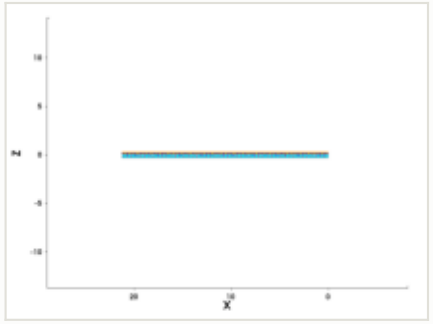
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



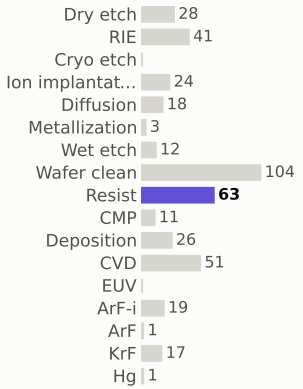
y cut at 57.351 µm



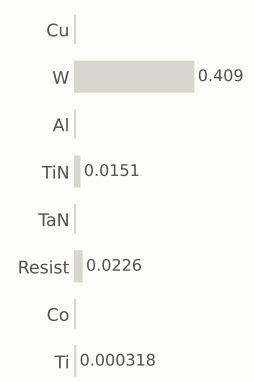
mask for this step

no mask for this step

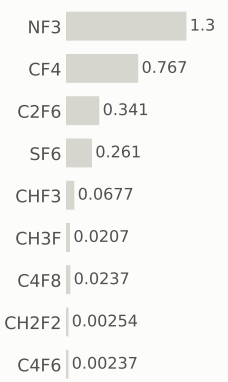
steps by process area



materials deposited (g)



gases emitted (g)



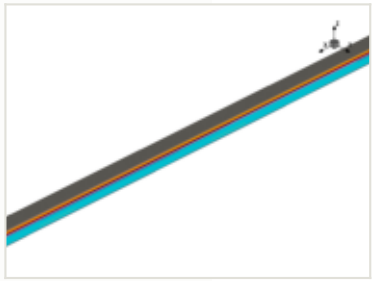
Step #5710: ArF-i

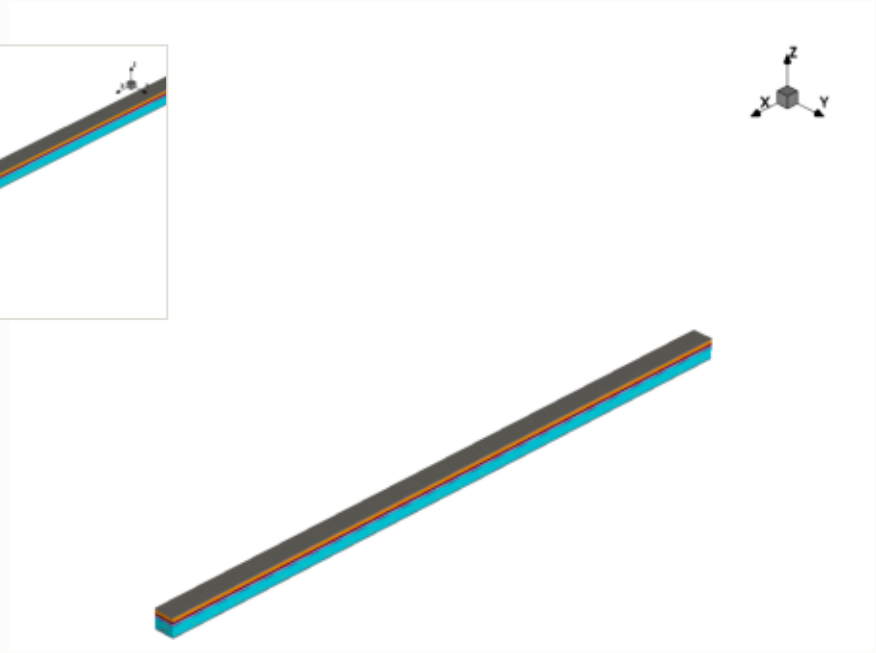
litho_patterning · ArF-i

0.217	252.4	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



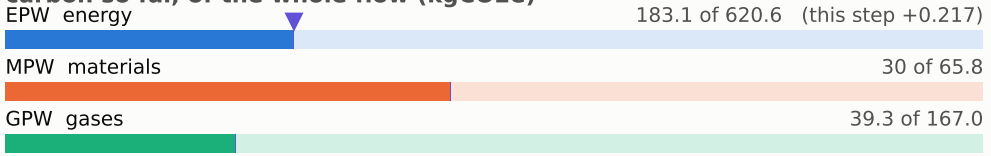


this step

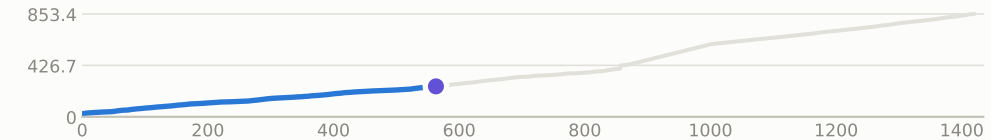
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

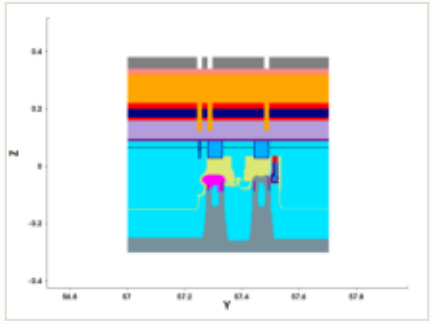
carbon so far, of the whole flow (kgCO2e)



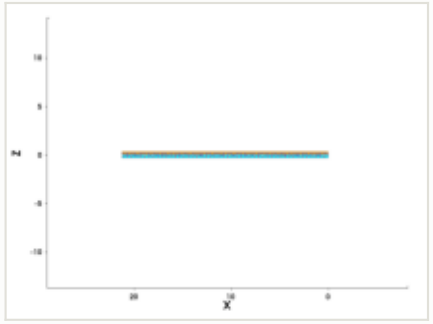
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

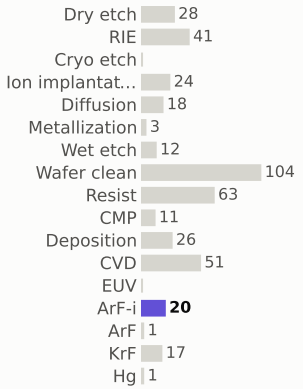


y cut at 57.351 µm

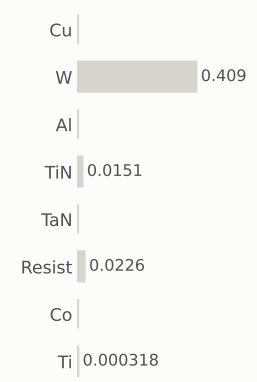


mask: layer 2018:0 (GDSII)

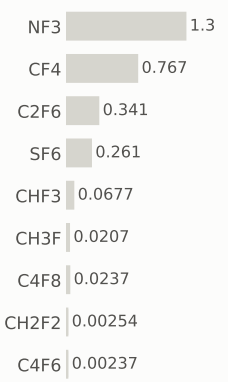
steps by process area



materials deposited (g)



gases emitted (g)



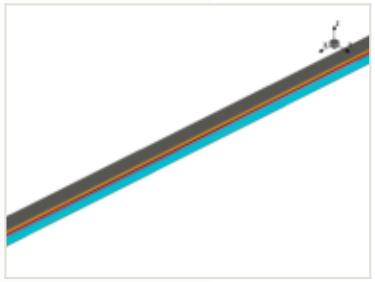
Step #5720: RIE

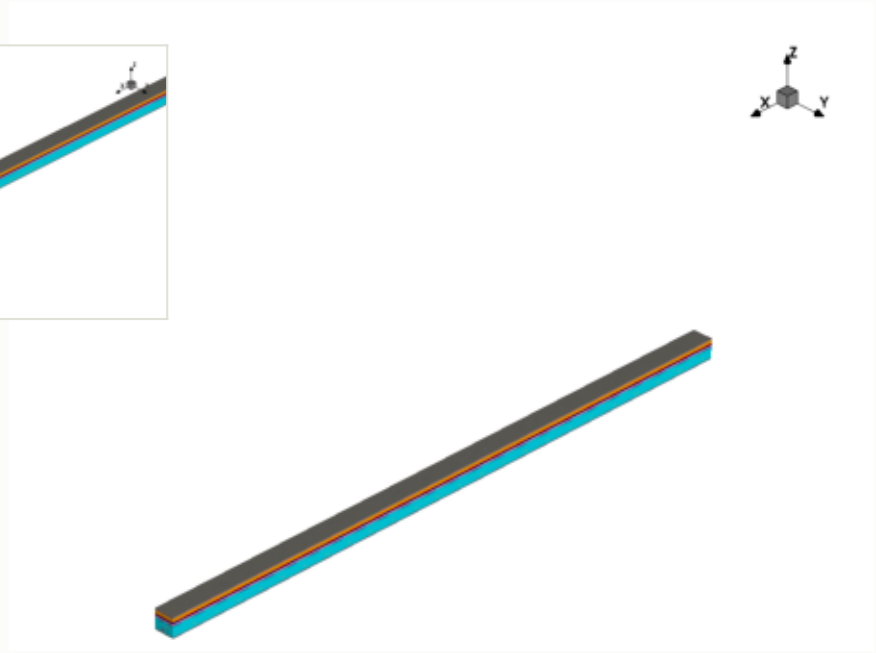
etch_anisotropic · RIE

1.04	253.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



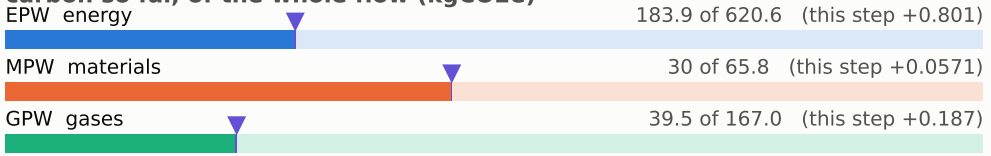


this step

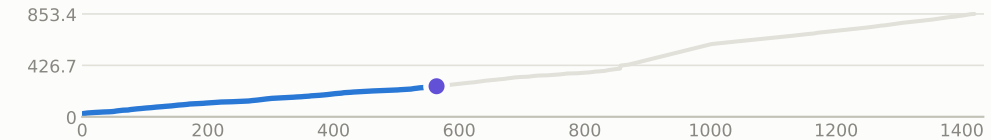
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

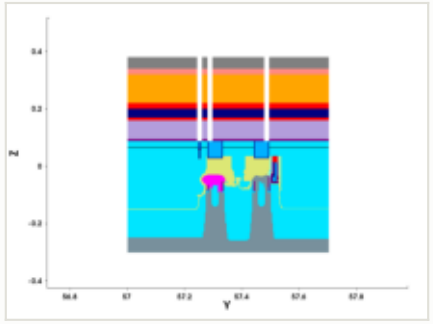
carbon so far, of the whole flow (kgCO2e)



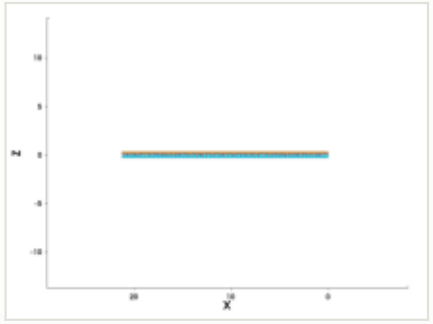
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



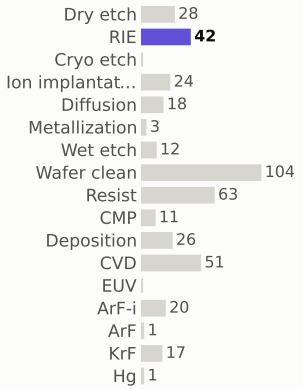
y cut at 57.351 µm



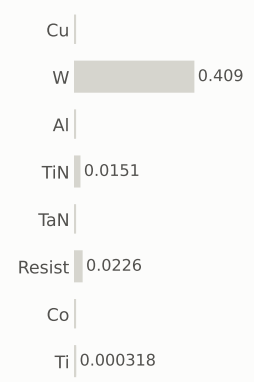
mask for this step

no mask for this step

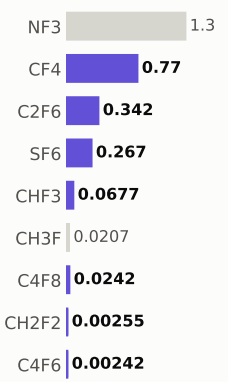
steps by process area



materials deposited (g)



gases emitted (g)



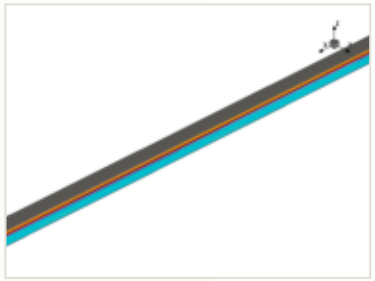
Step #5730: Wafer_clean

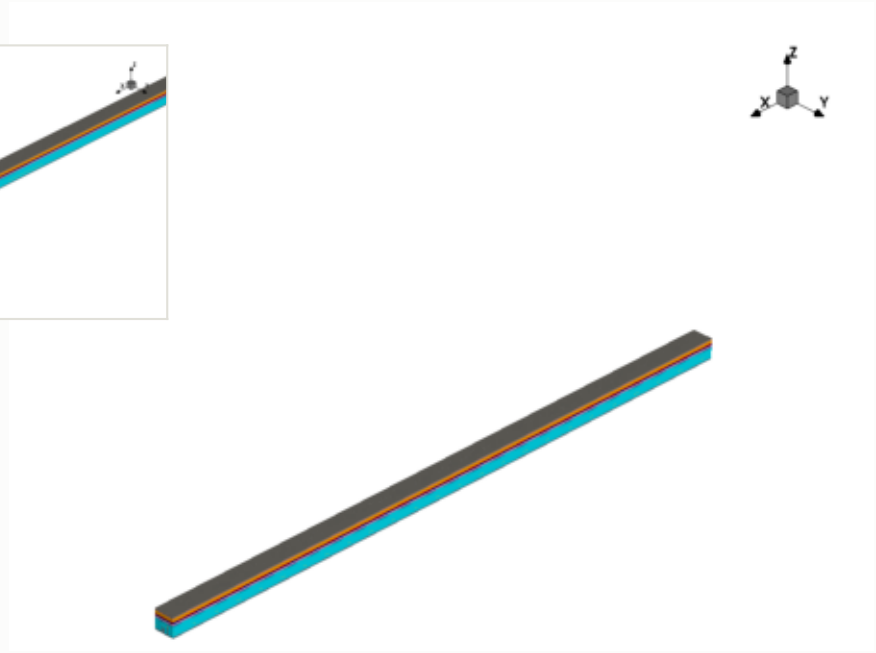
clean · Wafer_clean

0.169	253.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





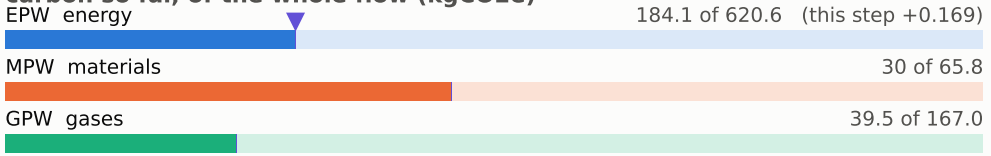
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

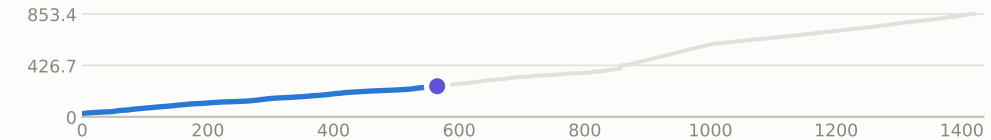
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

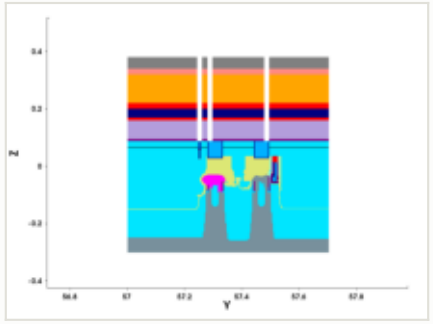
carbon so far, of the whole flow (kgCO2e)



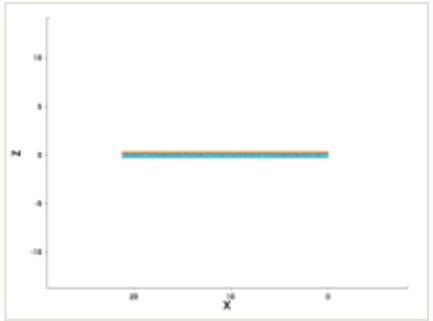
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



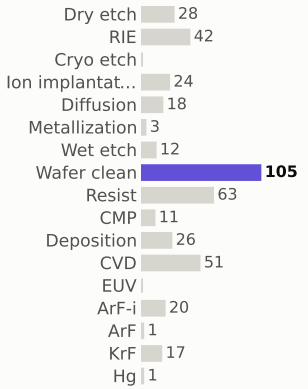
y cut at 57.351 µm



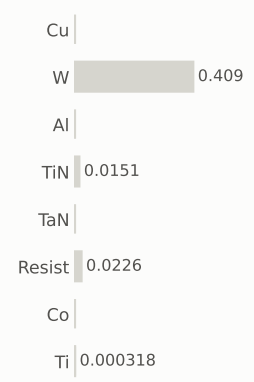
mask for this step

no mask for this step

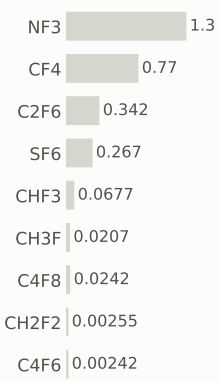
steps by process area



materials deposited (g)



gases emitted (g)



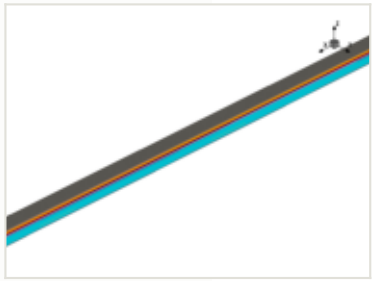
Step #5740: Resist

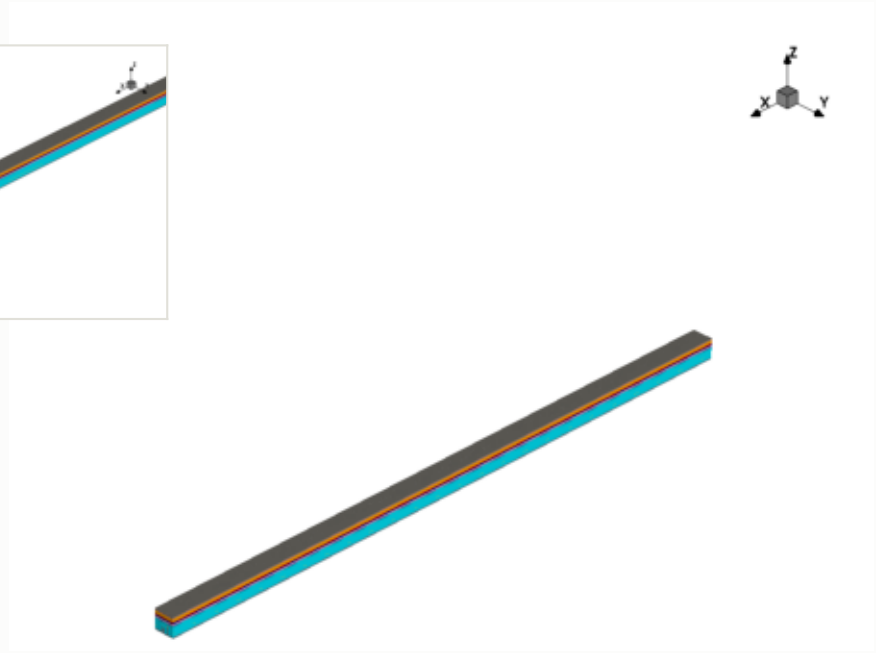
depo_fill · Resist

0.169	253.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



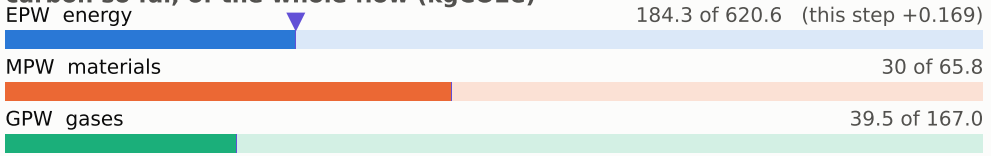


this step

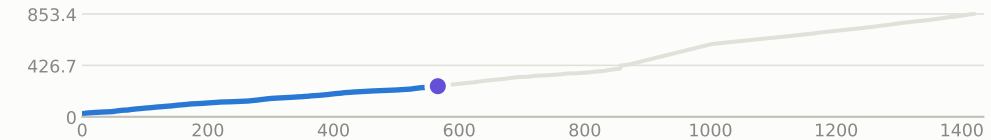
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

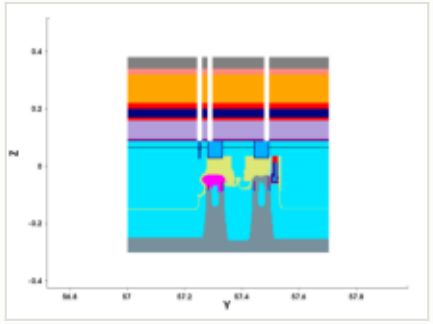
carbon so far, of the whole flow (kgCO2e)



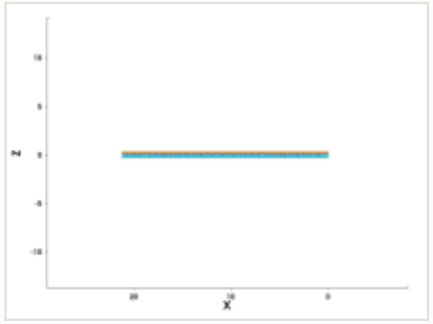
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



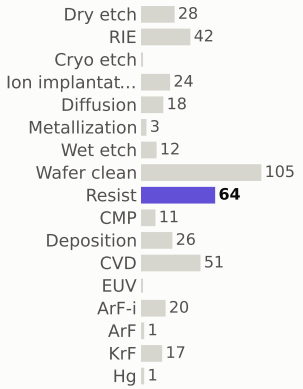
y cut at 57.351 µm



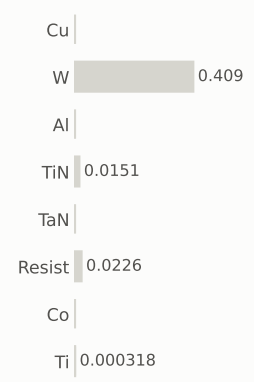
mask for this step

no mask for this step

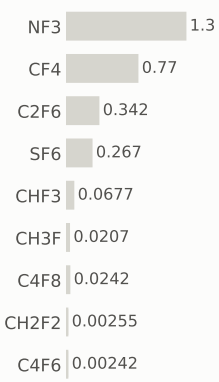
steps by process area



materials deposited (g)



gases emitted (g)



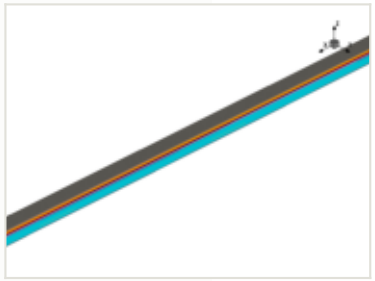
Step #5750: CVD

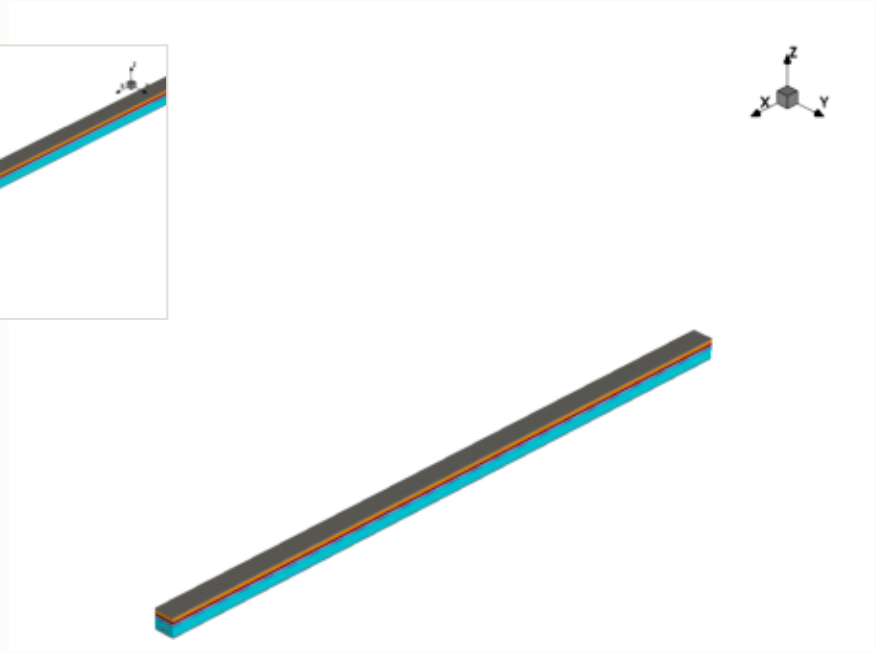
depo_isotropic · CVD

1.31	255.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



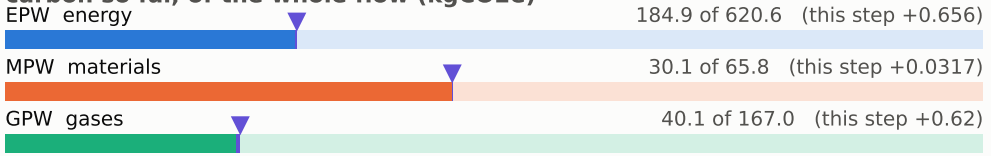


this step

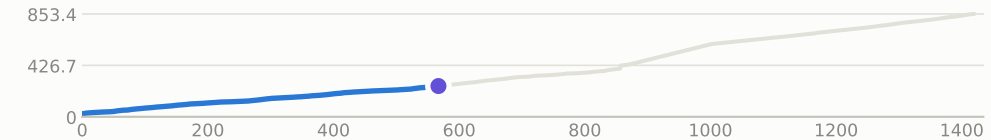
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

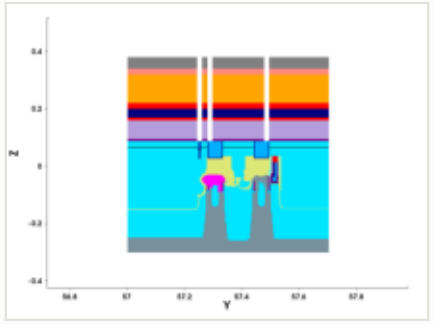
carbon so far, of the whole flow (kgCO2e)



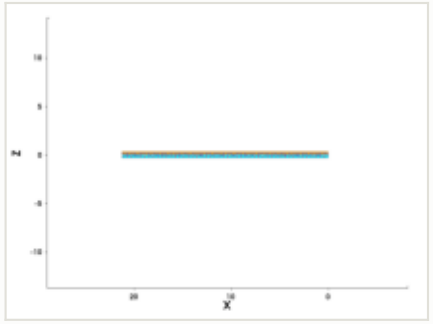
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



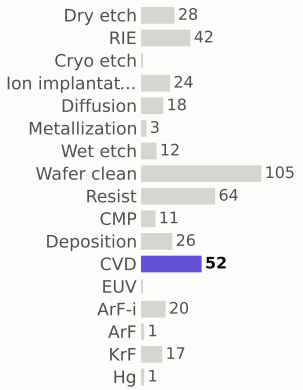
y cut at 57.351 µm



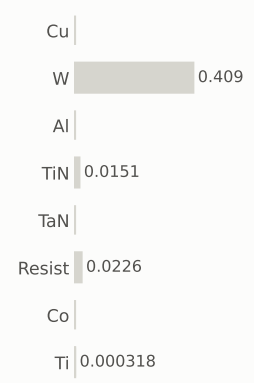
mask for this step

no mask for this step

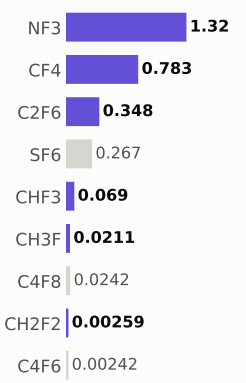
steps by process area



materials deposited (g)



gases emitted (g)



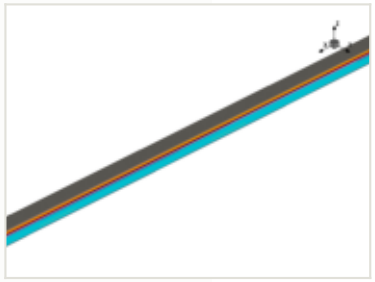
Step #5760: Wafer_clean

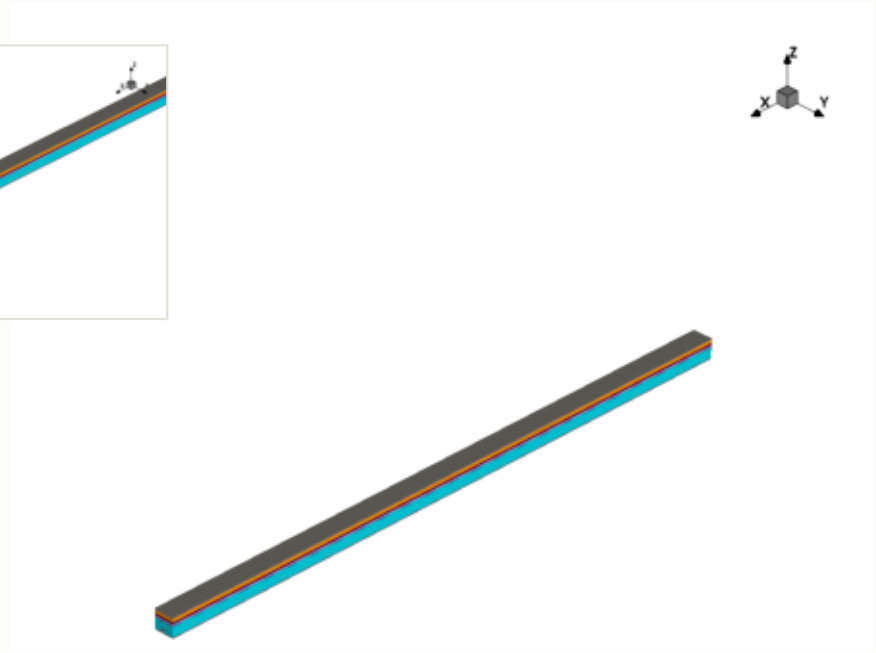
clean · Wafer_clean

0.169	255.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



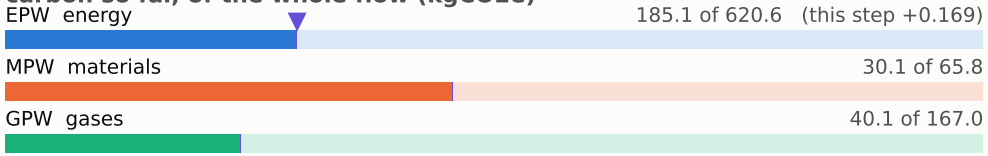


this step

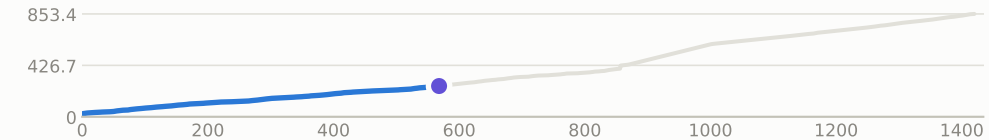
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

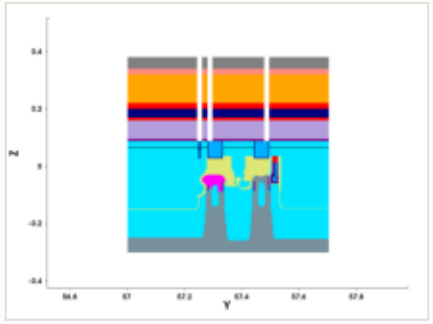
carbon so far, of the whole flow (kgCO2e)



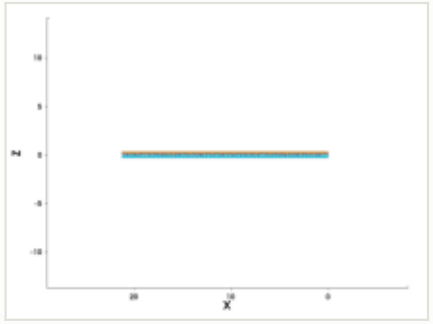
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



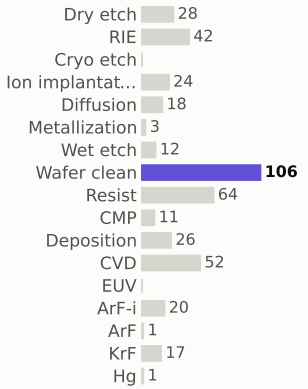
y cut at 57.351 µm



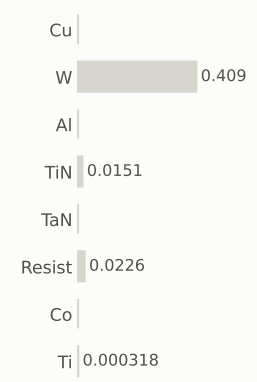
mask for this step

no mask for this step

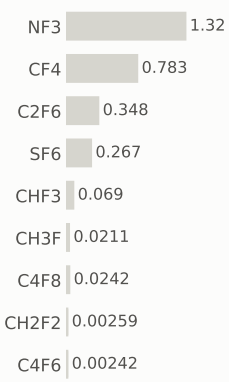
steps by process area



materials deposited (g)



gases emitted (g)



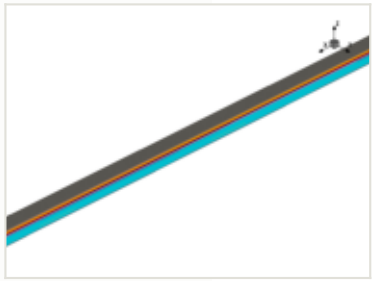
Step #5770: Resist

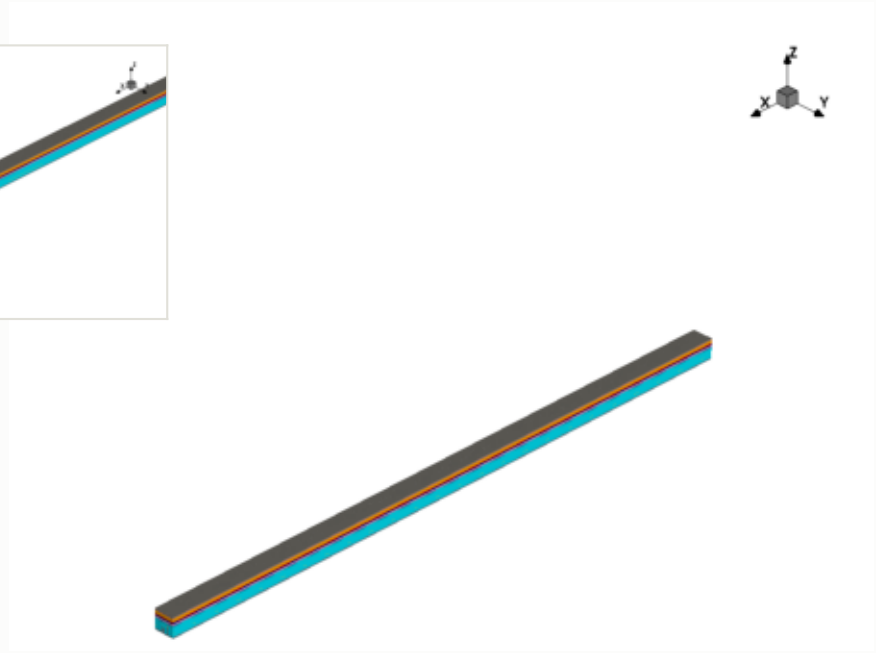
resist · Resist

0.169	255.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





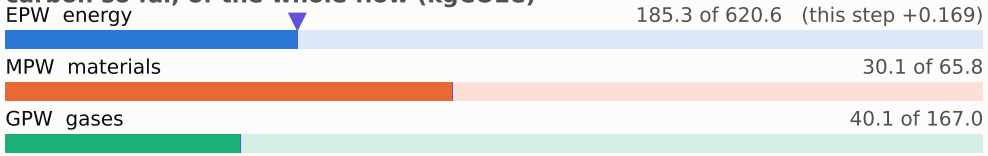
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

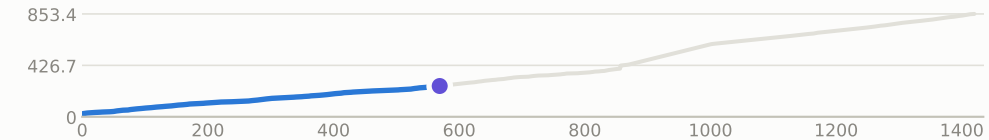
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

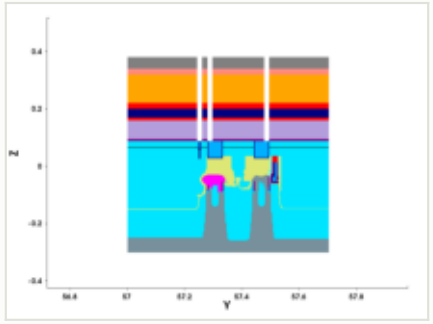
carbon so far, of the whole flow (kgCO2e)



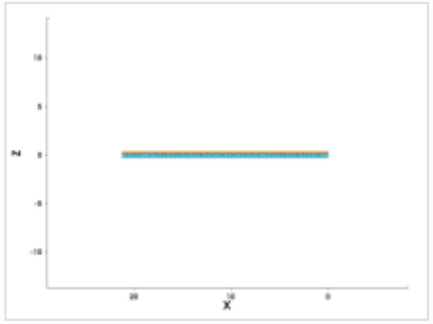
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



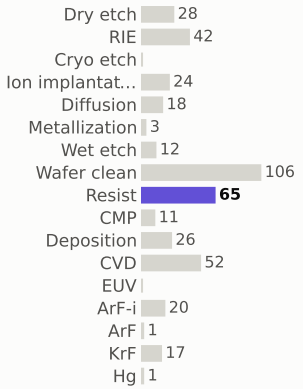
y cut at 57.351 µm



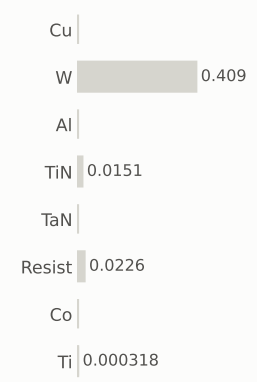
mask for this step

no mask for this step

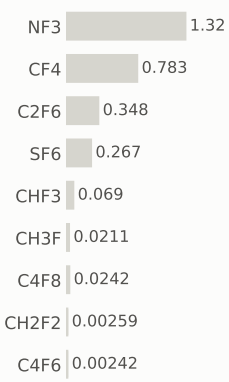
steps by process area



materials deposited (g)



gases emitted (g)



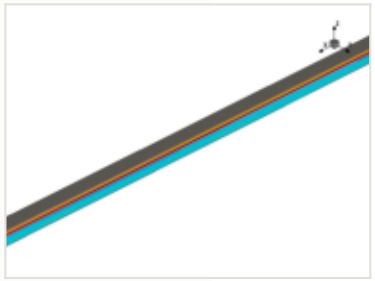
Step #5780: ArF-i

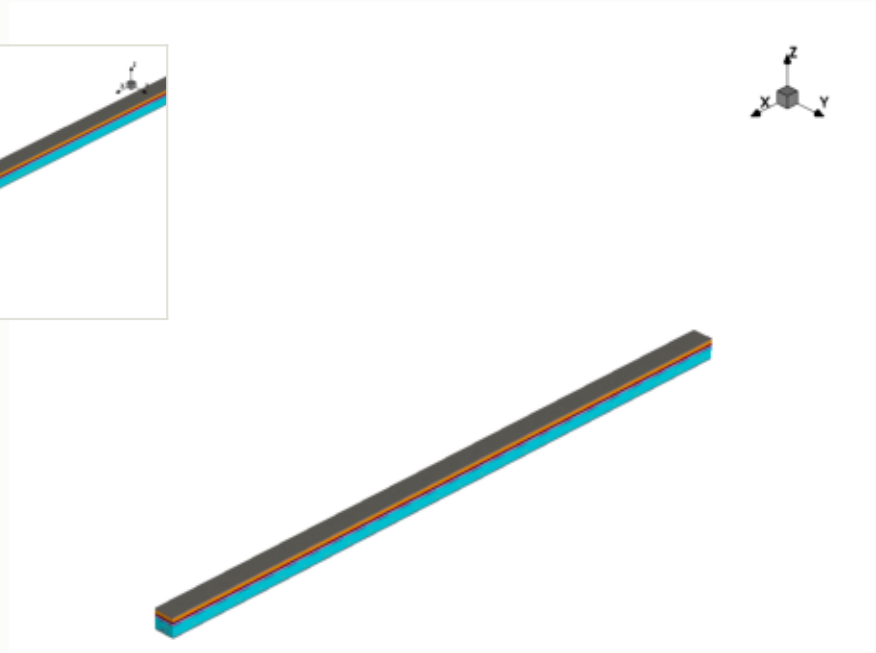
litho_patterning · ArF-i

0.217	255.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



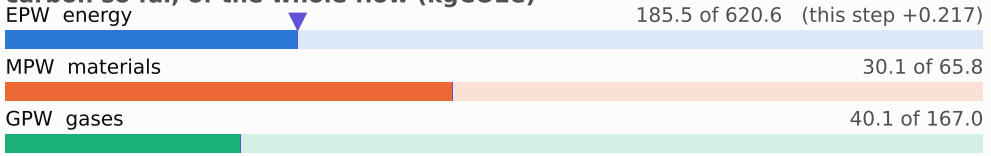


this step

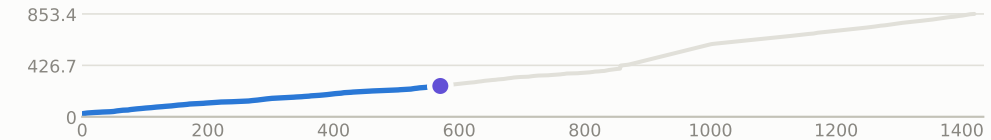
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

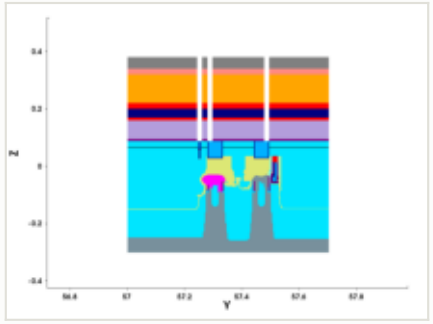
carbon so far, of the whole flow (kgCO2e)



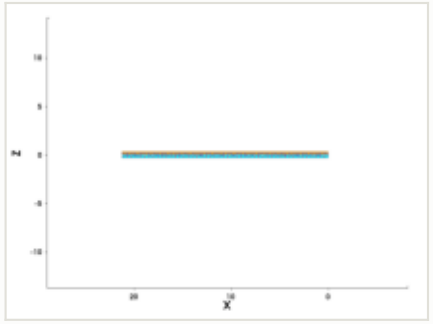
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

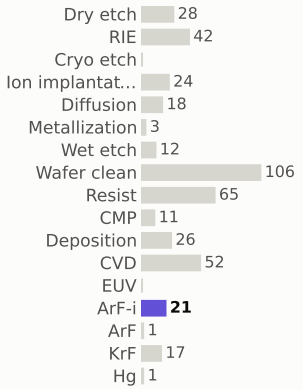


y cut at 57.351 µm

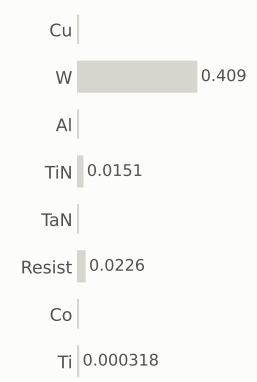


mask: layer 2100:0, positive (GDSII)

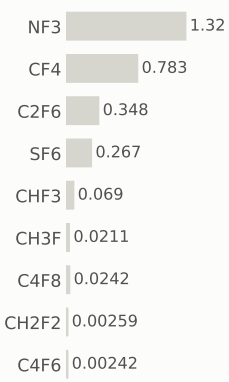
steps by process area



materials deposited (g)



gases emitted (g)



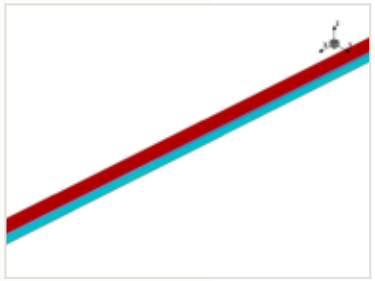
Step #5790: RIE

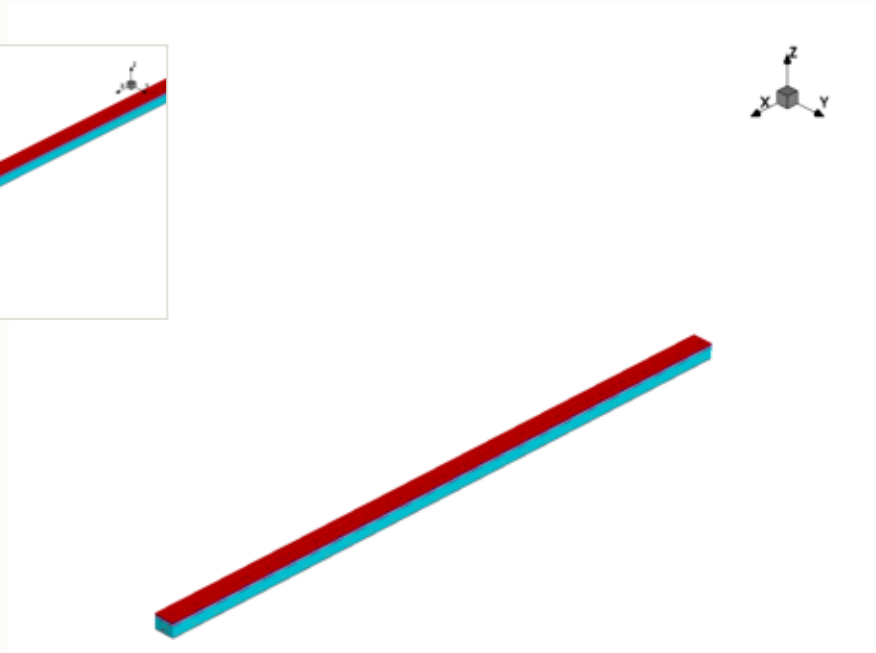
etch_strip · RIE

1.04	256.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



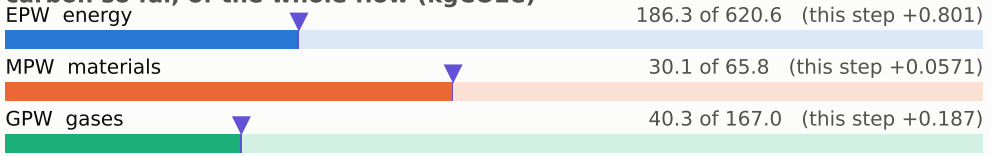


this step

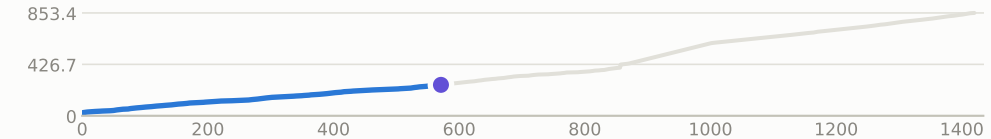
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

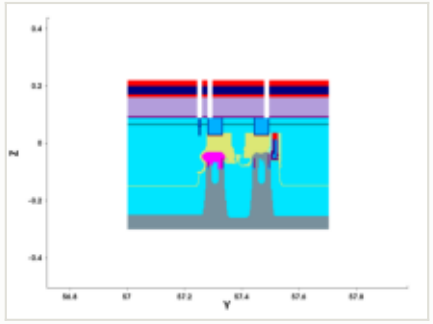
carbon so far, of the whole flow (kgCO2e)



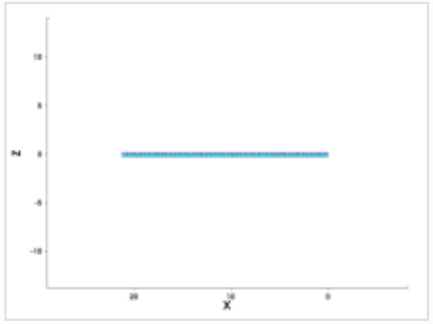
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



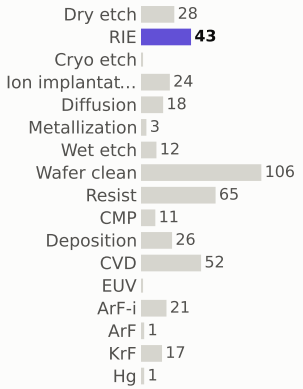
y cut at 57.351 µm



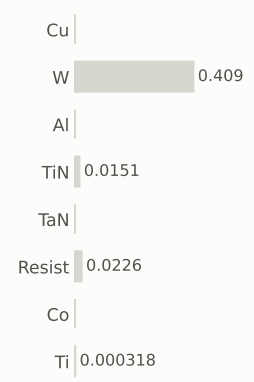
mask for this step

no mask for this step

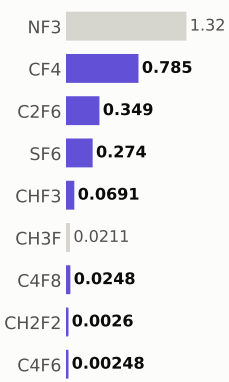
steps by process area



materials deposited (g)



gases emitted (g)



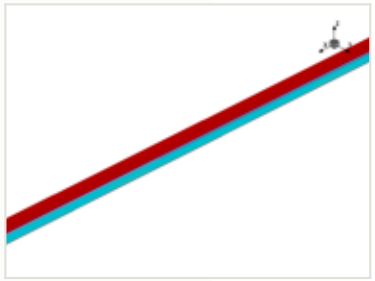
Step #5800: Wafer_clean

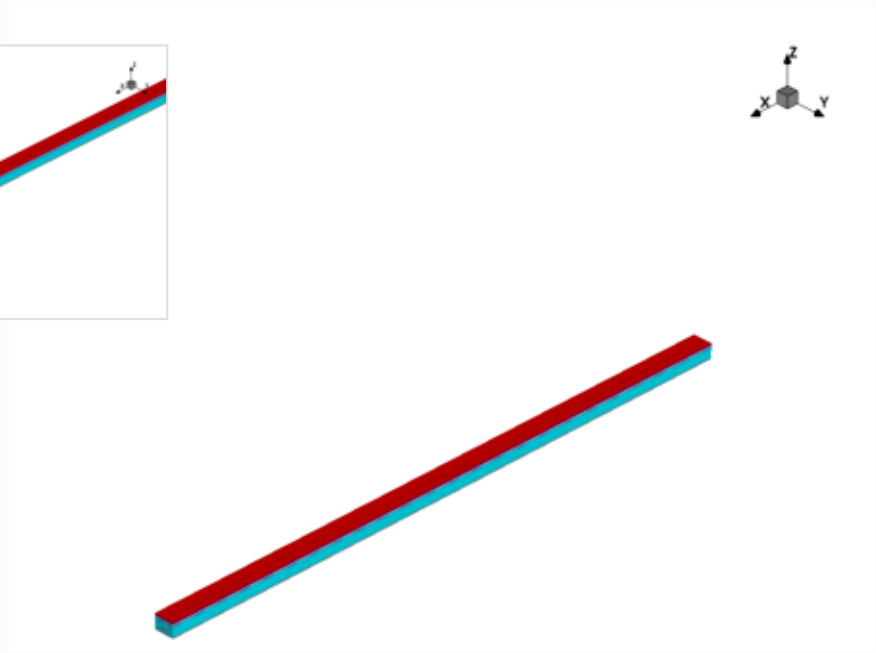
clean · Wafer_clean

0.169	256.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





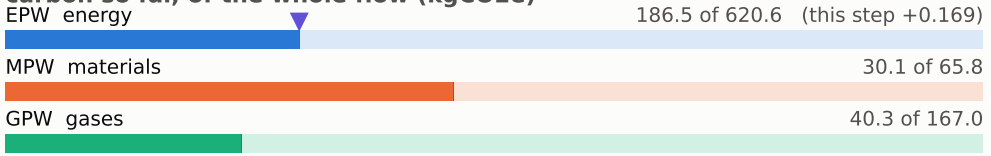
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

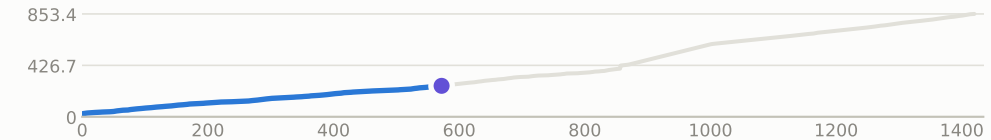
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

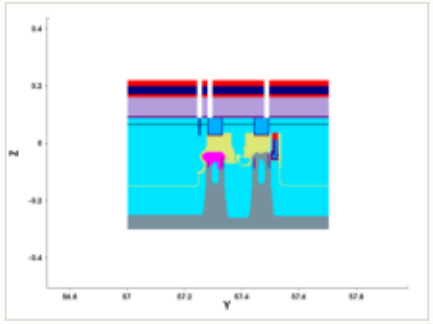
carbon so far, of the whole flow (kgCO2e)



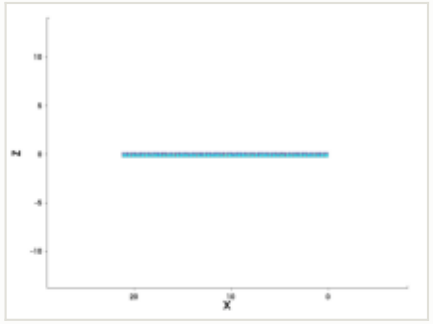
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



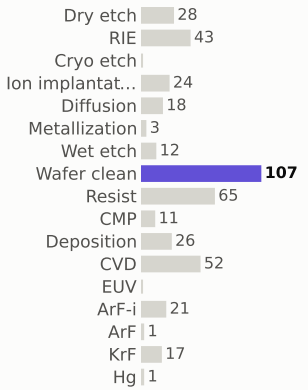
y cut at 57.351 µm



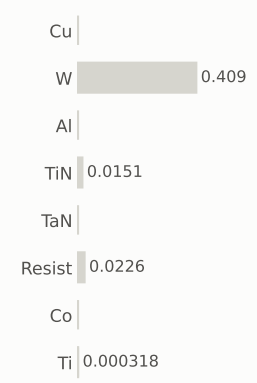
mask for this step

no mask for this step

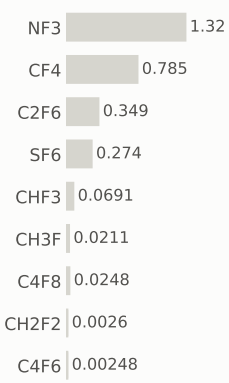
steps by process area



materials deposited (g)



gases emitted (g)



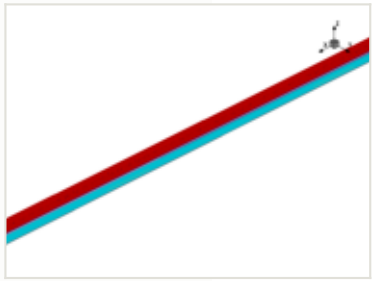
Step #5810: Resist

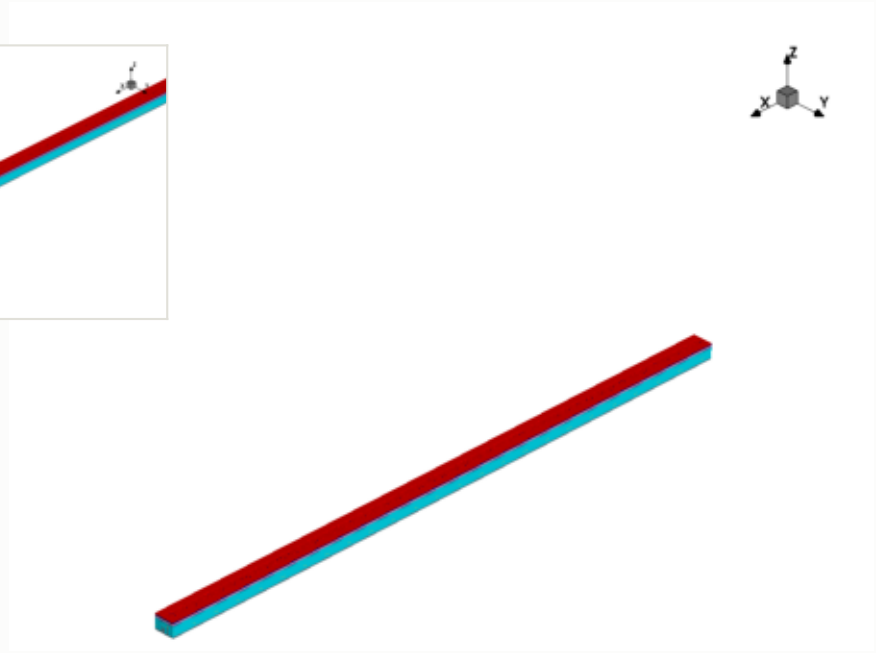
depo_fill · Resist

0.169	257.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



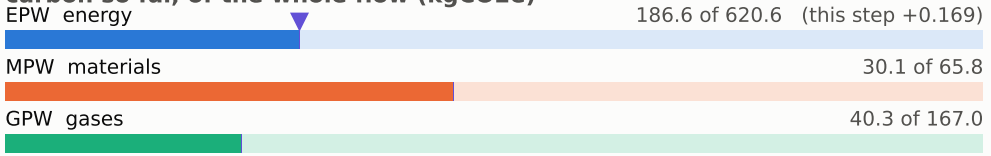


this step

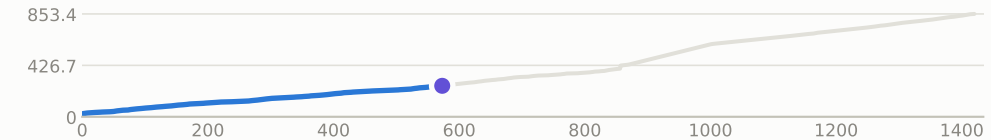
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

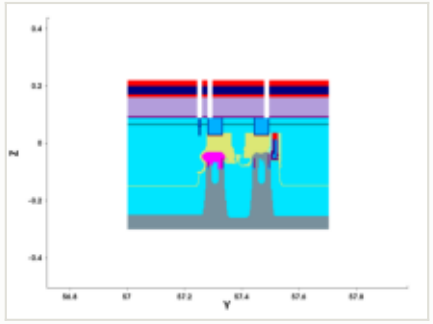
carbon so far, of the whole flow (kgCO2e)



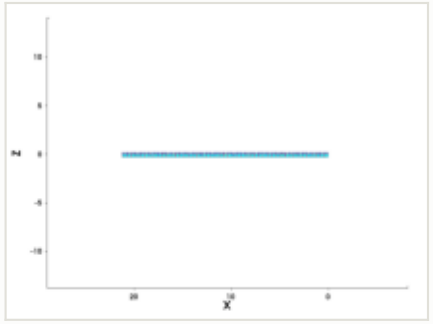
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



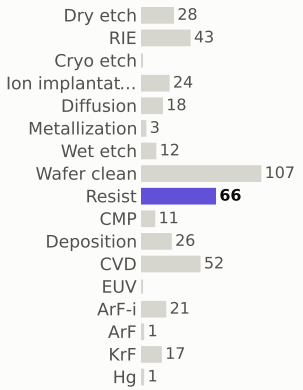
y cut at 57.351 µm



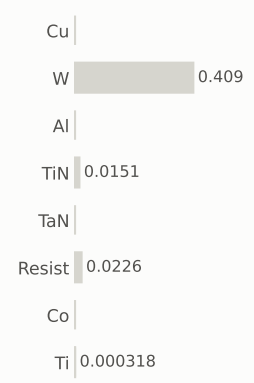
mask for this step

no mask for this step

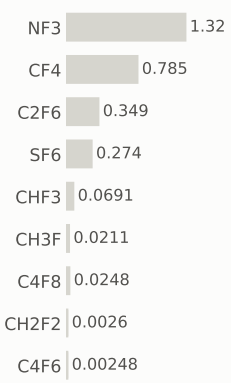
steps by process area



materials deposited (g)



gases emitted (g)



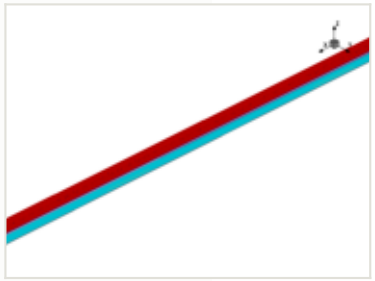
Step #5820: CVD

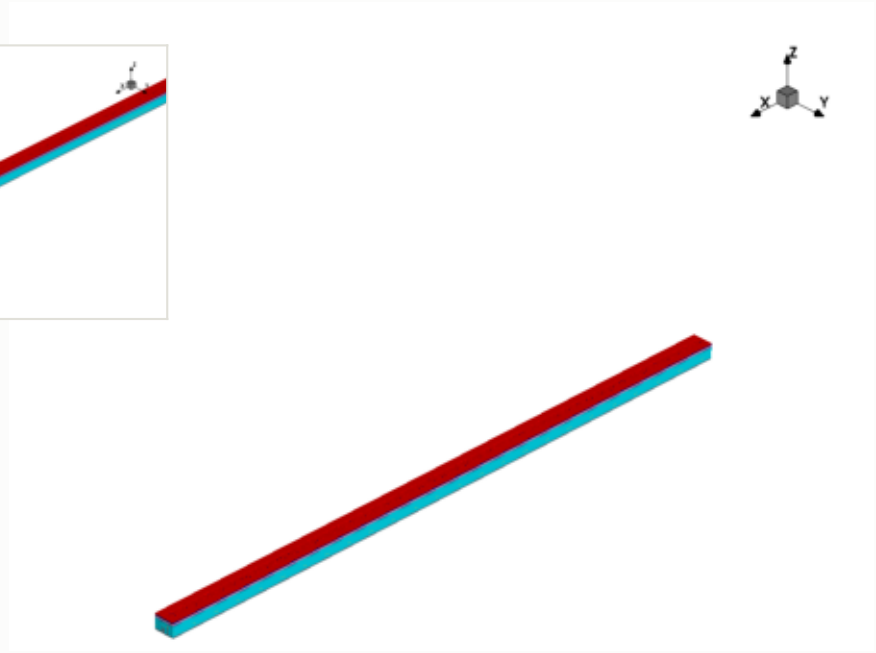
depo_isotropic · CVD

1.31	258.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



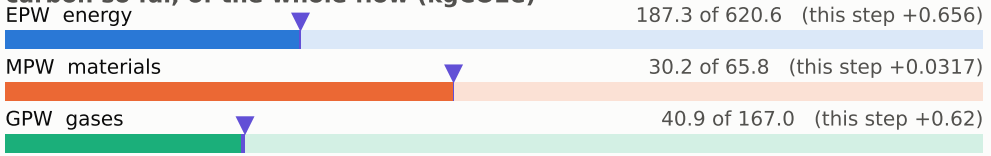


this step

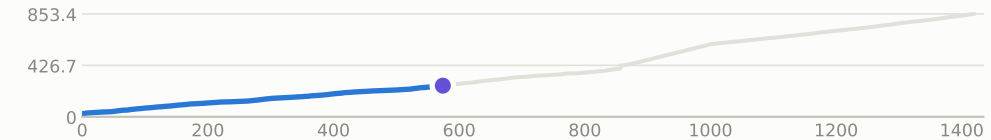
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

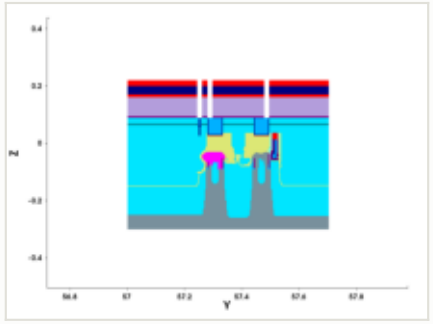
carbon so far, of the whole flow (kgCO2e)



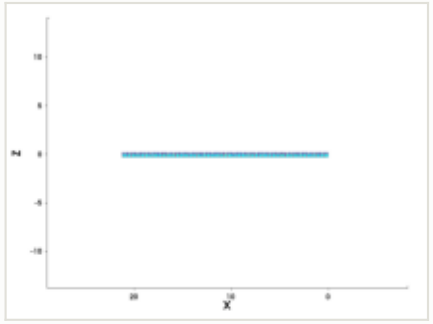
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



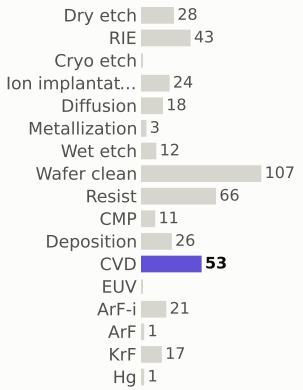
y cut at 57.351 µm



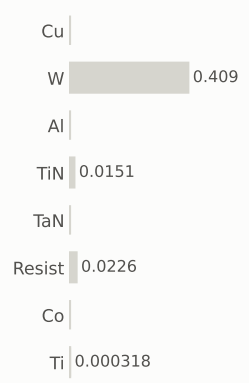
mask for this step

no mask for this step

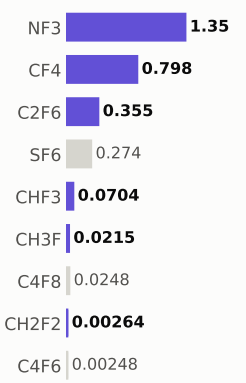
steps by process area



materials deposited (g)



gases emitted (g)



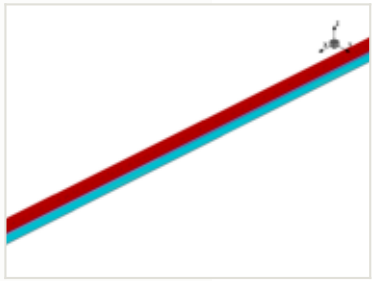
Step #5830: Wafer_clean

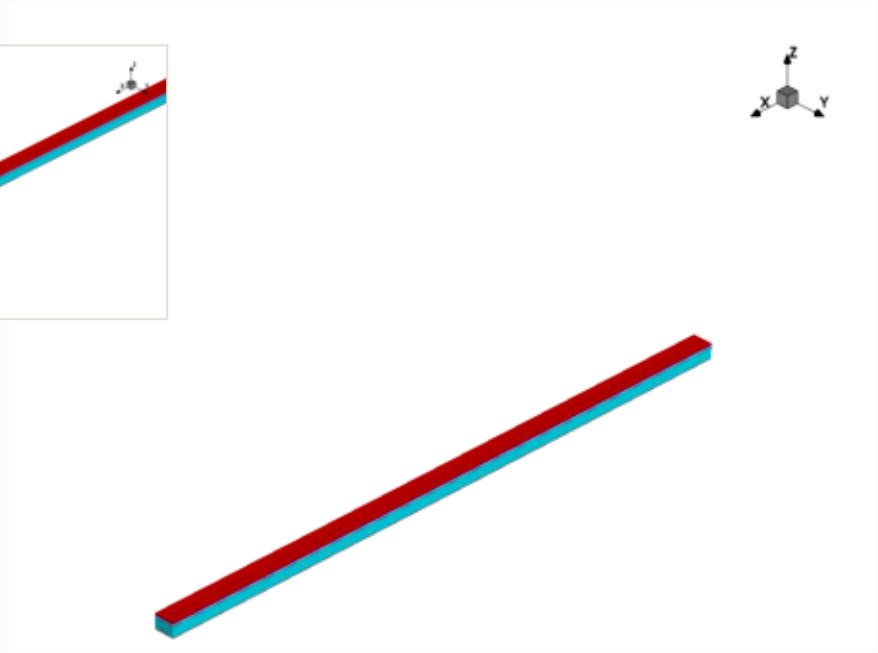
clean · Wafer_clean

0.169	258.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





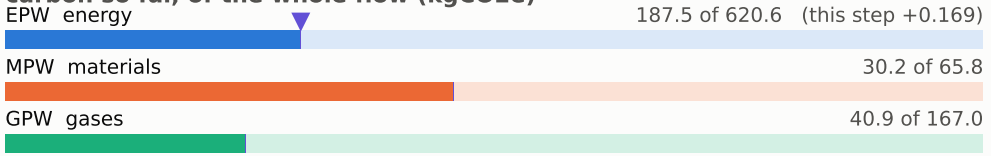
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

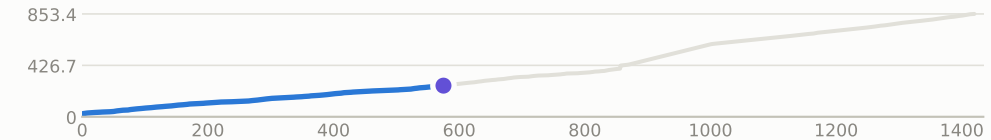
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

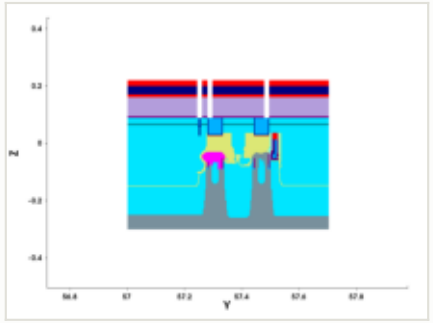
carbon so far, of the whole flow (kgCO2e)



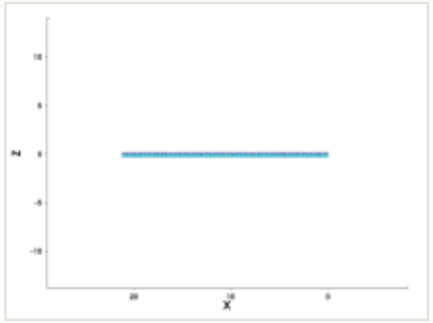
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



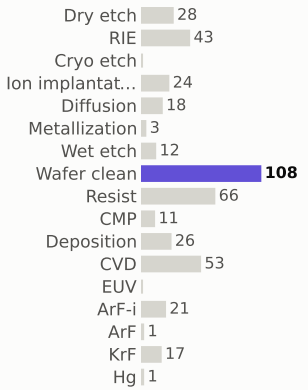
y cut at 57.351 µm



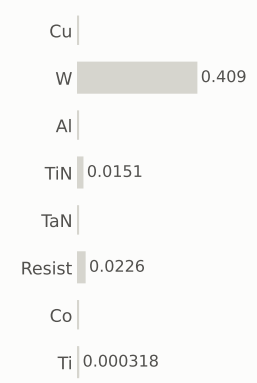
mask for this step

no mask for this step

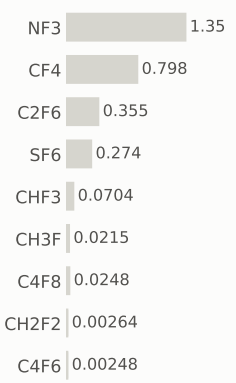
steps by process area



materials deposited (g)



gases emitted (g)



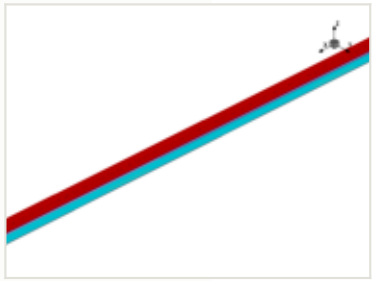
Step #5840: Resist

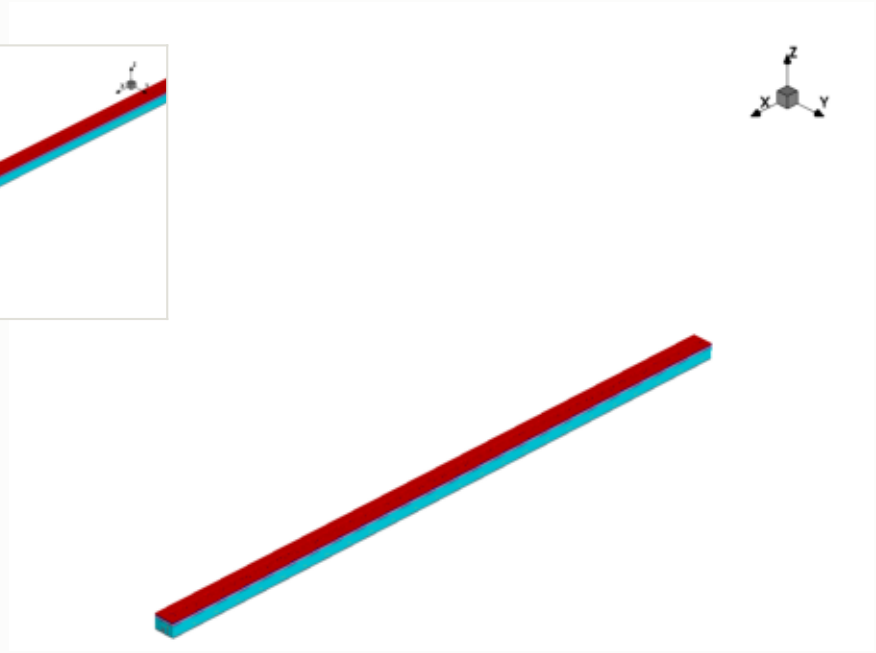
resist · Resist

0.169	258.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





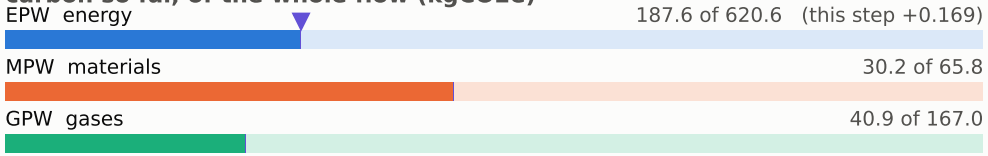
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

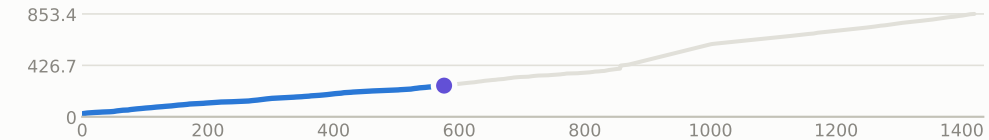
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

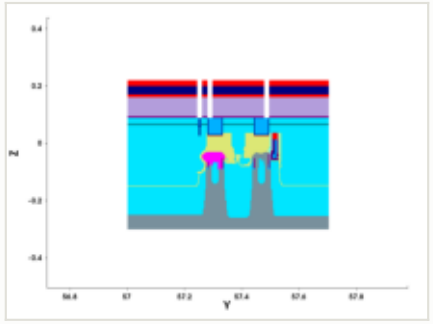
carbon so far, of the whole flow (kgCO2e)



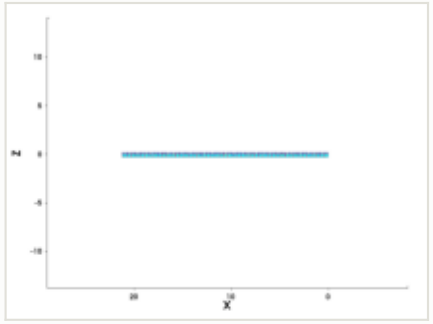
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



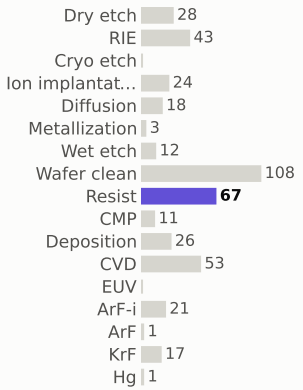
y cut at 57.351 µm



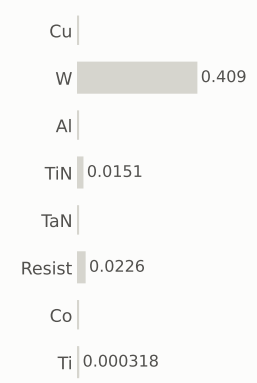
mask for this step

no mask for this step

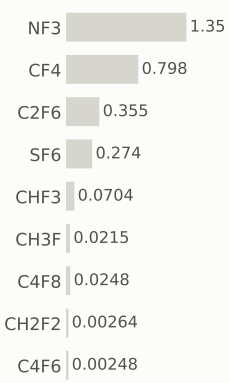
steps by process area



materials deposited (g)



gases emitted (g)



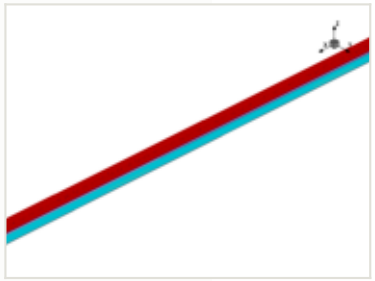
Step #5850: ArF-i

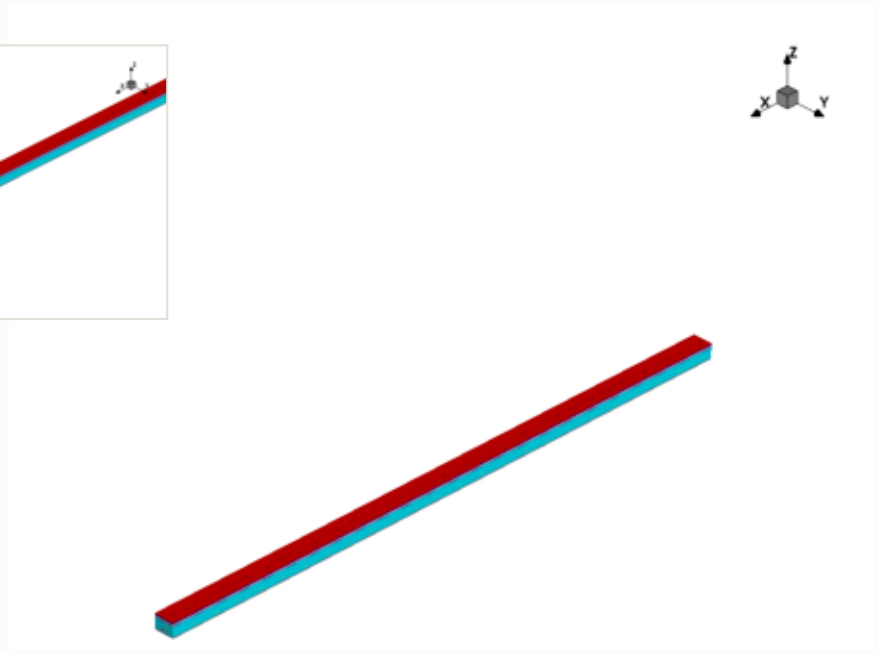
litho_patterning · ArF-i

0.217	258.9	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



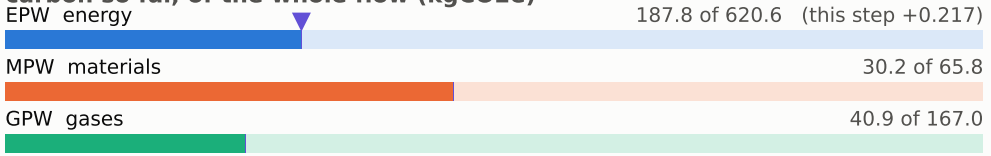


this step

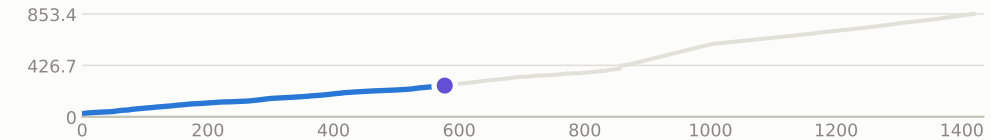
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

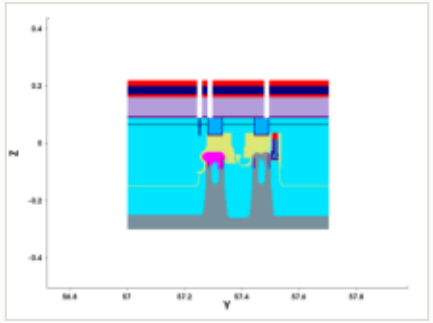
carbon so far, of the whole flow (kgCO2e)



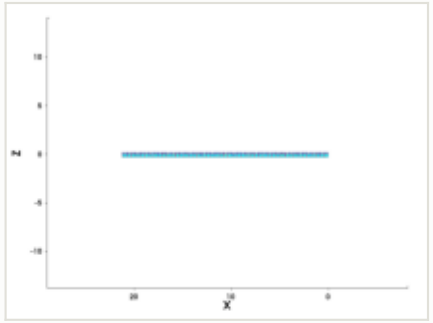
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

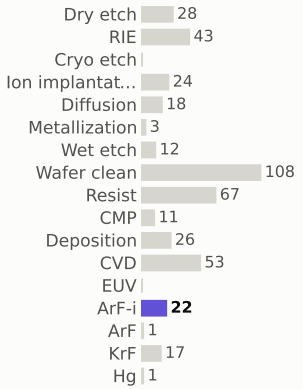


y cut at 57.351 µm

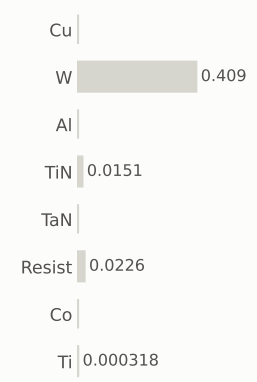


mask: layer 2100:0, positive (GDSII)

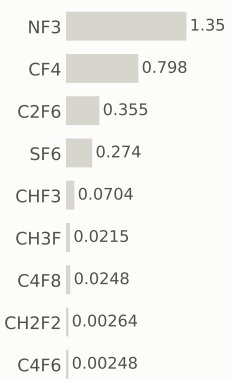
steps by process area



materials deposited (g)



gases emitted (g)



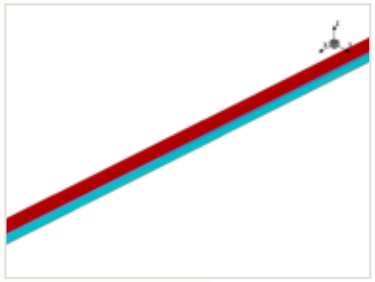
Step #5860: RIE

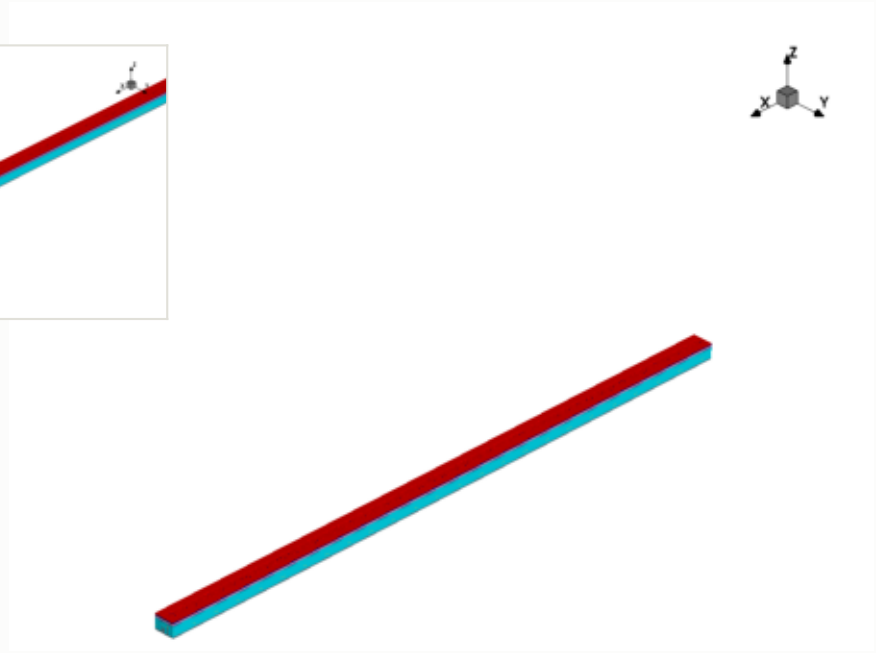
etch_anisotropic · RIE

1.04	260.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



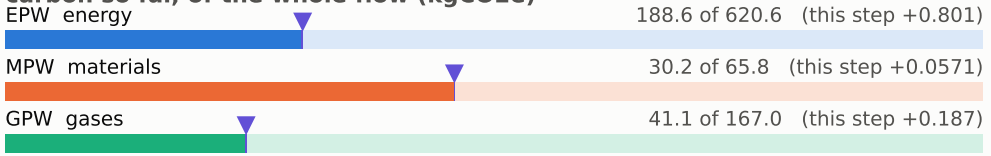


this step

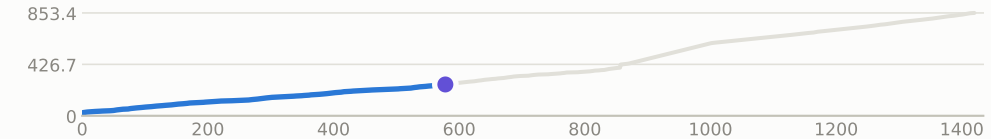
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

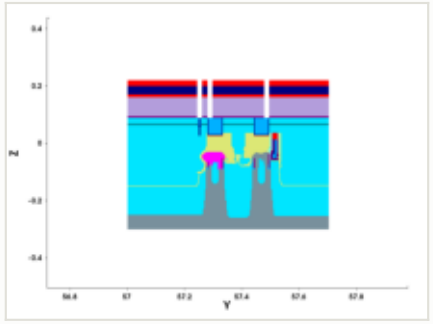
carbon so far, of the whole flow (kgCO2e)



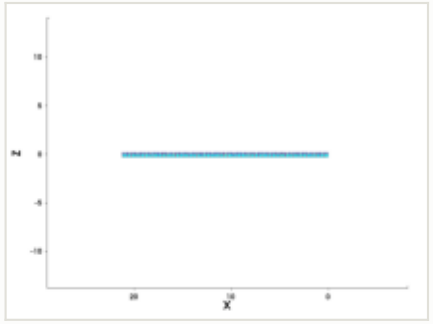
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



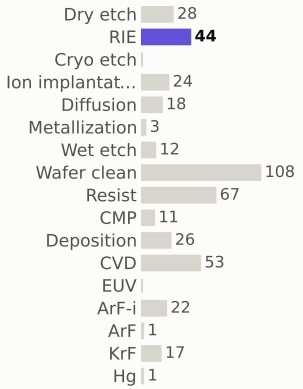
y cut at 57.351 µm



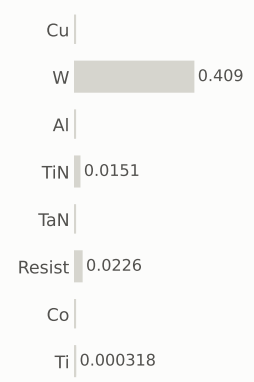
mask for this step

no mask for this step

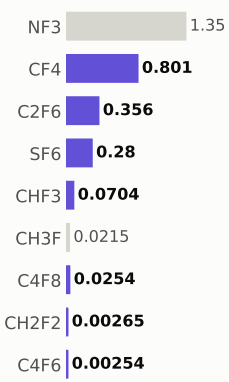
steps by process area



materials deposited (g)



gases emitted (g)



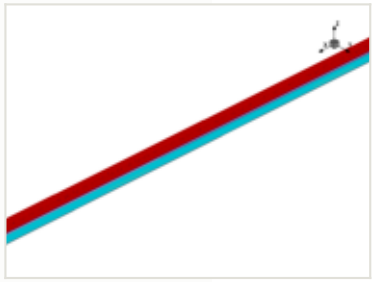
Step #5870: Wafer_clean

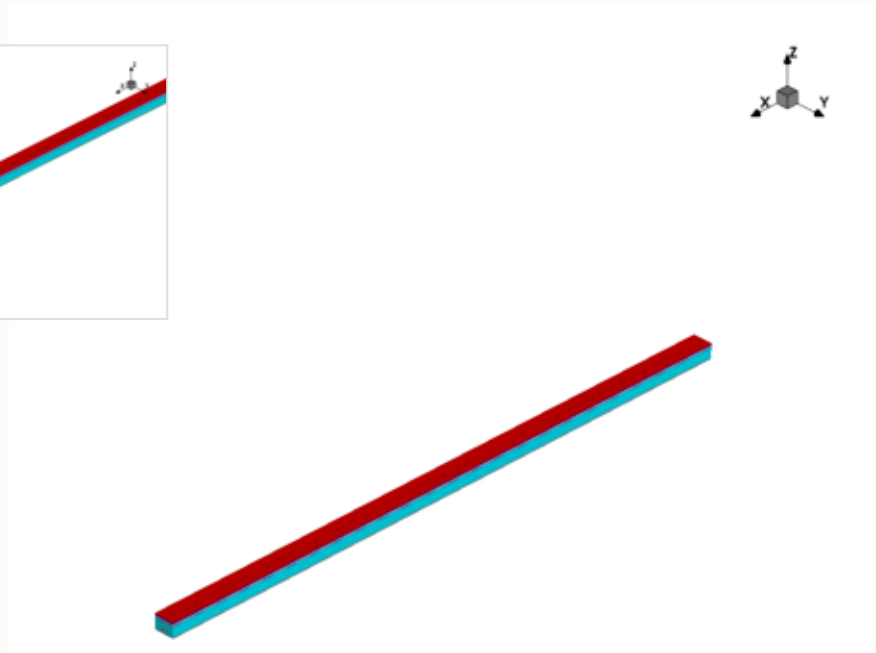
clean · Wafer_clean

0.169	260.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





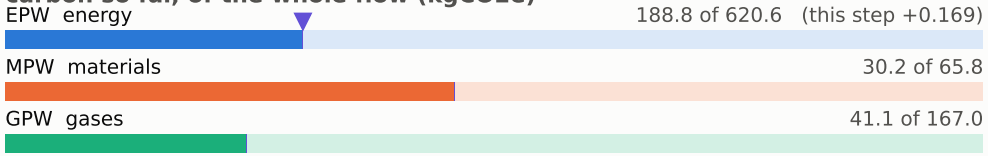
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

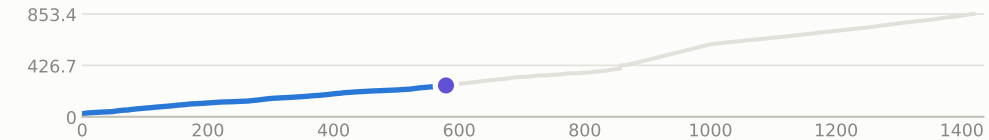
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

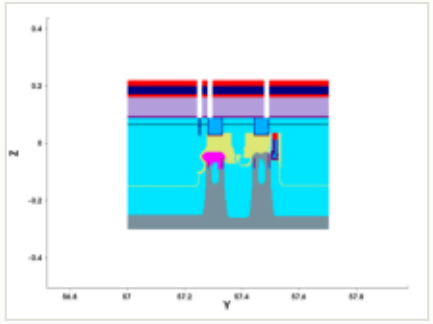
carbon so far, of the whole flow (kgCO2e)



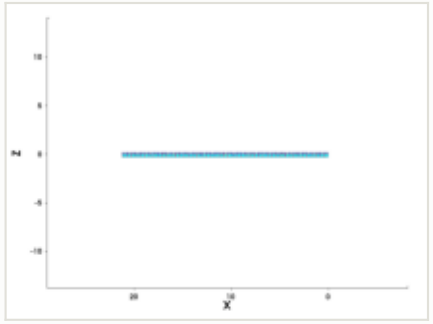
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



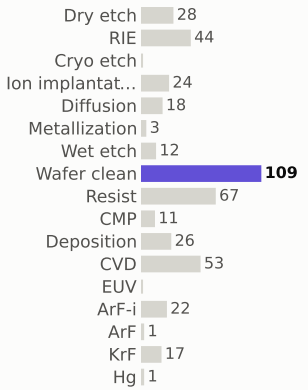
y cut at 57.351 µm



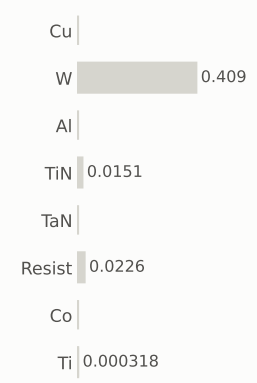
mask for this step

no mask for this step

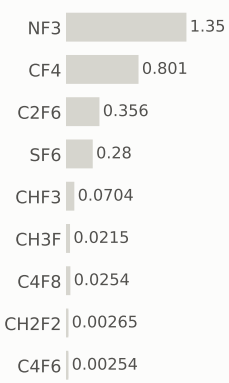
steps by process area



materials deposited (g)



gases emitted (g)



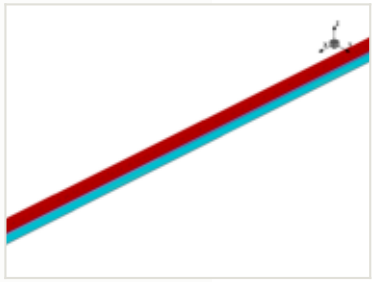
Step #5880: RIE

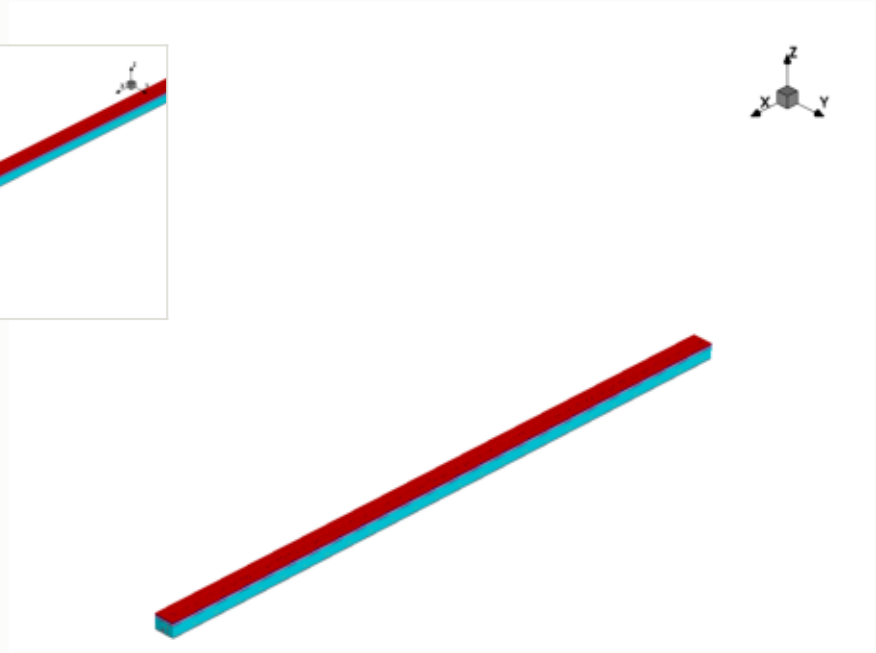
etch_anisotropic · RIE

1.04	261.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



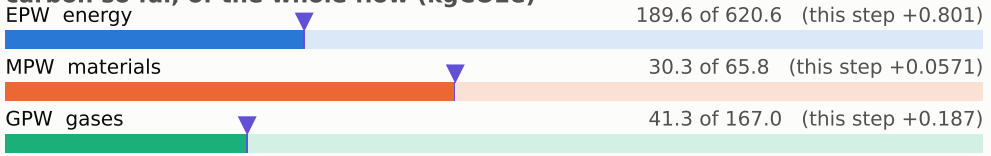


this step

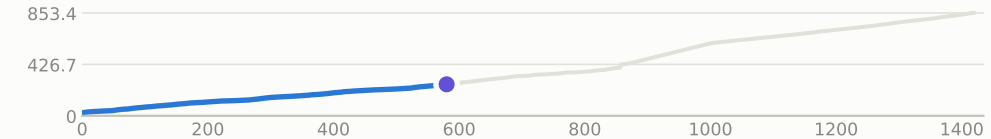
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

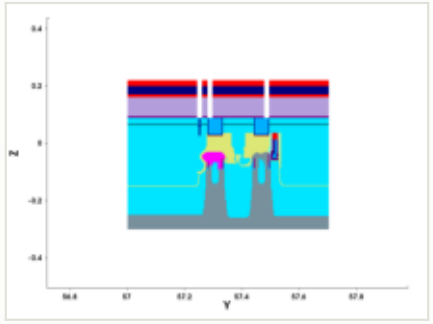
carbon so far, of the whole flow (kgCO2e)



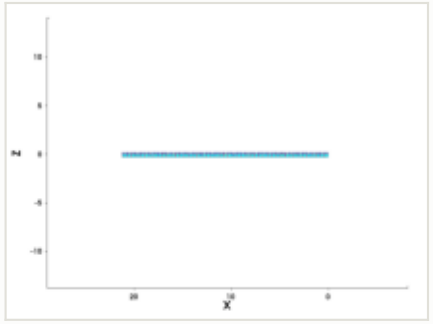
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



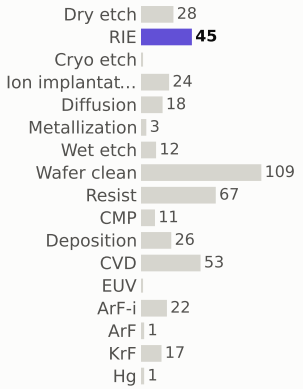
y cut at 57.351 µm



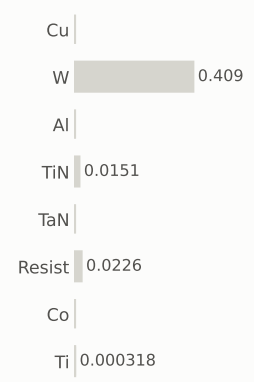
mask for this step

no mask for this step

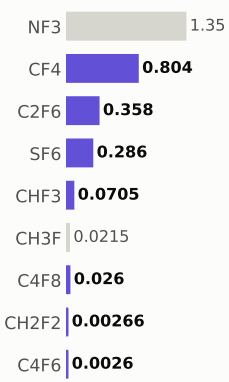
steps by process area



materials deposited (g)



gases emitted (g)



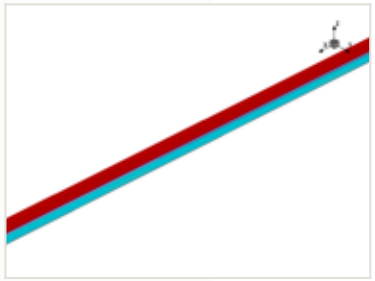
Step #5890: Wafer_clean

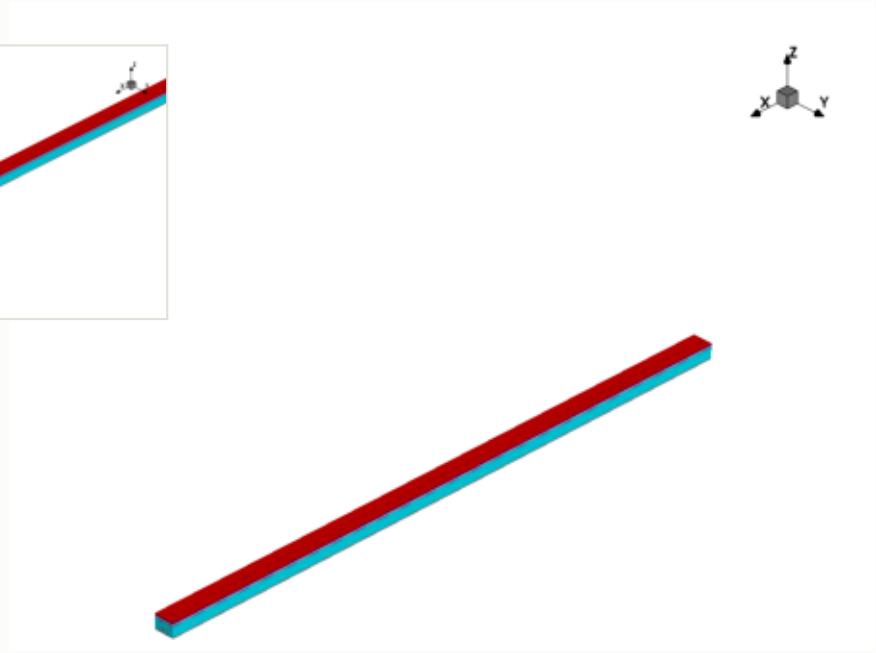
clean · Wafer_clean

0.169	261.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





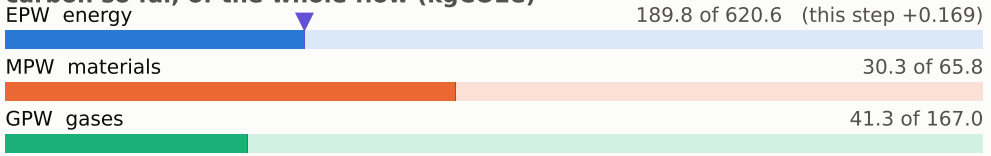
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

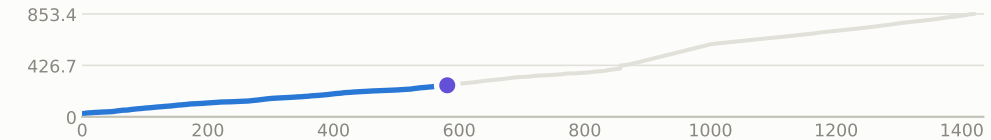
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

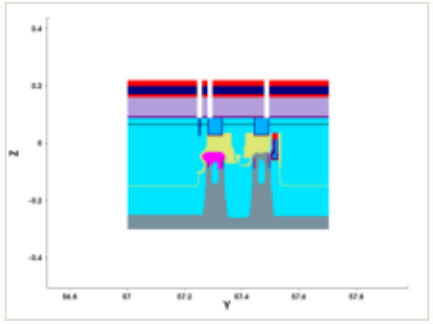
carbon so far, of the whole flow (kgCO2e)



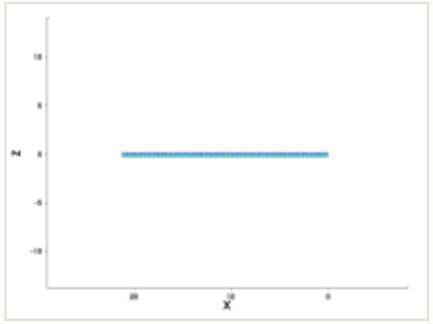
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



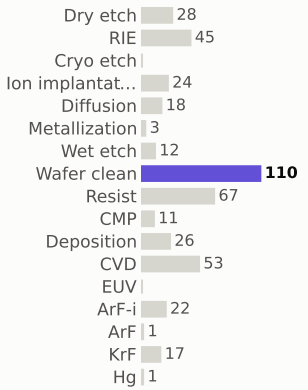
y cut at 57.351 µm



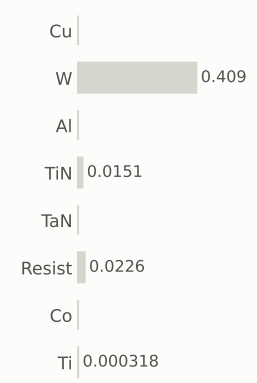
mask for this step

no mask for this step

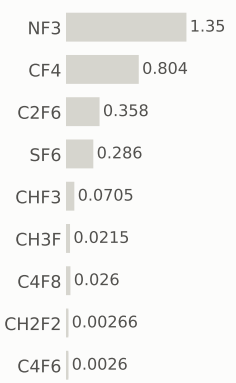
steps by process area



materials deposited (g)



gases emitted (g)



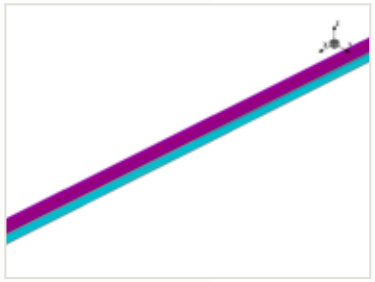
Step #5900: Metallization

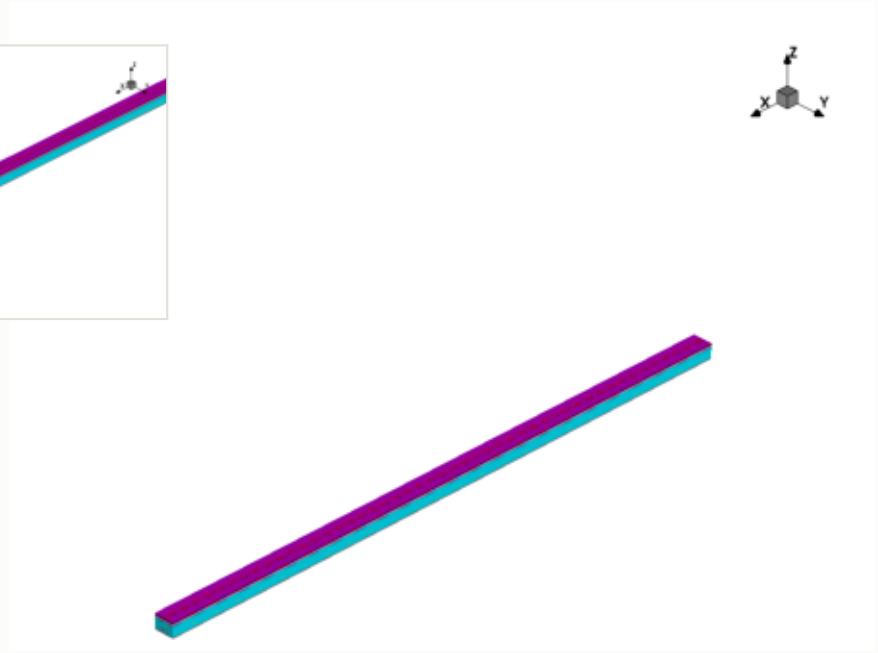
depo_ellipsoidal · Metallization

1.2	262.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



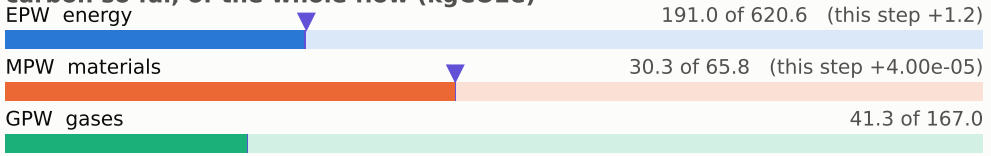


this step

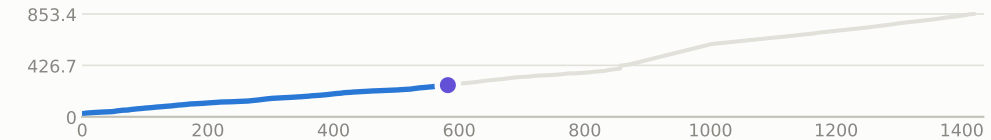
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

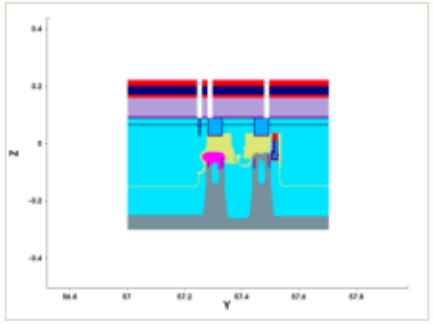
carbon so far, of the whole flow (kgCO2e)



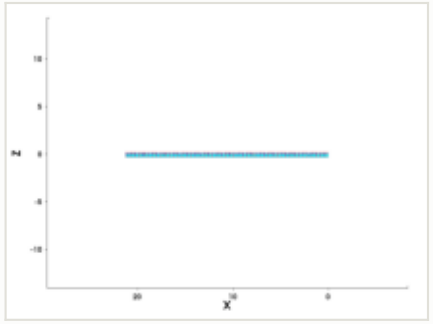
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



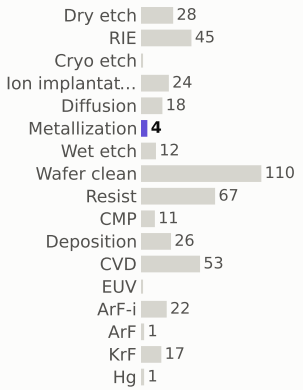
y cut at 57.351 µm



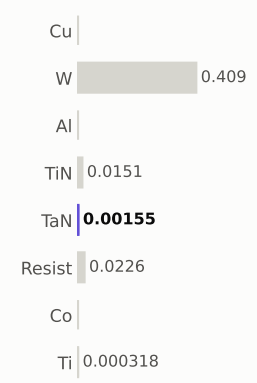
mask for this step

no mask for this step

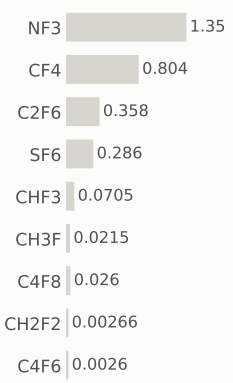
steps by process area



materials deposited (g)



gases emitted (g)



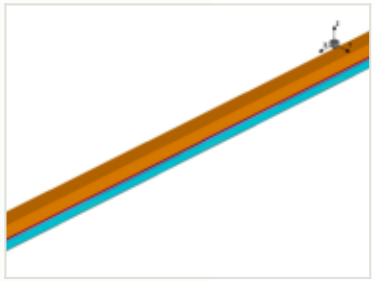
Step #5910: Metallization

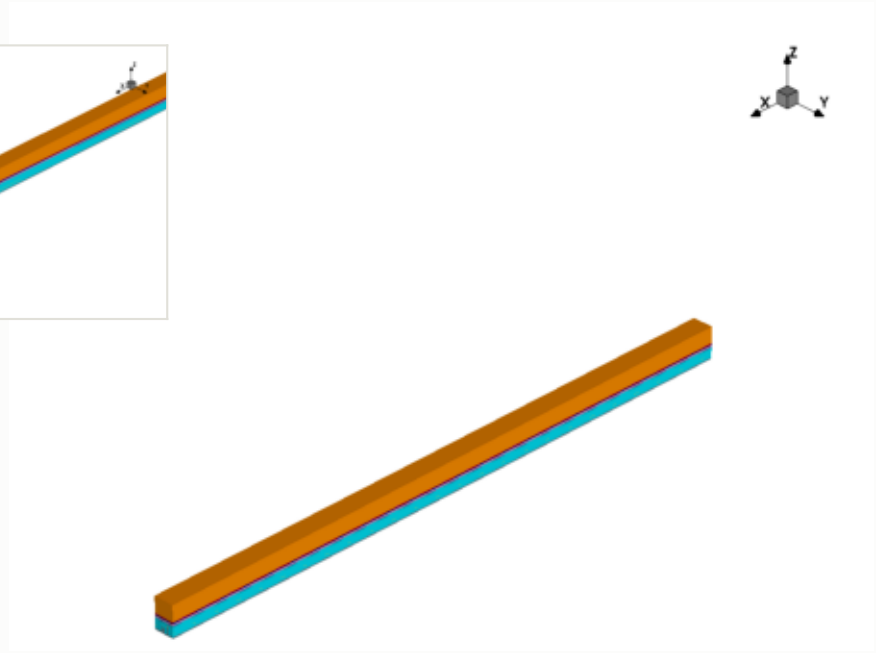
depo_fill · Metallization

1.21	263.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



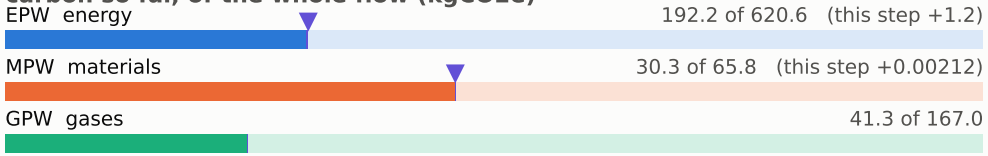


this step

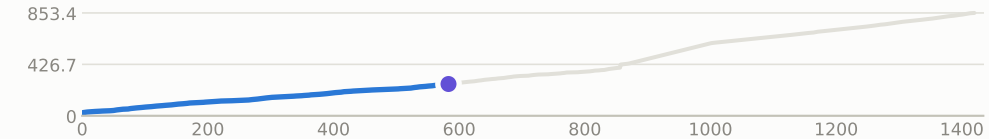
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

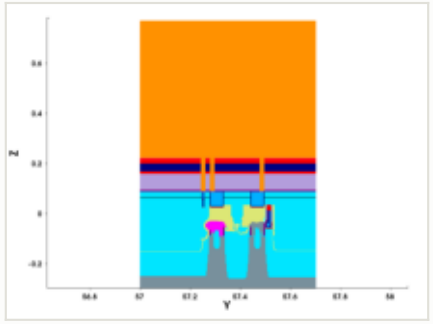
carbon so far, of the whole flow (kgCO2e)



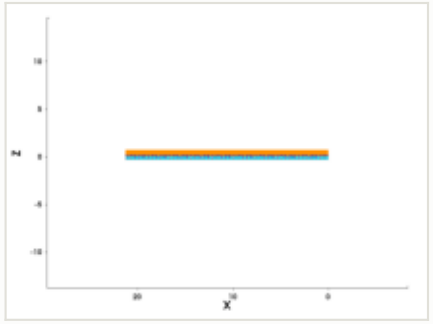
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



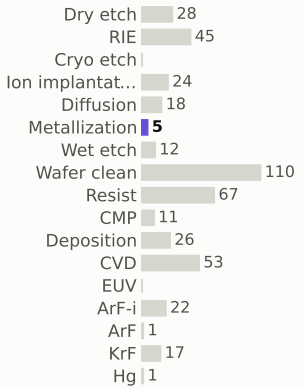
y cut at 57.351 µm



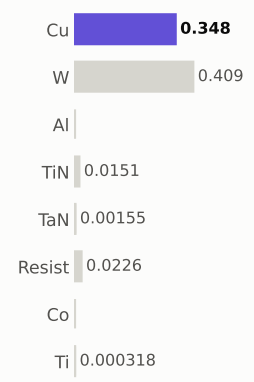
mask for this step

no mask for this step

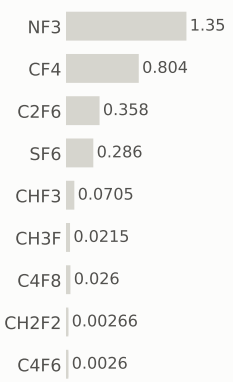
steps by process area



materials deposited (g)



gases emitted (g)



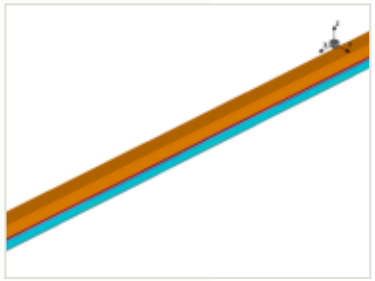
Step #5920: Diffusion

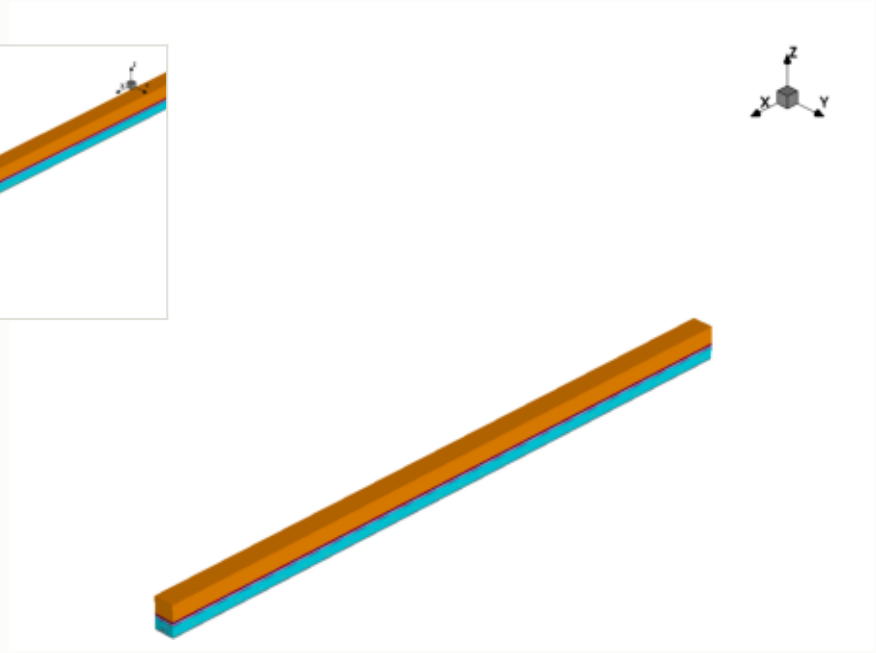
anneal · Diffusion

0.801	264.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





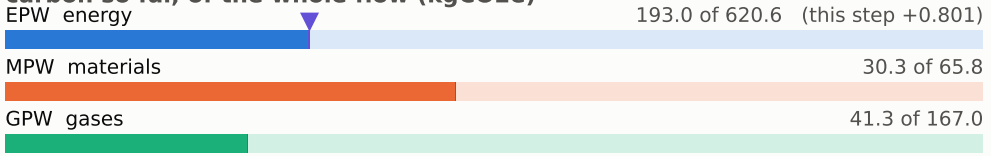
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

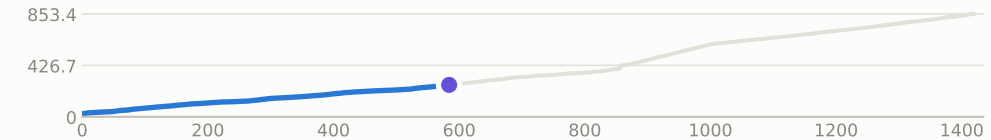
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

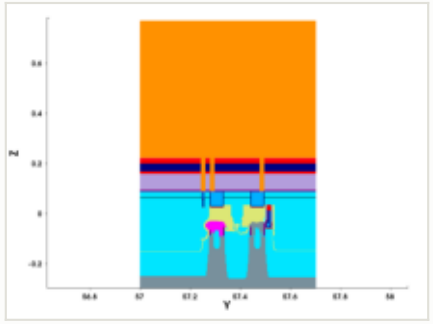
carbon so far, of the whole flow (kgCO2e)



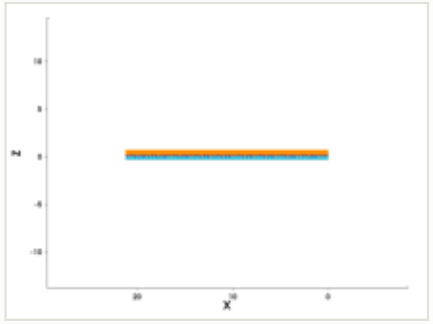
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



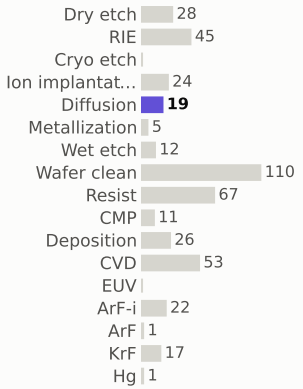
y cut at 57.351 µm



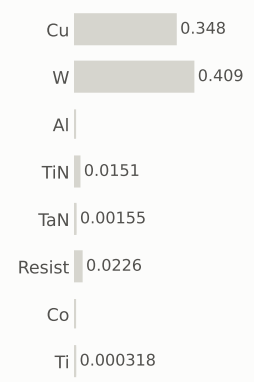
mask for this step

no mask for this step

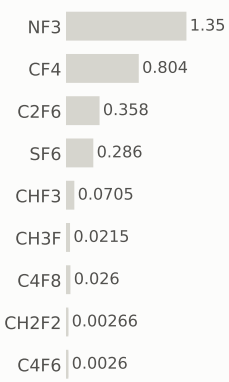
steps by process area



materials deposited (g)



gases emitted (g)



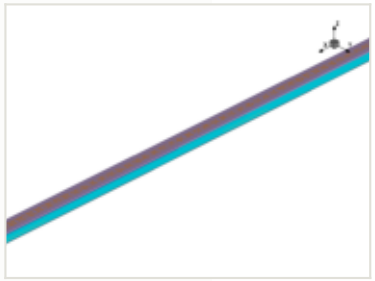
Step #5930: CMP

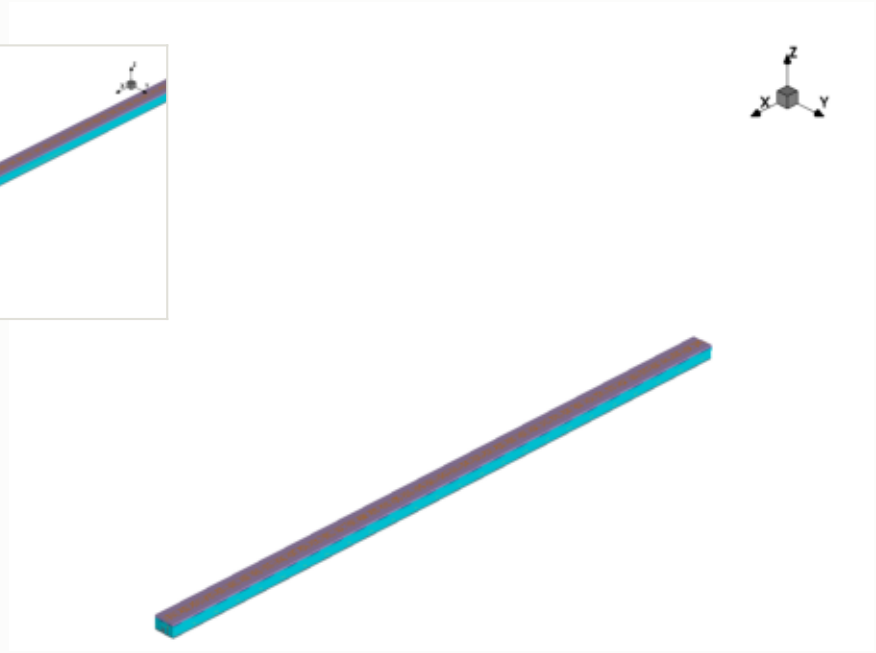
etch_cmp · CMP

0.169	264.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



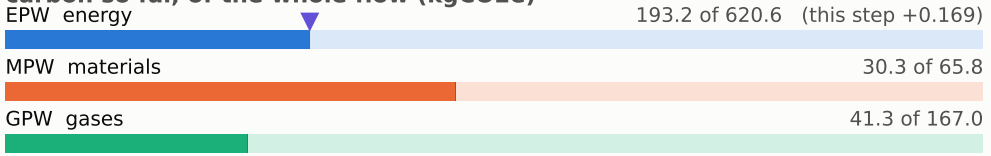


this step

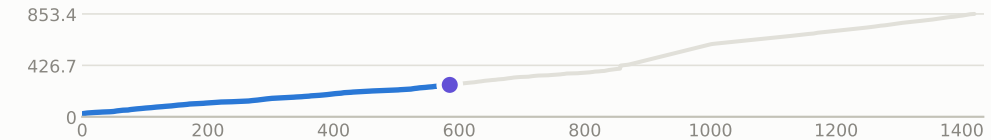
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

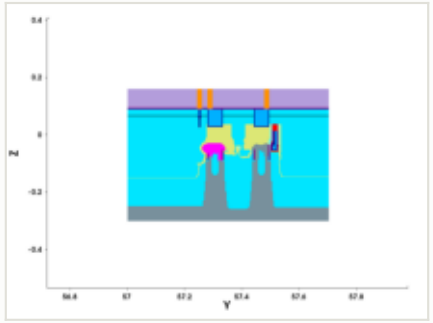
carbon so far, of the whole flow (kgCO2e)



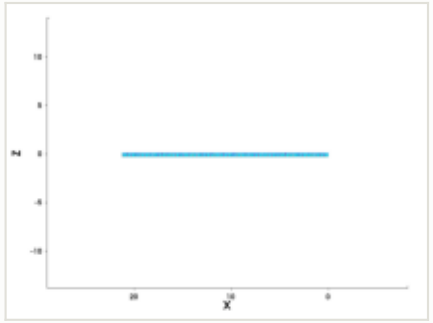
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



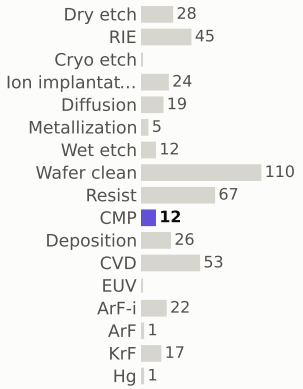
y cut at 57.351 µm



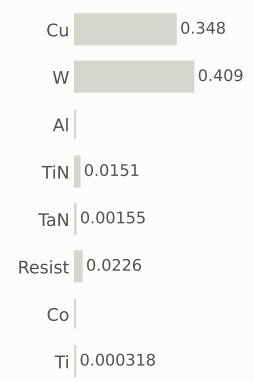
mask for this step

no mask for this step

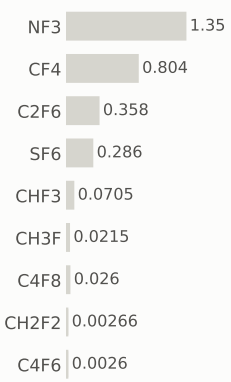
steps by process area



materials deposited (g)



gases emitted (g)



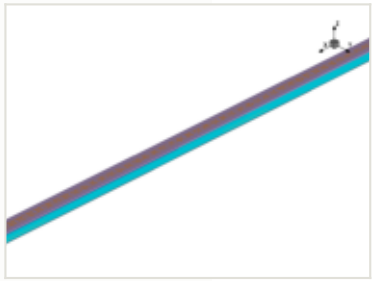
Step #5940: Wafer_clean

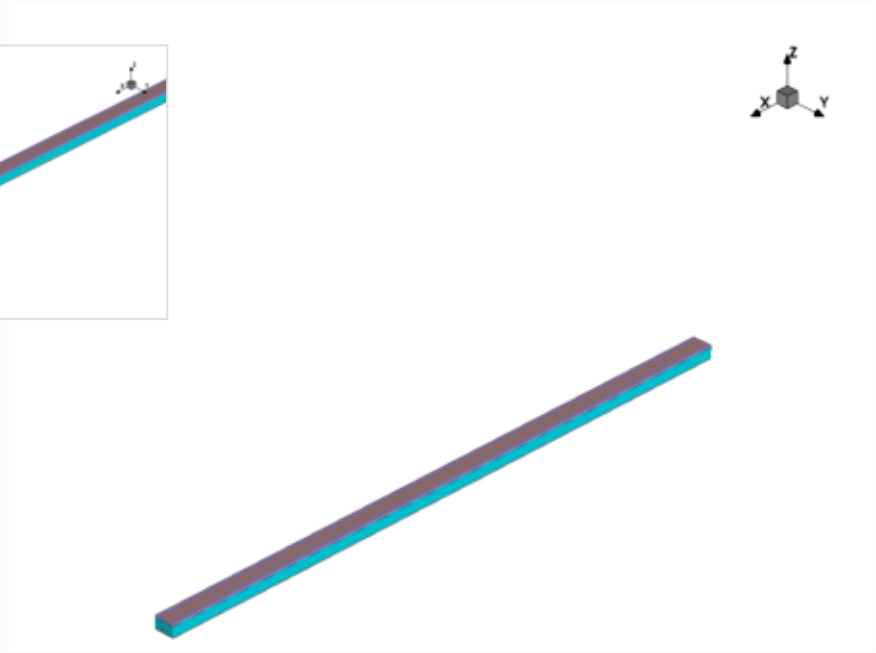
clean · Wafer_clean

0.169	264.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

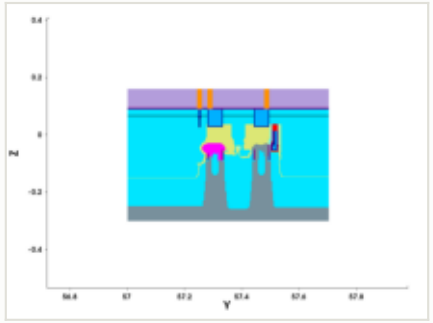
3D structure after this step

zoomed to fit

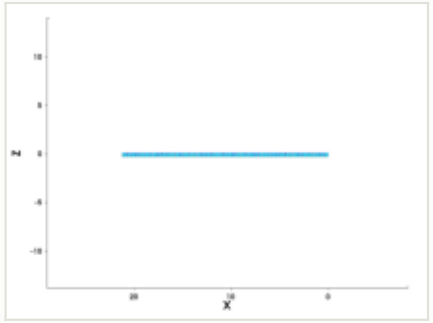




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

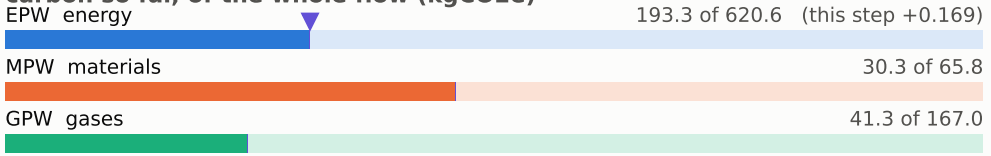
no mask for this step

this step

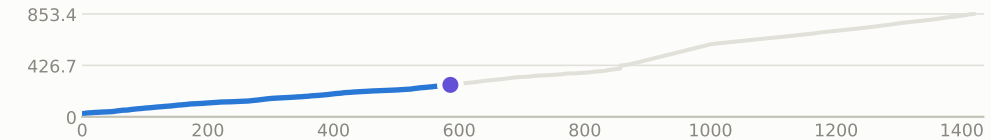
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

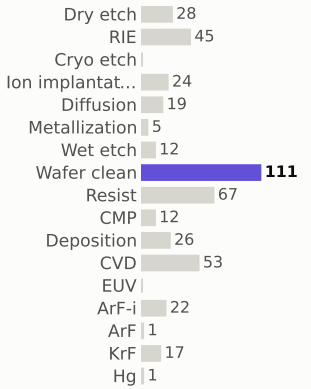
carbon so far, of the whole flow (kgCO2e)



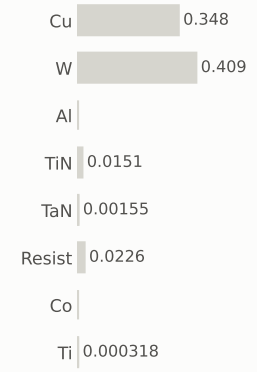
embodied carbon up to each step (kgCO2e per wafer)



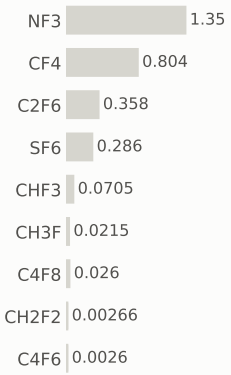
steps by process area



materials deposited (g)



gases emitted (g)



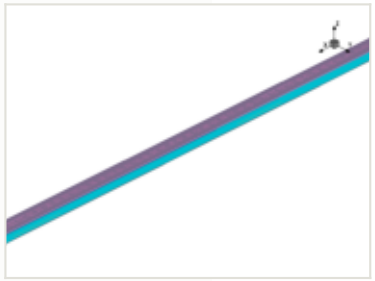
Step #5950: Metallization

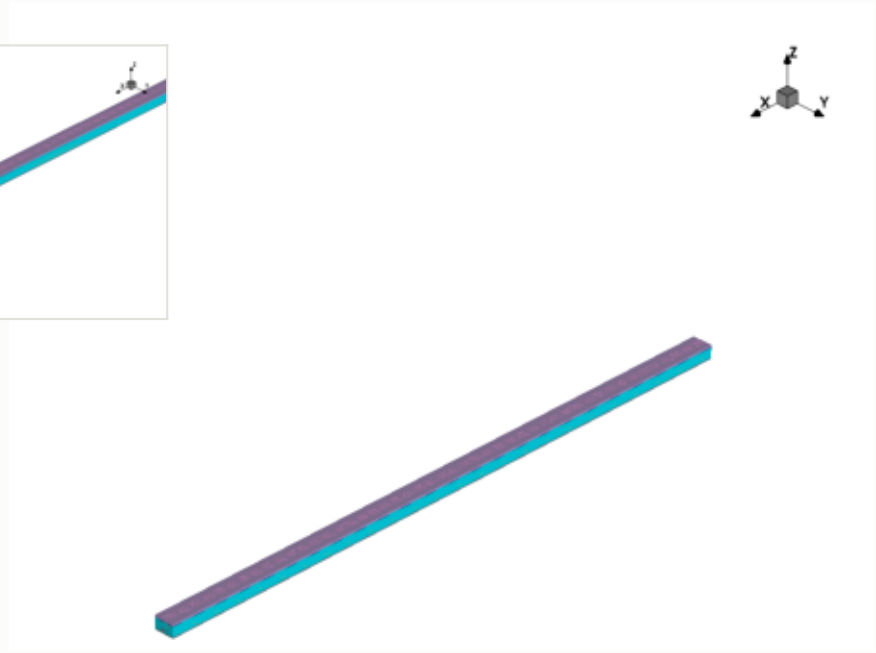
depo_isotropic · Metallization

1.2	266.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



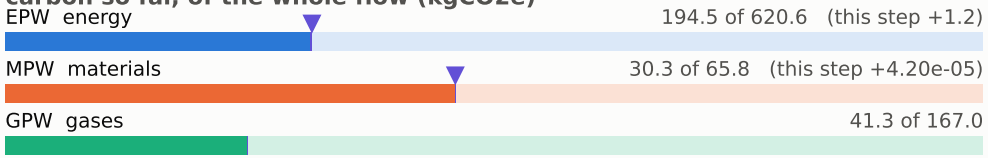


this step

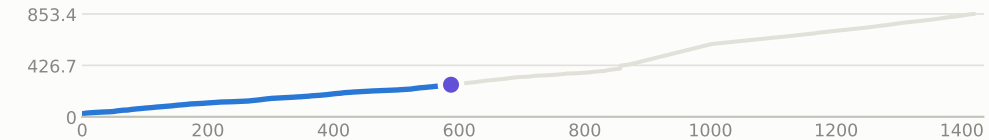
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

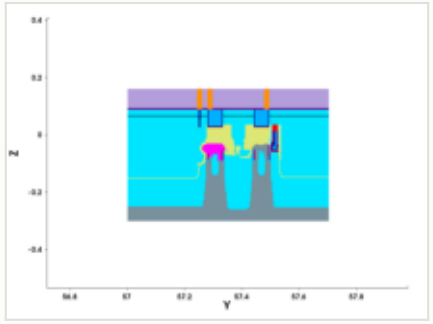
carbon so far, of the whole flow (kgCO2e)



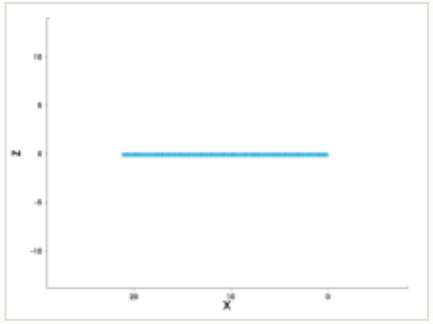
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



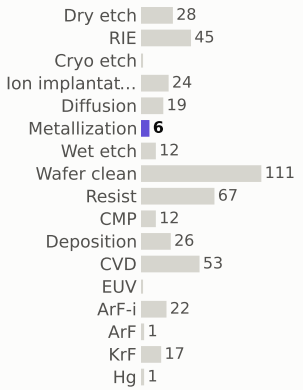
y cut at 57.351 µm



mask for this step

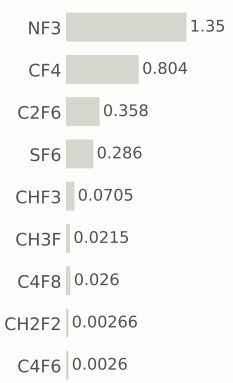
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



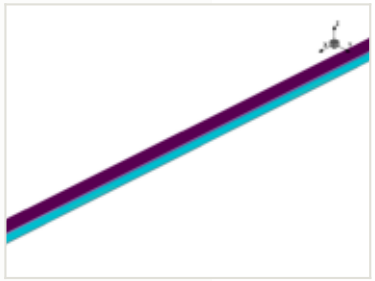
Step #5970: Deposition

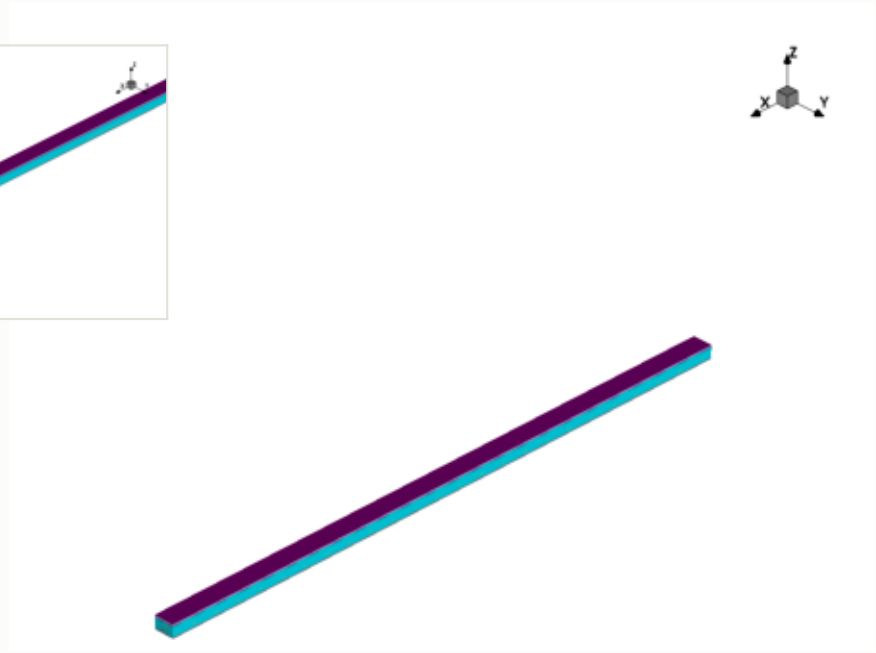
depo_fill · Deposition

0.656	266.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



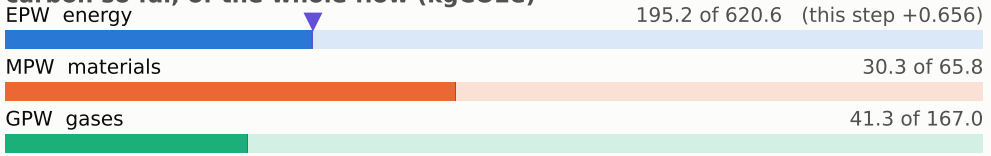


this step

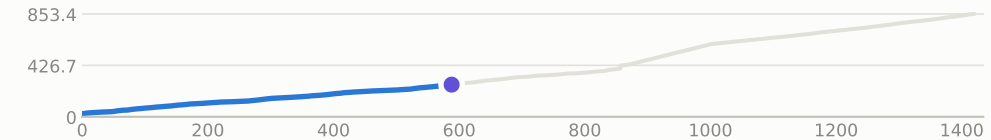
process area	Deposition
material	—
thickness	0.006 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

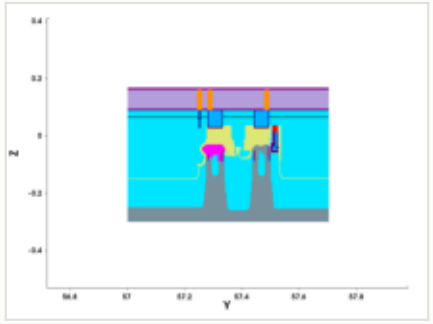
carbon so far, of the whole flow (kgCO2e)



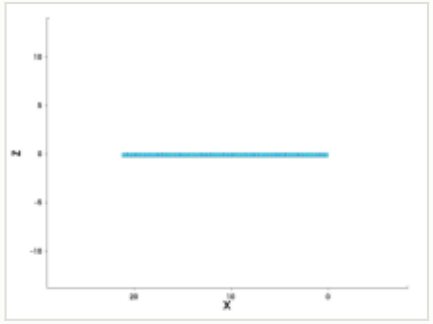
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



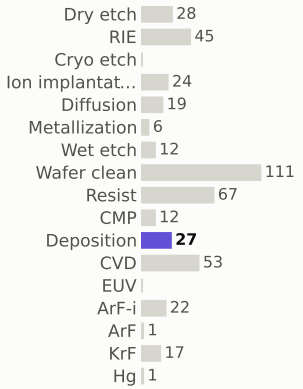
y cut at 57.351 µm



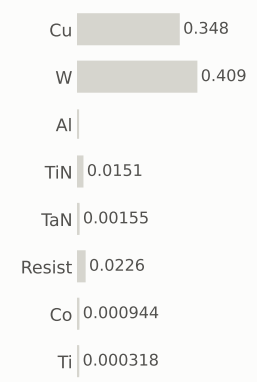
mask for this step

no mask for this step

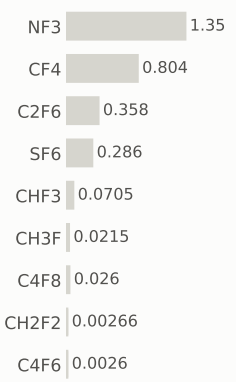
steps by process area



materials deposited (g)



gases emitted (g)



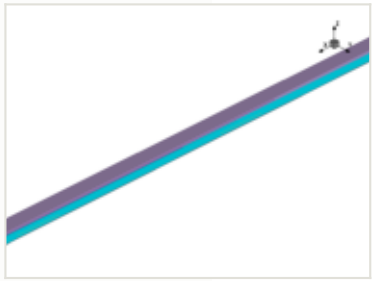
Step #5980: CVD

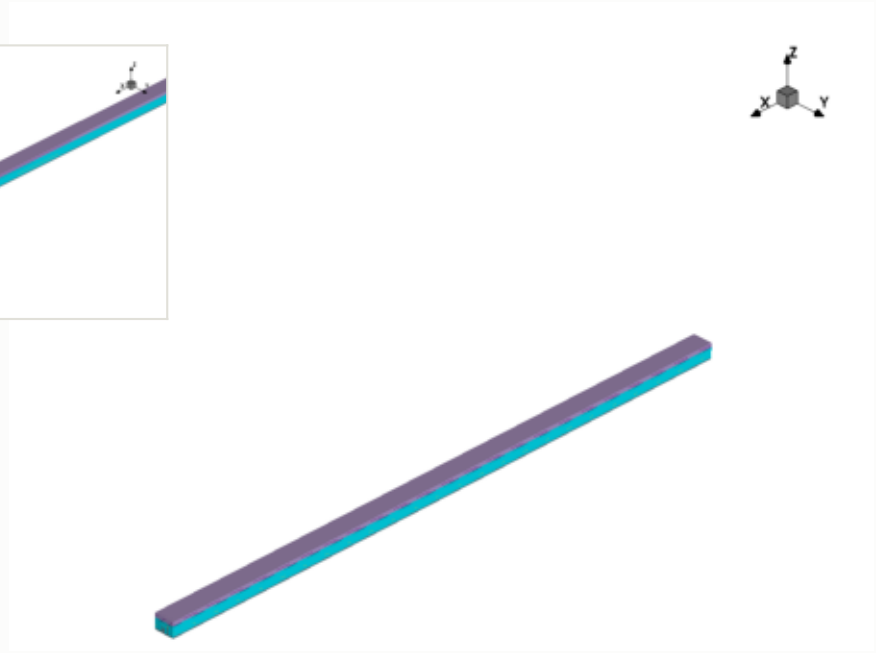
depo_fill · CVD

1.31	268.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



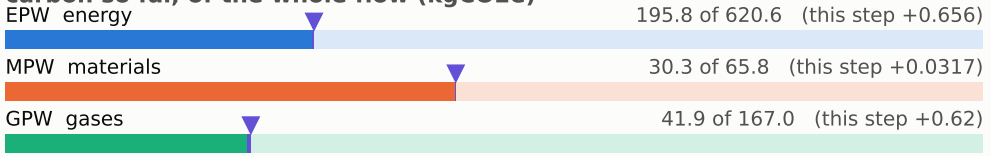


this step

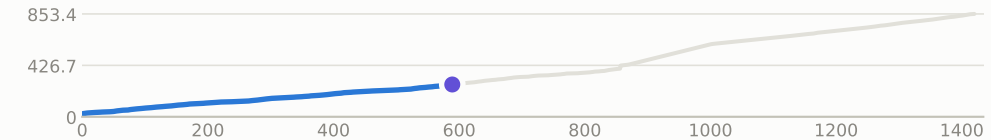
process area	CVD
material	—
thickness	0.066 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

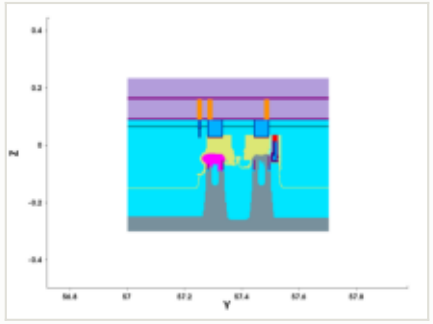
carbon so far, of the whole flow (kgCO2e)



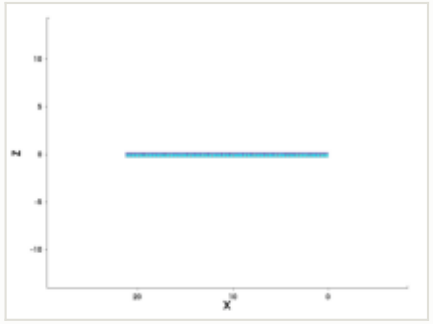
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



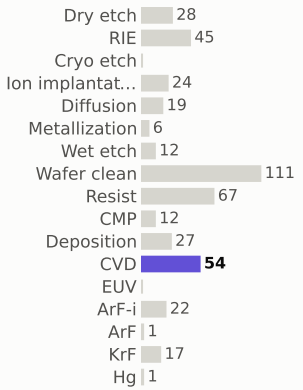
y cut at 57.351 µm



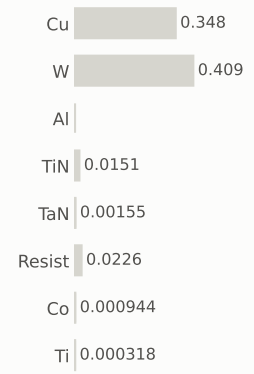
mask for this step

no mask for this step

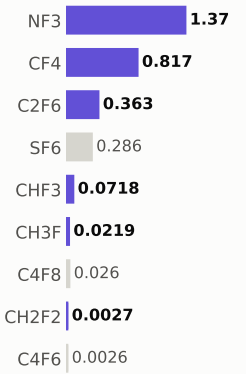
steps by process area



materials deposited (g)



gases emitted (g)



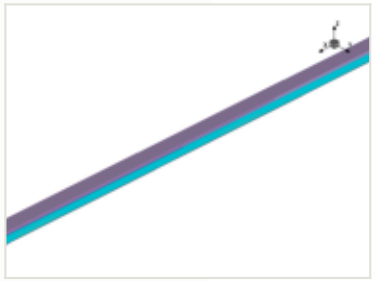
Step #5990: Diffusion

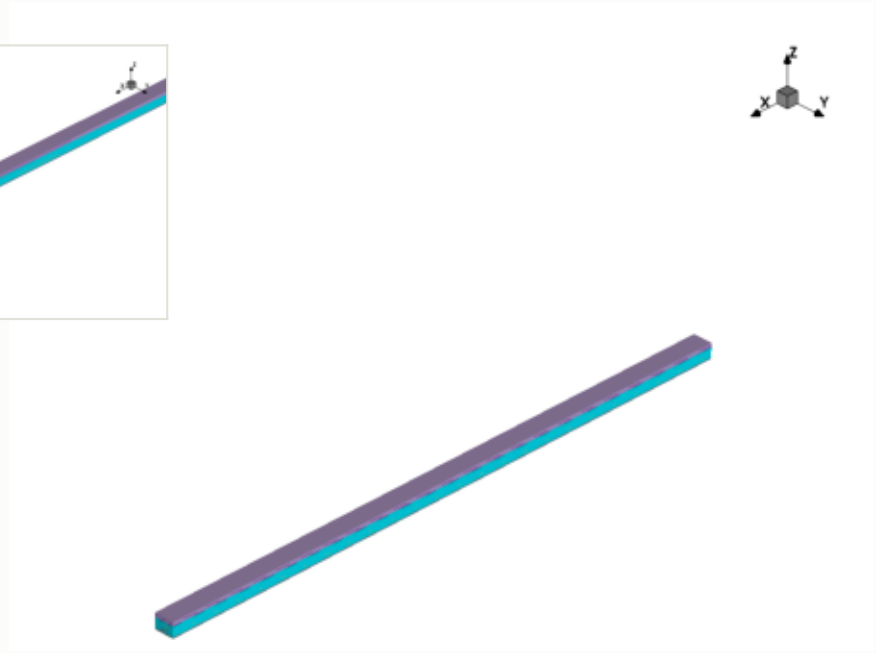
anneal · Diffusion

0.801	268.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

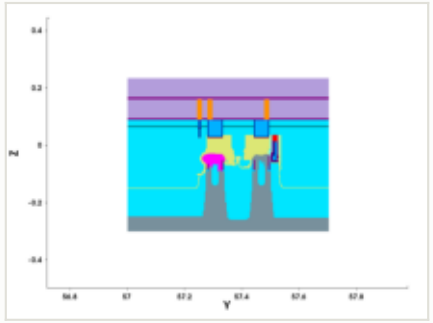
3D structure after this step

zoomed to fit

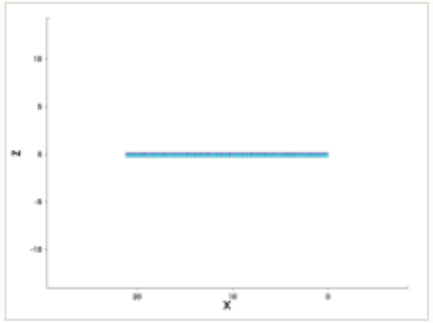




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

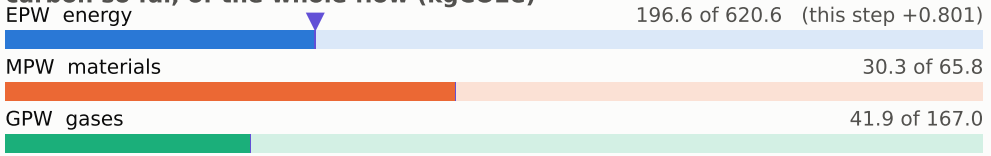
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

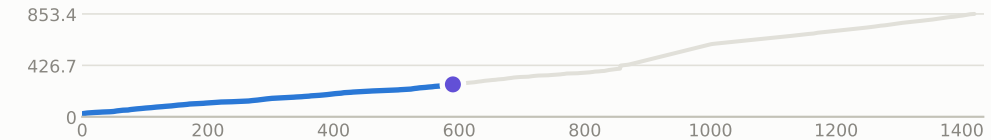
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

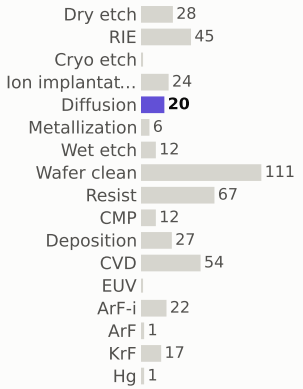
carbon so far, of the whole flow (kgCO2e)



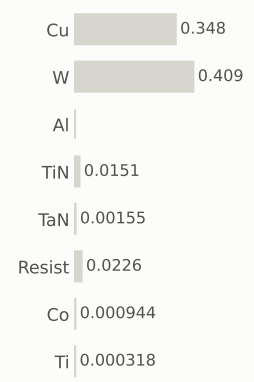
embodied carbon up to each step (kgCO2e per wafer)



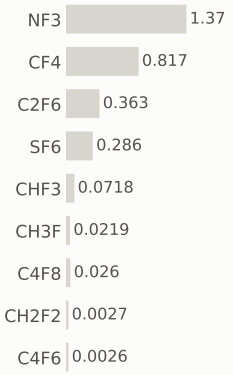
steps by process area



materials deposited (g)



gases emitted (g)



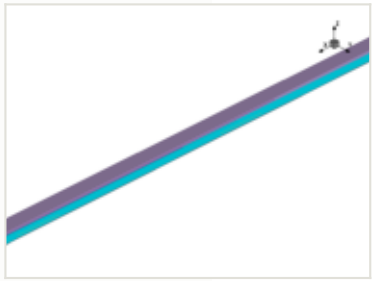
Step #6000: Wafer_clean

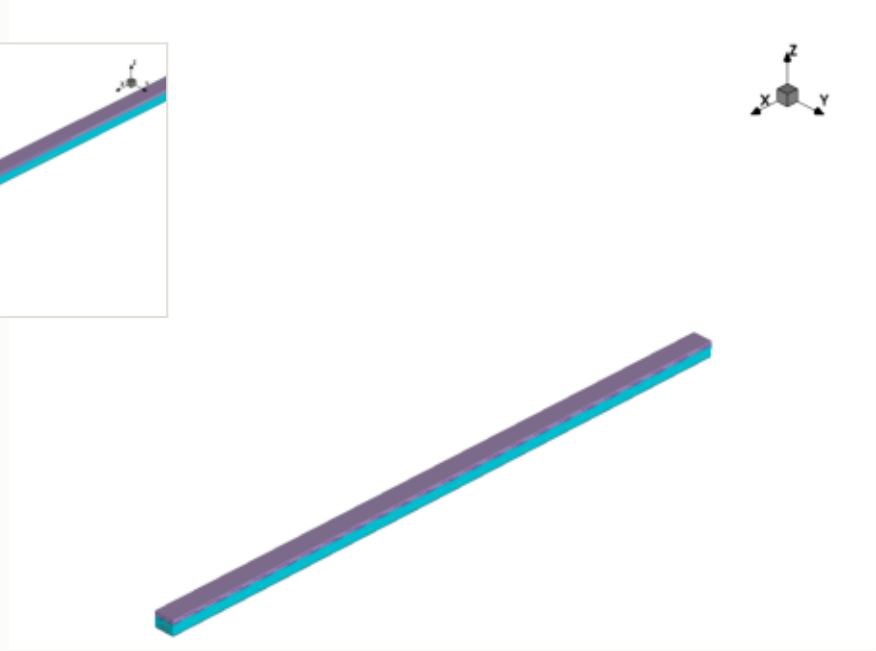
clean · Wafer_clean

0.169	269.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



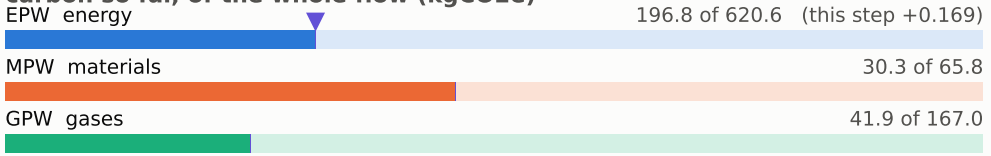


this step

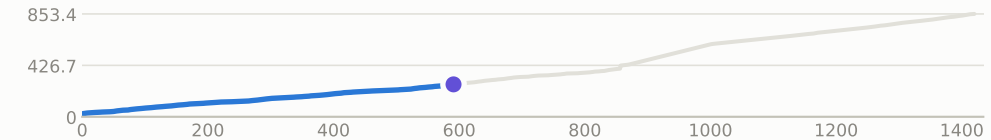
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

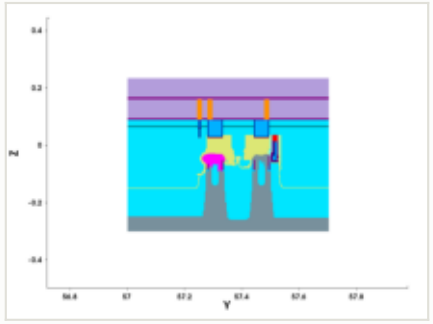
carbon so far, of the whole flow (kgCO2e)



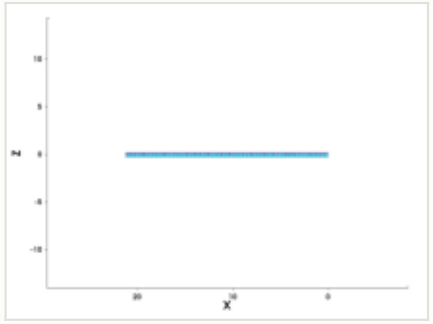
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



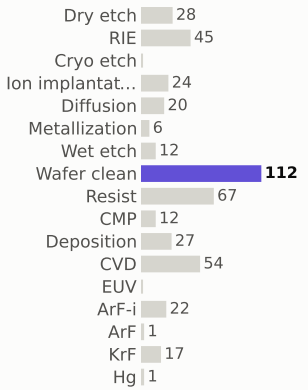
y cut at 57.351 µm



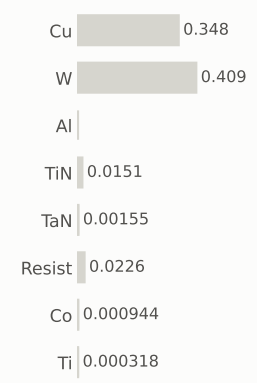
mask for this step

no mask for this step

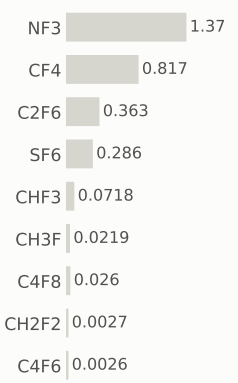
steps by process area



materials deposited (g)



gases emitted (g)



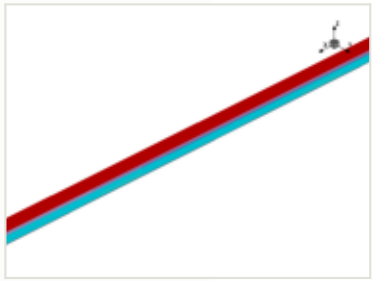
Step #6010: CVD

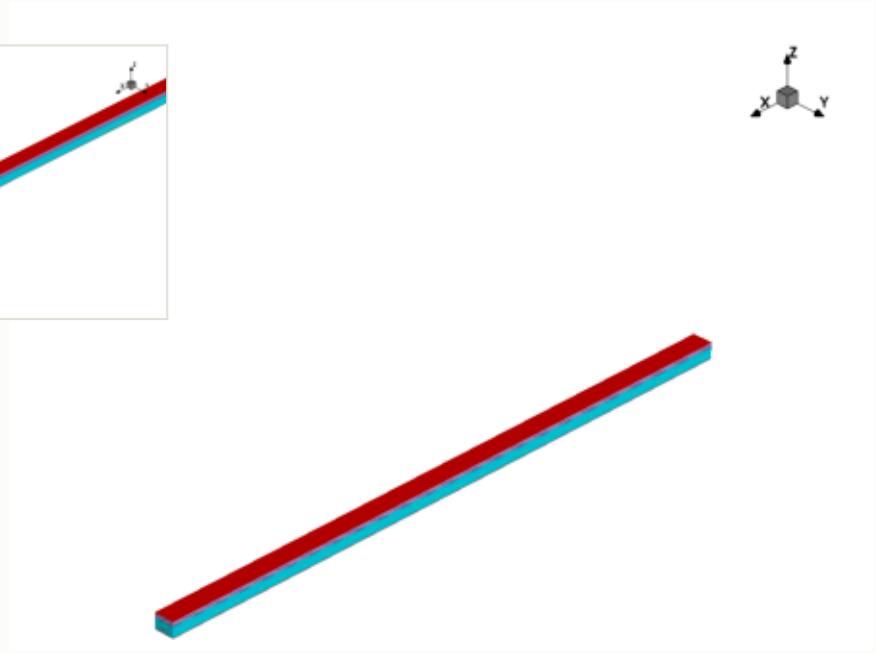
depo_fill · CVD

1.31	270.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



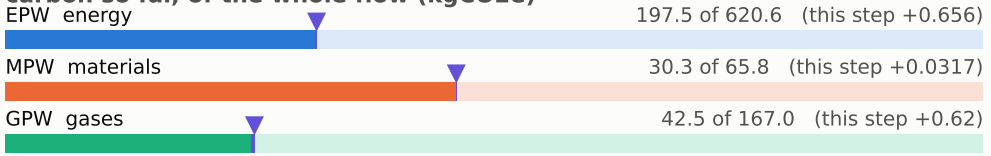


this step

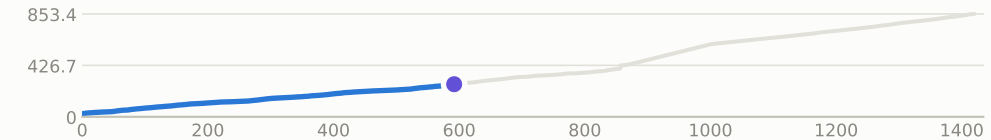
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

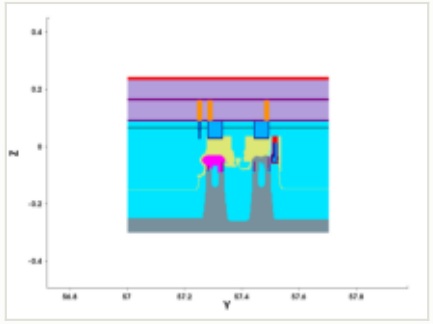
carbon so far, of the whole flow (kgCO2e)



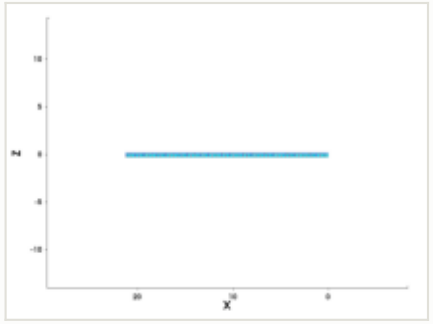
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



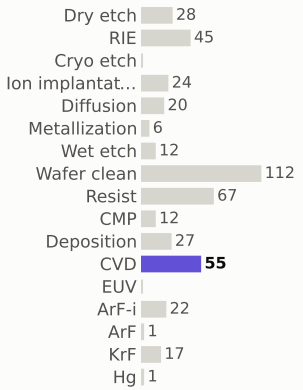
y cut at 57.351 µm



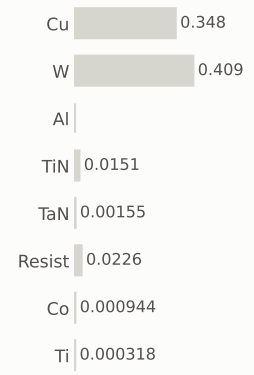
mask for this step

no mask for this step

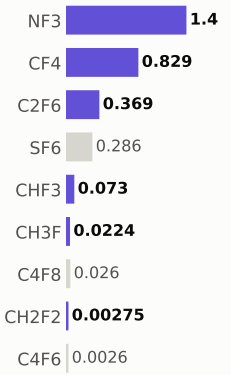
steps by process area



materials deposited (g)



gases emitted (g)



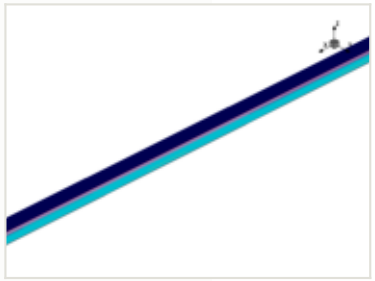
Step #6020: Deposition

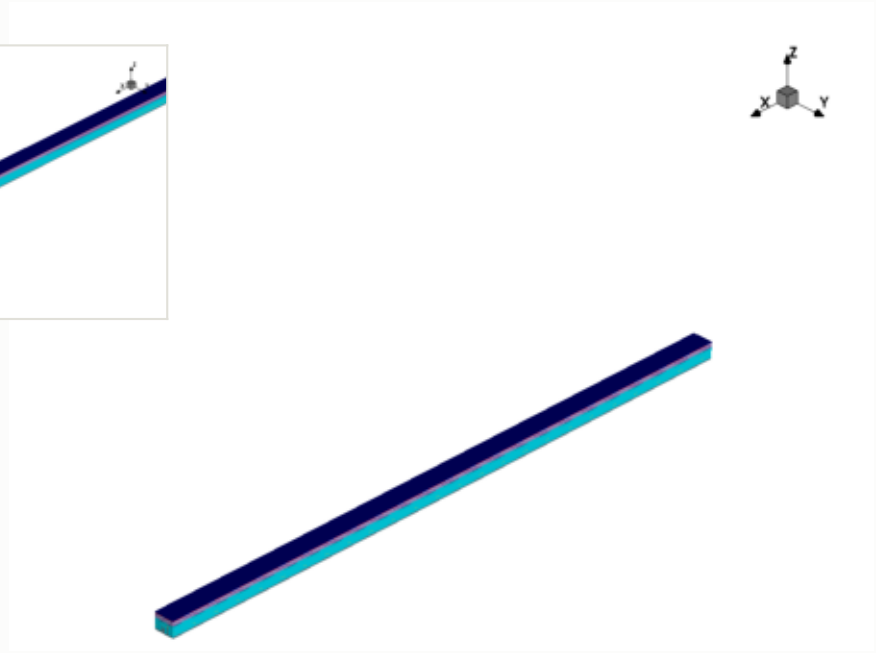
depo_fill · Deposition

0.657	271.0	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



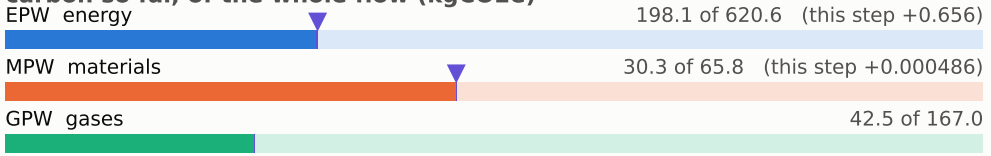


this step

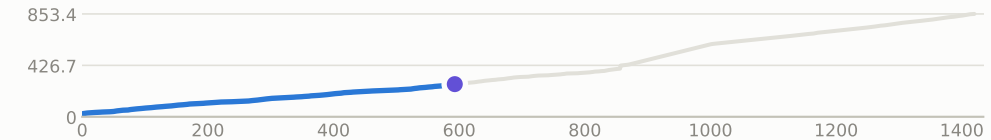
process area	Deposition
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000486 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

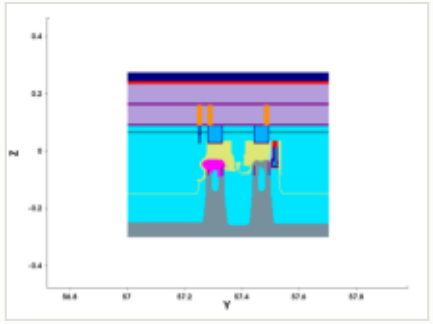
carbon so far, of the whole flow (kgCO2e)



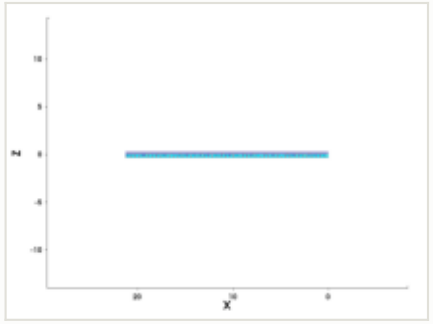
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



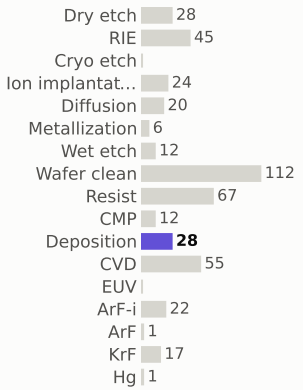
y cut at 57.351 µm



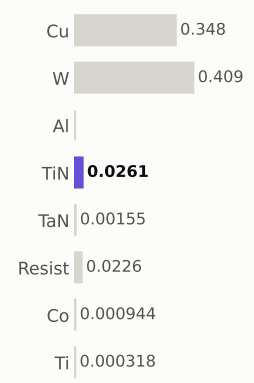
mask for this step

no mask for this step

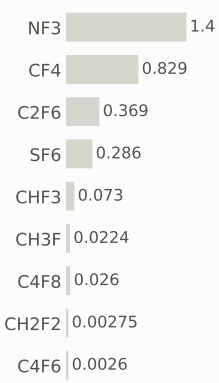
steps by process area



materials deposited (g)



gases emitted (g)



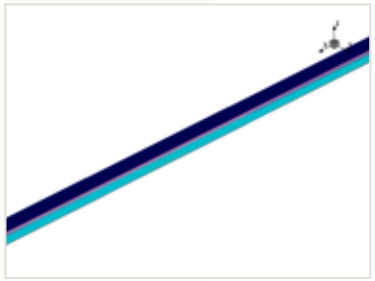
Step #6030: Wafer_clean

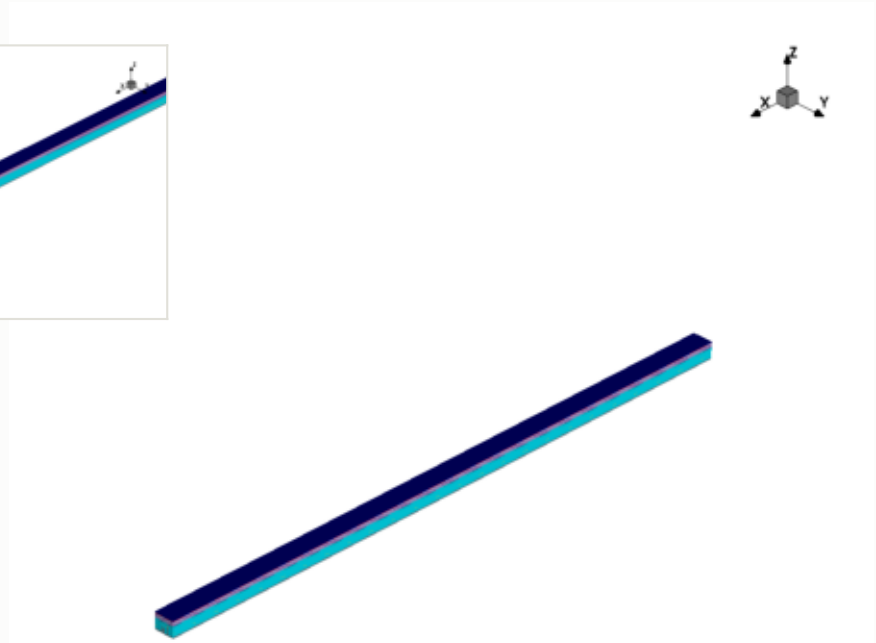
clean · Wafer_clean

0.169	271.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





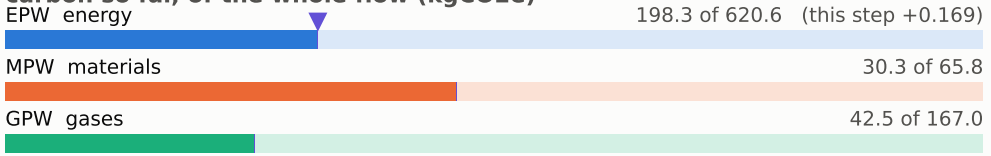
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

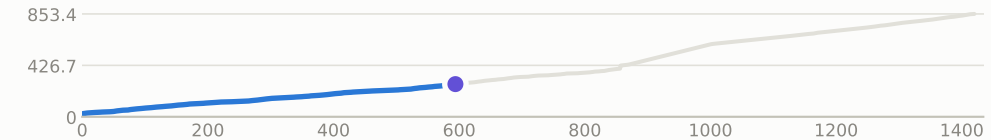
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

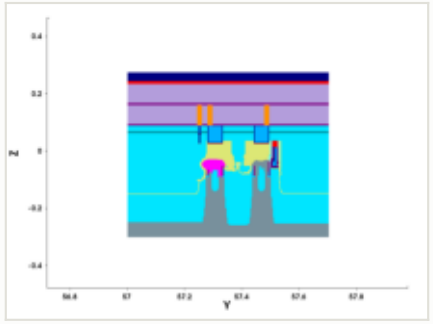
carbon so far, of the whole flow (kgCO2e)



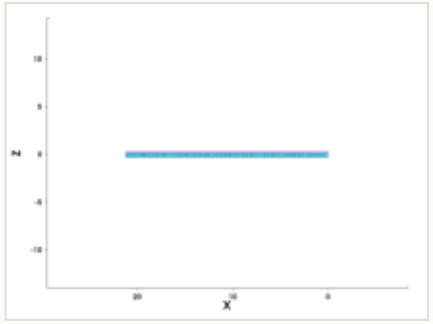
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



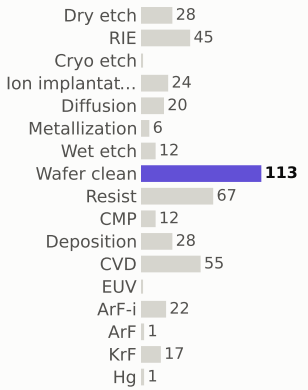
y cut at 57.351 µm



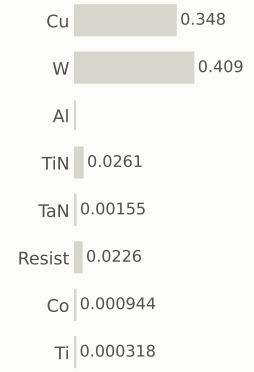
mask for this step

no mask for this step

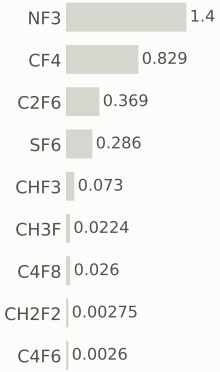
steps by process area



materials deposited (g)



gases emitted (g)



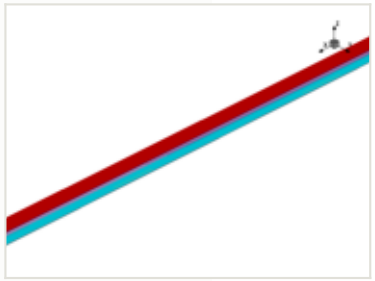
Step #6040: CVD

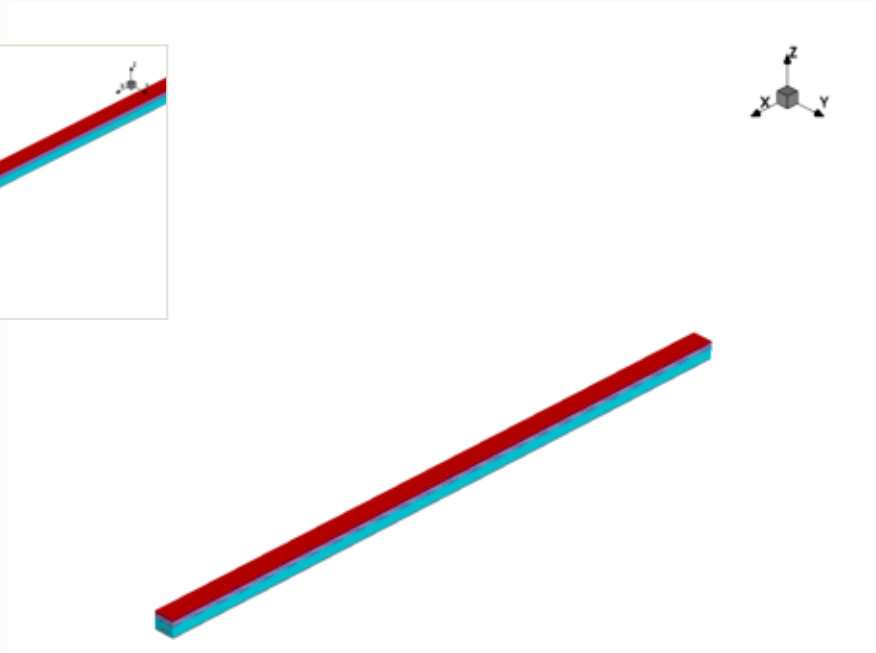
depo_fill · CVD

1.31	272.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



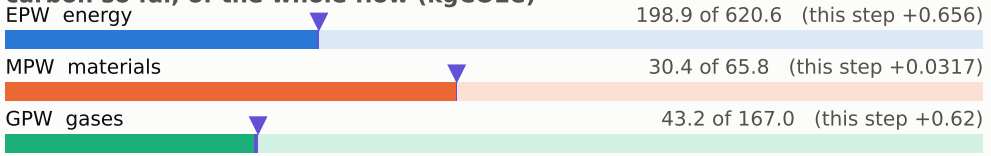


this step

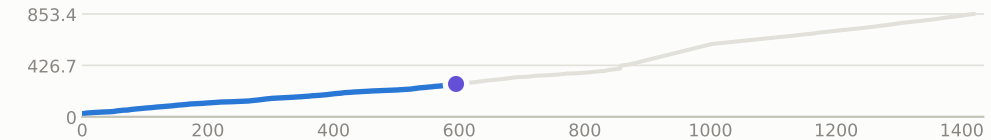
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

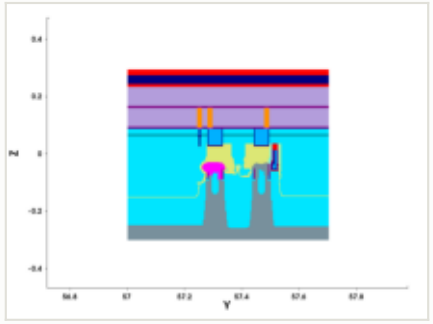
carbon so far, of the whole flow (kgCO2e)



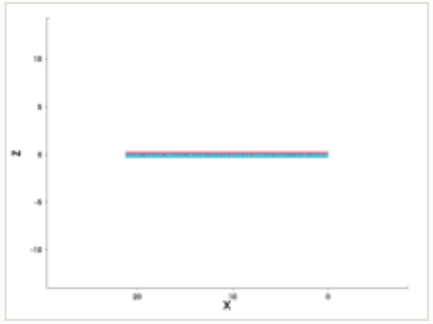
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



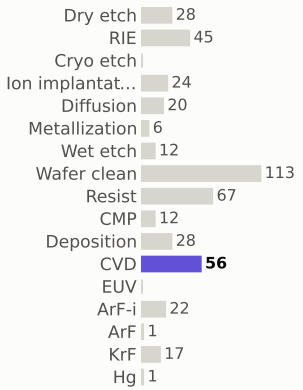
y cut at 57.351 µm



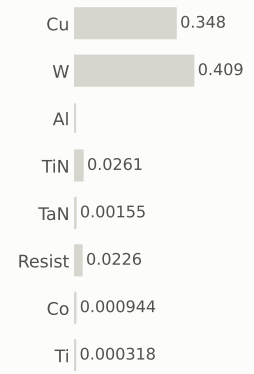
mask for this step

no mask for this step

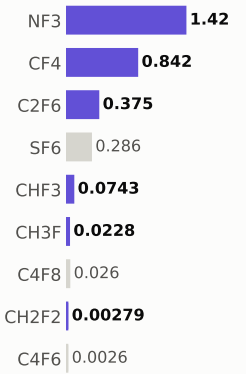
steps by process area



materials deposited (g)



gases emitted (g)



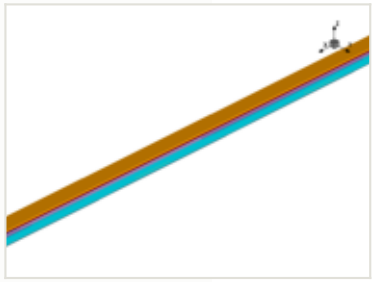
Step #6050: Resist

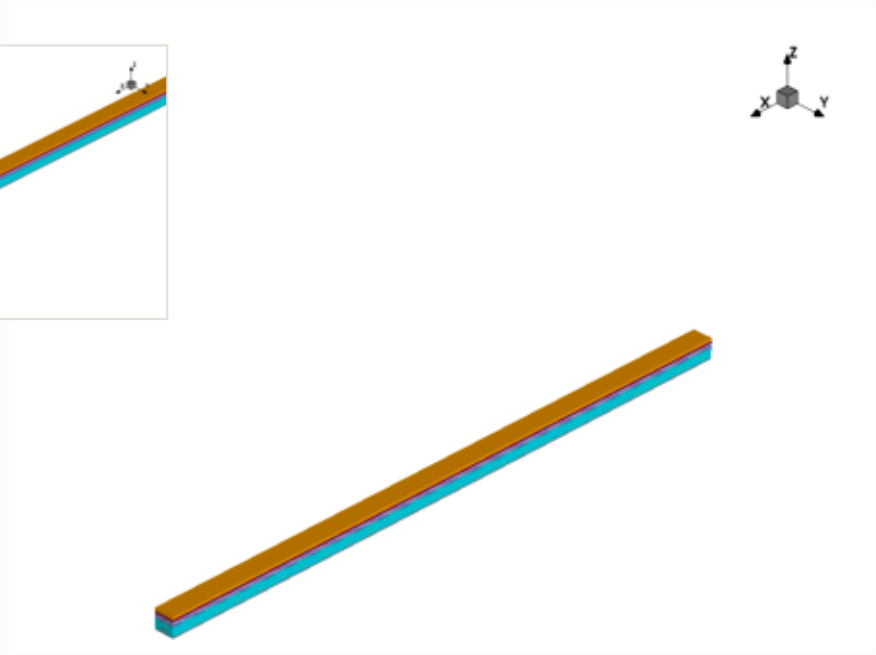
depo_fill · Resist

0.169	272.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



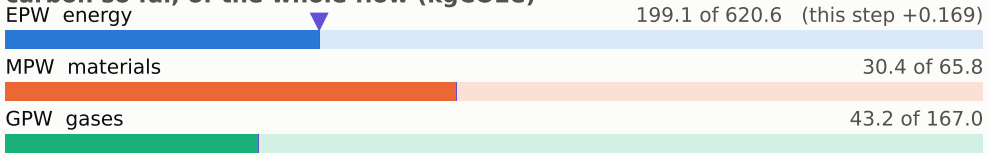


this step

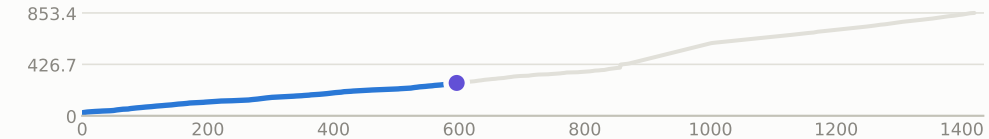
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

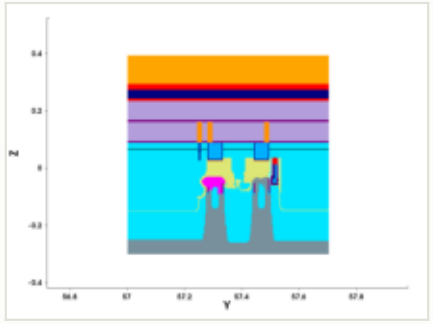
carbon so far, of the whole flow (kgCO2e)



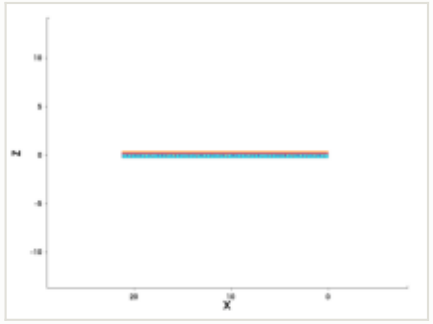
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



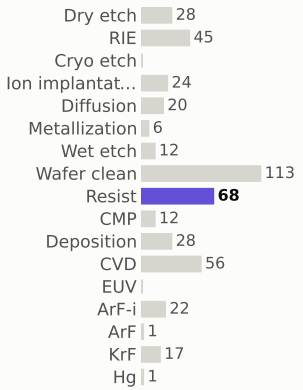
y cut at 57.351 µm



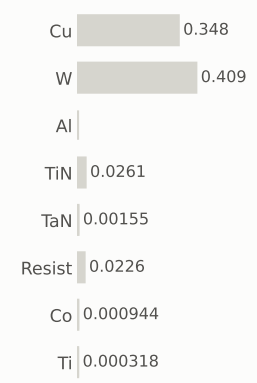
mask for this step

no mask for this step

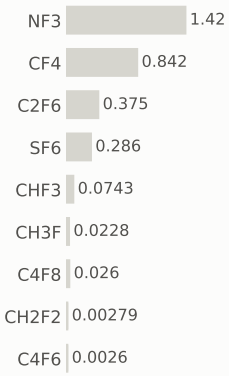
steps by process area



materials deposited (g)



gases emitted (g)



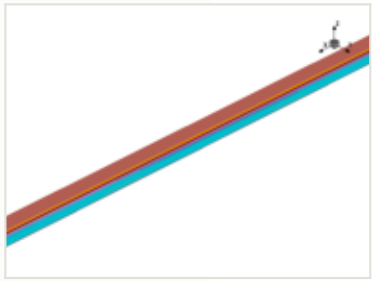
Step #6060: CVD

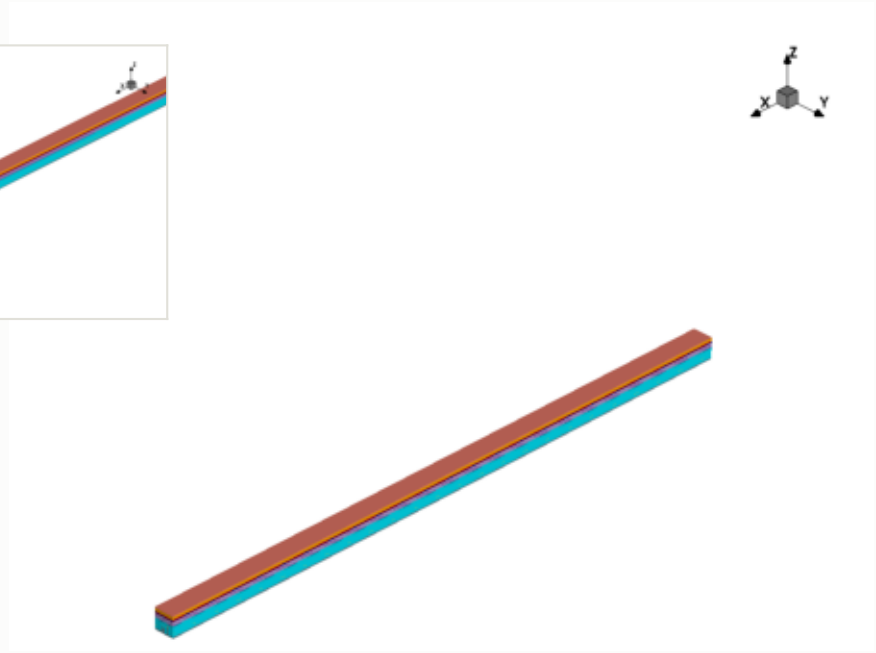
depo_fill · CVD

1.31	274.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



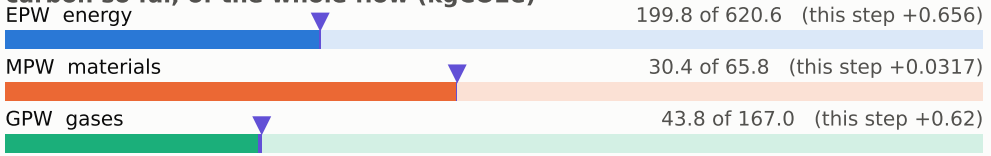


this step

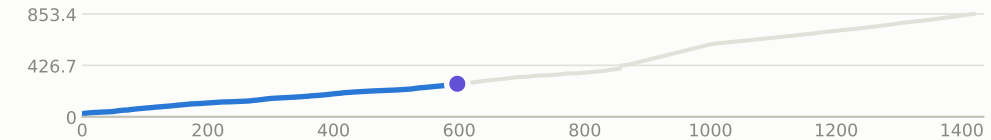
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

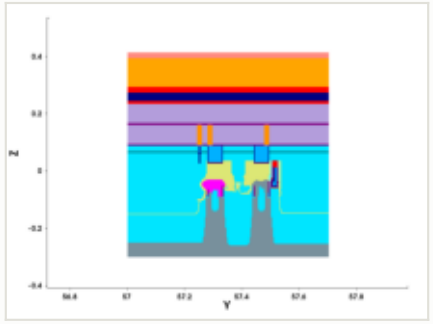
carbon so far, of the whole flow (kgCO2e)



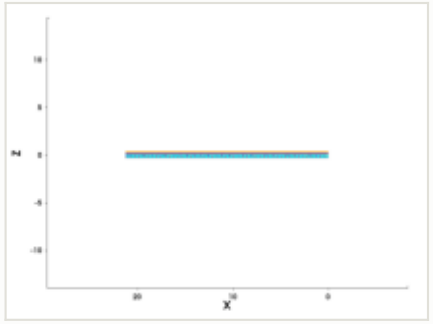
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



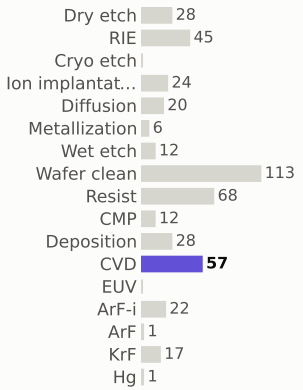
y cut at 57.351 µm



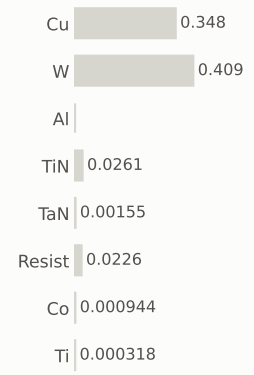
mask for this step

no mask for this step

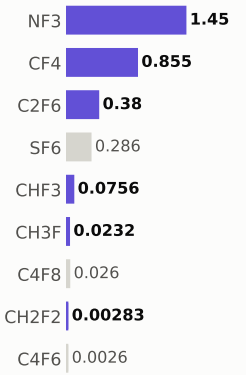
steps by process area



materials deposited (g)



gases emitted (g)



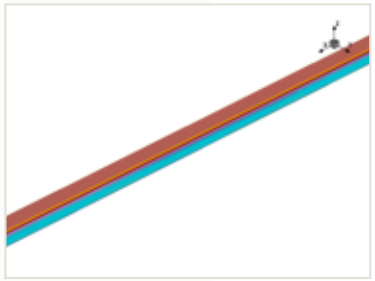
Step #6070: Wafer_clean

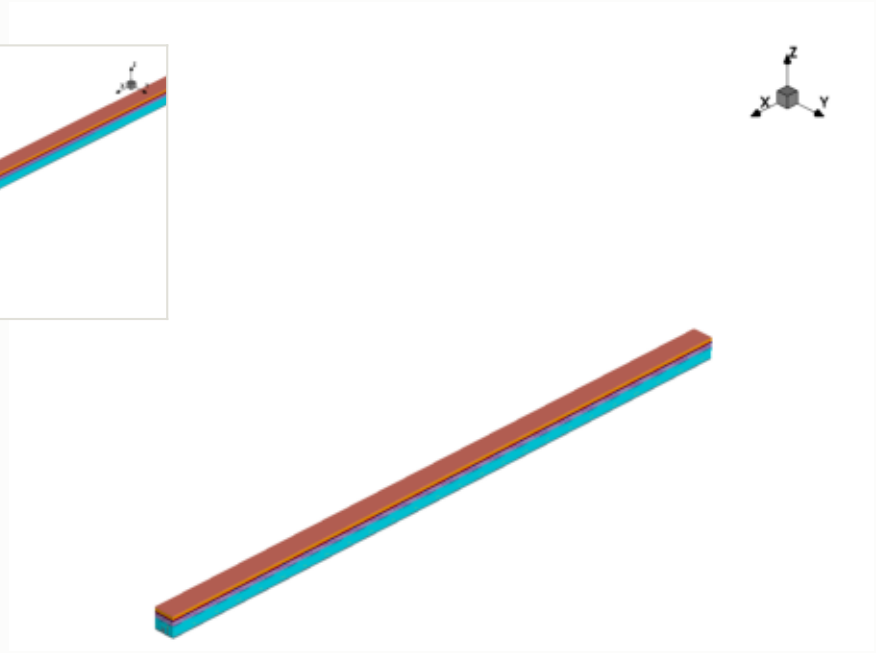
clean · Wafer_clean

0.169	274.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





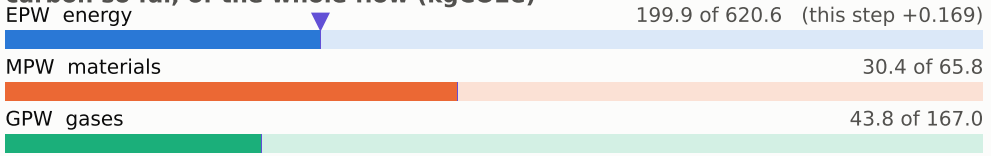
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

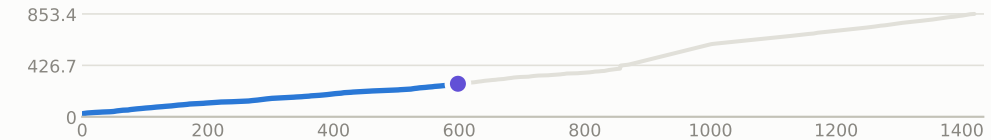
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

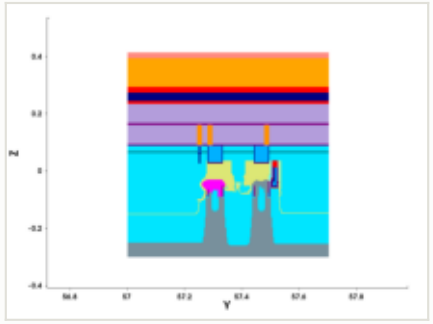
carbon so far, of the whole flow (kgCO2e)



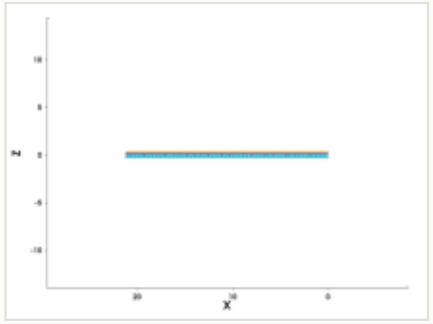
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



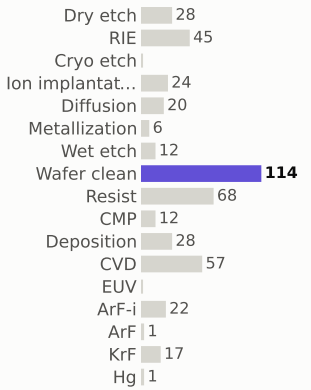
y cut at 57.351 µm



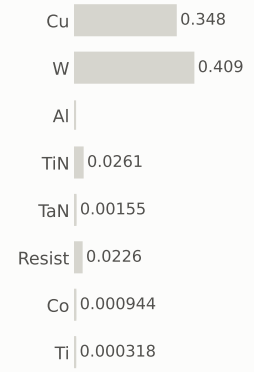
mask for this step

no mask for this step

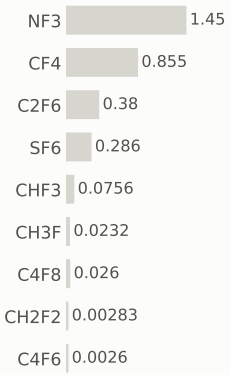
steps by process area



materials deposited (g)



gases emitted (g)



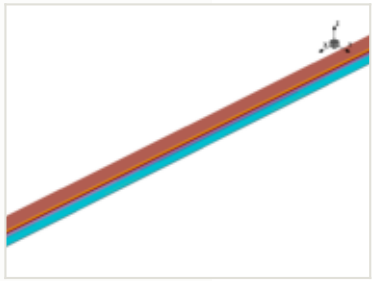
Step #6080: Resist

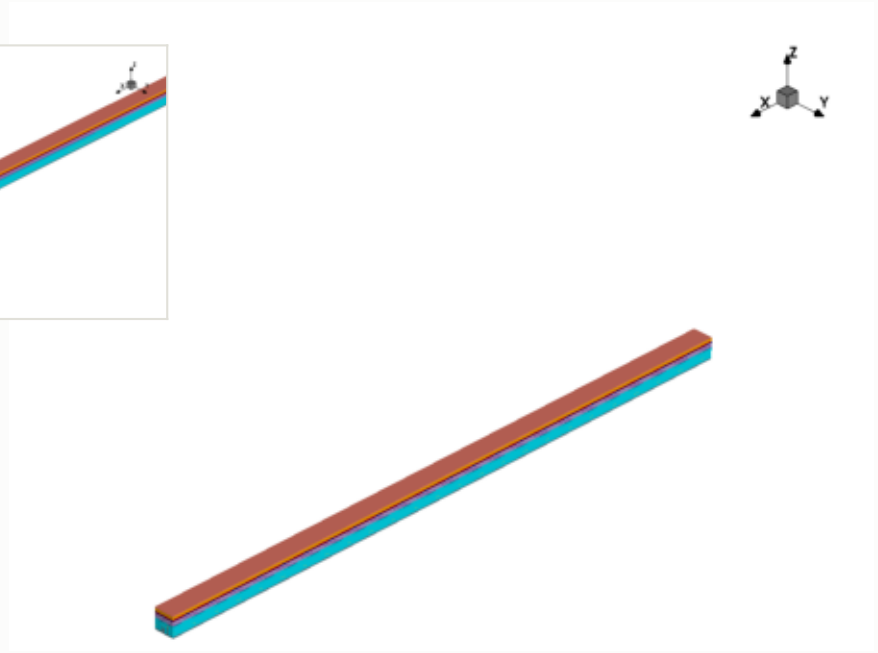
resist · Resist

0.169	274.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



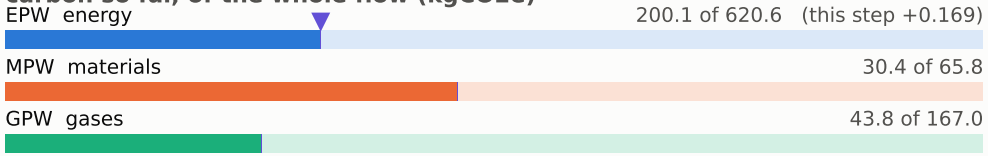


this step

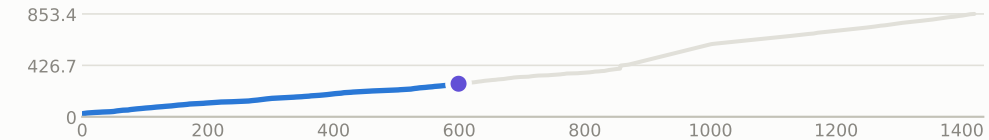
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

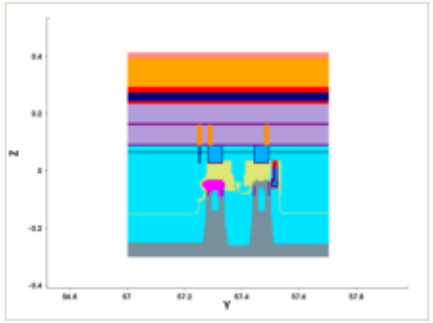
carbon so far, of the whole flow (kgCO2e)



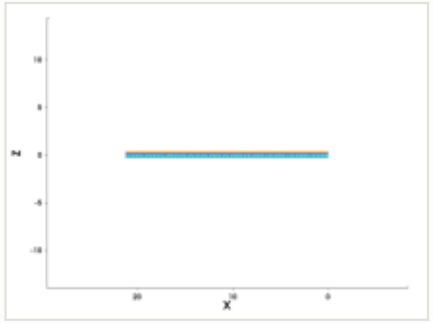
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



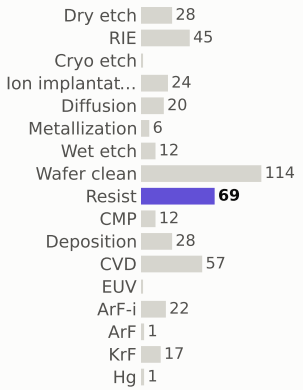
y cut at 57.351 µm



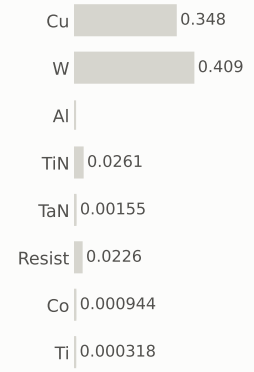
mask for this step

no mask for this step

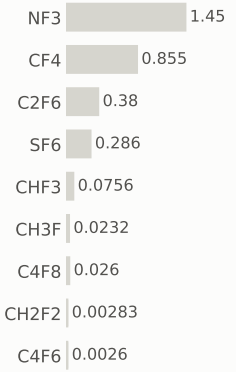
steps by process area



materials deposited (g)



gases emitted (g)



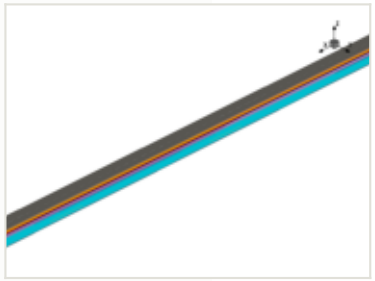
Step #6090: ArF-i

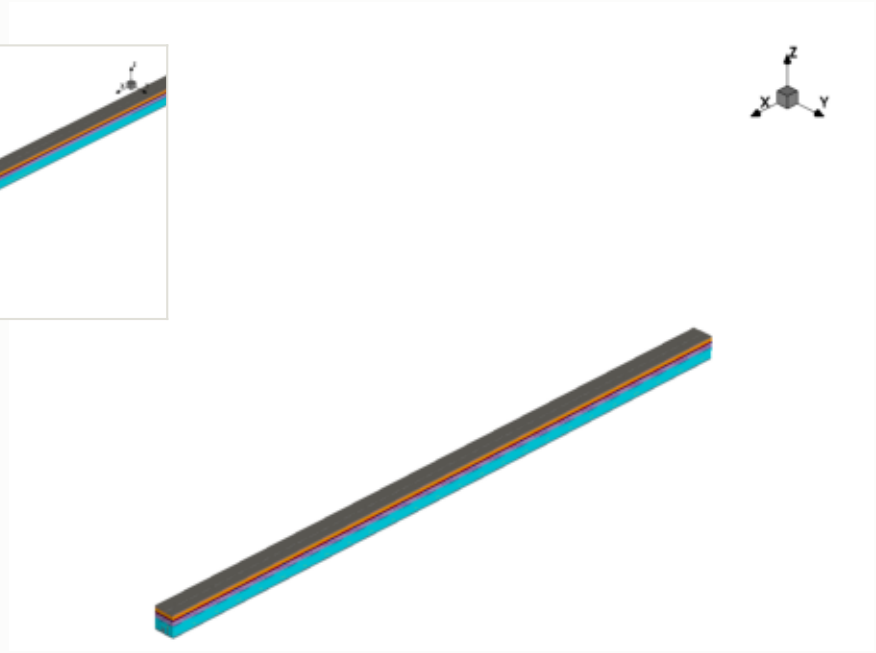
litho_patterning · ArF-i

0.217	274.5	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



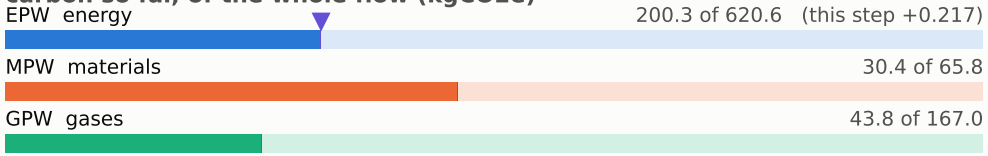


this step

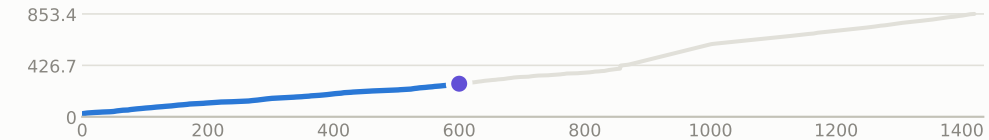
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

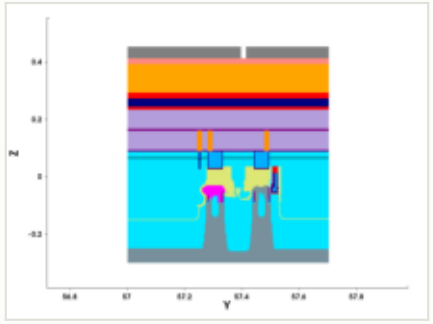
carbon so far, of the whole flow (kgCO2e)



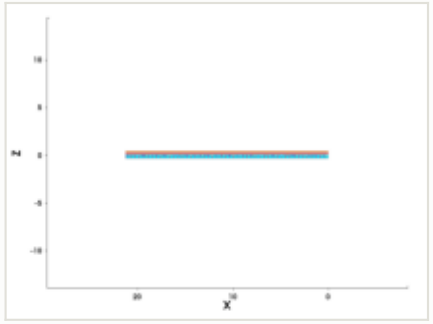
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

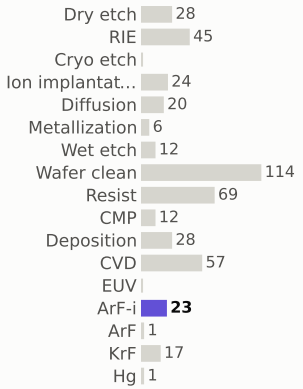


y cut at 57.351 µm

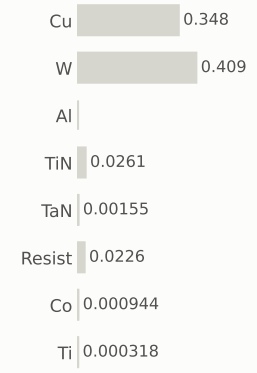


mask: layer 2020:0 (GDSII)

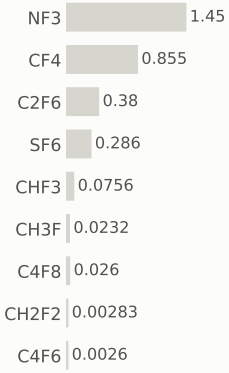
steps by process area



materials deposited (g)



gases emitted (g)



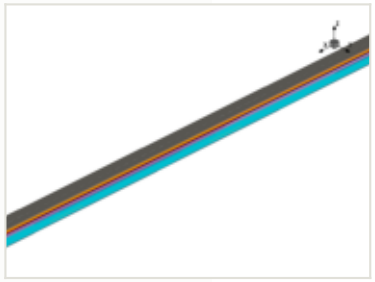
Step #6100: RIE

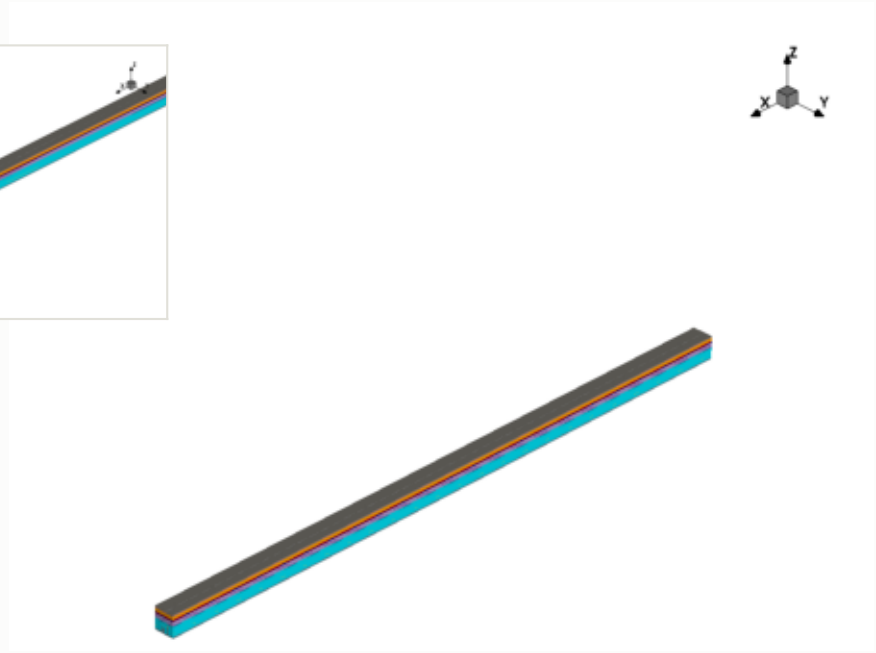
etch_anisotropic · RIE

1.04	275.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



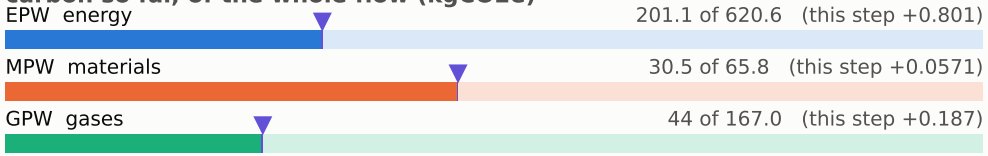


this step

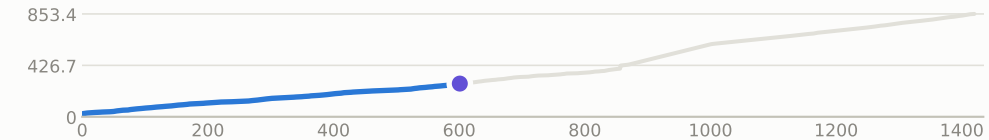
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

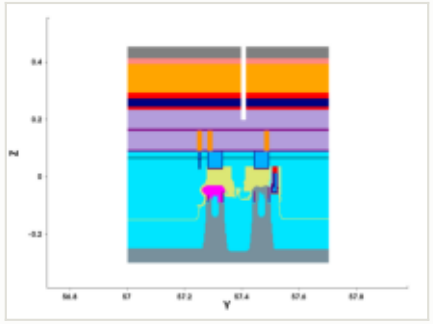
carbon so far, of the whole flow (kgCO2e)



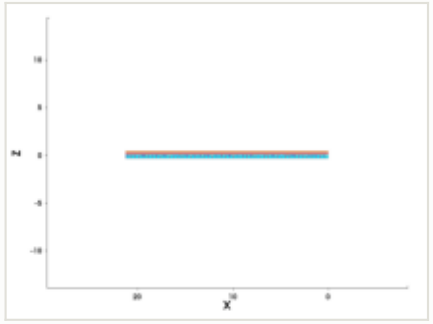
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



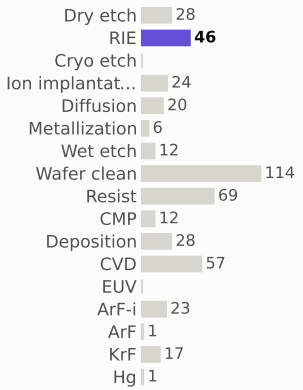
y cut at 57.351 µm



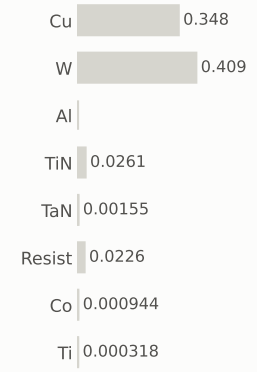
mask for this step

no mask for this step

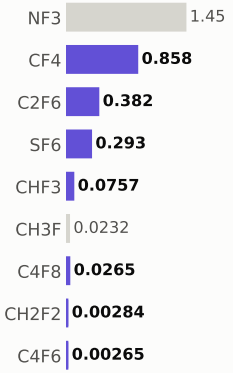
steps by process area



materials deposited (g)



gases emitted (g)



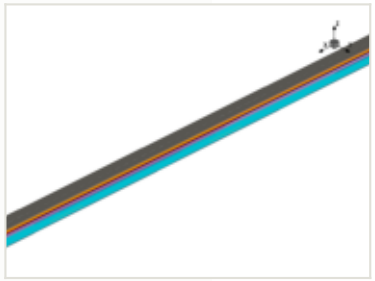
Step #6110: Wafer_clean

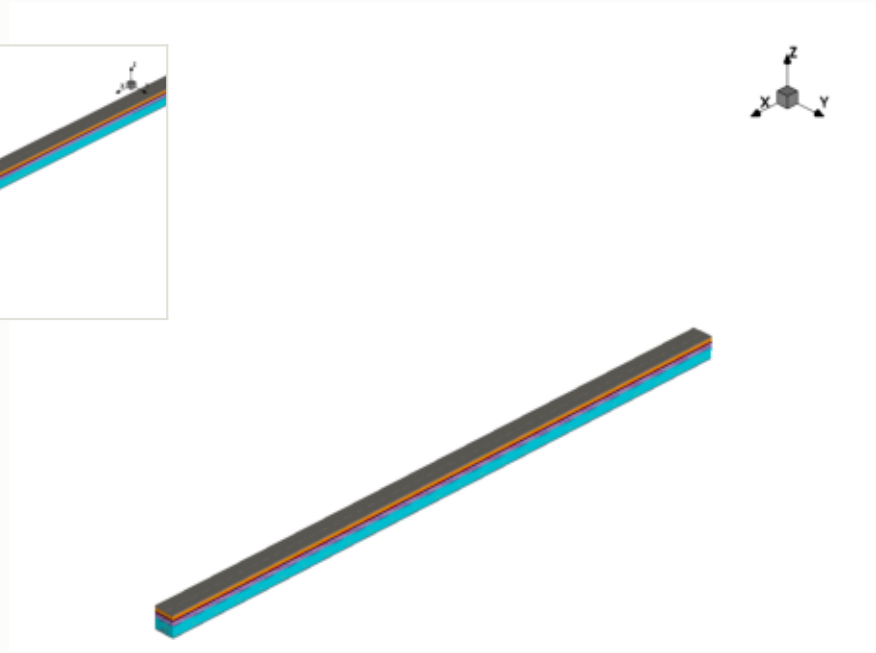
clean · Wafer_clean

0.169	275.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





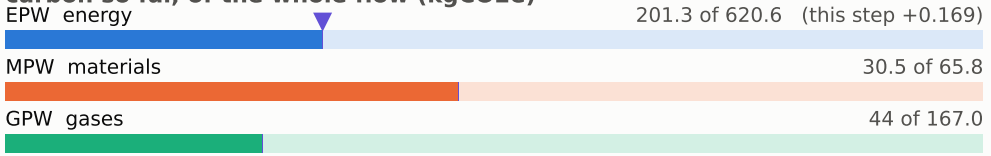
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

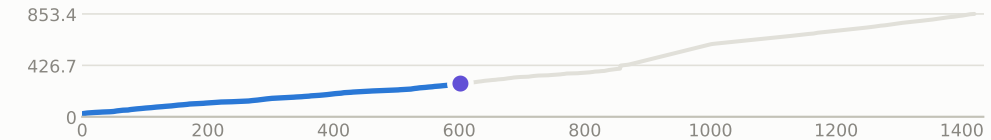
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

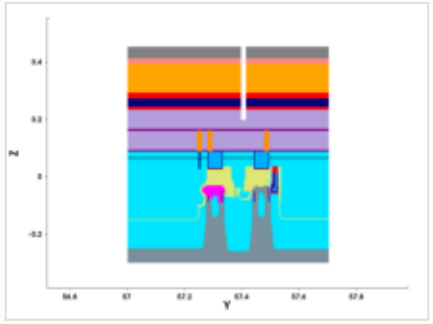
carbon so far, of the whole flow (kgCO2e)



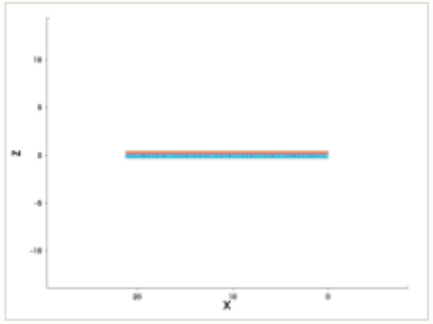
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



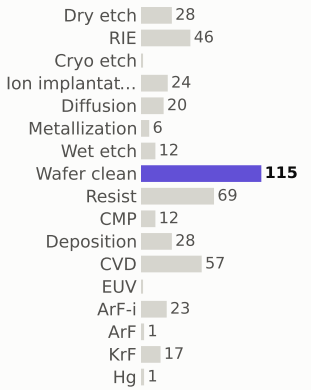
y cut at 57.351 µm



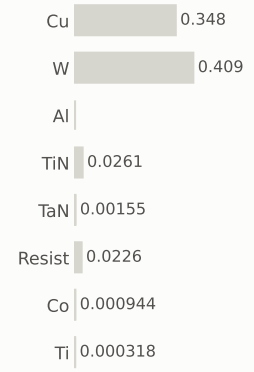
mask for this step

no mask for this step

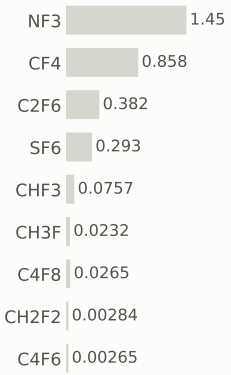
steps by process area



materials deposited (g)



gases emitted (g)



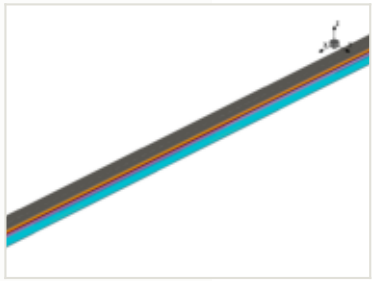
Step #6120: Resist

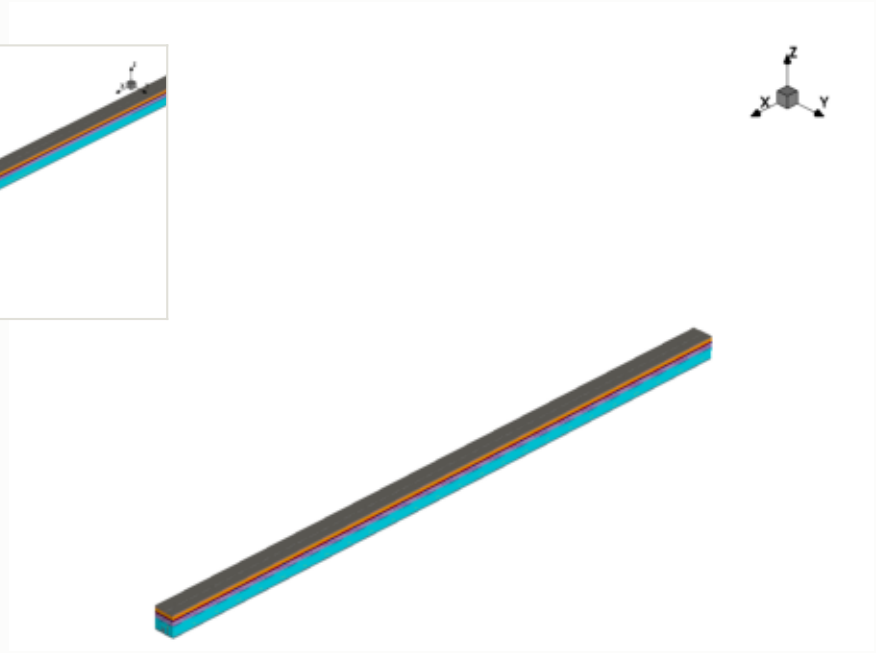
depo_fill · Resist

0.169	275.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



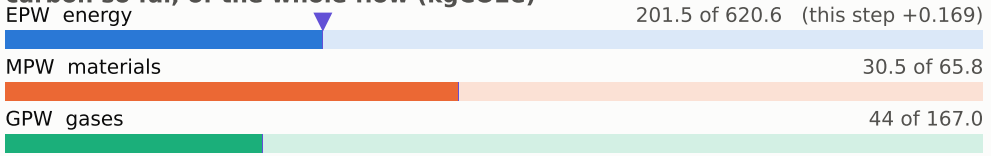


this step

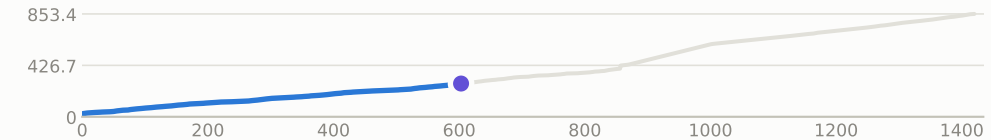
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

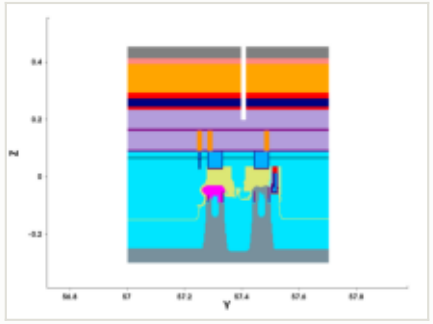
carbon so far, of the whole flow (kgCO2e)



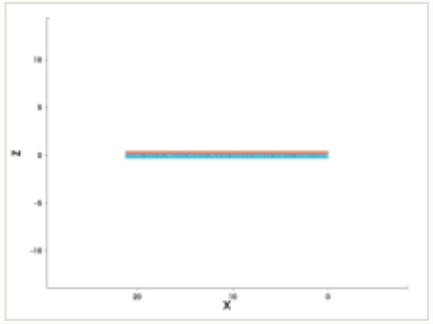
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



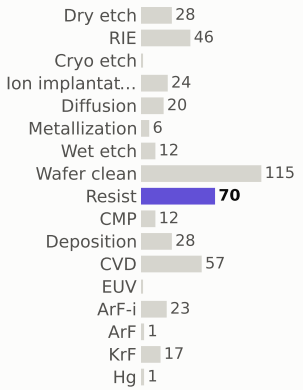
y cut at 57.351 µm



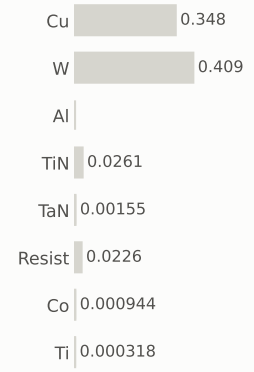
mask for this step

no mask for this step

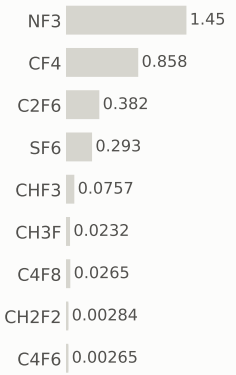
steps by process area



materials deposited (g)



gases emitted (g)



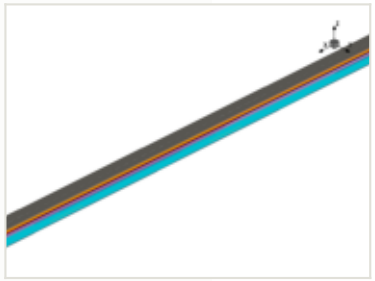
Step #6130: CVD

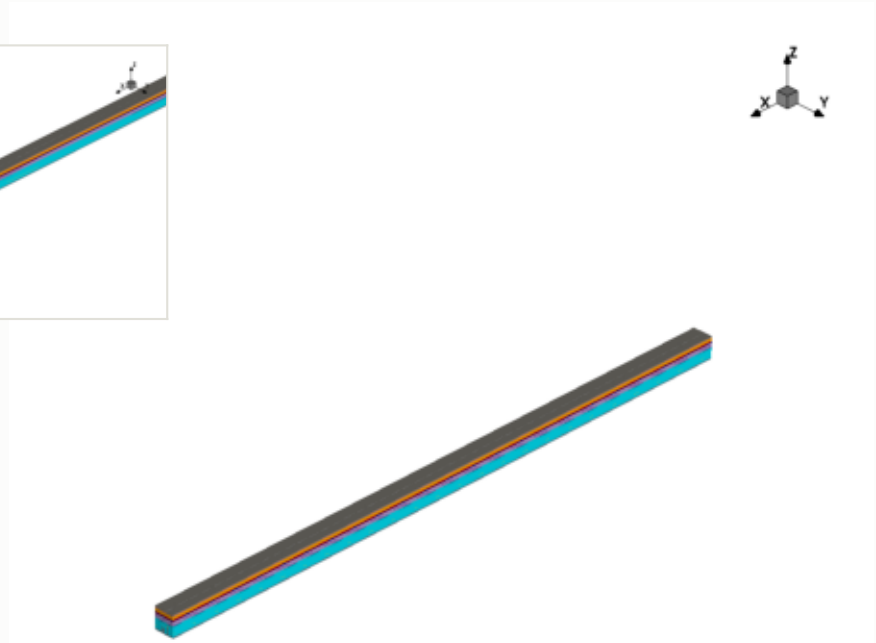
depo_isotropic · CVD

1.31	277.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



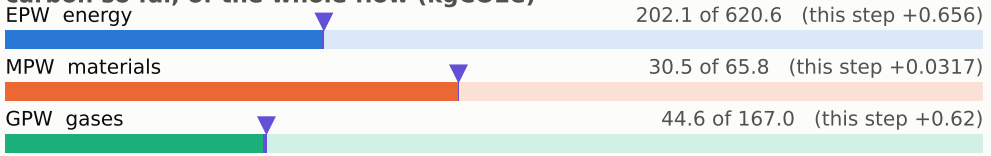


this step

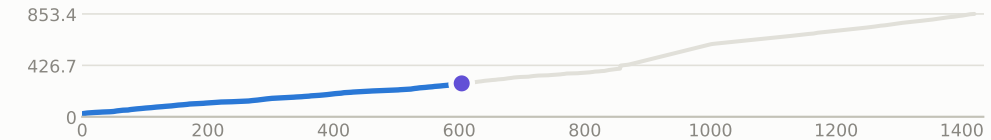
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

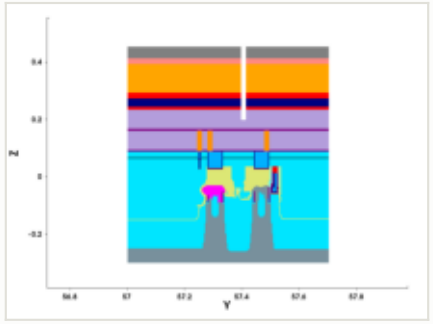
carbon so far, of the whole flow (kgCO2e)



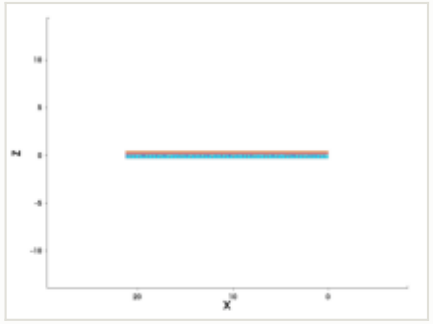
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



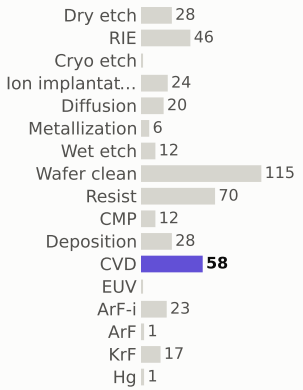
y cut at 57.351 µm



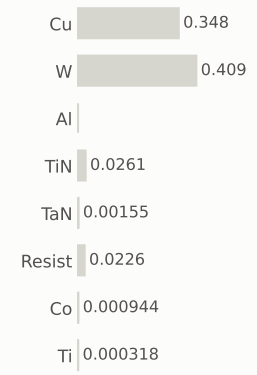
mask for this step

no mask for this step

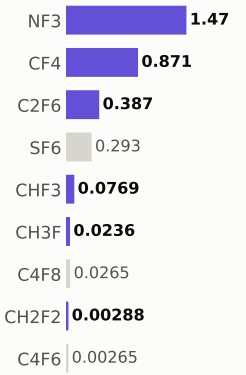
steps by process area



materials deposited (g)



gases emitted (g)



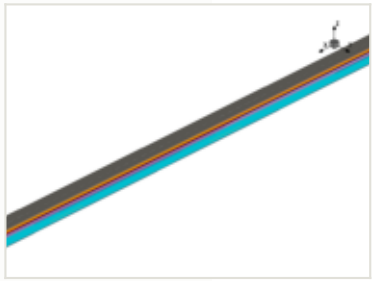
Step #6140: Wafer_clean

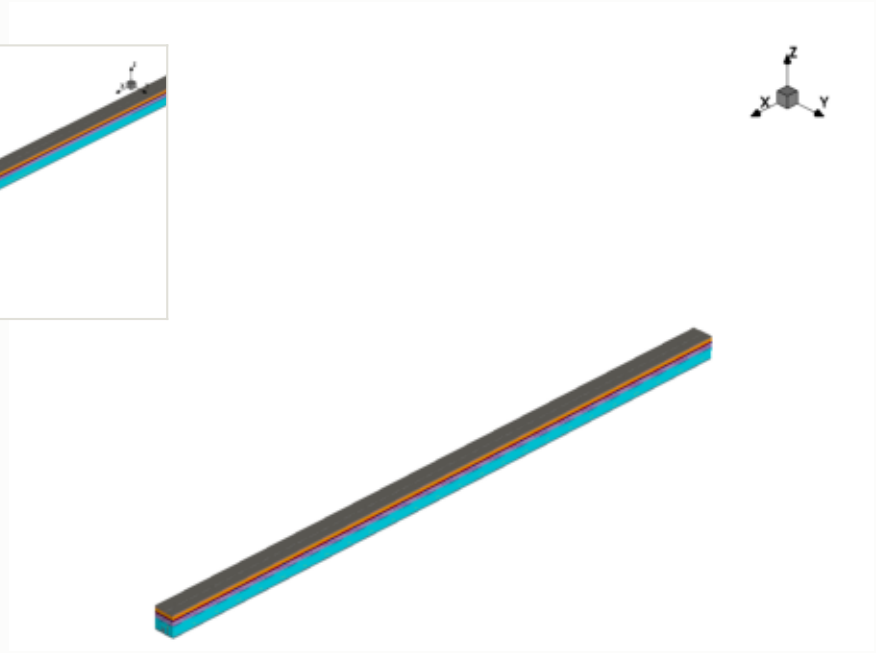
clean · Wafer_clean

0.169	277.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





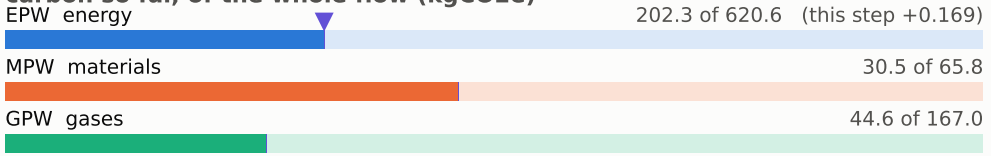
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

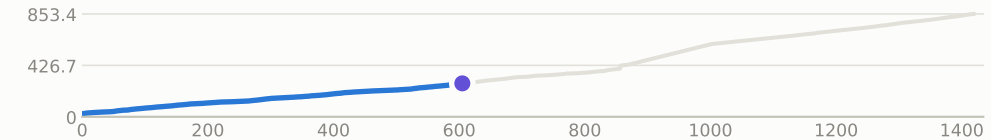
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

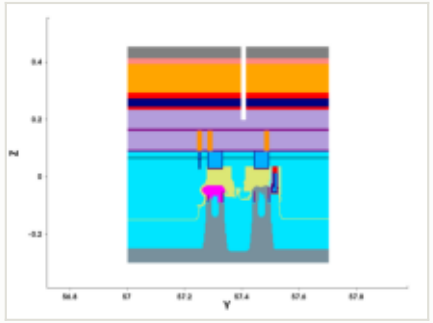
carbon so far, of the whole flow (kgCO2e)



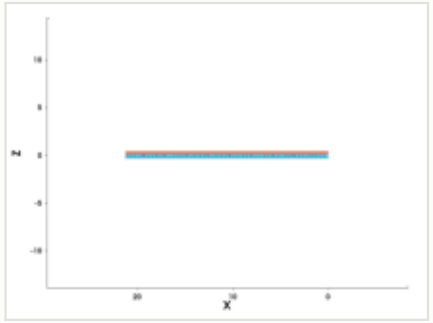
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



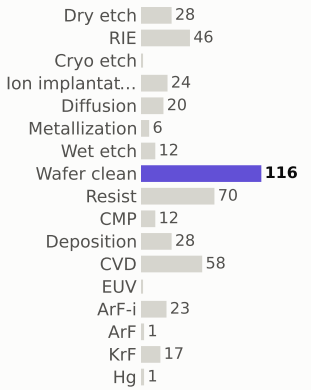
y cut at 57.351 µm



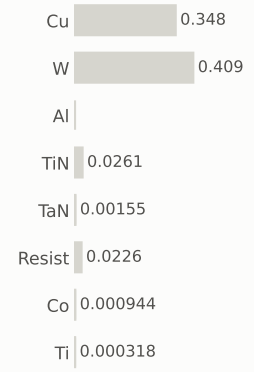
mask for this step

no mask for this step

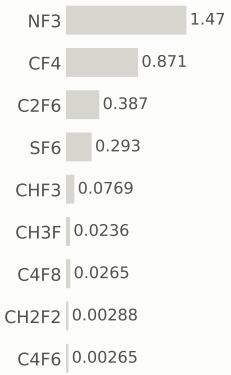
steps by process area



materials deposited (g)



gases emitted (g)



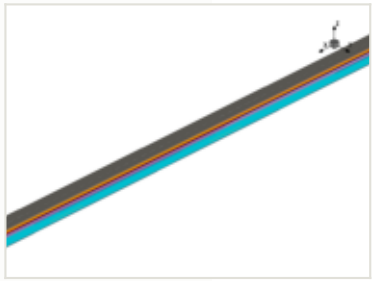
Step #6150: Resist

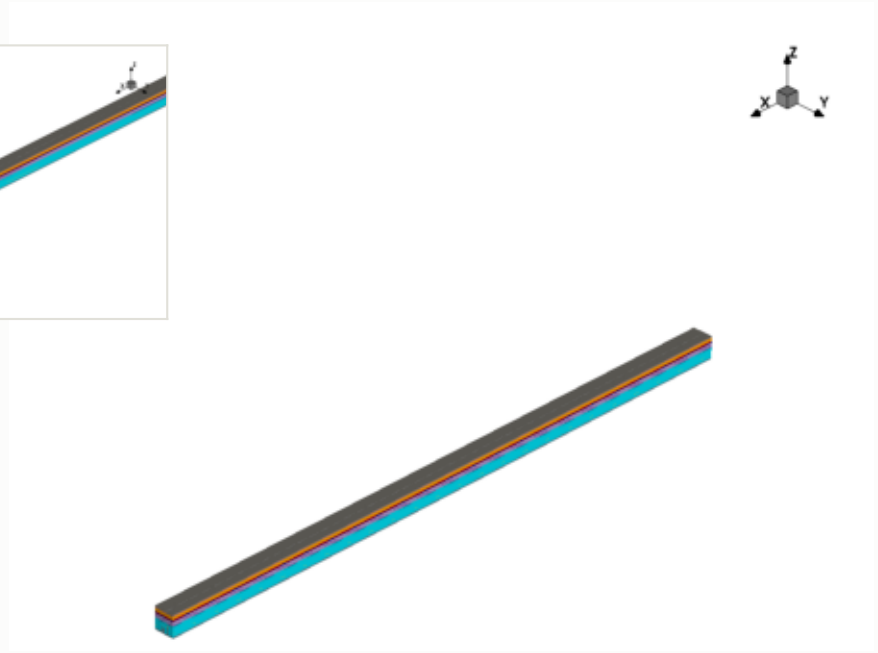
resist · Resist

0.169	277.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





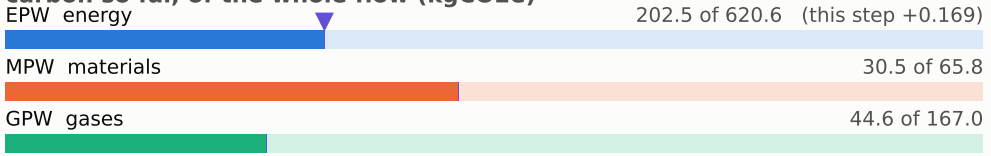
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

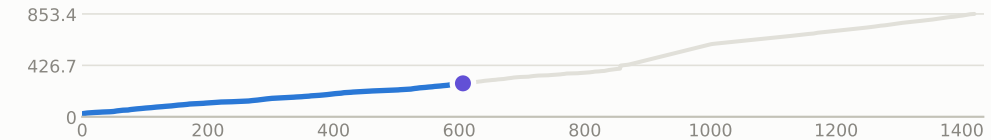
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

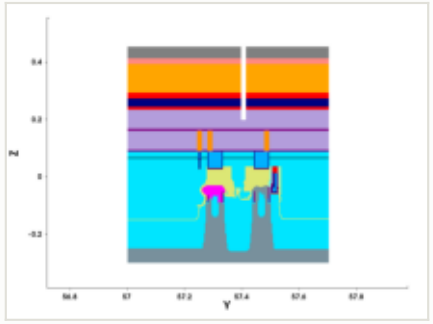
carbon so far, of the whole flow (kgCO2e)



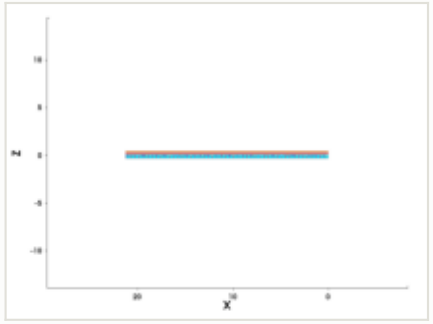
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



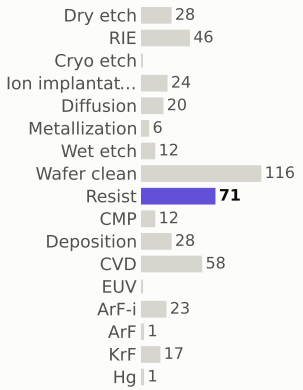
y cut at 57.351 µm



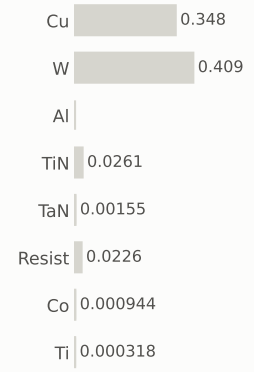
mask for this step

no mask for this step

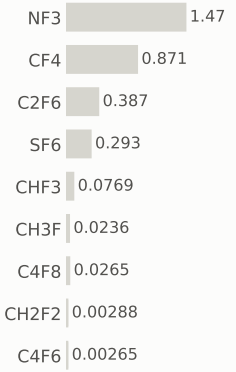
steps by process area



materials deposited (g)



gases emitted (g)



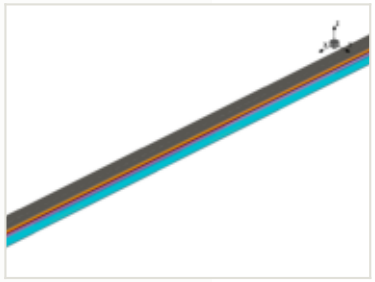
Step #6160: ArF-i

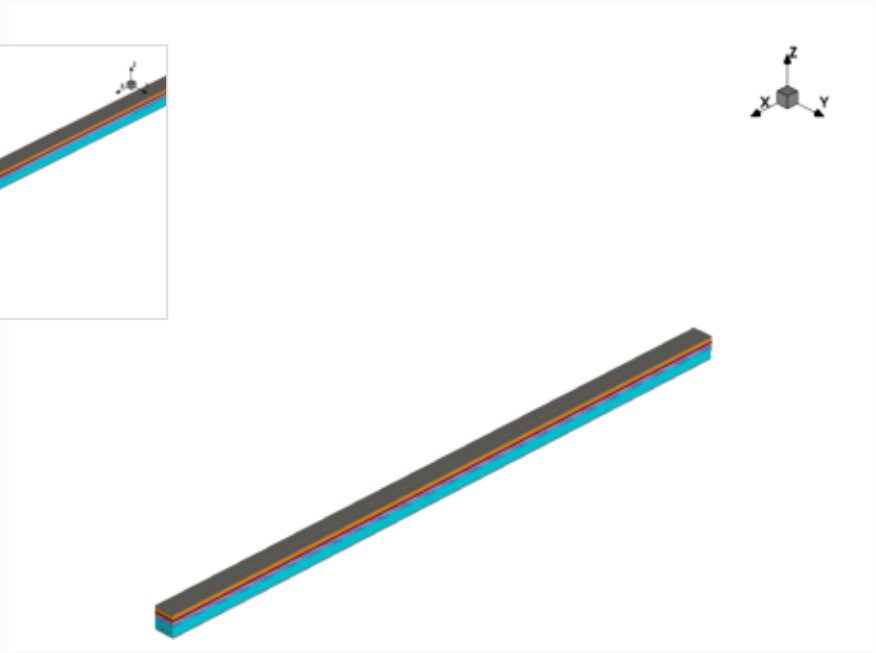
litho_patterning · ArF-i

0.217	277.8	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



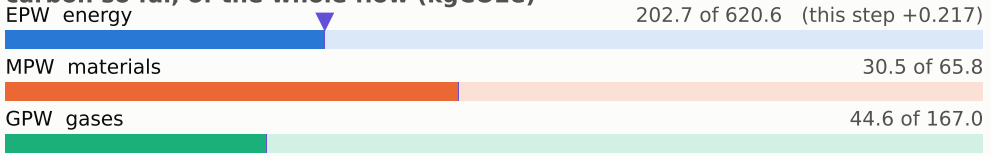


this step

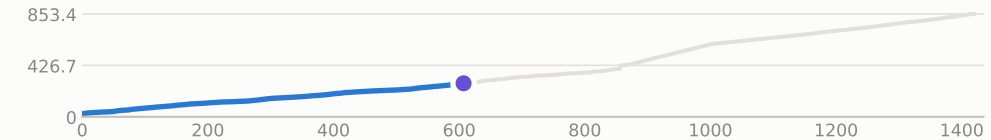
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

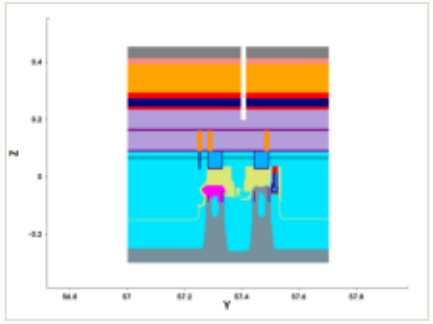
carbon so far, of the whole flow (kgCO2e)



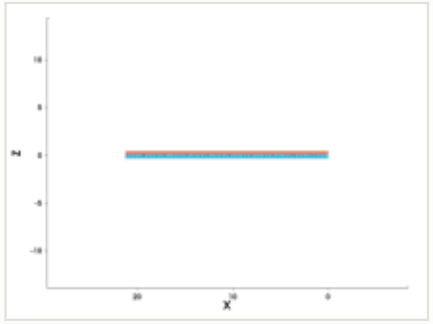
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

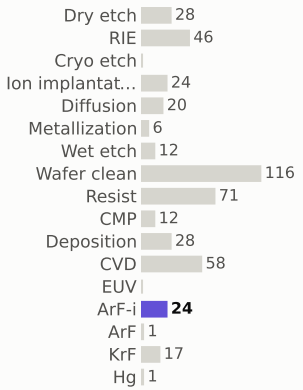


y cut at 57.351 µm

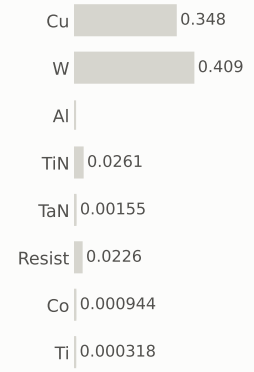


mask: layer 2100:0, positive (GDSII)

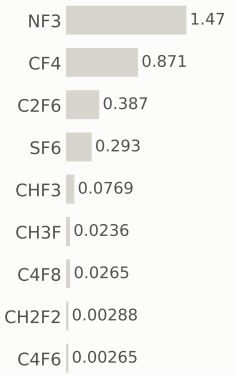
steps by process area



materials deposited (g)



gases emitted (g)



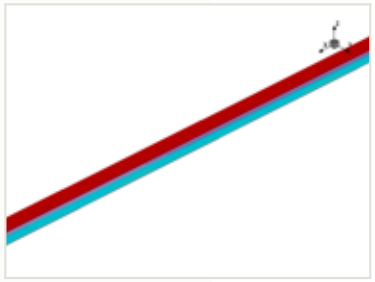
Step #6170: RIE

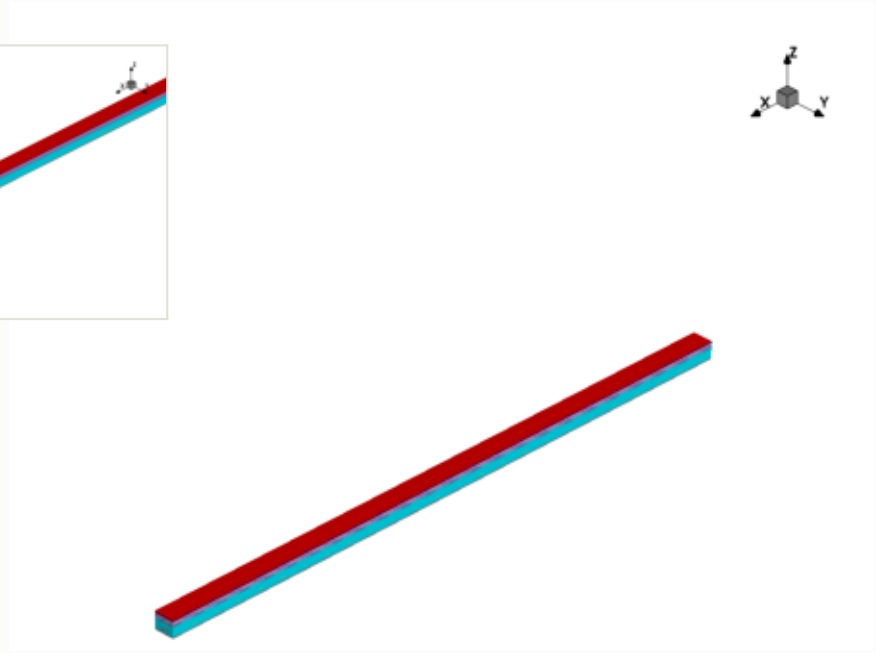
etch_strip · RIE

1.04	278.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



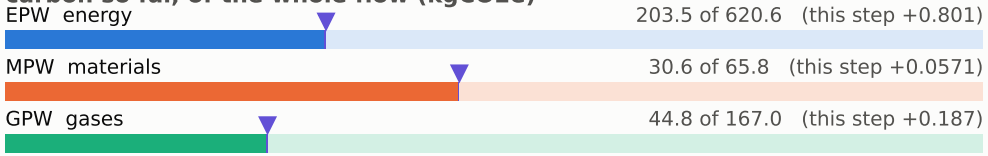


this step

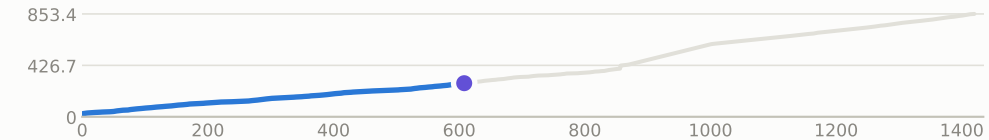
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

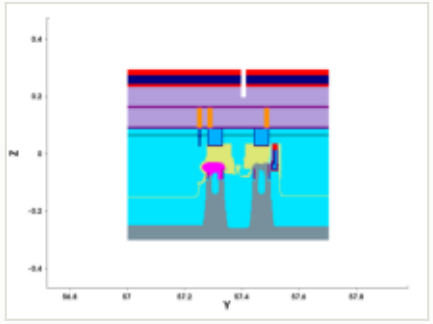
carbon so far, of the whole flow (kgCO2e)



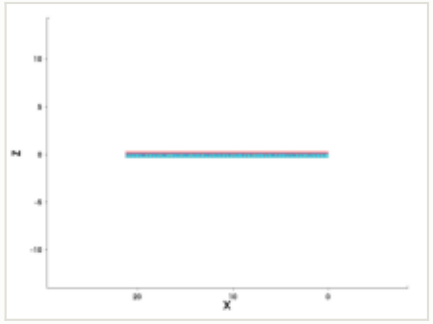
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



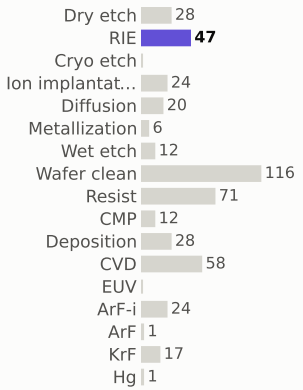
y cut at 57.351 µm



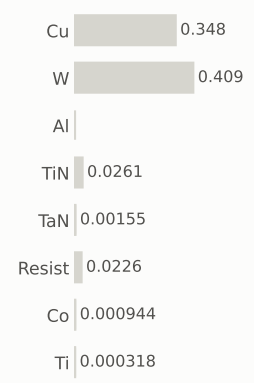
mask for this step

no mask for this step

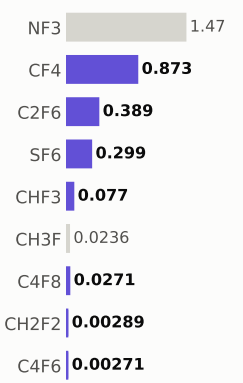
steps by process area



materials deposited (g)



gases emitted (g)



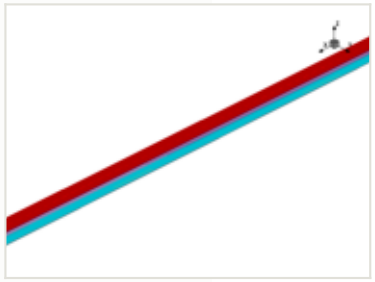
Step #6180: Wafer_clean

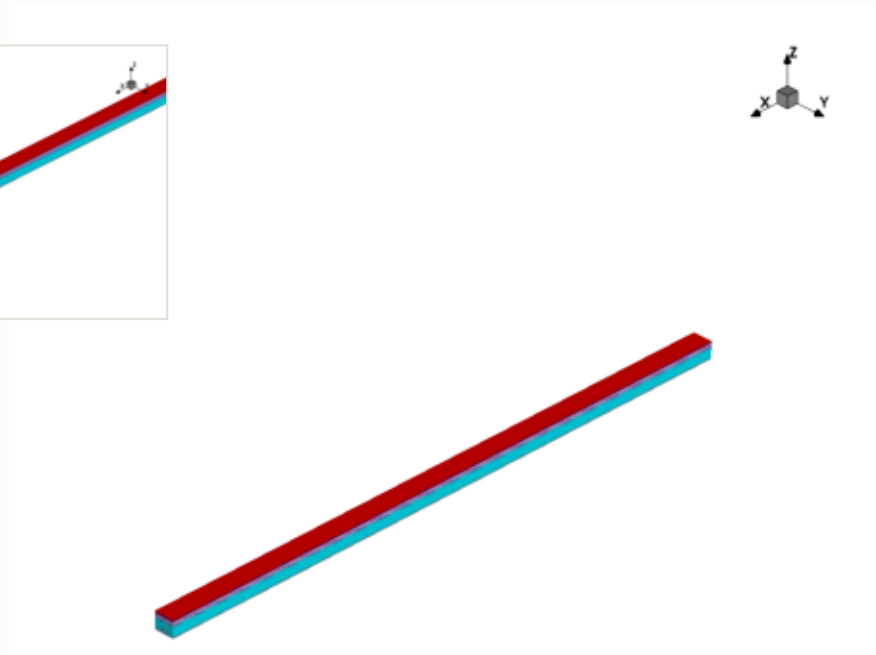
clean · Wafer_clean

0.169	279.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





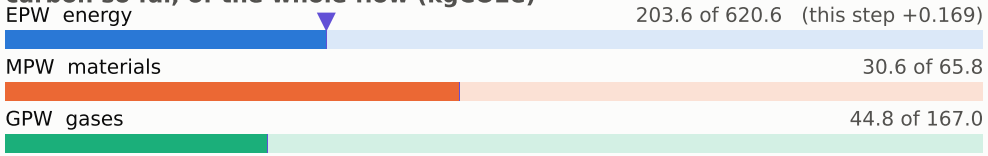
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

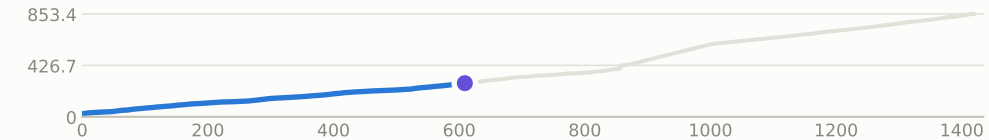
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

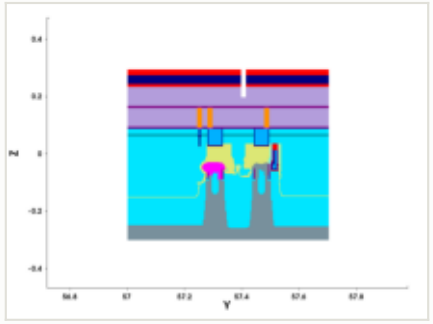
carbon so far, of the whole flow (kgCO2e)



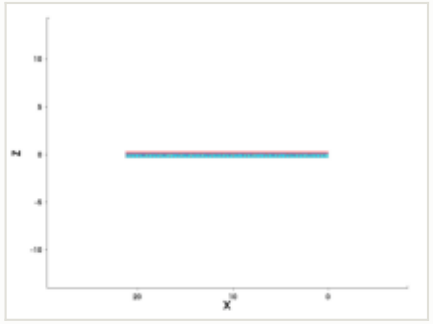
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



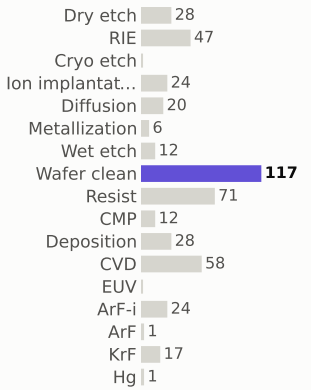
y cut at 57.351 µm



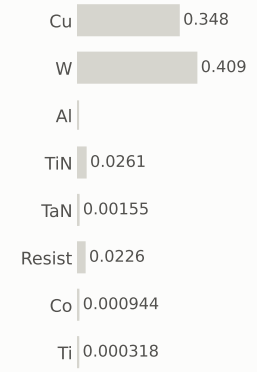
mask for this step

no mask for this step

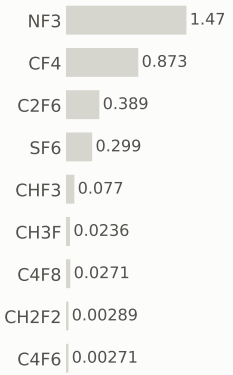
steps by process area



materials deposited (g)



gases emitted (g)



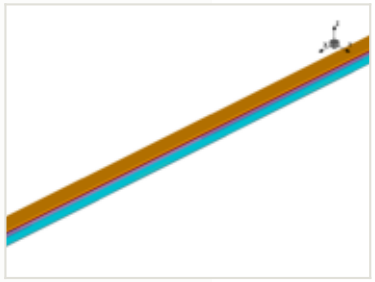
Step #6190: Resist

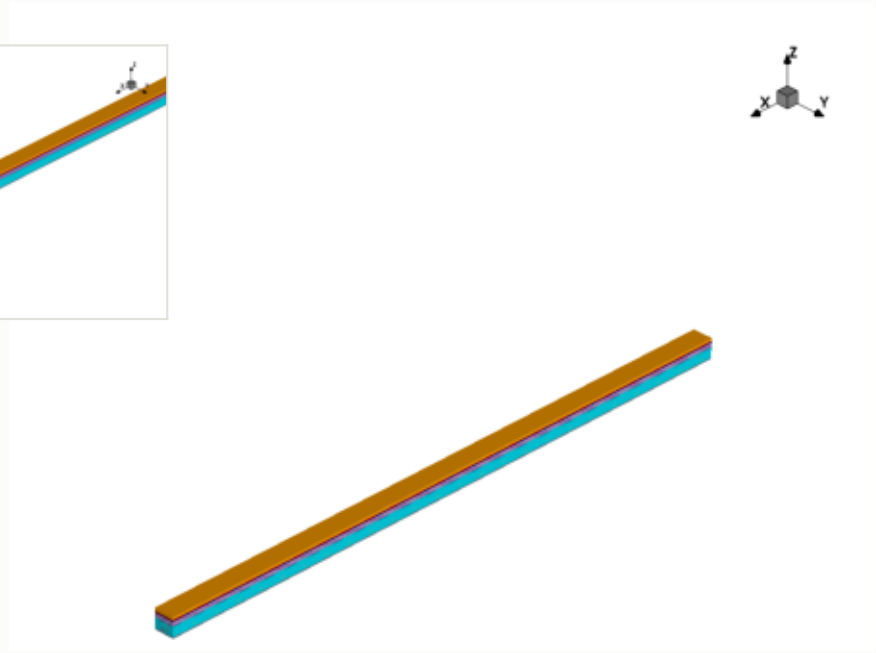
depo_fill · Resist

0.169	279.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



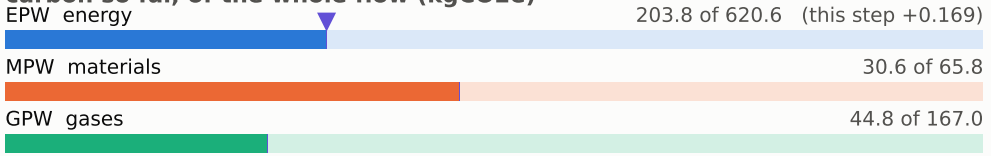


this step

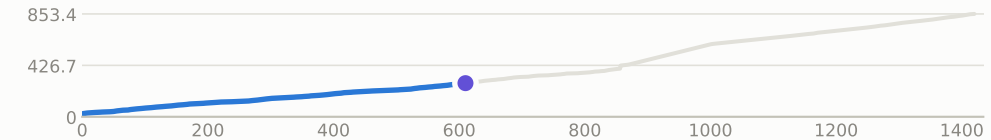
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

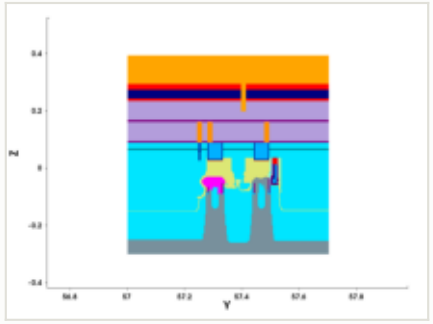
carbon so far, of the whole flow (kgCO2e)



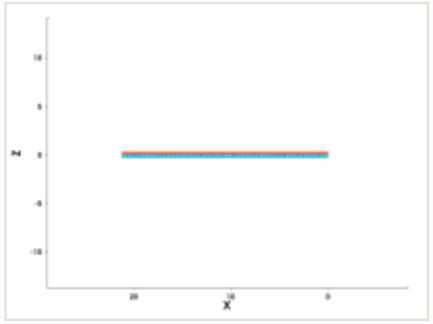
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



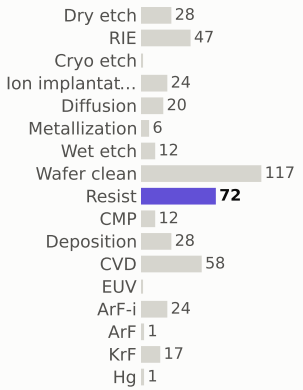
y cut at 57.351 µm



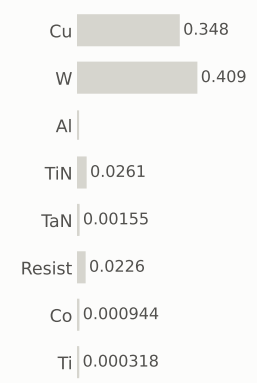
mask for this step

no mask for this step

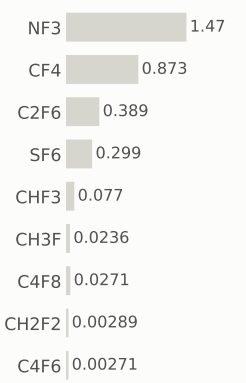
steps by process area



materials deposited (g)



gases emitted (g)



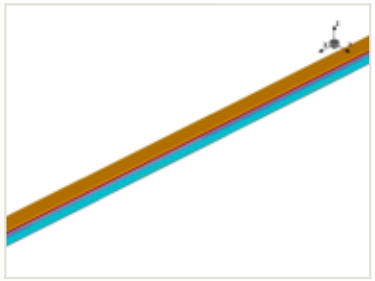
Step #6200: Deposition

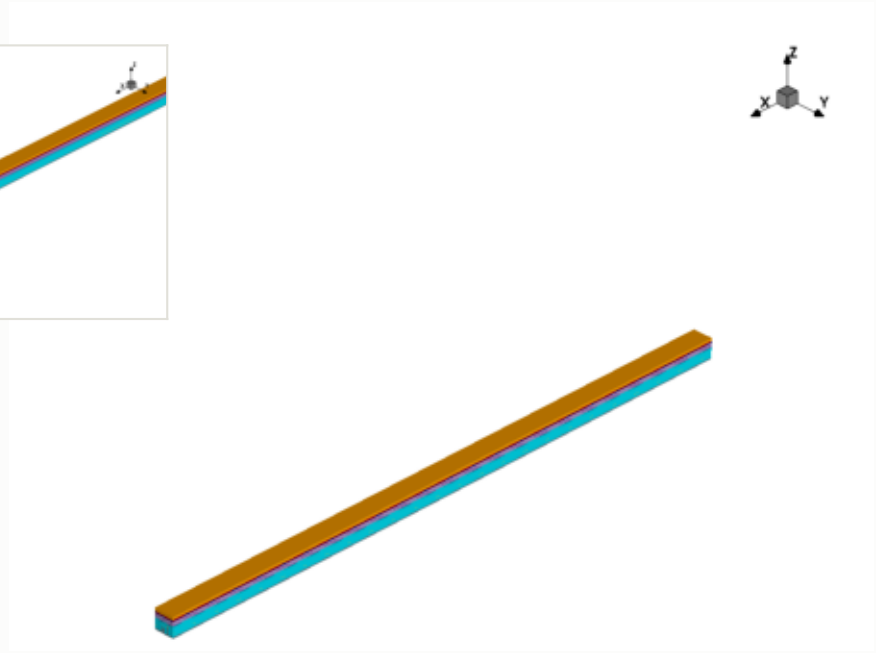
depo_isotropic · Deposition

0.656	279.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



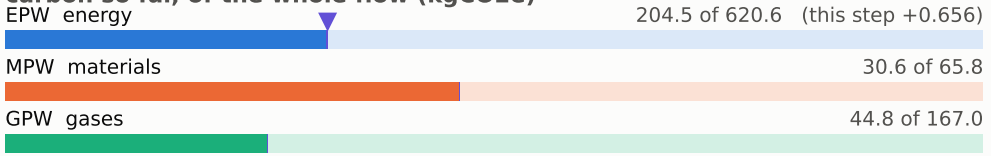


this step

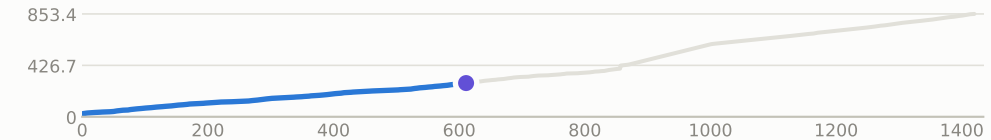
process area	Deposition
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

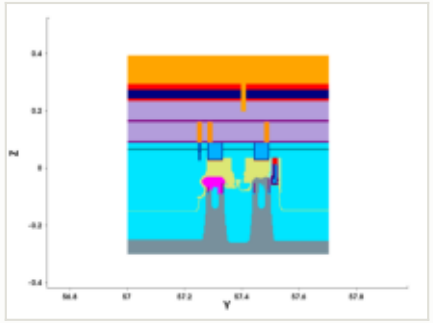
carbon so far, of the whole flow (kgCO2e)



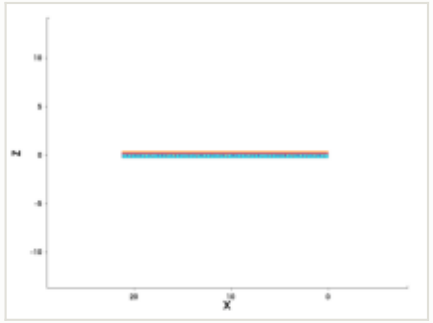
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



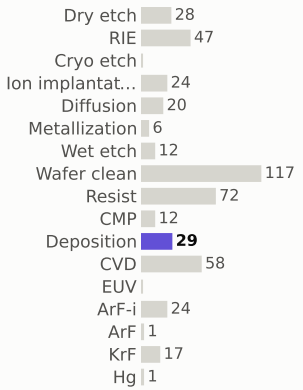
y cut at 57.351 µm



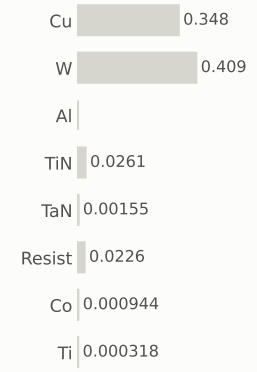
mask for this step

no mask for this step

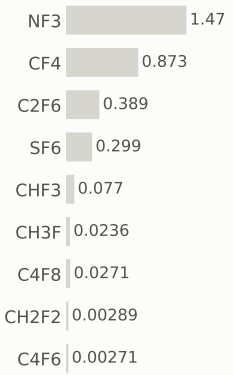
steps by process area



materials deposited (g)



gases emitted (g)



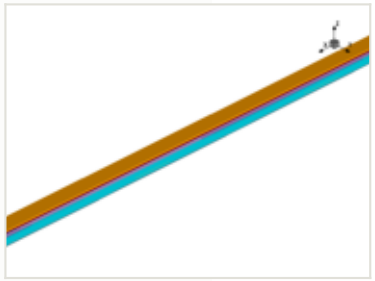
Step #6210: CVD

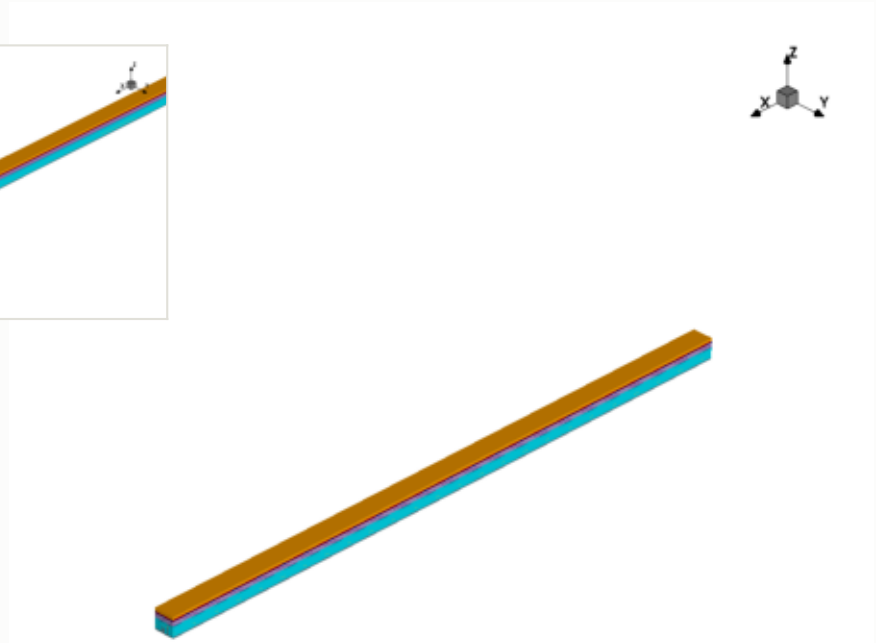
depo_isotropic · CVD

1.31	281.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



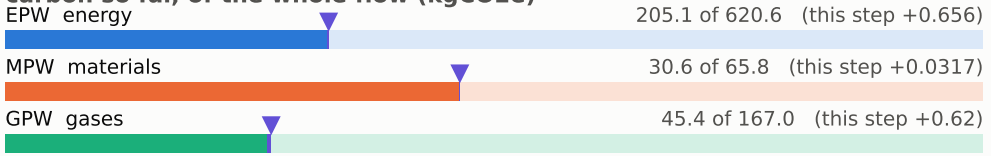


this step

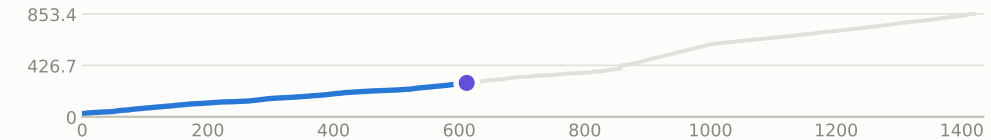
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

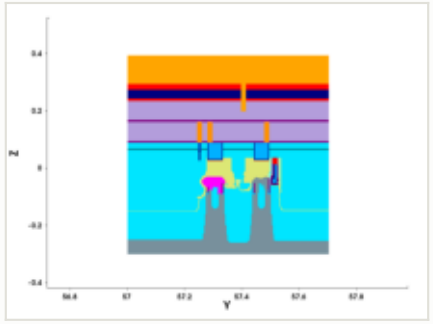
carbon so far, of the whole flow (kgCO2e)



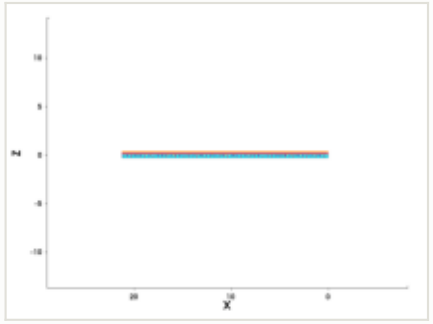
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



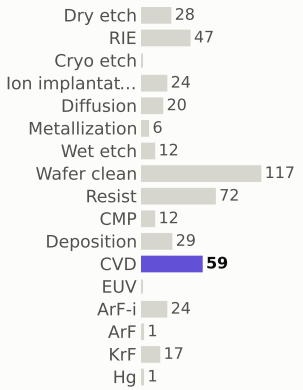
y cut at 57.351 µm



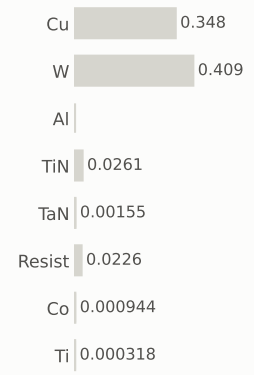
mask for this step

no mask for this step

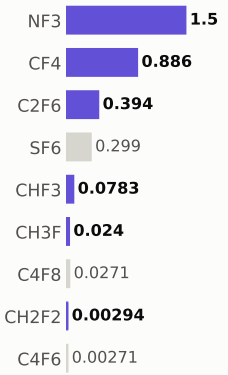
steps by process area



materials deposited (g)



gases emitted (g)



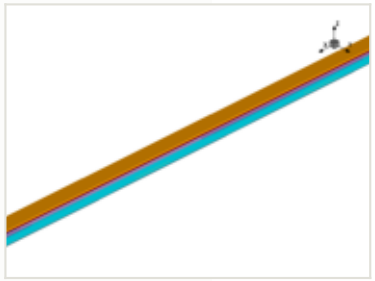
Step #6220: Resist

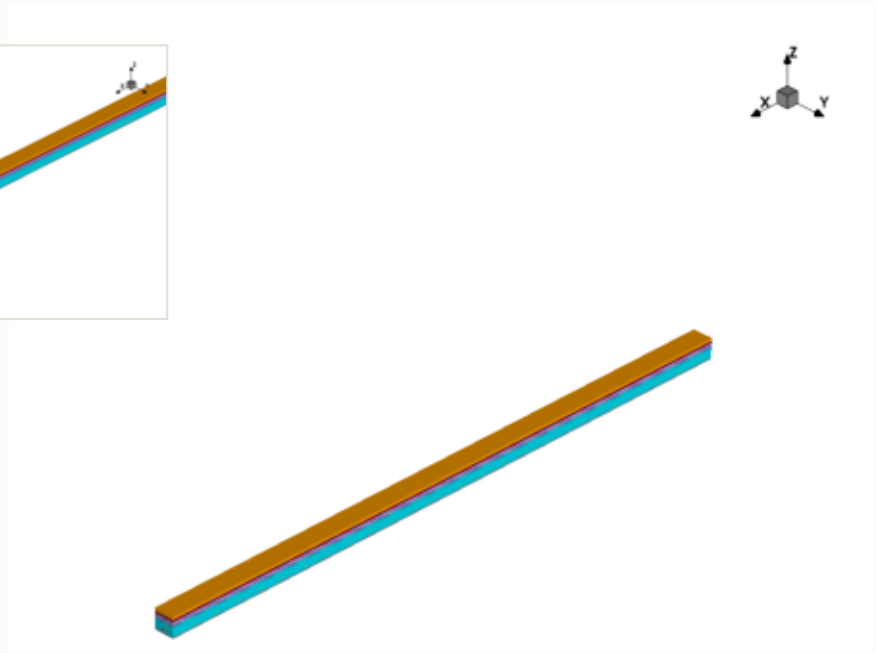
depo_fill · Resist

0.169	281.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



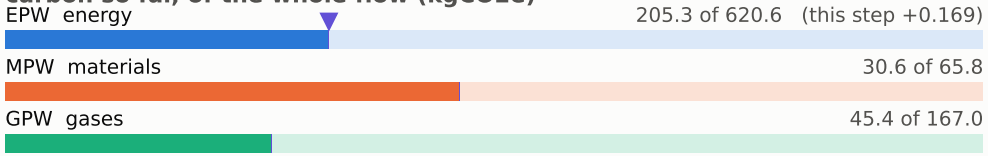


this step

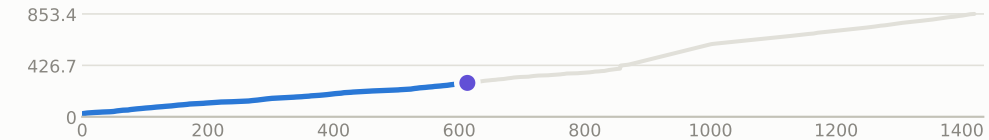
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

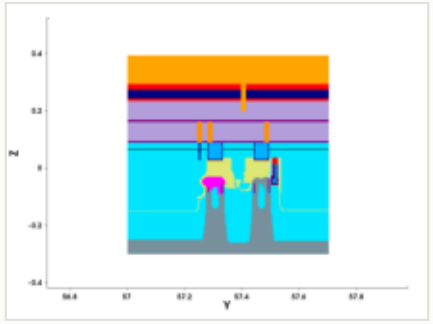
carbon so far, of the whole flow (kgCO2e)



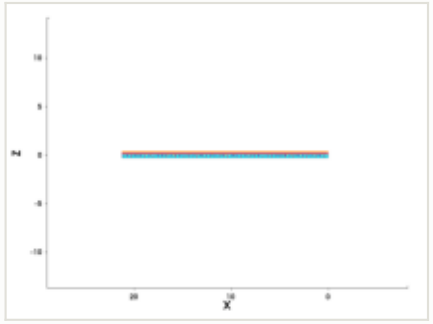
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



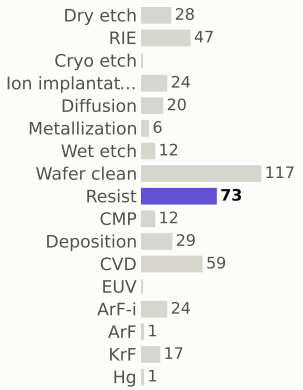
y cut at 57.351 µm



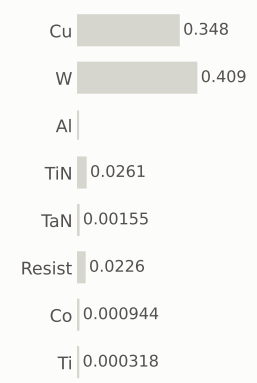
mask for this step

no mask for this step

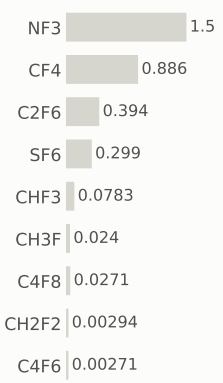
steps by process area



materials deposited (g)



gases emitted (g)



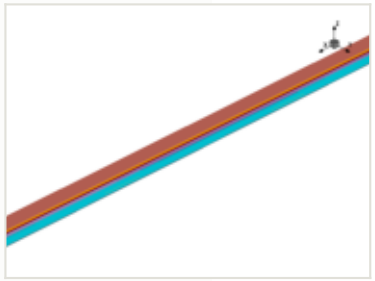
Step #6230: CVD

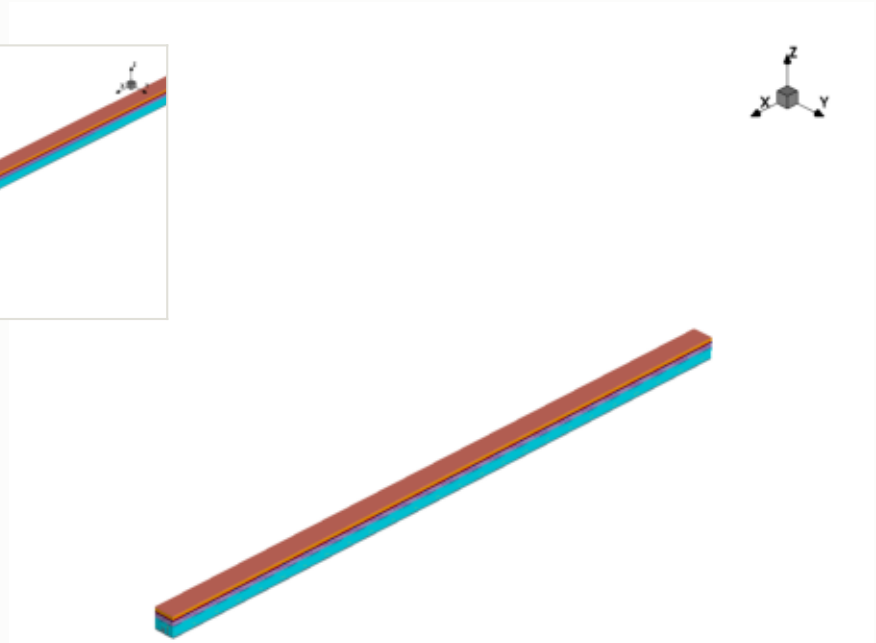
depo_fill · CVD

1.31	282.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



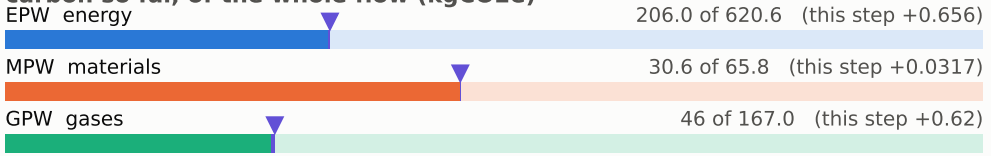


this step

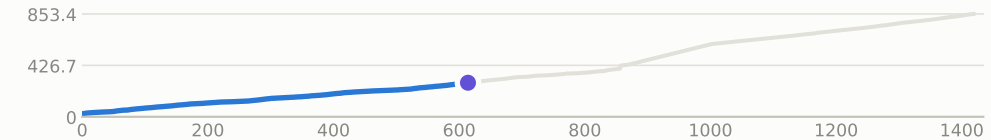
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

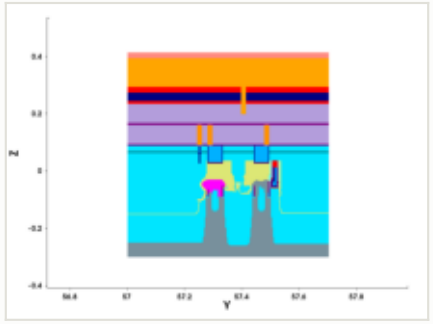
carbon so far, of the whole flow (kgCO2e)



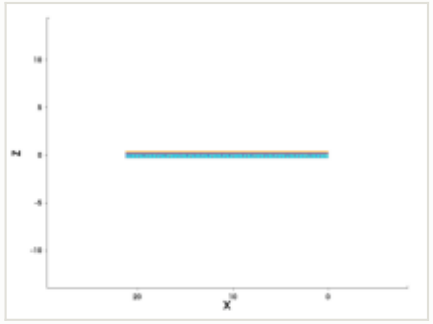
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



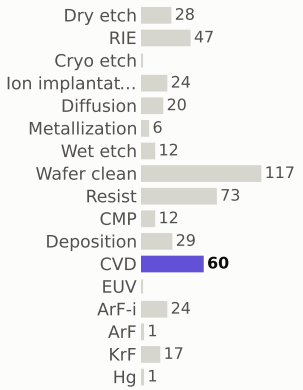
y cut at 57.351 µm



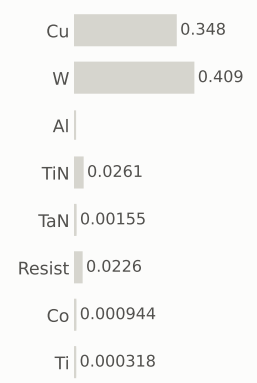
mask for this step

no mask for this step

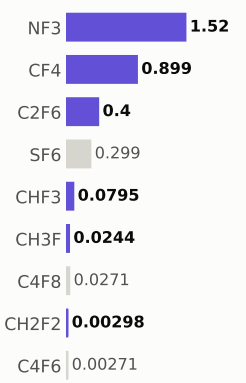
steps by process area



materials deposited (g)



gases emitted (g)



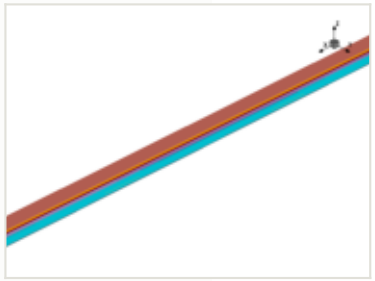
Step #6240: Wafer_clean

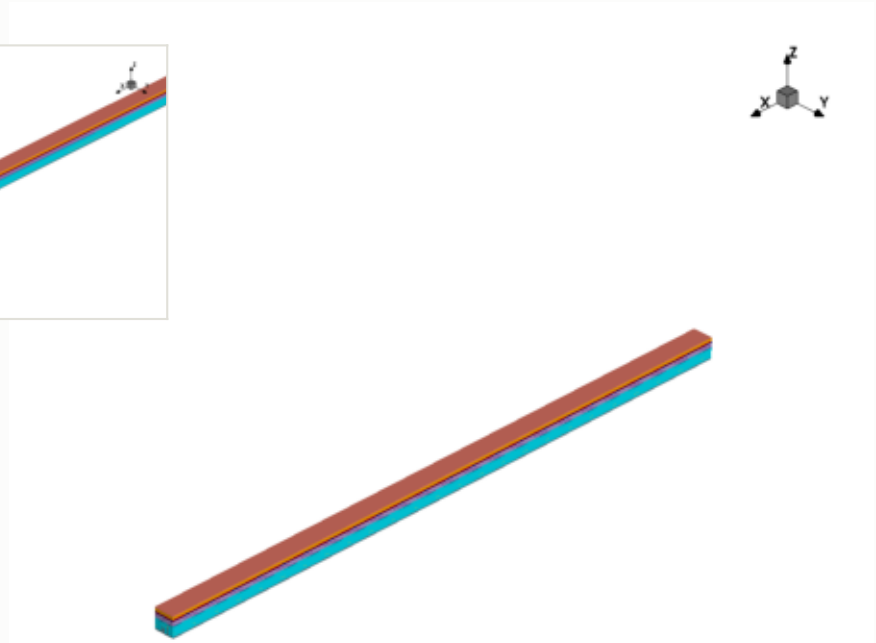
clean · Wafer_clean

0.169	282.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





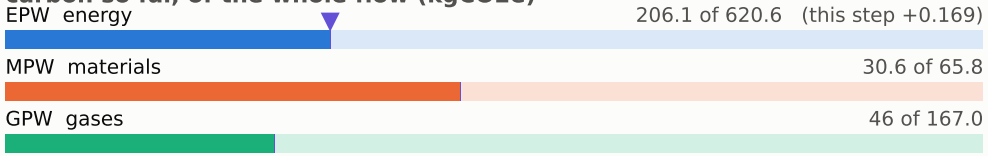
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

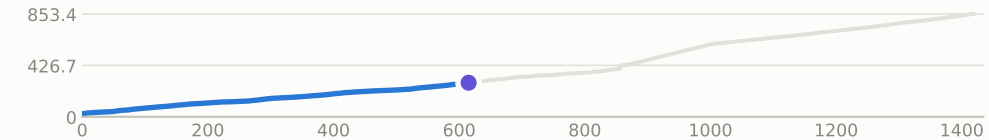
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

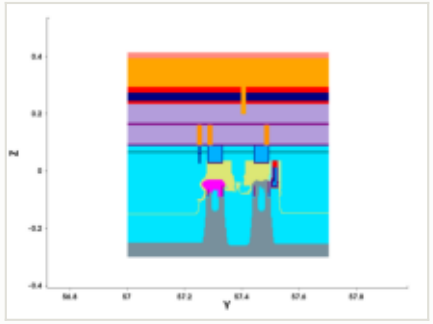
carbon so far, of the whole flow (kgCO2e)



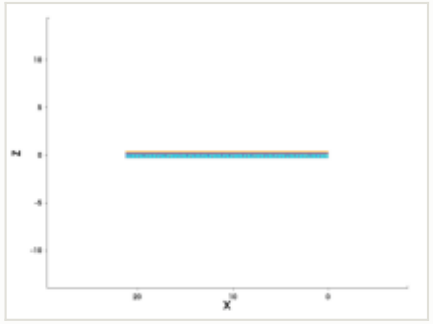
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



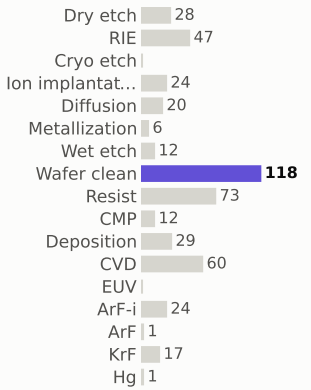
y cut at 57.351 µm



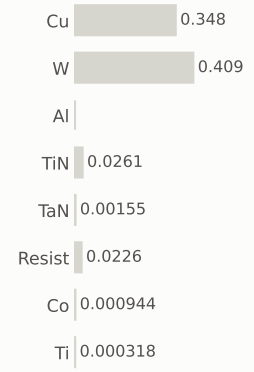
mask for this step

no mask for this step

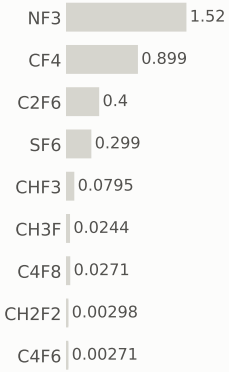
steps by process area



materials deposited (g)



gases emitted (g)



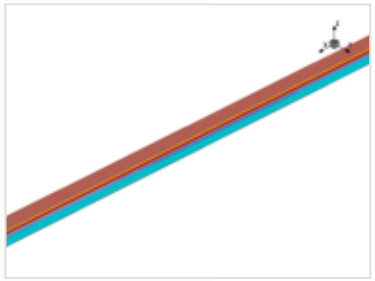
Step #6250: Resist

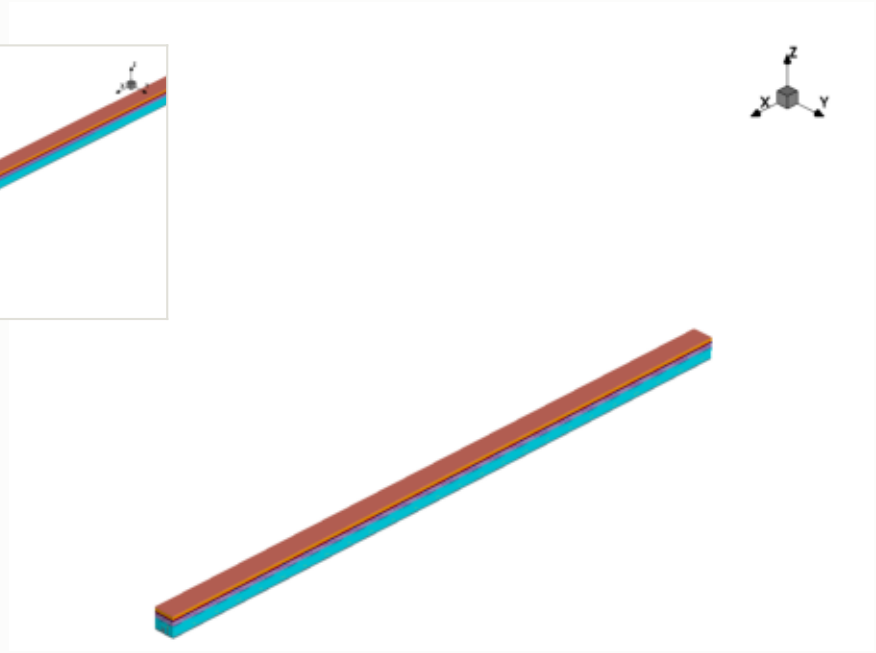
resist · Resist

0.169	282.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



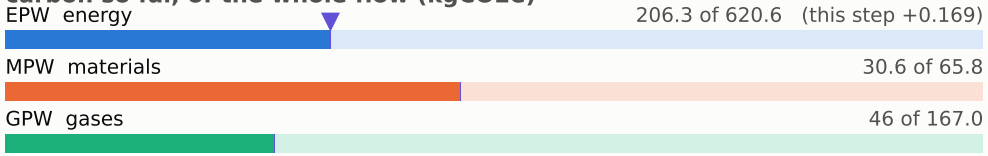


this step

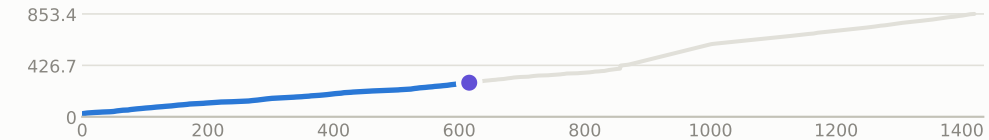
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

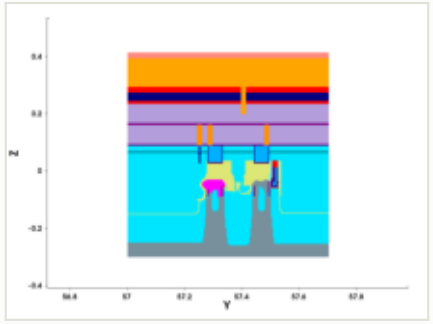
carbon so far, of the whole flow (kgCO2e)



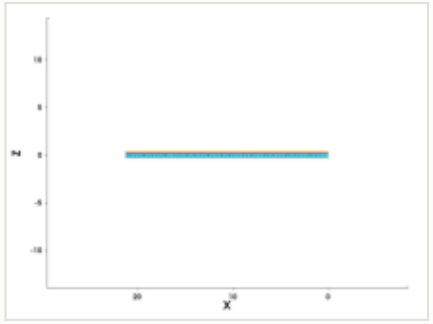
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



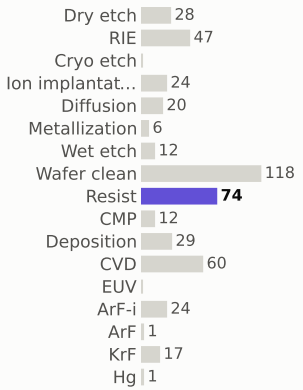
y cut at 57.351 µm



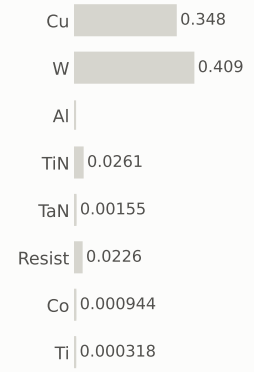
mask for this step

no mask for this step

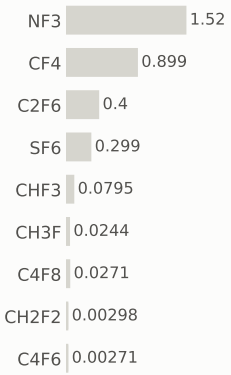
steps by process area



materials deposited (g)



gases emitted (g)



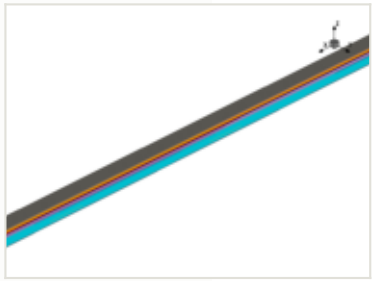
Step #6260: ArF-i

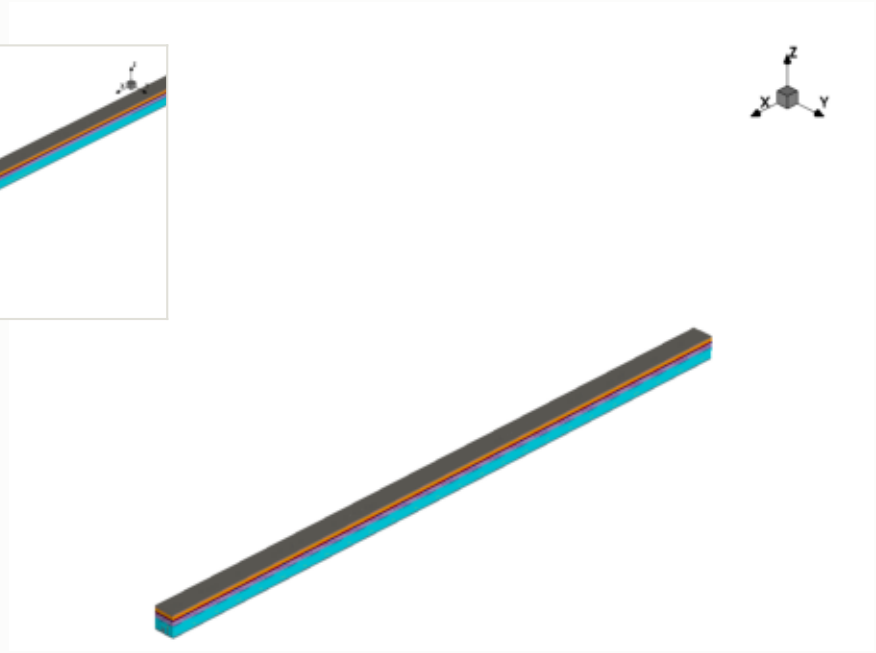
litho_patterning · ArF-i

0.217	283.1	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



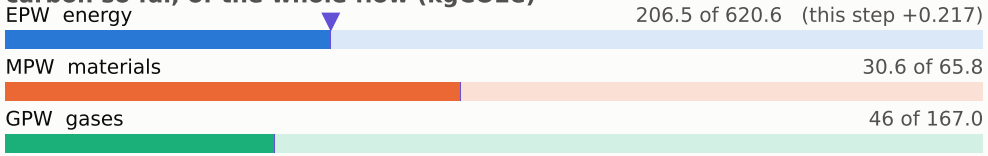


this step

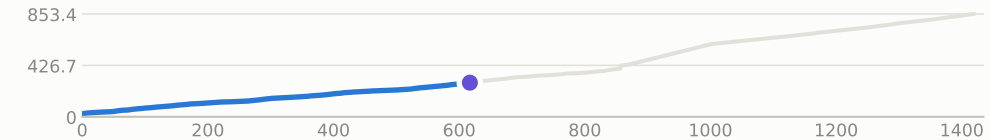
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

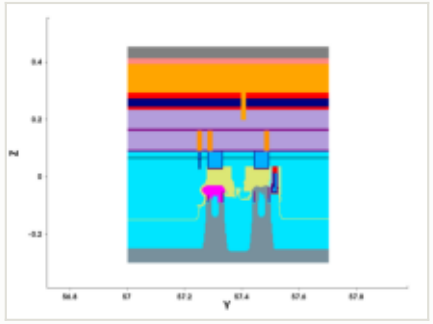
carbon so far, of the whole flow (kgCO2e)



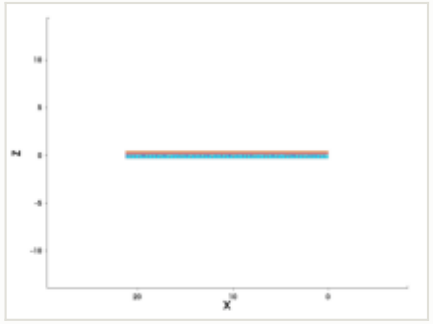
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

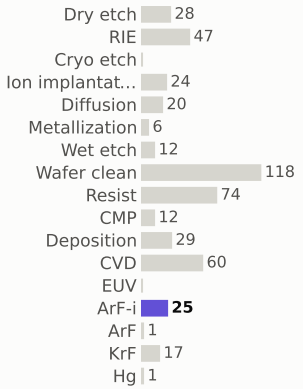


y cut at 57.351 µm

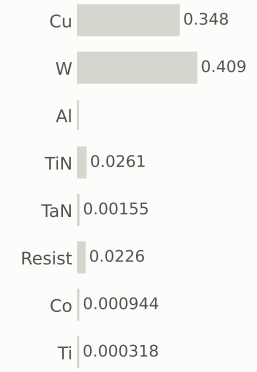


mask: layer 2021:0 (GDSII)

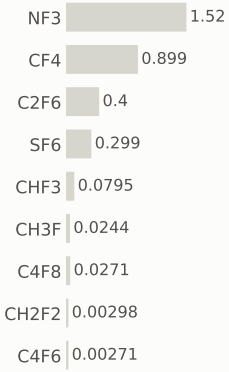
steps by process area



materials deposited (g)



gases emitted (g)



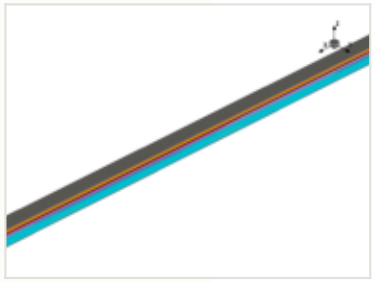
Step #6270: RIE

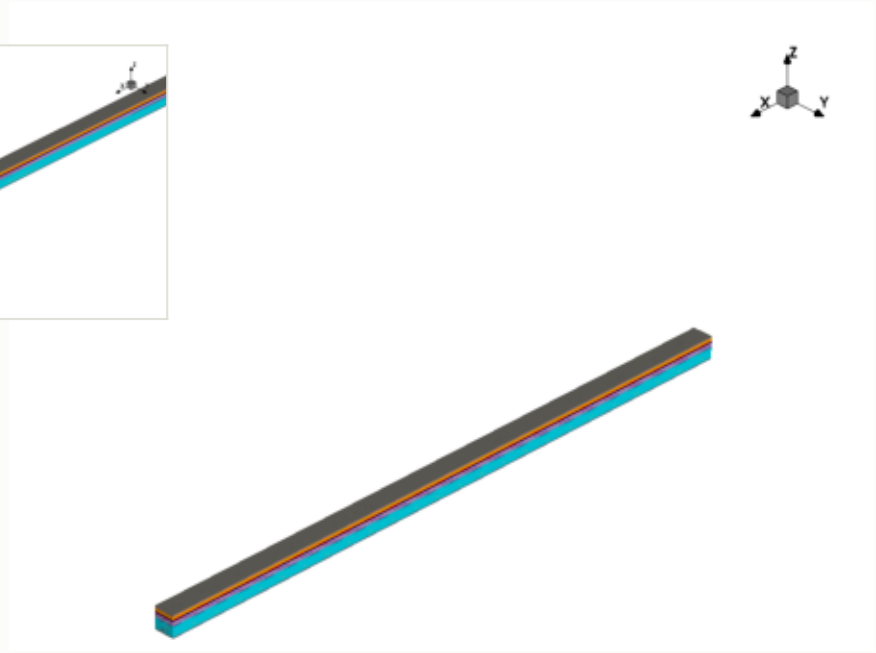
etch_anisotropic · RIE

1.04	284.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



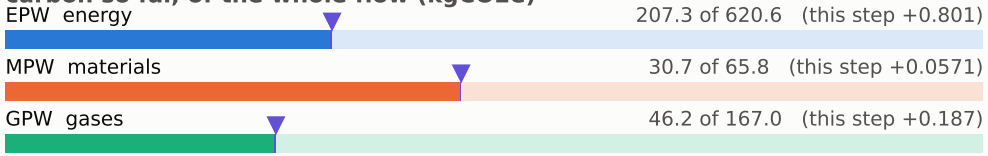


this step

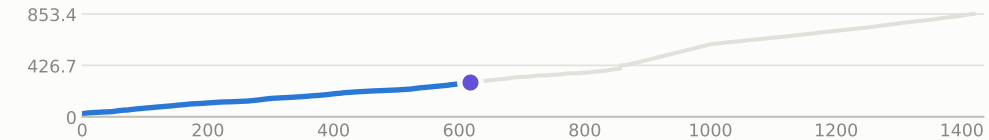
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

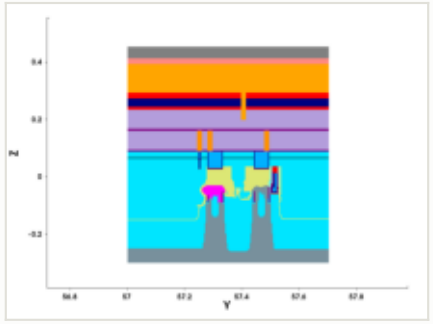
carbon so far, of the whole flow (kgCO2e)



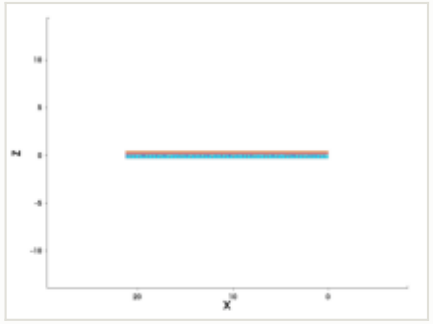
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



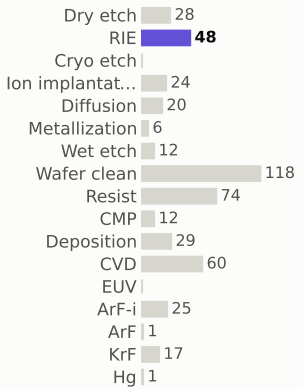
y cut at 57.351 µm



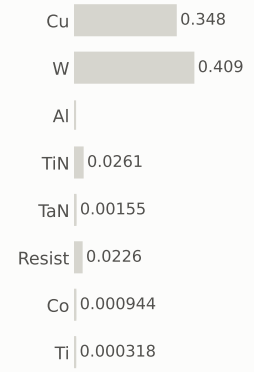
mask for this step

no mask for this step

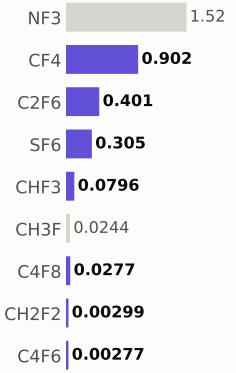
steps by process area



materials deposited (g)



gases emitted (g)



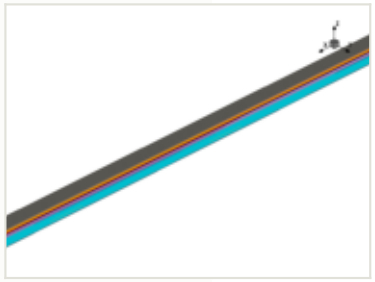
Step #6280: Wafer_clean

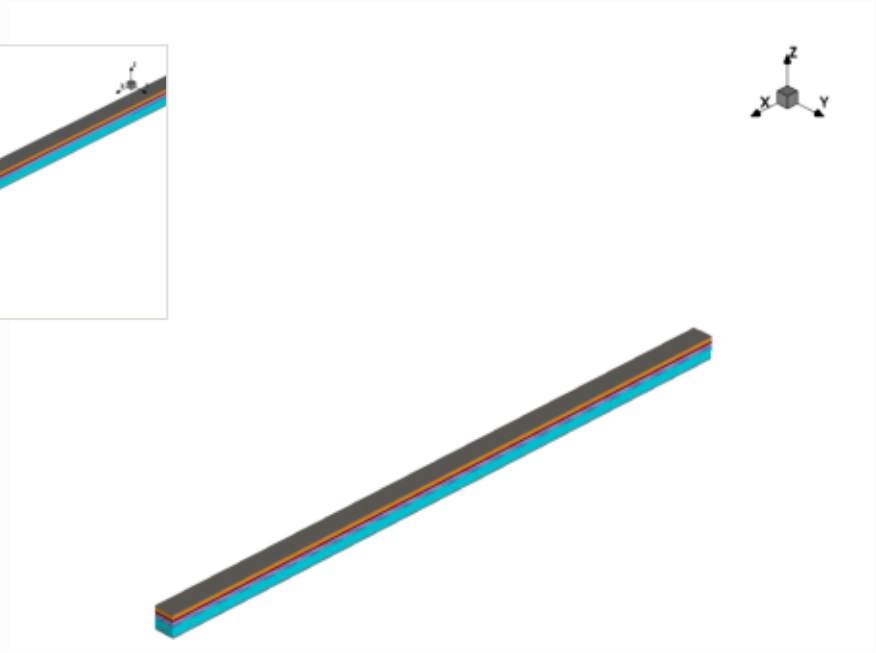
clean · Wafer_clean

0.169	284.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



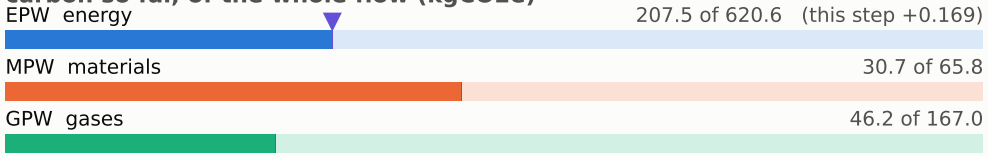


this step

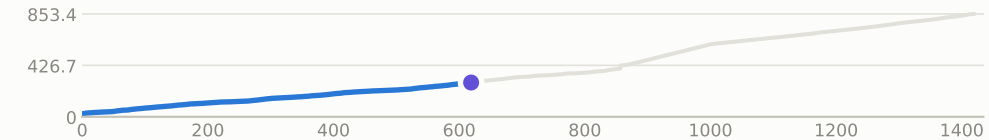
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

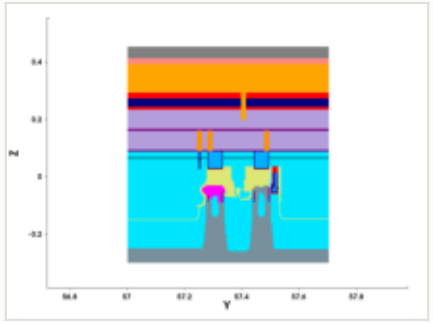
carbon so far, of the whole flow (kgCO2e)



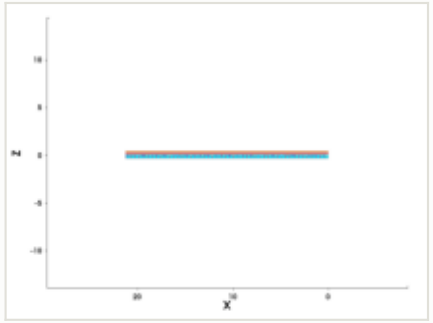
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



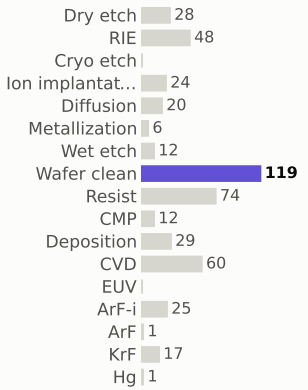
y cut at 57.351 µm



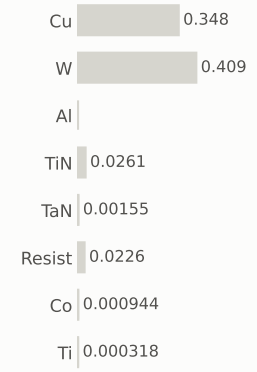
mask for this step

no mask for this step

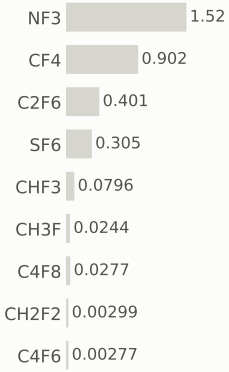
steps by process area



materials deposited (g)



gases emitted (g)



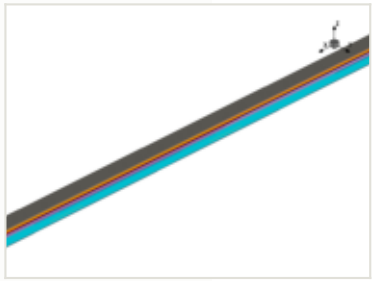
Step #6290: Resist

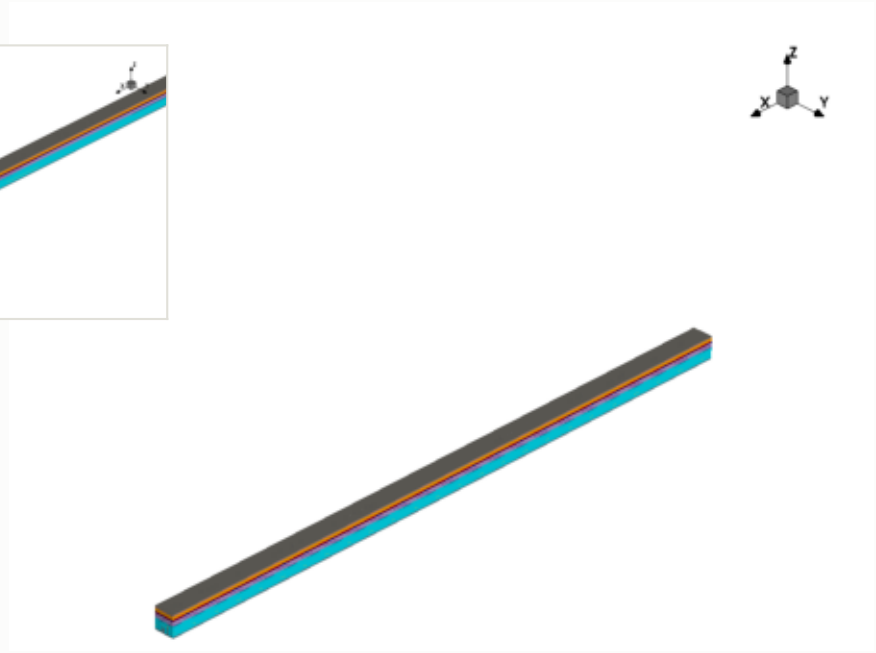
depo_fill · Resist

0.169	284.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



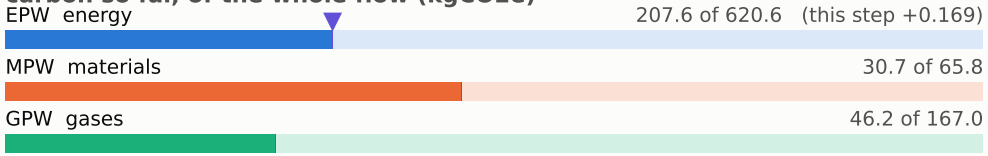


this step

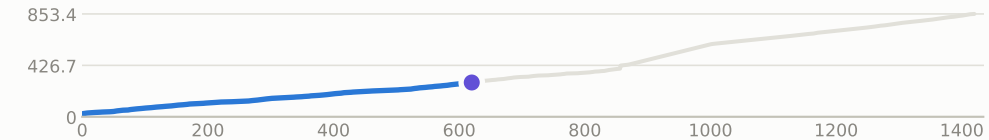
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

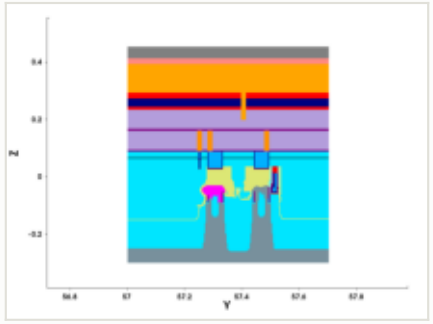
carbon so far, of the whole flow (kgCO2e)



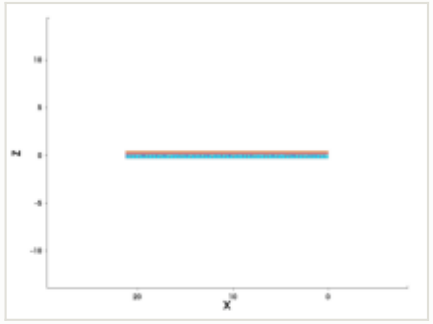
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



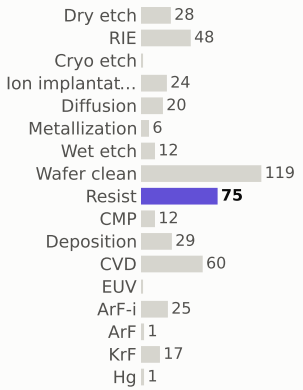
y cut at 57.351 µm



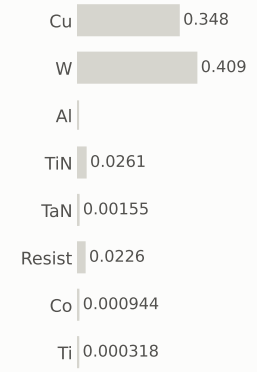
mask for this step

no mask for this step

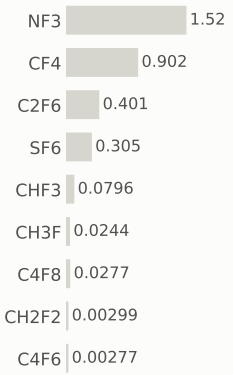
steps by process area



materials deposited (g)



gases emitted (g)



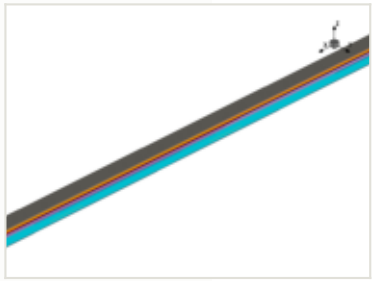
Step #6300: CVD

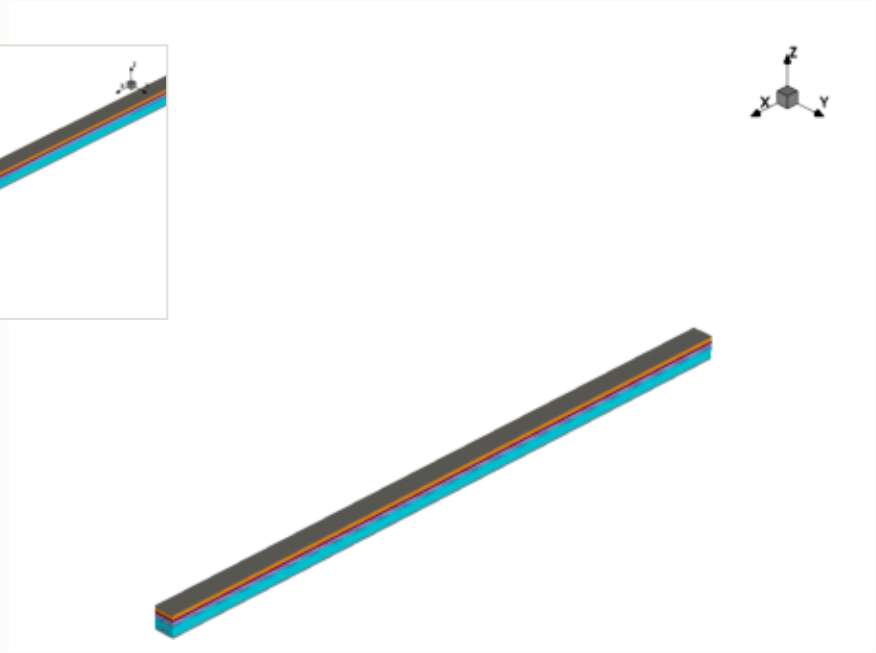
depo_isotropic · CVD

1.31	285.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



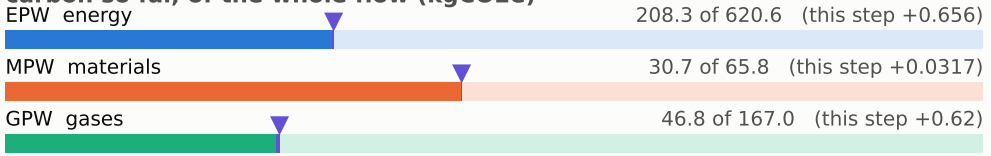


this step

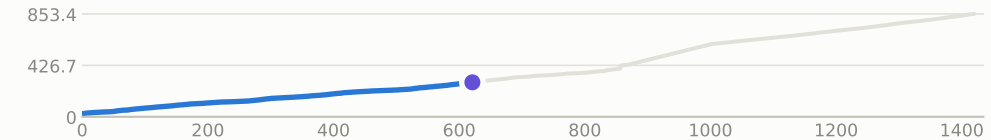
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

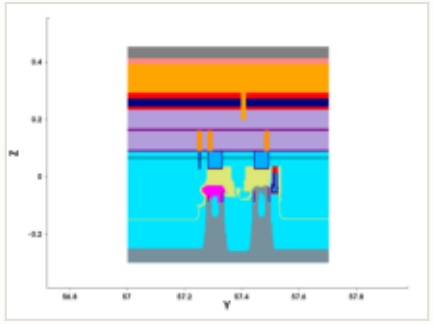
carbon so far, of the whole flow (kgCO2e)



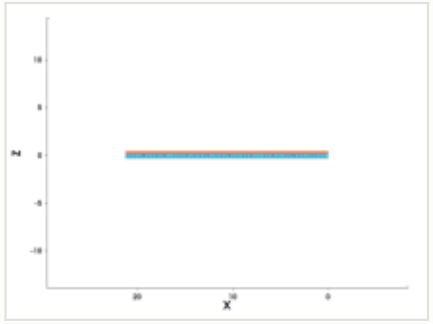
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



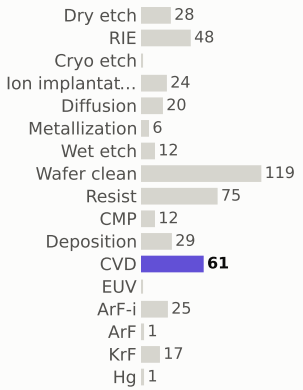
y cut at 57.351 µm



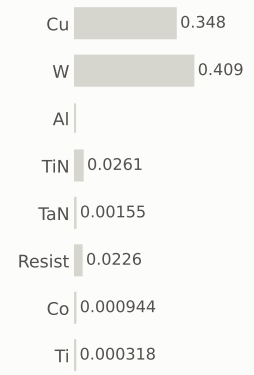
mask for this step

no mask for this step

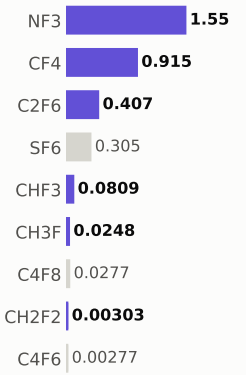
steps by process area



materials deposited (g)



gases emitted (g)



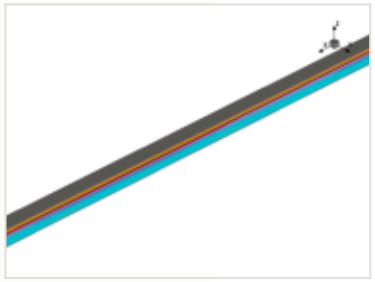
Step #6310: Wafer_clean

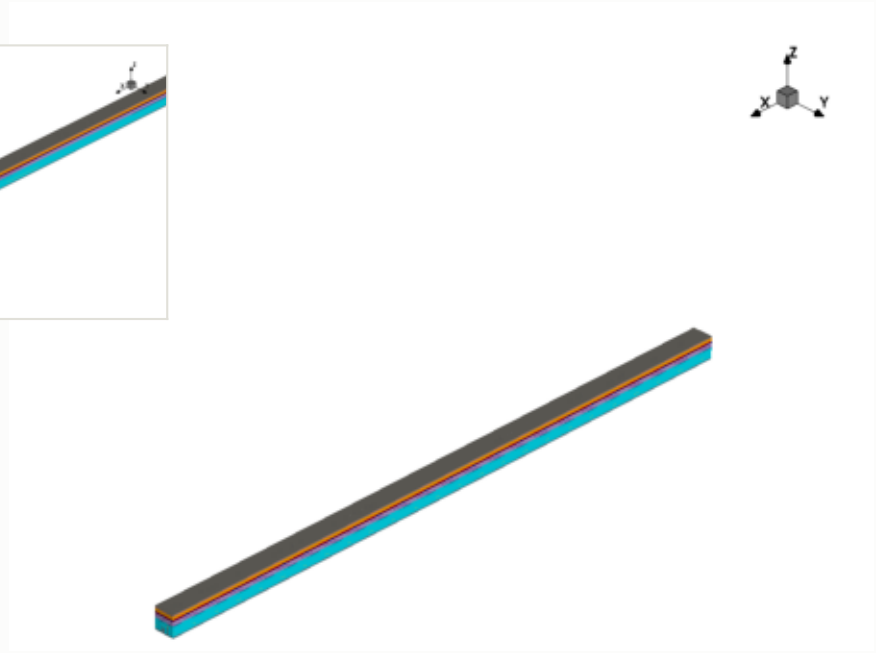
clean · Wafer_clean

0.169	286.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





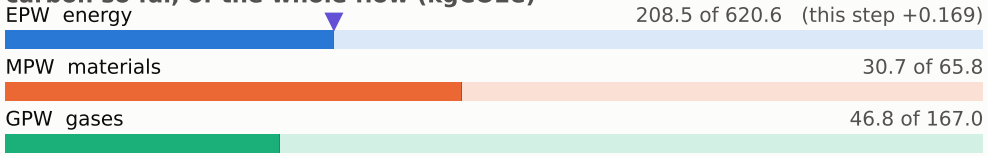
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

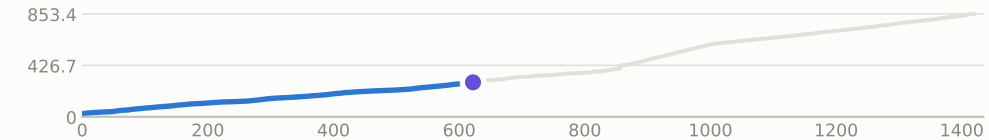
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

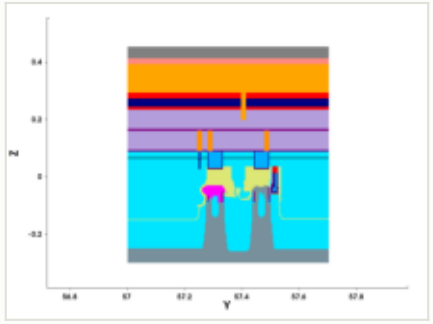
carbon so far, of the whole flow (kgCO2e)



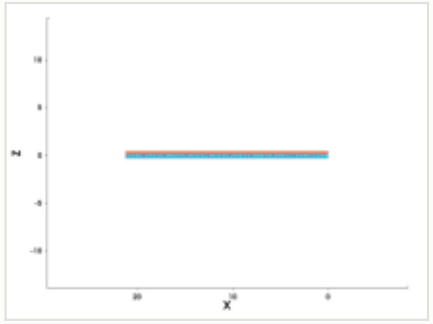
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



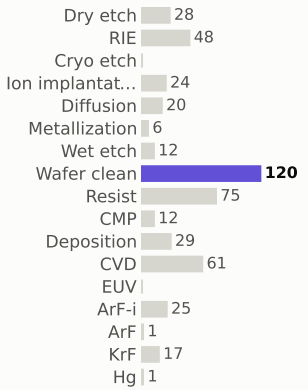
y cut at 57.351 µm



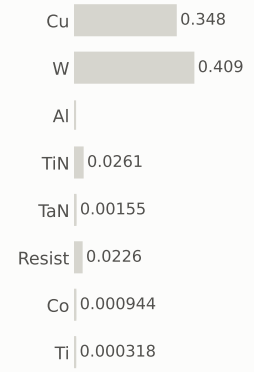
mask for this step

no mask for this step

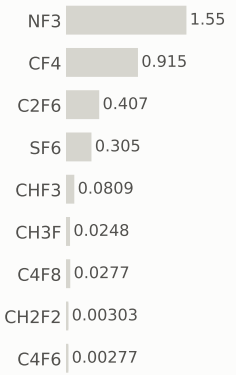
steps by process area



materials deposited (g)



gases emitted (g)



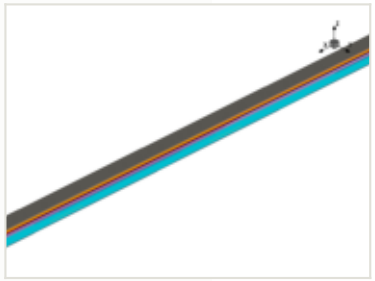
Step #6320: Resist

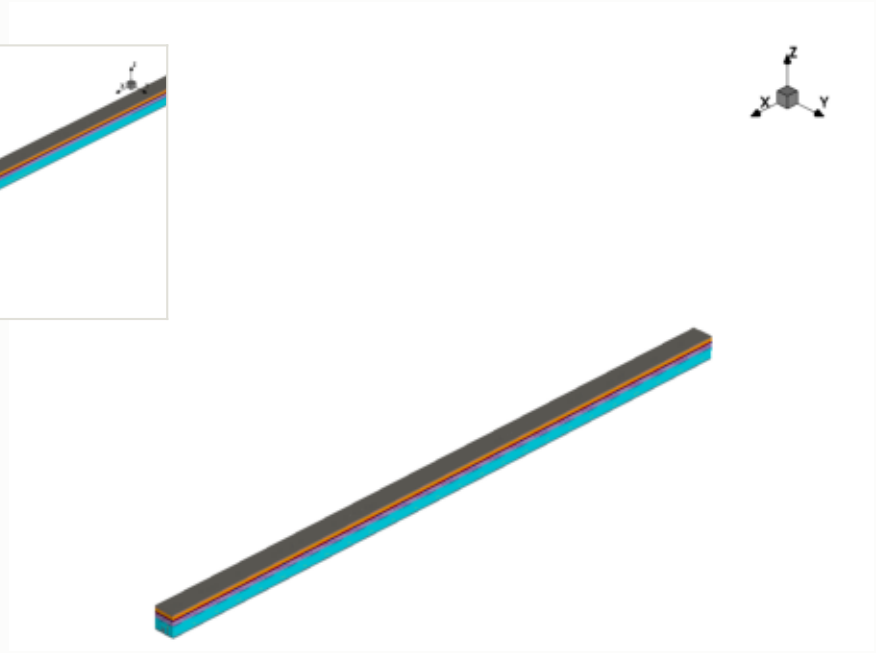
resist · Resist

0.169	286.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





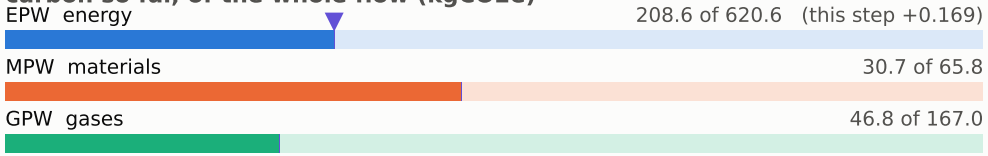
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

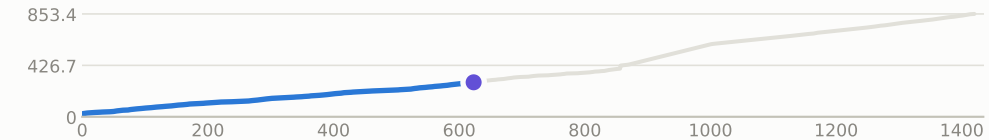
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

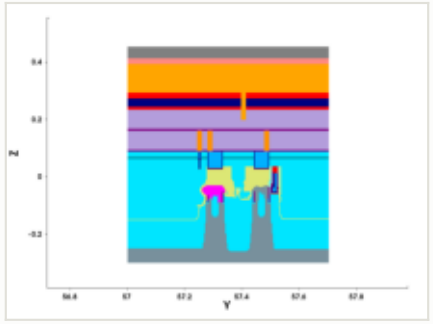
carbon so far, of the whole flow (kgCO2e)



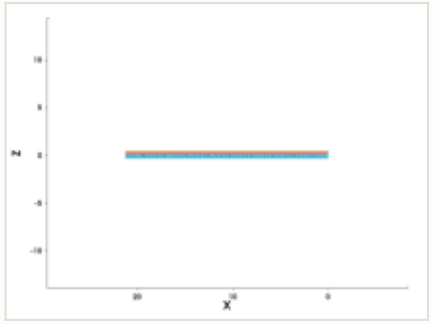
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



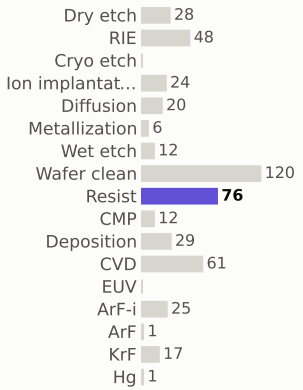
y cut at 57.351 µm



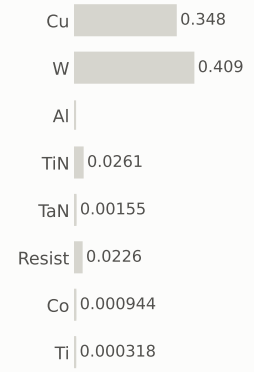
mask for this step

no mask for this step

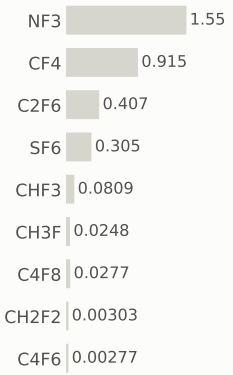
steps by process area



materials deposited (g)



gases emitted (g)



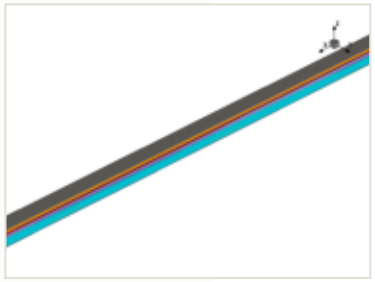
Step #6330: ArF-i

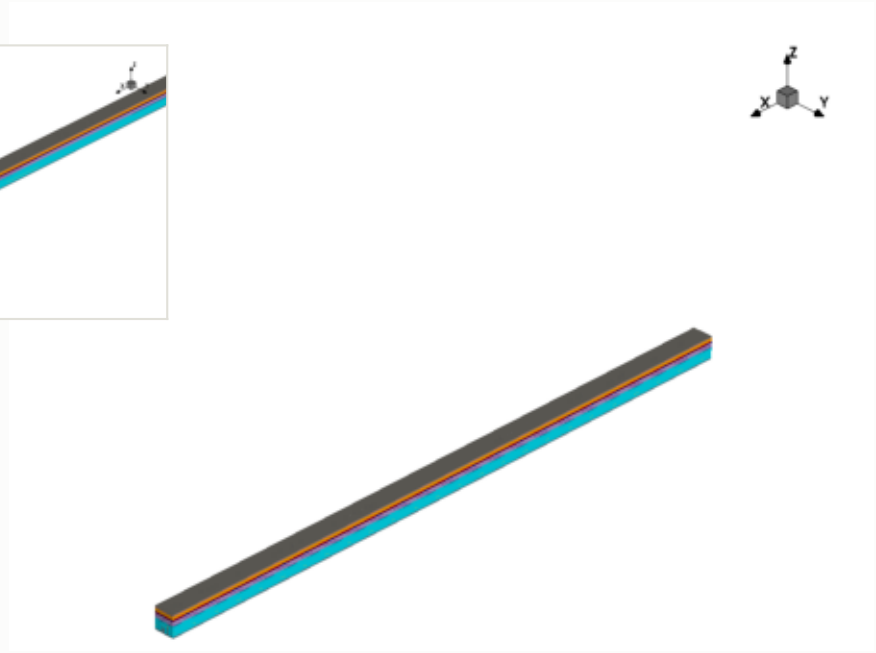
litho_patterning · ArF-i

0.217	286.4	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



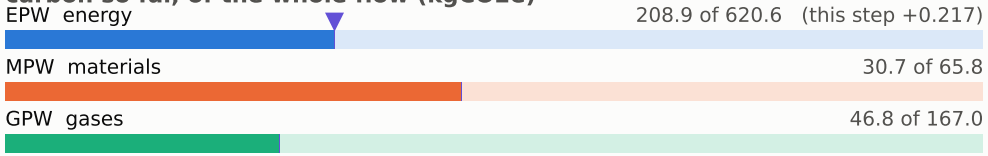


this step

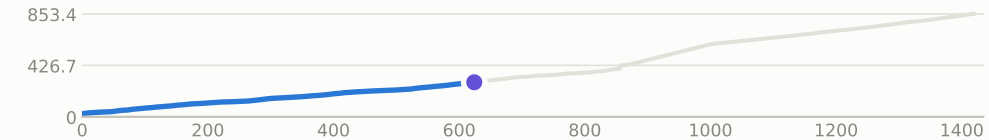
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

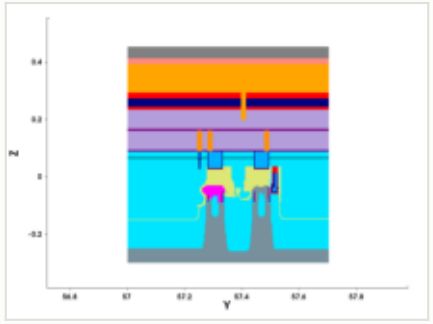
carbon so far, of the whole flow (kgCO2e)



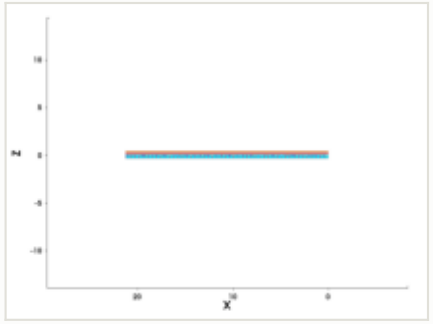
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

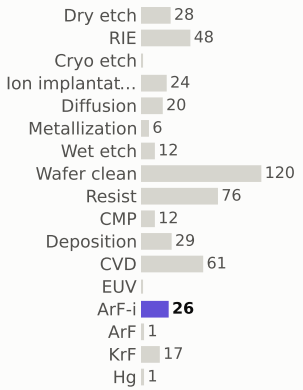


y cut at 57.351 µm

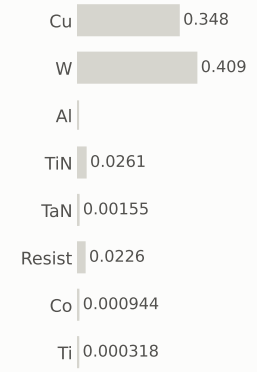


mask: layer 2100:0, positive (GDSII)

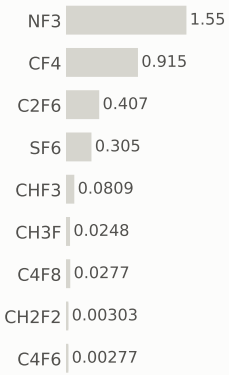
steps by process area



materials deposited (g)



gases emitted (g)



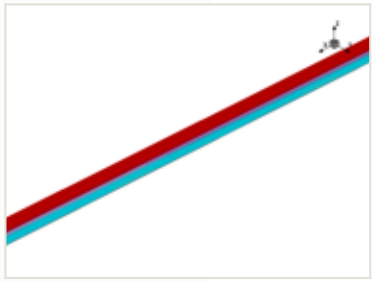
Step #6340: RIE

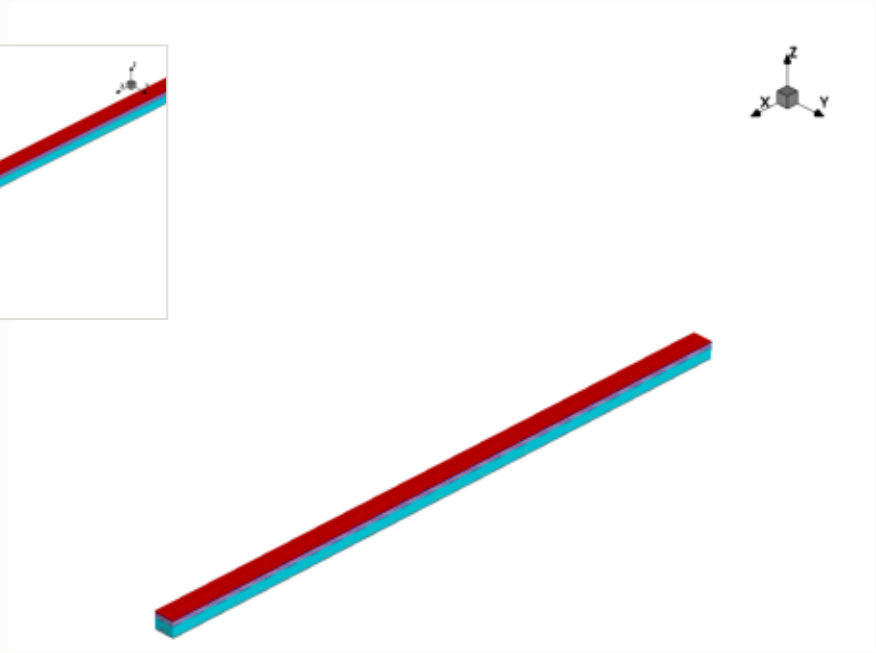
etch_strip · RIE

1.04	287.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



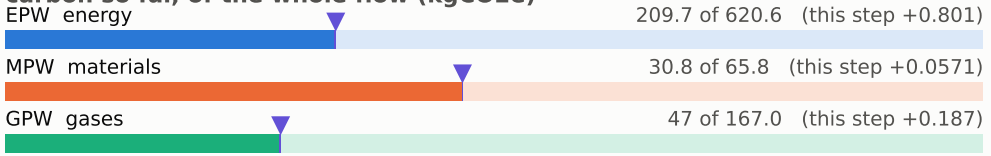


this step

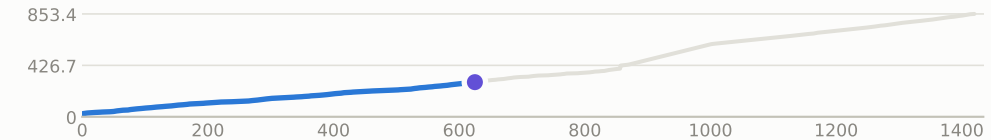
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

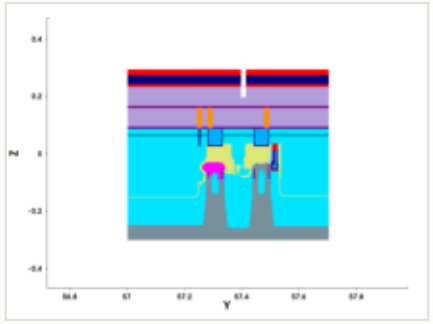
carbon so far, of the whole flow (kgCO2e)



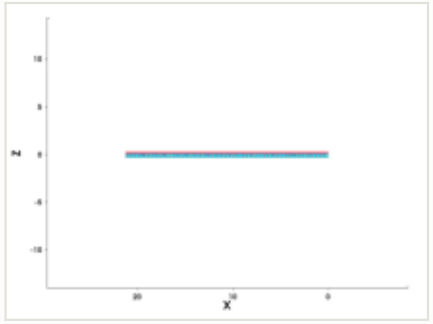
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



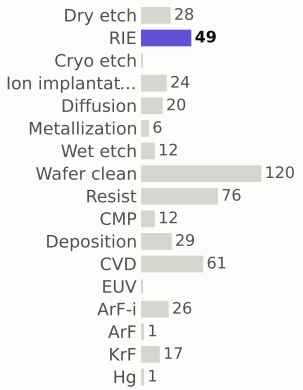
y cut at 57.351 µm



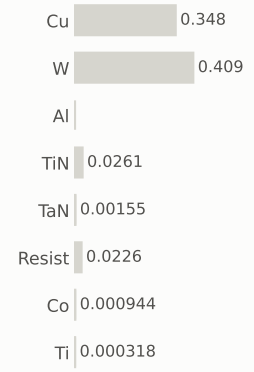
mask for this step

no mask for this step

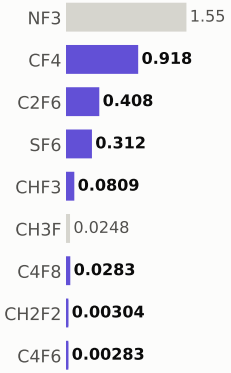
steps by process area



materials deposited (g)



gases emitted (g)



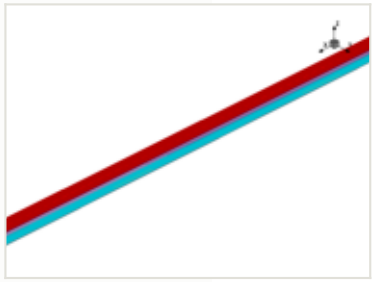
Step #6350: Wafer_clean

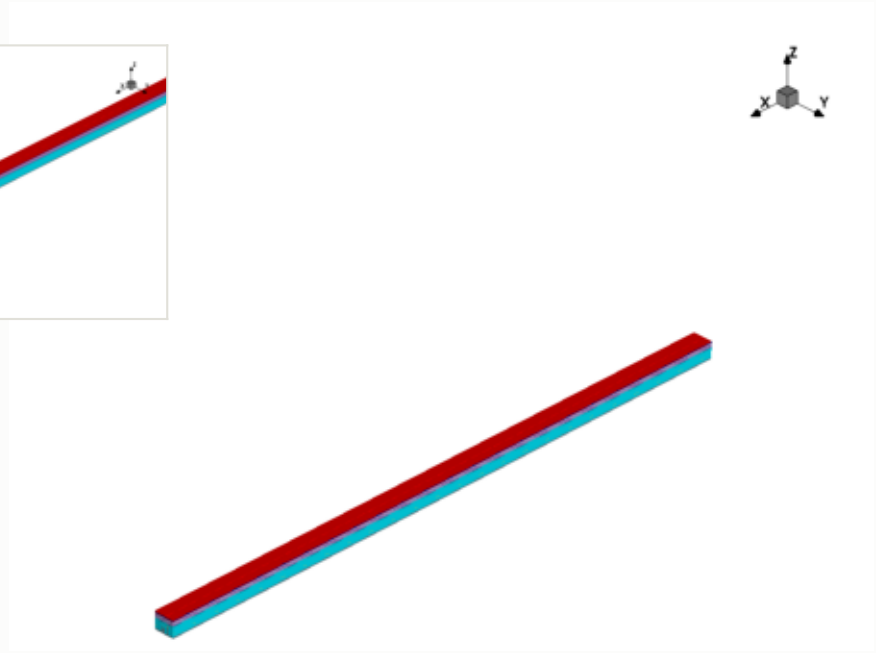
clean · Wafer_clean

0.169	287.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





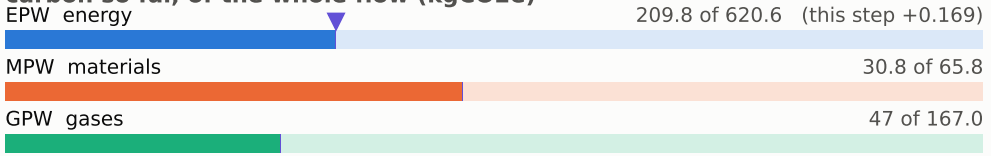
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

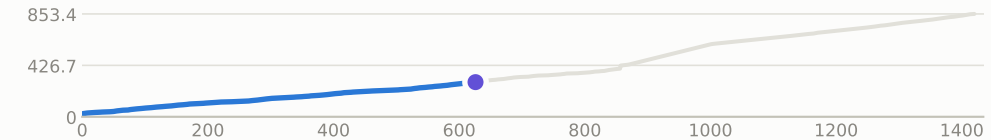
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

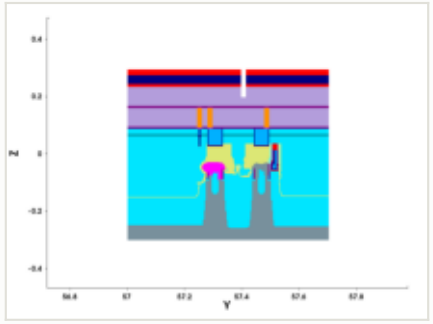
carbon so far, of the whole flow (kgCO2e)



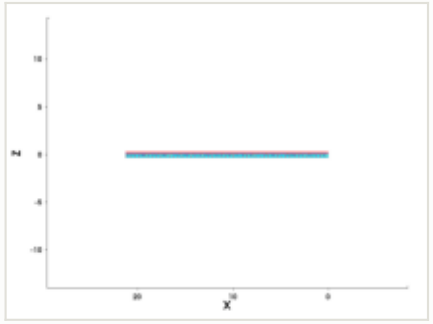
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



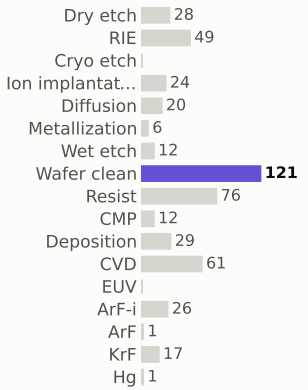
y cut at 57.351 µm



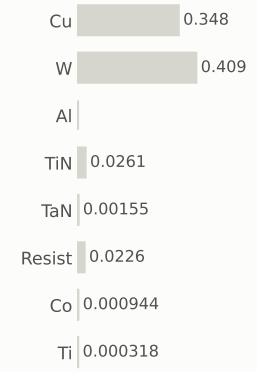
mask for this step

no mask for this step

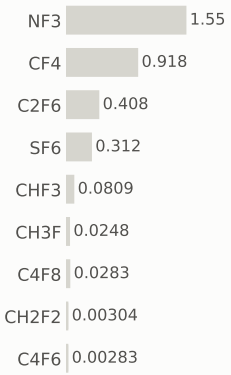
steps by process area



materials deposited (g)



gases emitted (g)



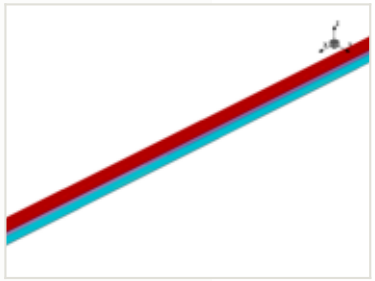
Step #6360: RIE

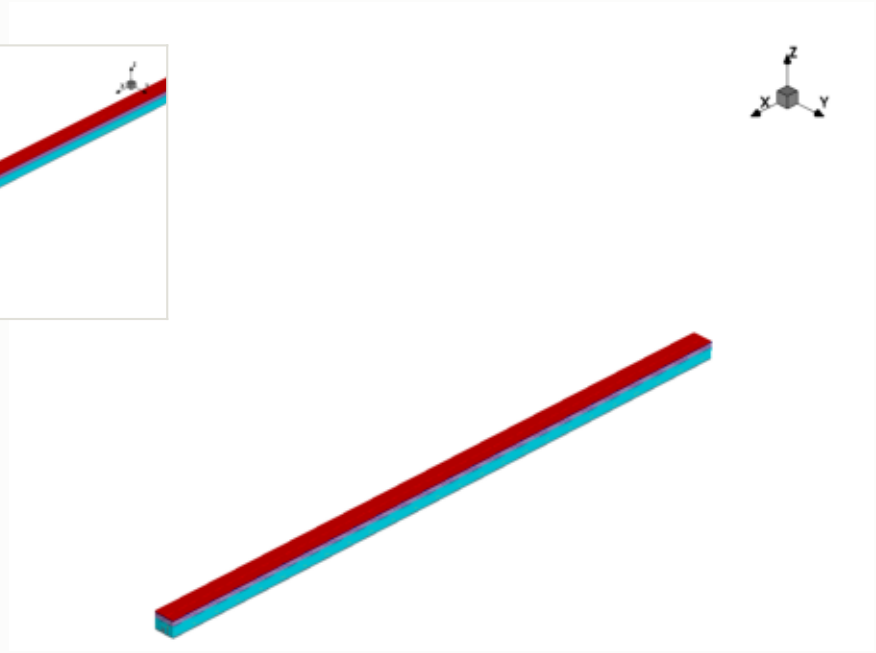
etch_anisotropic · RIE

1.04	288.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



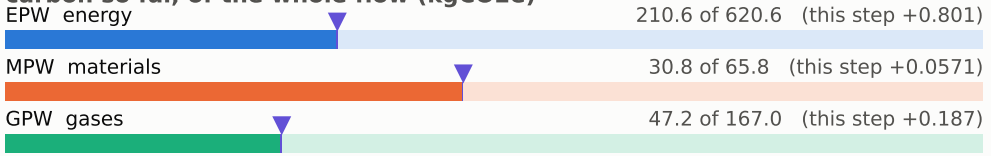


this step

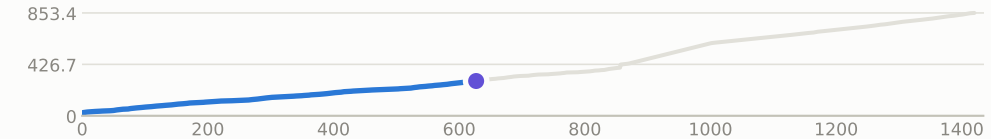
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

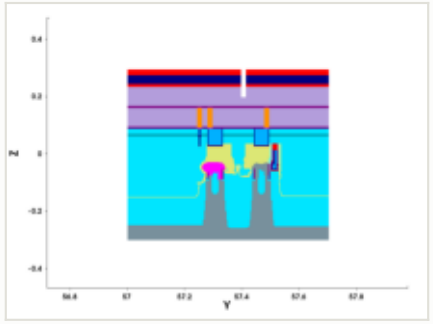
carbon so far, of the whole flow (kgCO2e)



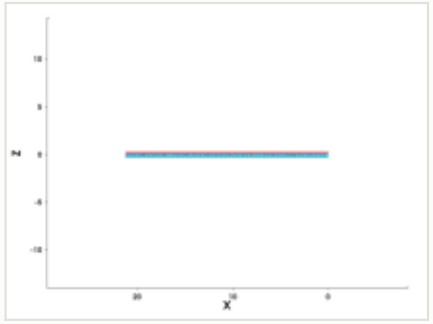
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



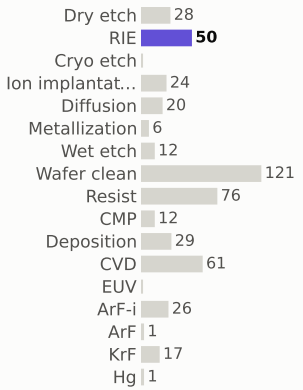
y cut at 57.351 µm



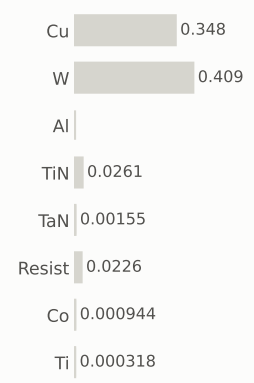
mask for this step

no mask for this step

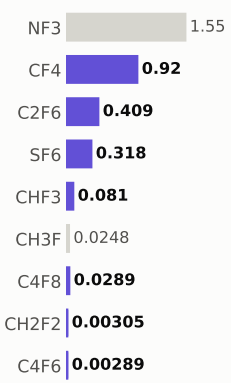
steps by process area



materials deposited (g)



gases emitted (g)



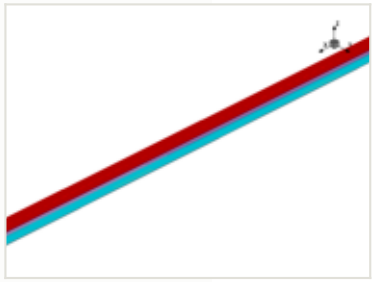
Step #6370: Wafer_clean

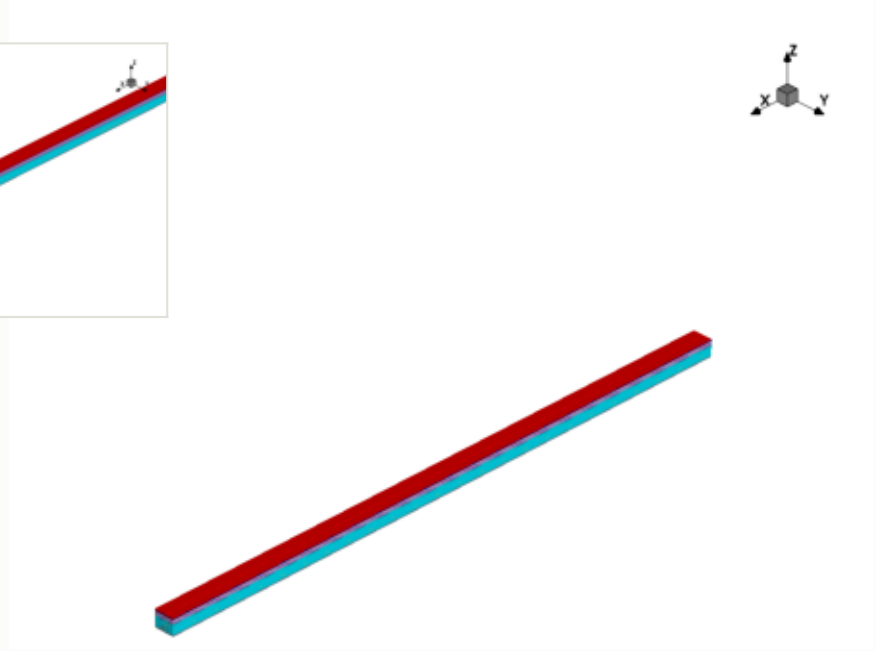
clean · Wafer_clean

0.169	288.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



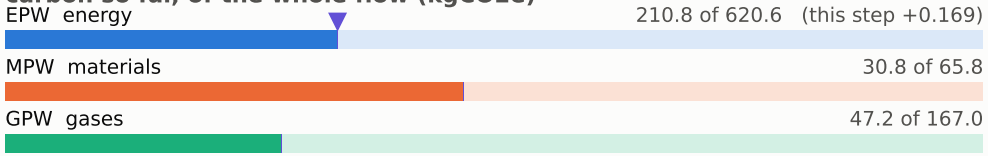


this step

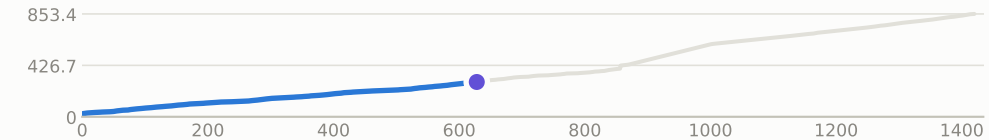
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

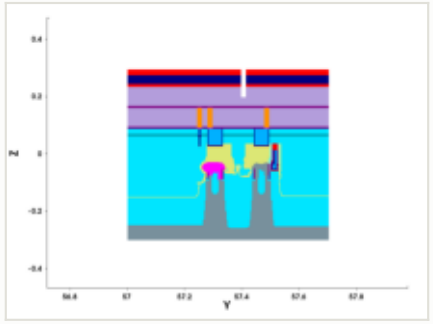
carbon so far, of the whole flow (kgCO2e)



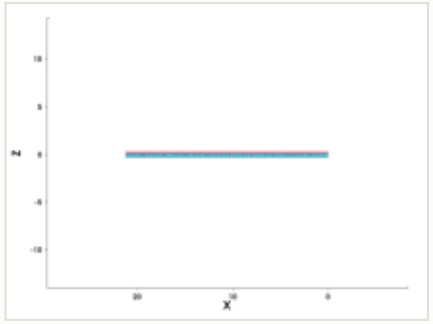
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



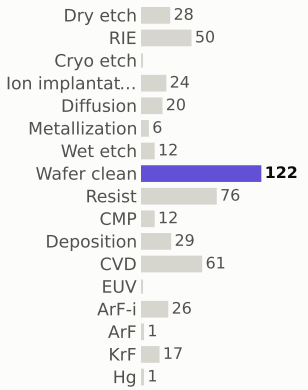
y cut at 57.351 µm



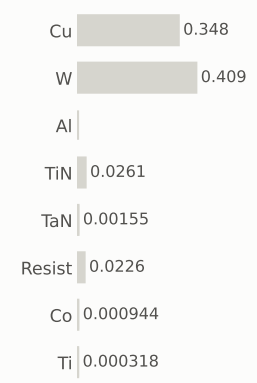
mask for this step

no mask for this step

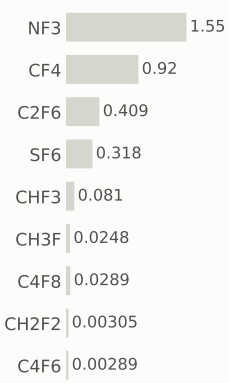
steps by process area



materials deposited (g)



gases emitted (g)



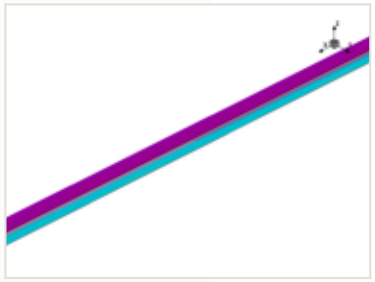
Step #6380: Metallization

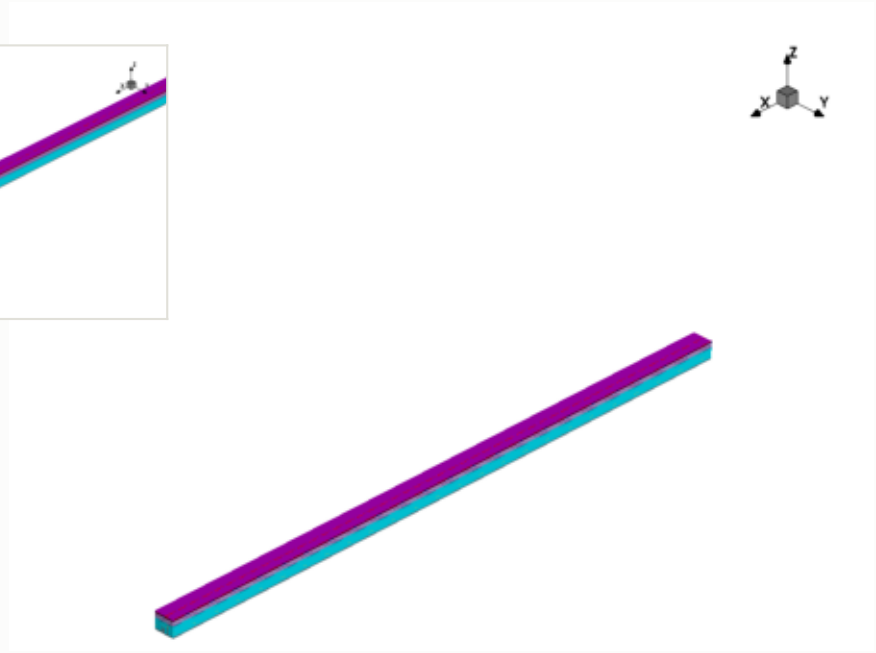
depo_ellipsoidal · Metallization

1.2	290.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



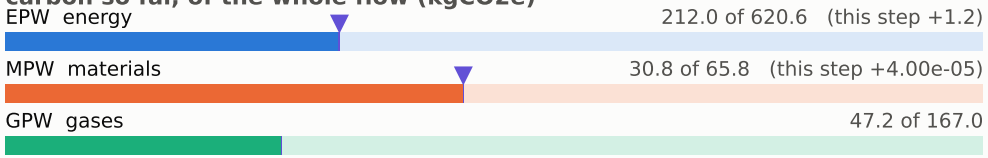


this step

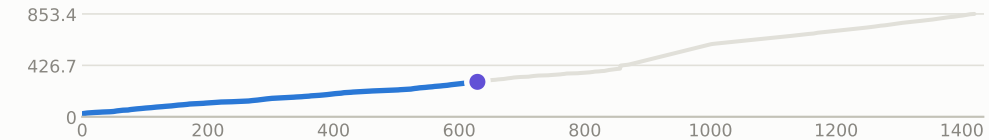
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

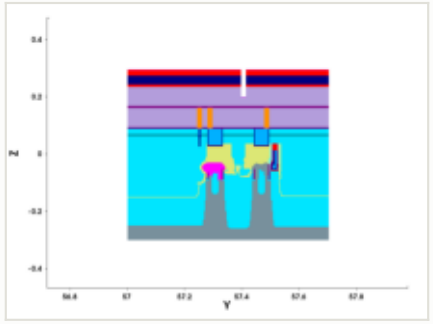
carbon so far, of the whole flow (kgCO2e)



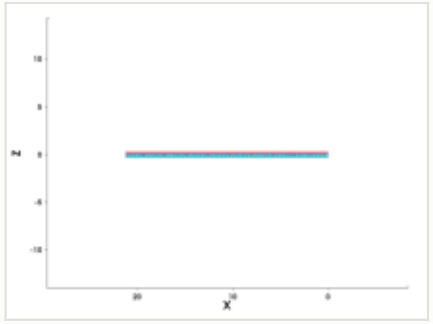
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



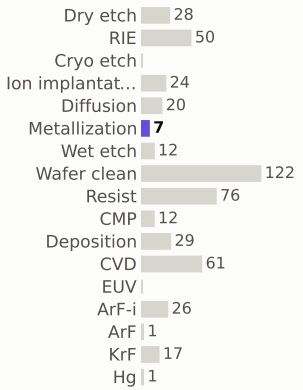
y cut at 57.351 µm



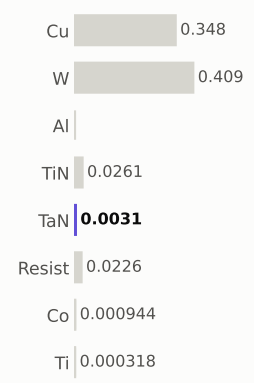
mask for this step

no mask for this step

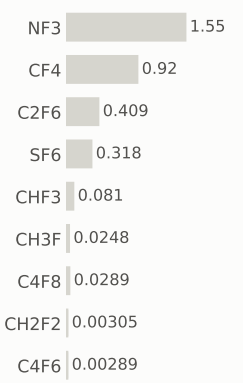
steps by process area



materials deposited (g)



gases emitted (g)



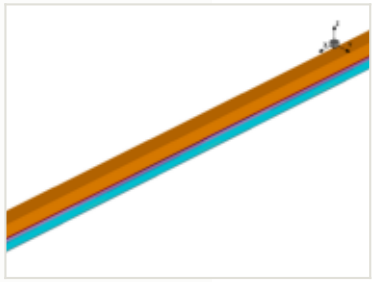
Step #6390: Metallization

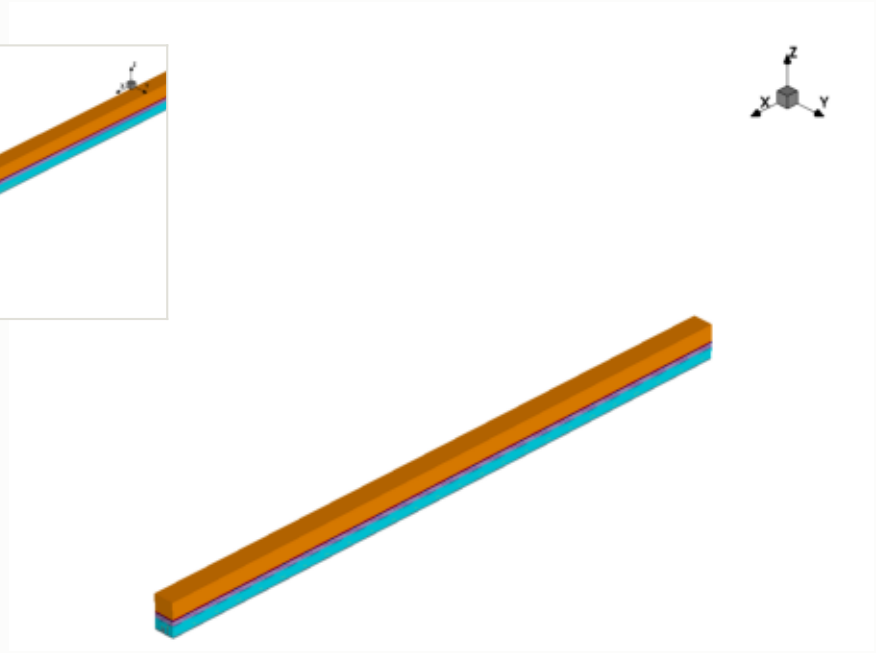
depo_fill · Metallization

1.21	291.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



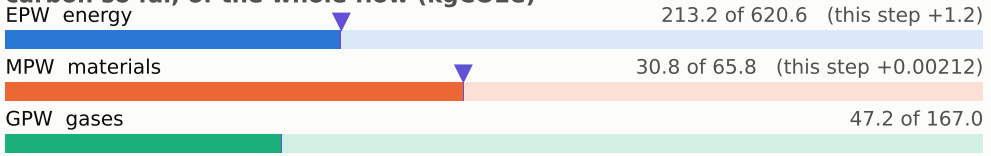


this step

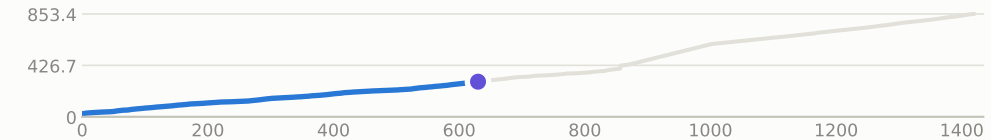
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

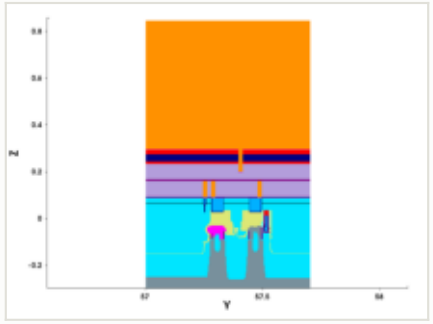
carbon so far, of the whole flow (kgCO2e)



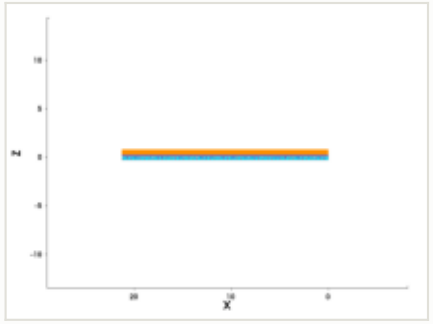
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



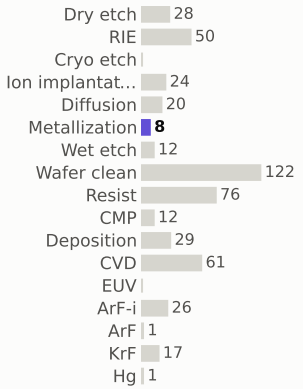
y cut at 57.351 µm



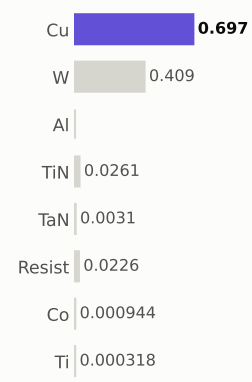
mask for this step

no mask for this step

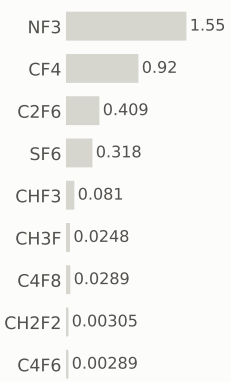
steps by process area



materials deposited (g)



gases emitted (g)



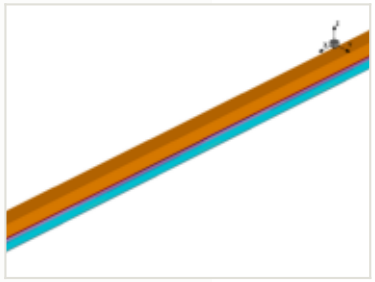
Step #6400: Diffusion

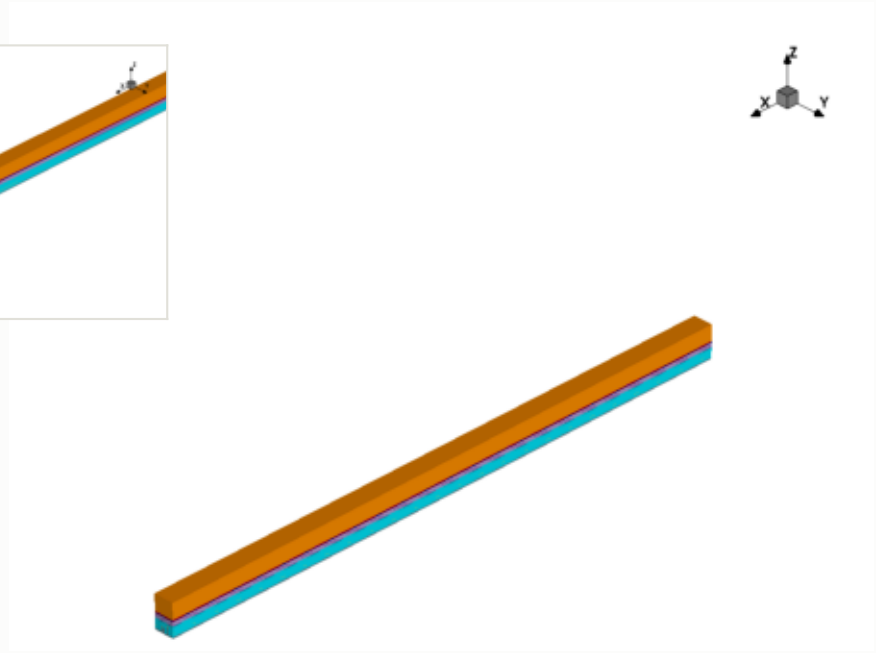
anneal · Diffusion

0.801	292.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





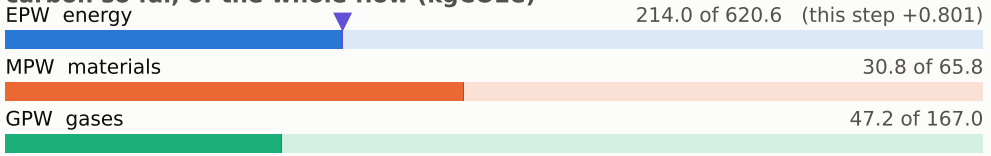
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

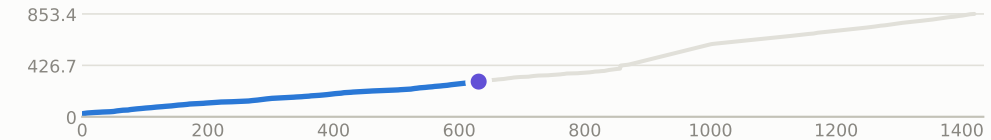
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

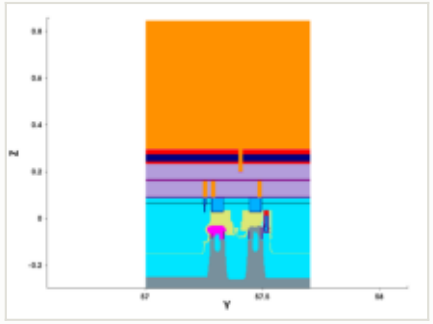
carbon so far, of the whole flow (kgCO2e)



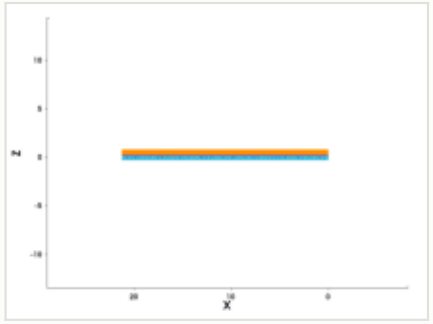
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



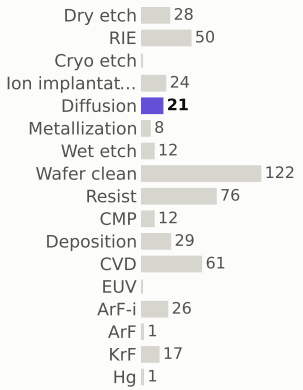
y cut at 57.351 µm



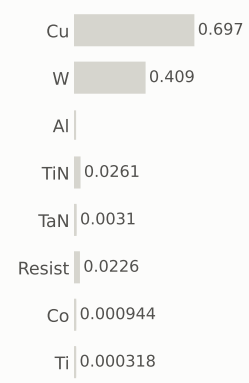
mask for this step

no mask for this step

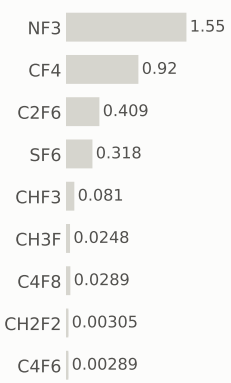
steps by process area



materials deposited (g)



gases emitted (g)



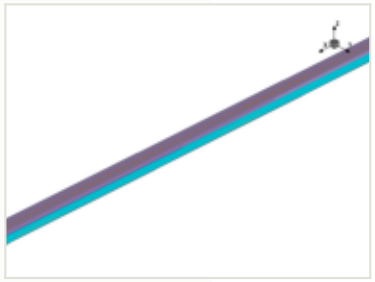
Step #6410: CMP

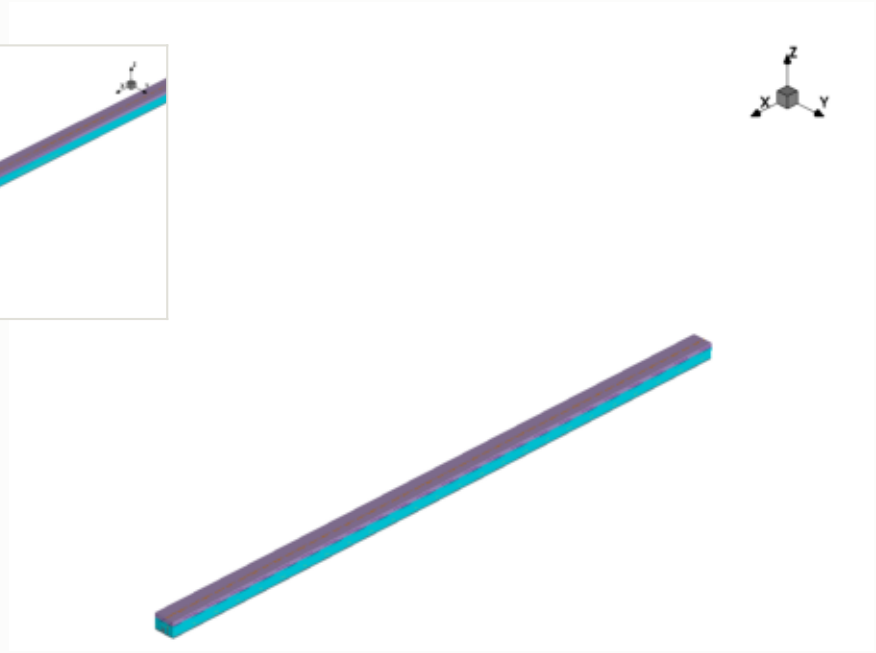
etch_cmp · CMP

0.169	292.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



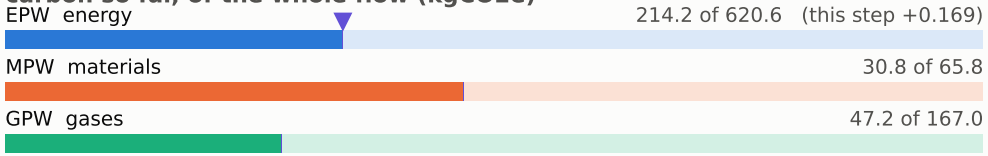


this step

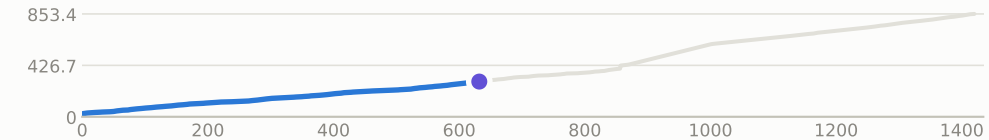
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

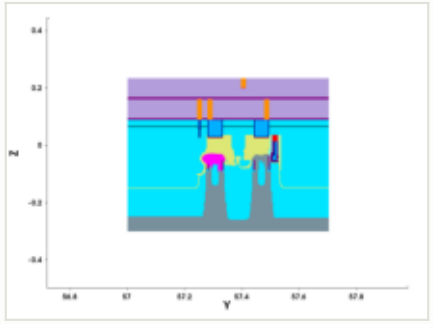
carbon so far, of the whole flow (kgCO2e)



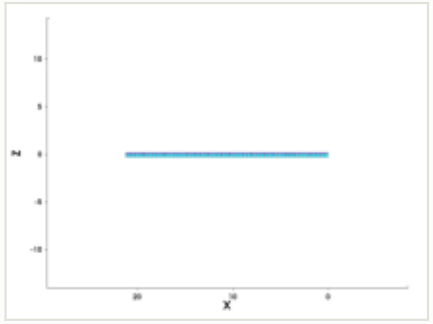
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



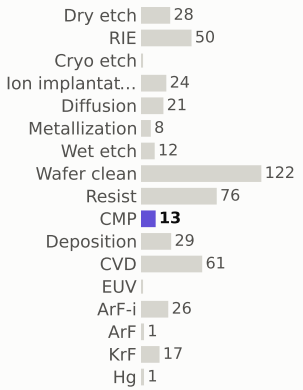
y cut at 57.351 µm



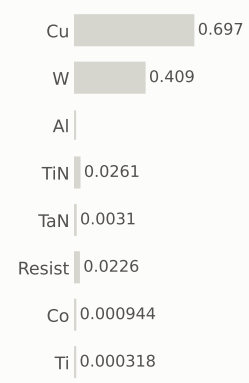
mask for this step

no mask for this step

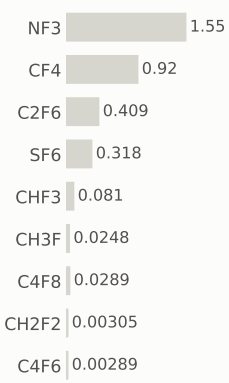
steps by process area



materials deposited (g)



gases emitted (g)



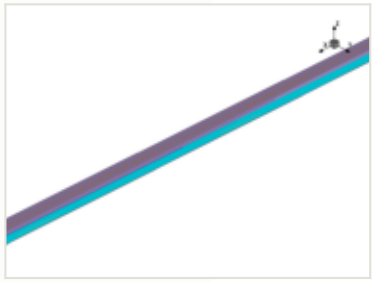
Step #6420: Wafer_clean

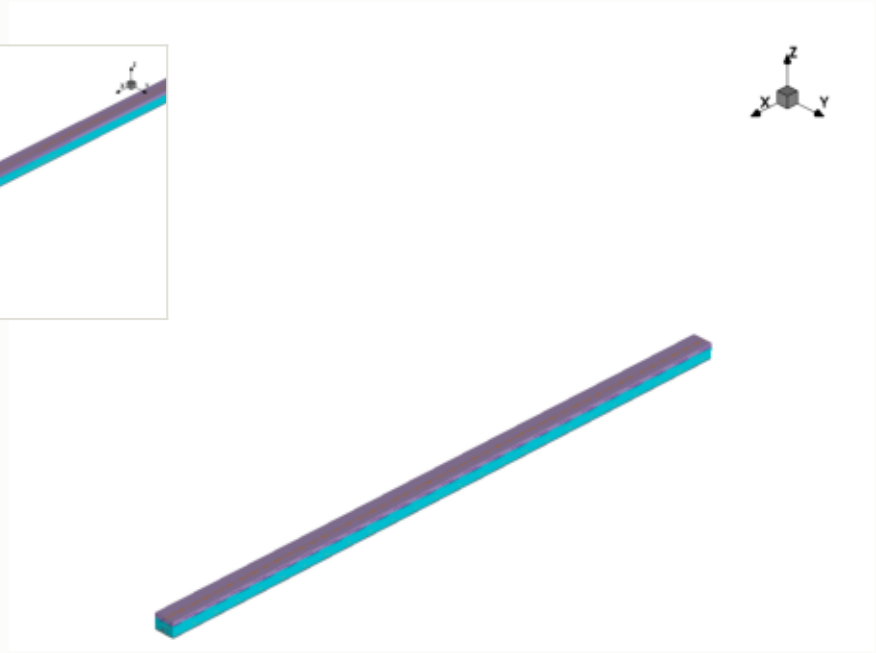
clean · Wafer_clean

0.169	292.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

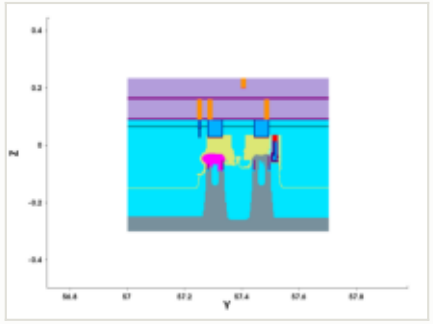
3D structure after this step

zoomed to fit

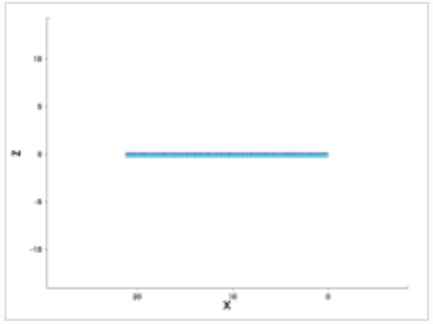




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

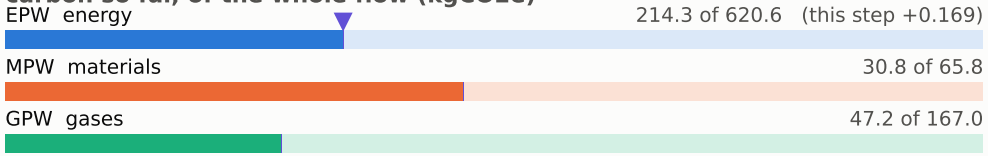
no mask for this step

this step

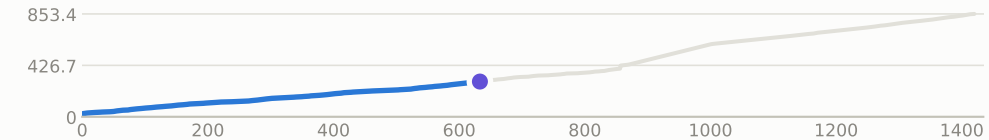
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

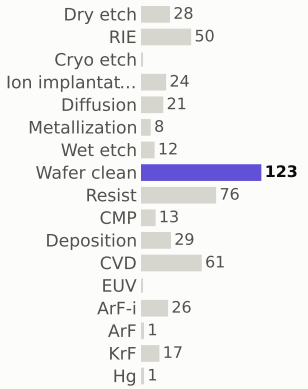
carbon so far, of the whole flow (kgCO2e)



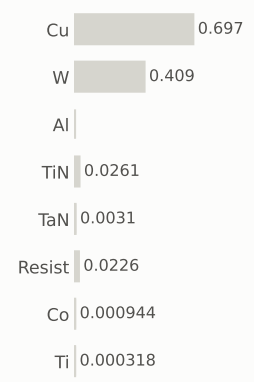
embodied carbon up to each step (kgCO2e per wafer)



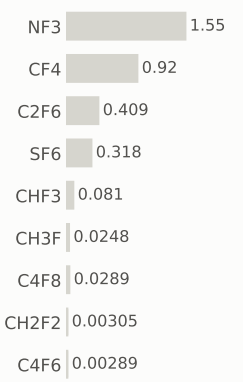
steps by process area



materials deposited (g)



gases emitted (g)



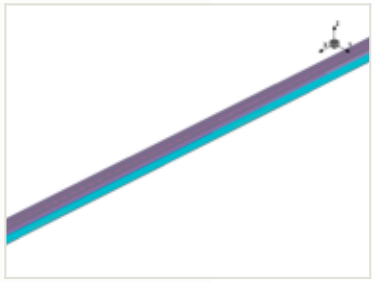
Step #6430: Metallization

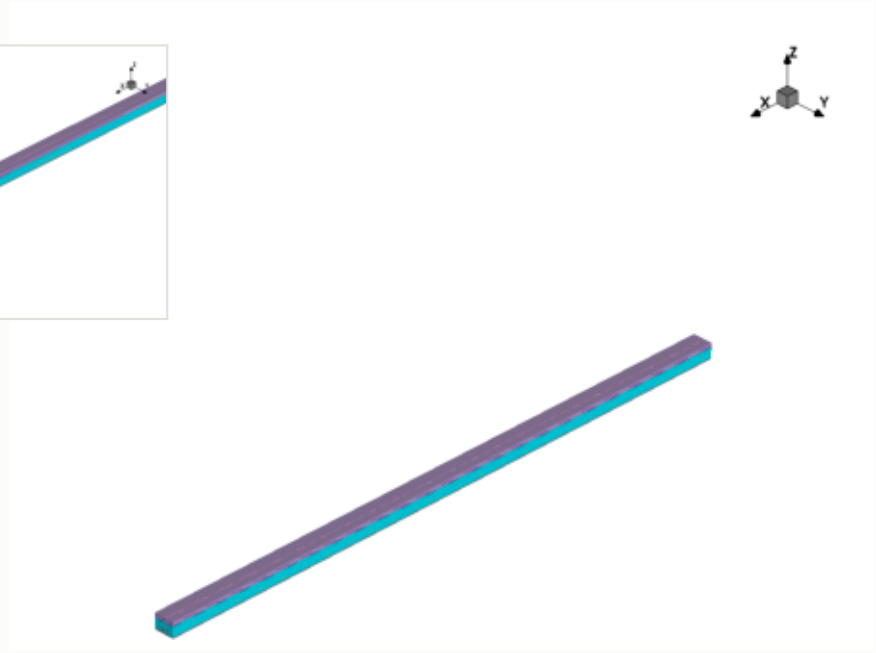
depo_isotropic · Metallization

1.2	293.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



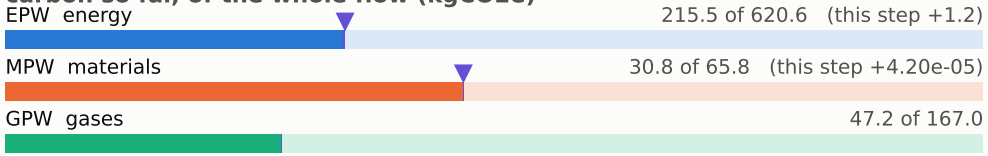


this step

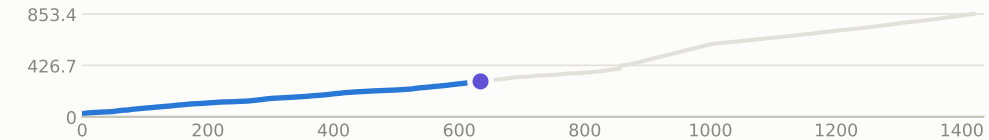
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

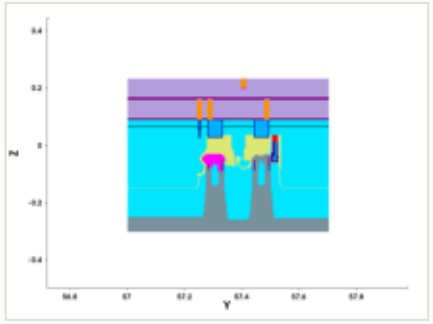
carbon so far, of the whole flow (kgCO2e)



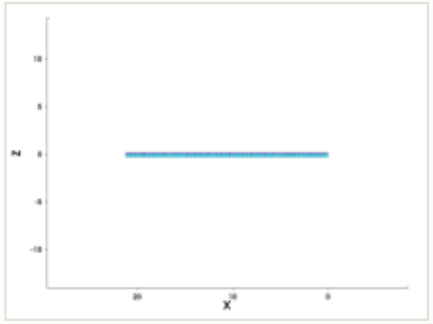
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



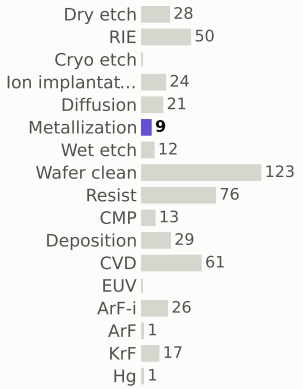
y cut at 57.351 µm



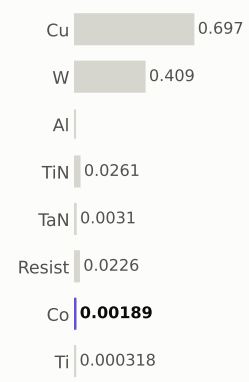
mask for this step

no mask for this step

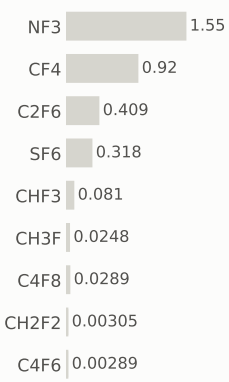
steps by process area



materials deposited (g)



gases emitted (g)



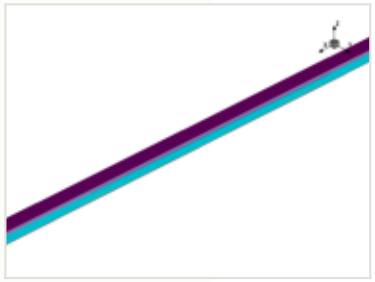
Step #6450: Deposition

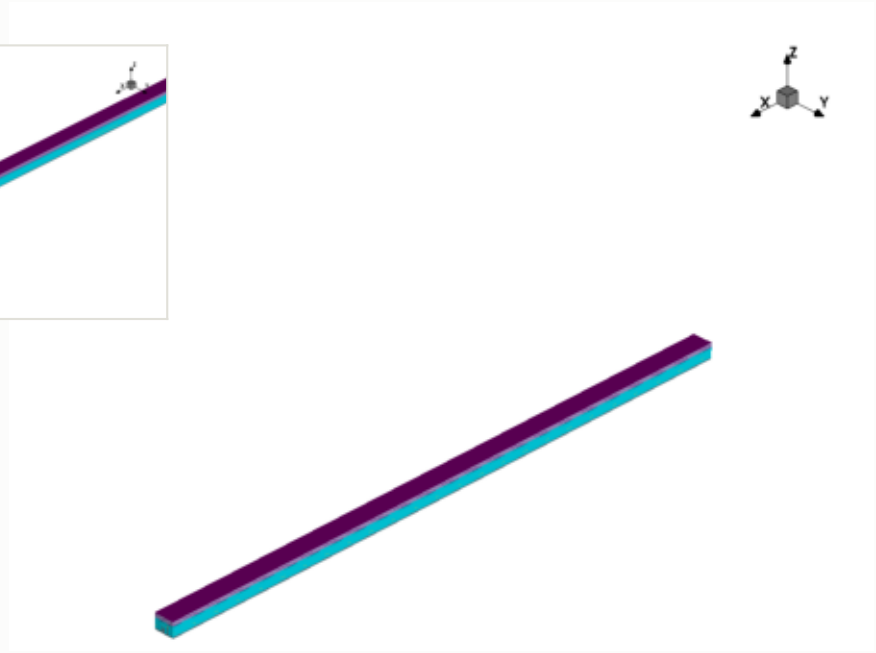
depo_fill · Deposition

0.656	294.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



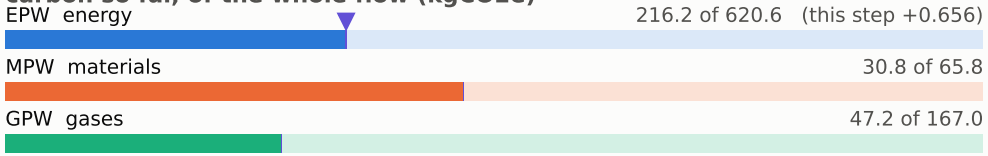


this step

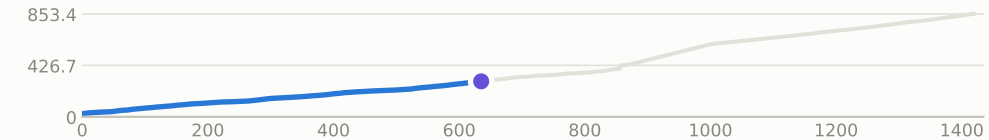
process area	Deposition
material	—
thickness	0.006 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

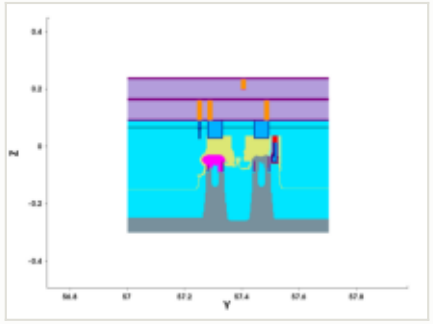
carbon so far, of the whole flow (kgCO2e)



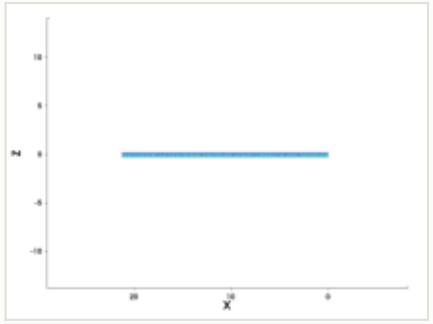
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



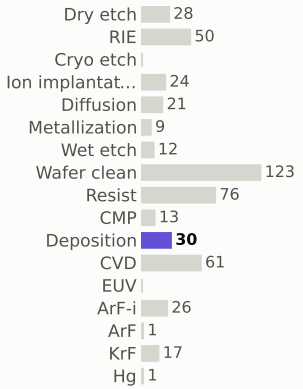
y cut at 57.351 µm



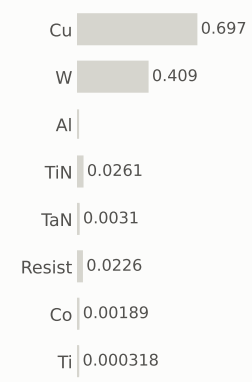
mask for this step

no mask for this step

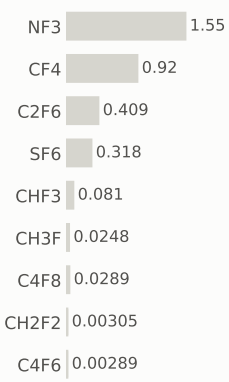
steps by process area



materials deposited (g)



gases emitted (g)



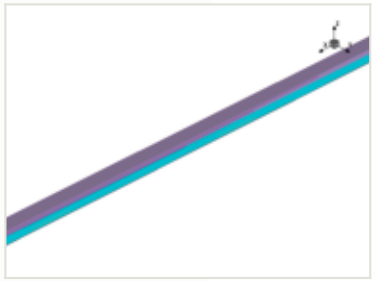
Step #6460: CVD

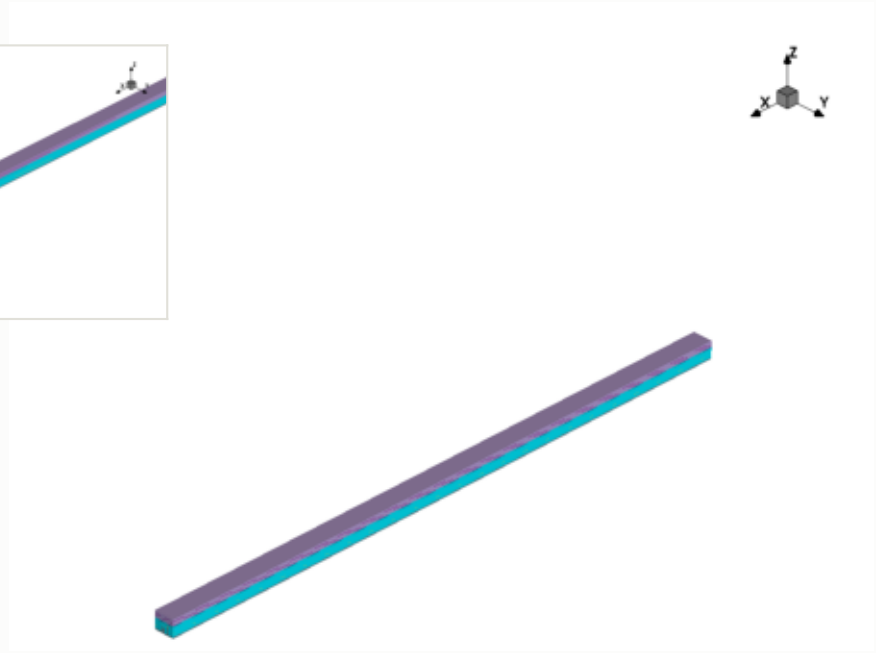
depo_fill · CVD

1.31	295.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



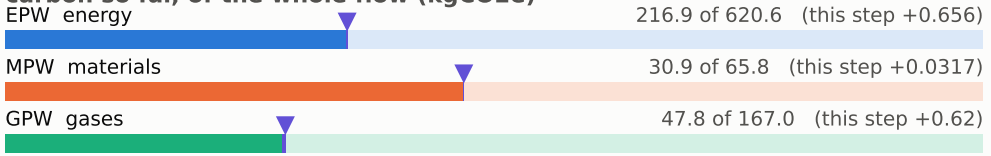


this step

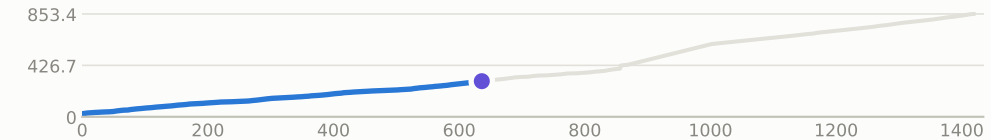
process area	CVD
material	—
thickness	0.066 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

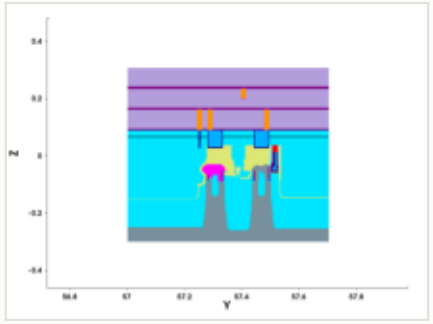
carbon so far, of the whole flow (kgCO2e)



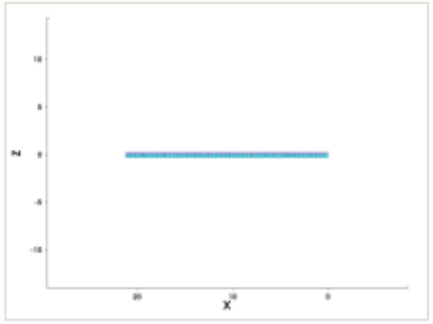
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



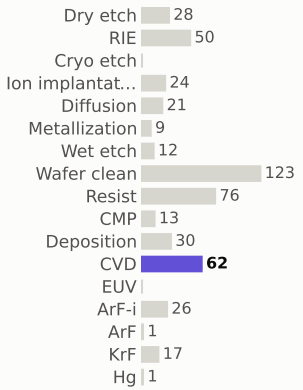
y cut at 57.351 µm



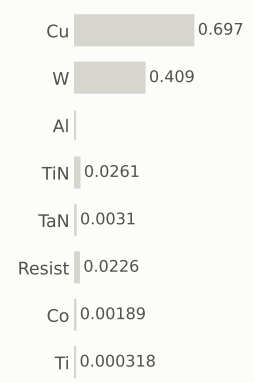
mask for this step

no mask for this step

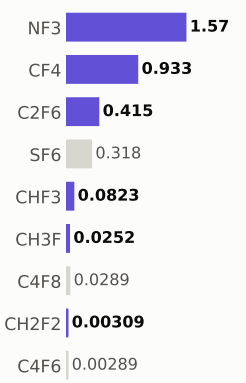
steps by process area



materials deposited (g)



gases emitted (g)



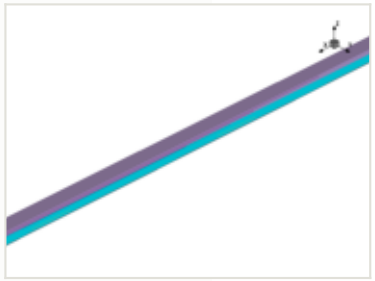
Step #6470: Diffusion

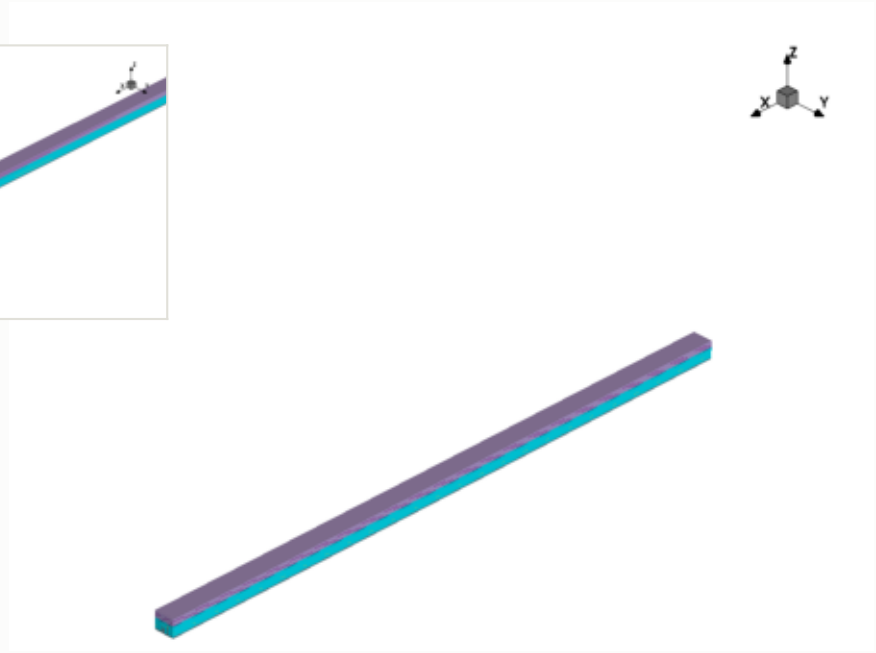
anneal · Diffusion

0.801	296.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





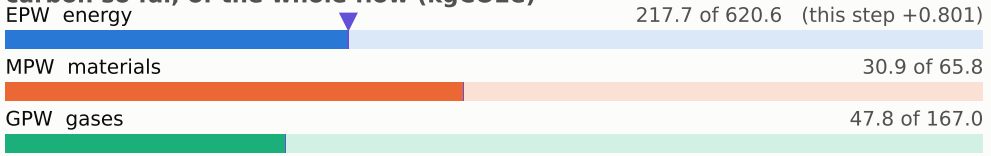
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

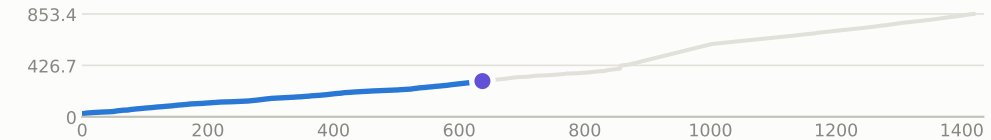
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

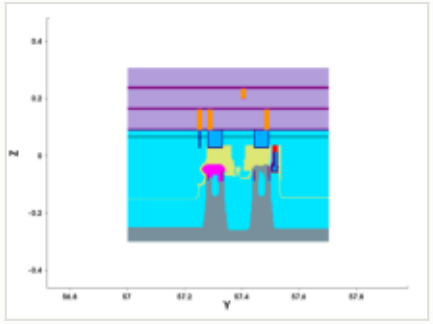
carbon so far, of the whole flow (kgCO2e)



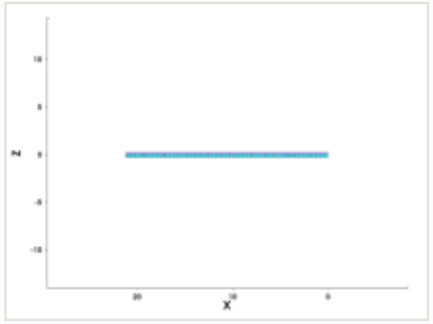
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



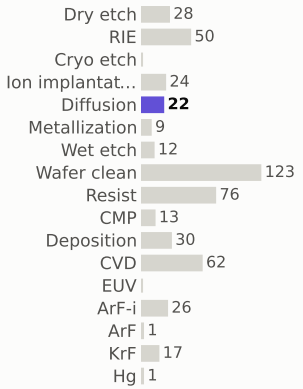
y cut at 57.351 µm



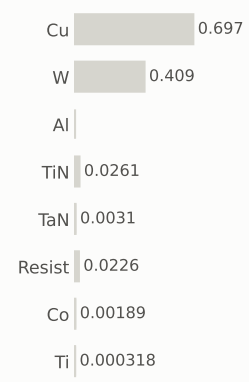
mask for this step

no mask for this step

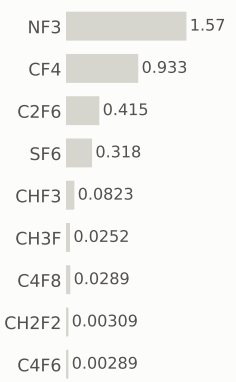
steps by process area



materials deposited (g)



gases emitted (g)



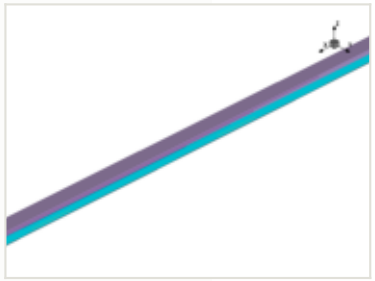
Step #6480: Wafer_clean

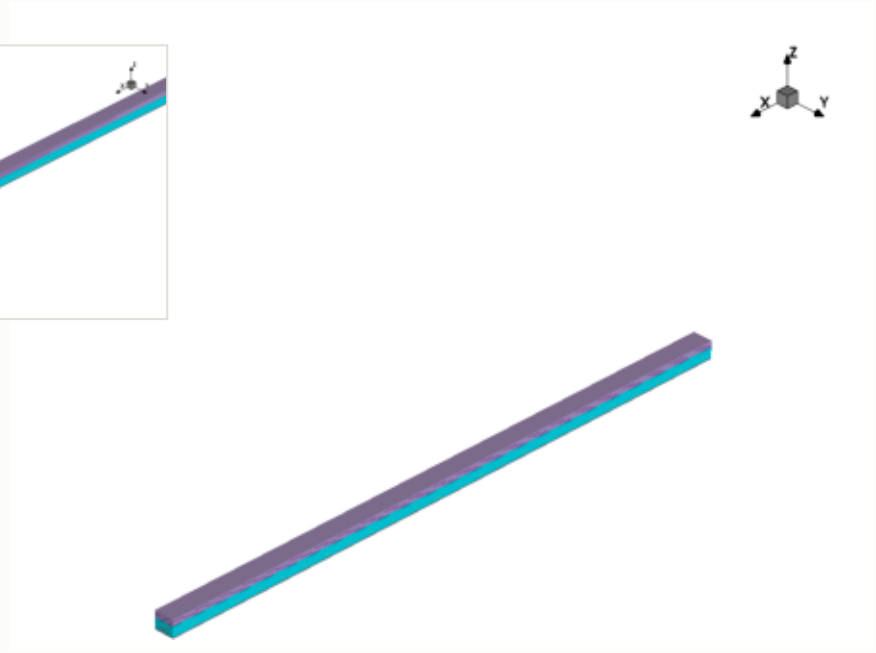
clean · Wafer_clean

0.169	296.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

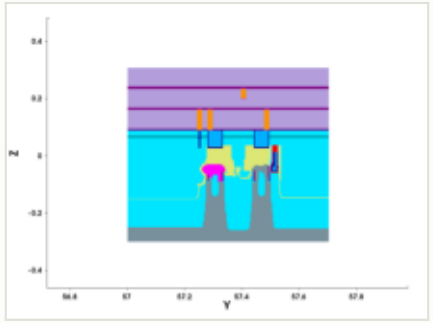
3D structure after this step

zoomed to fit

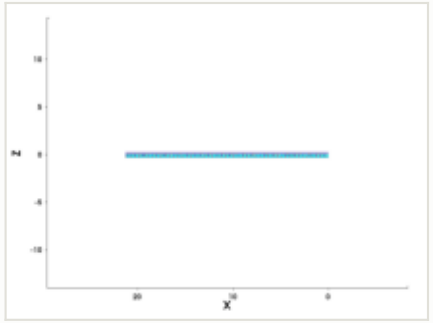




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

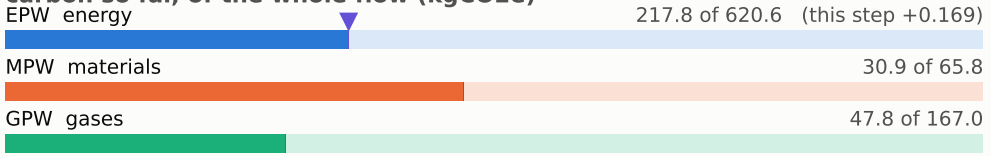
no mask for this step

this step

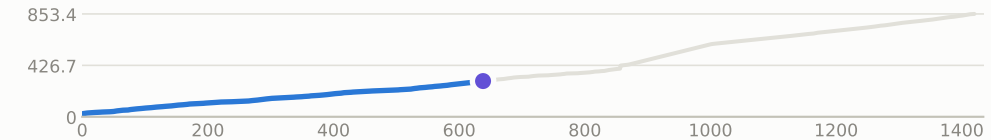
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

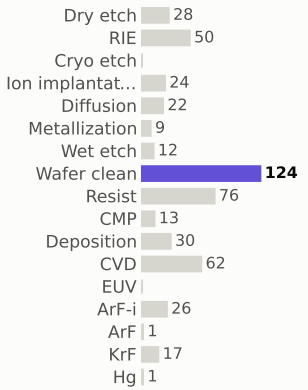
carbon so far, of the whole flow (kgCO2e)



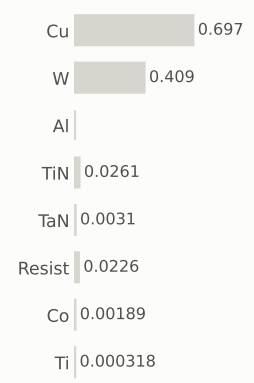
embodied carbon up to each step (kgCO2e per wafer)



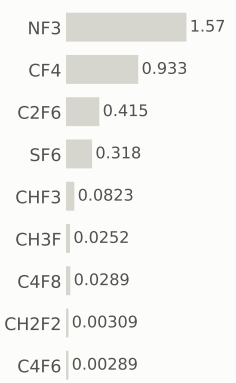
steps by process area



materials deposited (g)



gases emitted (g)



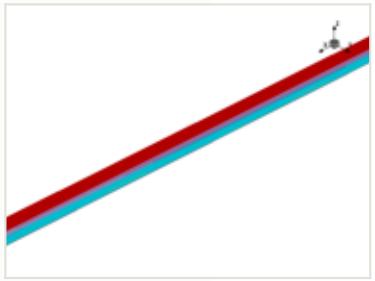
Step #6490: CVD

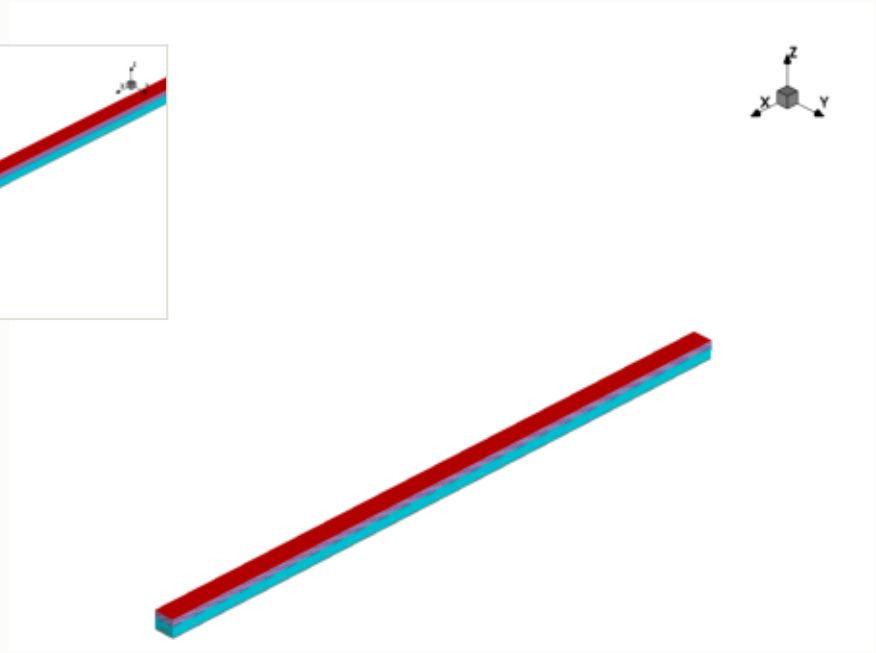
depo_fill · CVD

1.31	297.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



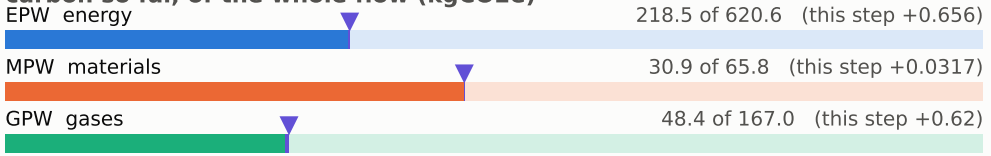


this step

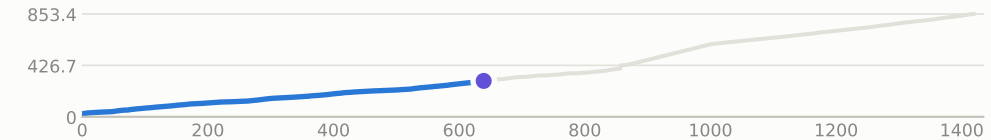
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

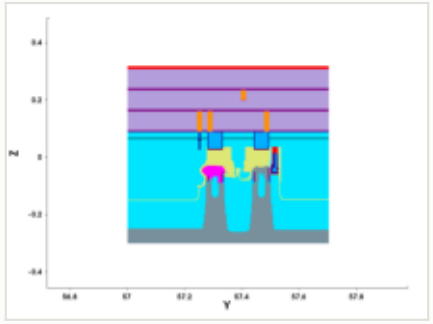
carbon so far, of the whole flow (kgCO2e)



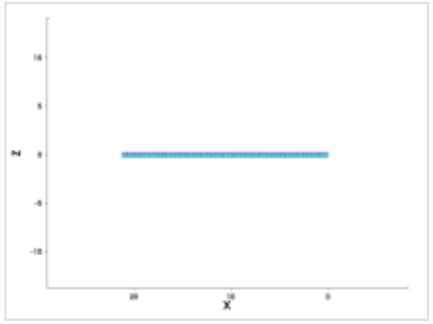
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



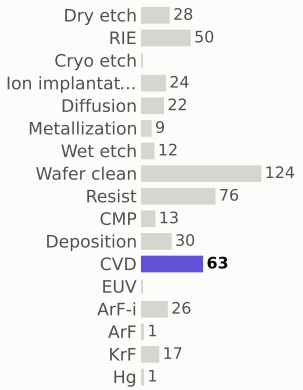
y cut at 57.351 µm



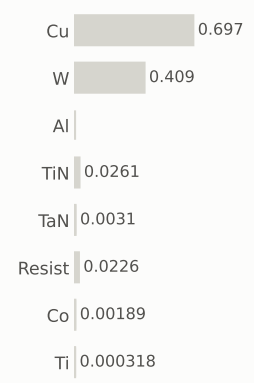
mask for this step

no mask for this step

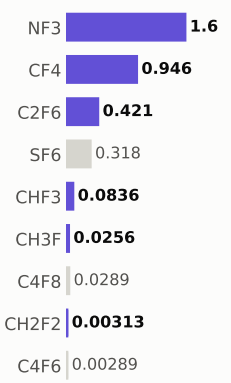
steps by process area



materials deposited (g)



gases emitted (g)



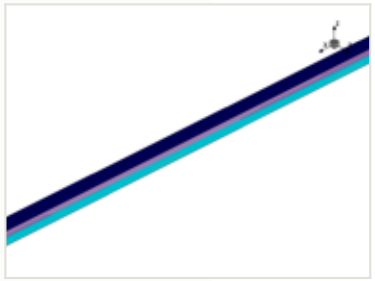
Step #6500: Deposition

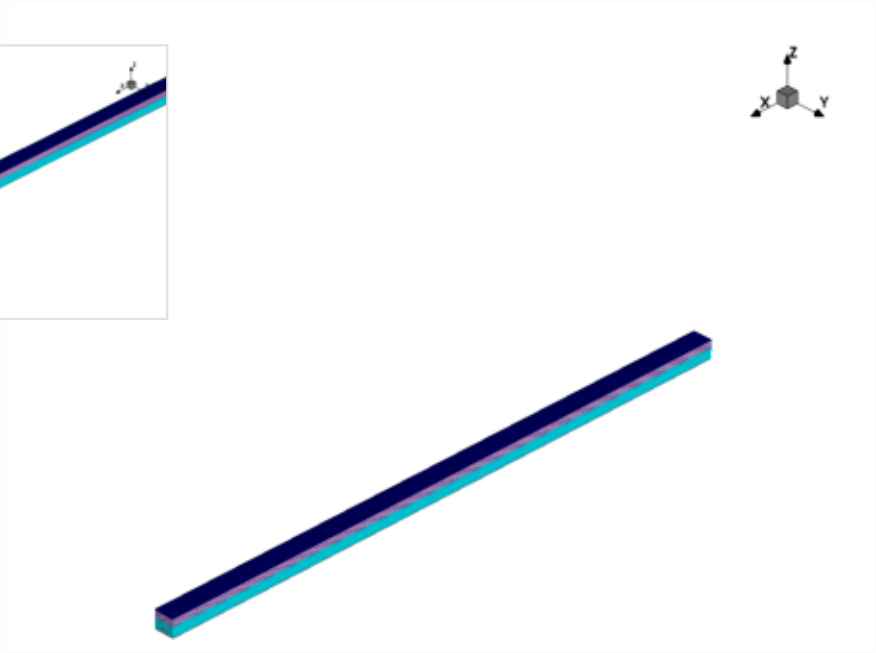
depo_fill · Deposition

0.657	298.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



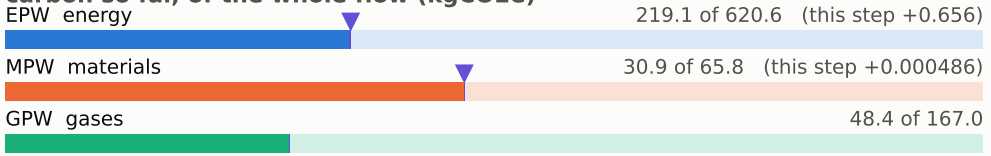


this step

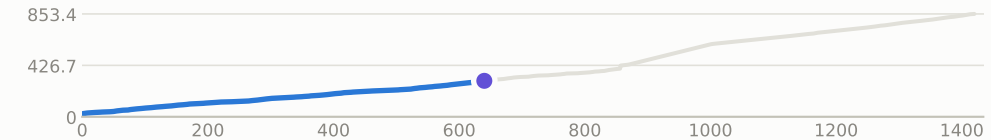
process area	Deposition
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000486 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

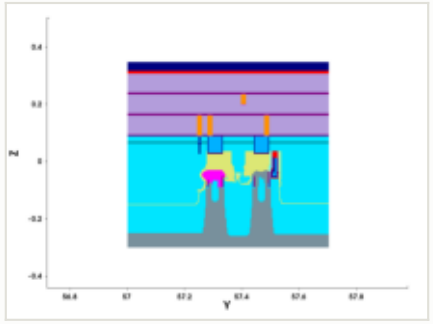
carbon so far, of the whole flow (kgCO2e)



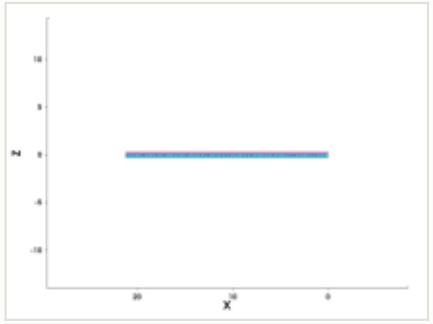
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



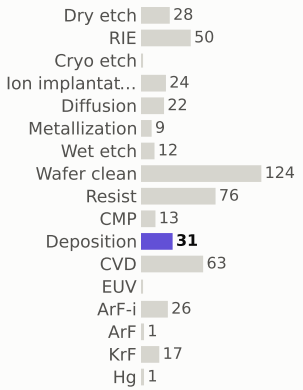
y cut at 57.351 µm



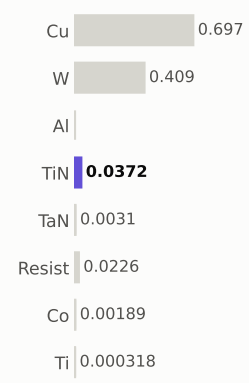
mask for this step

no mask for this step

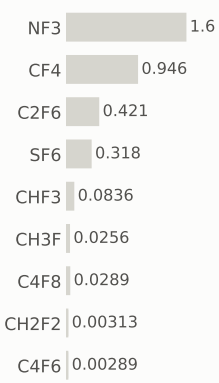
steps by process area



materials deposited (g)



gases emitted (g)



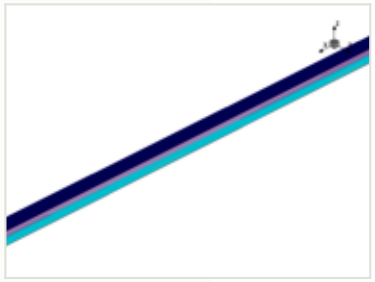
Step #6510: Wafer_clean

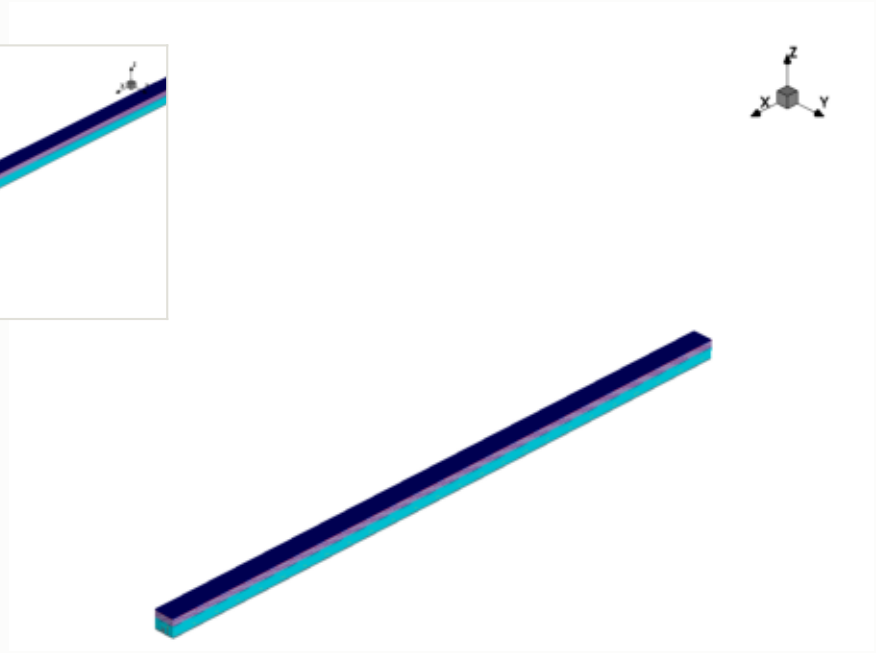
clean · Wafer_clean

0.169	298.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





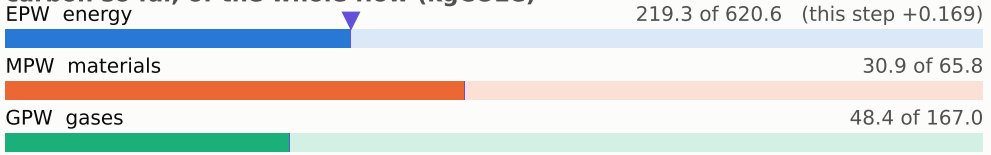
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

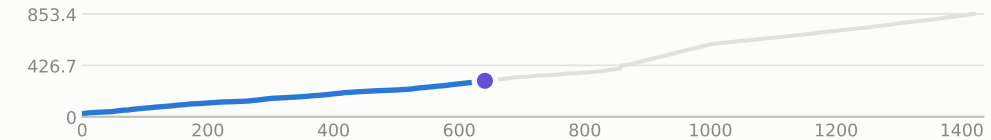
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

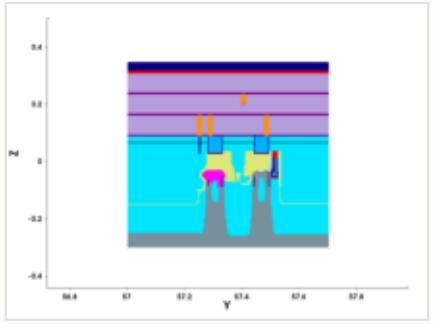
carbon so far, of the whole flow (kgCO2e)



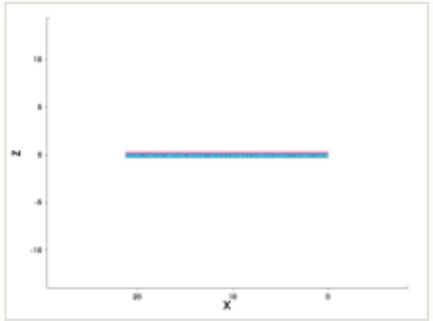
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



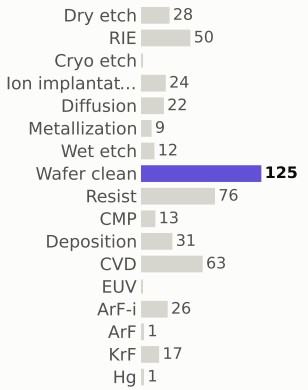
y cut at 57.351 µm



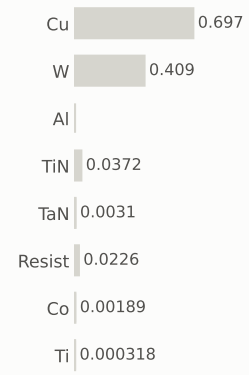
mask for this step

no mask for this step

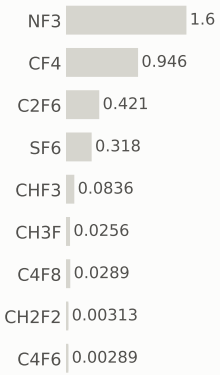
steps by process area



materials deposited (g)



gases emitted (g)



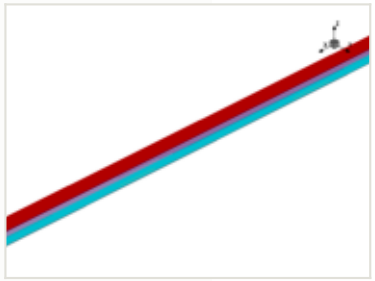
Step #6520: CVD

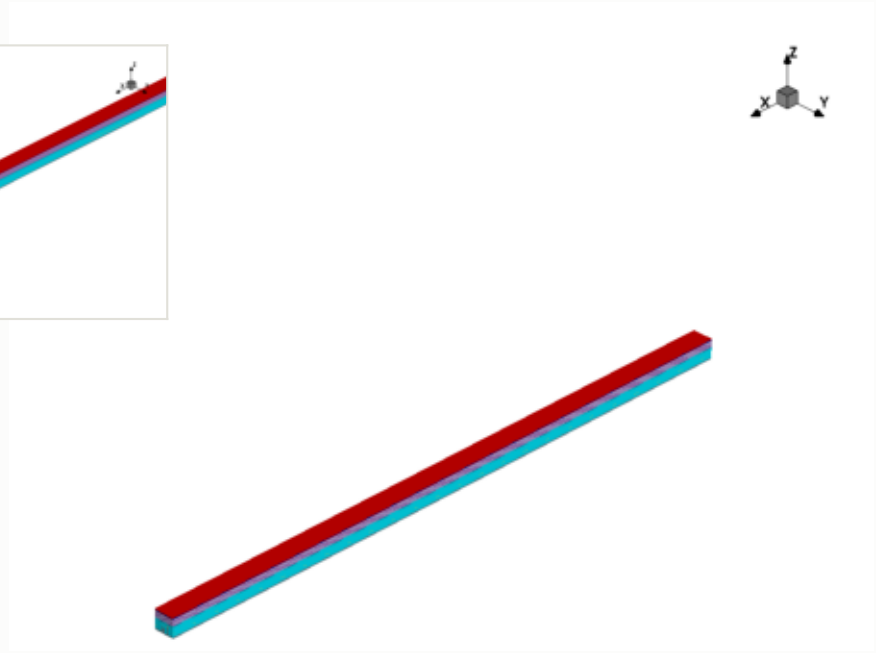
depo_fill · CVD

1.31	299.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



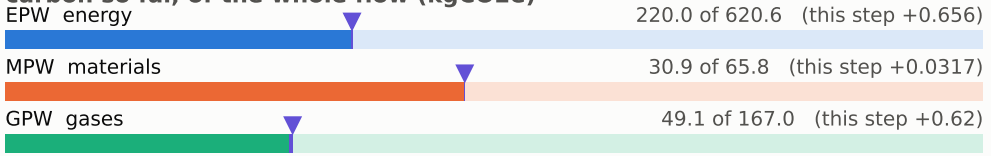


this step

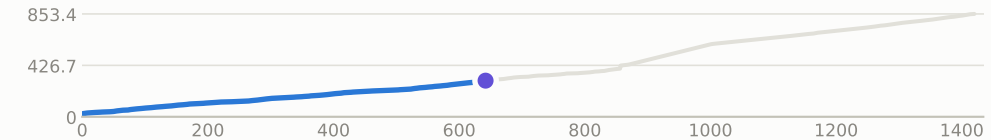
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

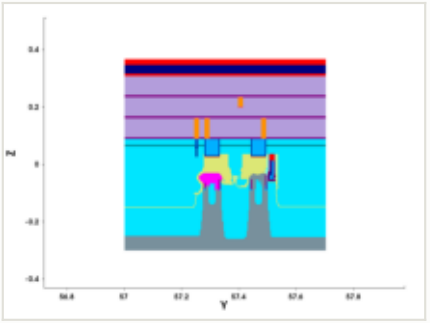
carbon so far, of the whole flow (kgCO2e)



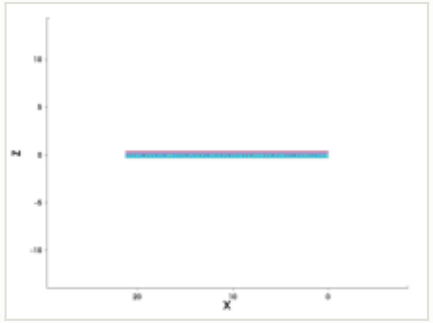
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



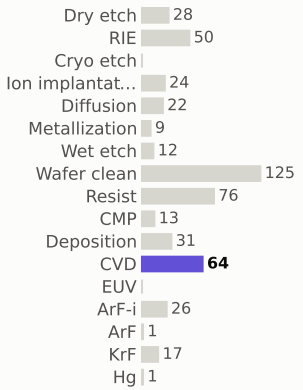
y cut at 57.351 µm



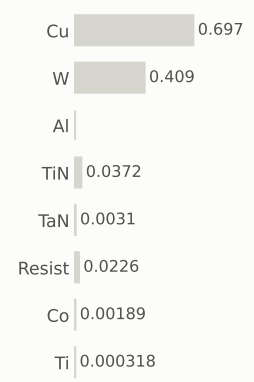
mask for this step

no mask for this step

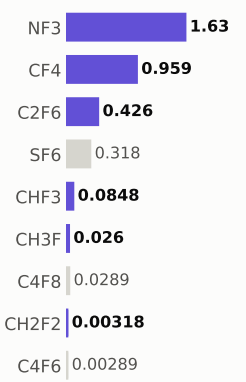
steps by process area



materials deposited (g)



gases emitted (g)



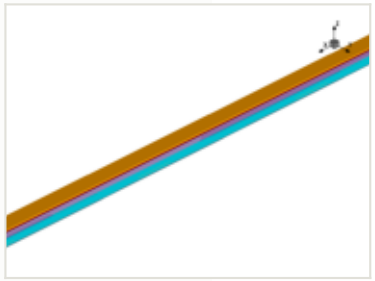
Step #6530: Resist

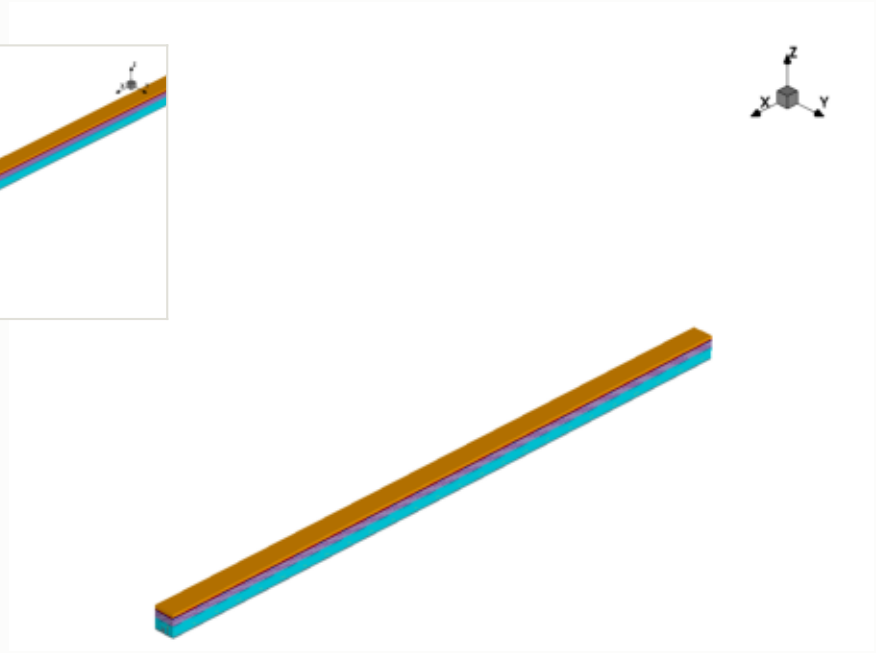
depo_fill · Resist

0.169	300.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



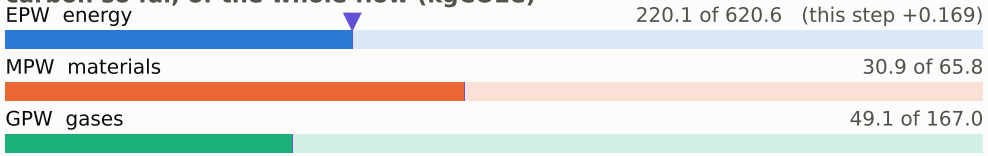


this step

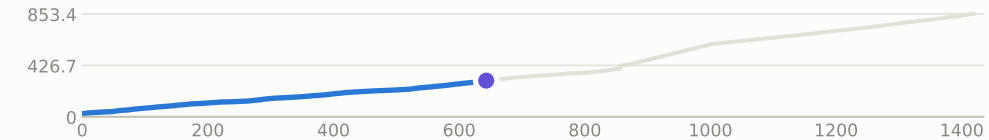
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

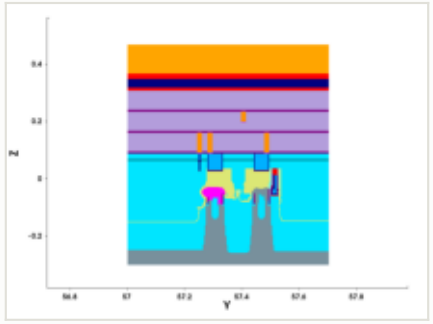
carbon so far, of the whole flow (kgCO2e)



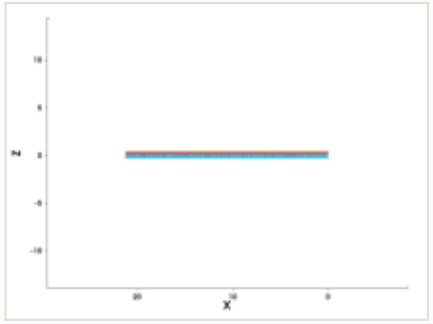
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



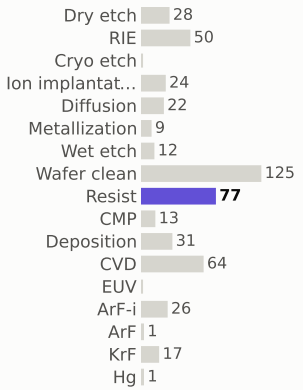
y cut at 57.351 µm



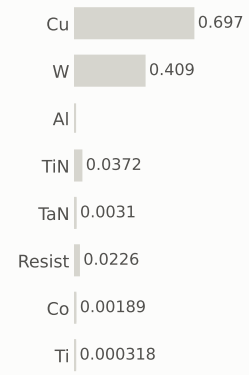
mask for this step

no mask for this step

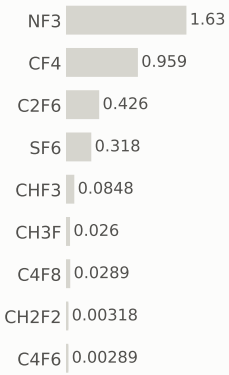
steps by process area



materials deposited (g)



gases emitted (g)



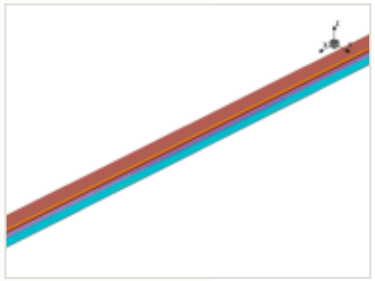
Step #6540: CVD

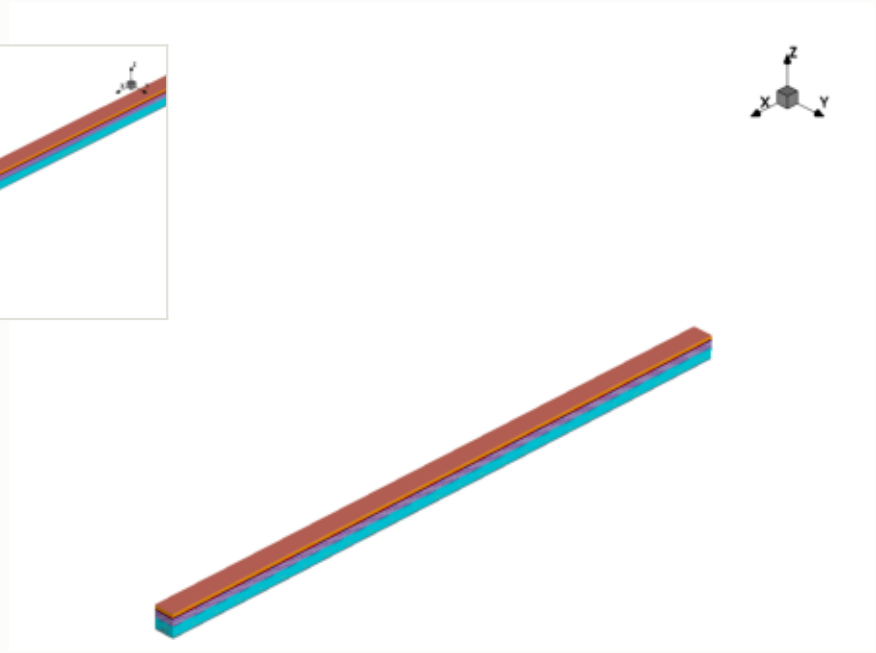
depo_fill · CVD

1.31	301.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



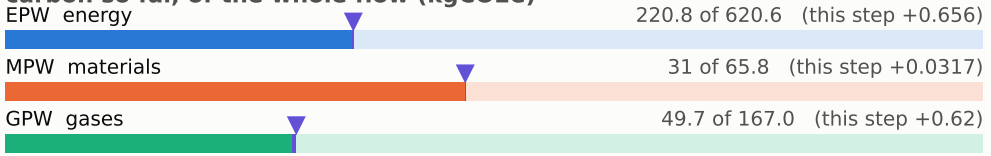


this step

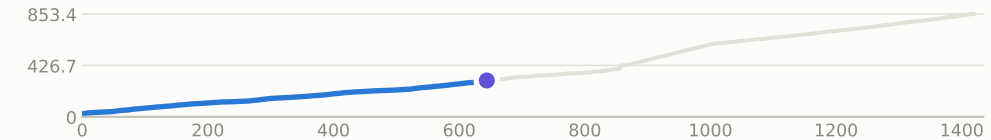
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

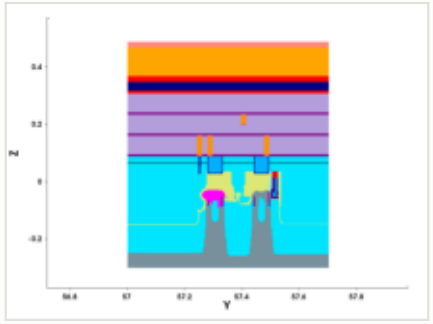
carbon so far, of the whole flow (kgCO2e)



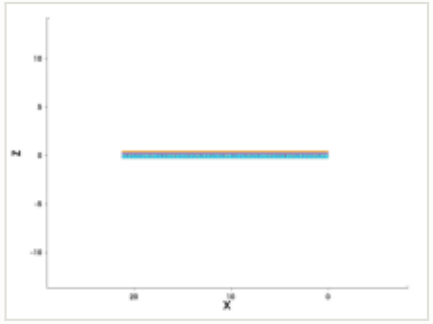
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



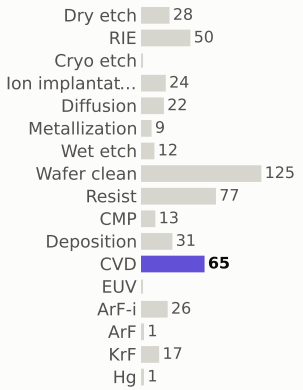
y cut at 57.351 µm



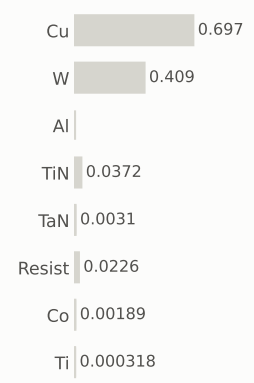
mask for this step

no mask for this step

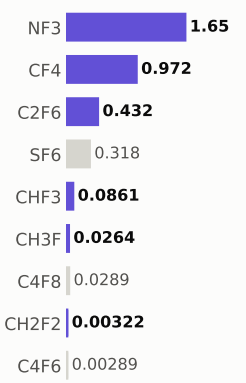
steps by process area



materials deposited (g)



gases emitted (g)



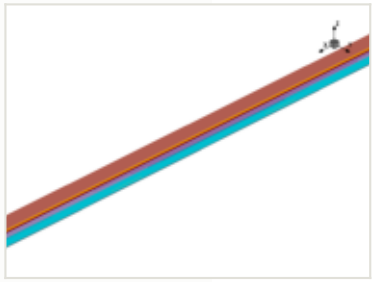
Step #6550: Wafer_clean

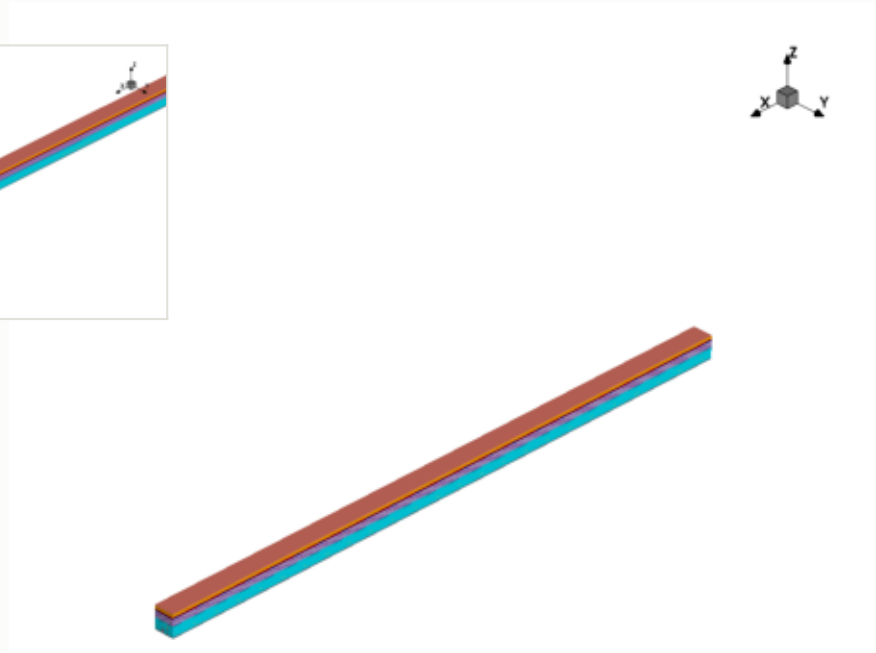
clean · Wafer_clean

0.169	301.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

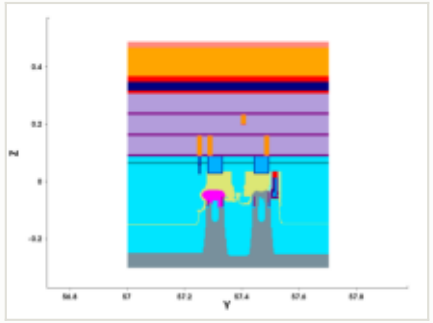
3D structure after this step

zoomed to fit

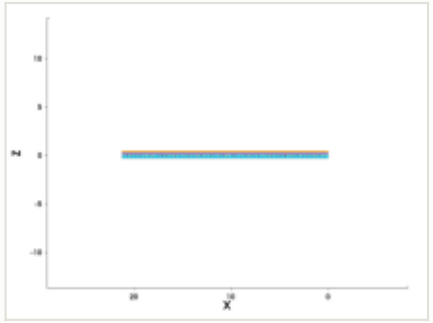




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

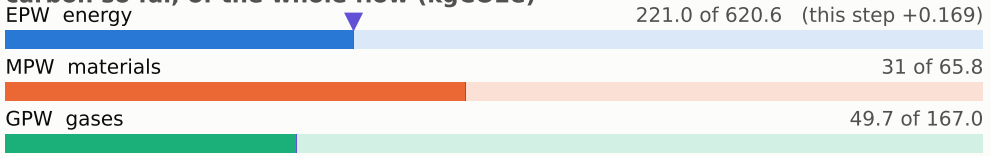
no mask for this step

this step

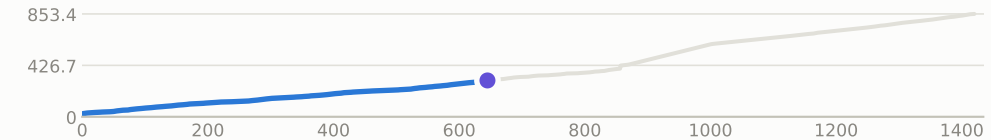
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

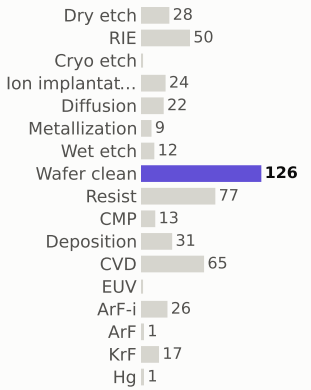
carbon so far, of the whole flow (kgCO2e)



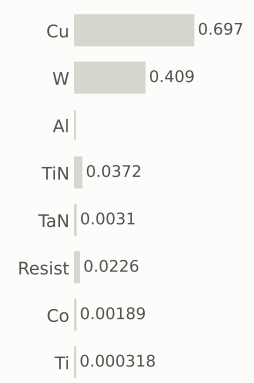
embodied carbon up to each step (kgCO2e per wafer)



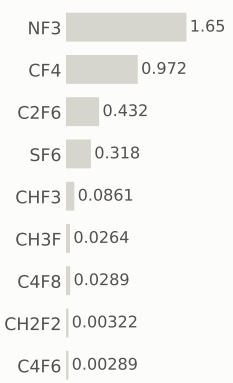
steps by process area



materials deposited (g)



gases emitted (g)



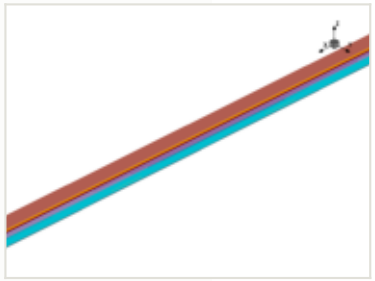
Step #6560: Resist

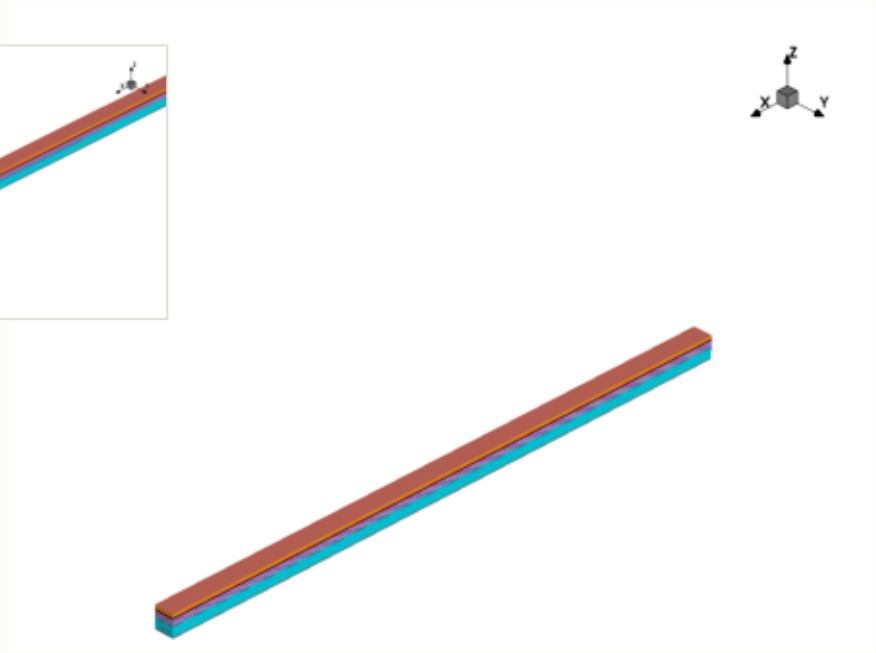
resist · Resist

0.169	301.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





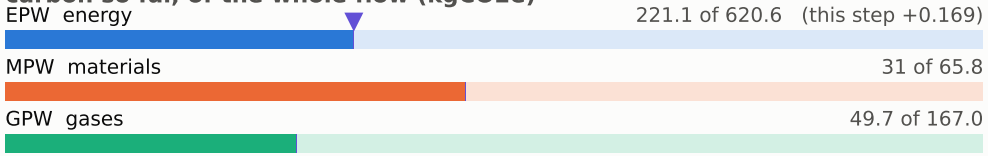
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

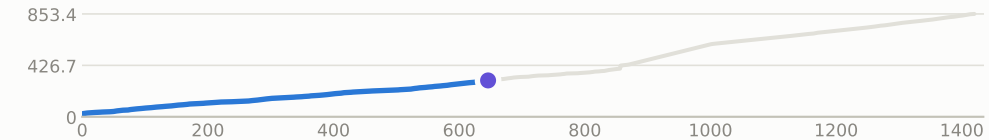
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

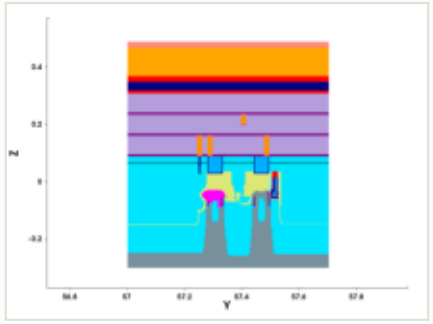
carbon so far, of the whole flow (kgCO2e)



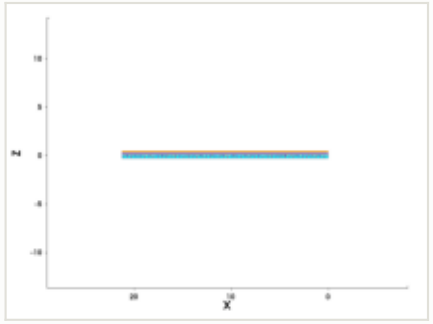
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



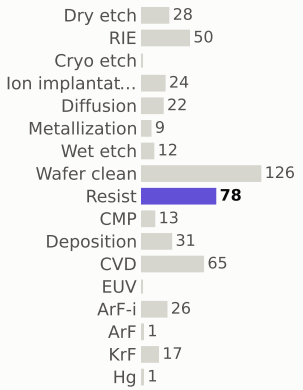
y cut at 57.351 µm



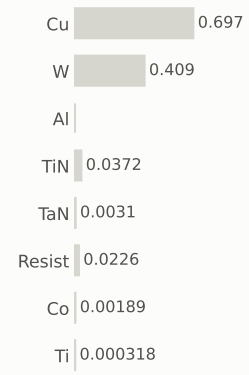
mask for this step

no mask for this step

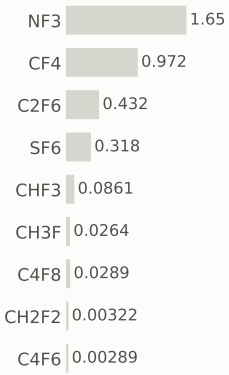
steps by process area



materials deposited (g)



gases emitted (g)



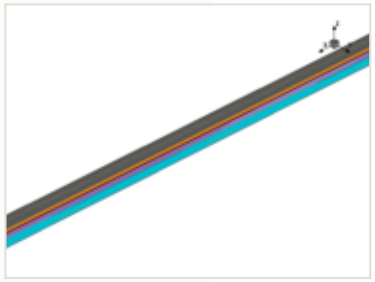
Step #6570: ArF-i

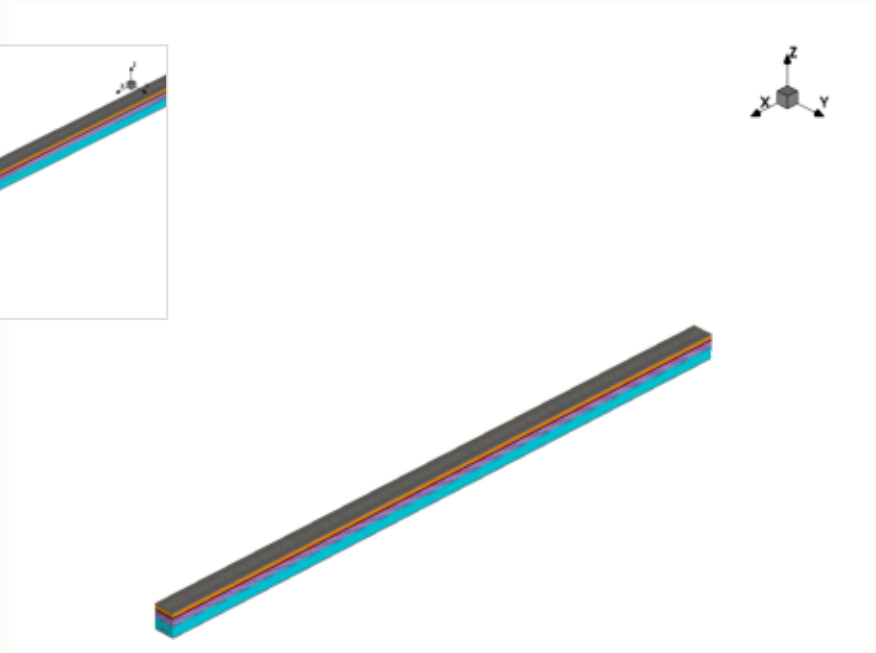
litho_patterning · ArF-i

0.217	302.0	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



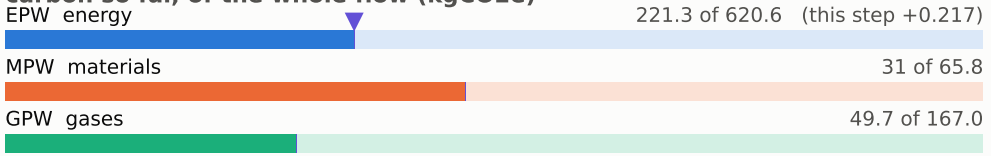


this step

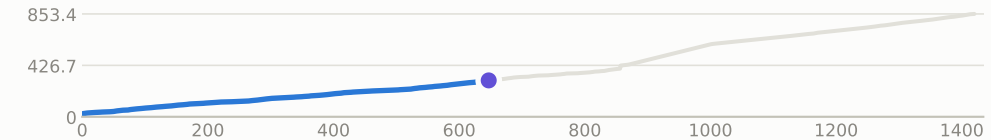
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

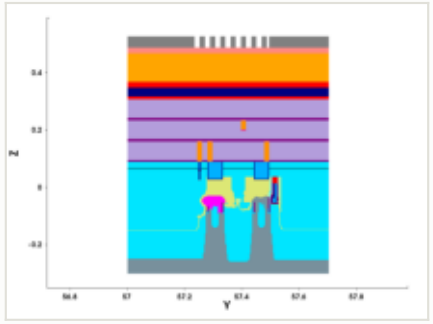
carbon so far, of the whole flow (kgCO2e)



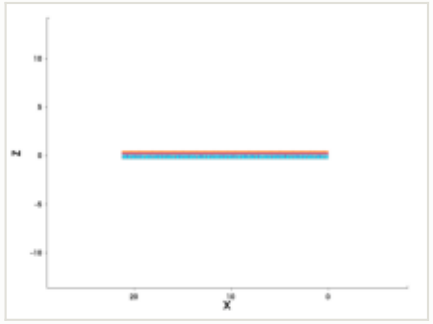
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

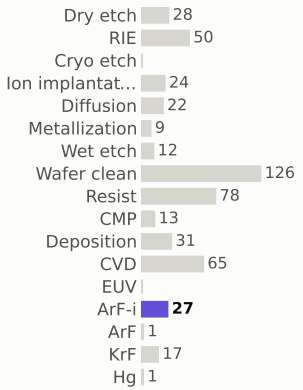


y cut at 57.351 µm

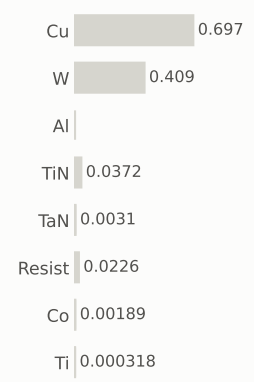


mask: layer 2030:0 (GDSII)

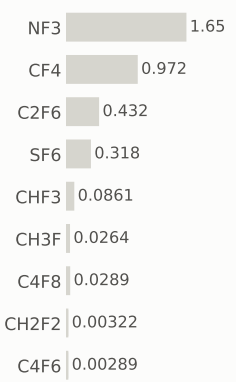
steps by process area



materials deposited (g)



gases emitted (g)



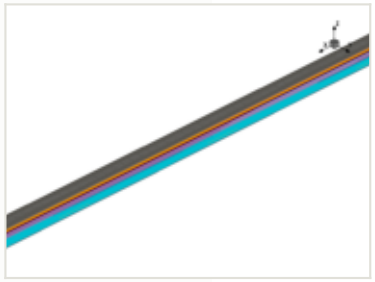
Step #6580: RIE

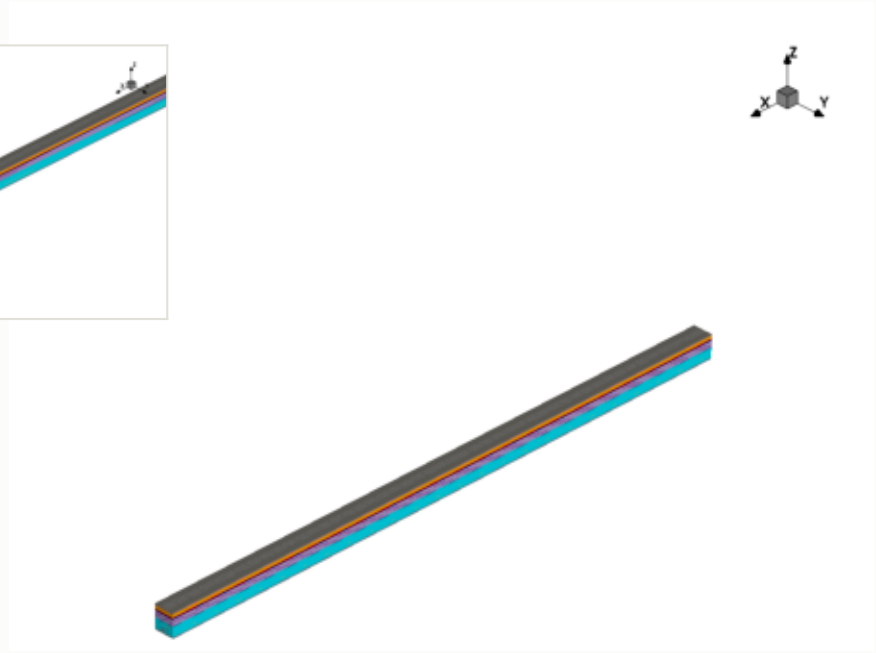
etch_anisotropic · RIE

1.04	303.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



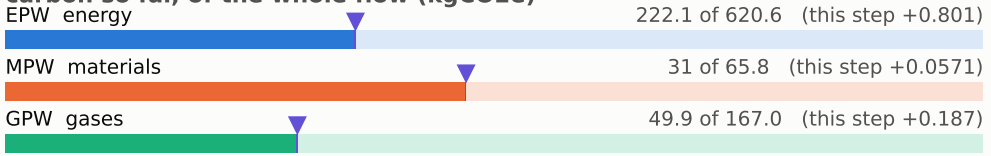


this step

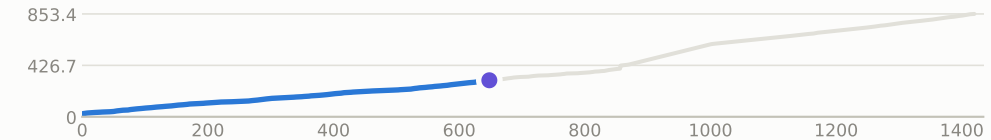
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

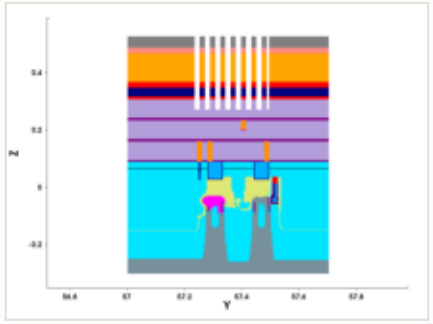
carbon so far, of the whole flow (kgCO2e)



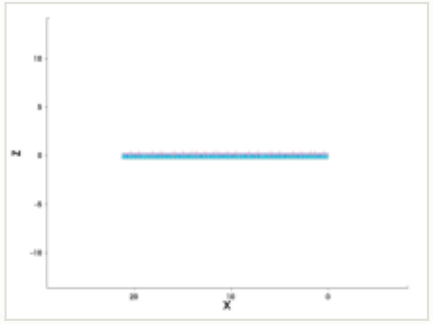
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



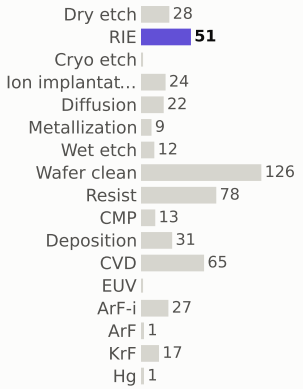
y cut at 57.351 µm



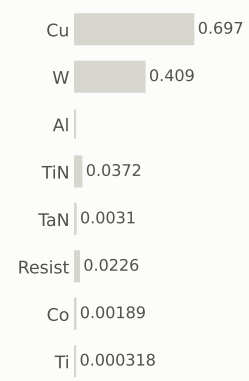
mask for this step

no mask for this step

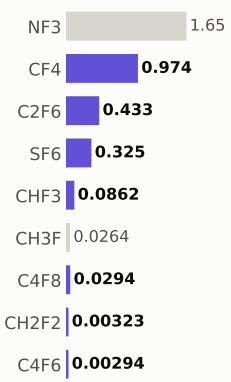
steps by process area



materials deposited (g)



gases emitted (g)



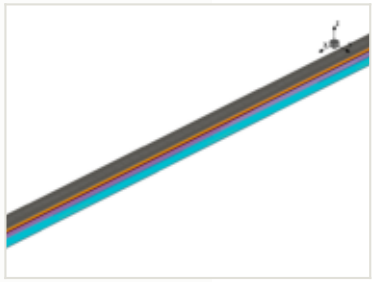
Step #6590: Wafer_clean

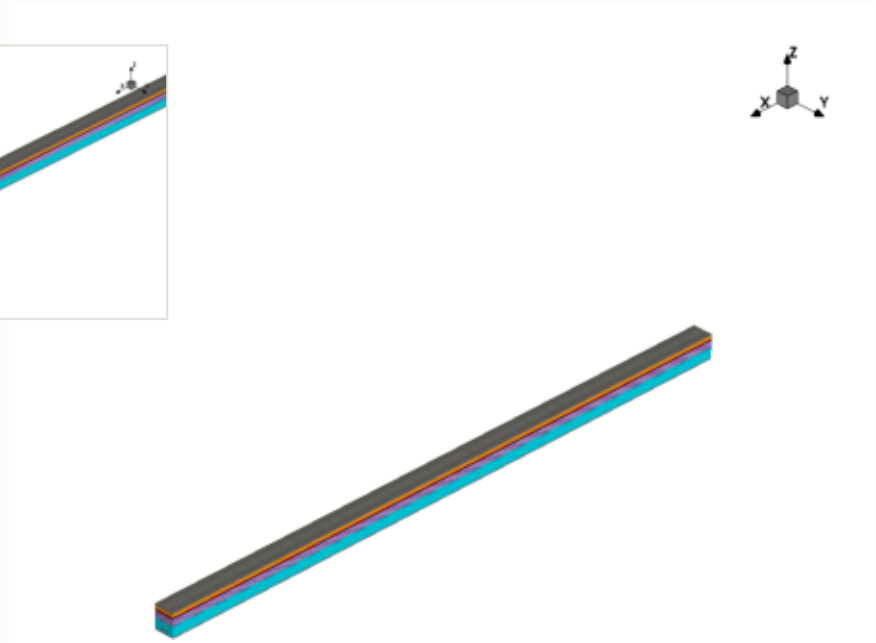
clean · Wafer_clean

0.169	303.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

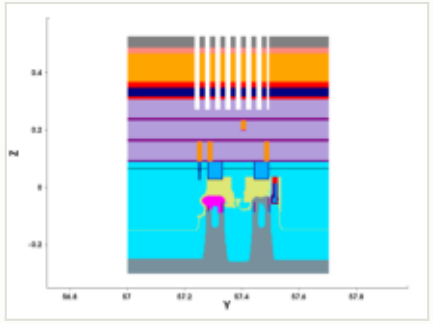
3D structure after this step

zoomed to fit

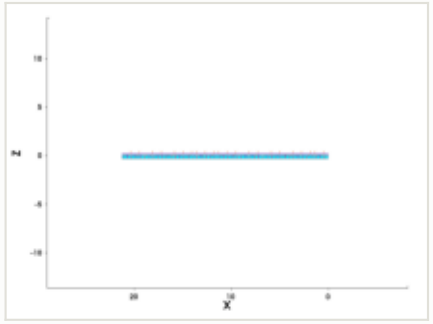




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

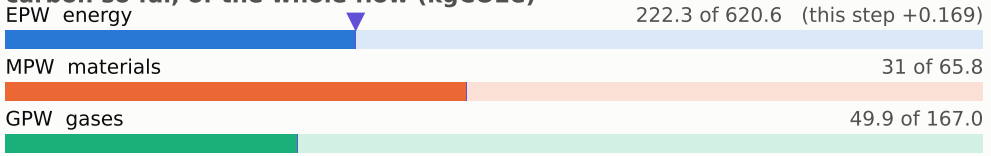
no mask for this step

this step

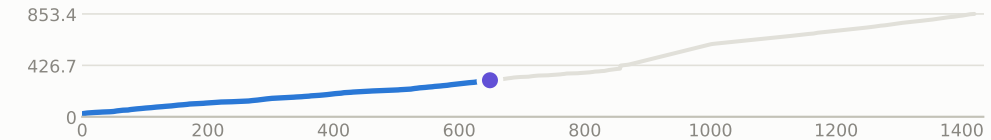
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

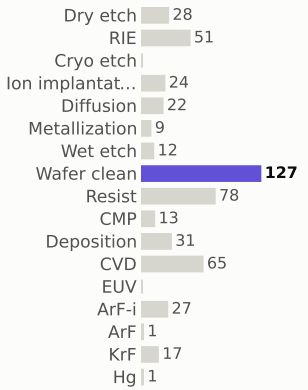
carbon so far, of the whole flow (kgCO2e)



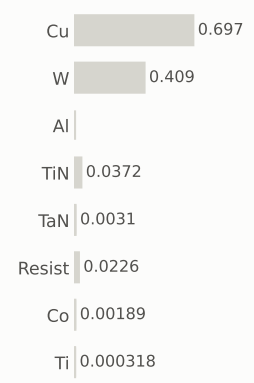
embodied carbon up to each step (kgCO2e per wafer)



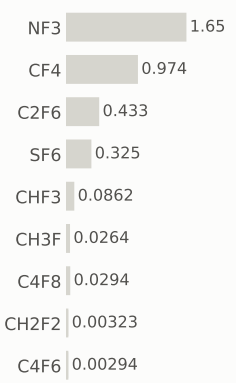
steps by process area



materials deposited (g)



gases emitted (g)



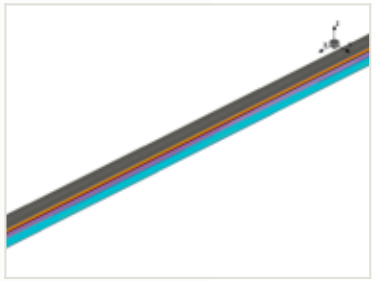
Step #6600: Resist

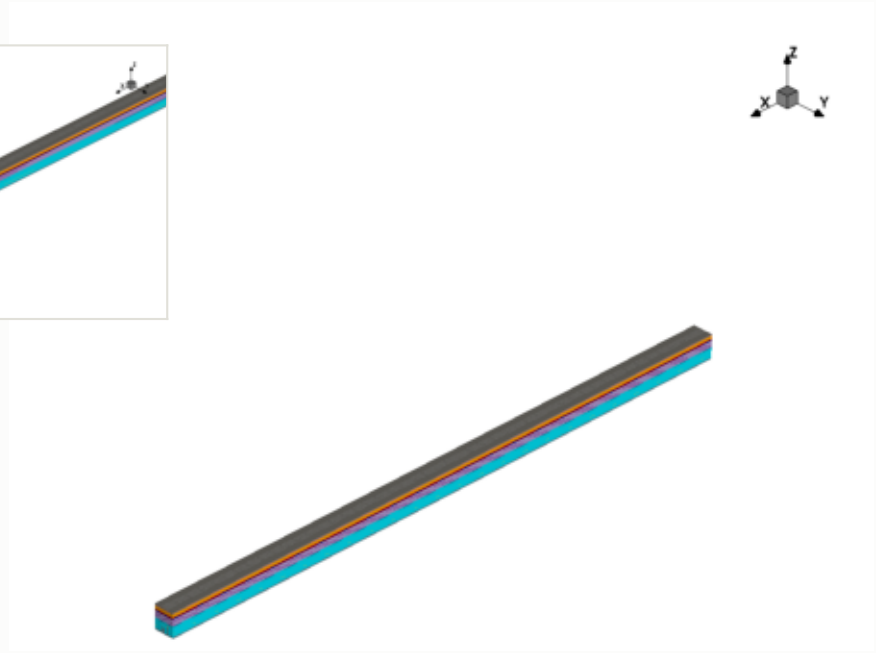
depo_fill · Resist

0.169	303.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



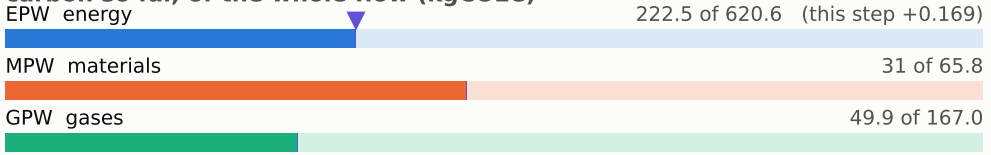


this step

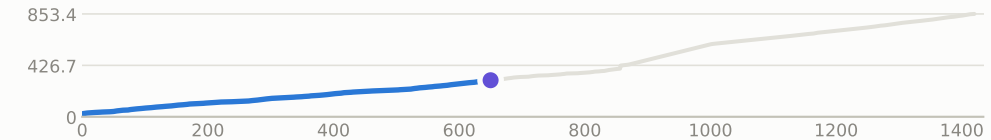
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

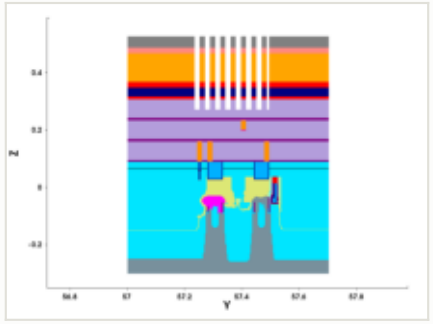
carbon so far, of the whole flow (kgCO2e)



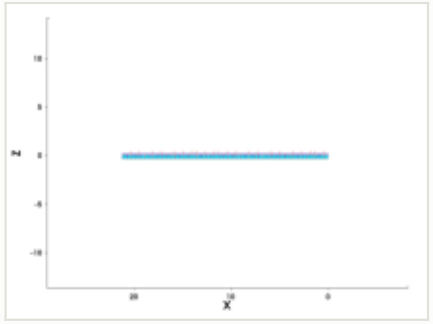
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



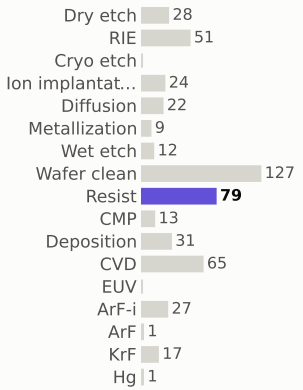
y cut at 57.351 µm



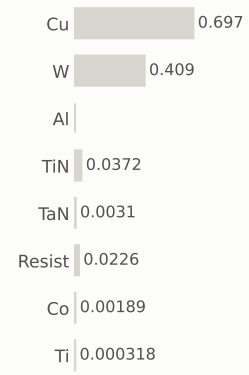
mask for this step

no mask for this step

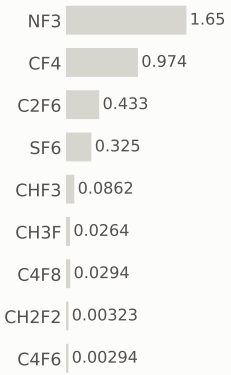
steps by process area



materials deposited (g)



gases emitted (g)



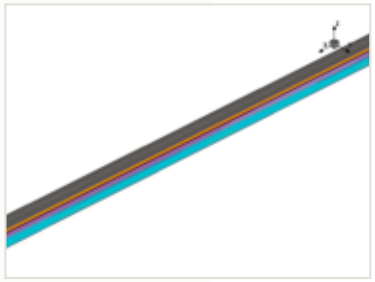
Step #6610: CVD

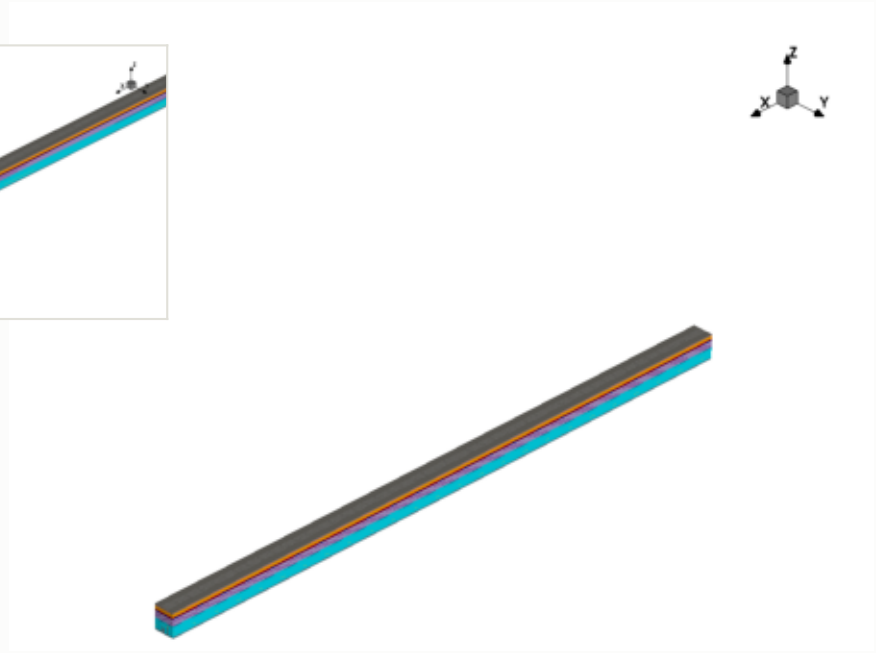
depo_isotropic · CVD

1.31	304.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



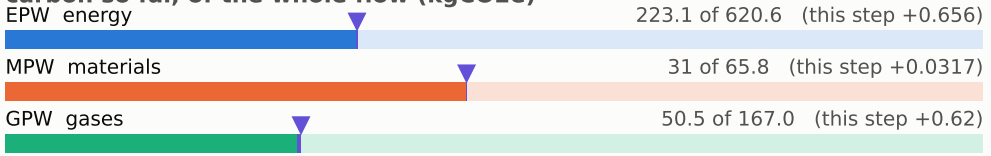


this step

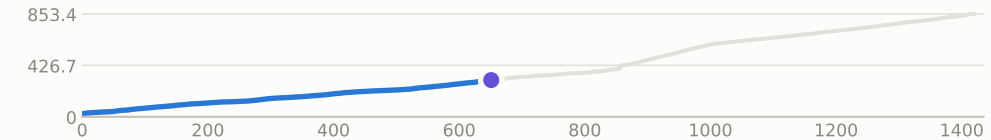
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

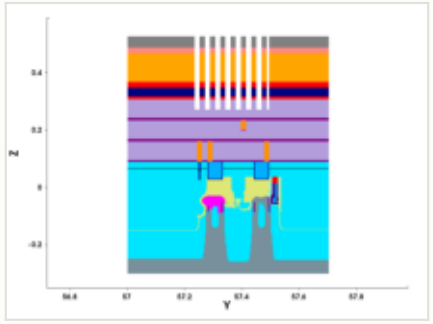
carbon so far, of the whole flow (kgCO2e)



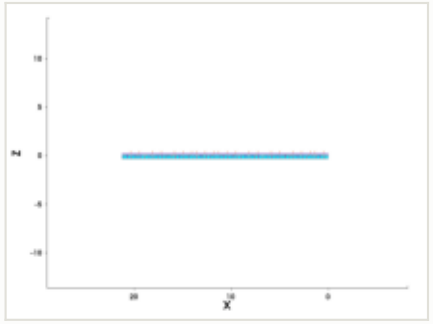
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



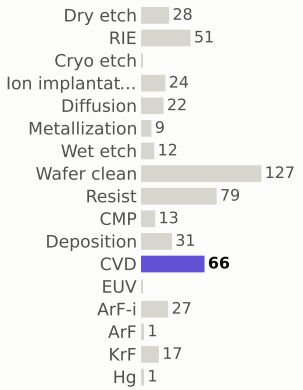
y cut at 57.351 µm



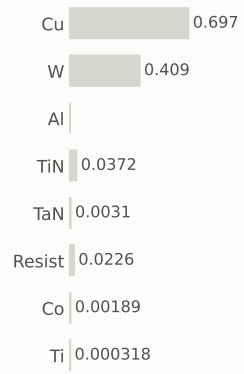
mask for this step

no mask for this step

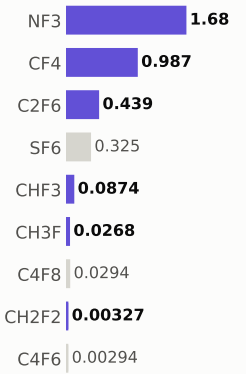
steps by process area



materials deposited (g)



gases emitted (g)



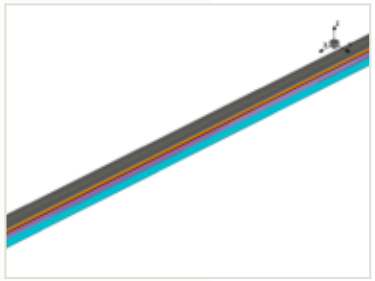
Step #6620: Wafer_clean

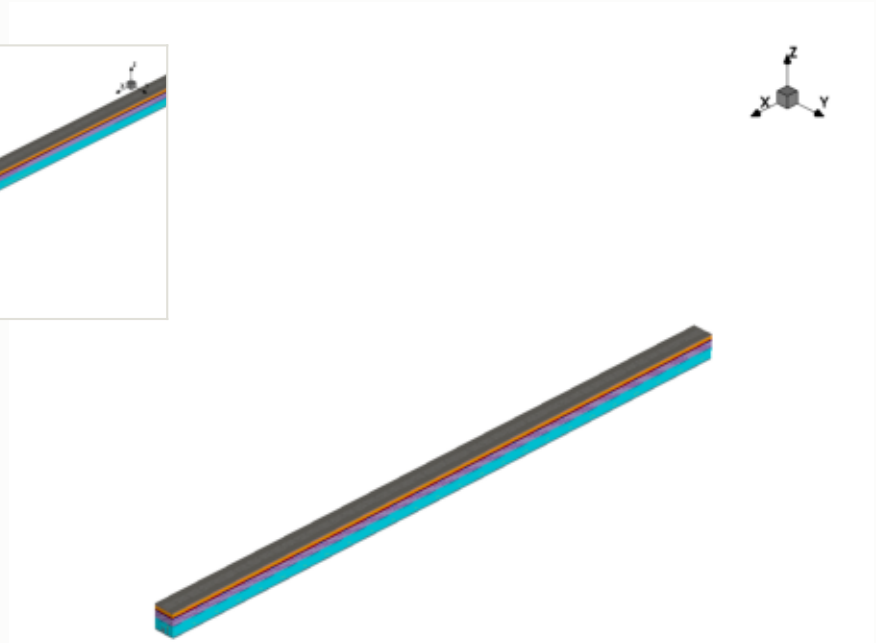
clean · Wafer_clean

0.169	304.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



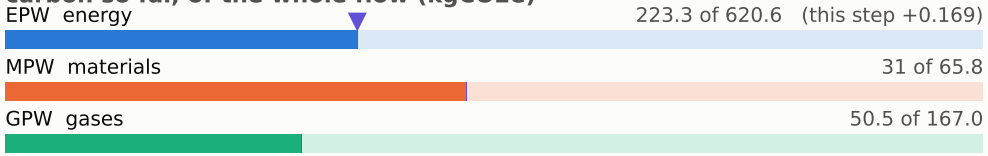


this step

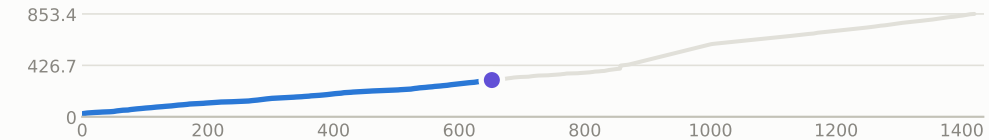
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

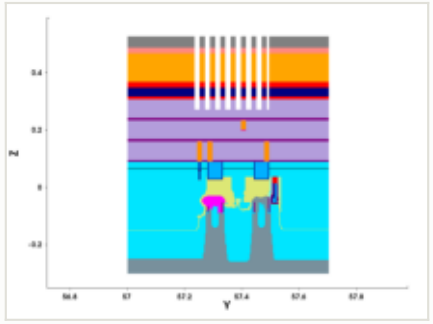
carbon so far, of the whole flow (kgCO2e)



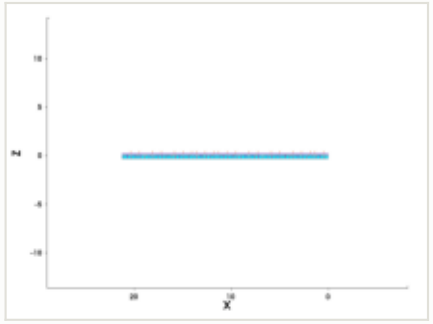
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



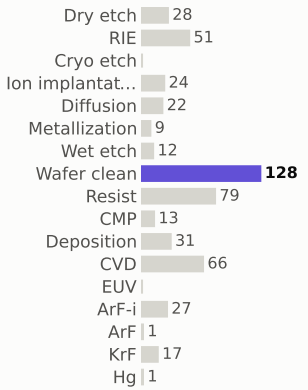
y cut at 57.351 µm



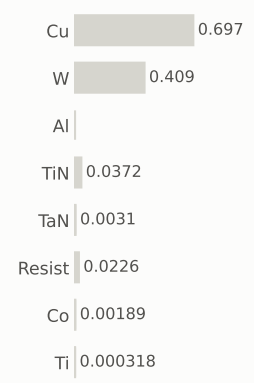
mask for this step

no mask for this step

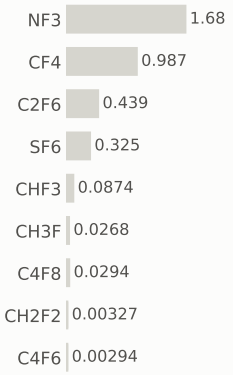
steps by process area



materials deposited (g)



gases emitted (g)



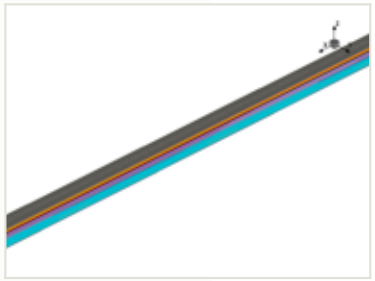
Step #6630: Resist

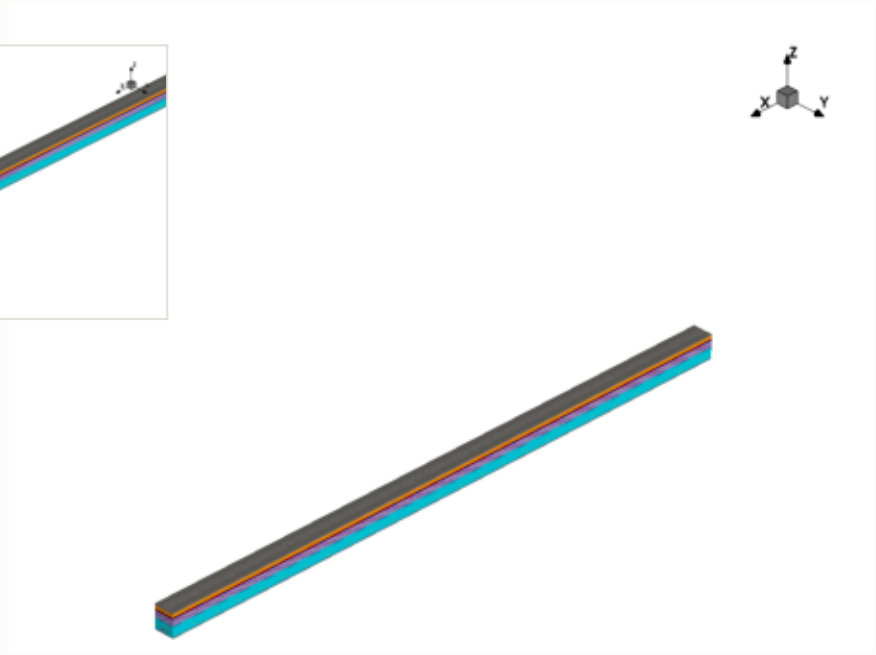
resist · Resist

0.169	305.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





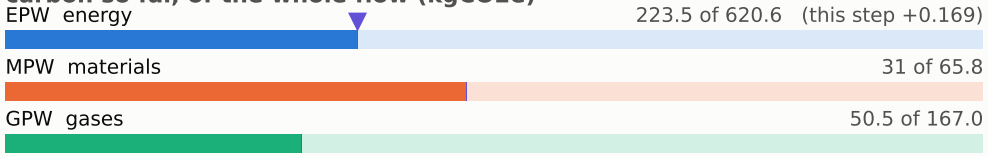
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

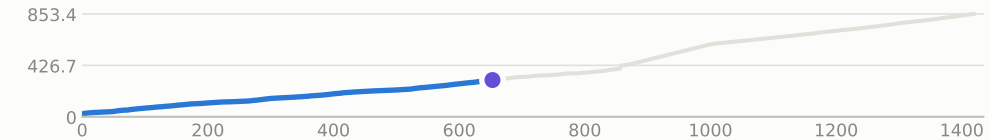
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

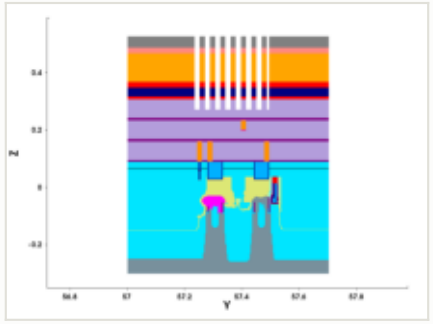
carbon so far, of the whole flow (kgCO2e)



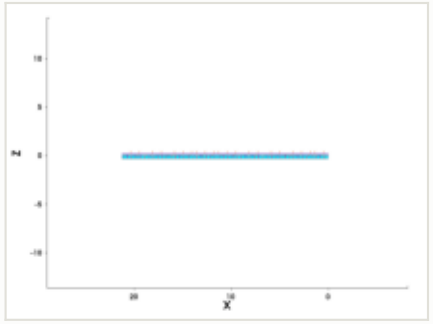
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



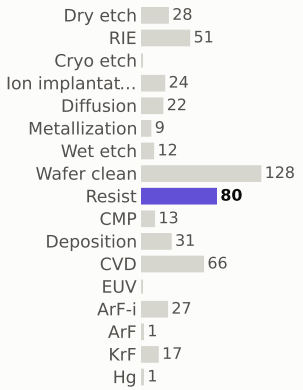
y cut at 57.351 µm



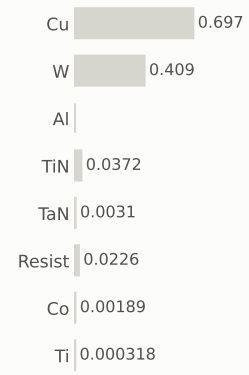
mask for this step

no mask for this step

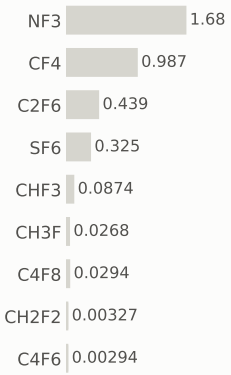
steps by process area



materials deposited (g)



gases emitted (g)



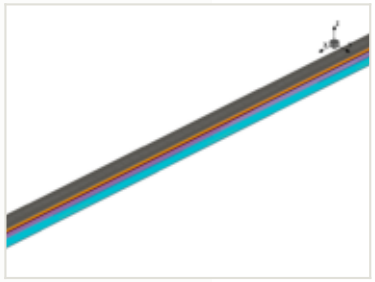
Step #6640: ArF-i

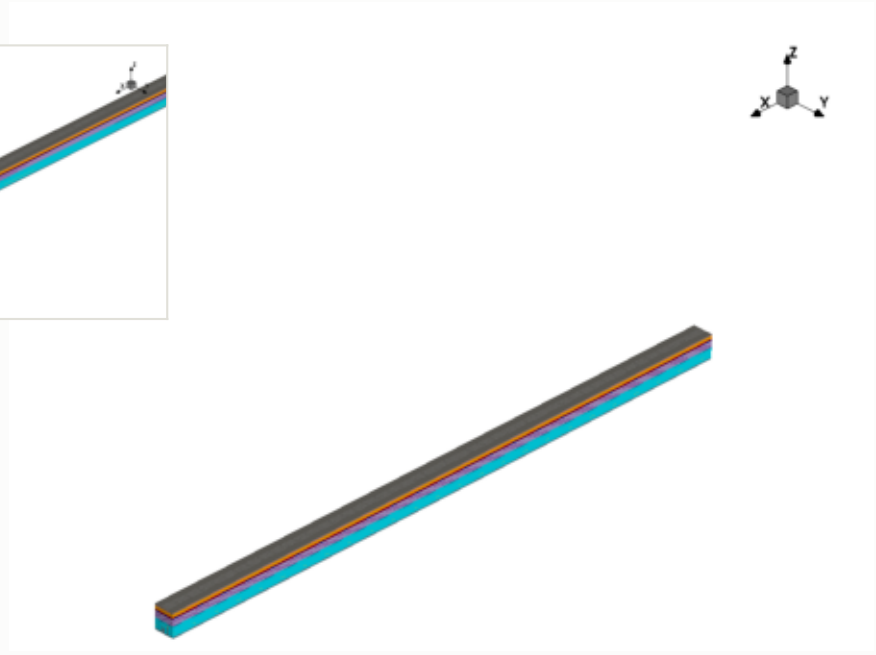
litho_patterning · ArF-i

0.217	305.2	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



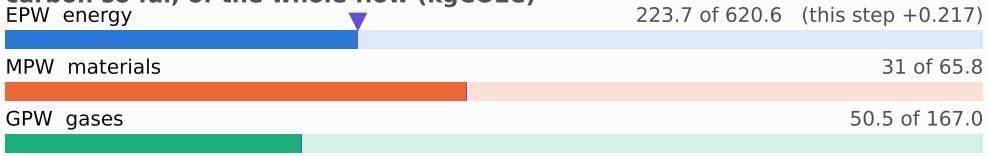


this step

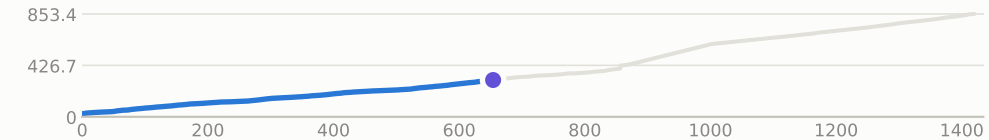
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

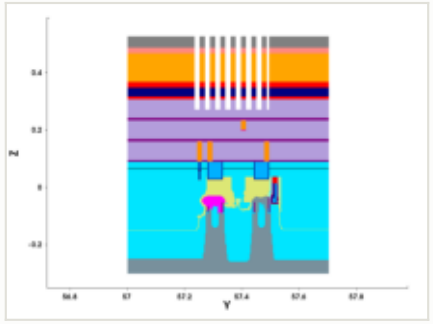
carbon so far, of the whole flow (kgCO2e)



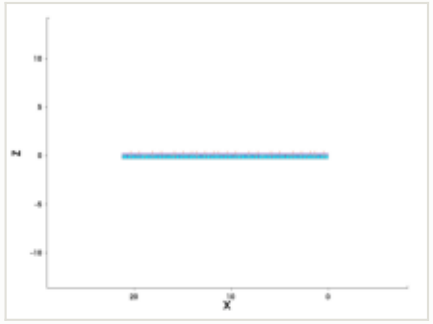
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

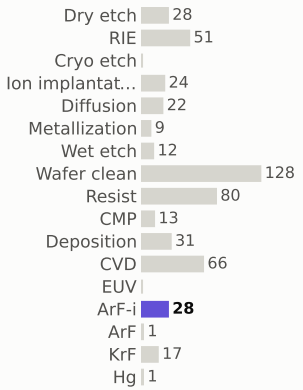


y cut at 57.351 µm

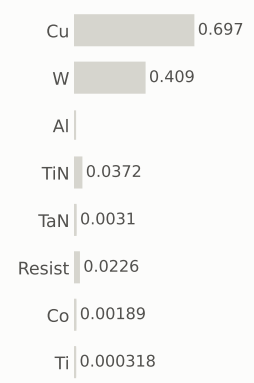


mask: layer 2100:0, positive (GDSII)

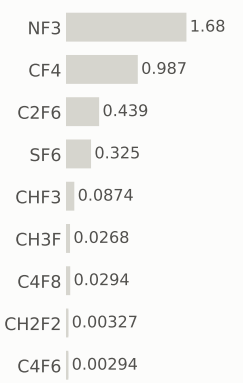
steps by process area



materials deposited (g)



gases emitted (g)



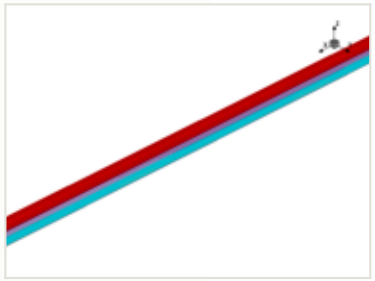
Step #6650: RIE

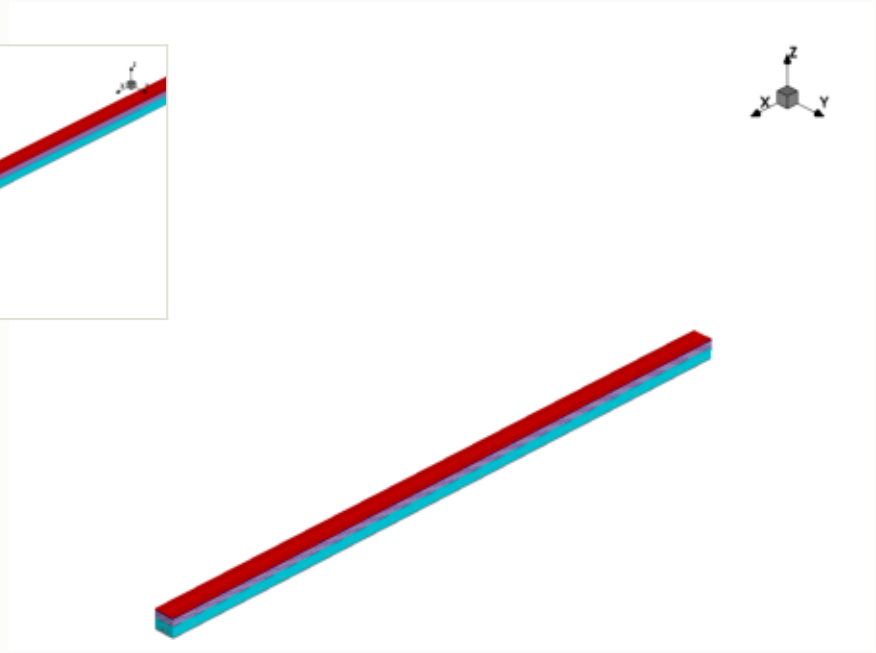
etch_strip · RIE

1.04	306.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



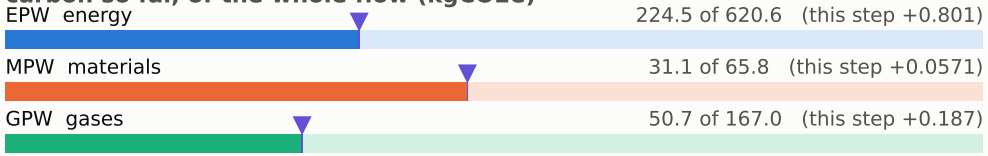


this step

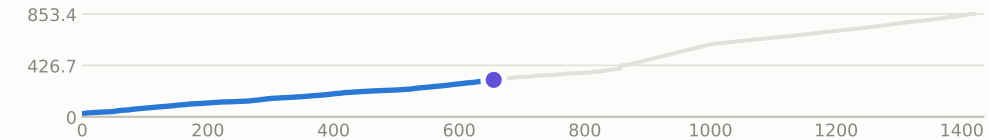
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

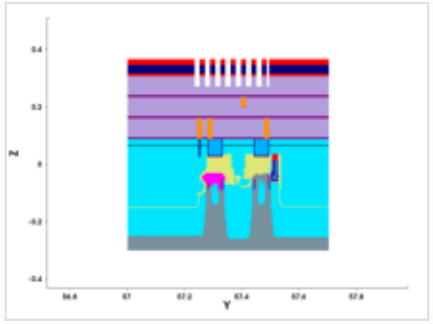
carbon so far, of the whole flow (kgCO2e)



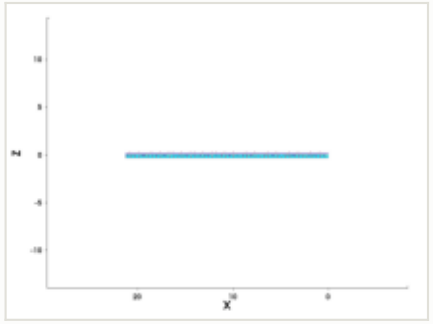
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



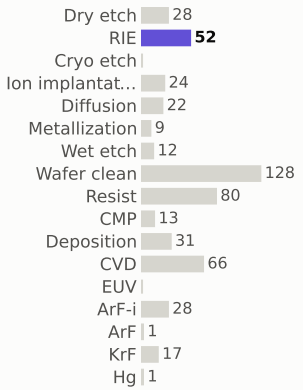
y cut at 57.351 µm



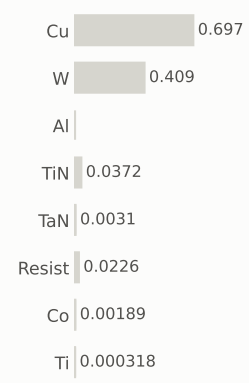
mask for this step

no mask for this step

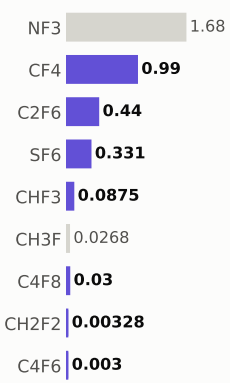
steps by process area



materials deposited (g)



gases emitted (g)



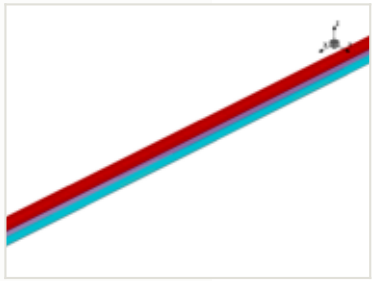
Step #6660: Wafer_clean

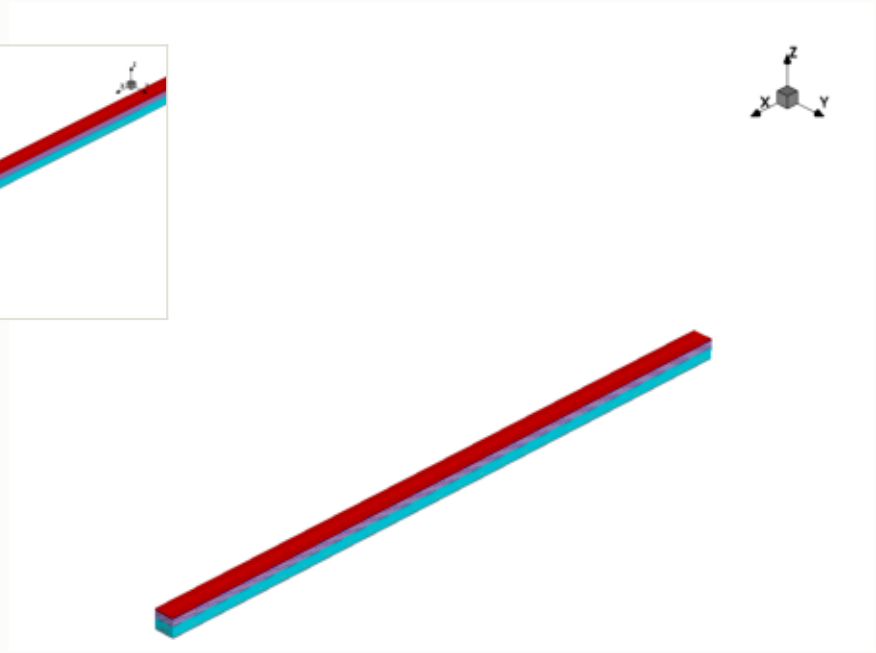
clean · Wafer_clean

0.169	306.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

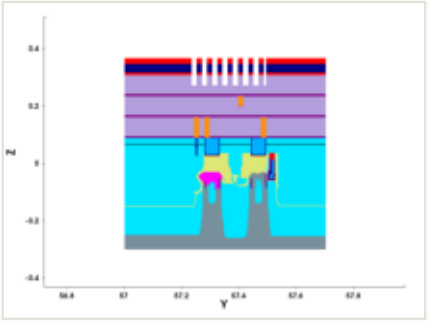
3D structure after this step

zoomed to fit

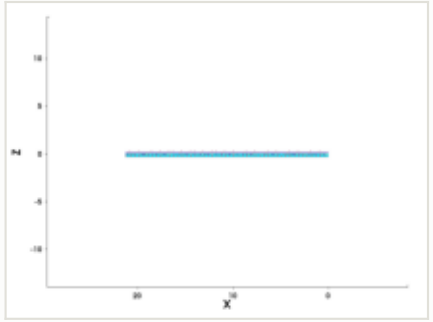




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

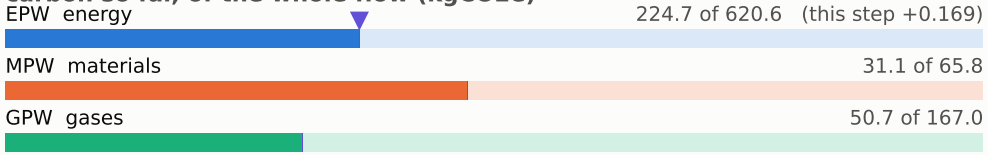
no mask for this step

this step

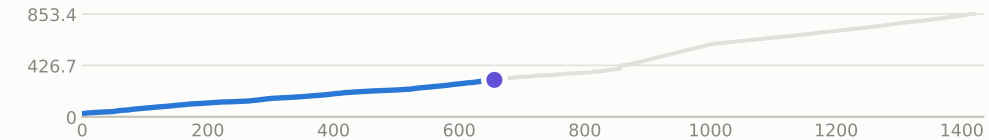
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

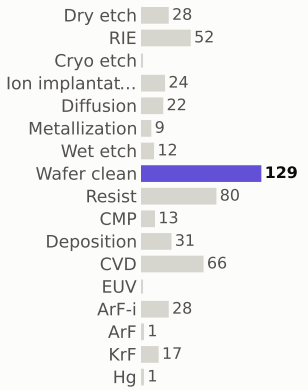
carbon so far, of the whole flow (kgCO2e)



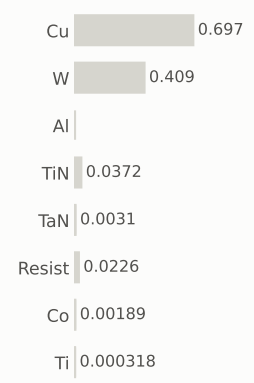
embodied carbon up to each step (kgCO2e per wafer)



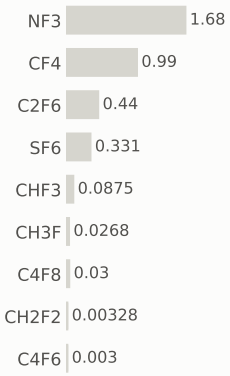
steps by process area



materials deposited (g)



gases emitted (g)



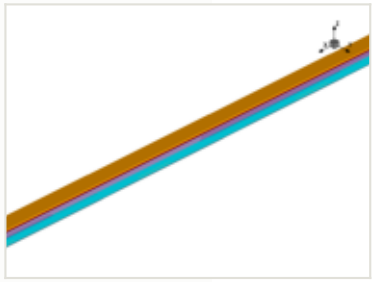
Step #6670: Resist

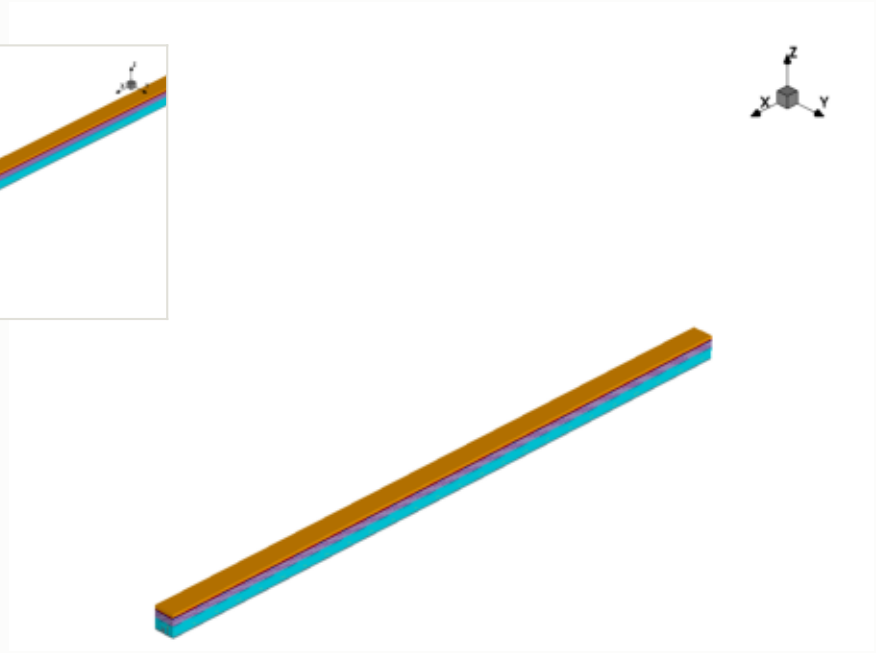
depo_fill · Resist

0.169	306.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



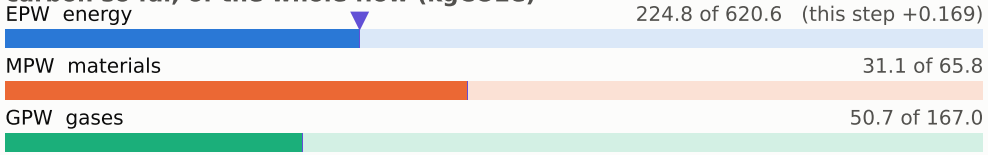


this step

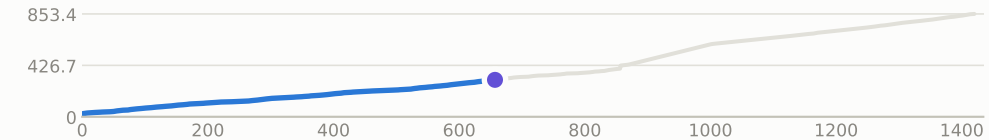
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

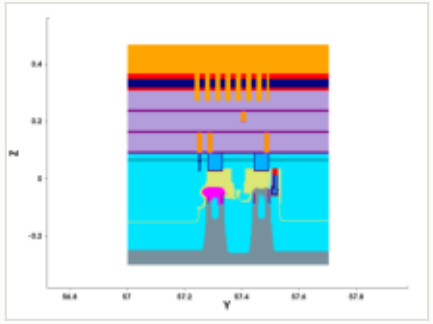
carbon so far, of the whole flow (kgCO2e)



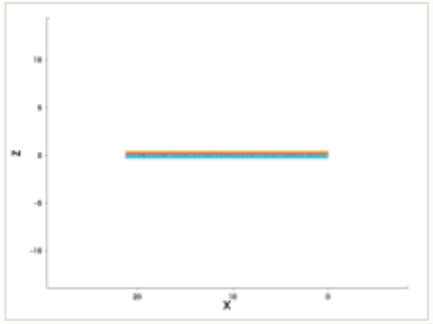
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



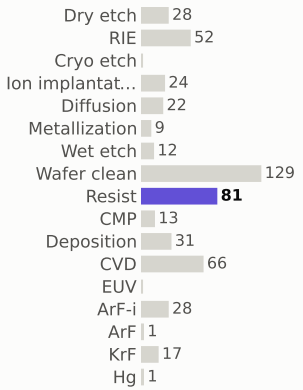
y cut at 57.351 µm



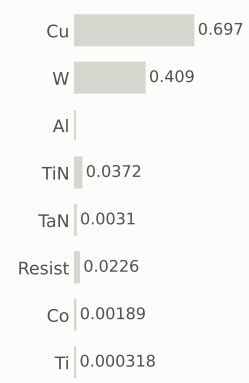
mask for this step

no mask for this step

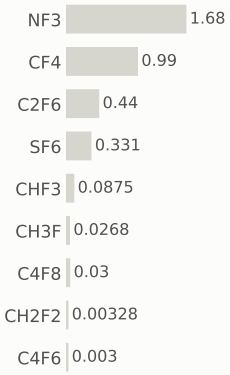
steps by process area



materials deposited (g)



gases emitted (g)



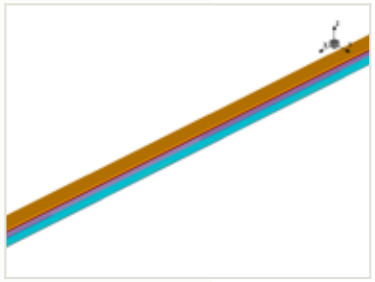
Step #6680: Deposition

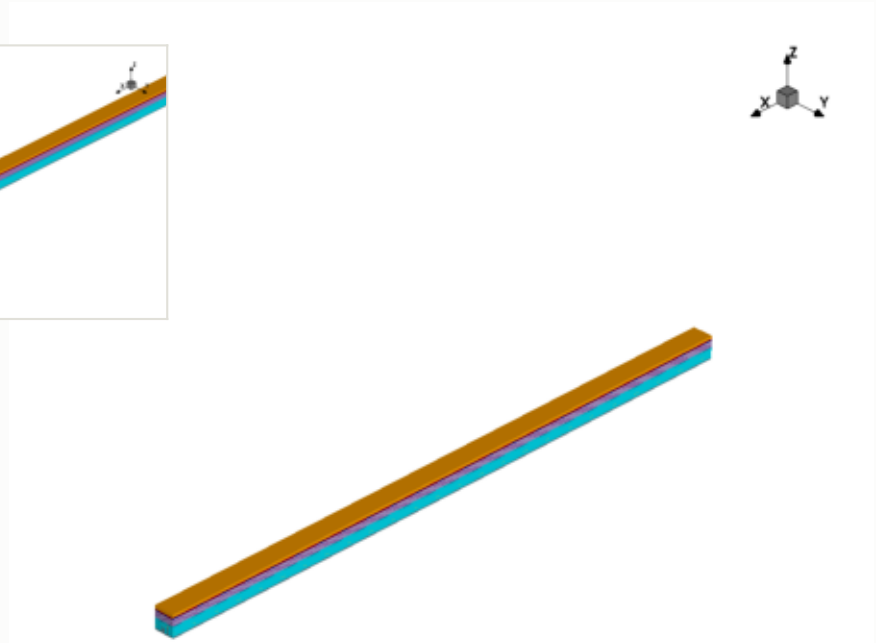
depo_isotropic · Deposition

0.656	307.3	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



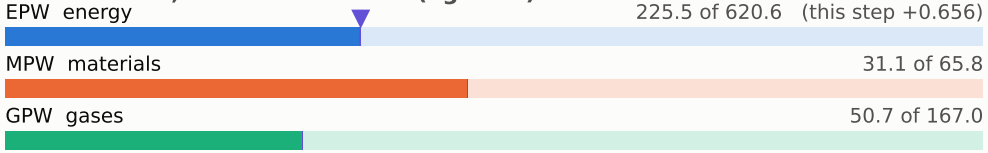


this step

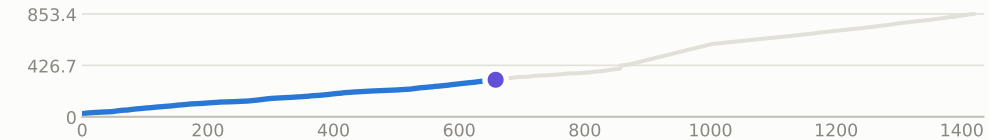
process area	Deposition
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

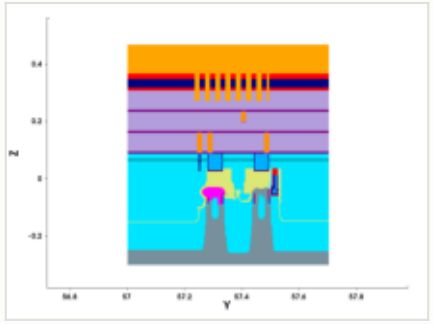
carbon so far, of the whole flow (kgCO2e)



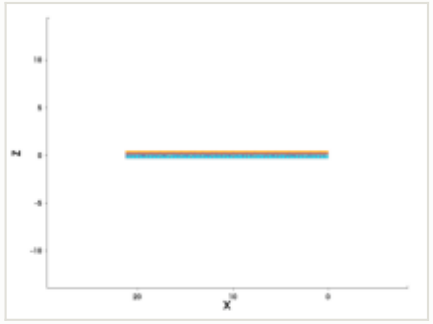
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



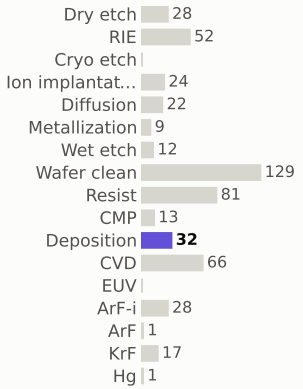
y cut at 57.351 µm



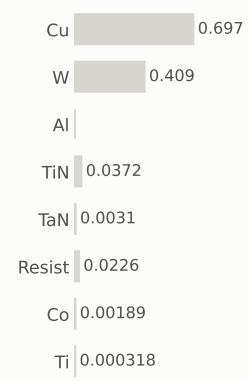
mask for this step

no mask for this step

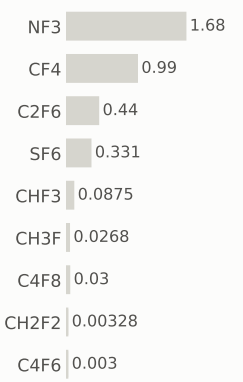
steps by process area



materials deposited (g)



gases emitted (g)



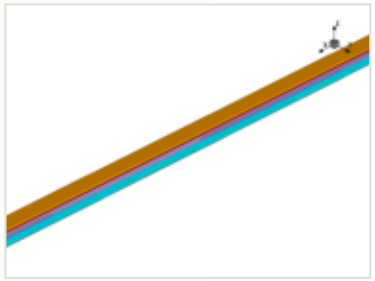
Step #6690: CVD

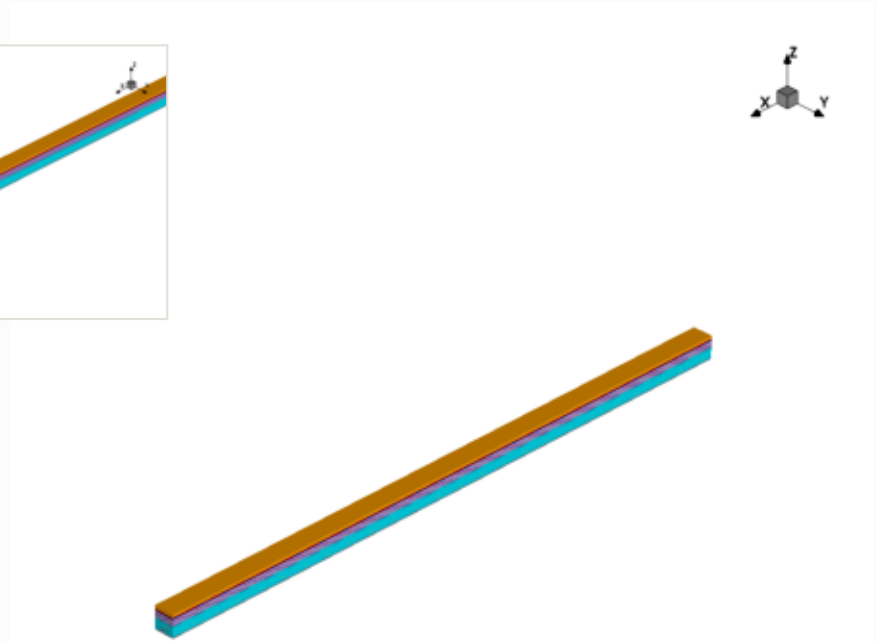
depo_isotropic · CVD

1.31	308.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



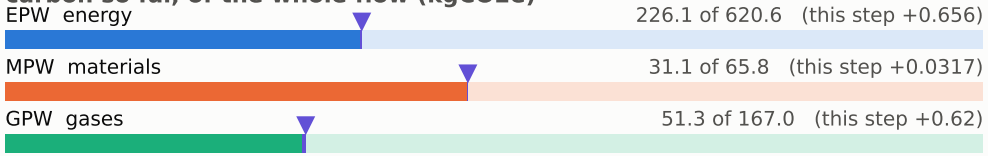


this step

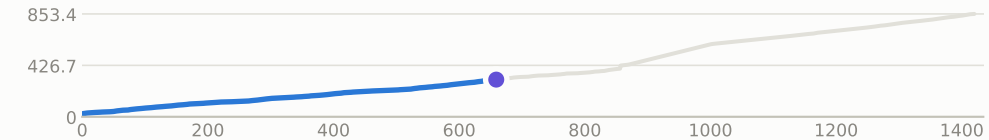
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

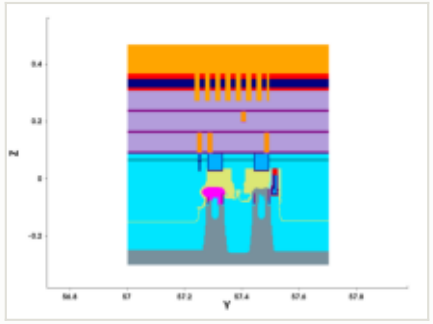
carbon so far, of the whole flow (kgCO2e)



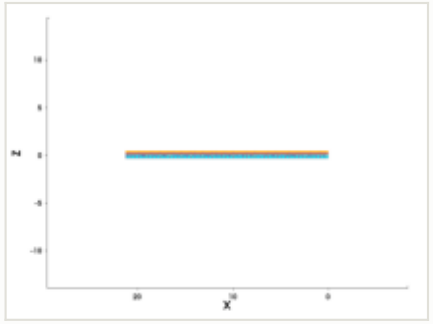
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



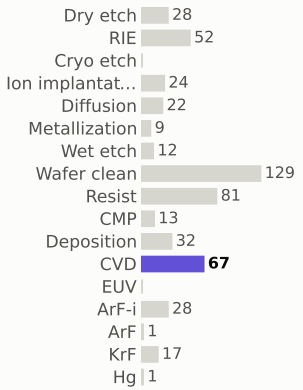
y cut at 57.351 µm



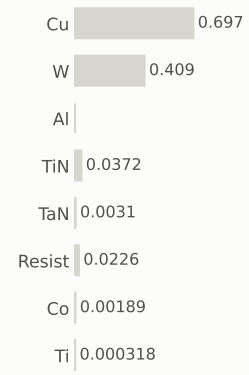
mask for this step

no mask for this step

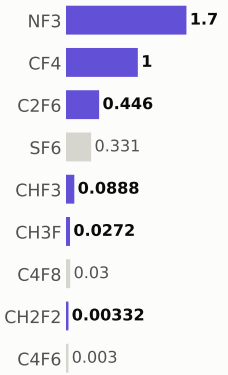
steps by process area



materials deposited (g)



gases emitted (g)



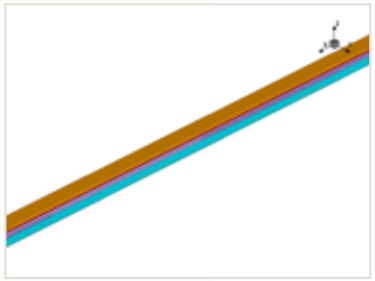
Step #6700: Resist

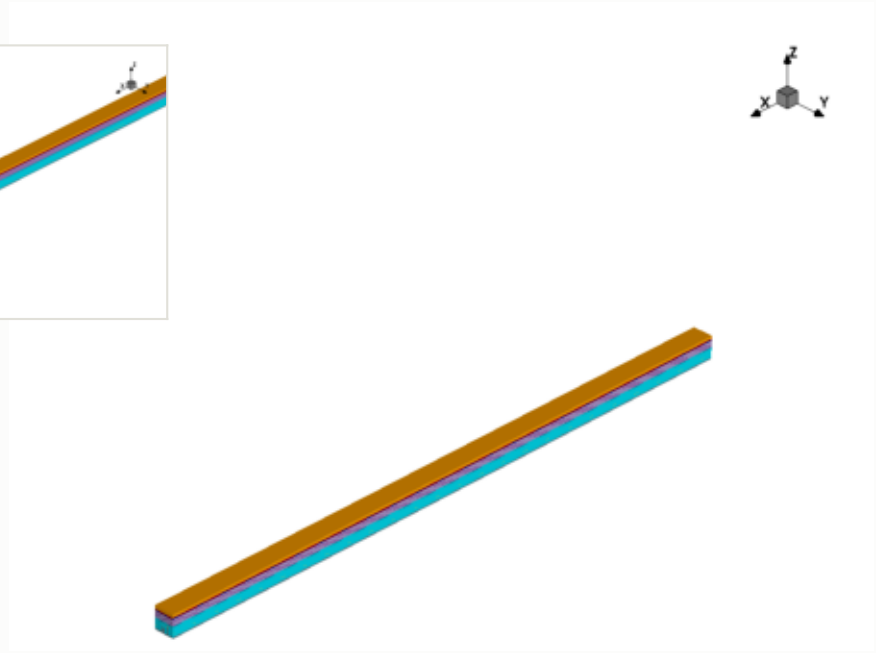
depo_fill · Resist

0.169	308.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



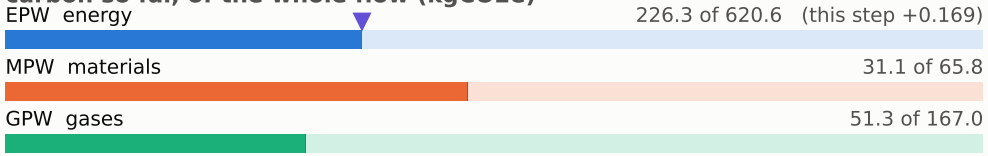


this step

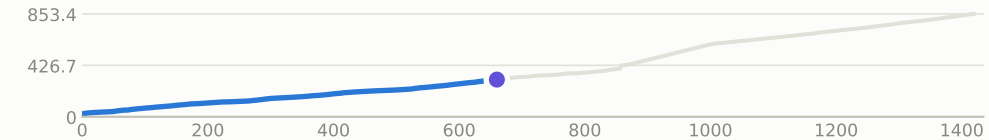
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

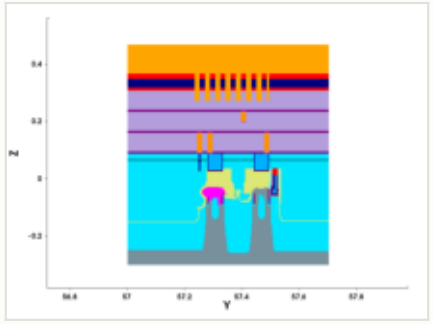
carbon so far, of the whole flow (kgCO2e)



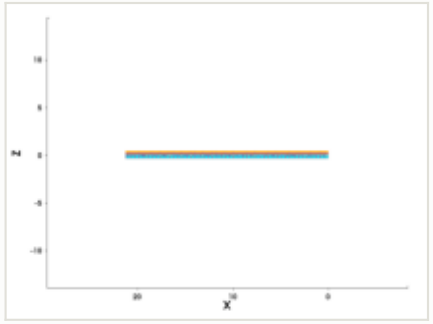
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



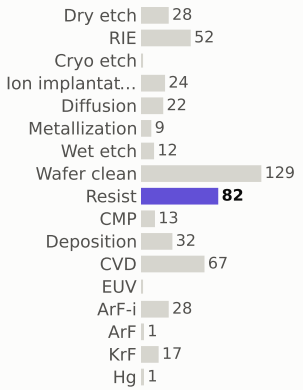
y cut at 57.351 µm



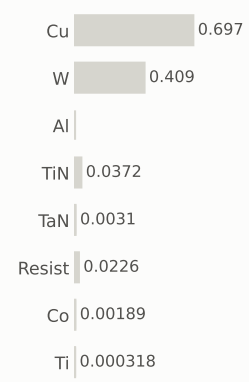
mask for this step

no mask for this step

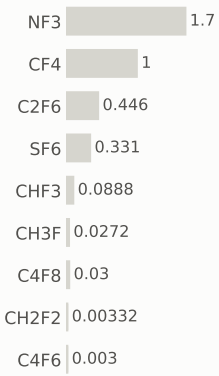
steps by process area



materials deposited (g)



gases emitted (g)



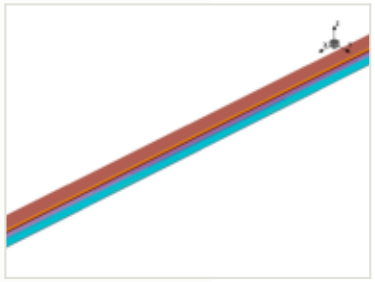
Step #6710: CVD

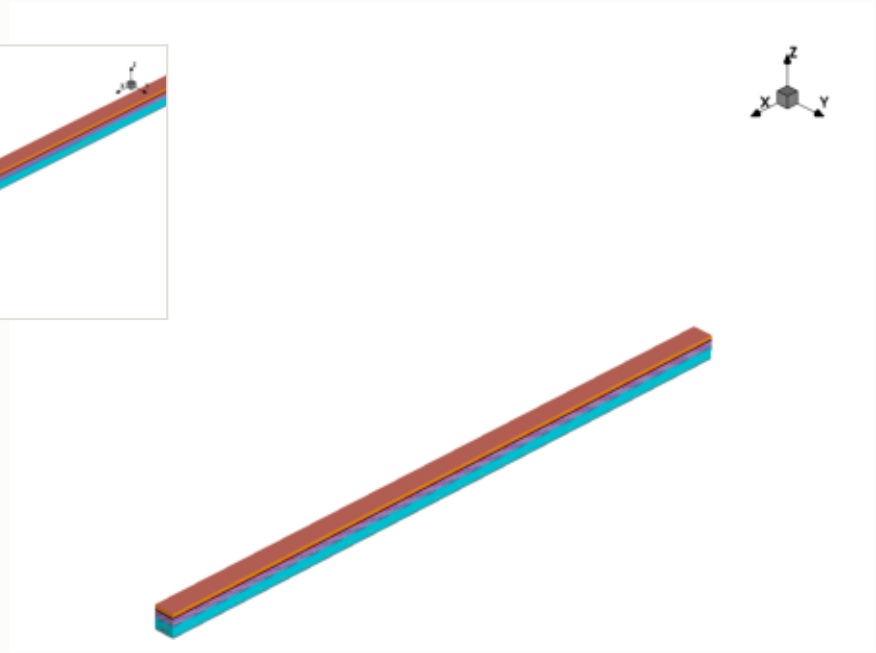
depo_fill · CVD

1.31	310.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



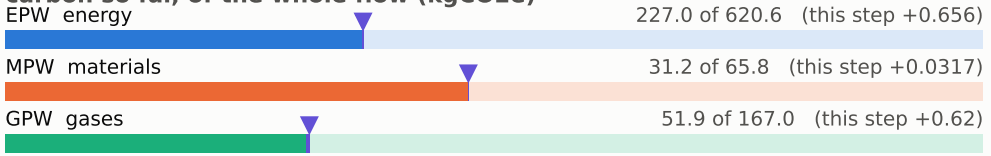


this step

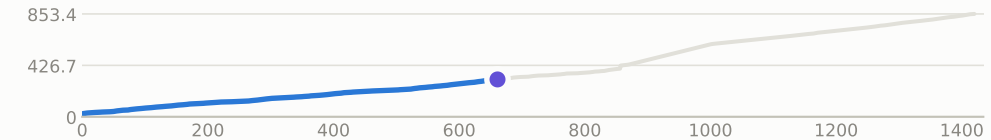
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

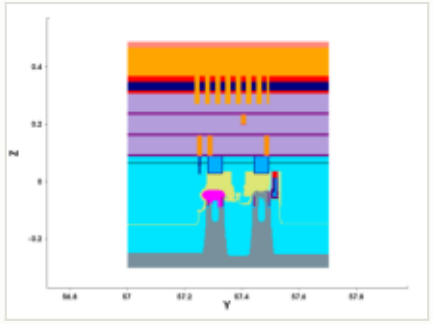
carbon so far, of the whole flow (kgCO2e)



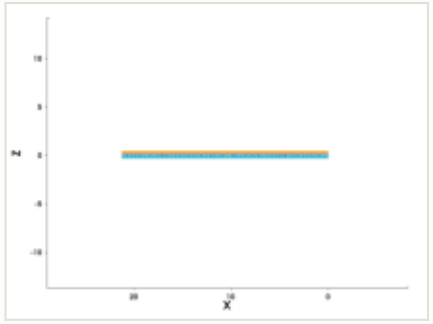
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



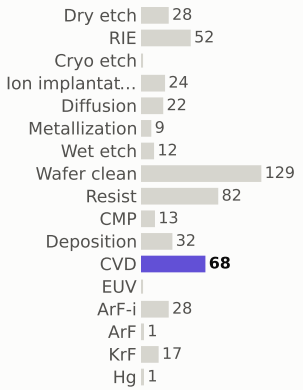
y cut at 57.351 µm



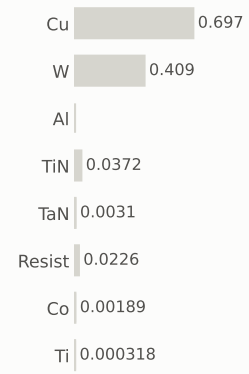
mask for this step

no mask for this step

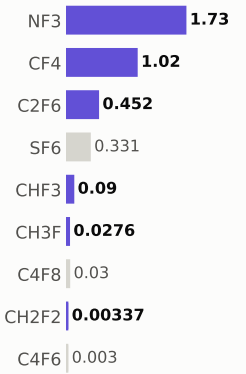
steps by process area



materials deposited (g)



gases emitted (g)



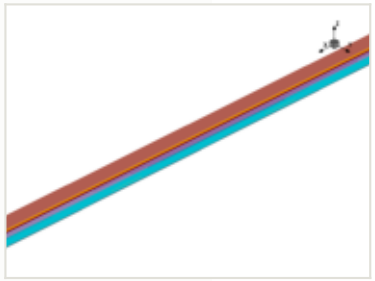
Step #6720: Wafer_clean

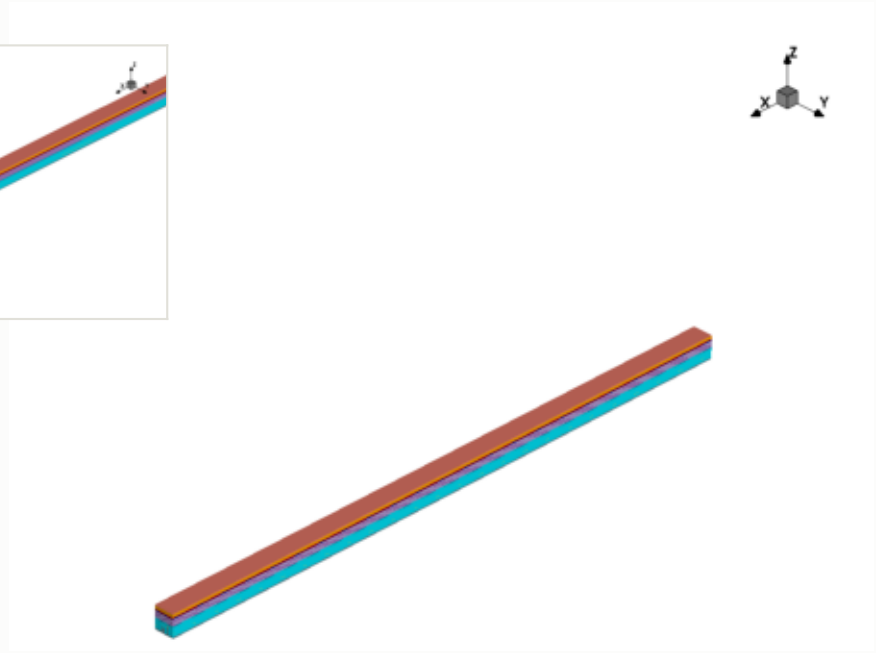
clean · Wafer_clean

0.169	310.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

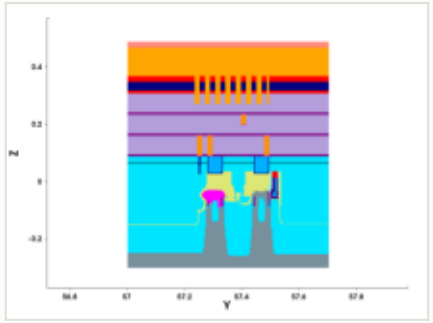
3D structure after this step

zoomed to fit

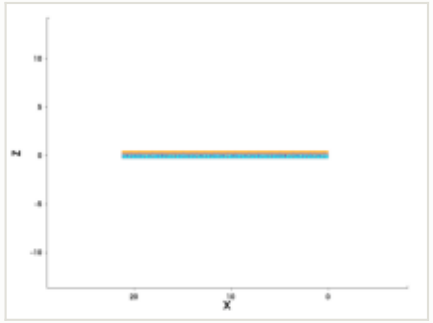




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

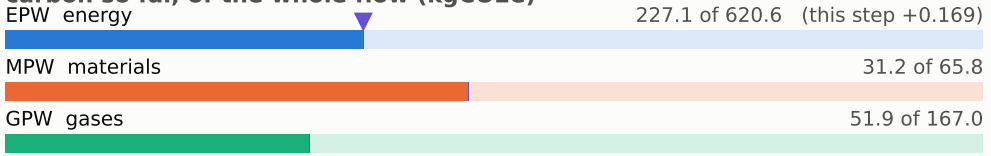
no mask for this step

this step

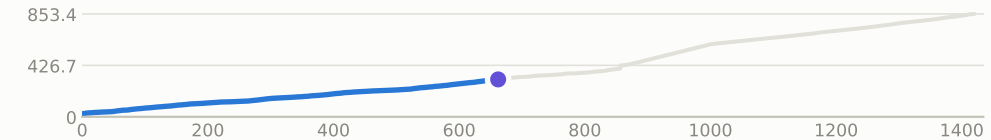
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

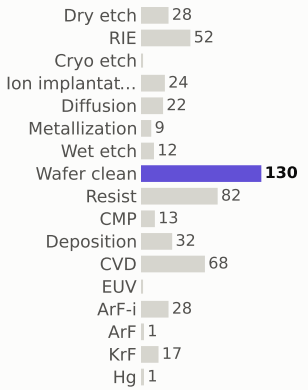
carbon so far, of the whole flow (kgCO2e)



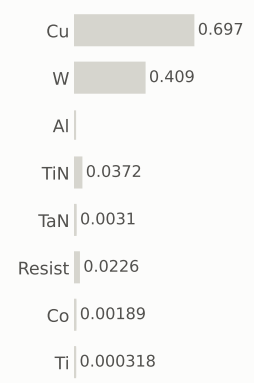
embodied carbon up to each step (kgCO2e per wafer)



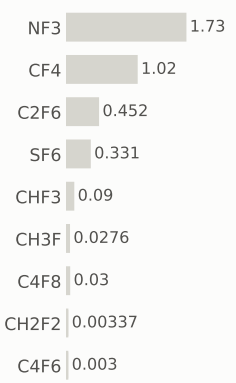
steps by process area



materials deposited (g)



gases emitted (g)



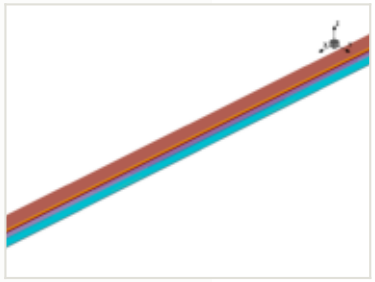
Step #6730: Resist

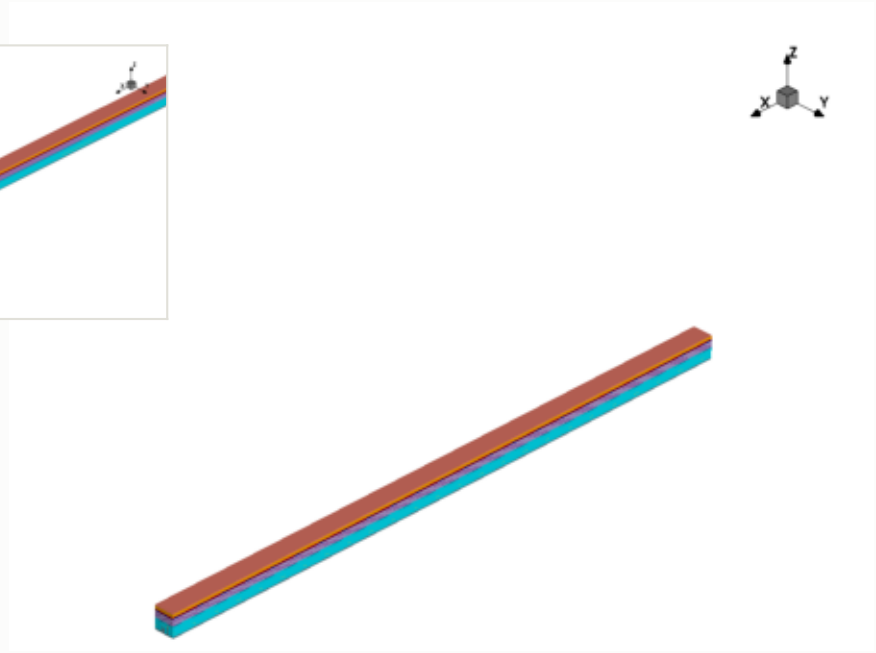
resist · Resist

0.169	310.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



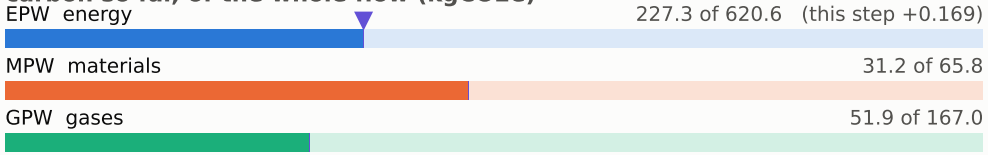


this step

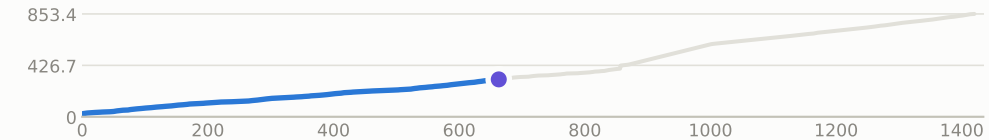
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

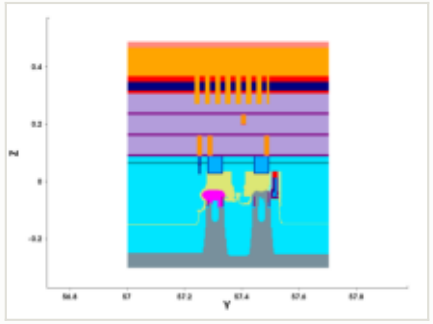
carbon so far, of the whole flow (kgCO2e)



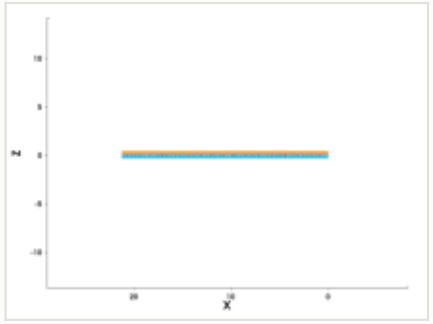
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



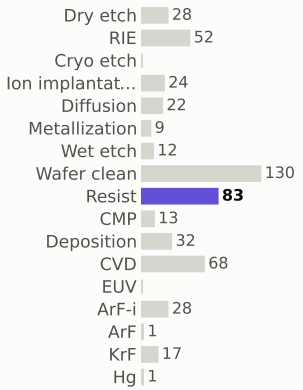
y cut at 57.351 µm



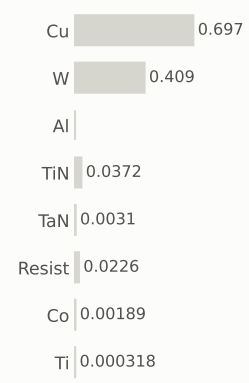
mask for this step

no mask for this step

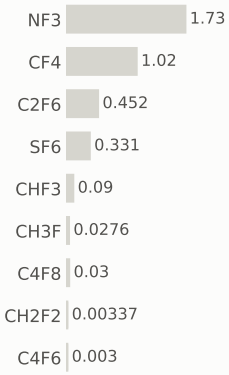
steps by process area



materials deposited (g)



gases emitted (g)



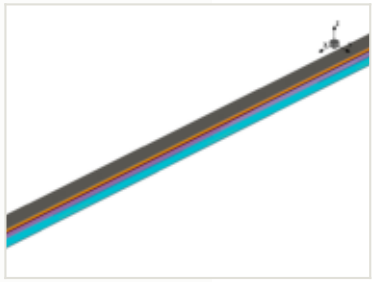
Step #6740: ArF-i

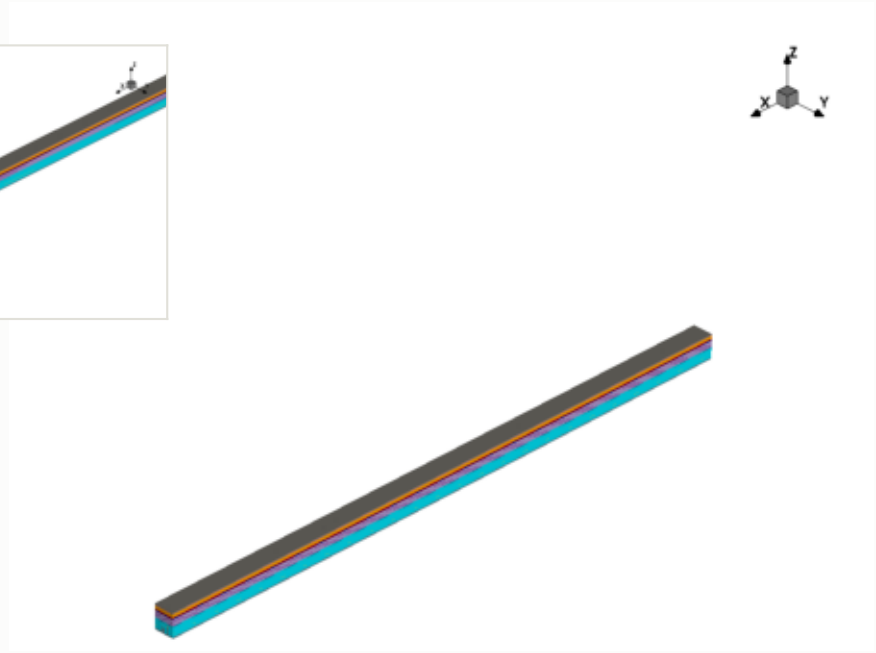
litho_patterning · ArF-i

0.217	310.6	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



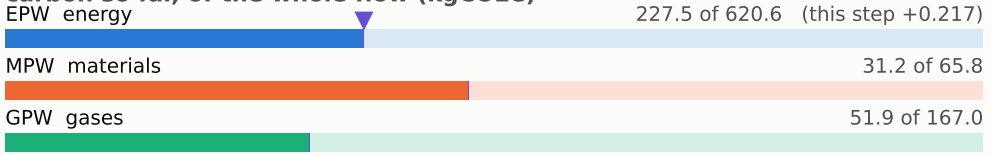


this step

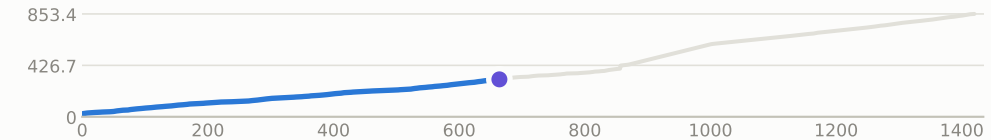
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

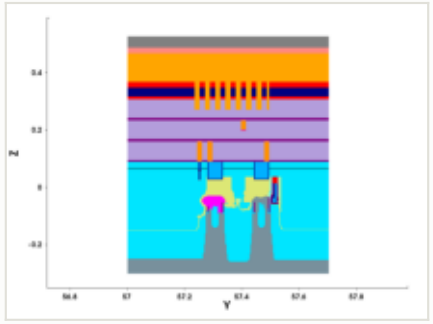
carbon so far, of the whole flow (kgCO2e)



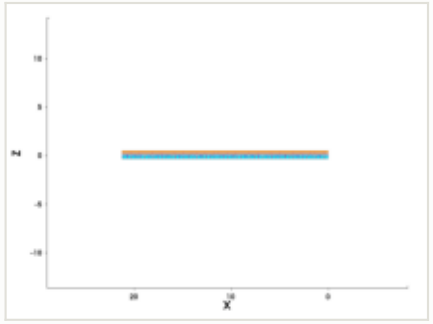
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

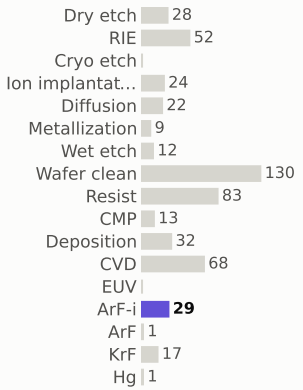


y cut at 57.351 µm

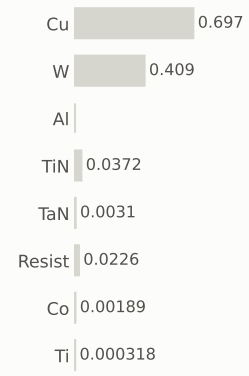


mask: layer 2025:0 (GDSII)

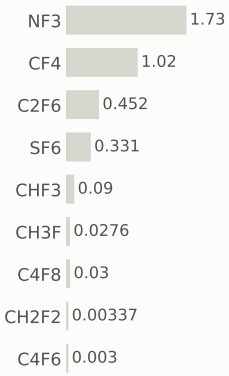
steps by process area



materials deposited (g)



gases emitted (g)



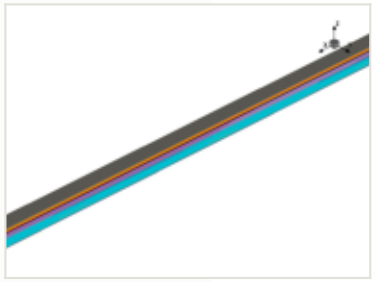
Step #6750: RIE

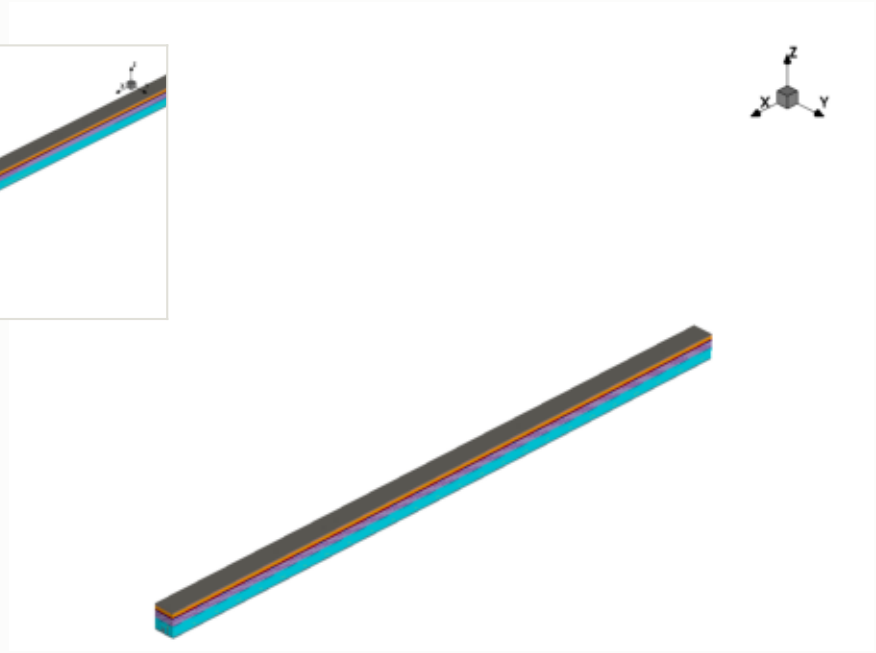
etch_anisotropic · RIE

1.04	311.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



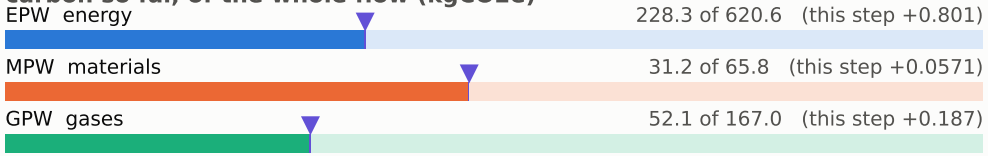


this step

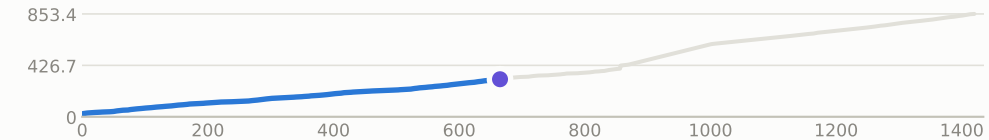
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

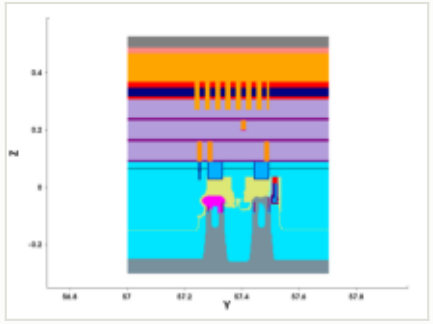
carbon so far, of the whole flow (kgCO2e)



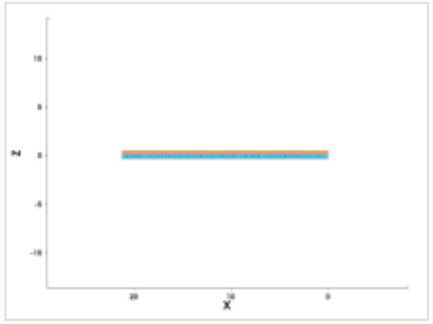
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



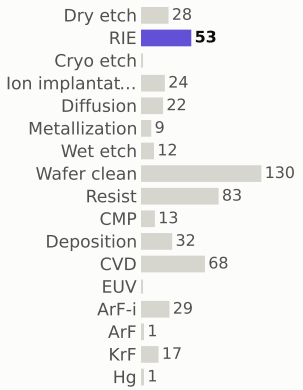
y cut at 57.351 µm



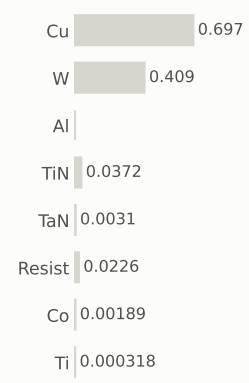
mask for this step

no mask for this step

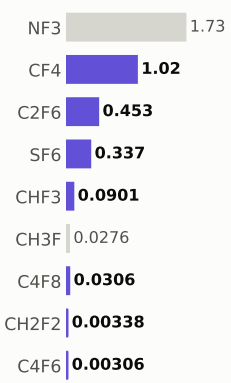
steps by process area



materials deposited (g)



gases emitted (g)



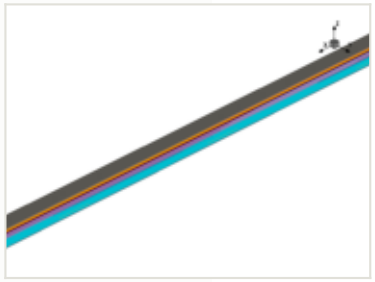
Step #6760: Wafer_clean

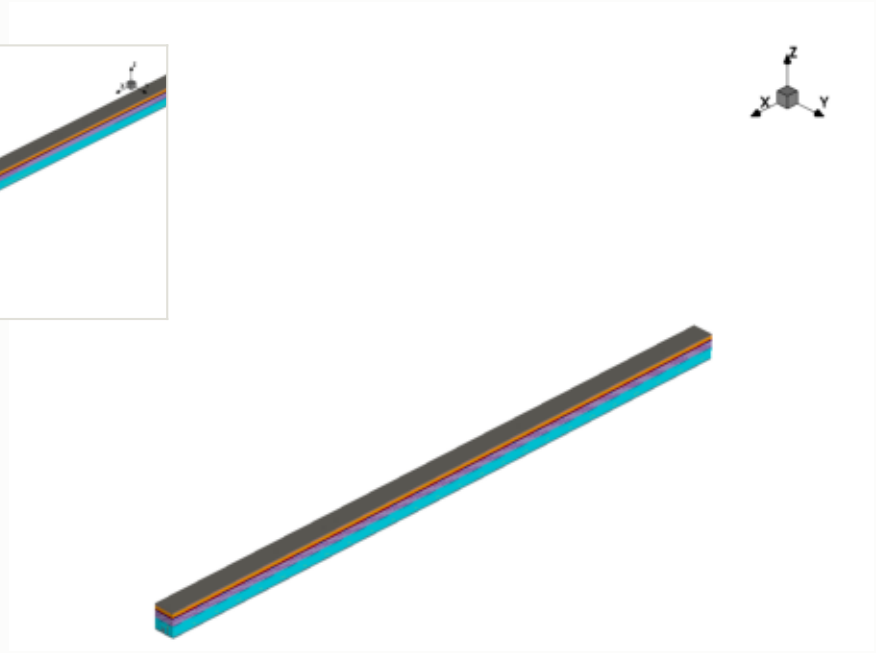
clean · Wafer_clean

0.169	311.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

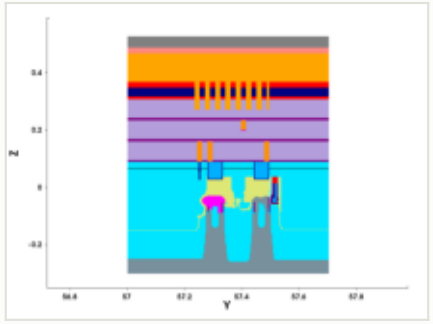
3D structure after this step

zoomed to fit

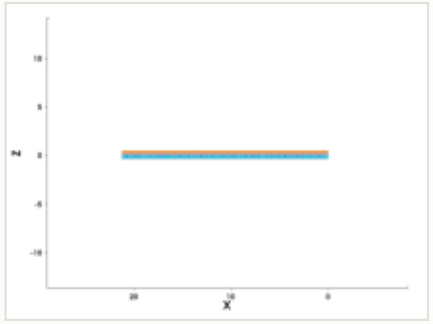




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

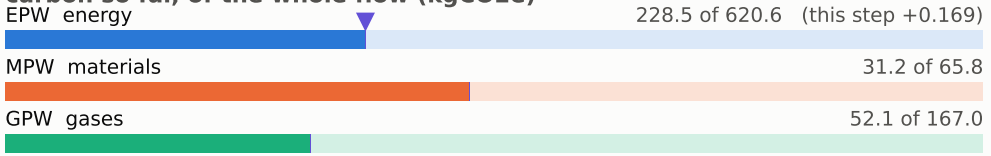
no mask for this step

this step

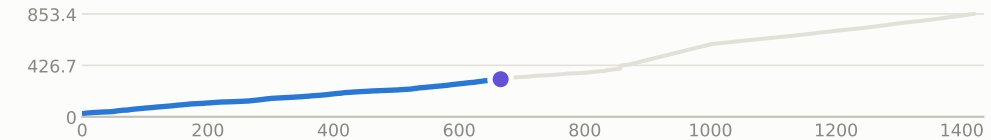
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

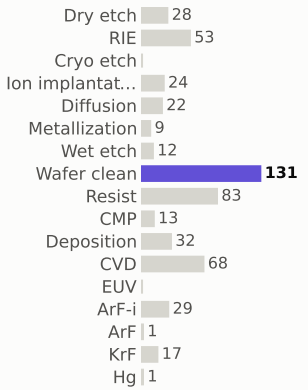
carbon so far, of the whole flow (kgCO2e)



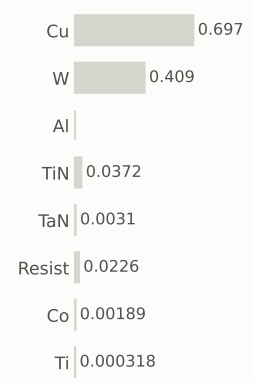
embodied carbon up to each step (kgCO2e per wafer)



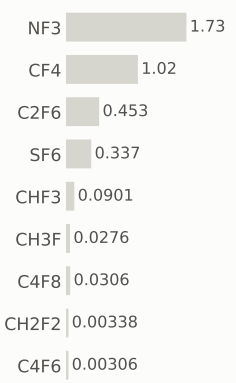
steps by process area



materials deposited (g)



gases emitted (g)



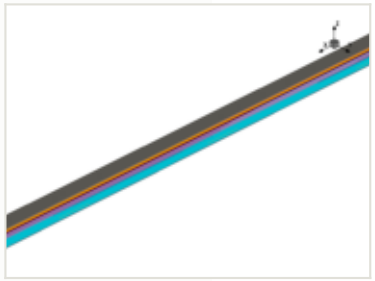
Step #6770: Resist

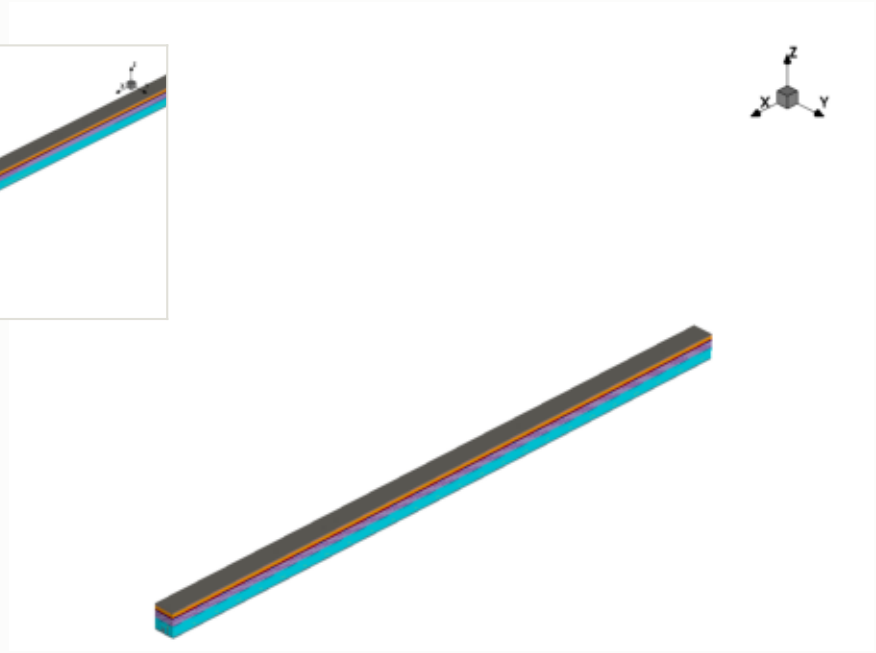
depo_fill · Resist

0.169	312.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



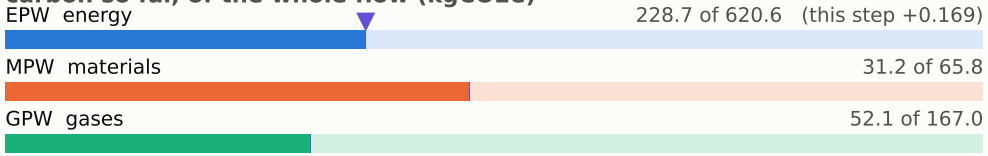


this step

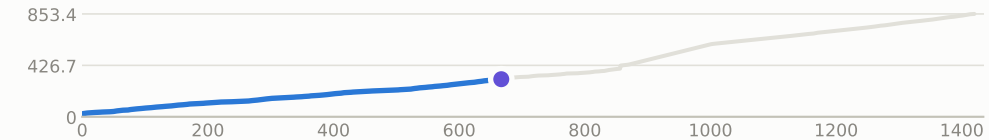
process area	Resist
material	—
thickness	0 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

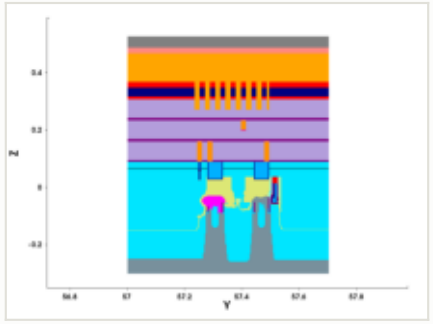
carbon so far, of the whole flow (kgCO2e)



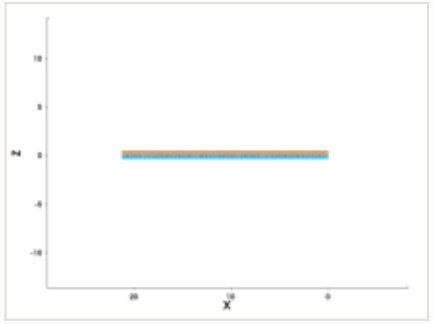
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



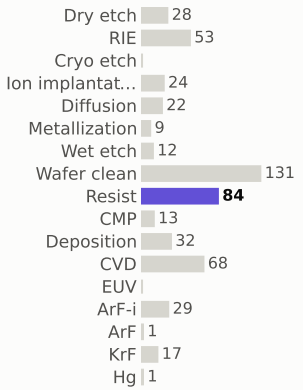
y cut at 57.351 µm



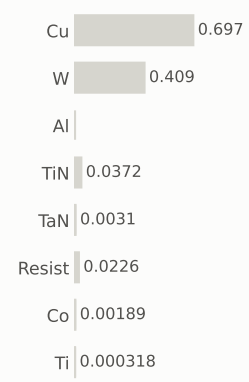
mask for this step

no mask for this step

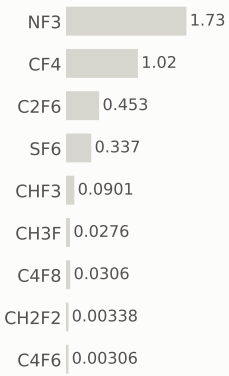
steps by process area



materials deposited (g)



gases emitted (g)



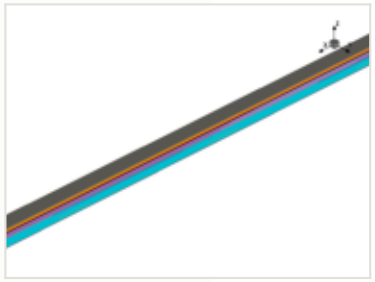
Step #6780: CVD

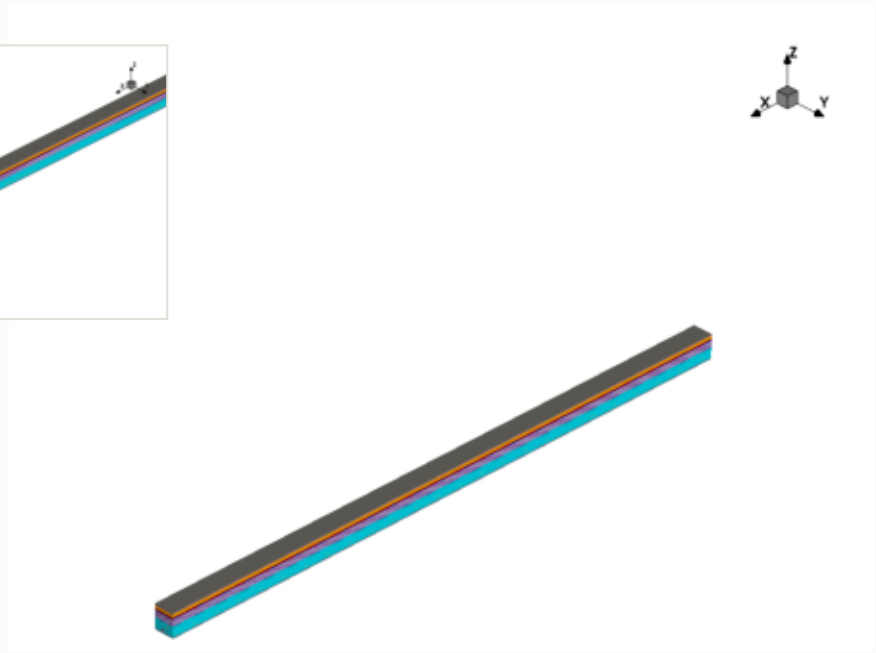
depo_isotropic · CVD

1.31	313.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



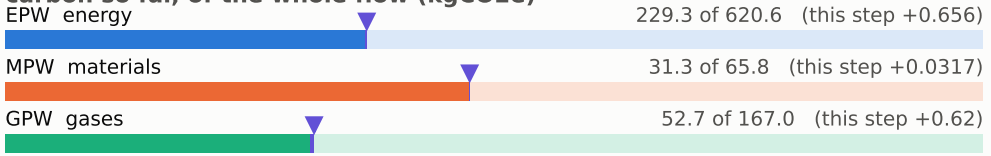


this step

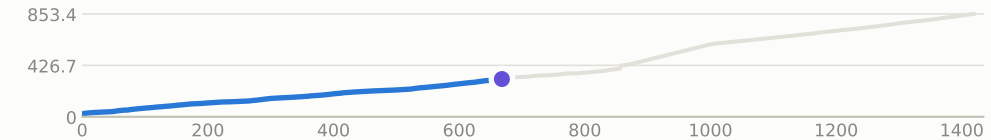
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

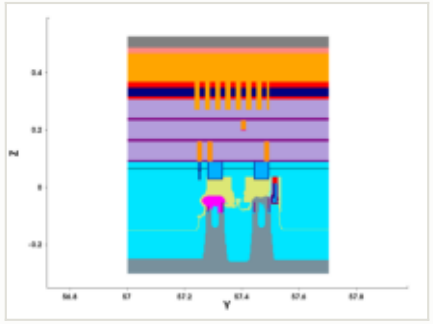
carbon so far, of the whole flow (kgCO2e)



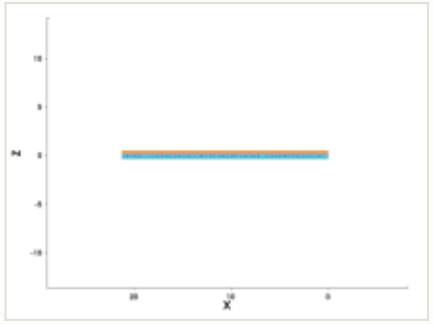
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



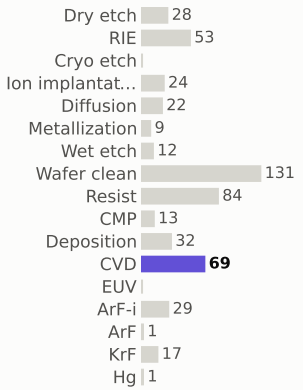
y cut at 57.351 µm



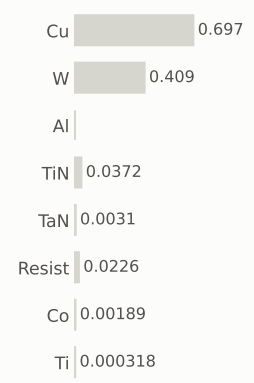
mask for this step

no mask for this step

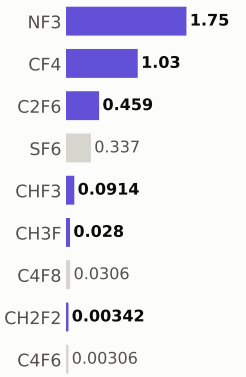
steps by process area



materials deposited (g)



gases emitted (g)



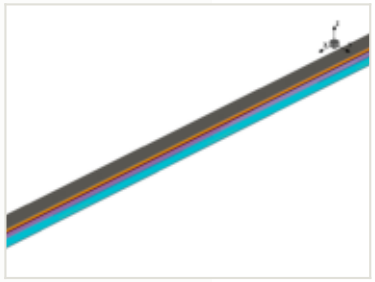
Step #6790: Wafer_clean

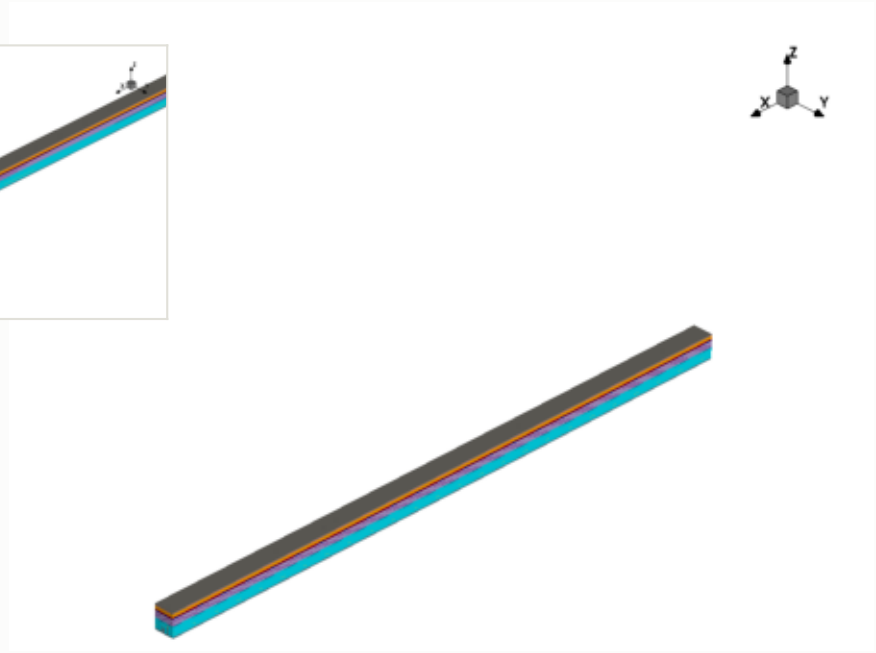
clean · Wafer_clean

0.169	313.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





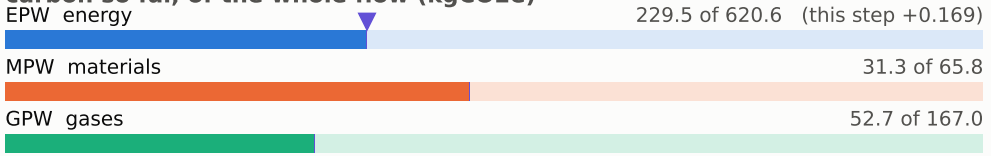
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

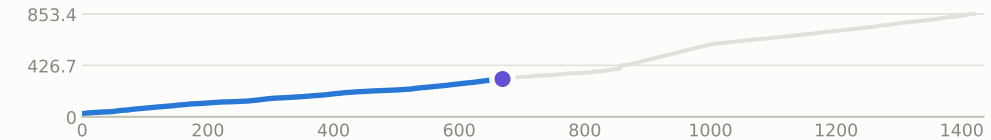
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

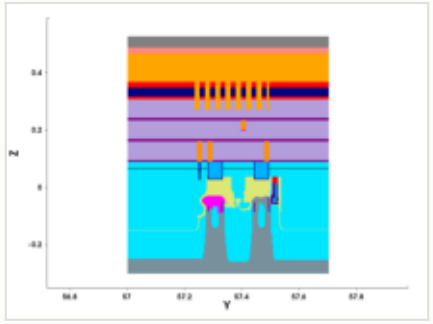
carbon so far, of the whole flow (kgCO2e)



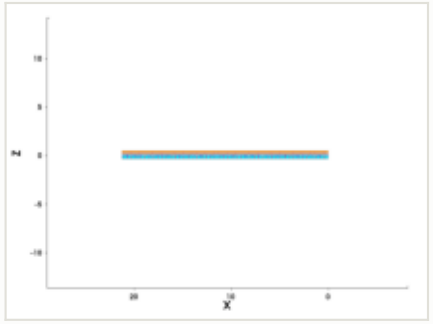
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



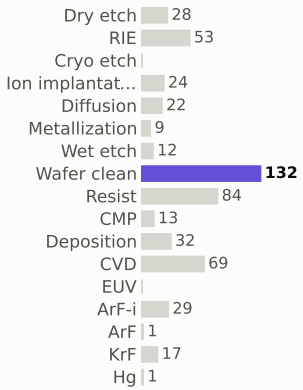
y cut at 57.351 µm



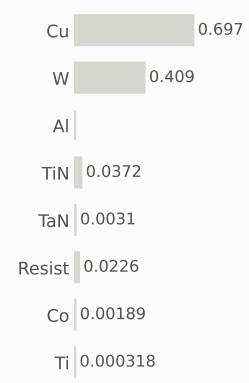
mask for this step

no mask for this step

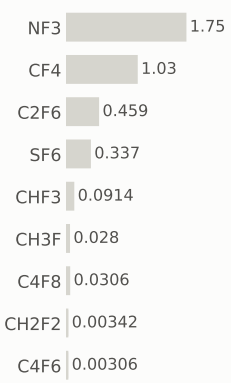
steps by process area



materials deposited (g)



gases emitted (g)



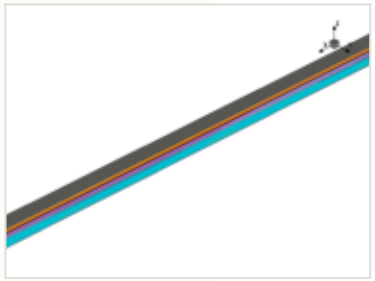
Step #6800: Resist

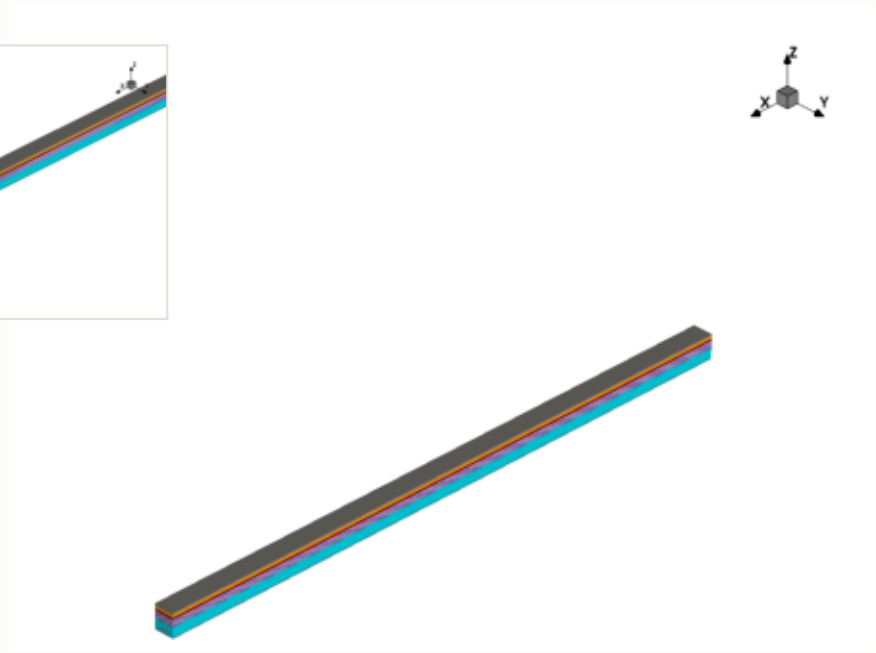
resist · Resist

0.169	313.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





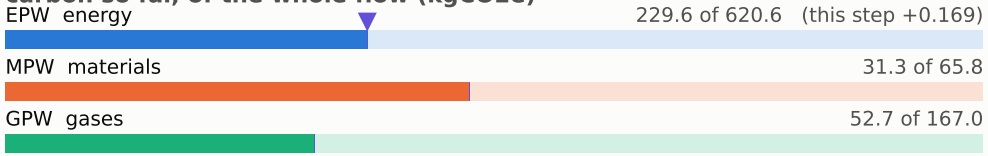
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

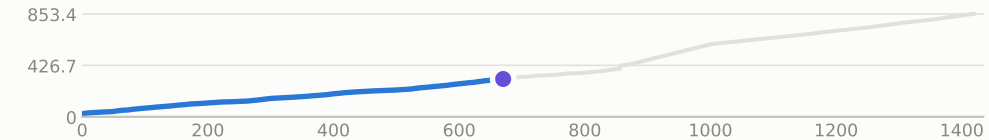
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

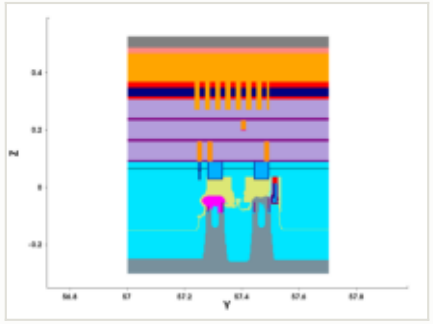
carbon so far, of the whole flow (kgCO2e)



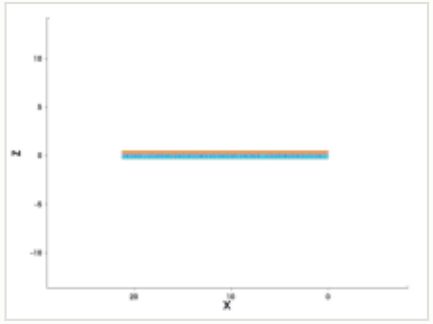
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



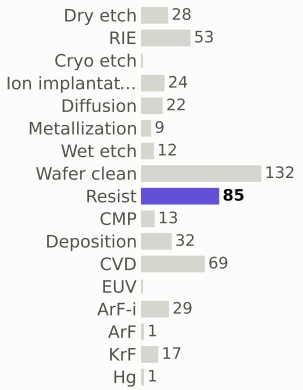
y cut at 57.351 µm



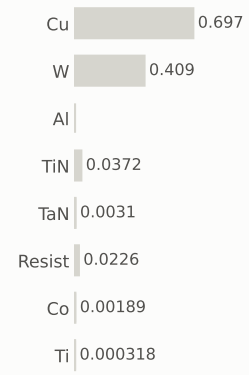
mask for this step

no mask for this step

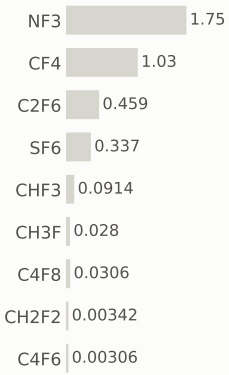
steps by process area



materials deposited (g)



gases emitted (g)



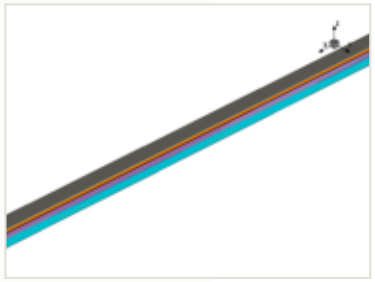
Step #6810: ArF-i

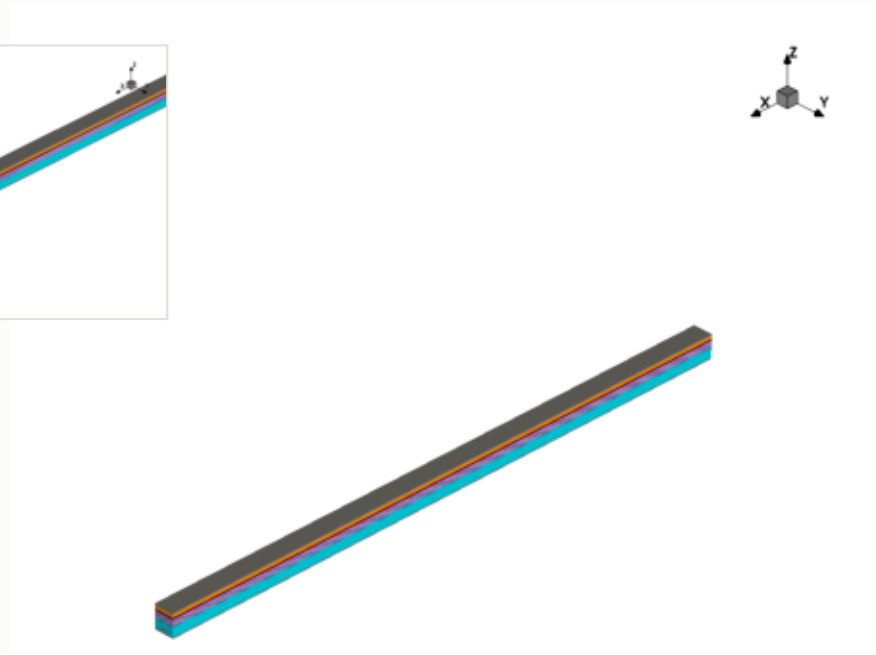
litho_patterning · ArF-i

0.217	313.8	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



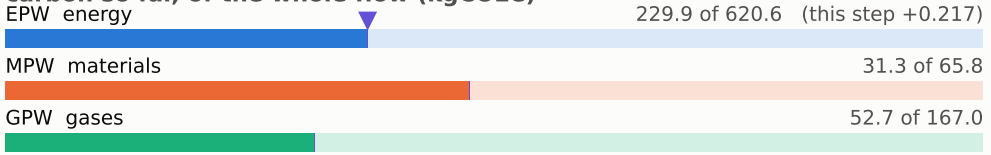


this step

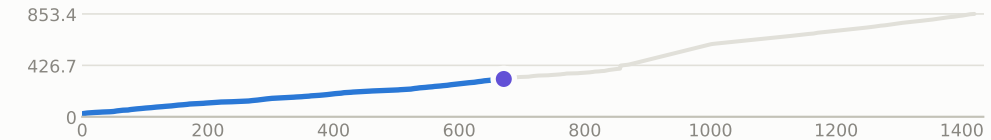
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

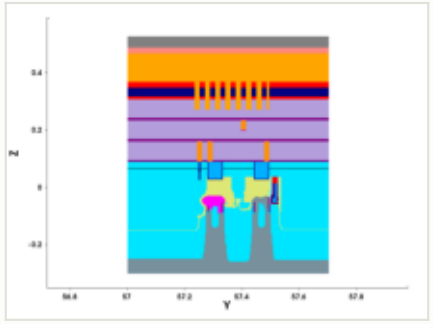
carbon so far, of the whole flow (kgCO2e)



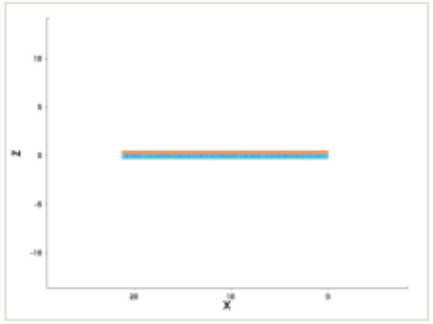
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

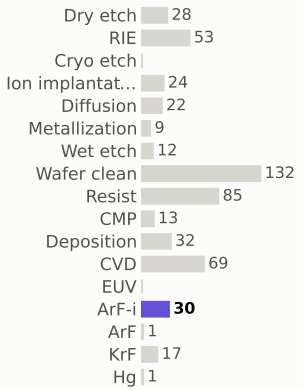


y cut at 57.351 µm

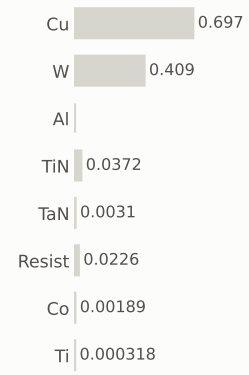


mask: layer 2100:0, positive (GDSII)

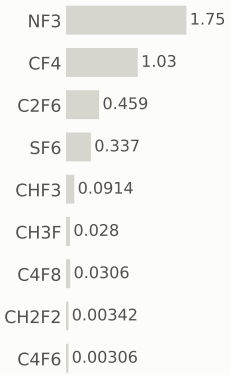
steps by process area



materials deposited (g)



gases emitted (g)



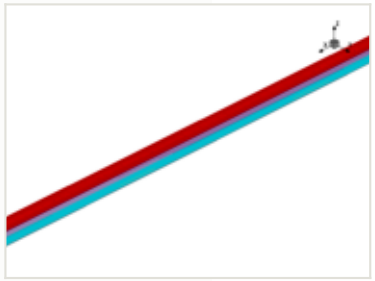
Step #6820: RIE

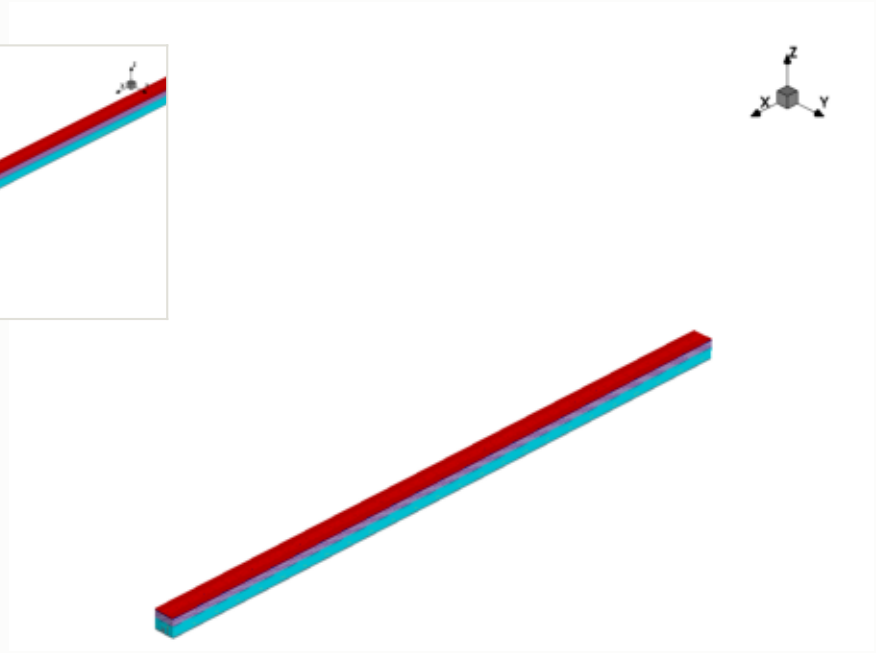
etch_strip · RIE

1.04	314.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



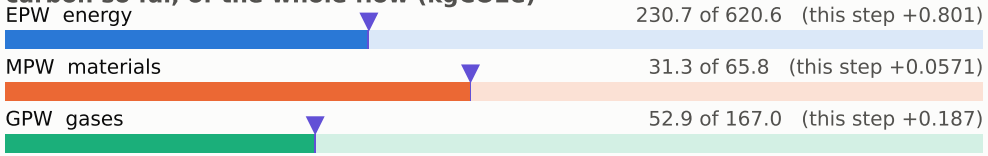


this step

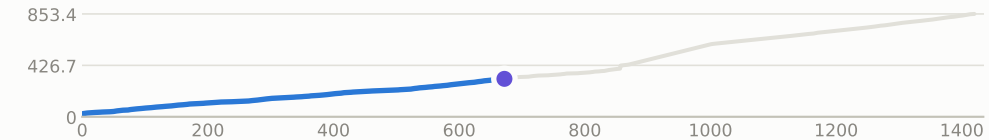
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

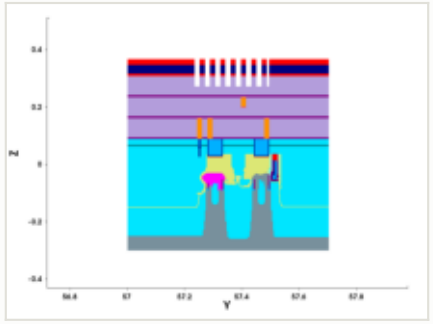
carbon so far, of the whole flow (kgCO2e)



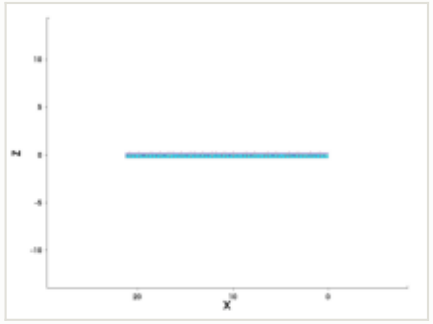
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



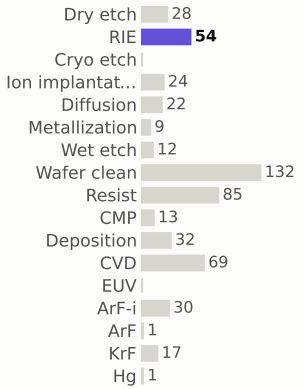
y cut at 57.351 µm



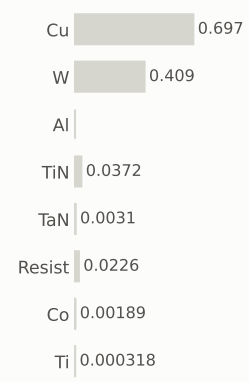
mask for this step

no mask for this step

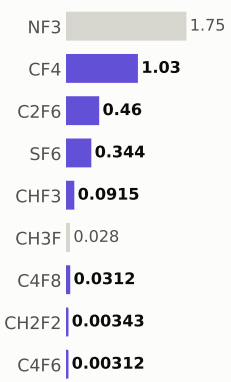
steps by process area



materials deposited (g)



gases emitted (g)



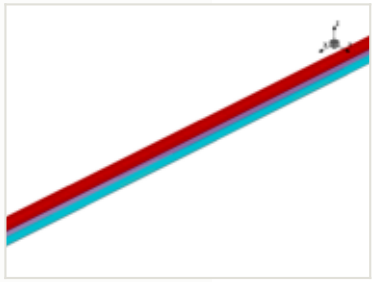
Step #6830: Wafer_clean

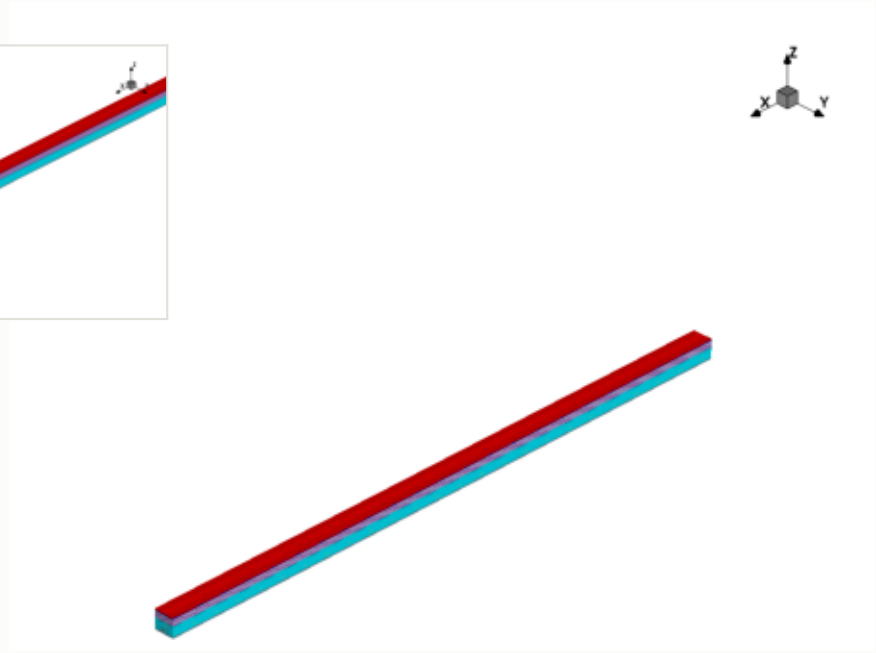
clean · Wafer_clean

0.169	315.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



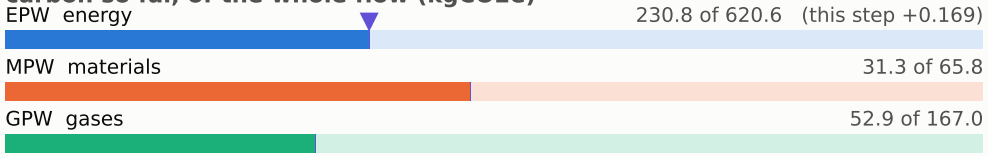


this step

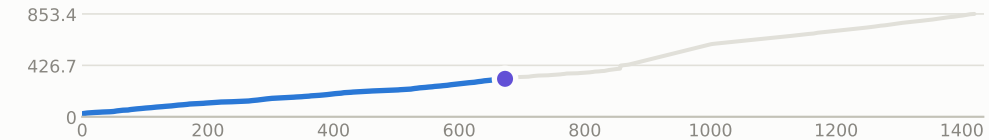
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

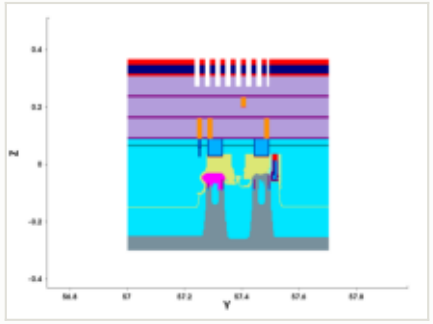
carbon so far, of the whole flow (kgCO2e)



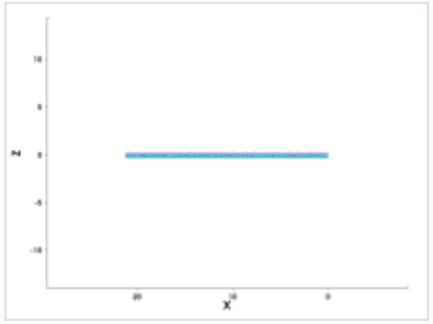
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



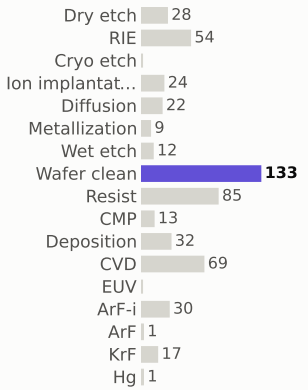
y cut at 57.351 µm



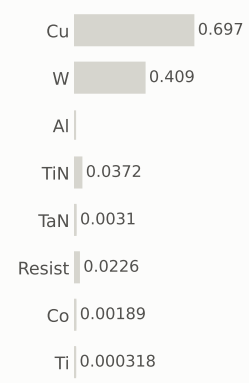
mask for this step

no mask for this step

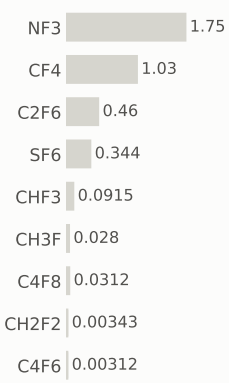
steps by process area



materials deposited (g)



gases emitted (g)



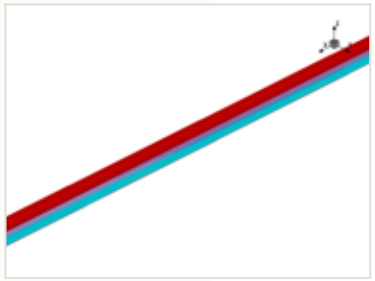
Step #6840: RIE

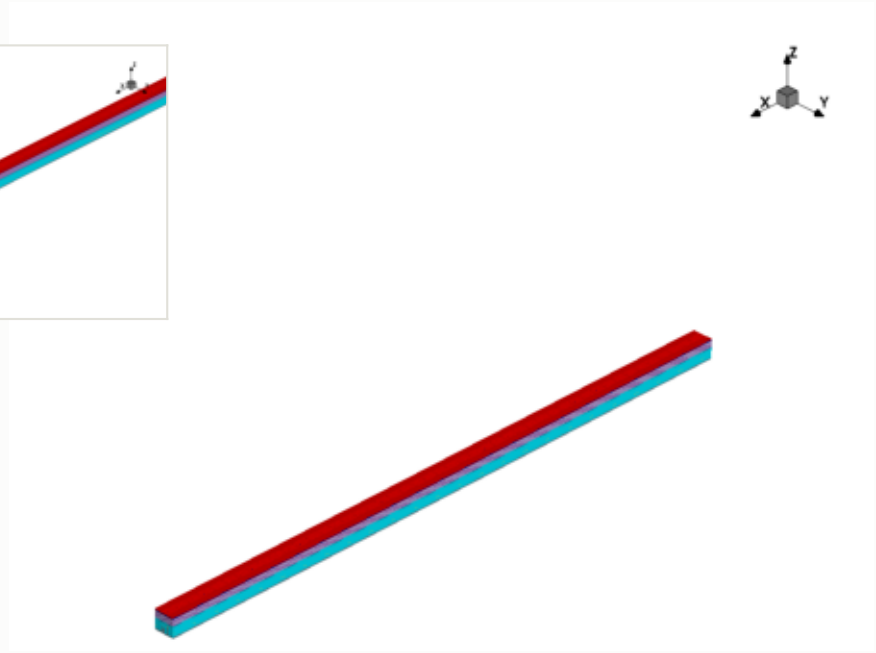
etch_anisotropic · RIE

1.04	316.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



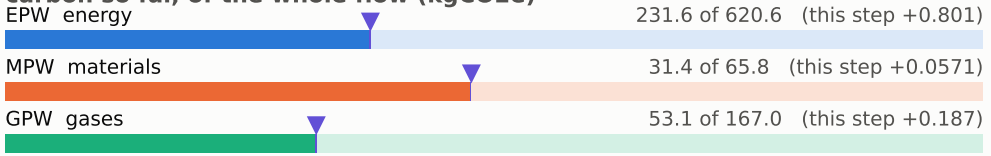


this step

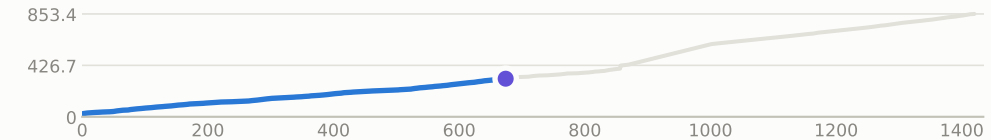
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

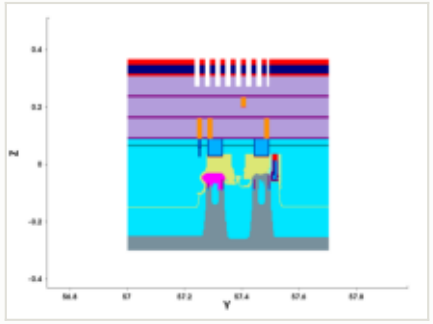
carbon so far, of the whole flow (kgCO2e)



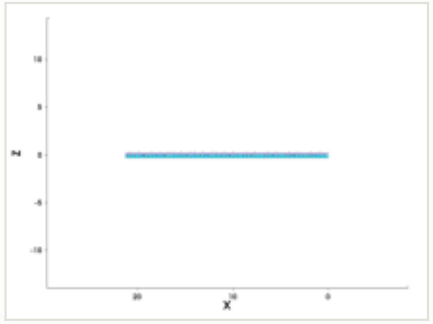
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



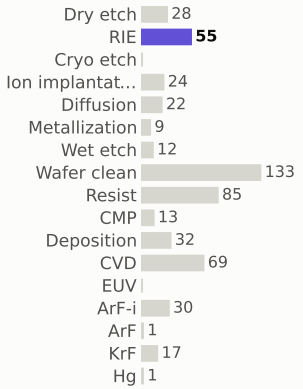
y cut at 57.351 µm



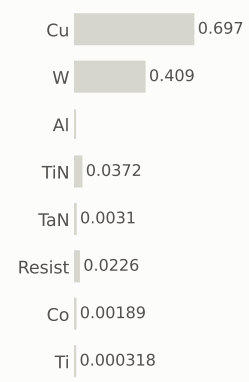
mask for this step

no mask for this step

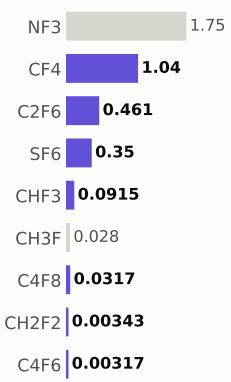
steps by process area



materials deposited (g)



gases emitted (g)



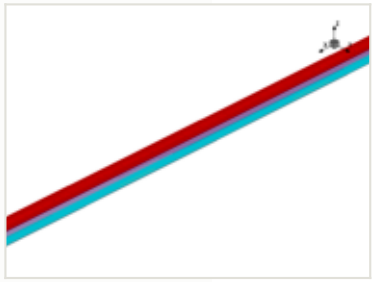
Step #6850: Wafer_clean

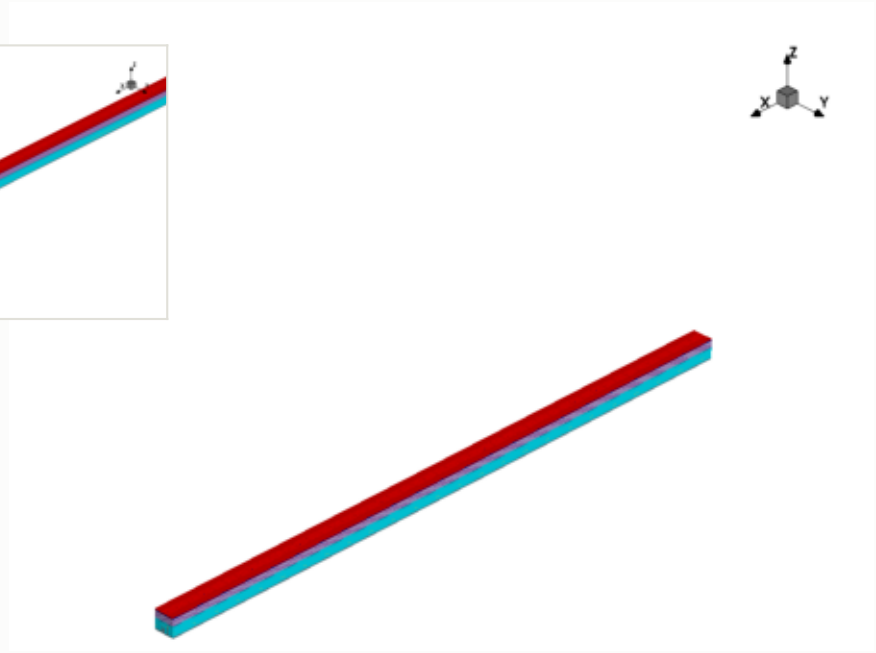
clean · Wafer_clean

0.169	316.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





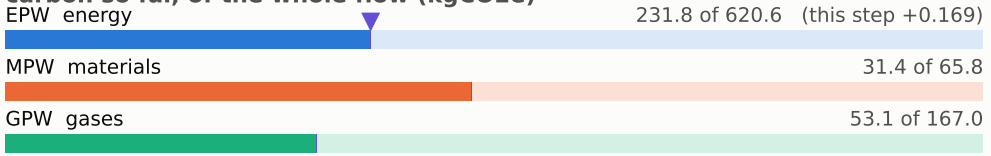
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

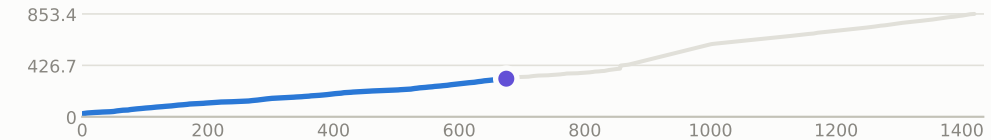
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

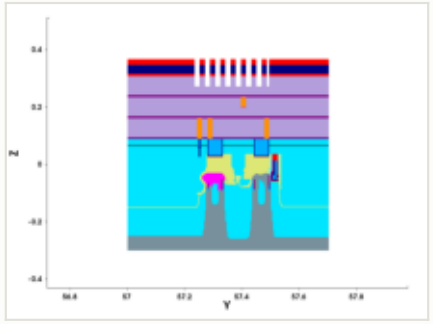
carbon so far, of the whole flow (kgCO2e)



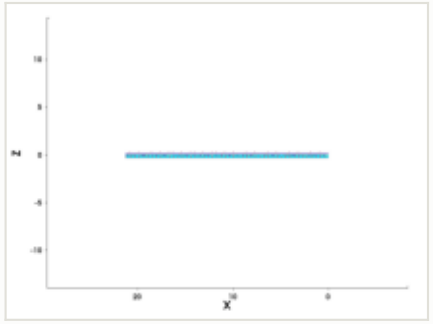
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



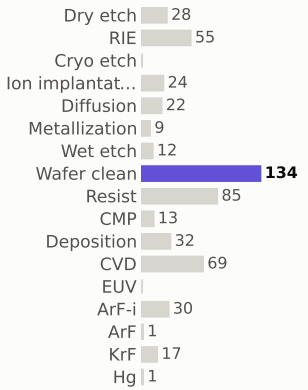
y cut at 57.351 µm



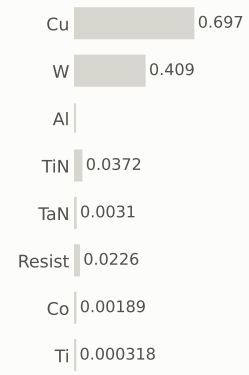
mask for this step

no mask for this step

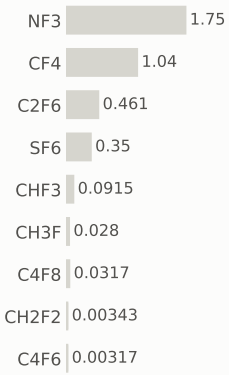
steps by process area



materials deposited (g)



gases emitted (g)



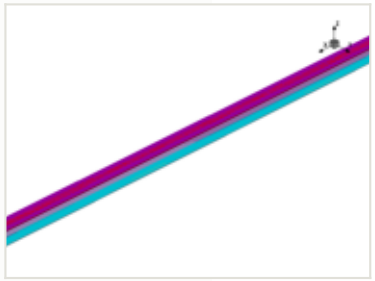
Step #6860: Metallization

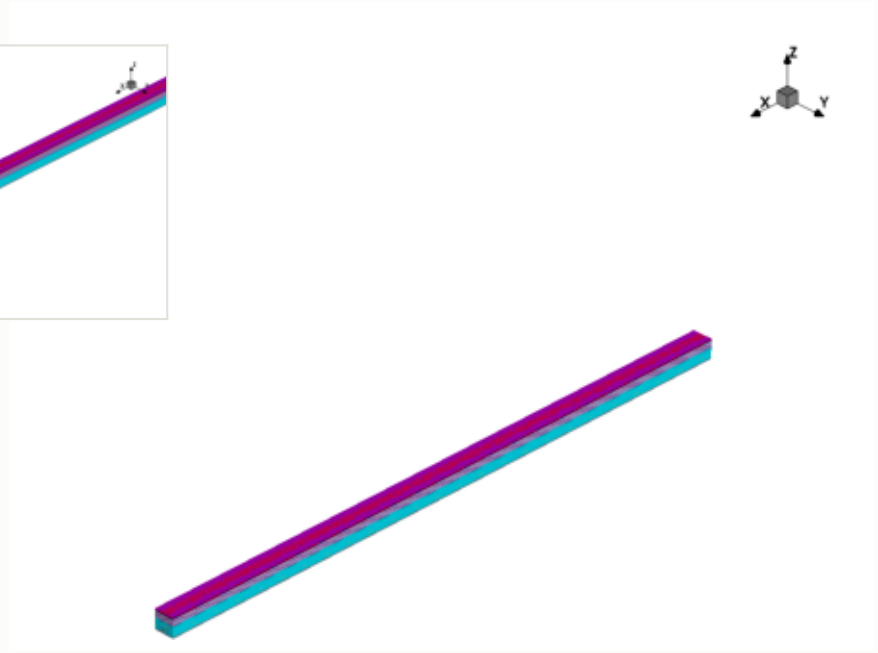
depo_ellipsoidal · Metallization

1.2	317.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



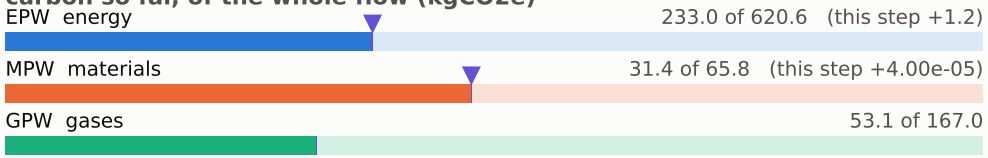


this step

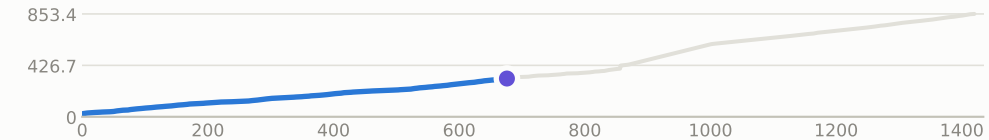
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

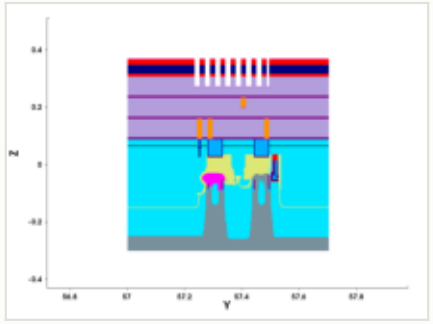
carbon so far, of the whole flow (kgCO2e)



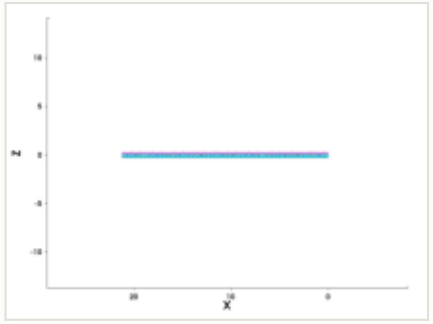
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



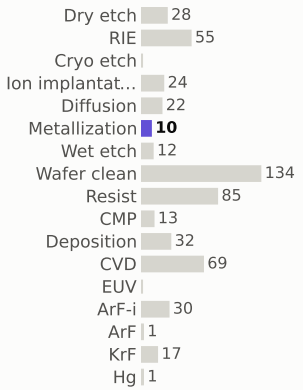
y cut at 57.351 µm



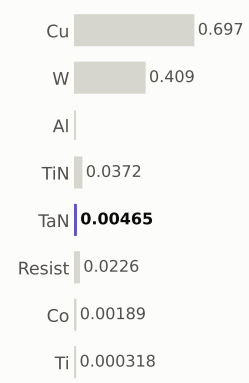
mask for this step

no mask for this step

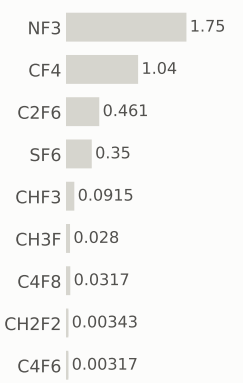
steps by process area



materials deposited (g)



gases emitted (g)



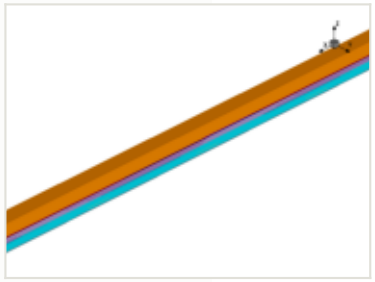
Step #6870: Metallization

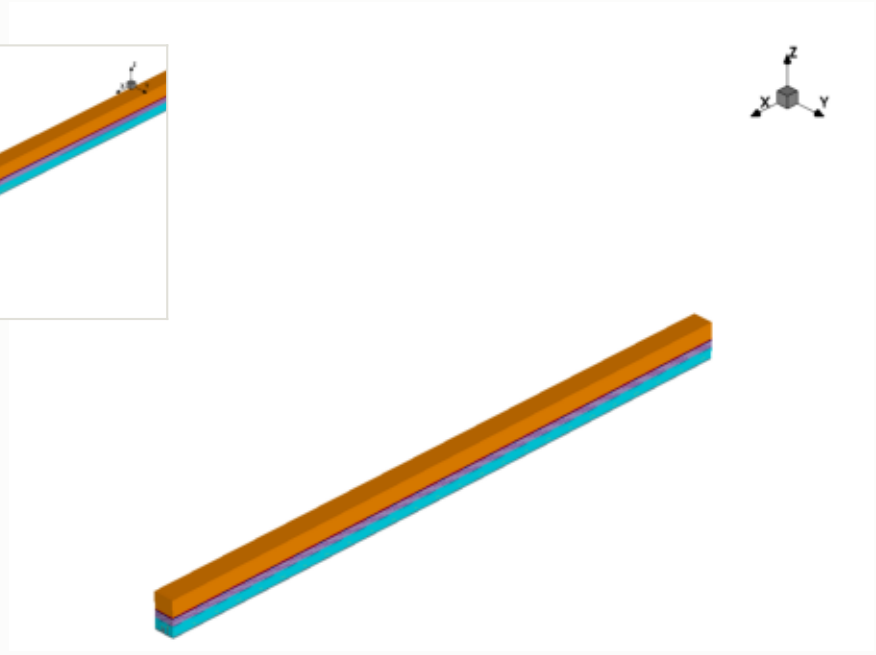
depo_fill · Metallization

1.21	318.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



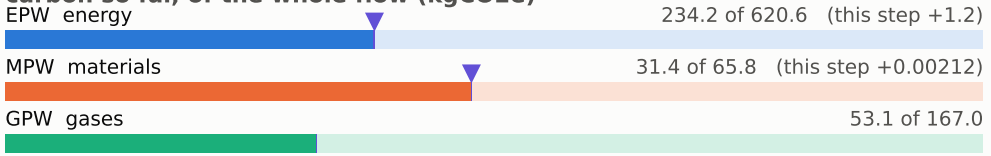


this step

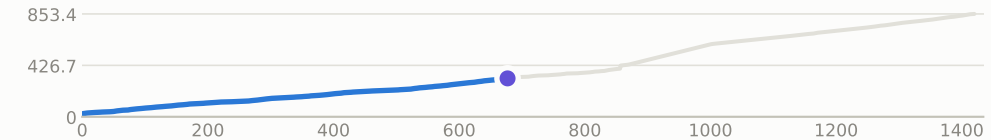
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

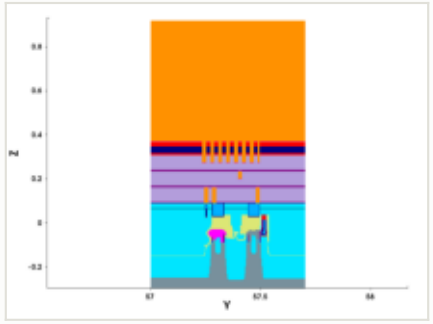
carbon so far, of the whole flow (kgCO2e)



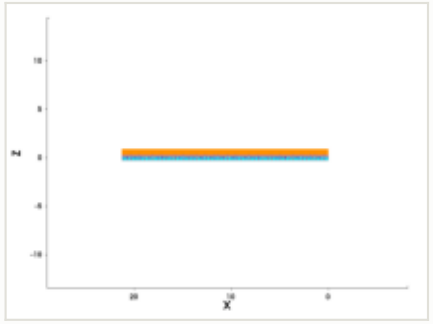
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



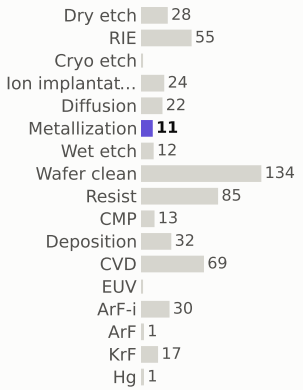
y cut at 57.351 µm



mask for this step

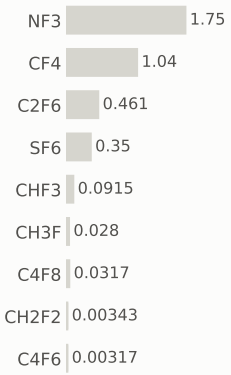
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



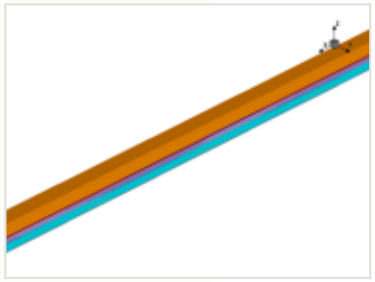
Step #6880: Diffusion

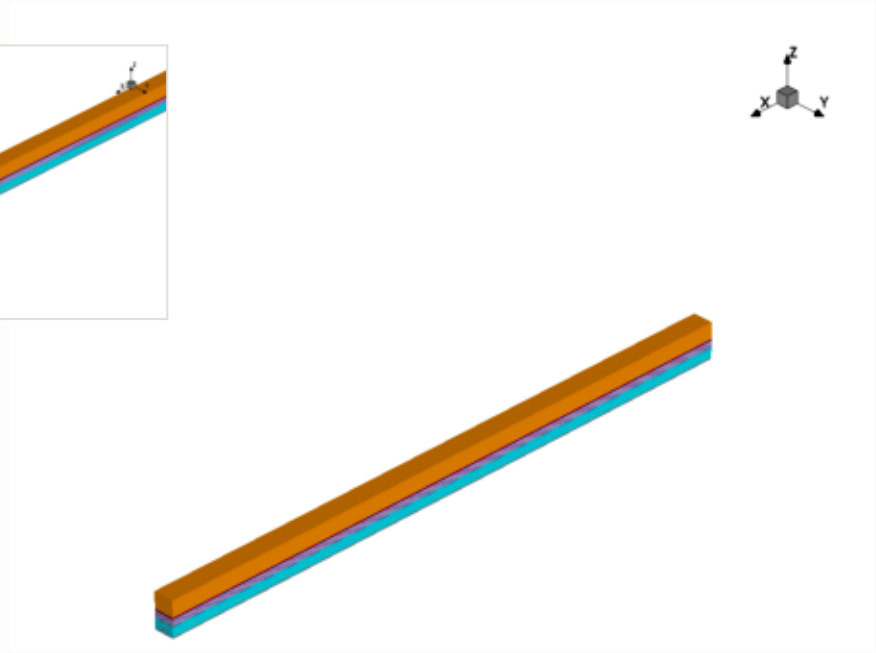
anneal · Diffusion

0.801	319.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





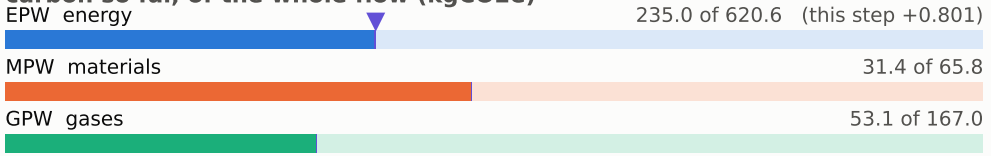
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

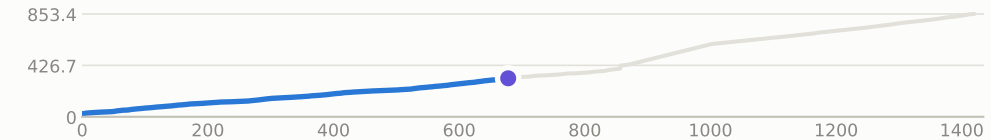
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

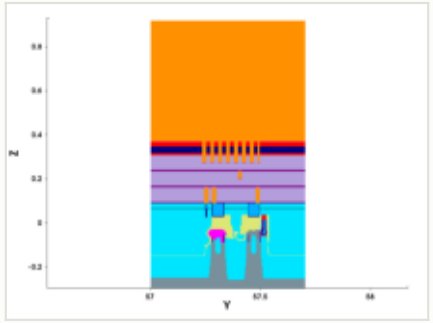
carbon so far, of the whole flow (kgCO2e)



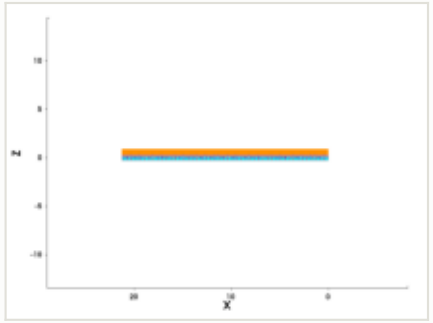
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



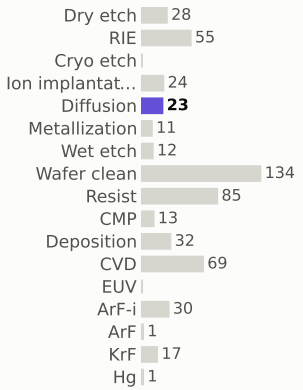
y cut at 57.351 µm



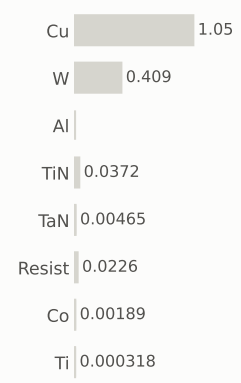
mask for this step

no mask for this step

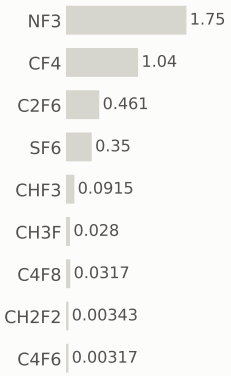
steps by process area



materials deposited (g)



gases emitted (g)



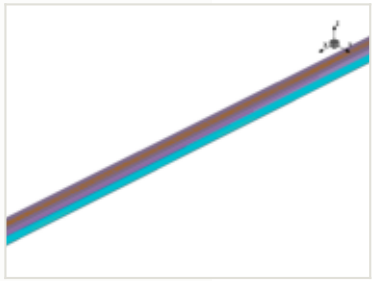
Step #6890: CMP

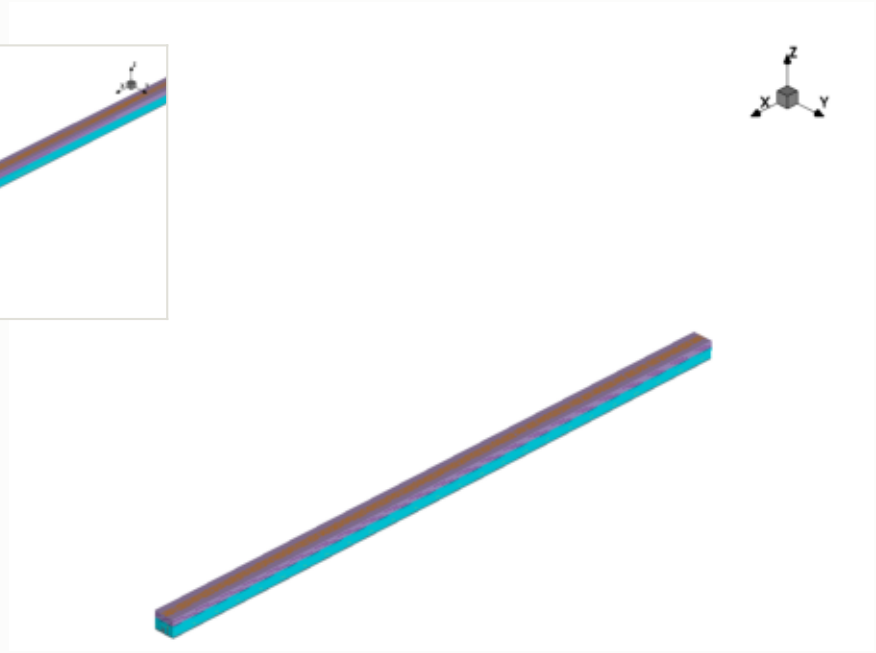
etch_cmp · CMP

0.169	319.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



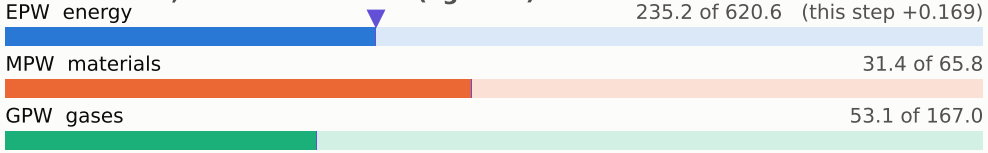


this step

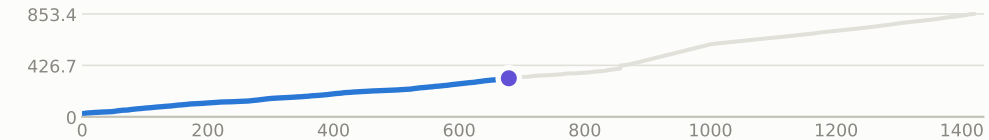
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

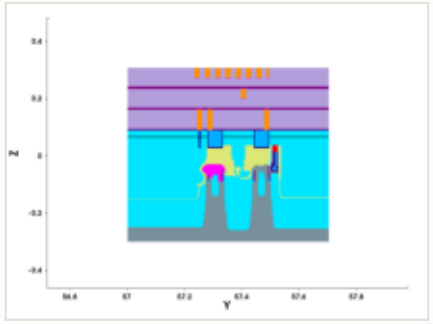
carbon so far, of the whole flow (kgCO2e)



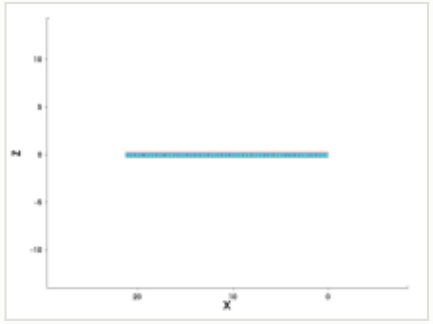
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



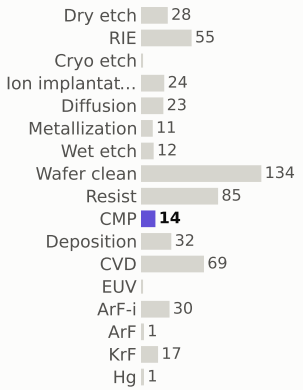
y cut at 57.351 µm



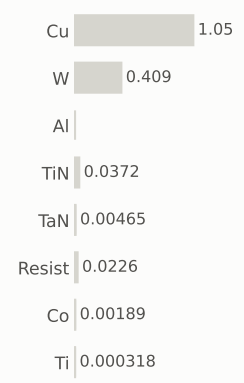
mask for this step

no mask for this step

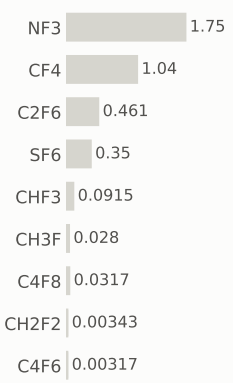
steps by process area



materials deposited (g)



gases emitted (g)



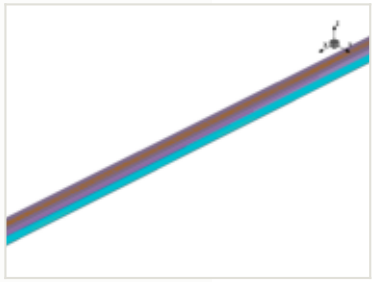
Step #6900: Wafer_clean

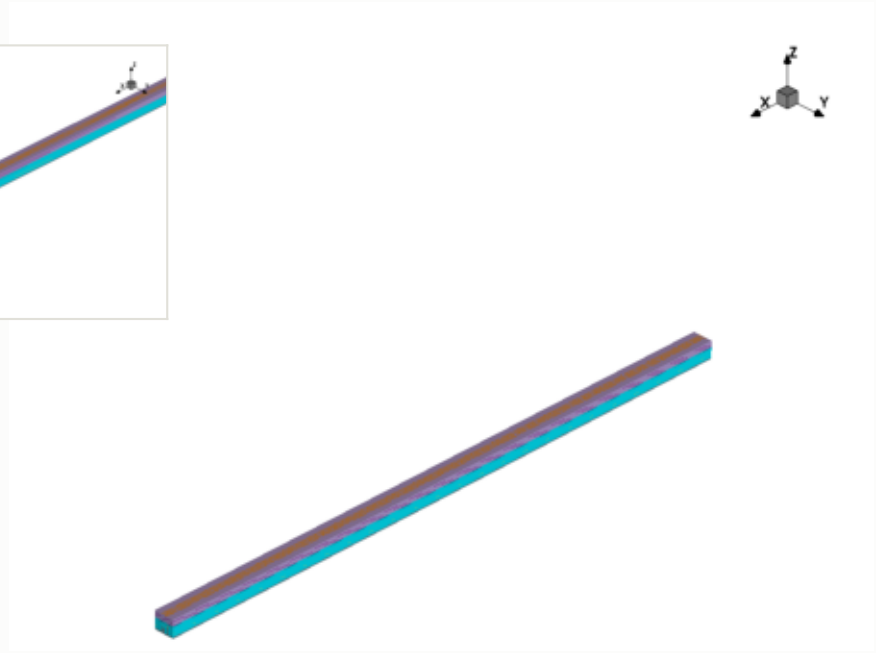
clean · Wafer_clean

0.169	319.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



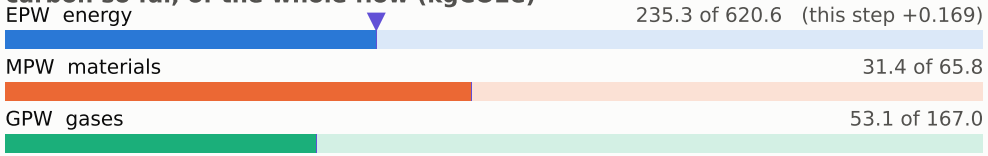


this step

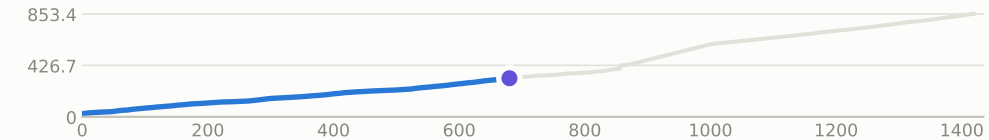
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

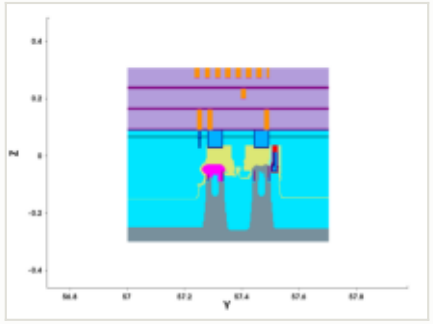
carbon so far, of the whole flow (kgCO2e)



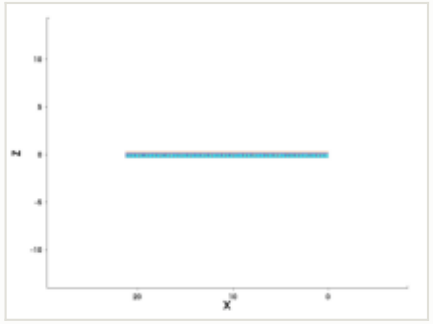
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



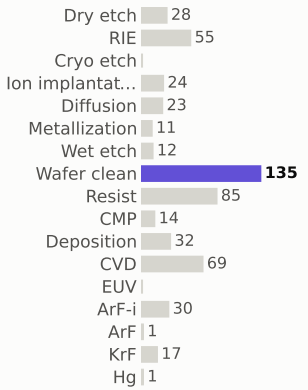
y cut at 57.351 µm



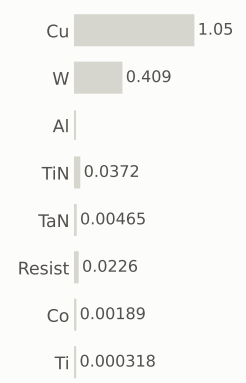
mask for this step

no mask for this step

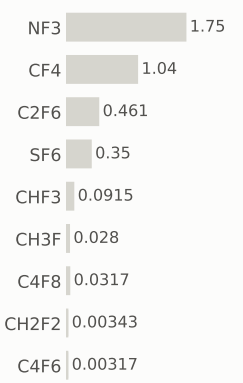
steps by process area



materials deposited (g)



gases emitted (g)



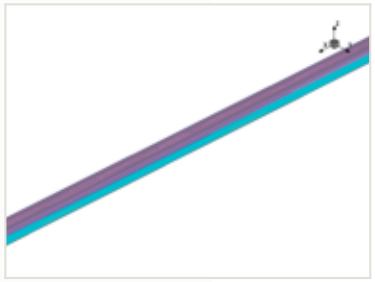
Step #6910: Metallization

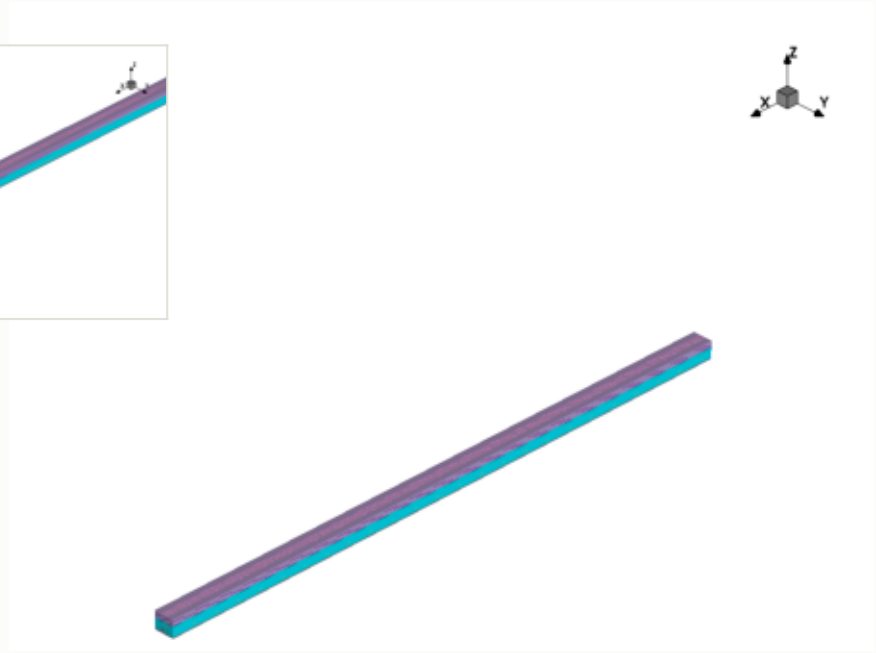
depo_isotropic · Metallization

1.2	321.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



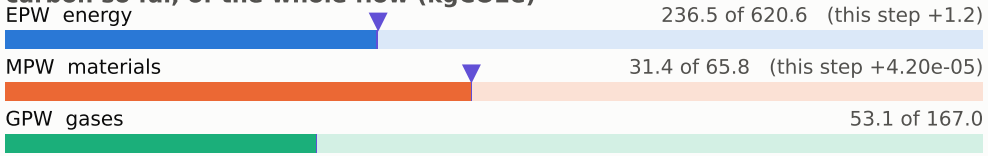


this step

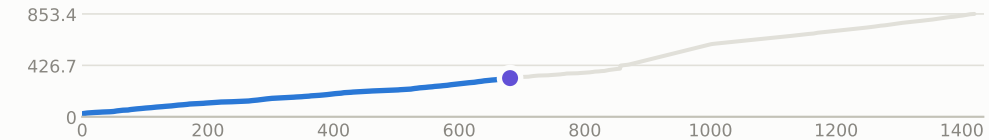
process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

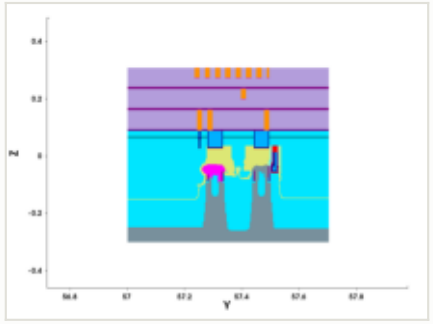
carbon so far, of the whole flow (kgCO2e)



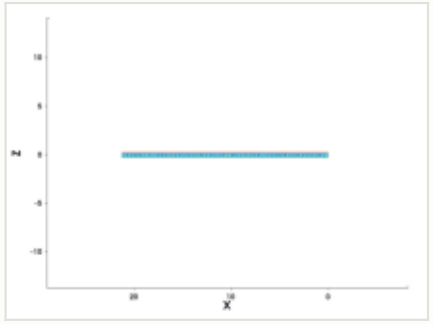
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



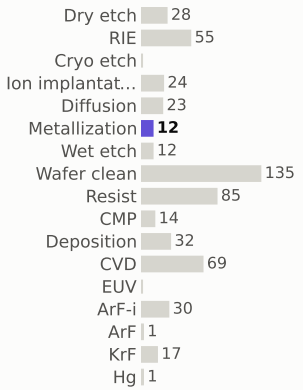
y cut at 57.351 µm



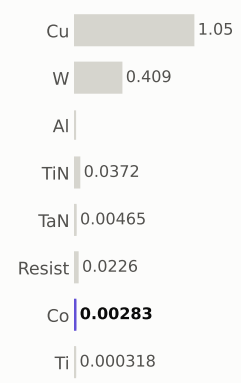
mask for this step

no mask for this step

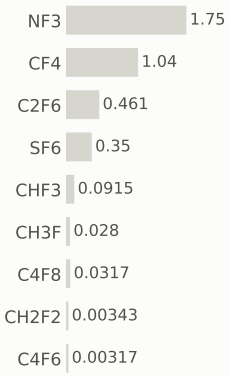
steps by process area



materials deposited (g)



gases emitted (g)



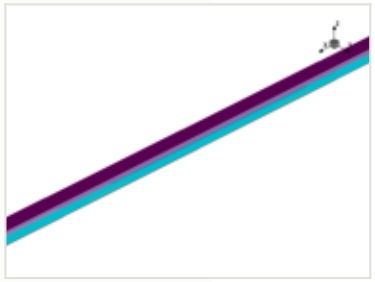
Step #6930: Deposition

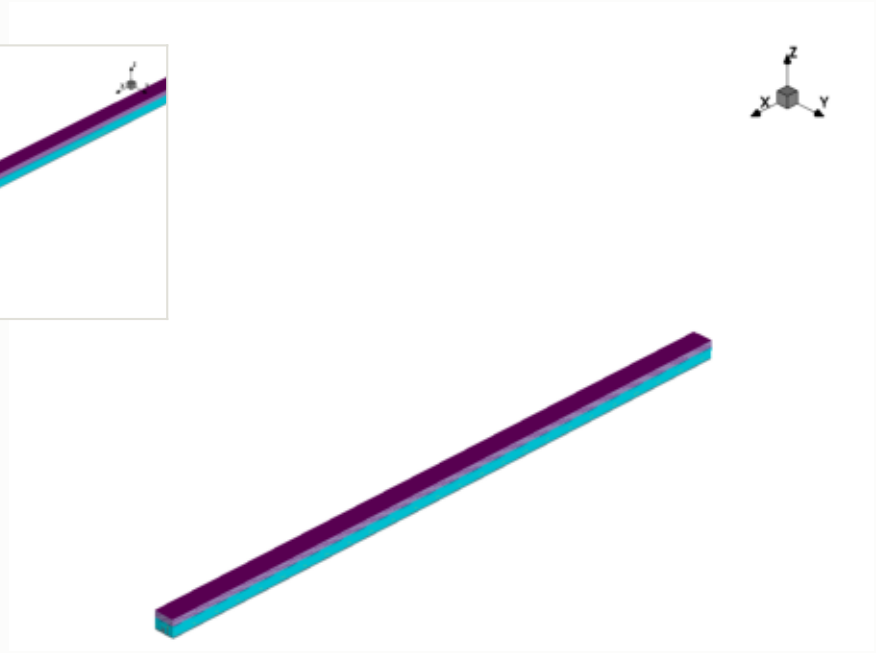
depo_fill · Deposition

0.656	321.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



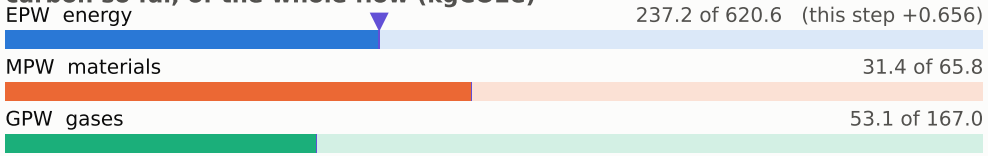


this step

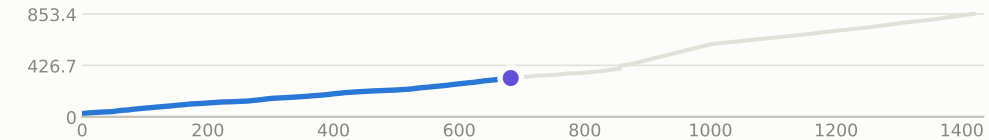
process area	Deposition
material	—
thickness	0.006 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

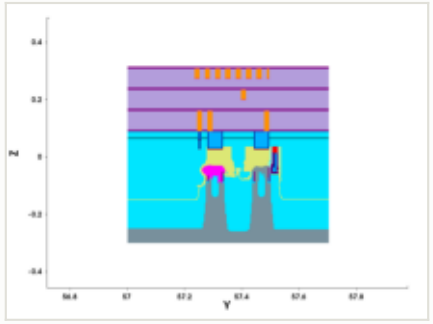
carbon so far, of the whole flow (kgCO2e)



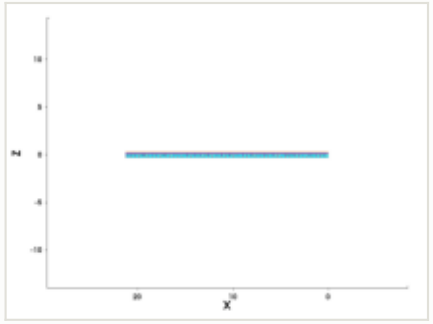
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



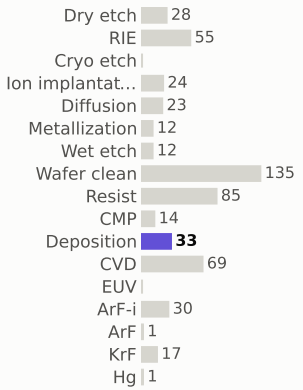
y cut at 57.351 µm



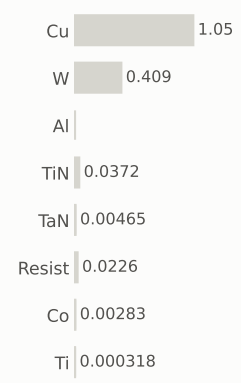
mask for this step

no mask for this step

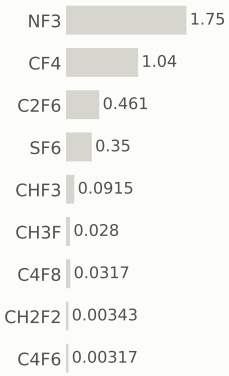
steps by process area



materials deposited (g)



gases emitted (g)



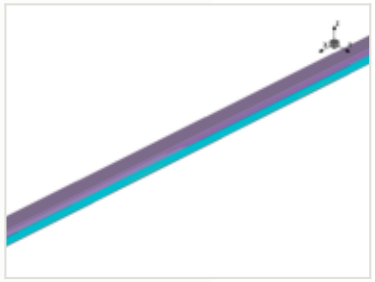
Step #6940: CVD

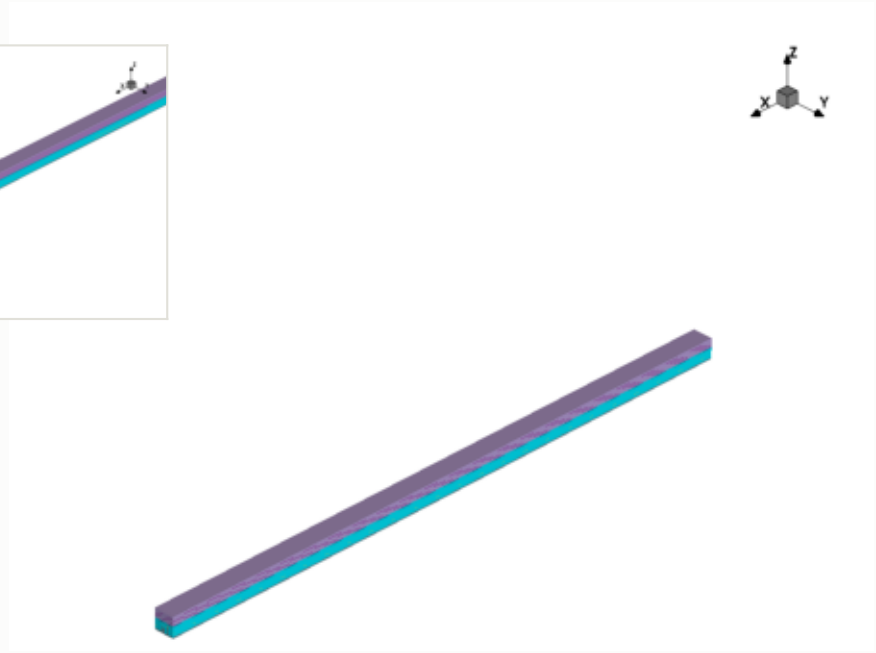
depo_fill · CVD

1.31	323.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



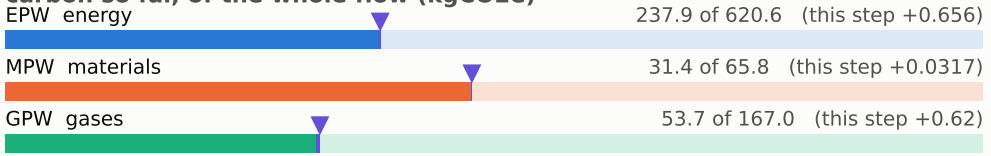


this step

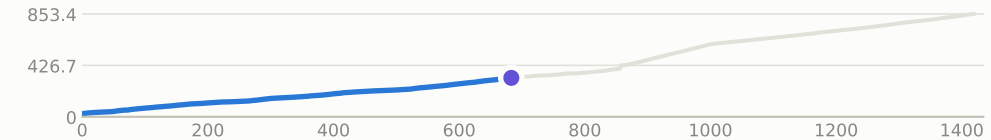
process area	CVD
material	—
thickness	0.078 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

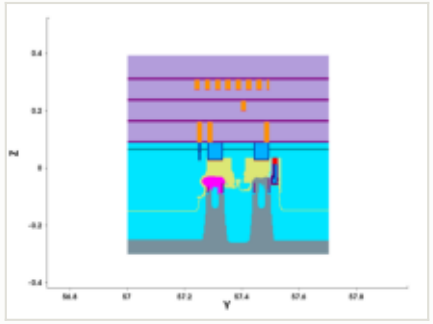
carbon so far, of the whole flow (kgCO2e)



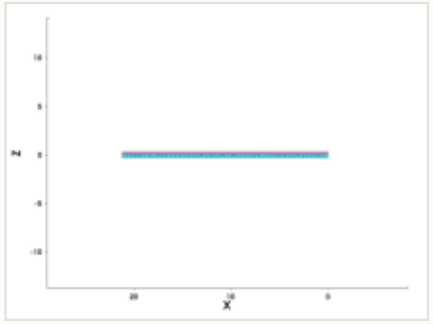
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



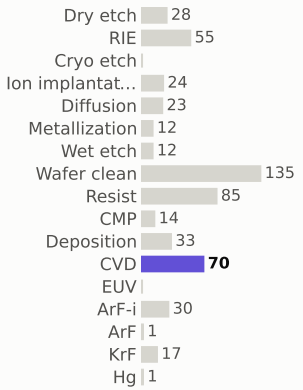
y cut at 57.351 µm



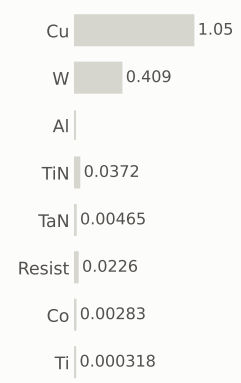
mask for this step

no mask for this step

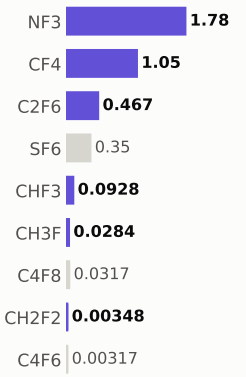
steps by process area



materials deposited (g)



gases emitted (g)



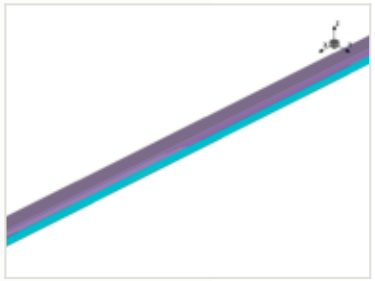
Step #6950: Diffusion

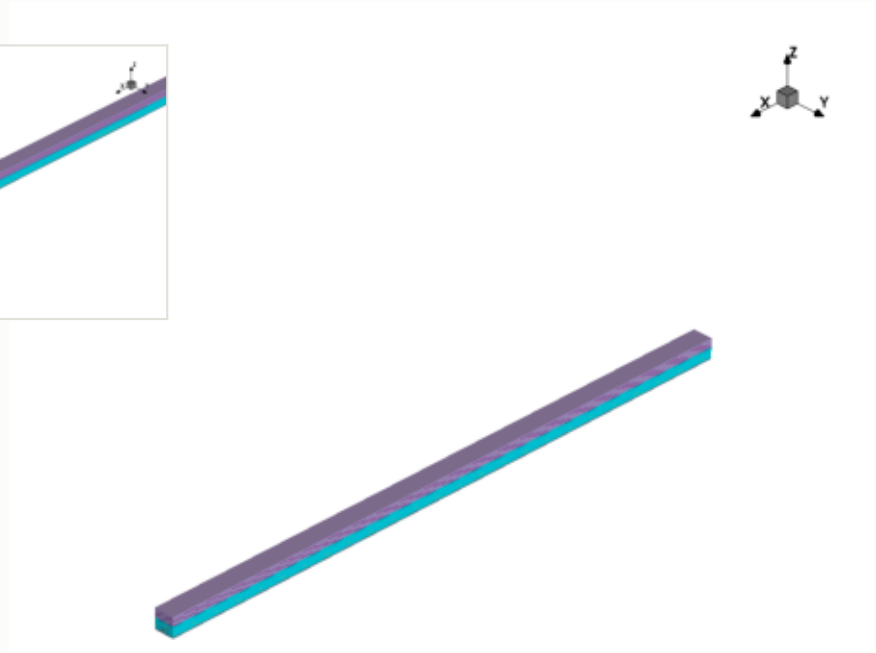
anneal · Diffusion

0.801	323.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

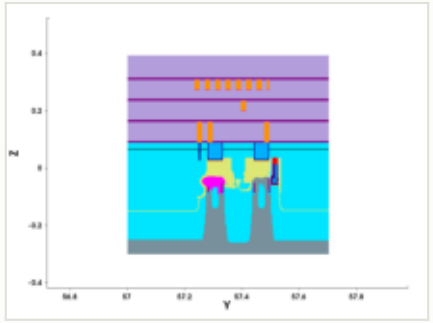
3D structure after this step

zoomed to fit

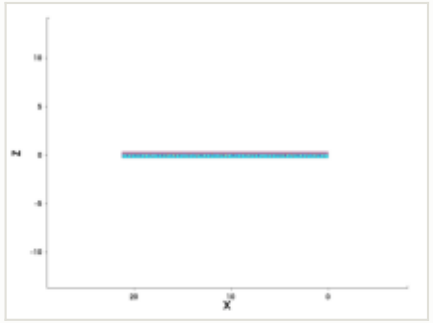




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

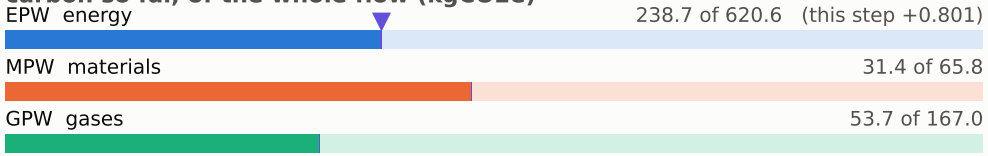
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

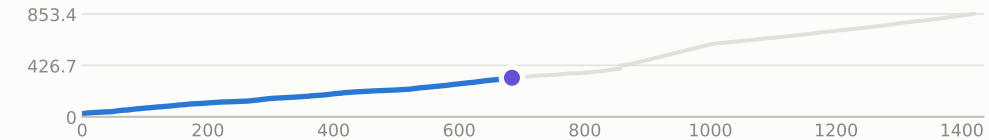
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

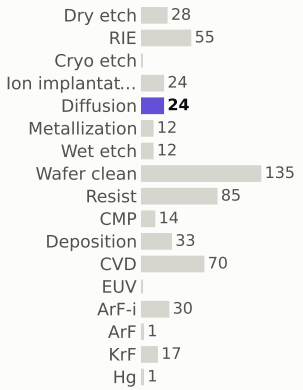
carbon so far, of the whole flow (kgCO2e)



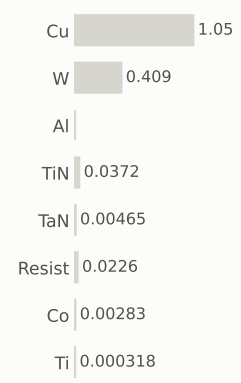
embodied carbon up to each step (kgCO2e per wafer)



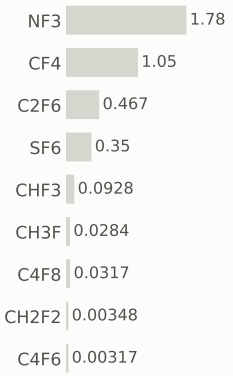
steps by process area



materials deposited (g)



gases emitted (g)



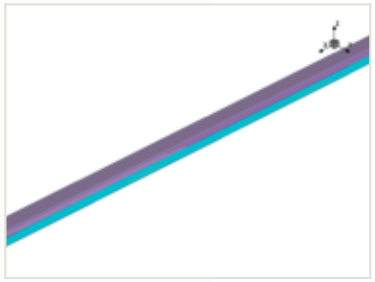
Step #6960: Wafer_clean

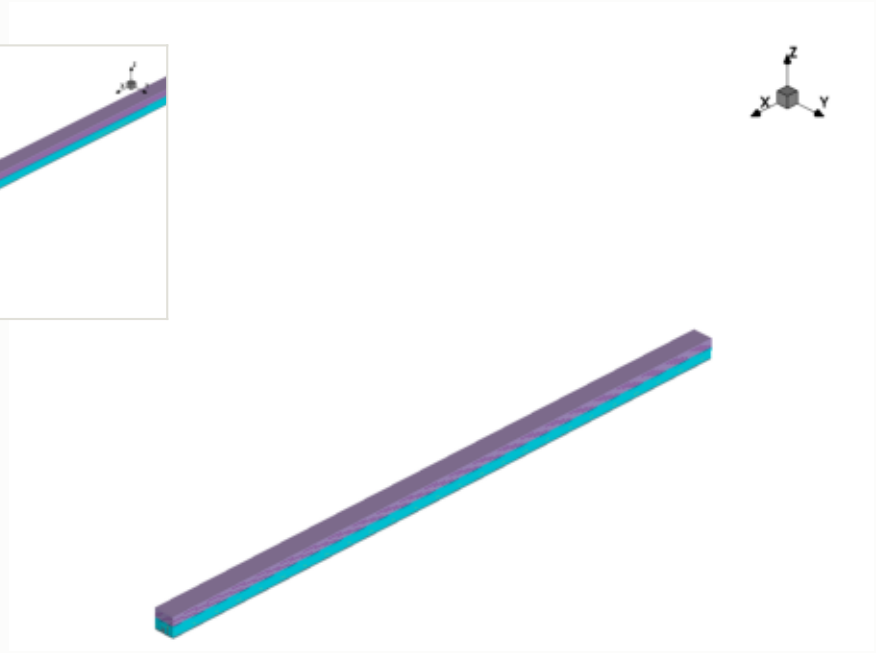
clean · Wafer_clean

0.169	324.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





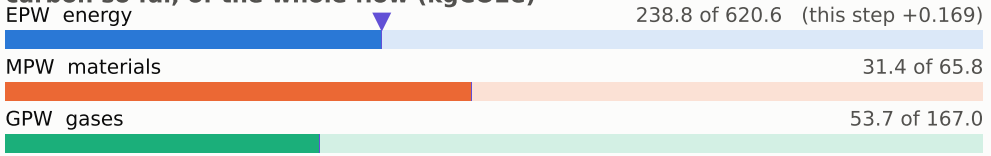
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

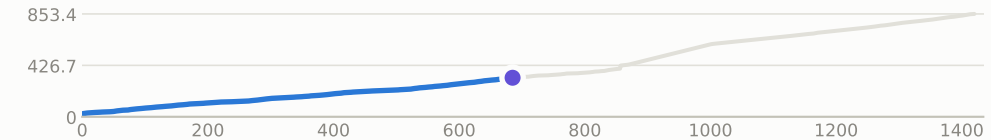
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

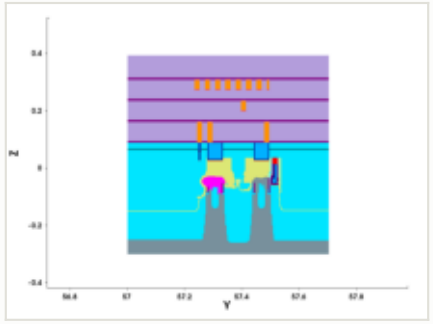
carbon so far, of the whole flow (kgCO2e)



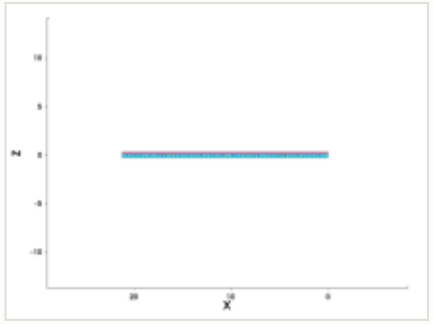
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



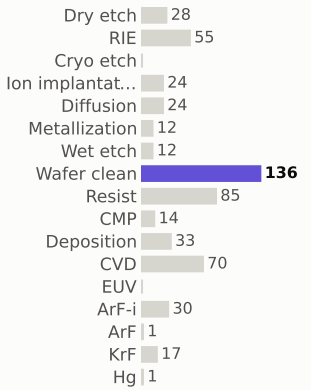
y cut at 57.351 µm



mask for this step

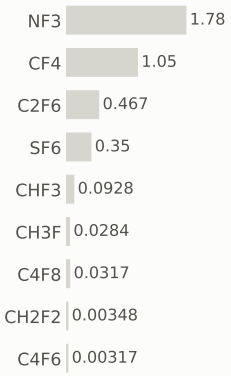
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



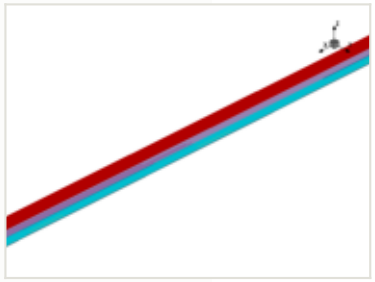
Step #6970: CVD

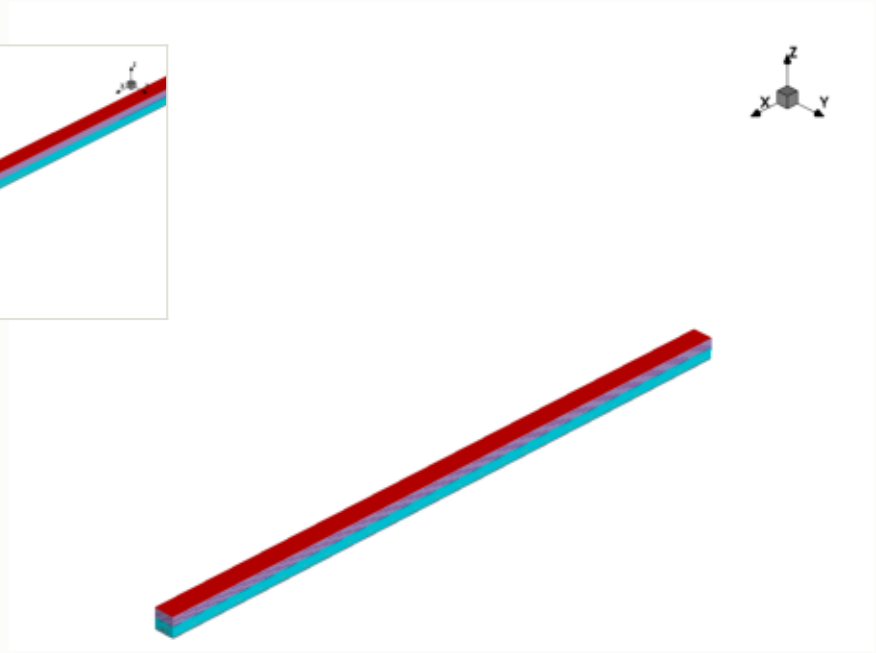
depo_fill · CVD

1.31	325.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



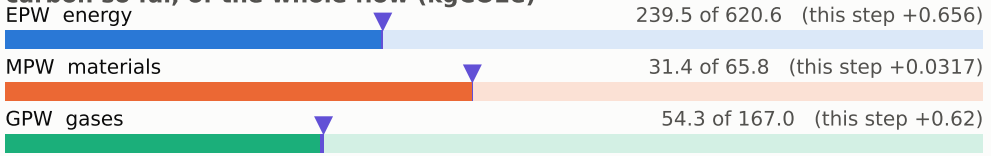


this step

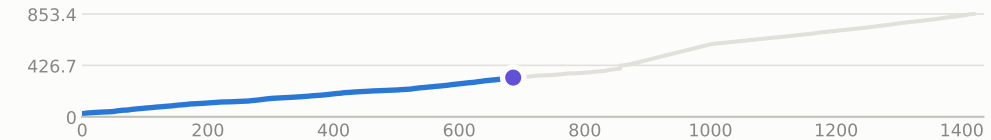
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

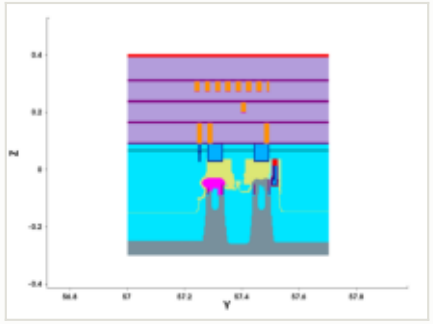
carbon so far, of the whole flow (kgCO2e)



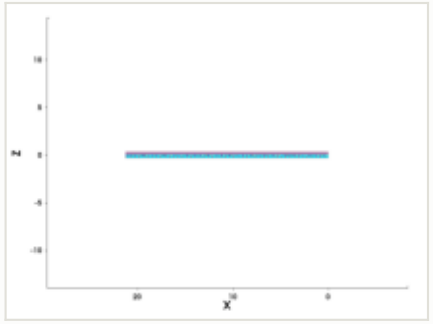
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



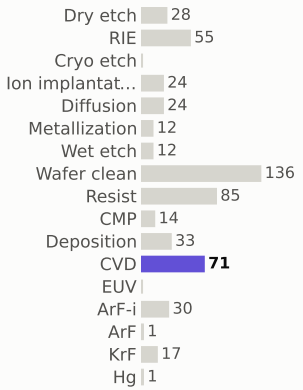
y cut at 57.351 µm



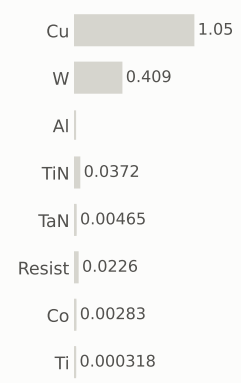
mask for this step

no mask for this step

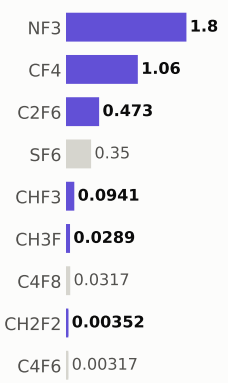
steps by process area



materials deposited (g)



gases emitted (g)



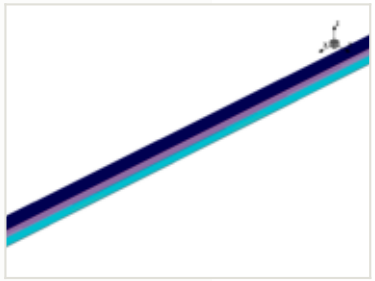
Step #6980: Deposition

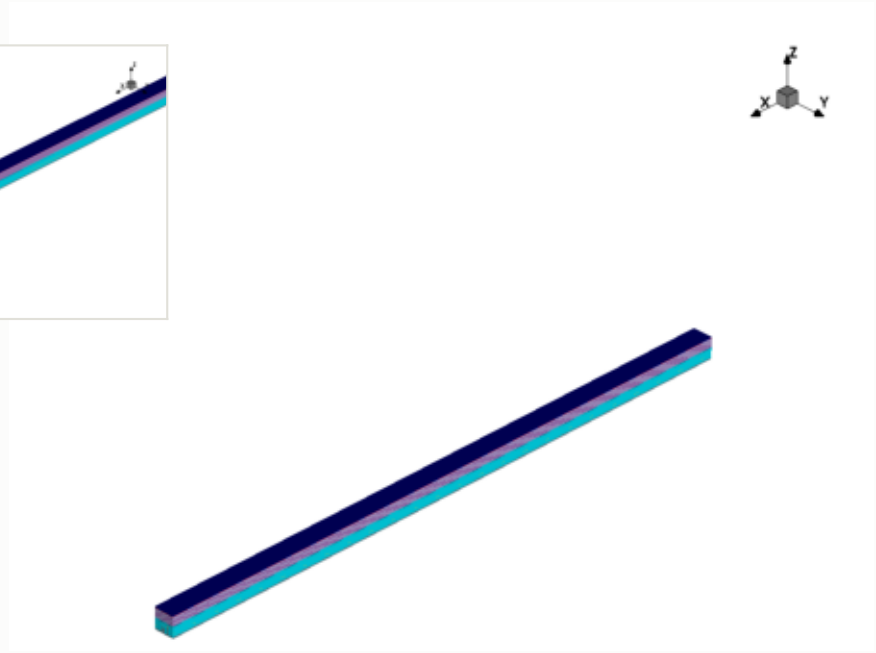
depo_fill · Deposition

0.657	325.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



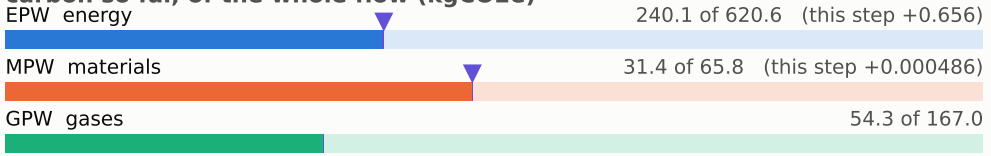


this step

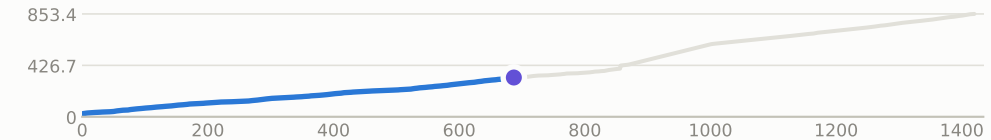
process area	Deposition
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000486 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

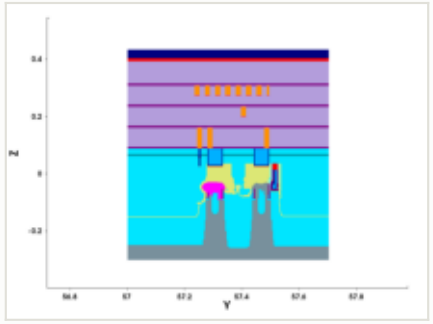
carbon so far, of the whole flow (kgCO2e)



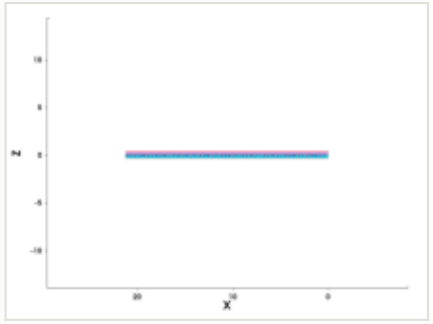
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



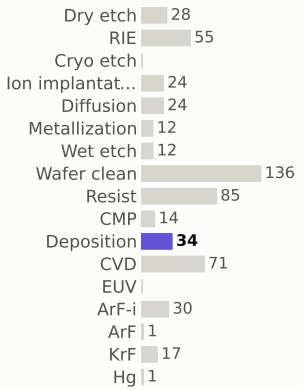
y cut at 57.351 µm



mask for this step

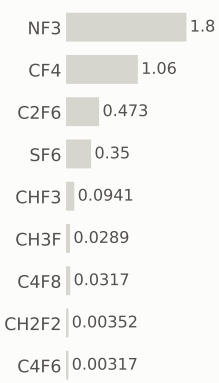
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



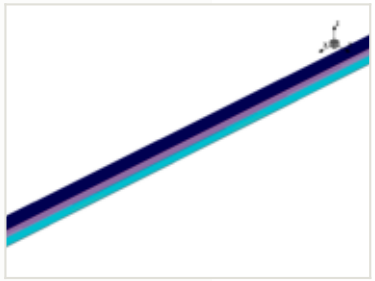
Step #6990: Wafer_clean

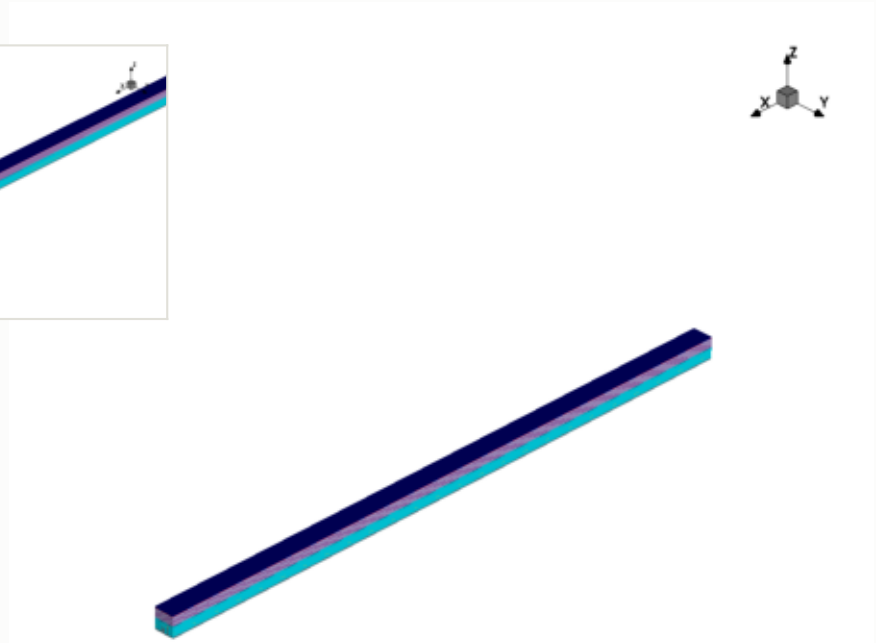
clean · Wafer_clean

0.169	326.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

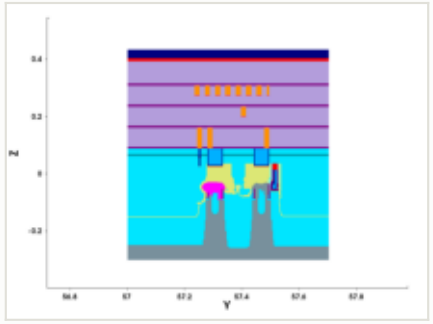
3D structure after this step

zoomed to fit

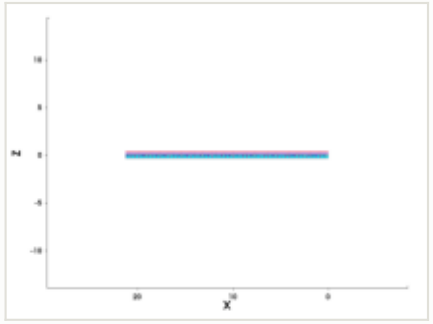




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

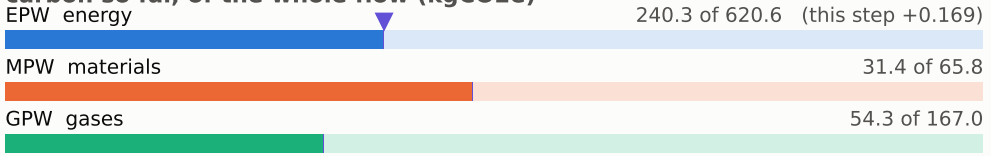
no mask for this step

this step

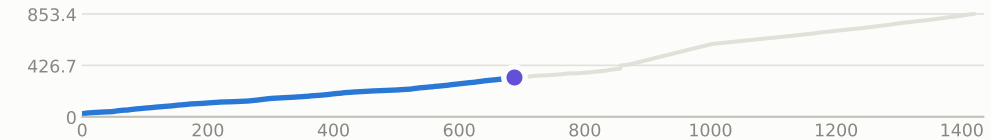
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

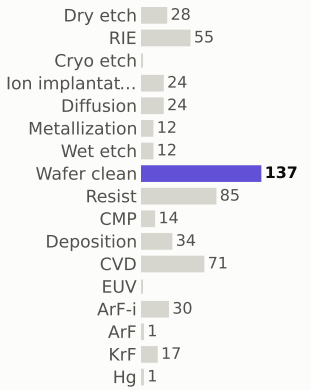
carbon so far, of the whole flow (kgCO2e)



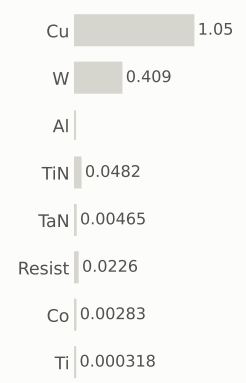
embodied carbon up to each step (kgCO2e per wafer)



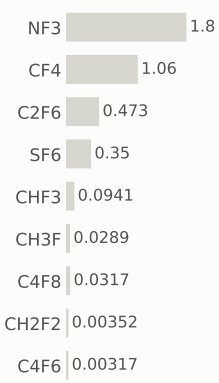
steps by process area



materials deposited (g)



gases emitted (g)



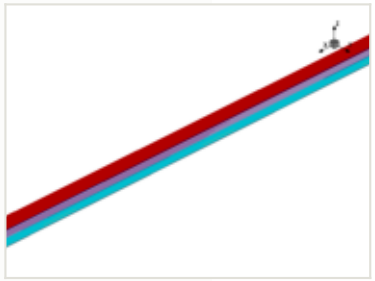
Step #7000: CVD

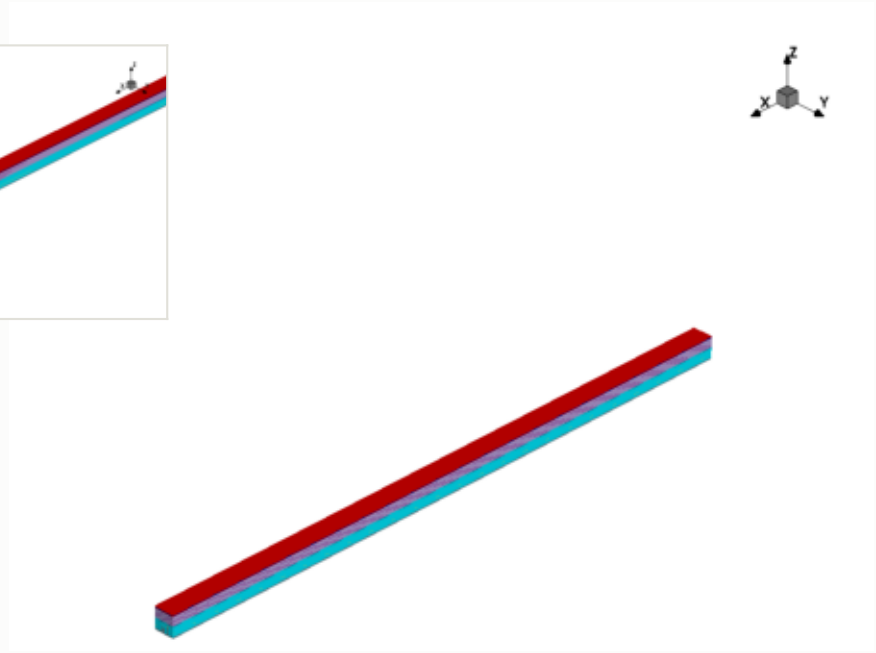
depo_fill · CVD

1.31	327.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



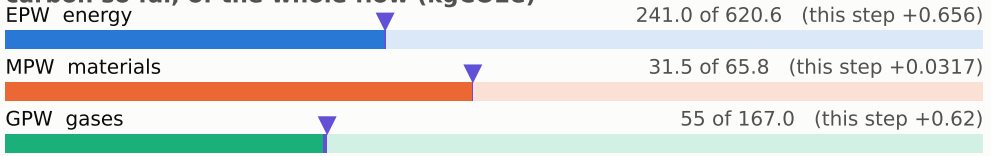


this step

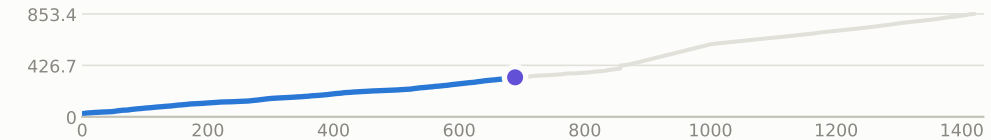
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

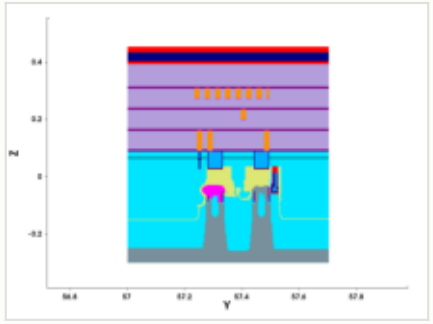
carbon so far, of the whole flow (kgCO2e)



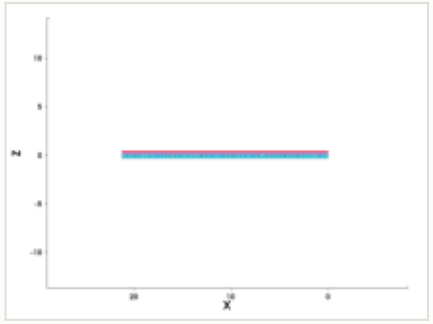
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



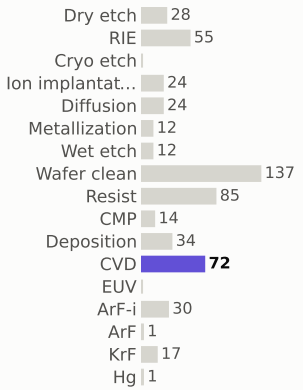
y cut at 57.351 µm



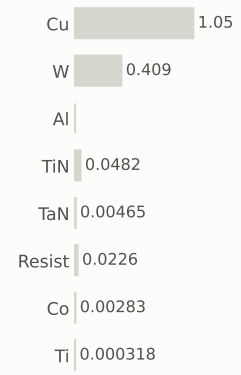
mask for this step

no mask for this step

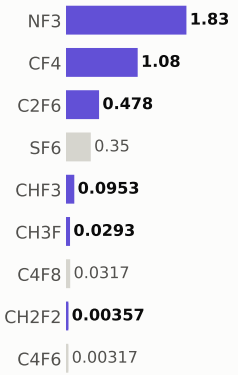
steps by process area



materials deposited (g)



gases emitted (g)



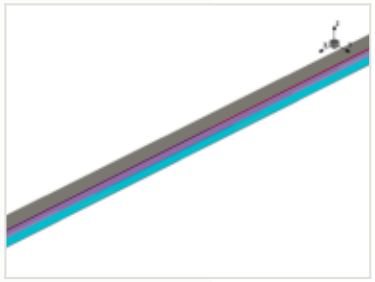
Step #7010: Spx_only

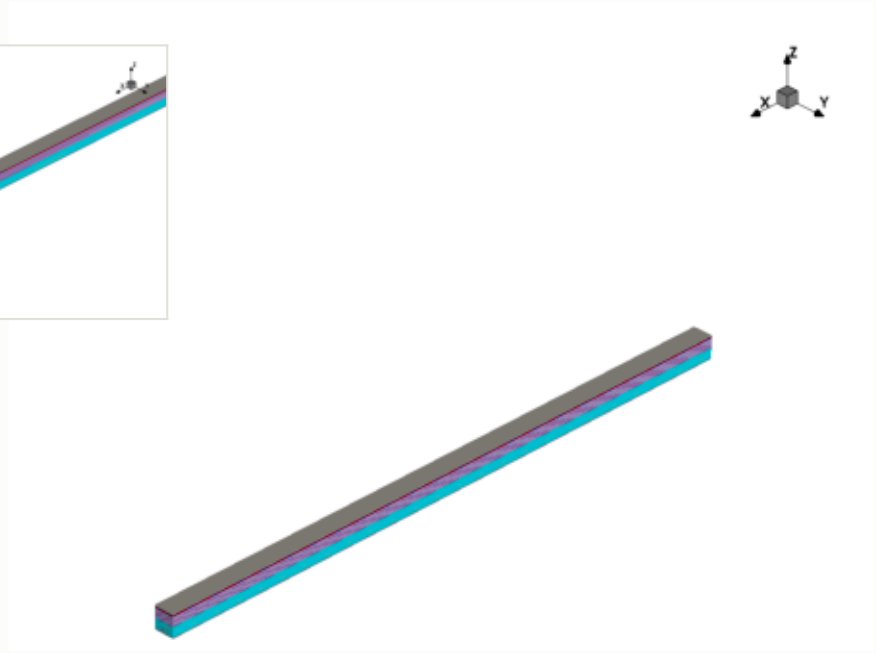
depo_fill · Spx_only

0	327.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





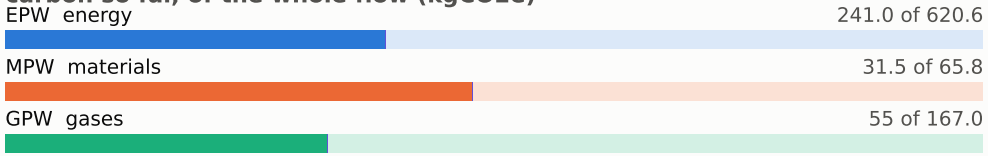
this step

process area	Spx_only
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

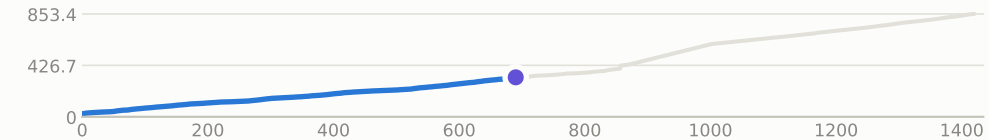
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

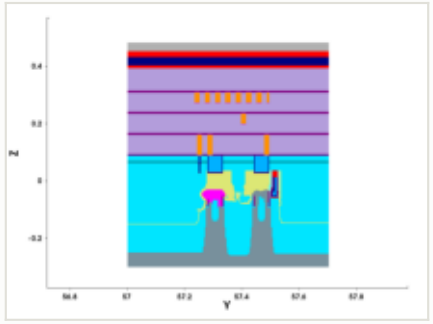
carbon so far, of the whole flow (kgCO2e)



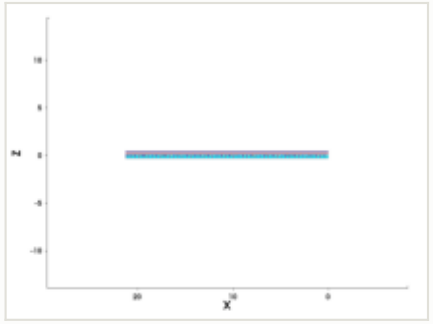
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



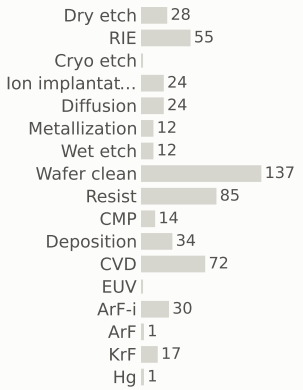
y cut at 57.351 µm



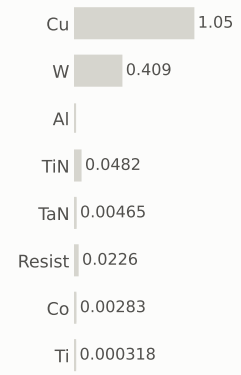
mask for this step

no mask for this step

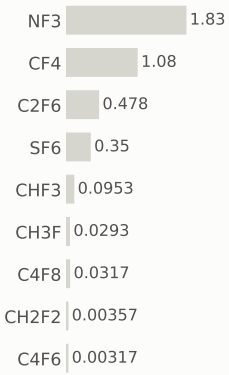
steps by process area



materials deposited (g)



gases emitted (g)



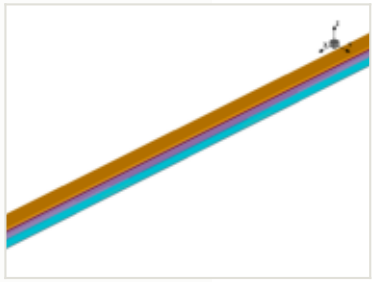
Step #7020: Resist

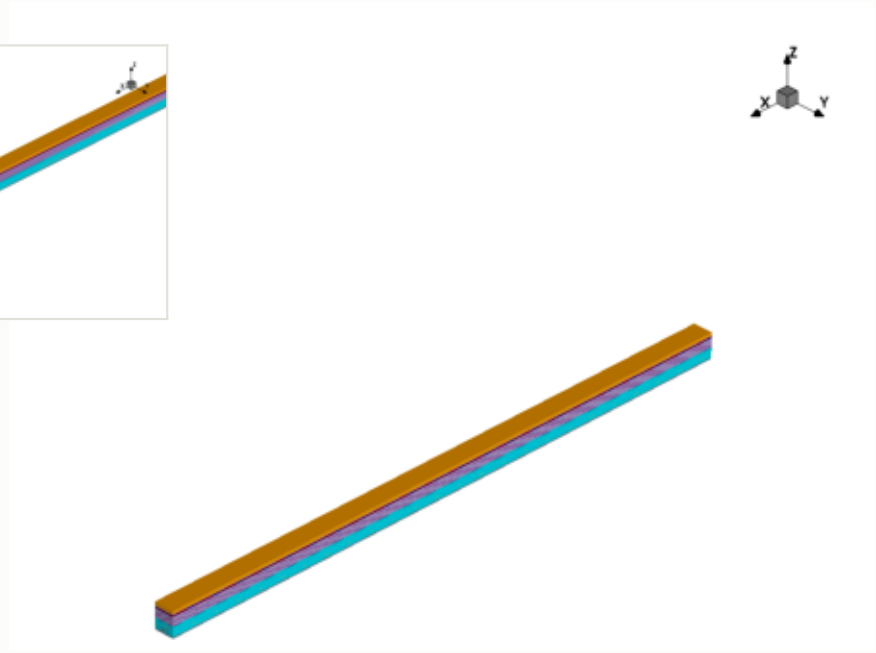
depo_fill · Resist

0.169	327.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



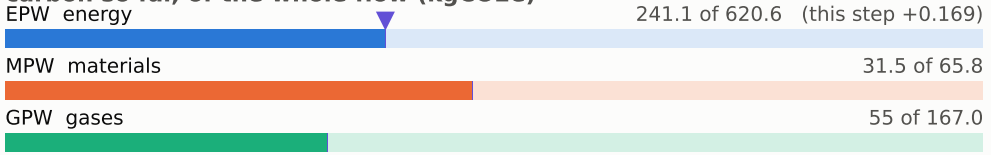


this step

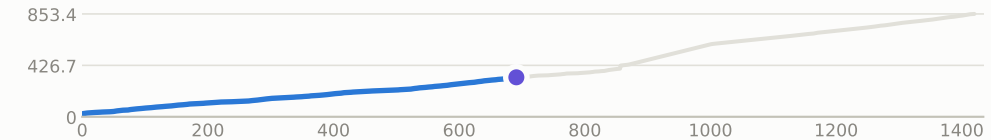
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

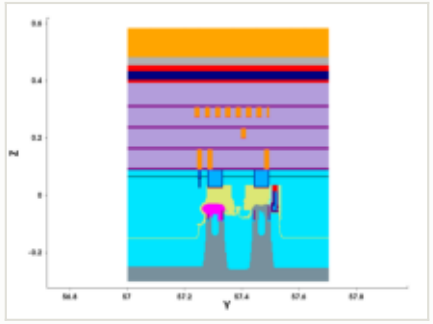
carbon so far, of the whole flow (kgCO2e)



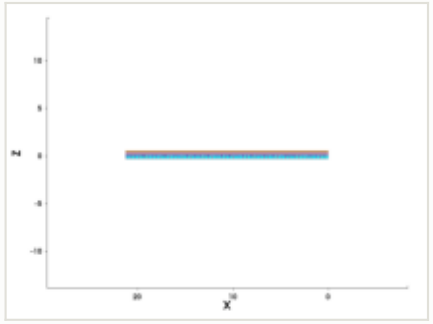
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



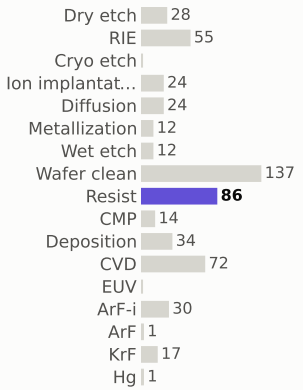
y cut at 57.351 µm



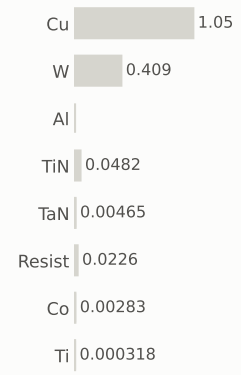
mask for this step

no mask for this step

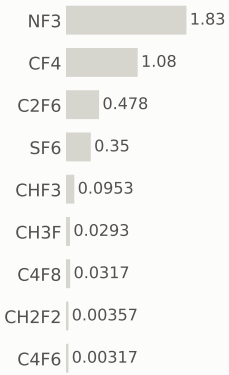
steps by process area



materials deposited (g)



gases emitted (g)



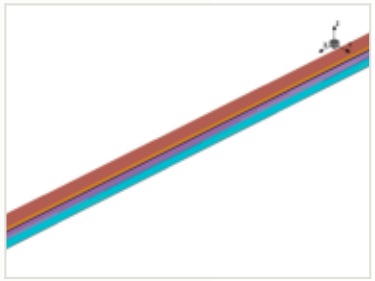
Step #7030: CVD

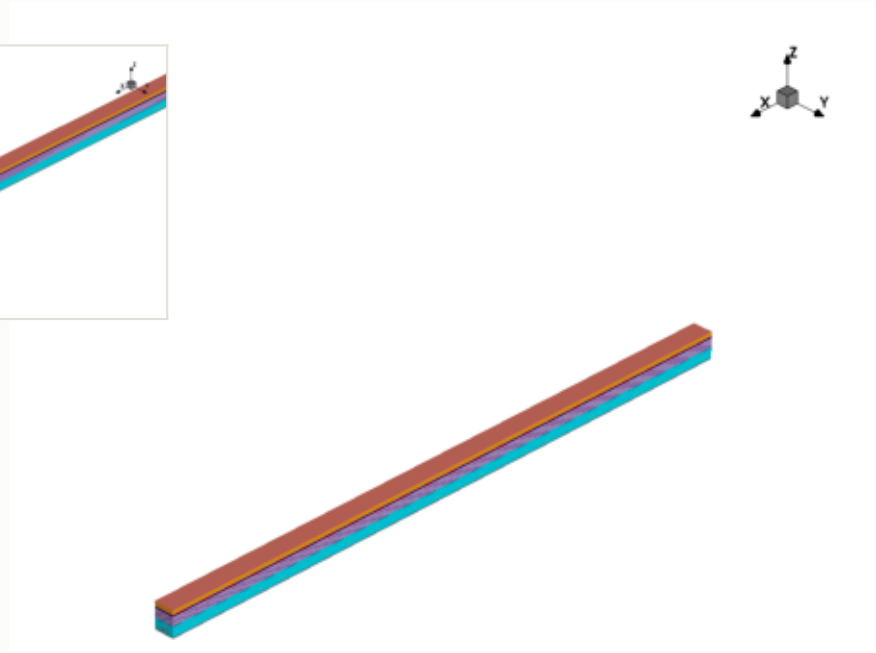
depo_fill · CVD

1.31	328.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



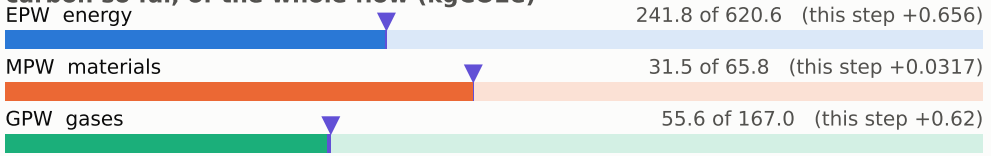


this step

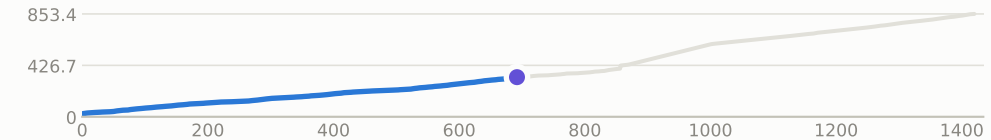
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

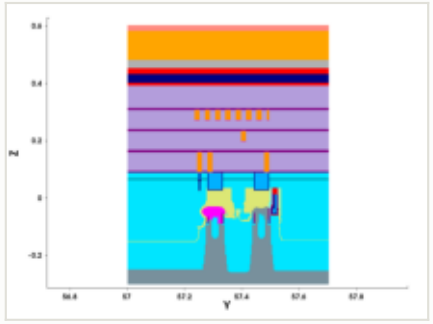
carbon so far, of the whole flow (kgCO2e)



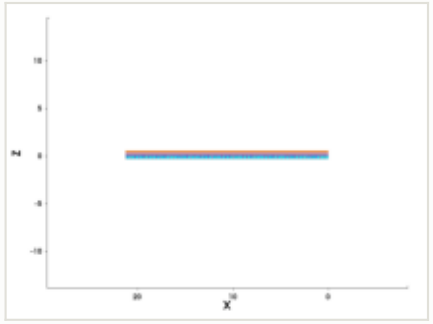
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



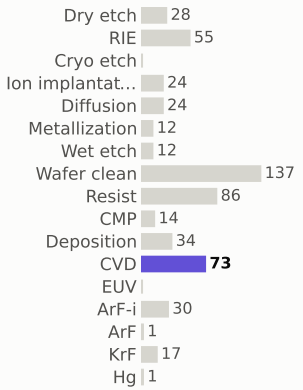
y cut at 57.351 µm



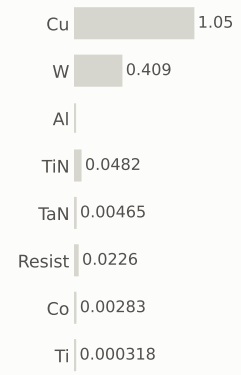
mask for this step

no mask for this step

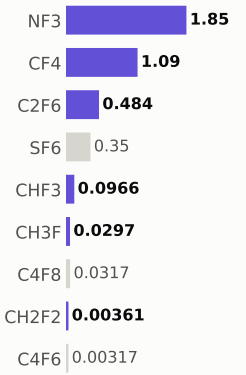
steps by process area



materials deposited (g)



gases emitted (g)



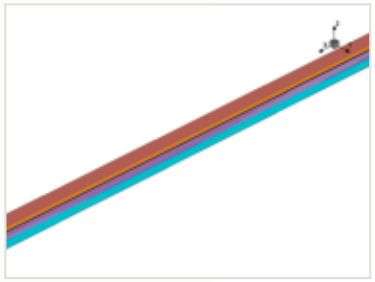
Step #7040: Wafer_clean

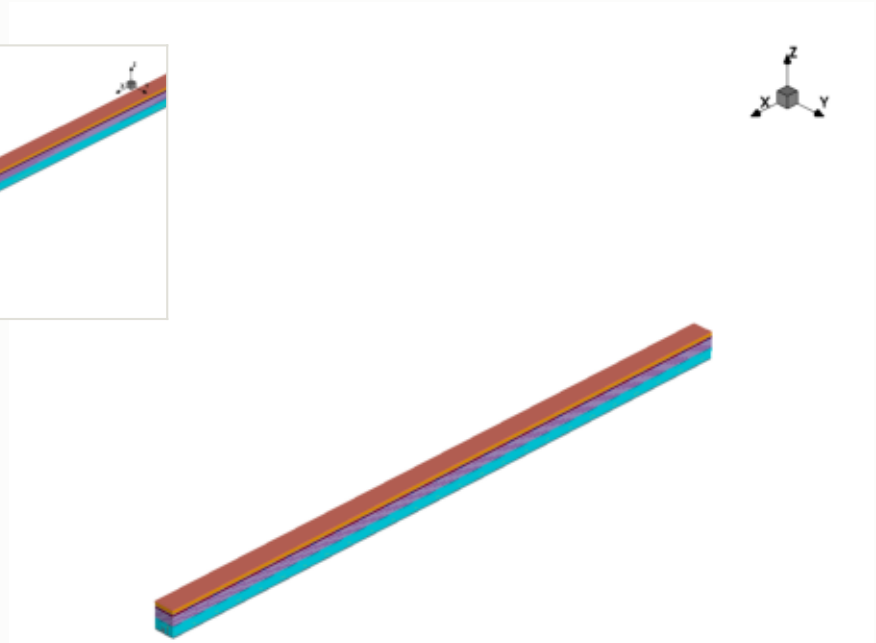
clean · Wafer_clean

0.169	329.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



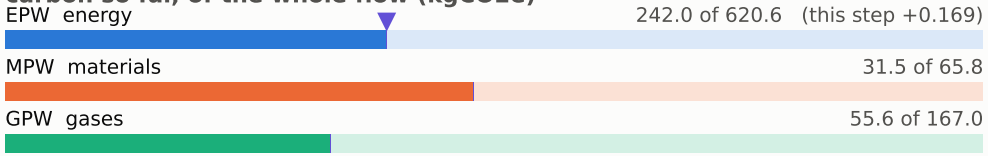


this step

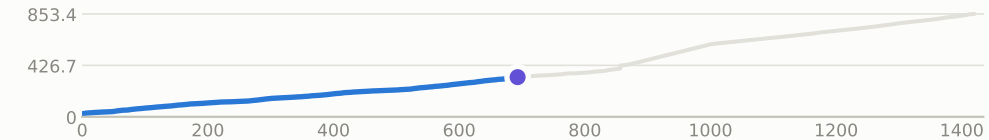
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

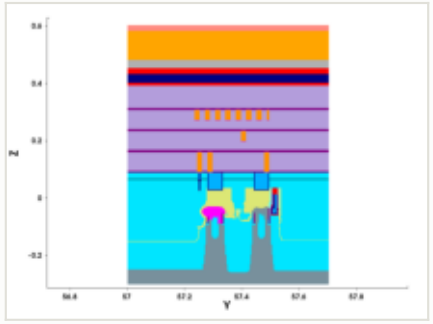
carbon so far, of the whole flow (kgCO2e)



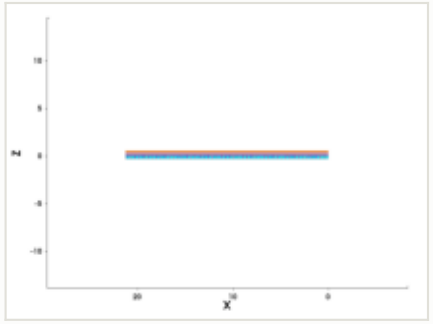
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



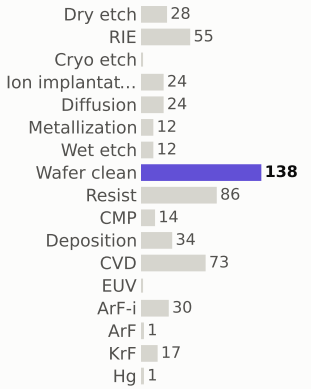
y cut at 57.351 µm



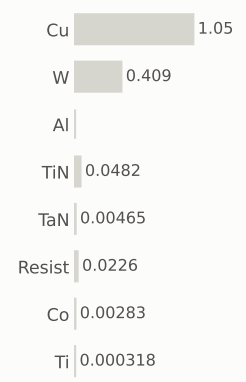
mask for this step

no mask for this step

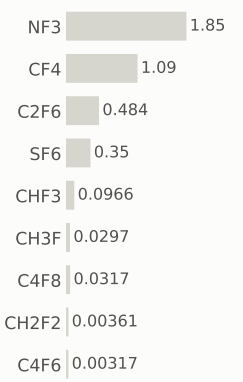
steps by process area



materials deposited (g)



gases emitted (g)



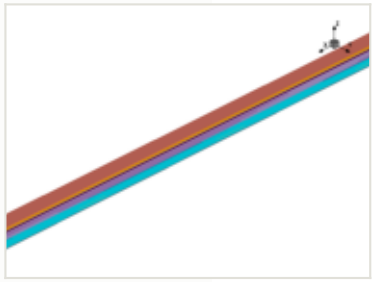
Step #7050: Resist

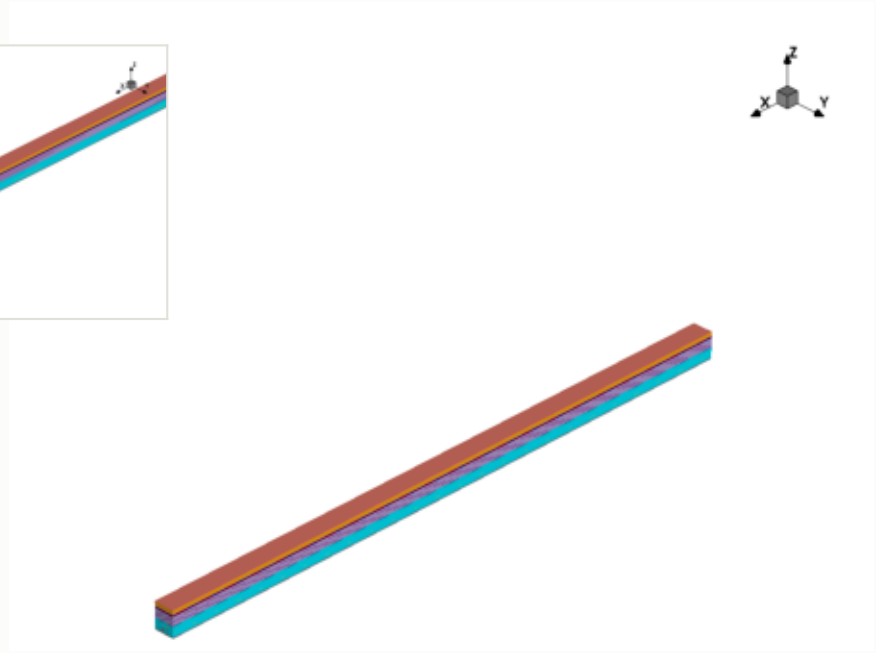
resist · Resist

0.169	329.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





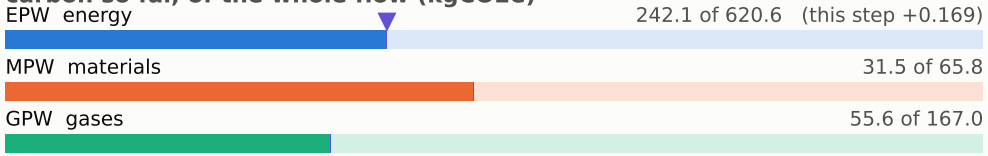
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

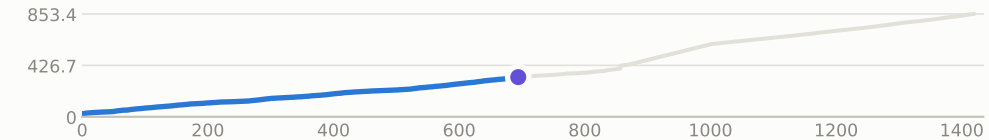
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

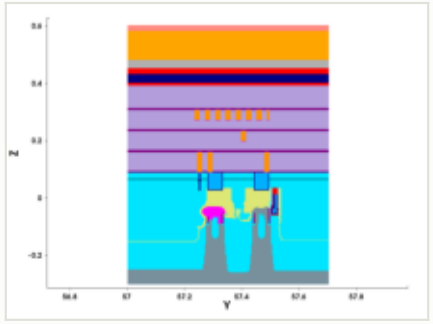
carbon so far, of the whole flow (kgCO2e)



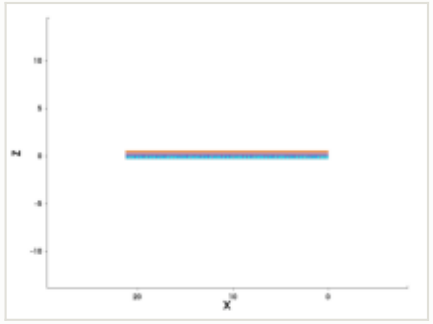
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



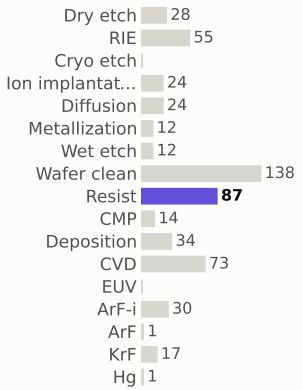
y cut at 57.351 µm



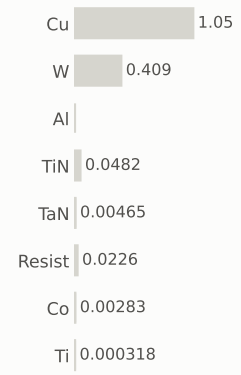
mask for this step

no mask for this step

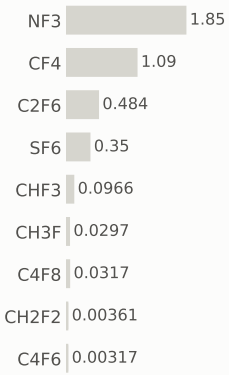
steps by process area



materials deposited (g)



gases emitted (g)



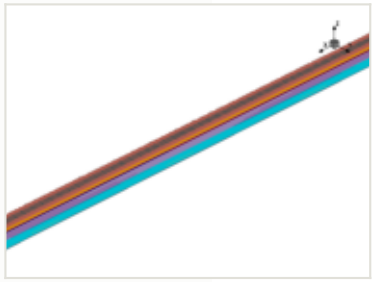
Step #7060: ArF-i

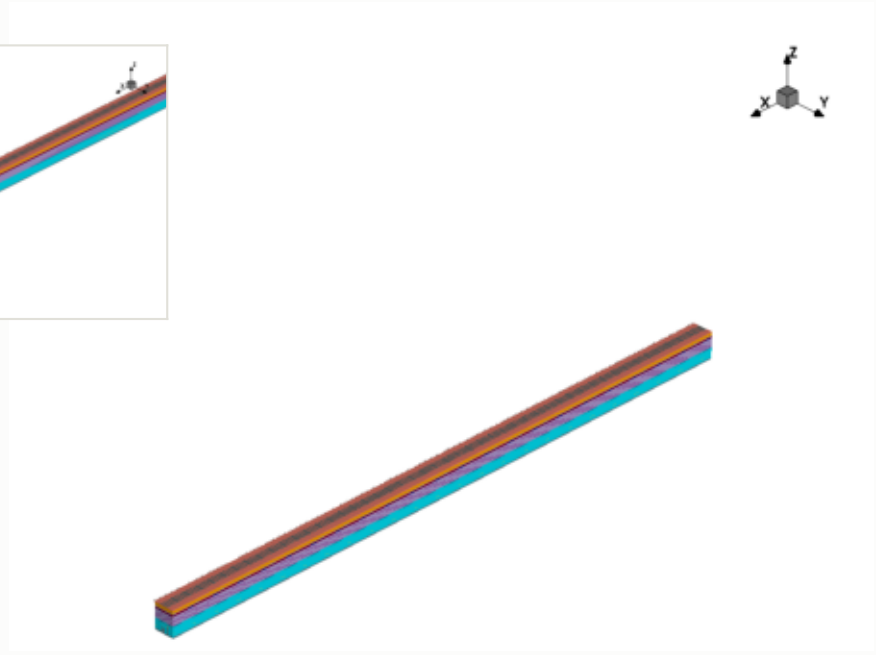
litho_patterning · ArF-i

0.217	329.4	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



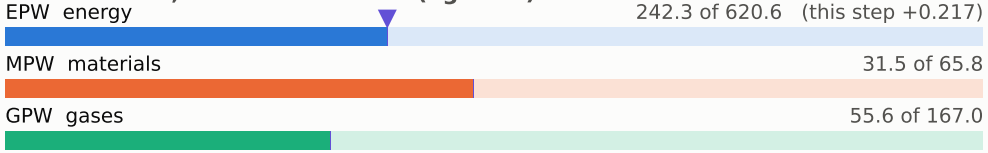


this step

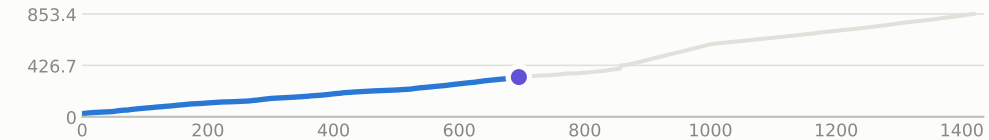
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

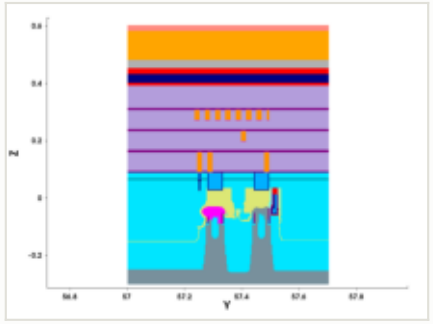
carbon so far, of the whole flow (kgCO2e)



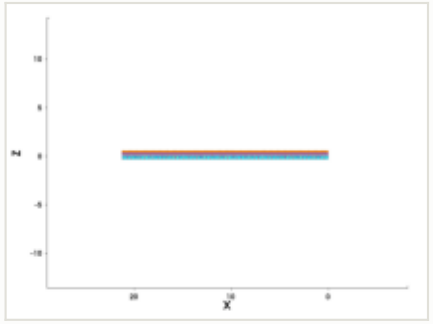
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

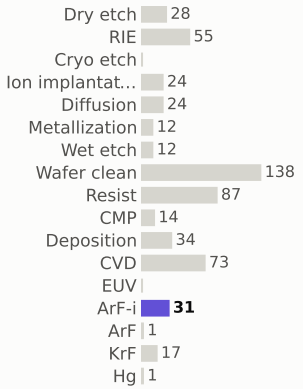


y cut at 57.351 µm

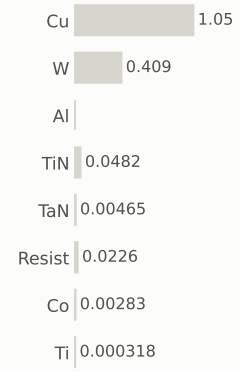


mask: layer 12401:0, positive (GDSII)

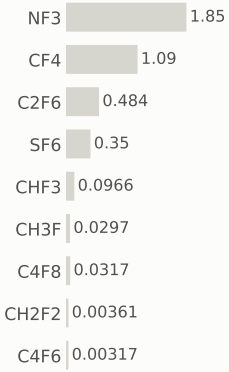
steps by process area



materials deposited (g)



gases emitted (g)



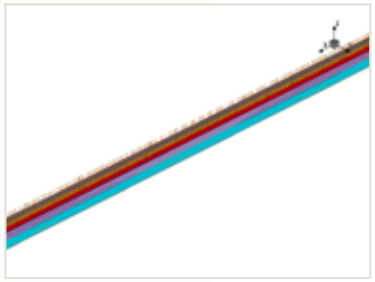
Step #7070: RIE

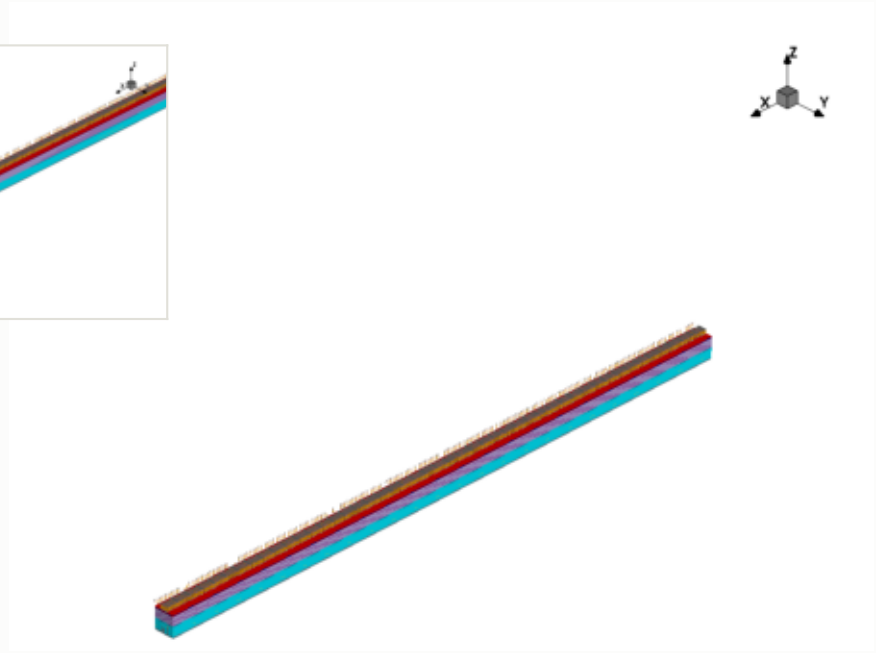
etch_anisotropic · RIE

1.04	330.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



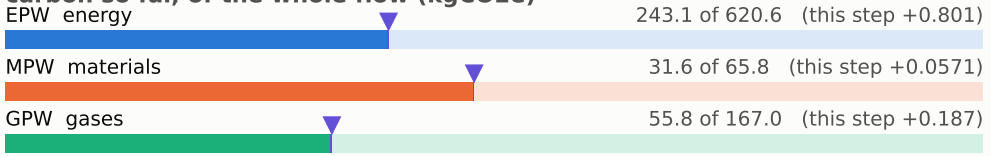


this step

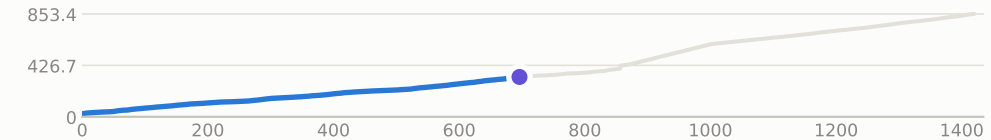
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

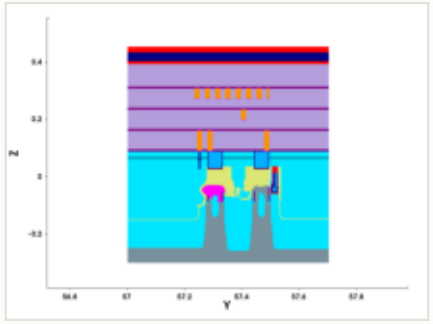
carbon so far, of the whole flow (kgCO2e)



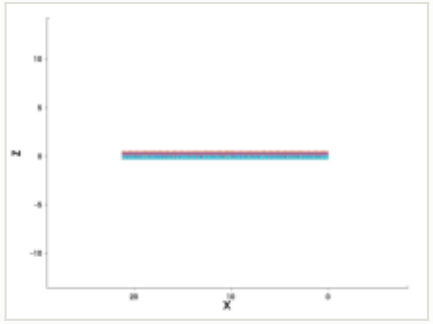
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



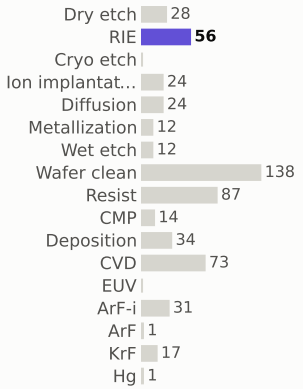
y cut at 57.351 µm



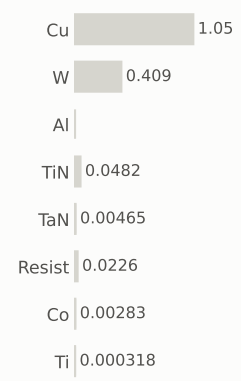
mask for this step

no mask for this step

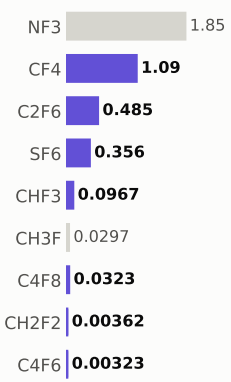
steps by process area



materials deposited (g)



gases emitted (g)



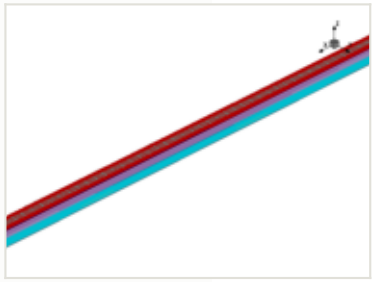
Step #7080: Spx_only

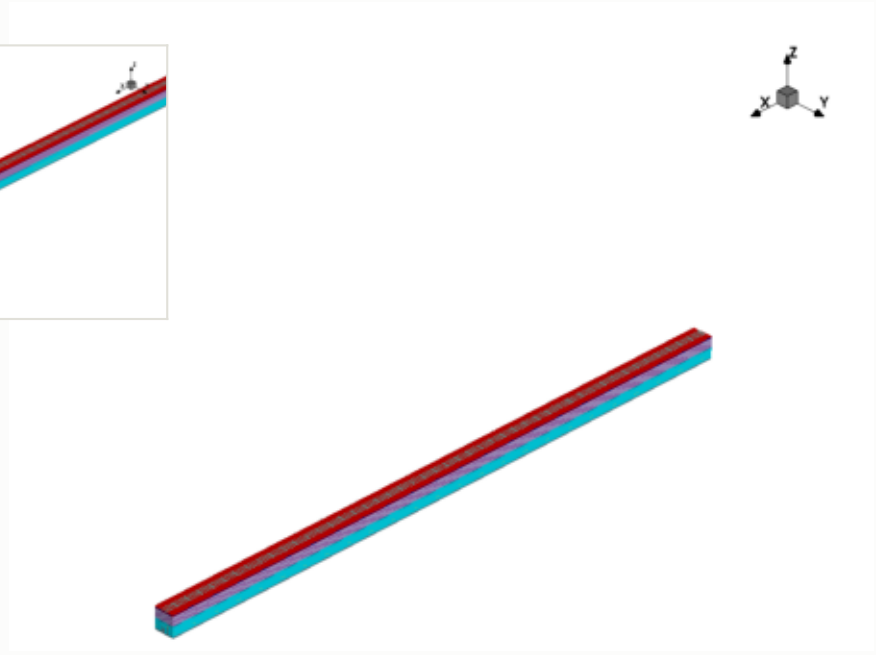
etch_strip · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





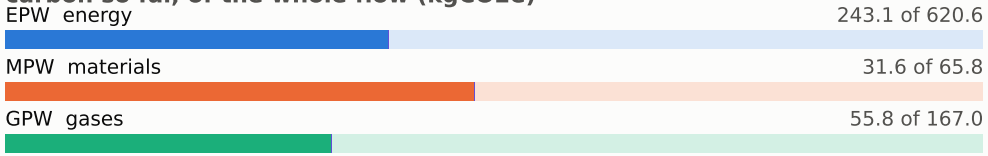
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

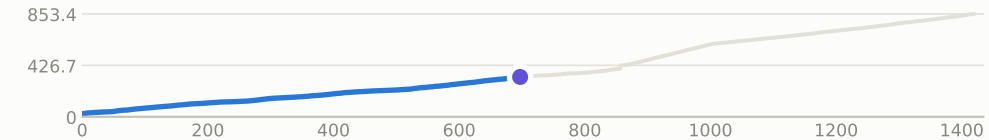
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

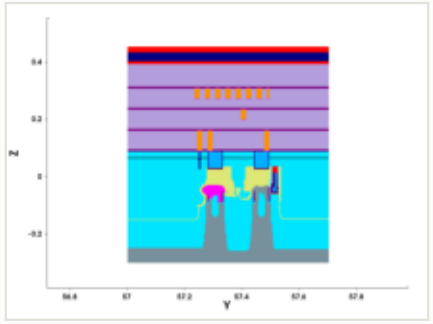
carbon so far, of the whole flow (kgCO2e)



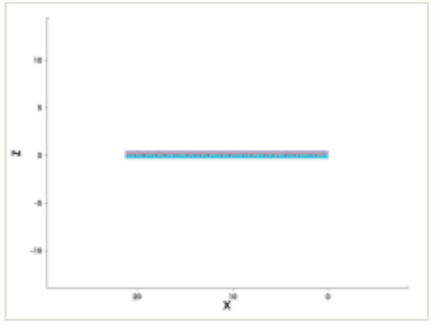
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



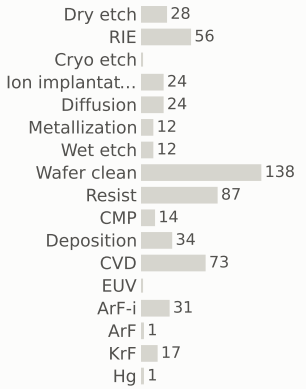
y cut at 57.351 µm



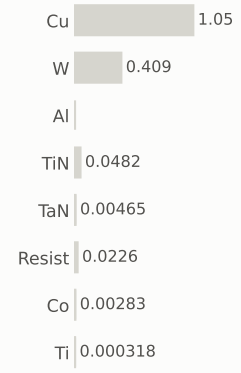
mask for this step

no mask for this step

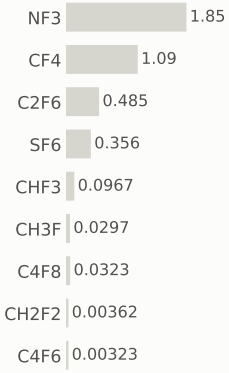
steps by process area



materials deposited (g)



gases emitted (g)



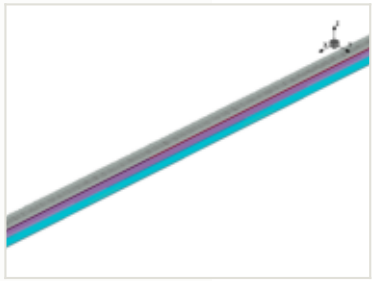
Step #7090: Spx_only

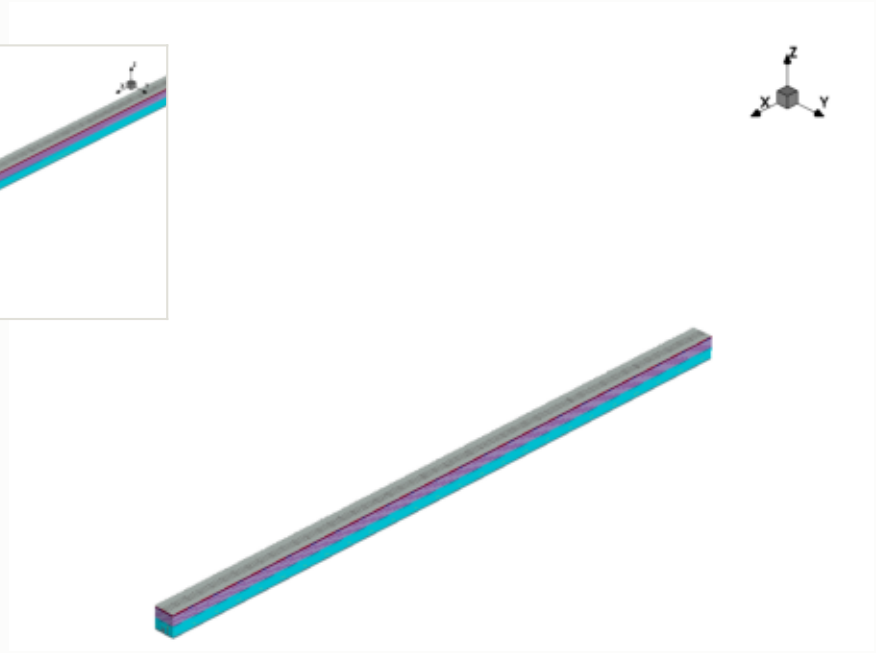
depo_isotropic · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





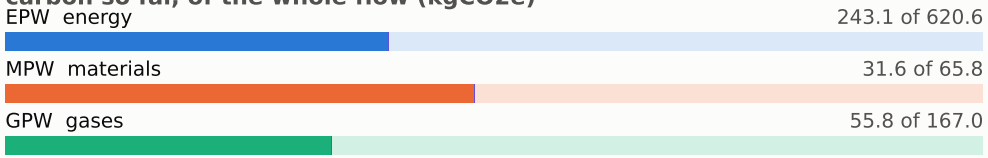
this step

process area	Spx_only
material	—
thickness	0.001 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

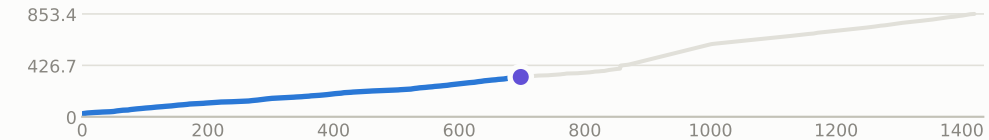
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

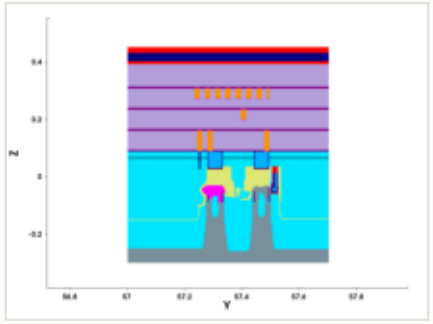
carbon so far, of the whole flow (kgCO2e)



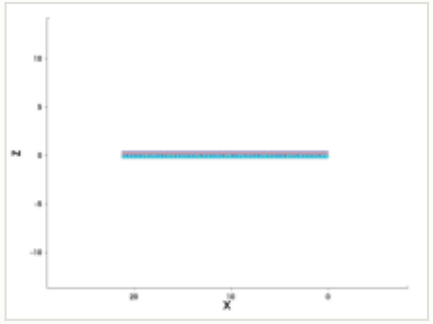
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



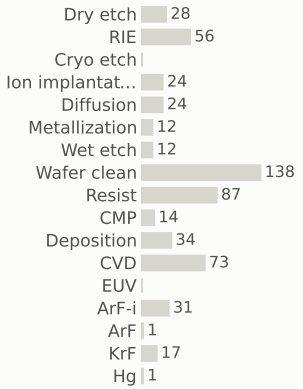
y cut at 57.351 µm



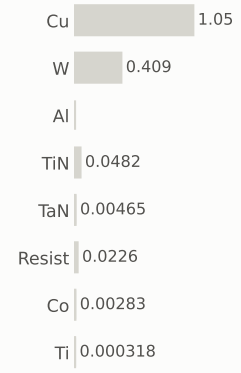
mask for this step

no mask for this step

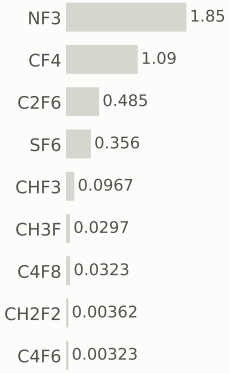
steps by process area



materials deposited (g)



gases emitted (g)



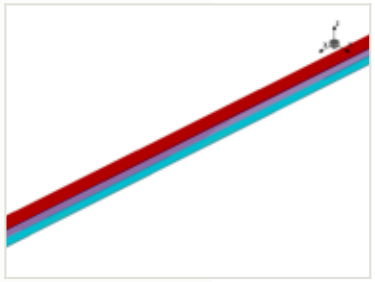
Step #7100: Spx_only

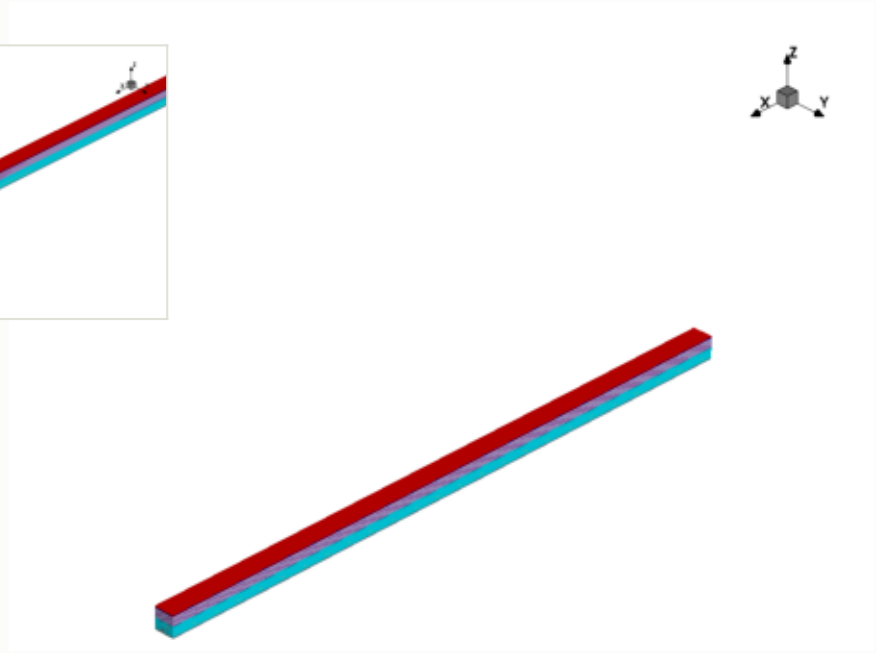
etch_anisotropic · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

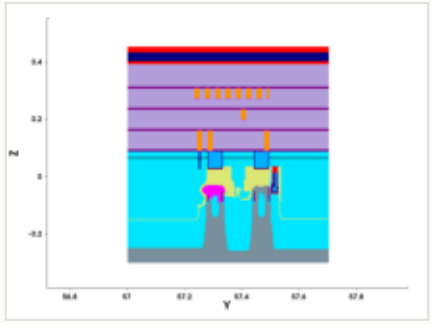
3D structure after this step

zoomed to fit

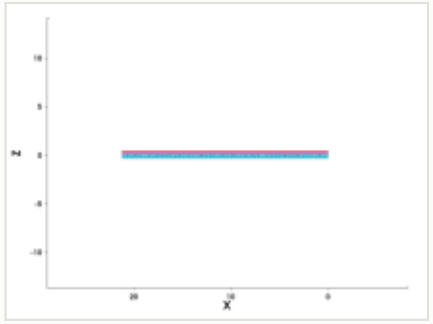




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

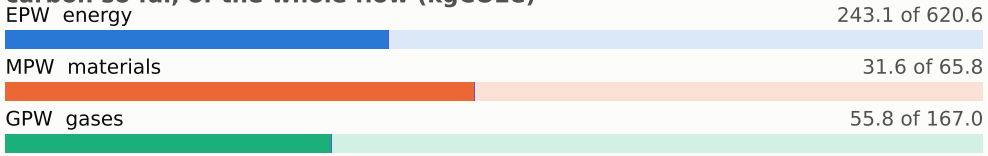
no mask for this step

this step

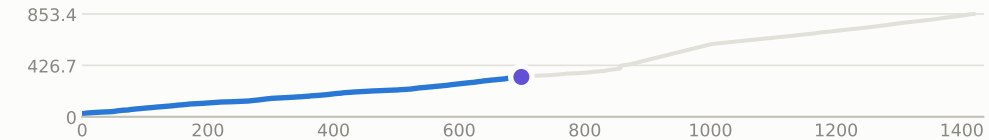
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

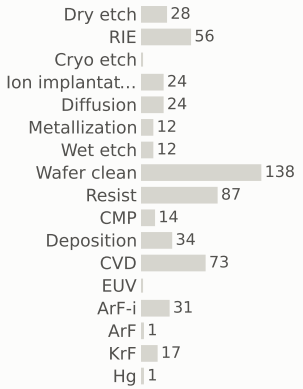
carbon so far, of the whole flow (kgCO2e)



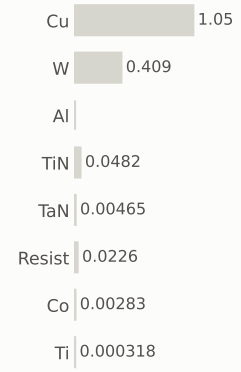
embodied carbon up to each step (kgCO2e per wafer)



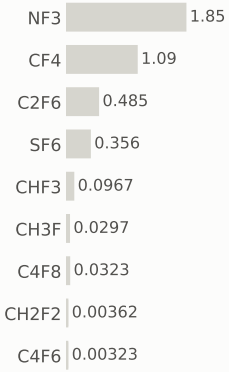
steps by process area



materials deposited (g)



gases emitted (g)



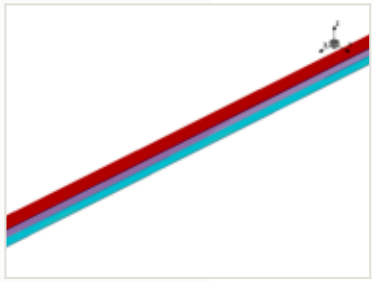
Step #7110: Spx_only

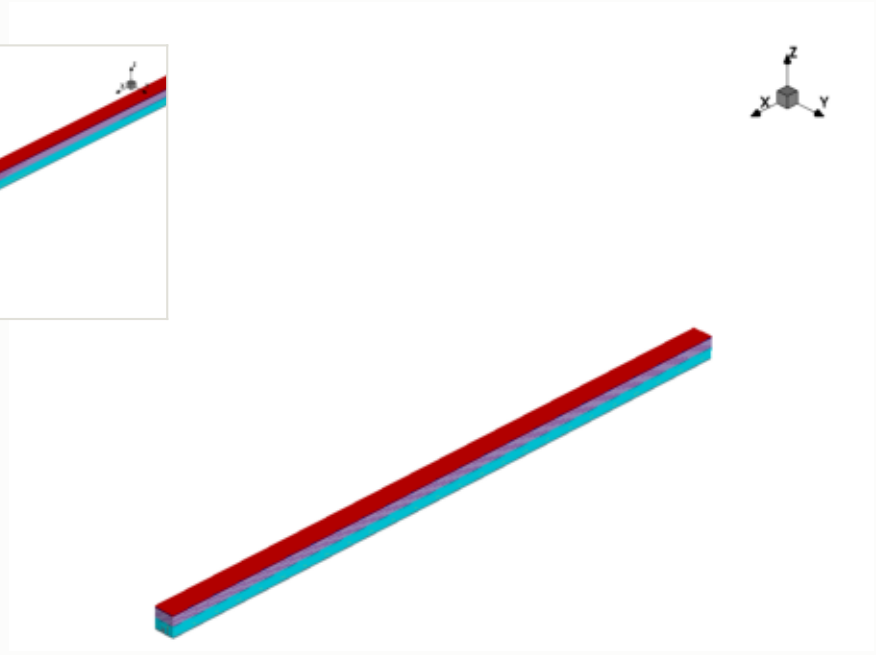
etch_strip · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





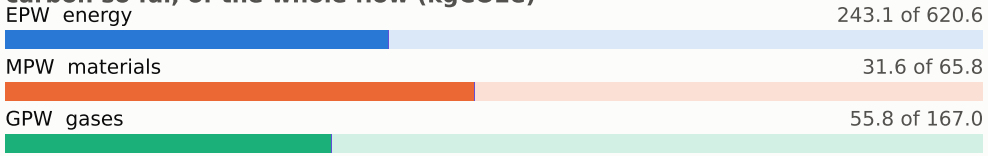
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

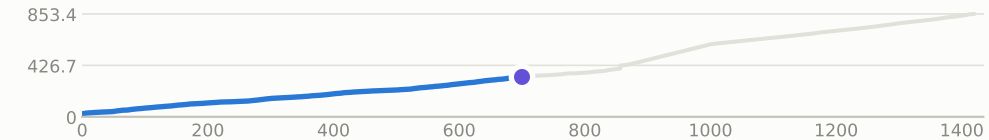
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

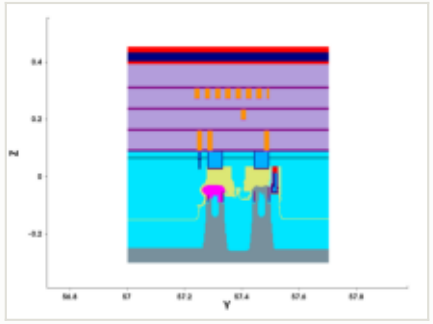
carbon so far, of the whole flow (kgCO2e)



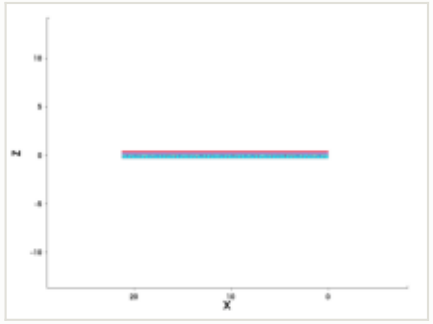
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



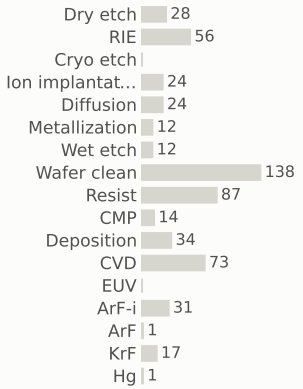
y cut at 57.351 µm



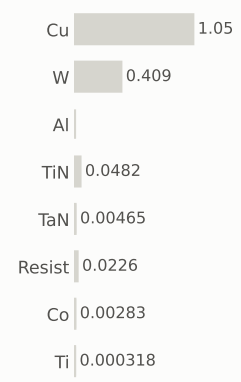
mask for this step

no mask for this step

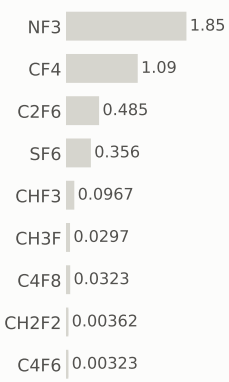
steps by process area



materials deposited (g)



gases emitted (g)



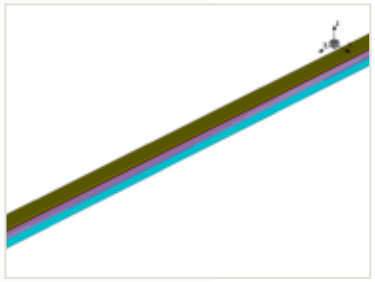
Step #7120: Spx_only

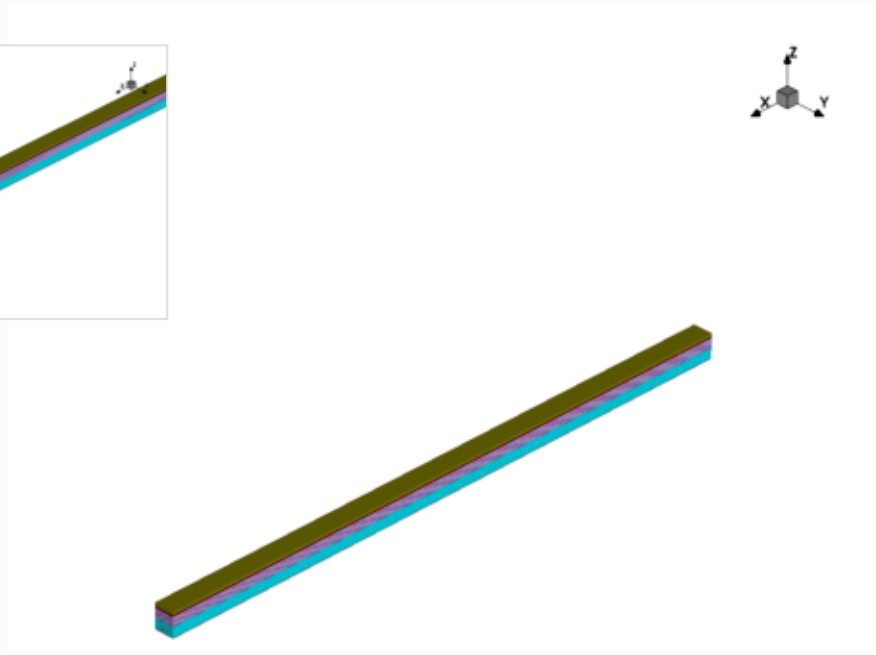
depo_fill · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





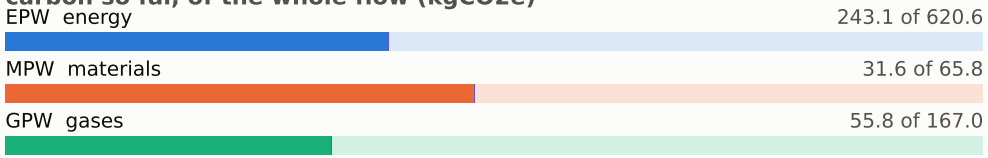
this step

process area	Spx_only
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

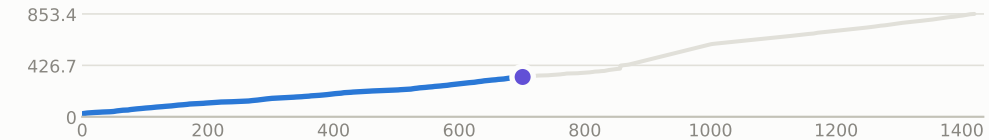
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

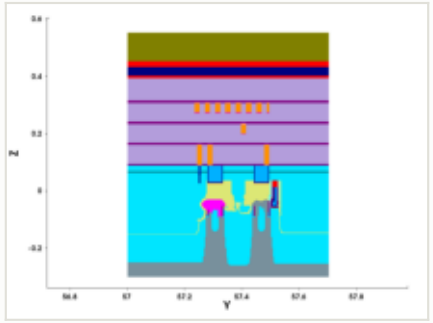
carbon so far, of the whole flow (kgCO2e)



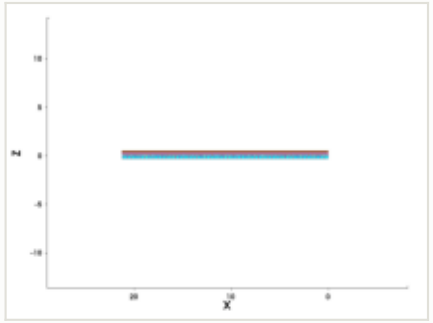
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



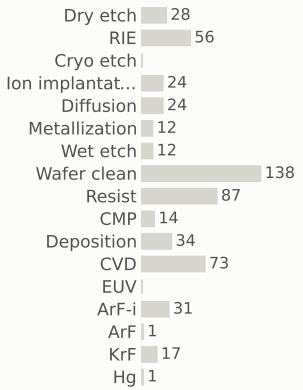
y cut at 57.351 µm



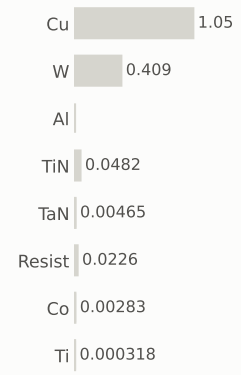
mask for this step

no mask for this step

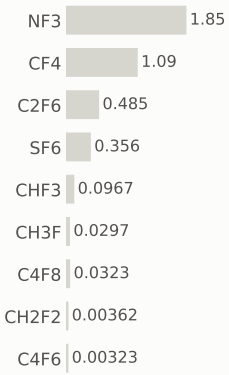
steps by process area



materials deposited (g)



gases emitted (g)



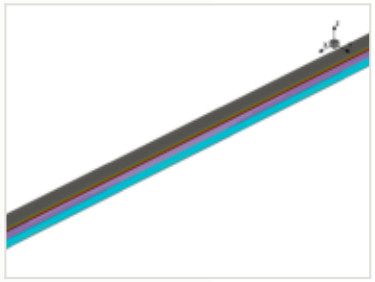
Step #7130: Spx_only

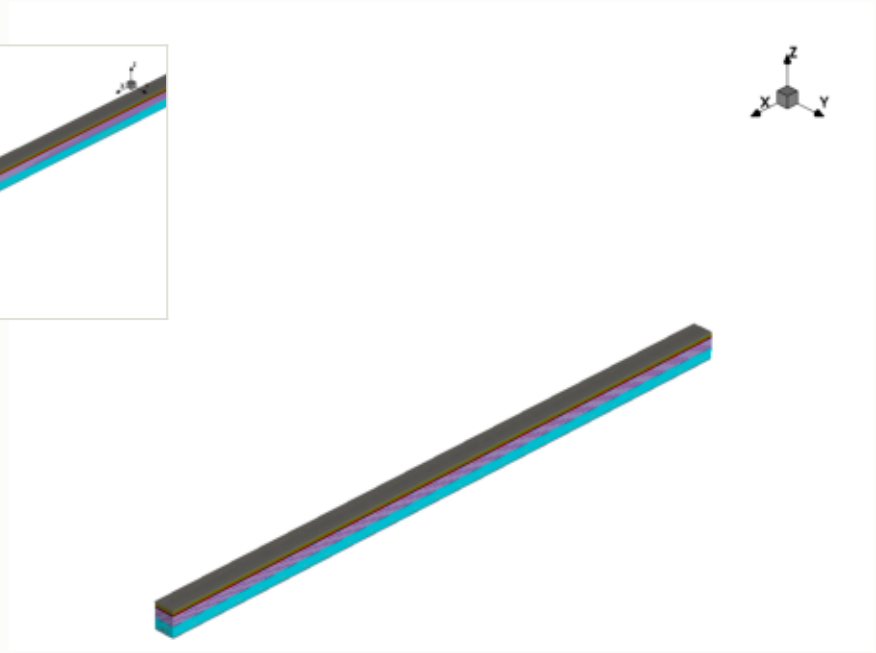
litho_patterning · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





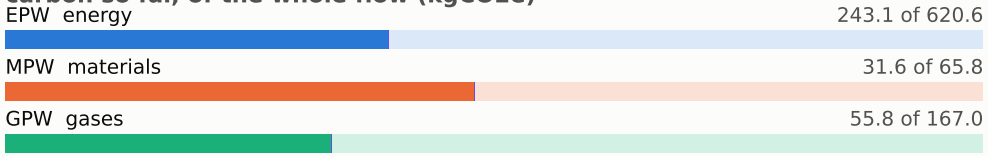
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

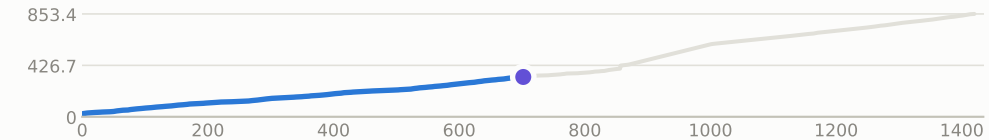
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

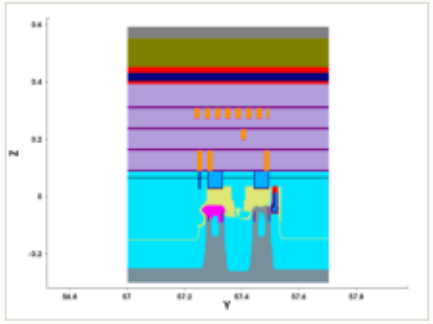
carbon so far, of the whole flow (kgCO2e)



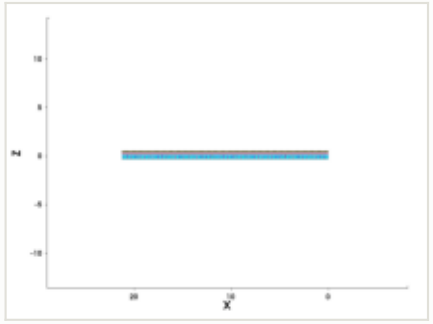
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

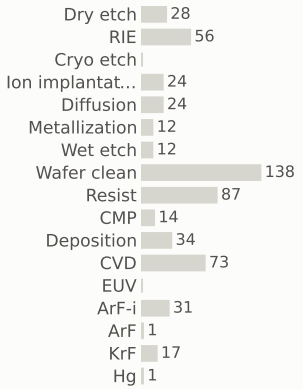


y cut at 57.351 µm

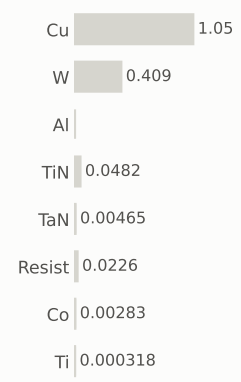


mask: layer 2040:0 (GDSII)

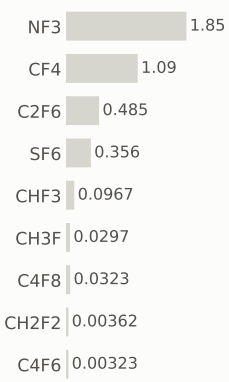
steps by process area



materials deposited (g)



gases emitted (g)



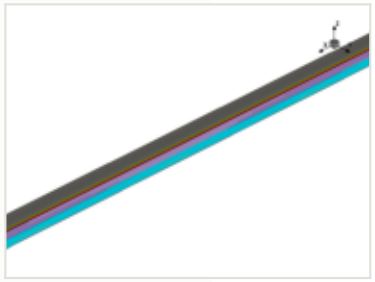
Step #7140: Spx_only

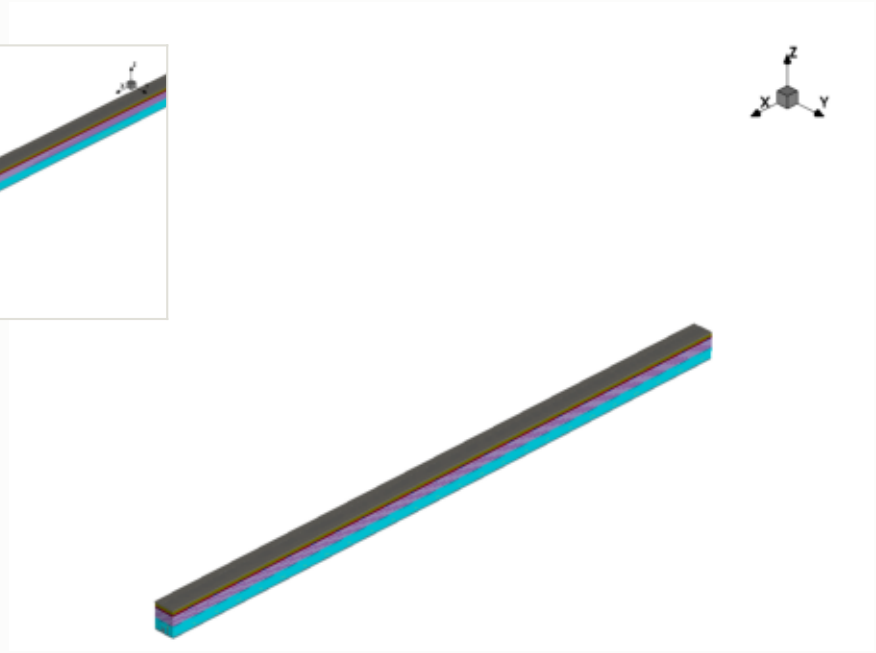
etch_anisotropic · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





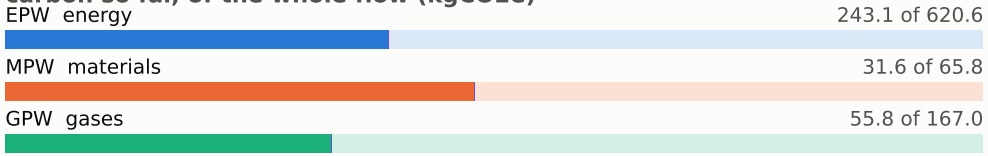
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

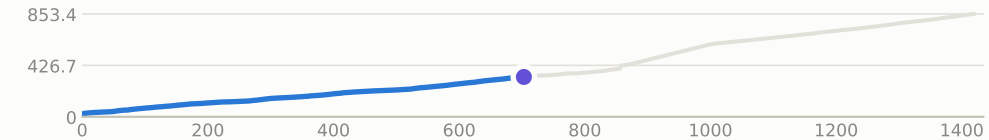
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

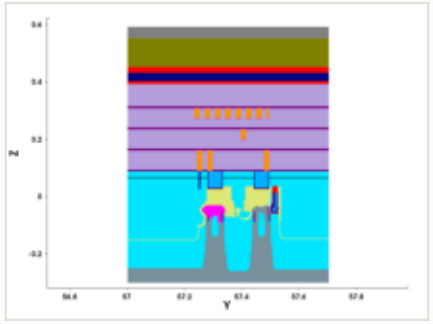
carbon so far, of the whole flow (kgCO2e)



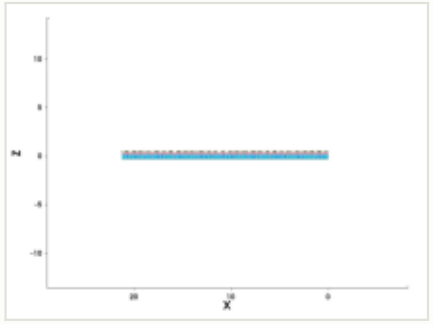
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



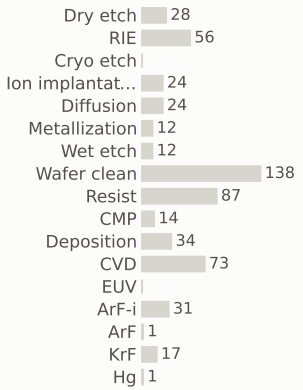
y cut at 57.351 µm



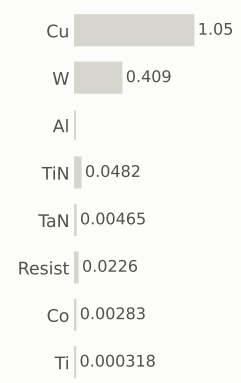
mask for this step

no mask for this step

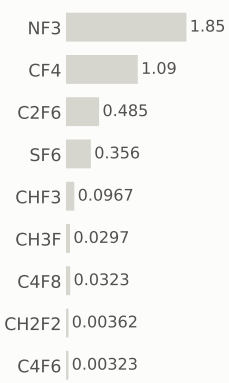
steps by process area



materials deposited (g)



gases emitted (g)



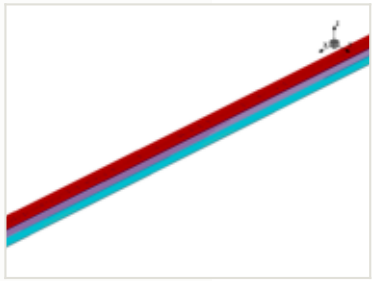
Step #7150: Spx_only

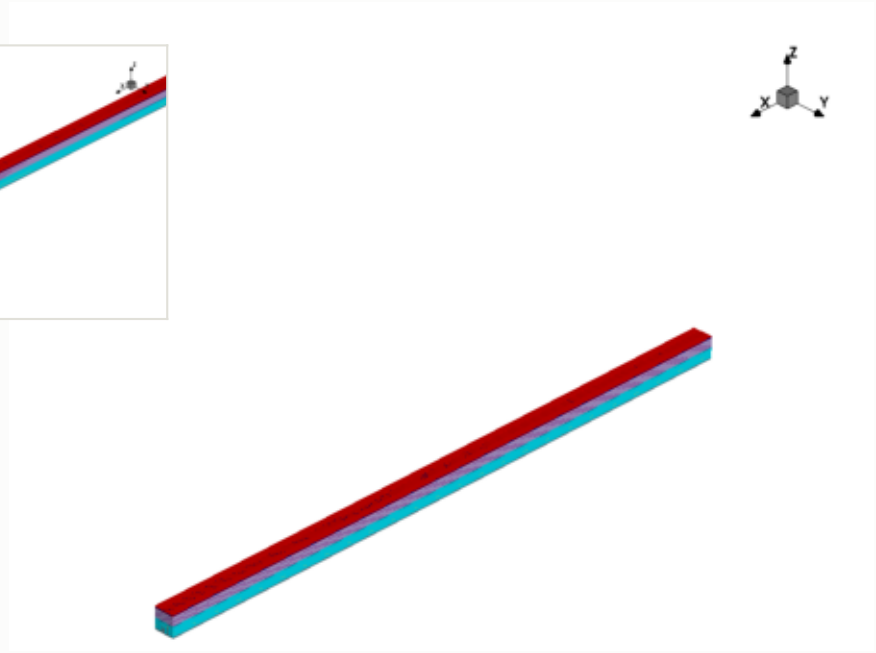
etch_strip · Spx_only

0	330.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

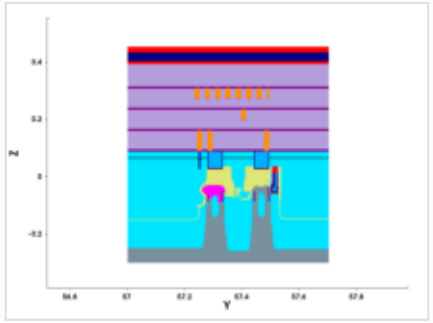
3D structure after this step

zoomed to fit

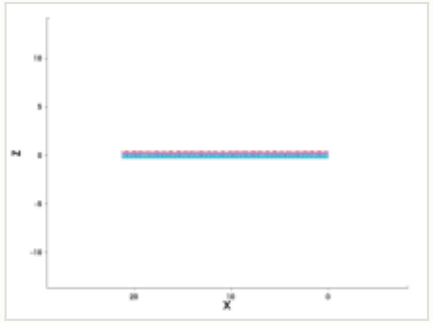




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

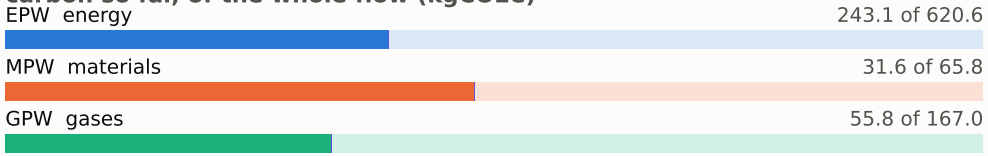
no mask for this step

this step

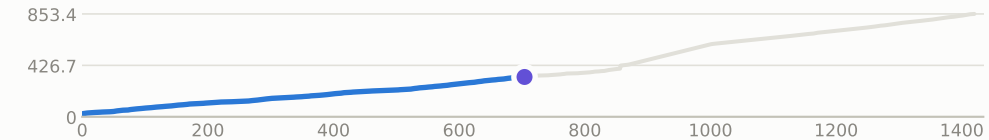
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

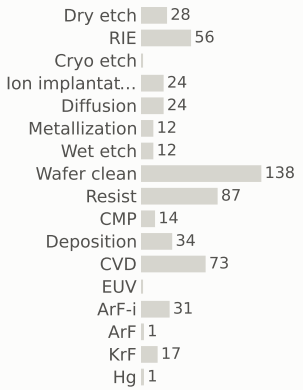
carbon so far, of the whole flow (kgCO2e)



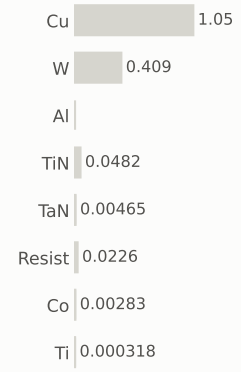
embodied carbon up to each step (kgCO2e per wafer)



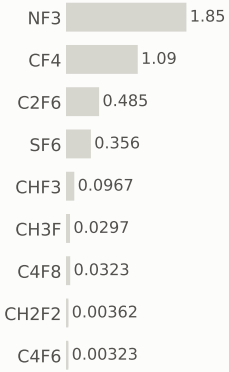
steps by process area



materials deposited (g)



gases emitted (g)



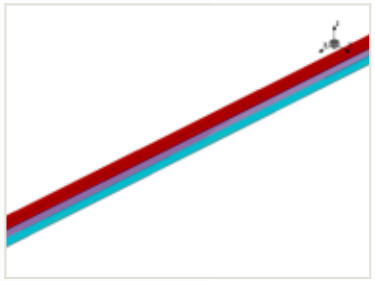
Step #7160: Wafer_clean

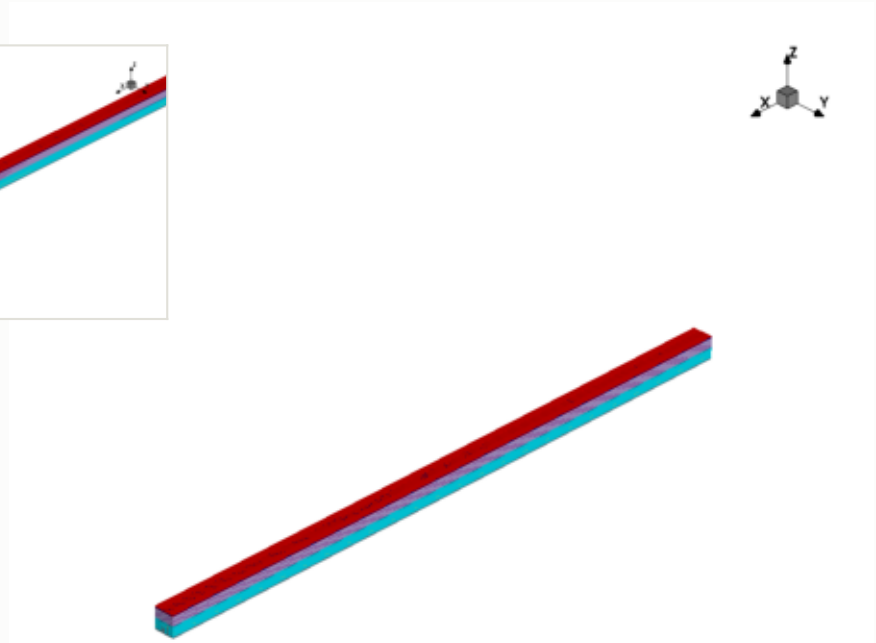
clean · Wafer_clean

0.169	330.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

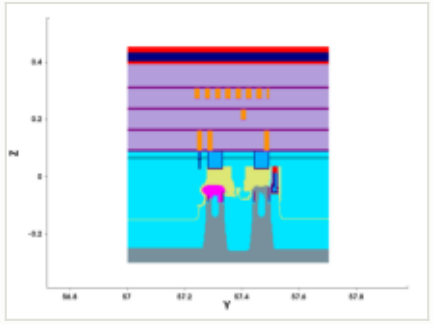
3D structure after this step

zoomed to fit

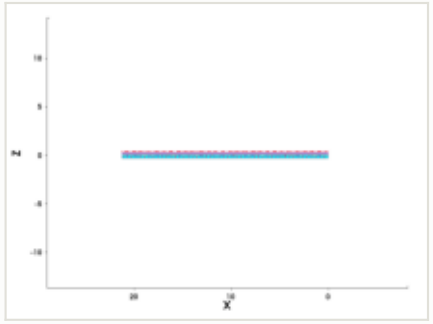




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

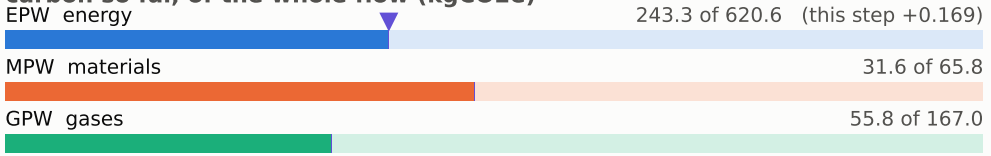
no mask for this step

this step

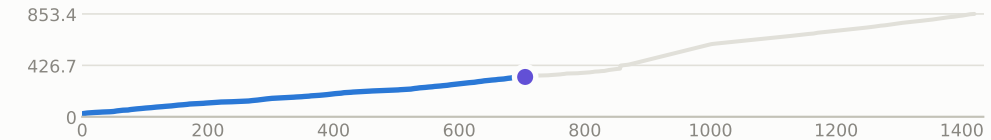
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

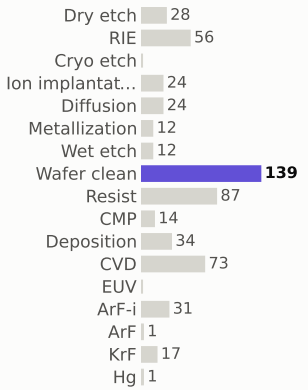
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

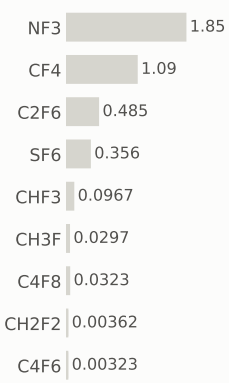


steps by process area



materials deposited (g)

gases emitted (g)



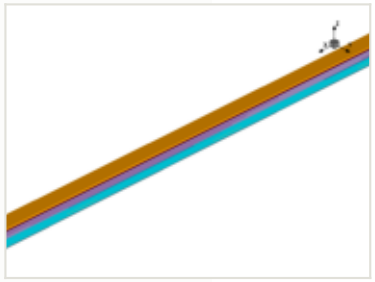
Step #7170: Resist

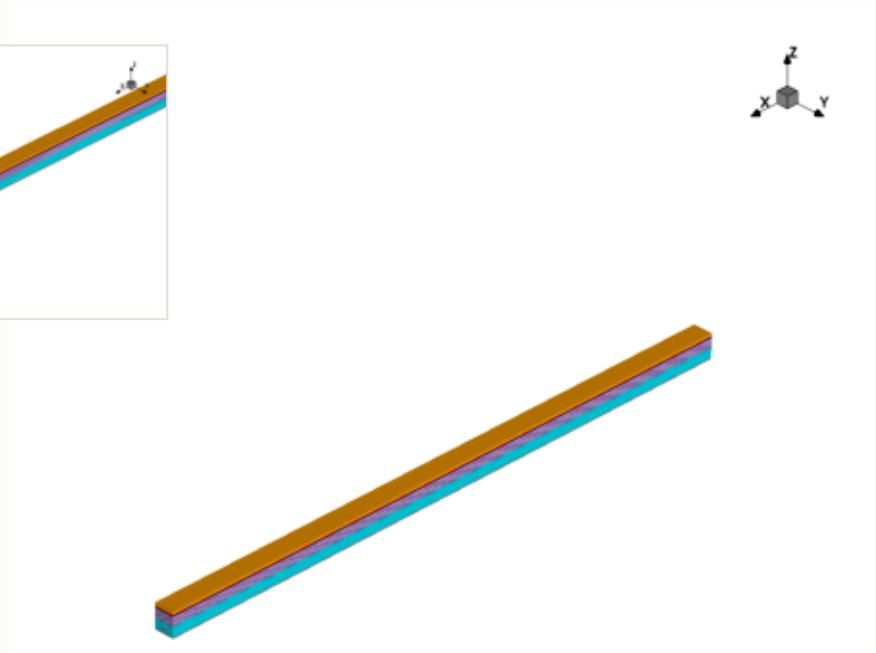
depo_fill · Resist

0.169	330.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



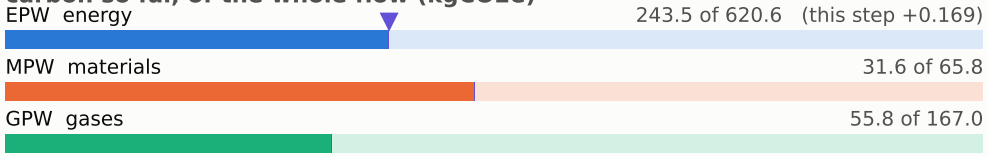


this step

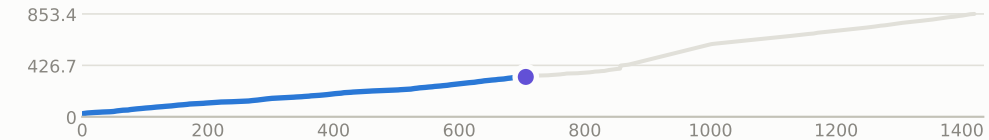
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

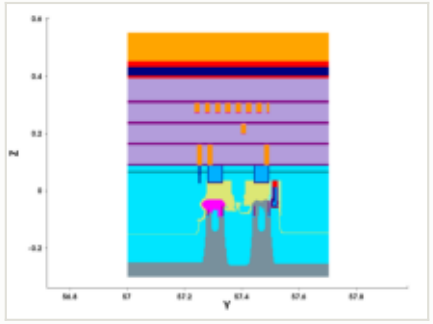
carbon so far, of the whole flow (kgCO2e)



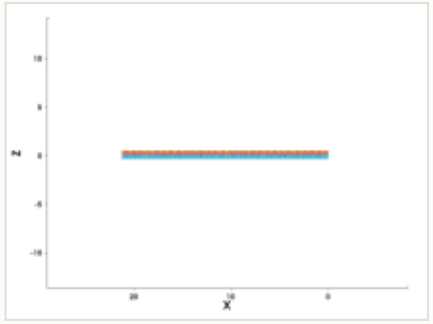
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



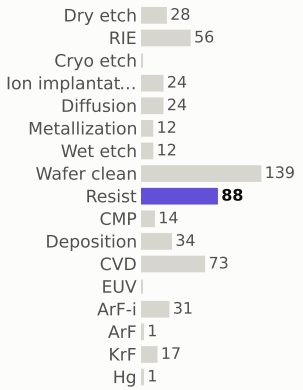
y cut at 57.351 µm



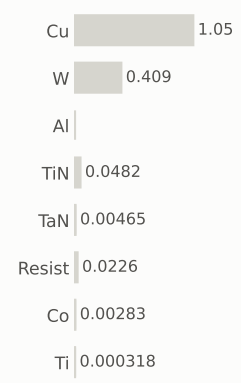
mask for this step

no mask for this step

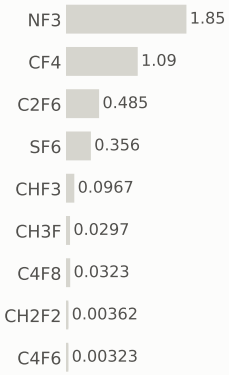
steps by process area



materials deposited (g)



gases emitted (g)



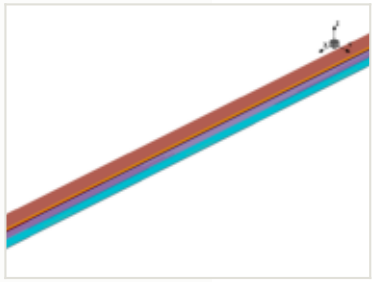
Step #7180: CVD

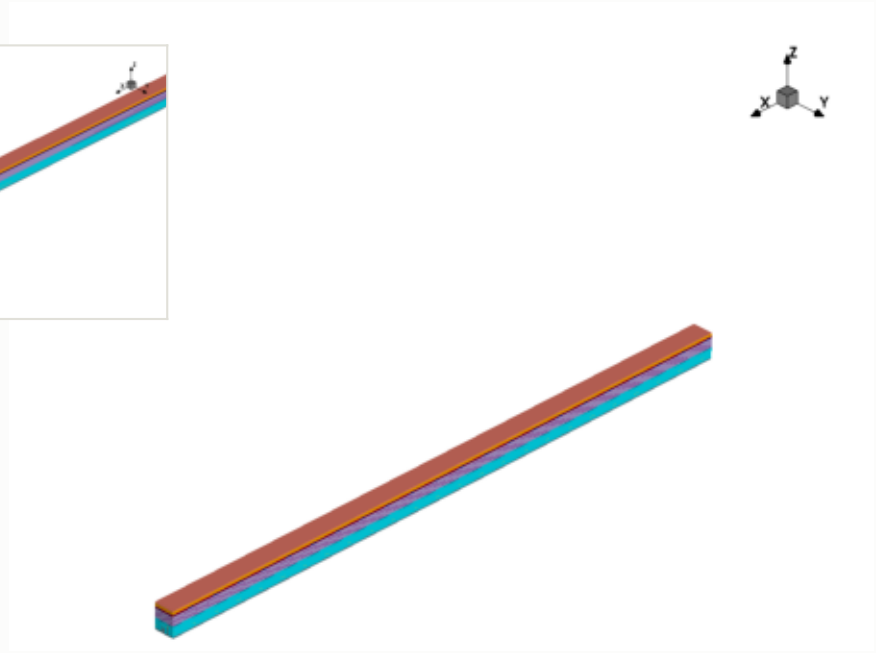
depo_fill · CVD

1.31	332.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



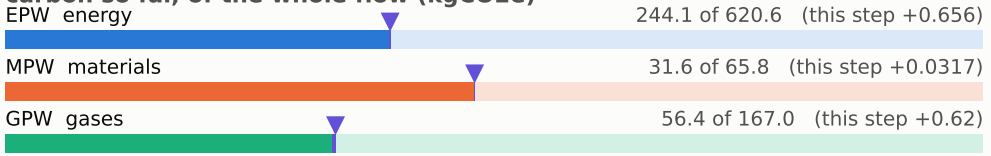


this step

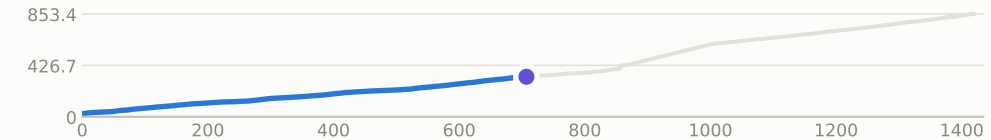
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

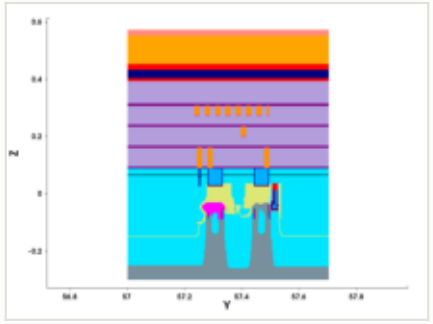
carbon so far, of the whole flow (kgCO2e)



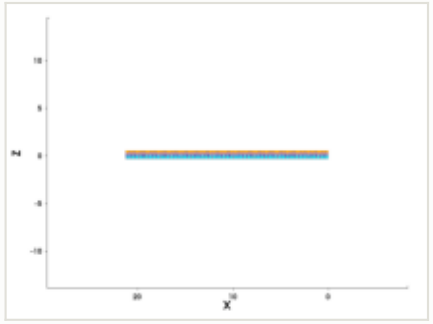
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



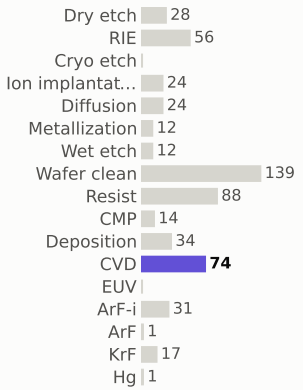
y cut at 57.351 µm



mask for this step

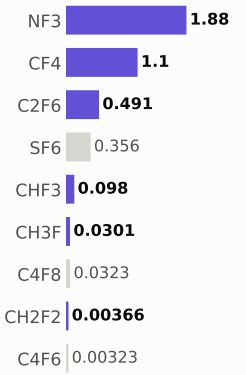
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



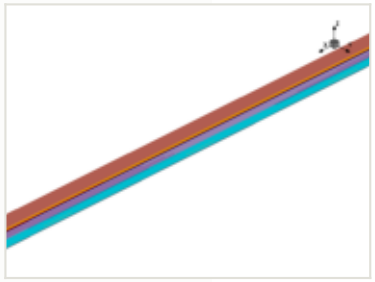
Step #7190: Wafer_clean

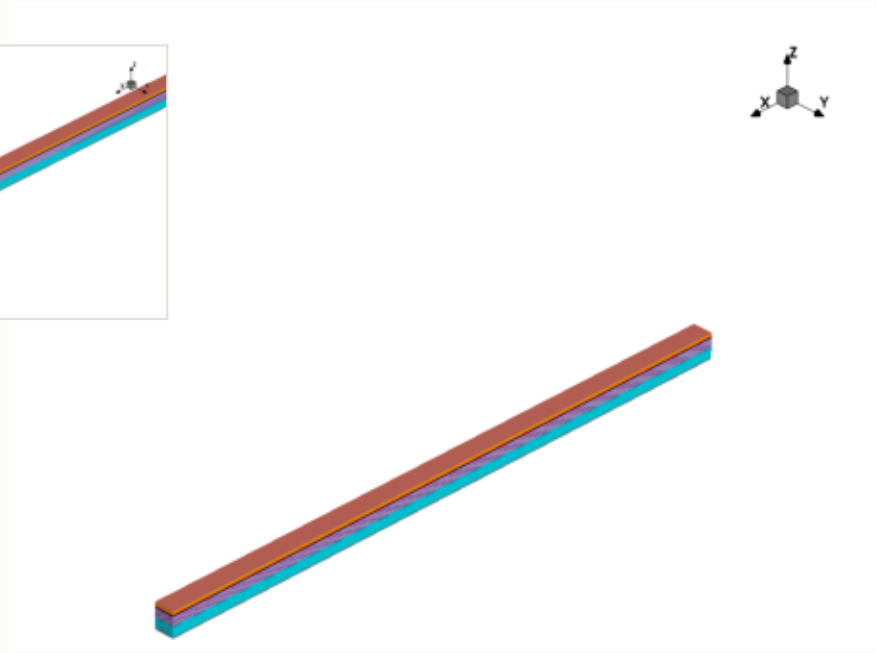
clean · Wafer_clean

0.169	332.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





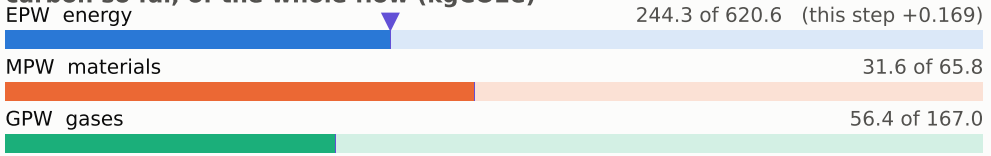
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

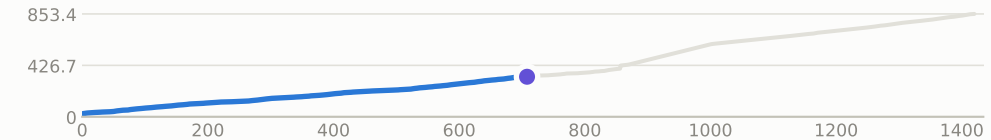
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

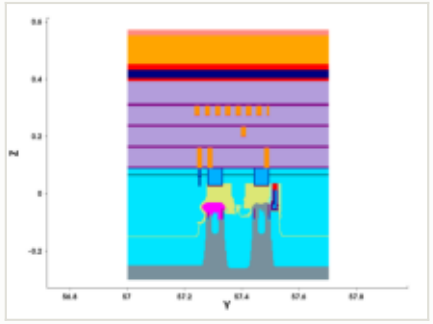
carbon so far, of the whole flow (kgCO2e)



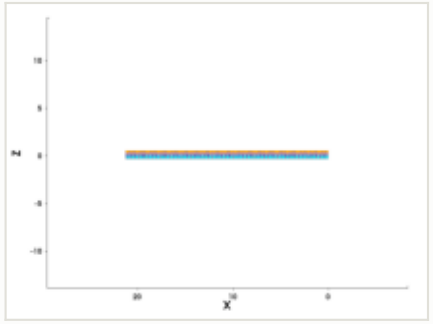
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



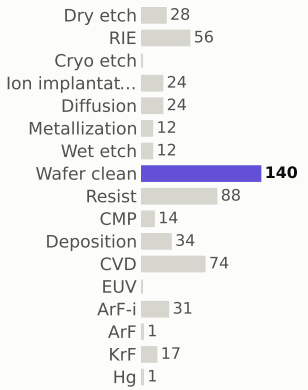
y cut at 57.351 µm



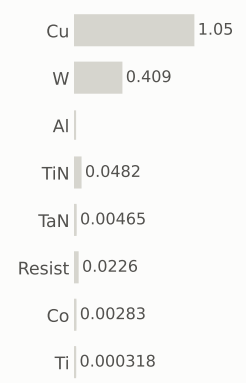
mask for this step

no mask for this step

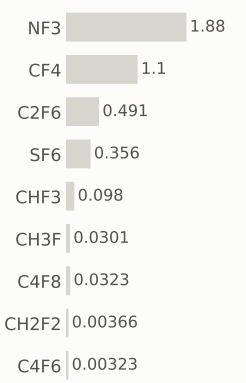
steps by process area



materials deposited (g)



gases emitted (g)



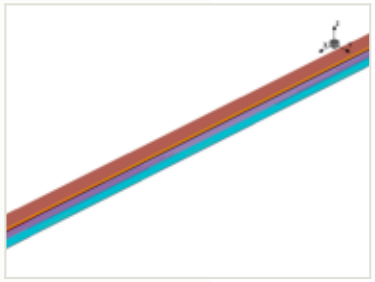
Step #7200: Resist

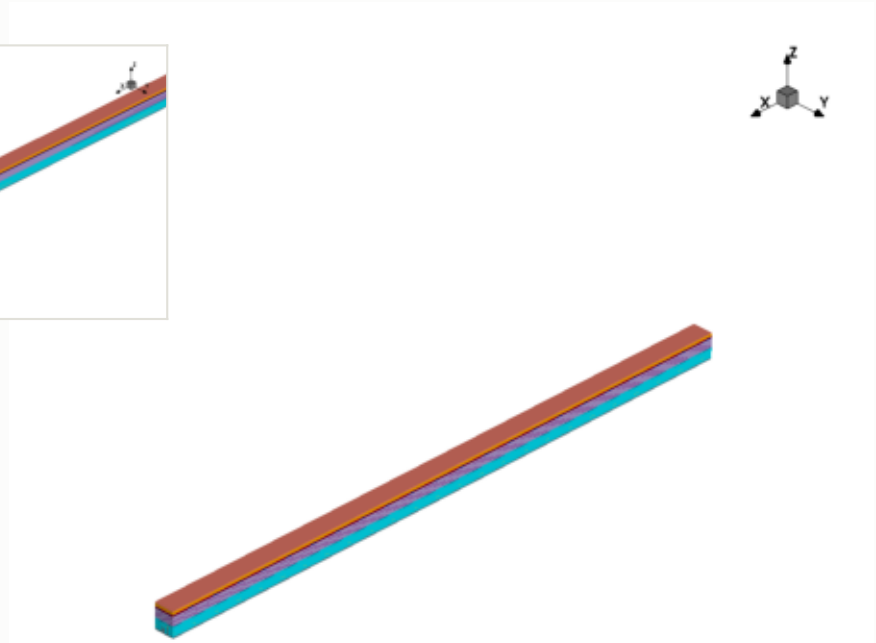
resist · Resist

0.169	332.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



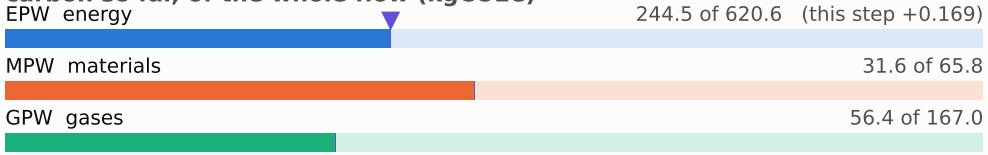


this step

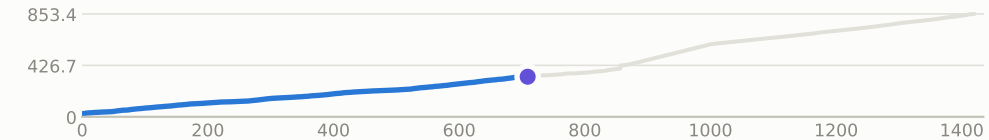
process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

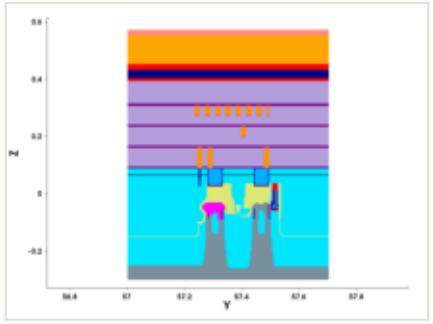
carbon so far, of the whole flow (kgCO2e)



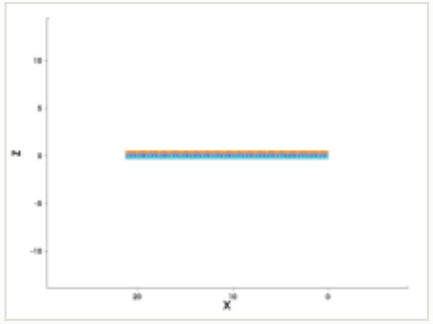
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



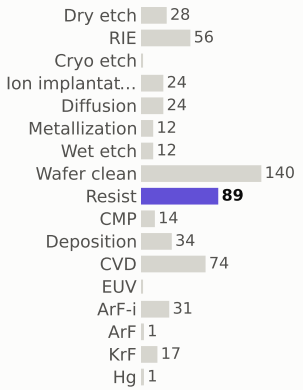
y cut at 57.351 µm



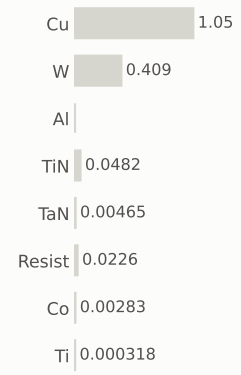
mask for this step

no mask for this step

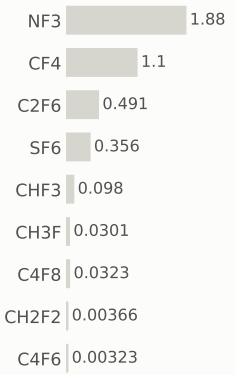
steps by process area



materials deposited (g)



gases emitted (g)



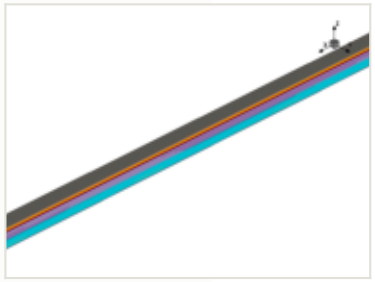
Step #7210: ArF-i

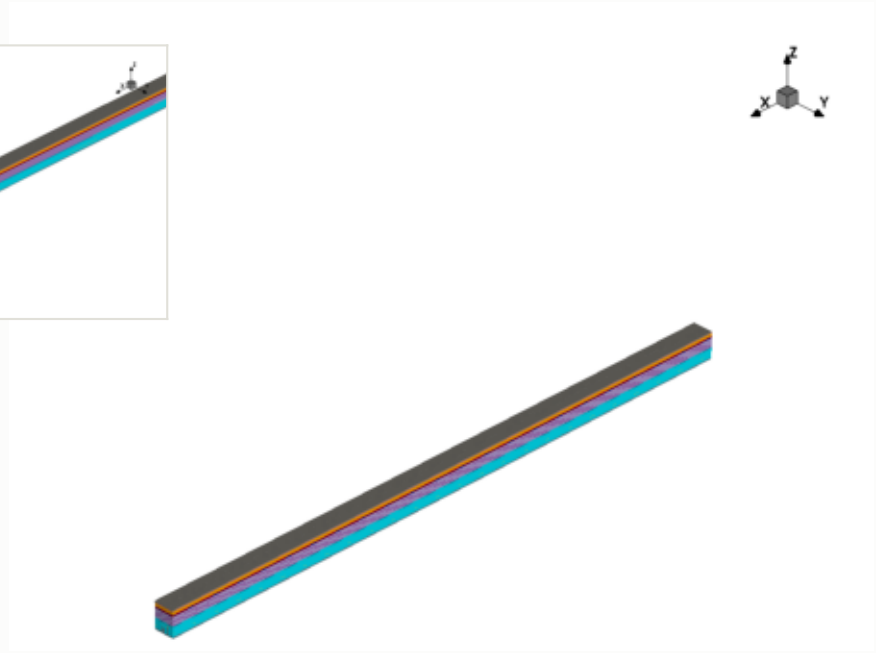
litho_patterning · ArF-i

0.217	332.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



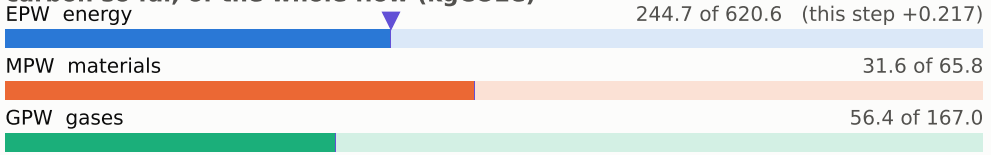


this step

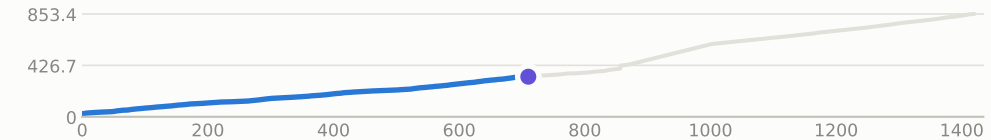
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

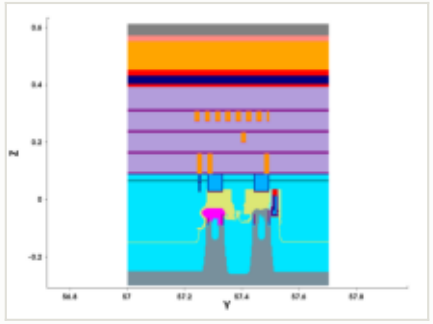
carbon so far, of the whole flow (kgCO2e)



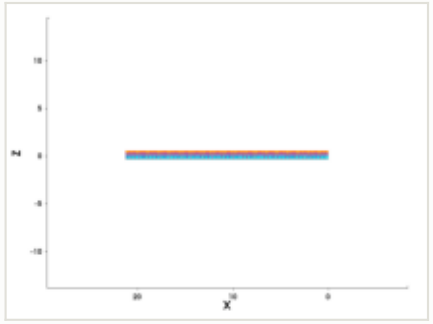
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

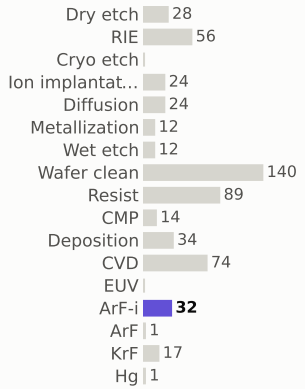


y cut at 57.351 µm

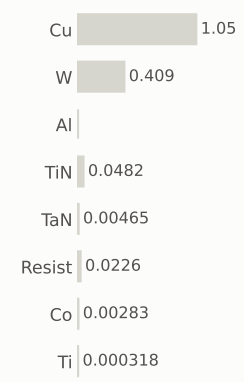


mask: layer 2035:0 (GDSII)

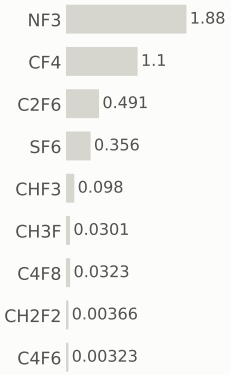
steps by process area



materials deposited (g)



gases emitted (g)



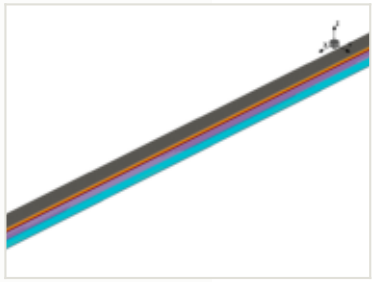
Step #7220: RIE

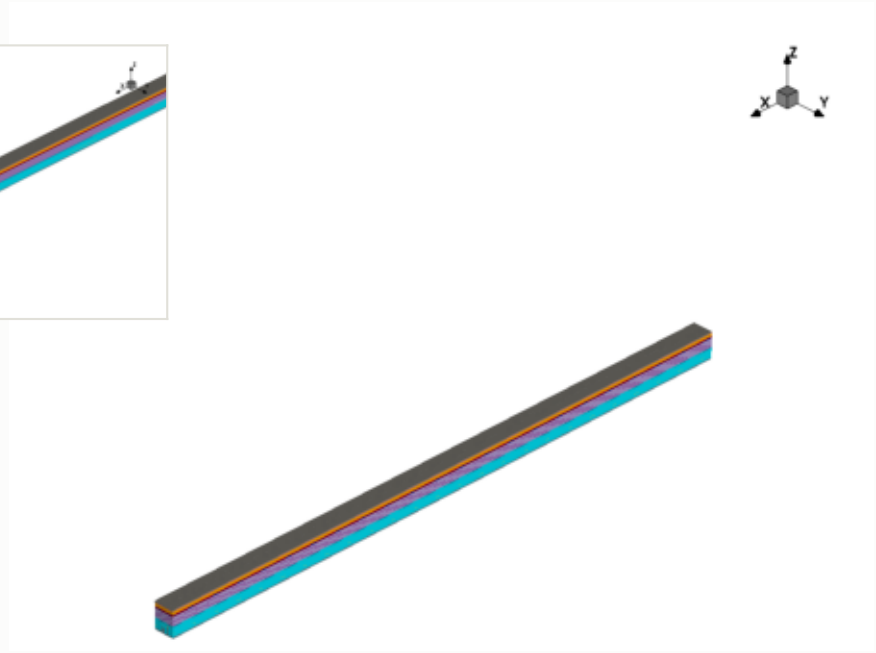
etch_anisotropic · RIE

1.04	333.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



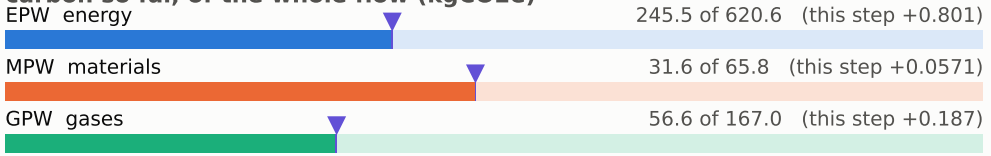


this step

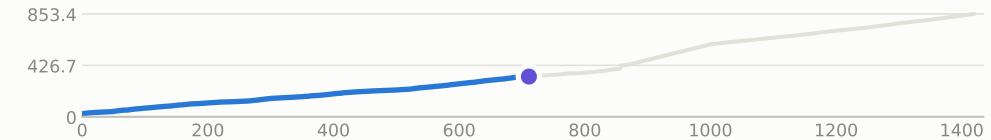
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

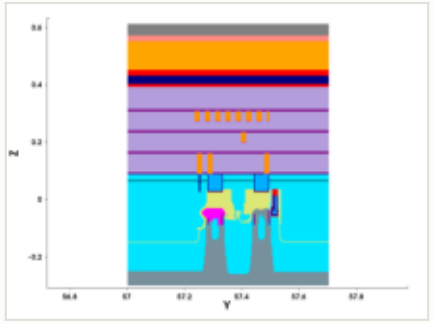
carbon so far, of the whole flow (kgCO2e)



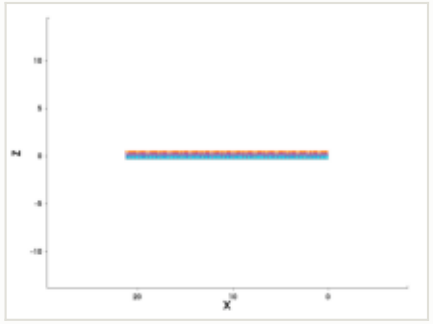
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



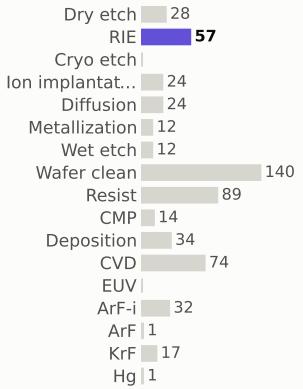
y cut at 57.351 µm



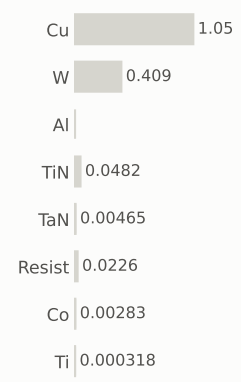
mask for this step

no mask for this step

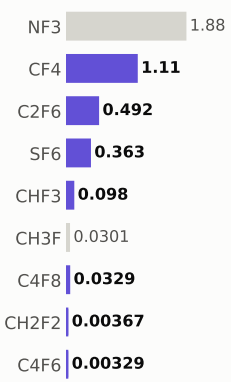
steps by process area



materials deposited (g)



gases emitted (g)



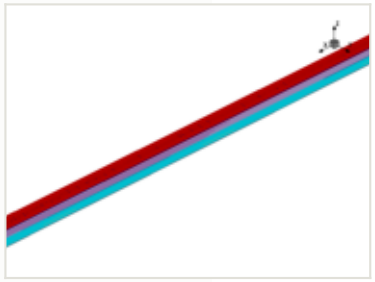
Step #7230: Spx_only

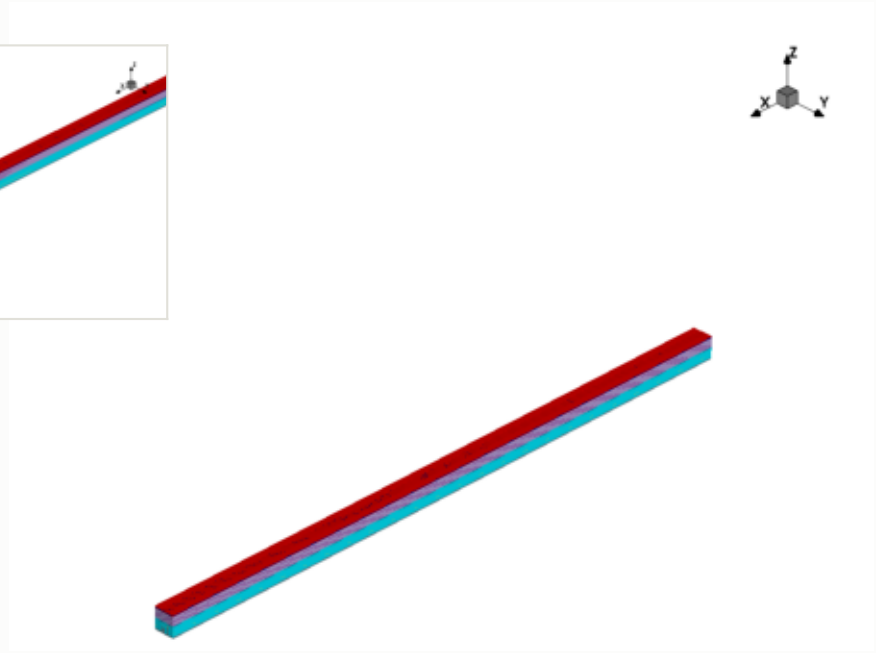
etch_strip · Spx_only

0	333.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



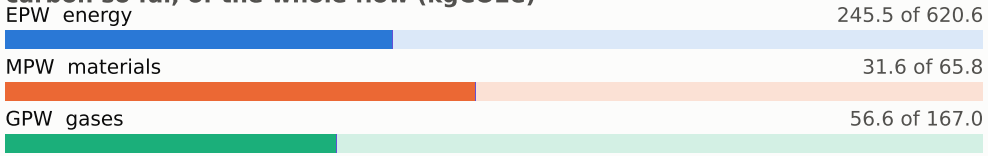


this step

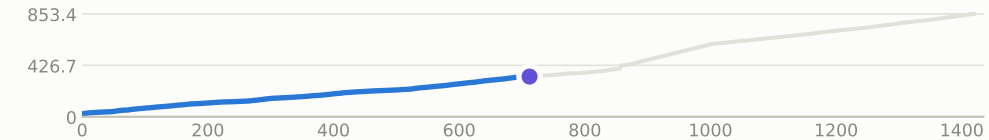
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

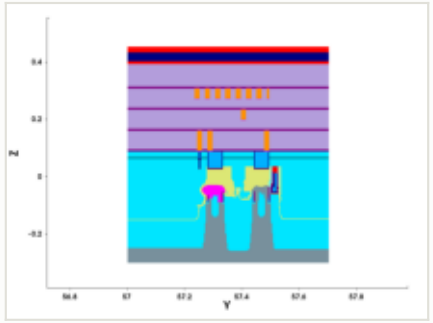
carbon so far, of the whole flow (kgCO2e)



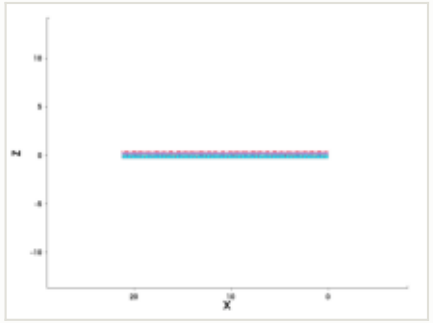
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



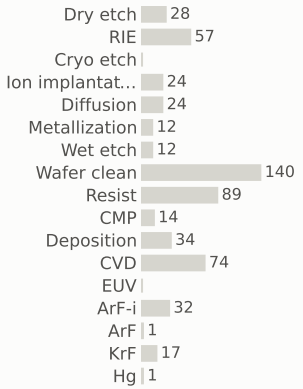
y cut at 57.351 µm



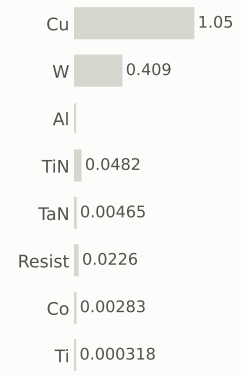
mask for this step

no mask for this step

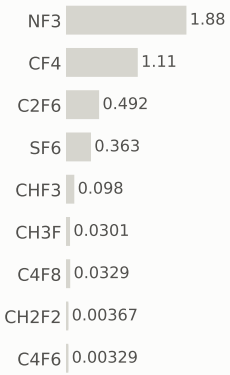
steps by process area



materials deposited (g)



gases emitted (g)



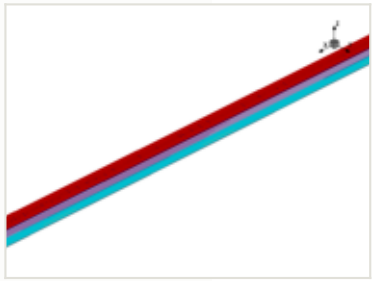
Step #7240: Wafer_clean

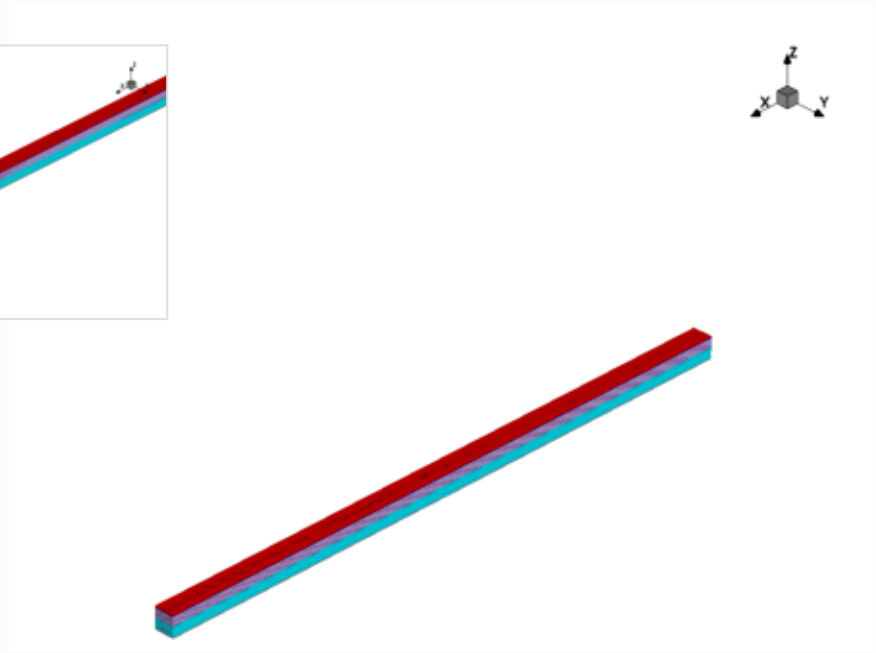
clean · Wafer_clean

0.169	333.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





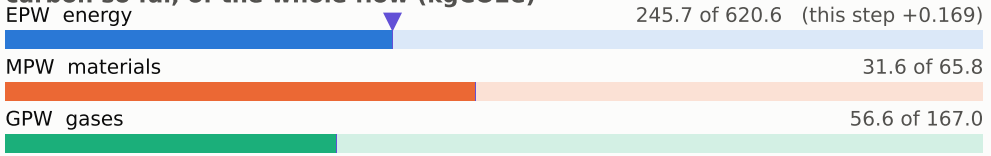
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

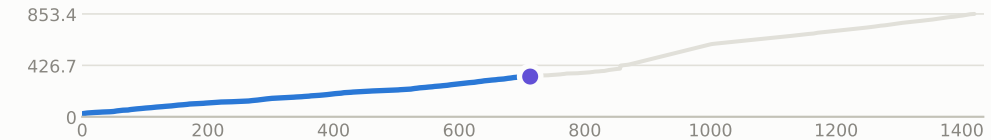
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

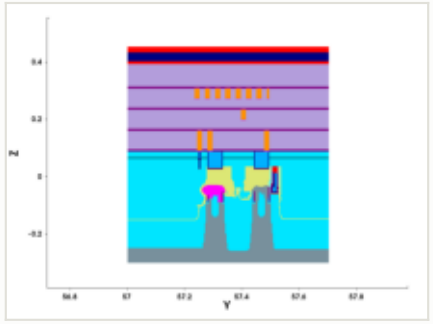
carbon so far, of the whole flow (kgCO2e)



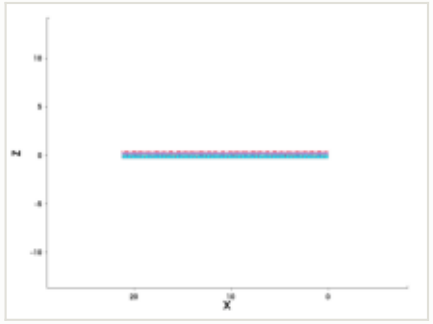
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



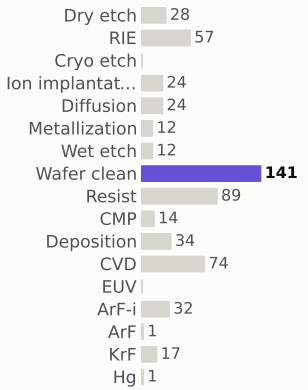
y cut at 57.351 µm



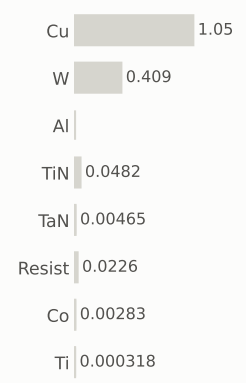
mask for this step

no mask for this step

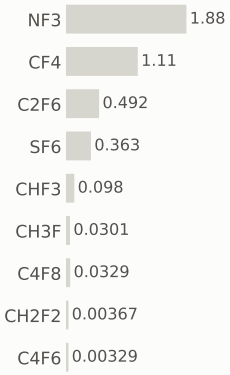
steps by process area



materials deposited (g)



gases emitted (g)



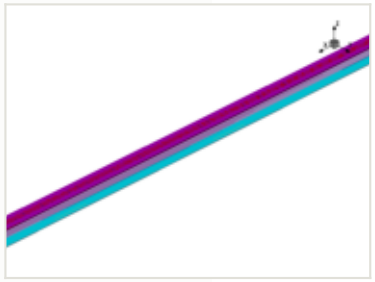
Step #7250: Metallization

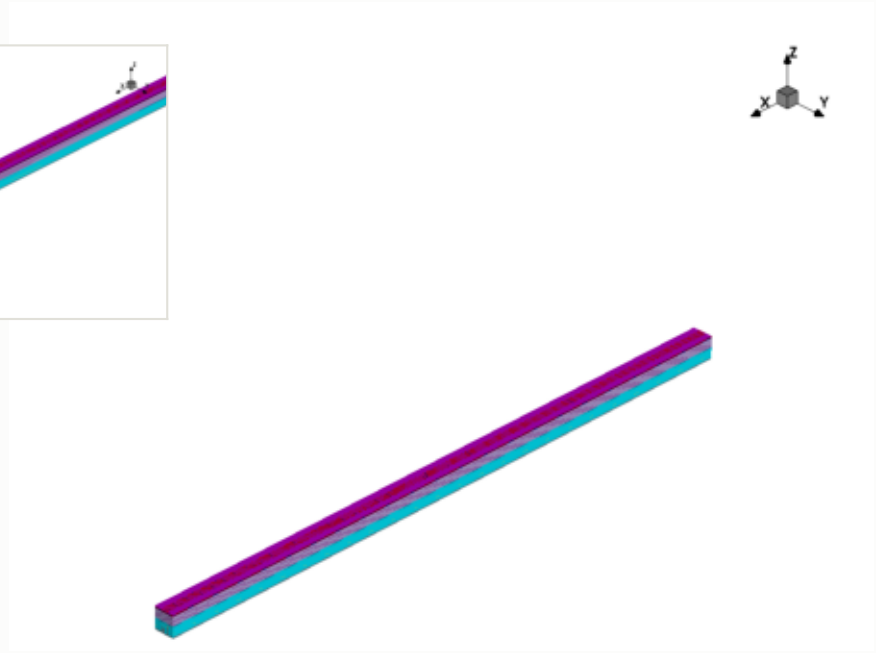
depo_ellipsoidal · Metallization

1.2	335.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



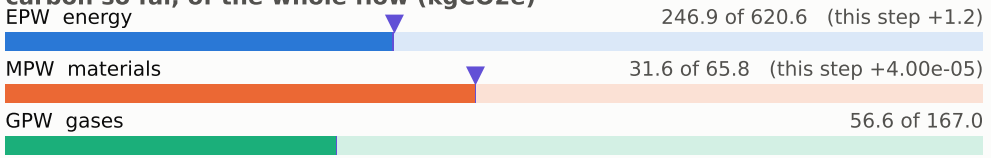


this step

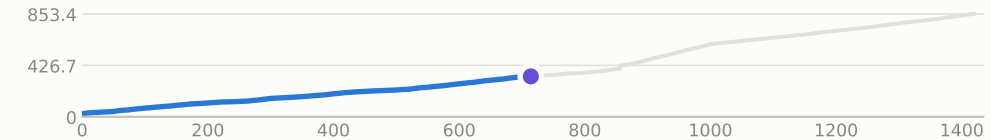
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

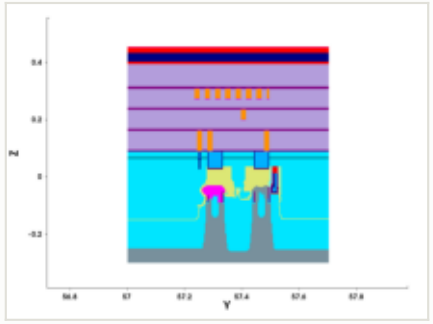
carbon so far, of the whole flow (kgCO2e)



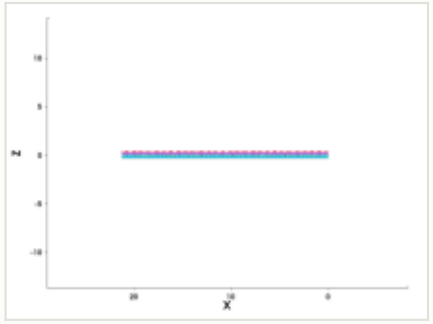
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



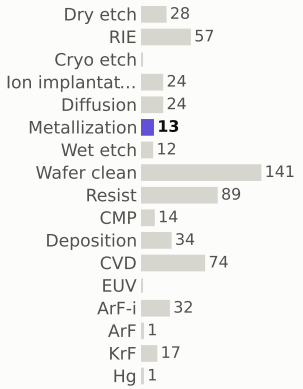
y cut at 57.351 µm



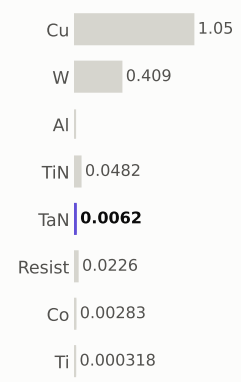
mask for this step

no mask for this step

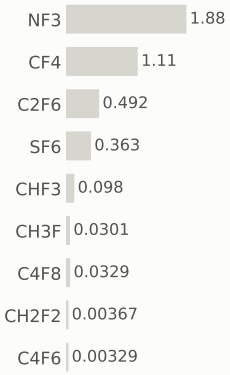
steps by process area



materials deposited (g)



gases emitted (g)



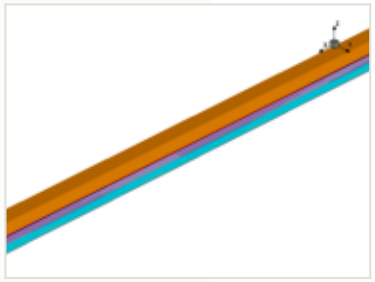
Step #7260: Metallization

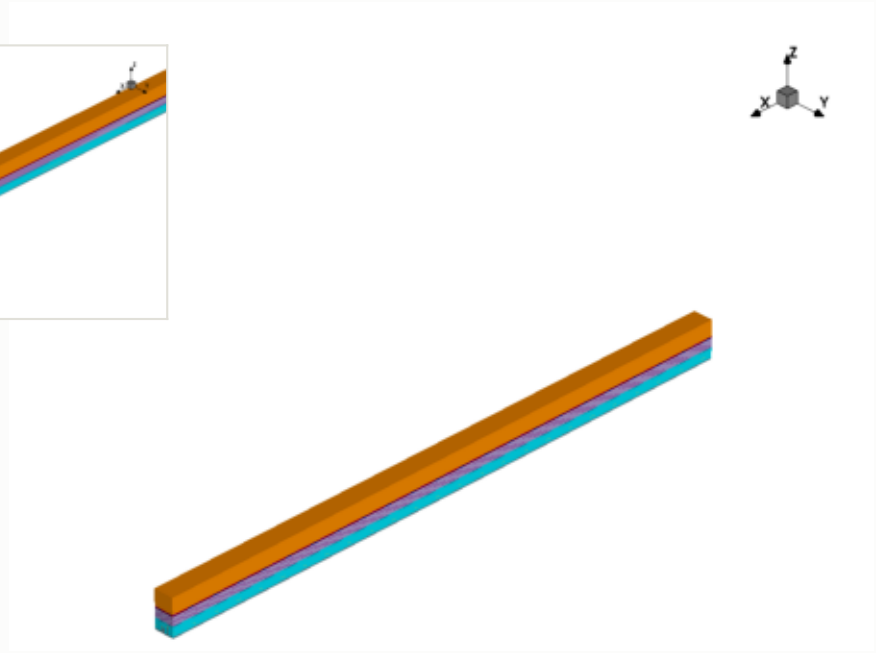
depo_fill · Metallization

1.21	336.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



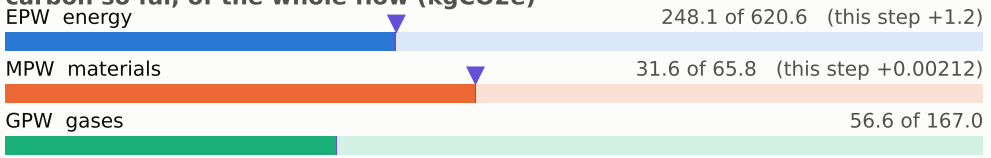


this step

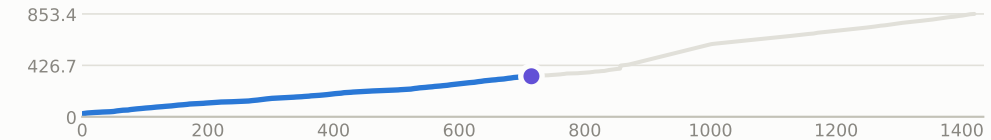
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

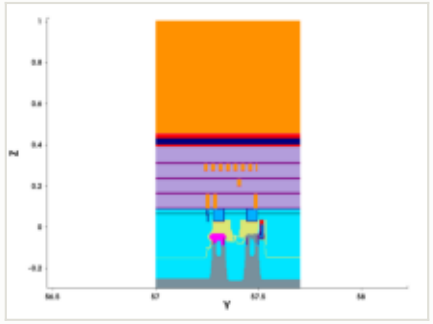
carbon so far, of the whole flow (kgCO2e)



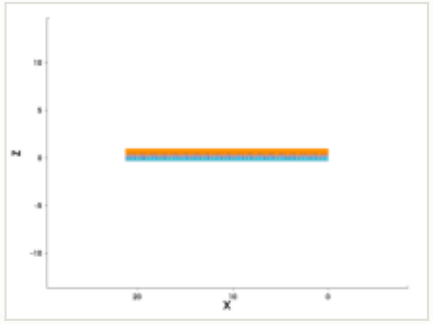
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



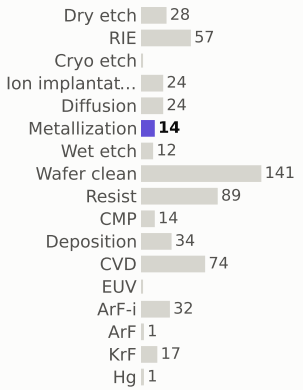
y cut at 57.351 µm



mask for this step

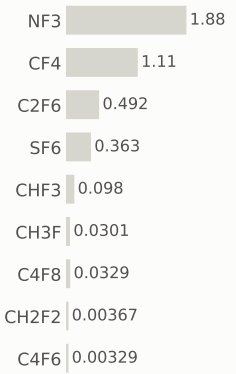
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



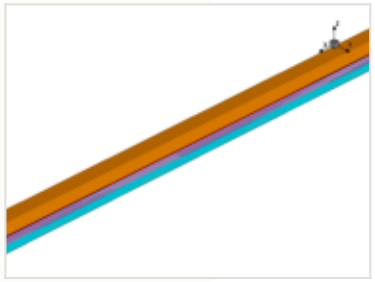
Step #7270: Diffusion

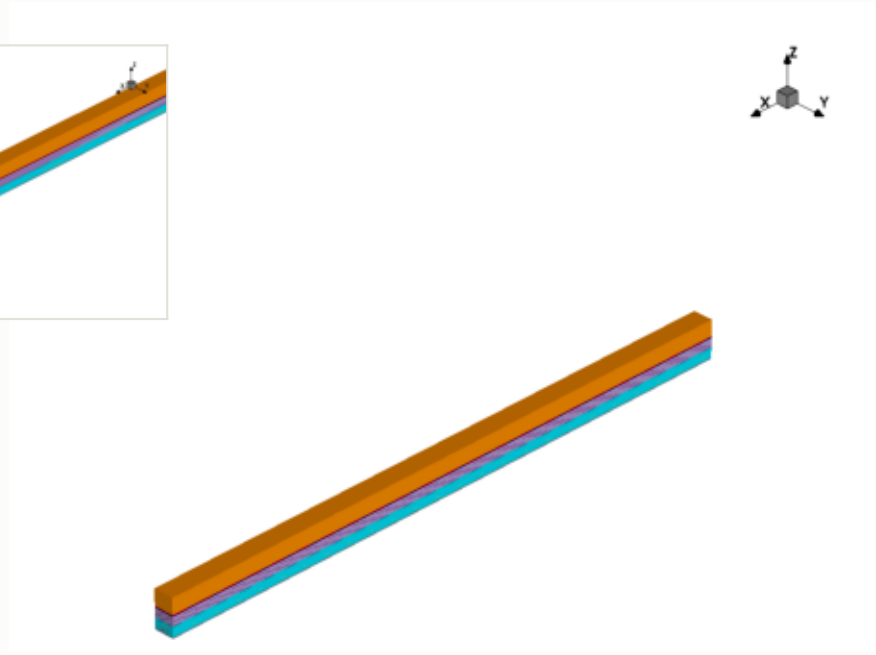
anneal · Diffusion

0.801	337.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





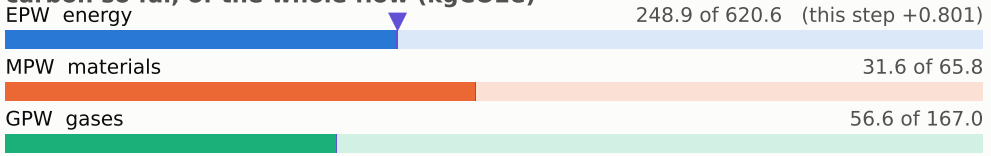
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

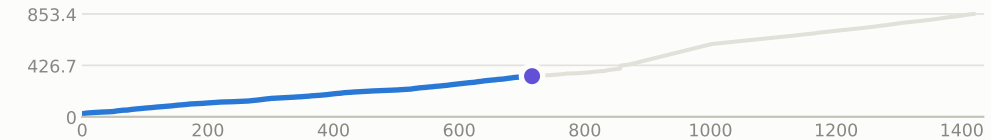
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

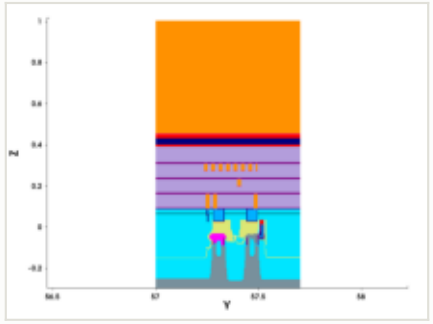
carbon so far, of the whole flow (kgCO2e)



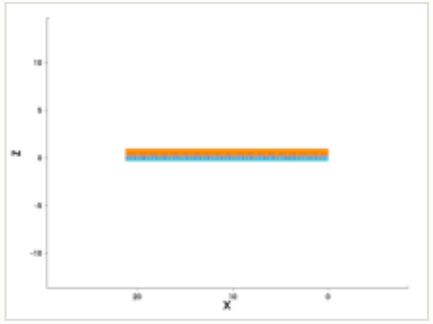
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



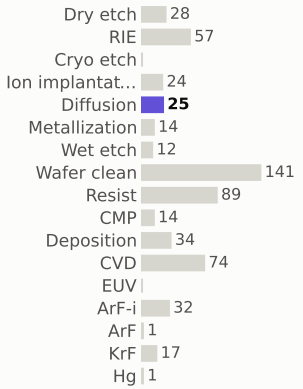
y cut at 57.351 µm



mask for this step

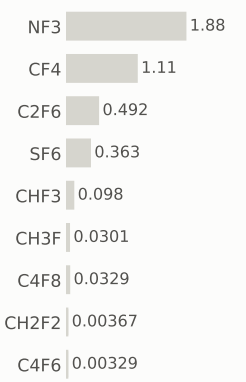
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



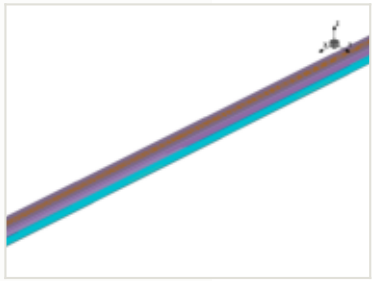
Step #7280: CMP

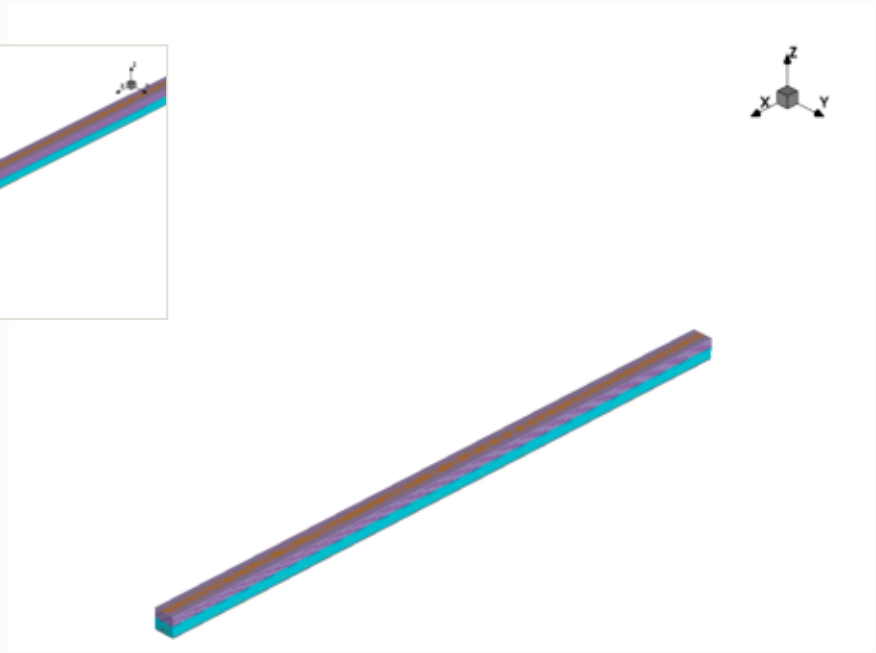
etch_cmp · CMP

0.169	337.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



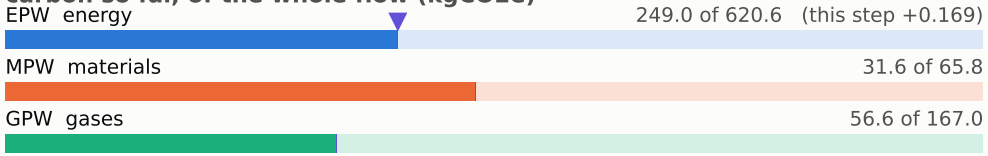


this step

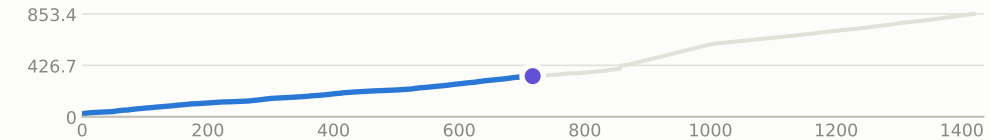
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

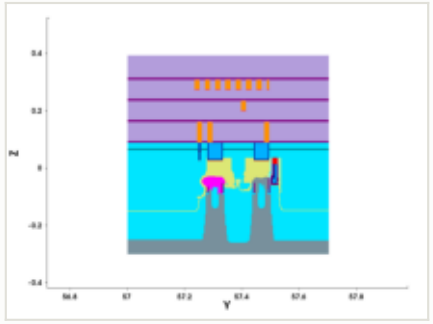
carbon so far, of the whole flow (kgCO2e)



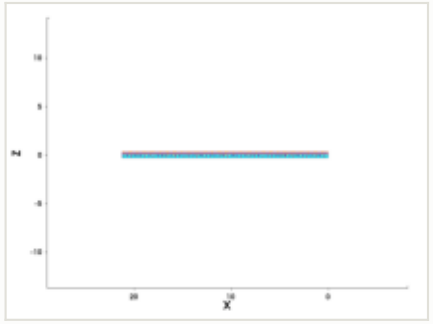
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



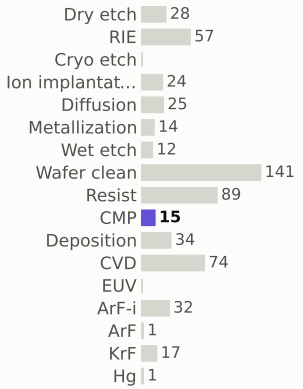
y cut at 57.351 µm



mask for this step

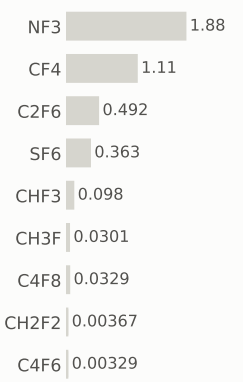
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



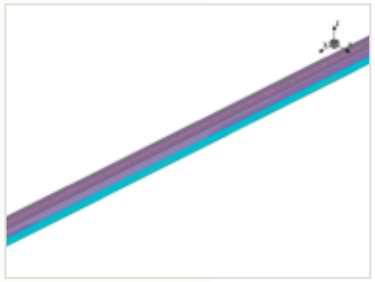
Step #7290: Spx_only

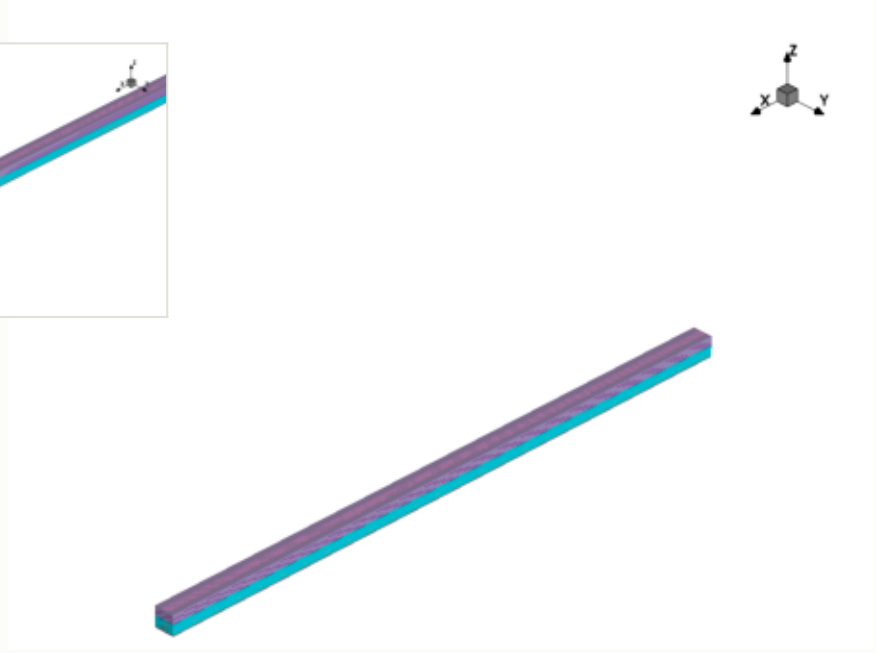
depo_isotropic · Spx_only

4.20e-05	337.3	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

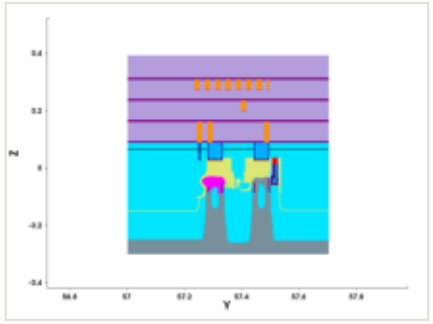
3D structure after this step

zoomed to fit

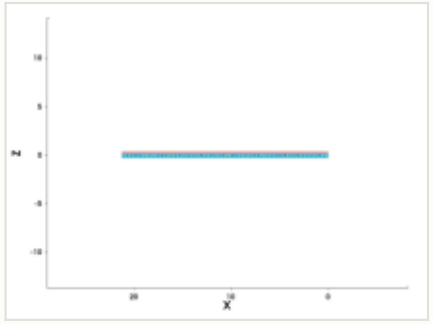




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

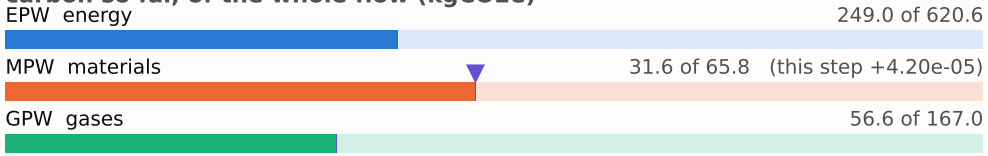
this step

process area	Spx_only
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

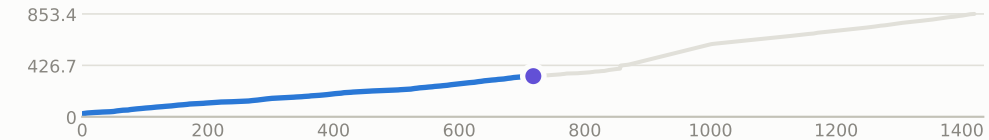
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**4.20e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

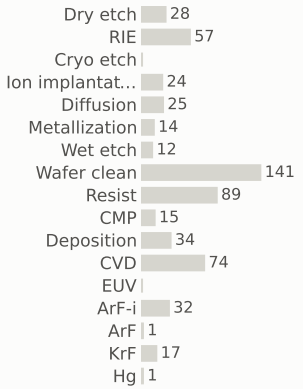
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

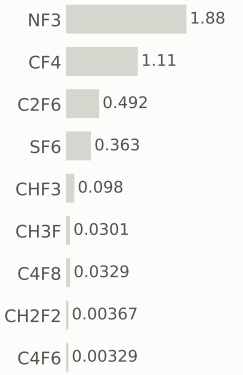


steps by process area



materials deposited (g)

gases emitted (g)



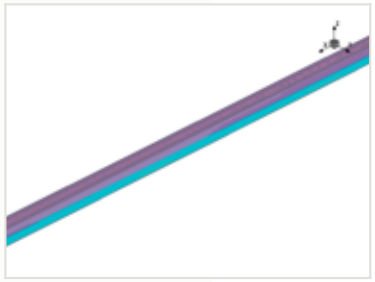
Step #7310: CVD

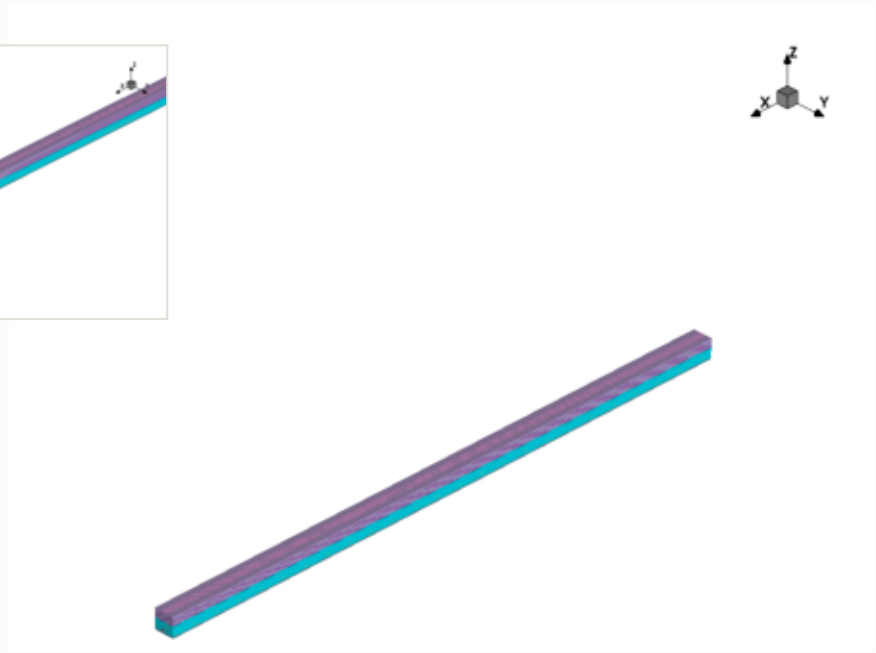
depo_isotropic · CVD

1.31	338.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



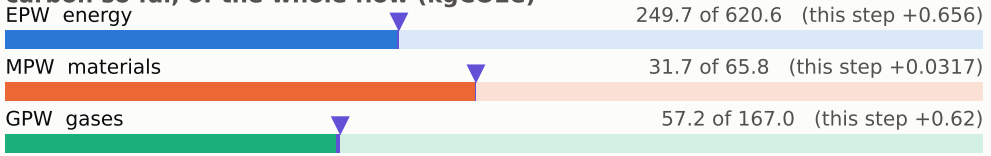


this step

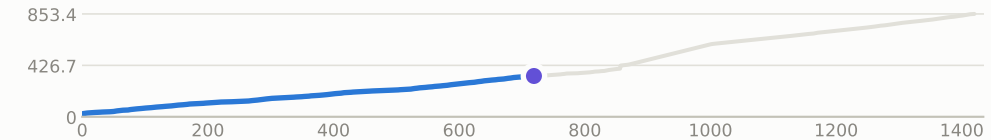
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

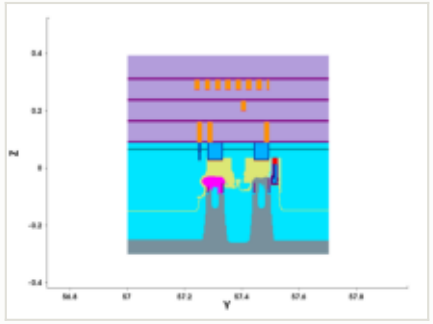
carbon so far, of the whole flow (kgCO2e)



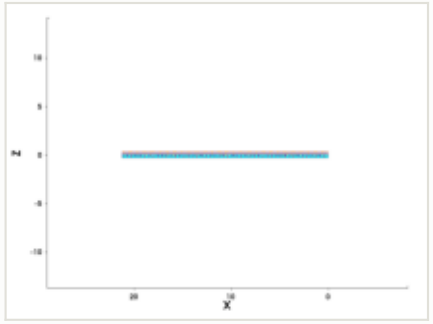
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



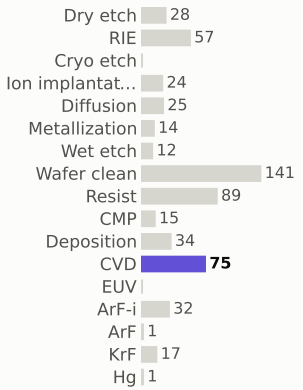
y cut at 57.351 µm



mask for this step

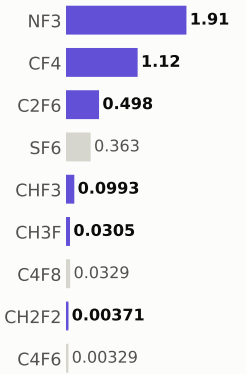
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



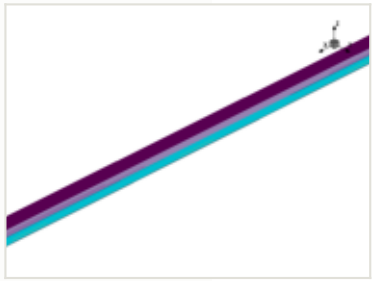
Step #7320: Spx_only

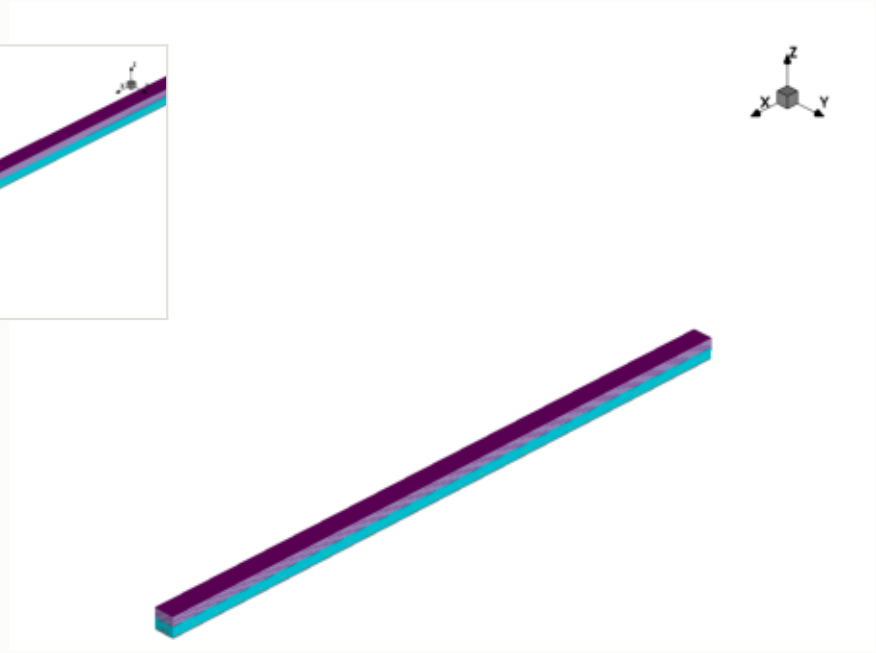
depo_fill · Spx_only

0	338.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





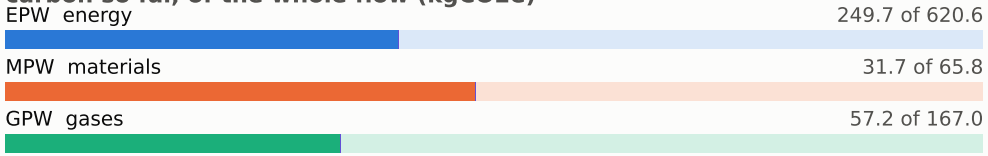
this step

process area	Spx_only
material	—
thickness	0.006 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

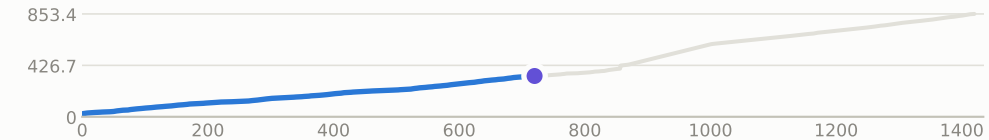
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

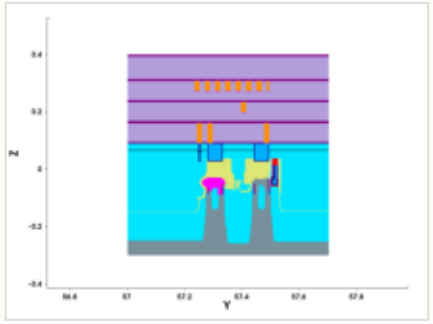
carbon so far, of the whole flow (kgCO2e)



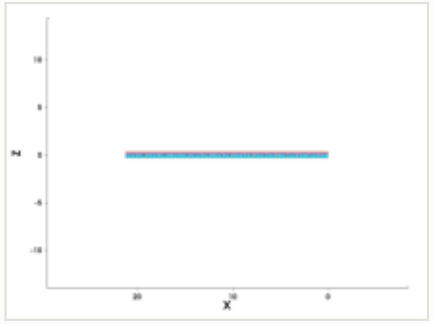
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



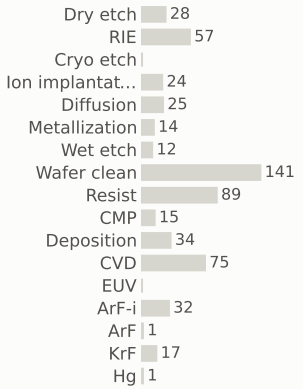
y cut at 57.351 µm



mask for this step

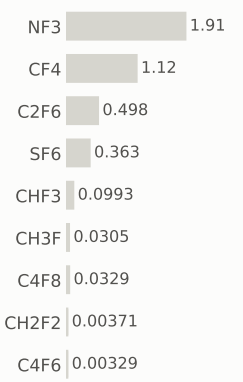
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



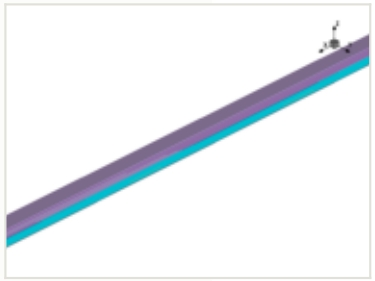
Step #7330: CVD

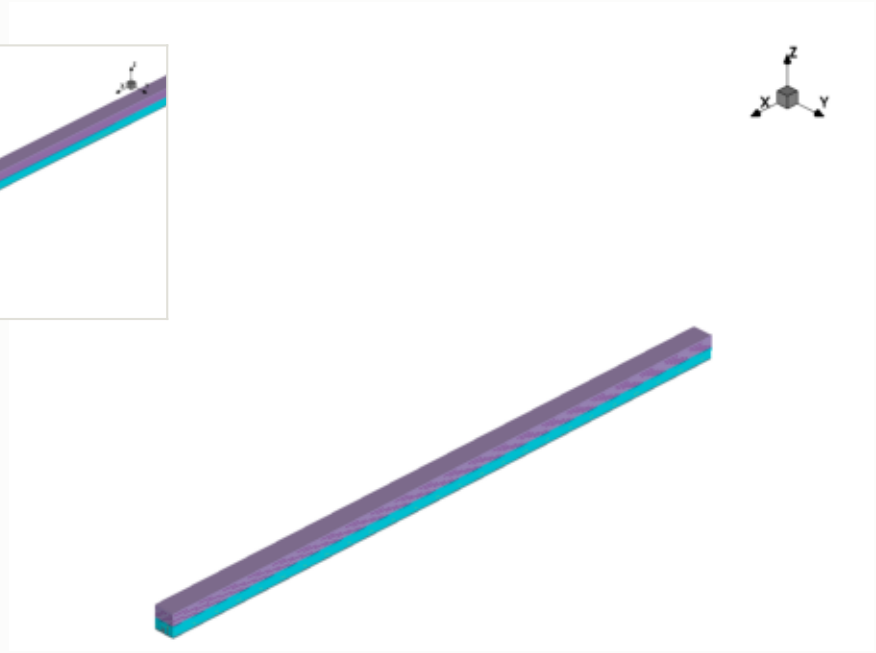
depo_fill · CVD

1.31	339.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



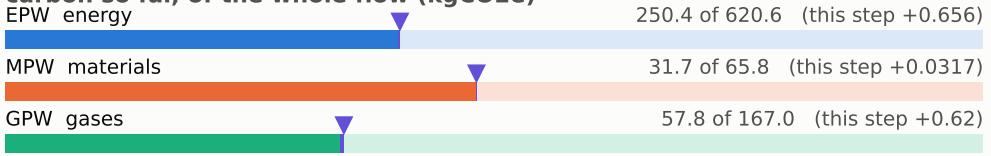


this step

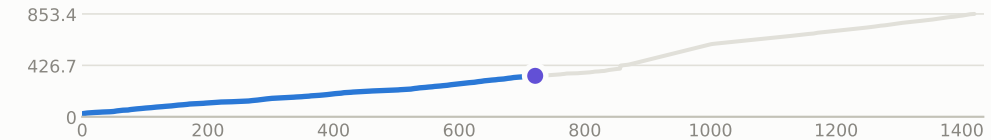
process area	CVD
material	—
thickness	0.09 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

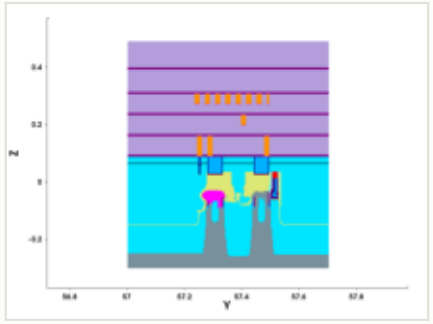
carbon so far, of the whole flow (kgCO2e)



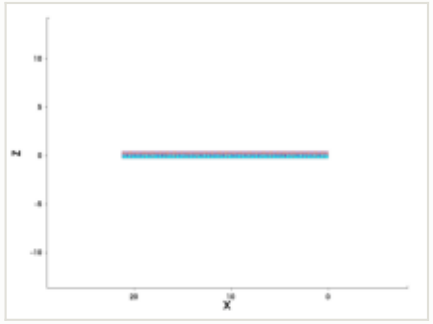
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



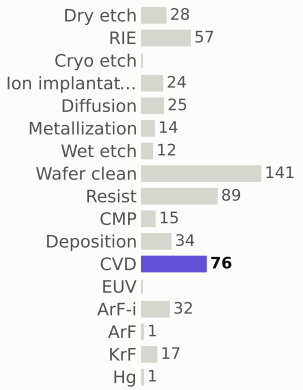
y cut at 57.351 µm



mask for this step

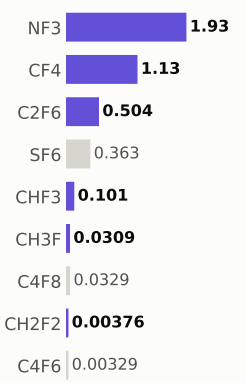
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



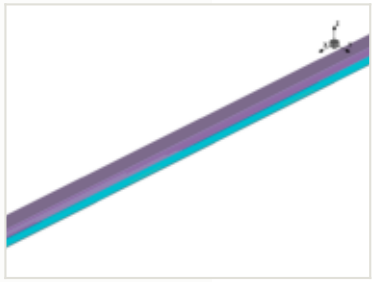
Step #7340: Diffusion

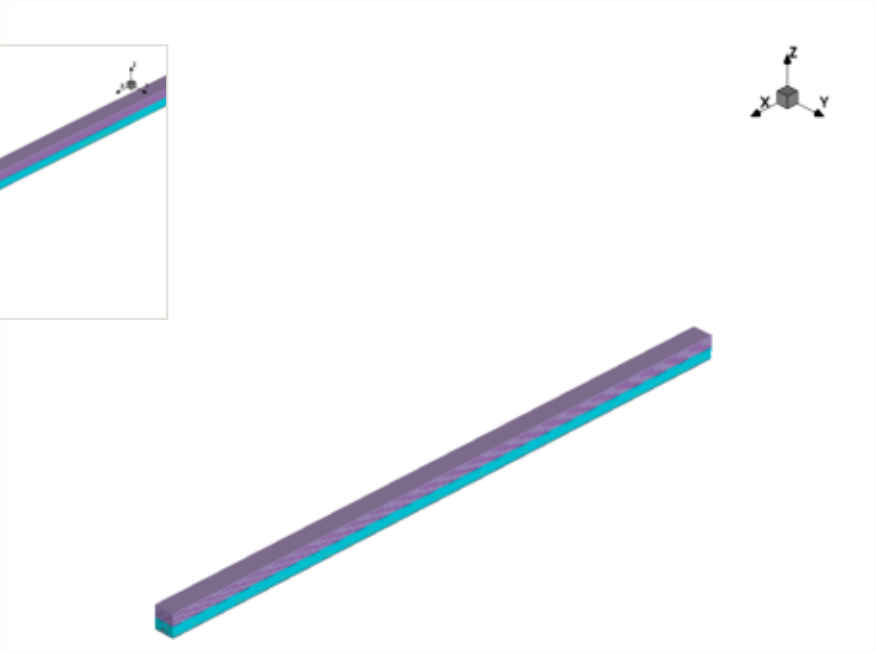
anneal · Diffusion

0.801	340.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





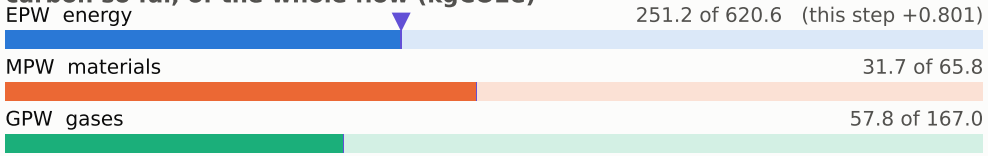
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

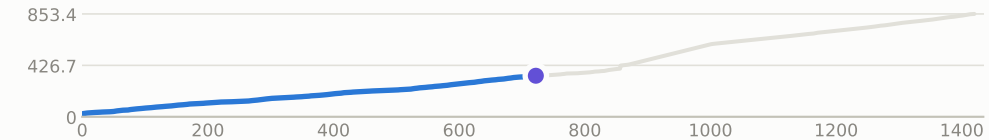
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

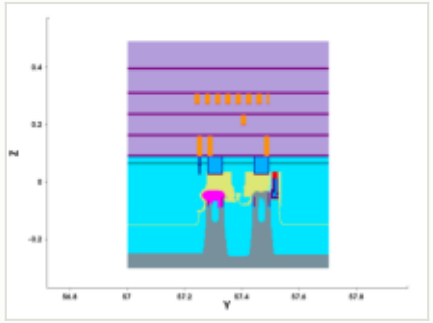
carbon so far, of the whole flow (kgCO2e)



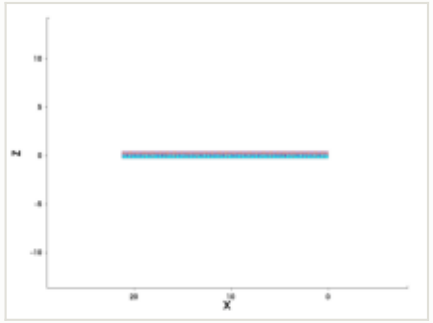
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



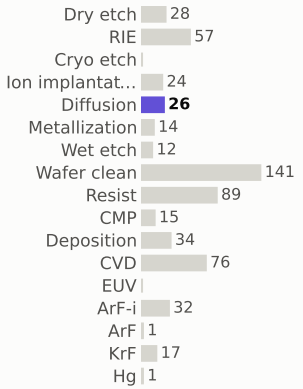
y cut at 57.351 µm



mask for this step

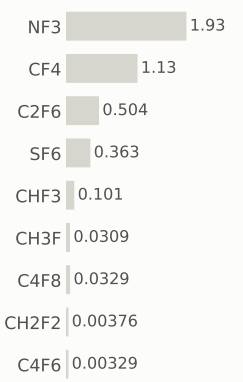
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



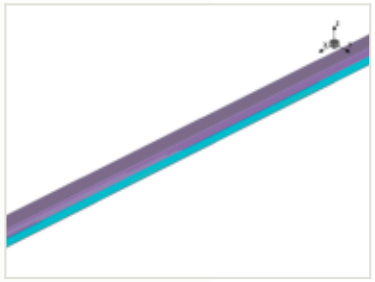
Step #7350: Wafer_clean

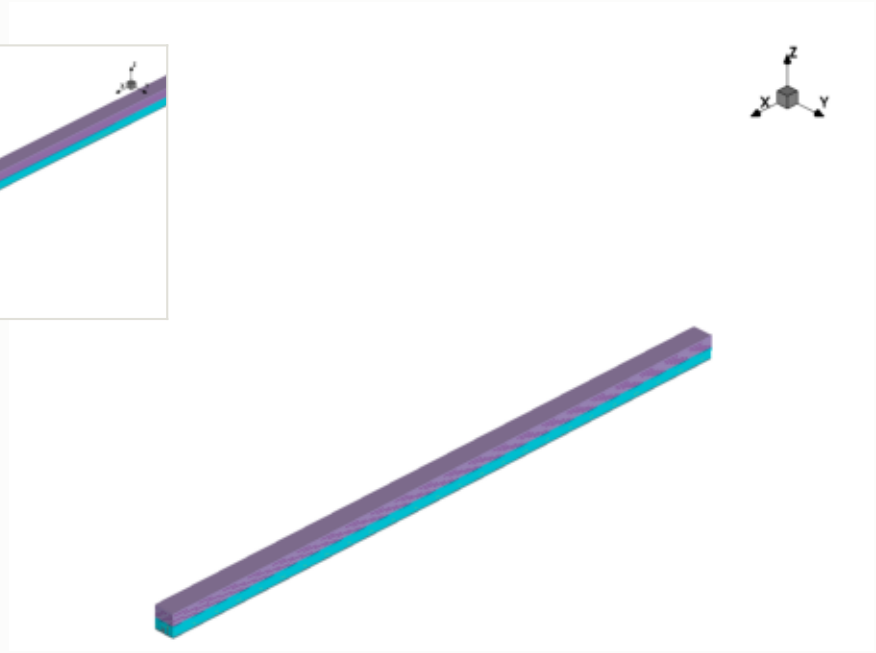
clean · Wafer_clean

0.169	340.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





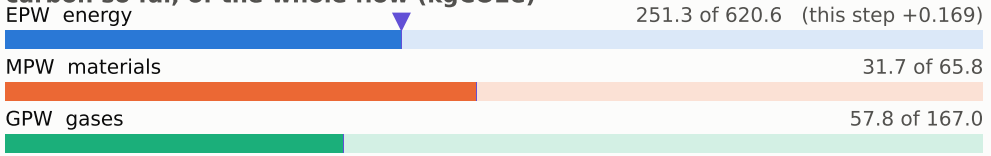
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

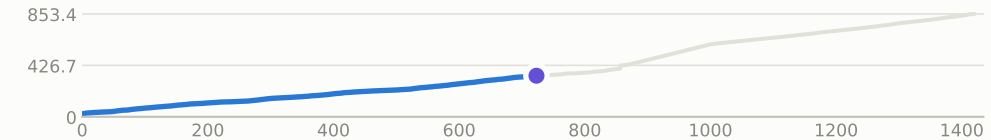
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

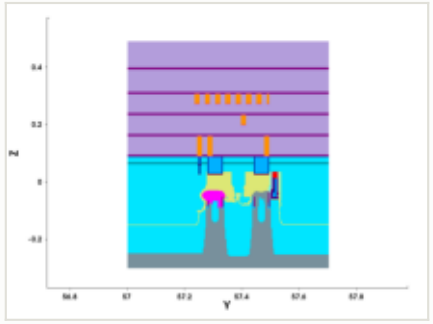
carbon so far, of the whole flow (kgCO2e)



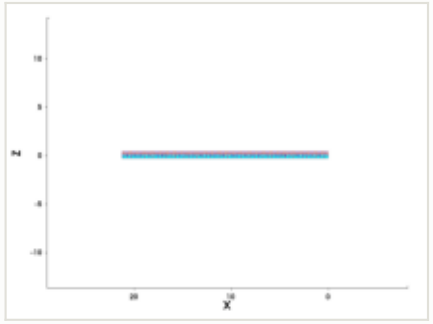
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



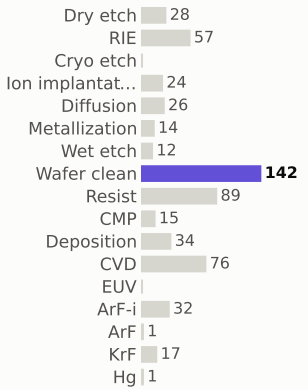
y cut at 57.351 µm



mask for this step

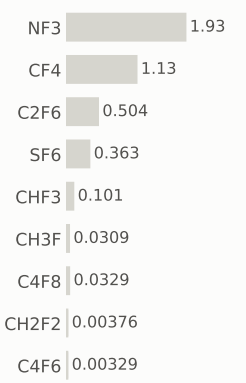
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



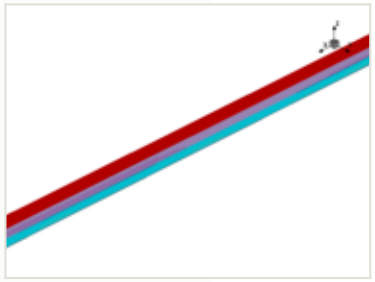
Step #7360: CVD

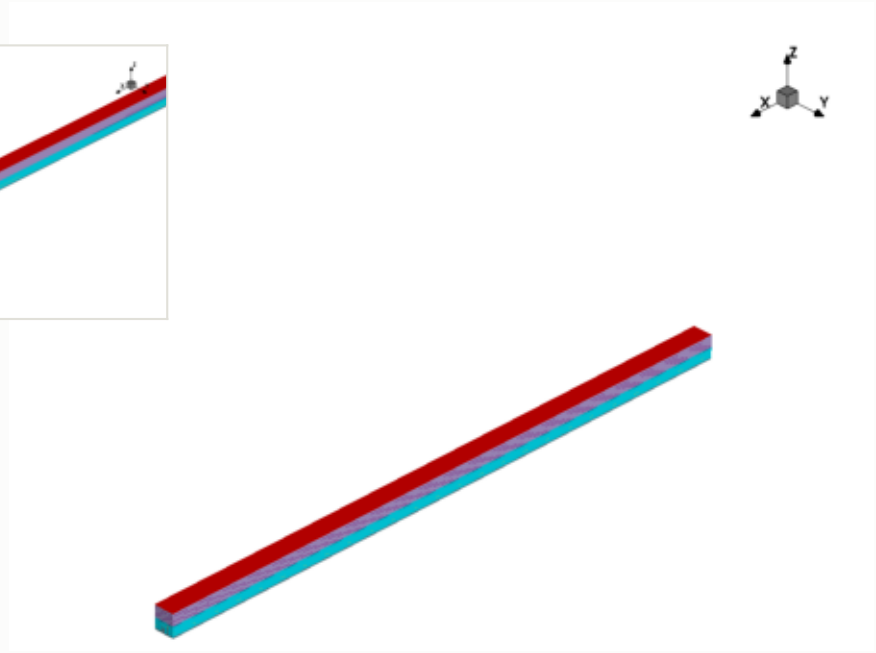
depo_fill · CVD

1.31	342.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



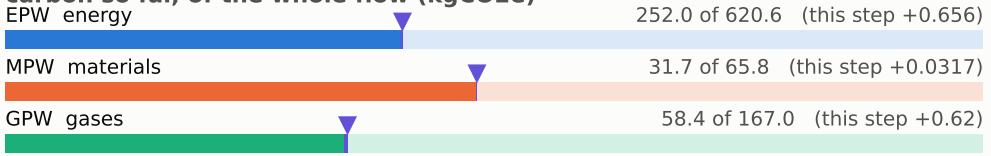


this step

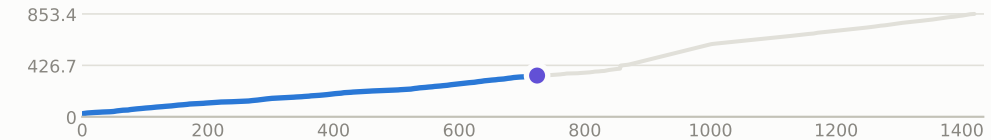
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

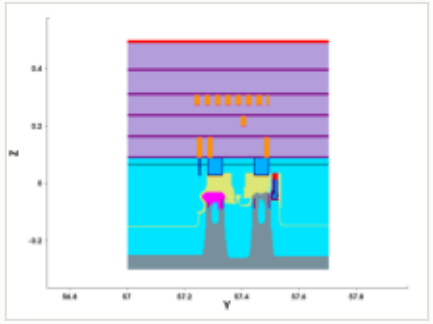
carbon so far, of the whole flow (kgCO2e)



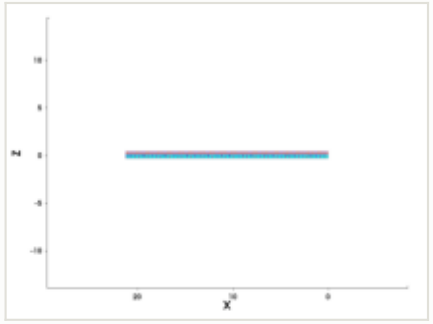
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



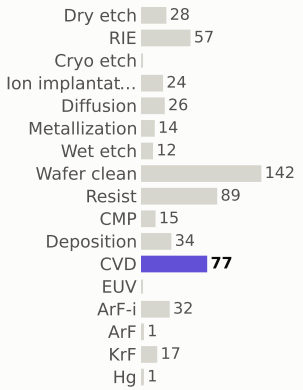
y cut at 57.351 µm



mask for this step

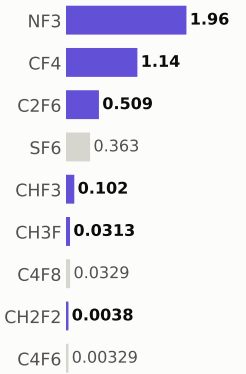
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



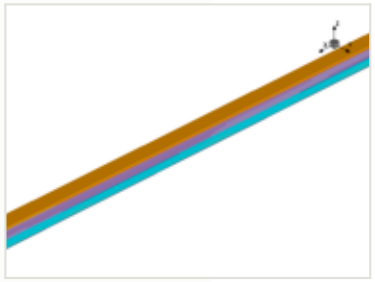
Step #7370: Spx_only

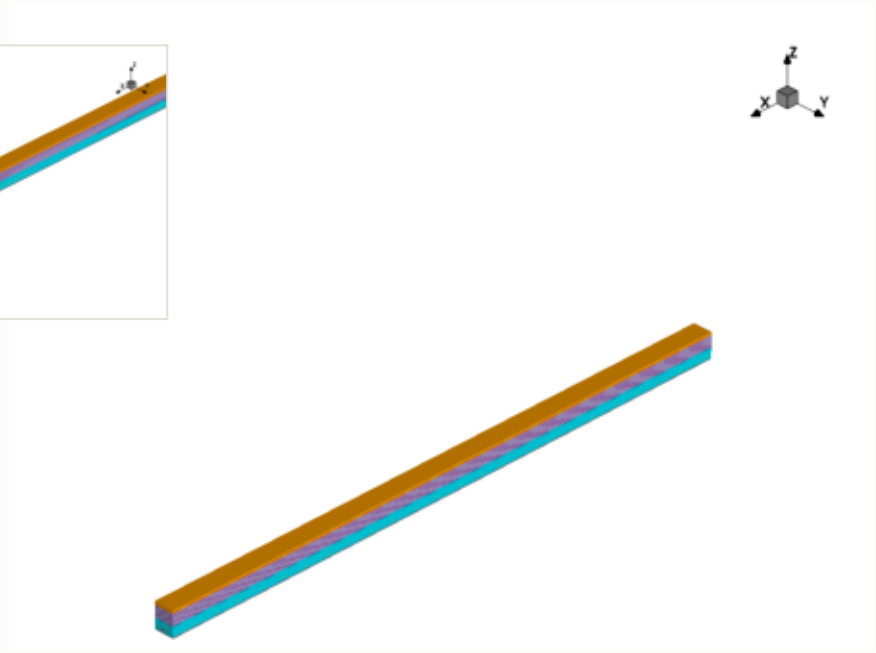
depo_fill · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





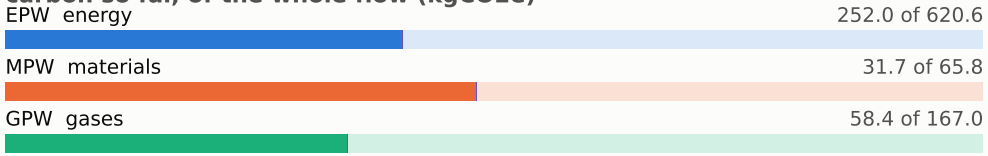
this step

process area	Spx_only
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

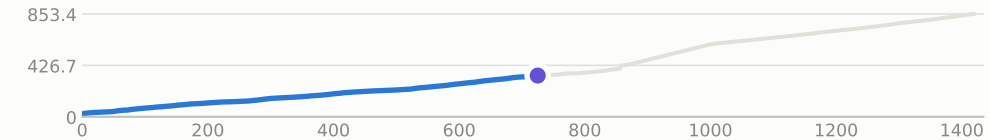
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

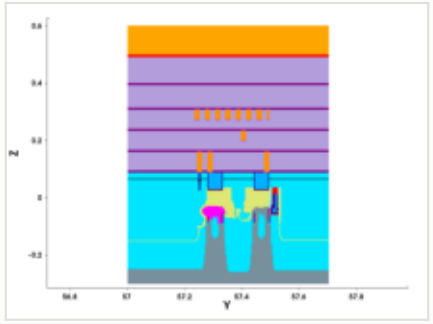
carbon so far, of the whole flow (kgCO2e)



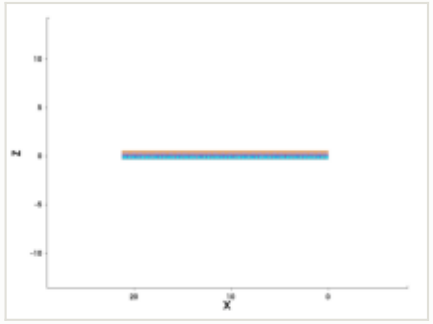
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



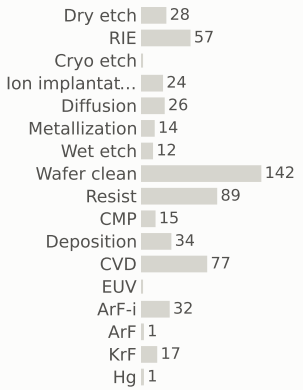
y cut at 57.351 µm



mask for this step

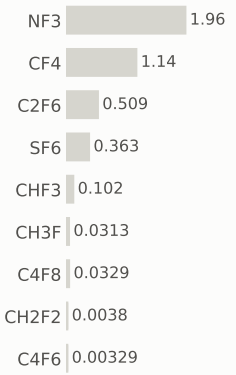
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



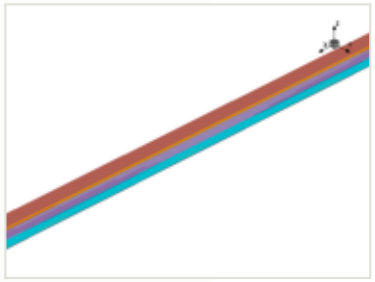
Step #7380: Spx_only

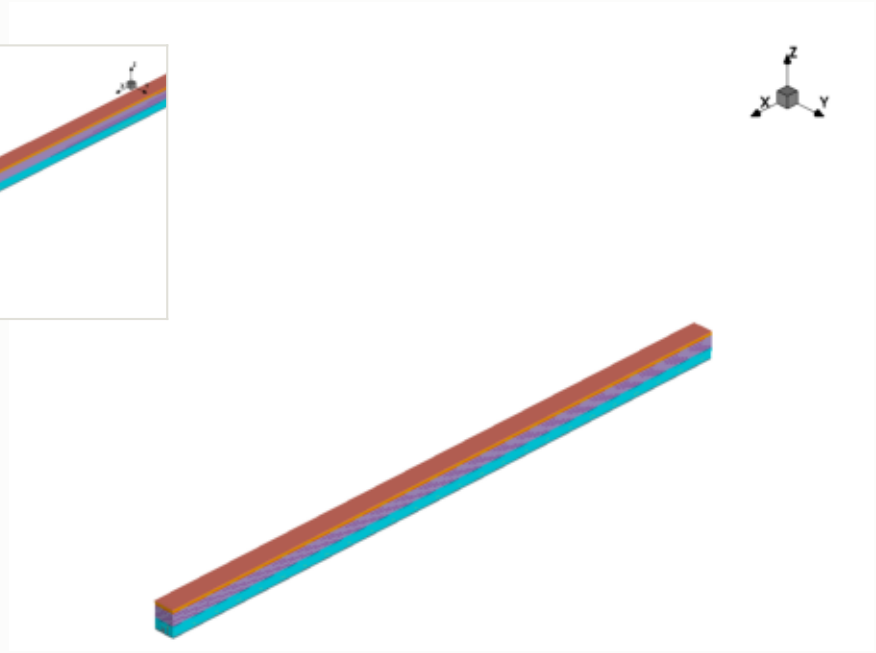
depo_fill · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





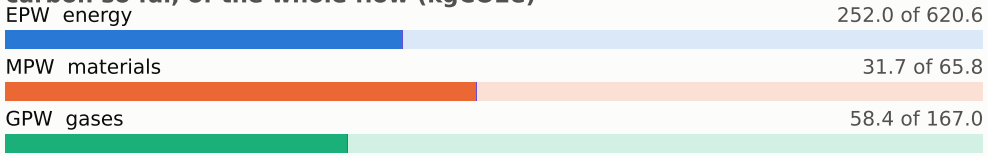
this step

process area	Spx_only
material	—
thickness	0.02 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

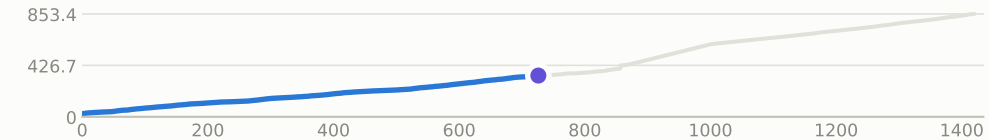
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

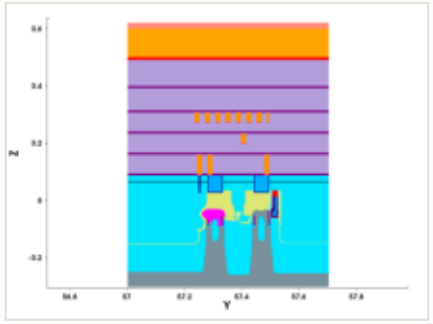
carbon so far, of the whole flow (kgCO2e)



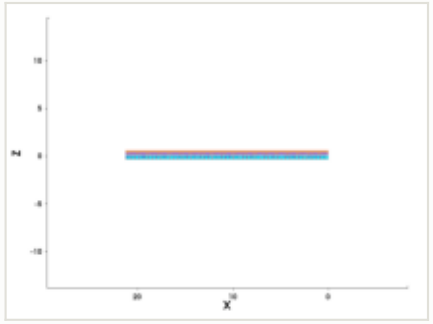
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



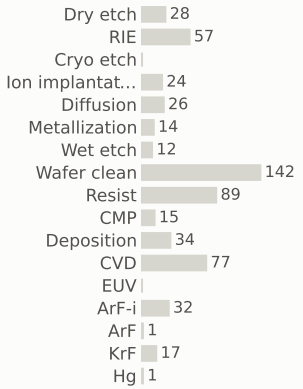
y cut at 57.351 µm



mask for this step

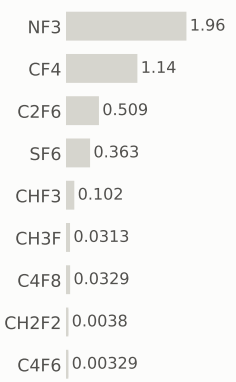
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



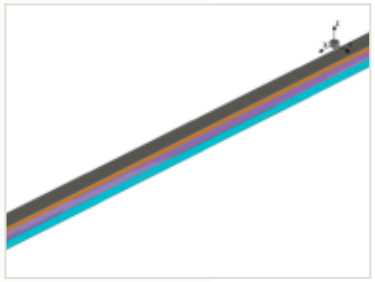
Step #7390: Spx_only

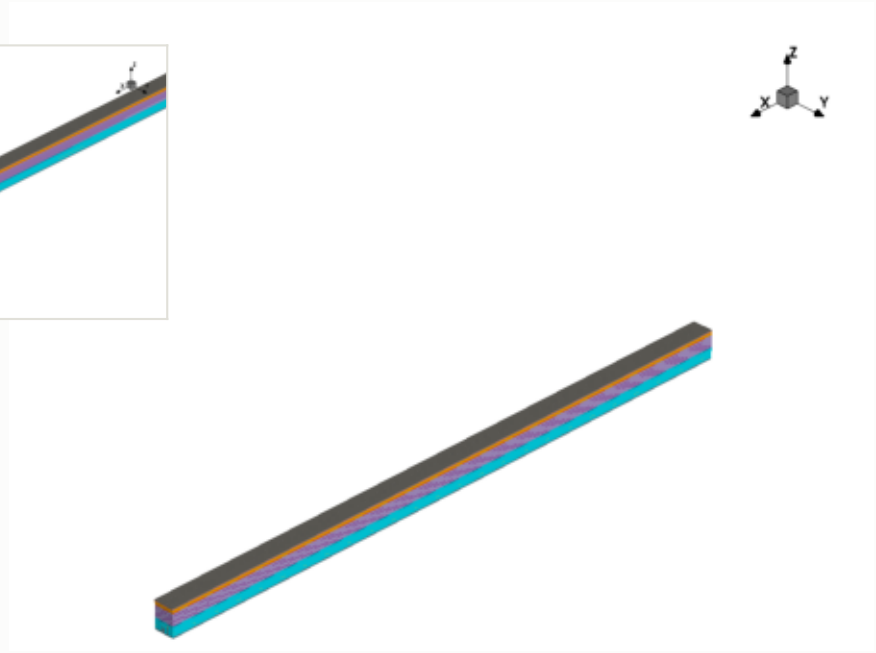
litho_patterning · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





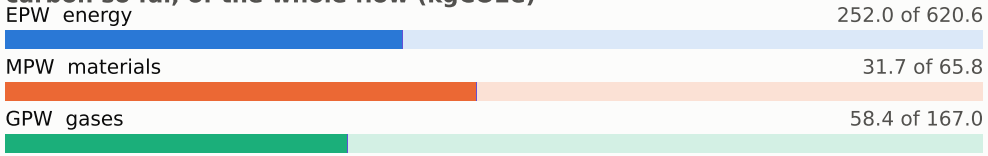
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

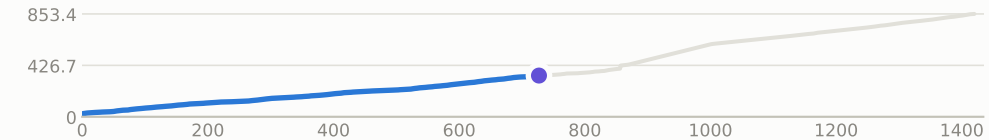
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

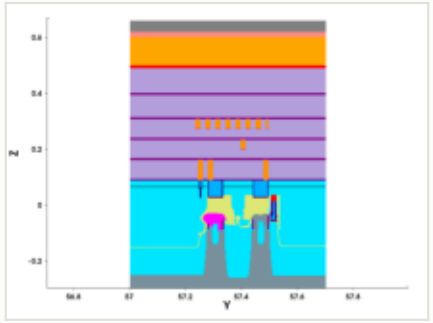
carbon so far, of the whole flow (kgCO2e)



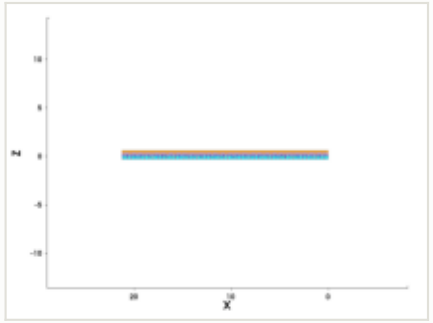
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

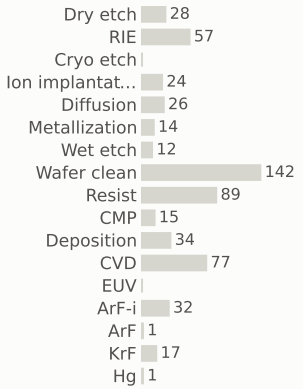


y cut at 57.351 µm

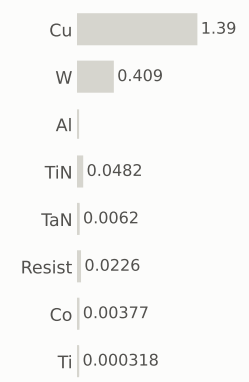


mask: layer 12451:0 (GDSII)

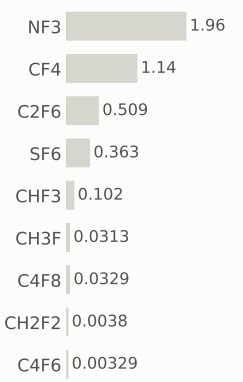
steps by process area



materials deposited (g)



gases emitted (g)



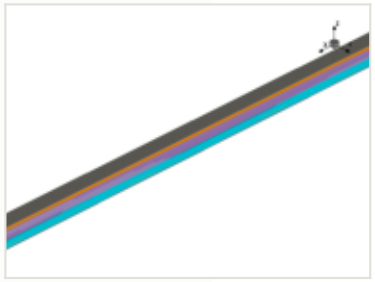
Step #7400: Spx_only

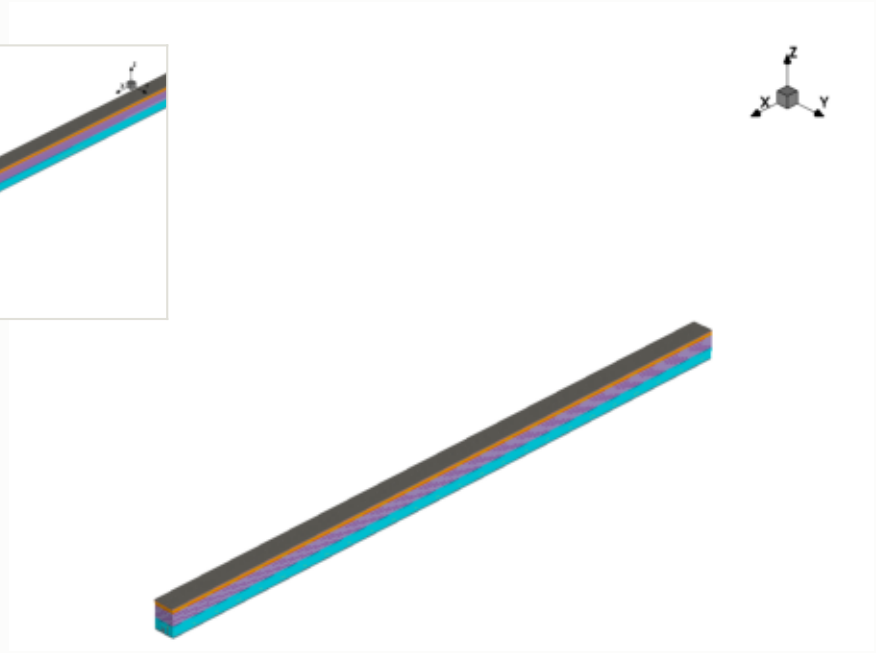
etch_anisotropic · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

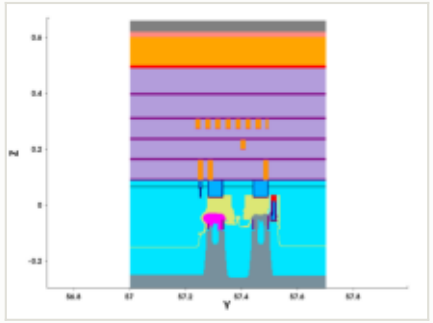
3D structure after this step

zoomed to fit

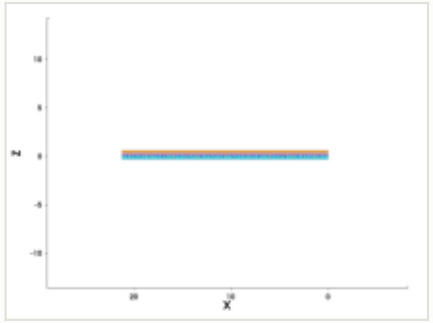




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

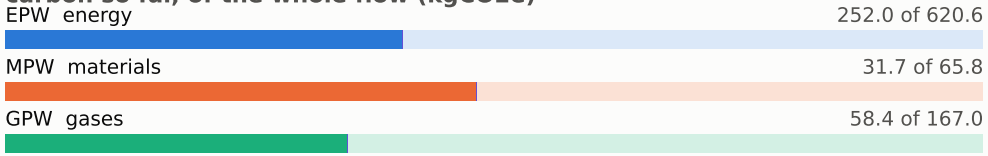
no mask for this step

this step

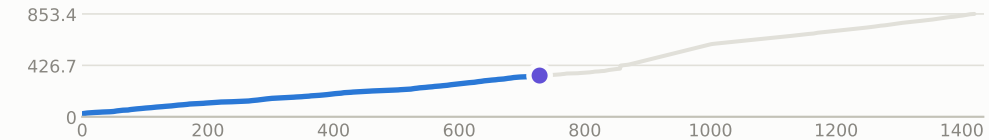
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

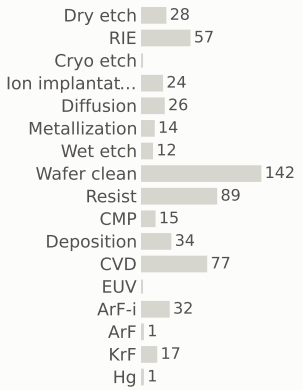
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

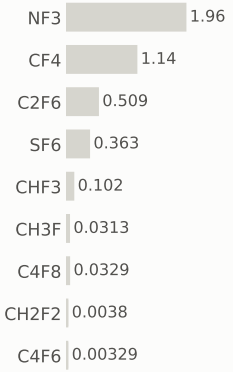


steps by process area



materials deposited (g)

gases emitted (g)



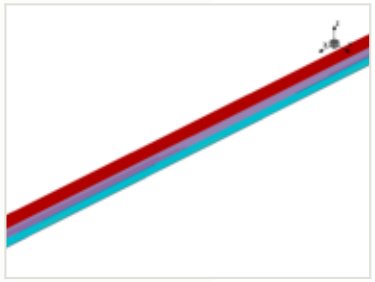
Step #7410: Spx_only

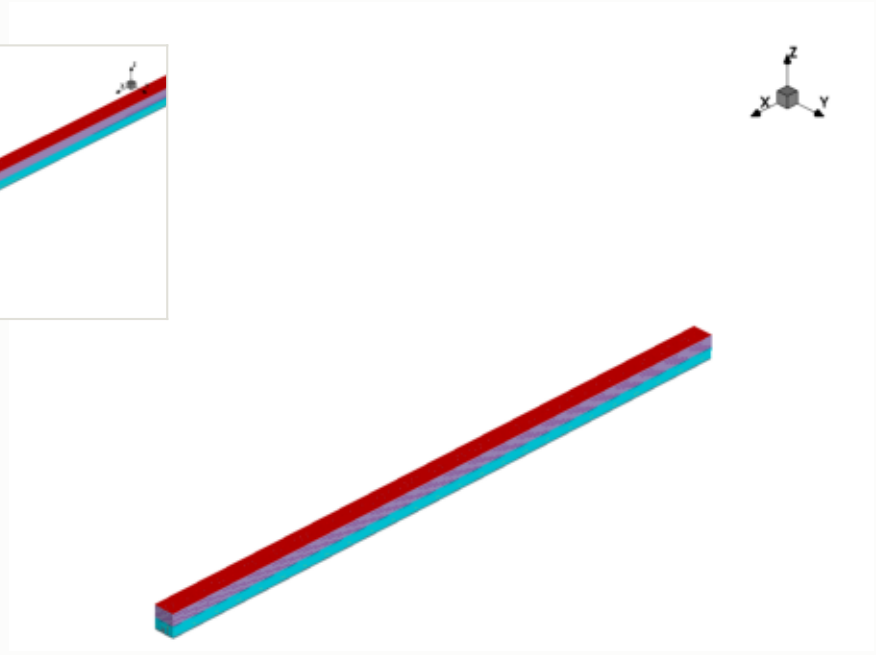
etch_strip · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





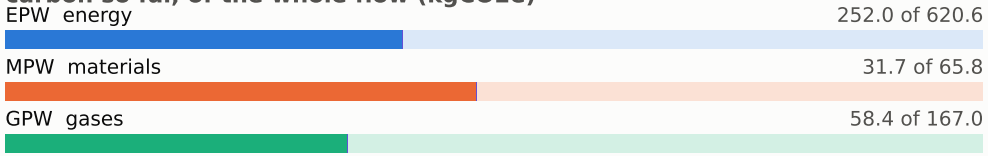
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

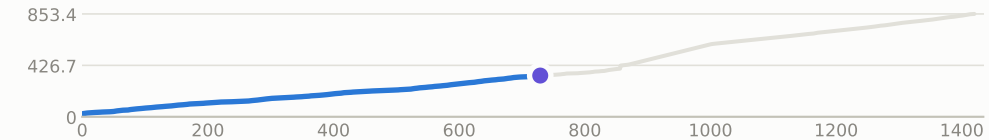
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

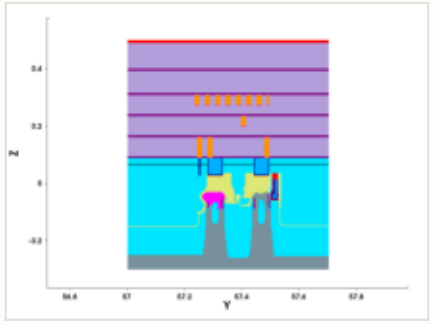
carbon so far, of the whole flow (kgCO2e)



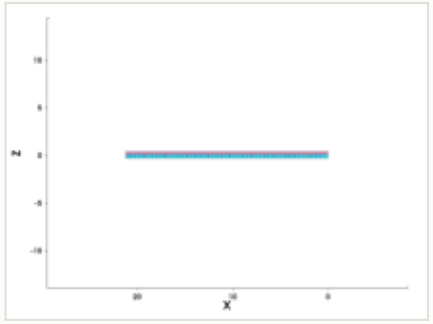
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



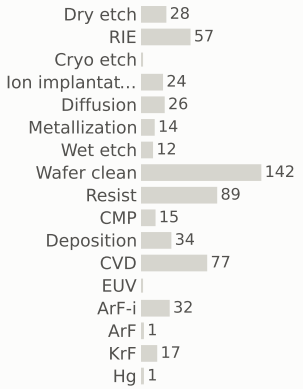
y cut at 57.351 µm



mask for this step

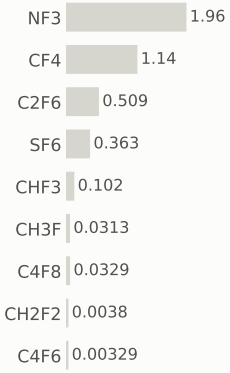
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



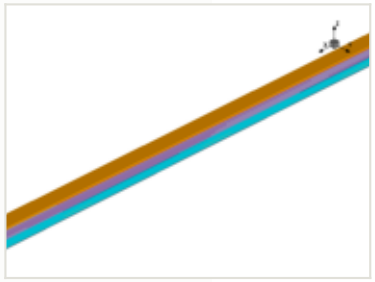
Step #7420: Spx_only

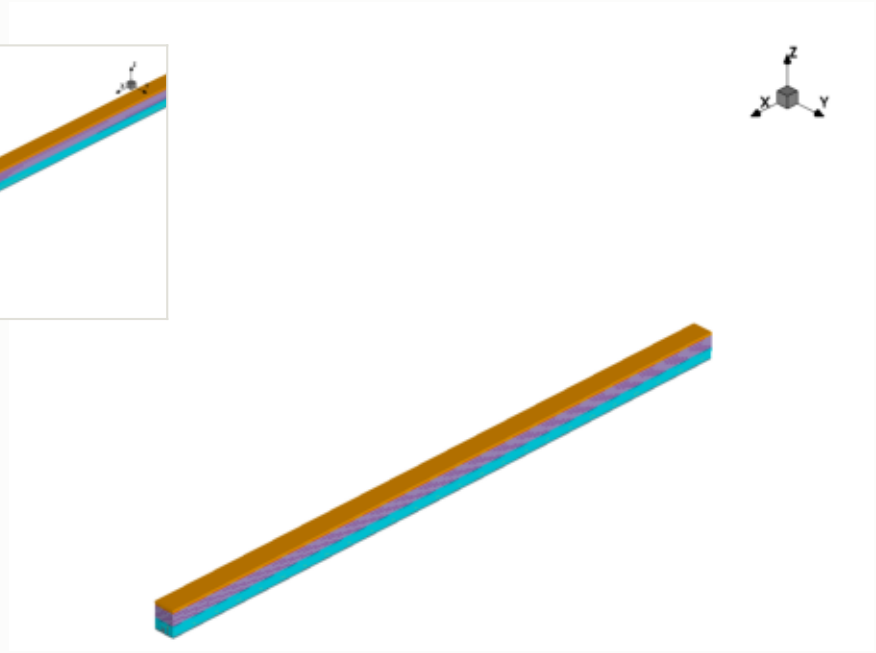
depo_fill · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





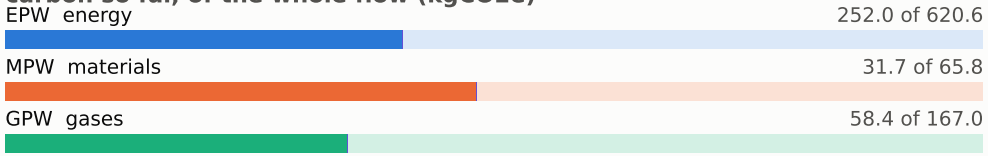
this step

process area	Spx_only
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

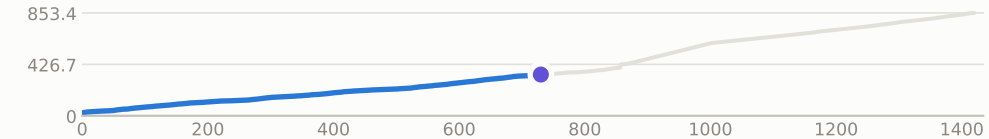
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

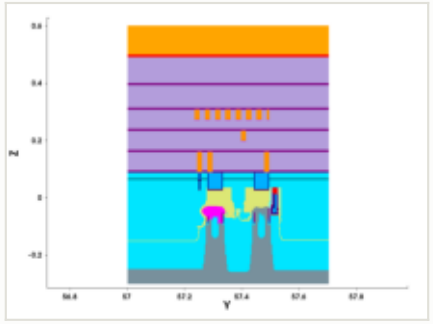
carbon so far, of the whole flow (kgCO2e)



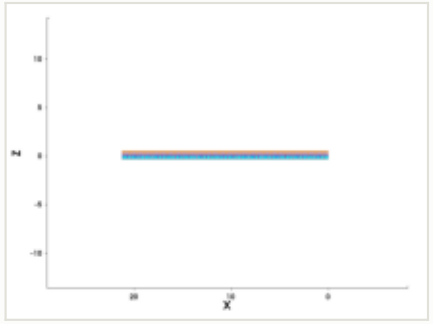
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



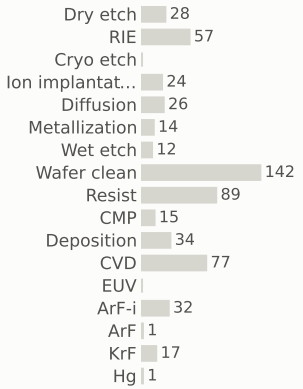
y cut at 57.351 µm



mask for this step

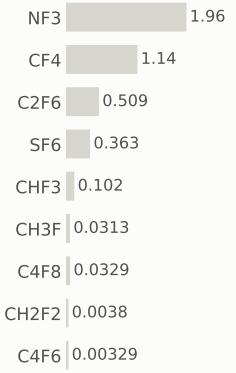
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



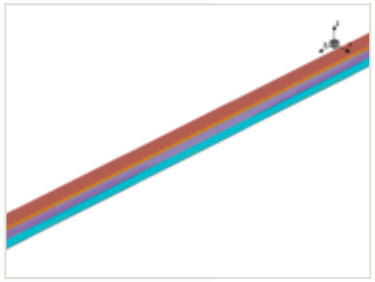
Step #7430: Spx_only

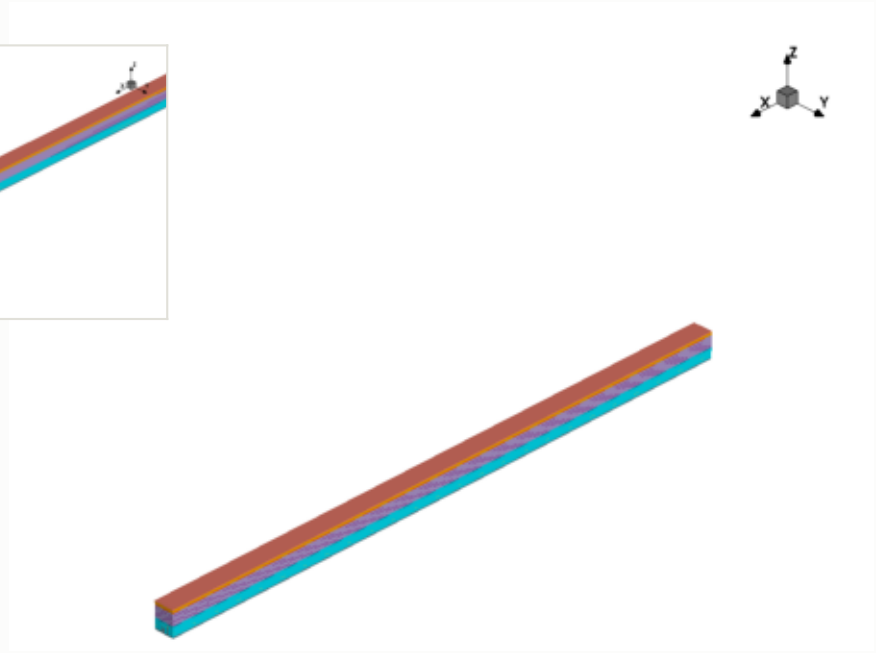
depo_fill · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

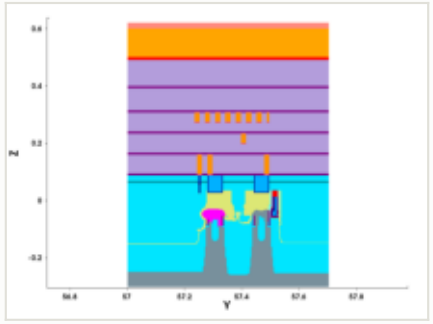
3D structure after this step

zoomed to fit

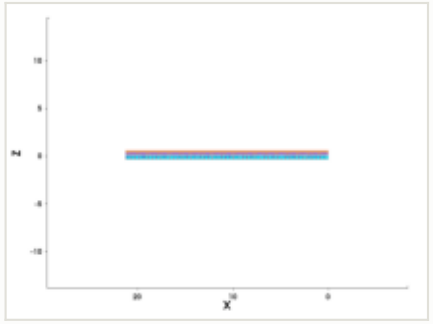




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

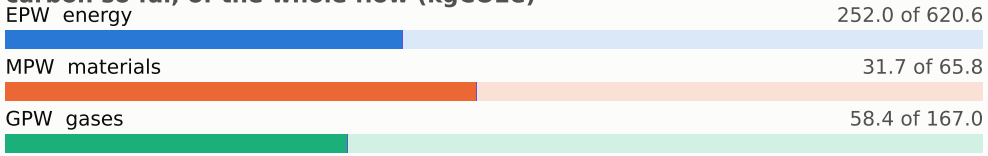
no mask for this step

this step

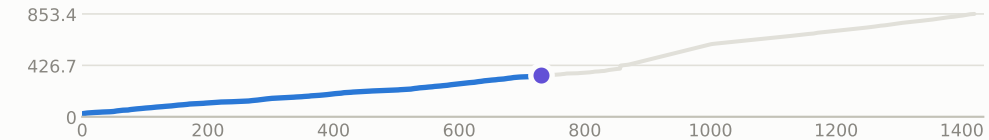
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	0.02 µm	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

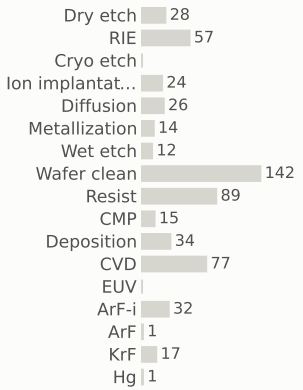
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

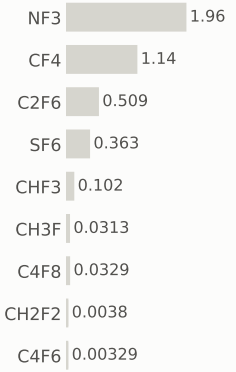


steps by process area



materials deposited (g)

gases emitted (g)



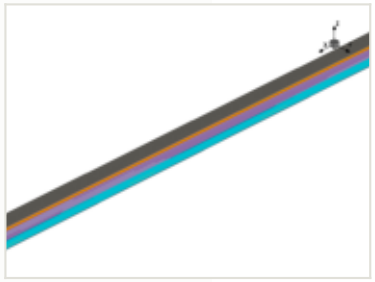
Step #7440: Spx_only

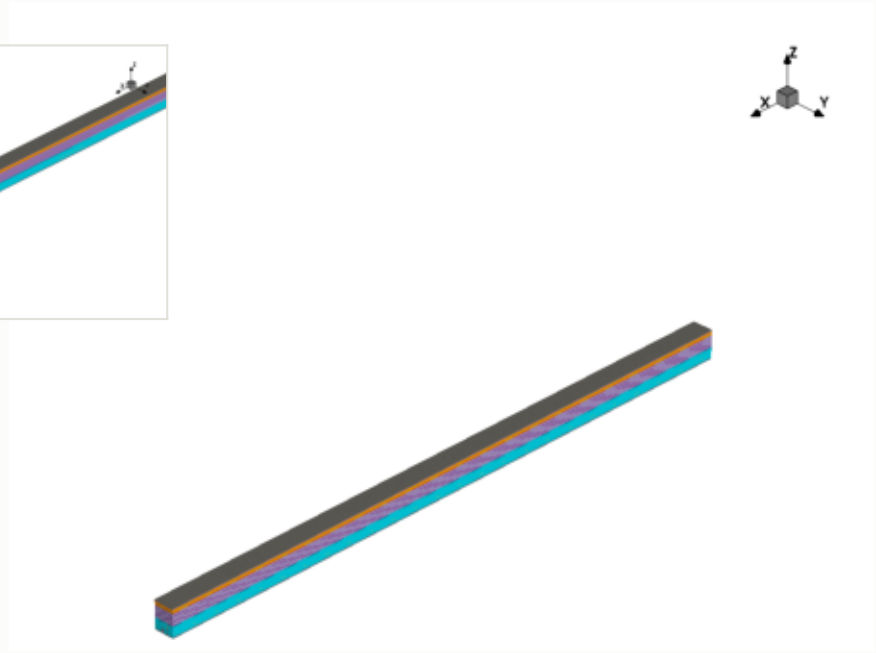
litho_patterning · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





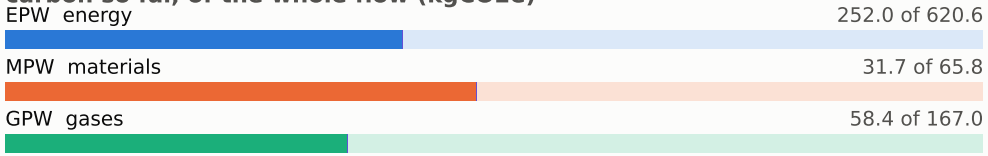
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

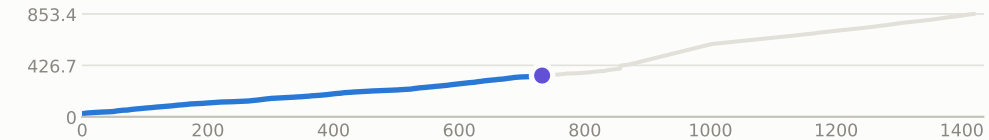
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

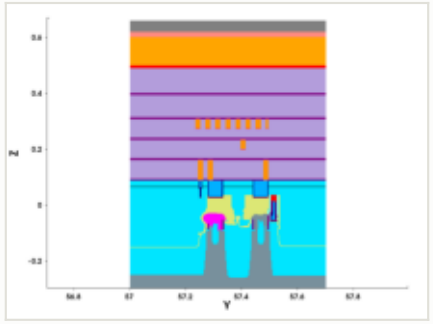
carbon so far, of the whole flow (kgCO2e)



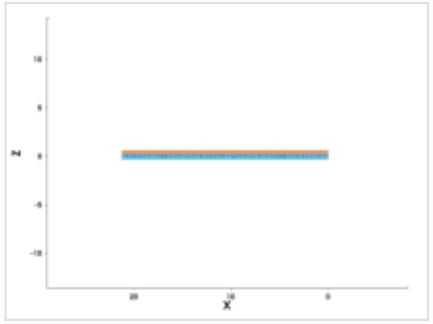
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

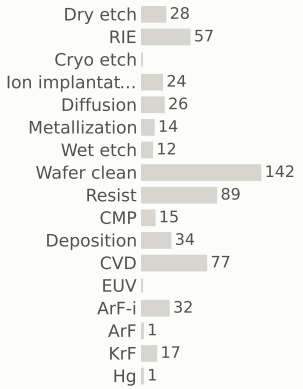


y cut at 57.351 µm

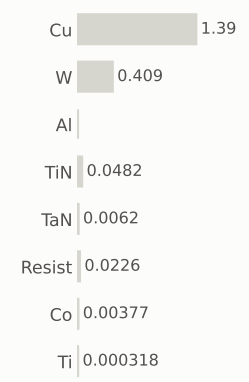


mask: layer 12452:0 (GDSII)

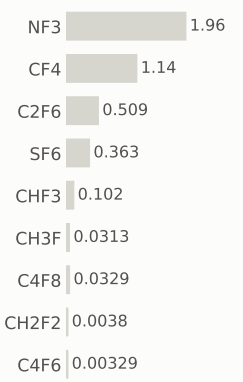
steps by process area



materials deposited (g)



gases emitted (g)



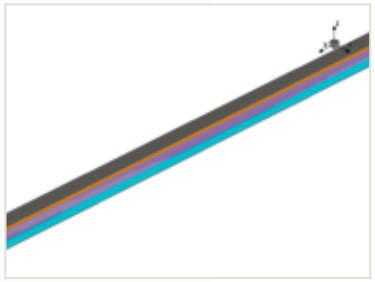
Step #7450: Spx_only

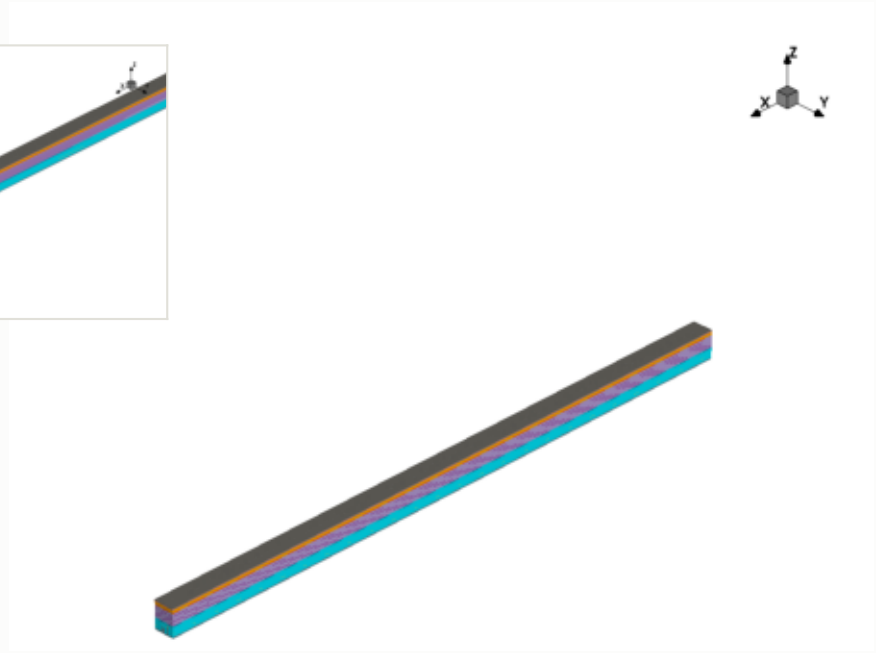
etch_anisotropic · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





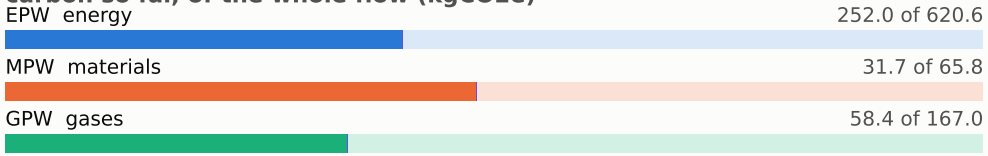
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

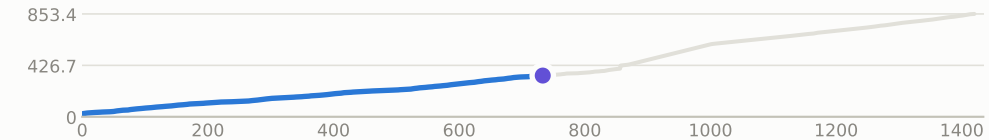
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

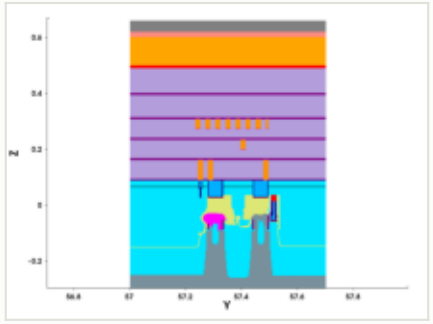
carbon so far, of the whole flow (kgCO2e)



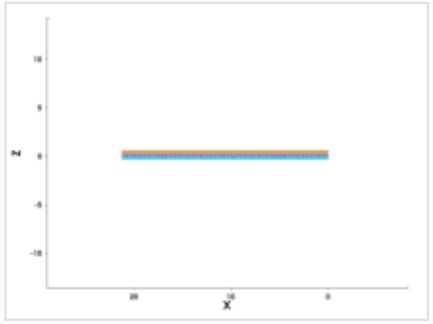
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



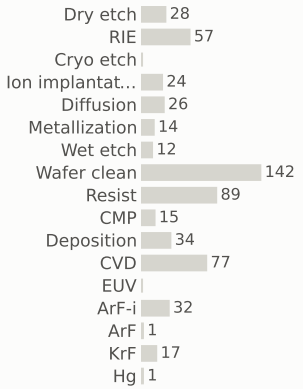
y cut at 57.351 µm



mask for this step

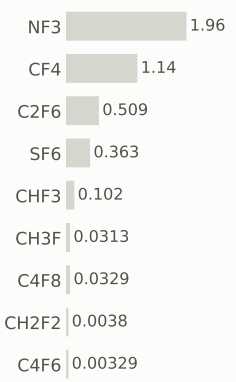
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



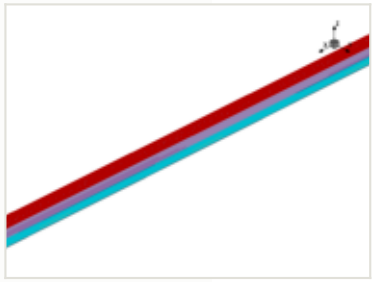
Step #7460: Spx_only

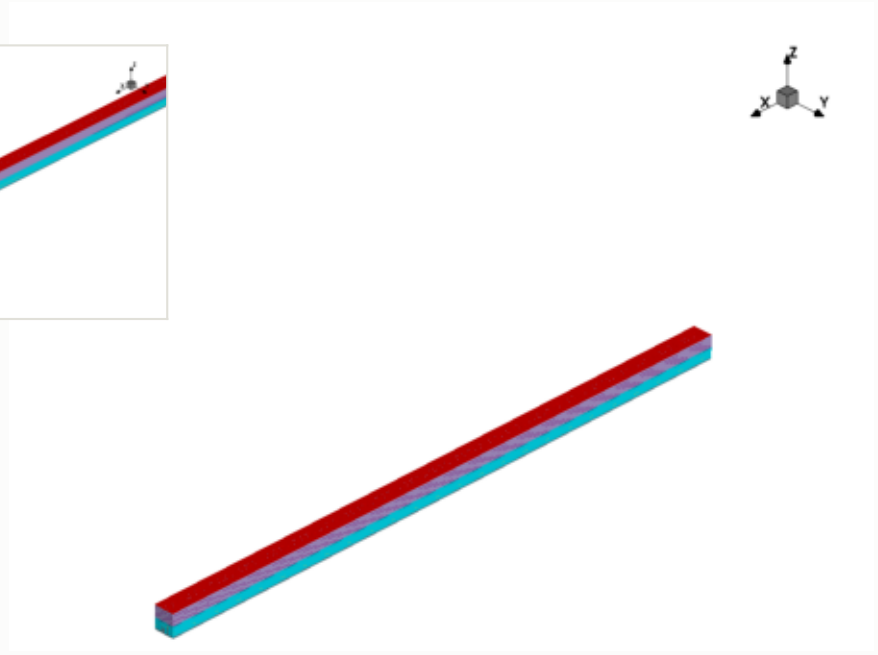
etch_strip · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





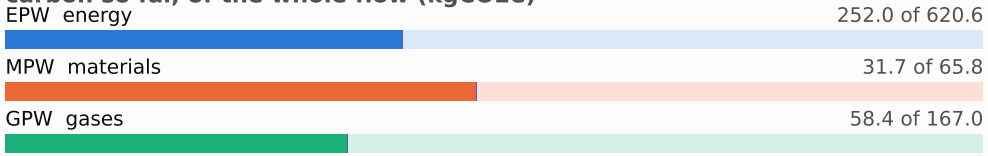
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

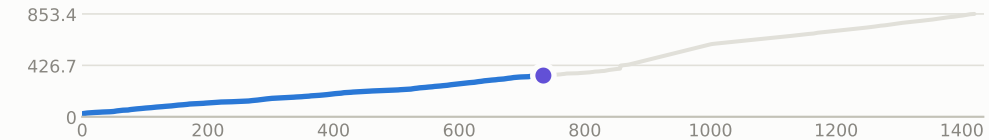
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

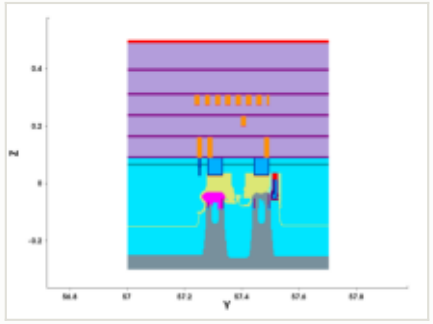
carbon so far, of the whole flow (kgCO2e)



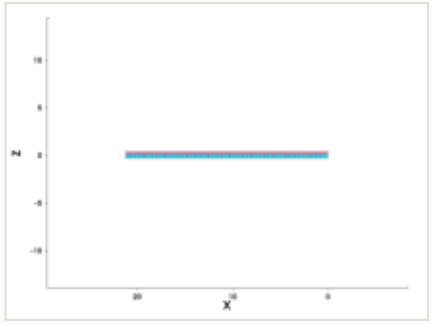
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



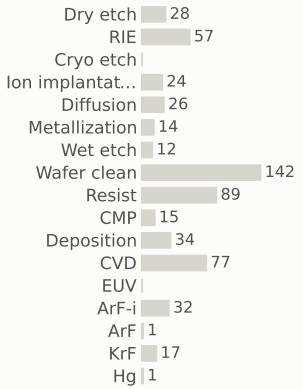
y cut at 57.351 µm



mask for this step

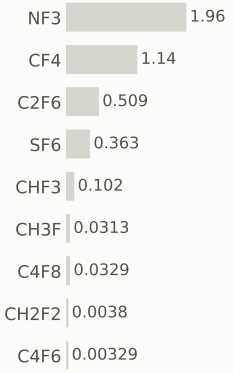
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



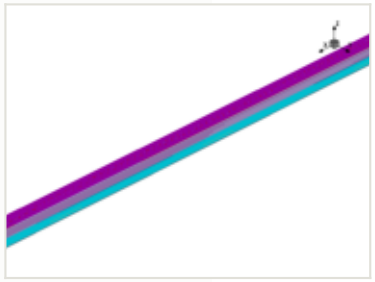
Step #7470: Spx_only

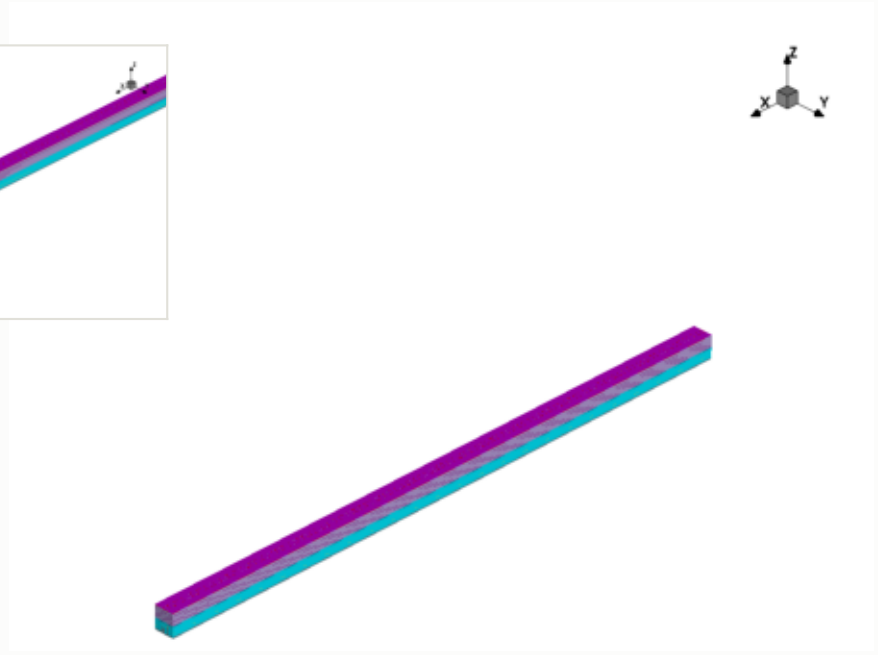
depo_ellipsoidal · Spx_only

4.00e-05	342.2	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



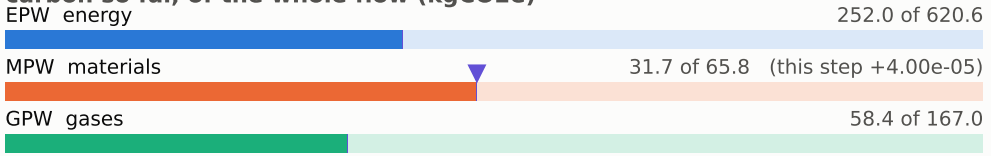


this step

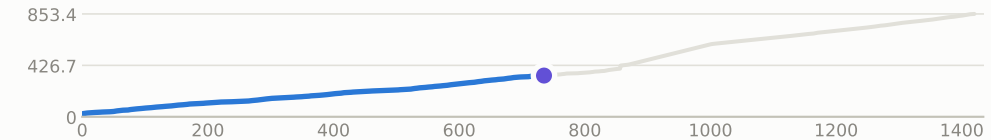
process area	Spx_only	EPW	0 kgCO2e
material	TaN	EPW tool	0 kgCO2e
thickness	0.0016 µm	EPW overhead	0 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**4.00e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

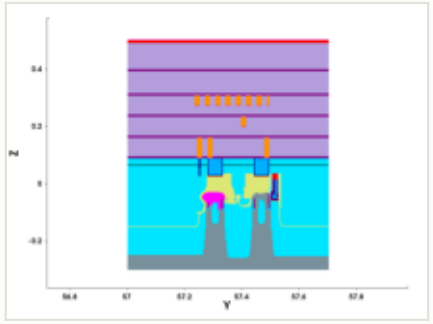
carbon so far, of the whole flow (kgCO2e)



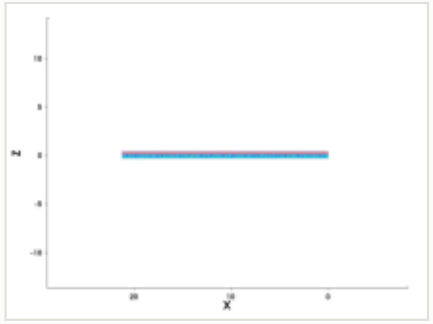
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



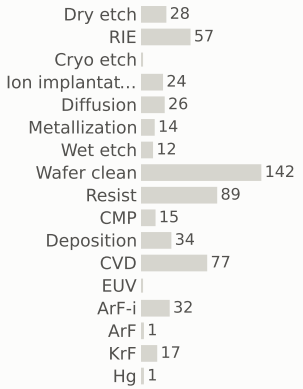
y cut at 57.351 µm



mask for this step

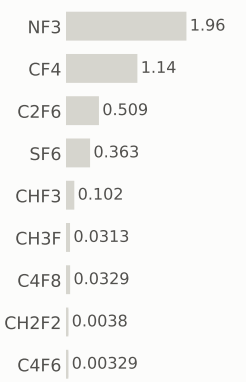
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



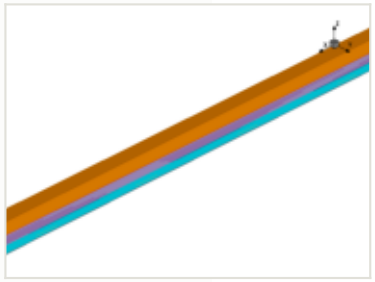
Step #7480: Spx_only

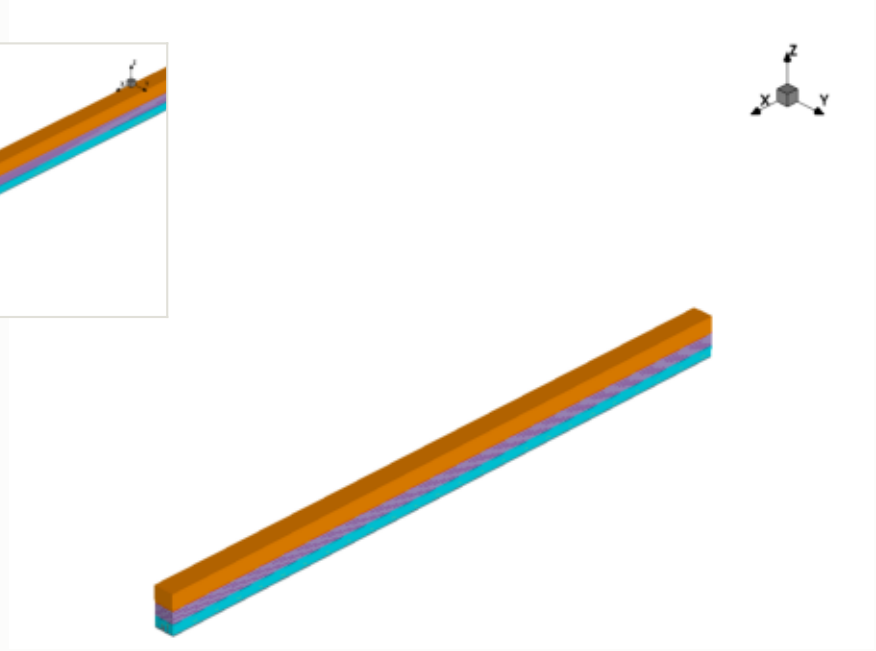
depo_fill · Spx_only

0.00212	342.2	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





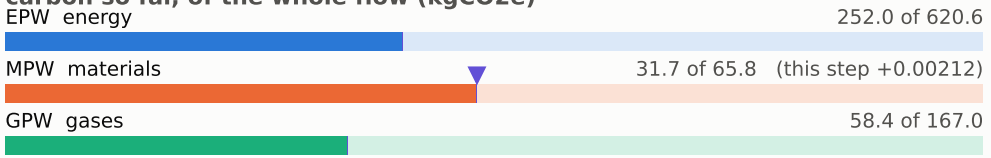
this step

process area	Spx_only
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

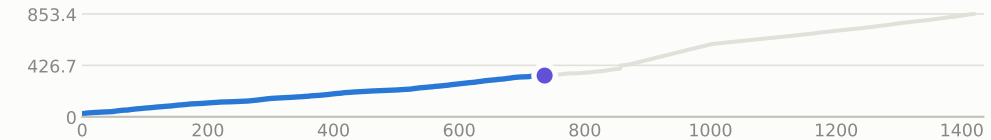
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.00212 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

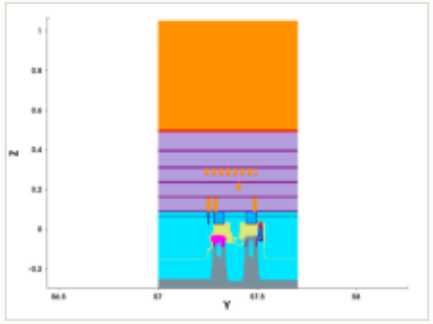
carbon so far, of the whole flow (kgCO2e)



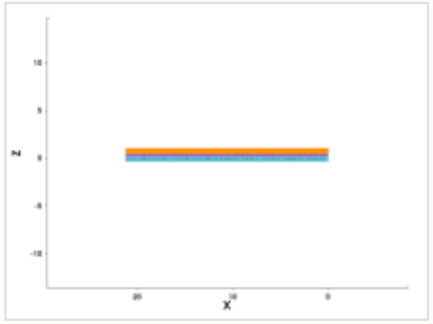
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



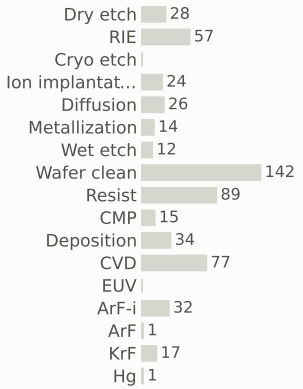
y cut at 57.351 µm



mask for this step

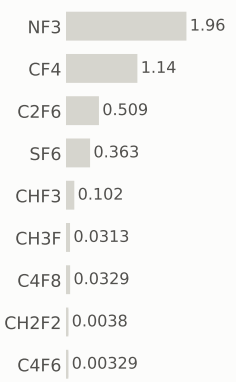
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



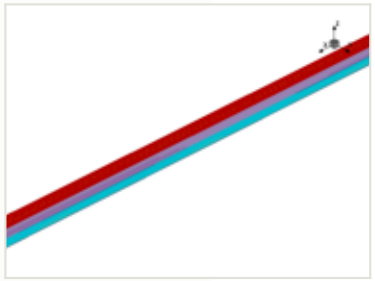
Step #7490: Spx_only

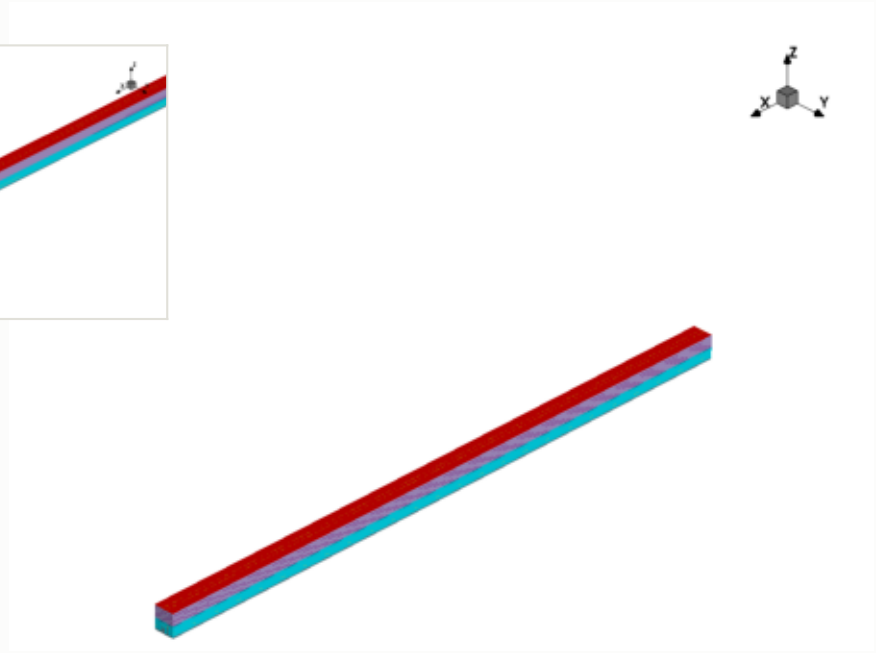
etch_cmp · Spx_only

0	342.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





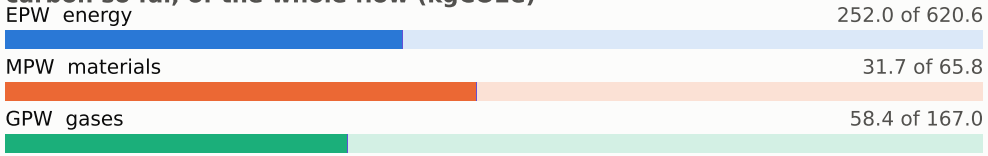
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

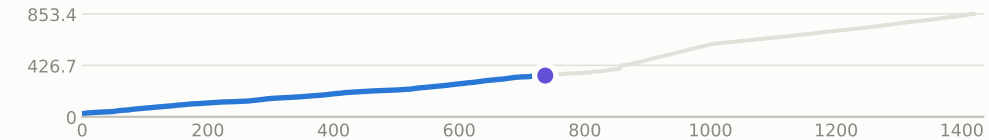
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

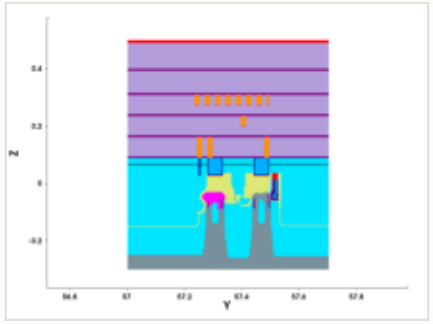
carbon so far, of the whole flow (kgCO2e)



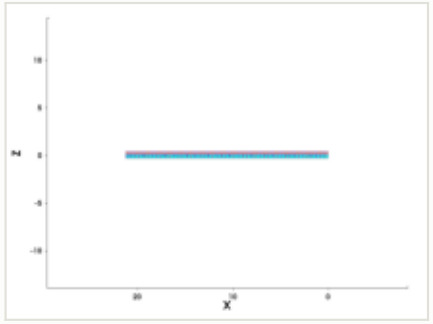
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



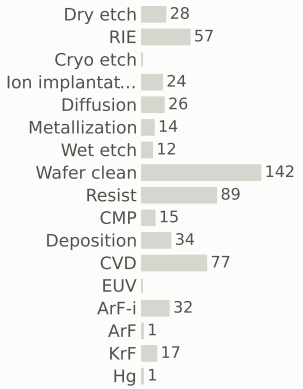
y cut at 57.351 µm



mask for this step

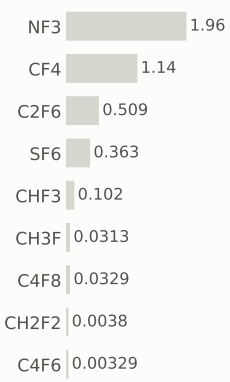
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



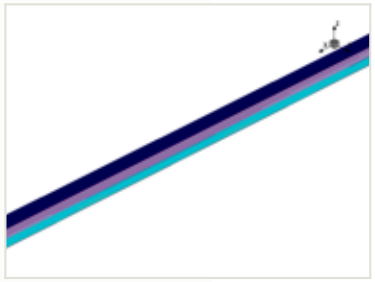
Step #7500: Deposition

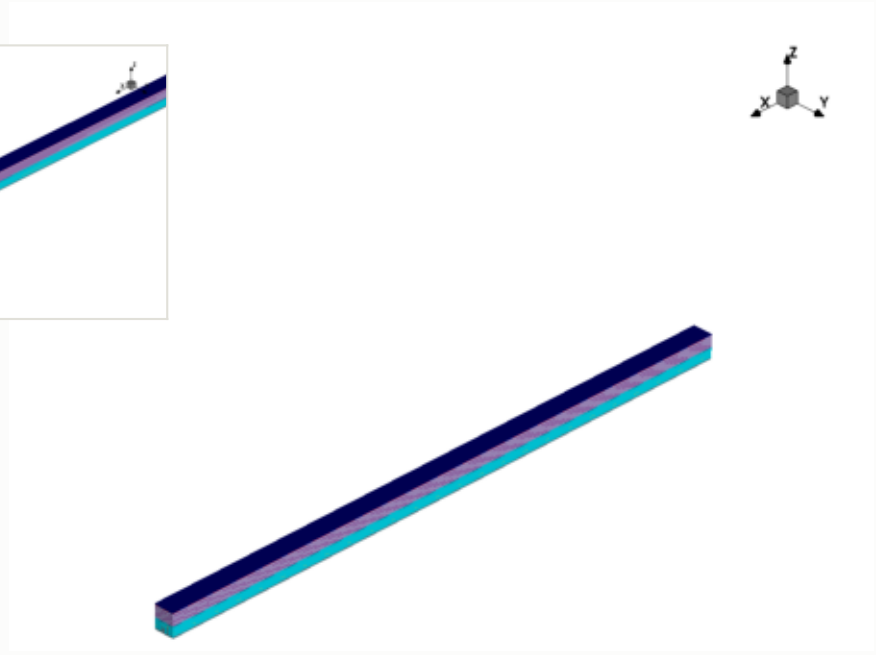
depo_fill · Deposition

0.657	342.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



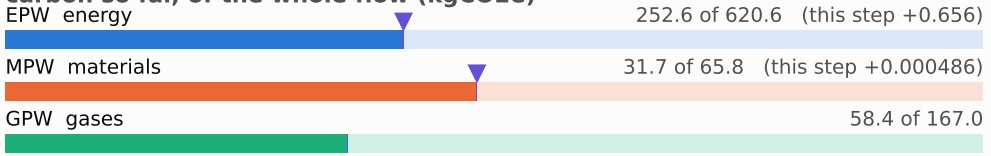


this step

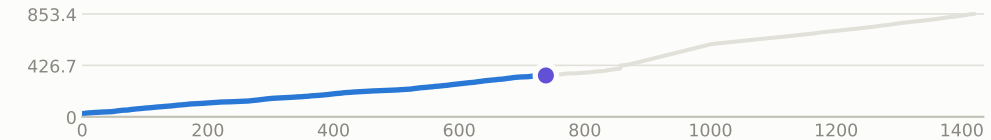
process area	Deposition
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000486 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

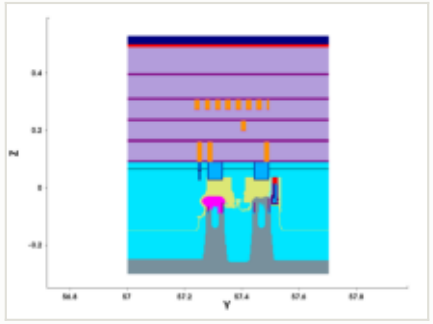
carbon so far, of the whole flow (kgCO2e)



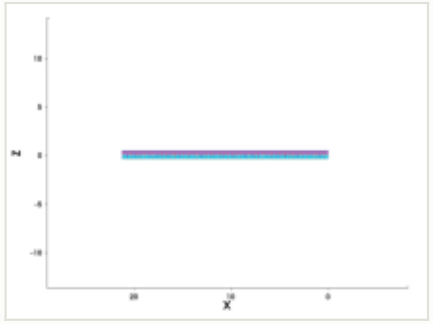
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



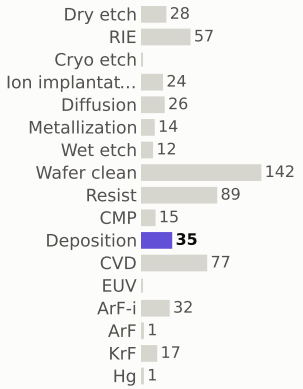
y cut at 57.351 µm



mask for this step

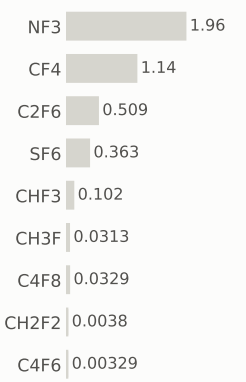
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



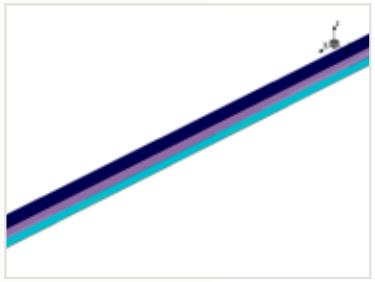
Step #7510: Wafer_clean

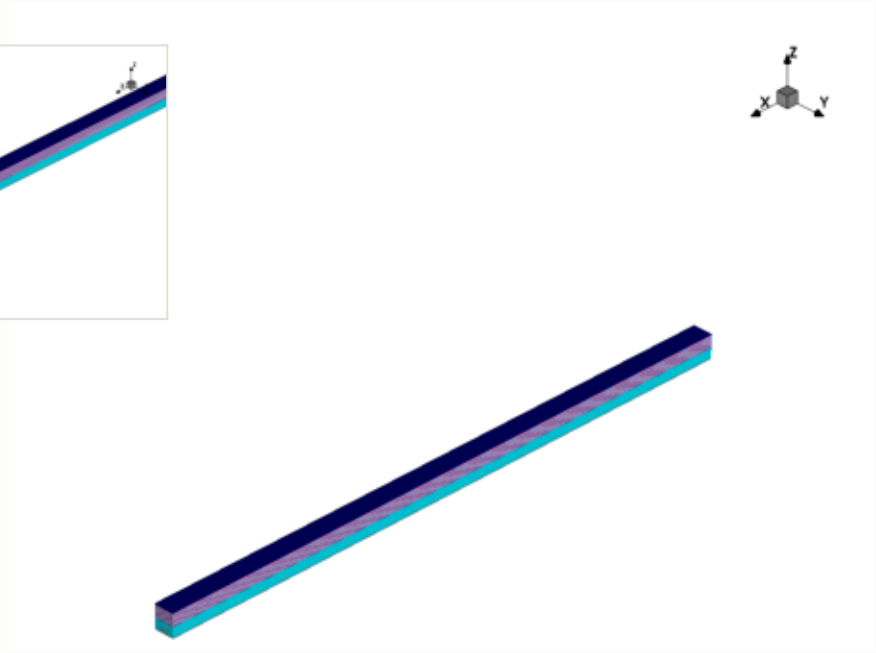
clean · Wafer_clean

0.169	343.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





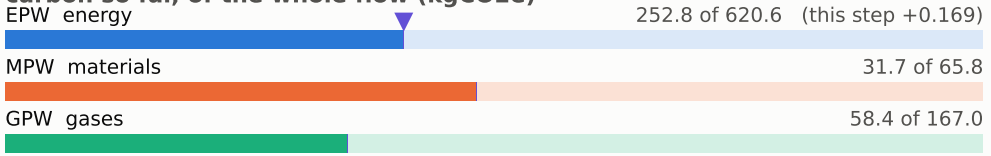
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

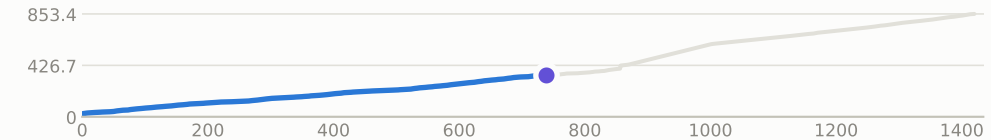
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

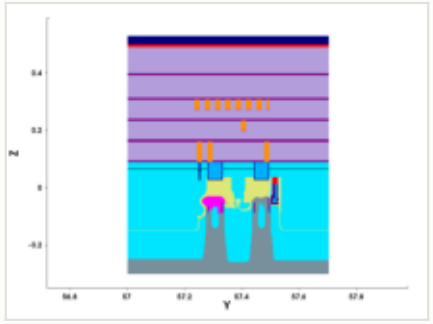
carbon so far, of the whole flow (kgCO2e)



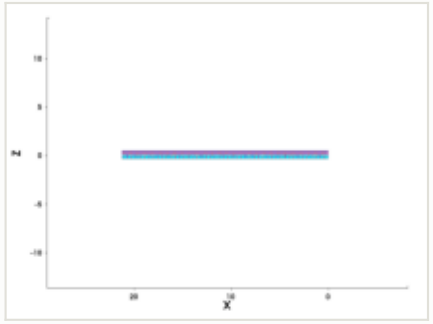
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



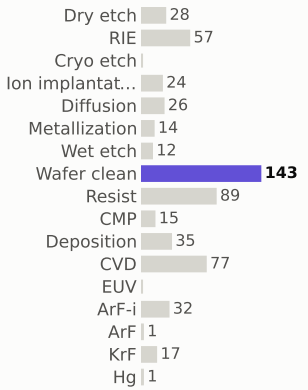
y cut at 57.351 µm



mask for this step

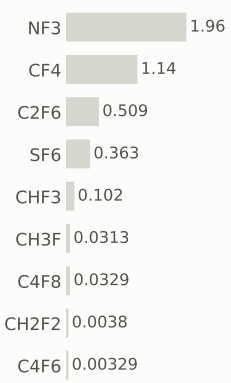
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



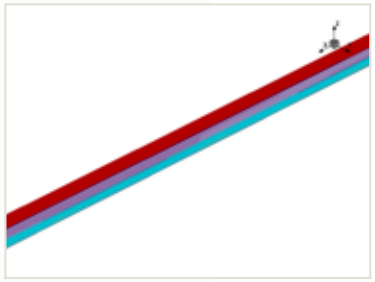
Step #7520: CVD

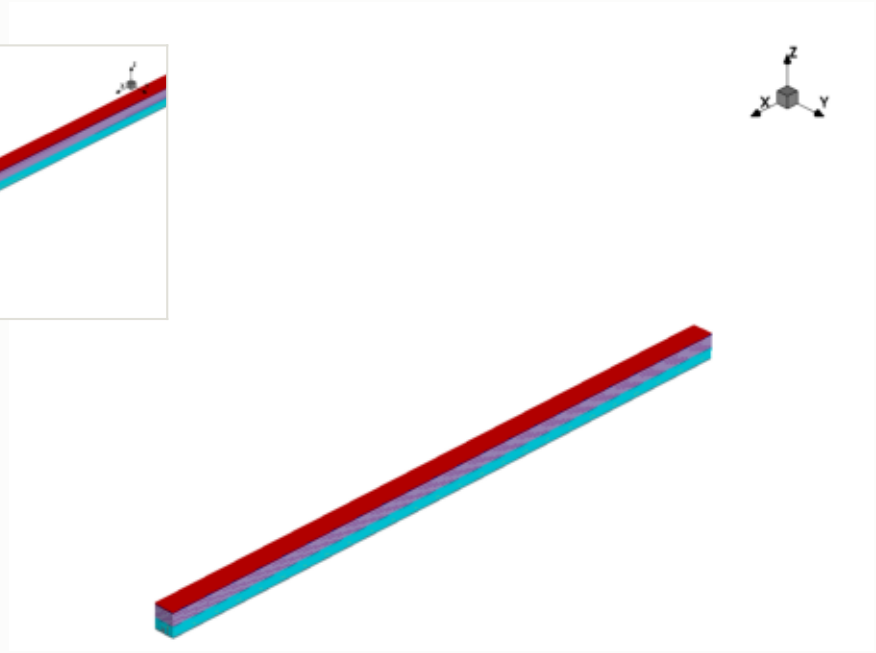
depo_fill · CVD

1.31	344.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



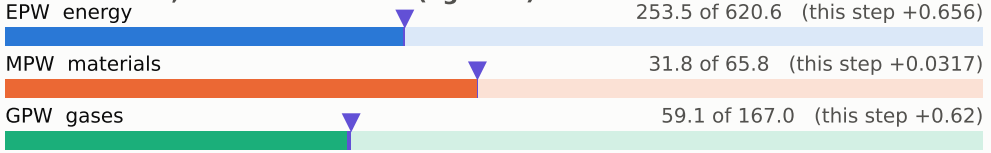


this step

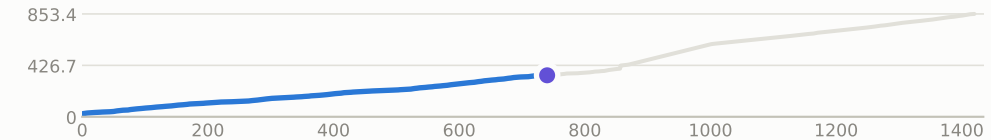
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

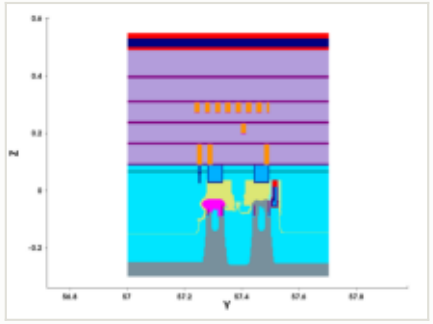
carbon so far, of the whole flow (kgCO2e)



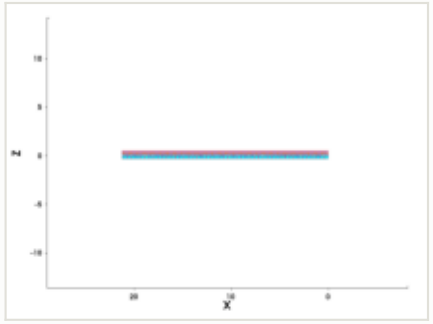
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



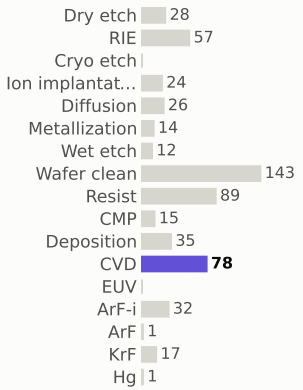
y cut at 57.351 µm



mask for this step

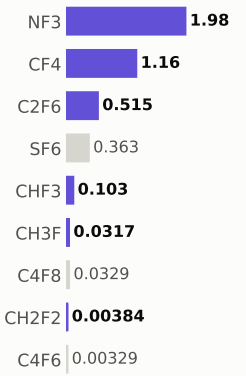
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



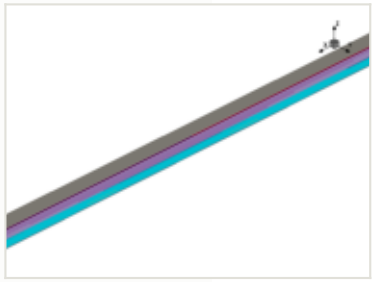
Step #7530: Spx_only

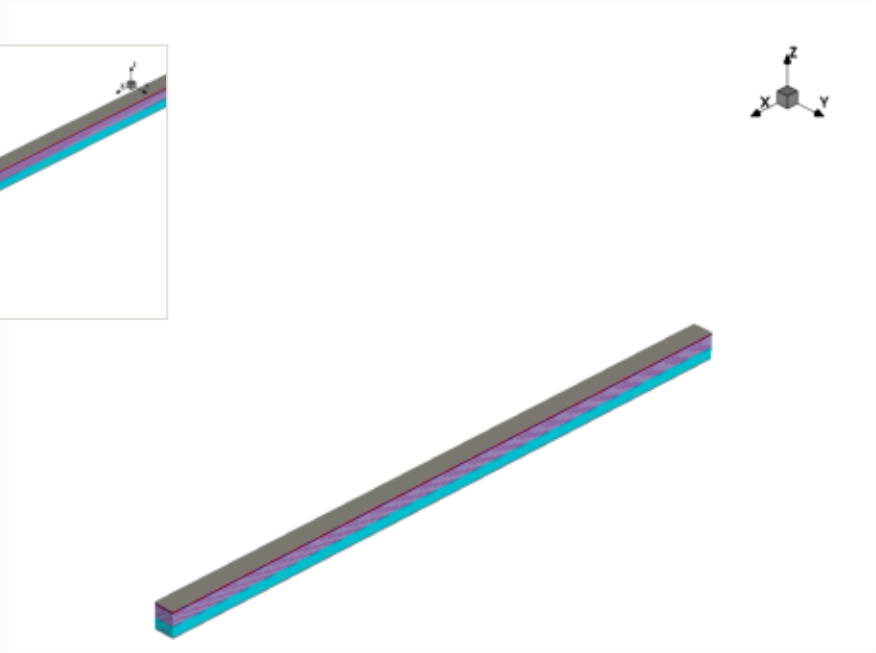
depo_fill · Spx_only

0	344.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





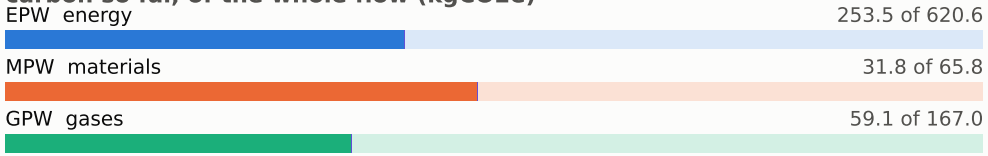
this step

process area	Spx_only
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

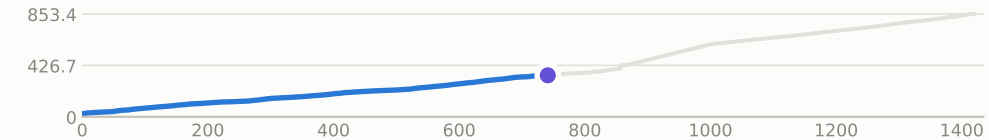
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

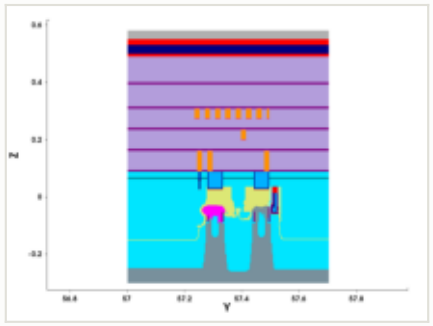
carbon so far, of the whole flow (kgCO2e)



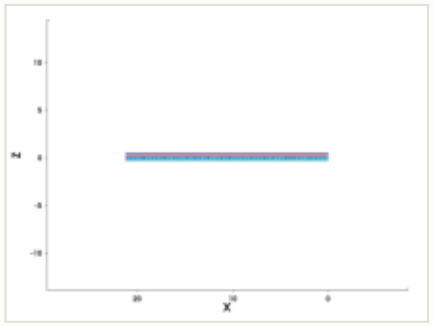
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



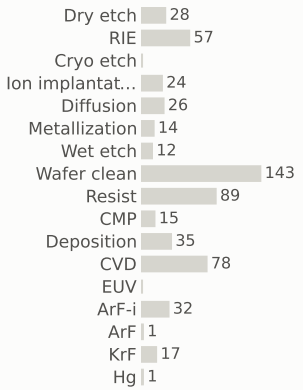
y cut at 57.351 µm



mask for this step

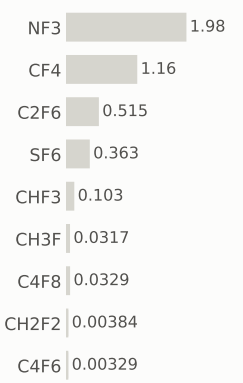
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



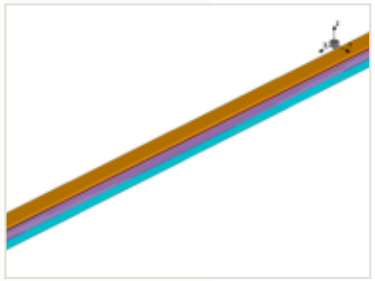
Step #7540: Resist

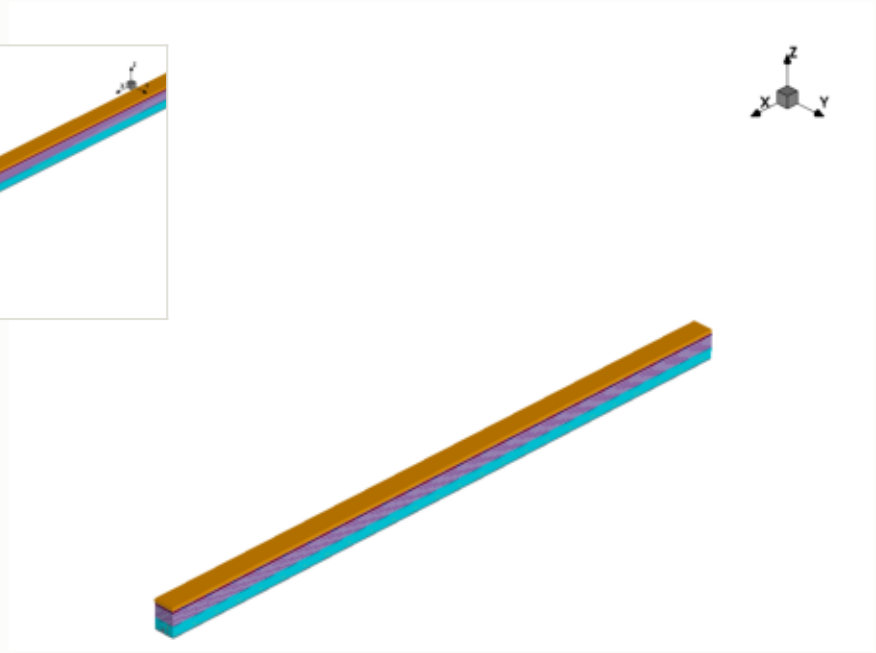
depo_fill · Resist

0.169	344.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



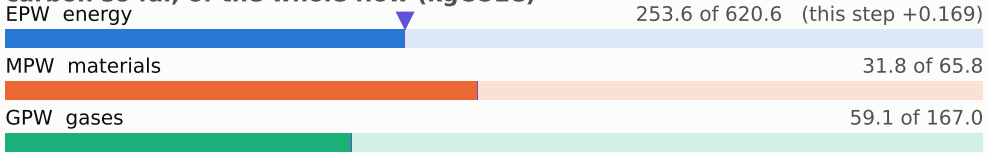


this step

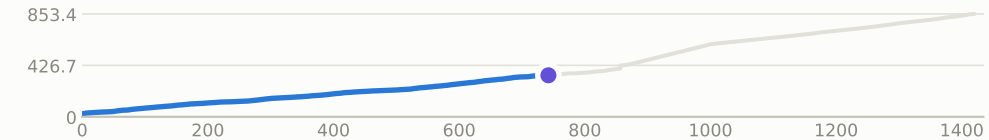
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

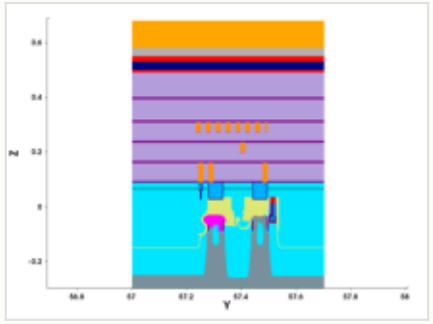
carbon so far, of the whole flow (kgCO2e)



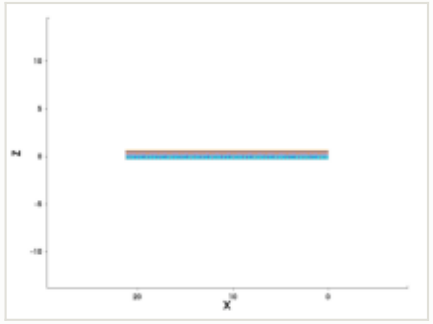
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



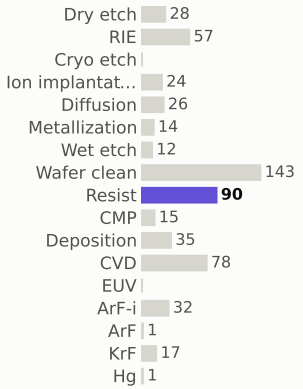
y cut at 57.351 µm



mask for this step

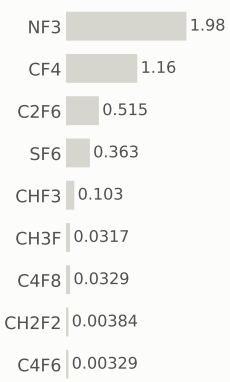
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



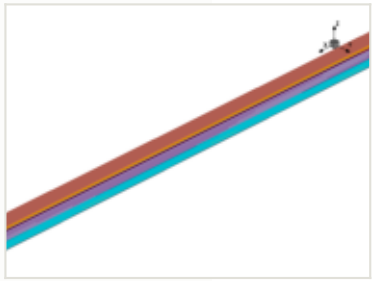
Step #7550: CVD

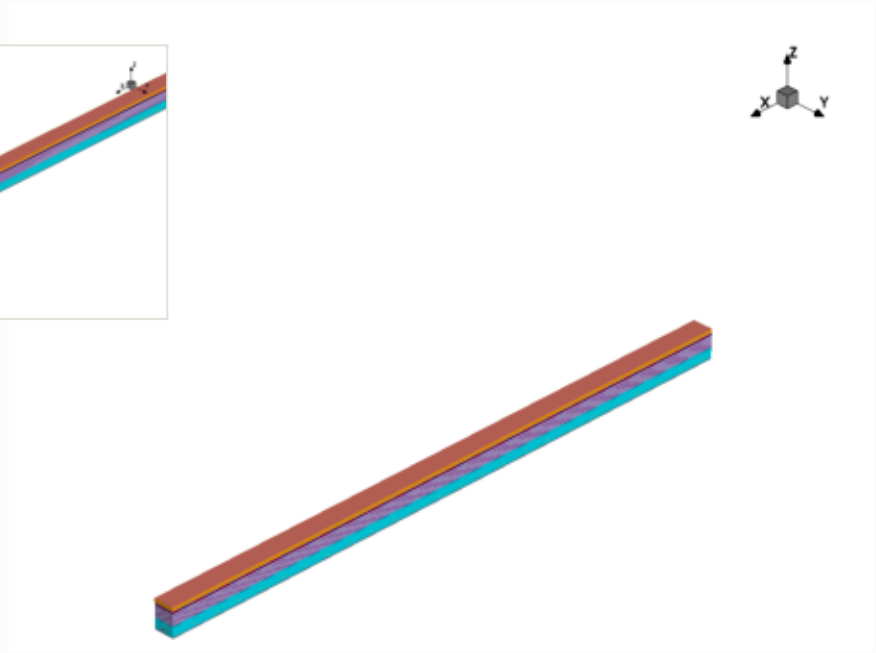
depo_fill · CVD

1.31	345.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



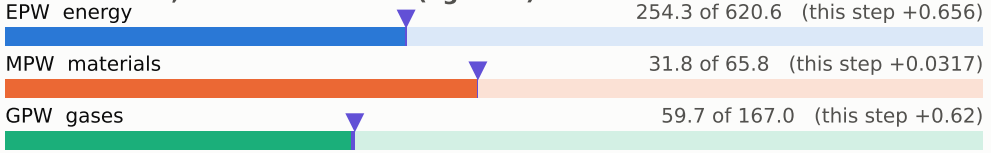


this step

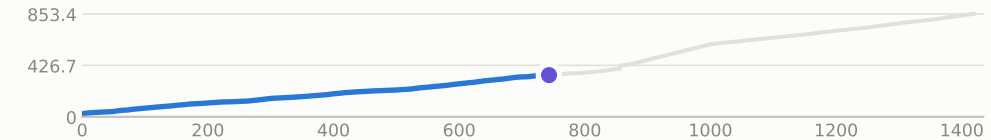
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

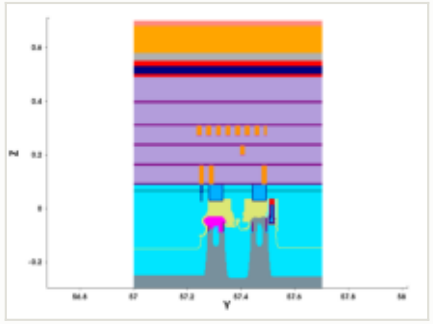
carbon so far, of the whole flow (kgCO2e)



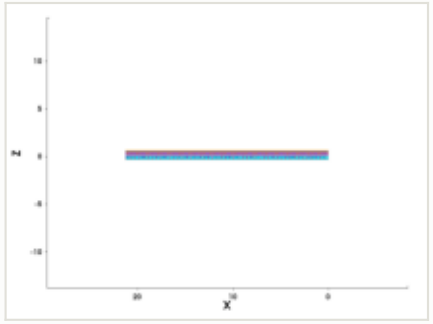
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



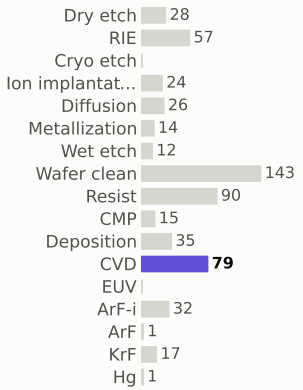
y cut at 57.351 µm



mask for this step

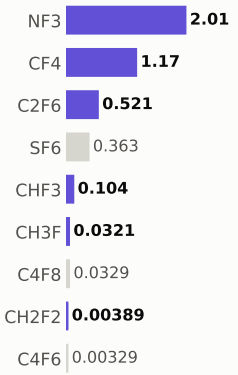
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



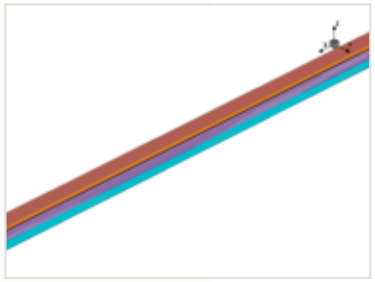
Step #7560: Wafer_clean

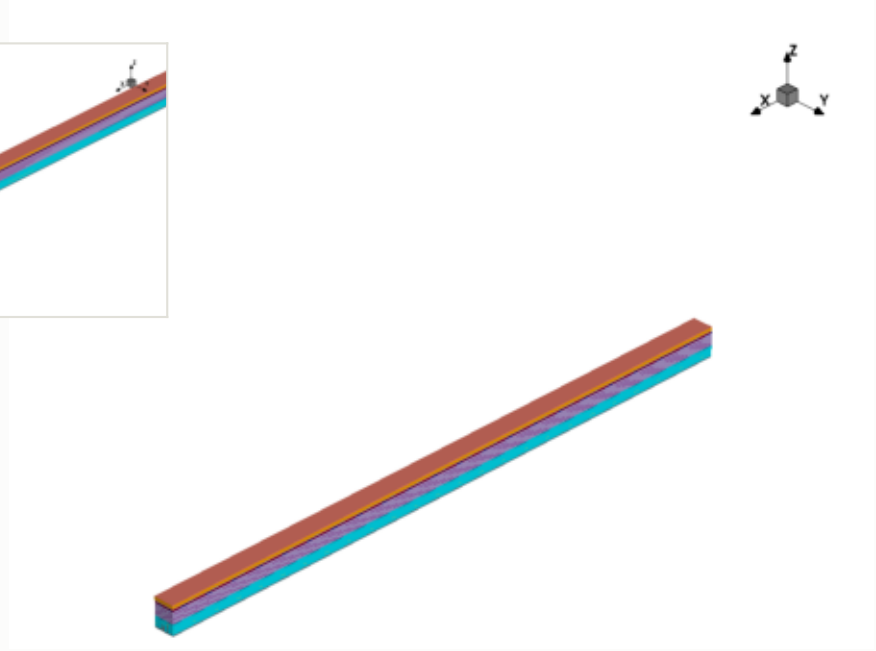
clean · Wafer_clean

0.169	345.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





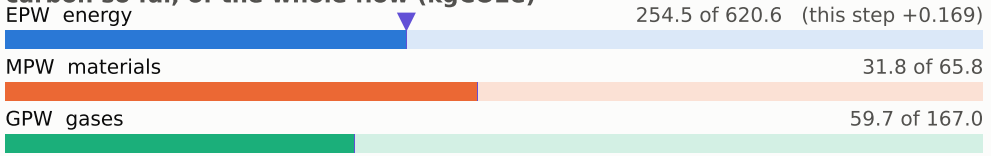
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

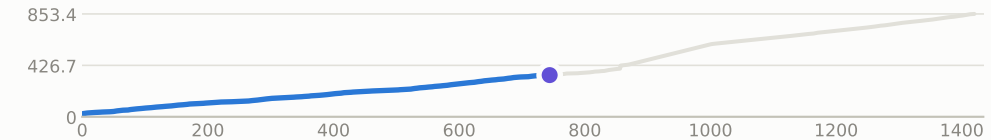
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

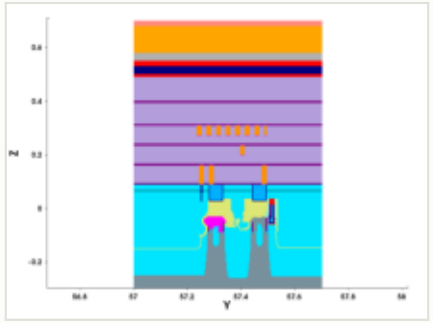
carbon so far, of the whole flow (kgCO2e)



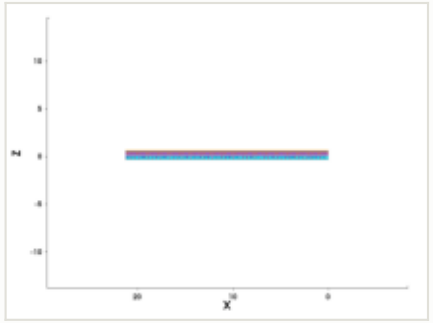
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



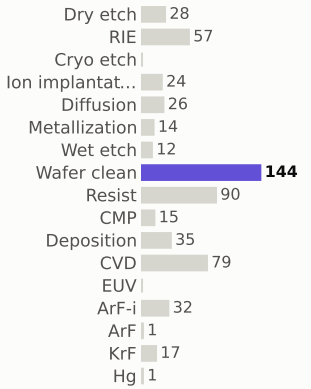
y cut at 57.351 µm



mask for this step

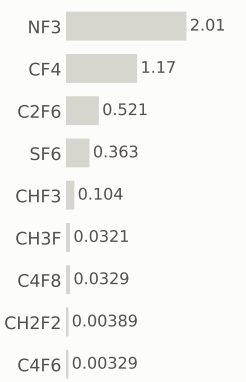
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



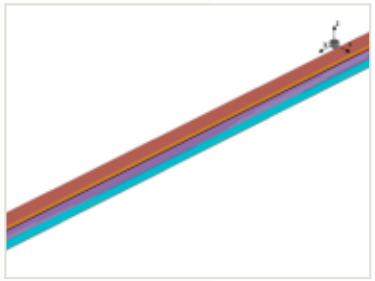
Step #7570: Resist

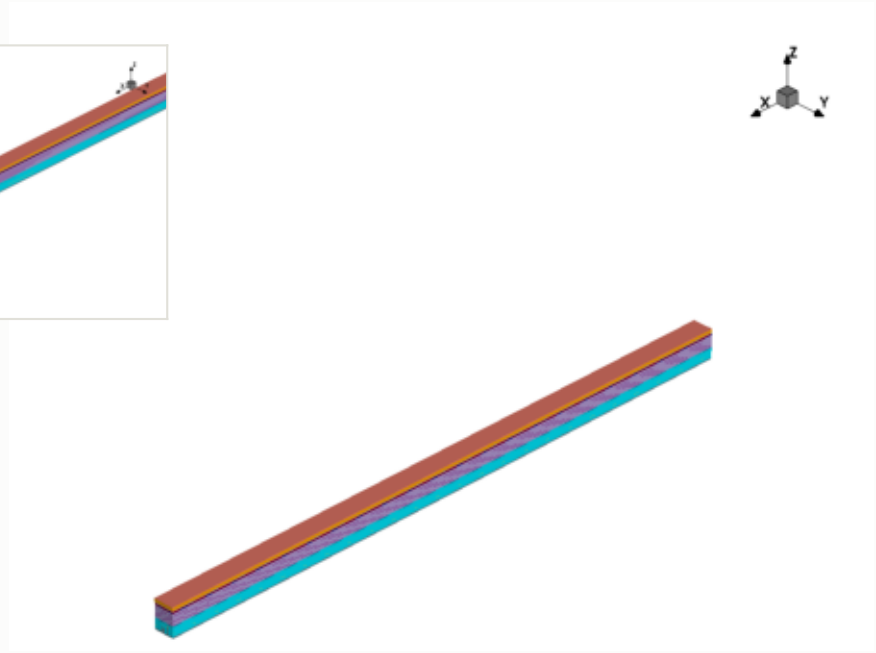
resist · Resist

0.169	346.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





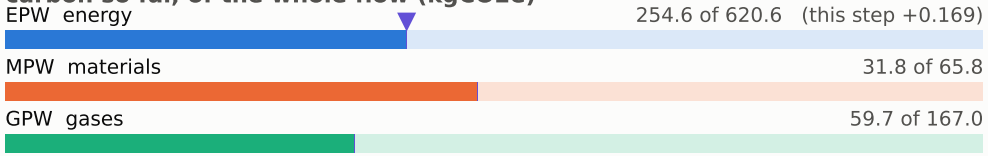
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

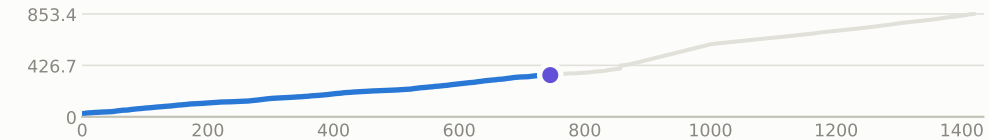
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

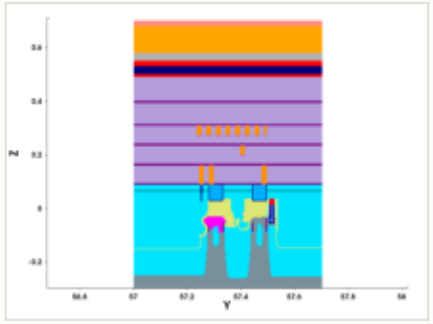
carbon so far, of the whole flow (kgCO2e)



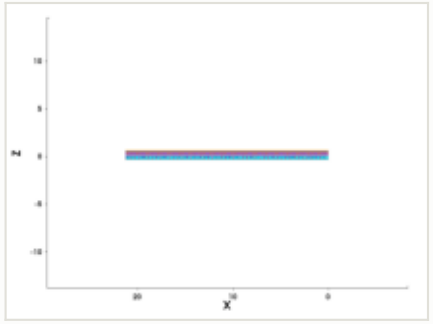
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



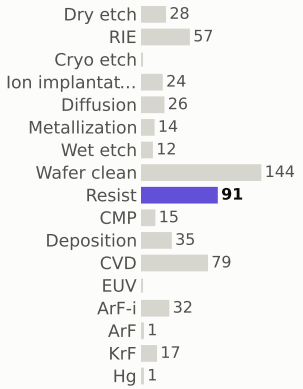
y cut at 57.351 µm



mask for this step

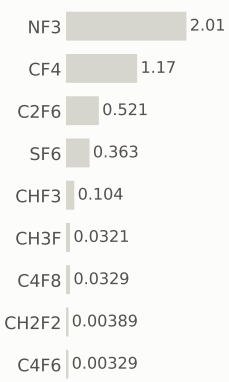
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



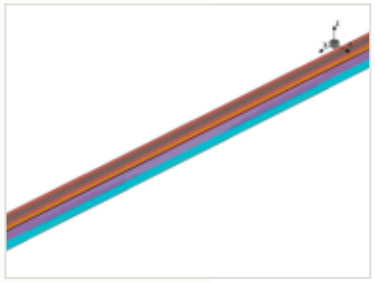
Step #7580: ArF-i

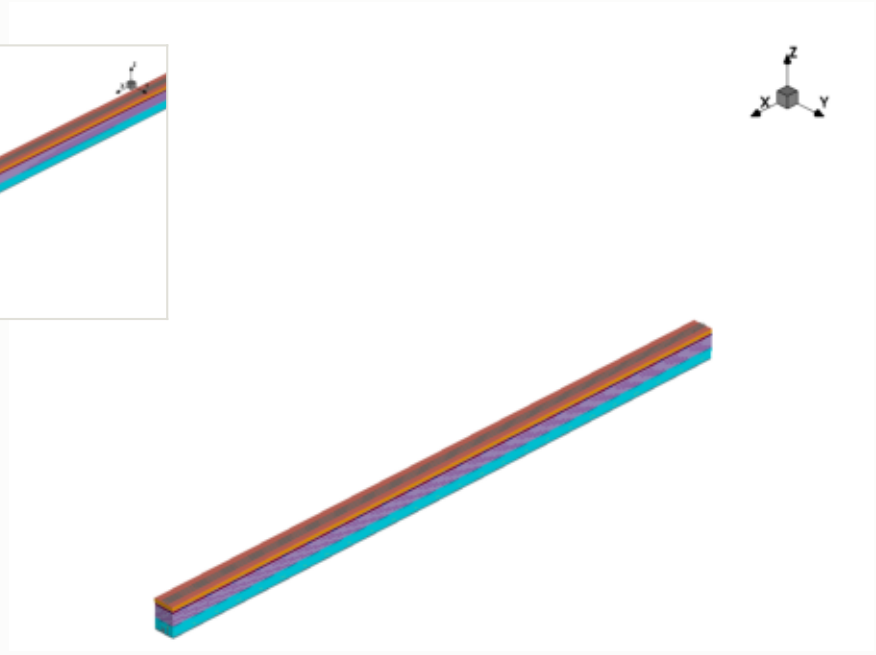
litho_patterning · ArF-i

0.217	346.3	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



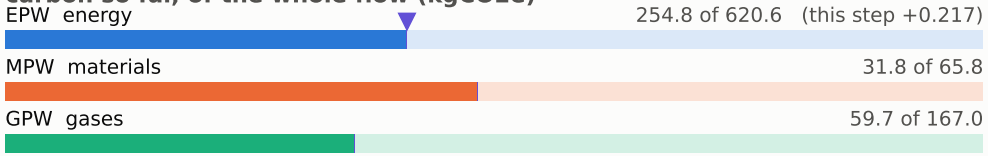


this step

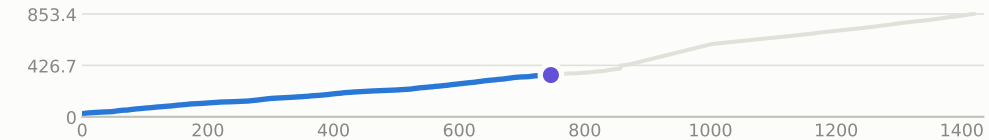
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

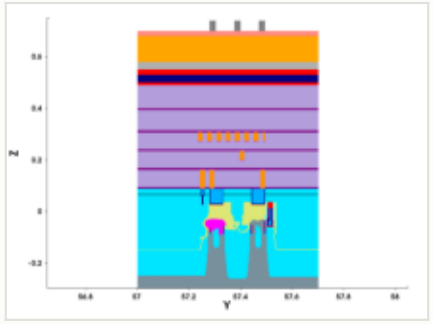
carbon so far, of the whole flow (kgCO2e)



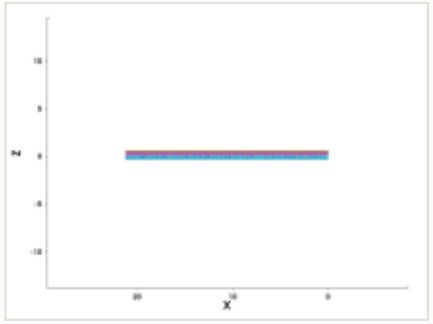
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

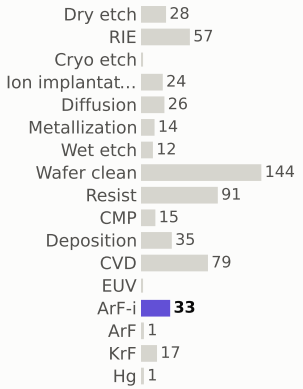


y cut at 57.351 µm

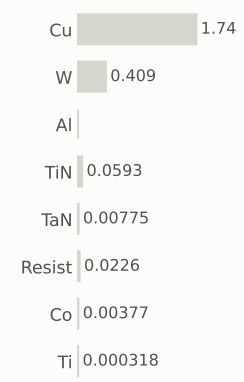


mask: layer 12501:0, positive (GDSII)

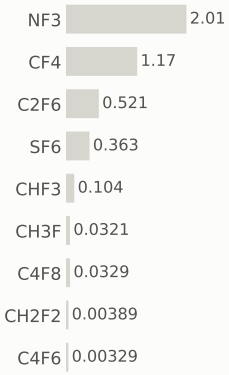
steps by process area



materials deposited (g)



gases emitted (g)



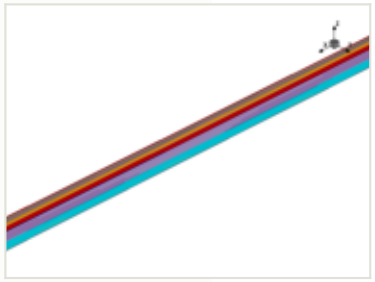
Step #7590: RIE

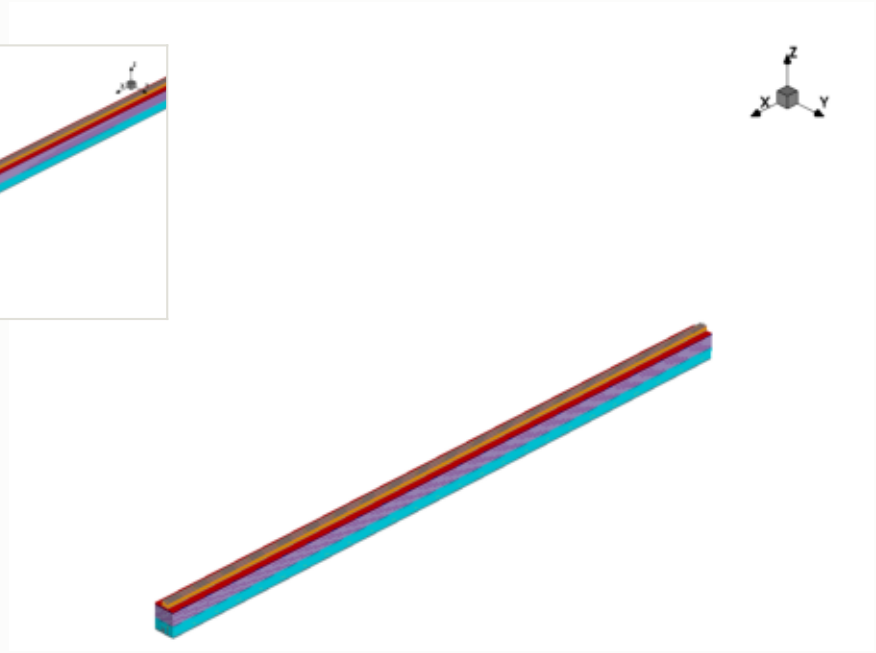
etch_anisotropic · RIE

1.04	347.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



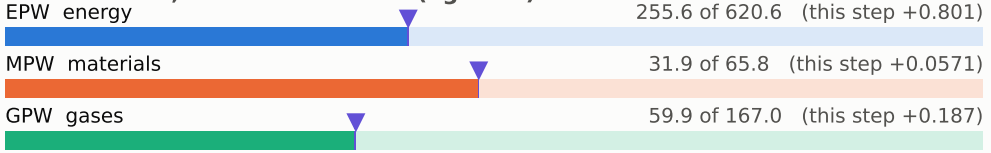


this step

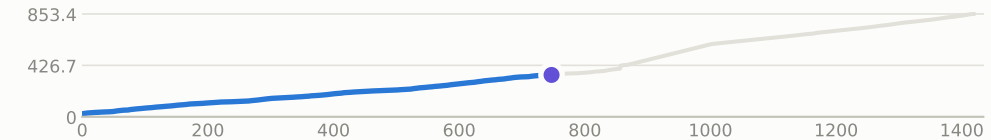
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

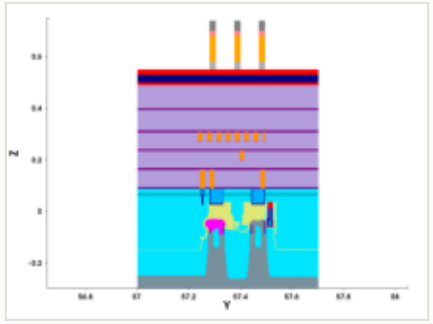
carbon so far, of the whole flow (kgCO2e)



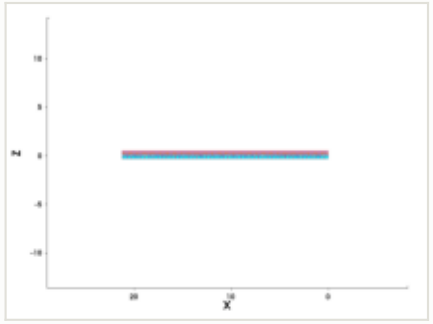
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



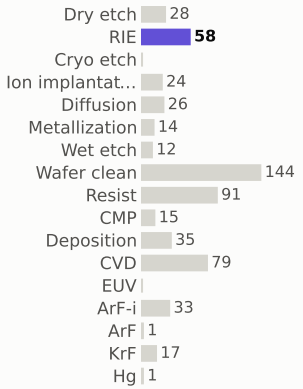
y cut at 57.351 µm



mask for this step

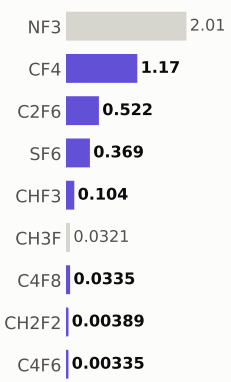
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



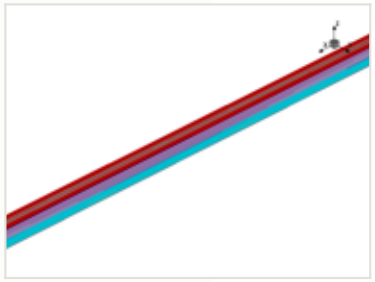
Step #7600: Spx_only

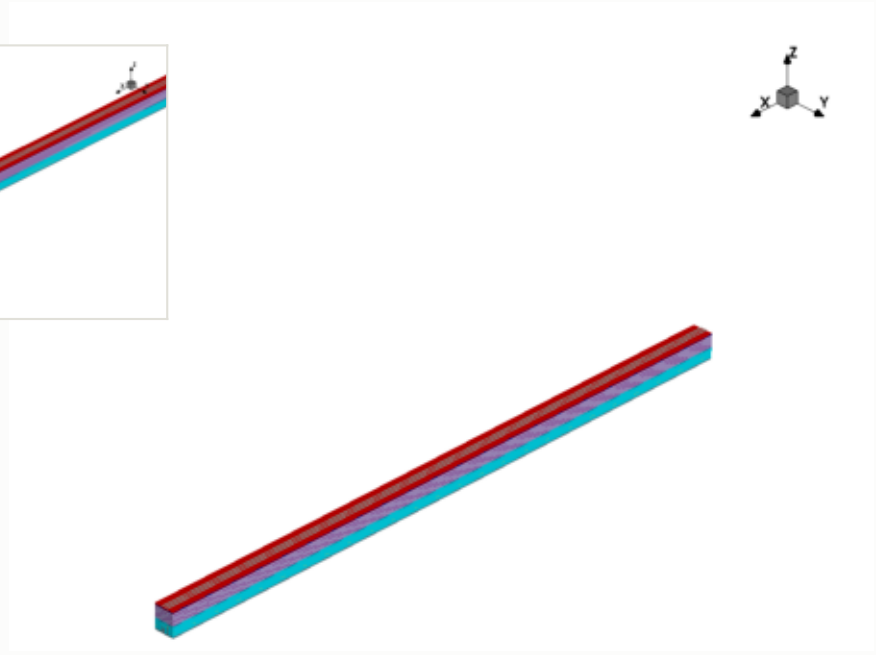
etch_strip · Spx_only

0	347.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

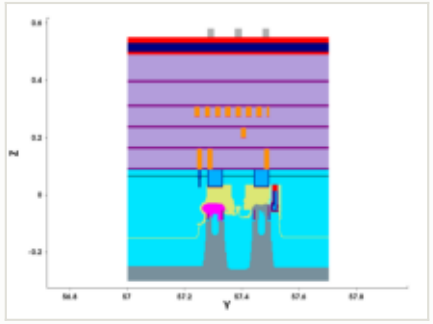
3D structure after this step

zoomed to fit

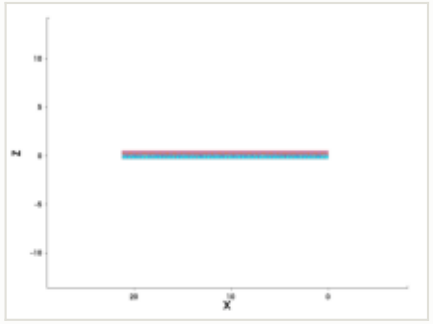




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

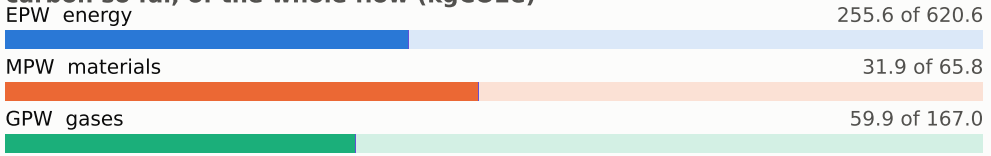
no mask for this step

this step

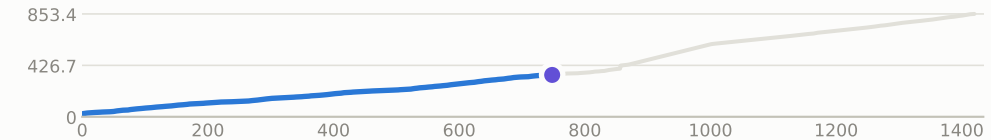
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

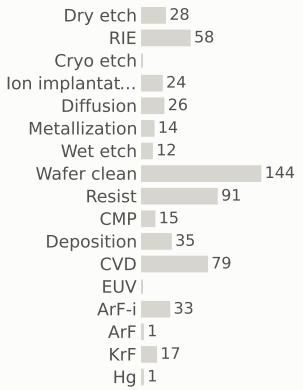
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

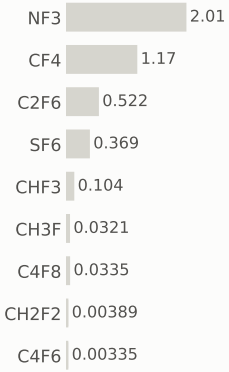


steps by process area



materials deposited (g)

gases emitted (g)



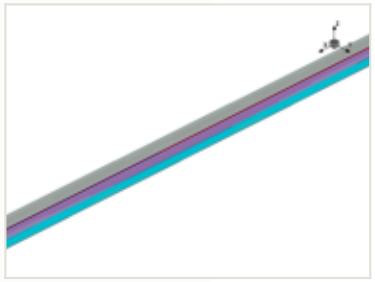
Step #7610: Spx_only

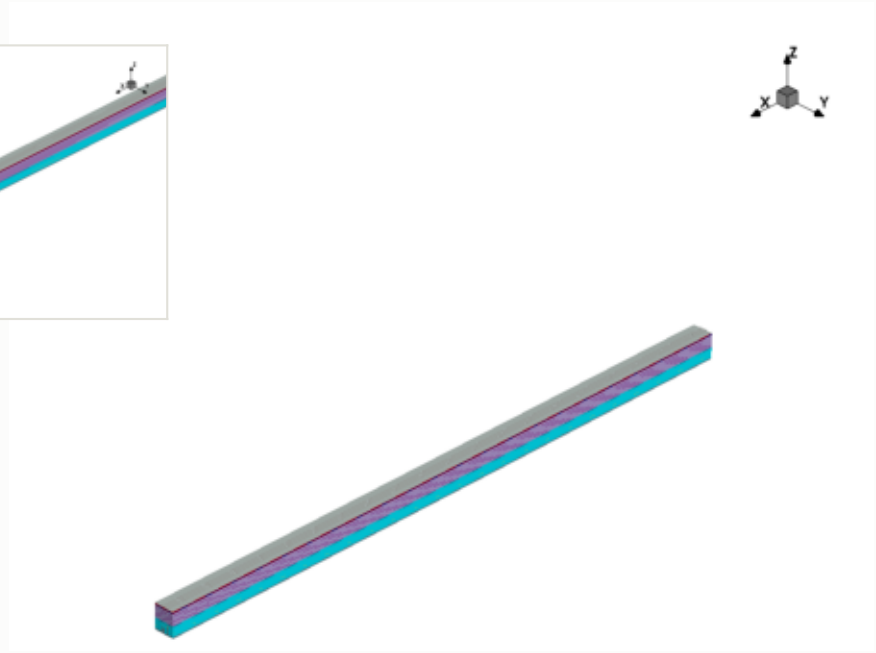
depo_isotropic · Spx_only

0	347.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





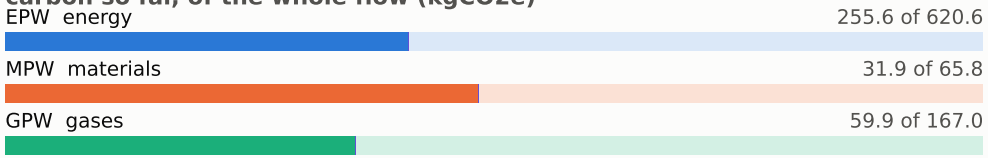
this step

process area	Spx_only
material	—
thickness	0.001 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

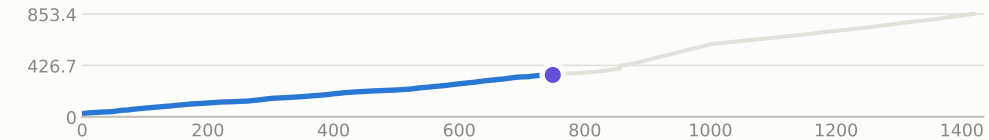
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

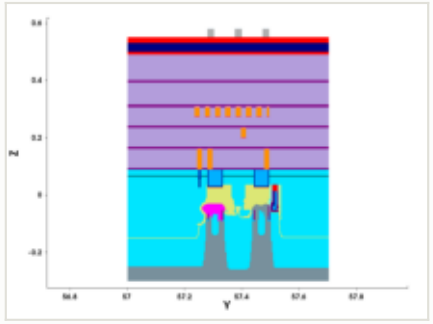
carbon so far, of the whole flow (kgCO2e)



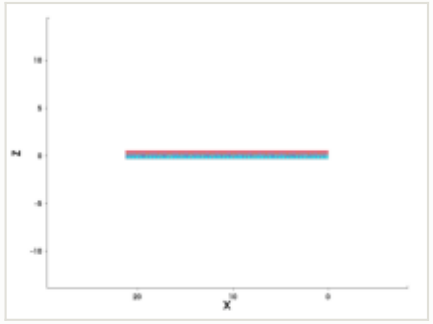
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



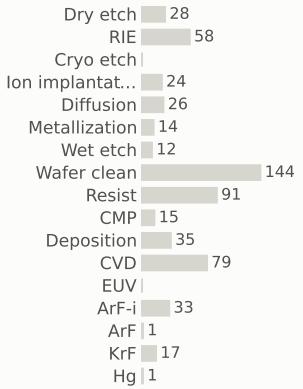
y cut at 57.351 µm



mask for this step

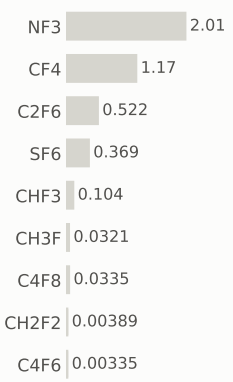
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



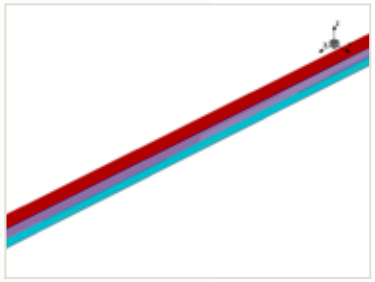
Step #7620: Spx_only

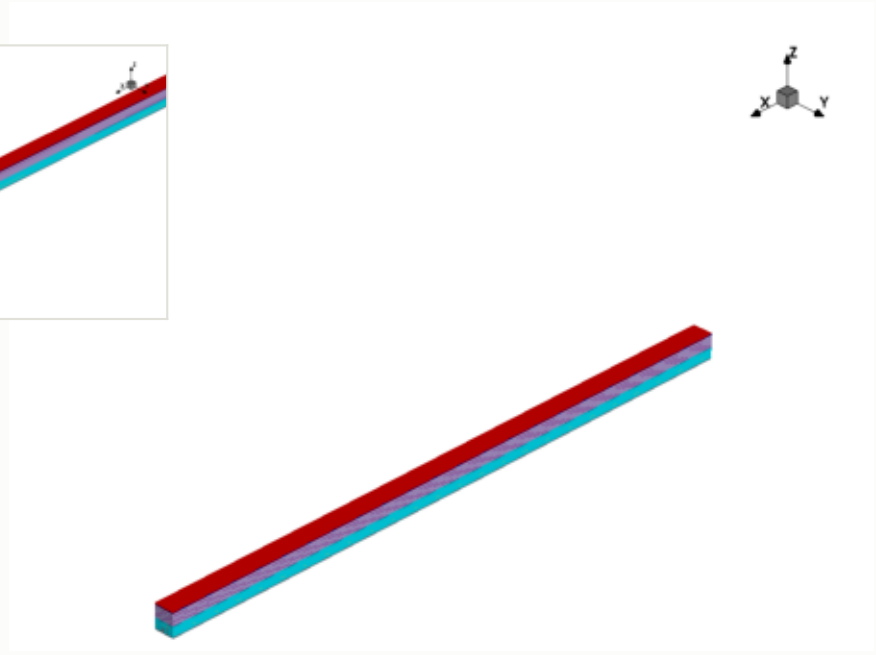
etch_anisotropic · Spx_only

0	347.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





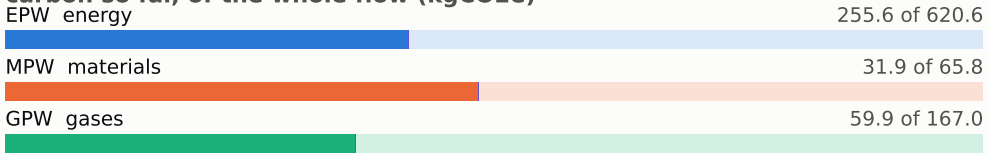
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

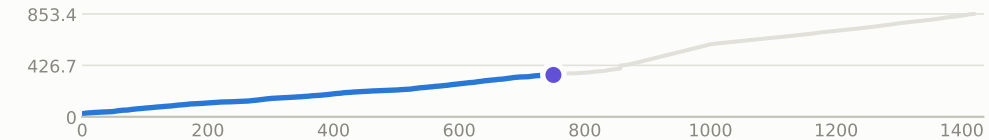
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

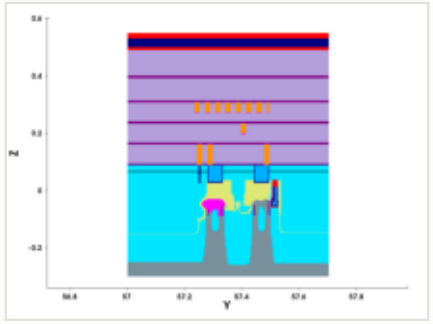
carbon so far, of the whole flow (kgCO2e)



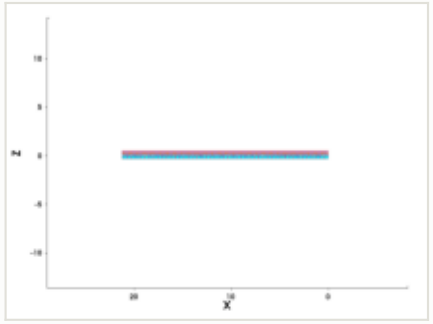
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



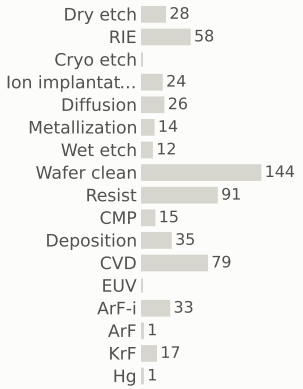
y cut at 57.351 µm



mask for this step

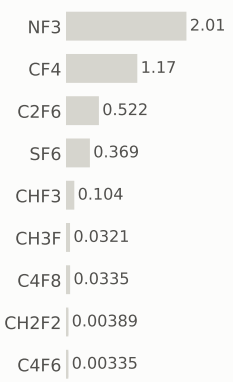
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



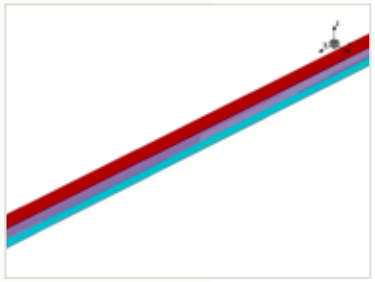
Step #7630: Spx_only

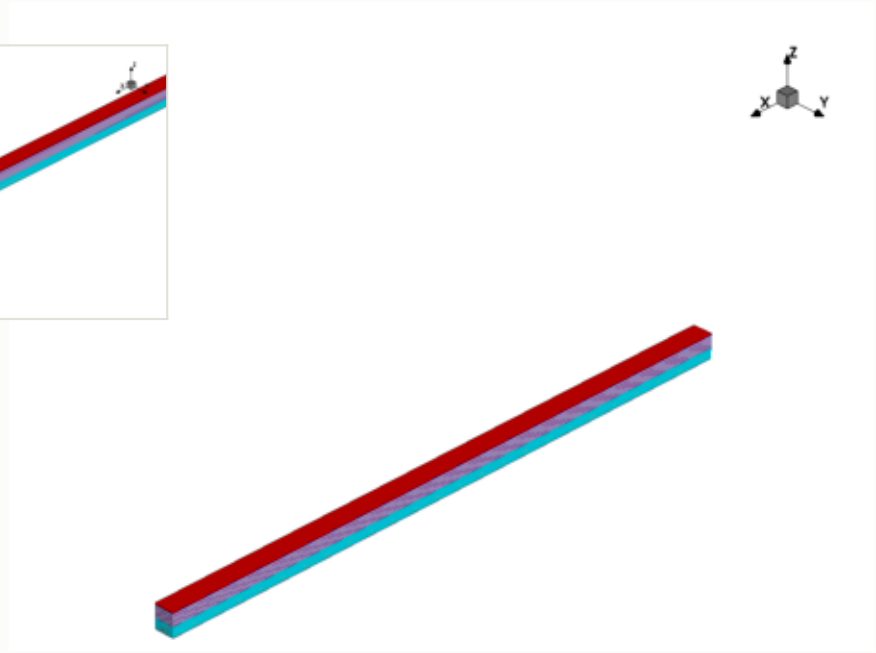
etch_strip · Spx_only

0	347.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





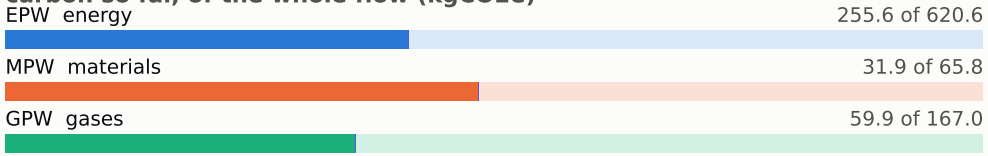
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

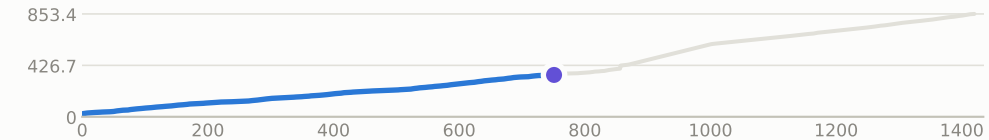
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

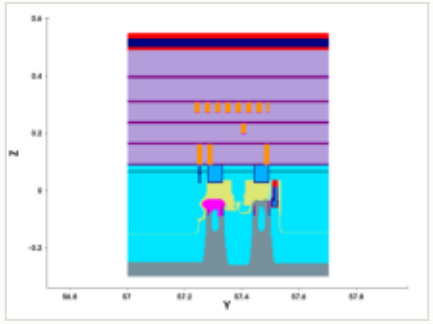
carbon so far, of the whole flow (kgCO2e)



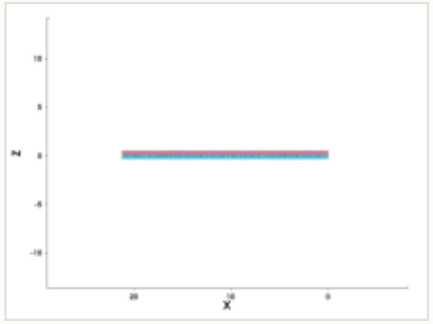
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



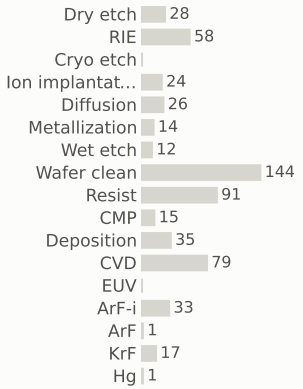
y cut at 57.351 µm



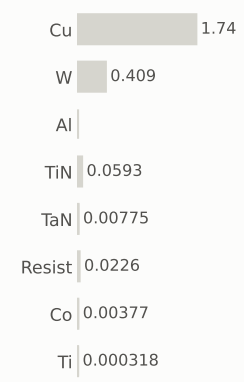
mask for this step

no mask for this step

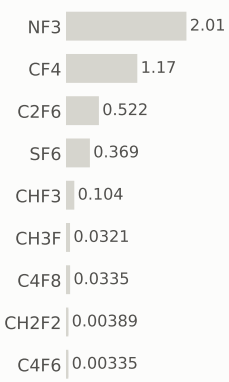
steps by process area



materials deposited (g)



gases emitted (g)



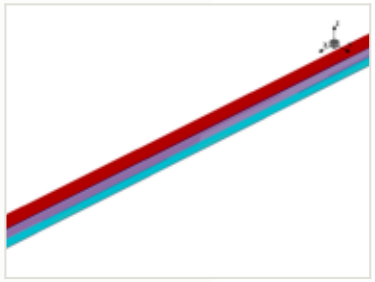
Step #7640: Wafer_clean

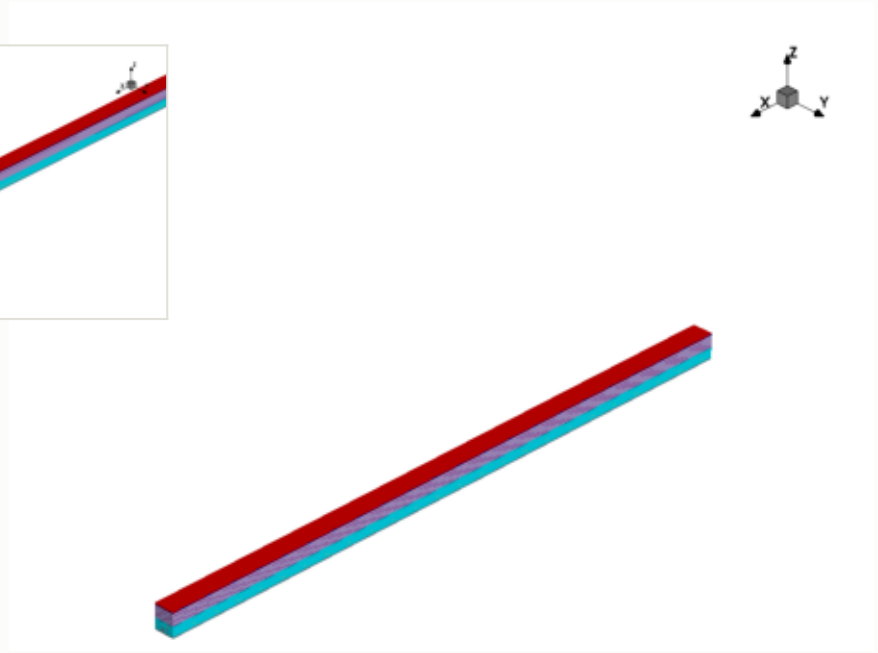
clean · Wafer_clean

0.169	347.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





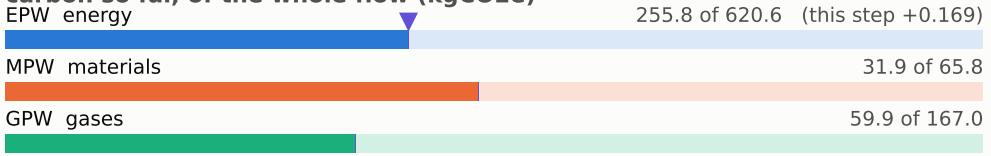
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

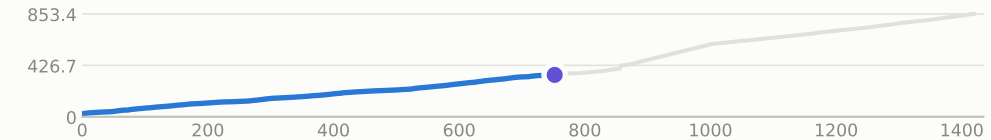
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

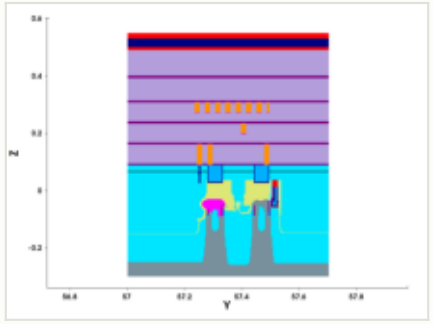
carbon so far, of the whole flow (kgCO2e)



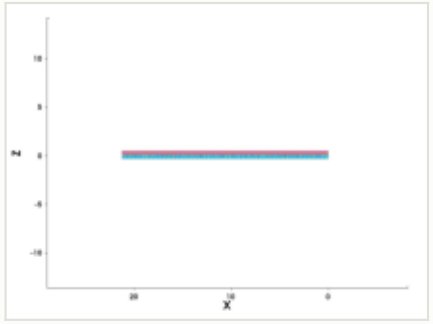
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



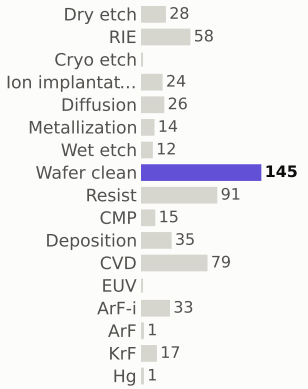
y cut at 57.351 µm



mask for this step

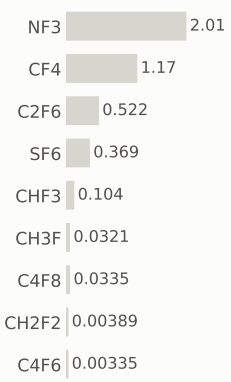
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



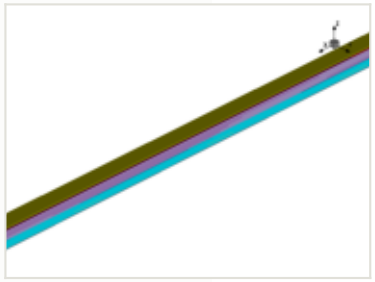
Step #7650: Resist

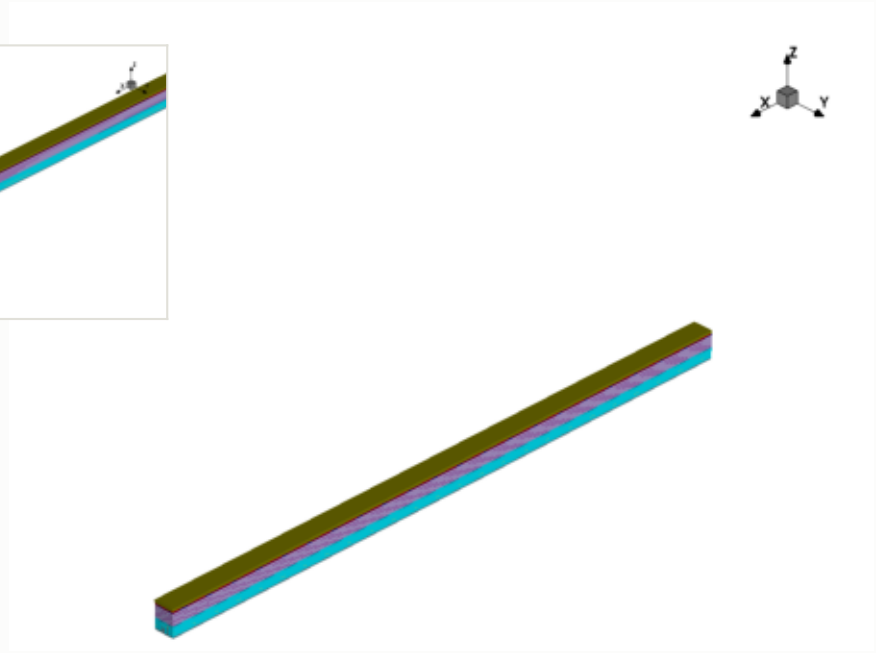
depo_fill · Resist

0.169	347.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



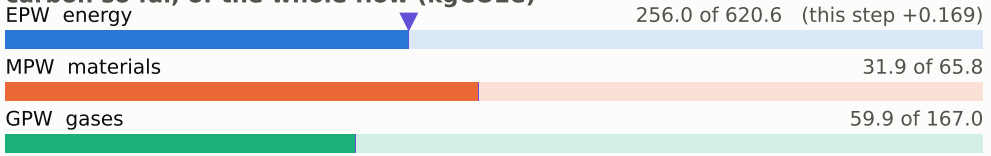


this step

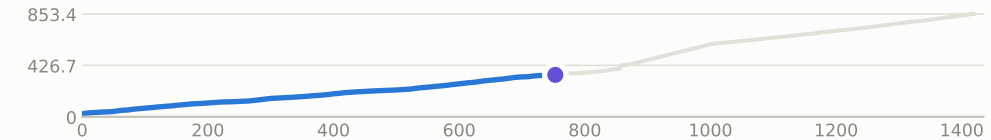
process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

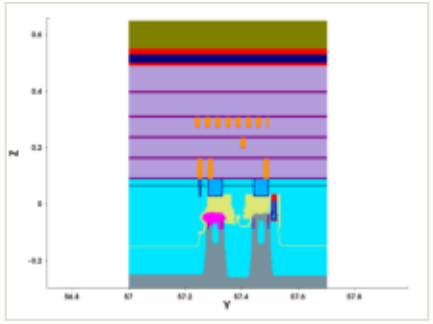
carbon so far, of the whole flow (kgCO2e)



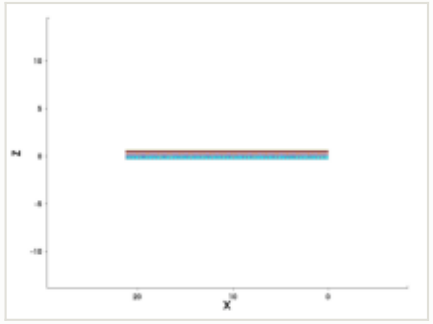
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



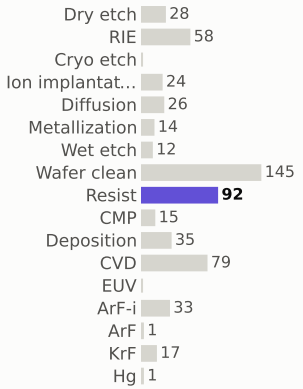
y cut at 57.351 µm



mask for this step

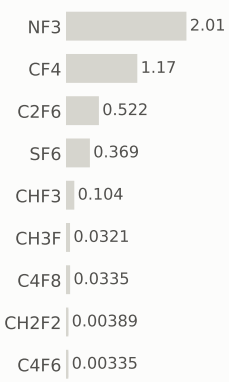
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



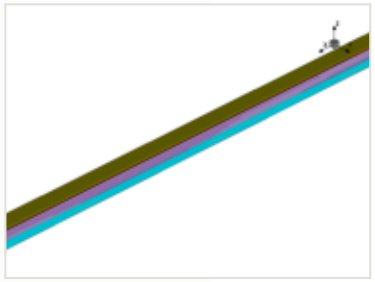
Step #7660: CVD

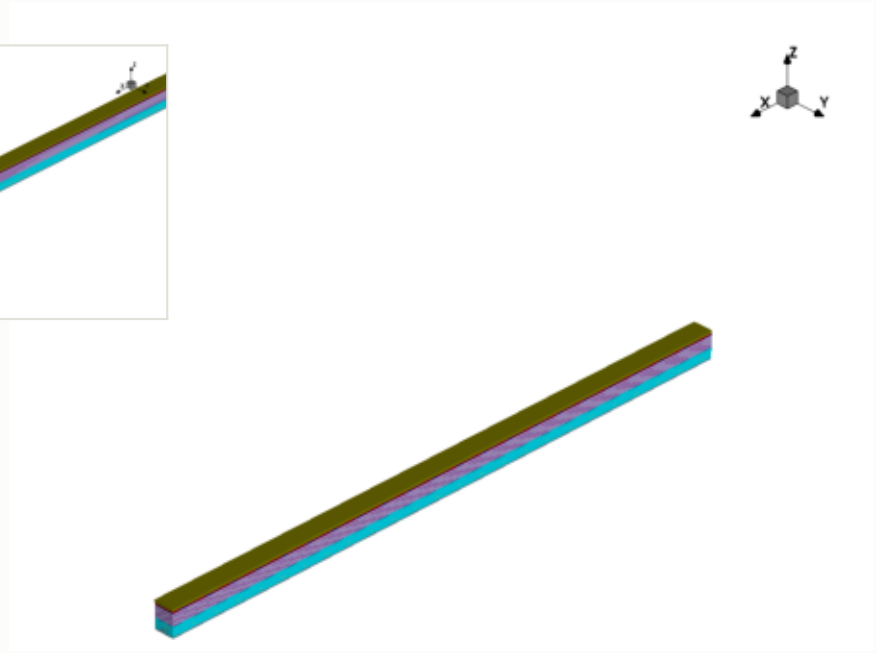
depo_isotropic · CVD

1.31	349.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



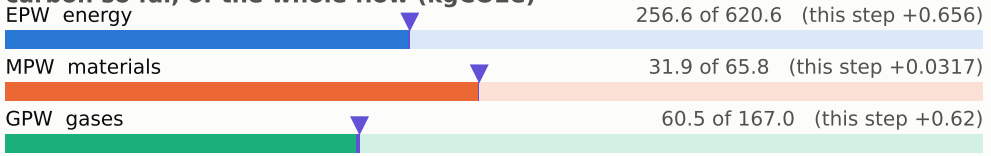


this step

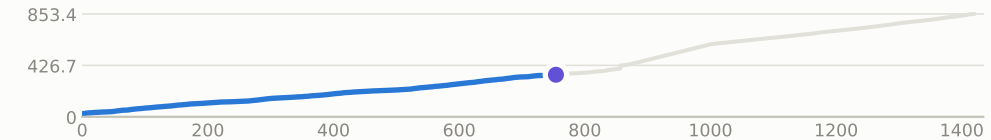
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

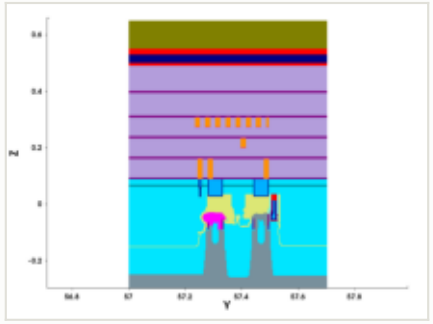
carbon so far, of the whole flow (kgCO2e)



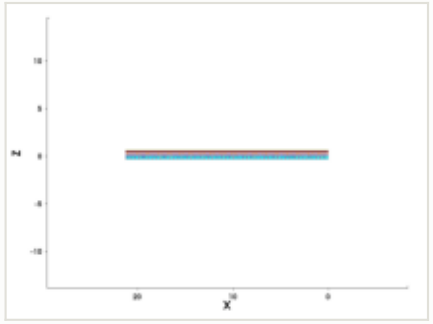
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



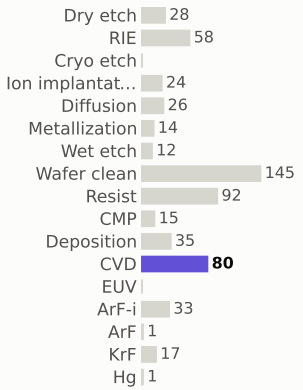
y cut at 57.351 µm



mask for this step

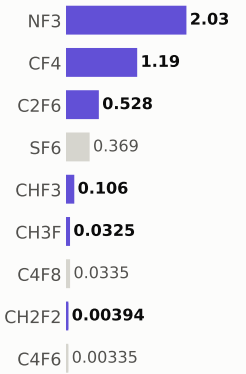
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



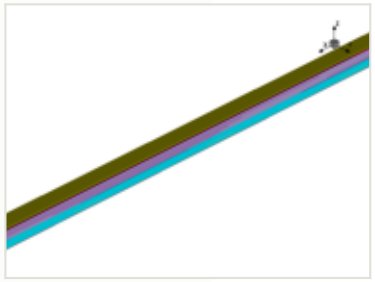
Step #7670: Wafer_clean

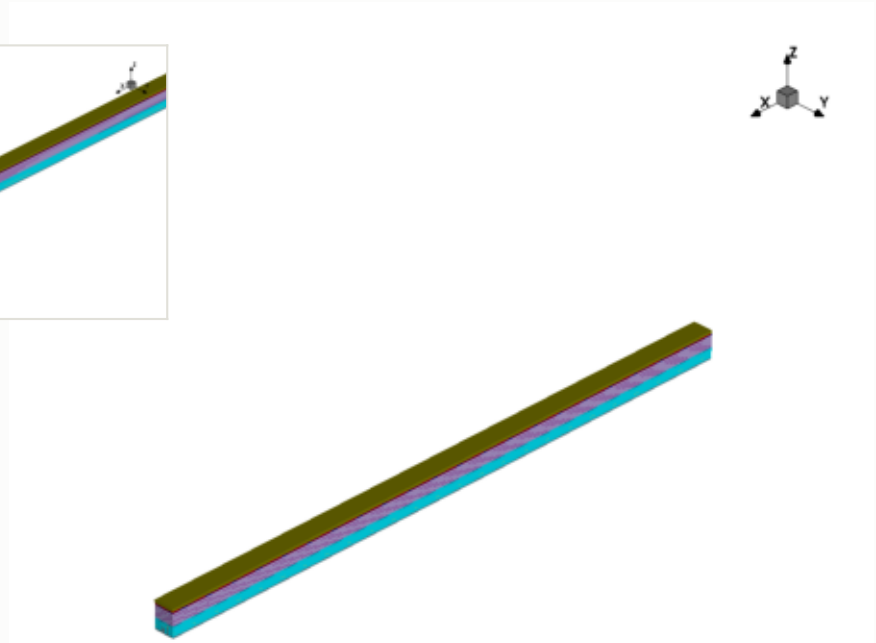
clean · Wafer_clean

0.169	349.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

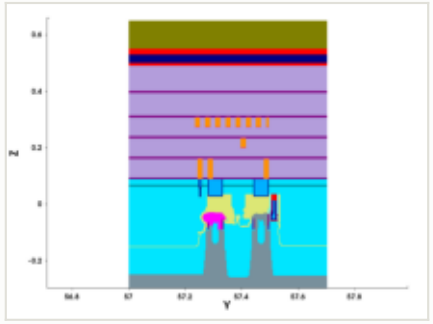
3D structure after this step

zoomed to fit

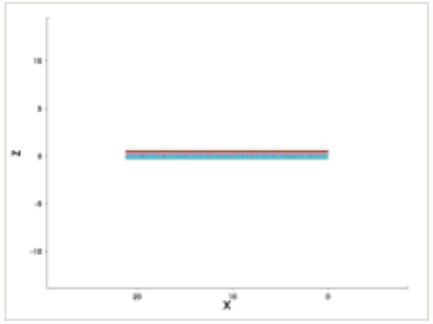




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

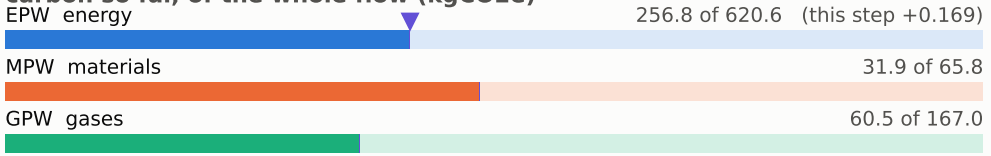
no mask for this step

this step

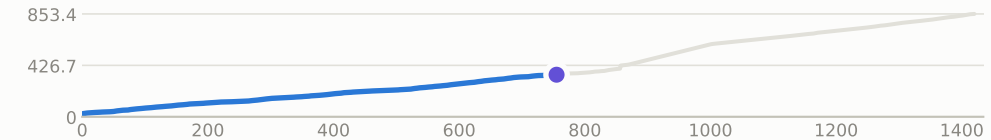
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

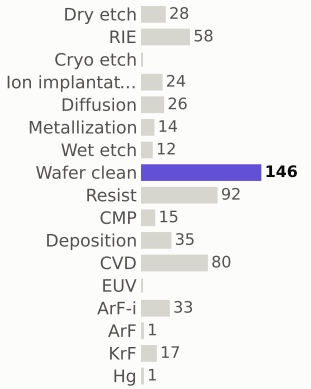
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

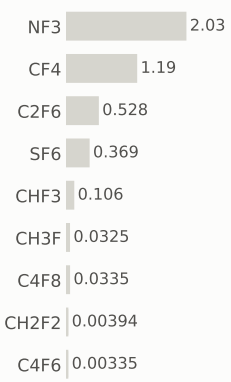


steps by process area



materials deposited (g)

gases emitted (g)



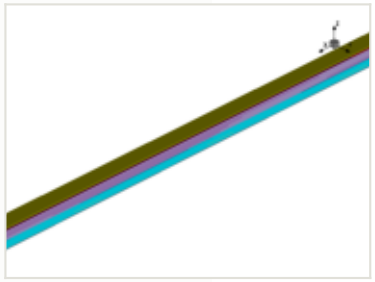
Step #7680: Resist

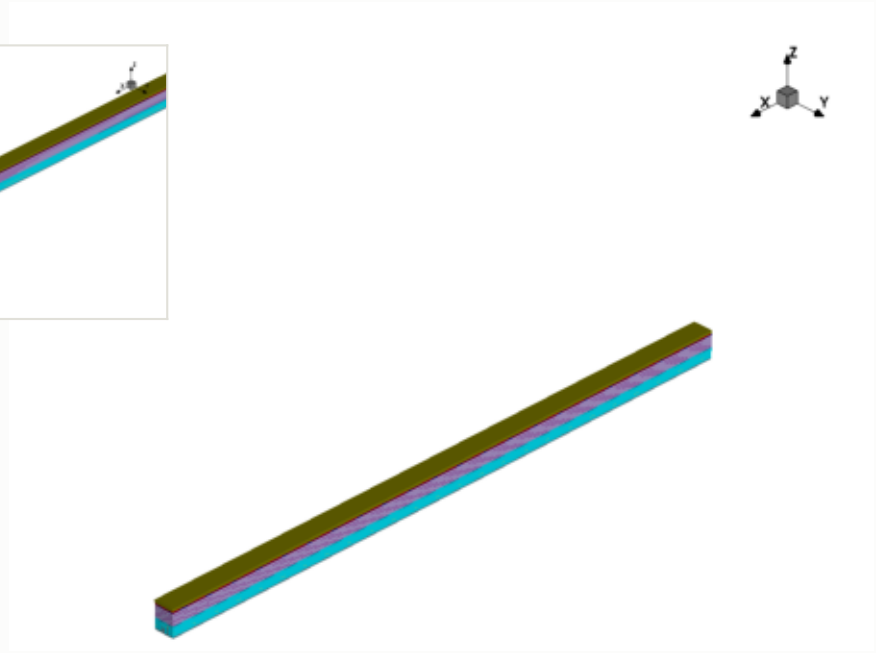
resist · Resist

0.169	349.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





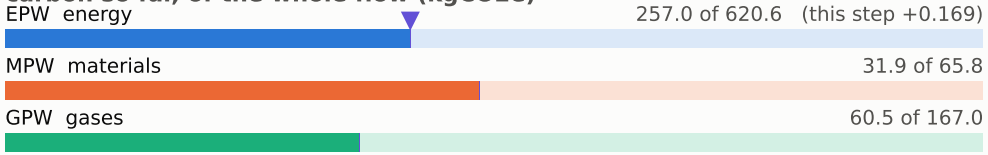
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

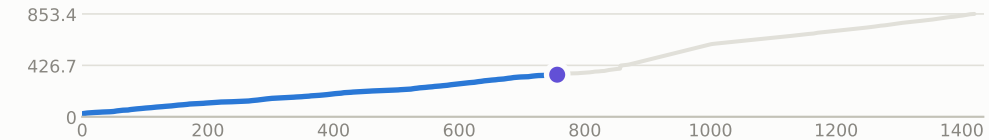
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

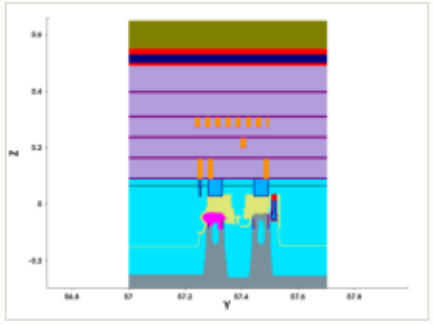
carbon so far, of the whole flow (kgCO2e)



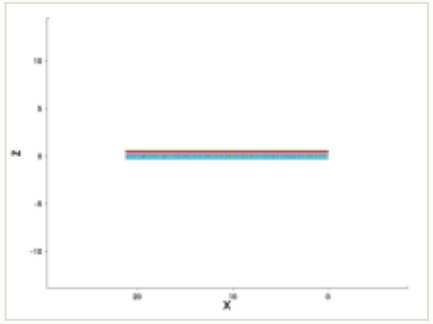
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



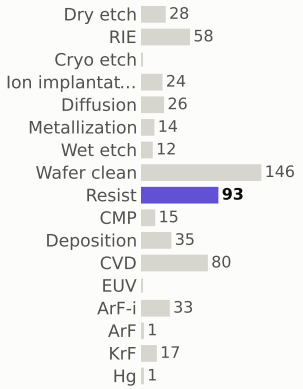
y cut at 57.351 µm



mask for this step

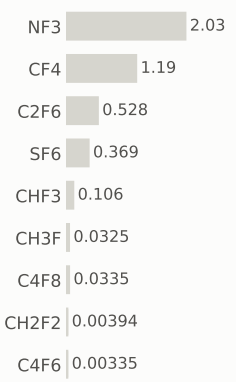
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



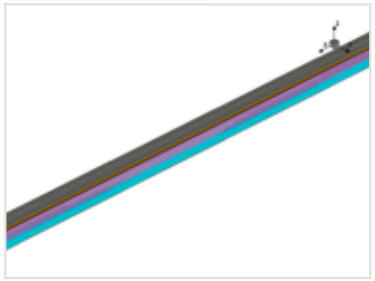
Step #7690: ArF-i

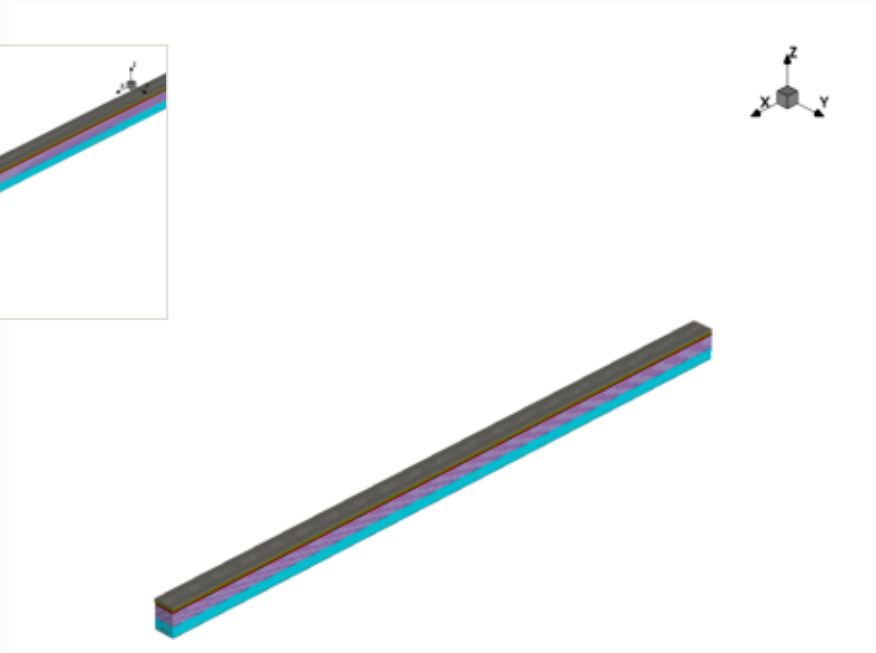
litho_patterning · ArF-i

0.217	349.6	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



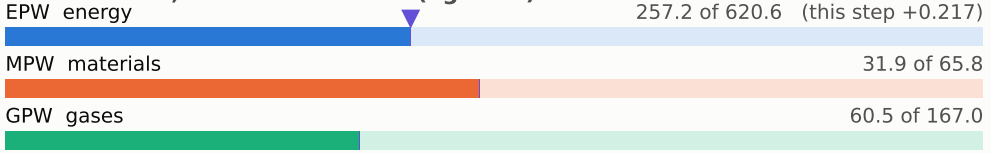


this step

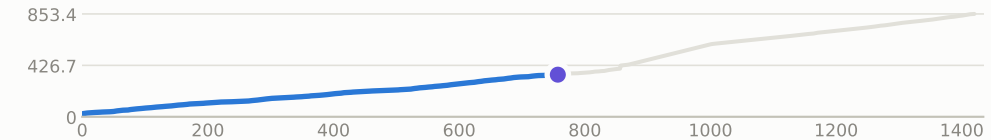
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

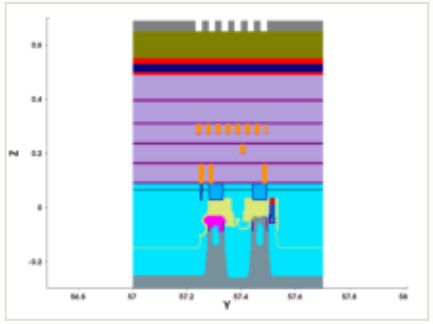
carbon so far, of the whole flow (kgCO2e)



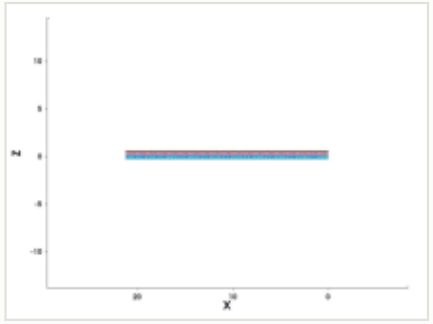
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

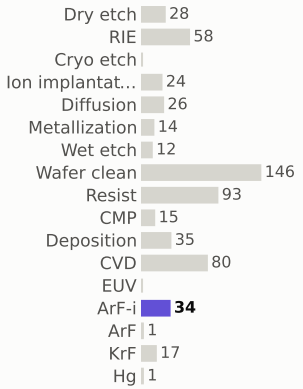


y cut at 57.351 µm

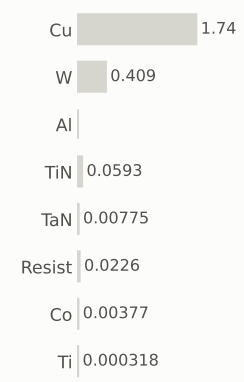


mask: layer 2050:0 (GDSII)

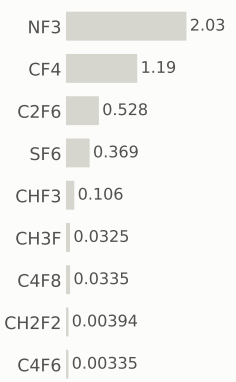
steps by process area



materials deposited (g)



gases emitted (g)



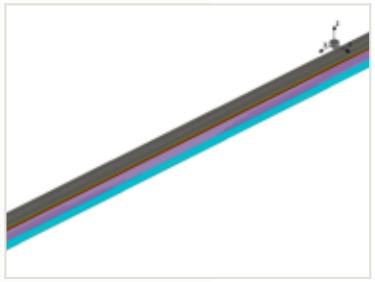
Step #7700: RIE

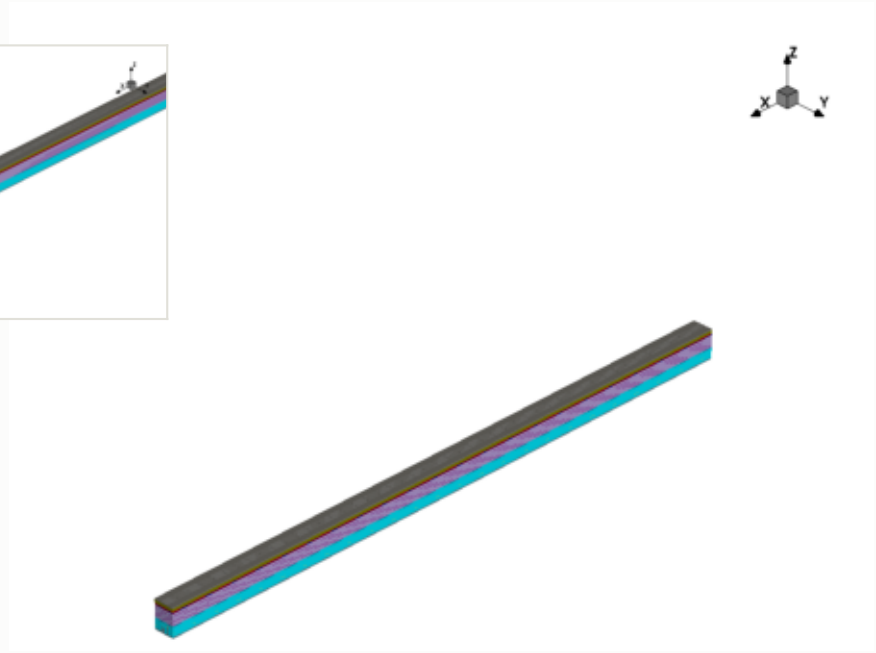
etch_anisotropic · RIE

1.04	350.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



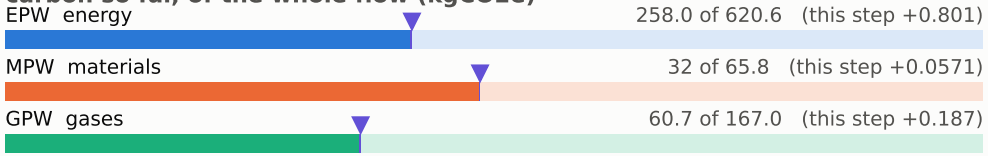


this step

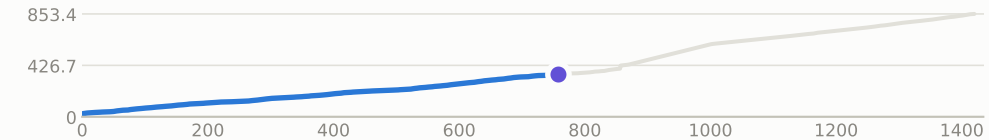
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

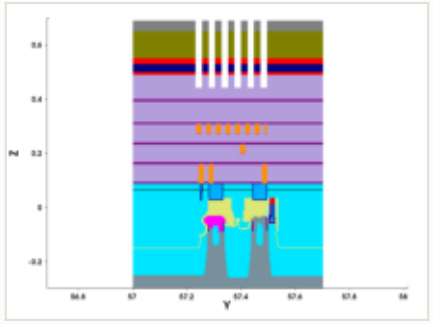
carbon so far, of the whole flow (kgCO2e)



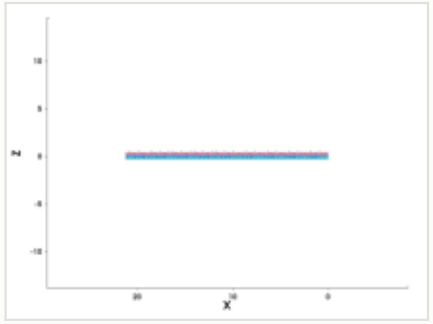
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



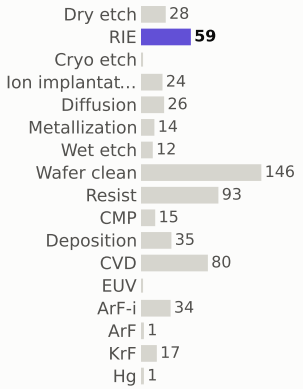
y cut at 57.351 µm



mask for this step

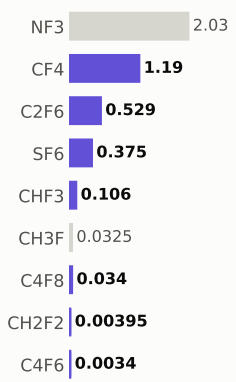
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



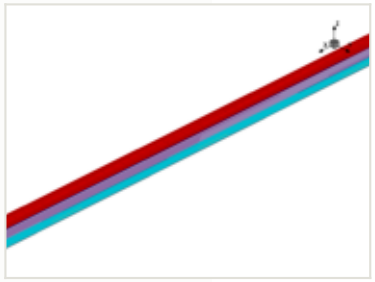
Step #7710: Spx_only

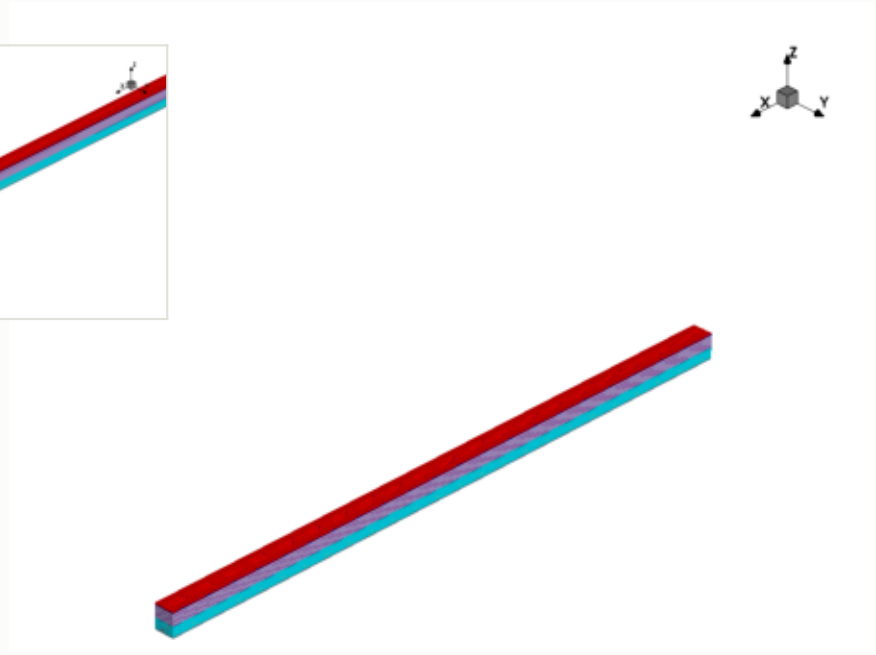
etch_strip · Spx_only

0	350.6	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





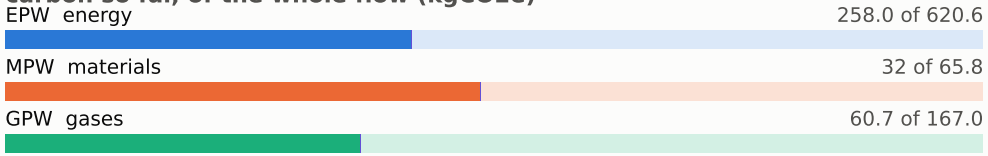
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

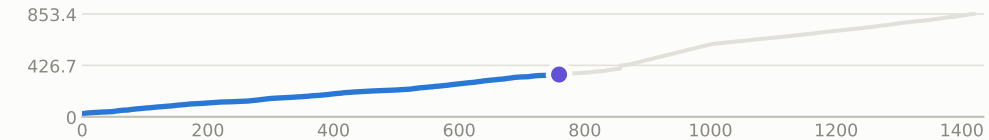
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

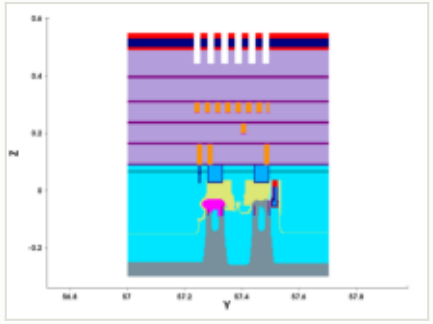
carbon so far, of the whole flow (kgCO2e)



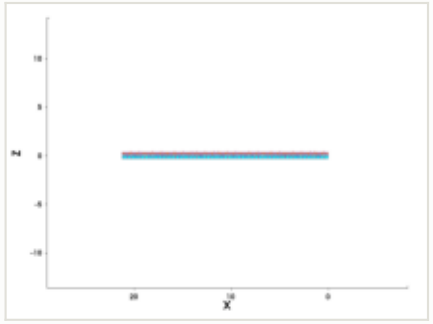
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



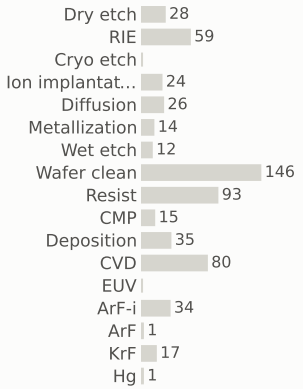
y cut at 57.351 µm



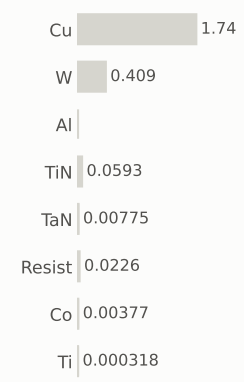
mask for this step

no mask for this step

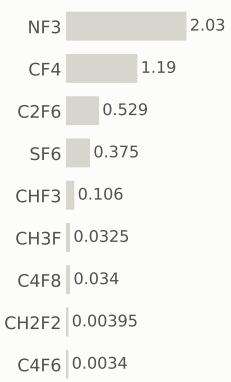
steps by process area



materials deposited (g)



gases emitted (g)



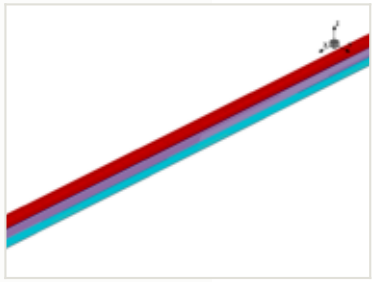
Step #7720: Wafer_clean

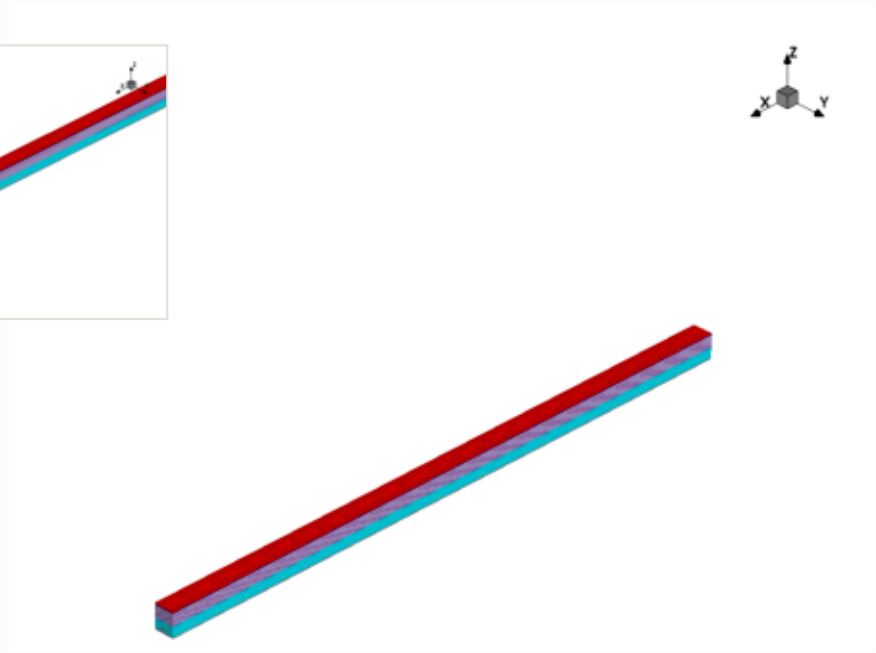
clean · Wafer_clean

0.169	350.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





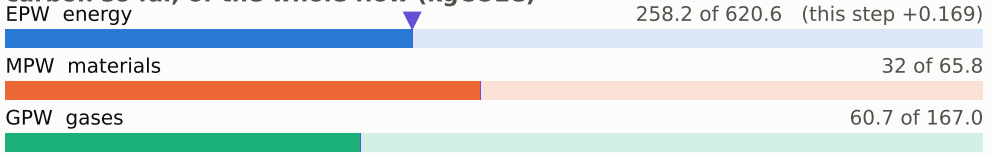
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

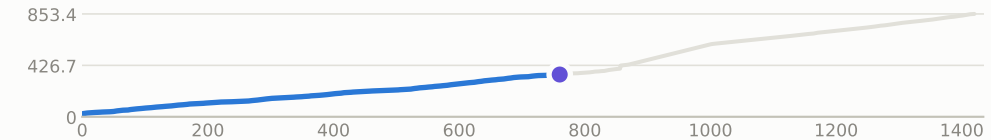
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

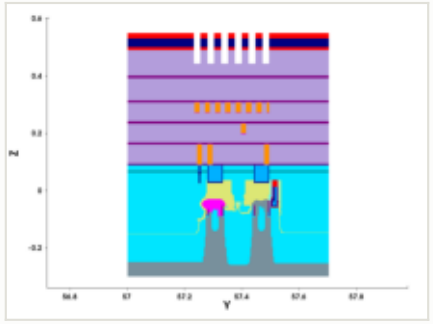
carbon so far, of the whole flow (kgCO2e)



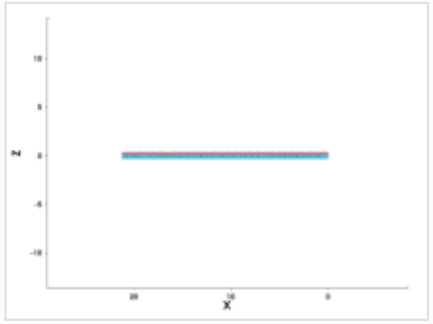
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



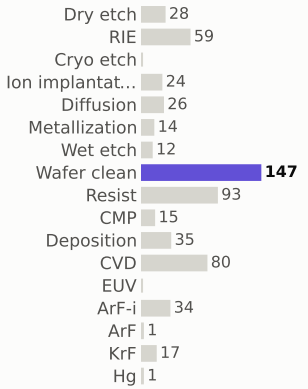
y cut at 57.351 µm



mask for this step

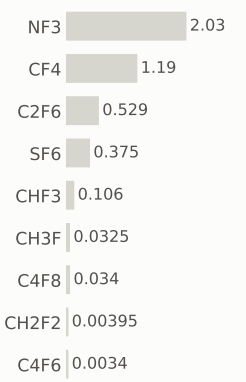
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



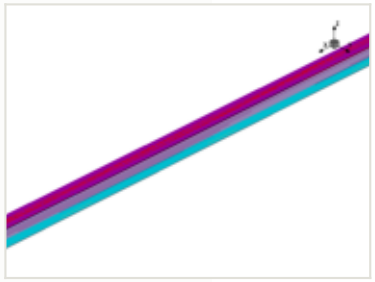
Step #7730: Metallization

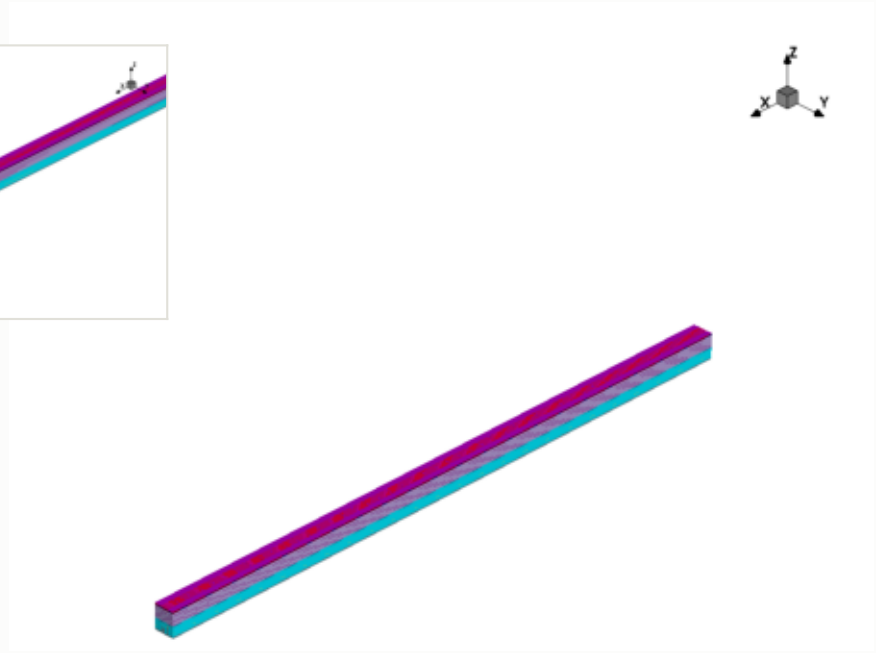
depo_ellipsoidal · Metallization

1.2	352.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



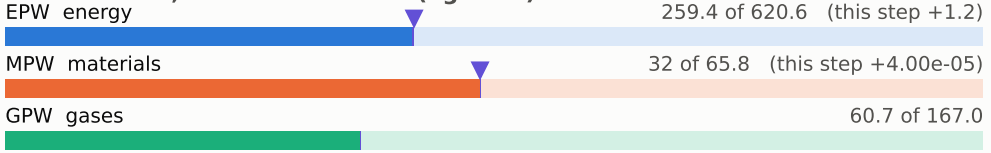


this step

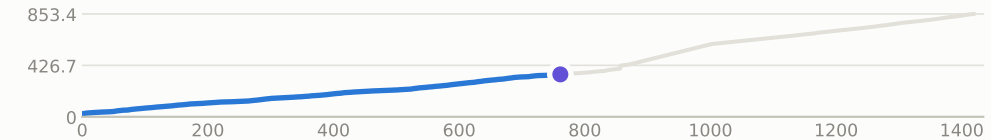
process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

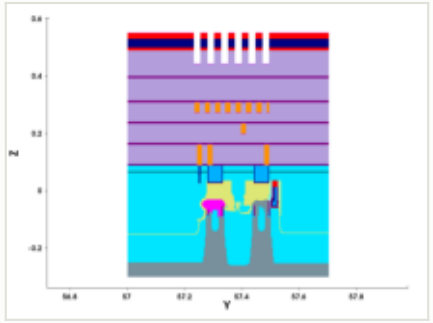
carbon so far, of the whole flow (kgCO2e)



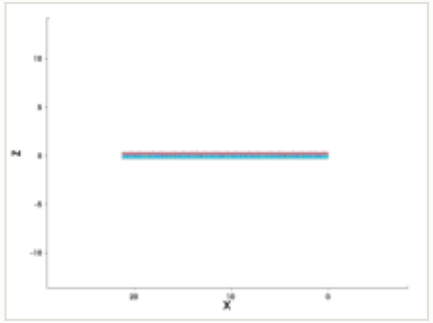
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



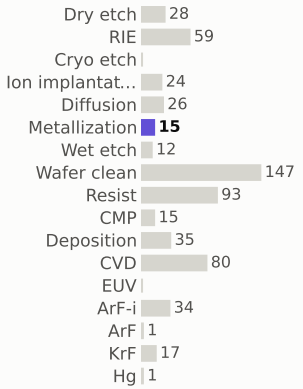
y cut at 57.351 µm



mask for this step

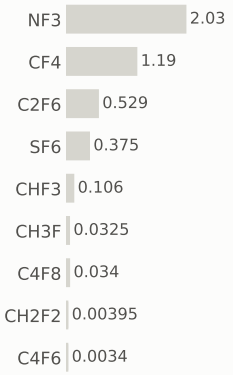
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



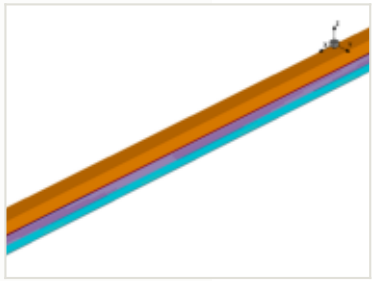
Step #7740: Metallization

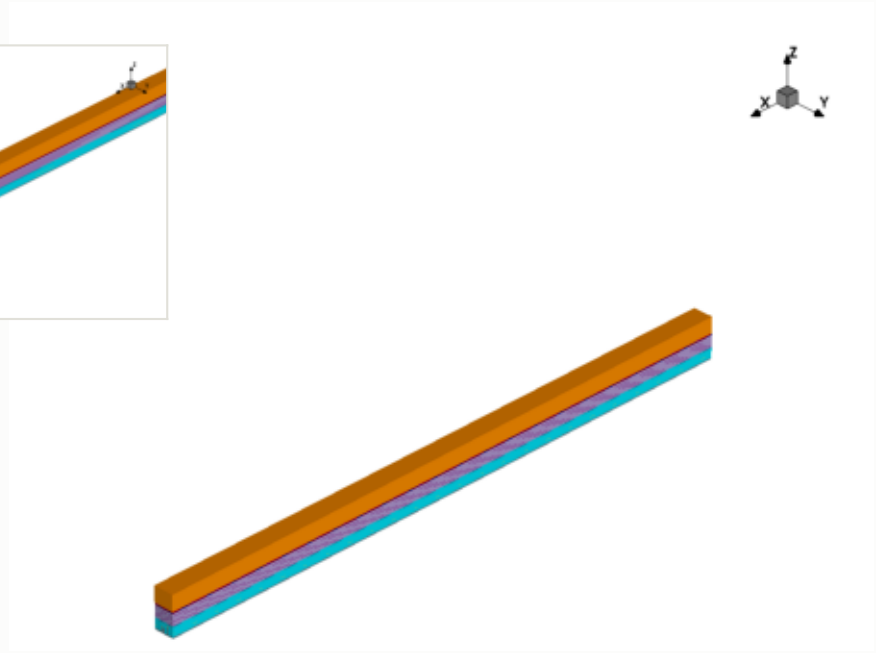
depo_fill · Metallization

1.21	353.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



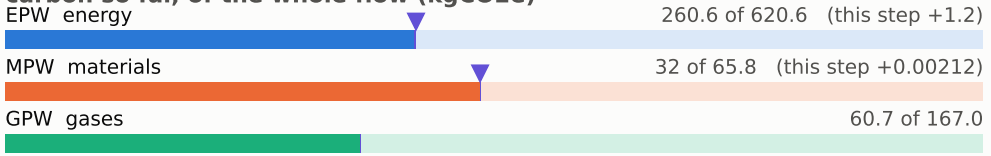


this step

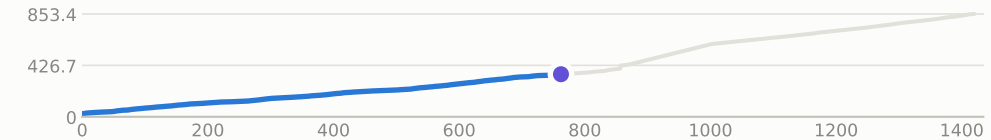
process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

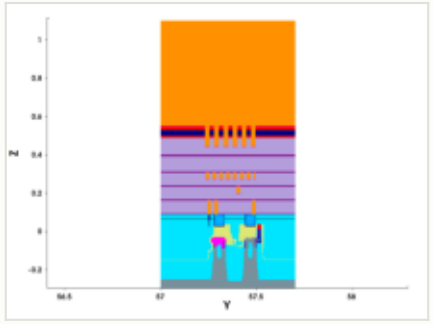
carbon so far, of the whole flow (kgCO2e)



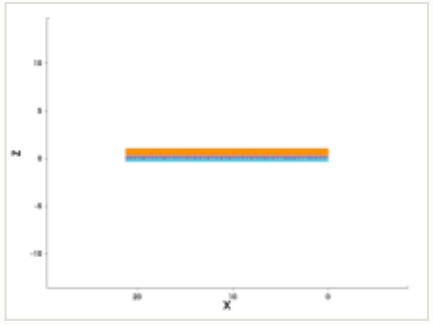
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



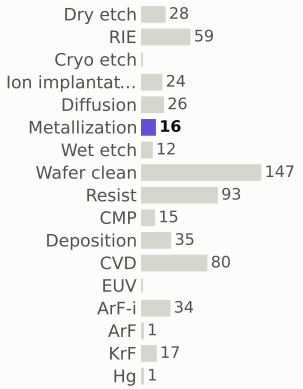
y cut at 57.351 µm



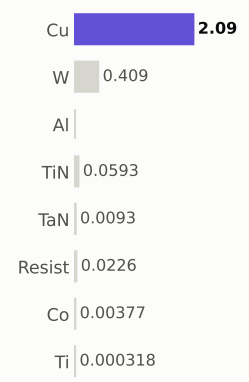
mask for this step

no mask for this step

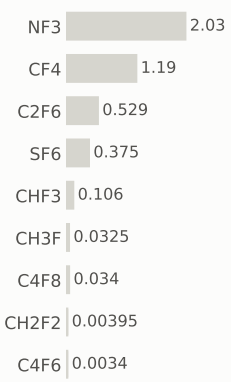
steps by process area



materials deposited (g)



gases emitted (g)



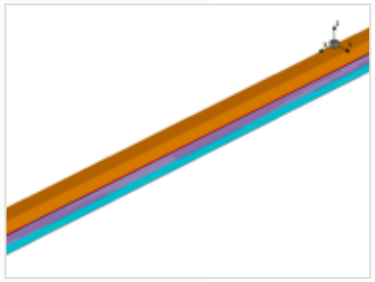
Step #7750: Diffusion

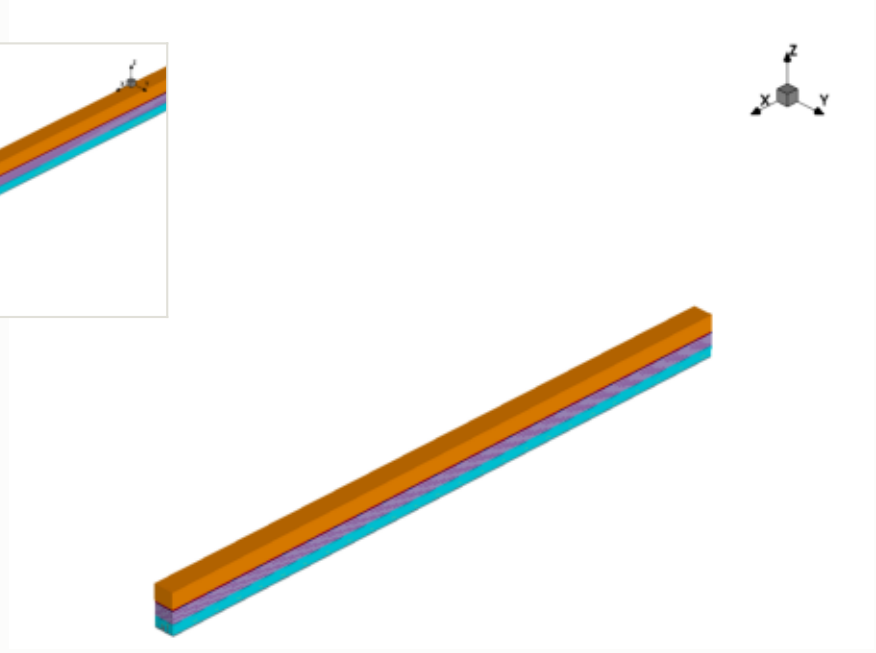
anneal · Diffusion

0.801	354.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





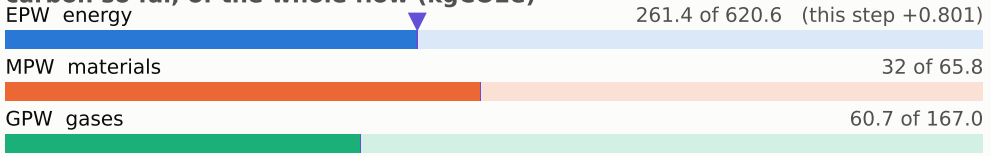
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

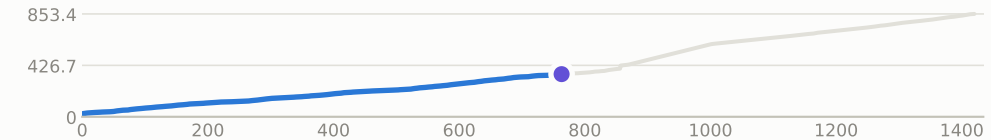
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

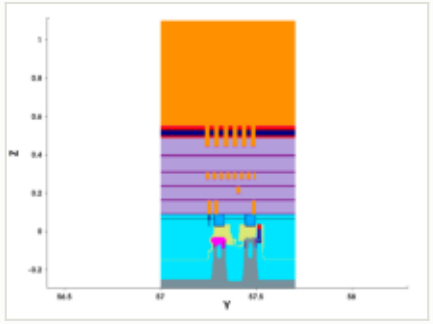
carbon so far, of the whole flow (kgCO2e)



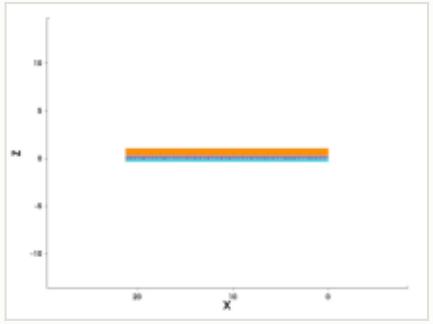
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



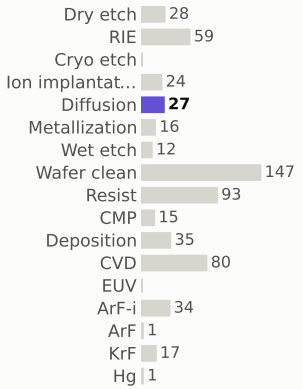
y cut at 57.351 µm



mask for this step

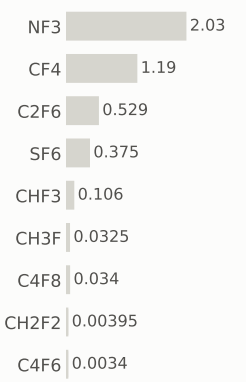
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



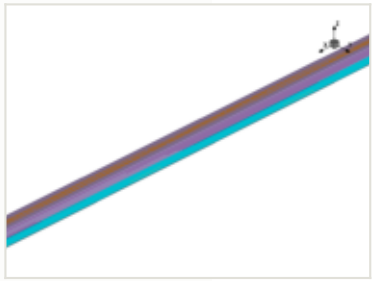
Step #7760: CMP

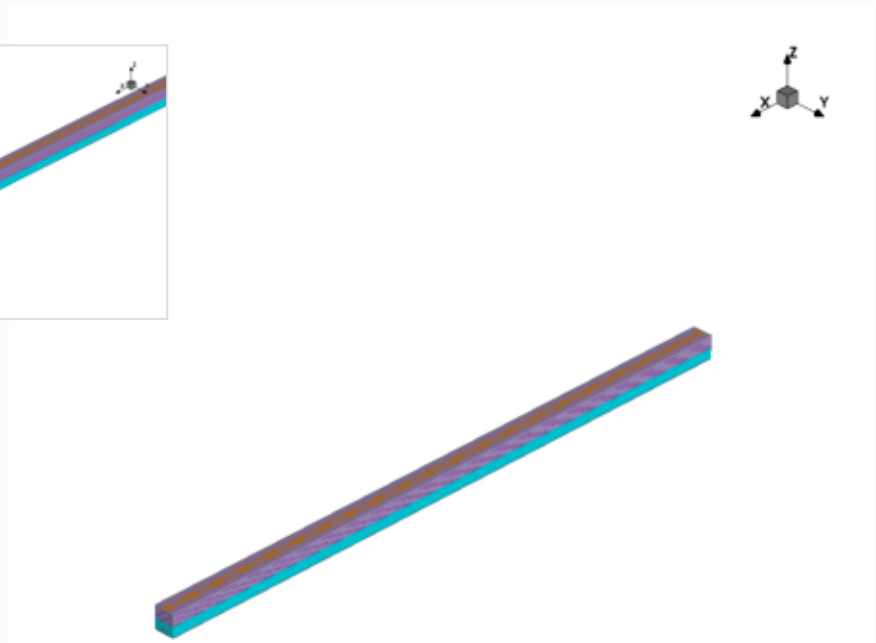
etch_cmp · CMP

0.169	354.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



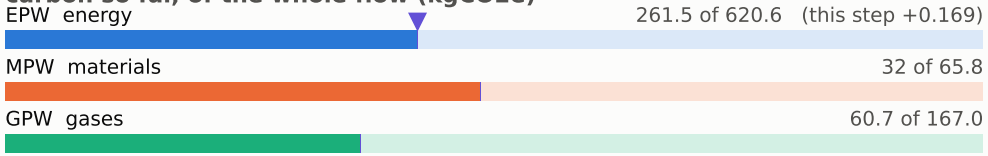


this step

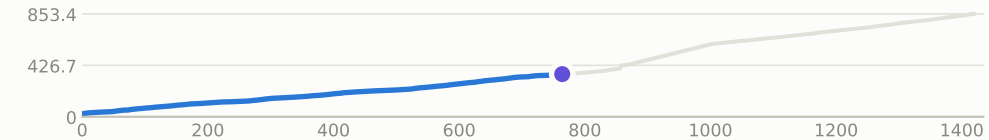
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

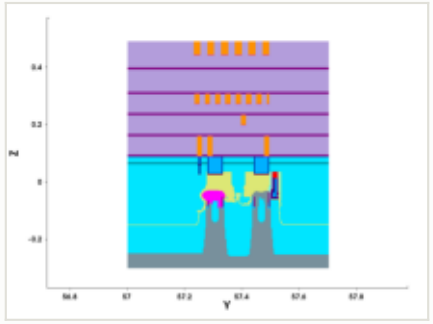
carbon so far, of the whole flow (kgCO2e)



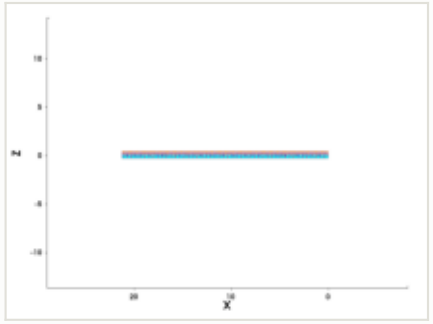
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



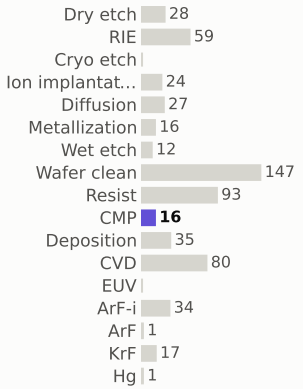
y cut at 57.351 µm



mask for this step

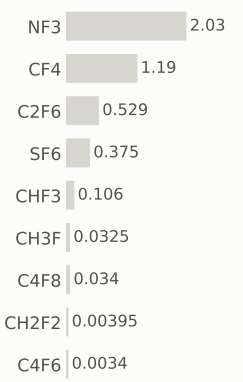
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



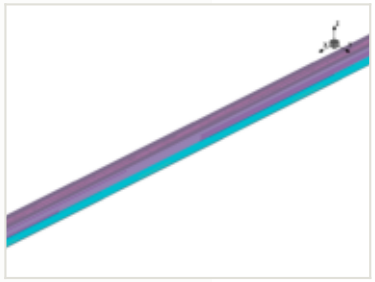
Step #7770: Spx_only

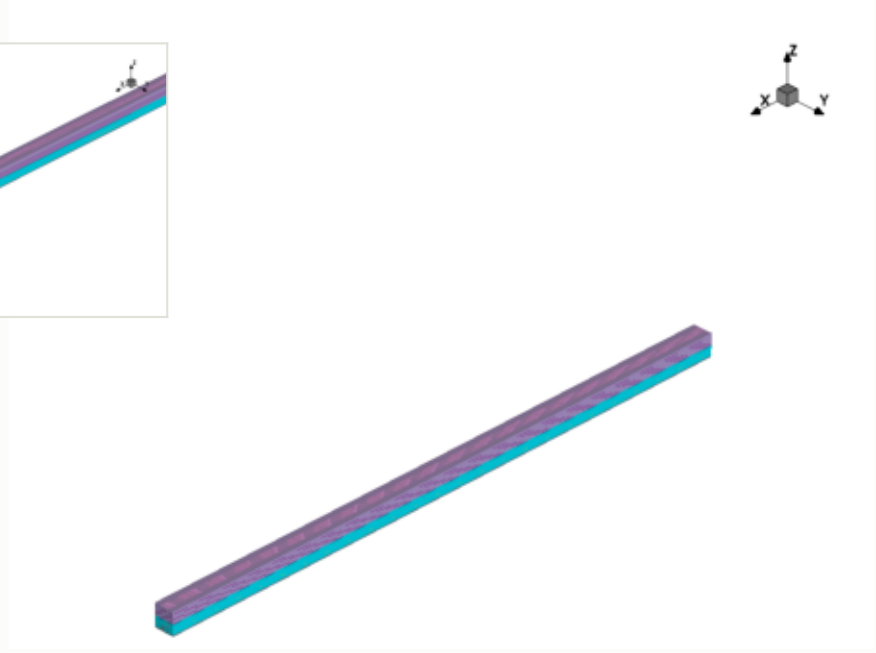
depo_isotropic · Spx_only

4.20e-05	354.2	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

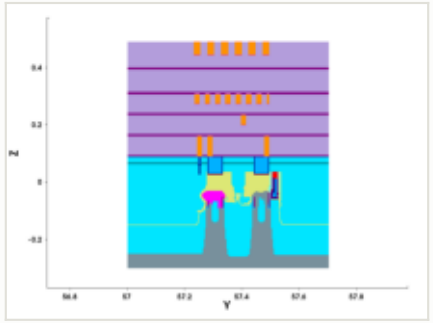
3D structure after this step

zoomed to fit

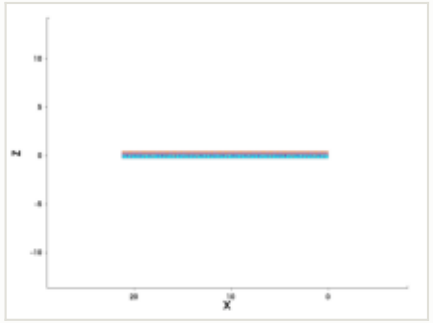




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

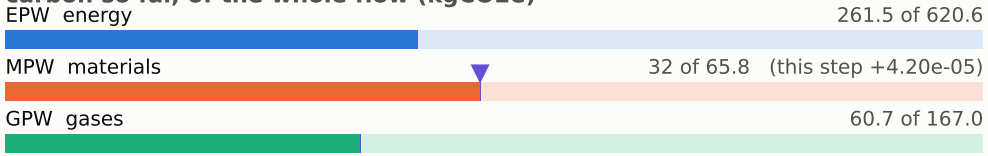
this step

process area	Spx_only
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

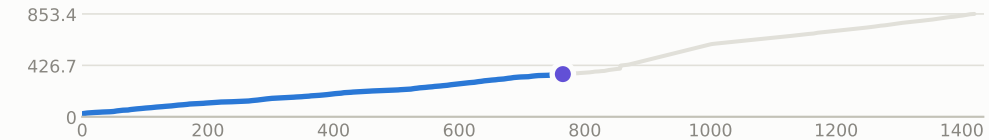
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**4.20e-05 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

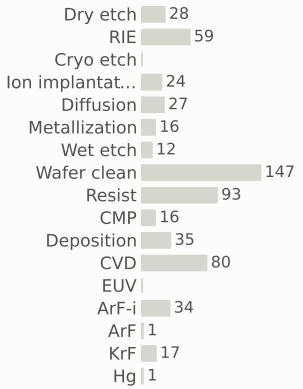
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

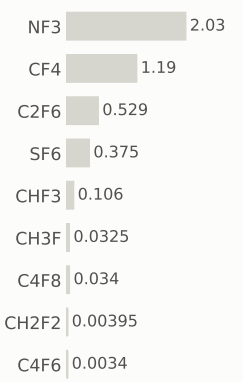


steps by process area



materials deposited (g)

gases emitted (g)



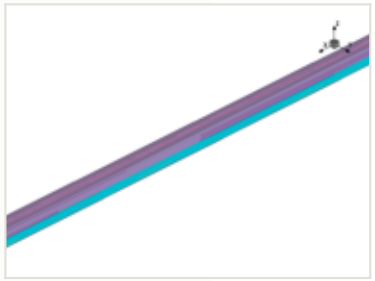
Step #7790: CVD

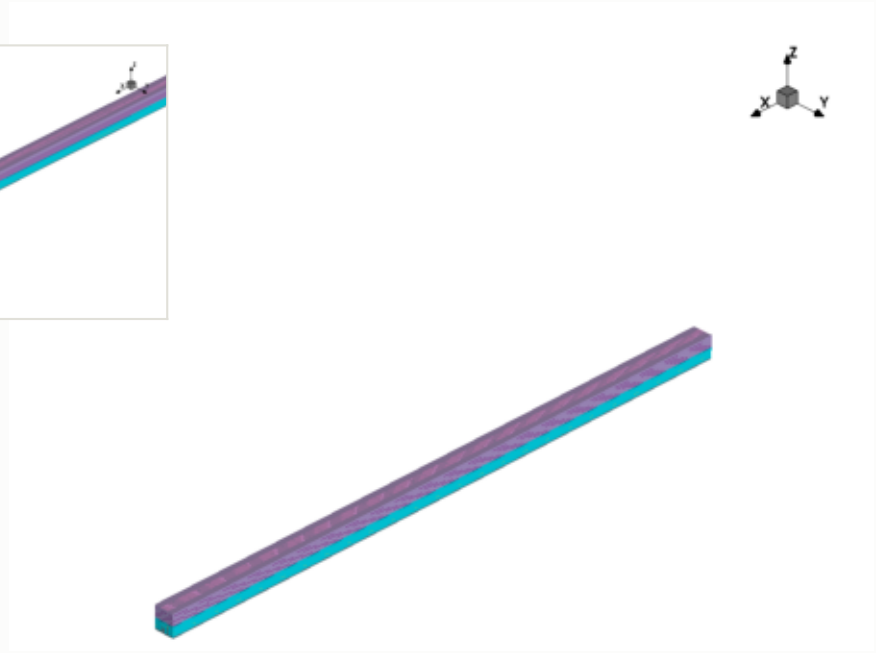
depo_isotropic · CVD

1.31	355.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



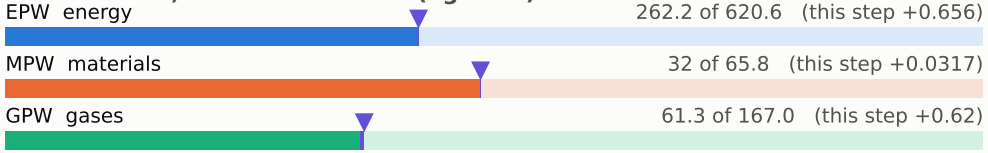


this step

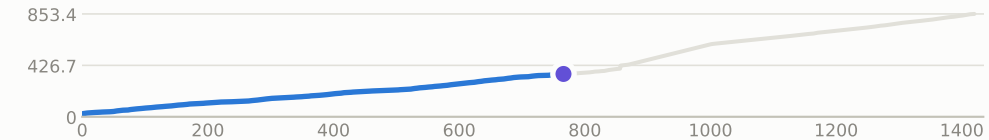
process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

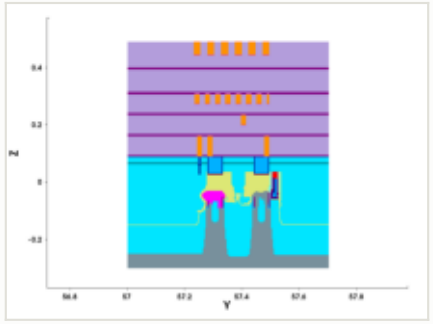
carbon so far, of the whole flow (kgCO2e)



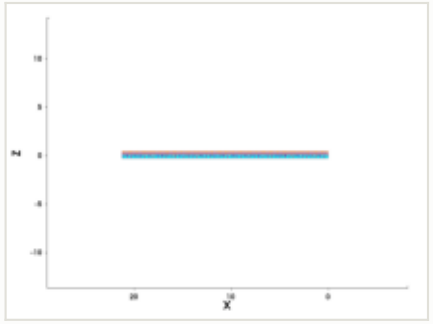
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



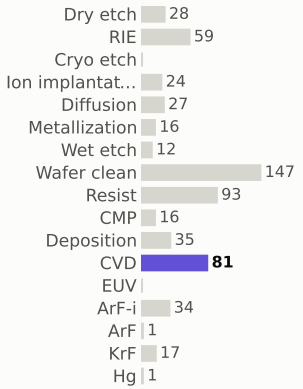
y cut at 57.351 µm



mask for this step

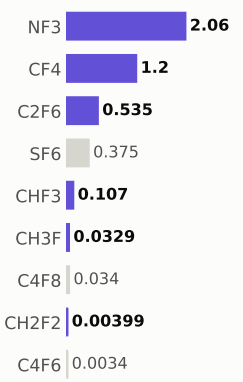
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



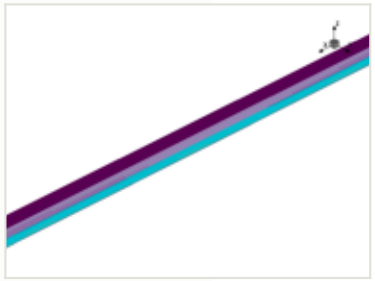
Step #7800: Spx_only

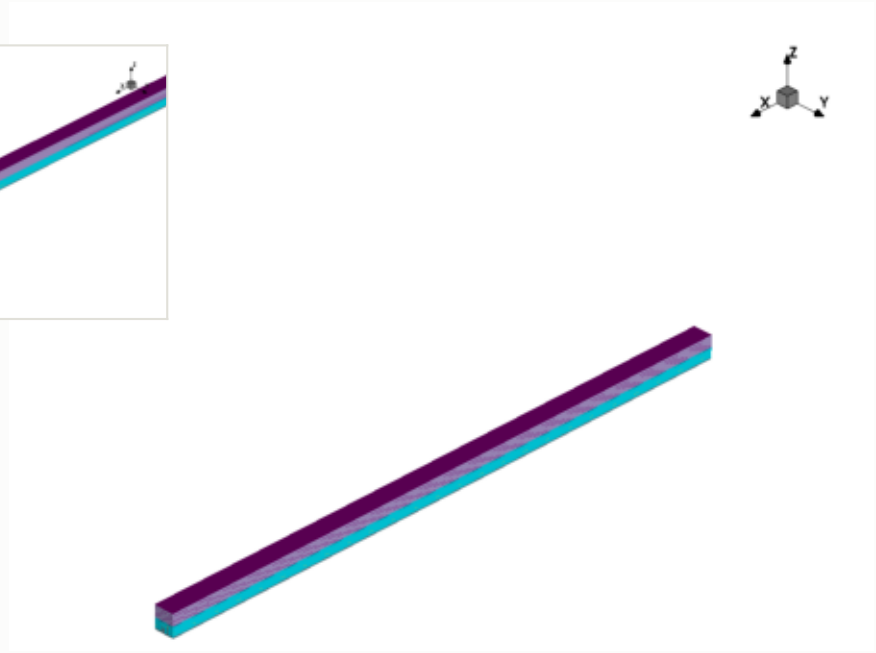
depo_fill · Spx_only

0	355.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





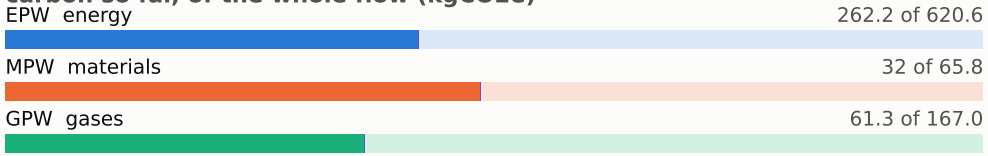
this step

process area	Spx_only
material	—
thickness	0.006 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

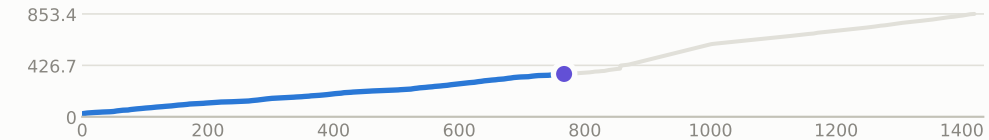
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

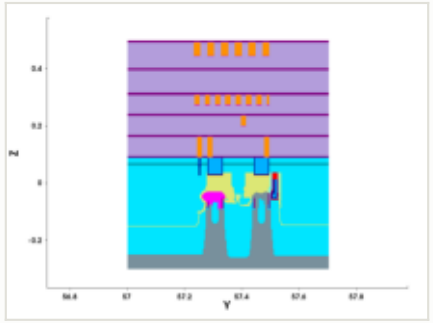
carbon so far, of the whole flow (kgCO2e)



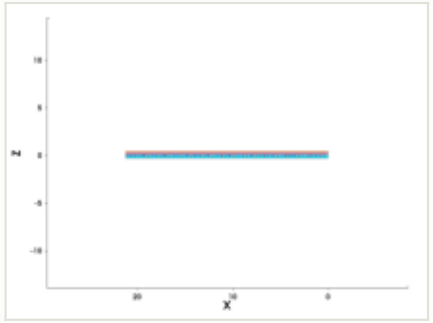
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



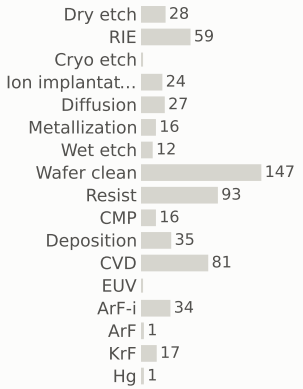
y cut at 57.351 µm



mask for this step

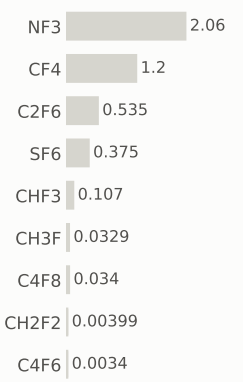
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



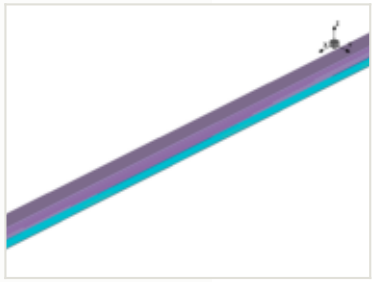
Step #7810: CVD

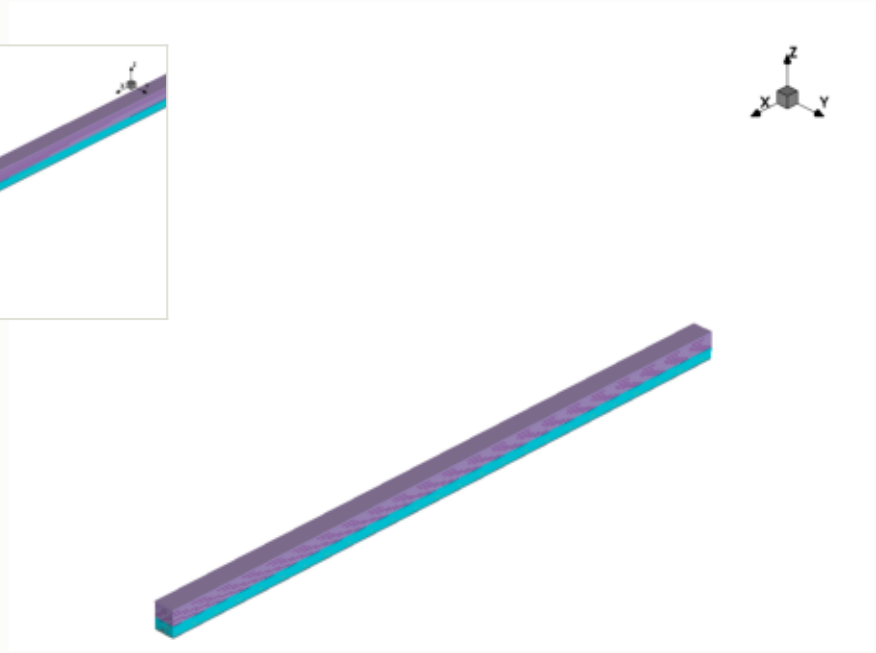
depo_fill · CVD

1.31	356.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



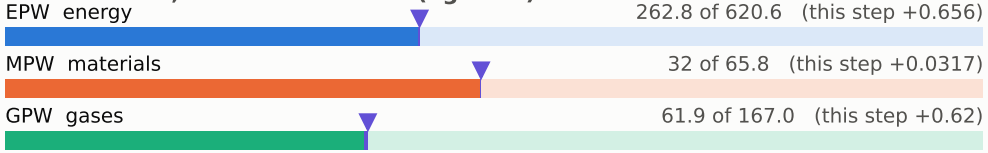


this step

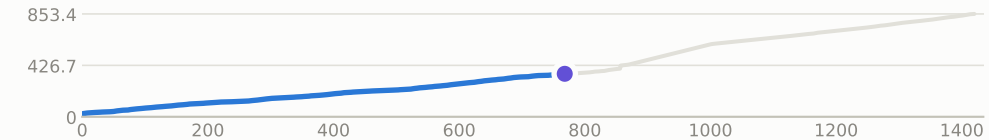
process area	CVD
material	—
thickness	0.106 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

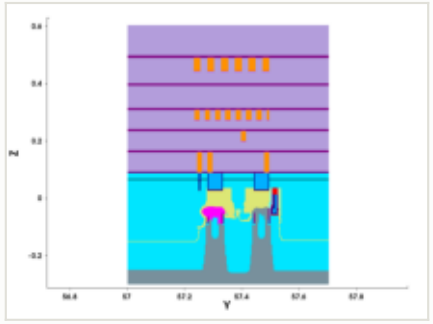
carbon so far, of the whole flow (kgCO2e)



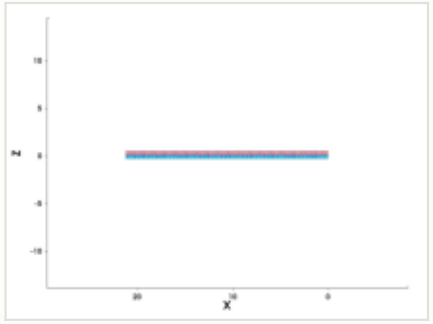
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



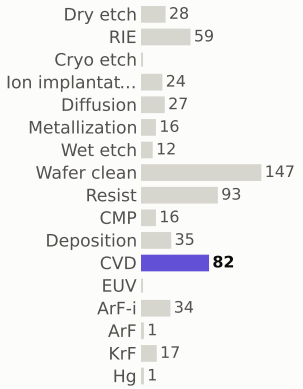
y cut at 57.351 µm



mask for this step

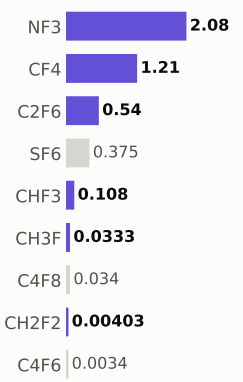
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



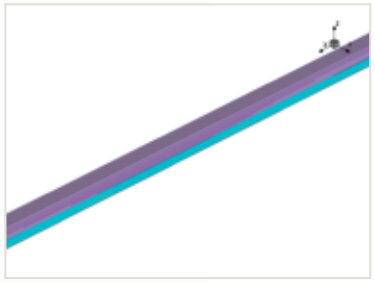
Step #7820: Diffusion

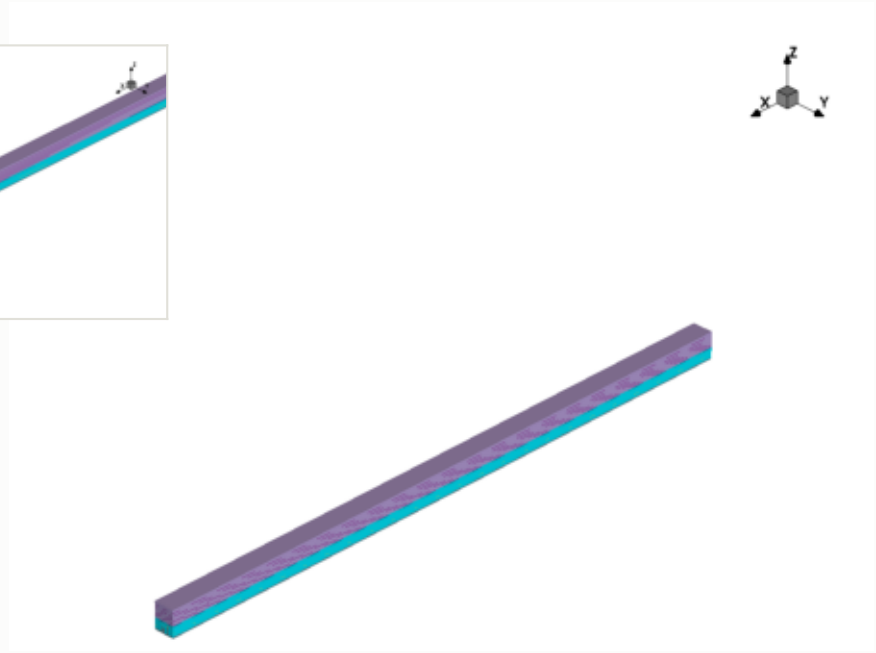
anneal · Diffusion

0.801	357.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





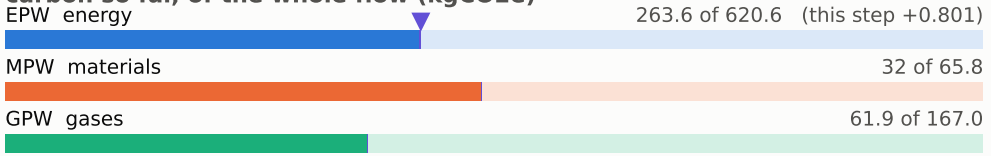
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

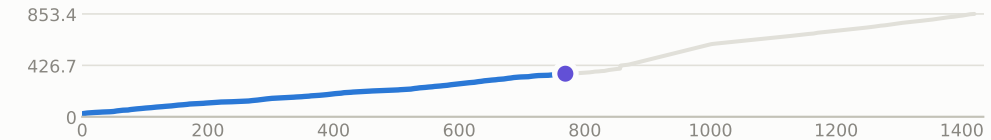
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

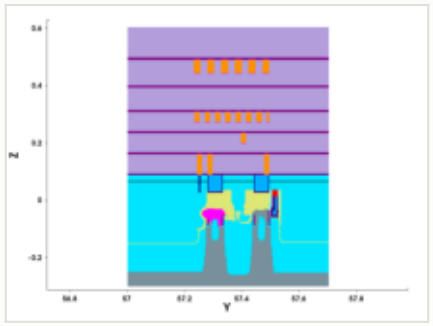
carbon so far, of the whole flow (kgCO2e)



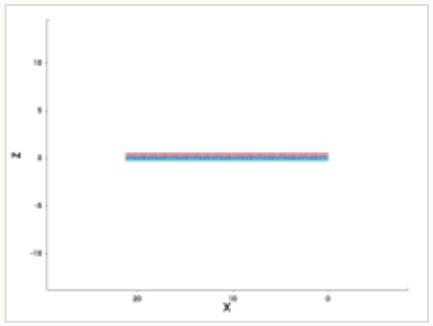
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



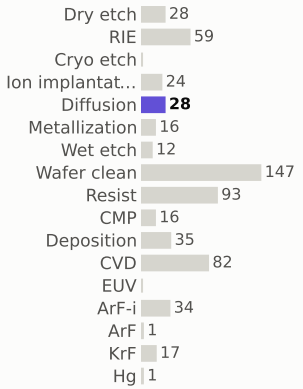
y cut at 57.351 µm



mask for this step

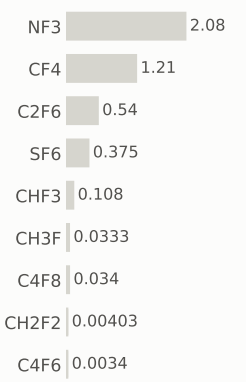
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



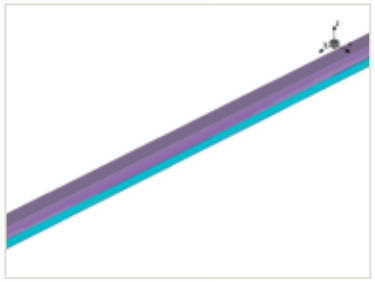
Step #7830: Wafer_clean

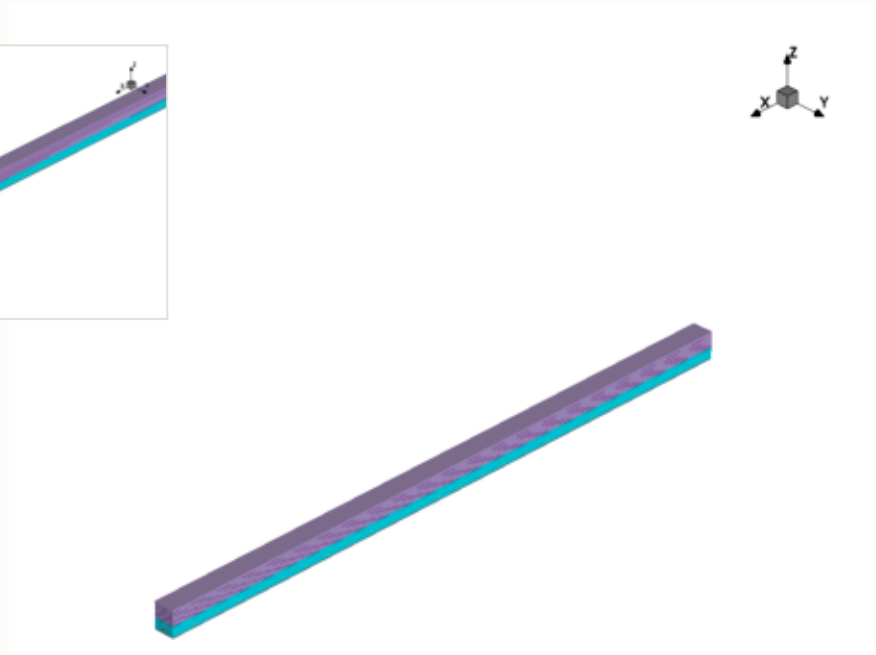
clean · Wafer_clean

0.169	357.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





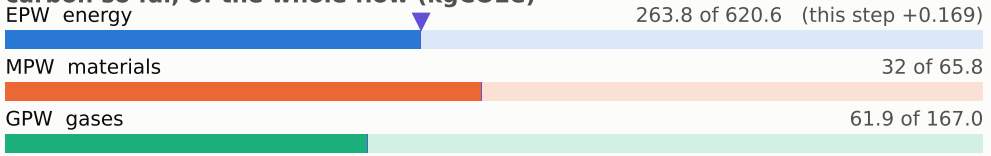
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

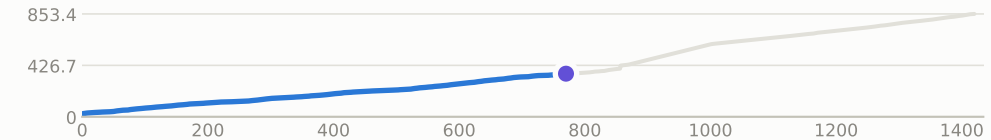
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

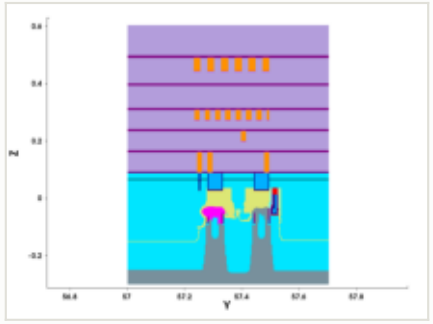
carbon so far, of the whole flow (kgCO2e)



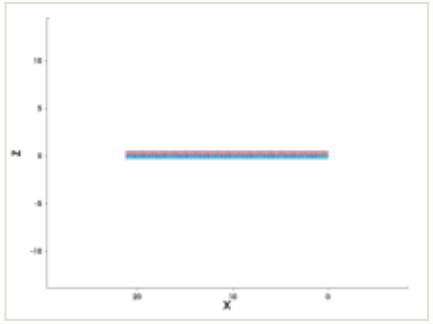
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



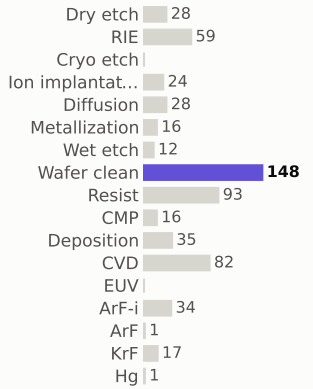
y cut at 57.351 µm



mask for this step

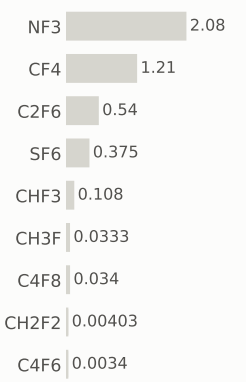
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



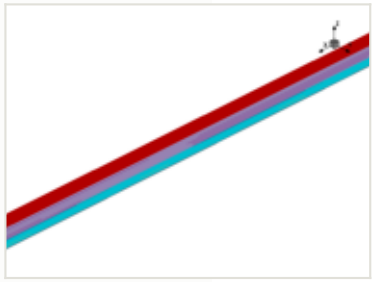
Step #7840: CVD

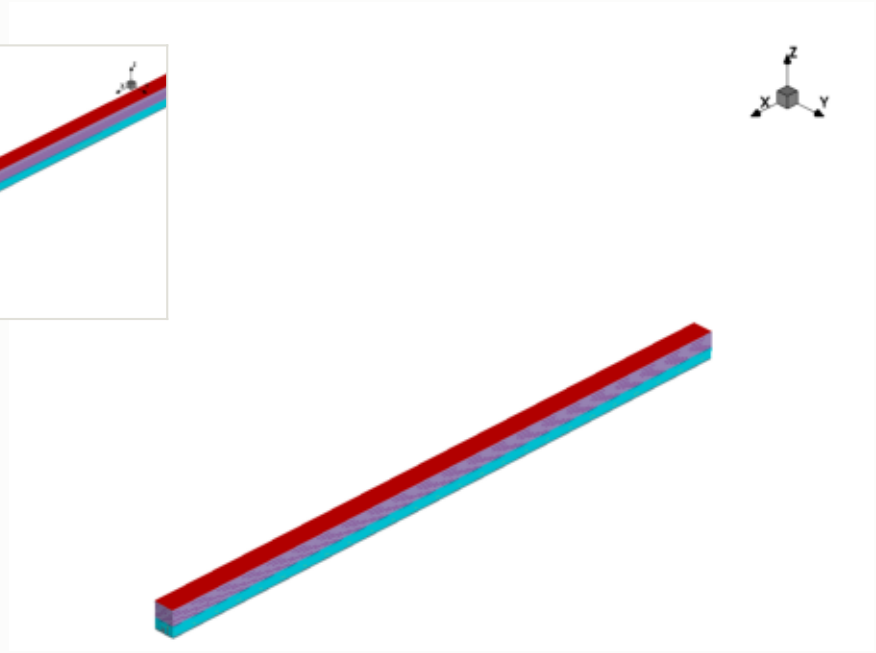
depo_fill · CVD

1.31	359.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



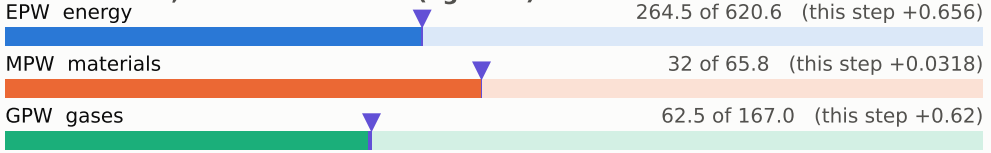


this step

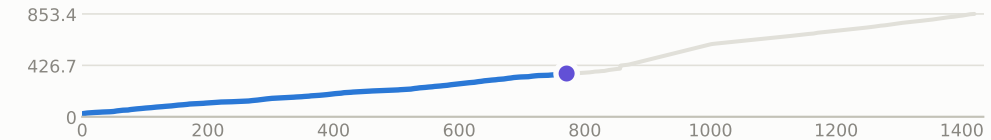
process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

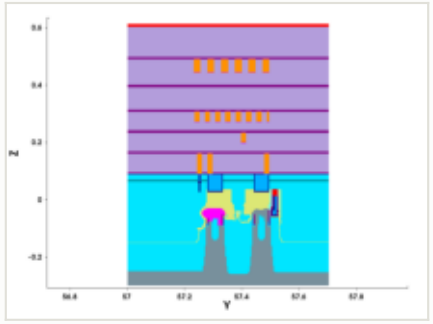
carbon so far, of the whole flow (kgCO2e)



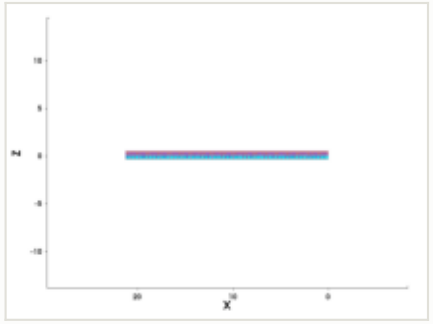
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



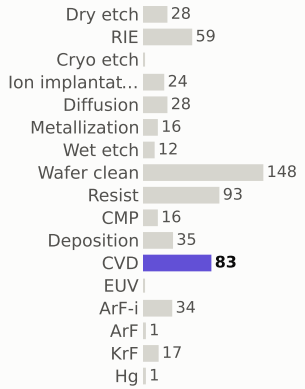
y cut at 57.351 µm



mask for this step

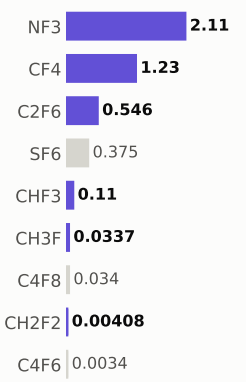
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



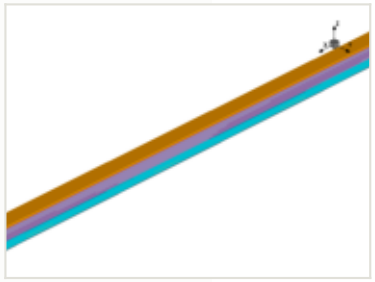
Step #7850: Spx_only

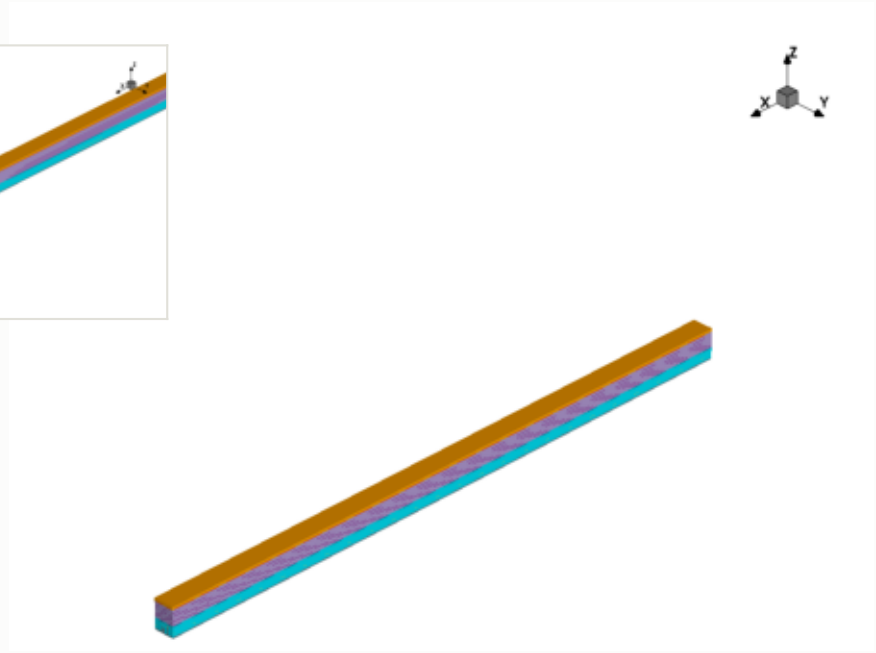
depo_fill · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

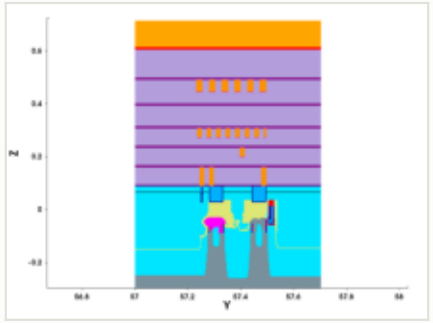
3D structure after this step

zoomed to fit

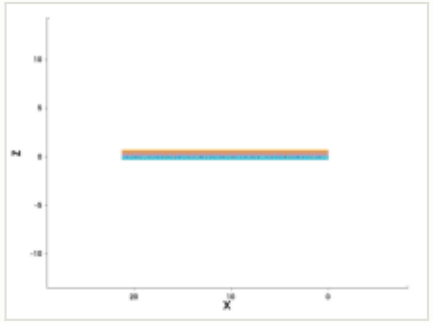




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

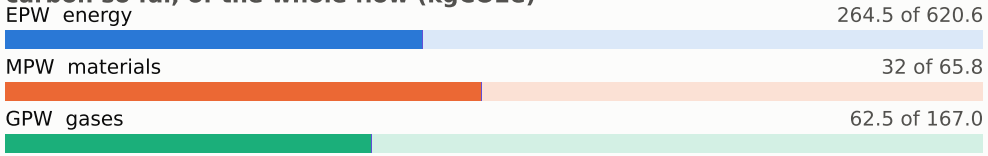
this step

process area	Spx_only
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

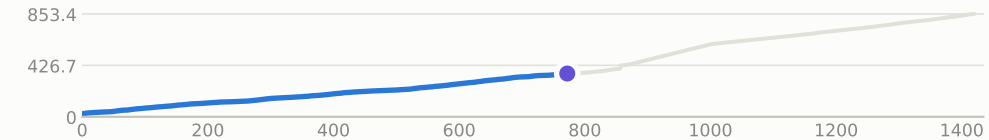
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

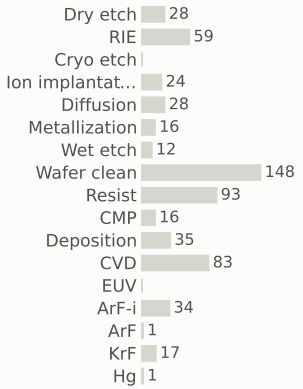
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

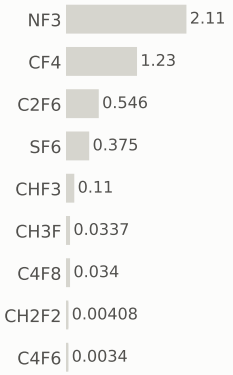


steps by process area



materials deposited (g)

gases emitted (g)



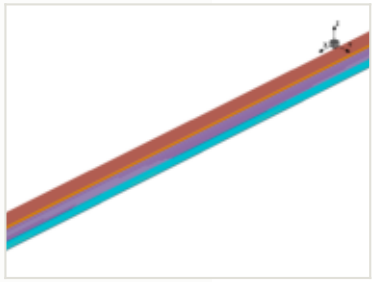
Step #7860: Spx_only

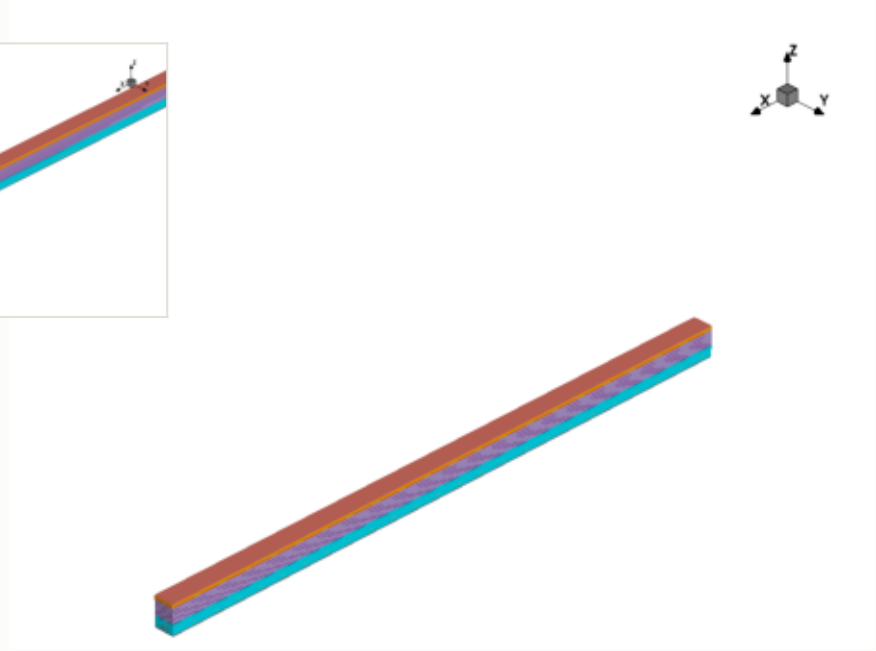
depo_fill · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



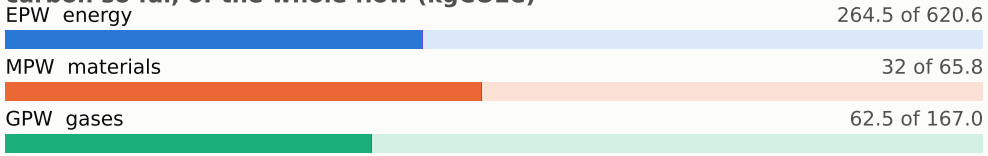


this step

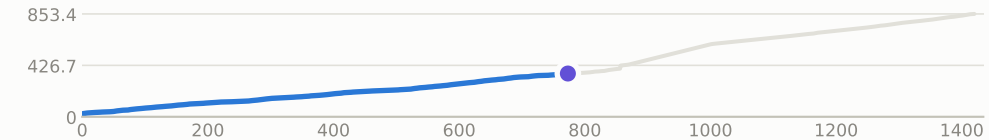
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	0.02 µm	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

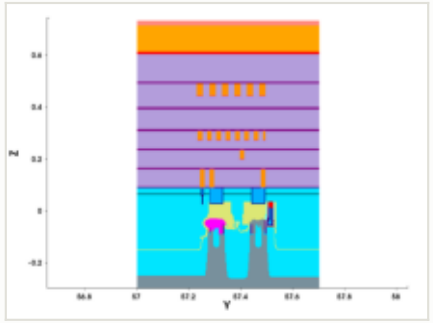
carbon so far, of the whole flow (kgCO2e)



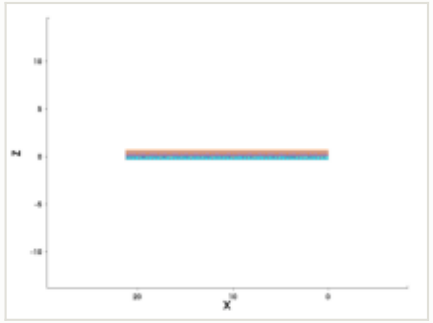
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



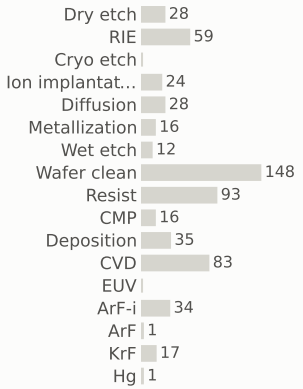
y cut at 57.351 µm



mask for this step

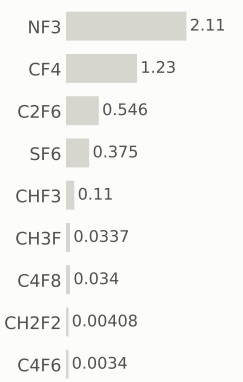
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



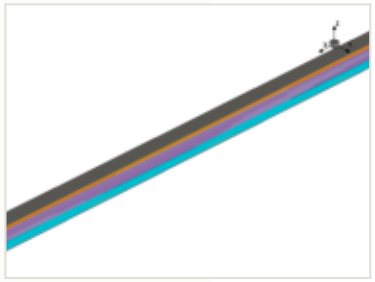
Step #7870: Spx_only

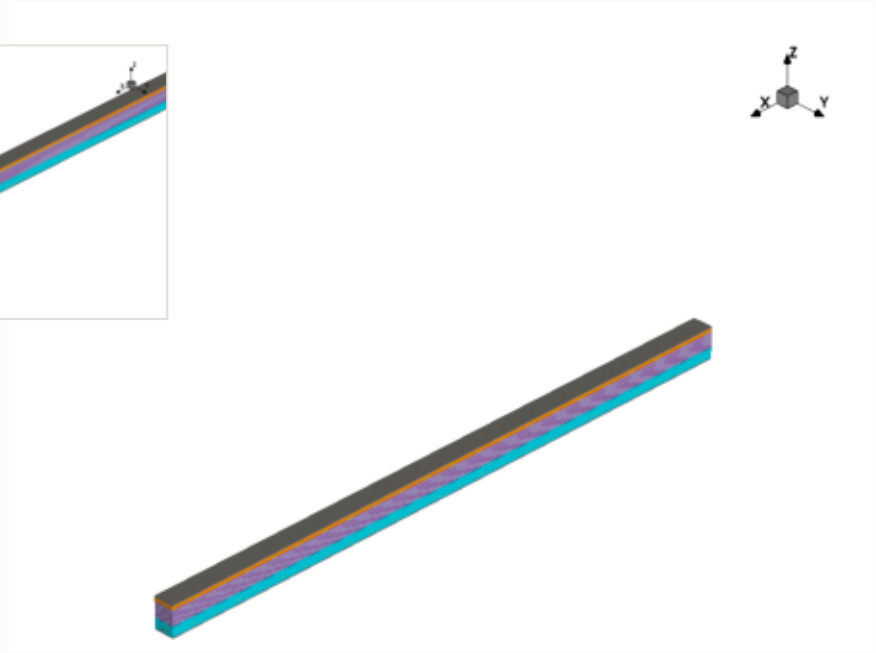
litho_patterning · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





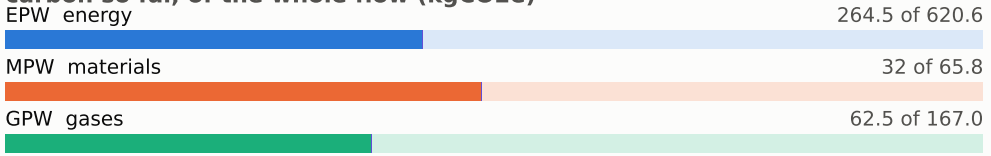
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

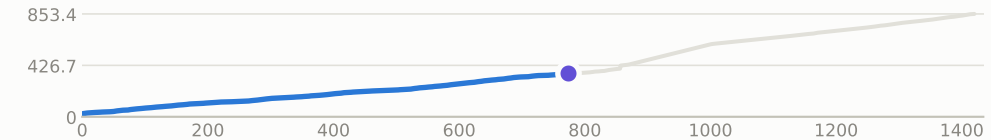
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

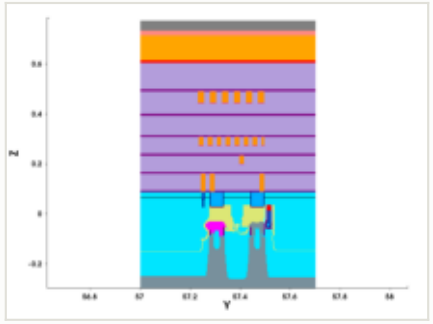
carbon so far, of the whole flow (kgCO2e)



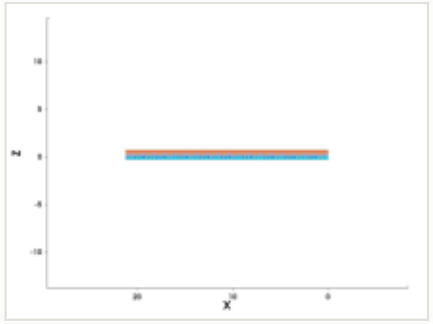
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm

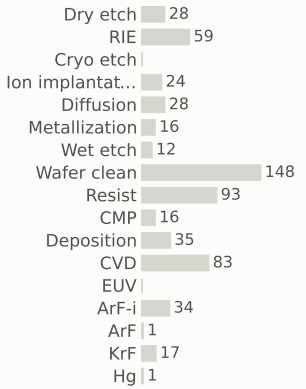


y cut at 57.351 µm



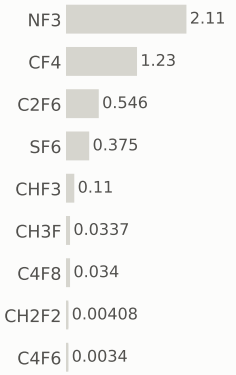
mask: layer 12551:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



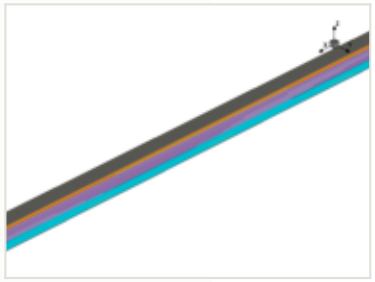
Step #7880: Spx_only

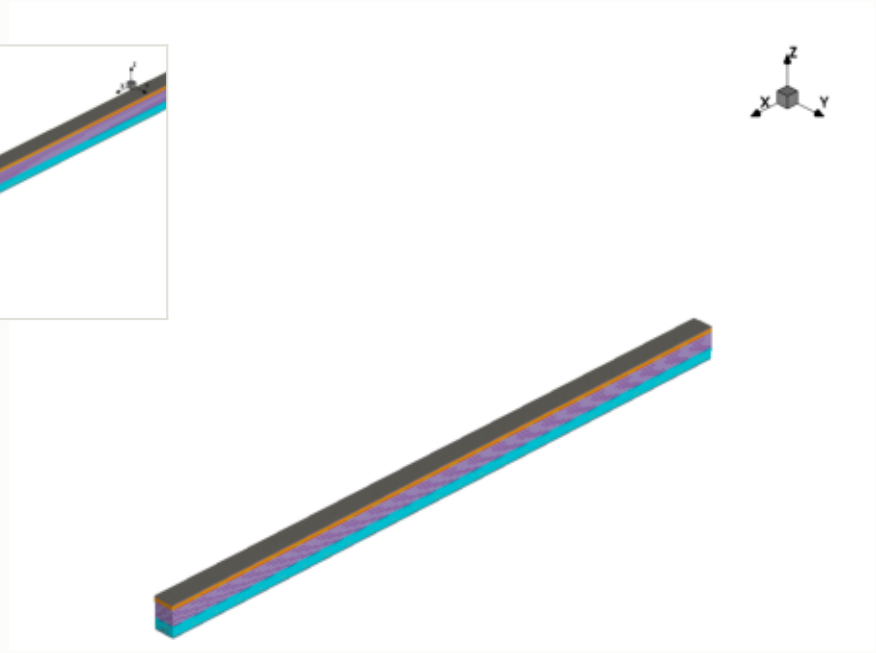
etch_anisotropic · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





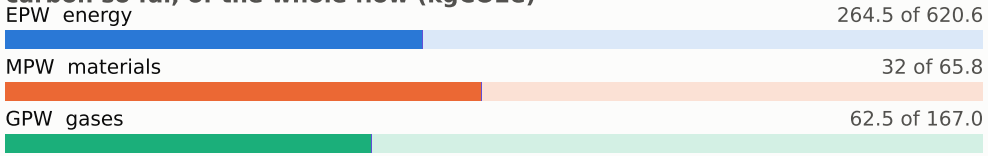
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

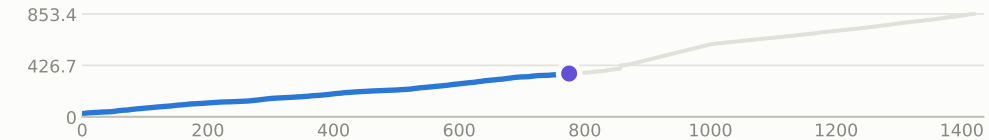
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

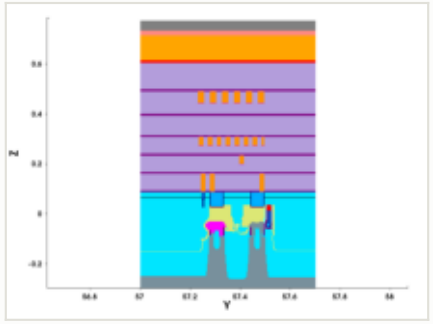
carbon so far, of the whole flow (kgCO2e)



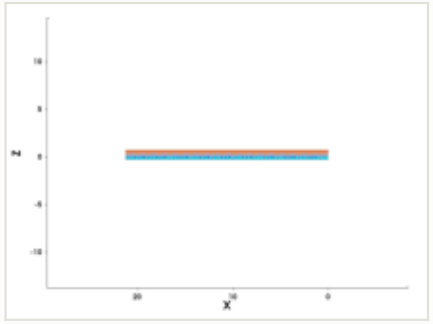
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



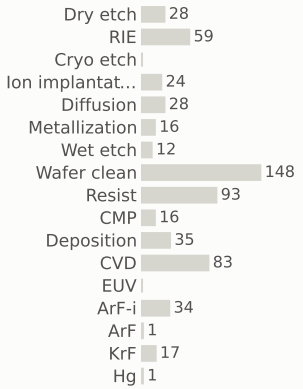
y cut at 57.351 µm



mask for this step

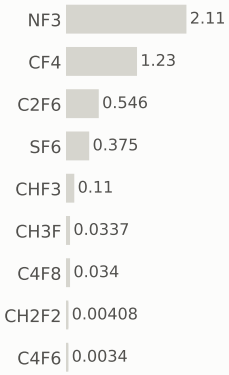
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



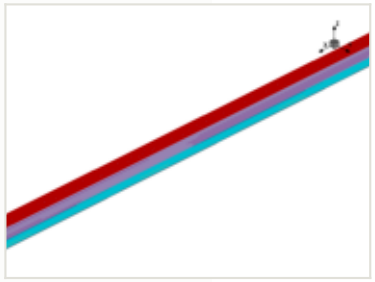
Step #7890: Spx_only

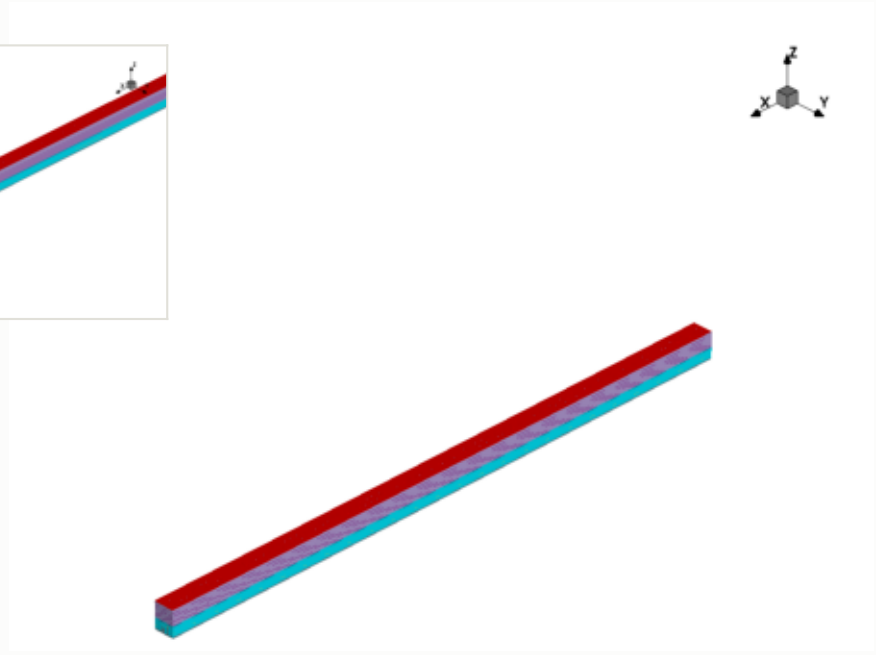
etch_strip · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





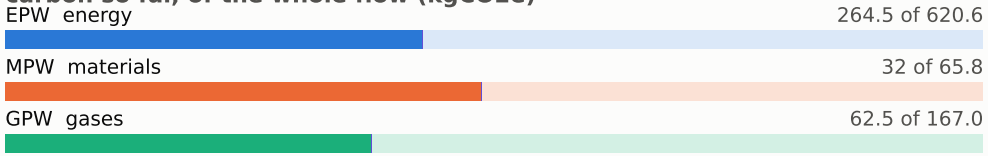
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

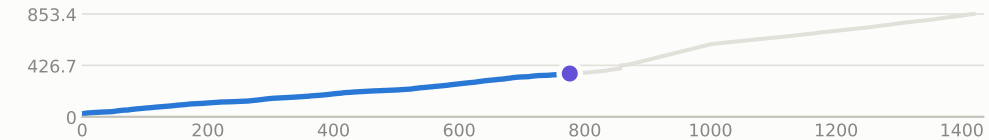
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

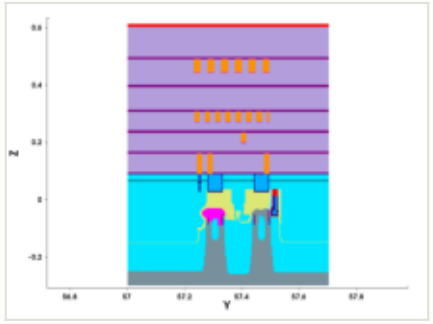
carbon so far, of the whole flow (kgCO2e)



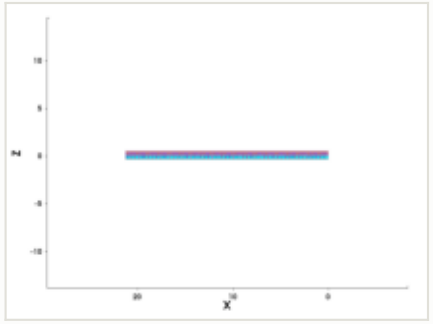
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



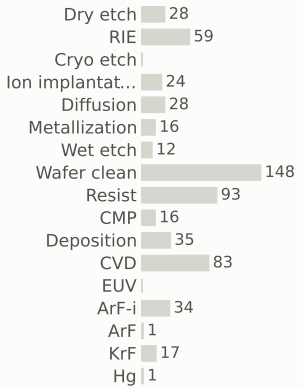
y cut at 57.351 µm



mask for this step

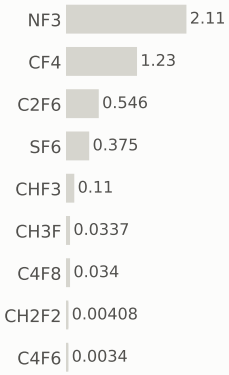
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



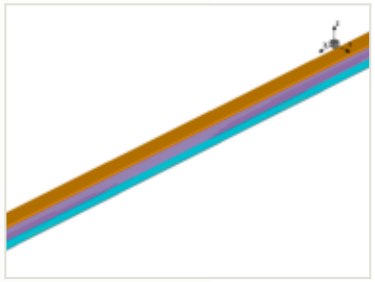
Step #7900: Spx_only

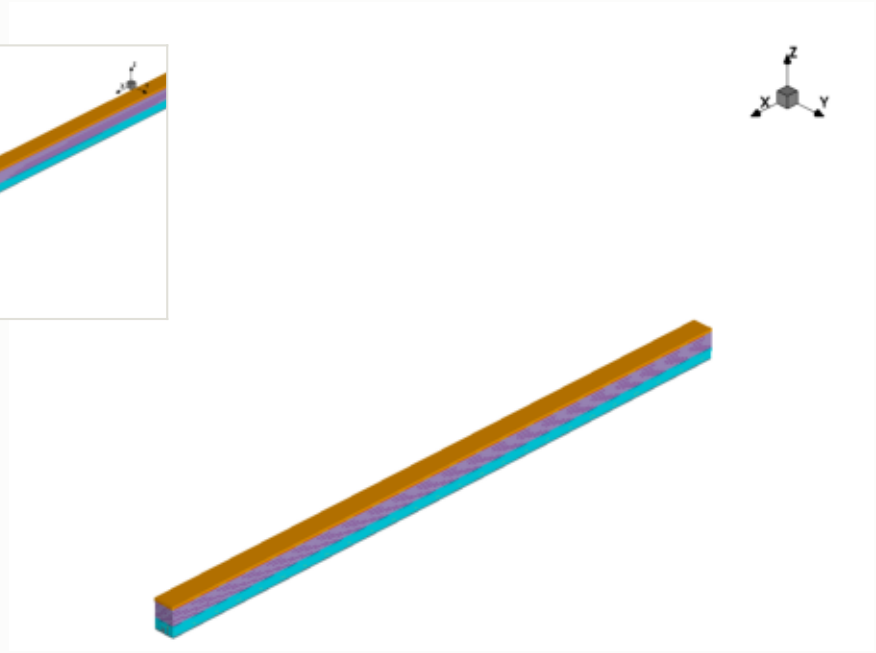
depo_fill · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





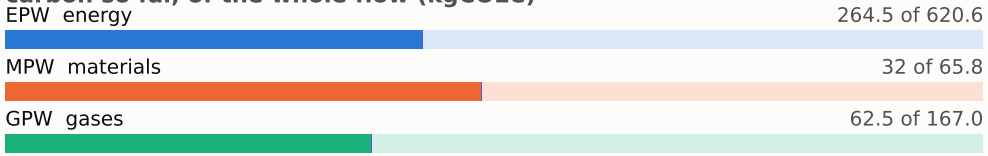
this step

process area	Spx_only
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

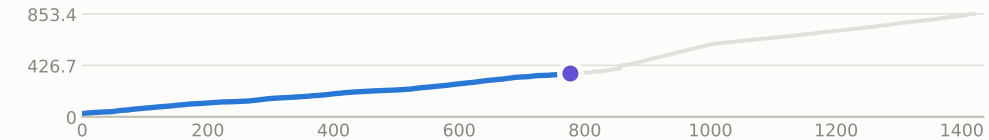
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

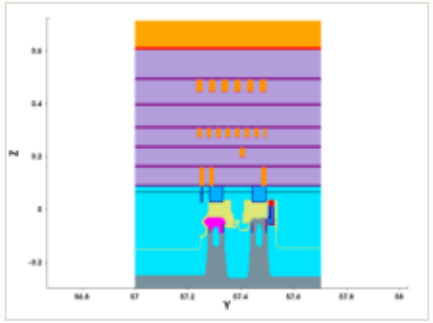
carbon so far, of the whole flow (kgCO2e)



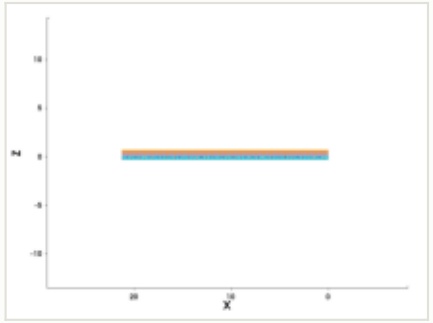
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



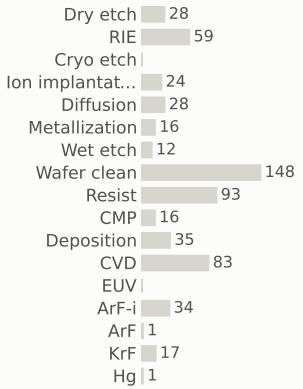
y cut at 57.351 µm



mask for this step

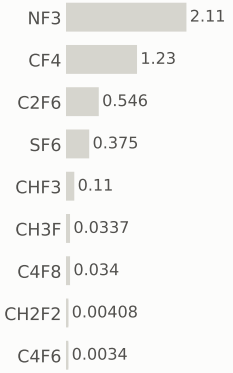
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



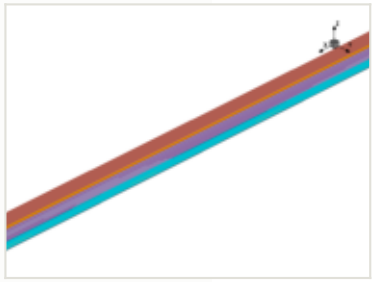
Step #7910: Spx_only

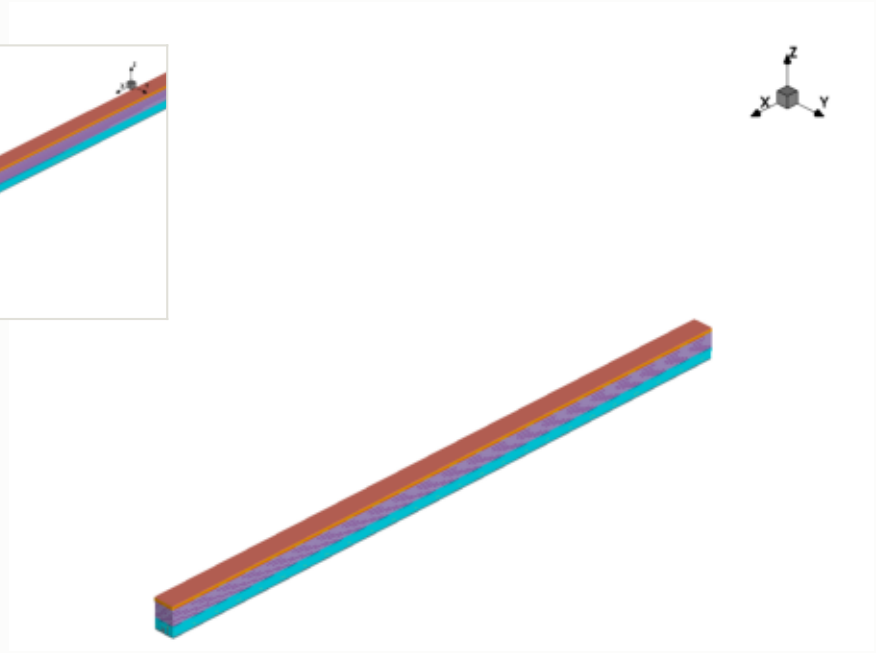
depo_fill · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





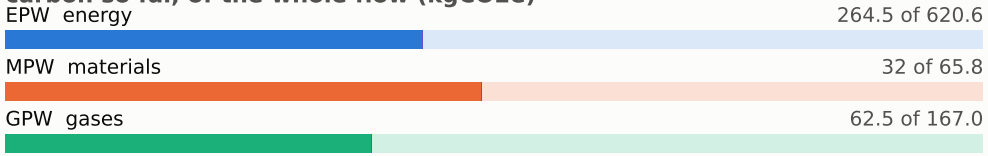
this step

process area	Spx_only
material	—
thickness	0.02 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

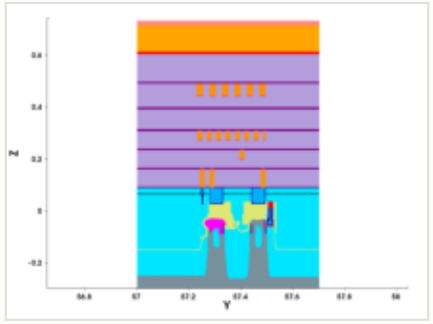
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

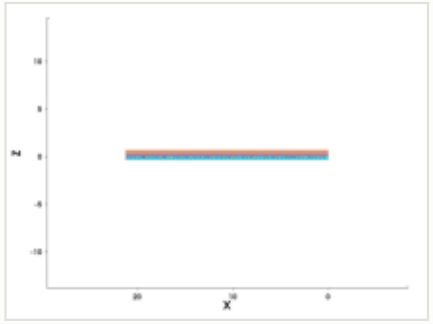


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



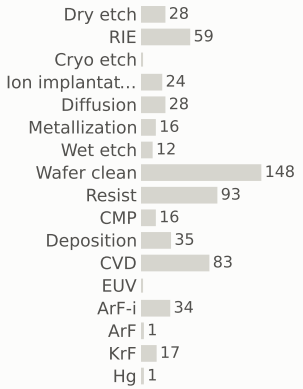
y cut at 57.351 µm



mask for this step

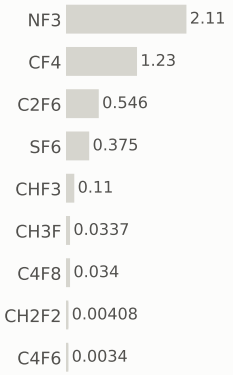
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



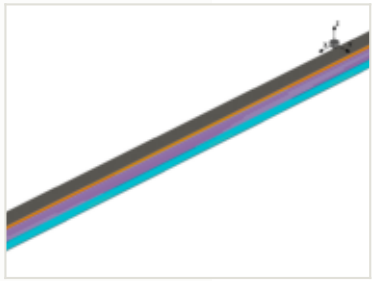
Step #7920: Spx_only

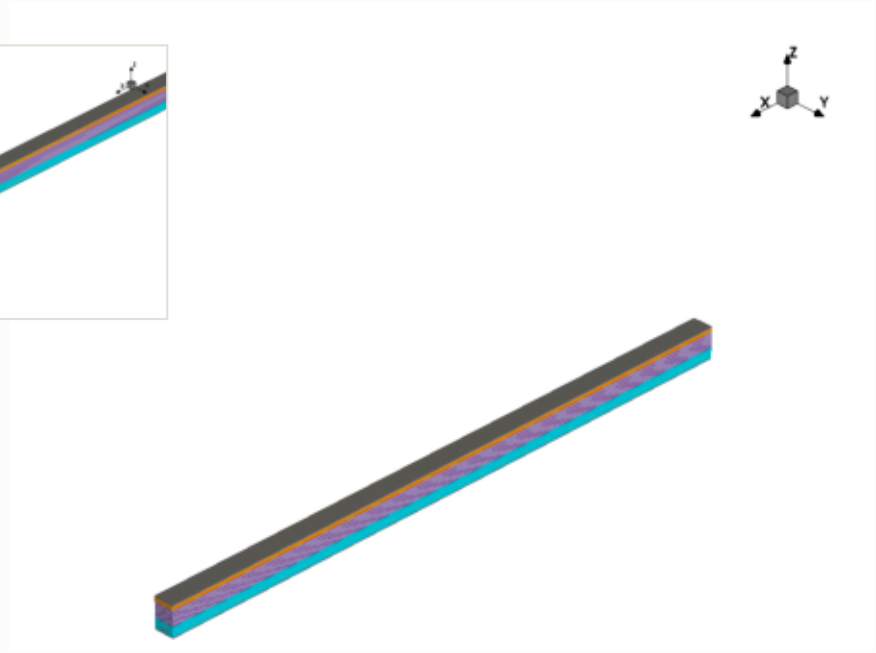
litho_patterning · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





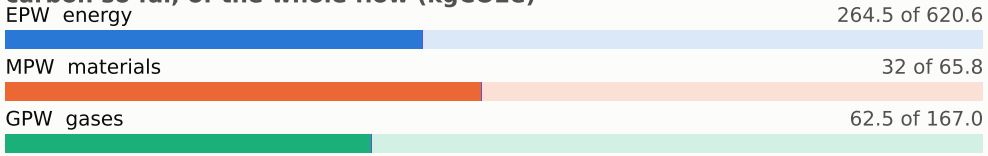
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

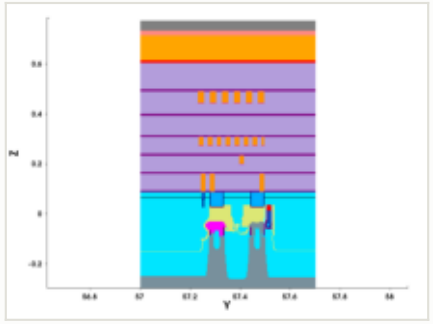
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

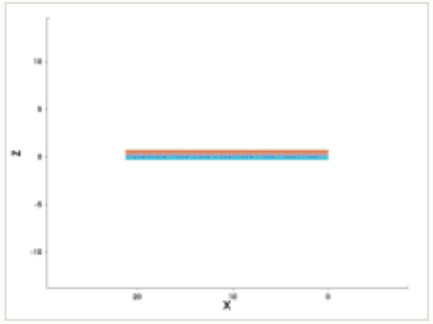


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

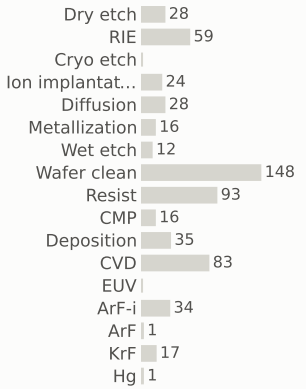


y cut at 57.351 µm



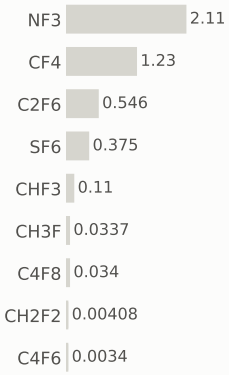
mask: layer 12552:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



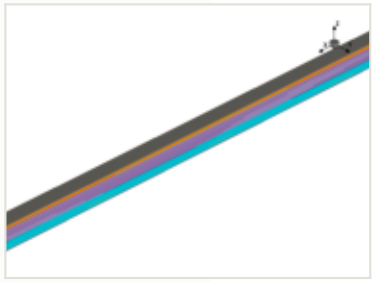
Step #7930: Spx_only

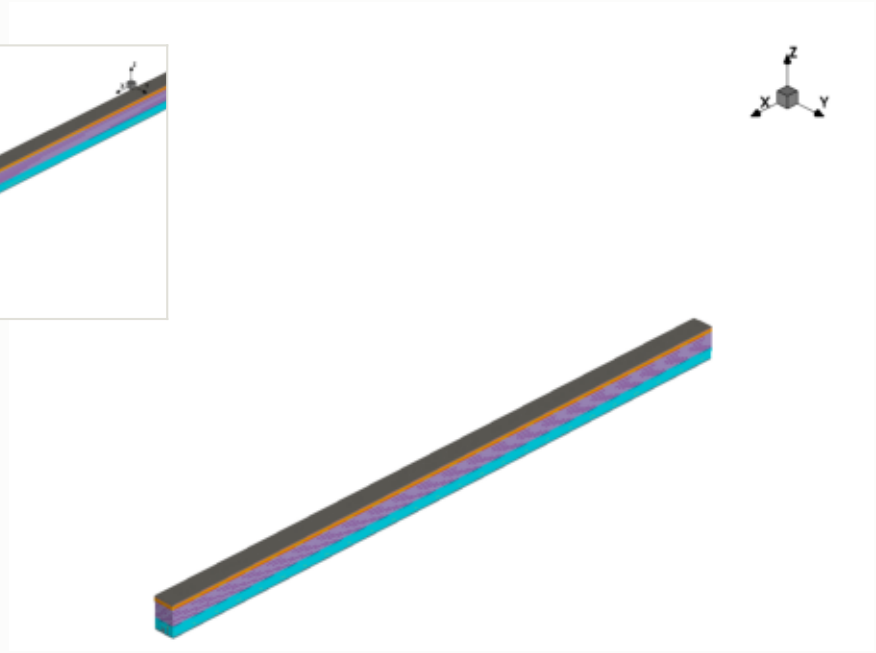
etch_anisotropic · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





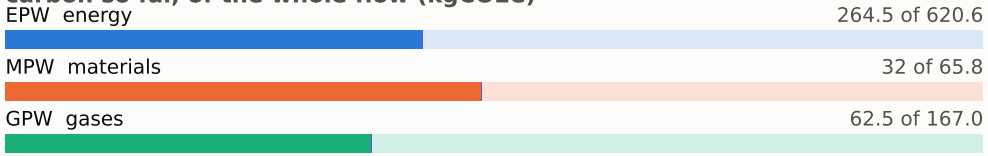
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

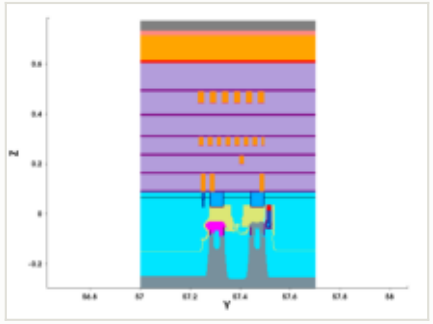
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

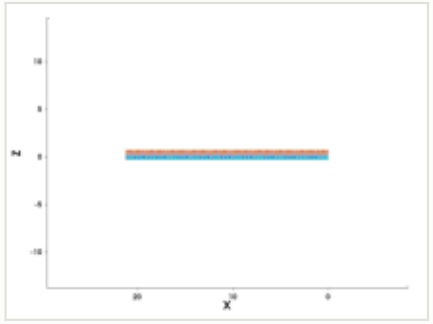


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



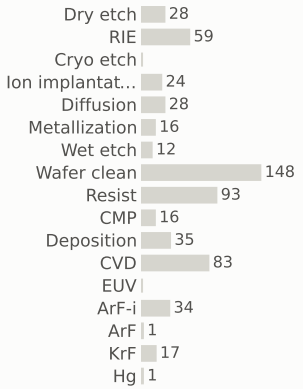
y cut at 57.351 µm



mask for this step

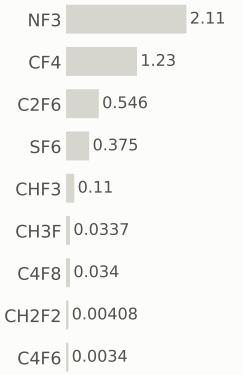
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



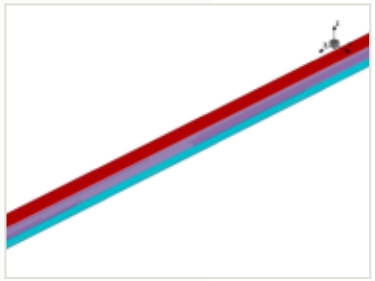
Step #7940: Spx_only

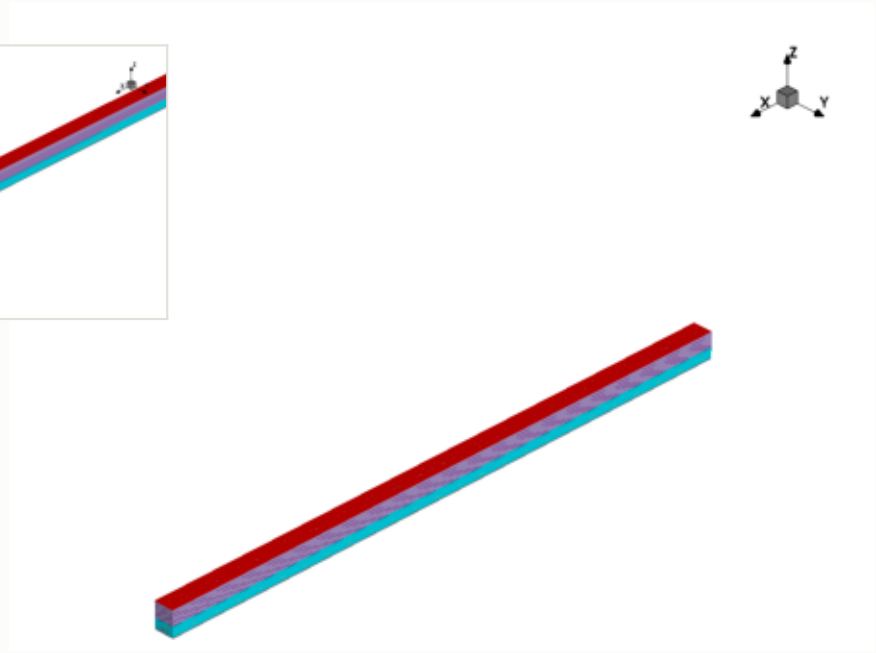
etch_strip · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





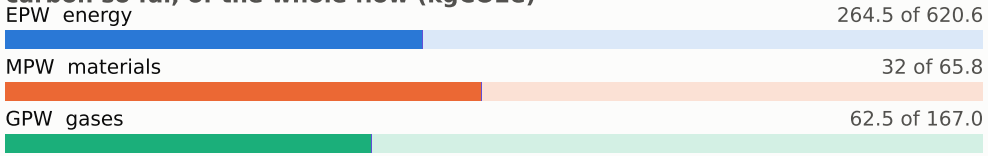
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

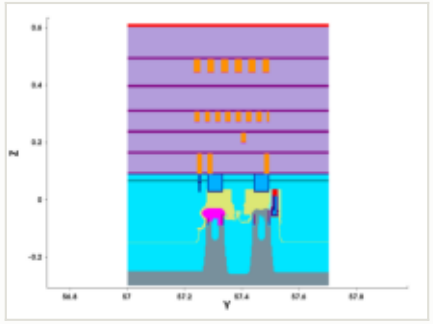
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

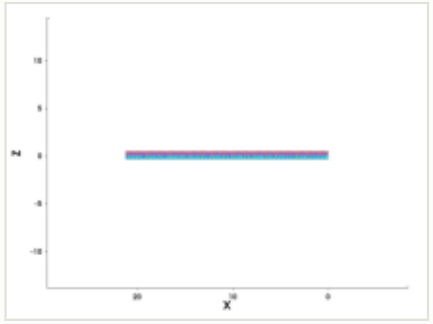


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



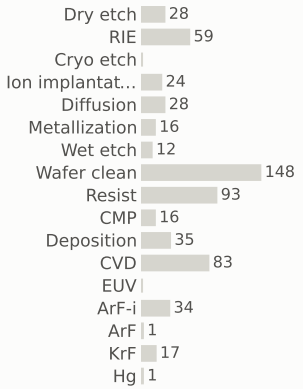
y cut at 57.351 µm



mask for this step

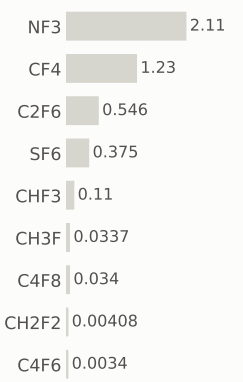
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



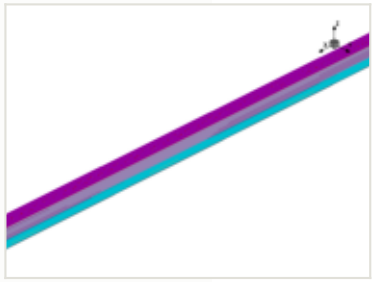
Step #7950: Spx_only

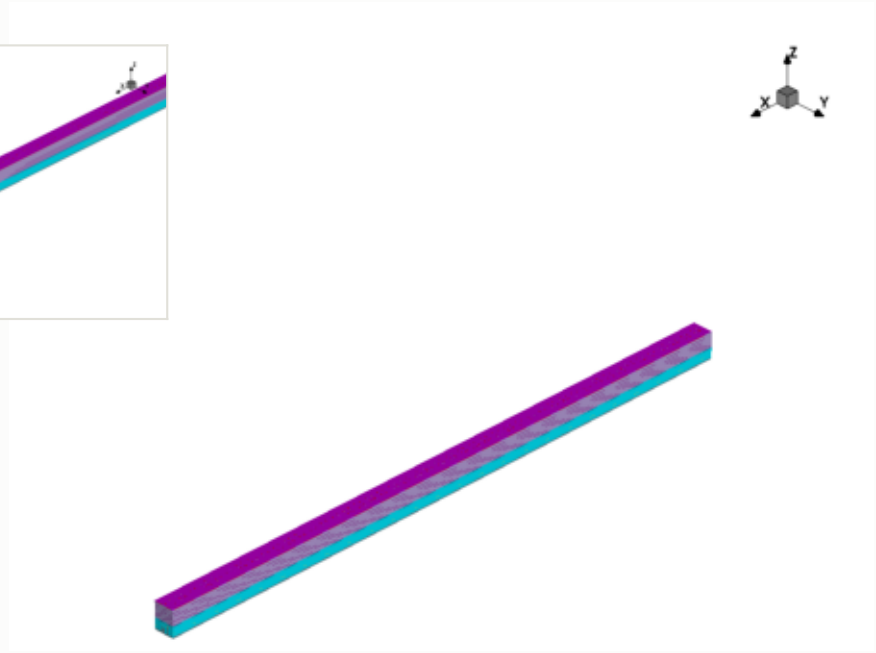
depo_ellipsoidal · Spx_only

4.00e-05	359.1	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



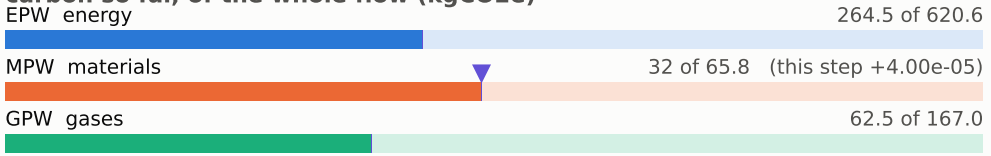


this step

process area	Spx_only	EPW	0 kgCO2e
material	TaN	EPW tool	0 kgCO2e
thickness	0.0016 µm	EPW overhead	0 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	4.00e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**4.00e-05 kgCO2e**

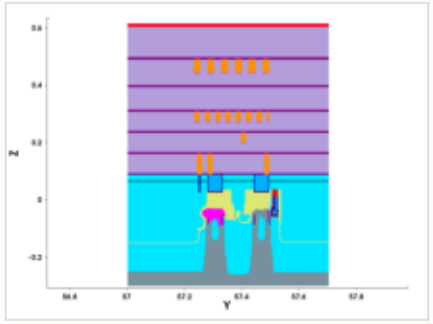
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

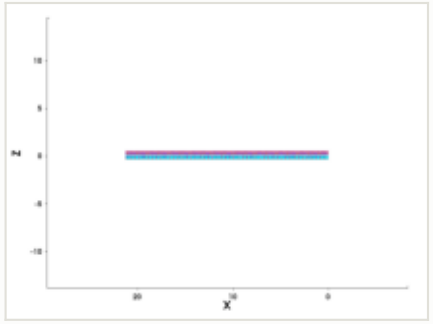


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



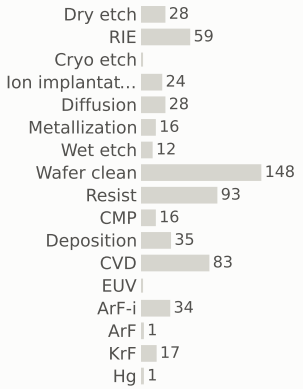
y cut at 57.351 µm



mask for this step

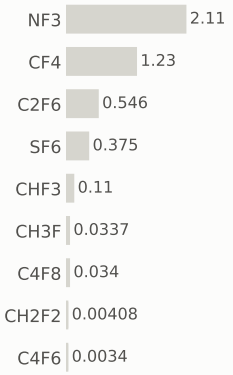
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



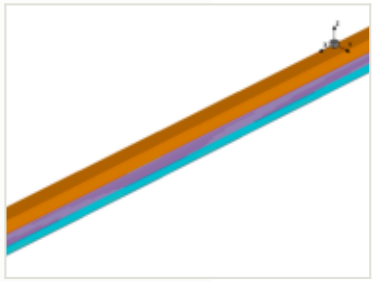
Step #7960: Spx_only

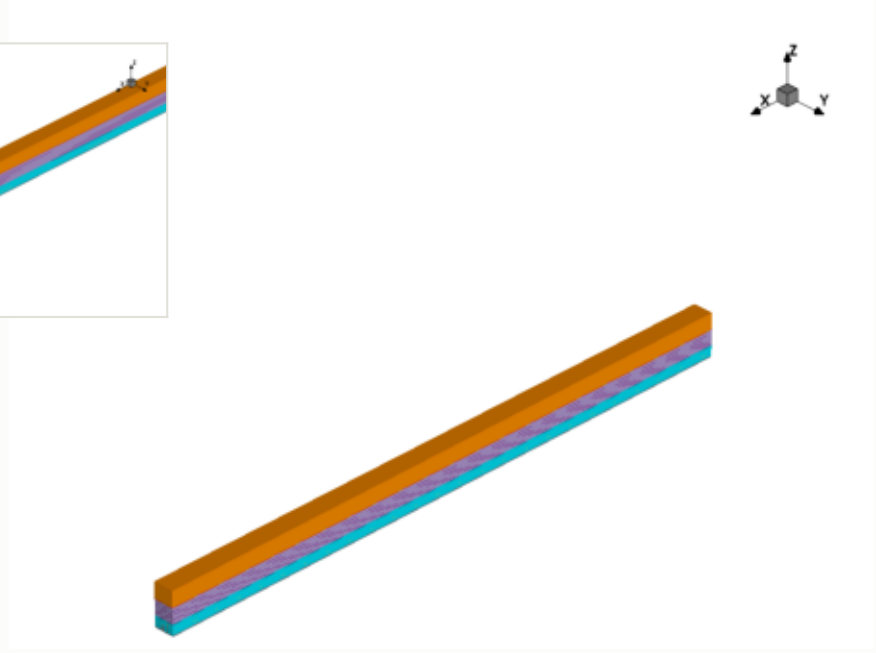
depo_fill · Spx_only

0.00212	359.1	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





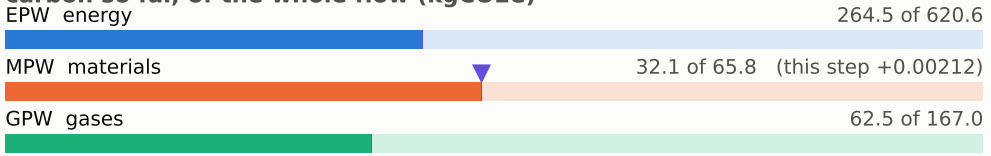
this step

process area	Spx_only
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.00212 kgCO2e**

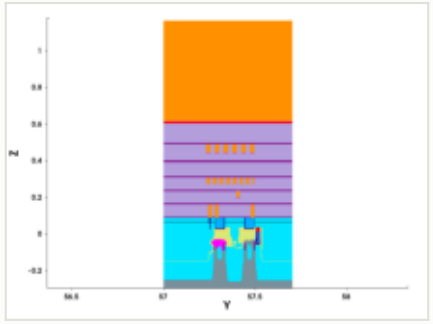
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

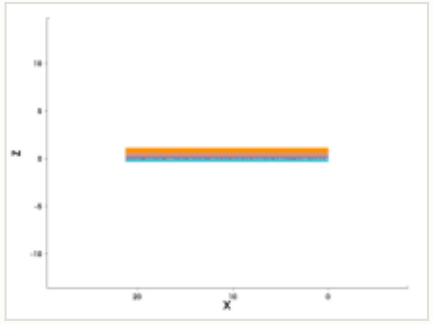


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



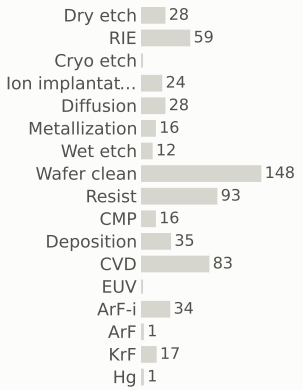
y cut at 57.351 µm



mask for this step

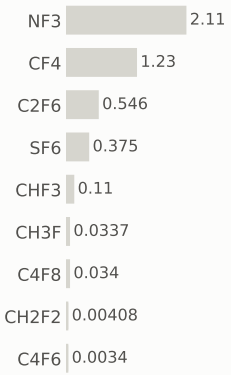
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



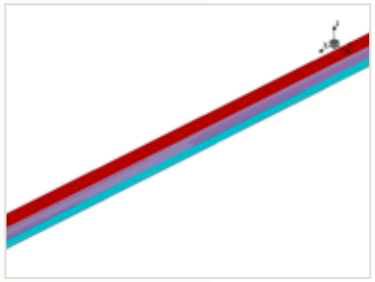
Step #7970: Spx_only

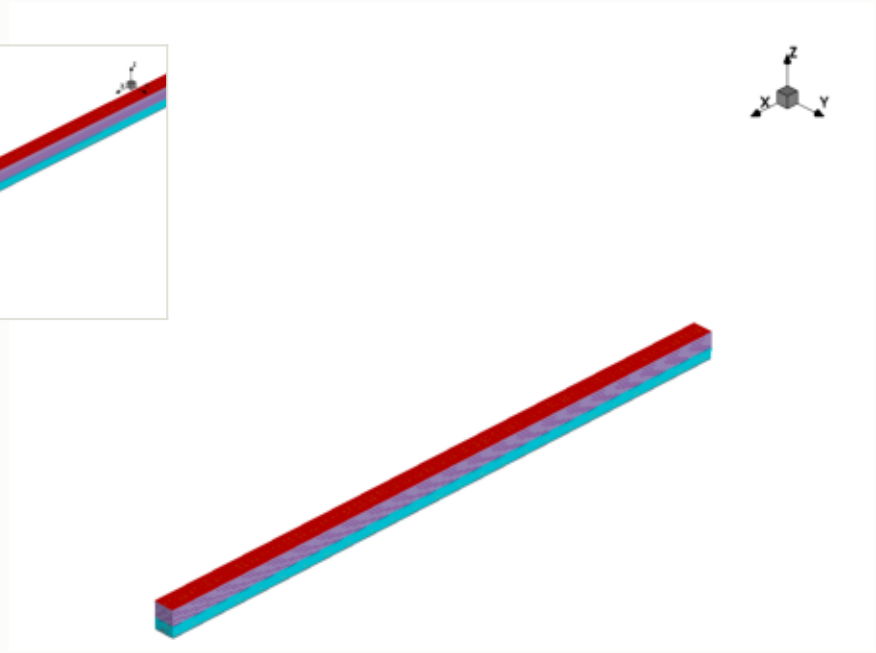
etch_cmp · Spx_only

0	359.1	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





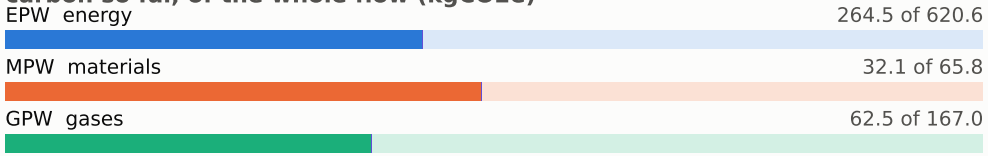
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

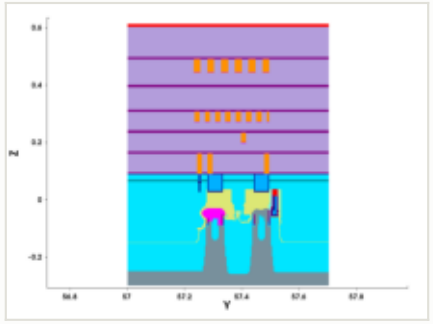
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

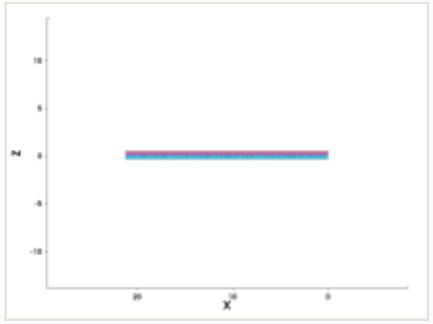


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



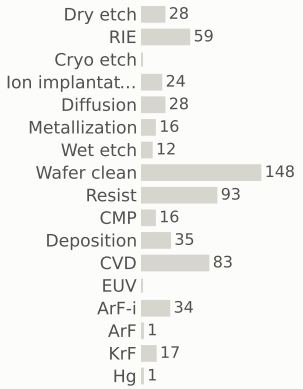
y cut at 57.351 µm



mask for this step

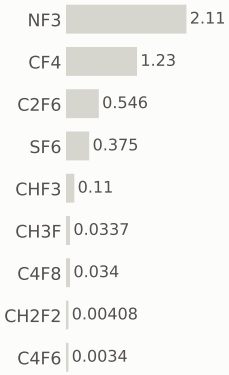
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



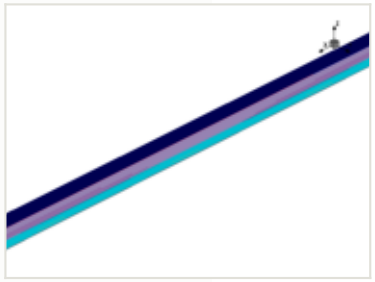
Step #7980: Deposition

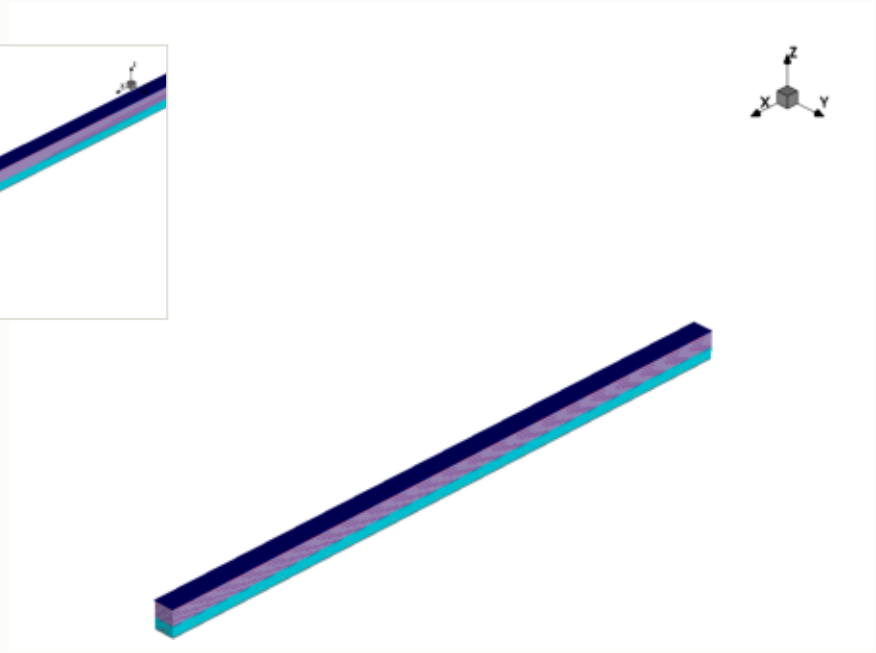
depo_fill · Deposition

0.657	359.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



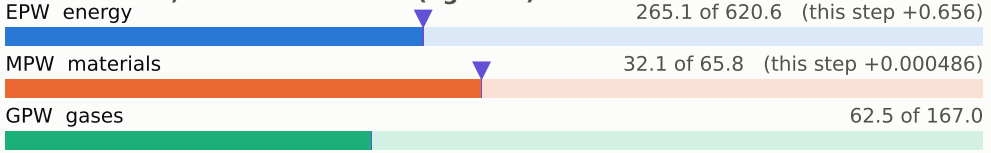


this step

process area	Deposition
material	TiN
thickness	0.03 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

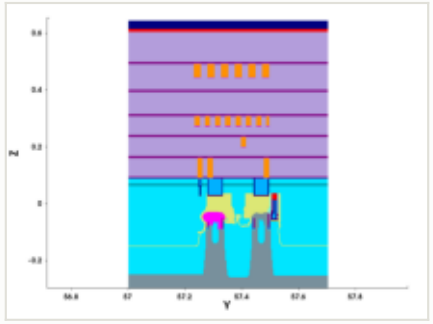
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000486 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

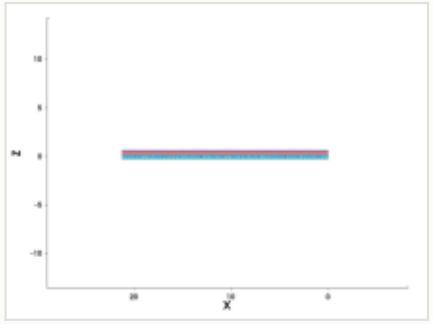


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



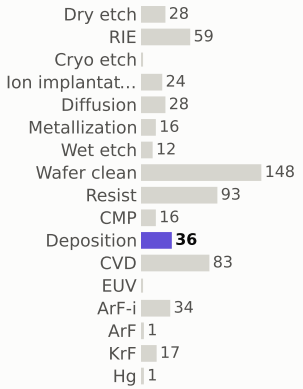
y cut at 57.351 µm



mask for this step

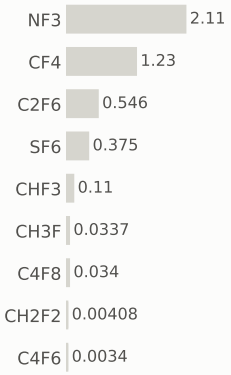
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



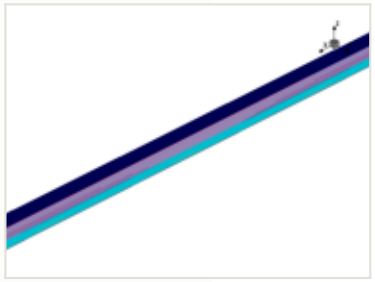
Step #7990: Wafer_clean

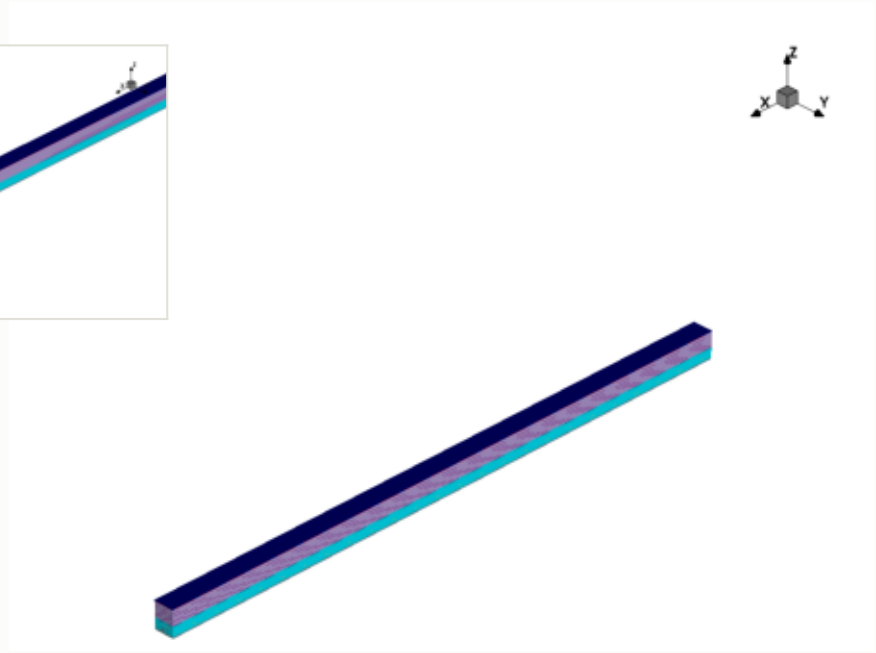
clean · Wafer_clean

0.169	359.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





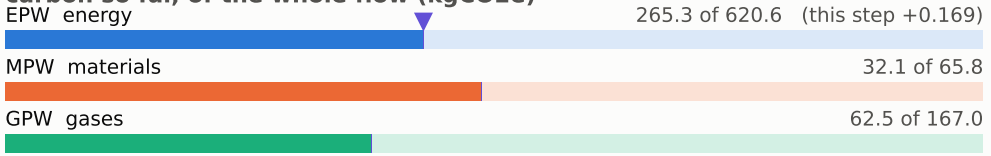
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

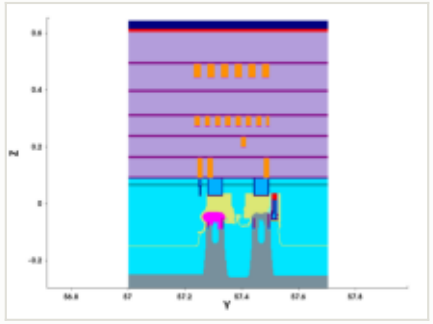
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

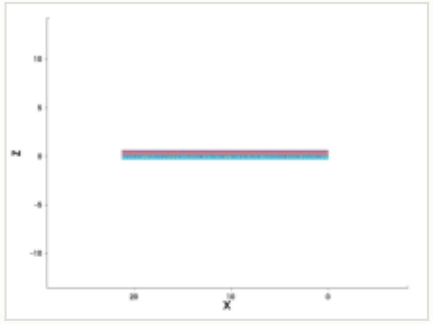


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



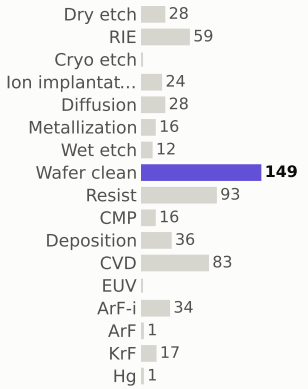
y cut at 57.351 µm



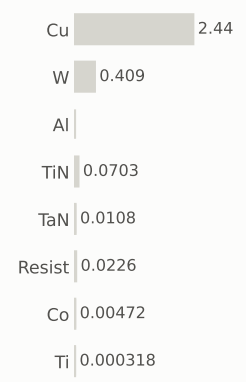
mask for this step

no mask for this step

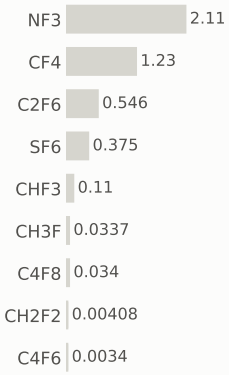
steps by process area



materials deposited (g)



gases emitted (g)



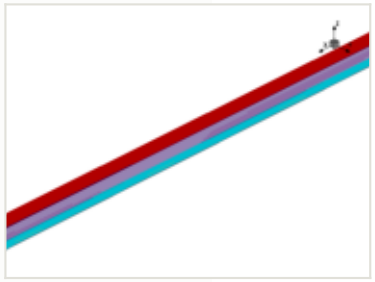
Step #8000: CVD

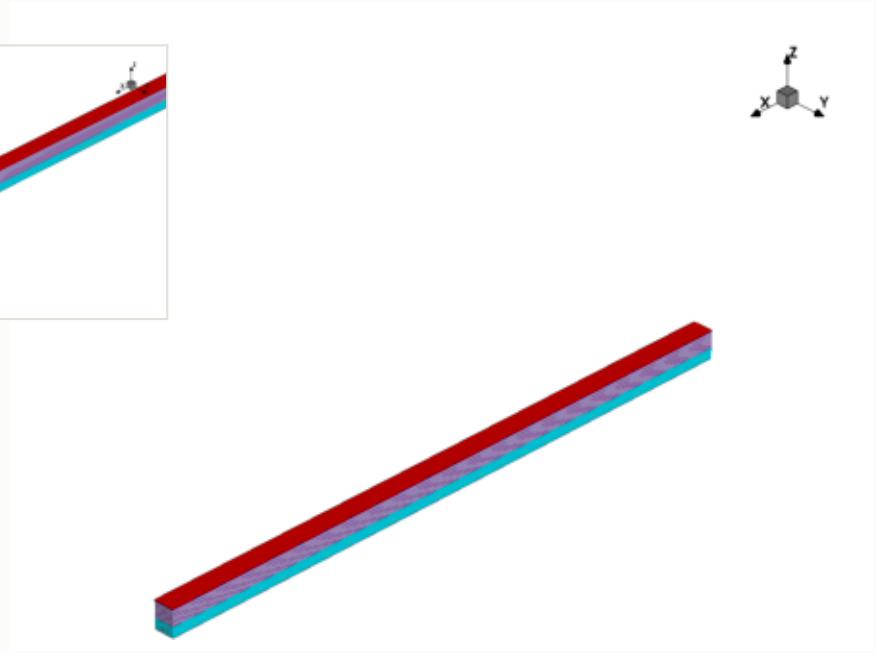
depo_fill · CVD

1.31	361.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



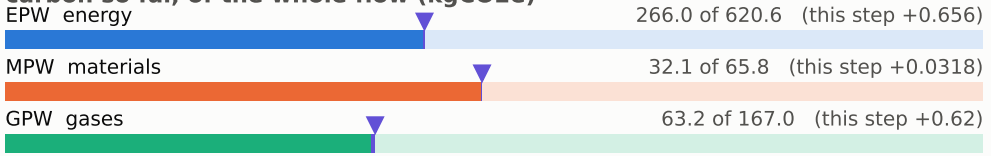


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

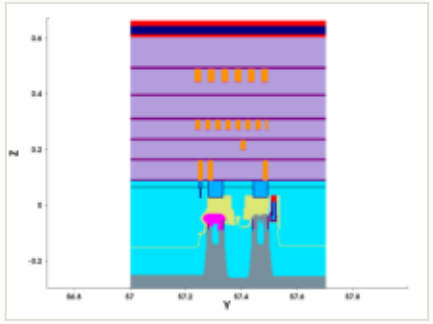
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

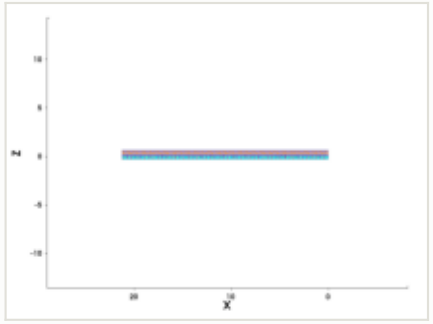


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



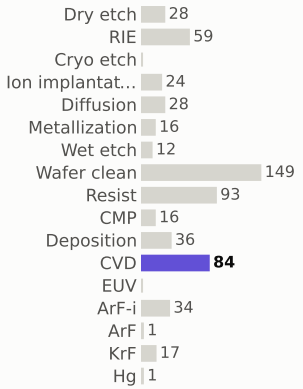
y cut at 57.351 µm



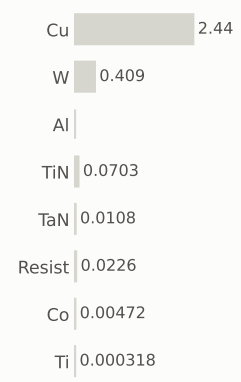
mask for this step

no mask for this step

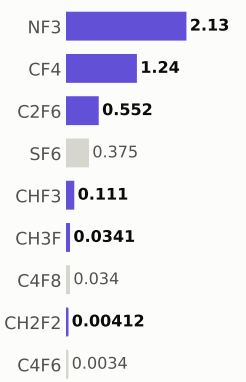
steps by process area



materials deposited (g)



gases emitted (g)



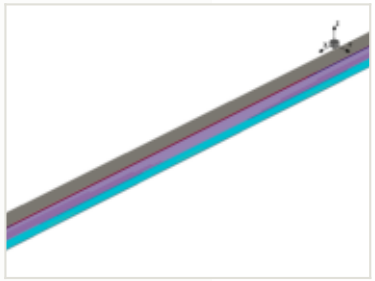
Step #8010: Spx_only

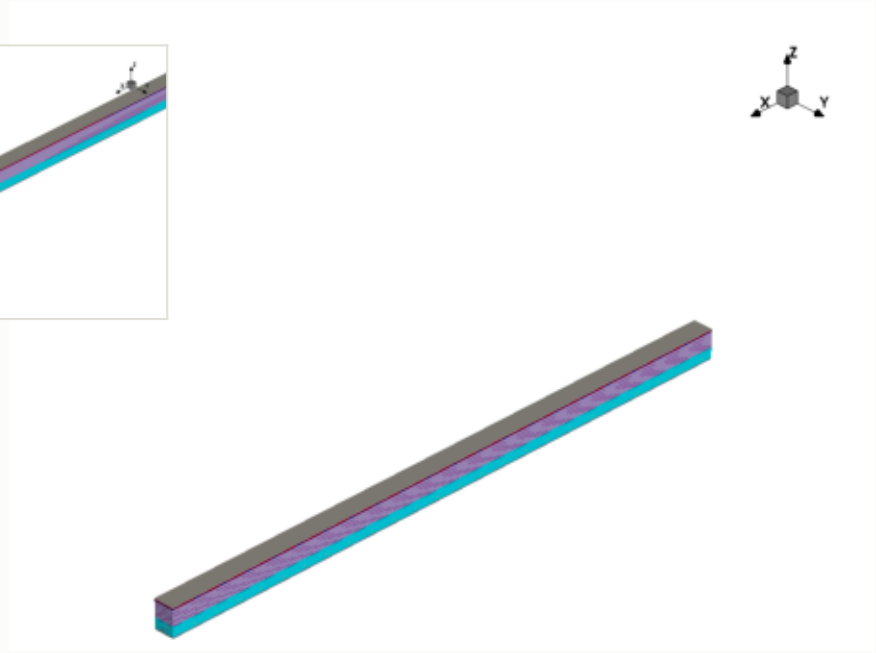
depo_fill · Spx_only

0	361.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





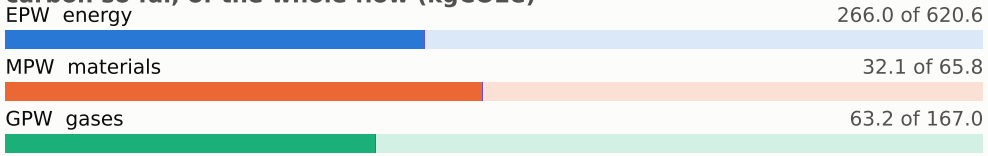
this step

process area	Spx_only
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

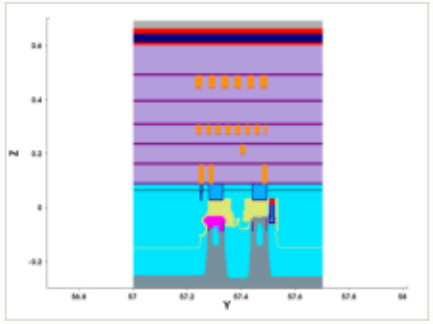
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

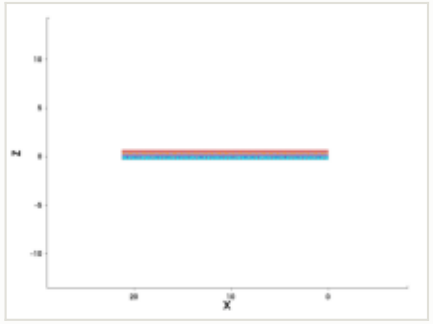


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



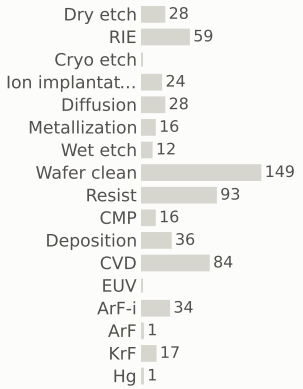
y cut at 57.351 µm



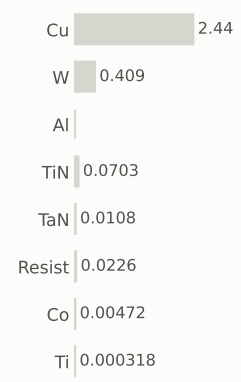
mask for this step

no mask for this step

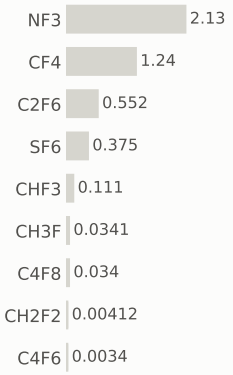
steps by process area



materials deposited (g)



gases emitted (g)



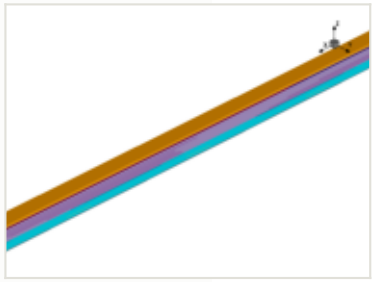
Step #8020: Resist

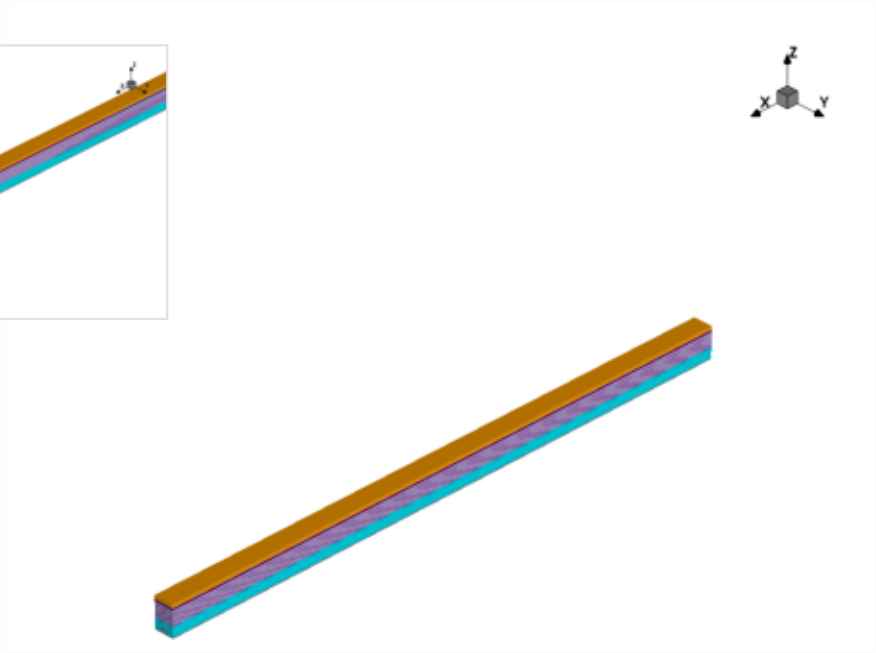
depo_fill · Resist

0.169	361.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



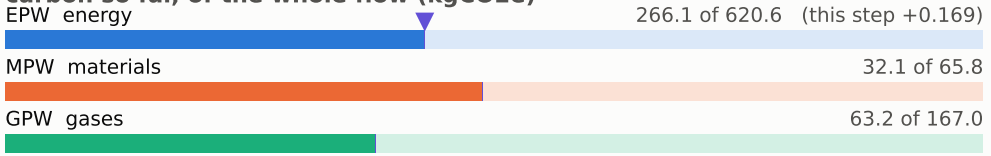


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

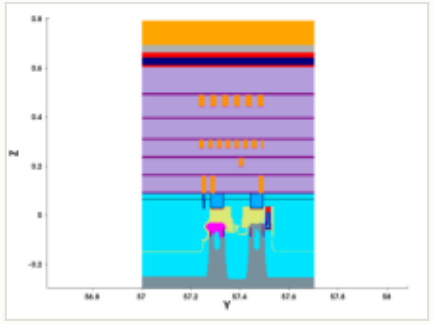
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

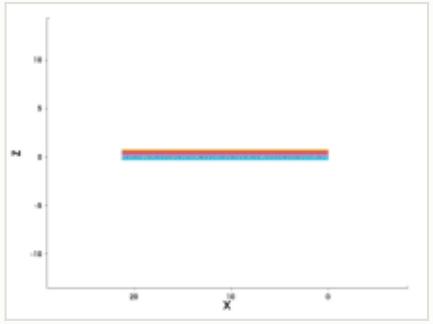


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



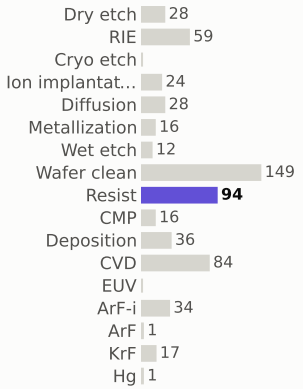
y cut at 57.351 µm



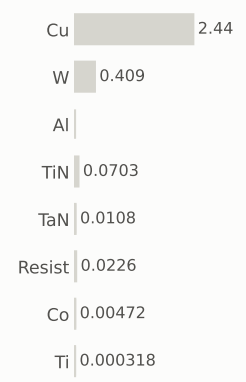
mask for this step

no mask for this step

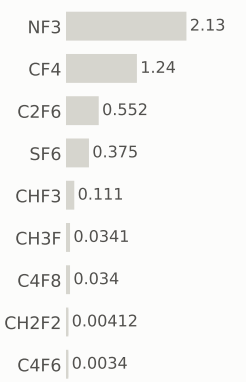
steps by process area



materials deposited (g)



gases emitted (g)



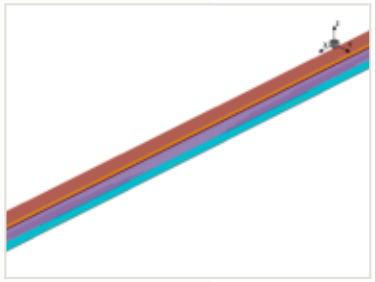
Step #8030: CVD

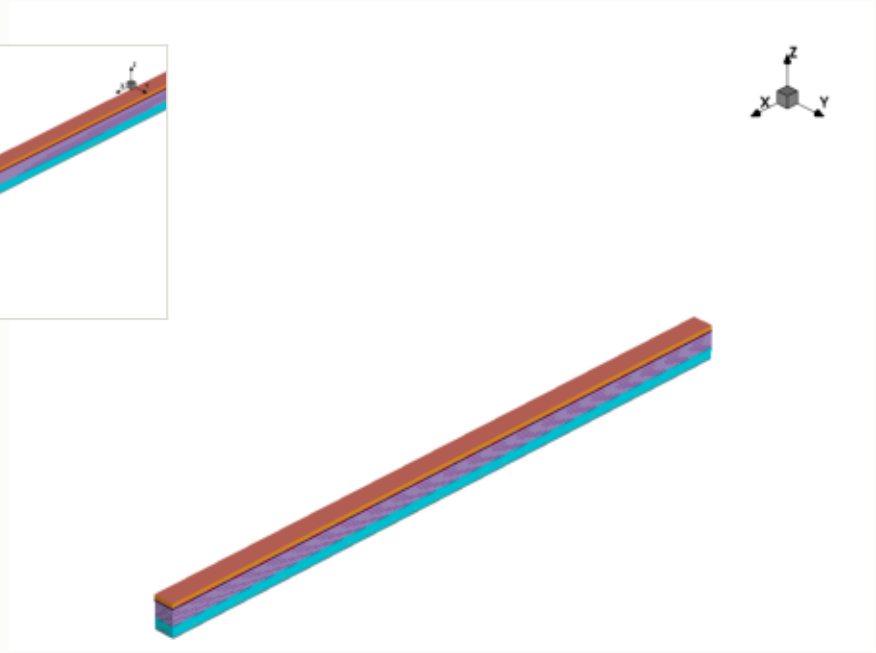
depo_fill · CVD

1.31	362.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



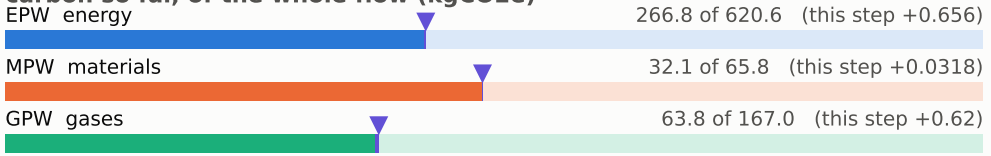


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

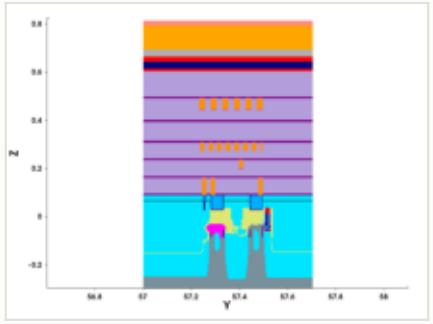
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

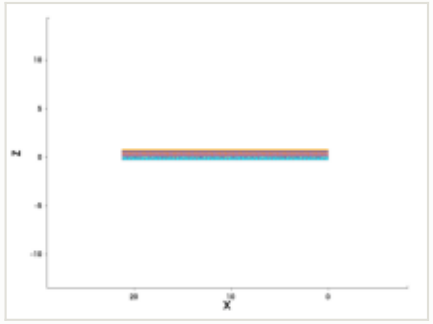


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



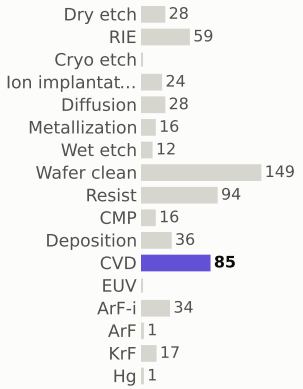
y cut at 57.351 µm



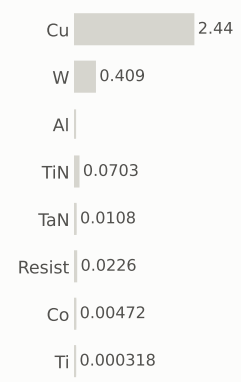
mask for this step

no mask for this step

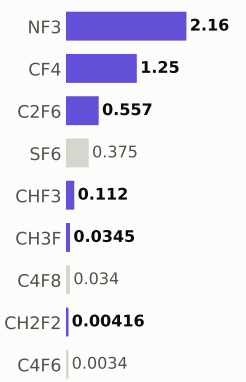
steps by process area



materials deposited (g)



gases emitted (g)



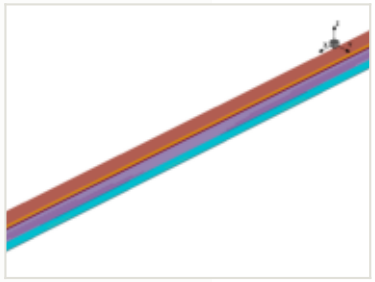
Step #8040: Wafer_clean

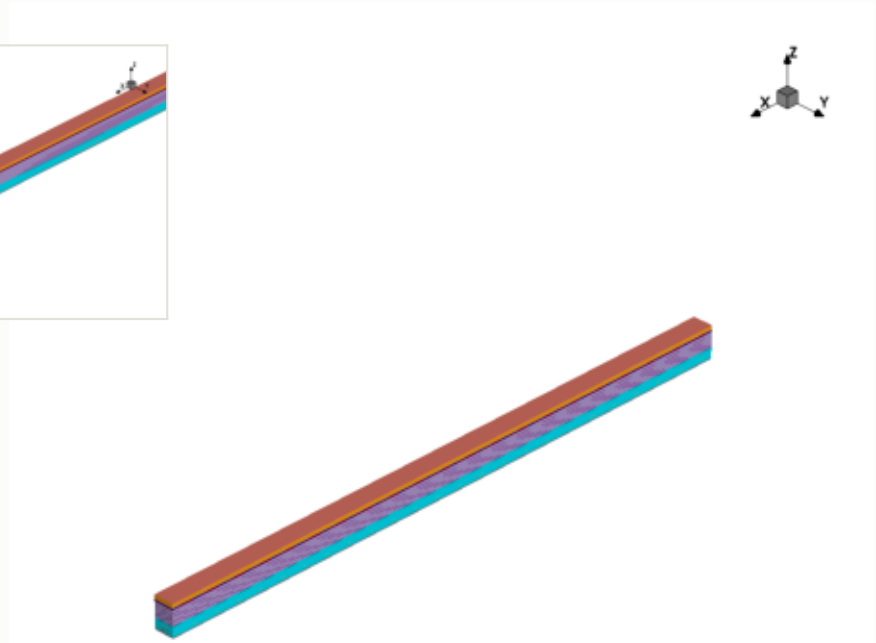
clean · Wafer_clean

0.169	362.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



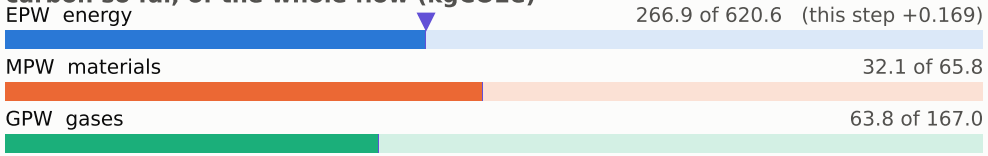


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

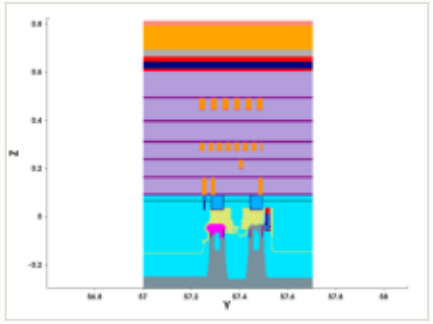
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

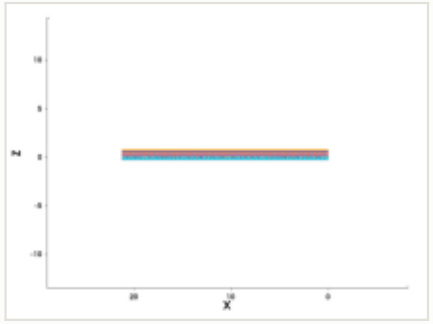


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



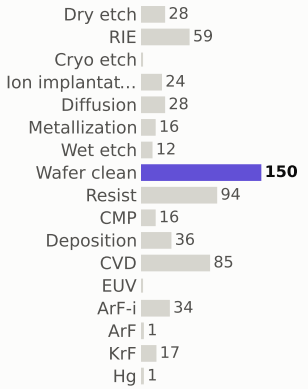
y cut at 57.351 µm



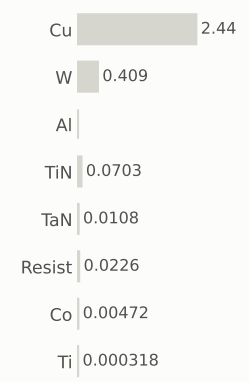
mask for this step

no mask for this step

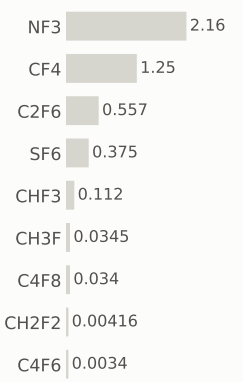
steps by process area



materials deposited (g)



gases emitted (g)



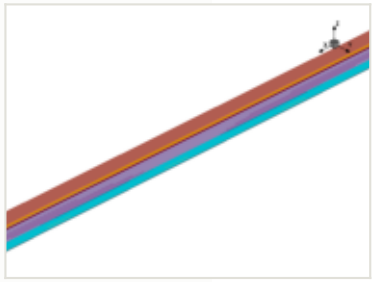
Step #8050: Resist

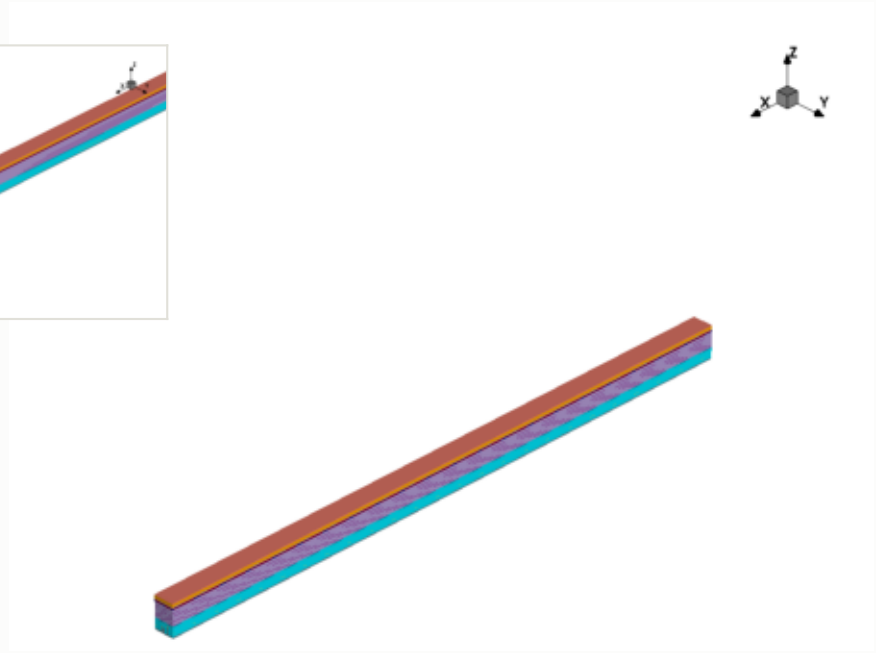
resist · Resist

0.169	363.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





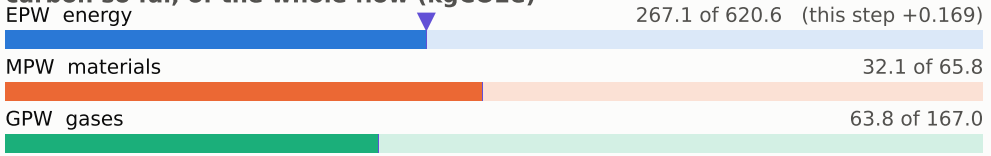
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

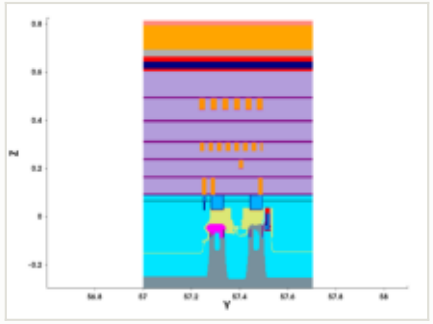
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

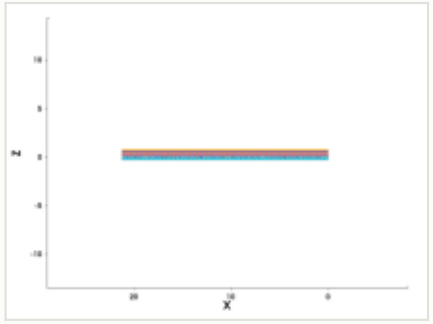


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



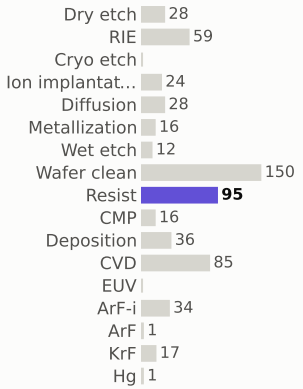
y cut at 57.351 µm



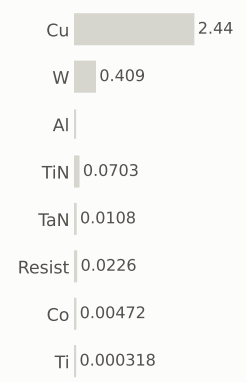
mask for this step

no mask for this step

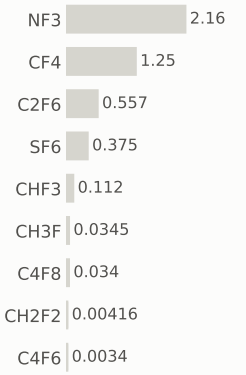
steps by process area



materials deposited (g)



gases emitted (g)



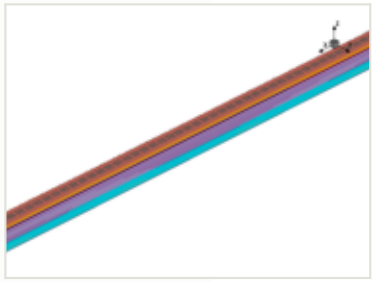
Step #8060: ArF-i

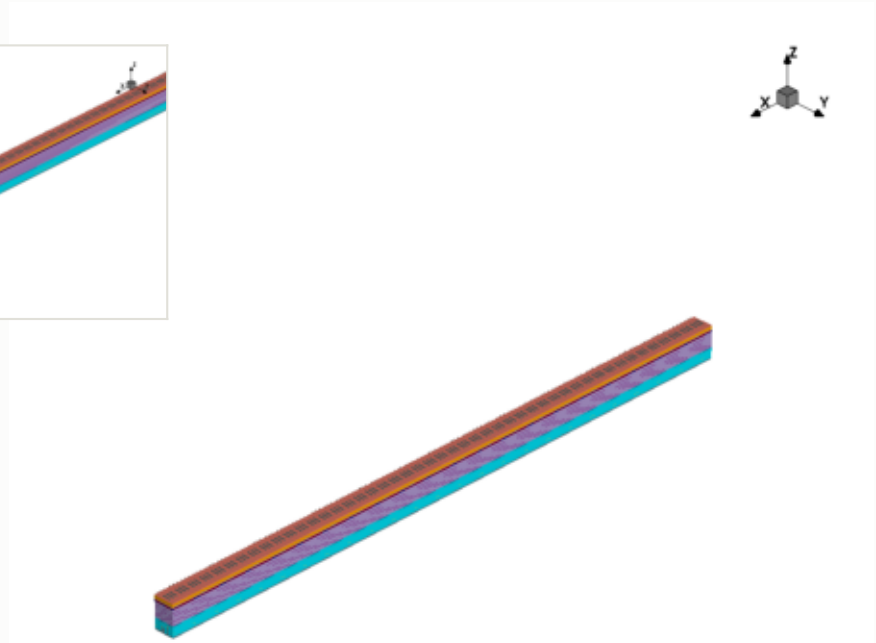
litho_patterning · ArF-i

0.217	363.2	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



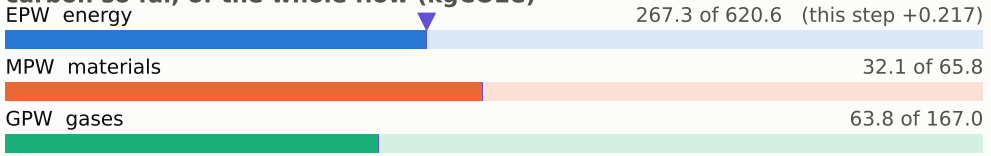


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

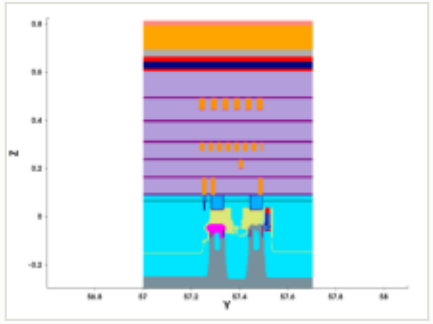
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

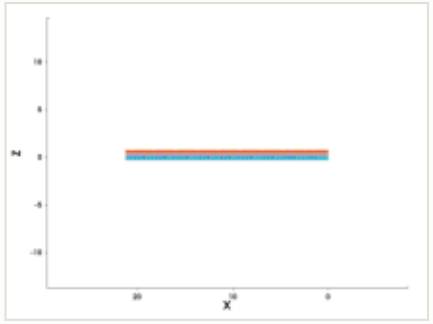


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

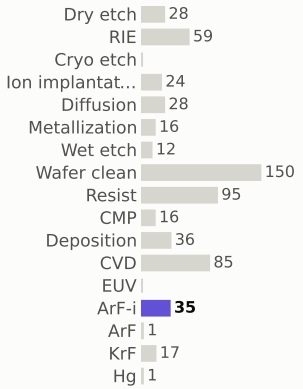


y cut at 57.351 µm



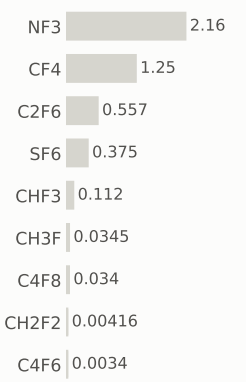
mask: layer 12601:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



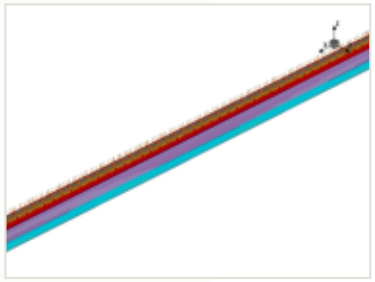
Step #8070: RIE

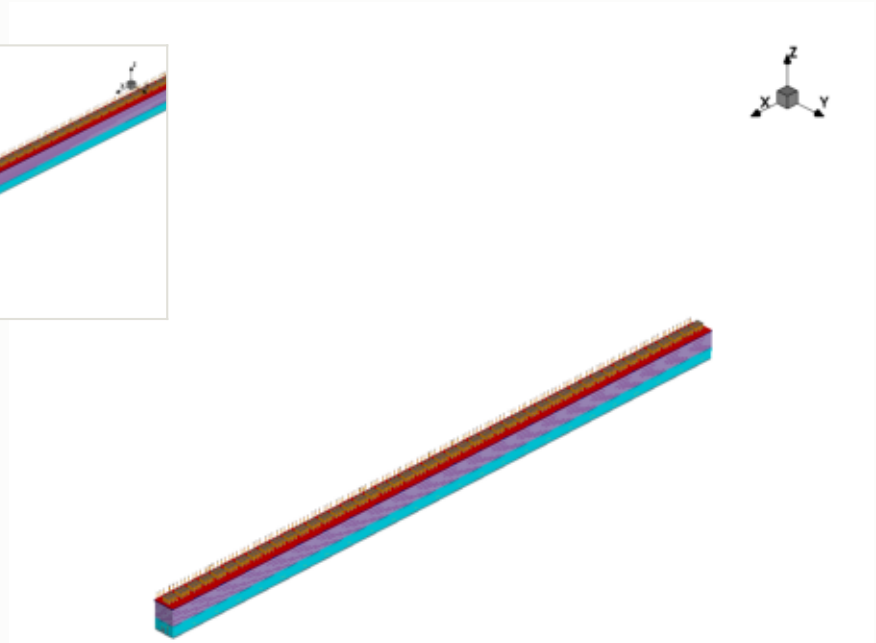
etch_anisotropic · RIE

1.04	364.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



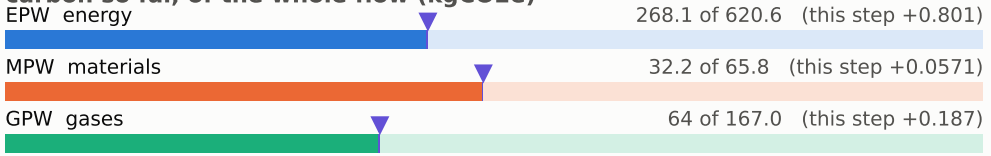


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

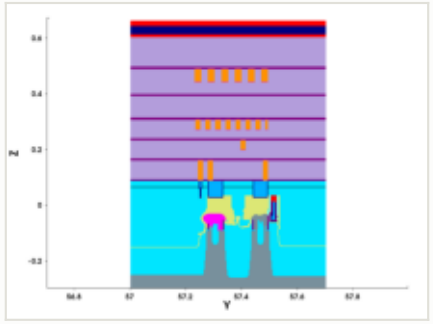
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

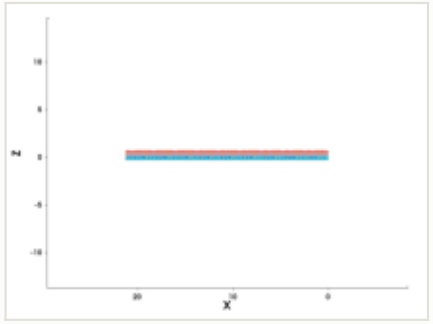


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



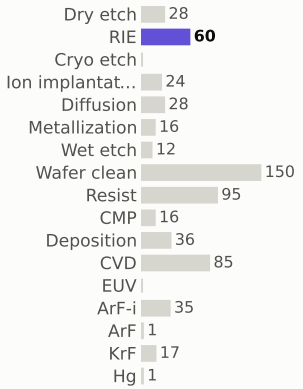
y cut at 57.351 µm



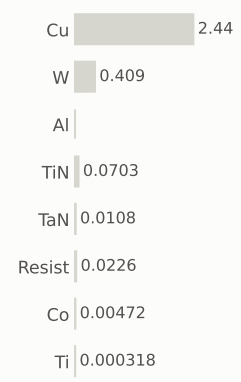
mask for this step

no mask for this step

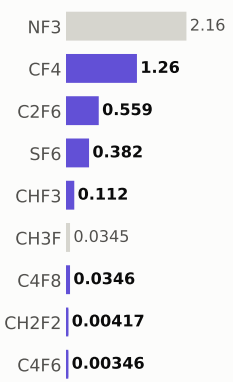
steps by process area



materials deposited (g)



gases emitted (g)



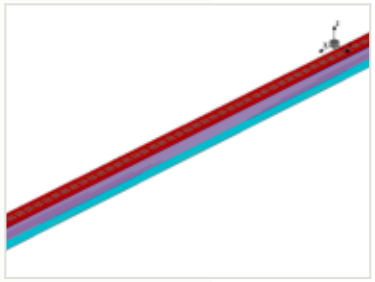
Step #8080: Spx_only

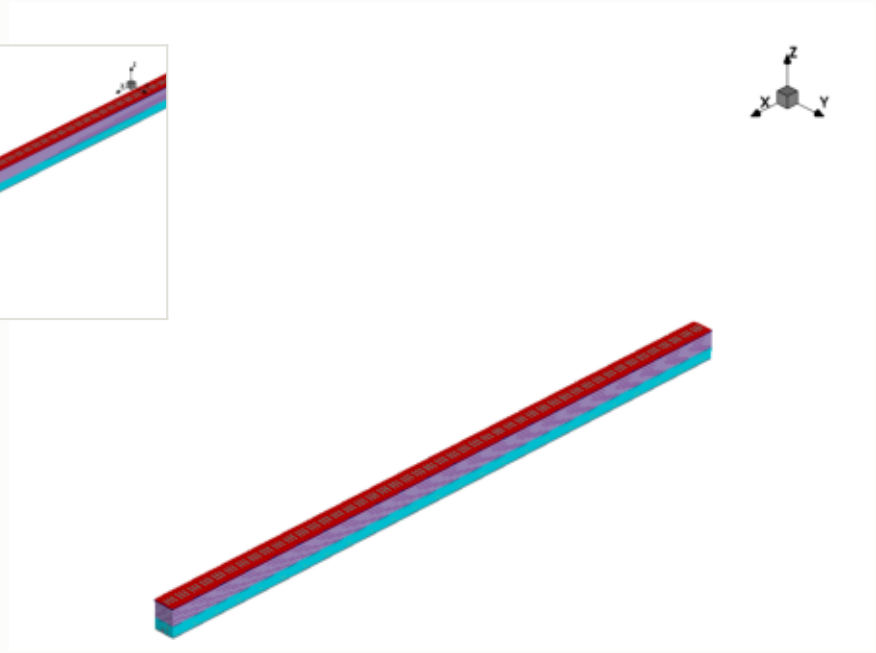
etch_strip · Spx_only

0	364.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





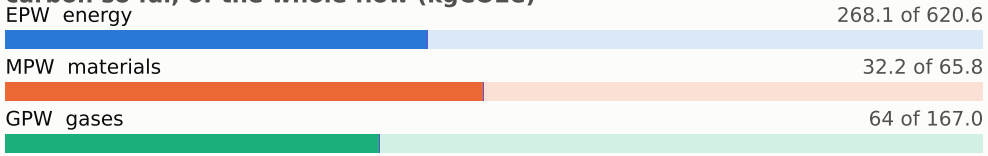
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

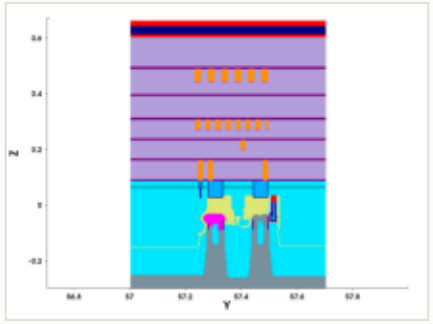
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

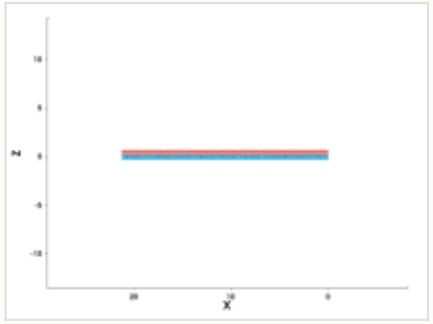


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



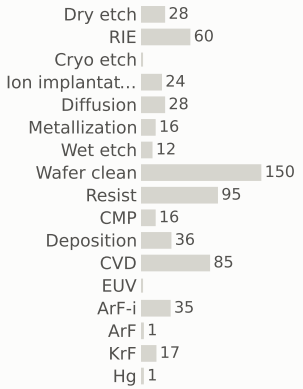
y cut at 57.351 µm



mask for this step

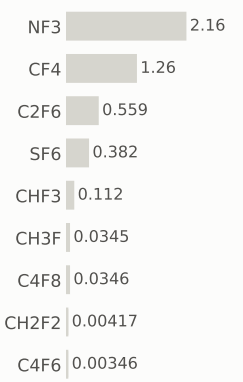
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



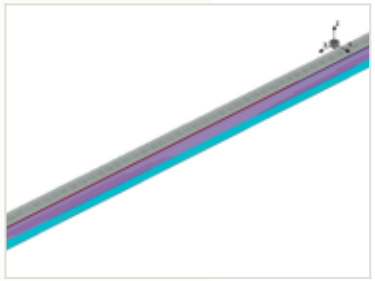
Step #8090: Spx_only

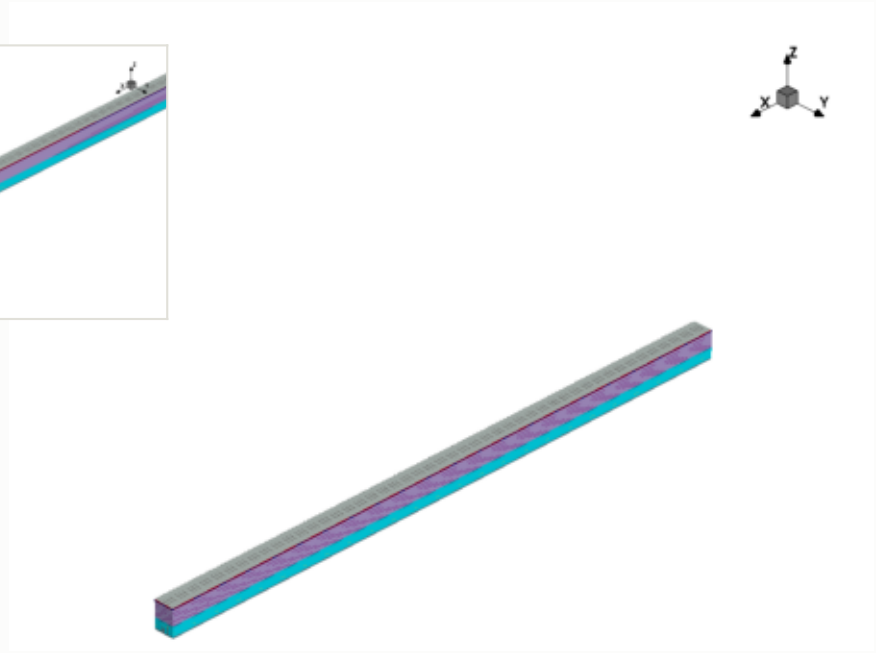
depo_isotropic · Spx_only

0	364.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

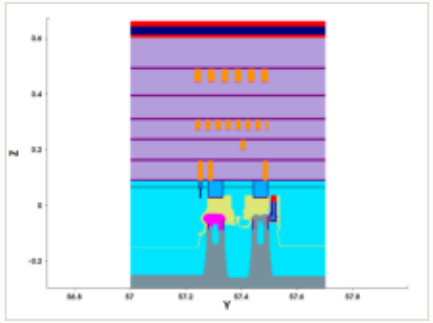
3D structure after this step

zoomed to fit

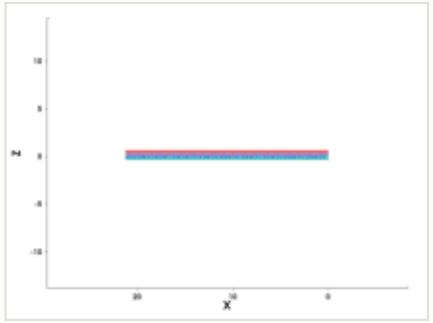




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

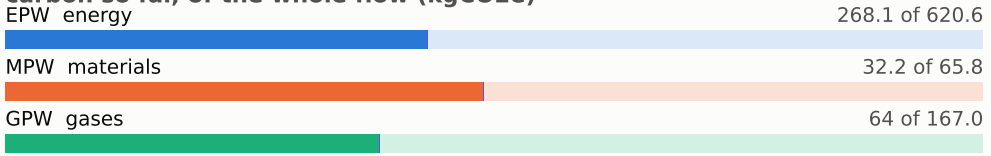
no mask for this step

this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	0.001 µm	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

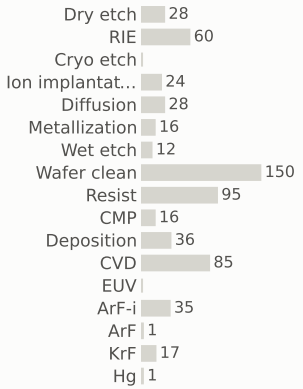
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

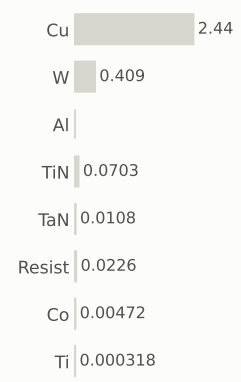


embodied carbon up to each step (kgCO2e per wafer)

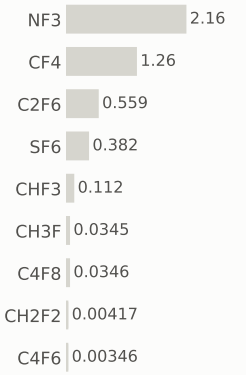
steps by process area



materials deposited (g)



gases emitted (g)



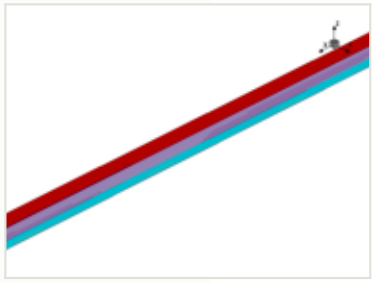
Step #8100: Spx_only

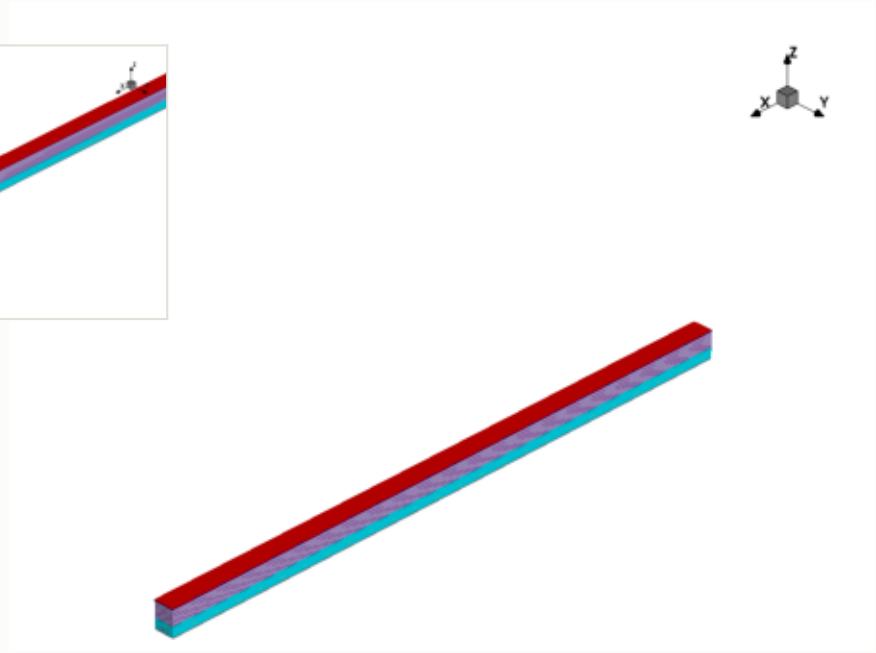
etch_anisotropic · Spx_only

0	364.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





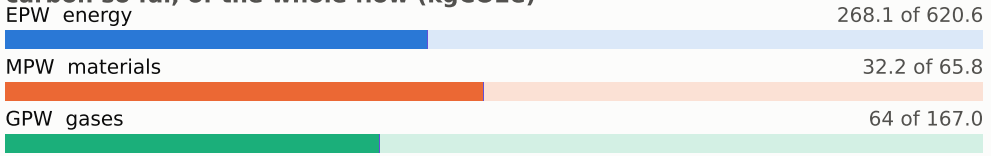
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

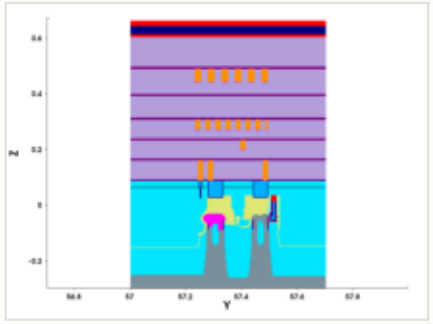
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

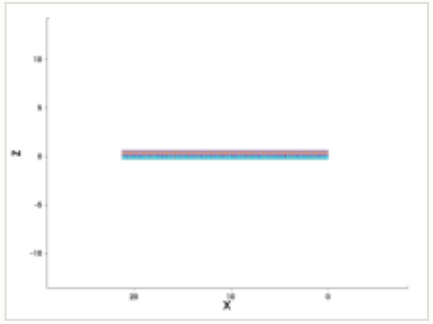


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



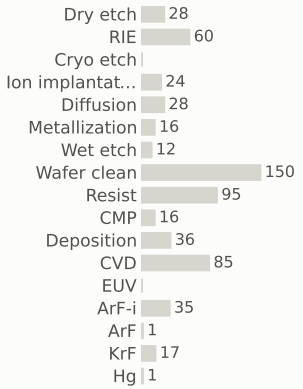
y cut at 57.351 µm



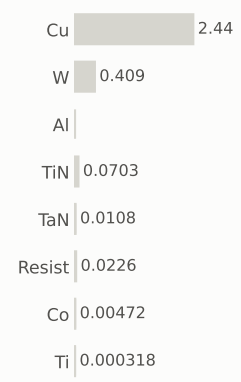
mask for this step

no mask for this step

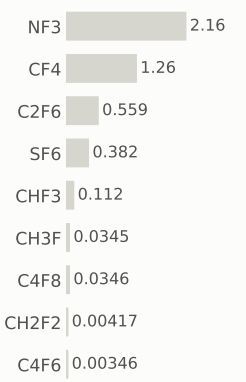
steps by process area



materials deposited (g)



gases emitted (g)



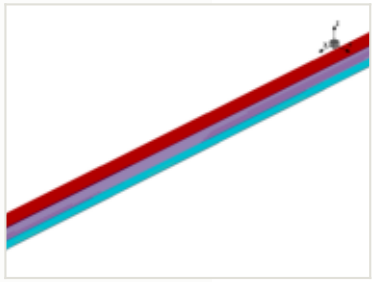
Step #8110: Spx_only

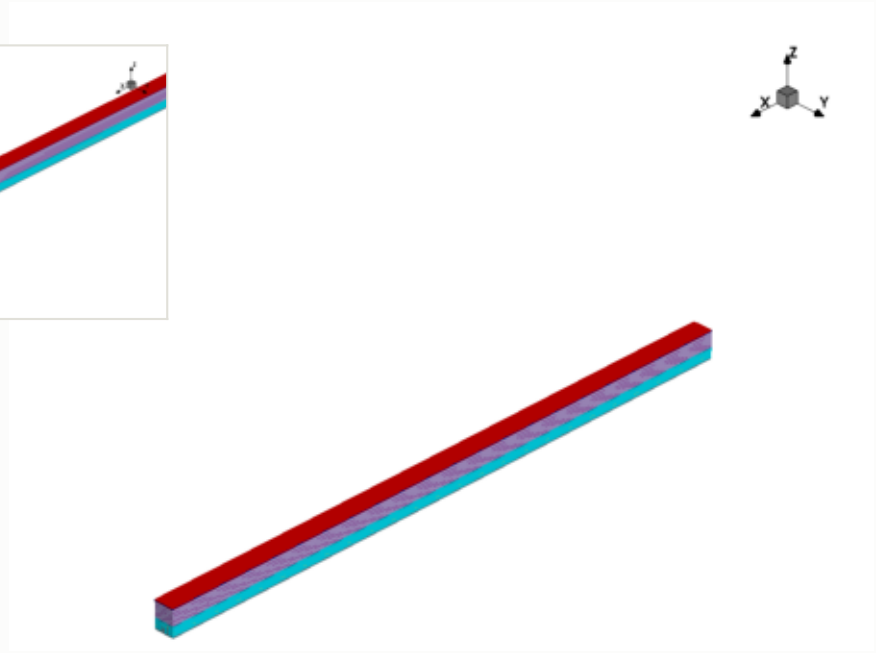
etch_strip · Spx_only

0	364.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

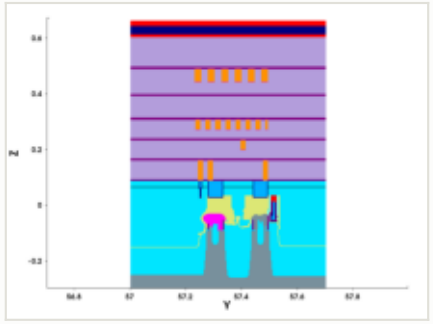
3D structure after this step

zoomed to fit

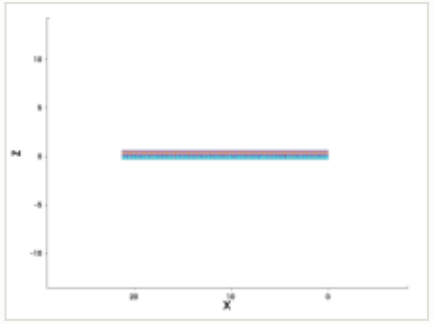




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

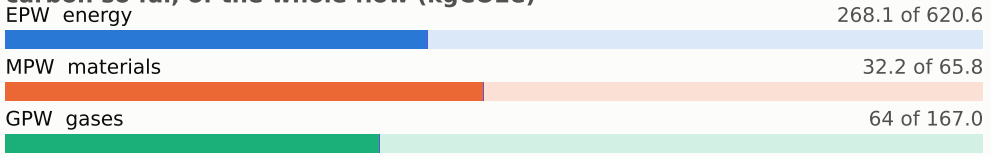
no mask for this step

this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

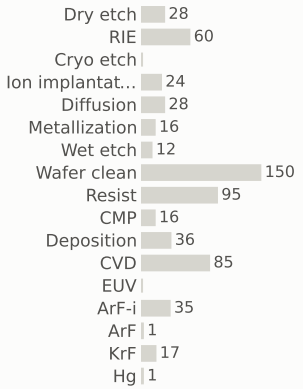
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

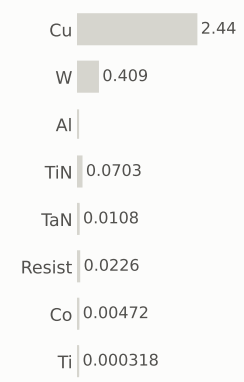


embodied carbon up to each step (kgCO2e per wafer)

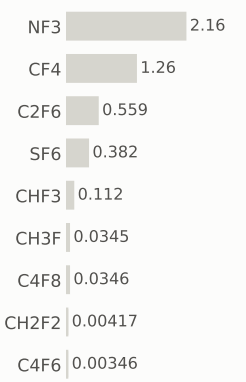
steps by process area



materials deposited (g)



gases emitted (g)



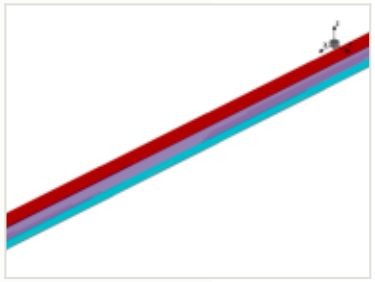
Step #8120: Wafer_clean

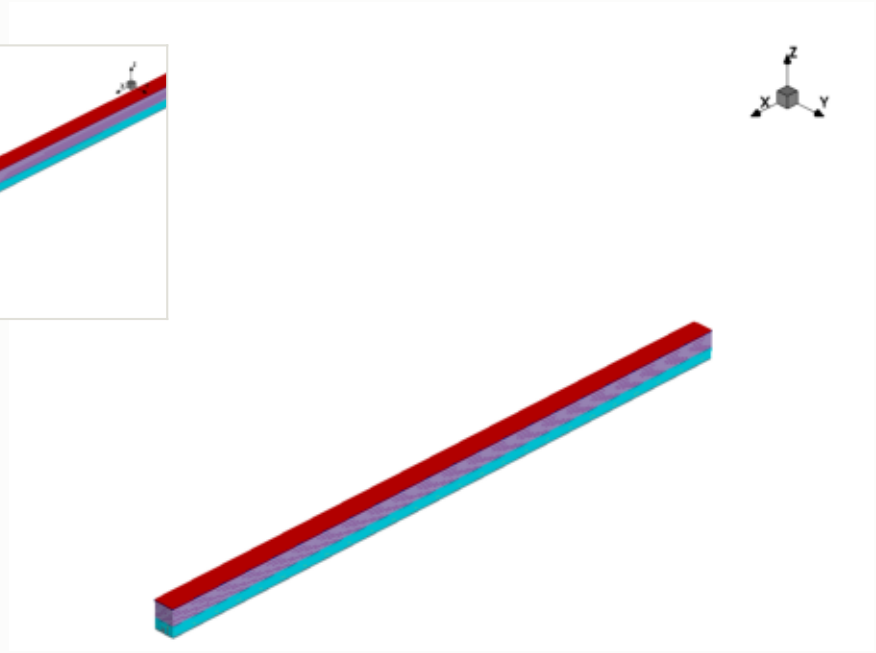
clean · Wafer_clean

0.169	364.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

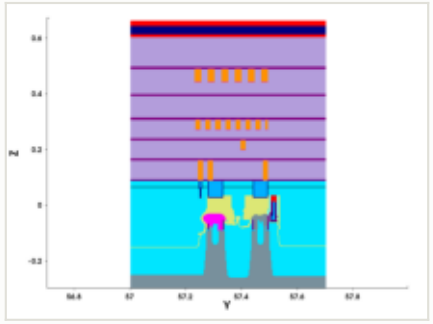
3D structure after this step

zoomed to fit

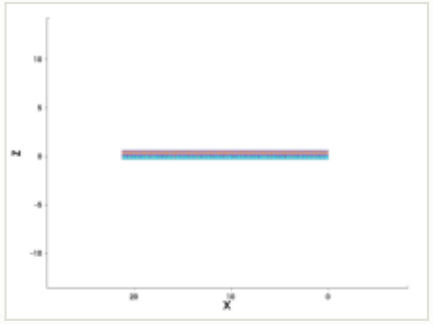




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

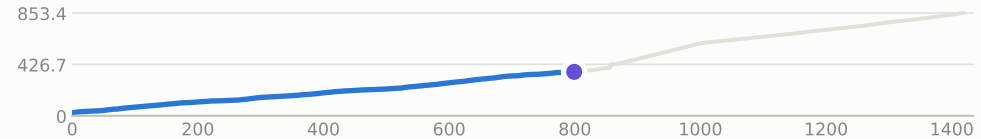
carbon so far, of the whole flow (kgCO2e)

EPW energy — 268.3 of 620.6 (this step +0.169)

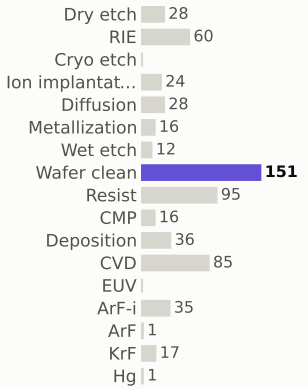
MPW materials — 32.2 of 65.8

GPW gases — 64 of 167.0

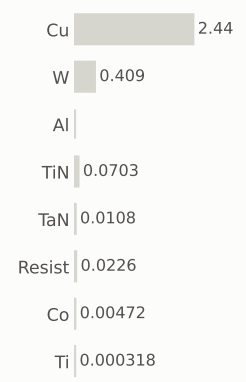
embodied carbon up to each step (kgCO2e per wafer)



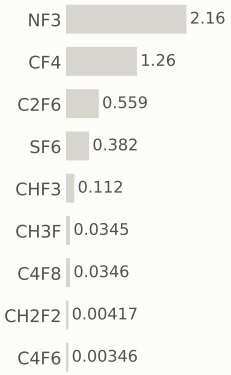
steps by process area



materials deposited (g)



gases emitted (g)



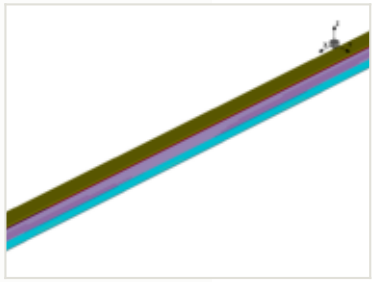
Step #8130: Resist

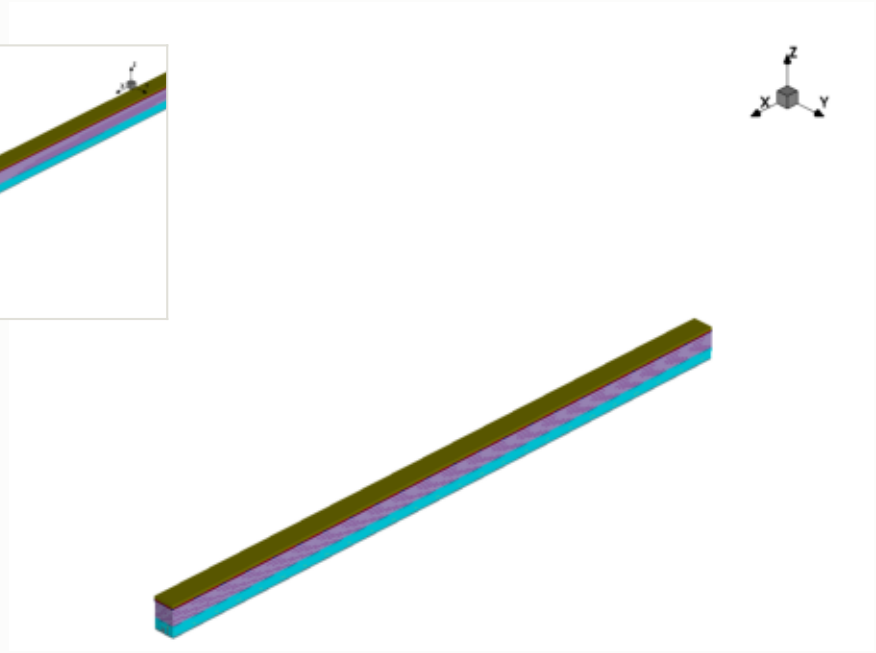
depo_fill · Resist

0.169	364.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



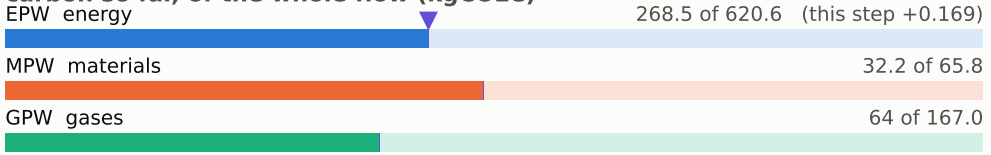


this step

process area	Resist
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

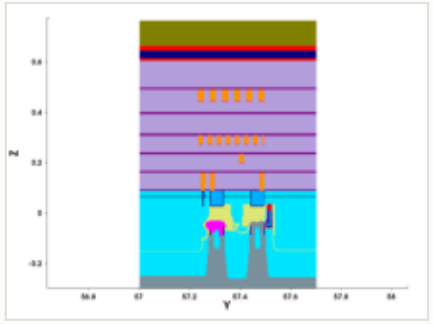
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

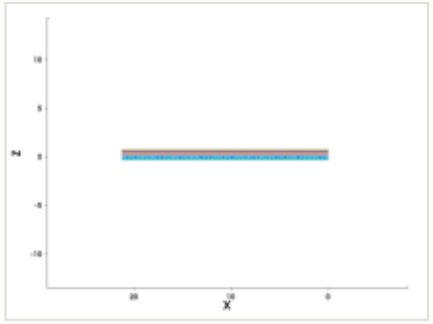


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



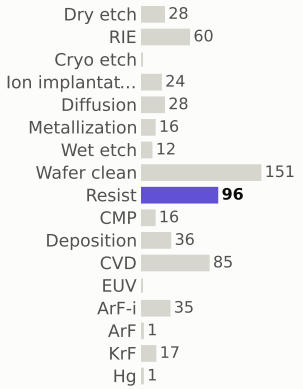
y cut at 57.351 µm



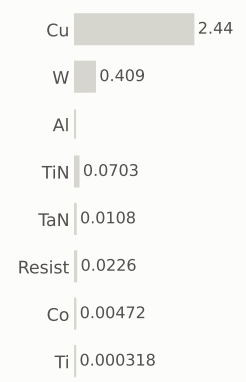
mask for this step

no mask for this step

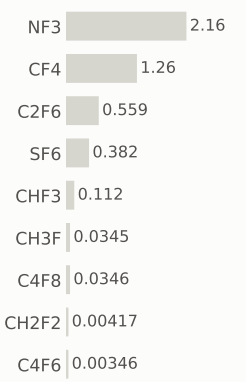
steps by process area



materials deposited (g)



gases emitted (g)



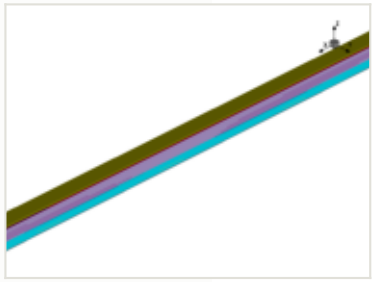
Step #8140: CVD

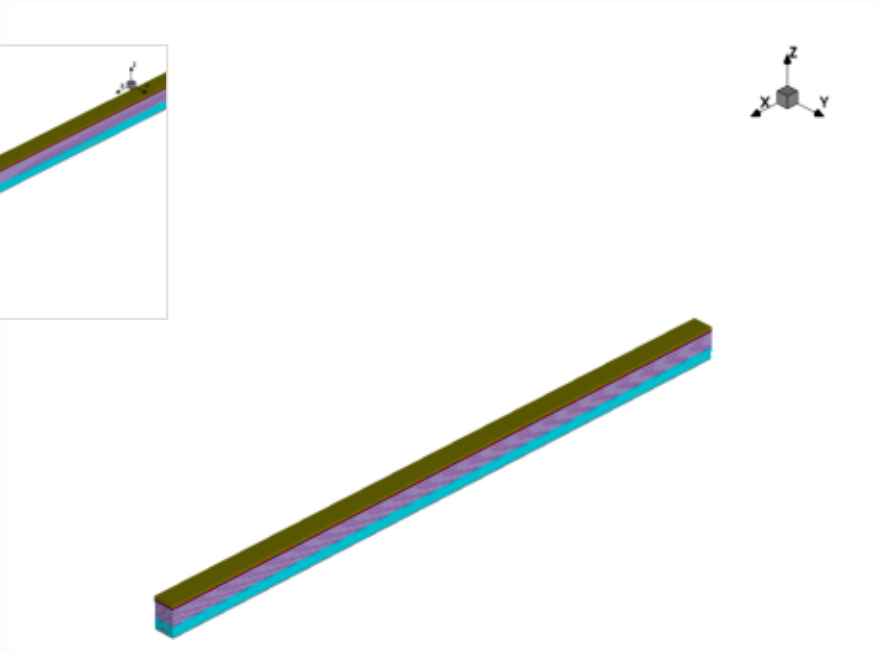
depo_isotropic · CVD

1.31	365.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



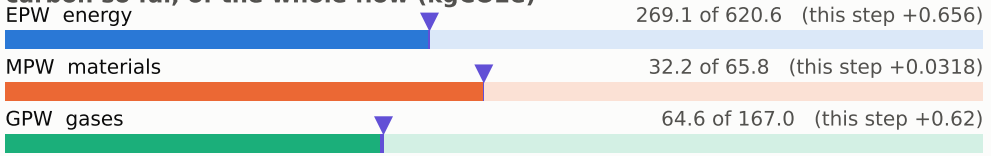


this step

process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

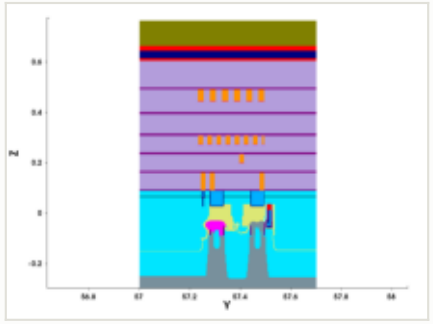
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

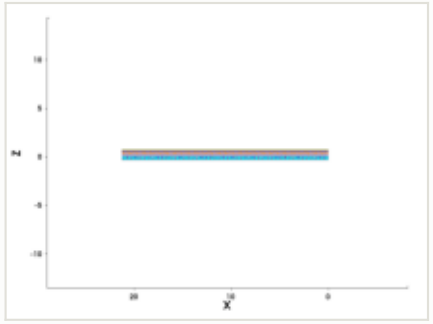


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



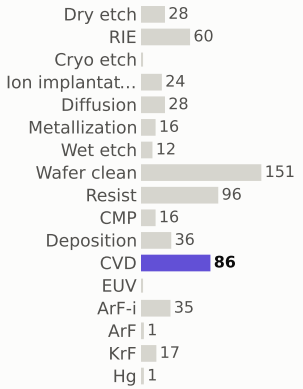
y cut at 57.351 µm



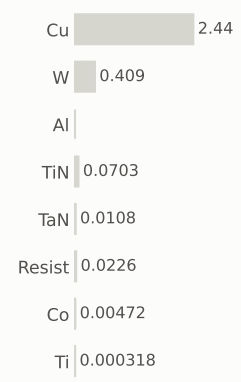
mask for this step

no mask for this step

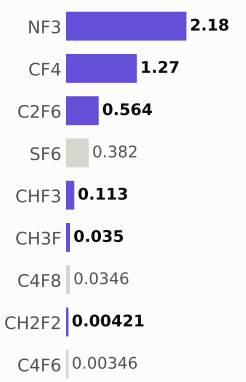
steps by process area



materials deposited (g)



gases emitted (g)



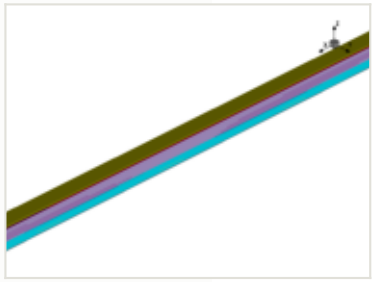
Step #8150: Wafer_clean

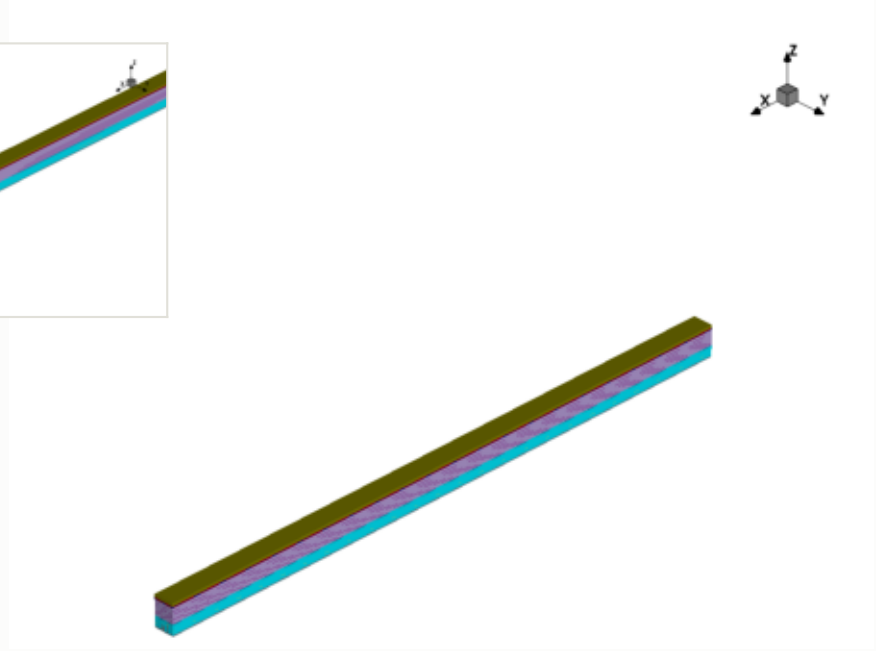
clean · Wafer_clean

0.169	366.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





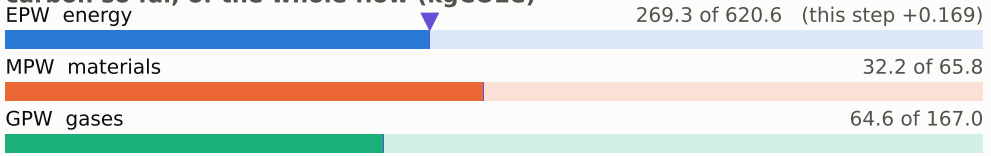
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

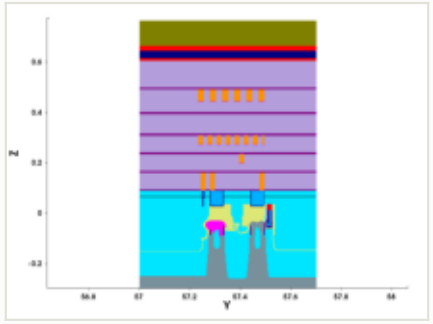
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

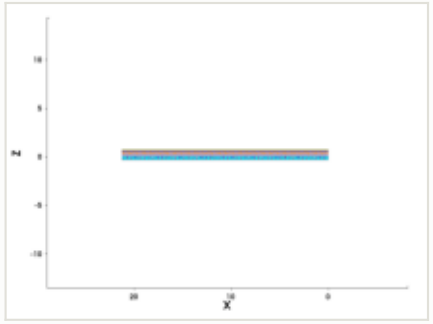


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



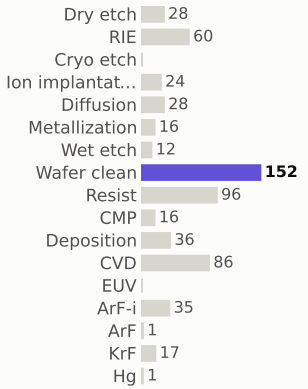
y cut at 57.351 µm



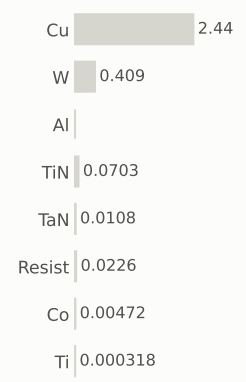
mask for this step

no mask for this step

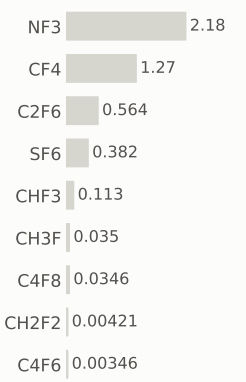
steps by process area



materials deposited (g)



gases emitted (g)



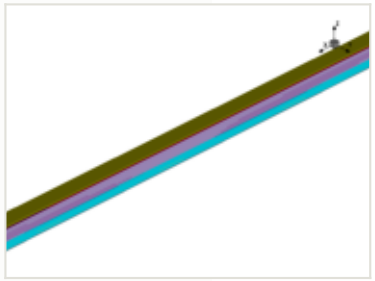
Step #8160: Resist

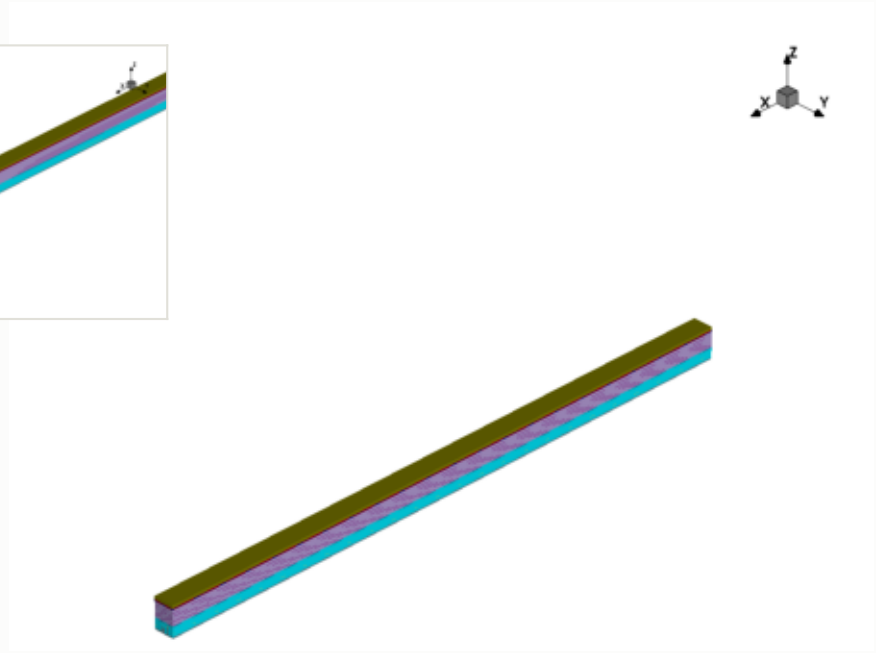
resist · Resist

0.169	366.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





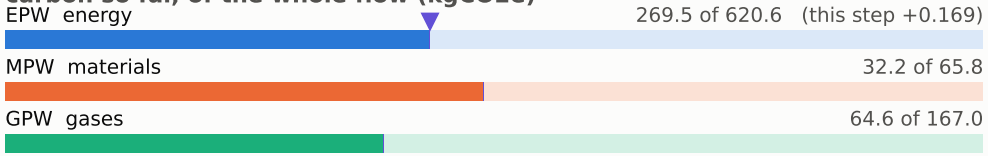
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

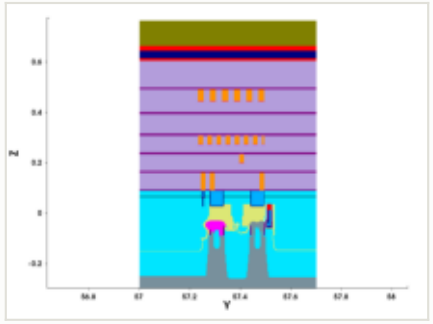
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

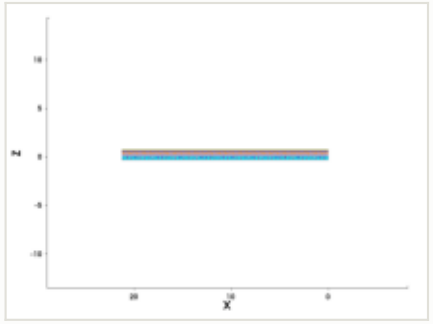


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



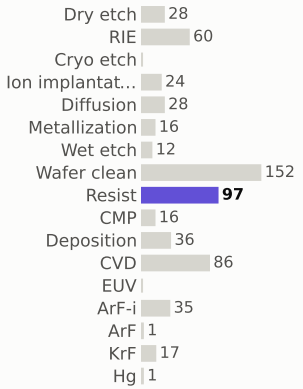
y cut at 57.351 µm



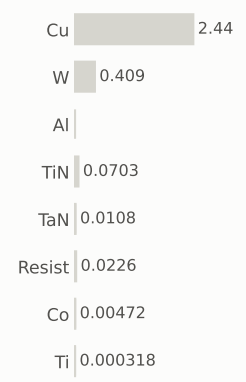
mask for this step

no mask for this step

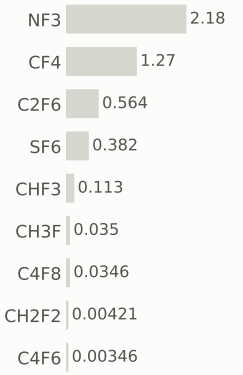
steps by process area



materials deposited (g)



gases emitted (g)



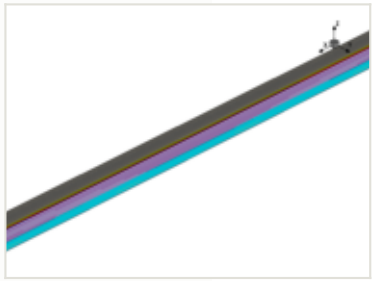
Step #8170: ArF-i

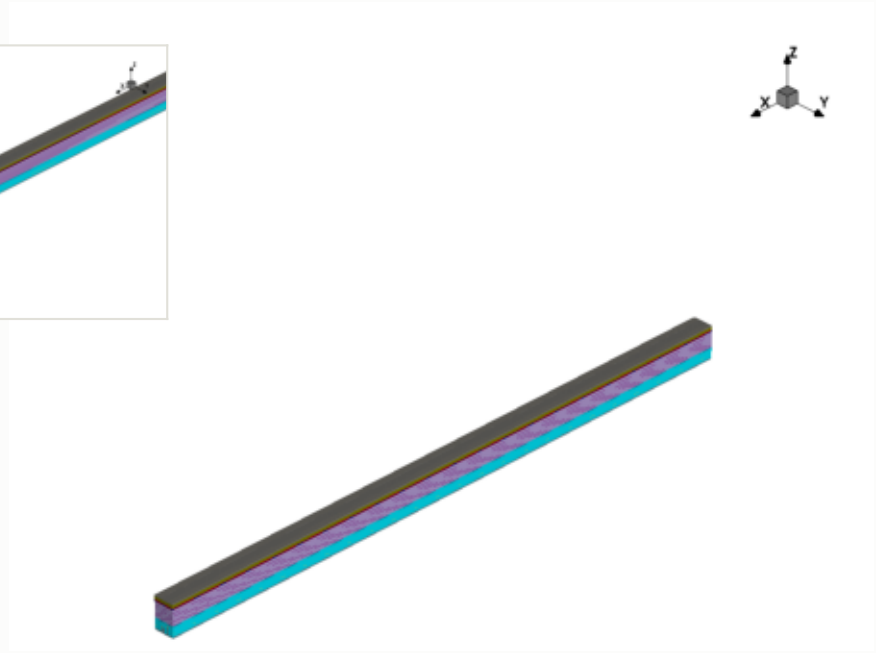
litho_patterning · ArF-i

0.217	366.5	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



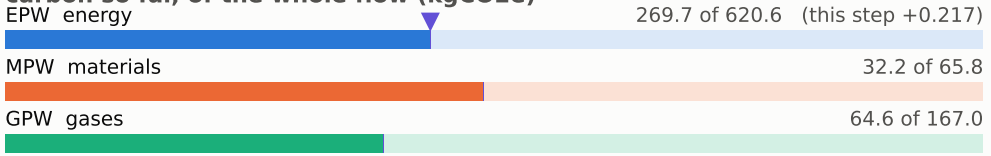


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

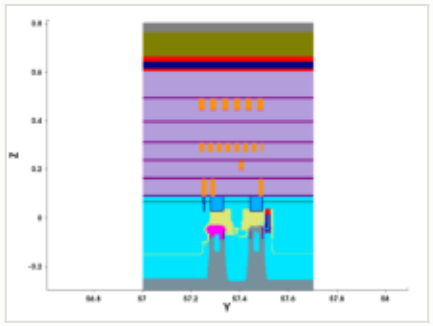
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

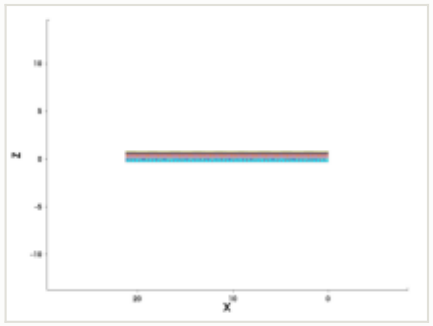


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

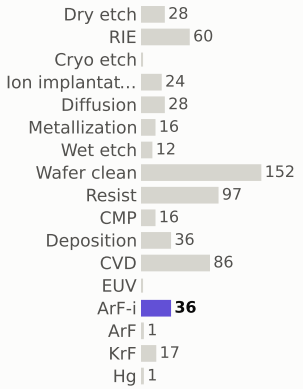


y cut at 57.351 µm

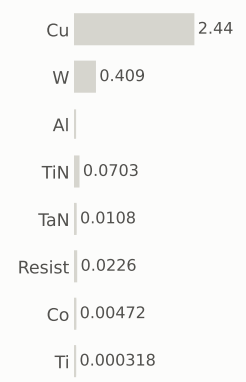


mask: layer 2060:0 (GDSII)

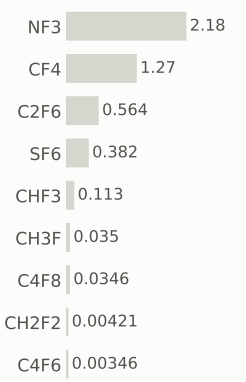
steps by process area



materials deposited (g)



gases emitted (g)



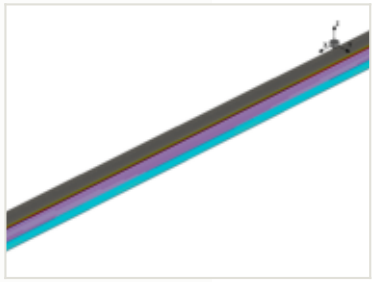
Step #8180: RIE

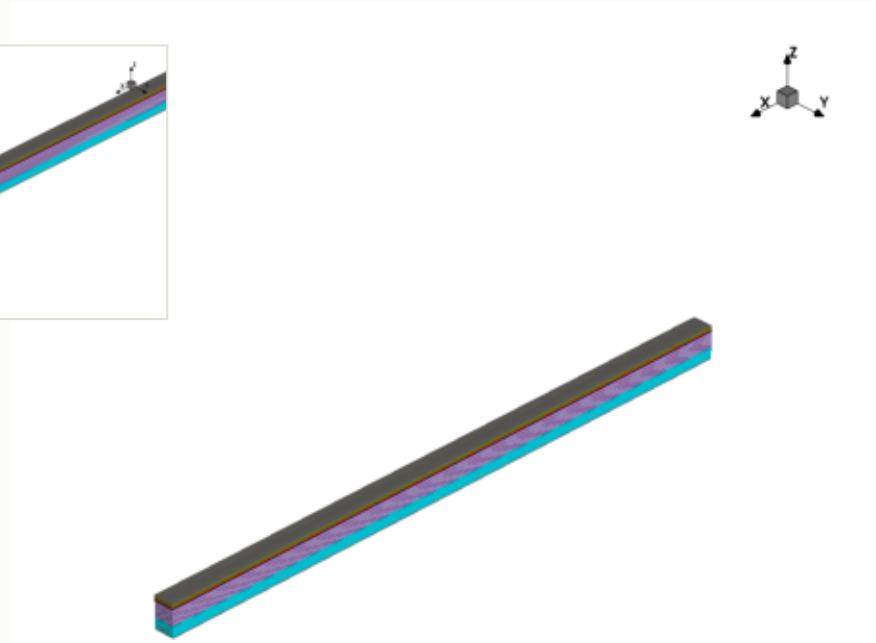
etch_anisotropic · RIE

1.04	367.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



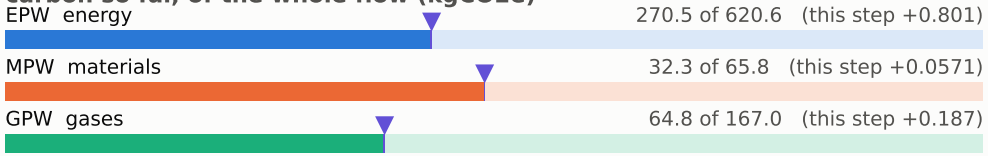


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

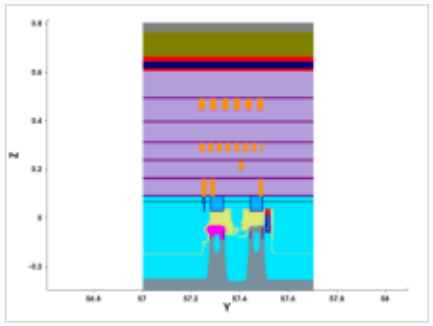
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

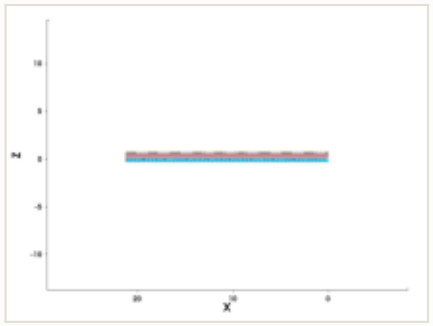


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



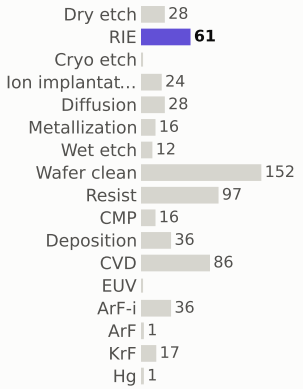
y cut at 57.351 µm



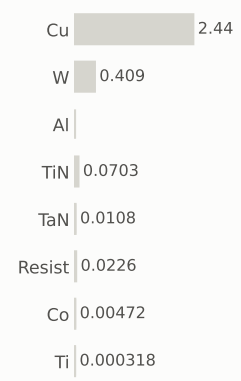
mask for this step

no mask for this step

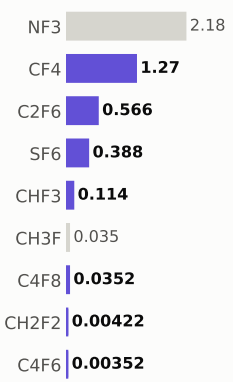
steps by process area



materials deposited (g)



gases emitted (g)



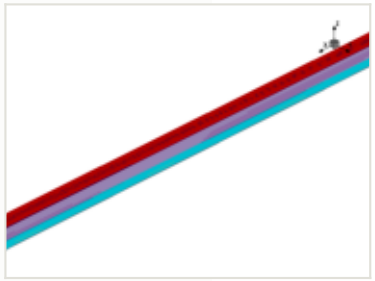
Step #8190: Spx_only

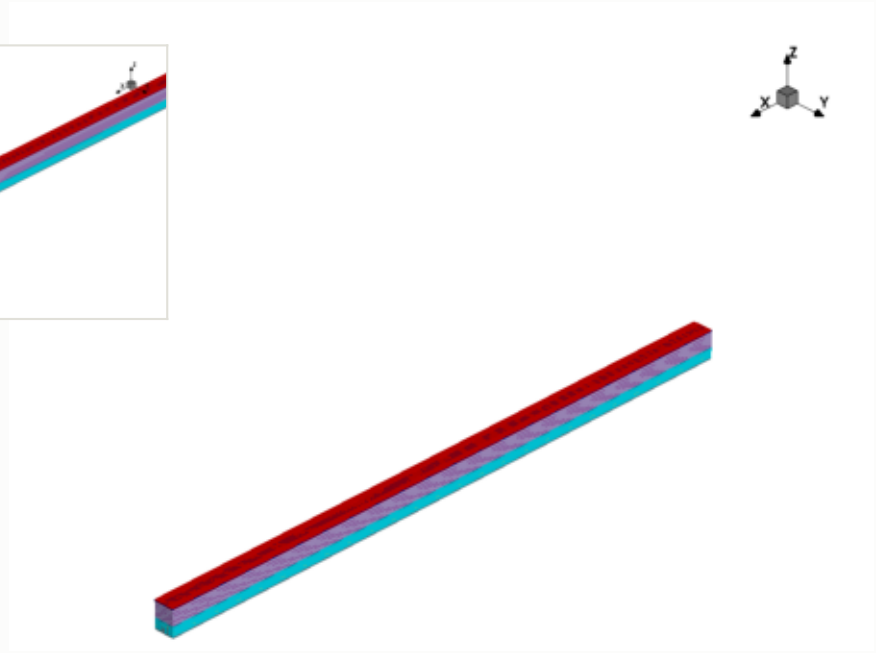
etch_strip · Spx_only

0	367.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

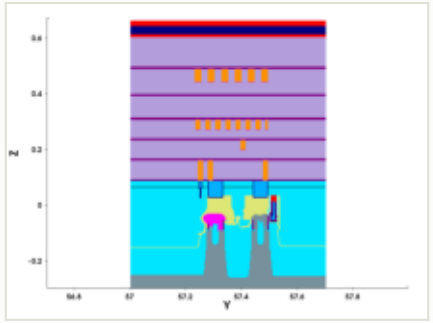
3D structure after this step

zoomed to fit

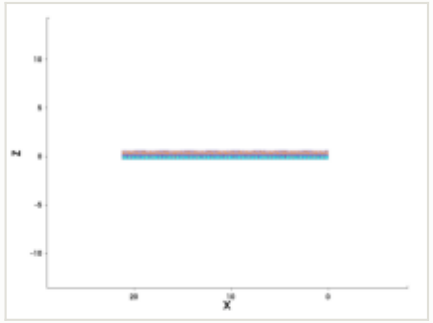




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

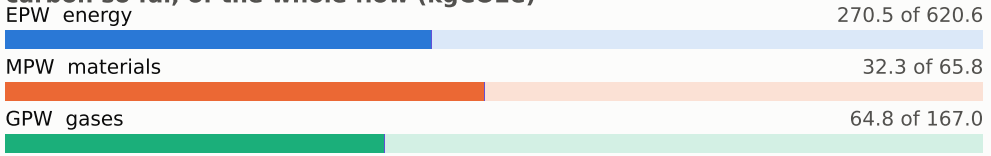
no mask for this step

this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

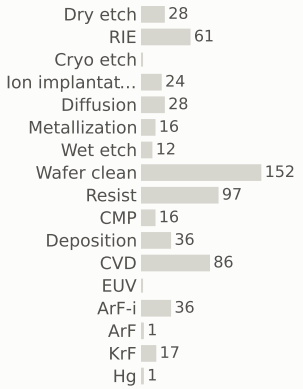
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

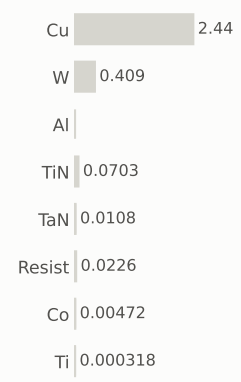


embodied carbon up to each step (kgCO2e per wafer)

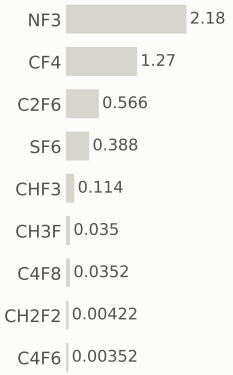
steps by process area



materials deposited (g)



gases emitted (g)



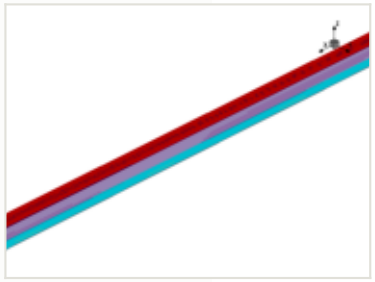
Step #8200: Wafer_clean

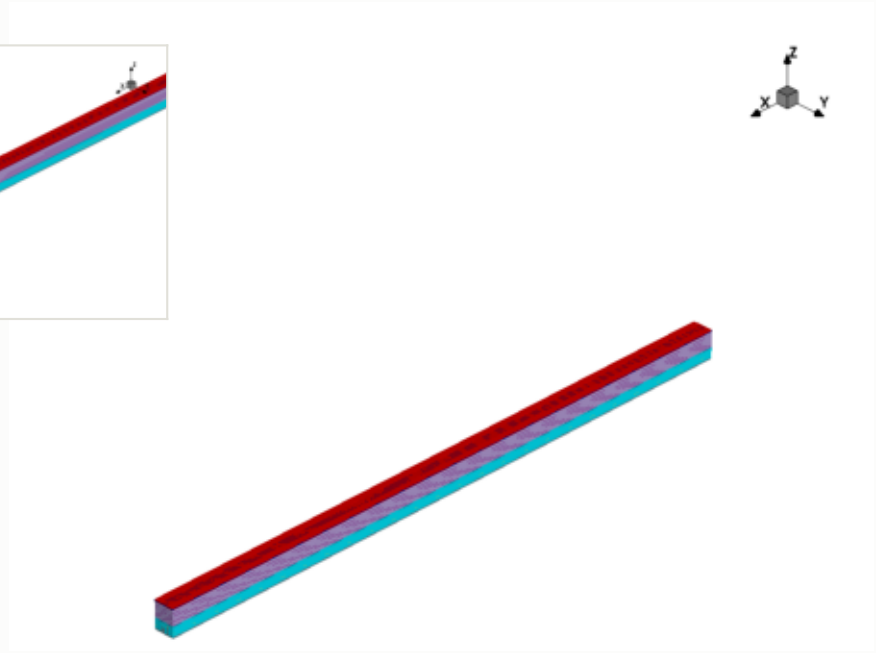
clean · Wafer_clean

0.169	367.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





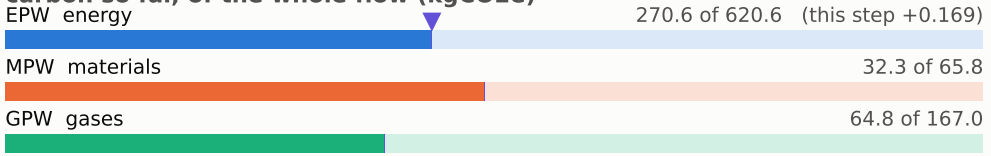
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

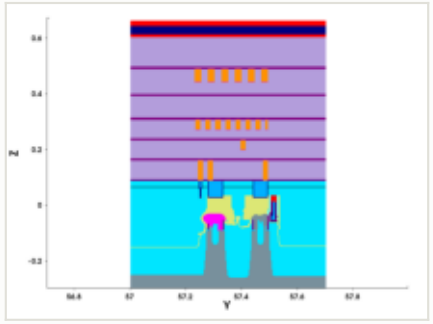
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

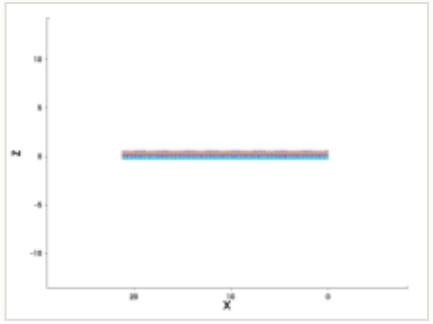


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



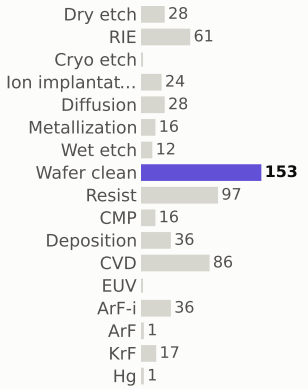
y cut at 57.351 µm



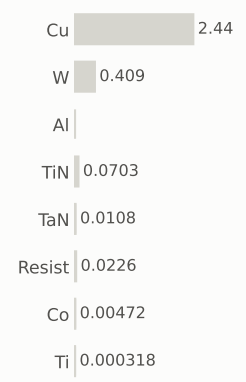
mask for this step

no mask for this step

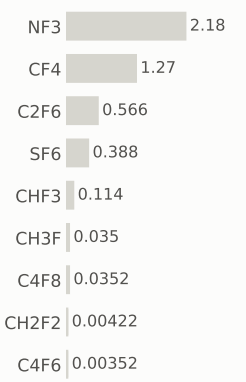
steps by process area



materials deposited (g)



gases emitted (g)



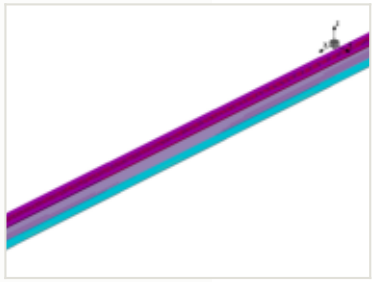
Step #8210: Metallization

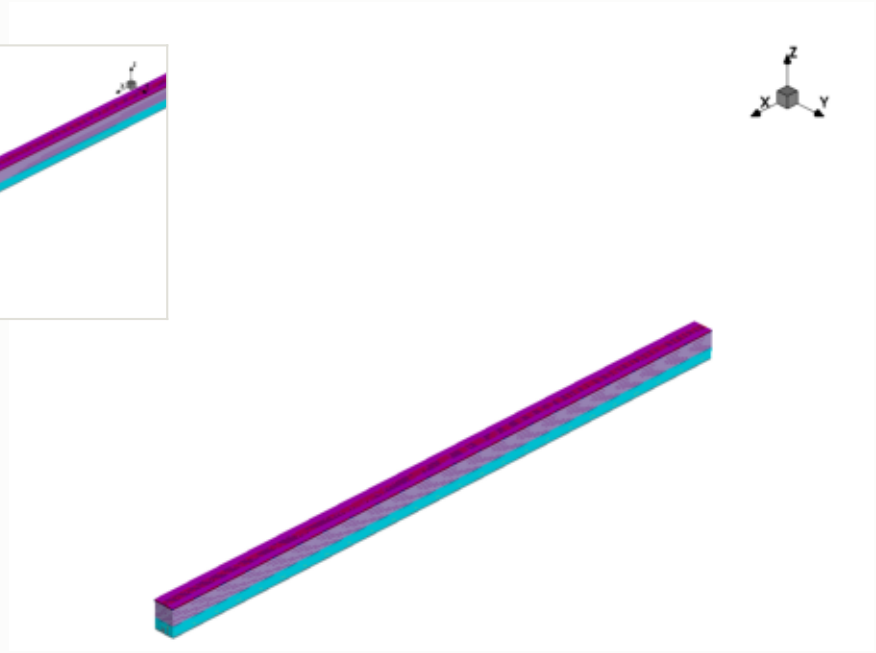
depo_ellipsoidal · Metallization

1.2	368.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



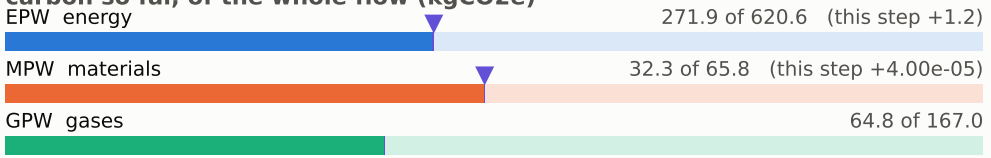


this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

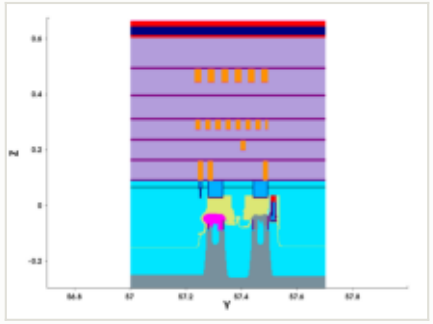
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

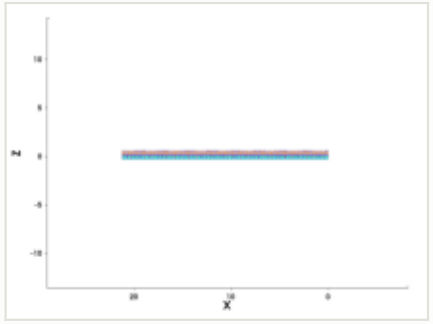


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



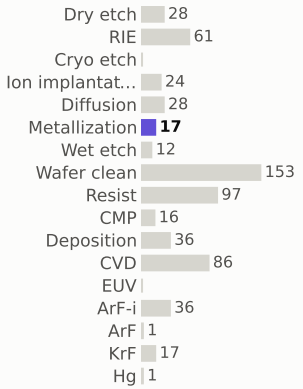
y cut at 57.351 µm



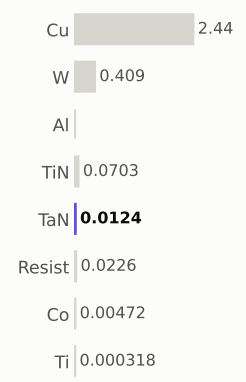
mask for this step

no mask for this step

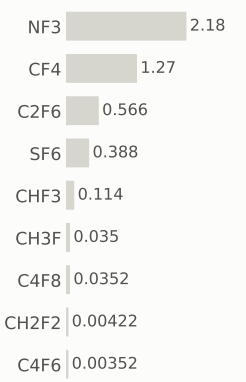
steps by process area



materials deposited (g)



gases emitted (g)



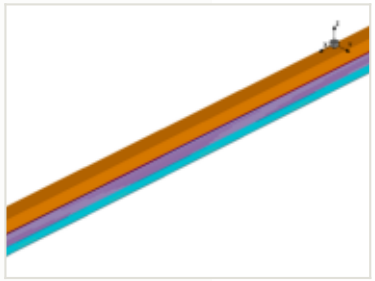
Step #8220: Metallization

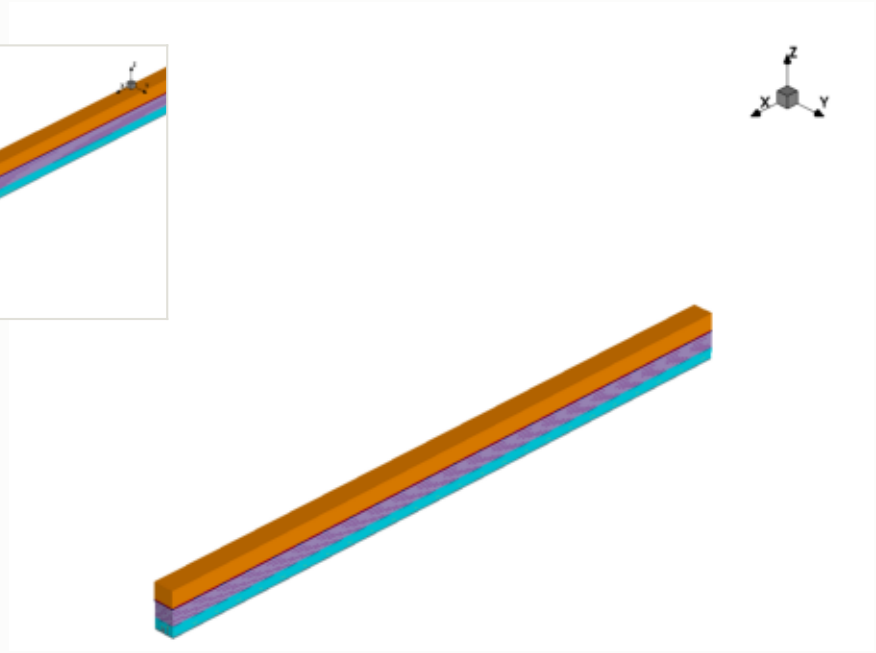
depo_fill · Metallization

1.21	370.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



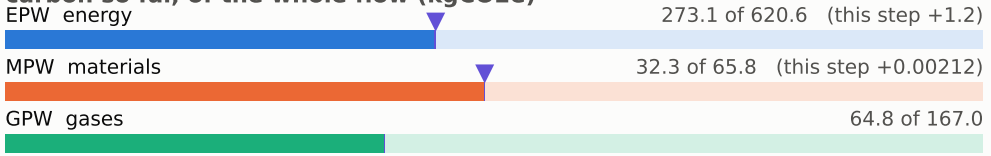


this step

process area	Metallization
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

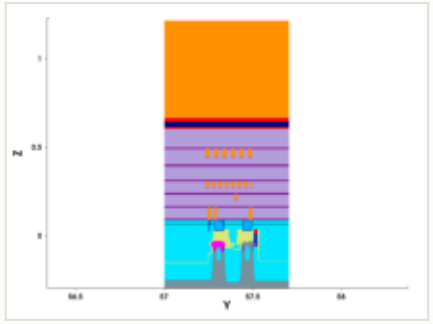
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

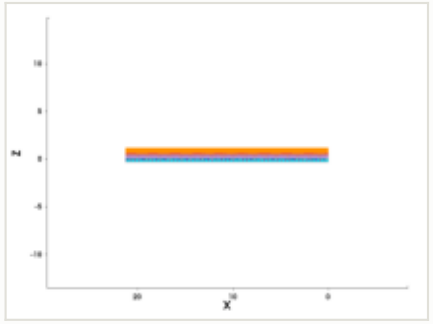


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



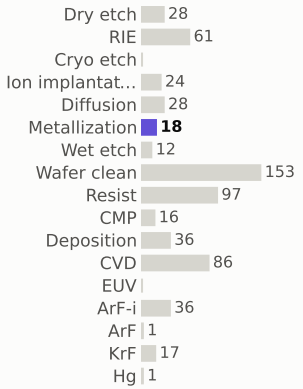
y cut at 57.351 µm



mask for this step

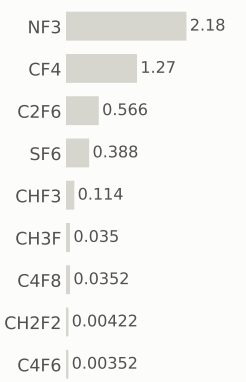
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



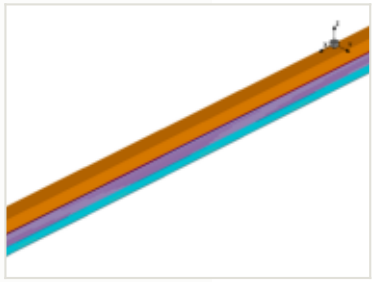
Step #8230: Diffusion

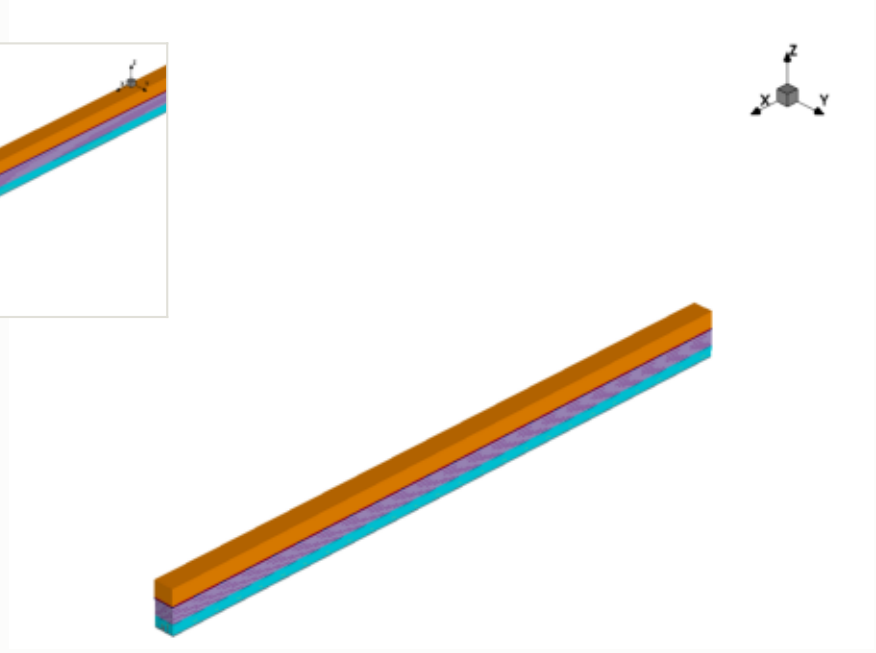
anneal · Diffusion

0.801	370.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





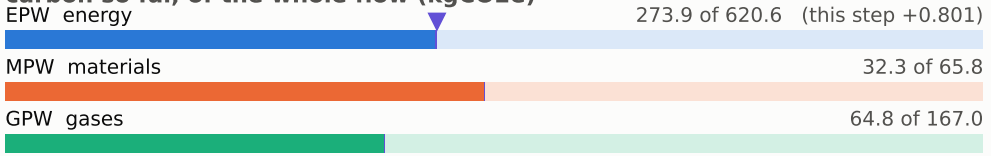
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

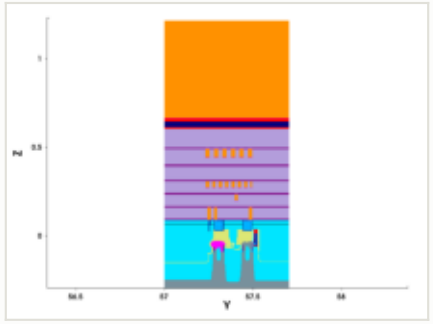
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

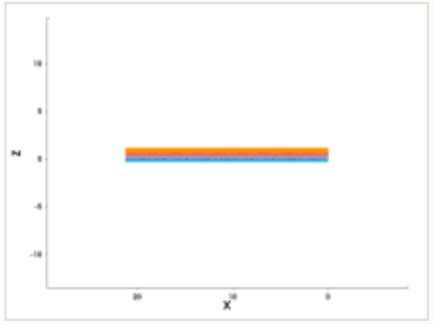


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



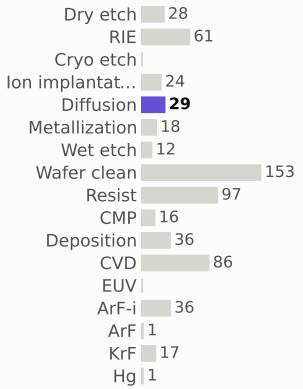
y cut at 57.351 µm



mask for this step

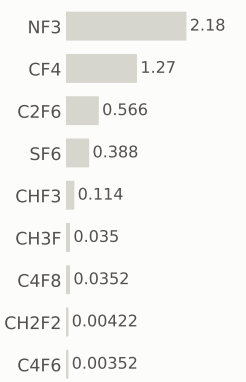
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



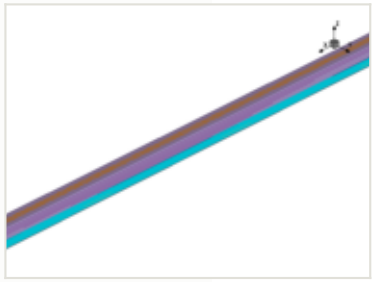
Step #8240: CMP

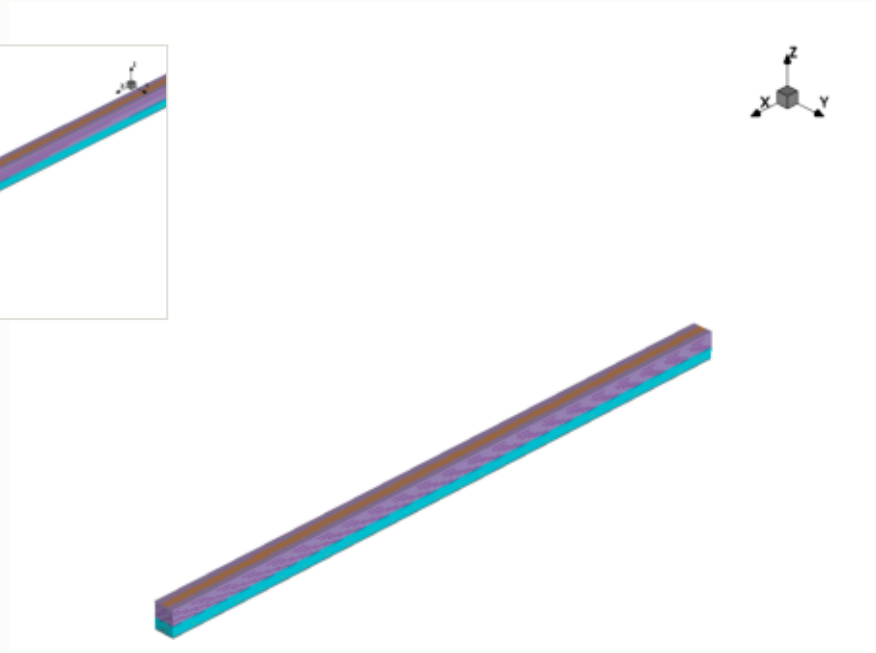
etch_cmp · CMP

0.169	371.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



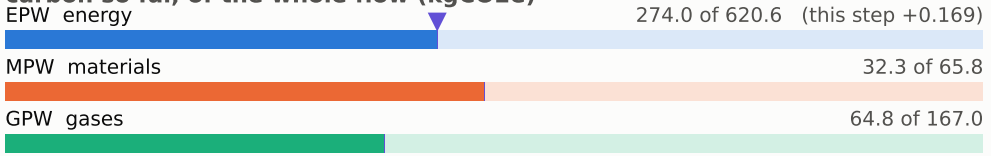


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

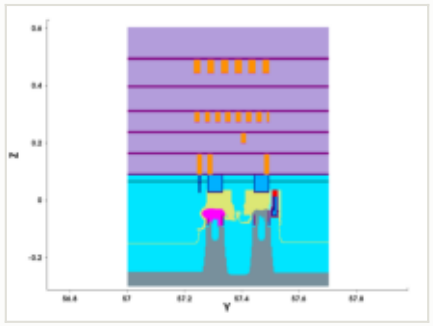
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

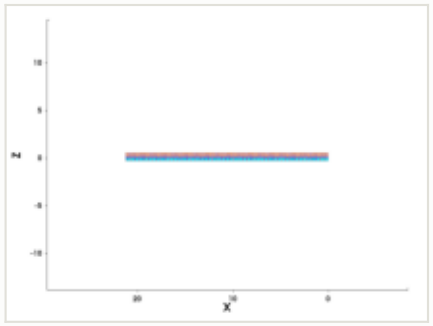


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



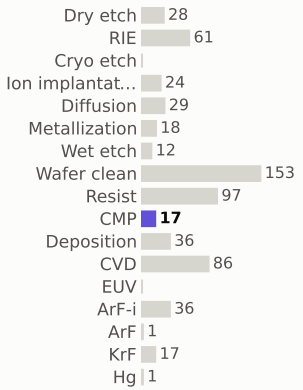
y cut at 57.351 µm



mask for this step

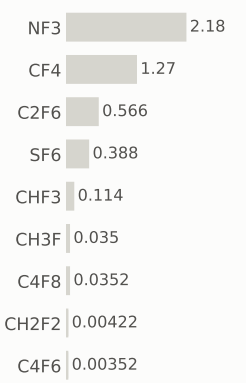
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



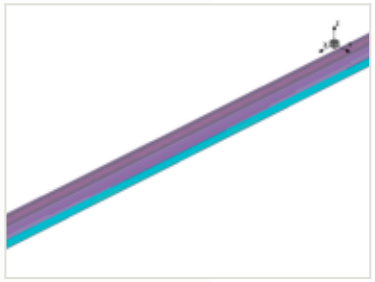
Step #8250: Spx_only

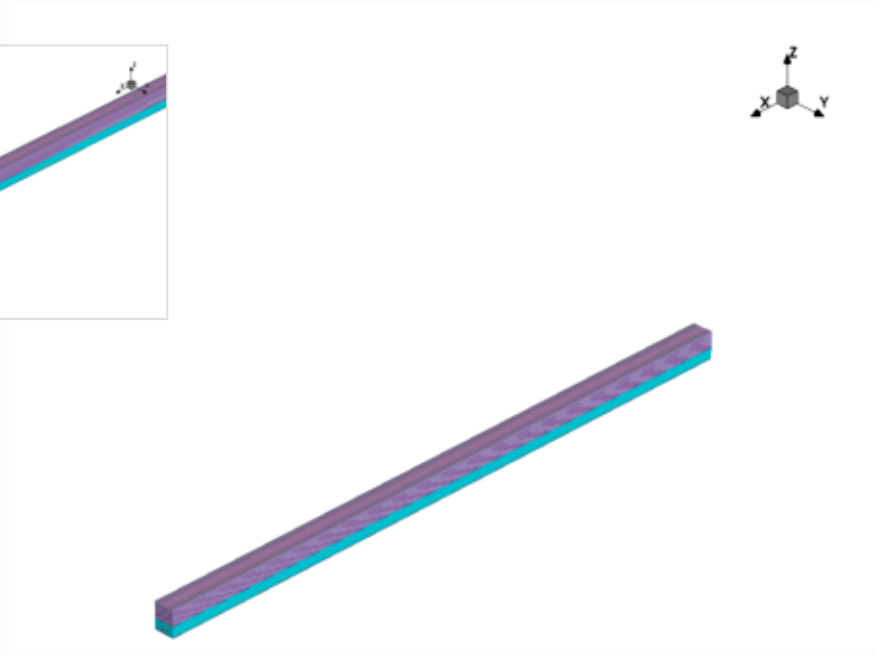
depo_isotropic · Spx_only

4.20e-05	371.1	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





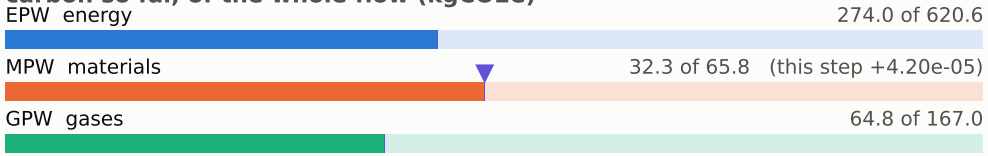
this step

process area	Spx_only
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**4.20e-05 kgCO2e**

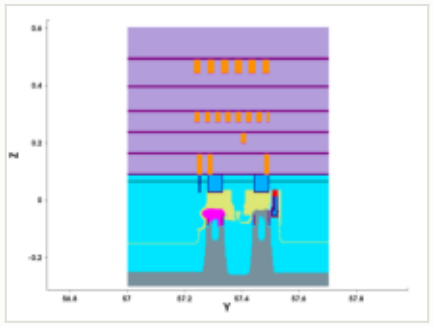
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

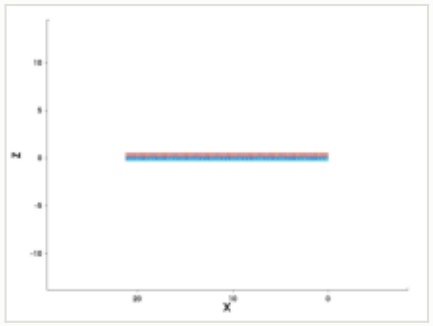


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



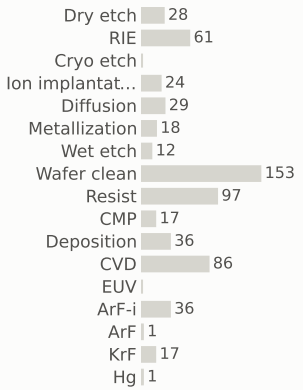
y cut at 57.351 µm



mask for this step

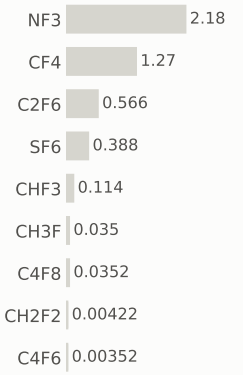
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



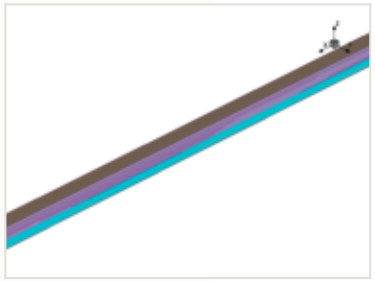
Step #8270: CVD

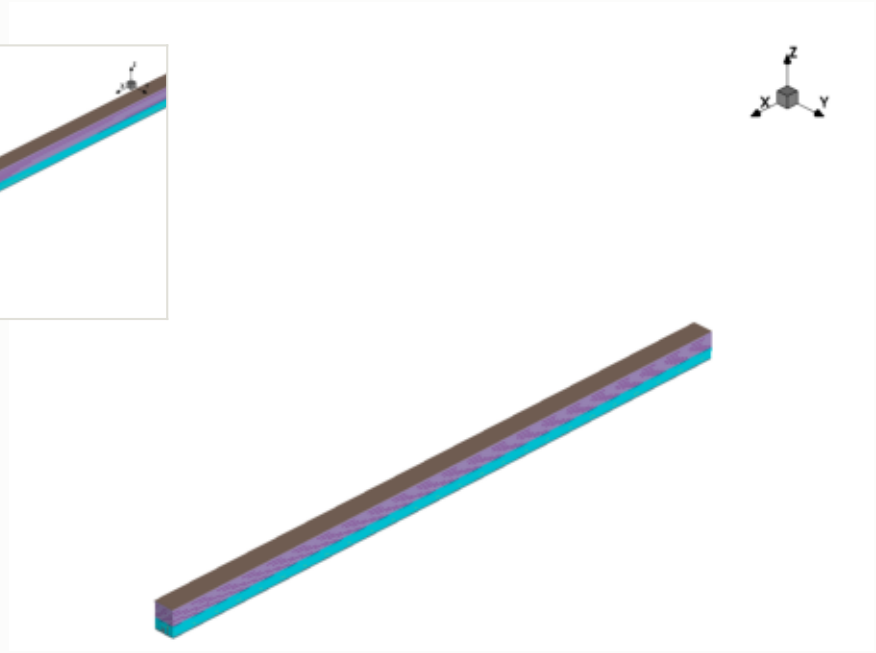
depo_fill · CVD

1.31	372.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



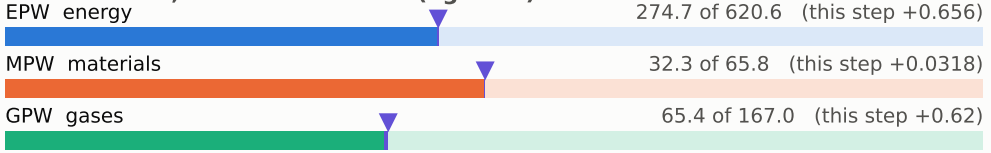


this step

process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

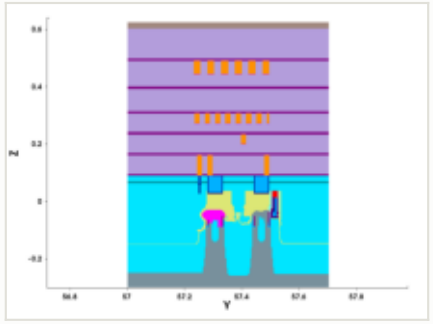
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

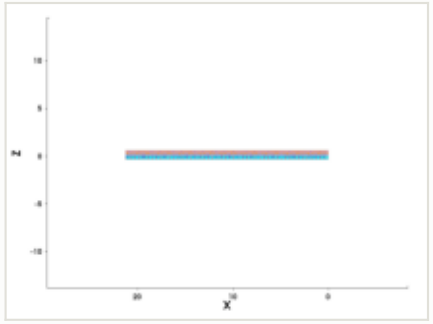


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



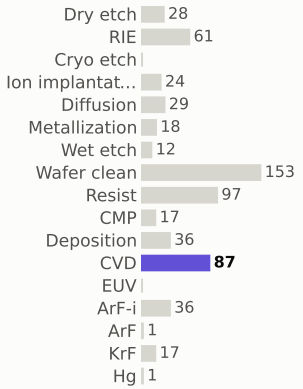
y cut at 57.351 µm



mask for this step

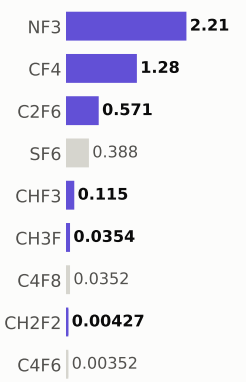
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



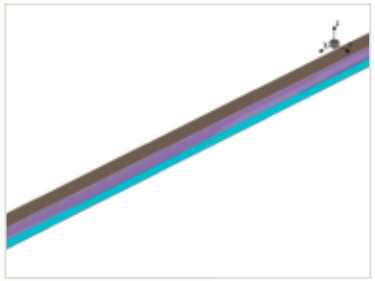
Step #8280: CVD

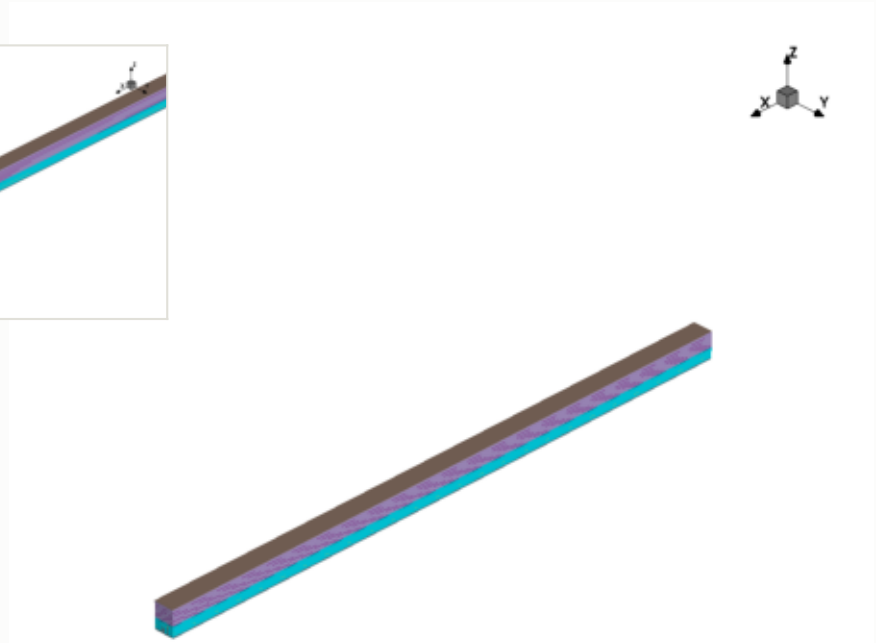
depo_isotropic · CVD

1.31	373.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



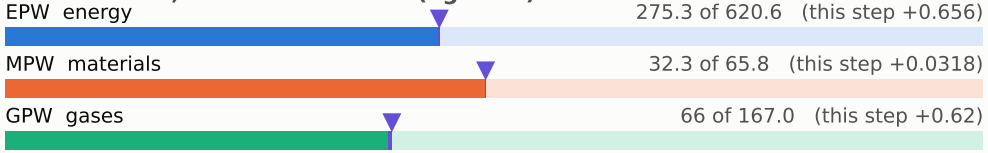


this step

process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

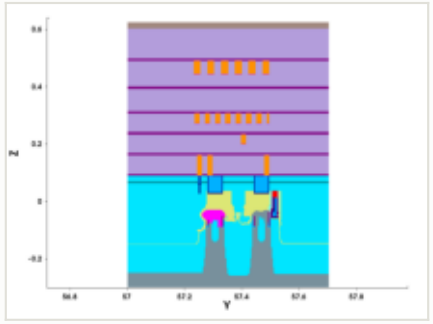
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

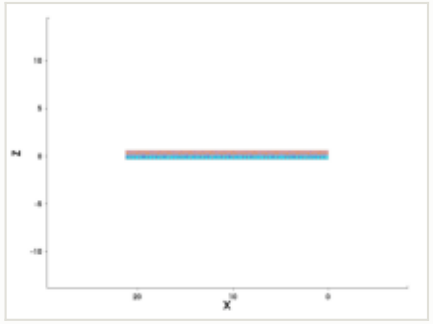


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



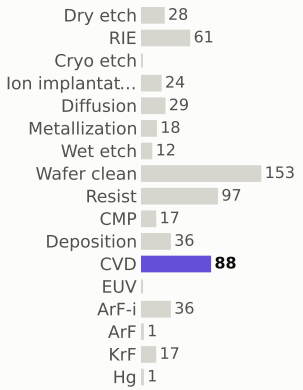
y cut at 57.351 µm



mask for this step

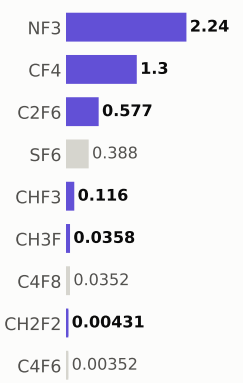
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



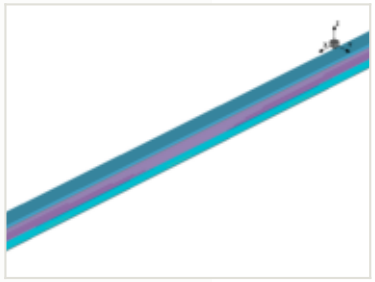
Step #8290: CVD

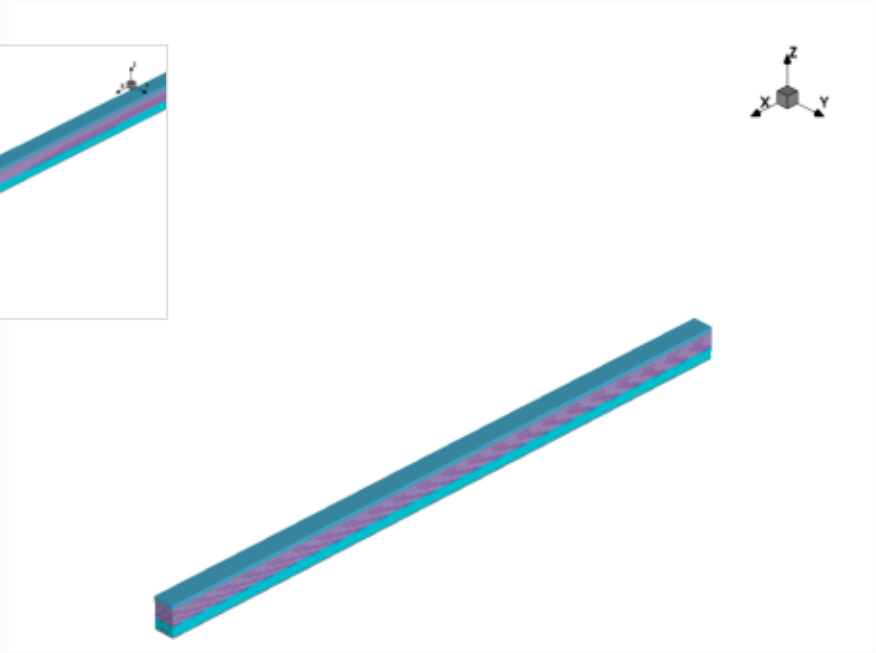
depo_fill · CVD

1.31	375.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



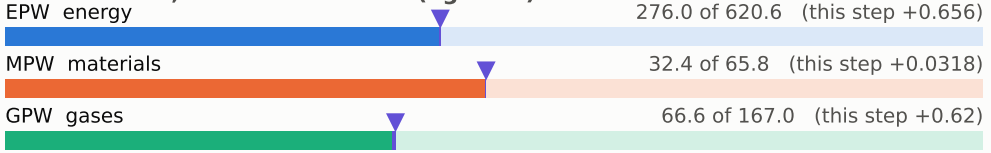


this step

process area	CVD
material	—
thickness	0.14 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

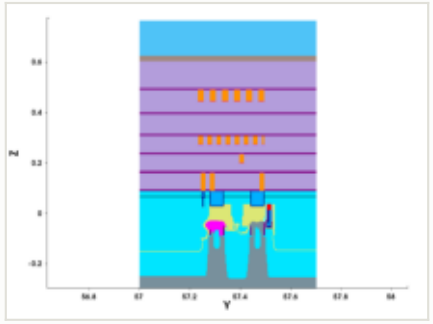
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

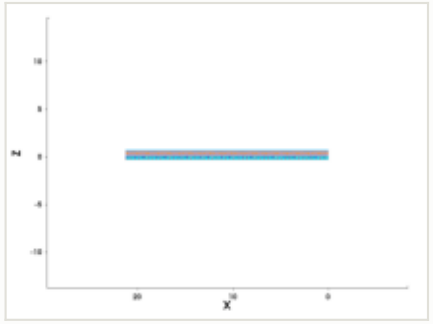


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



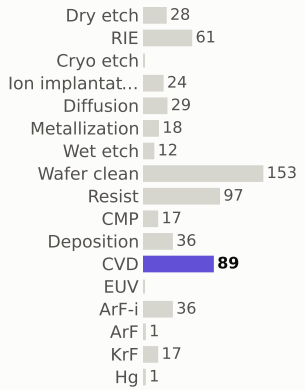
y cut at 57.351 µm



mask for this step

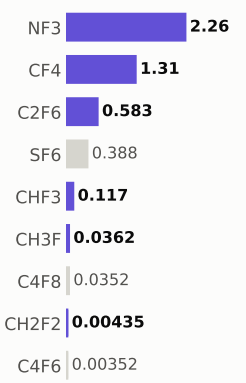
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



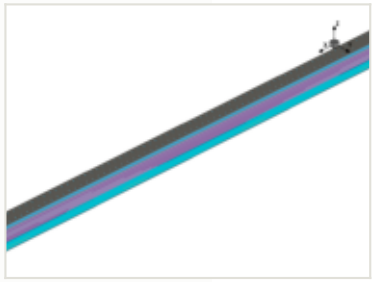
Step #8300: Spx_only

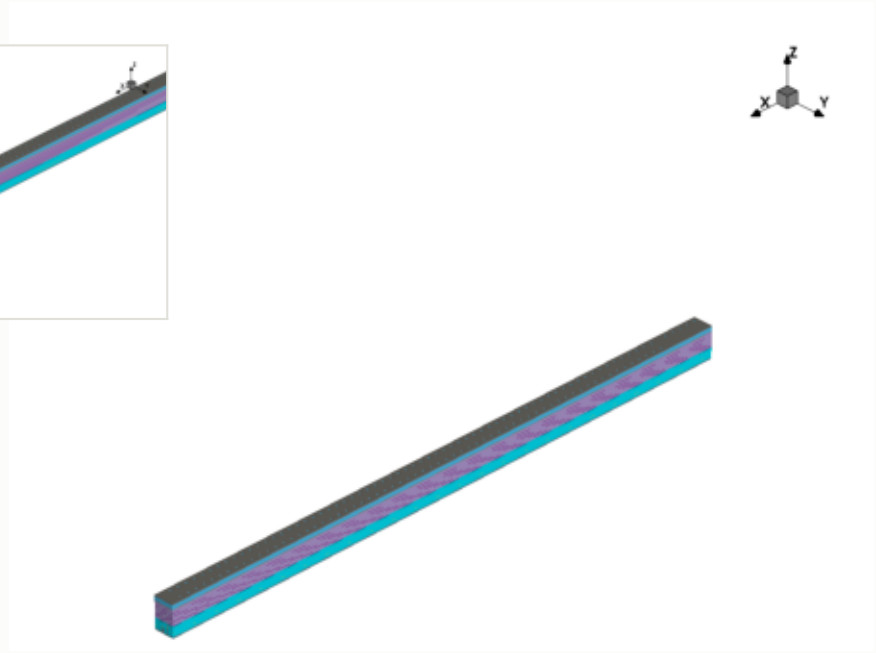
litho_patterning · Spx_only

0	375.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





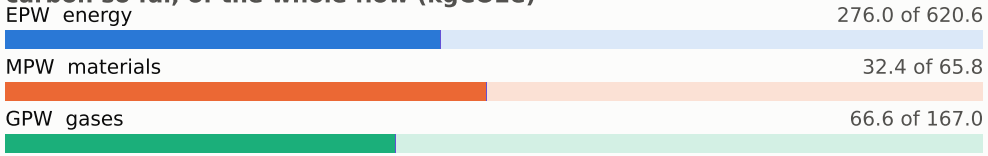
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

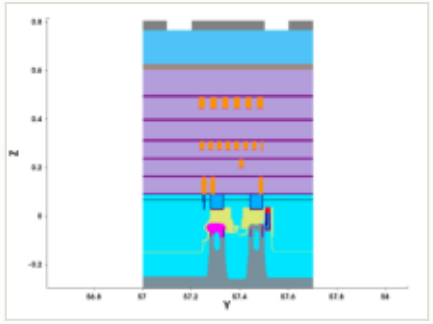
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

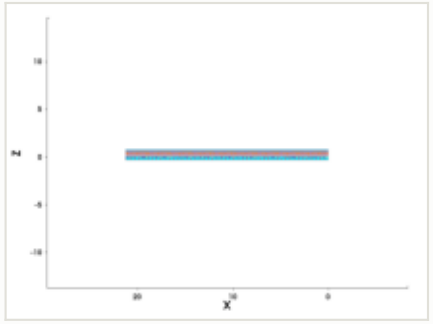


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

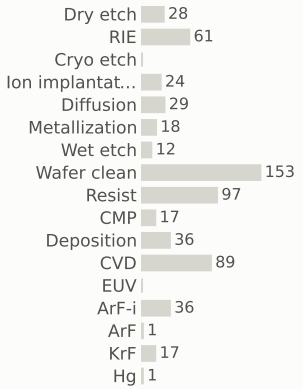


y cut at 57.351 µm



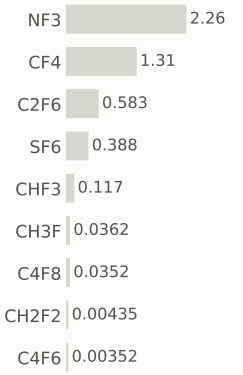
mask: layer 2900:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



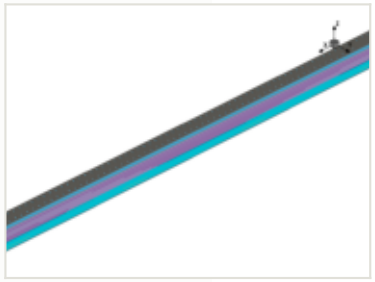
Step #8310: Spx_only

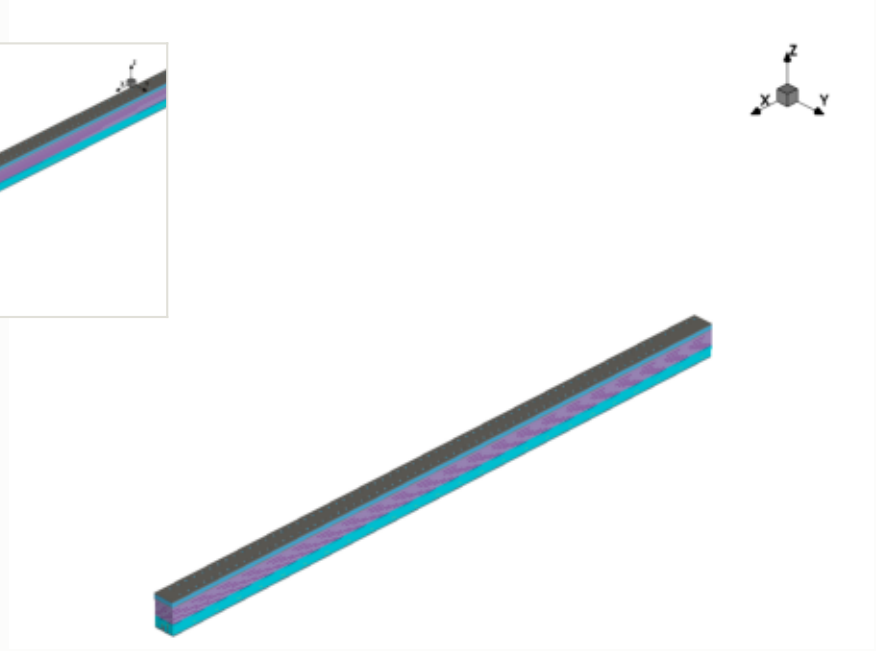
etch_anisotropic · Spx_only

0	375.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





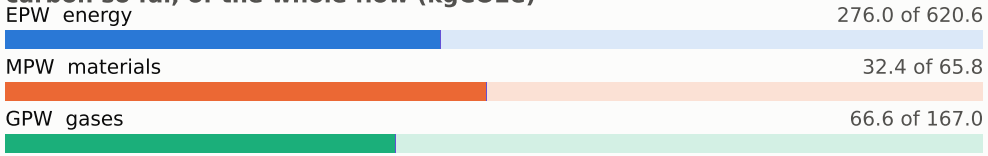
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

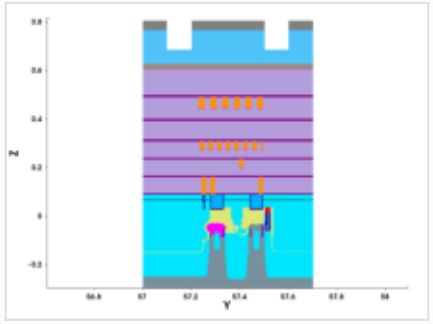
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

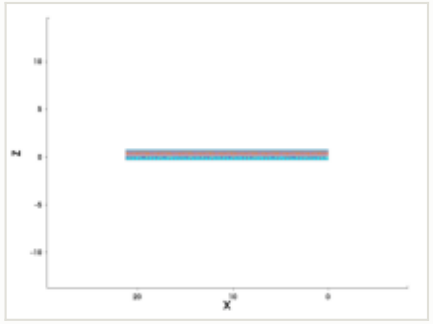


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



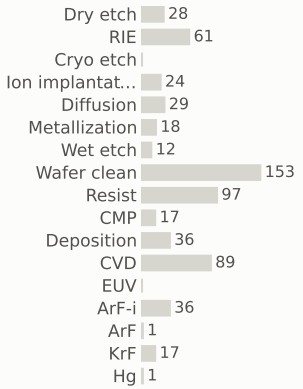
y cut at 57.351 µm



mask for this step

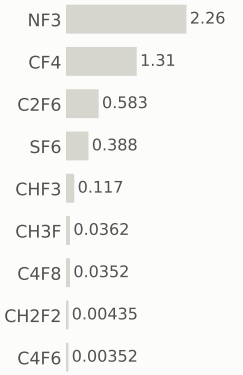
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



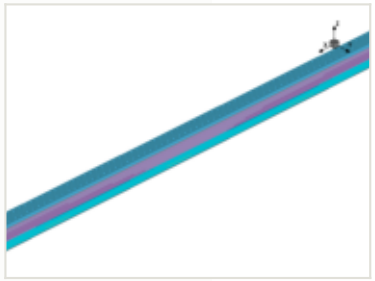
Step #8320: Spx_only

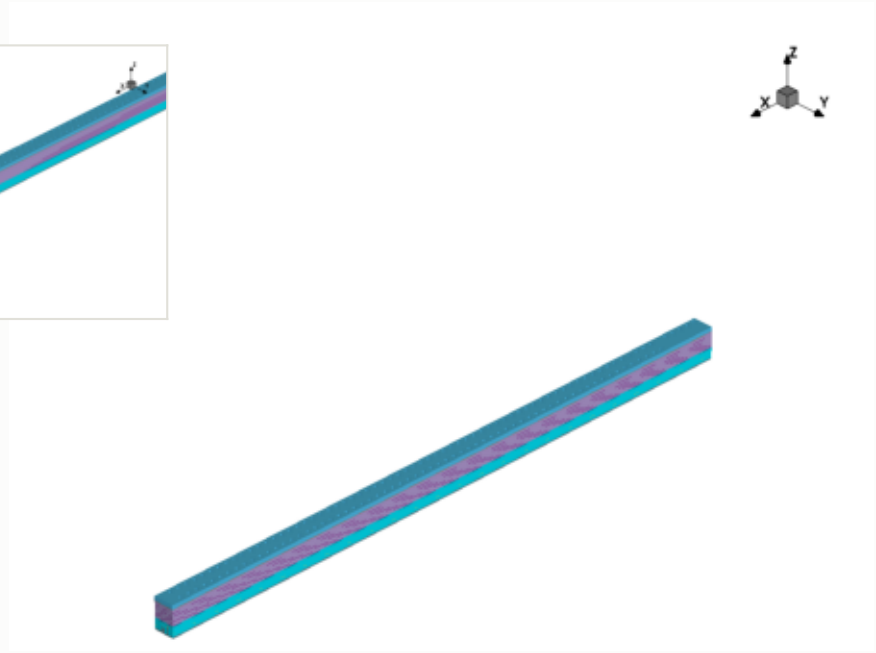
etch_strip · Spx_only

0	375.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



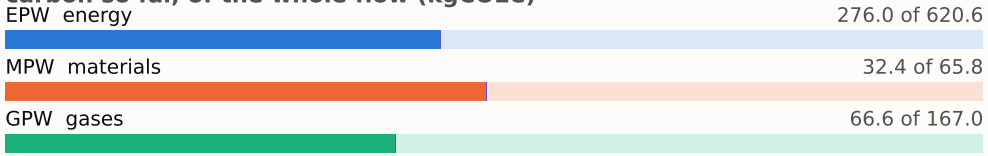


this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

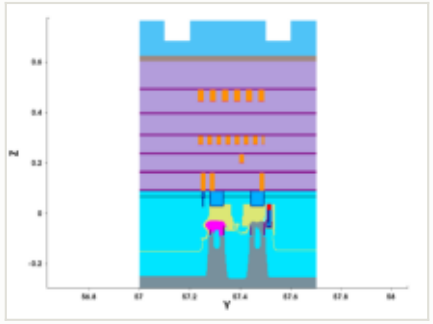
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

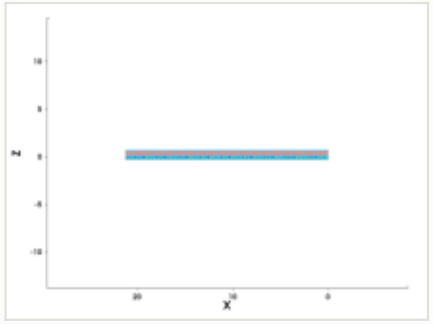


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



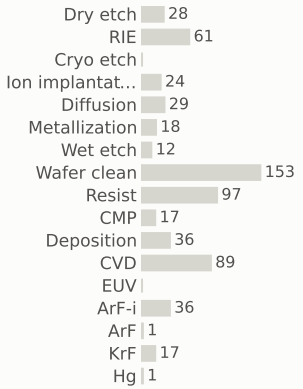
y cut at 57.351 µm



mask for this step

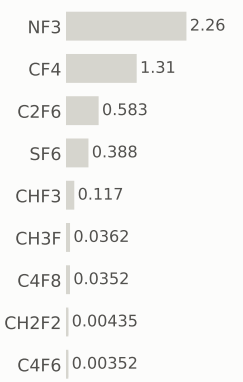
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



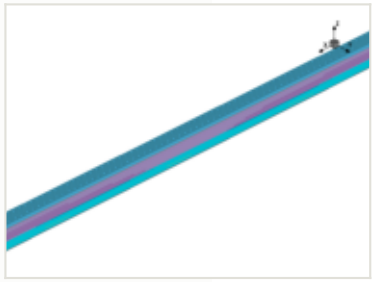
Step #8330: Wafer_clean

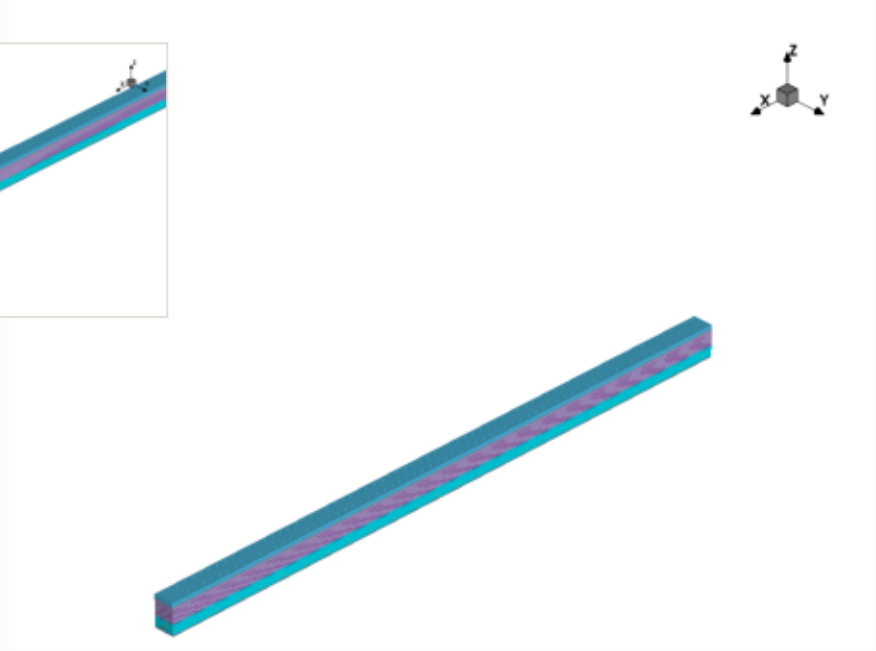
clean · Wafer_clean

0.169	375.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





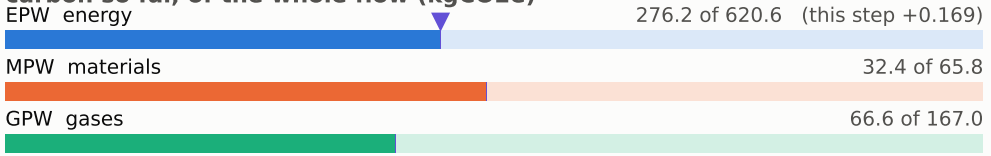
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

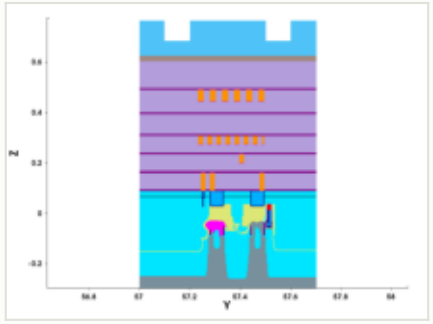
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

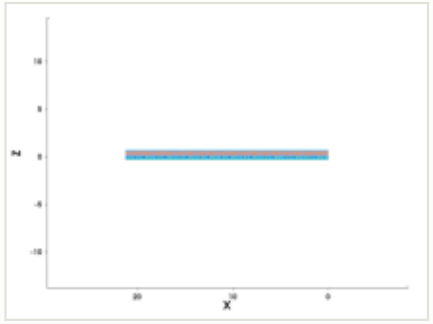


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



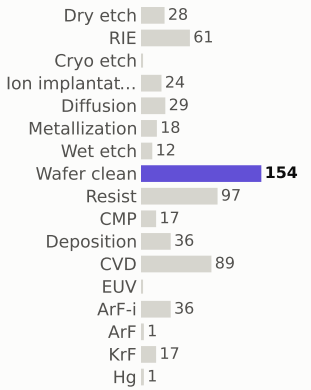
y cut at 57.351 µm



mask for this step

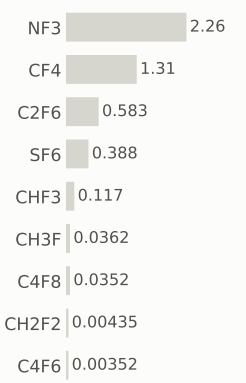
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



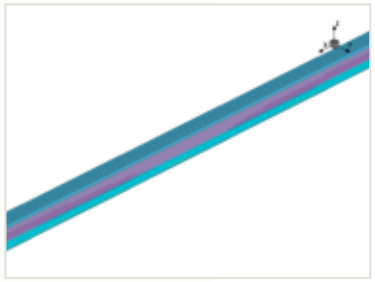
Step #8340: Resist

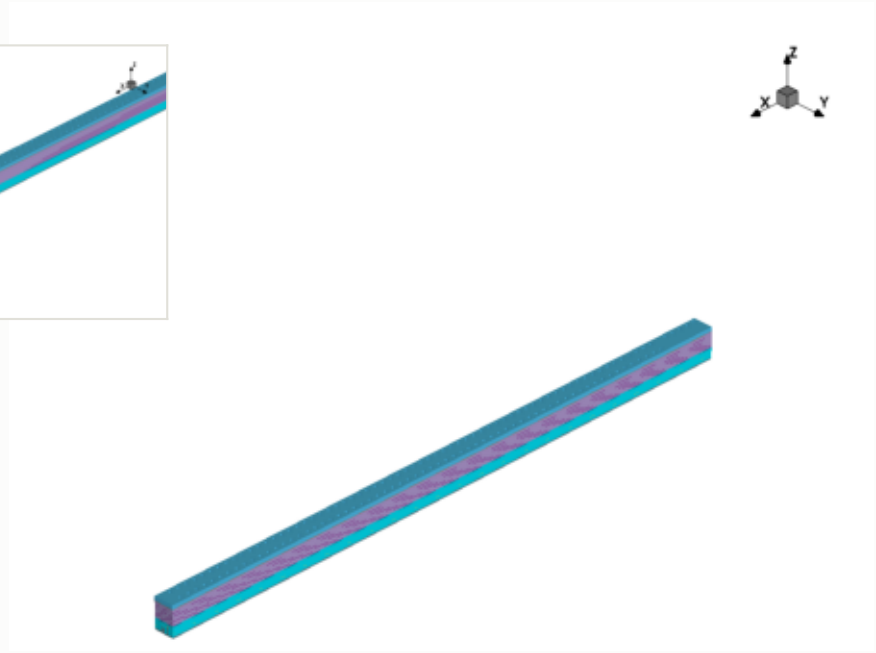
resist · Resist

0.169	375.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





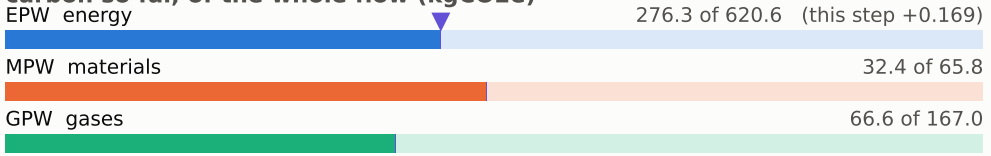
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

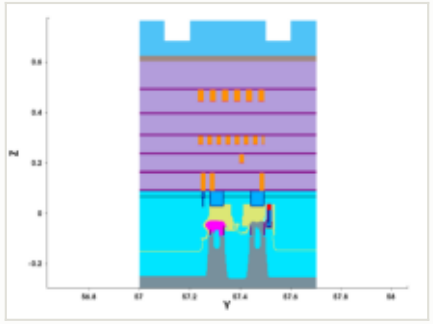
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

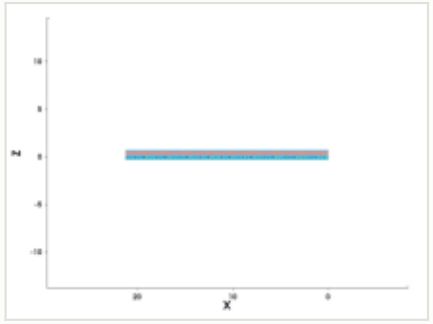


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



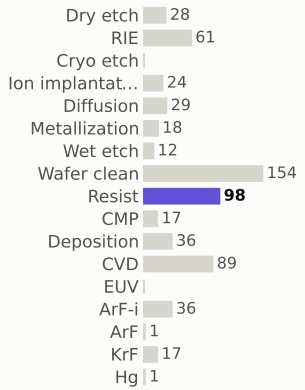
y cut at 57.351 µm



mask for this step

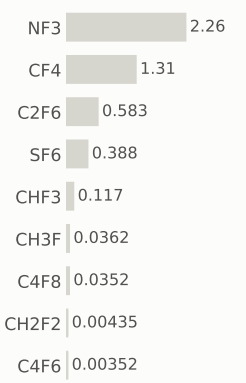
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



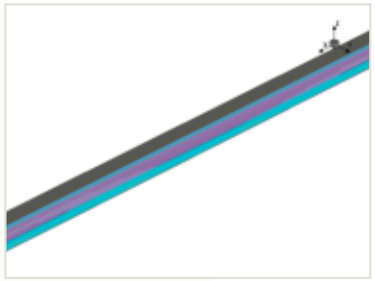
Step #8350: Hg

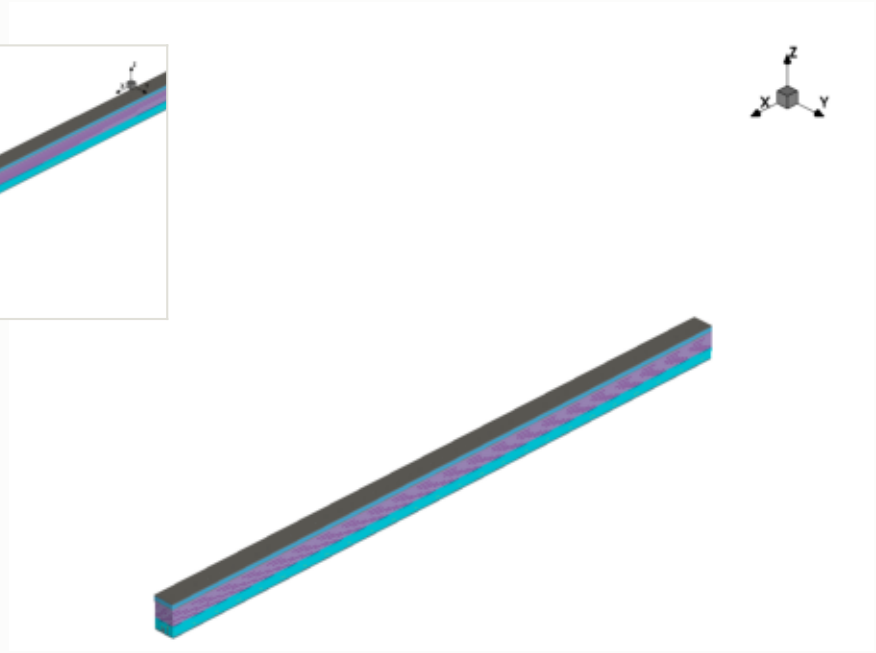
litho_patterning · Hg

0.199	375.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



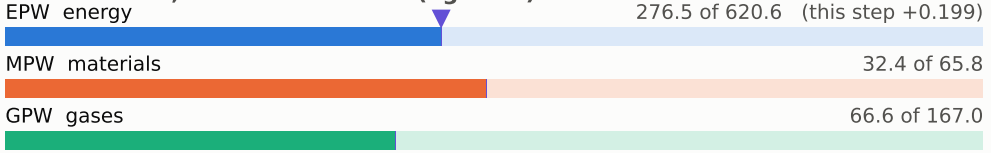


this step

process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

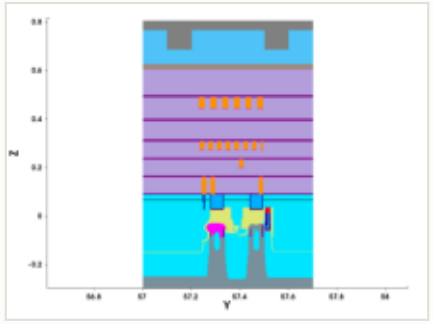
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

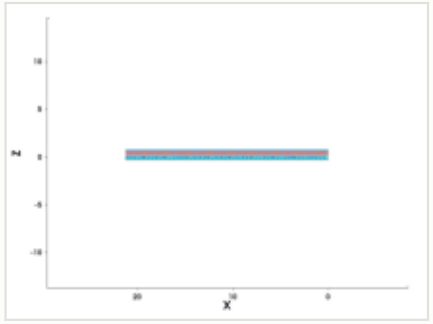


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

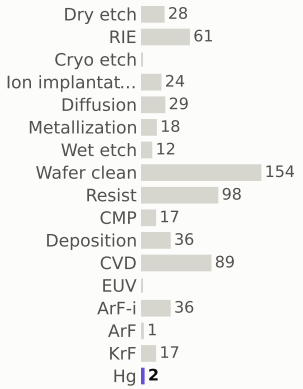


y cut at 57.351 µm



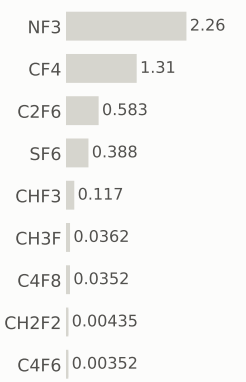
mask: layer 2085:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



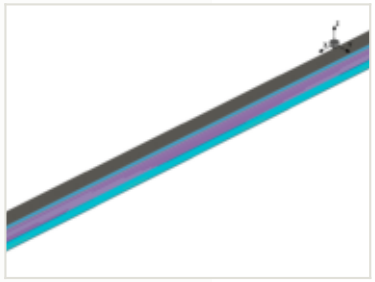
Step #8360: RIE

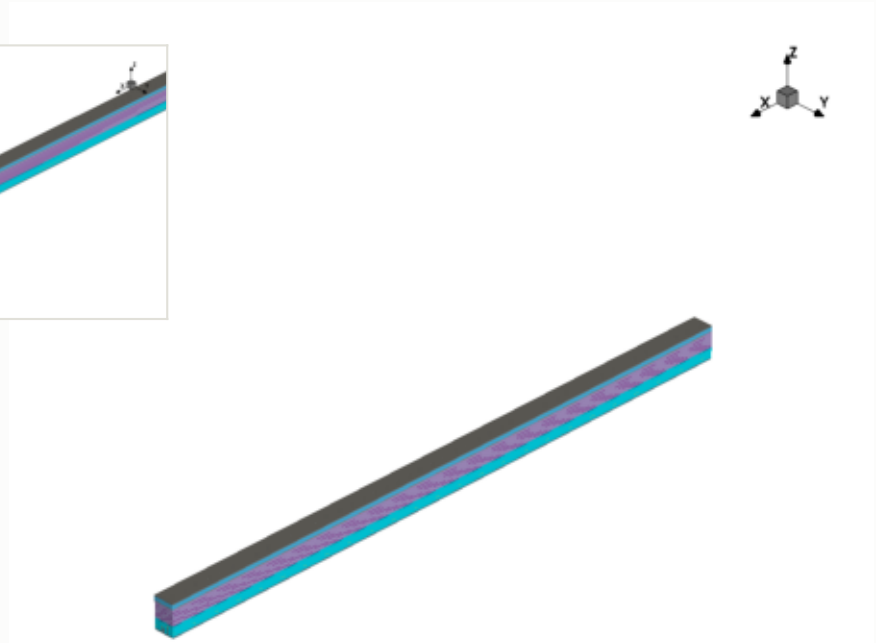
etch_anisotropic · RIE

1.04	376.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



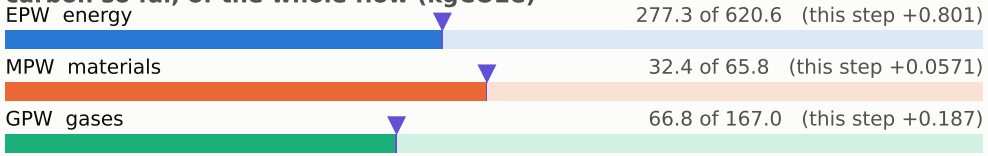


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

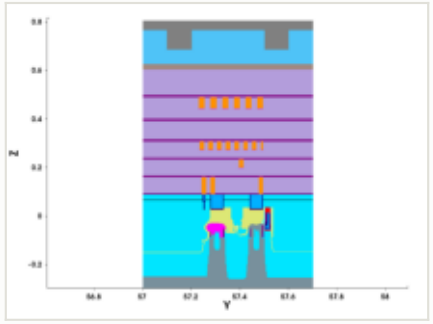
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

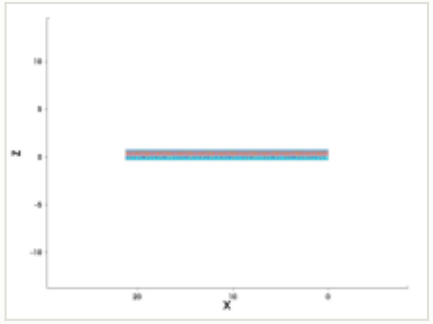


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



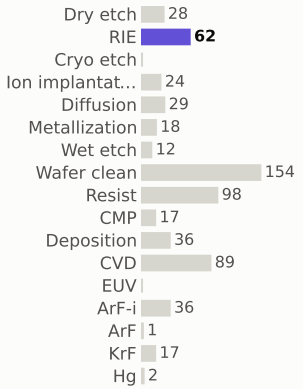
y cut at 57.351 µm



mask for this step

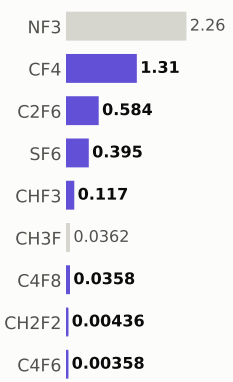
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



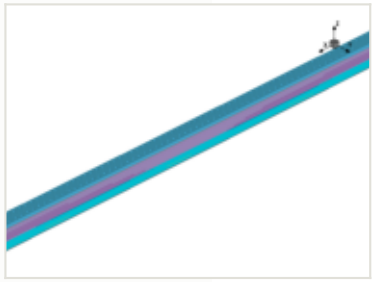
Step #8370: RIE

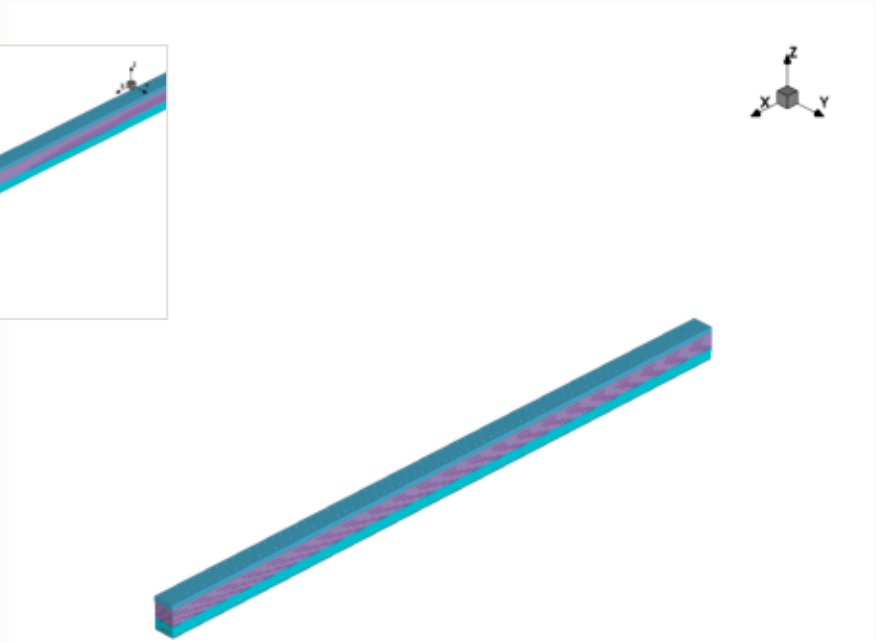
etch_strip · RIE

1.04	377.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



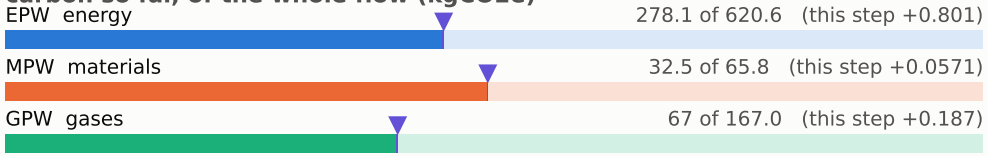


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

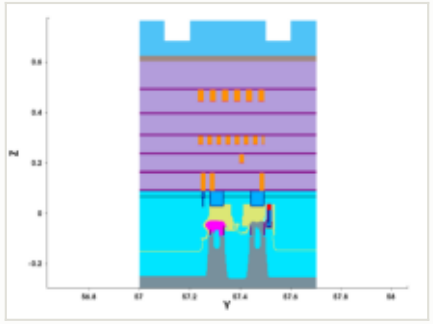
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

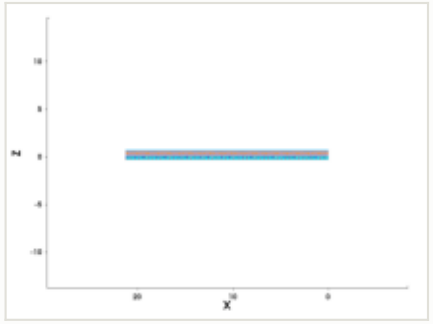


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



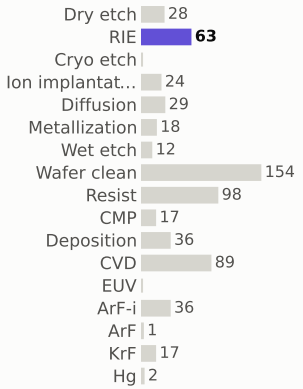
y cut at 57.351 µm



mask for this step

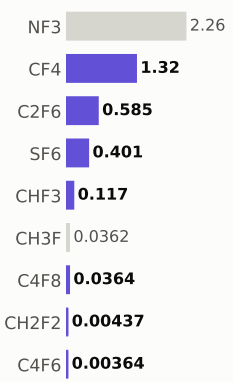
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



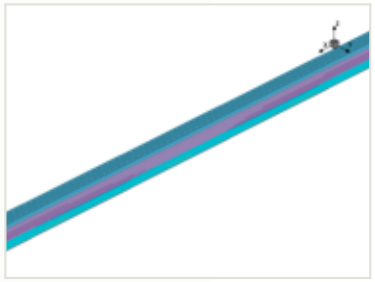
Step #8380: Wafer_clean

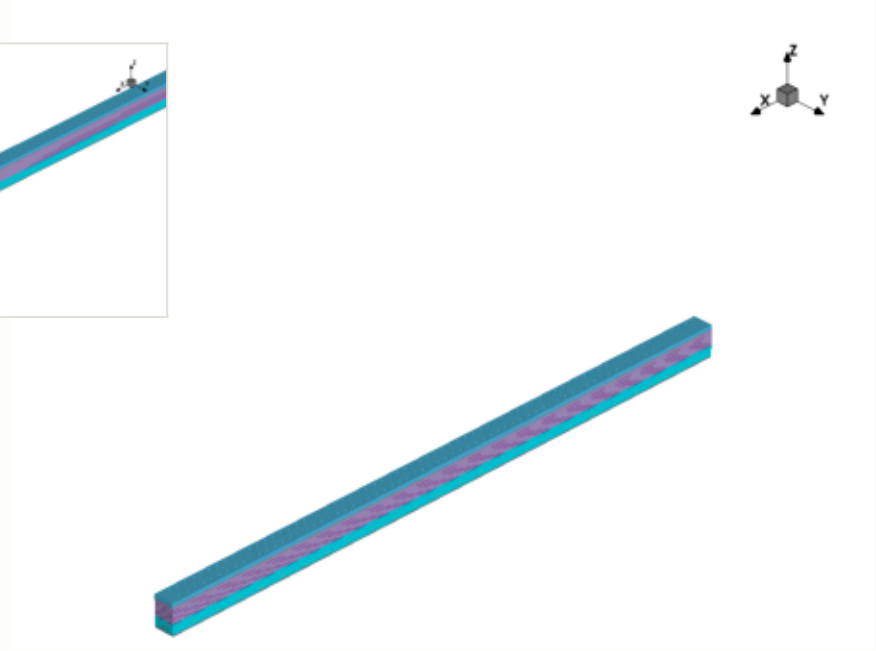
clean · Wafer_clean

0.169	377.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





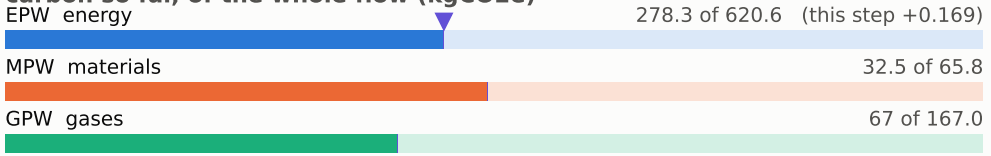
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

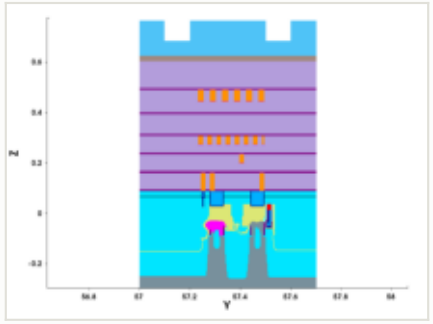
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

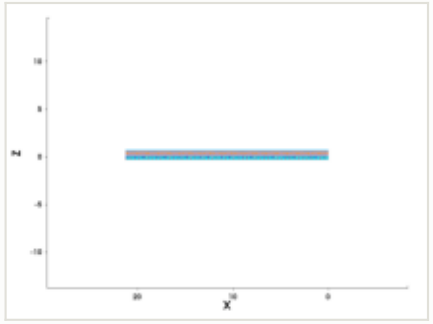


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



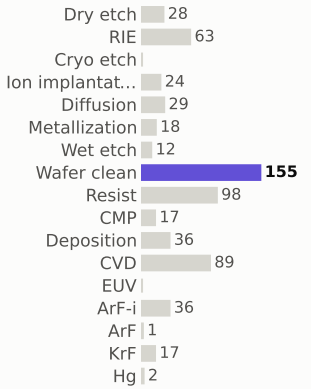
y cut at 57.351 µm



mask for this step

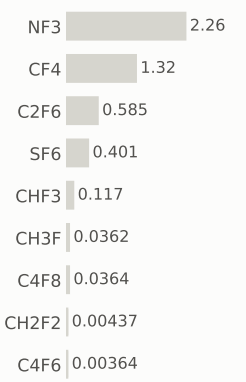
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



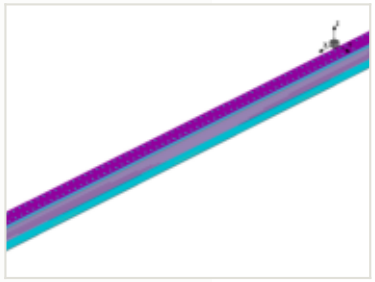
Step #8390: Metallization

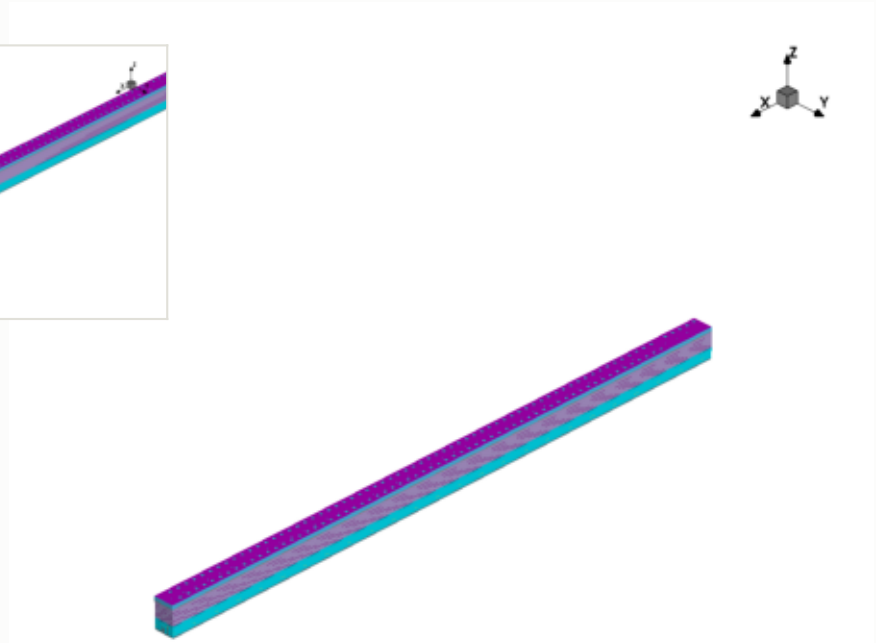
depo_ellipsoidal · Metallization

1.2	379.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Metallization
material	TaN
thickness	0.0016 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.00e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

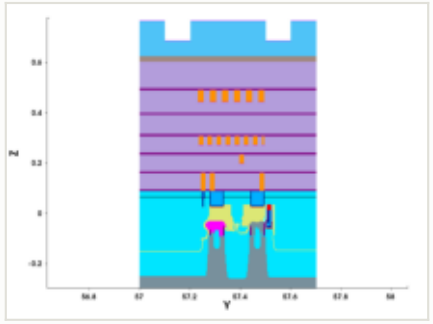
EPW energy — 279.5 of 620.6 (this step +1.2)

MPW materials — 32.5 of 65.8 (this step +4.00e-05)

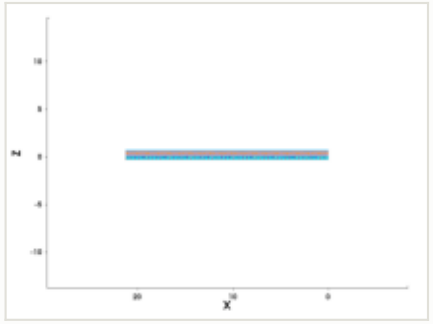
GPW gases — 67 of 167.0

embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



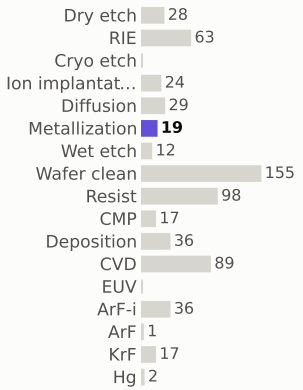
y cut at 57.351 µm



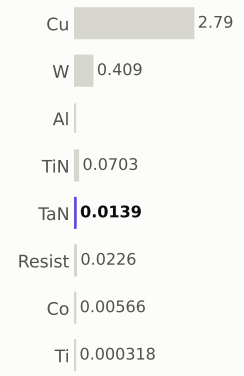
mask for this step

no mask for this step

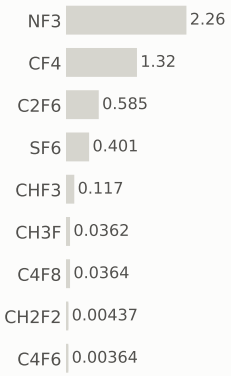
steps by process area



materials deposited (g)



gases emitted (g)



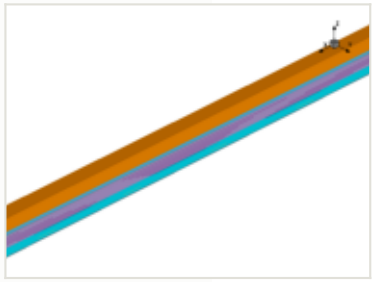
Step #8400: Spx_only

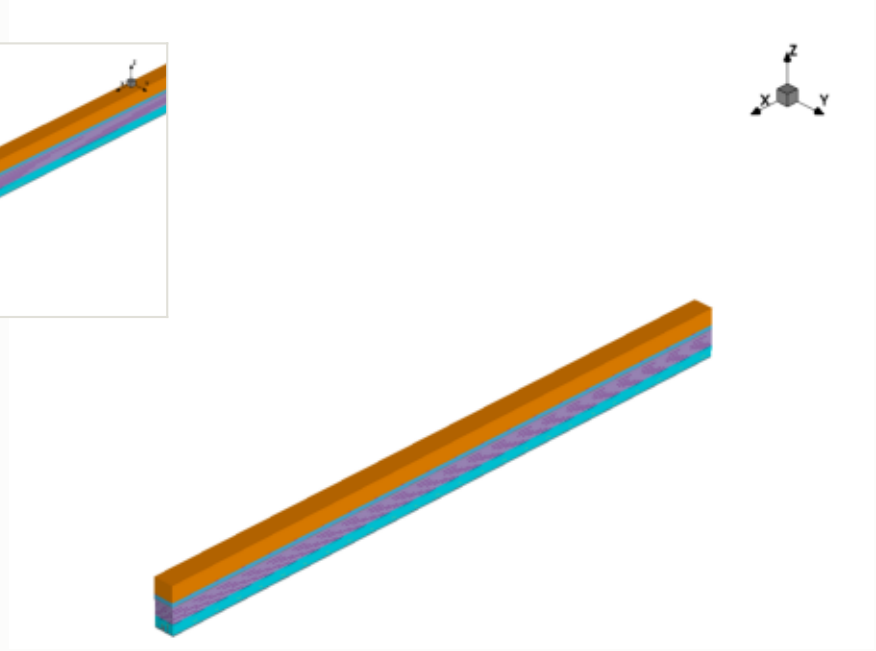
depo_fill · Spx_only

0.00212	379.0	<0.01%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





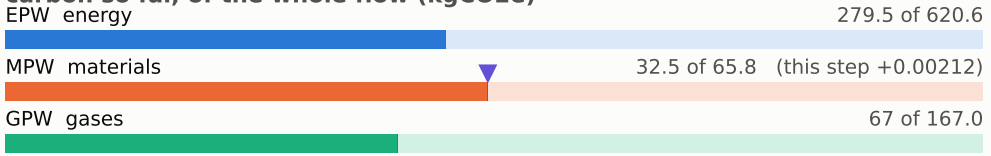
this step

process area	Spx_only
material	Cu
thickness	0.55 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0.00212 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.00212 kgCO2e**

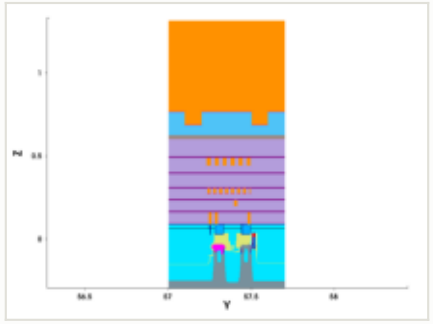
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

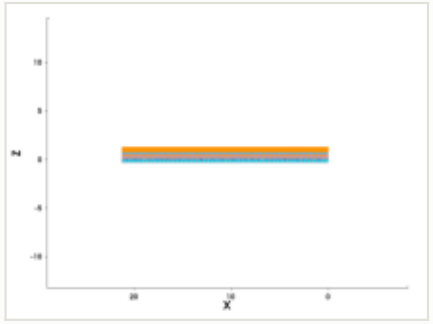


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



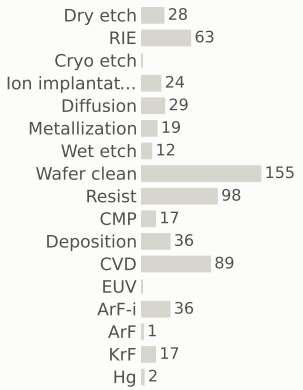
y cut at 57.351 µm



mask for this step

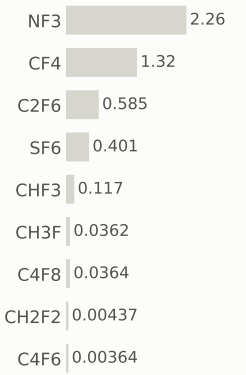
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



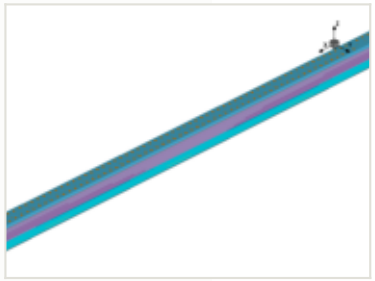
Step #8410: Spx_only

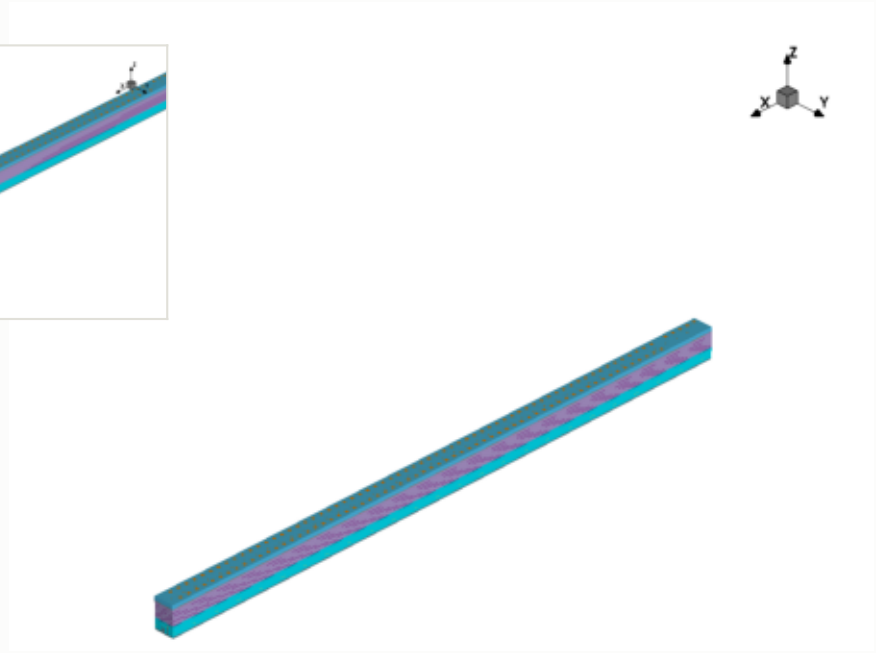
etch_cmp · Spx_only

0	379.0	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





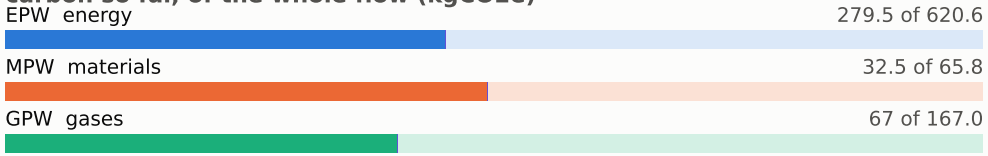
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

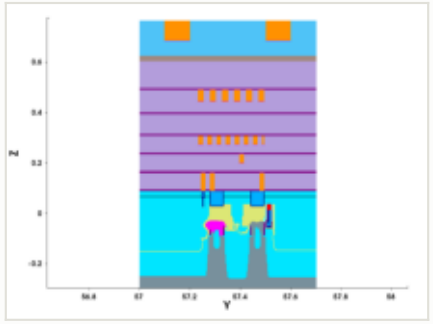
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

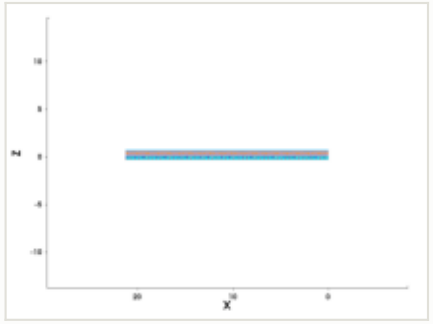


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



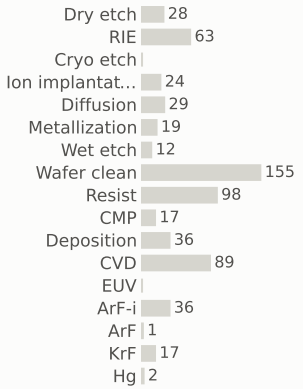
y cut at 57.351 µm



mask for this step

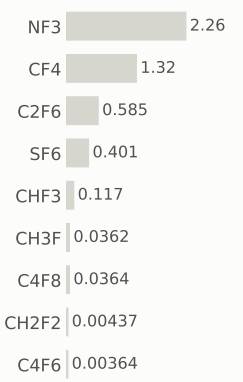
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



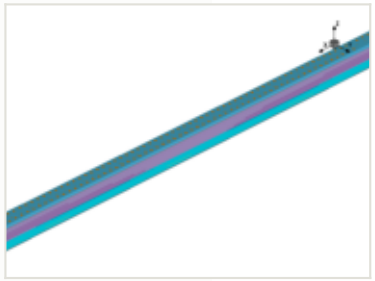
Step #8420: Resist

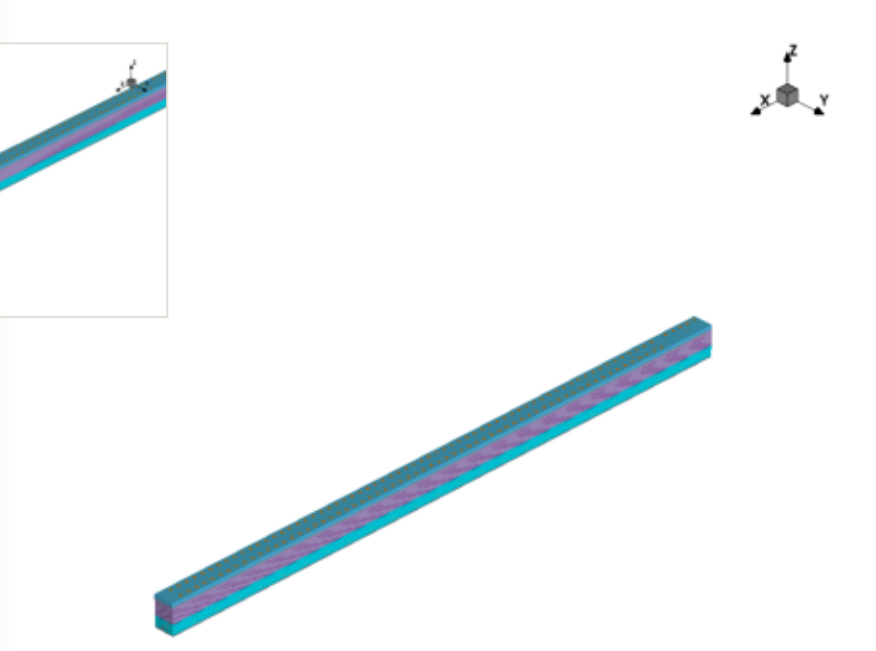
resist · Resist

0.169	379.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





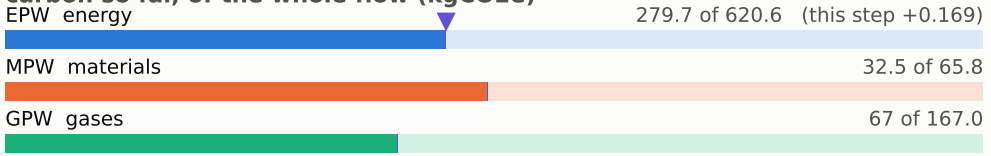
this step

process area	Resist
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

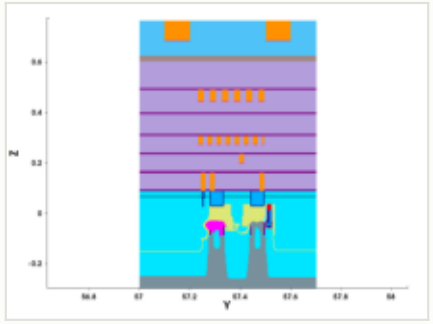
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

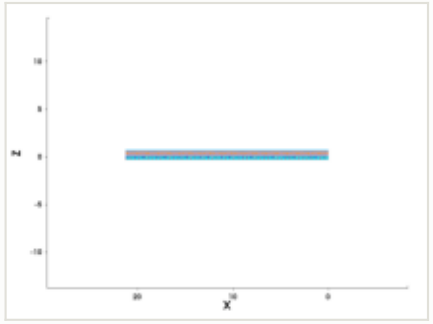


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



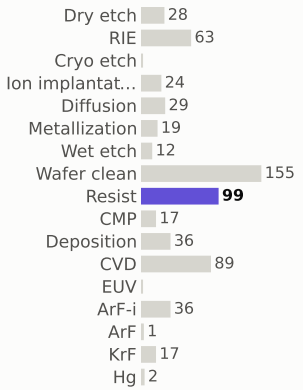
y cut at 57.351 µm



mask for this step

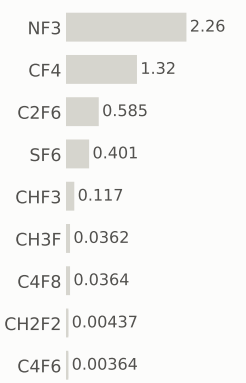
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



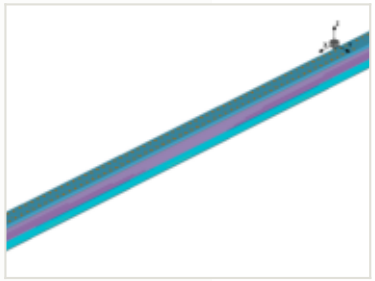
Step #8430: Hg

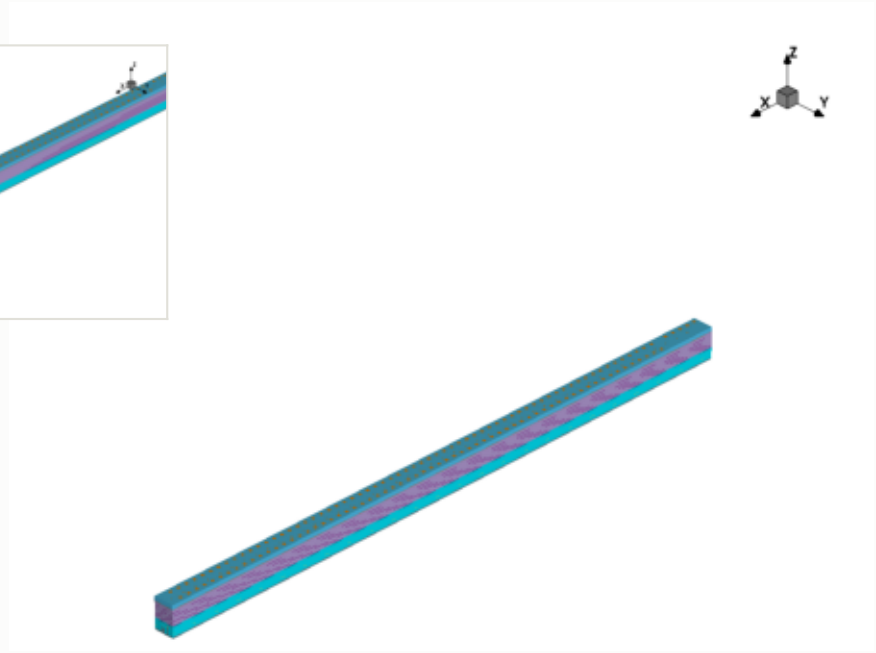
litho_patterning · Hg

0.199	379.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



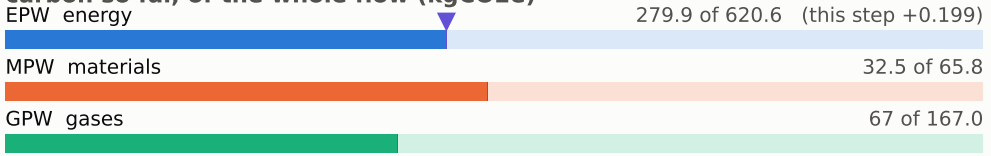


this step

process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

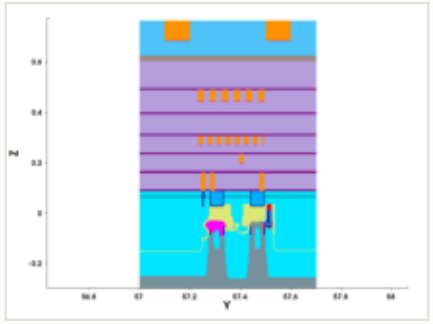
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

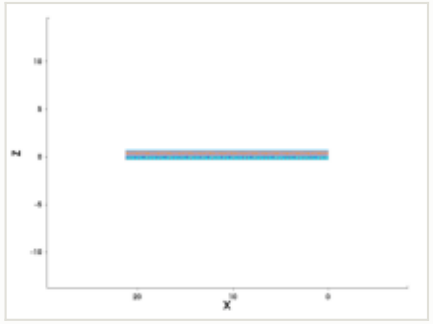


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

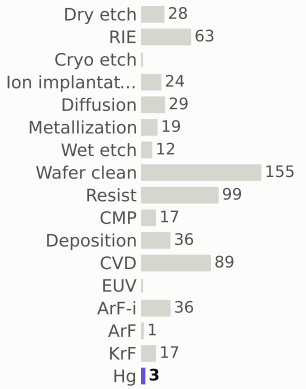


y cut at 57.351 µm



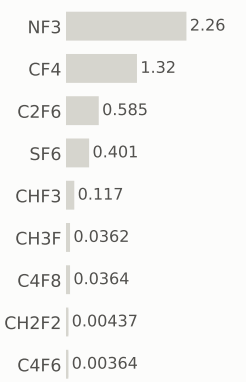
mask: layer 2900:0, positive (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



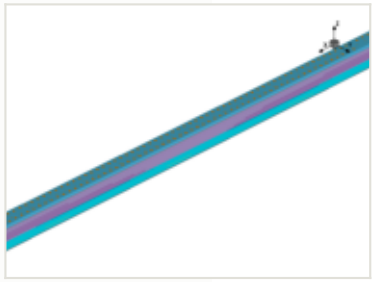
Step #8440: RIE

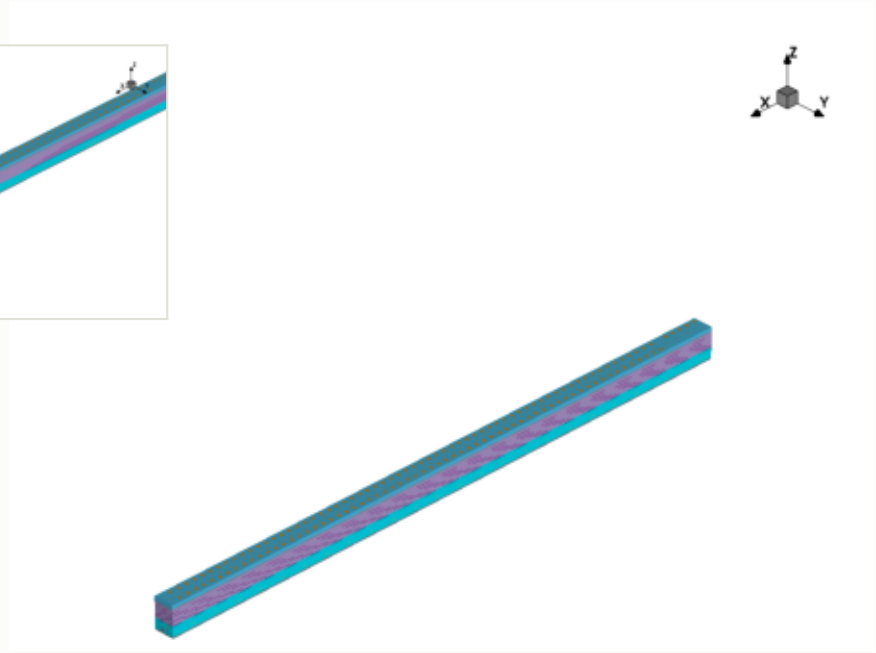
etch_anisotropic · RIE

1.04	380.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



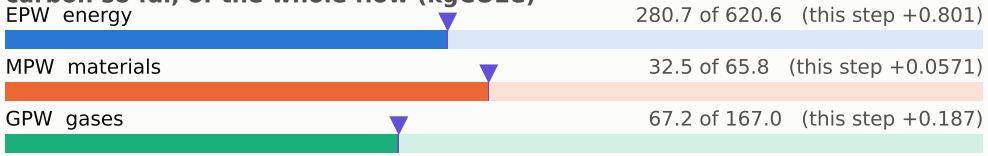


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

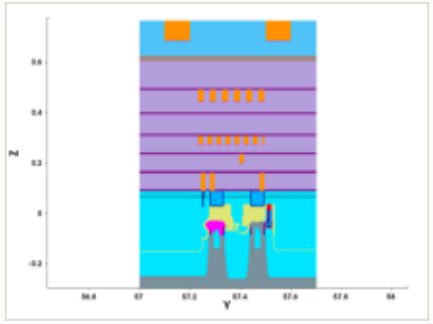
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

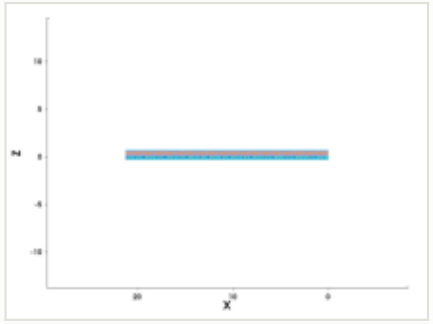


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



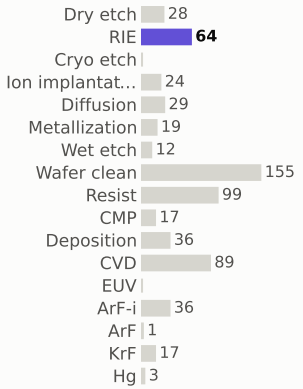
y cut at 57.351 µm



mask for this step

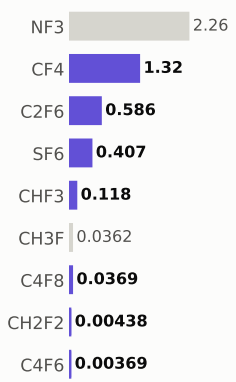
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



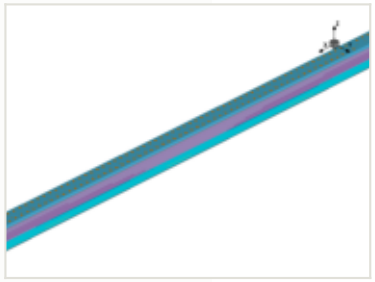
Step #8450: Wafer_clean

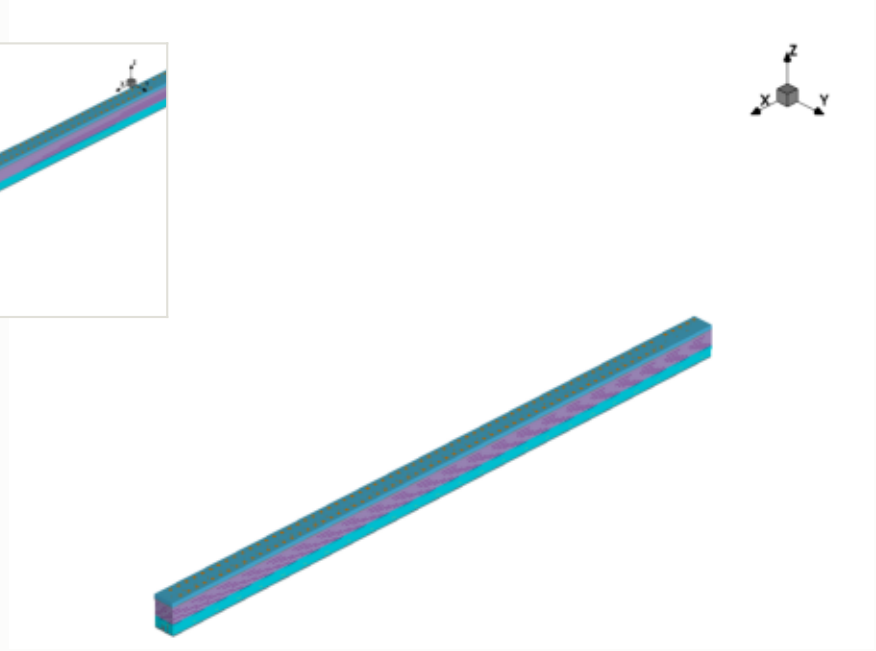
clean · Wafer_clean

0.169	380.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





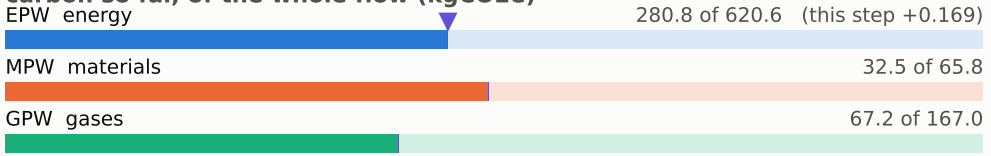
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

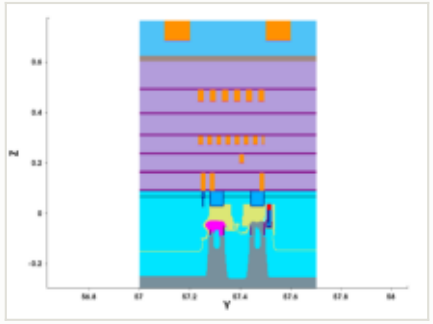
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

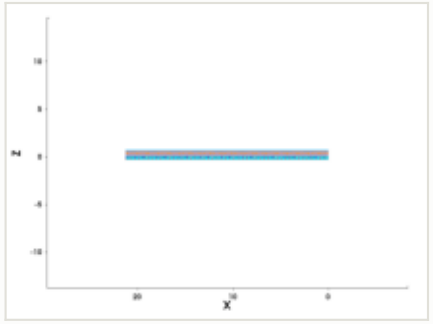


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



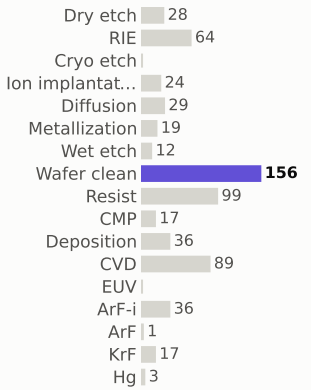
y cut at 57.351 µm



mask for this step

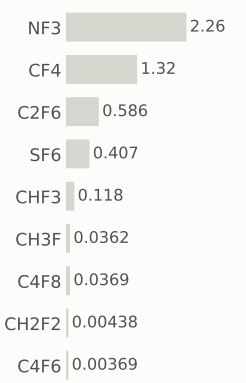
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



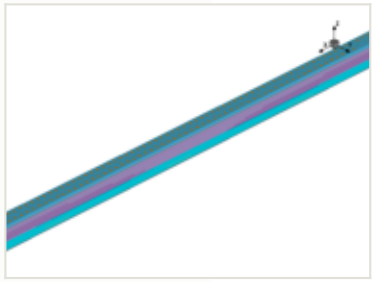
Step #8460: CVD

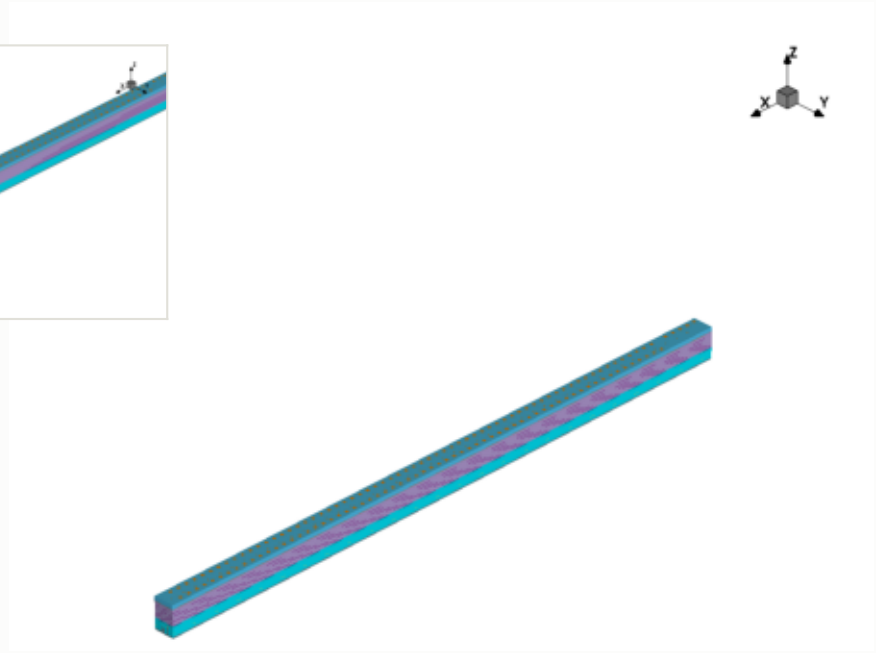
depo_isotropic · CVD

1.31	381.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



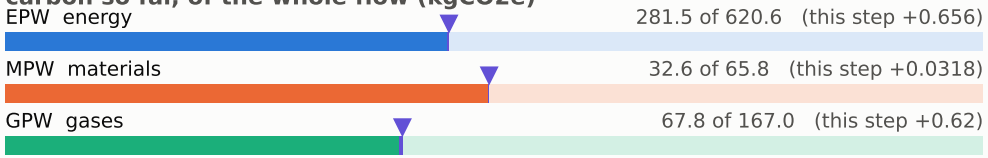


this step

process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

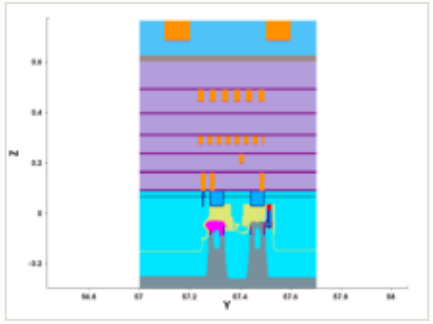
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

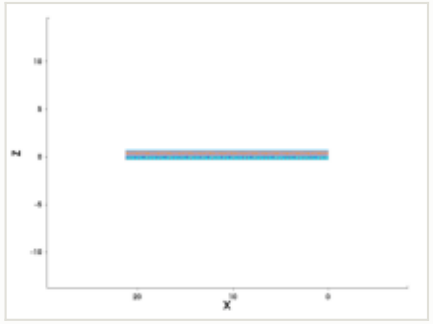


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



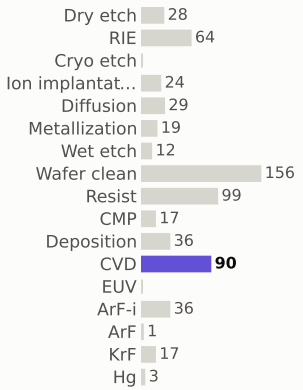
y cut at 57.351 µm



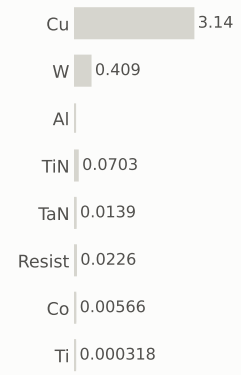
mask for this step

no mask for this step

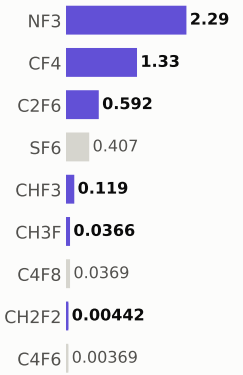
steps by process area



materials deposited (g)



gases emitted (g)



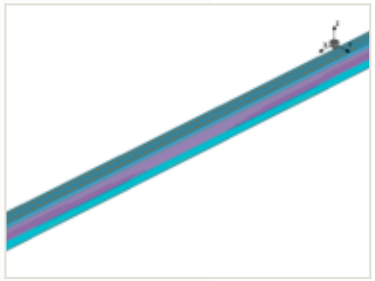
Step #8470: Diffusion

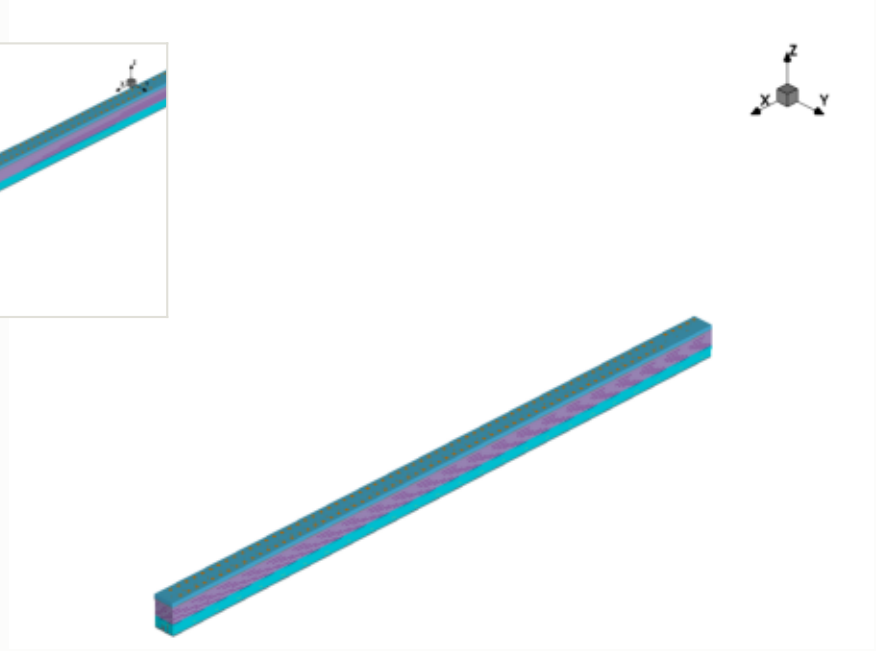
anneal · Diffusion

0.801	382.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





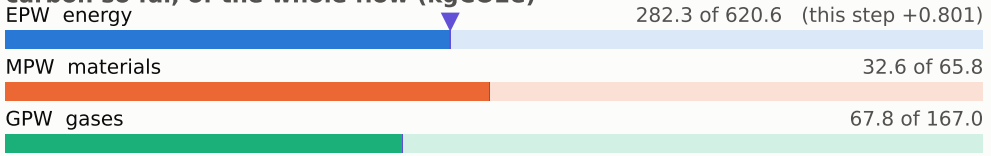
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

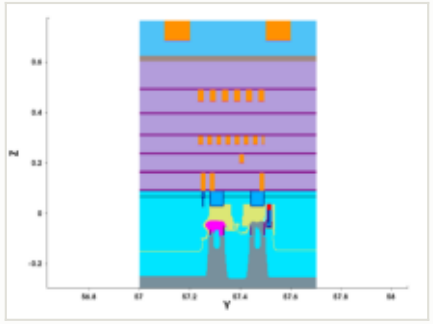
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

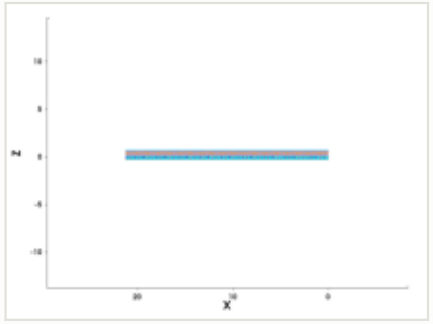


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



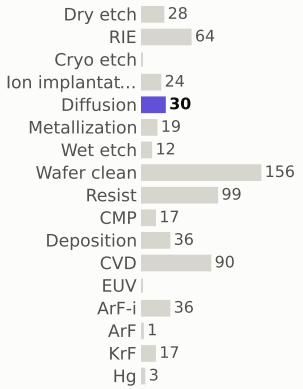
y cut at 57.351 µm



mask for this step

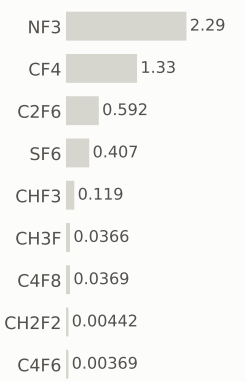
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



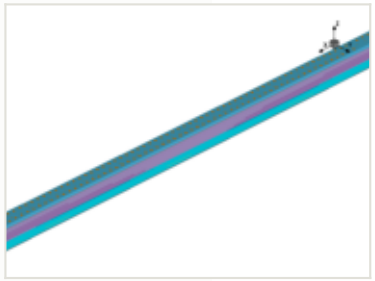
Step #8480: CVD

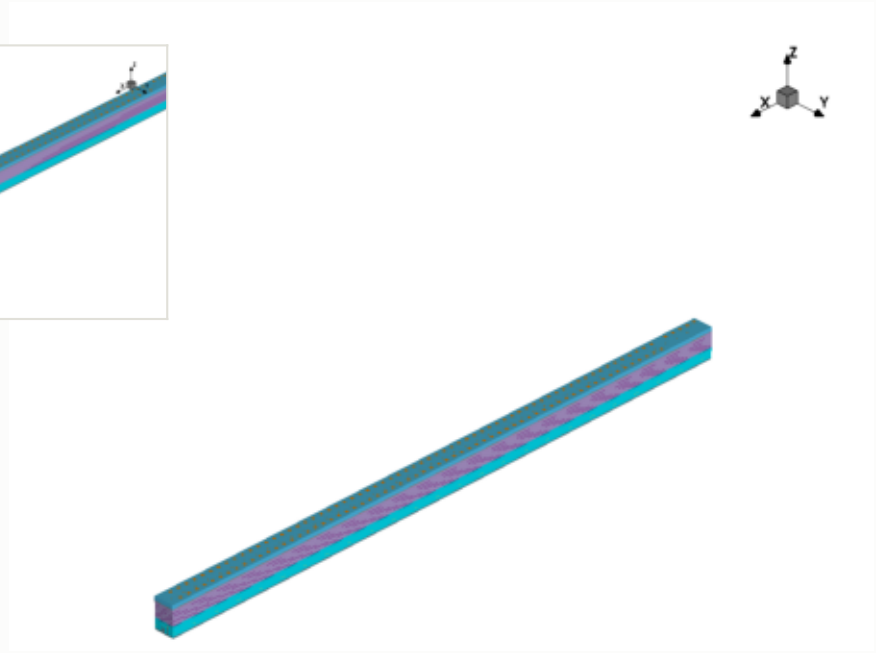
depo_isotropic · CVD

1.31	384.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



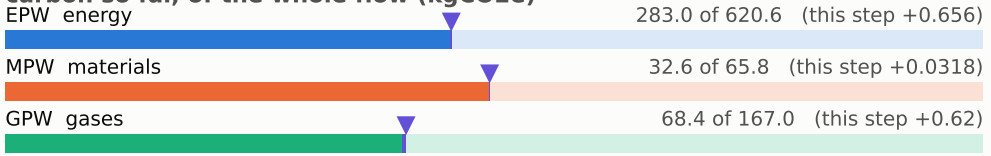


this step

process area	CVD
material	—
thickness	0 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

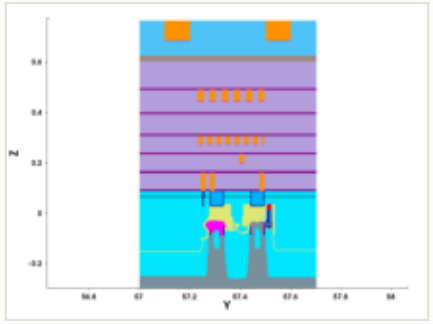
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

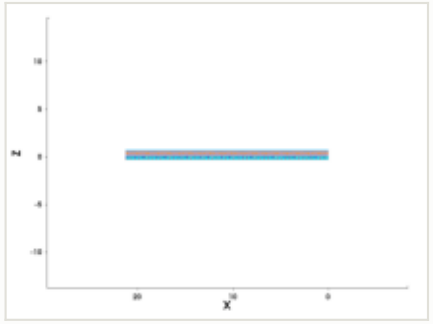


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



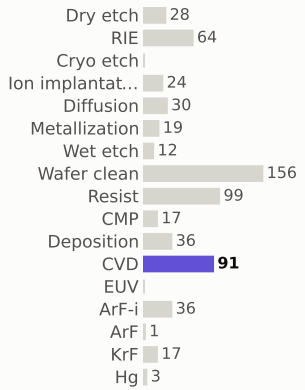
y cut at 57.351 µm



mask for this step

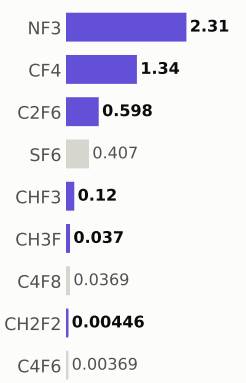
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



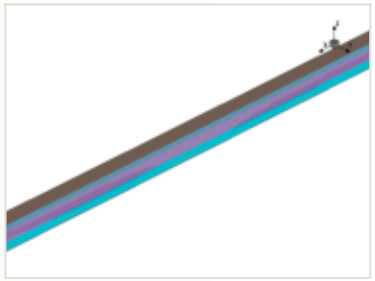
Step #8500: CVD

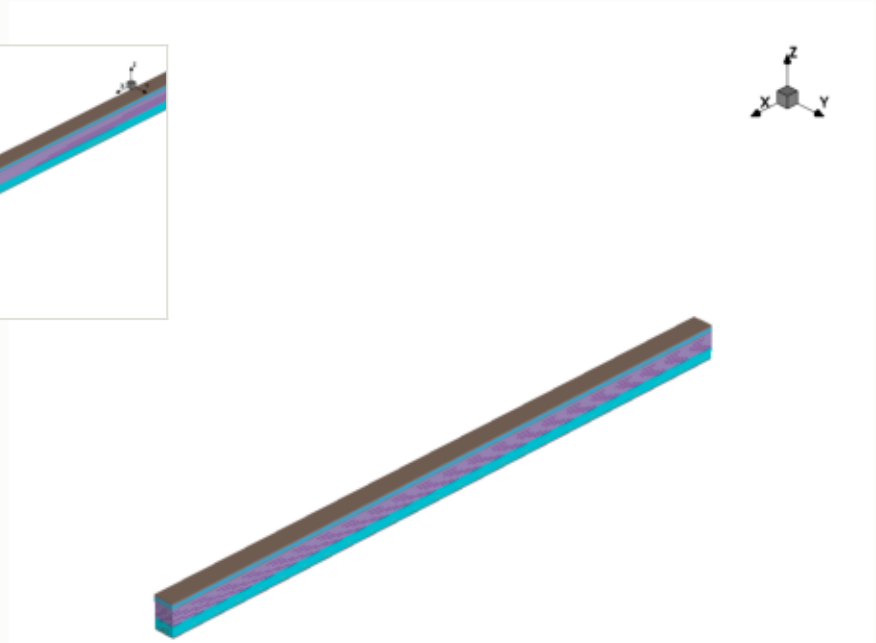
depo_isotropic · CVD

1.31	385.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



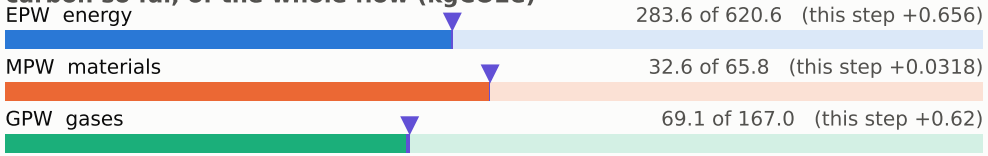


this step

process area	CVD
material	—
thickness	0.05 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

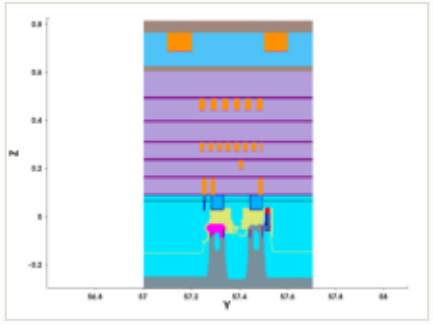
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

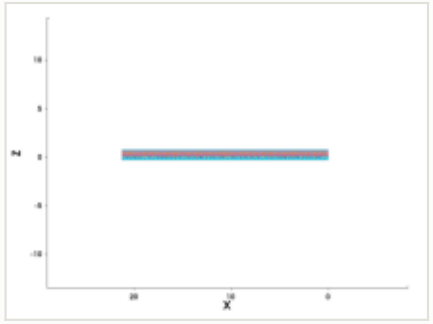


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



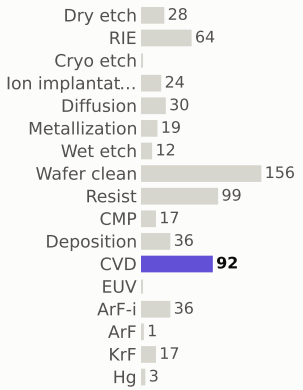
y cut at 57.351 µm



mask for this step

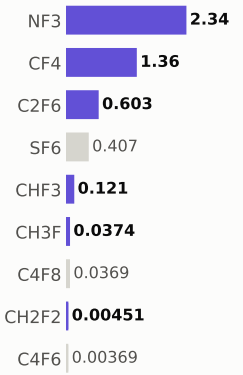
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



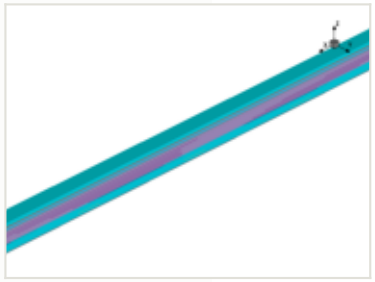
Step #8510: CVD

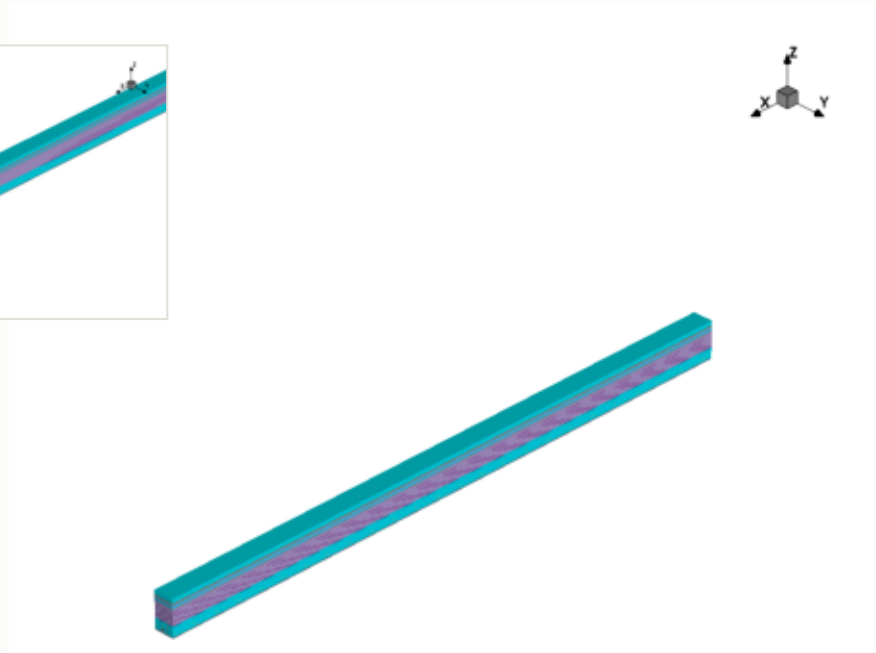
depo_isotropic · CVD

1.31	386.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



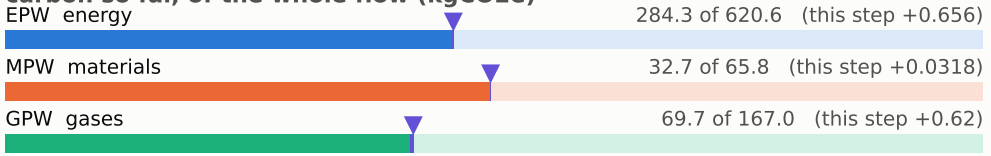


this step

process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

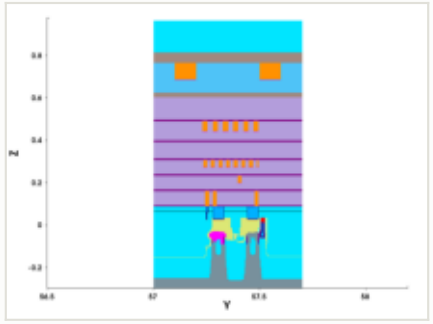
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

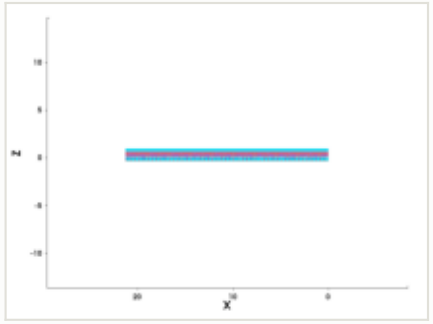


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



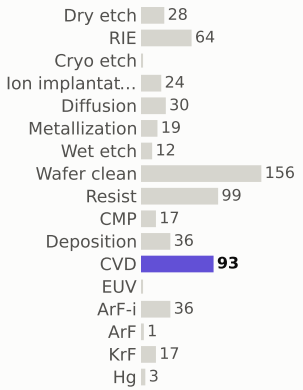
y cut at 57.351 µm



mask for this step

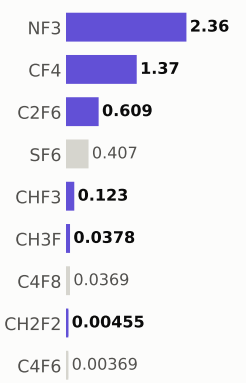
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



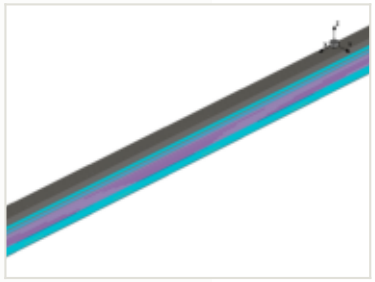
Step #8520: ArF

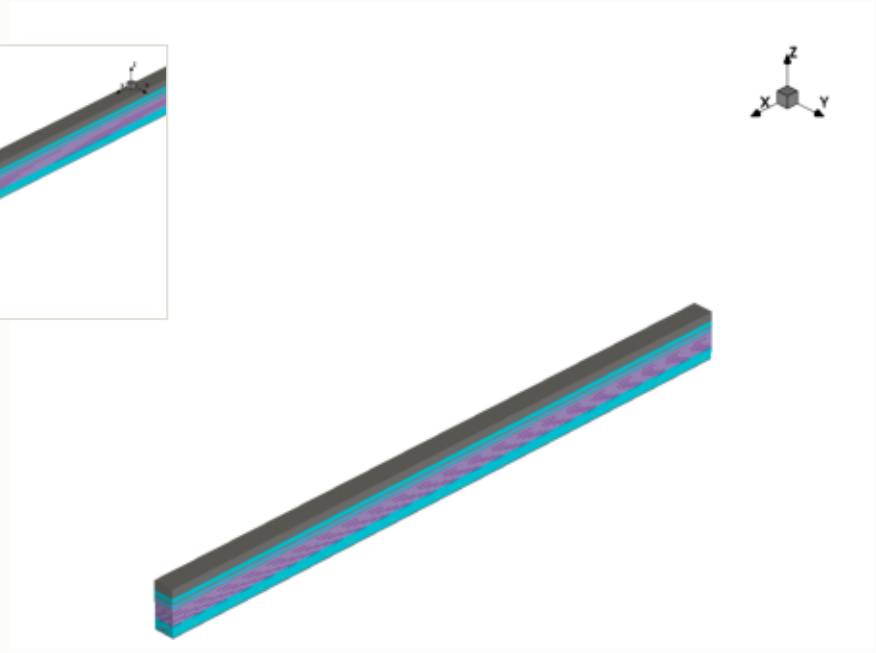
litho_patterning · ArF

0.199	386.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



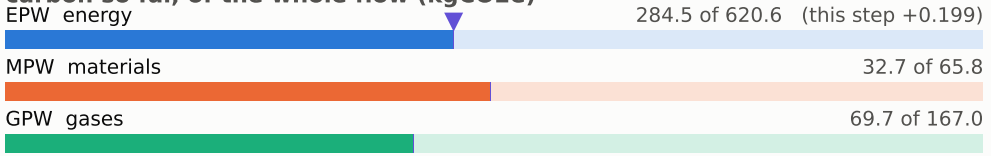


this step

process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

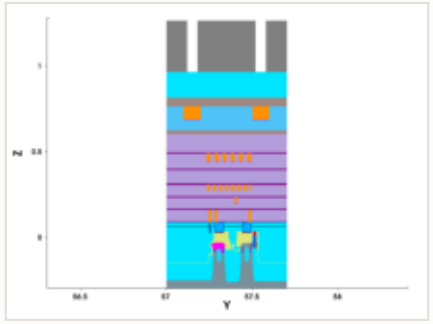
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

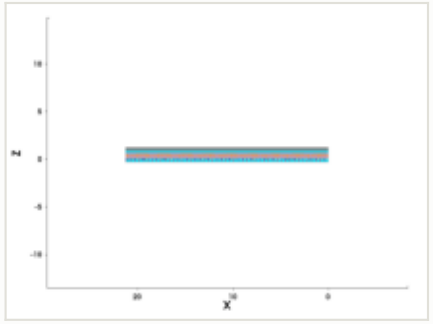


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

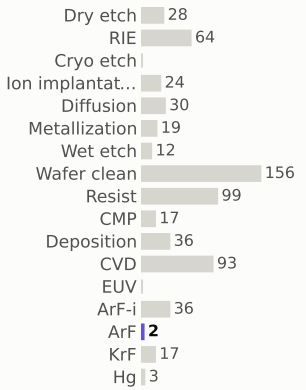


y cut at 57.351 µm



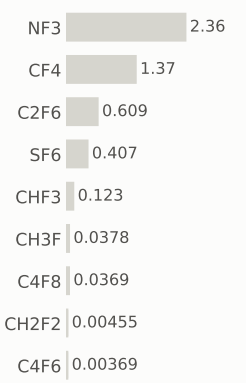
mask: layer 2905:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



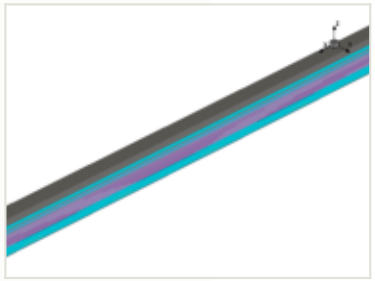
Step #8530: RIE

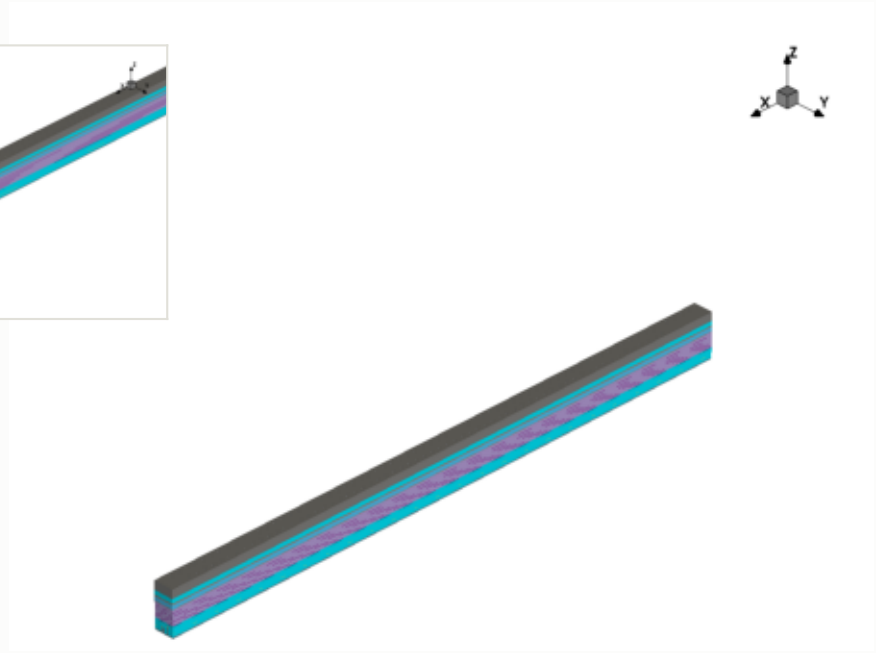
etch_anisotropic · RIE

1.04	387.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



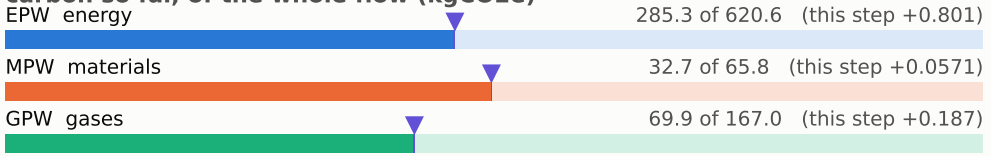


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

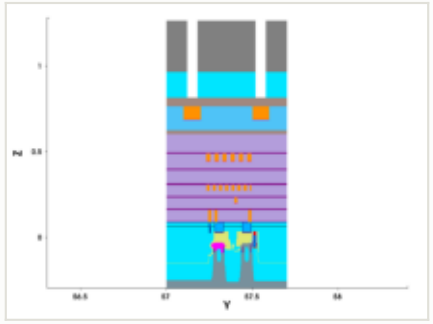
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

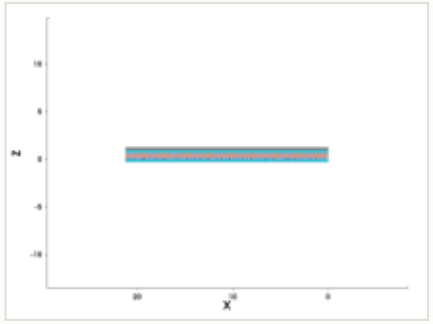


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



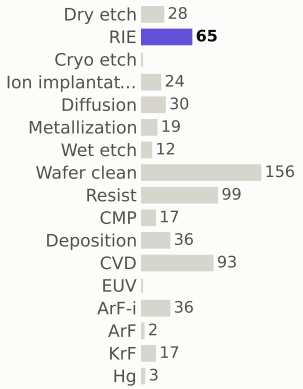
y cut at 57.351 µm



mask for this step

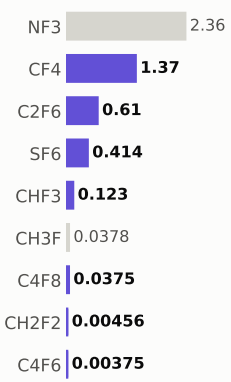
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



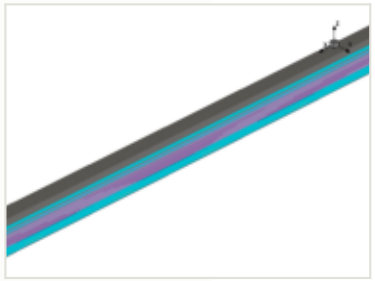
Step #8540: RIE

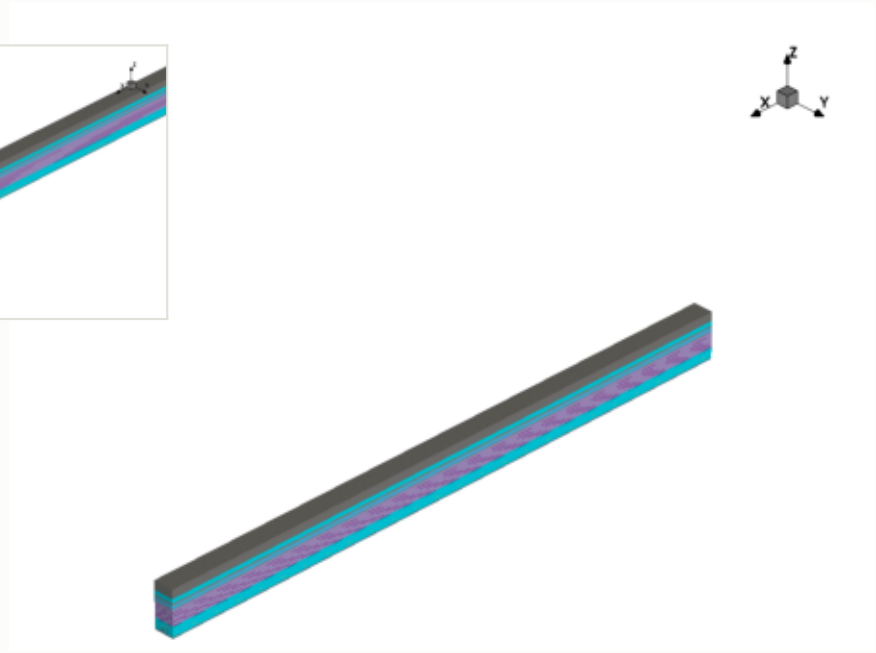
etch_anisotropic · RIE

1.04	388.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



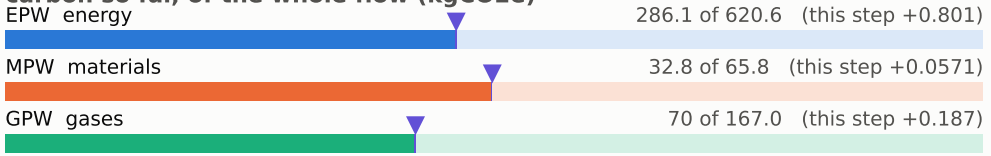


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

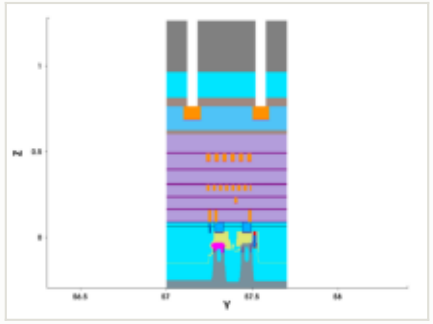
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

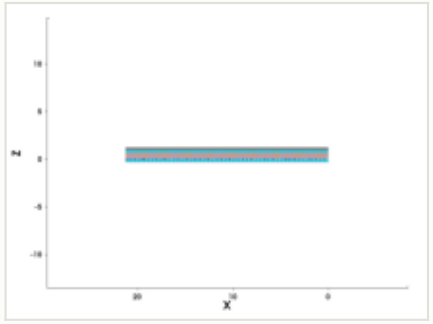


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



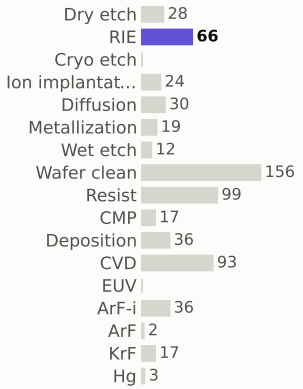
y cut at 57.351 µm



mask for this step

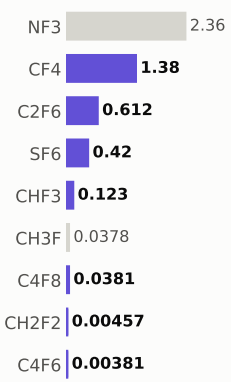
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



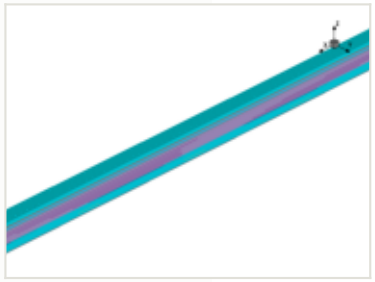
Step #8550: Dry_etch

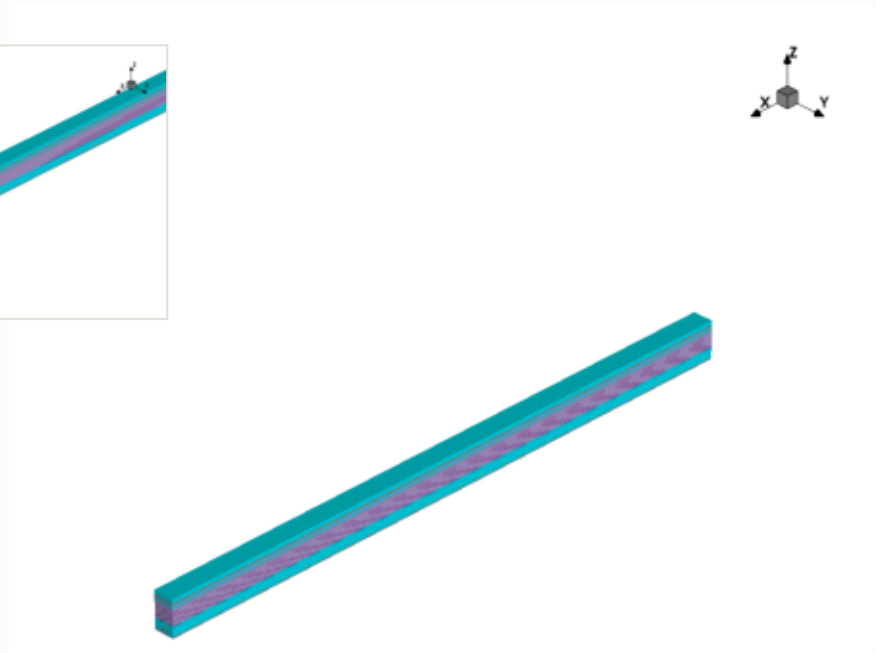
etch_strip · Dry_etch

0.801	389.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



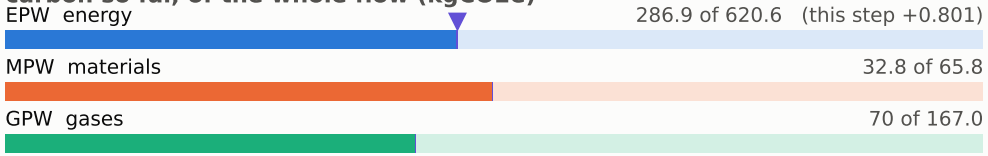


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

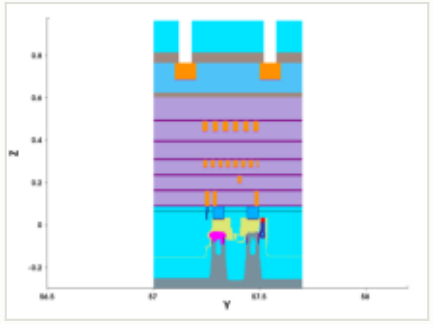
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

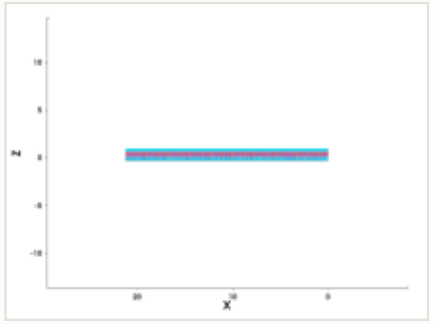


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



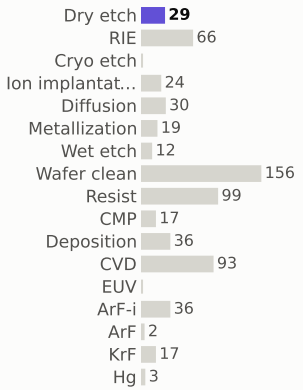
y cut at 57.351 µm



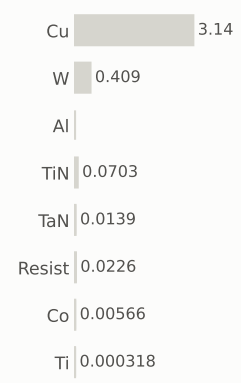
mask for this step

no mask for this step

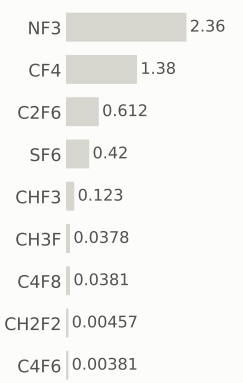
steps by process area



materials deposited (g)



gases emitted (g)



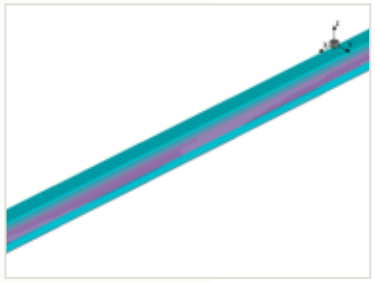
Step #8555: Wafer_clean

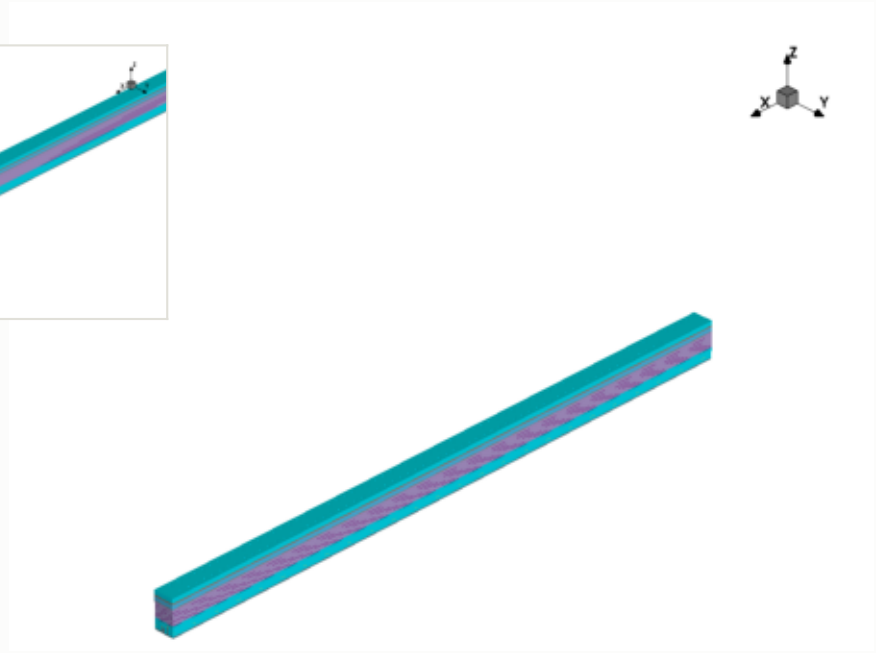
clean · Wafer_clean

0.169	389.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





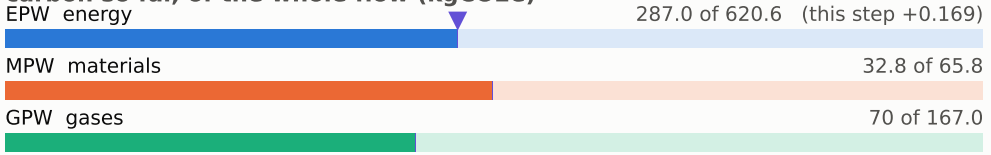
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

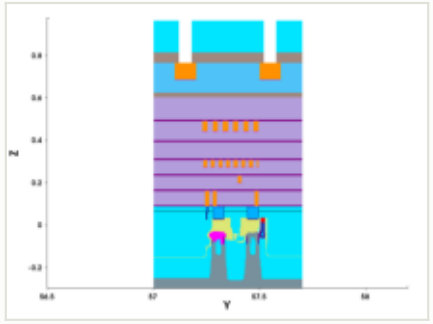
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

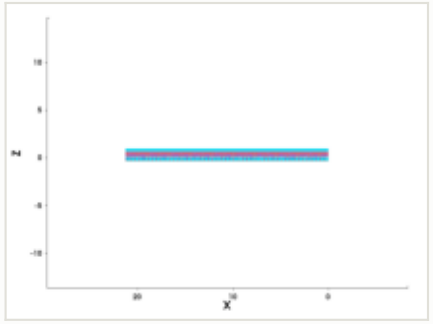


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



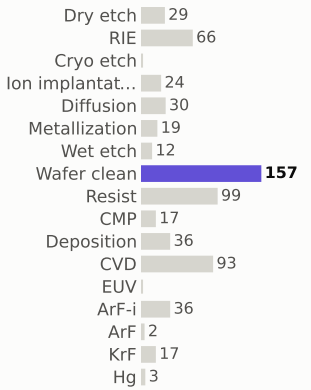
y cut at 57.351 µm



mask for this step

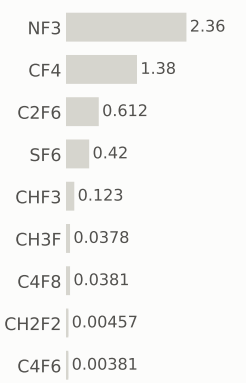
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



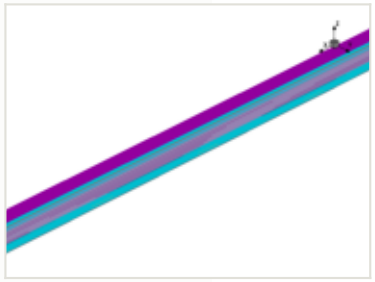
Step #8560: Deposition

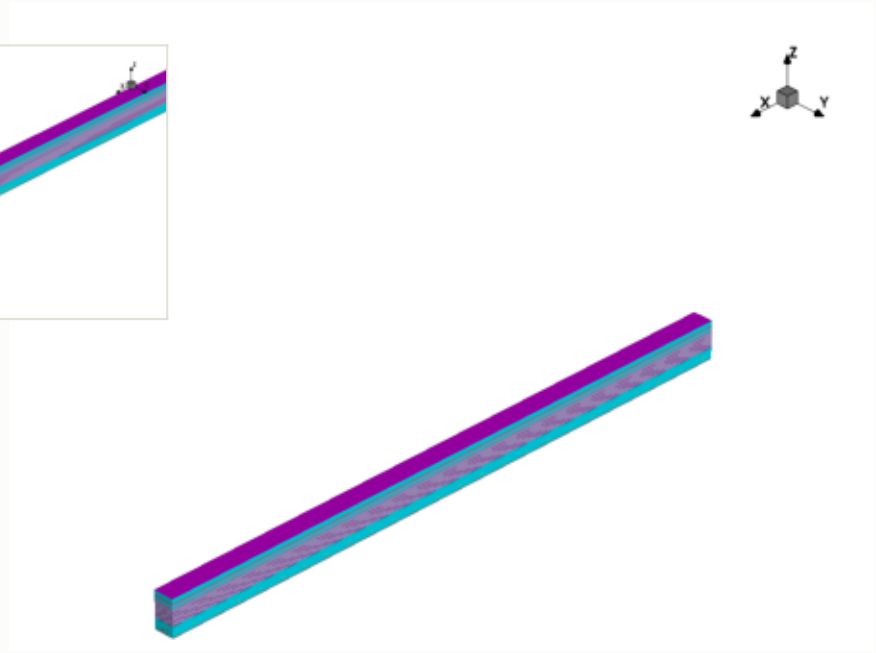
depo_isotropic · Deposition

0.656	390.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



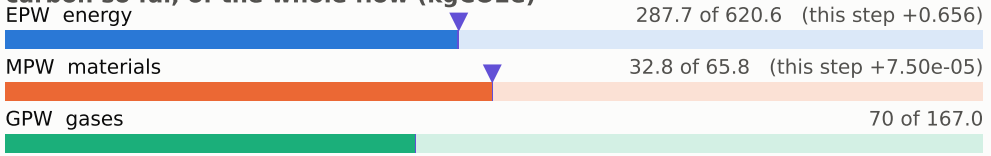


this step

process area	Deposition
material	TaN
thickness	0.003 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

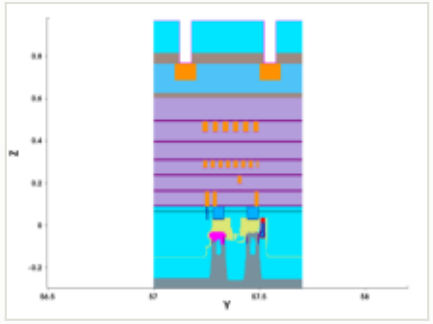
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	7.50e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

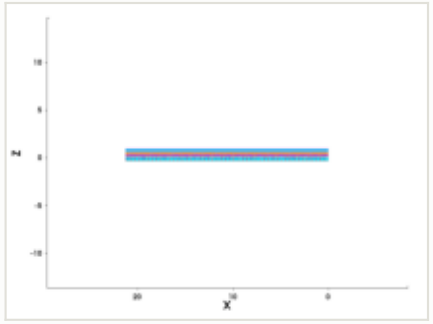


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



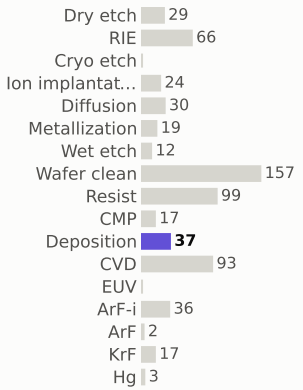
y cut at 57.351 µm



mask for this step

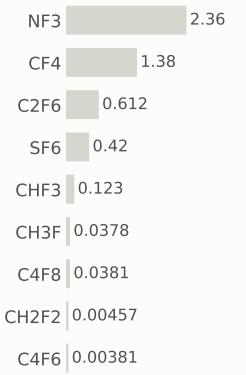
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



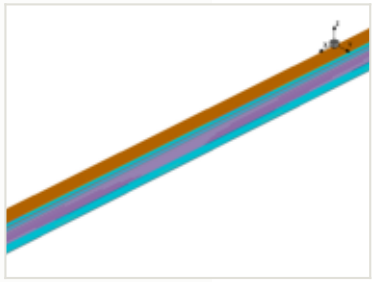
Step #8570: Metallization

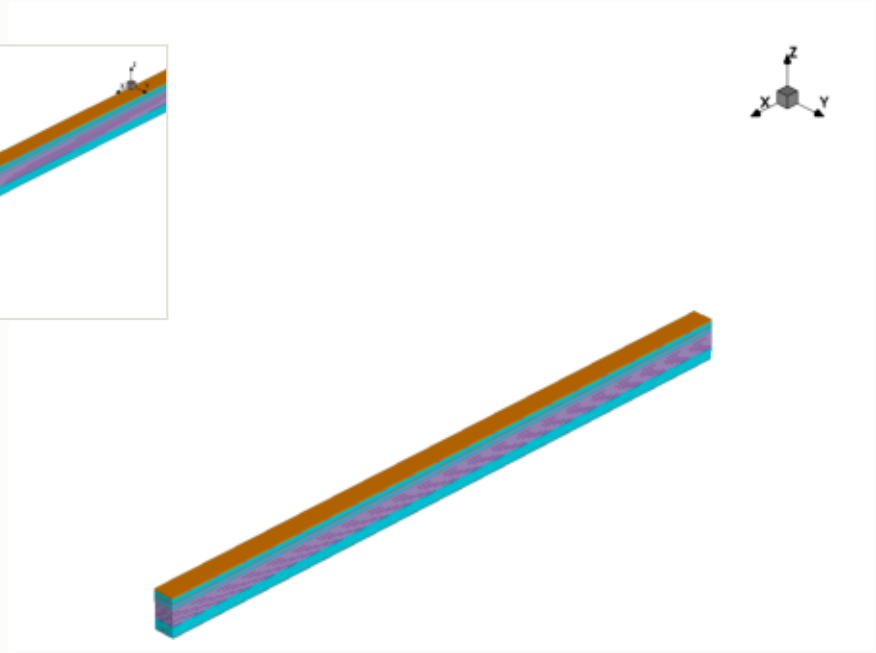
depo_isotropic · Metallization

1.2	391.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



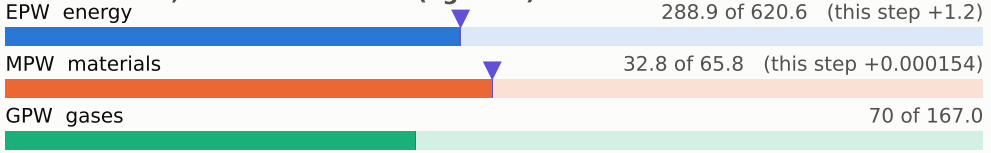


this step

process area	Metallization
material	Cu
thickness	0.04 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

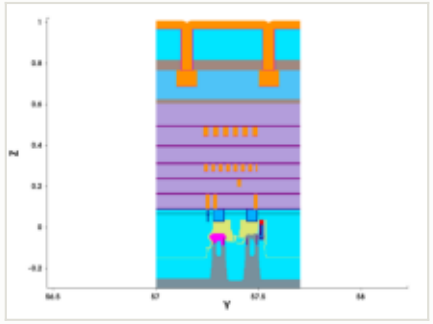
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000154 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

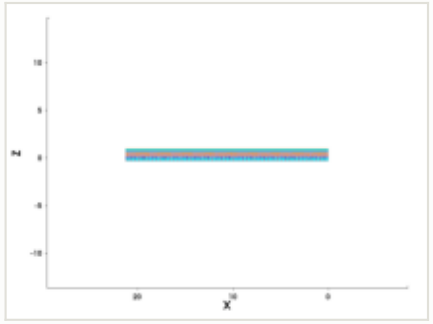


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



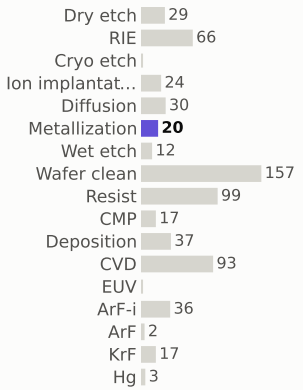
y cut at 57.351 µm



mask for this step

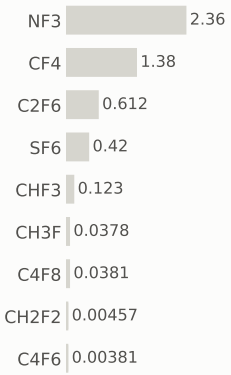
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



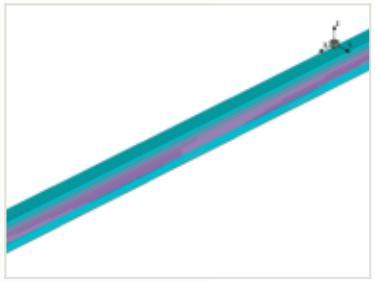
Step #8580: CMP

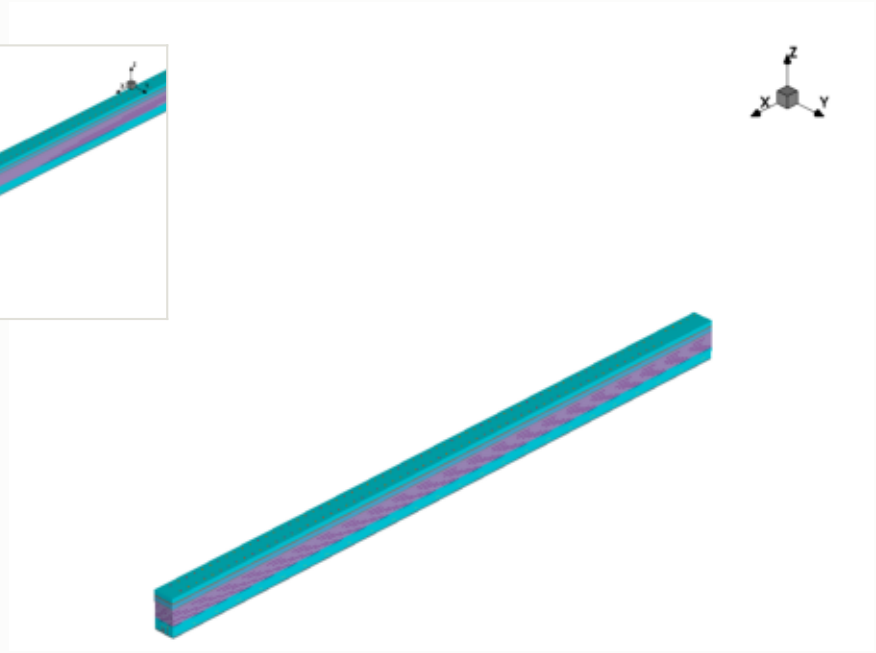
etch_cmp · CMP

0.169	391.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



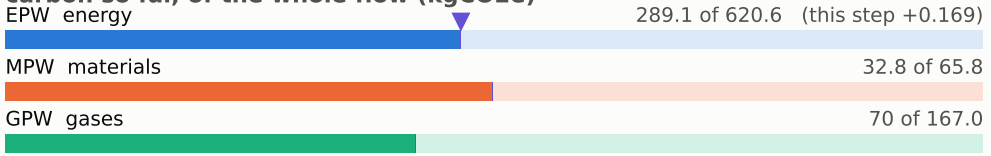


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

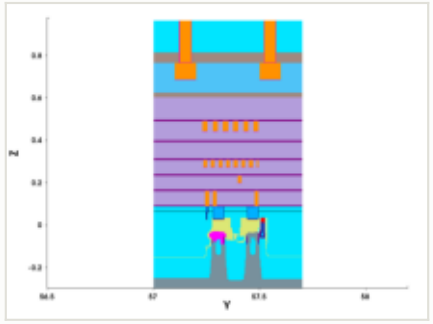
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

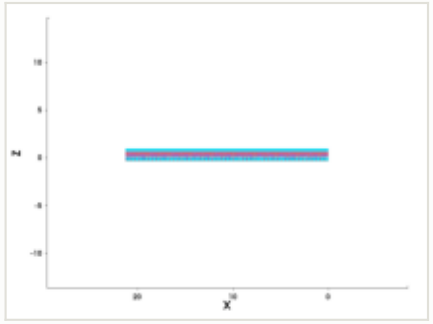


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



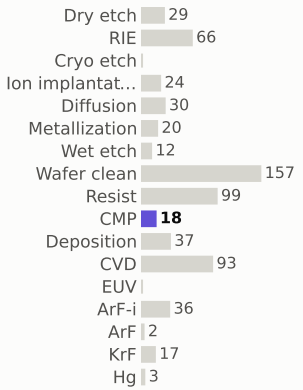
y cut at 57.351 µm



mask for this step

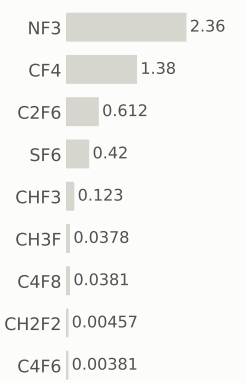
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



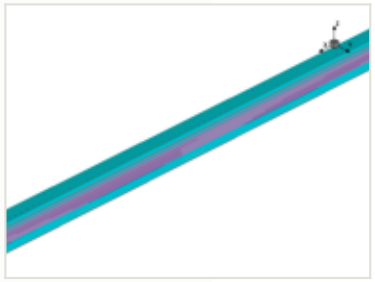
Step #8585: Wafer_clean

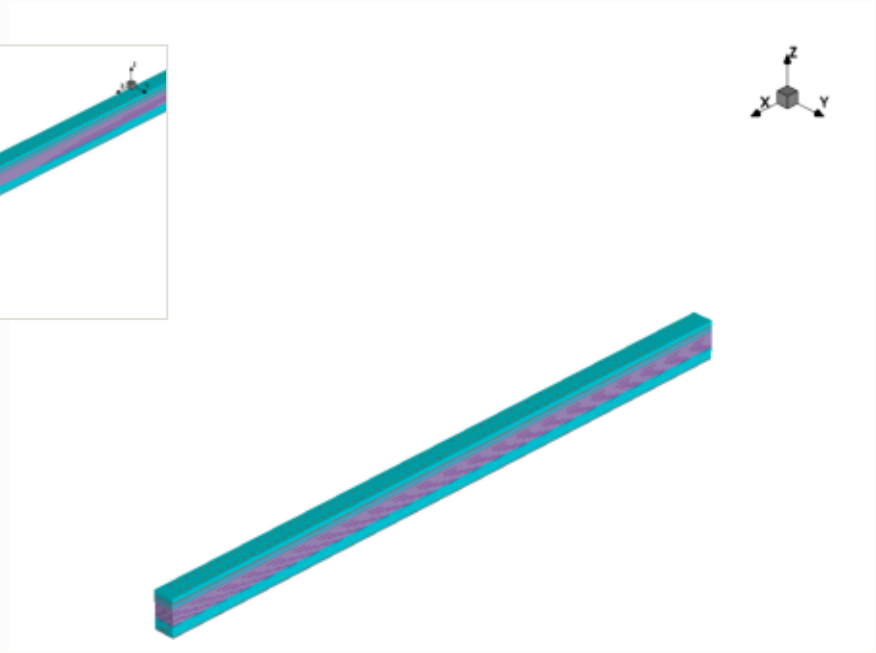
clean · Wafer_clean

0.169	392.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





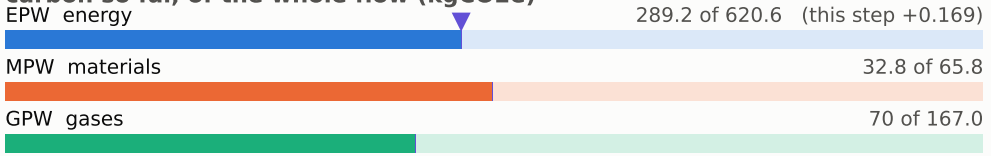
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

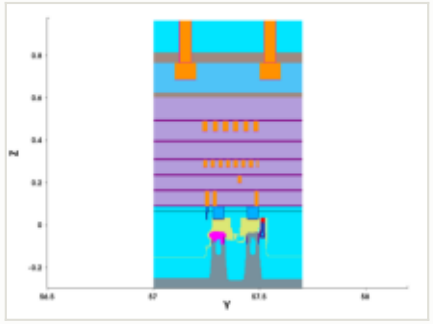
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

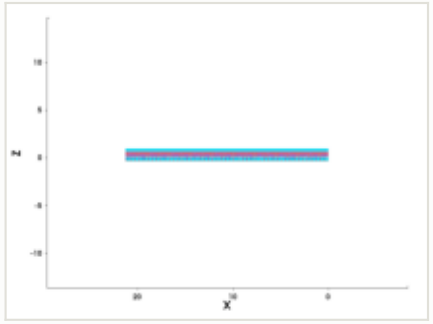


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



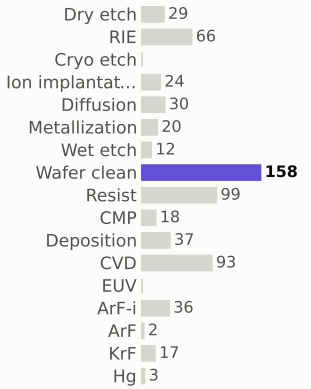
y cut at 57.351 µm



mask for this step

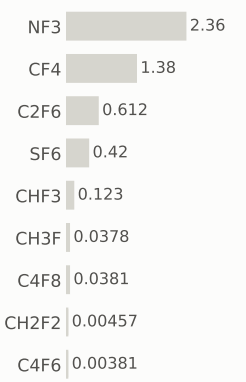
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



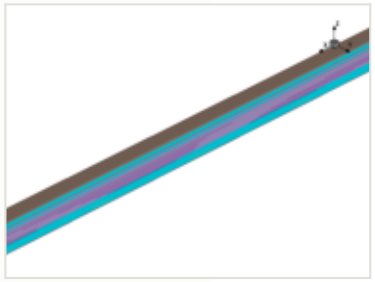
Step #8590: CVD

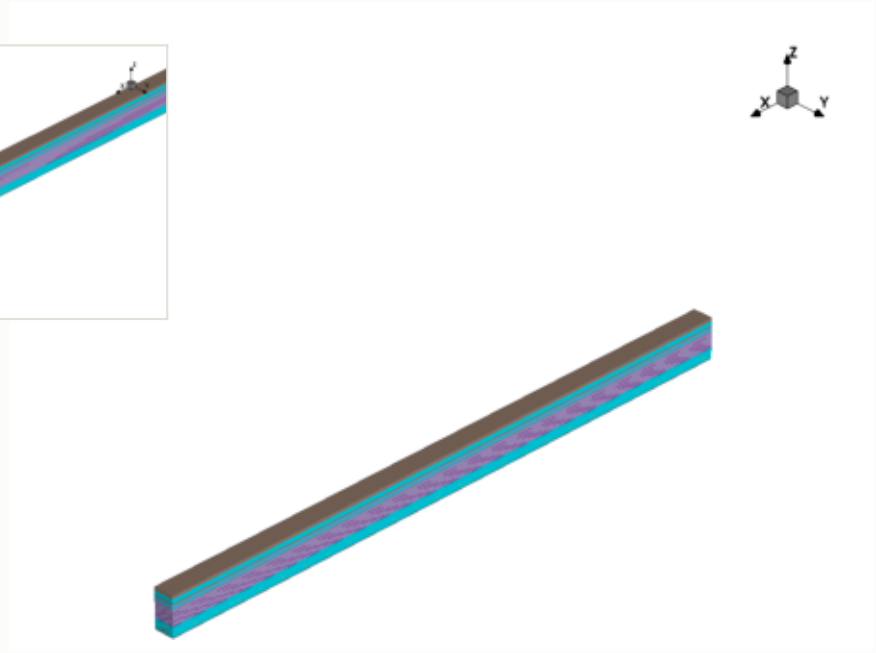
depo_isotropic · CVD

1.31	393.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



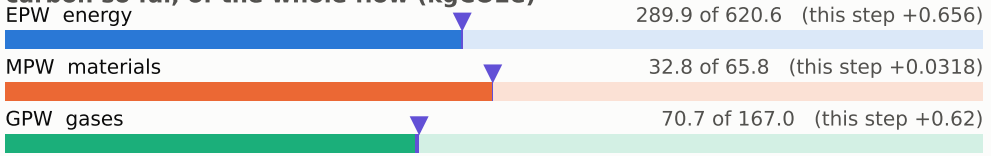


this step

process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

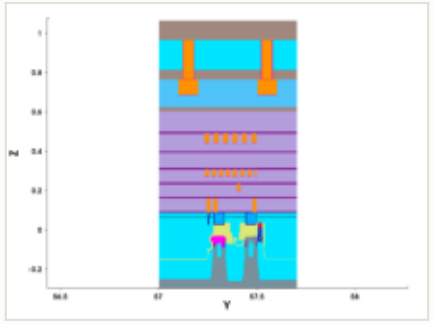
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

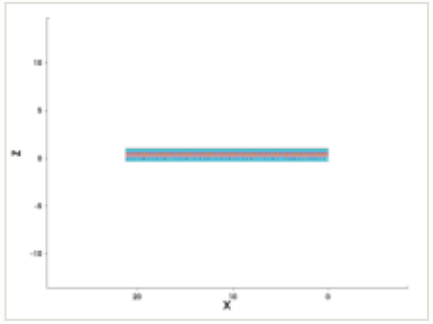


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



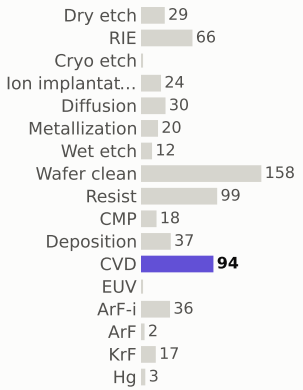
y cut at 57.351 µm



mask for this step

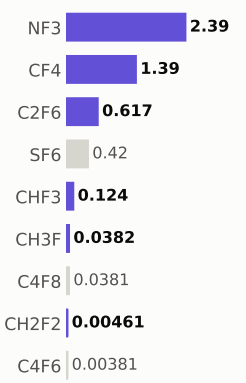
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



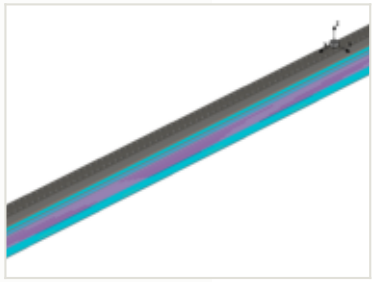
Step #8600: KrF

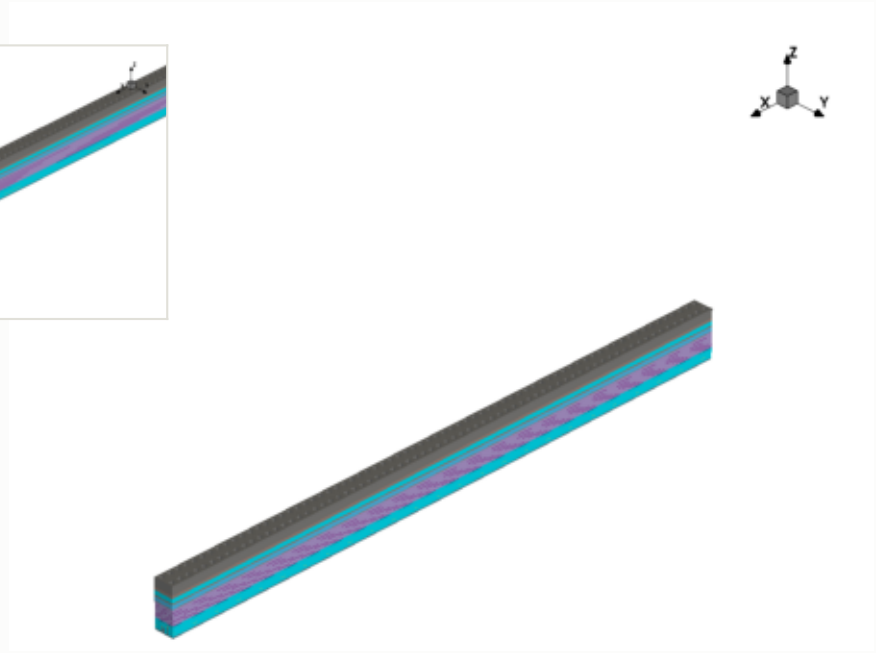
litho_patterning · KrF

0.199	393.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



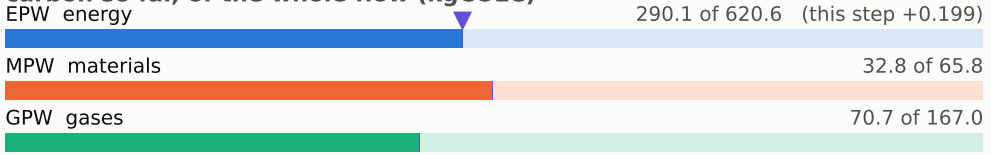


this step

process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

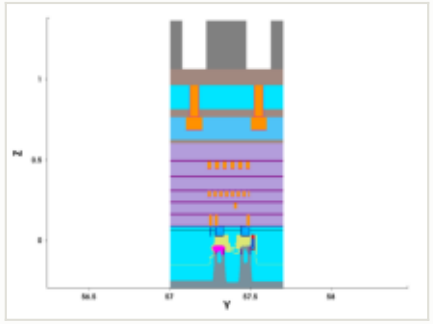
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

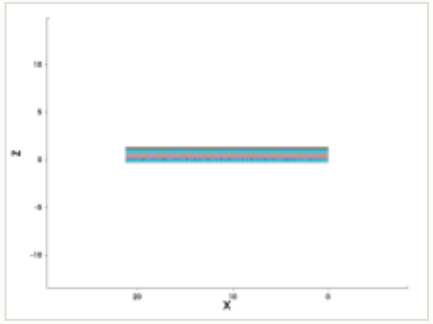


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

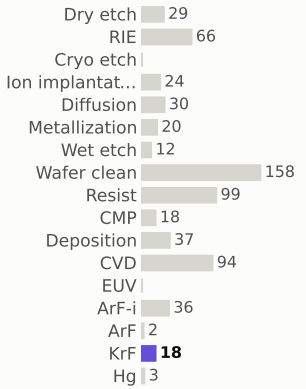


y cut at 57.351 µm

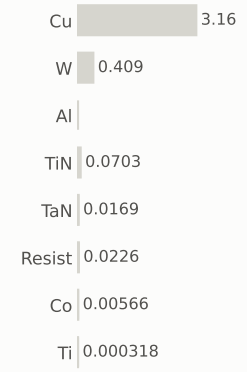


mask: layer 2910:0, negative (GDSII)

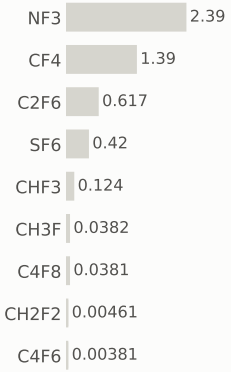
steps by process area



materials deposited (g)



gases emitted (g)



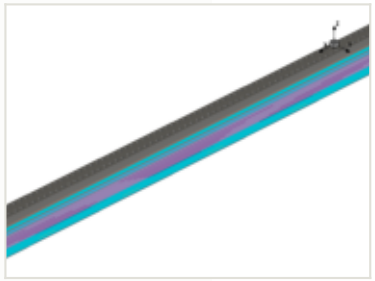
Step #8610: RIE

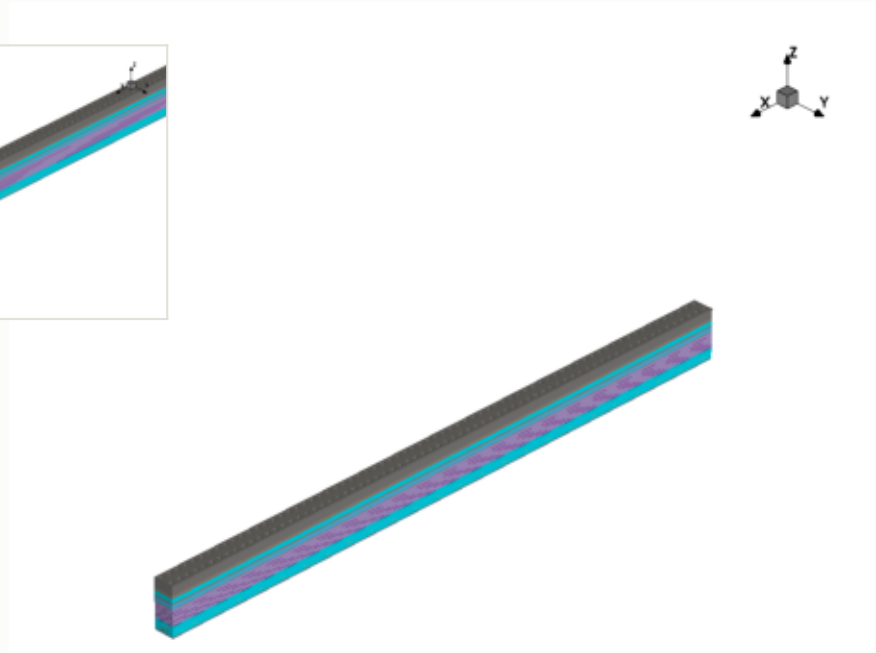
etch_anisotropic · RIE

1.04	394.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



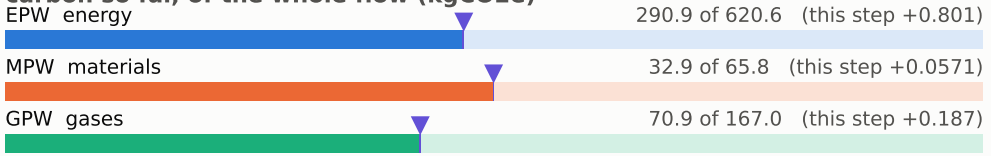


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

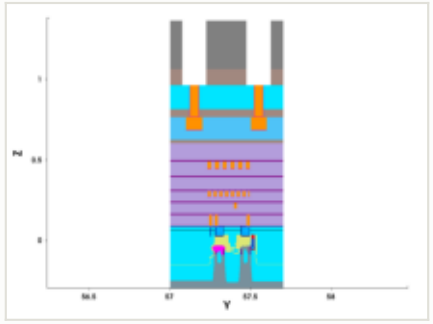
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

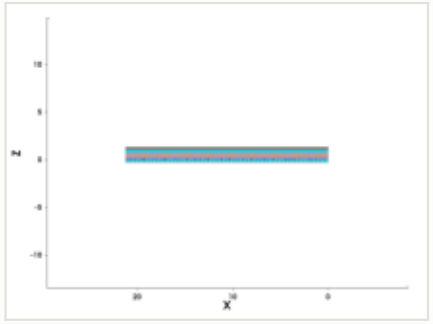


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



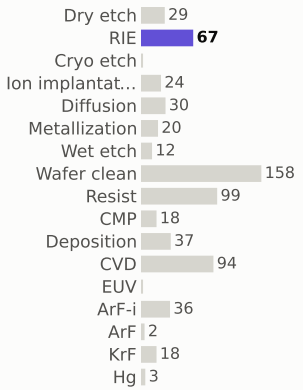
y cut at 57.351 µm



mask for this step

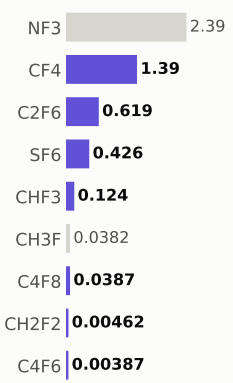
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



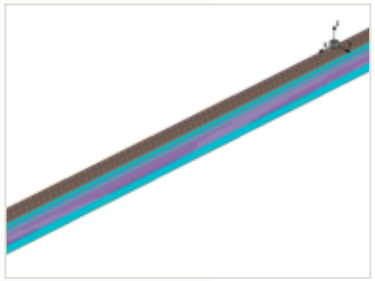
Step #8620: Dry_etch

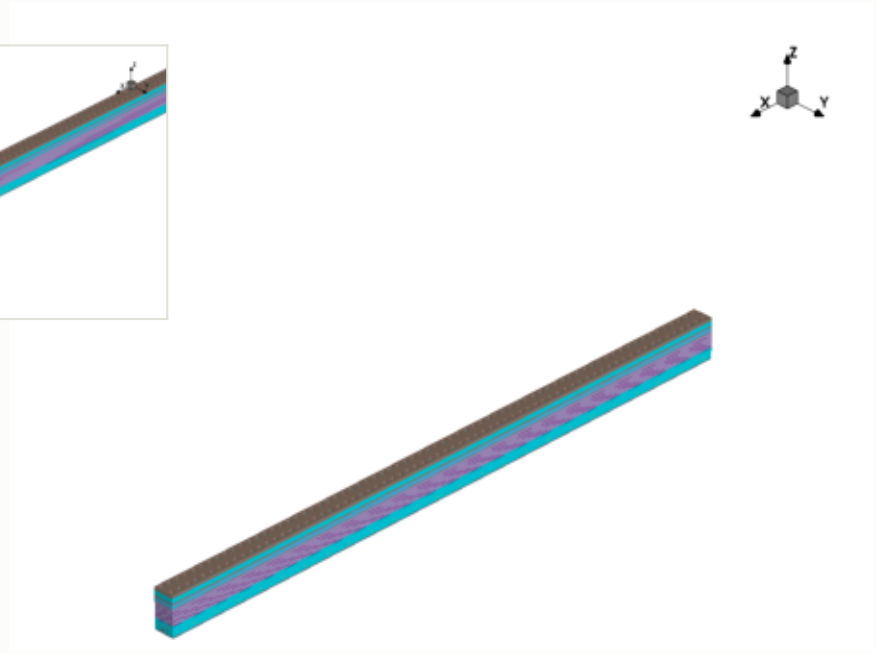
etch_strip · Dry_etch

0.801	395.4	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



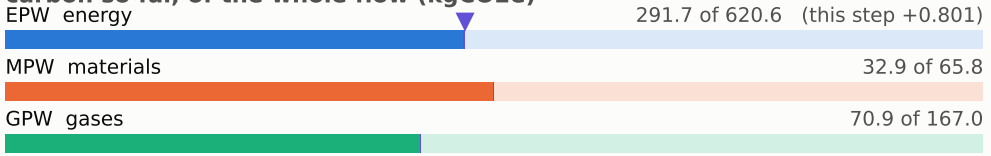


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

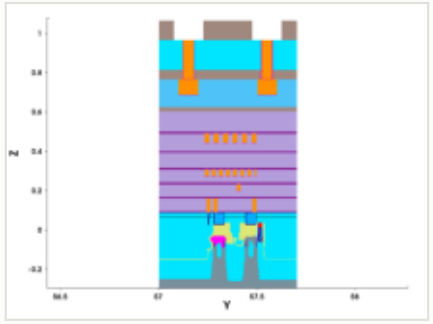
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

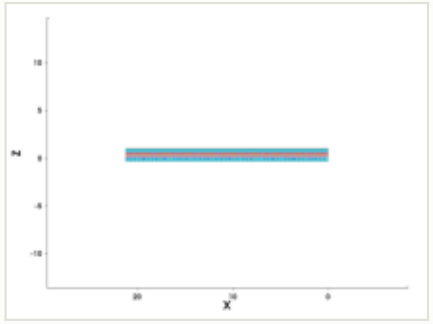


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



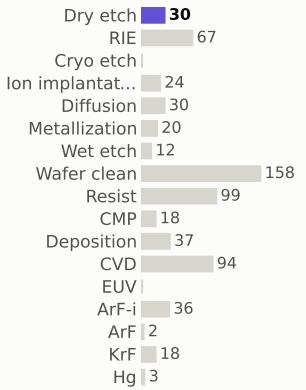
y cut at 57.351 µm



mask for this step

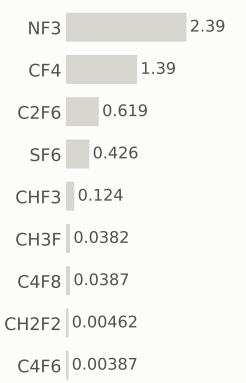
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



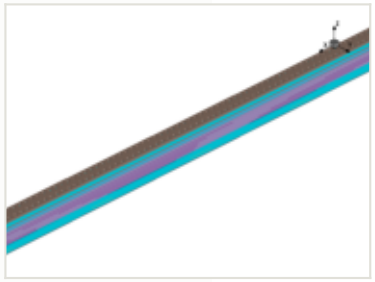
Step #8625: Wafer_clean

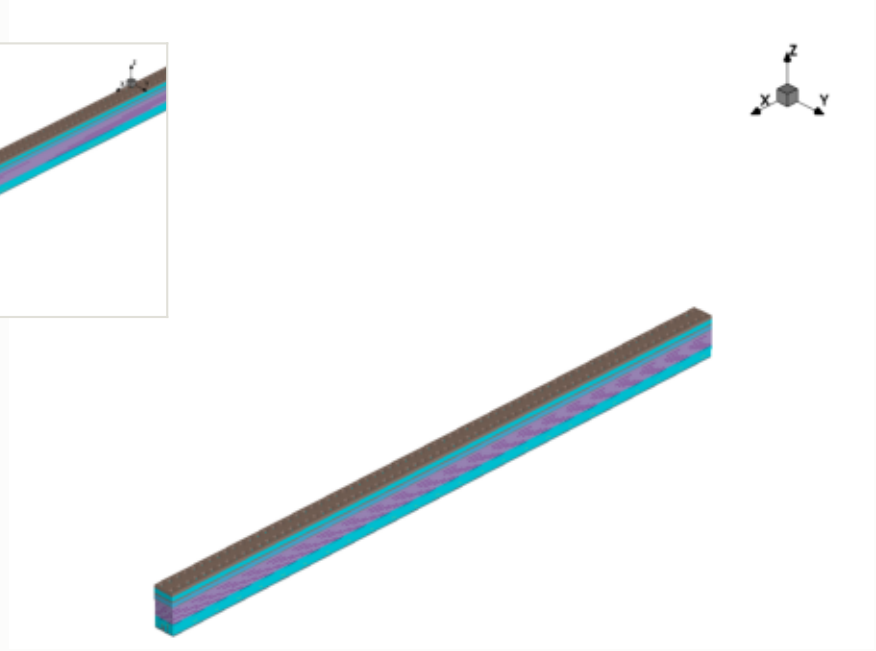
clean · Wafer_clean

0.169	395.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





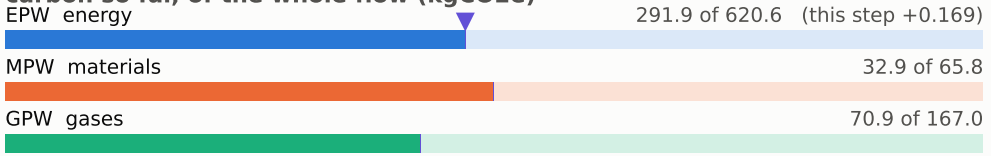
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

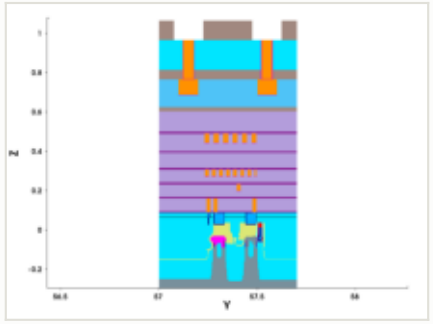
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

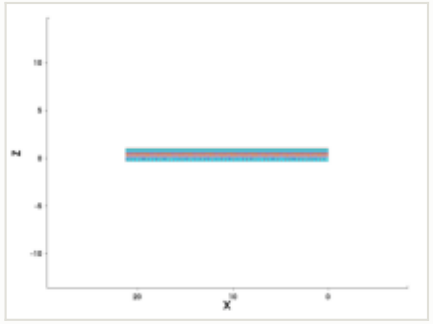


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



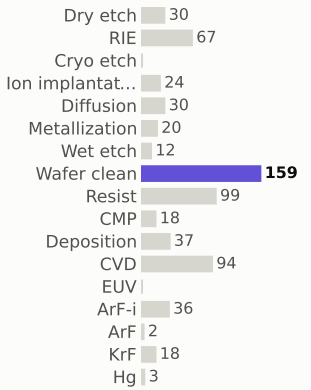
y cut at 57.351 µm



mask for this step

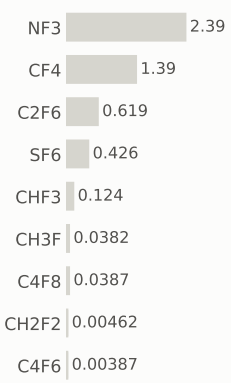
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



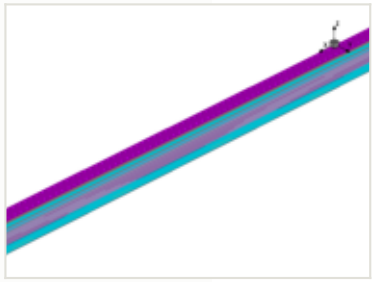
Step #8630: Deposition

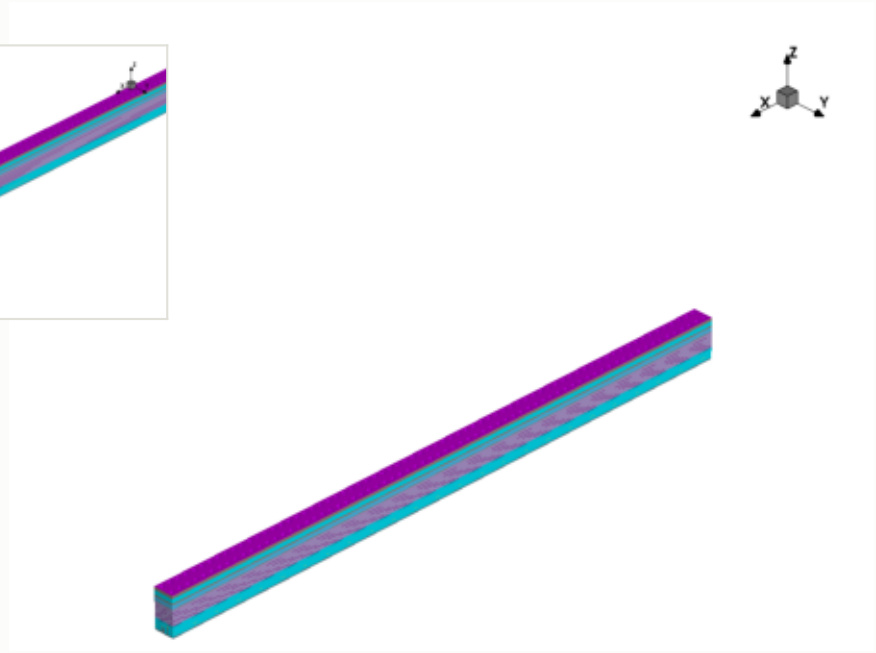
depo_isotropic · Deposition

0.656	396.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



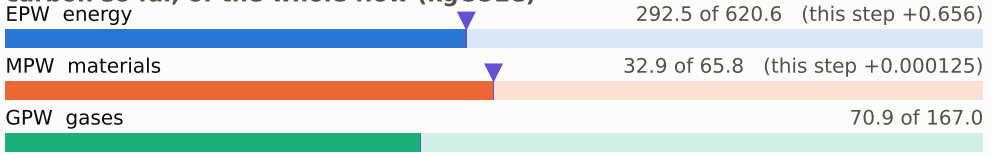


this step

process area	Deposition
material	TaN
thickness	0.005 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

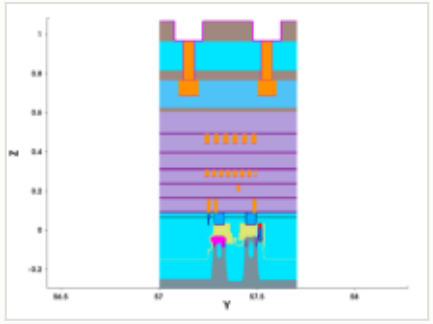
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000125 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

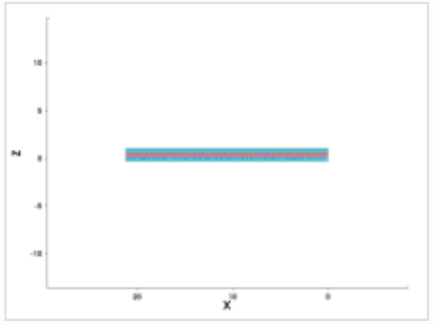


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



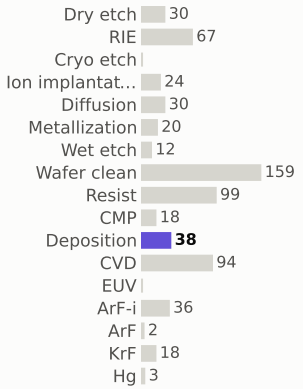
y cut at 57.351 µm



mask for this step

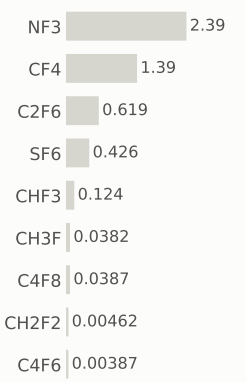
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



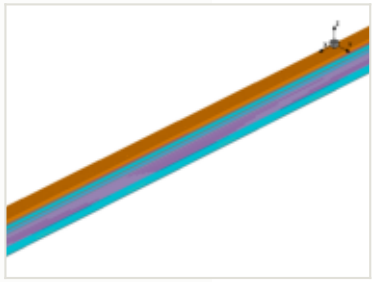
Step #8640: Metallization

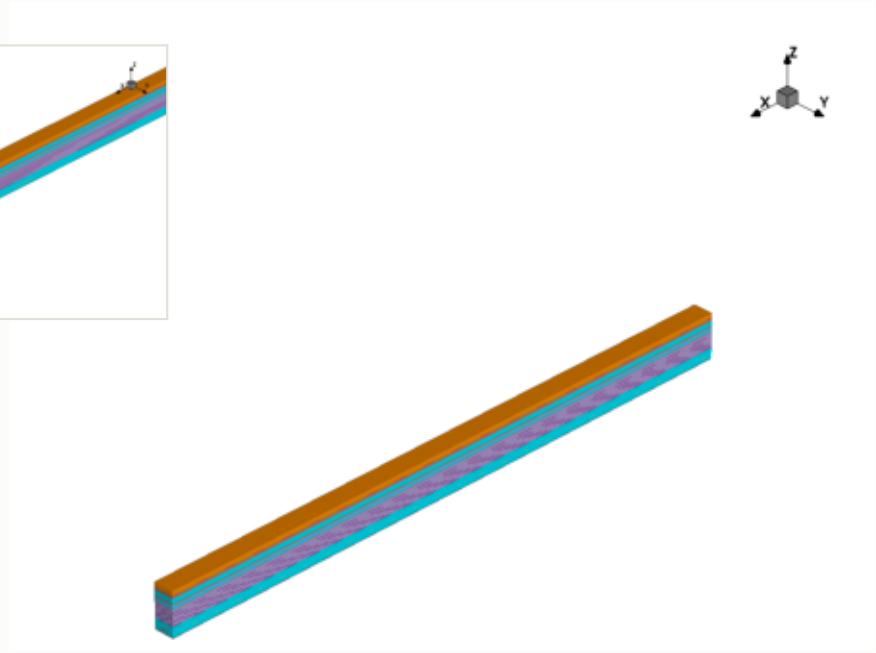
depo_isotropic · Metallization

1.2	397.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



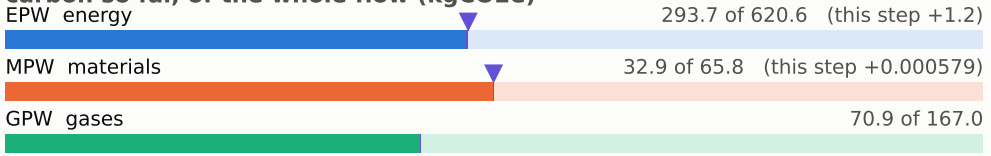


this step

process area	Metallization
material	Cu
thickness	0.15 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

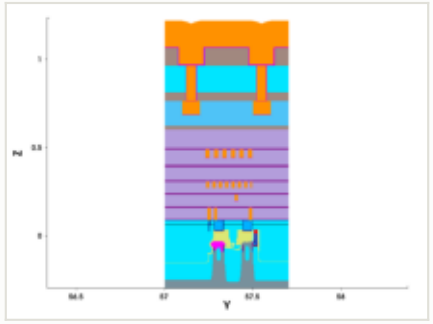
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000579 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

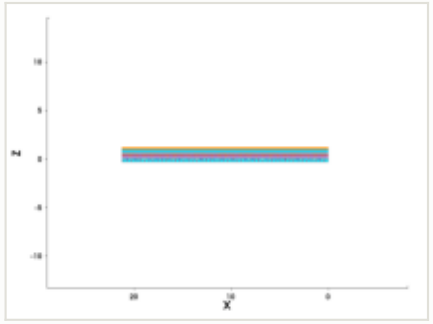


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



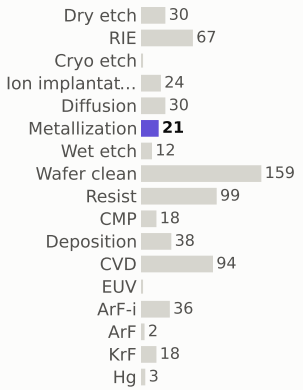
y cut at 57.351 µm



mask for this step

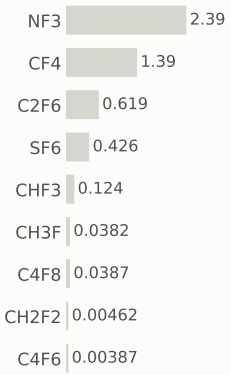
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



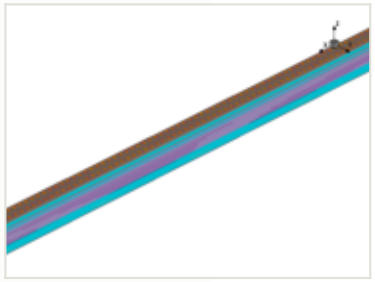
Step #8650: CMP

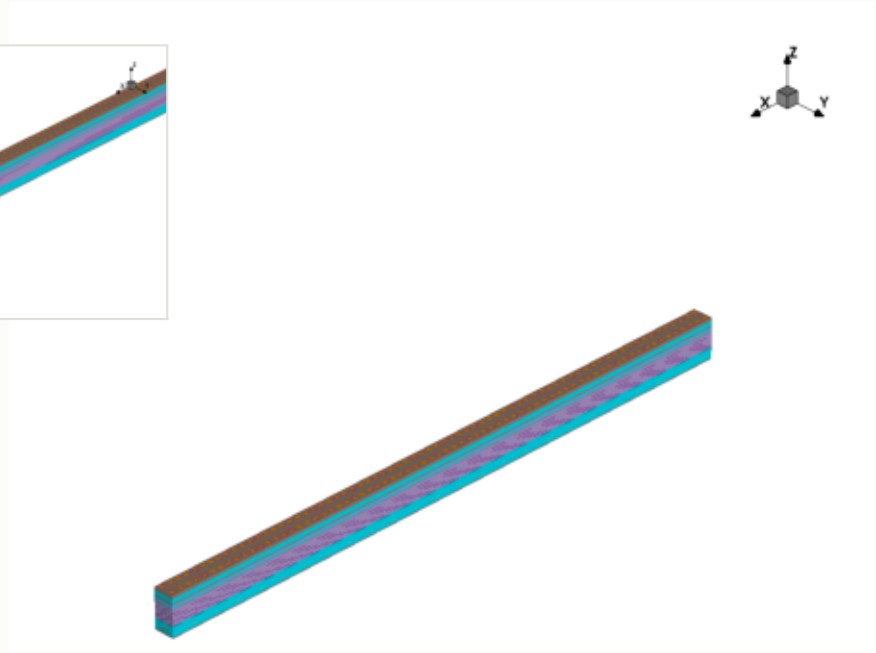
etch_cmp · CMP

0.169	397.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



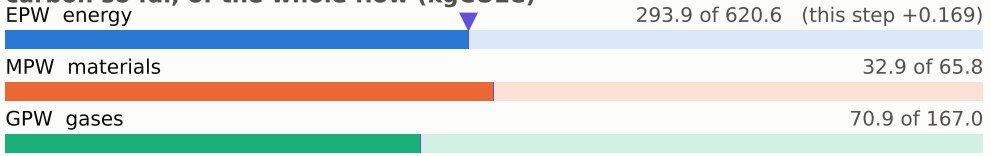


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

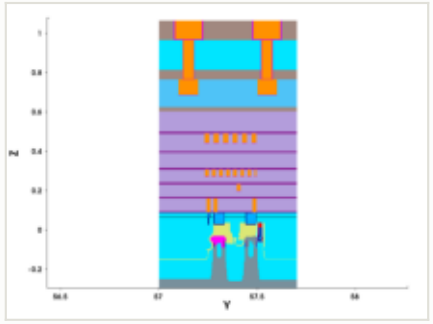
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

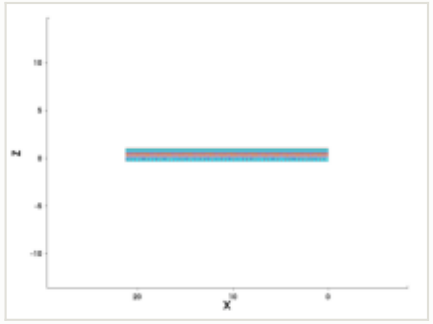


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



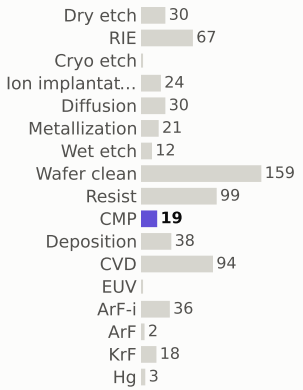
y cut at 57.351 µm



mask for this step

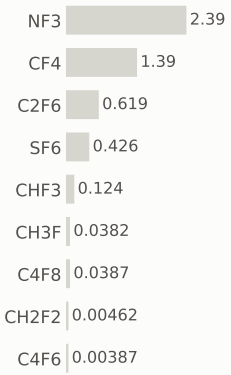
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



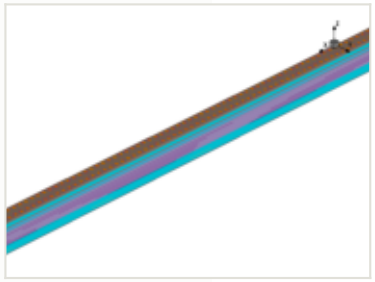
Step #8655: Wafer_clean

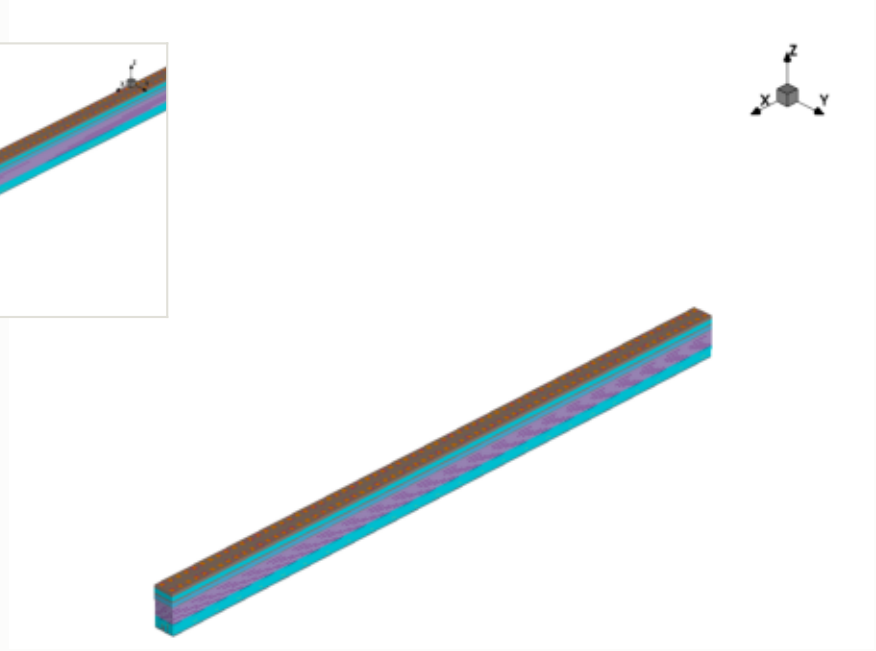
clean · Wafer_clean

0.169	397.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





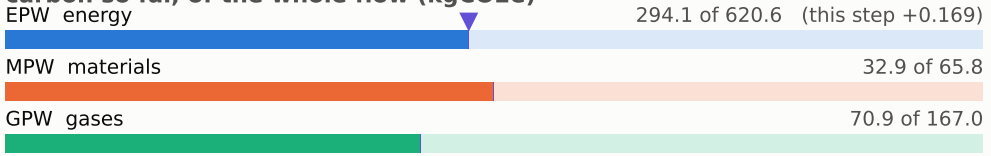
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

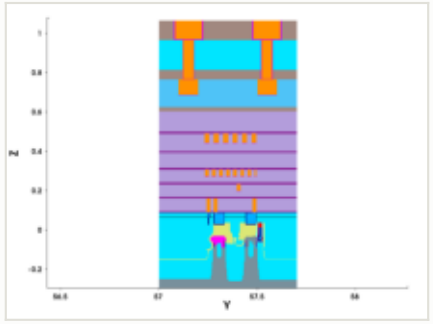
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

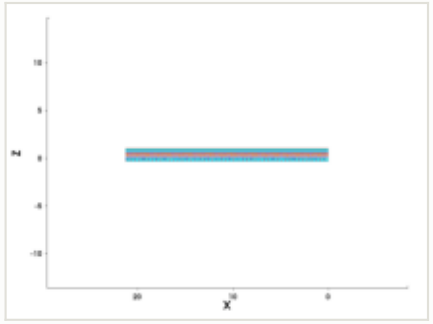


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



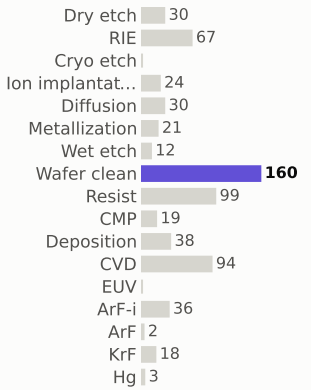
y cut at 57.351 µm



mask for this step

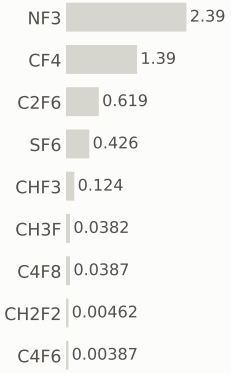
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



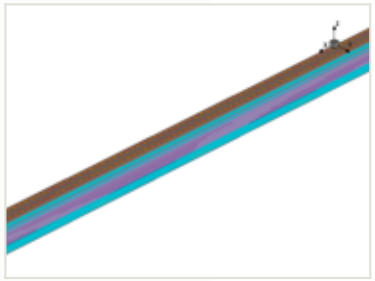
Step #8670: Diffusion

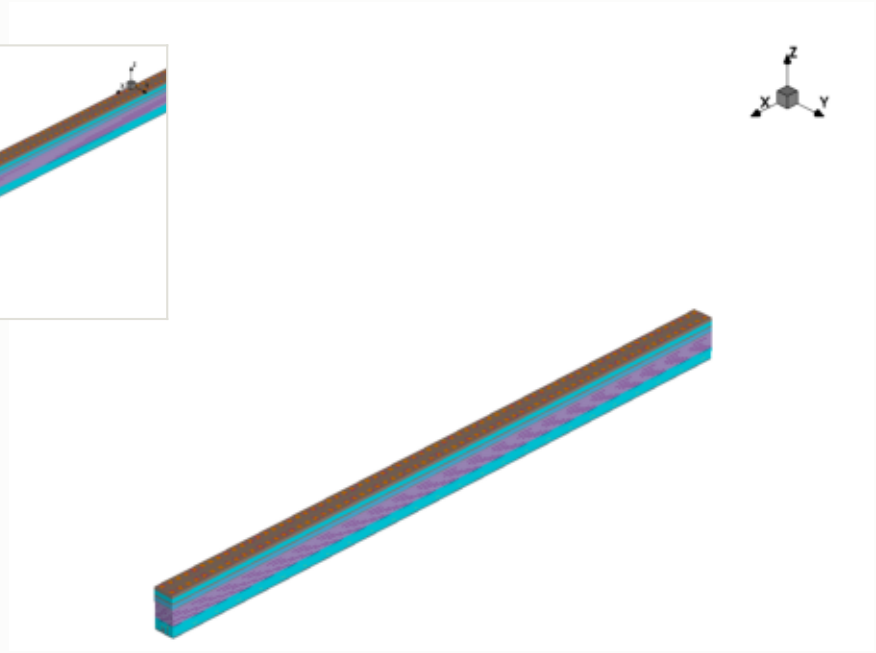
treatment · Diffusion

0.801	398.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





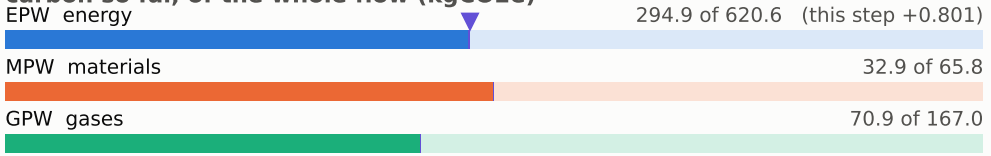
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

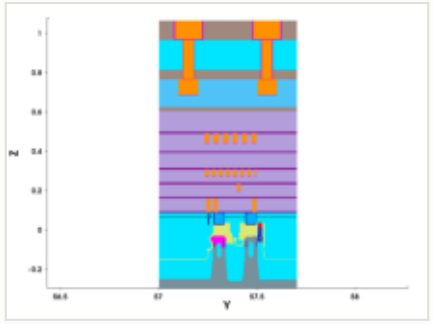
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

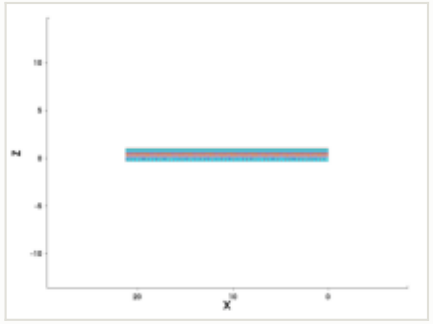


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



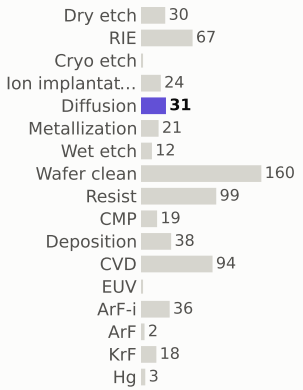
y cut at 57.351 µm



mask for this step

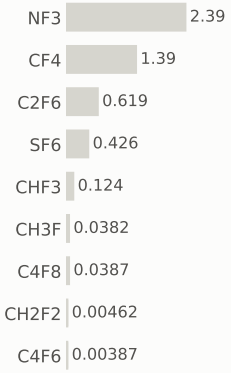
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



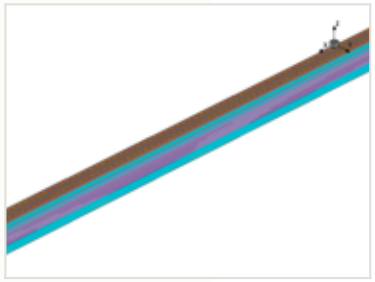
Step #8680: Deposition

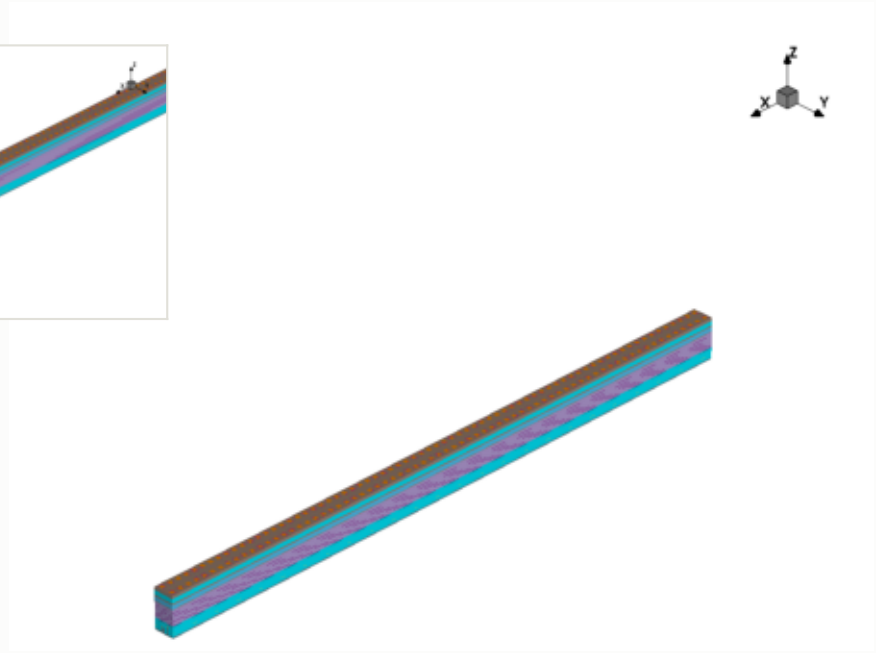
treatment · Deposition

0.656	399.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





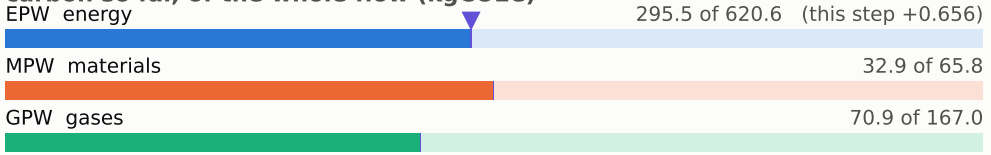
this step

process area	Deposition
material	—
thickness	—
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

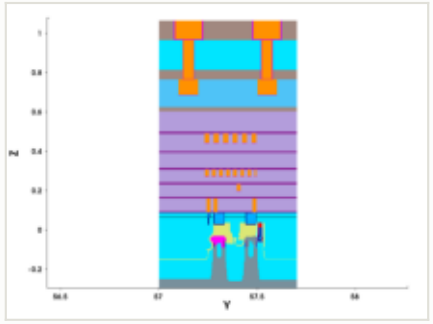
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

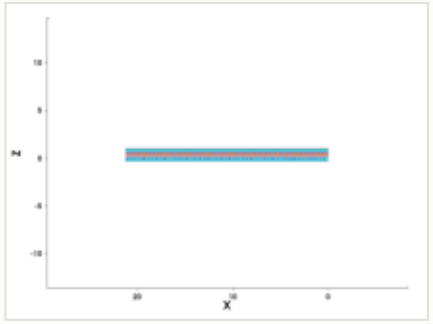


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



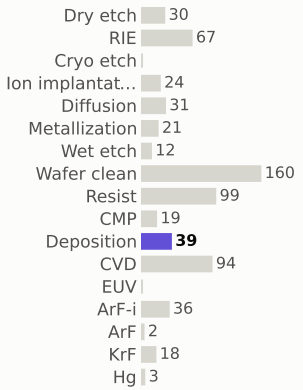
y cut at 57.351 µm



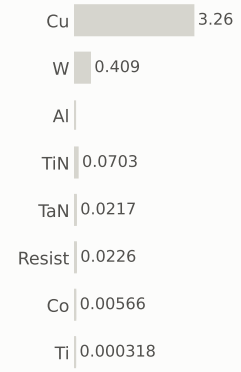
mask for this step

no mask for this step

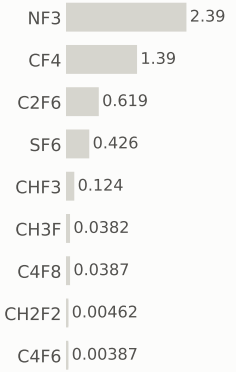
steps by process area



materials deposited (g)



gases emitted (g)



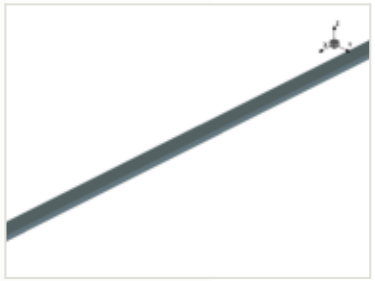
Step #9120: Wafer

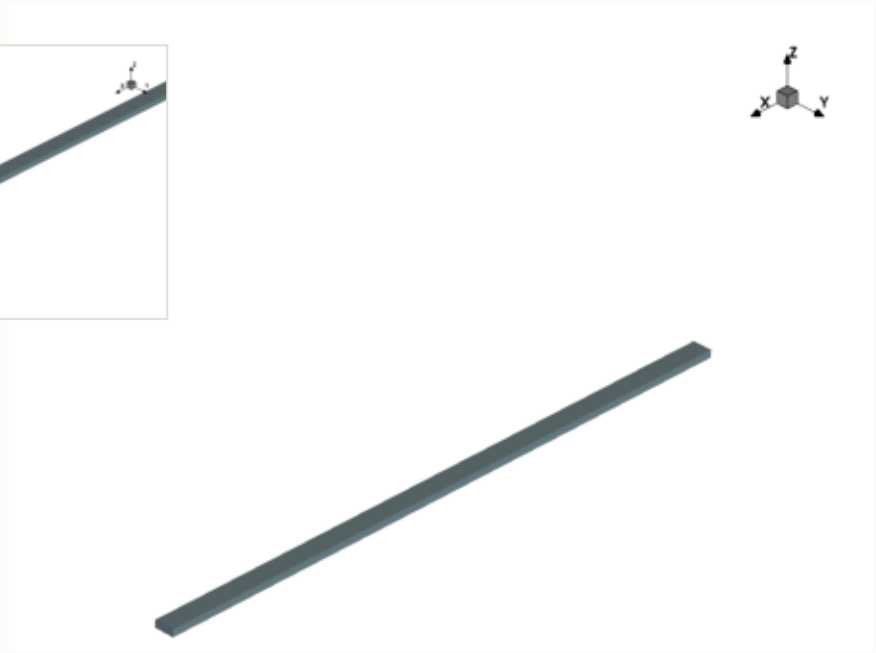
substrate · Wafer

26	425.2	3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





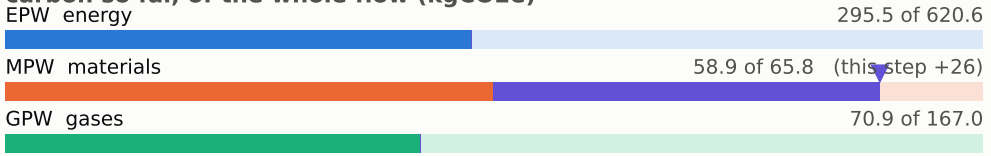
this step

process area	Wafer
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	26 kgCO2e
GPW	0 kgCO2e
embodied carbon	**26 kgCO2e**

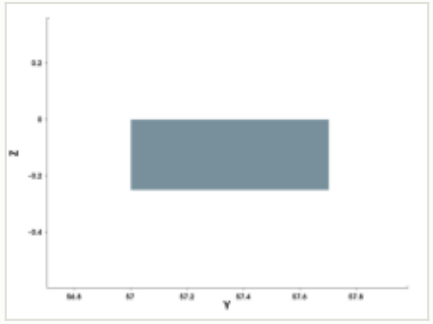
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

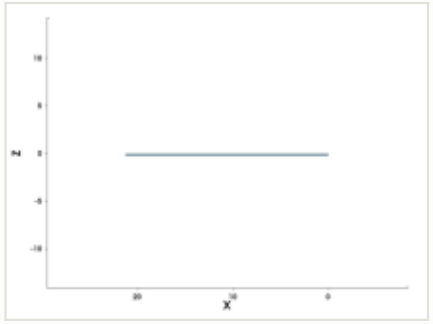


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



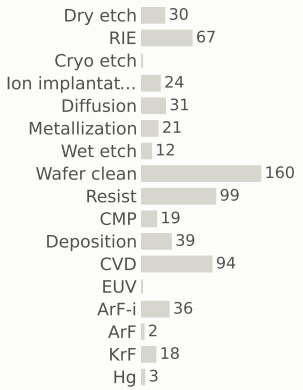
y cut at 57.351 µm



mask for this step

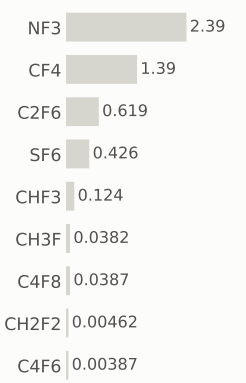
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



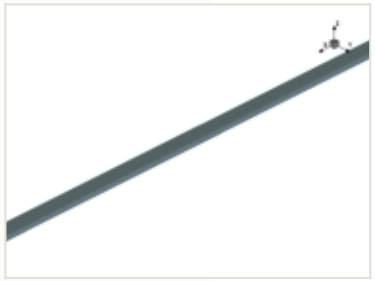
Step #9130: Dry_etch

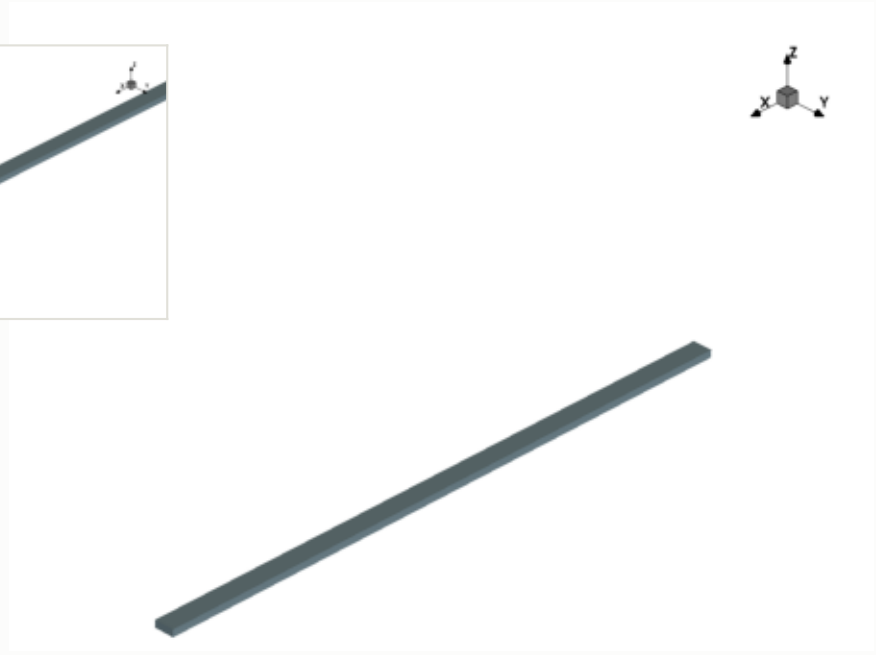
marking · Dry_etch

0.801	426.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



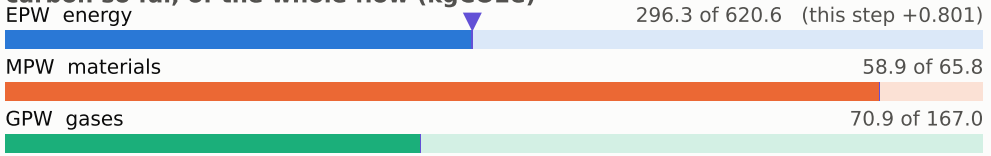


this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

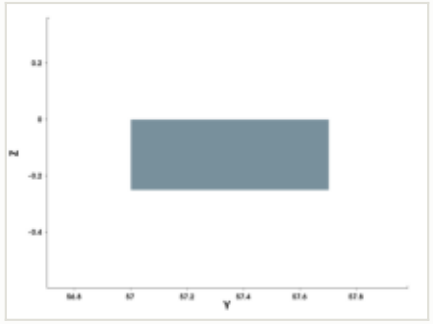
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

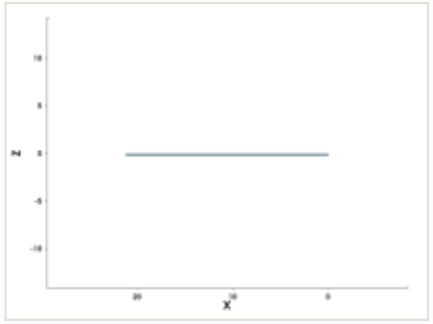


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



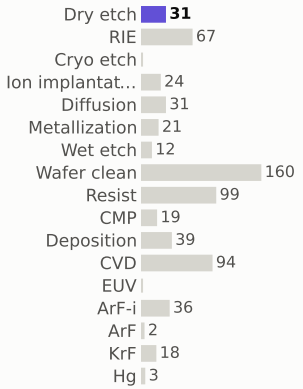
y cut at 57.351 µm



mask for this step

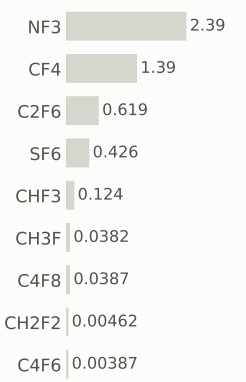
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



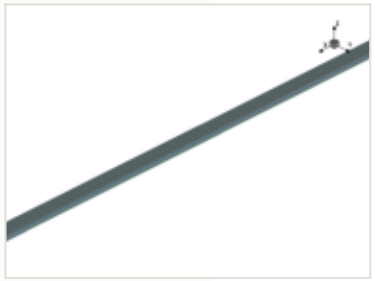
Step #9150: Wafer_clean

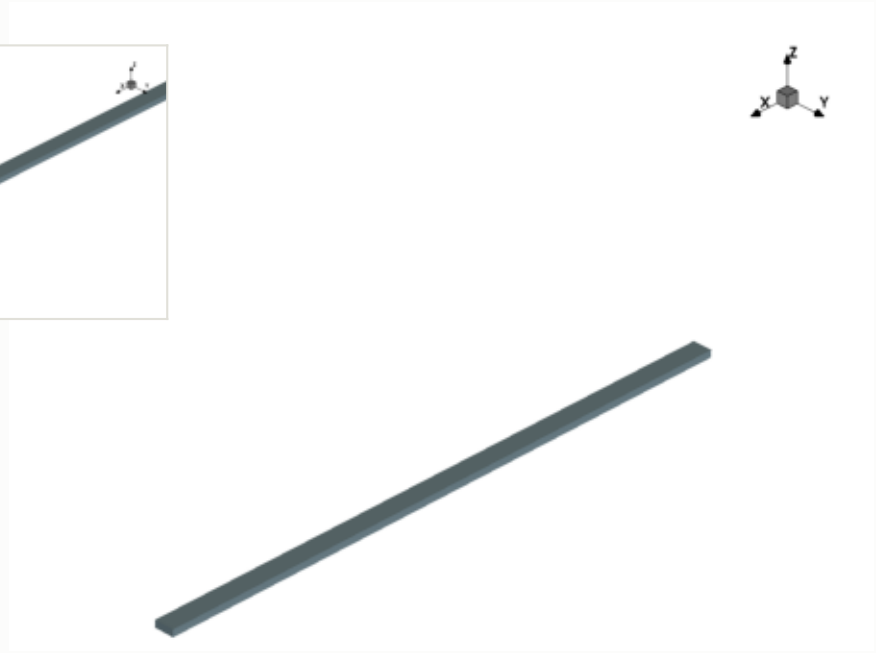
clean · Wafer_clean

0.169	426.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





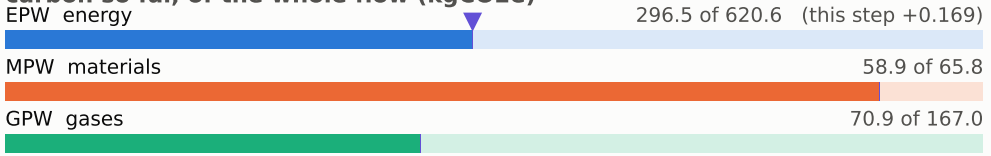
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

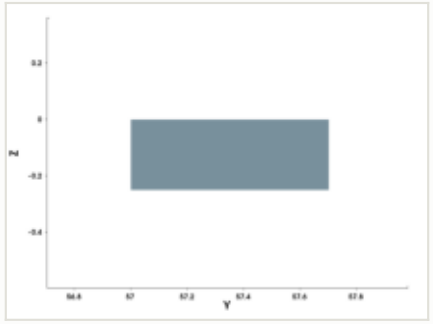
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

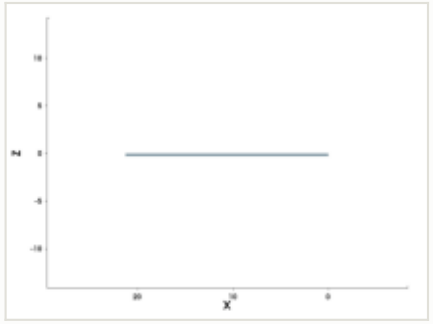


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



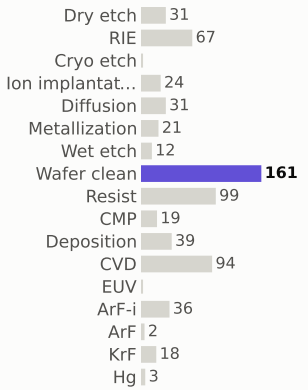
y cut at 57.351 µm



mask for this step

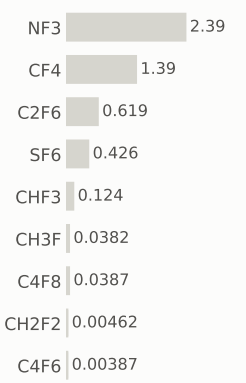
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



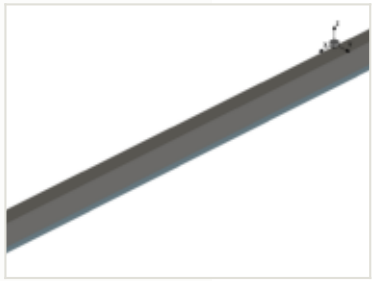
Step #9160: Hg

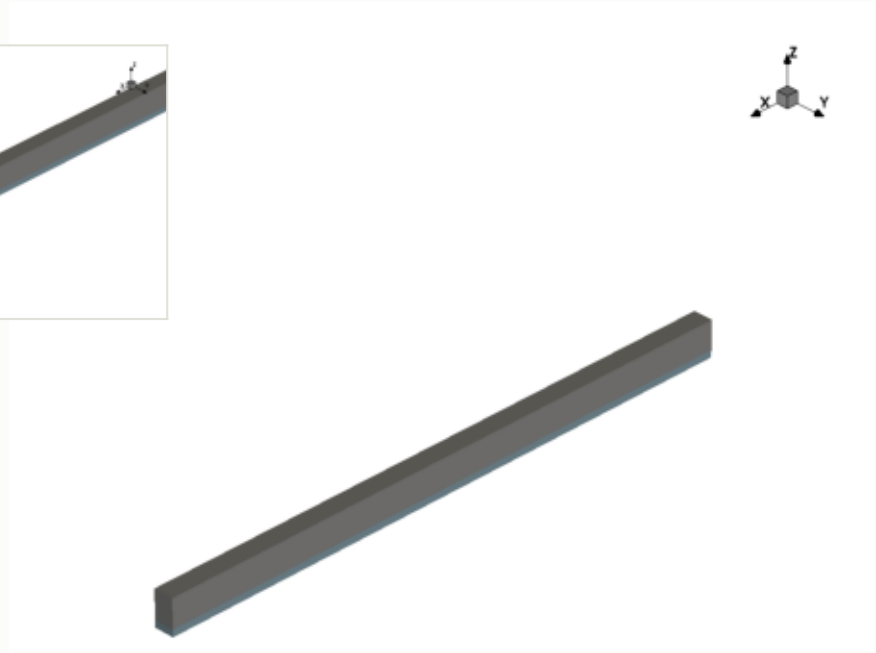
litho_patterning · Hg

0.199	426.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



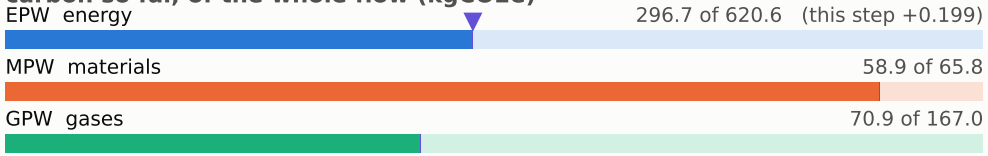


this step

process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

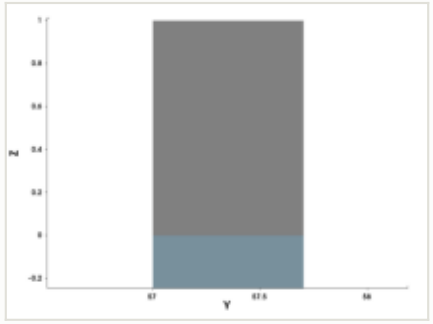
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

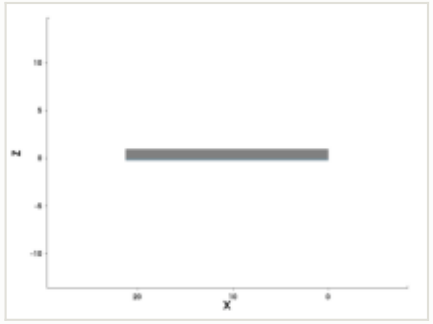


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



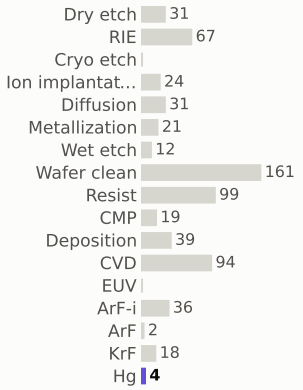
y cut at 57.351 µm



mask: layer 50:0, negative (GDSII)

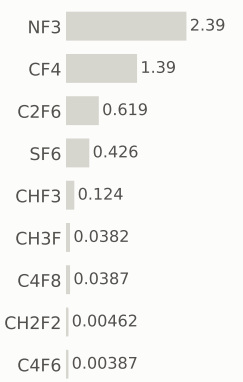
nothing in the clip window

steps by process area



materials deposited (g)

gases emitted (g)



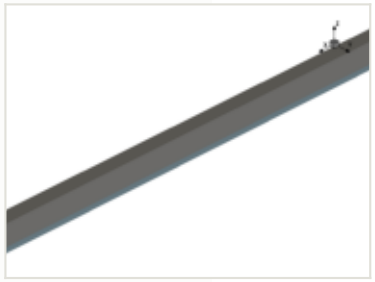
Step #9170: RIE

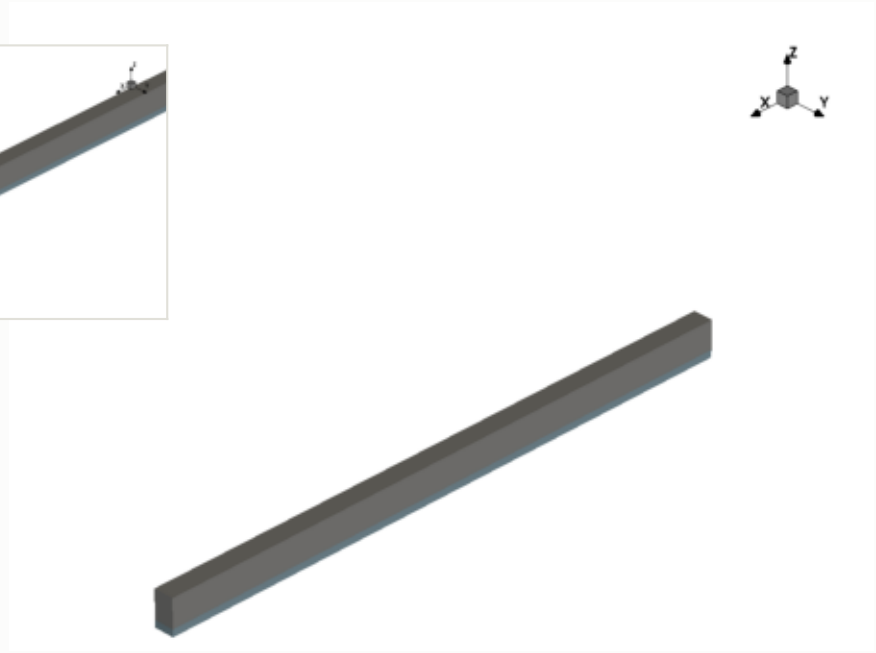
etch_anisotropic · RIE

1.04	427.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



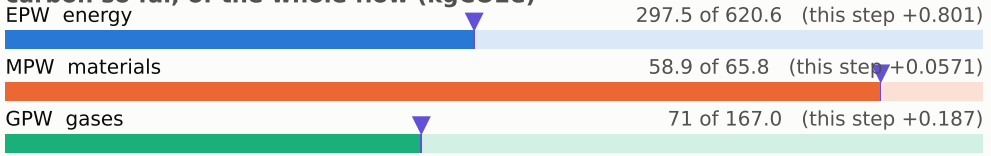


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

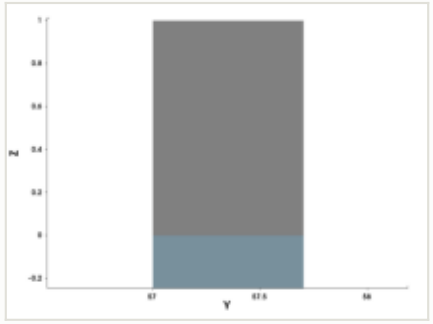
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

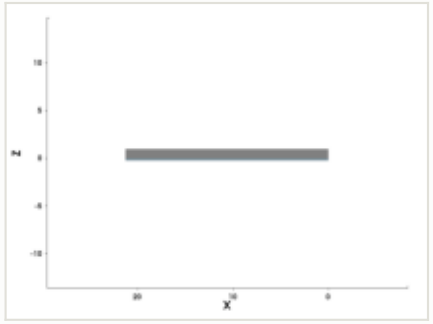


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



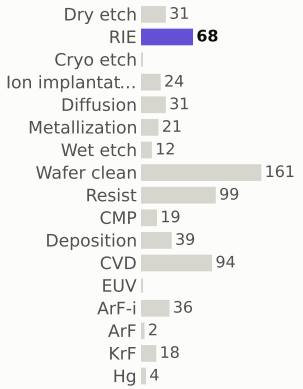
y cut at 57.351 µm



mask for this step

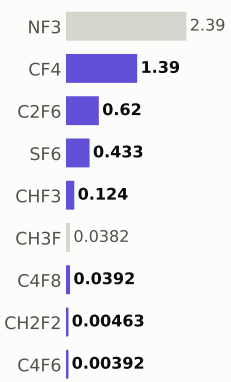
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



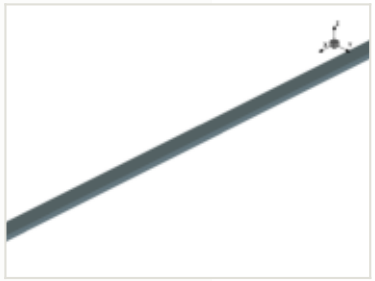
Step #9180: Dry_etch

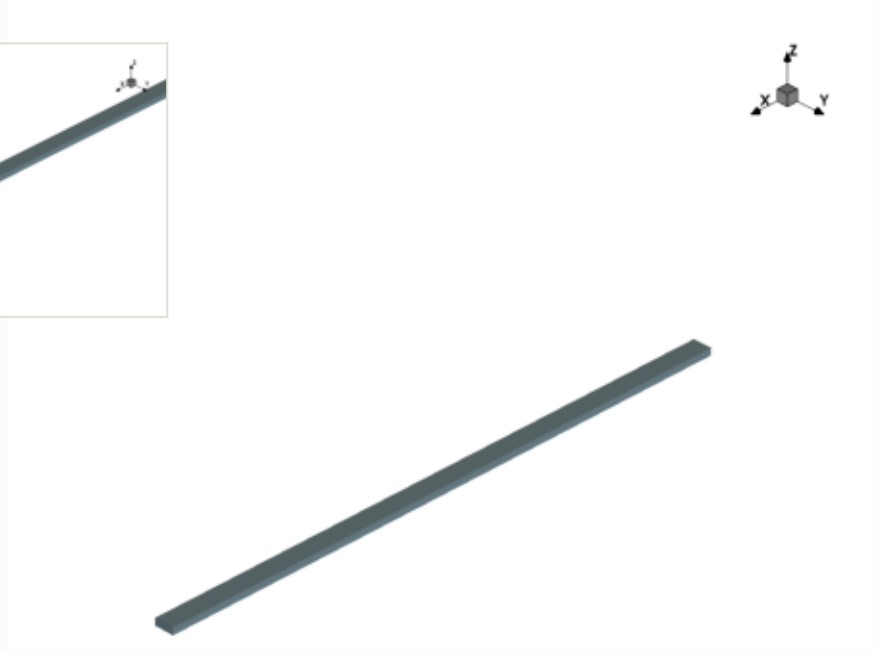
etch_strip · Dry_etch

0.801	428.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



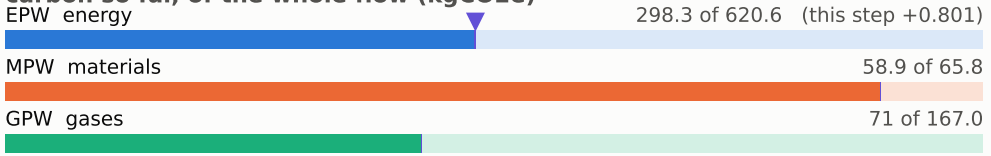


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

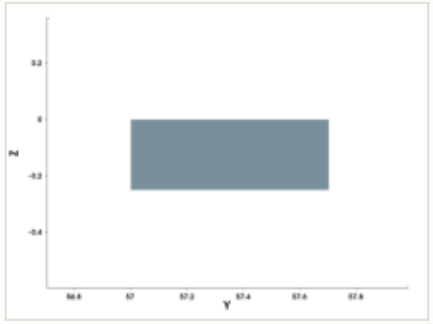
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

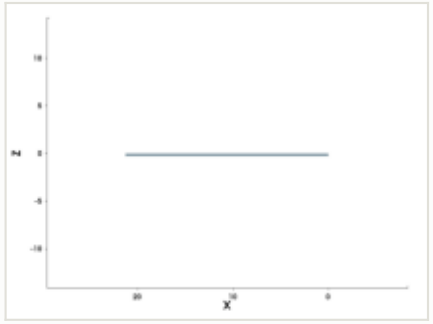


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



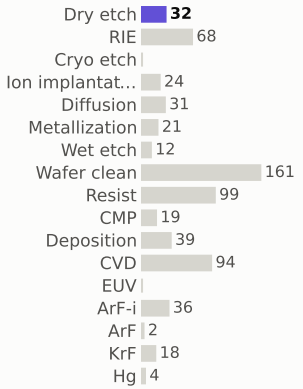
y cut at 57.351 µm



mask for this step

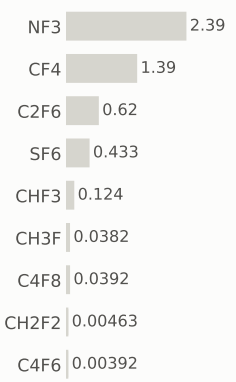
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



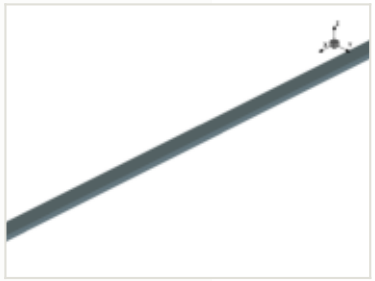
Step #9190: Wafer_clean

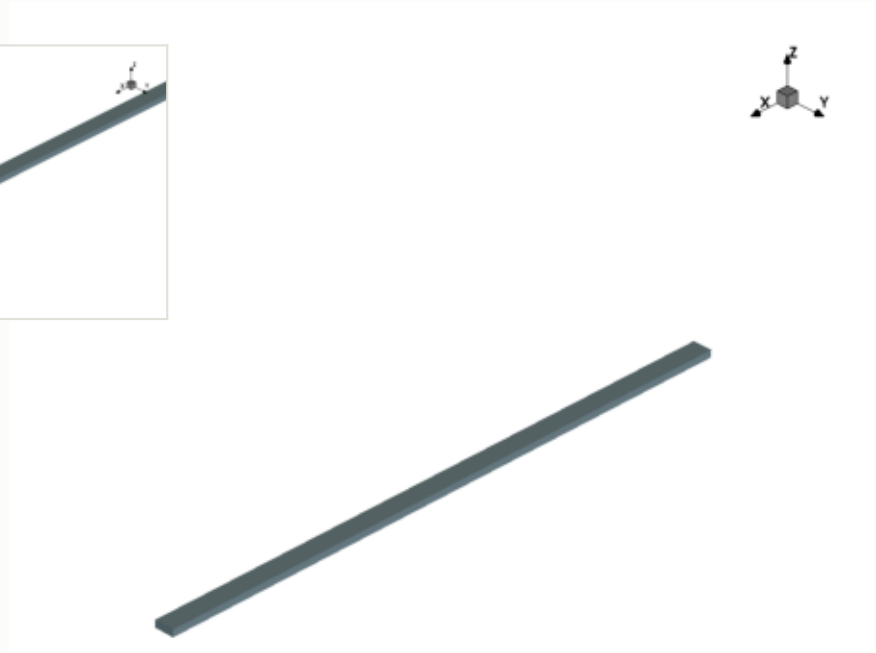
clean · Wafer_clean

0.169	428.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





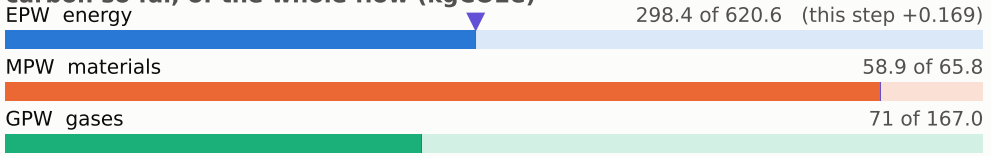
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

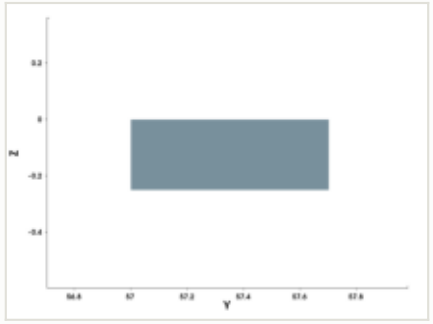
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

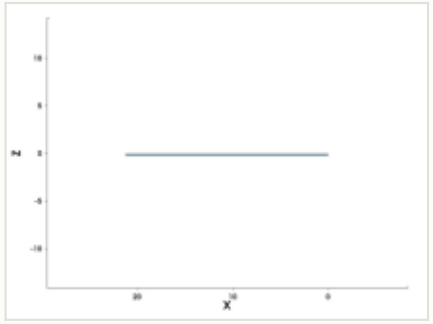


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



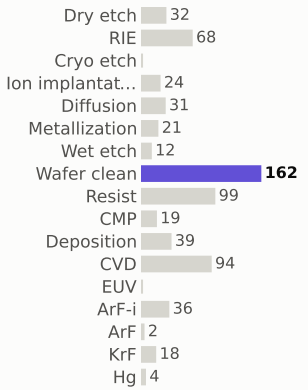
y cut at 57.351 µm



mask for this step

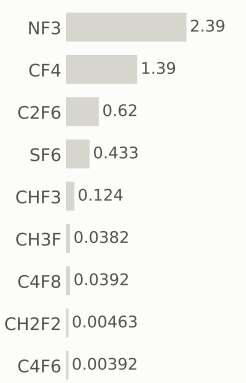
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



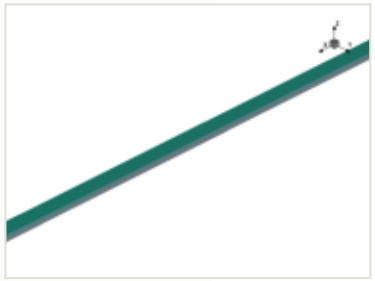
Step #9210: Diffusion

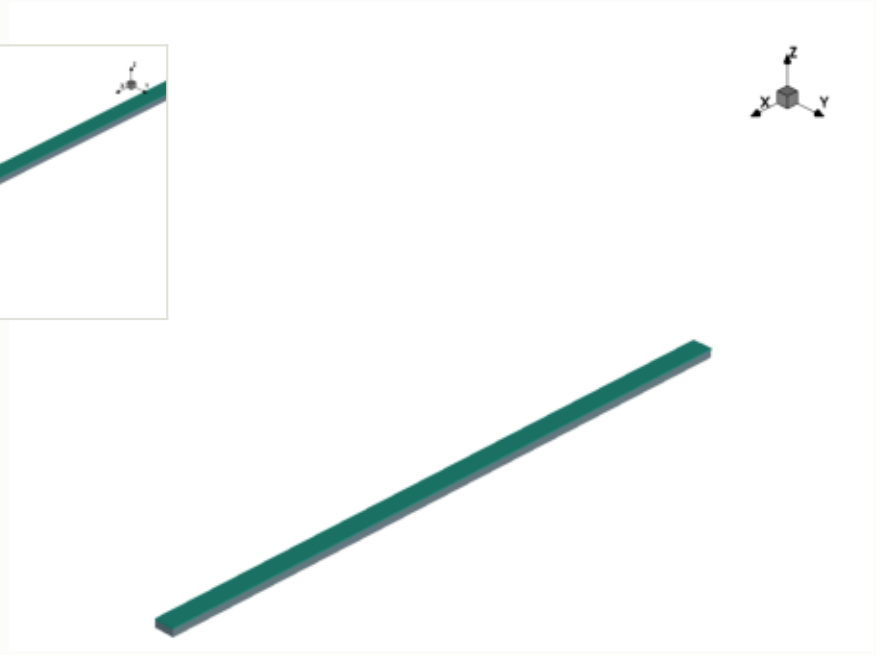
depo_isotropic · Diffusion

0.801	429.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



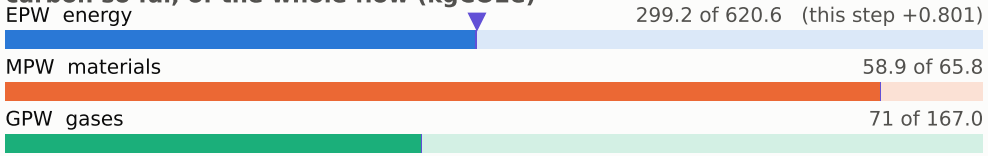


this step

process area	Diffusion
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

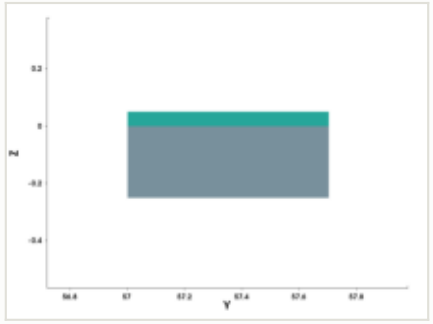
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

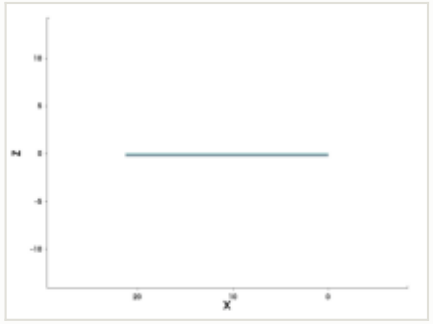


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



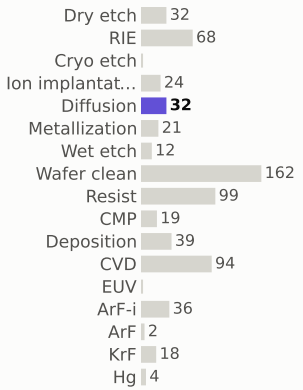
y cut at 57.351 µm



mask for this step

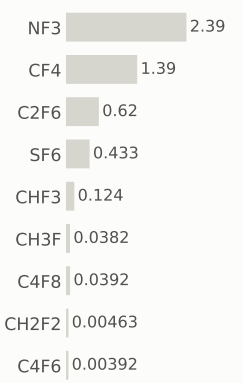
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



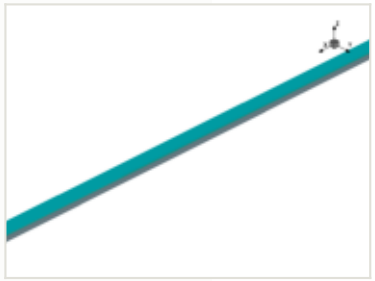
Step #9220: Diffusion

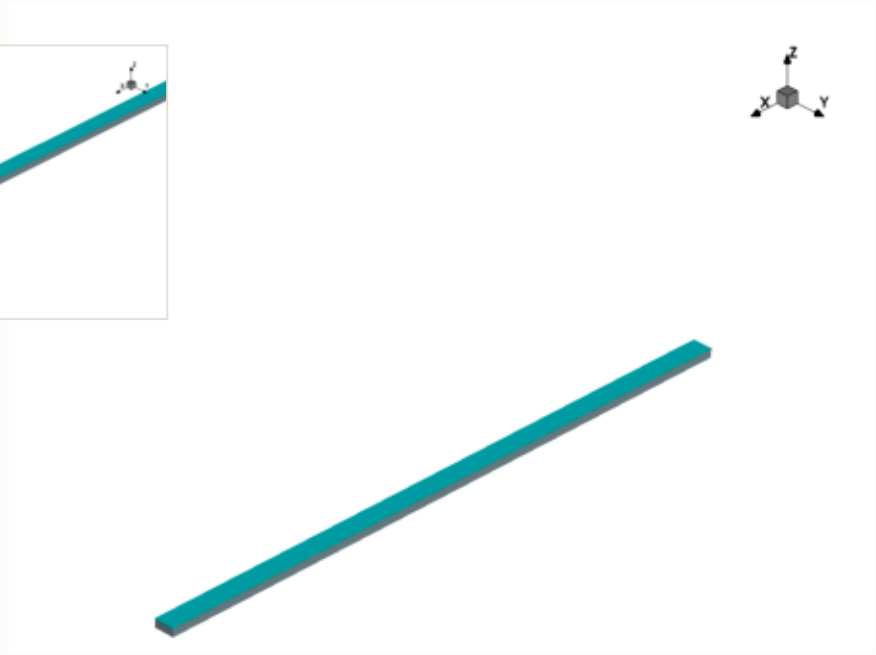
depo_isotropic · Diffusion

0.801	430.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	0.005 µm	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

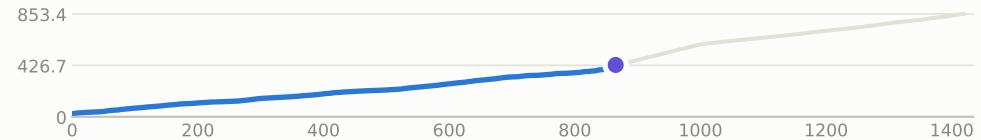
carbon so far, of the whole flow (kgCO2e)

EPW energy — 300.0 of 620.6 (this step +0.801)

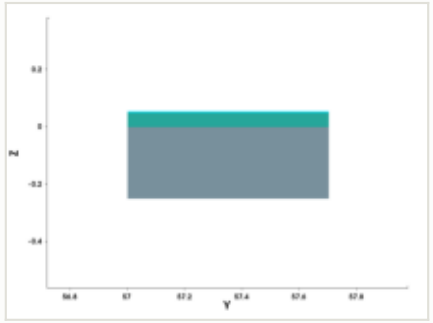
MPW materials — 58.9 of 65.8

GPW gases — 71 of 167.0

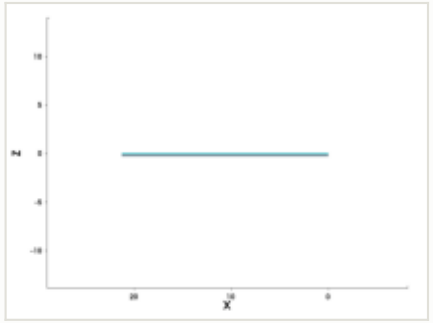
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



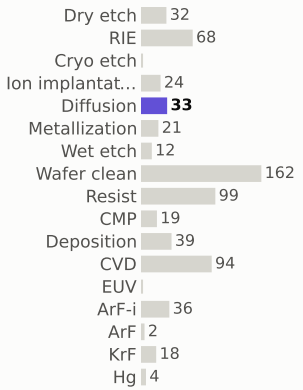
y cut at 57.351 µm



mask for this step

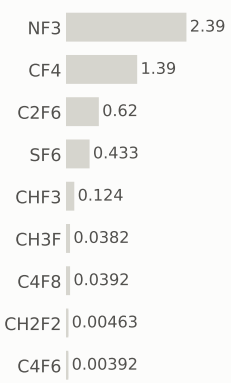
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



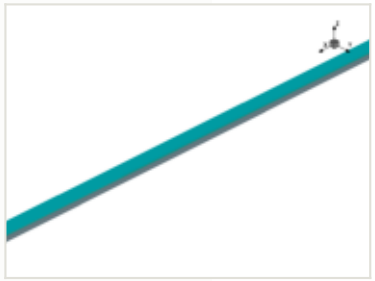
Step #9230: Ion_implantation

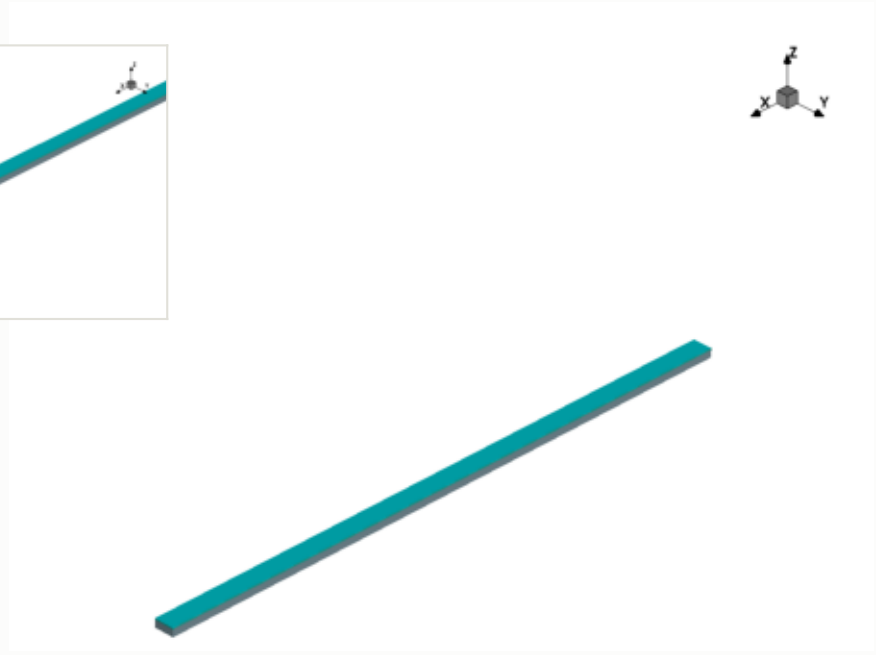
implant_ellipsoid · Ion_implantation

0.801	430.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

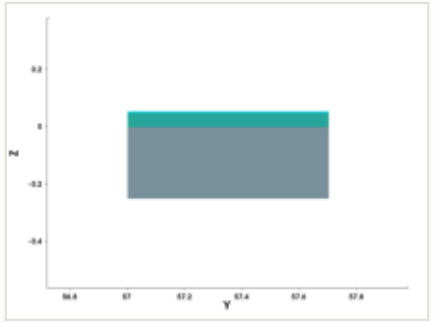
3D structure after this step

zoomed to fit

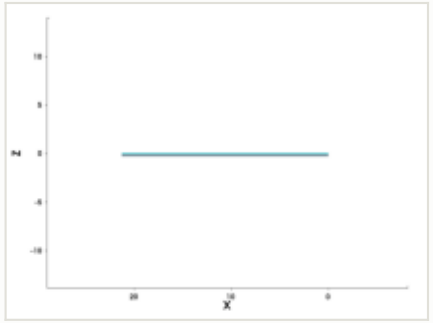




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

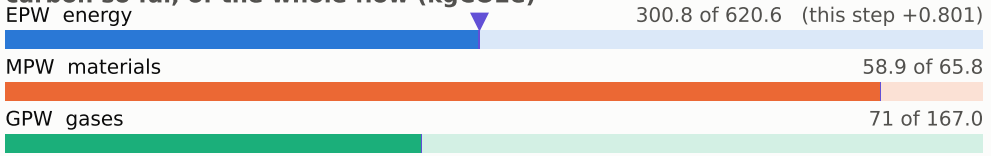
no mask for this step

this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

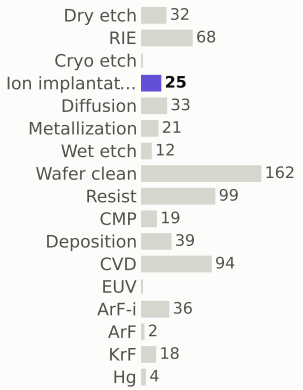
spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

carbon so far, of the whole flow (kgCO2e)



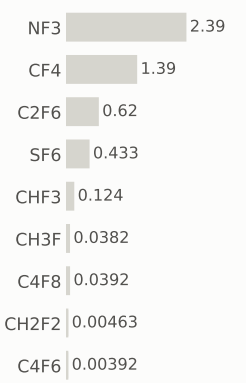
embodied carbon up to each step (kgCO2e per wafer)

steps by process area



materials deposited (g)

gases emitted (g)



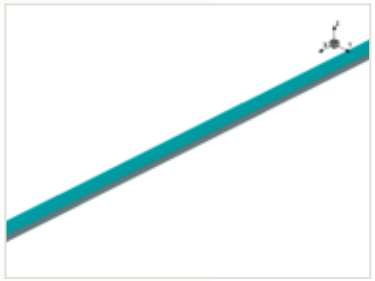
Step #9240: Diffusion

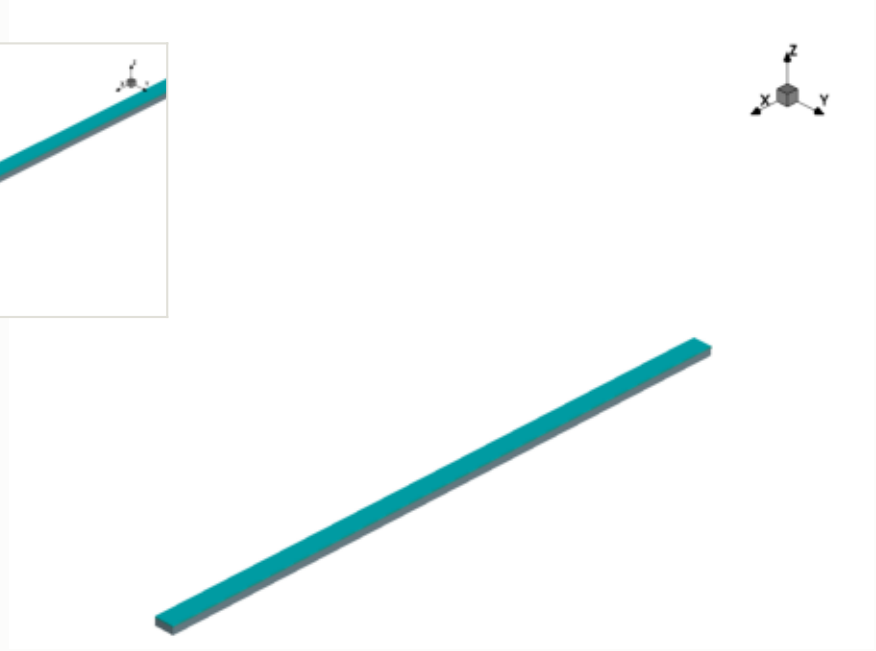
anneal · Diffusion

0.801	431.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





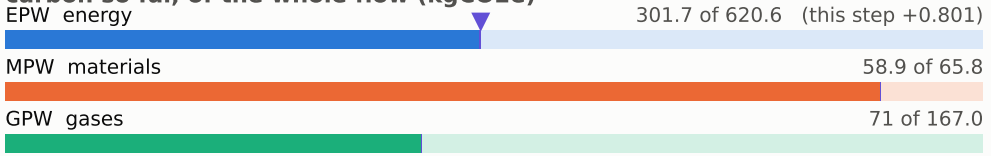
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

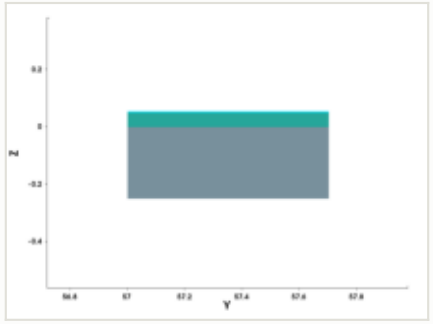
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

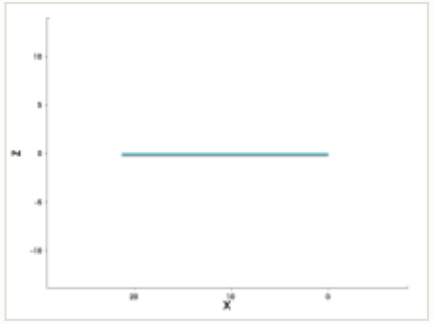


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



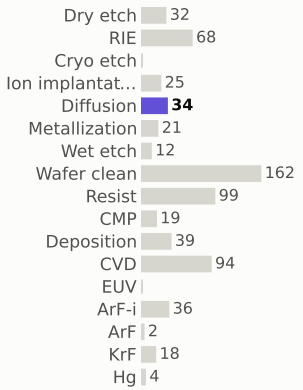
y cut at 57.351 µm



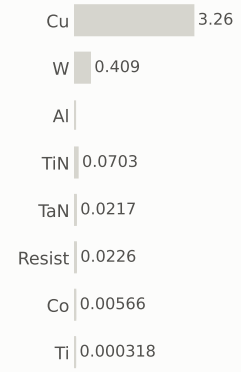
mask for this step

no mask for this step

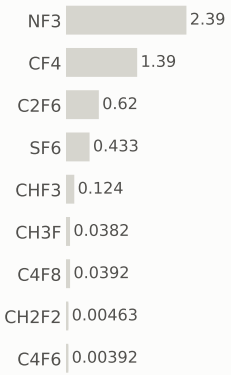
steps by process area



materials deposited (g)



gases emitted (g)



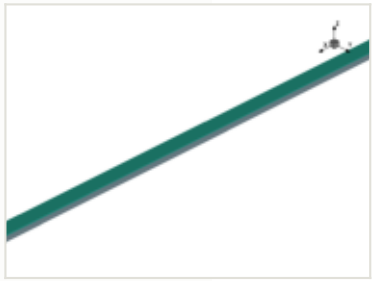
Step #9250: Wet_etch

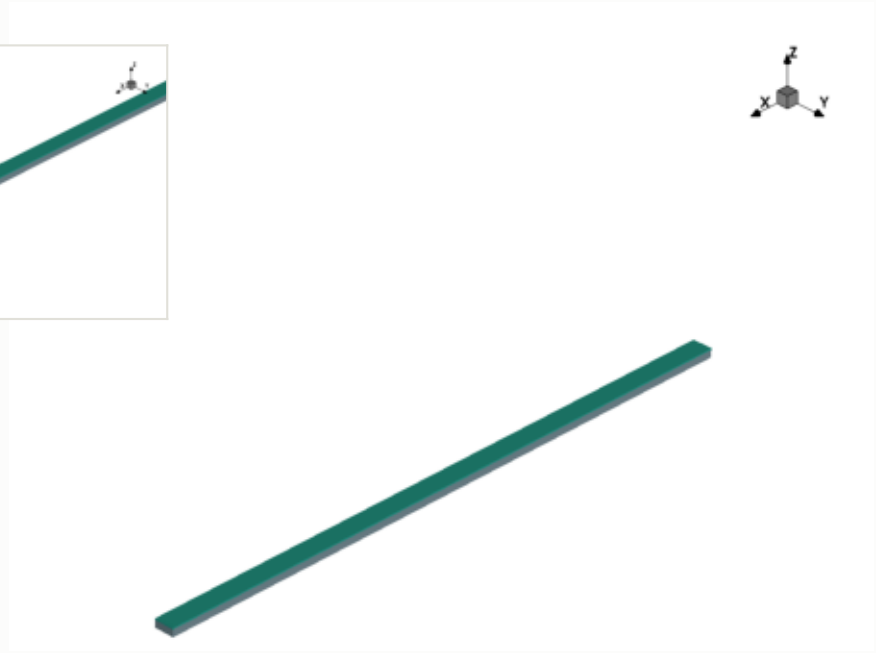
etch_isotropic · Wet_etch

0.169	431.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

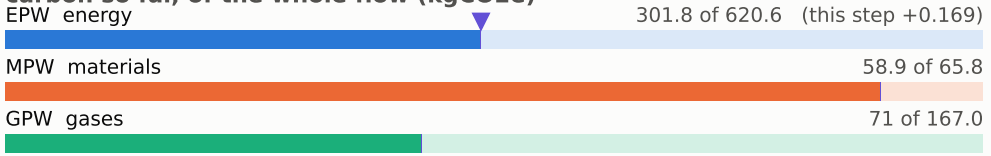




this step

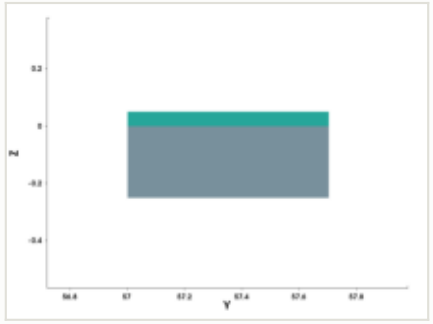
process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

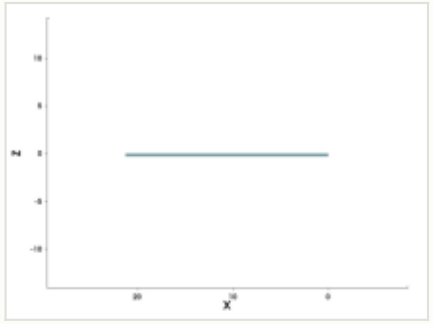


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



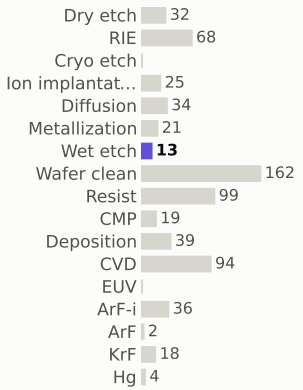
y cut at 57.351 µm



mask for this step

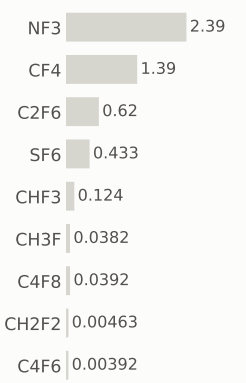
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



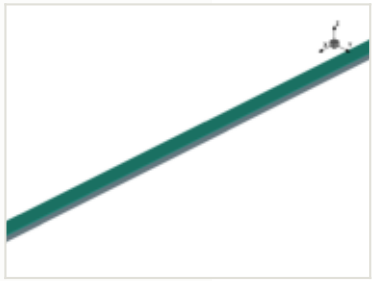
Step #9270: Wafer_clean

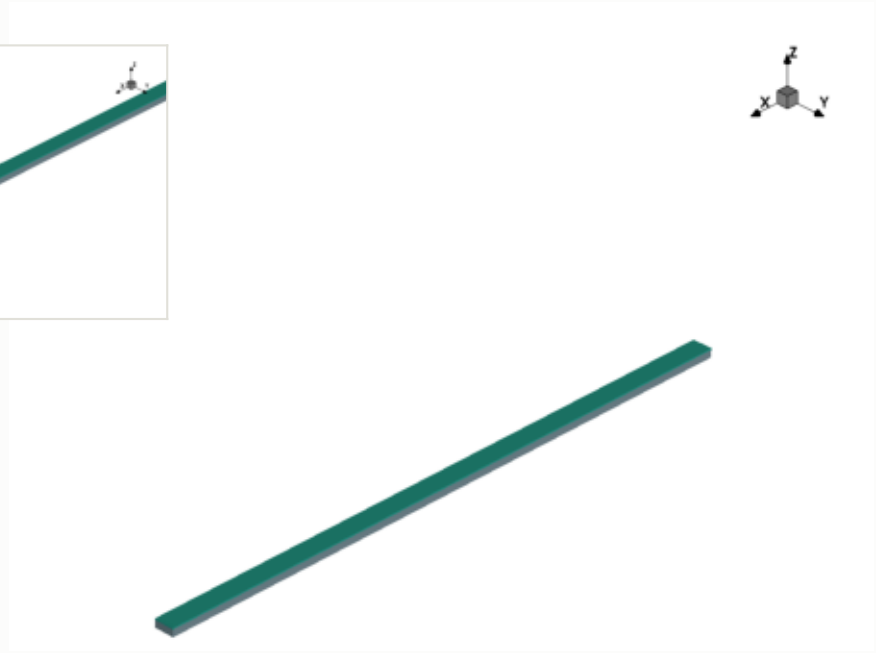
clean · Wafer_clean

0.169	431.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



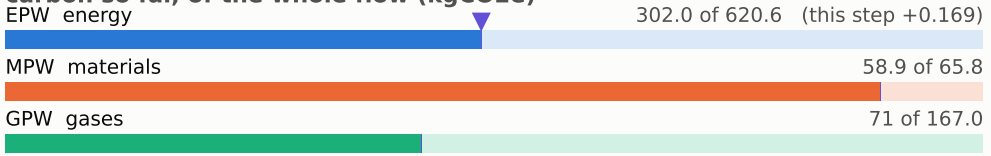


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

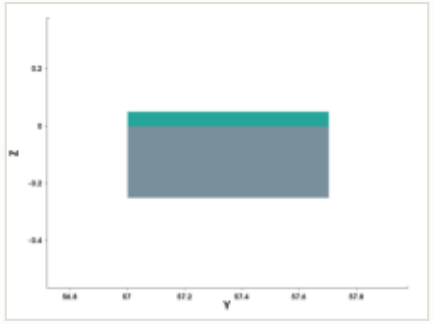
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

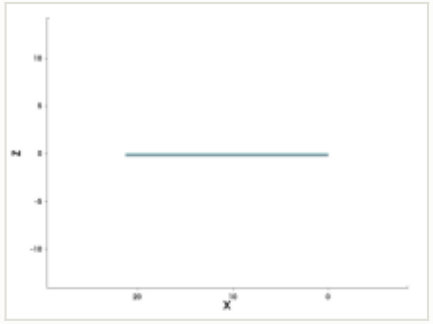


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



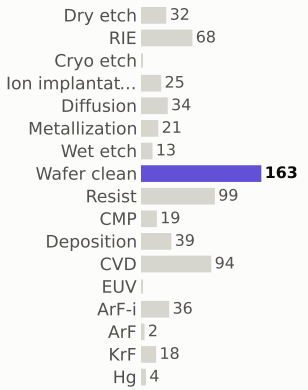
y cut at 57.351 µm



mask for this step

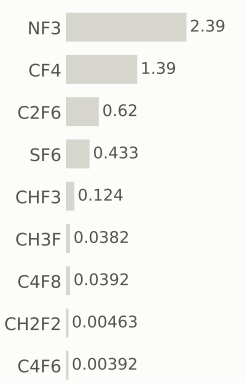
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



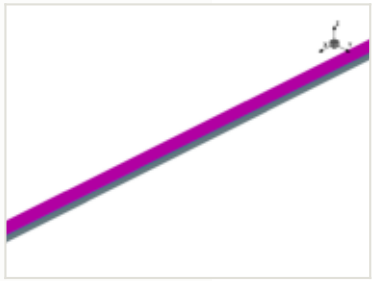
Step #9280: CVD

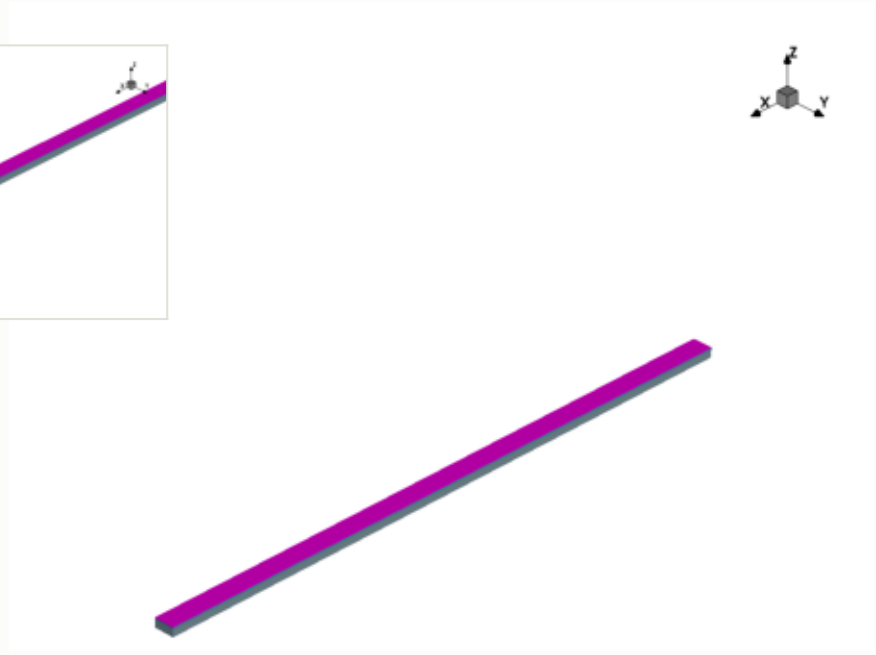
depo_isotropic · CVD

1.31	433.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



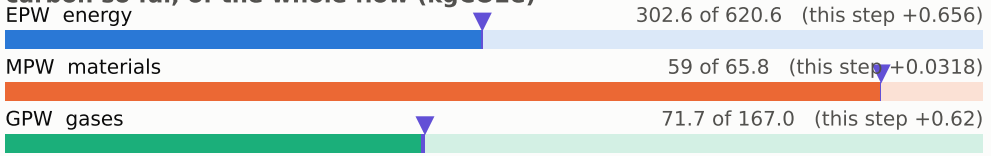


this step

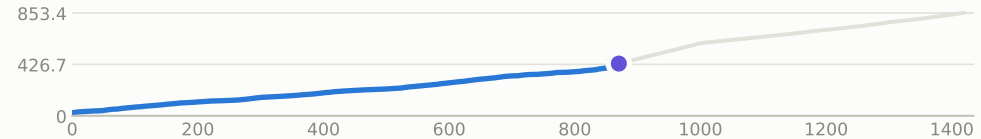
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

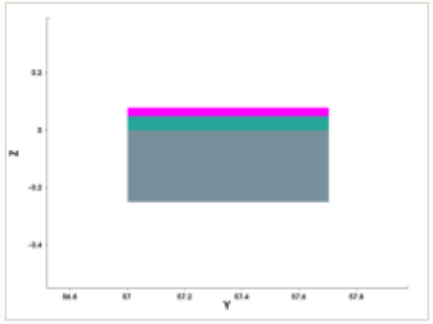
carbon so far, of the whole flow (kgCO2e)



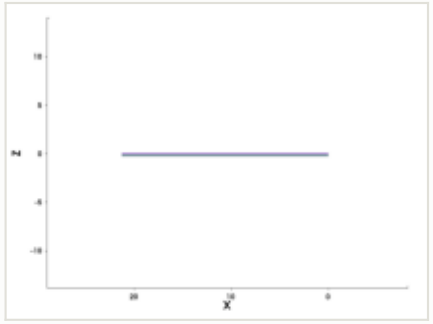
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



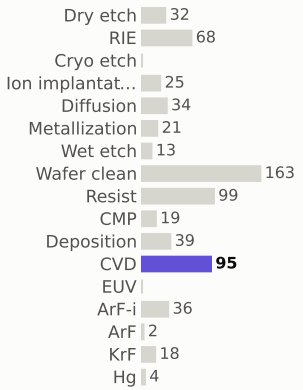
y cut at 57.351 µm



mask for this step

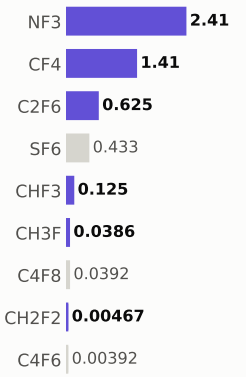
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



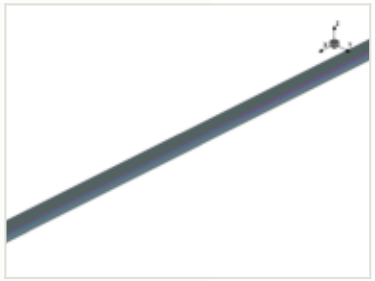
Step #9290: CVD

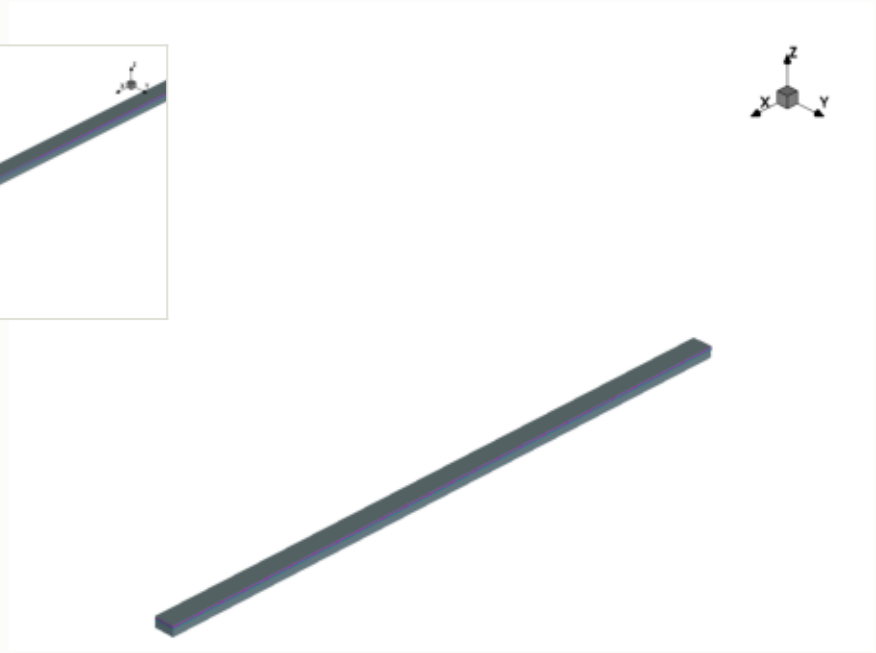
depo_isotropic · CVD

1.31	434.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



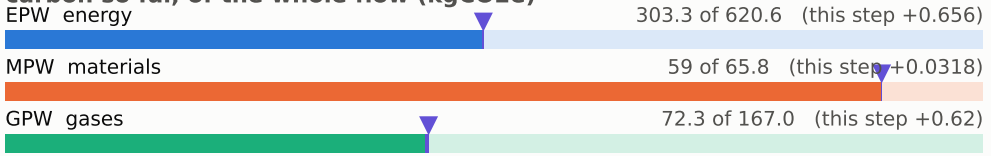


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

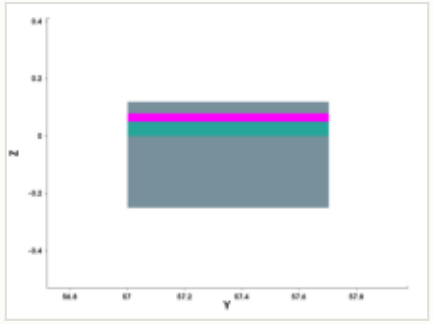
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

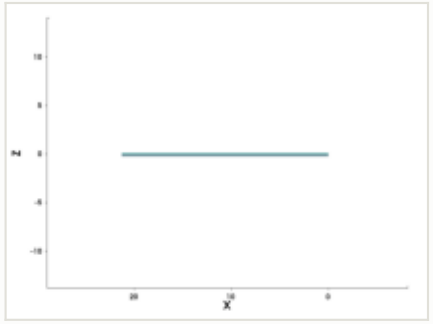


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



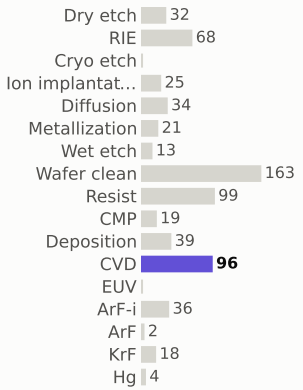
y cut at 57.351 µm



mask for this step

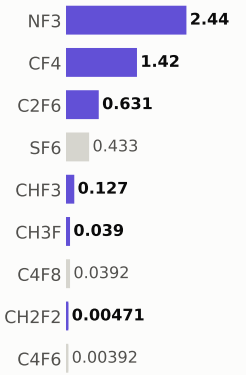
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



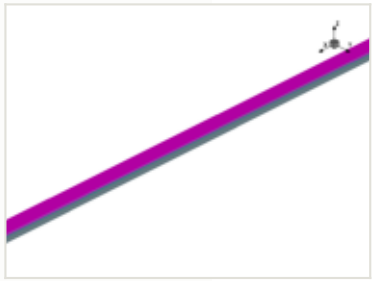
Step #9300: CVD

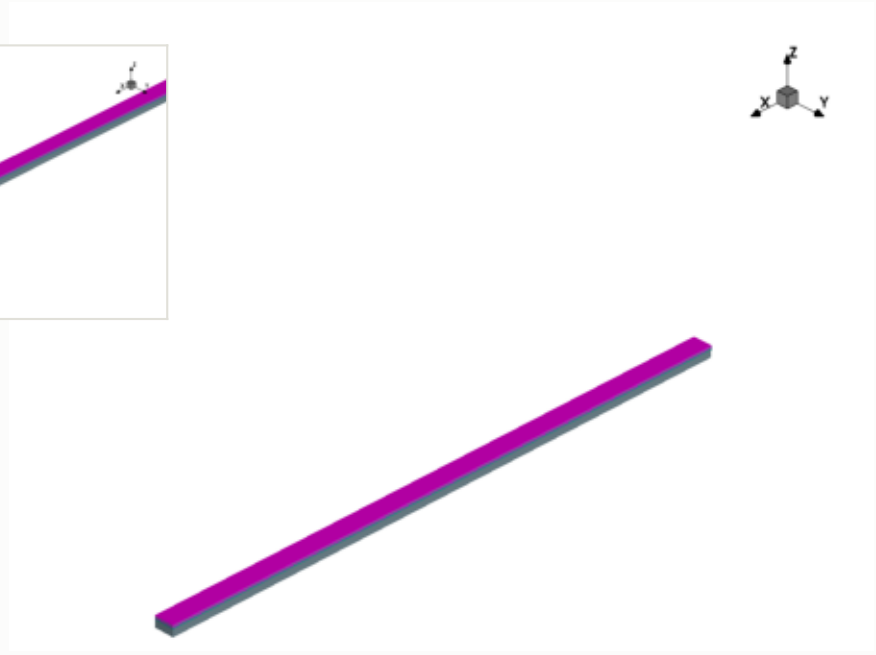
depo_isotropic · CVD

1.31	435.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



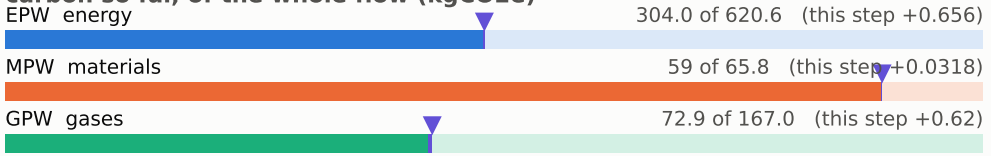


this step

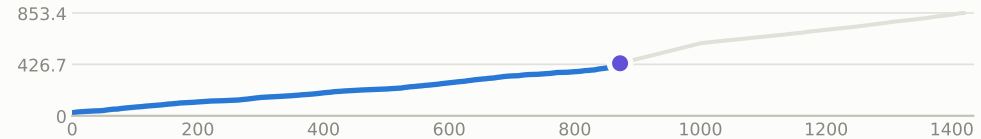
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

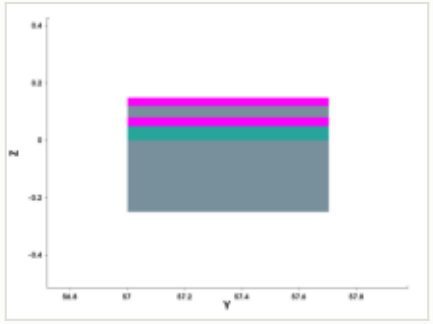
carbon so far, of the whole flow (kgCO2e)



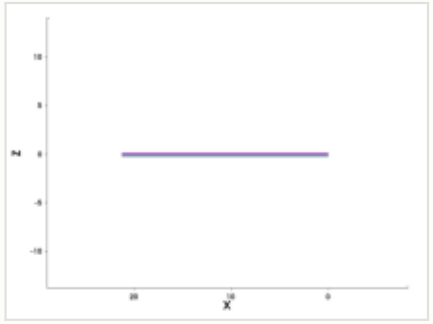
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



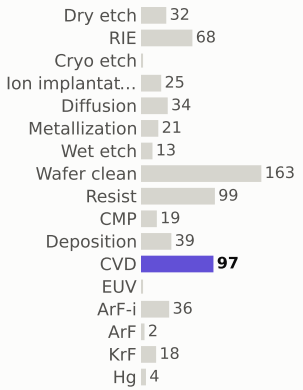
y cut at 57.351 µm



mask for this step

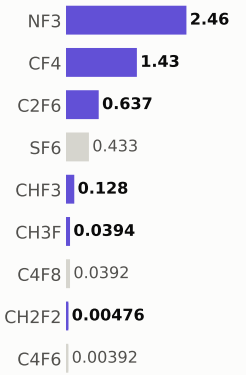
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



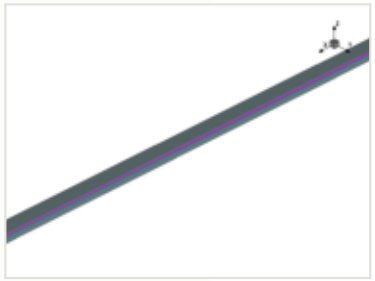
Step #9310: CVD

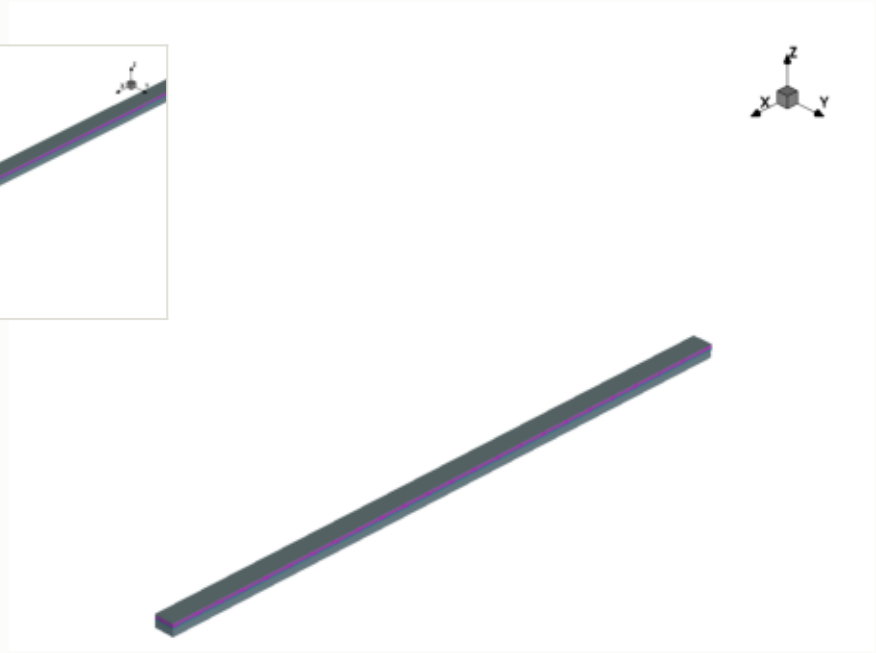
depo_isotropic · CVD

1.31	437.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



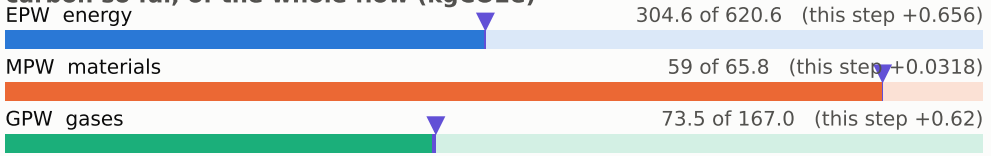


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

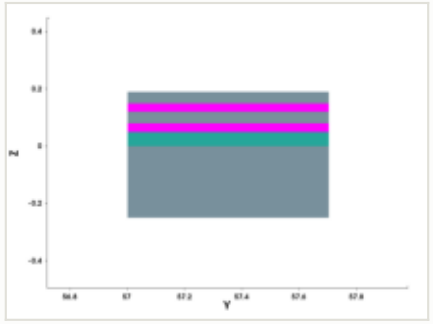
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

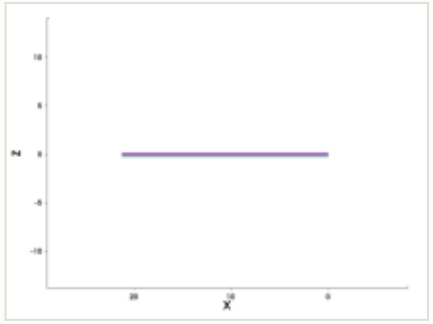


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



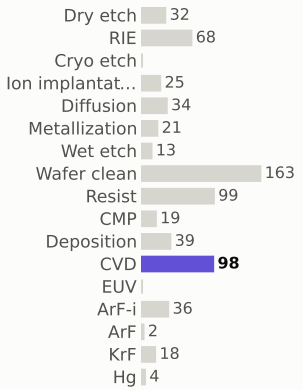
y cut at 57.351 µm



mask for this step

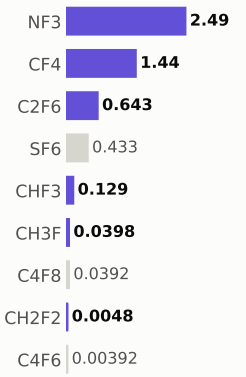
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



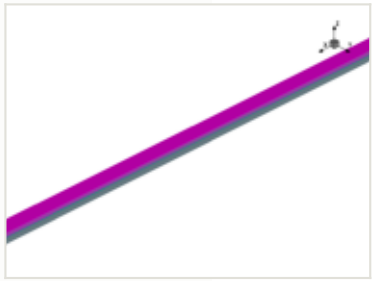
Step #9320: CVD

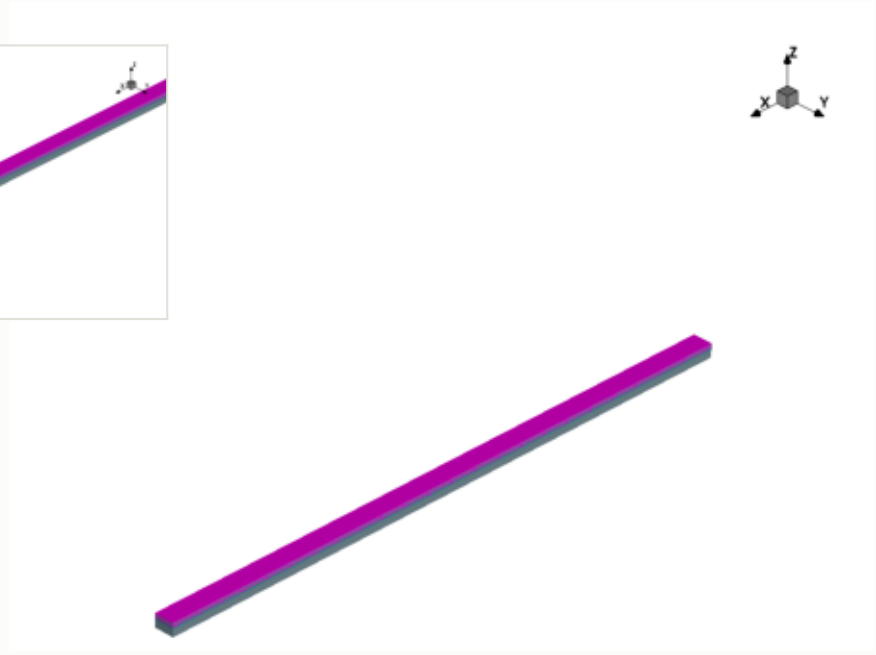
depo_isotropic · CVD

1.31	438.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



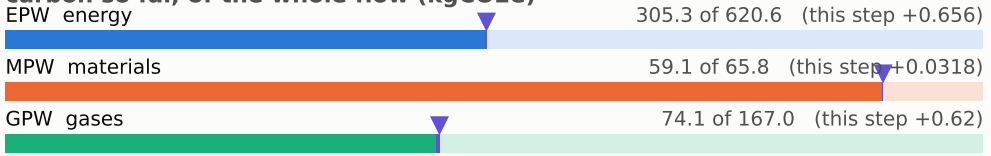


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

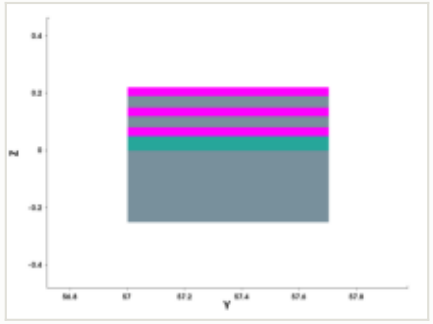
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

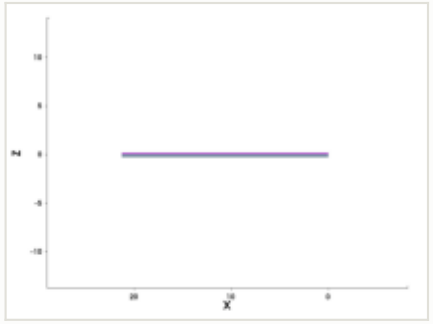


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



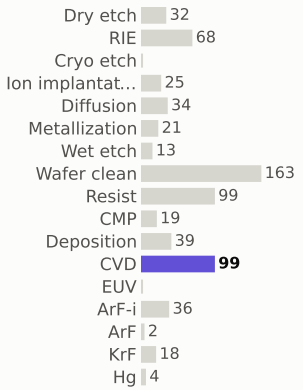
y cut at 57.351 µm



mask for this step

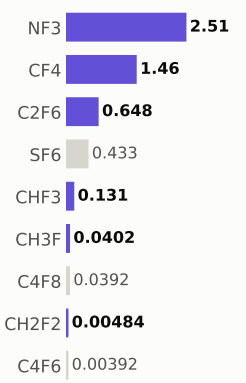
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



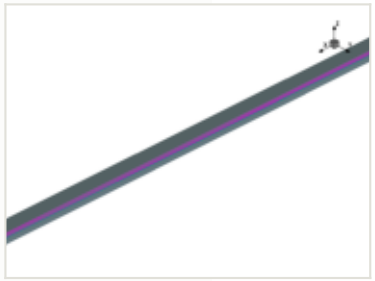
Step #9330: CVD

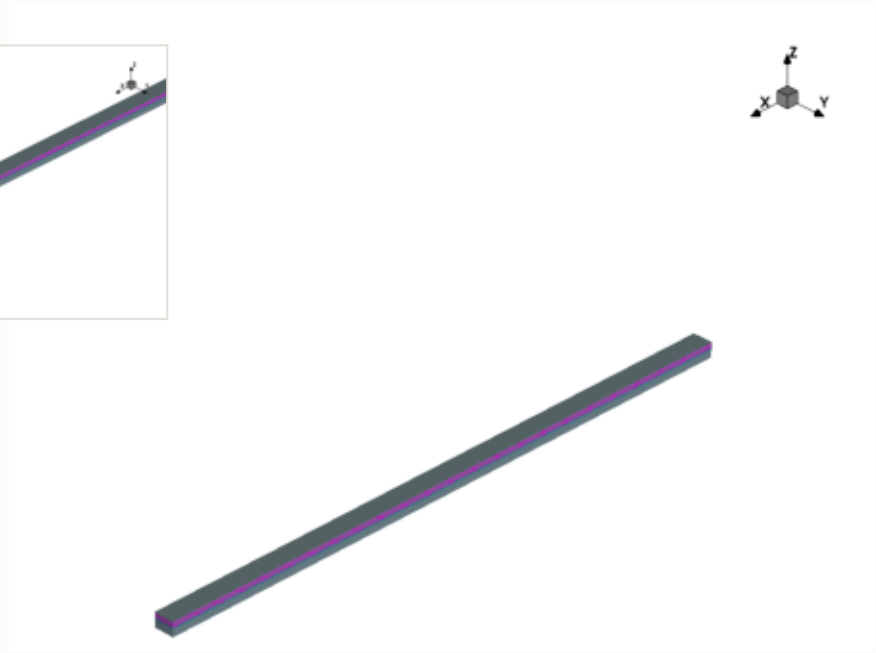
depo_isotropic · CVD

1.31	439.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



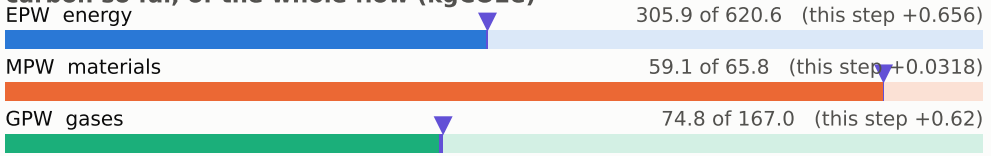


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

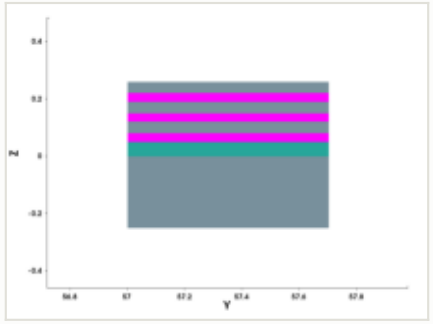
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

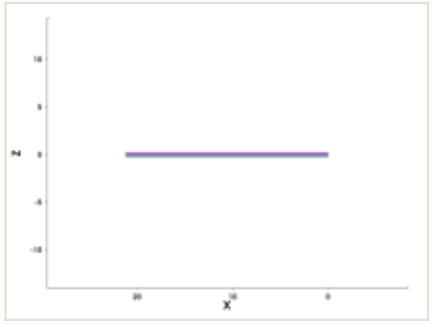


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



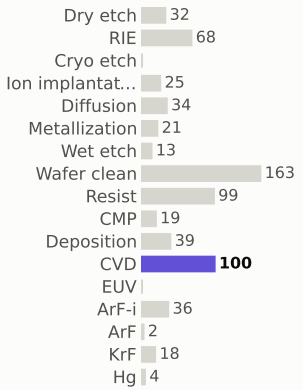
y cut at 57.351 µm



mask for this step

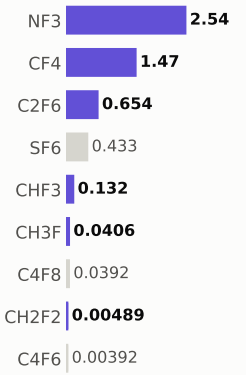
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



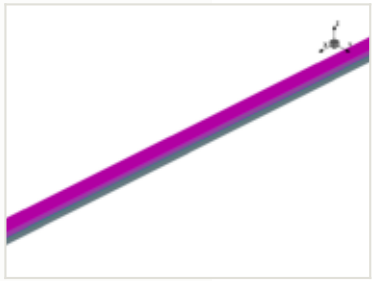
Step #9340: CVD

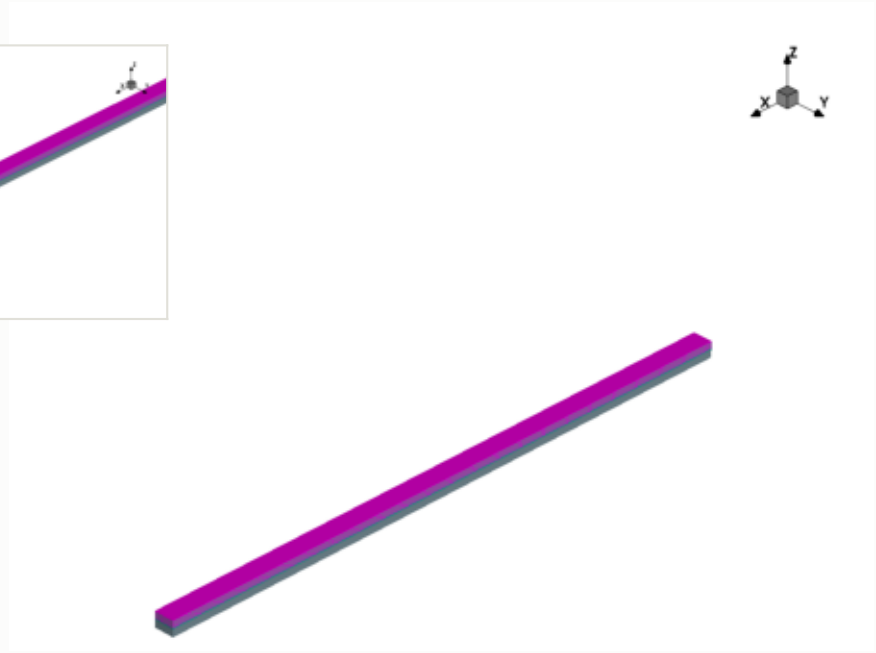
depo_isotropic · CVD

1.31	441.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



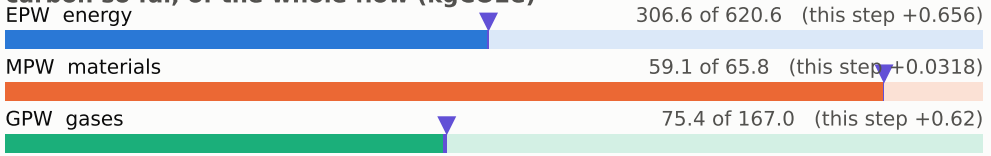


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

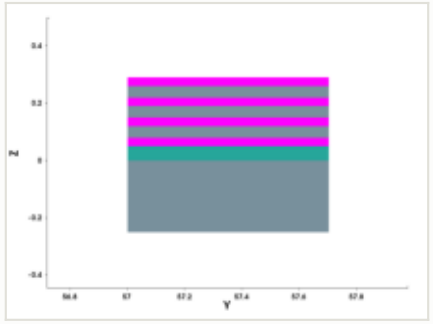
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

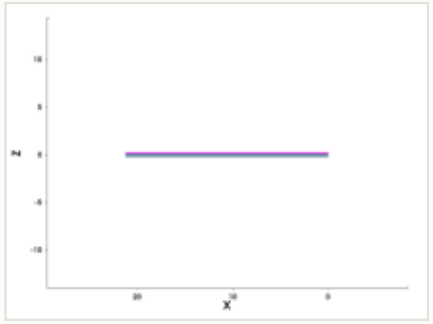


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



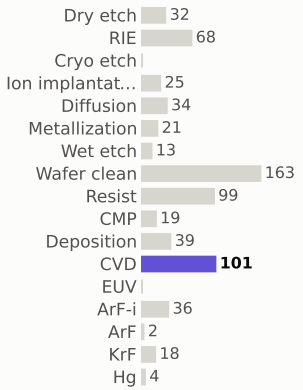
y cut at 57.351 µm



mask for this step

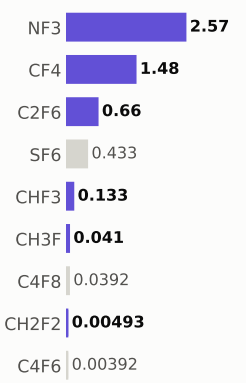
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



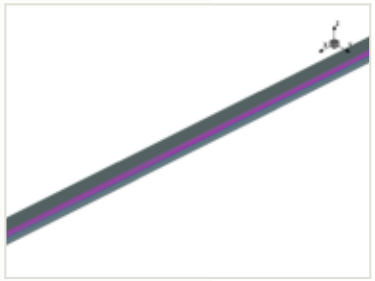
Step #9350: CVD

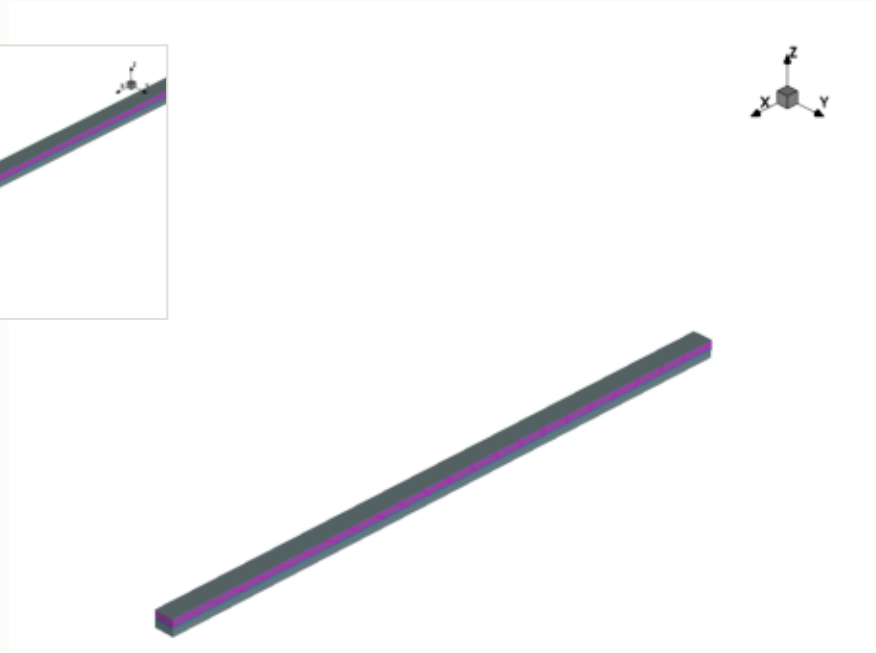
depo_isotropic · CVD

1.31	442.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



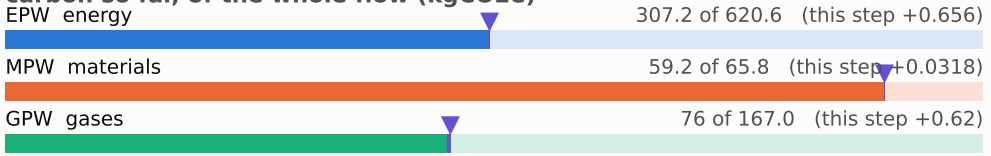


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

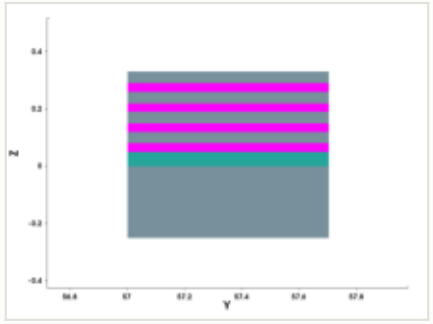
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

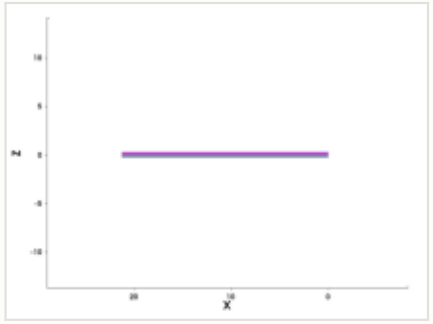


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



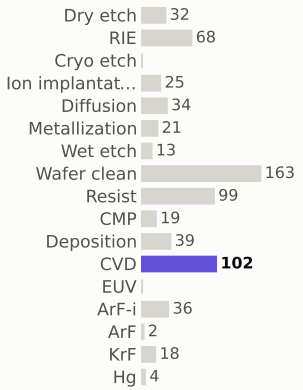
y cut at 57.351 µm



mask for this step

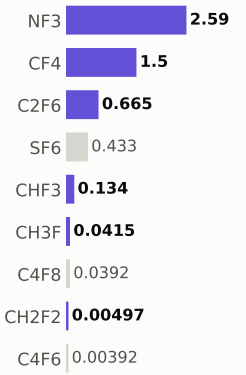
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



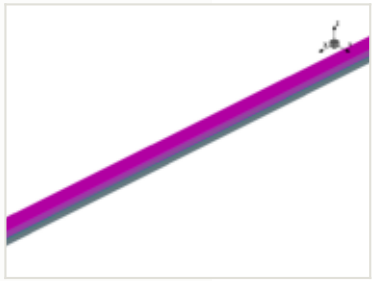
Step #9360: CVD

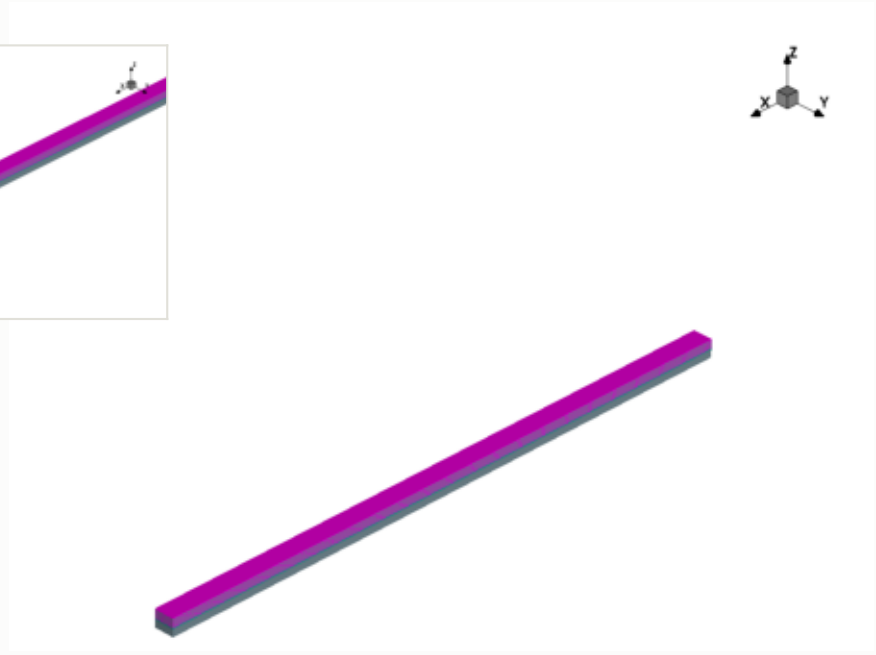
depo_isotropic · CVD

1.31	443.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



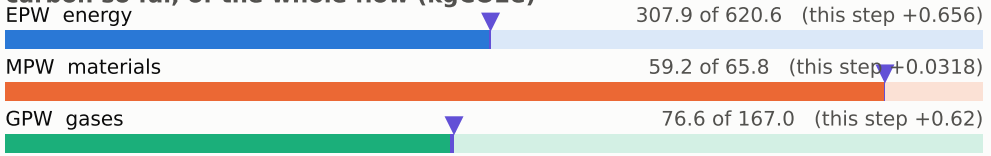


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

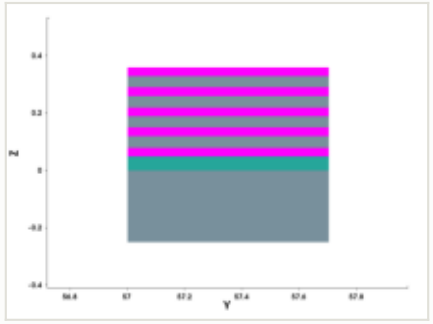
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

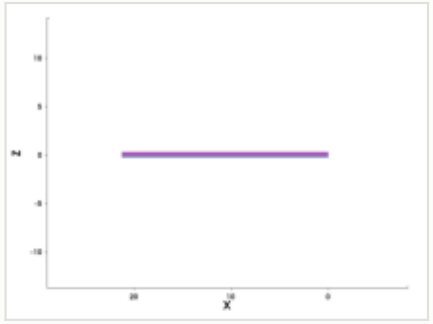


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



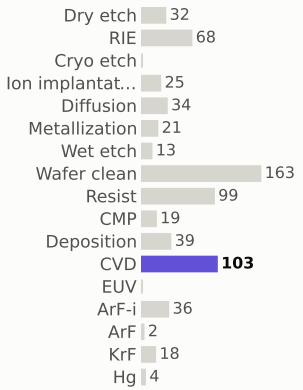
y cut at 57.351 µm



mask for this step

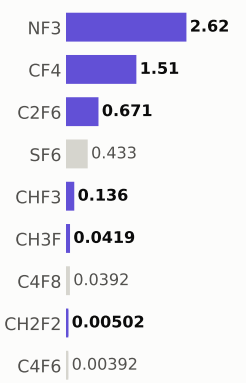
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



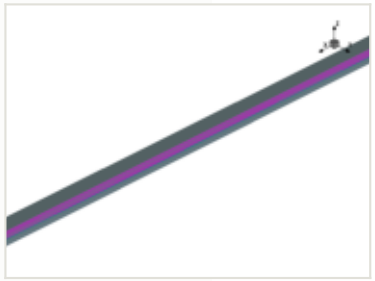
Step #9370: CVD

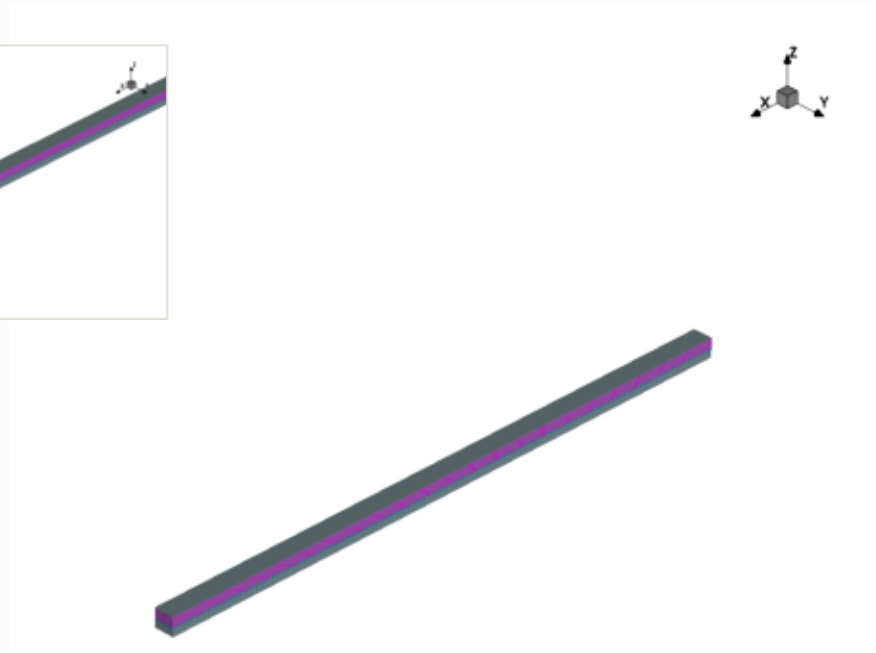
depo_isotropic · CVD

1.31	445.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



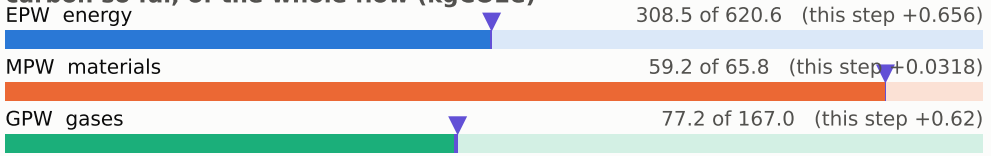


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

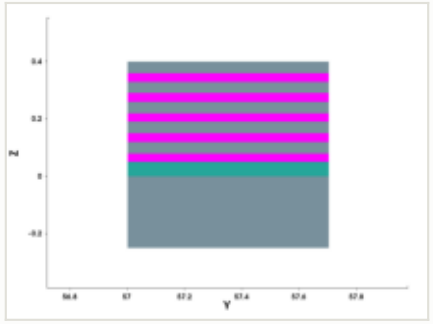
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

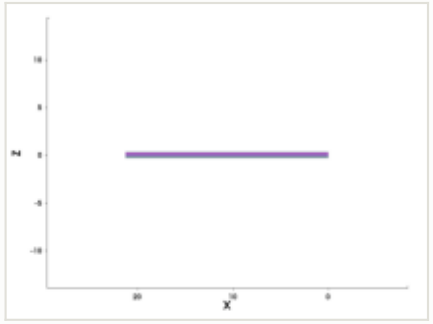


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



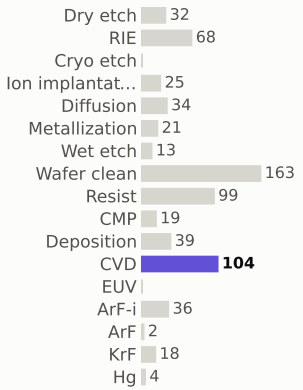
y cut at 57.351 µm



mask for this step

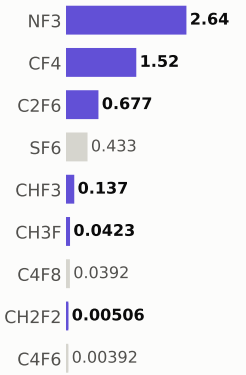
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



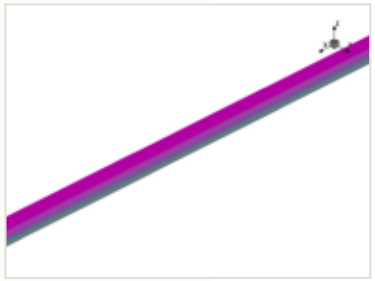
Step #9380: CVD

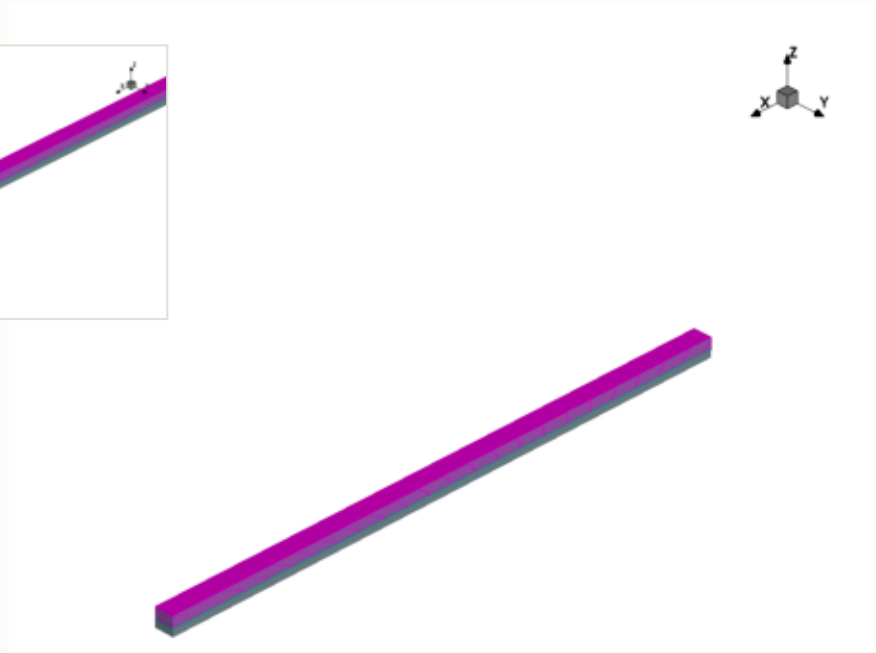
depo_isotropic · CVD

1.31	446.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



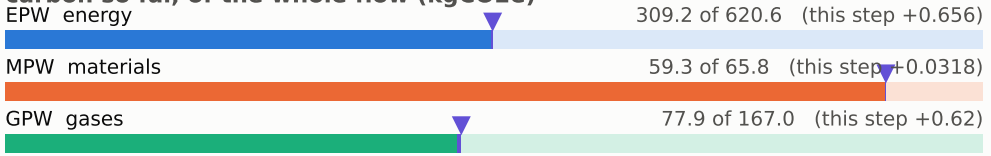


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

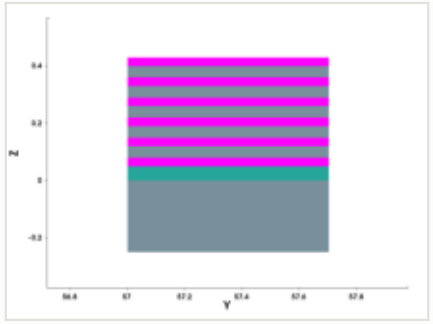
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

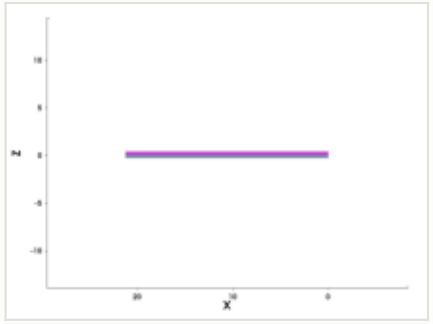


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



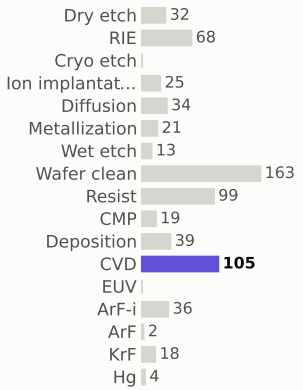
y cut at 57.351 µm



mask for this step

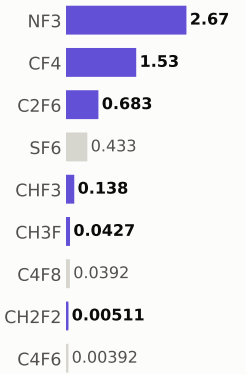
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



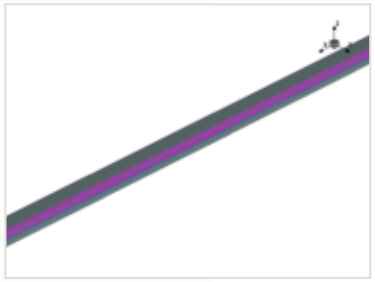
Step #9390: CVD

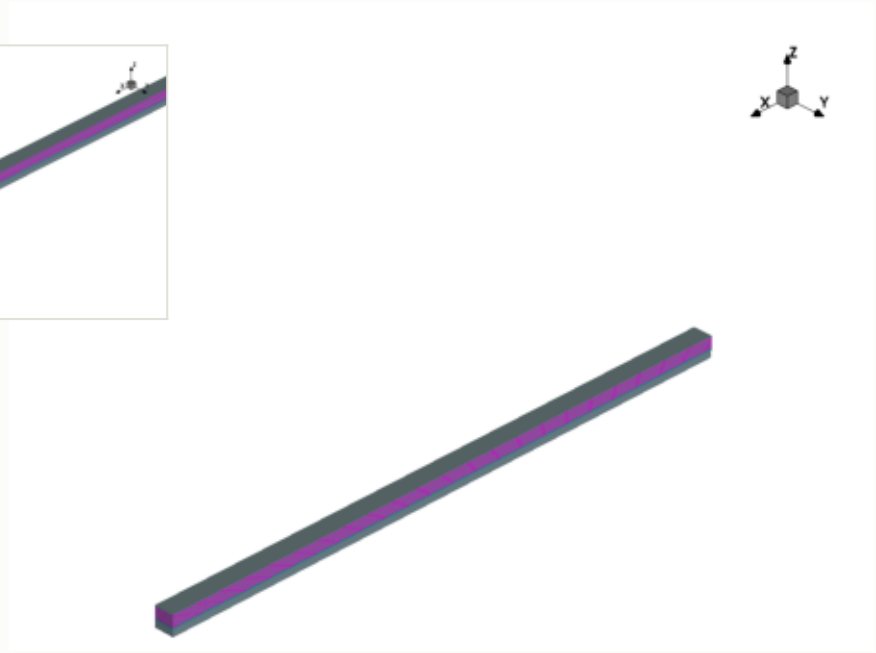
depo_isotropic · CVD

1.31	447.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



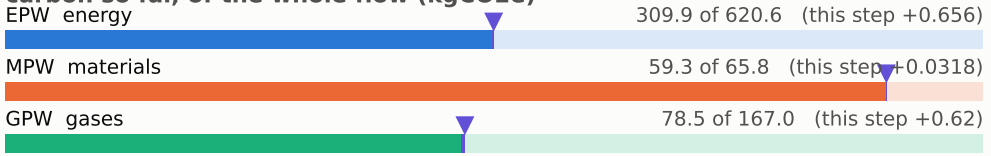


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

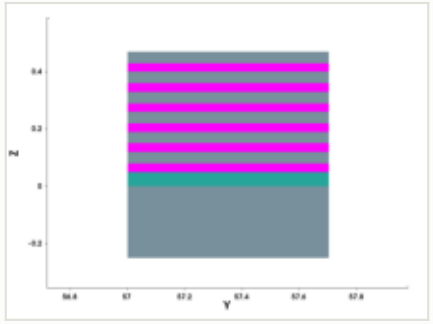
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

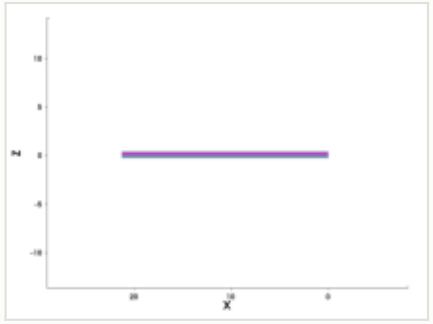


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



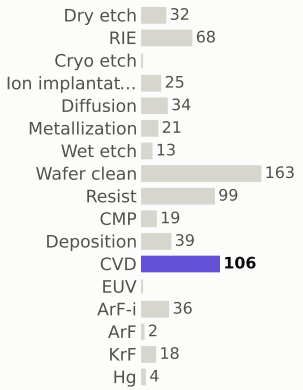
y cut at 57.351 µm



mask for this step

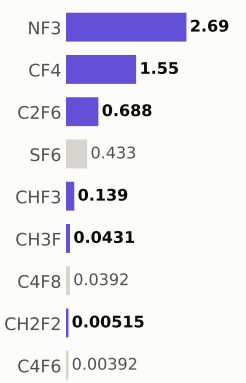
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



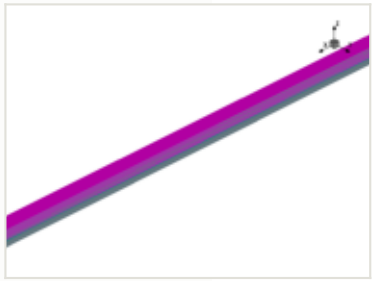
Step #9400: CVD

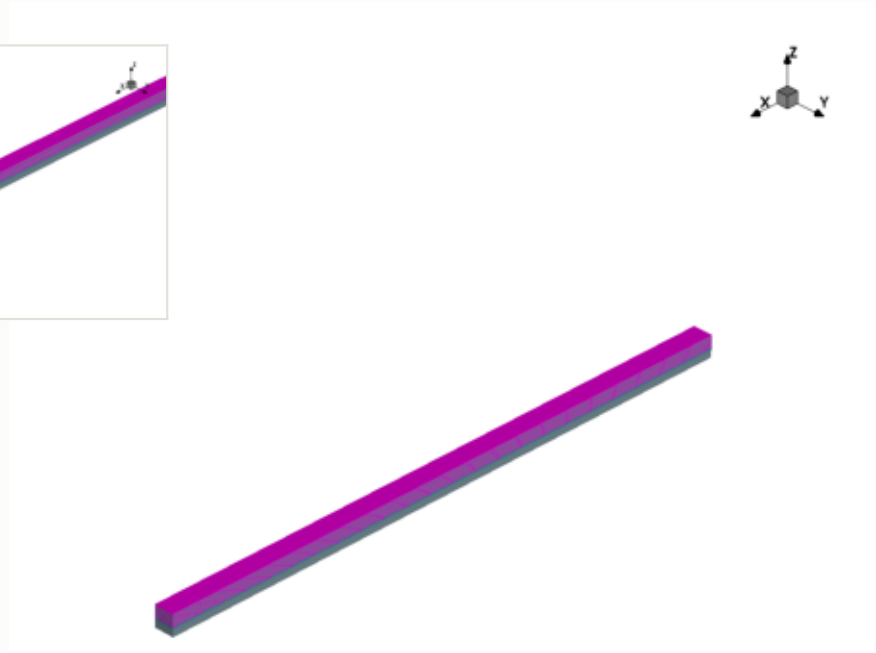
depo_isotropic · CVD

1.31	449.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



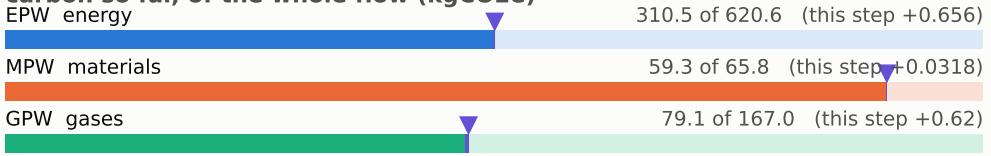


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

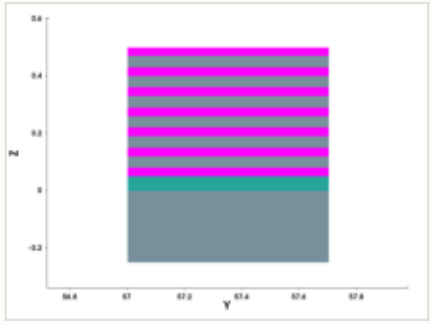
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

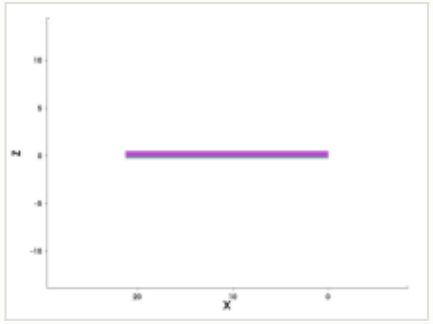


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



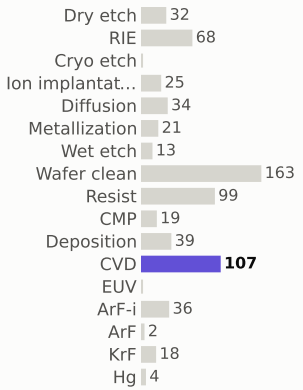
y cut at 57.351 µm



mask for this step

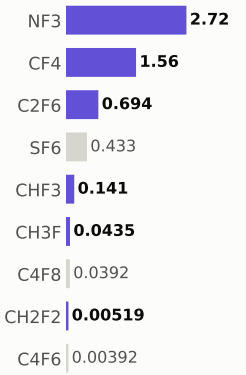
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



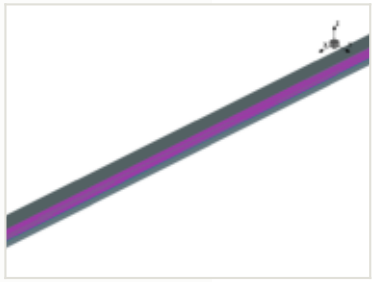
Step #9410: CVD

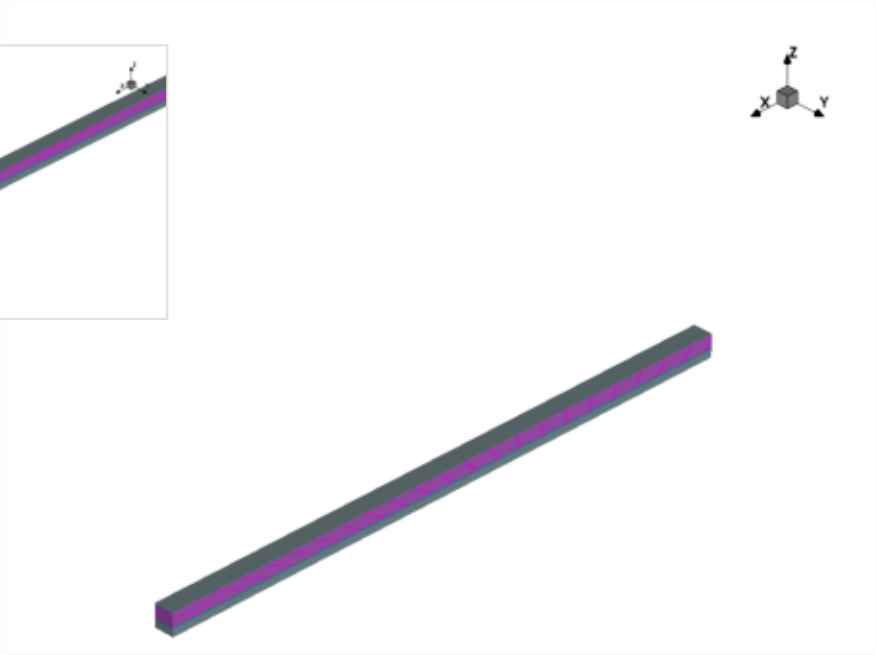
depo_isotropic · CVD

1.31	450.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



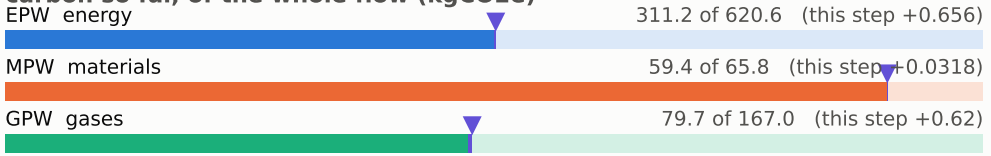


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

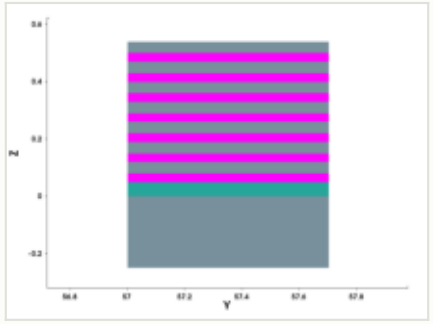
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

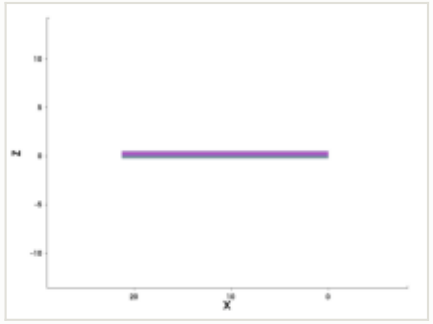


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



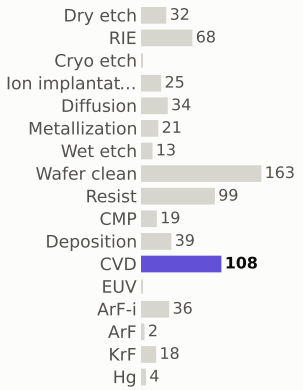
y cut at 57.351 µm



mask for this step

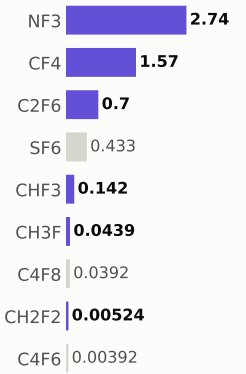
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



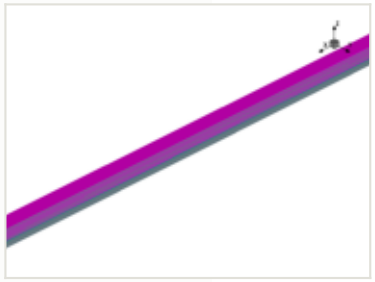
Step #9420: CVD

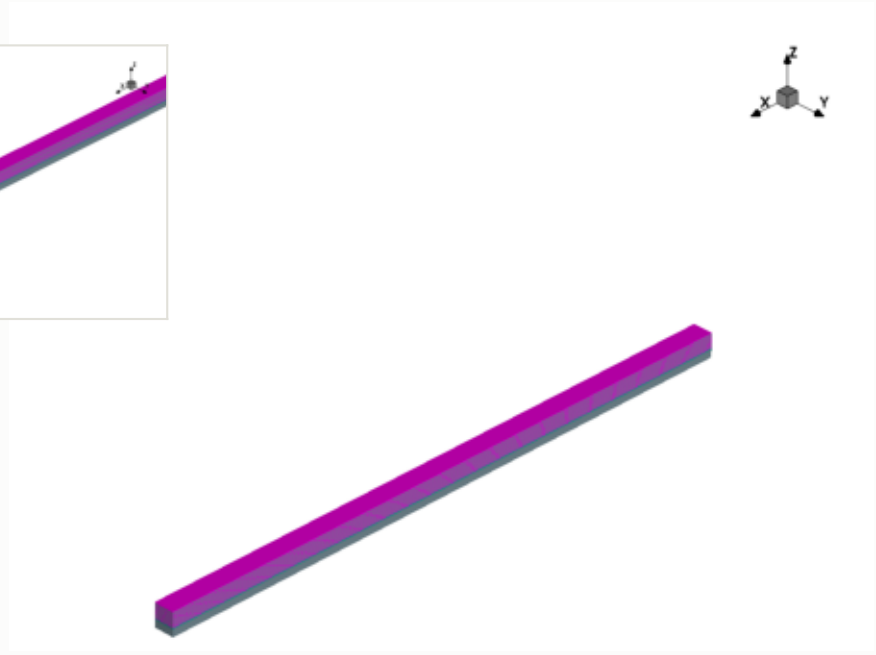
depo_isotropic · CVD

1.31	451.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



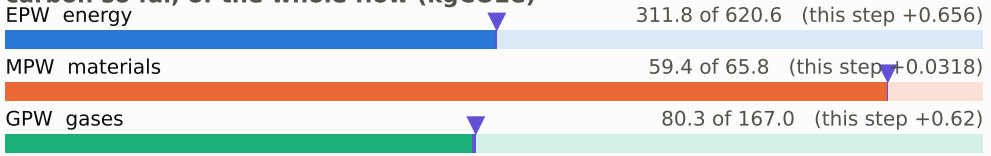


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

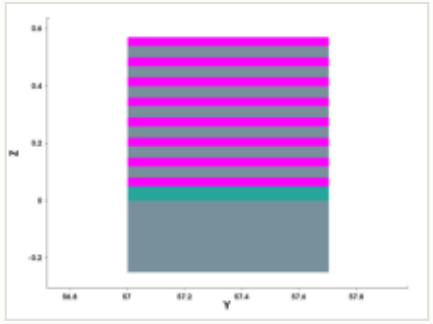
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

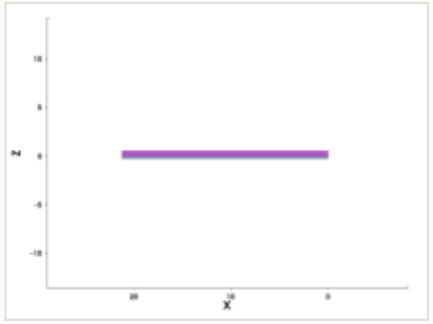


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



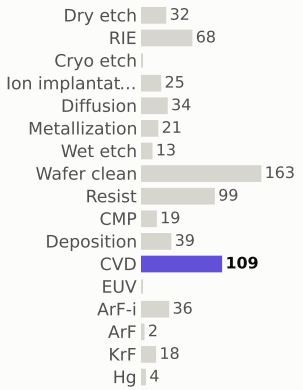
y cut at 57.351 µm



mask for this step

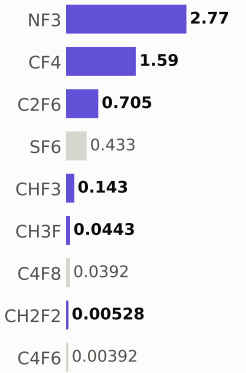
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



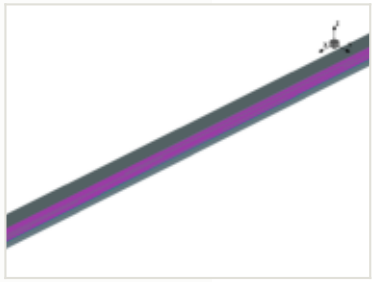
Step #9430: CVD

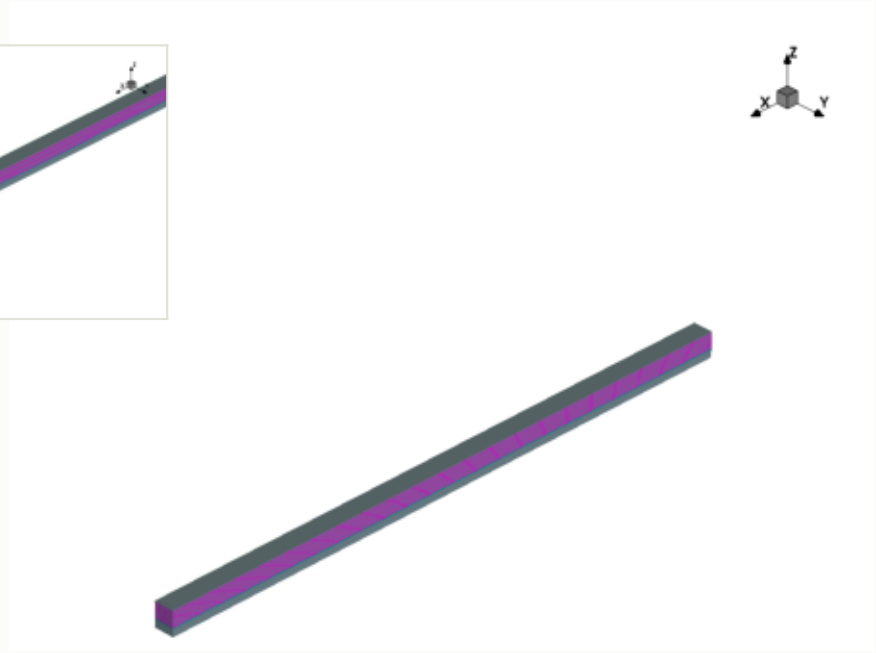
depo_isotropic · CVD

1.31	452.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



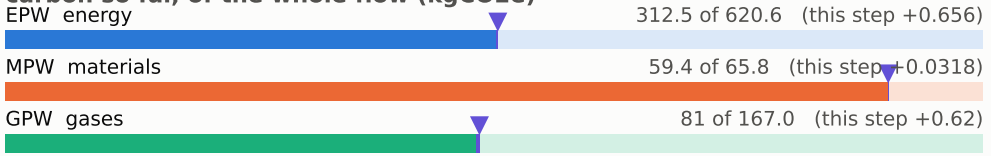


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

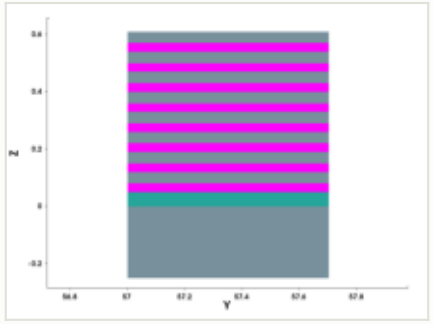
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

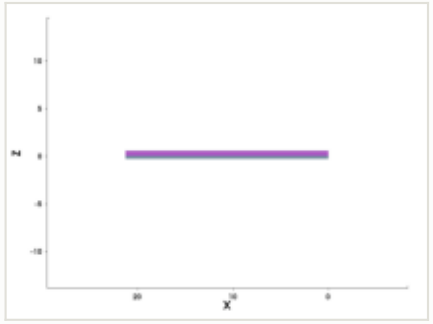


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



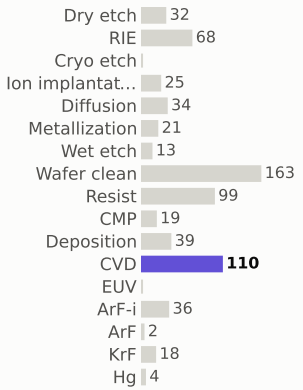
y cut at 57.351 µm



mask for this step

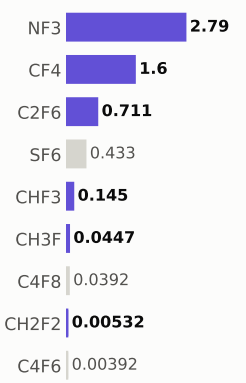
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



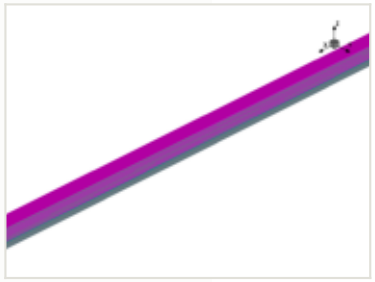
Step #9440: CVD

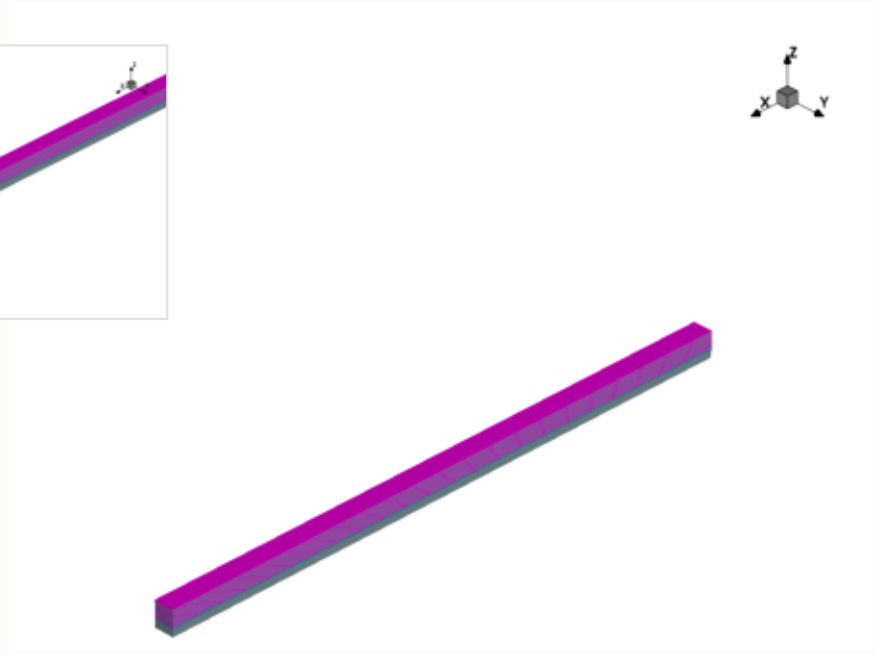
depo_isotropic · CVD

1.31	454.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



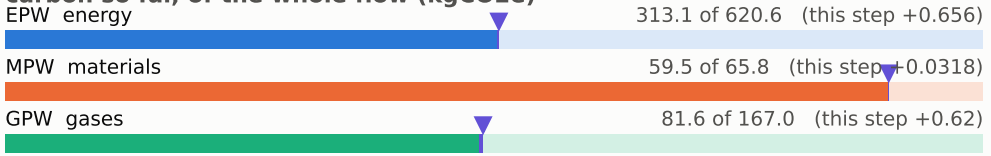


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

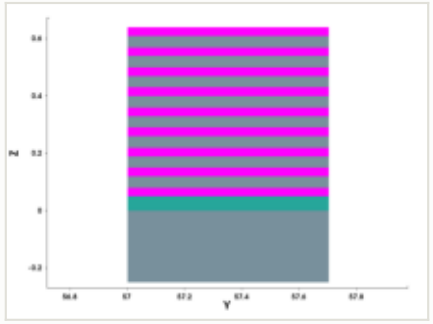
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

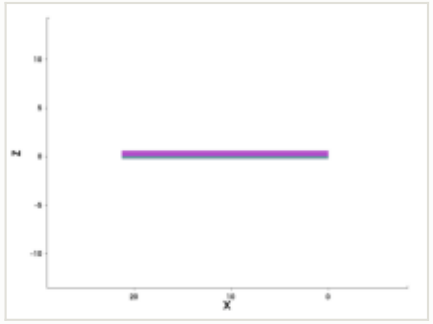


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



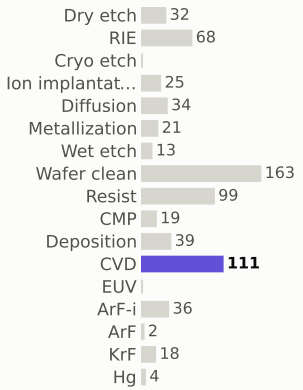
y cut at 57.351 µm



mask for this step

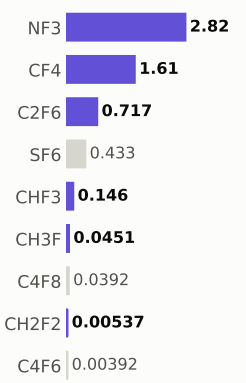
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



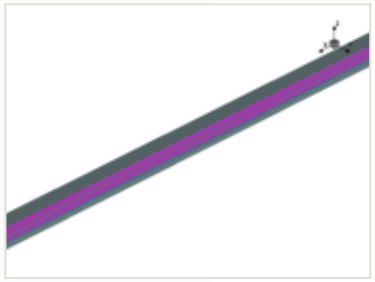
Step #9450: CVD

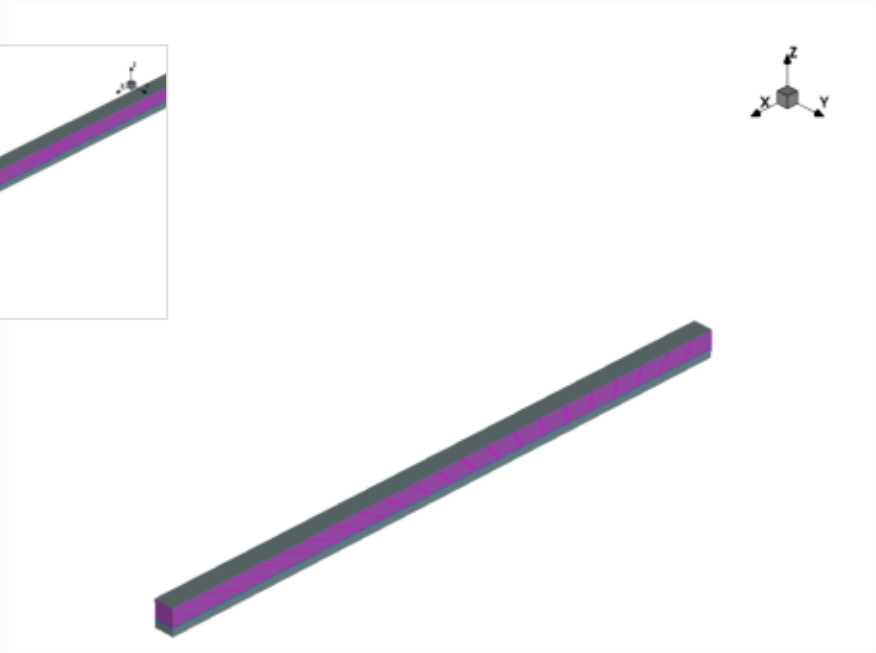
depo_isotropic · CVD

1.31	455.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



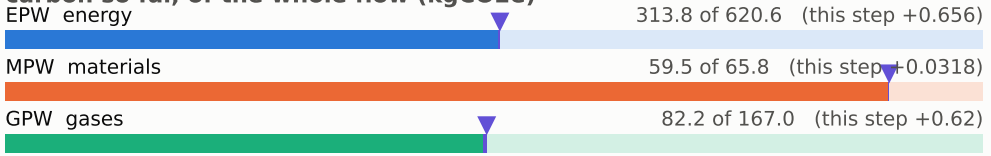


this step

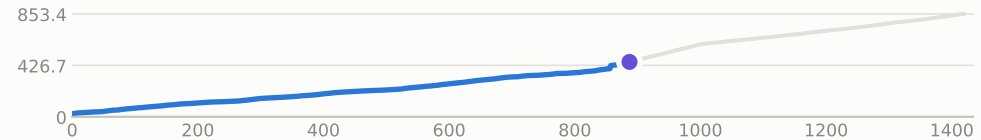
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

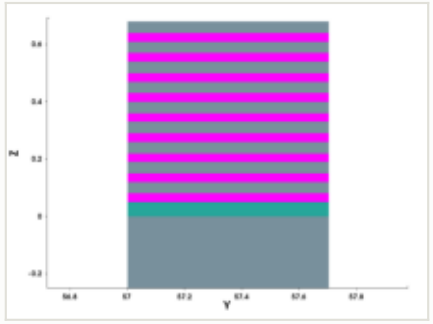
carbon so far, of the whole flow (kgCO2e)



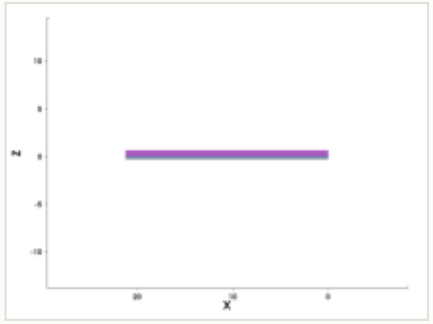
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



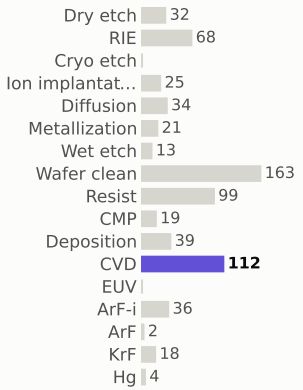
y cut at 57.351 µm



mask for this step

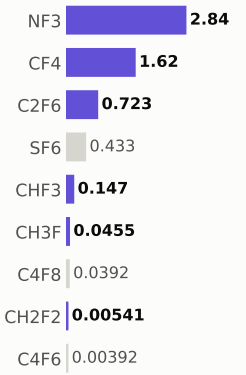
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



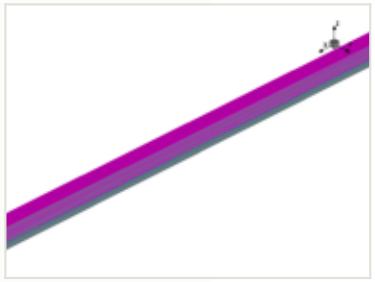
Step #9460: CVD

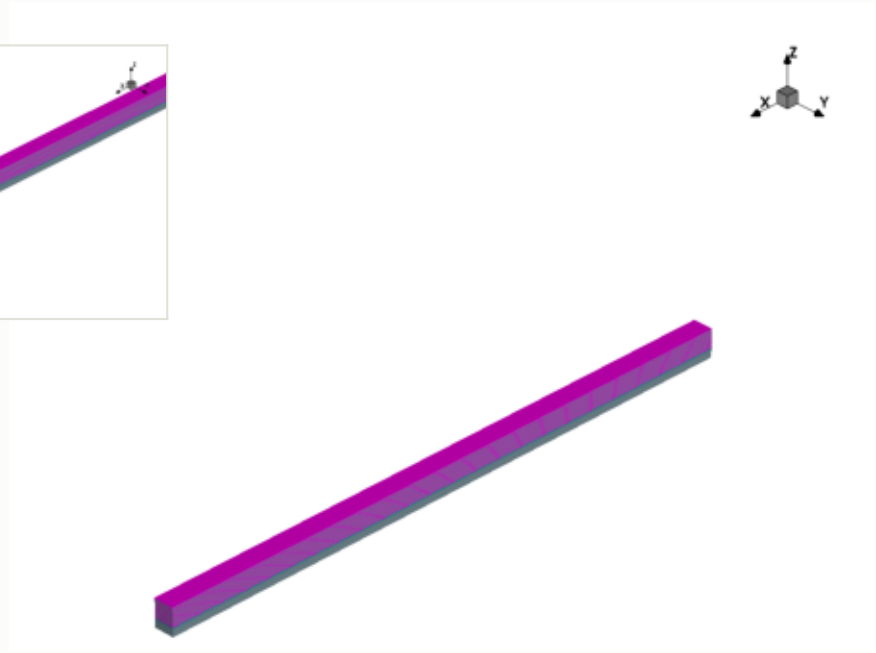
depo_isotropic · CVD

1.31	456.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



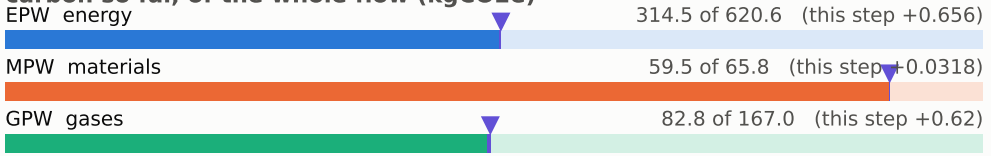


this step

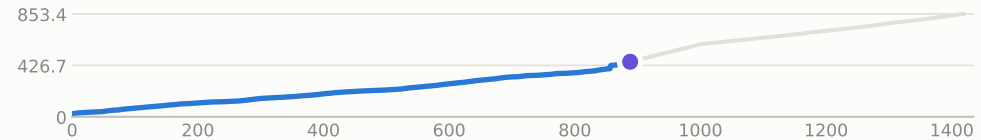
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

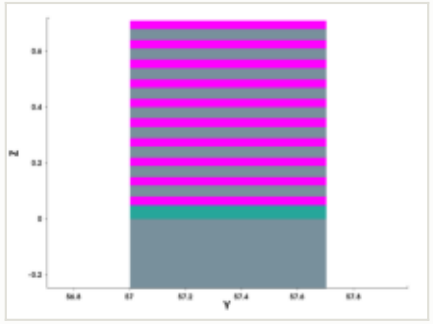
carbon so far, of the whole flow (kgCO2e)



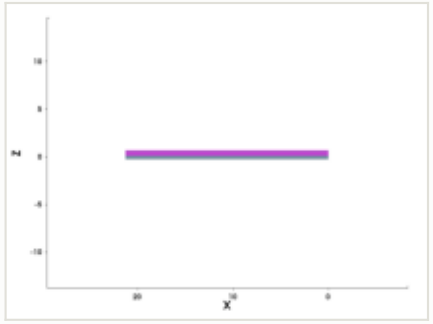
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



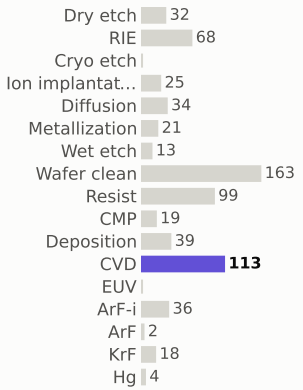
y cut at 57.351 µm



mask for this step

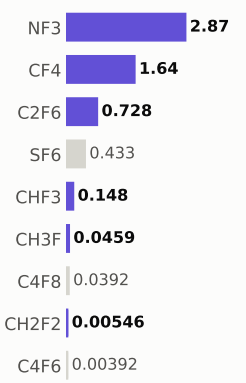
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



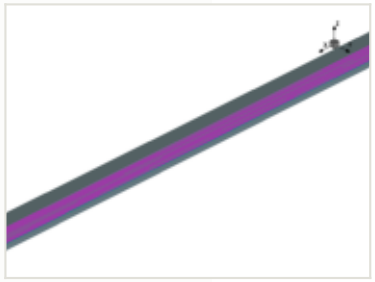
Step #9470: CVD

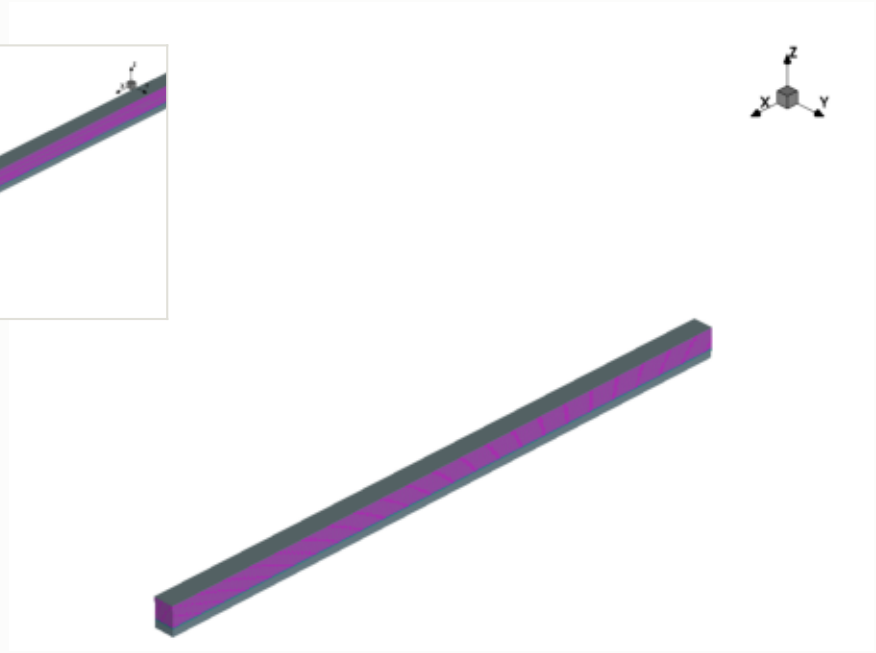
depo_isotropic · CVD

1.31	458.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



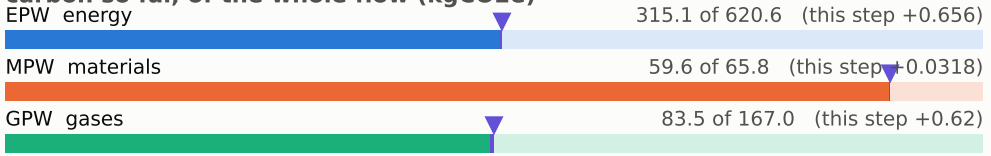


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

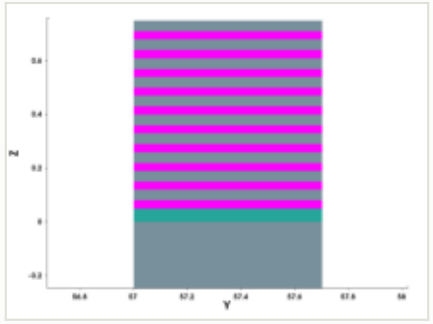
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

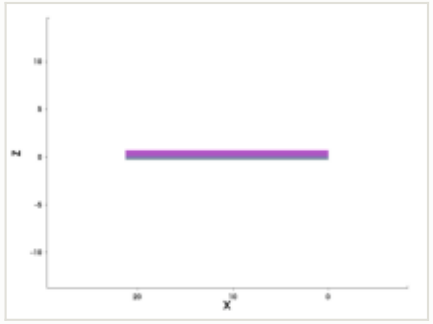


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



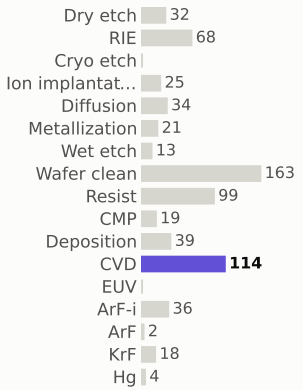
y cut at 57.351 µm



mask for this step

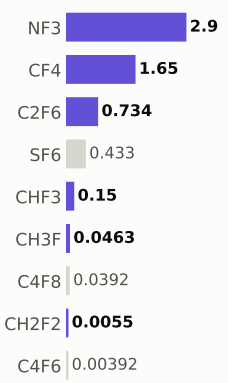
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



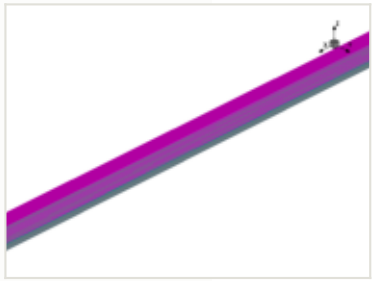
Step #9480: CVD

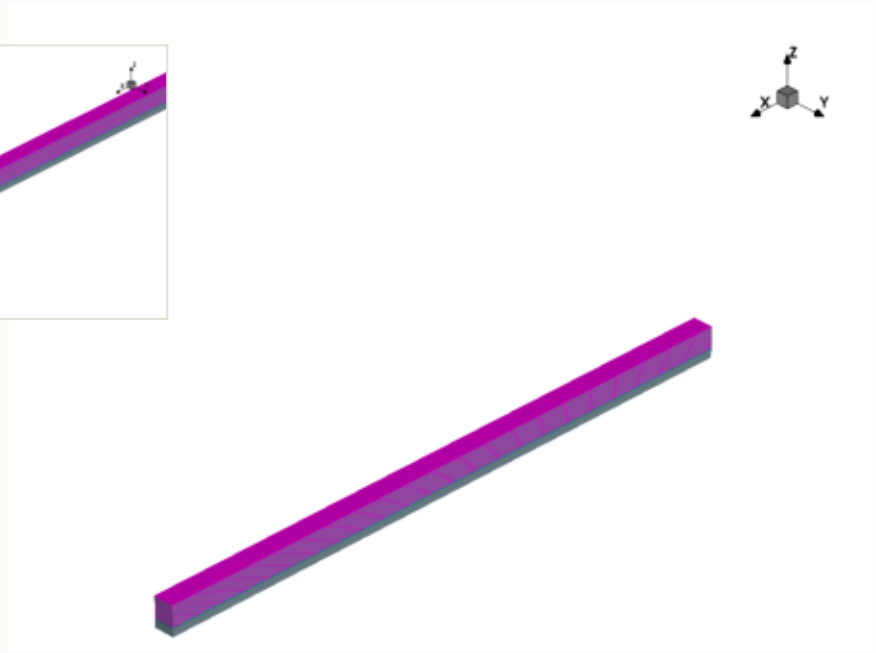
depo_isotropic · CVD

1.31	459.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



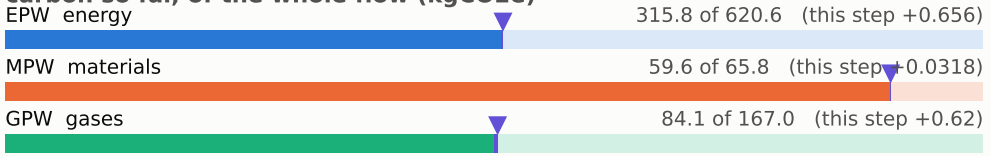


this step

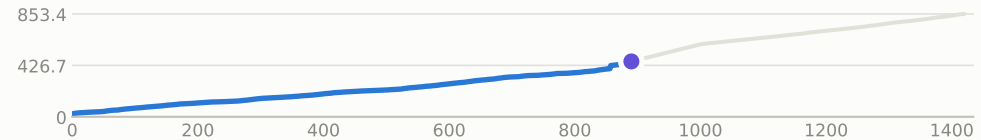
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

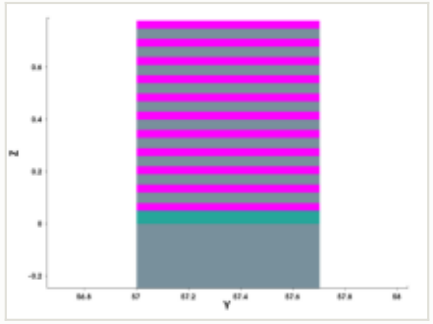
carbon so far, of the whole flow (kgCO2e)



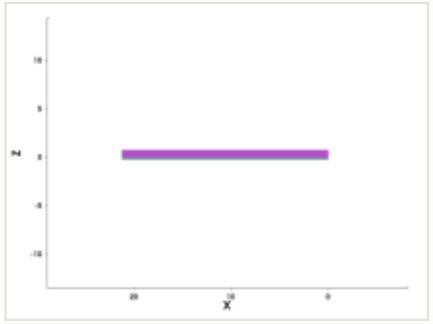
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



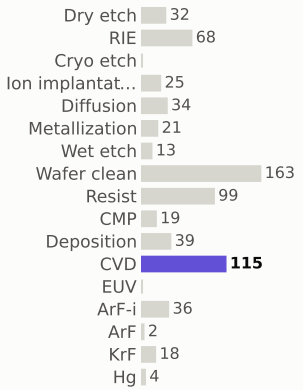
y cut at 57.351 µm



mask for this step

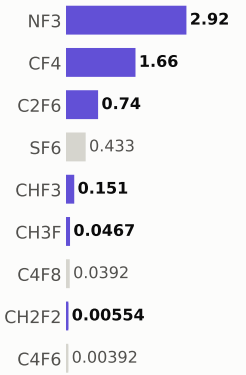
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



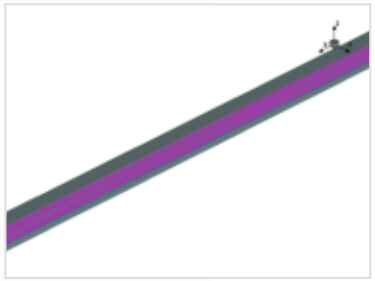
Step #9490: CVD

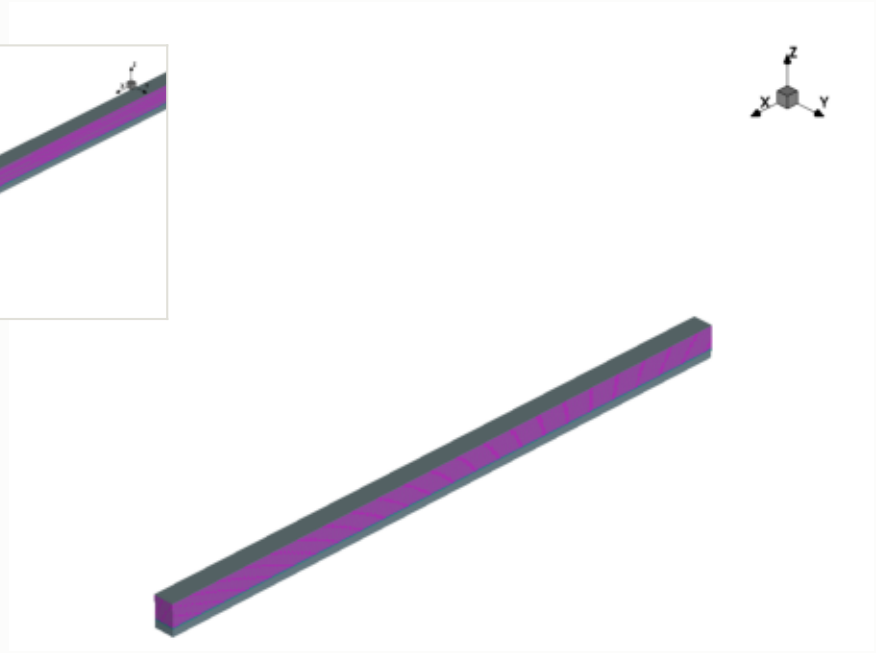
depo_isotropic · CVD

1.31	460.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



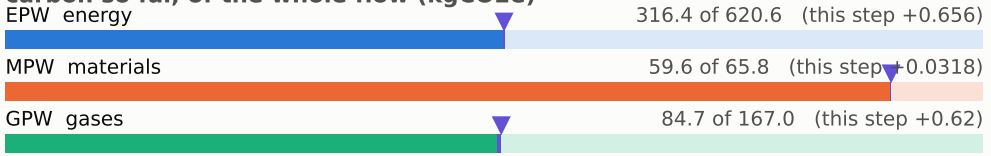


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

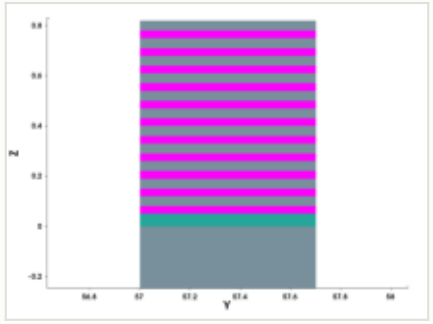
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

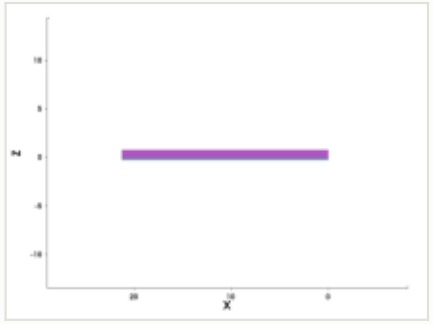


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



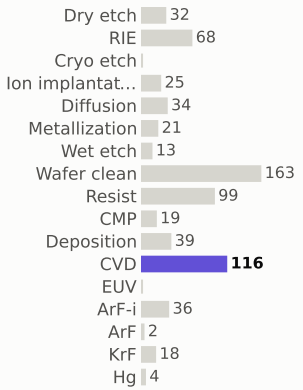
y cut at 57.351 µm



mask for this step

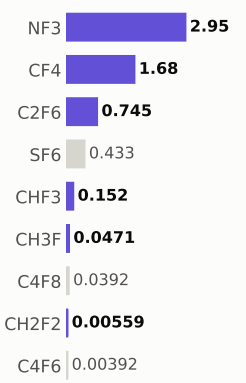
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



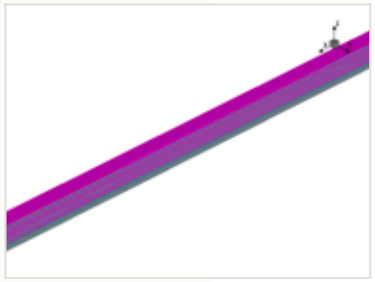
Step #9500: CVD

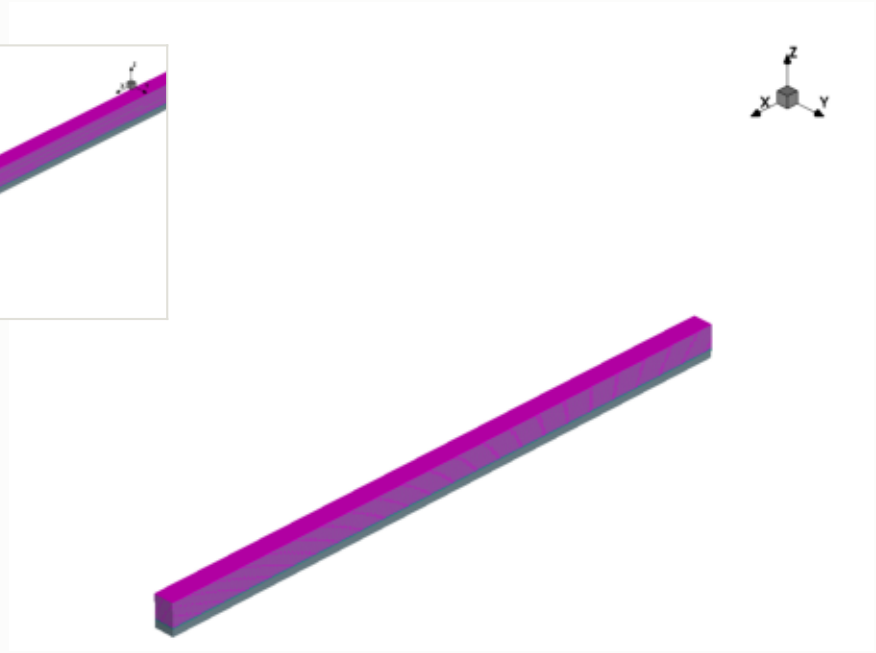
depo_isotropic · CVD

1.31	462.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



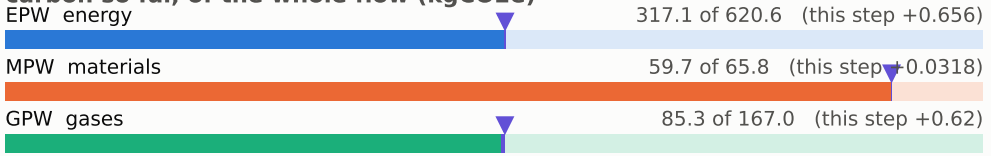


this step

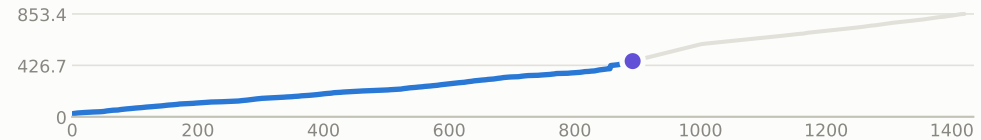
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

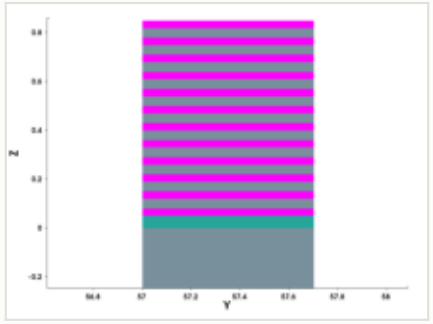
carbon so far, of the whole flow (kgCO2e)



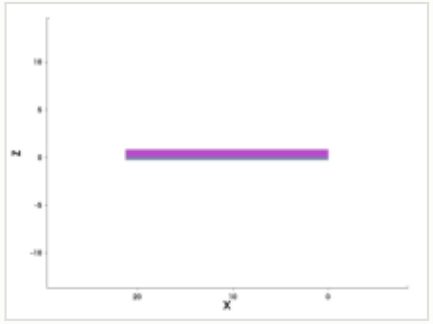
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



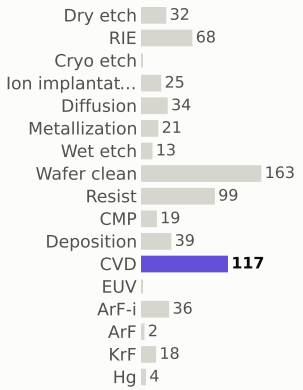
y cut at 57.351 µm



mask for this step

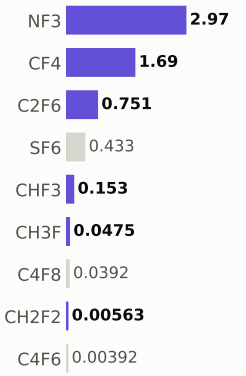
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



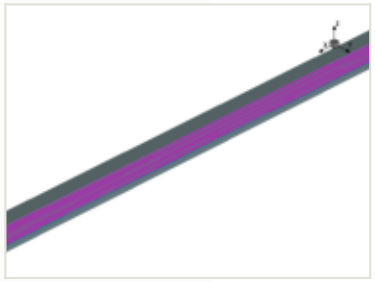
Step #9510: CVD

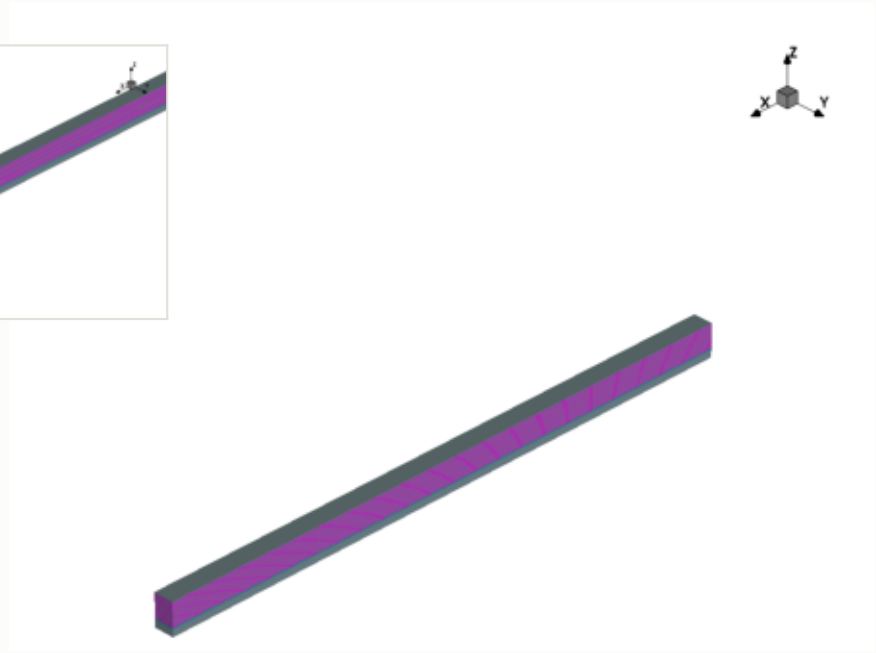
depo_isotropic · CVD

1.31	463.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



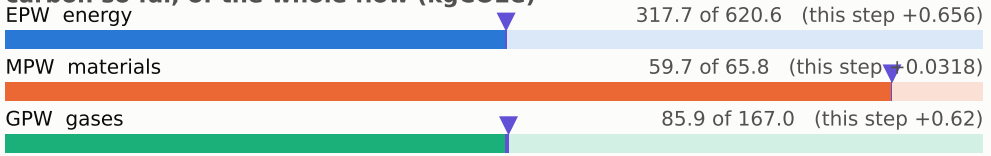


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

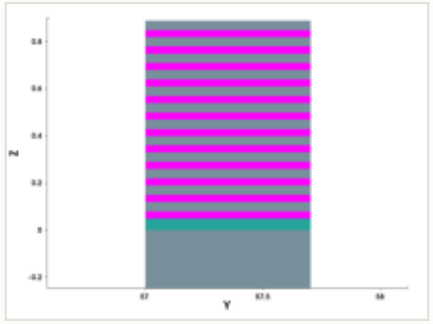
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

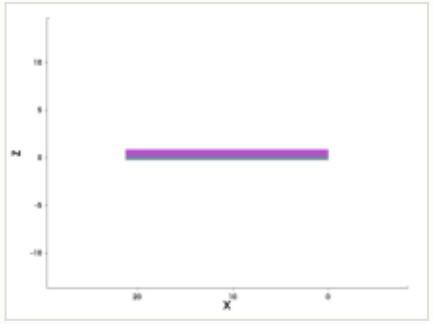


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



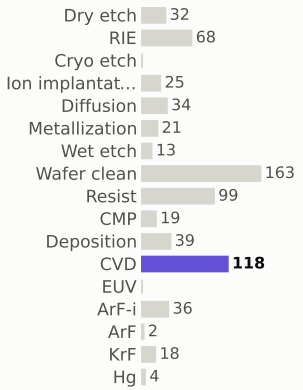
y cut at 57.351 µm



mask for this step

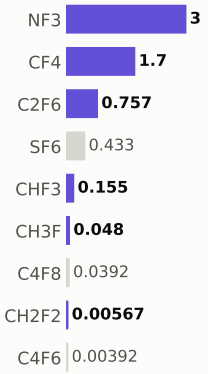
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



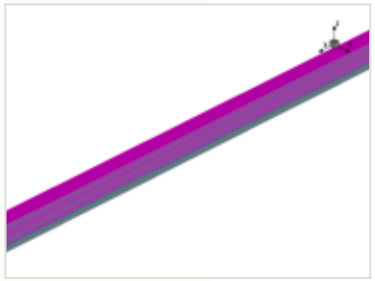
Step #9520: CVD

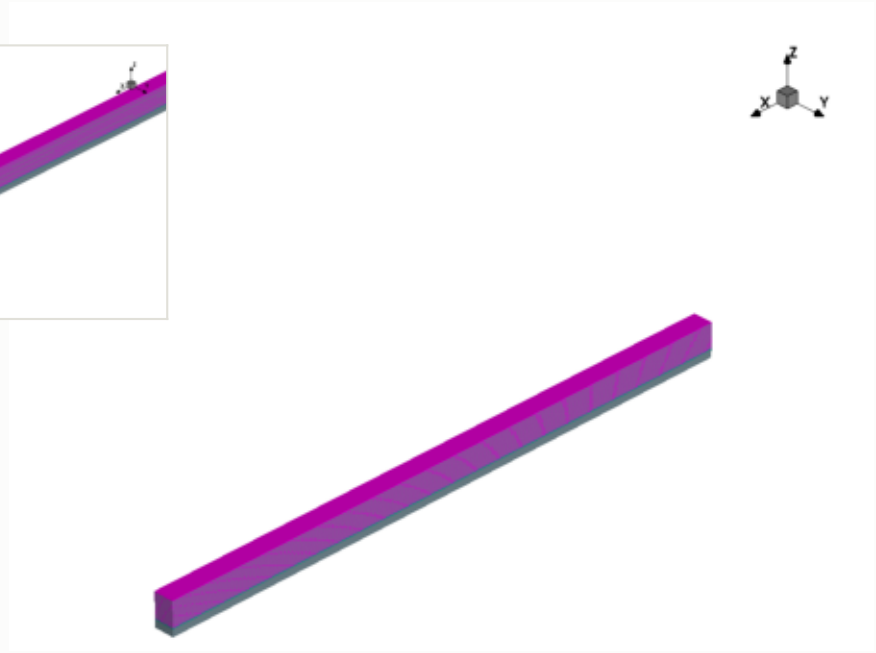
depo_isotropic · CVD

1.31	464.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



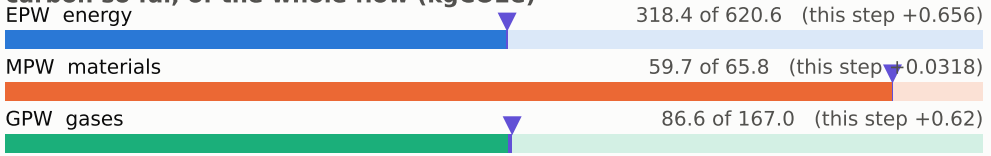


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

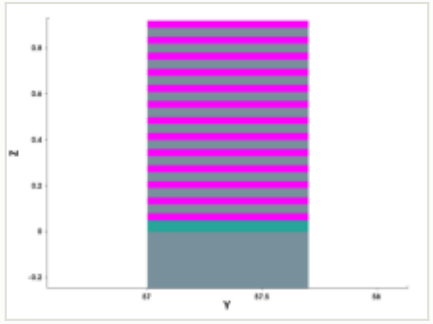
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

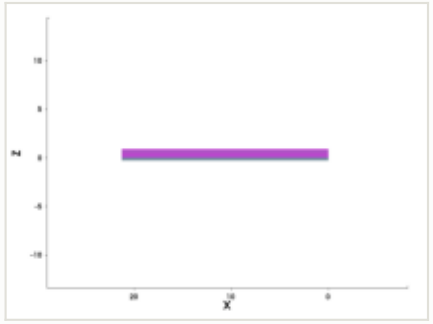


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



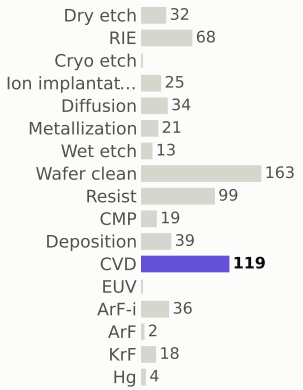
y cut at 57.351 µm



mask for this step

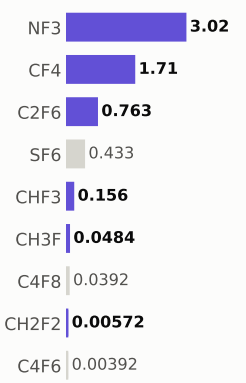
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



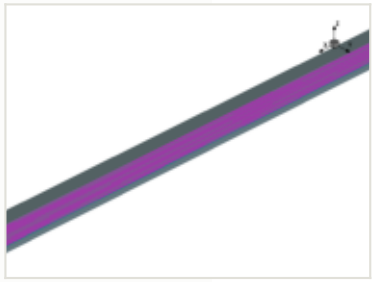
Step #9530: CVD

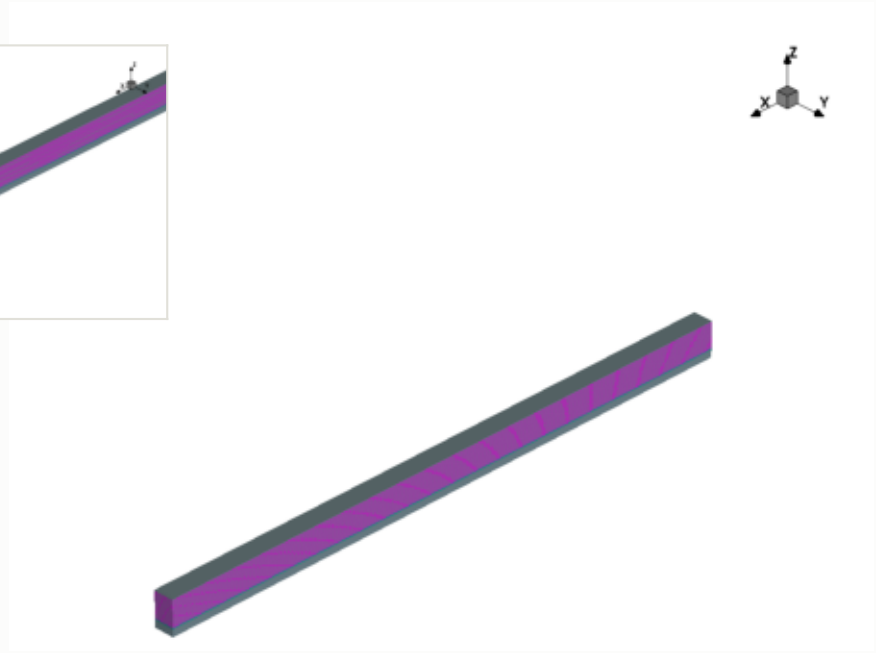
depo_isotropic · CVD

1.31	466.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



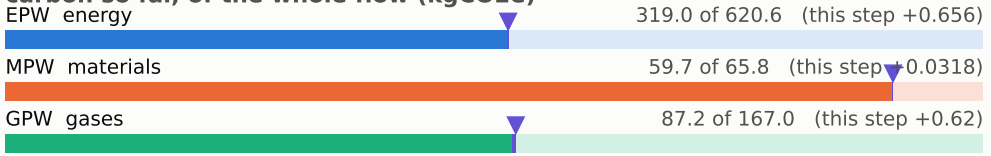


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

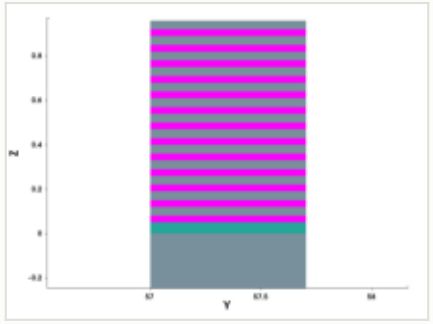
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

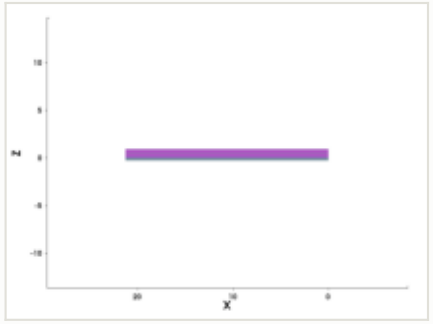


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



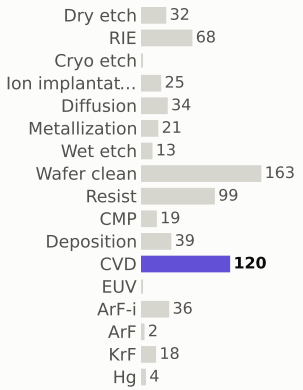
y cut at 57.351 µm



mask for this step

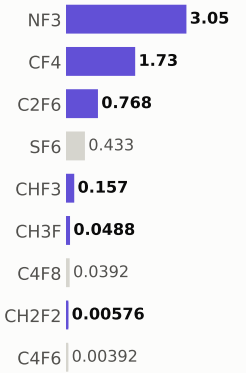
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



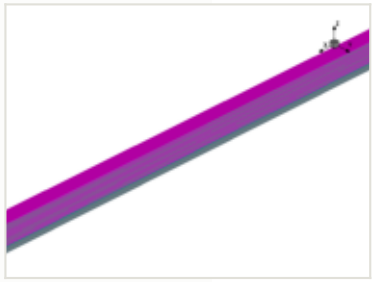
Step #9540: CVD

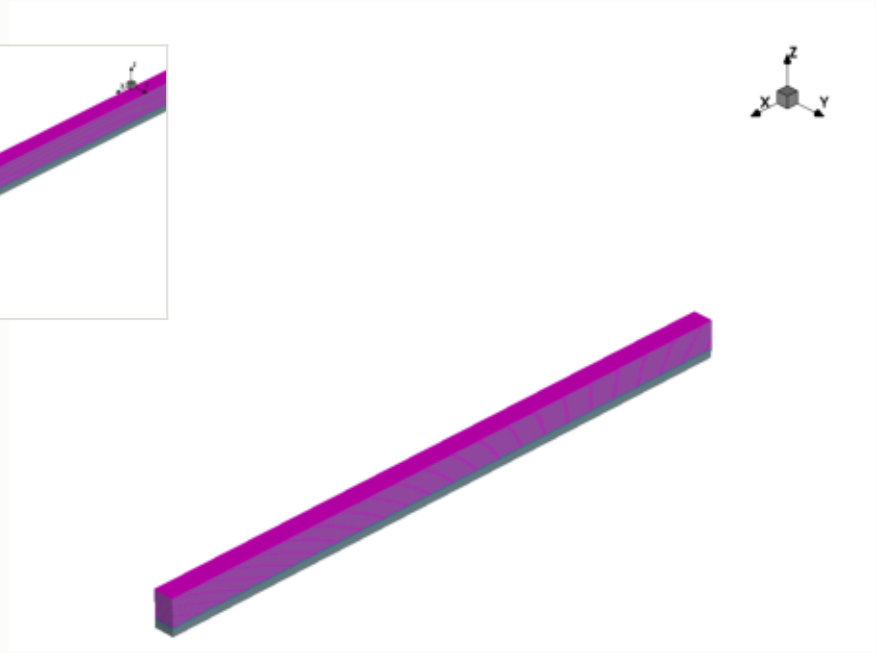
depo_isotropic · CVD

1.31	467.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



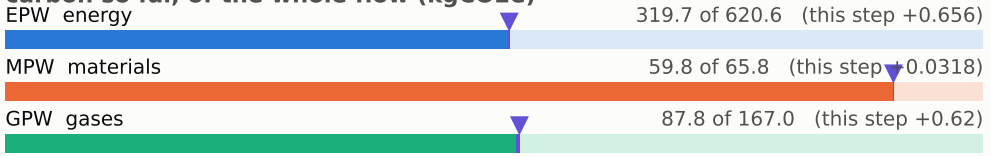


this step

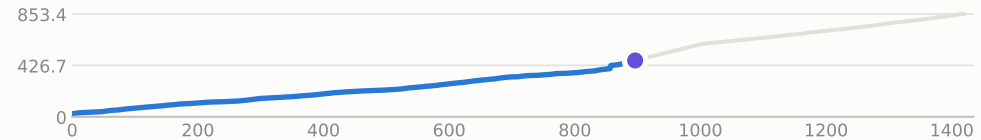
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

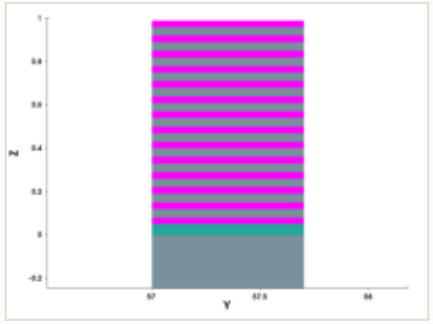
carbon so far, of the whole flow (kgCO2e)



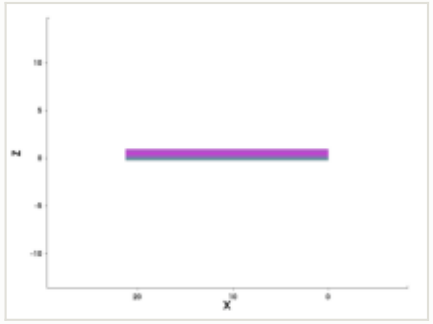
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



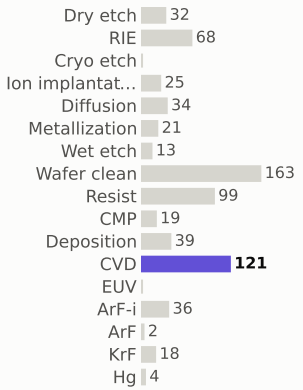
y cut at 57.351 µm



mask for this step

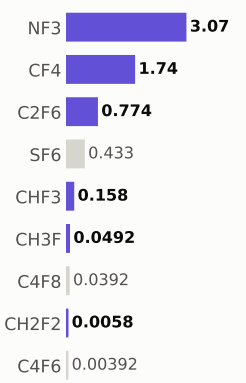
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



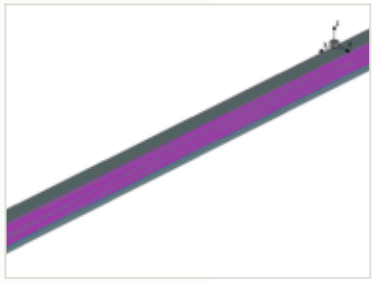
Step #9550: CVD

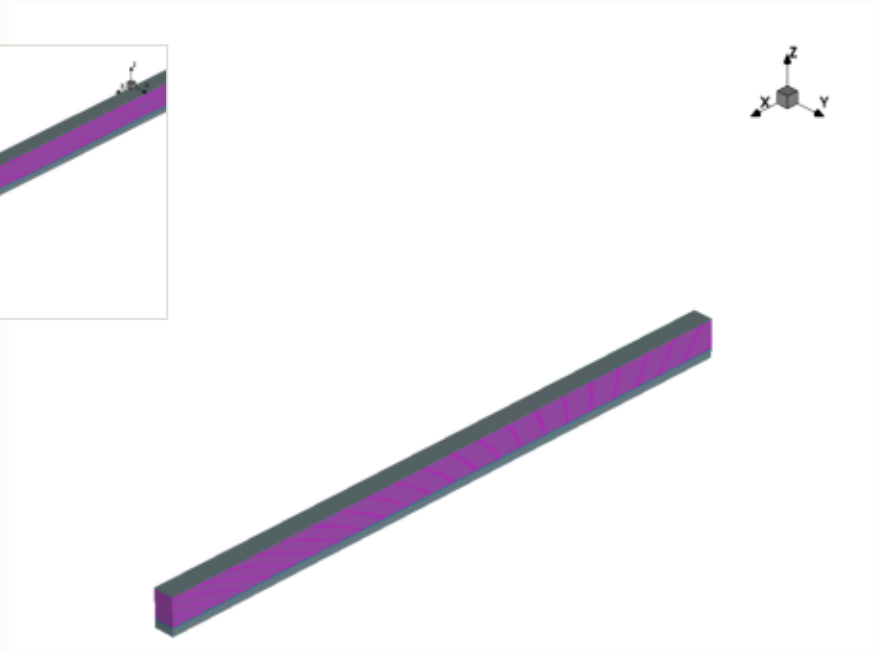
depo_isotropic · CVD

1.31	468.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



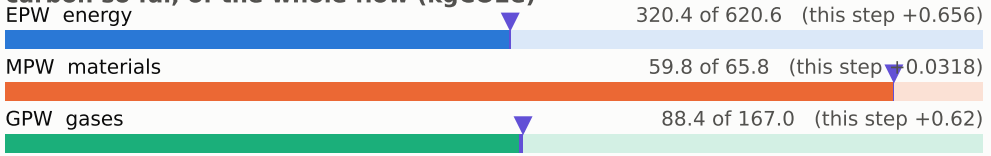


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

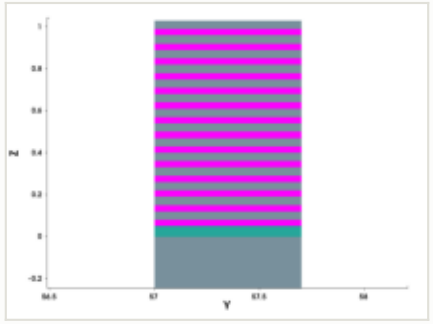
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

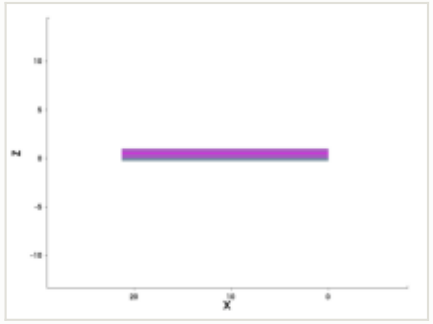


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



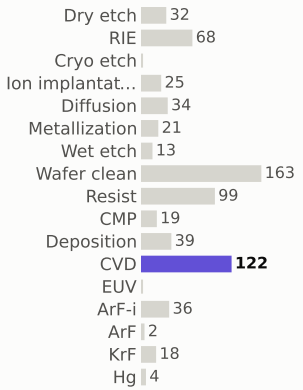
y cut at 57.351 µm



mask for this step

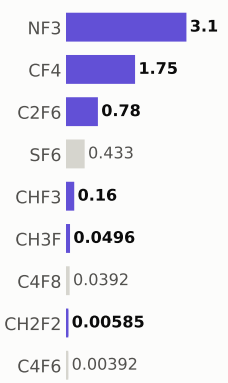
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



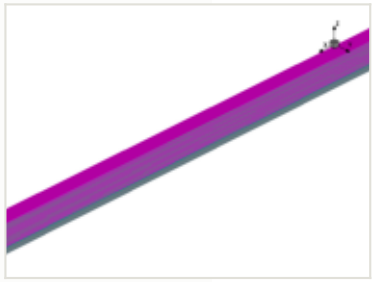
Step #9560: CVD

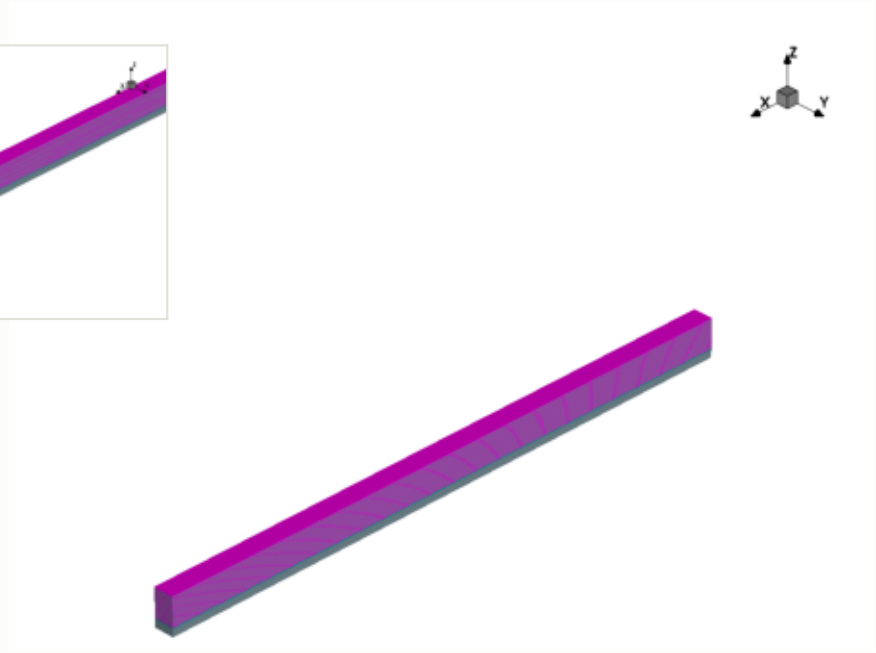
depo_isotropic · CVD

1.31	469.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



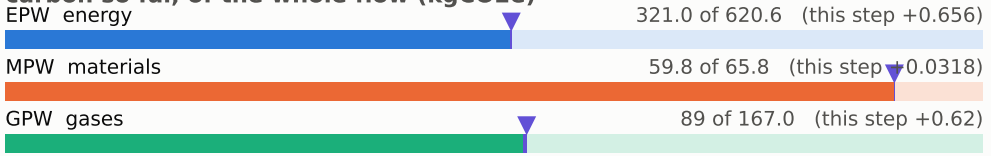


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

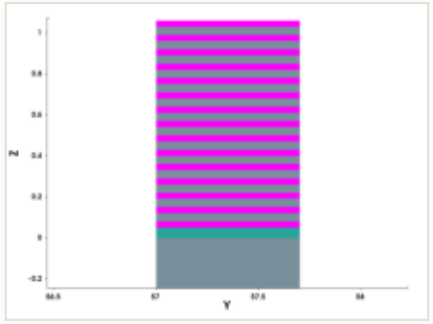
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

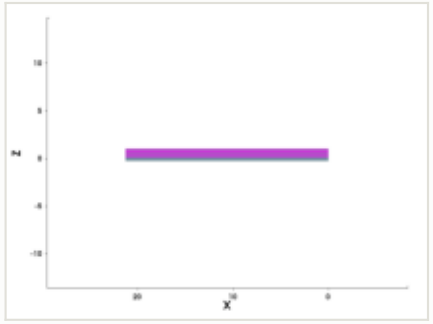


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



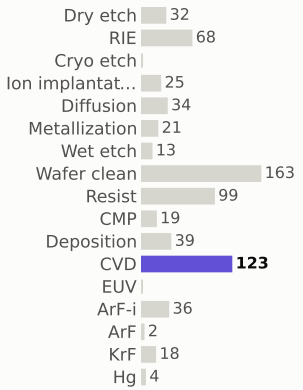
y cut at 57.351 µm



mask for this step

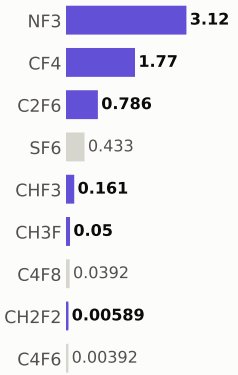
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



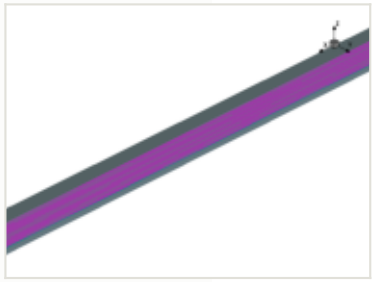
Step #9570: CVD

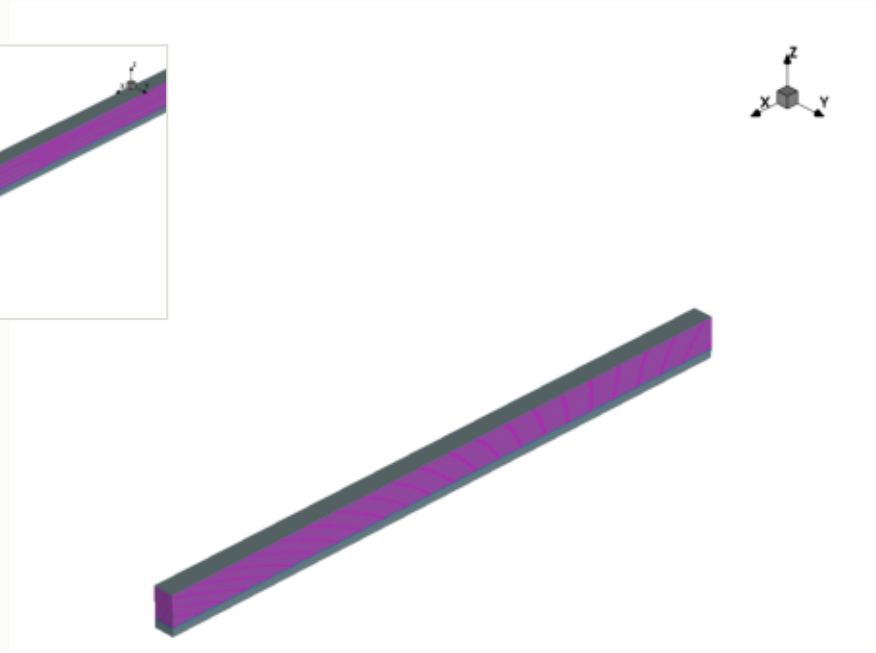
depo_isotropic · CVD

1.31	471.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



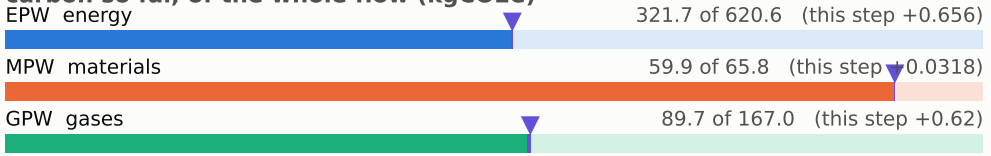


this step

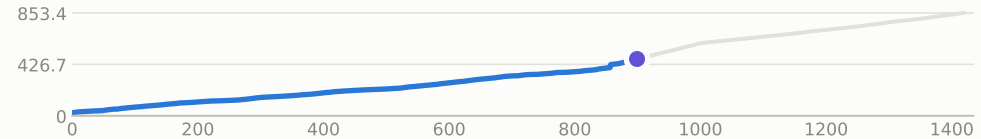
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

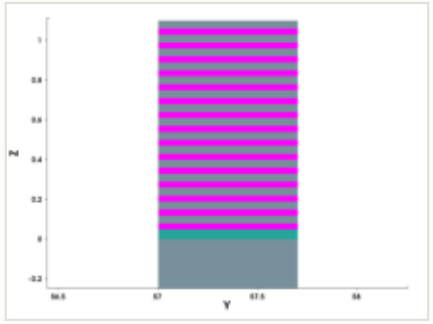
carbon so far, of the whole flow (kgCO2e)



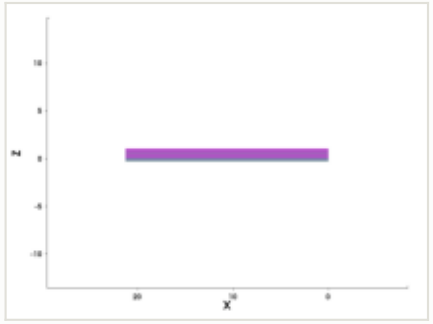
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



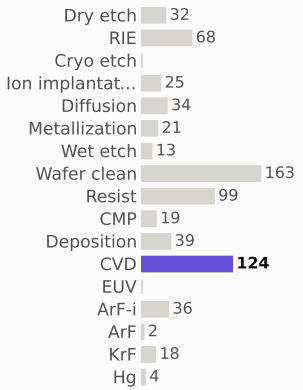
y cut at 57.351 µm



mask for this step

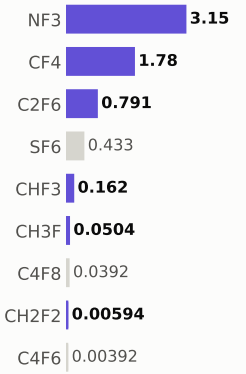
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



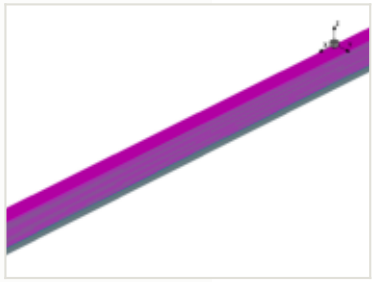
Step #9580: CVD

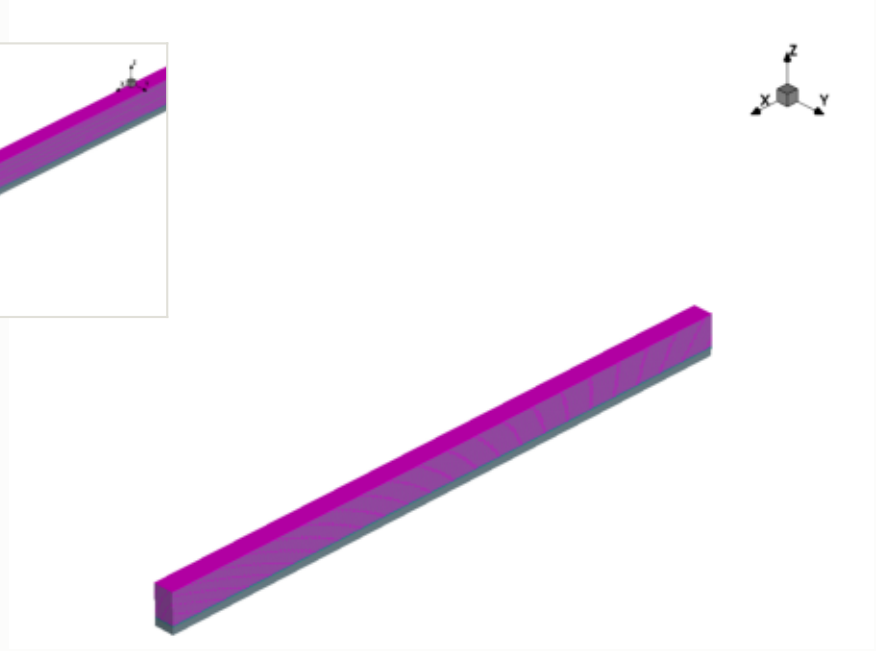
depo_isotropic · CVD

1.31	472.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



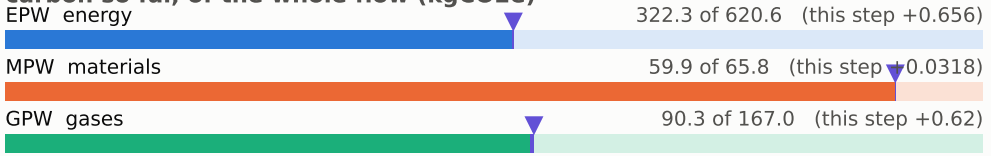


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

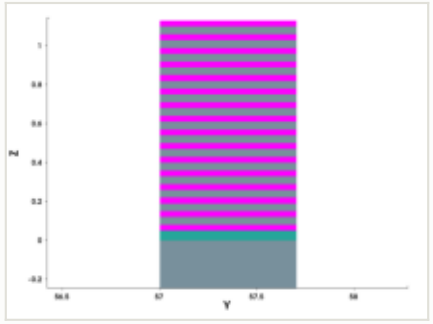
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

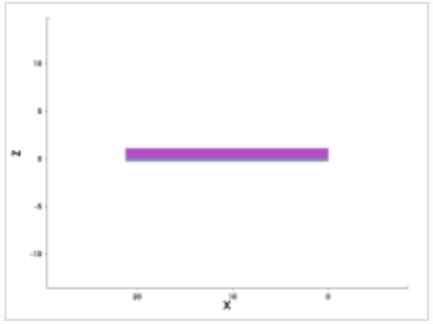


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



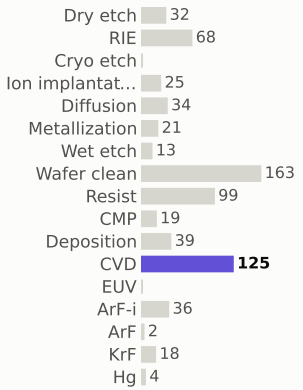
y cut at 57.351 µm



mask for this step

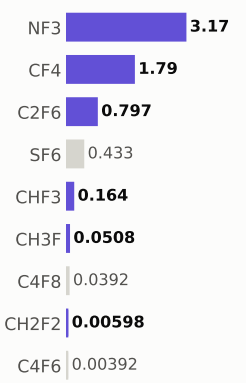
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



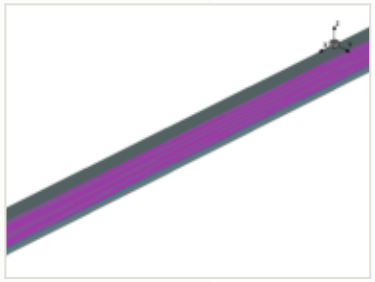
Step #9590: CVD

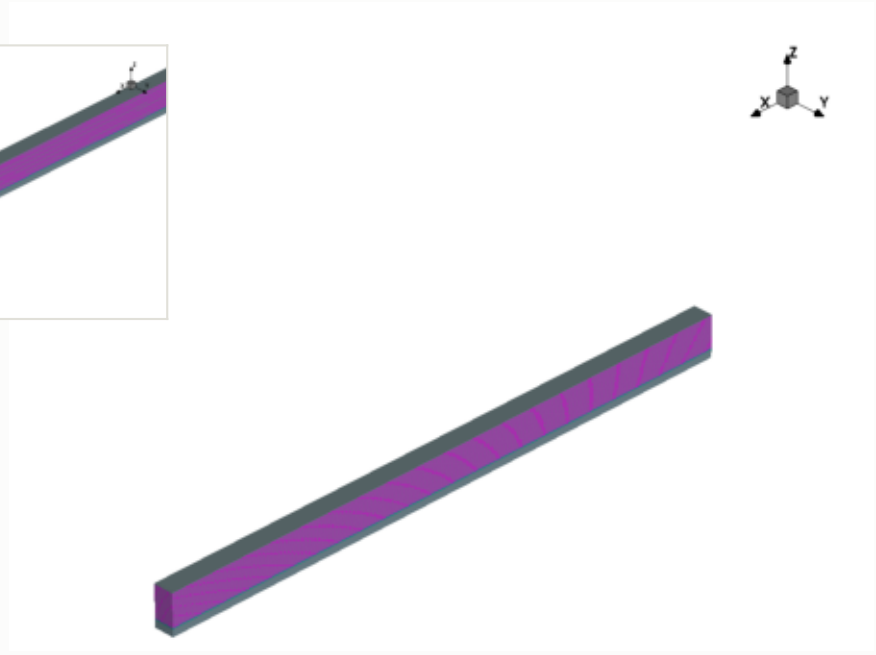
depo_isotropic · CVD

1.31	473.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



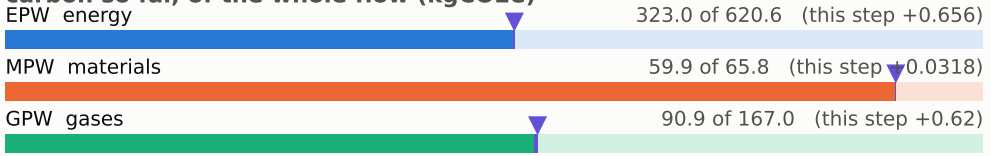


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

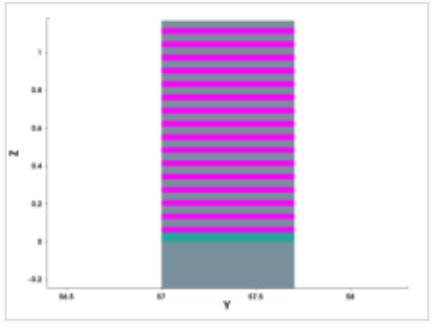
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

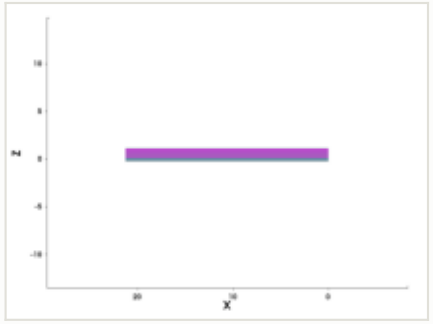


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



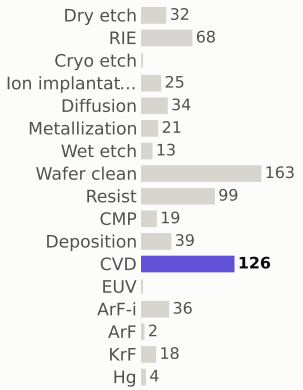
y cut at 57.351 µm



mask for this step

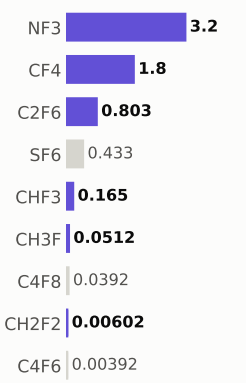
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



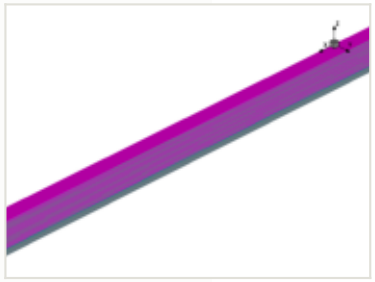
Step #9600: CVD

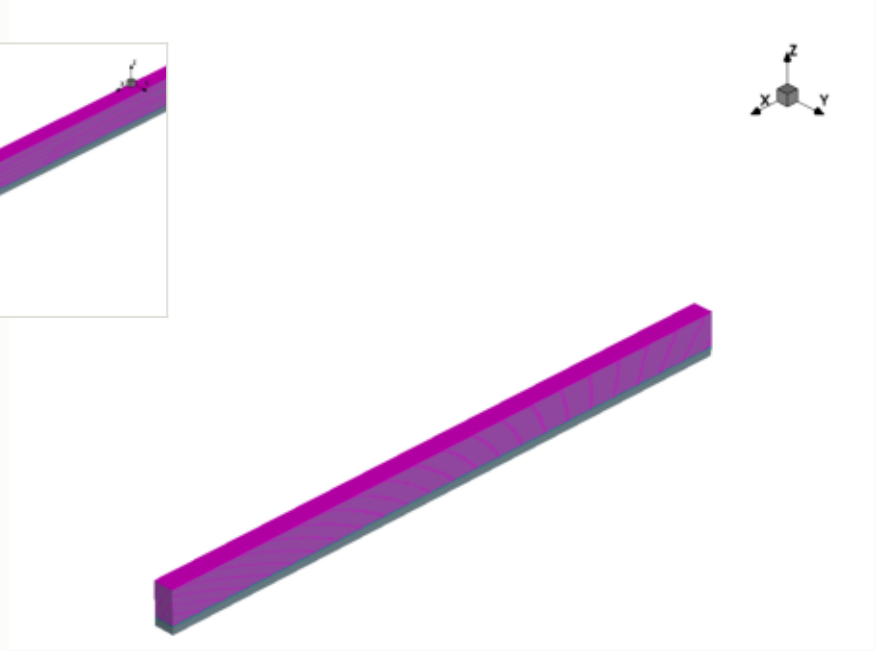
depo_isotropic · CVD

1.31	475.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



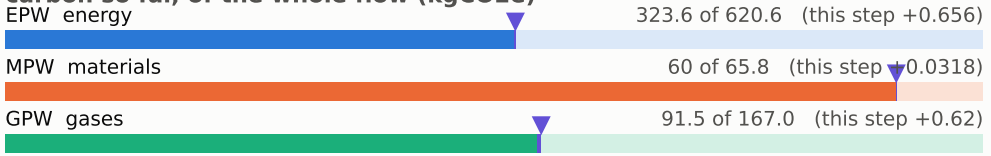


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

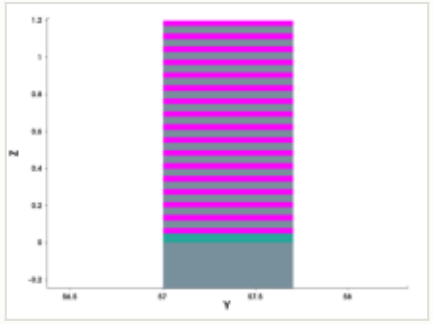
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

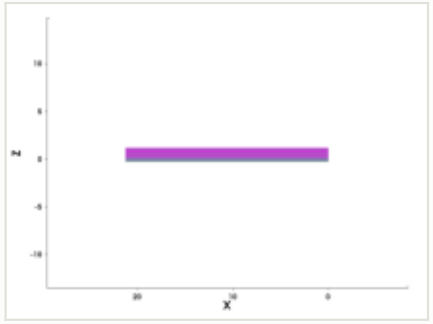


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



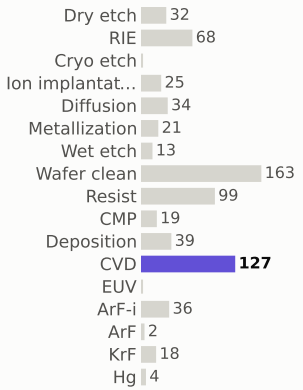
y cut at 57.351 µm



mask for this step

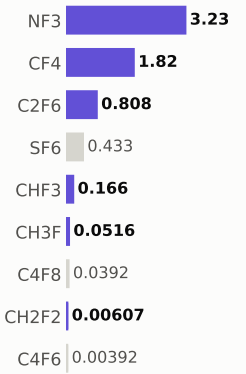
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



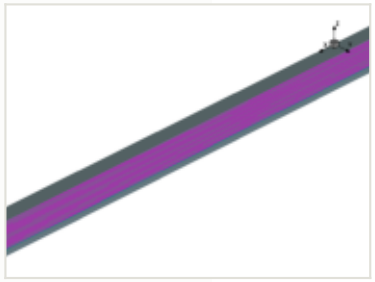
Step #9610: CVD

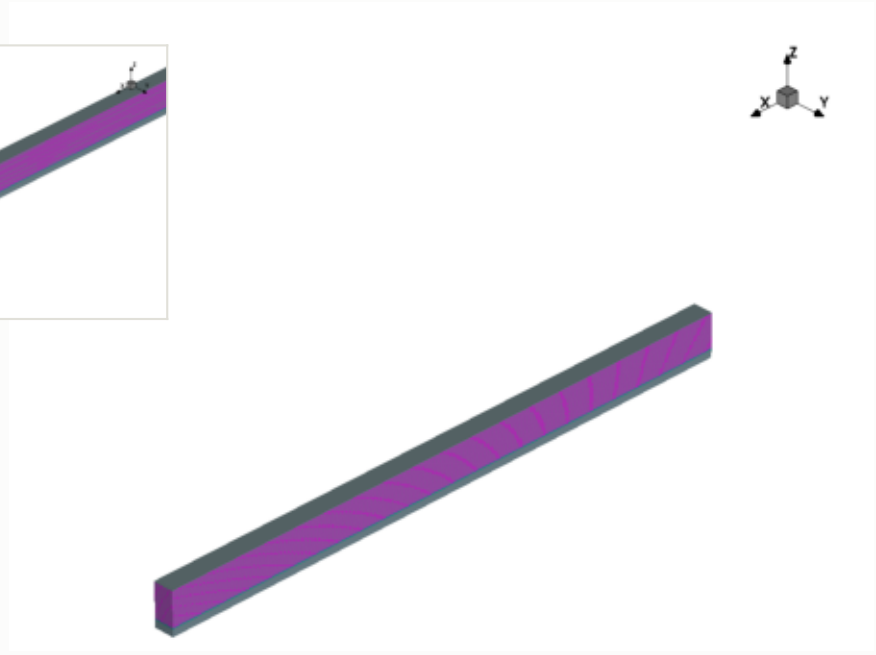
depo_isotropic · CVD

1.31	476.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



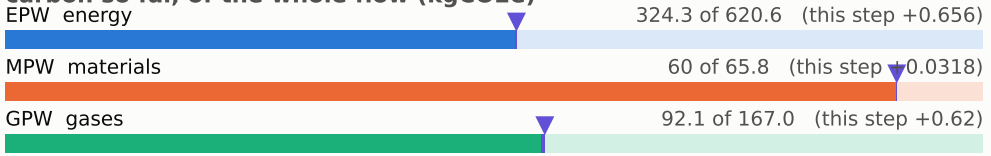


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

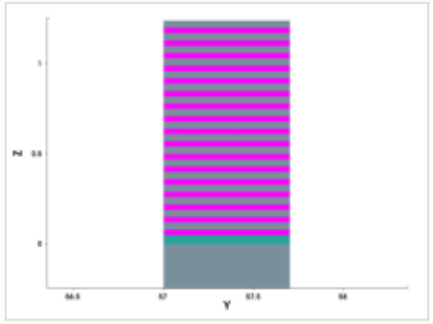
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

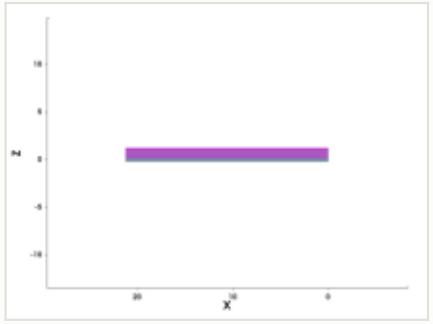


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



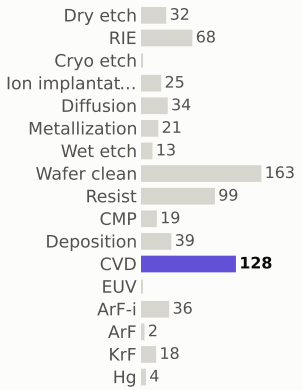
y cut at 57.351 µm



mask for this step

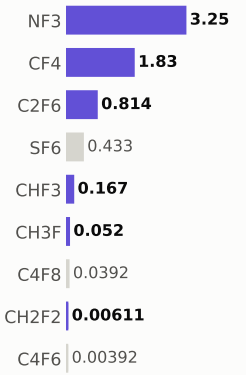
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



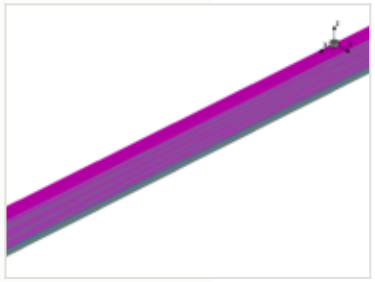
Step #9620: CVD

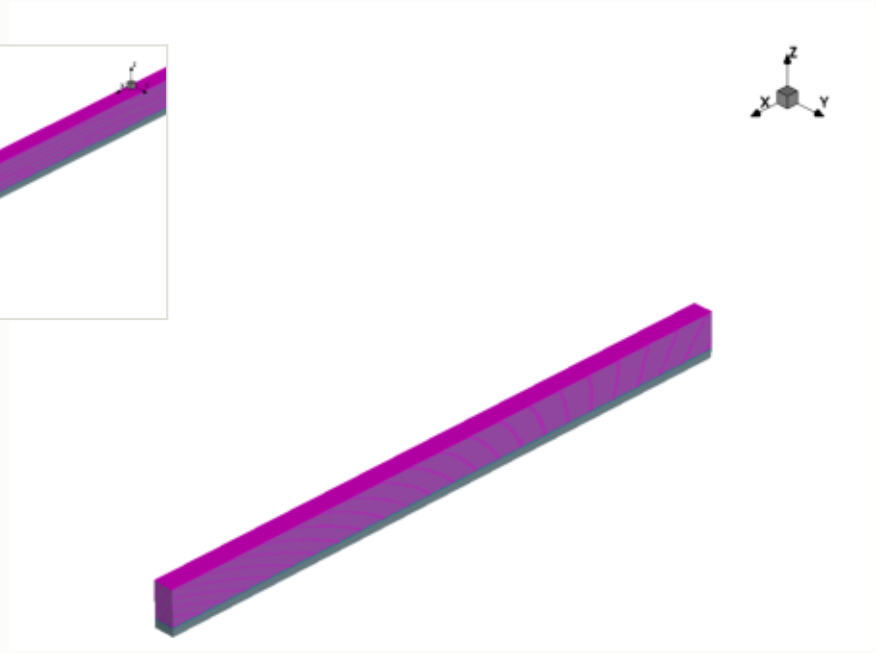
depo_isotropic · CVD

1.31	477.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



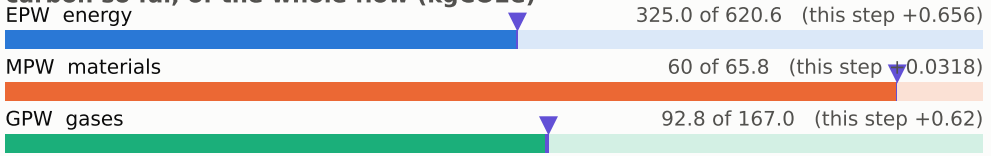


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

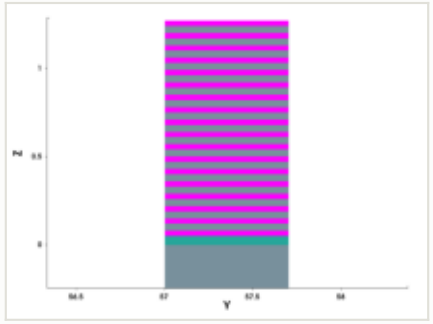
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

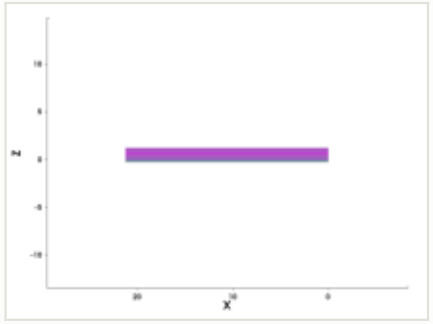


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



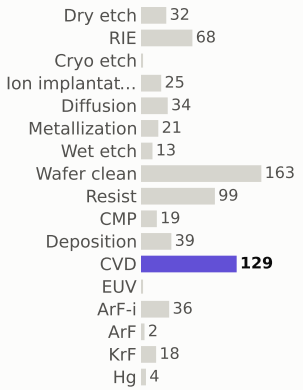
y cut at 57.351 µm



mask for this step

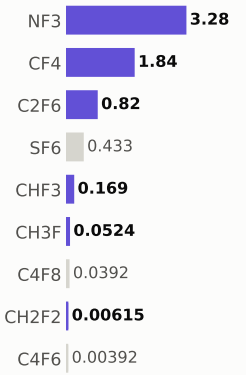
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



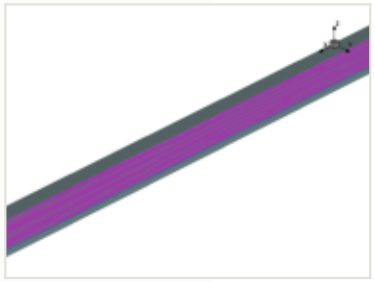
Step #9630: CVD

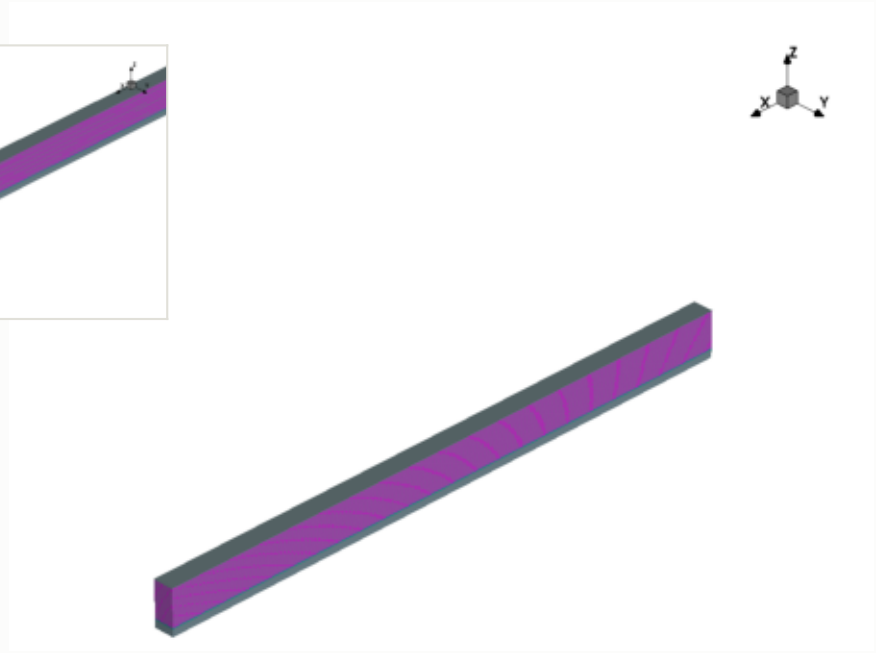
depo_isotropic · CVD

1.31	479.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



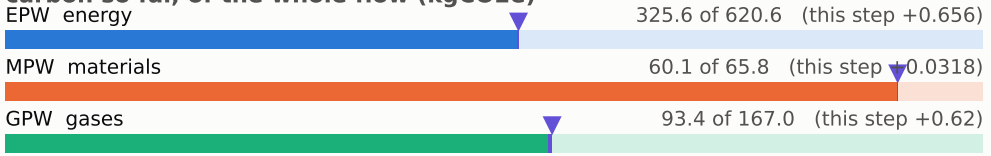


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

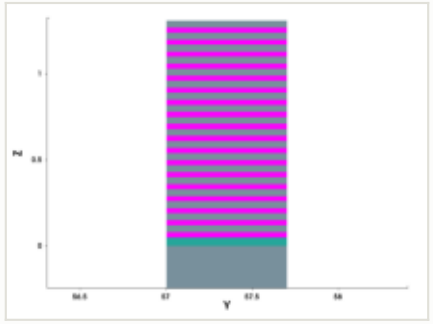
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

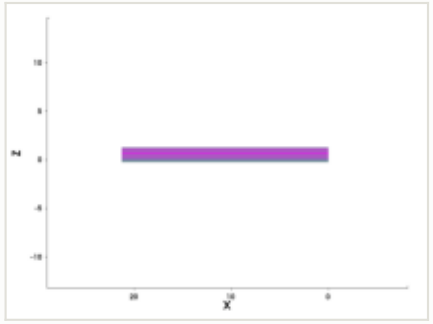


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



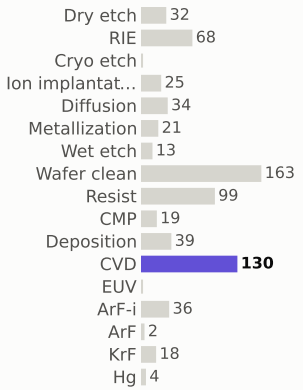
y cut at 57.351 µm



mask for this step

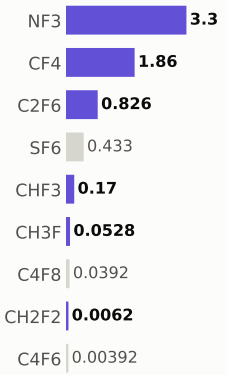
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



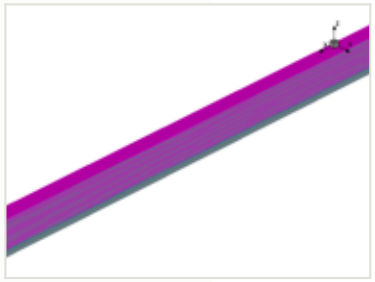
Step #9640: CVD

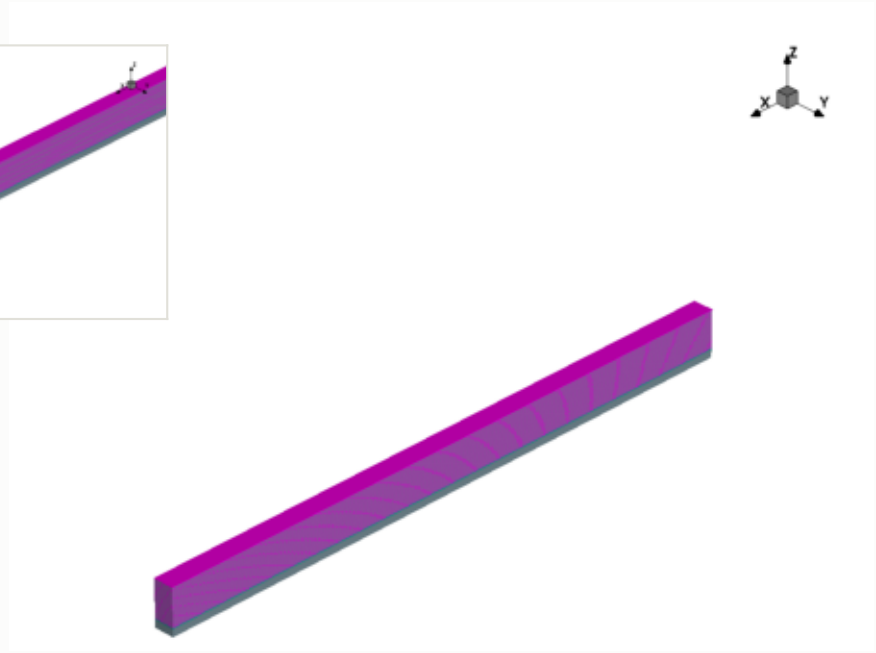
depo_isotropic · CVD

1.31	480.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



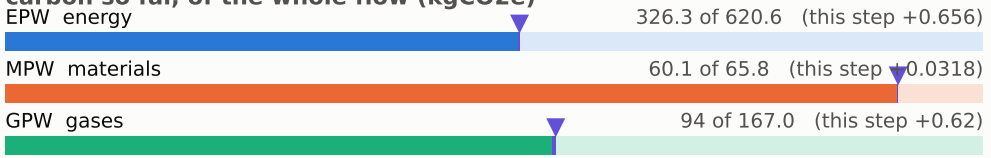


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

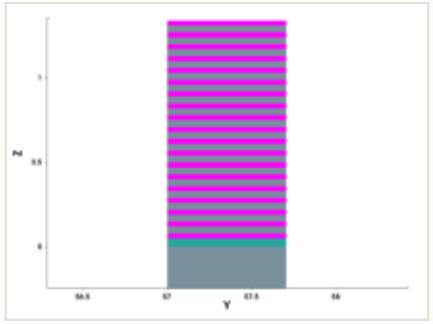
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

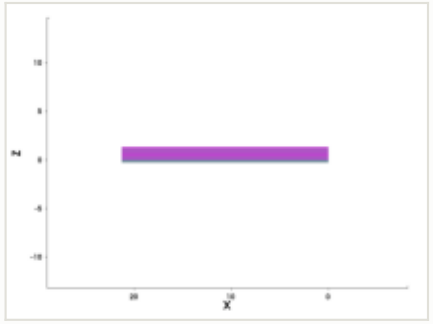


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



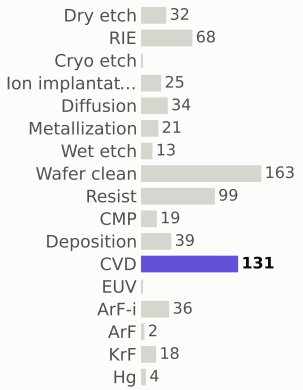
y cut at 57.351 µm



mask for this step

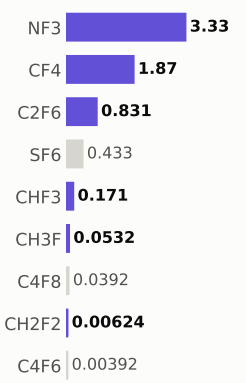
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



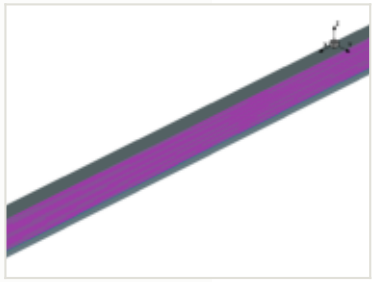
Step #9650: CVD

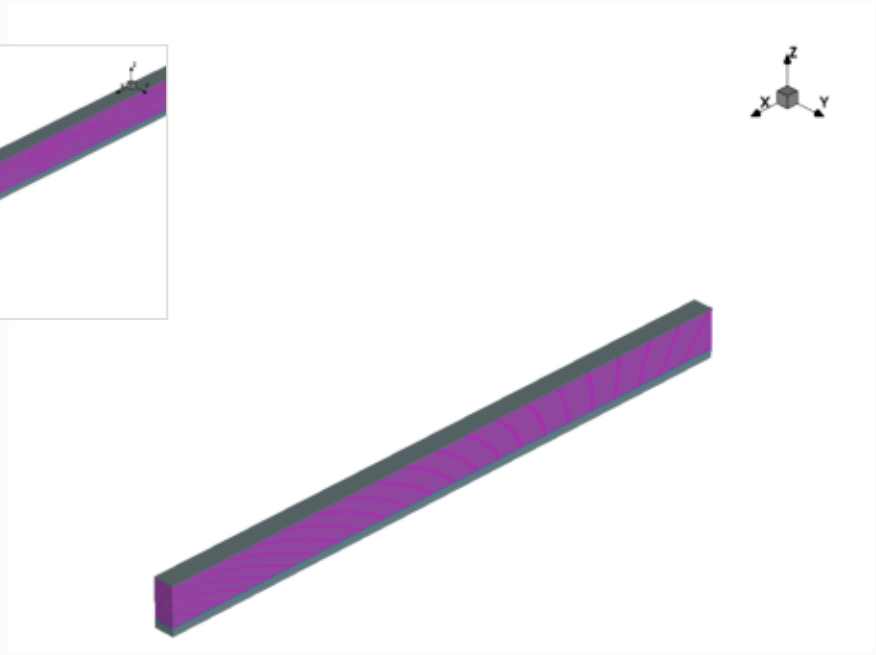
depo_isotropic · CVD

1.31	481.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



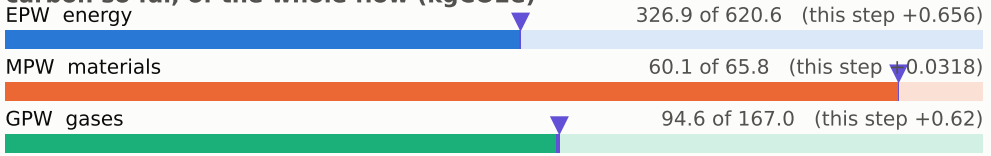


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

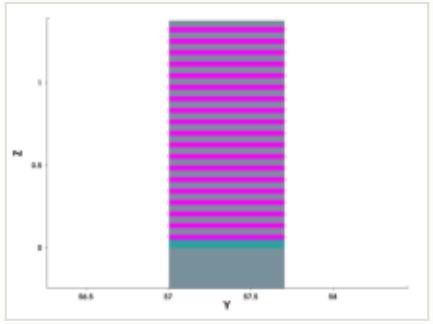
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

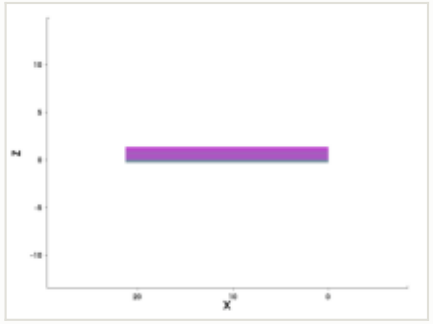


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



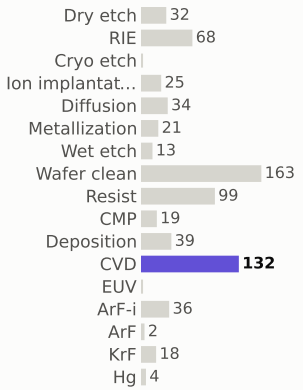
y cut at 57.351 µm



mask for this step

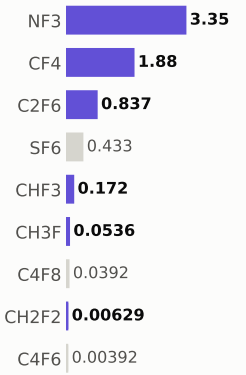
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



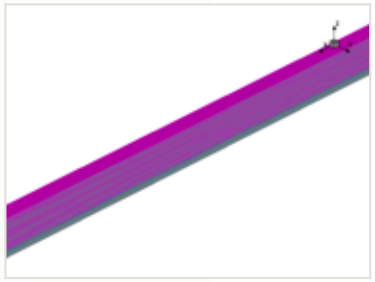
Step #9660: CVD

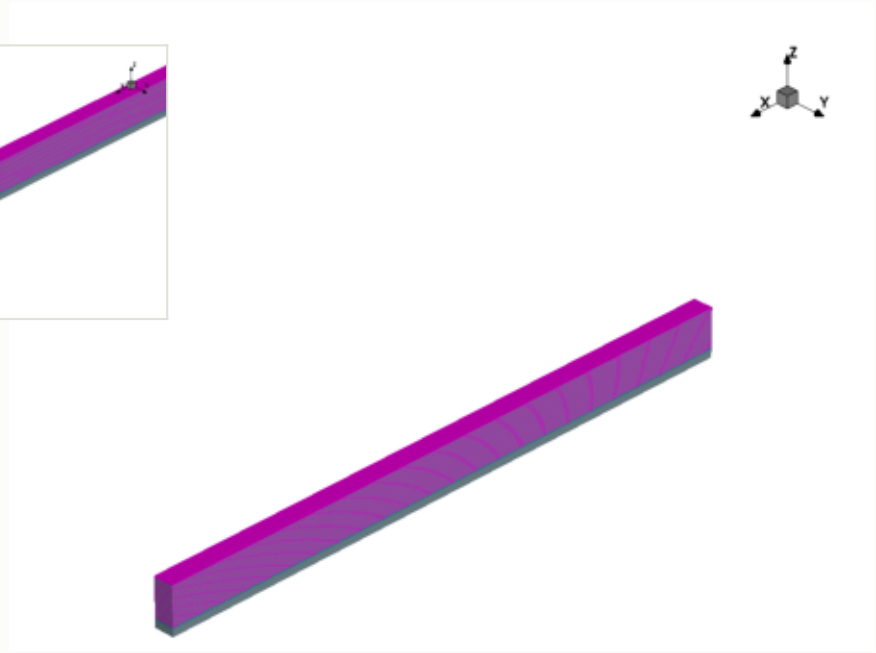
depo_isotropic · CVD

1.31	483.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



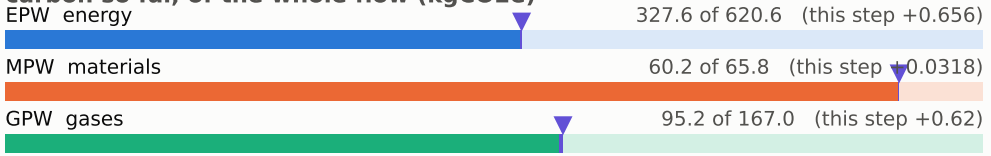


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

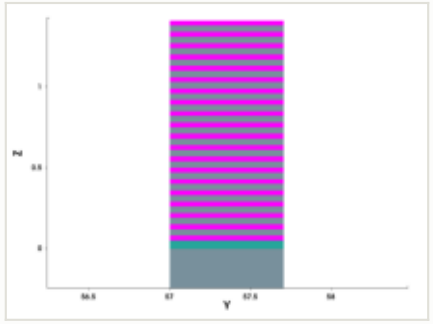
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

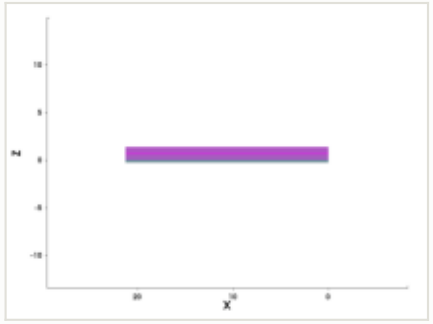


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



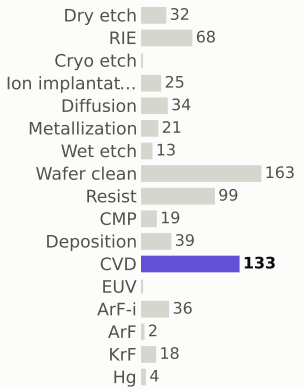
y cut at 57.351 µm



mask for this step

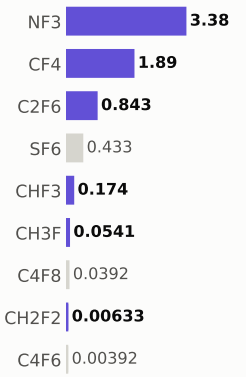
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



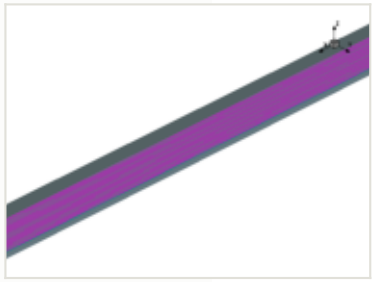
Step #9670: CVD

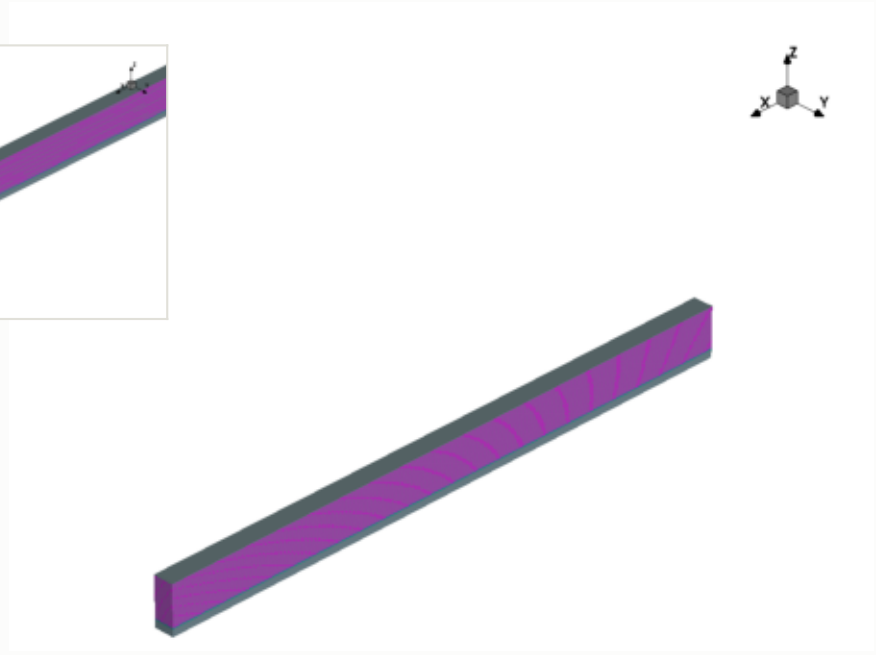
depo_isotropic · CVD

1.31	484.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



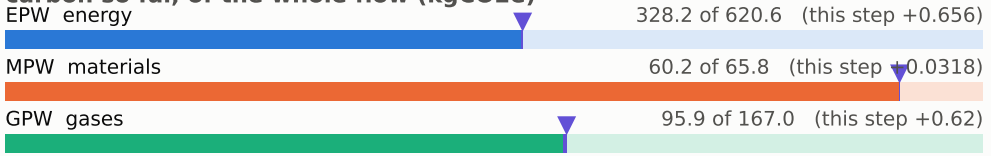


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

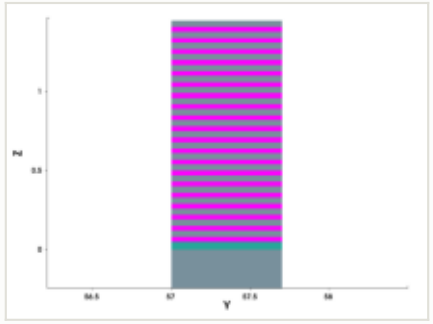
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

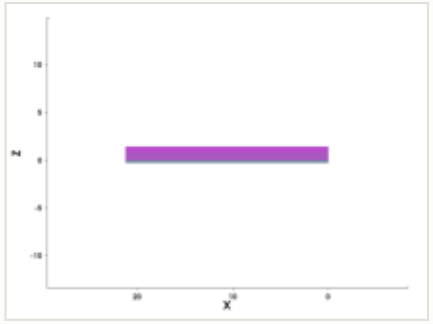


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



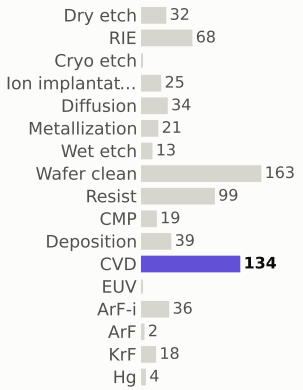
y cut at 57.351 µm



mask for this step

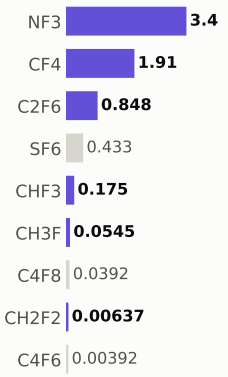
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



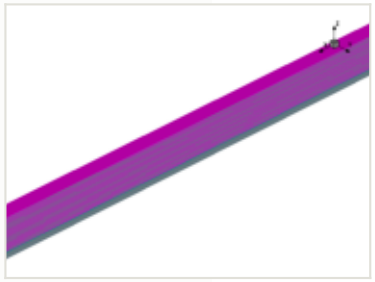
Step #9680: CVD

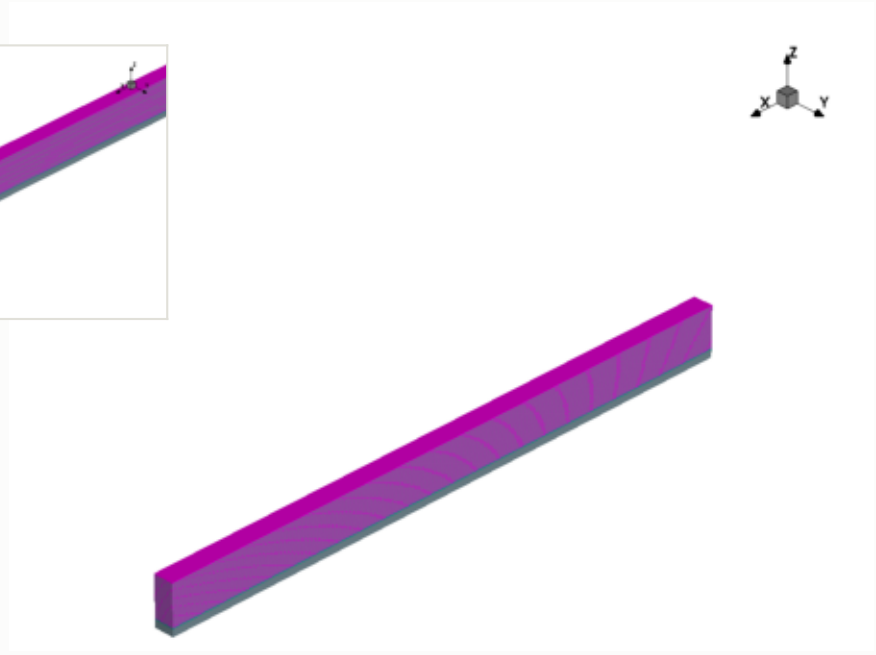
depo_isotropic · CVD

1.31	485.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



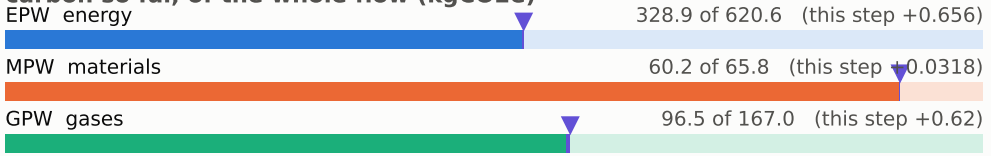


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

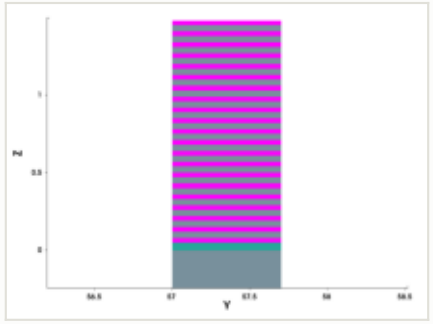
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

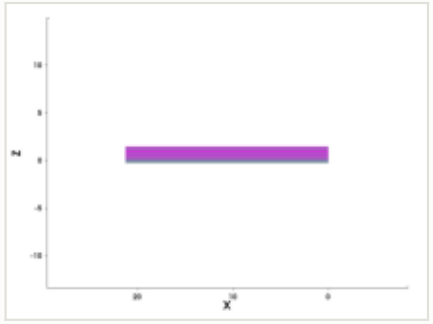


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



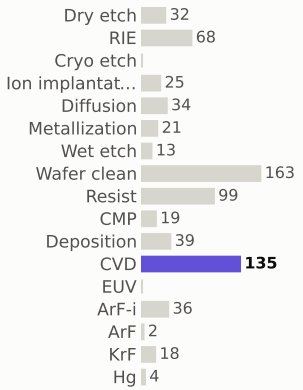
y cut at 57.351 µm



mask for this step

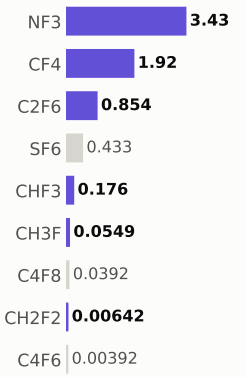
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



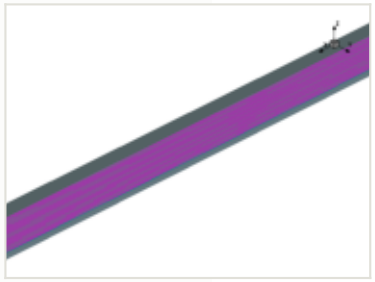
Step #9690: CVD

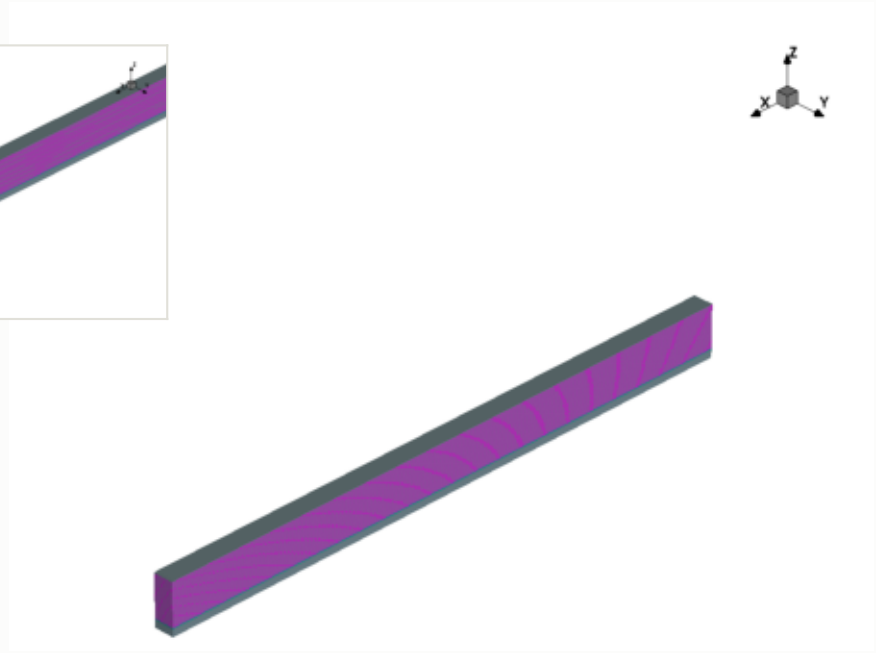
depo_isotropic · CVD

1.31	486.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

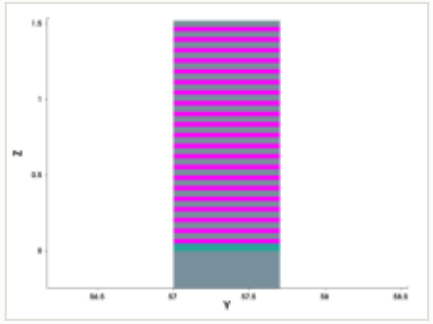
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

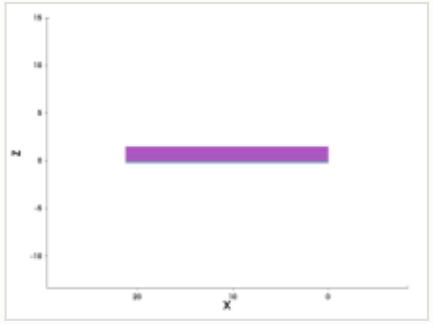
carbon so far, of the whole flow (kgCO2e)

embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



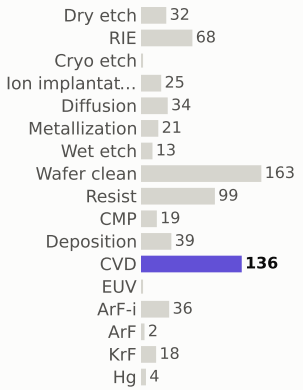
y cut at 57.351 µm



mask for this step

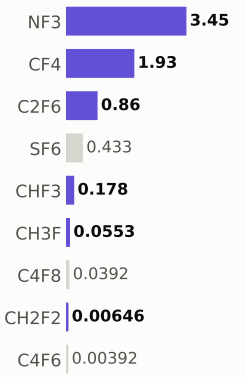
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



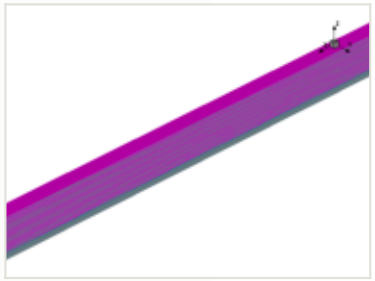
Step #9700: CVD

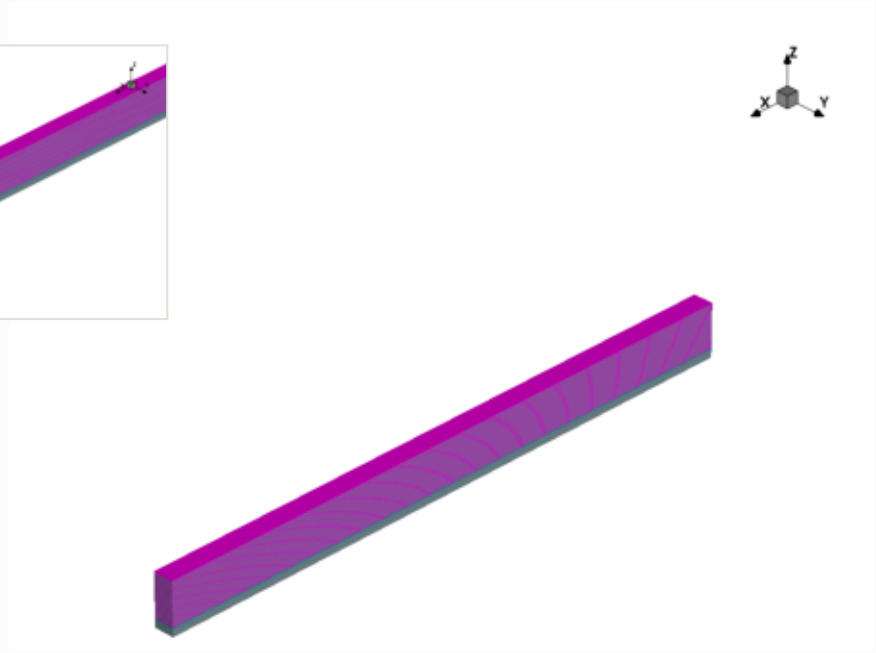
depo_isotropic · CVD

1.31	488.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



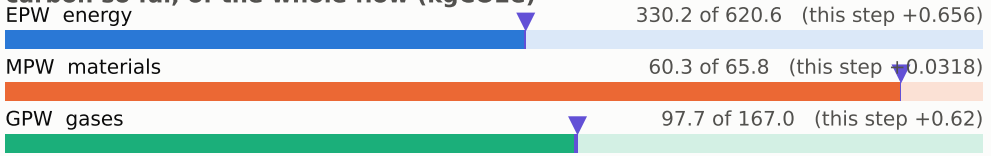


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

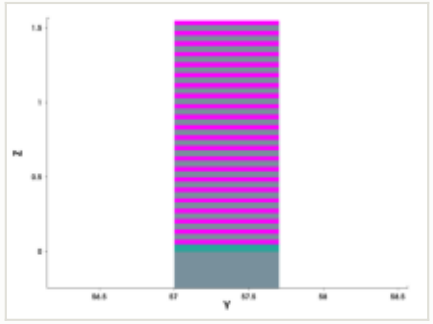
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

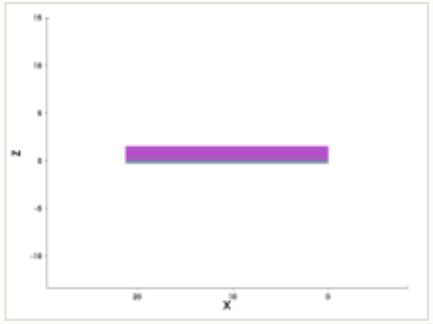


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



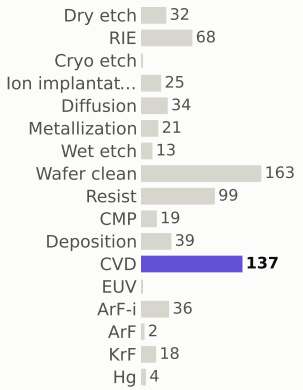
y cut at 57.351 µm



mask for this step

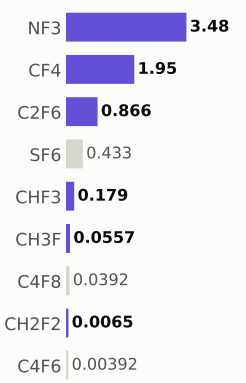
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



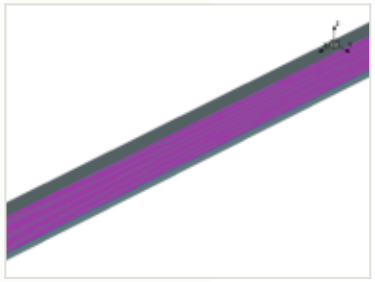
Step #9710: CVD

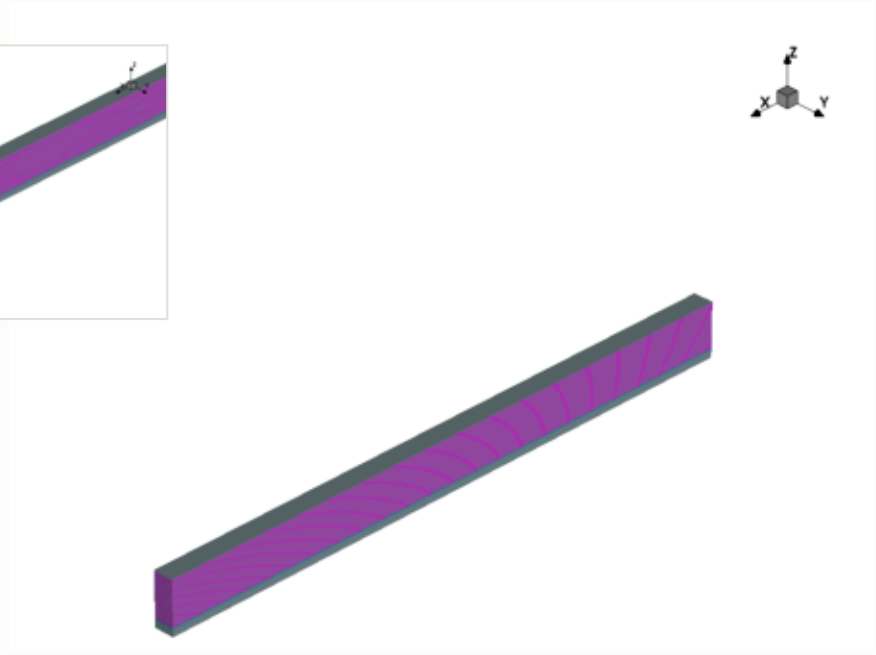
depo_isotropic · CVD

1.31	489.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



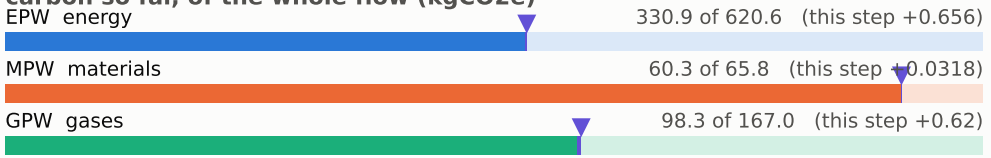


this step

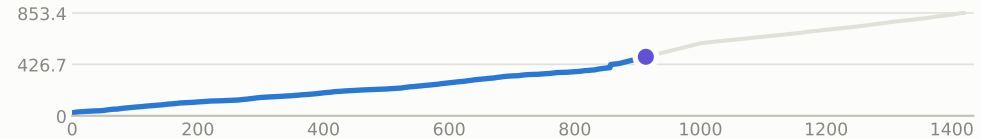
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

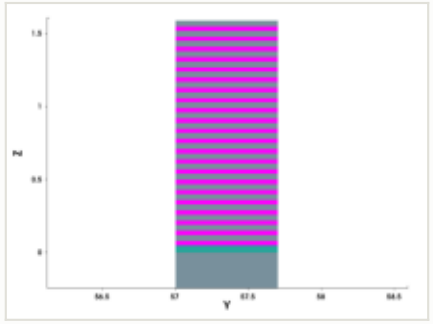
carbon so far, of the whole flow (kgCO2e)



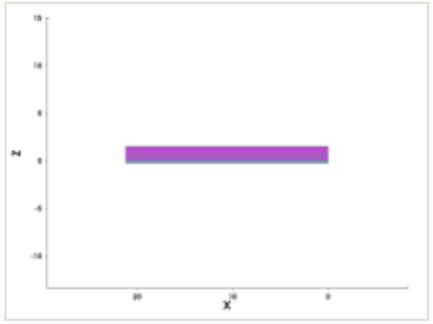
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



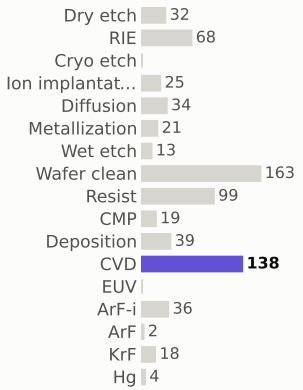
y cut at 57.351 µm



mask for this step

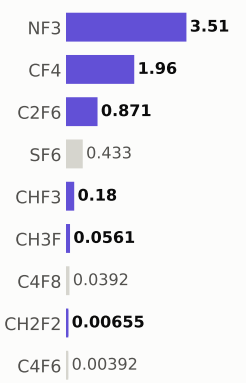
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



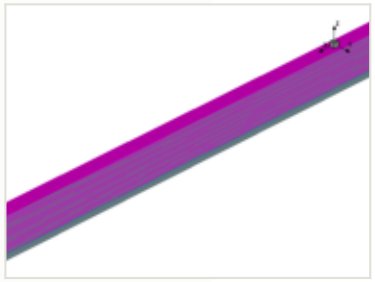
Step #9720: CVD

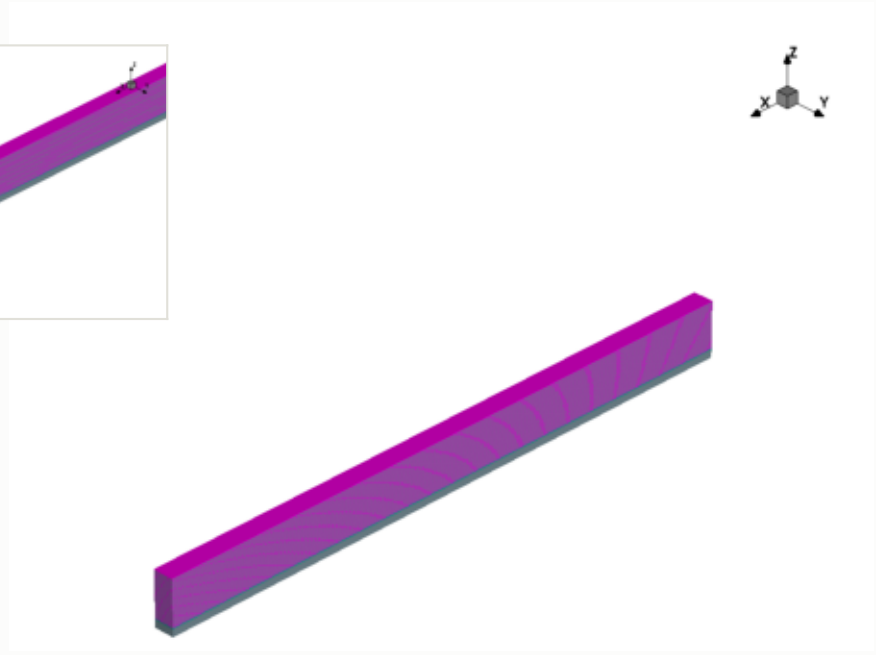
depo_isotropic · CVD

1.31	490.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



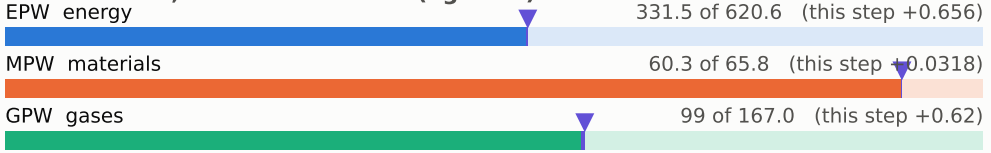


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

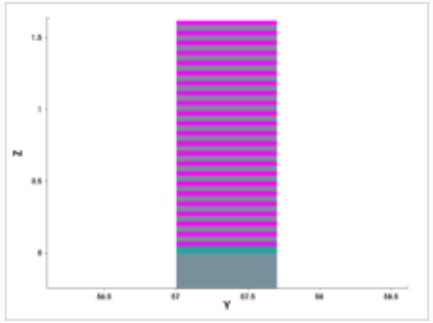
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

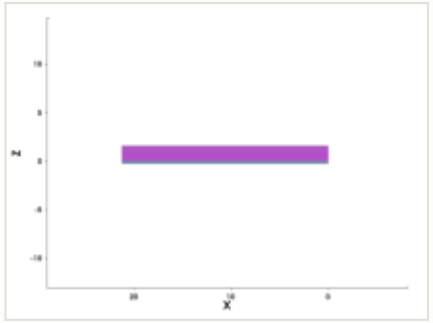


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



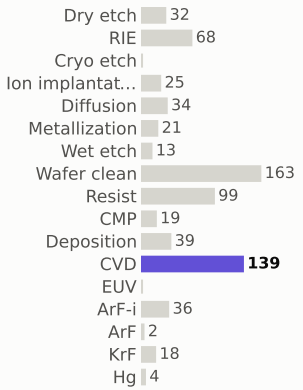
y cut at 57.351 µm



mask for this step

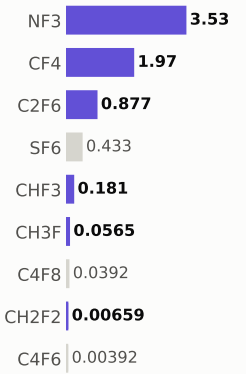
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



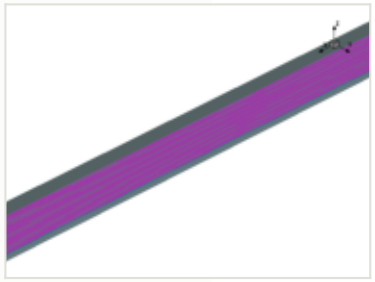
Step #9730: CVD

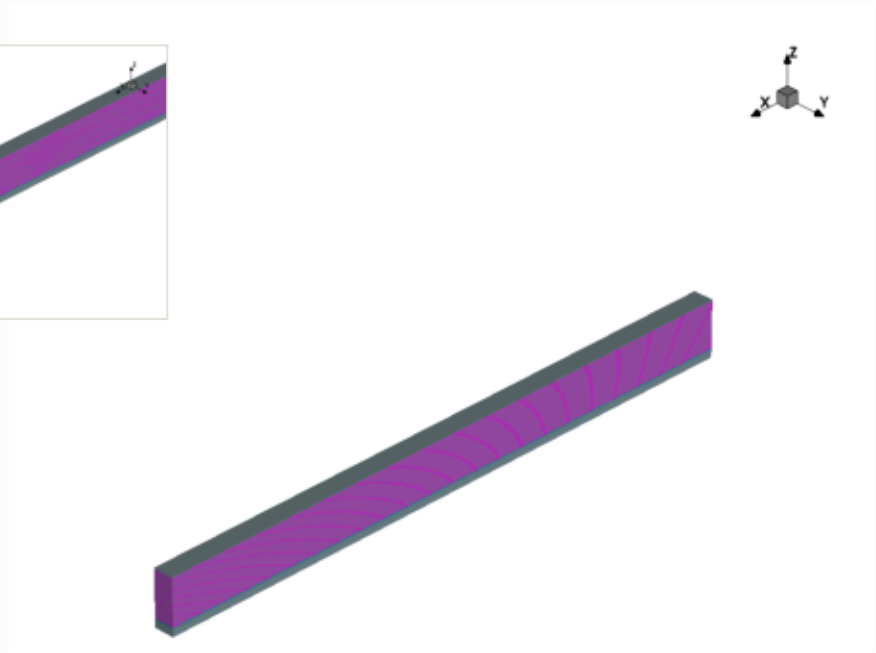
depo_isotropic · CVD

1.31	492.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



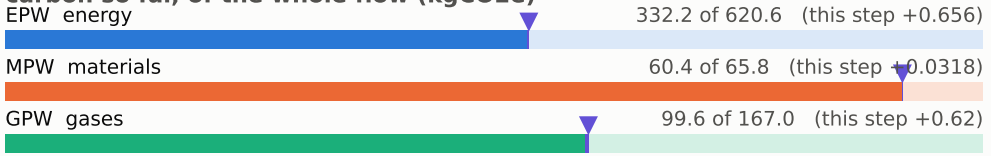


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

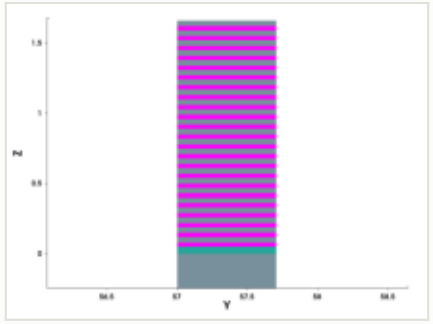
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

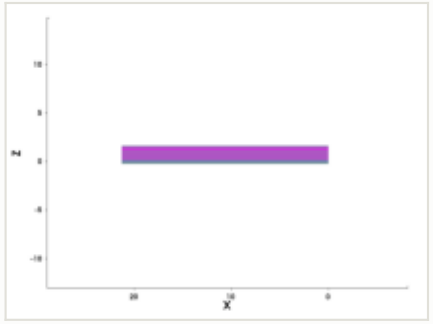


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



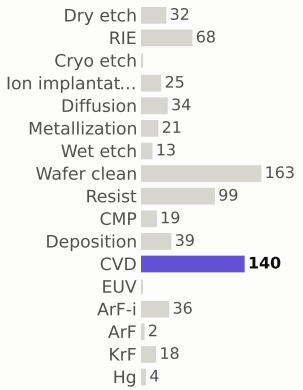
y cut at 57.351 µm



mask for this step

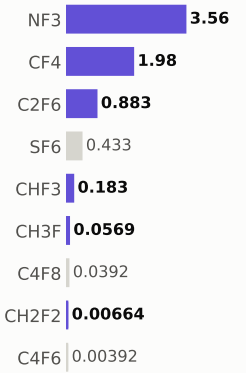
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



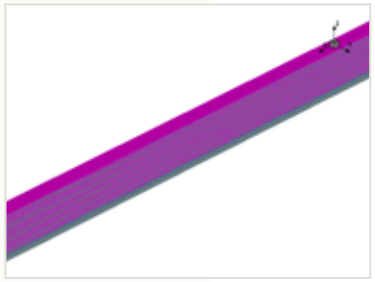
Step #9740: CVD

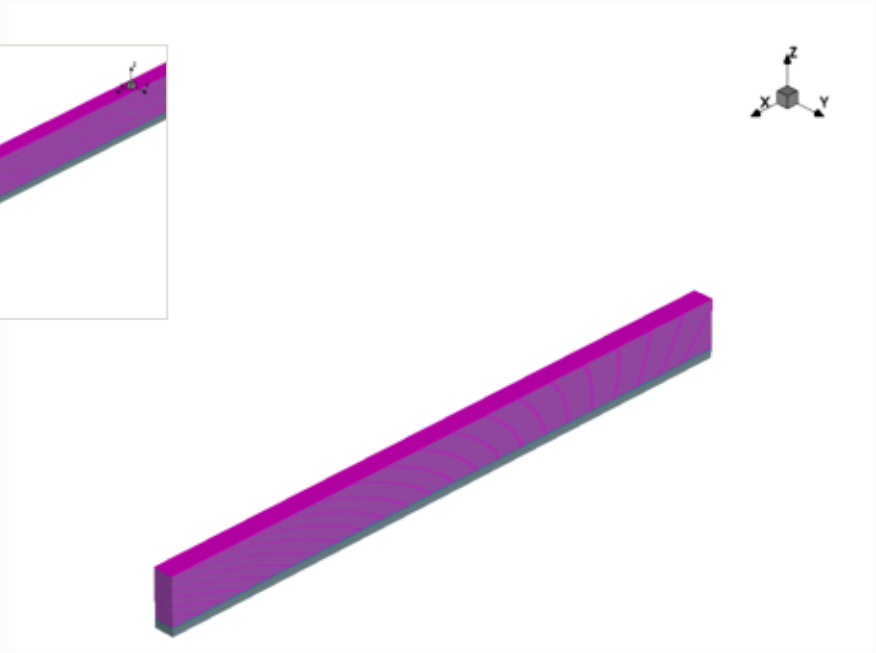
depo_isotropic · CVD

1.31	493.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



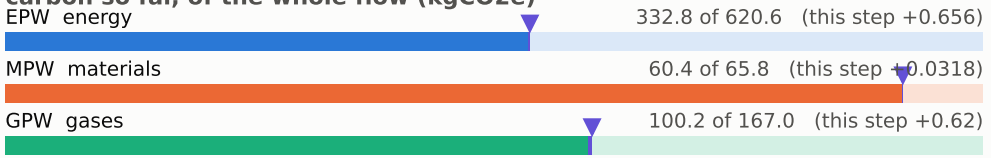


this step

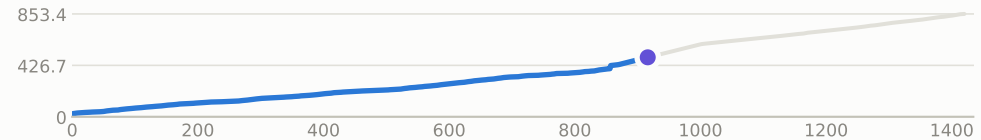
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

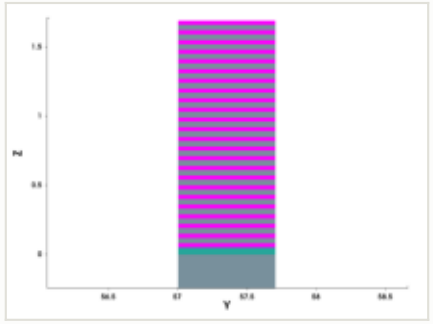
carbon so far, of the whole flow (kgCO2e)



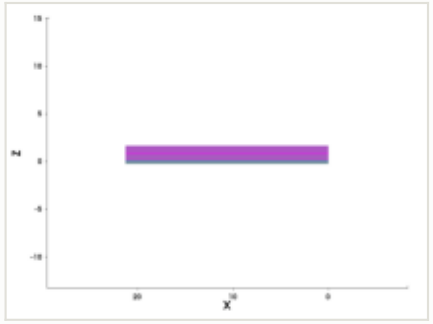
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



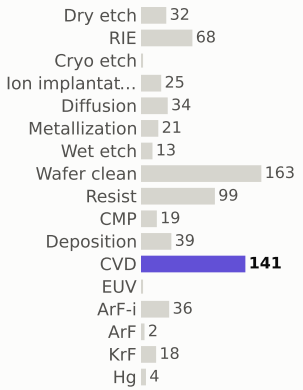
y cut at 57.351 µm



mask for this step

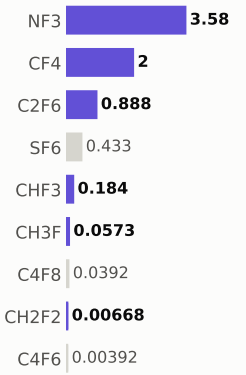
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



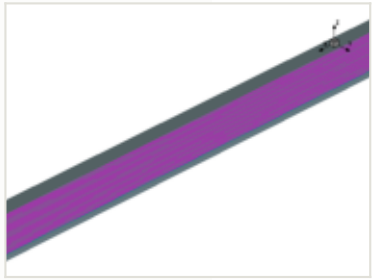
Step #9750: CVD

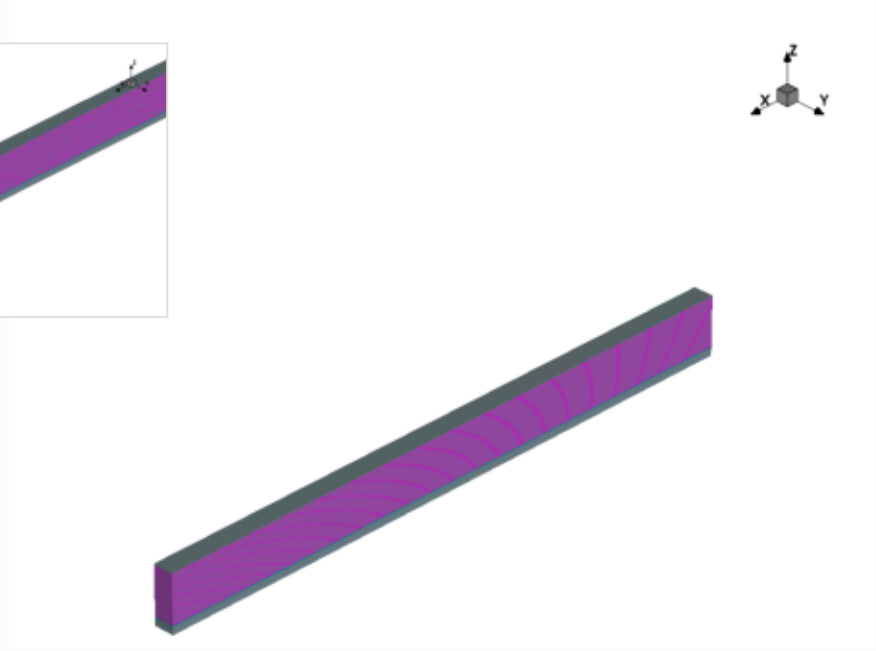
depo_isotropic · CVD

1.31	494.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



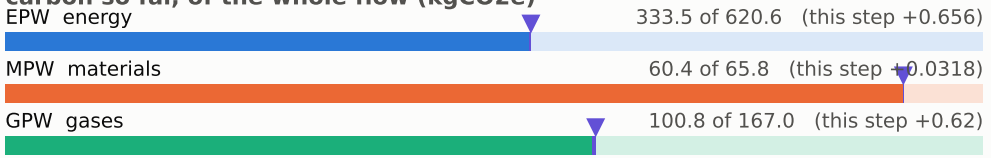


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

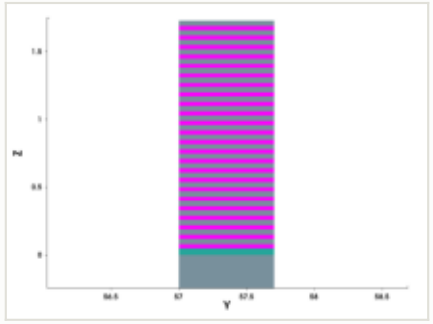
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

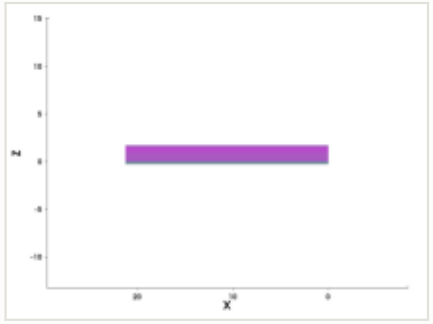


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



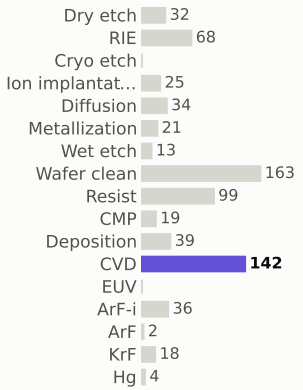
y cut at 57.351 µm



mask for this step

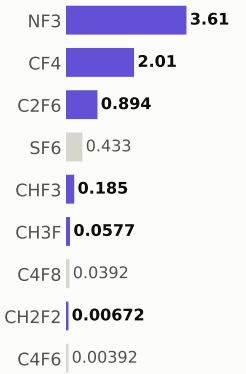
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



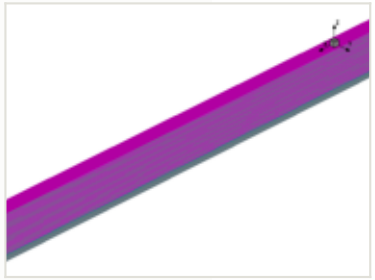
Step #9760: CVD

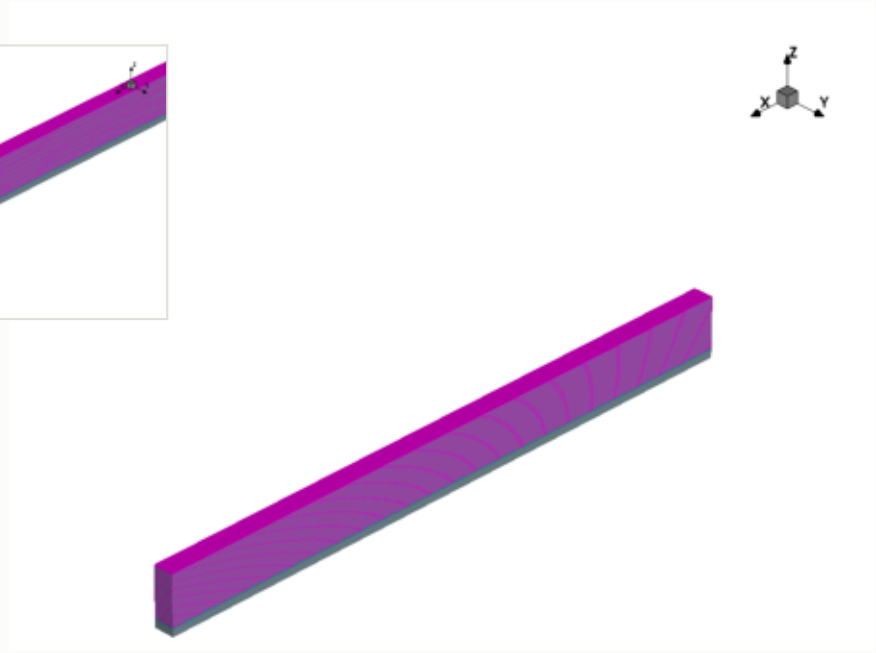
depo_isotropic · CVD

1.31	496.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



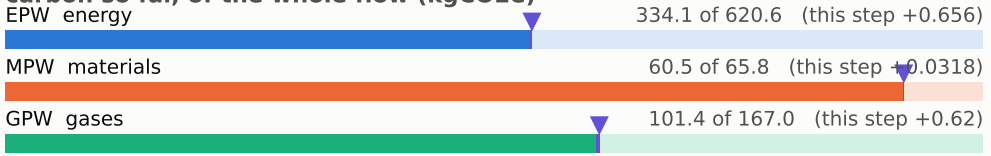


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

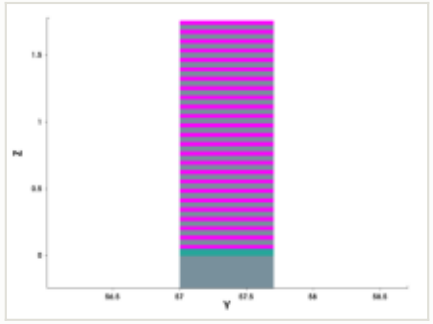
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

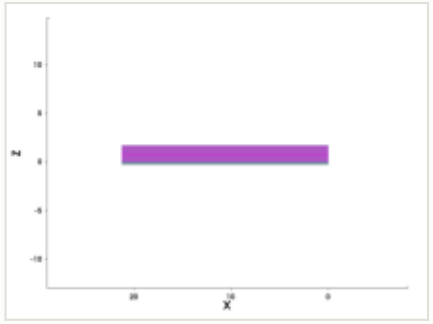


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



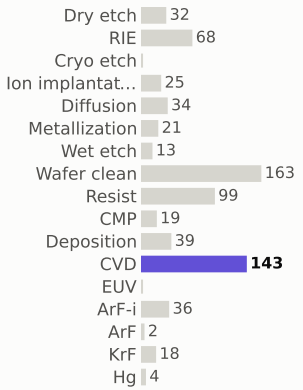
y cut at 57.351 µm



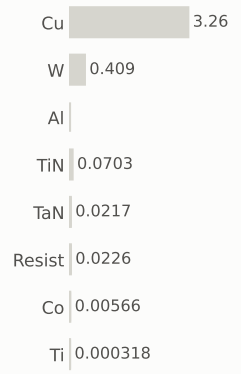
mask for this step

no mask for this step

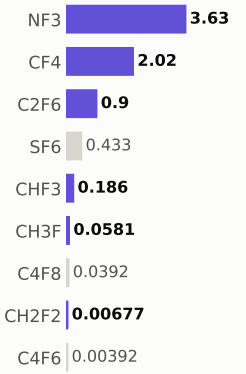
steps by process area



materials deposited (g)



gases emitted (g)



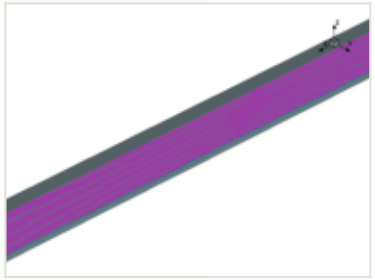
Step #9770: CVD

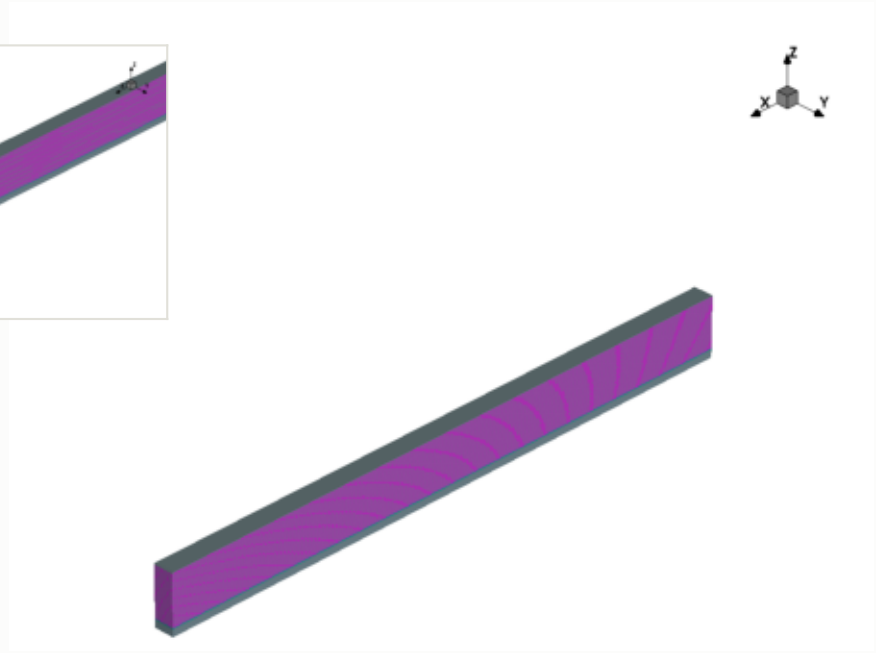
depo_isotropic · CVD

1.31	497.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



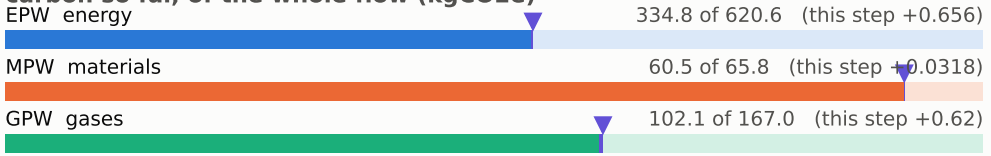


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

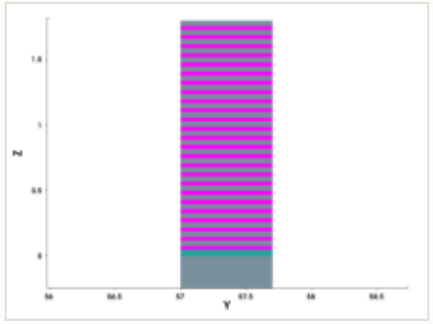
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

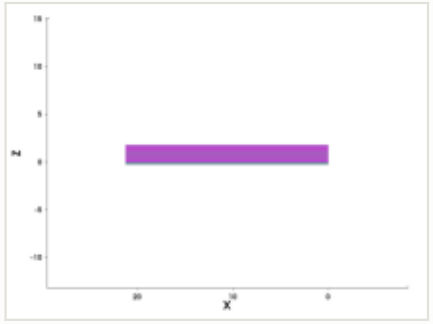


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



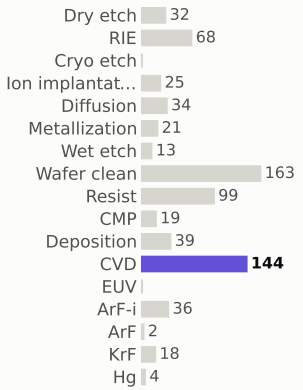
y cut at 57.351 µm



mask for this step

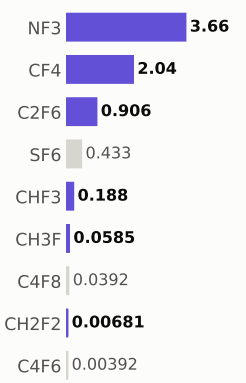
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



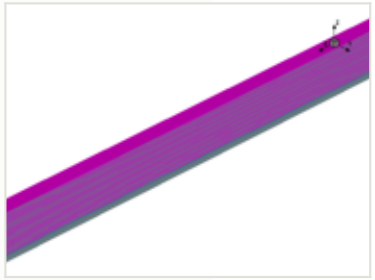
Step #9780: CVD

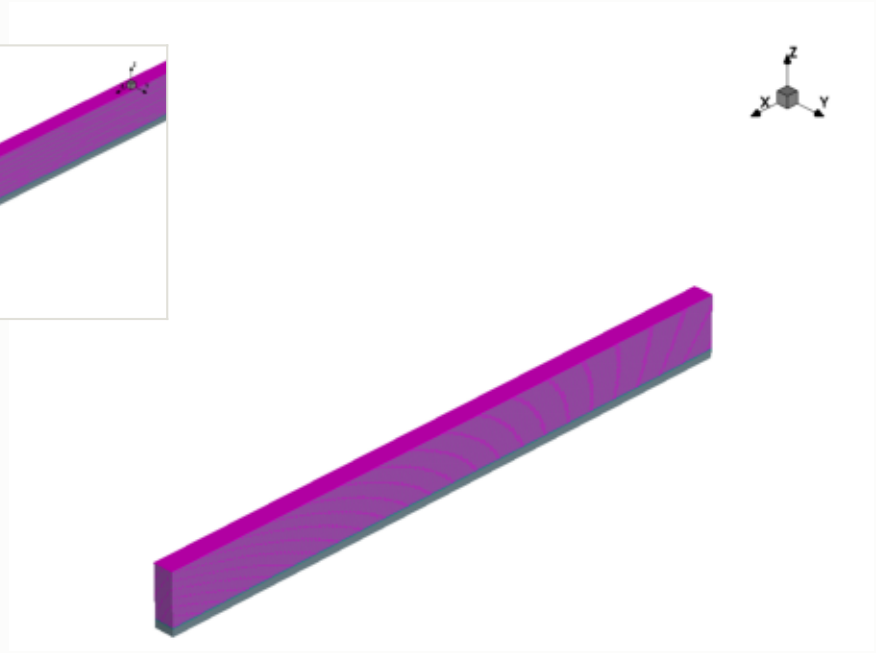
depo_isotropic · CVD

1.31	498.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



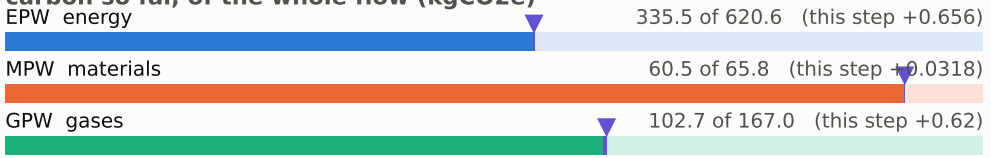


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

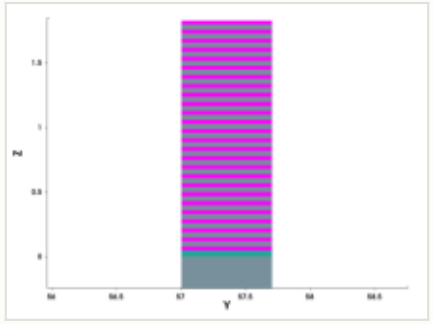
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

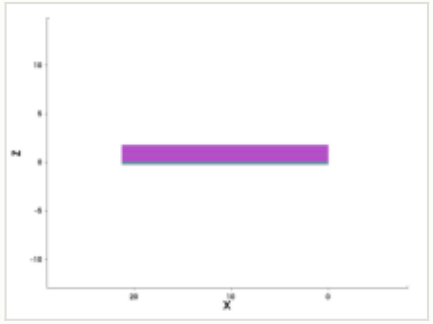


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



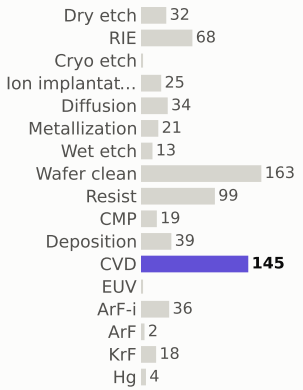
y cut at 57.351 µm



mask for this step

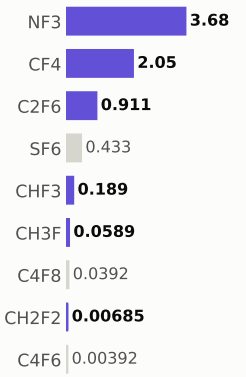
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



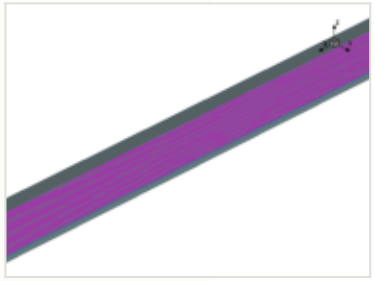
Step #9790: CVD

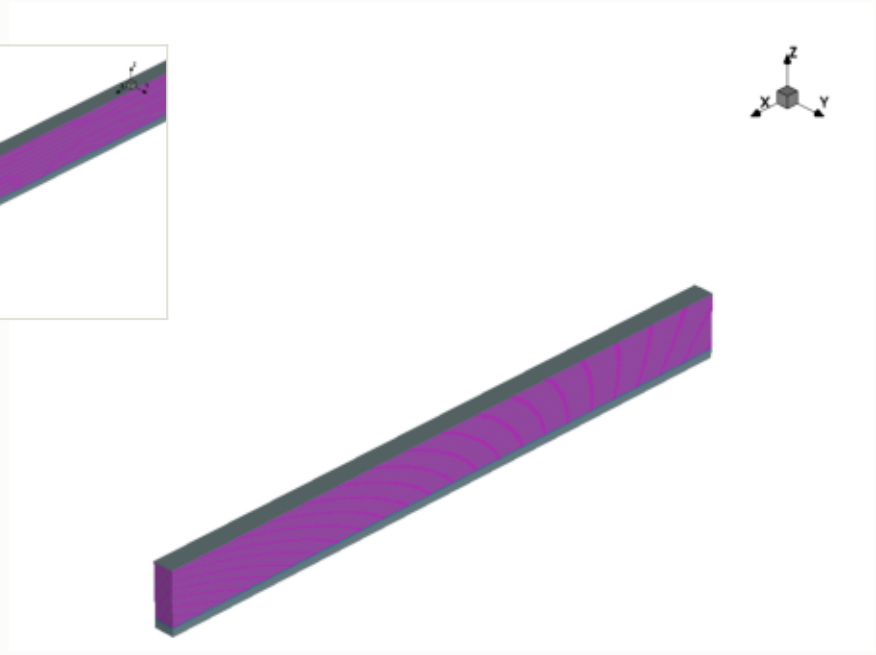
depo_isotropic · CVD

1.31	500.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



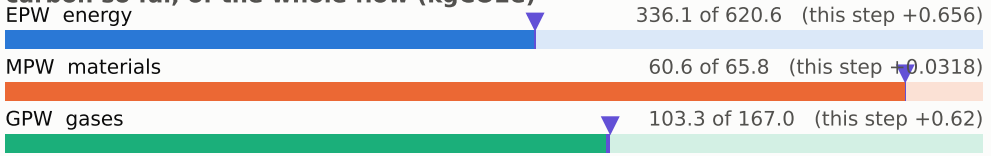


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

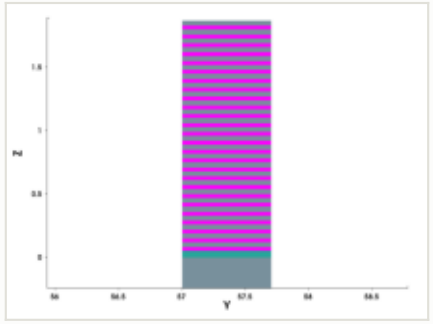
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

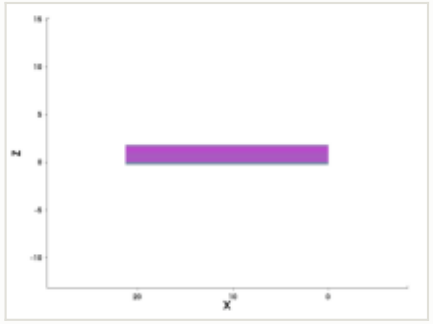


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



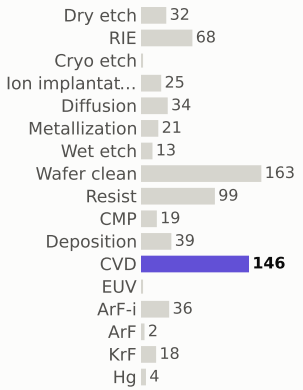
y cut at 57.351 µm



mask for this step

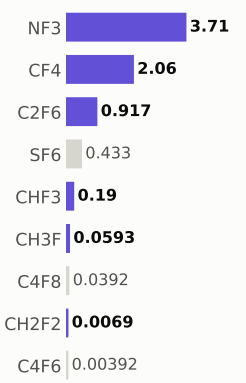
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



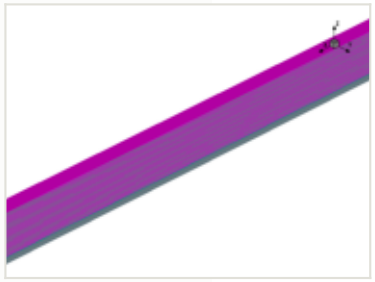
Step #9800: CVD

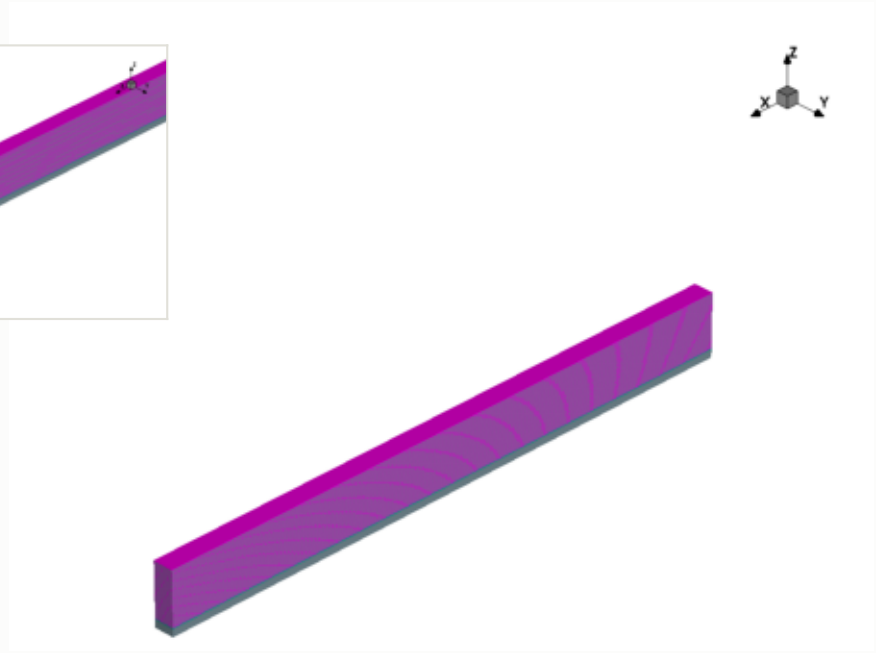
depo_isotropic · CVD

1.31	501.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



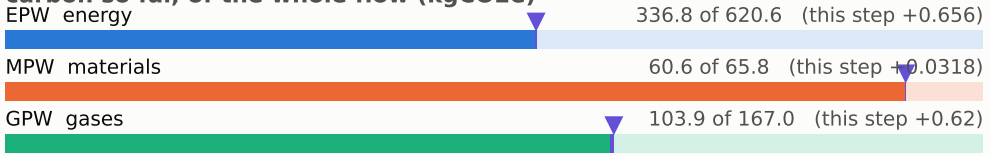


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

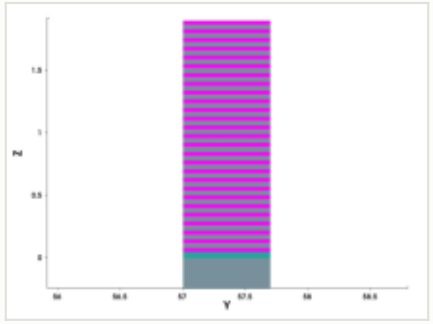
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

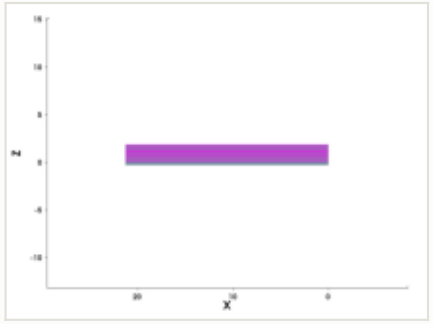


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



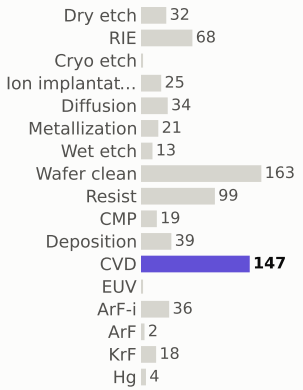
y cut at 57.351 µm



mask for this step

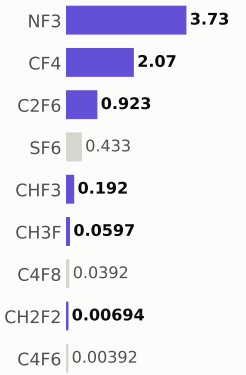
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



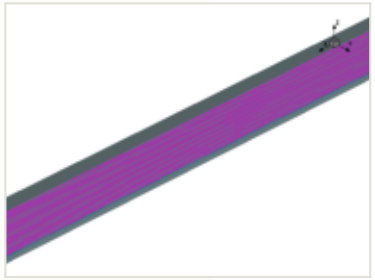
Step #9810: CVD

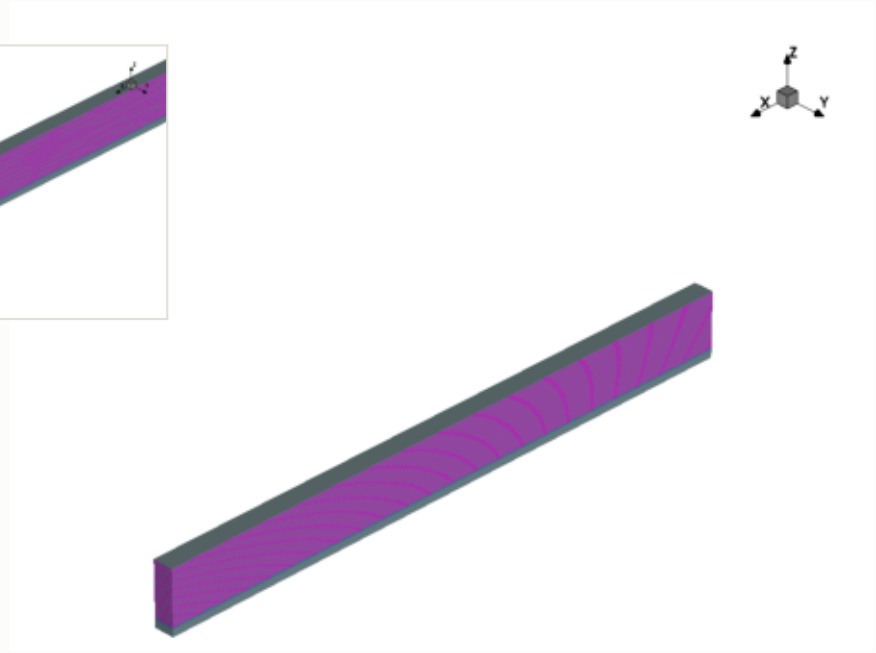
depo_isotropic · CVD

1.31	502.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



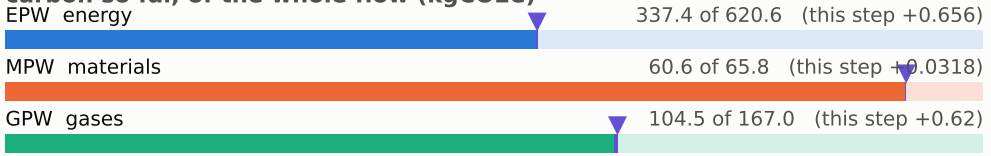


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

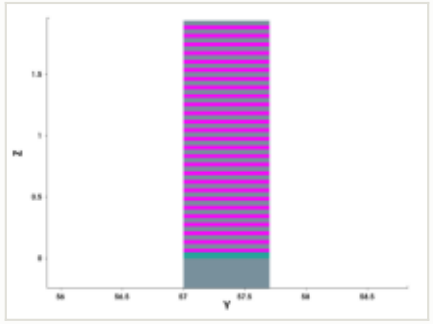
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

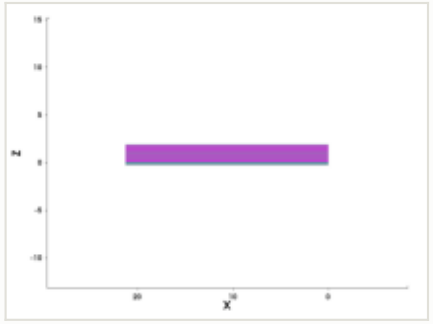


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



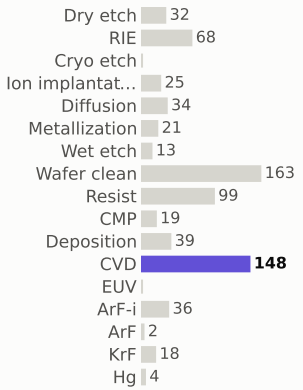
y cut at 57.351 µm



mask for this step

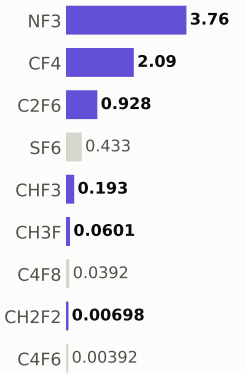
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



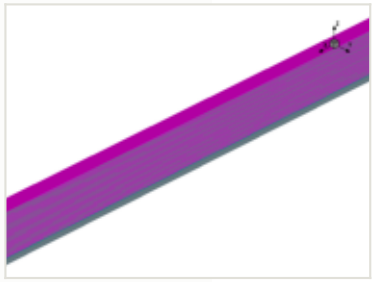
Step #9820: CVD

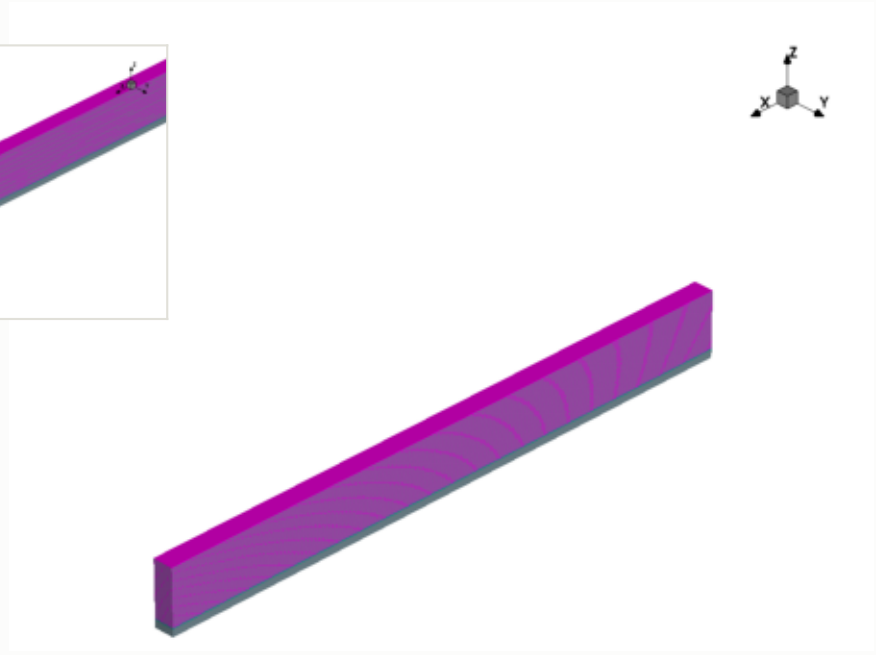
depo_isotropic · CVD

1.31	503.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



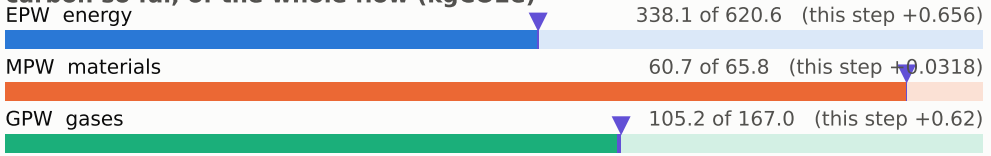


this step

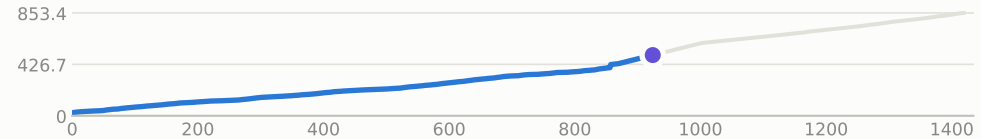
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

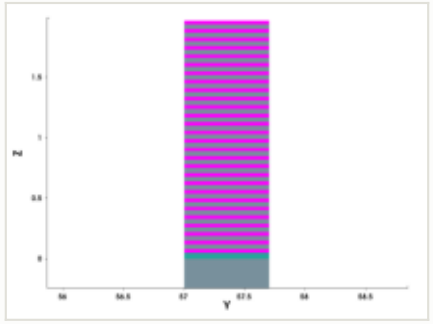
carbon so far, of the whole flow (kgCO2e)



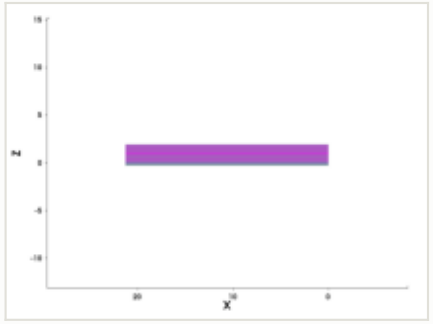
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



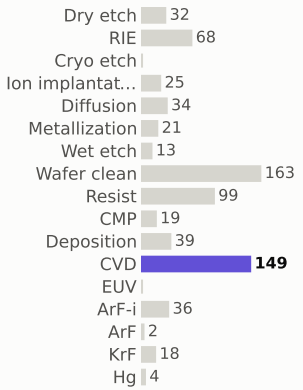
y cut at 57.351 µm



mask for this step

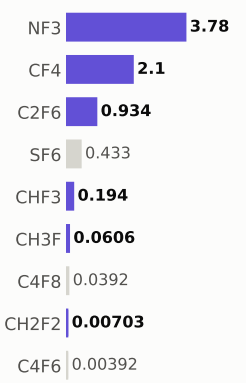
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



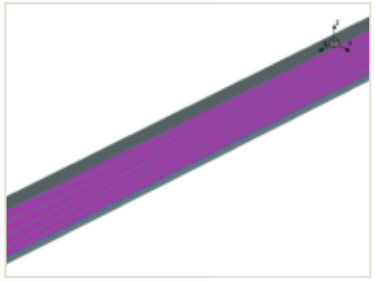
Step #9830: CVD

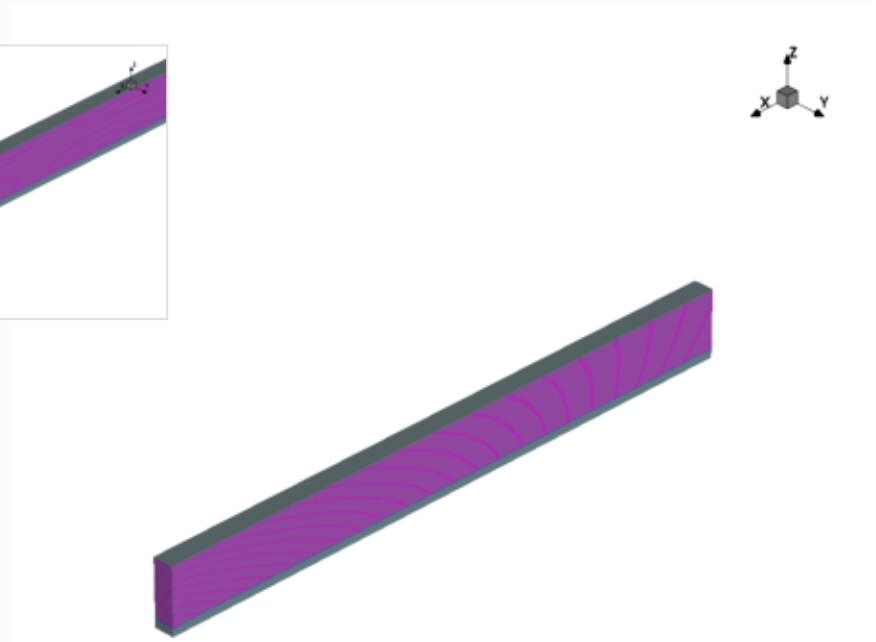
depo_isotropic · CVD

1.31	505.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



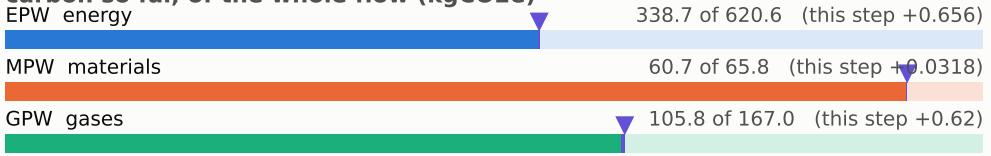


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

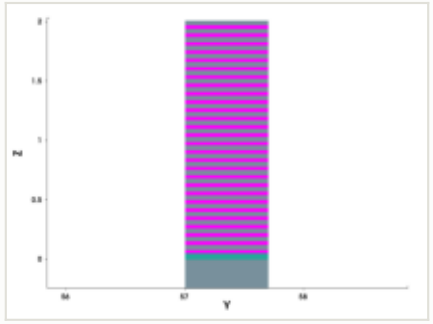
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

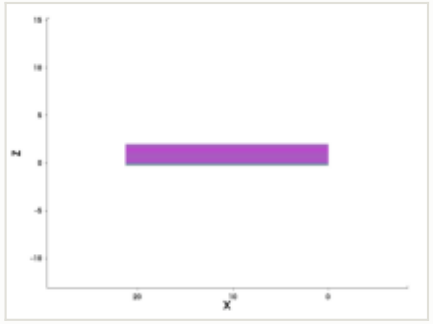


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



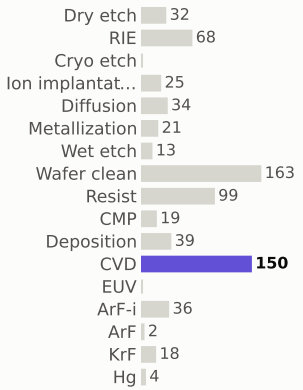
y cut at 57.351 µm



mask for this step

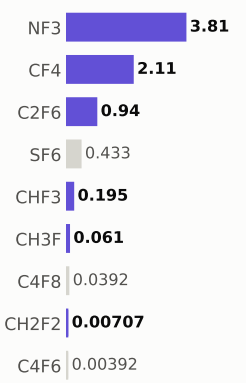
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



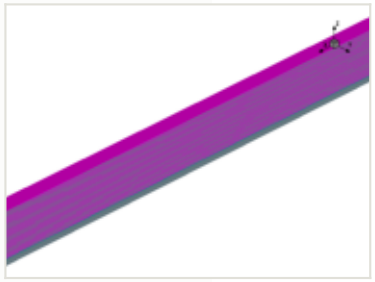
Step #9840: CVD

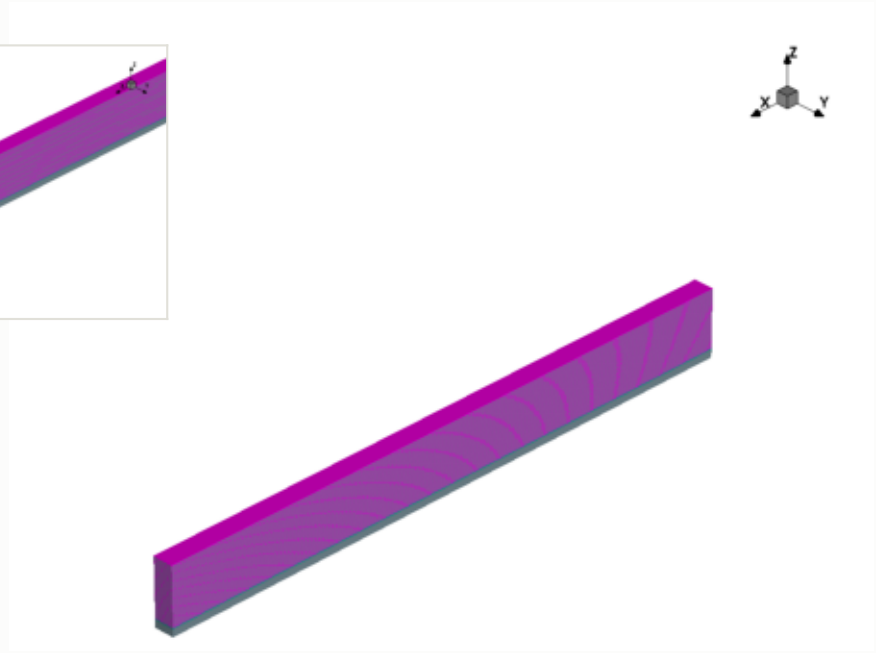
depo_isotropic · CVD

1.31	506.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



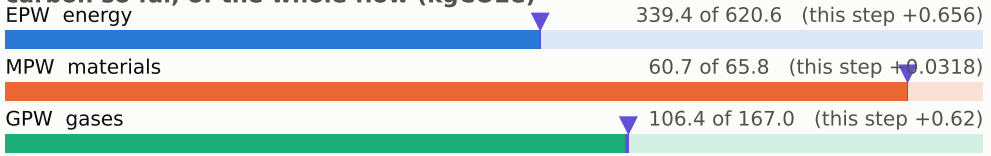


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

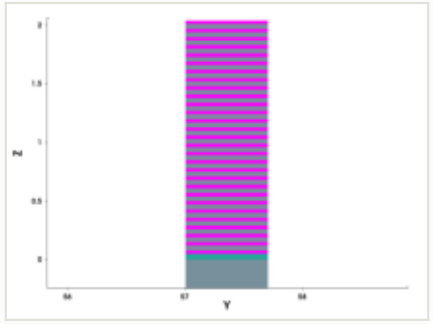
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

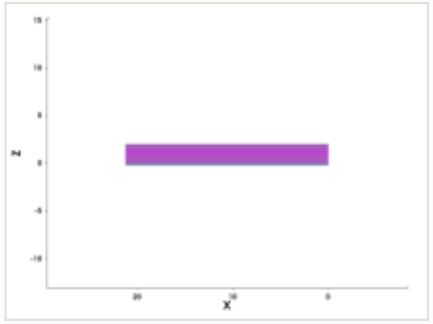


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



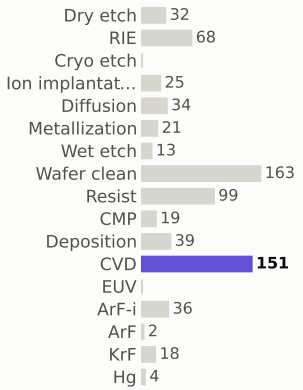
y cut at 57.351 µm



mask for this step

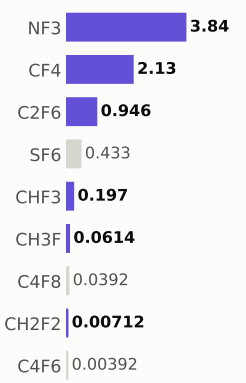
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



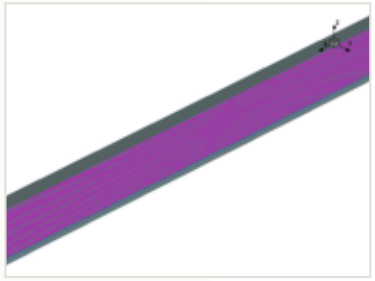
Step #9850: CVD

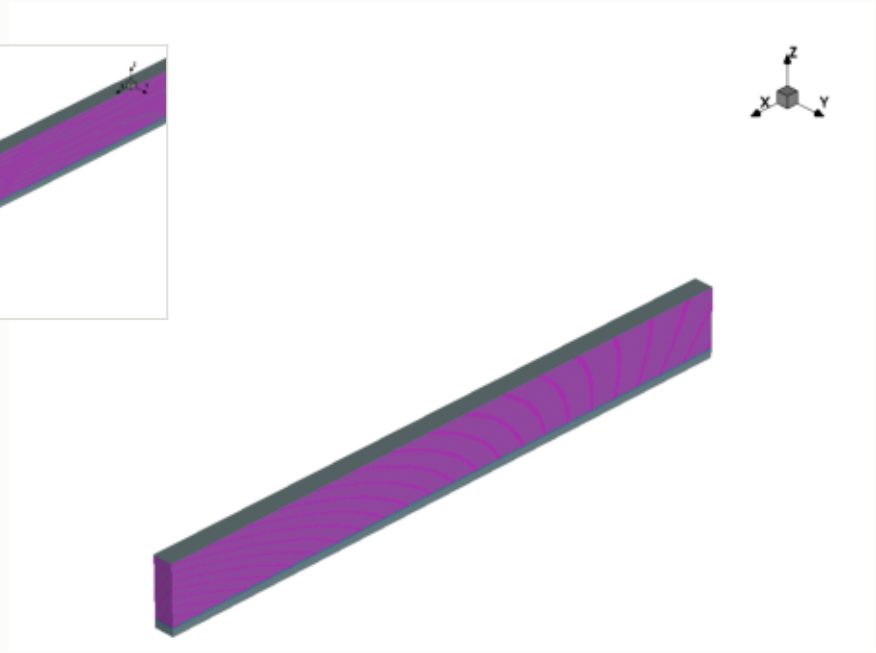
depo_isotropic · CVD

1.31	507.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



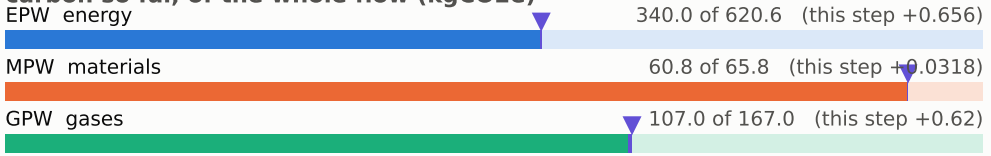


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

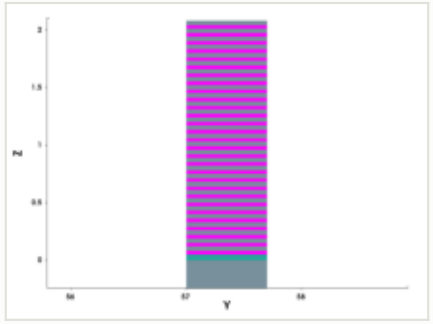
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

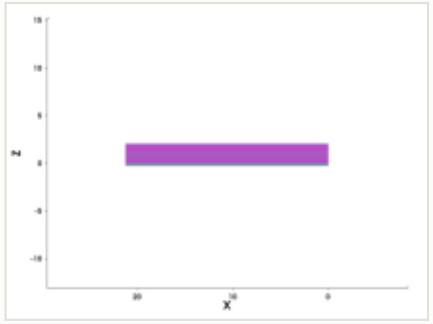


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



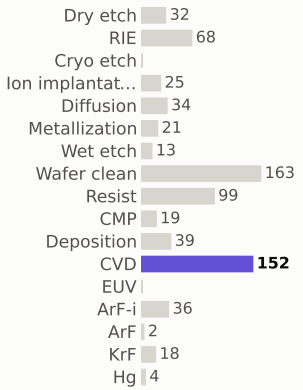
y cut at 57.351 µm



mask for this step

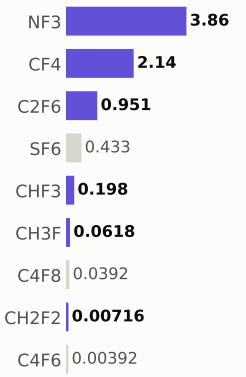
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



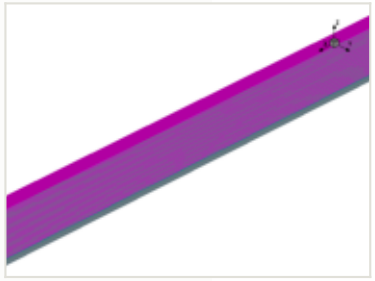
Step #9860: CVD

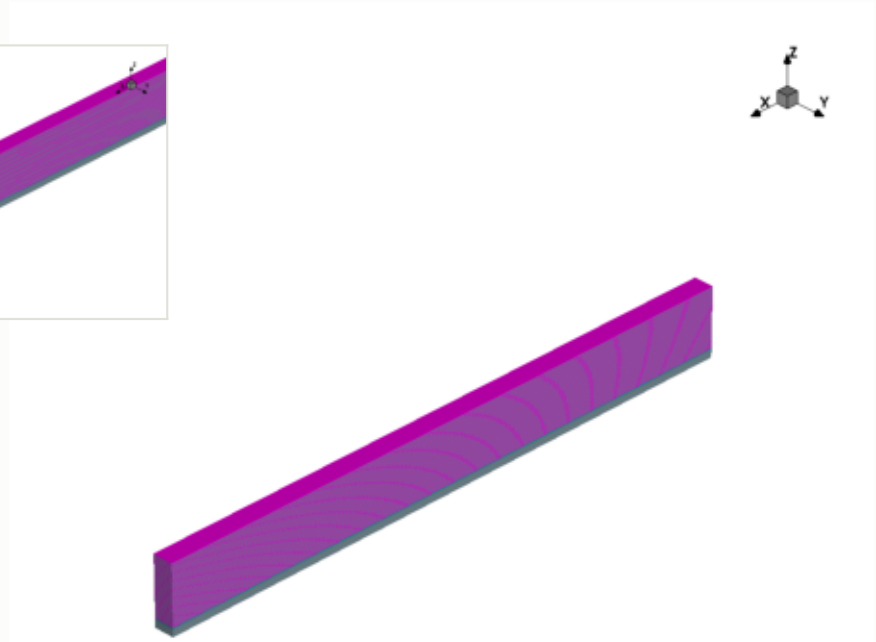
depo_isotropic · CVD

1.31	509.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



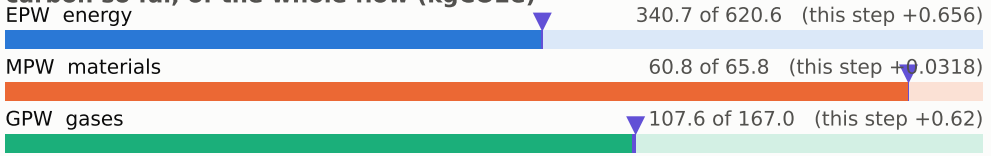


this step

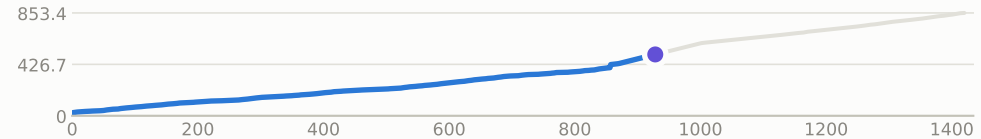
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

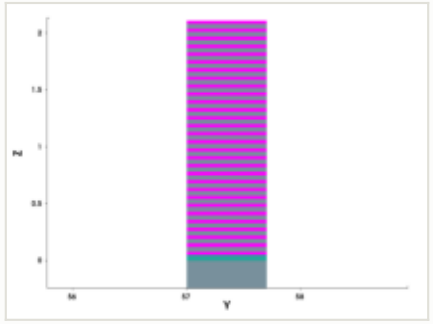
carbon so far, of the whole flow (kgCO2e)



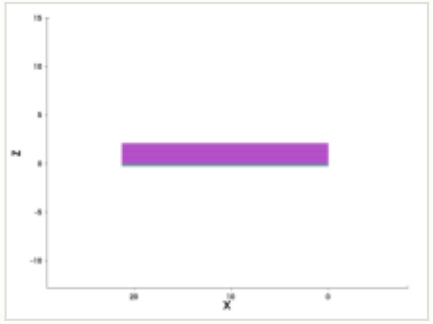
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



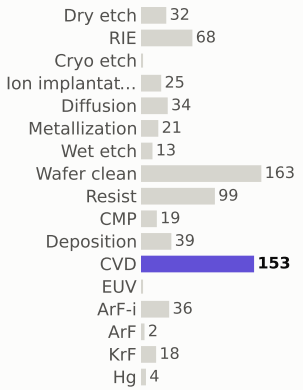
y cut at 57.351 µm



mask for this step

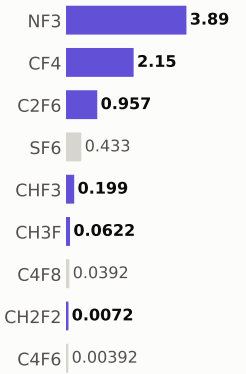
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



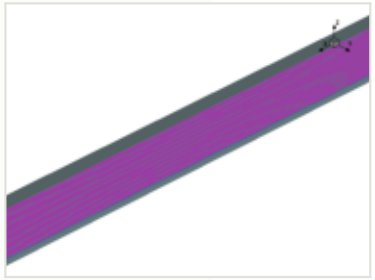
Step #9870: CVD

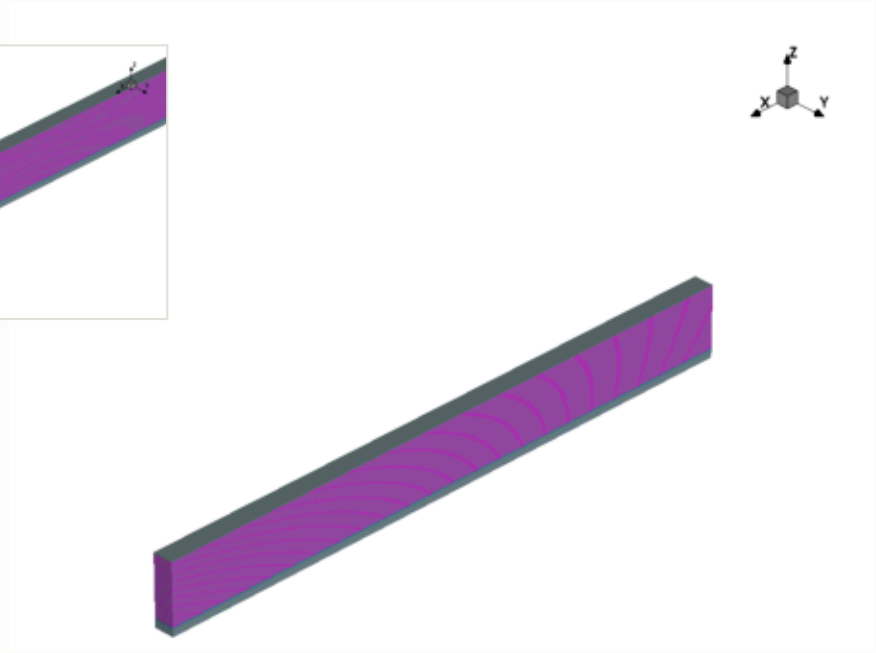
depo_isotropic · CVD

1.31	510.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



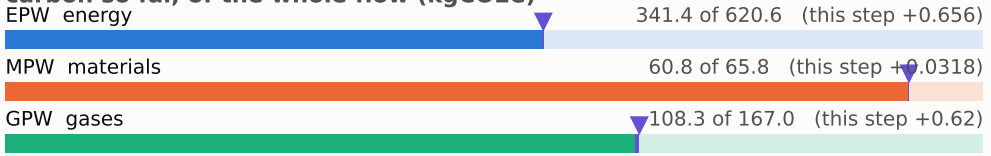


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

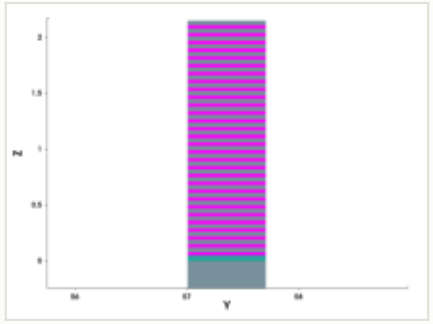
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

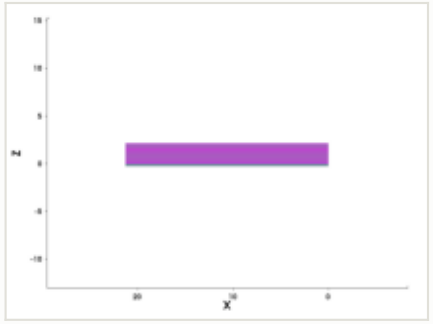


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



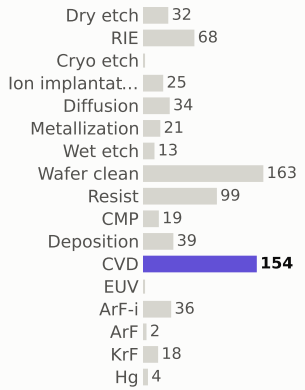
y cut at 57.351 µm



mask for this step

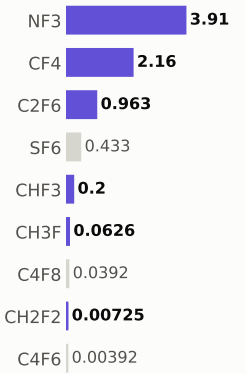
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



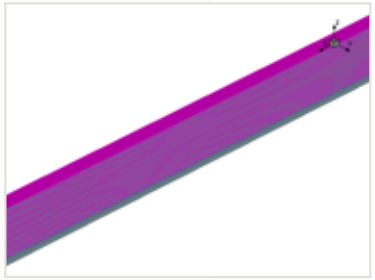
Step #9880: CVD

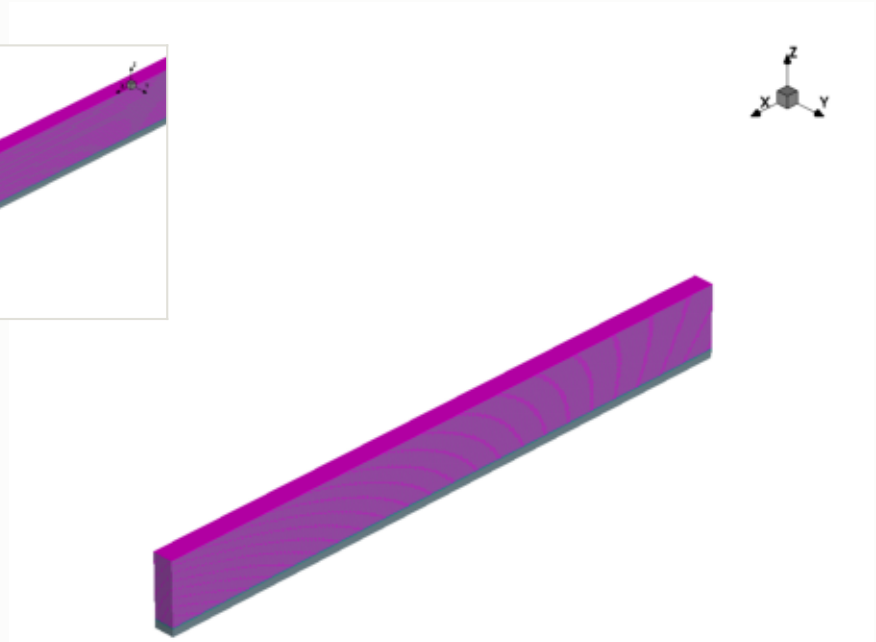
depo_isotropic · CVD

1.31	511.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



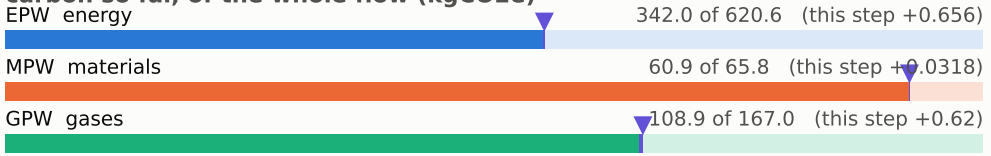


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

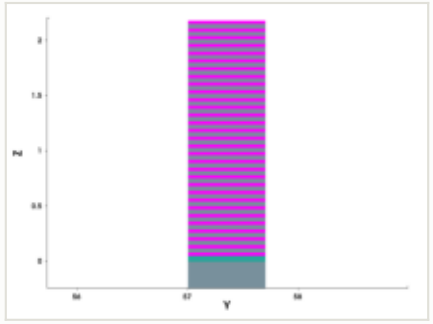
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

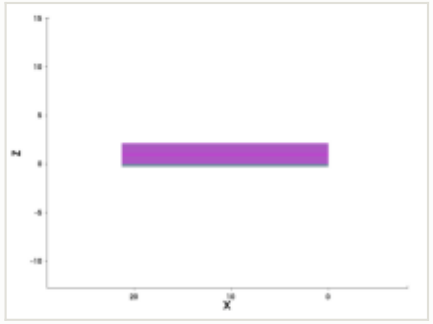


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



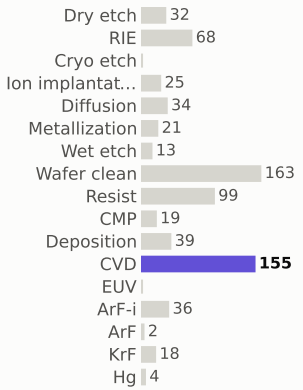
y cut at 57.351 µm



mask for this step

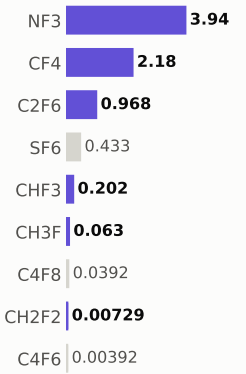
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



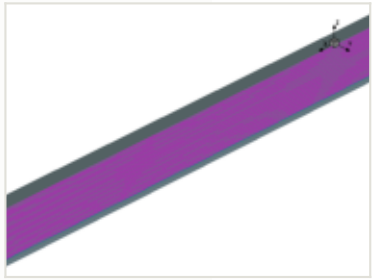
Step #9890: CVD

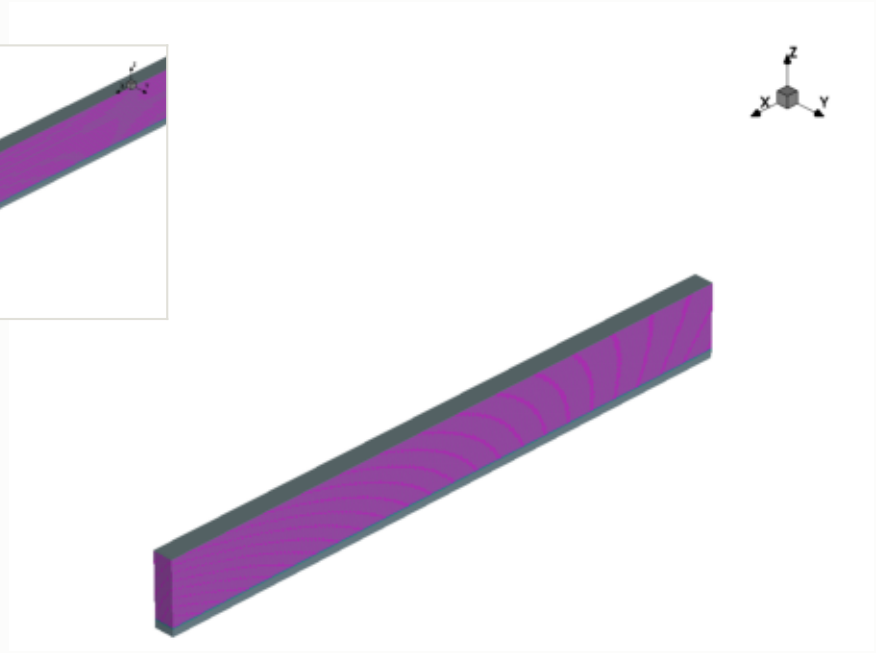
depo_isotropic · CVD

1.31	513.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



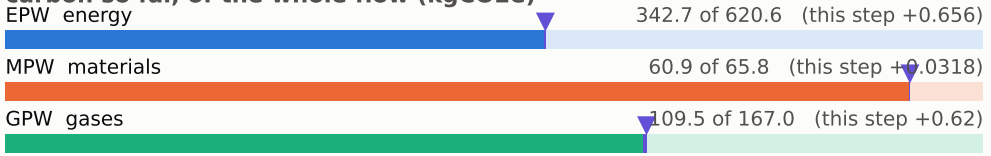


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

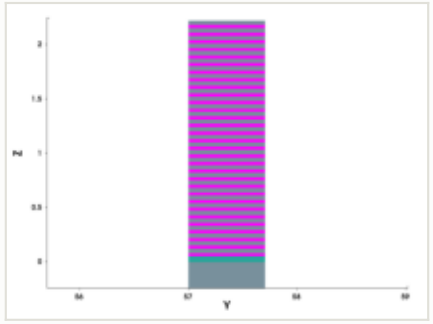
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

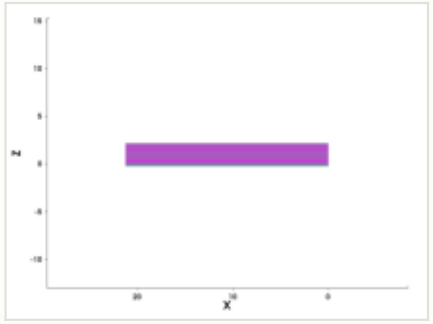


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



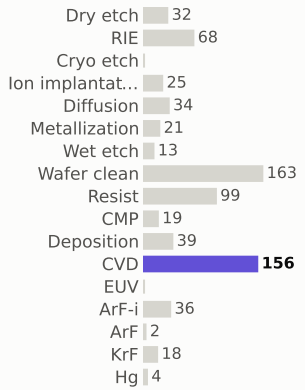
y cut at 57.351 µm



mask for this step

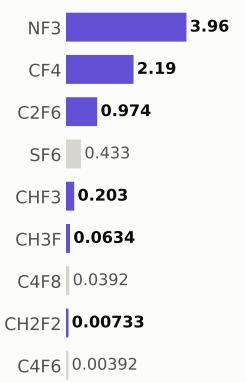
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



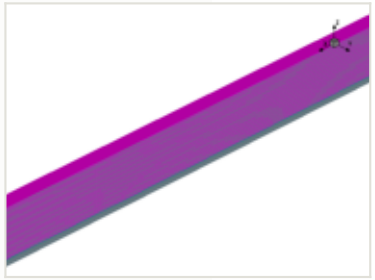
Step #9900: CVD

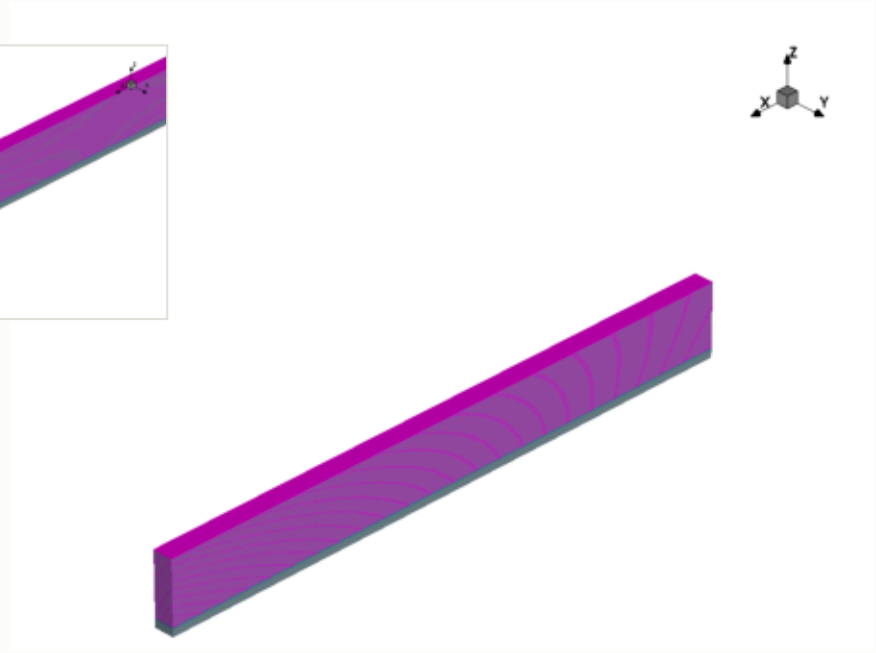
depo_isotropic · CVD

1.31	514.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



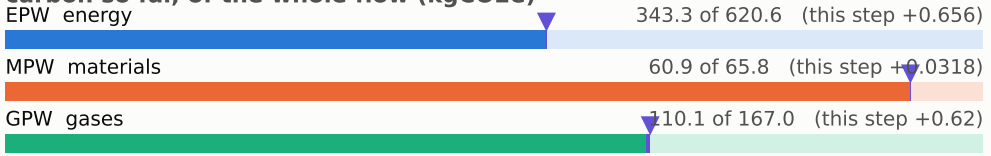


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

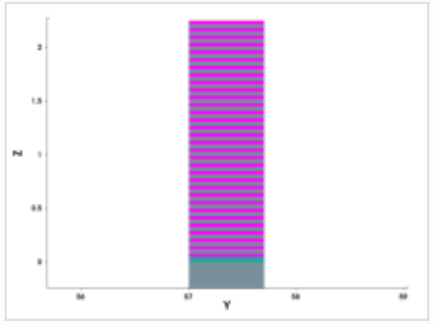
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

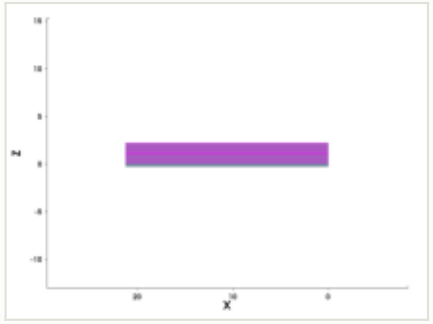


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



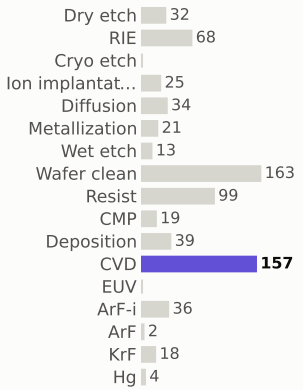
y cut at 57.351 µm



mask for this step

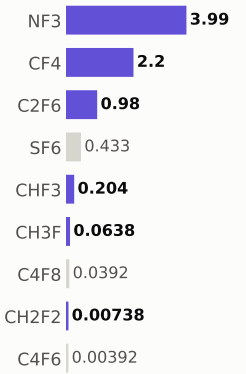
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



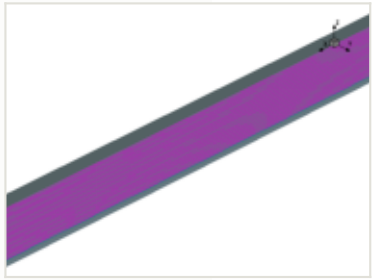
Step #9910: CVD

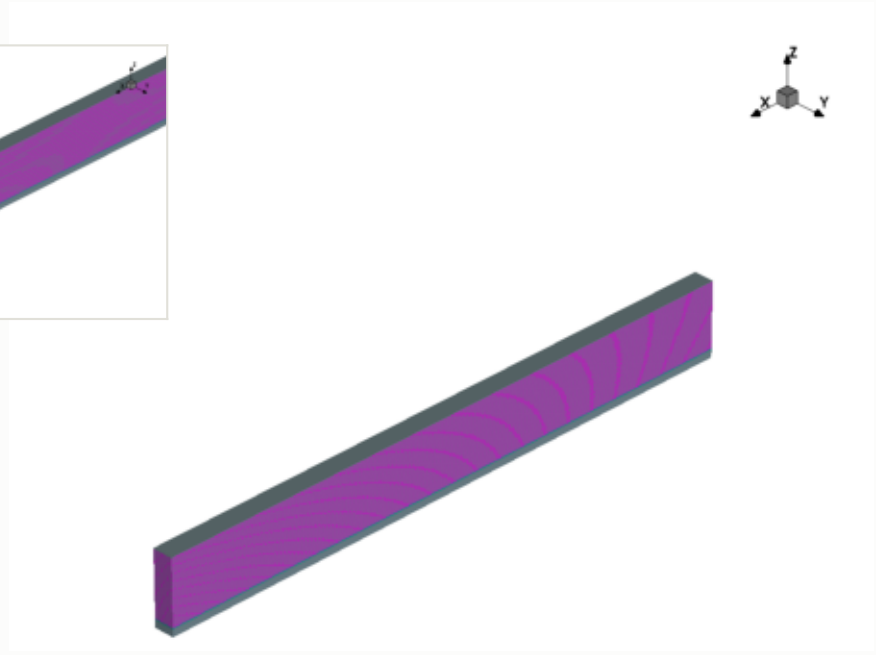
depo_isotropic · CVD

1.31	515.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



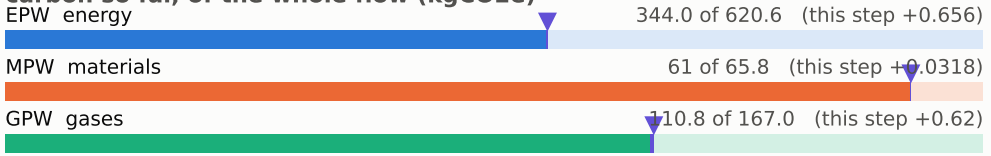


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

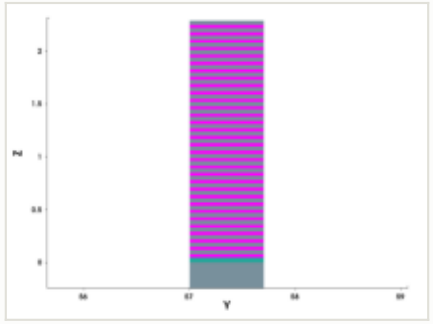
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

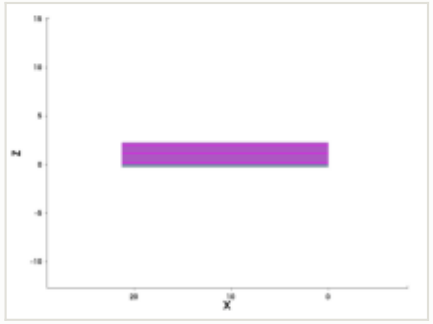


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



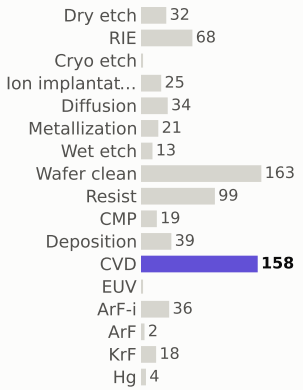
y cut at 57.351 µm



mask for this step

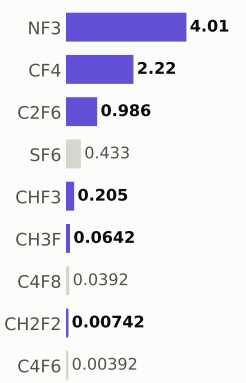
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



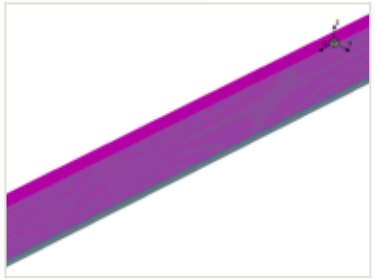
Step #9920: CVD

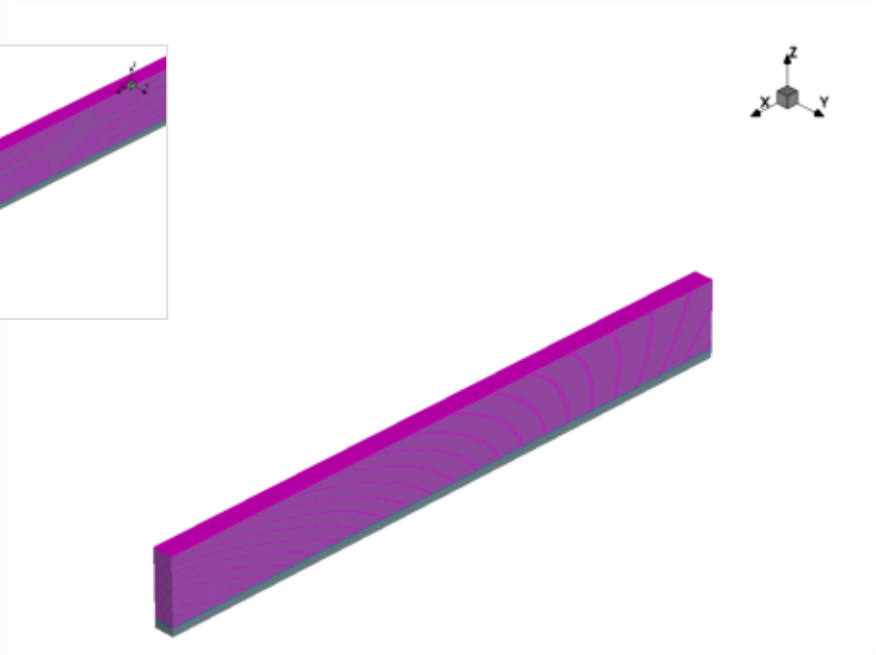
depo_isotropic · CVD

1.31	517.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



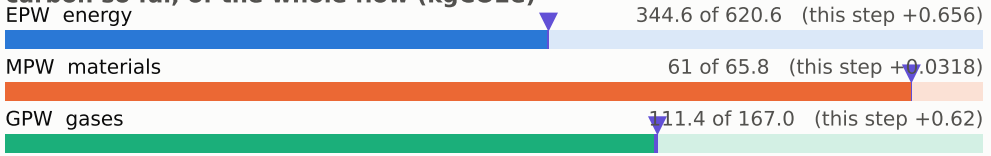


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

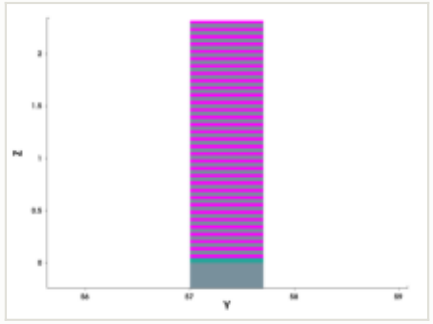
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

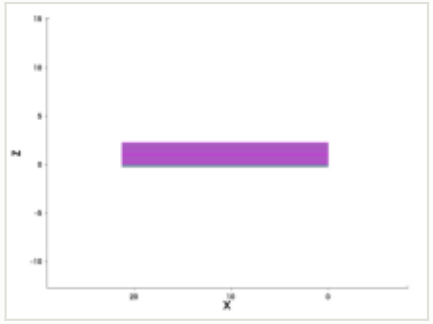


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



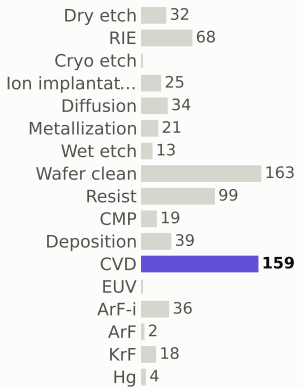
y cut at 57.351 µm



mask for this step

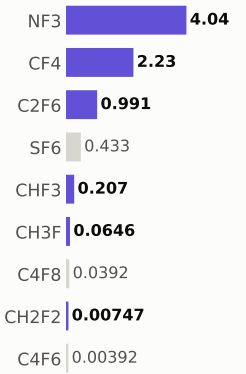
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



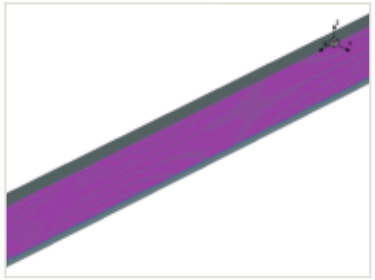
Step #9930: CVD

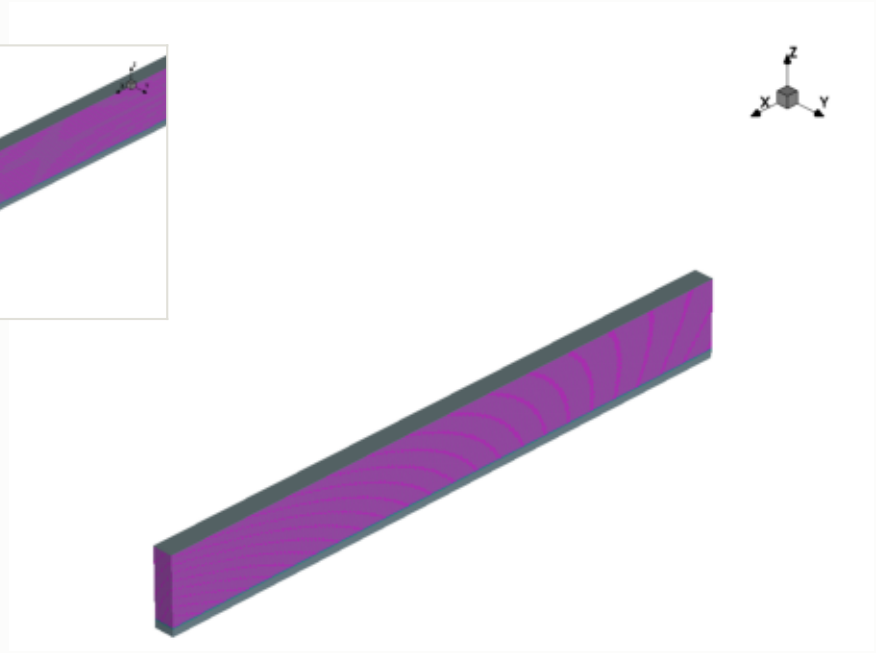
depo_isotropic · CVD

1.31	518.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



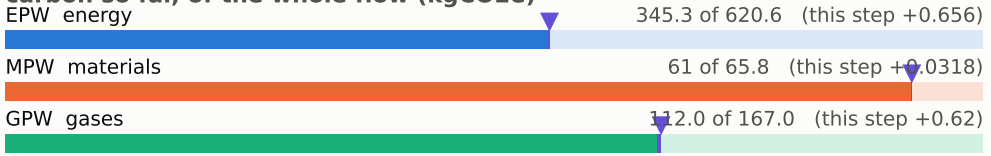


this step

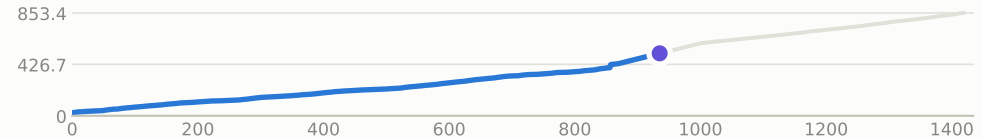
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

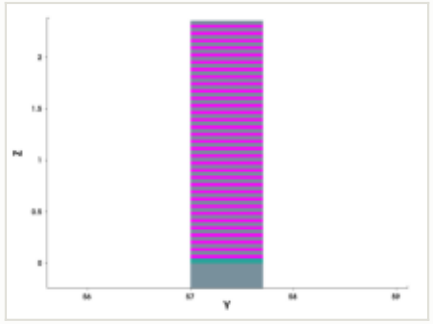
carbon so far, of the whole flow (kgCO2e)



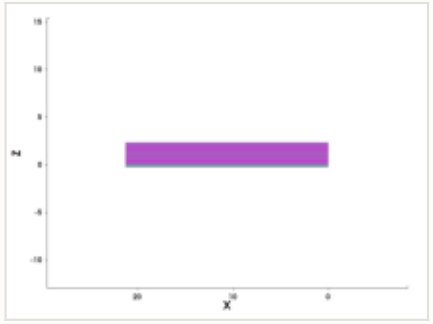
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



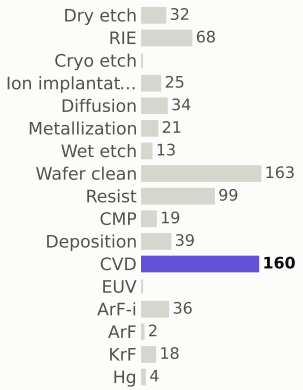
y cut at 57.351 µm



mask for this step

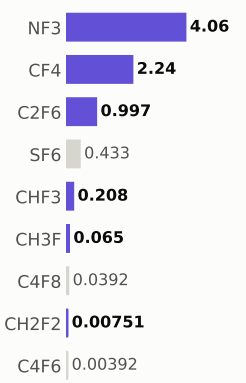
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



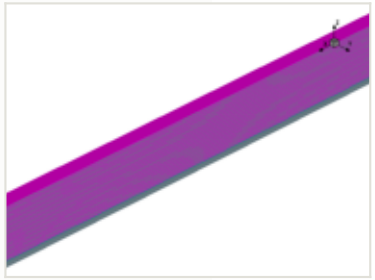
Step #9940: CVD

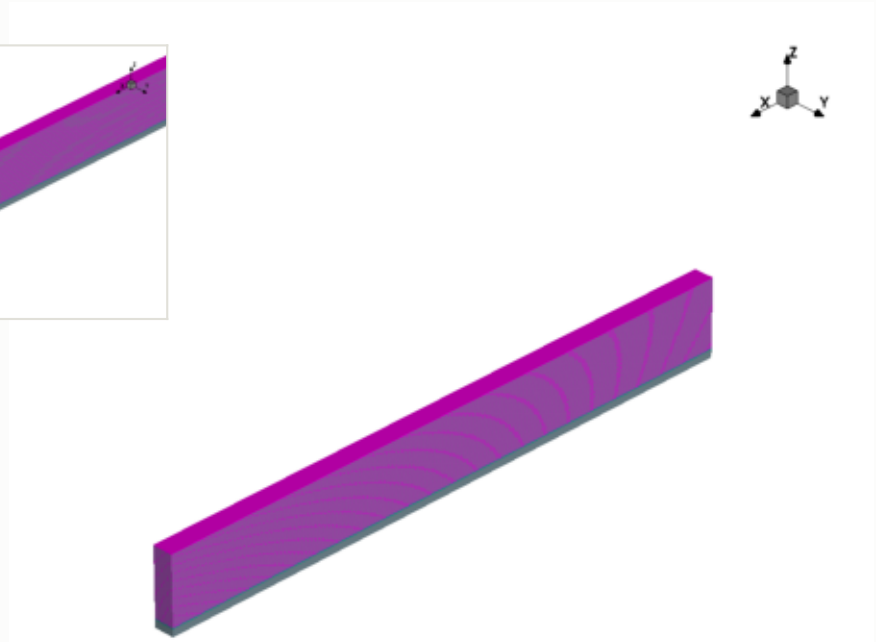
depo_isotropic · CVD

1.31	519.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



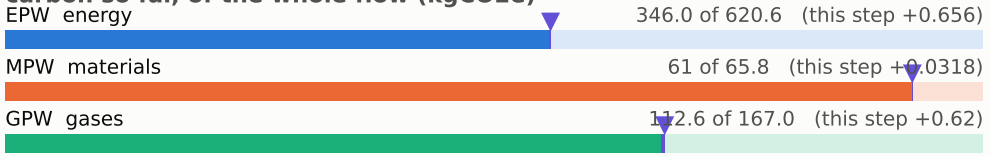


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

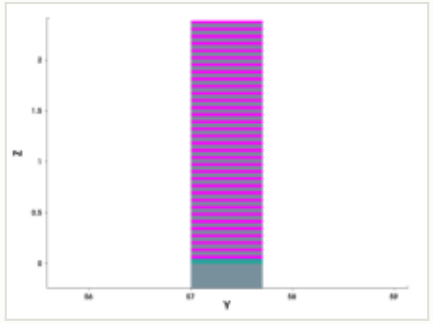
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

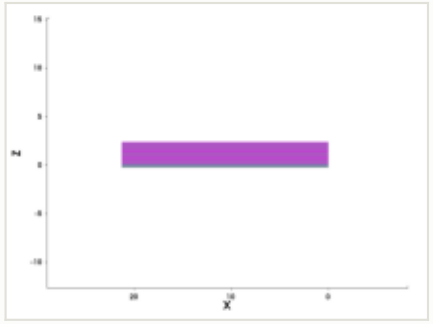


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



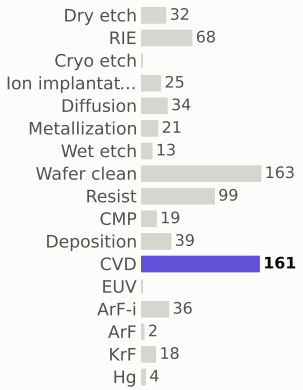
y cut at 57.351 µm



mask for this step

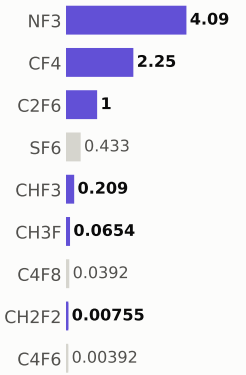
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



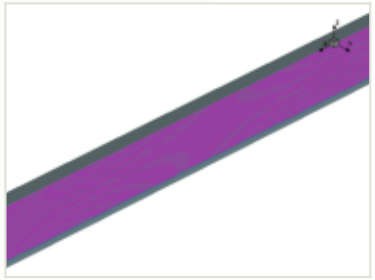
Step #9950: CVD

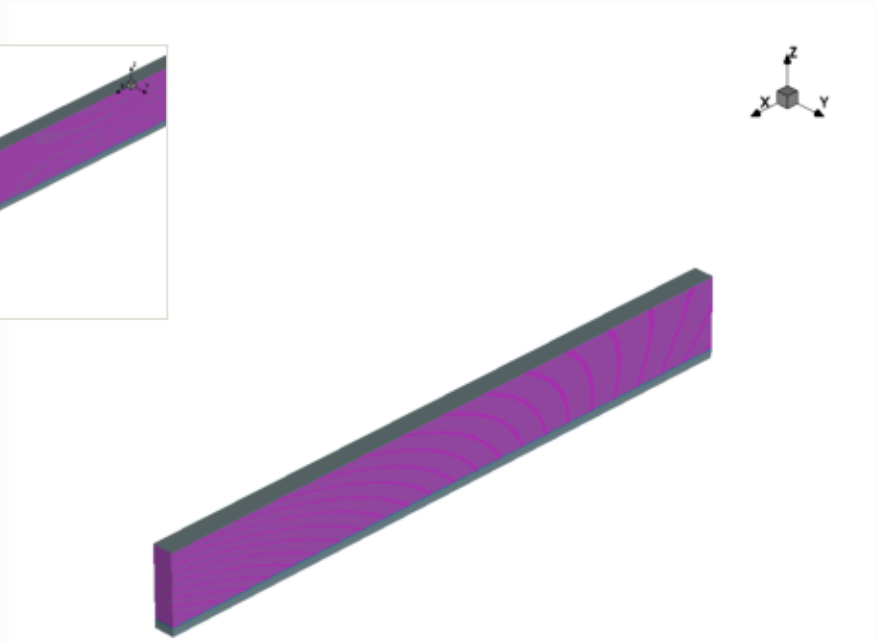
depo_isotropic · CVD

1.31	520.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



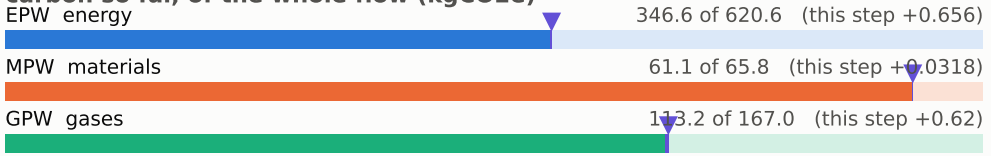


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

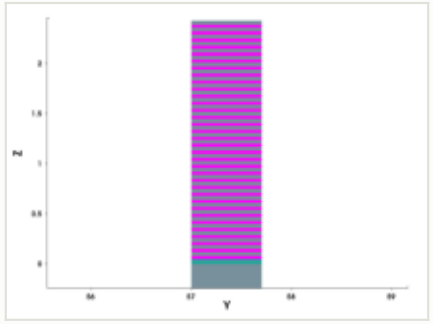
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

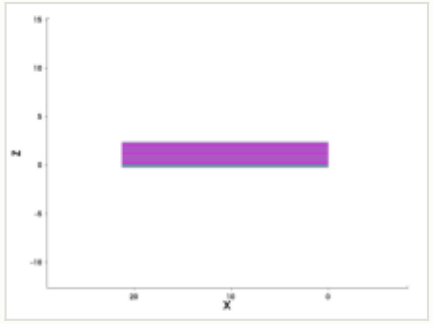


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



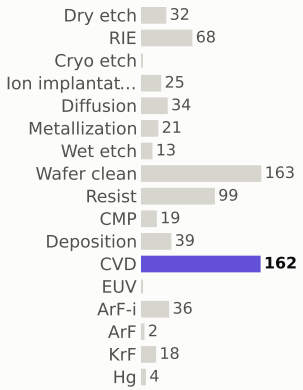
y cut at 57.351 µm



mask for this step

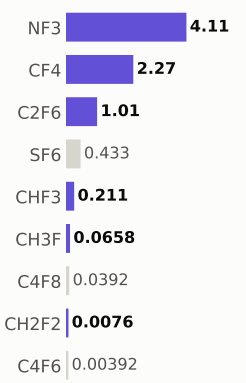
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



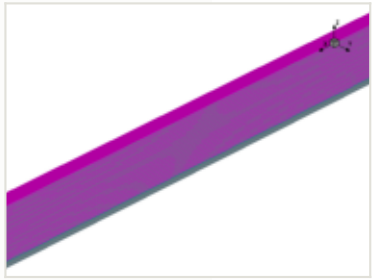
Step #9960: CVD

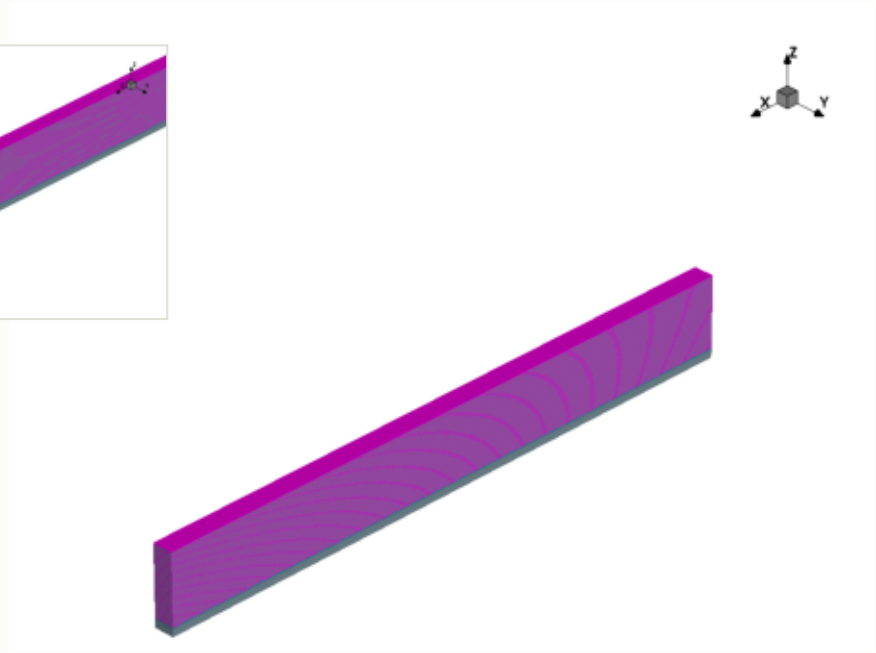
depo_isotropic · CVD

1.31	522.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



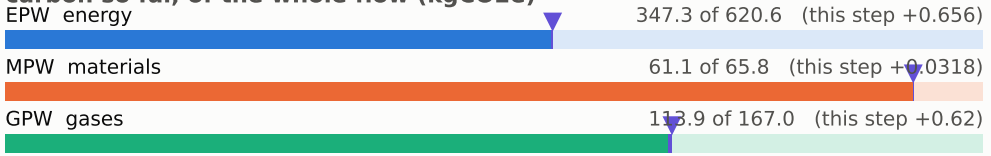


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

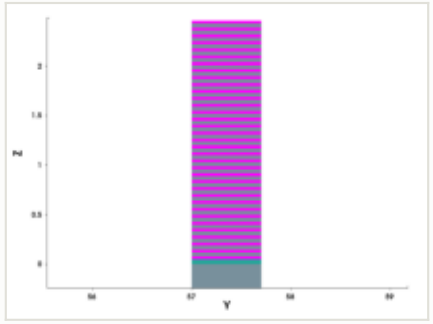
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

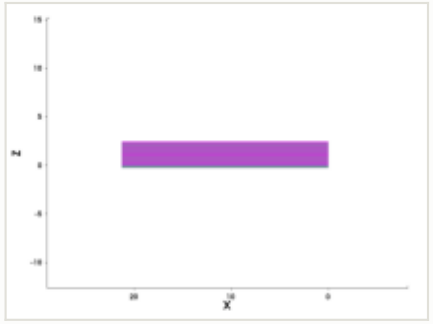


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



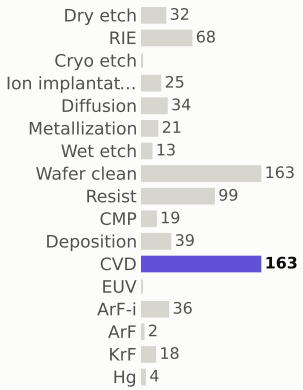
y cut at 57.351 µm



mask for this step

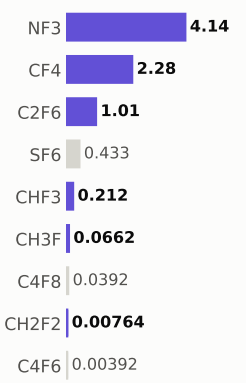
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



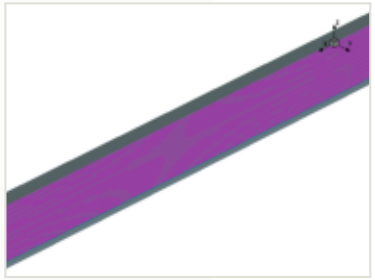
Step #9970: CVD

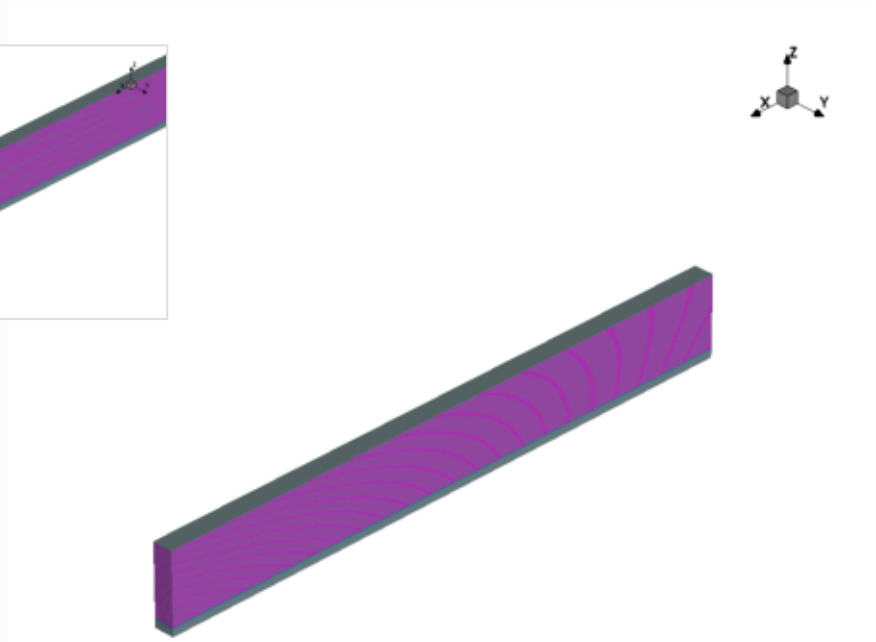
depo_isotropic · CVD

1.31	523.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



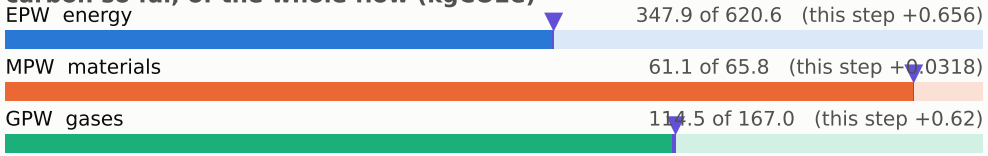


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

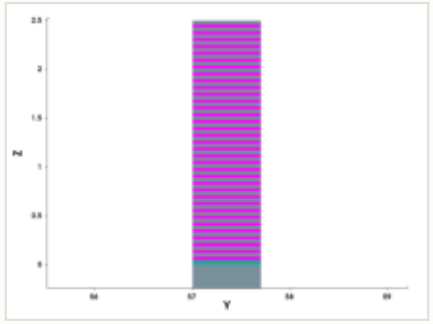
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

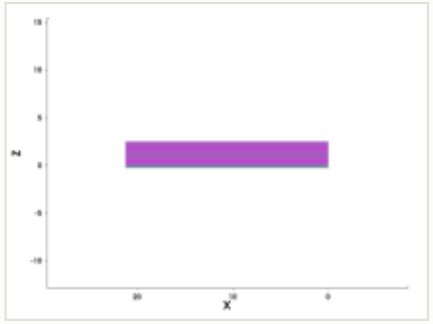


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



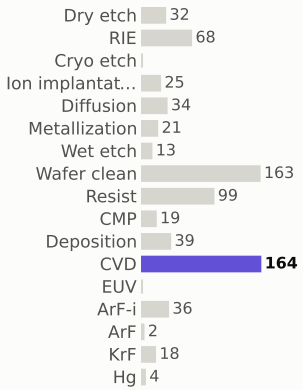
y cut at 57.351 µm



mask for this step

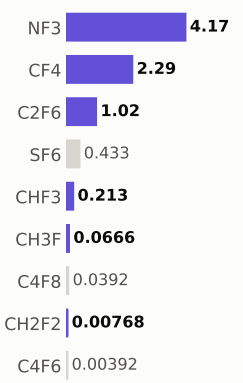
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



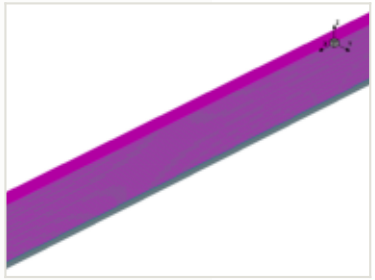
Step #9980: CVD

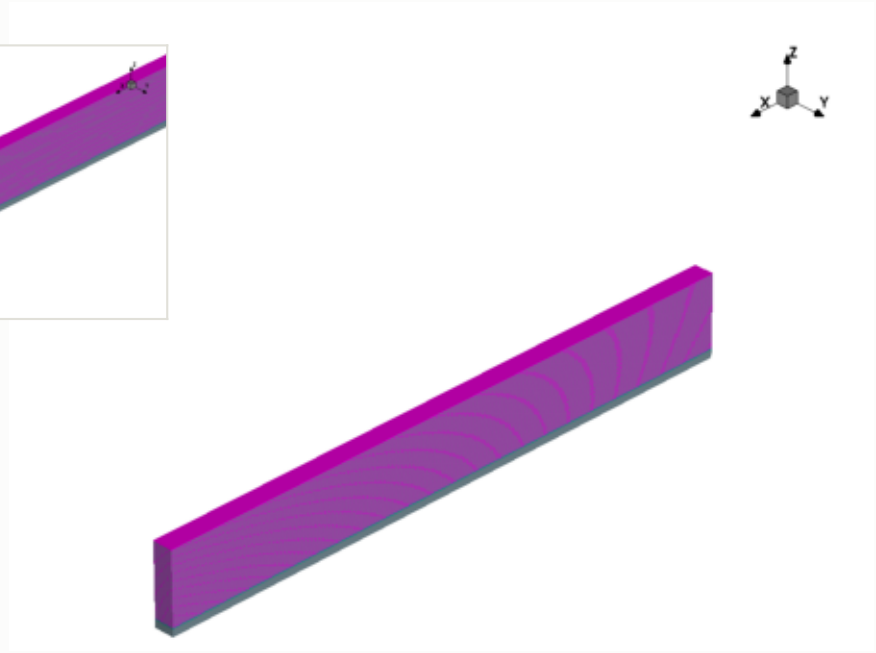
depo_isotropic · CVD

1.31	524.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



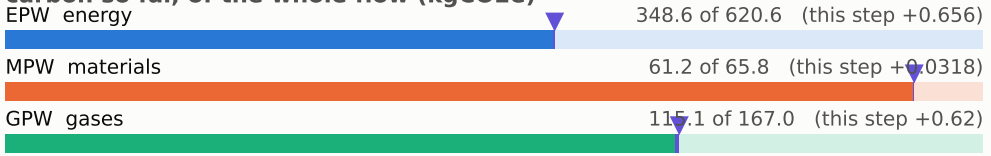


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

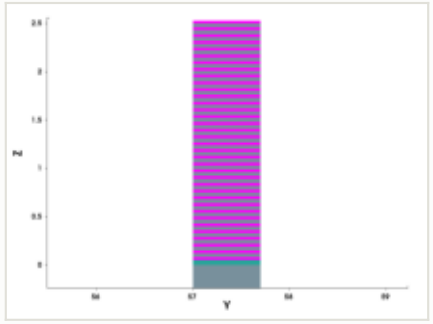
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

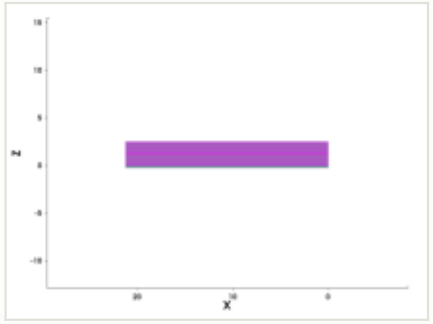


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



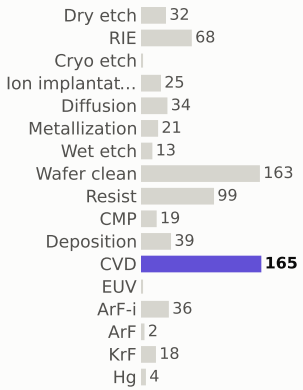
y cut at 57.351 µm



mask for this step

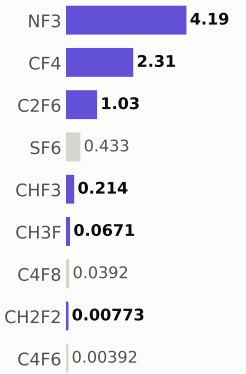
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



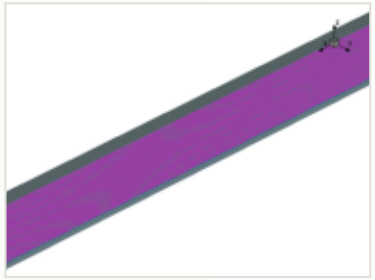
Step #9990: CVD

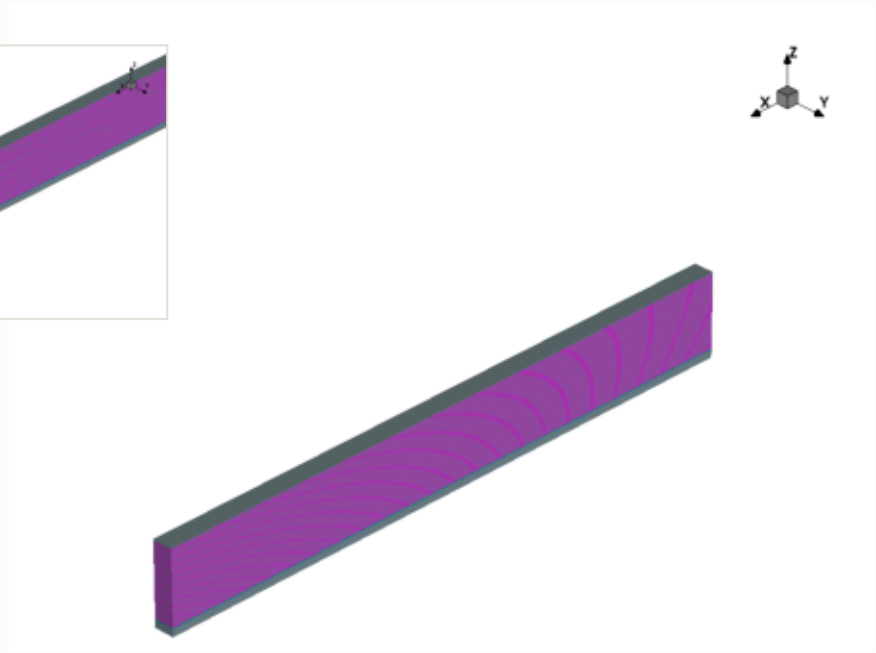
depo_isotropic · CVD

1.31	526.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



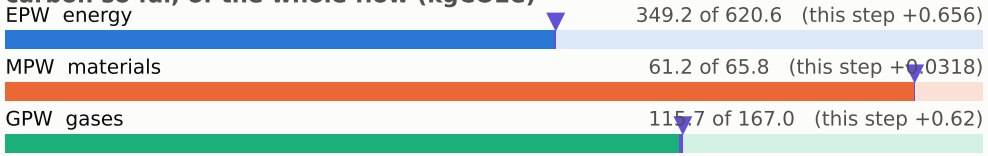


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

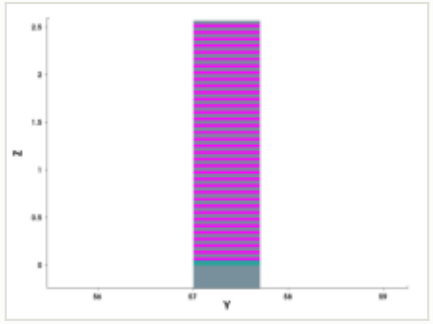
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

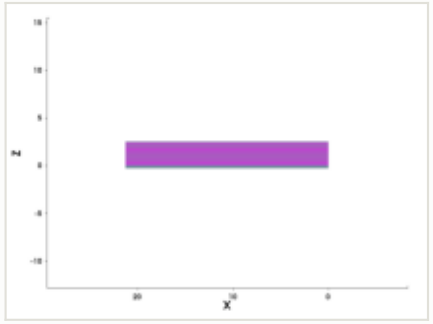


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



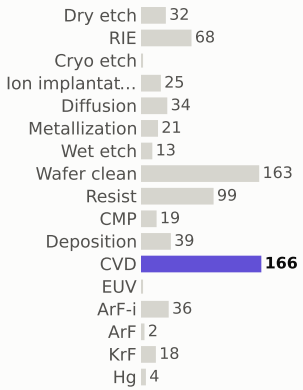
y cut at 57.351 µm



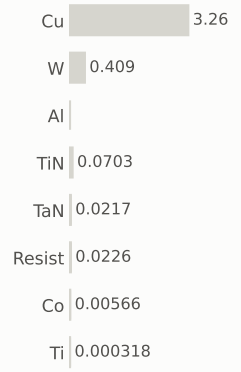
mask for this step

no mask for this step

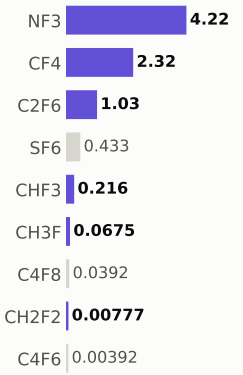
steps by process area



materials deposited (g)



gases emitted (g)



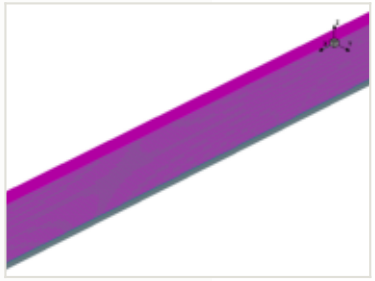
Step #10000: CVD

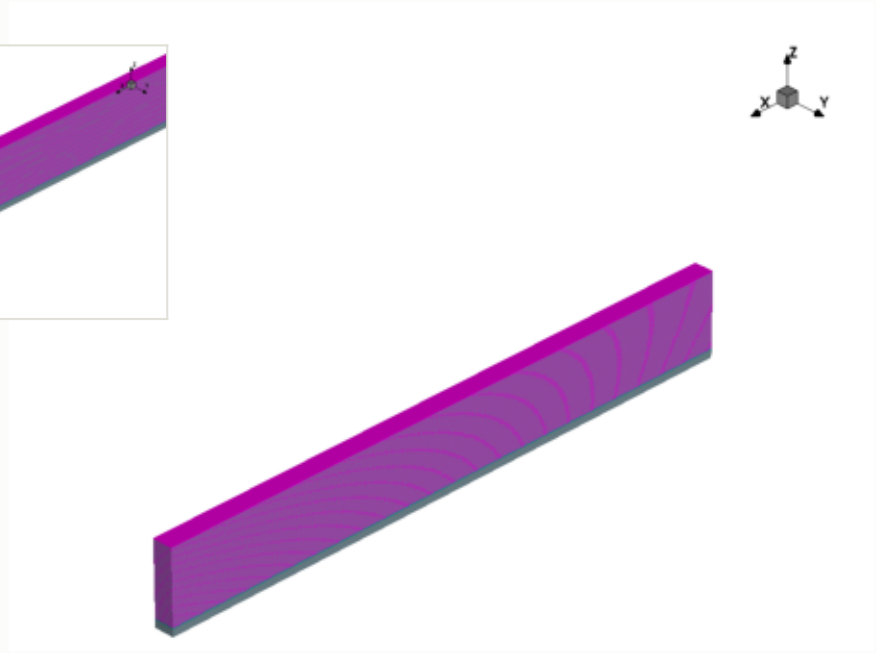
depo_isotropic · CVD

1.31	527.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



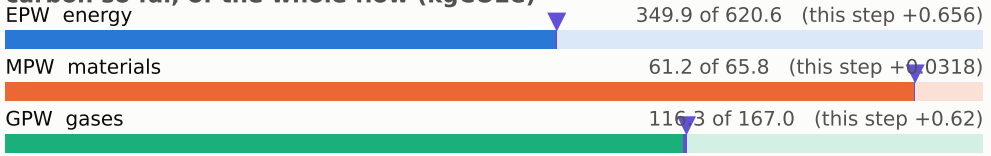


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

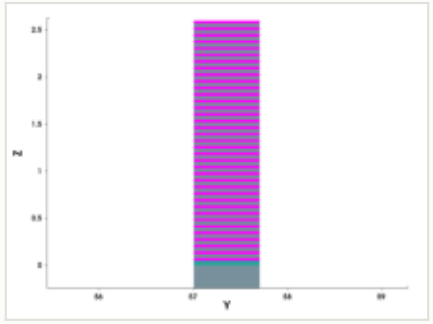
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

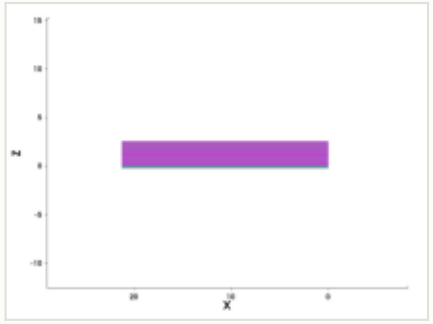


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



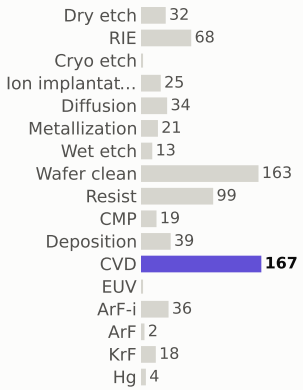
y cut at 57.351 µm



mask for this step

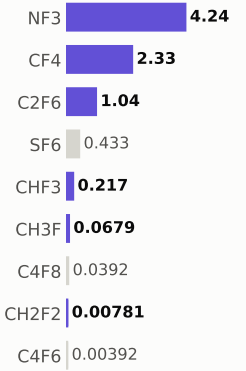
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



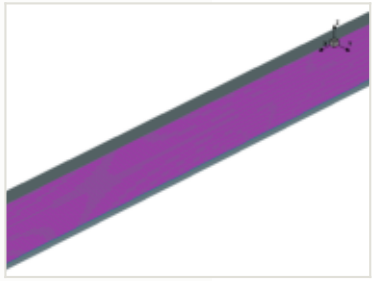
Step #10010: CVD

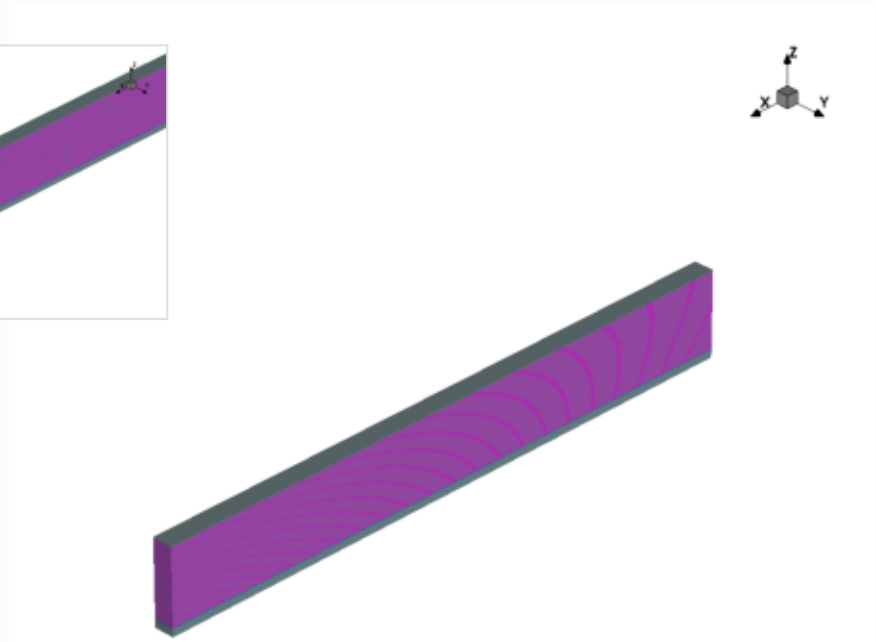
depo_isotropic · CVD

1.31	528.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



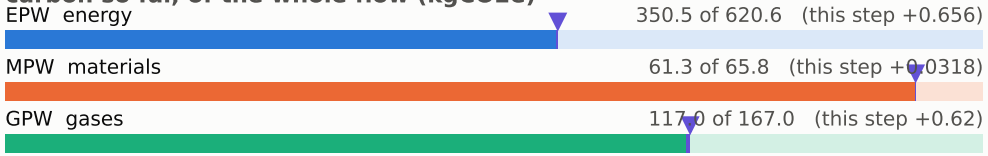


this step

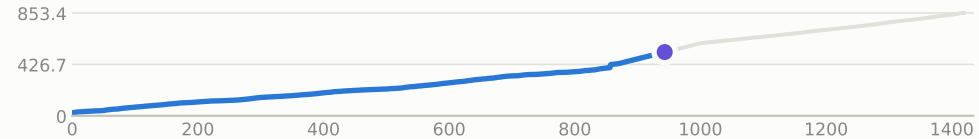
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

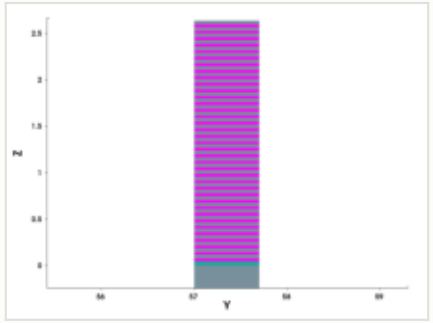
carbon so far, of the whole flow (kgCO2e)



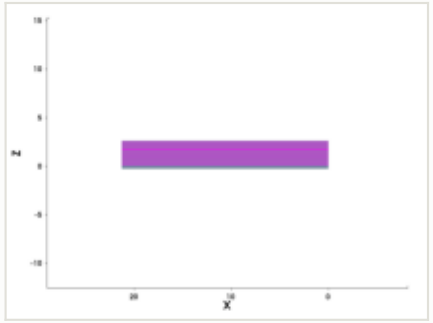
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



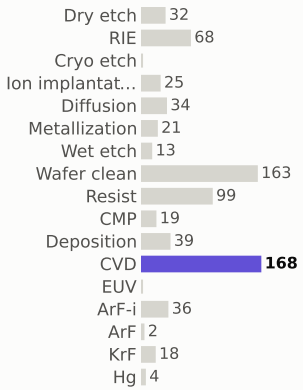
y cut at 57.351 µm



mask for this step

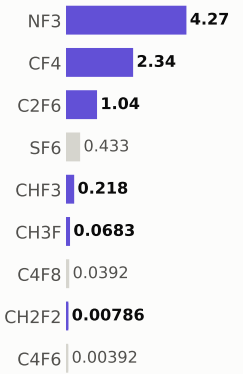
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



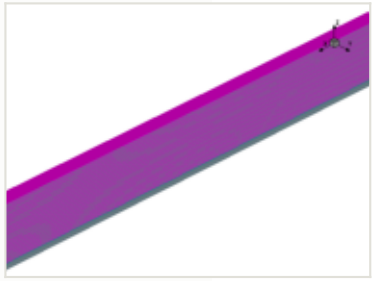
Step #10020: CVD

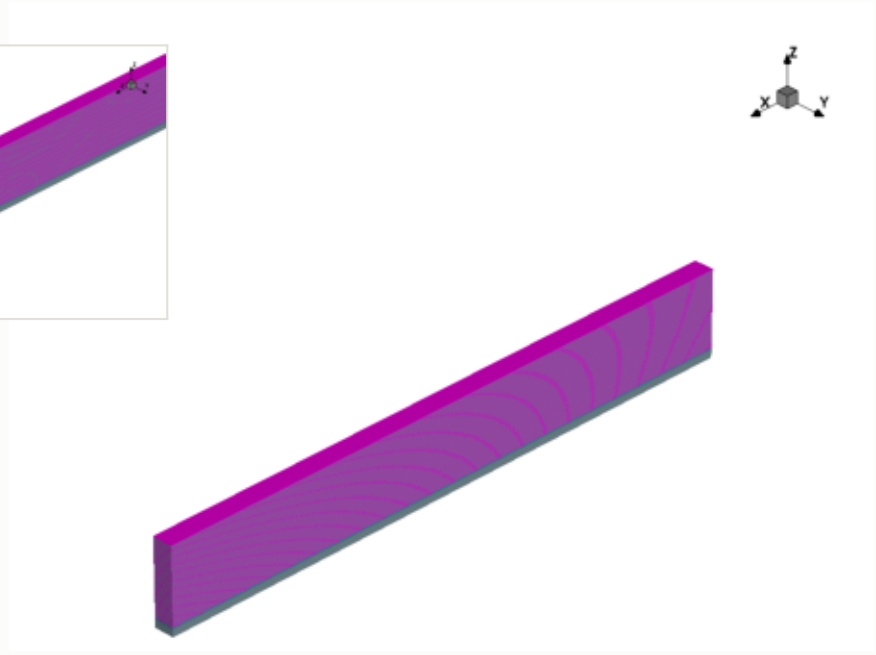
depo_isotropic · CVD

1.31	530.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



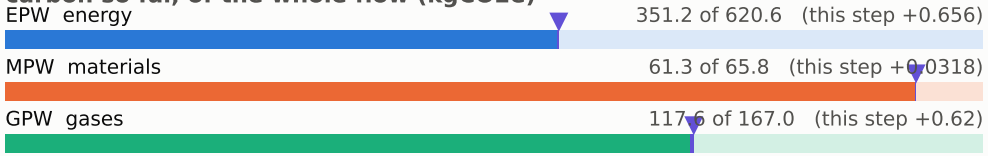


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

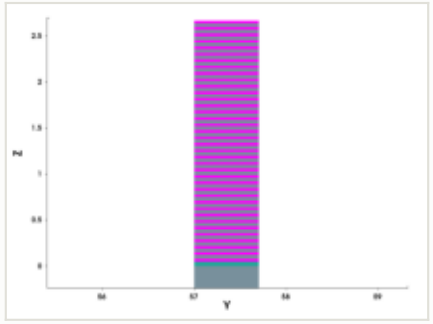
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

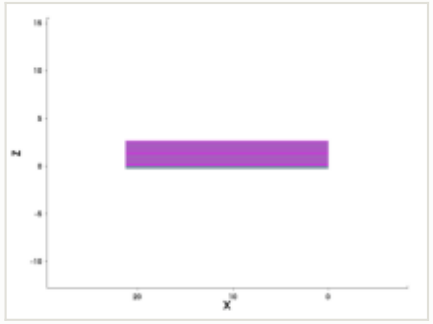


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



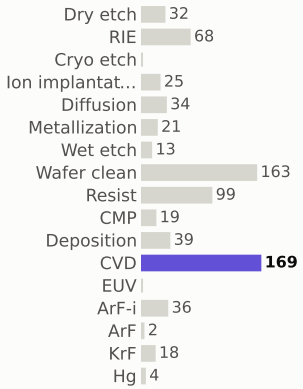
y cut at 57.351 µm



mask for this step

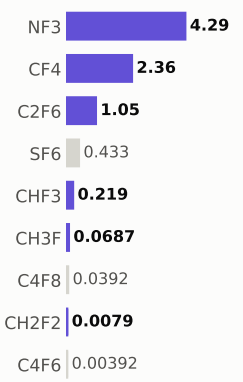
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



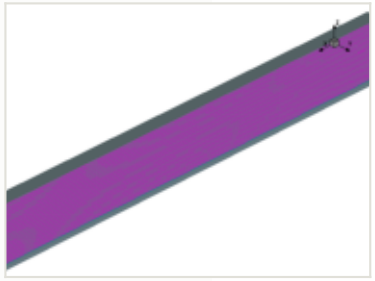
Step #10030: CVD

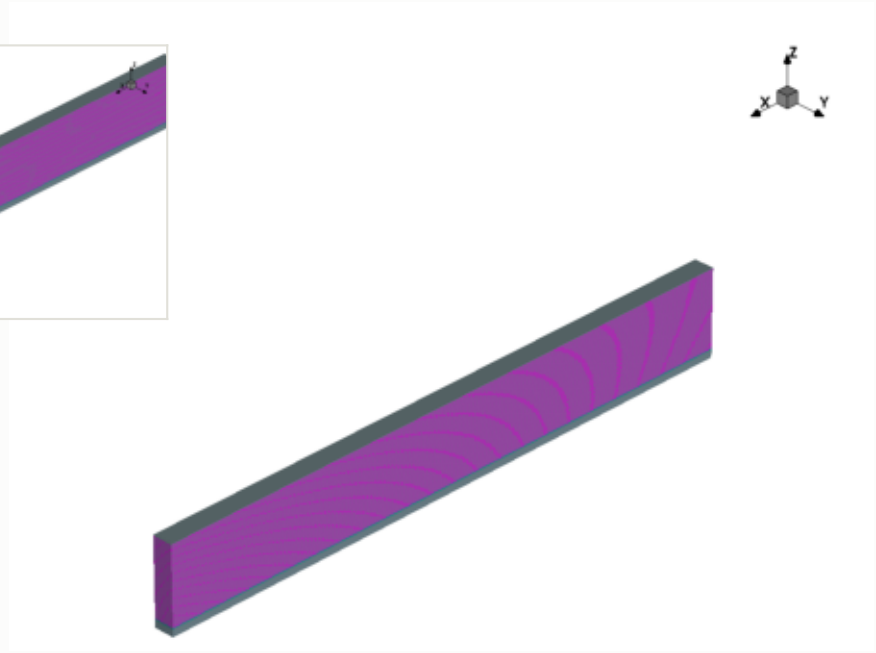
depo_isotropic · CVD

1.31	531.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



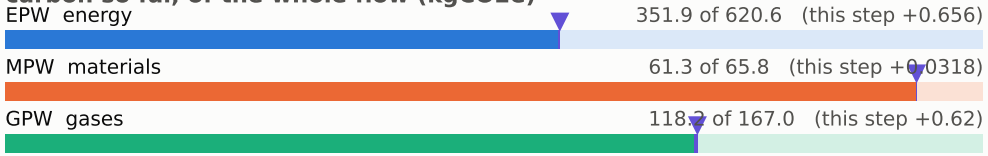


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

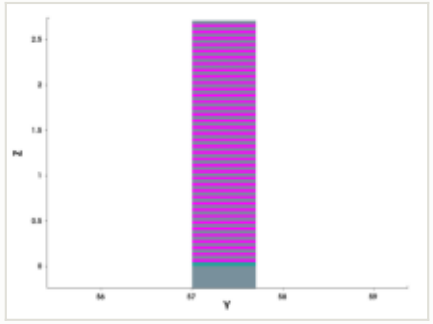
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

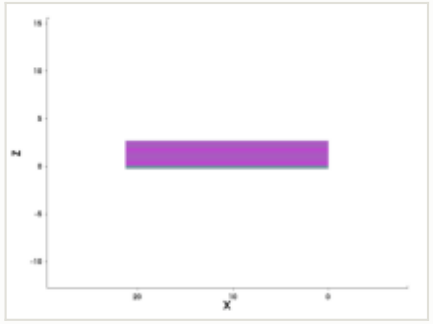


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



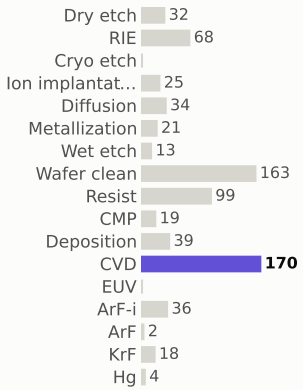
y cut at 57.351 µm



mask for this step

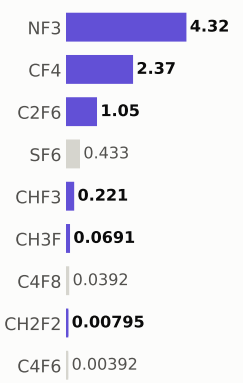
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



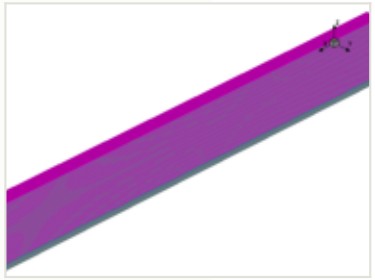
Step #10040: CVD

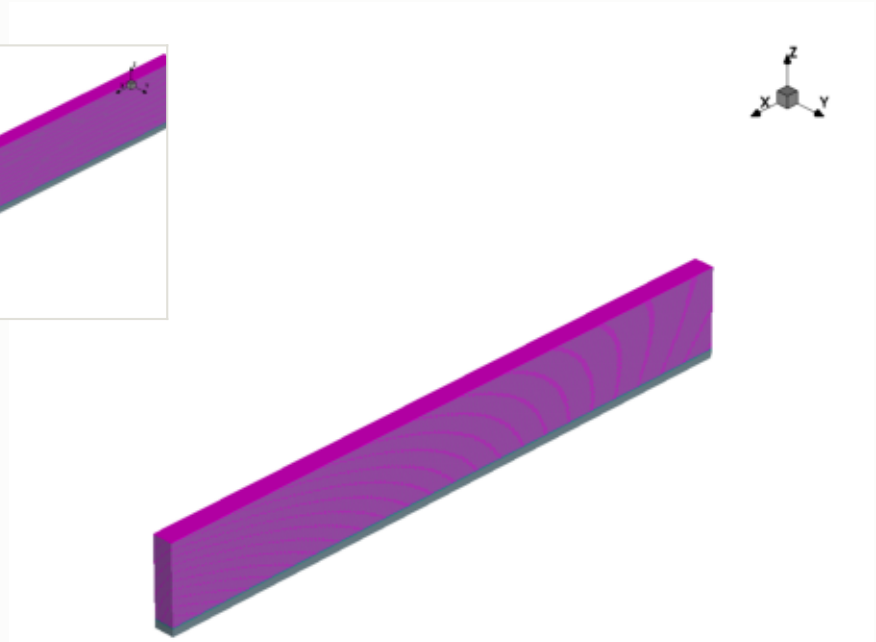
depo_isotropic · CVD

1.31	532.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



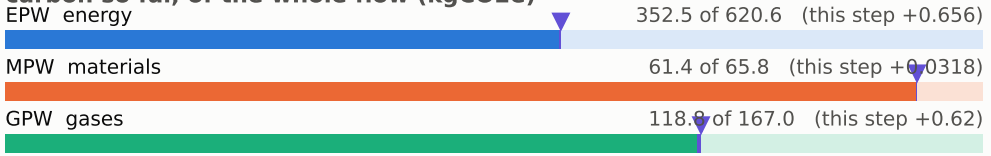


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

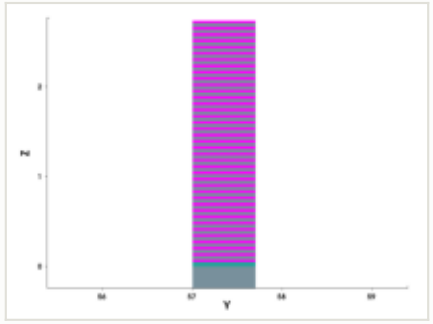
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

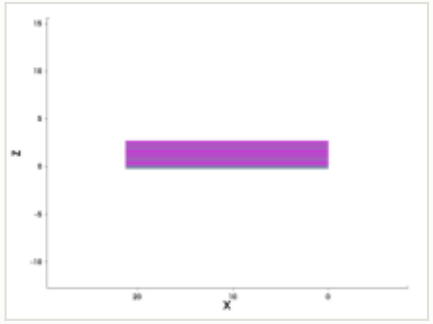


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



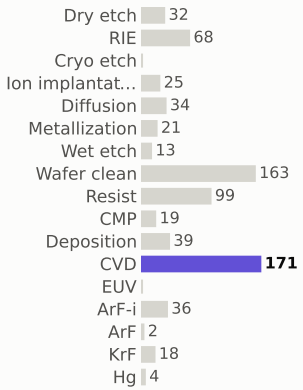
y cut at 57.351 µm



mask for this step

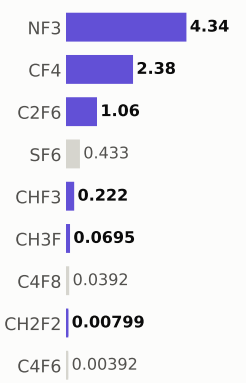
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



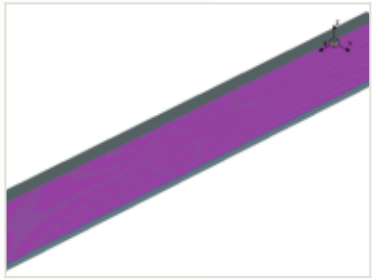
Step #10050: CVD

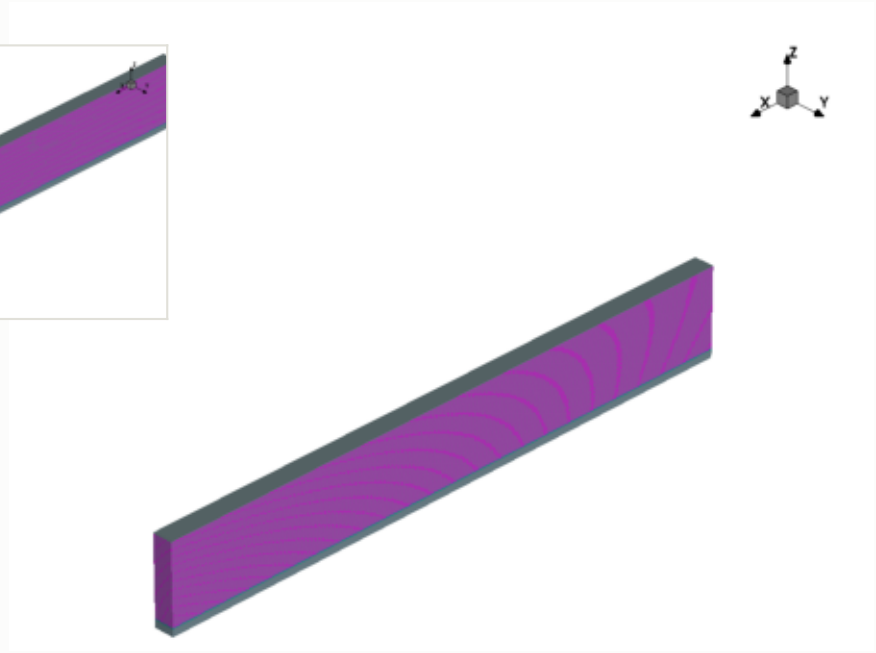
depo_isotropic · CVD

1.31	534.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



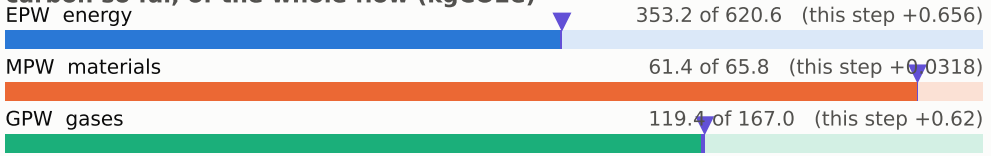


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

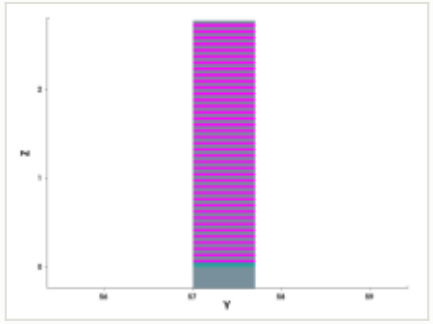
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

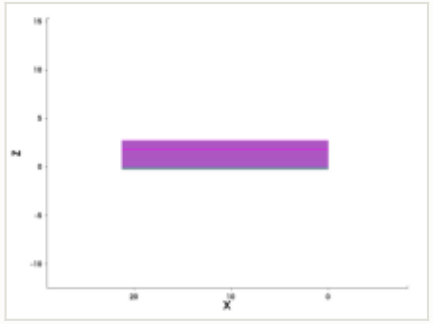


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



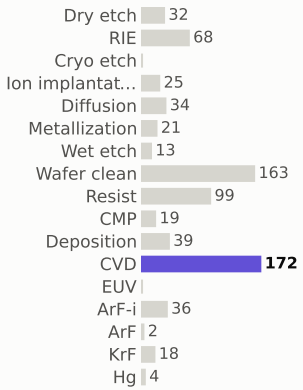
y cut at 57.351 µm



mask for this step

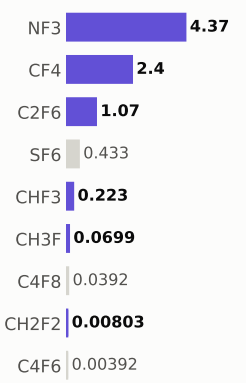
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



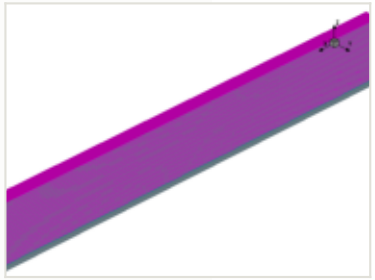
Step #10060: CVD

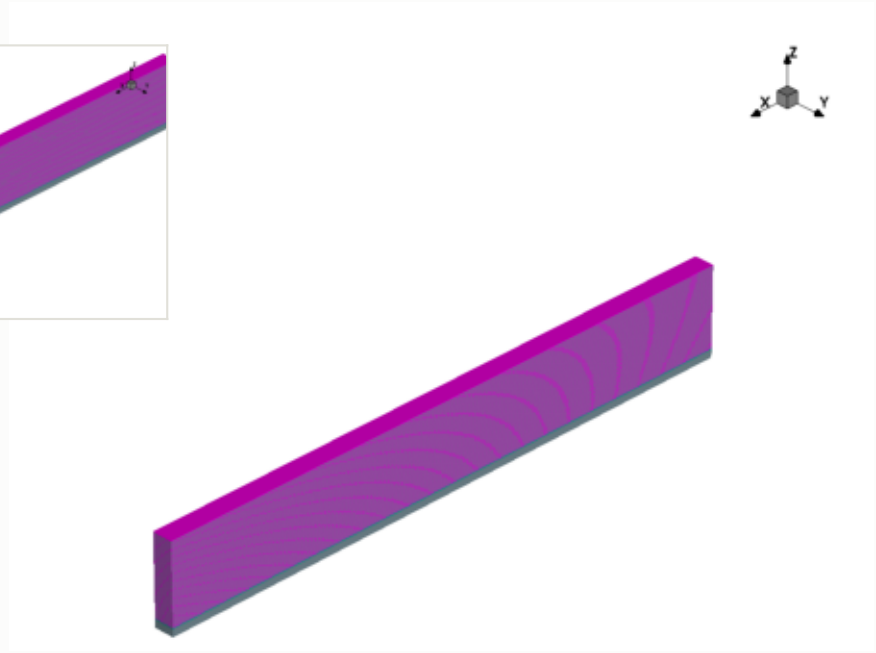
depo_isotropic · CVD

1.31	535.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



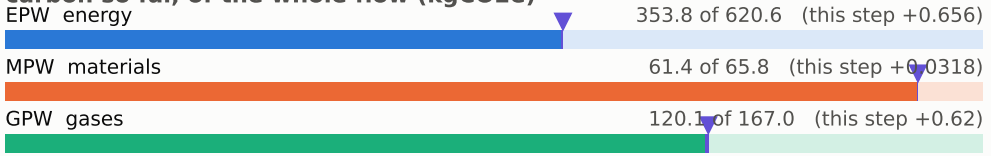


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

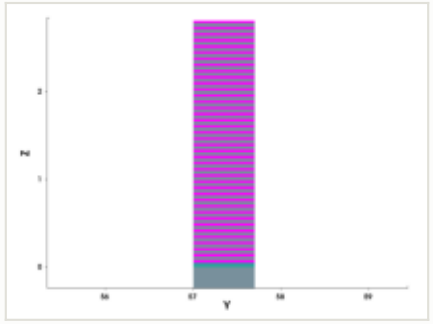
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

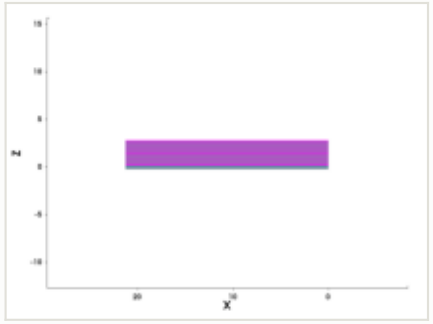


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



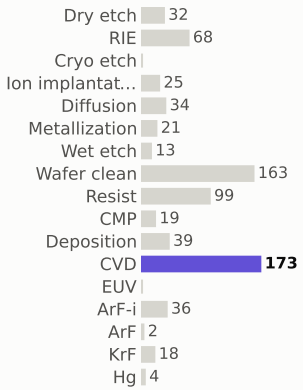
y cut at 57.351 µm



mask for this step

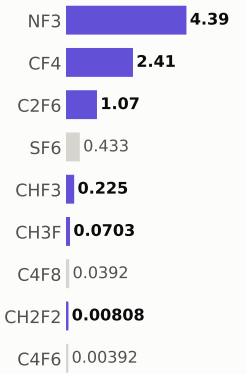
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



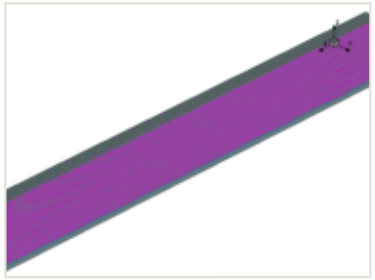
Step #10070: CVD

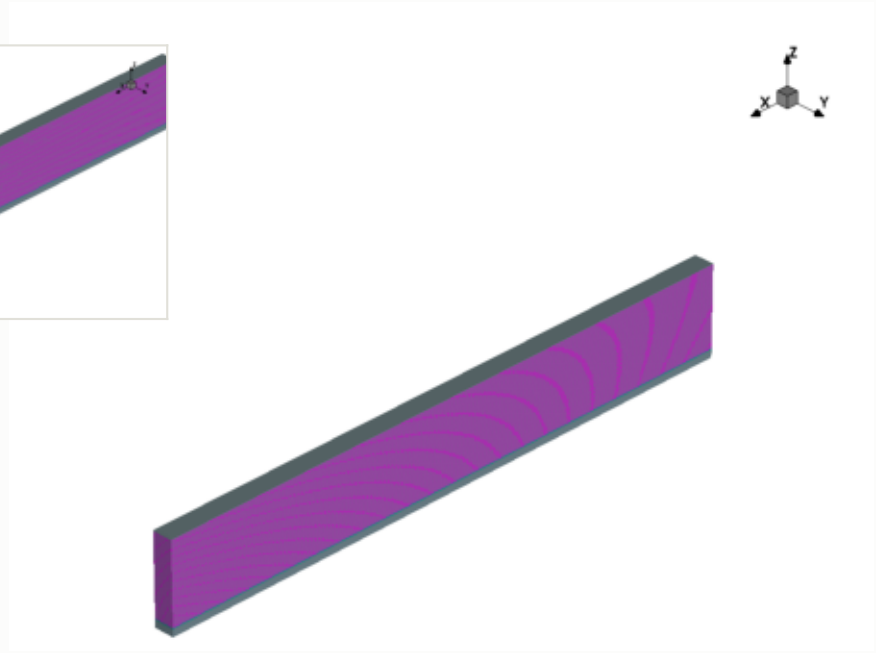
depo_isotropic · CVD

1.31	536.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



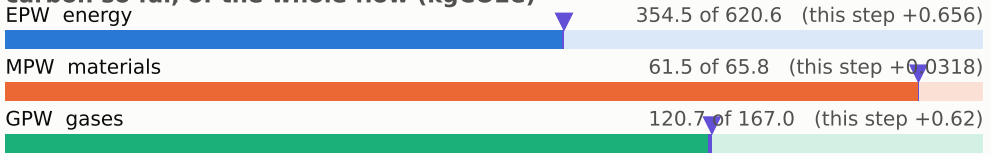


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

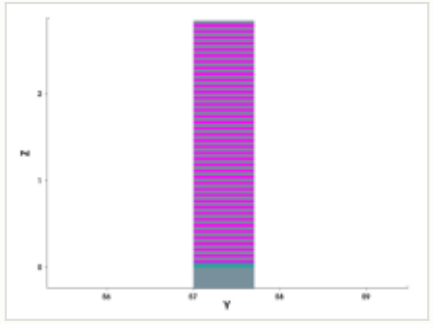
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

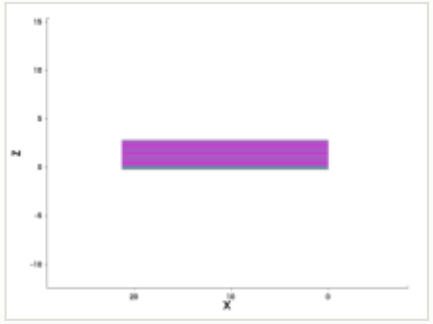


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



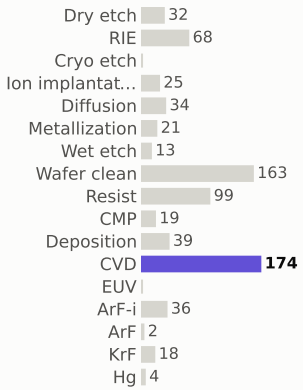
y cut at 57.351 µm



mask for this step

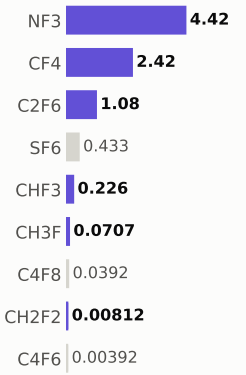
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



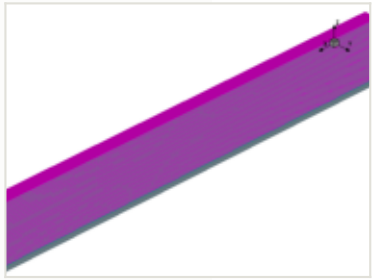
Step #10080: CVD

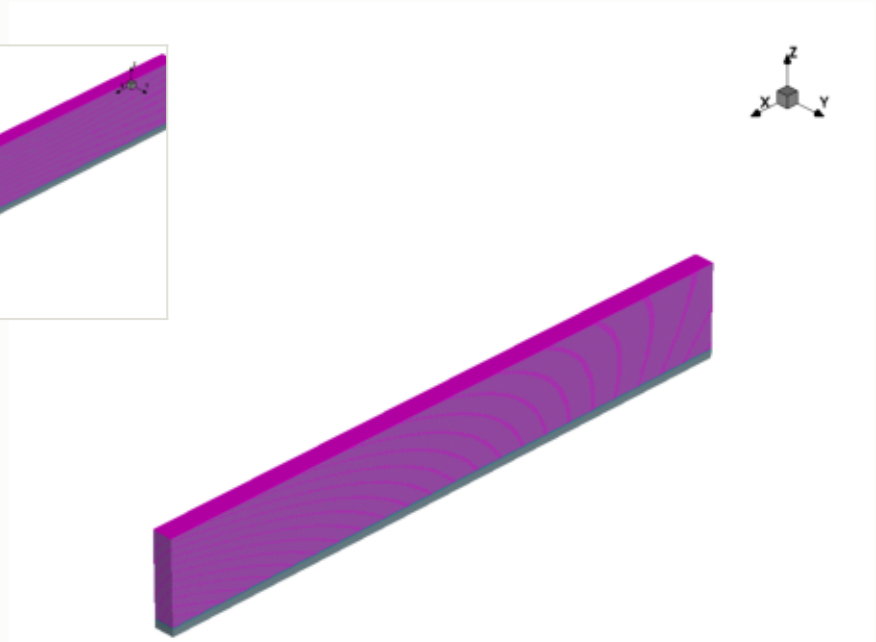
depo_isotropic · CVD

1.31	537.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

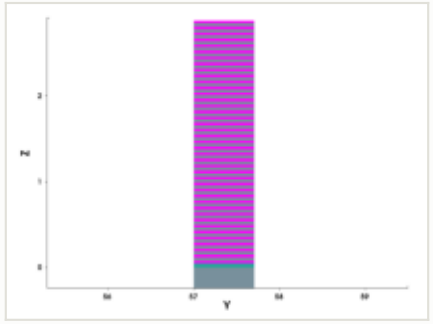
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

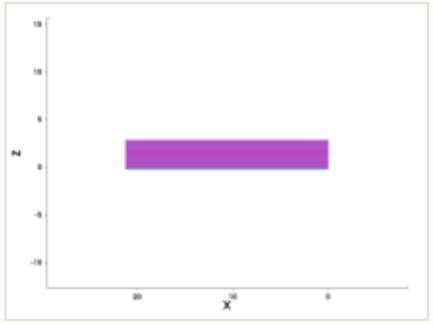
carbon so far, of the whole flow (kgCO2e)

embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



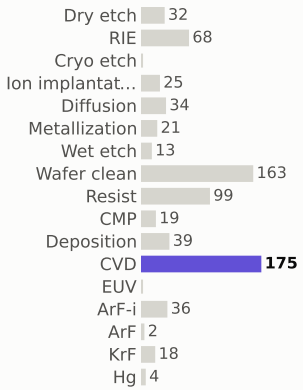
y cut at 57.351 µm



mask for this step

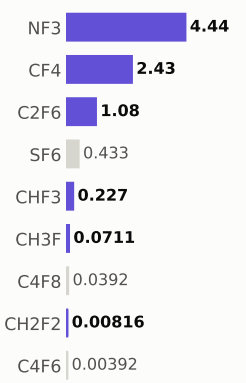
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



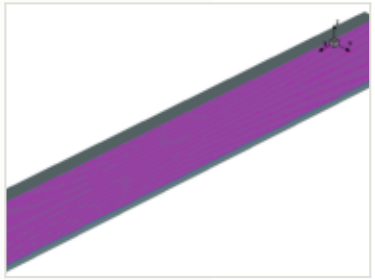
Step #10090: CVD

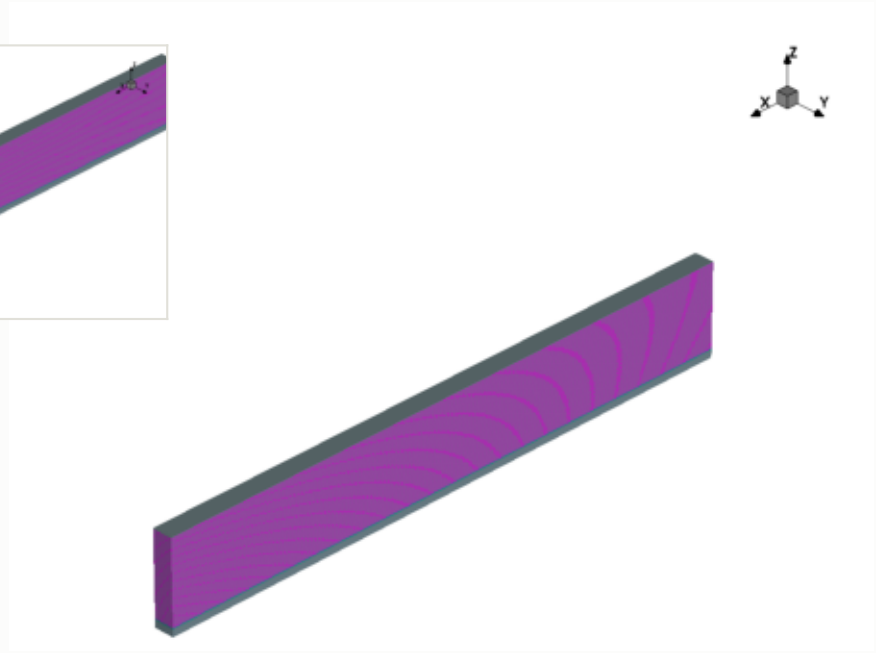
depo_isotropic · CVD

1.31	539.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



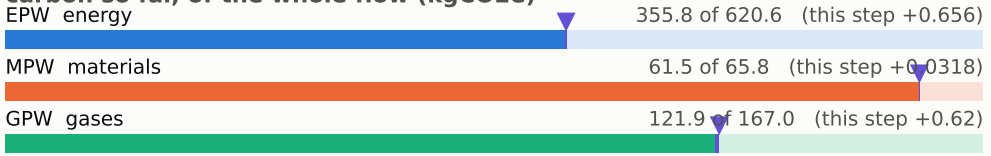


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

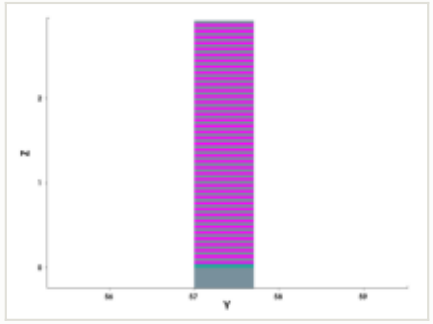
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

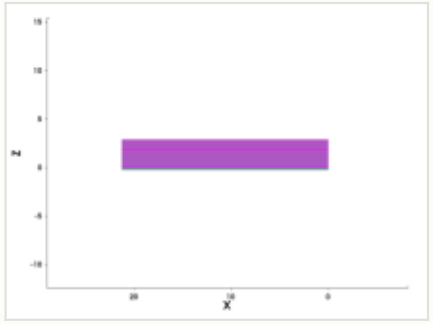


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



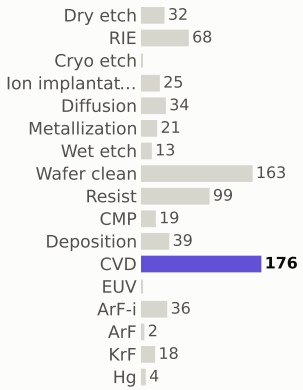
y cut at 57.351 µm



mask for this step

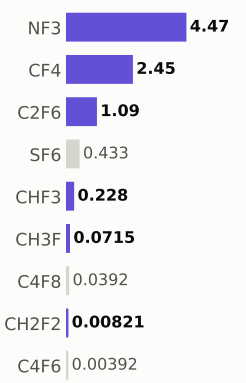
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



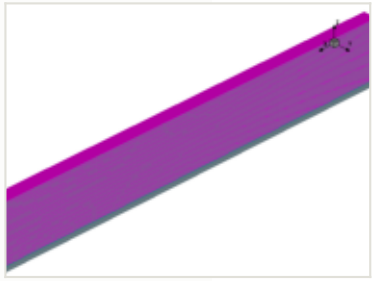
Step #10100: CVD

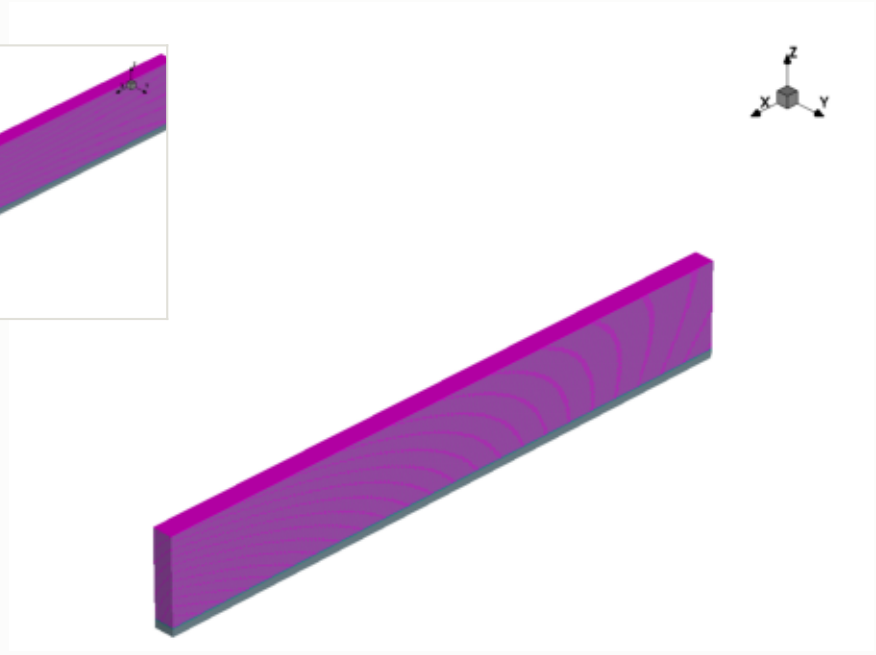
depo_isotropic · CVD

1.31	540.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



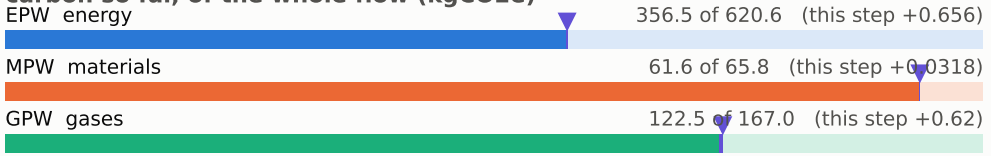


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

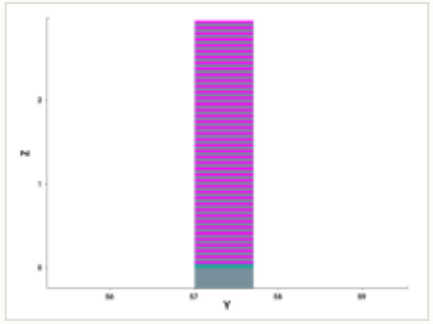
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

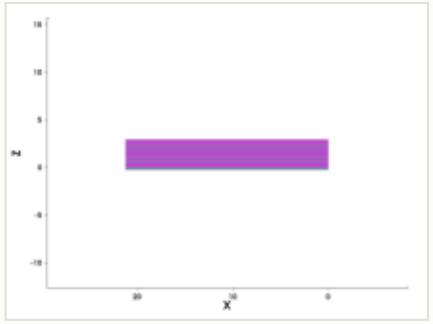


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



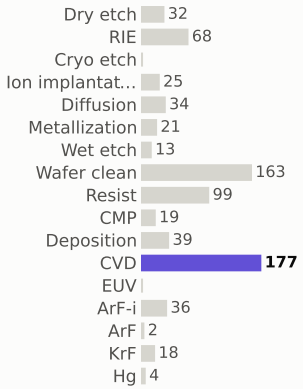
y cut at 57.351 µm



mask for this step

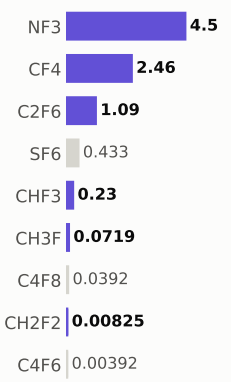
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



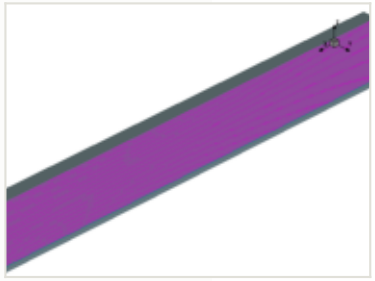
Step #10110: CVD

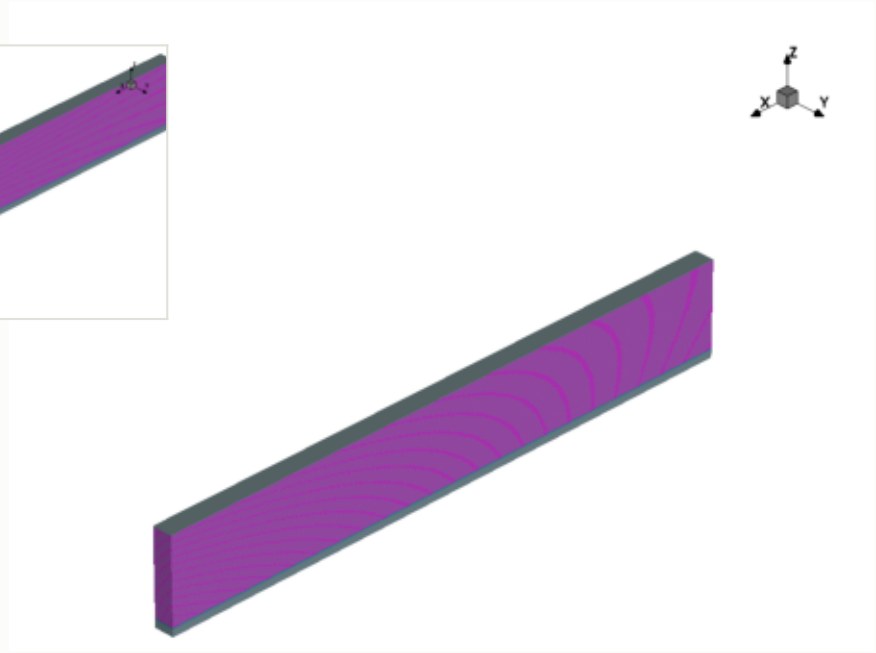
depo_isotropic · CVD

1.31	541.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



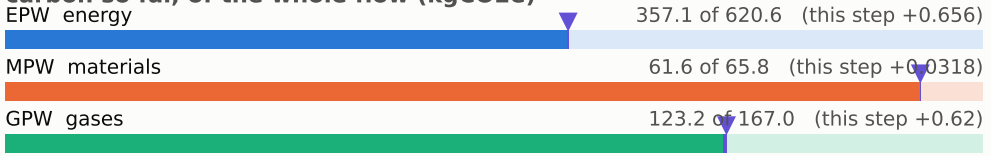


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

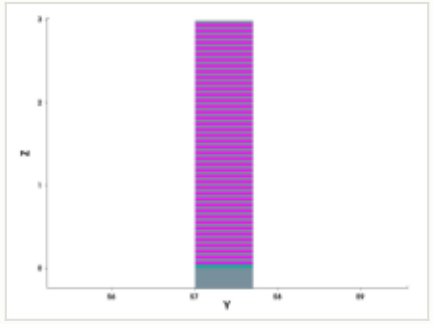
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

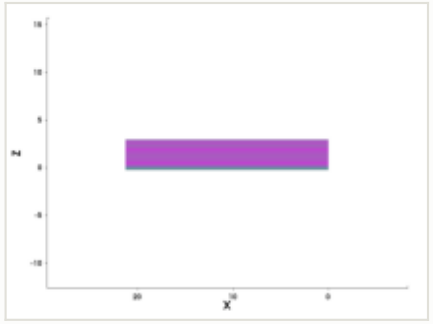


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



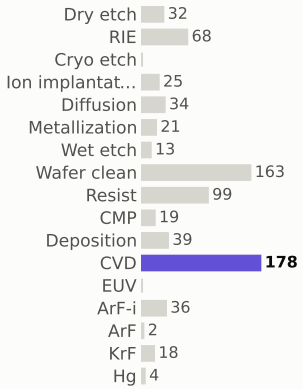
y cut at 57.351 µm



mask for this step

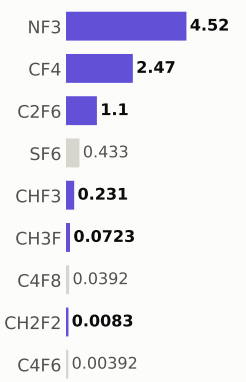
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



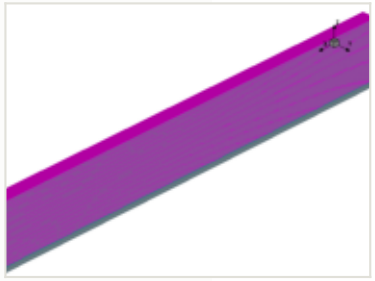
Step #10120: CVD

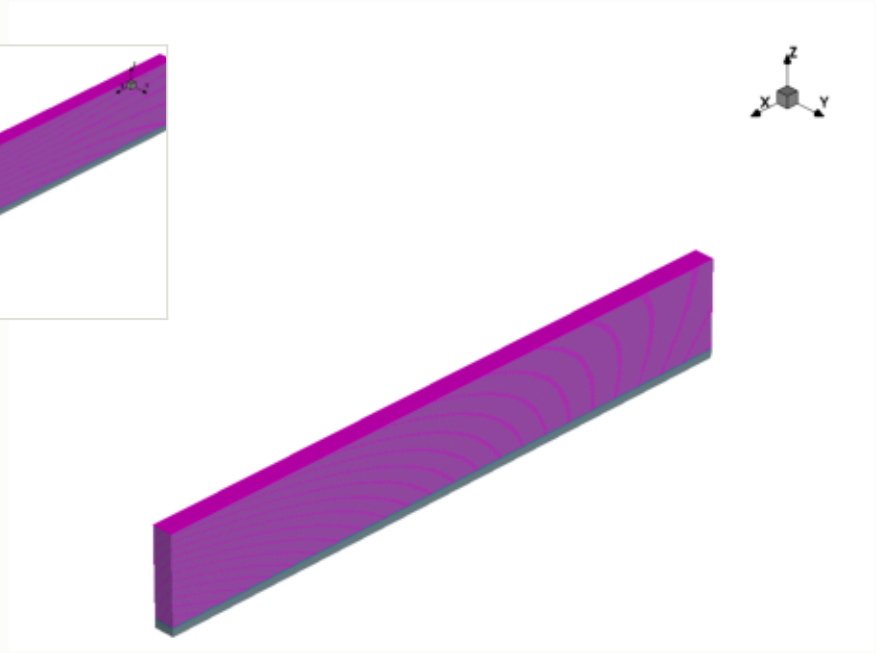
depo_isotropic · CVD

1.31	543.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



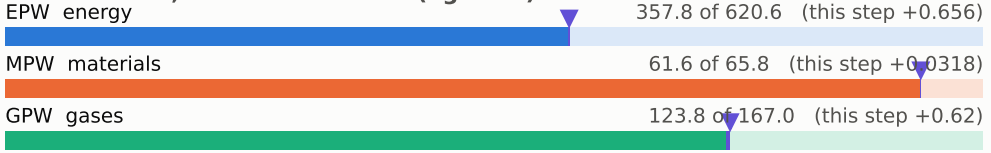


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

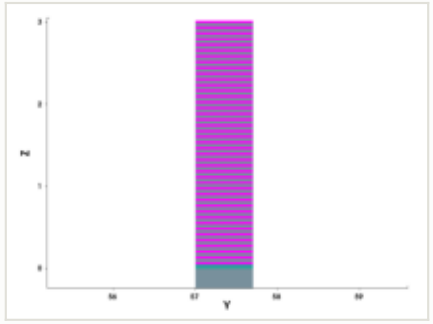
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

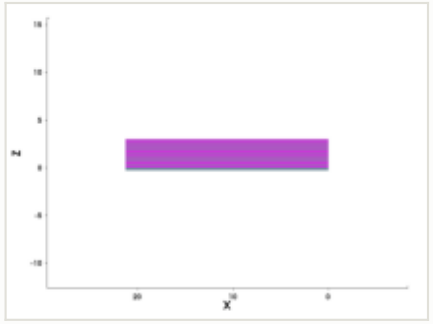


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



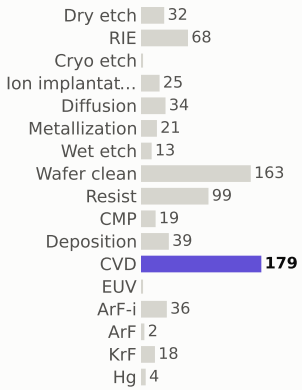
y cut at 57.351 µm



mask for this step

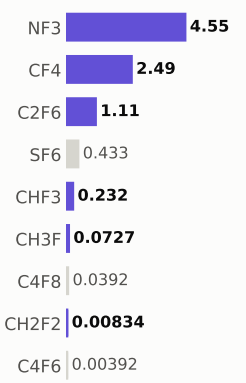
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



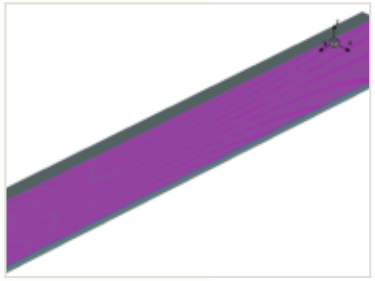
Step #10130: CVD

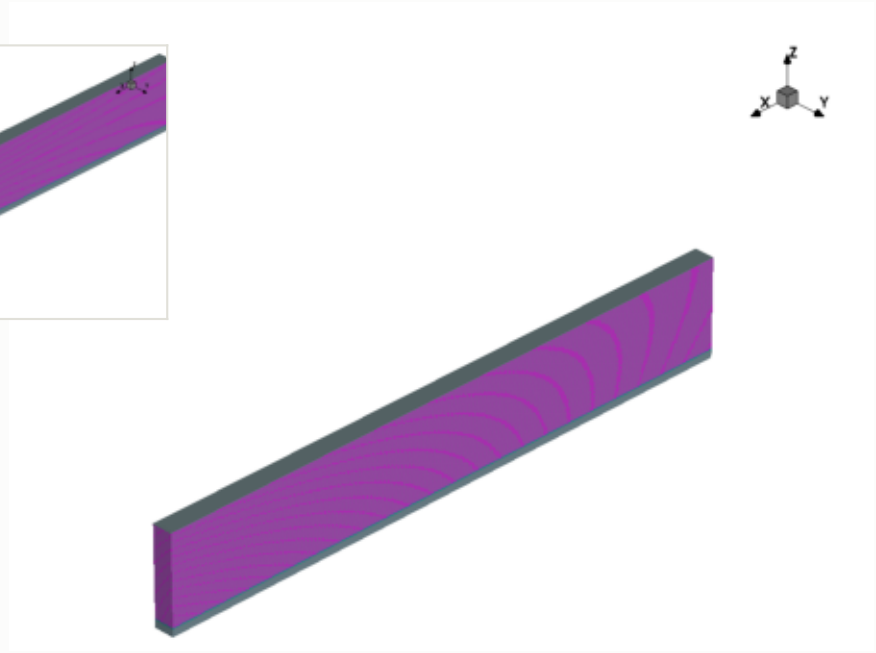
depo_isotropic · CVD

1.31	544.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



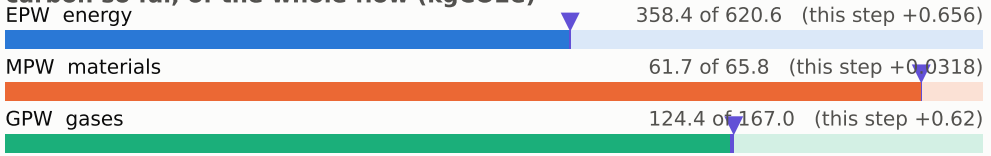


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

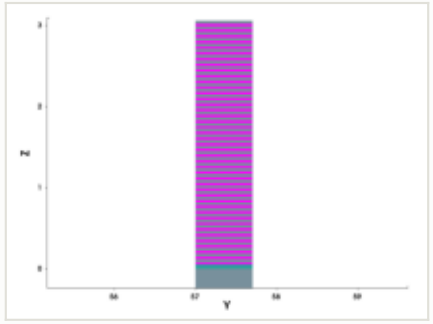
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

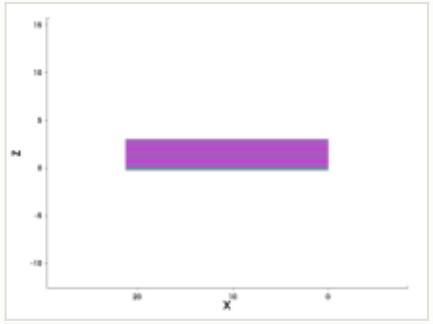


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



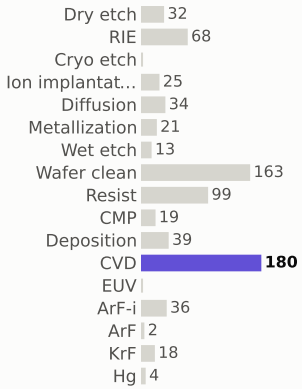
y cut at 57.351 µm



mask for this step

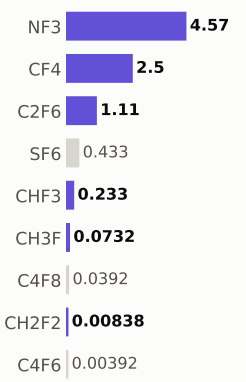
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



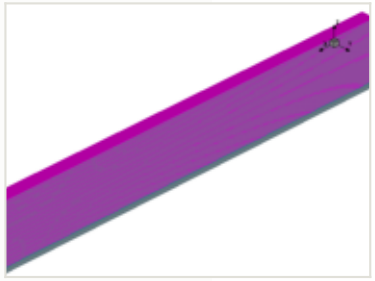
Step #10140: CVD

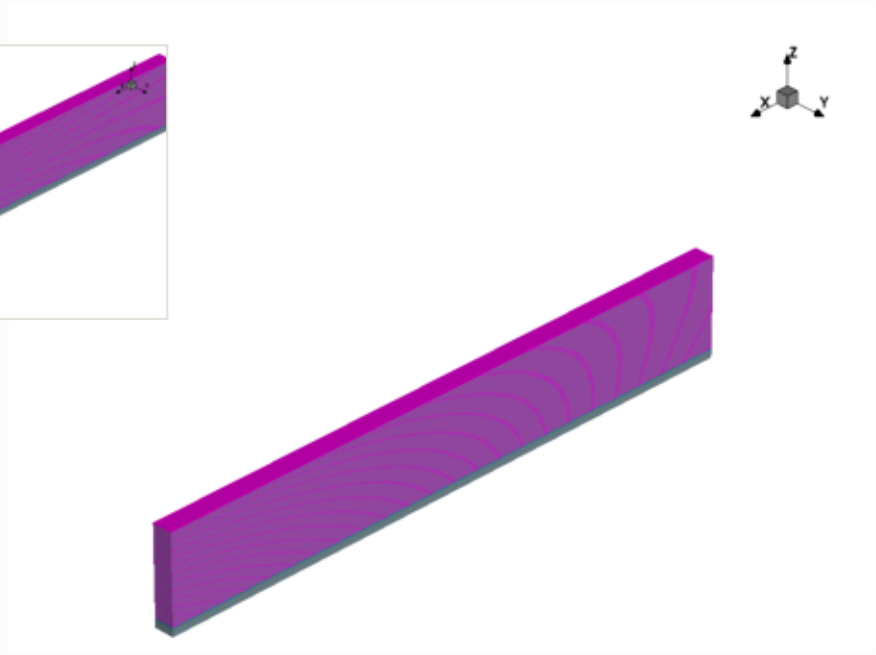
depo_isotropic · CVD

1.31	545.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



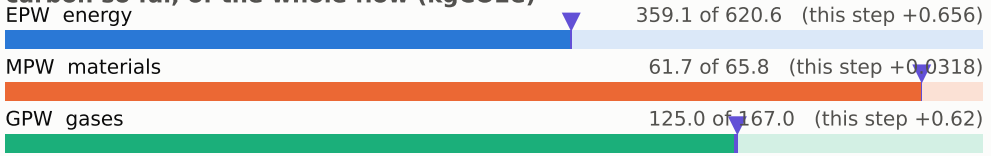


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

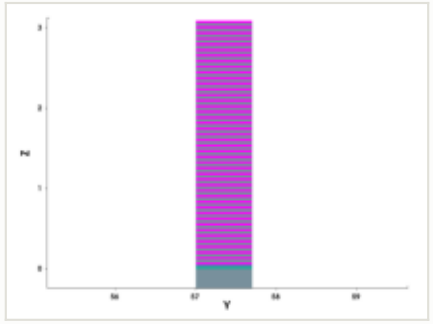
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

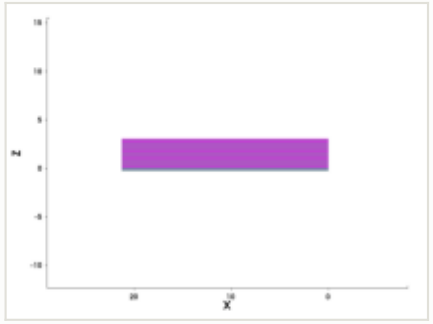


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



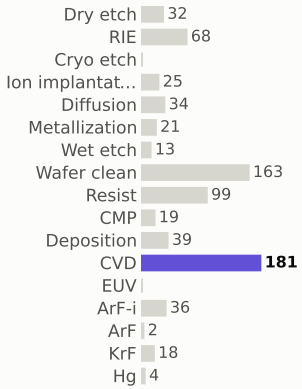
y cut at 57.351 µm



mask for this step

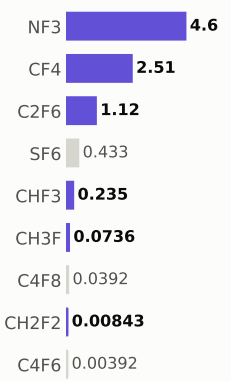
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



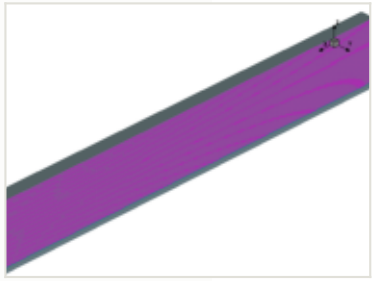
Step #10150: CVD

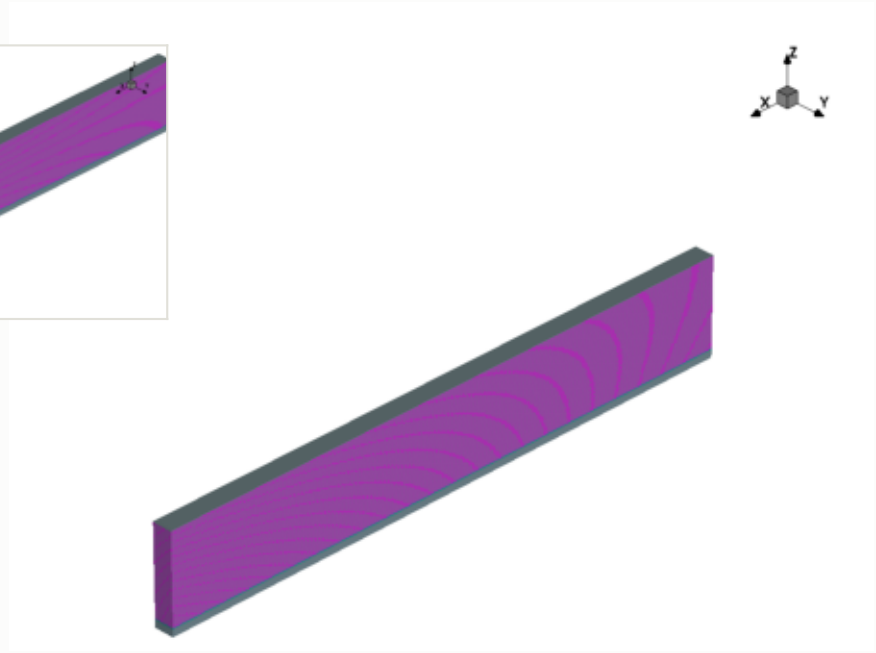
depo_isotropic · CVD

1.31	547.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



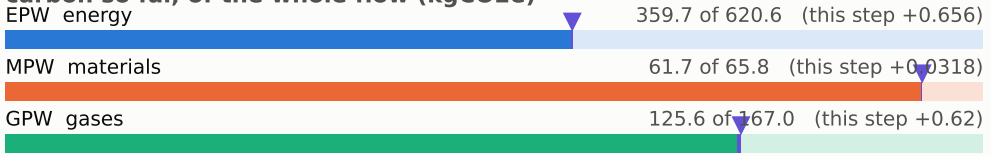


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

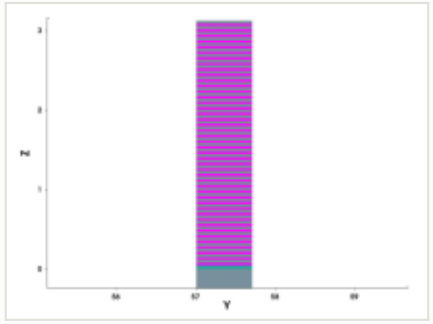
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

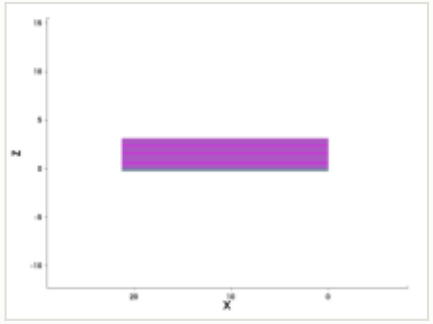


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



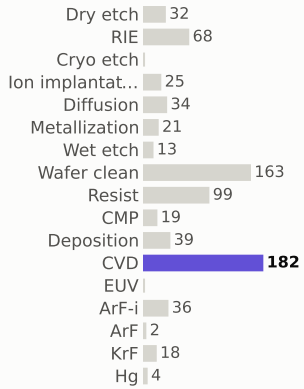
y cut at 57.351 µm



mask for this step

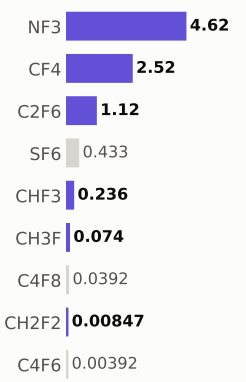
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



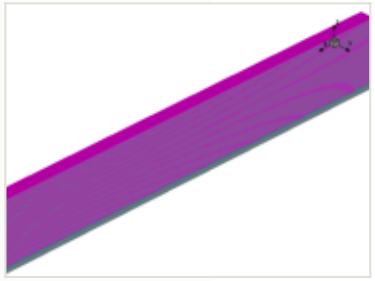
Step #10160: CVD

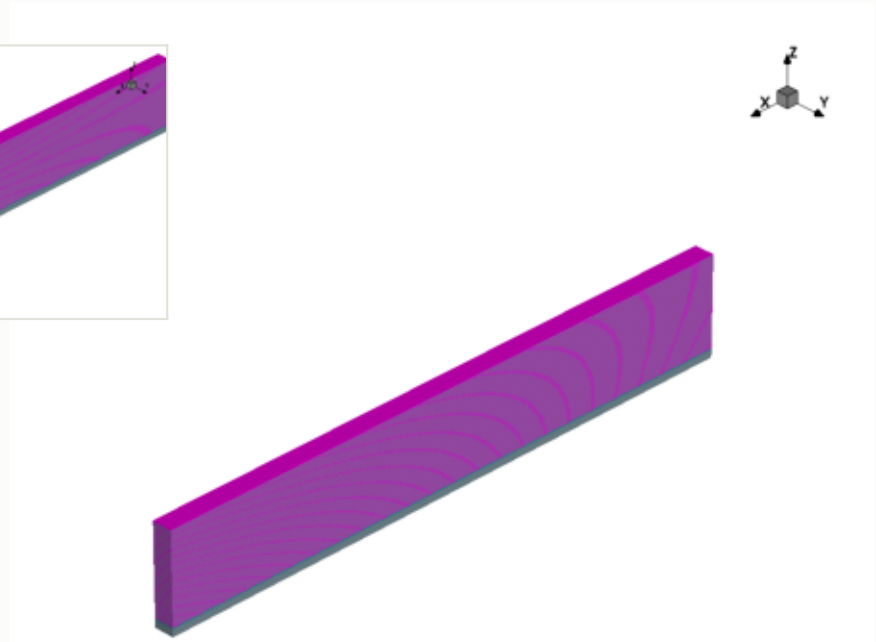
depo_isotropic · CVD

1.31	548.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



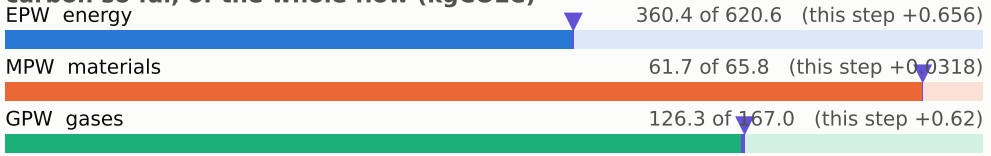


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

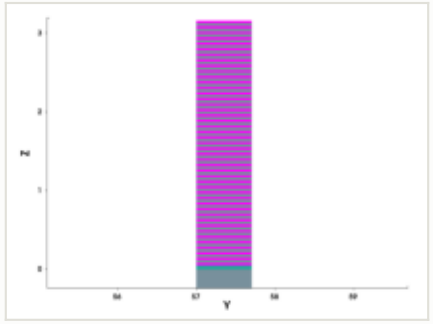
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

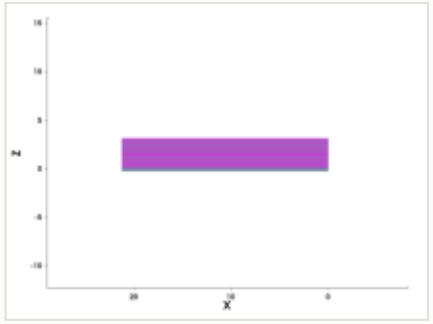


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



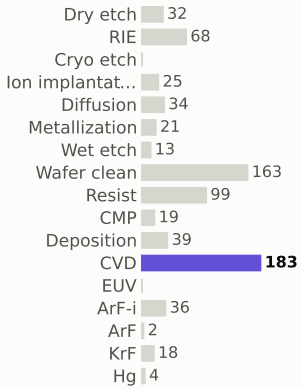
y cut at 57.351 µm



mask for this step

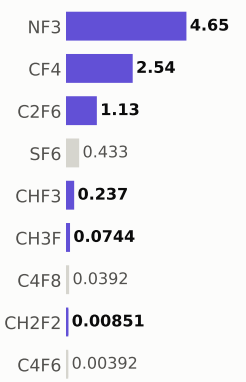
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



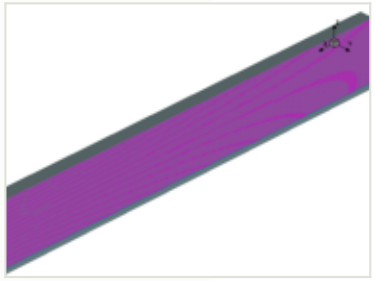
Step #10170: CVD

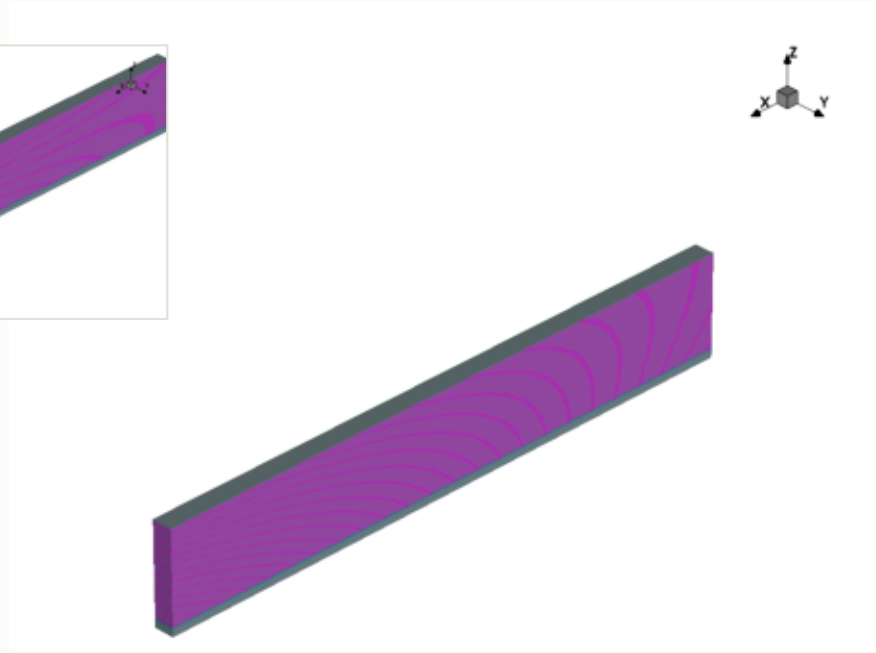
depo_isotropic · CVD

1.31	549.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



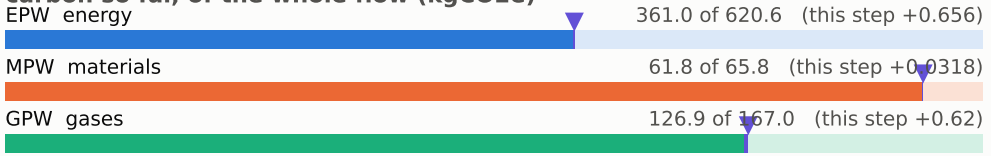


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

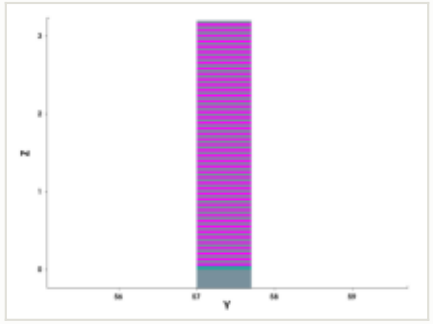
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

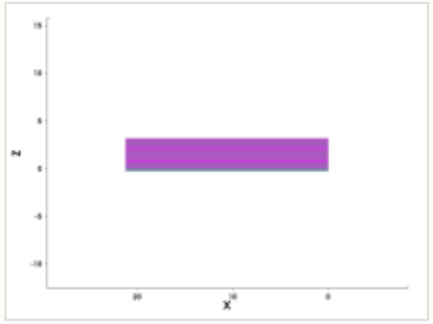


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



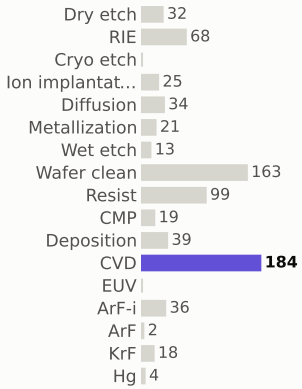
y cut at 57.351 µm



mask for this step

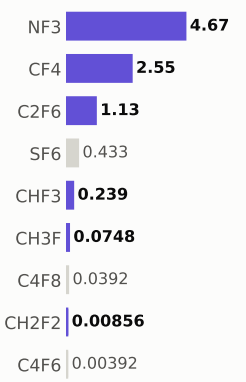
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



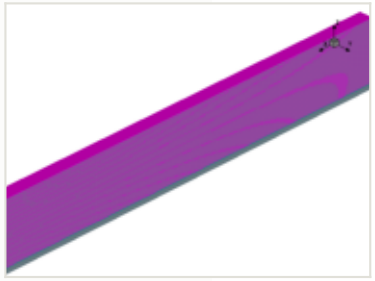
Step #10180: CVD

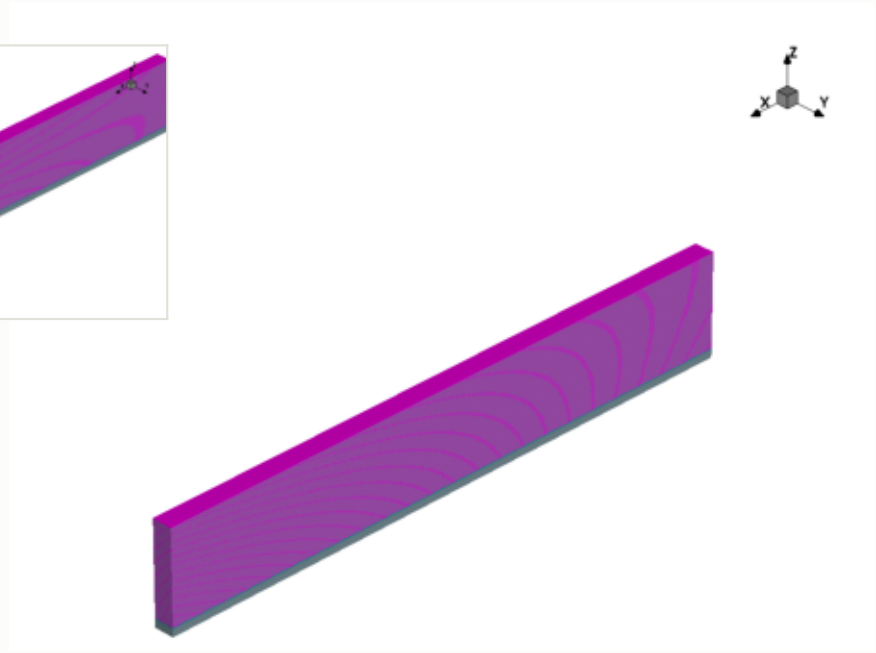
depo_isotropic · CVD

1.31	551.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



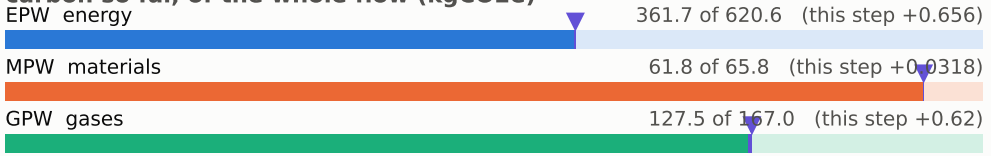


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

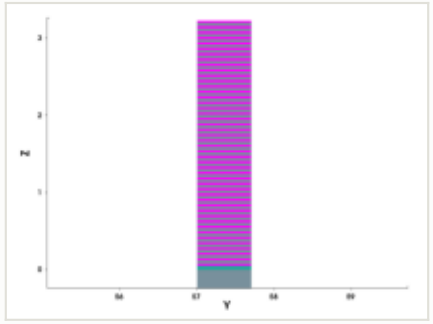
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

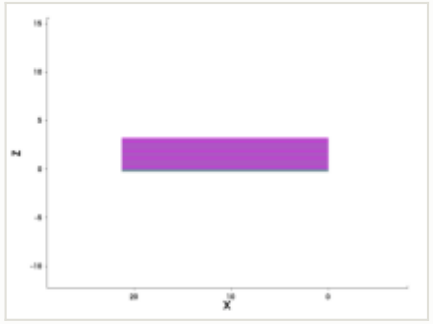


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



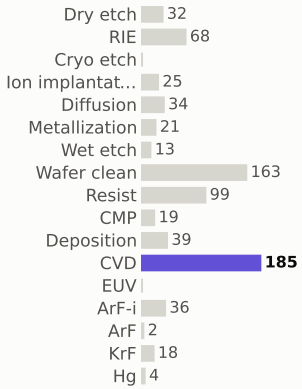
y cut at 57.351 µm



mask for this step

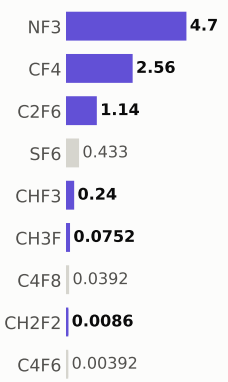
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



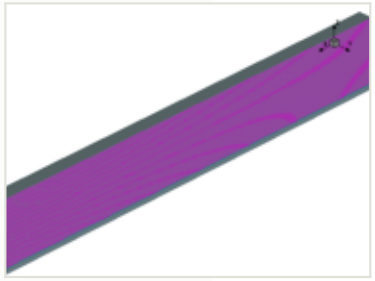
Step #10190: CVD

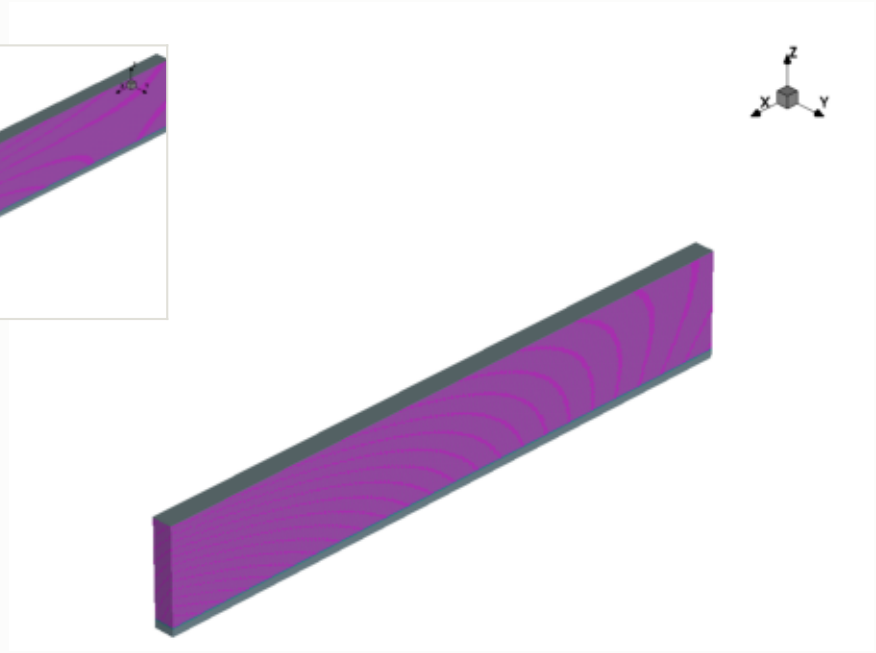
depo_isotropic · CVD

1.31	552.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



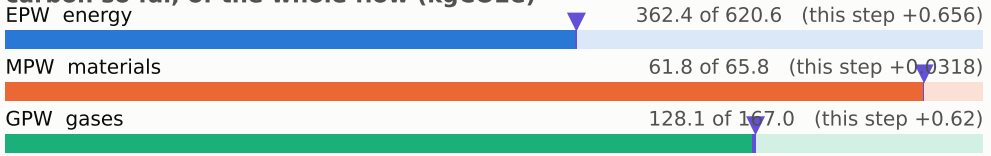


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

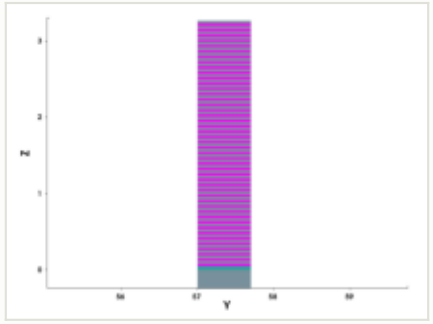
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

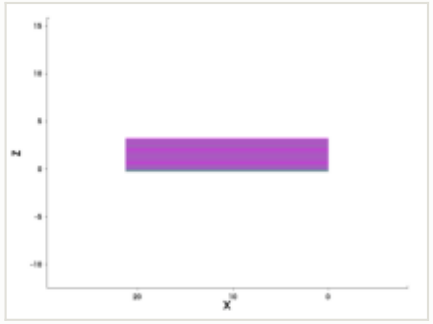


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



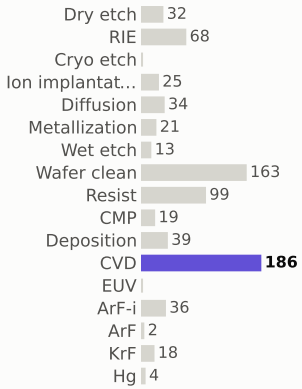
y cut at 57.351 µm



mask for this step

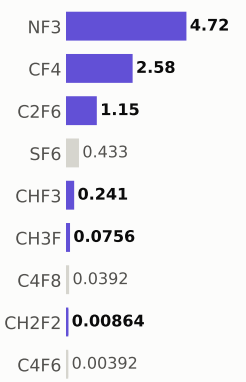
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



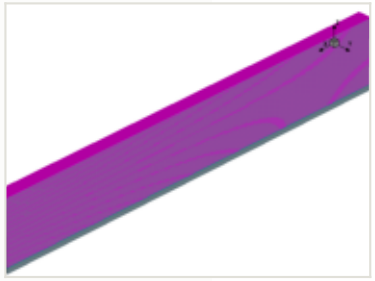
Step #10200: CVD

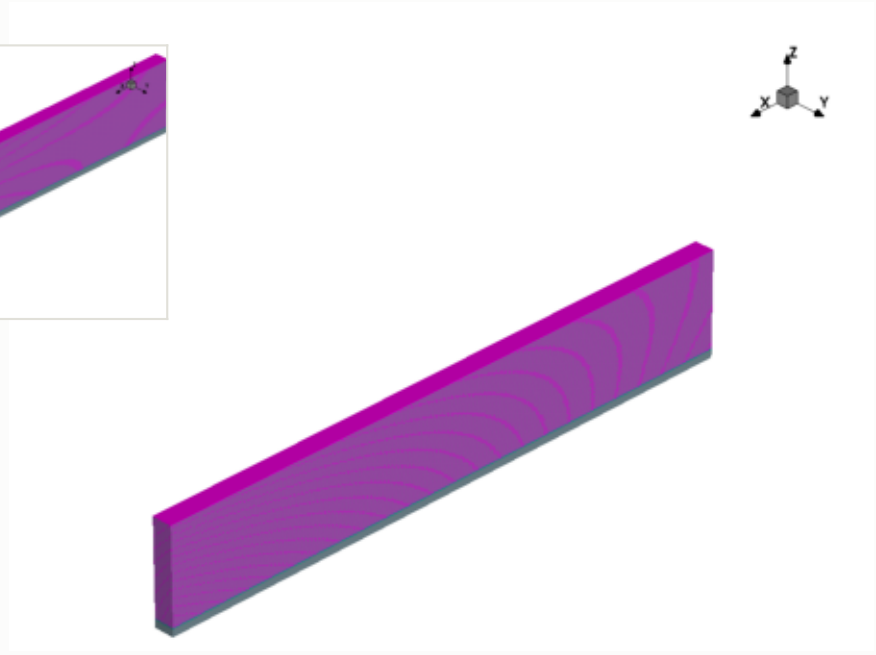
depo_isotropic · CVD

1.31	553.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



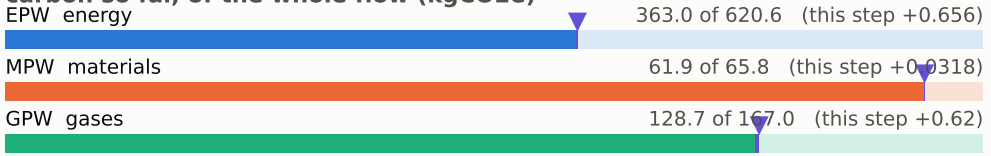


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

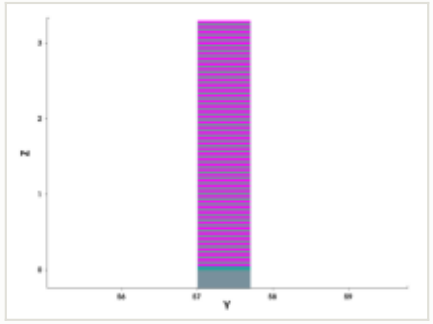
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

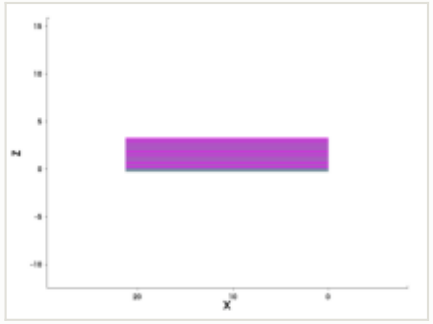


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



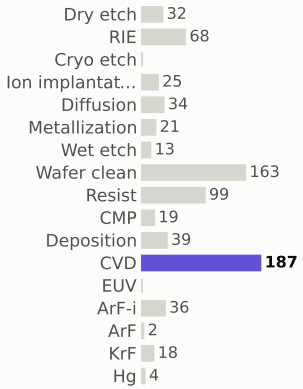
y cut at 57.351 µm



mask for this step

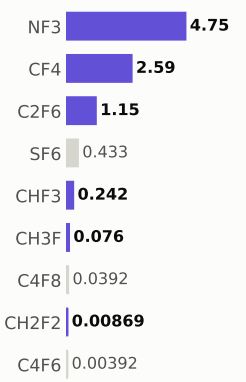
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



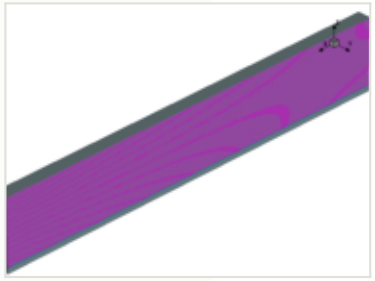
Step #10210: CVD

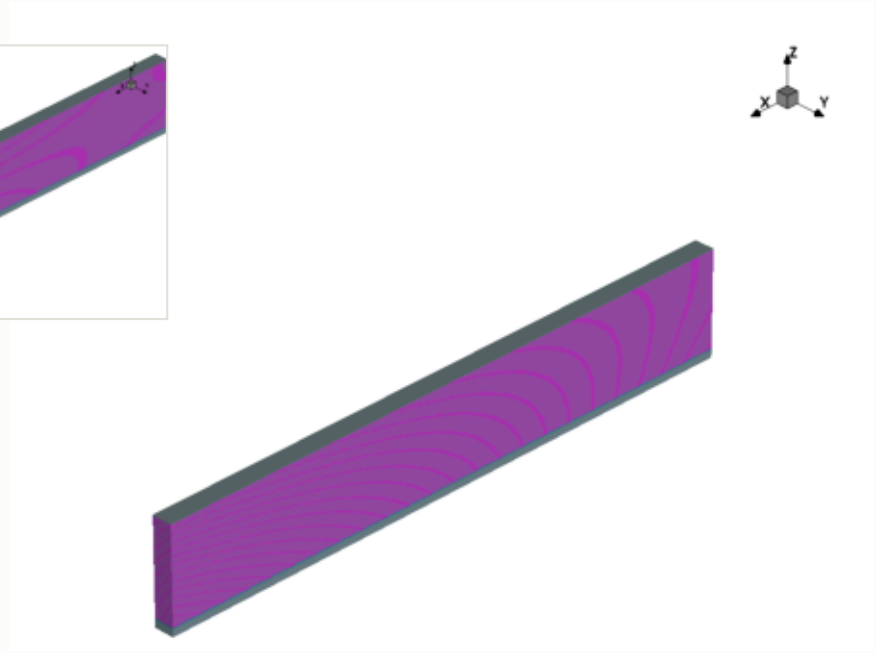
depo_isotropic · CVD

1.31	554.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



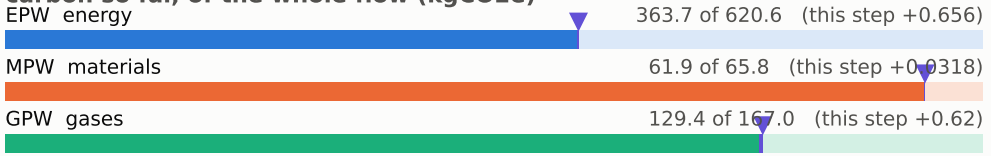


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

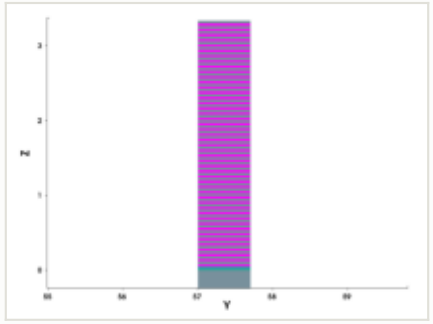
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

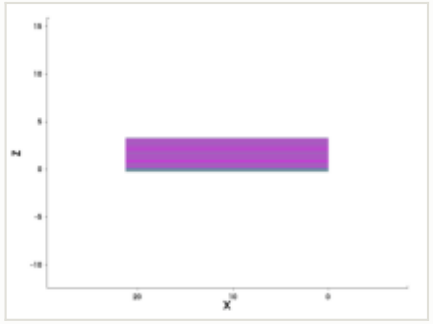


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



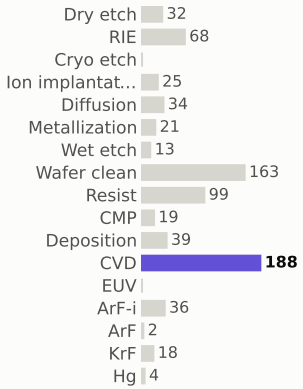
y cut at 57.351 µm



mask for this step

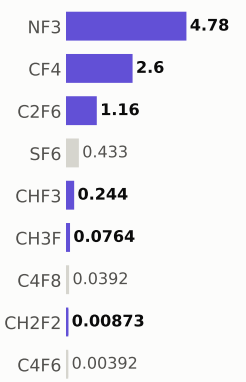
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



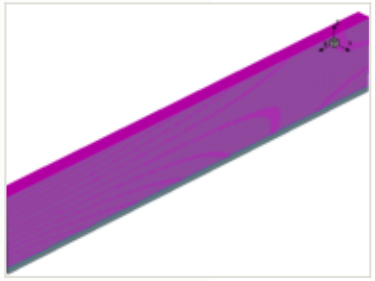
Step #10220: CVD

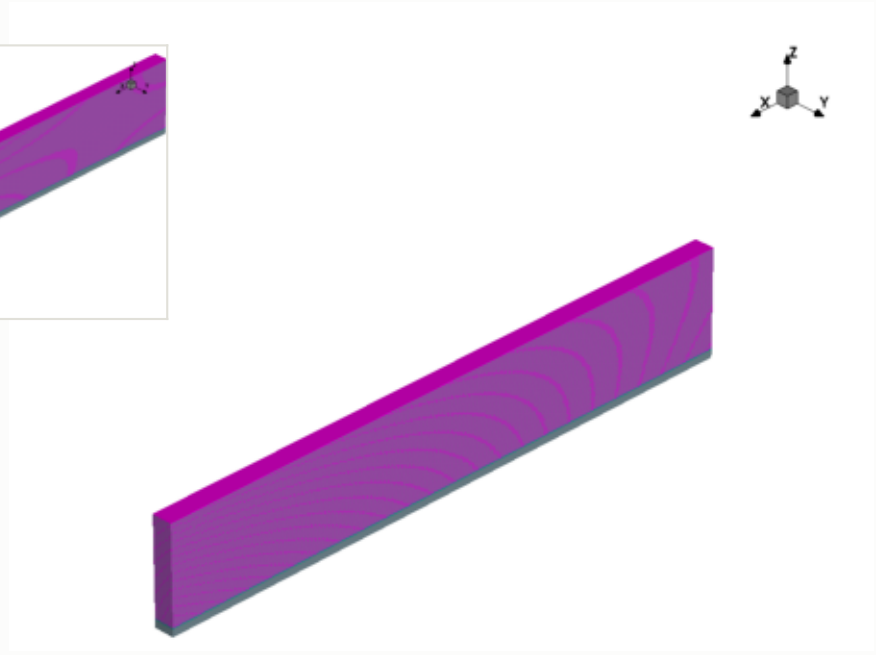
depo_isotropic · CVD

1.31	556.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



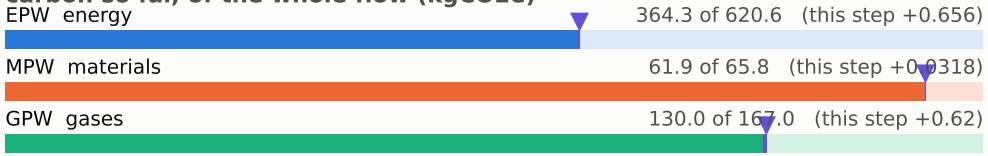


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

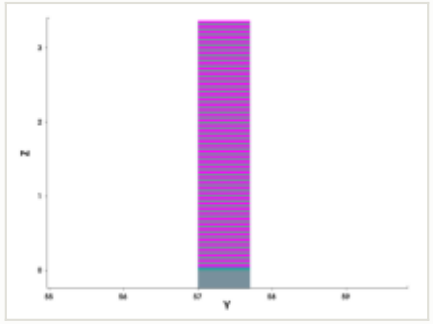
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

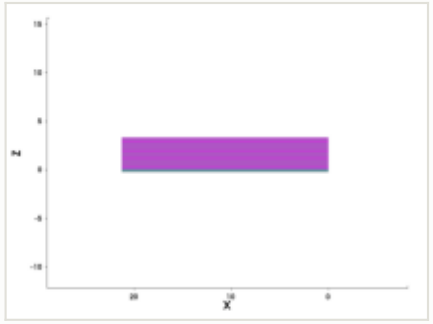


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



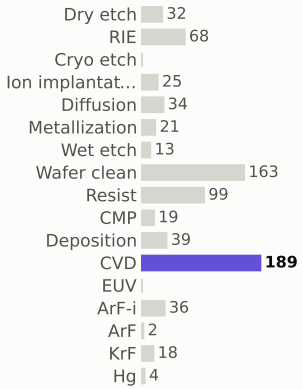
y cut at 57.351 µm



mask for this step

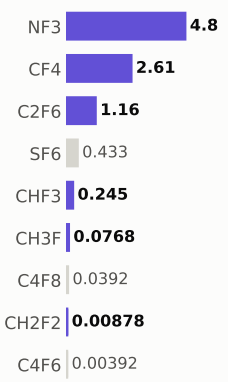
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



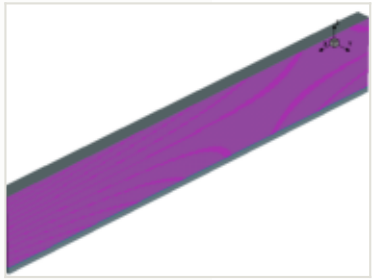
Step #10230: CVD

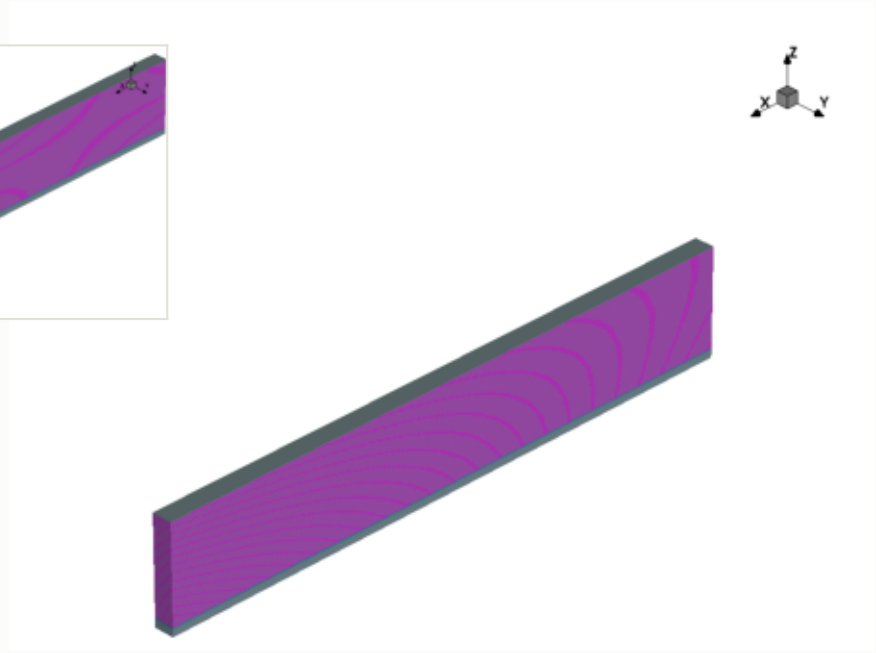
depo_isotropic · CVD

1.31	557.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



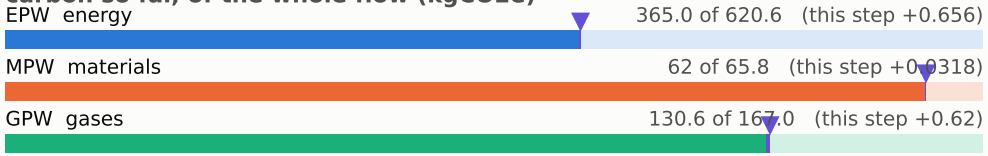


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

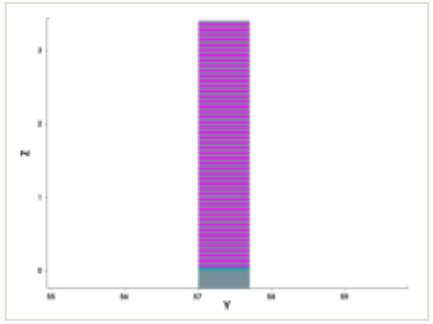
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

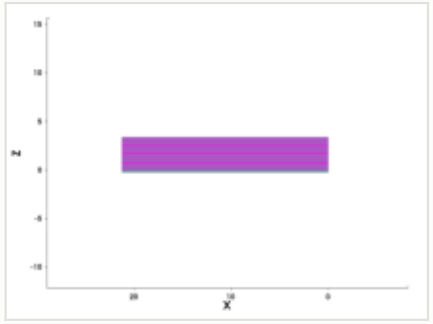


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



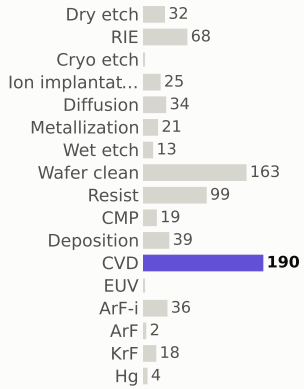
y cut at 57.351 µm



mask for this step

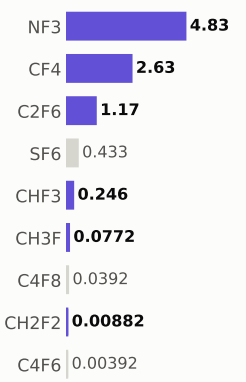
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



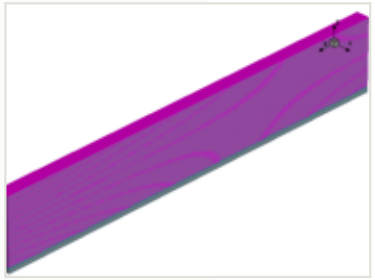
Step #10240: CVD

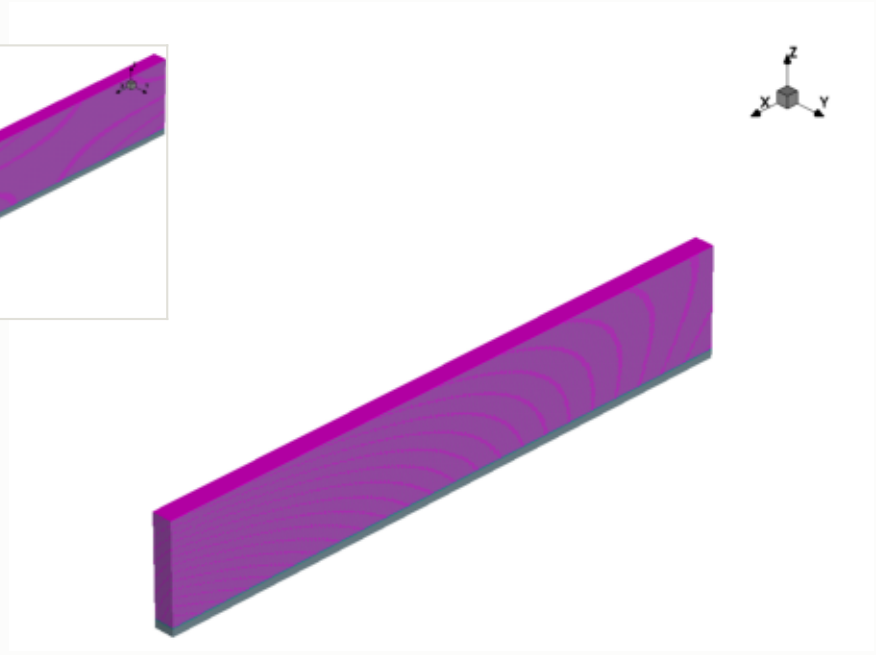
depo_isotropic · CVD

1.31	558.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



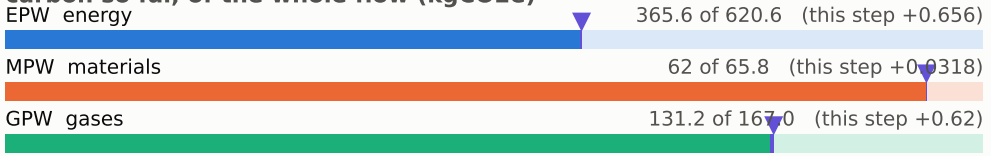


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

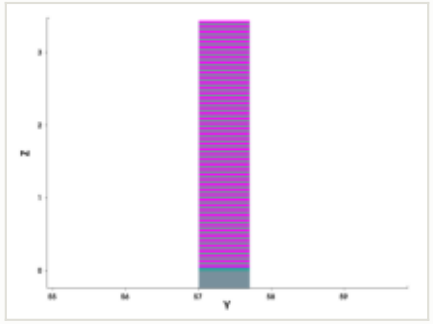
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

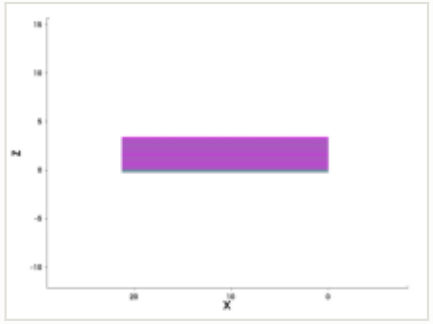


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



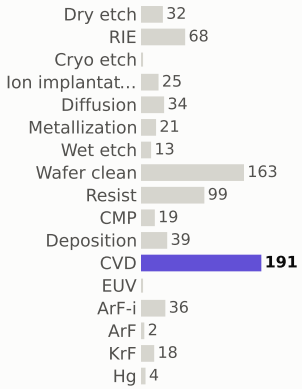
y cut at 57.351 µm



mask for this step

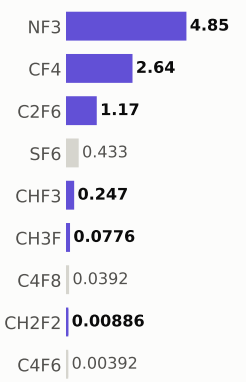
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



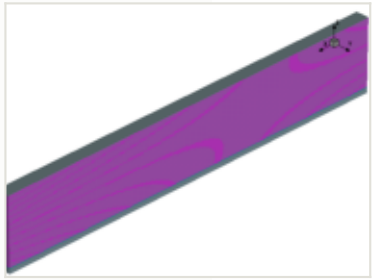
Step #10250: CVD

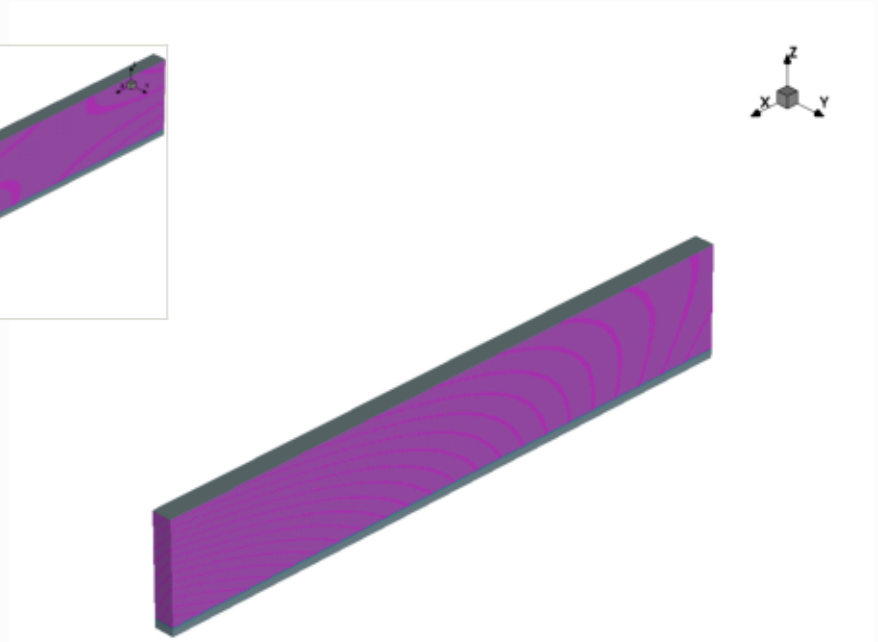
depo_isotropic · CVD

1.31	560.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



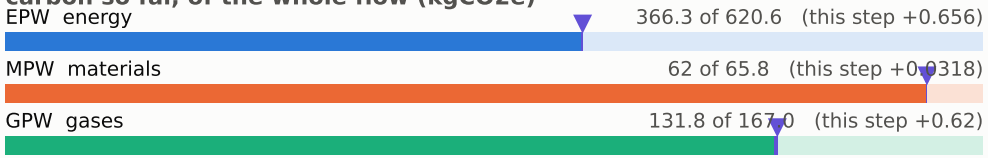


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

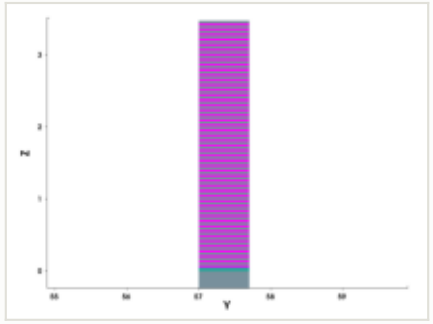
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

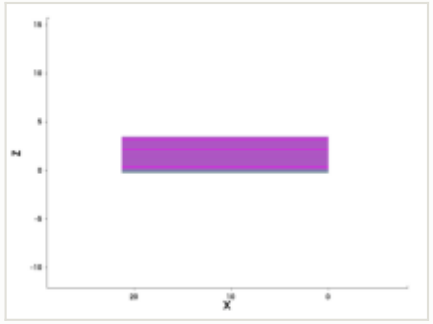


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



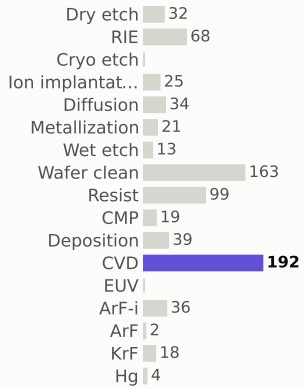
y cut at 57.351 µm



mask for this step

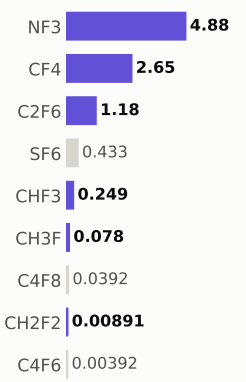
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



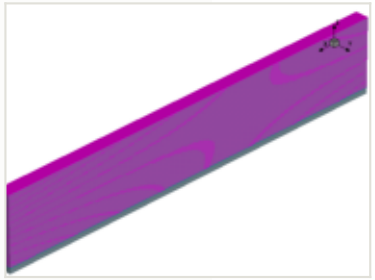
Step #10260: CVD

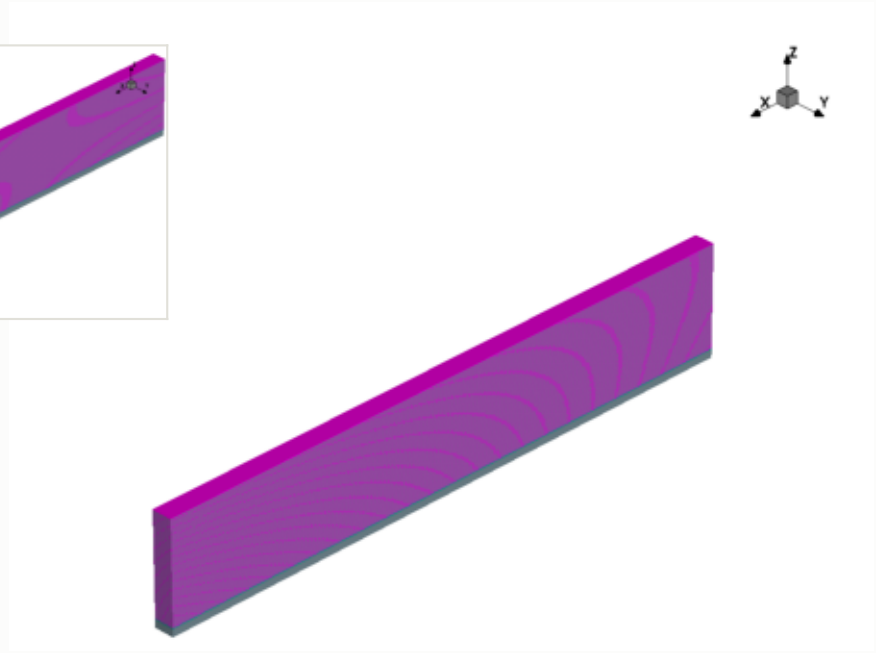
depo_isotropic · CVD

1.31	561.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



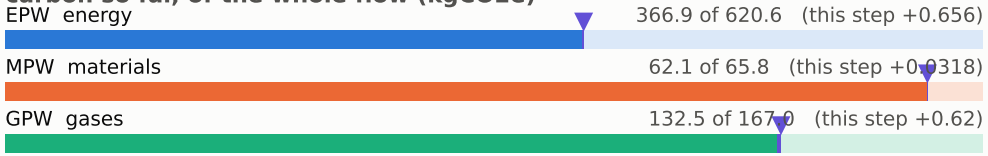


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

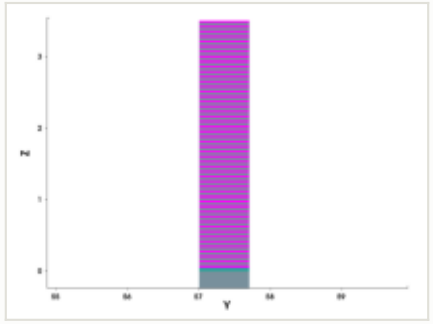
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

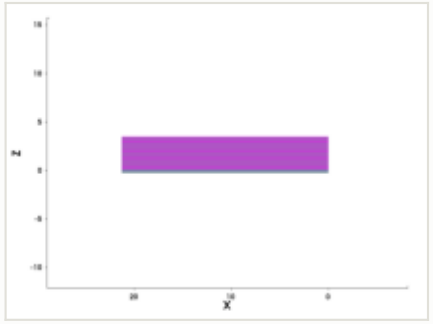


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



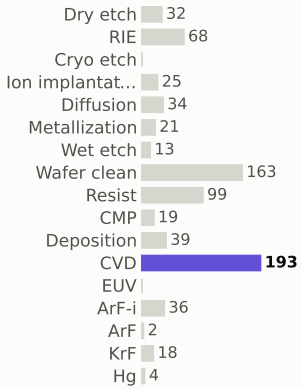
y cut at 57.351 µm



mask for this step

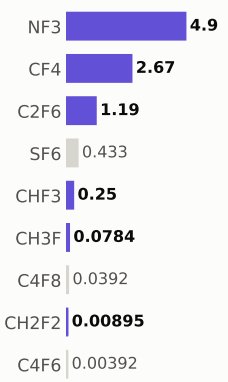
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



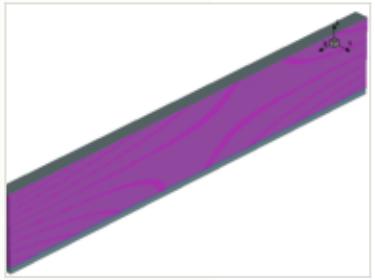
Step #10270: CVD

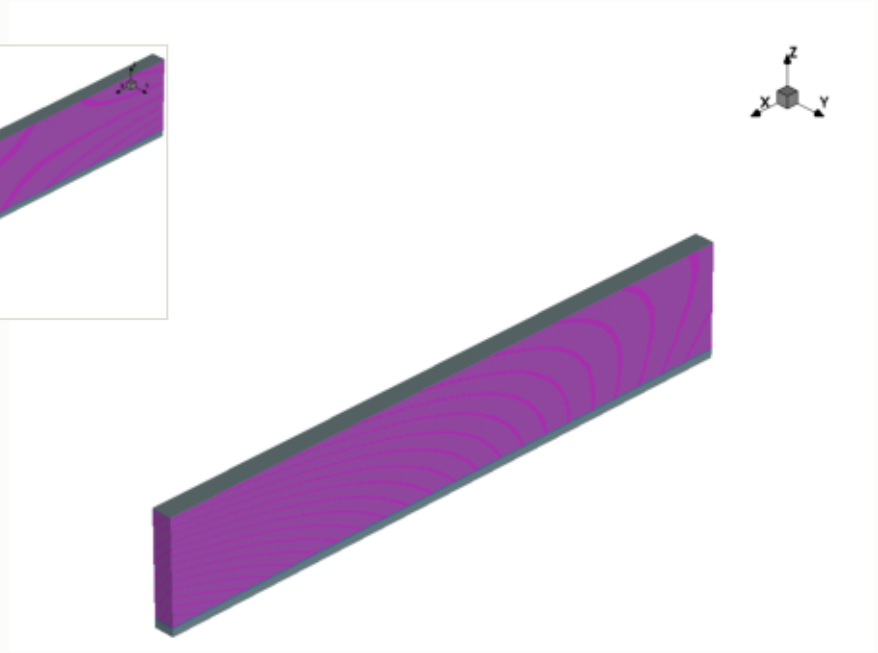
depo_isotropic · CVD

1.31	562.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



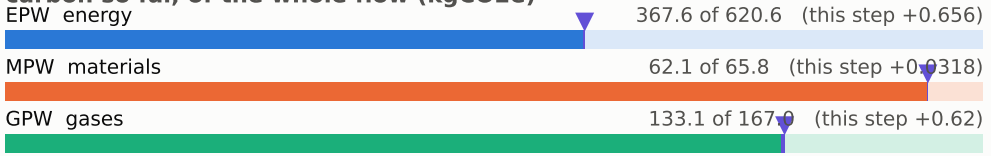


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

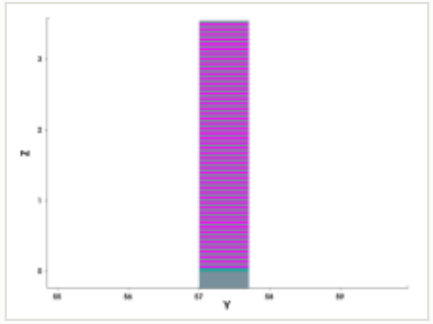
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

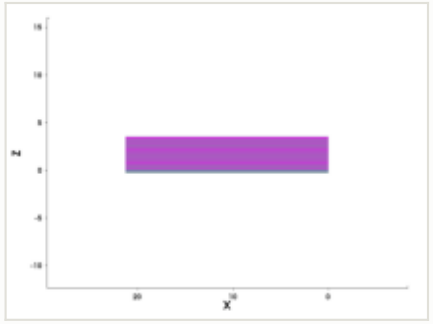


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



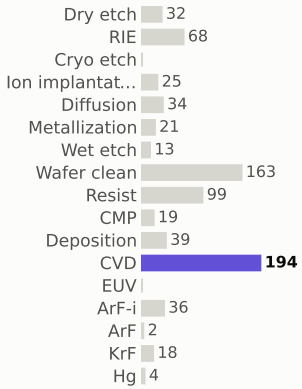
y cut at 57.351 µm



mask for this step

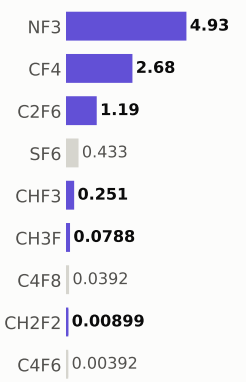
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



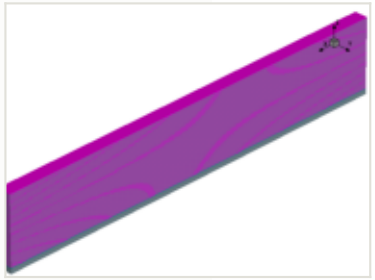
Step #10280: CVD

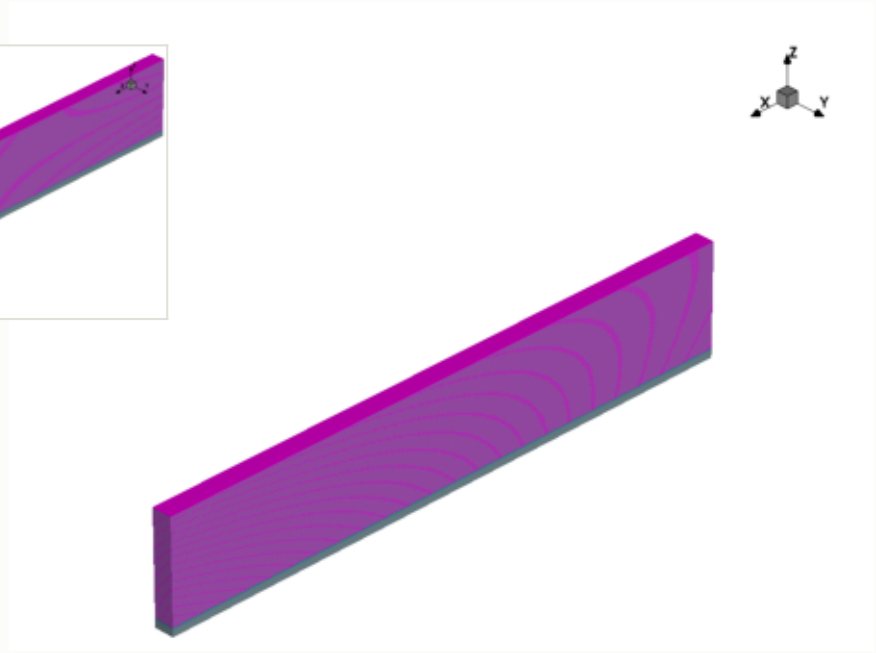
depo_isotropic · CVD

1.31	564.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



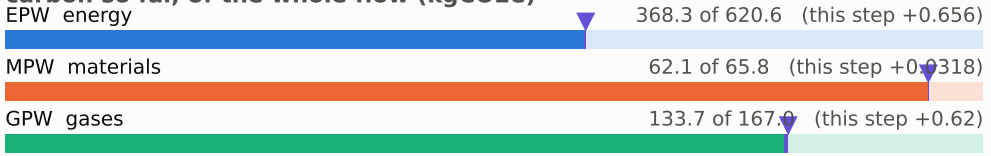


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

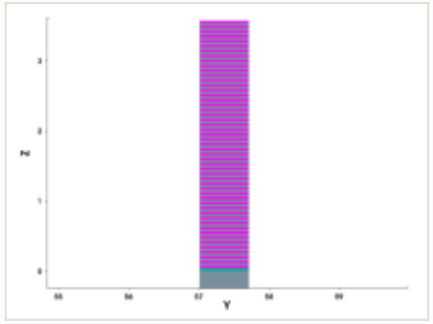
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

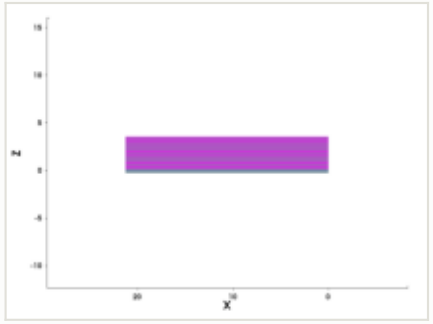


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



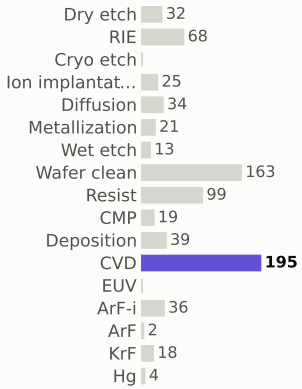
y cut at 57.351 µm



mask for this step

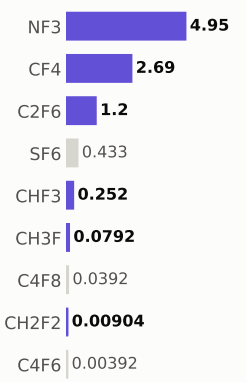
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



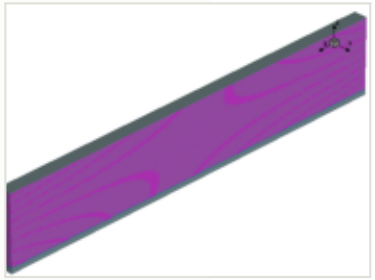
Step #10290: CVD

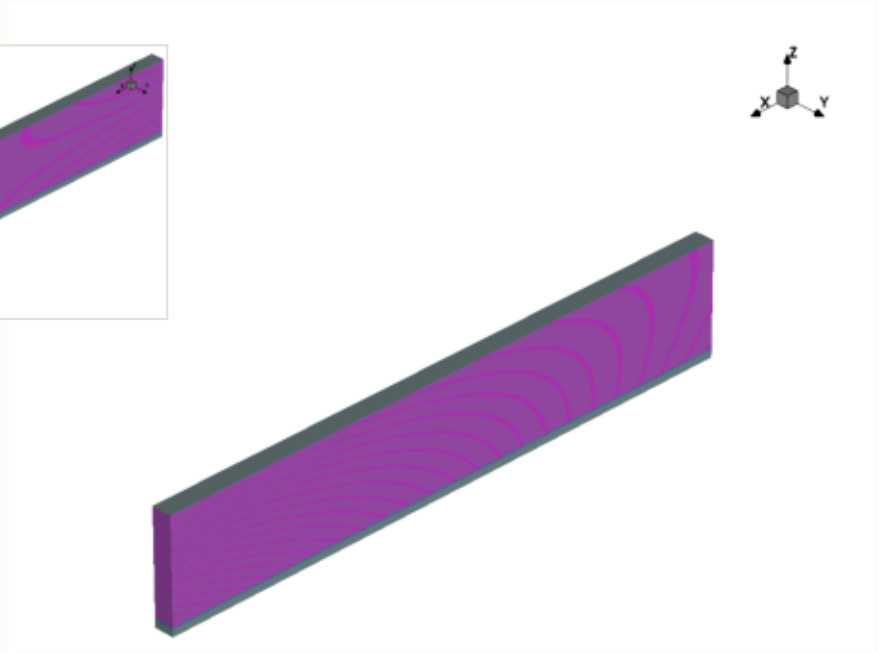
depo_isotropic · CVD

1.31	565.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



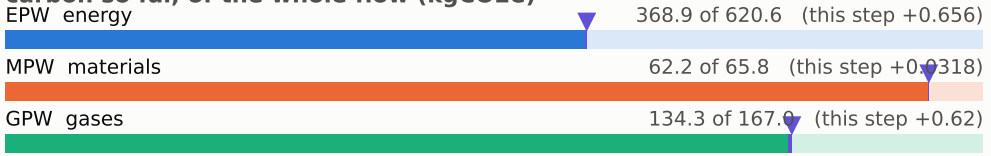


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

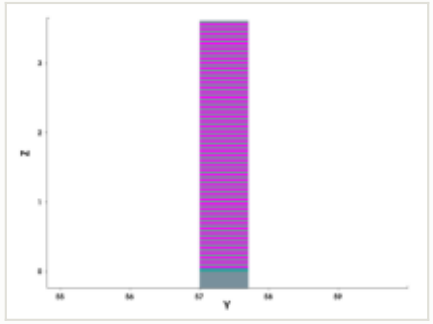
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

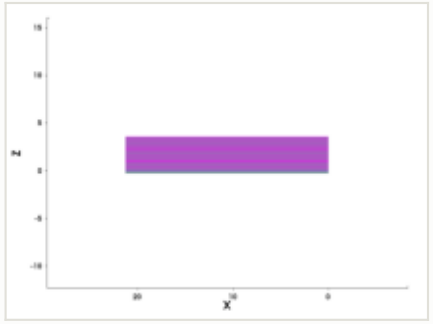


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



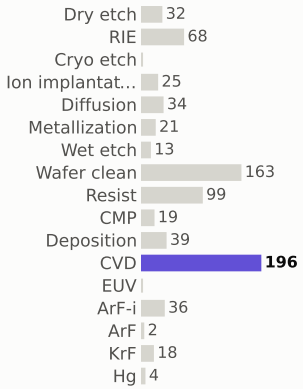
y cut at 57.351 µm



mask for this step

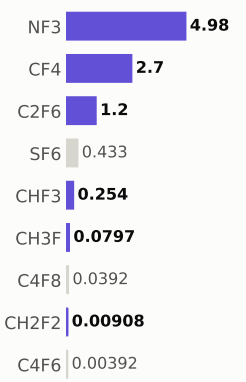
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



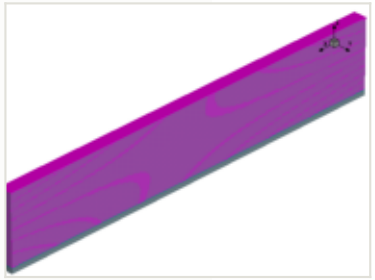
Step #10300: CVD

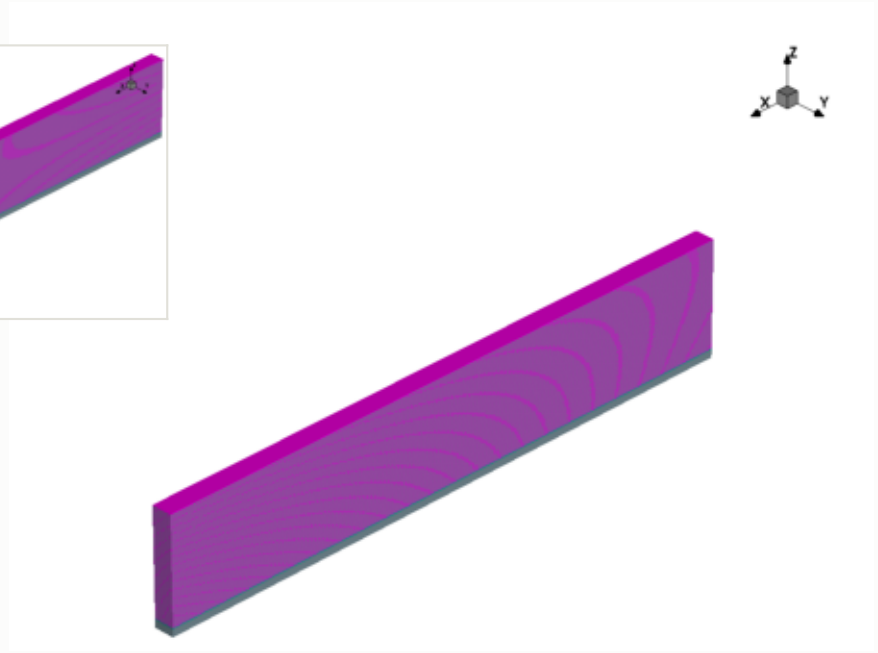
depo_isotropic · CVD

1.31	566.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



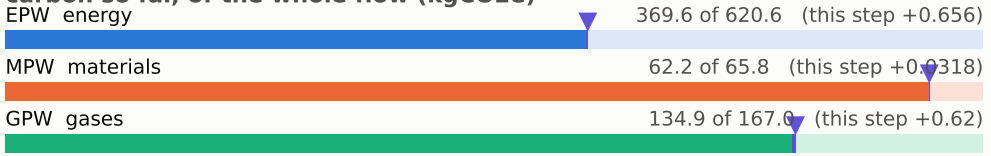


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

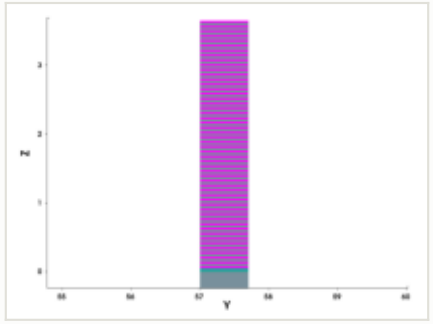
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

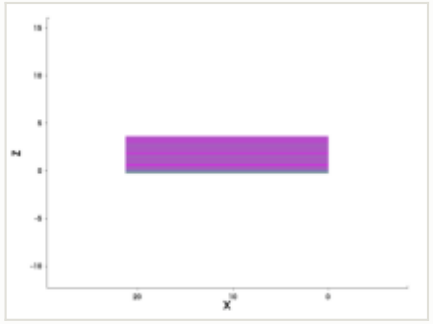


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



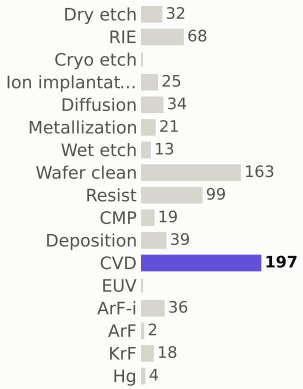
y cut at 57.351 µm



mask for this step

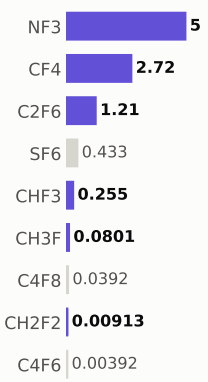
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



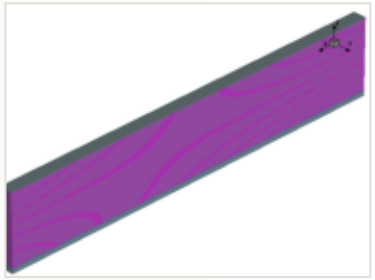
Step #10310: CVD

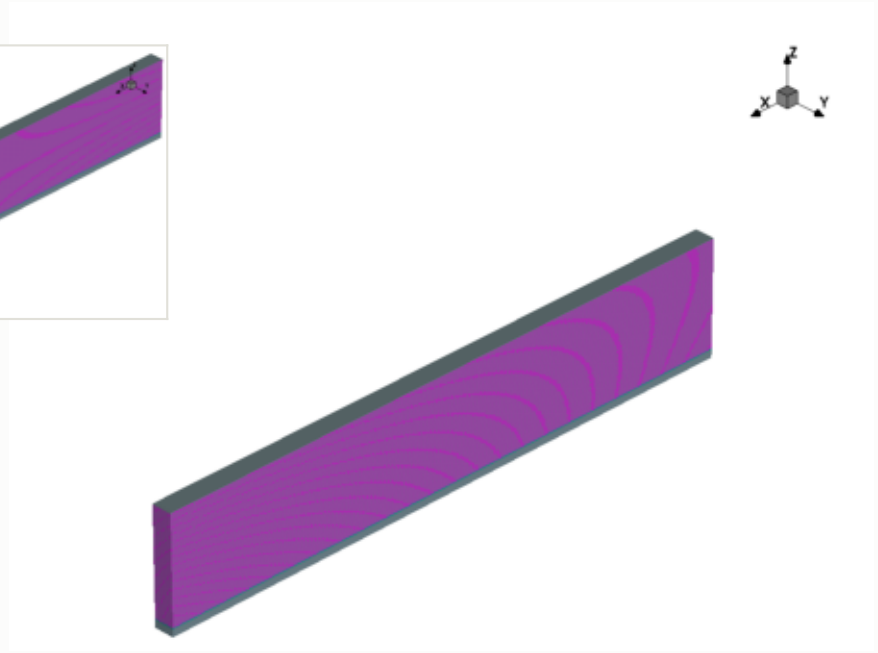
depo_isotropic · CVD

1.31	568.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



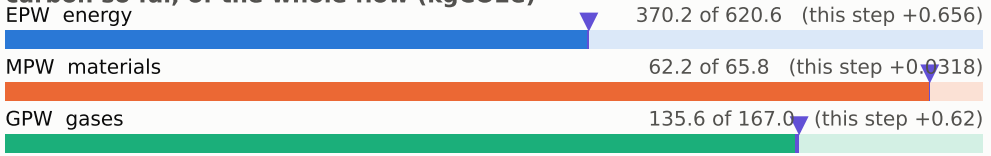


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

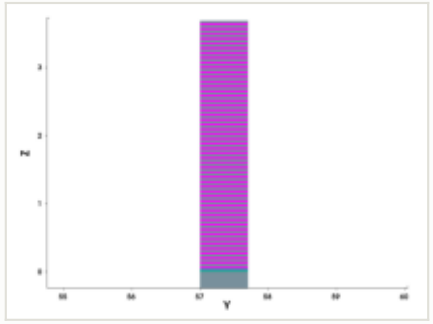
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

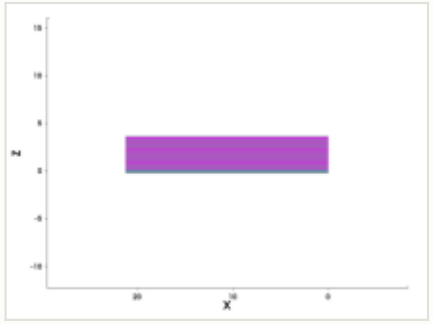


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



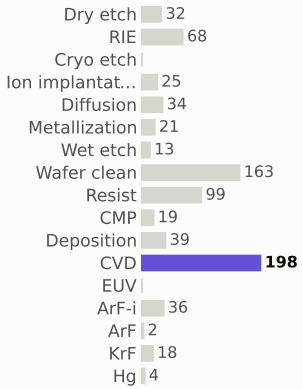
y cut at 57.351 µm



mask for this step

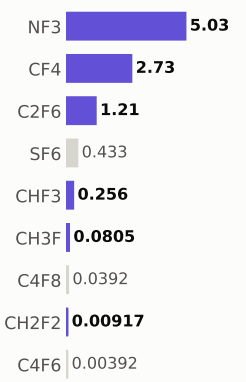
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



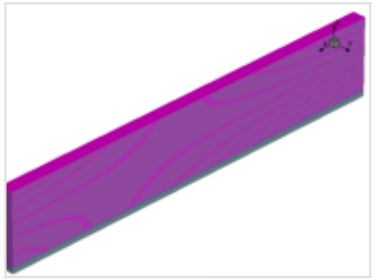
Step #10320: CVD

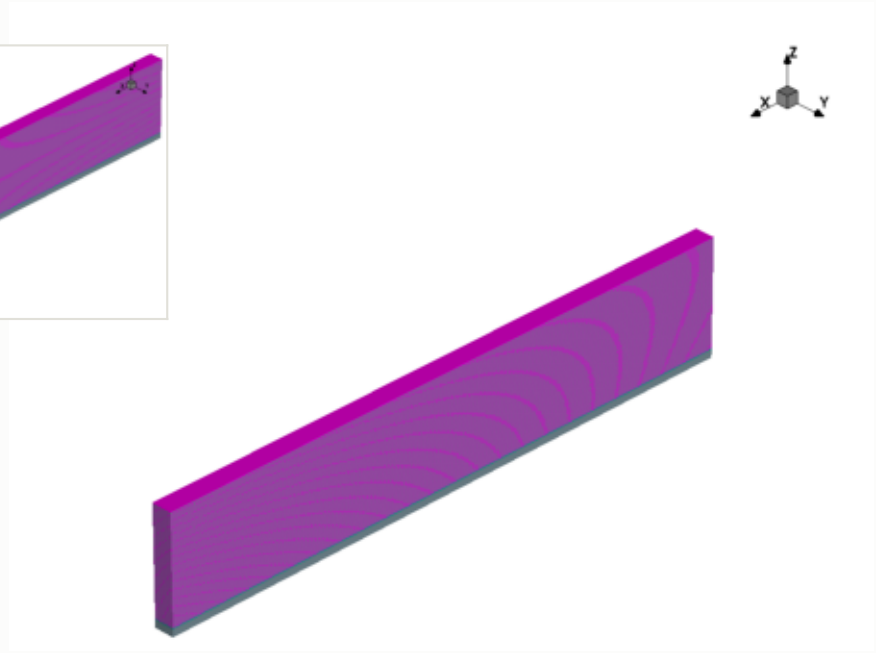
depo_isotropic · CVD

1.31	569.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



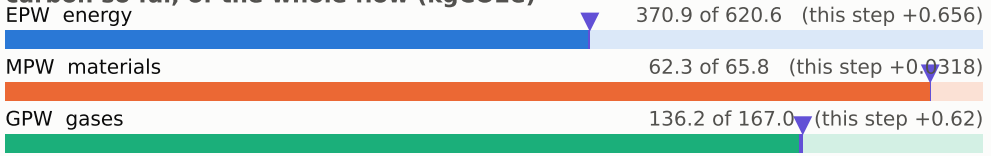


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

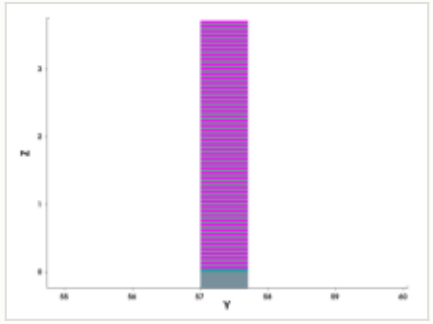
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

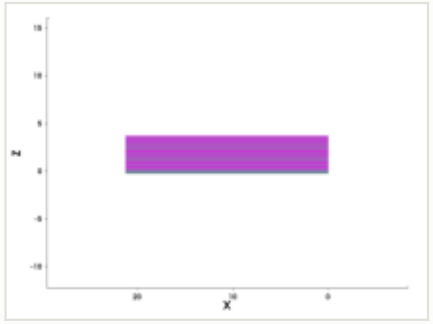


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



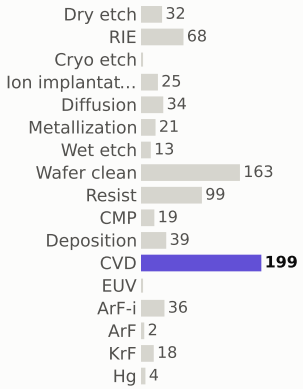
y cut at 57.351 µm



mask for this step

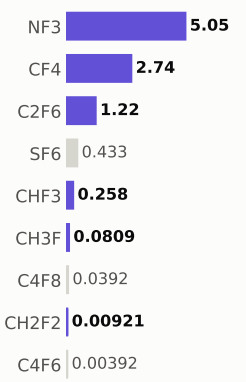
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



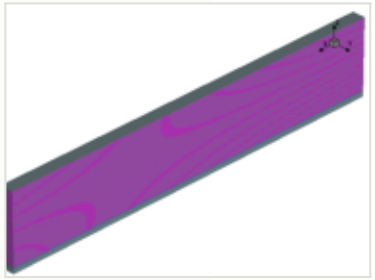
Step #10330: CVD

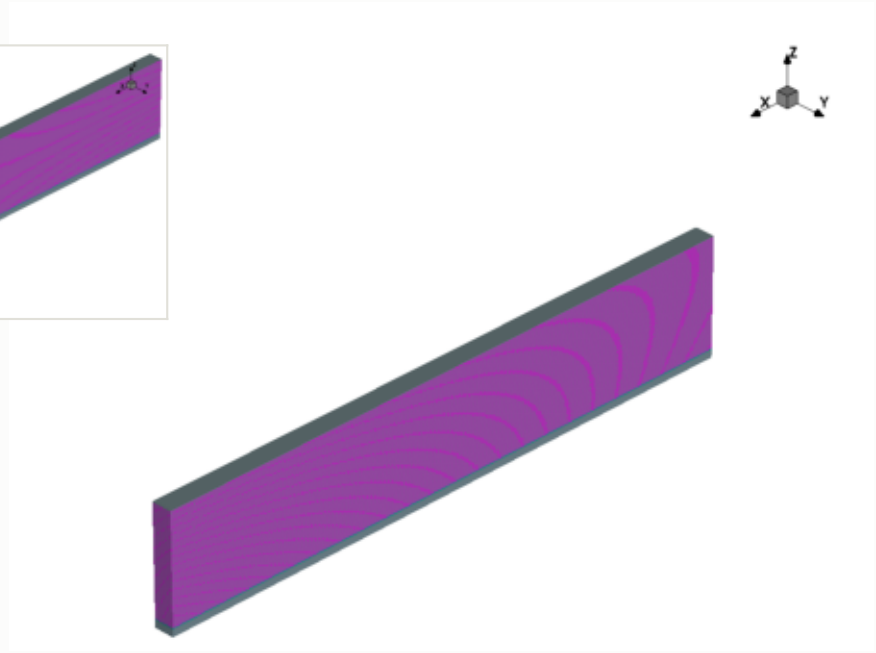
depo_isotropic · CVD

1.31	570.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



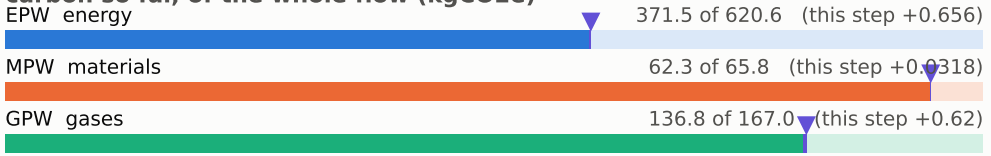


this step

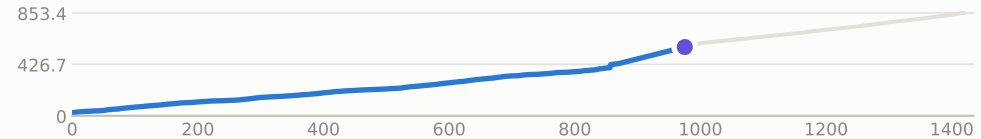
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

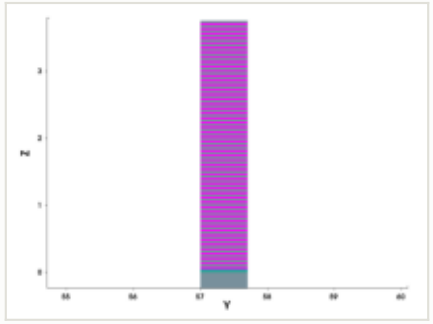
carbon so far, of the whole flow (kgCO2e)



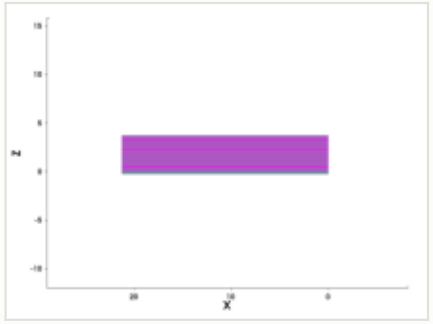
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



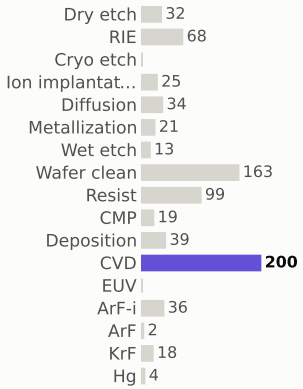
y cut at 57.351 µm



mask for this step

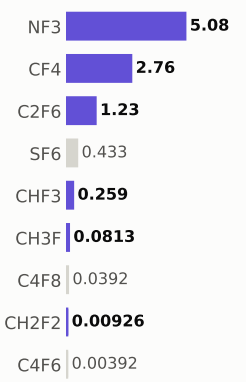
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



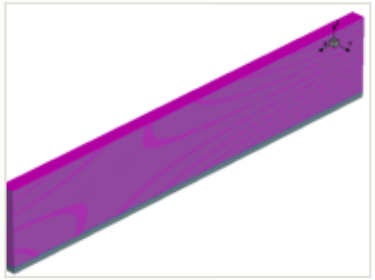
Step #10340: CVD

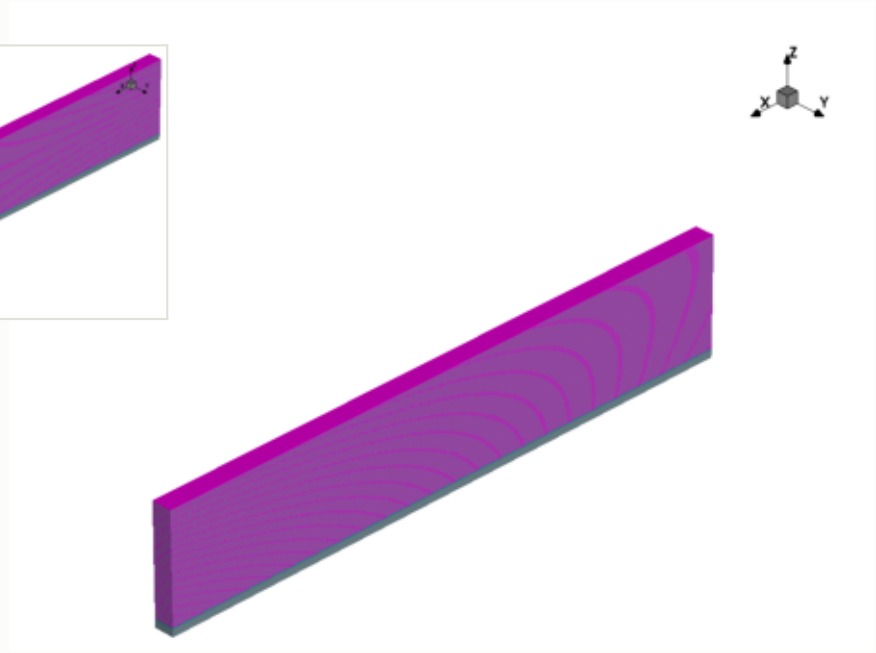
depo_isotropic · CVD

1.31	571.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



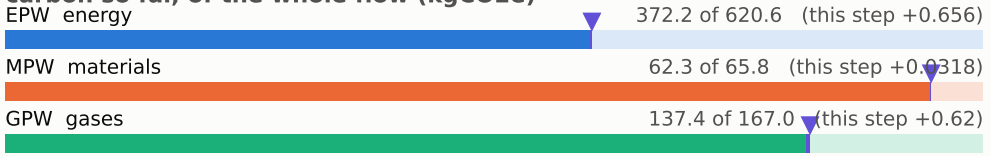


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

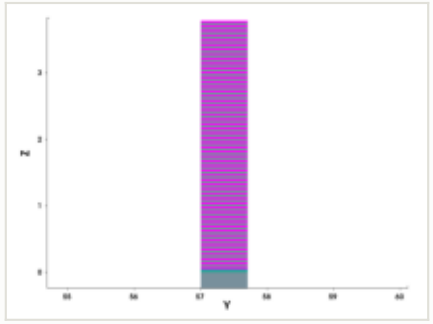
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

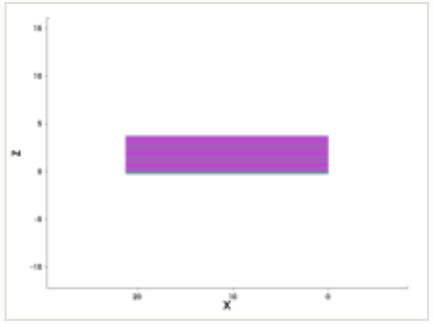


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



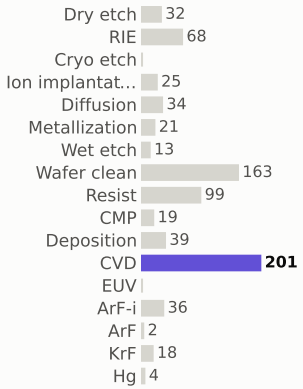
y cut at 57.351 µm



mask for this step

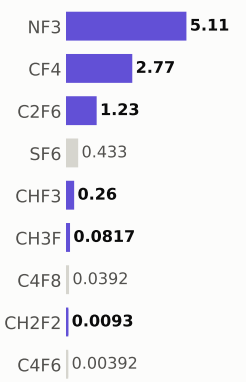
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



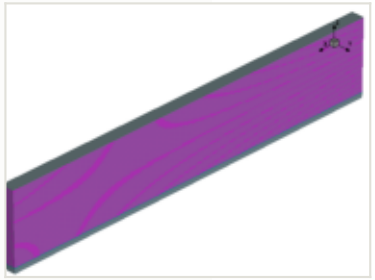
Step #10350: CVD

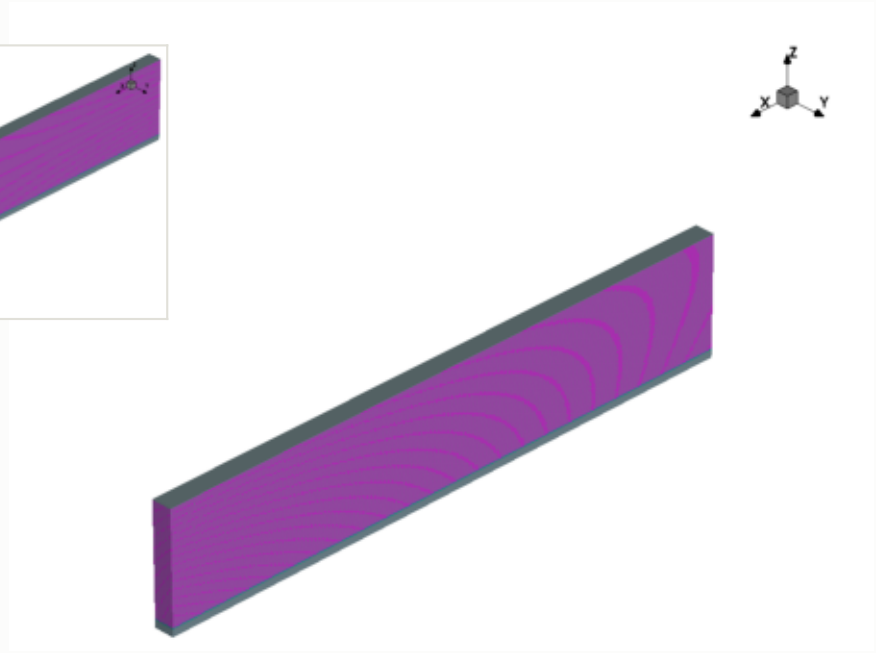
depo_isotropic · CVD

1.31	573.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



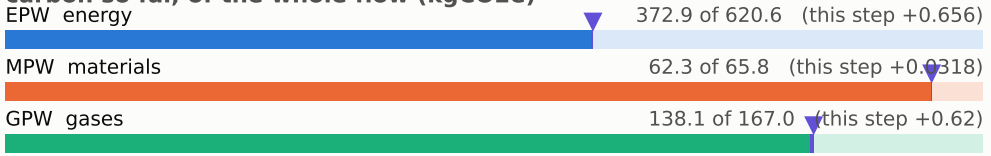


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

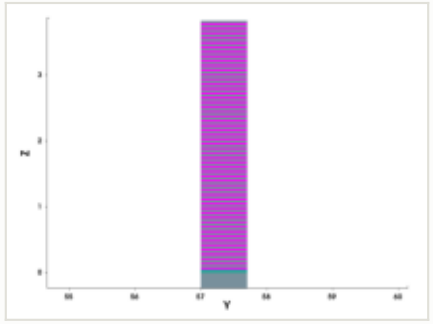
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

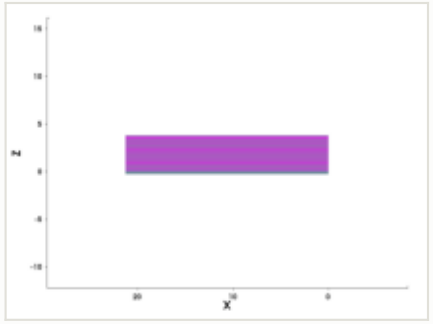


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



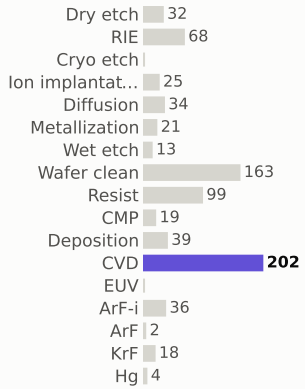
y cut at 57.351 µm



mask for this step

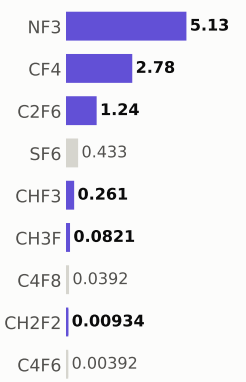
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



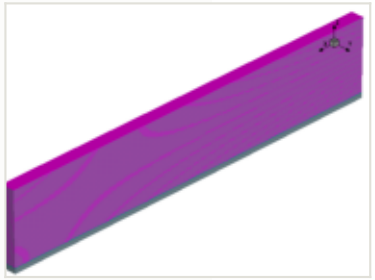
Step #10360: CVD

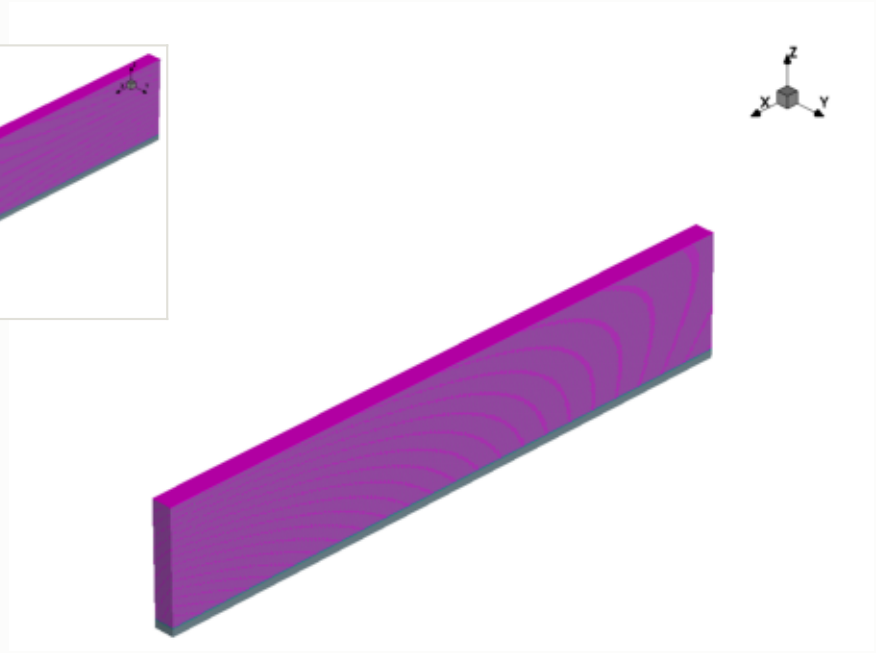
depo_isotropic · CVD

1.31	574.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



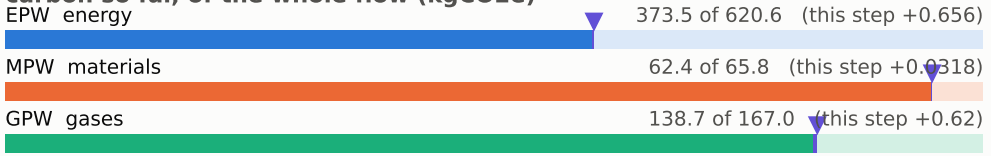


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

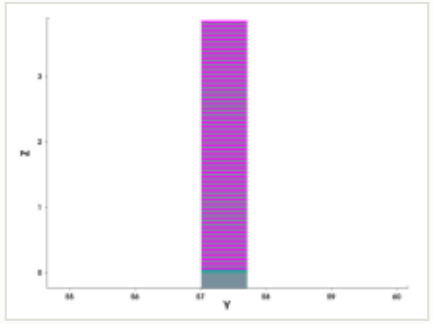
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

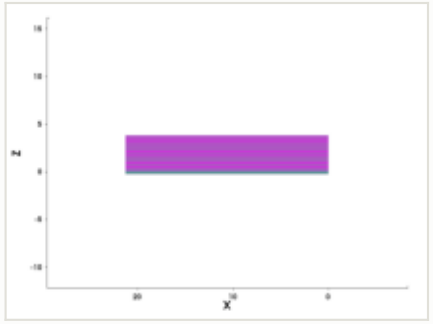


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



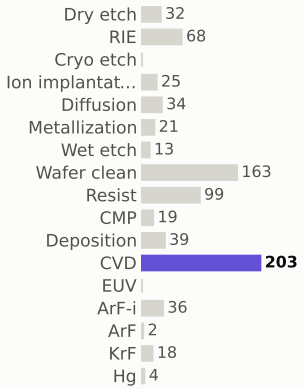
y cut at 57.351 µm



mask for this step

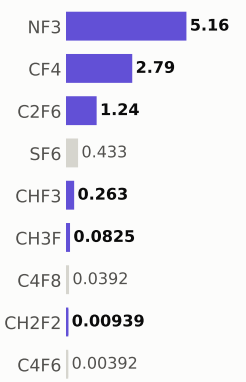
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



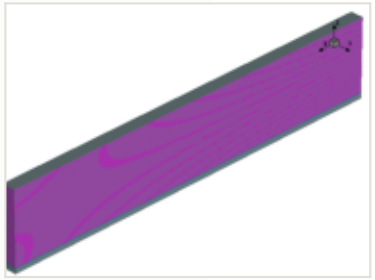
Step #10370: CVD

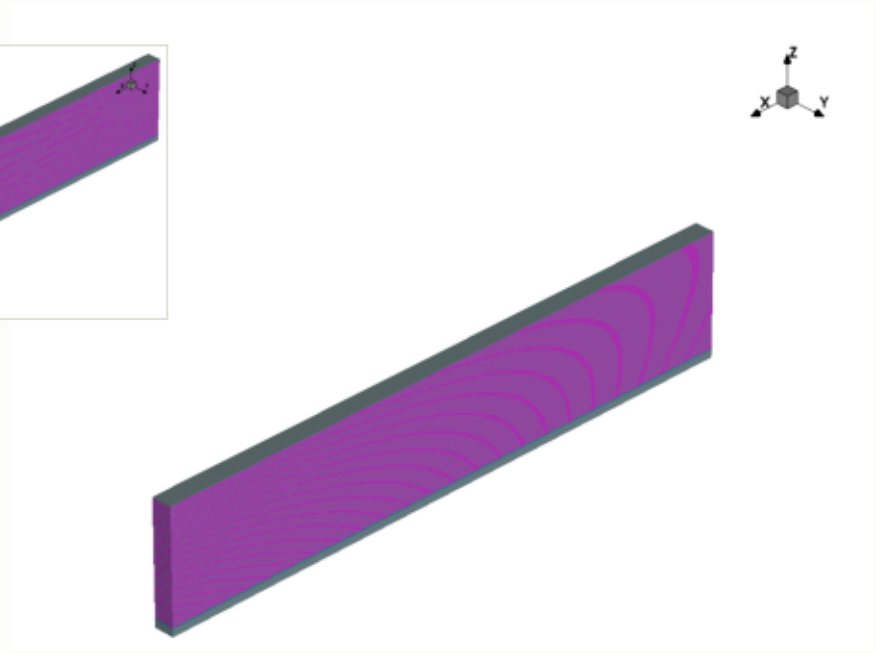
depo_isotropic · CVD

1.31	575.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



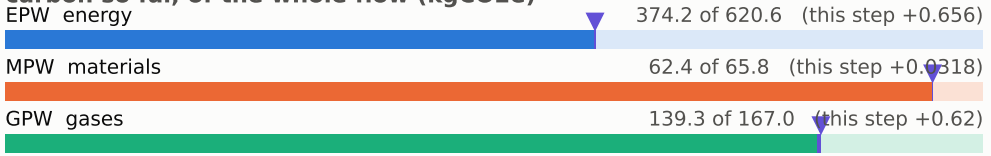


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

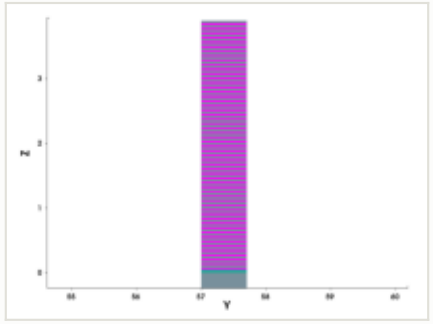
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

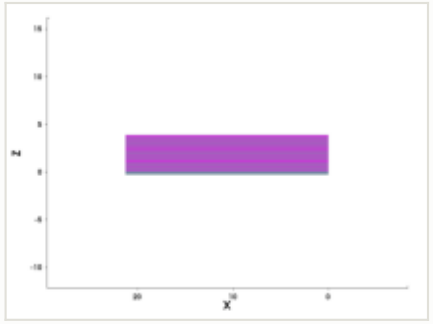


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



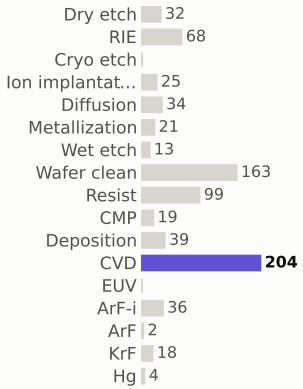
y cut at 57.351 µm



mask for this step

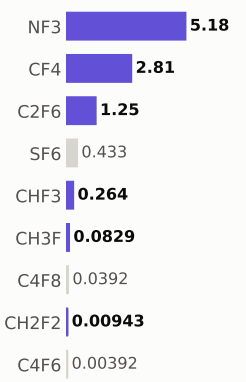
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



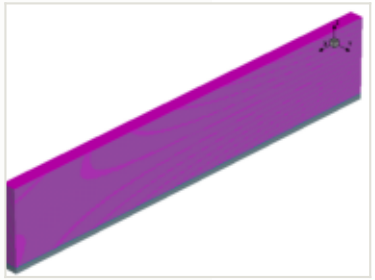
Step #10380: CVD

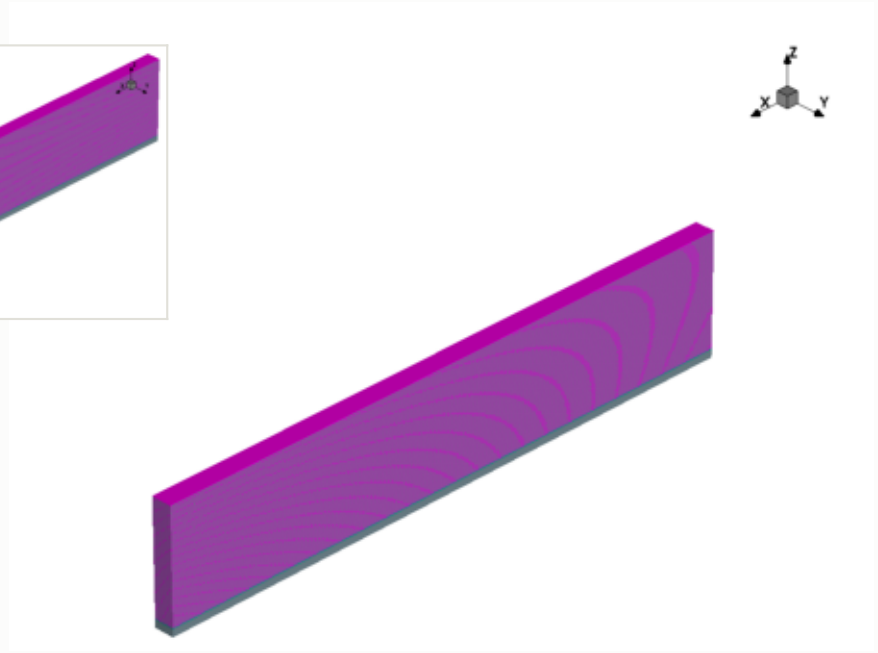
depo_isotropic · CVD

1.31	577.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



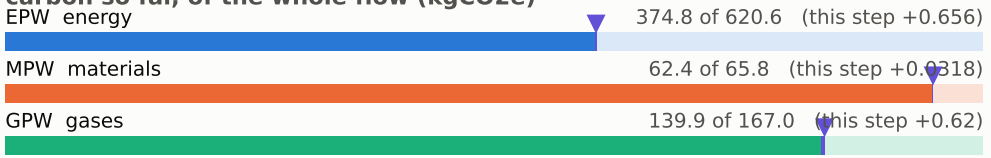


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

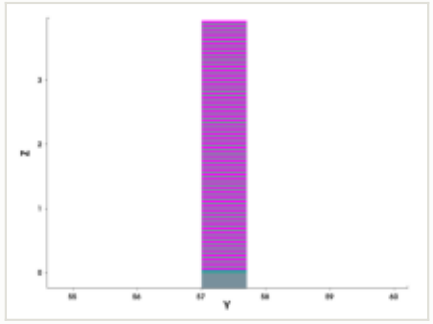
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

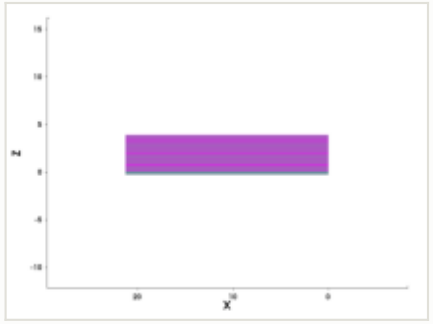


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



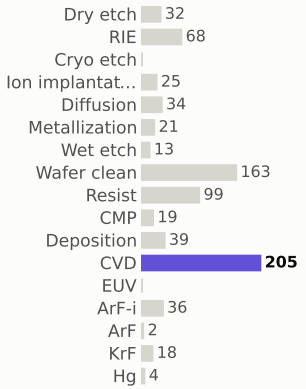
y cut at 57.351 µm



mask for this step

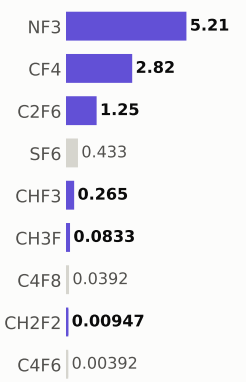
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



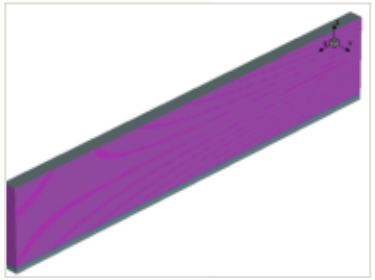
Step #10390: CVD

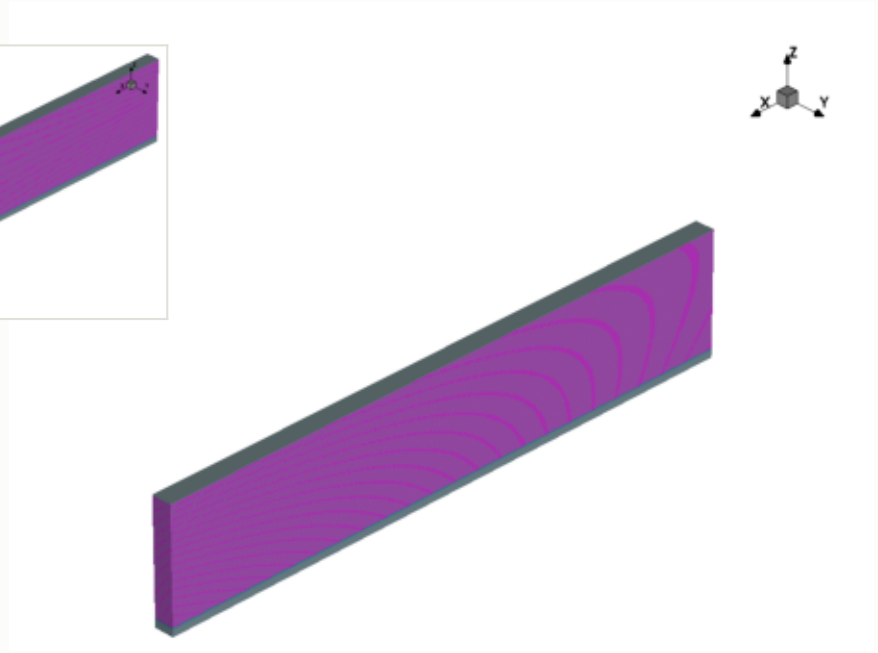
depo_isotropic · CVD

1.31	578.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



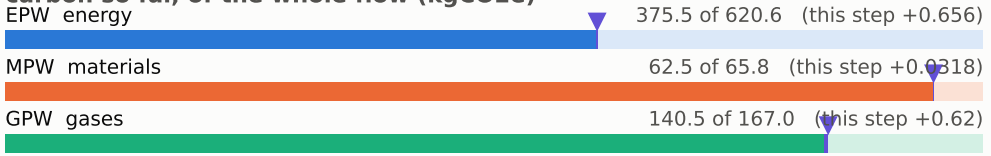


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

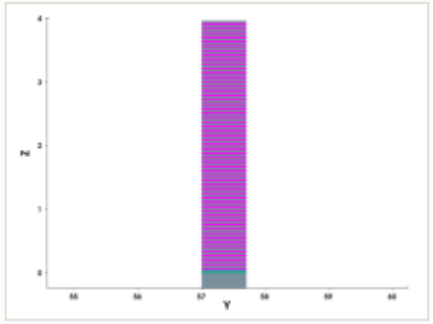
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

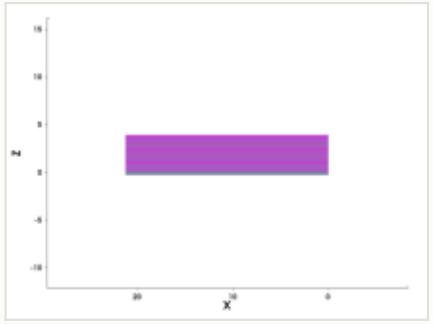


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



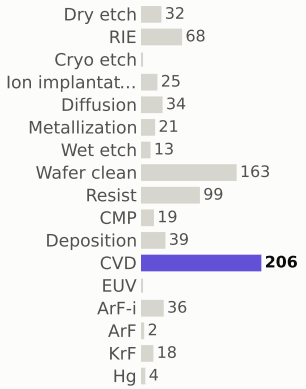
y cut at 57.351 µm



mask for this step

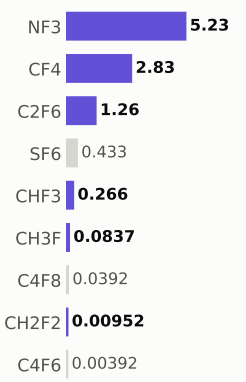
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



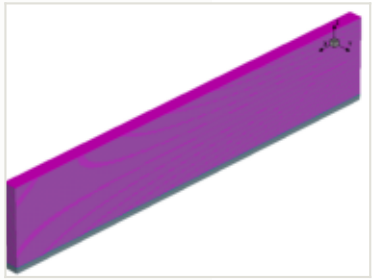
Step #10400: CVD

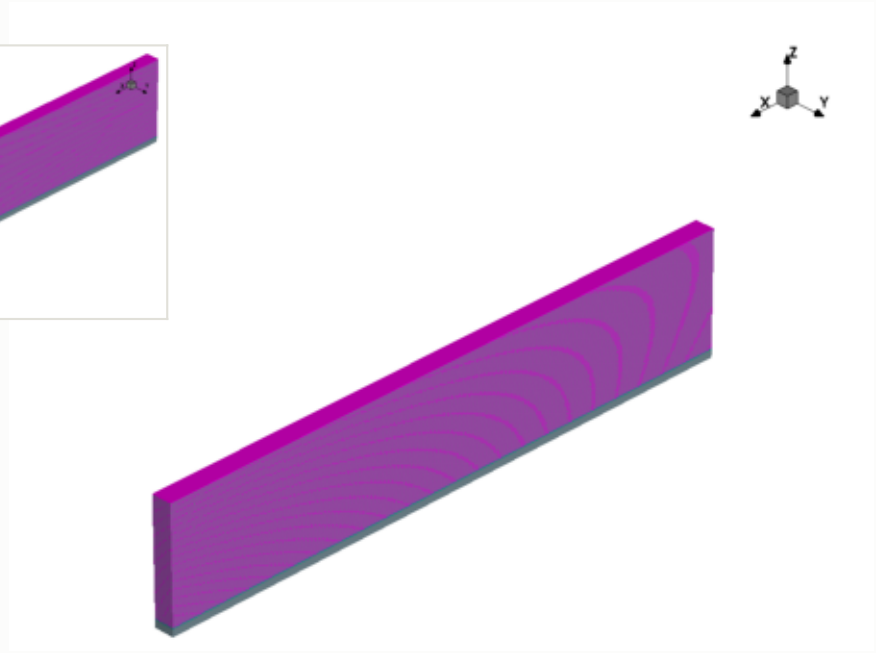
depo_isotropic · CVD

1.31	579.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



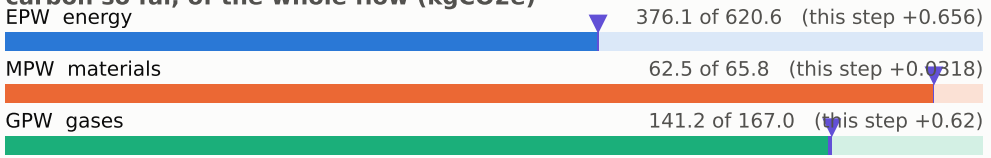


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

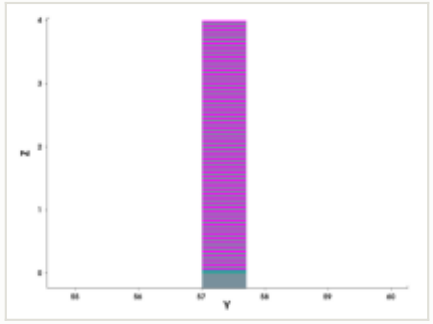
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

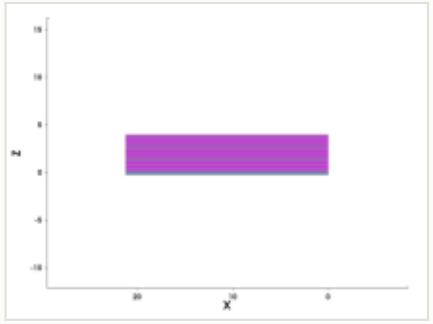


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



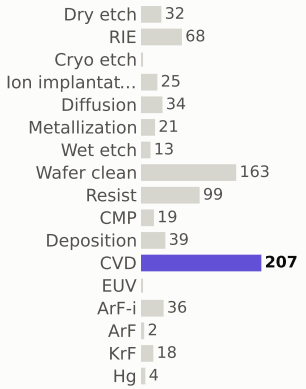
y cut at 57.351 µm



mask for this step

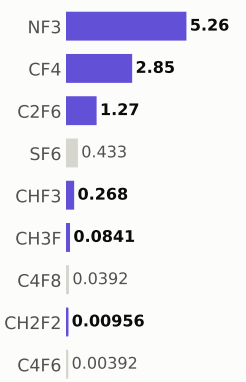
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



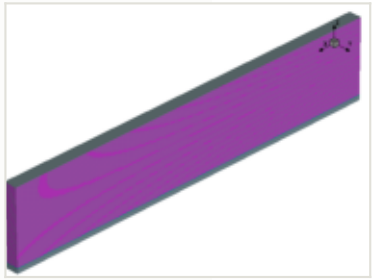
Step #10410: CVD

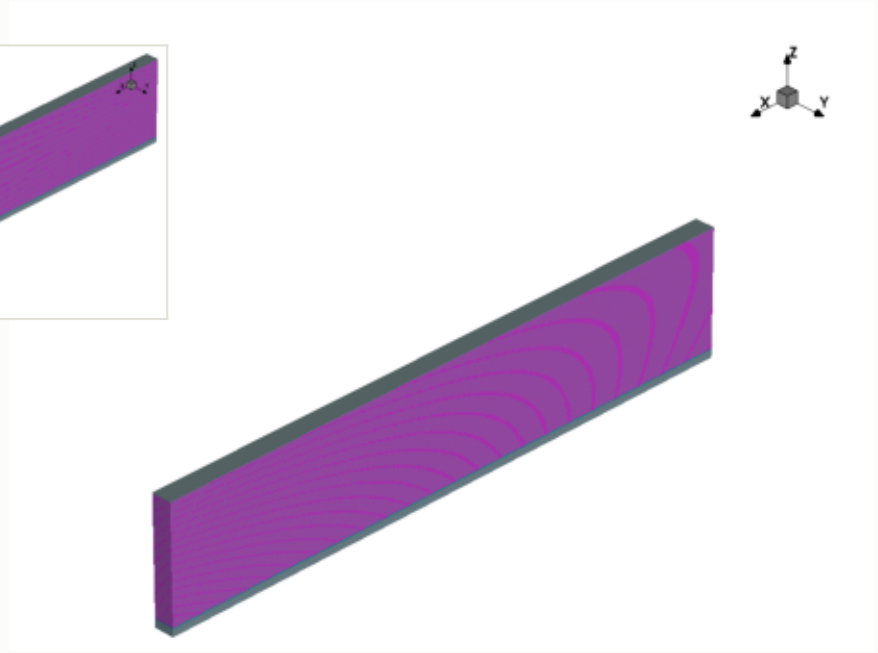
depo_isotropic · CVD

1.31	581.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



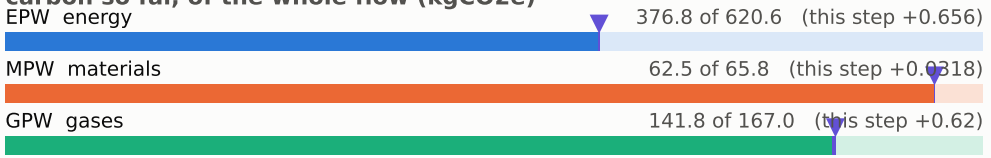


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

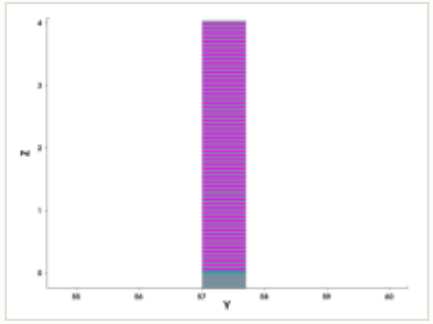
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

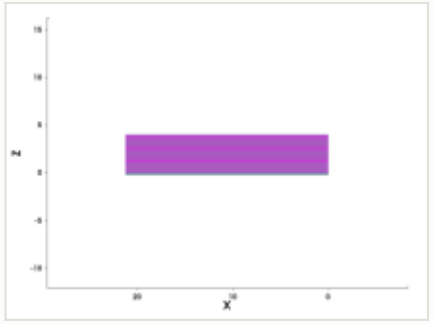


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



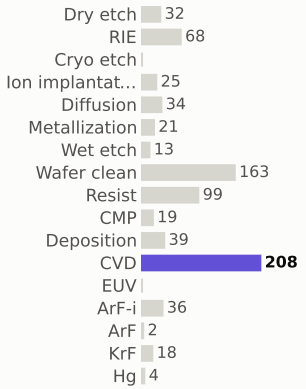
y cut at 57.351 µm



mask for this step

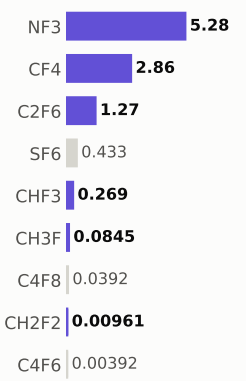
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



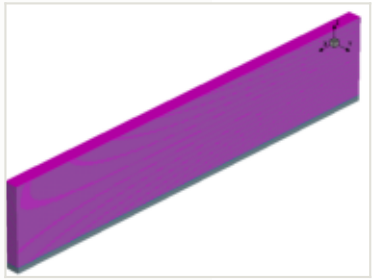
Step #10420: CVD

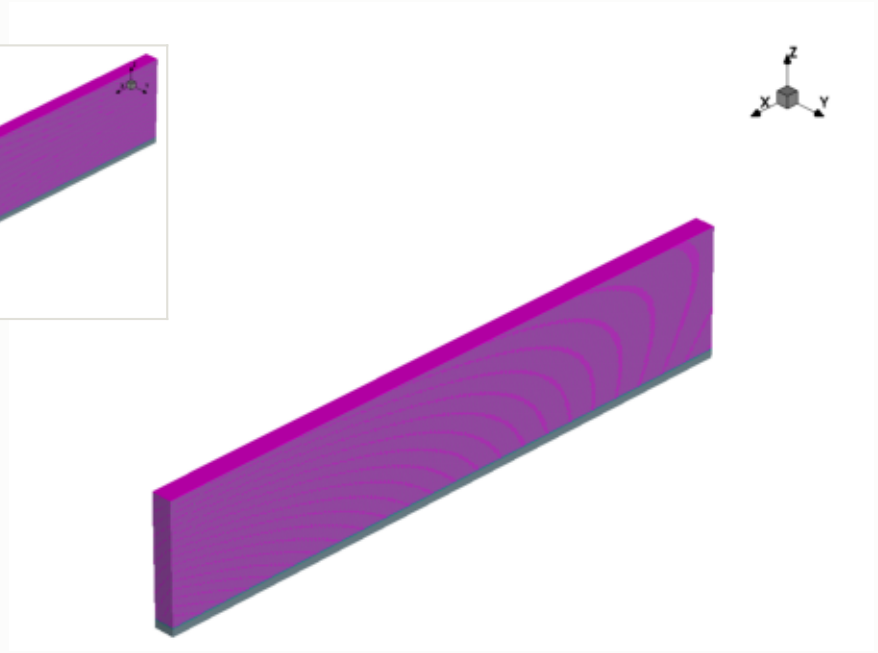
depo_isotropic · CVD

1.31	582.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



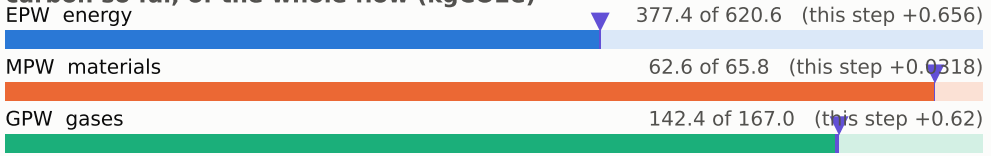


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

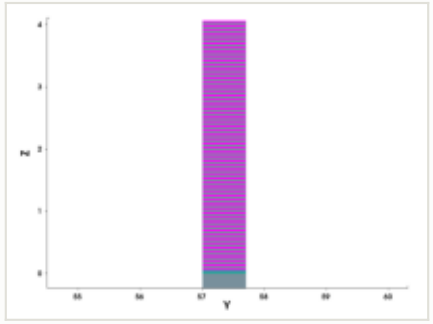
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

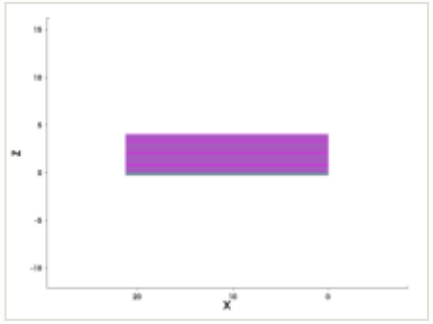


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



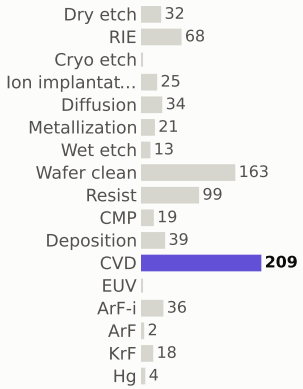
y cut at 57.351 µm



mask for this step

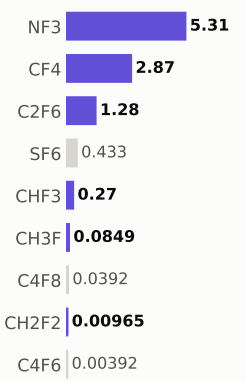
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



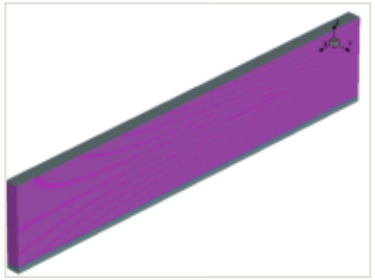
Step #10430: CVD

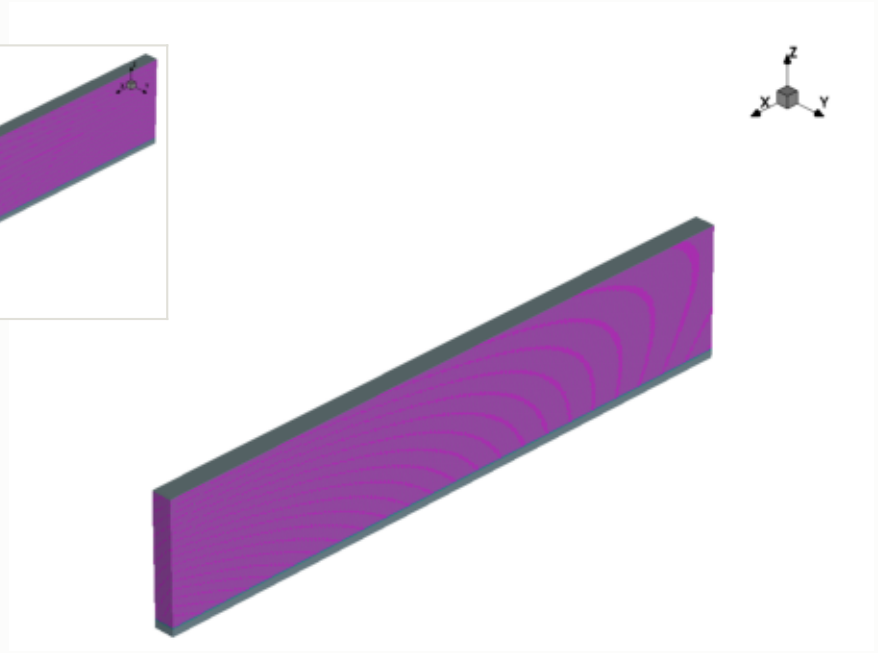
depo_isotropic · CVD

1.31	583.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



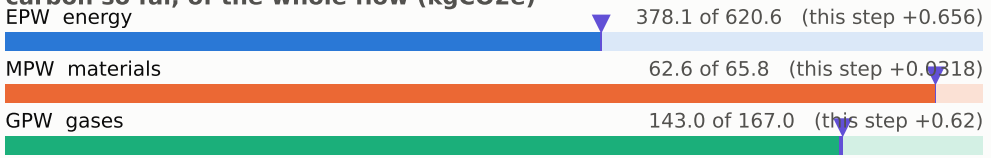


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

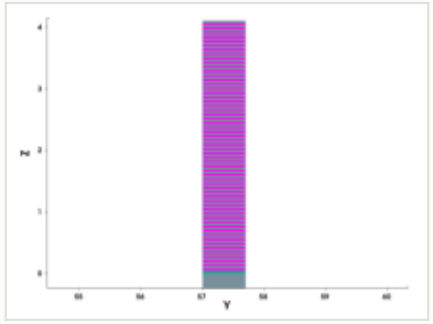
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

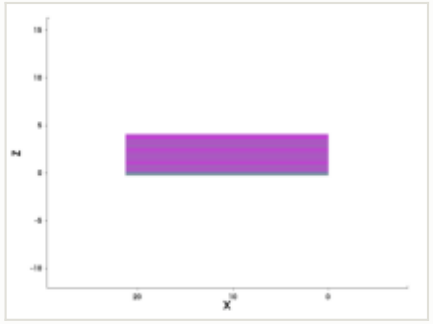


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



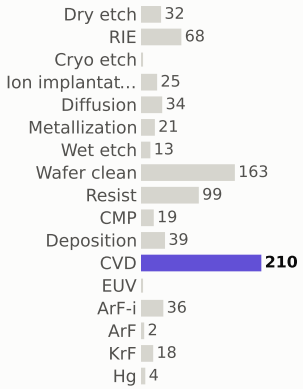
y cut at 57.351 µm



mask for this step

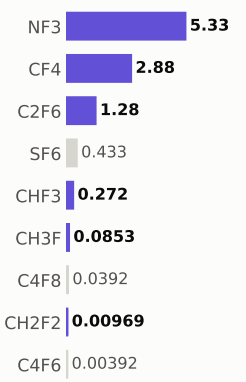
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



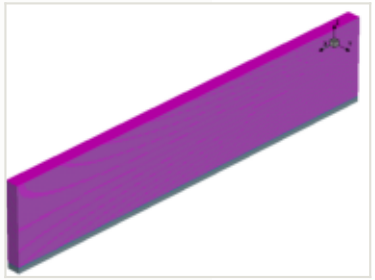
Step #10440: CVD

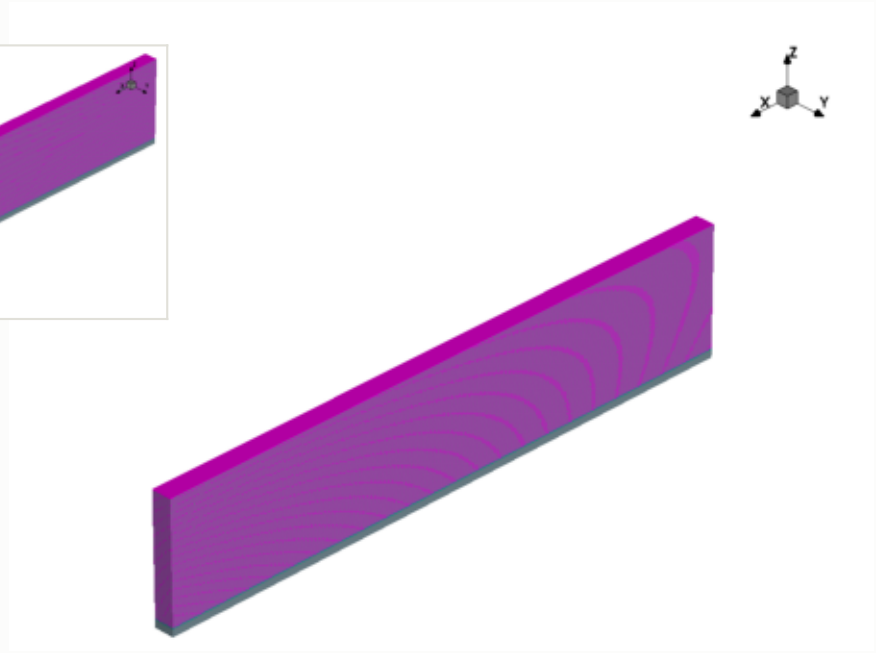
depo_isotropic · CVD

1.31	585.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



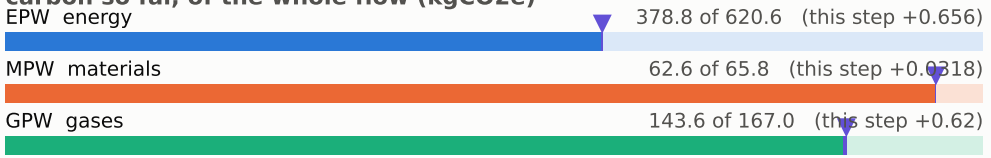


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

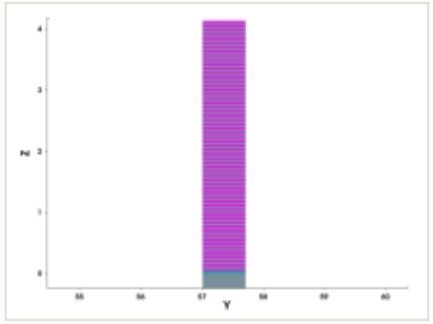
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

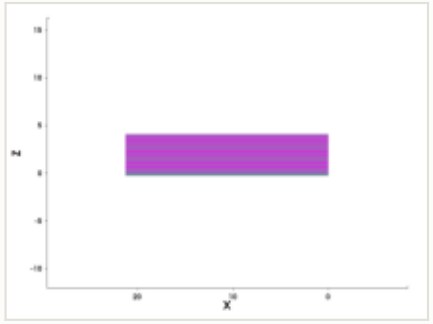


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



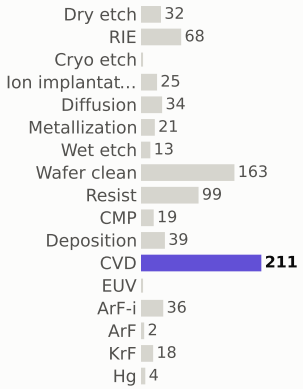
y cut at 57.351 µm



mask for this step

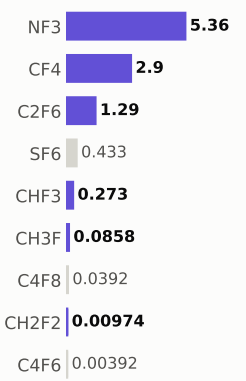
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



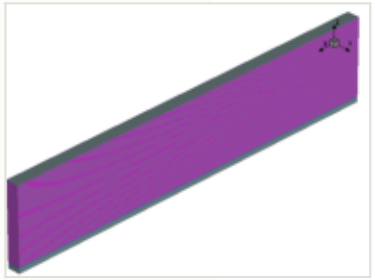
Step #10450: CVD

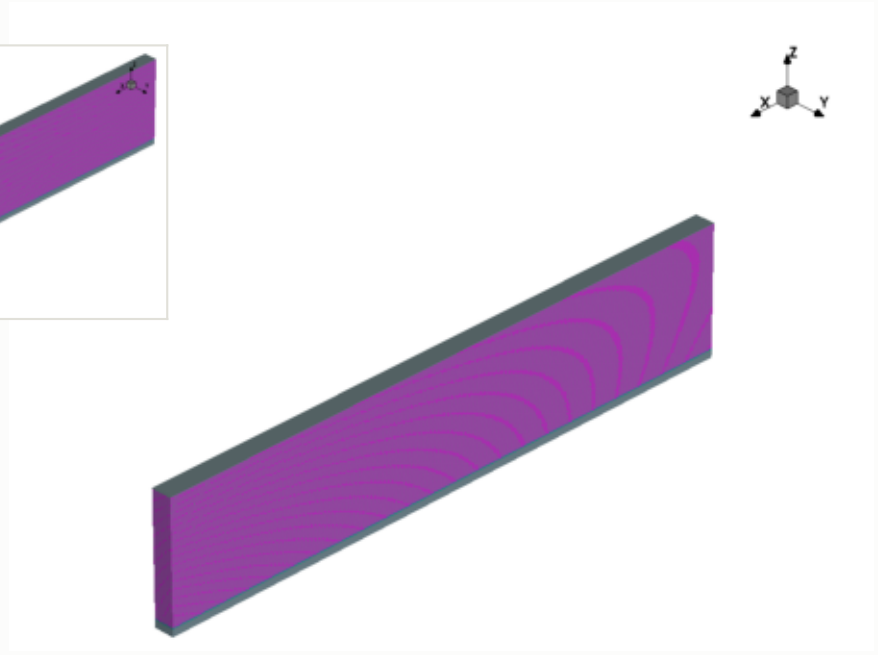
depo_isotropic · CVD

1.31	586.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



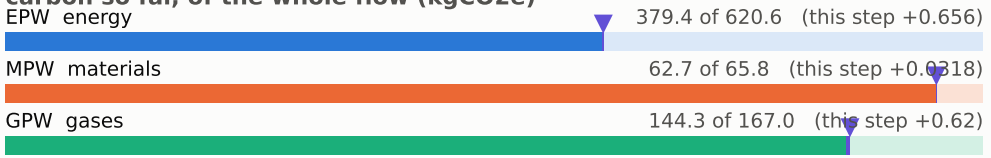


this step

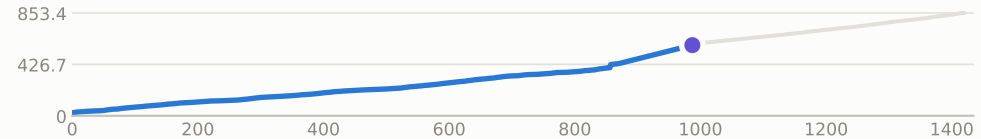
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

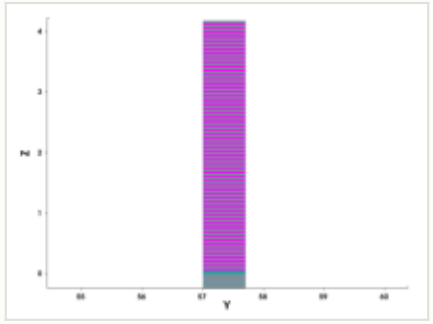
carbon so far, of the whole flow (kgCO2e)



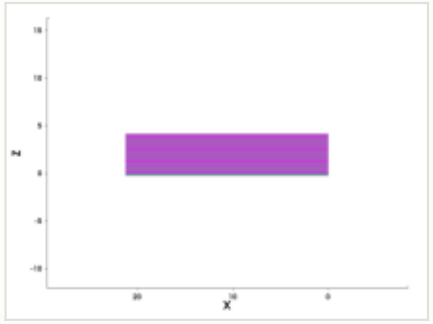
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



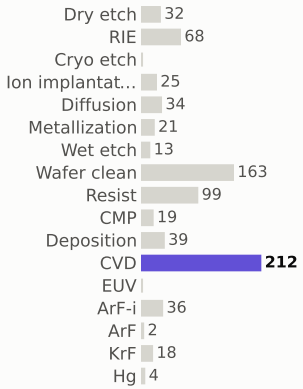
y cut at 57.351 µm



mask for this step

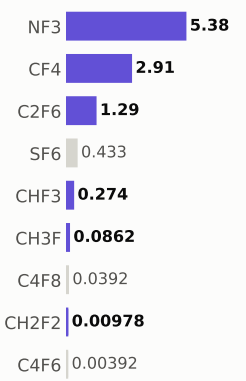
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



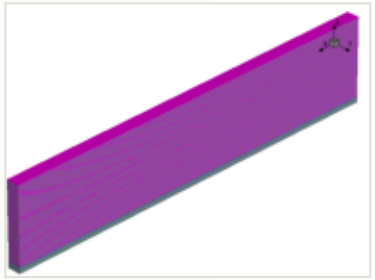
Step #10460: CVD

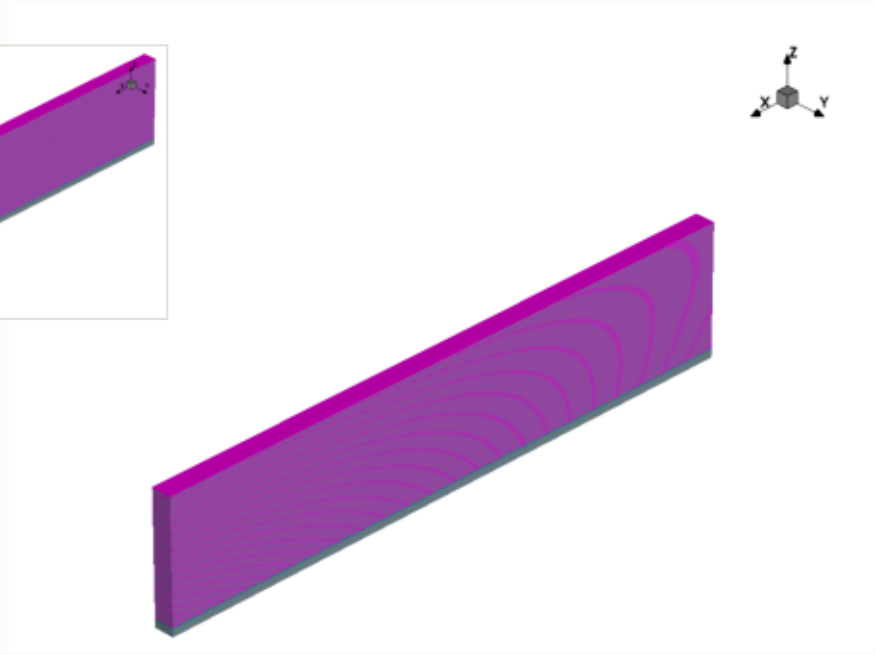
depo_isotropic · CVD

1.31	587.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



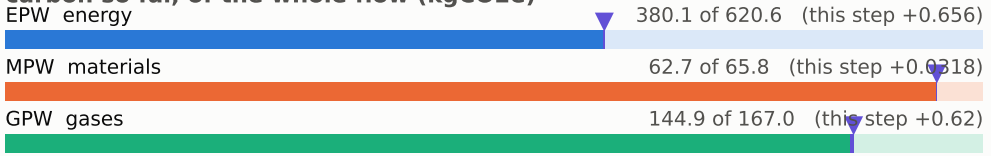


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

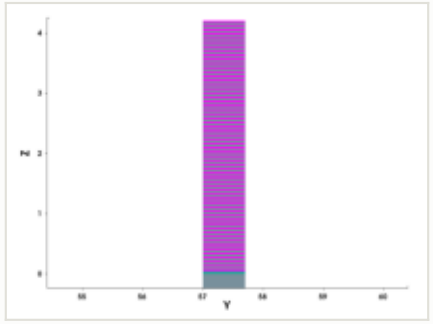
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

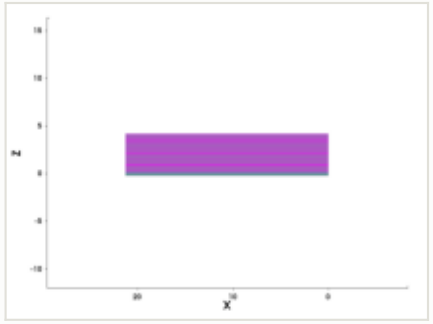


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



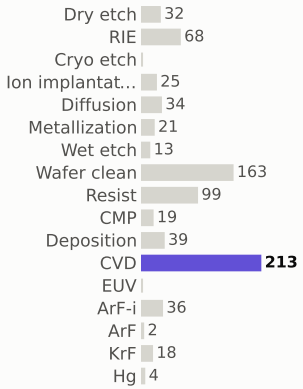
y cut at 57.351 µm



mask for this step

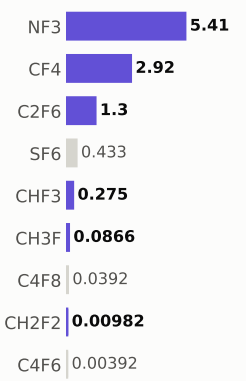
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



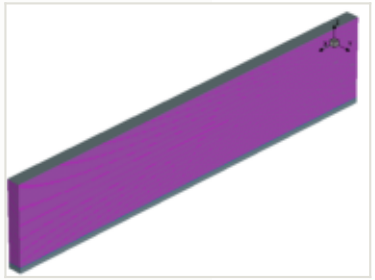
Step #10470: CVD

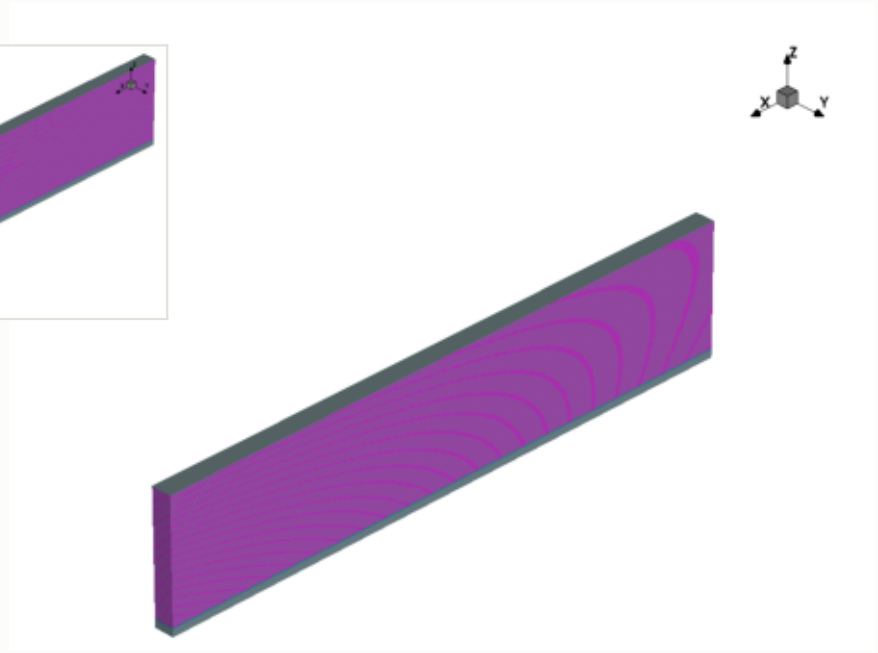
depo_isotropic · CVD

1.31	589.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



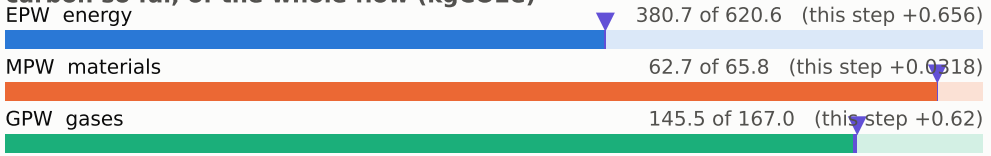


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

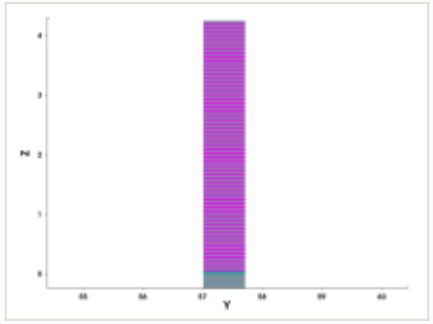
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

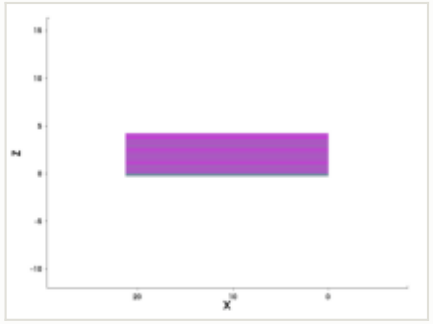


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



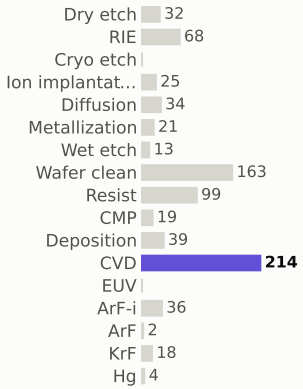
y cut at 57.351 µm



mask for this step

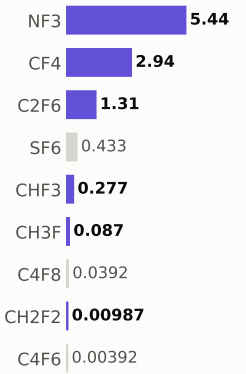
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



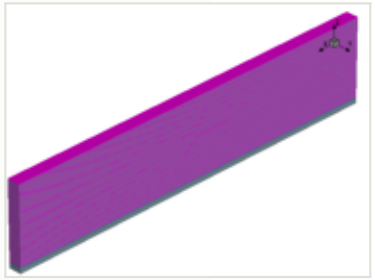
Step #10480: CVD

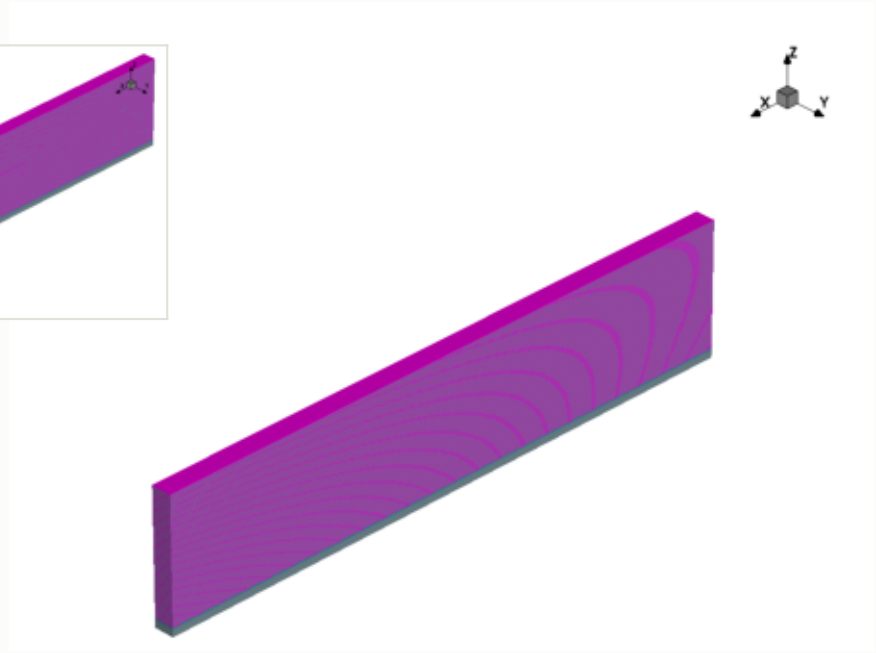
depo_isotropic · CVD

1.31	590.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



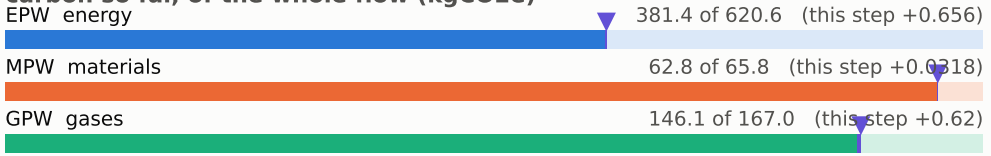


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

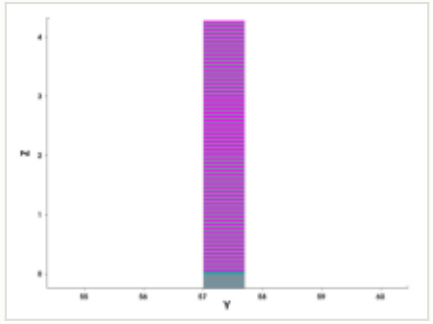
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

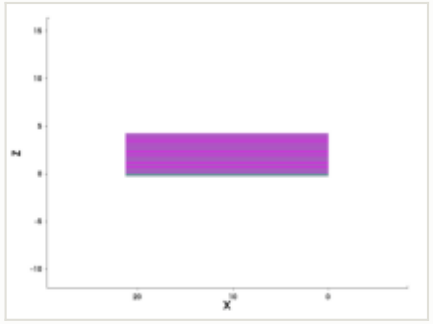


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



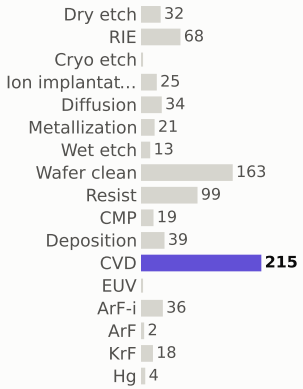
y cut at 57.351 µm



mask for this step

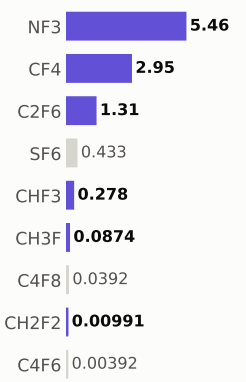
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



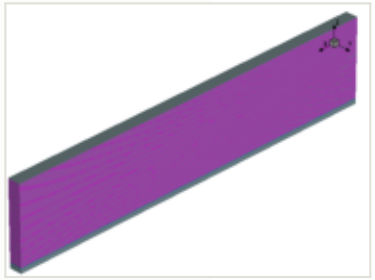
Step #10490: CVD

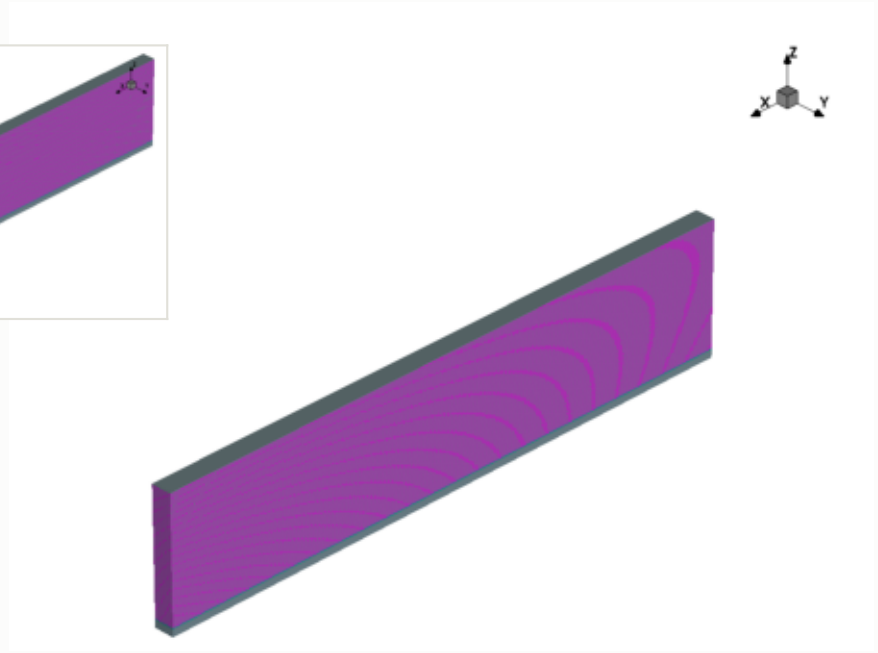
depo_isotropic · CVD

1.31	591.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



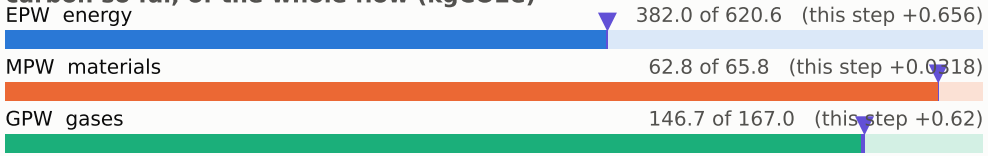


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

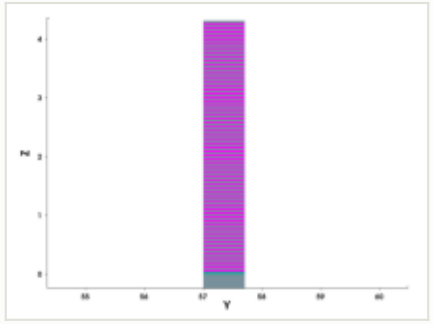
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

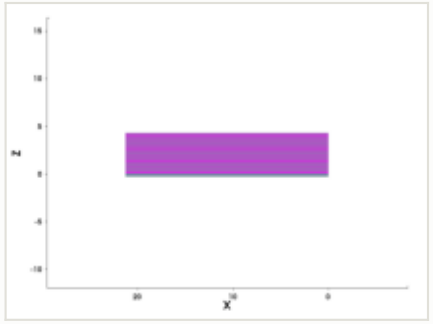


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



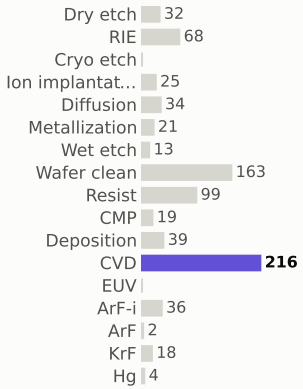
y cut at 57.351 µm



mask for this step

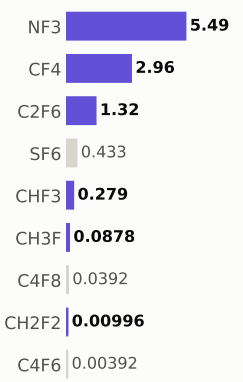
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



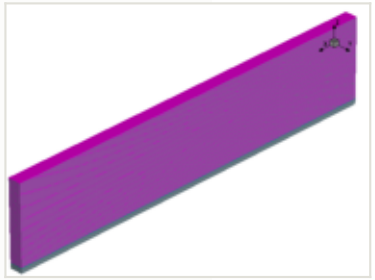
Step #10500: CVD

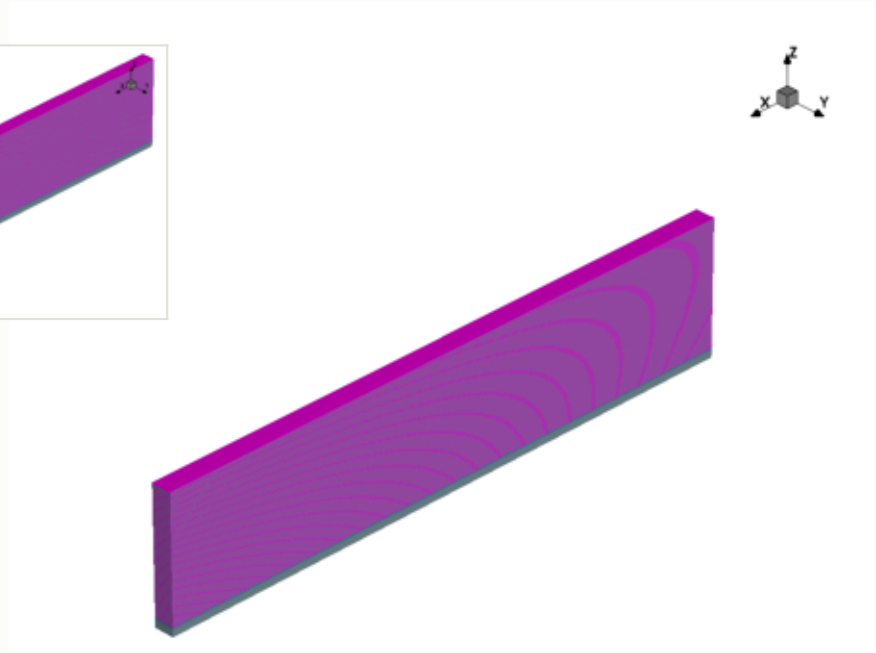
depo_isotropic · CVD

1.31	592.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



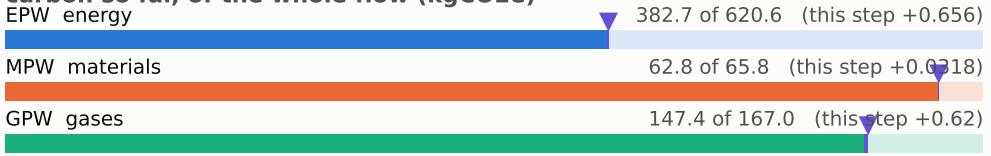


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

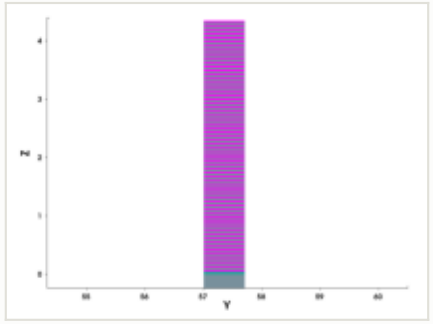
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

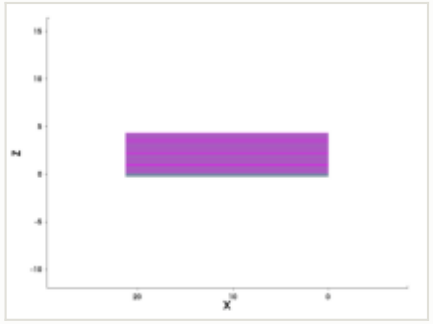


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



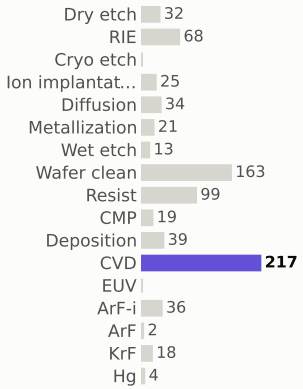
y cut at 57.351 µm



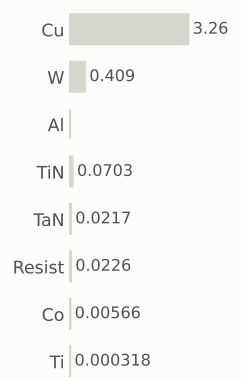
mask for this step

no mask for this step

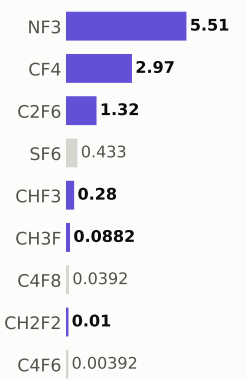
steps by process area



materials deposited (g)



gases emitted (g)



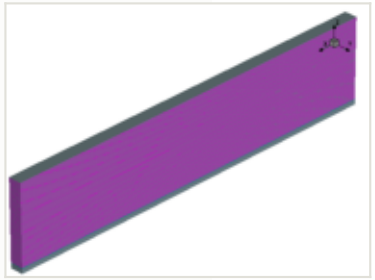
Step #10510: CVD

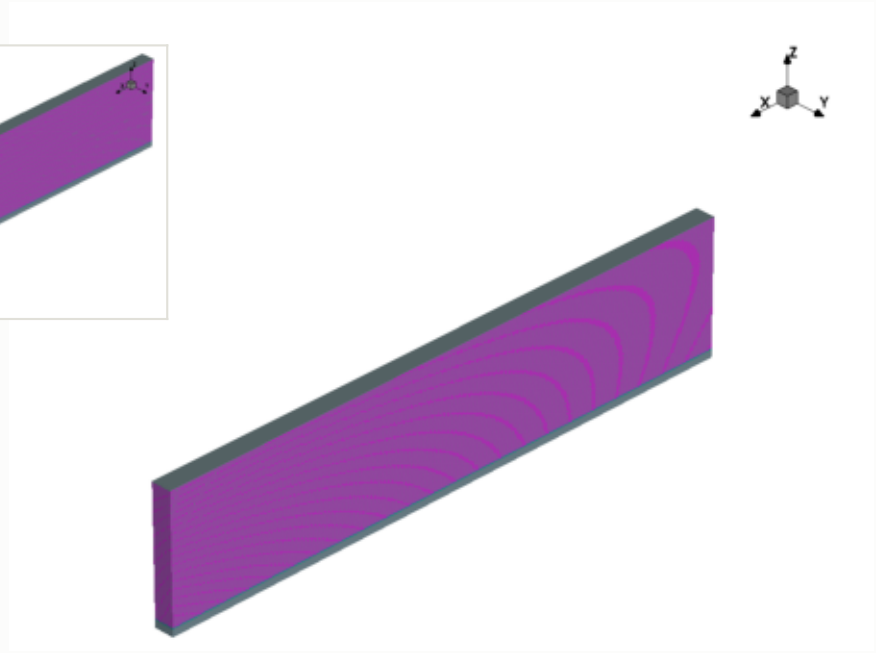
depo_isotropic · CVD

1.31	594.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



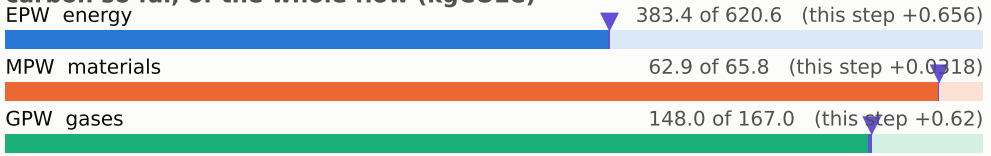


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

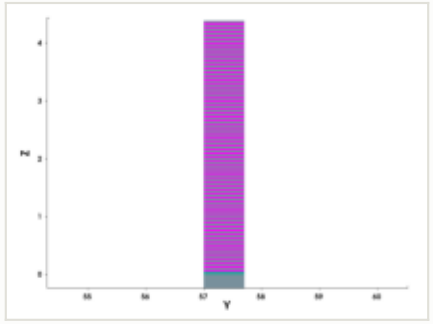
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

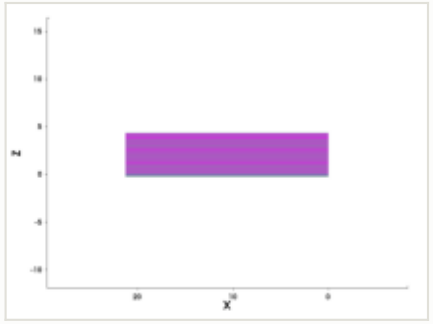


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



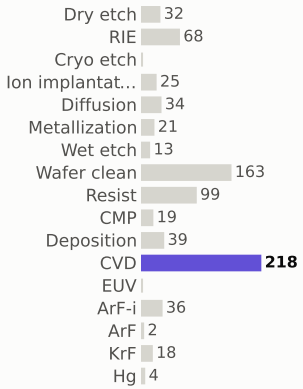
y cut at 57.351 µm



mask for this step

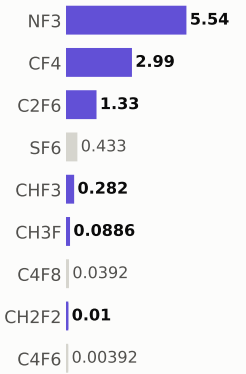
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



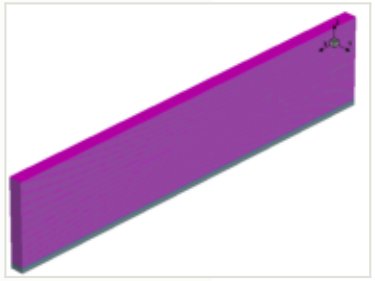
Step #10520: CVD

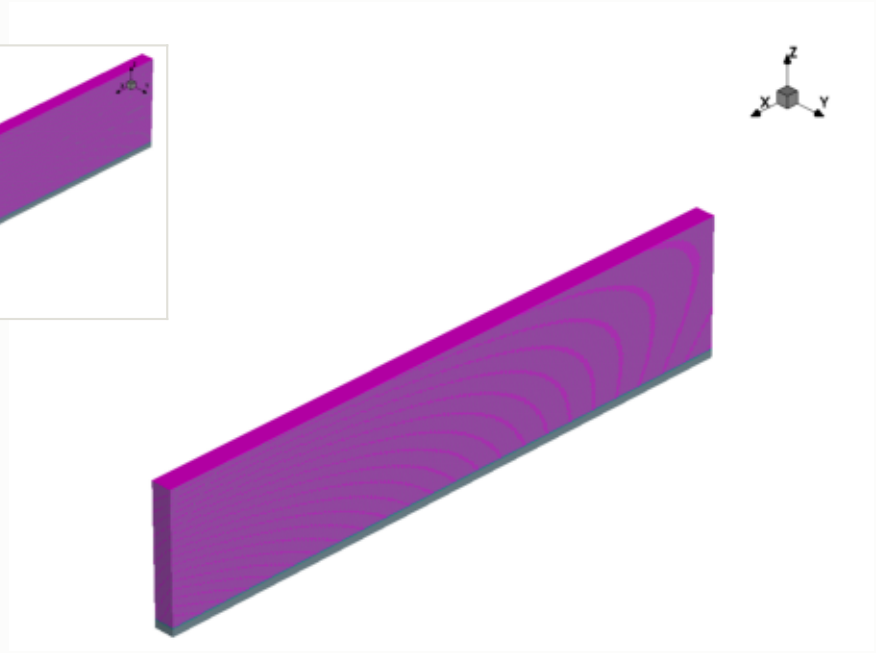
depo_isotropic · CVD

1.31	595.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



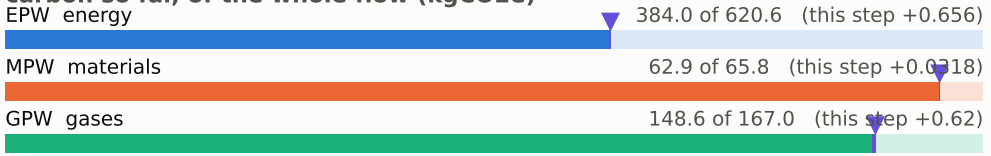


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

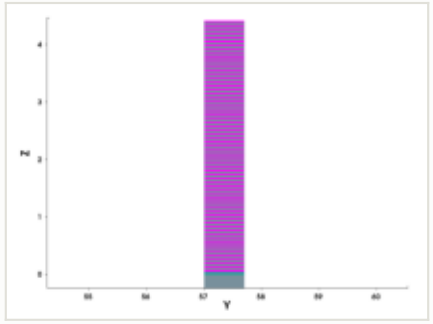
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

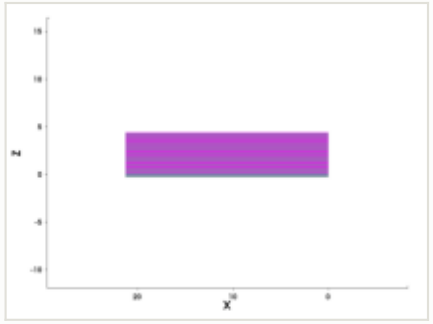


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



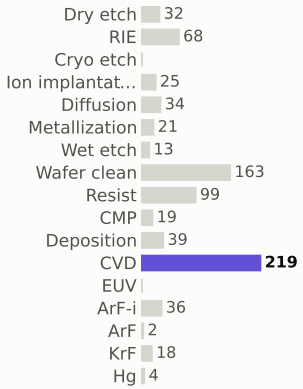
y cut at 57.351 µm



mask for this step

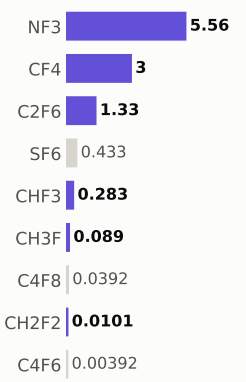
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



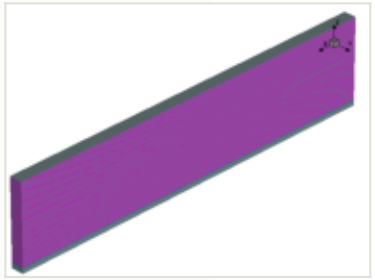
Step #10530: CVD

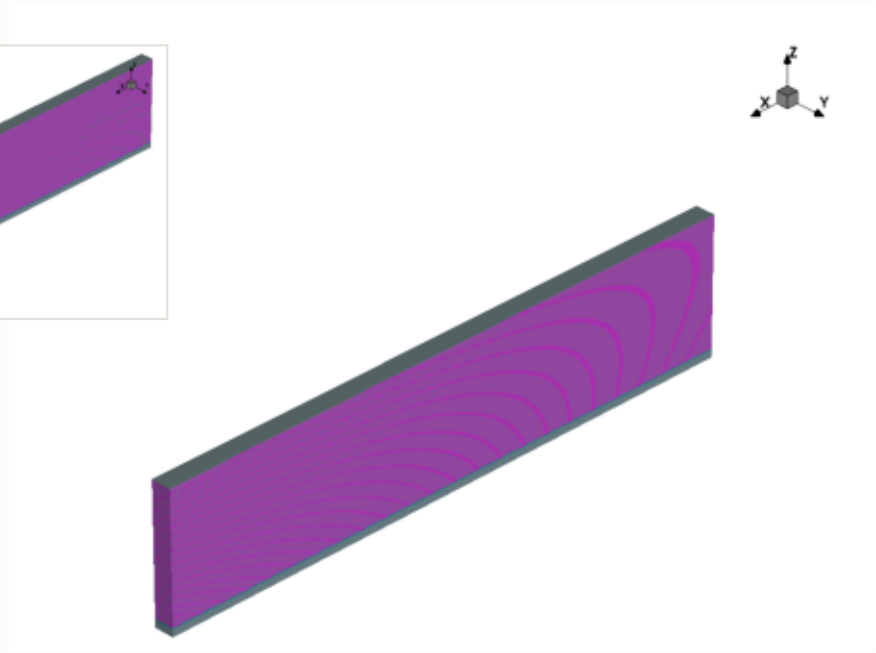
depo_isotropic · CVD

1.31	596.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



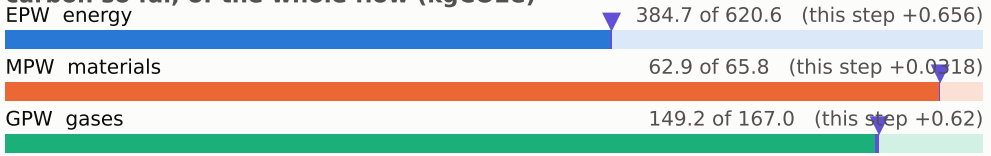


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

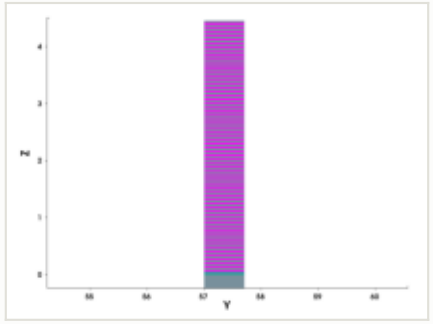
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

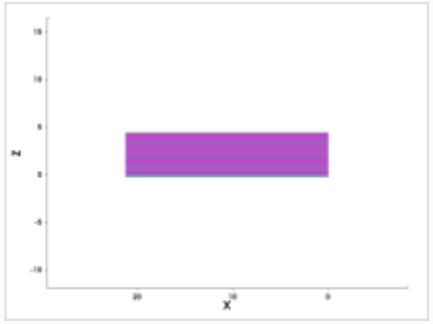


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



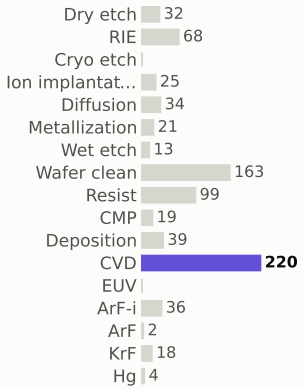
y cut at 57.351 µm



mask for this step

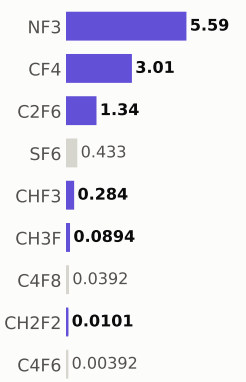
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



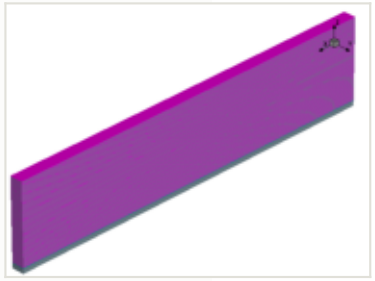
Step #10540: CVD

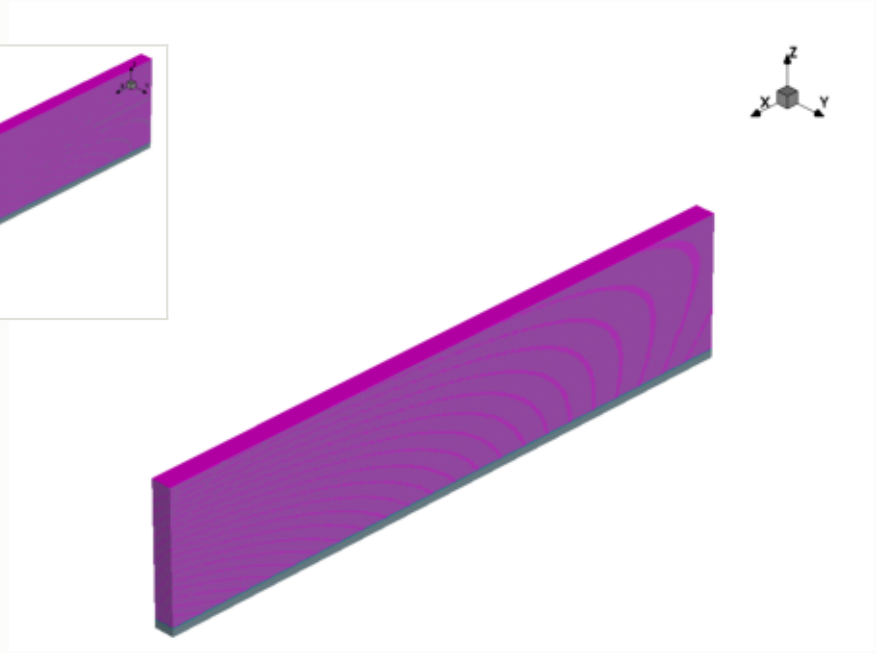
depo_isotropic · CVD

1.31	598.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



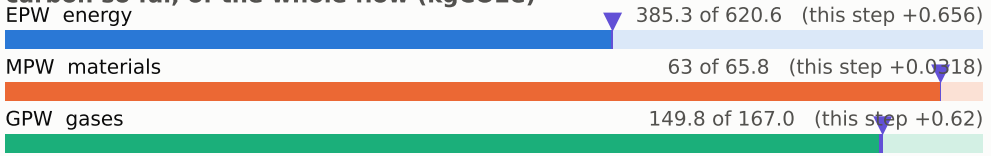


this step

process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

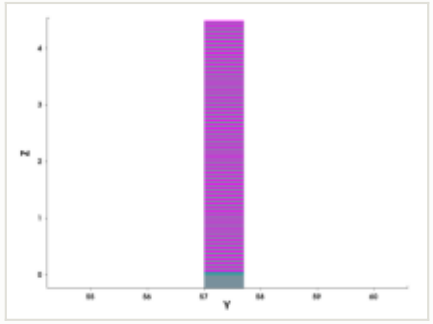
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

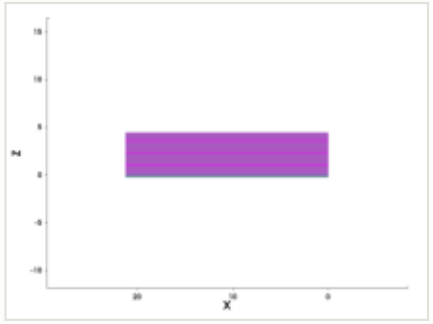


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



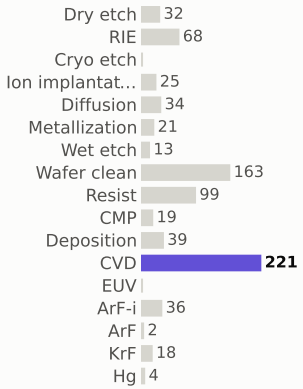
y cut at 57.351 µm



mask for this step

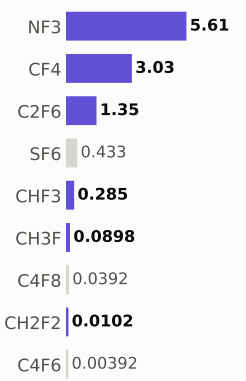
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



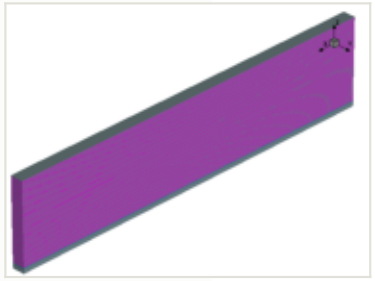
Step #10550: CVD

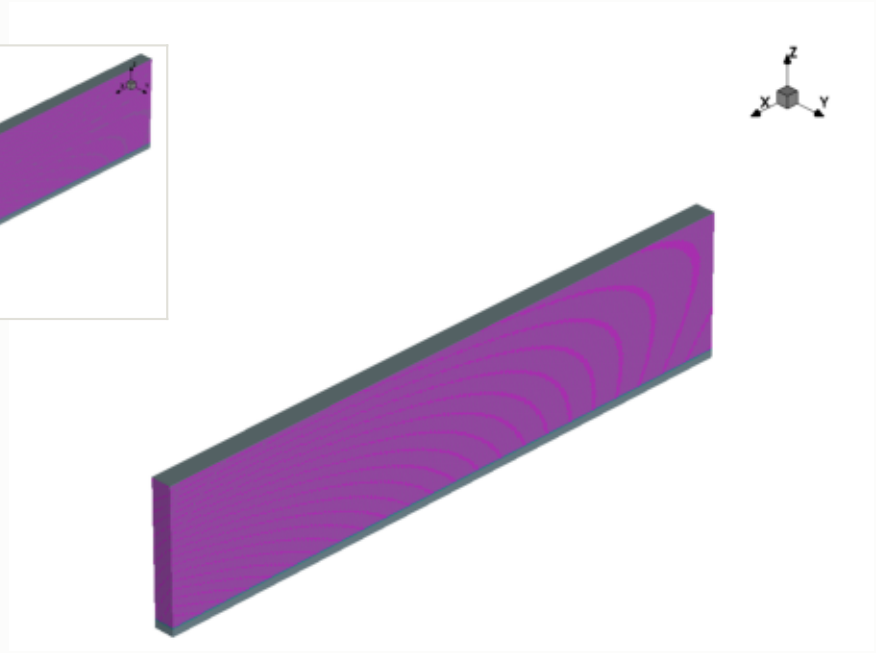
depo_isotropic · CVD

1.31	599.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



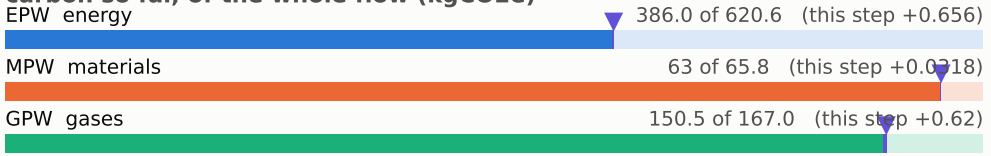


this step

process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

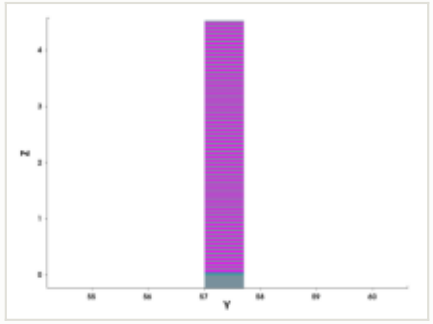
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

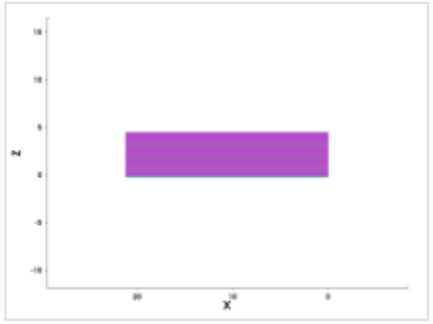


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



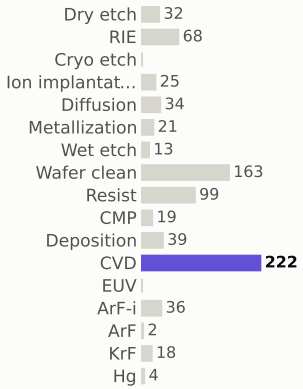
y cut at 57.351 µm



mask for this step

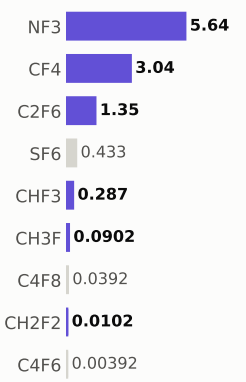
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



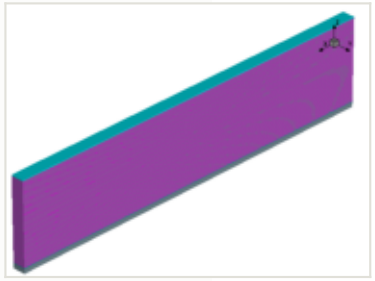
Step #10560: Deposition

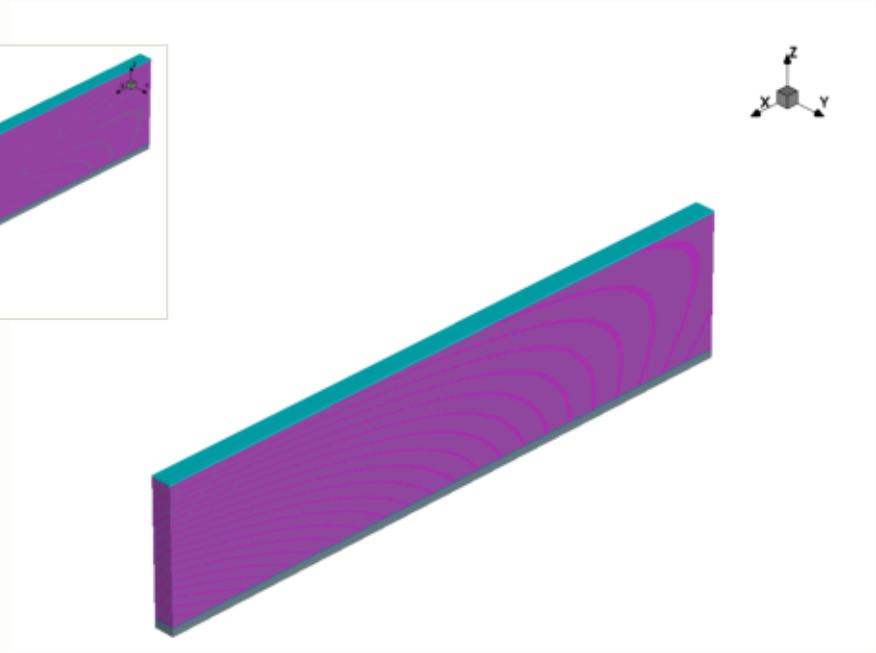
depo_isotropic · Deposition

0.656	600.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



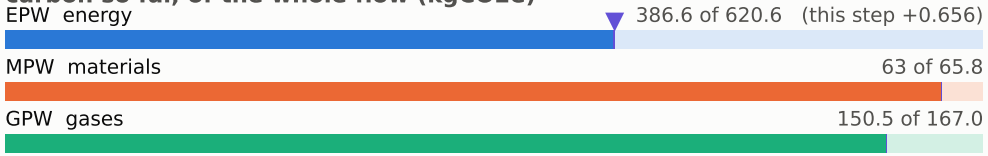


this step

process area	Deposition
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

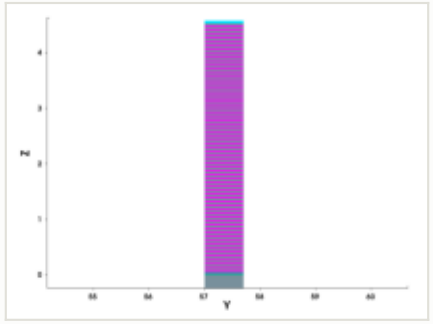
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

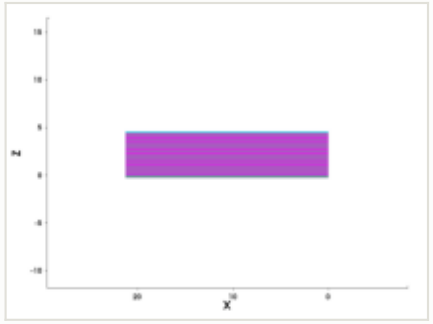


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



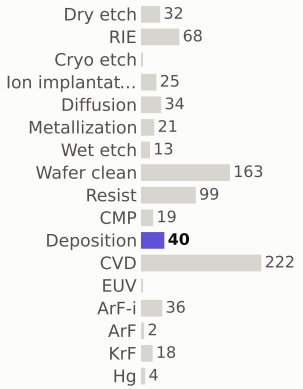
y cut at 57.351 µm



mask for this step

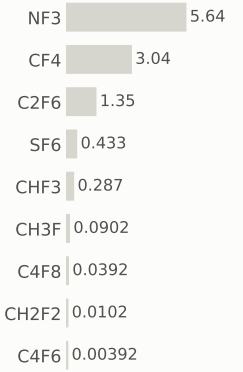
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



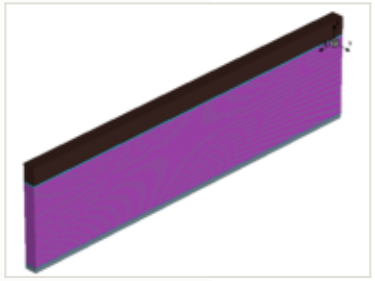
Step #10580: Deposition

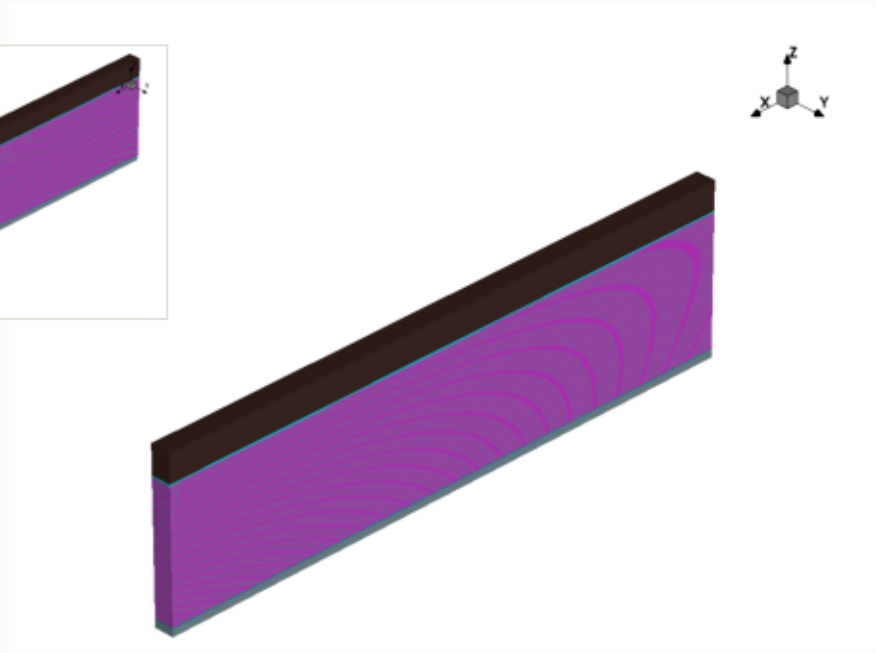
depo_isotropic · Deposition

0.656	600.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



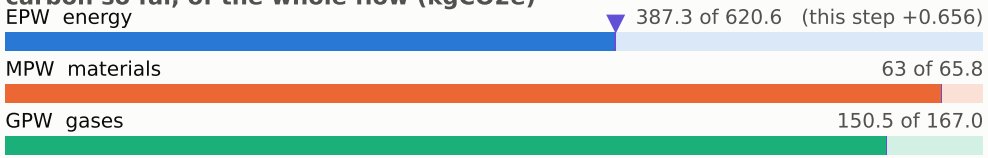


this step

process area	Deposition
material	—
thickness	1 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

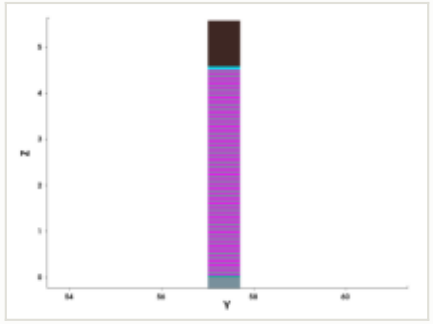
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

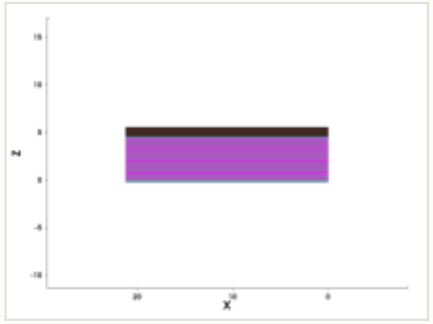


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



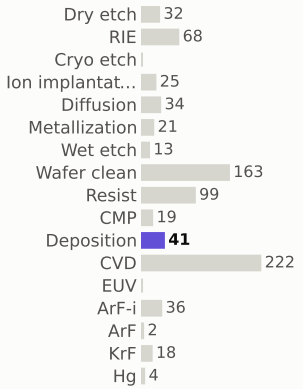
y cut at 57.351 µm



mask for this step

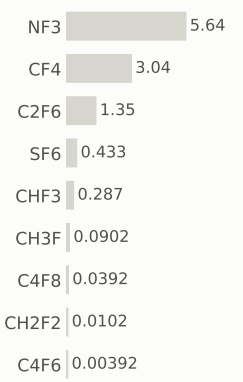
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



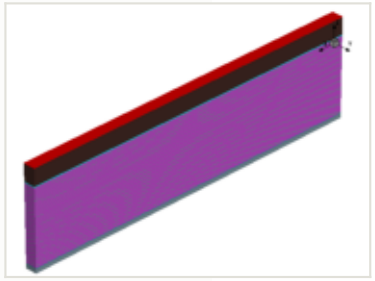
Step #10590: Deposition

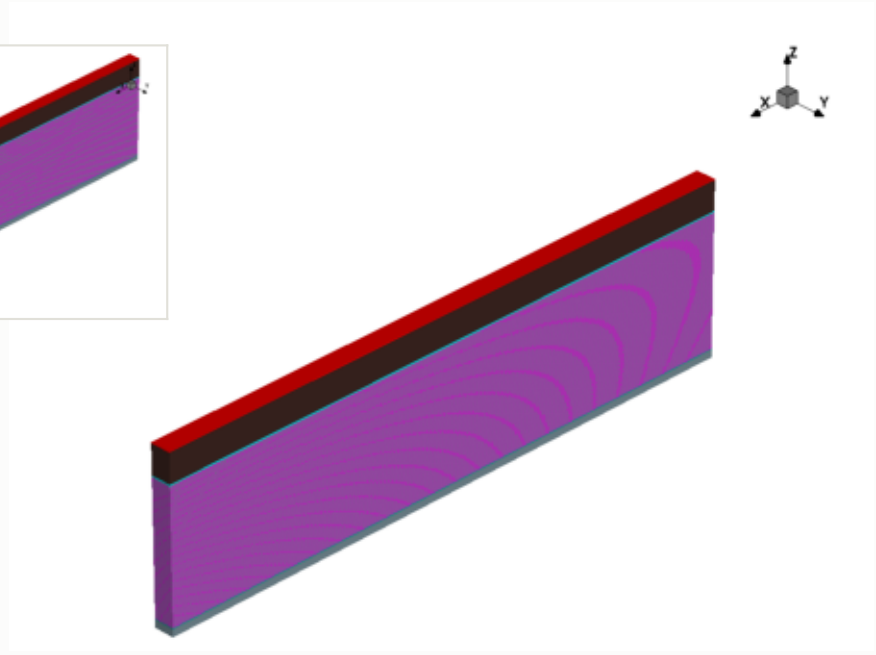
depo_isotropic · Deposition

0.656	601.4	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



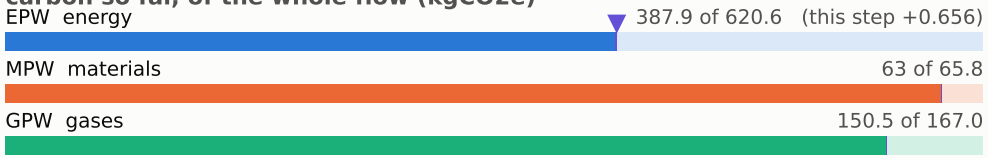


this step

process area	Deposition
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

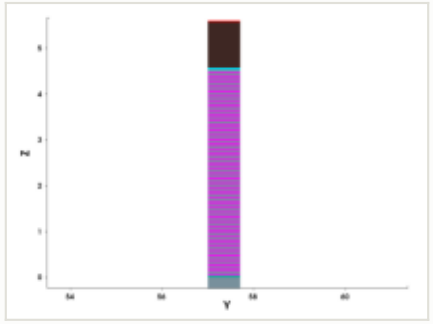
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

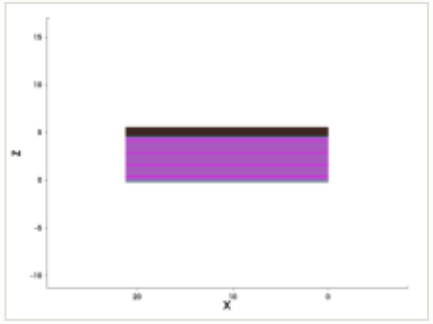


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



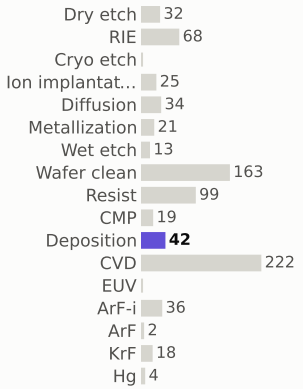
y cut at 57.351 µm



mask for this step

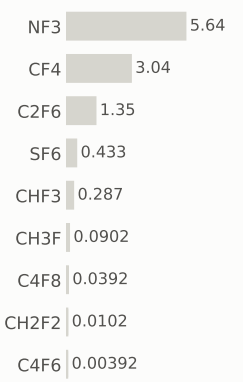
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



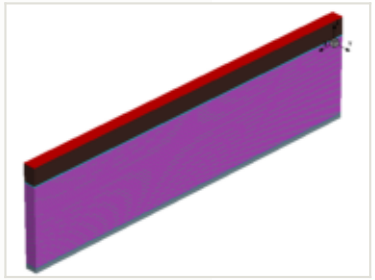
Step #10600: Wafer_clean

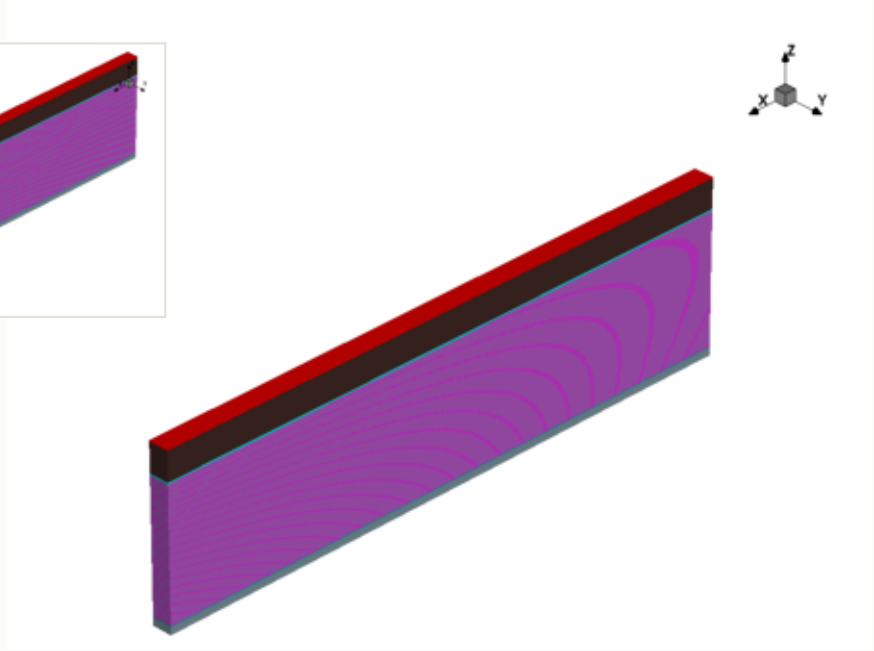
clean · Wafer_clean

0.169	601.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





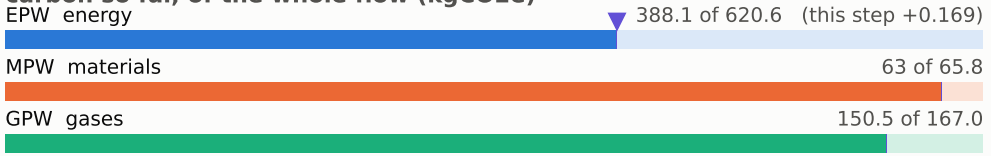
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

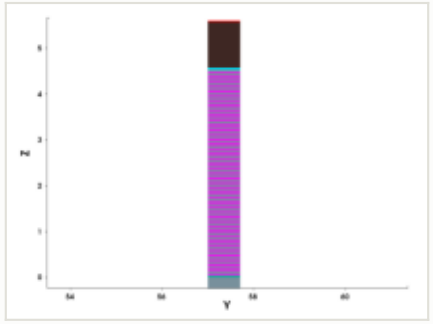
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

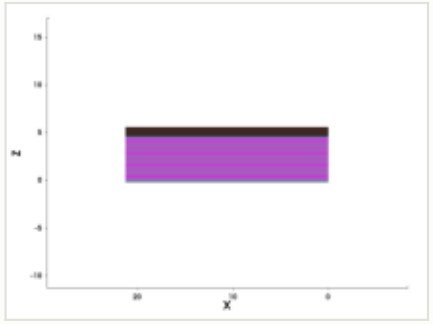


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



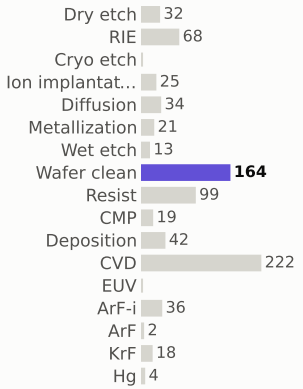
y cut at 57.351 µm



mask for this step

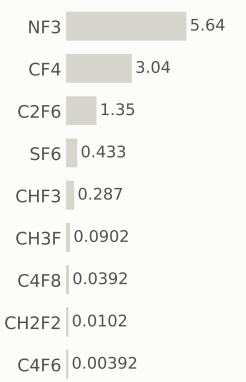
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



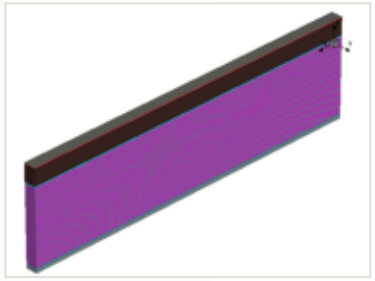
Step #10610: EUV

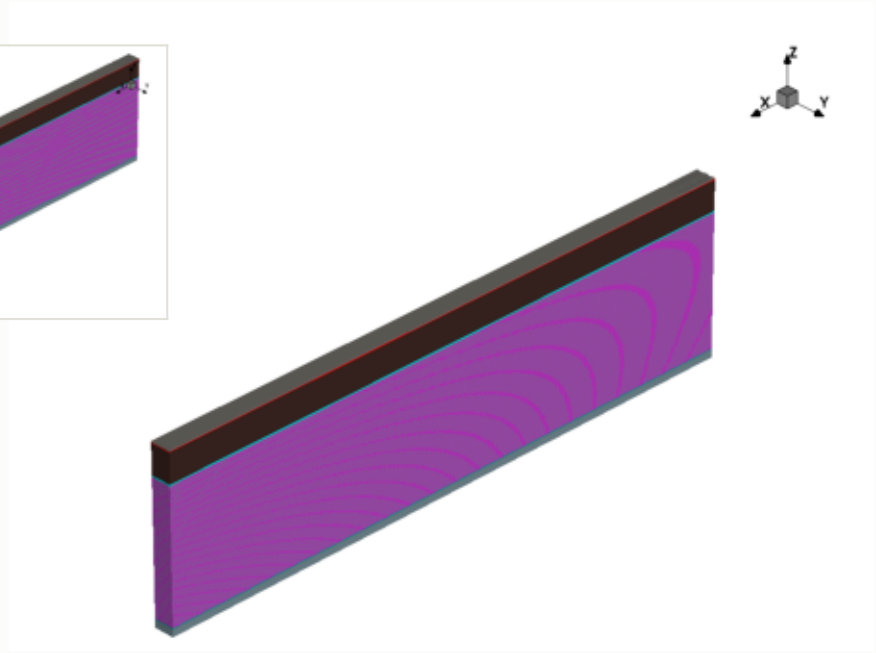
litho_patterning · EUV

1.96	603.5	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



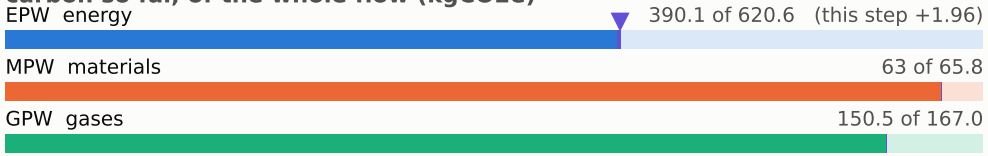


this step

process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

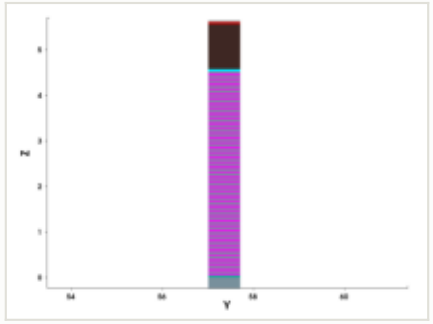
EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

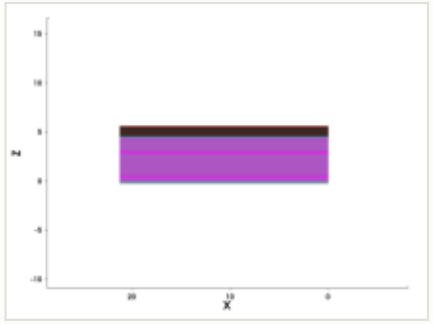


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

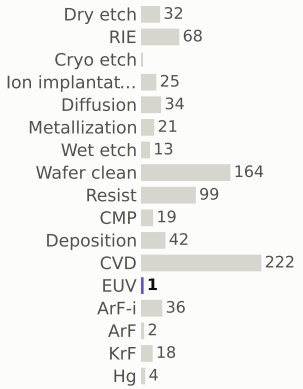


y cut at 57.351 µm



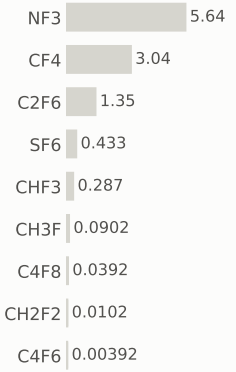
mask: layer 1:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



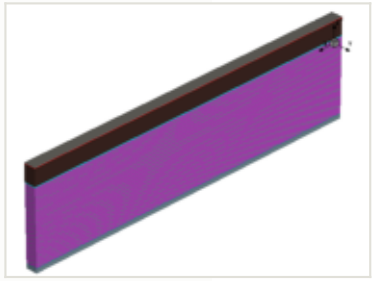
Step #10620: RIE

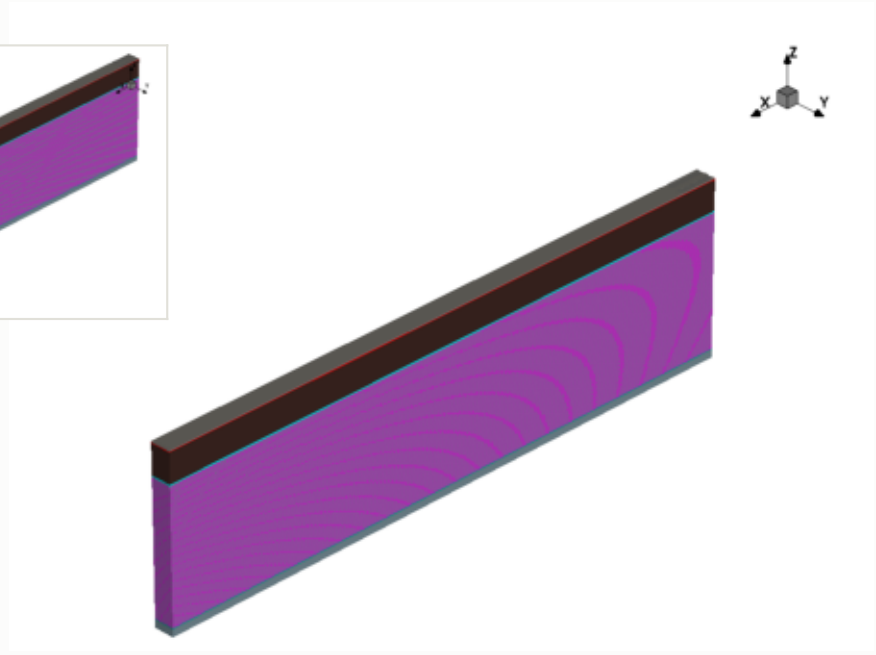
etch_anisotropic · RIE

1.04	604.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



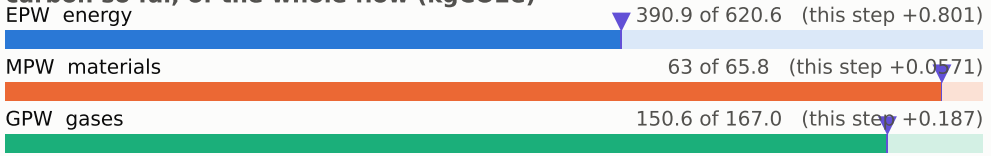


this step

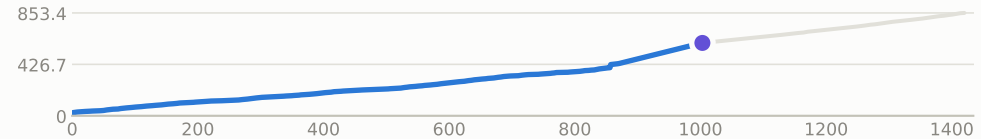
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

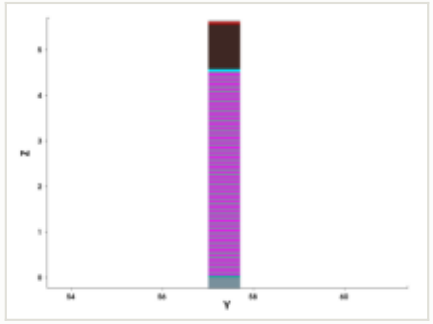
carbon so far, of the whole flow (kgCO2e)



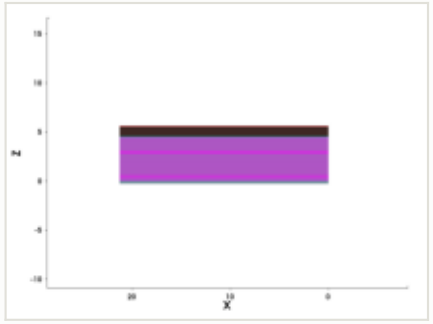
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



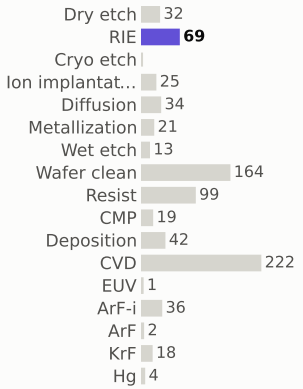
y cut at 57.351 µm



mask for this step

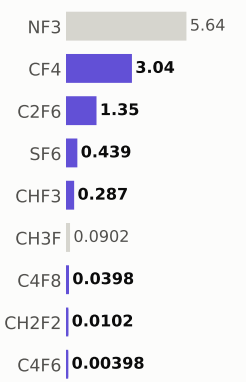
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



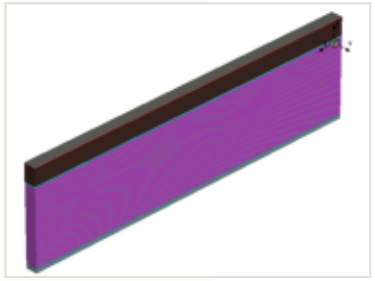
Step #10630: Cryo_etch

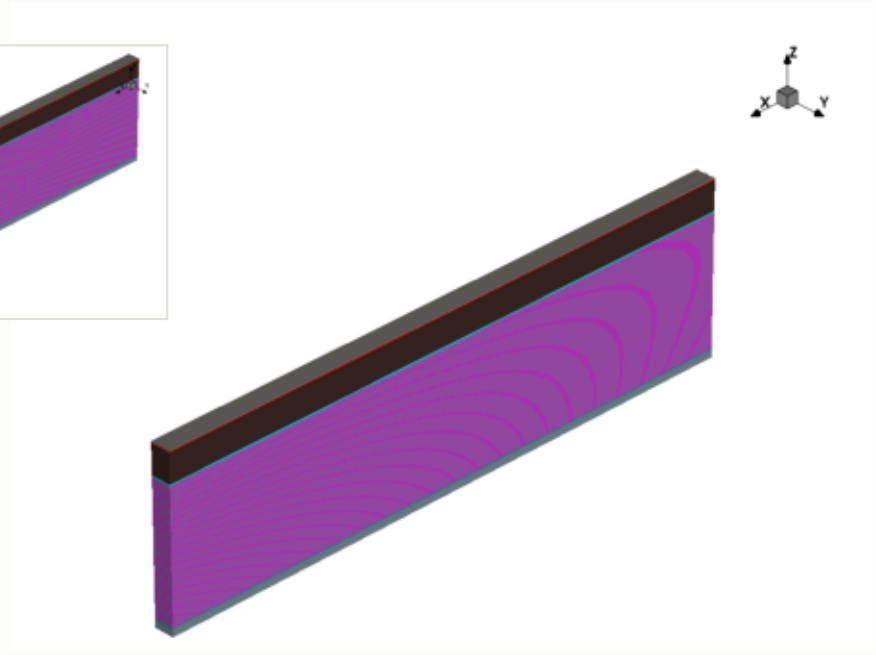
etch_anisotropic · Cryo_etch

0.517	605.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



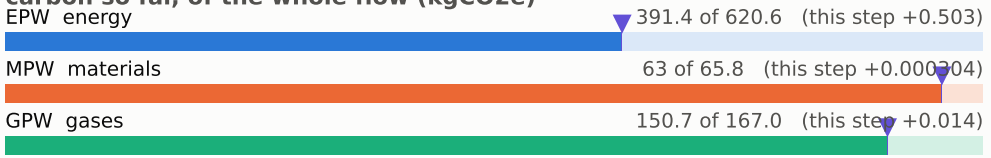


this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

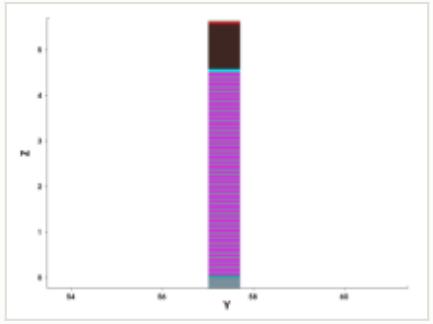
EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

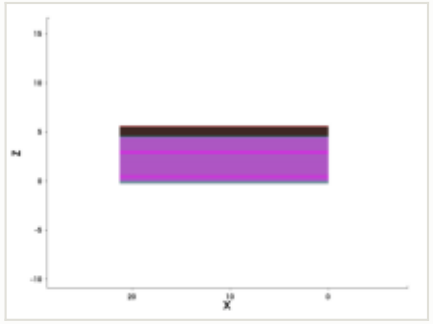


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



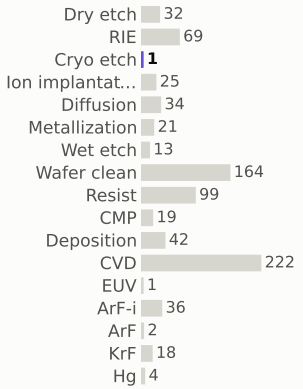
y cut at 57.351 µm



mask for this step

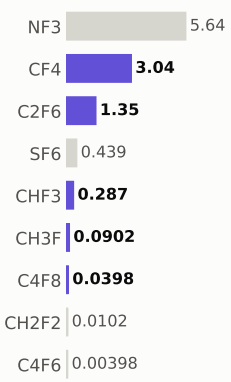
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



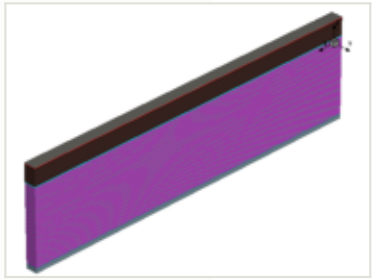
Step #10630.3846: Cryo_etch

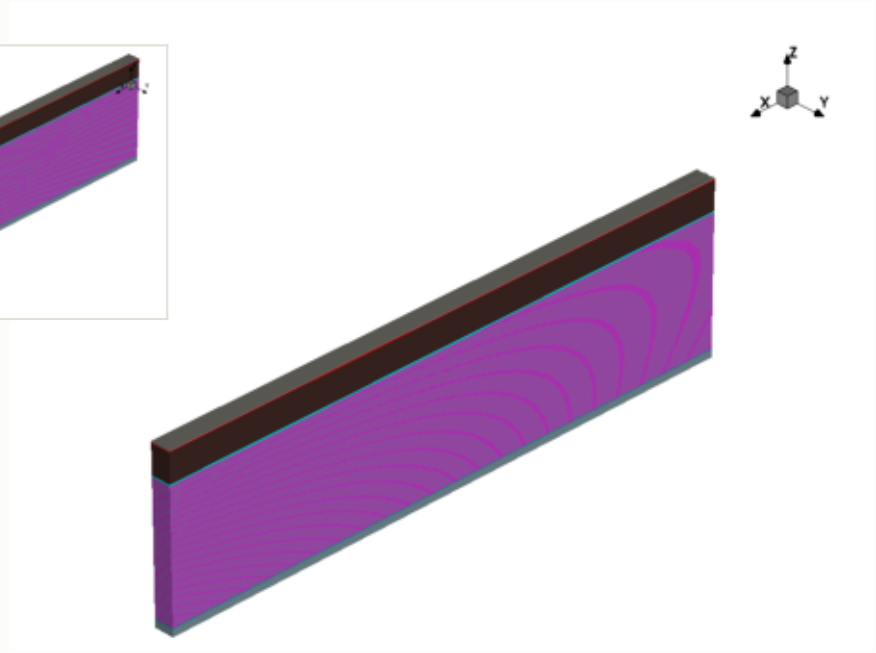
etch_noop · Cryo_etch

0.517	605.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





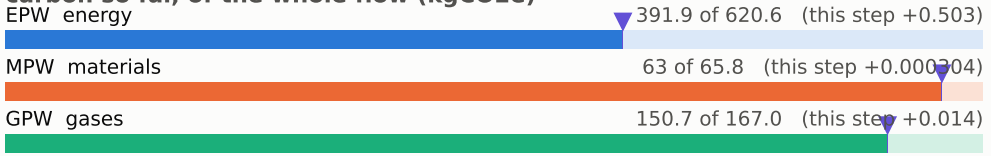
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

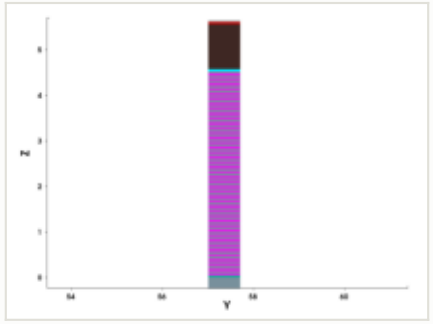
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

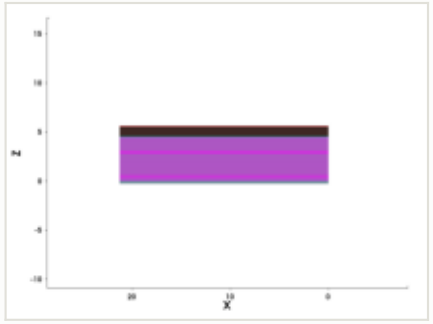


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



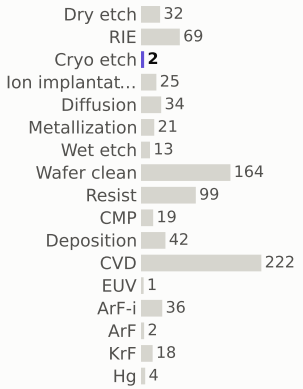
y cut at 57.351 µm



mask for this step

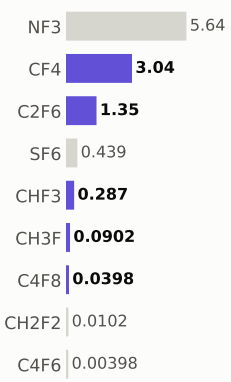
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



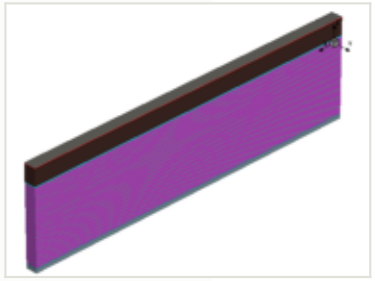
Step #10630.7692: Cryo_etch

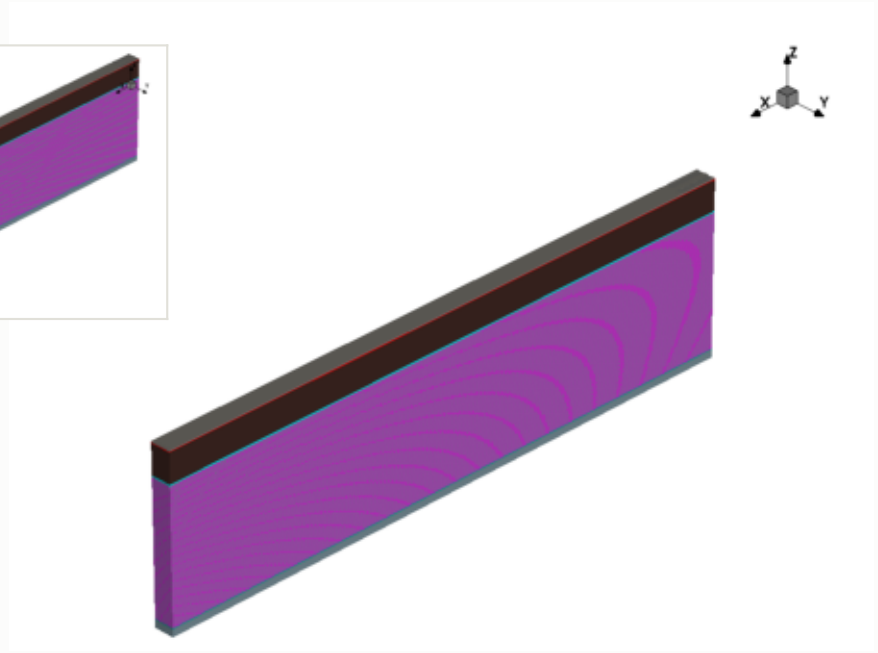
etch_noop · Cryo_etch

0.517	606.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





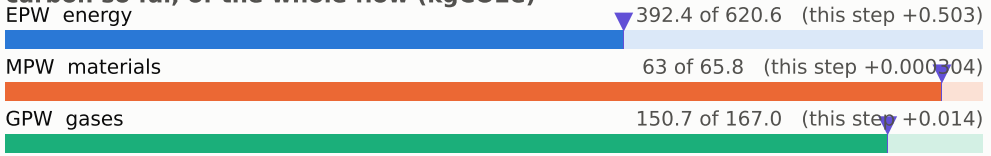
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

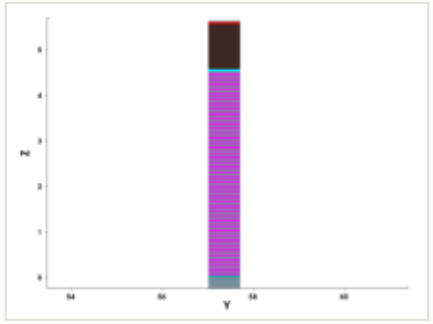
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

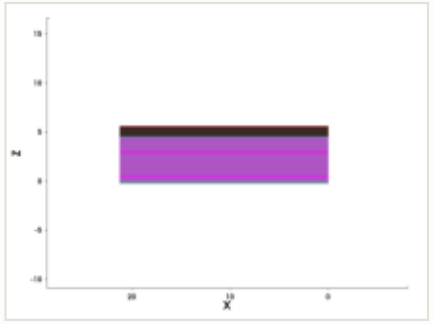


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



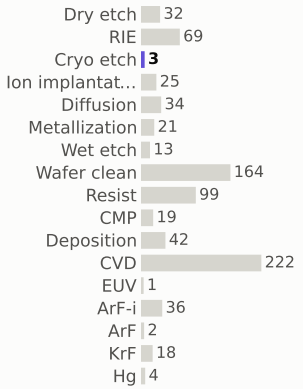
y cut at 57.351 µm



mask for this step

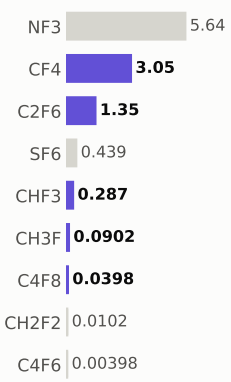
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



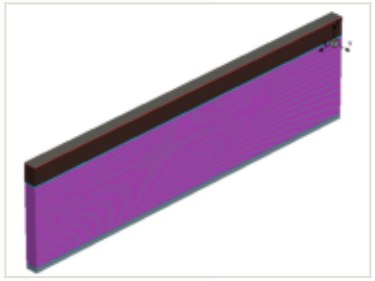
Step #10631.1538: Cryo_etch

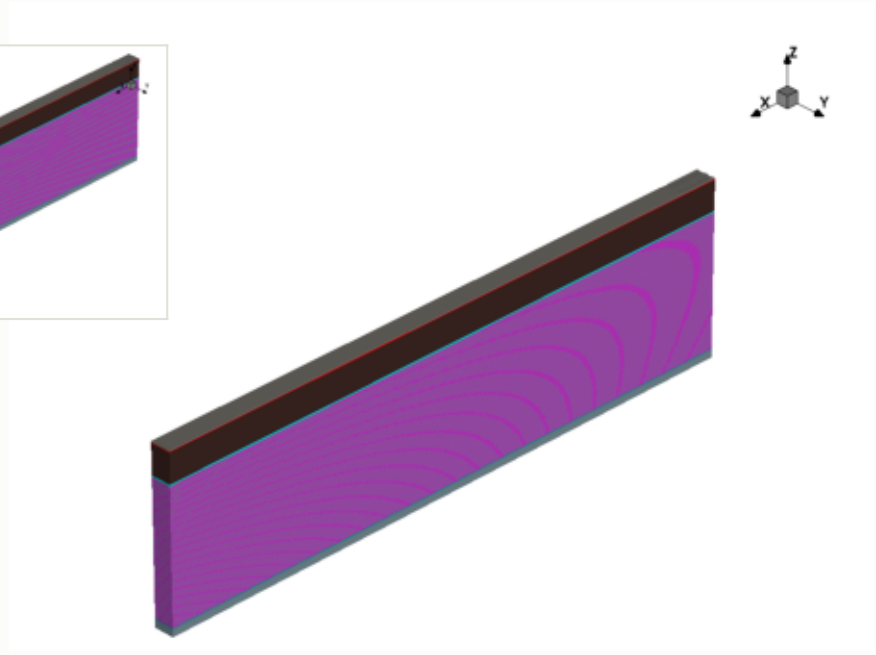
etch_noop · Cryo_etch

0.517	606.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





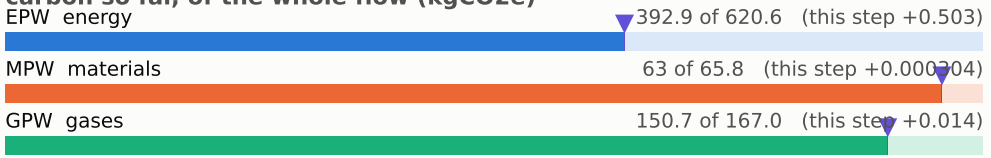
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

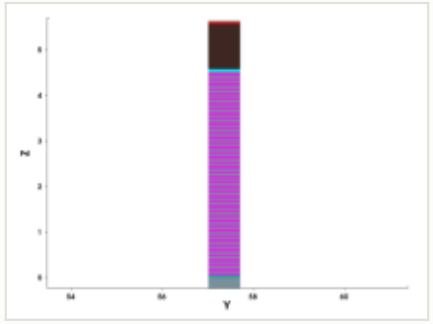
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

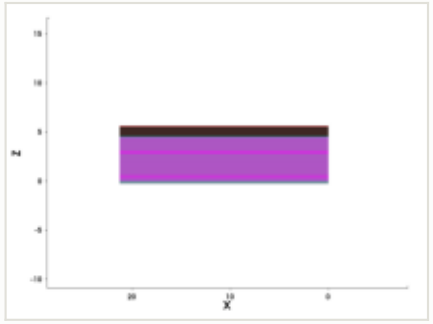


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



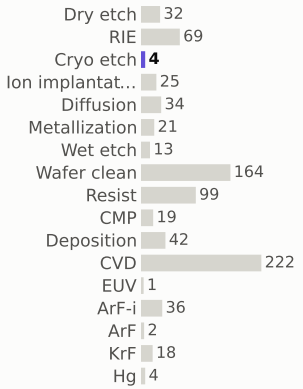
y cut at 57.351 µm



mask for this step

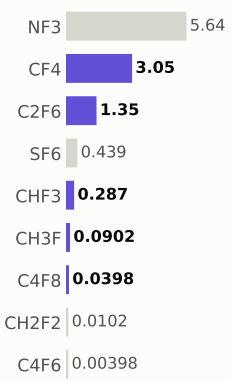
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



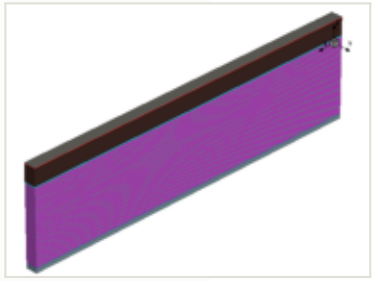
Step #10631.5385: Cryo_etch

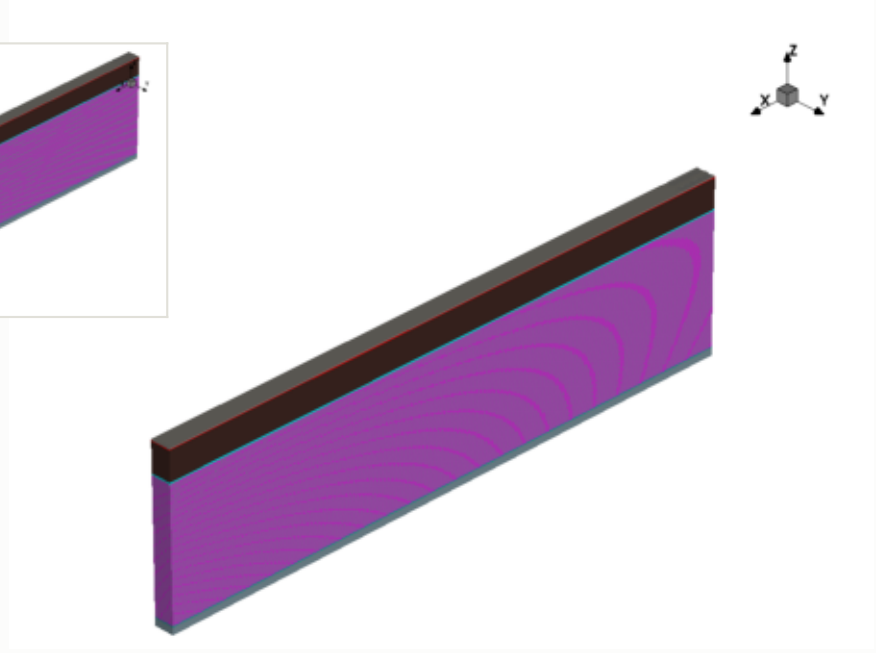
etch_noop · Cryo_etch

0.517	607.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





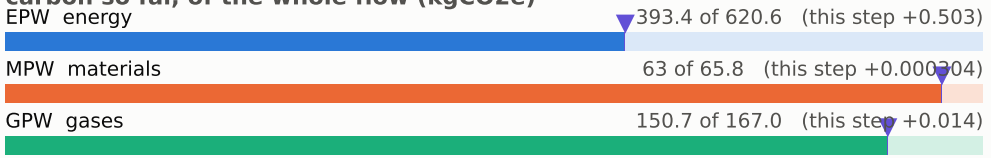
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

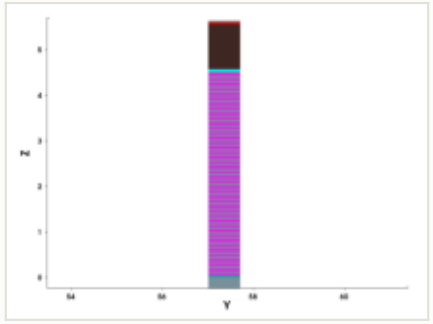
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

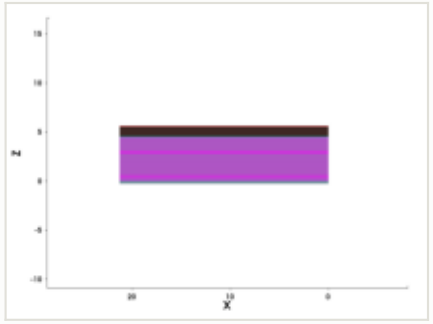


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



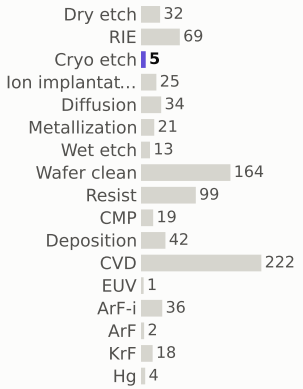
y cut at 57.351 µm



mask for this step

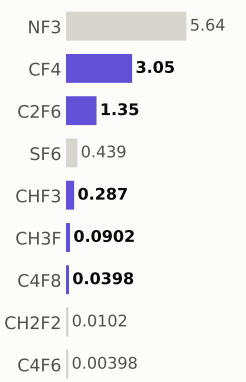
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



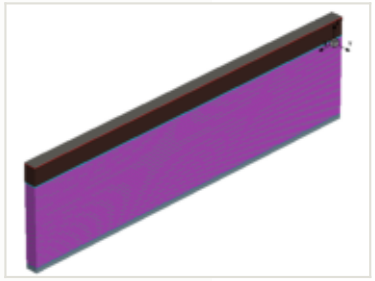
Step #10631.9231: Cryo_etch

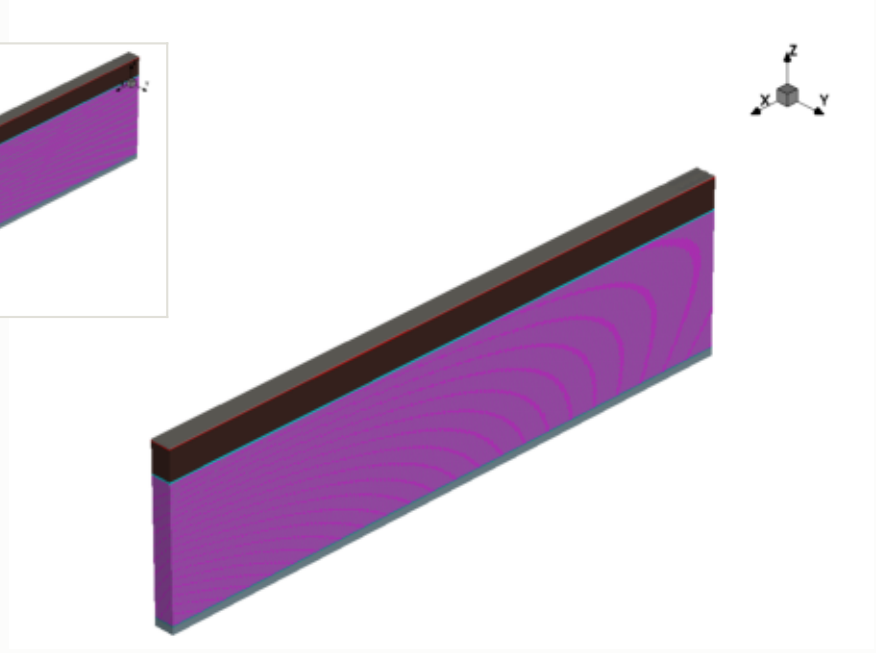
etch_noop · Cryo_etch

0.517	607.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





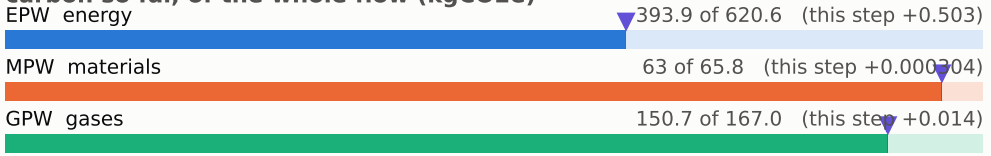
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

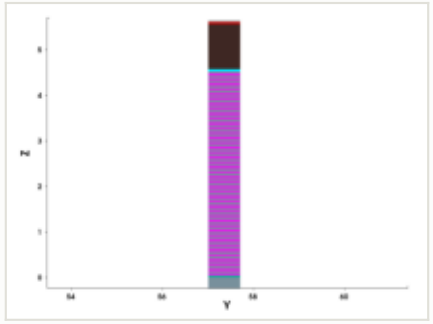
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

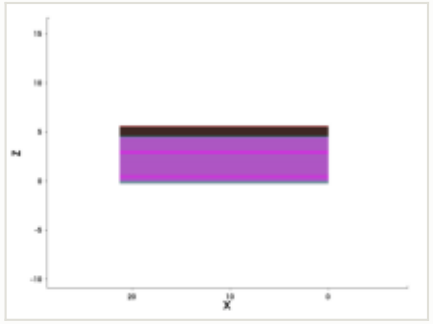


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



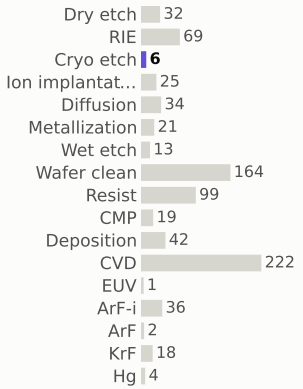
y cut at 57.351 µm



mask for this step

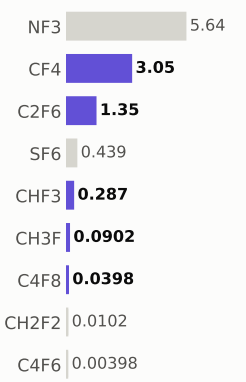
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



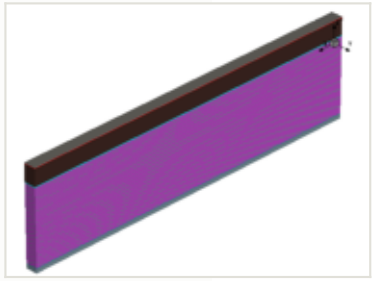
Step #10632.3077: Cryo_etch

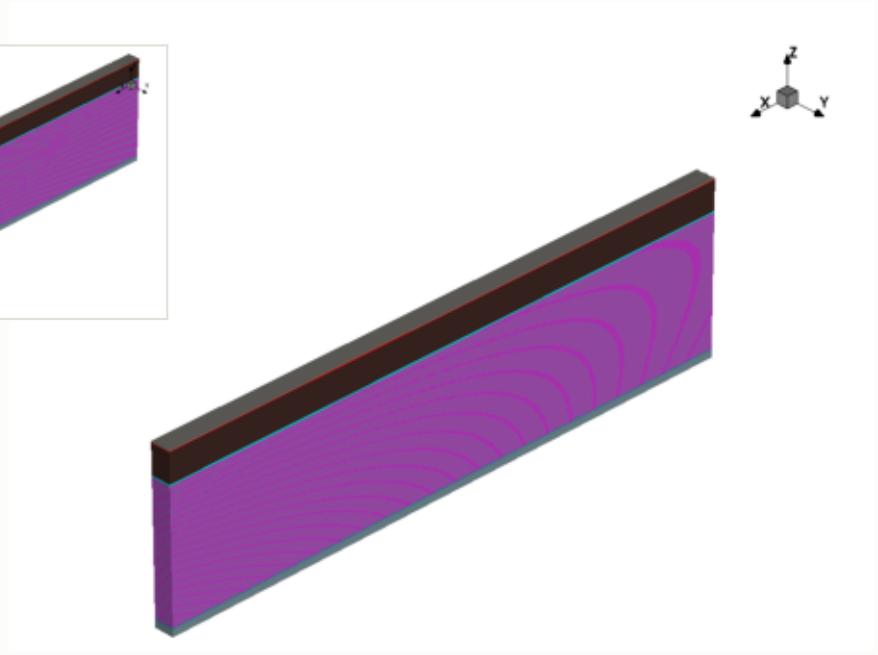
etch_noop · Cryo_etch

0.517	608.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





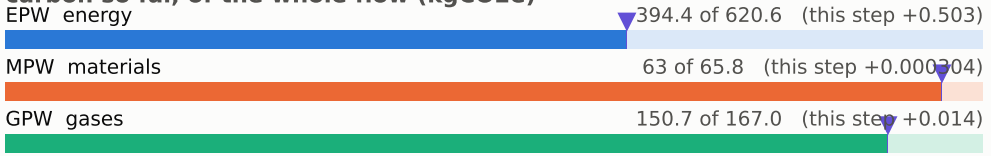
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

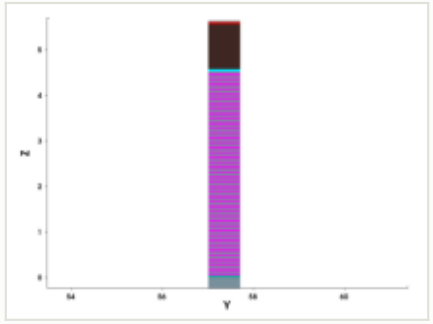
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

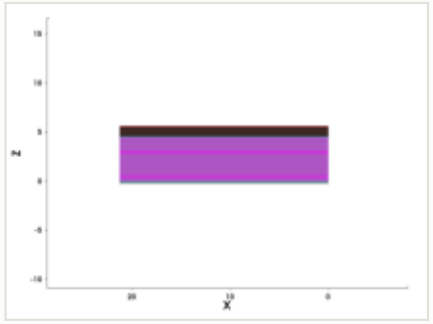


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



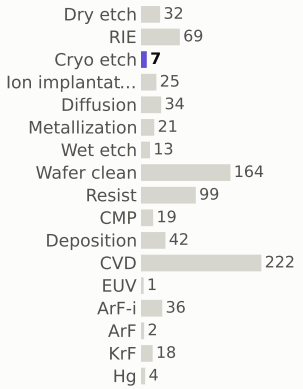
y cut at 57.351 µm



mask for this step

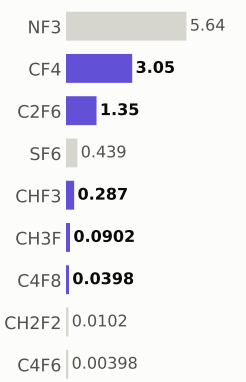
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



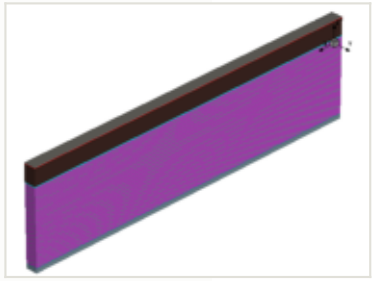
Step #10632.6923: Cryo_etch

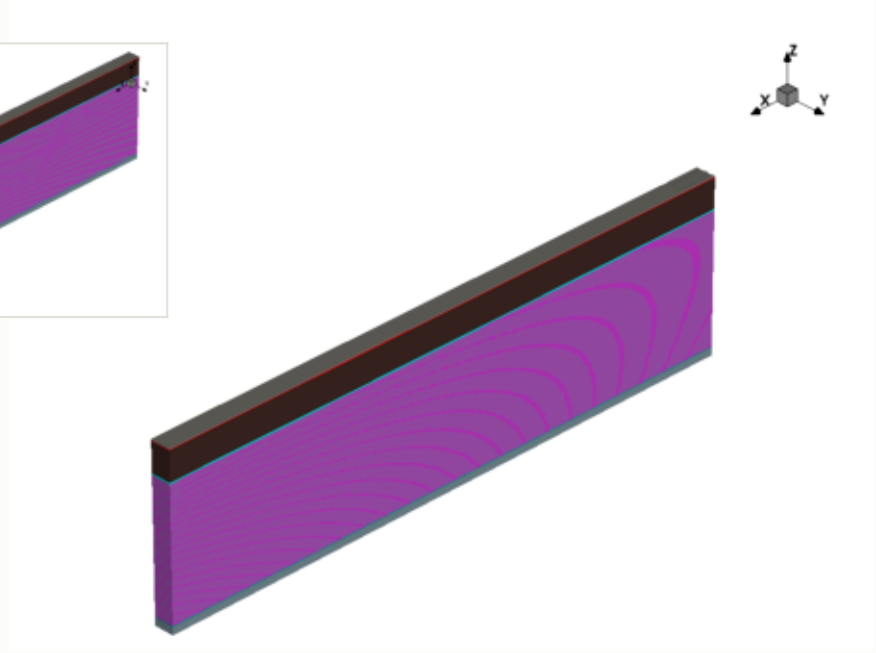
etch_noop · Cryo_etch

0.517	608.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





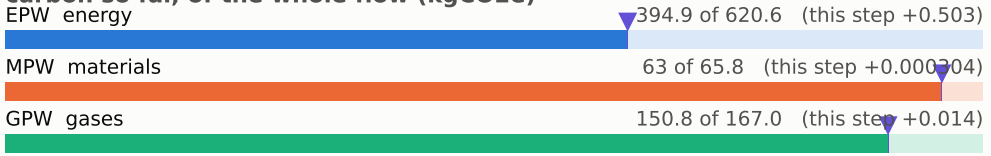
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

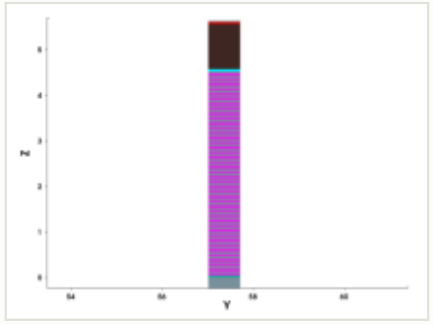
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

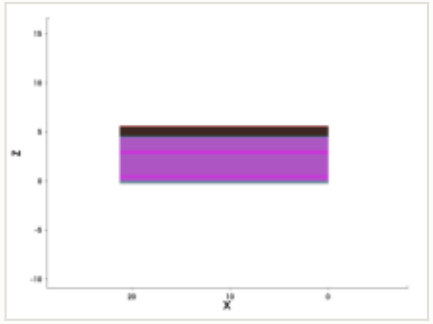


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



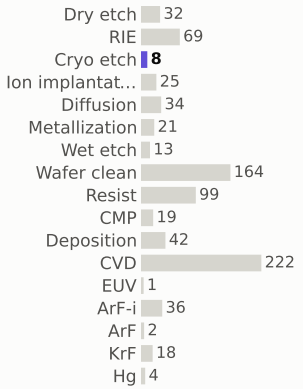
y cut at 57.351 µm



mask for this step

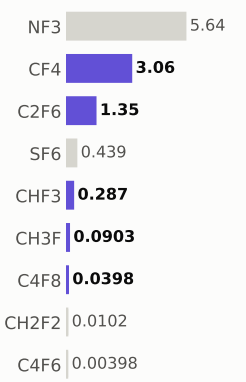
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



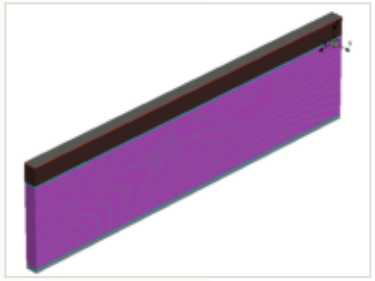
Step #10633.0769: Cryo_etch

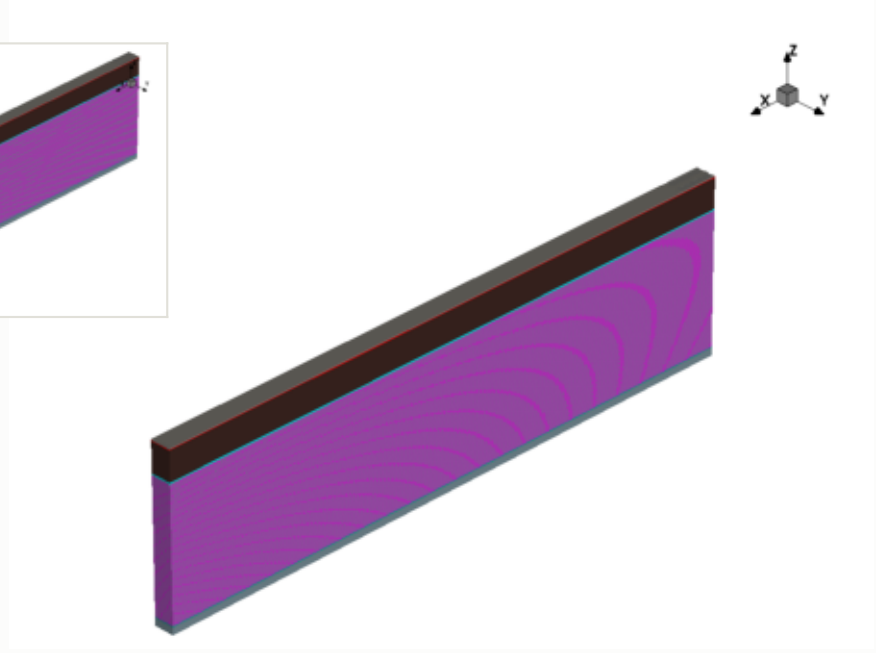
etch_noop · Cryo_etch

0.517	609.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





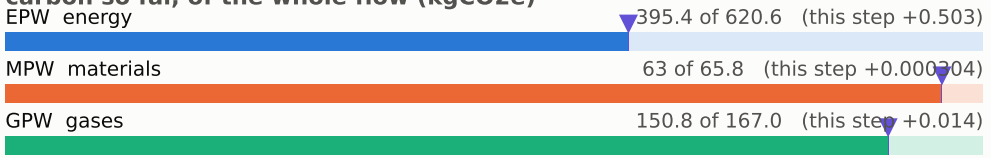
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

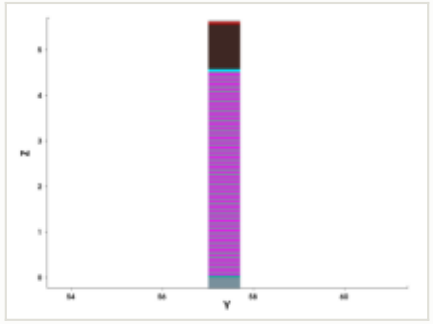
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

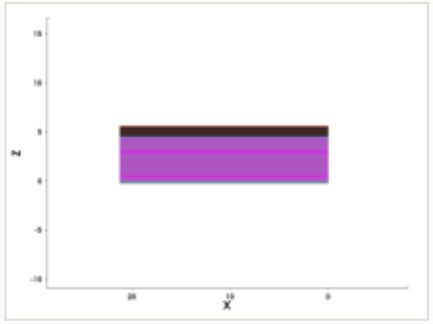


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



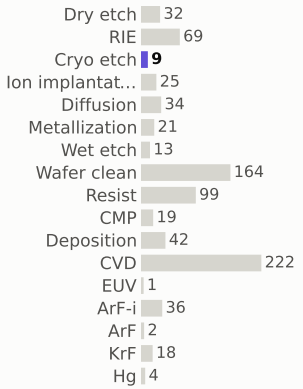
y cut at 57.351 µm



mask for this step

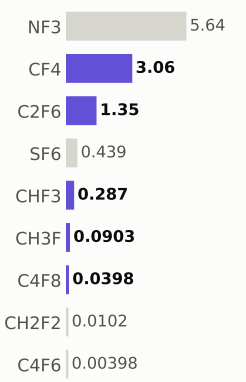
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



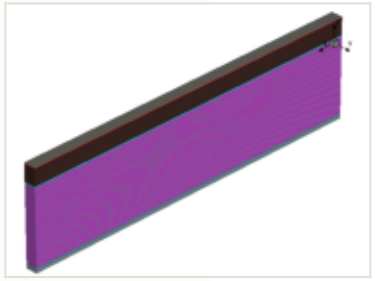
Step #10633.4615: Cryo_etch

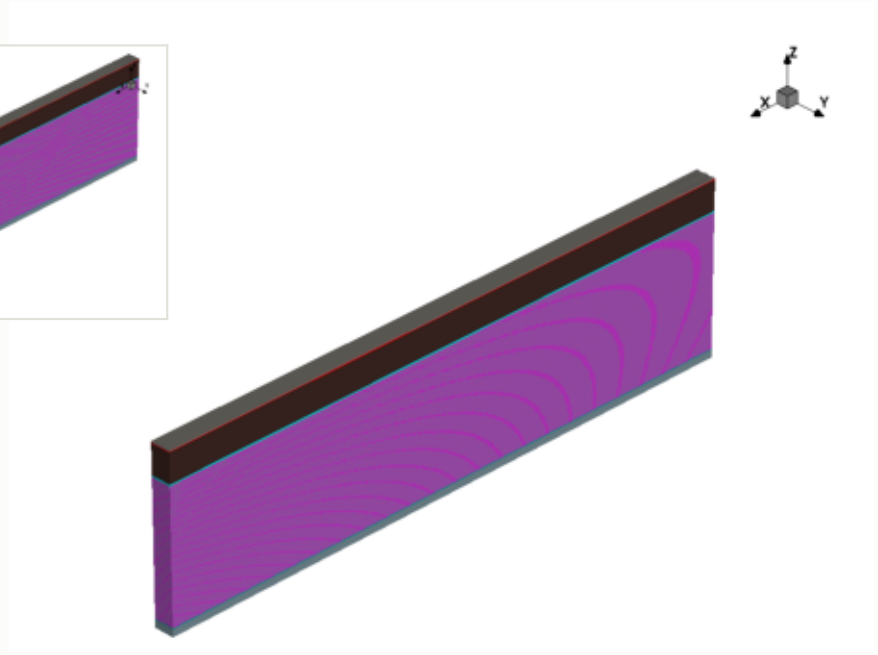
etch_noop · Cryo_etch

0.517	609.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





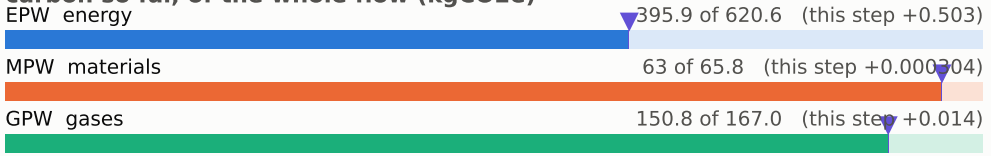
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

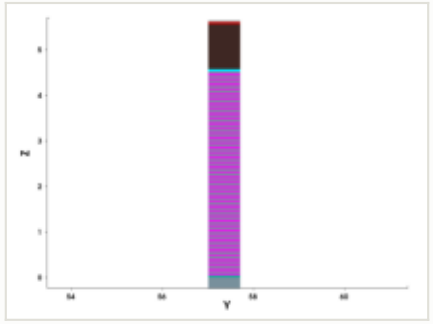
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

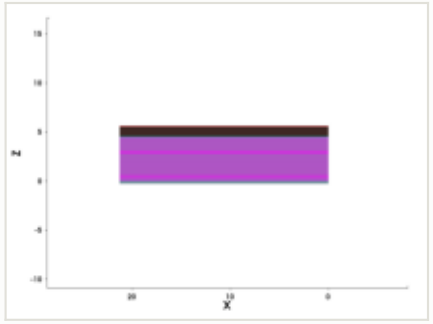


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



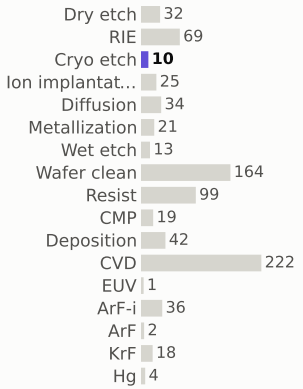
y cut at 57.351 µm



mask for this step

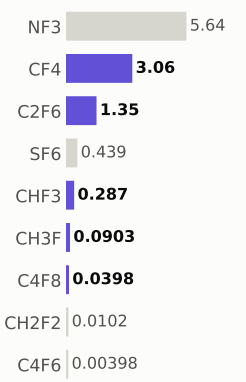
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



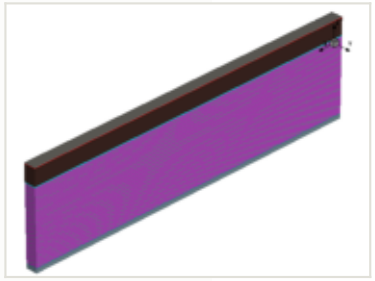
Step #10633.8462: Cryo_etch

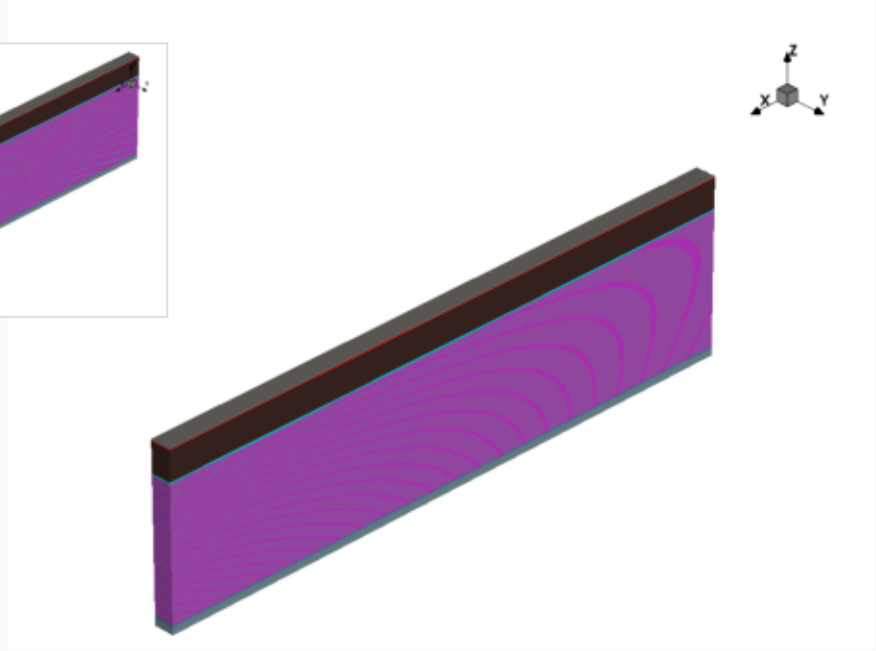
etch_noop · Cryo_etch

0.517	610.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





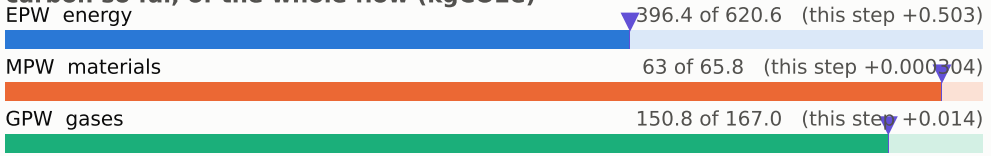
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

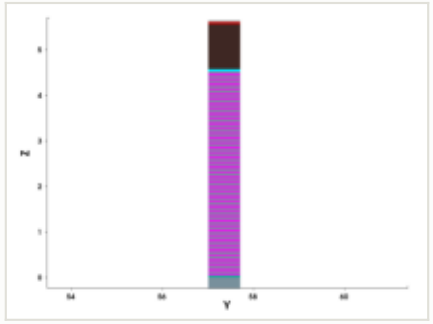
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

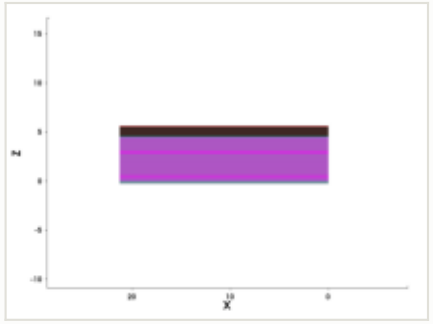


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



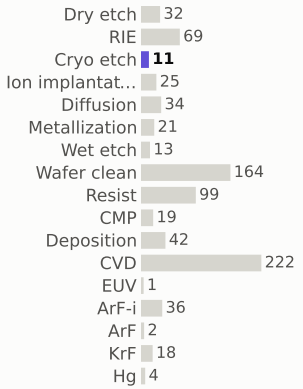
y cut at 57.351 µm



mask for this step

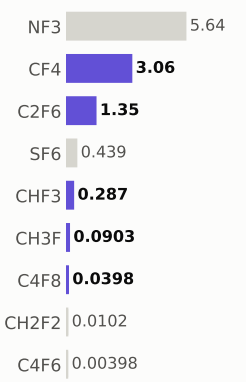
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



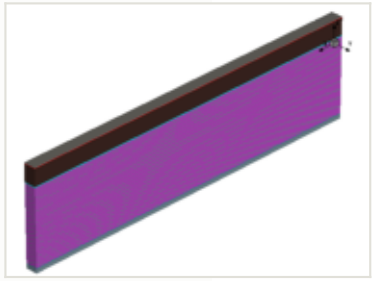
Step #10634.2308: Cryo_etch

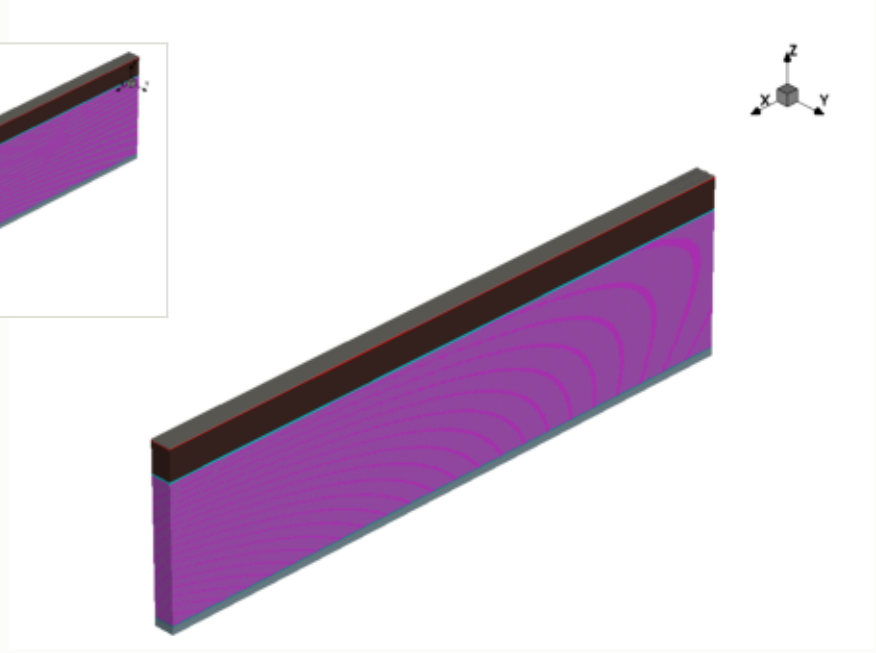
etch_noop · Cryo_etch

0.517	610.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





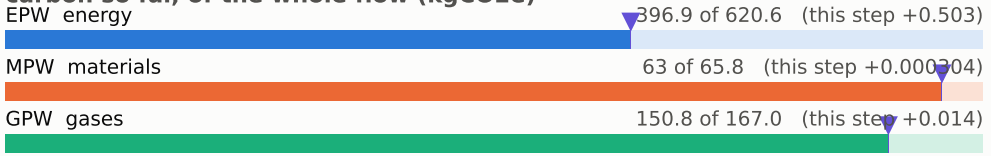
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

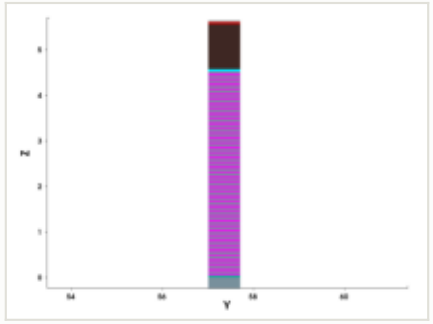
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

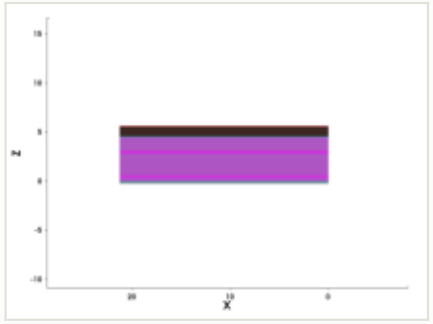


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



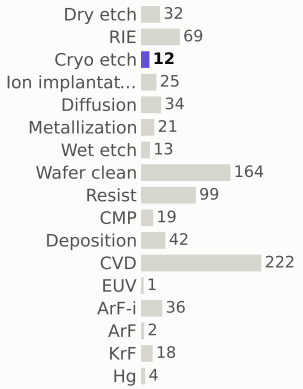
y cut at 57.351 µm



mask for this step

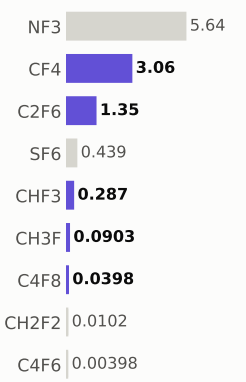
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



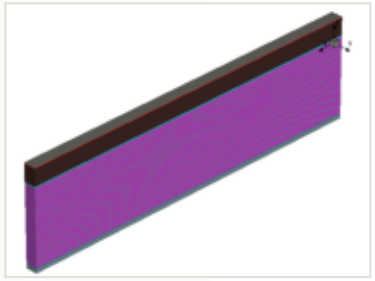
Step #10634.6154: Cryo_etch

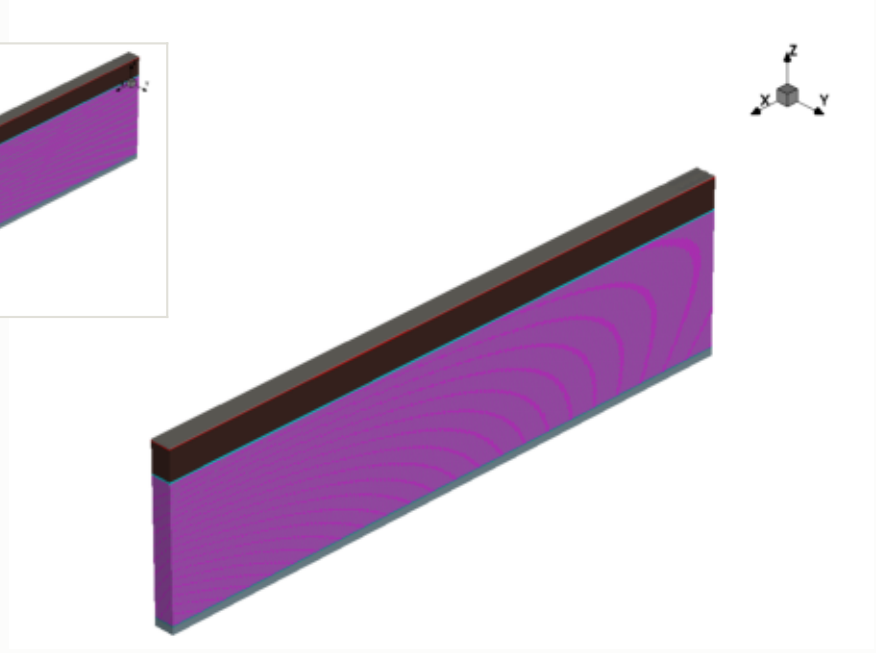
etch_noop · Cryo_etch

0.517	611.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





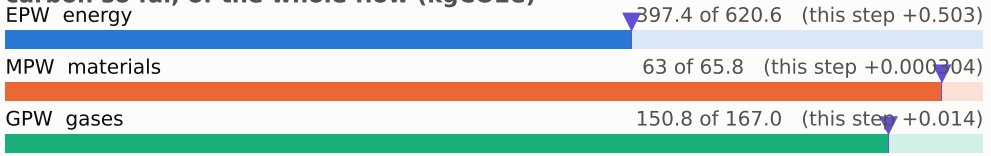
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

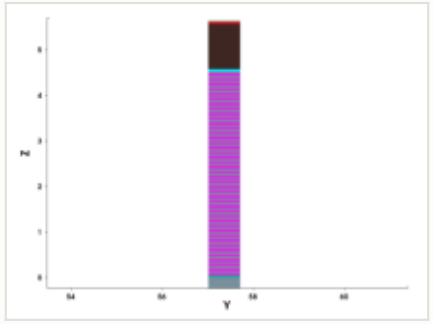
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

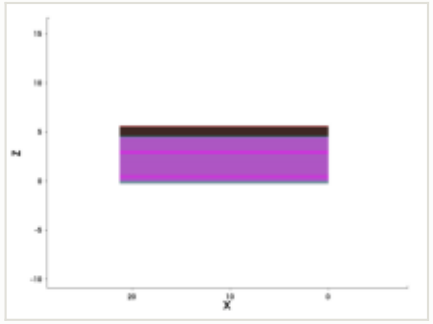


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



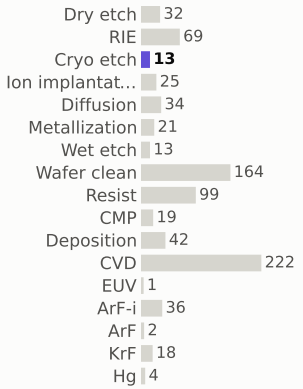
y cut at 57.351 µm



mask for this step

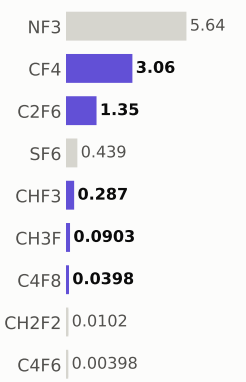
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



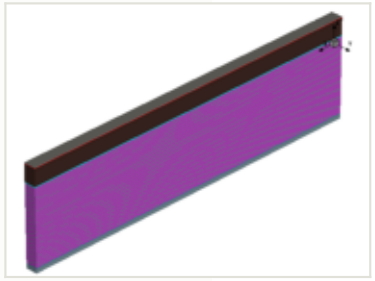
Step #10635.0000: Cryo_etch

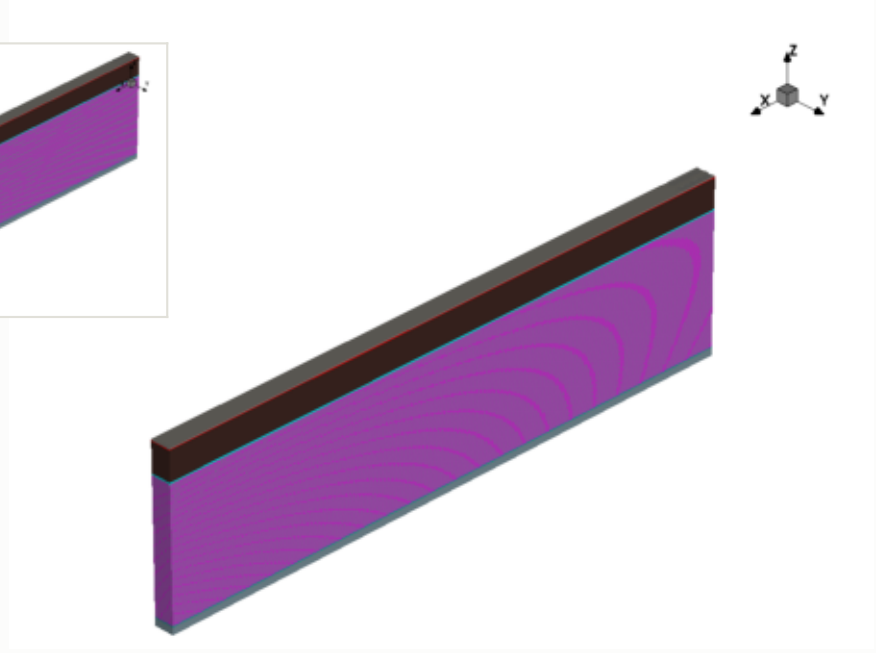
etch_noop · Cryo_etch

0.517	611.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





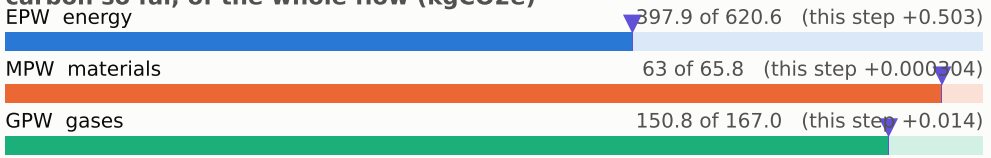
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

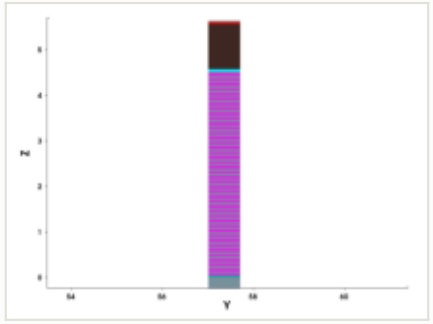
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

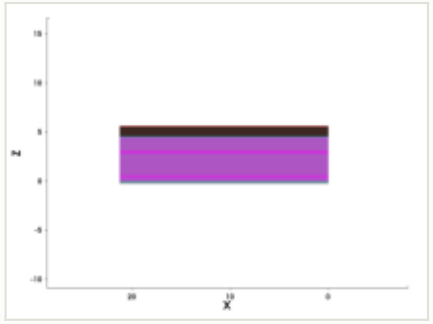


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



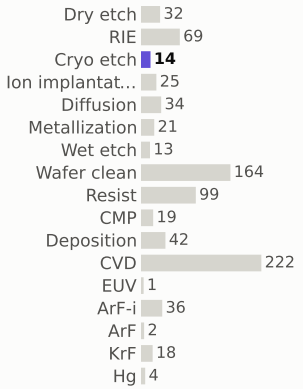
y cut at 57.351 µm



mask for this step

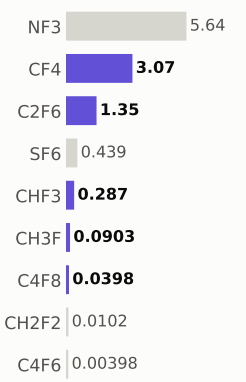
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



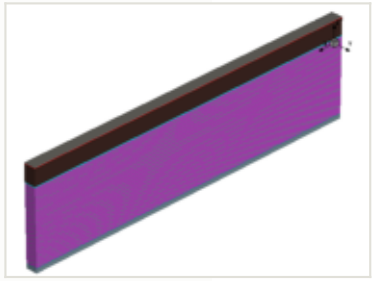
Step #10635.3846: Cryo_etch

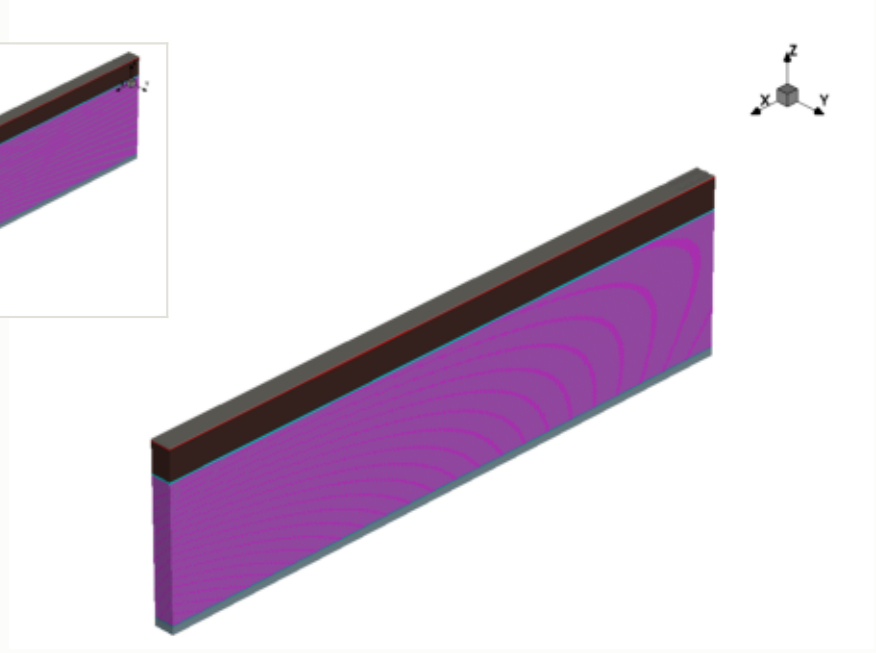
etch_noop · Cryo_etch

0.517	612.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





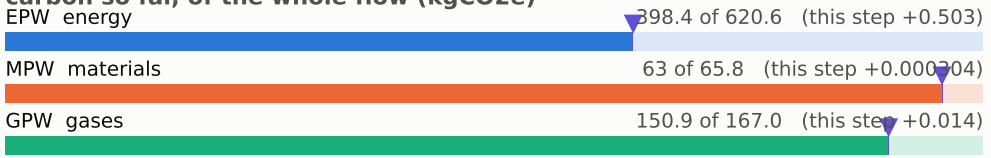
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

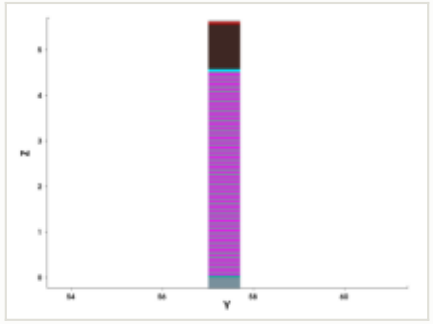
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

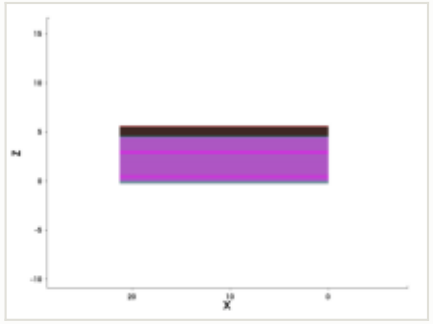


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



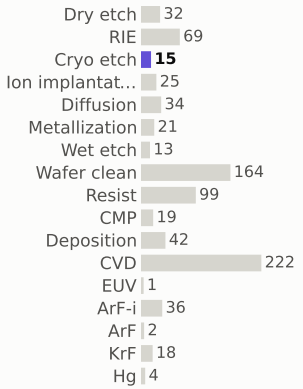
y cut at 57.351 µm



mask for this step

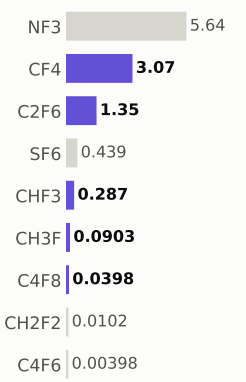
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



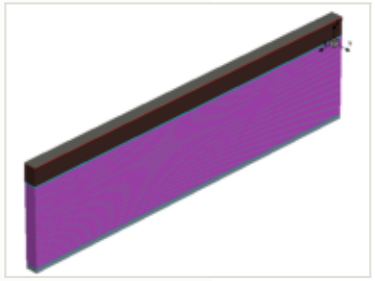
Step #10635.7692: Cryo_etch

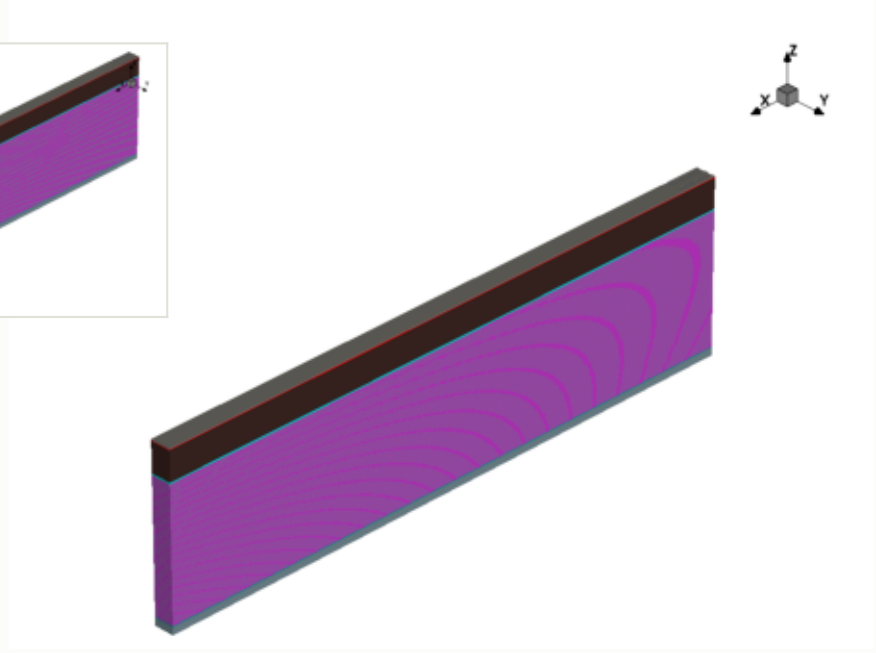
etch_noop · Cryo_etch

0.517	612.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





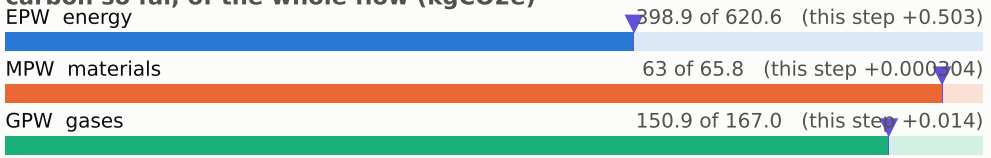
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

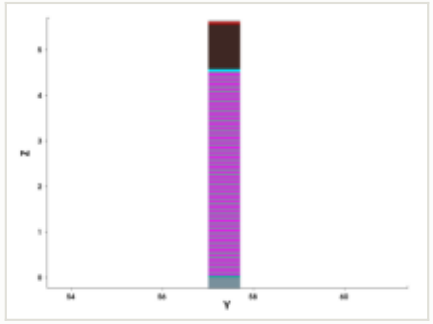
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

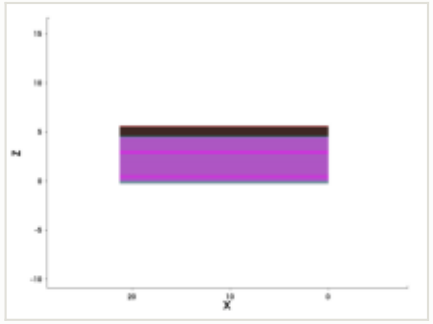


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



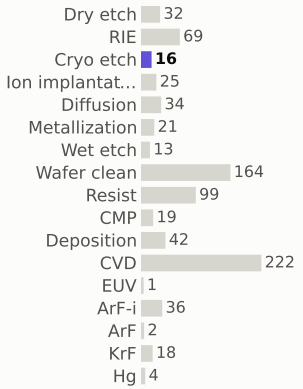
y cut at 57.351 µm



mask for this step

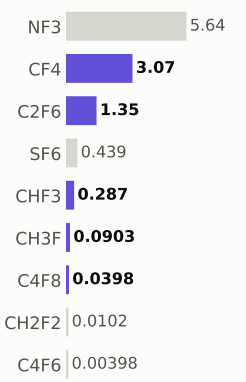
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



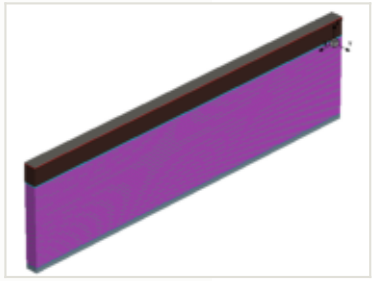
Step #10636.1538: Cryo_etch

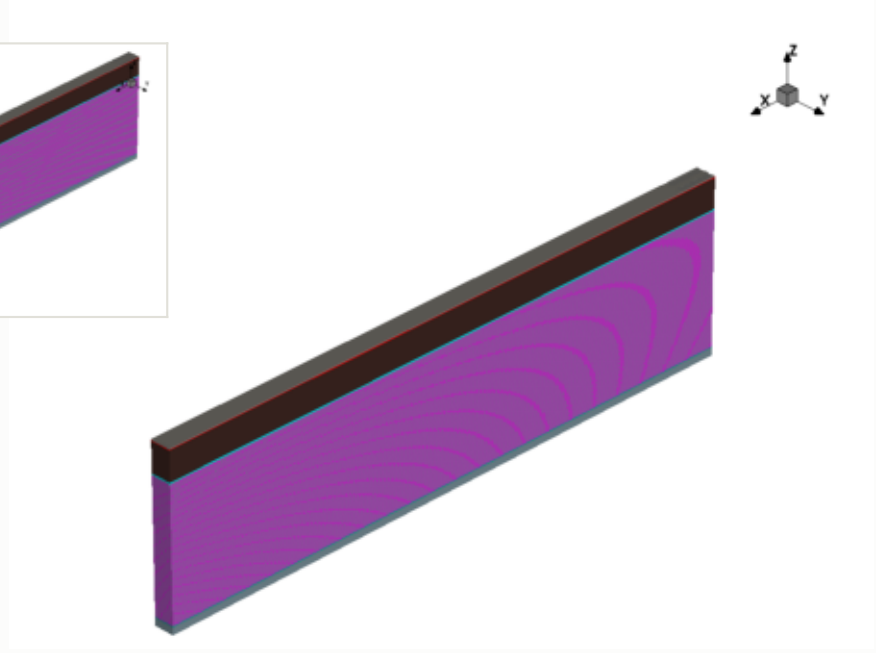
etch_noop · Cryo_etch

0.517	613.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





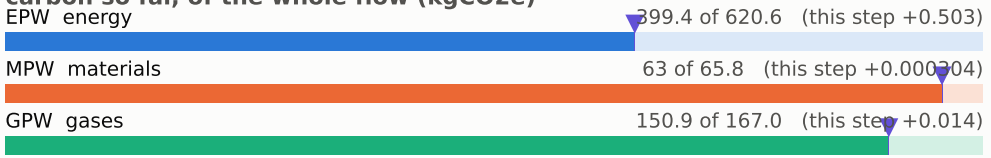
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

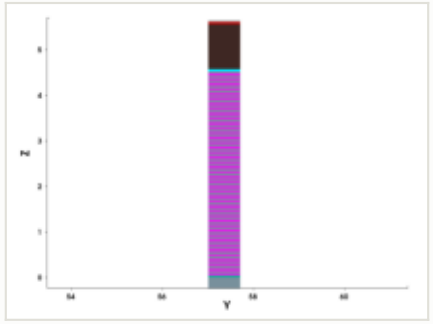
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

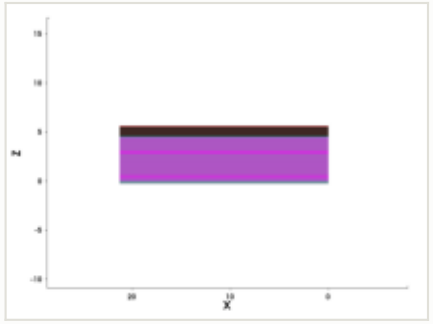


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



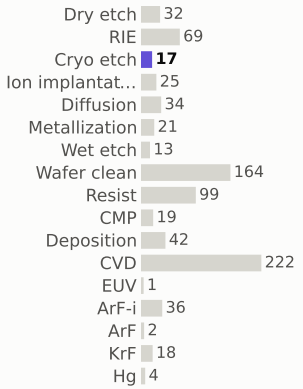
y cut at 57.351 µm



mask for this step

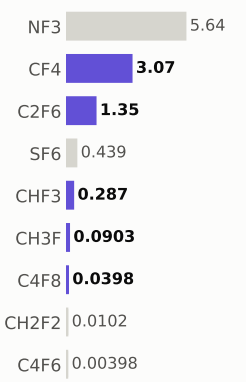
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



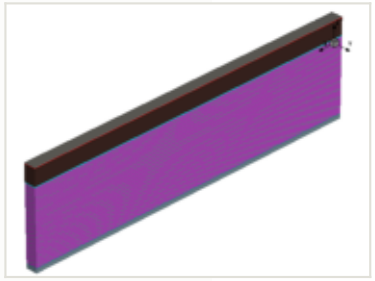
Step #10636.5385: Cryo_etch

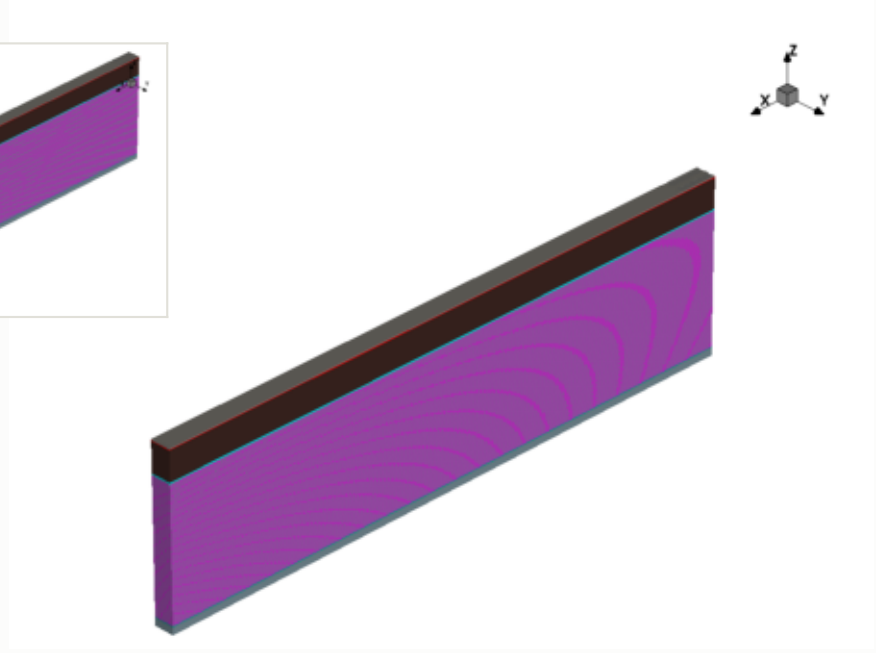
etch_noop · Cryo_etch

0.517	613.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





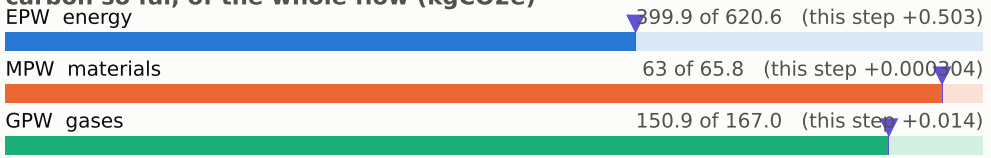
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

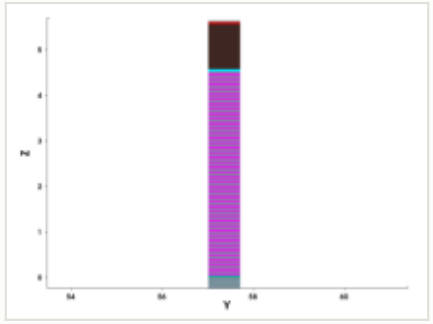
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

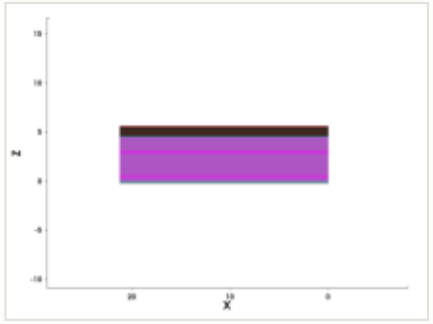


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



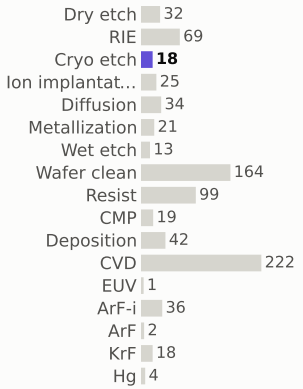
y cut at 57.351 µm



mask for this step

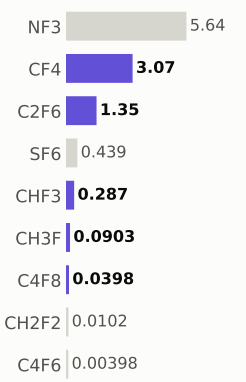
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



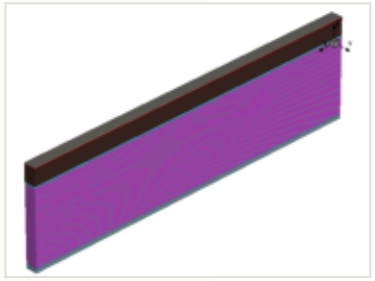
Step #10636.9231: Cryo_etch

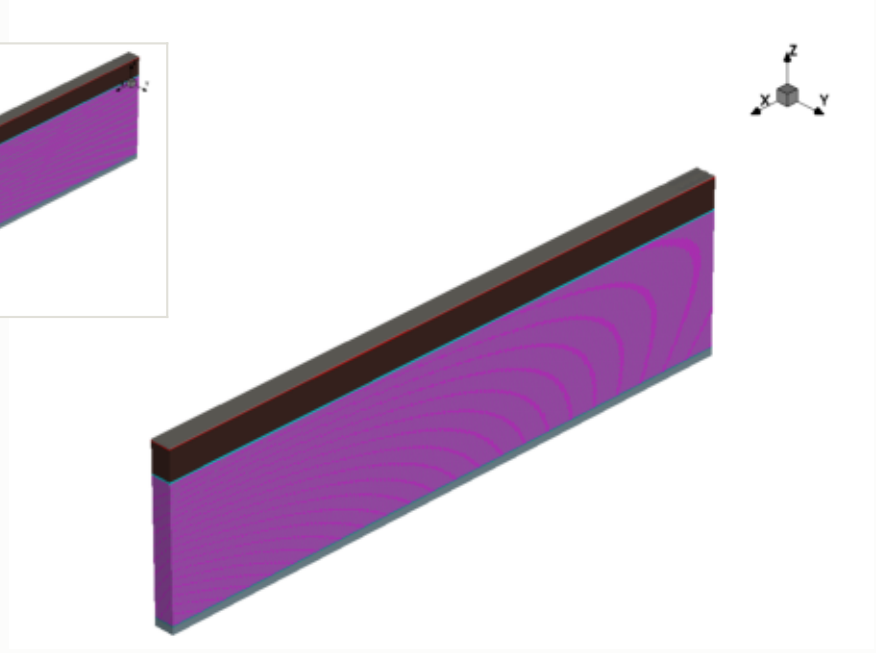
etch_noop · Cryo_etch

0.517	614.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





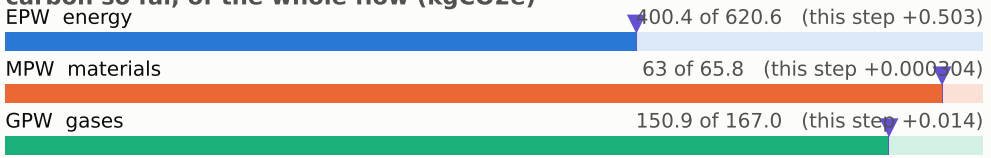
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

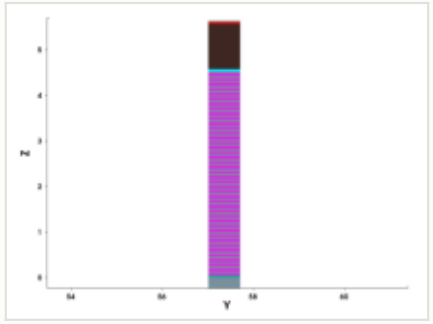
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

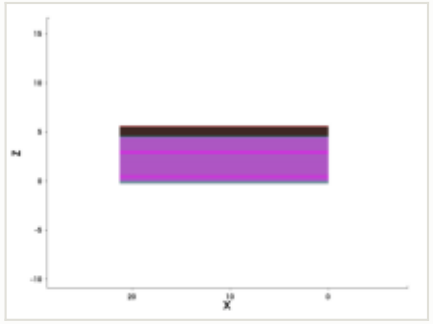


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



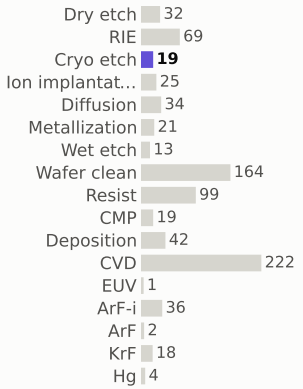
y cut at 57.351 µm



mask for this step

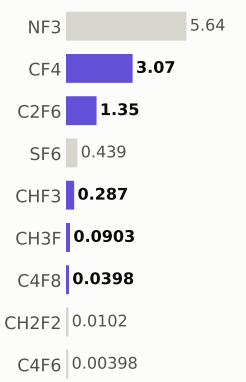
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



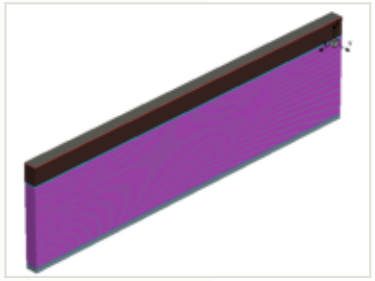
Step #10637.3077: Cryo_etch

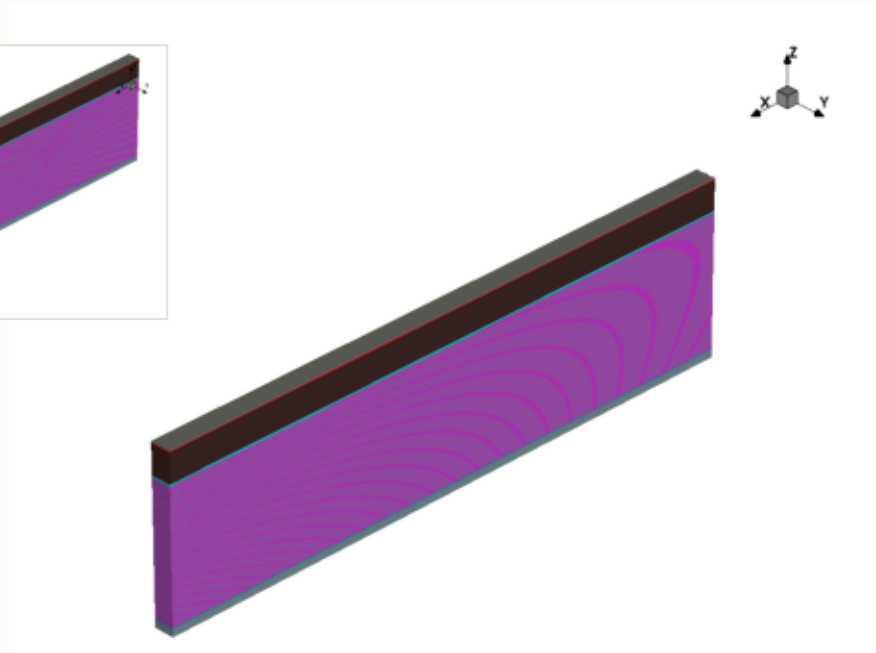
etch_noop · Cryo_etch

0.517	614.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





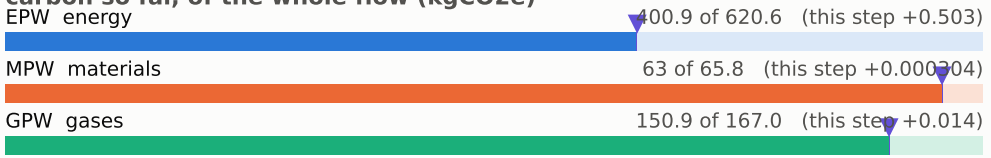
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

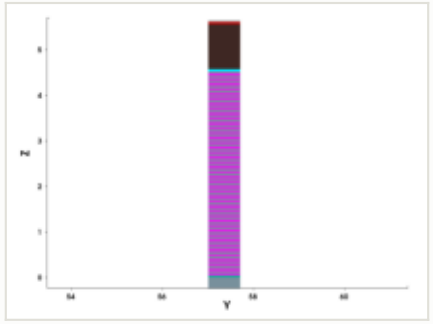
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

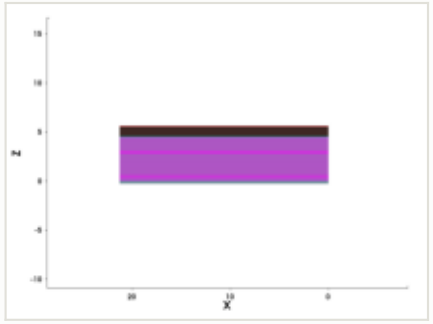


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



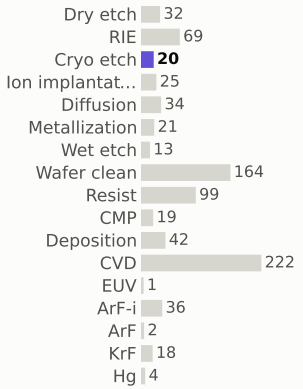
y cut at 57.351 µm



mask for this step

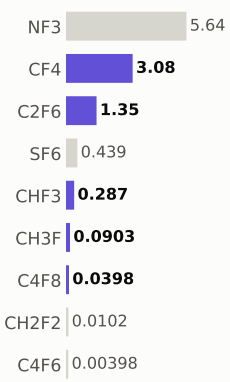
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



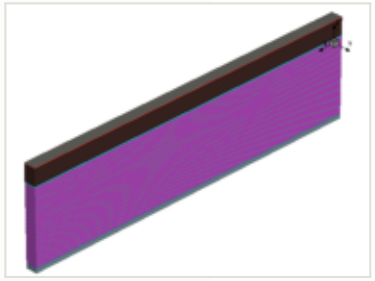
Step #10637.6923: Cryo_etch

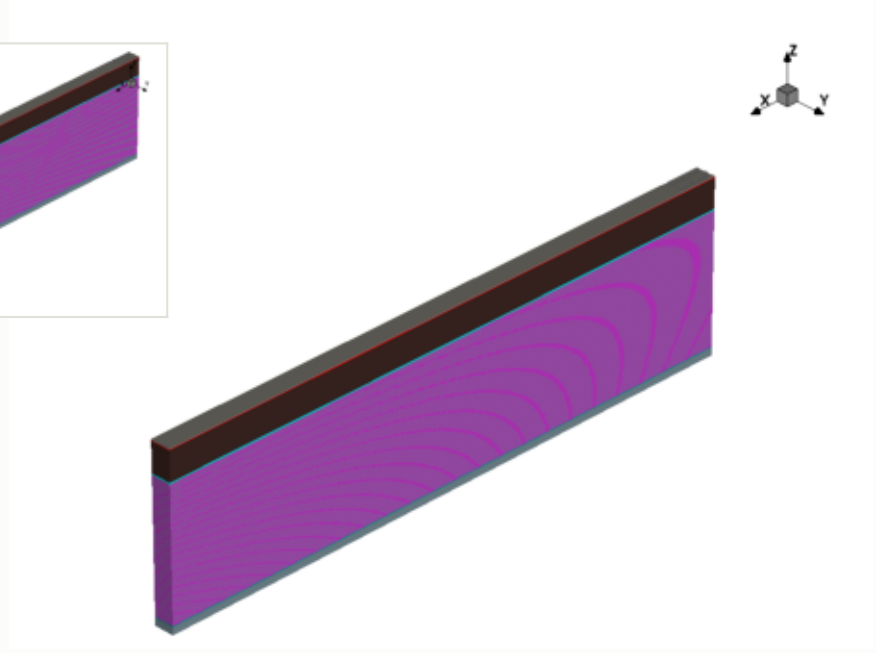
etch_noop · Cryo_etch

0.517	615.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





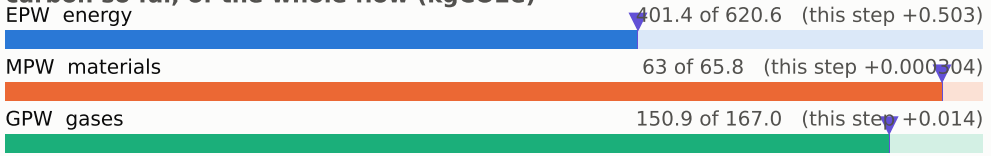
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

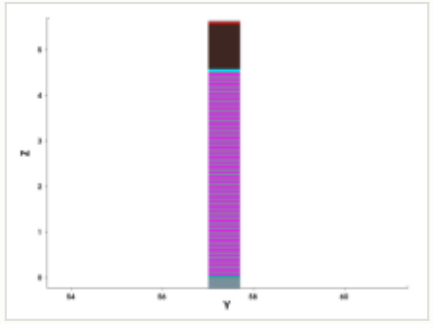
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

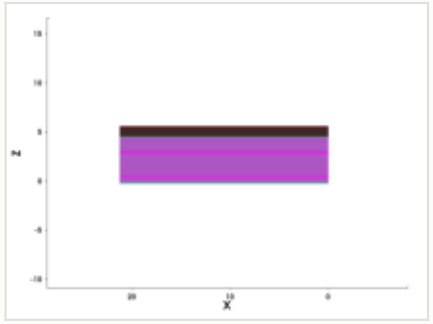


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



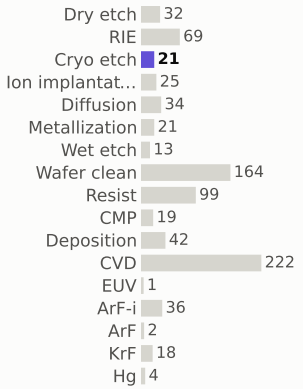
y cut at 57.351 µm



mask for this step

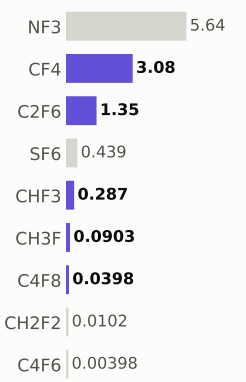
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



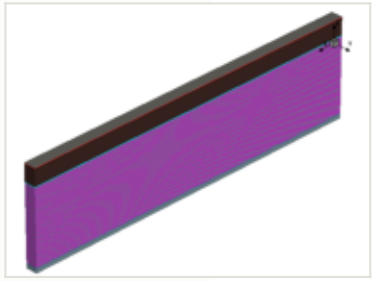
Step #10638.0769: Cryo_etch

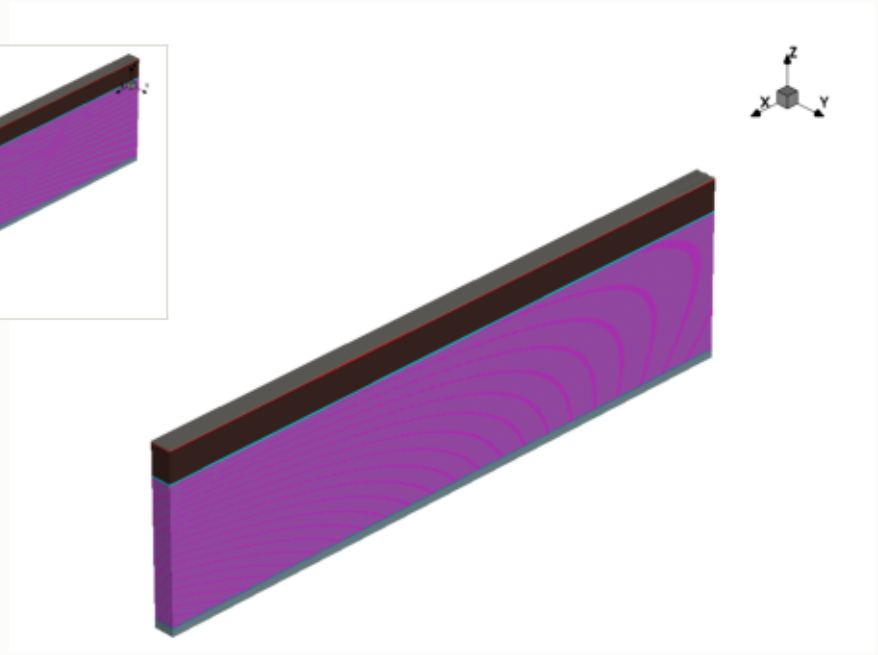
etch_noop · Cryo_etch

0.517	615.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





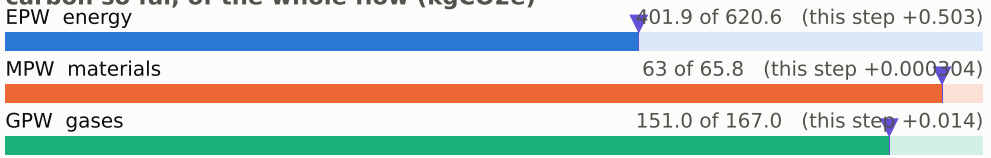
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

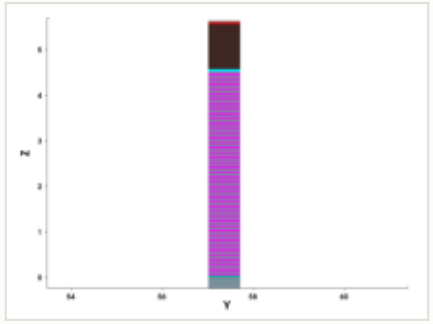
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

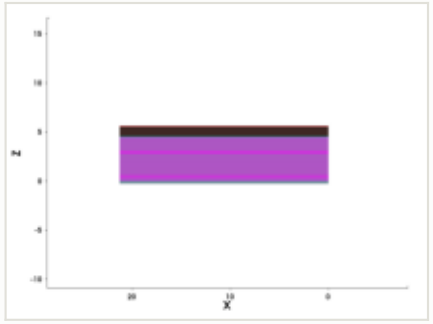


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



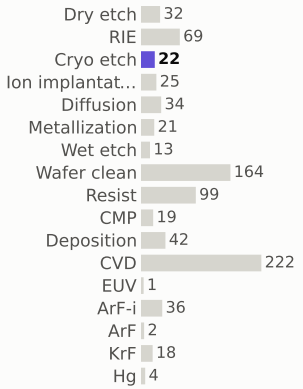
y cut at 57.351 µm



mask for this step

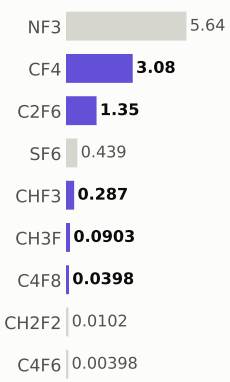
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



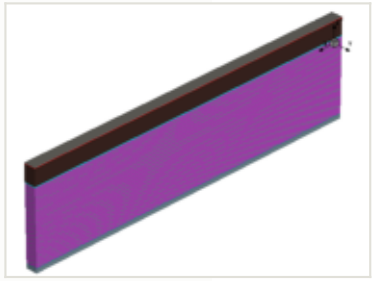
Step #10638.4615: Cryo_etch

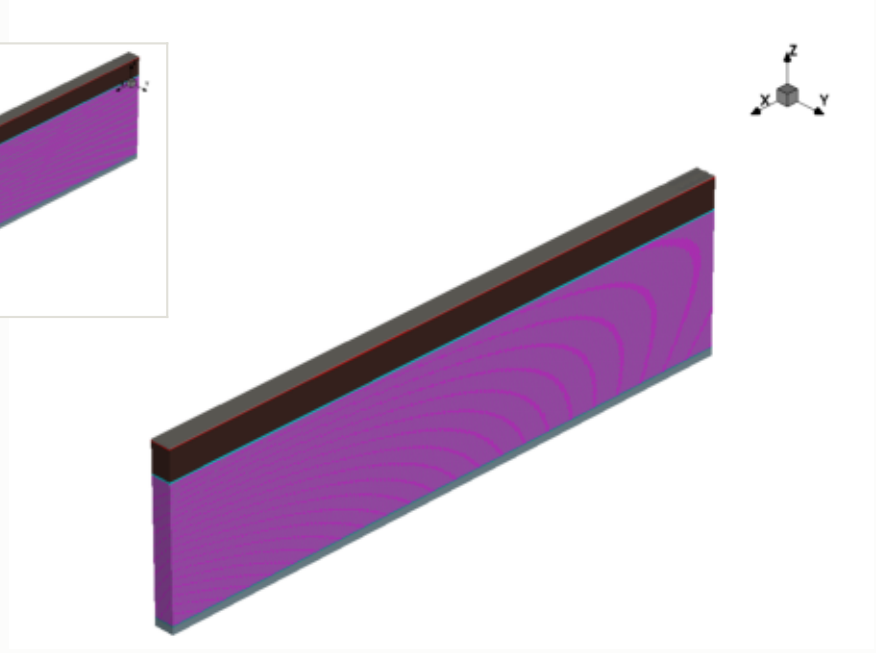
etch_noop · Cryo_etch

0.517	616.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





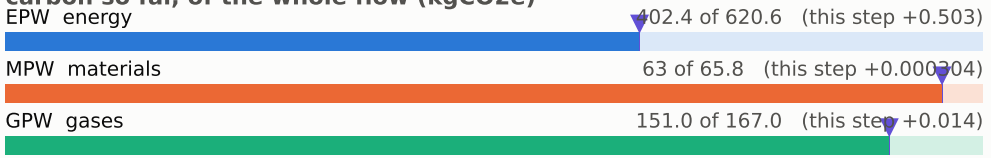
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

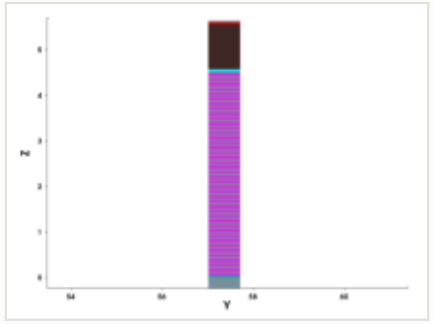
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

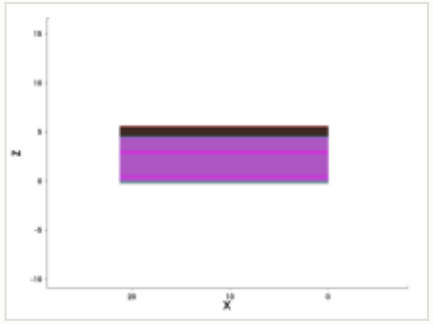


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



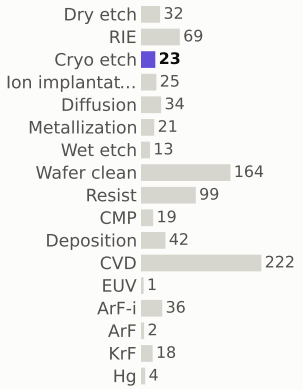
y cut at 57.351 µm



mask for this step

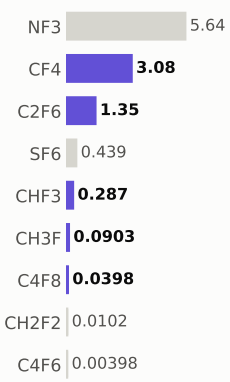
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



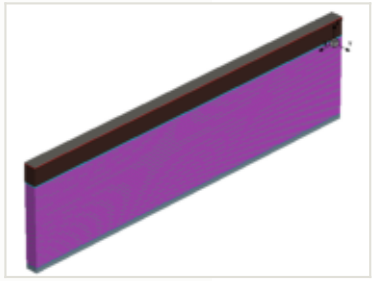
Step #10638.8462: Cryo_etch

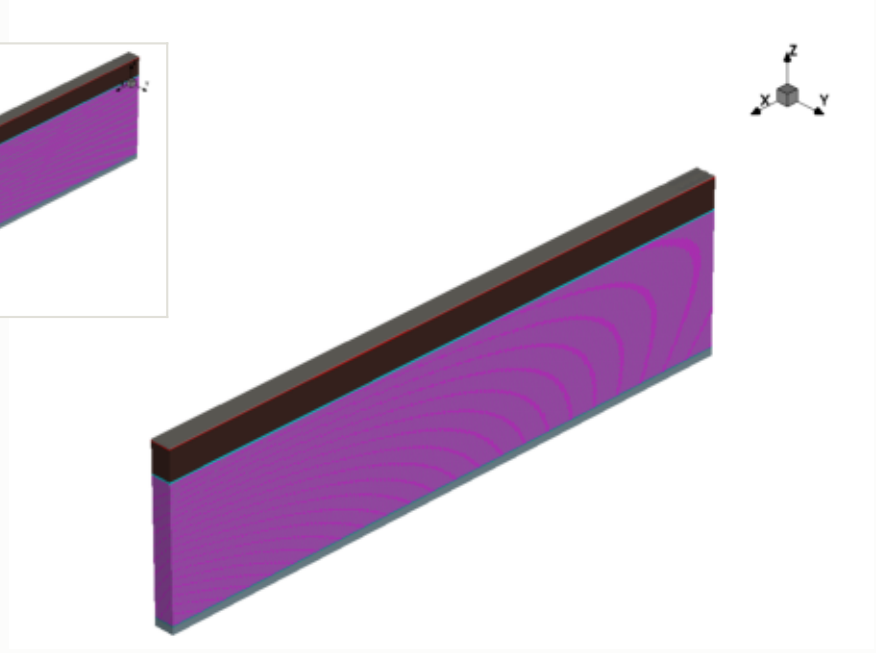
etch_noop · Cryo_etch

0.517	617.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





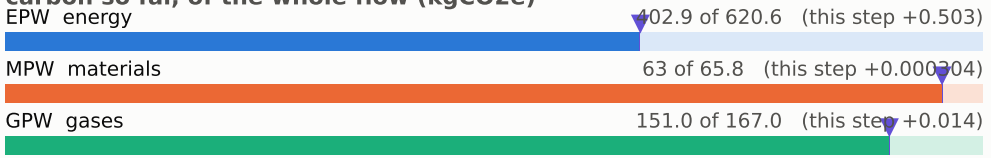
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

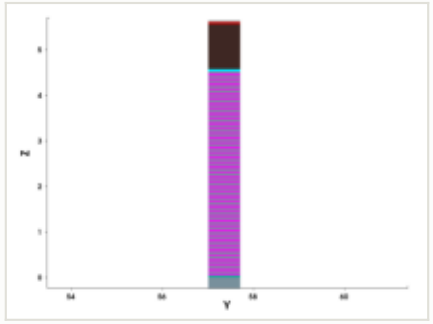
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

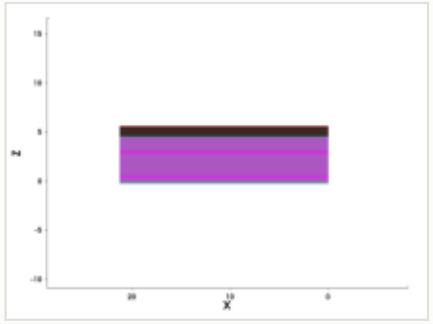


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



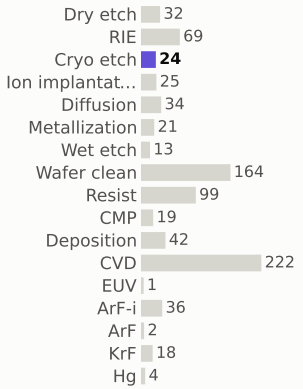
y cut at 57.351 µm



mask for this step

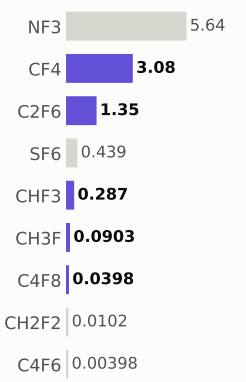
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



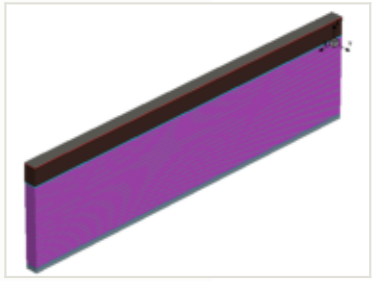
Step #10639.2308: Cryo_etch

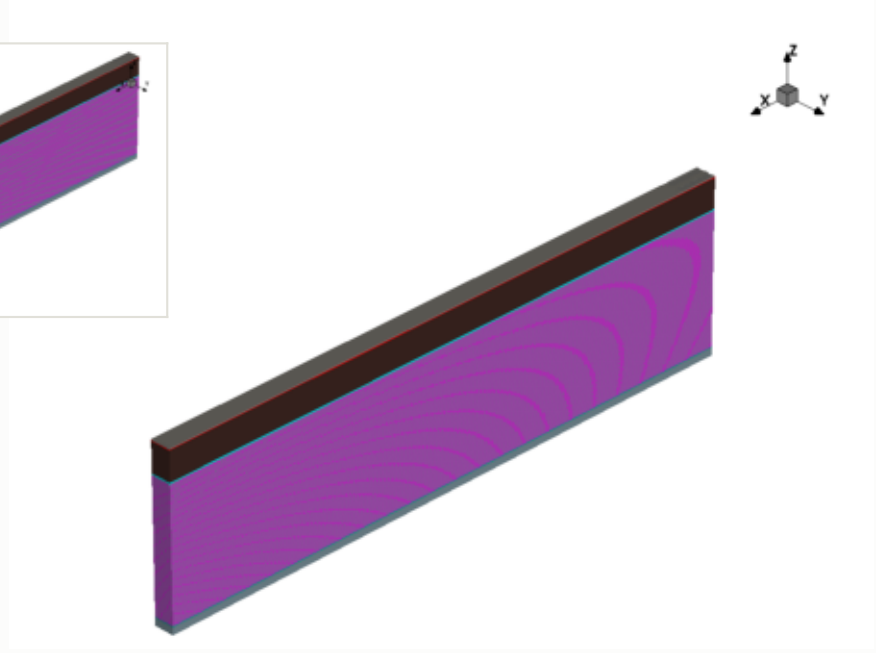
etch_noop · Cryo_etch

0.517	617.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





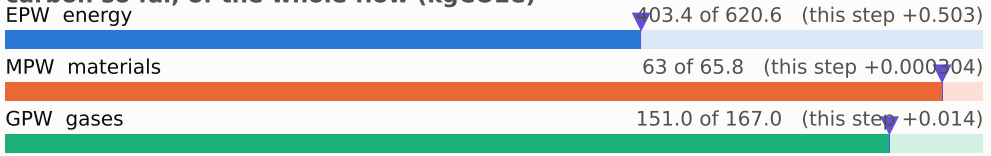
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

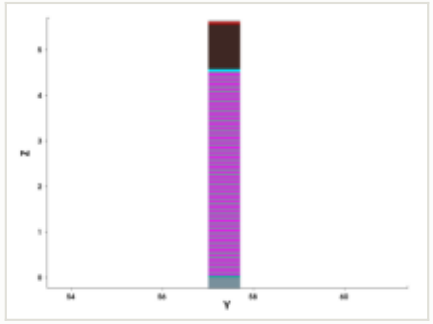
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

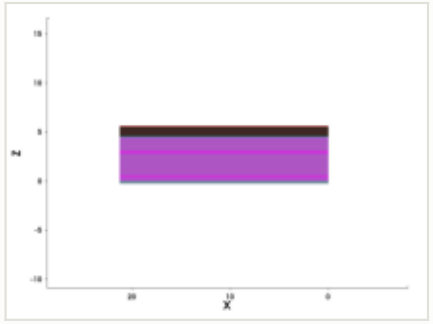


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



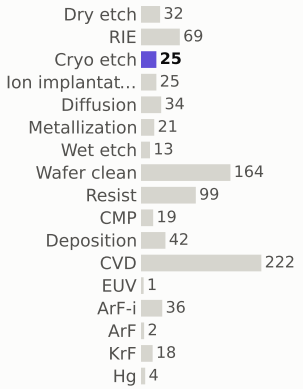
y cut at 57.351 µm



mask for this step

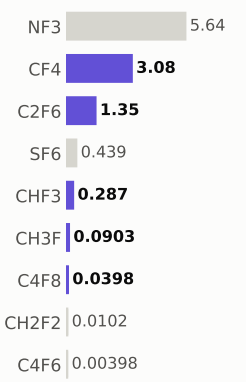
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



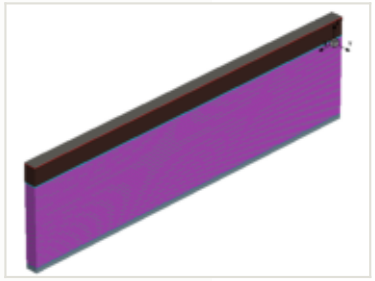
Step #10639.6154: Cryo_etch

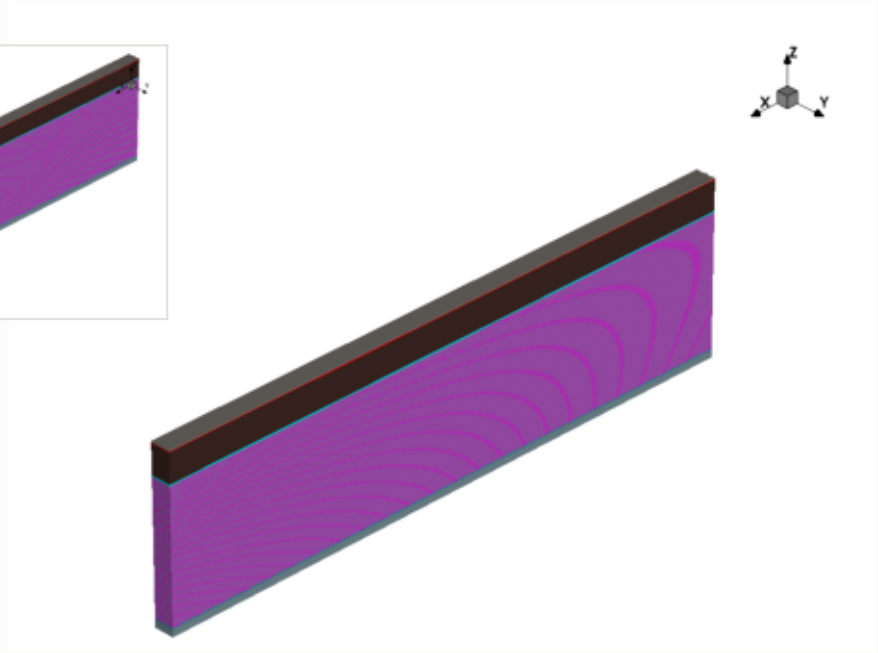
etch_noop · Cryo_etch

0.517	618.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





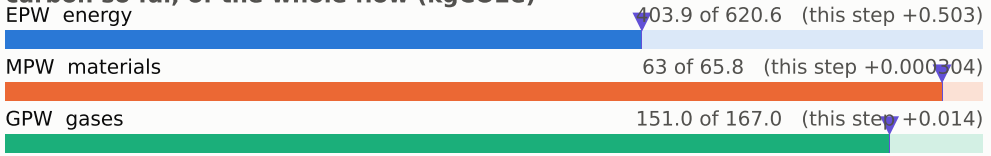
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

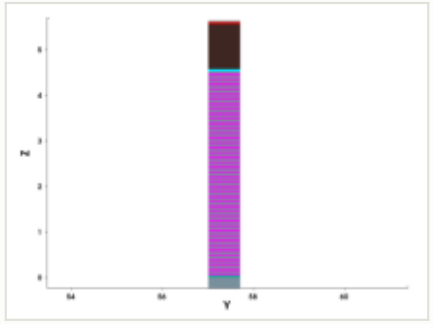
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

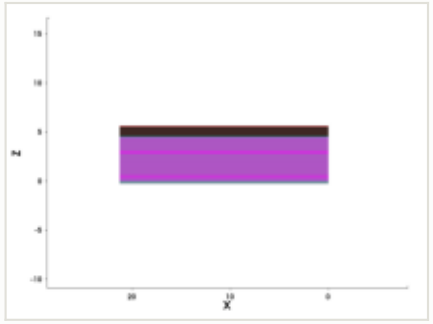


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



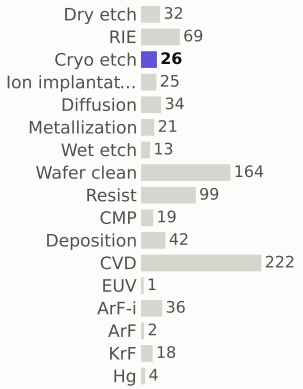
y cut at 57.351 µm



mask for this step

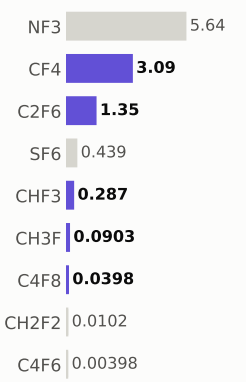
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



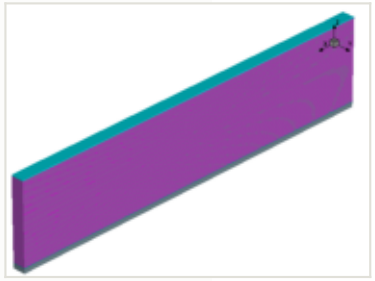
Step #10640: Dry_etch

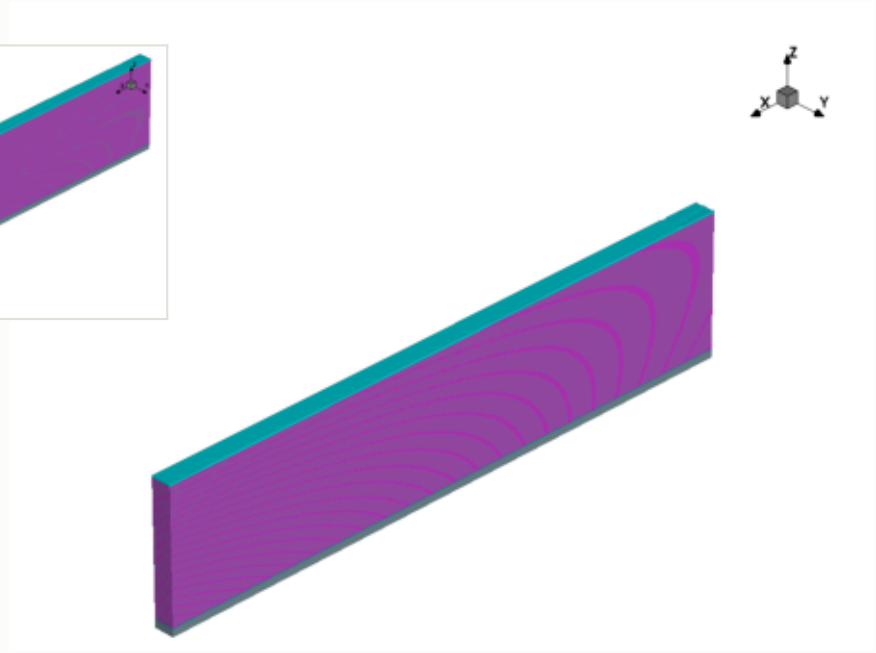
etch_strip · Dry_etch

0.801	618.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

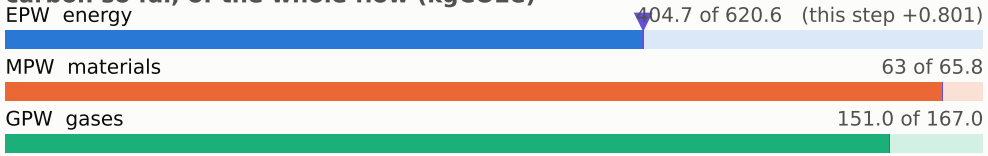




this step

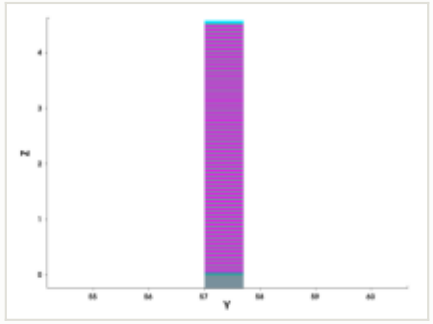
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

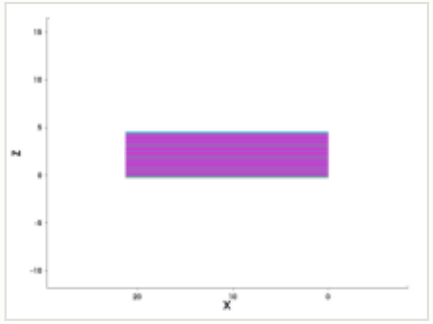


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



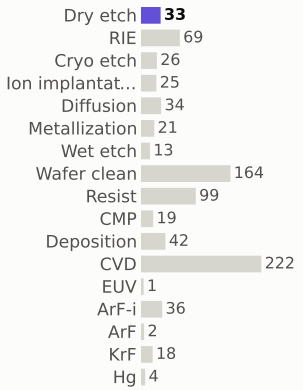
y cut at 57.351 µm



mask for this step

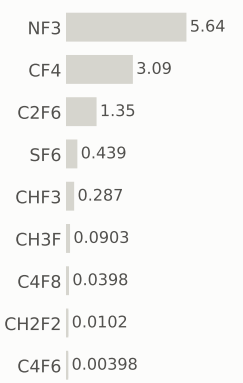
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



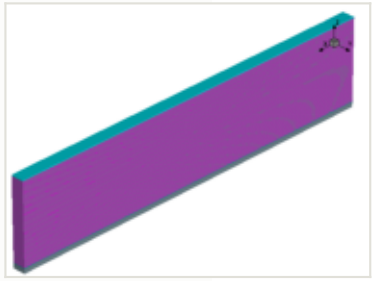
Step #10650: Wafer_clean

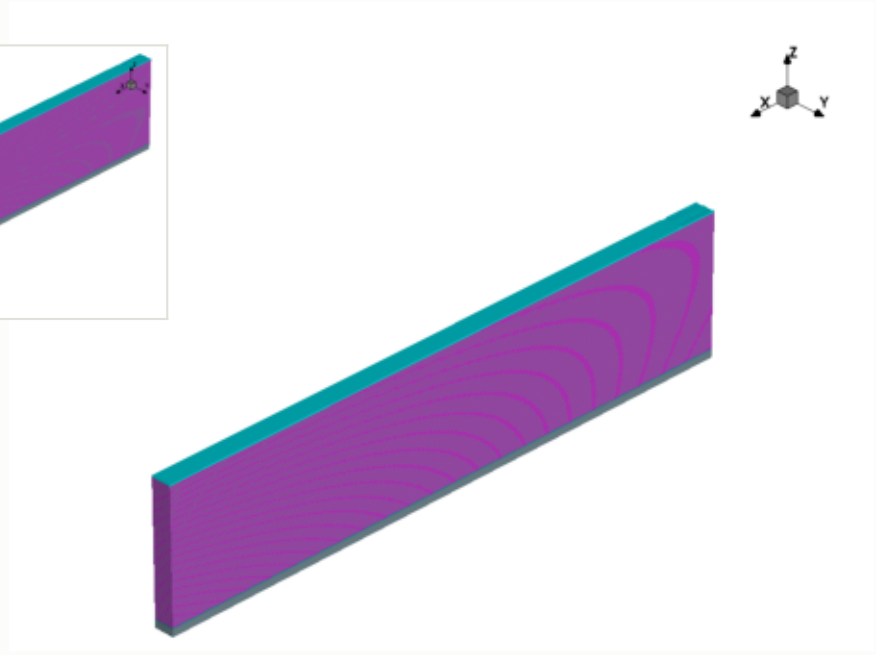
clean · Wafer_clean

0.169	619.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





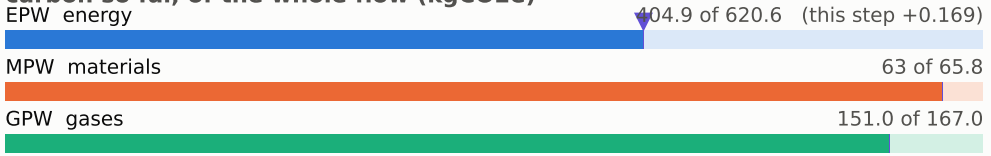
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

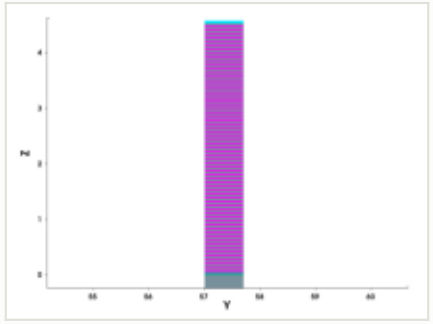
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

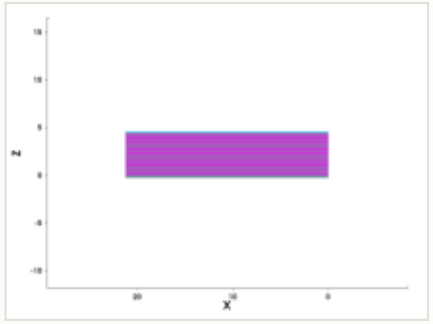


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



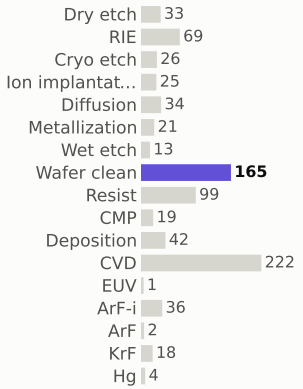
y cut at 57.351 µm



mask for this step

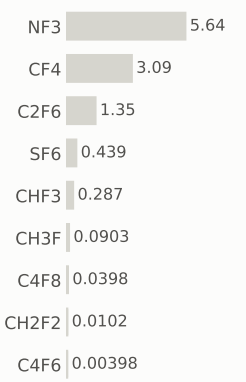
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



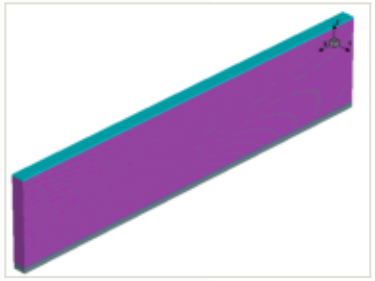
Step #10660: Deposition

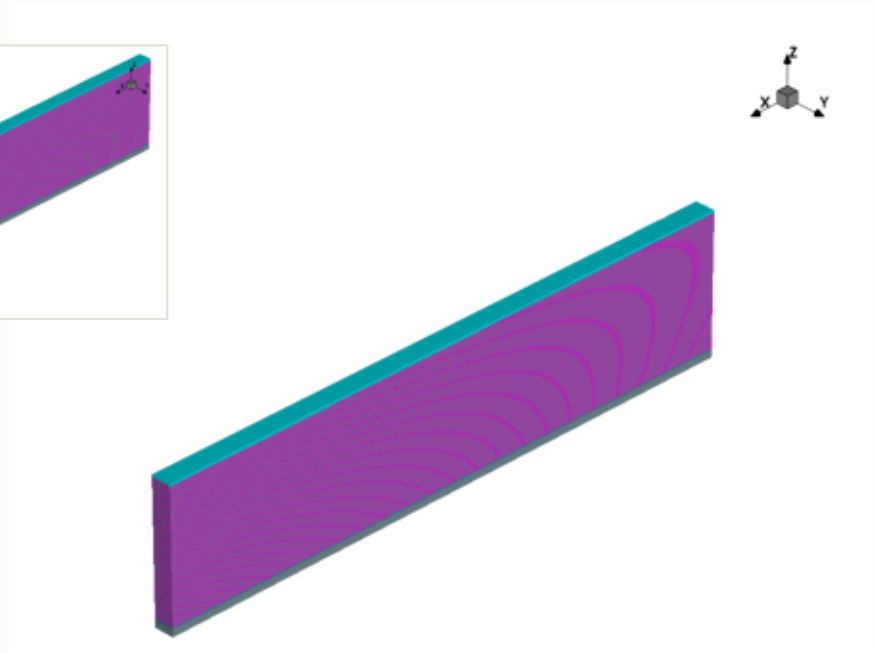
depo_isotropic · Deposition

0.656	619.6	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



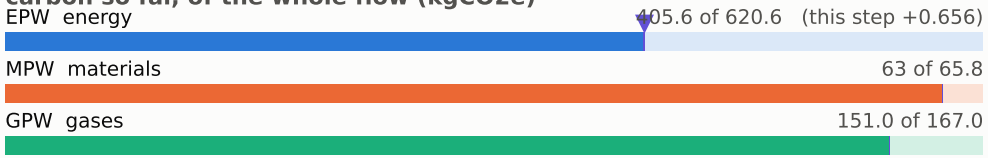


this step

process area	Deposition
material	—
thickness	0.025 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

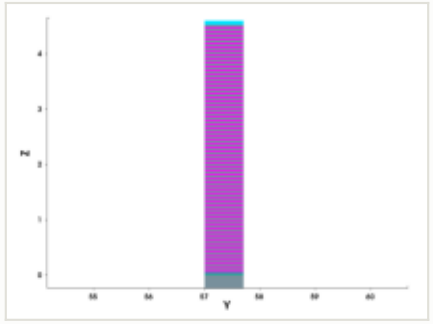
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

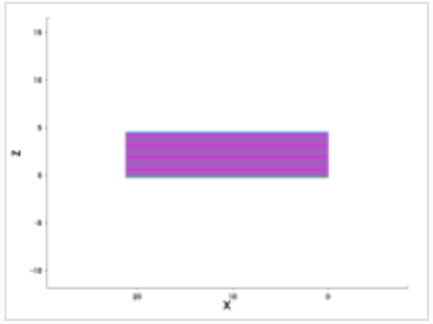


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



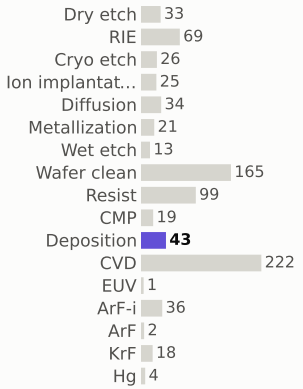
y cut at 57.351 µm



mask for this step

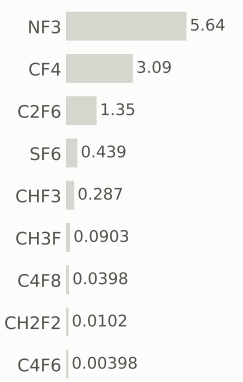
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



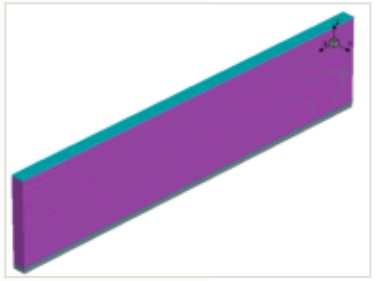
Step #10670: CMP

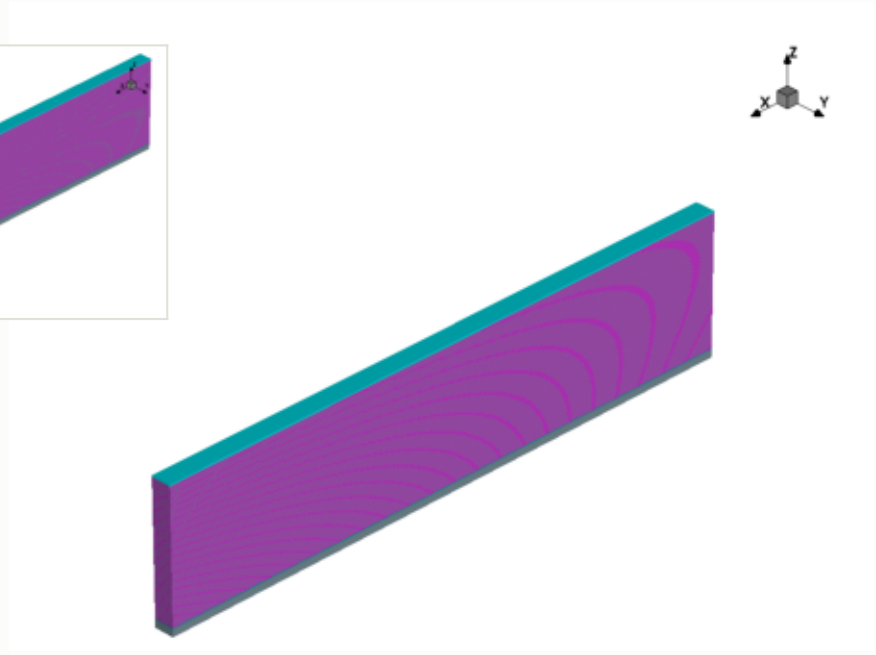
etch_cmp · CMP

0.169	619.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



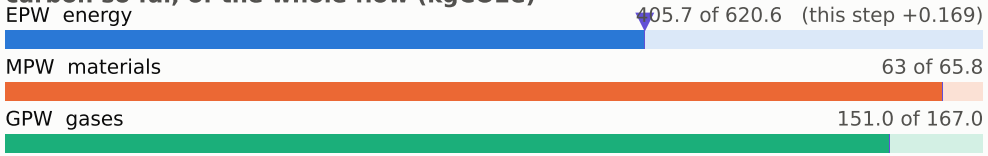


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

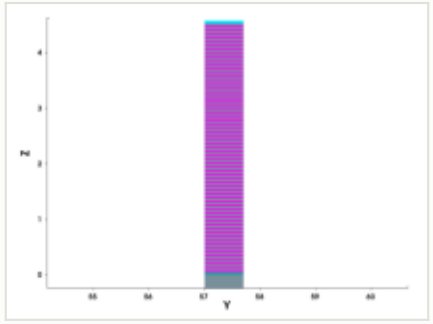
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

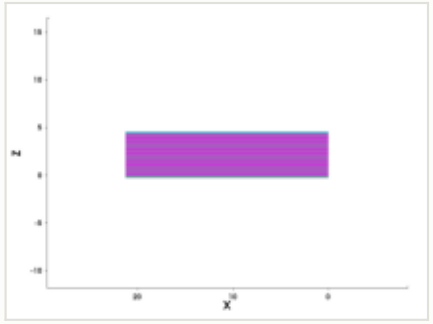


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



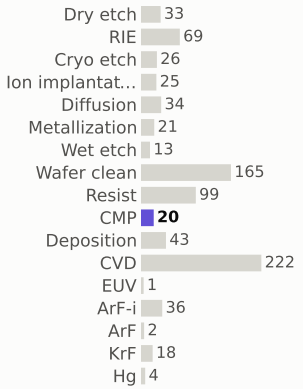
y cut at 57.351 µm



mask for this step

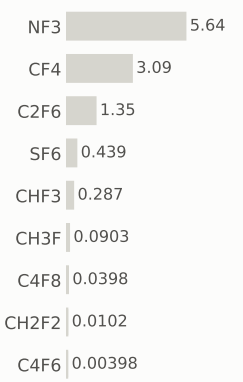
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



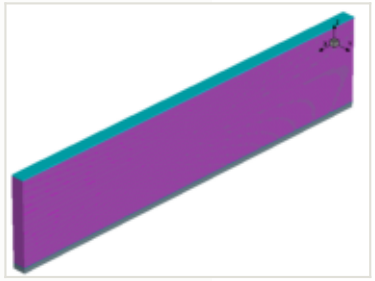
Step #10680: Wafer_clean

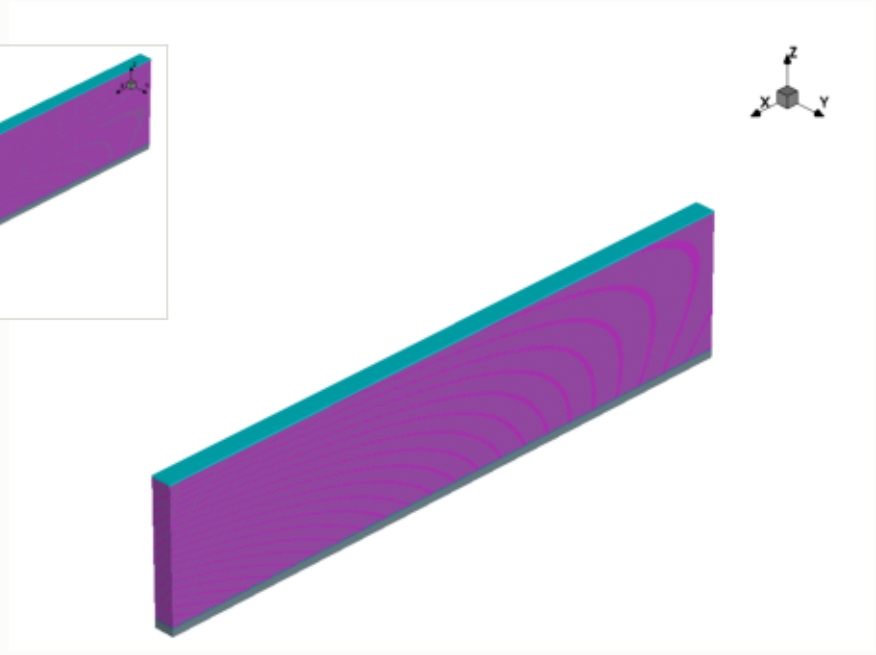
clean · Wafer_clean

0.169	620.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





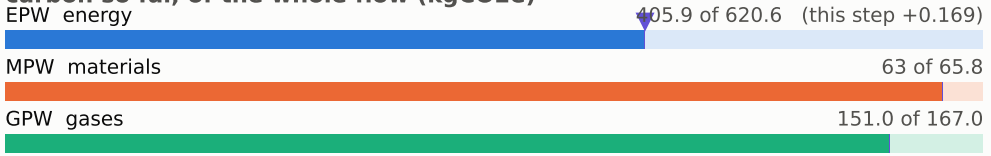
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

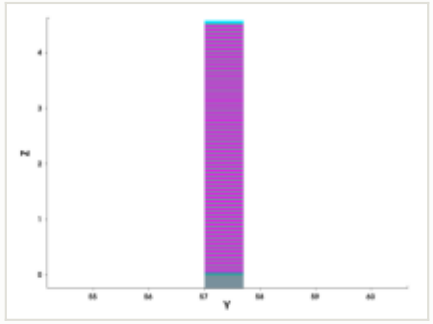
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

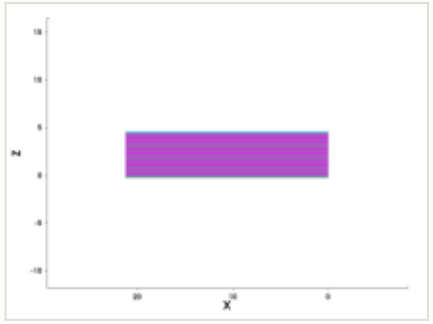


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



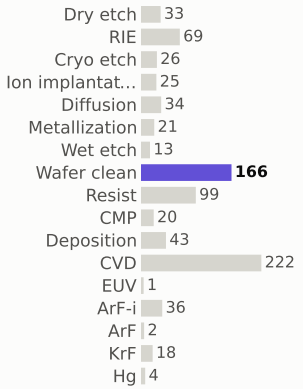
y cut at 57.351 µm



mask for this step

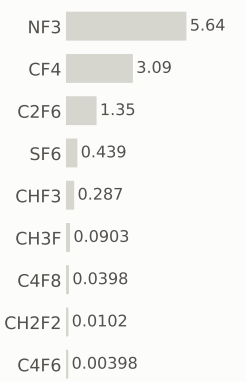
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



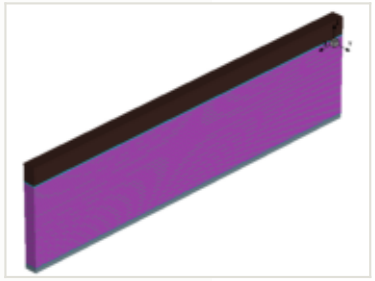
Step #10700: Deposition

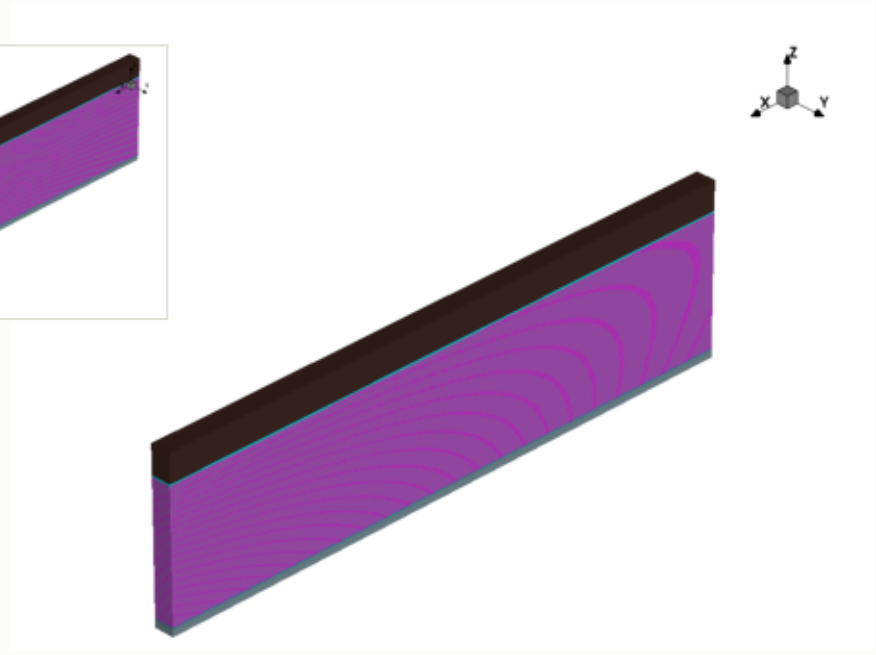
depo_isotropic · Deposition

0.656	620.6	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



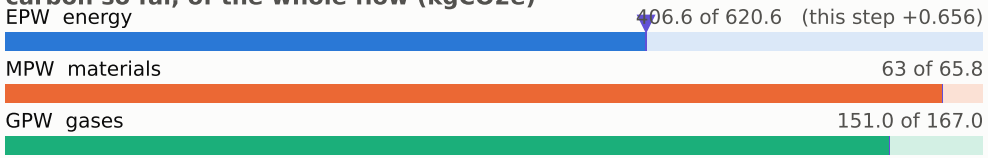


this step

process area	Deposition
material	—
thickness	1 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

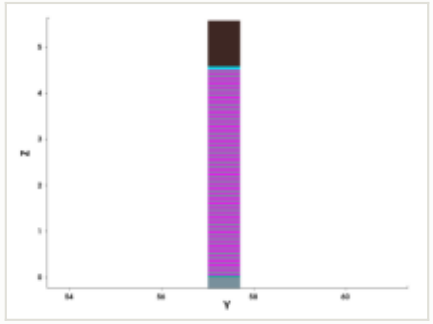
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

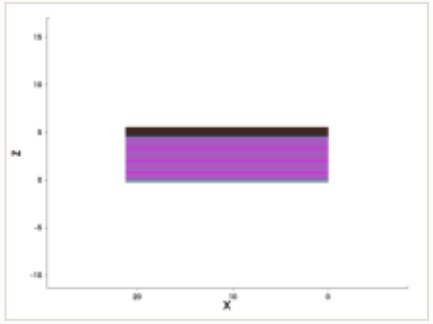


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



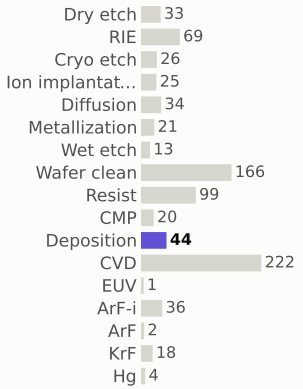
y cut at 57.351 µm



mask for this step

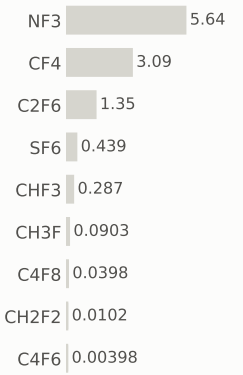
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



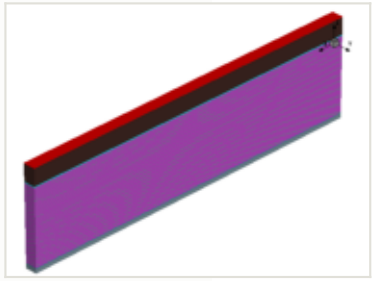
Step #10710: Deposition

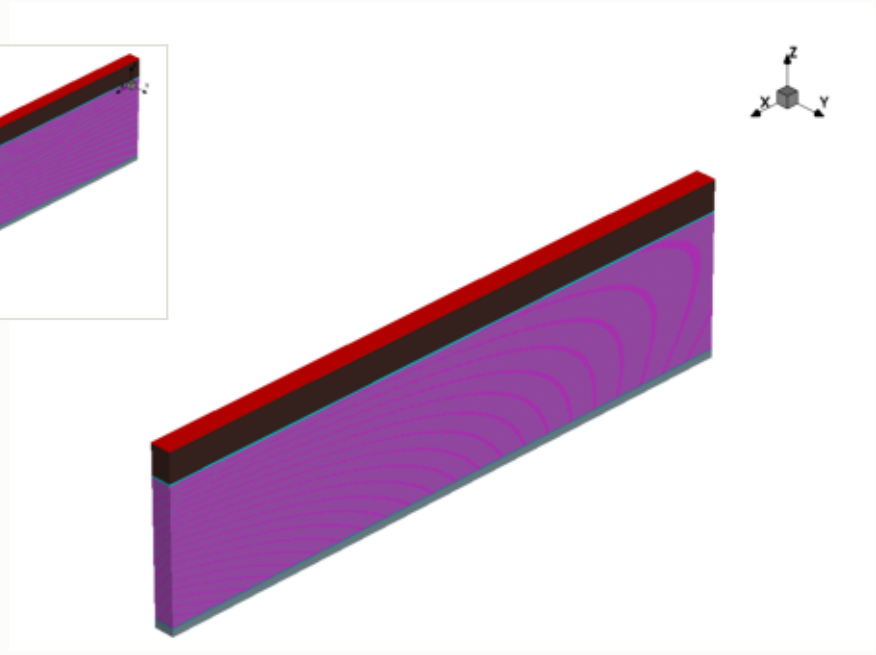
depo_isotropic · Deposition

0.656	621.3	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



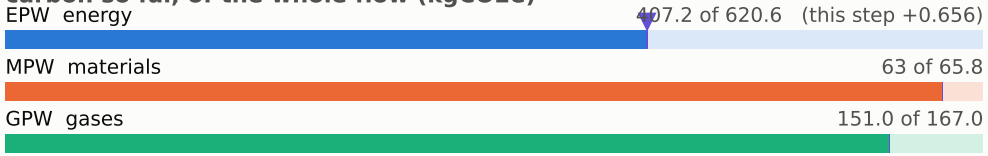


this step

process area	Deposition
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

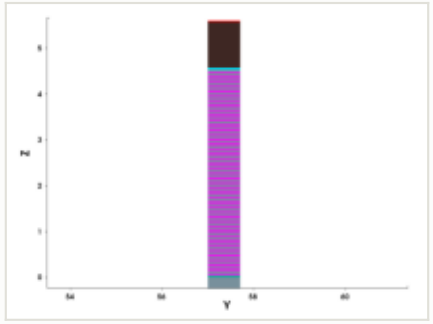
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

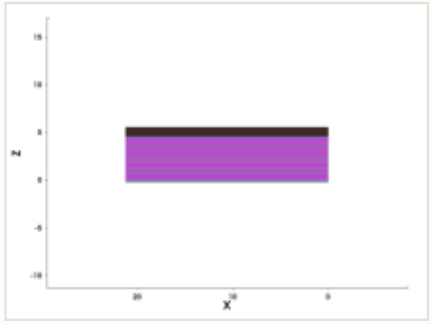


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



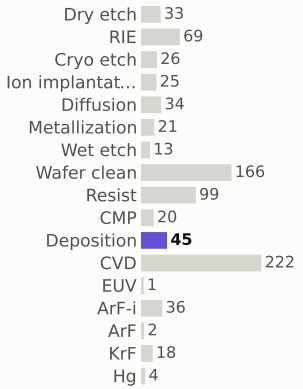
y cut at 57.351 µm



mask for this step

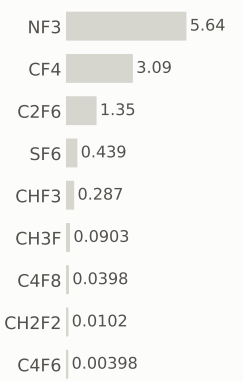
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



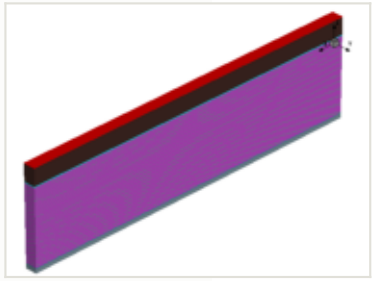
Step #10720: Wafer_clean

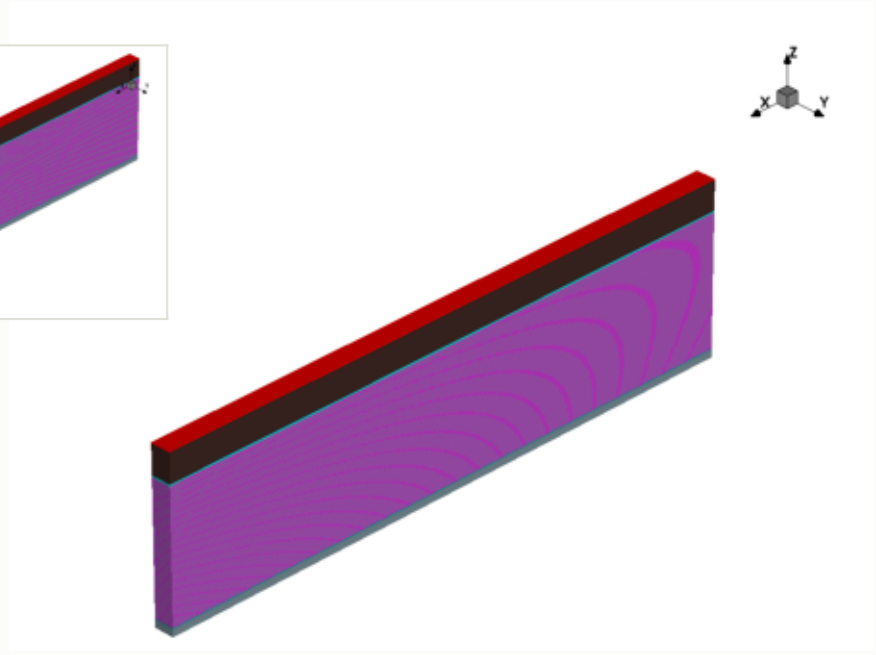
clean · Wafer_clean

0.169	621.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





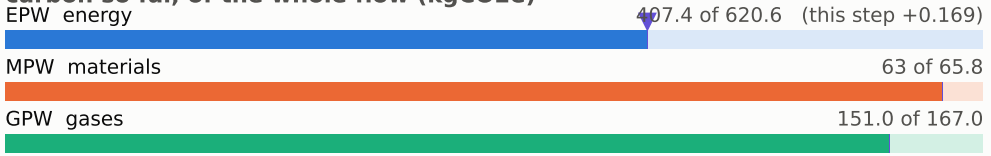
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

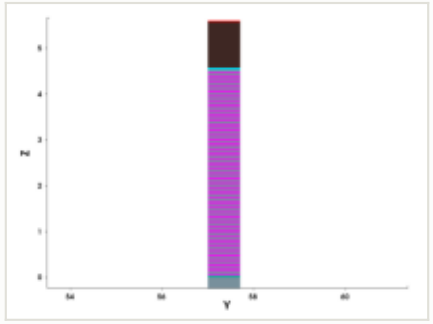
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

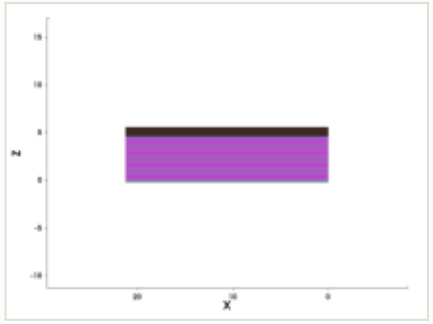


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



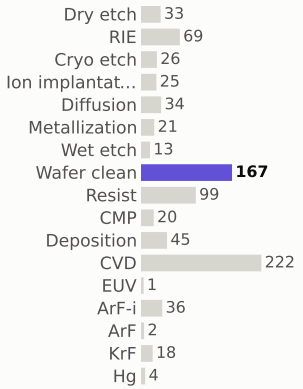
y cut at 57.351 µm



mask for this step

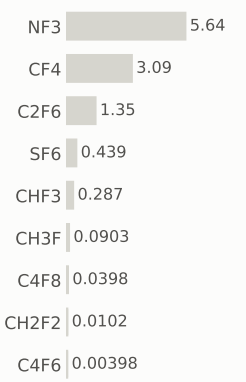
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



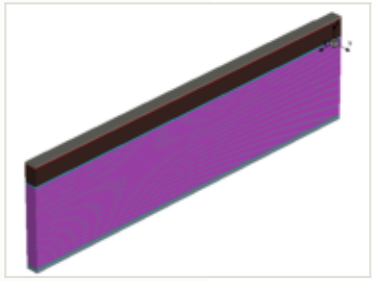
Step #10730: ArF-i

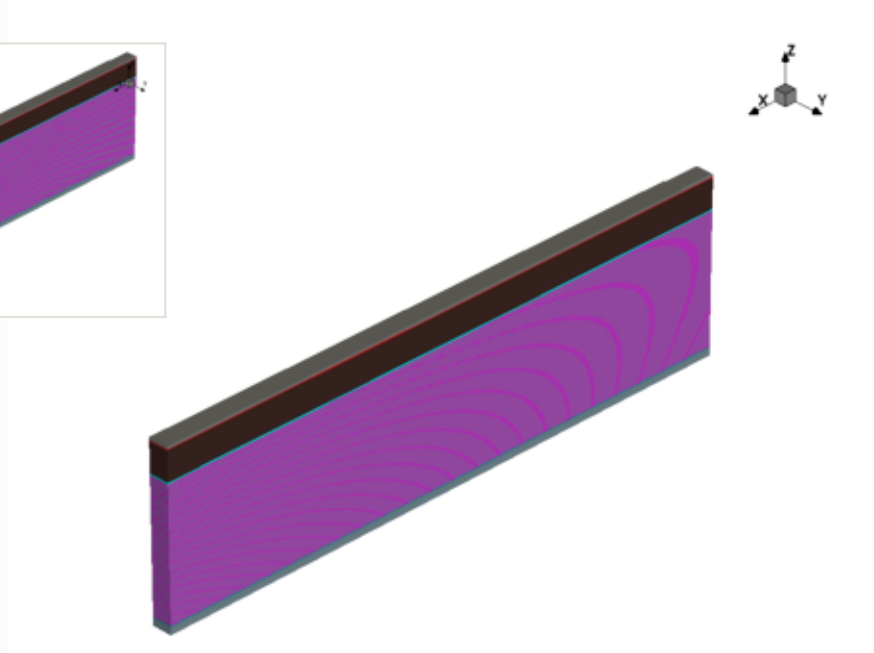
litho_patterning · ArF-i

0.217	621.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



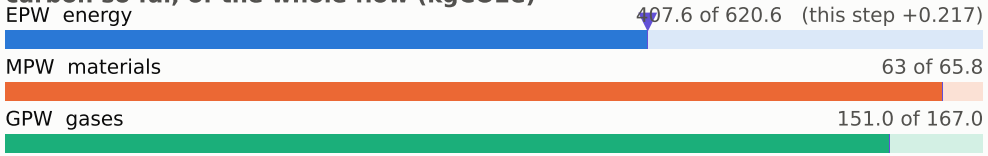


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

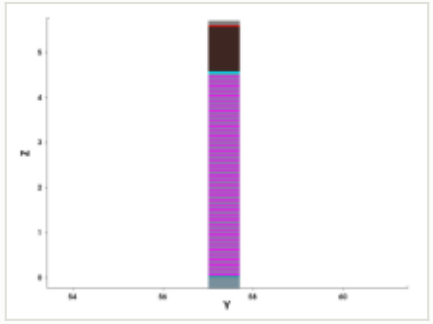
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

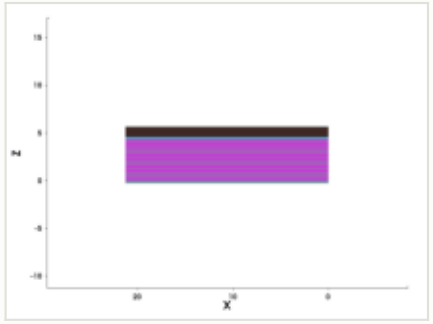


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

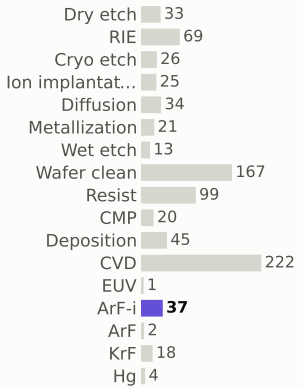


y cut at 57.351 µm



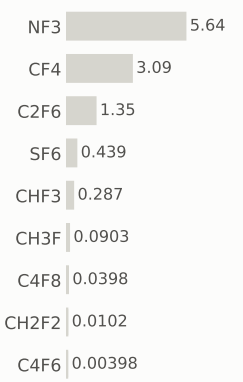
mask: layer 2:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



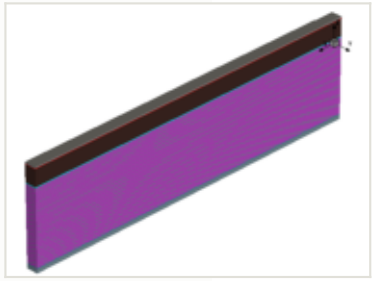
Step #10740: RIE

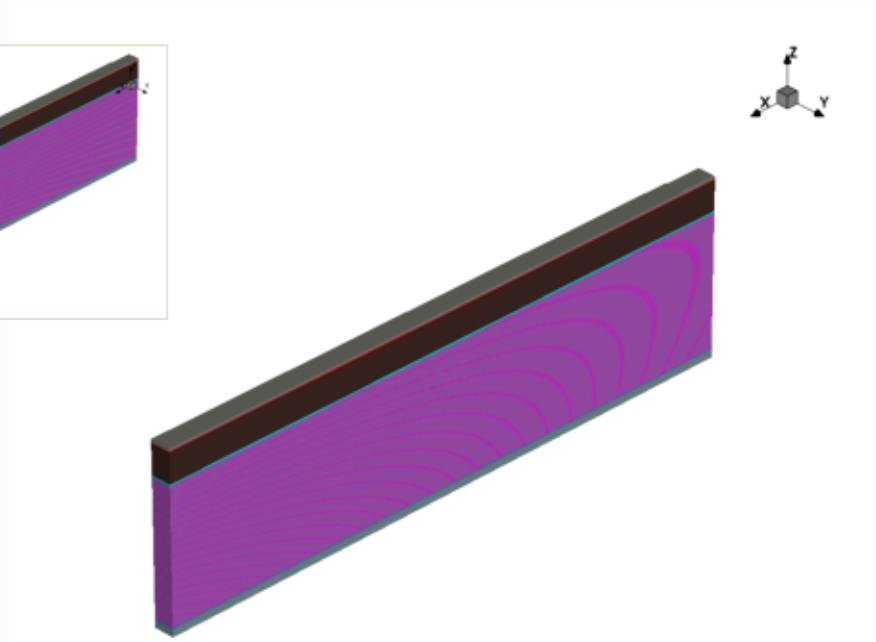
etch_anisotropic · RIE

1.04	622.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



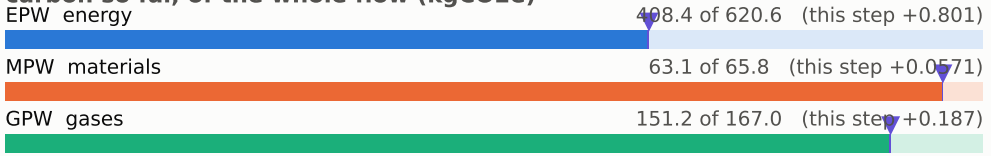


this step

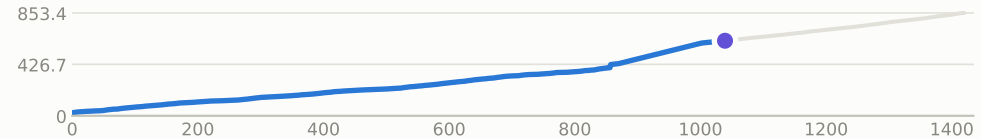
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

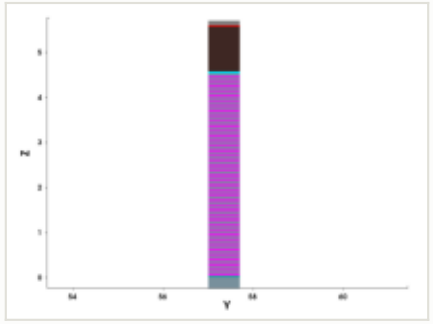
carbon so far, of the whole flow (kgCO2e)



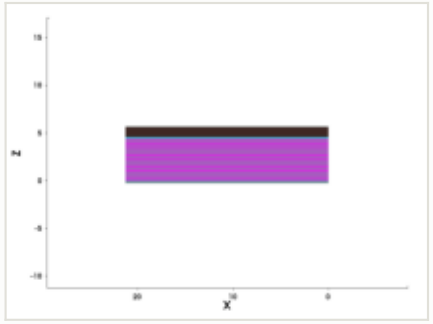
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



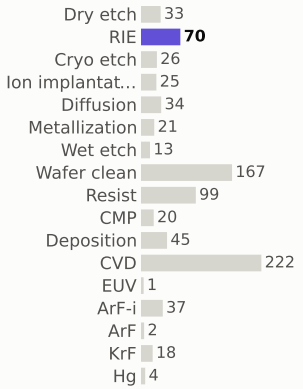
y cut at 57.351 µm



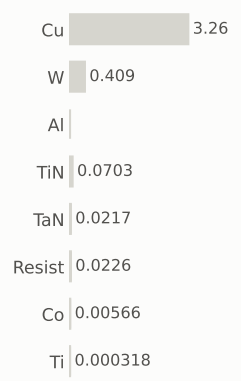
mask for this step

no mask for this step

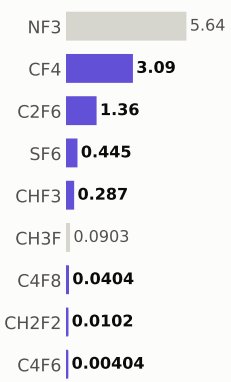
steps by process area



materials deposited (g)



gases emitted (g)



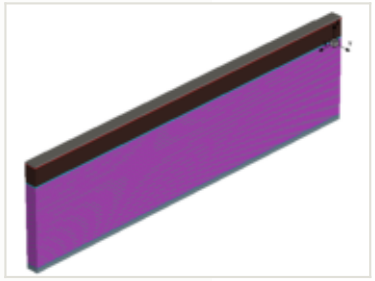
Step #10750: Cryo_etch

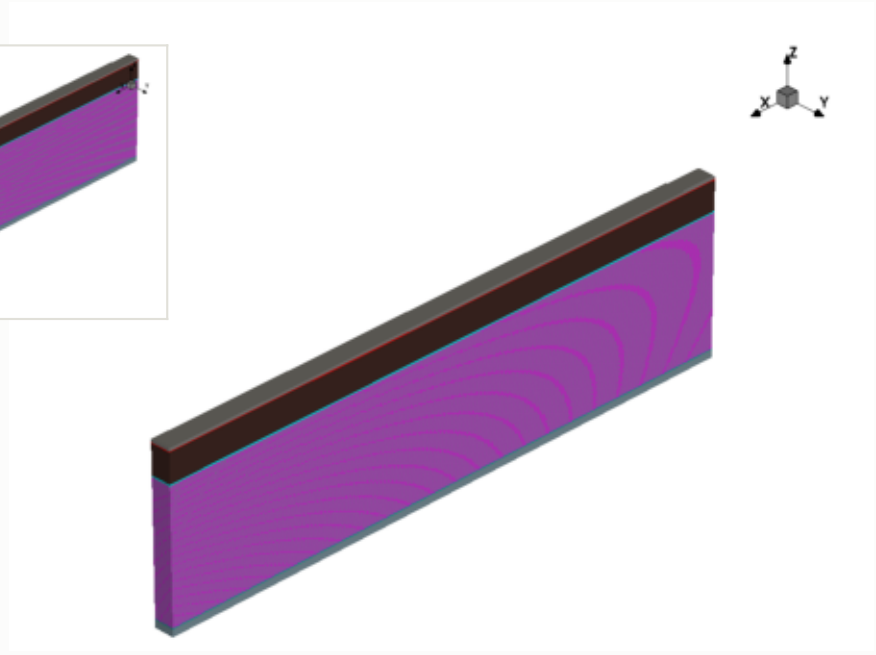
etch_anisotropic · Cryo_etch

0.517	623.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



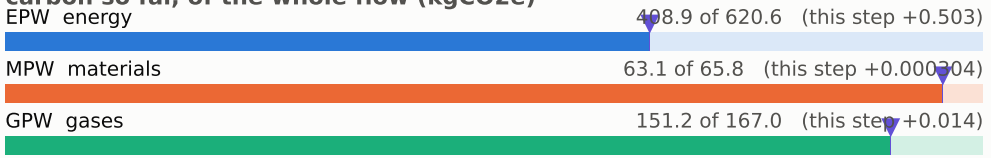


this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

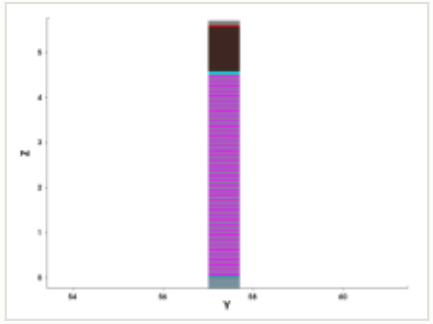
EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

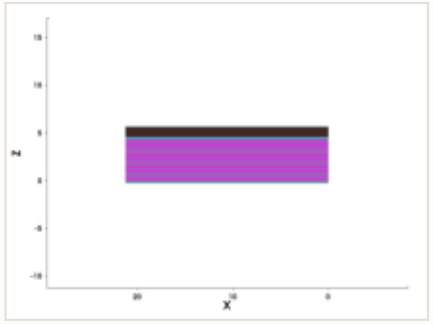


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



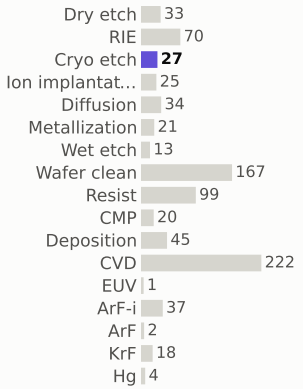
y cut at 57.351 µm



mask for this step

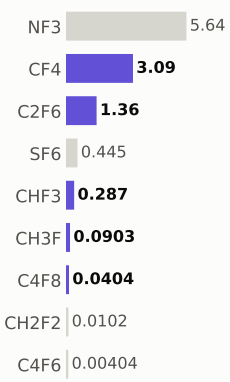
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



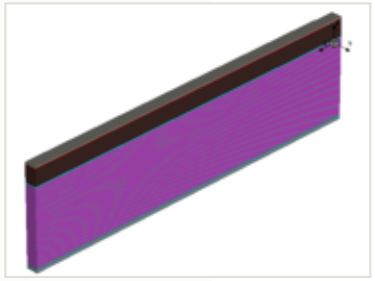
Step #10750.3846: Cryo_etch

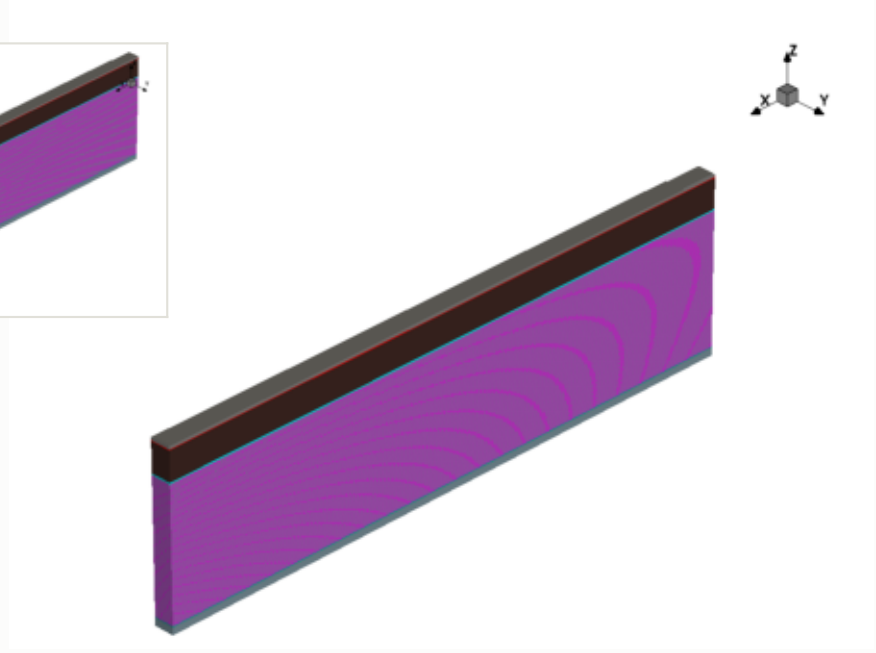
etch_noop · Cryo_etch

0.517	623.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





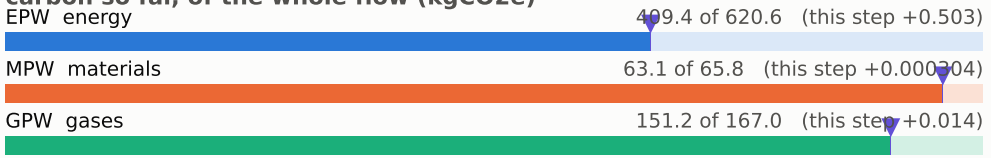
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

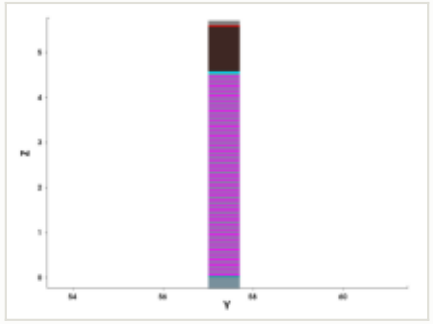
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

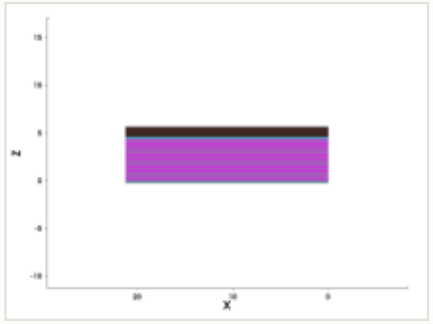


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



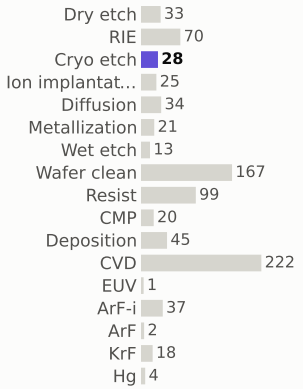
y cut at 57.351 µm



mask for this step

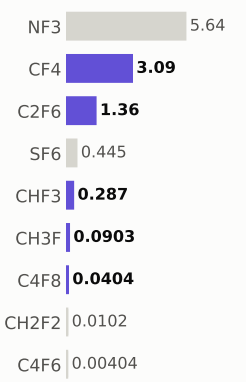
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



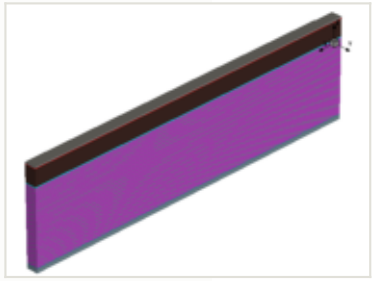
Step #10750.7692: Cryo_etch

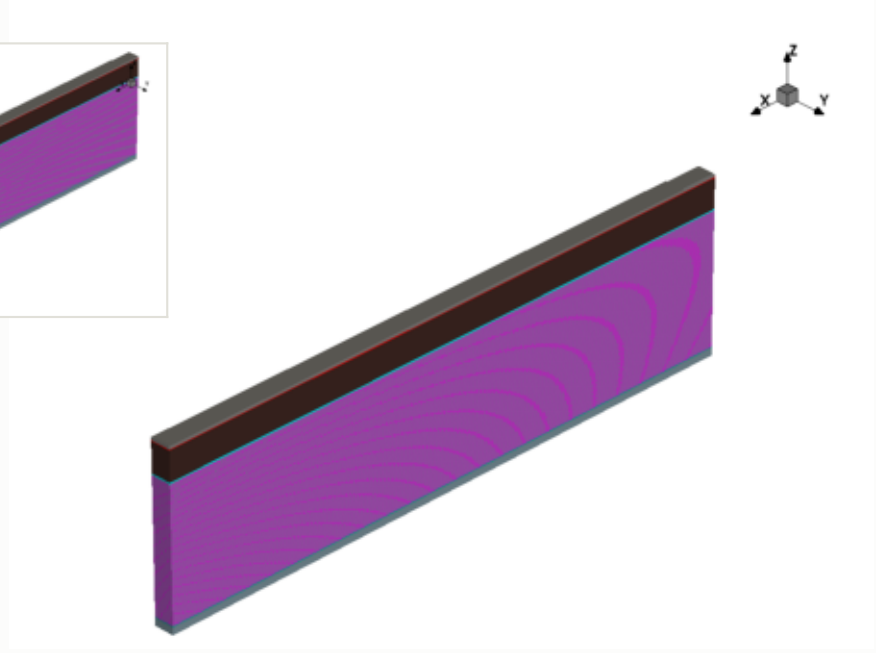
etch_noop · Cryo_etch

0.517	624.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





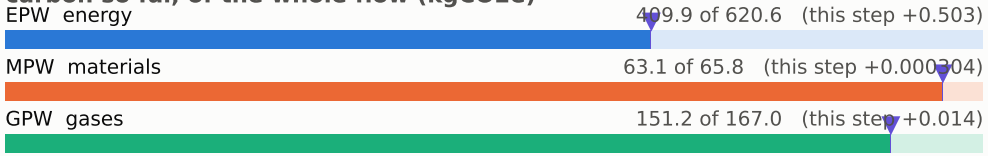
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

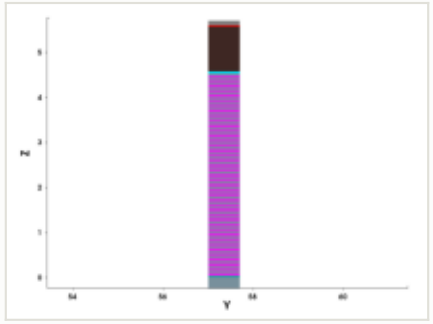
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

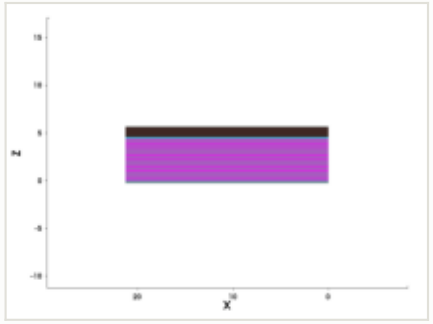


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



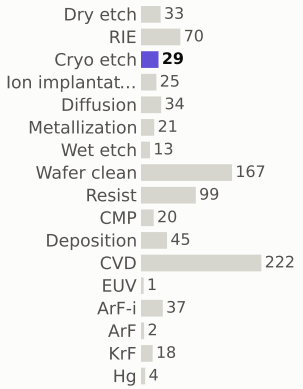
y cut at 57.351 µm



mask for this step

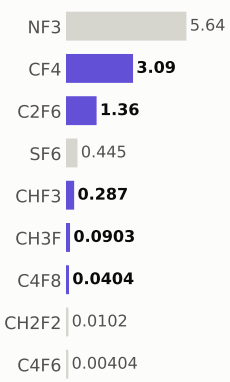
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



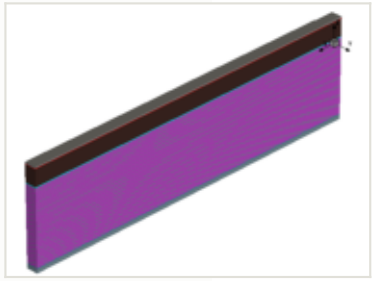
Step #10751.1538: Cryo_etch

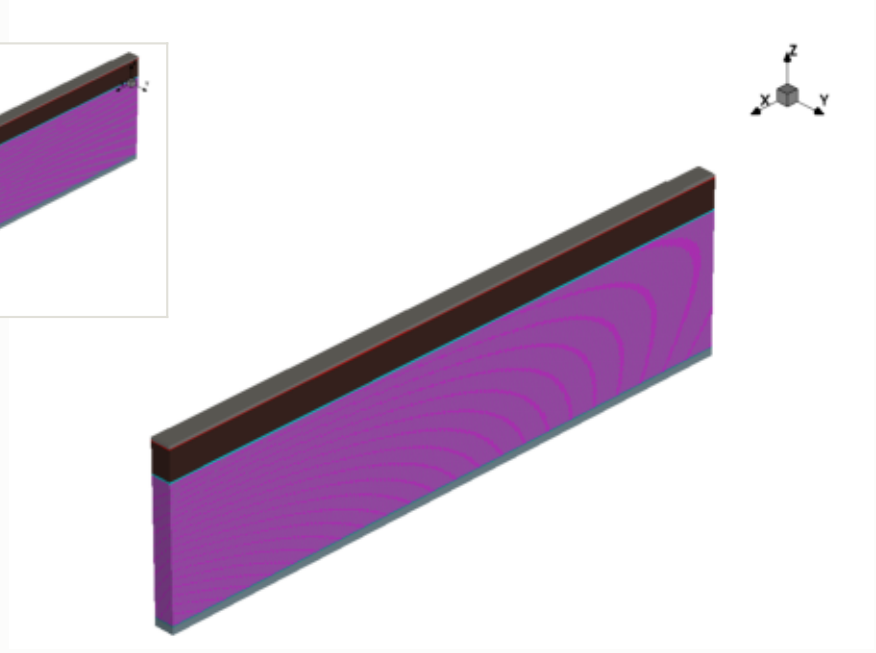
etch_noop · Cryo_etch

0.517	624.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





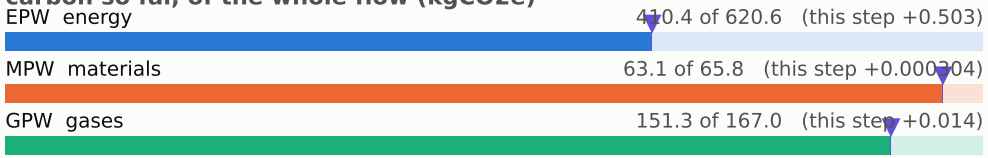
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

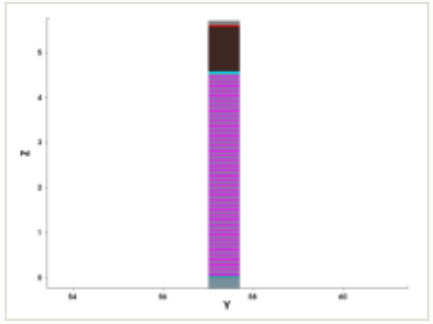
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

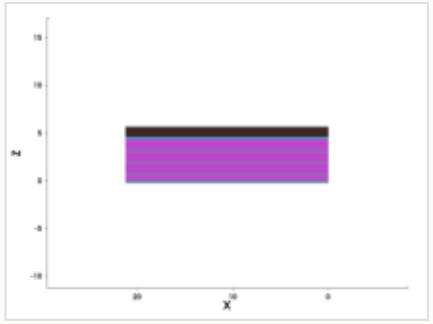


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



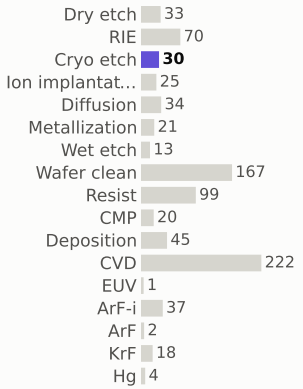
y cut at 57.351 µm



mask for this step

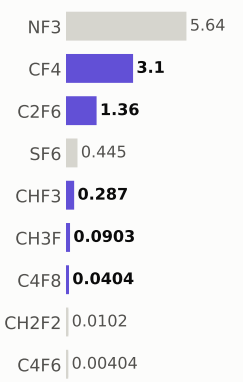
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



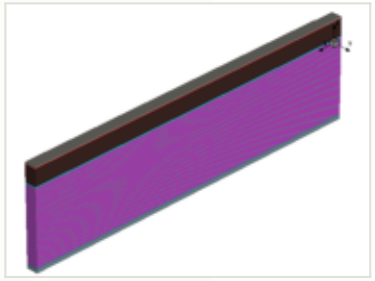
Step #10751.5385: Cryo_etch

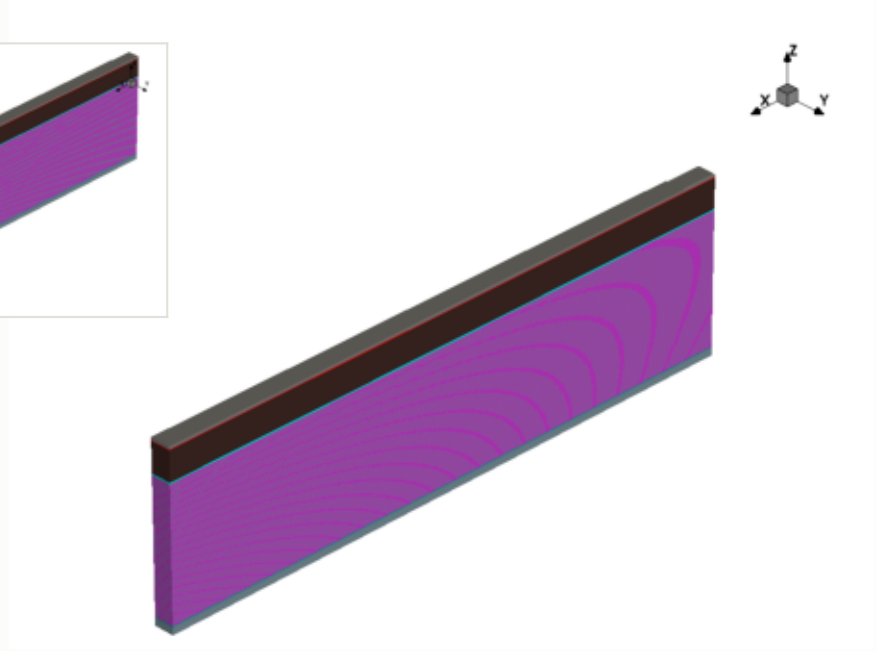
etch_noop · Cryo_etch

0.517	625.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





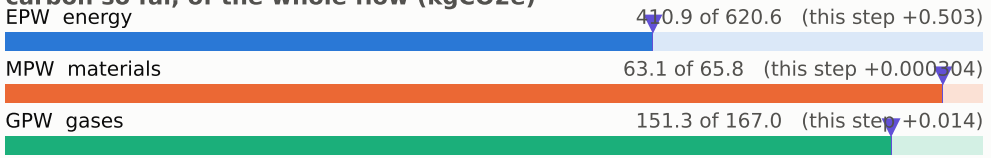
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

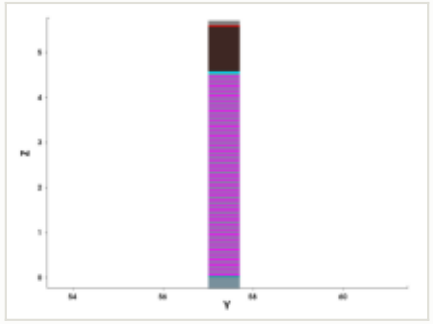
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

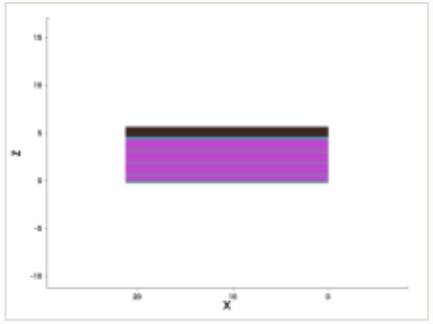


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



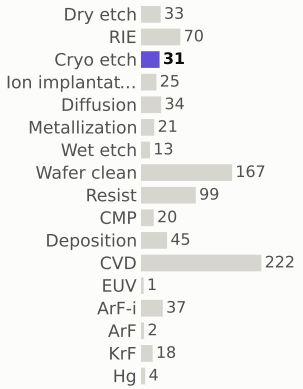
y cut at 57.351 µm



mask for this step

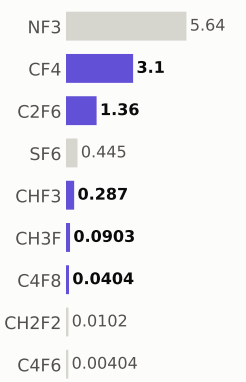
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



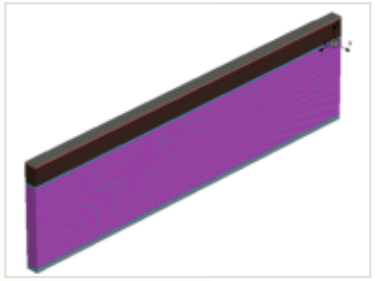
Step #10751.9231: Cryo_etch

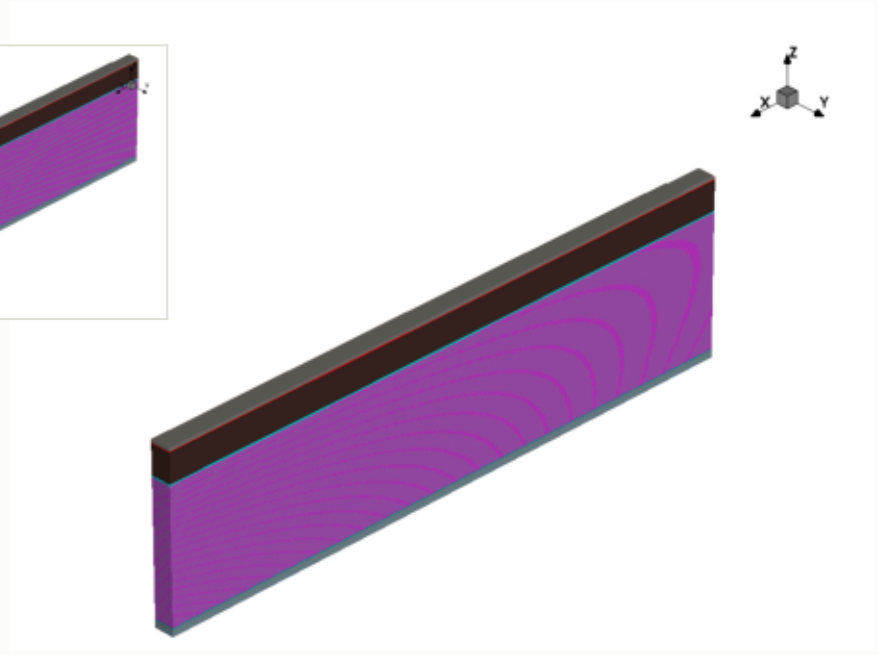
etch_noop · Cryo_etch

0.517	625.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





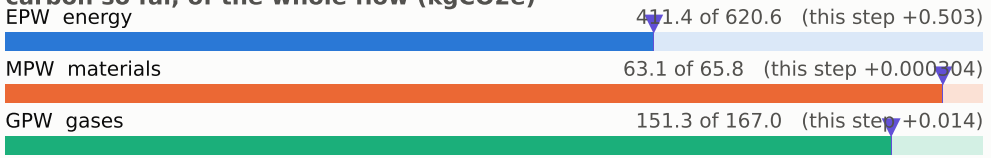
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

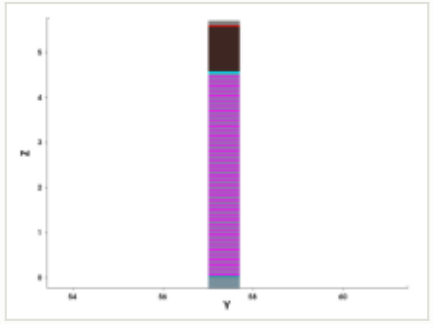
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

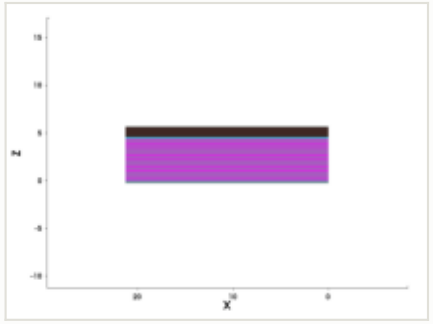


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



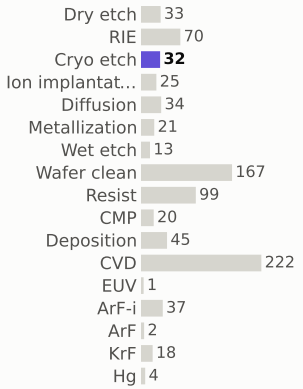
y cut at 57.351 µm



mask for this step

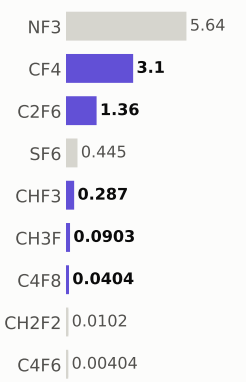
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



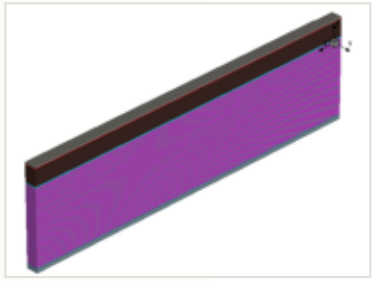
Step #10752.3077: Cryo_etch

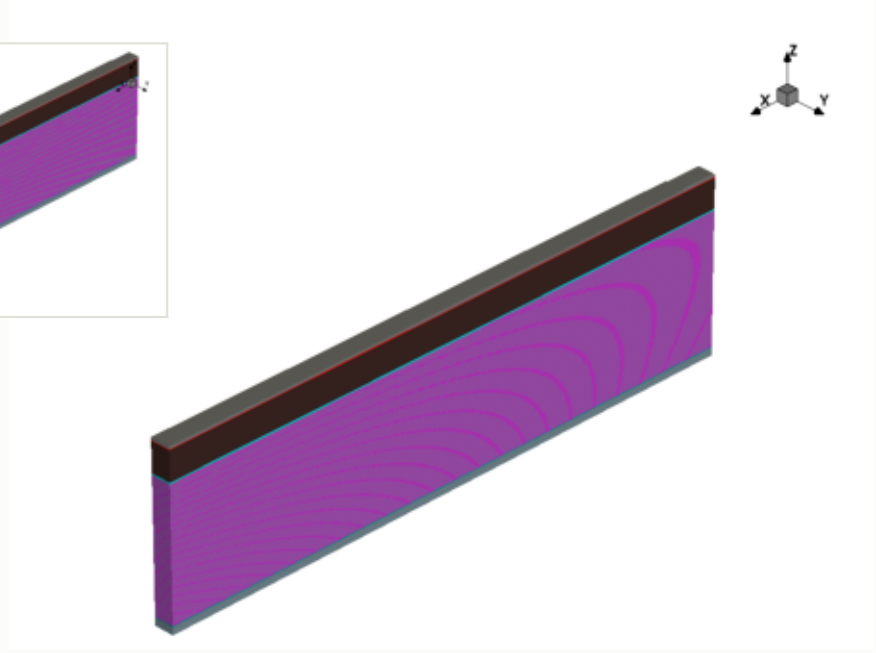
etch_noop · Cryo_etch

0.517	626.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





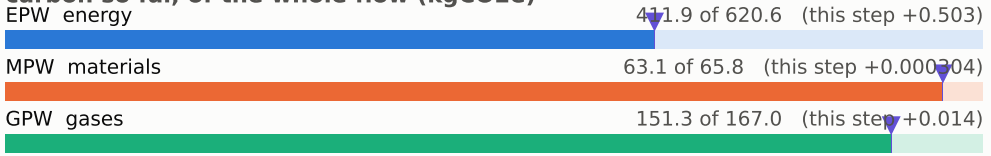
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

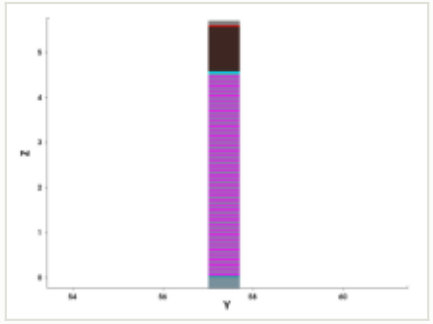
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

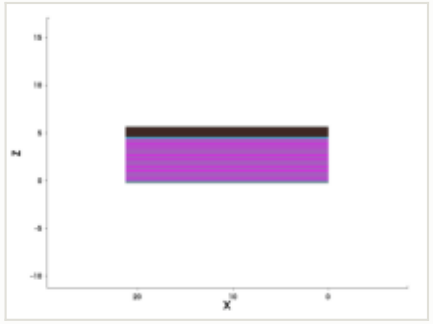


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



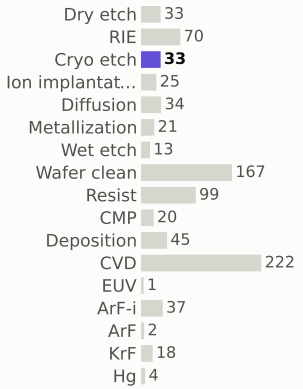
y cut at 57.351 µm



mask for this step

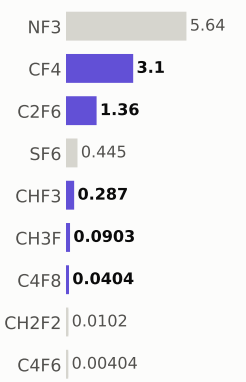
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



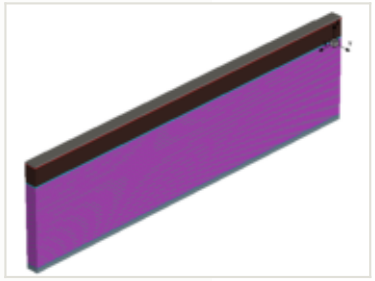
Step #10752.6923: Cryo_etch

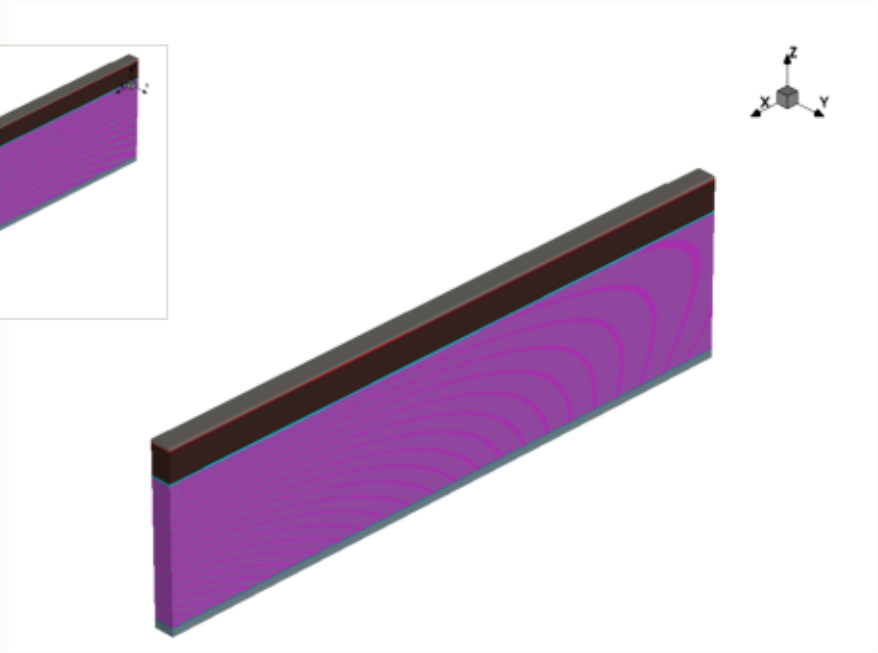
etch_noop · Cryo_etch

0.517	626.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





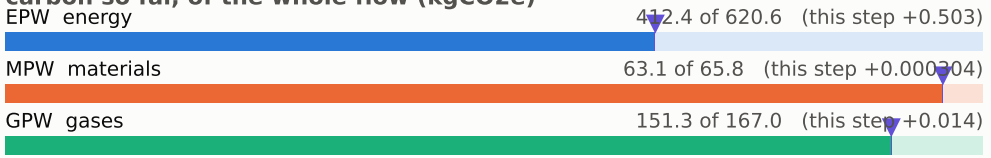
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

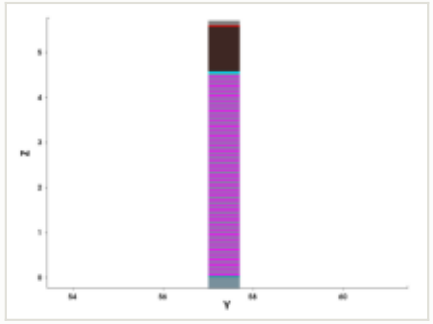
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

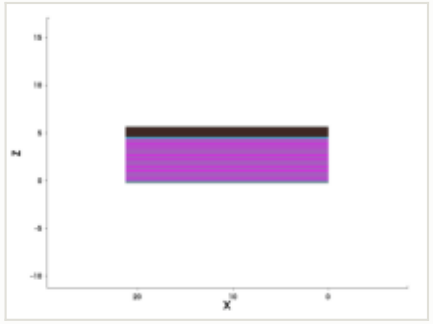


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



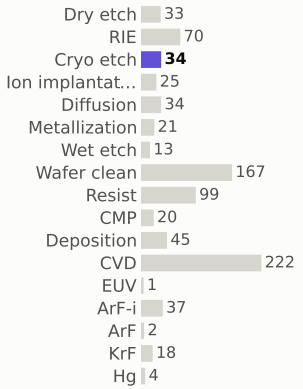
y cut at 57.351 µm



mask for this step

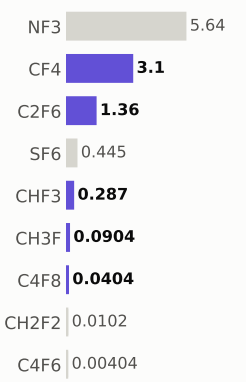
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



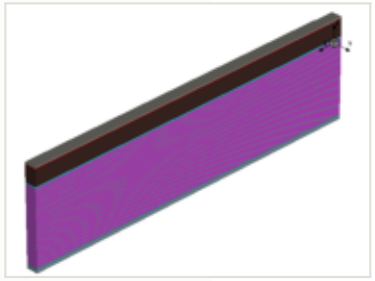
Step #10753.0769: Cryo_etch

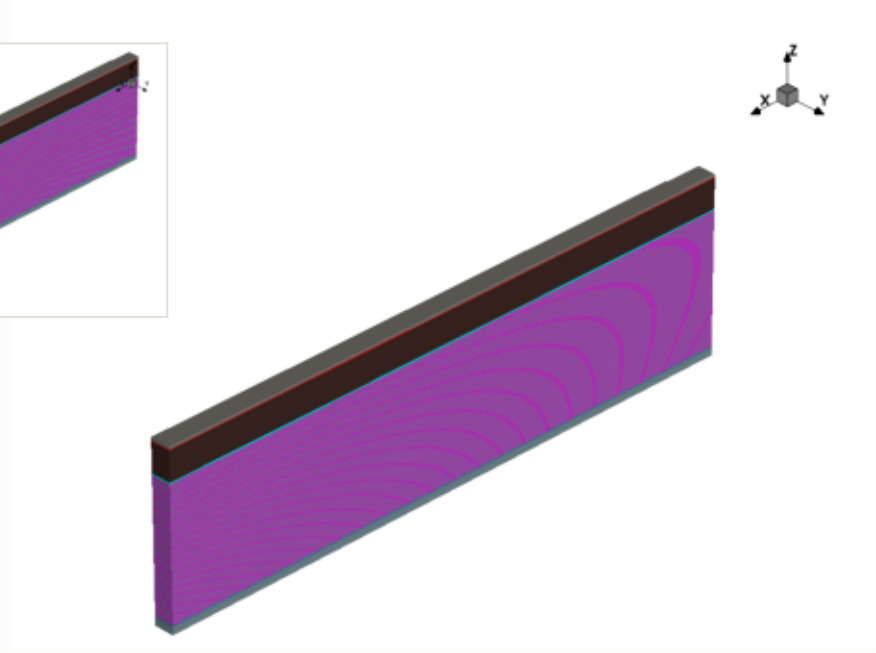
etch_noop · Cryo_etch

0.517	627.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





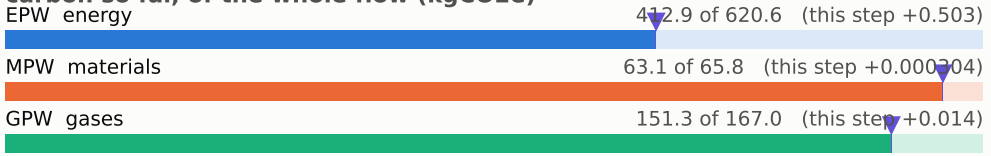
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

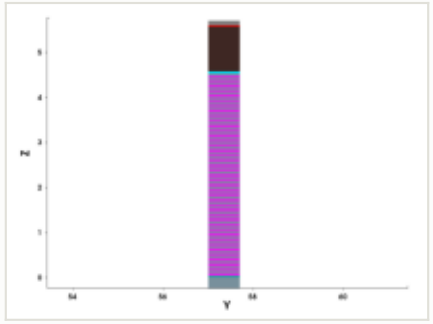
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

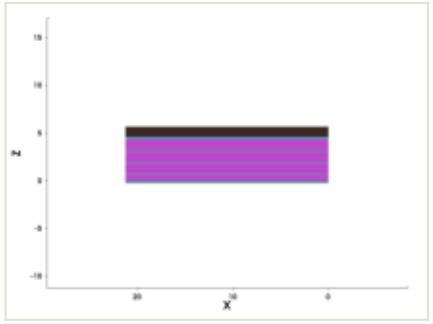


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



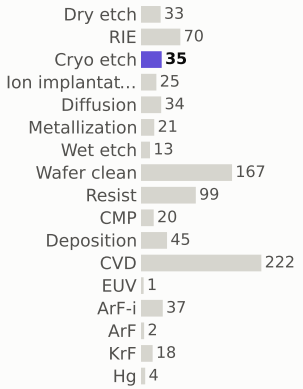
y cut at 57.351 µm



mask for this step

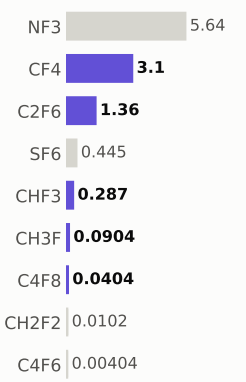
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



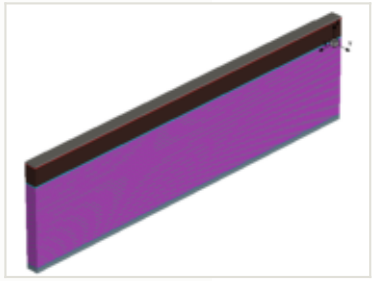
Step #10753.4615: Cryo_etch

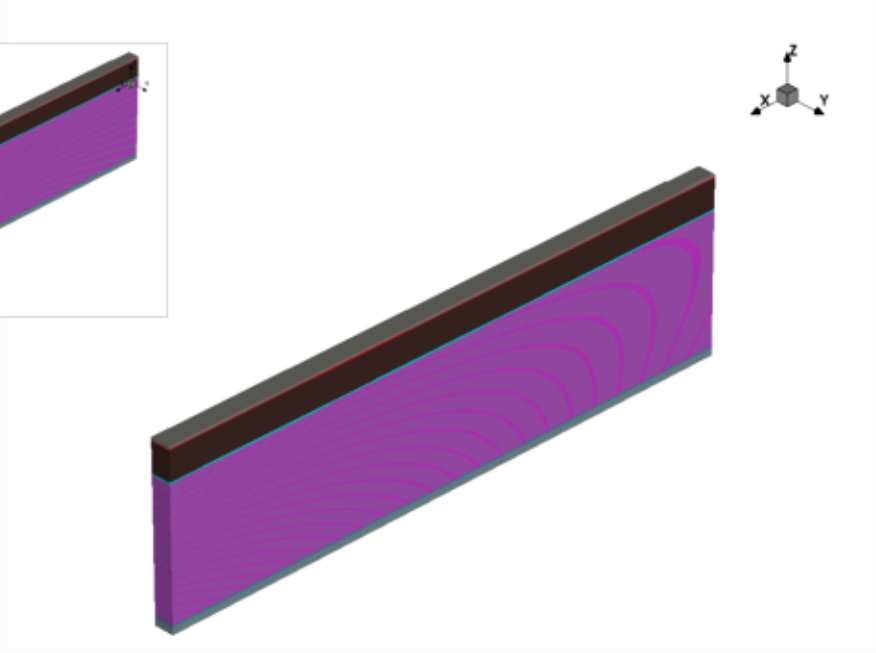
etch_noop · Cryo_etch

0.517	627.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





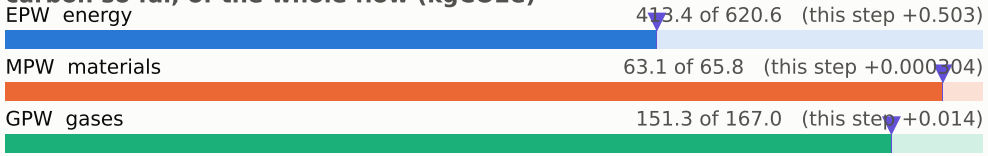
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

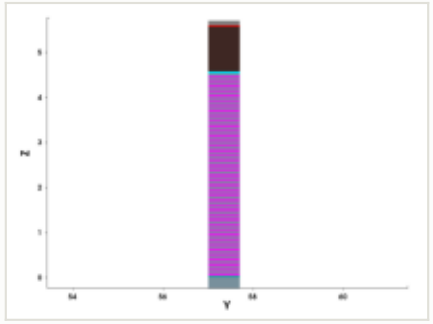
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

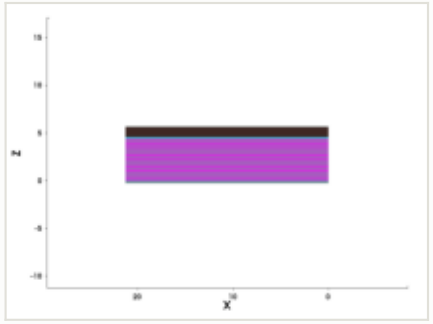


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



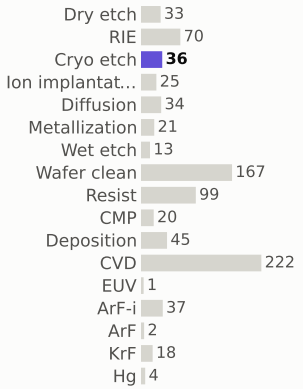
y cut at 57.351 µm



mask for this step

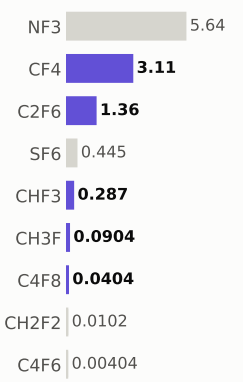
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



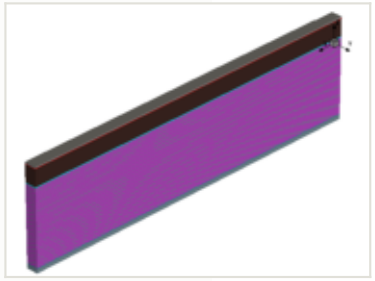
Step #10753.8462: Cryo_etch

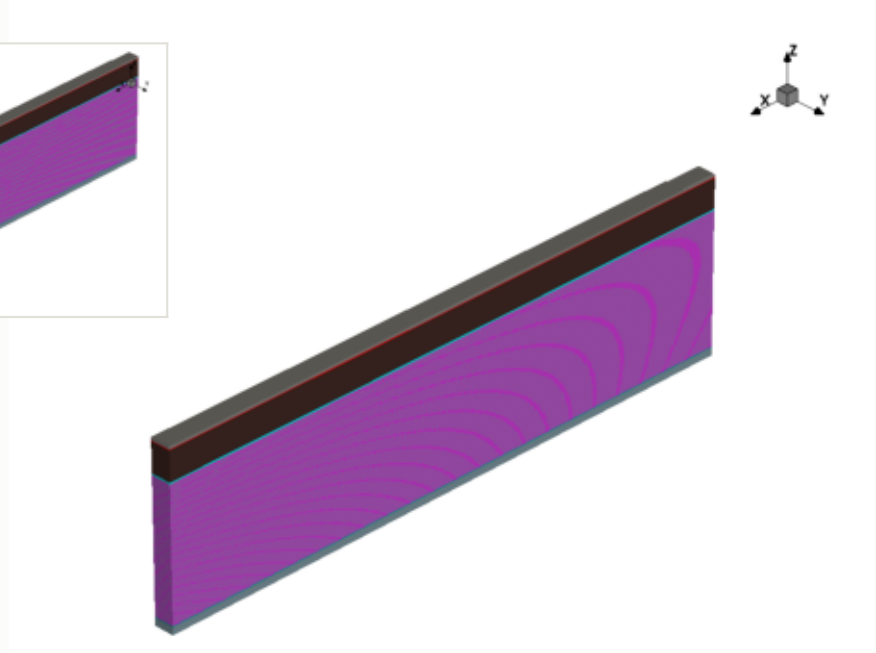
etch_noop · Cryo_etch

0.517	628.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





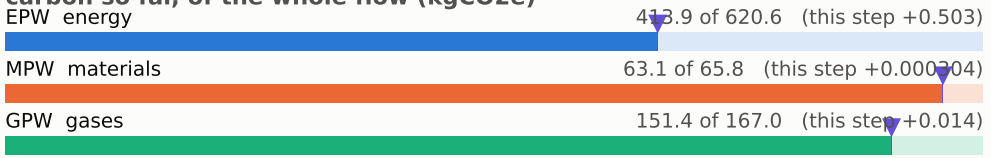
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

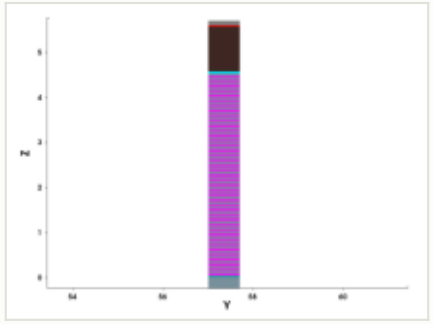
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

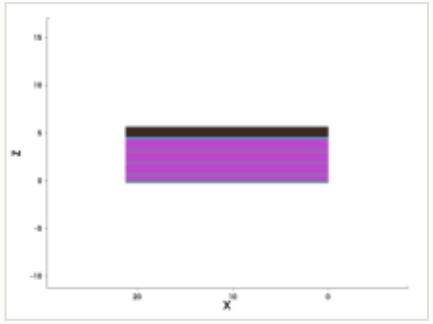


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



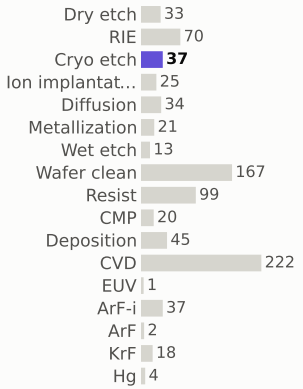
y cut at 57.351 µm



mask for this step

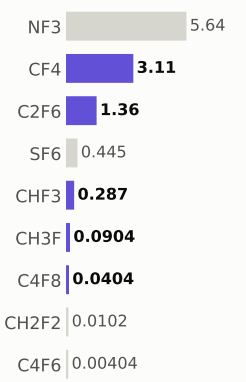
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



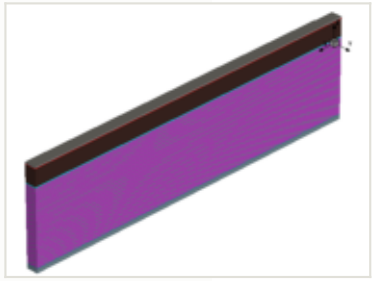
Step #10754.2308: Cryo_etch

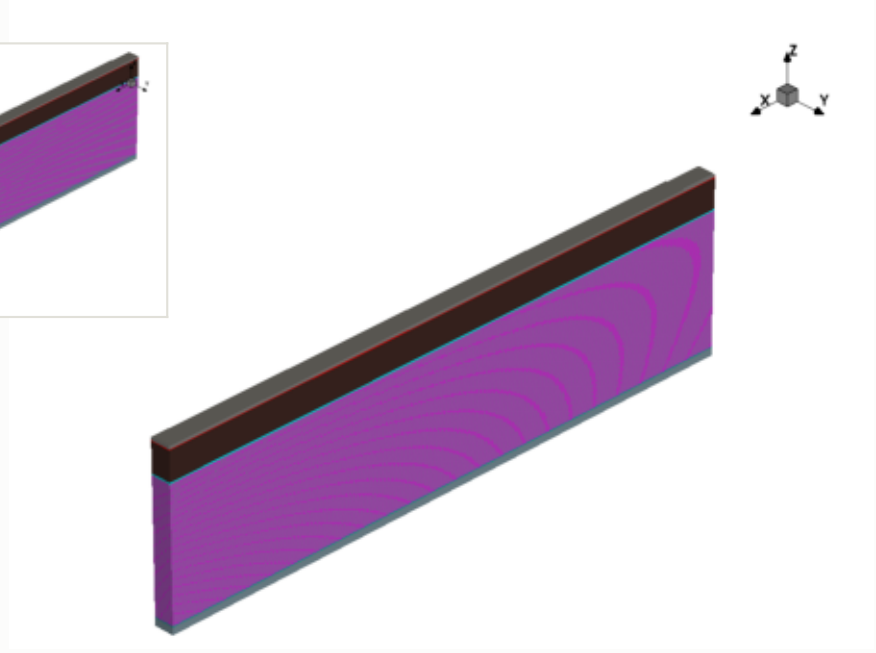
etch_noop · Cryo_etch

0.517	628.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





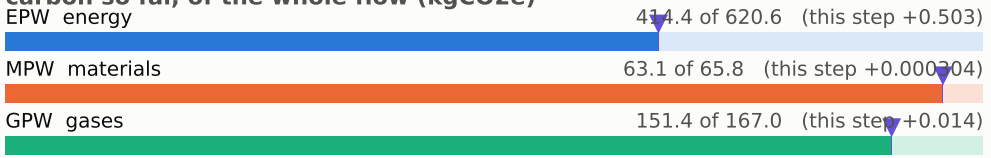
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

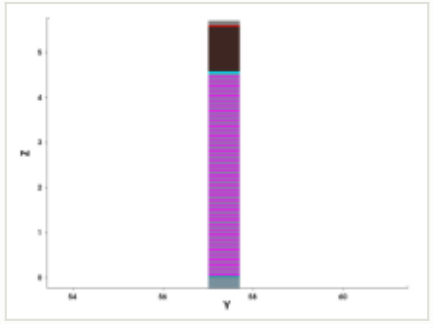
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

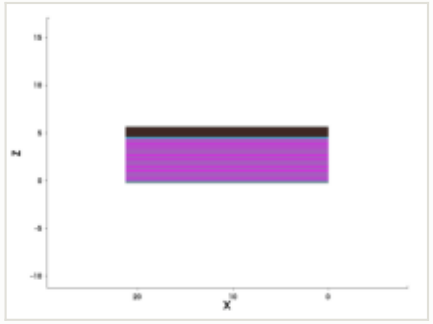


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



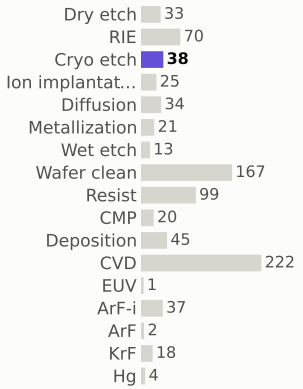
y cut at 57.351 µm



mask for this step

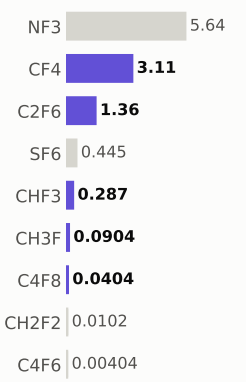
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



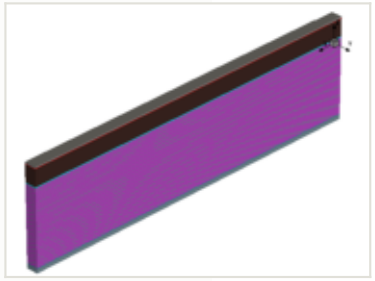
Step #10754.6154: Cryo_etch

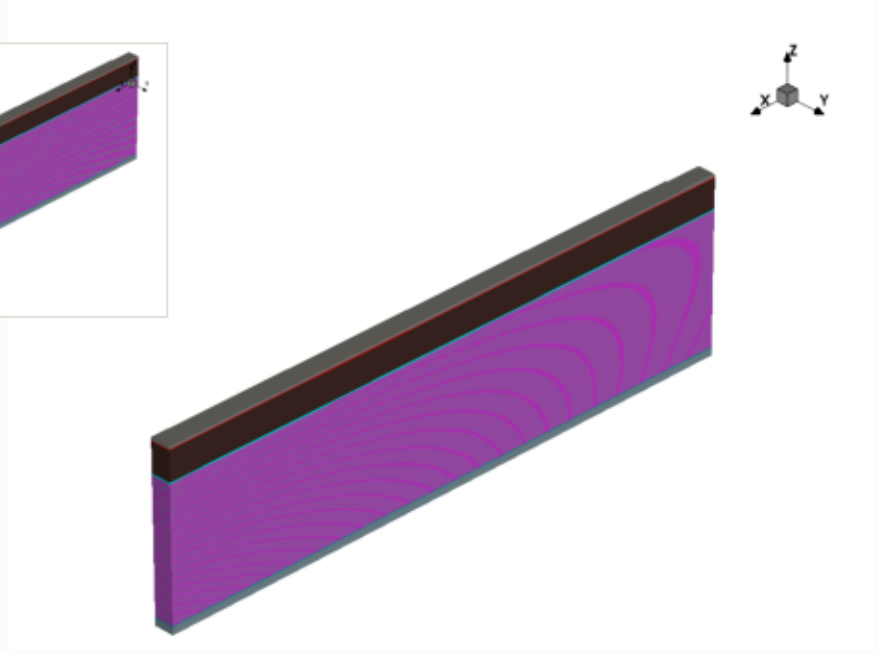
etch_noop · Cryo_etch

0.517	629.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





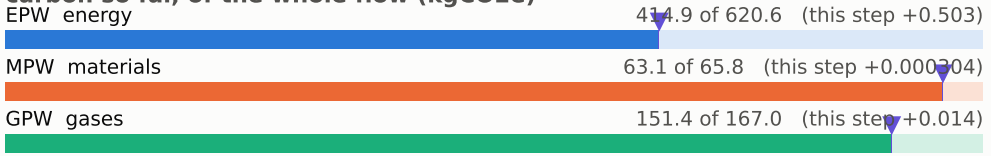
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

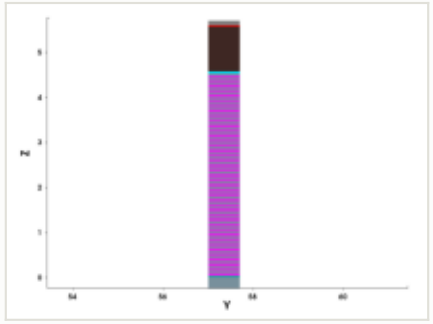
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

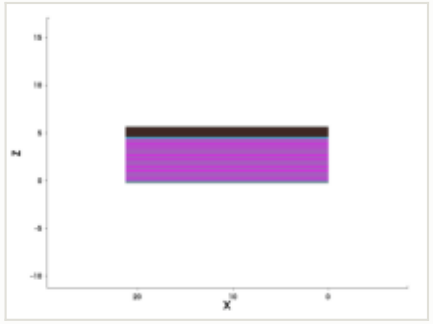


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



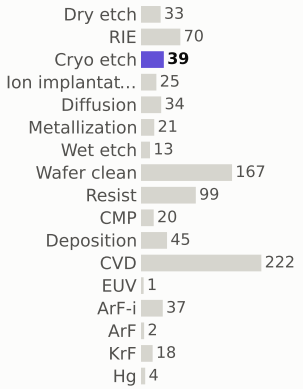
y cut at 57.351 µm



mask for this step

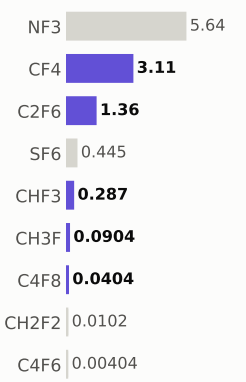
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



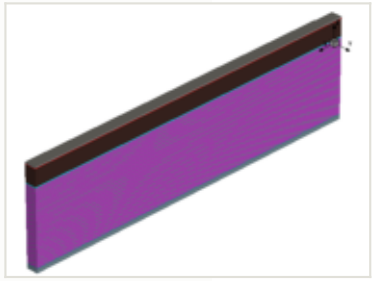
Step #10755.0000: Cryo_etch

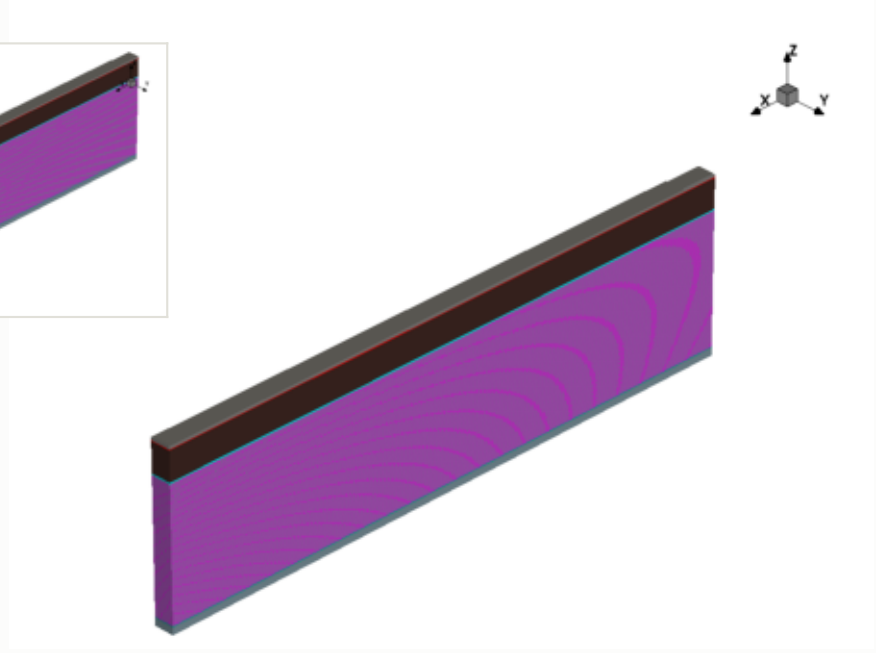
etch_noop · Cryo_etch

0.517	629.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





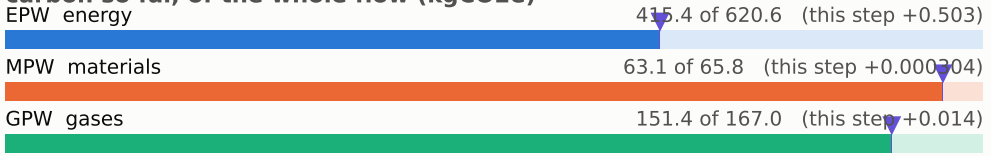
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

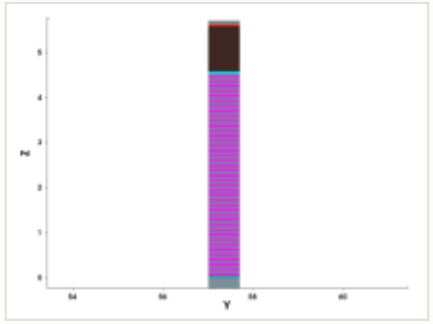
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

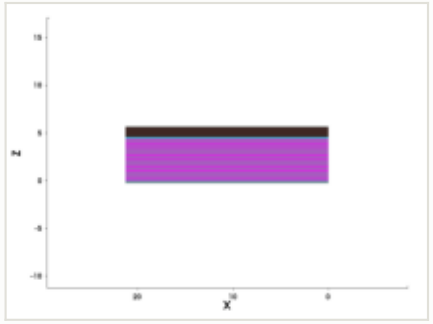


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



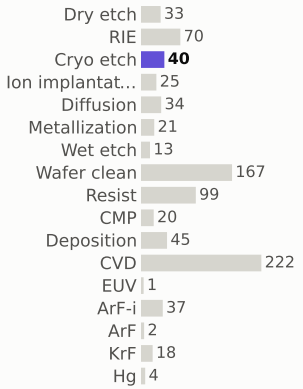
y cut at 57.351 µm



mask for this step

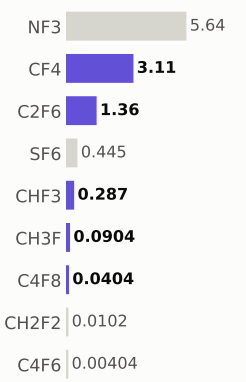
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



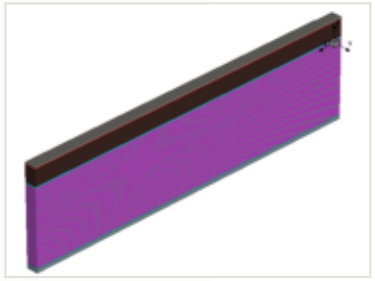
Step #10755.3846: Cryo_etch

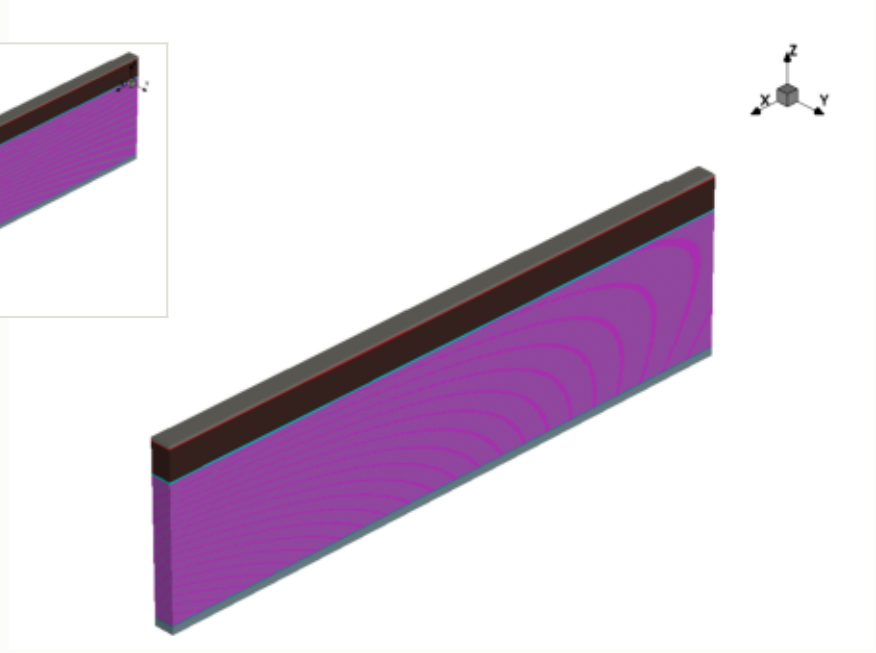
etch_noop · Cryo_etch

0.517	630.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





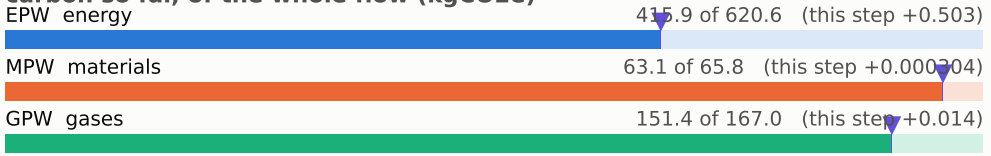
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

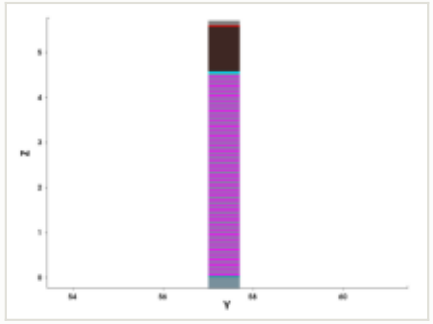
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

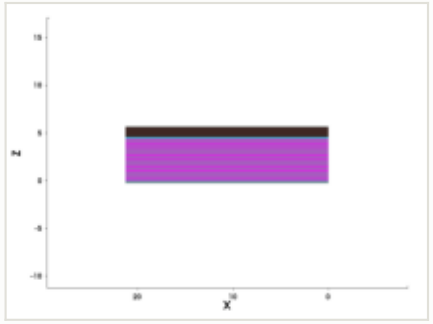


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



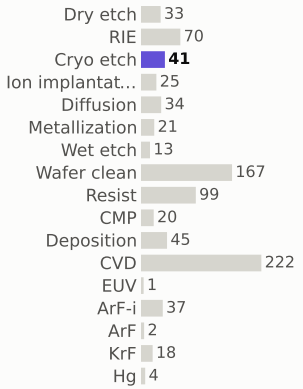
y cut at 57.351 µm



mask for this step

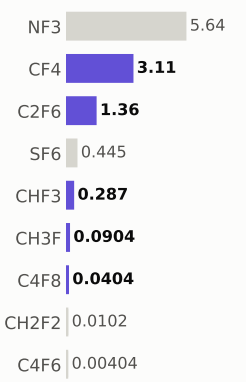
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



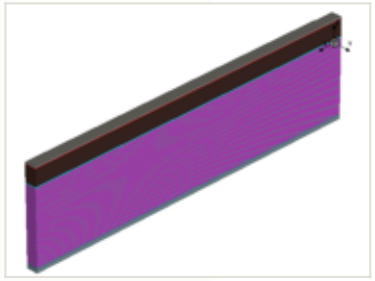
Step #10755.7692: Cryo_etch

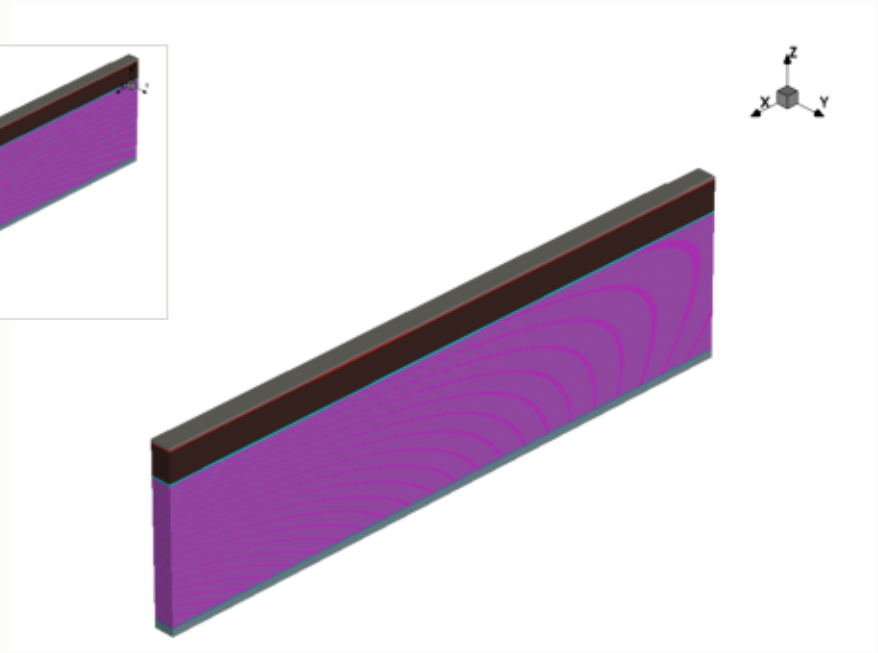
etch_noop · Cryo_etch

0.517	631.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





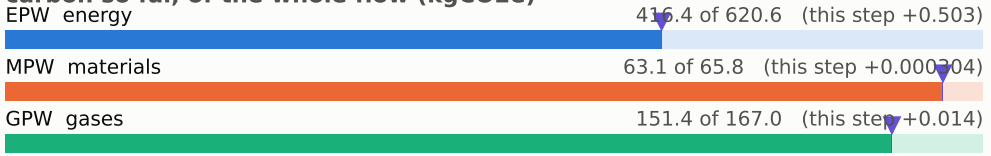
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

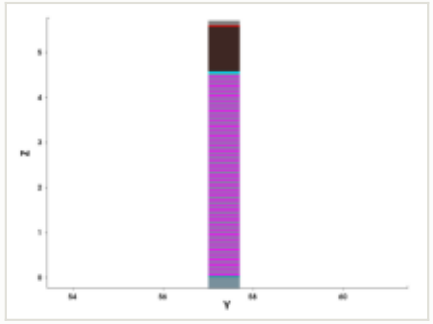
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

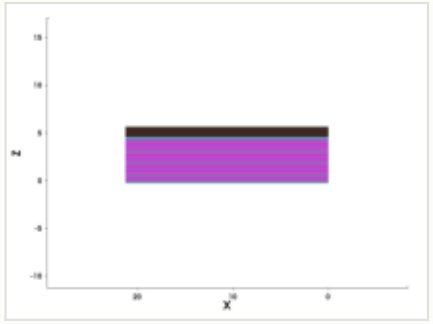


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



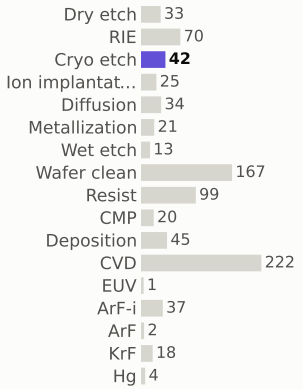
y cut at 57.351 µm



mask for this step

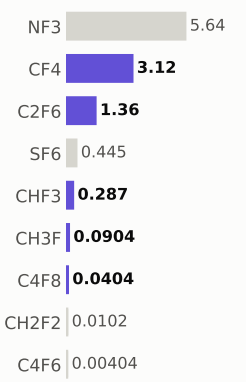
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



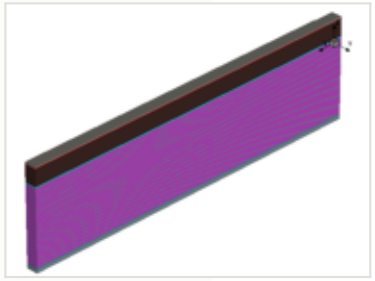
Step #10756.1538: Cryo_etch

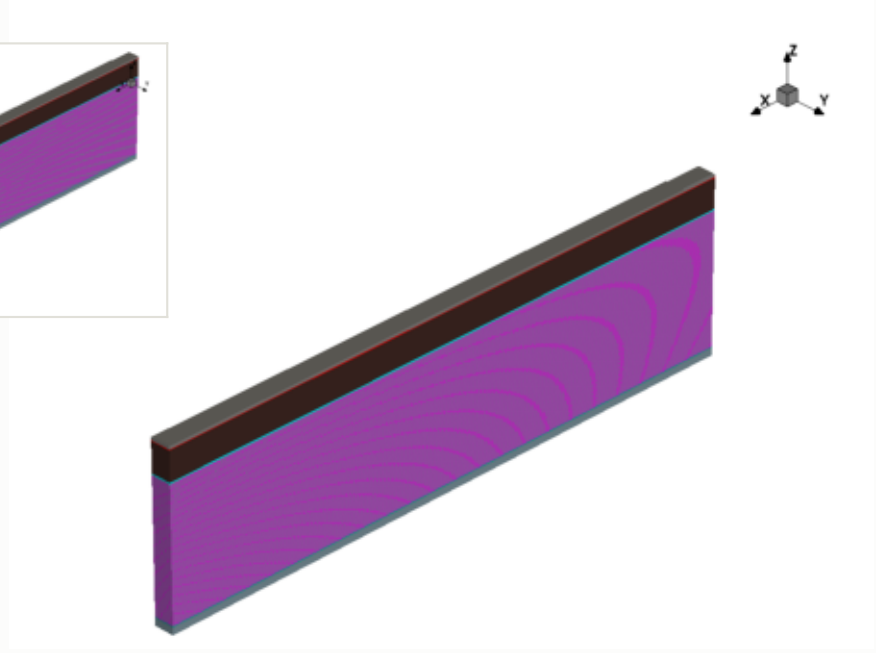
etch_noop · Cryo_etch

0.517	631.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





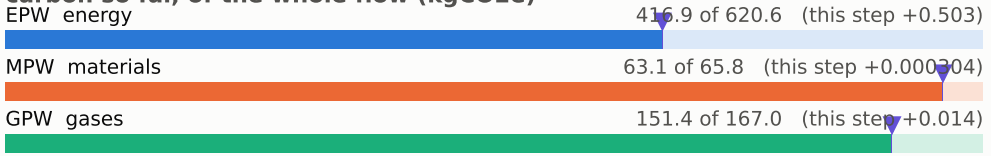
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

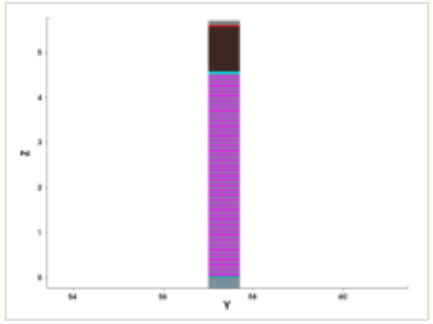
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

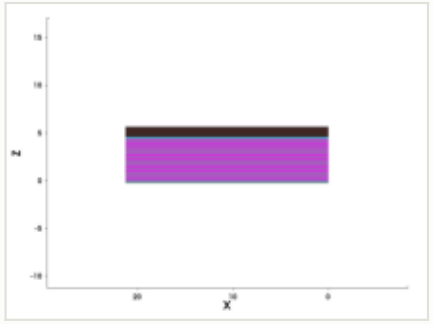


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



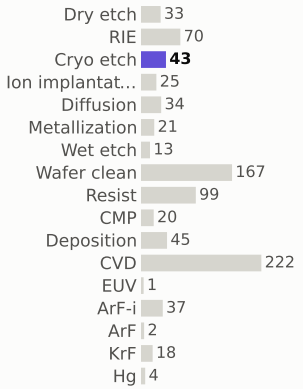
y cut at 57.351 µm



mask for this step

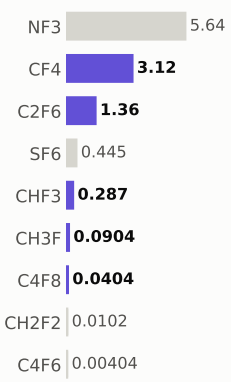
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



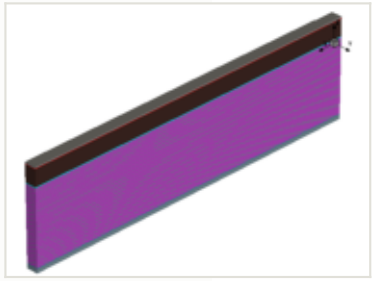
Step #10756.5385: Cryo_etch

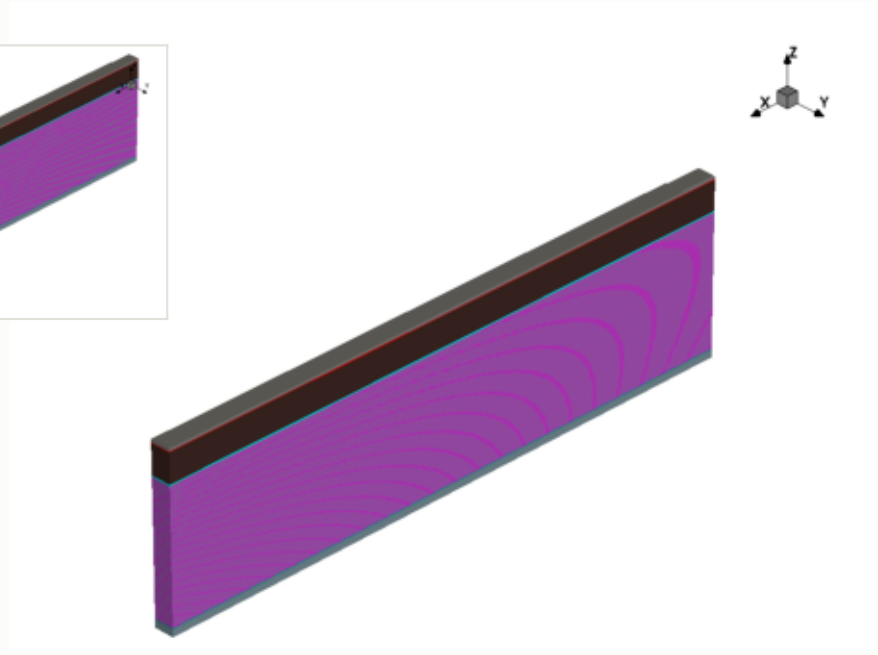
etch_noop · Cryo_etch

0.517	632.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





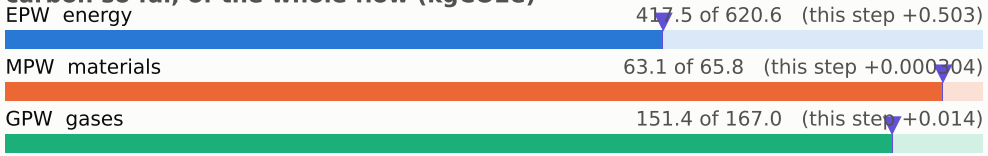
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

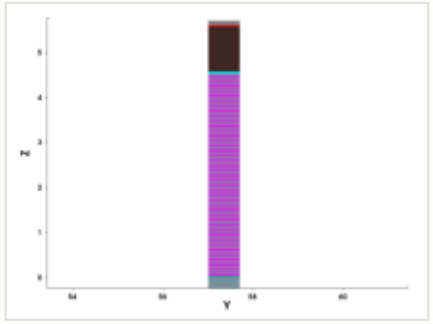
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

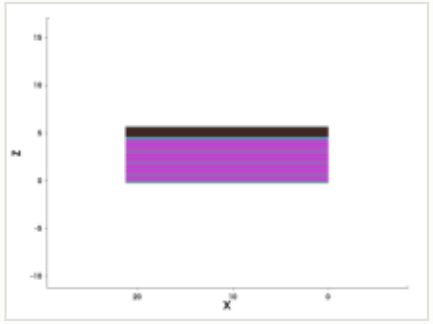


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



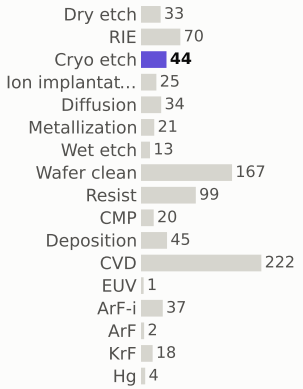
y cut at 57.351 µm



mask for this step

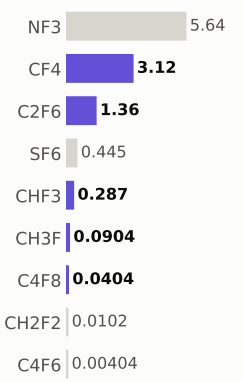
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



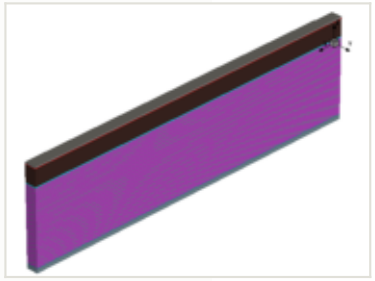
Step #10756.9231: Cryo_etch

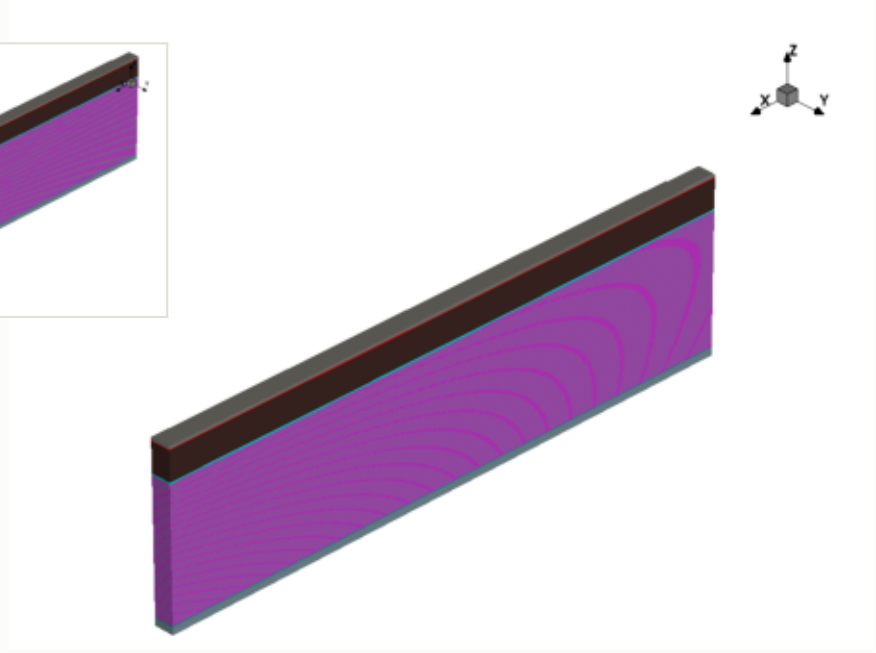
etch_noop · Cryo_etch

0.517	632.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





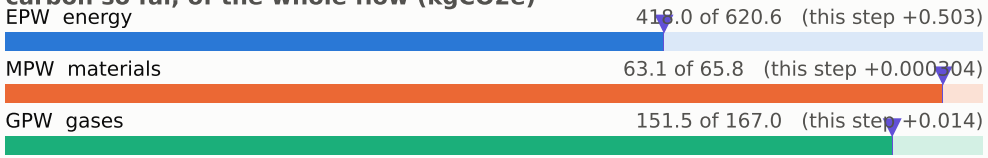
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

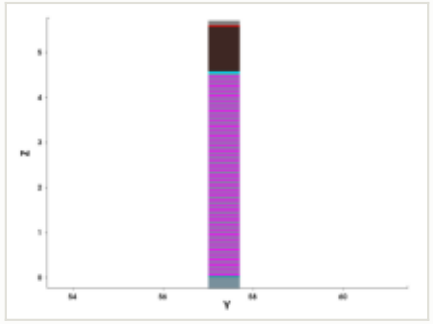
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

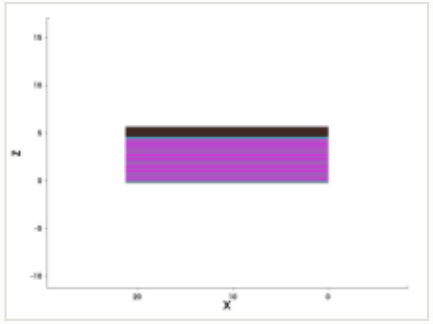


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



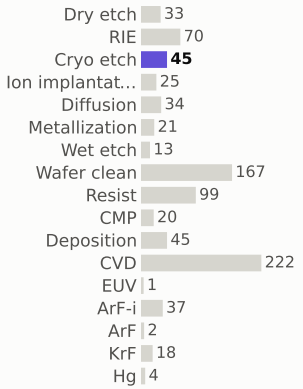
y cut at 57.351 µm



mask for this step

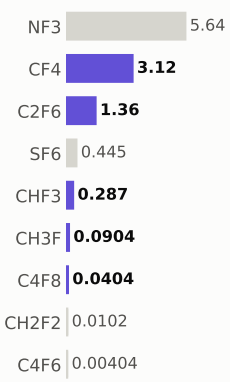
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



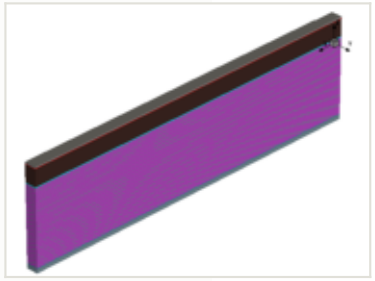
Step #10757.3077: Cryo_etch

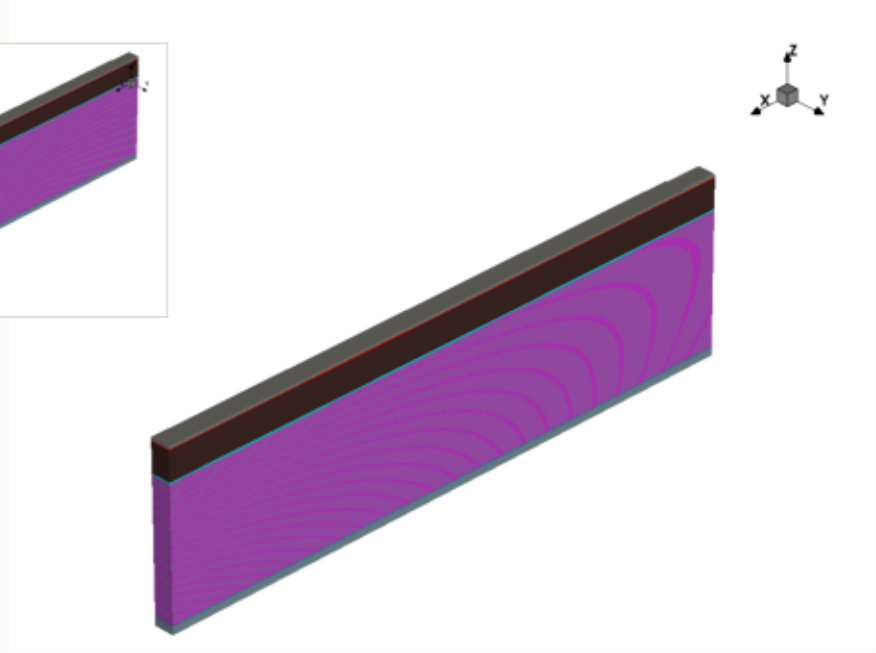
etch_noop · Cryo_etch

0.517	633.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





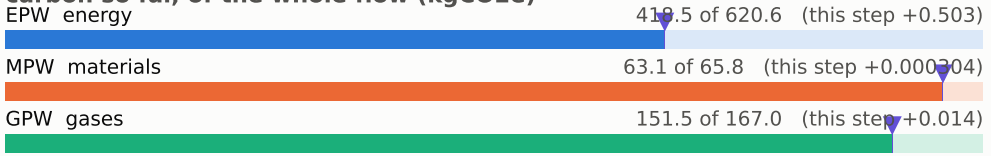
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

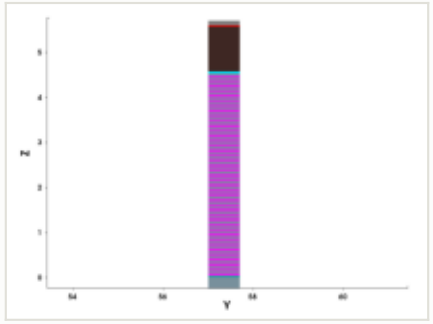
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

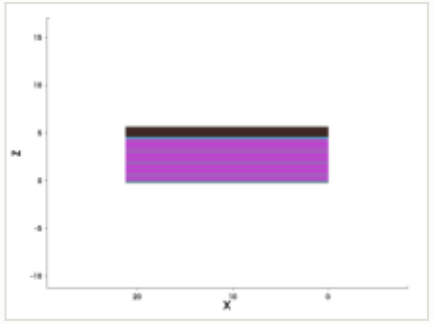


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



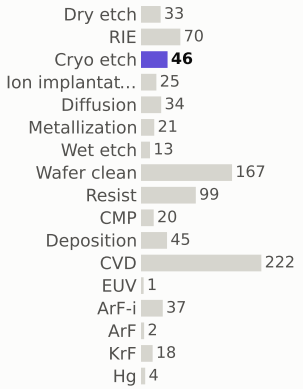
y cut at 57.351 µm



mask for this step

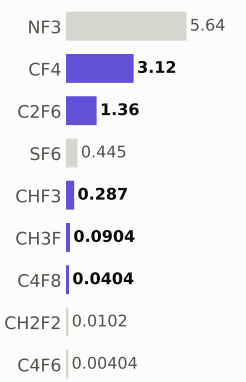
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



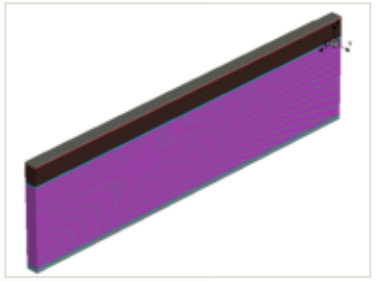
Step #10757.6923: Cryo_etch

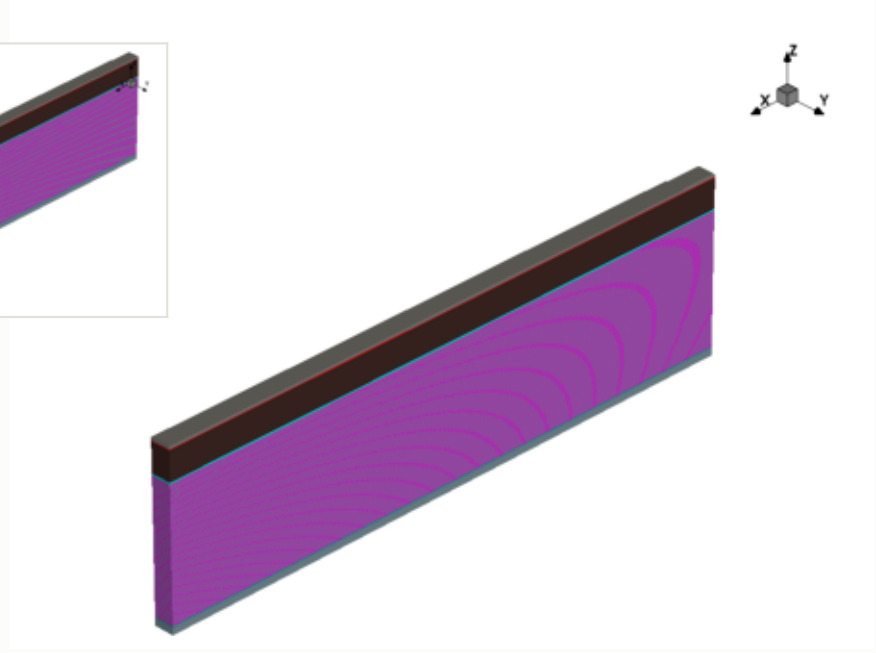
etch_noop · Cryo_etch

0.517	633.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





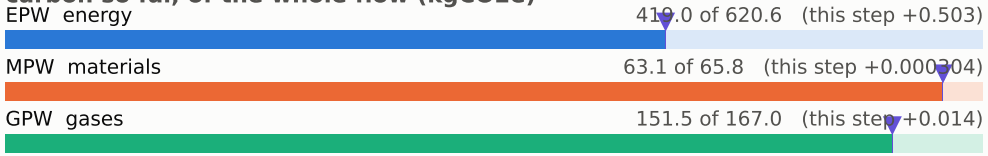
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

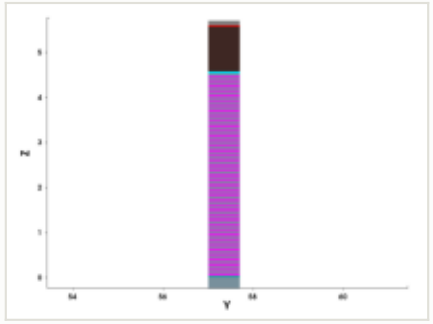
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

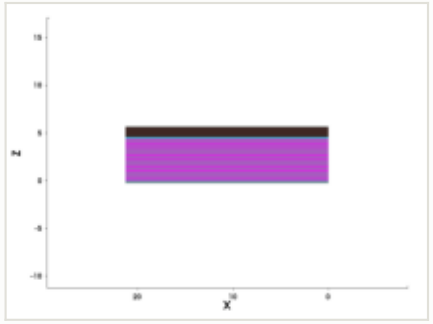


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



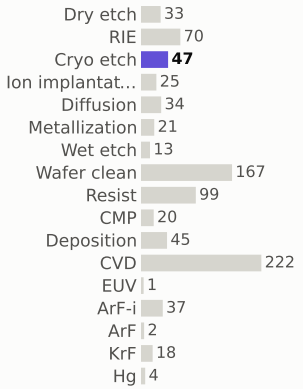
y cut at 57.351 µm



mask for this step

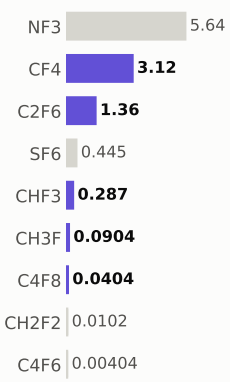
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



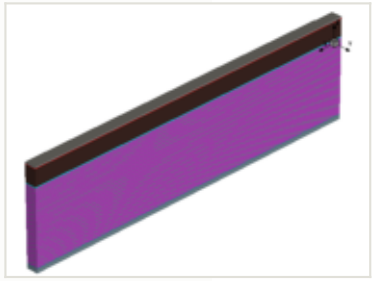
Step #10758.0769: Cryo_etch

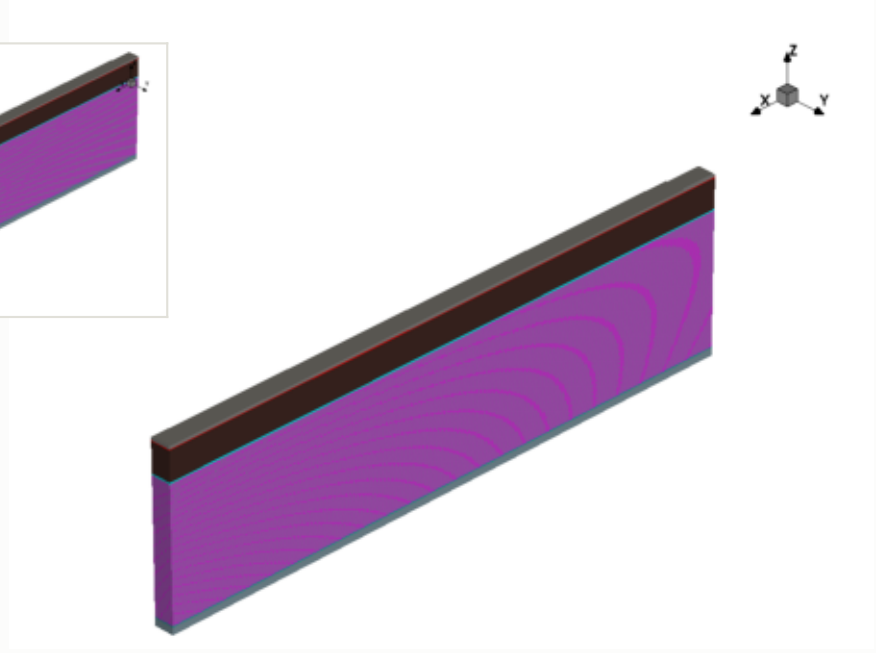
etch_noop · Cryo_etch

0.517	634.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





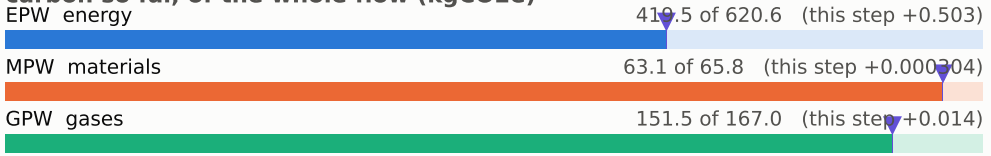
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

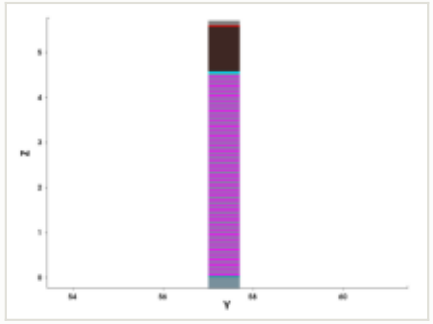
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

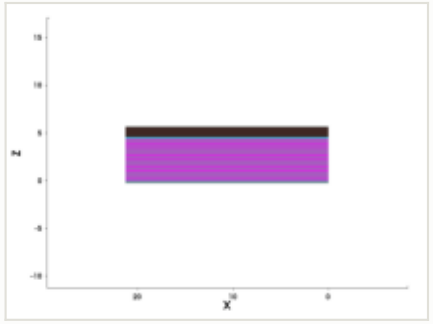


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



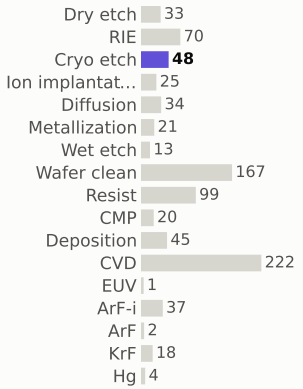
y cut at 57.351 µm



mask for this step

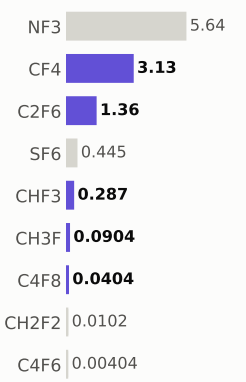
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



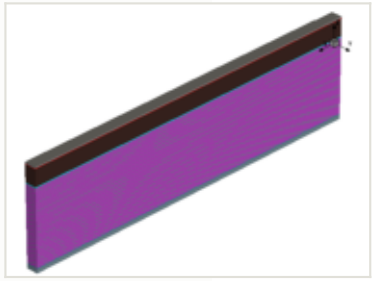
Step #10758.4615: Cryo_etch

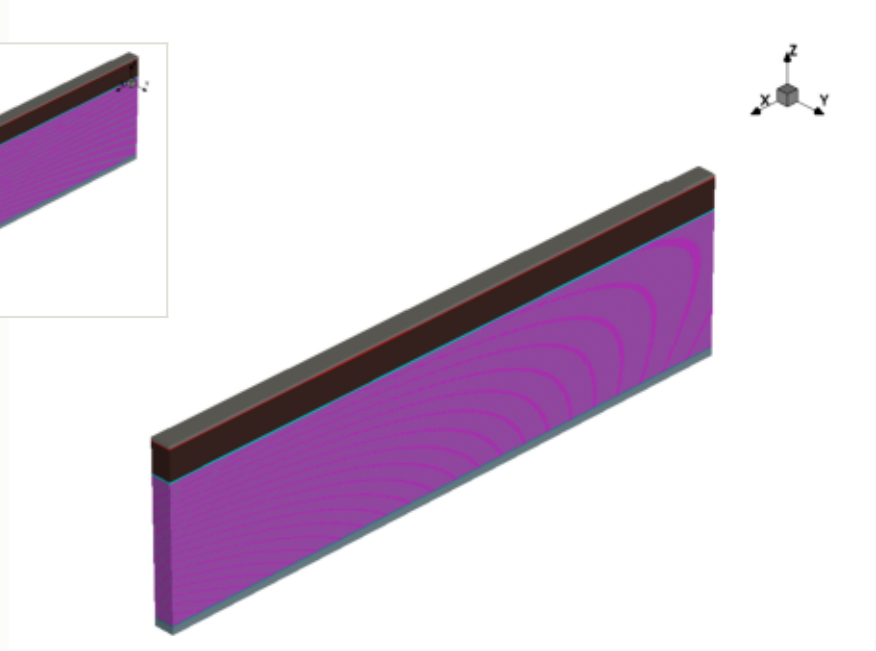
etch_noop · Cryo_etch

0.517	634.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





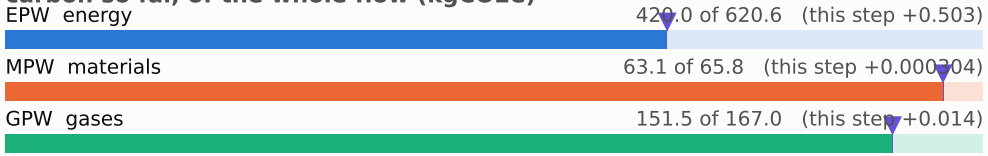
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

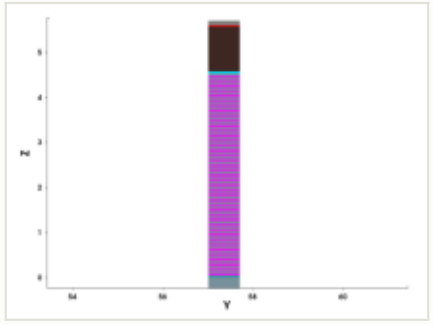
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

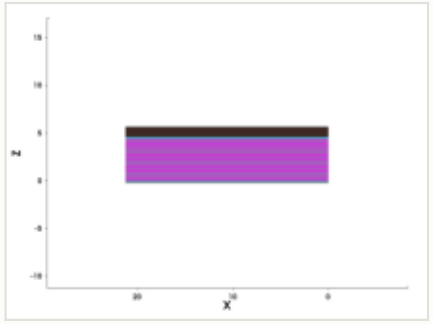


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



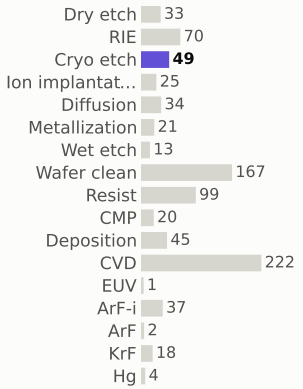
y cut at 57.351 µm



mask for this step

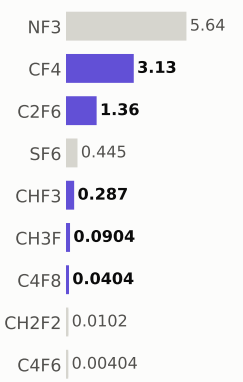
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



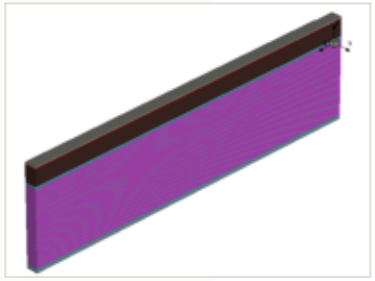
Step #10758.8462: Cryo_etch

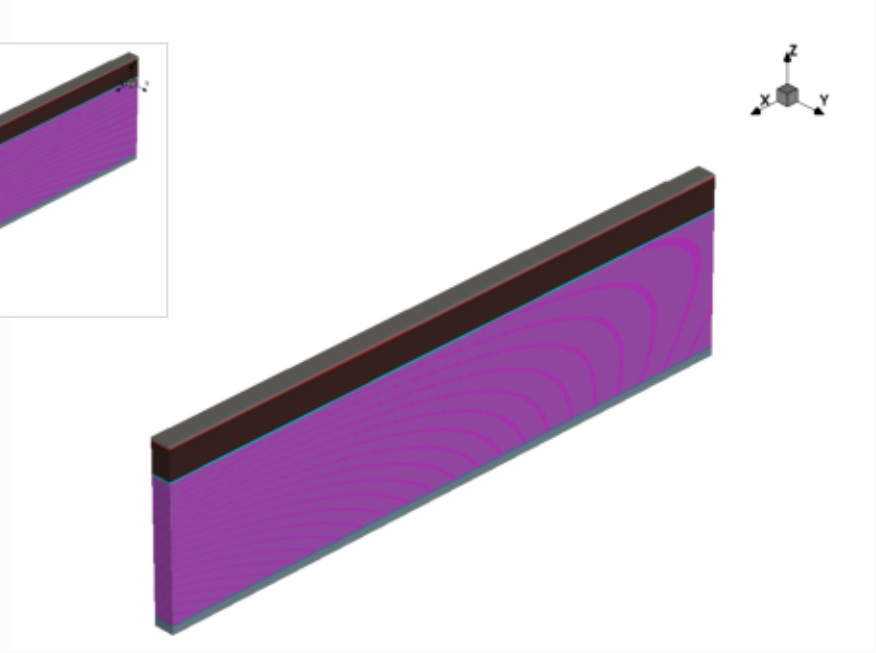
etch_noop · Cryo_etch

0.517	635.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





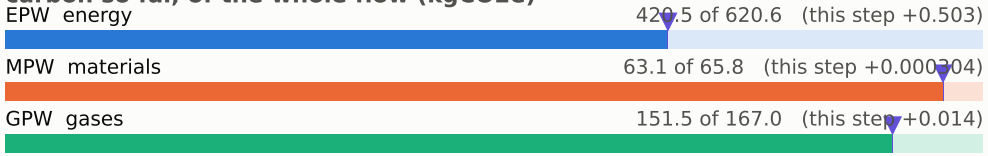
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

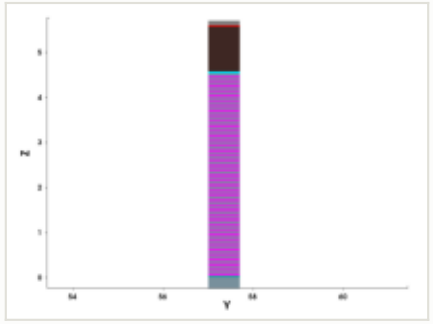
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

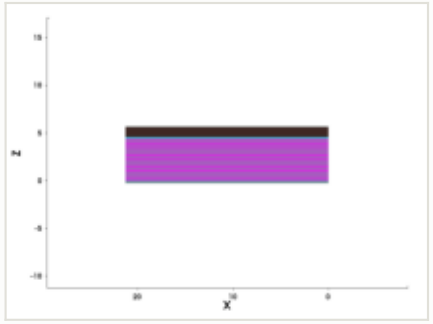


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



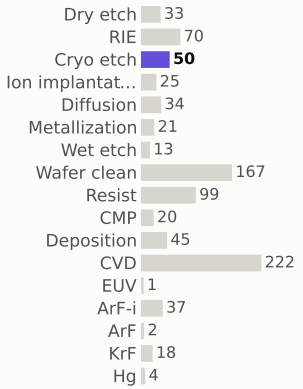
y cut at 57.351 µm



mask for this step

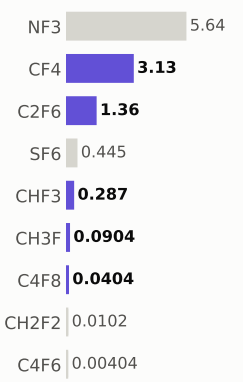
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



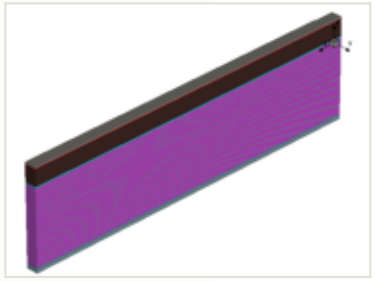
Step #10759.2308: Cryo_etch

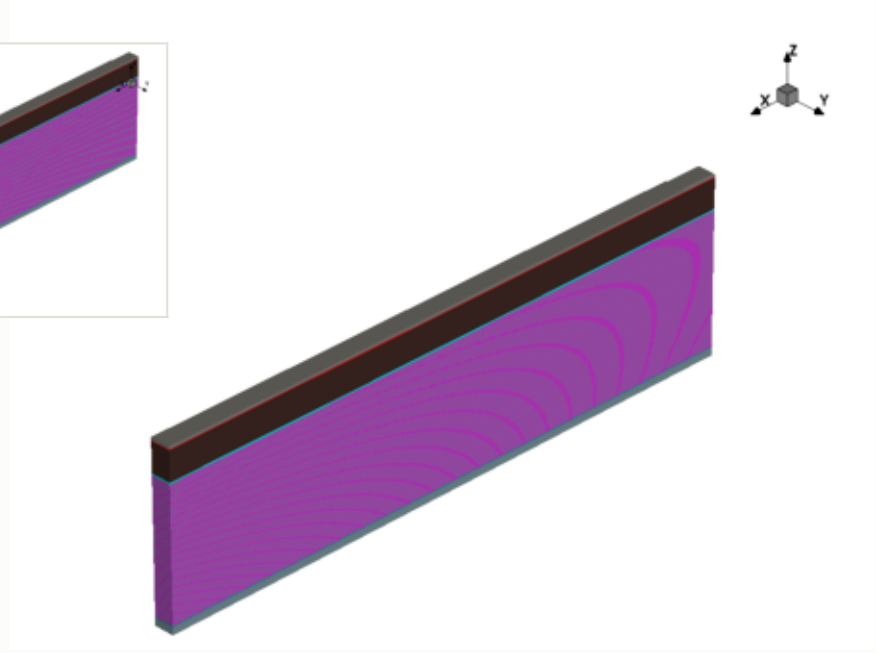
etch_noop · Cryo_etch

0.517	635.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





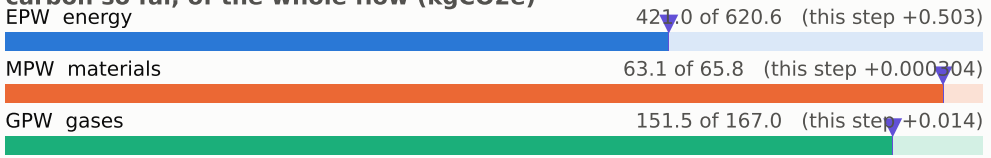
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

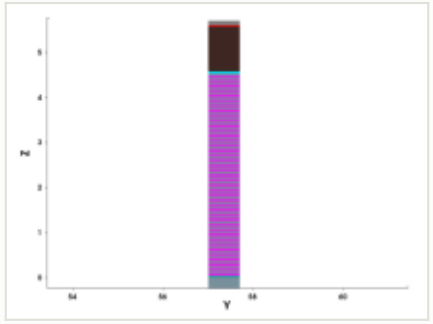
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

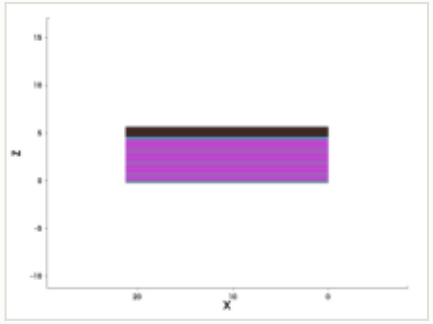


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



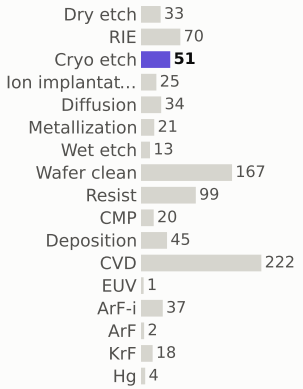
y cut at 57.351 µm



mask for this step

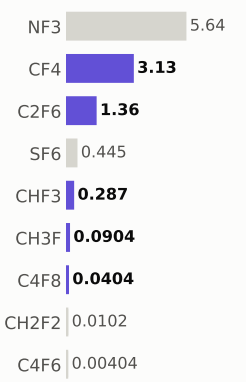
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



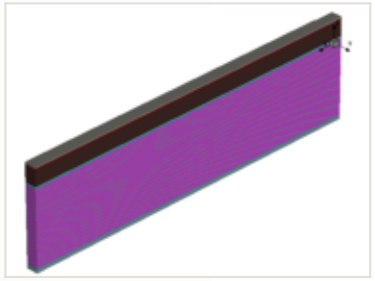
Step #10759.6154: Cryo_etch

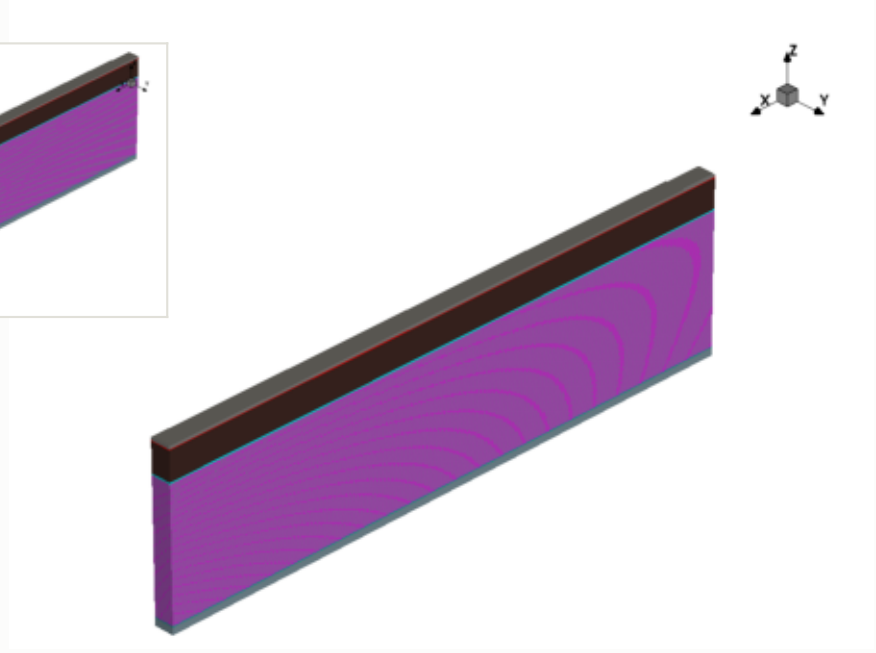
etch_noop · Cryo_etch

0.517	636.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





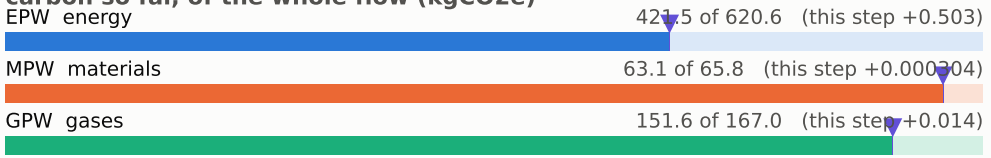
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

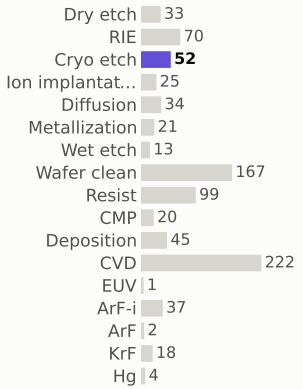
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)



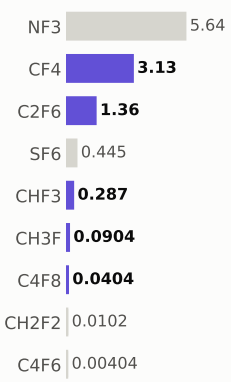
embodied carbon up to each step (kgCO2e per wafer)

steps by process area

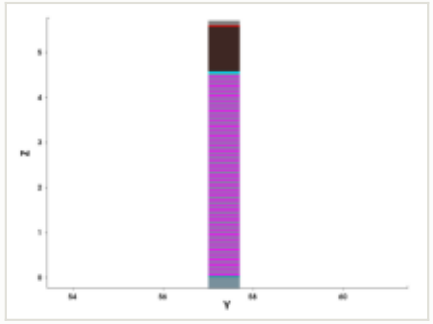


materials deposited (g)

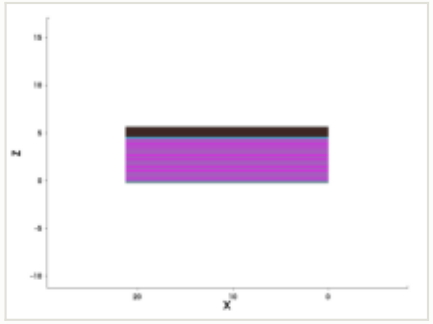
gases emitted (g)



x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

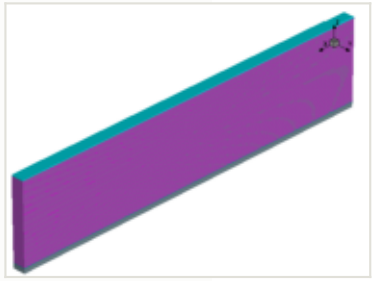
Step #10760: Dry_etch

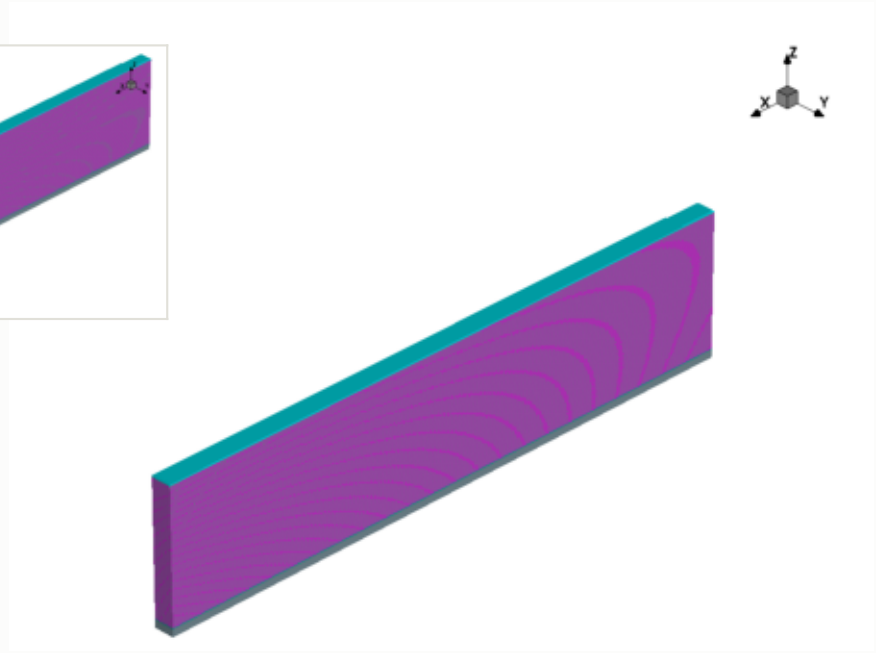
etch_strip · Dry_etch

0.801	636.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

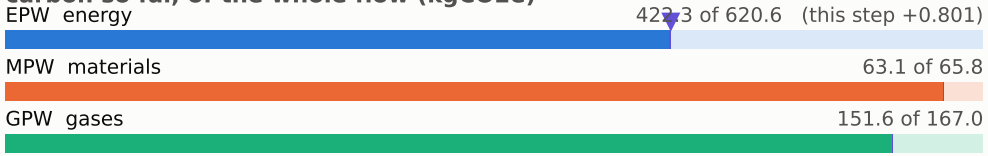




this step

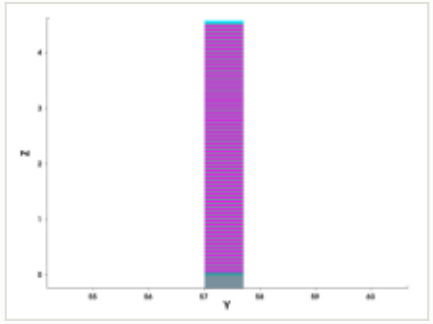
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

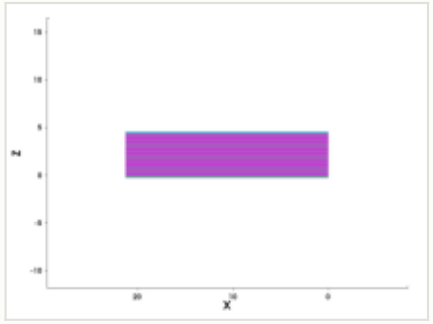


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



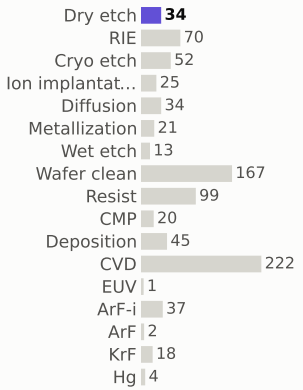
y cut at 57.351 µm



mask for this step

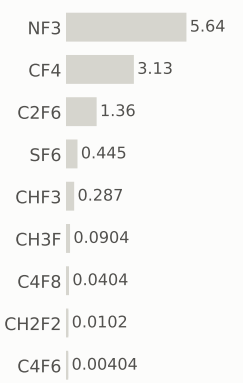
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



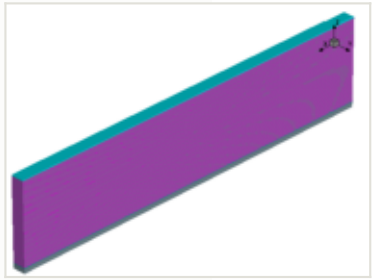
Step #10770: Wafer_clean

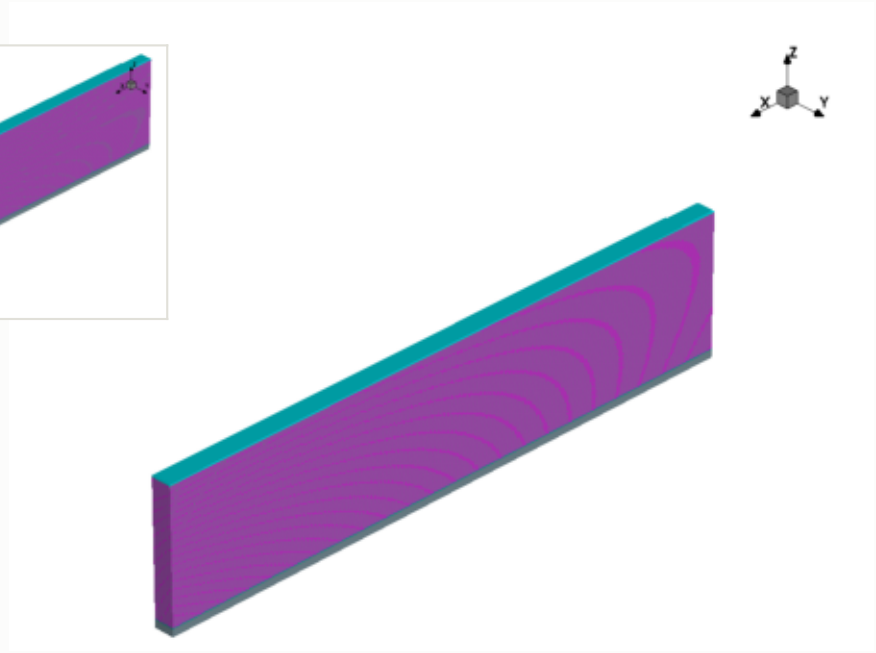
clean · Wafer_clean

0.169	637.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





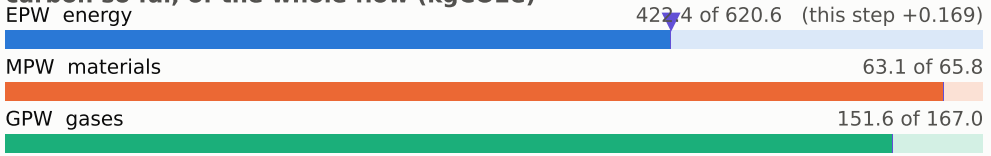
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

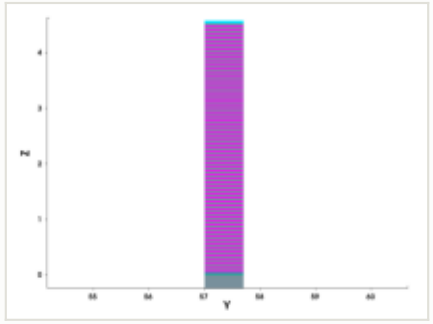
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

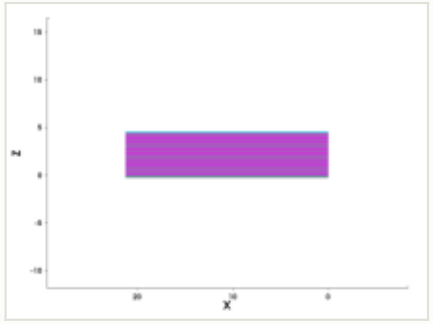


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



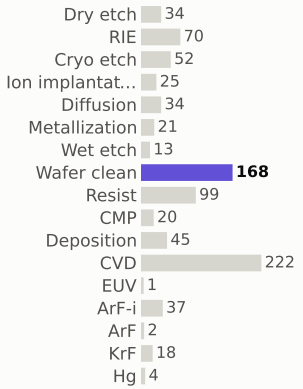
y cut at 57.351 µm



mask for this step

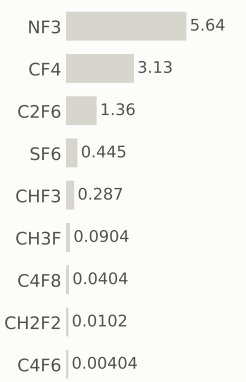
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



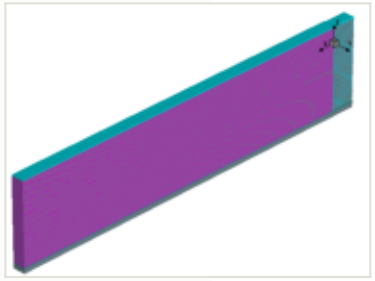
Step #10780: Wet_etch

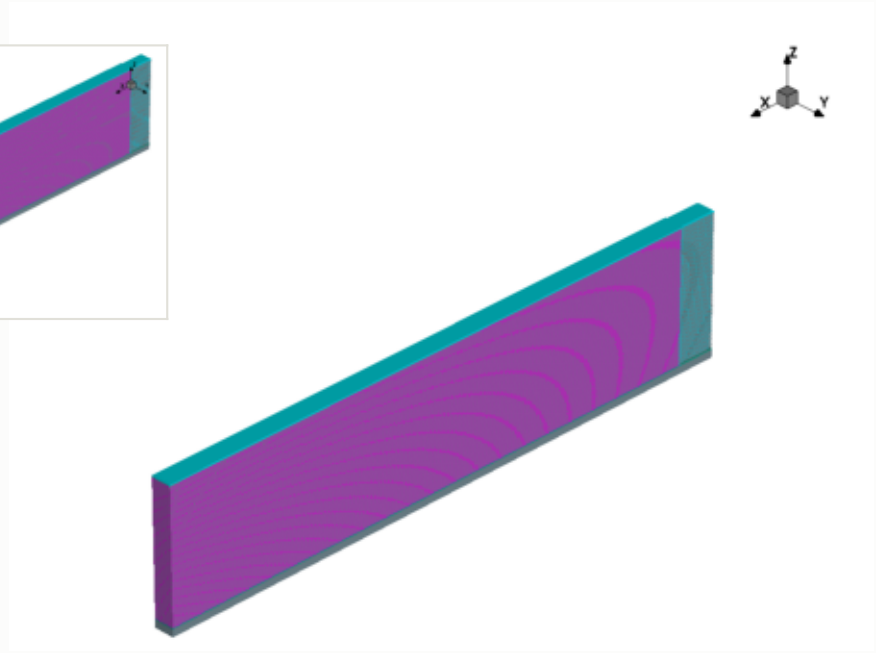
etch_isotropic · Wet_etch

0.169	637.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



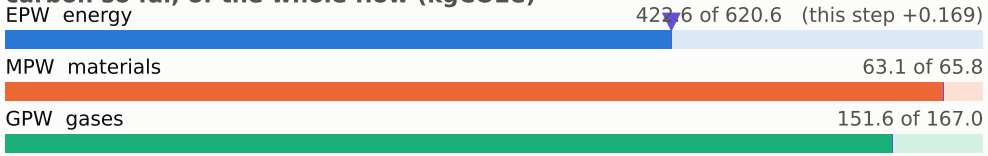


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

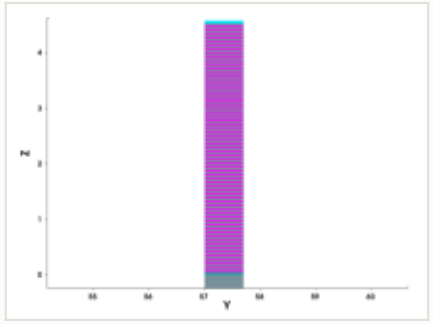
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

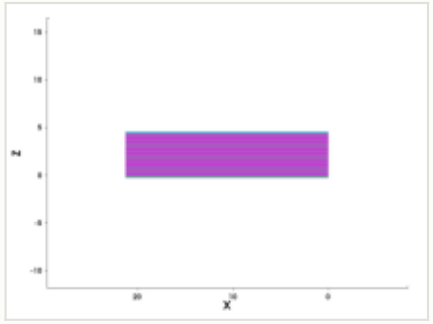


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



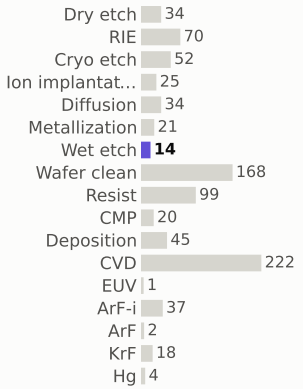
y cut at 57.351 µm



mask for this step

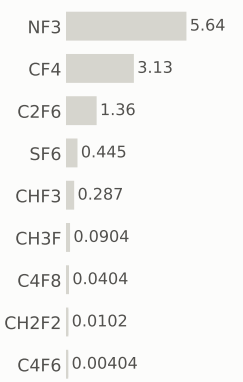
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



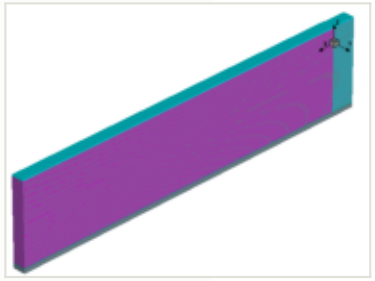
Step #10790: Deposition

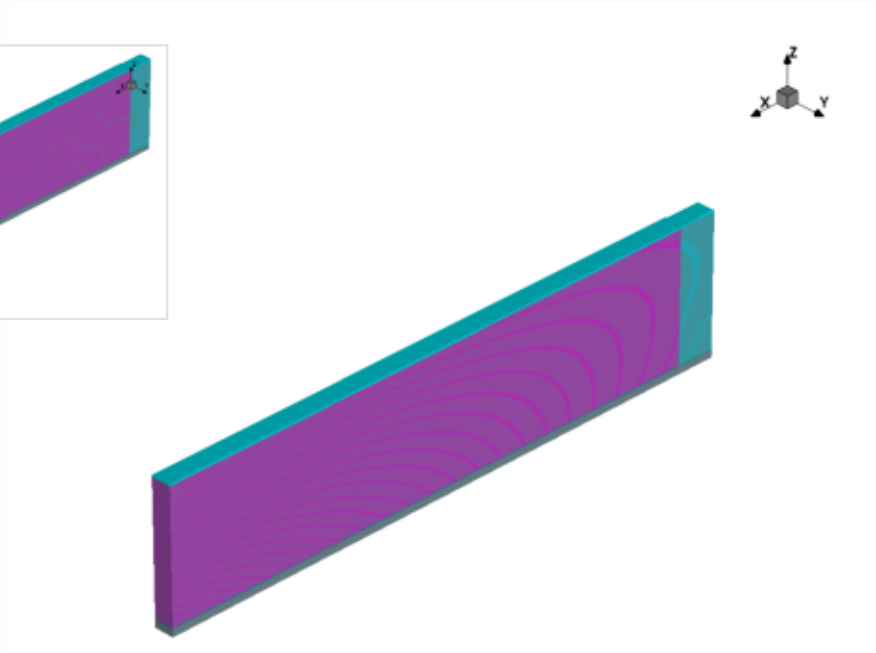
depo_isotropic · Deposition

0.656	637.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



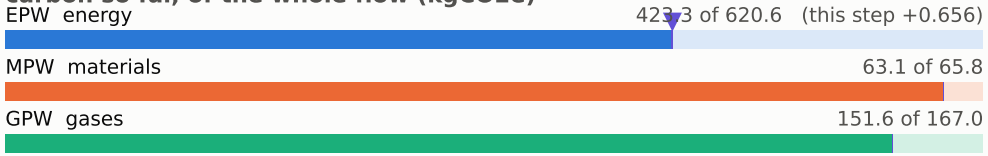


this step

process area	Deposition
material	—
thickness	0.015 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

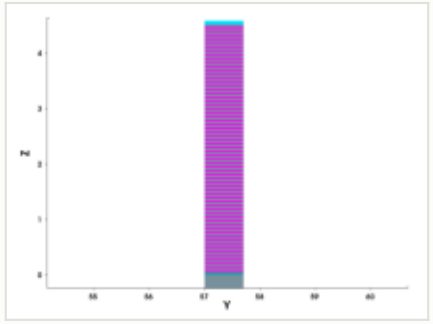
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

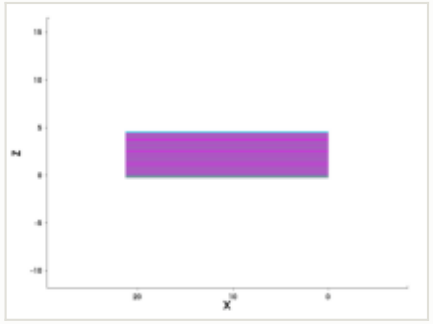


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



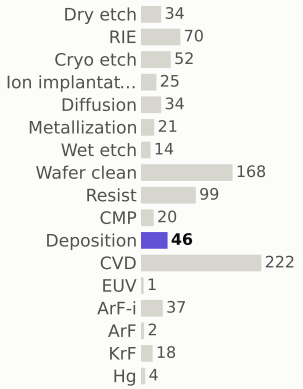
y cut at 57.351 µm



mask for this step

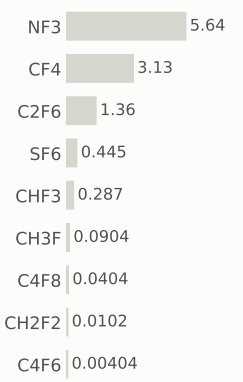
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



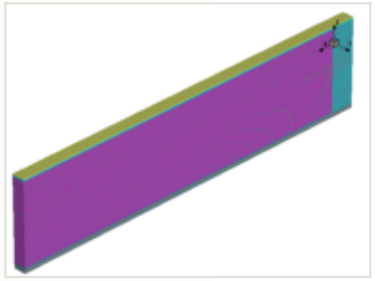
Step #10800: CVD

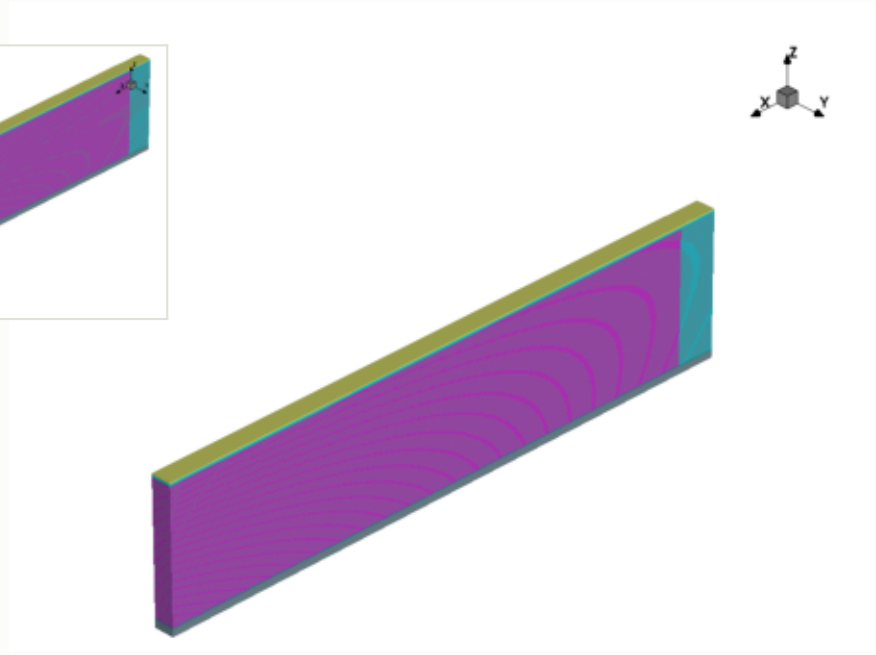
depo_isotropic · CVD

1.31	639.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



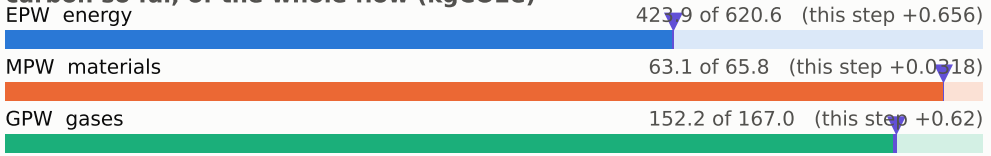


this step

process area	CVD
material	—
thickness	0.05 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

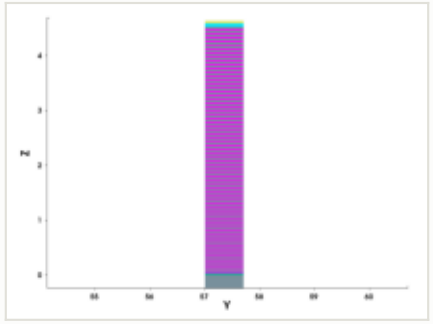
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

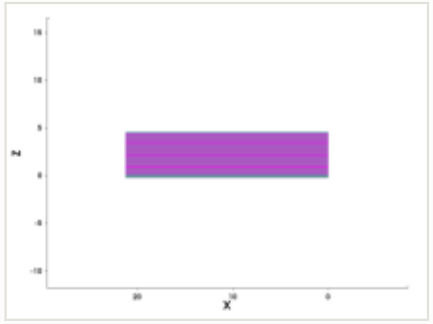


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



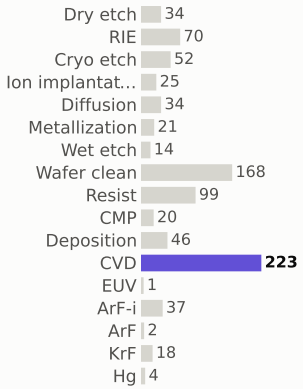
y cut at 57.351 µm



mask for this step

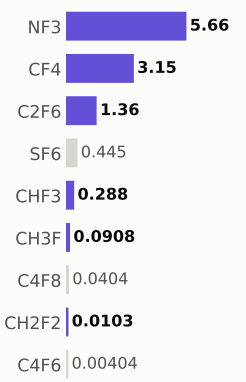
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



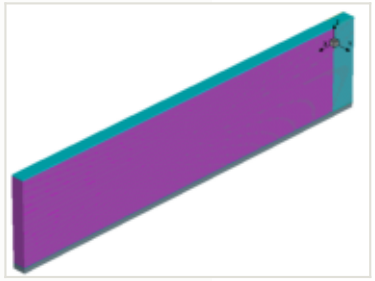
Step #10810: CMP

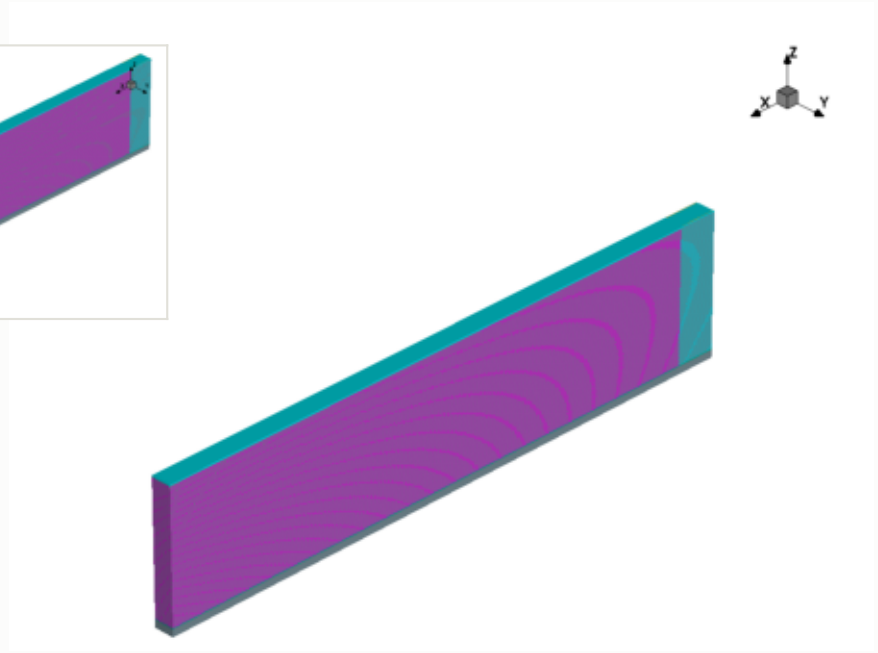
etch_cmp · CMP

0.169	639.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



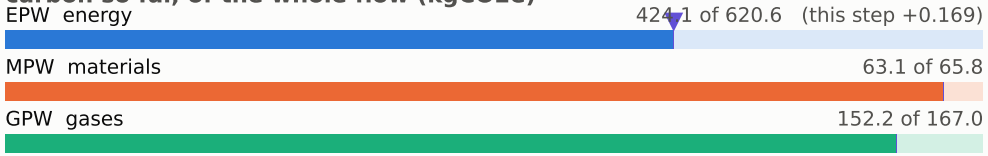


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

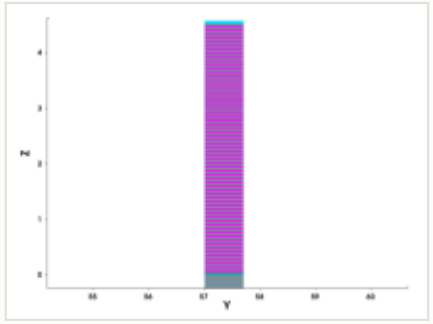
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

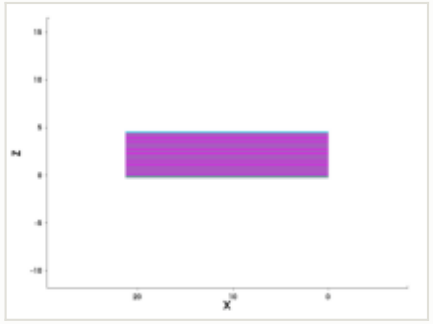


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



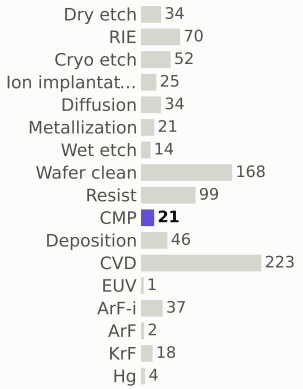
y cut at 57.351 µm



mask for this step

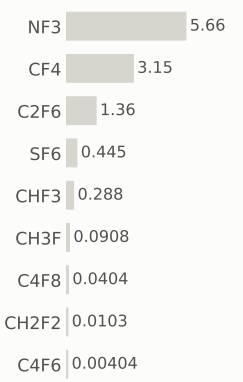
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



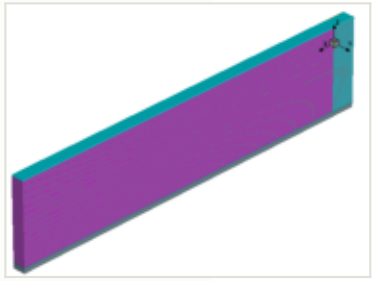
Step #10820: Wafer_clean

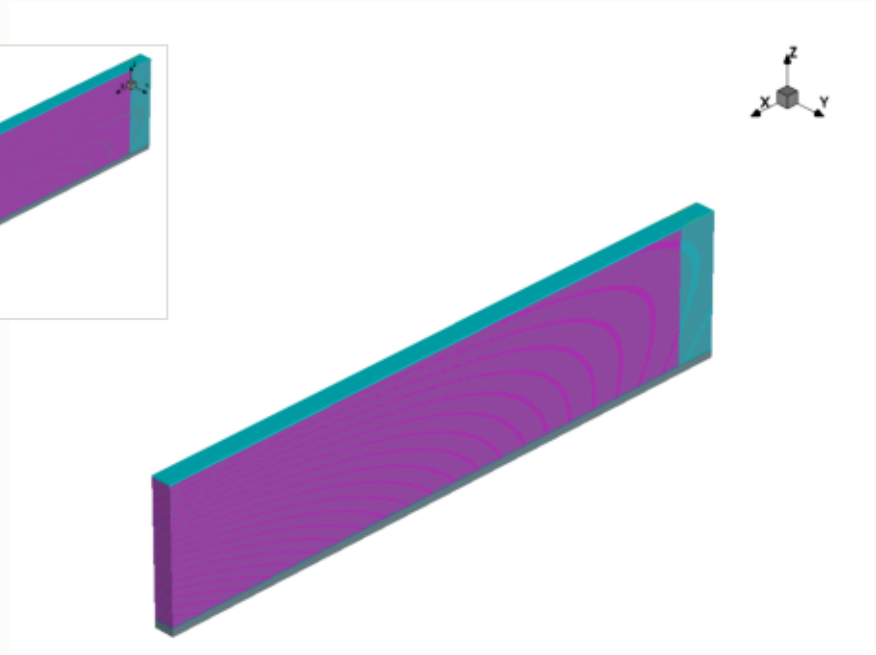
clean · Wafer_clean

0.169	639.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





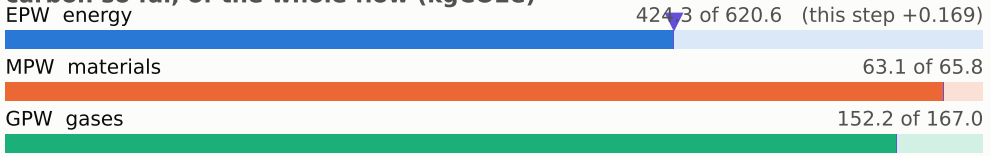
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

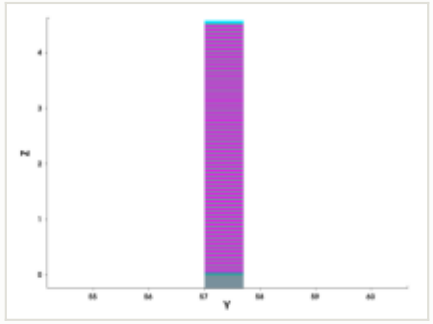
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

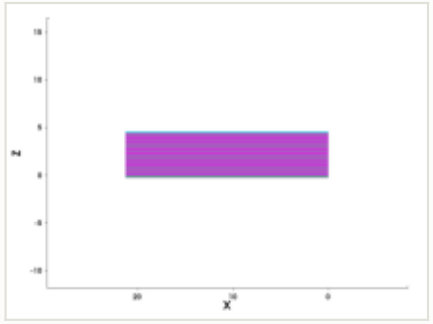


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



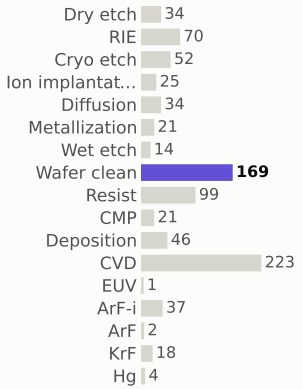
y cut at 57.351 µm



mask for this step

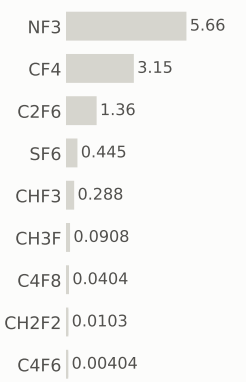
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



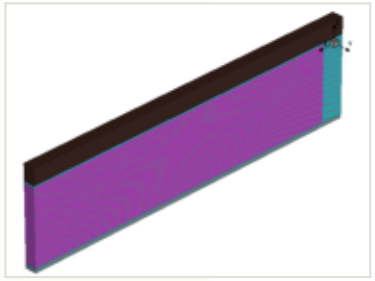
Step #10830: Deposition

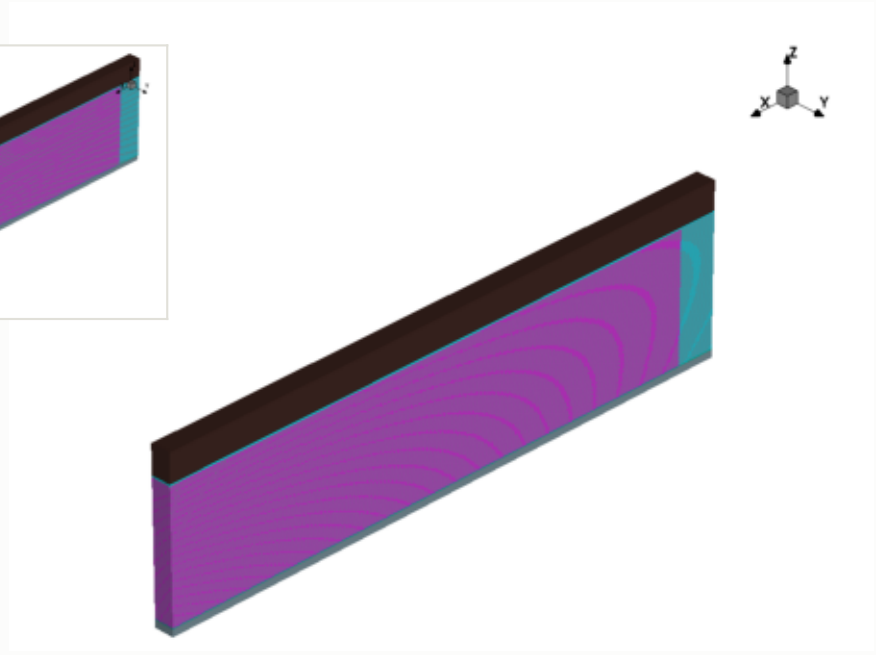
depo_isotropic · Deposition

0.656	640.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



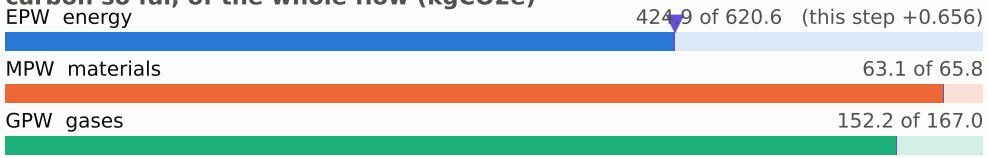


this step

process area	Deposition
material	—
thickness	1 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

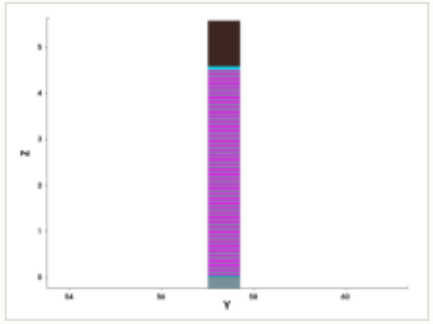
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

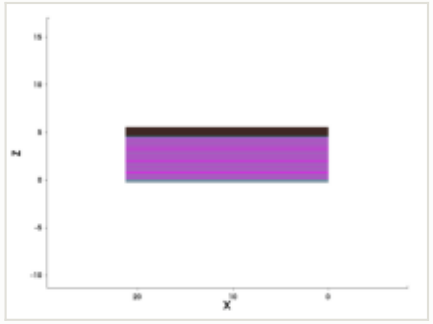


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



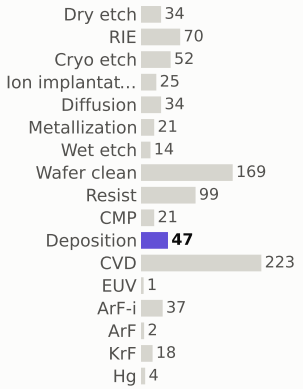
y cut at 57.351 µm



mask for this step

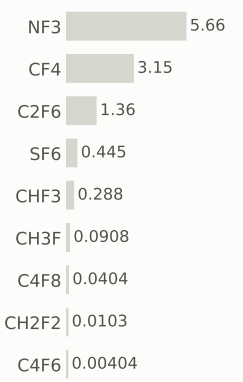
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



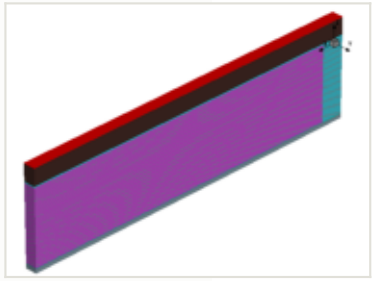
Step #10840: Deposition

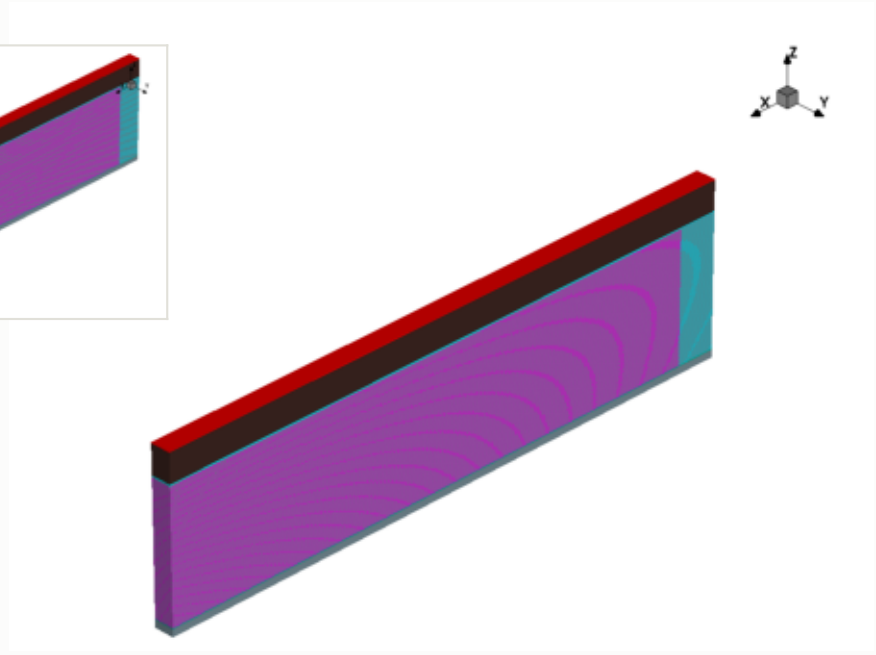
depo_isotropic · Deposition

0.656	640.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



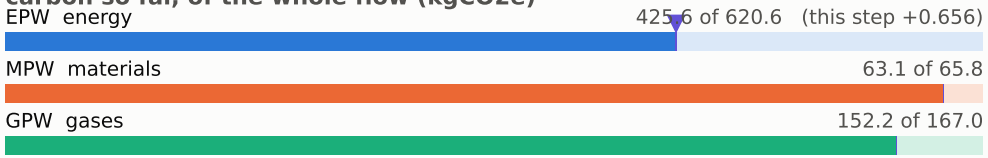


this step

process area	Deposition
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

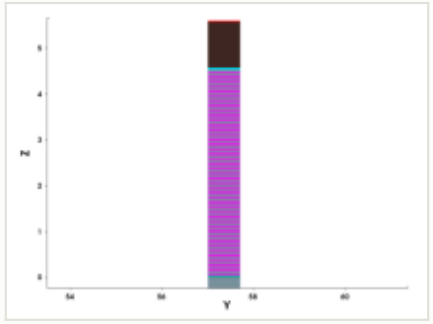
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

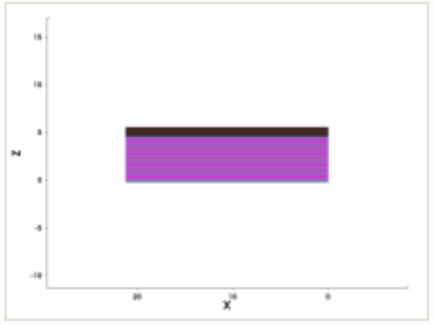


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



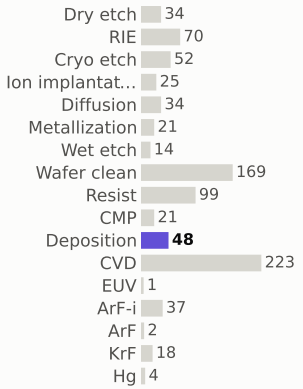
y cut at 57.351 µm



mask for this step

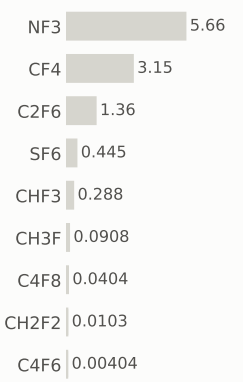
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



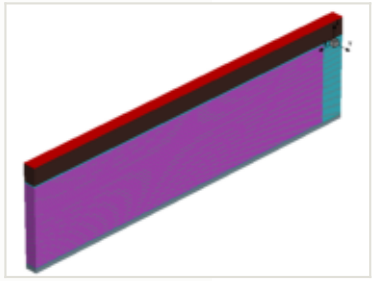
Step #10850: Wafer_clean

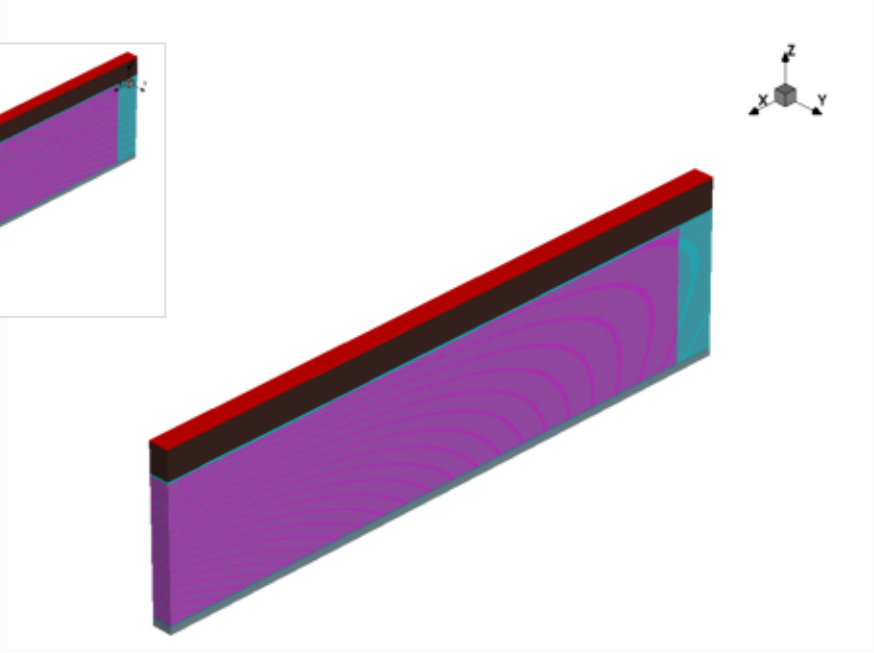
clean · Wafer_clean

0.169	641.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





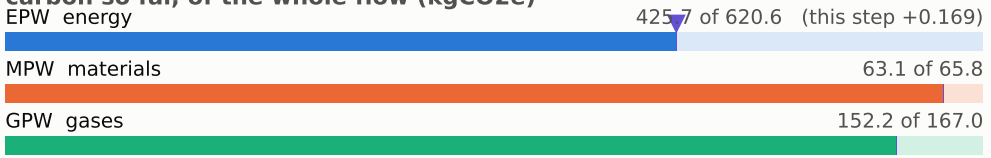
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

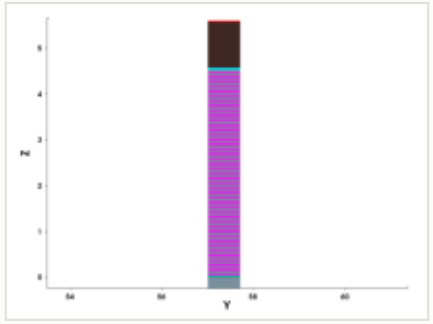
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

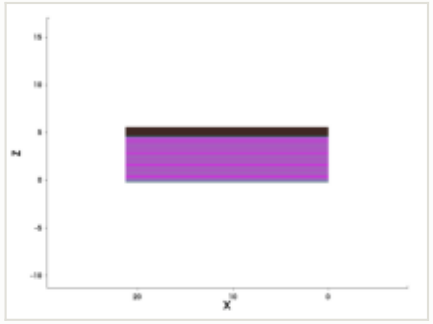


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



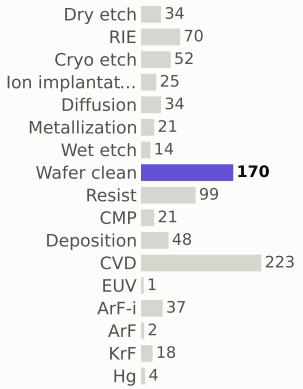
y cut at 57.351 µm



mask for this step

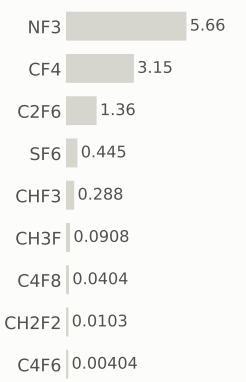
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



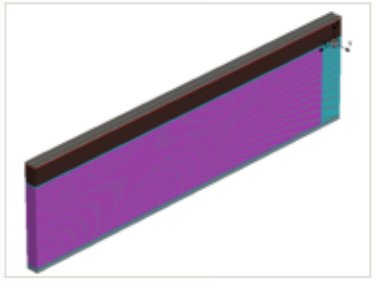
Step #10860: ArF-i

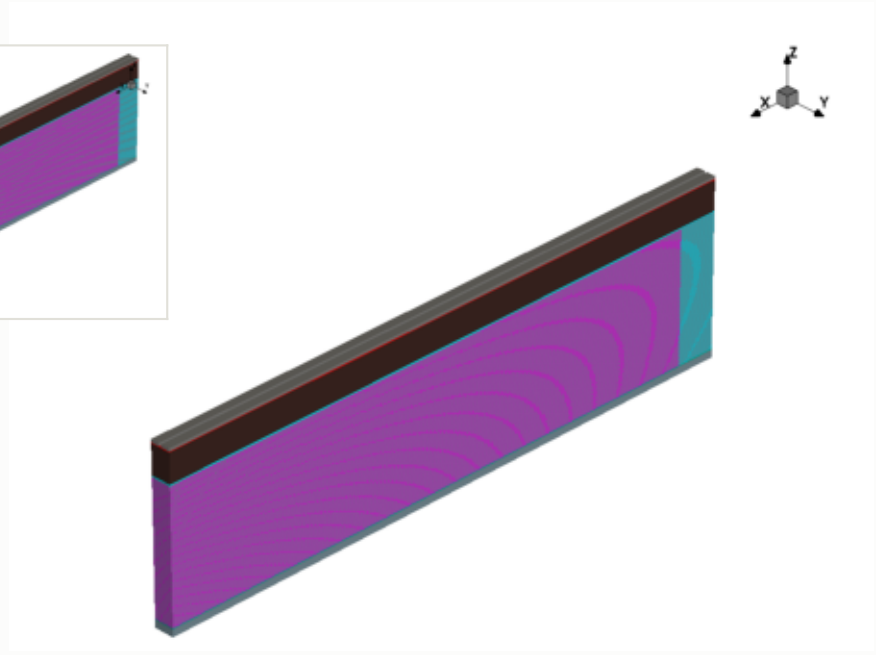
litho_patterning · ArF-i

0.217	641.3	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



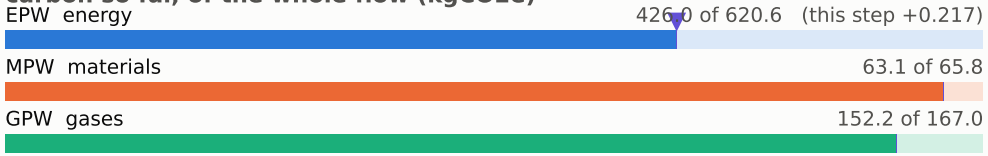


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

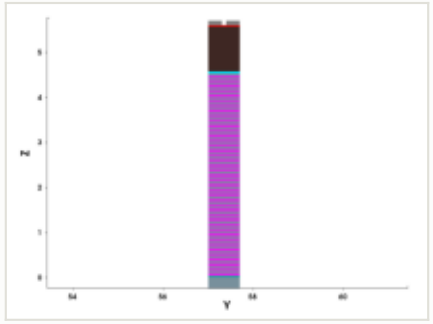
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

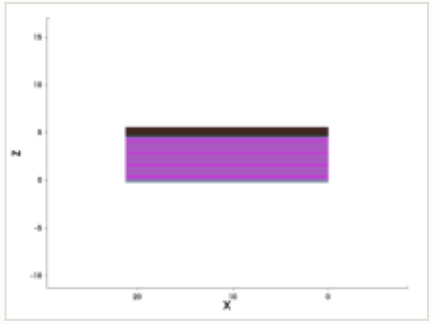


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

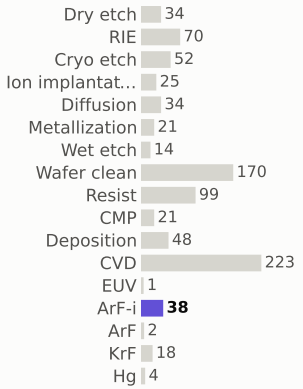


y cut at 57.351 µm



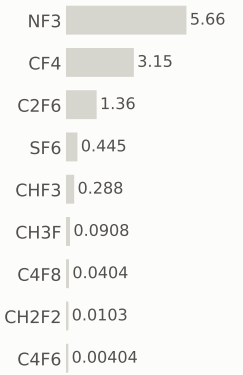
mask: layer 3:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



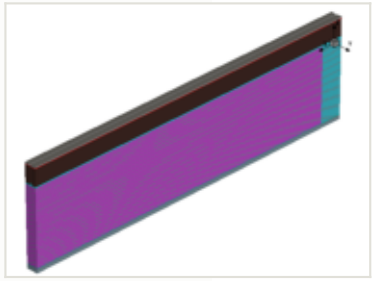
Step #10870: RIE

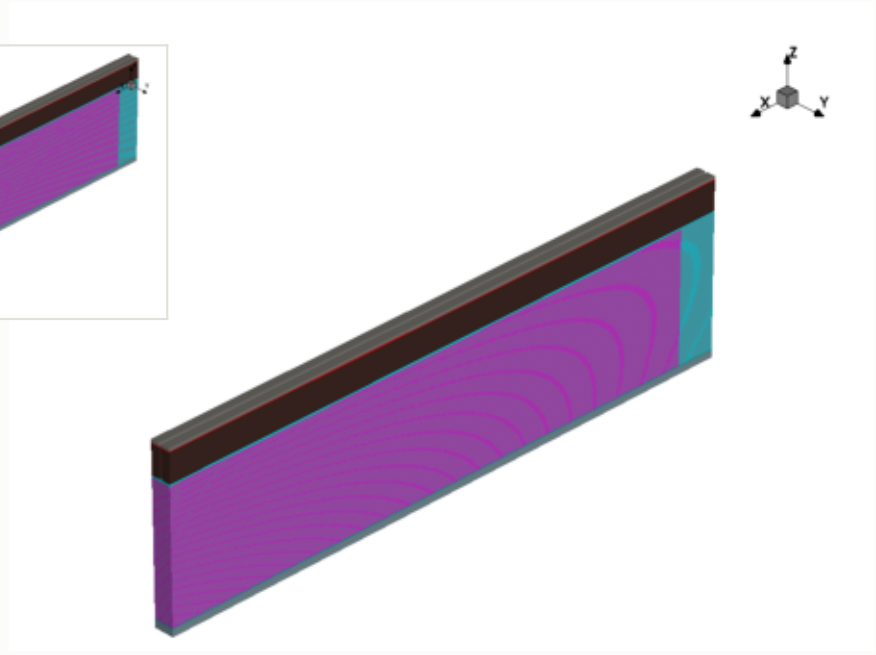
etch_anisotropic · RIE

1.04	642.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



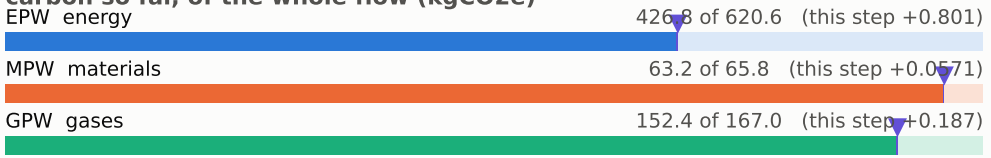


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

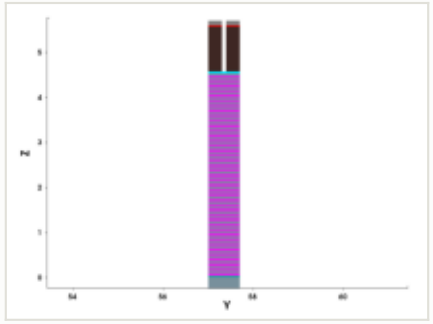
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

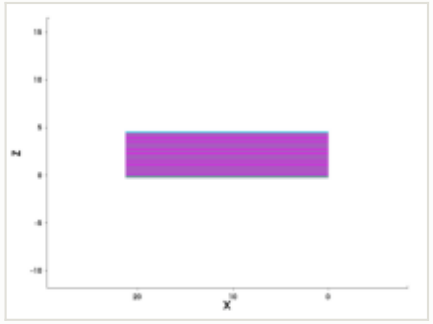


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



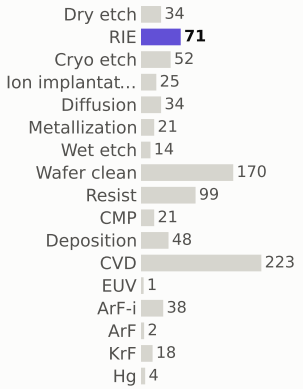
y cut at 57.351 µm



mask for this step

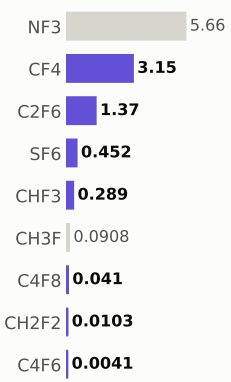
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



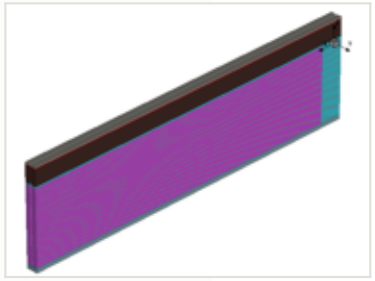
Step #10880: Cryo_etch

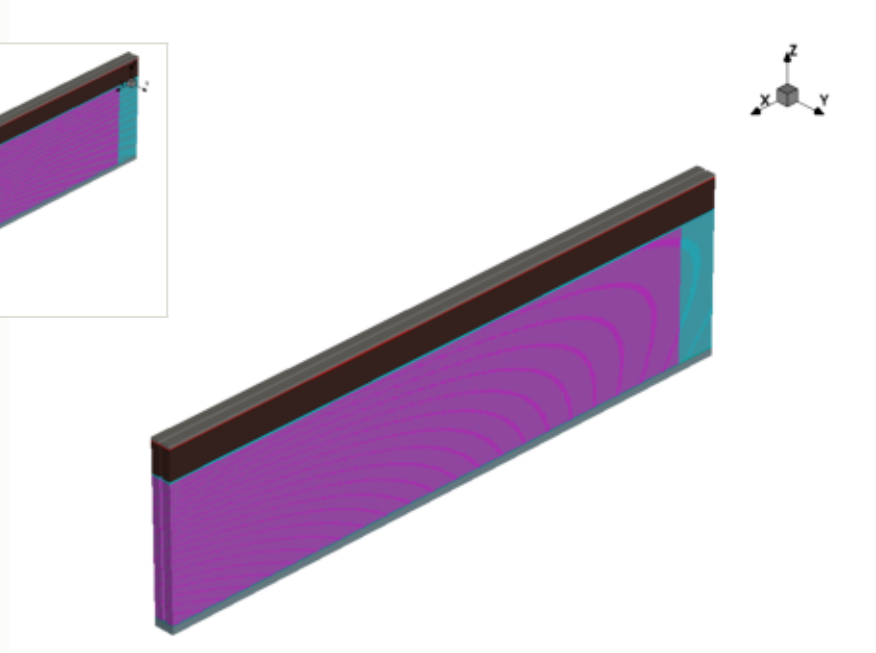
etch_anisotropic · Cryo_etch

0.517	642.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



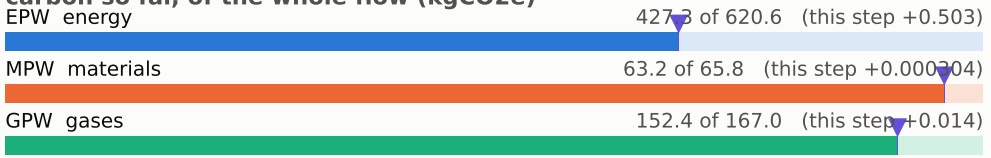


this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

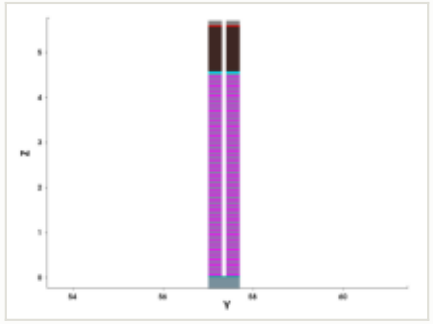
EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

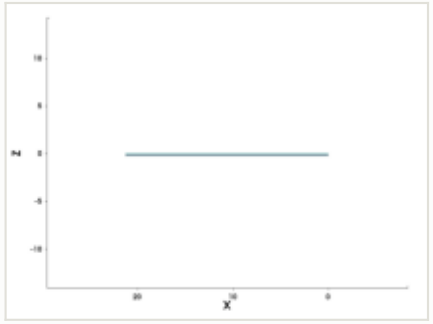


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



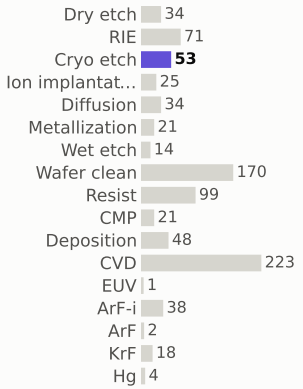
y cut at 57.351 µm



mask for this step

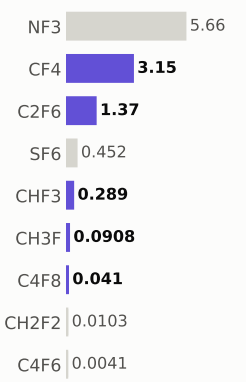
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



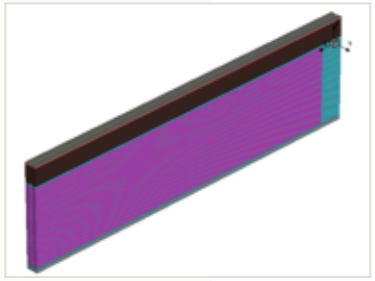
Step #10880.3846: Cryo_etch

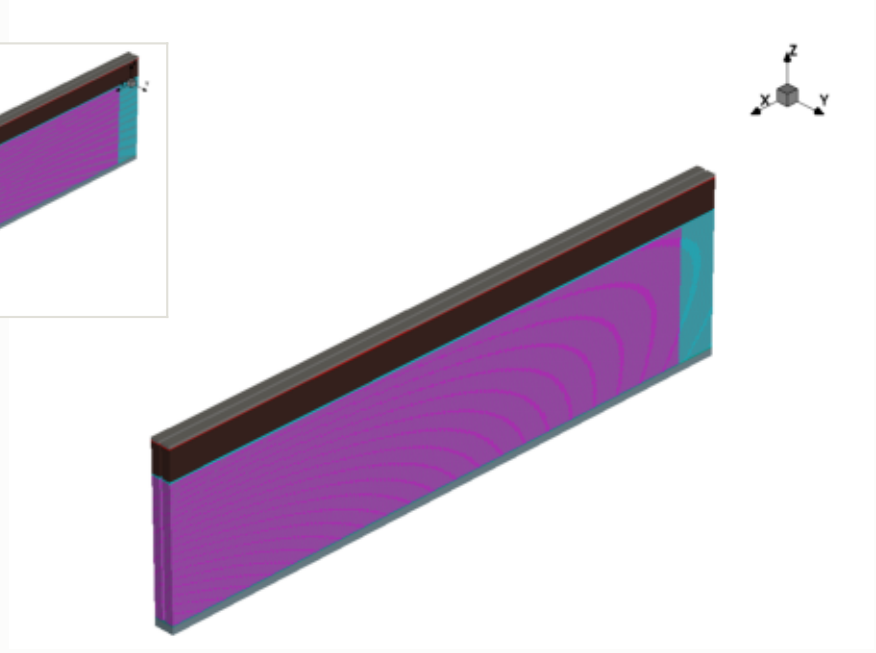
etch_noop · Cryo_etch

0.517	643.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





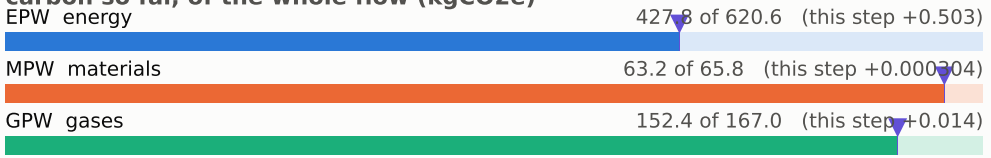
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

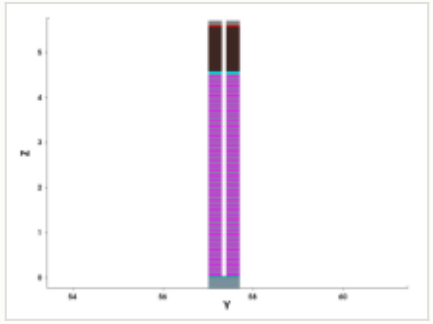
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

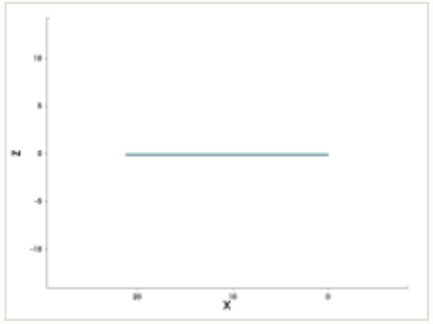


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



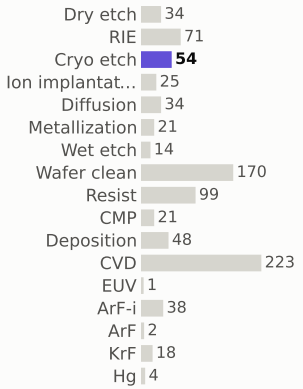
y cut at 57.351 µm



mask for this step

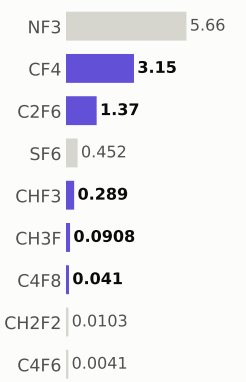
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



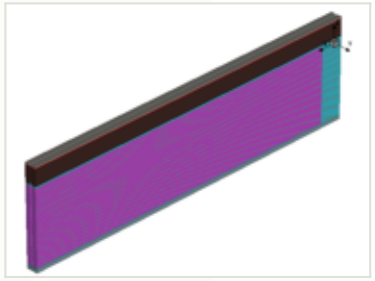
Step #10880.7692: Cryo_etch

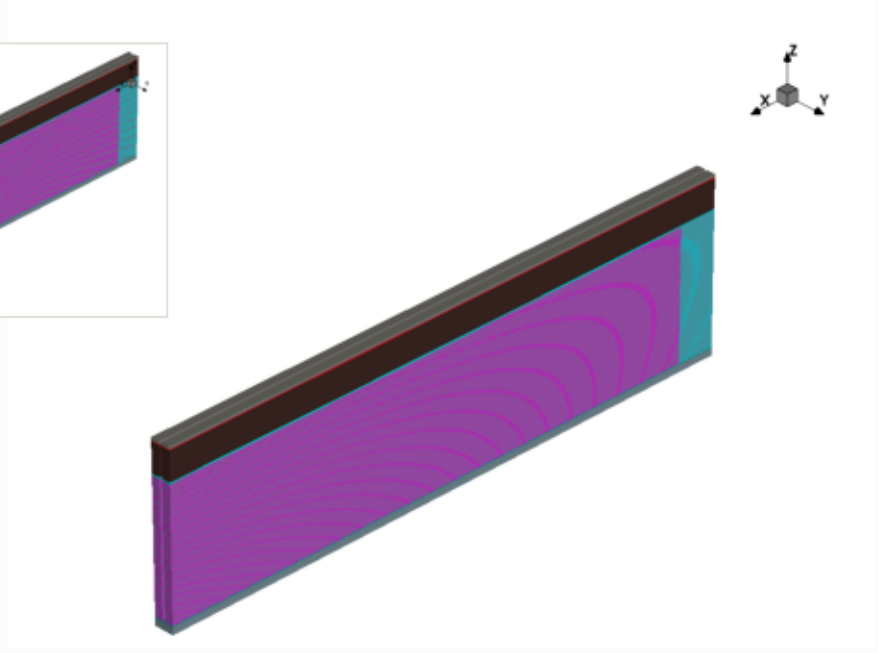
etch_noop · Cryo_etch

0.517	643.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





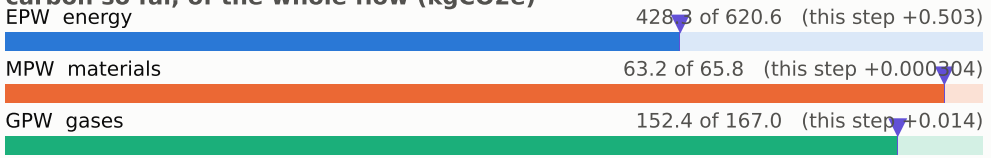
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

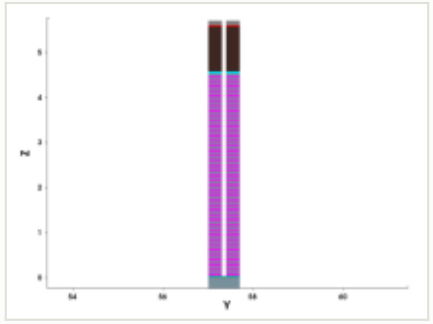
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

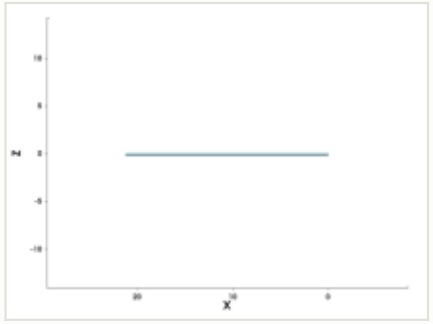


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



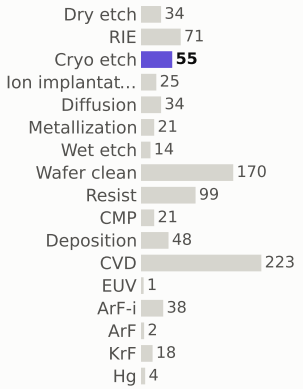
y cut at 57.351 µm



mask for this step

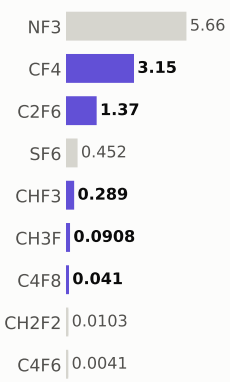
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



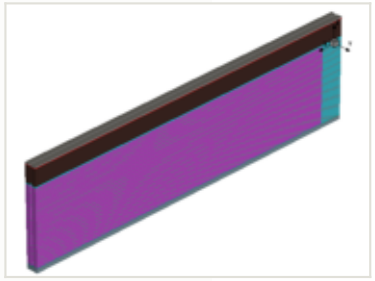
Step #10881.1538: Cryo_etch

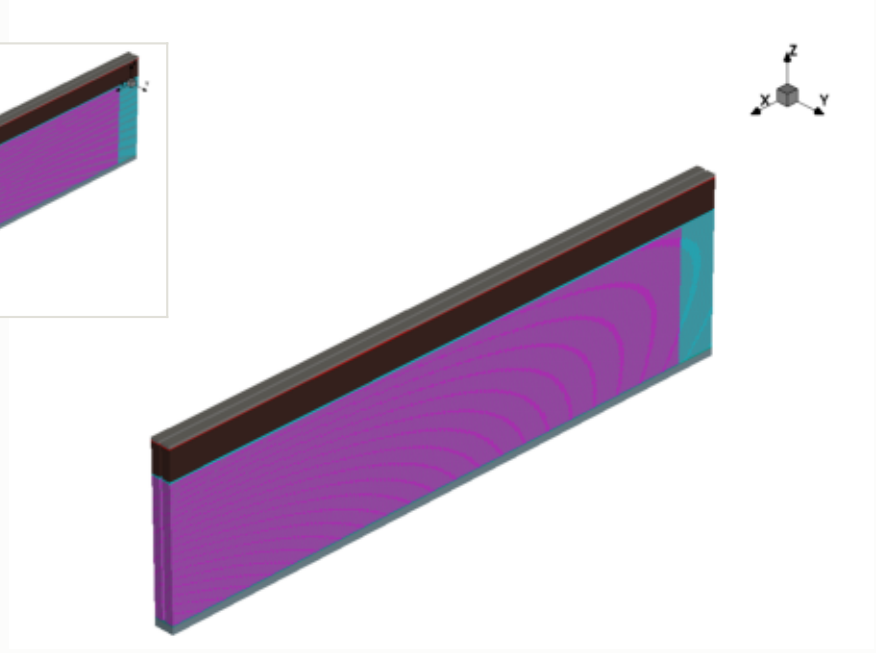
etch_noop · Cryo_etch

0.517	644.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





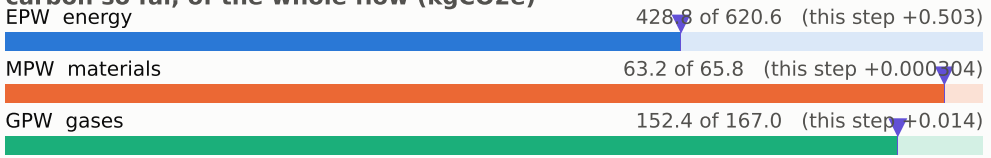
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

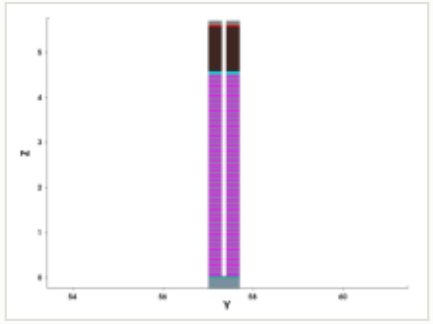
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

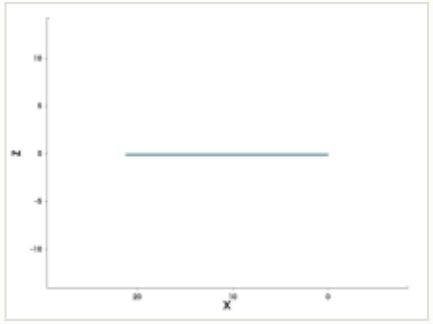


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



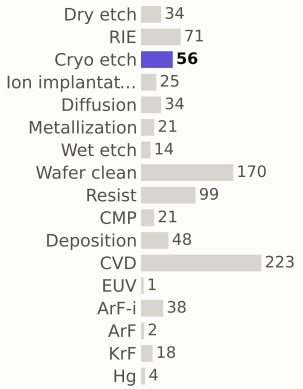
y cut at 57.351 µm



mask for this step

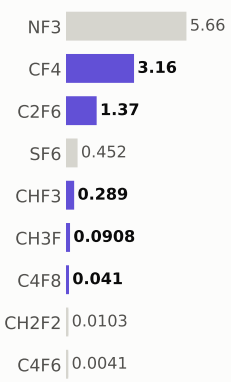
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



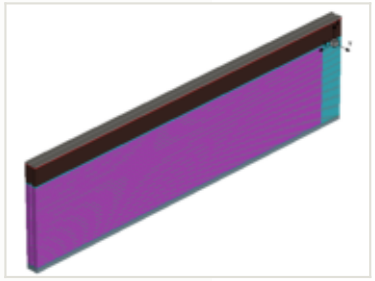
Step #10881.5385: Cryo_etch

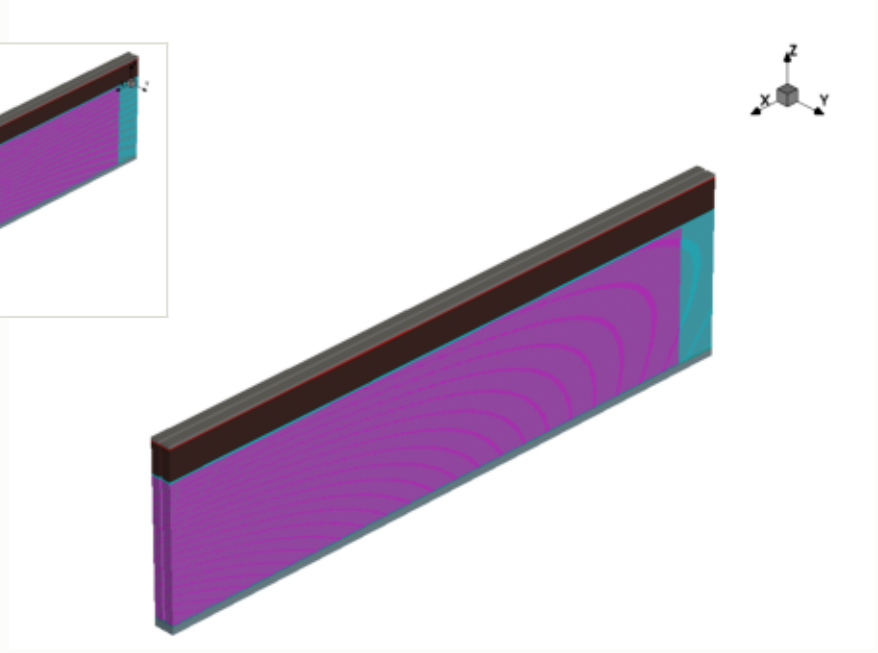
etch_noop · Cryo_etch

0.517	644.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





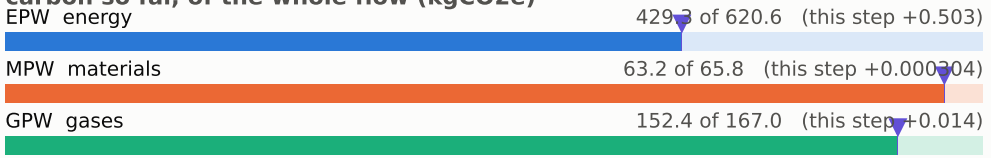
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

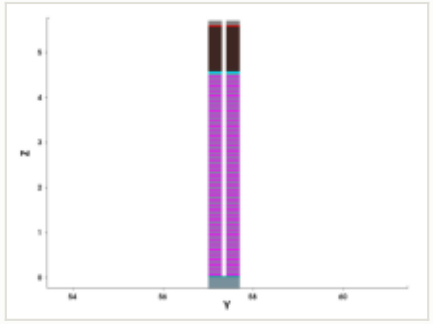
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

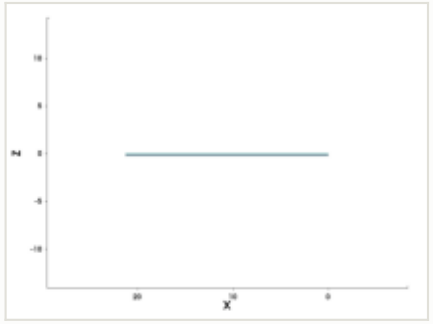


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



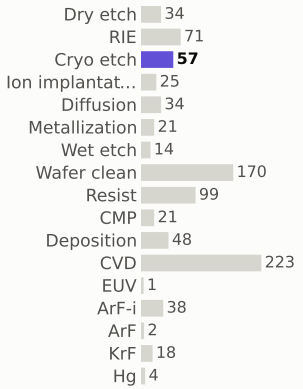
y cut at 57.351 µm



mask for this step

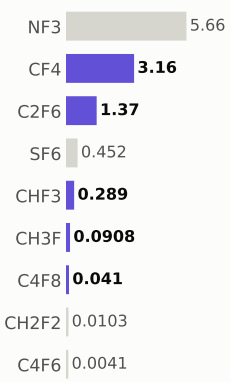
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



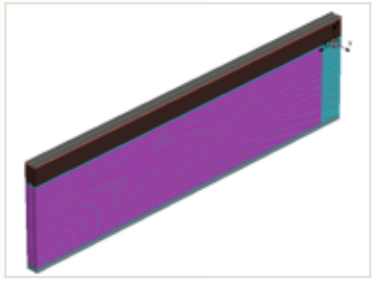
Step #10881.9231: Cryo_etch

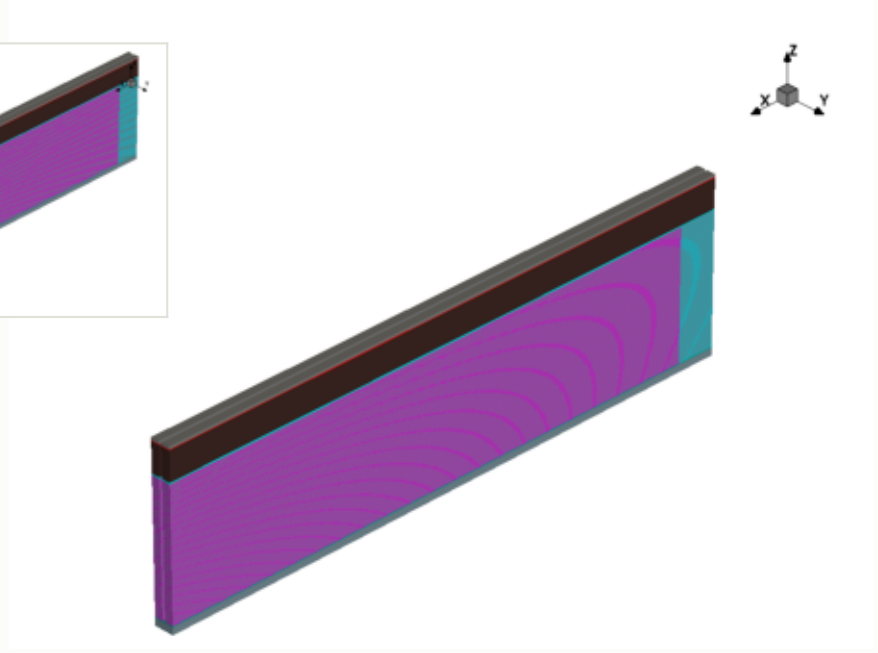
etch_noop · Cryo_etch

0.517	645.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





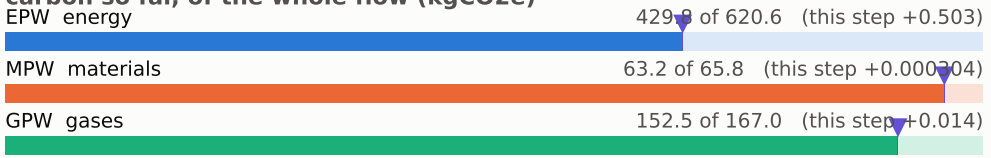
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

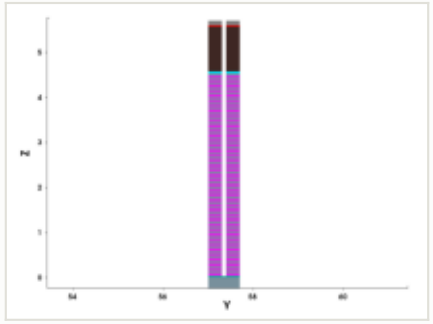
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

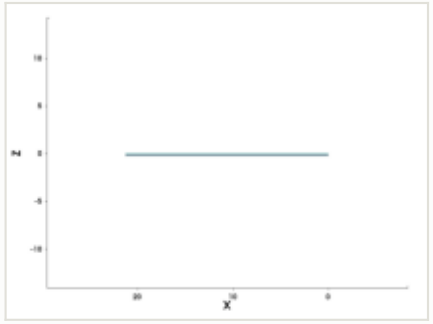


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



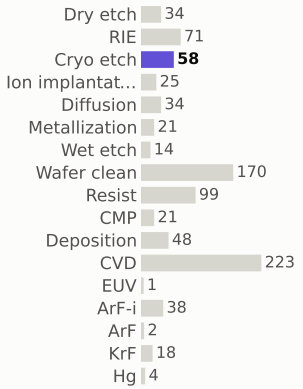
y cut at 57.351 µm



mask for this step

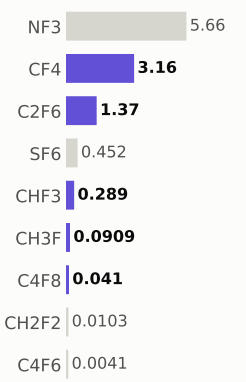
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



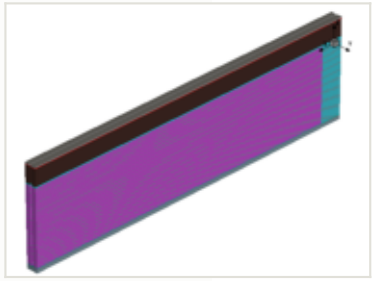
Step #10882.3077: Cryo_etch

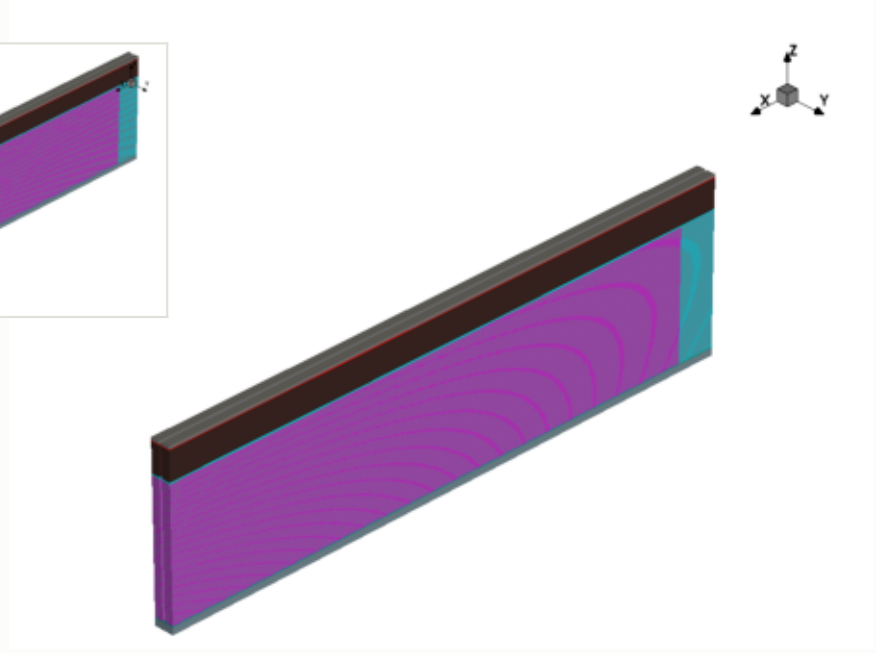
etch_noop · Cryo_etch

0.517	645.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





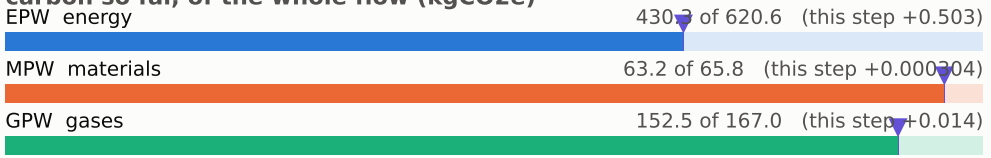
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

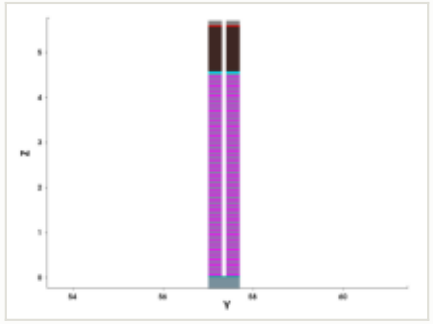
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

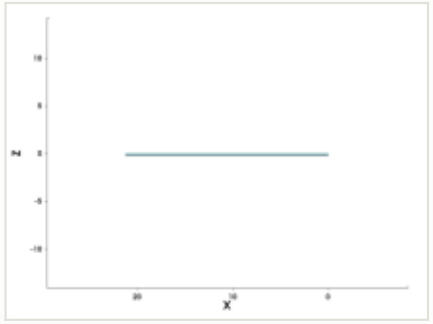


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



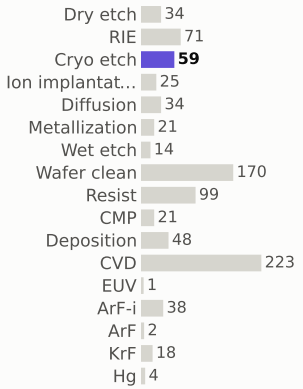
y cut at 57.351 µm



mask for this step

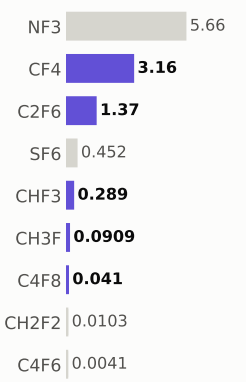
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



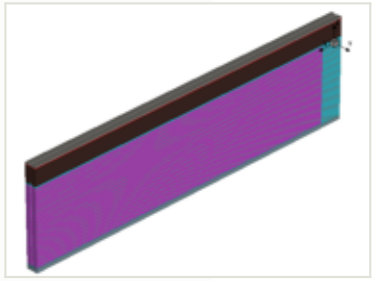
Step #10882.6923: Cryo_etch

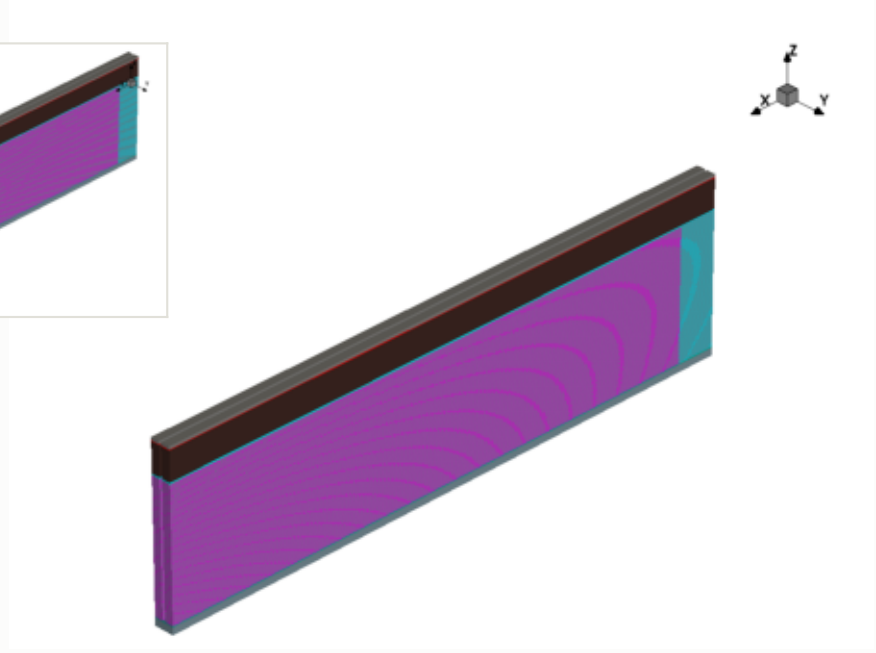
etch_noop · Cryo_etch

0.517	646.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





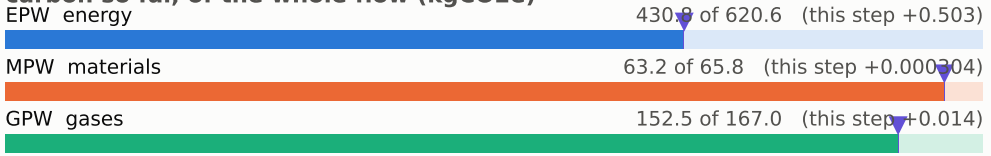
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

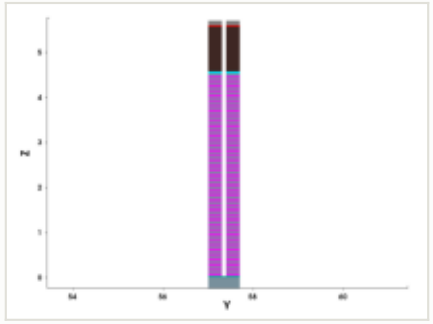
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

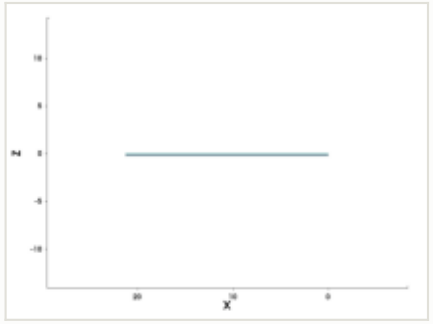


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



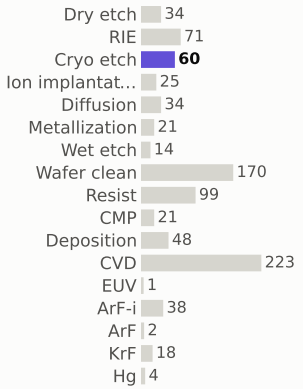
y cut at 57.351 µm



mask for this step

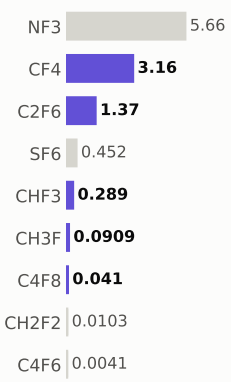
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



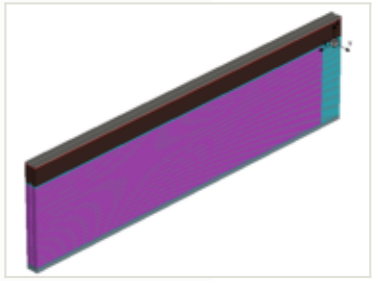
Step #10883.0769: Cryo_etch

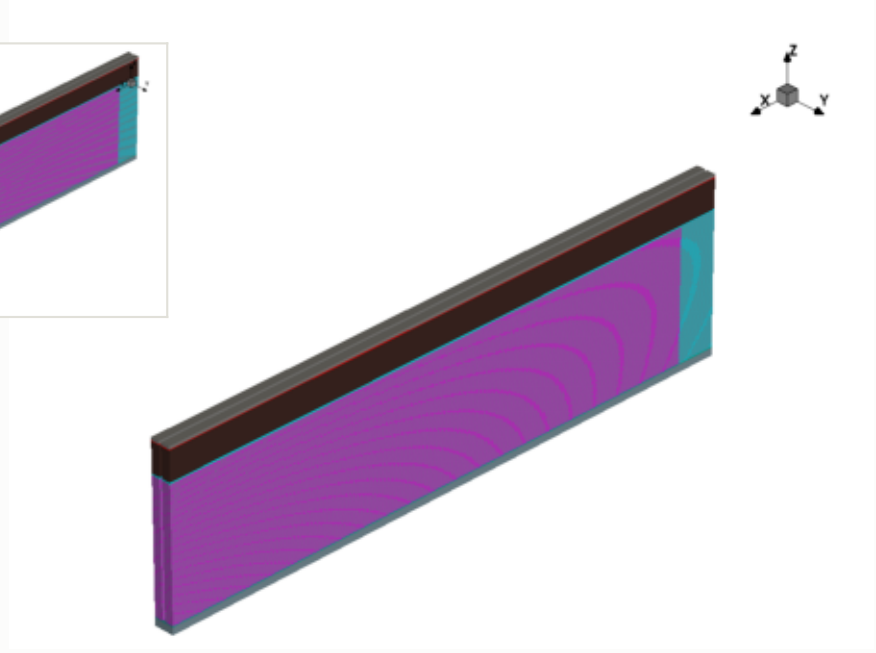
etch_noop · Cryo_etch

0.517	647.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





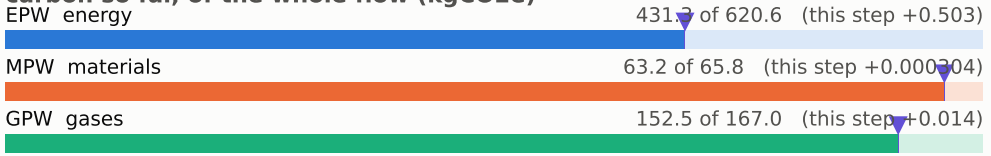
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

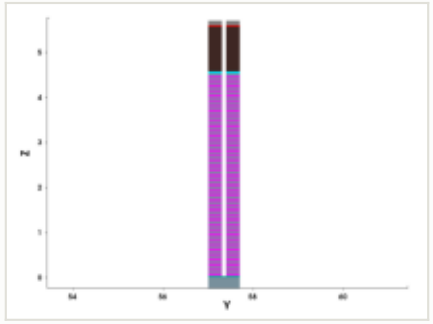
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

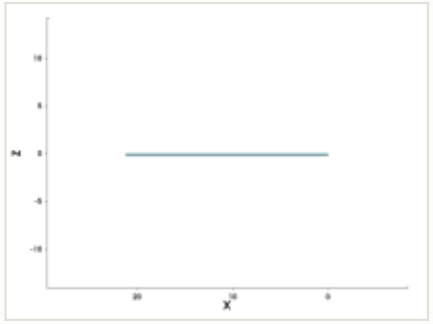


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



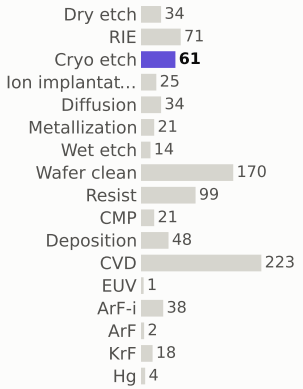
y cut at 57.351 µm



mask for this step

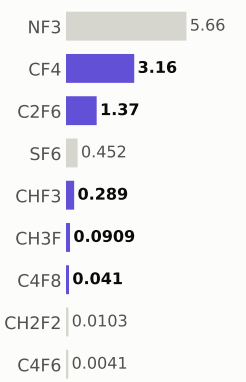
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



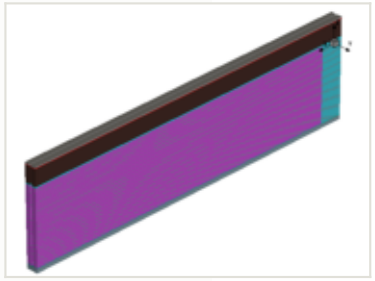
Step #10883.4615: Cryo_etch

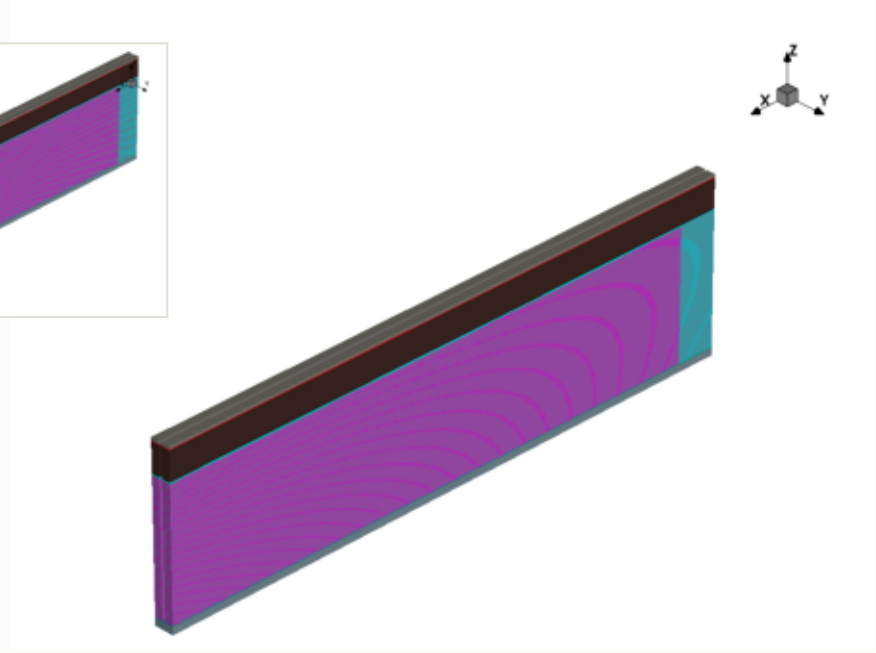
etch_noop · Cryo_etch

0.517	647.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





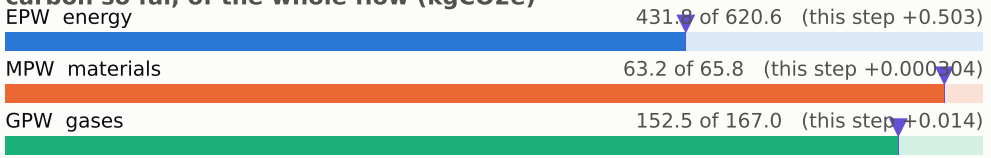
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

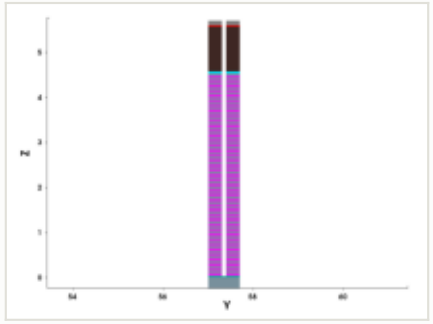
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

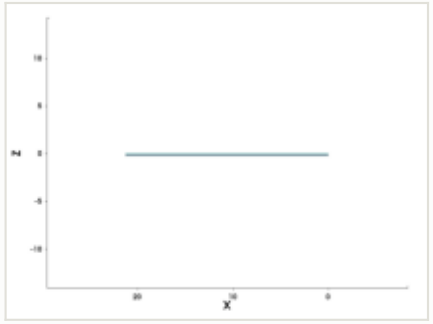


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



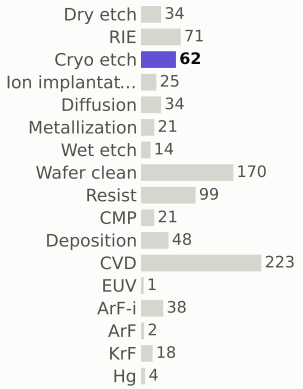
y cut at 57.351 µm



mask for this step

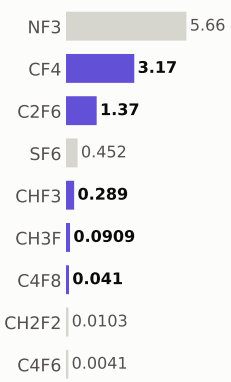
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



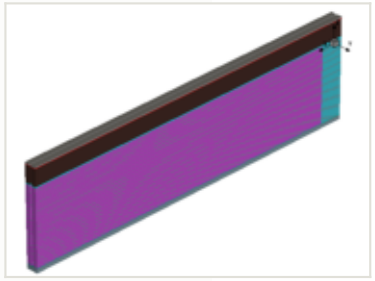
Step #10883.8462: Cryo_etch

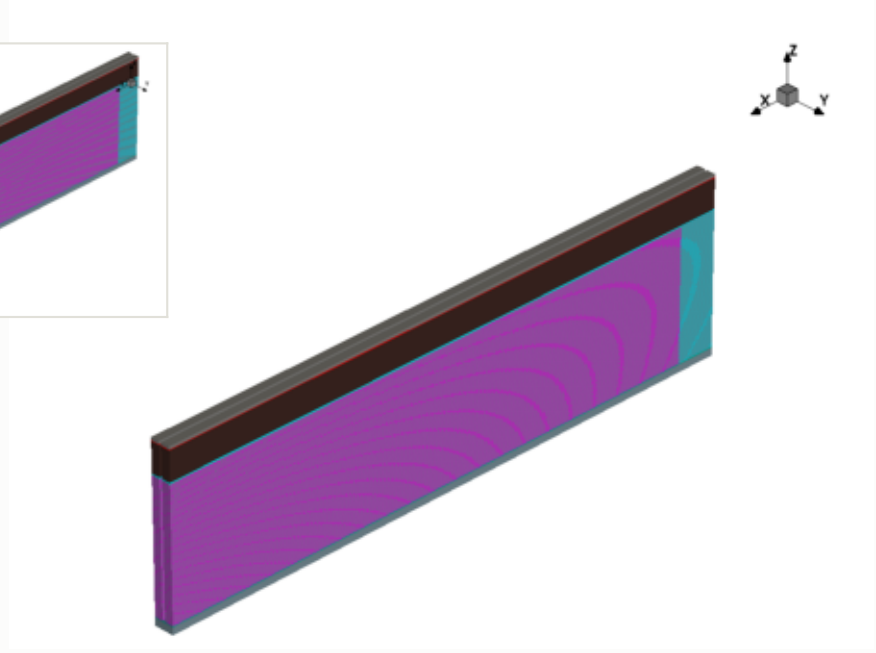
etch_noop · Cryo_etch

0.517	648.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



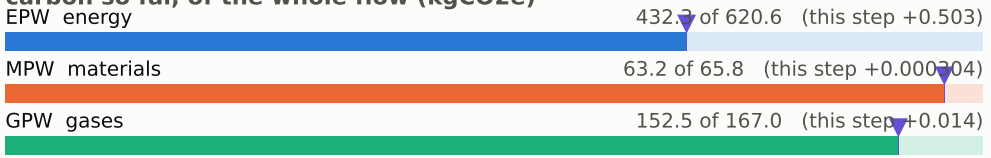


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

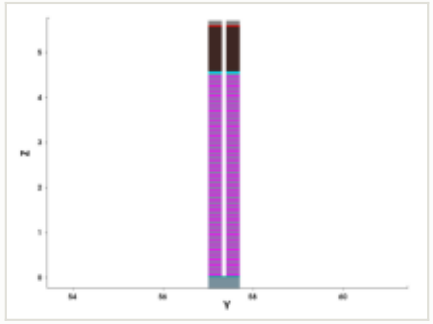
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

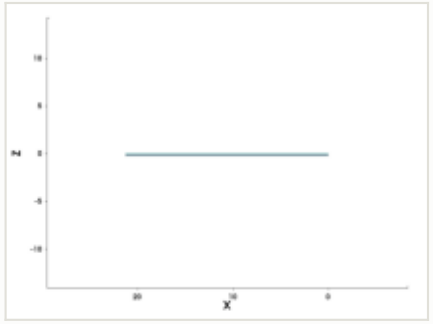


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



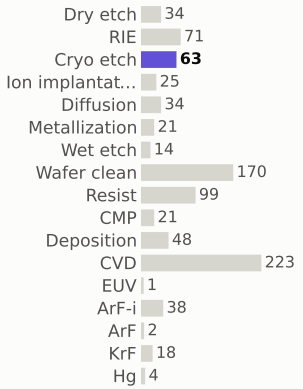
y cut at 57.351 µm



mask for this step

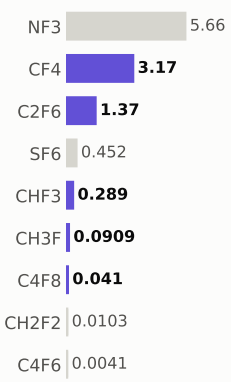
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



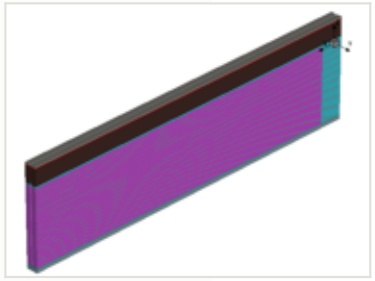
Step #10884.2308: Cryo_etch

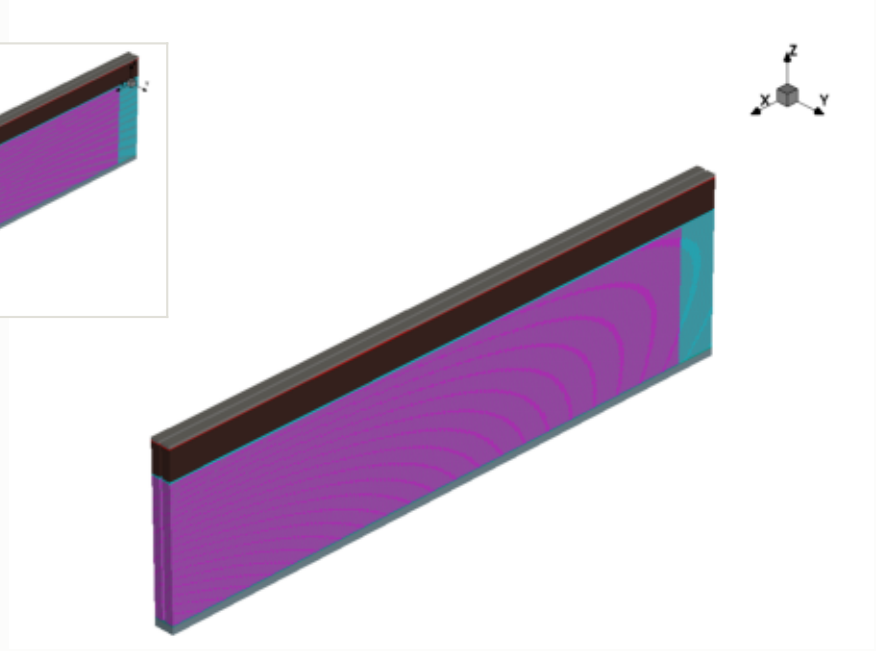
etch_noop · Cryo_etch

0.517	648.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





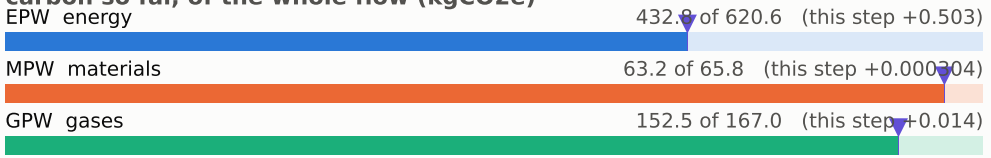
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

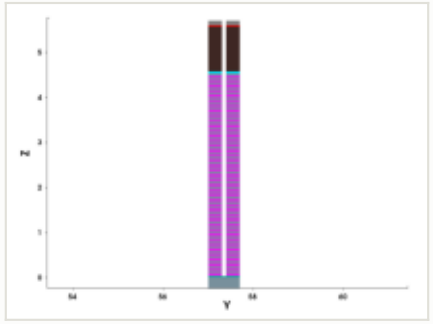
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

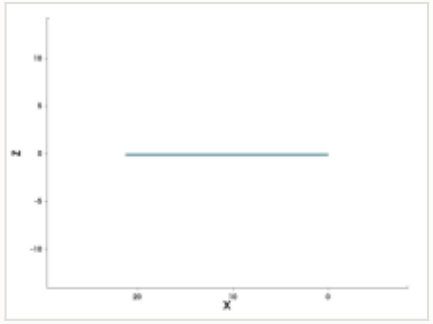


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



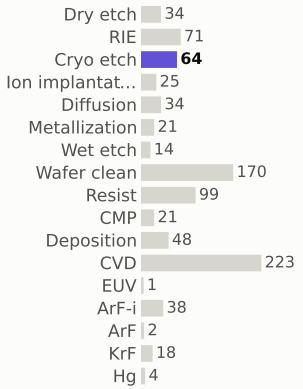
y cut at 57.351 µm



mask for this step

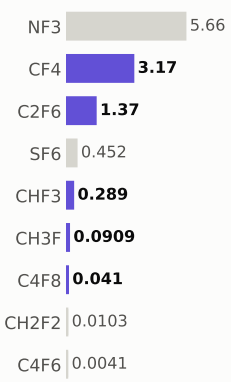
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



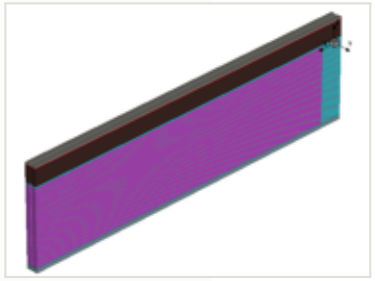
Step #10884.6154: Cryo_etch

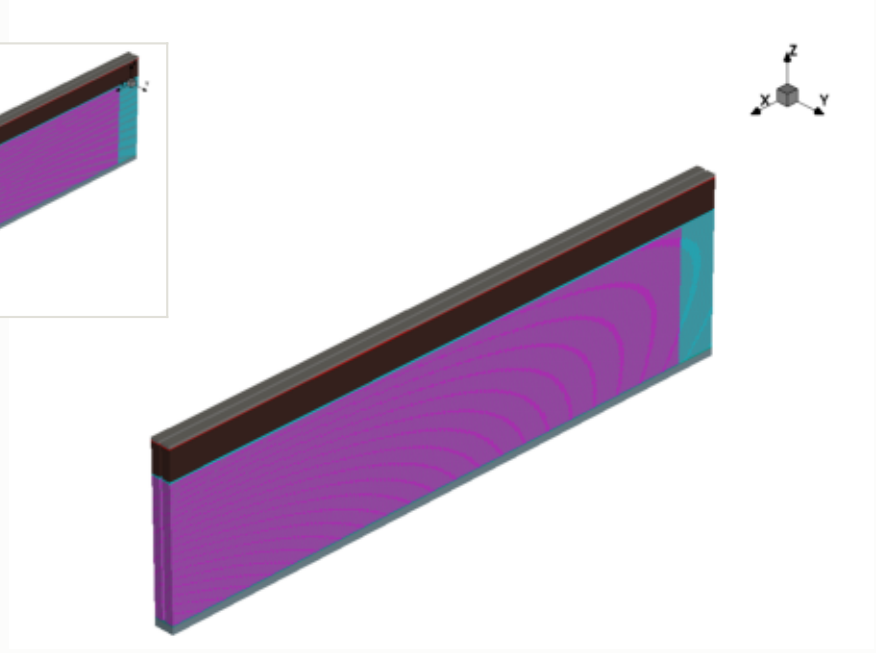
etch_noop · Cryo_etch

0.517	649.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





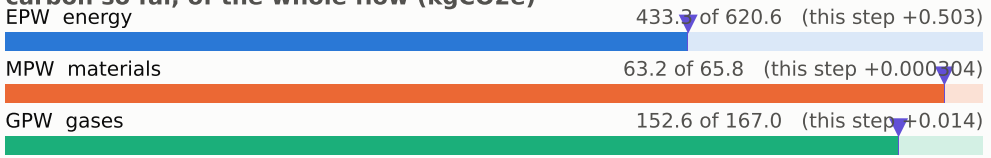
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

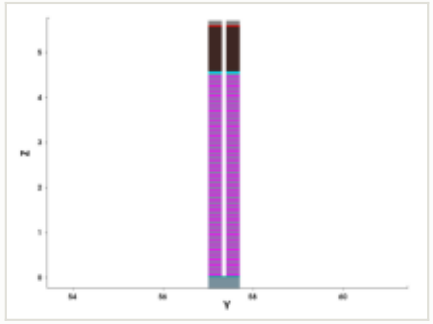
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

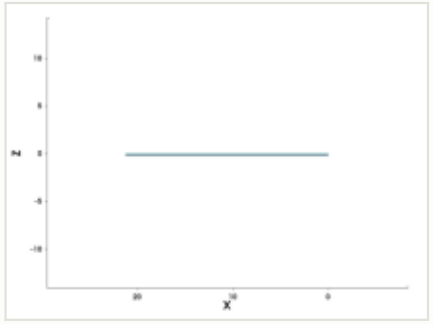


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



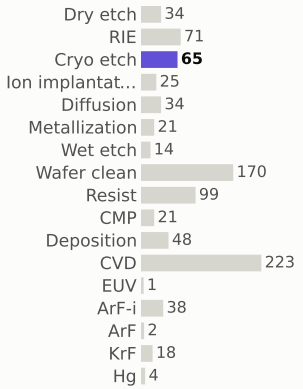
y cut at 57.351 µm



mask for this step

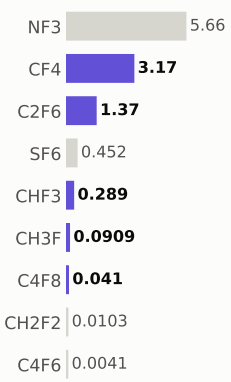
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



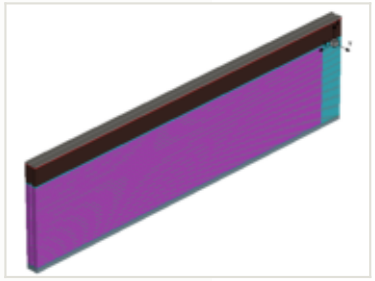
Step #10885.0000: Cryo_etch

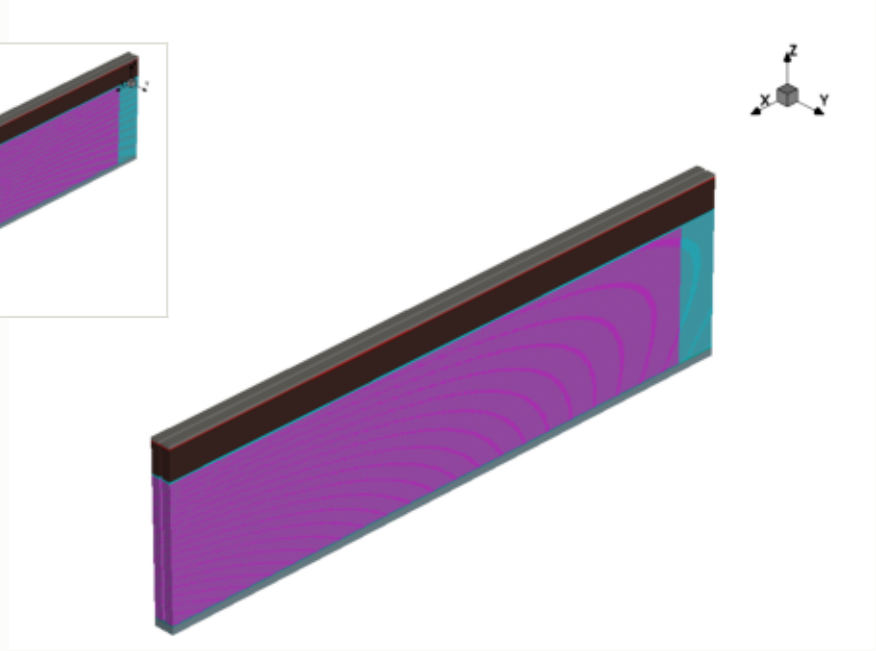
etch_noop · Cryo_etch

0.517	649.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





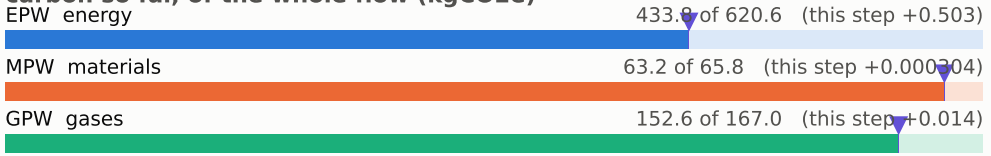
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

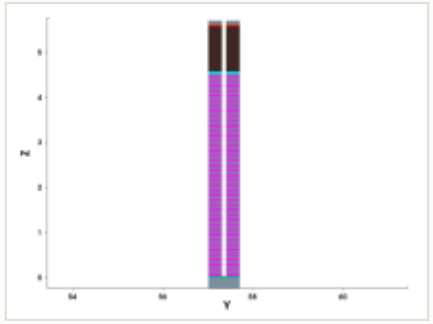
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

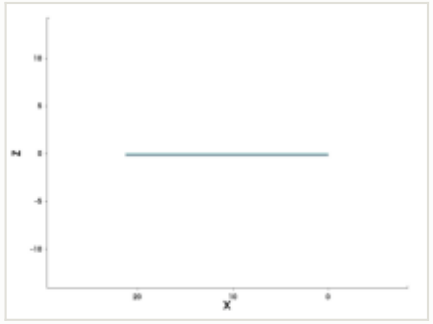


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



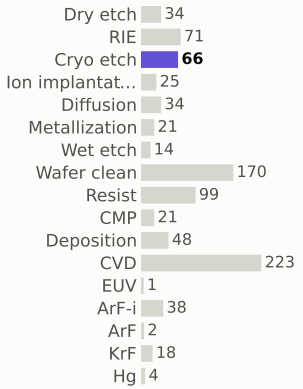
y cut at 57.351 µm



mask for this step

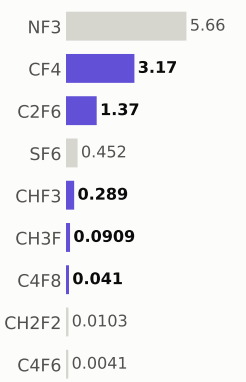
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



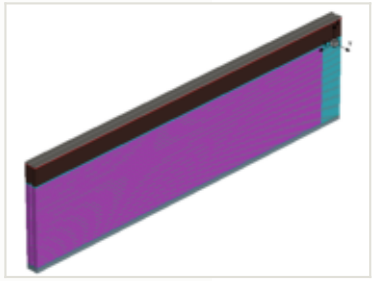
Step #10885.3846: Cryo_etch

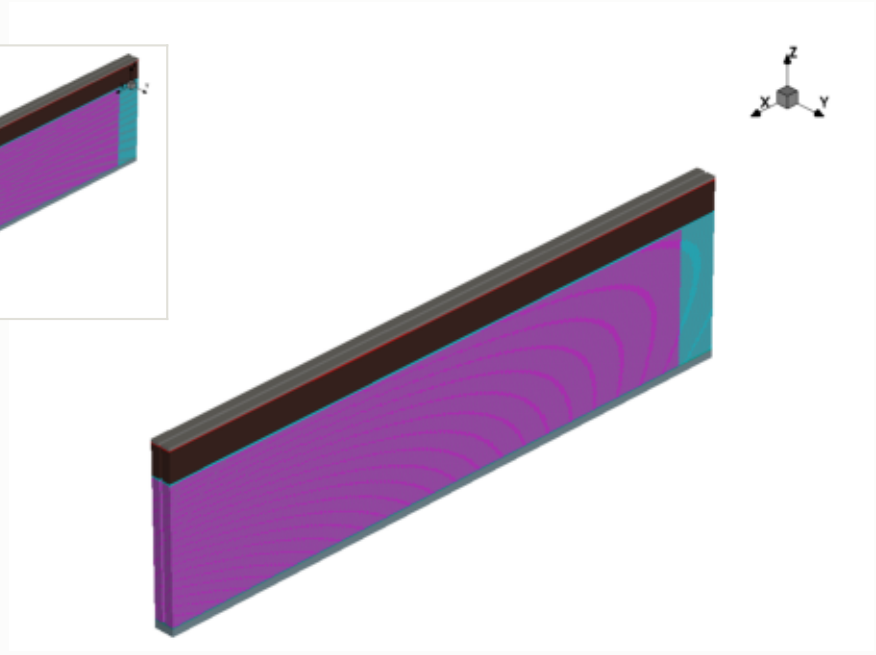
etch_noop · Cryo_etch

0.517	650.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





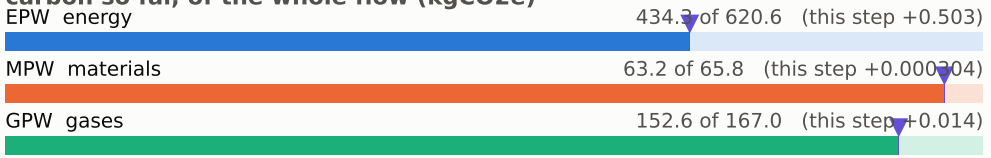
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

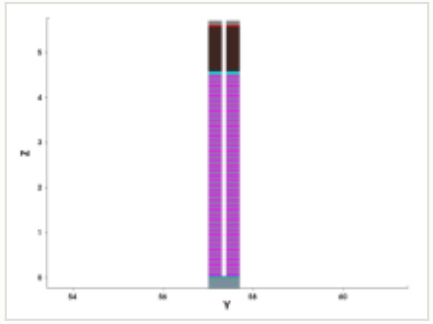
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

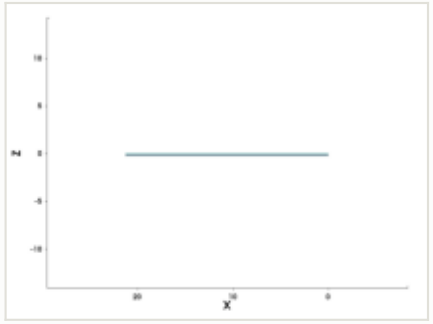


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



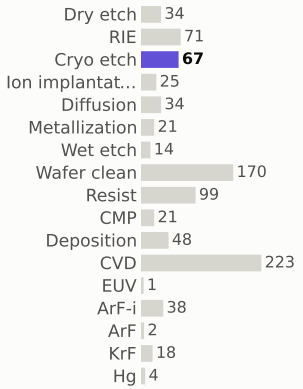
y cut at 57.351 µm



mask for this step

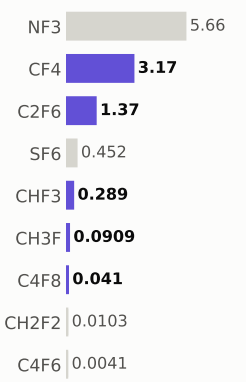
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



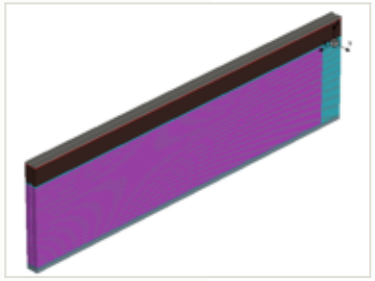
Step #10885.7692: Cryo_etch

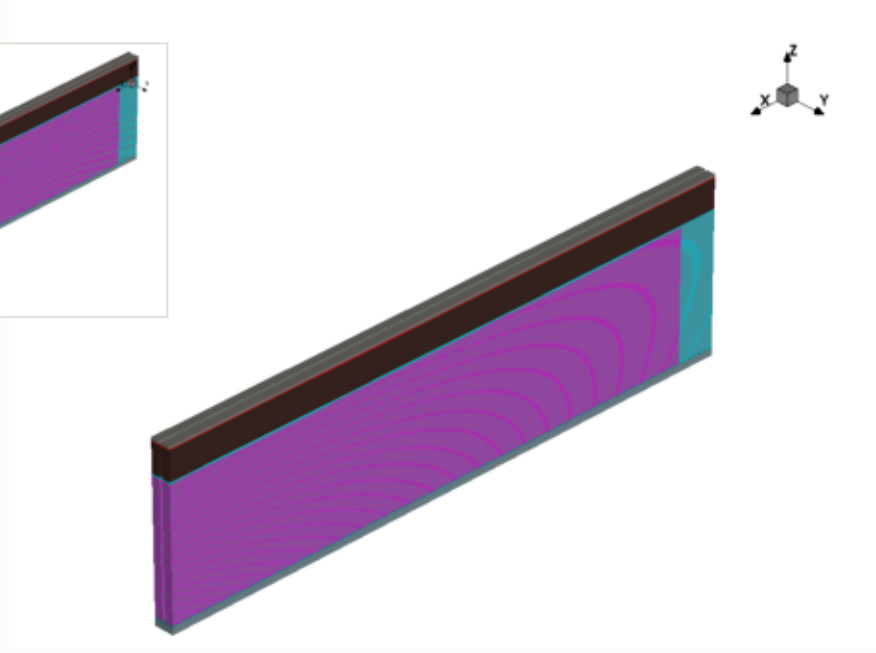
etch_noop · Cryo_etch

0.517	650.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





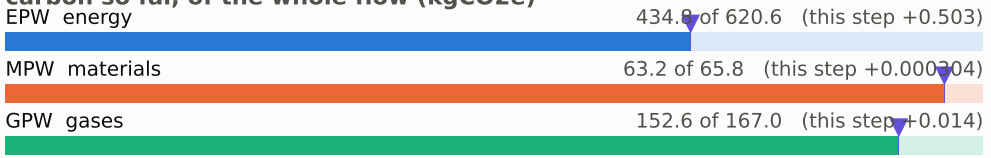
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

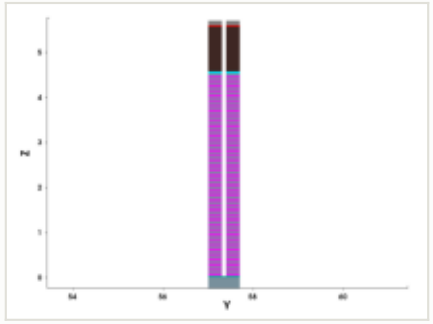
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

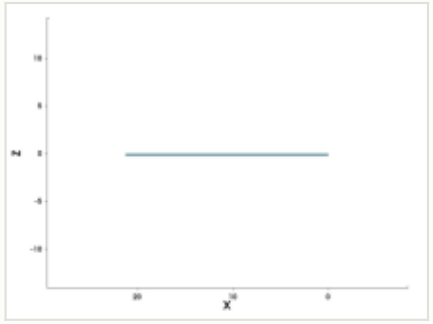


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



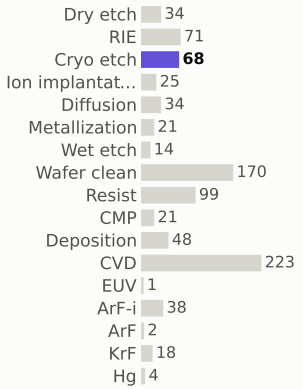
y cut at 57.351 µm



mask for this step

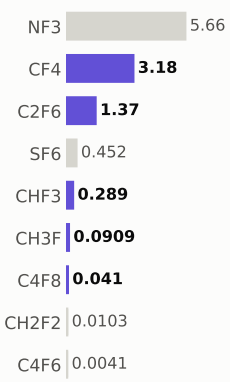
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



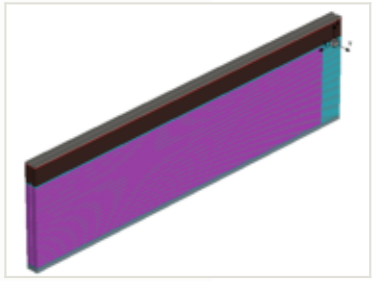
Step #10886.1538: Cryo_etch

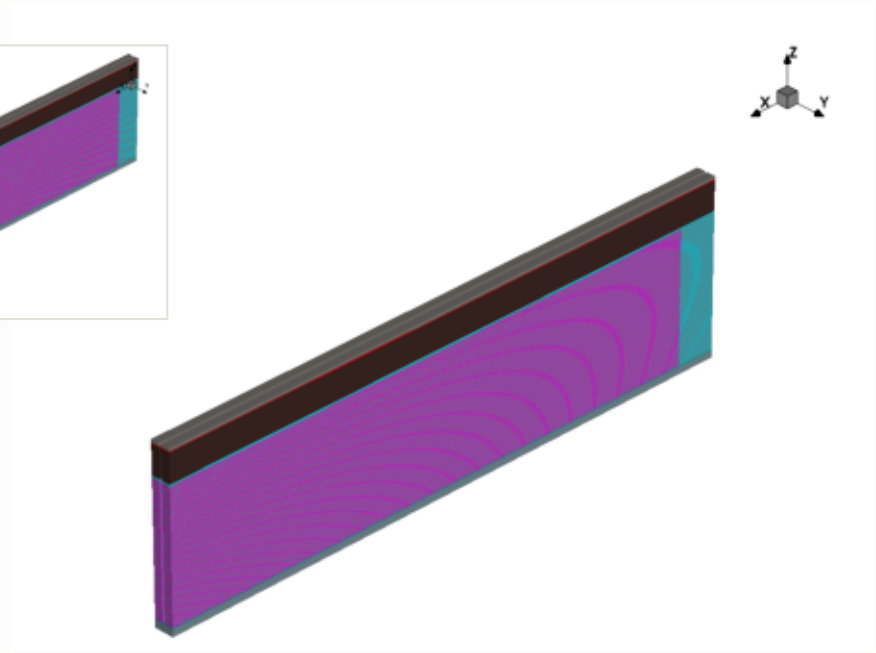
etch_noop · Cryo_etch

0.517	651.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





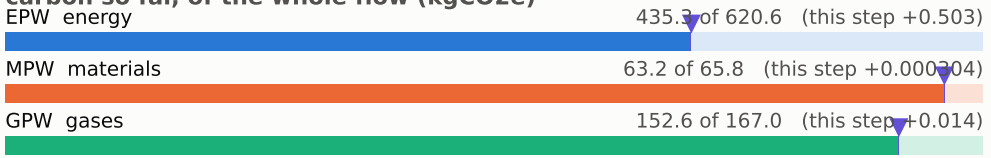
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

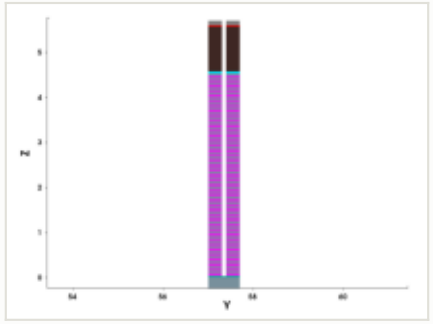
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

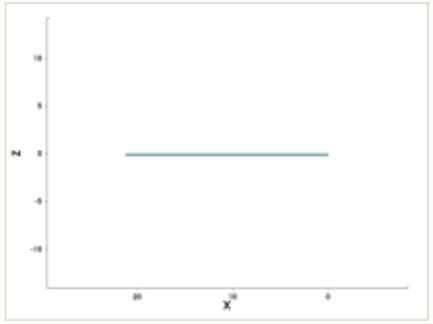


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



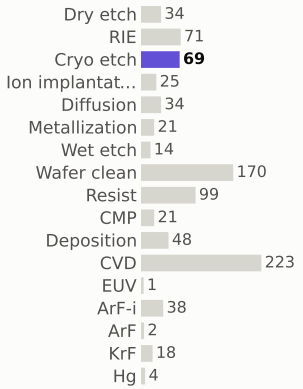
y cut at 57.351 µm



mask for this step

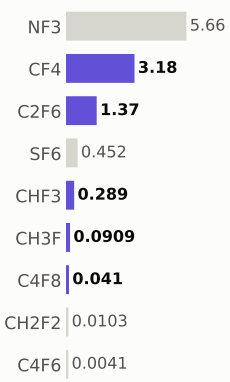
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



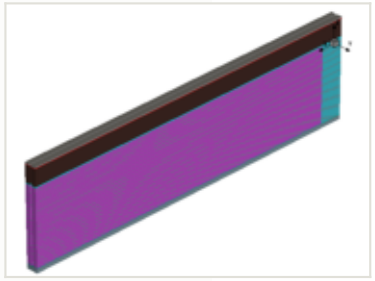
Step #10886.5385: Cryo_etch

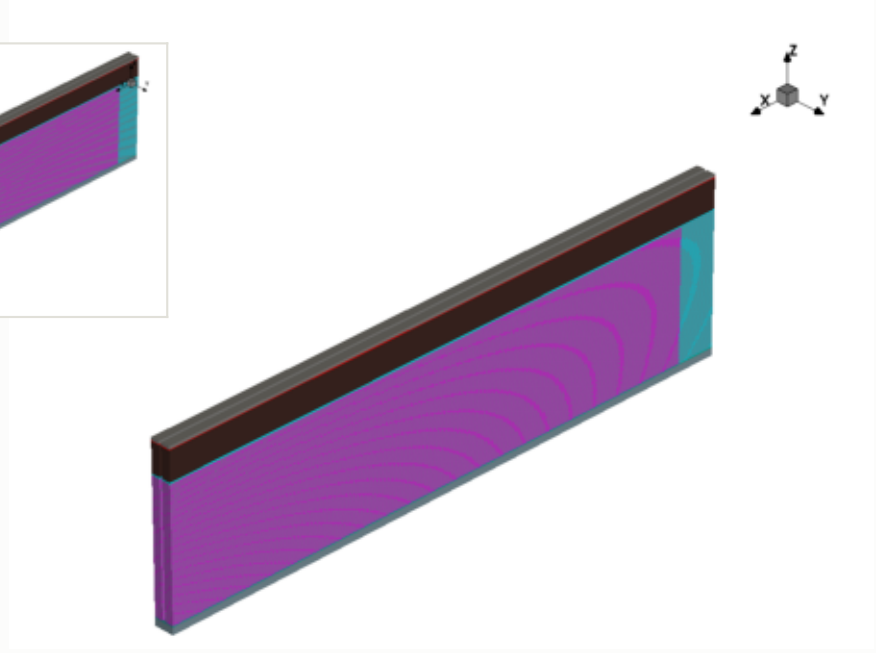
etch_noop · Cryo_etch

0.517	651.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





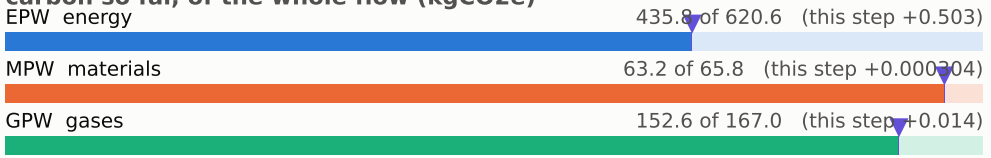
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

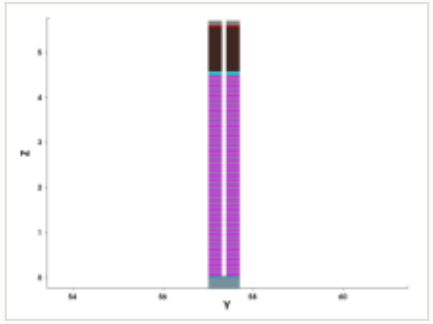
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

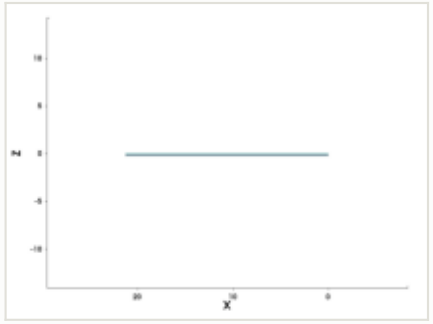


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



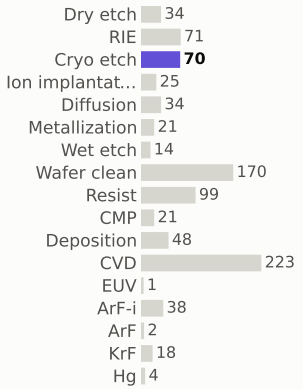
y cut at 57.351 µm



mask for this step

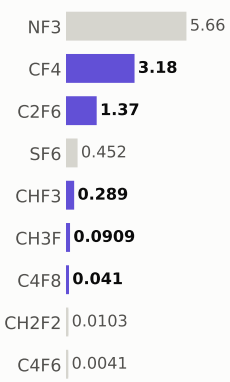
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



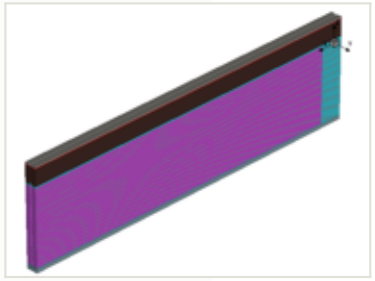
Step #10886.9231: Cryo_etch

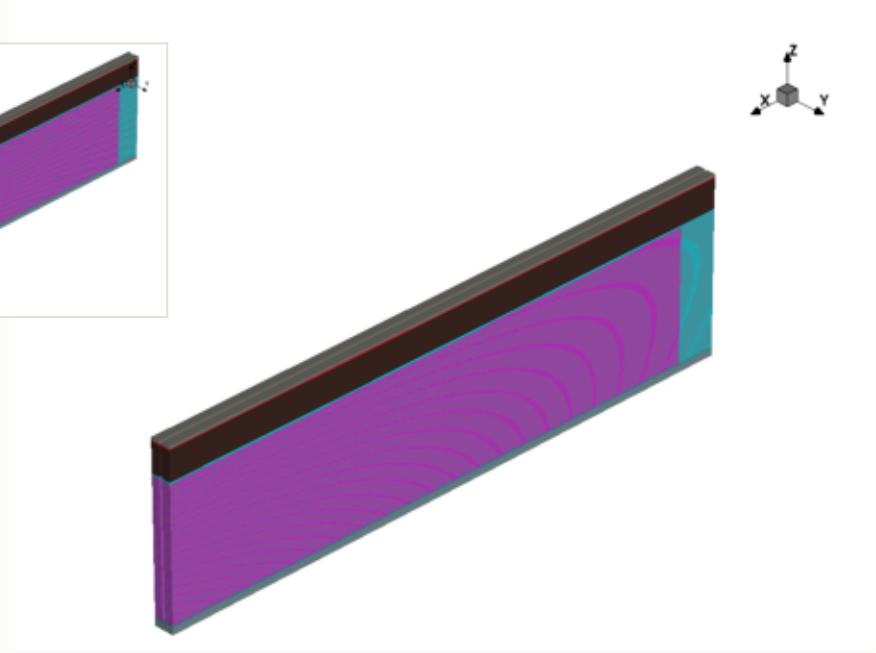
etch_noop · Cryo_etch

0.517	652.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





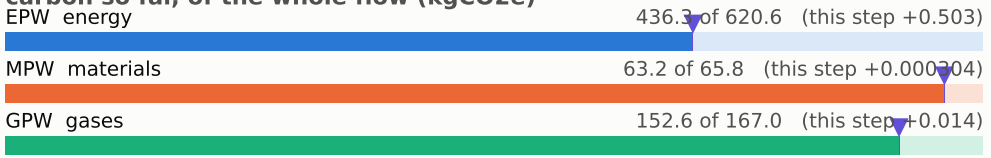
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

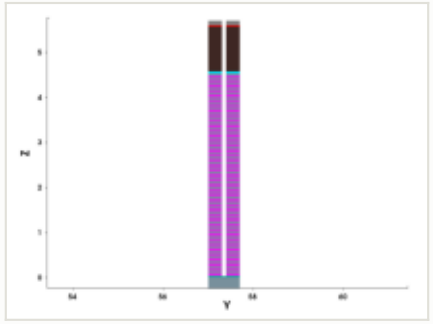
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

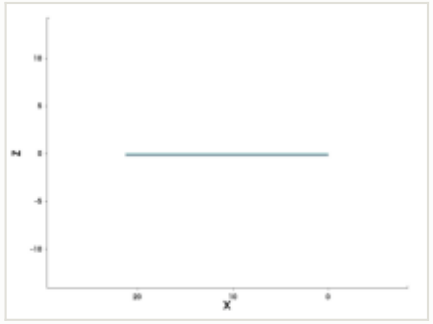


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



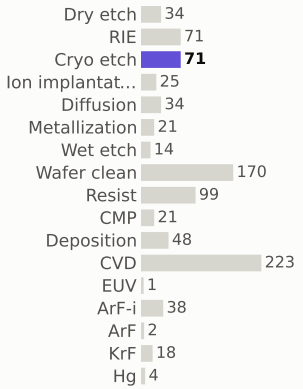
y cut at 57.351 µm



mask for this step

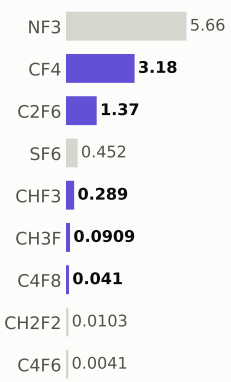
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



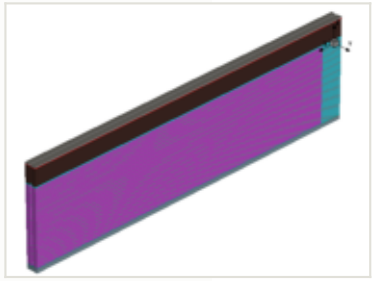
Step #10887.3077: Cryo_etch

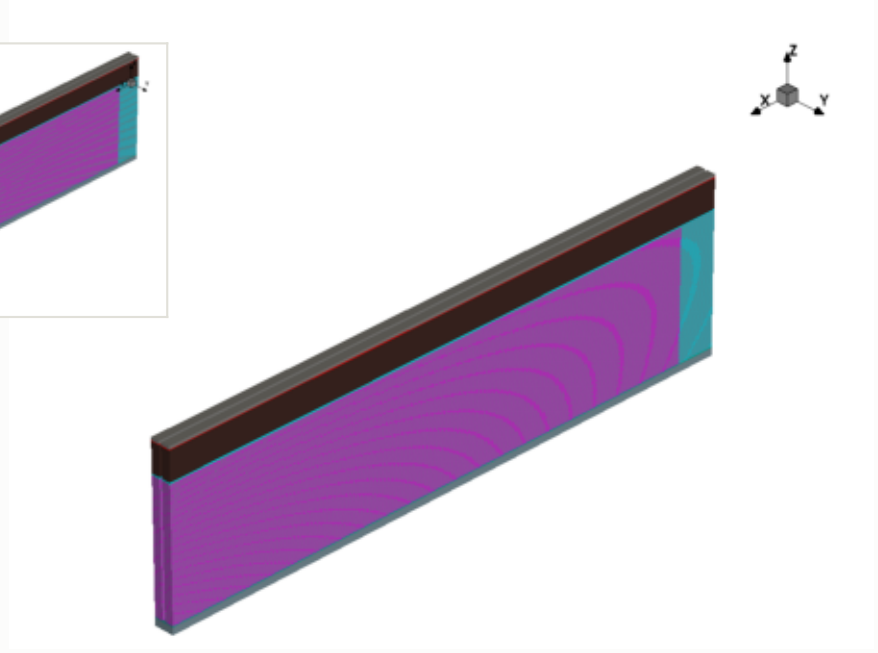
etch_noop · Cryo_etch

0.517	652.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





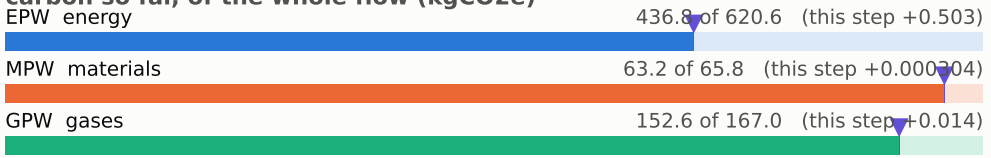
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

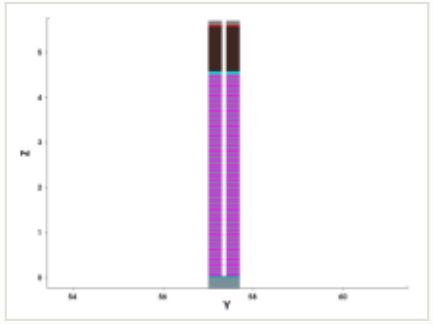
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

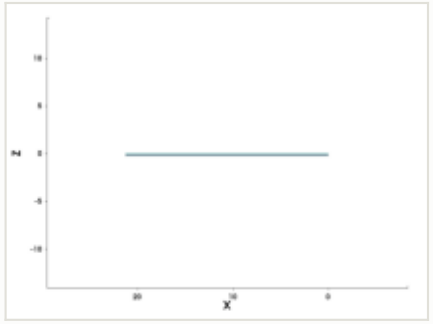


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



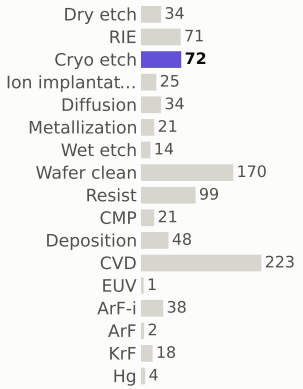
y cut at 57.351 µm



mask for this step

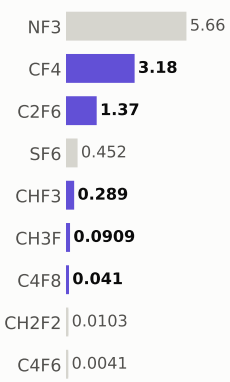
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



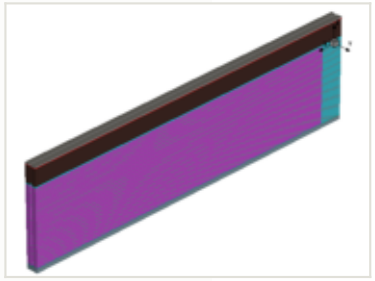
Step #10887.6923: Cryo_etch

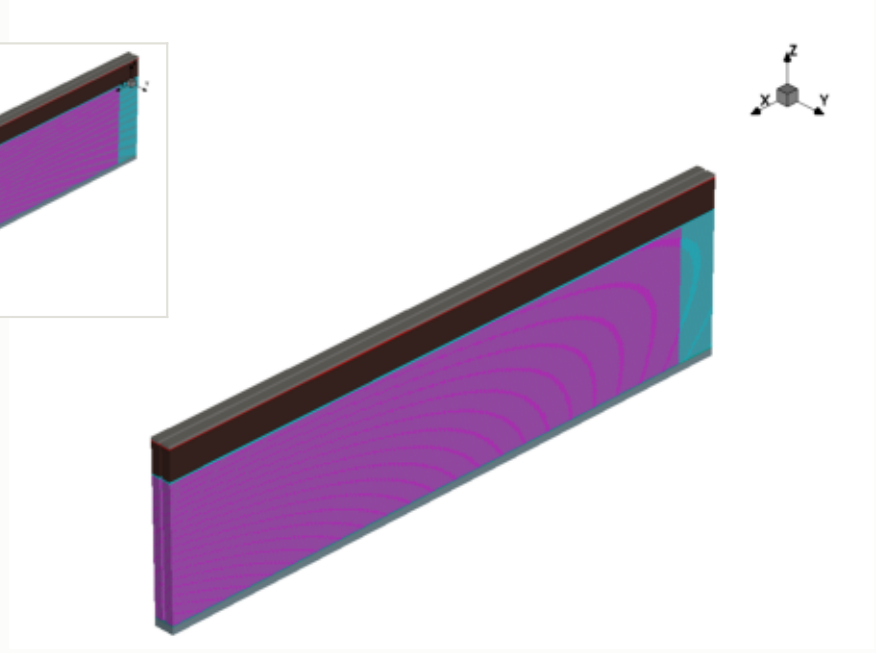
etch_noop · Cryo_etch

0.517	653.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





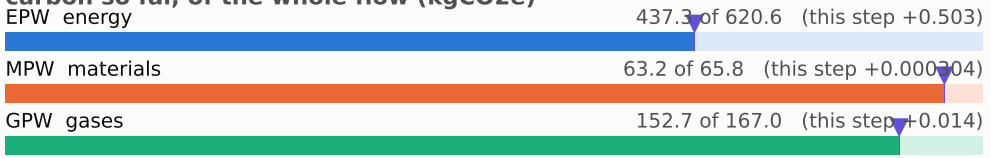
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

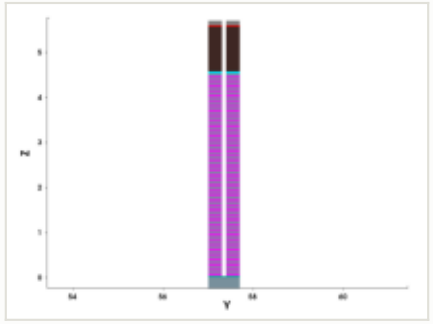
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

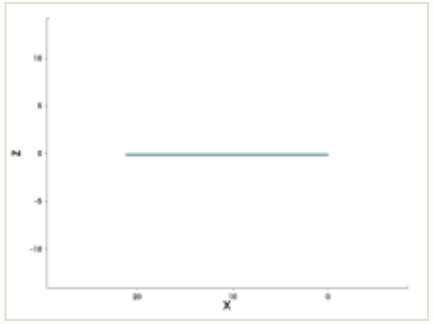


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



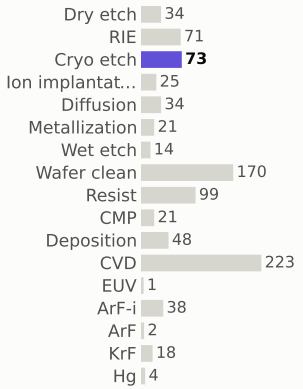
y cut at 57.351 µm



mask for this step

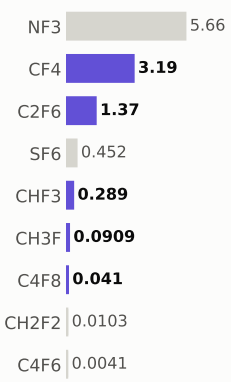
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



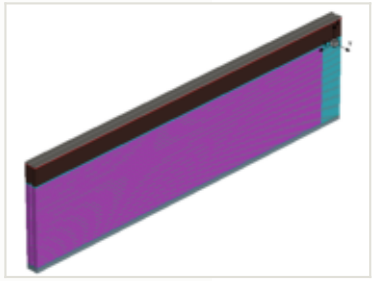
Step #10888.0769: Cryo_etch

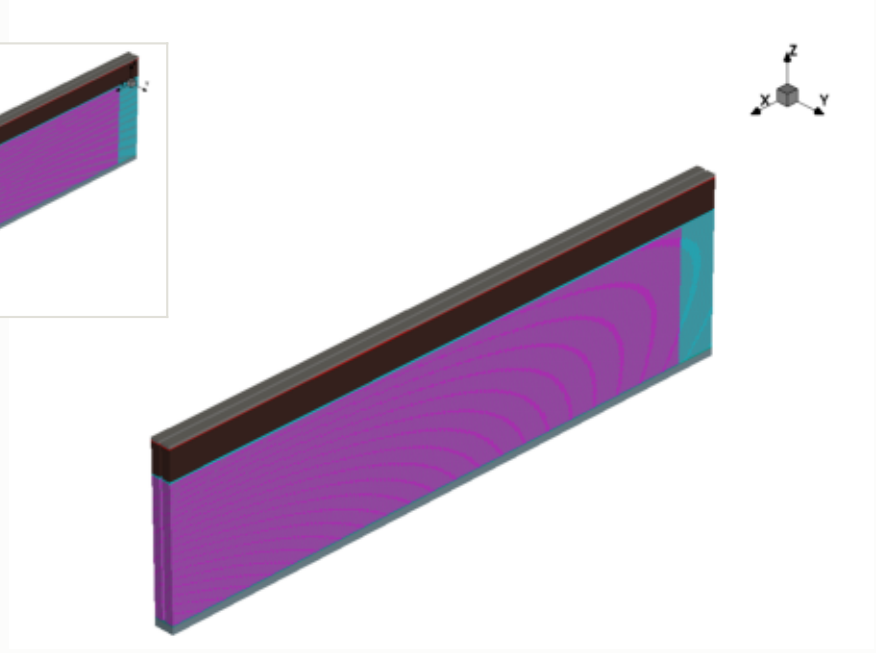
etch_noop · Cryo_etch

0.517	653.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





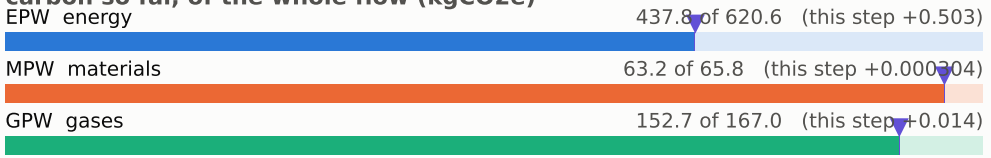
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

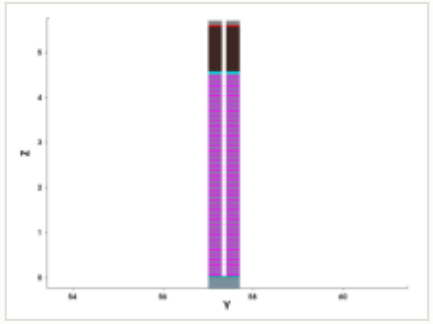
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

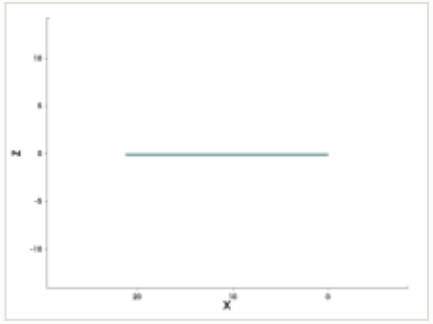


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



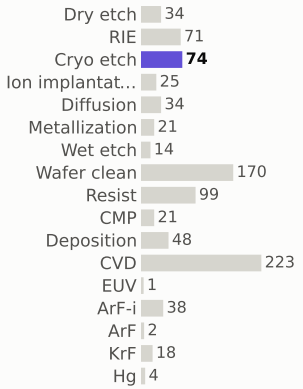
y cut at 57.351 µm



mask for this step

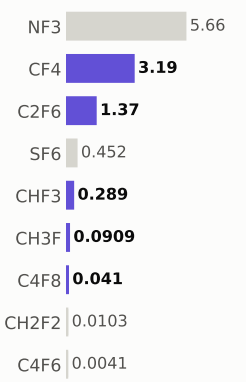
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



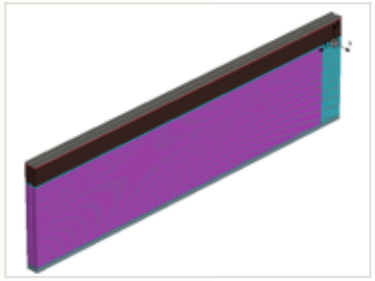
Step #10888.4615: Cryo_etch

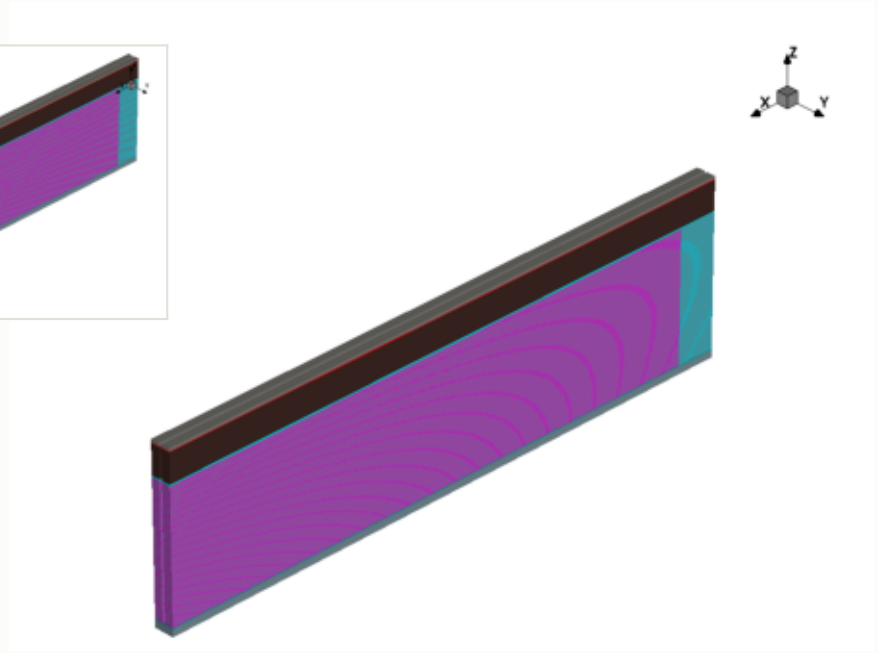
etch_noop · Cryo_etch

0.517	654.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





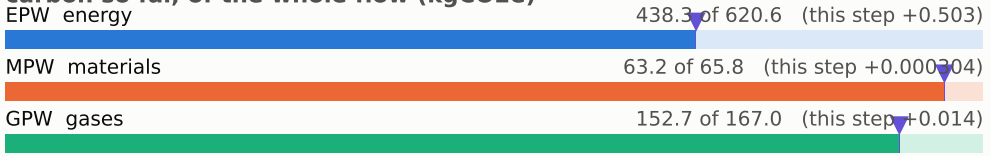
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

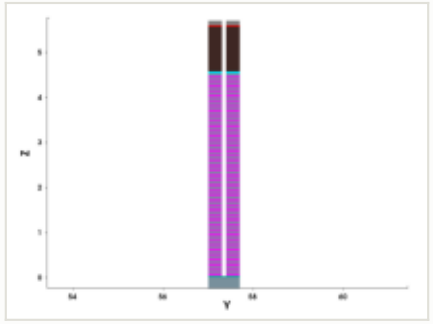
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

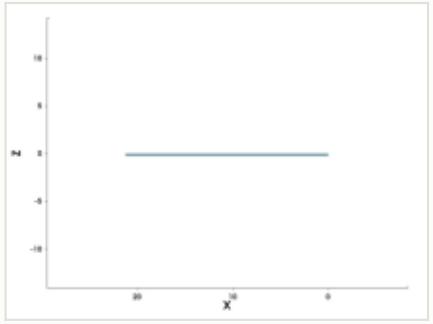


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



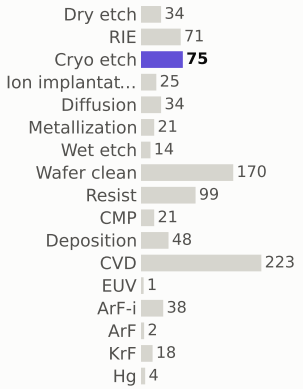
y cut at 57.351 µm



mask for this step

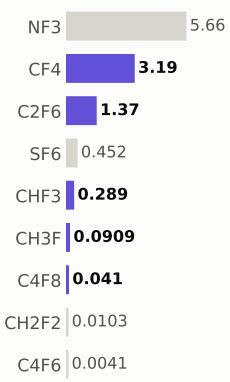
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



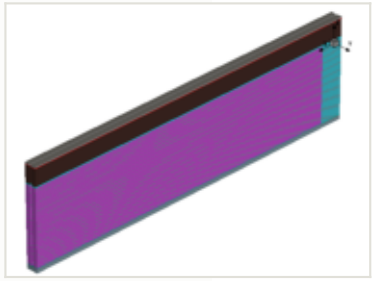
Step #10888.8462: Cryo_etch

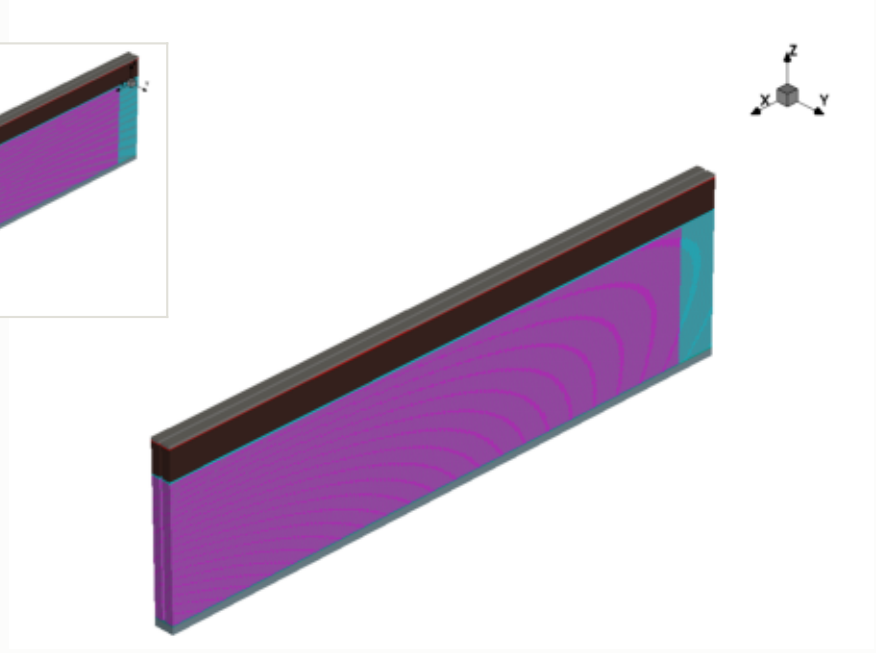
etch_noop · Cryo_etch

0.517	654.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





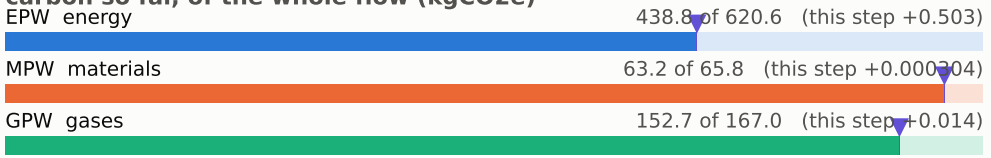
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

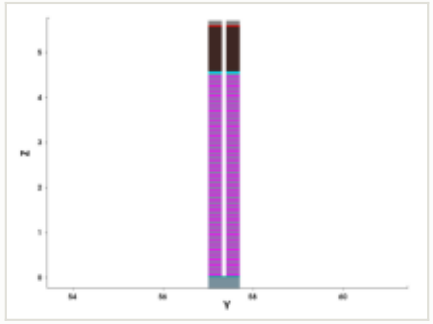
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

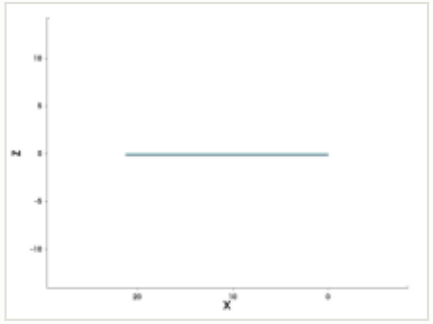


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



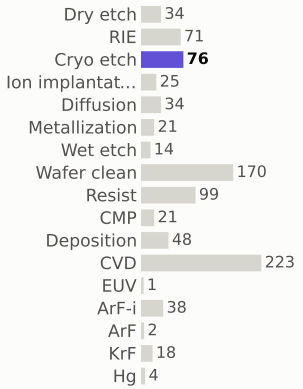
y cut at 57.351 µm



mask for this step

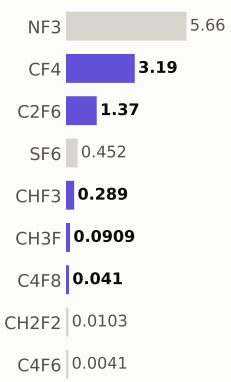
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



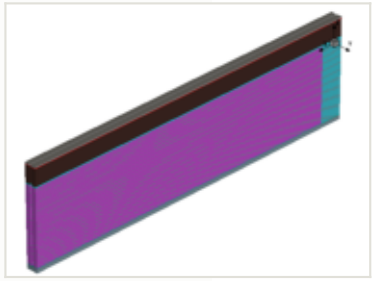
Step #10889.2308: Cryo_etch

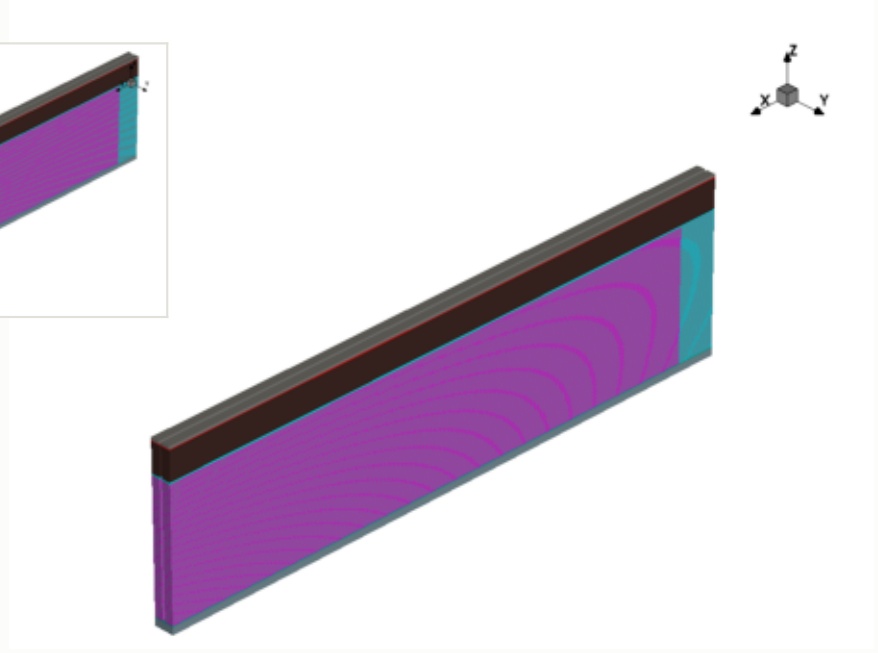
etch_noop · Cryo_etch

0.517	655.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





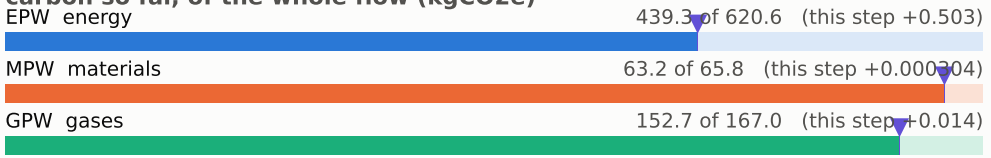
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

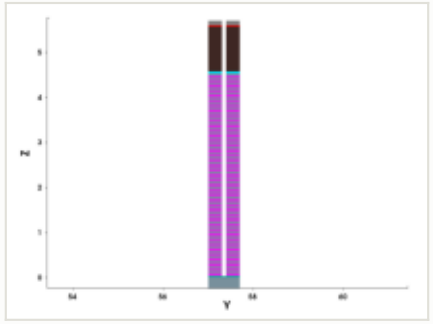
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

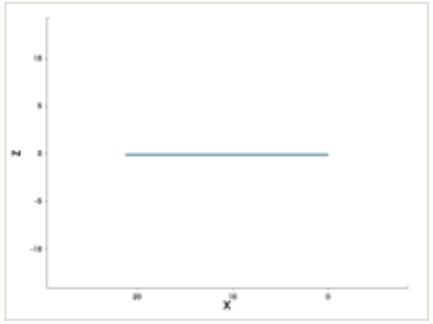


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



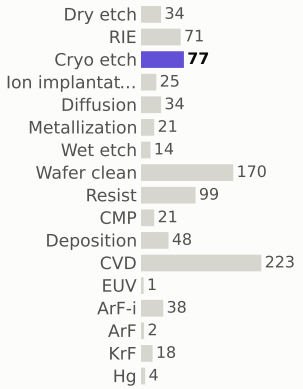
y cut at 57.351 µm



mask for this step

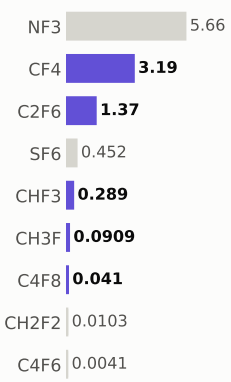
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



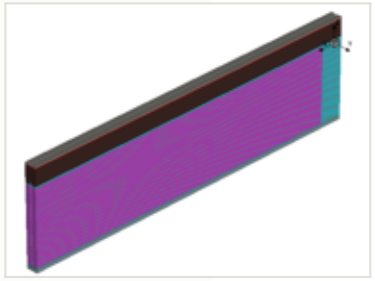
Step #10889.6154: Cryo_etch

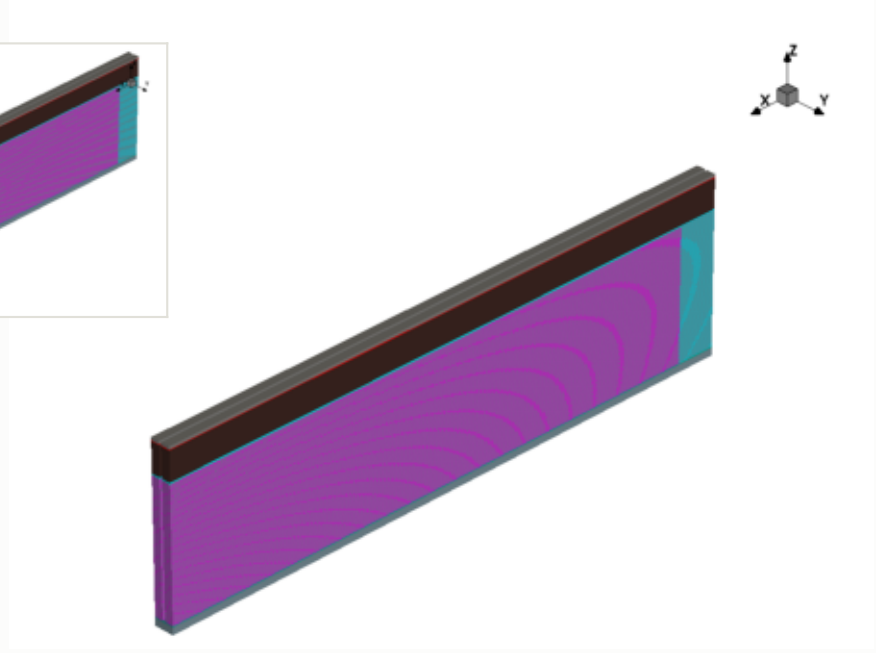
etch_noop · Cryo_etch

0.517	655.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





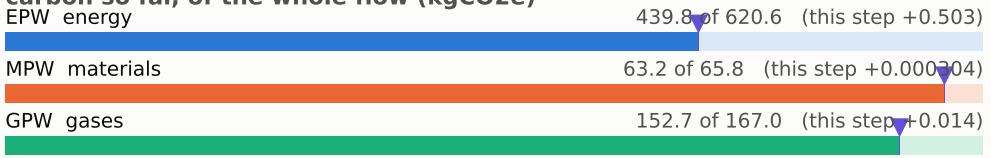
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

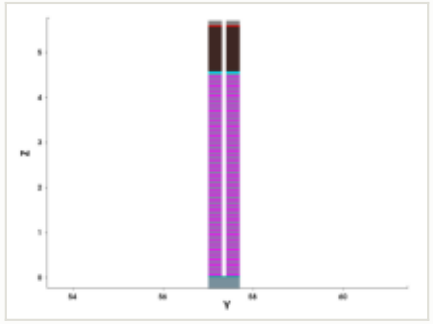
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

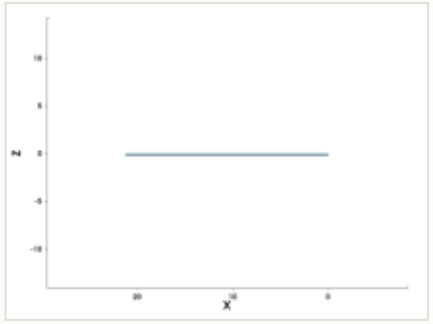


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



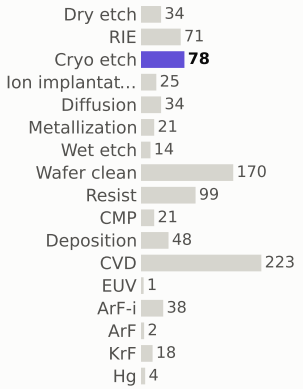
y cut at 57.351 µm



mask for this step

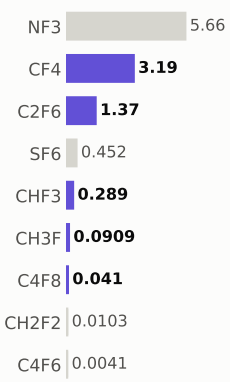
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



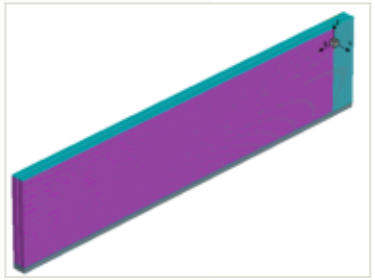
Step #10890: Dry_etch

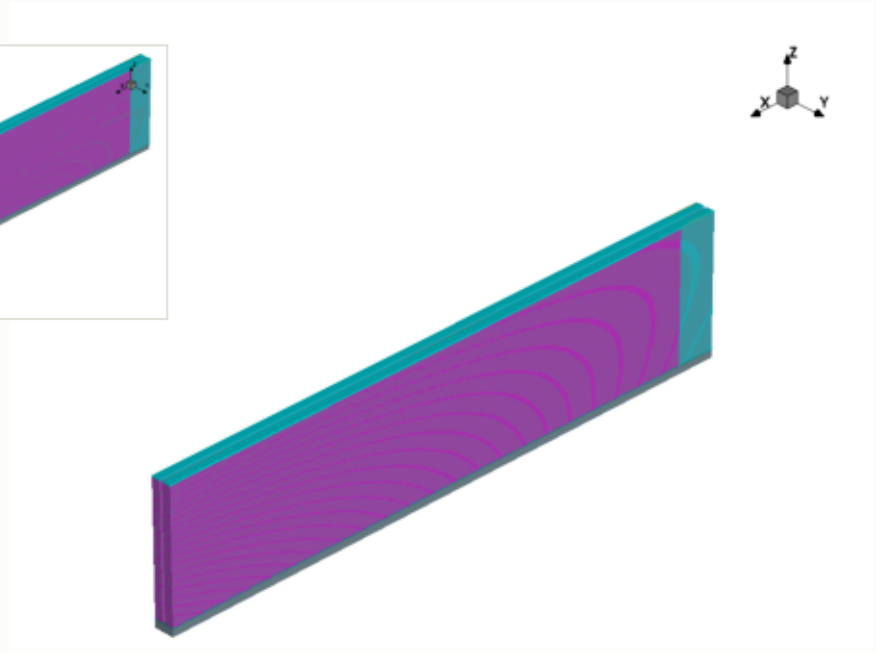
etch_strip · Dry_etch

0.801	656.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



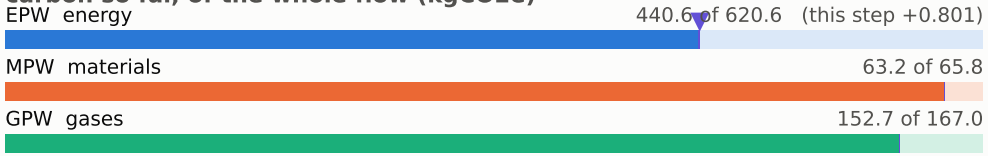


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

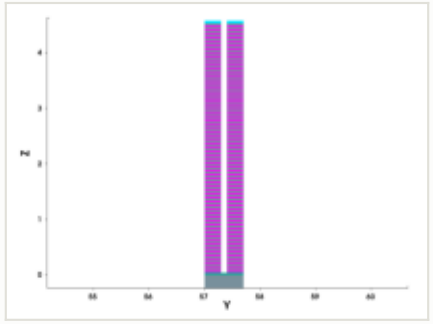
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

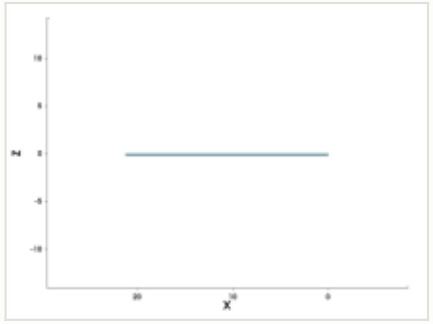


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



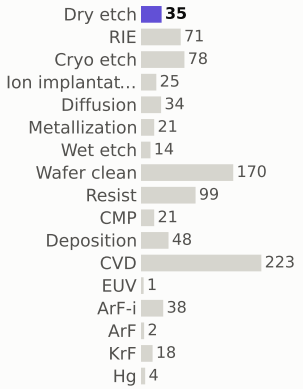
y cut at 57.351 µm



mask for this step

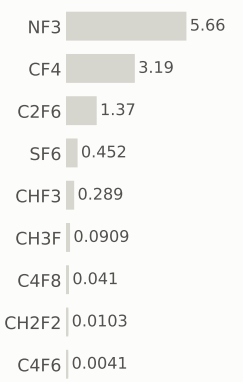
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



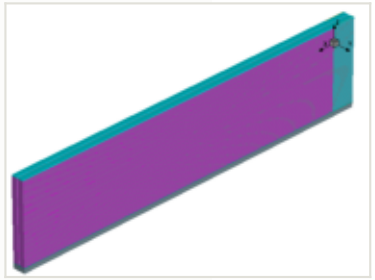
Step #10900: Wafer_clean

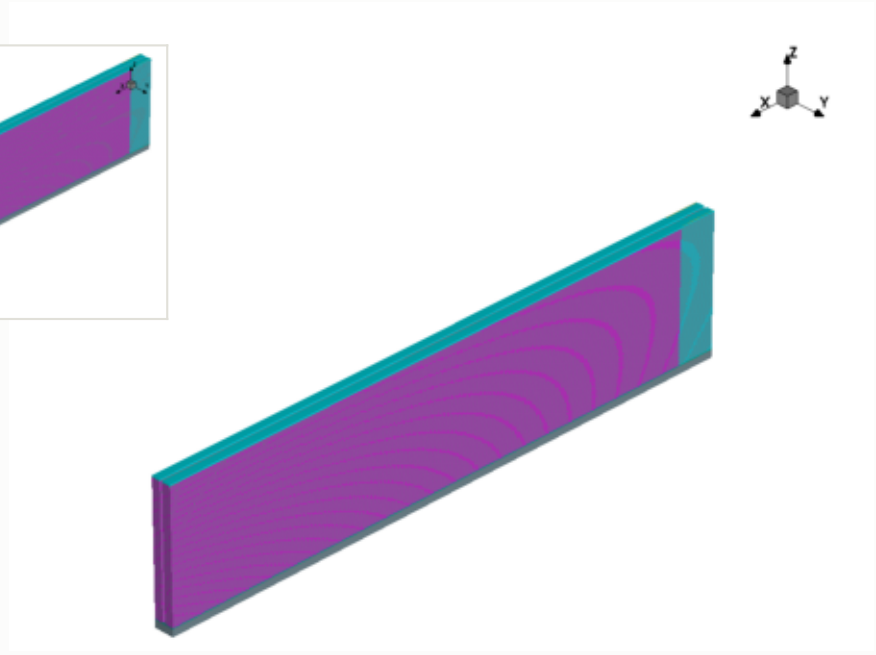
clean · Wafer_clean

0.169	656.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





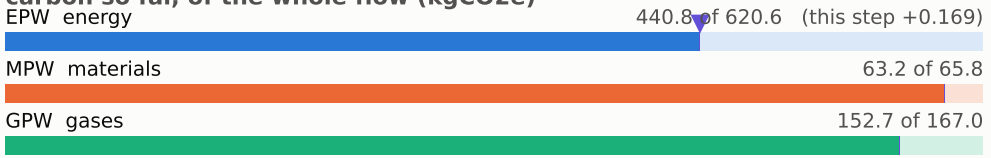
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

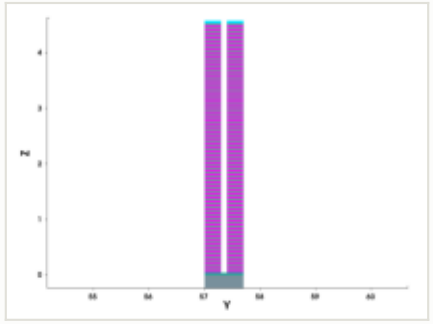
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

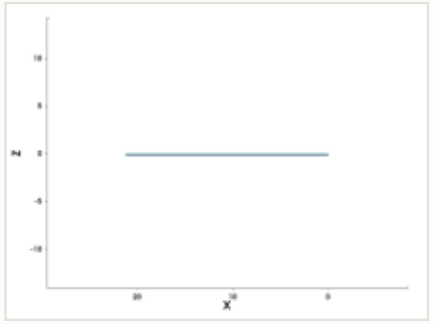


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



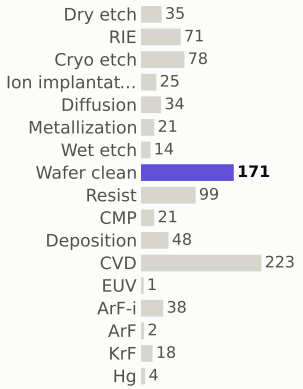
y cut at 57.351 µm



mask for this step

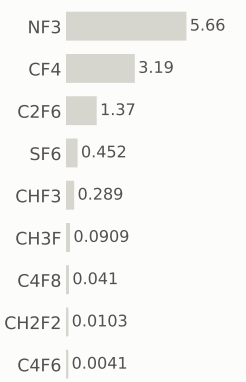
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



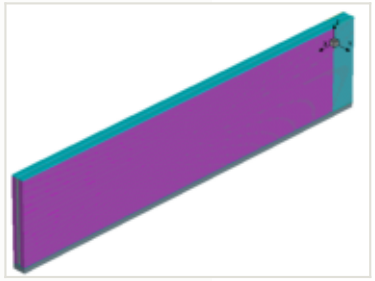
Step #10910: Wet_etch

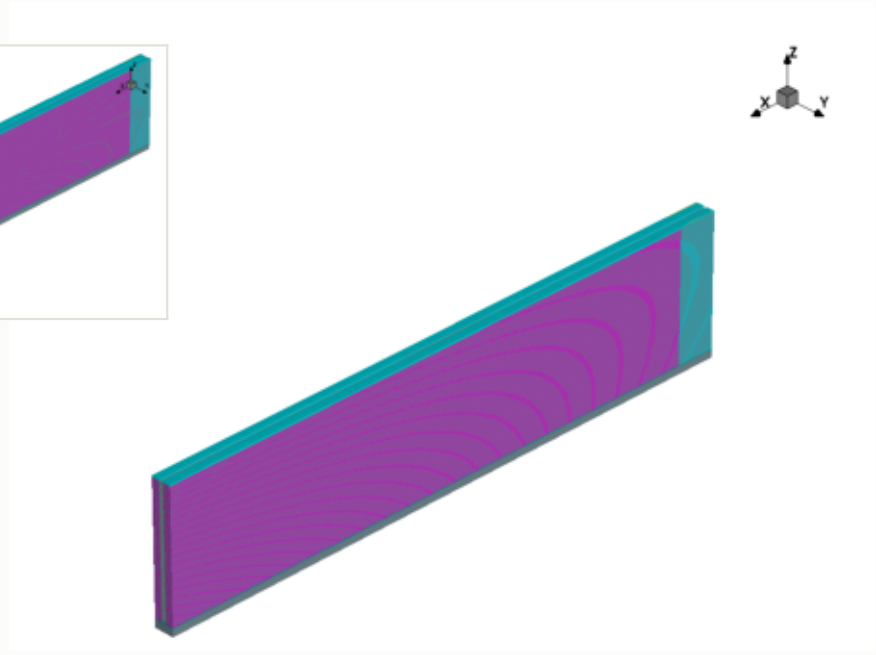
etch_isotropic · Wet_etch

0.169	656.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



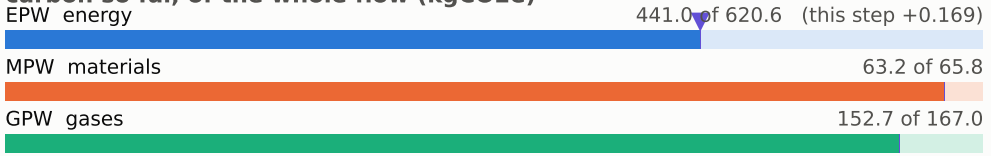


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

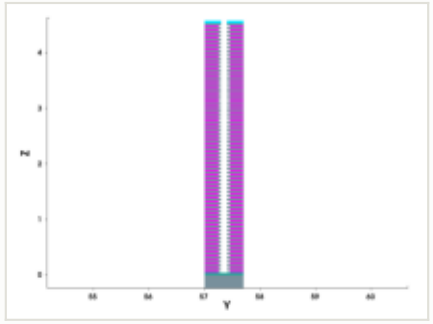
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

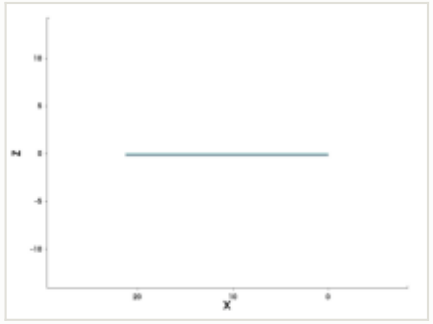


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



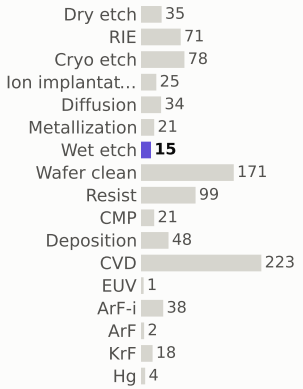
y cut at 57.351 µm



mask for this step

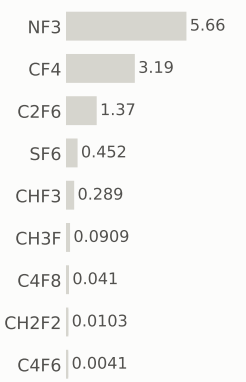
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



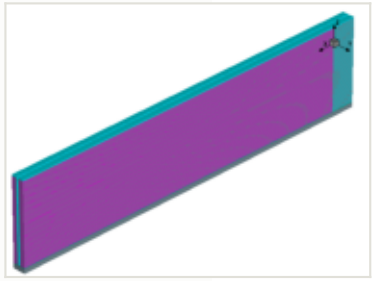
Step #10920: Deposition

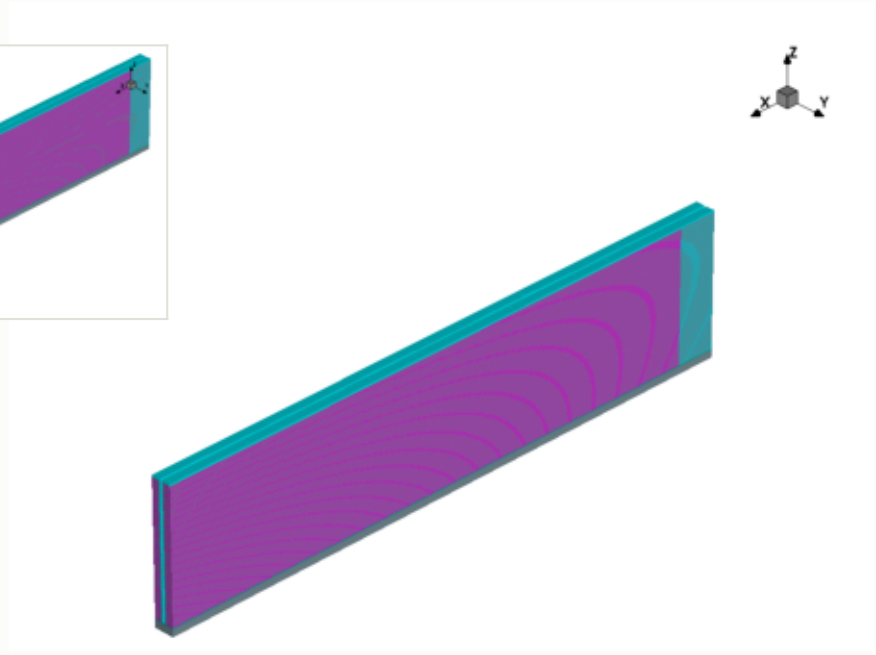
depo_isotropic · Deposition

0.656	657.6	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



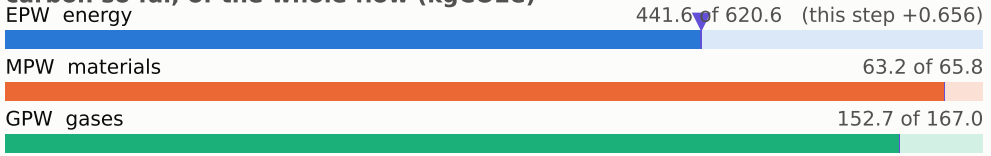


this step

process area	Deposition
material	—
thickness	0.015 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

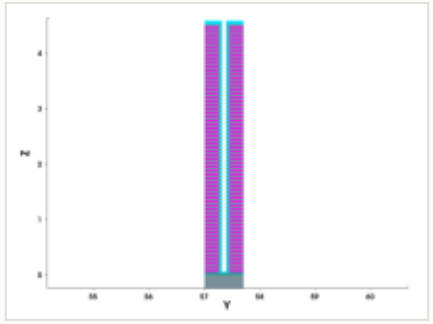
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

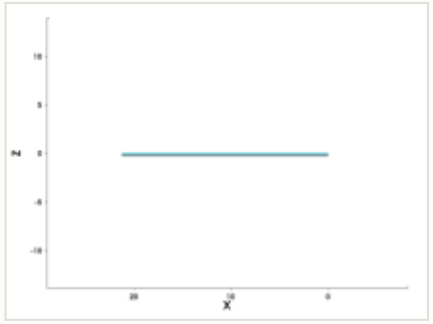


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



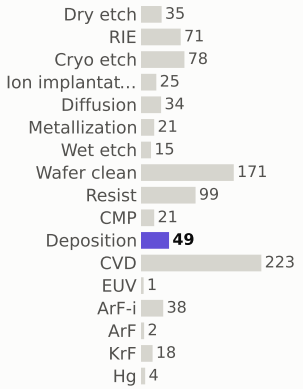
y cut at 57.351 µm



mask for this step

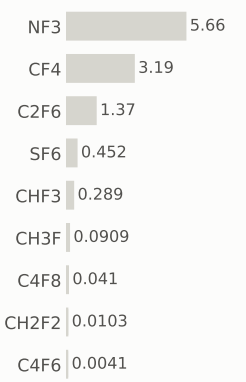
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



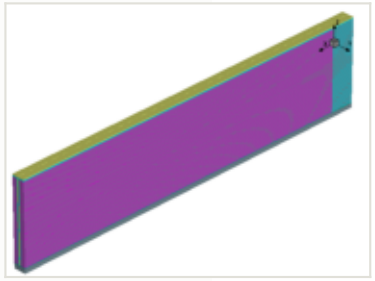
Step #10930: CVD

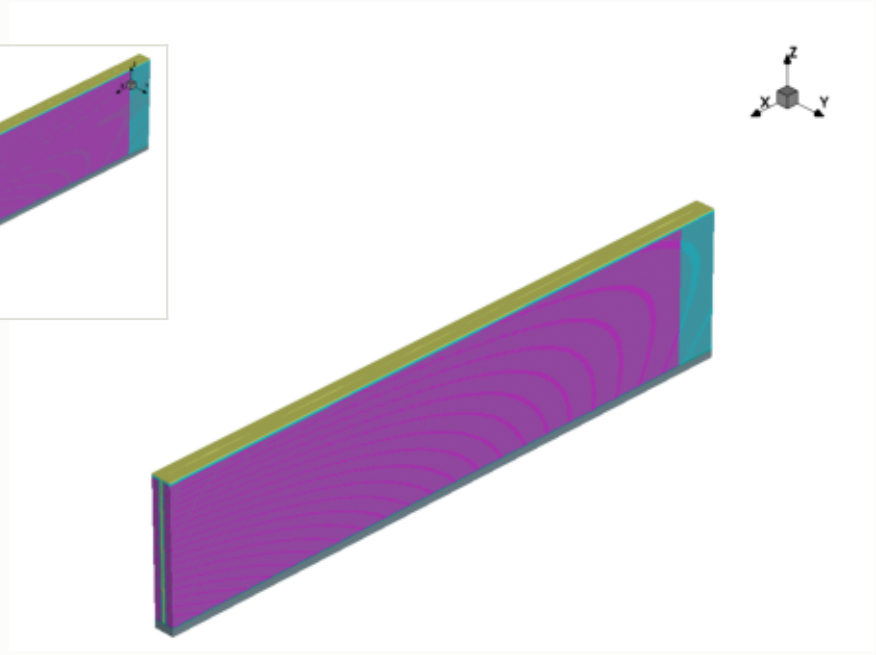
depo_isotropic · CVD

1.31	658.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



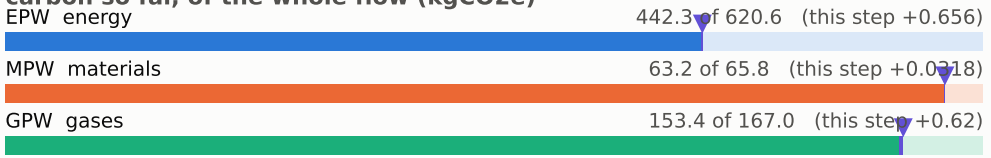


this step

process area	CVD
material	—
thickness	0.05 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

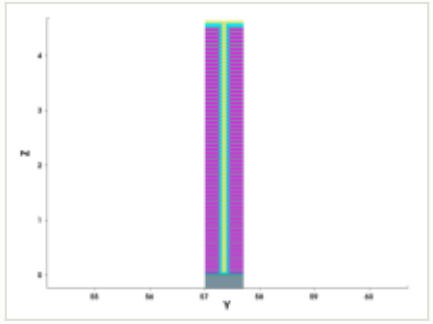
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

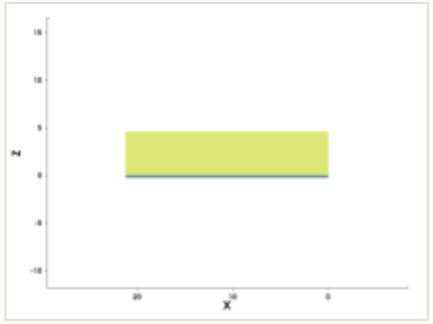


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



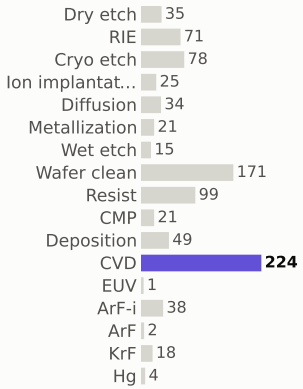
y cut at 57.351 µm



mask for this step

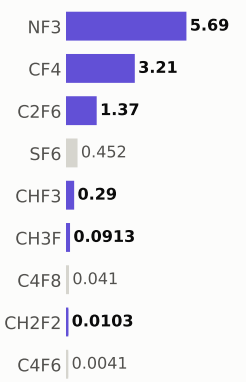
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



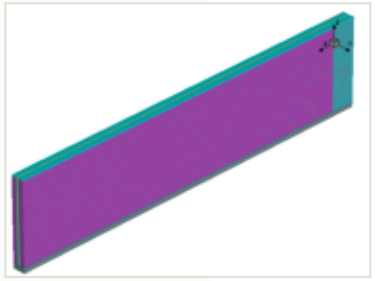
Step #10940: CMP

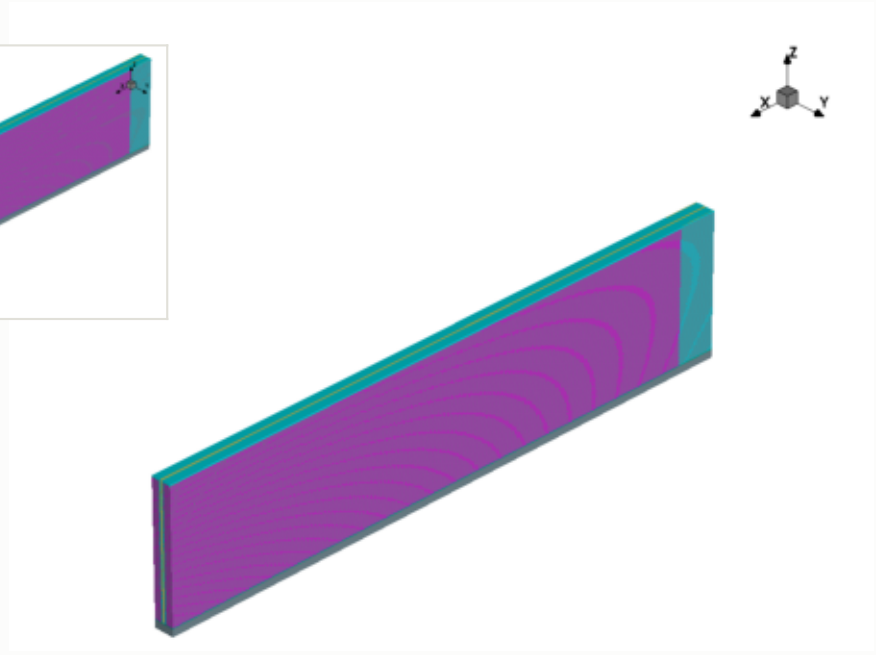
etch_cmp · CMP

0.169	659.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



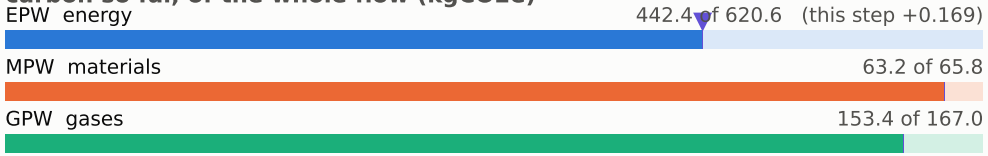


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

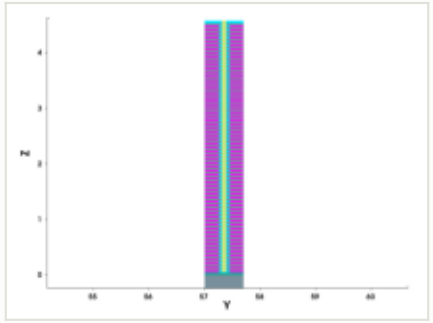
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

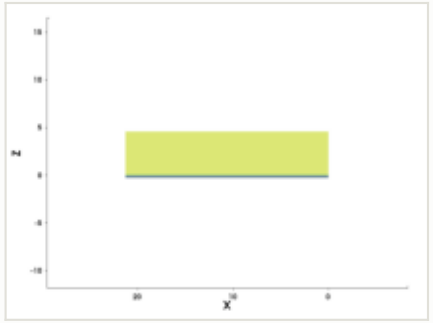


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



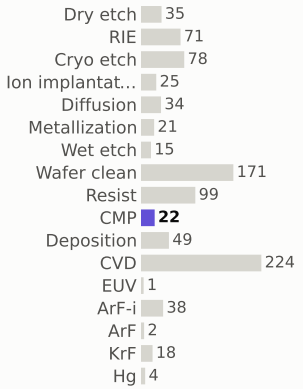
y cut at 57.351 µm



mask for this step

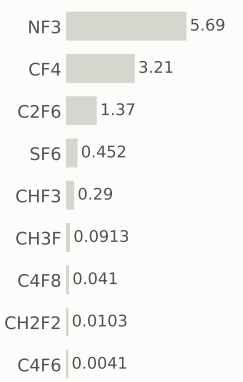
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



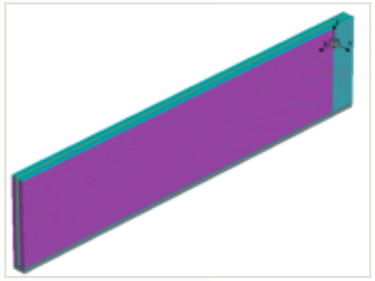
Step #10950: Wafer_clean

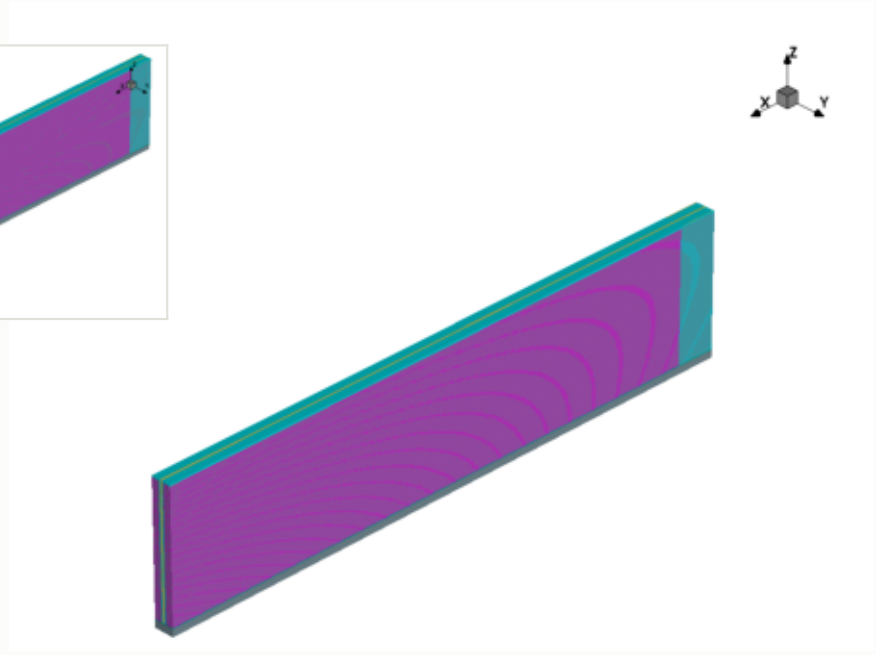
clean · Wafer_clean

0.169	659.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





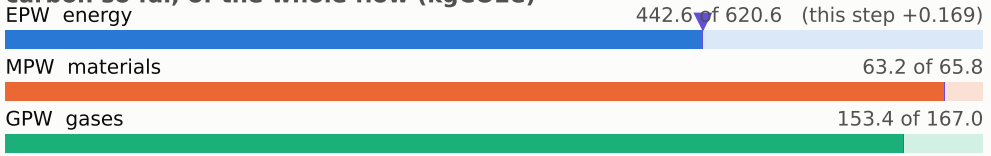
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

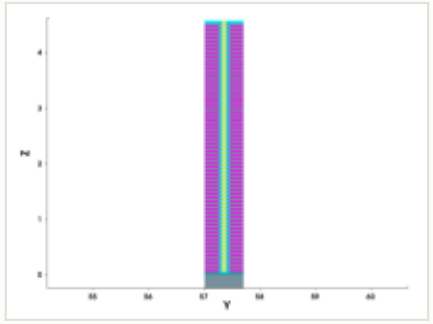
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

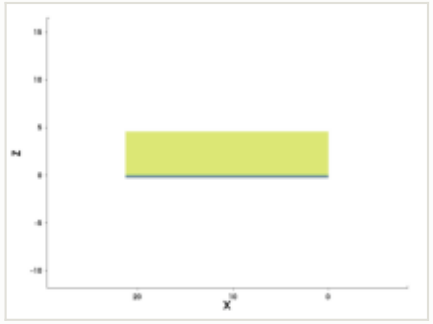


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



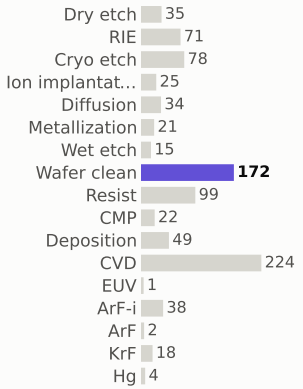
y cut at 57.351 µm



mask for this step

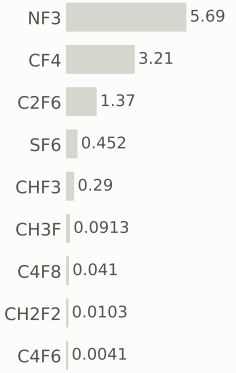
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



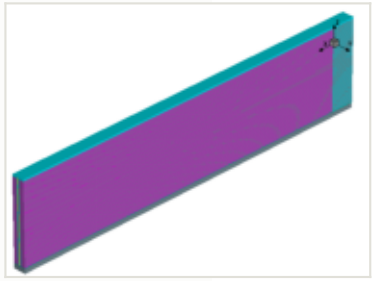
Step #10960: Deposition

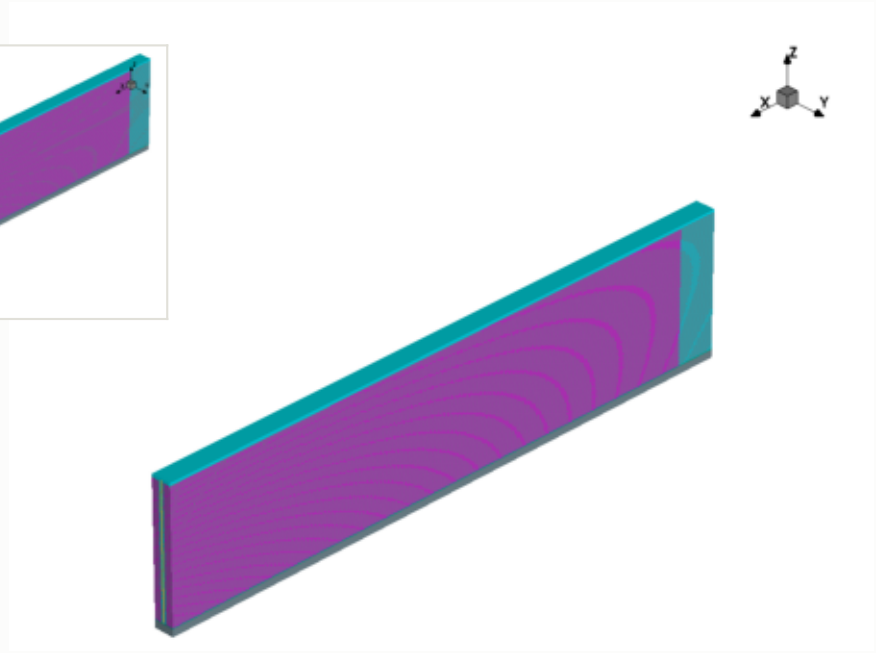
depo_isotropic · Deposition

0.656	659.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



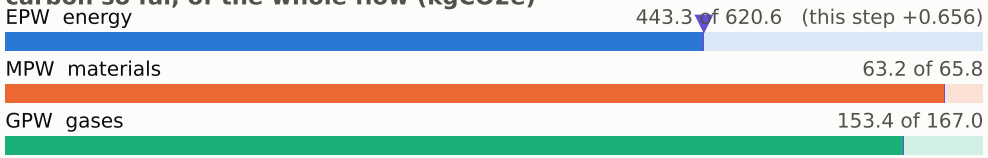


this step

process area	Deposition
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

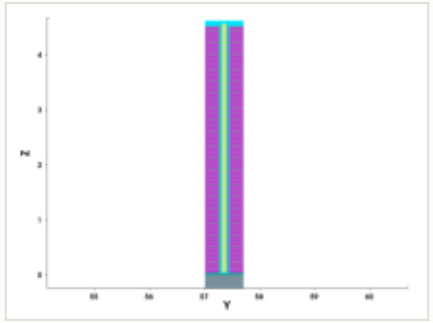
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

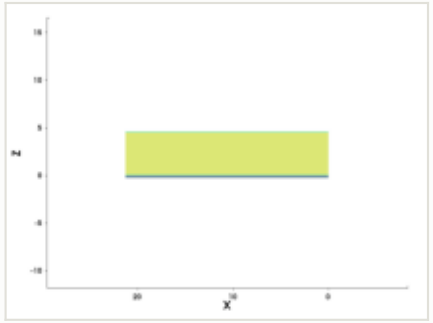


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



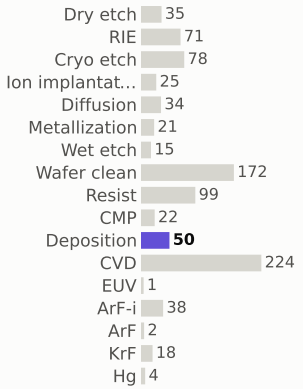
y cut at 57.351 µm



mask for this step

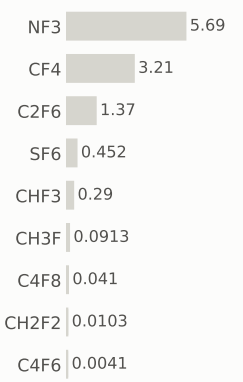
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



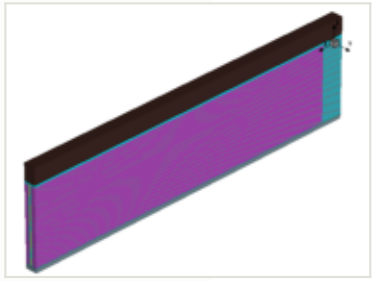
Step #10980: Deposition

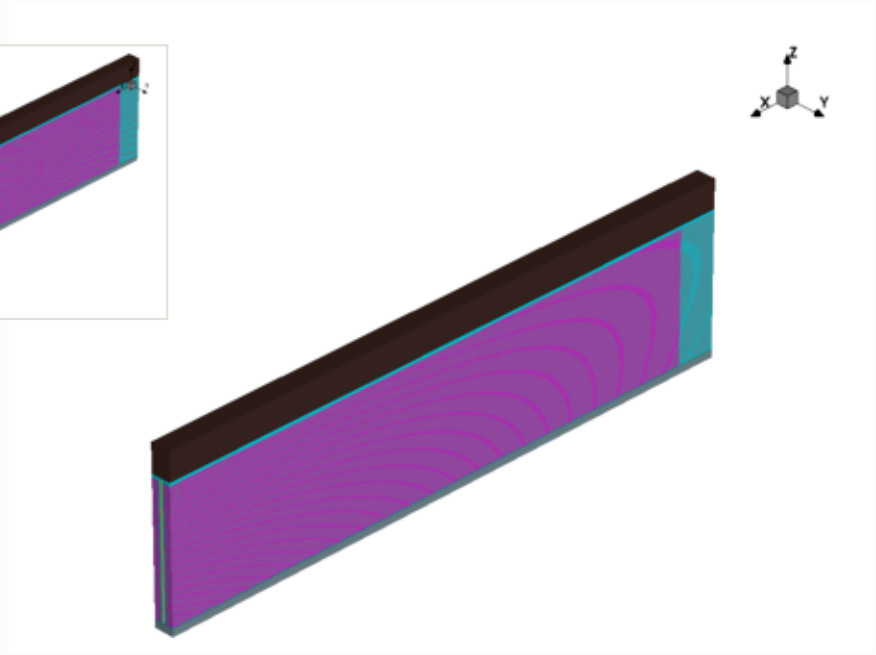
depo_isotropic · Deposition

0.656	660.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



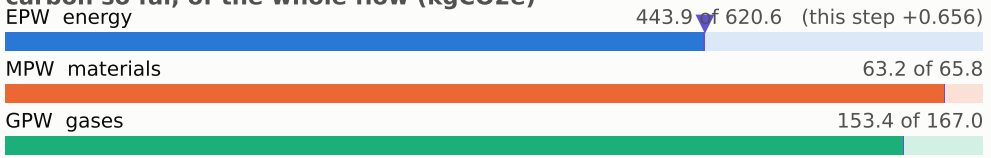


this step

process area	Deposition
material	—
thickness	1 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

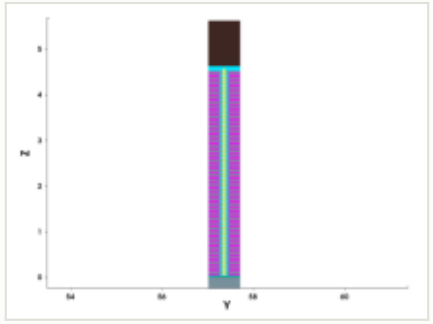
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

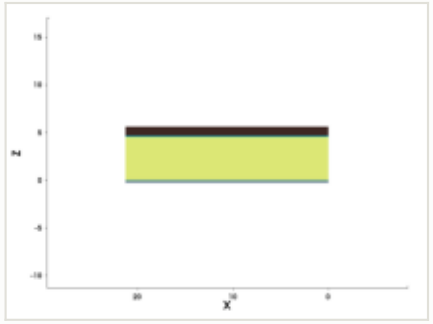


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



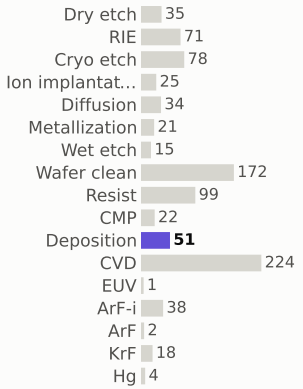
y cut at 57.351 µm



mask for this step

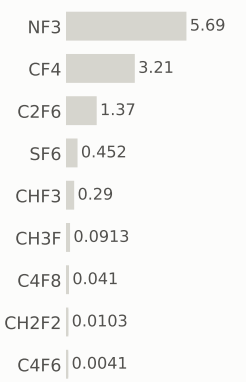
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



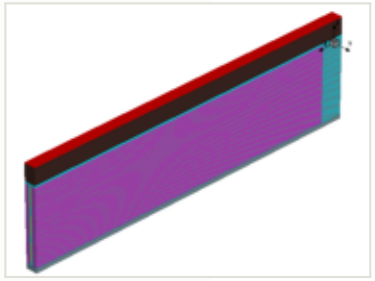
Step #10990: Deposition

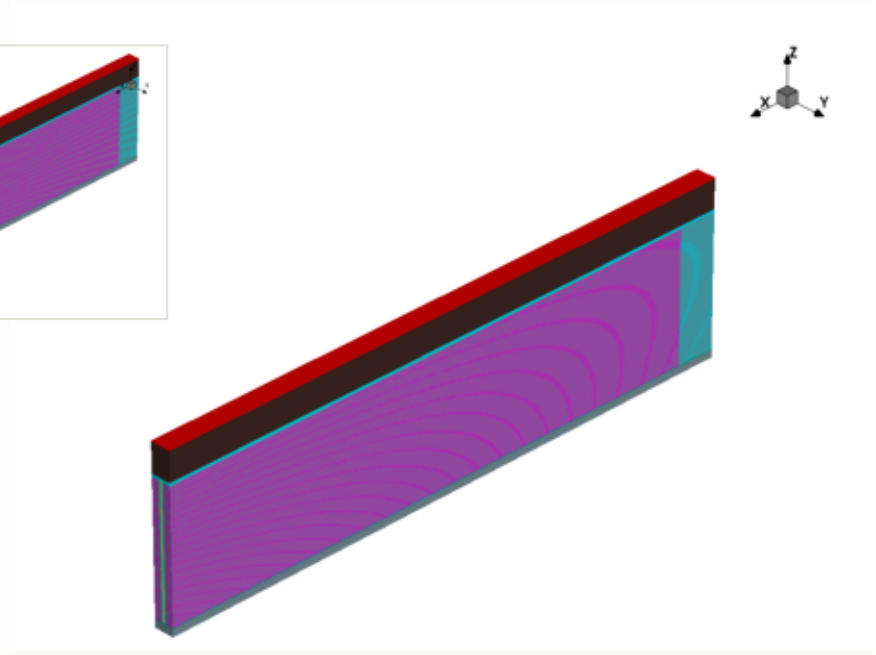
depo_isotropic · Deposition

0.656	661.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



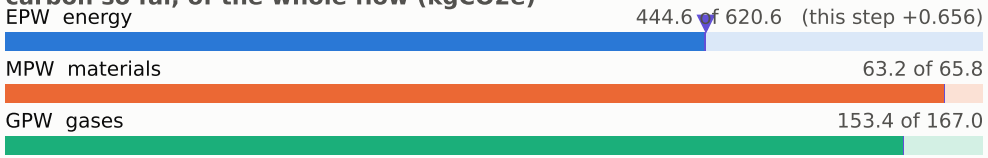


this step

process area	Deposition
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

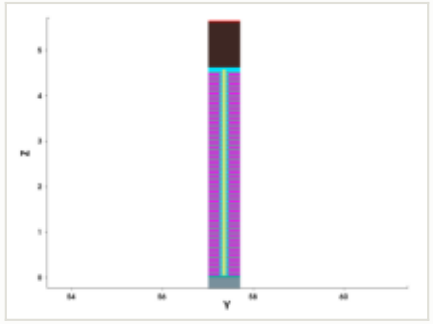
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

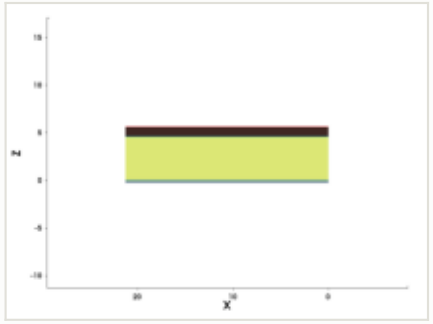


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



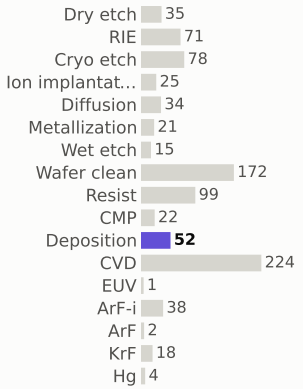
y cut at 57.351 µm



mask for this step

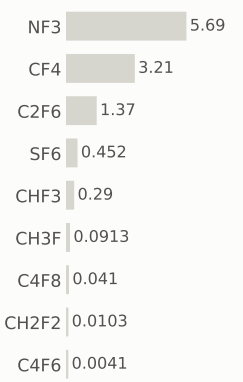
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



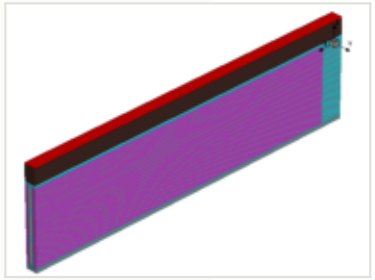
Step #11000: Wafer_clean

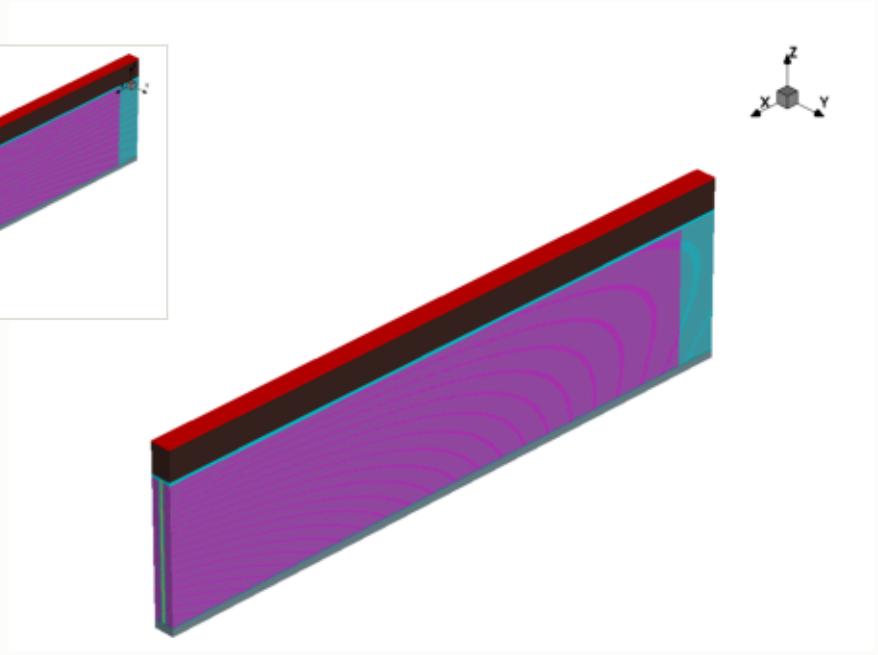
clean · Wafer_clean

0.169	661.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





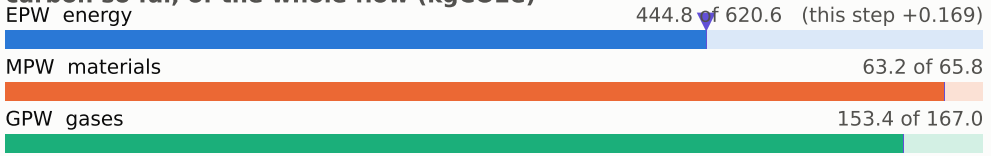
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

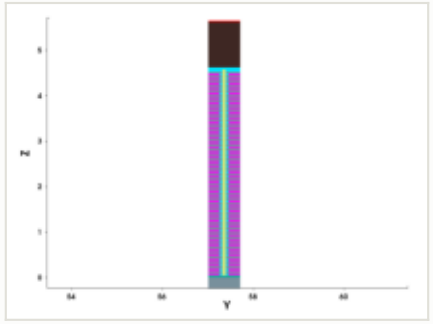
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

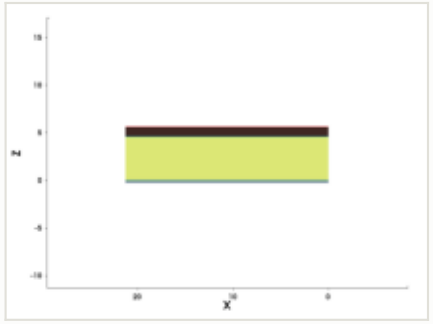


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



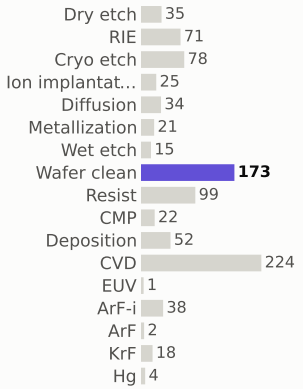
y cut at 57.351 µm



mask for this step

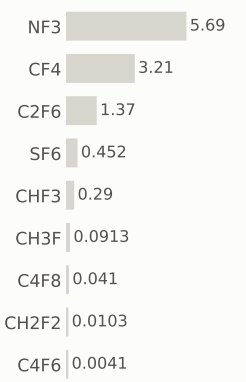
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



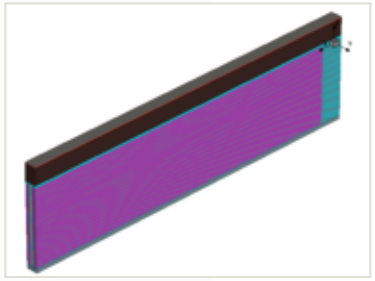
Step #11010: EUV

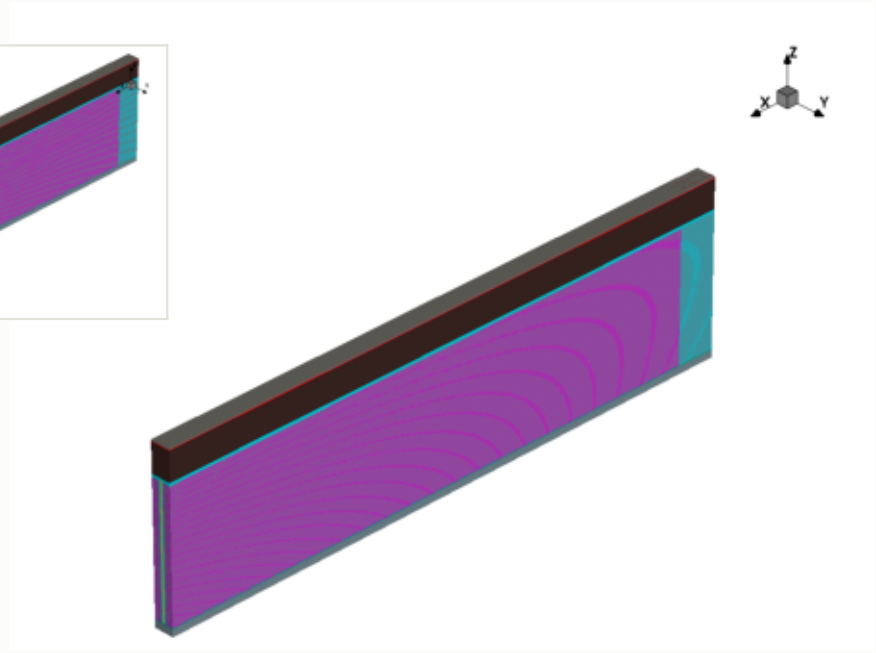
litho_patterning · EUV

1.96	663.3	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



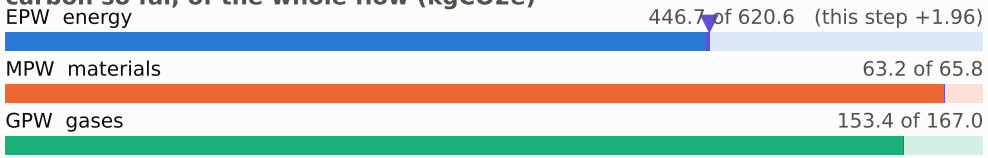


this step

process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

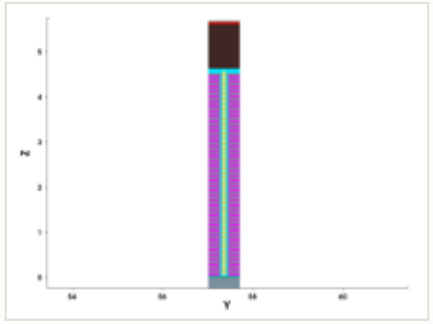
EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

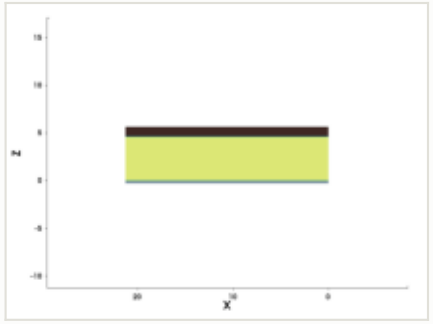


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

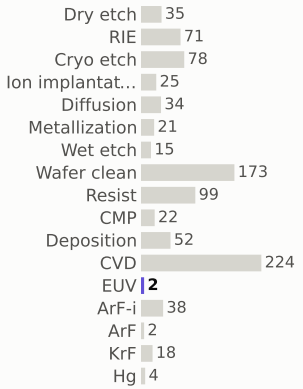


y cut at 57.351 µm



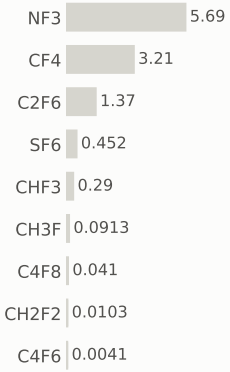
mask: layer 4:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



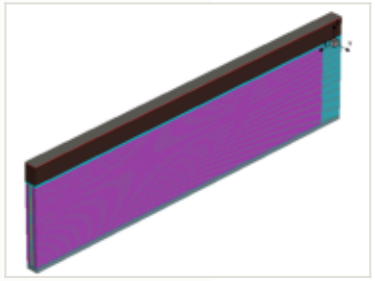
Step #11020: RIE

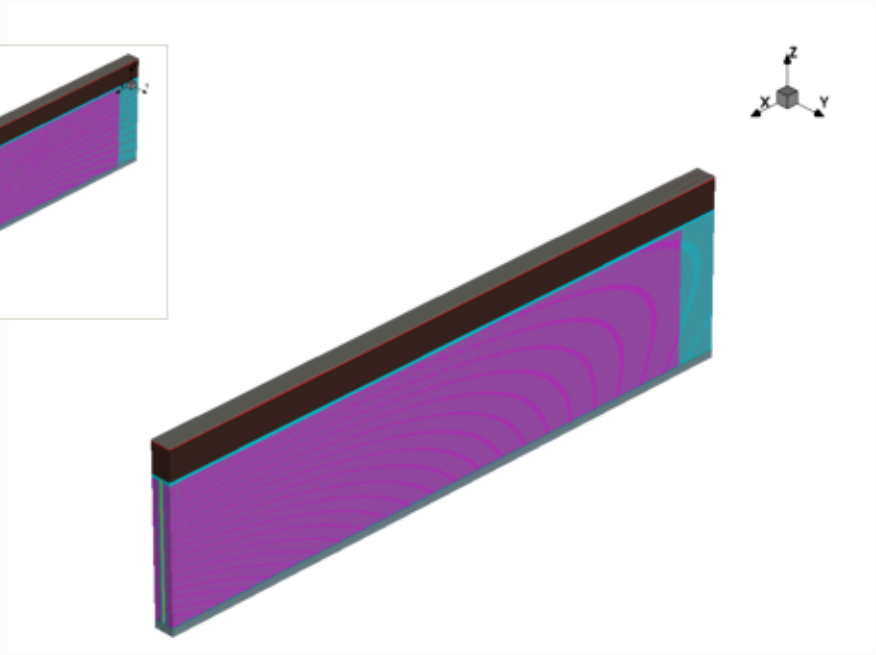
etch_anisotropic · RIE

1.04	664.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



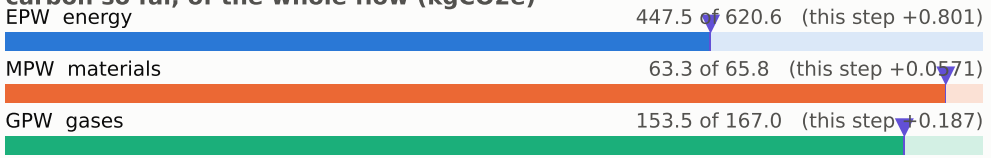


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

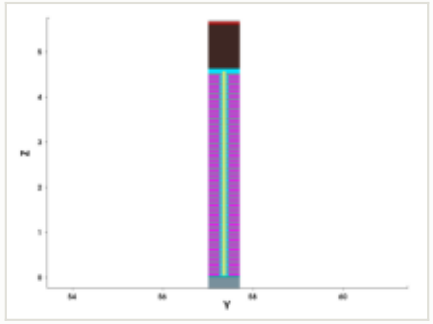
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

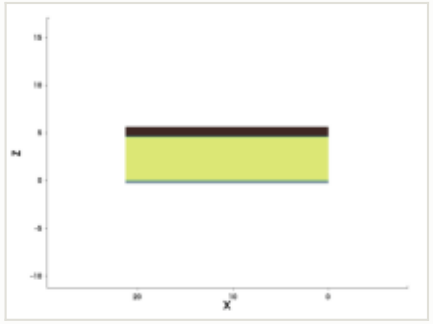


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



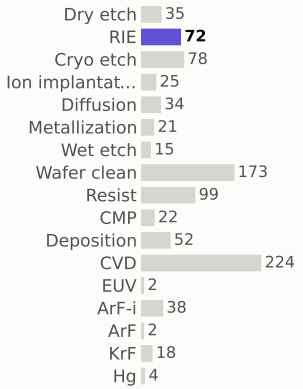
y cut at 57.351 µm



mask for this step

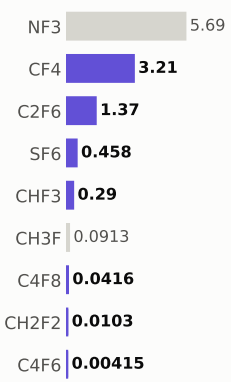
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



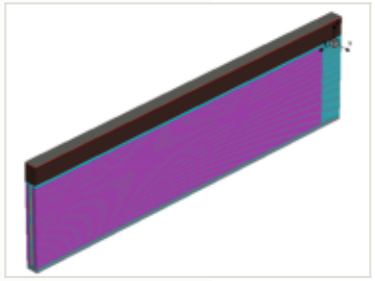
Step #11030: RIE

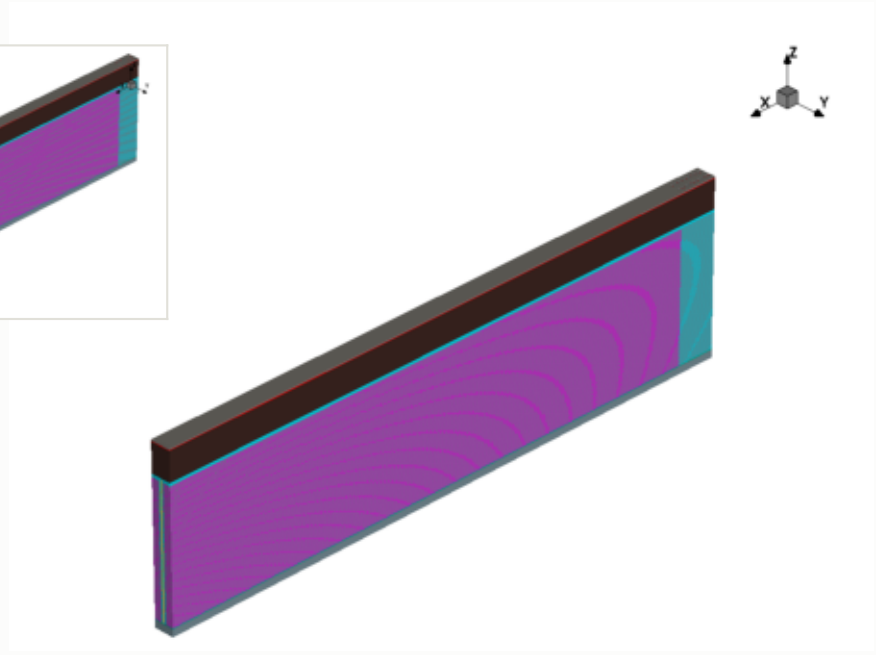
etch_anisotropic · RIE

1.04	665.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



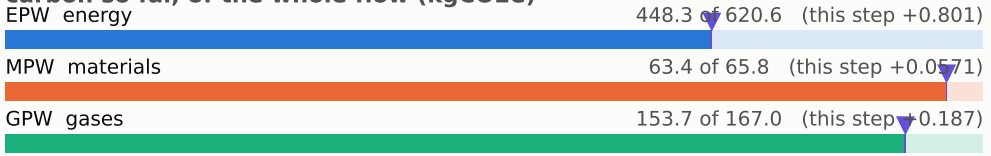


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

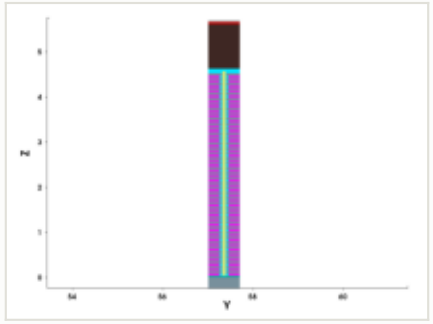
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

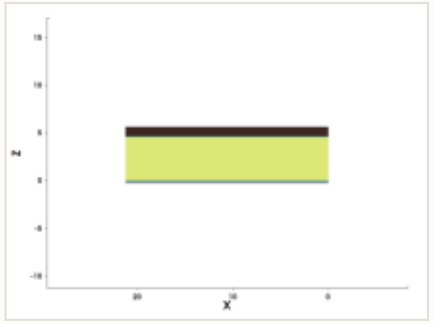


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



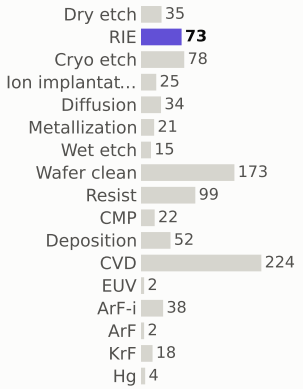
y cut at 57.351 µm



mask for this step

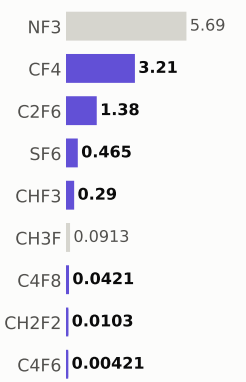
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



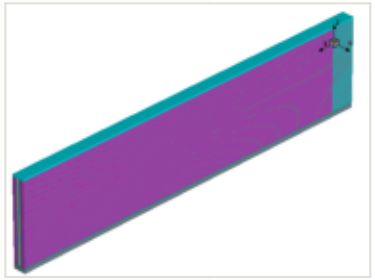
Step #11040: Dry_etch

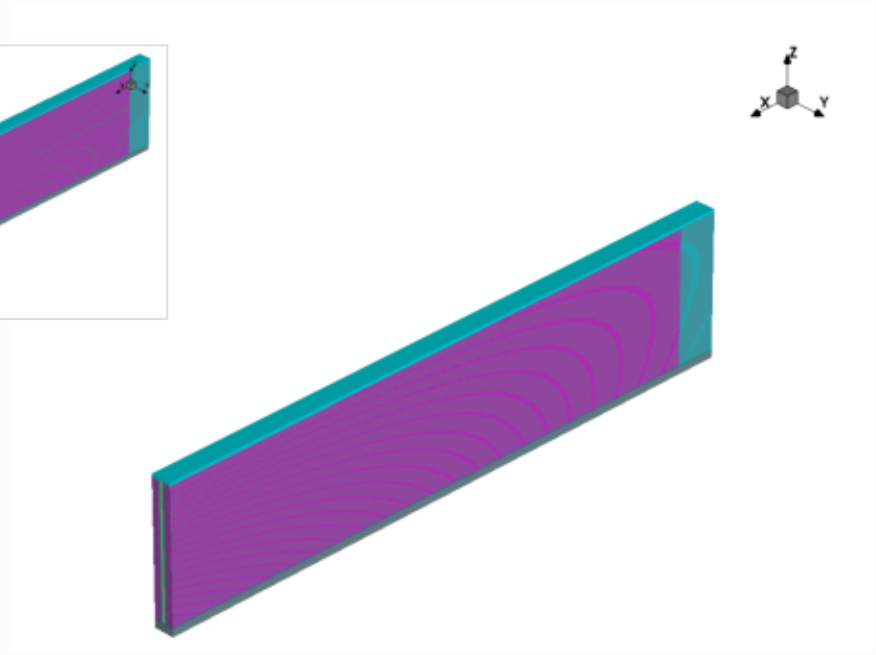
etch_strip · Dry_etch

0.801	666.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



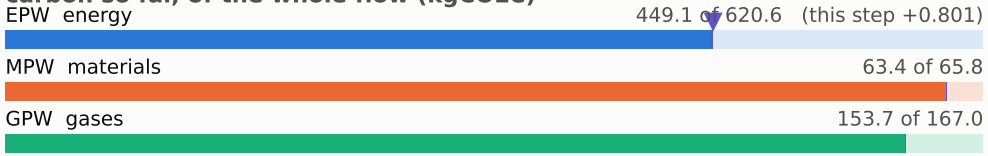


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

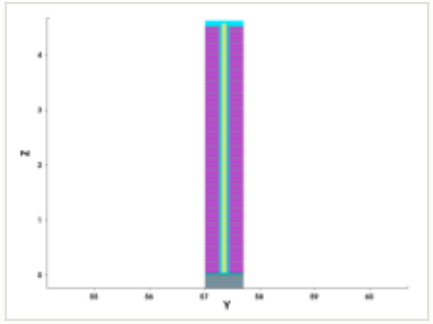
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

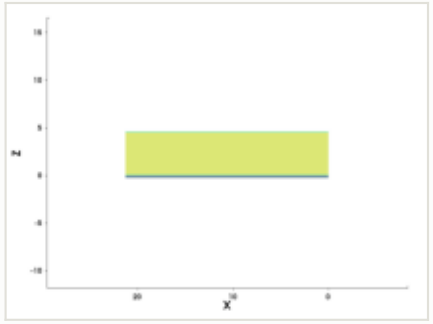


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



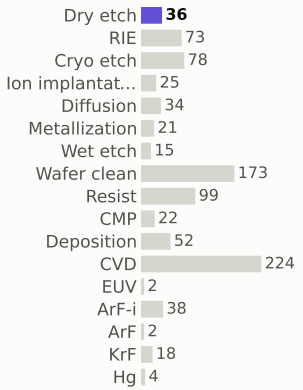
y cut at 57.351 µm



mask for this step

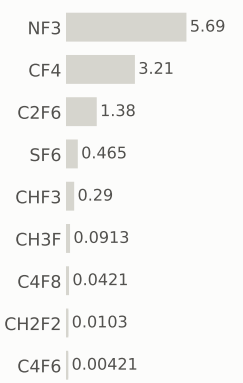
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



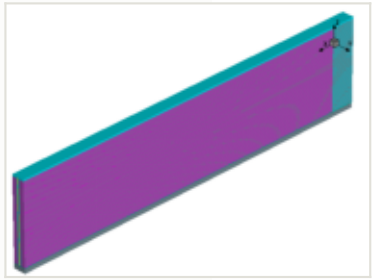
Step #11050: Wafer_clean

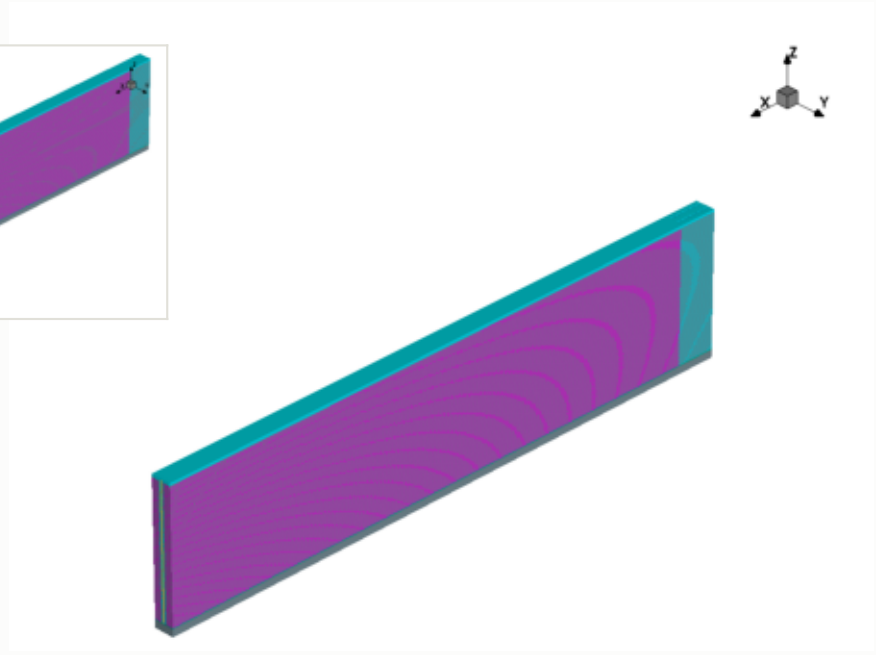
clean · Wafer_clean

0.169	666.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





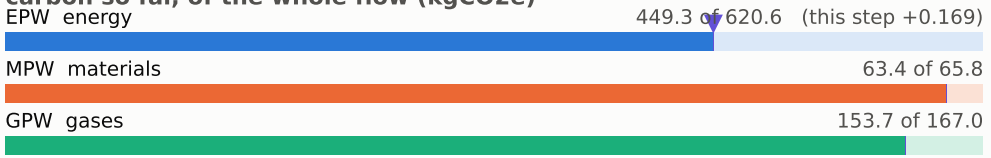
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

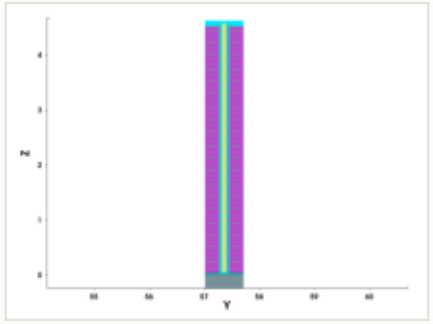
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

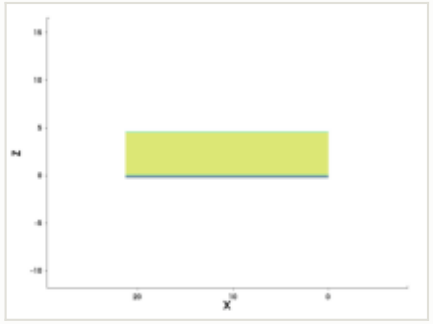


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



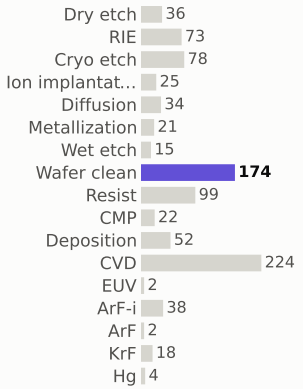
y cut at 57.351 µm



mask for this step

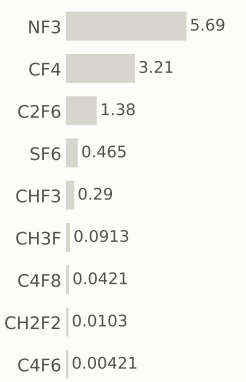
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



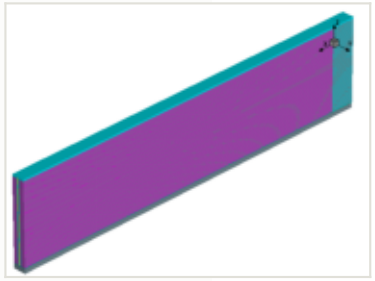
Step #11060: Wet_etch

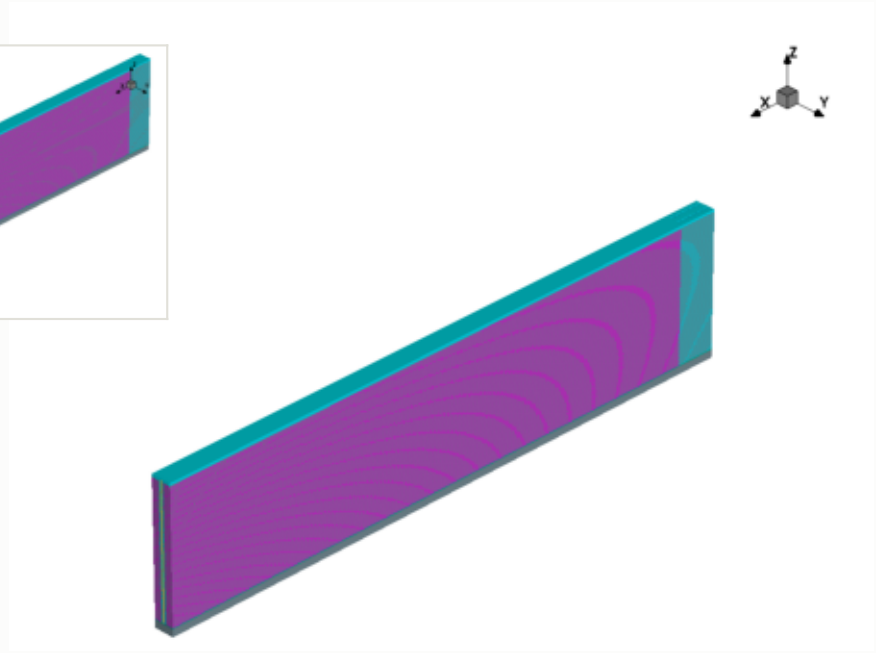
etch_isotropic · Wet_etch

0.169	666.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



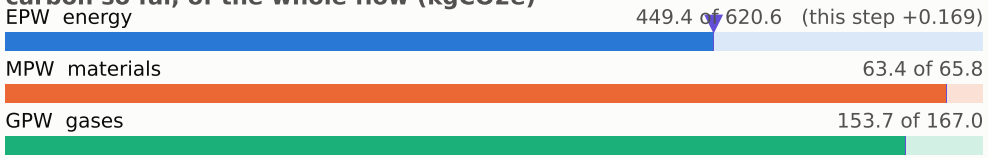


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

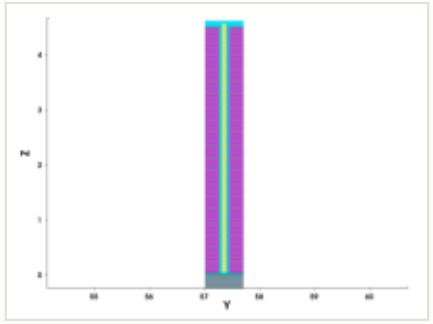
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

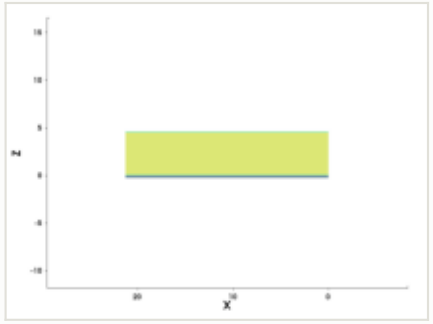


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



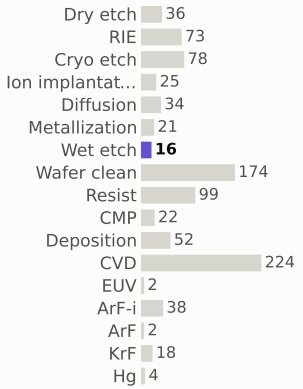
y cut at 57.351 µm



mask for this step

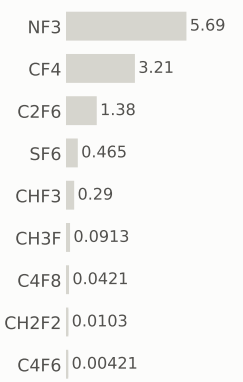
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



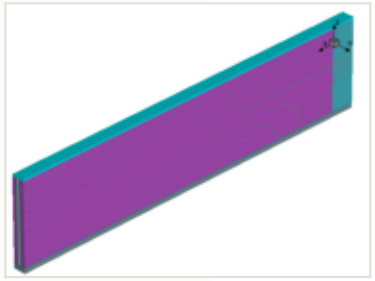
Step #11070: Wet_etch

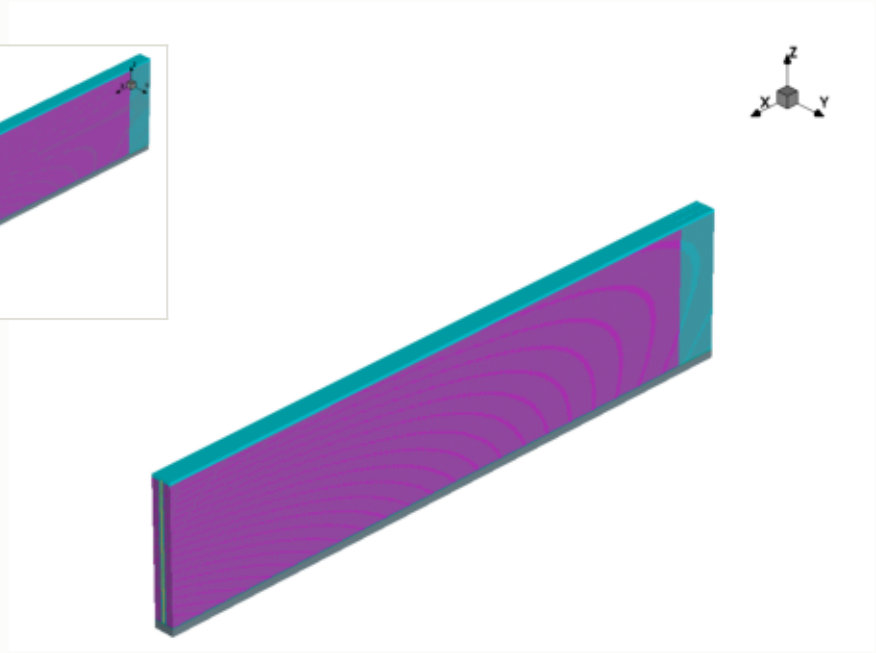
etch_isotropic · Wet_etch

0.169	666.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



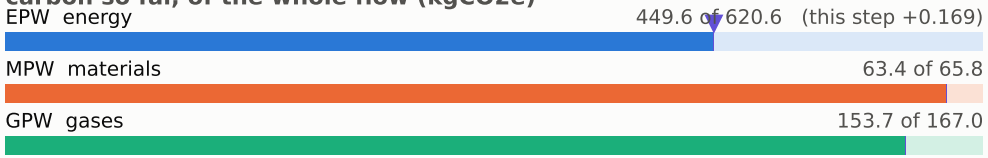


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

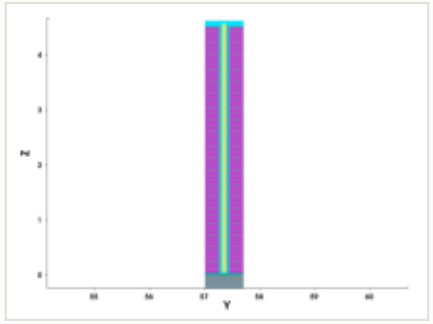
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

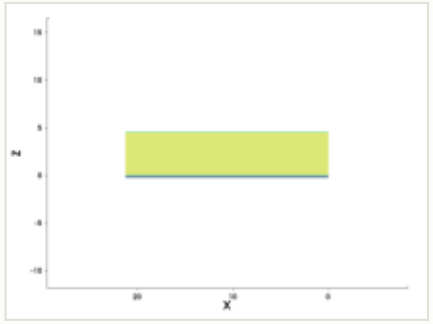


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



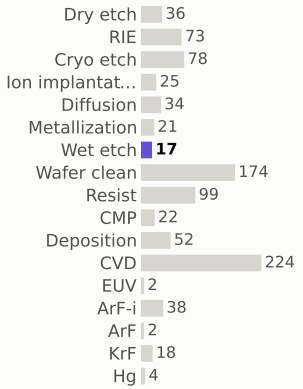
y cut at 57.351 µm



mask for this step

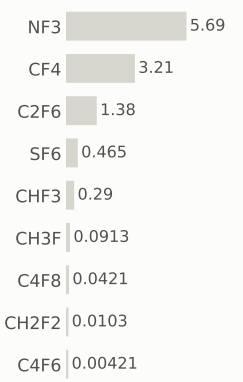
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



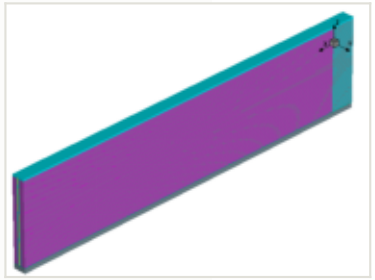
Step #11080: Wafer_clean

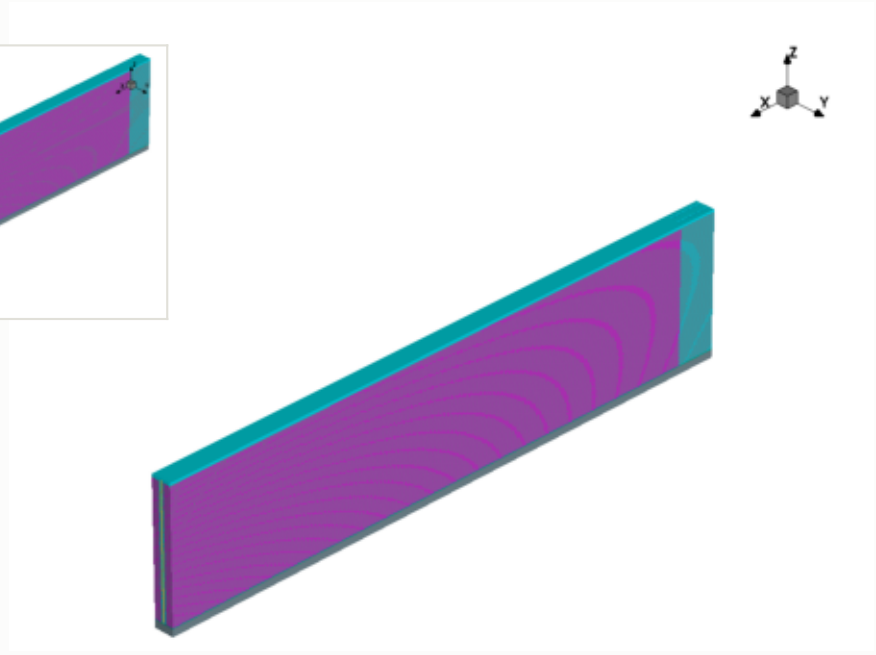
clean · Wafer_clean

0.169	666.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





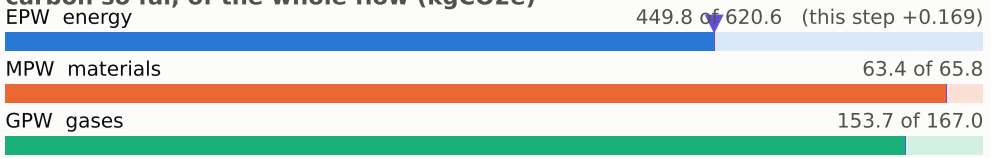
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

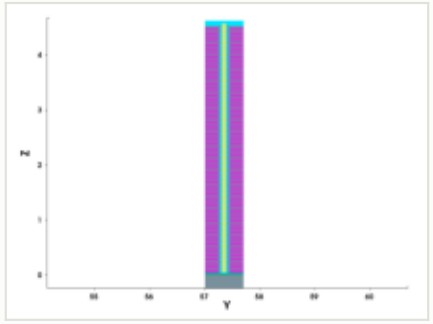
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

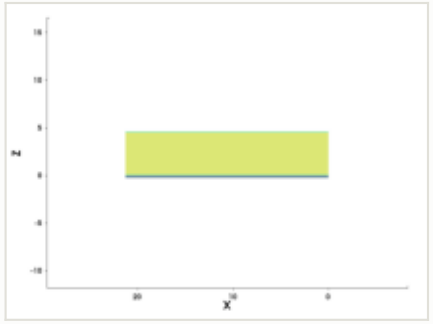


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



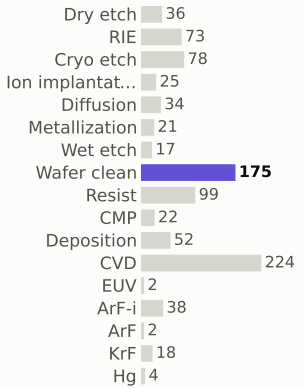
y cut at 57.351 µm



mask for this step

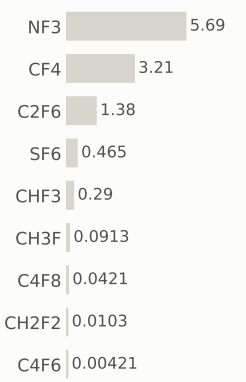
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



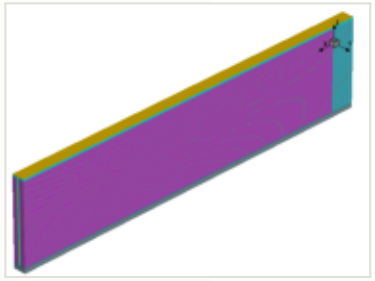
Step #11090: Deposition

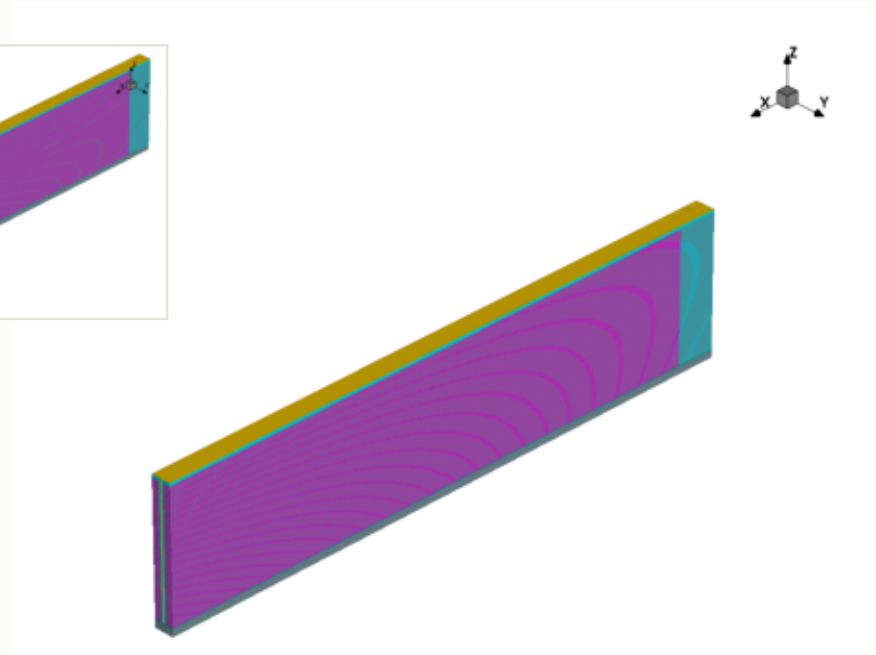
depo_isotropic · Deposition

0.656	667.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



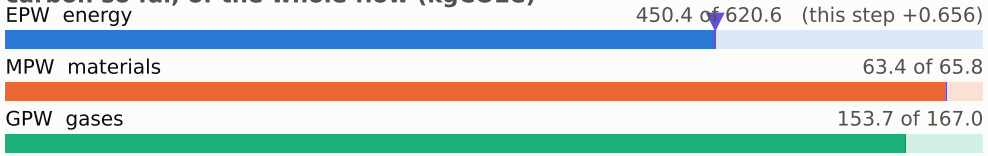


this step

process area	Deposition
material	—
thickness	0.005 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

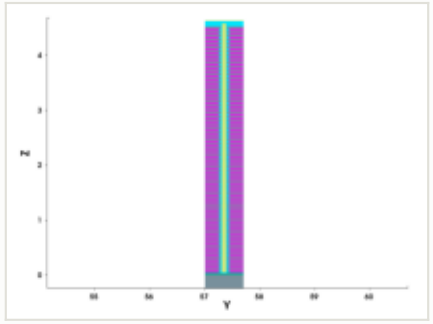
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

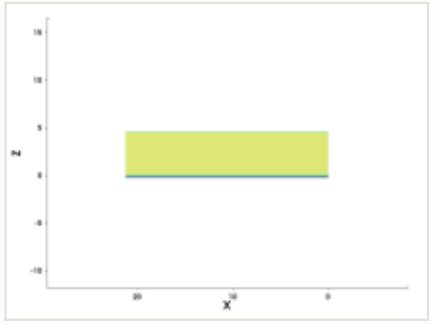


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



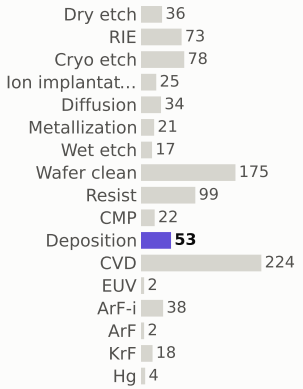
y cut at 57.351 µm



mask for this step

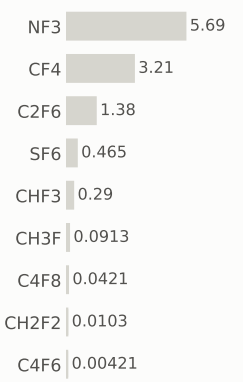
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



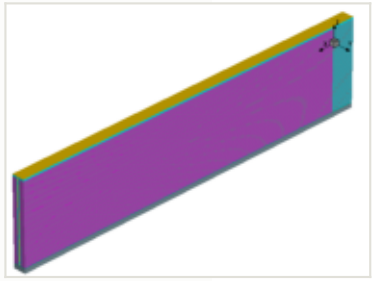
Step #11100: Diffusion

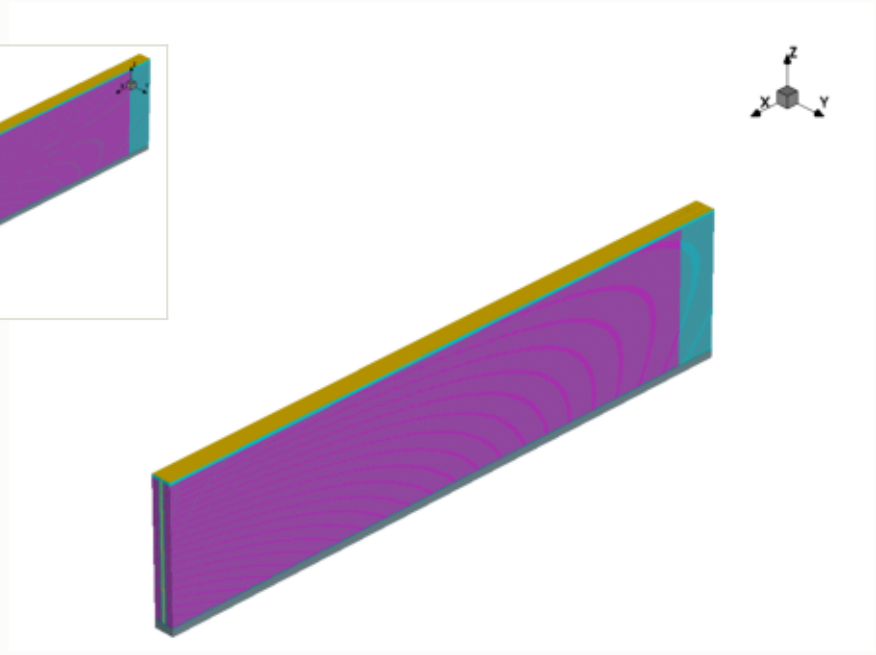
anneal · Diffusion

0.801	668.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





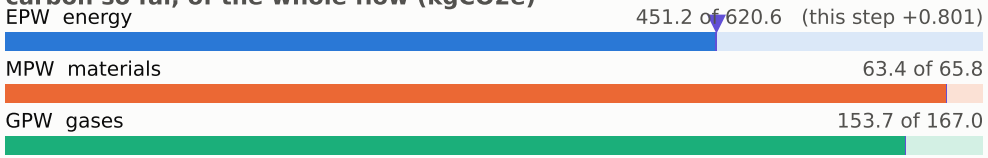
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

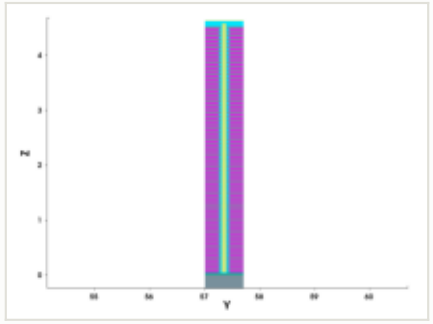
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

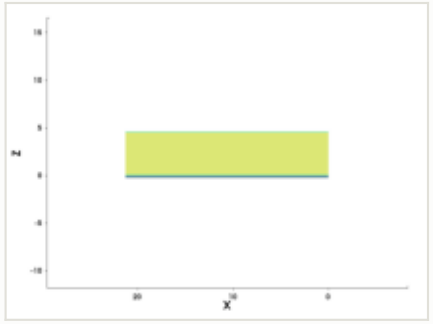


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



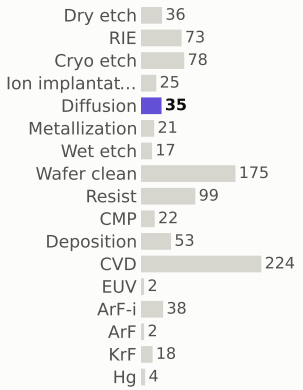
y cut at 57.351 µm



mask for this step

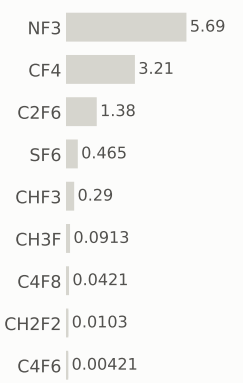
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



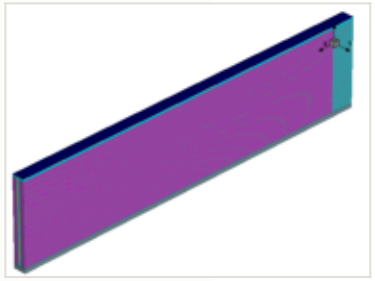
Step #11110: Metallization

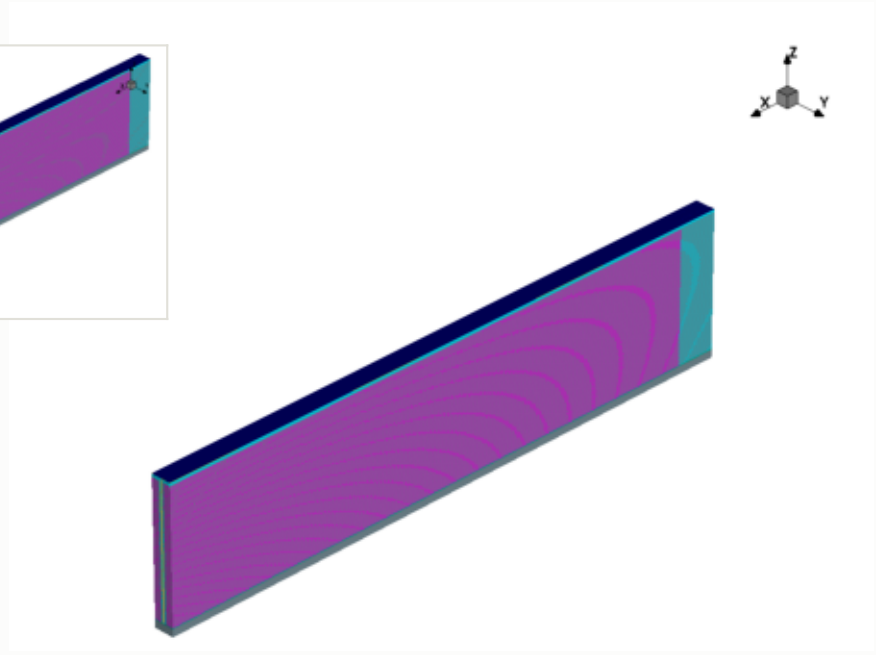
depo_isotropic · Metallization

1.2	669.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



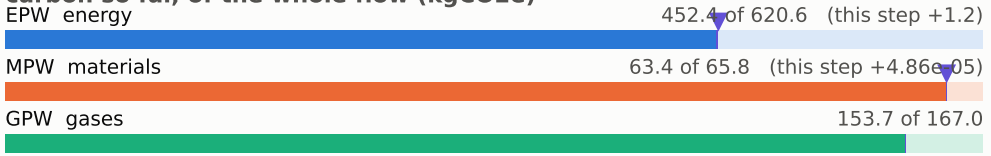


this step

process area	Metallization
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

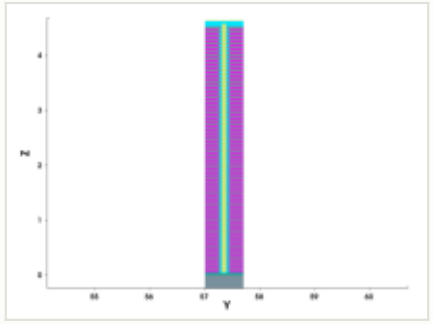
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

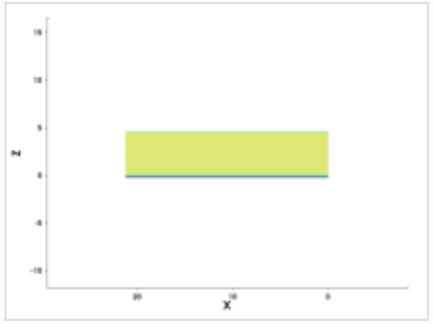


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



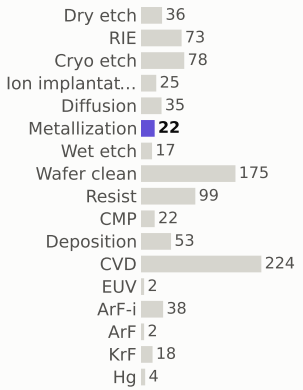
y cut at 57.351 µm



mask for this step

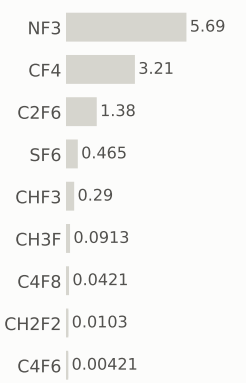
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



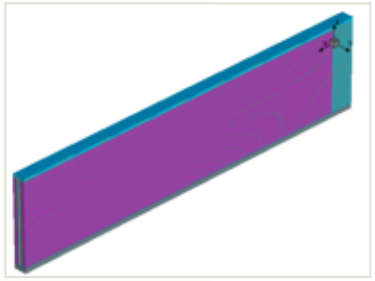
Step #11120: Metallization

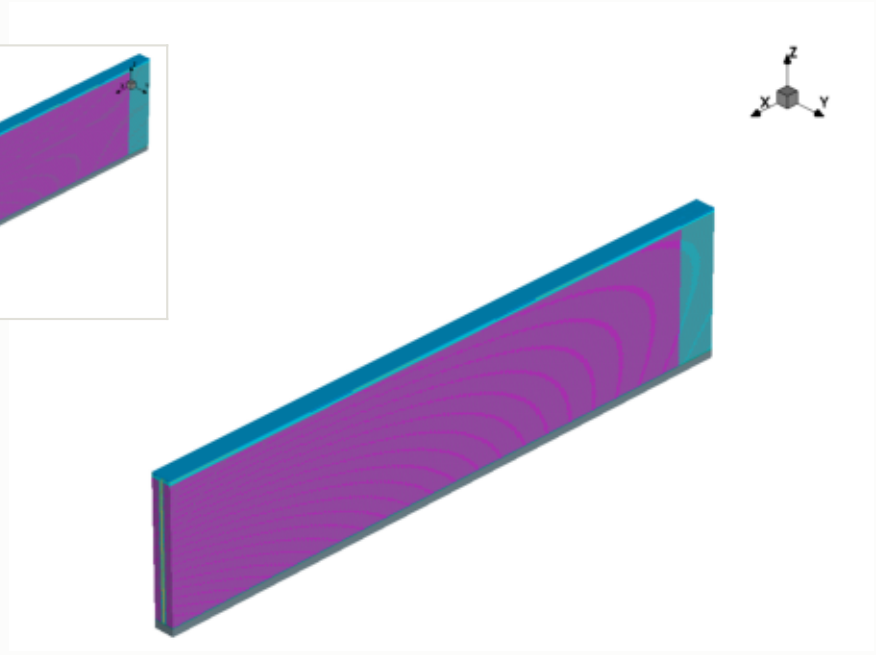
depo_isotropic · Metallization

1.21	670.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



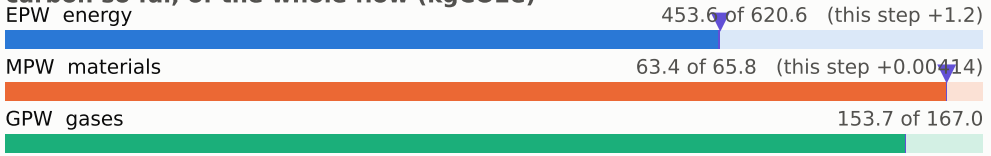


this step

process area	Metallization
material	W
thickness	0.04 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

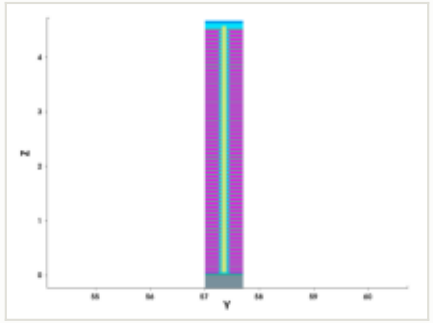
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00414 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

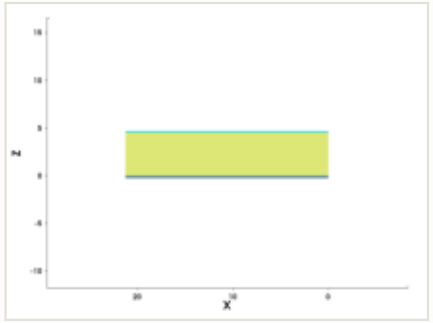


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



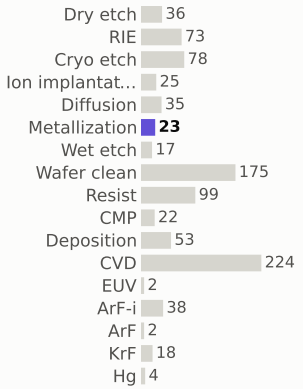
y cut at 57.351 µm



mask for this step

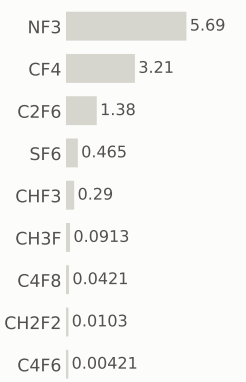
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



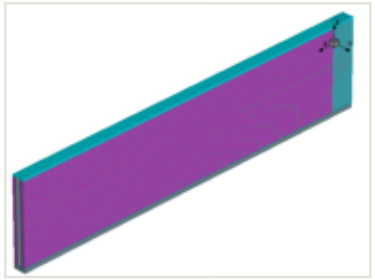
Step #11130: CMP

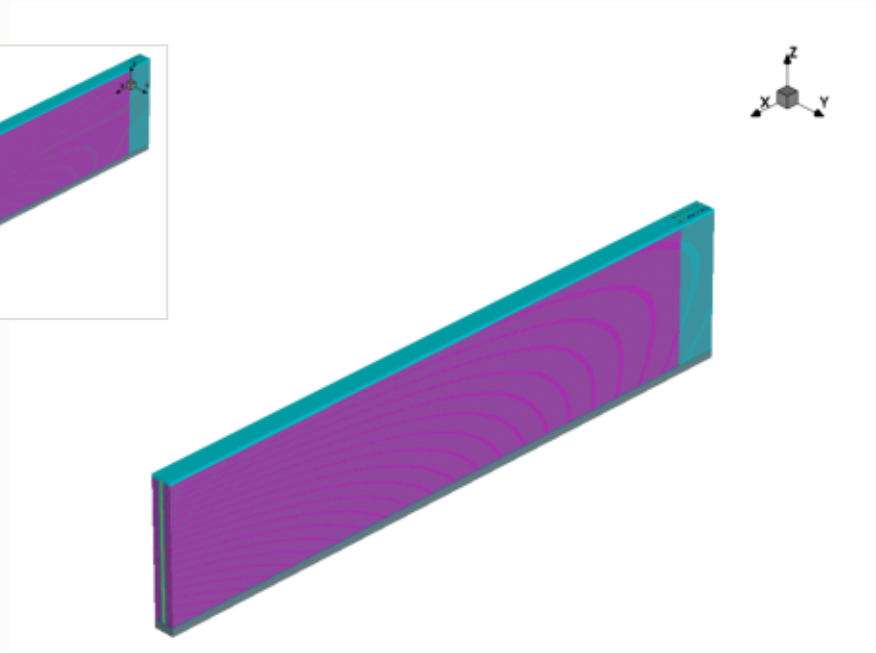
etch_cmp · CMP

0.169	670.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



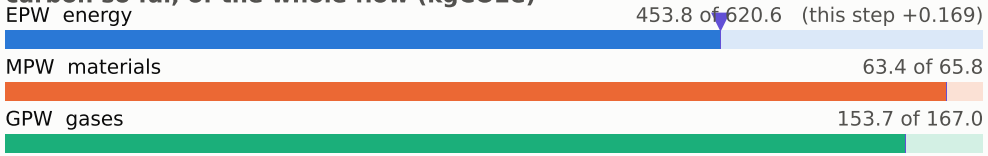


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

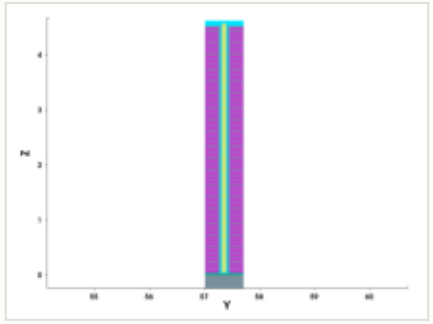
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

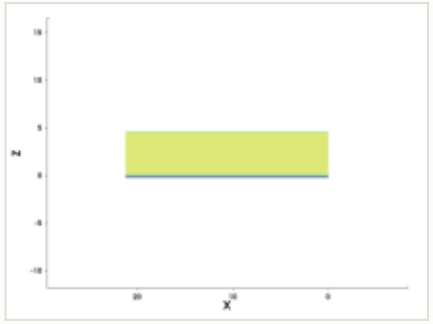


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



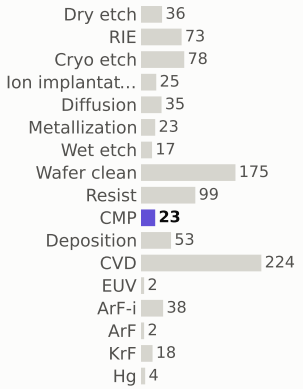
y cut at 57.351 µm



mask for this step

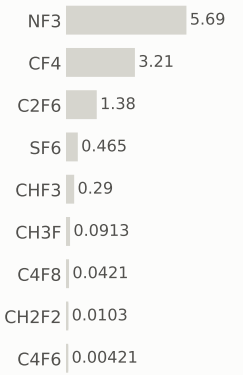
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



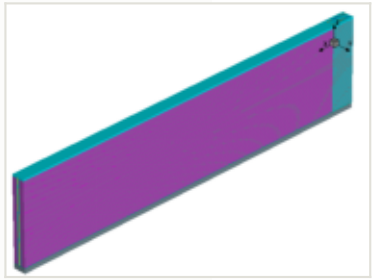
Step #11140: Wafer_clean

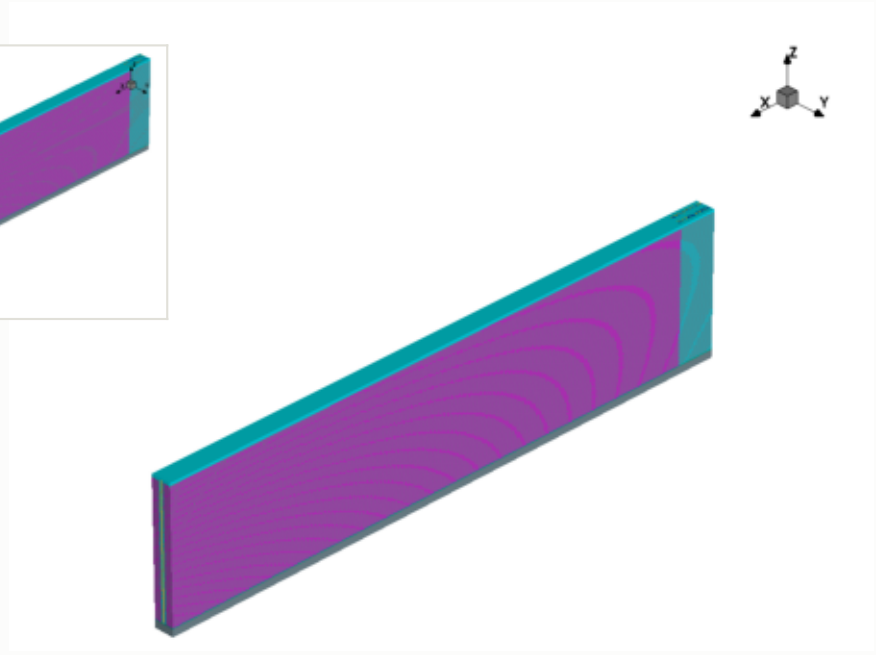
clean · Wafer_clean

0.169	671.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





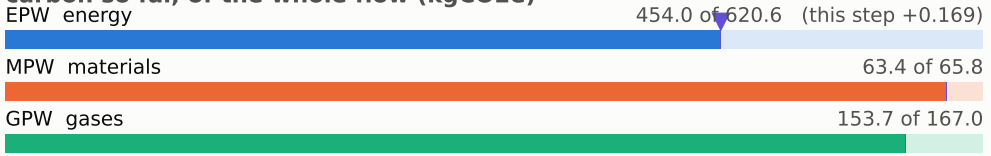
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

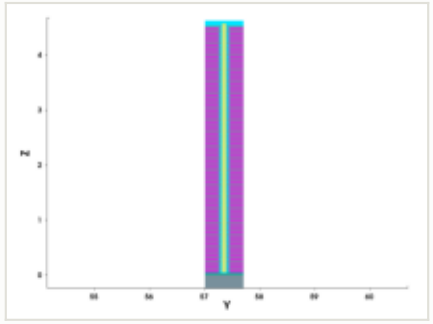
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

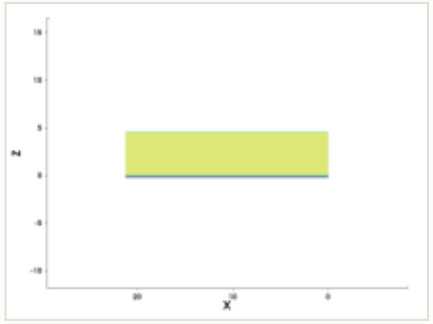


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



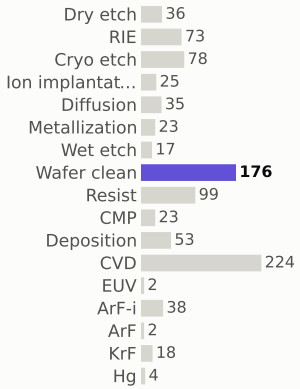
y cut at 57.351 µm



mask for this step

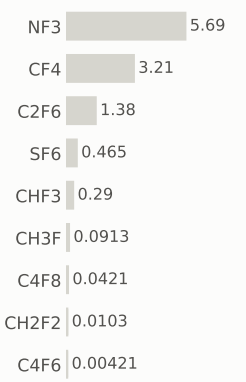
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



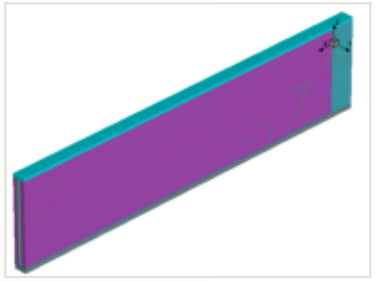
Step #11150: Deposition

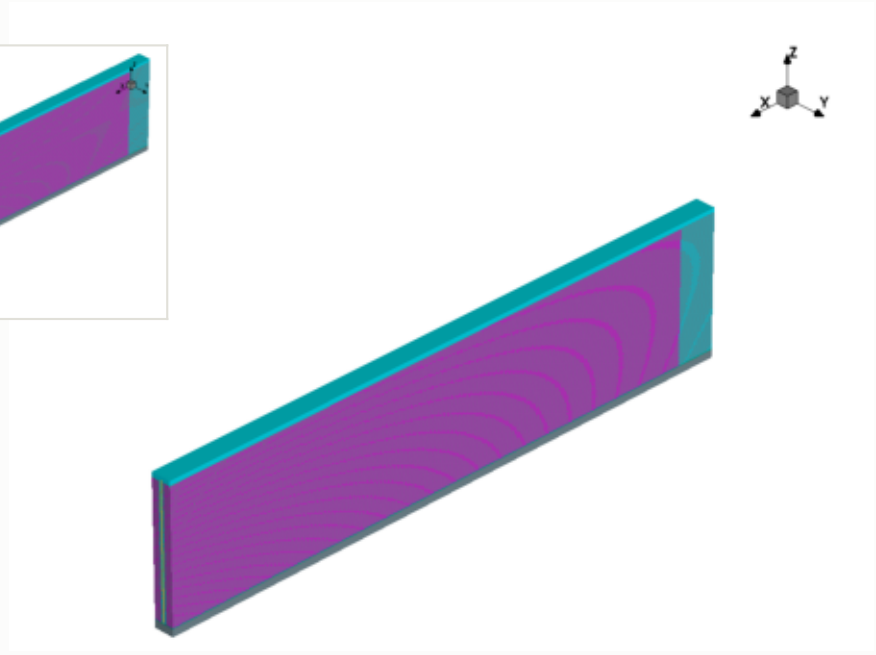
depo_isotropic · Deposition

0.656	671.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



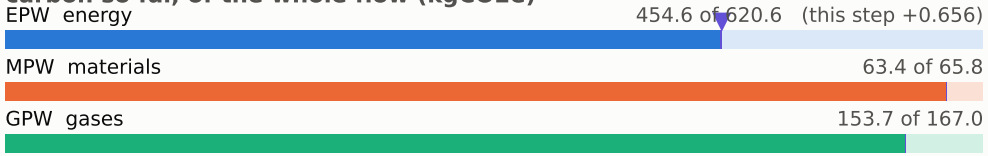


this step

process area	Deposition
material	—
thickness	0.06 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

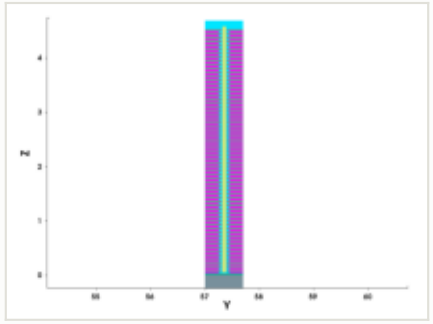
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

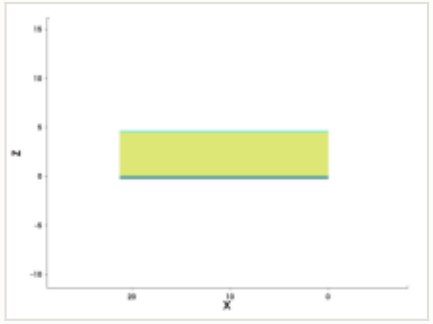


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



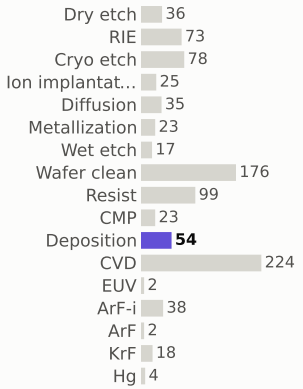
y cut at 57.351 µm



mask for this step

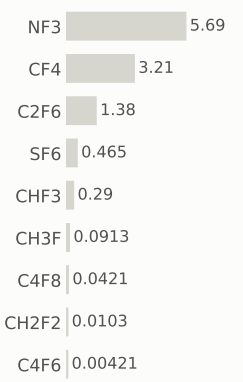
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



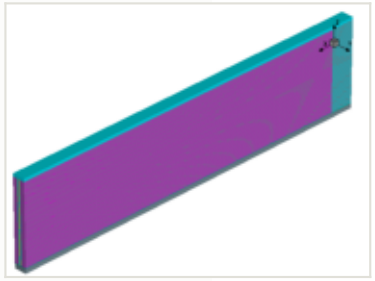
Step #11170: Wafer_clean

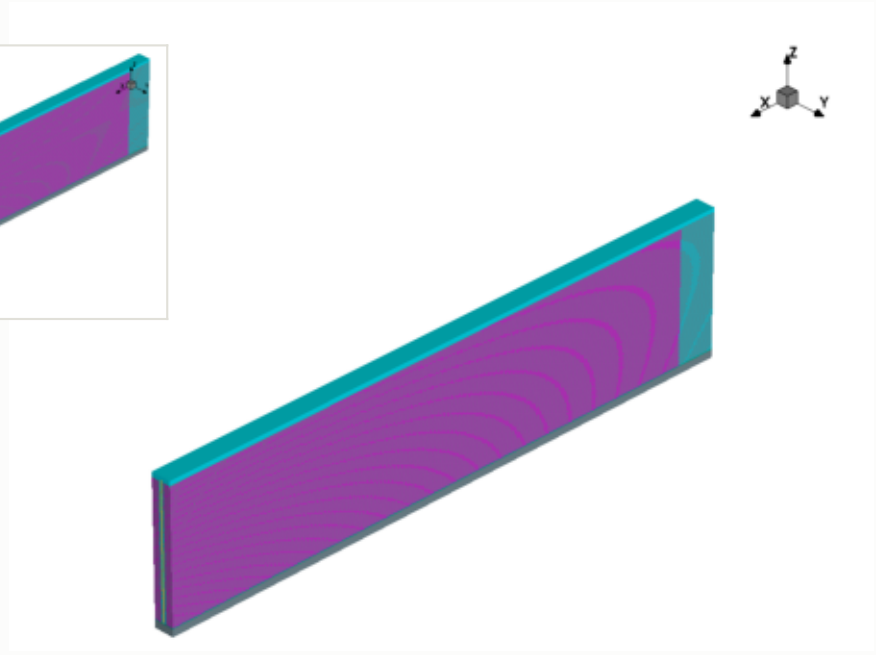
clean · Wafer_clean

0.169	671.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





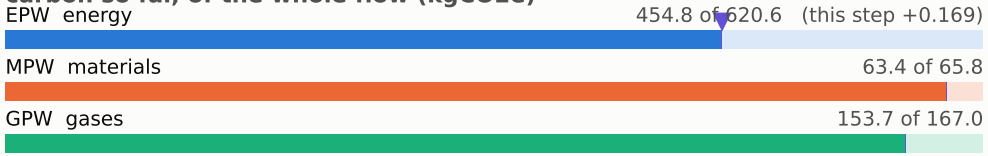
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

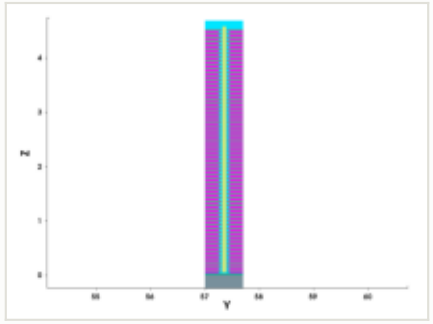
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

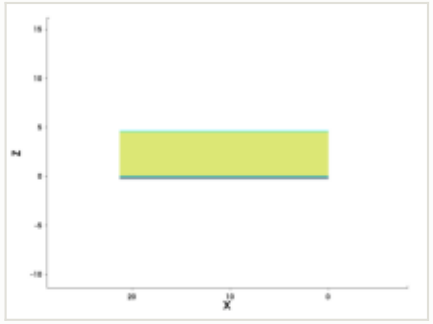


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



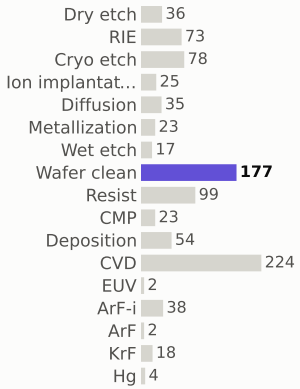
y cut at 57.351 µm



mask for this step

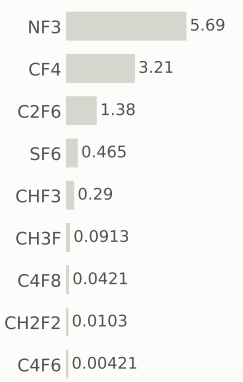
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



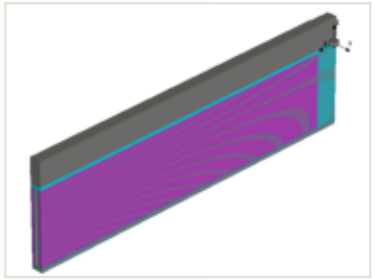
Step #11180: KrF

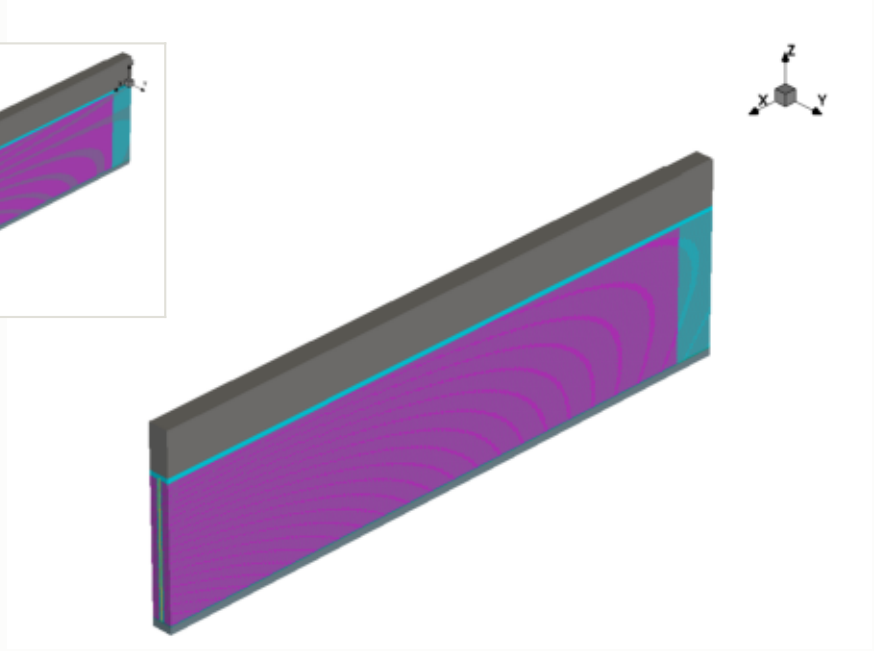
litho_patterning · KrF

0.199	672.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

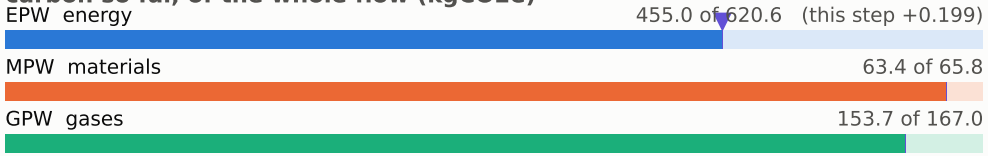




this step

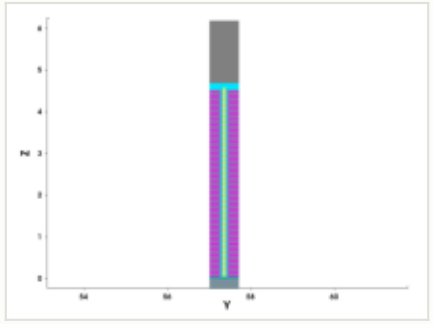
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

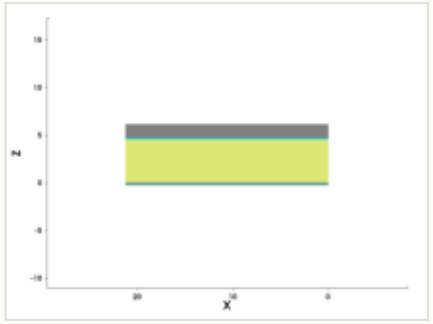


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

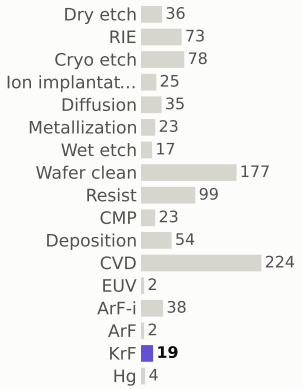


y cut at 57.351 µm



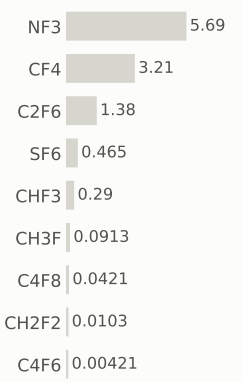
mask: layer 5:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



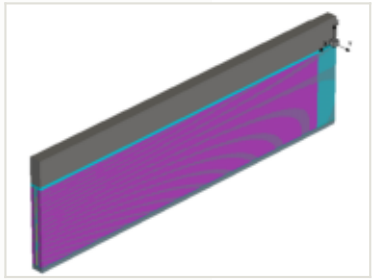
Step #11190: RIE

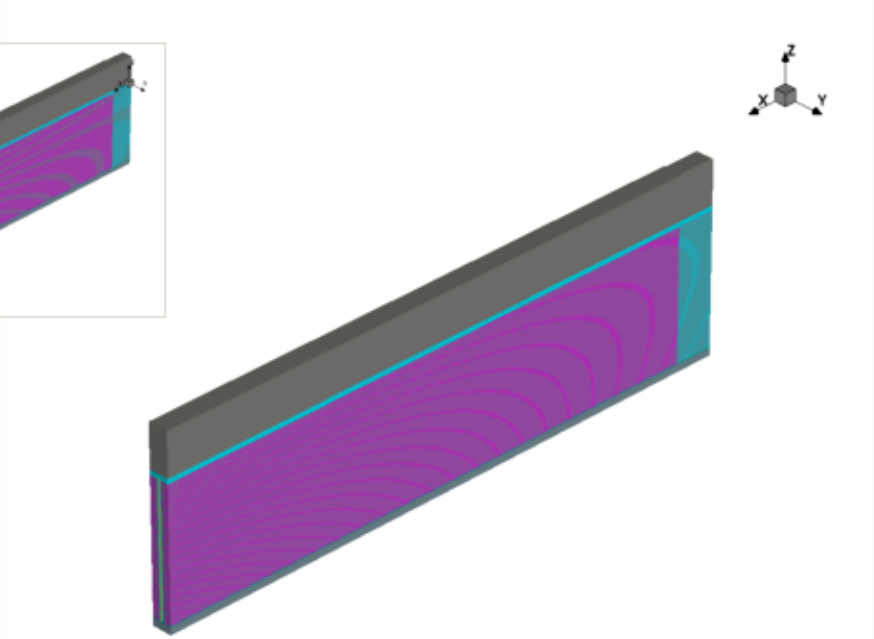
etch_anisotropic · RIE

1.04	673.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



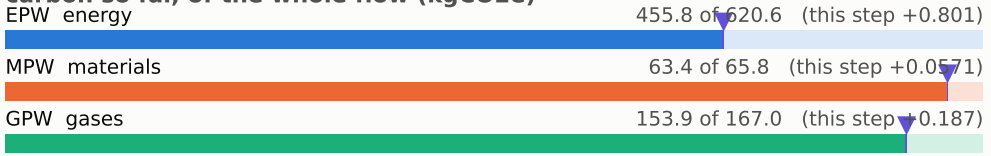


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

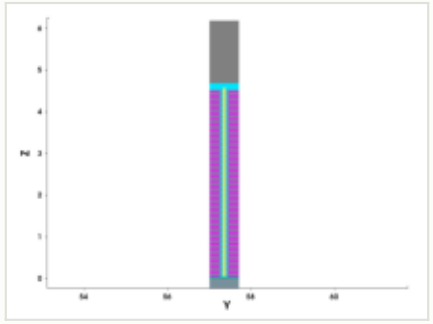
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

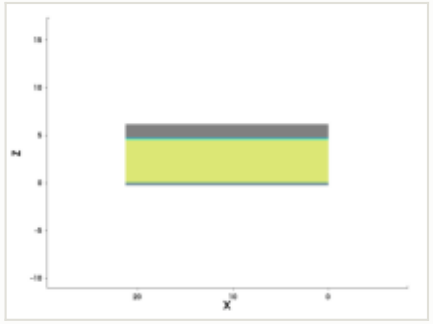


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



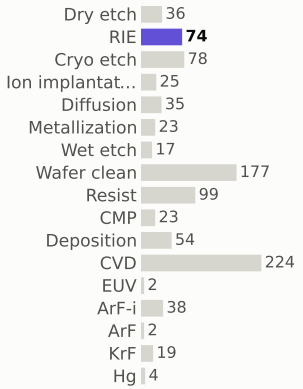
y cut at 57.351 µm



mask for this step

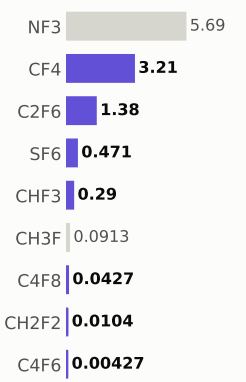
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



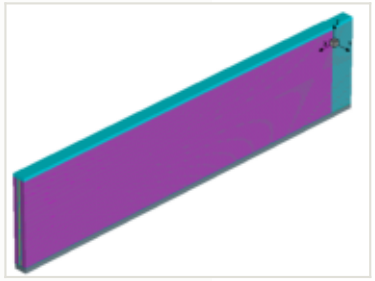
Step #11200: Dry_etch

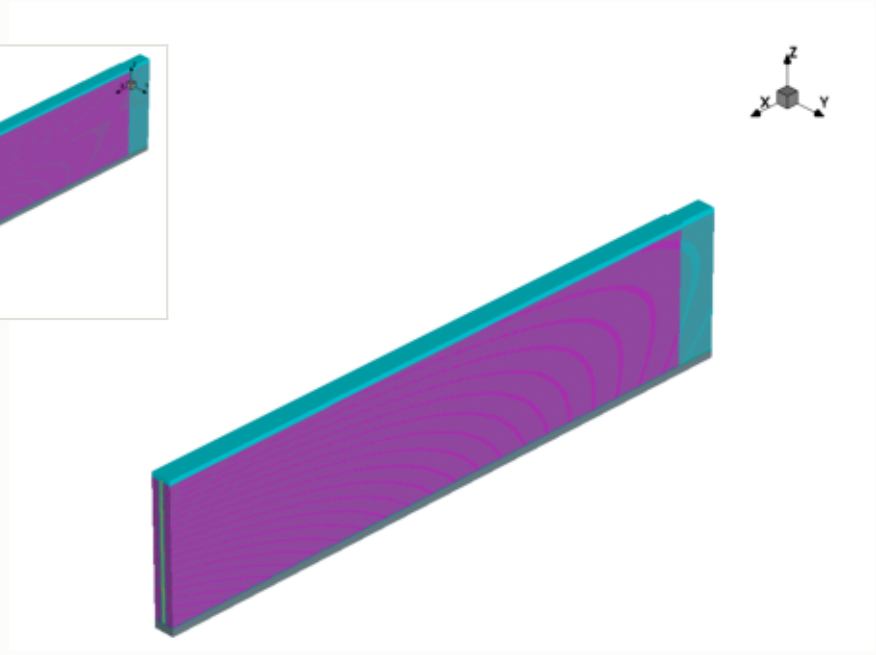
etch_strip · Dry_etch

0.801	673.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



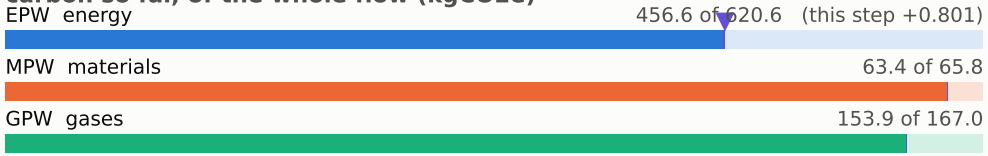


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

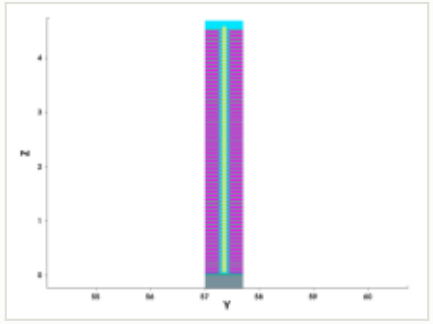
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

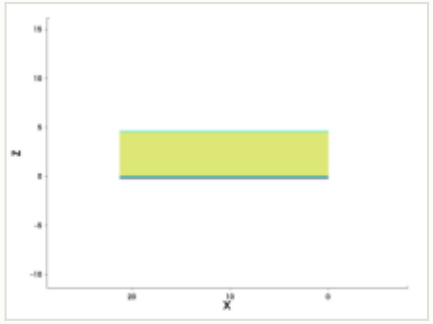


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



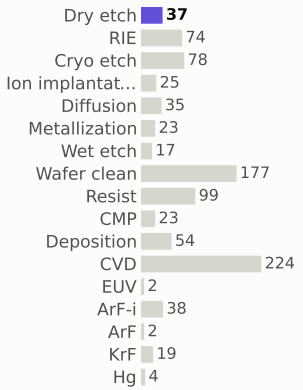
y cut at 57.351 µm



mask for this step

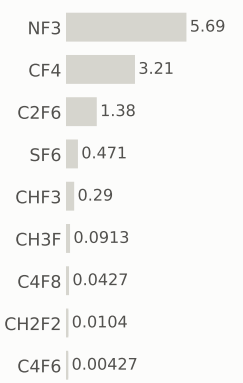
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



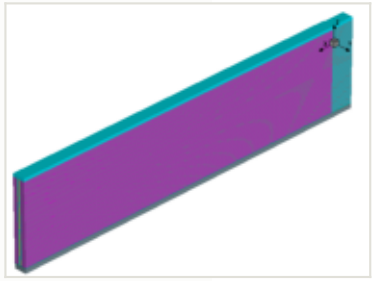
Step #11210: Wafer_clean

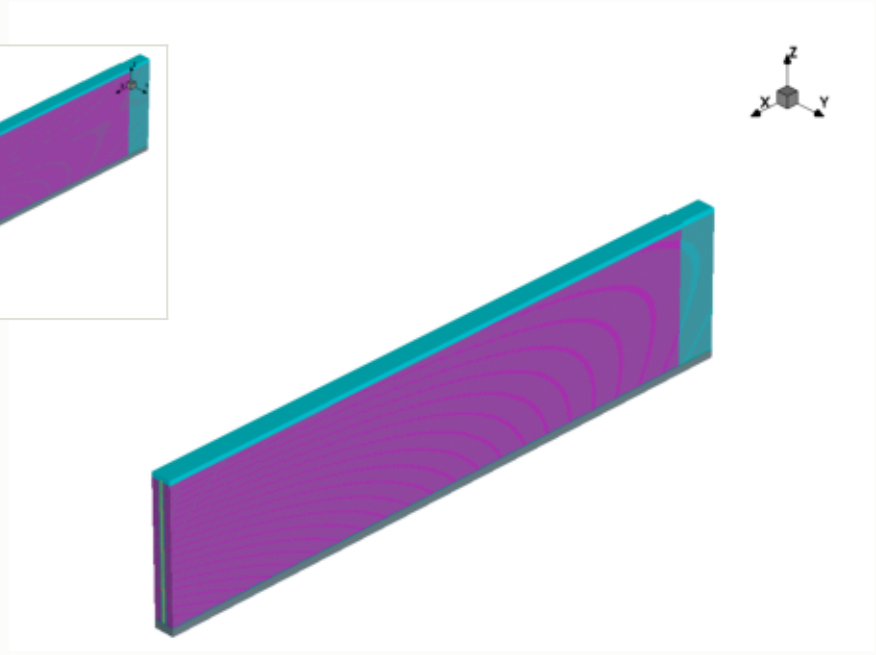
clean · Wafer_clean

0.169	674.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





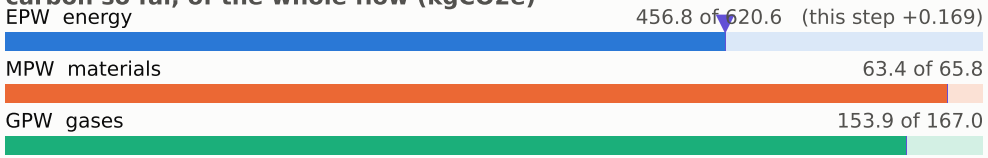
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

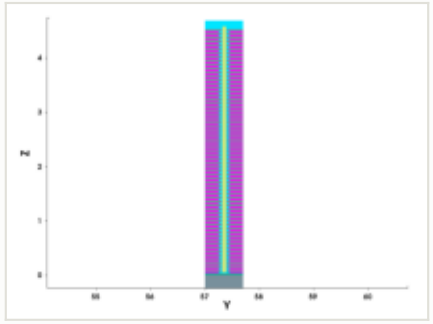
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

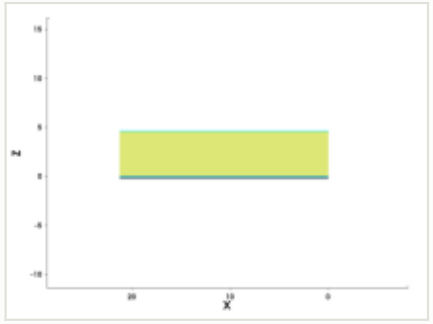


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



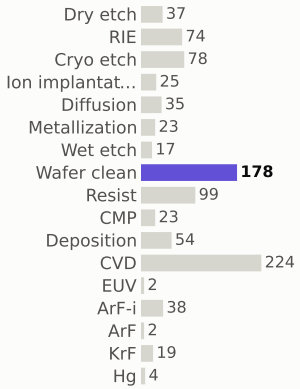
y cut at 57.351 µm



mask for this step

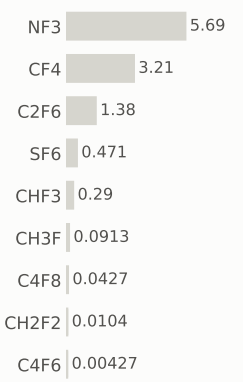
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



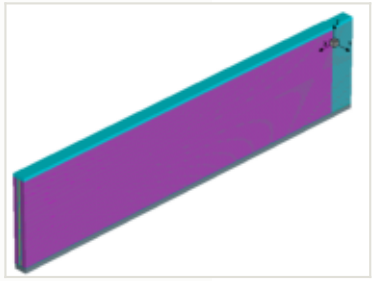
Step #11220: Wet_etch

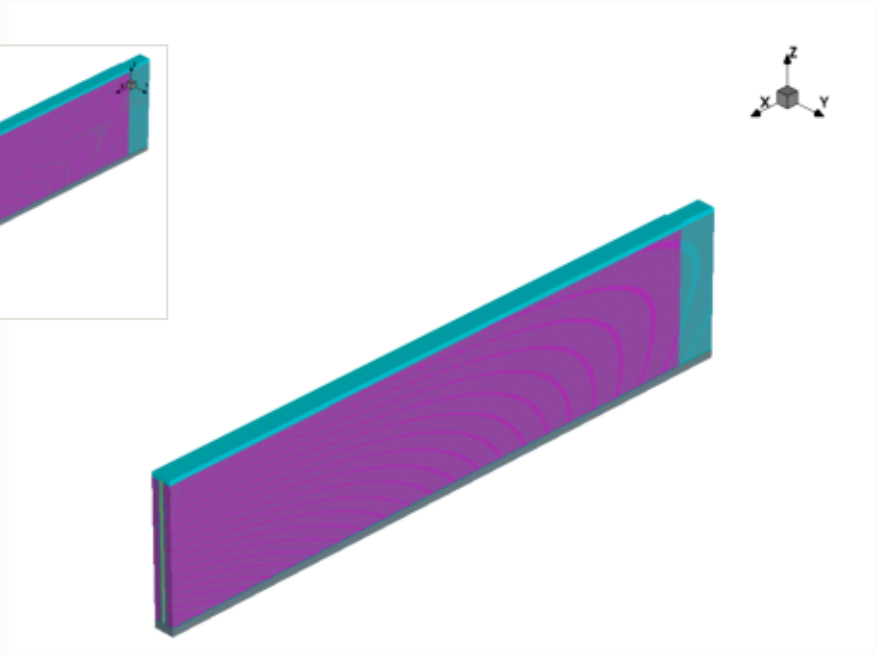
etch_isotropic · Wet_etch

0.169	674.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

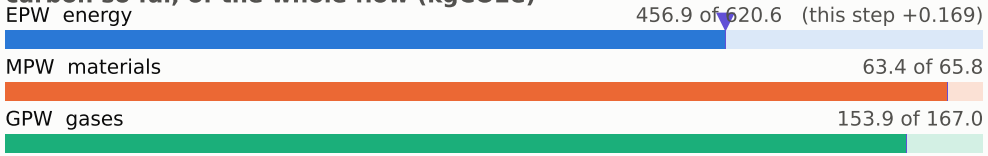




this step

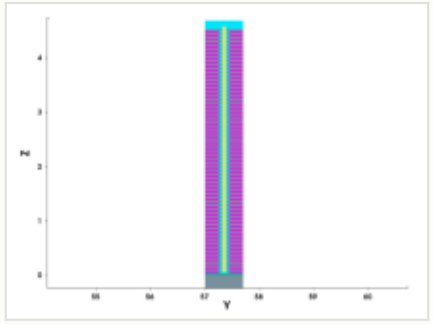
process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

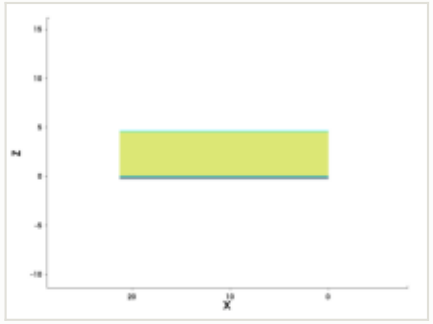


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



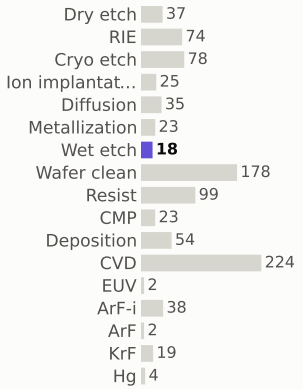
y cut at 57.351 µm



mask for this step

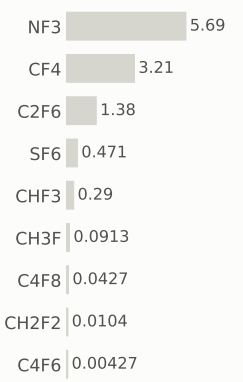
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



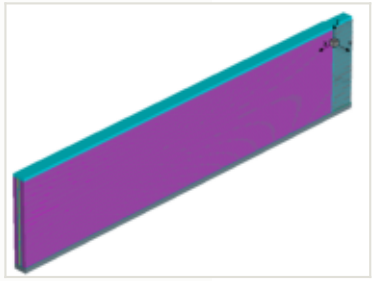
Step #11230: Wet_etch

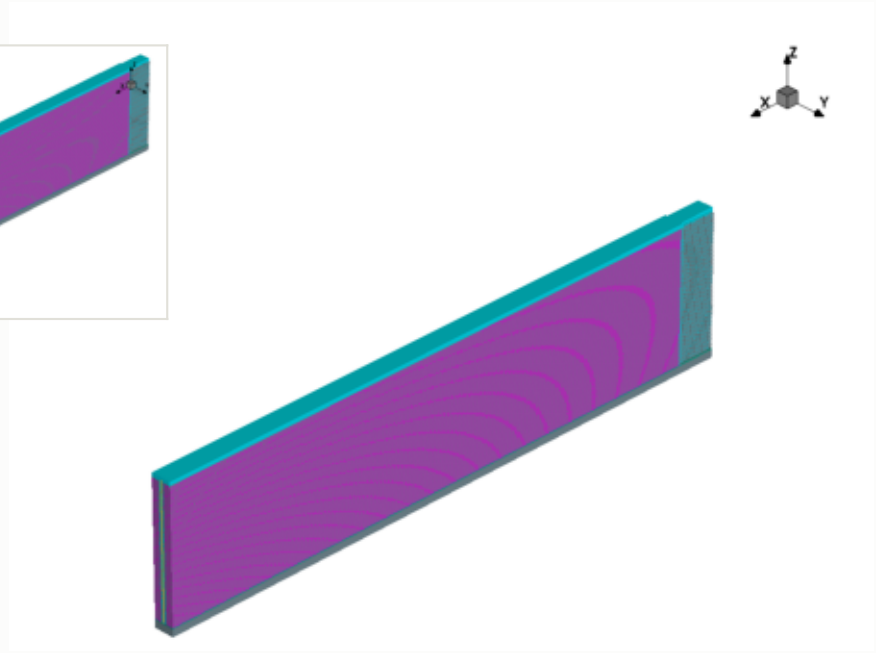
etch_isotropic · Wet_etch

0.169	674.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



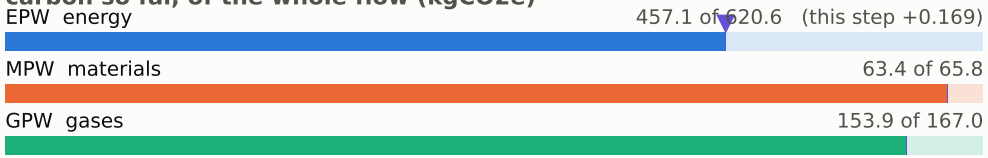


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

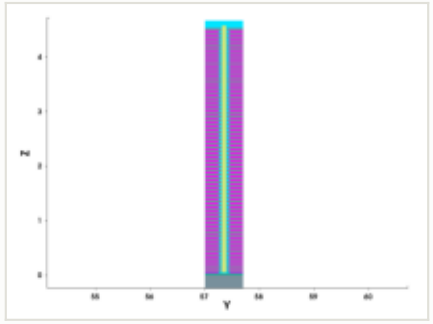
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

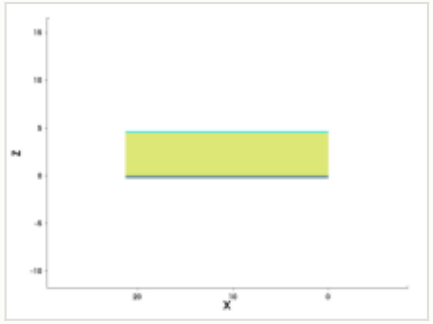


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



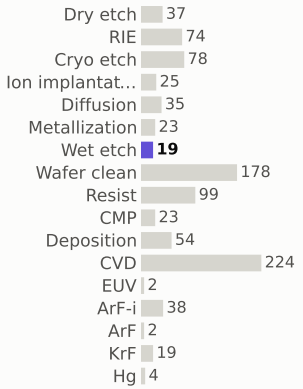
y cut at 57.351 µm



mask for this step

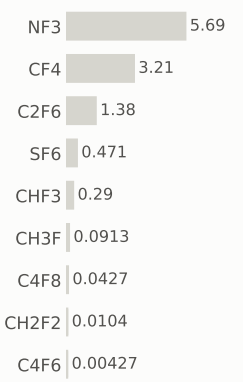
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



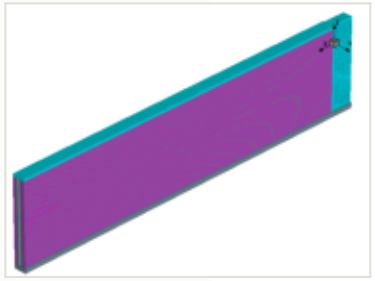
Step #11240: Wet_etch

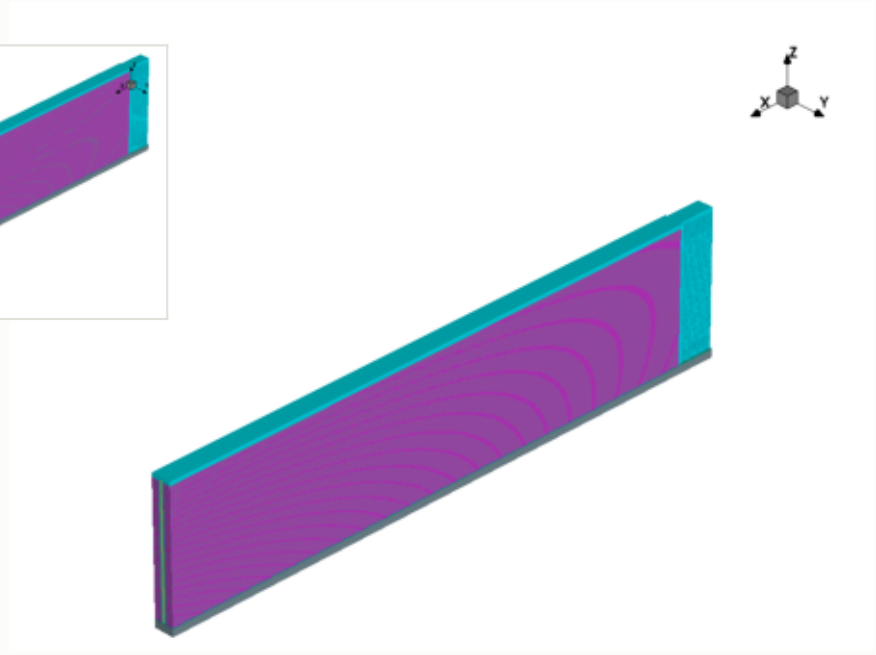
etch_isotropic · Wet_etch

0.169	674.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



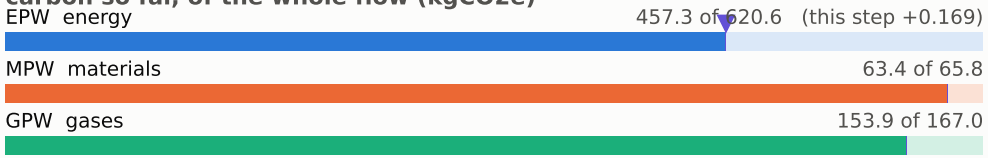


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

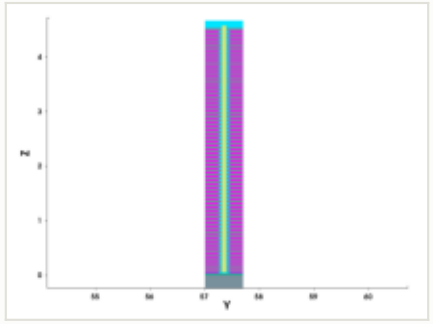
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

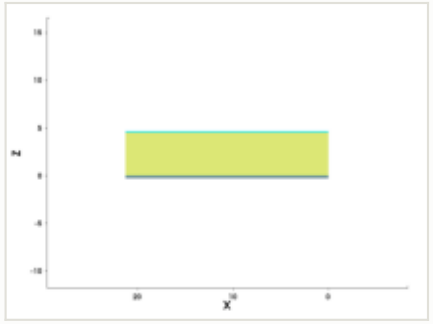


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



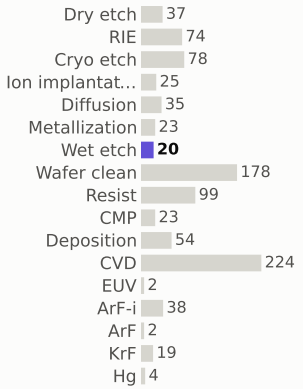
y cut at 57.351 µm



mask for this step

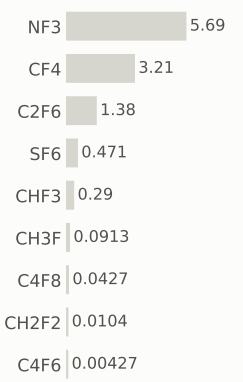
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



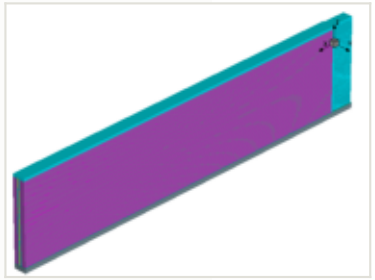
Step #11250: Dry_etch

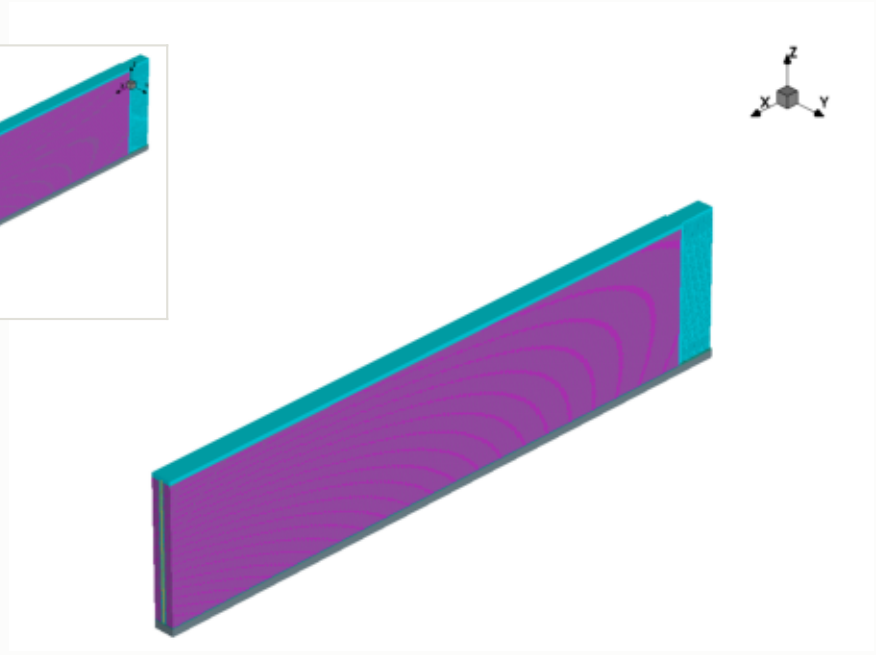
clean · Dry_etch

0.801	675.4	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





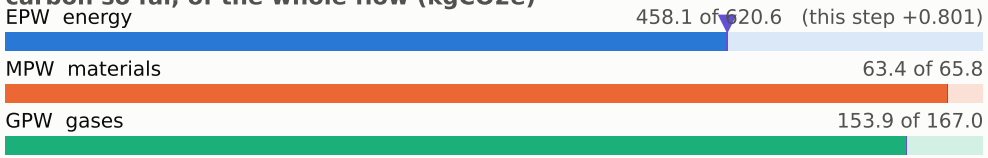
this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

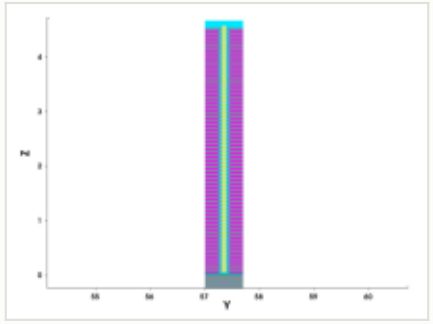
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

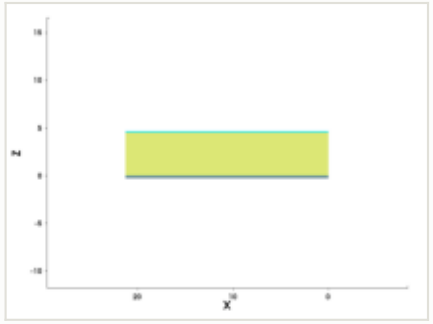


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



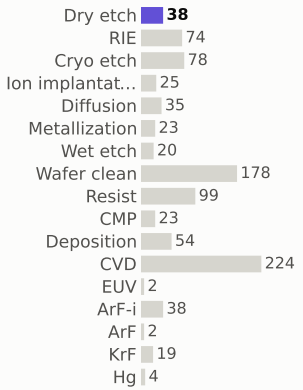
y cut at 57.351 µm



mask for this step

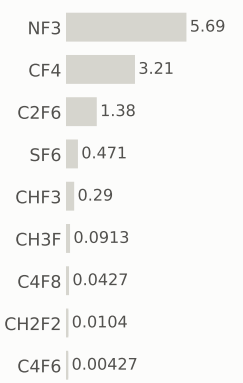
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



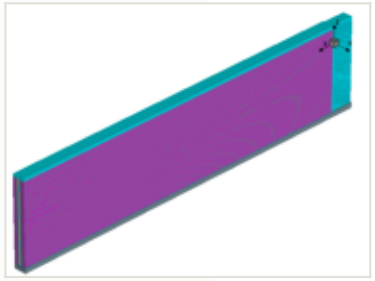
Step #11260: CVD

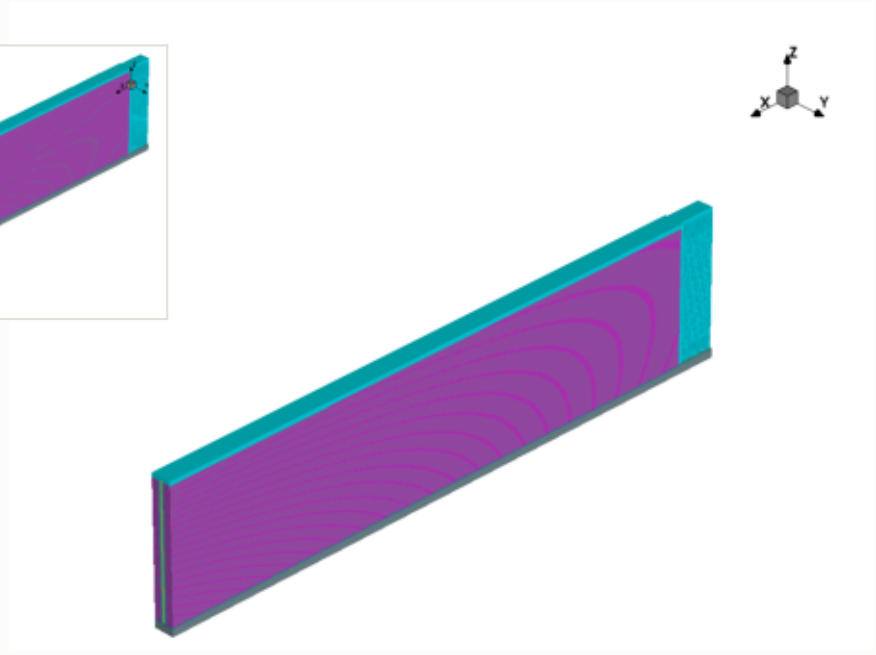
depo_isotropic · CVD

1.31	676.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



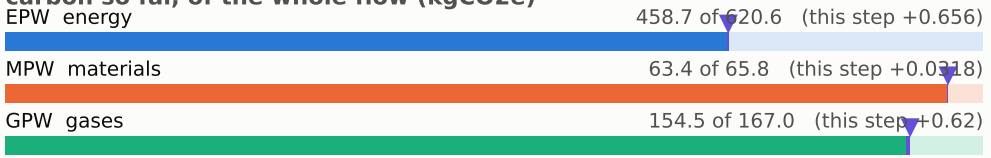


this step

process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

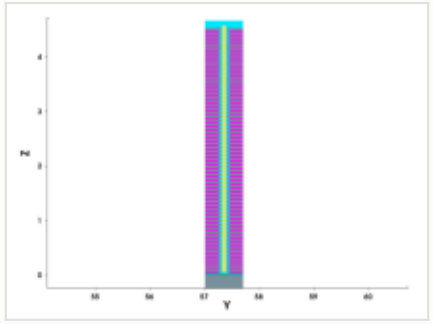
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

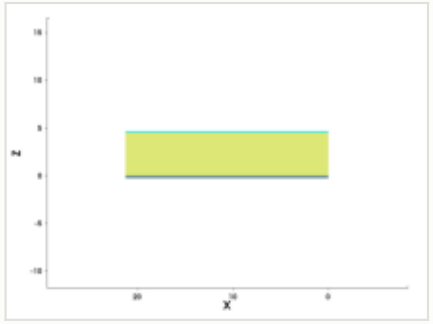


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



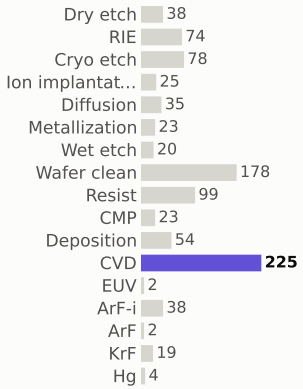
y cut at 57.351 µm



mask for this step

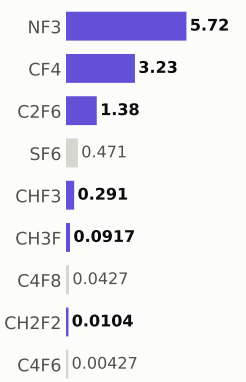
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



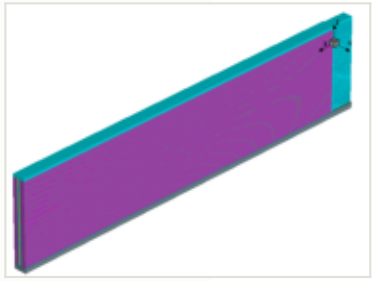
Step #11270: Metallization

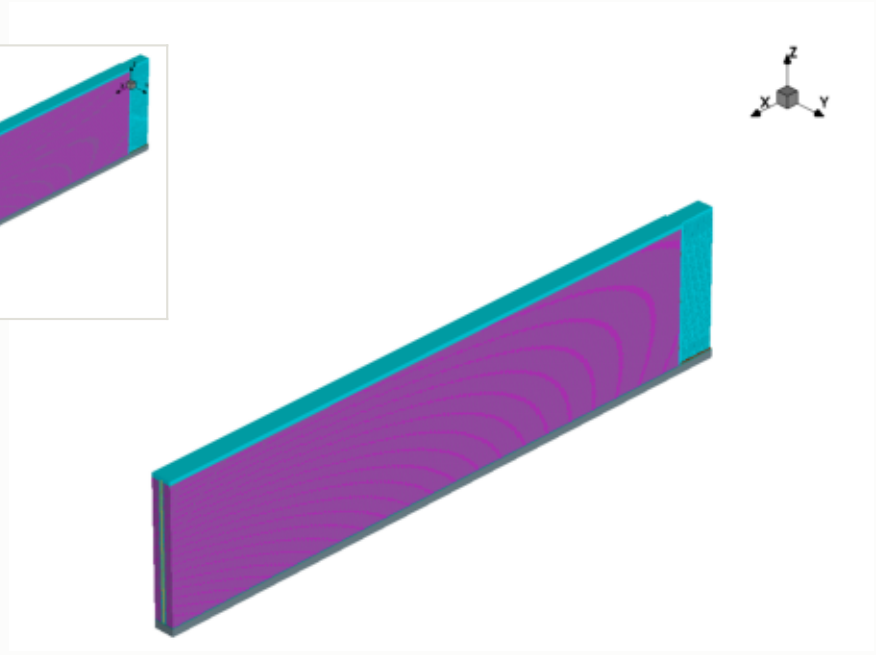
depo_isotropic · Metallization

1.2	677.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



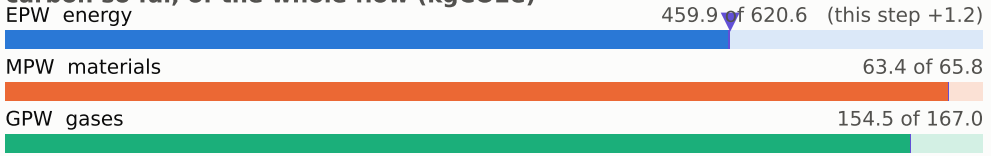


this step

process area	Metallization
material	—
thickness	0.003 µm
material GWP	—
input gases	—
energy	2 kWh + 0.8 overhead

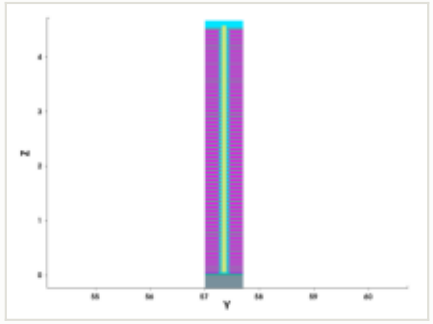
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

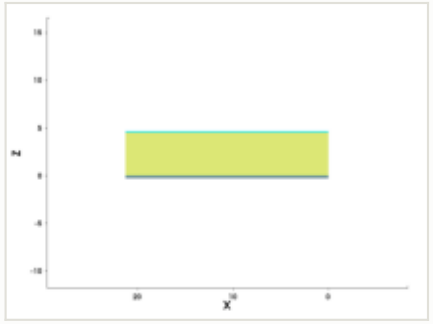


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



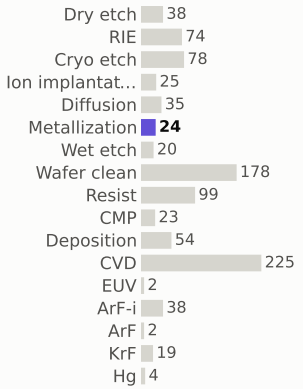
y cut at 57.351 µm



mask for this step

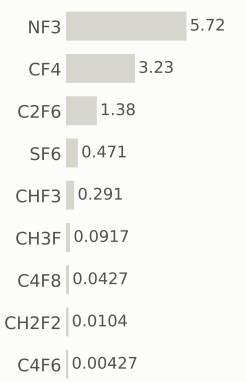
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



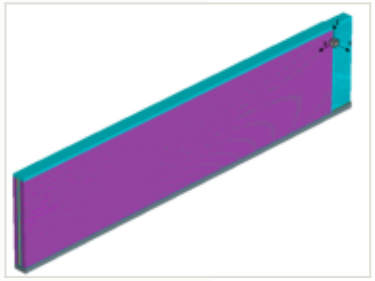
Step #11280: Diffusion

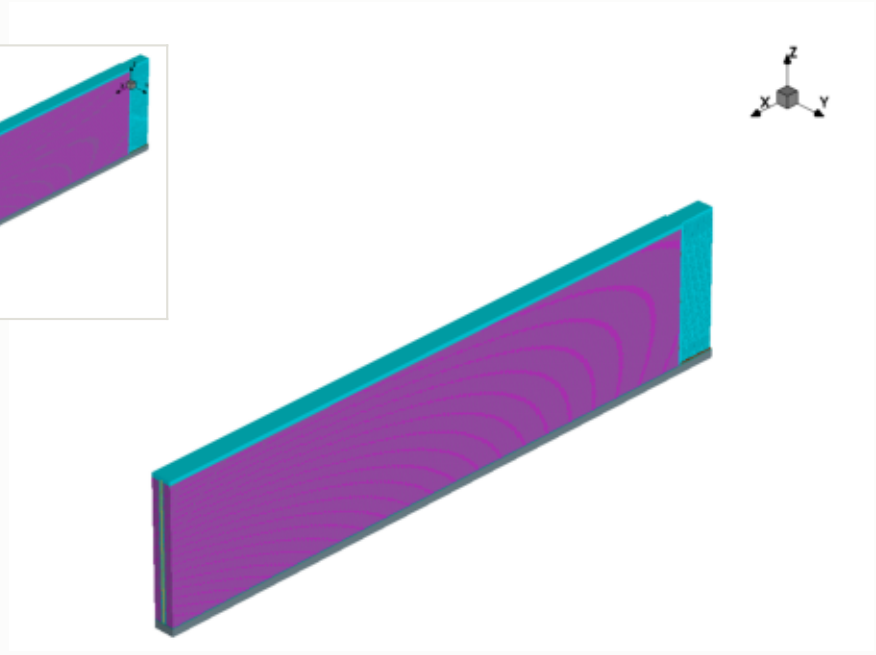
anneal · Diffusion

0.801	678.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





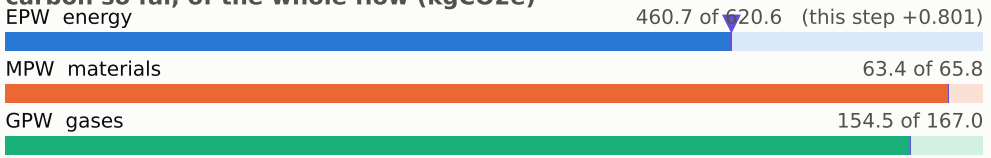
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

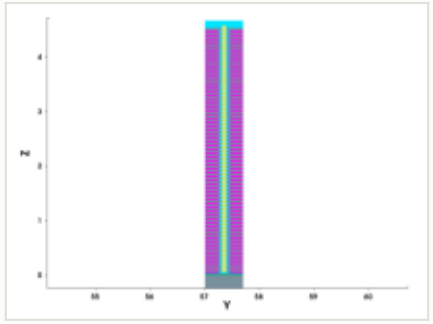
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

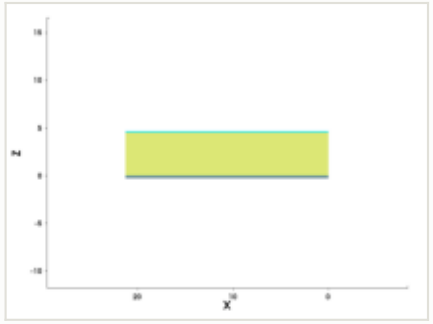


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



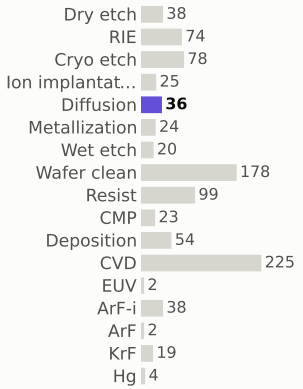
y cut at 57.351 µm



mask for this step

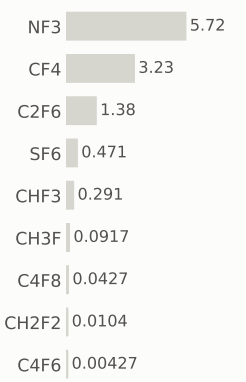
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



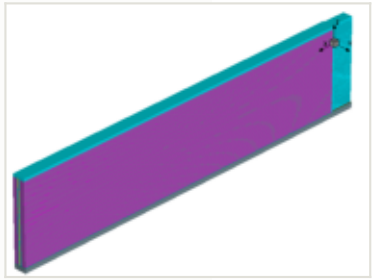
Step #11290: Wafer_clean

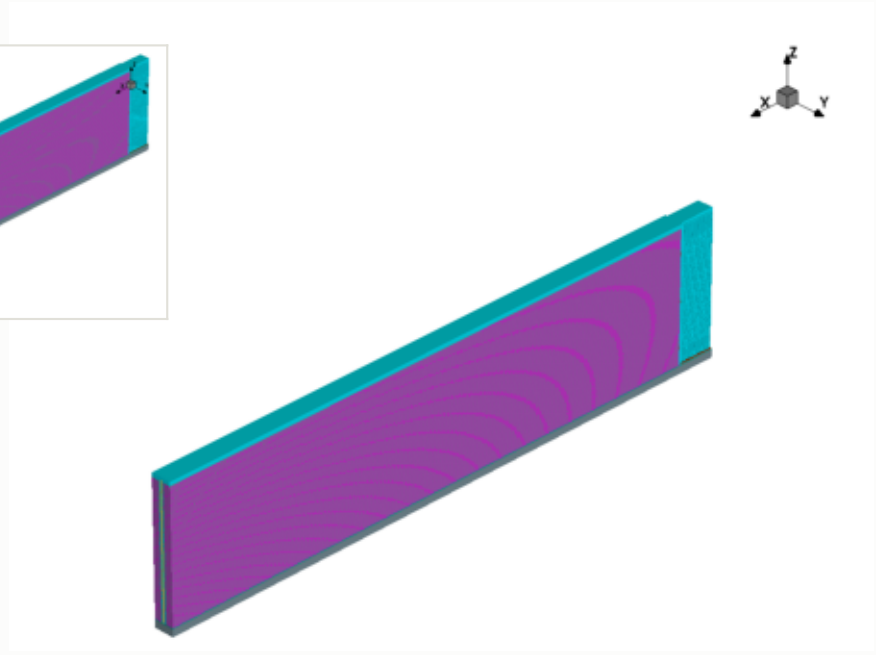
clean · Wafer_clean

0.169	678.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





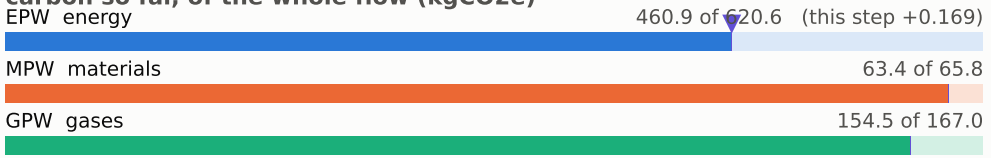
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

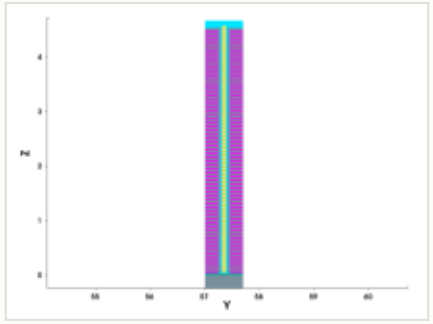
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

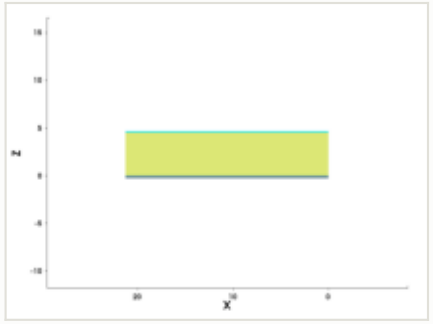


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



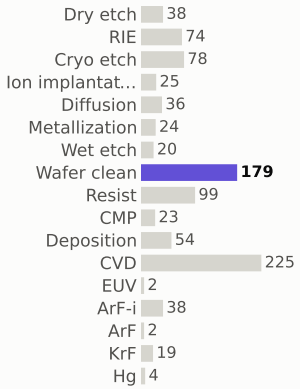
y cut at 57.351 µm



mask for this step

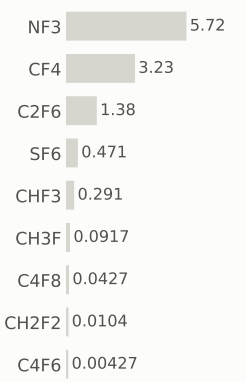
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



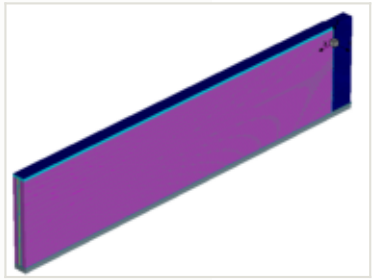
Step #11300: Metallization

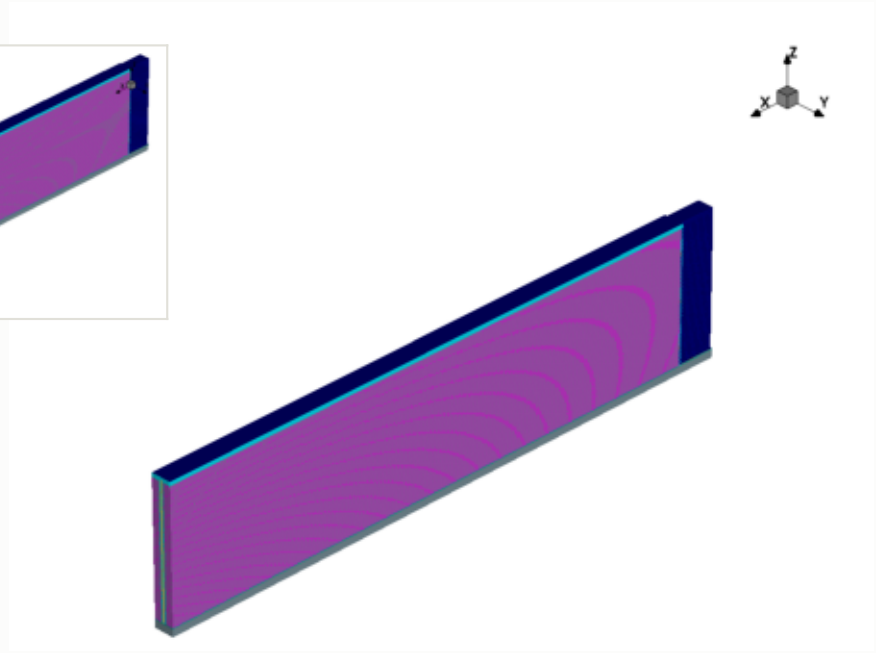
depo_isotropic · Metallization

1.2	680.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



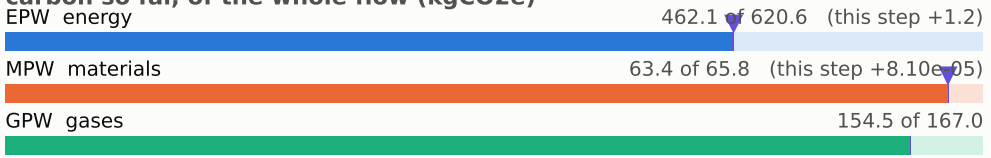


this step

process area	Metallization
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

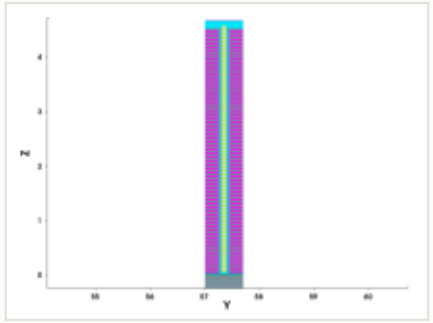
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	8.10e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

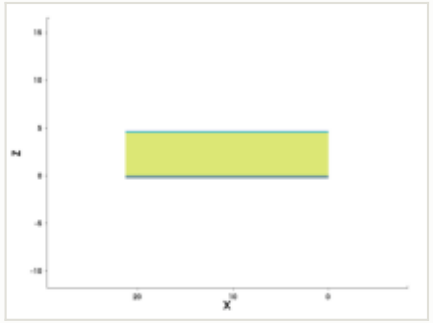


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



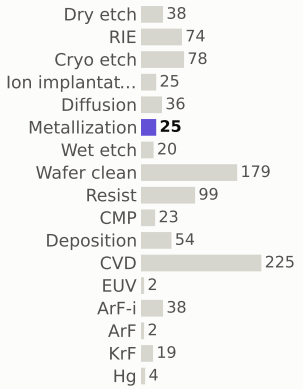
y cut at 57.351 µm



mask for this step

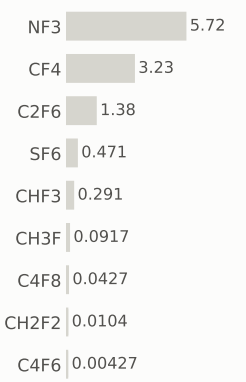
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



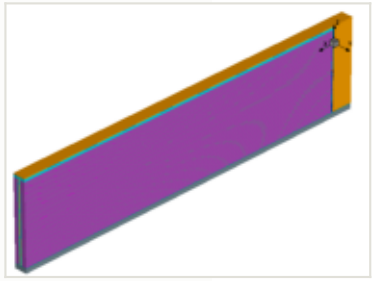
Step #11310: Resist

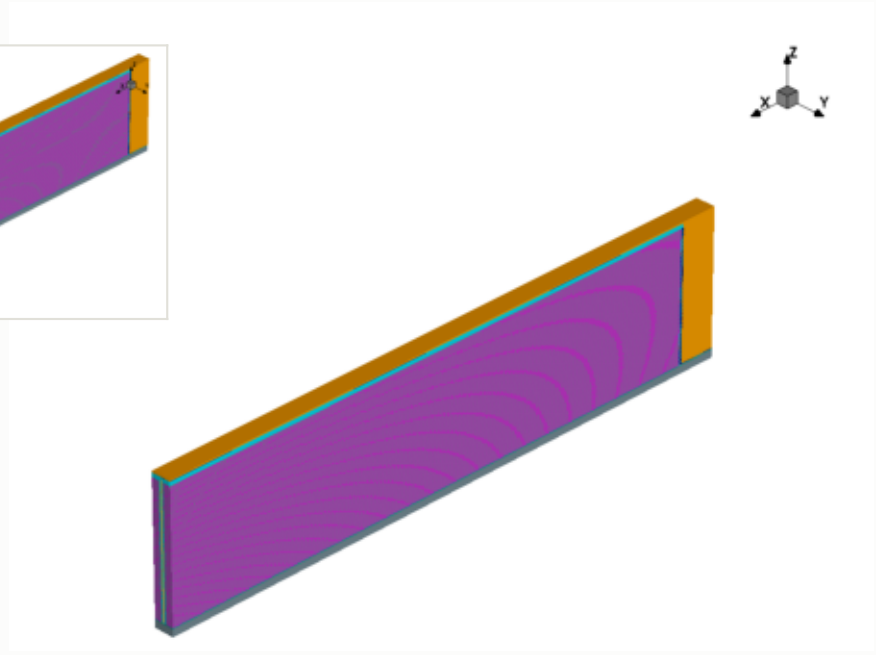
depo_fill · Resist

0.169	680.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



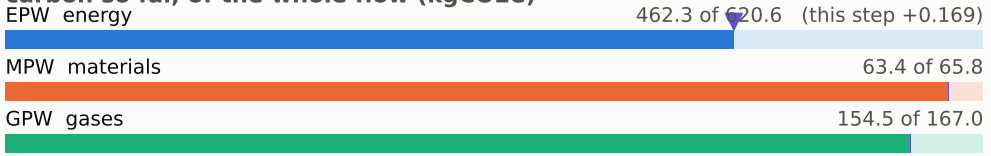


this step

process area	Resist
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

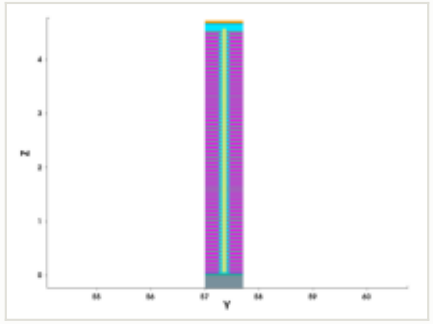
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

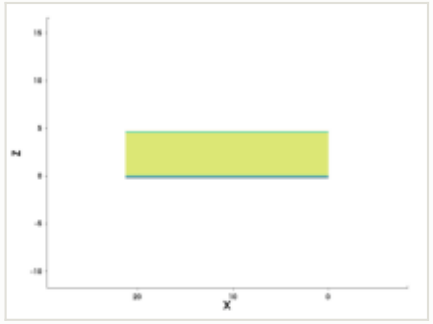


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



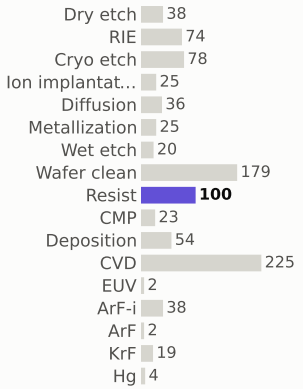
y cut at 57.351 µm



mask for this step

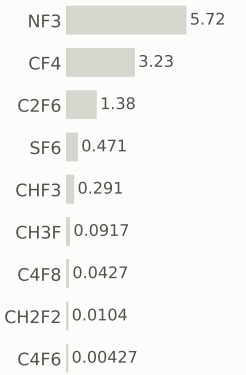
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



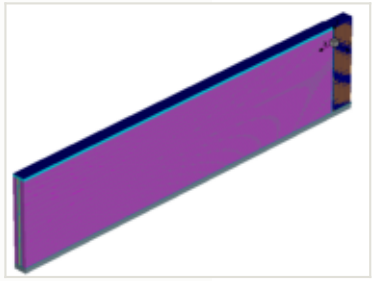
Step #11320: RIE

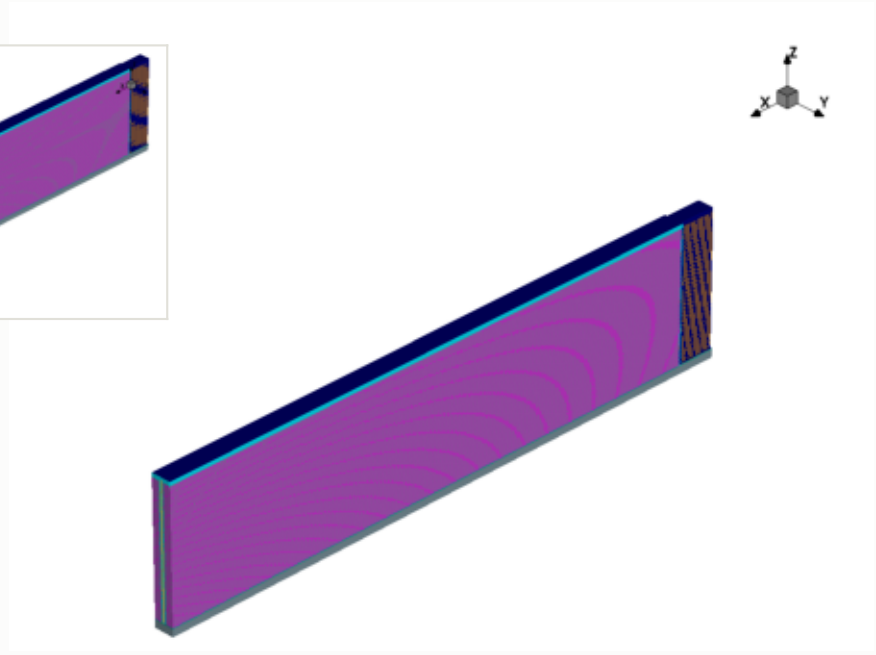
etch_anisotropic · RIE

1.04	681.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



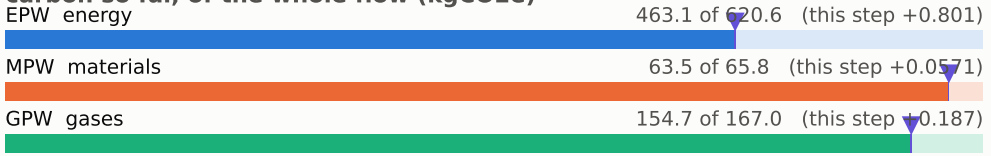


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

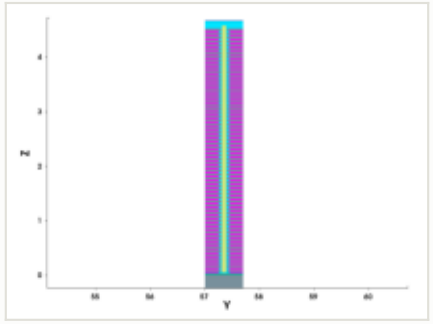
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

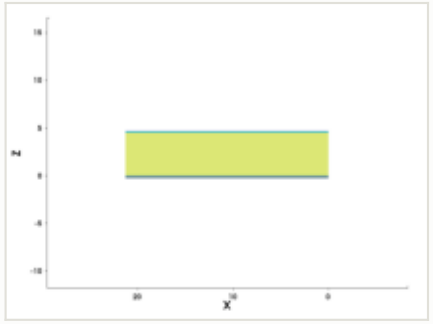


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



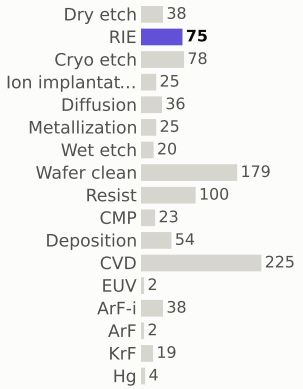
y cut at 57.351 µm



mask for this step

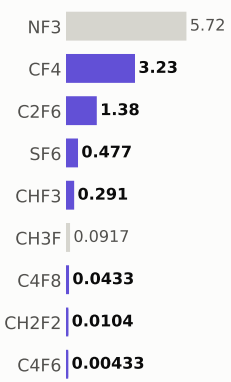
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



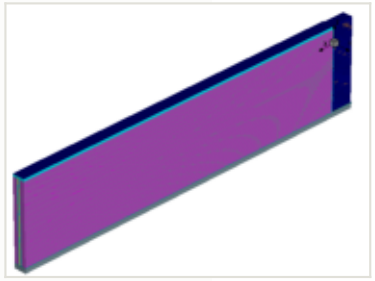
Step #11330: Spx_only

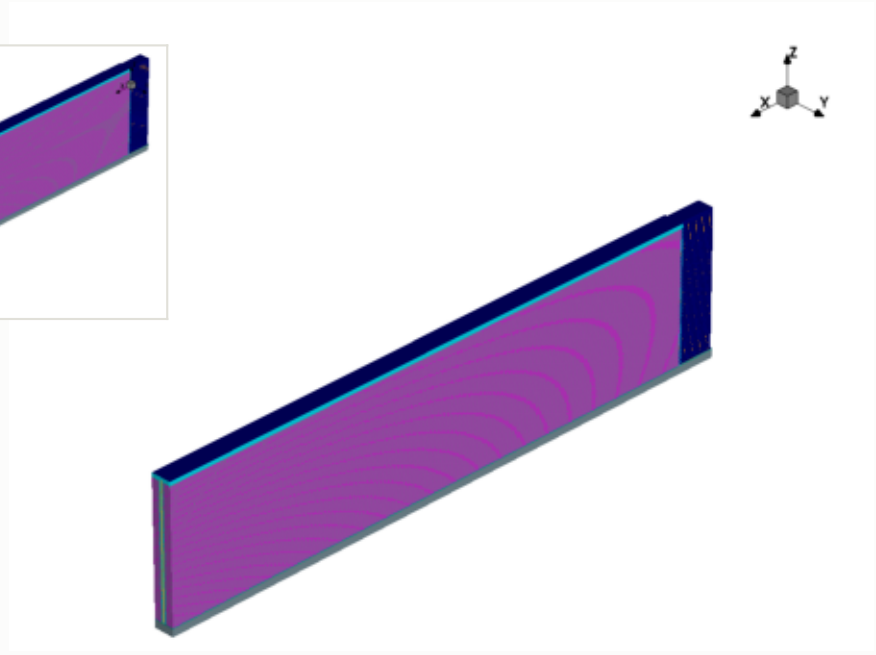
etch_isotropic · Spx_only

0	681.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





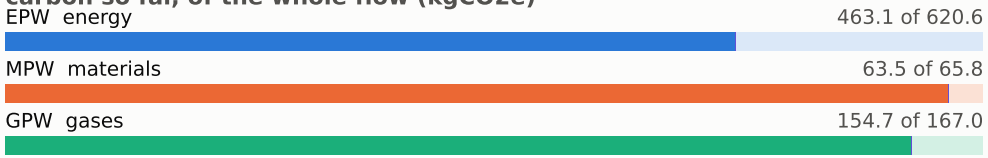
this step

process area	Spx_only
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

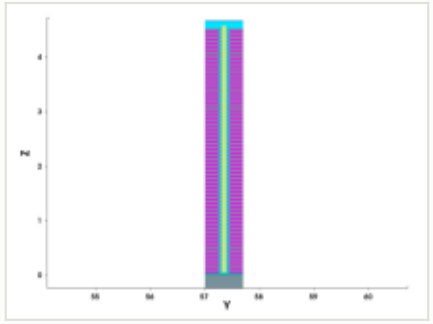
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

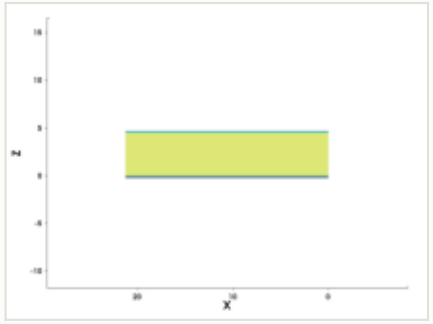


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



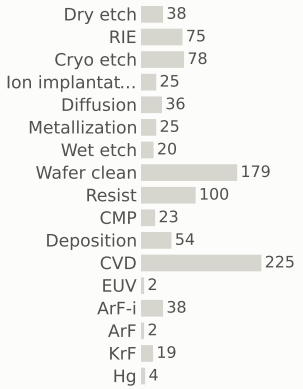
y cut at 57.351 µm



mask for this step

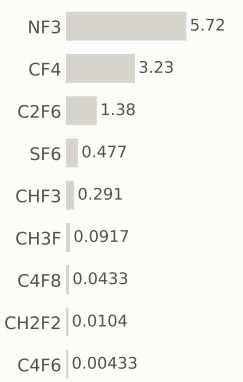
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



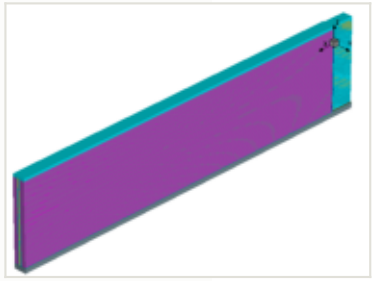
Step #11340: Wet_etch

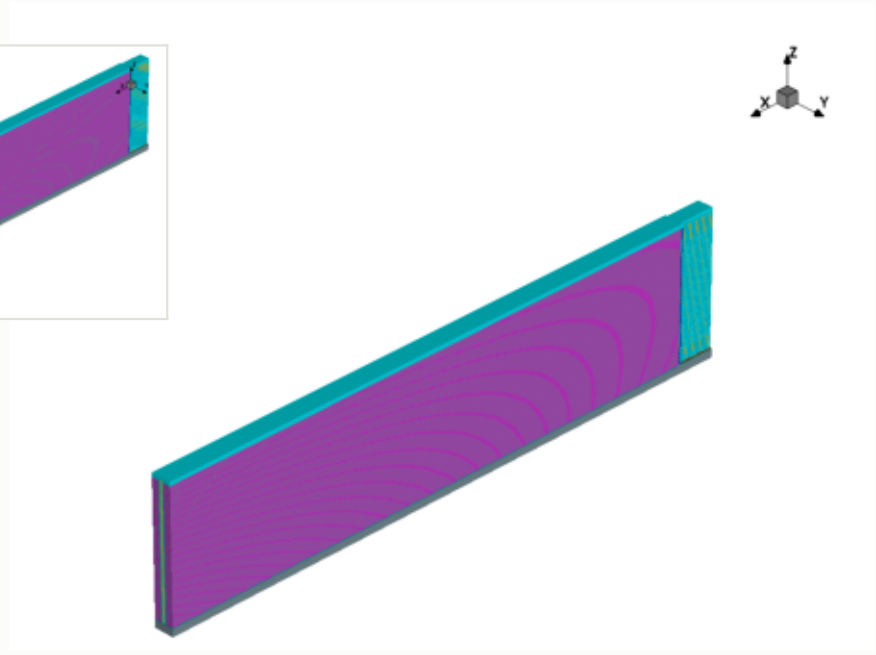
etch_isotropic · Wet_etch

0.169	681.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



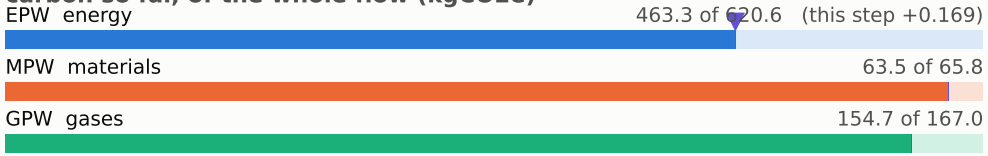


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

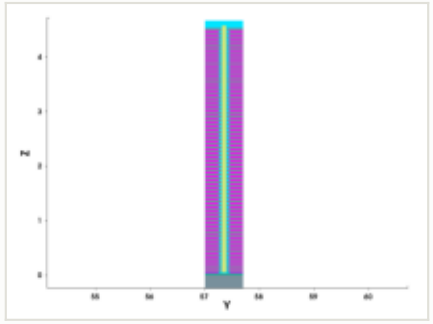
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

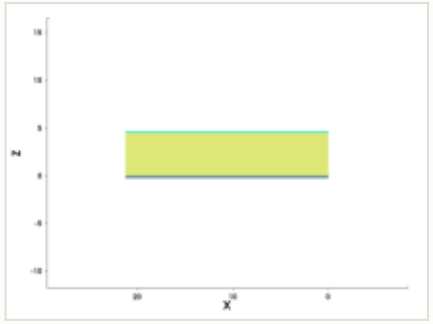


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



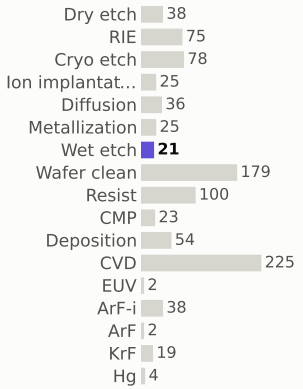
y cut at 57.351 µm



mask for this step

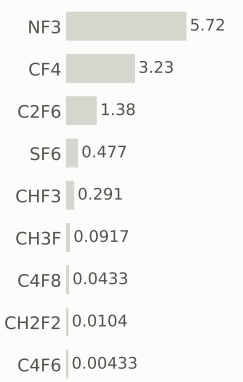
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



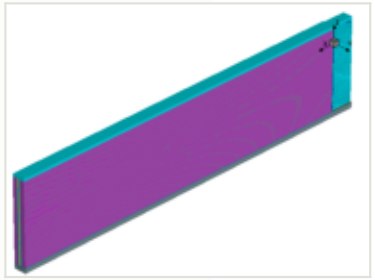
Step #11350: Dry_etch

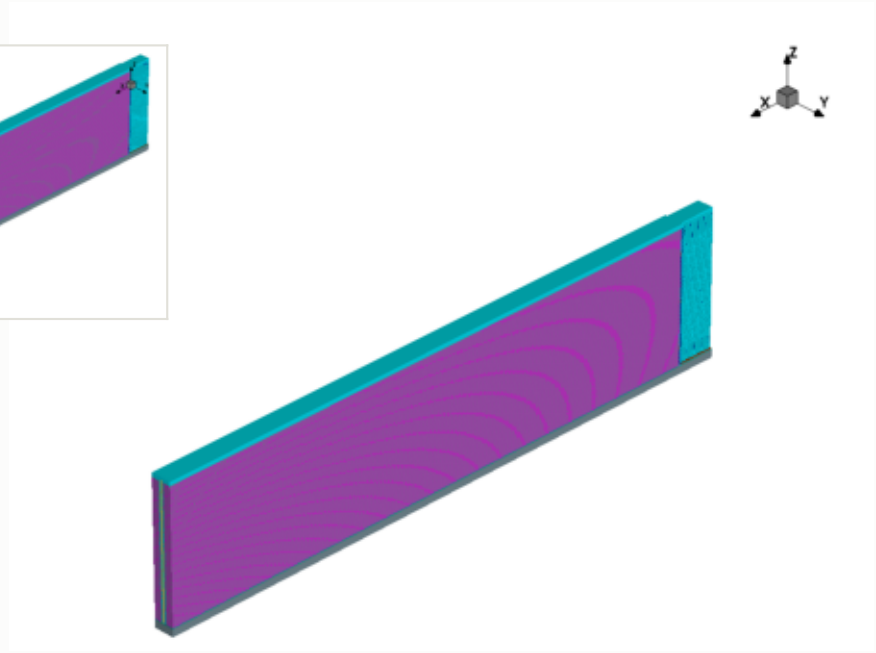
etch_strip · Dry_etch

0.801	682.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



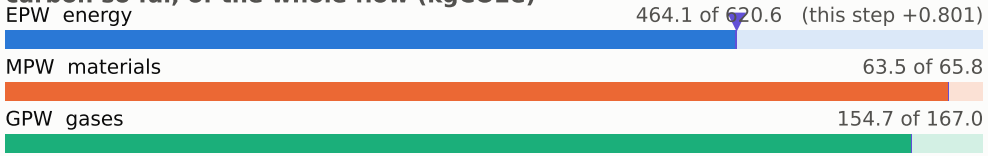


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

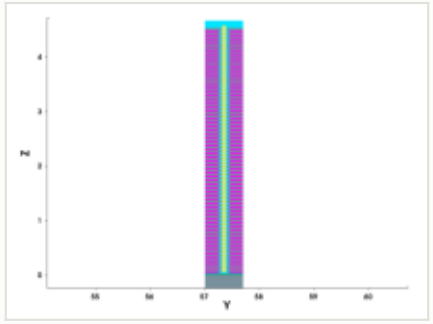
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

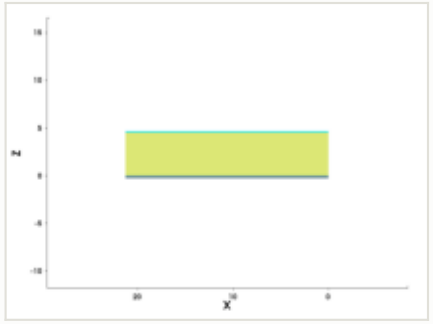


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



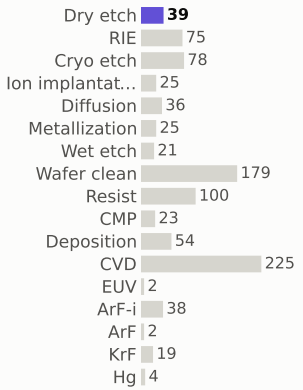
y cut at 57.351 µm



mask for this step

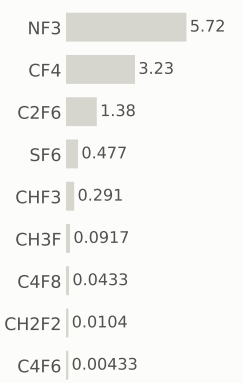
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



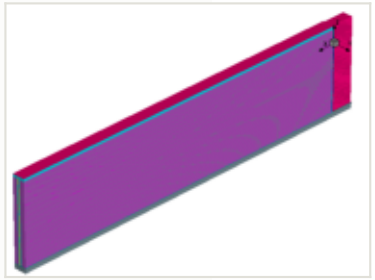
Step #11360: Deposition

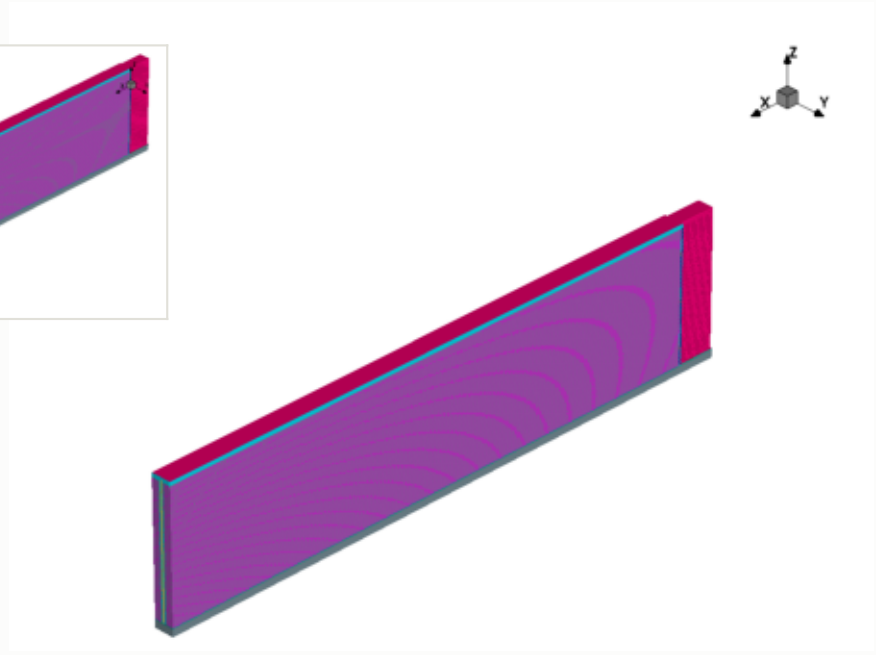
depo_isotropic · Deposition

0.656	682.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



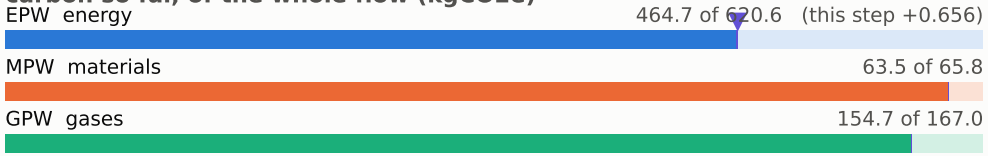


this step

process area	Deposition
material	—
thickness	0.005 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

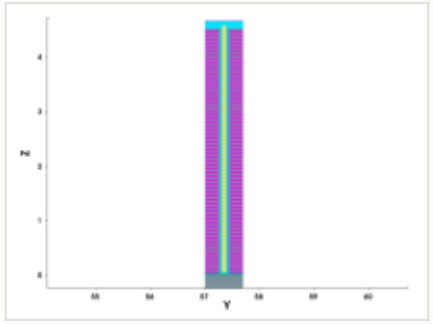
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

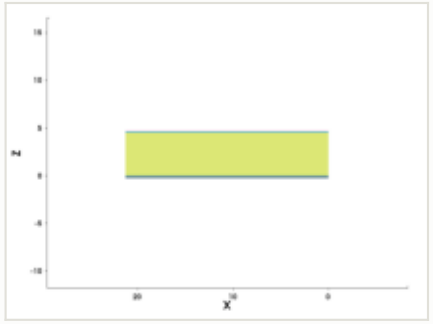


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



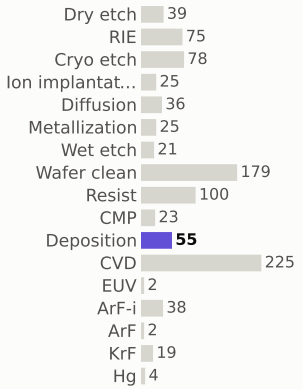
y cut at 57.351 µm



mask for this step

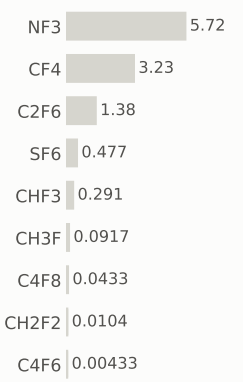
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



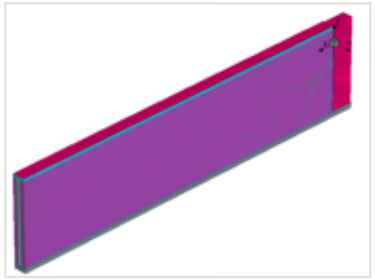
Step #11370: Diffusion

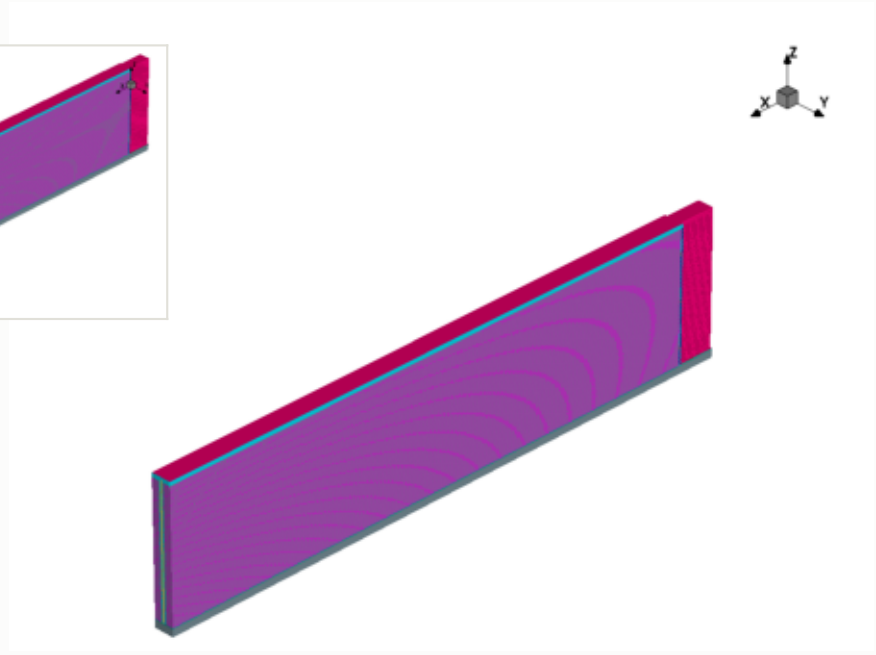
anneal · Diffusion

0.801	683.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





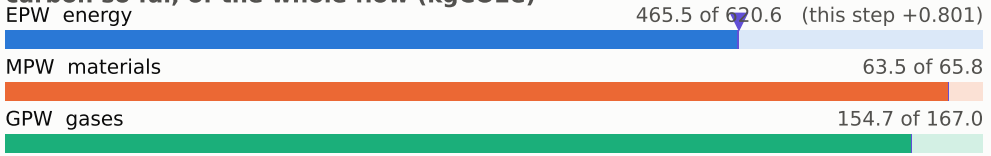
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

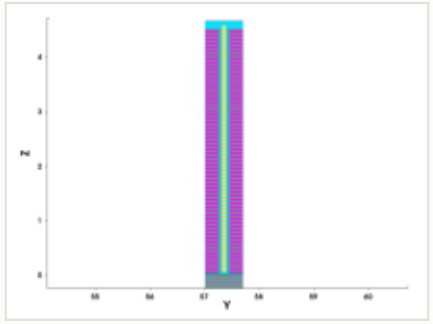
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

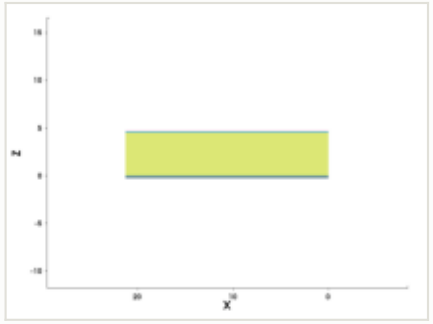


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



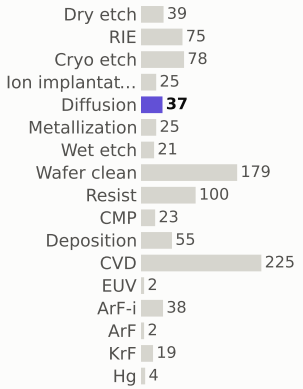
y cut at 57.351 µm



mask for this step

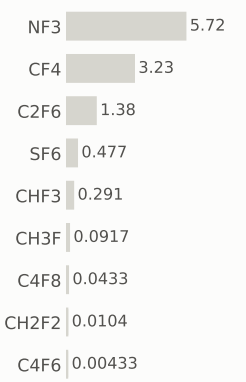
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



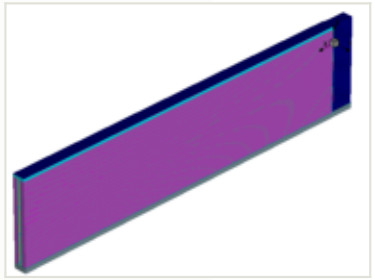
Step #11380: Metallization

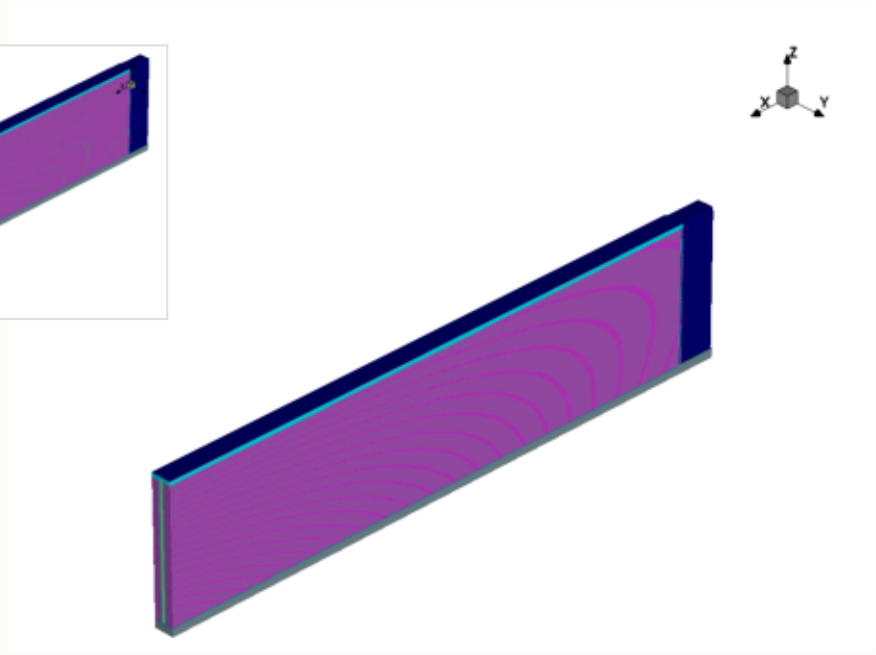
depo_isotropic · Metallization

1.2	684.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



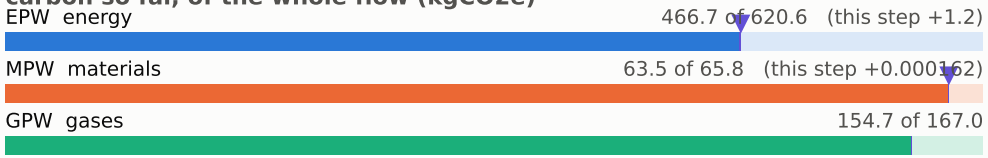


this step

process area	Metallization
material	TiN
thickness	0.01 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

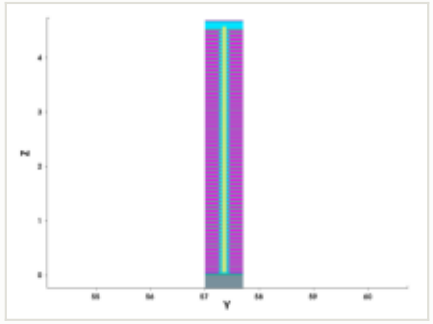
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000162 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

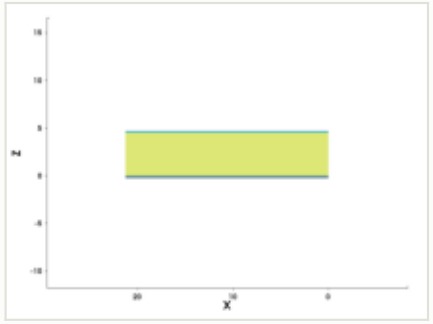


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



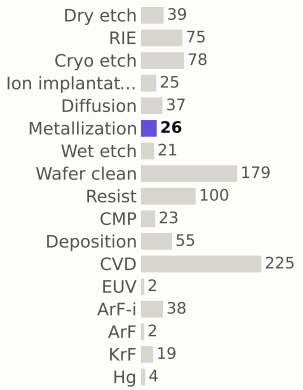
y cut at 57.351 µm



mask for this step

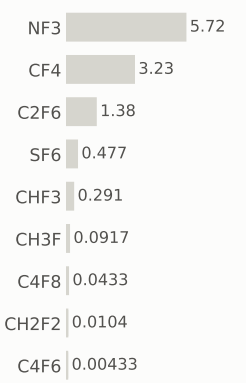
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



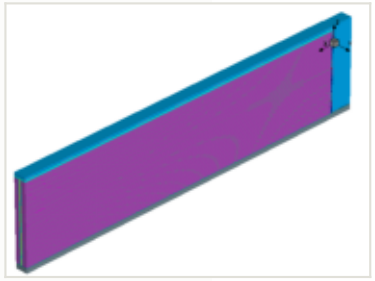
Step #11390: Metallization

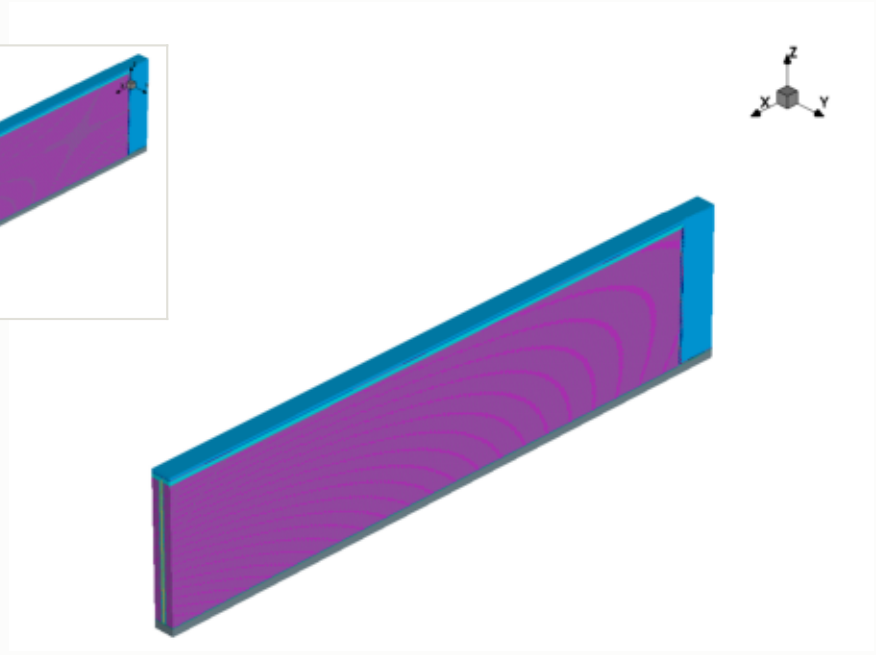
depo_isotropic · Metallization

1.21	686.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



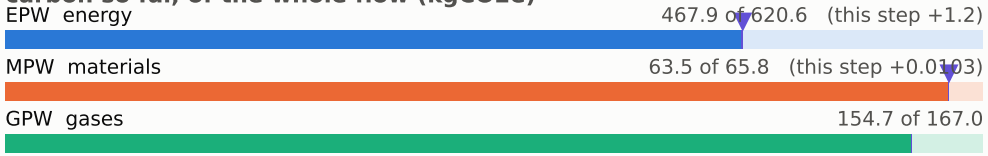


this step

process area	Metallization
material	W
thickness	0.1 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

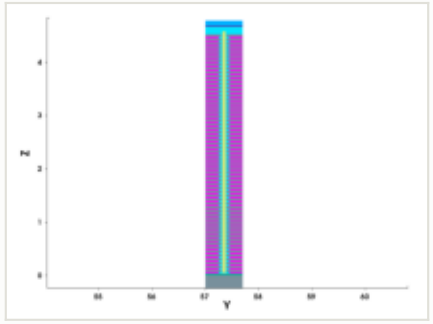
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0103 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

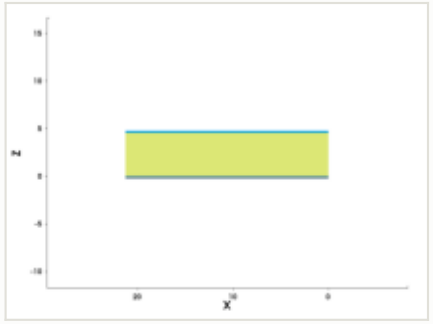


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



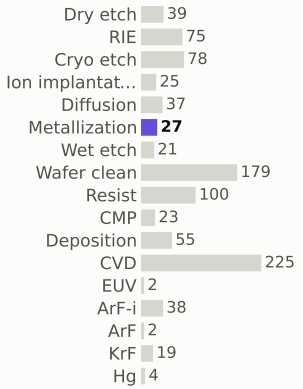
y cut at 57.351 µm



mask for this step

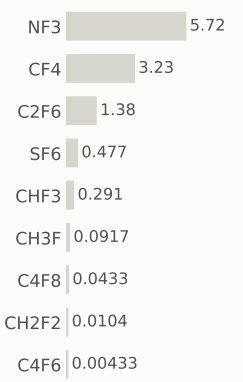
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



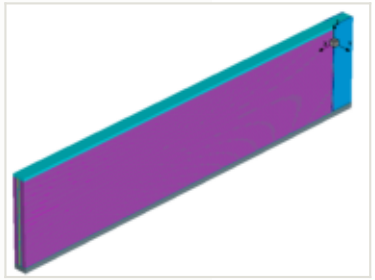
Step #11400: CMP

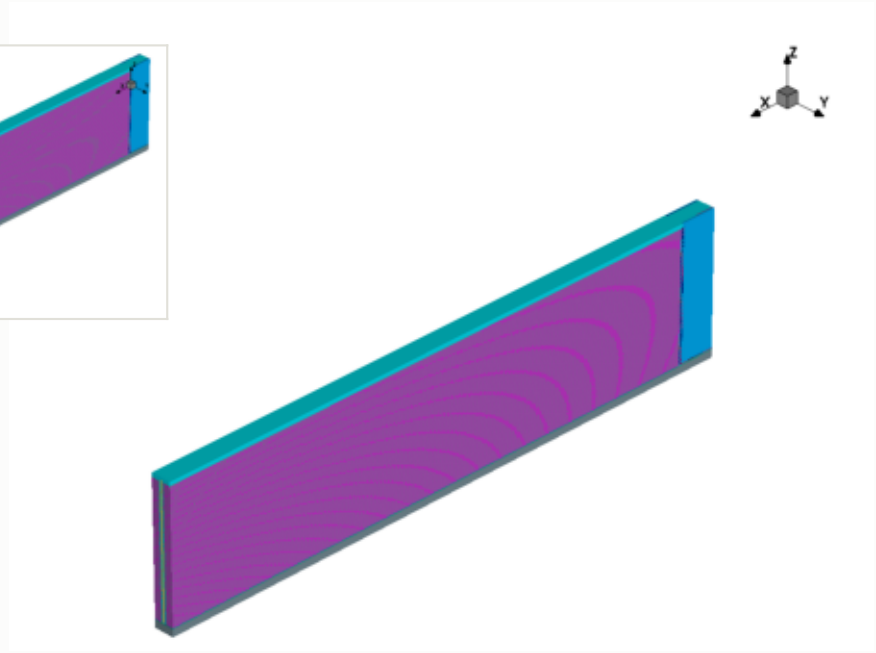
etch_cmp · CMP

0.169	686.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



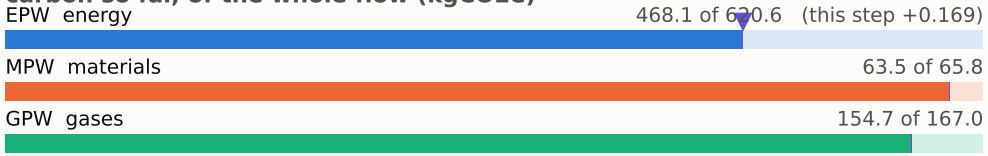


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

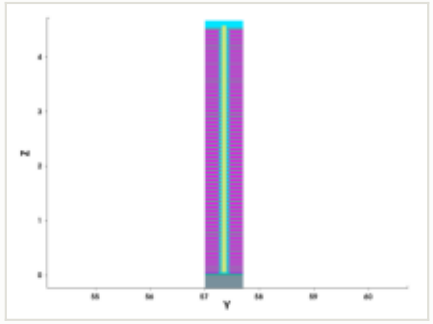
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

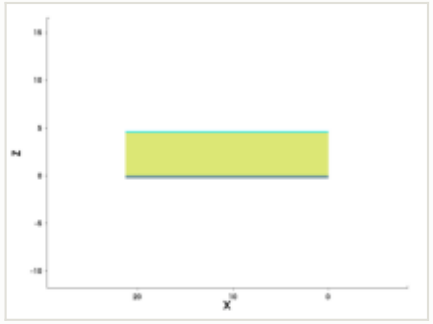


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



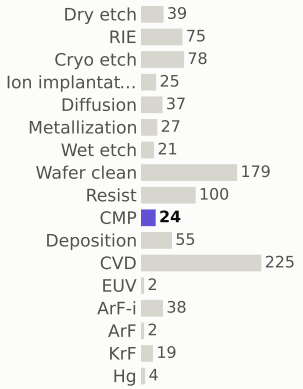
y cut at 57.351 µm



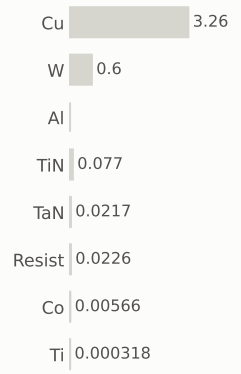
mask for this step

no mask for this step

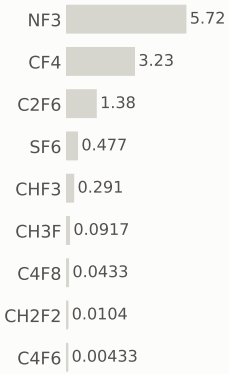
steps by process area



materials deposited (g)



gases emitted (g)



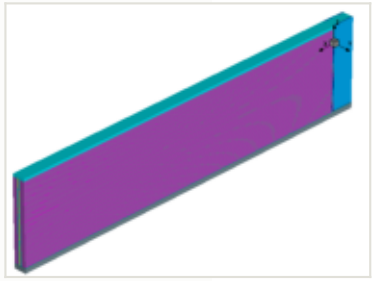
Step #11410: Wafer_clean

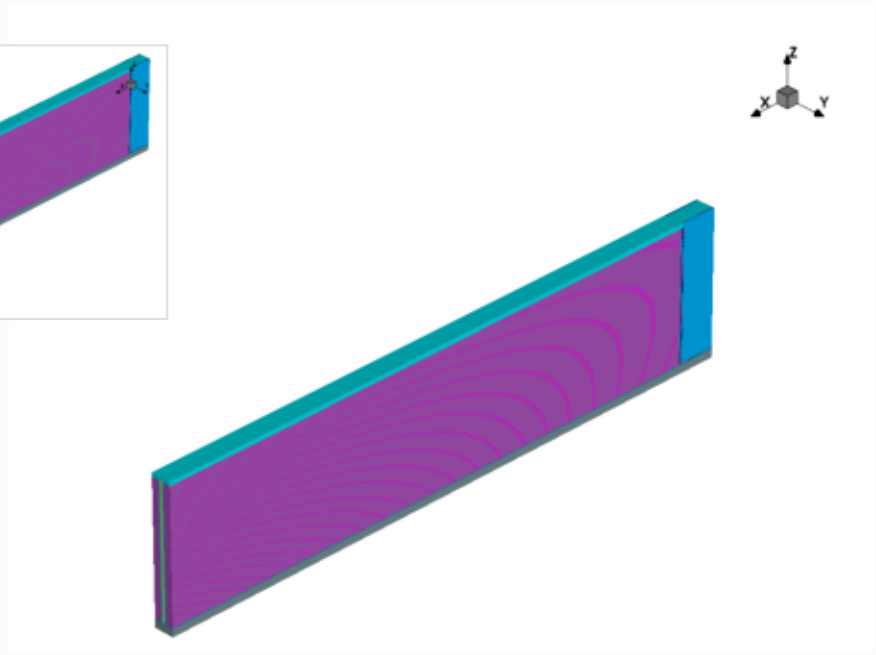
clean · Wafer_clean

0.169	686.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





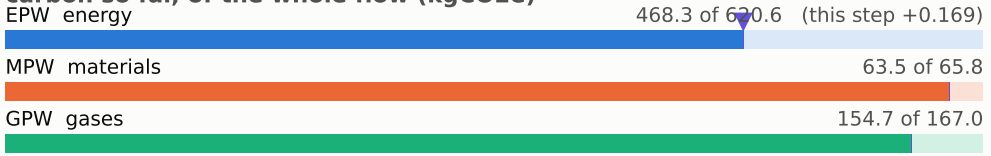
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

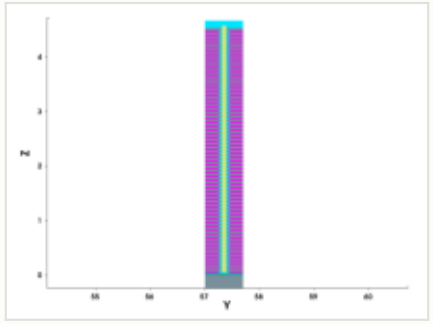
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

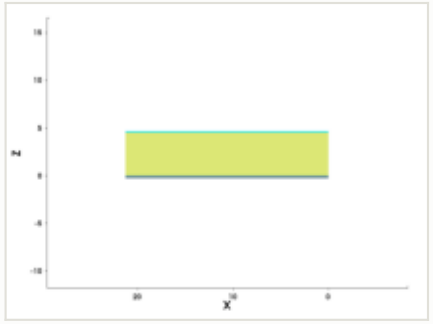


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



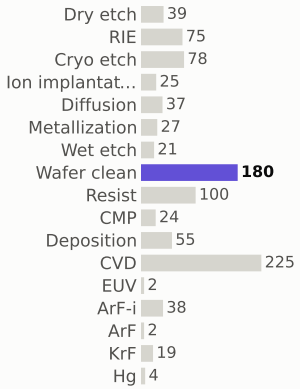
y cut at 57.351 µm



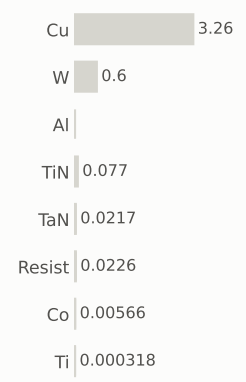
mask for this step

no mask for this step

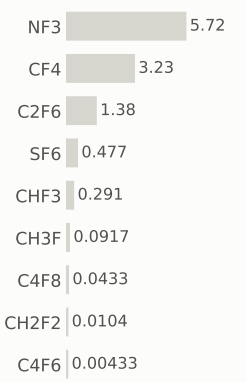
steps by process area



materials deposited (g)



gases emitted (g)



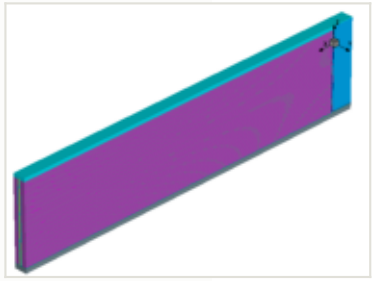
Step #11420: Deposition

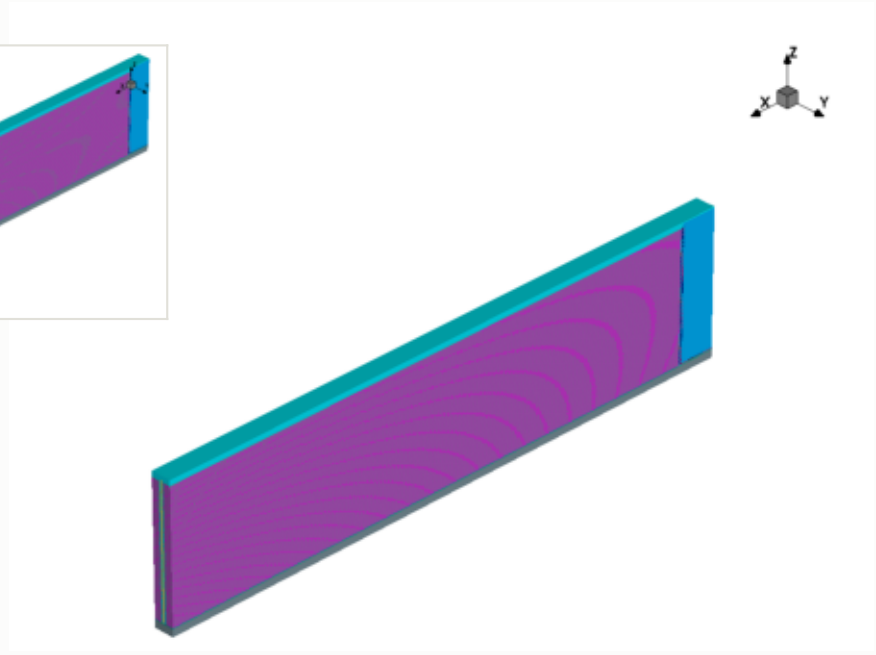
depo_isotropic · Deposition

0.656	687.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



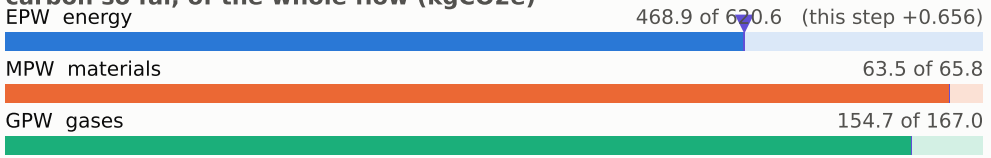


this step

process area	Deposition
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

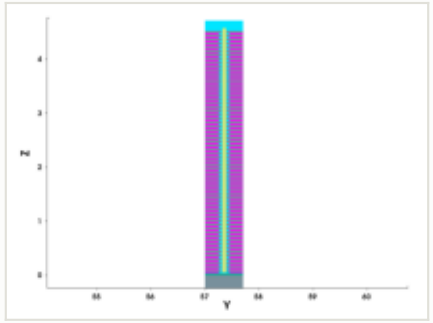
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

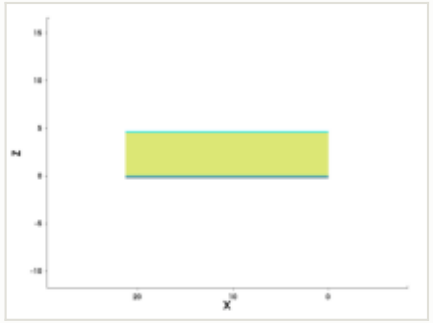


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



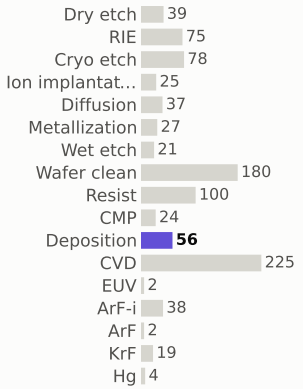
y cut at 57.351 µm



mask for this step

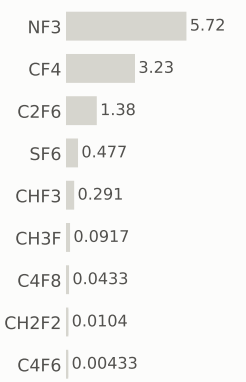
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



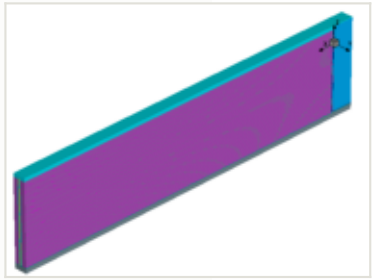
Step #11440: Wafer_clean

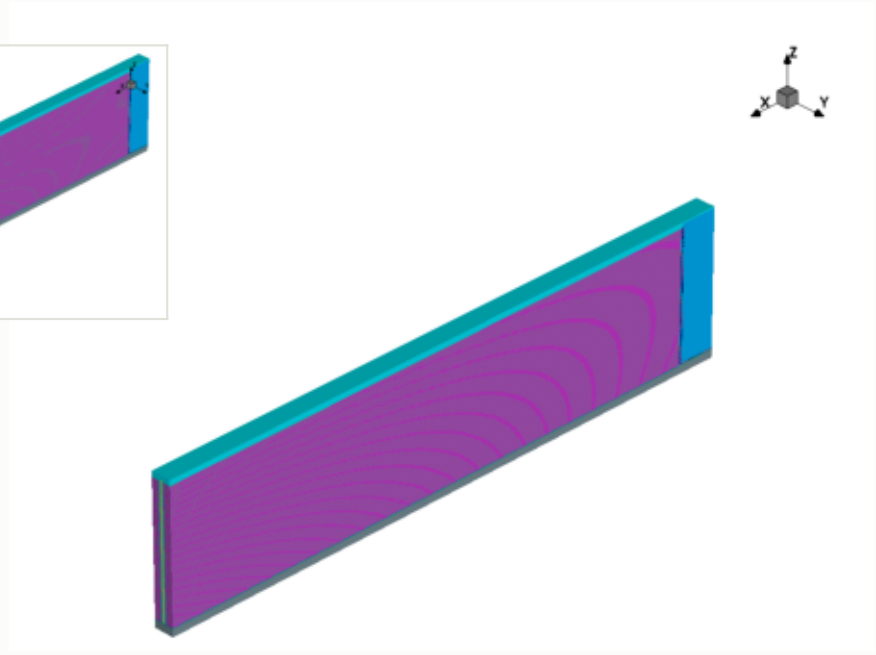
clean · Wafer_clean

0.169	687.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





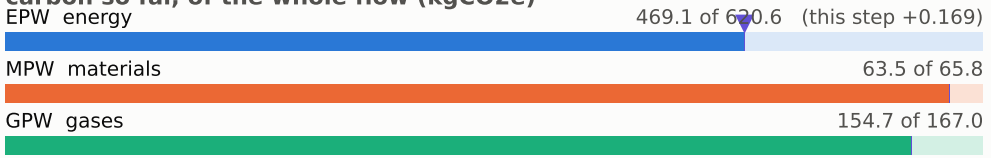
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

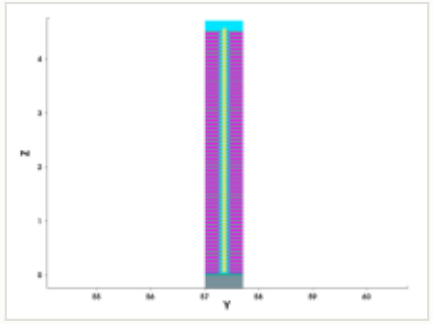
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

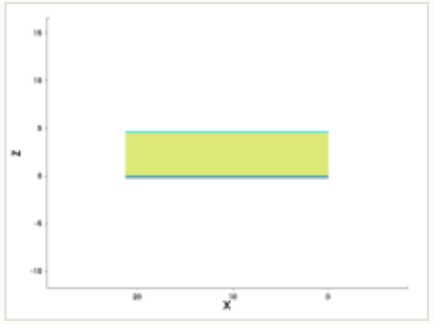


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



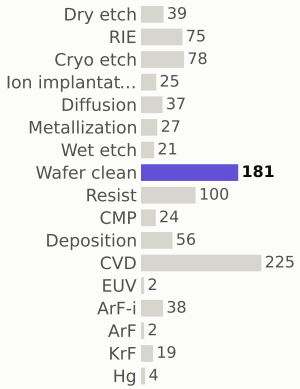
y cut at 57.351 µm



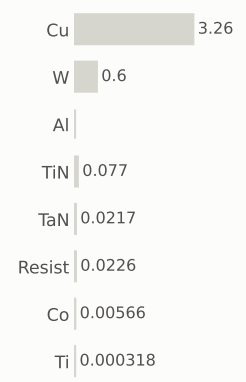
mask for this step

no mask for this step

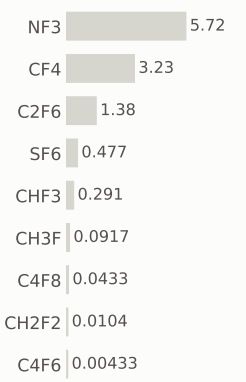
steps by process area



materials deposited (g)



gases emitted (g)



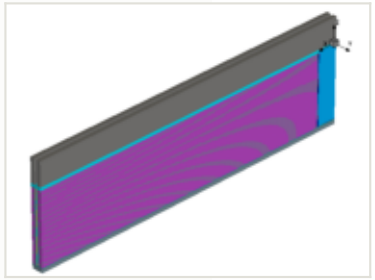
Step #11450: KrF

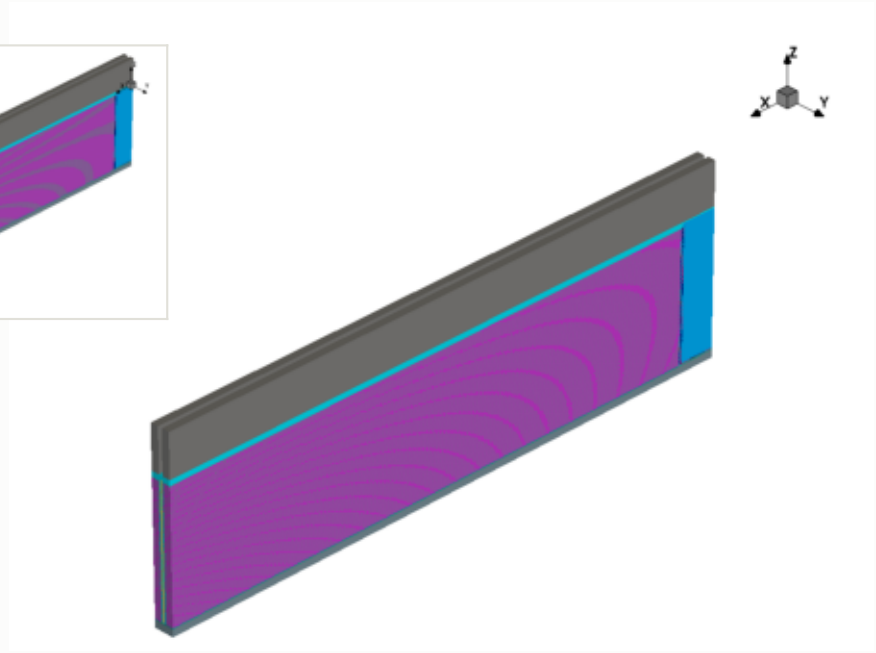
litho_patterning · KrF

0.199	687.5	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



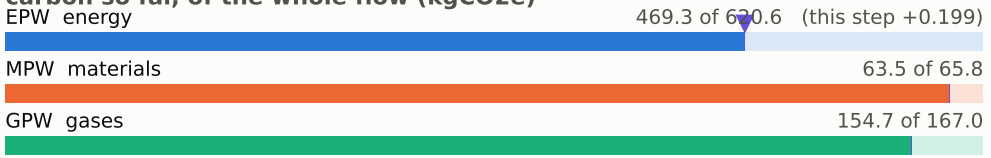


this step

process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

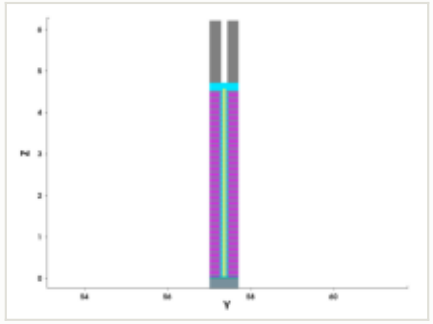
EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

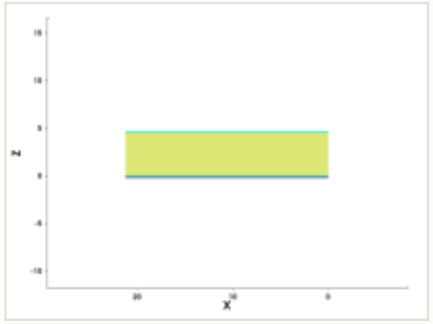


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

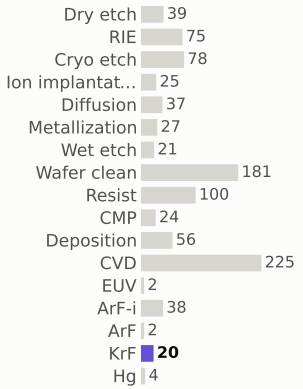


y cut at 57.351 µm

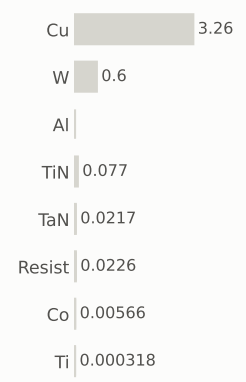


mask: layer 6:0, negative (GDSII)

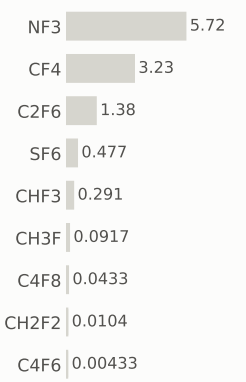
steps by process area



materials deposited (g)



gases emitted (g)



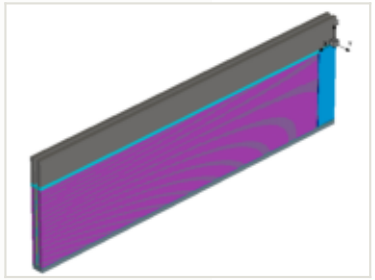
Step #11460: RIE

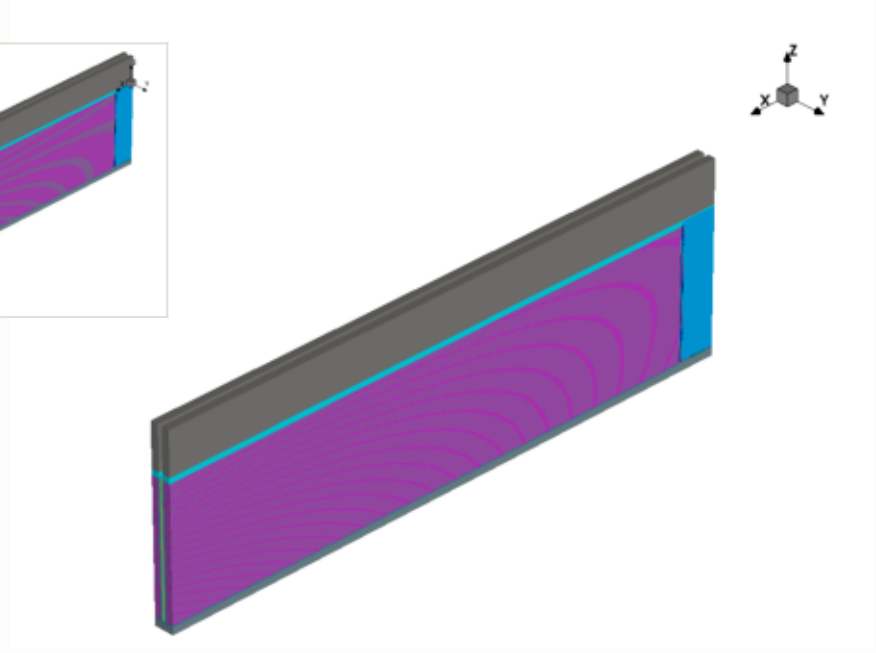
etch_anisotropic · RIE

1.04	688.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



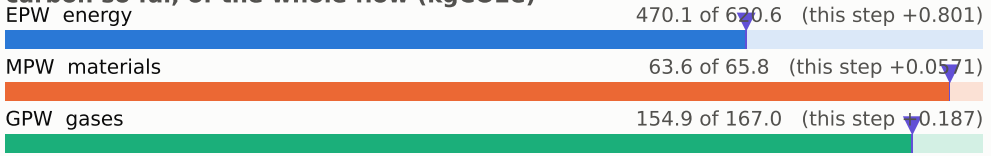


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

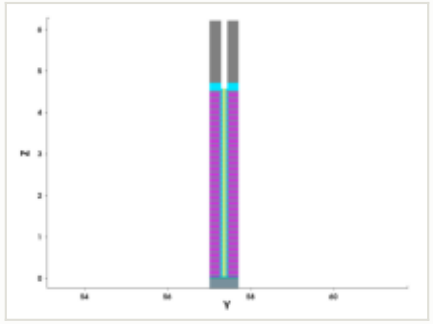
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

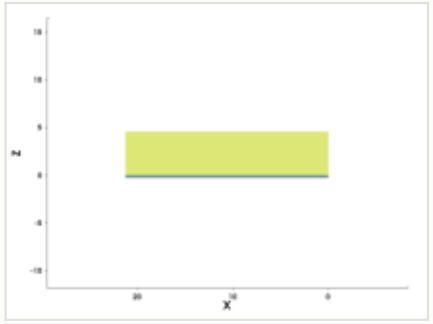


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



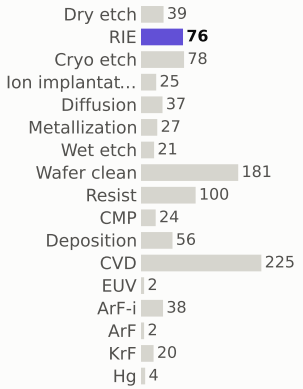
y cut at 57.351 µm



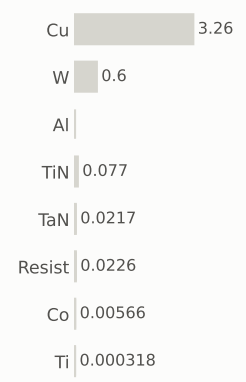
mask for this step

no mask for this step

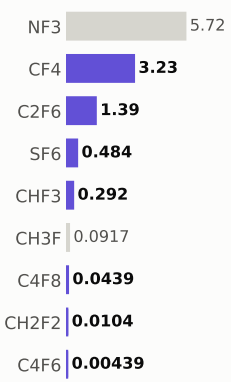
steps by process area



materials deposited (g)



gases emitted (g)



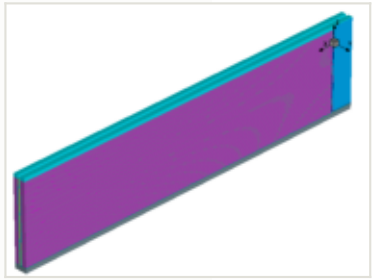
Step #11470: Dry_etch

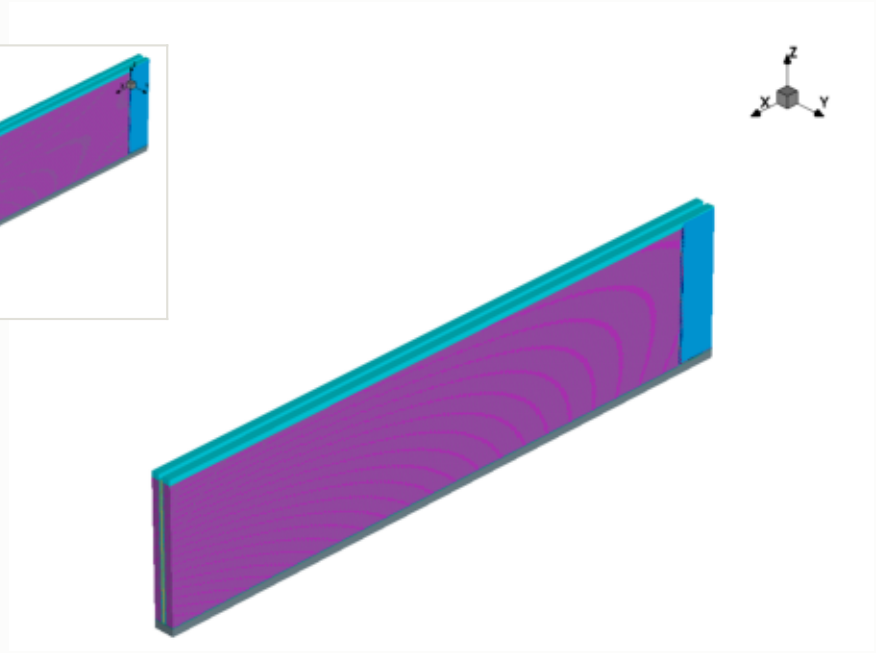
etch_strip · Dry_etch

0.801	689.4	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



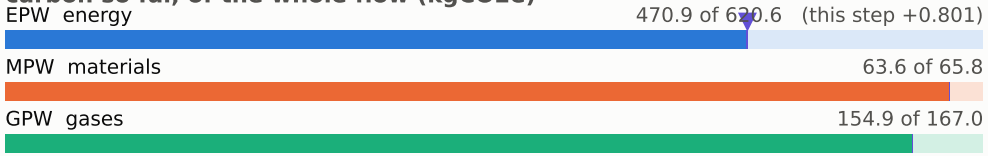


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

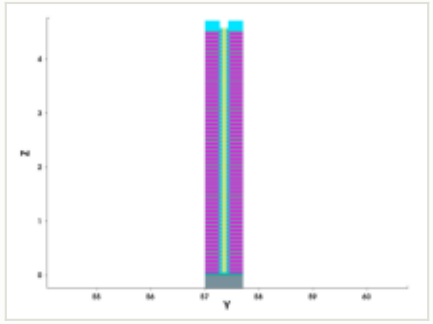
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

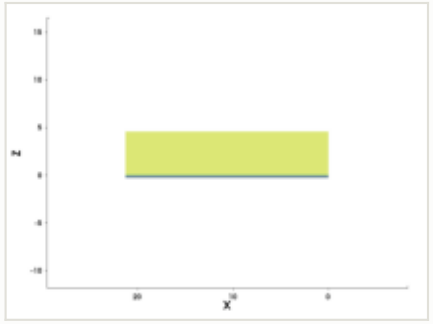


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



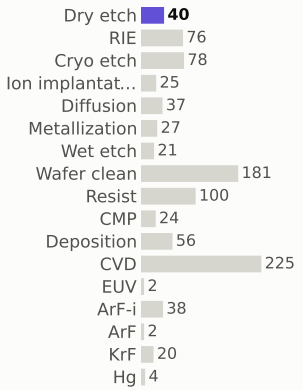
y cut at 57.351 µm



mask for this step

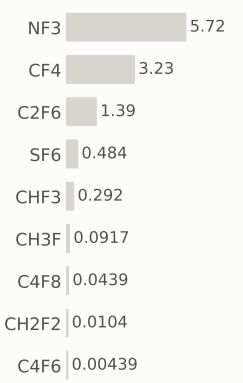
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



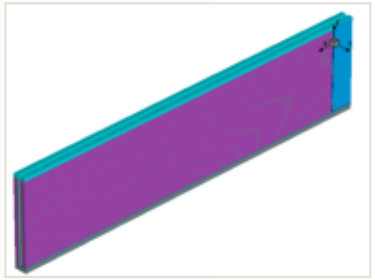
Step #11480: Wafer_clean

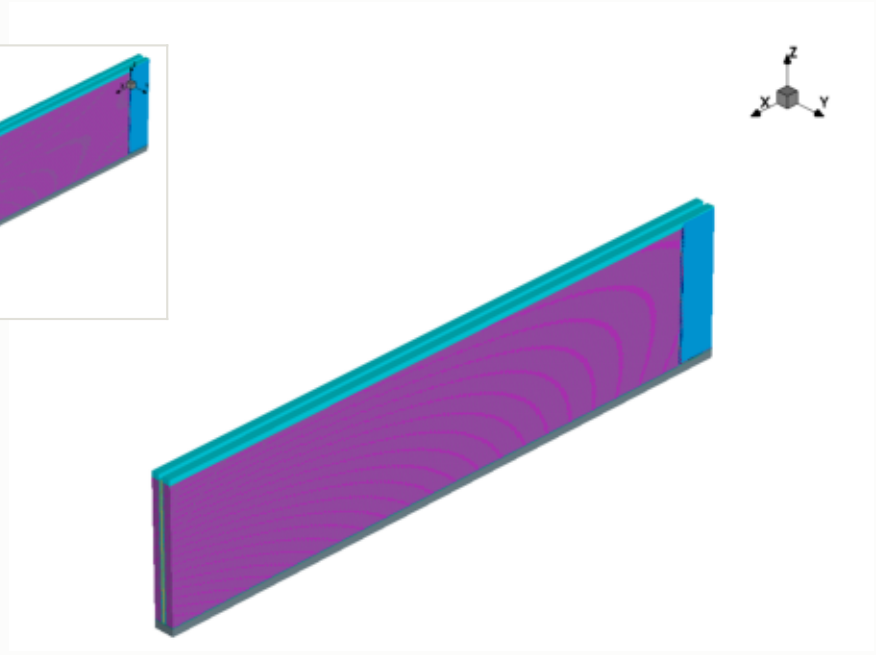
clean · Wafer_clean

0.169	689.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





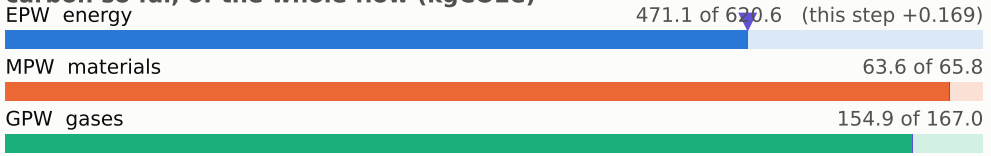
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

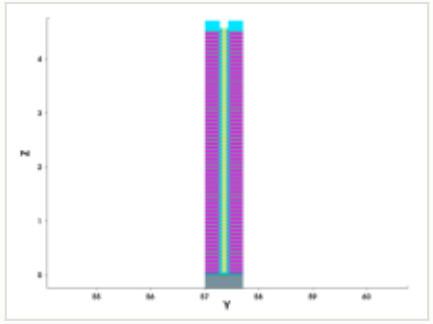
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

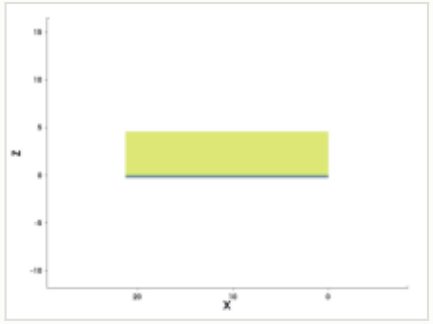


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



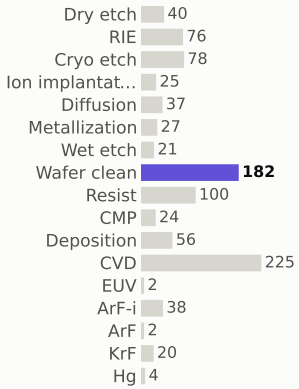
y cut at 57.351 µm



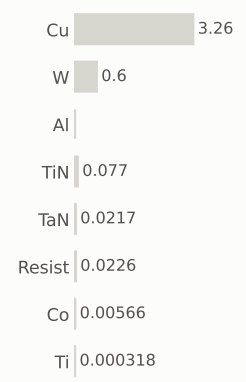
mask for this step

no mask for this step

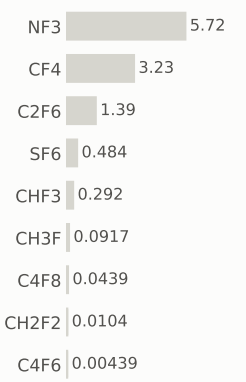
steps by process area



materials deposited (g)



gases emitted (g)



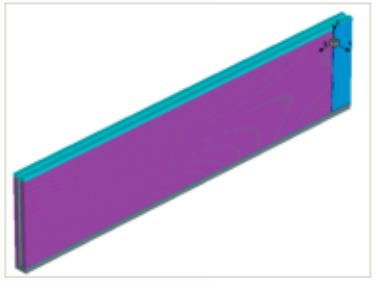
Step #11490: Wet_etch

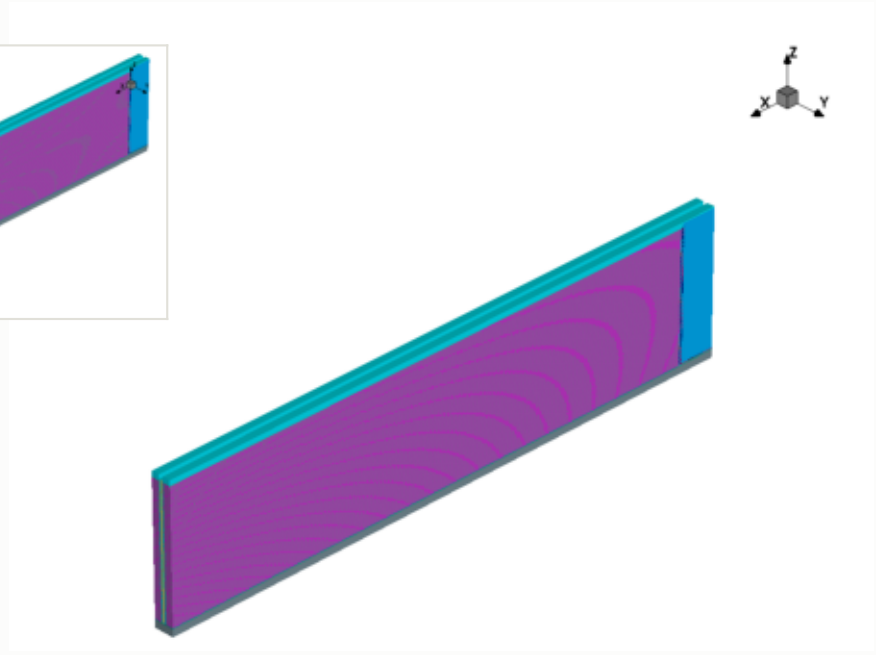
etch_isotropic · Wet_etch

0.169	689.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



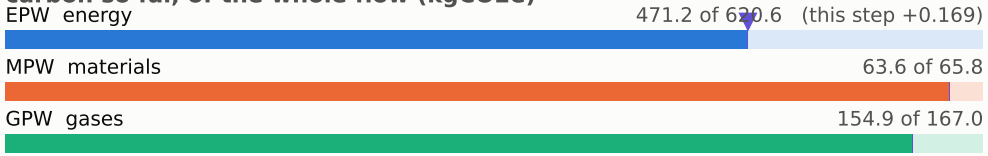


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

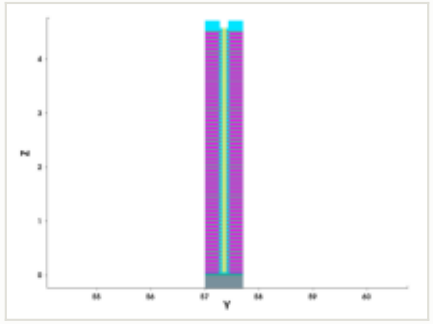
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

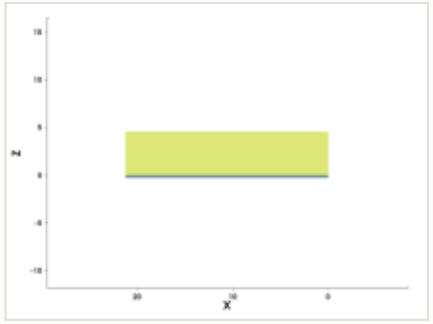


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



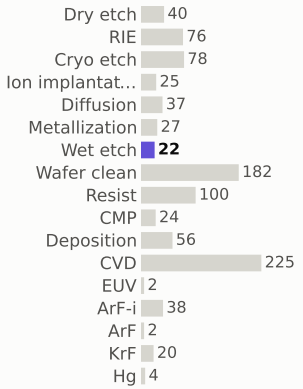
y cut at 57.351 µm



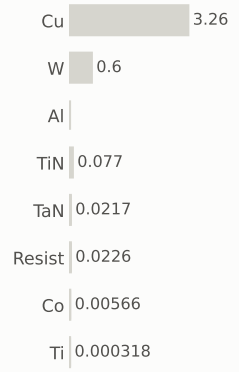
mask for this step

no mask for this step

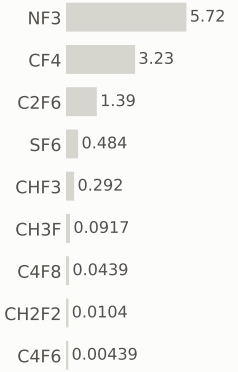
steps by process area



materials deposited (g)



gases emitted (g)



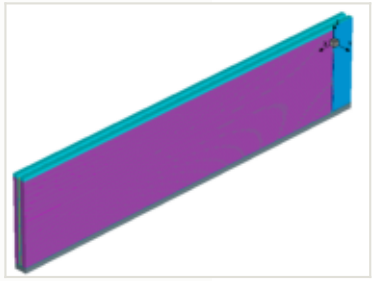
Step #11500: Wet_etch

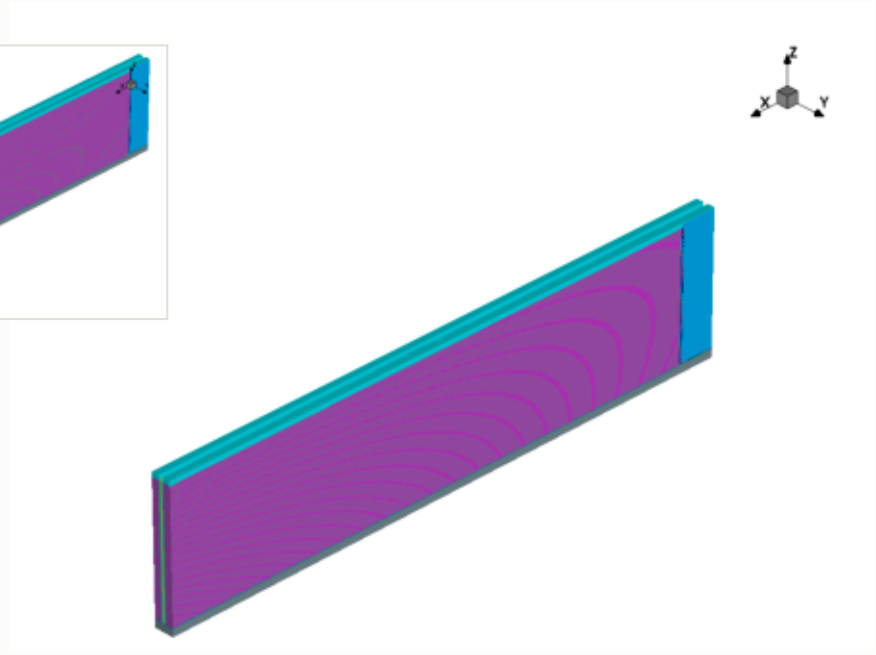
etch_isotropic · Wet_etch

0.169	689.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



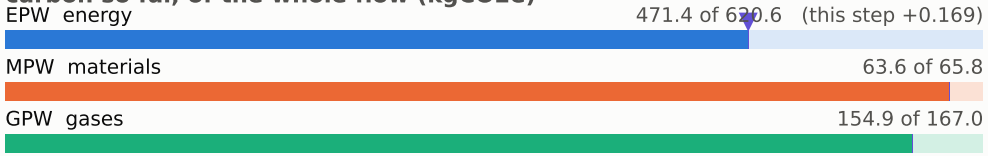


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

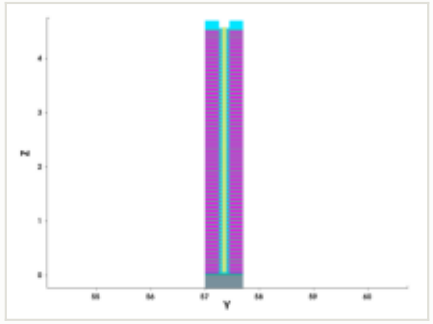
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

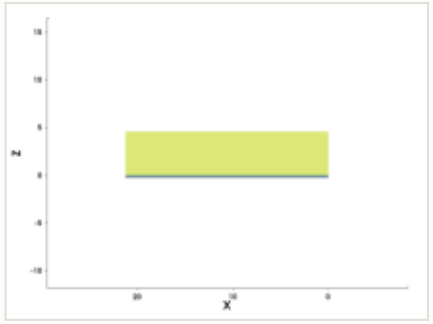


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



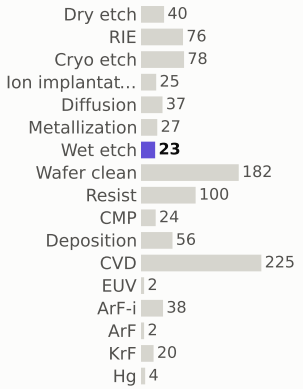
y cut at 57.351 µm



mask for this step

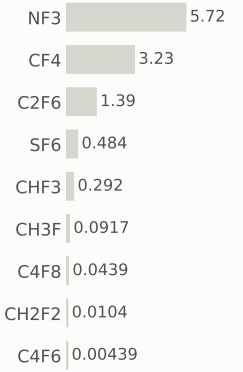
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



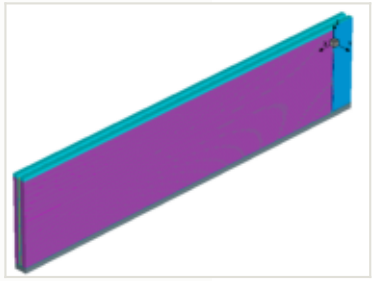
Step #11510: Wet_etch

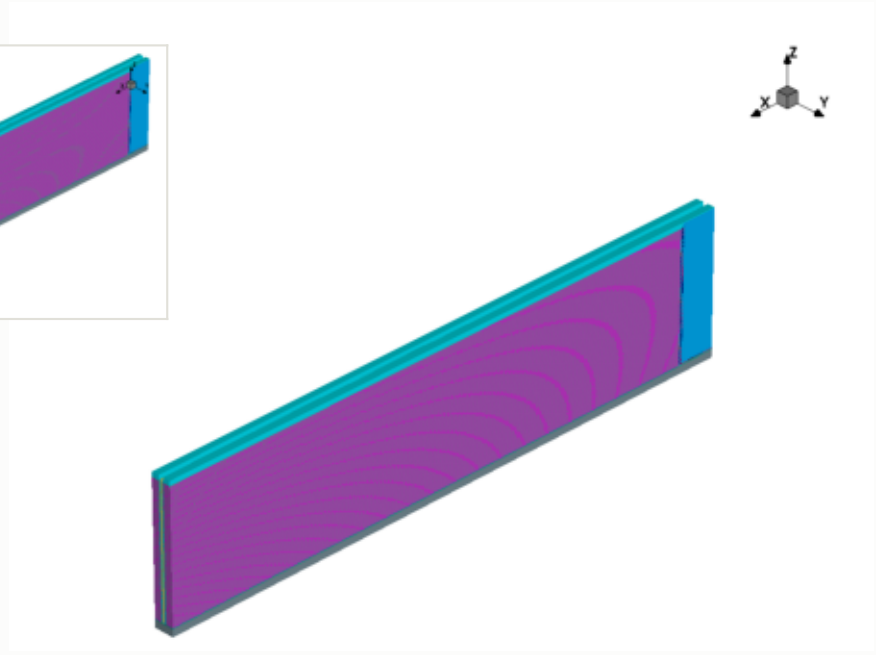
etch_isotropic · Wet_etch

0.169	690.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



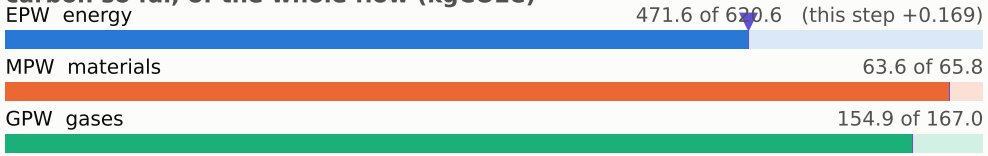


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

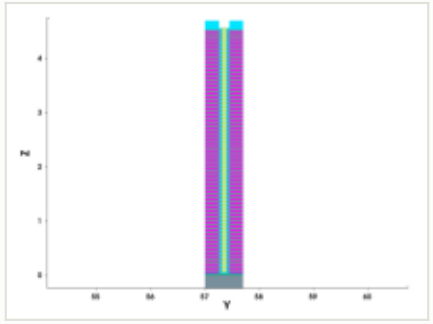
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

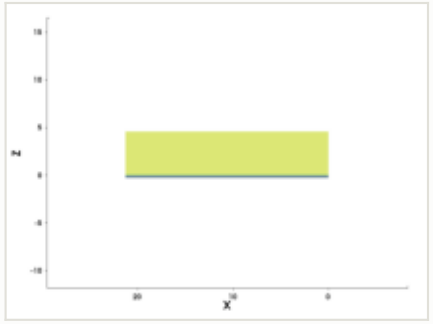


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



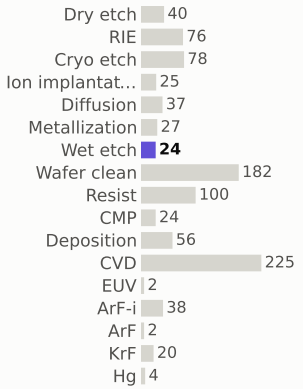
y cut at 57.351 µm



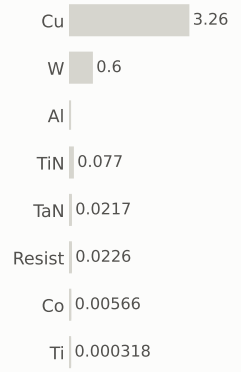
mask for this step

no mask for this step

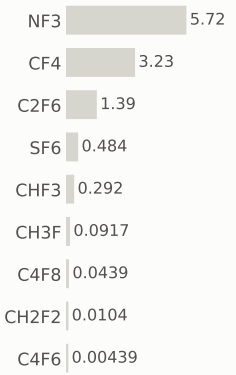
steps by process area



materials deposited (g)



gases emitted (g)



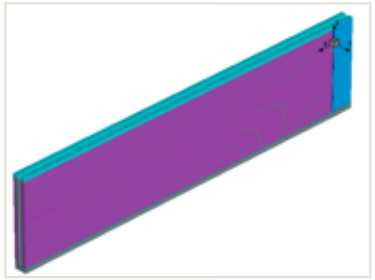
Step #11520: Dry_etch

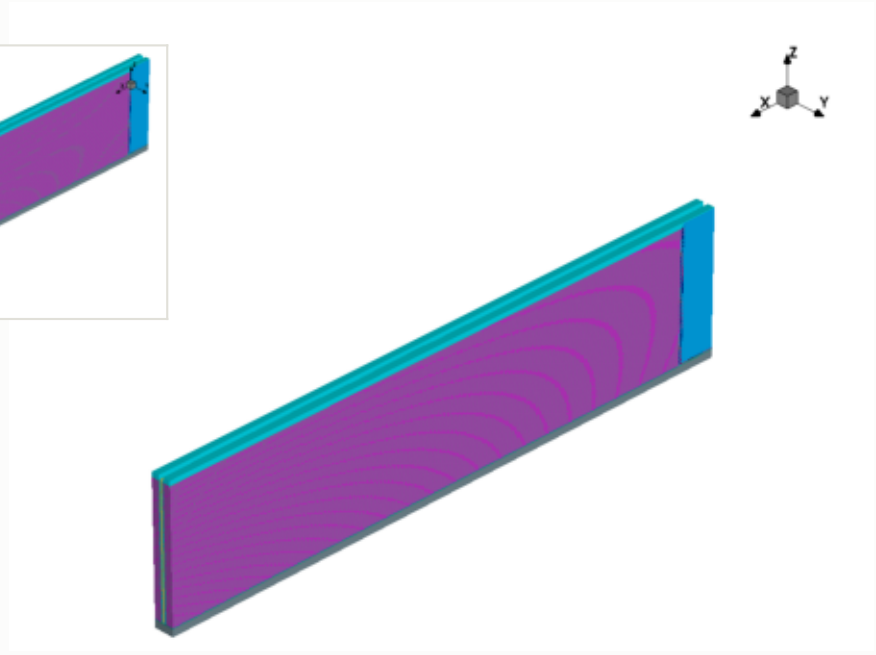
clean · Dry_etch

0.801	690.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





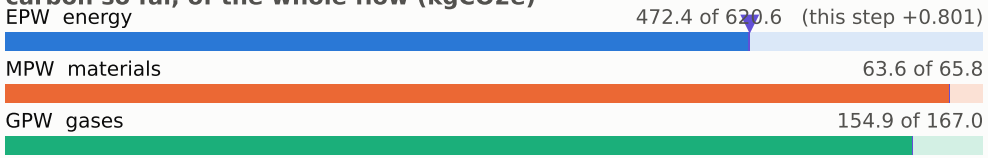
this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

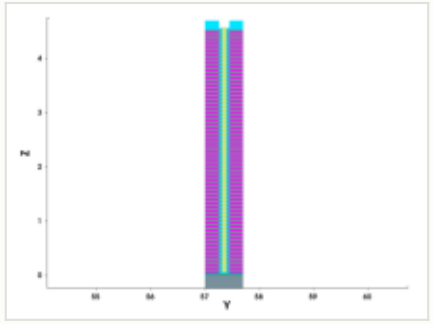
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

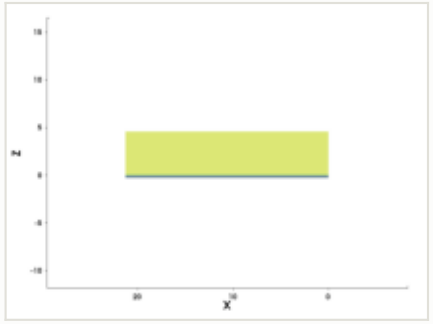


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



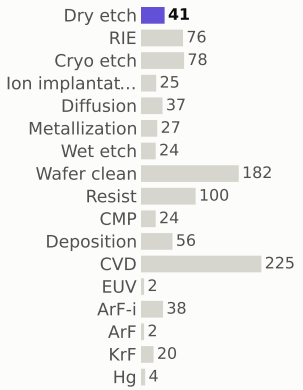
y cut at 57.351 µm



mask for this step

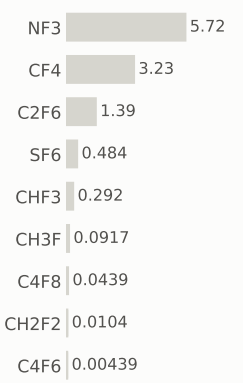
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



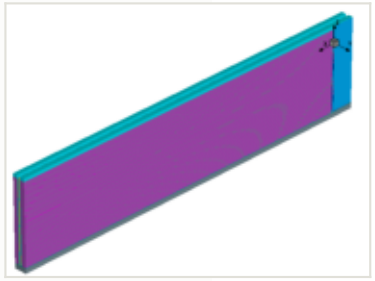
Step #11530: CVD

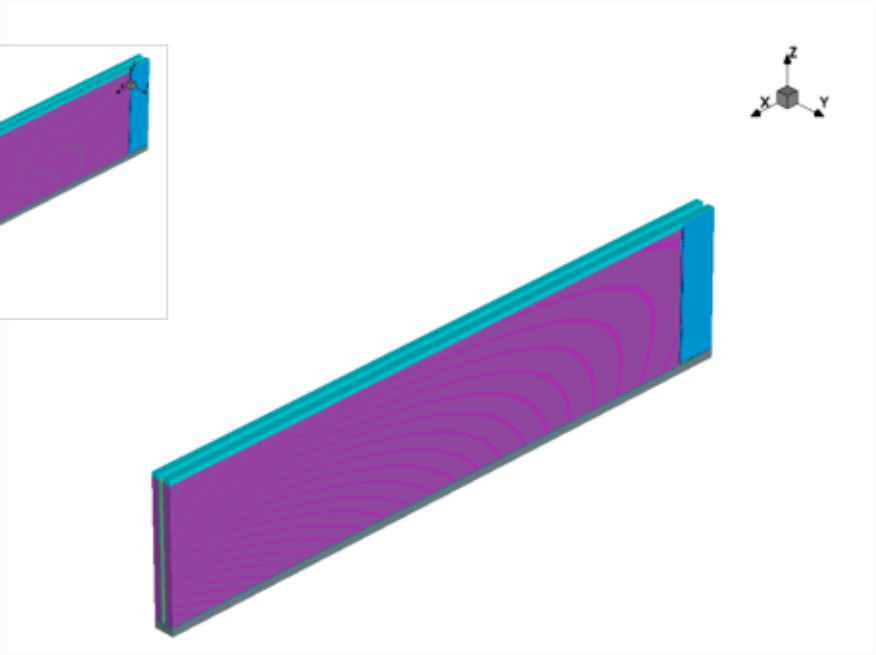
depo_isotropic · CVD

1.31	692.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



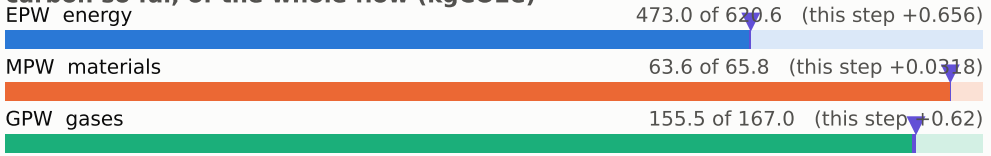


this step

process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

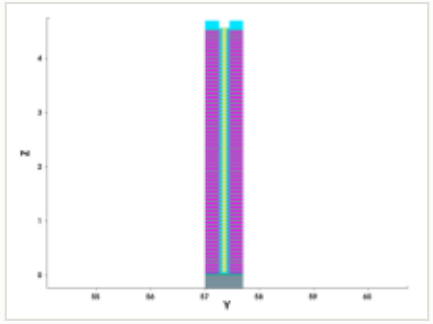
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

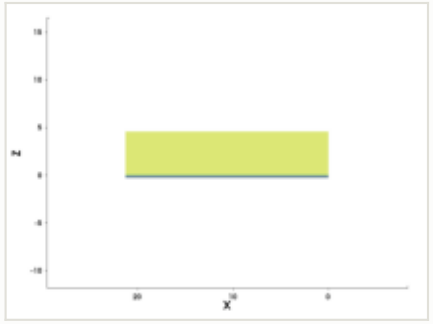


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



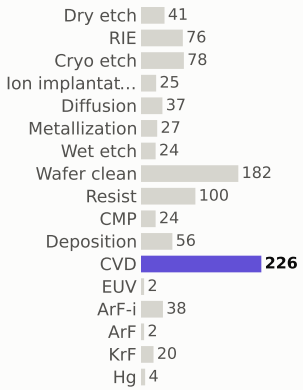
y cut at 57.351 µm



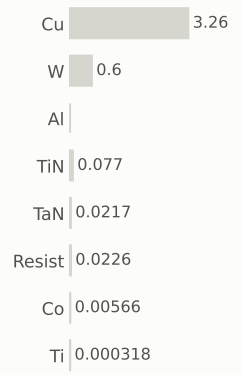
mask for this step

no mask for this step

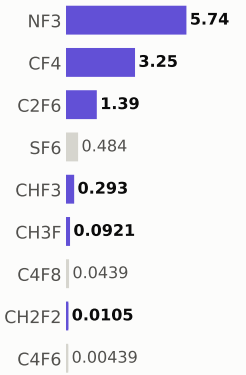
steps by process area



materials deposited (g)



gases emitted (g)



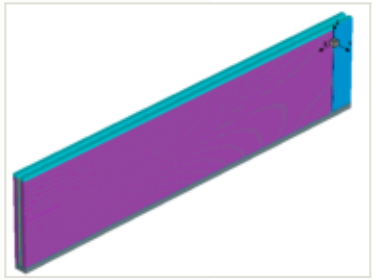
Step #11540: Metallization

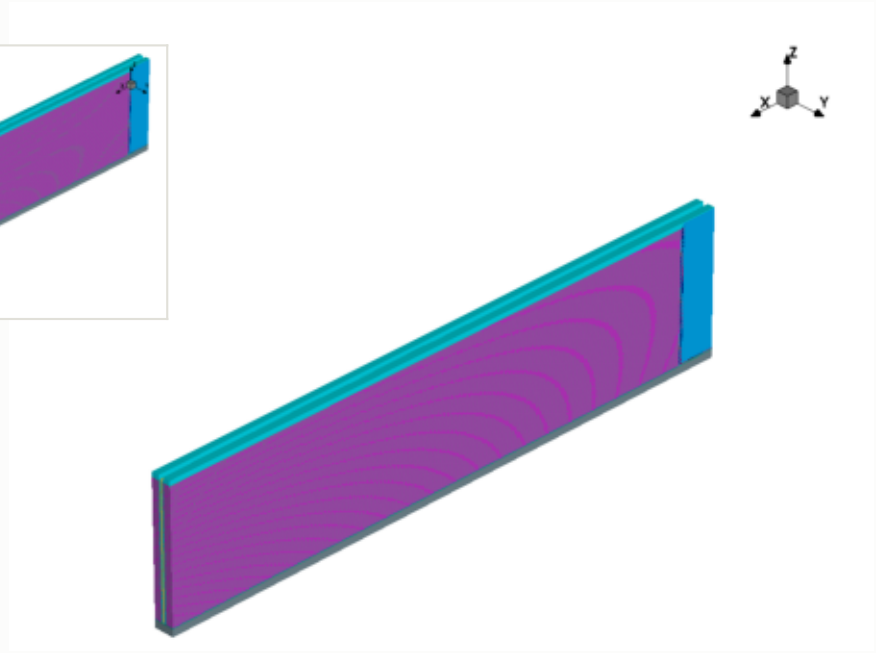
depo_isotropic · Metallization

1.2	693.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



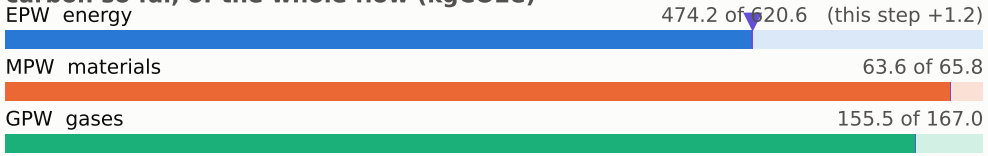


this step

process area	Metallization
material	—
thickness	0.003 µm
material GWP	—
input gases	—
energy	2 kWh + 0.8 overhead

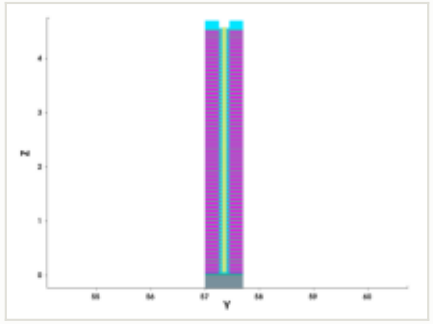
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

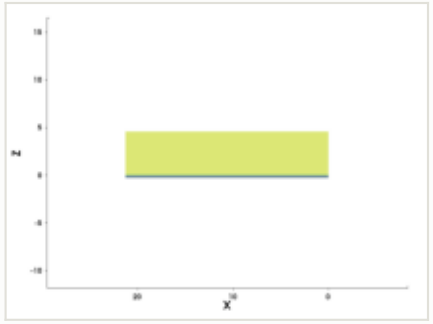


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



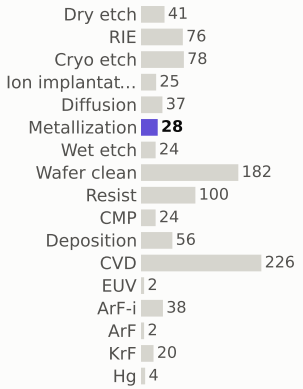
y cut at 57.351 µm



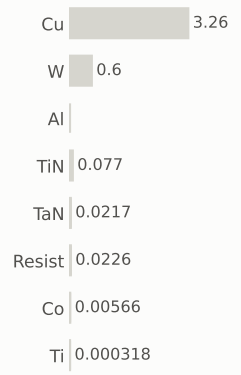
mask for this step

no mask for this step

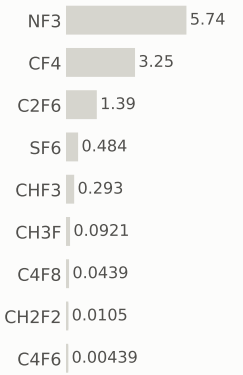
steps by process area



materials deposited (g)



gases emitted (g)



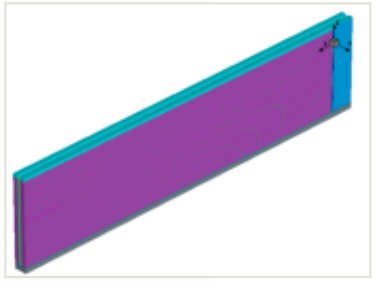
Step #11550: Diffusion

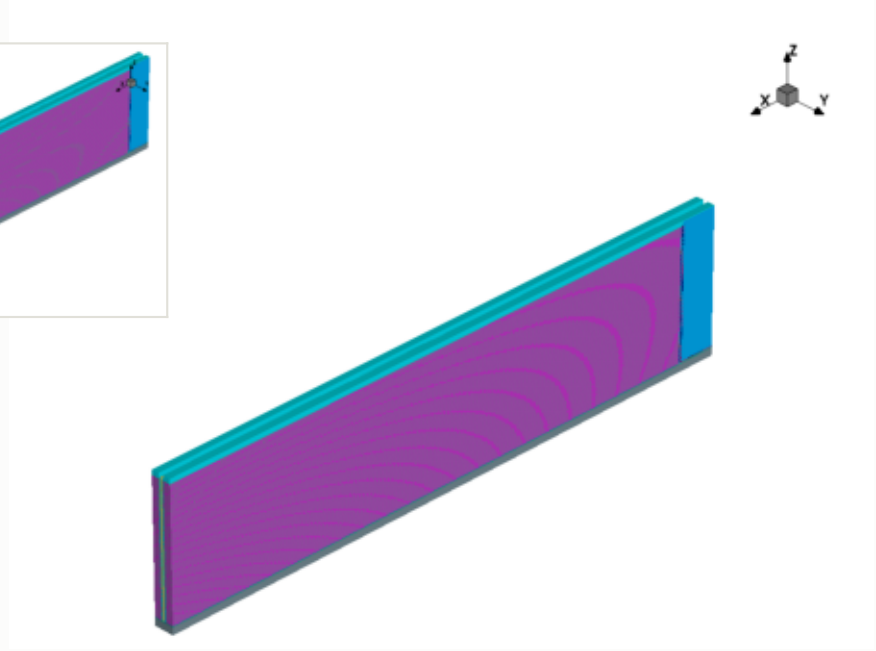
anneal · Diffusion

0.801	694.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





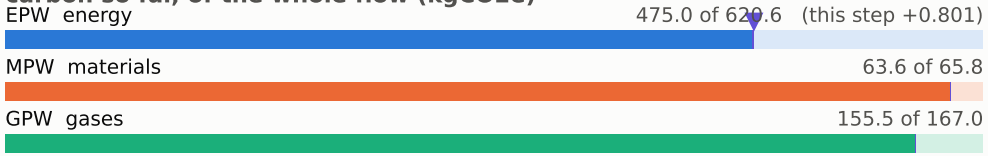
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

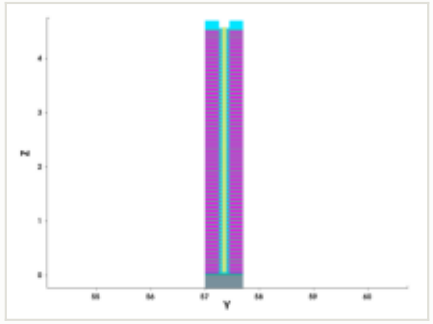
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

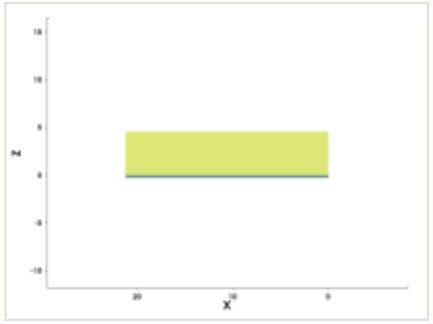


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



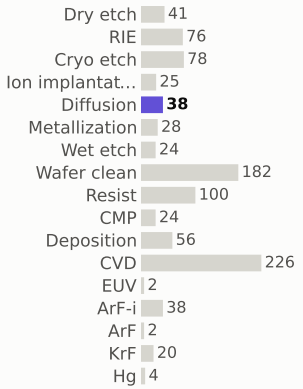
y cut at 57.351 µm



mask for this step

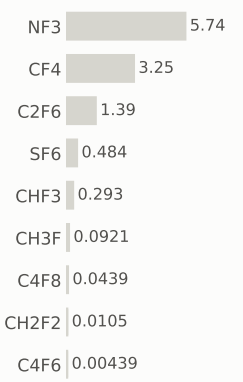
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



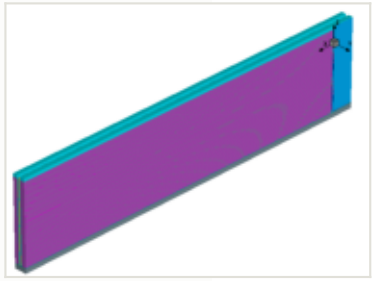
Step #11560: Wafer_clean

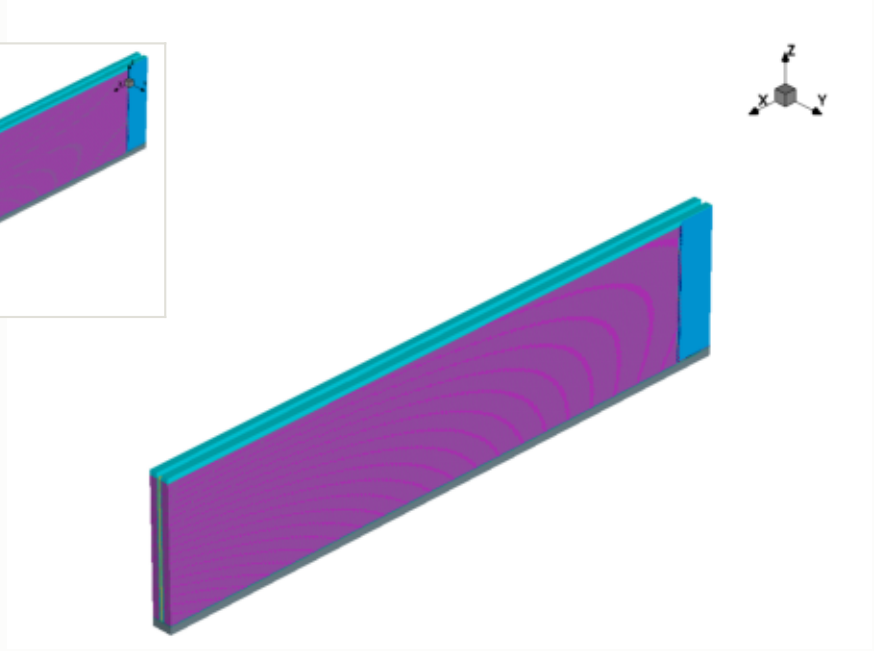
clean · Wafer_clean

0.169	694.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





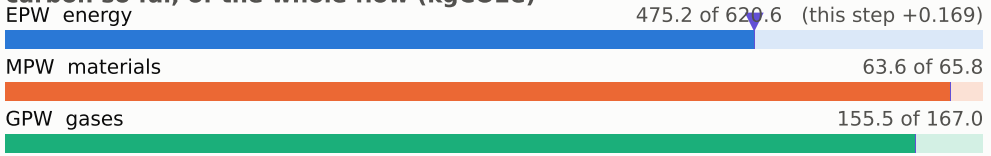
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

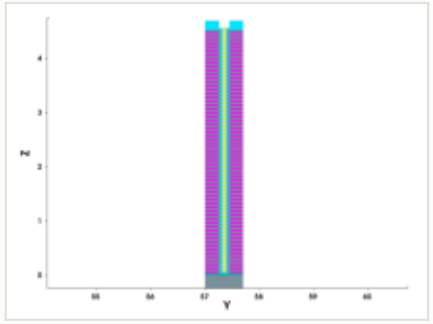
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

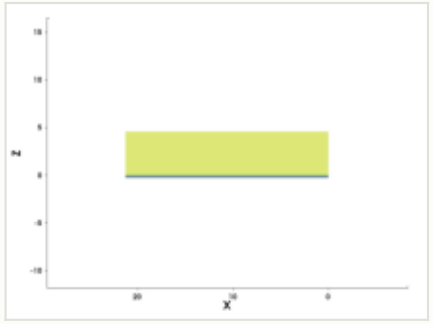


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



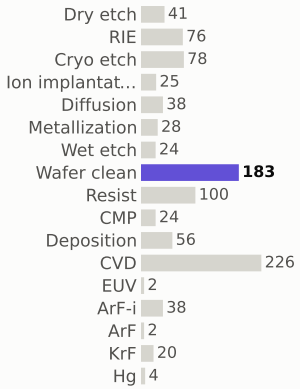
y cut at 57.351 µm



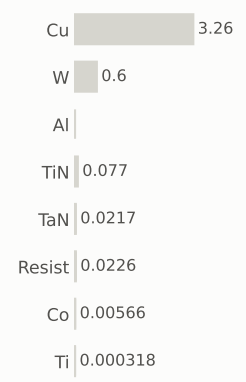
mask for this step

no mask for this step

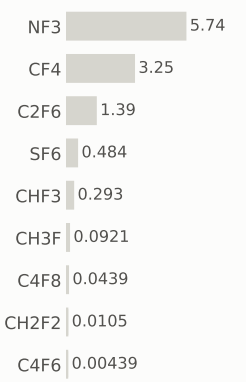
steps by process area



materials deposited (g)



gases emitted (g)



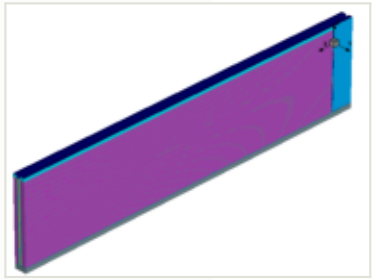
Step #11570: Metallization

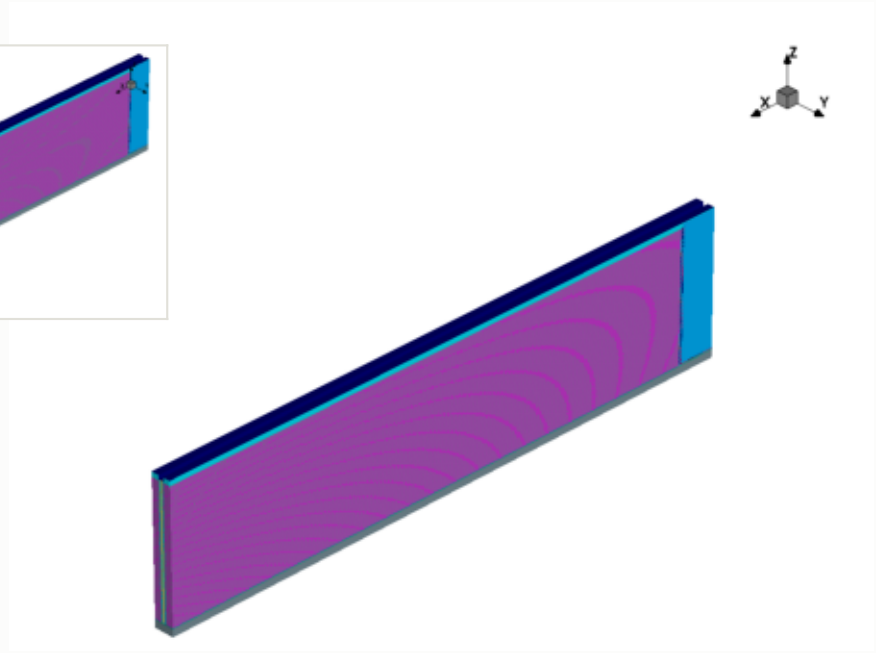
depo_isotropic · Metallization

1.2	695.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



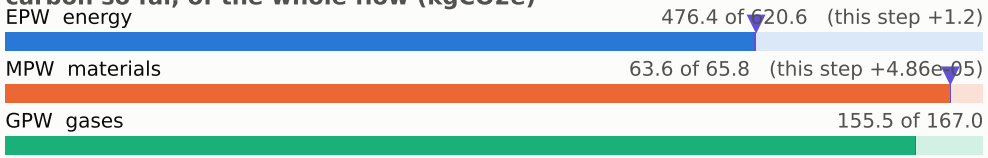


this step

process area	Metallization
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

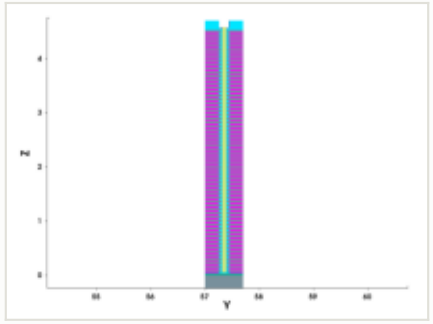
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

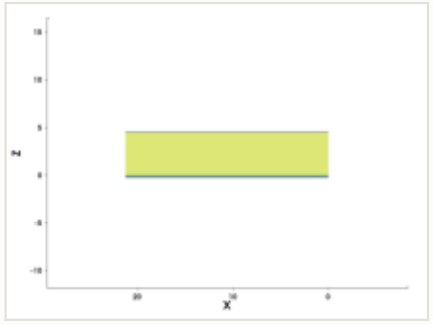


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



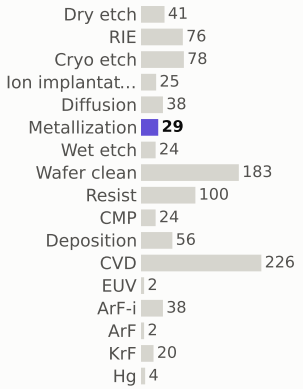
y cut at 57.351 µm



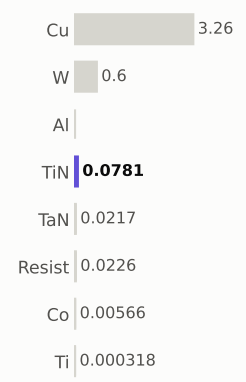
mask for this step

no mask for this step

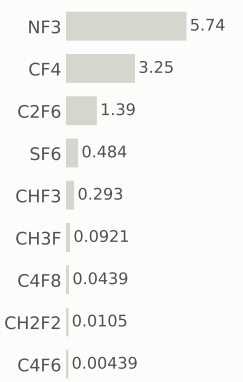
steps by process area



materials deposited (g)



gases emitted (g)



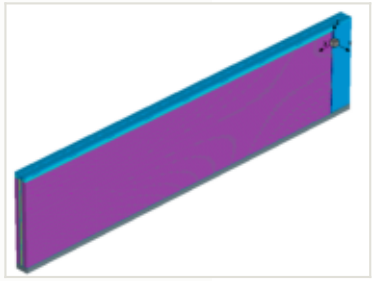
Step #11580: Metallization

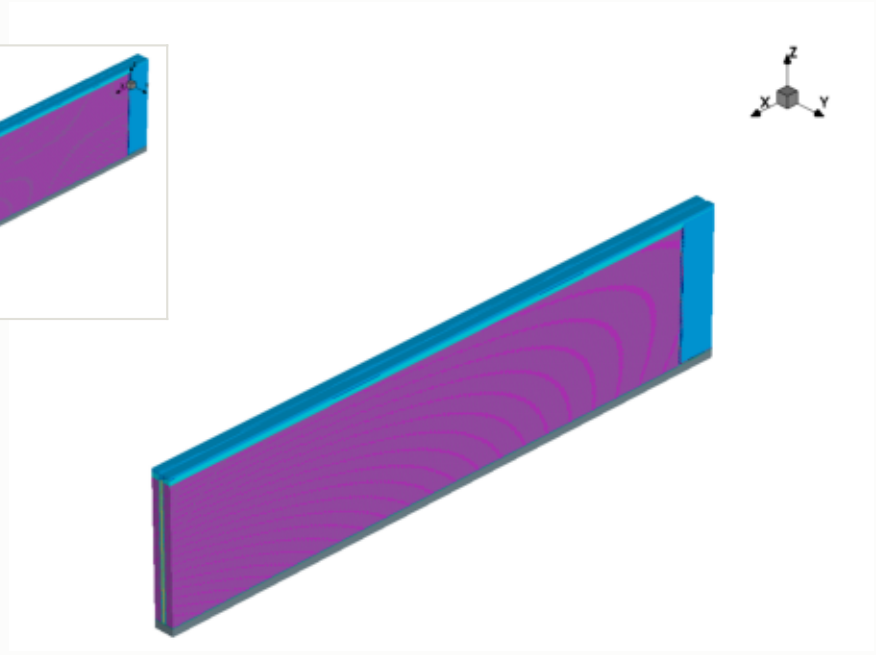
depo_isotropic · Metallization

1.21	696.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



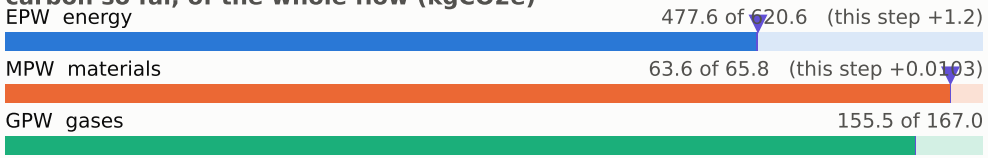


this step

process area	Metallization
material	W
thickness	0.1 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

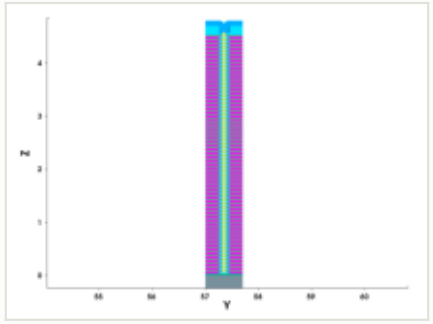
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0103 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

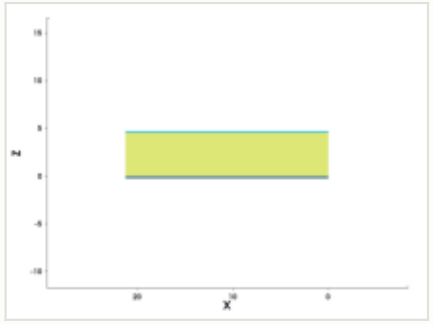


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



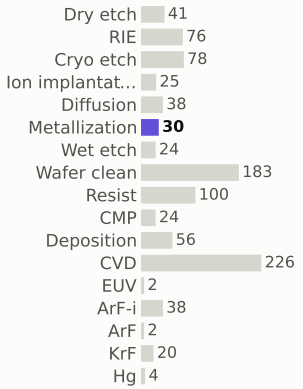
y cut at 57.351 µm



mask for this step

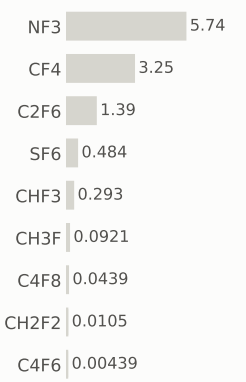
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



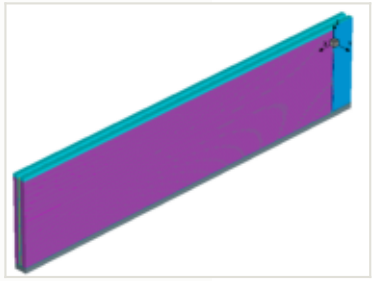
Step #11590: RIE

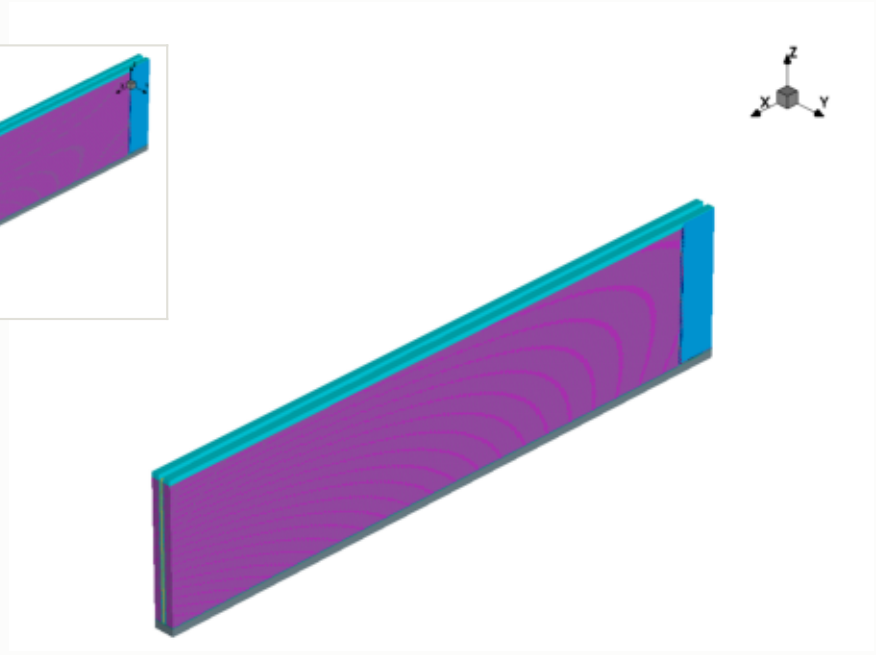
etch_anisotropic · RIE

1.04	697.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



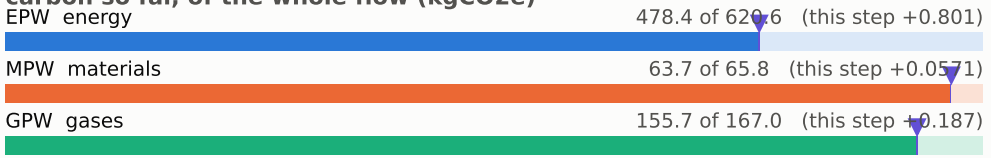


this step

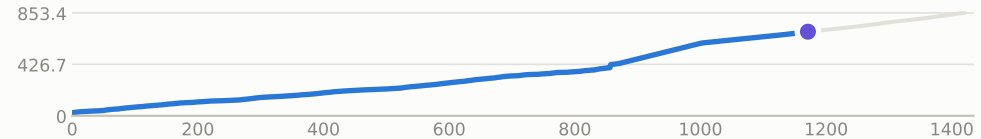
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

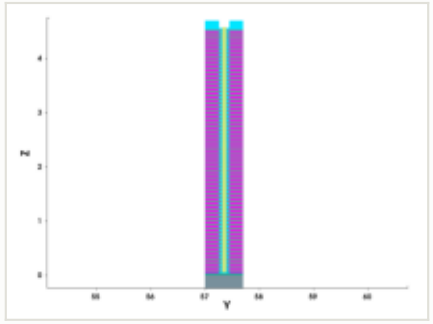
carbon so far, of the whole flow (kgCO2e)



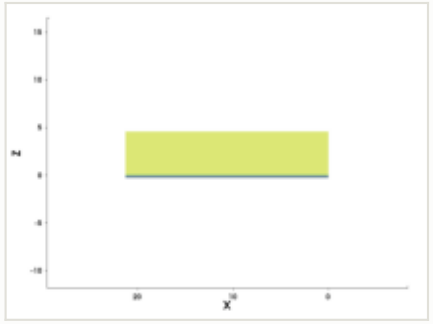
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



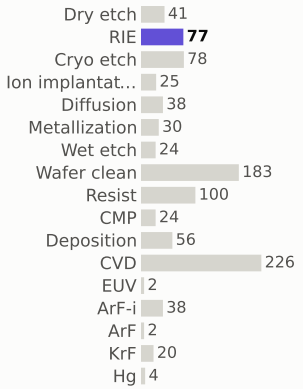
y cut at 57.351 µm



mask for this step

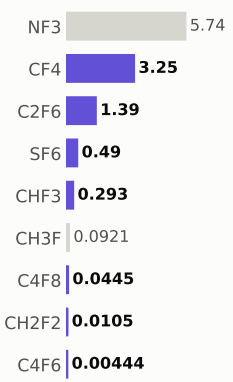
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



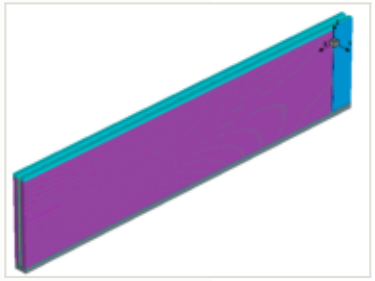
Step #11600: Dry_etch

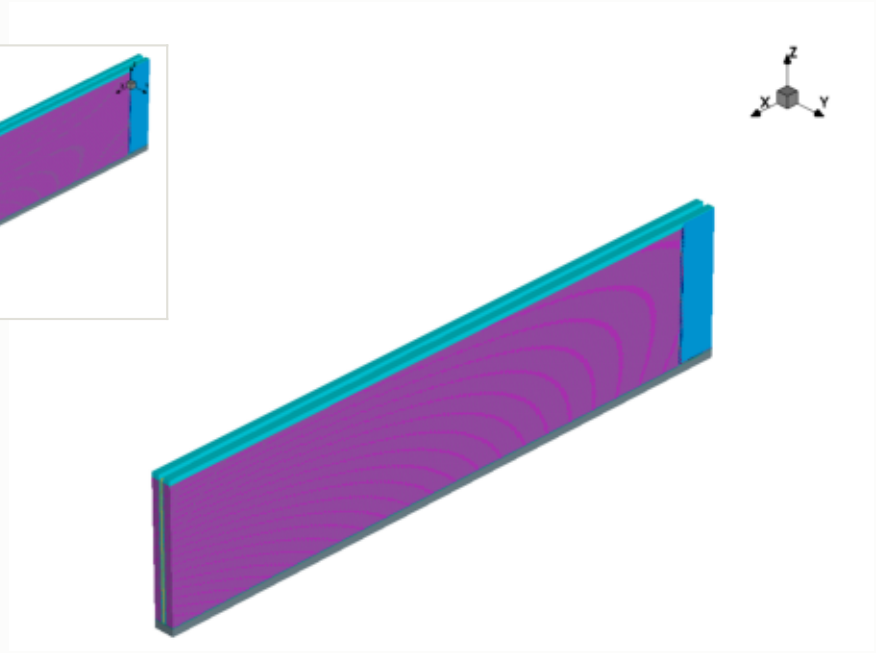
etch_isotropic · Dry_etch

0.801	698.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



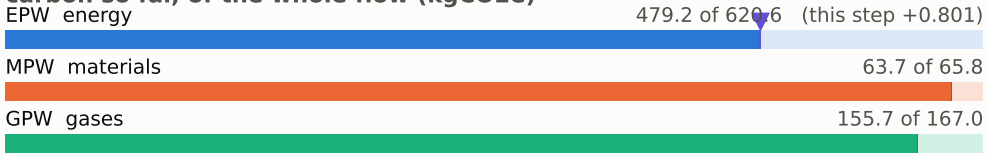


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

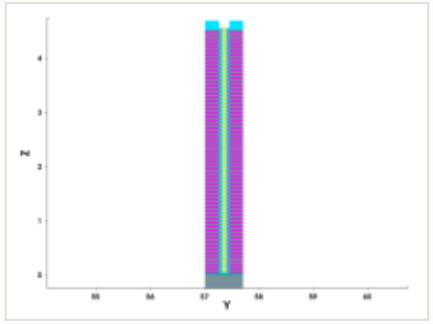
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

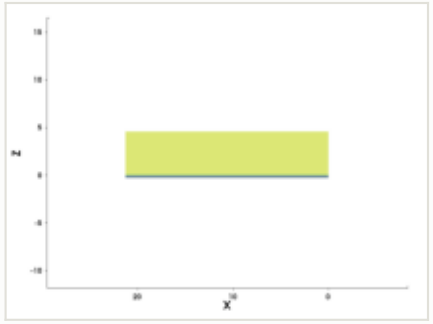


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



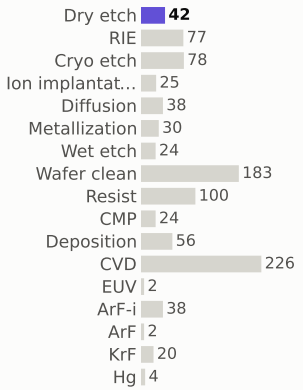
y cut at 57.351 µm



mask for this step

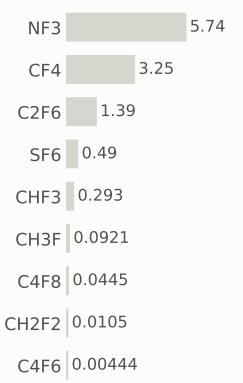
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



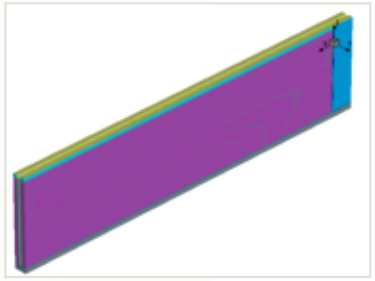
Step #11610: Deposition

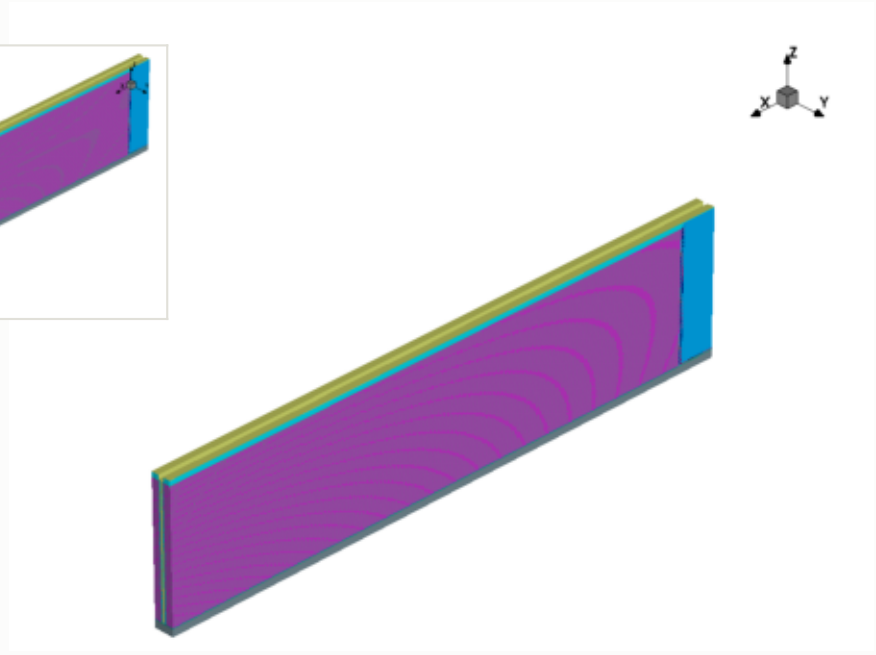
depo_isotropic · Deposition

0.656	699.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



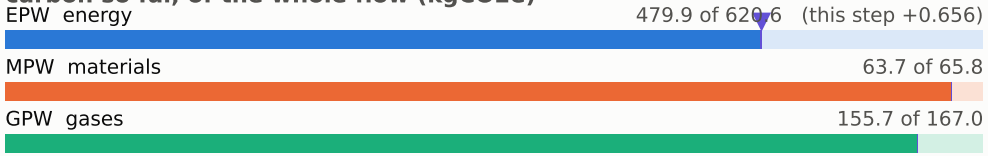


this step

process area	Deposition
material	—
thickness	0.005 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

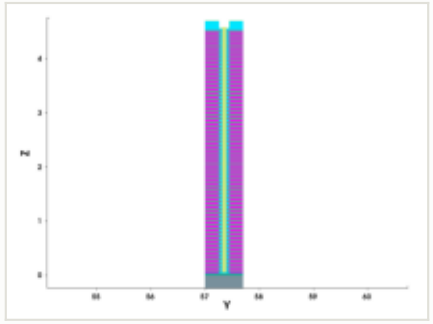
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

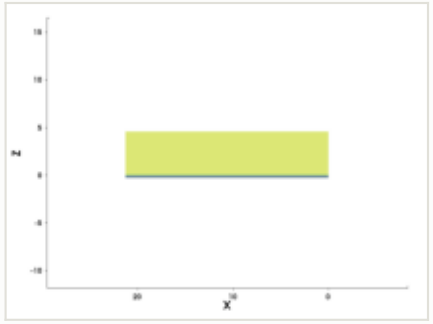


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



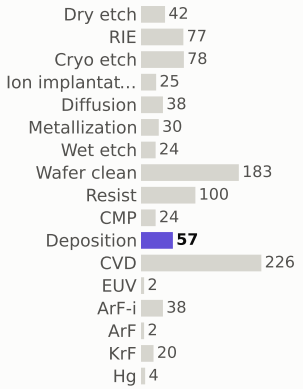
y cut at 57.351 µm



mask for this step

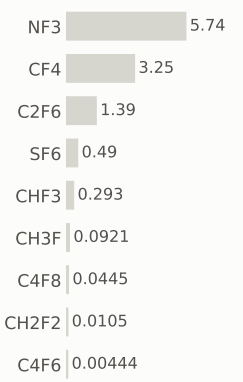
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



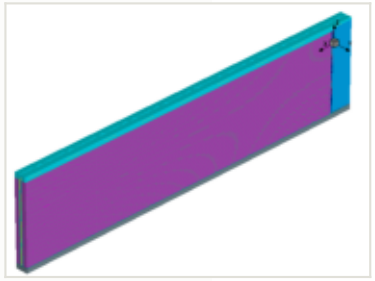
Step #11620: Deposition

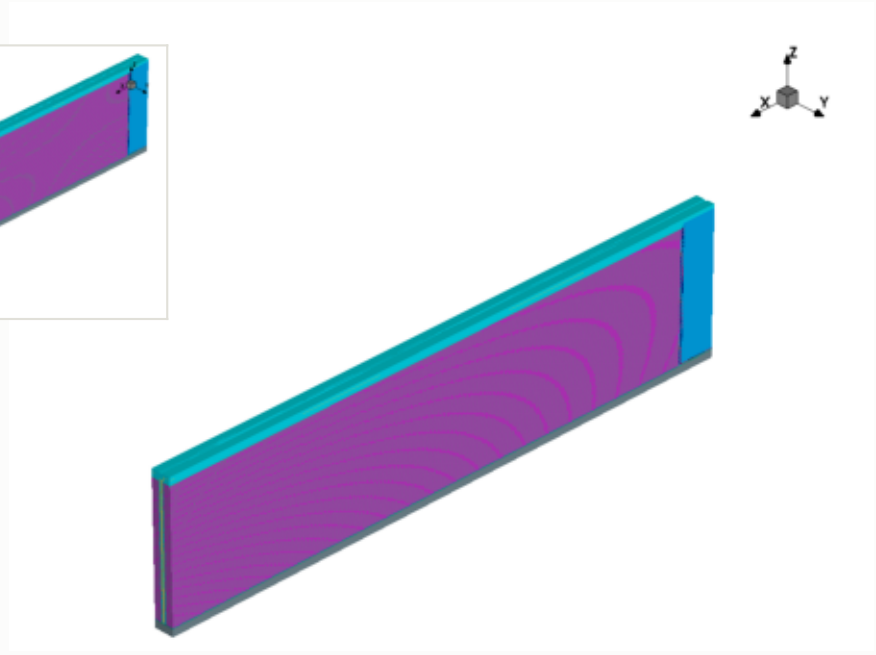
depo_isotropic · Deposition

0.656	699.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



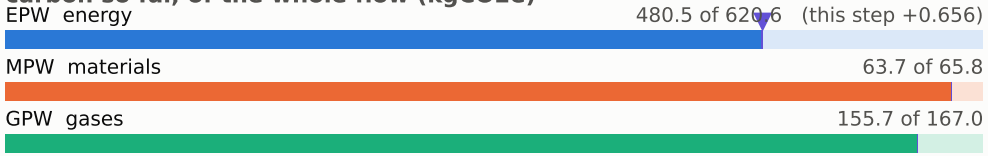


this step

process area	Deposition
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

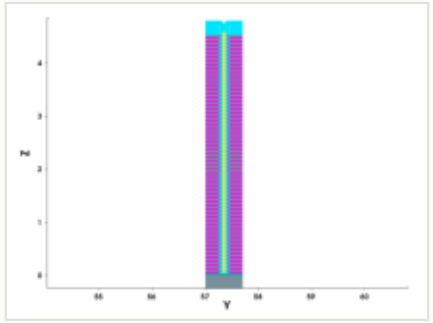
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

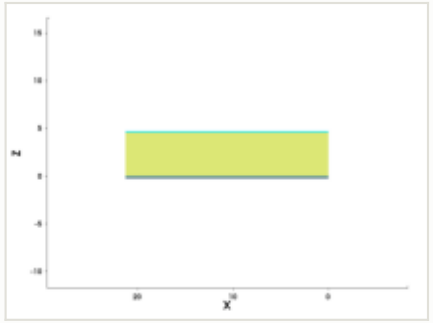


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



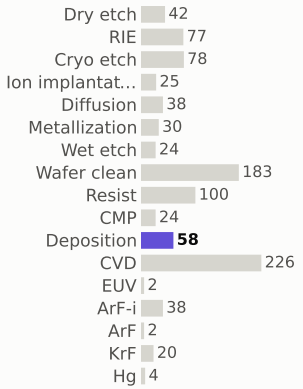
y cut at 57.351 µm



mask for this step

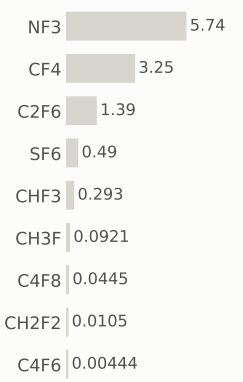
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



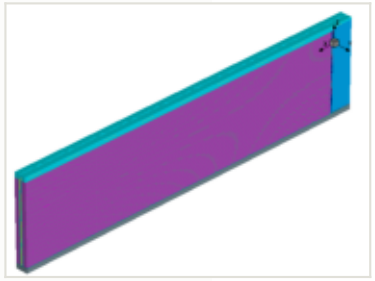
Step #11630: Dry_etch

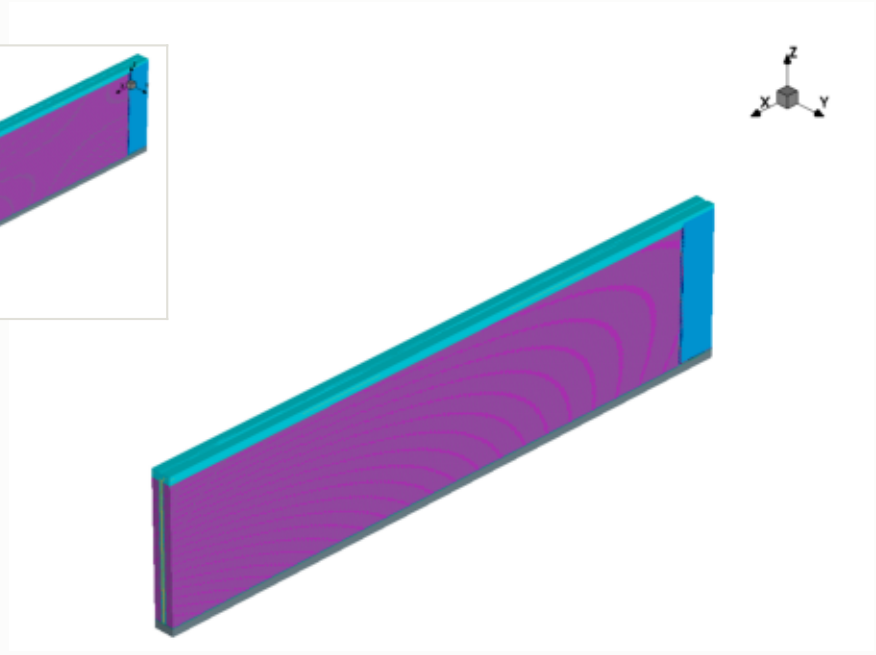
treatment · Dry_etch

0.801	700.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





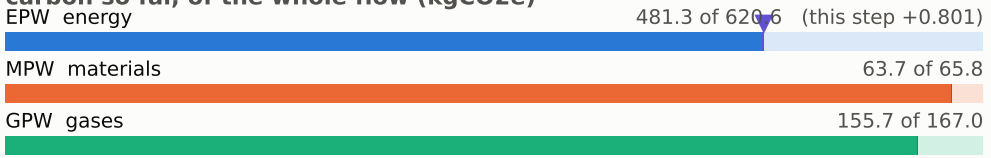
this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

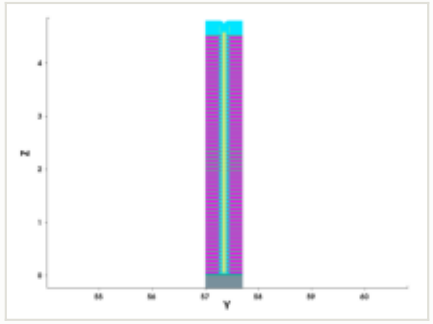
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

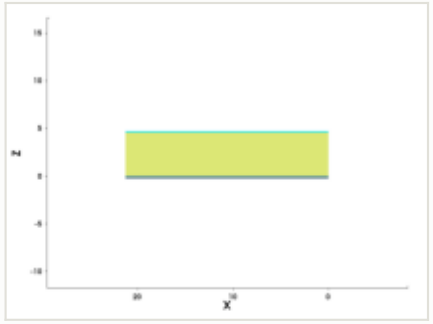


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



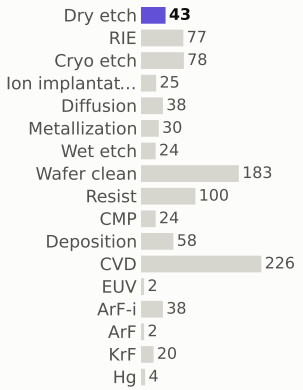
y cut at 57.351 µm



mask for this step

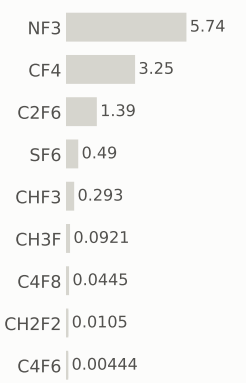
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



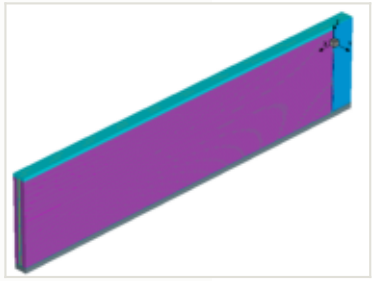
Step #11640: CMP

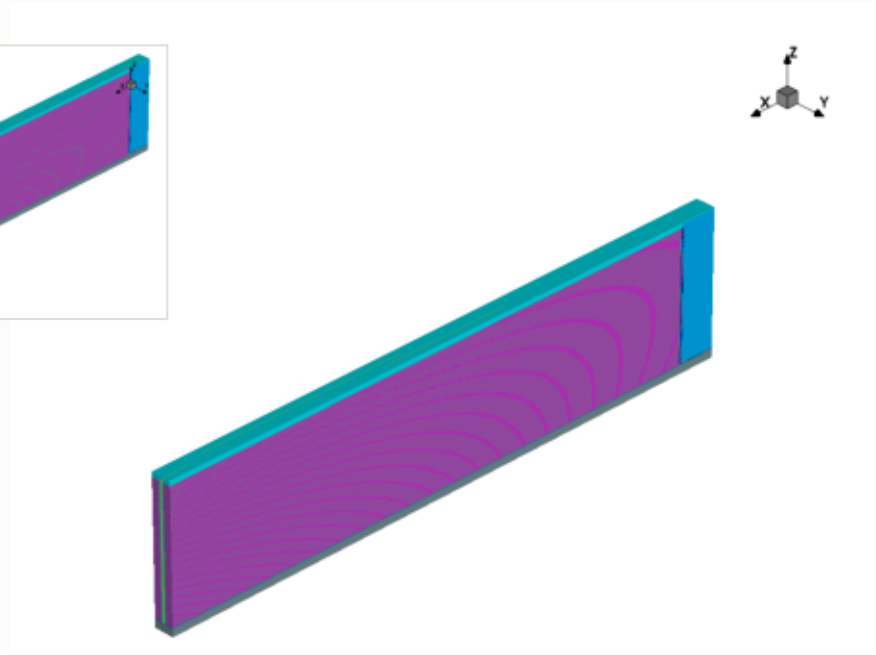
etch_cmp · CMP

0.169	700.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



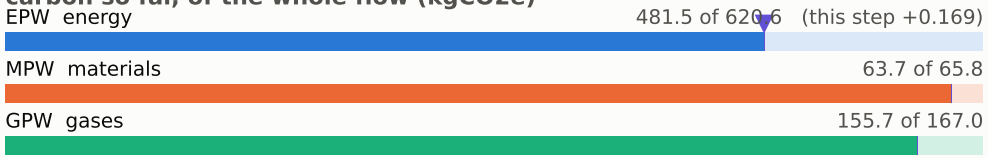


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

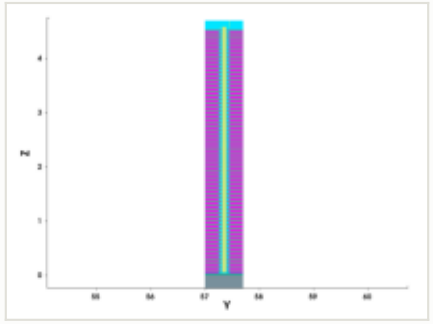
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

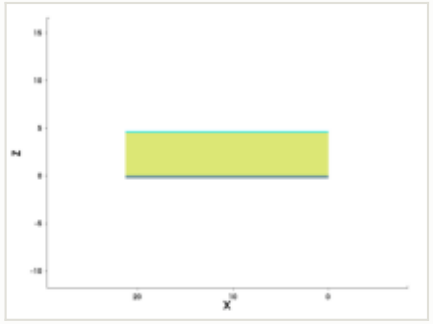


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



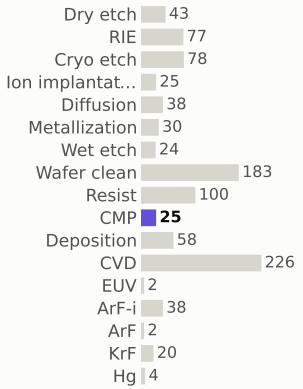
y cut at 57.351 µm



mask for this step

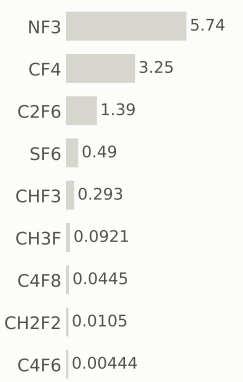
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



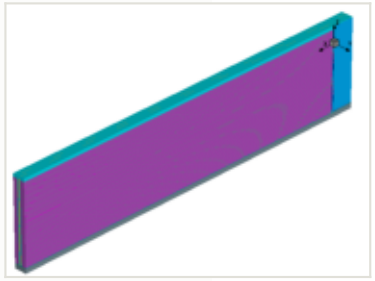
Step #11650: Wafer_clean

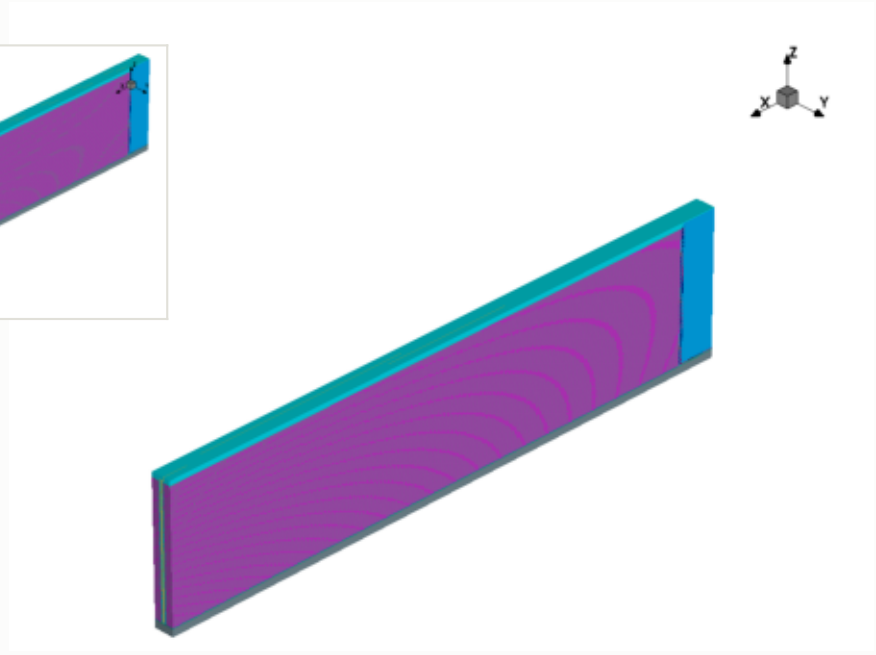
clean · Wafer_clean

0.169	701.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





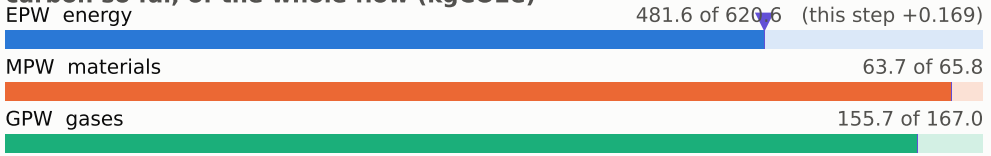
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

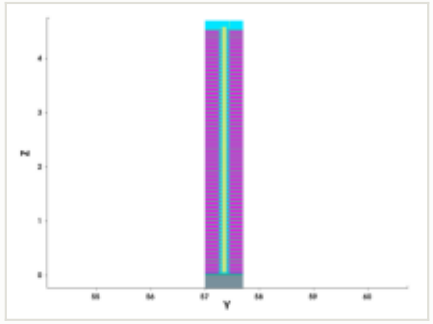
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

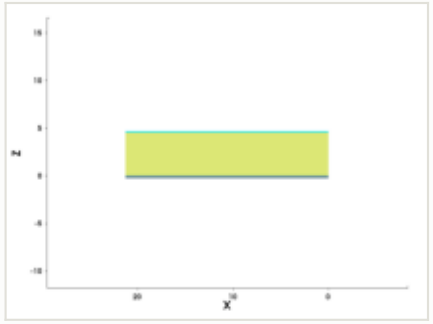


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



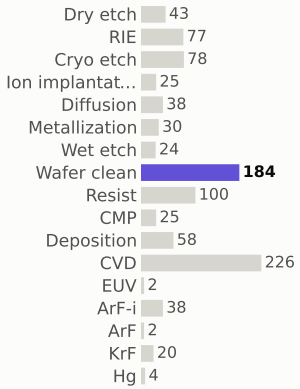
y cut at 57.351 µm



mask for this step

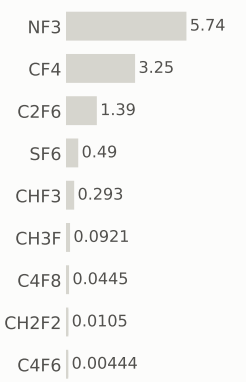
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



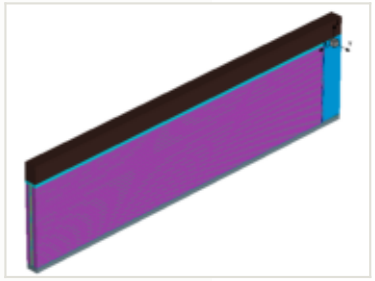
Step #11670: Deposition

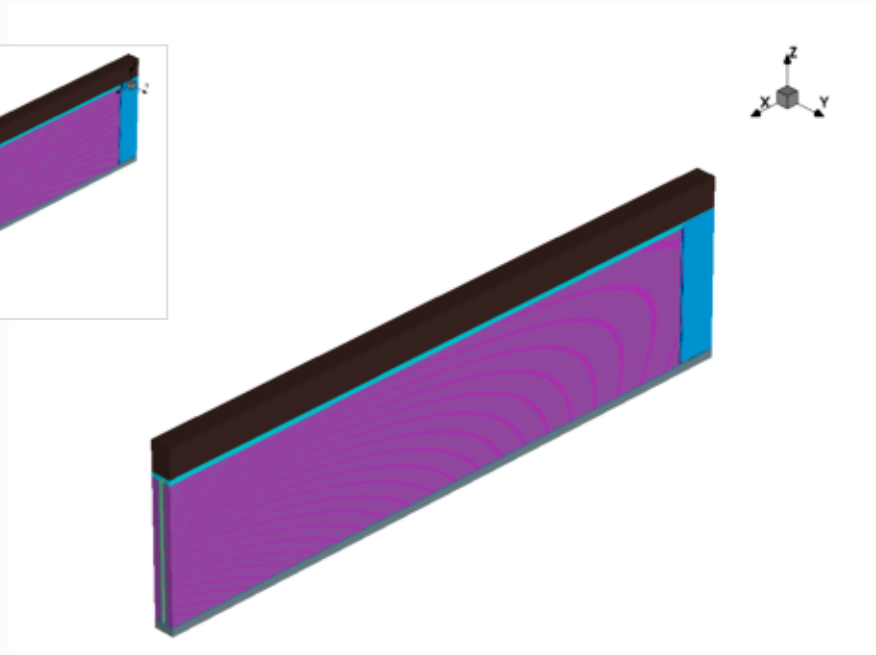
depo_isotropic · Deposition

0.656	701.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

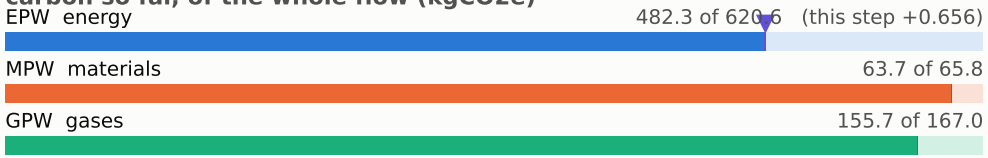




this step

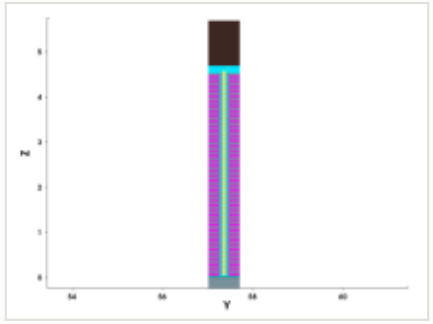
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	1 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

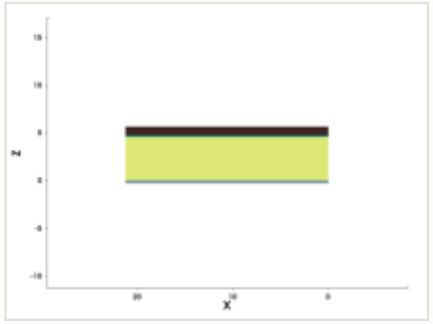


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



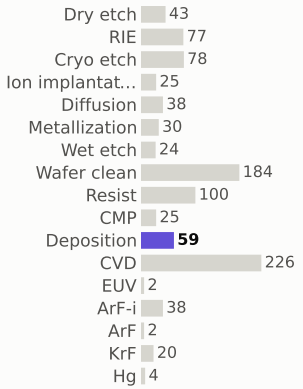
y cut at 57.351 µm



mask for this step

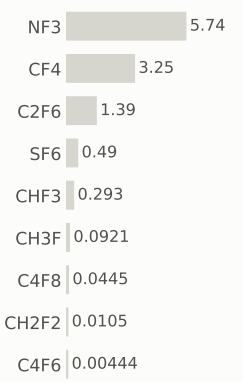
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



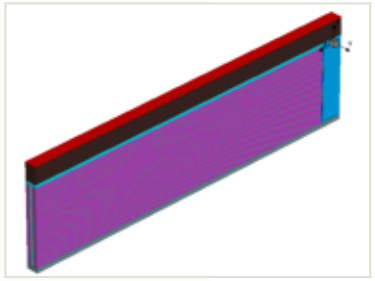
Step #11680: Deposition

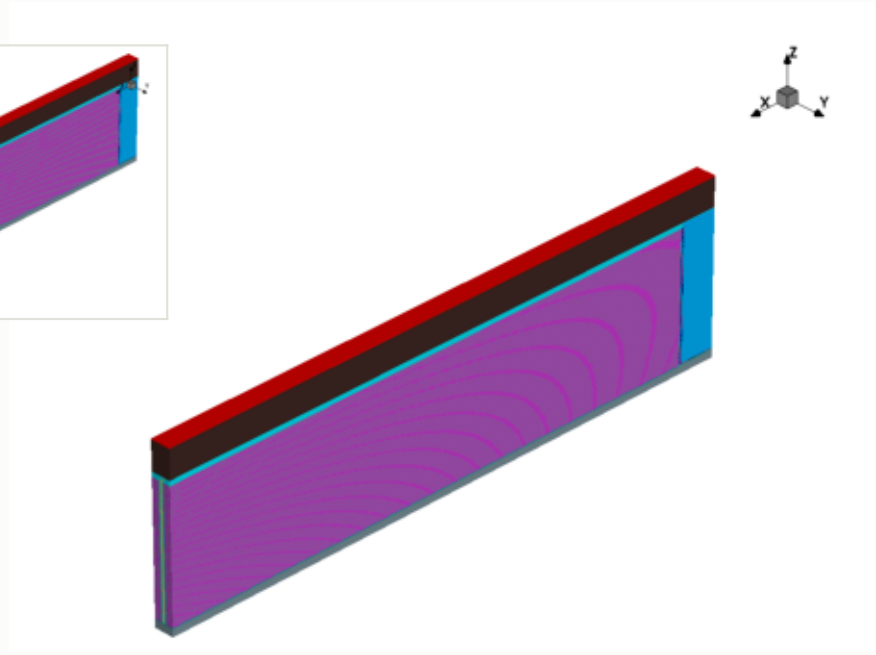
depo_isotropic · Deposition

0.656	702.3	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



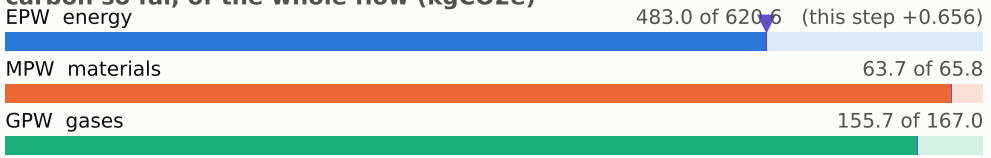


this step

process area	Deposition
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

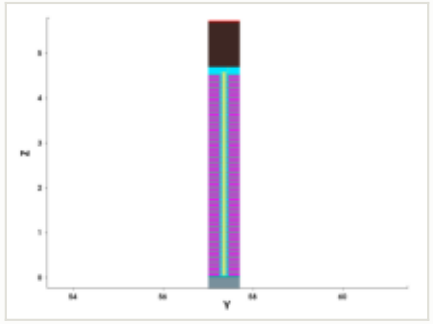
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

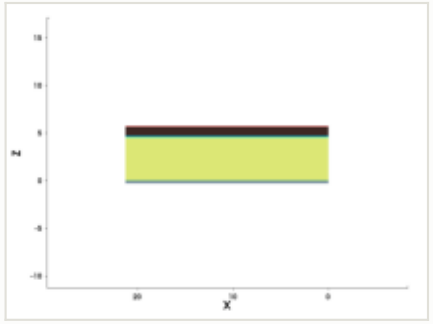


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



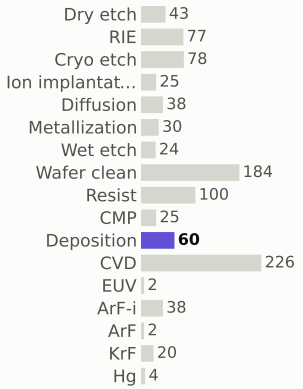
y cut at 57.351 µm



mask for this step

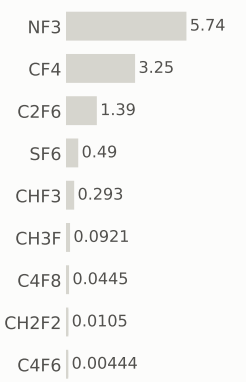
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



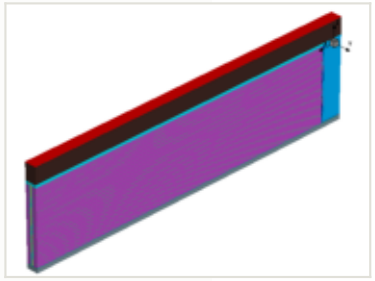
Step #11690: Wafer_clean

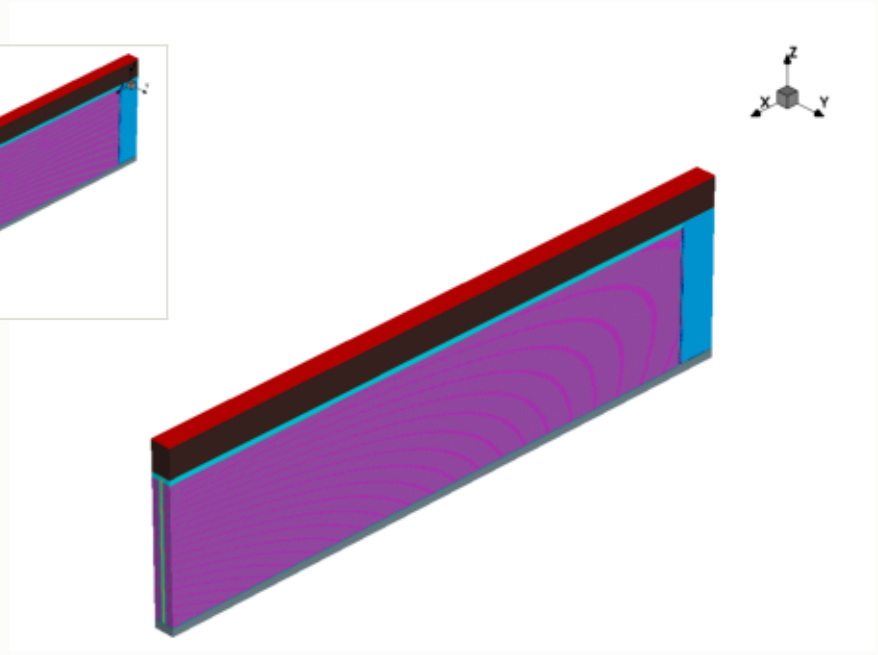
clean · Wafer_clean

0.169	702.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





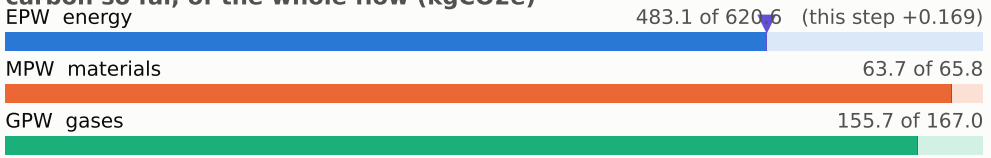
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

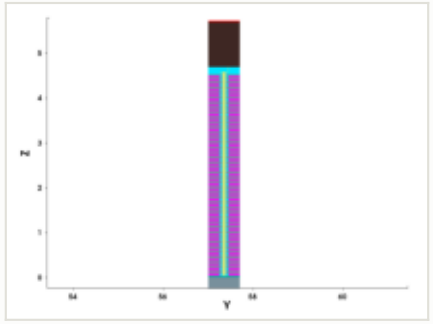
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

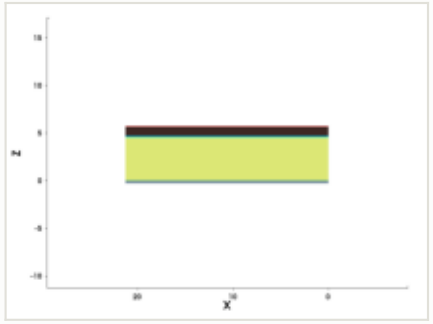


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



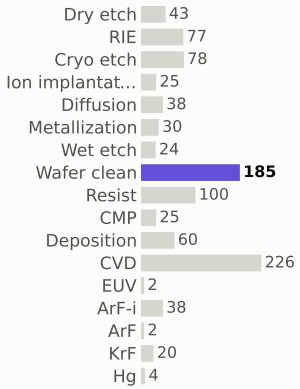
y cut at 57.351 µm



mask for this step

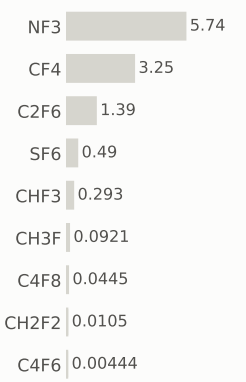
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



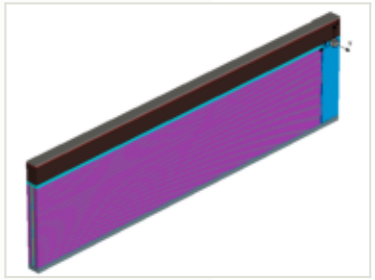
Step #11700: ArF-i

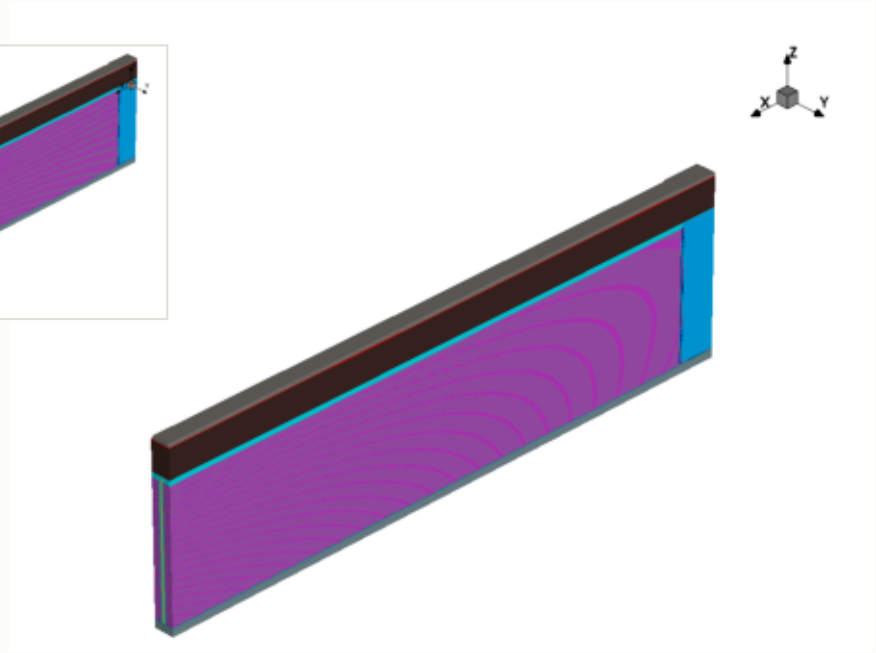
litho_patterning · ArF-i

0.217	702.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



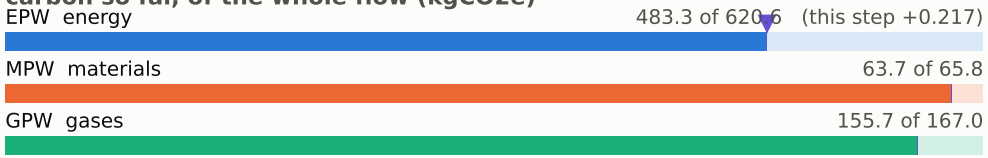


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

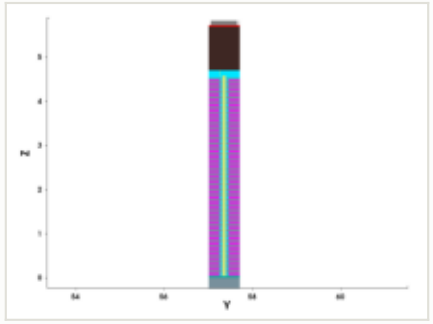
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

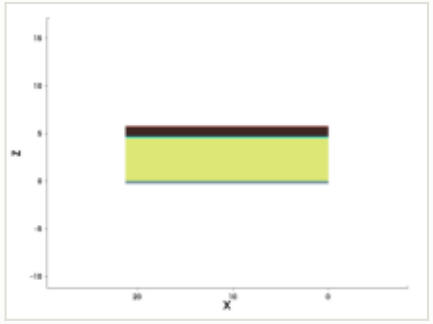


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

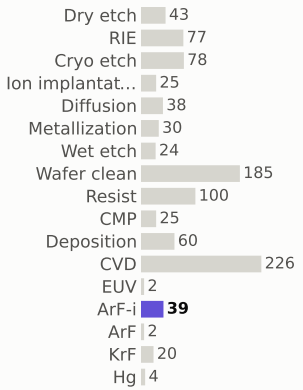


y cut at 57.351 µm



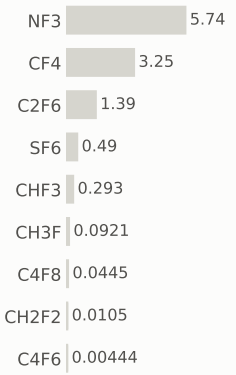
mask: layer 16:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



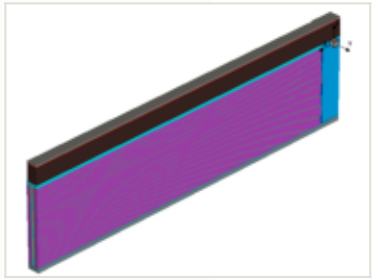
Step #11710: RIE

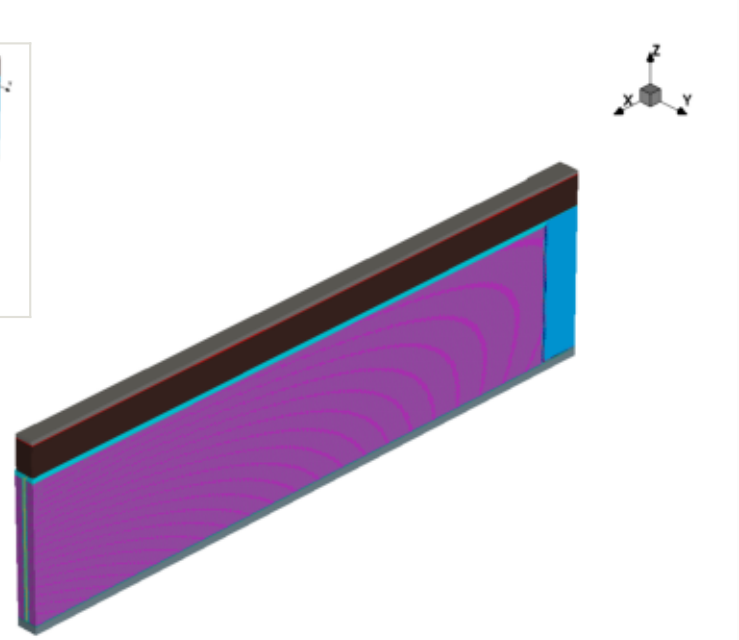
etch_anisotropic · RIE

1.04	703.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



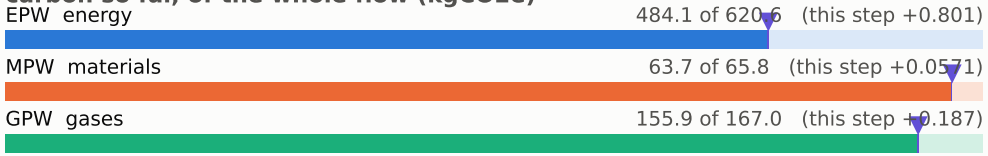


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

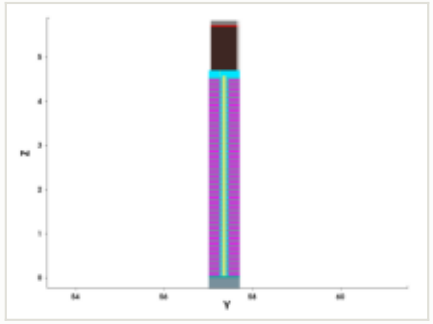
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

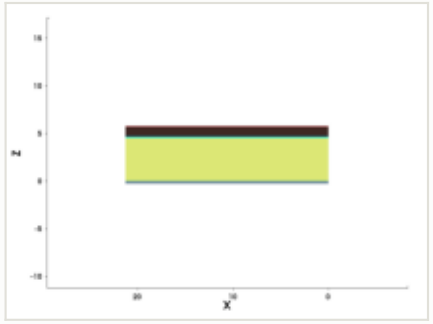


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



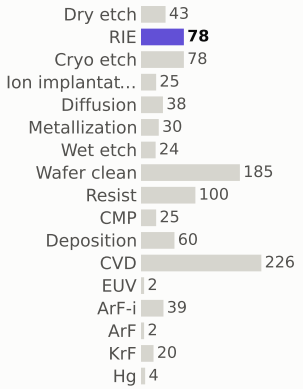
y cut at 57.351 µm



mask for this step

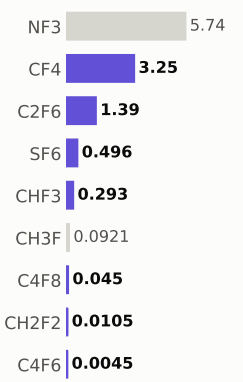
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



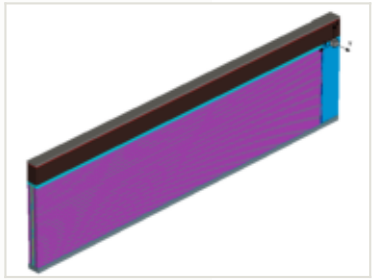
Step #11720: Cryo_etch

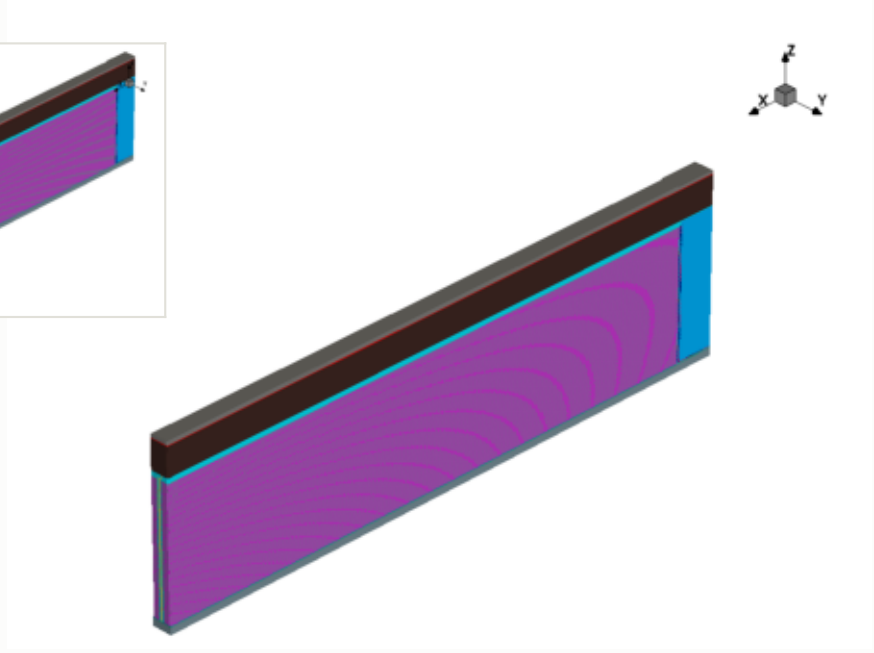
etch_anisotropic · Cryo_etch

0.517	704.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

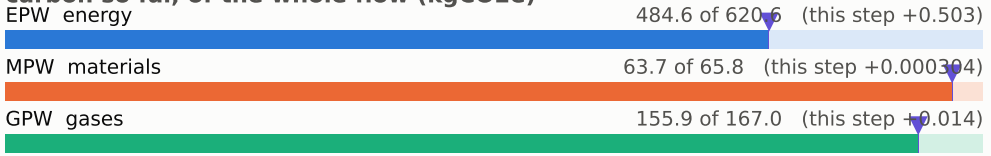




this step

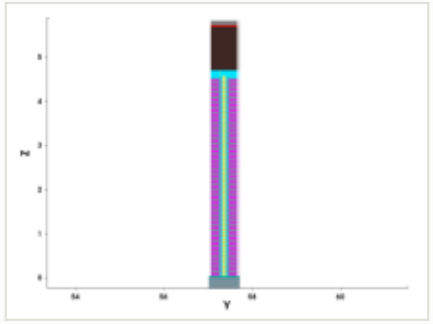
process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

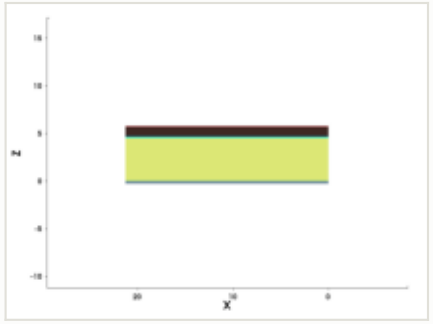


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



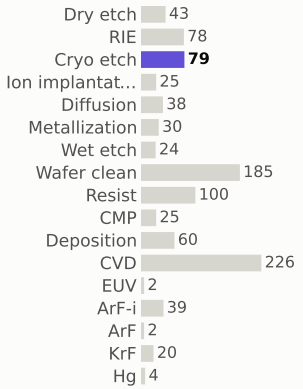
y cut at 57.351 µm



mask for this step

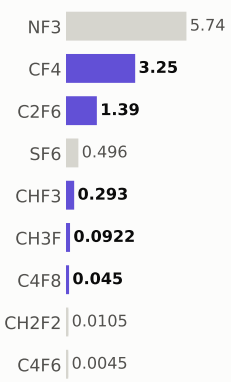
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



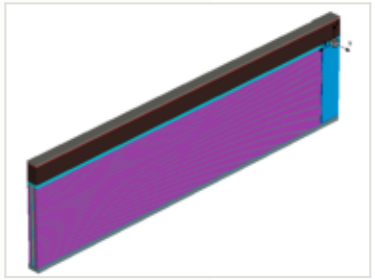
Step #11720.3846: Cryo_etch

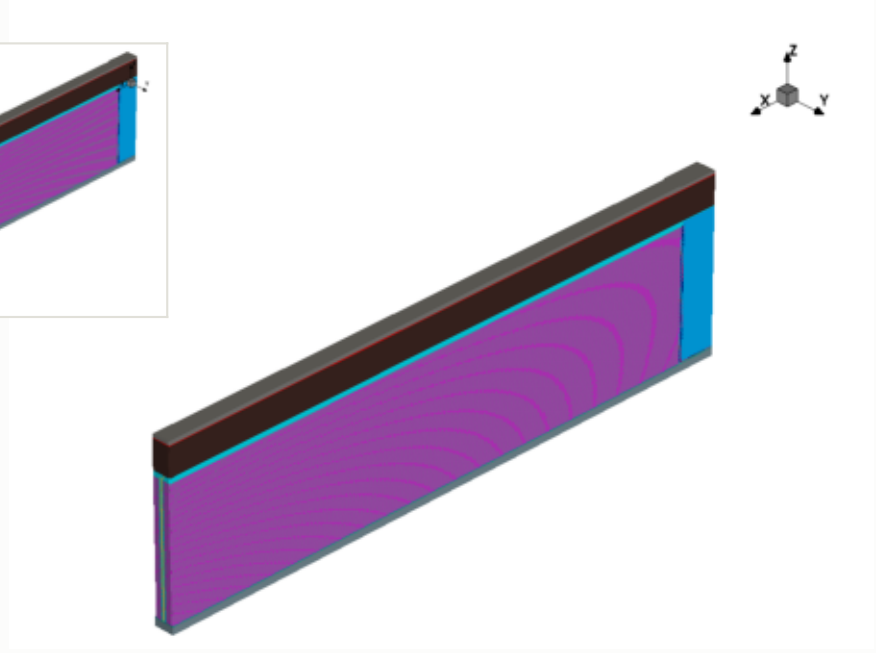
etch_noop · Cryo_etch

0.517	704.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





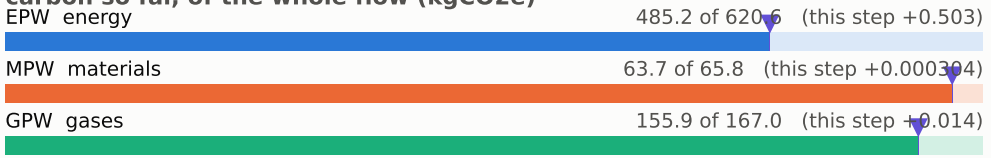
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

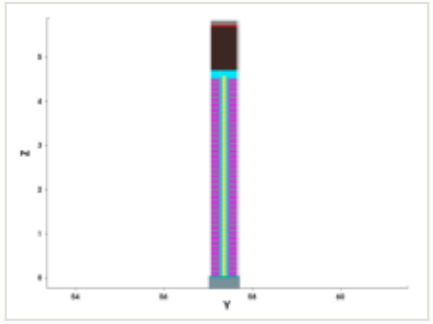
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

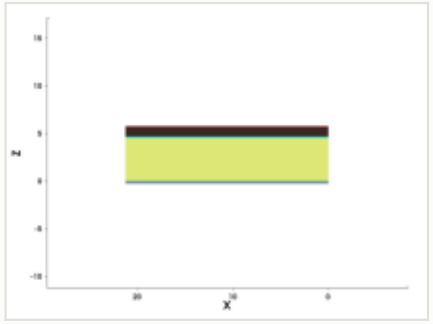


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



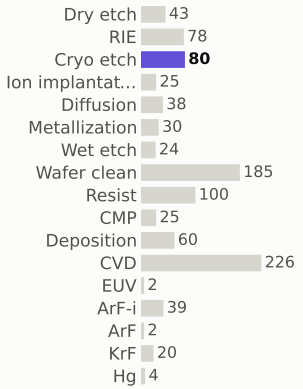
y cut at 57.351 µm



mask for this step

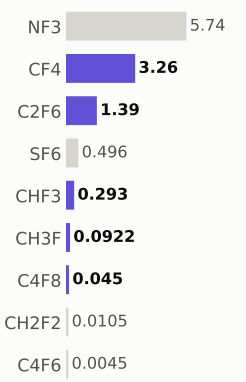
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



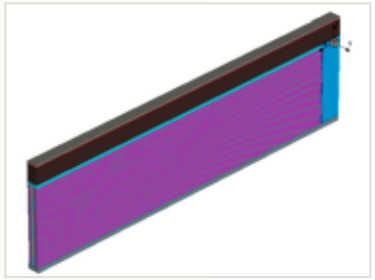
Step #11720.7692: Cryo_etch

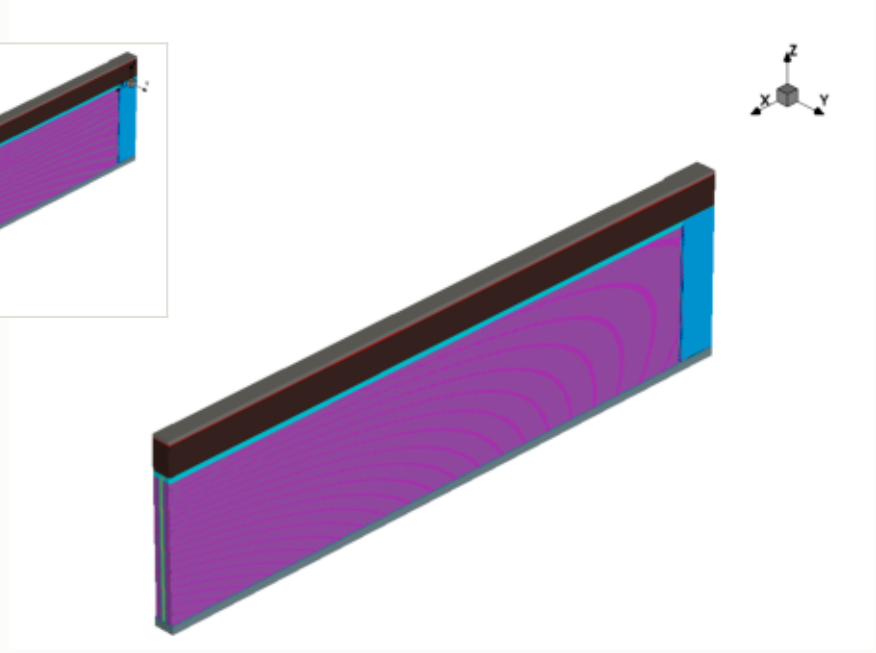
etch_noop · Cryo_etch

0.517	705.3	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





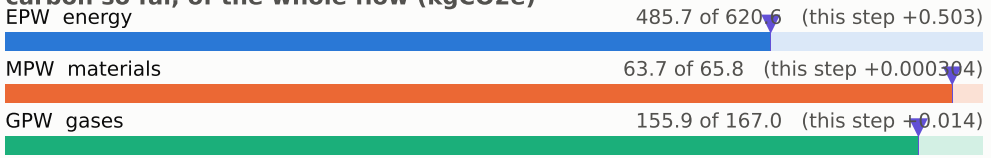
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

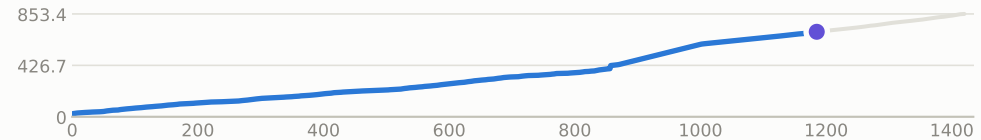
EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

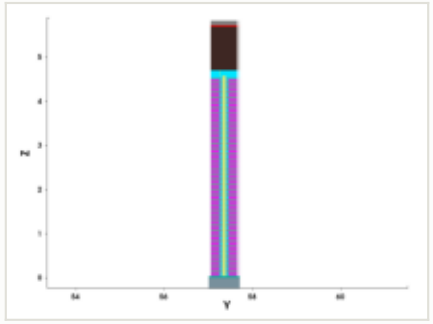
carbon so far, of the whole flow (kgCO2e)



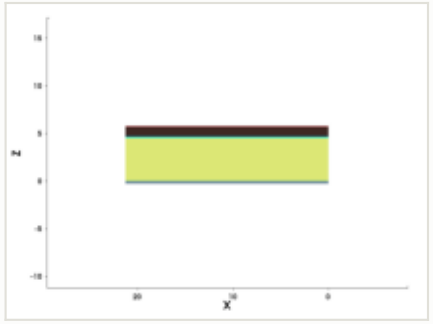
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



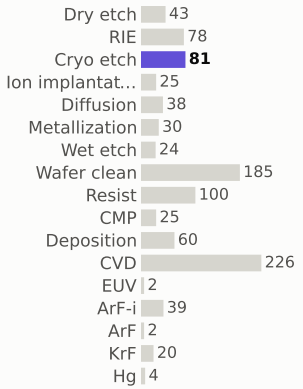
y cut at 57.351 µm



mask for this step

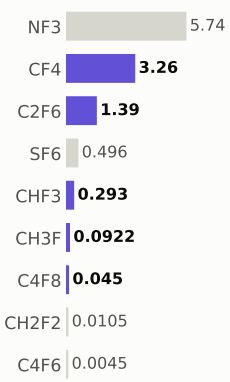
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



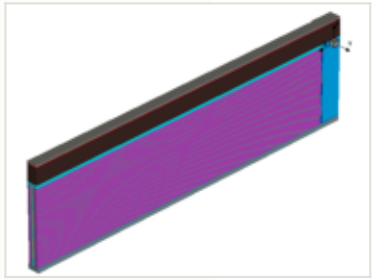
Step #11721.1538: Cryo_etch

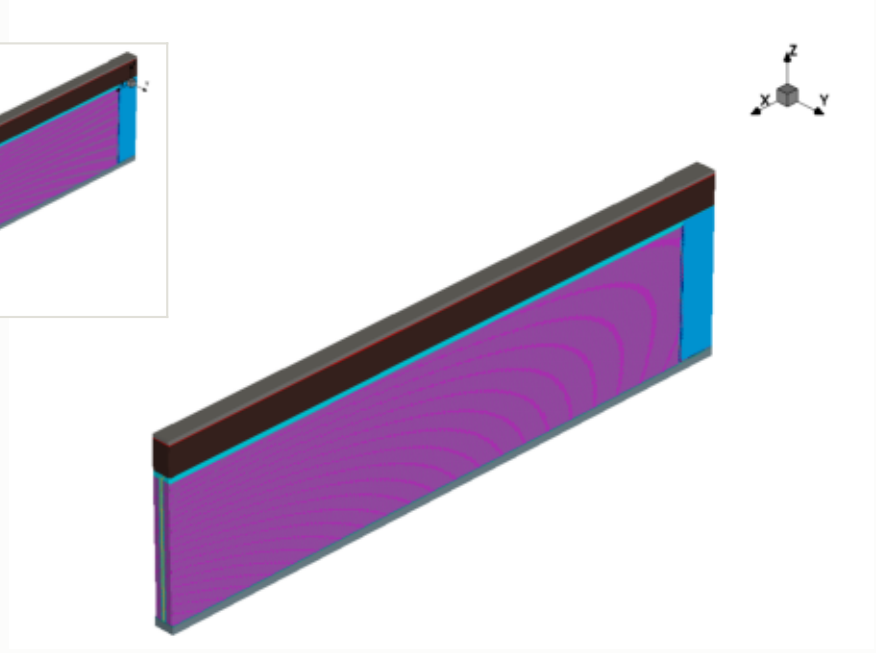
etch_noop · Cryo_etch

0.517	705.8	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



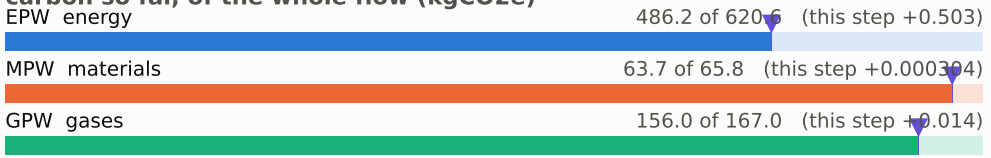


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

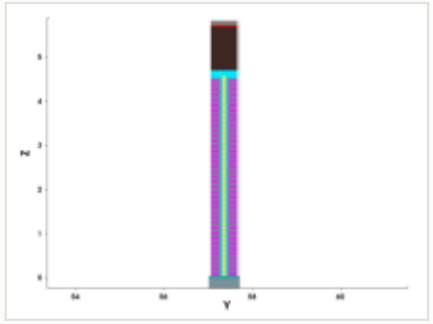
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

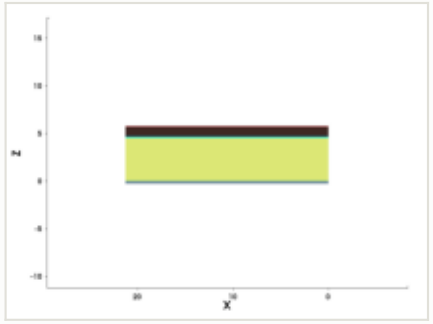


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



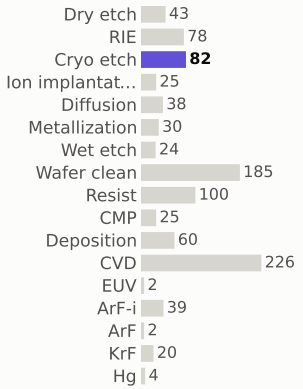
y cut at 57.351 µm



mask for this step

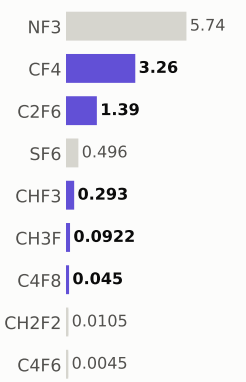
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



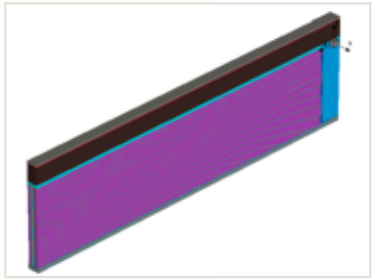
Step #11721.5385: Cryo_etch

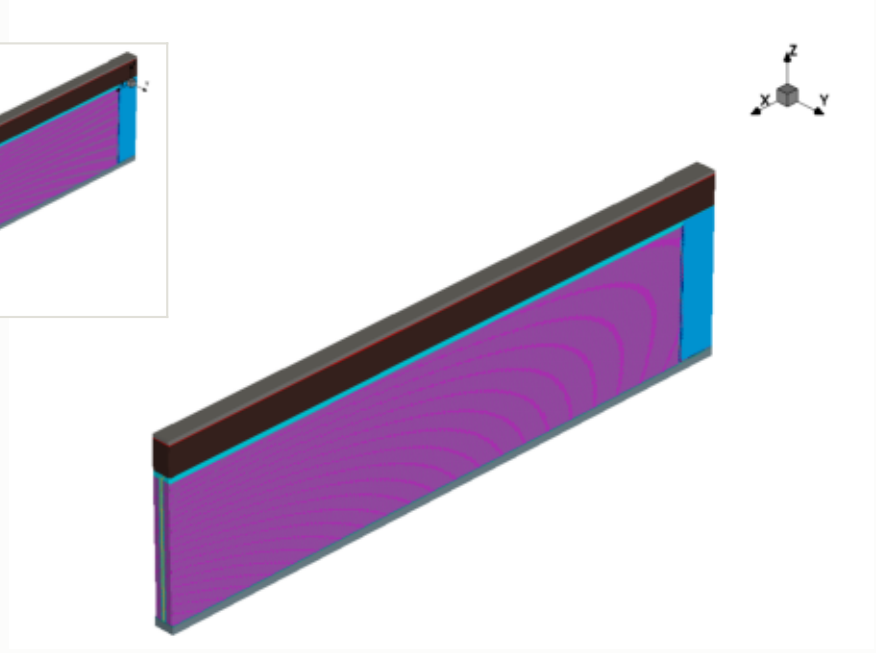
etch_noop · Cryo_etch

0.517	706.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



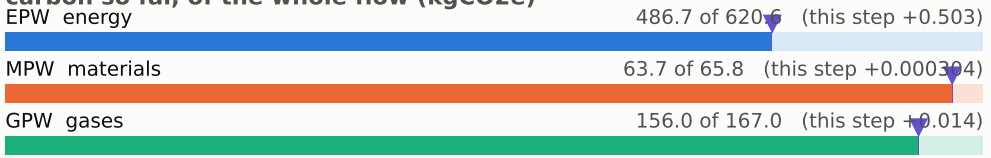


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

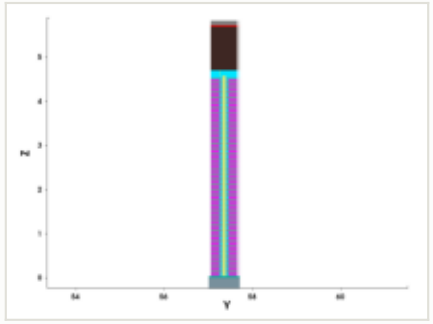
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

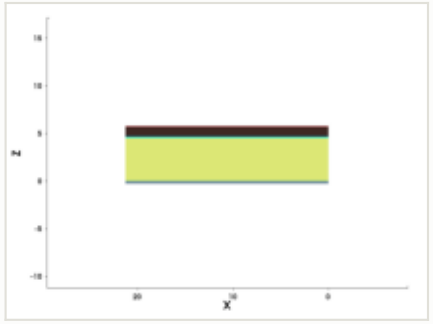


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



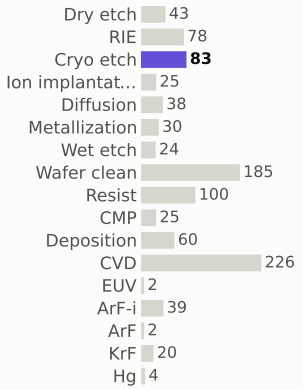
y cut at 57.351 µm



mask for this step

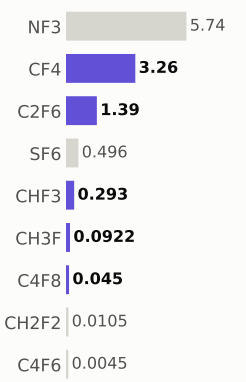
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



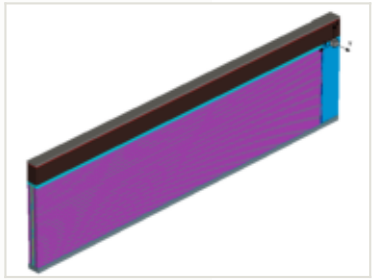
Step #11721.9231: Cryo_etch

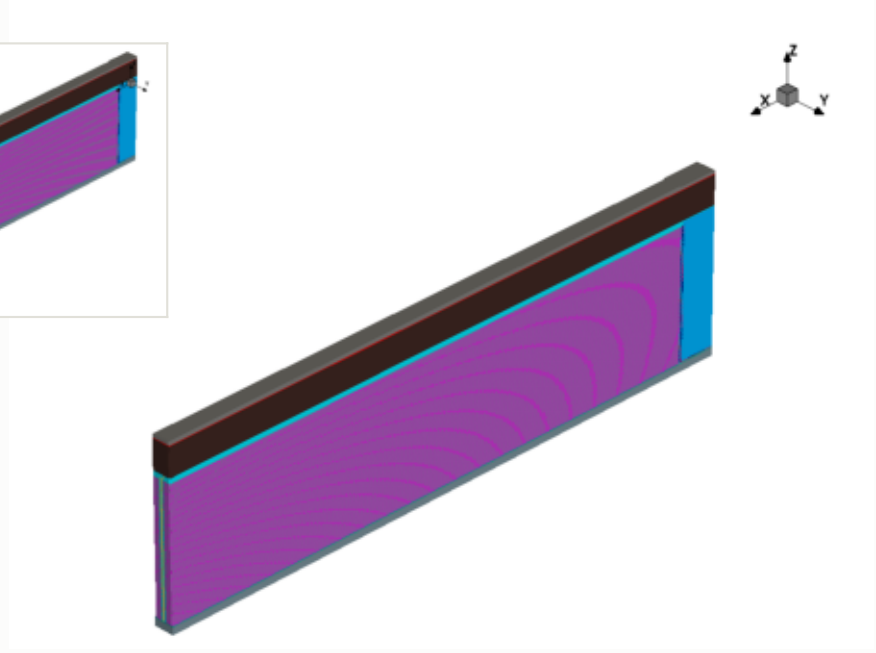
etch_noop · Cryo_etch

0.517	706.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





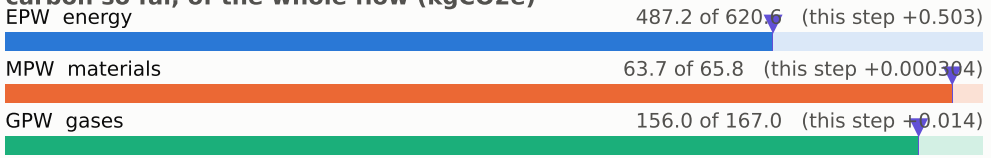
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

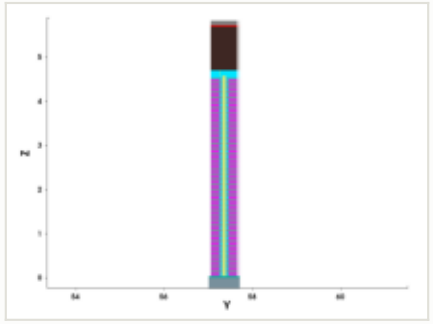
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

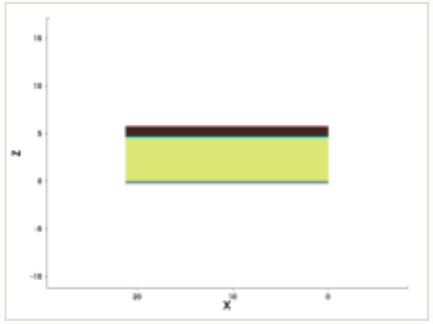


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



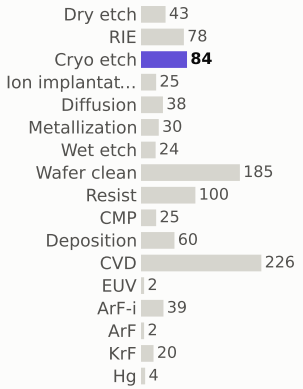
y cut at 57.351 µm



mask for this step

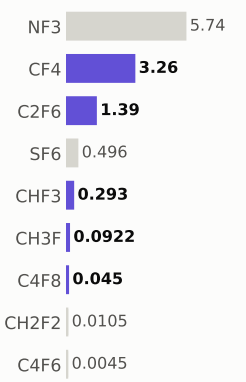
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



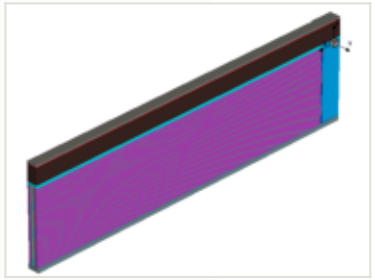
Step #11722.3077: Cryo_etch

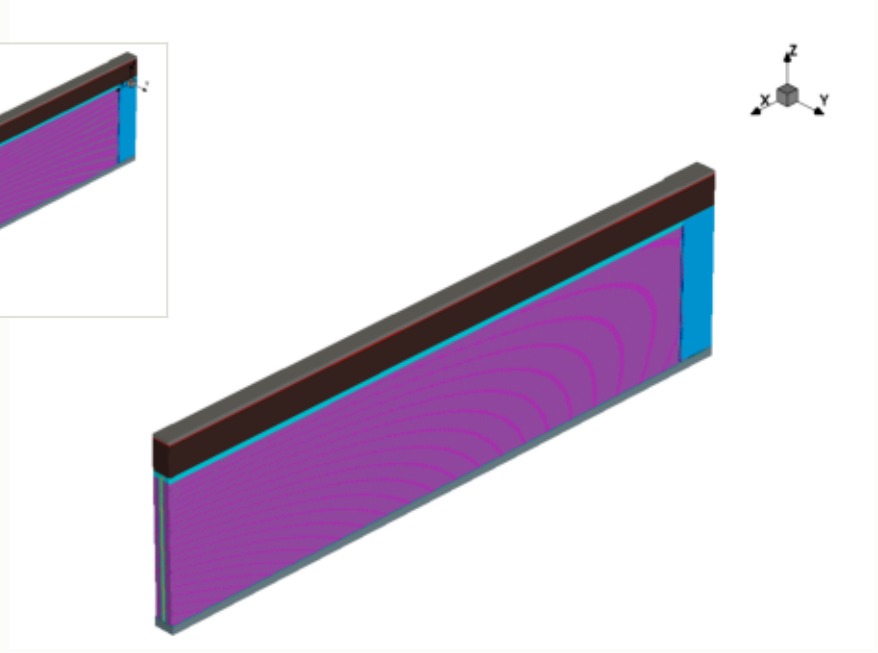
etch_noop · Cryo_etch

0.517	707.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





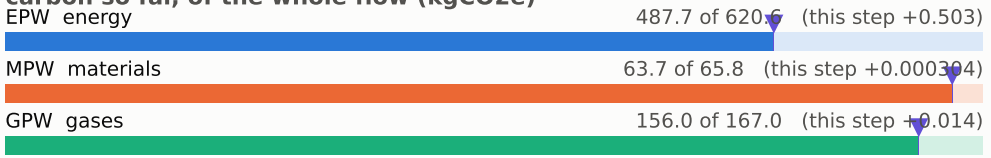
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

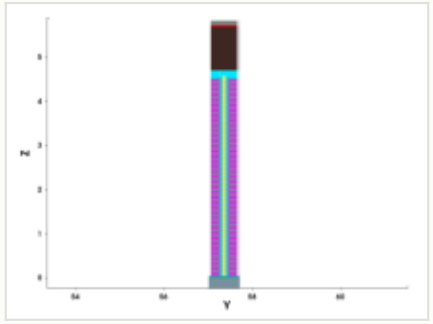
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

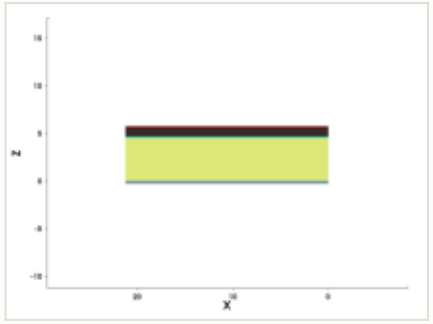


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



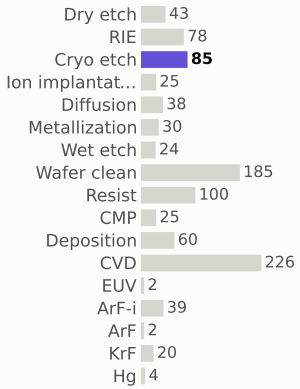
y cut at 57.351 µm



mask for this step

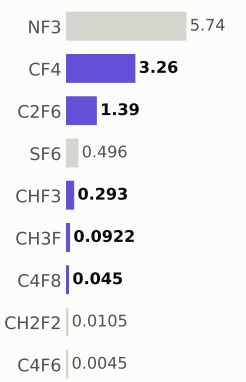
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



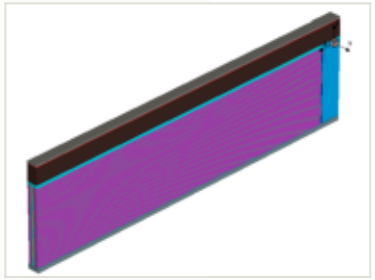
Step #11722.6923: Cryo_etch

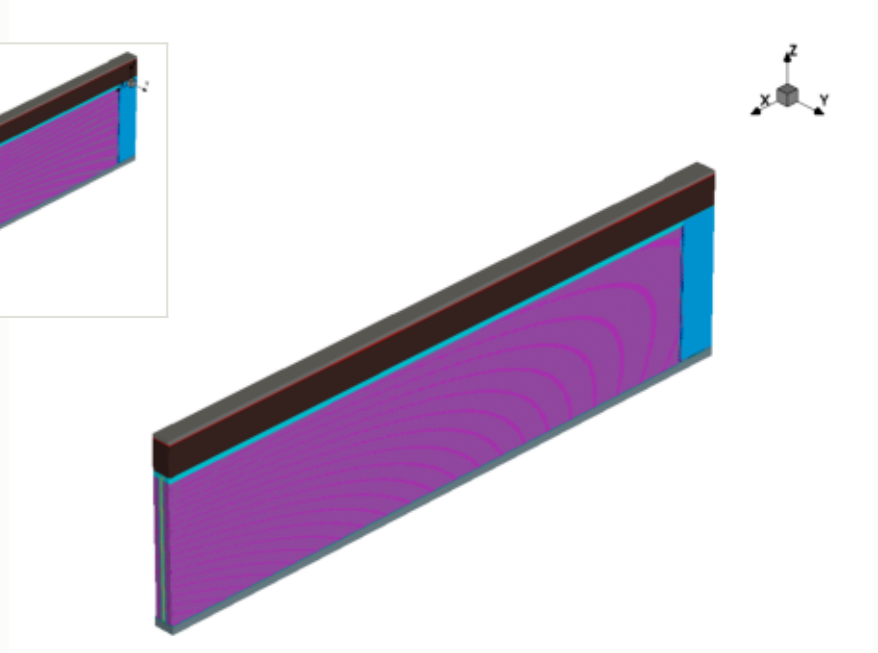
etch_noop · Cryo_etch

0.517	707.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



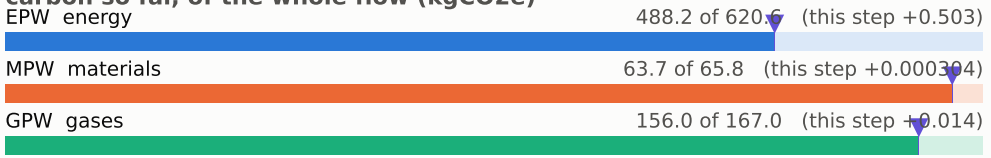


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

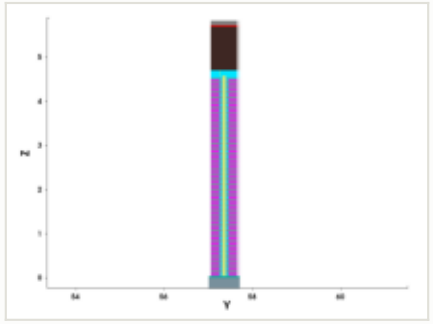
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

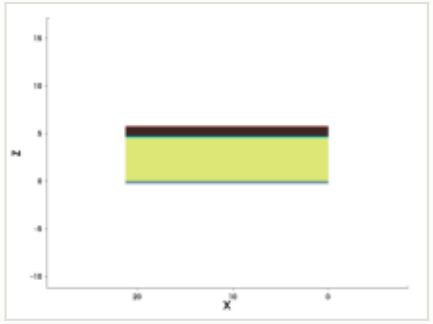


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



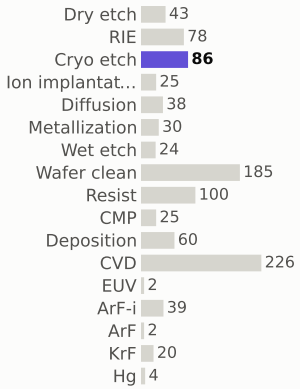
y cut at 57.351 µm



mask for this step

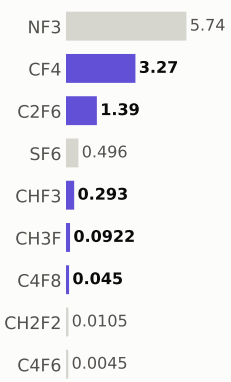
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



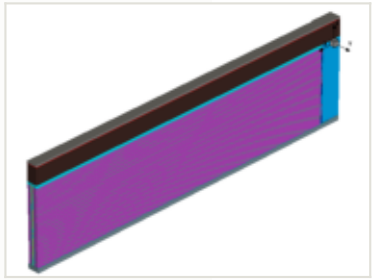
Step #11723.0769: Cryo_etch

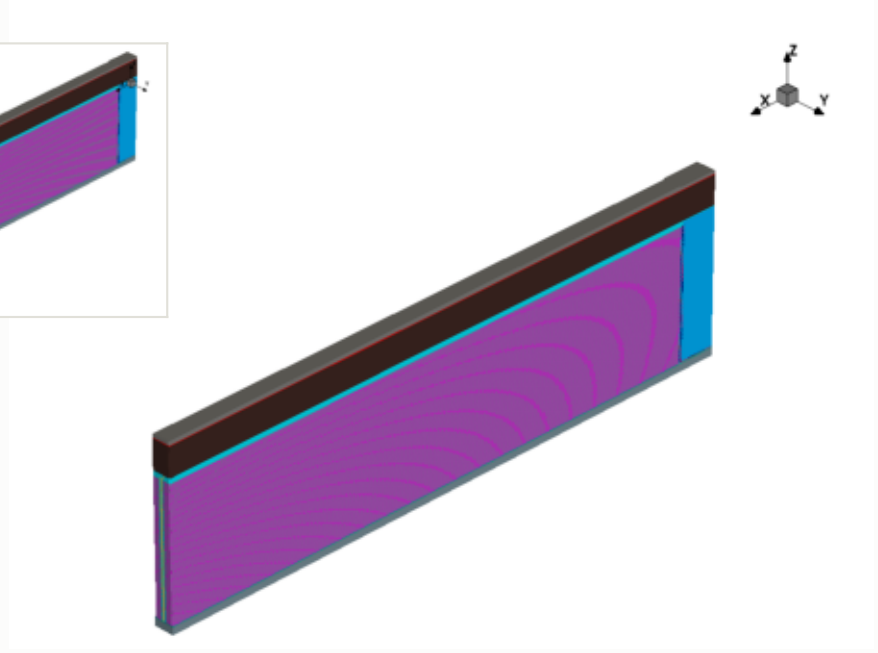
etch_noop · Cryo_etch

0.517	708.4	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



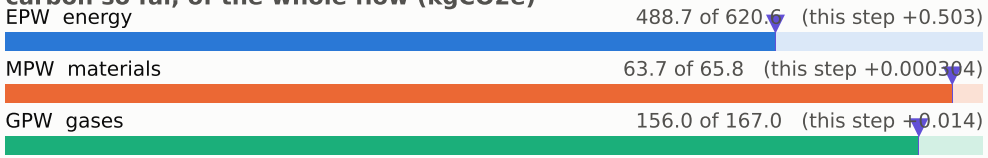


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

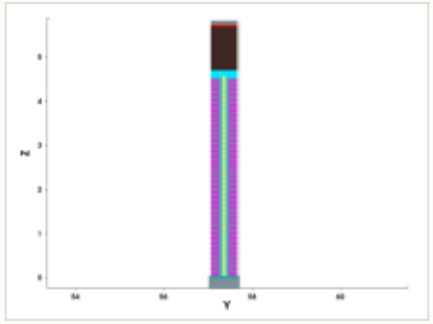
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

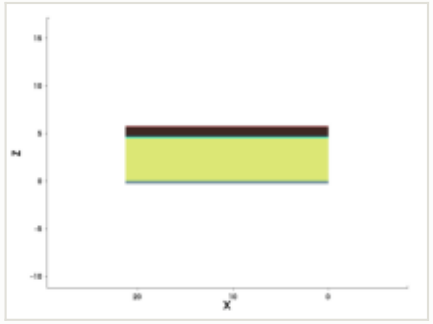


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



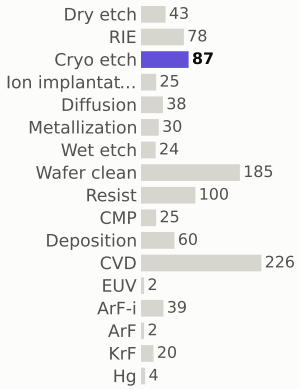
y cut at 57.351 µm



mask for this step

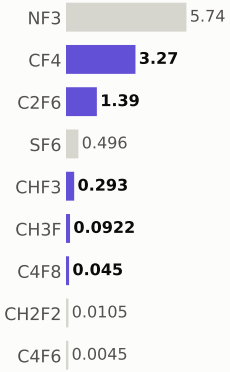
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



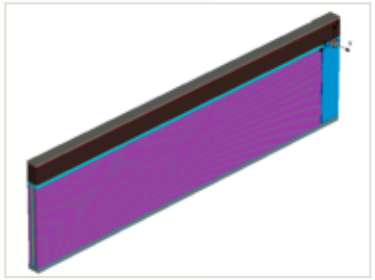
Step #11723.4615: Cryo_etch

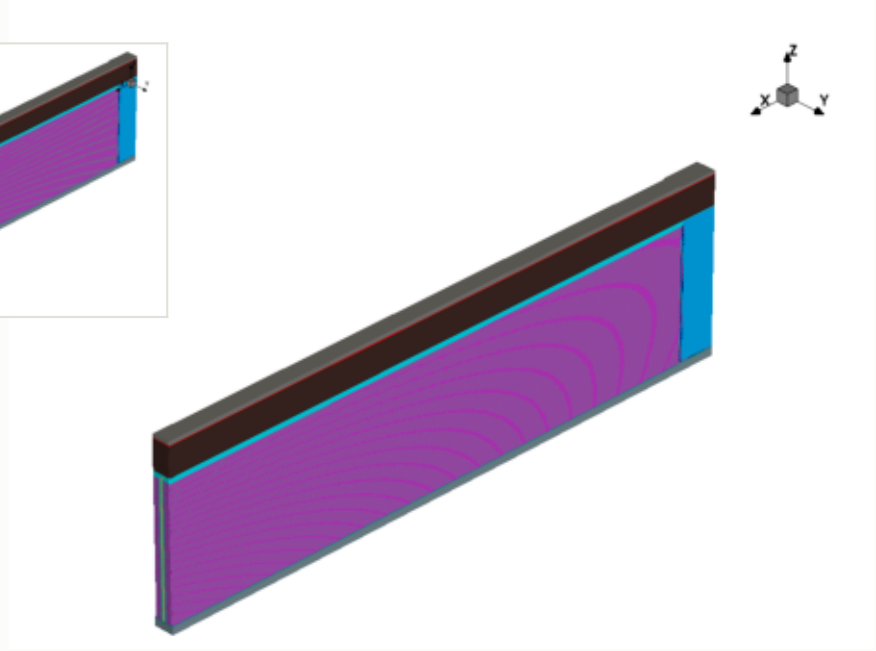
etch_noop · Cryo_etch

0.517	708.9	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





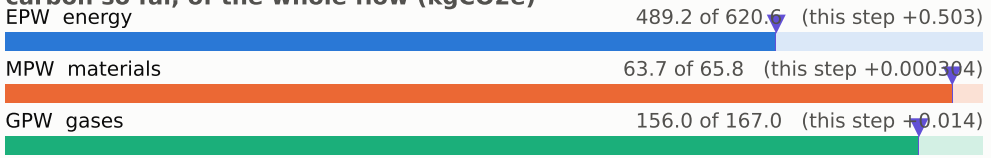
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

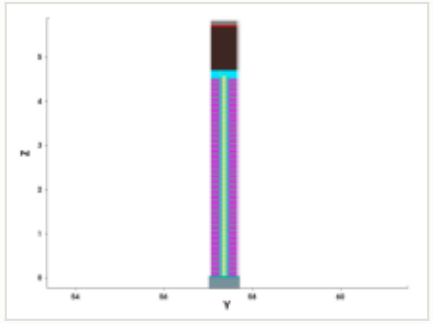
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

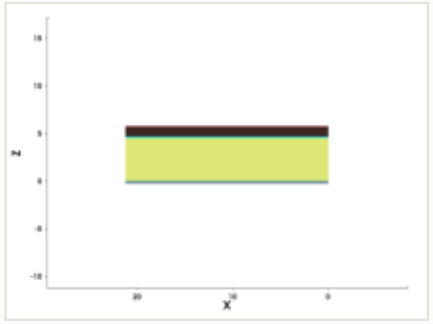


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



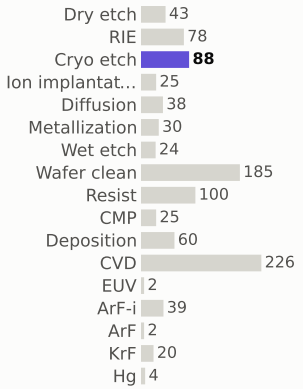
y cut at 57.351 µm



mask for this step

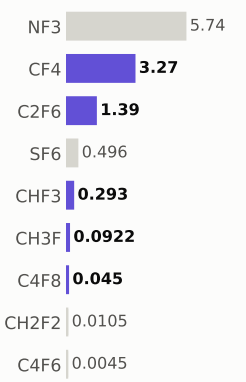
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



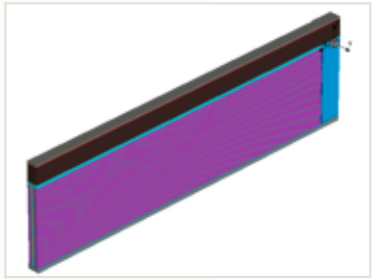
Step #11723.8462: Cryo_etch

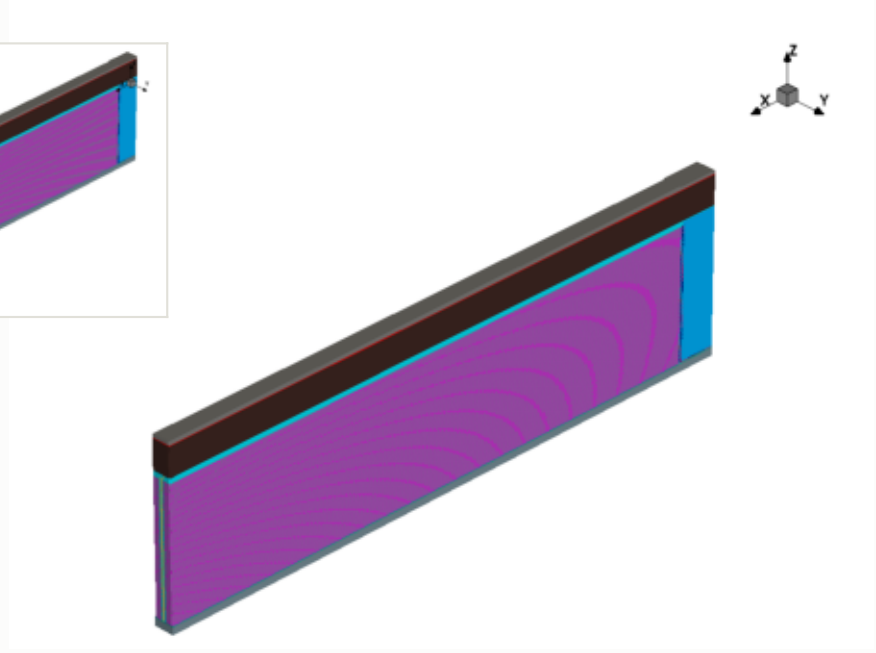
etch_noop · Cryo_etch

0.517	709.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





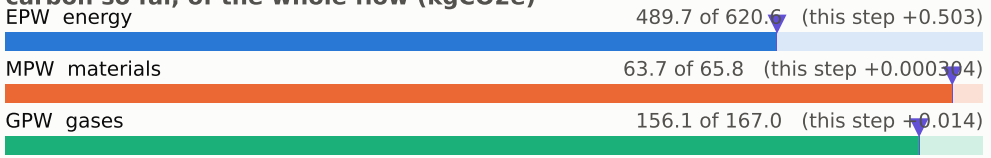
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

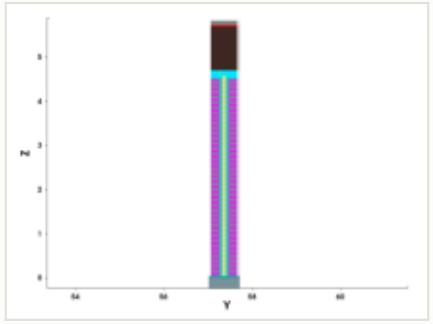
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

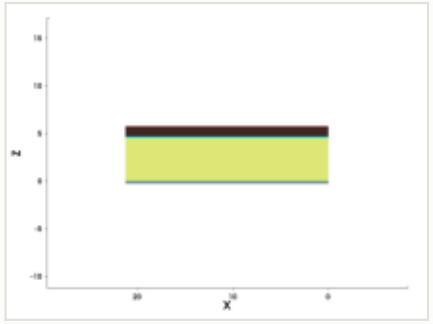


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



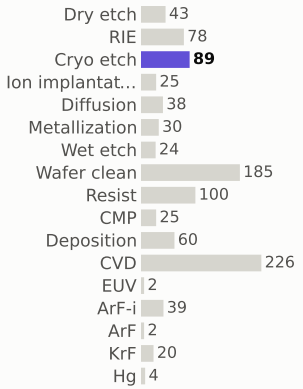
y cut at 57.351 µm



mask for this step

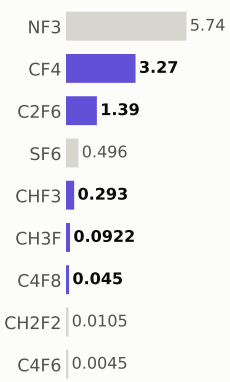
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



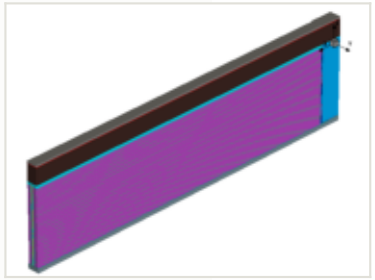
Step #11724.2308: Cryo_etch

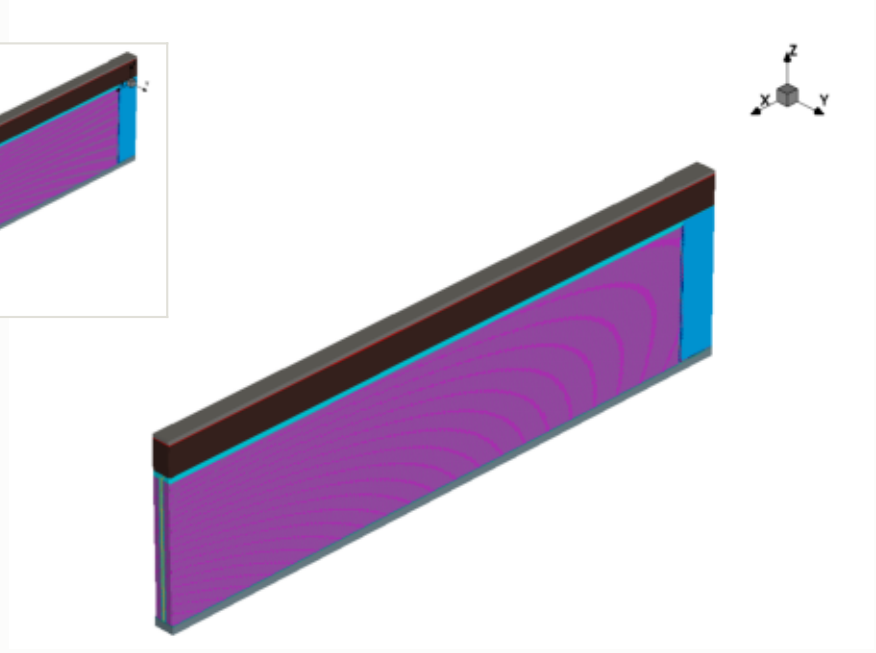
etch_noop · Cryo_etch

0.517	710.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





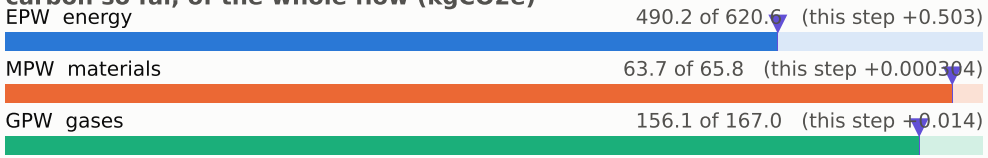
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

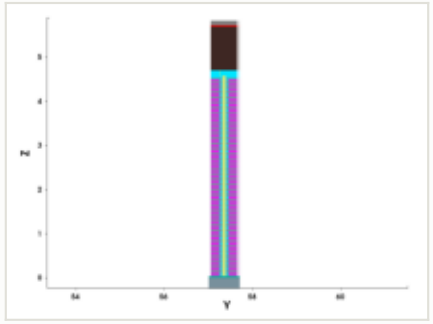
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

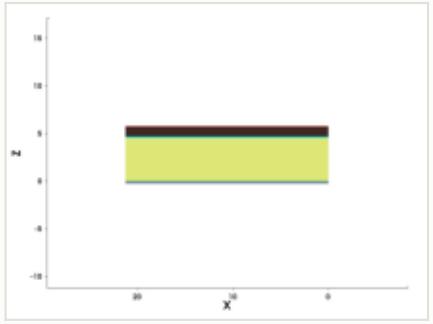


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



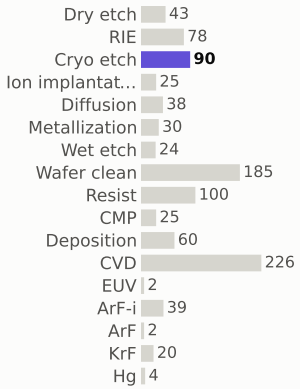
y cut at 57.351 µm



mask for this step

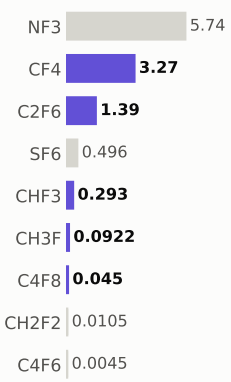
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



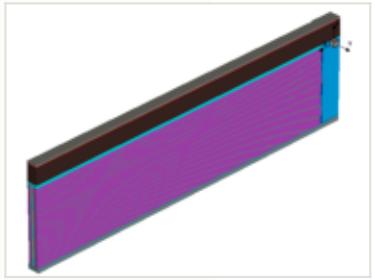
Step #11724.6154: Cryo_etch

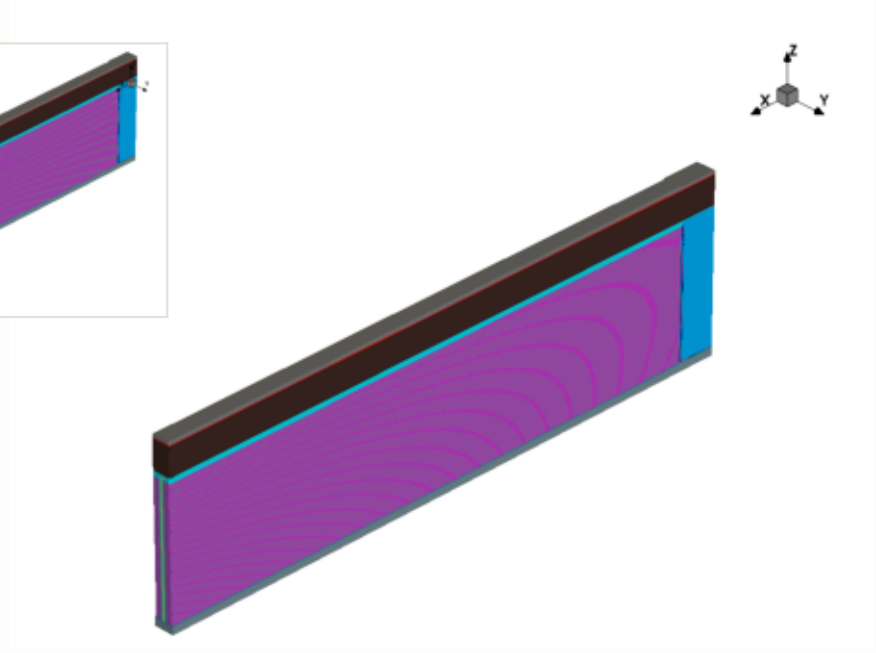
etch_noop · Cryo_etch

0.517	710.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



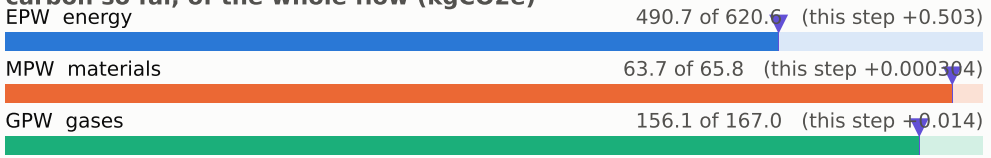


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

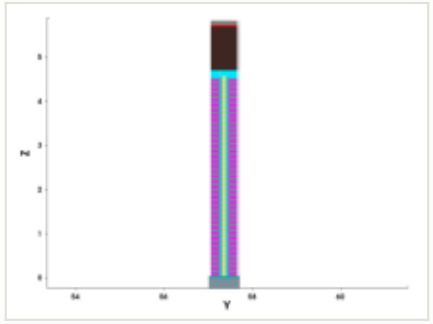
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

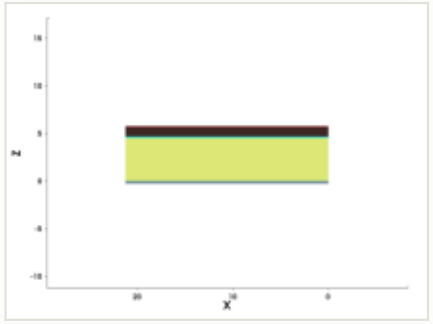


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



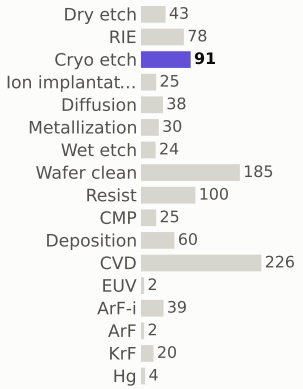
y cut at 57.351 µm



mask for this step

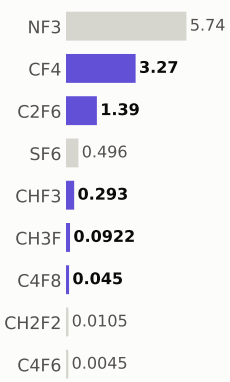
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



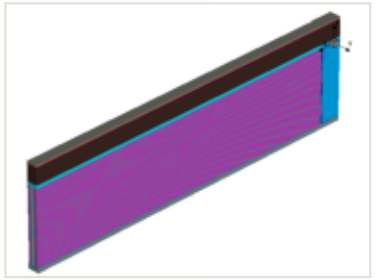
Step #11725.0000: Cryo_etch

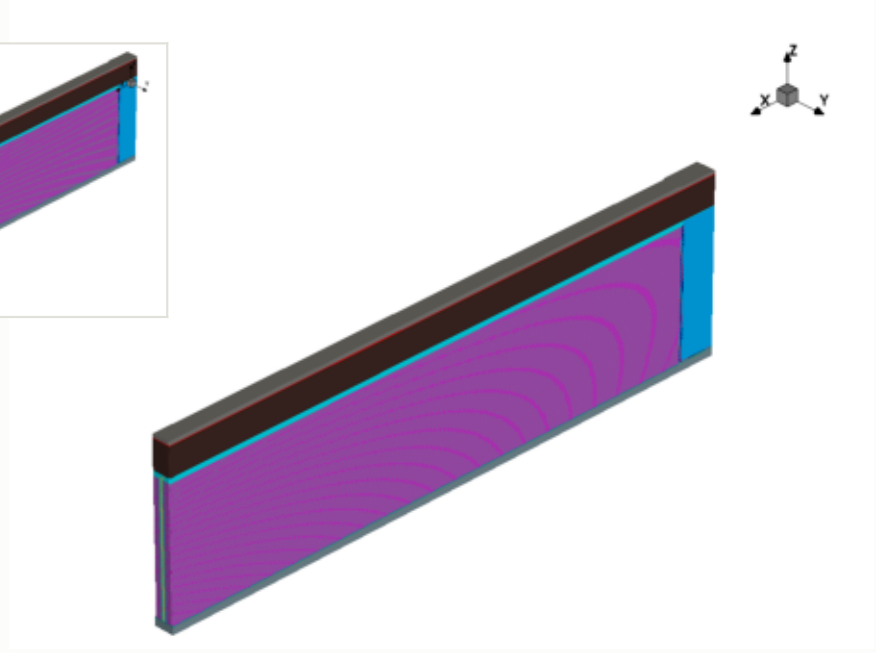
etch_noop · Cryo_etch

0.517	711.0	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



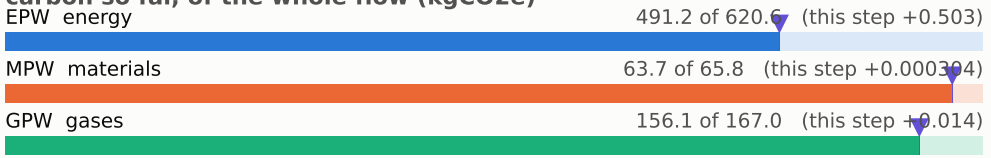


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

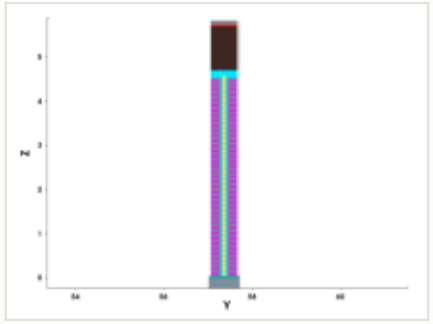
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

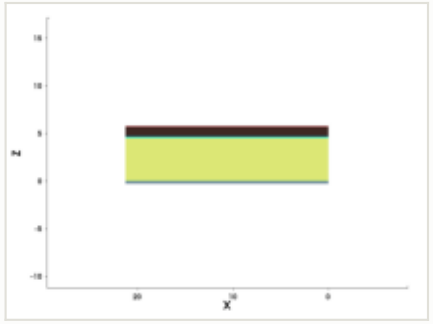


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



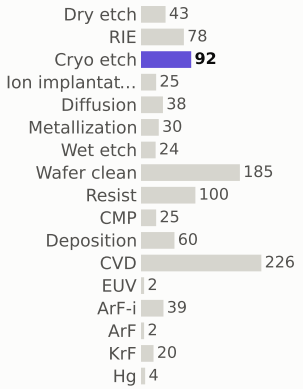
y cut at 57.351 µm



mask for this step

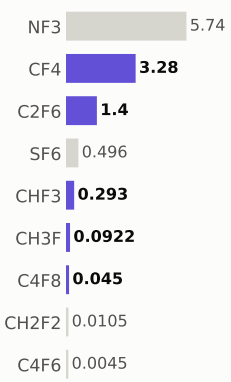
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



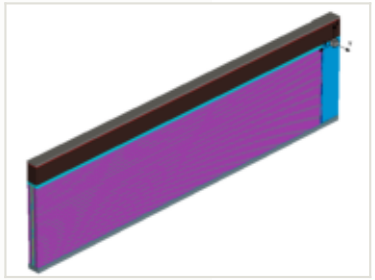
Step #11725.3846: Cryo_etch

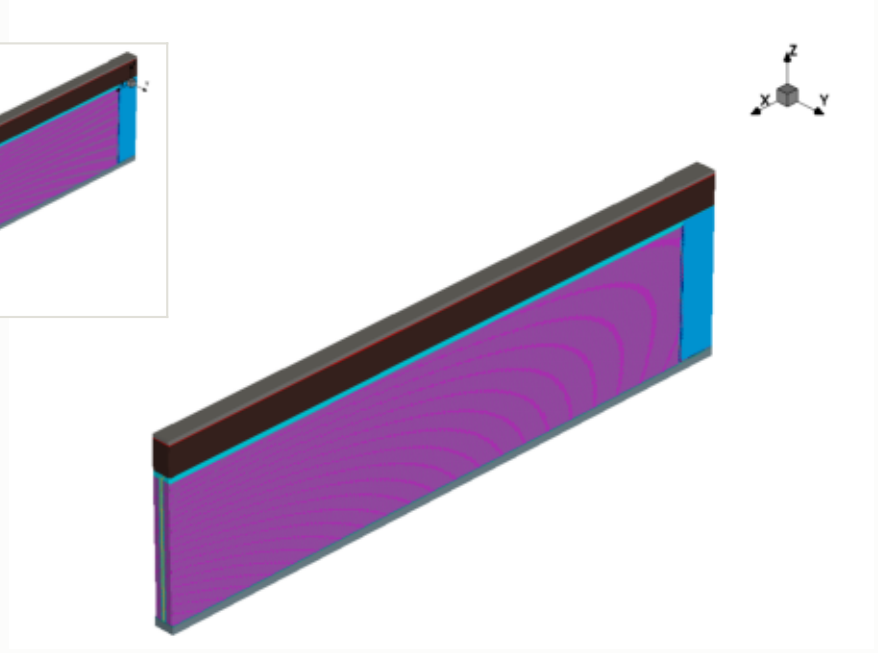
etch_noop · Cryo_etch

0.517	711.5	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





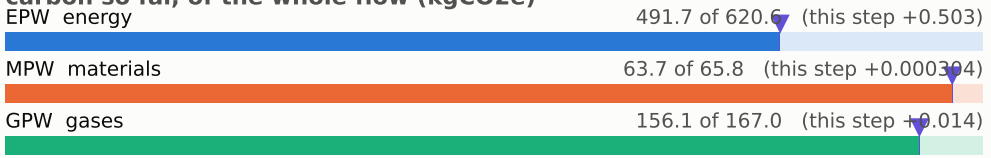
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

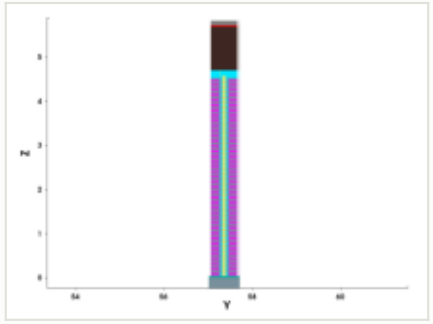
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

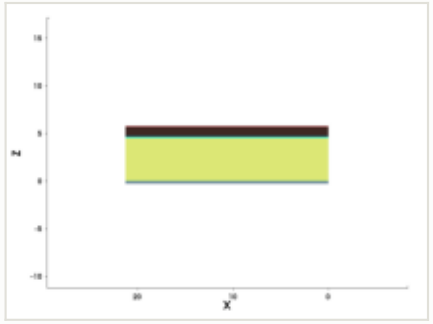


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



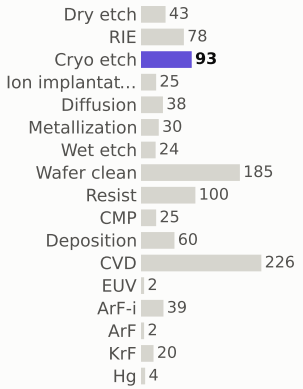
y cut at 57.351 µm



mask for this step

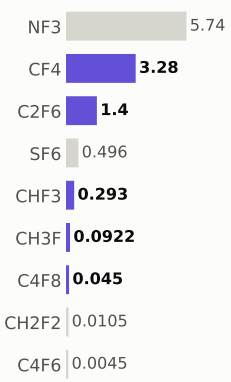
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



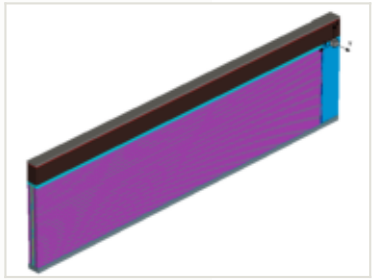
Step #11725.7692: Cryo_etch

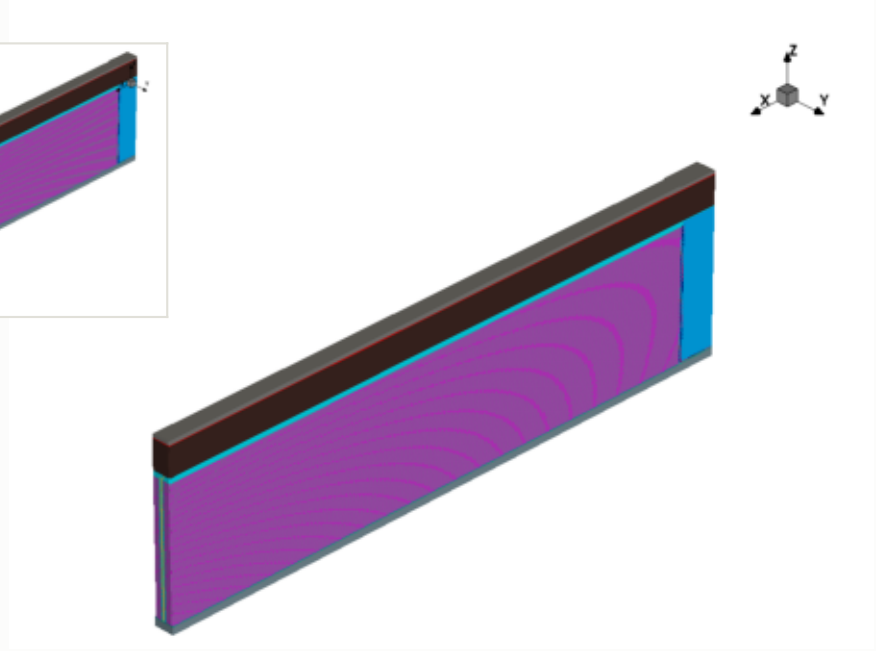
etch_noop · Cryo_etch

0.517	712.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





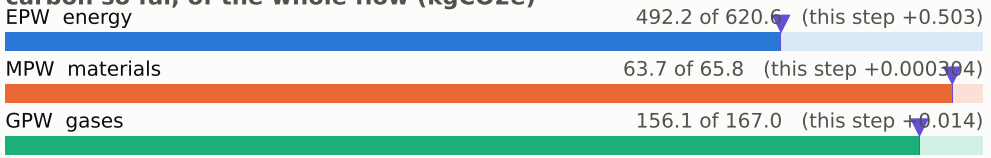
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

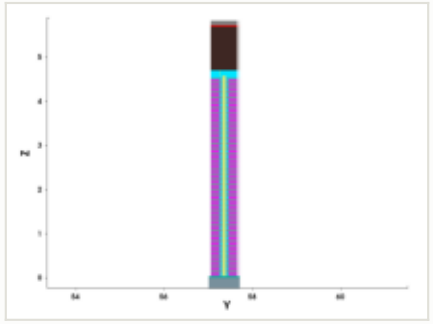
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

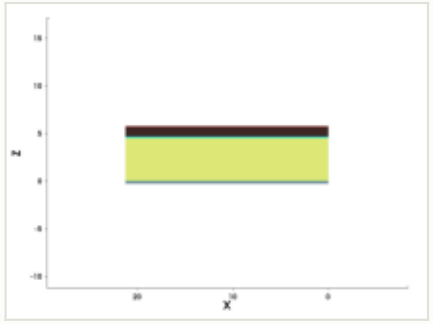


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



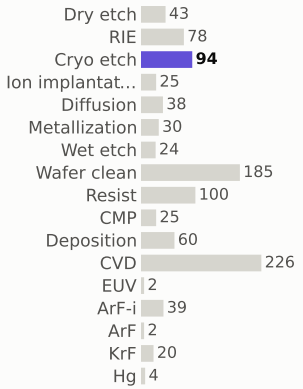
y cut at 57.351 µm



mask for this step

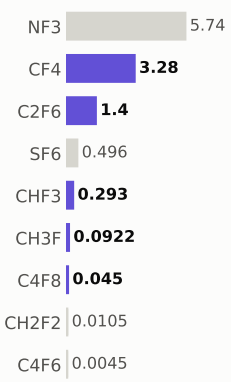
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



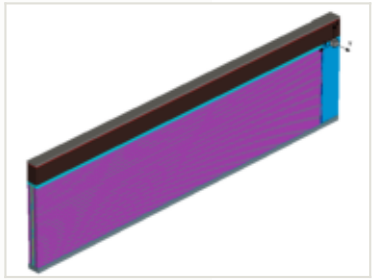
Step #11726.1538: Cryo_etch

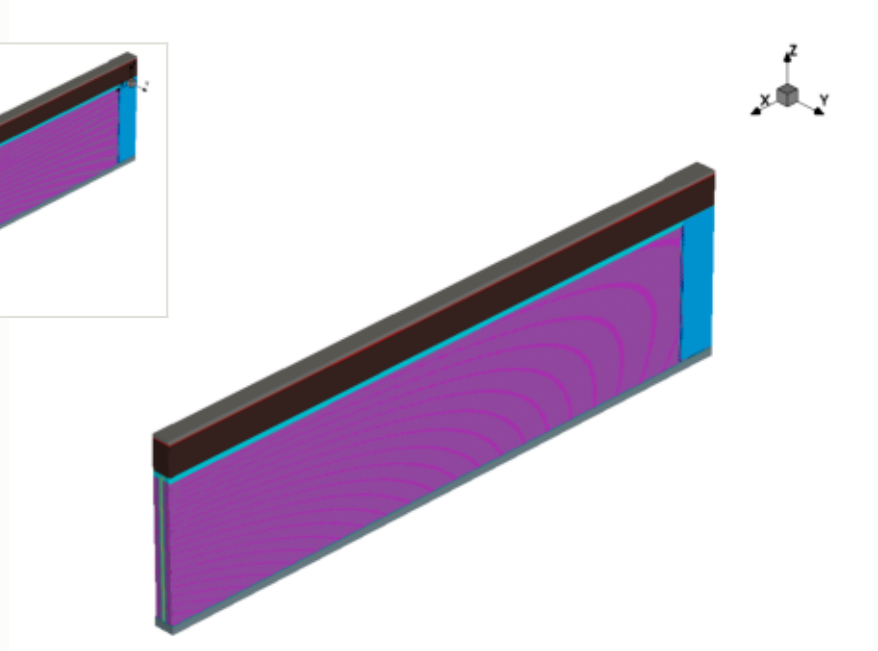
etch_noop · Cryo_etch

0.517	712.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





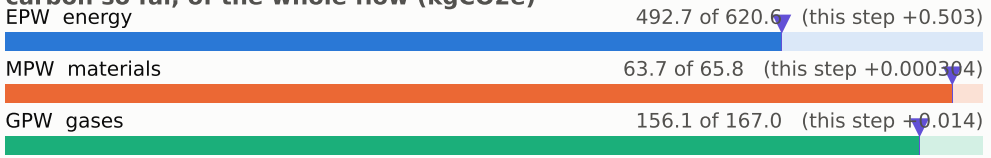
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

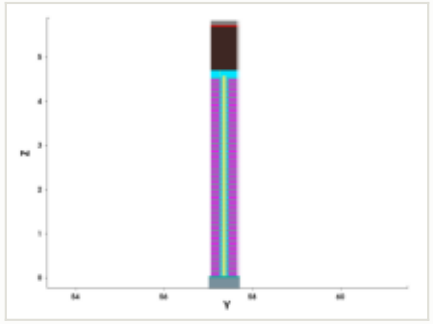
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

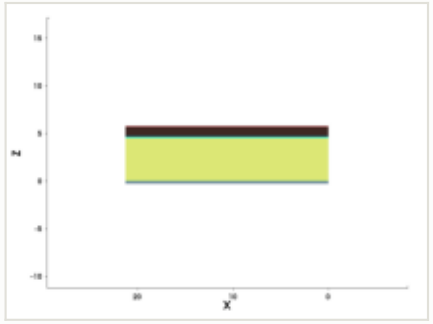


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



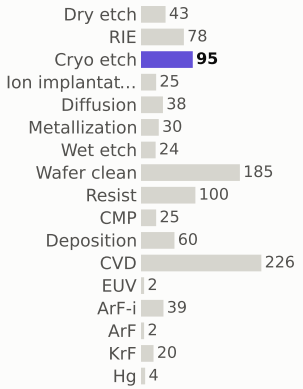
y cut at 57.351 µm



mask for this step

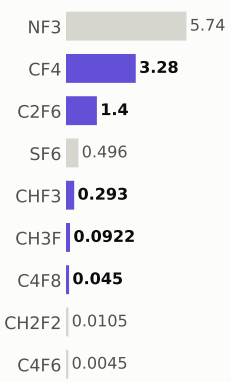
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



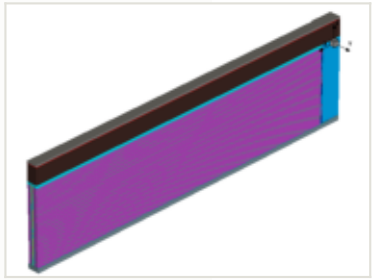
Step #11726.5385: Cryo_etch

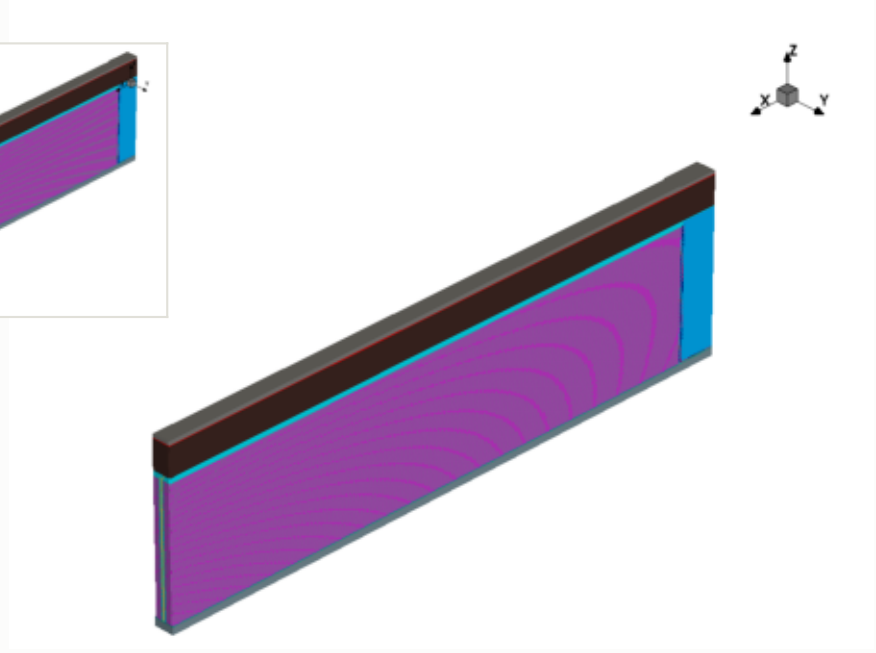
etch_noop · Cryo_etch

0.517	713.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





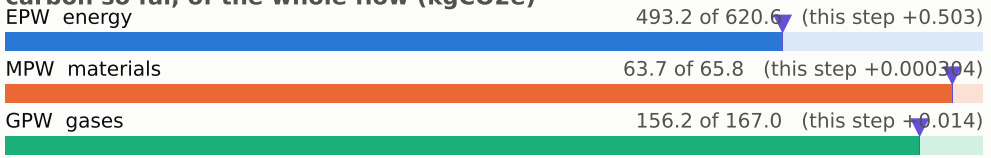
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

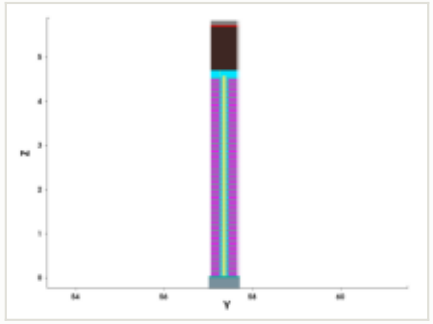
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

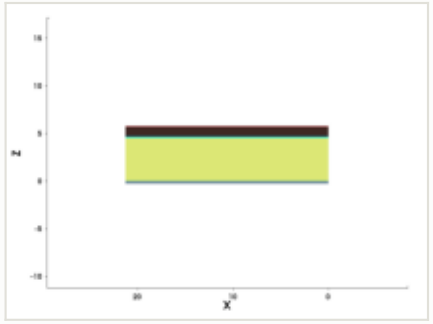


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



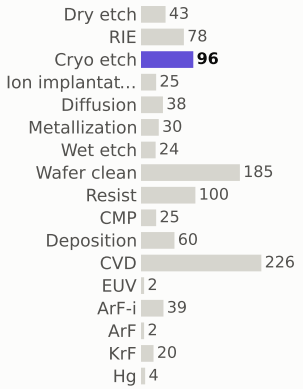
y cut at 57.351 µm



mask for this step

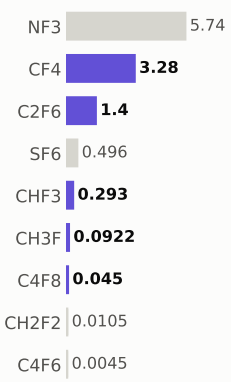
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



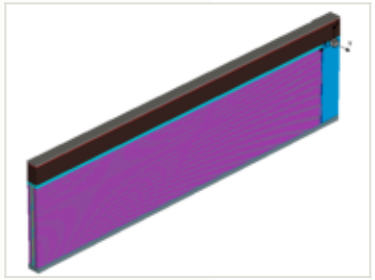
Step #11726.9231: Cryo_etch

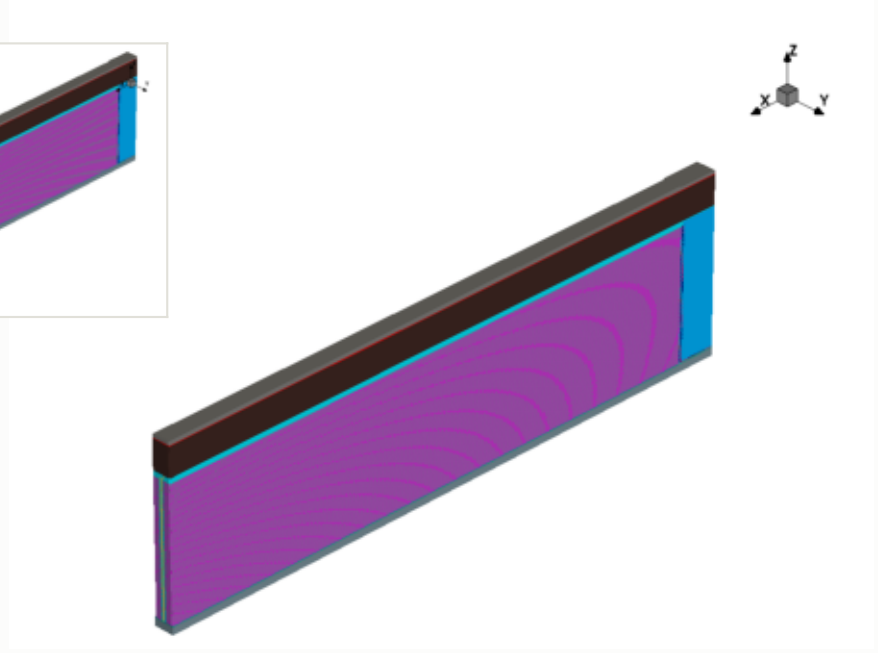
etch_noop · Cryo_etch

0.517	713.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





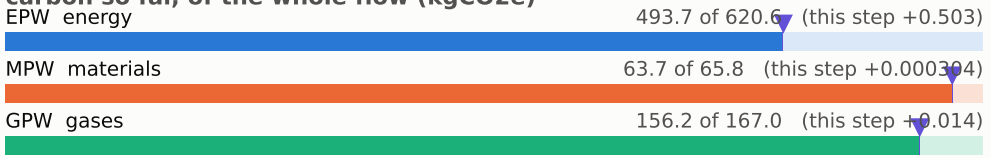
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

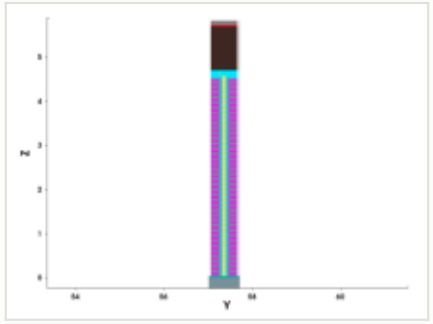
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

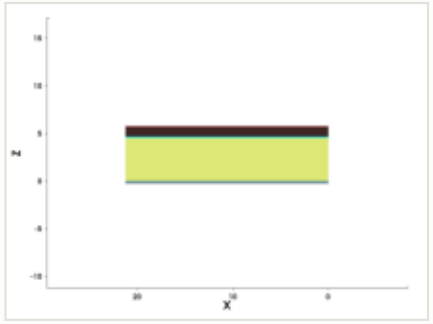


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



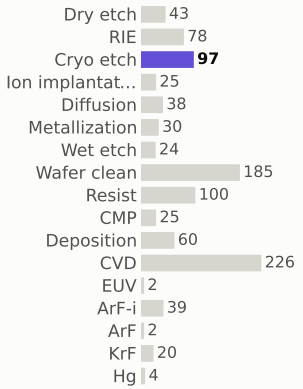
y cut at 57.351 µm



mask for this step

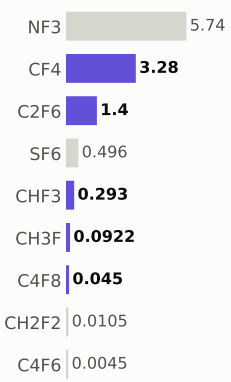
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



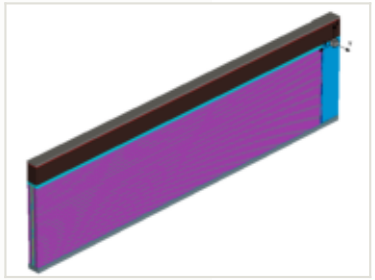
Step #11727.3077: Cryo_etch

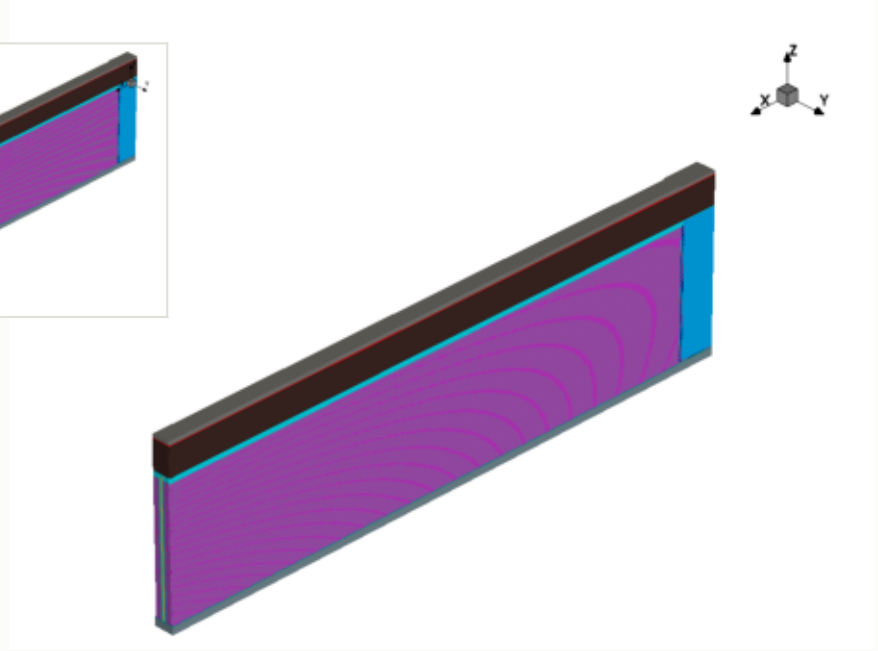
etch_noop · Cryo_etch

0.517	714.1	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





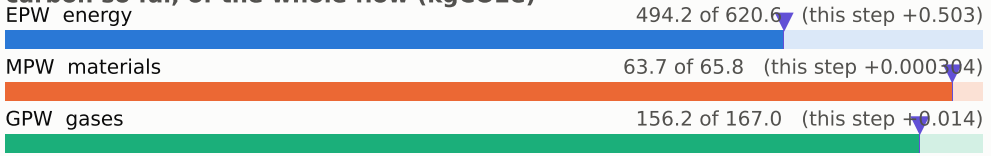
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

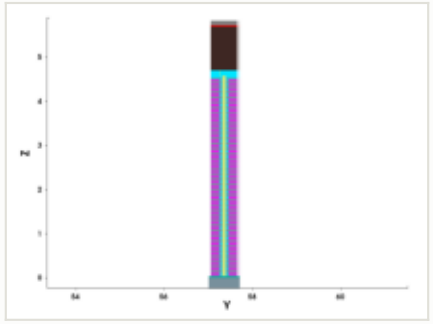
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

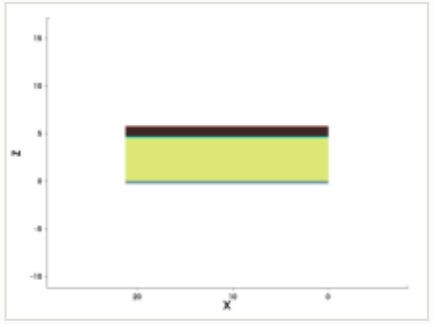


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



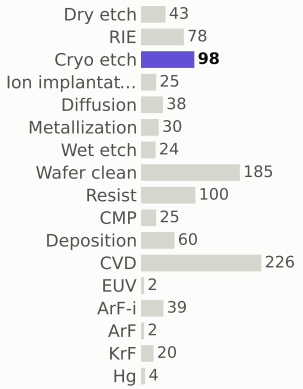
y cut at 57.351 µm



mask for this step

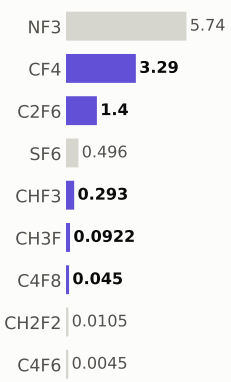
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



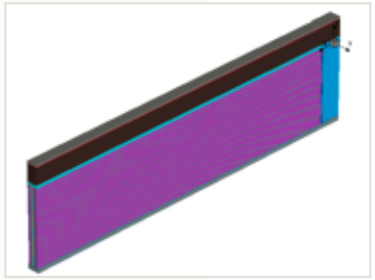
Step #11727.6923: Cryo_etch

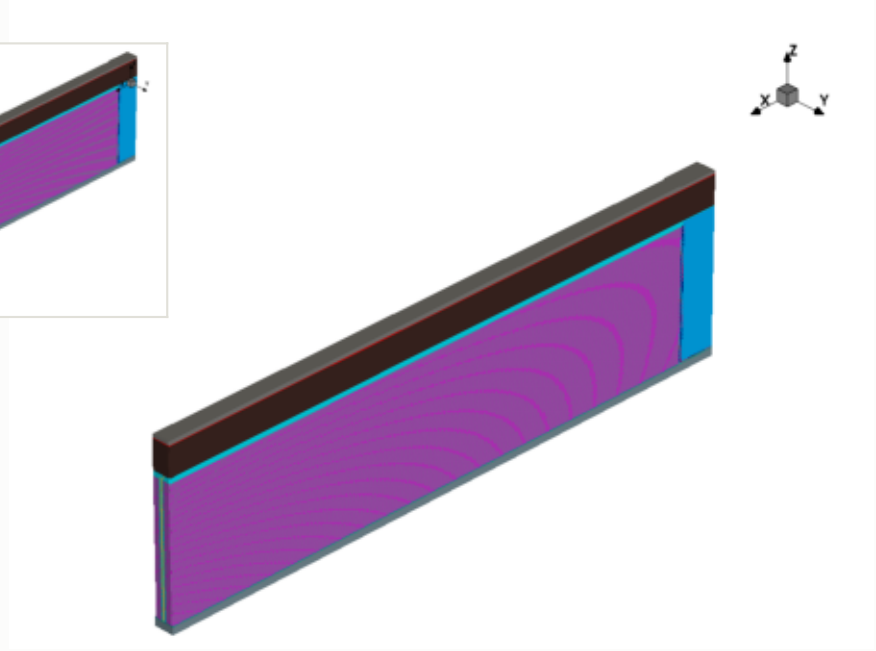
etch_noop · Cryo_etch

0.517	714.6	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



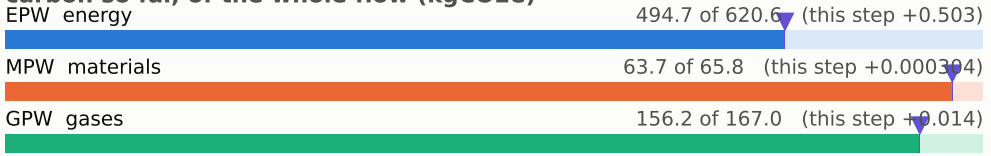


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

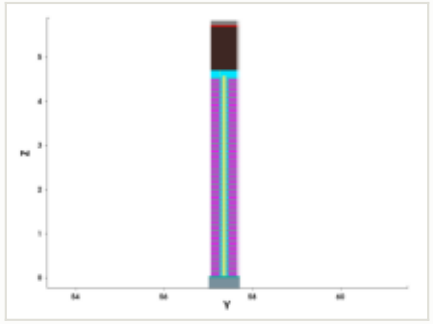
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

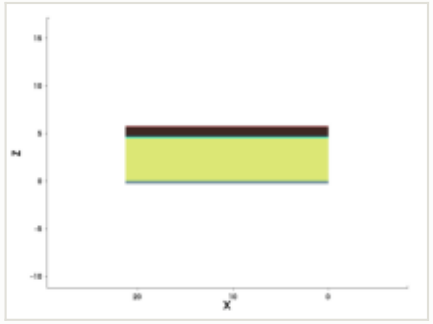


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



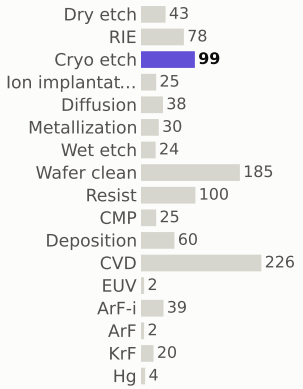
y cut at 57.351 µm



mask for this step

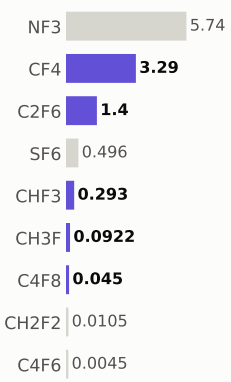
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



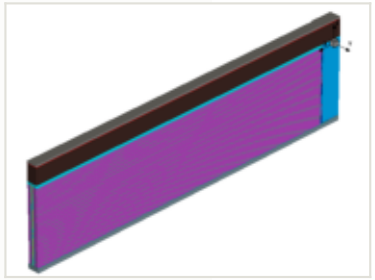
Step #11728.0769: Cryo_etch

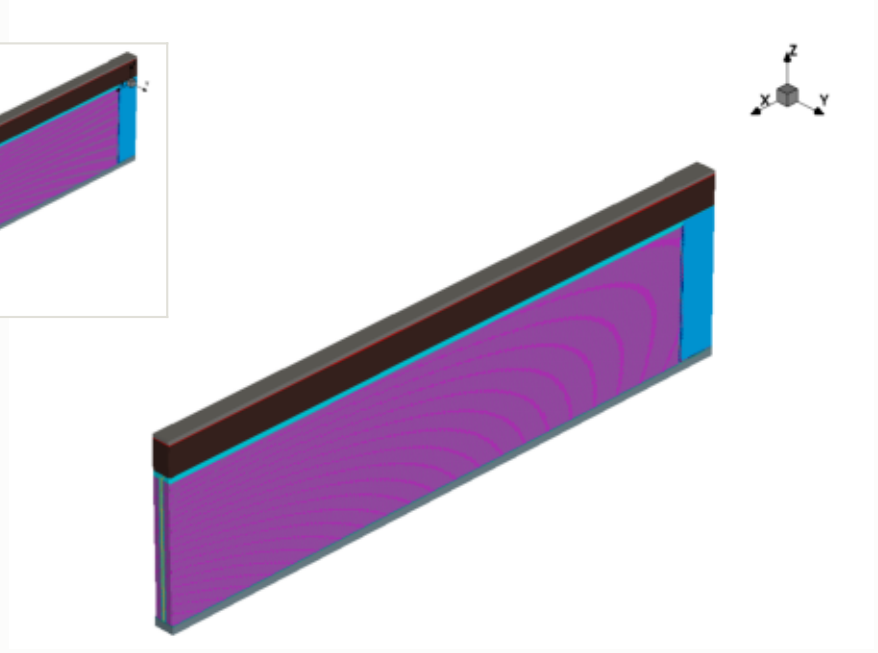
etch_noop · Cryo_etch

0.517	715.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



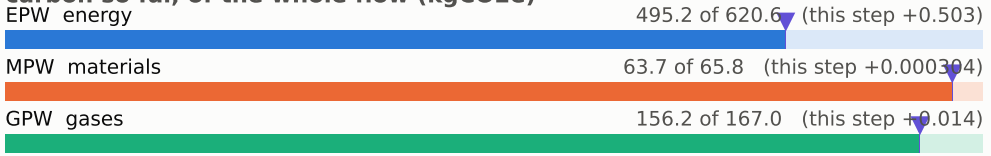


this step

process area	Cryo_etch	EPW	0.503 kgCO2e
material	—	EPW tool	0.359 kgCO2e
thickness	—	EPW overhead	0.144 kgCO2e
material GWP	—	MPW	0.000304 kgCO2e (gas 0.000304)
input gases	CF4 0.0241 g	GPW	0.014 kgCO2e
energy	0.835 kWh + 0.334 overhead	embodied carbon	**0.517 kgCO2e**

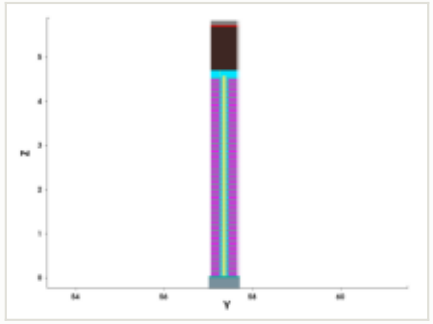
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

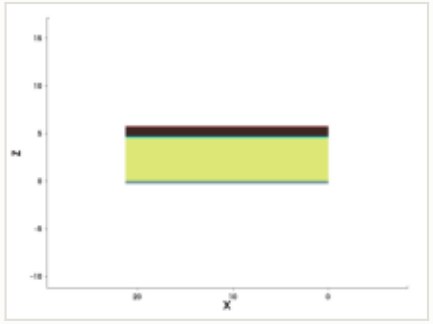


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



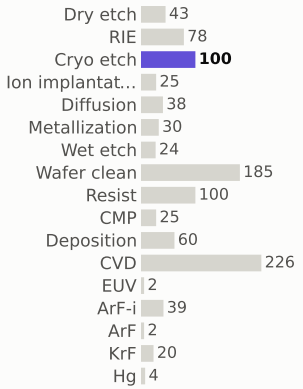
y cut at 57.351 µm



mask for this step

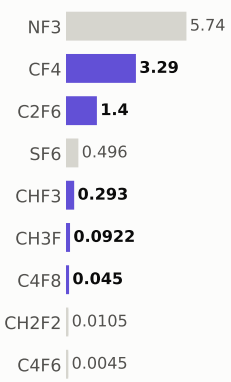
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



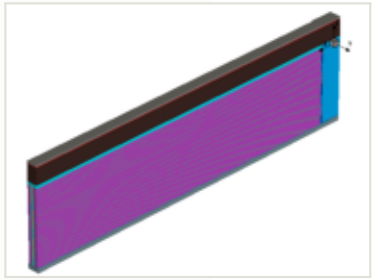
Step #11728.4615: Cryo_etch

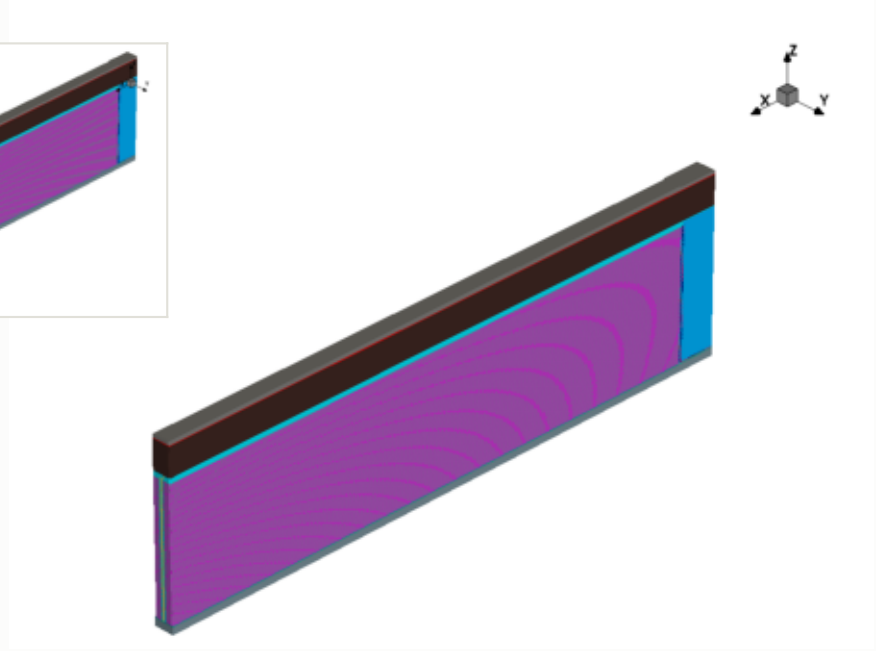
etch_noop · Cryo_etch

0.517	715.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





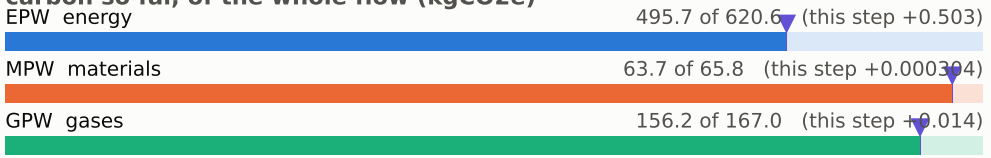
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

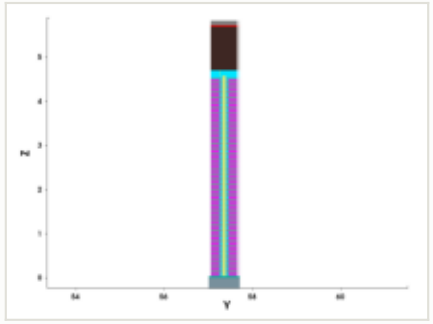
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

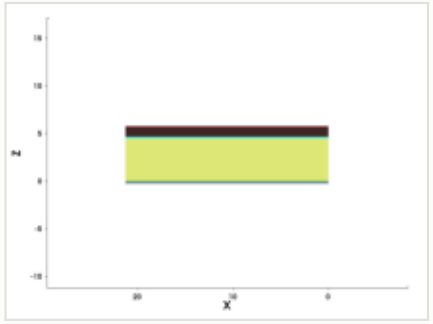


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



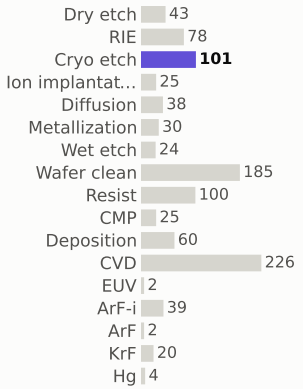
y cut at 57.351 µm



mask for this step

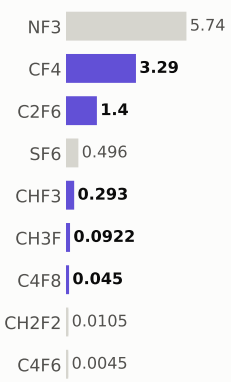
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



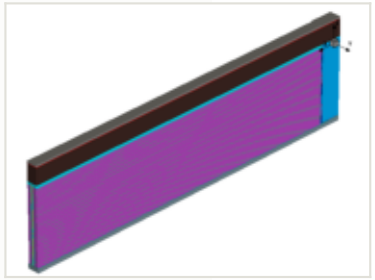
Step #11728.8462: Cryo_etch

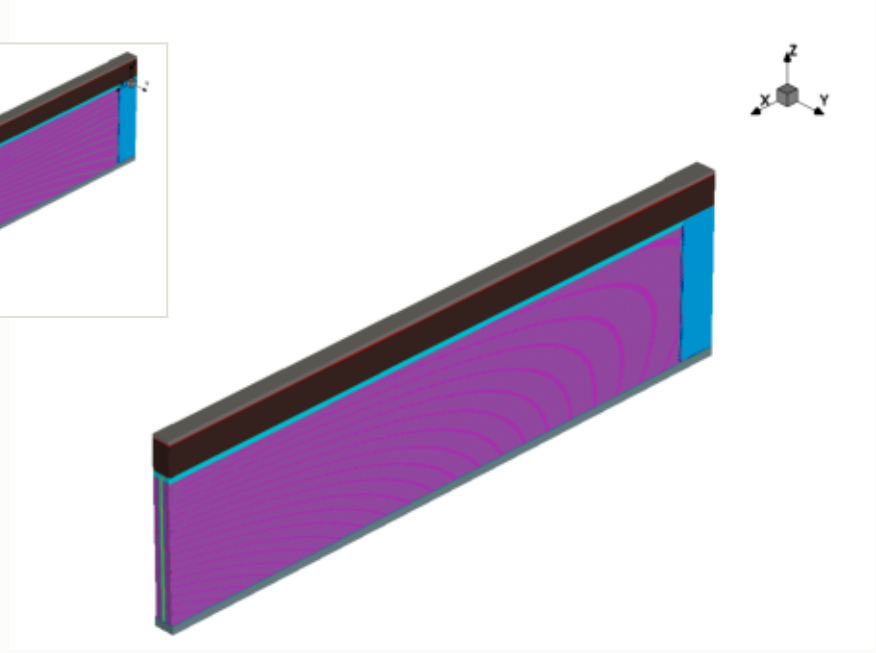
etch_noop · Cryo_etch

0.517	716.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





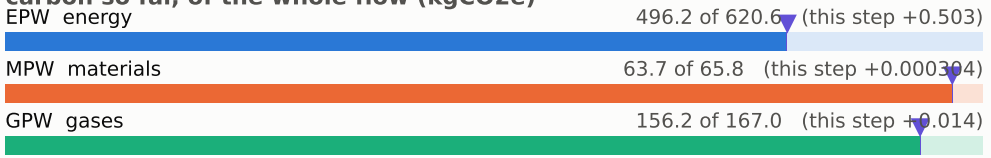
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

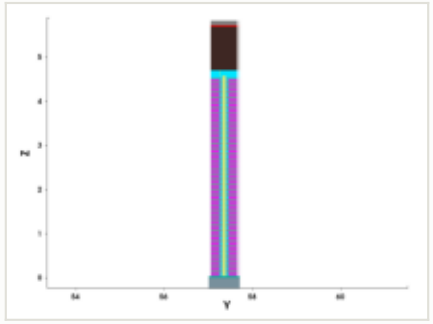
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

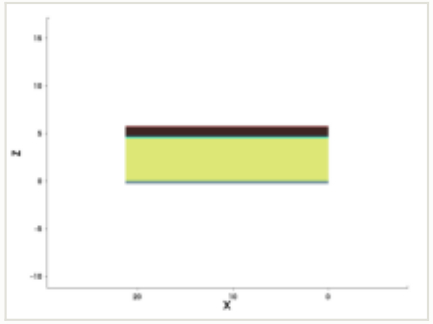


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



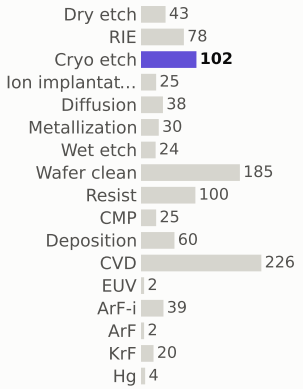
y cut at 57.351 µm



mask for this step

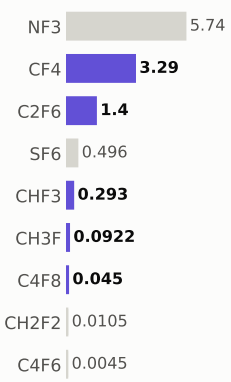
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



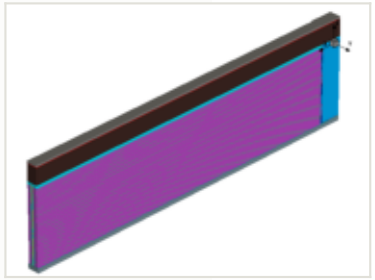
Step #11729.2308: Cryo_etch

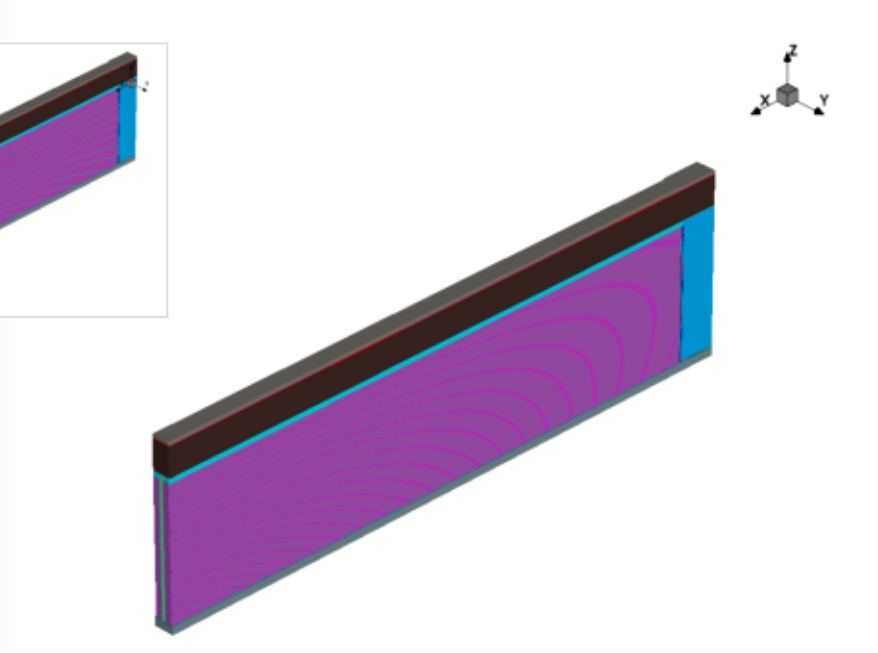
etch_noop · Cryo_etch

0.517	716.7	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





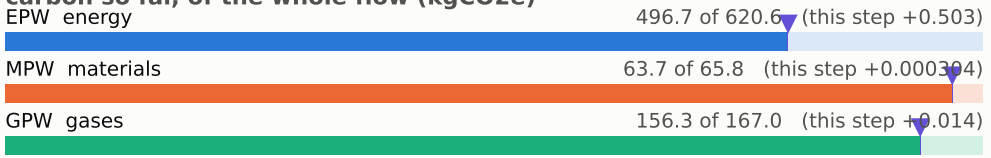
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

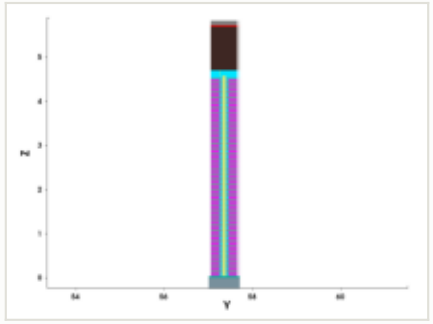
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

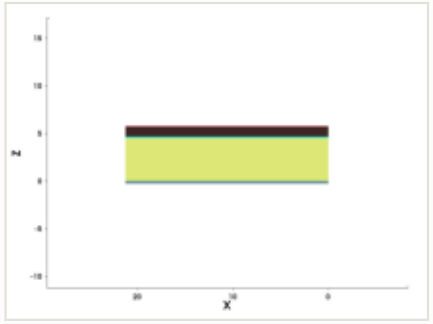


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



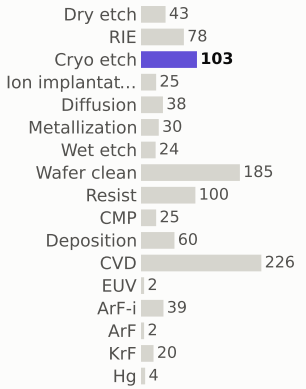
y cut at 57.351 µm



mask for this step

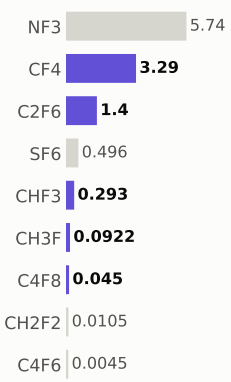
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



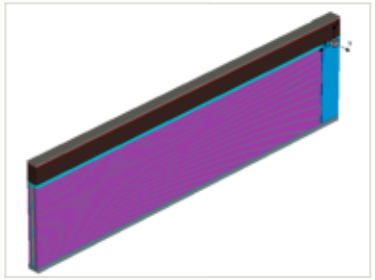
Step #11729.6154: Cryo_etch

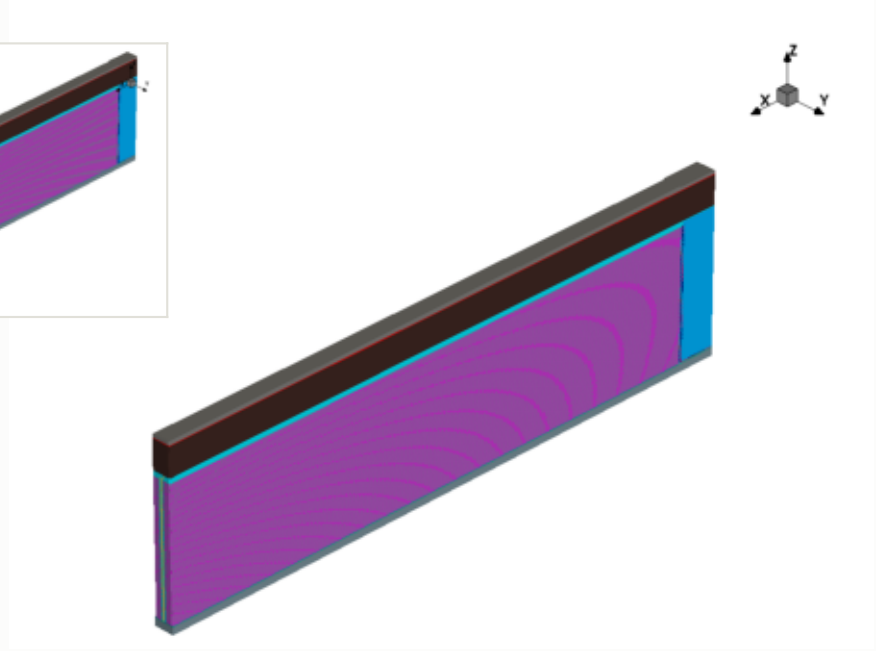
etch_noop · Cryo_etch

0.517	717.2	0.061%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





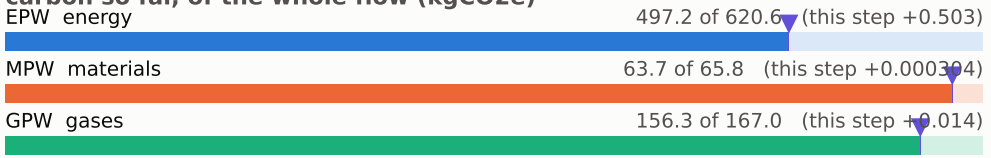
this step

process area	Cryo_etch
material	—
thickness	—
material GWP	—
input gases	CF4 0.0241 g
energy	0.835 kWh + 0.334 overhead

EPW	0.503 kgCO2e
EPW tool	0.359 kgCO2e
EPW overhead	0.144 kgCO2e
MPW	0.000304 kgCO2e (gas 0.000304)
GPW	0.014 kgCO2e
embodied carbon	**0.517 kgCO2e**

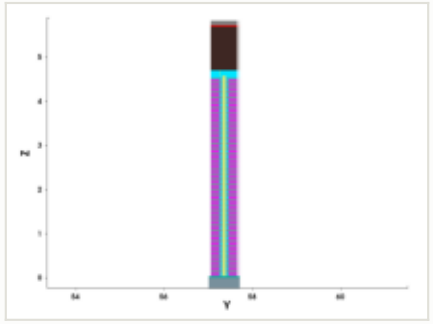
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

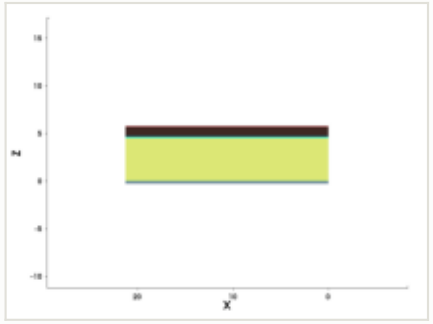


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



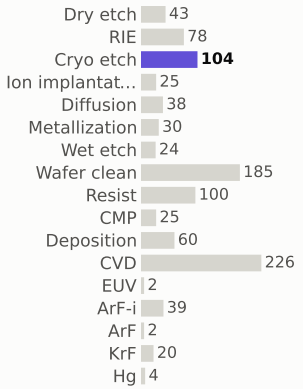
y cut at 57.351 µm



mask for this step

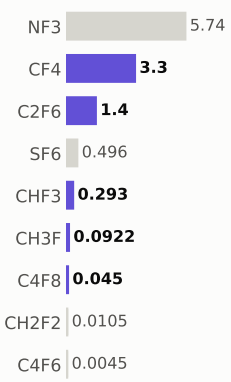
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



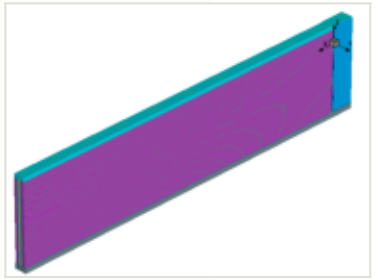
Step #11730: Dry_etch

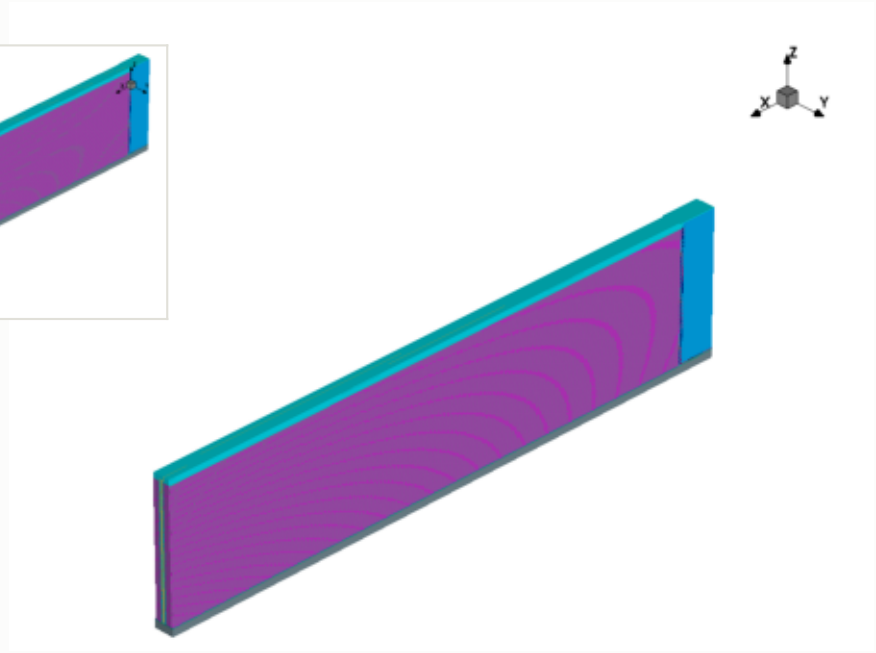
etch_strip · Dry_etch

0.801	718.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



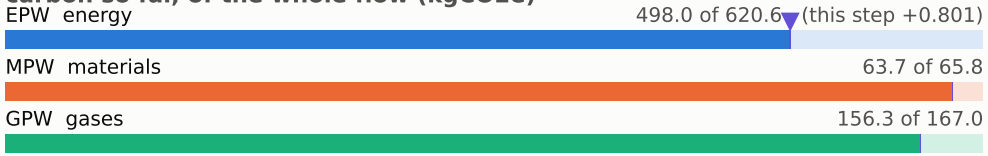


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

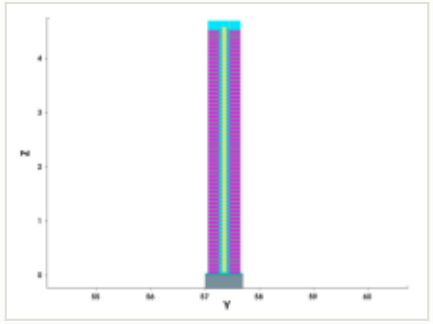
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

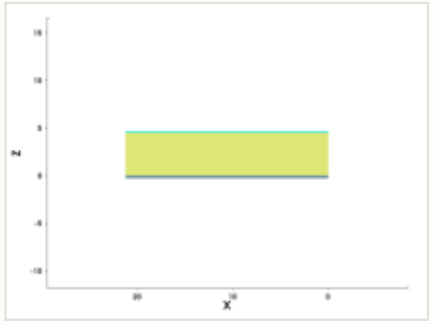


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



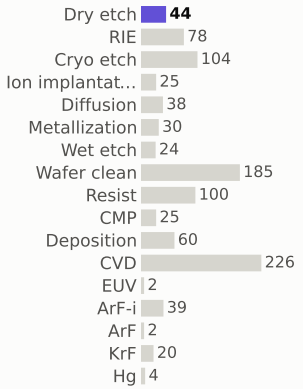
y cut at 57.351 µm



mask for this step

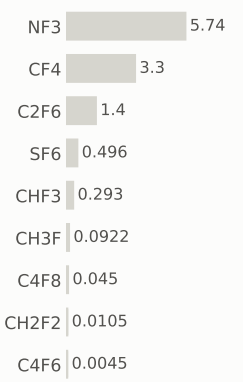
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



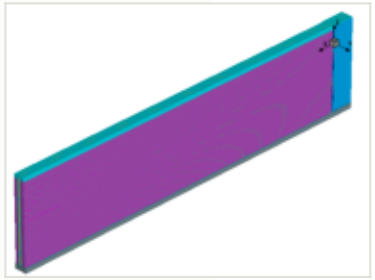
Step #11740: Wafer_clean

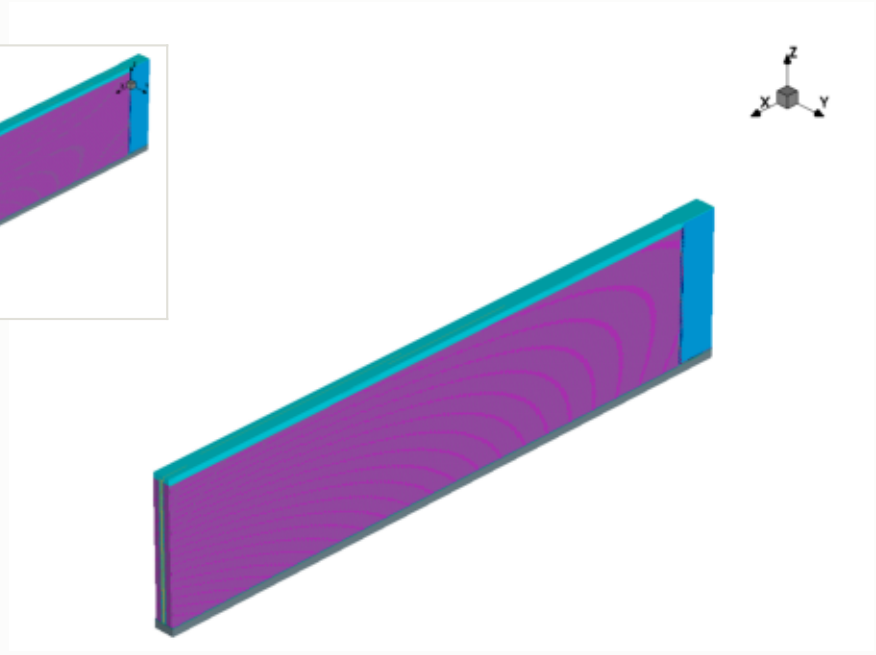
clean · Wafer_clean

0.169	718.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





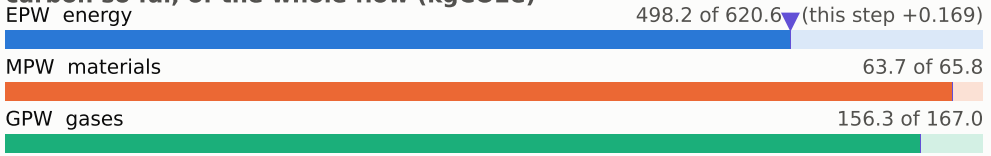
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

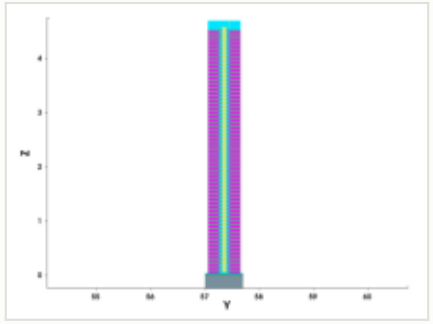
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

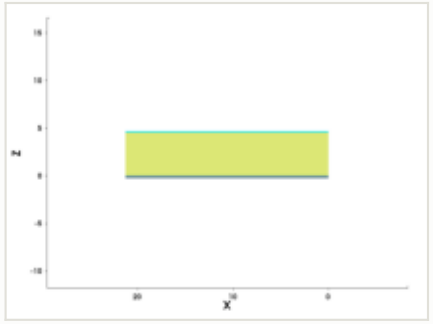


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



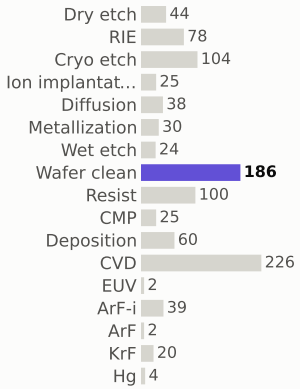
y cut at 57.351 µm



mask for this step

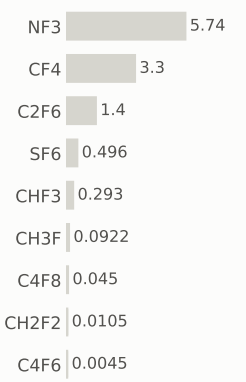
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



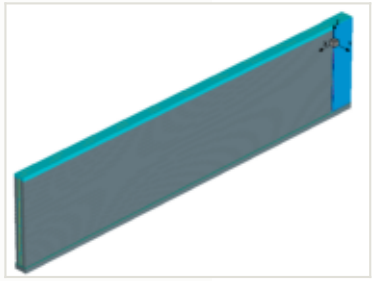
Step #11750: Wet_etch

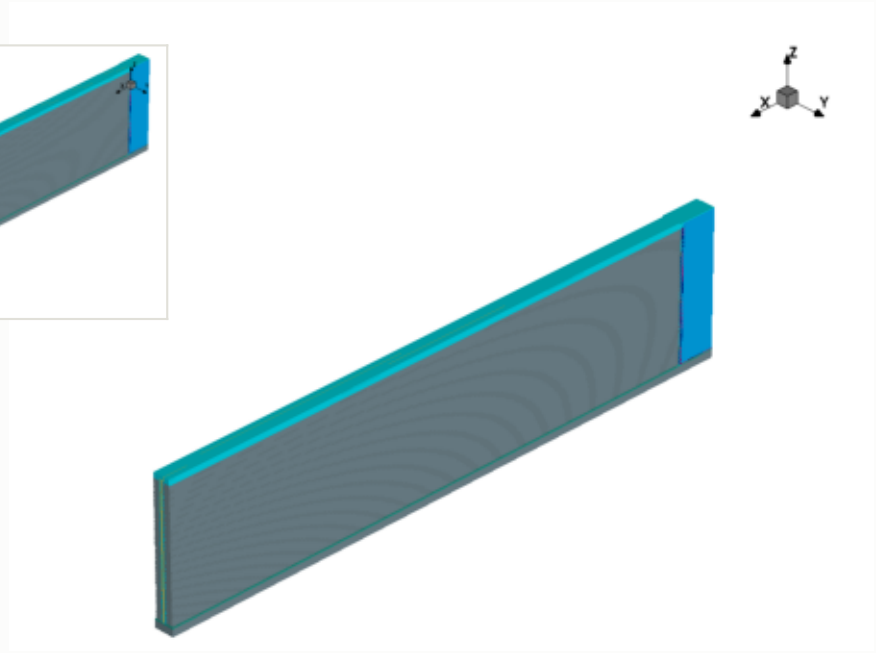
etch_isotropic · Wet_etch

0.169	718.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



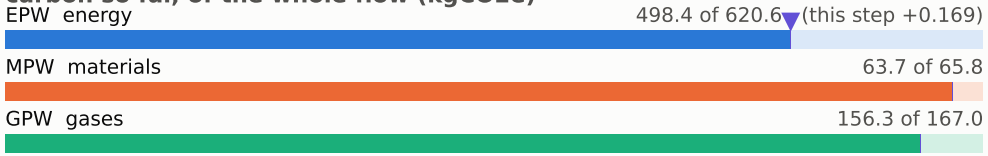


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

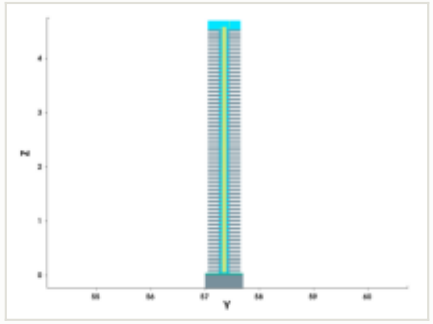
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

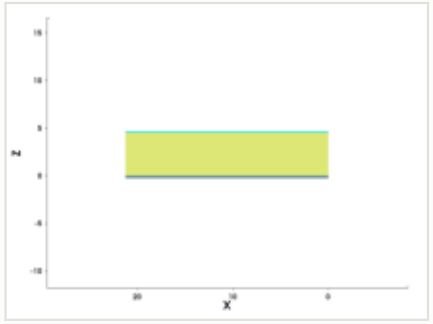


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



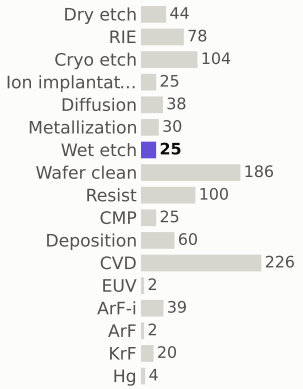
y cut at 57.351 µm



mask for this step

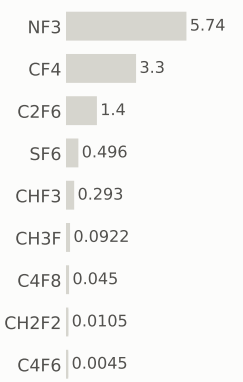
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



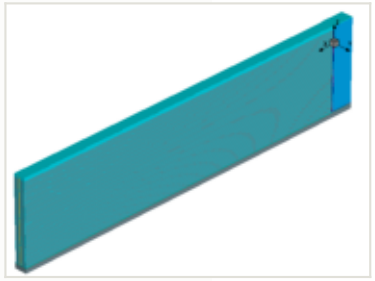
Step #11760: Deposition

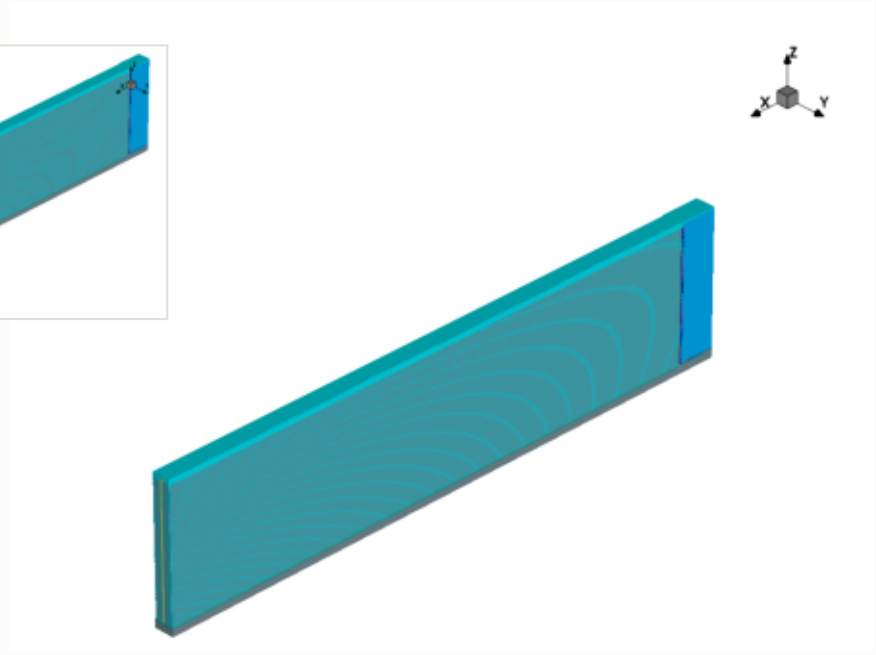
depo_isotropic · Deposition

0.656	719.0	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



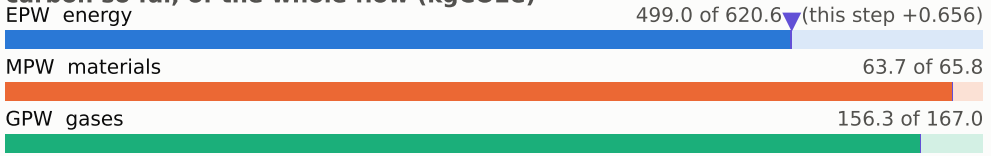


this step

process area	Deposition
material	—
thickness	0.015 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

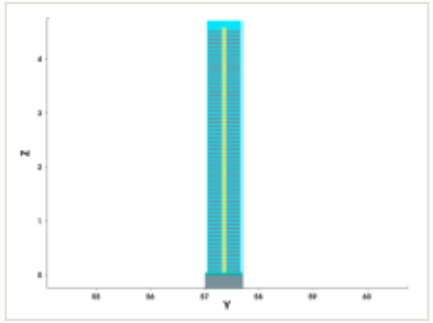
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

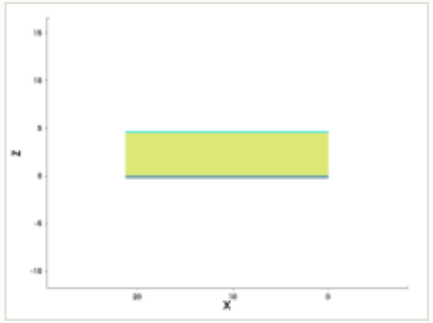


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



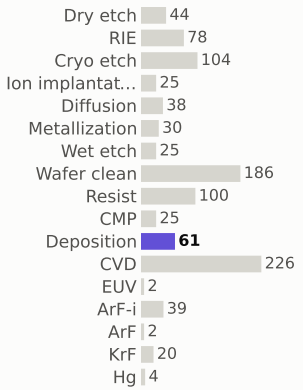
y cut at 57.351 µm



mask for this step

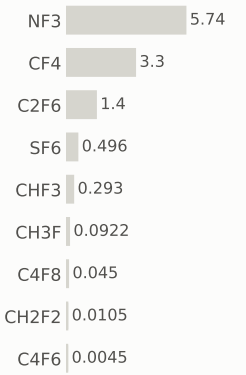
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



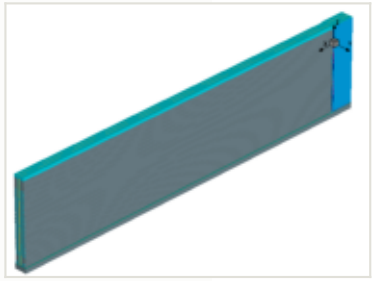
Step #11770: Wet_etch

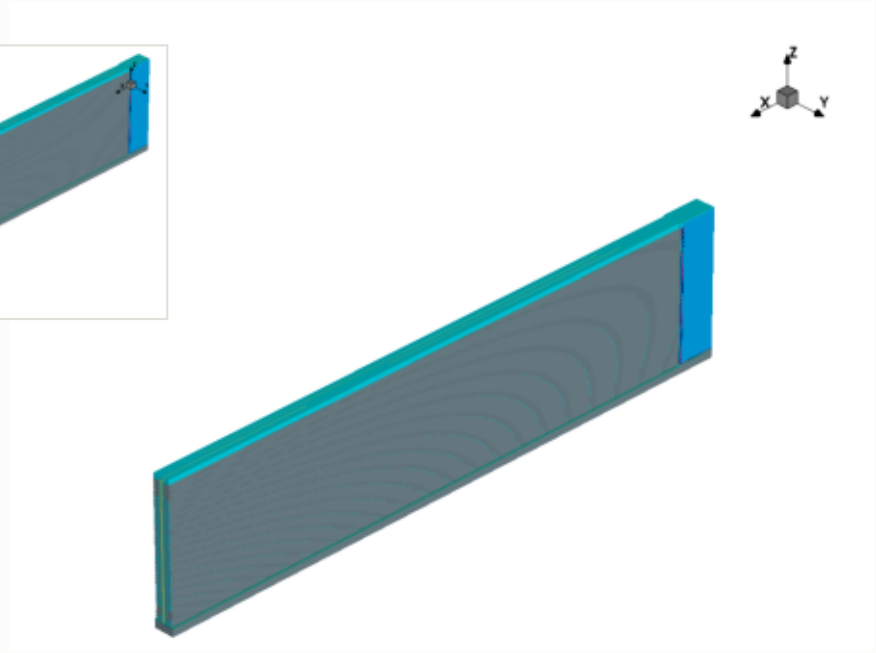
etch_isotropic · Wet_etch

0.169	719.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



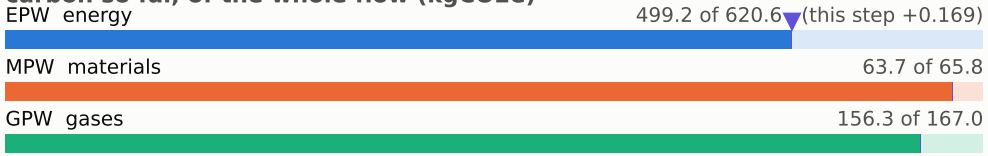


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

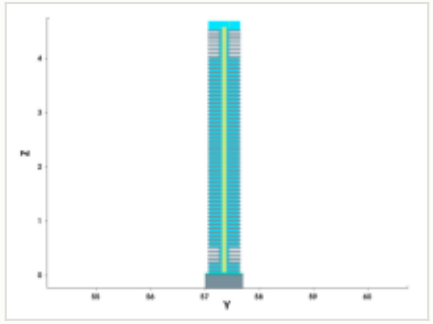
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

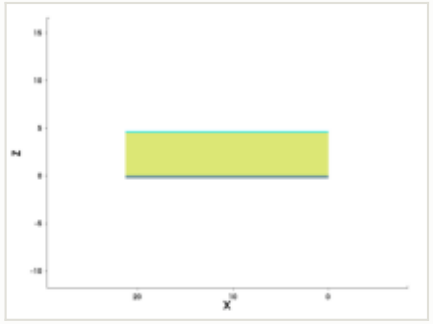


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



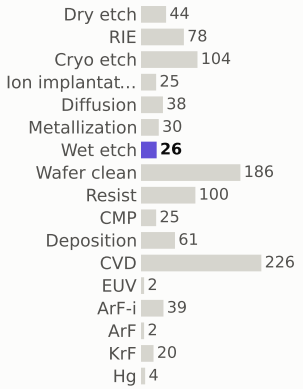
y cut at 57.351 µm



mask for this step

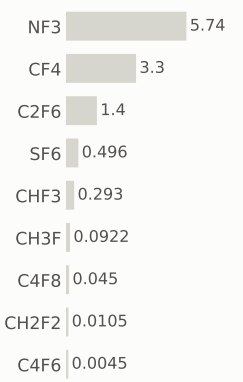
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



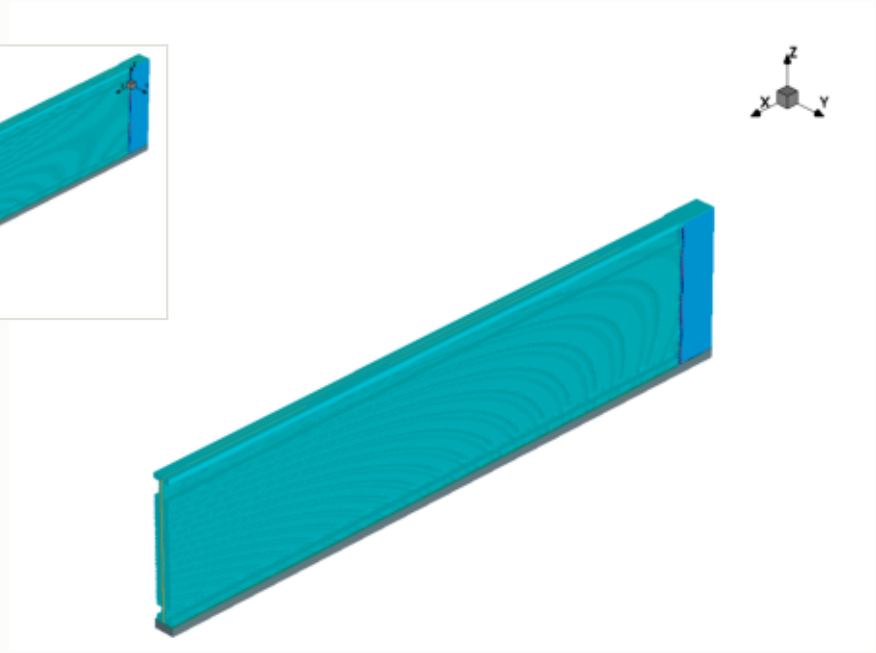
Step #11780: Wet_etch

etch_isotropic · Wet_etch

0.169	719.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

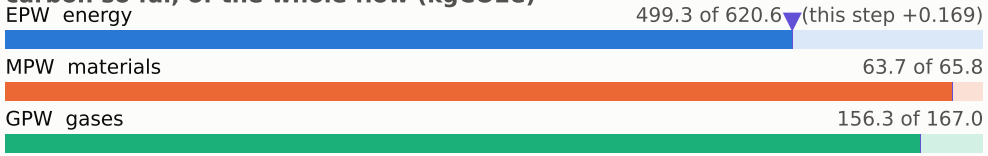
zoomed to fit



this step

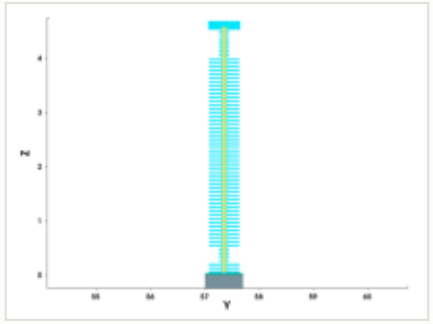
process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

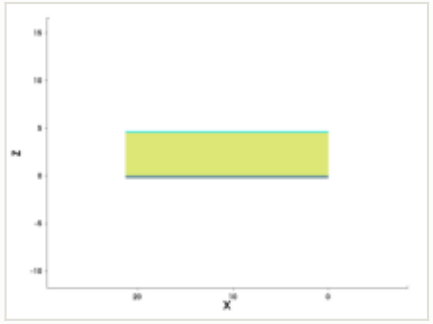


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



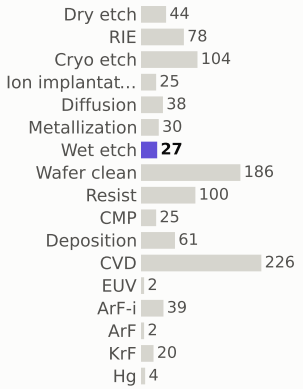
y cut at 57.351 µm



mask for this step

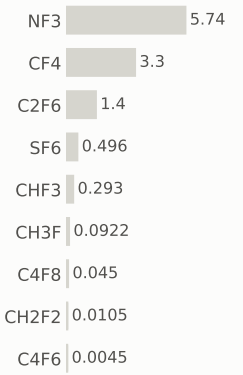
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



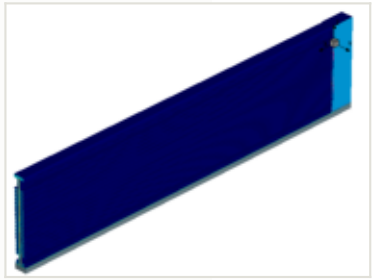
Step #11790: Metallization

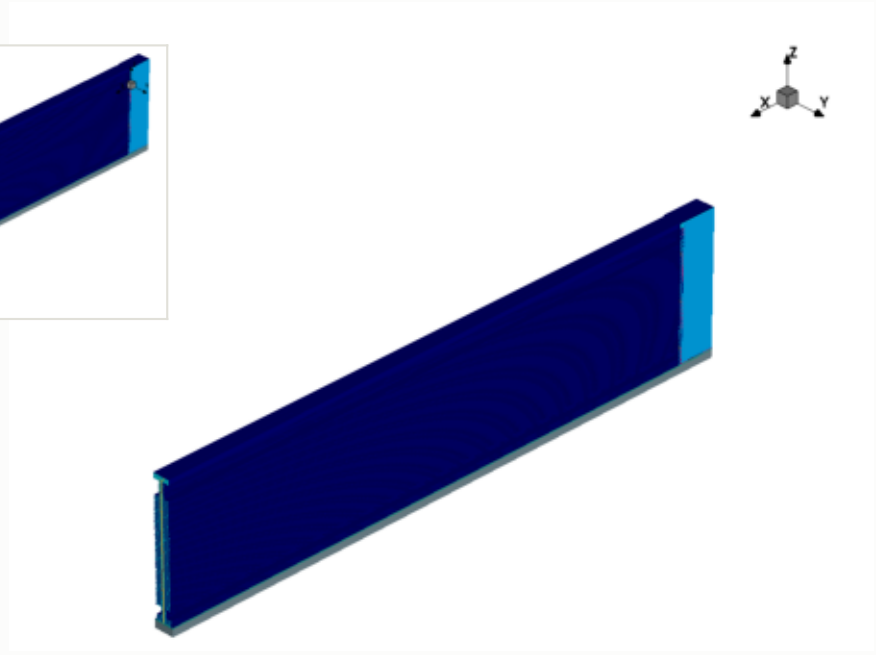
depo_isotropic · Metallization

1.2	720.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



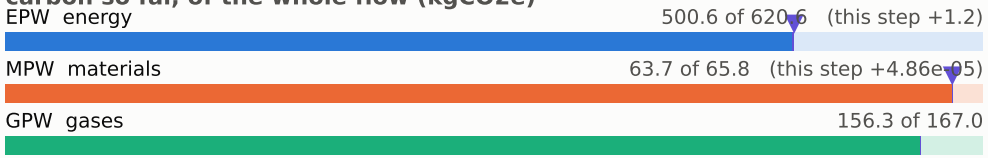


this step

process area	Metallization
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

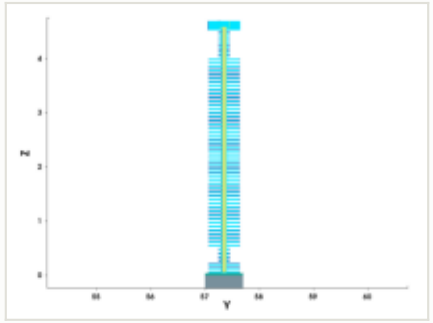
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

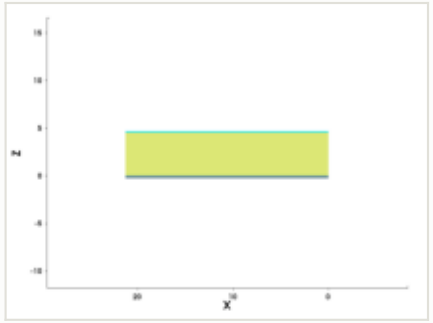


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



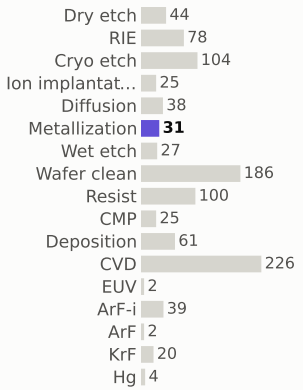
y cut at 57.351 µm



mask for this step

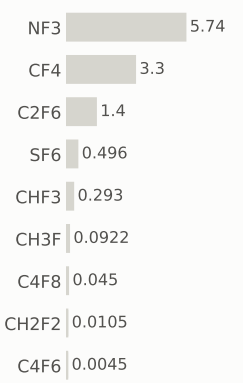
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



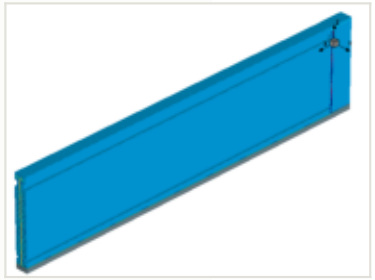
Step #11800: Metallization

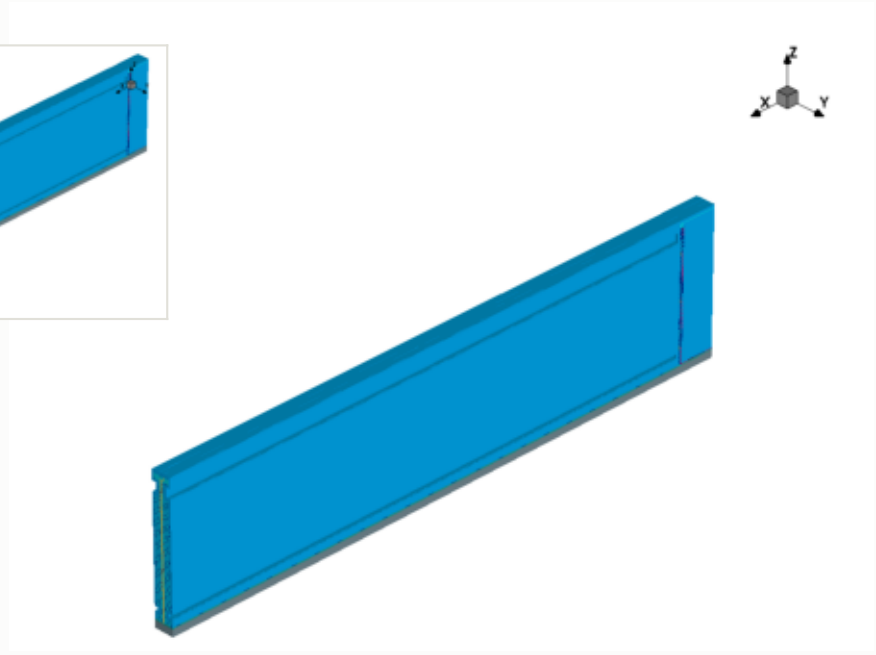
depo_isotropic · Metallization

1.21	721.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



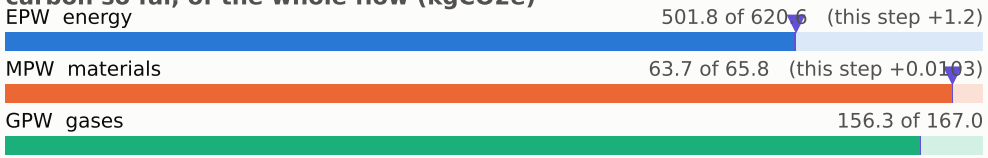


this step

process area	Metallization
material	W
thickness	0.1 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

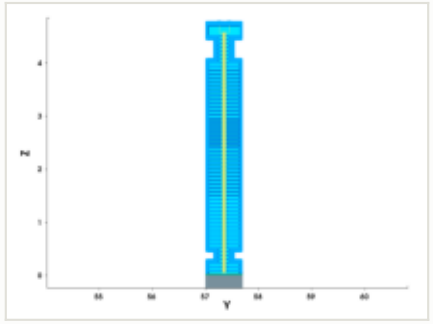
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0103 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

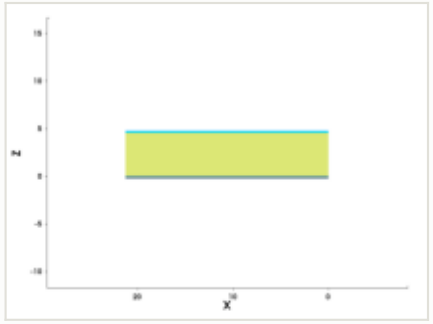


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



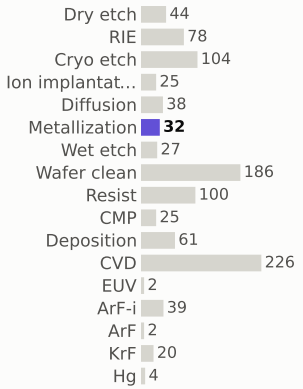
y cut at 57.351 µm



mask for this step

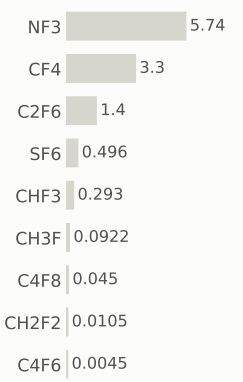
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



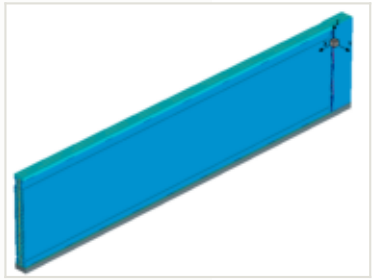
Step #11810: RIE

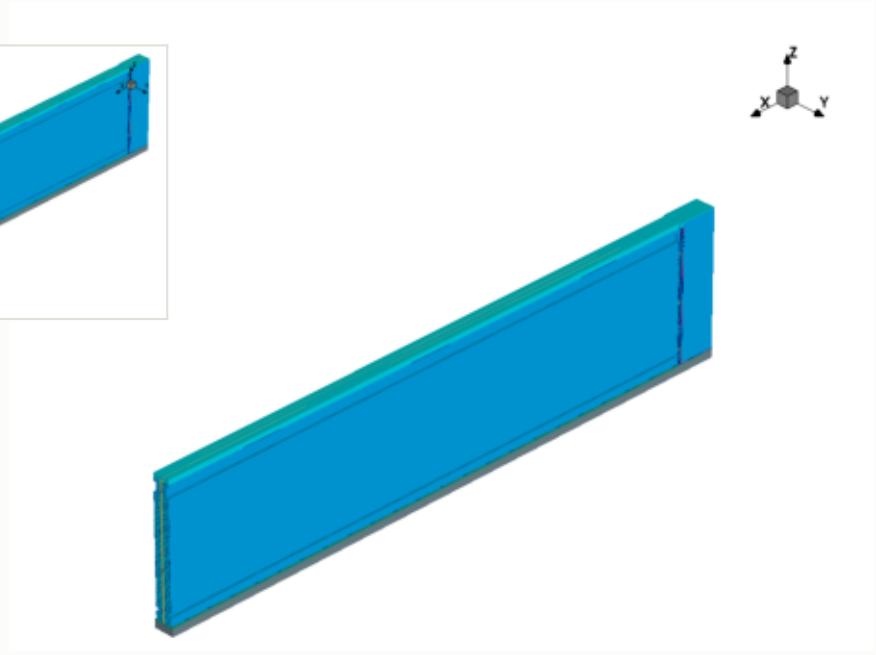
etch_anisotropic · RIE

1.04	722.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



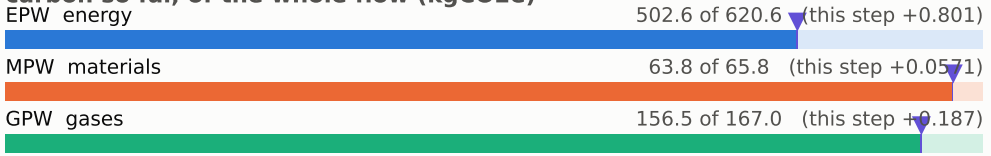


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

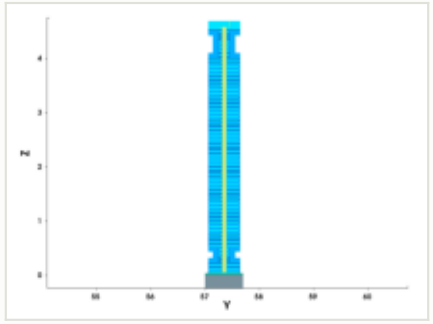
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

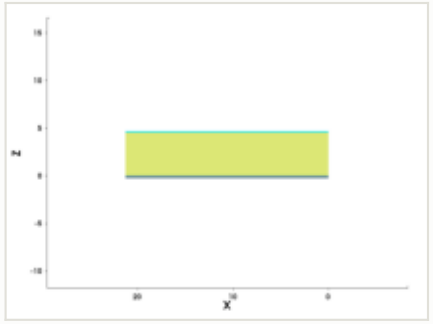


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



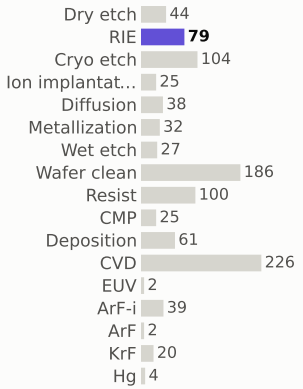
y cut at 57.351 µm



mask for this step

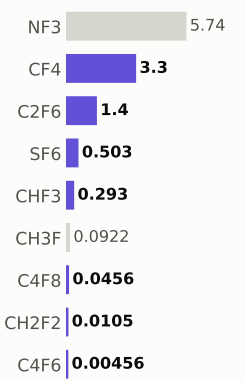
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



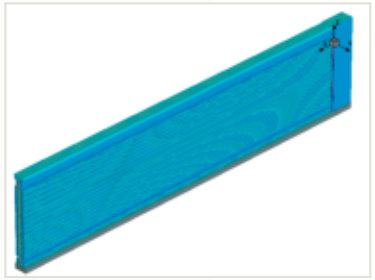
Step #11820: Dry_etch

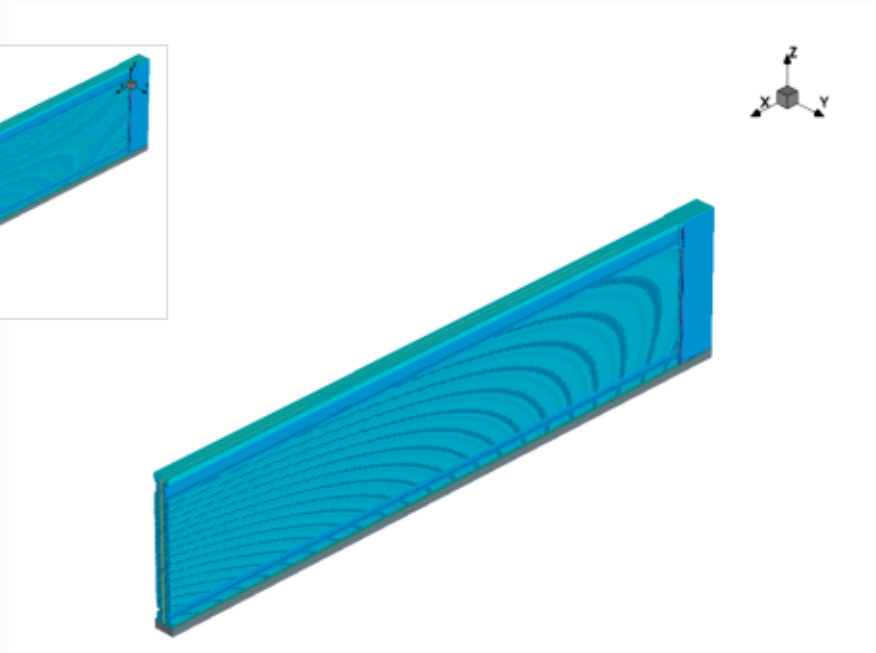
etch_isotropic · Dry_etch

0.801	723.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



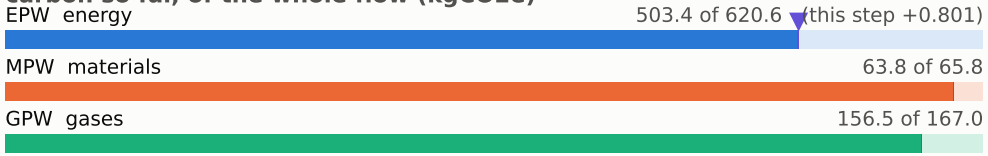


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

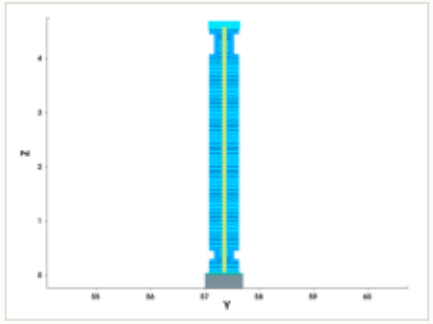
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

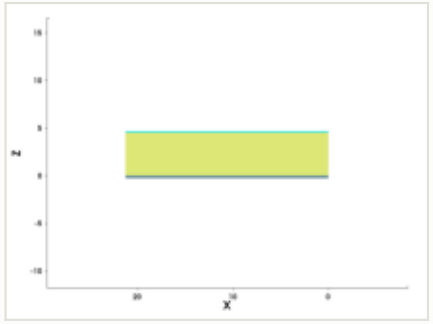


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



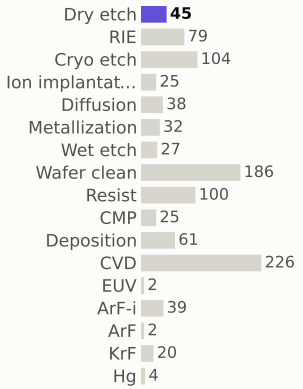
y cut at 57.351 µm



mask for this step

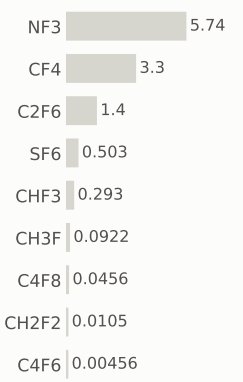
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



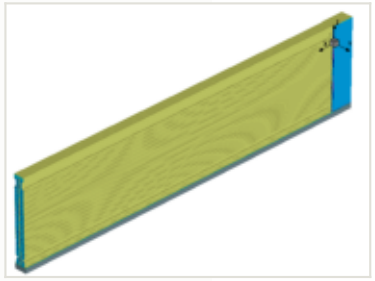
Step #11830: Deposition

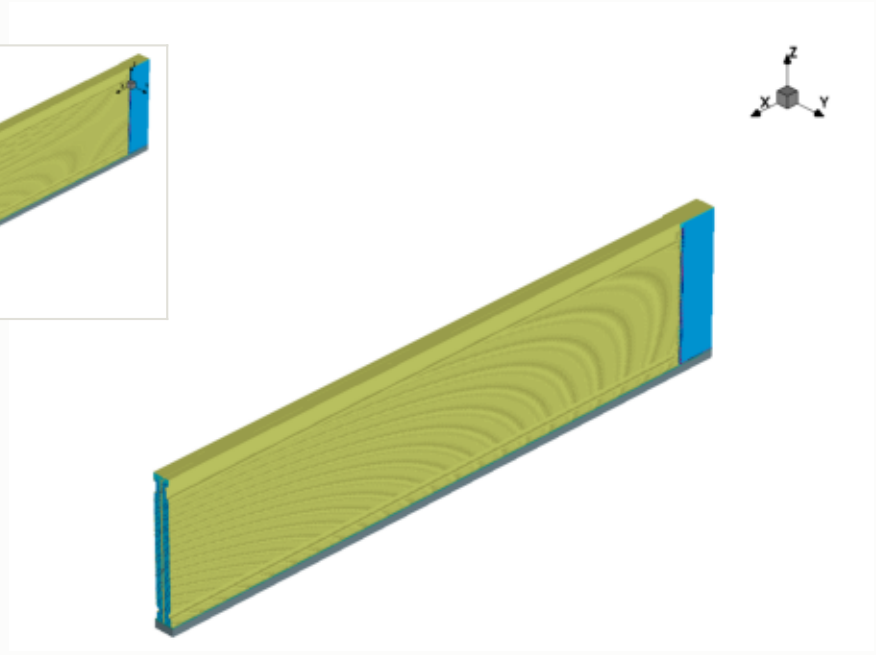
depo_isotropic · Deposition

0.656	724.3	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



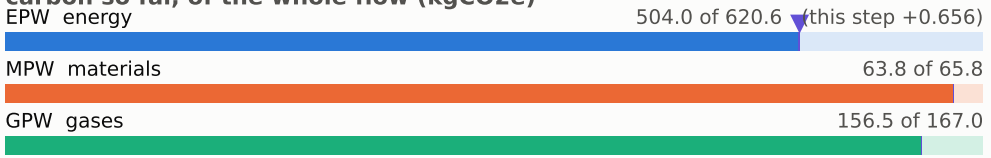


this step

process area	Deposition
material	—
thickness	0.005 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

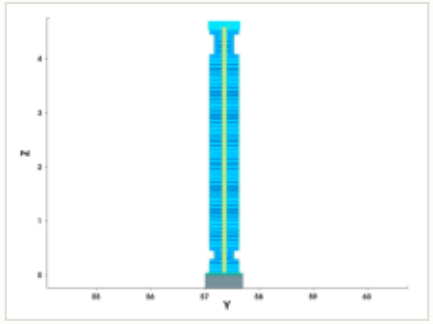
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

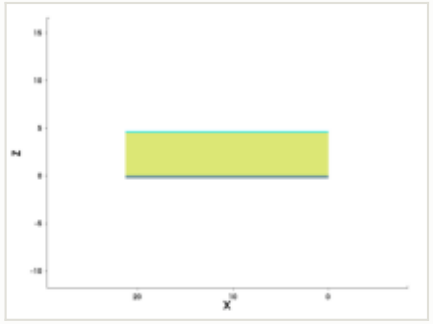


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



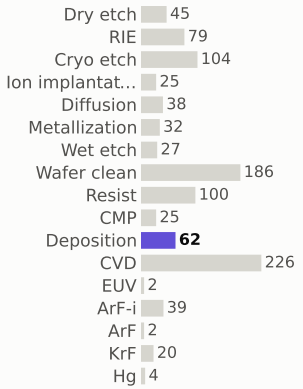
y cut at 57.351 µm



mask for this step

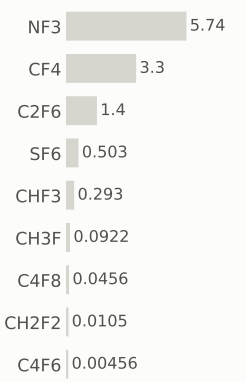
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



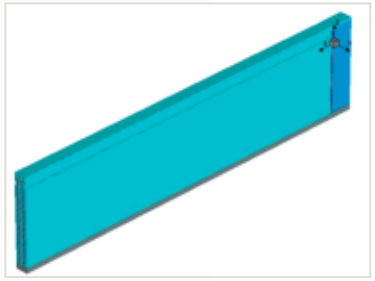
Step #11840: Deposition

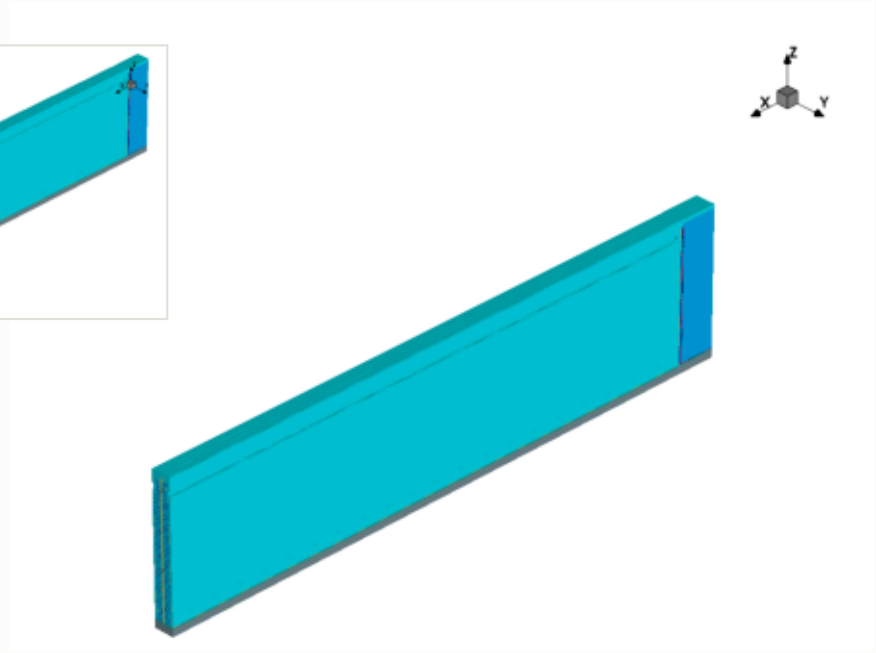
depo_isotropic · Deposition

0.656	724.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



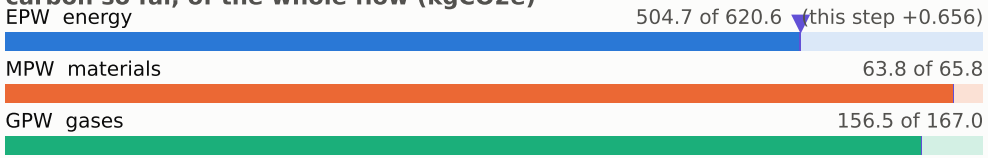


this step

process area	Deposition
material	—
thickness	0.1 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

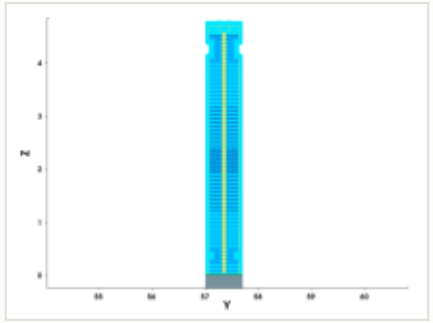
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

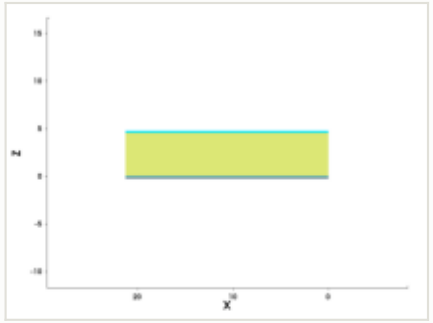


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



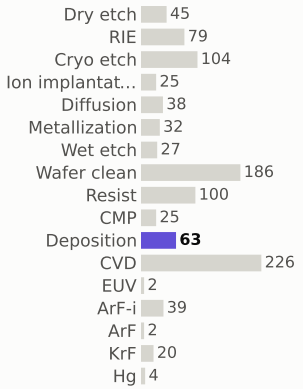
y cut at 57.351 µm



mask for this step

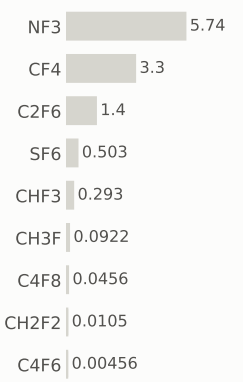
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



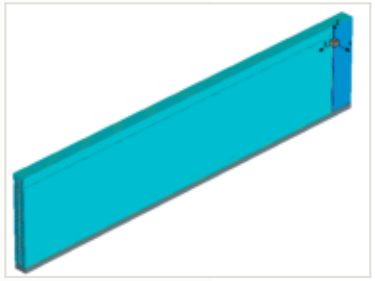
Step #11850: CMP

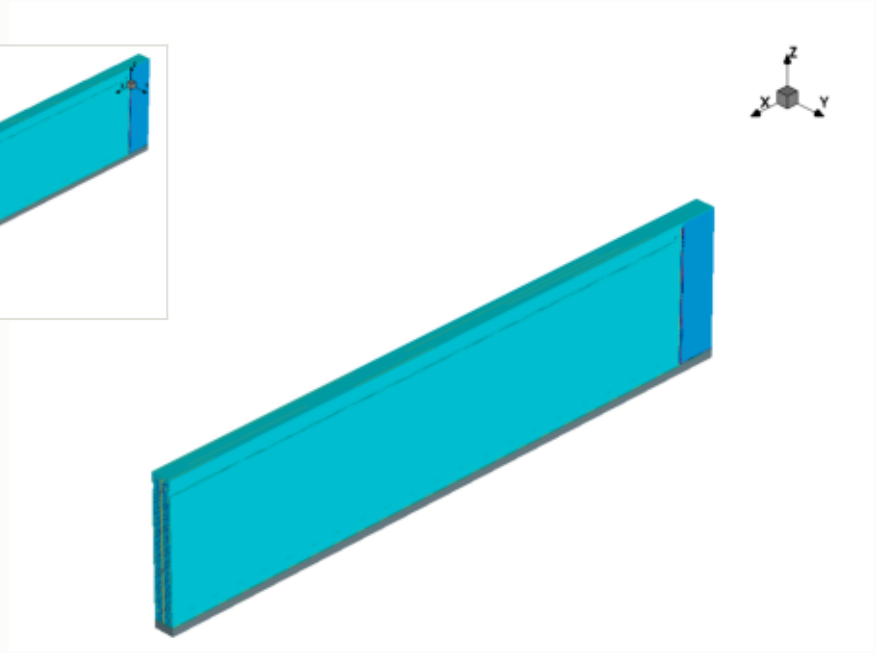
etch_cmp · CMP

0.169	725.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



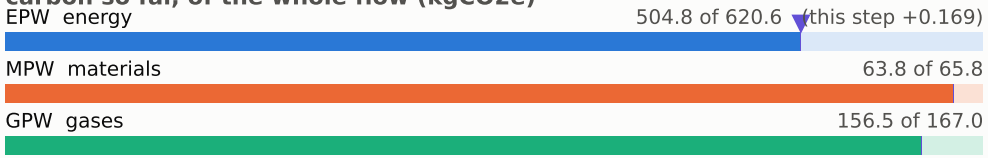


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

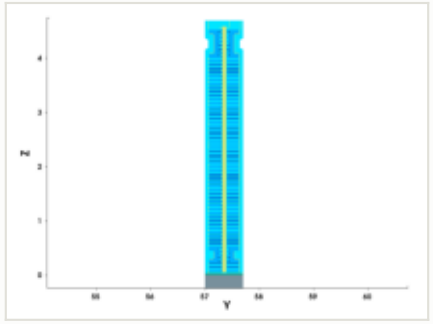
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

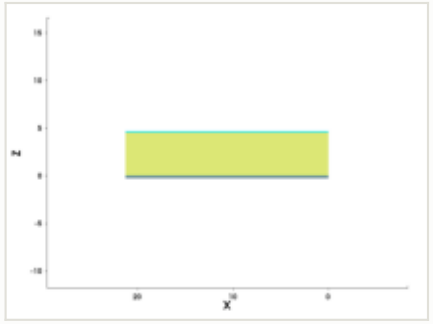


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



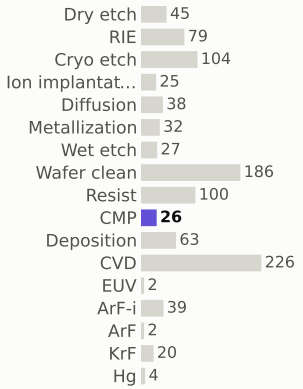
y cut at 57.351 µm



mask for this step

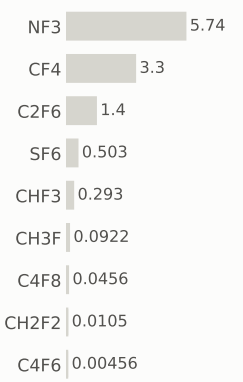
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



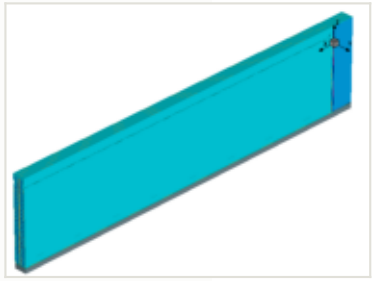
Step #11860: Wafer_clean

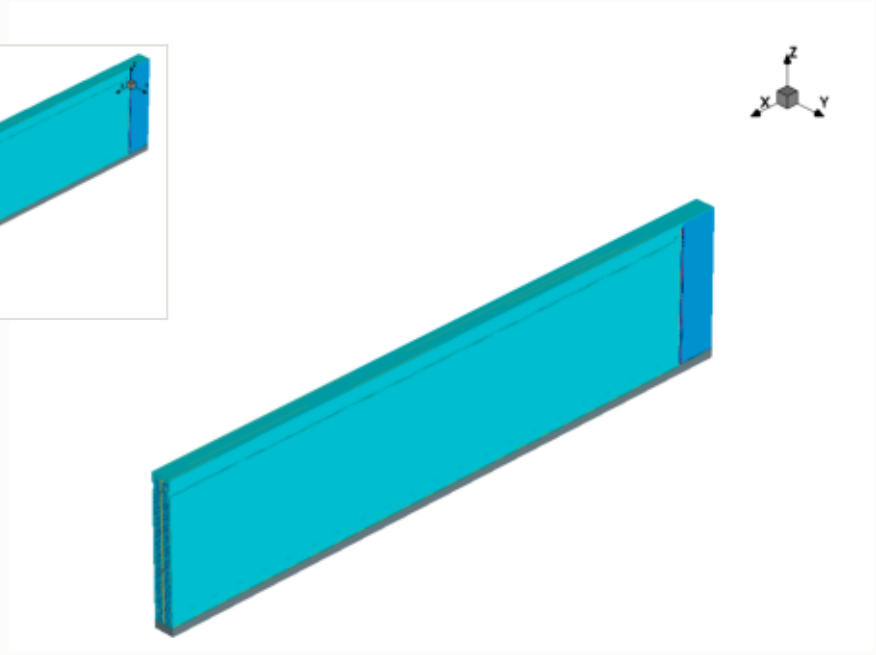
clean · Wafer_clean

0.169	725.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





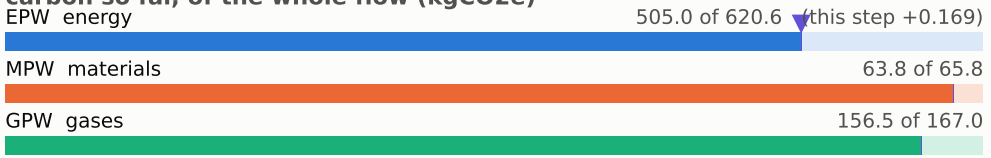
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

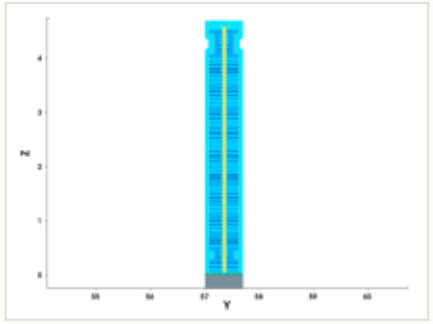
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

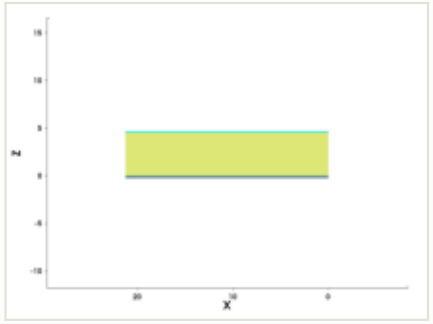


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



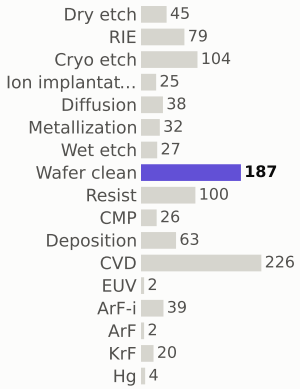
y cut at 57.351 µm



mask for this step

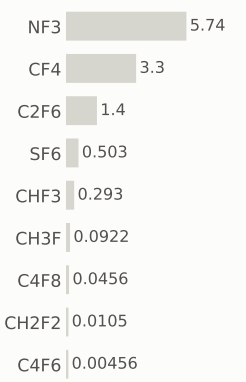
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



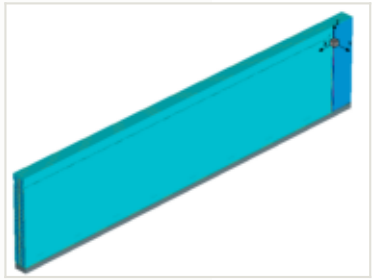
Step #11880: Diffusion

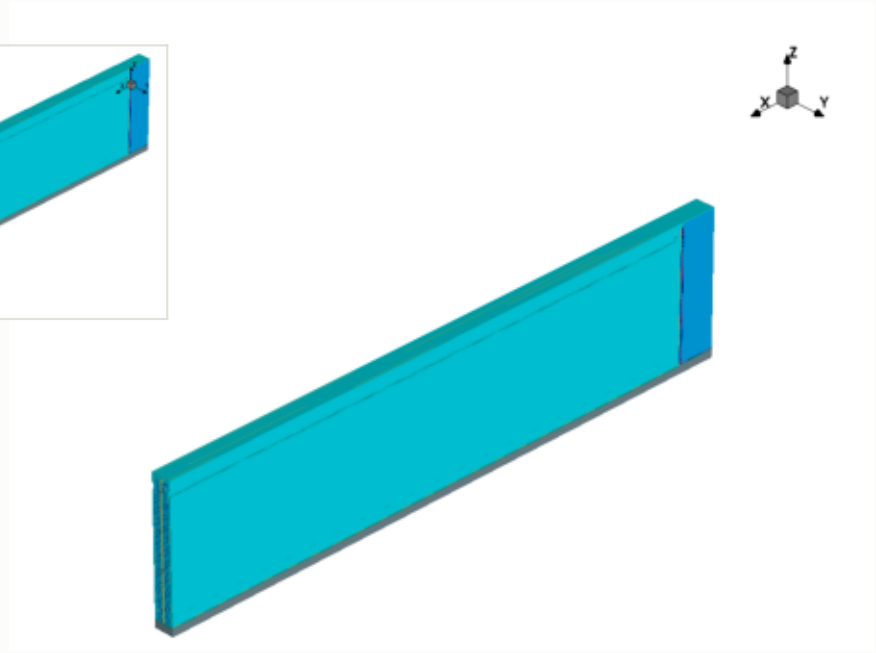
anneal · Diffusion

0.801	726.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





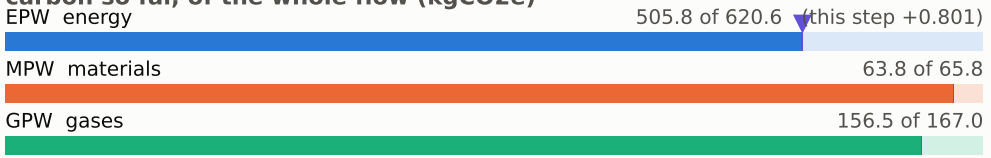
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

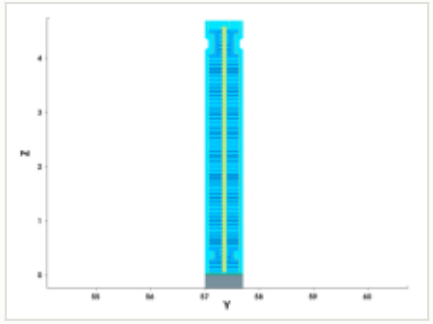
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

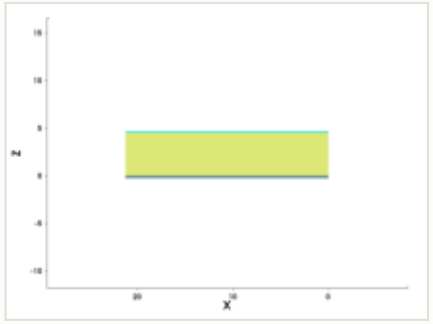


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



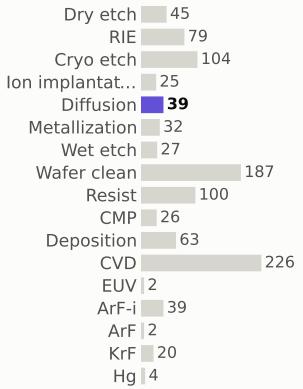
y cut at 57.351 µm



mask for this step

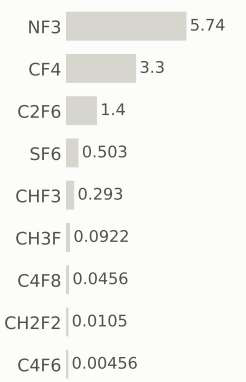
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



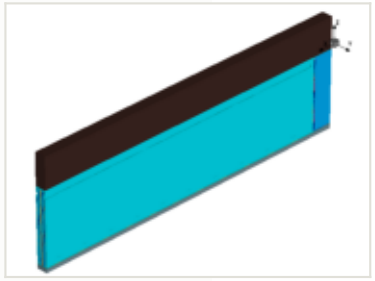
Step #11900: Deposition

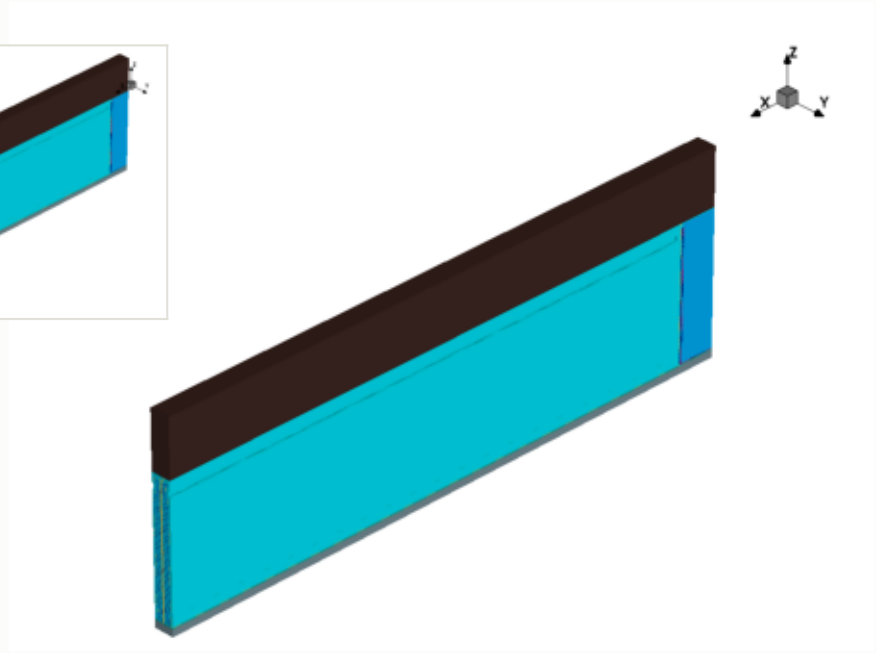
depo_isotropic · Deposition

0.656	726.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

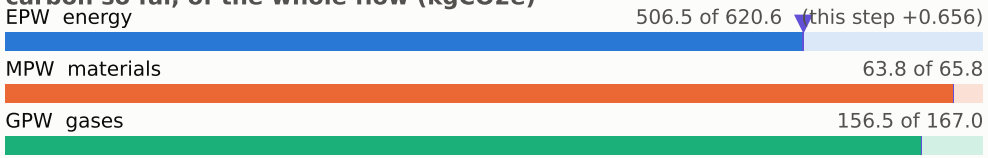




this step

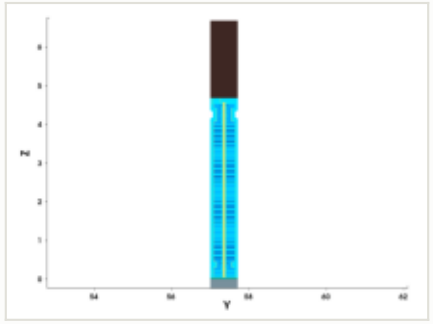
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	2 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

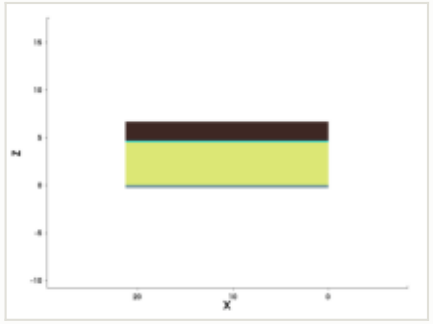


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



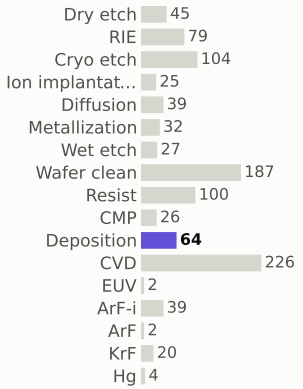
y cut at 57.351 µm



mask for this step

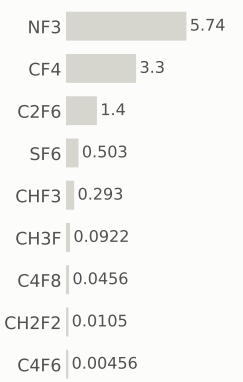
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



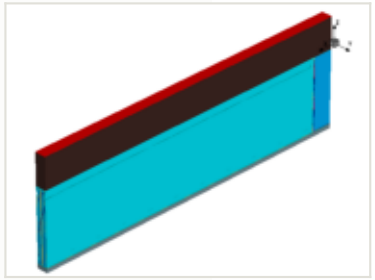
Step #11910: Deposition

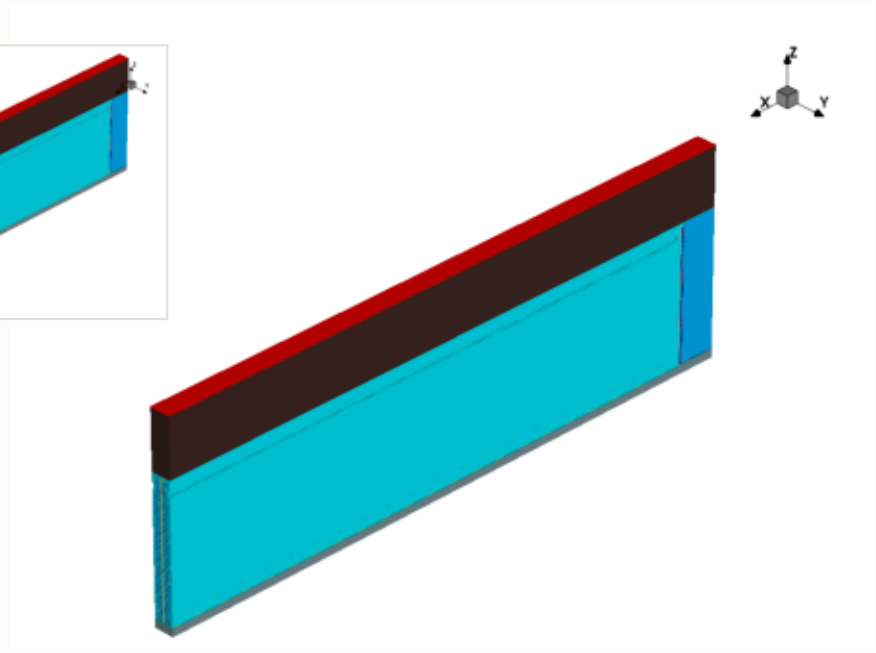
depo_isotropic · Deposition

0.656	727.4	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

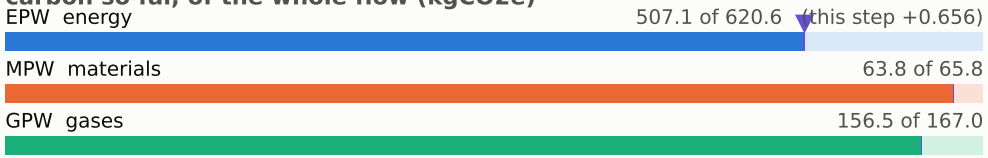




this step

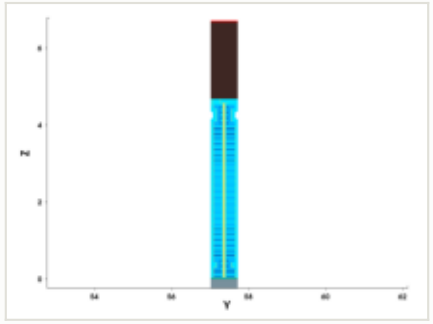
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.03 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

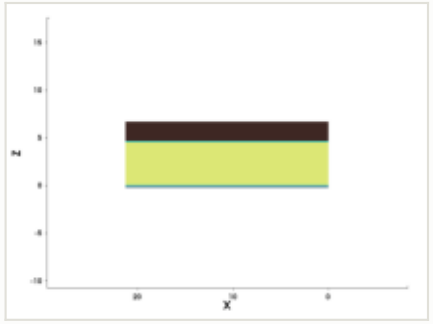


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



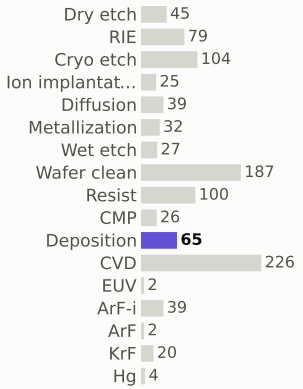
y cut at 57.351 µm



mask for this step

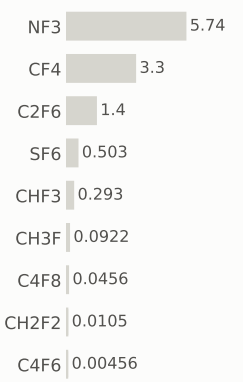
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



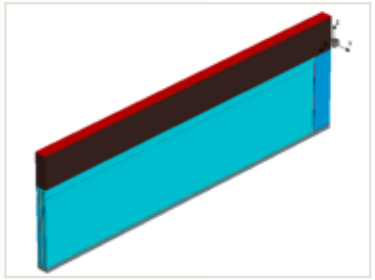
Step #11920: Wafer_clean

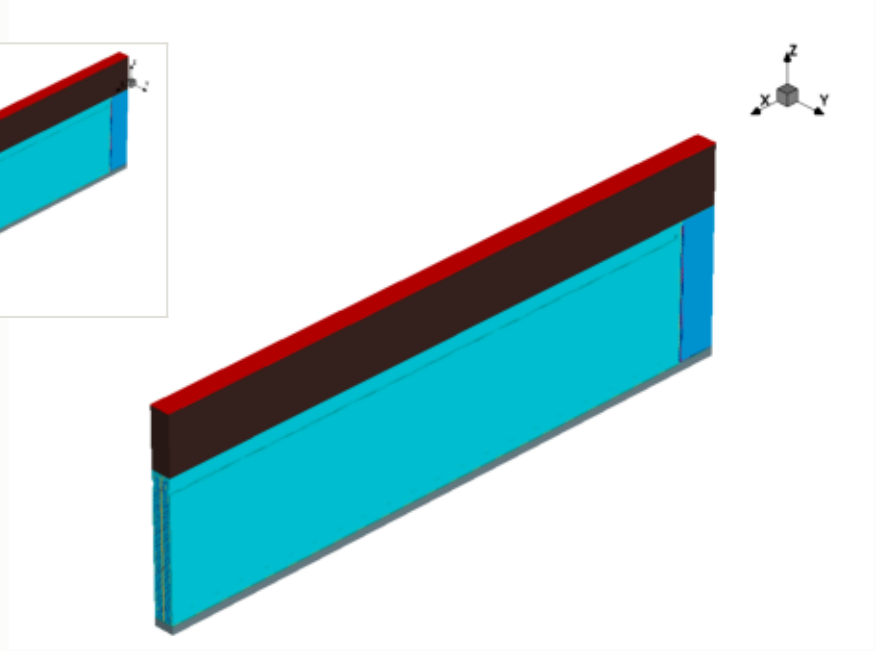
clean · Wafer_clean

0.169	727.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



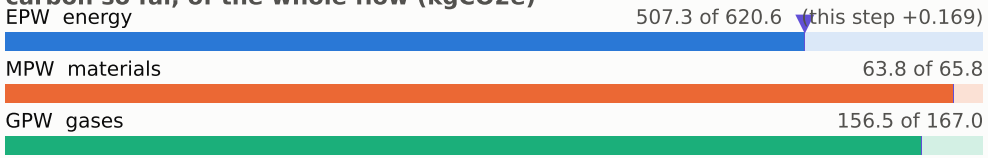


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

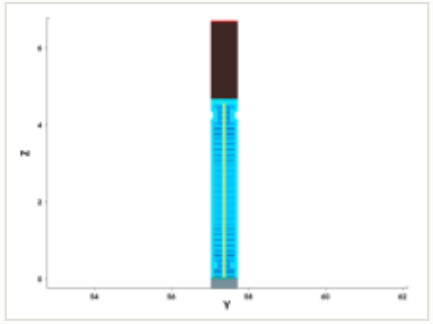
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

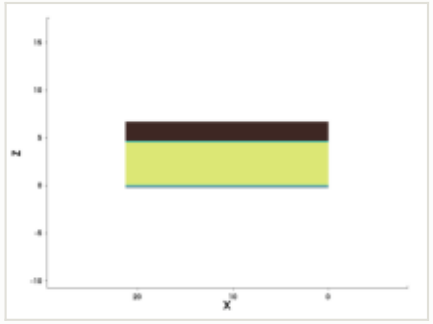


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



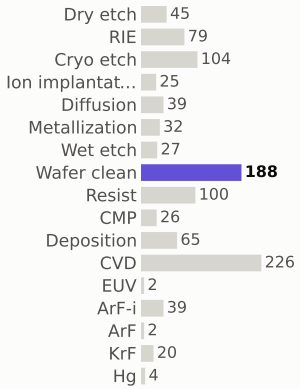
y cut at 57.351 µm



mask for this step

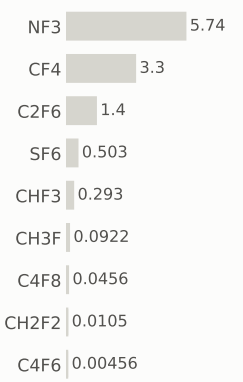
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



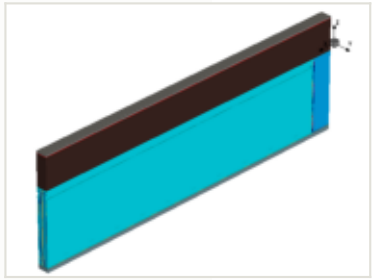
Step #11930: ArF-i

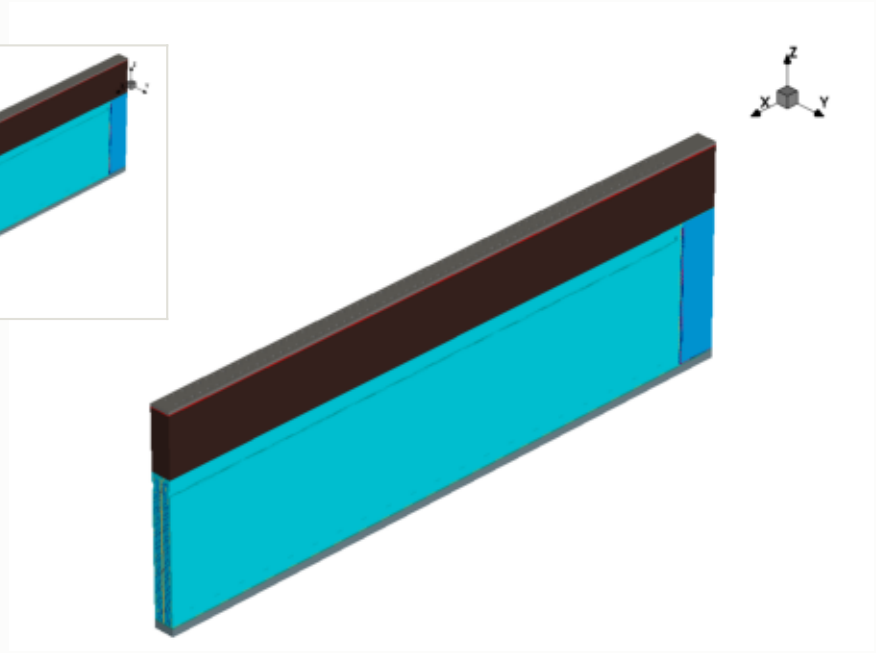
litho_patterning · ArF-i

0.217	727.8	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



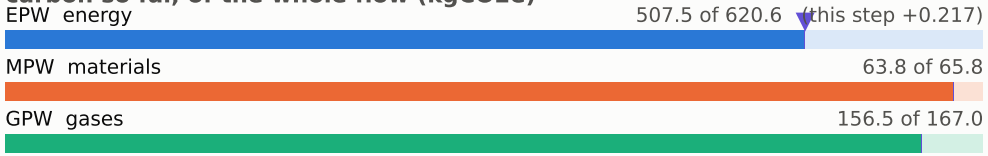


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

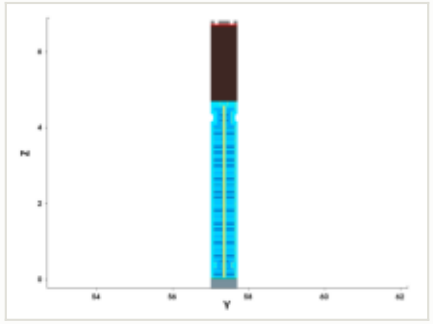
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

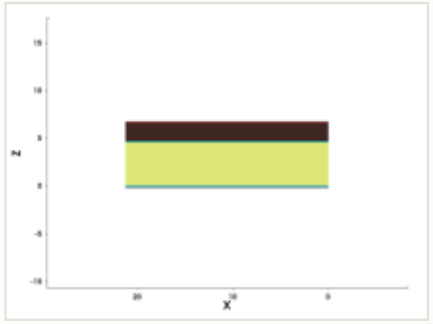


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

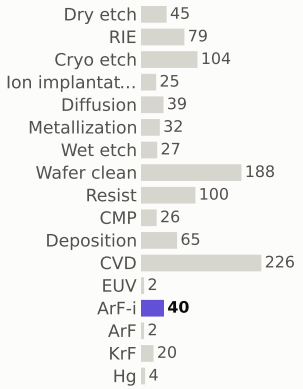


y cut at 57.351 µm



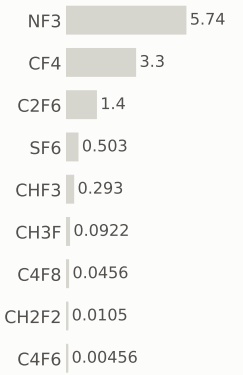
mask: layer 7:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



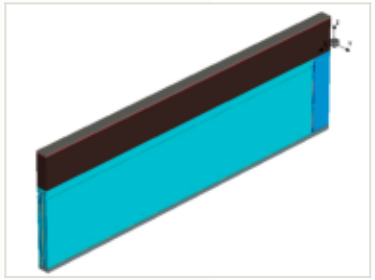
Step #11940: RIE

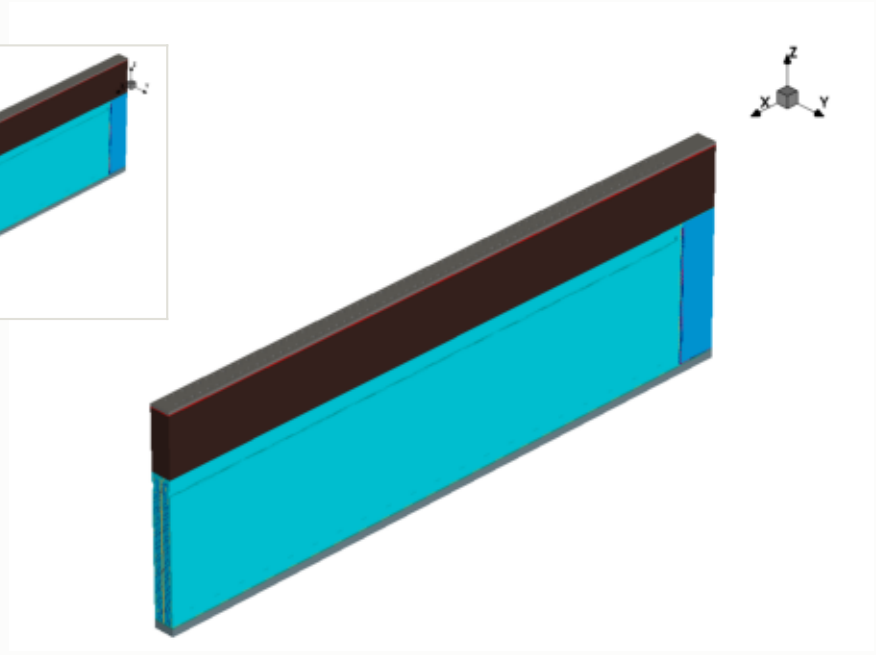
etch_anisotropic · RIE

1.04	728.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



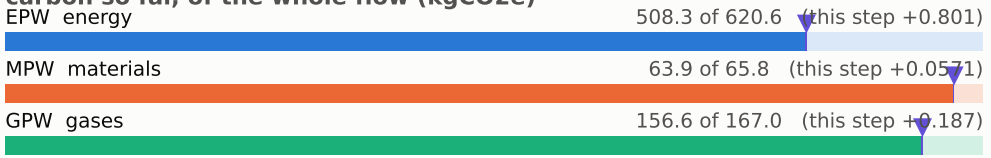


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

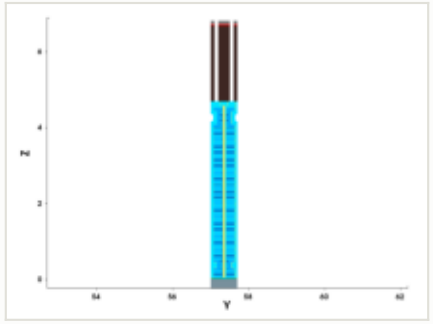
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

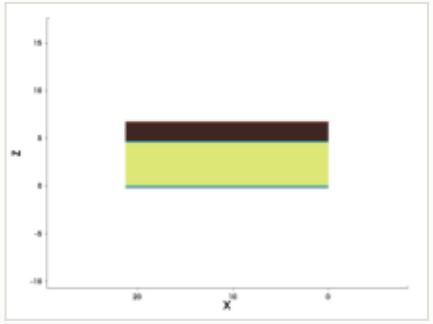


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



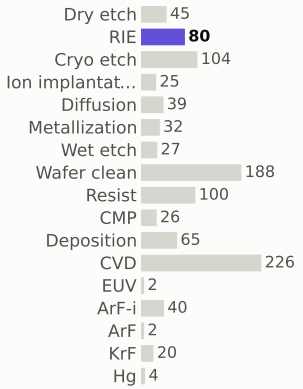
y cut at 57.351 µm



mask for this step

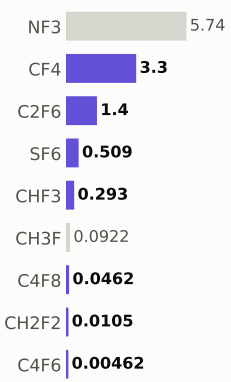
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



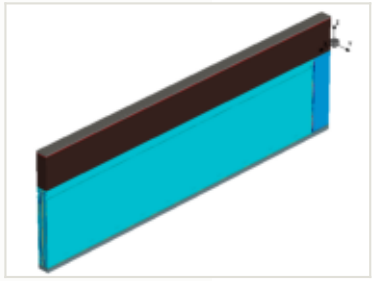
Step #11950: RIE

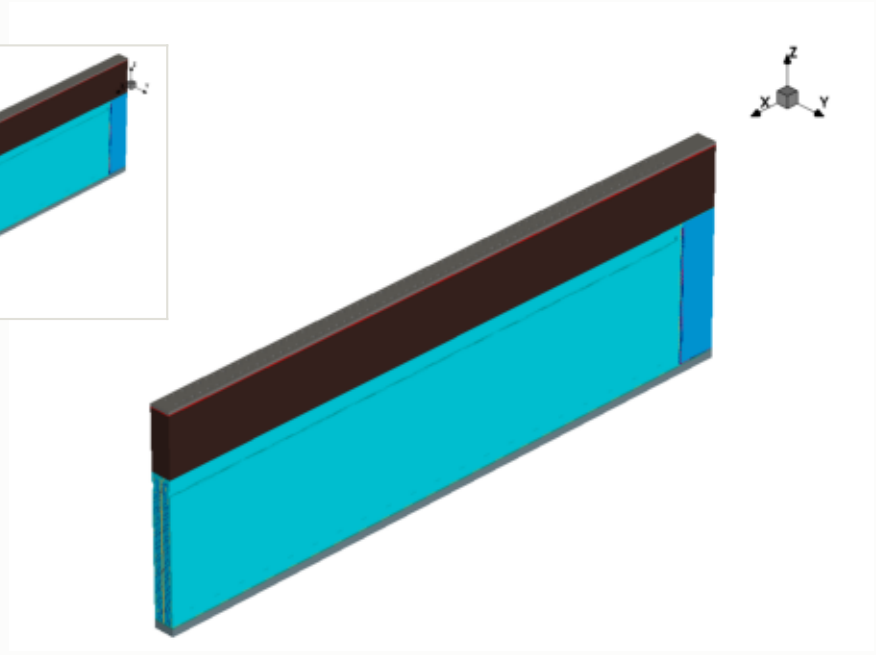
etch_anisotropic · RIE

1.04	729.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



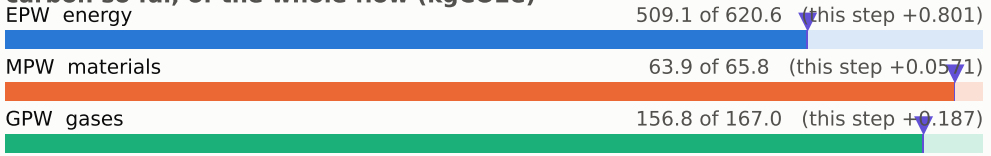


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

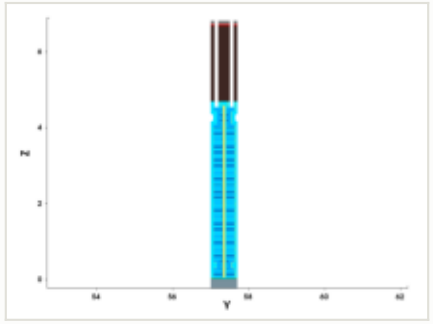
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

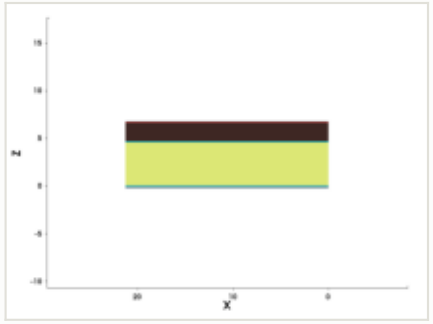


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



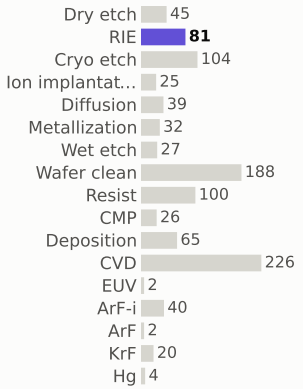
y cut at 57.351 µm



mask for this step

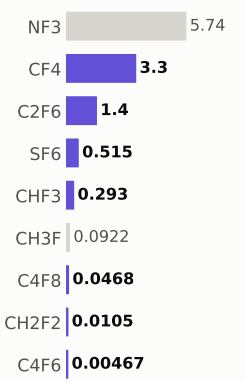
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



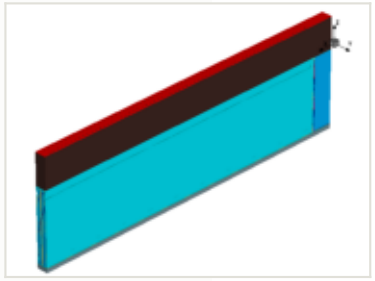
Step #11960: Wafer_clean

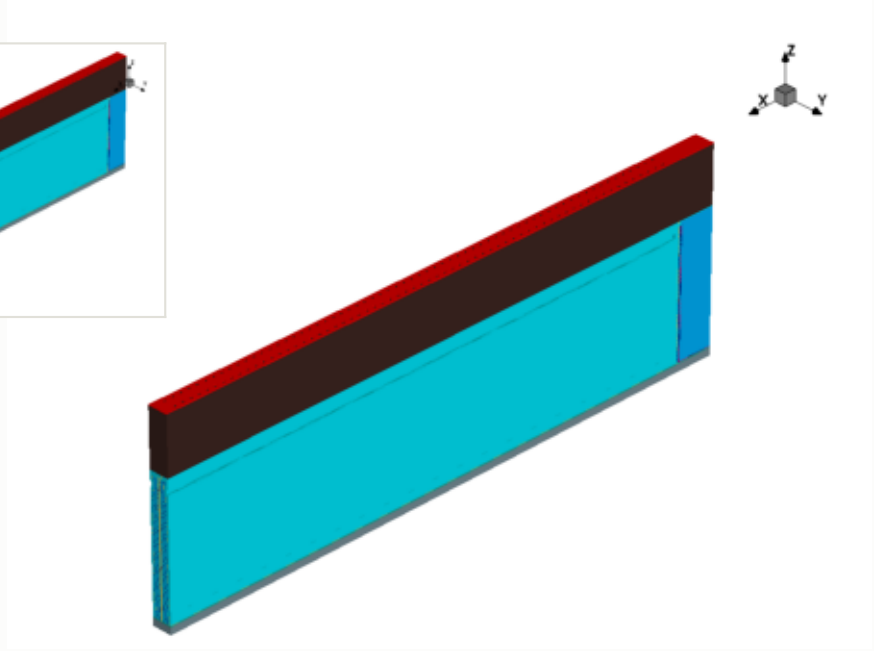
etch_strip · Wafer_clean

0.169	730.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

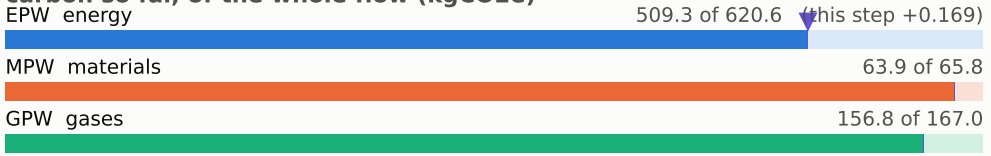




this step

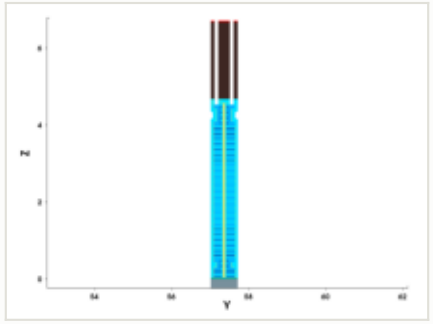
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

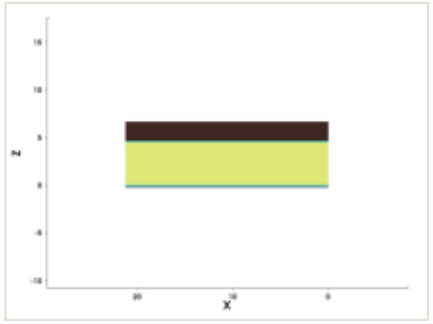


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



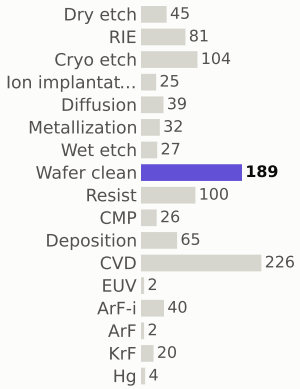
y cut at 57.351 µm



mask for this step

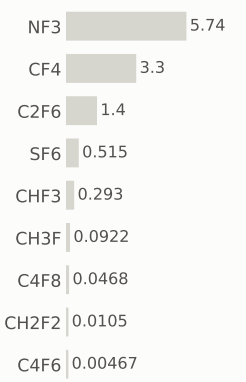
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



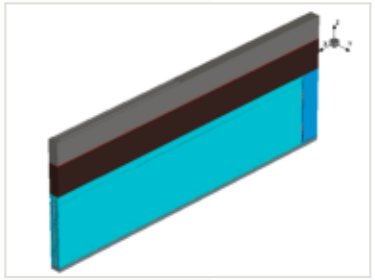
Step #11970: KrF

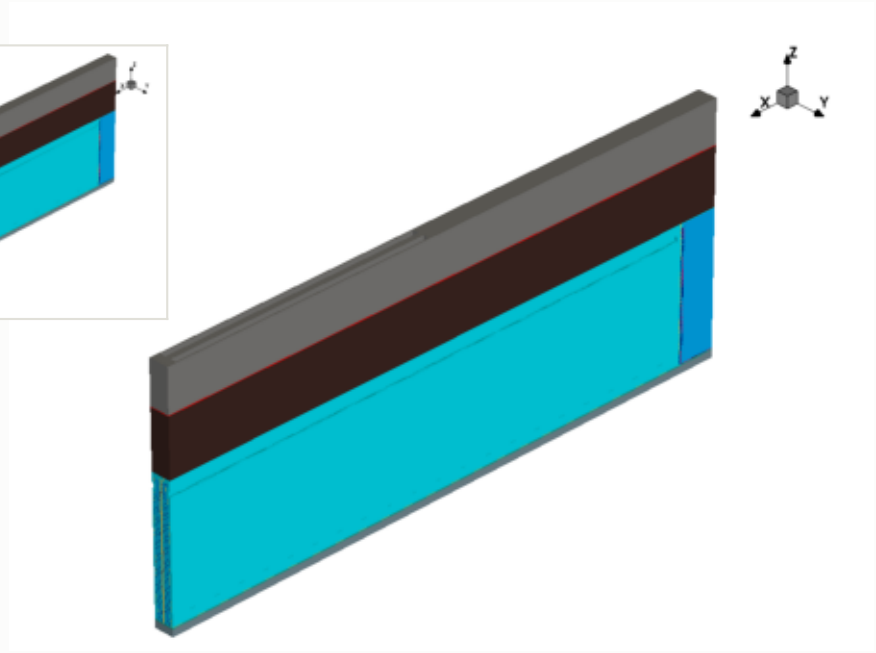
litho_patterning · KrF

0.199	730.2	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

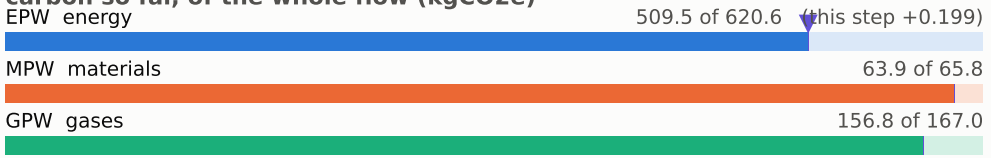




this step

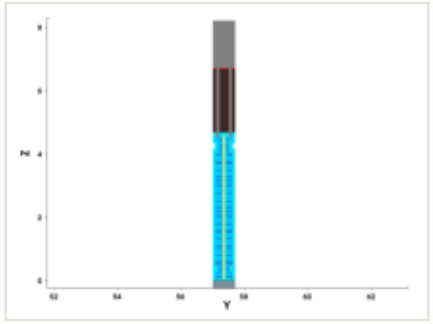
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

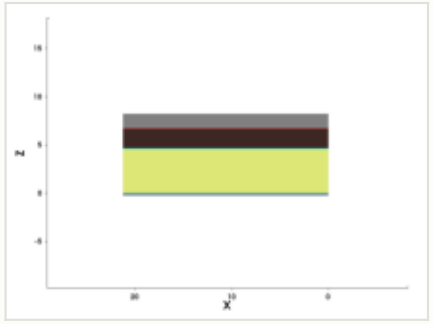


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

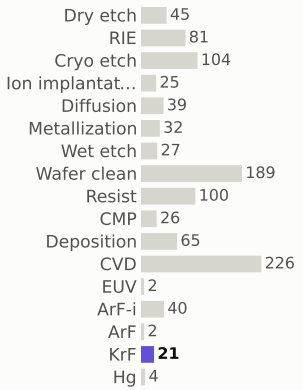


y cut at 57.351 µm



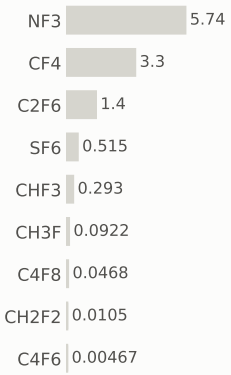
mask: layer 8:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



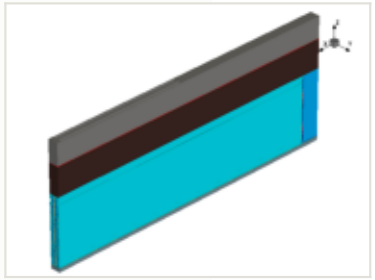
Step #11980: RIE

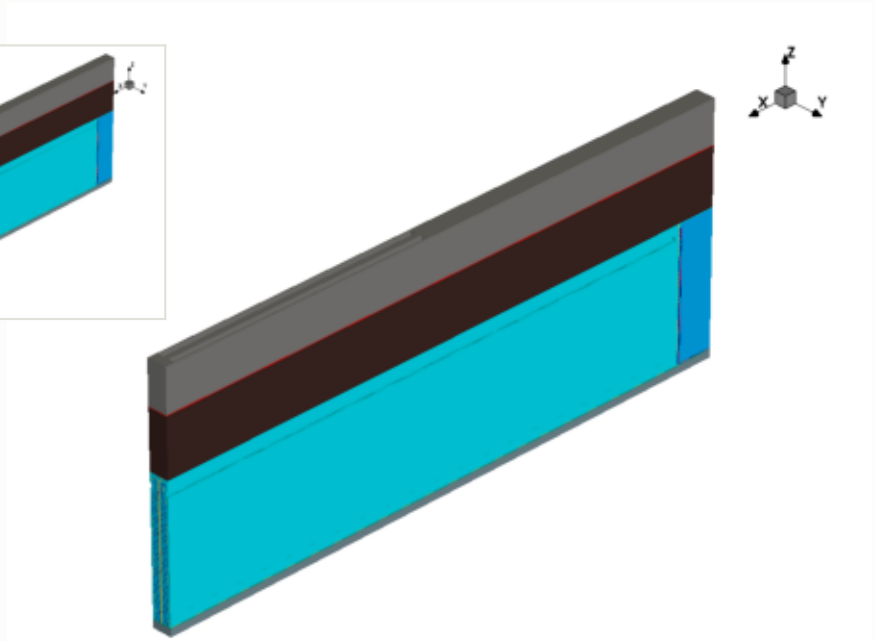
etch_anisotropic · RIE

1.04	731.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



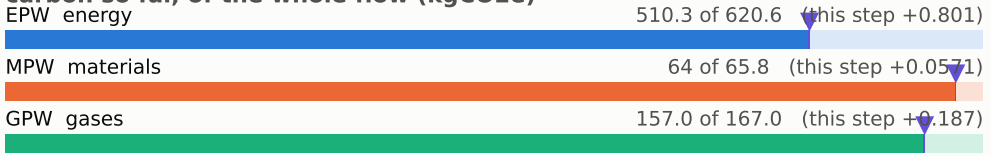


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

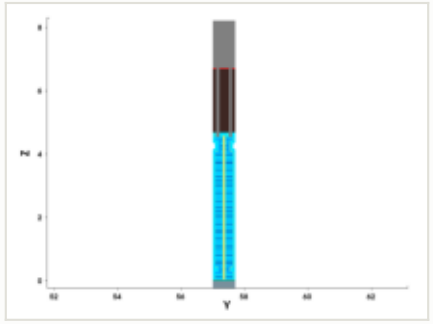
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

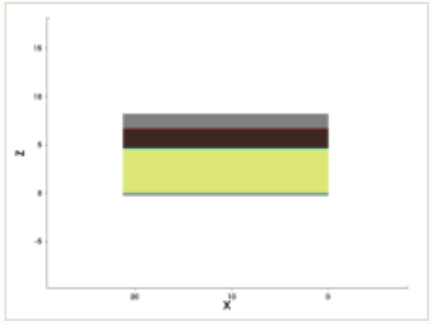


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



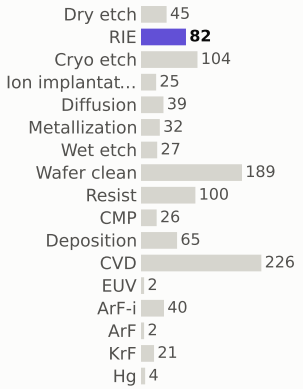
y cut at 57.351 µm



mask for this step

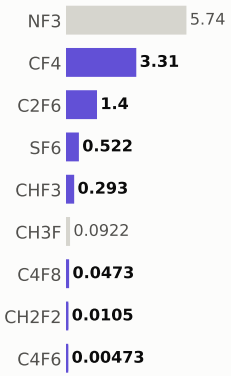
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



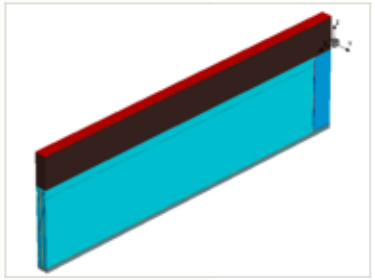
Step #11990: Dry_etch

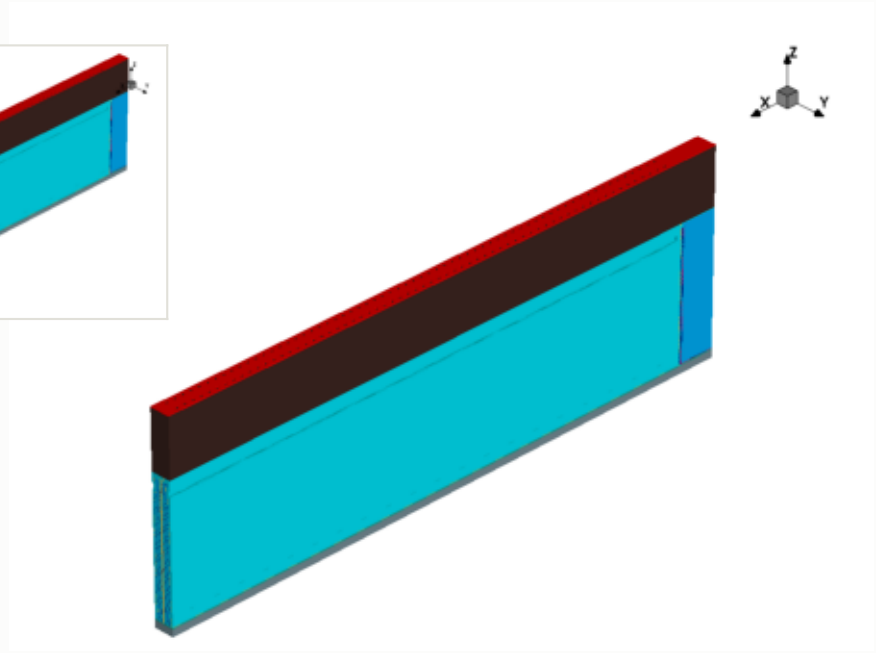
etch_strip · Dry_etch

0.801	732.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

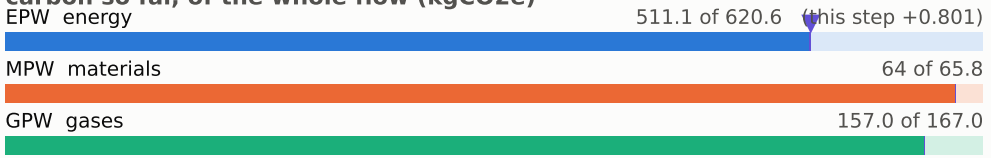




this step

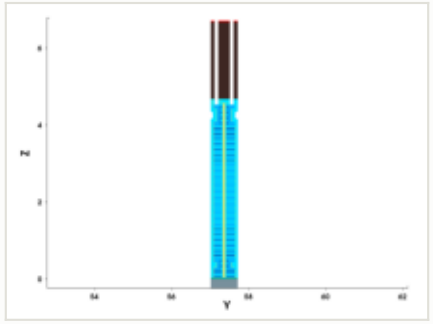
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

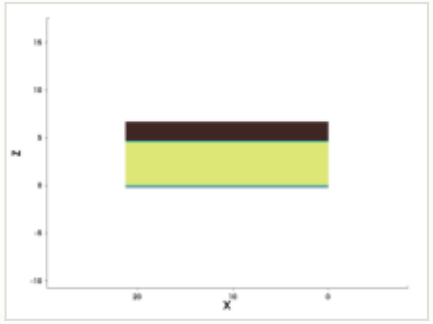


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



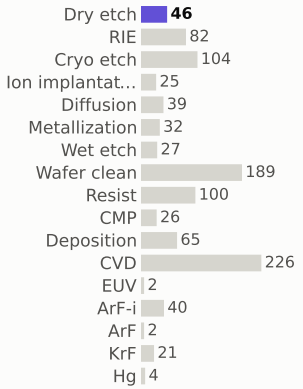
y cut at 57.351 µm



mask for this step

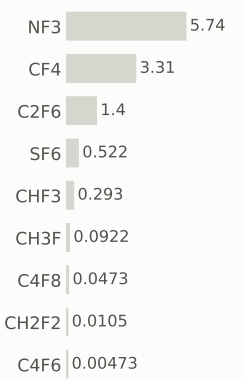
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



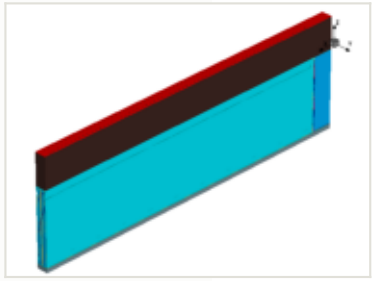
Step #12000: Wafer_clean

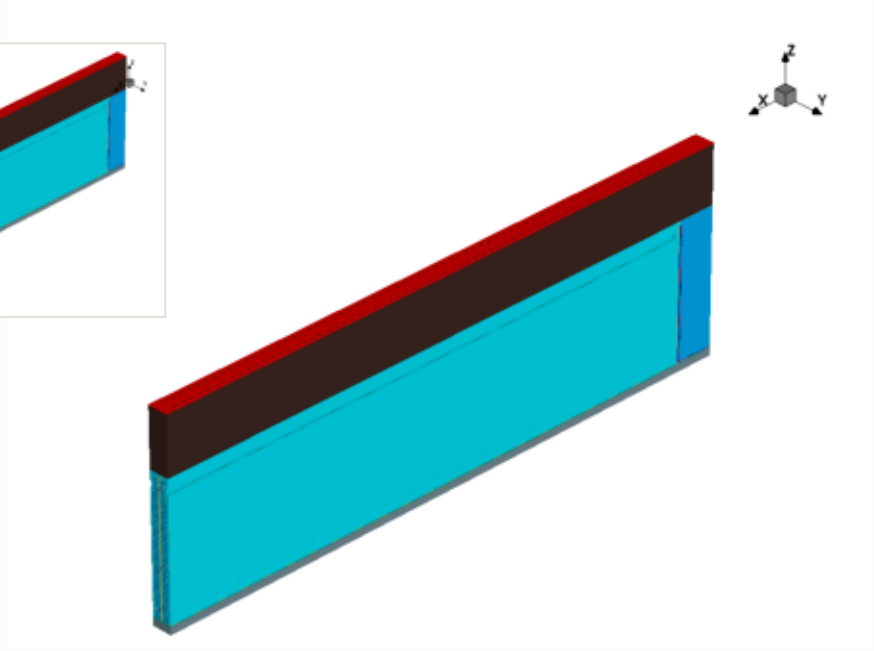
clean · Wafer_clean

0.169	732.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





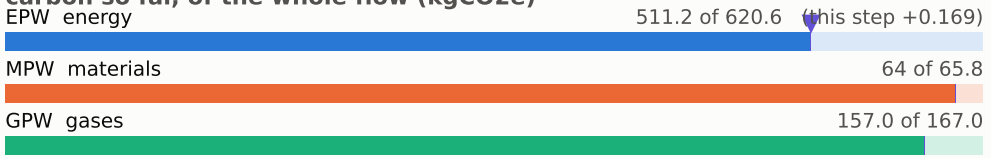
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

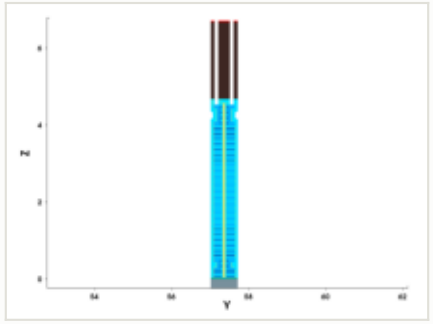
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

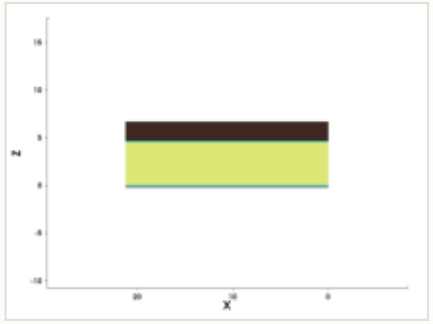


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



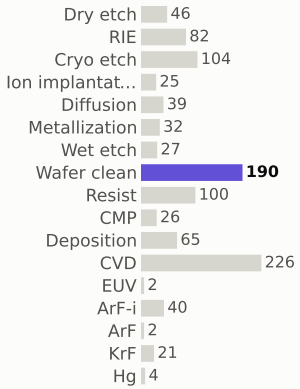
y cut at 57.351 µm



mask for this step

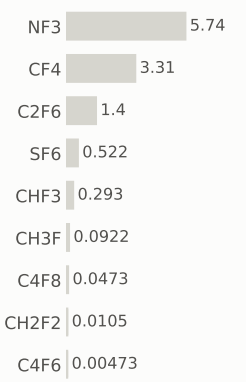
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



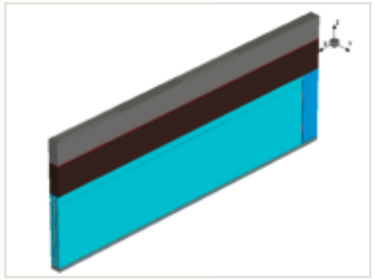
Step #12010: KrF

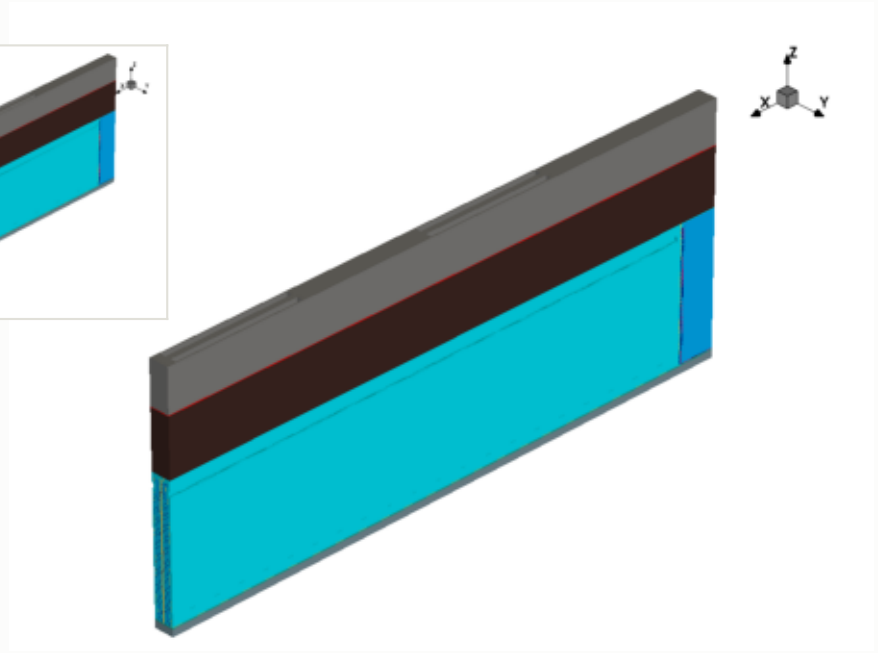
litho_patterning · KrF

0.199	732.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

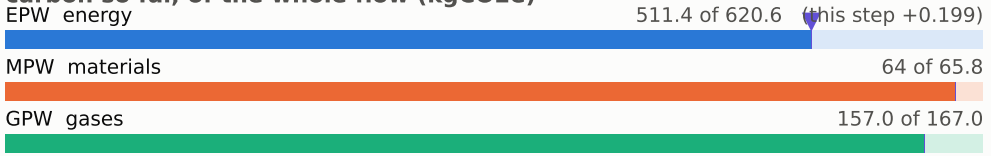




this step

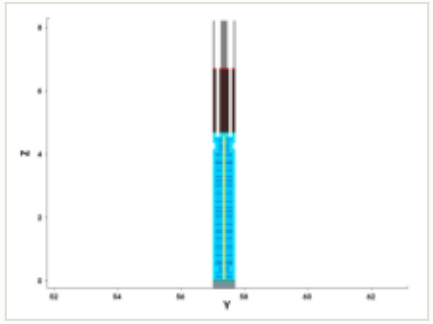
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

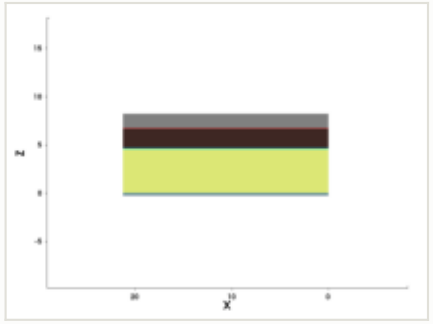


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

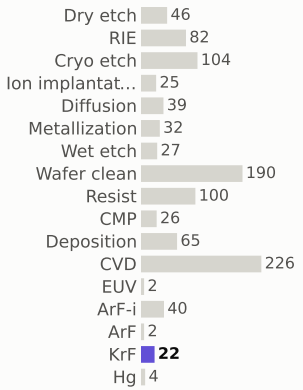


y cut at 57.351 µm



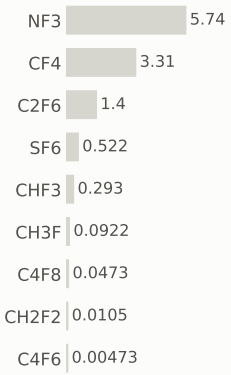
mask: layer 9:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



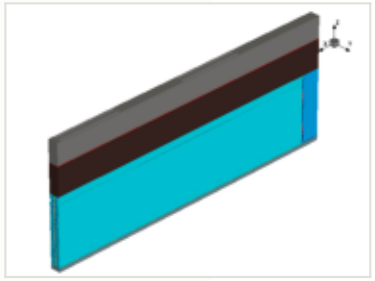
Step #12020: RIE

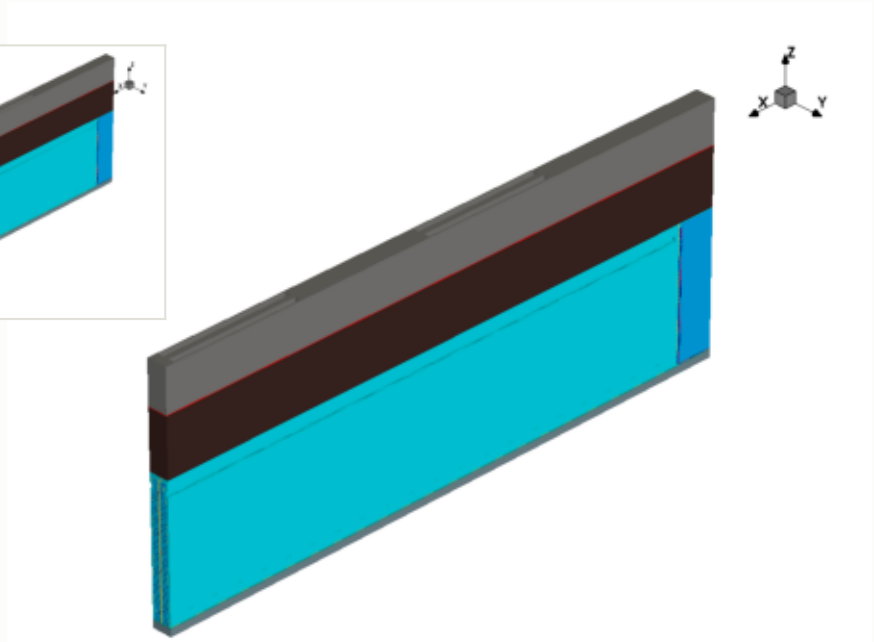
etch_anisotropic · RIE

1.04	733.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



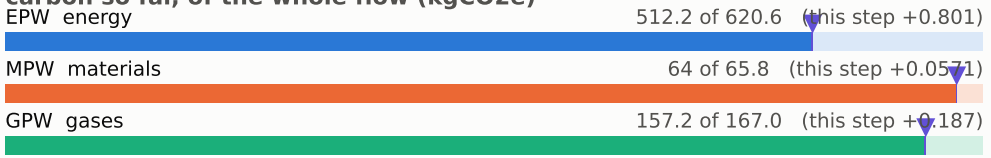


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

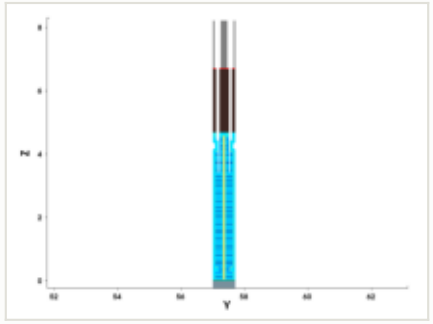
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

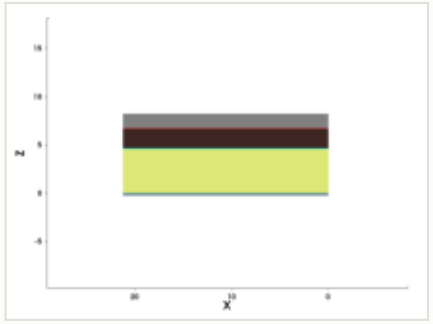


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



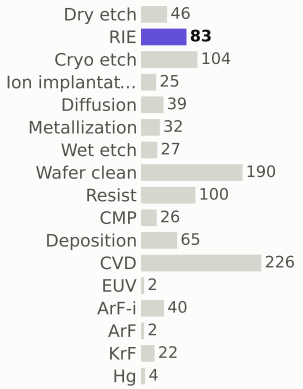
y cut at 57.351 µm



mask for this step

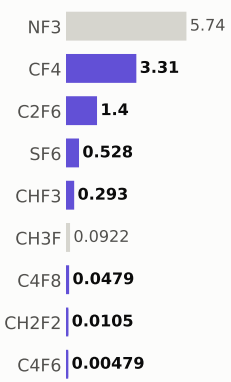
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



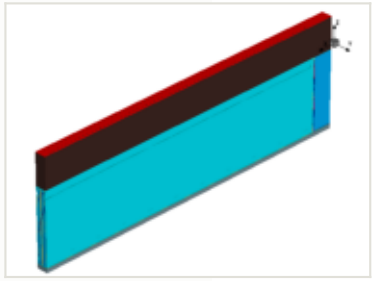
Step #12030: Dry_etch

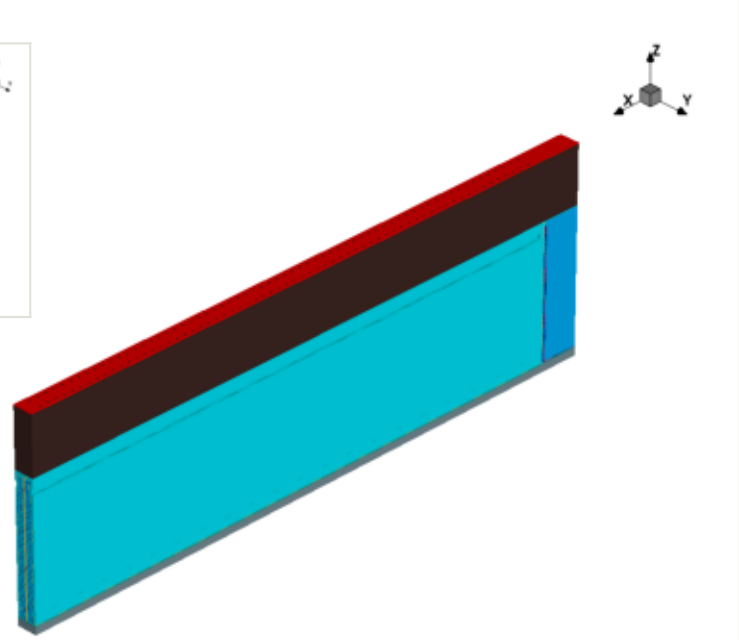
etch_strip · Dry_etch

0.801	734.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

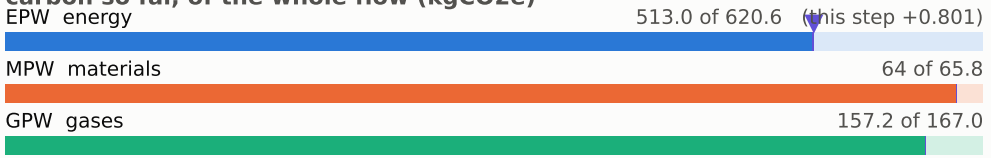




this step

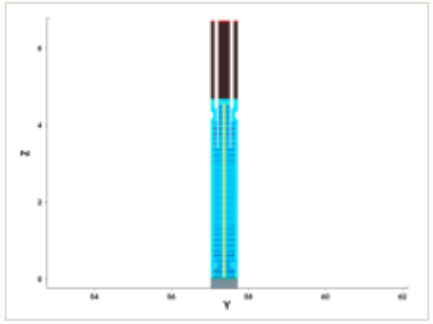
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

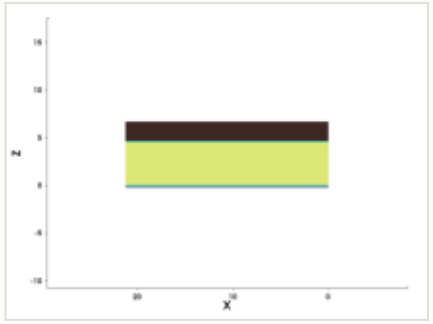


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



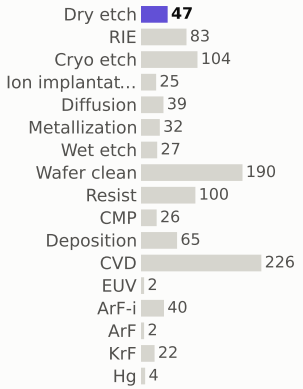
y cut at 57.351 µm



mask for this step

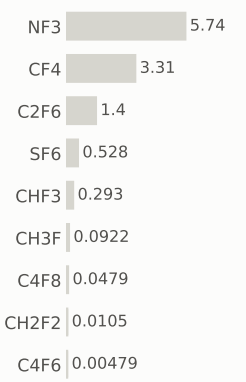
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



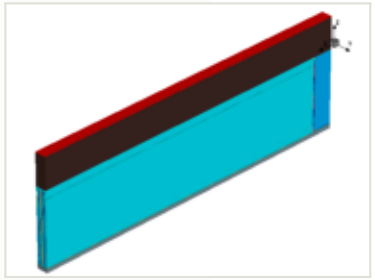
Step #12040: Wafer_clean

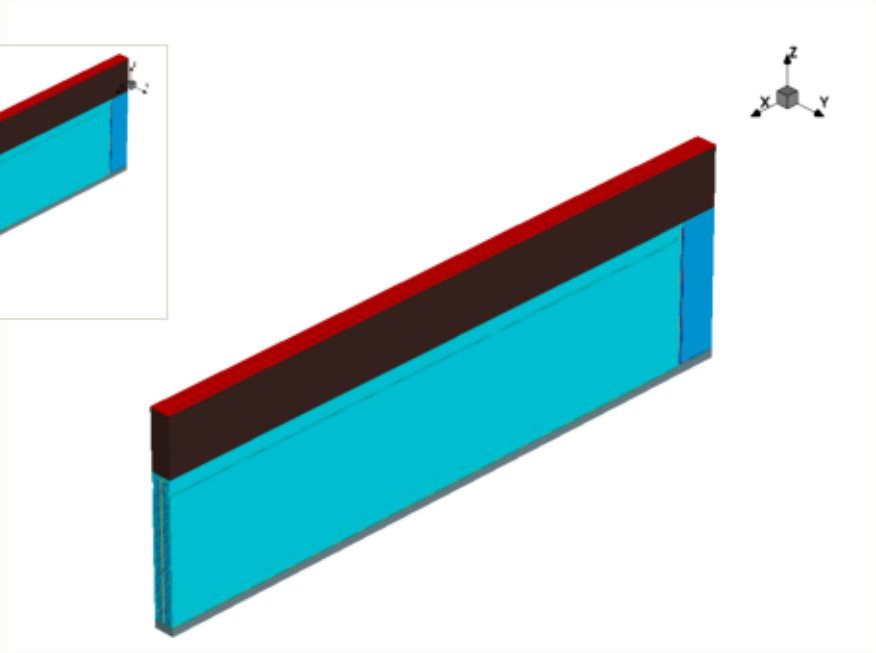
clean · Wafer_clean

0.169	734.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



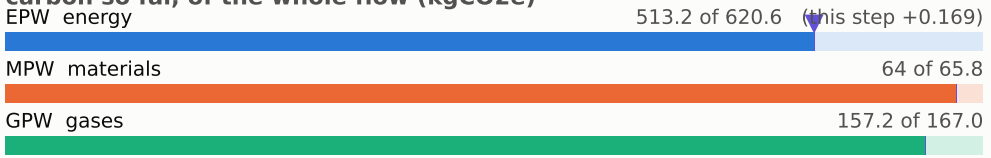


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

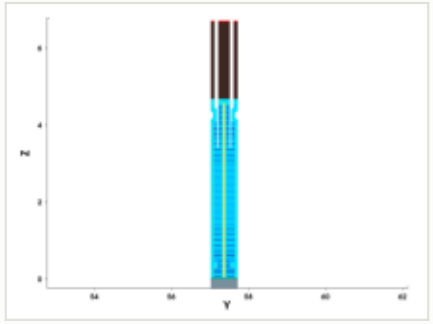
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

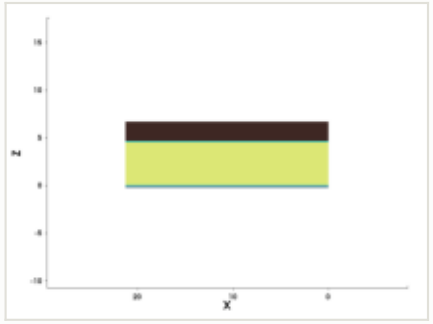


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



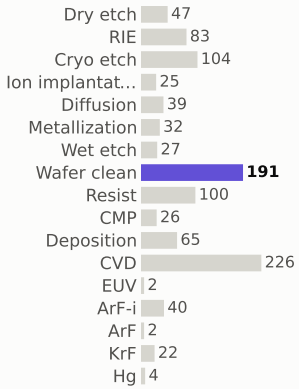
y cut at 57.351 µm



mask for this step

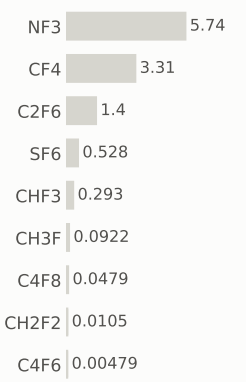
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



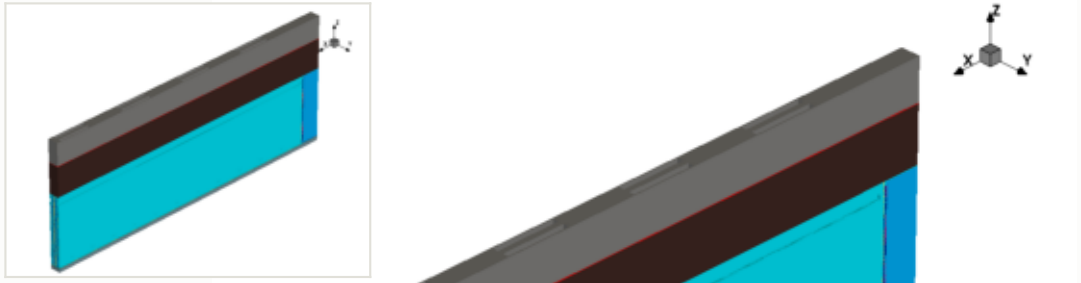
Step #12050: KrF

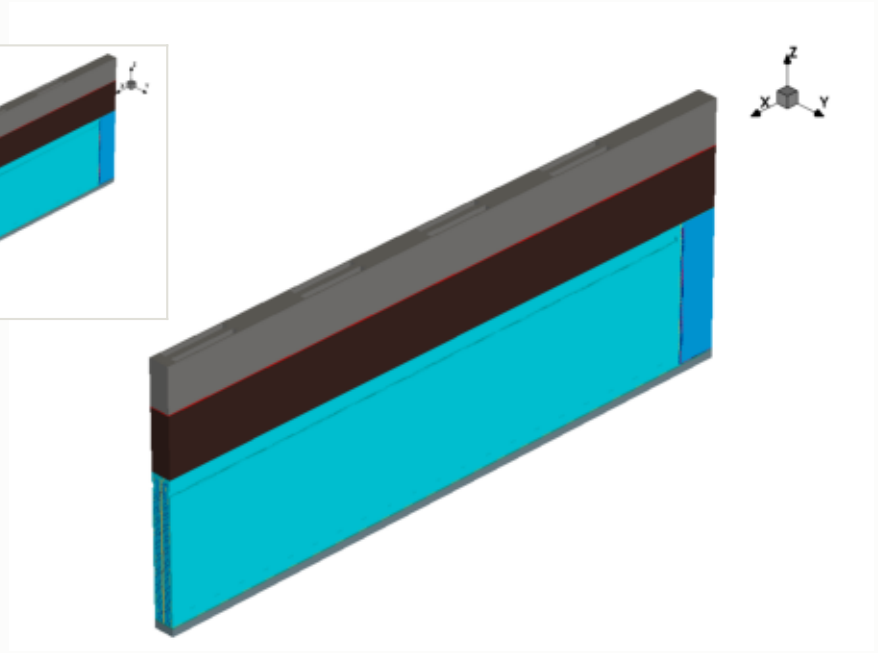
litho_patterning · KrF

0.199	734.6	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

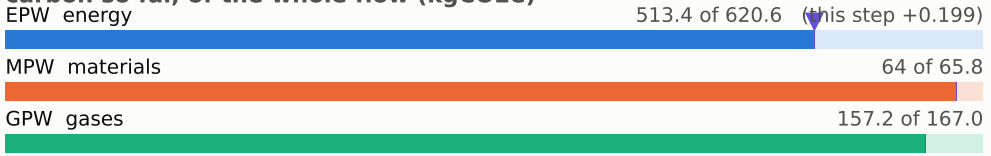




this step

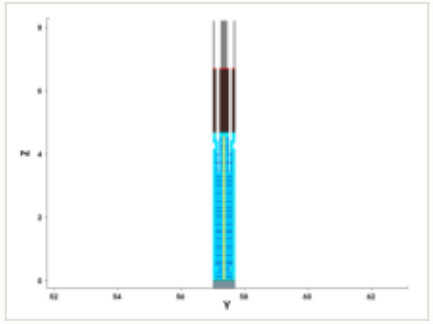
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

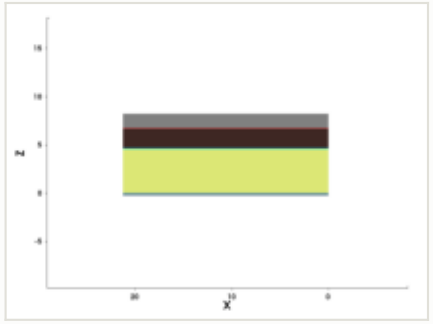


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

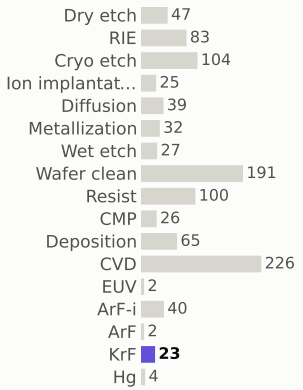


y cut at 57.351 µm



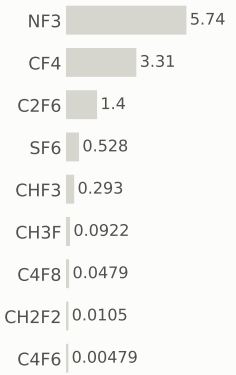
mask: layer 10:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



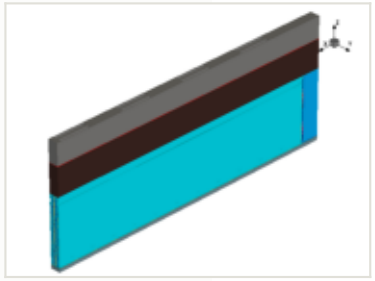
Step #12060: RIE

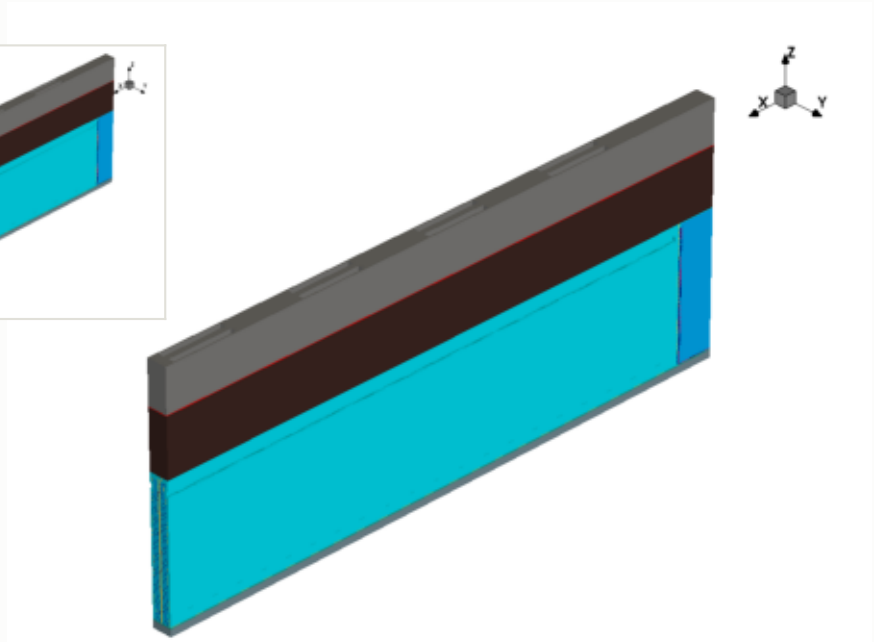
etch_anisotropic · RIE

1.04	735.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



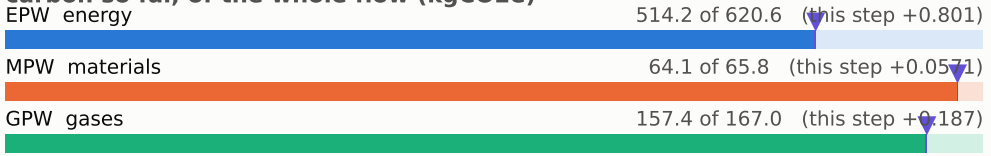


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

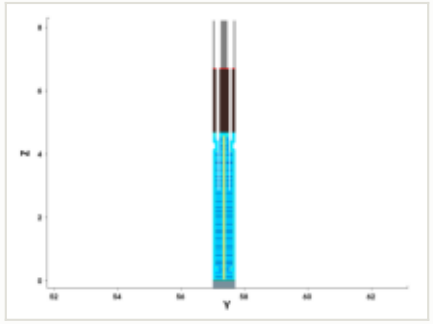
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

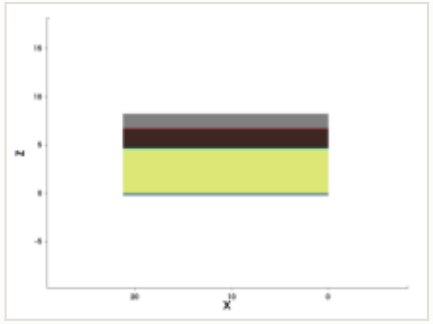


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



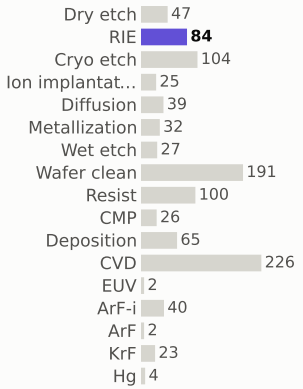
y cut at 57.351 µm



mask for this step

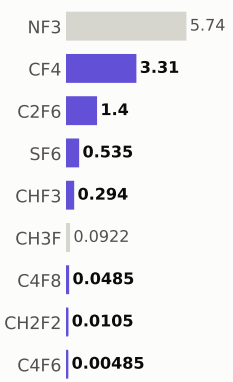
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



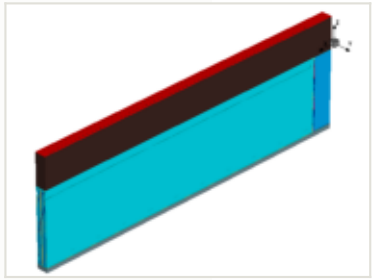
Step #12070: Dry_etch

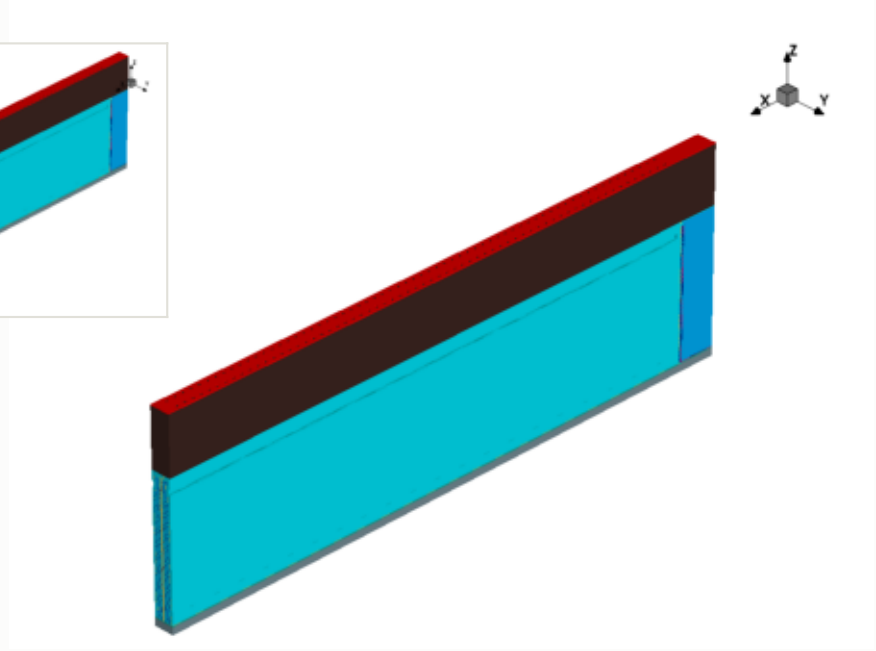
etch_strip · Dry_etch

0.801	736.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

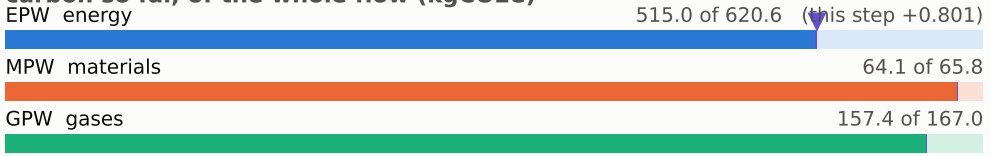




this step

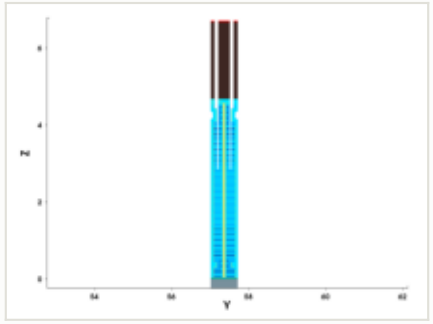
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

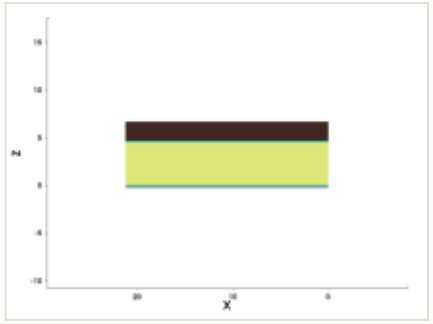


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



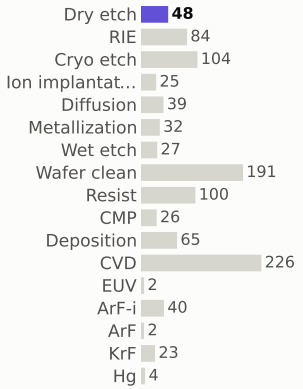
y cut at 57.351 µm



mask for this step

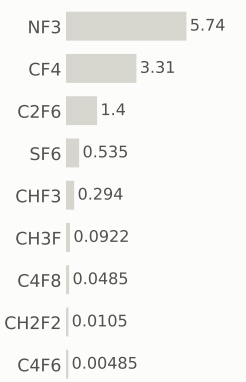
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



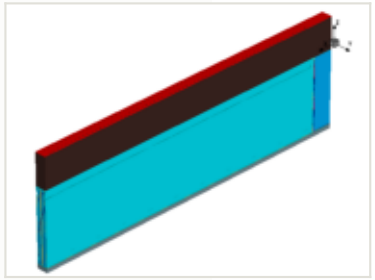
Step #12080: Wafer_clean

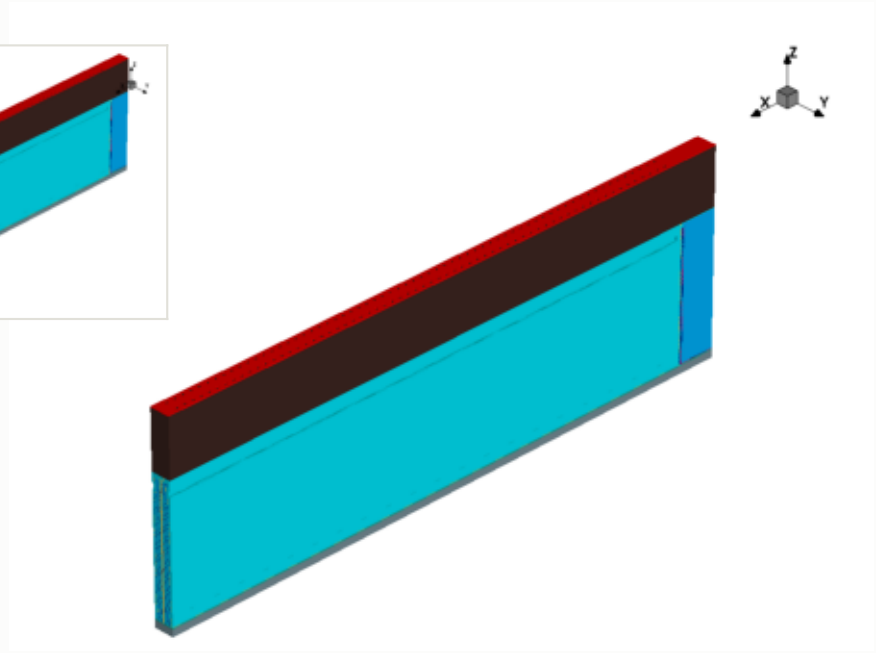
clean · Wafer_clean

0.169	736.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



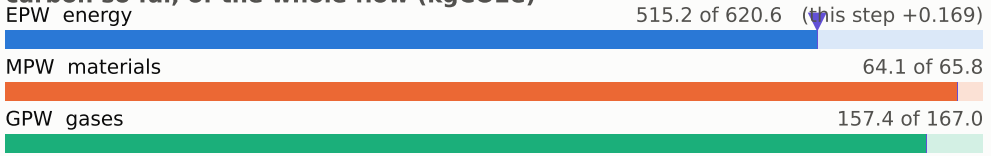


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

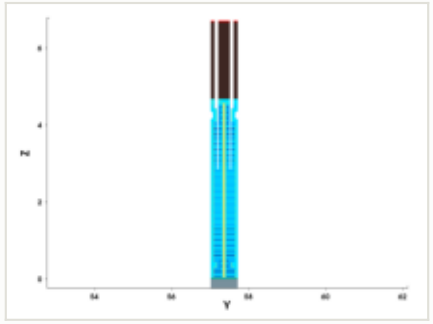
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

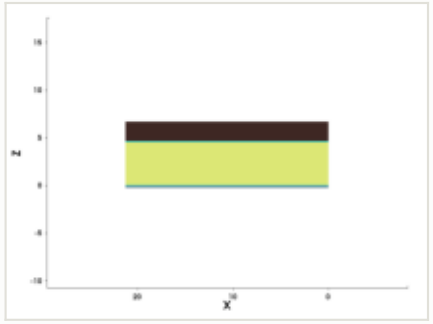


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



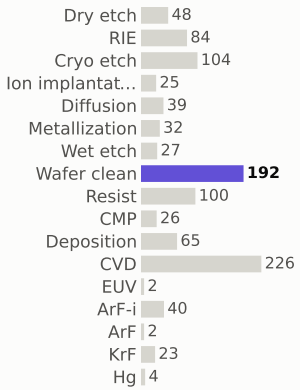
y cut at 57.351 µm



mask for this step

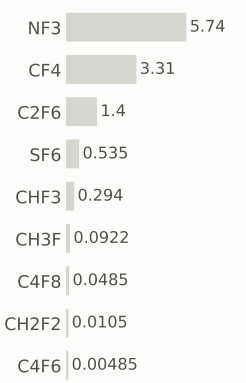
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



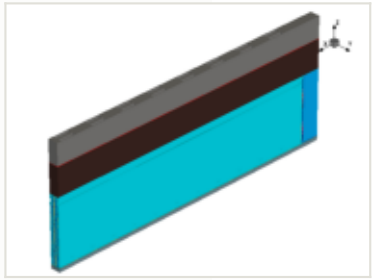
Step #12090: KrF

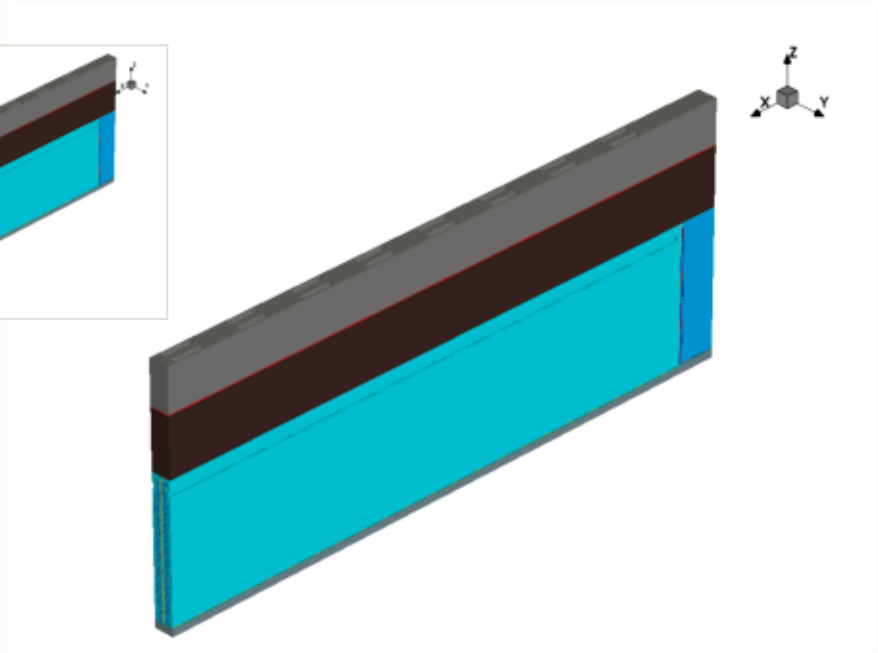
litho_patterning · KrF

0.199	736.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

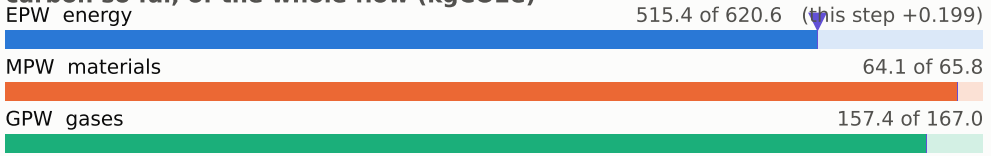




this step

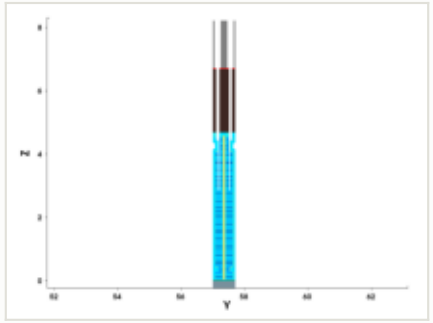
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

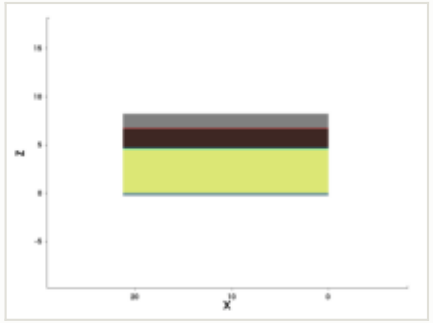


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

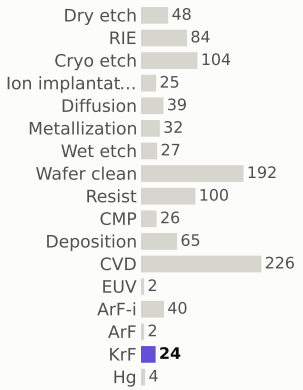


y cut at 57.351 µm



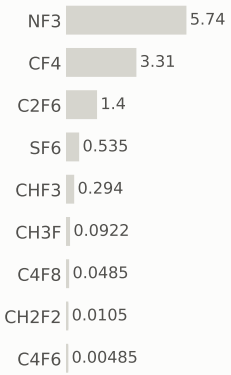
mask: layer 11:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



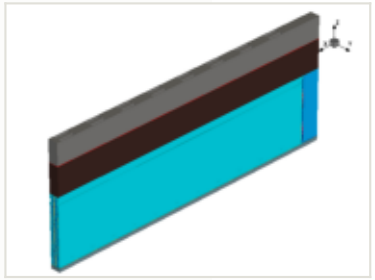
Step #12100: RIE

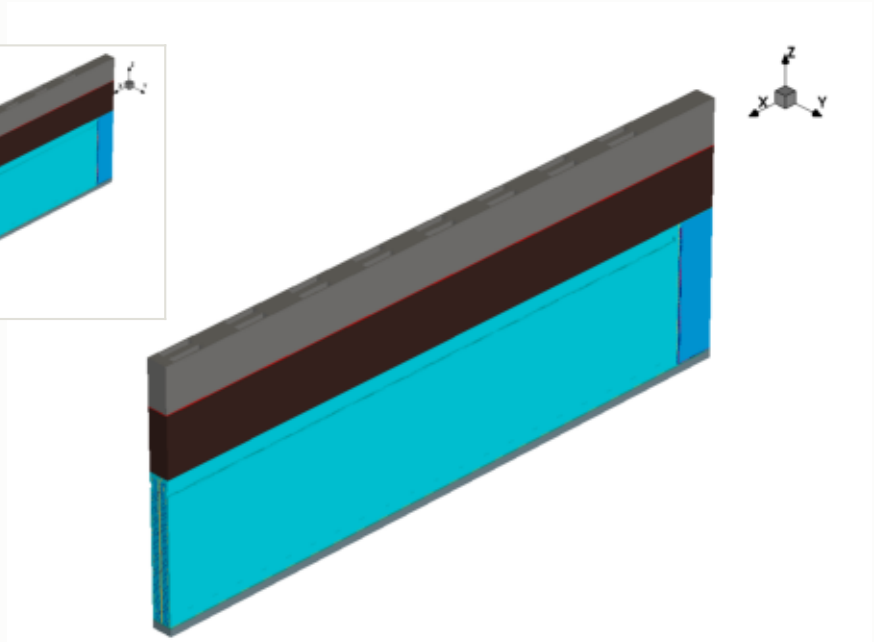
etch_anisotropic · RIE

1.04	737.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



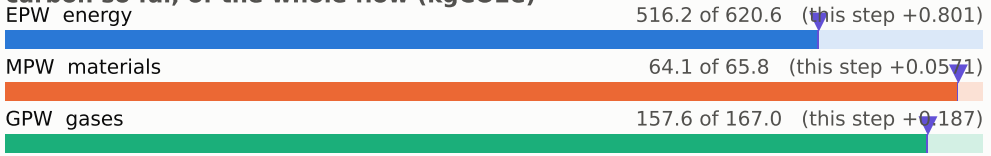


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

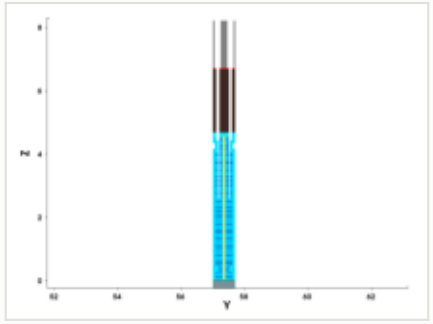
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

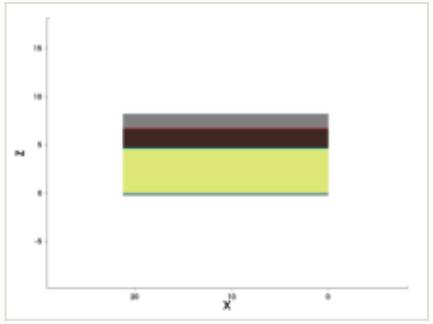


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



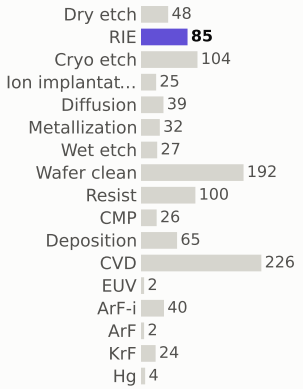
y cut at 57.351 µm



mask for this step

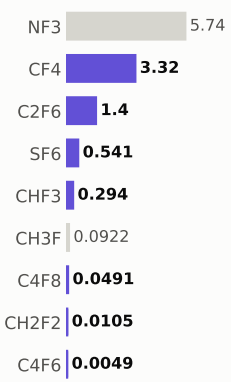
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



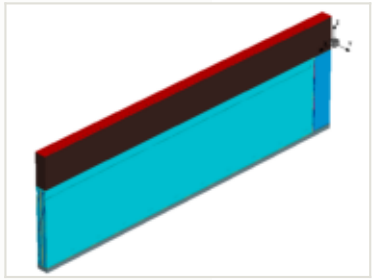
Step #12110: Dry_etch

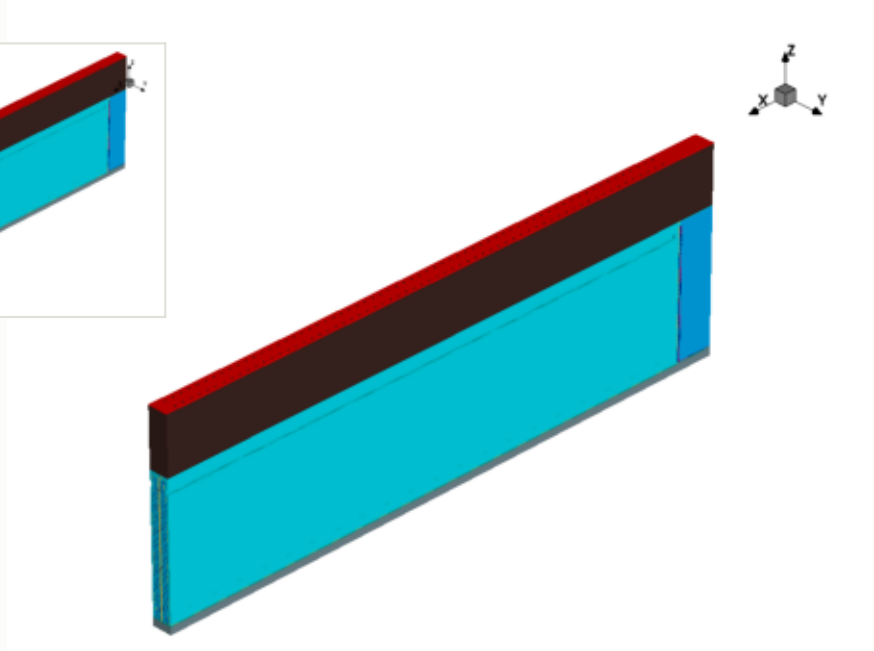
etch_strip · Dry_etch

0.801	738.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



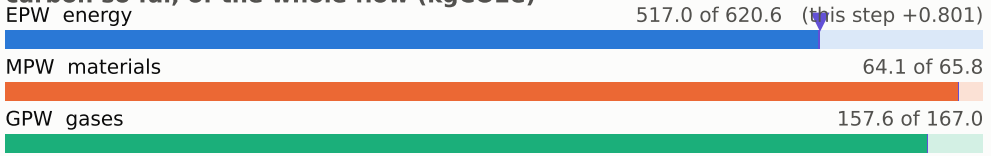


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

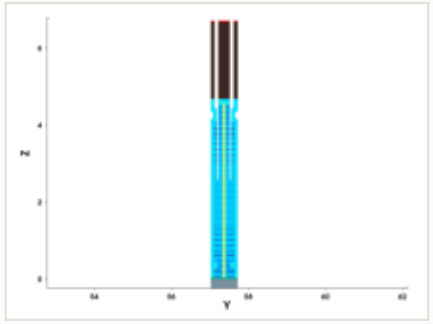
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

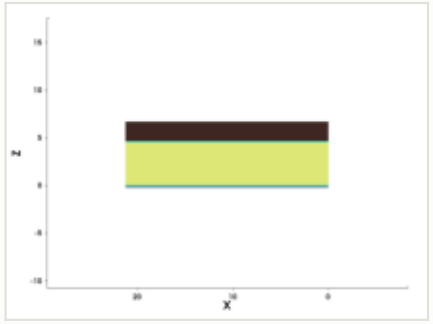


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



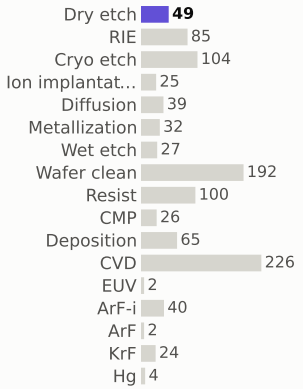
y cut at 57.351 µm



mask for this step

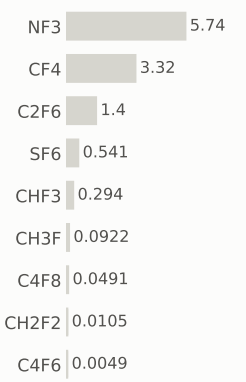
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



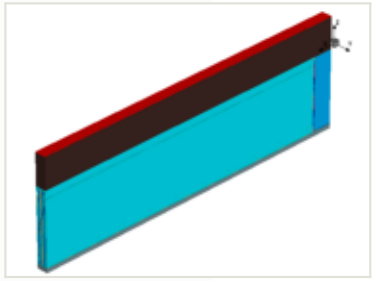
Step #12120: Wafer_clean

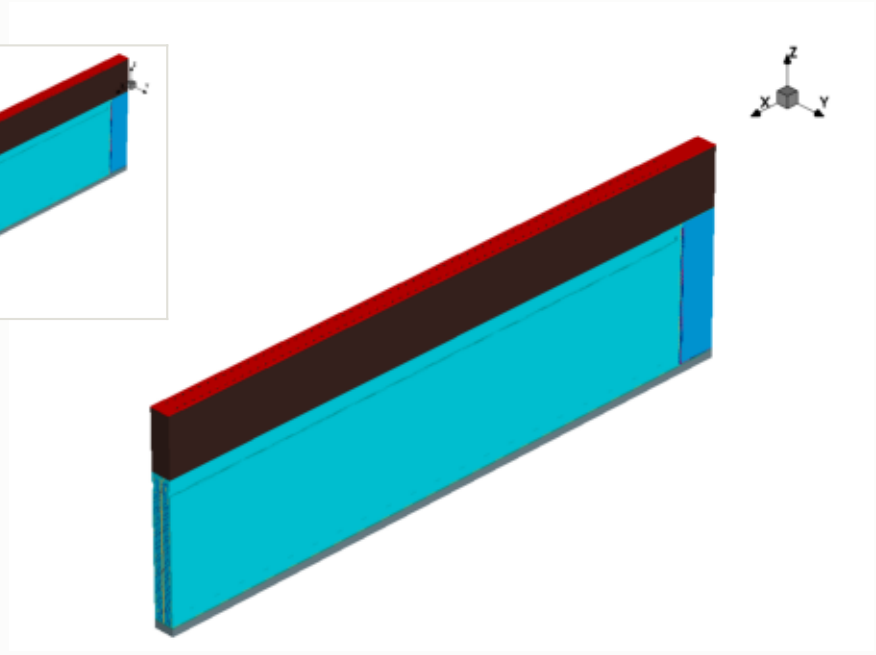
clean · Wafer_clean

0.169	738.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





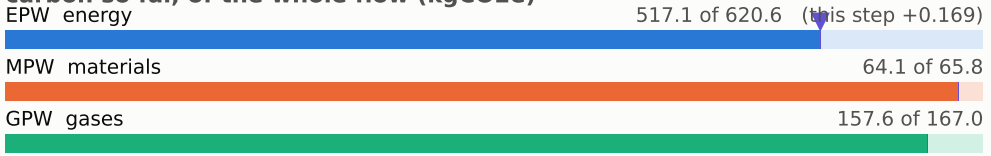
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

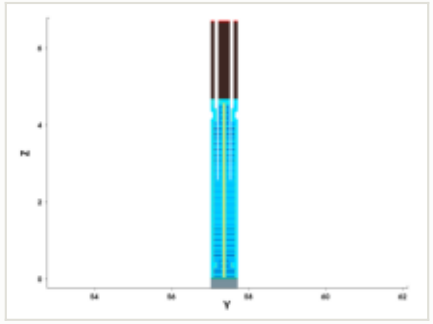
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

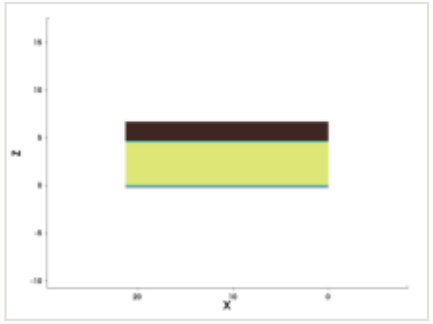


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



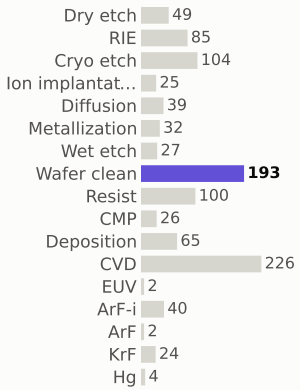
y cut at 57.351 µm



mask for this step

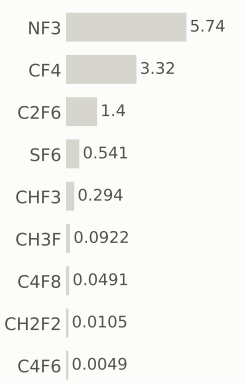
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



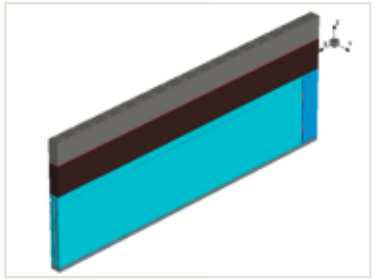
Step #12130: KrF

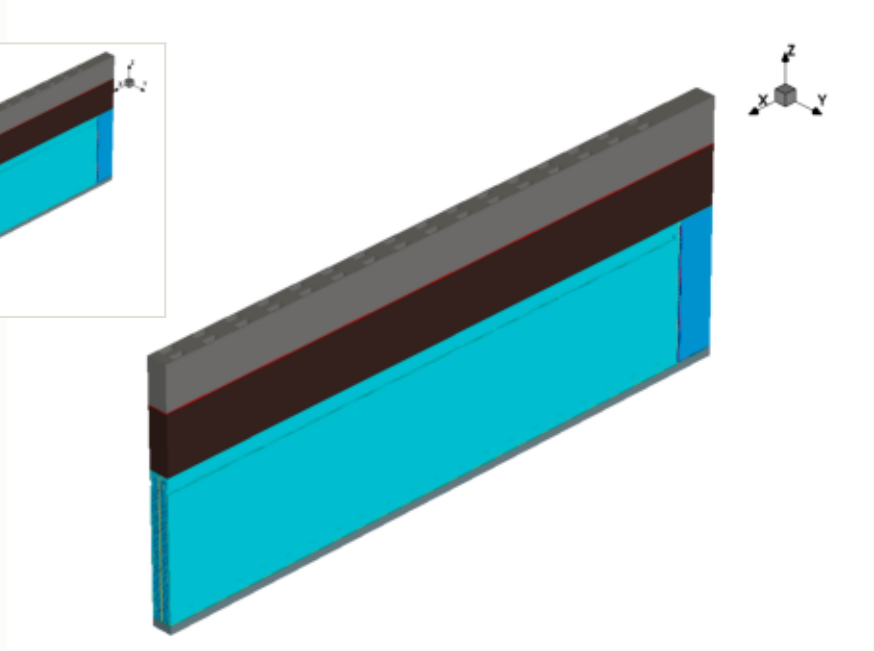
litho_patterning · KrF

0.199	739.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

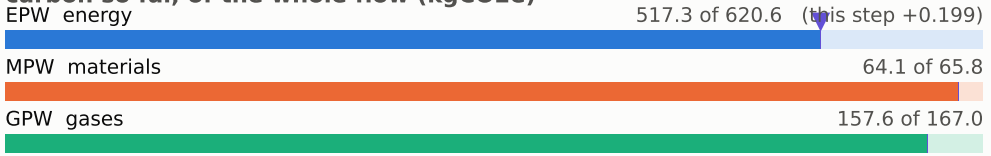




this step

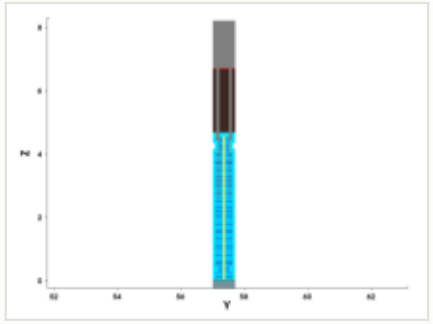
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

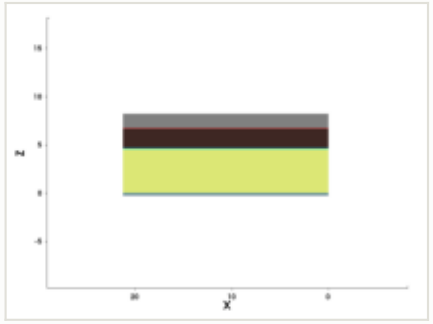


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

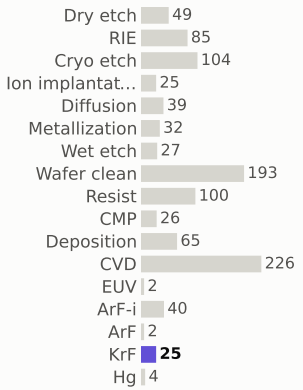


y cut at 57.351 µm



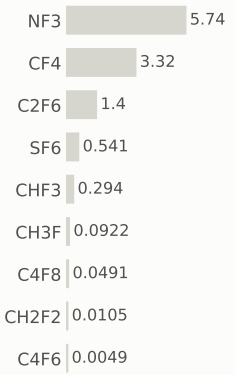
mask: layer 12:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



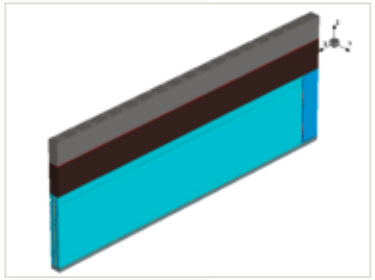
Step #12140: RIE

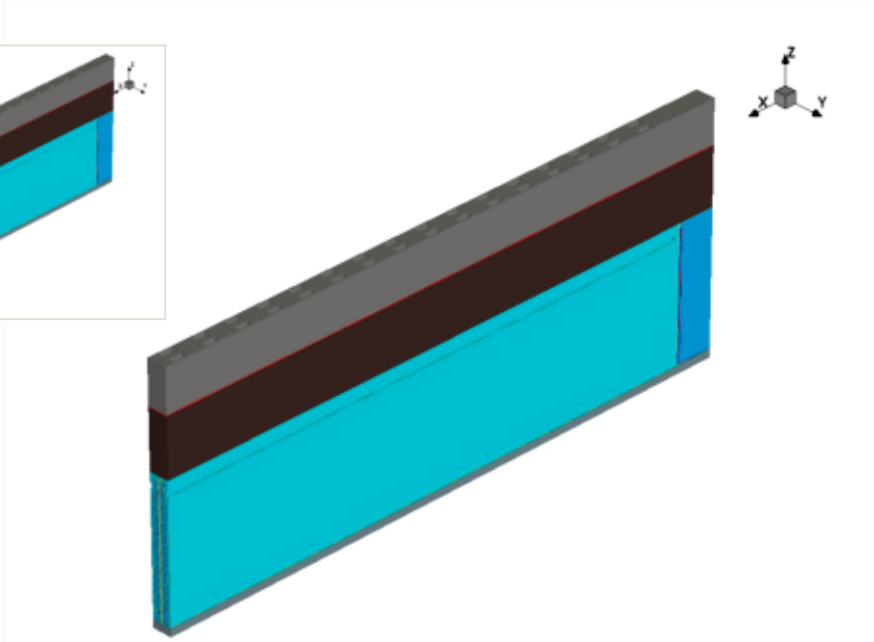
etch_anisotropic · RIE

1.04	740.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



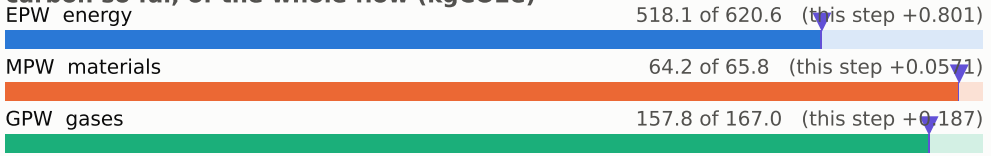


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

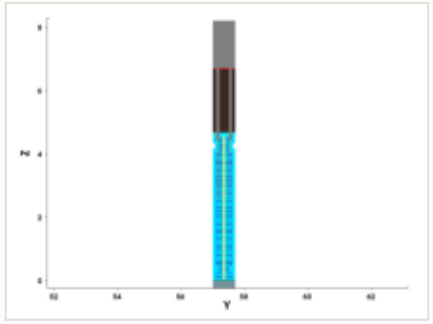
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

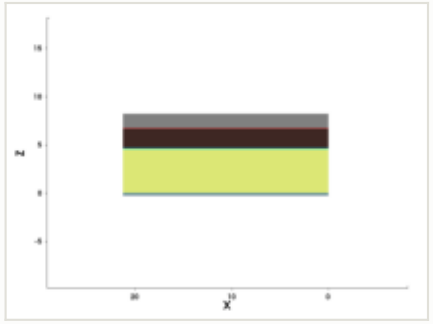


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



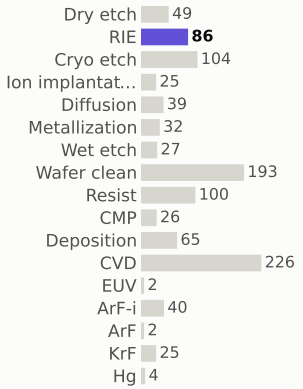
y cut at 57.351 µm



mask for this step

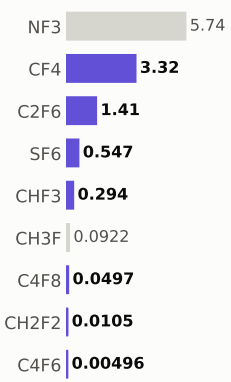
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



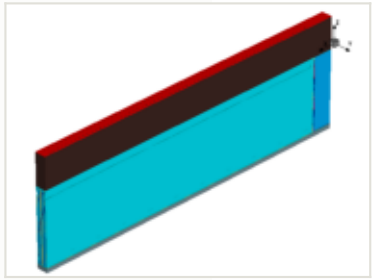
Step #12150: Dry_etch

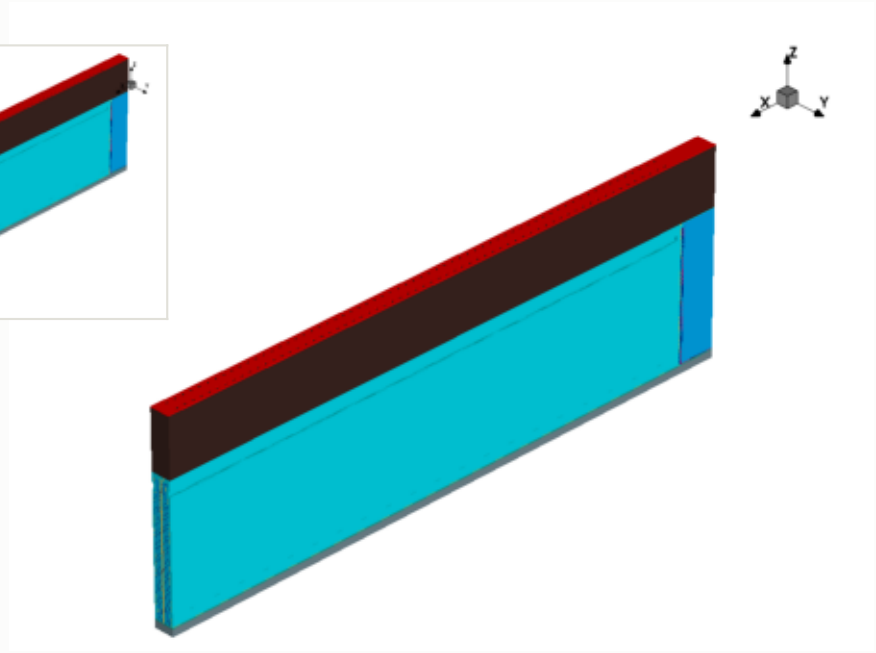
etch_strip · Dry_etch

0.801	740.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

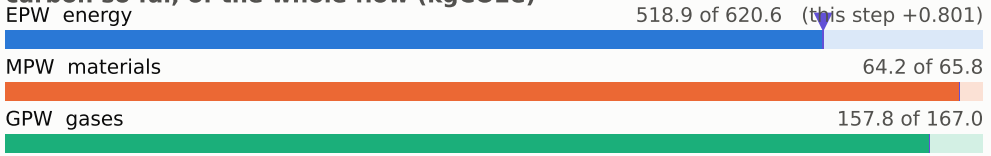




this step

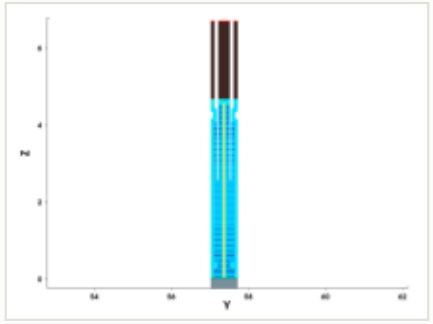
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

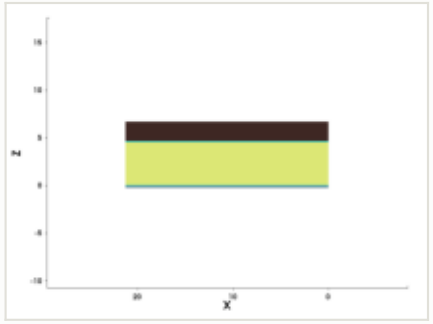


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



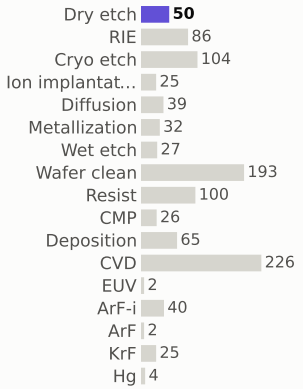
y cut at 57.351 µm



mask for this step

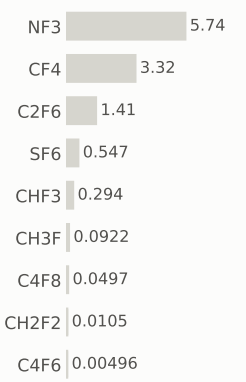
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



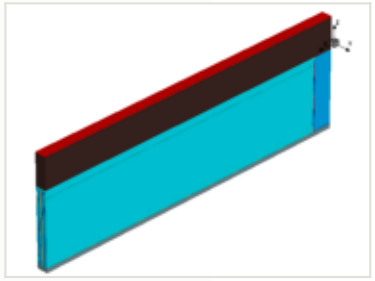
Step #12160: Wafer_clean

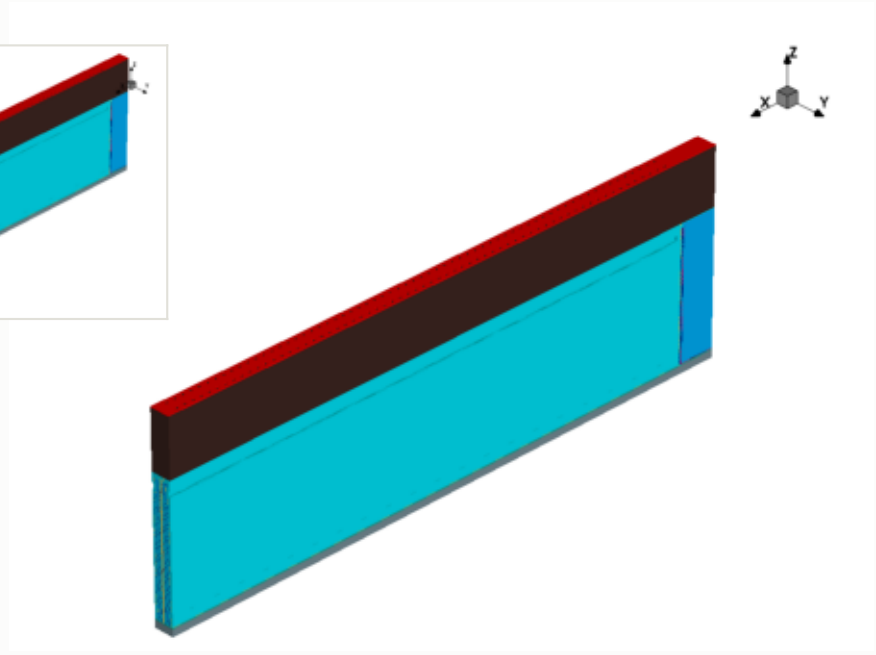
clean · Wafer_clean

0.169	741.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



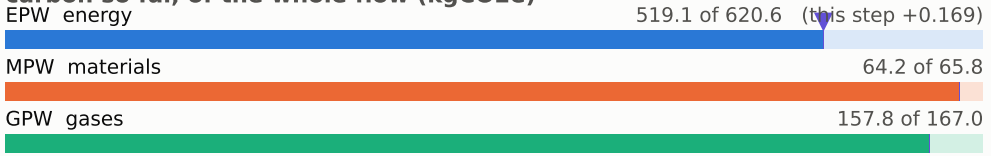


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

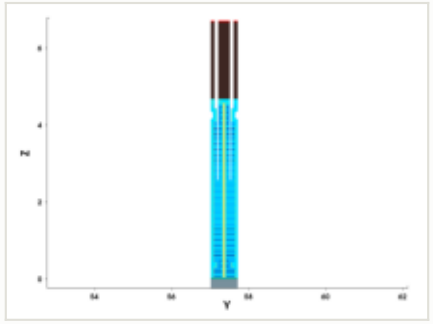
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

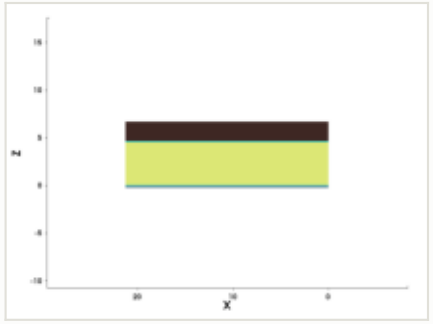


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



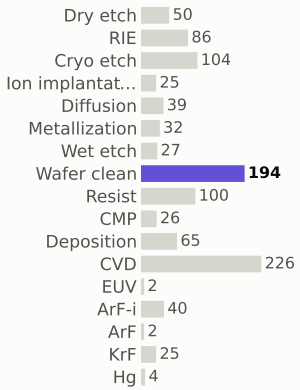
y cut at 57.351 µm



mask for this step

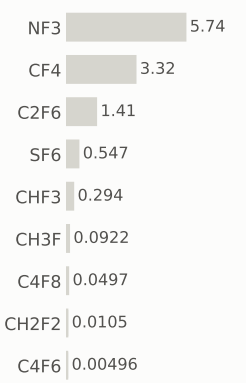
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



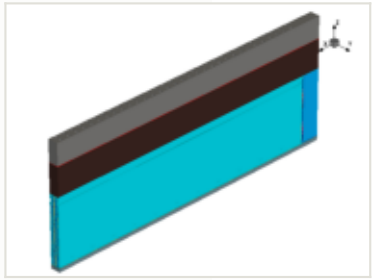
Step #12170: KrF

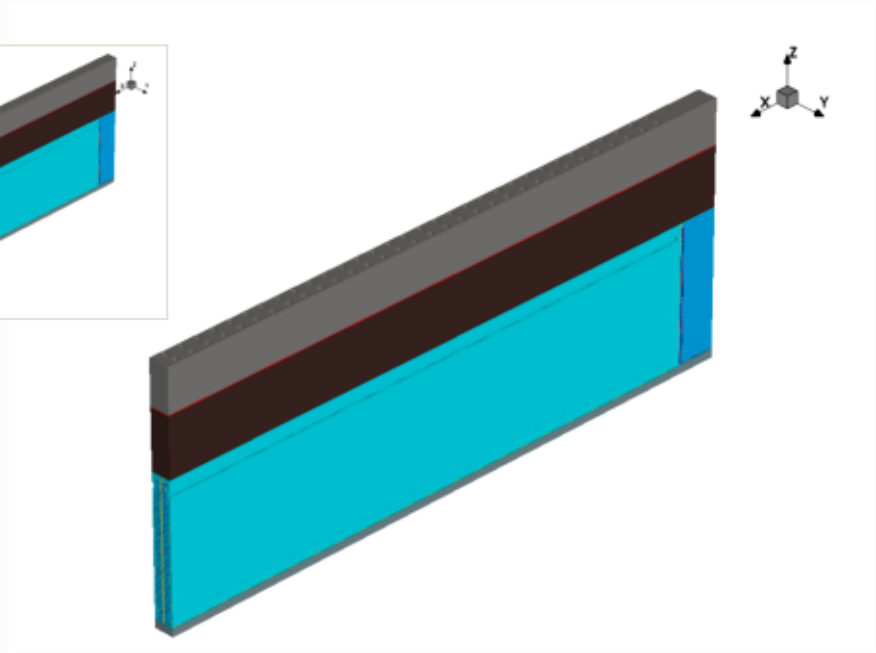
litho_patterning · KrF

0.199	741.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

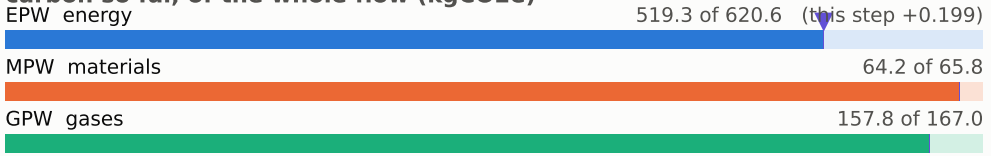




this step

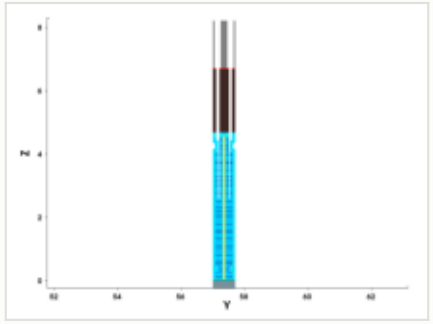
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

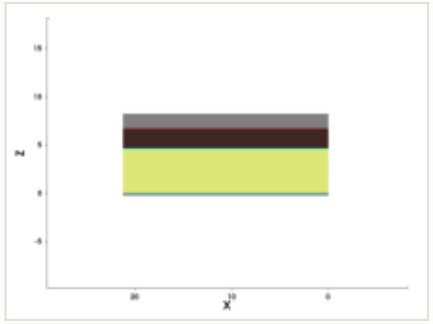


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

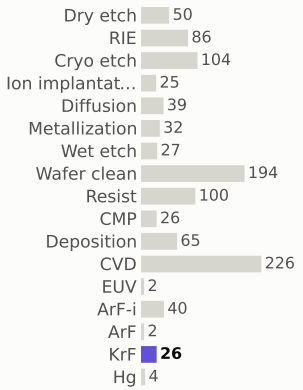


y cut at 57.351 µm



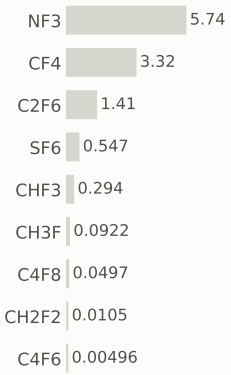
mask: layer 13:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



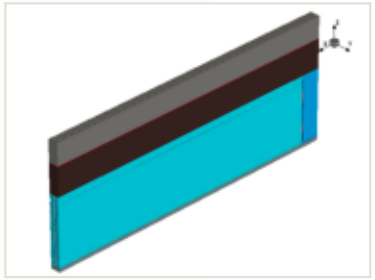
Step #12180: RIE

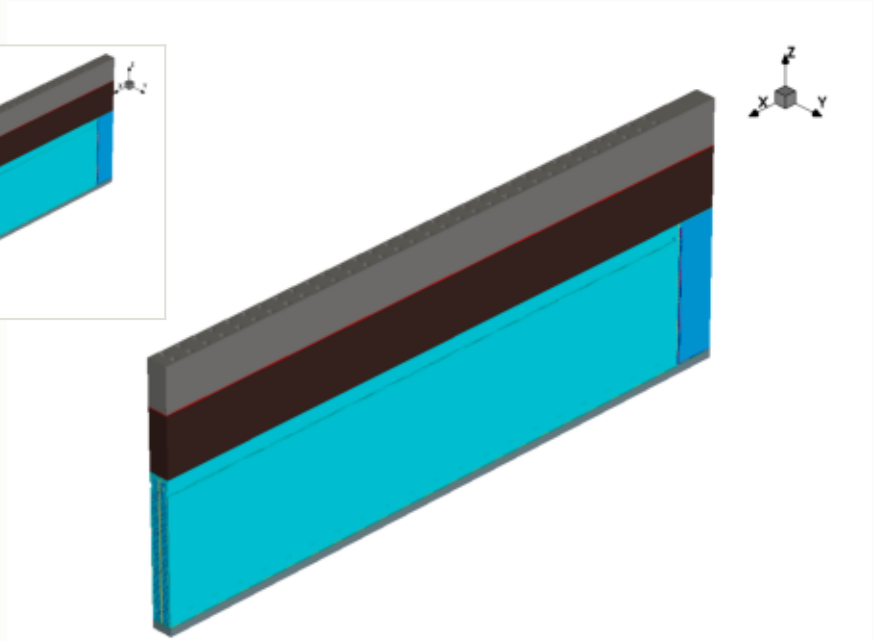
etch_anisotropic · RIE

1.04	742.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

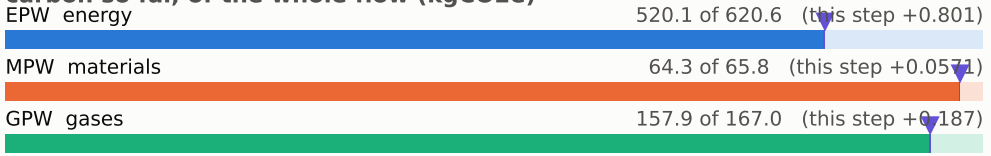




this step

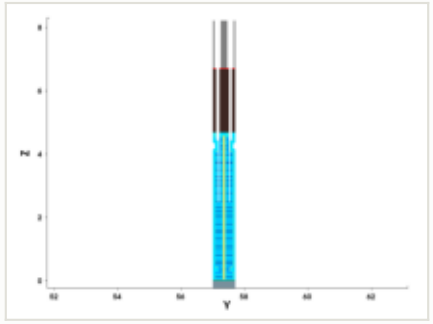
process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

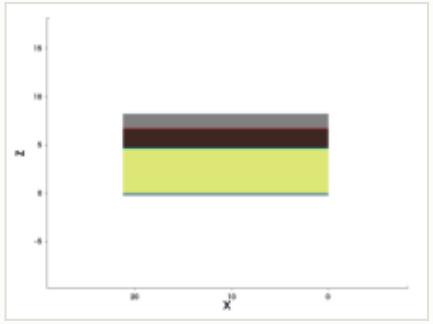


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



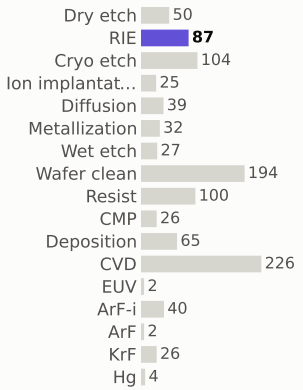
y cut at 57.351 µm



mask for this step

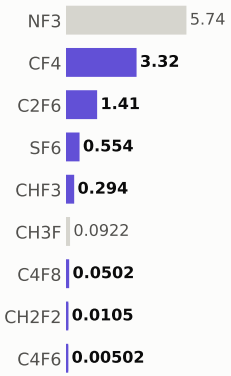
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



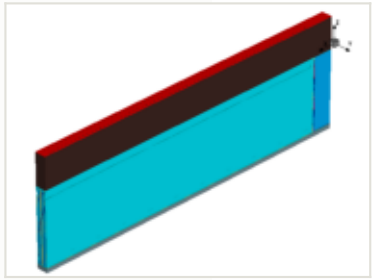
Step #12190: Dry_etch

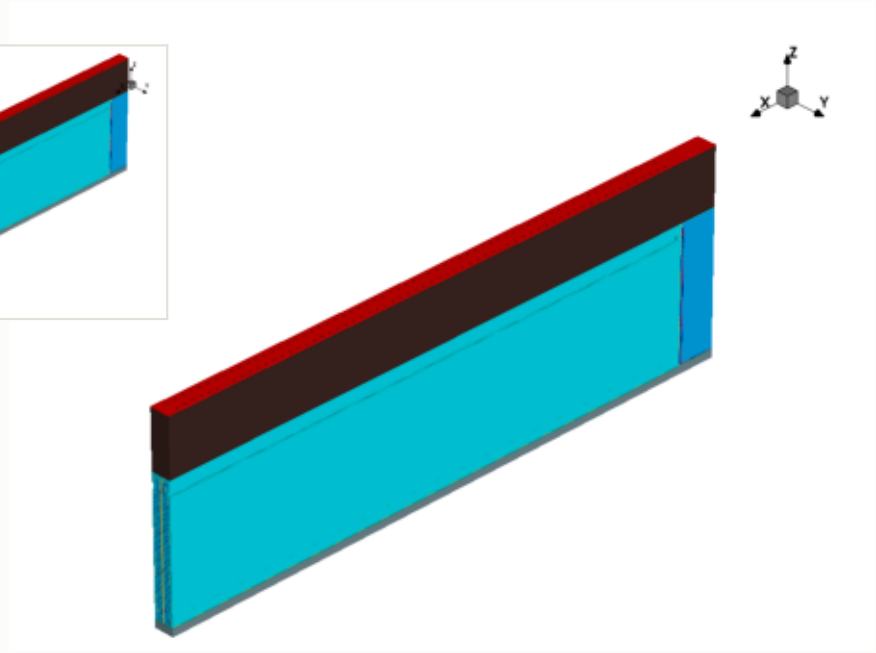
etch_strip · Dry_etch

0.801	743.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



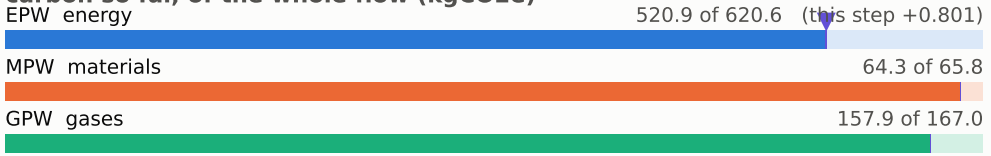


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

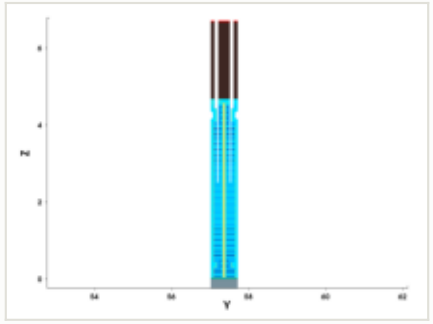
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

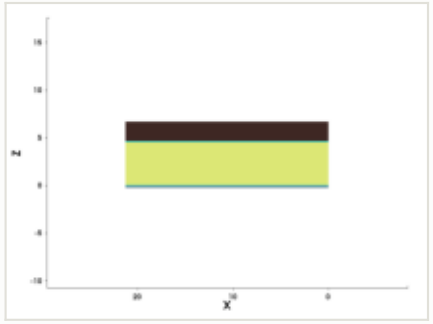


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



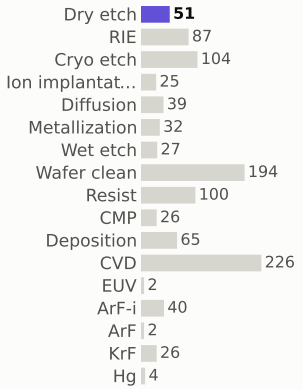
y cut at 57.351 µm



mask for this step

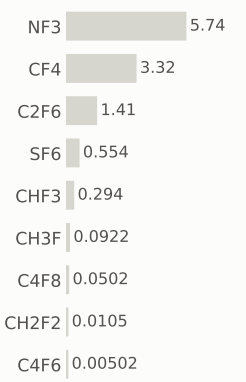
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



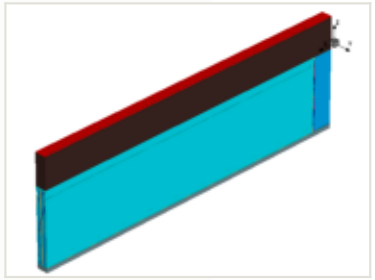
Step #12200: Wafer_clean

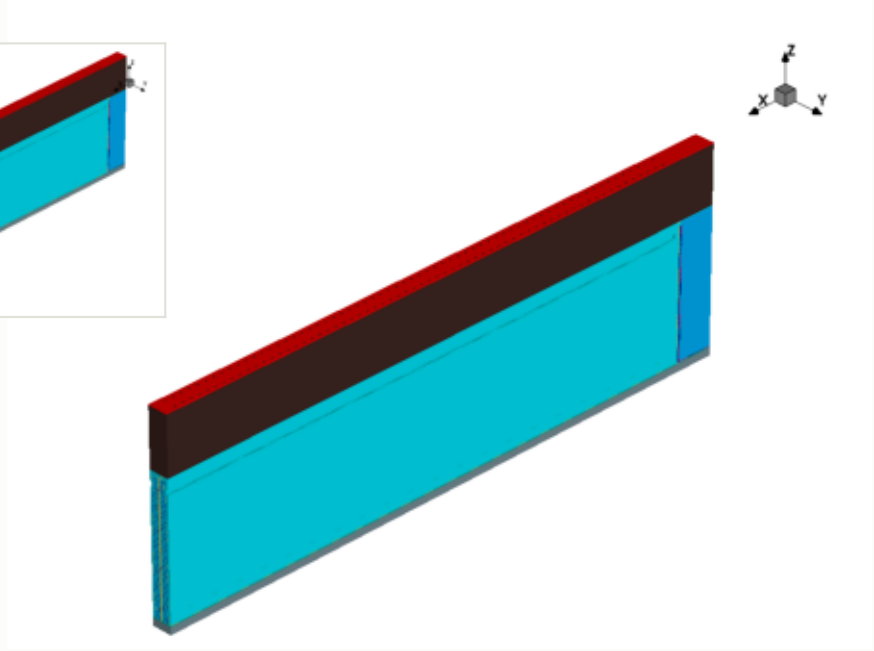
clean · Wafer_clean

0.169	743.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



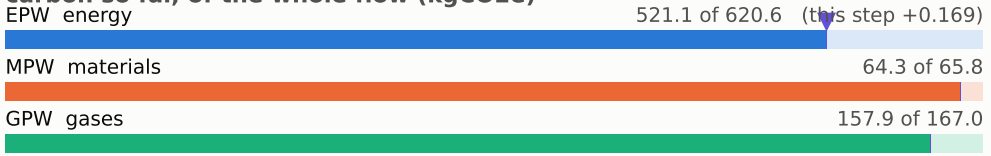


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

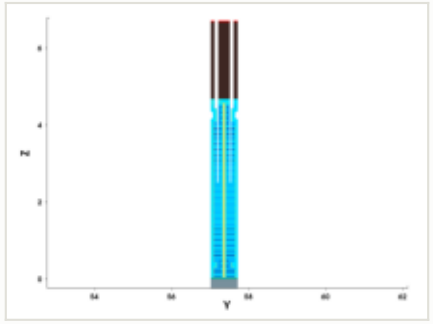
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

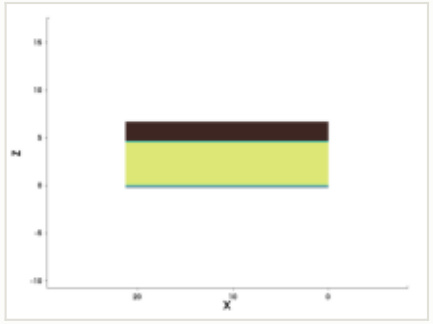


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



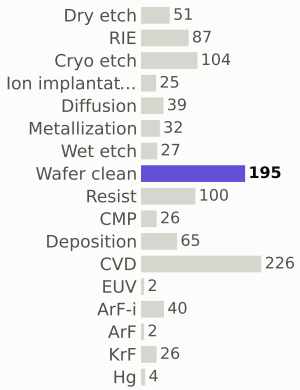
y cut at 57.351 µm



mask for this step

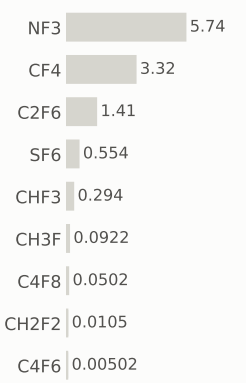
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



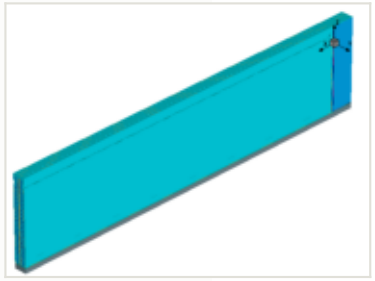
Step #12210: Dry_etch

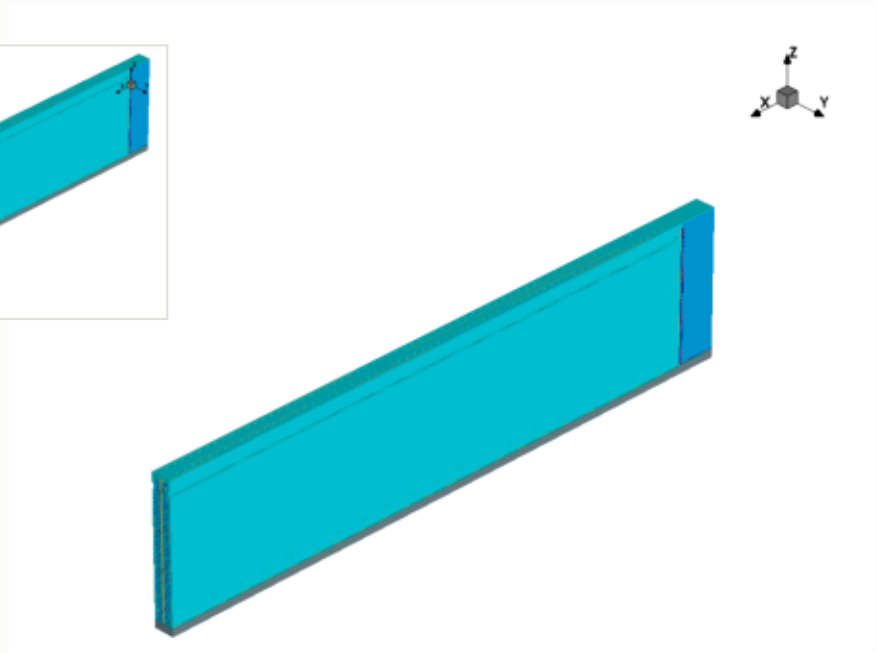
etch_strip · Dry_etch

0.801	744.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

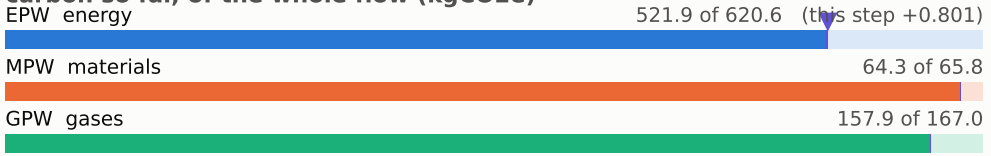




this step

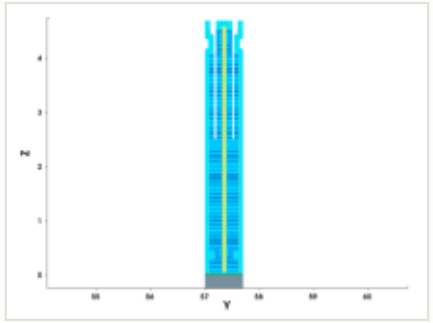
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

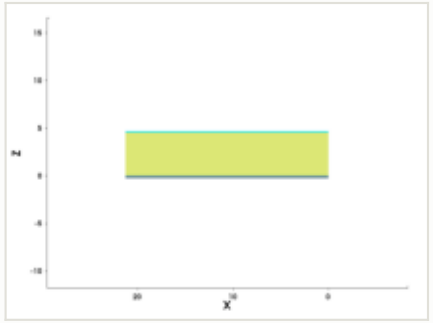


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



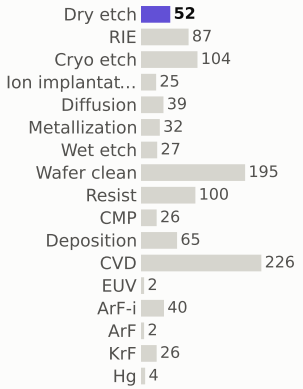
y cut at 57.351 µm



mask for this step

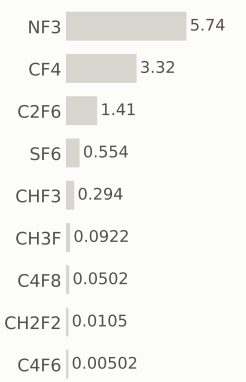
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



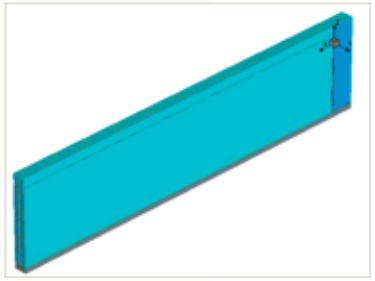
Step #12220: Wet_etch

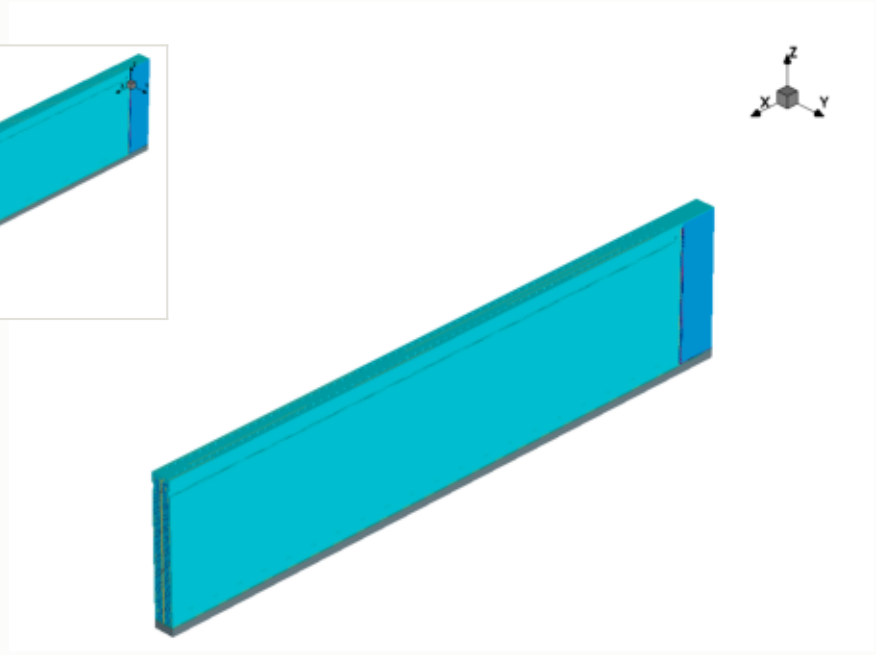
etch_isotropic · Wet_etch

0.169	744.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



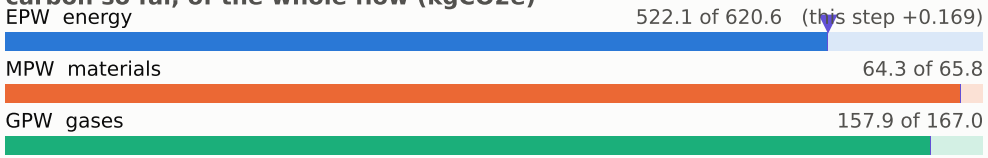


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

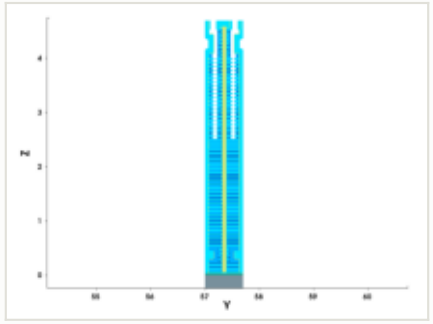
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

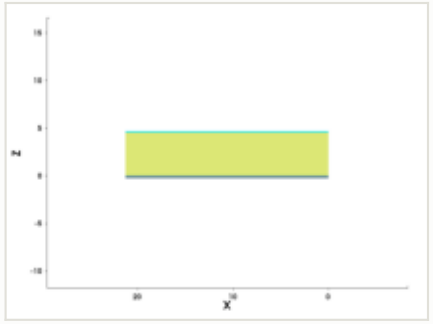


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



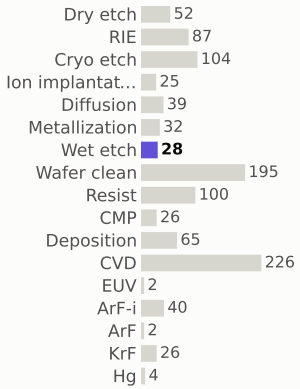
y cut at 57.351 µm



mask for this step

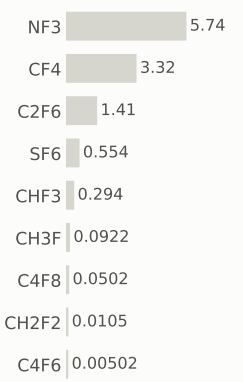
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



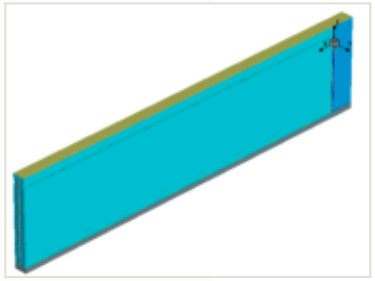
Step #12230: Deposition

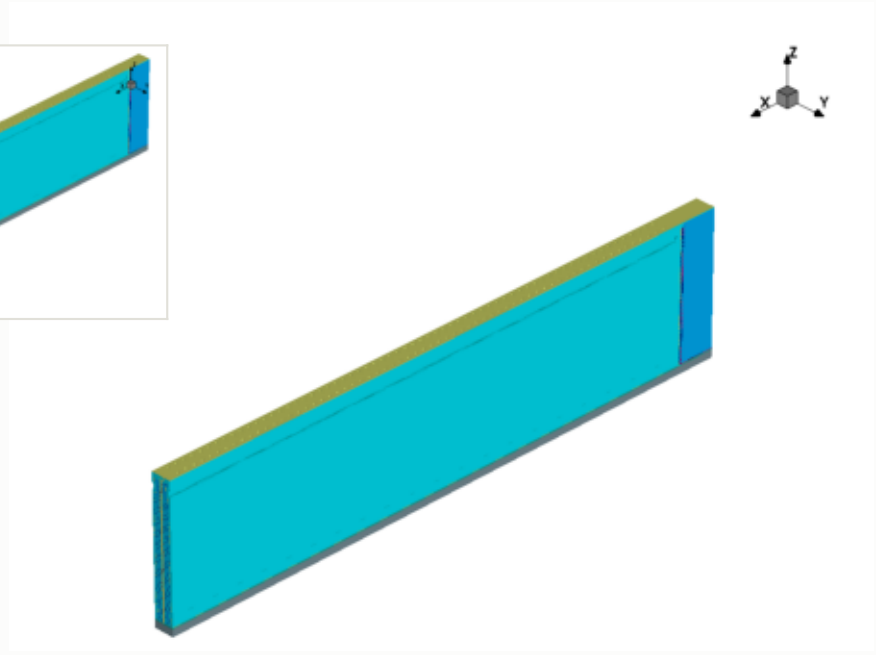
depo_isotropic · Deposition

0.656	744.9	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



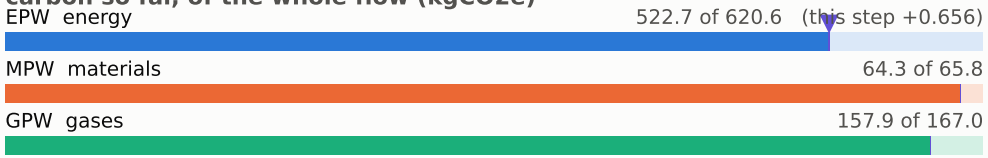


this step

process area	Deposition
material	—
thickness	0.01 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

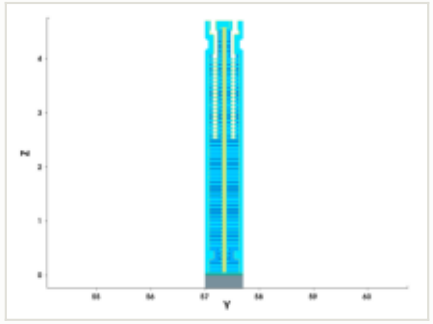
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

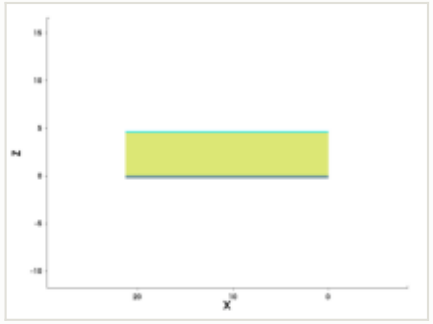


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



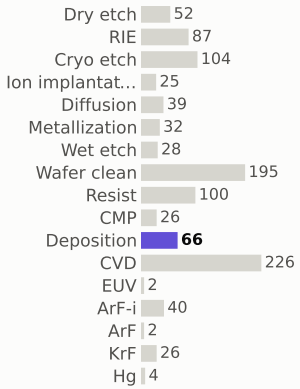
y cut at 57.351 µm



mask for this step

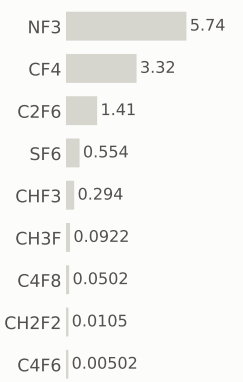
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



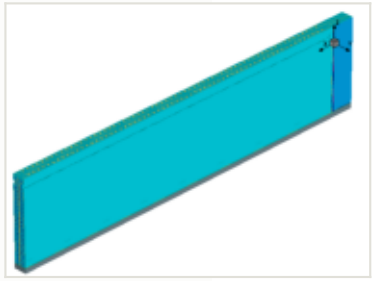
Step #12240: RIE

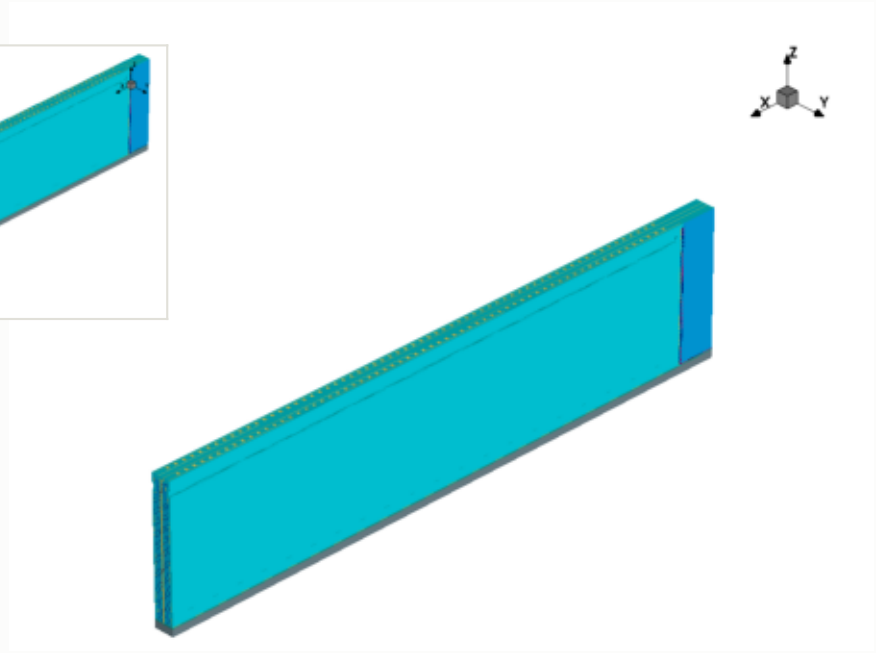
etch_anisotropic · RIE

1.04	746.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



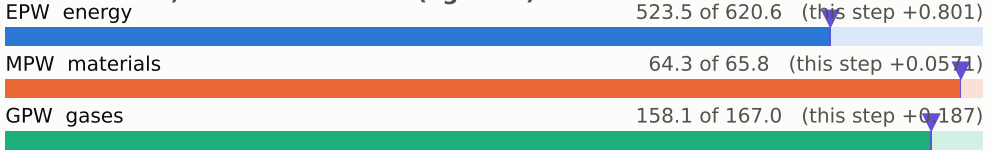


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

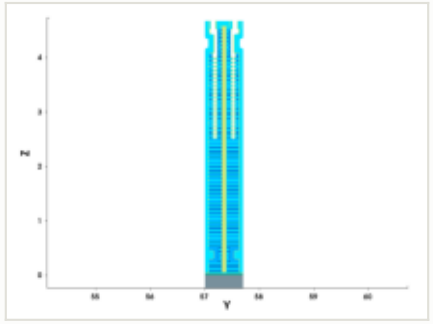
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

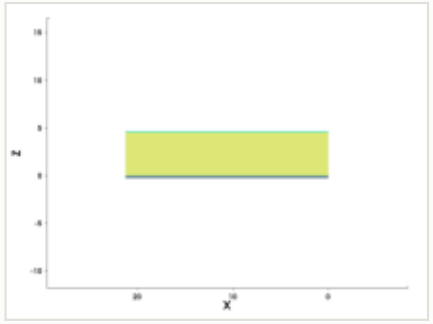


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



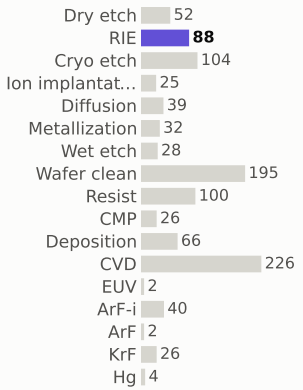
y cut at 57.351 µm



mask for this step

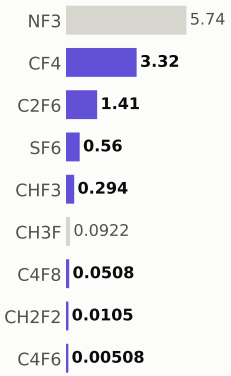
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



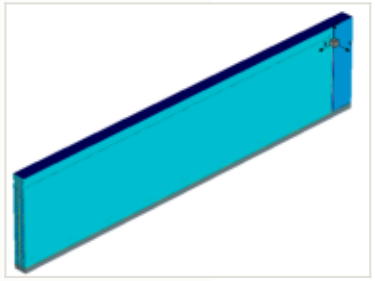
Step #12250: Metallization

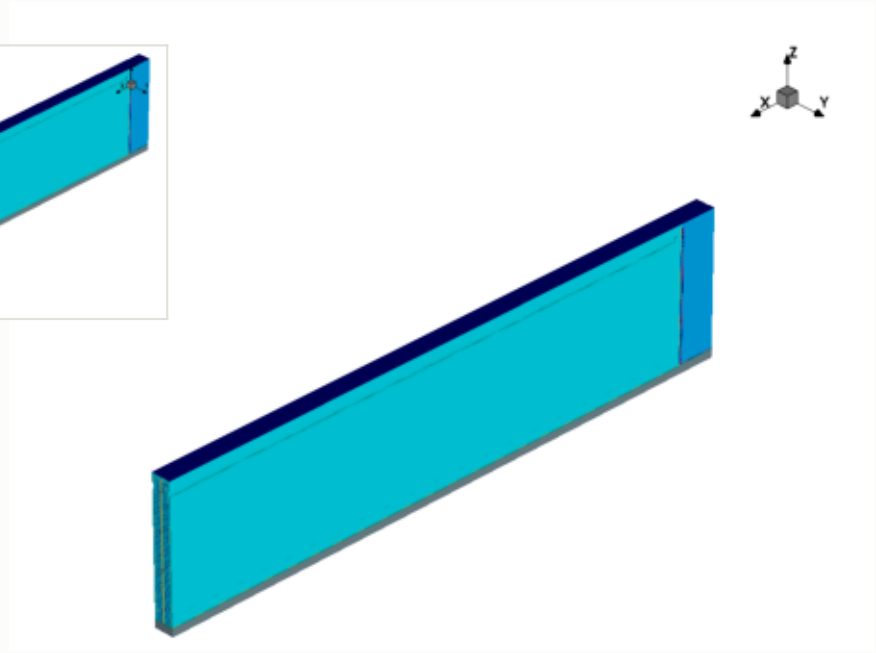
depo_isotropic · Metallization

1.2	747.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



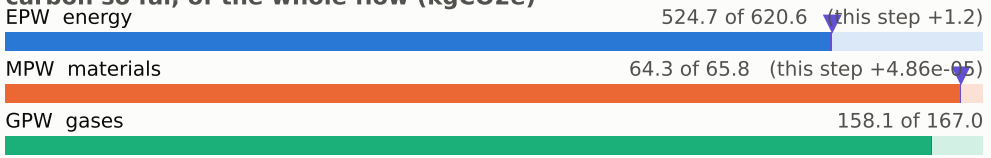


this step

process area	Metallization
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

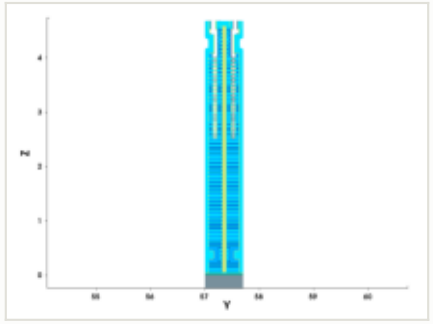
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

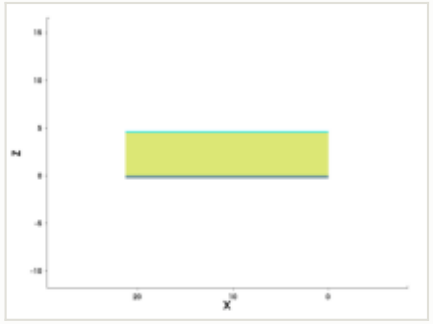


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



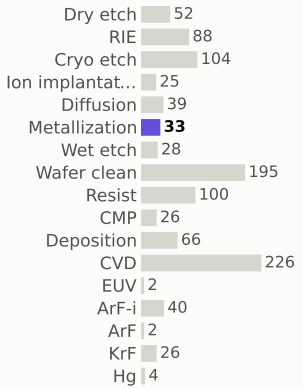
y cut at 57.351 µm



mask for this step

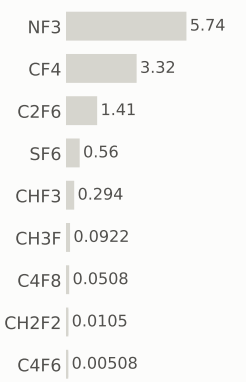
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



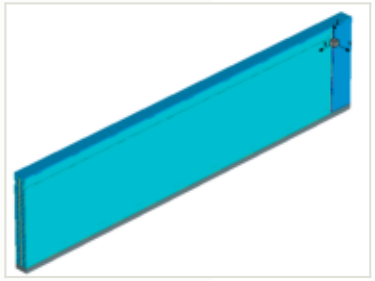
Step #12260: Metallization

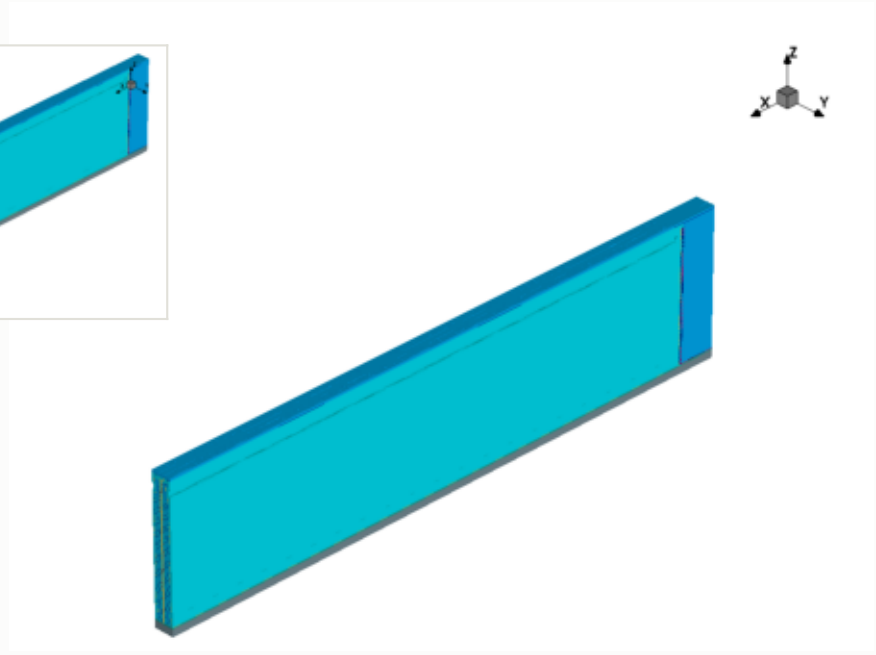
depo_isotropic · Metallization

1.21	748.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



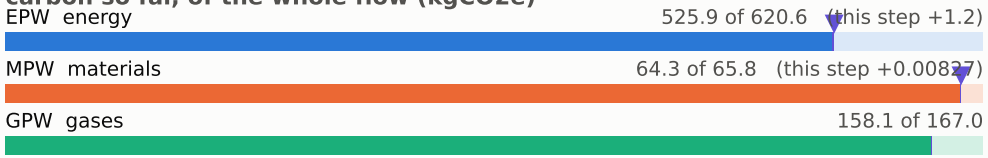


this step

process area	Metallization
material	W
thickness	0.08 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

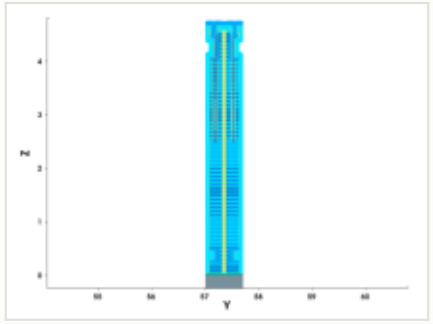
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00827 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

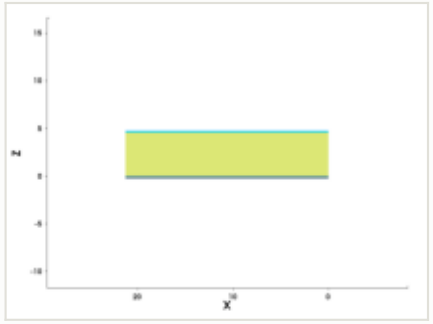


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



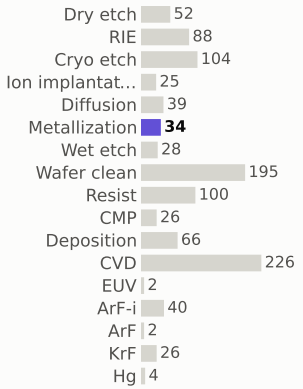
y cut at 57.351 µm



mask for this step

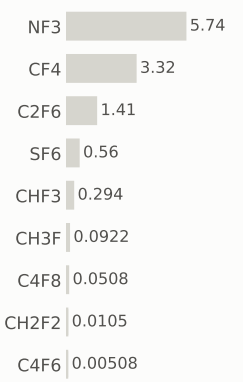
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



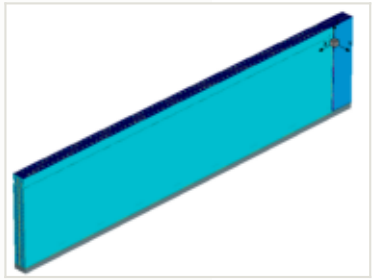
Step #12270: CMP

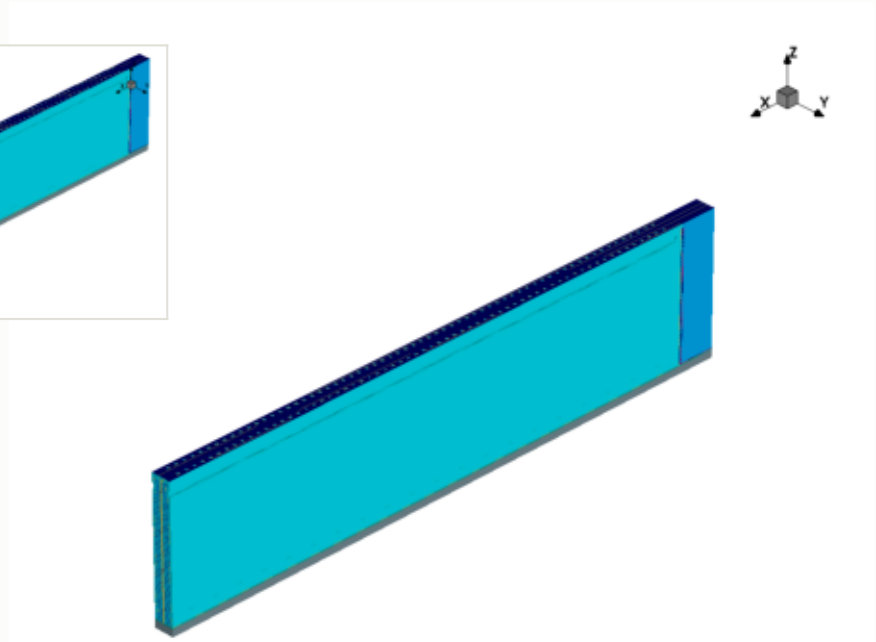
etch_cmp · CMP

0.169	748.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



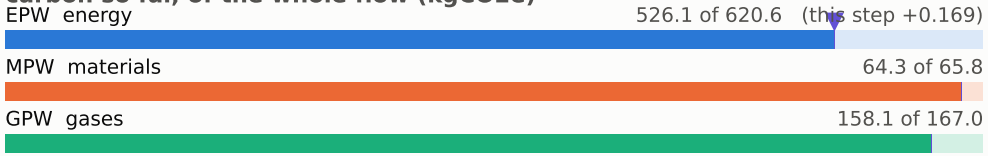


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

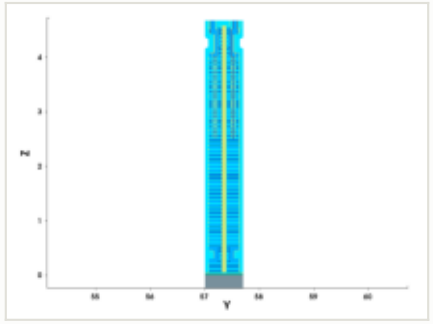
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

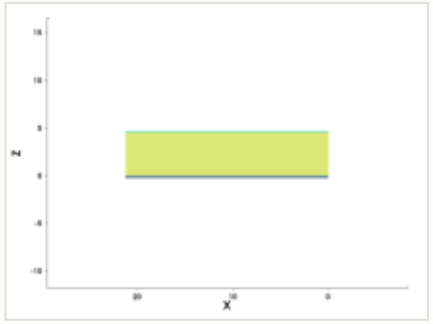


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



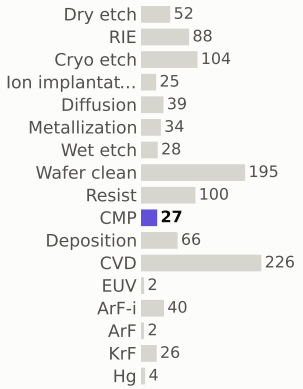
y cut at 57.351 µm



mask for this step

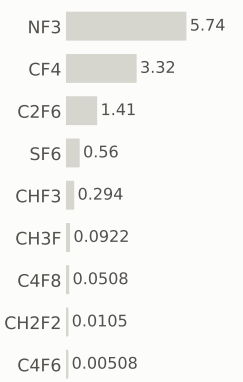
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



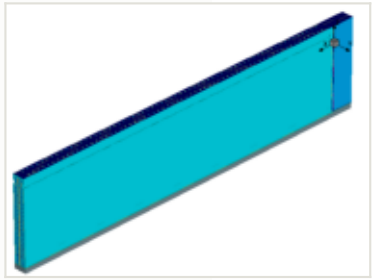
Step #12280: Wafer_clean

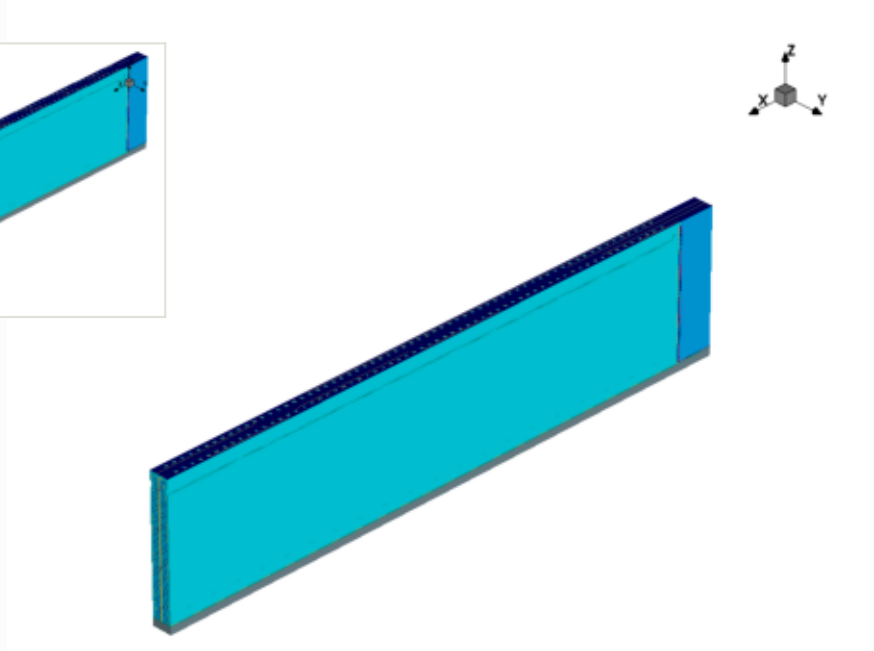
clean · Wafer_clean

0.169	748.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





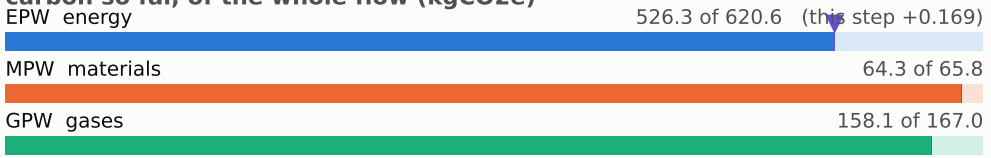
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

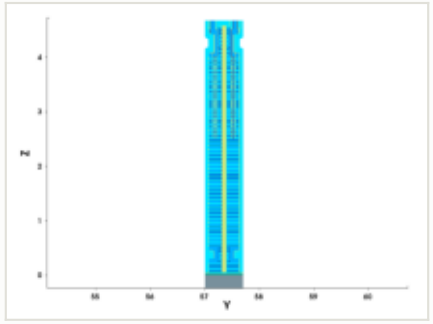
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

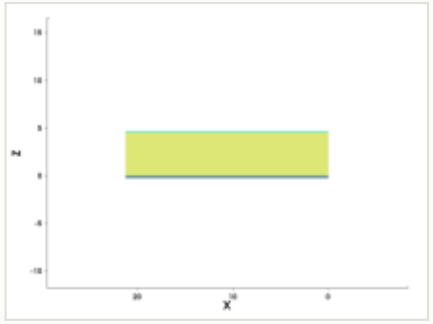


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



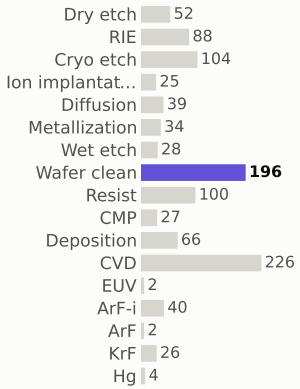
y cut at 57.351 µm



mask for this step

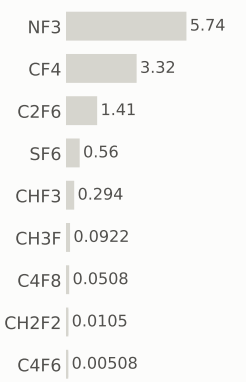
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



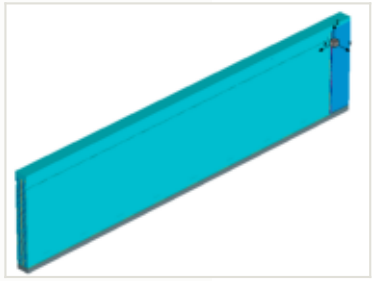
Step #12300: Deposition

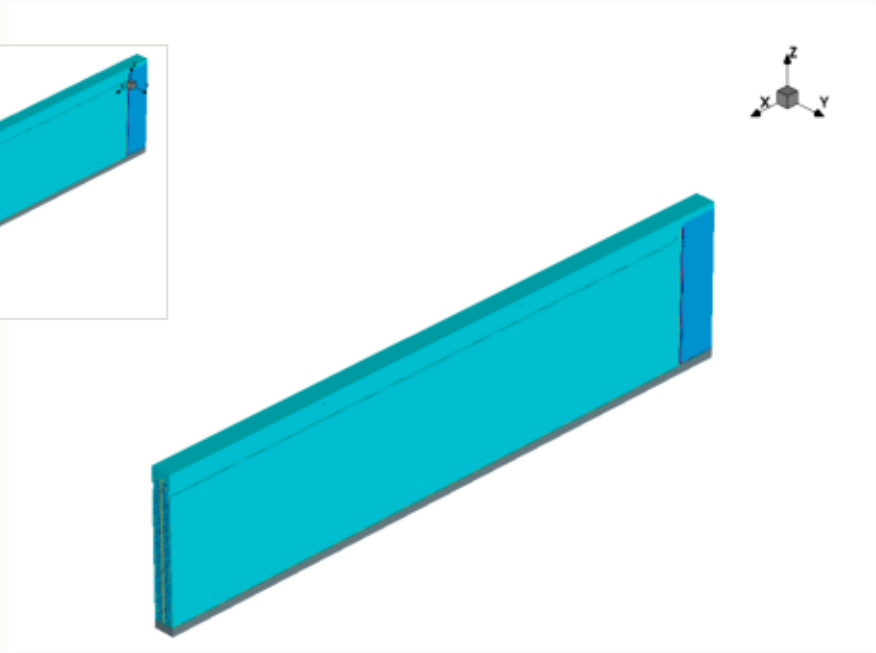
depo_isotropic · Deposition

0.656	749.4	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



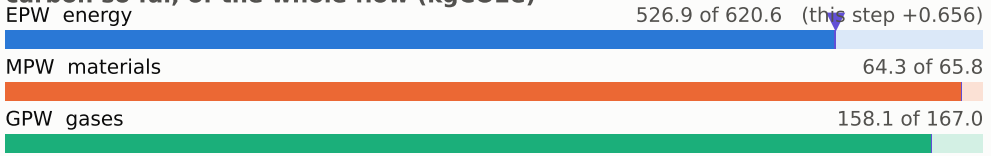


this step

process area	Deposition
material	—
thickness	0.2 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

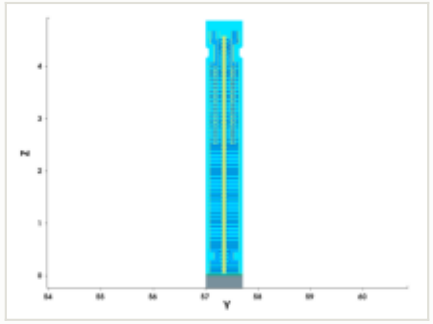
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

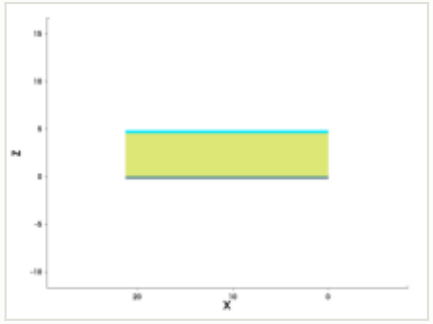


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



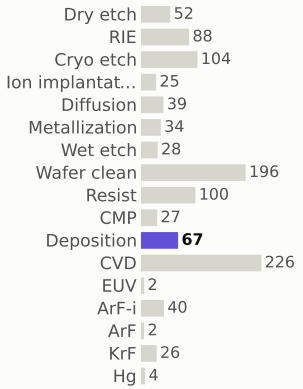
y cut at 57.351 µm



mask for this step

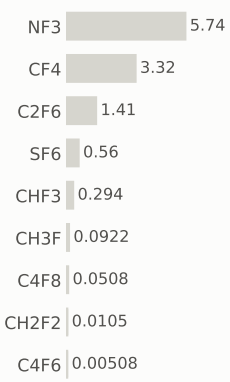
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



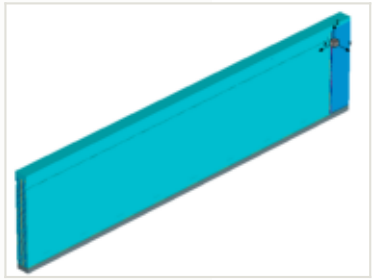
Step #12310: Wafer_clean

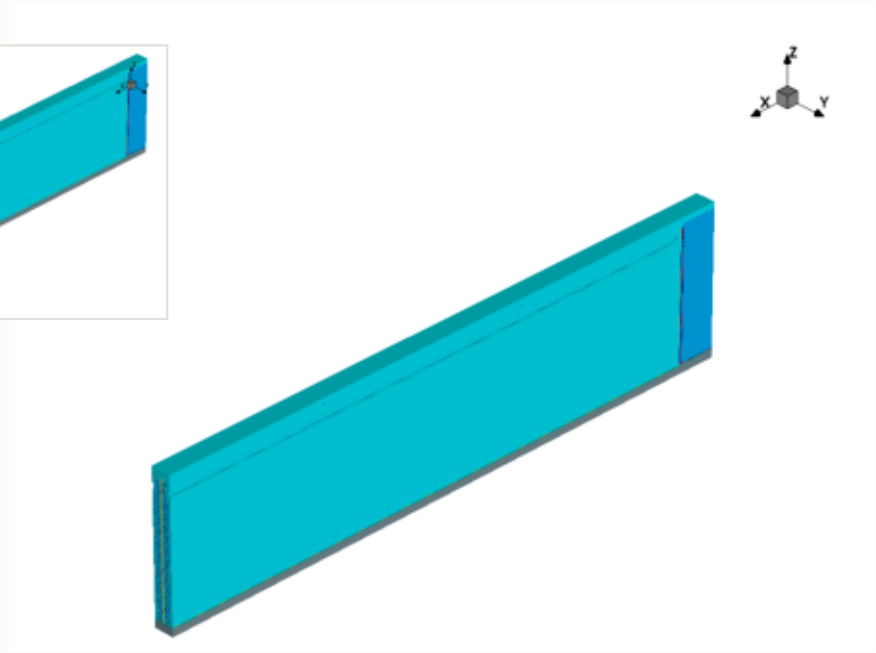
clean · Wafer_clean

0.169	749.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





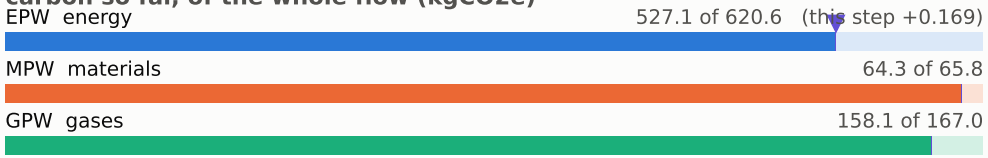
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

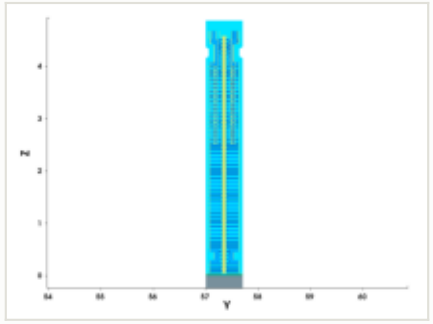
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

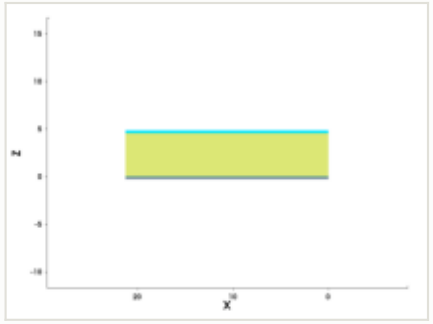


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



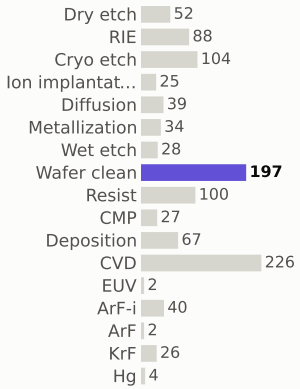
y cut at 57.351 µm



mask for this step

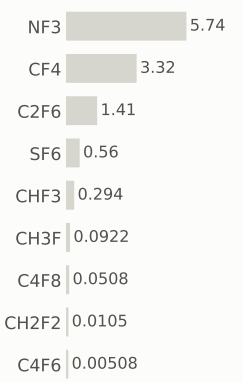
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



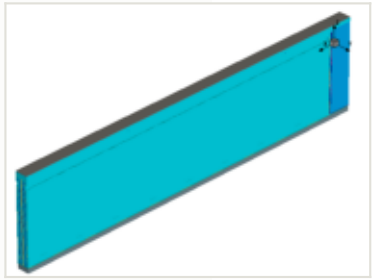
Step #12320: EUV

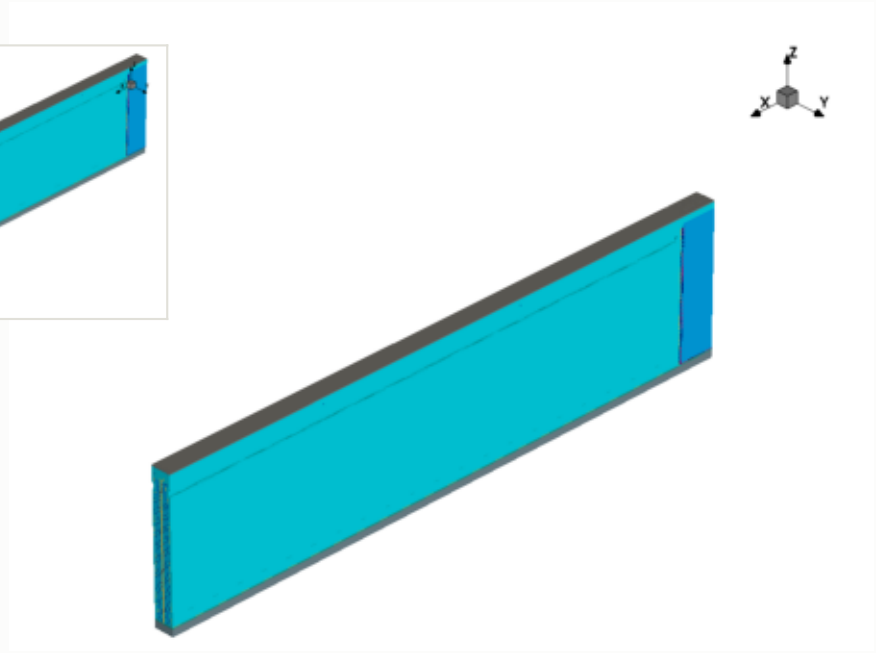
litho_patterning · EUV

1.96	751.5	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



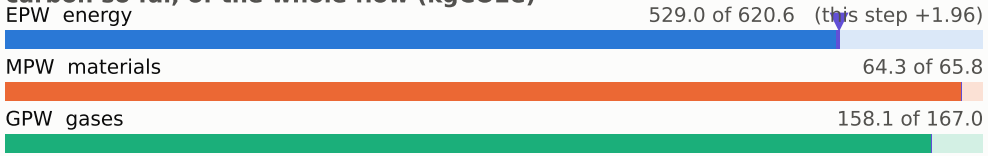


this step

process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

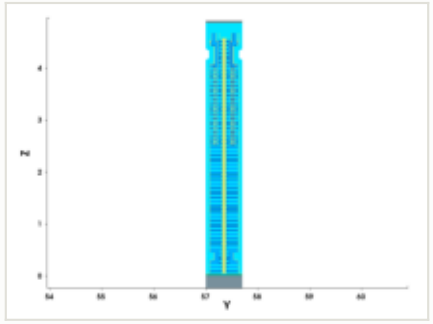
EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

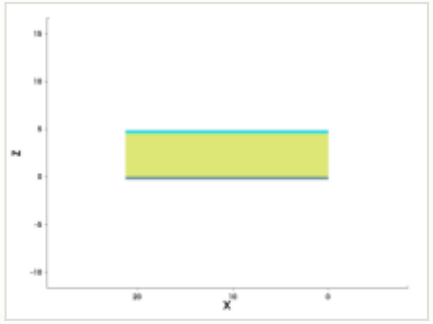


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

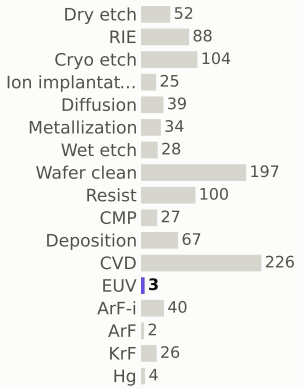


y cut at 57.351 µm



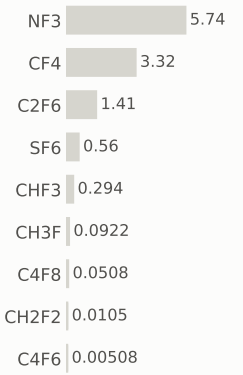
mask: layer 14:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



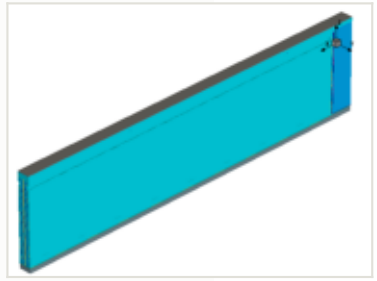
Step #12330: RIE

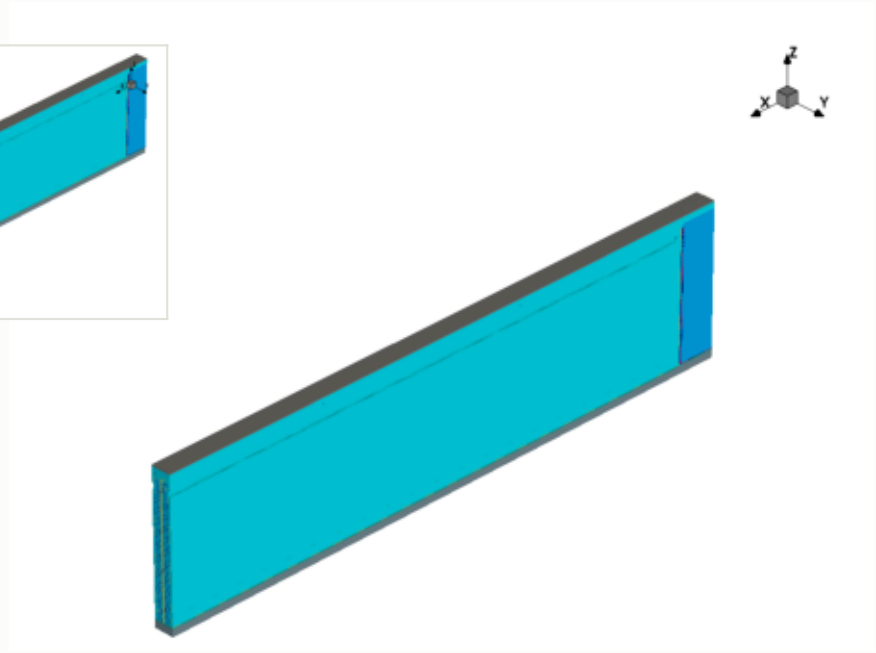
etch_anisotropic · RIE

1.04	752.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



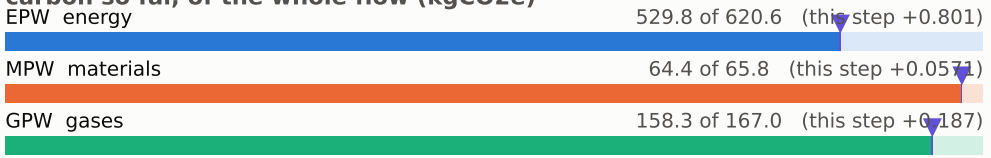


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

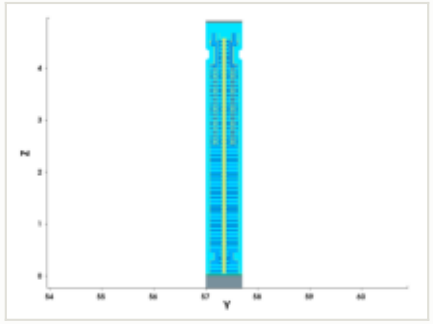
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

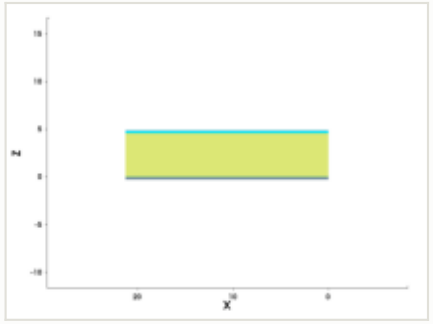


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



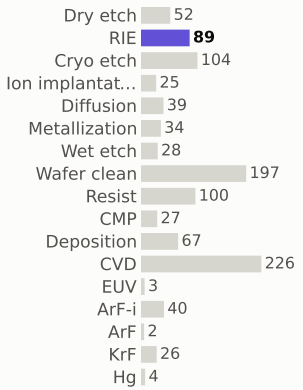
y cut at 57.351 µm



mask for this step

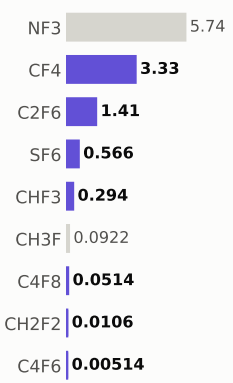
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



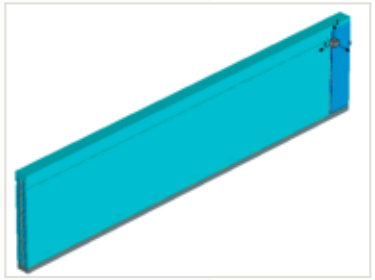
Step #12340: Dry_etch

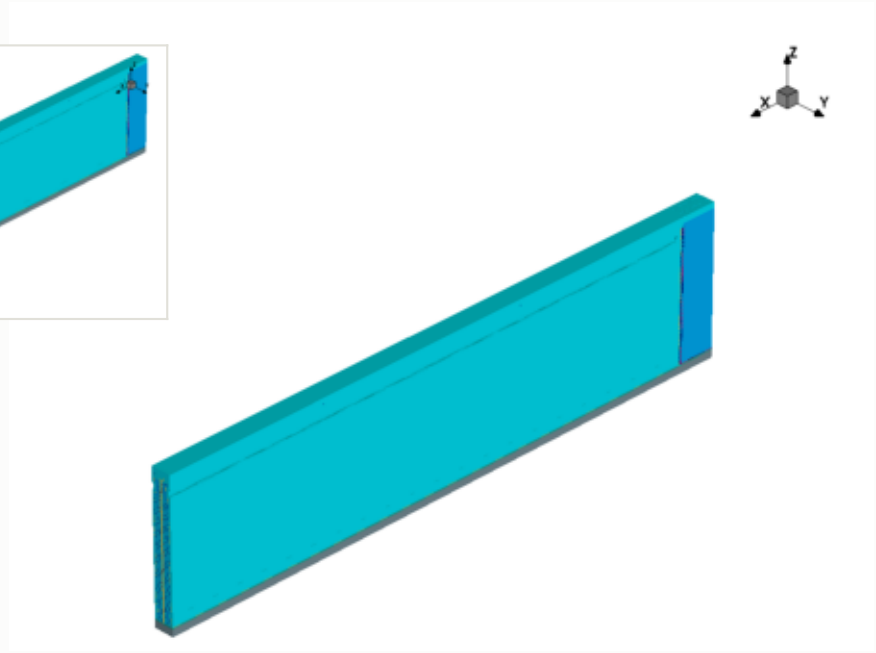
etch_strip · Dry_etch

0.801	753.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

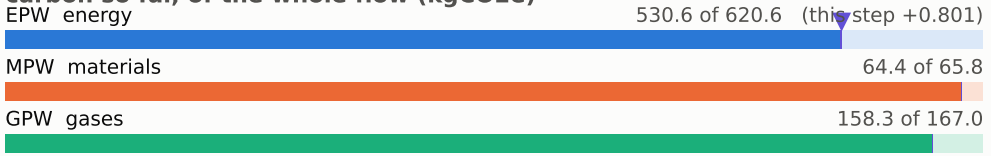




this step

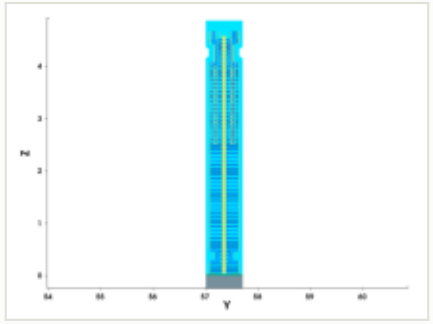
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

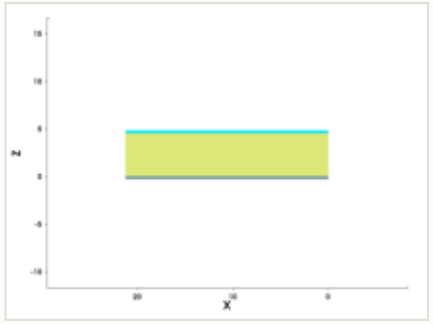


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



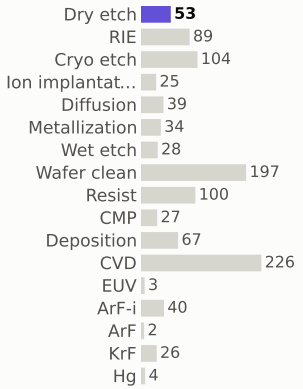
y cut at 57.351 µm



mask for this step

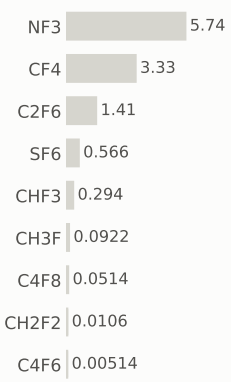
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



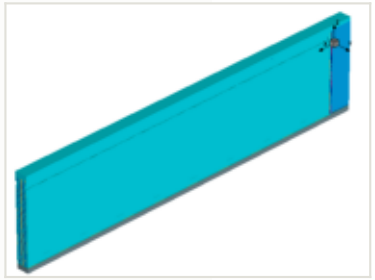
Step #12350: Wafer_clean

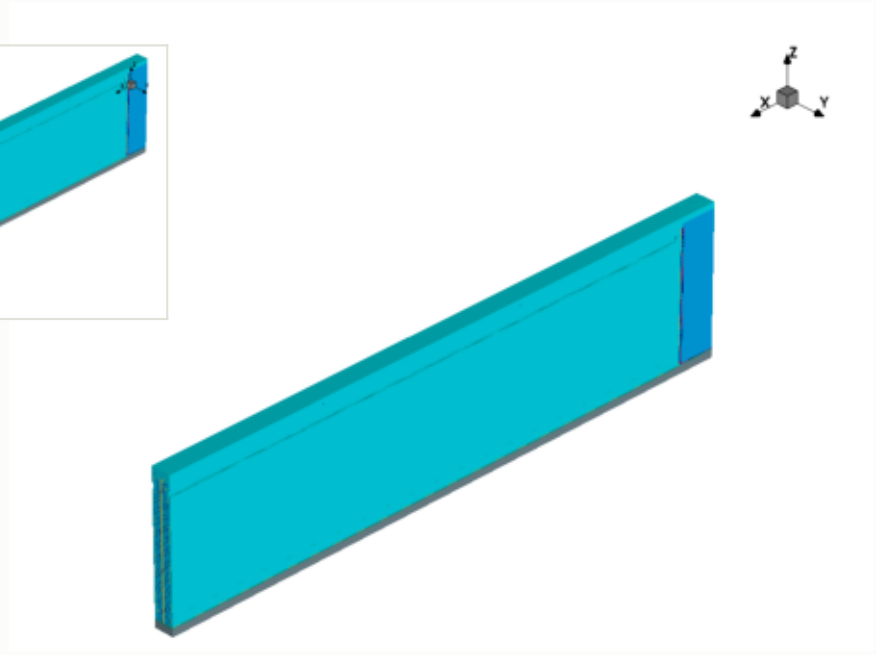
clean · Wafer_clean

0.169	753.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





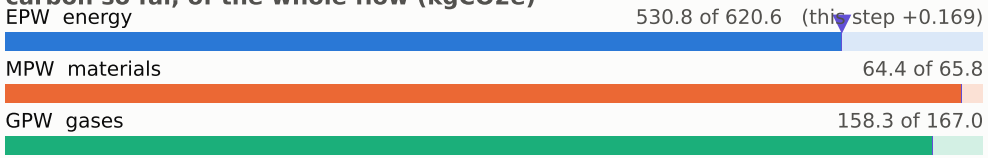
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

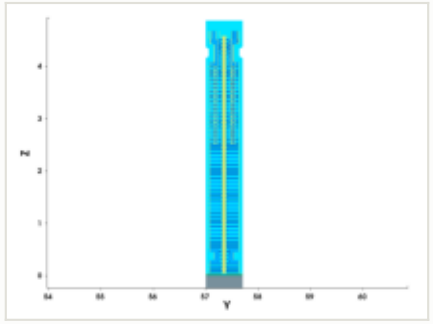
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

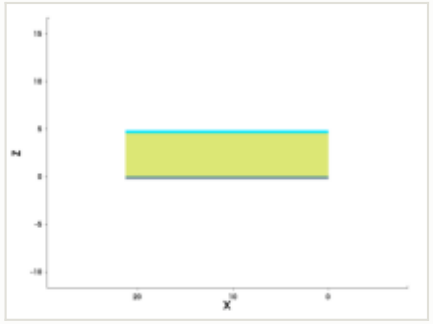


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



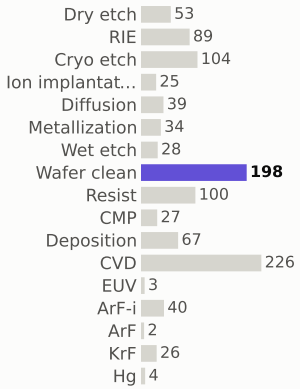
y cut at 57.351 µm



mask for this step

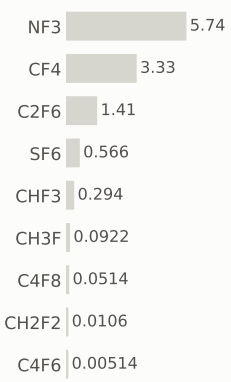
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



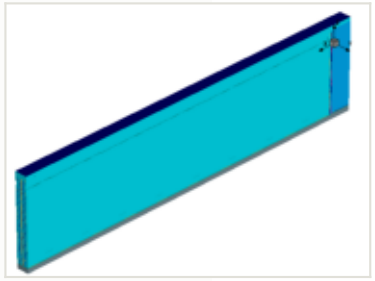
Step #12360: Metallization

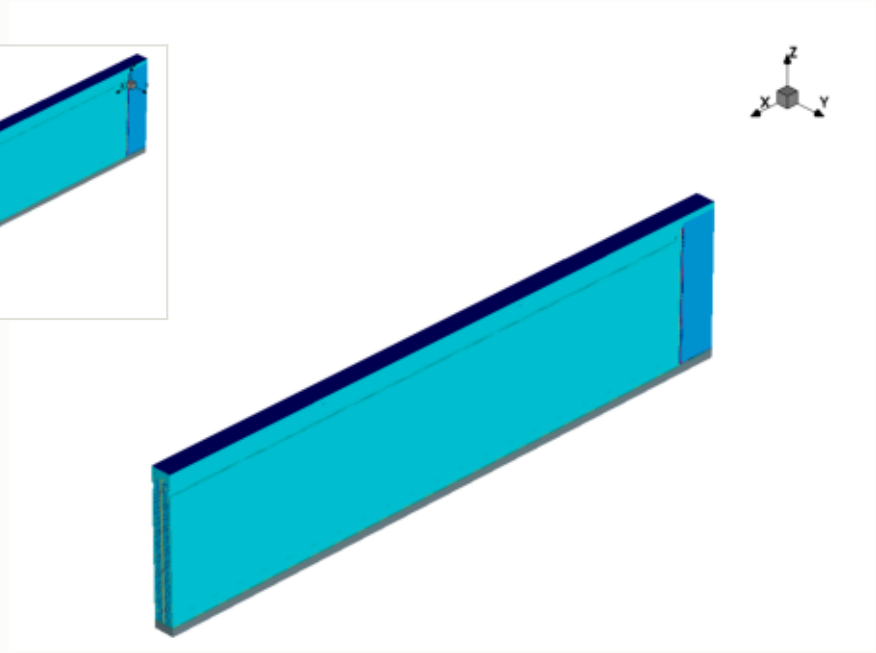
depo_isotropic · Metallization

1.2	754.7	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



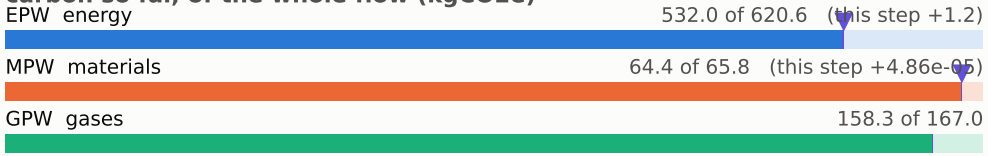


this step

process area	Metallization
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

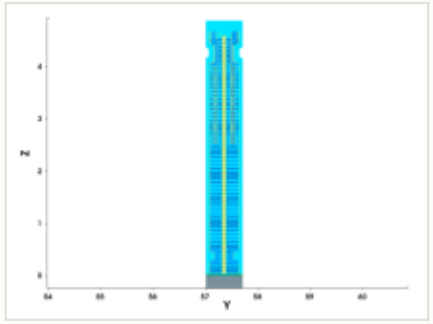
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

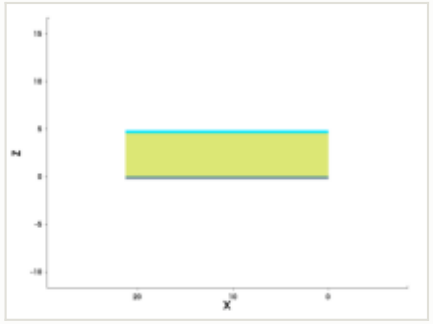


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



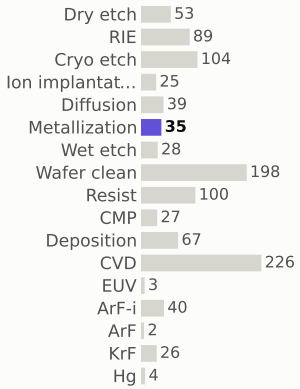
y cut at 57.351 µm



mask for this step

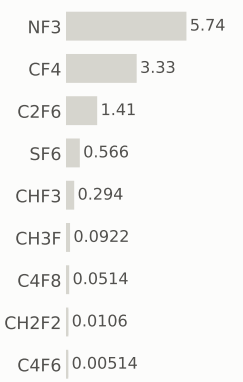
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



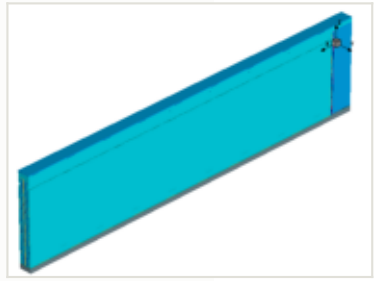
Step #12370: Metallization

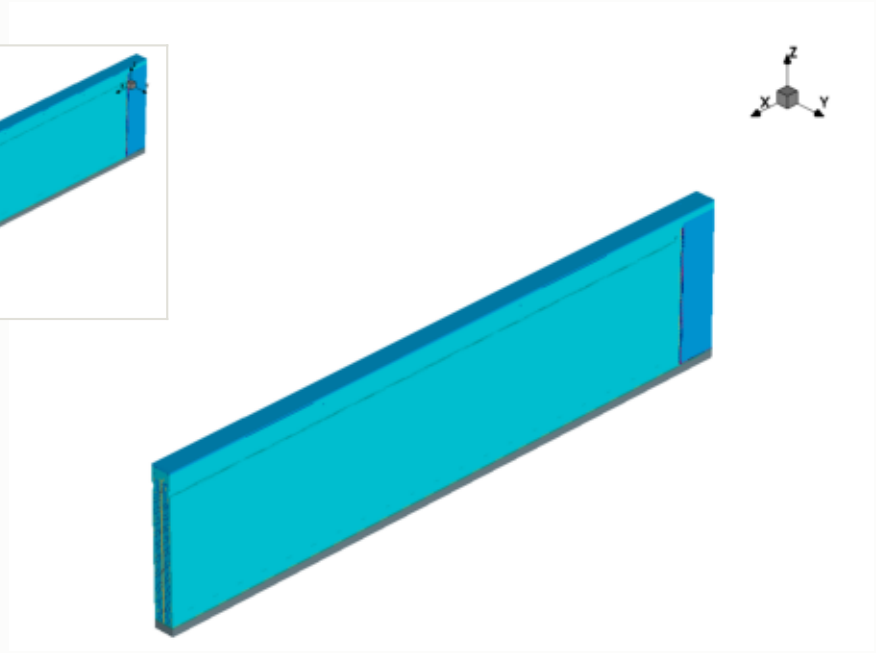
depo_isotropic · Metallization

1.21	755.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



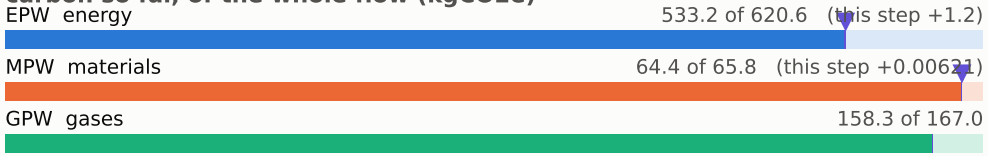


this step

process area	Metallization
material	W
thickness	0.06 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

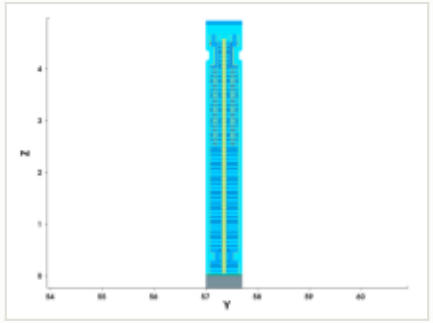
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00621 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

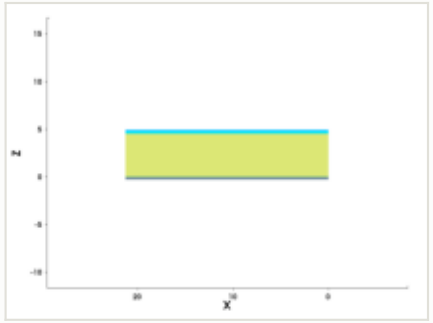


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



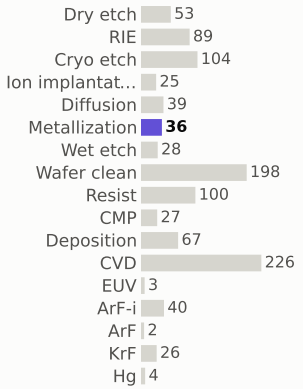
y cut at 57.351 µm



mask for this step

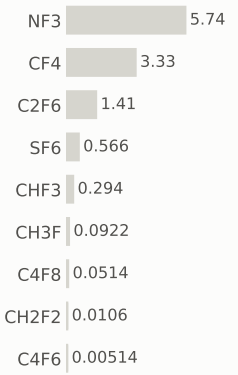
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



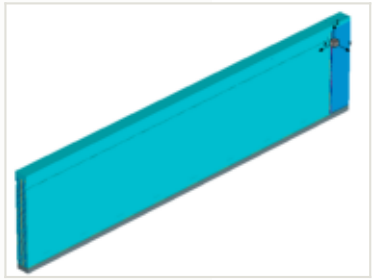
Step #12380: CMP

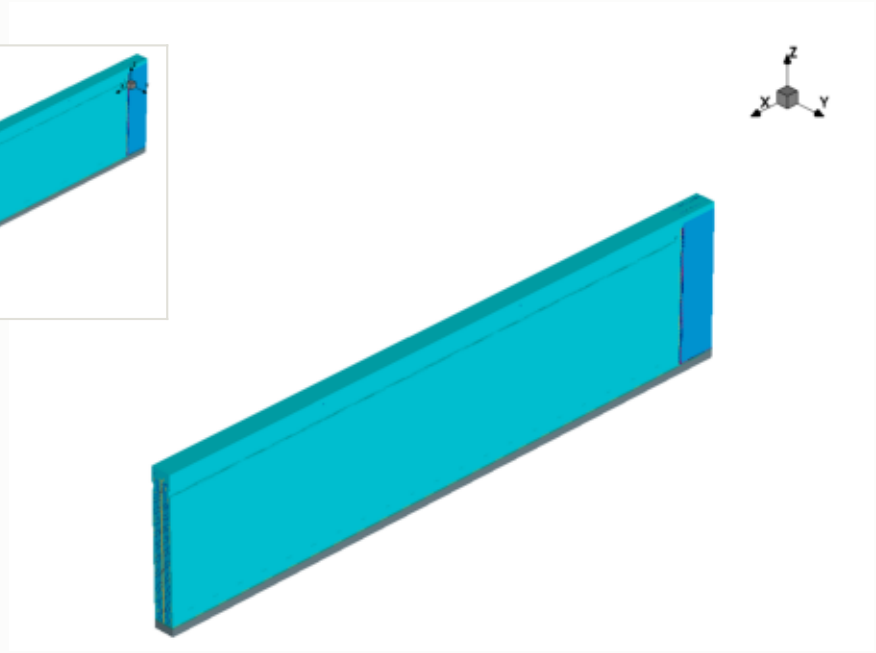
etch_cmp · CMP

0.169	756.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



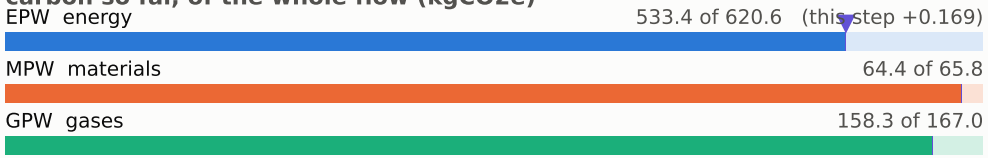


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

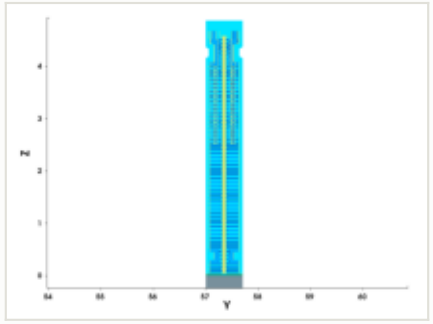
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

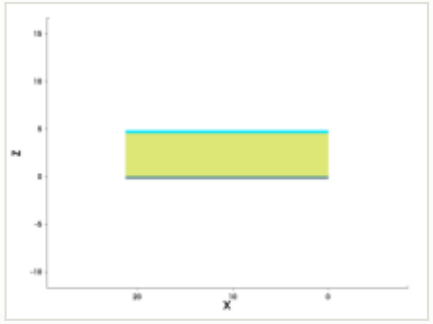


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



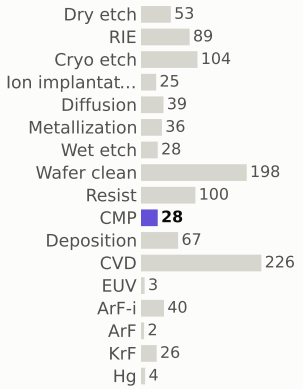
y cut at 57.351 µm



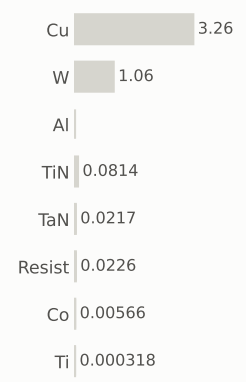
mask for this step

no mask for this step

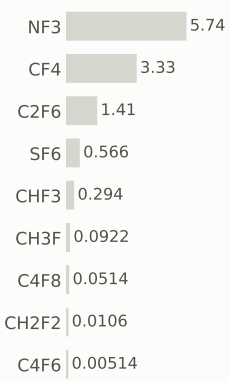
steps by process area



materials deposited (g)



gases emitted (g)



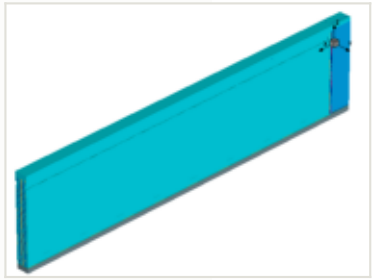
Step #12390: Wafer_clean

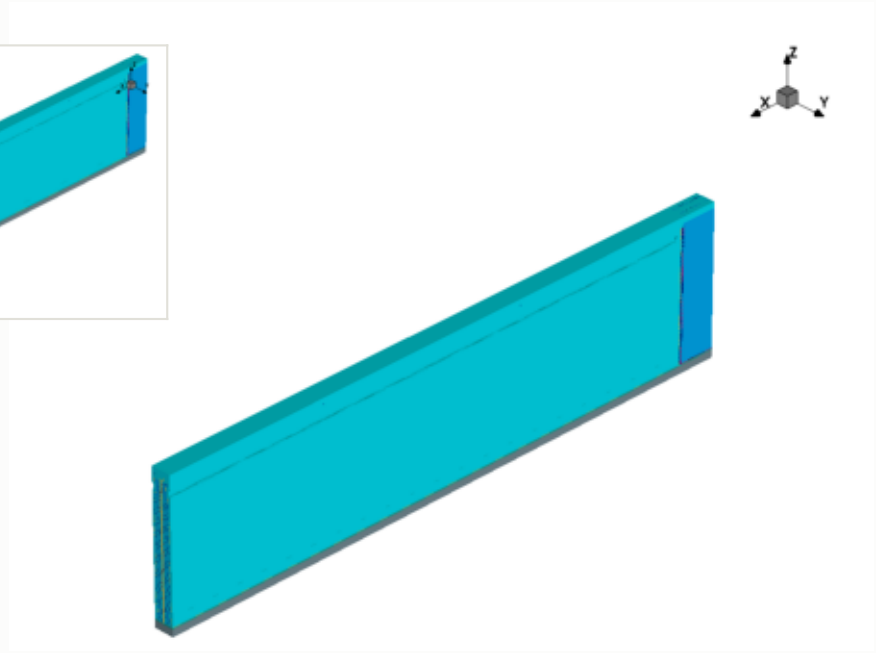
clean · Wafer_clean

0.169	756.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





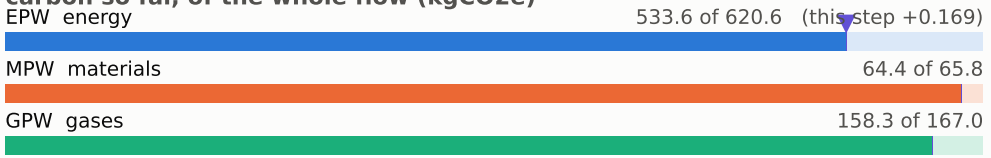
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

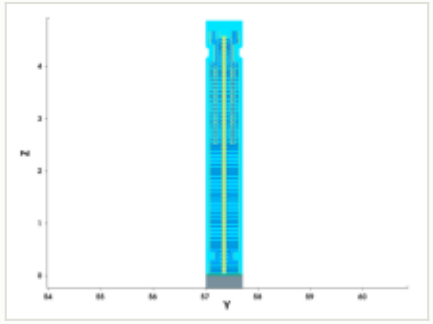
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

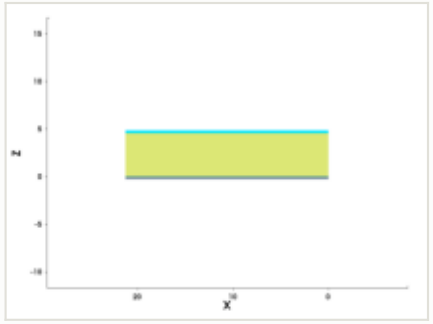


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



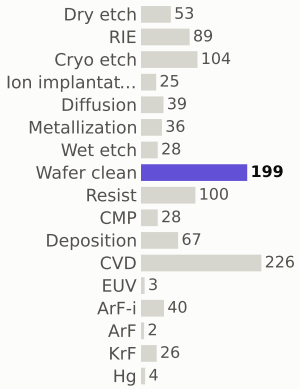
y cut at 57.351 µm



mask for this step

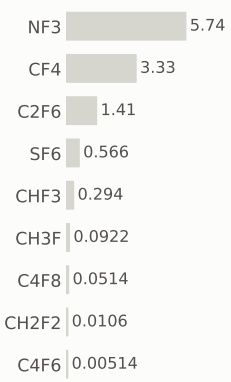
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



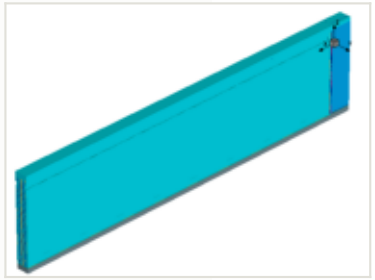
Step #12410: Wafer_clean

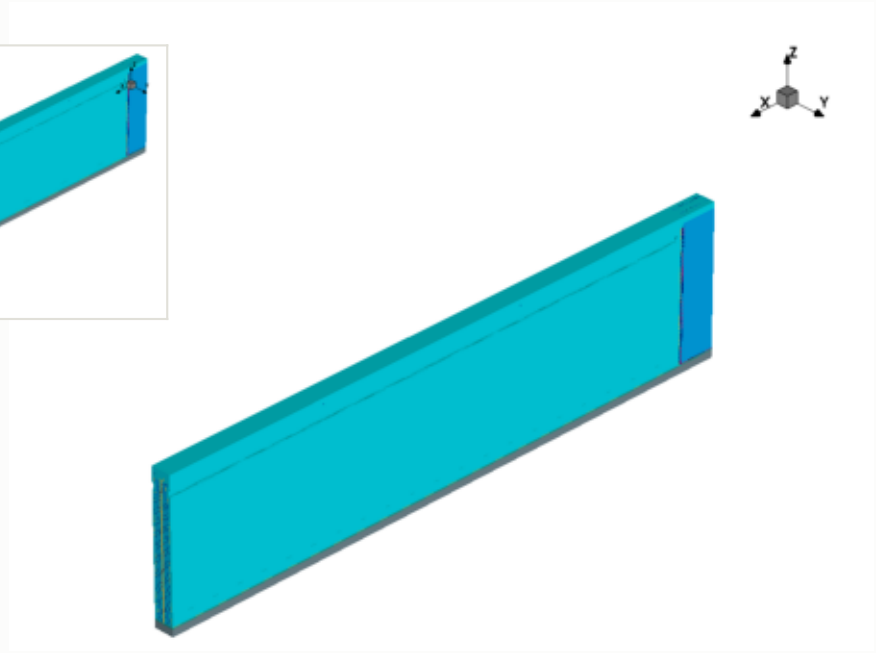
clean · Wafer_clean

0.169	756.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





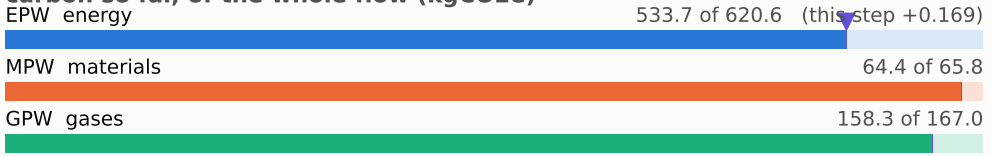
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

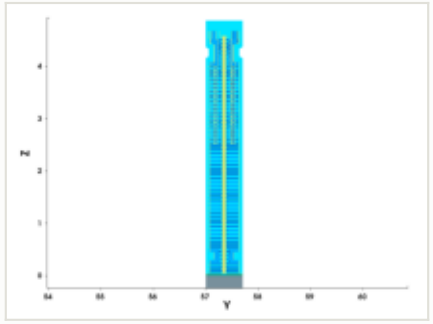
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

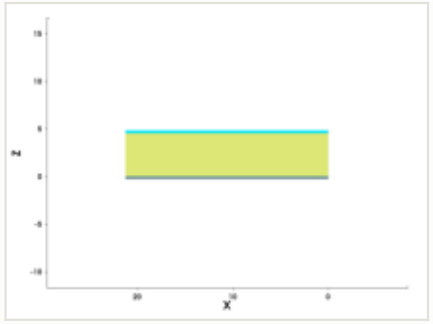


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



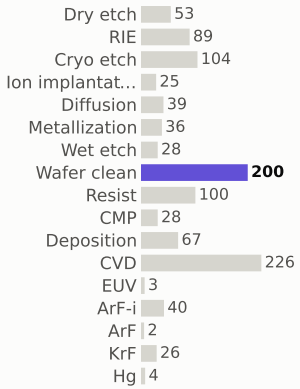
y cut at 57.351 µm



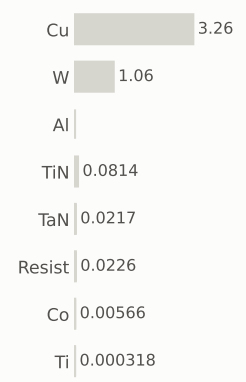
mask for this step

no mask for this step

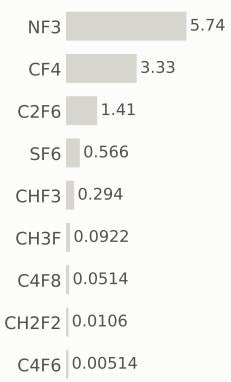
steps by process area



materials deposited (g)



gases emitted (g)



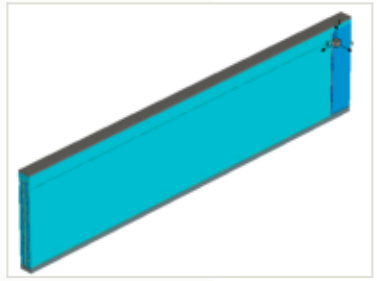
Step #12420: ArF-i

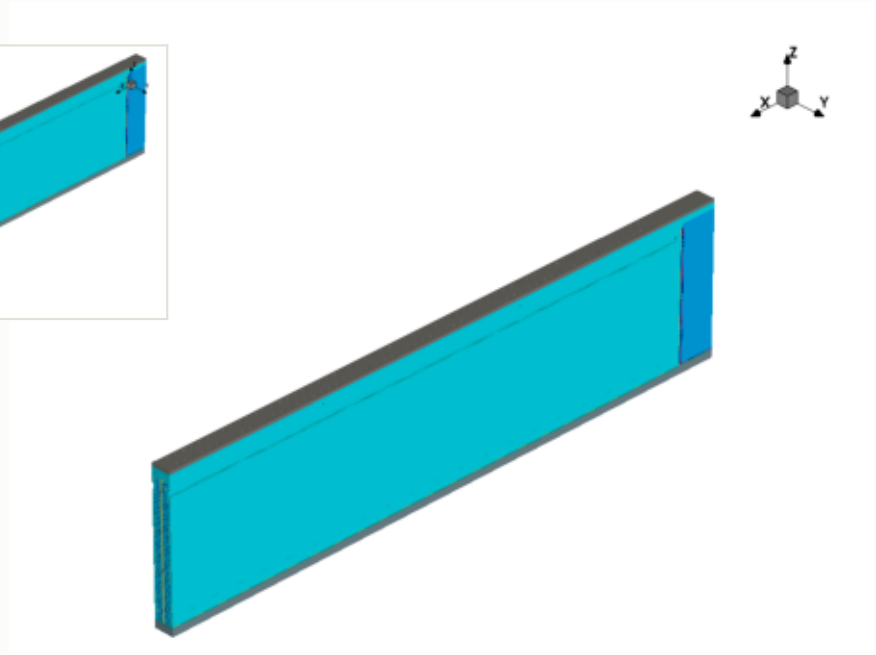
litho_patterning · ArF-i

0.217	756.7	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



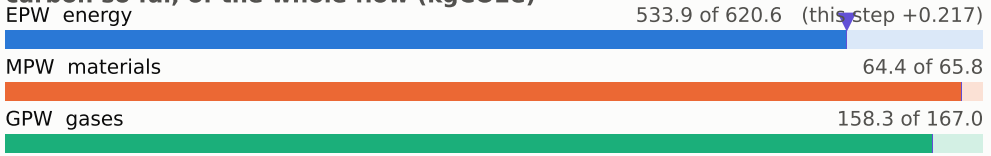


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

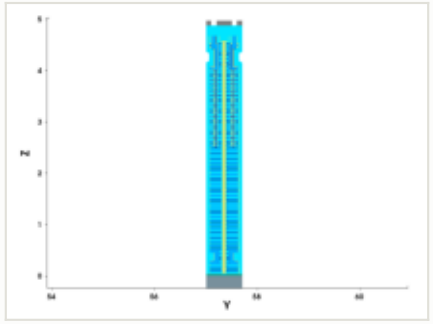
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

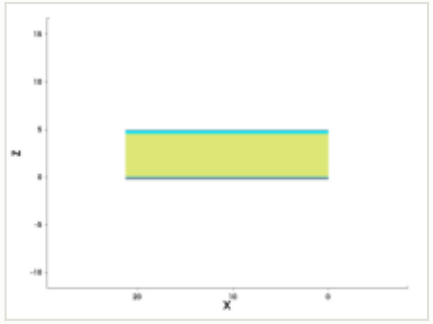


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

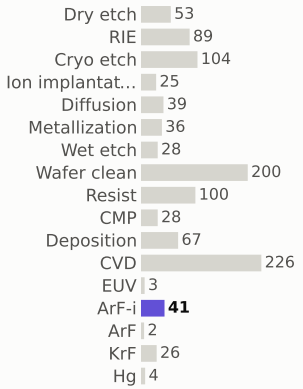


y cut at 57.351 µm

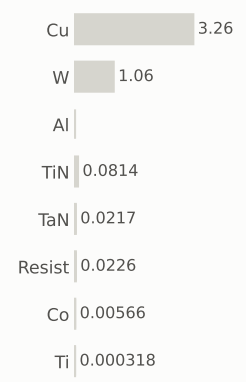


mask: layer 15:0, negative (GDSII)

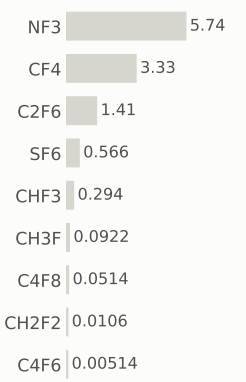
steps by process area



materials deposited (g)



gases emitted (g)



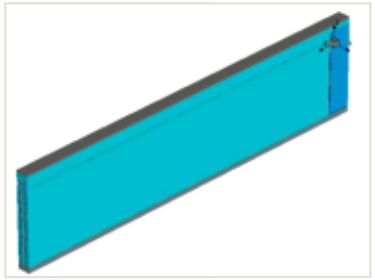
Step #12430: RIE

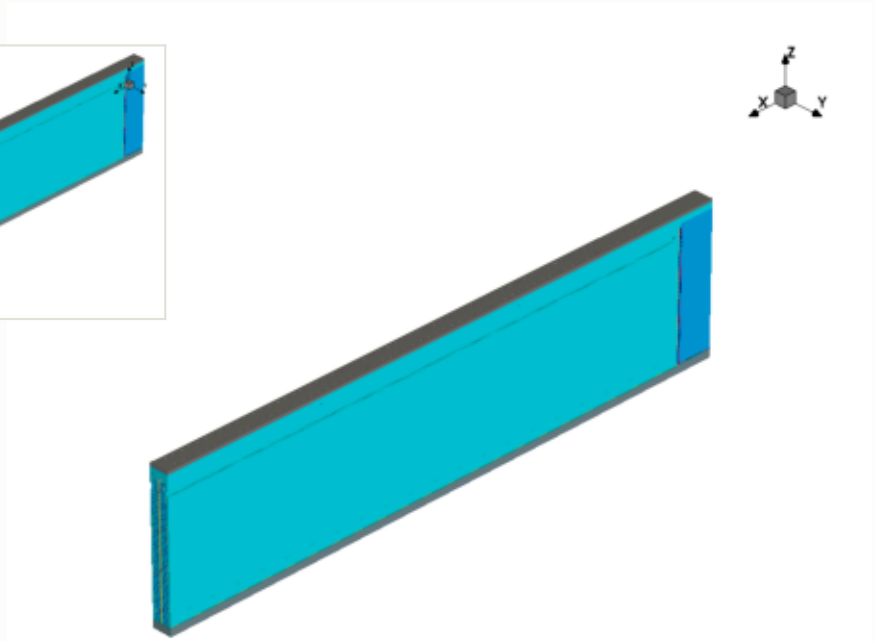
etch_anisotropic · RIE

1.04	757.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



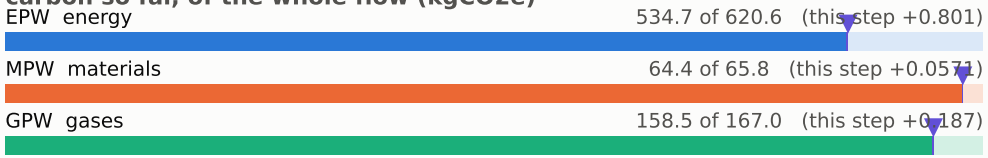


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

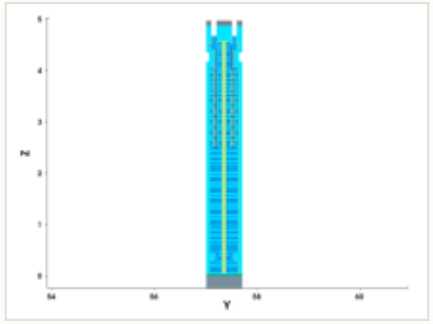
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

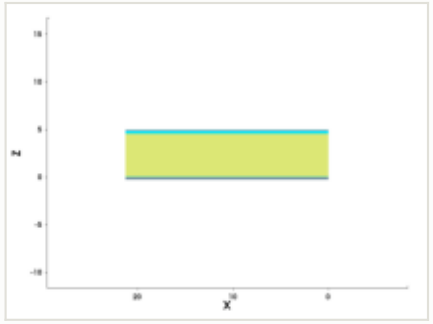


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



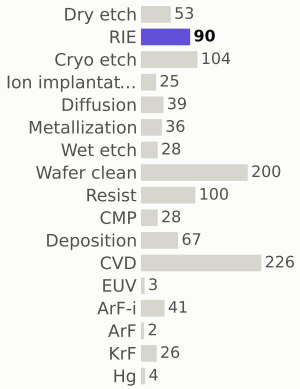
y cut at 57.351 µm



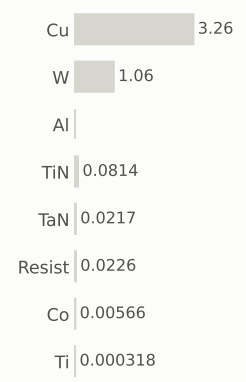
mask for this step

no mask for this step

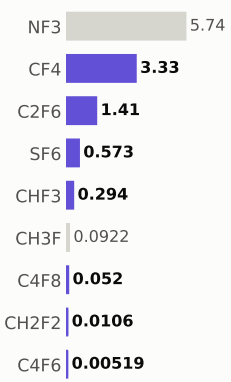
steps by process area



materials deposited (g)



gases emitted (g)



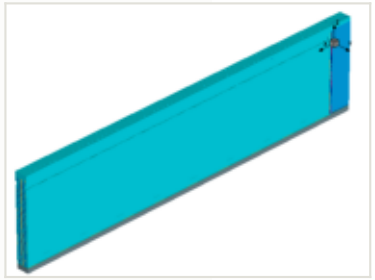
Step #12440: Dry_etch

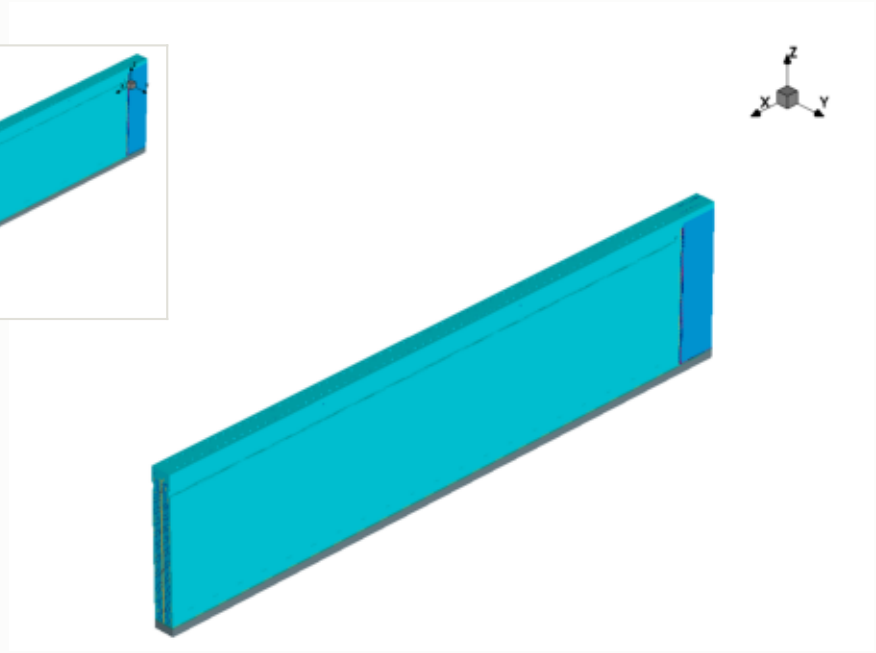
etch_strip · Dry_etch

0.801	758.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



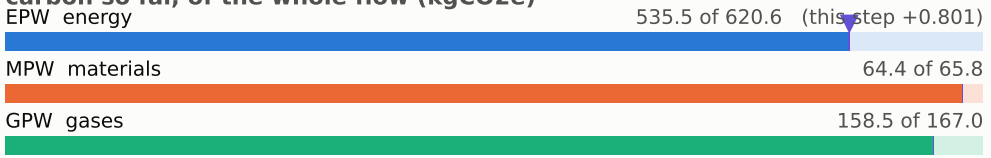


this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

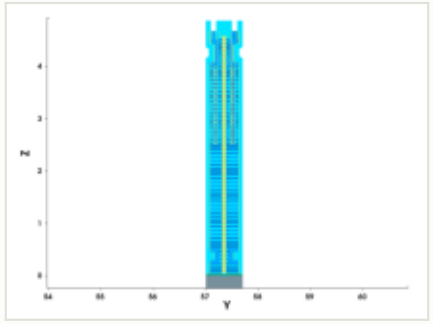
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

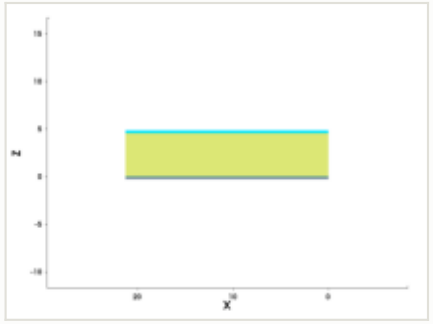


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



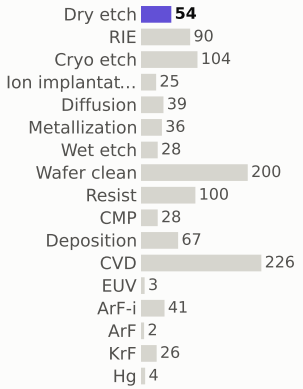
y cut at 57.351 µm



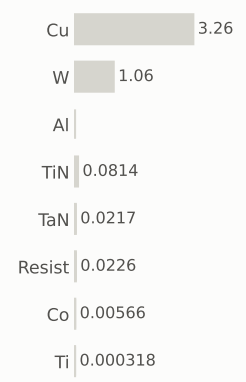
mask for this step

no mask for this step

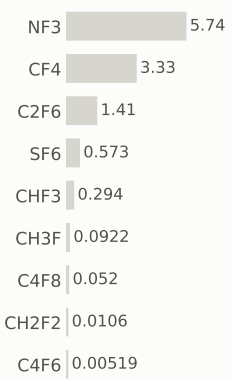
steps by process area



materials deposited (g)



gases emitted (g)



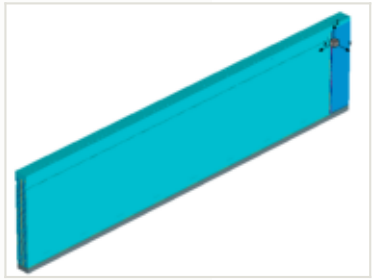
Step #12450: Wafer_clean

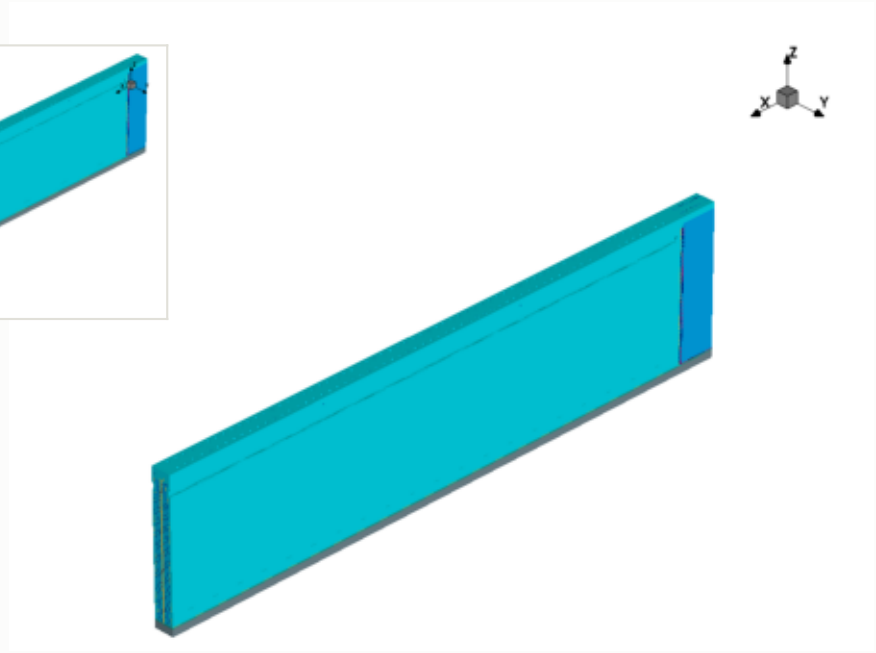
clean · Wafer_clean

0.169	758.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





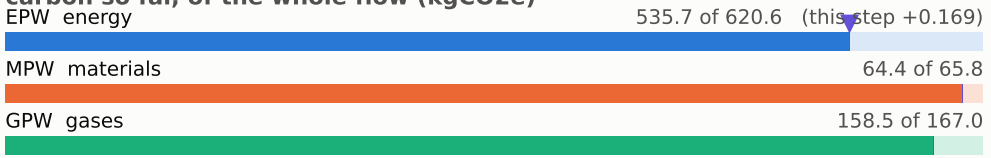
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

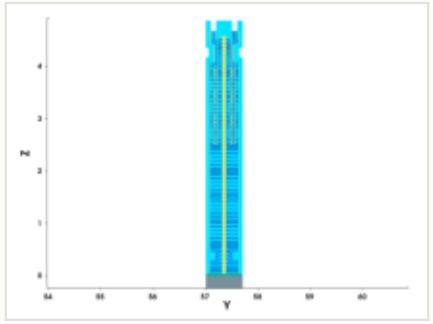
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

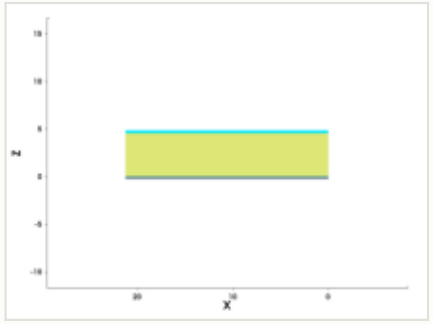


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



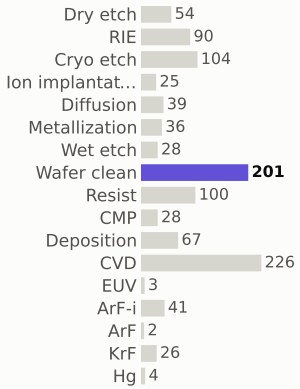
y cut at 57.351 µm



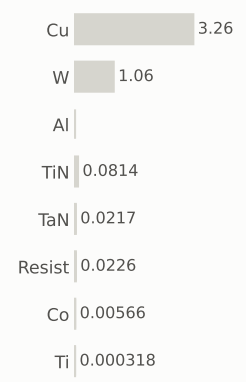
mask for this step

no mask for this step

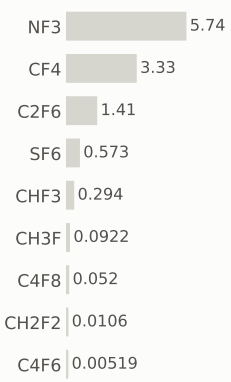
steps by process area



materials deposited (g)



gases emitted (g)



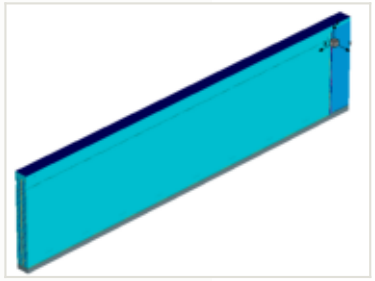
Step #12460: Metallization

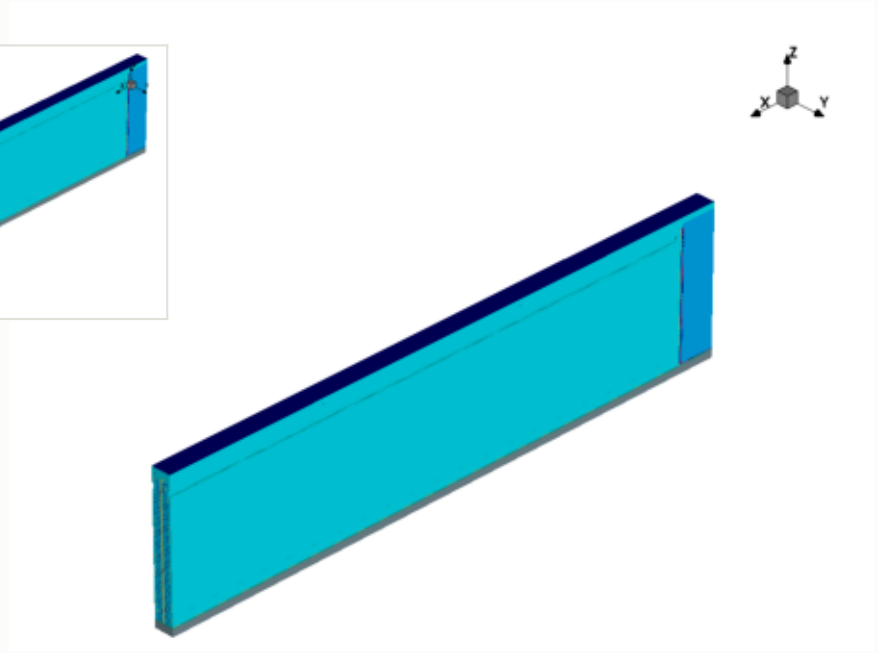
depo_isotropic · Metallization

1.2	759.9	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



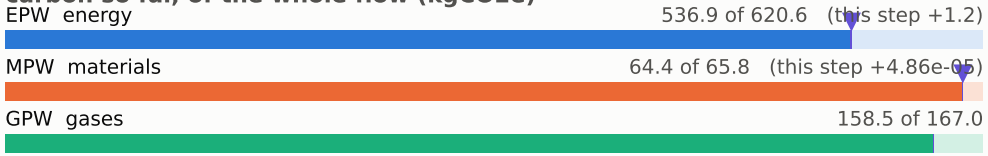


this step

process area	Metallization
material	TiN
thickness	0.003 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

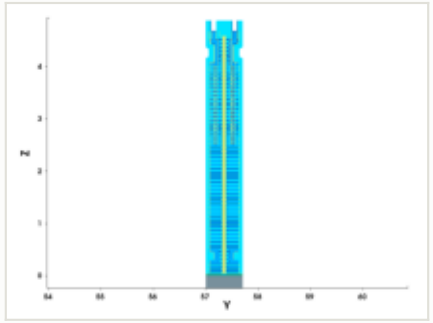
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.86e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

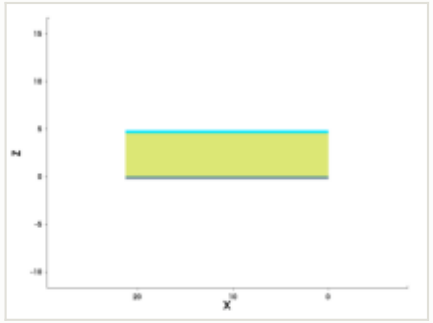


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



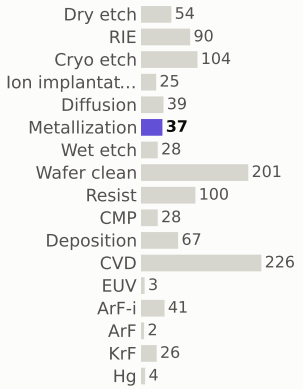
y cut at 57.351 µm



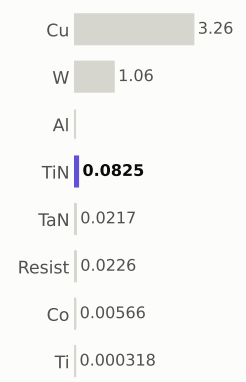
mask for this step

no mask for this step

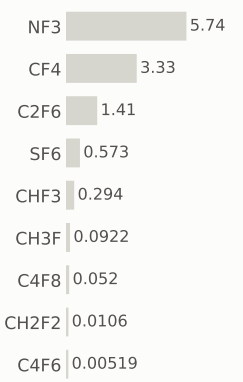
steps by process area



materials deposited (g)



gases emitted (g)



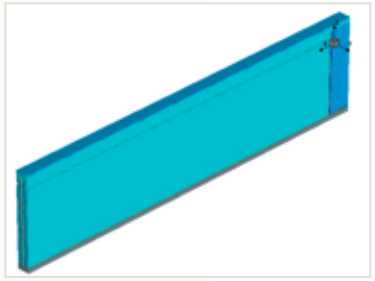
Step #12470: Metallization

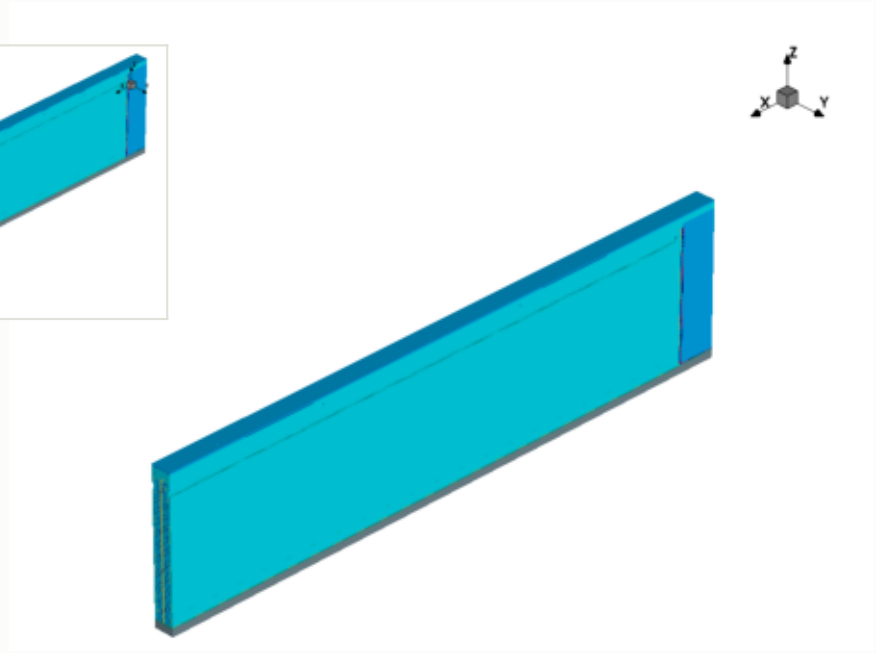
depo_isotropic · Metallization

1.21	761.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



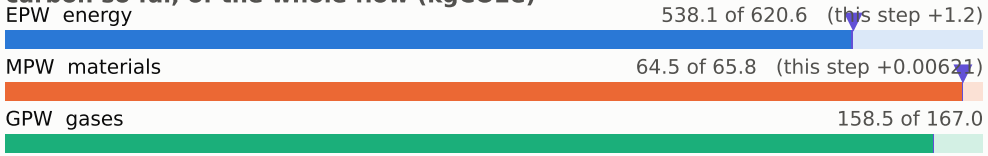


this step

process area	Metallization
material	W
thickness	0.06 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

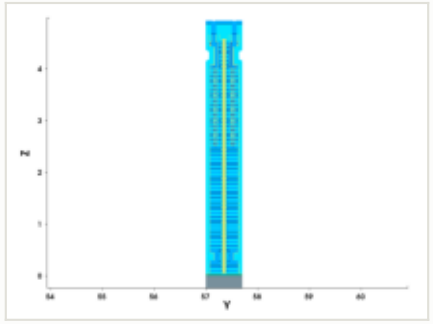
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00621 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

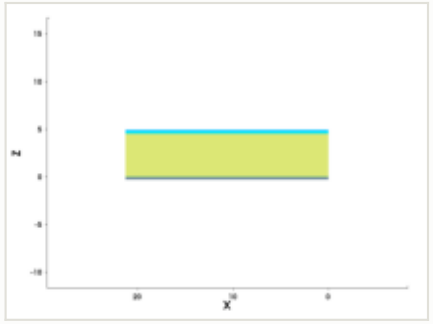


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



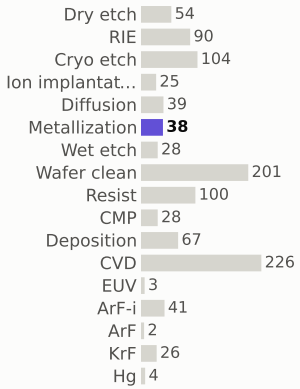
y cut at 57.351 µm



mask for this step

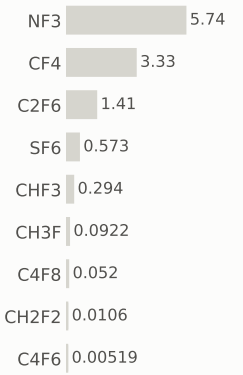
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



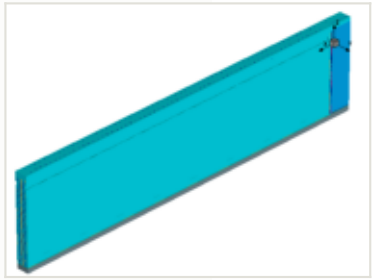
Step #12480: CMP

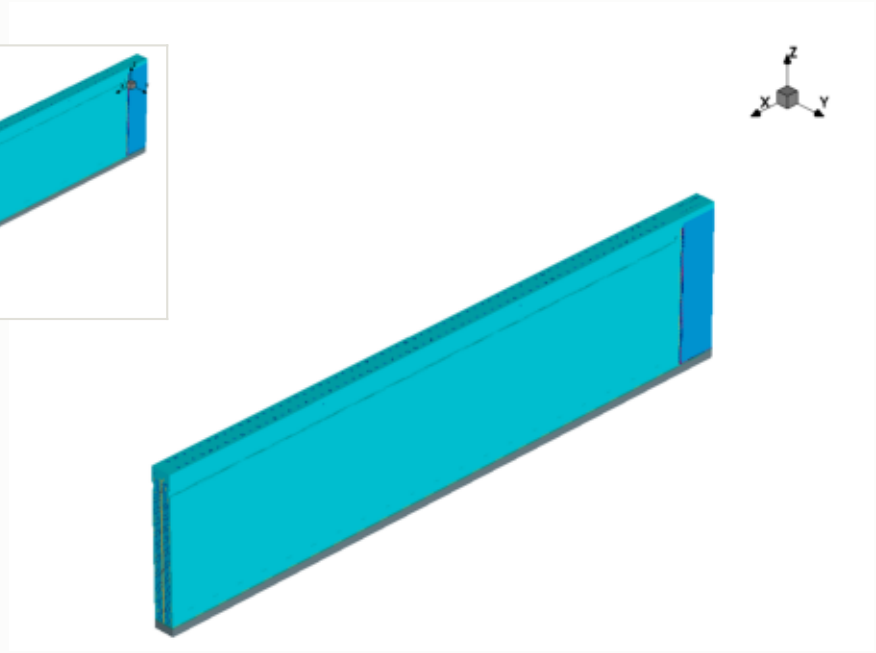
etch_cmp · CMP

0.169	761.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



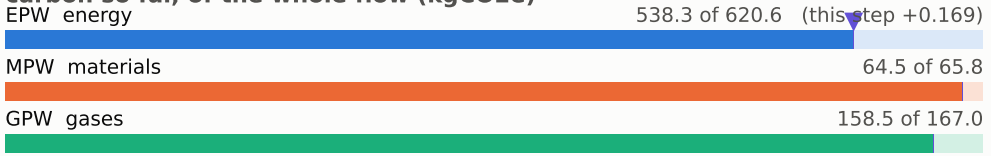


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

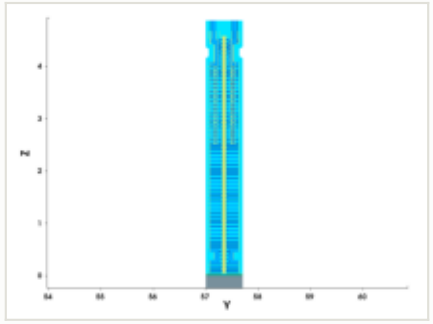
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

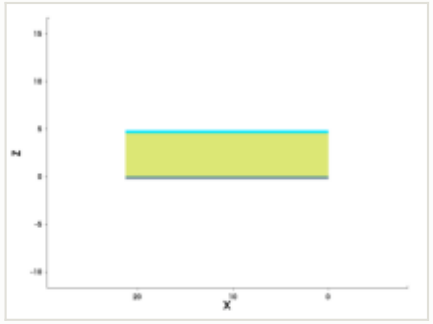


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



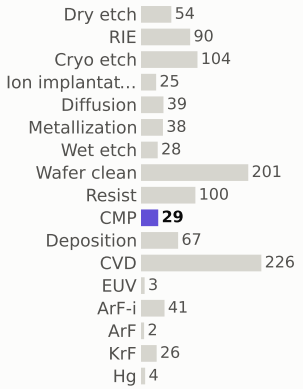
y cut at 57.351 µm



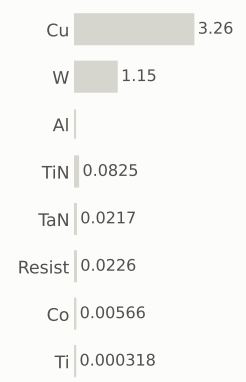
mask for this step

no mask for this step

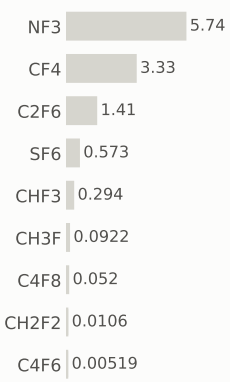
steps by process area



materials deposited (g)



gases emitted (g)



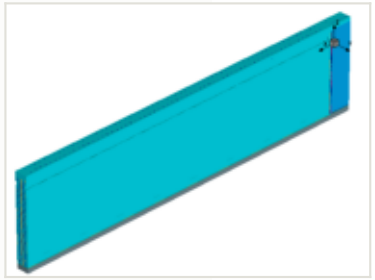
Step #12490: Wafer_clean

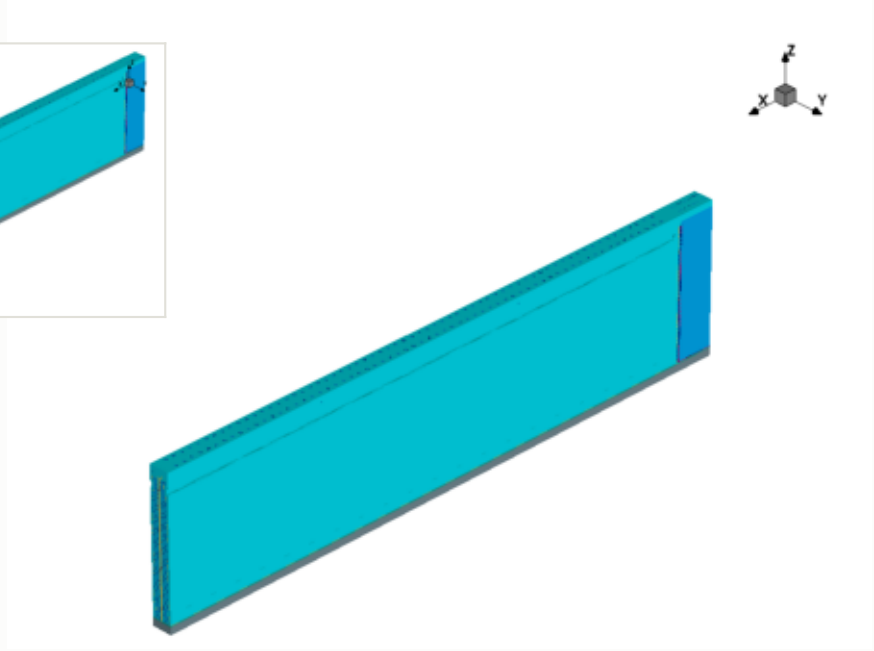
clean · Wafer_clean

0.169	761.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





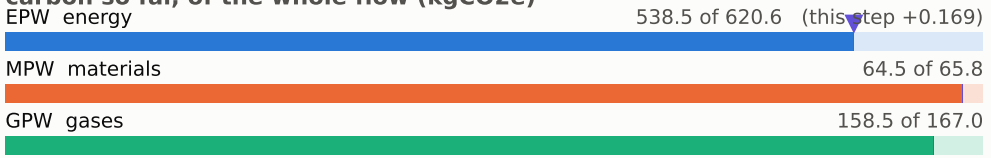
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

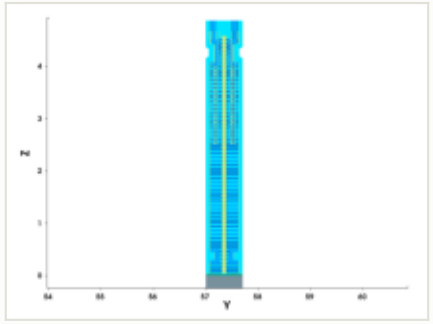
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

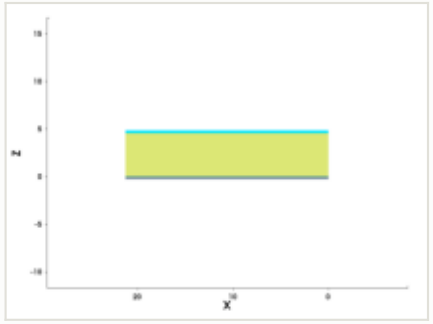


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



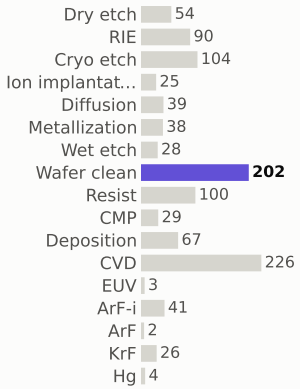
y cut at 57.351 µm



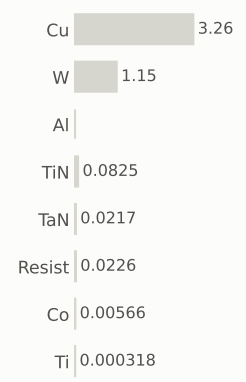
mask for this step

no mask for this step

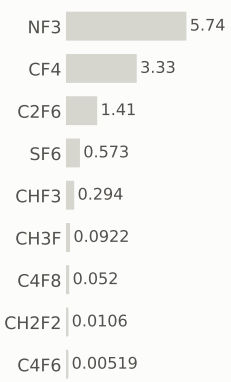
steps by process area



materials deposited (g)



gases emitted (g)



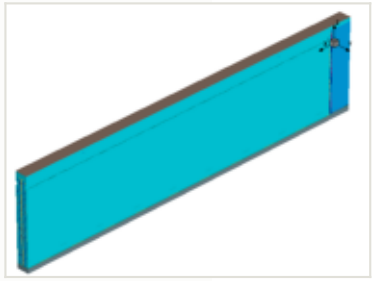
Step #12510: Deposition

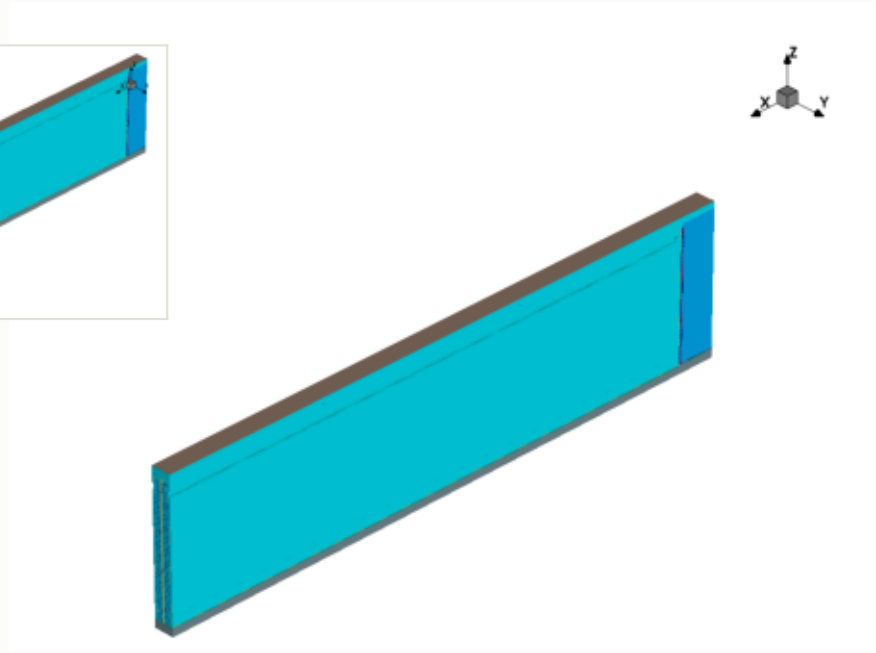
depo_isotropic · Deposition

0.656	762.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

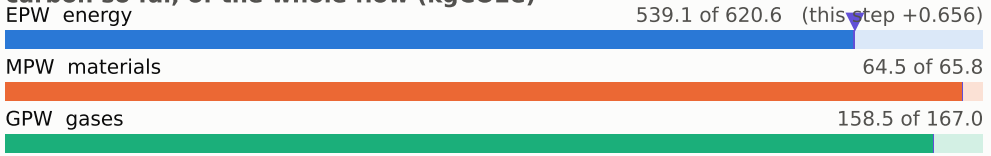




this step

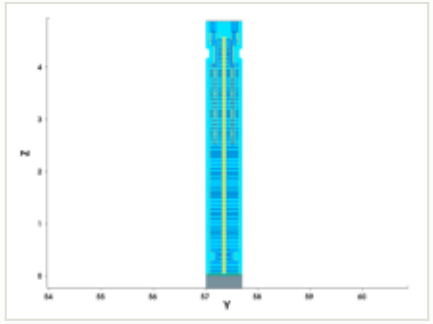
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

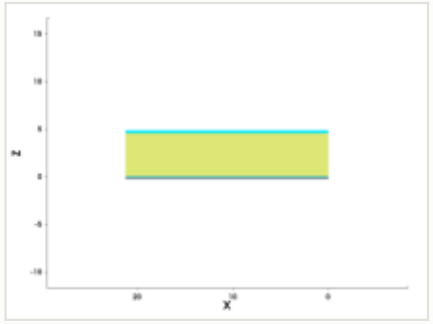


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



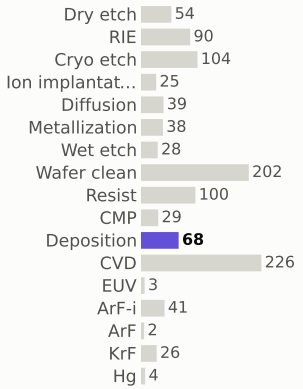
y cut at 57.351 µm



mask for this step

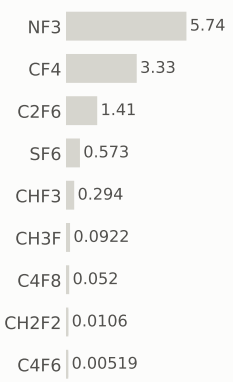
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



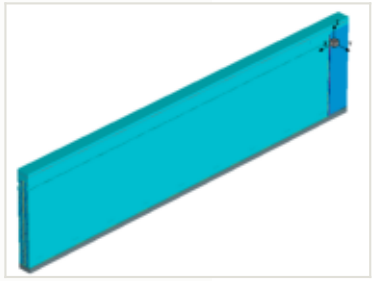
Step #12520: Deposition

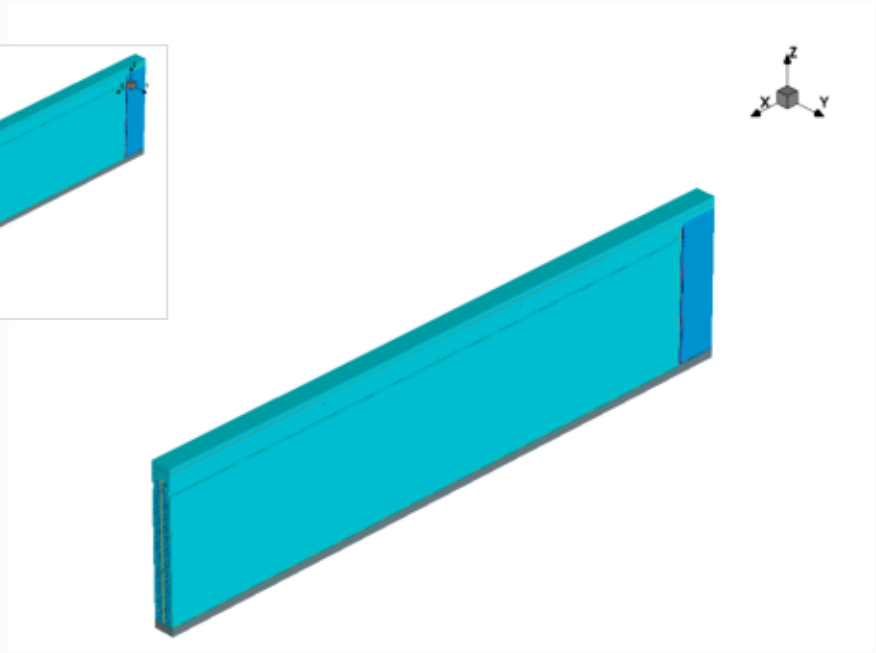
depo_isotropic · Deposition

0.656	762.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



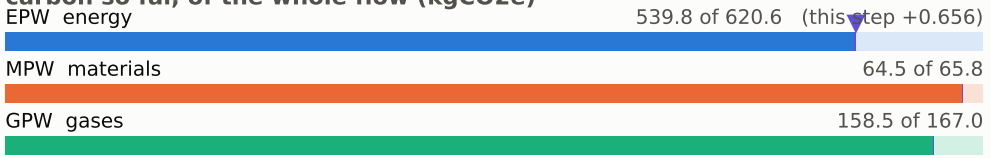


this step

process area	Deposition
material	—
thickness	0.15 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

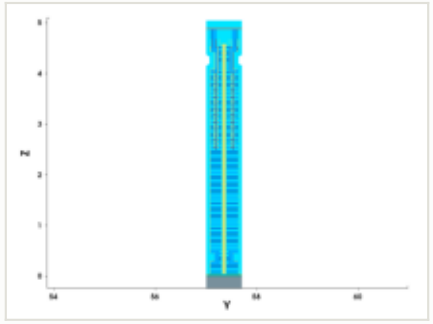
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

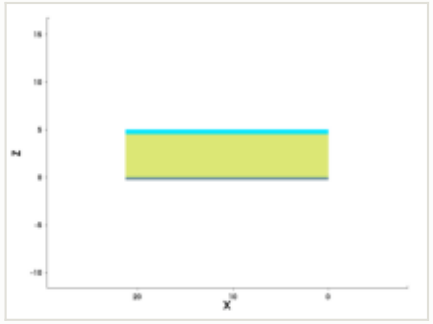


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



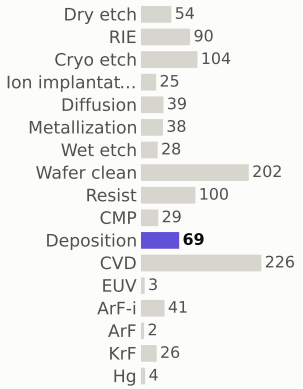
y cut at 57.351 µm



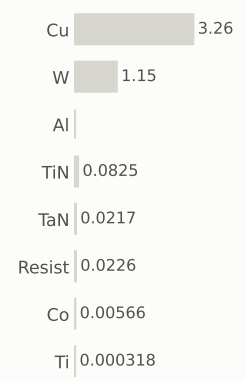
mask for this step

no mask for this step

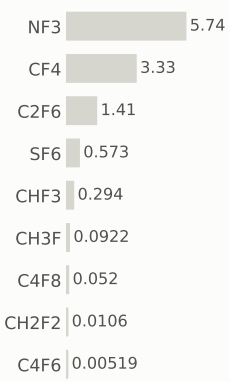
steps by process area



materials deposited (g)



gases emitted (g)



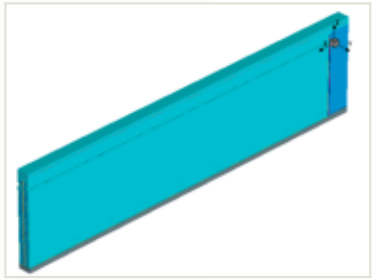
Step #12530: Wafer_clean

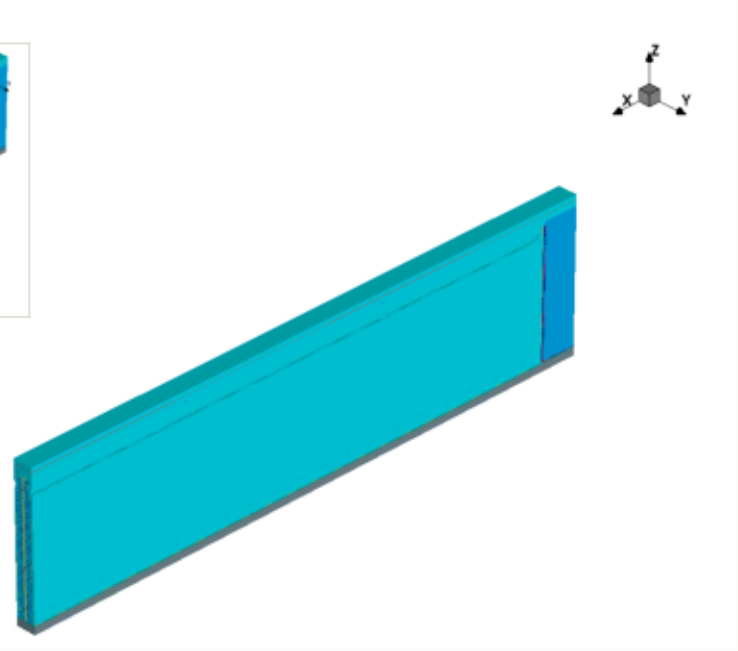
clean · Wafer_clean

0.169	762.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





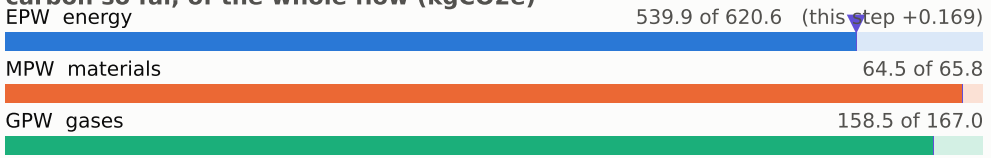
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

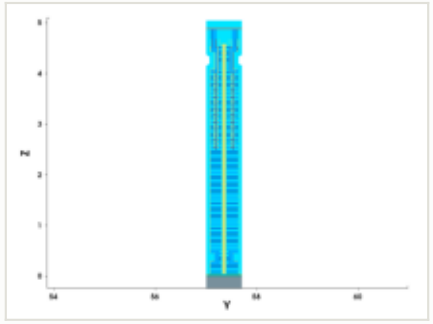
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

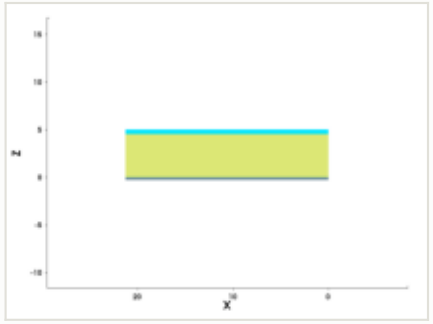


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



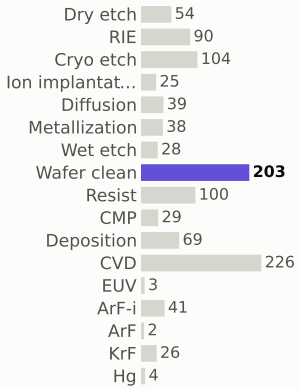
y cut at 57.351 µm



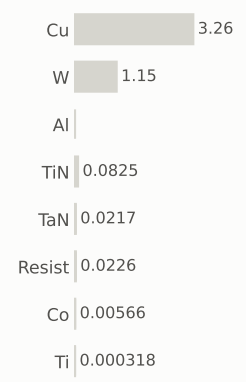
mask for this step

no mask for this step

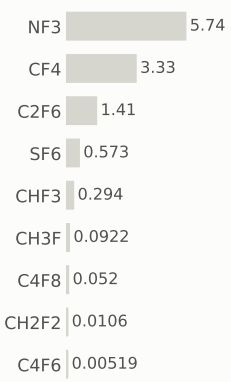
steps by process area



materials deposited (g)



gases emitted (g)



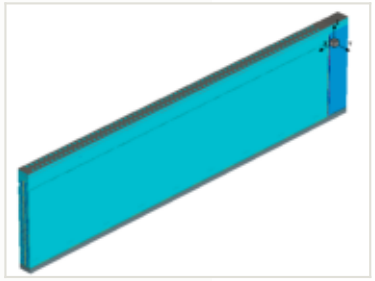
Step #12540: EUV

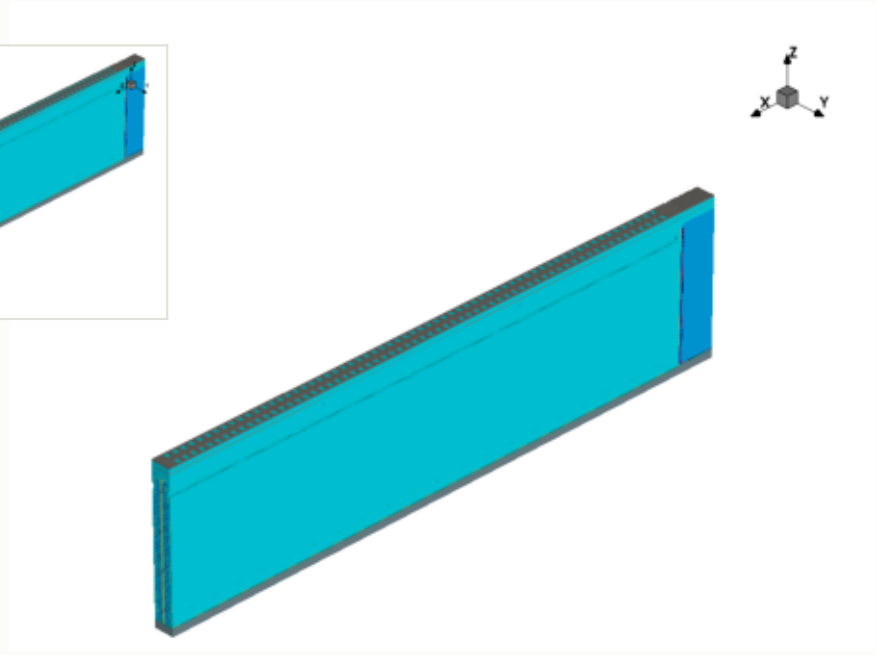
litho_patterning · EUV

1.96	764.9	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



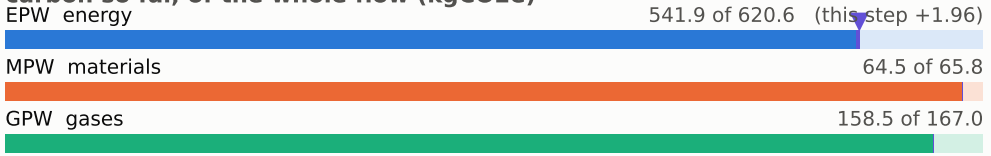


this step

process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

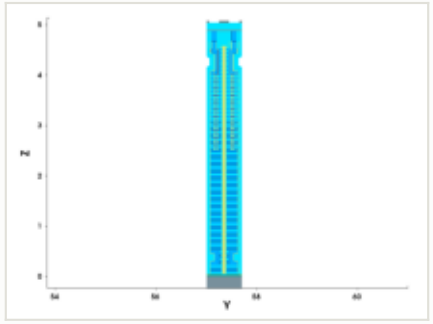
EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

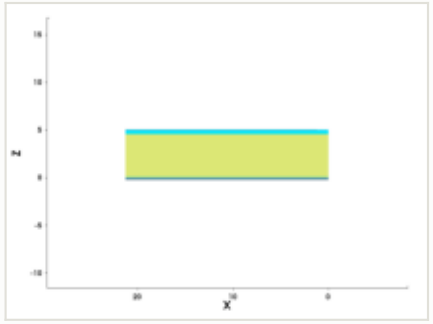


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

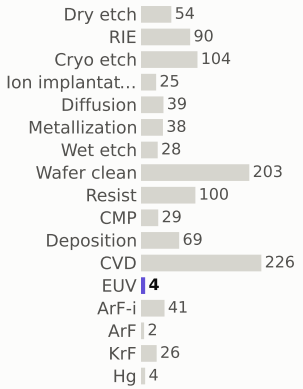


y cut at 57.351 µm

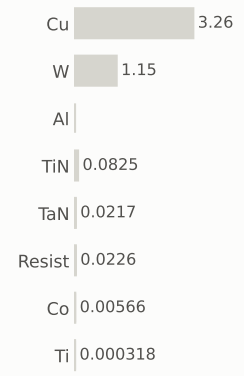


mask: layer 19:0, negative (GDSII)

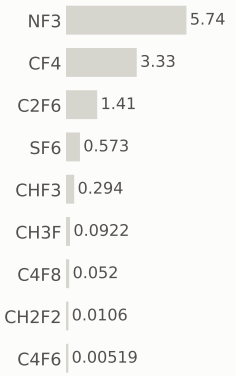
steps by process area



materials deposited (g)



gases emitted (g)



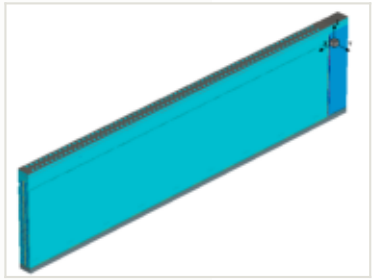
Step #12550: RIE

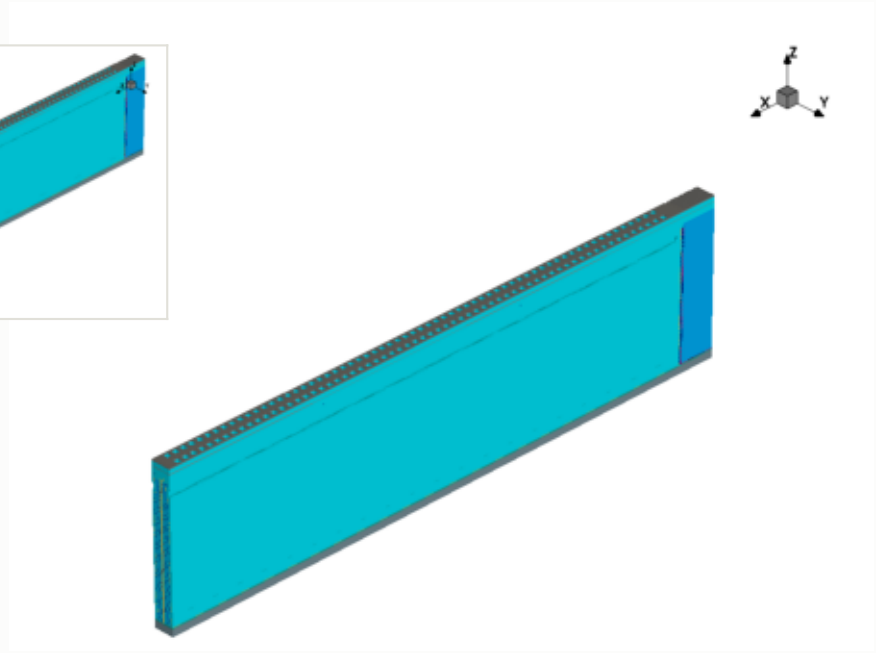
etch_anisotropic · RIE

1.04	765.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



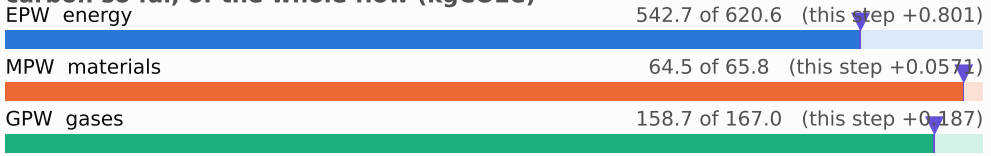


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

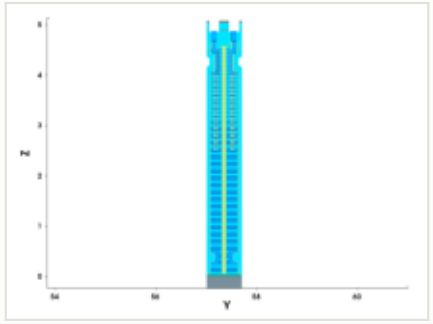
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

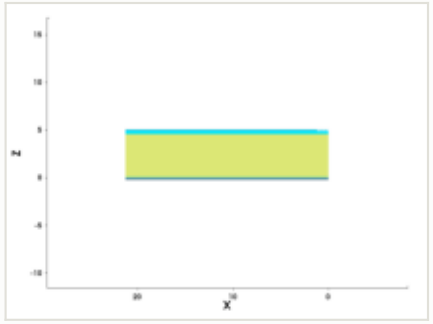


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



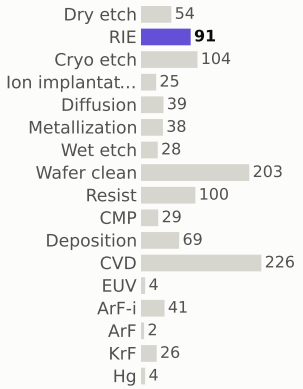
y cut at 57.351 µm



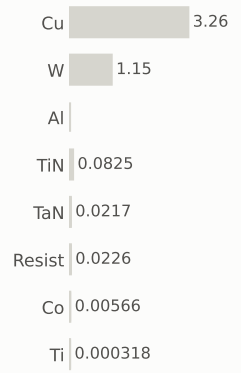
mask for this step

no mask for this step

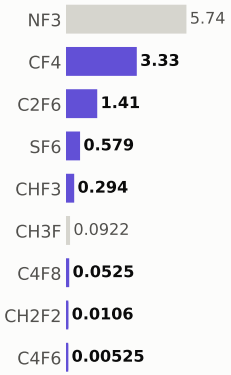
steps by process area



materials deposited (g)



gases emitted (g)



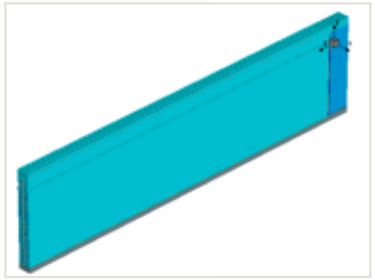
Step #12560: Dry_etch

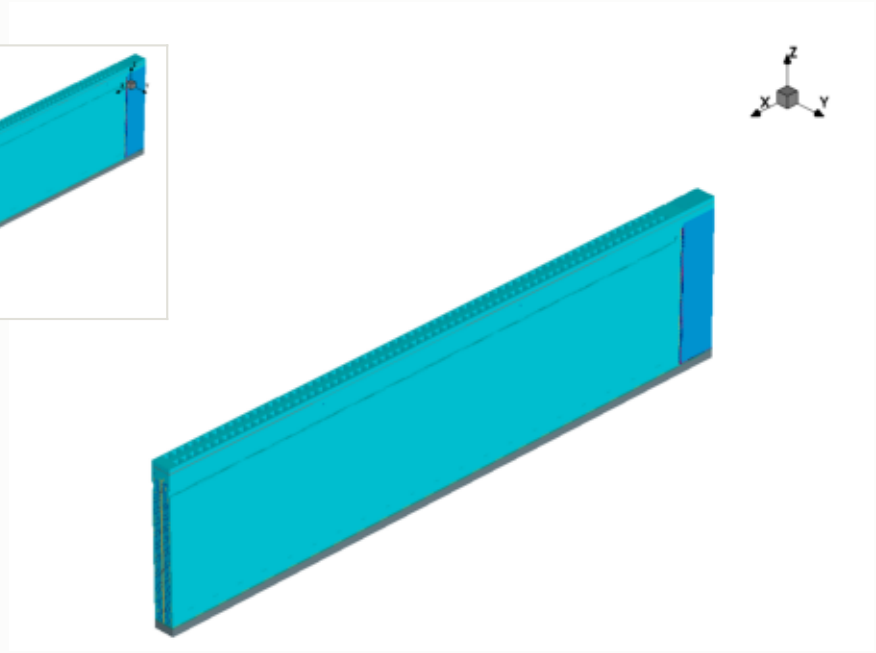
etch_strip · Dry_etch

0.801	766.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

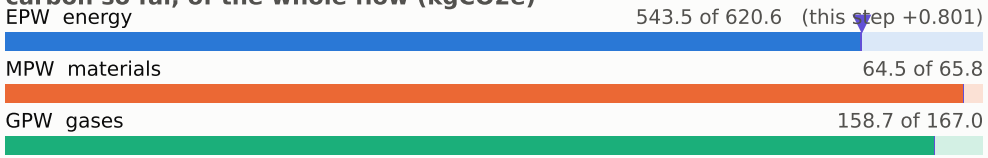




this step

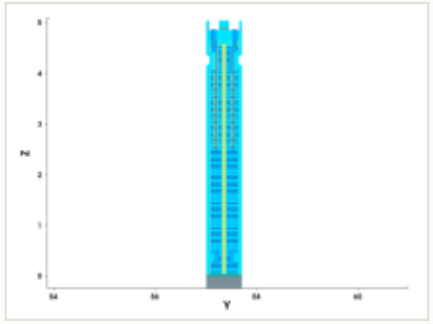
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

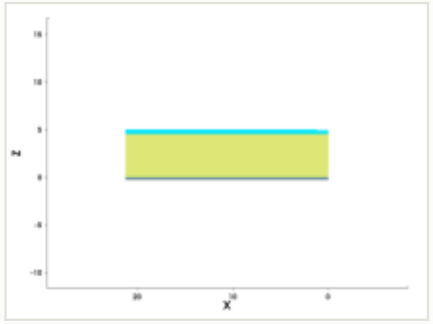


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



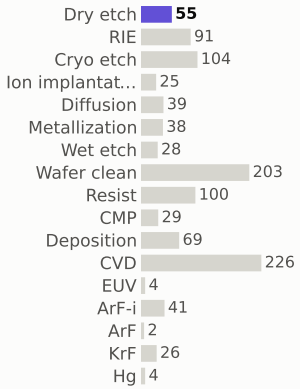
y cut at 57.351 µm



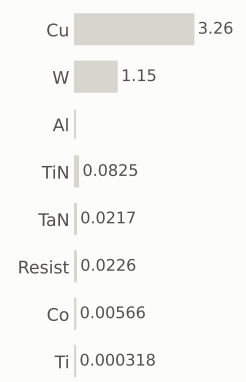
mask for this step

no mask for this step

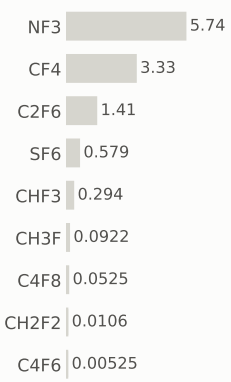
steps by process area



materials deposited (g)



gases emitted (g)



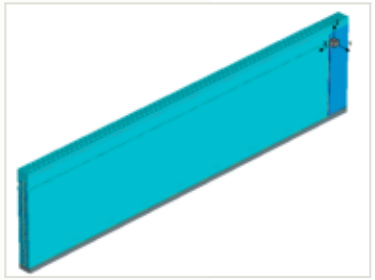
Step #12570: Wafer_clean

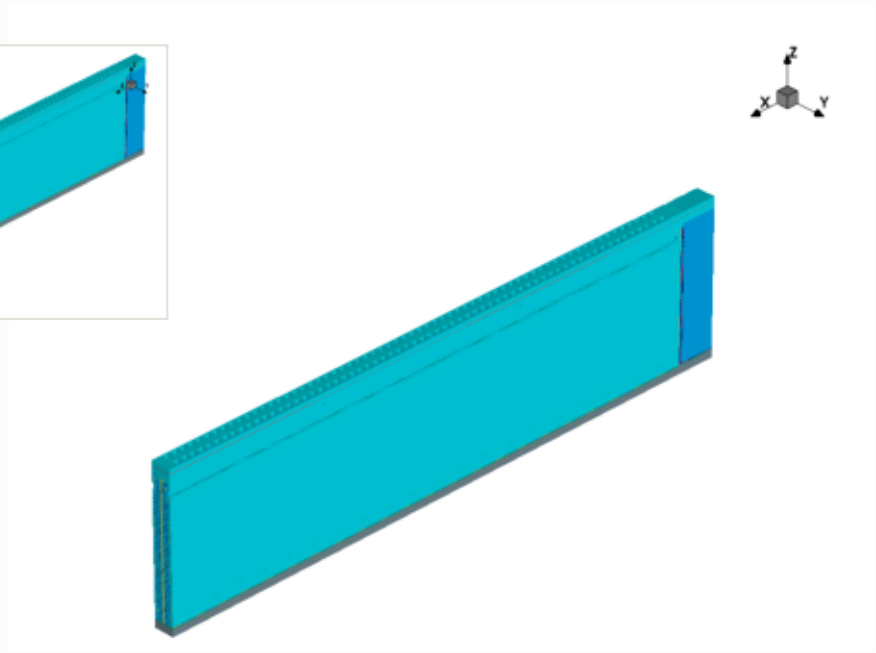
clean · Wafer_clean

0.169	766.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





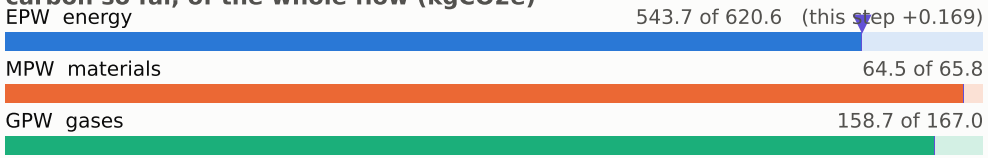
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

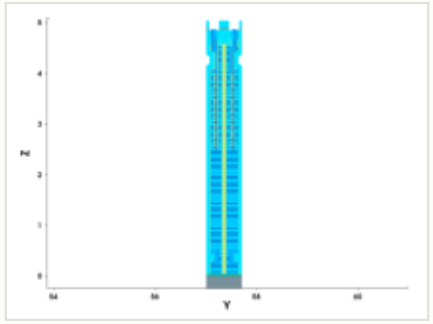
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

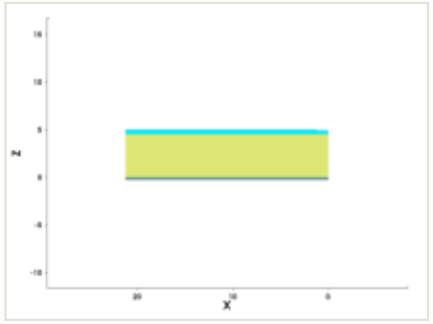


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



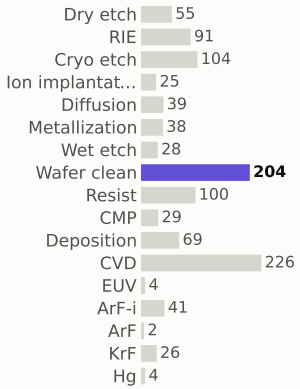
y cut at 57.351 µm



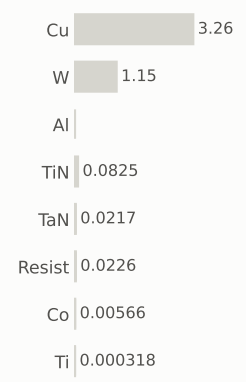
mask for this step

no mask for this step

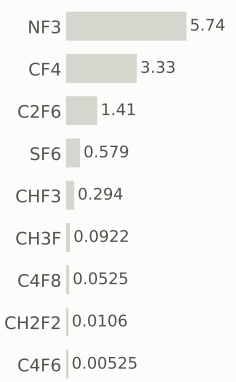
steps by process area



materials deposited (g)



gases emitted (g)



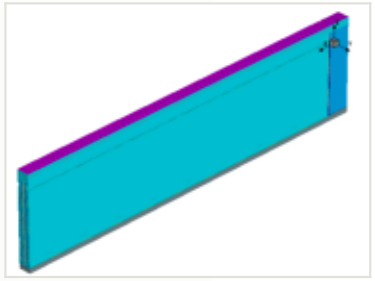
Step #12580: Metallization

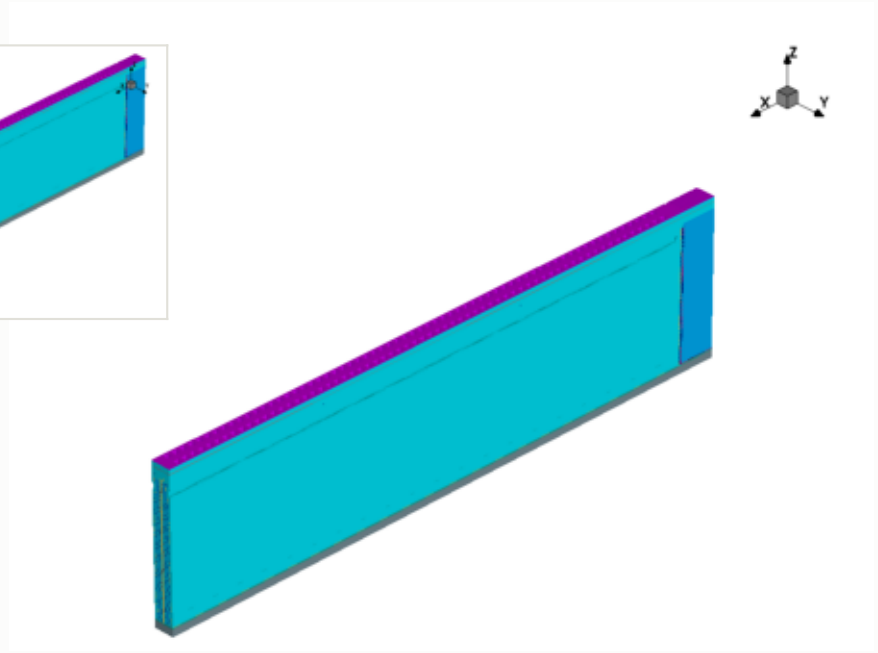
depo_isotropic · Metallization

1.2	768.1	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



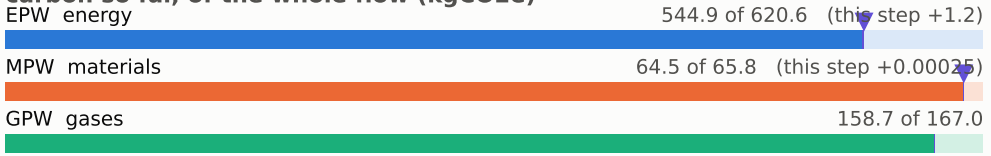


this step

process area	Metallization
material	TaN
thickness	0.01 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

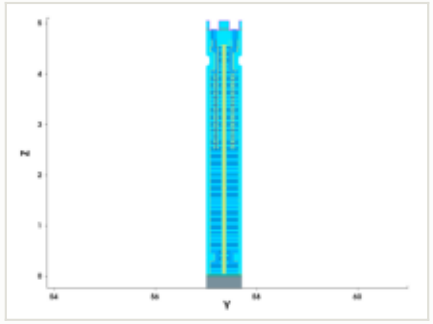
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00025 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

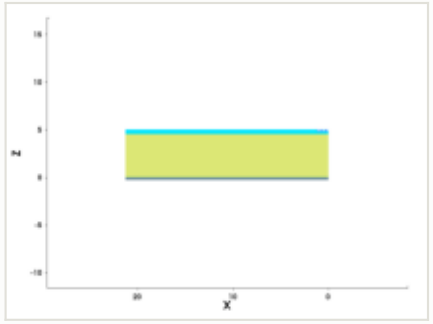


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



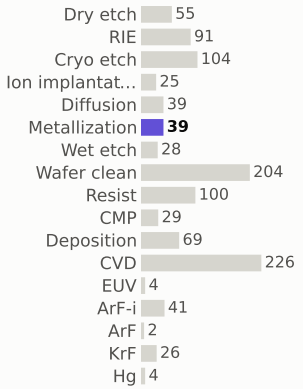
y cut at 57.351 µm



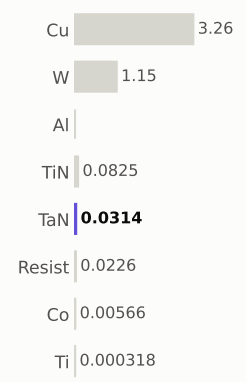
mask for this step

no mask for this step

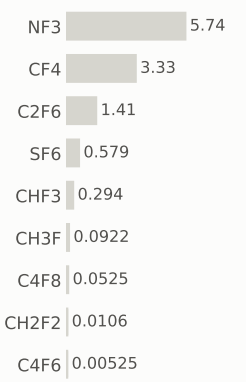
steps by process area



materials deposited (g)



gases emitted (g)



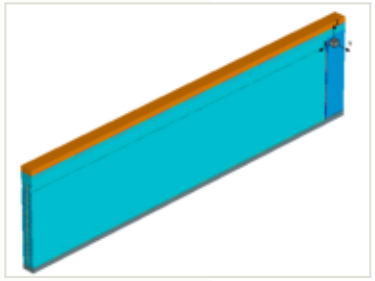
Step #12590: Metallization

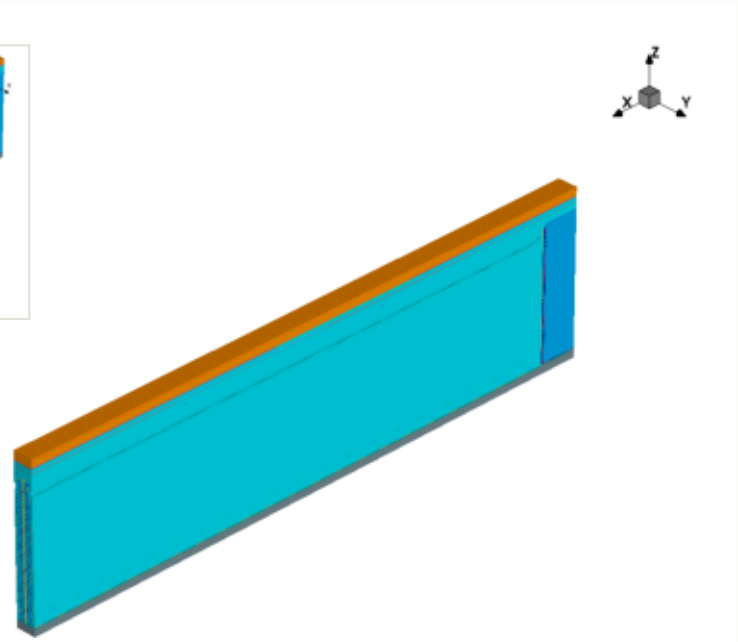
depo_isotropic · Metallization

1.21	769.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



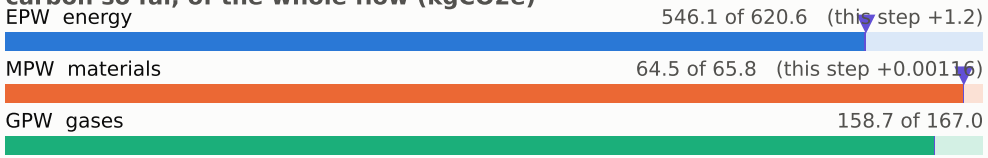


this step

process area	Metallization
material	Cu
thickness	0.3 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

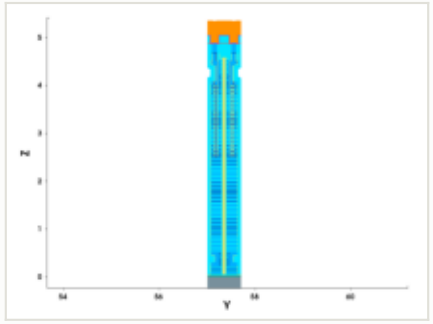
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00116 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

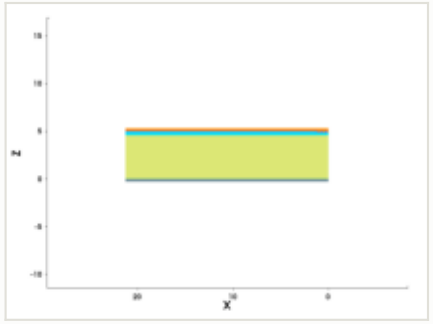


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



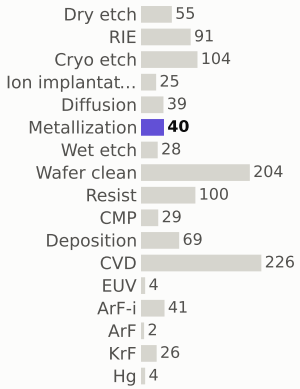
y cut at 57.351 µm



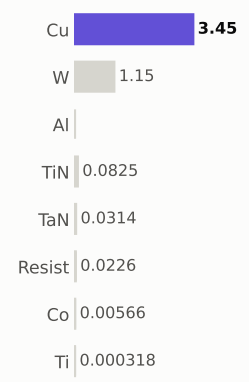
mask for this step

no mask for this step

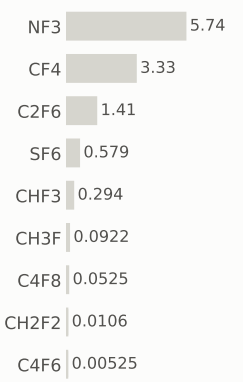
steps by process area



materials deposited (g)



gases emitted (g)



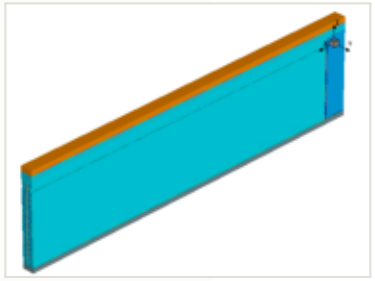
Step #12600: Diffusion

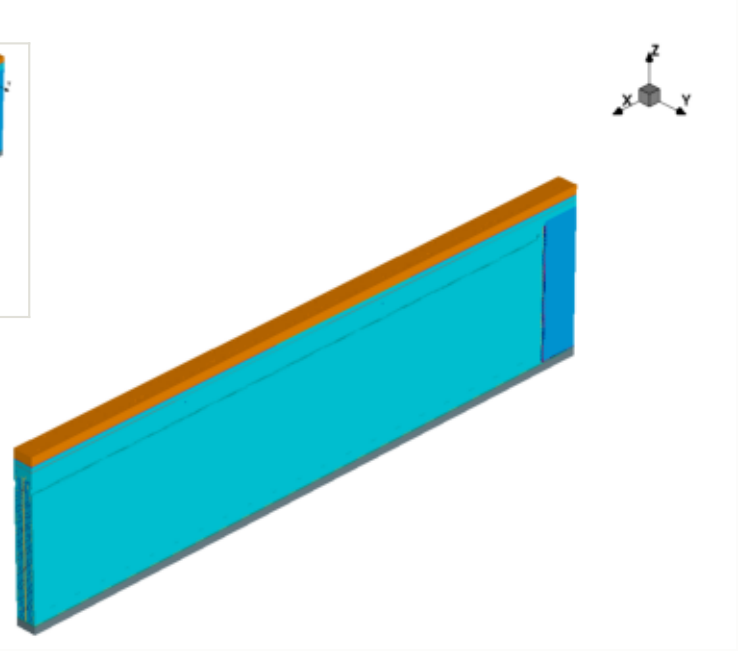
anneal · Diffusion

0.801	770.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





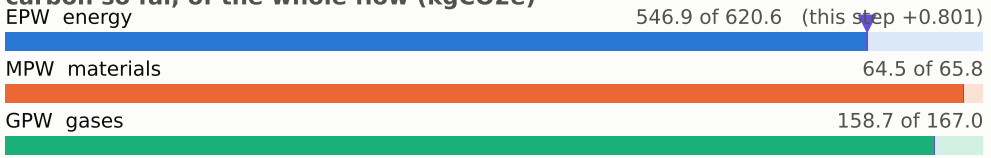
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

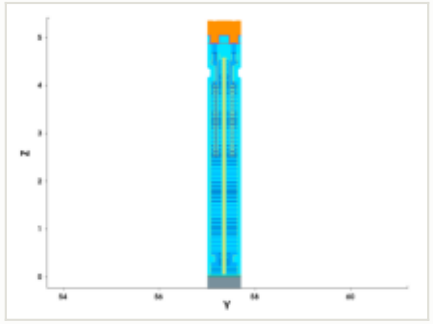
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

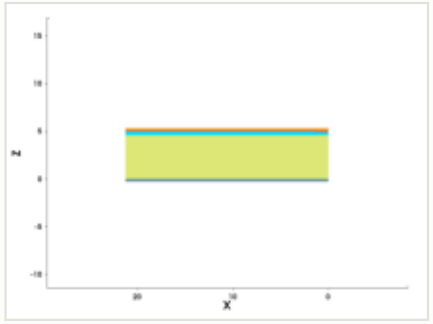


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



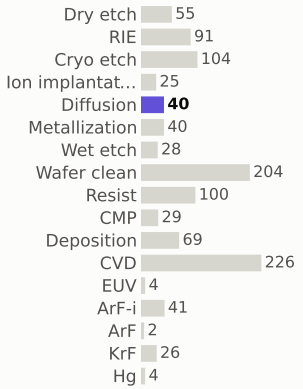
y cut at 57.351 µm



mask for this step

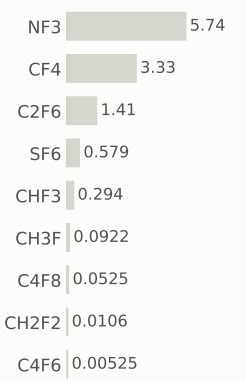
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



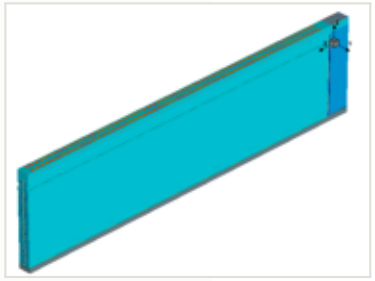
Step #12610: CMP

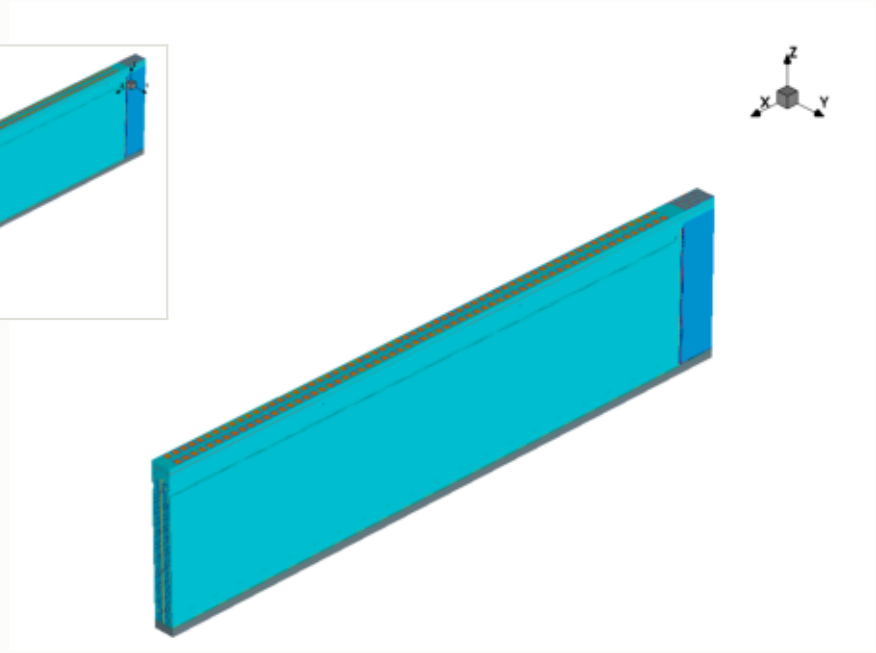
etch_cmp · CMP

0.169	770.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



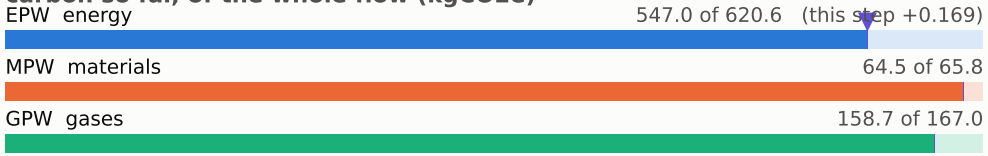


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

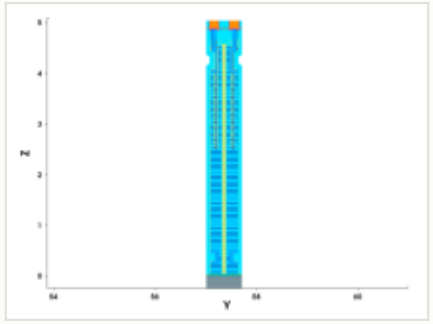
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

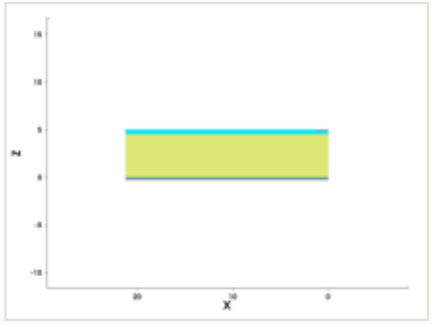


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



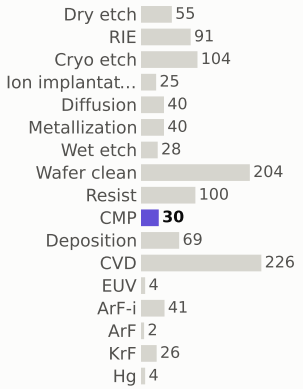
y cut at 57.351 µm



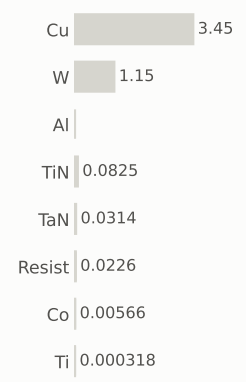
mask for this step

no mask for this step

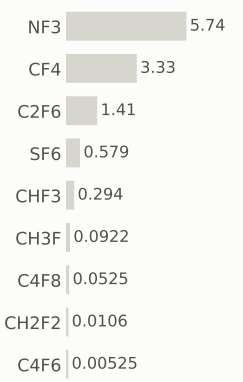
steps by process area



materials deposited (g)



gases emitted (g)



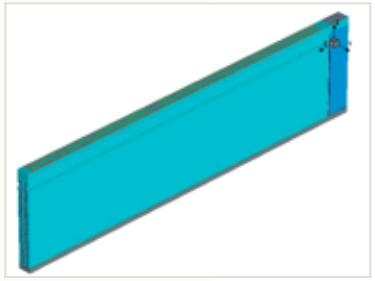
Step #12620: Wafer_clean

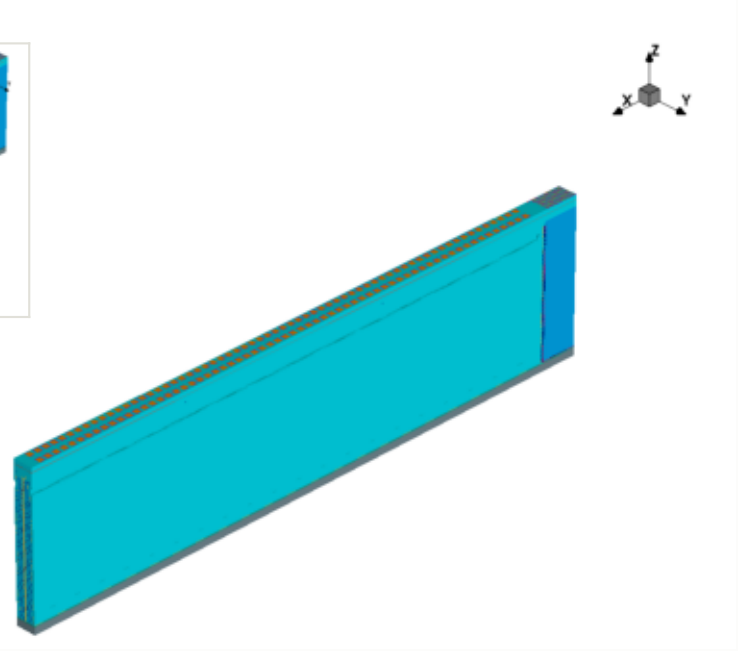
clean · Wafer_clean

0.169	770.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





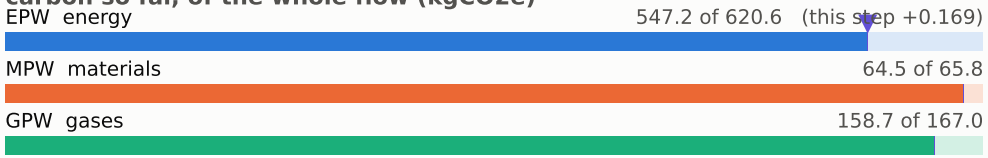
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

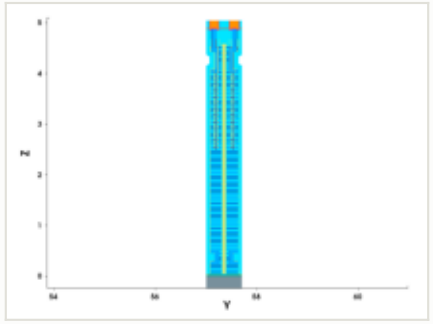
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

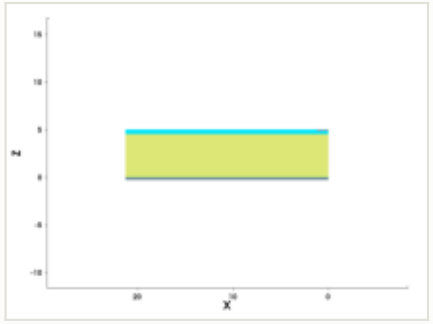


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



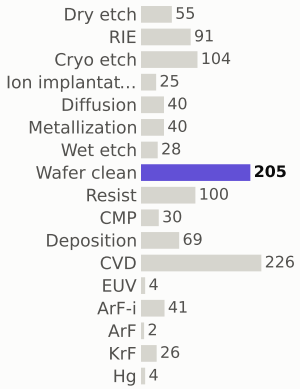
y cut at 57.351 µm



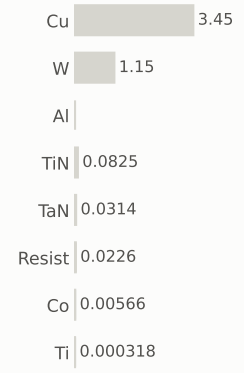
mask for this step

no mask for this step

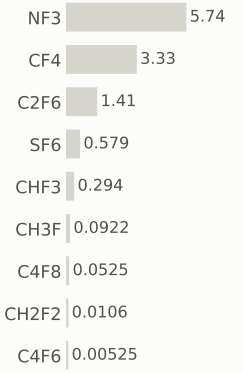
steps by process area



materials deposited (g)



gases emitted (g)



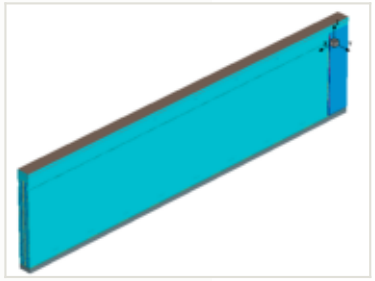
Step #12640: Deposition

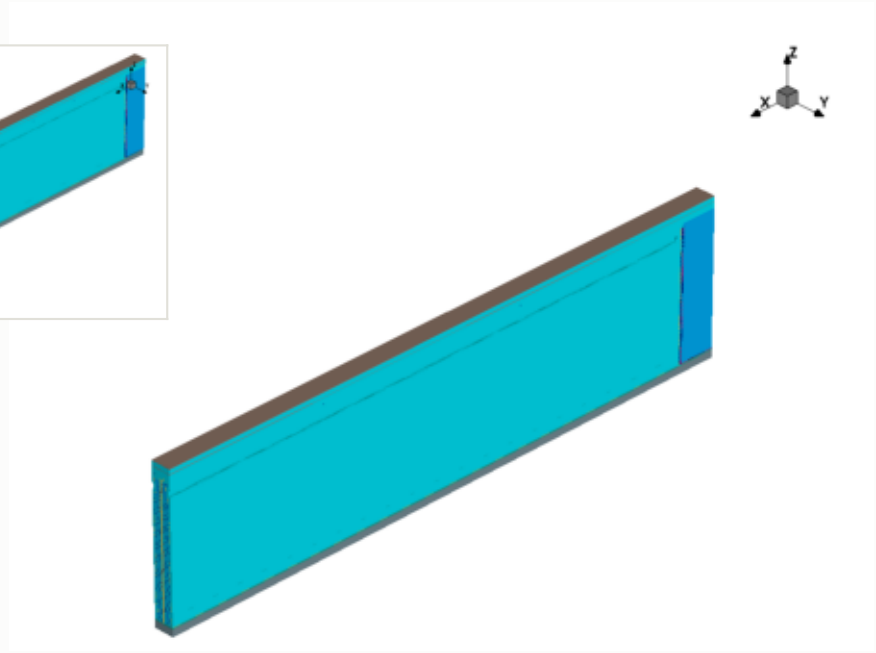
depo_isotropic · Deposition

0.656	771.1	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



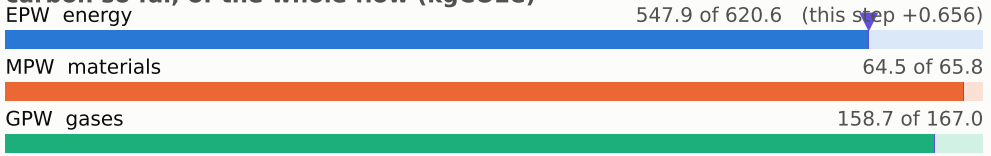


this step

process area	Deposition
material	—
thickness	0.02 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

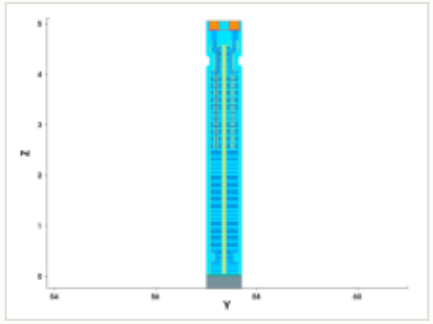
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

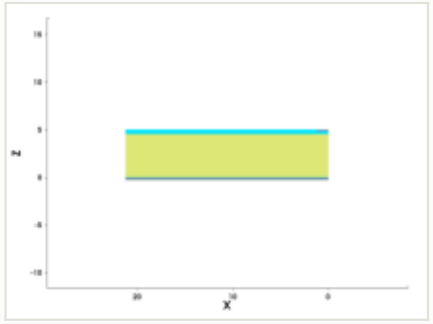


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



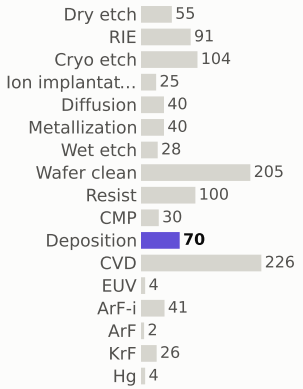
y cut at 57.351 µm



mask for this step

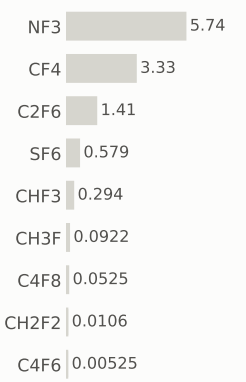
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



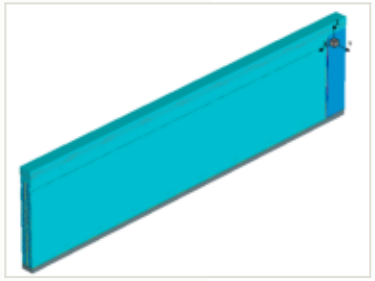
Step #12650: Deposition

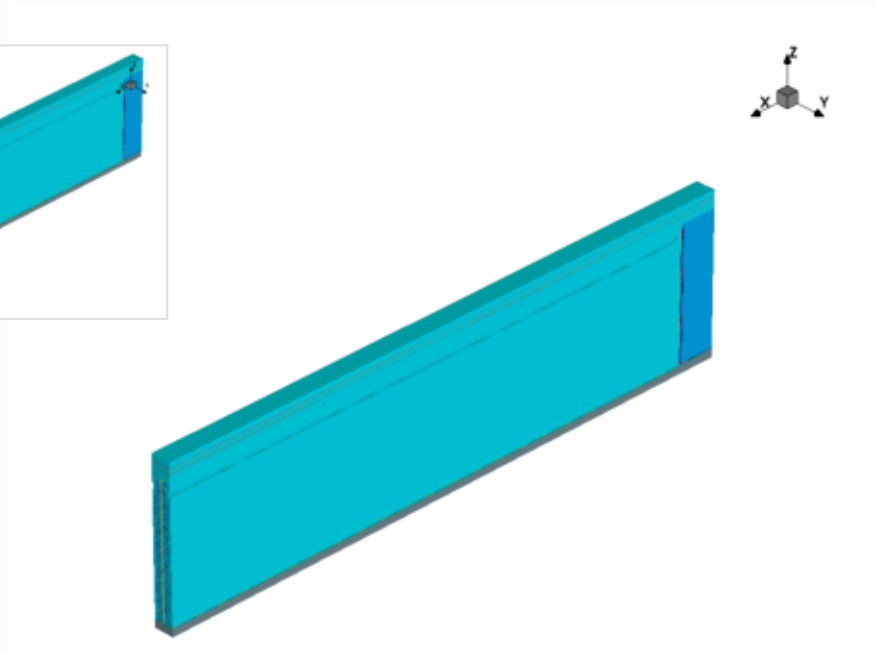
depo_isotropic · Deposition

0.656	771.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



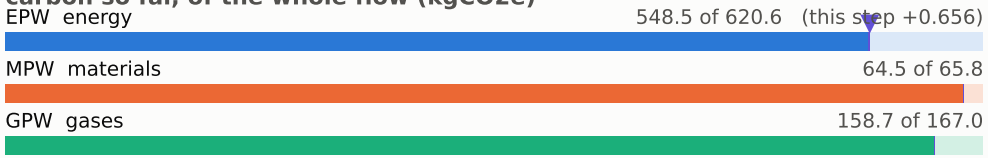


this step

process area	Deposition
material	—
thickness	0.2 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

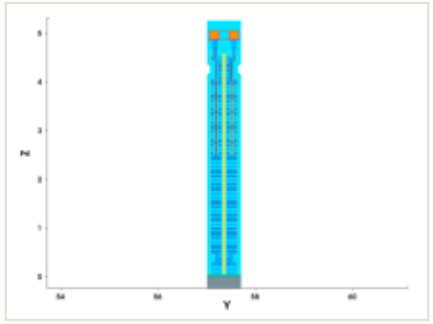
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

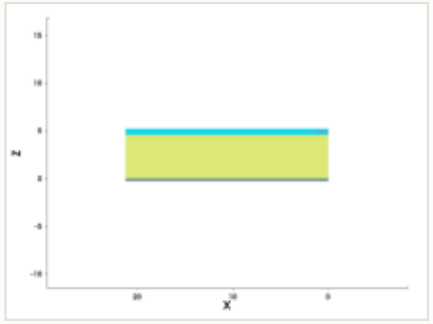


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



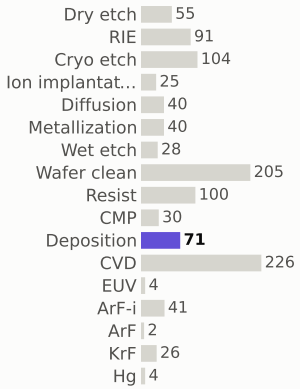
y cut at 57.351 µm



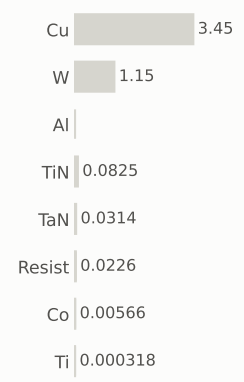
mask for this step

no mask for this step

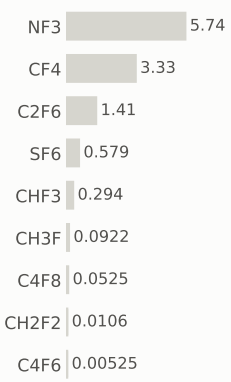
steps by process area



materials deposited (g)



gases emitted (g)



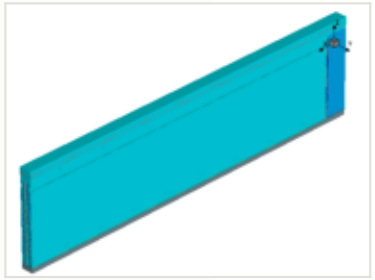
Step #12660: Wafer_clean

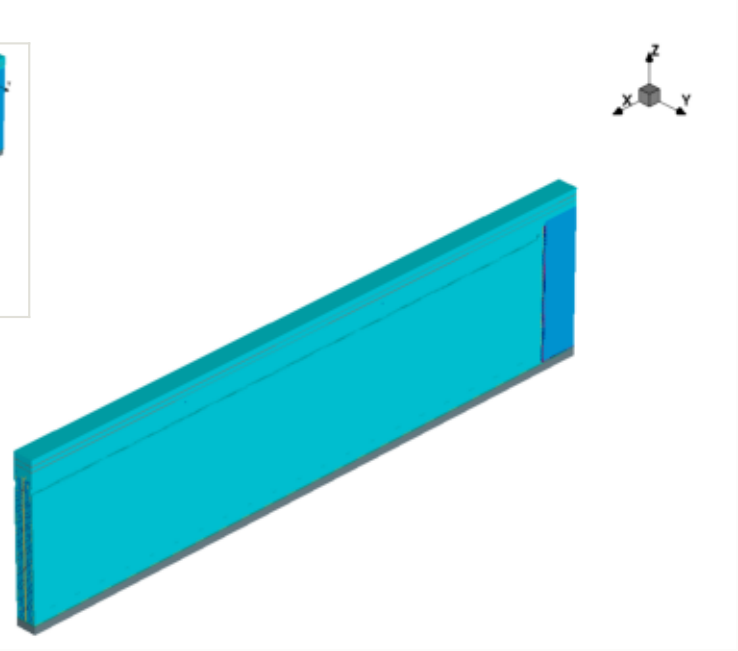
clean · Wafer_clean

0.169	771.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





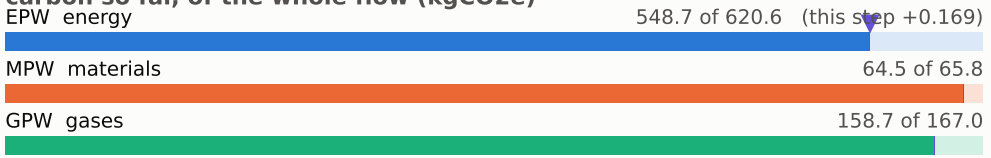
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

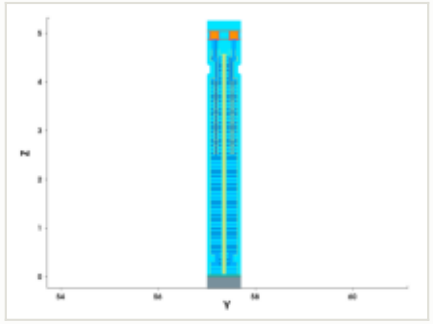
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

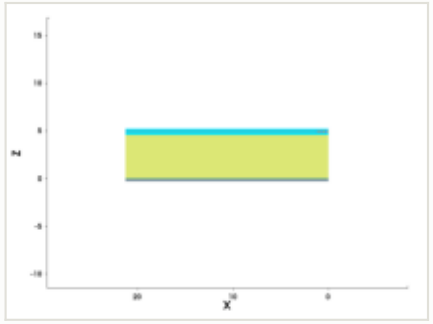


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



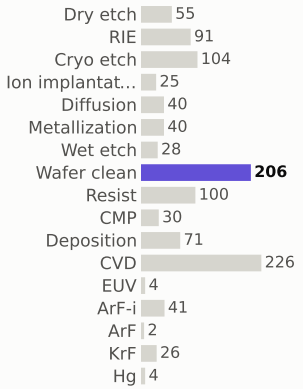
y cut at 57.351 µm



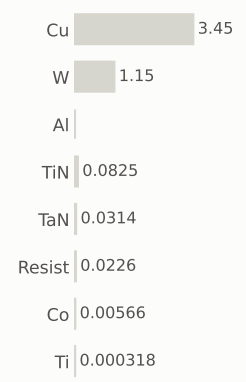
mask for this step

no mask for this step

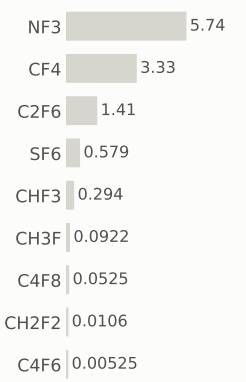
steps by process area



materials deposited (g)



gases emitted (g)



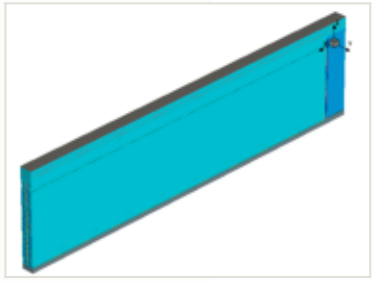
Step #12670: EUV

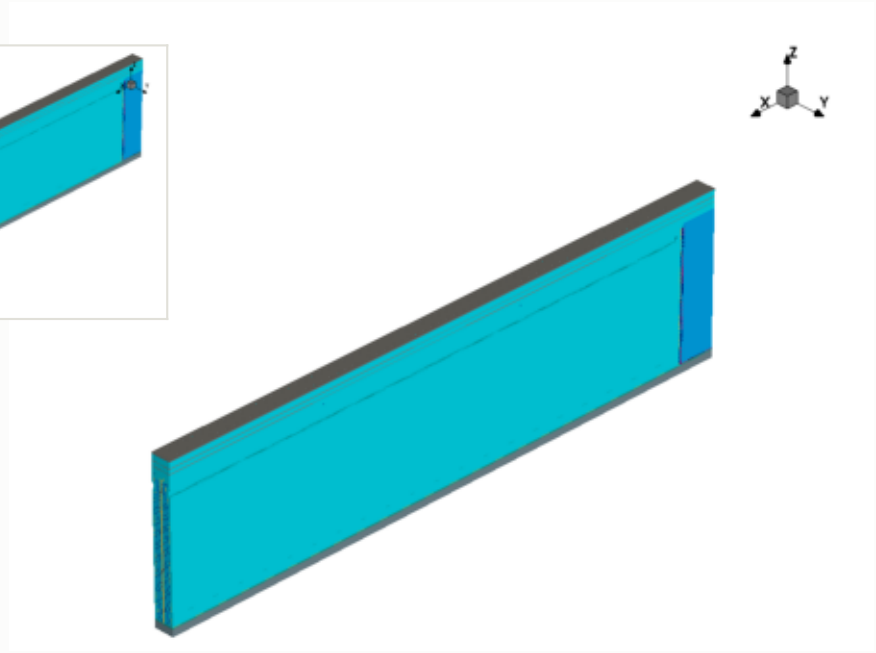
litho_patterning · EUV

1.96	773.9	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



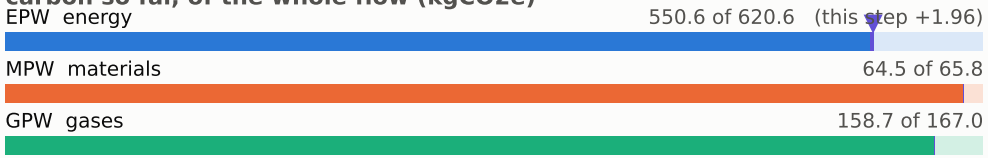


this step

process area	EUV
material	—
thickness	—
material GWP	—
input gases	—
energy	3.25 kWh + 1.3 overhead

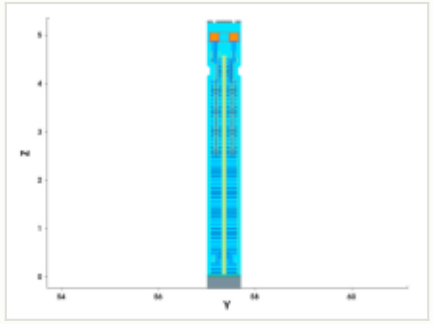
EPW	1.96 kgCO2e
EPW tool	1.4 kgCO2e
EPW overhead	0.559 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.96 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

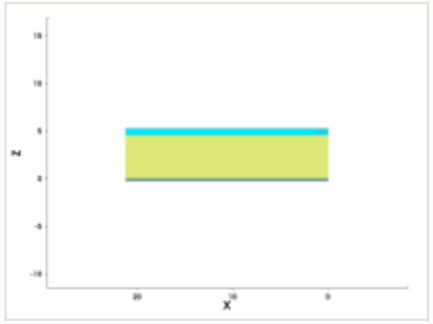


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

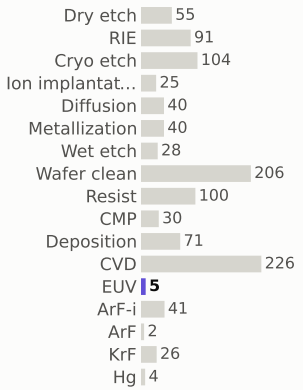


y cut at 57.351 µm

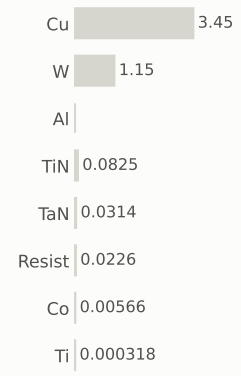


mask: layer 17:0, negative (GDSII)

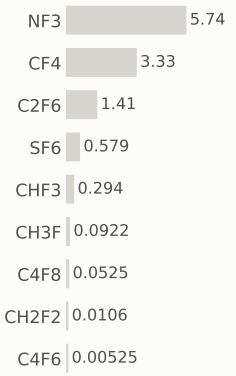
steps by process area



materials deposited (g)



gases emitted (g)



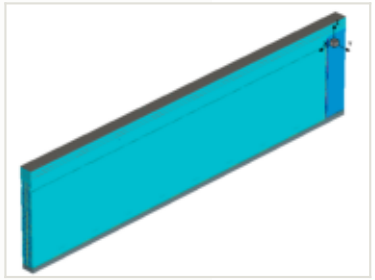
Step #12680: RIE

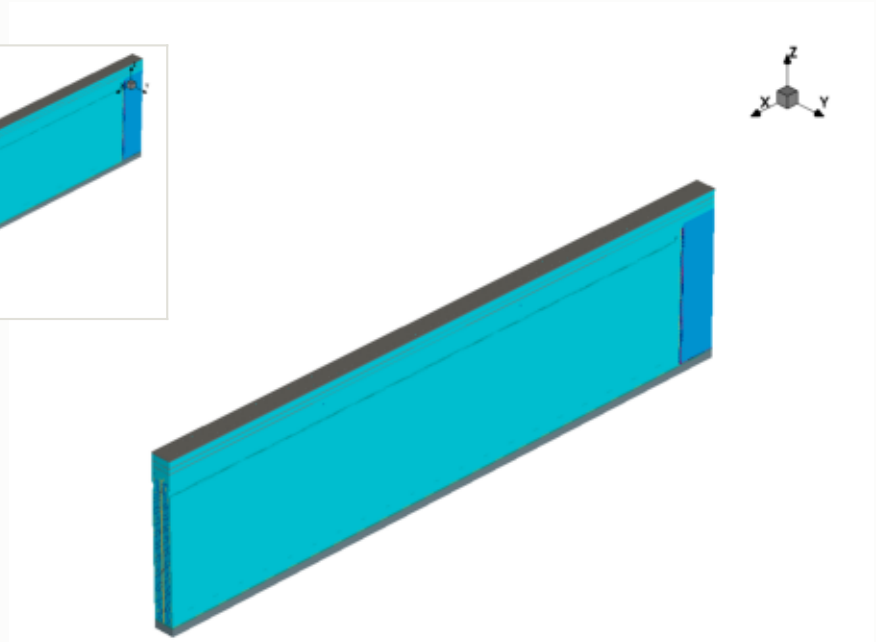
etch_anisotropic · RIE

1.04	774.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



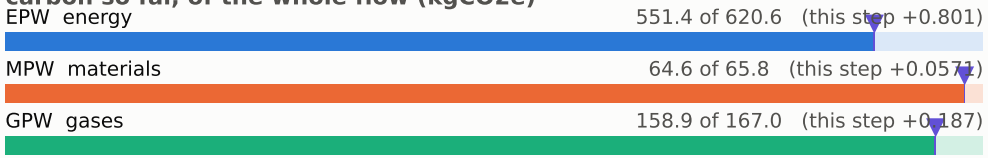


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

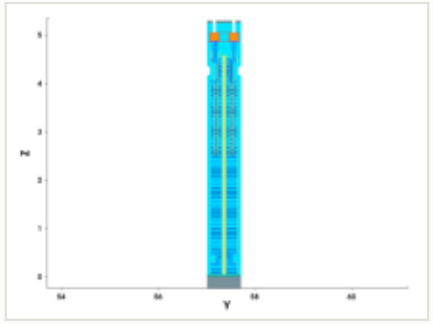
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

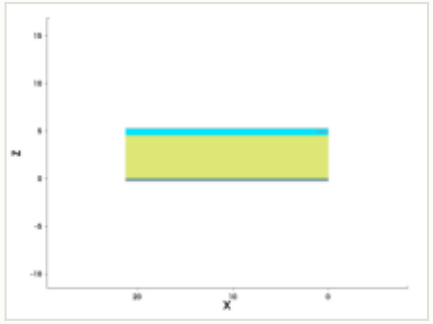


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



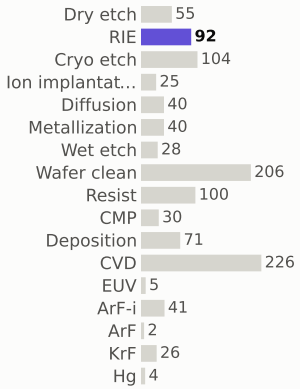
y cut at 57.351 µm



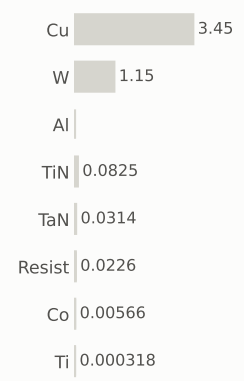
mask for this step

no mask for this step

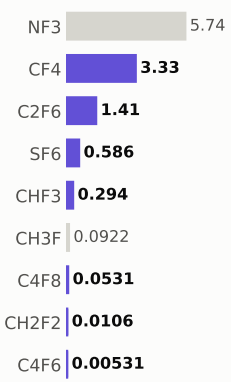
steps by process area



materials deposited (g)



gases emitted (g)



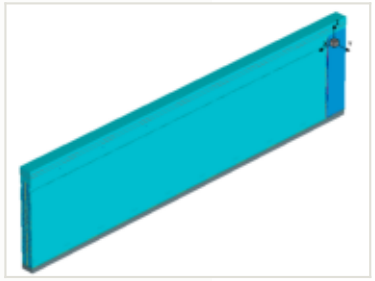
Step #12690: Dry_etch

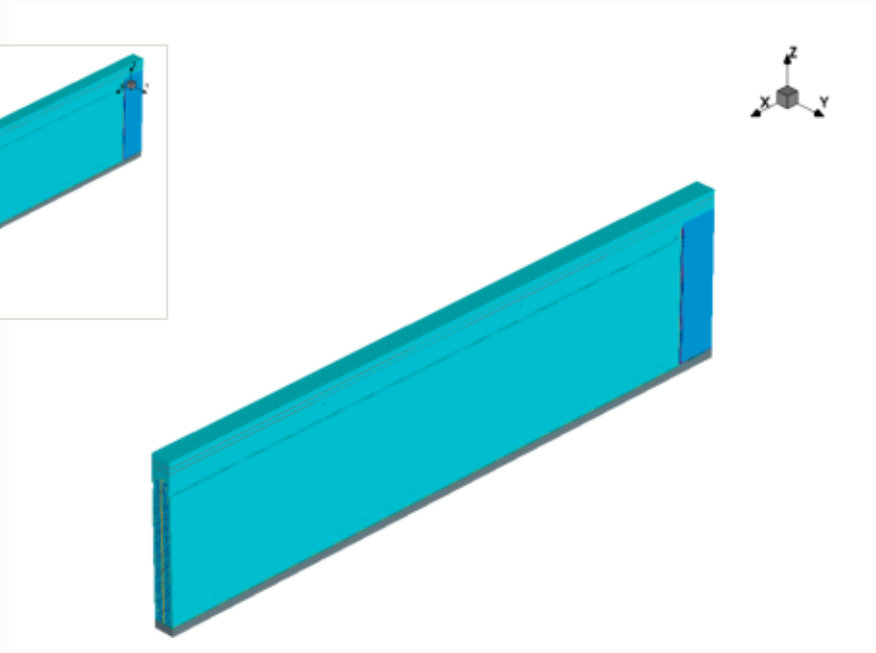
etch_strip · Dry_etch

0.801	775.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

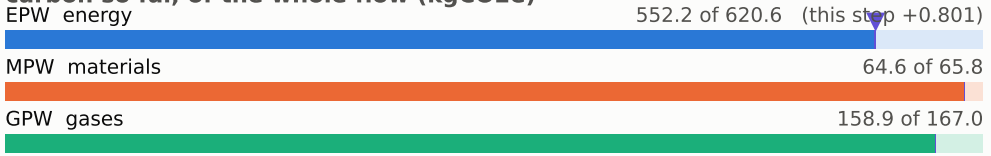




this step

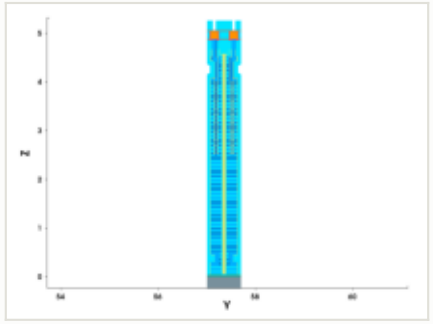
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

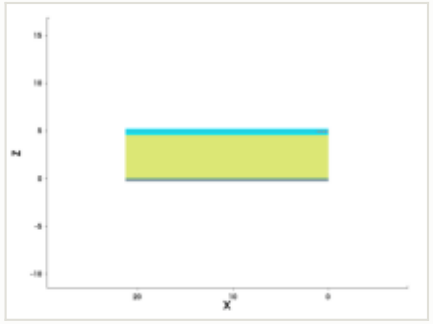


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



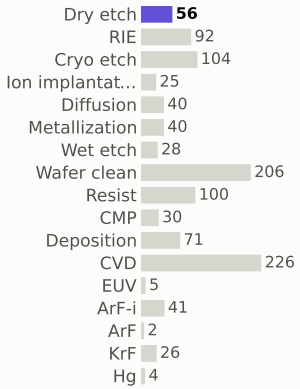
y cut at 57.351 µm



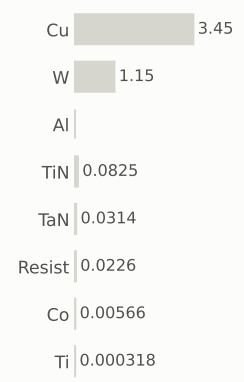
mask for this step

no mask for this step

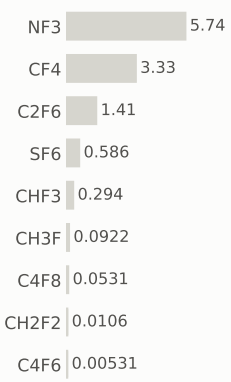
steps by process area



materials deposited (g)



gases emitted (g)



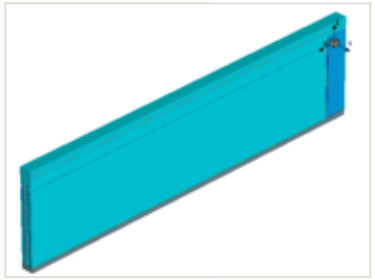
Step #12700: Wafer_clean

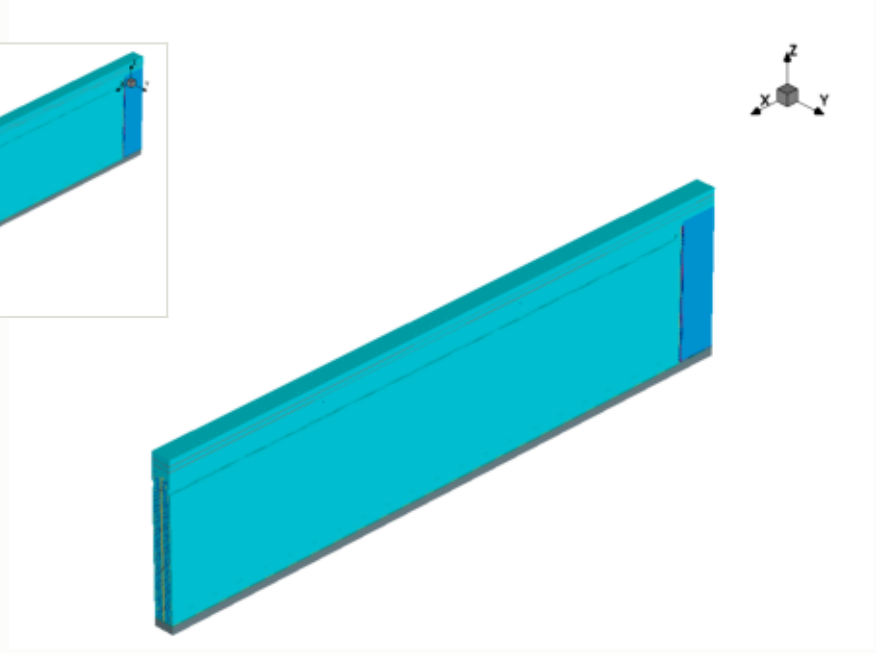
clean · Wafer_clean

0.169	775.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





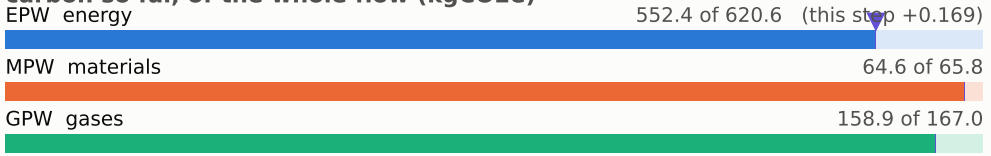
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

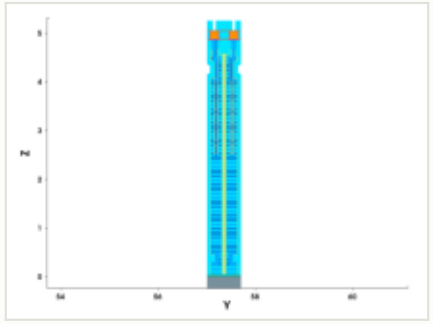
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

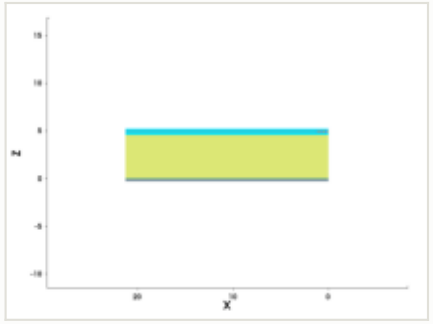


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



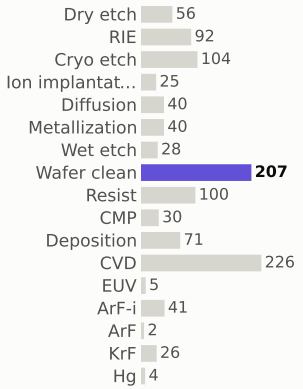
y cut at 57.351 µm



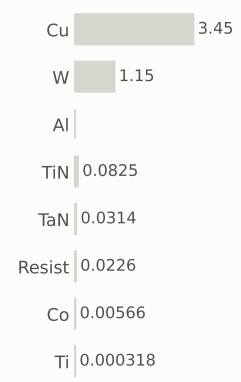
mask for this step

no mask for this step

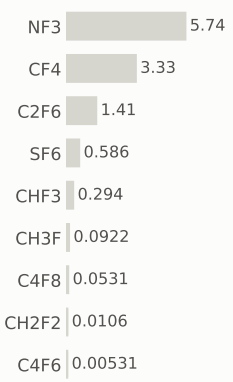
steps by process area



materials deposited (g)



gases emitted (g)



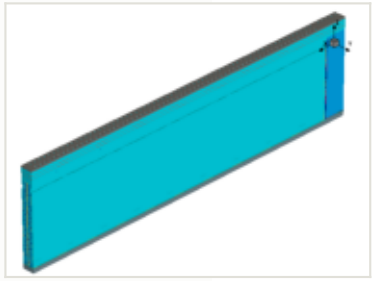
Step #12710: ArF-i

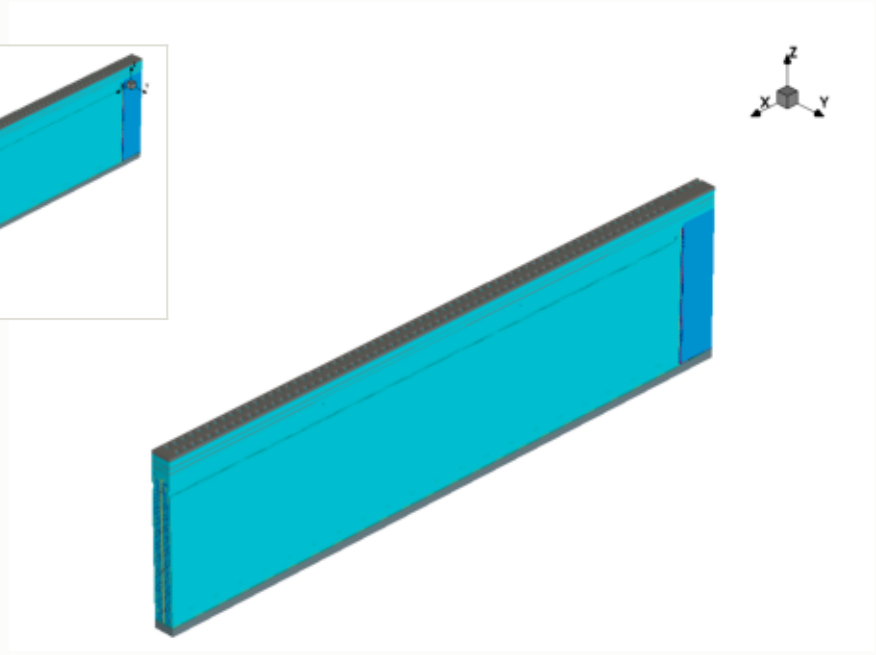
litho_patterning · ArF-i

0.217	776.1	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



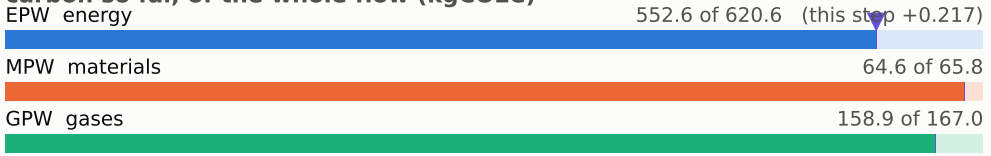


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

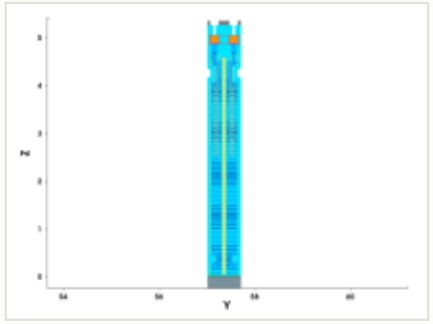
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

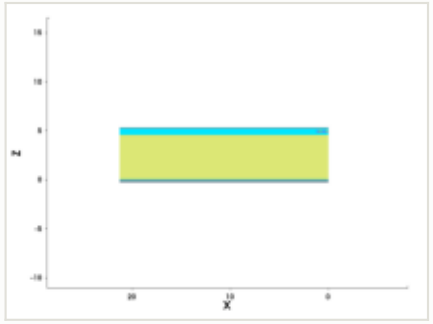


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

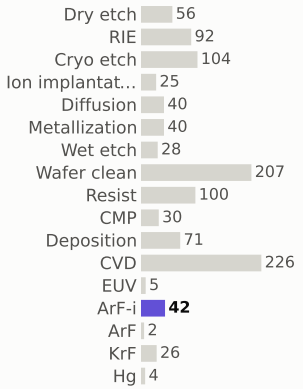


y cut at 57.351 µm

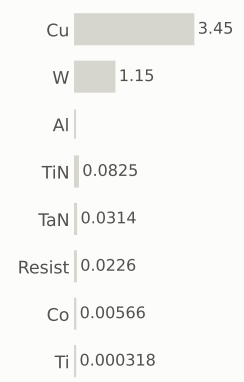


mask: layer 18:0, negative (GDSII)

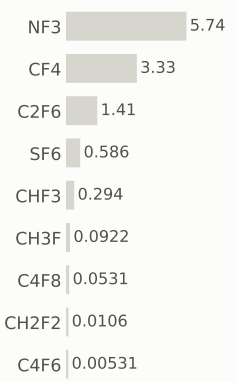
steps by process area



materials deposited (g)



gases emitted (g)



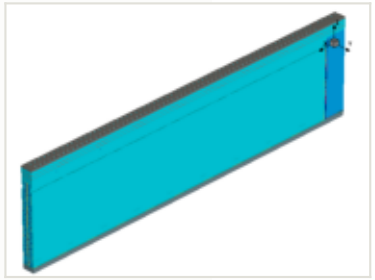
Step #12720: RIE

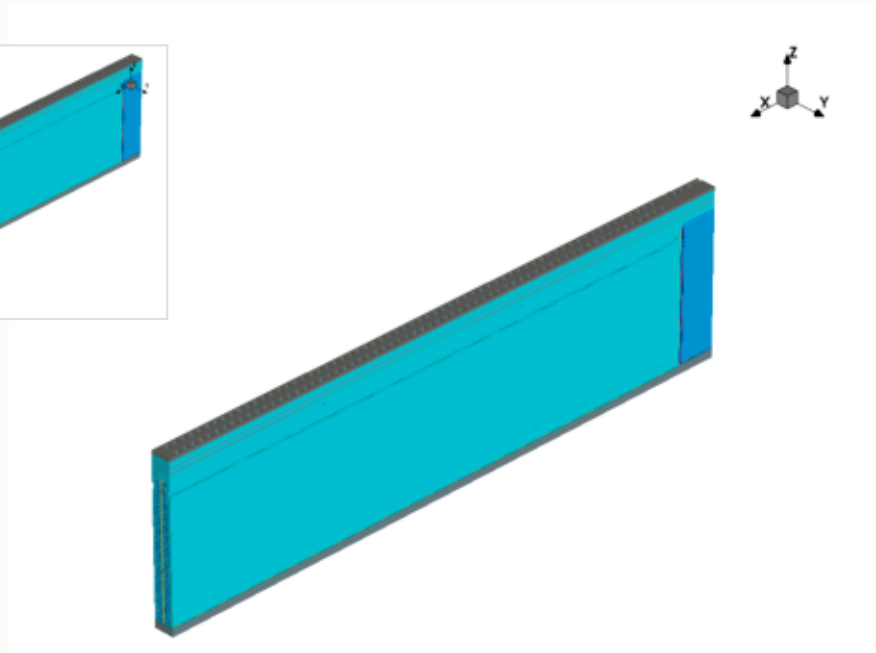
etch_anisotropic · RIE

1.04	777.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



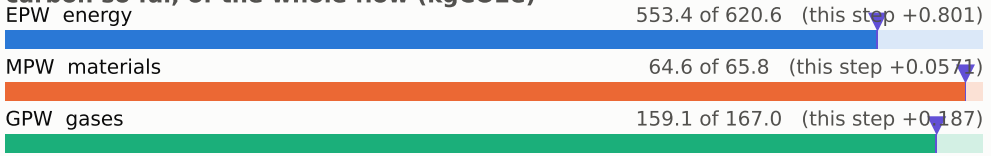


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

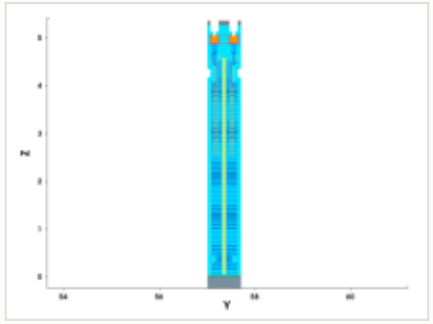
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

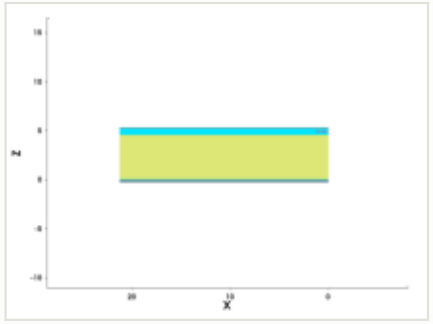


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



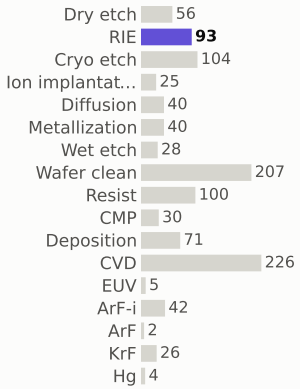
y cut at 57.351 µm



mask for this step

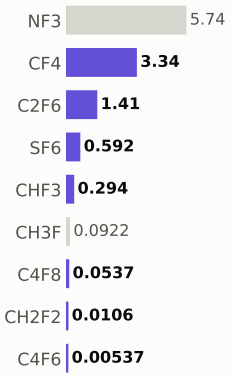
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



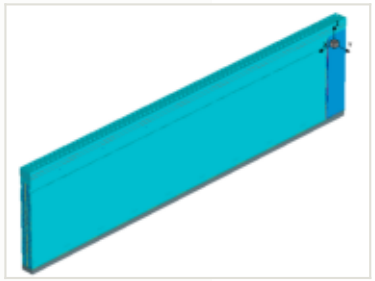
Step #12730: Dry_etch

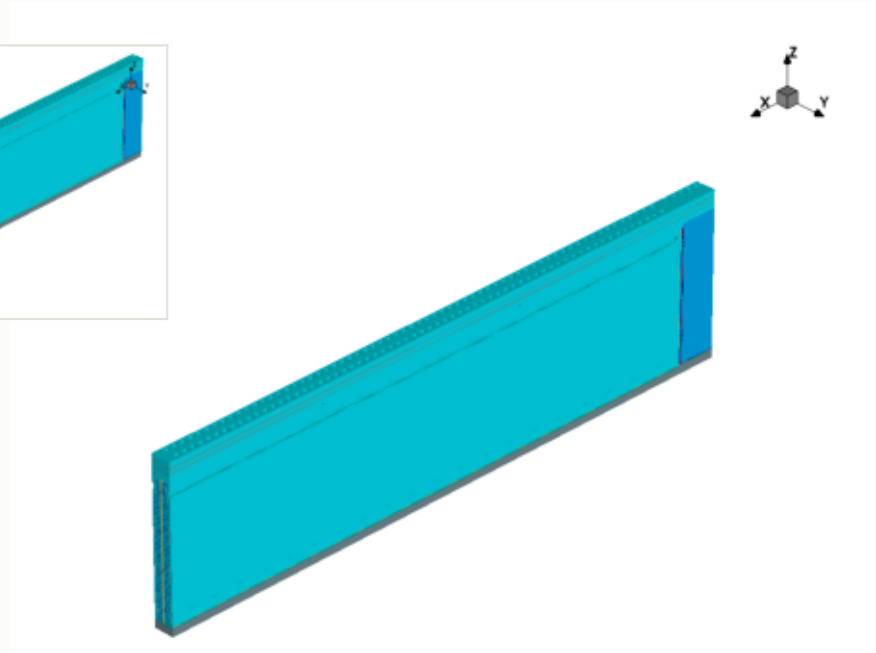
etch_strip · Dry_etch

0.801	777.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

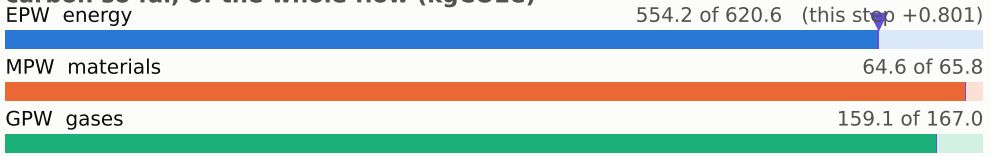




this step

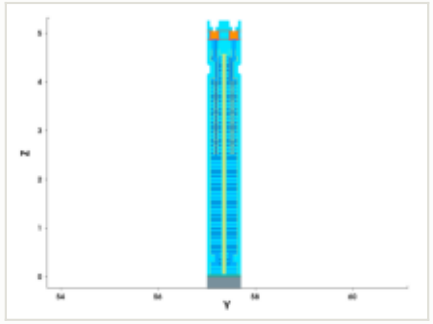
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

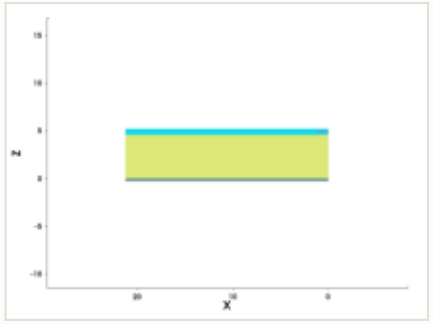


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



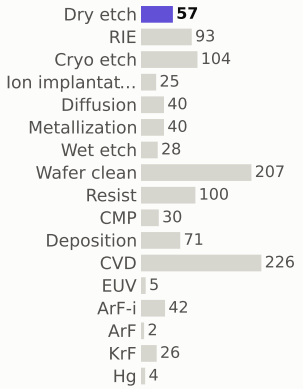
y cut at 57.351 µm



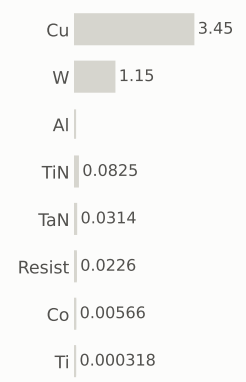
mask for this step

no mask for this step

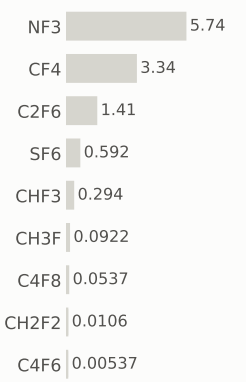
steps by process area



materials deposited (g)



gases emitted (g)



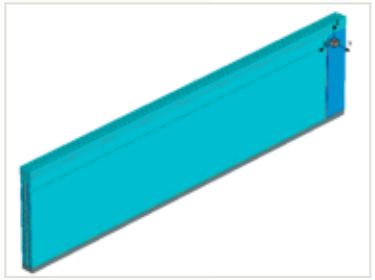
Step #12740: Wafer_clean

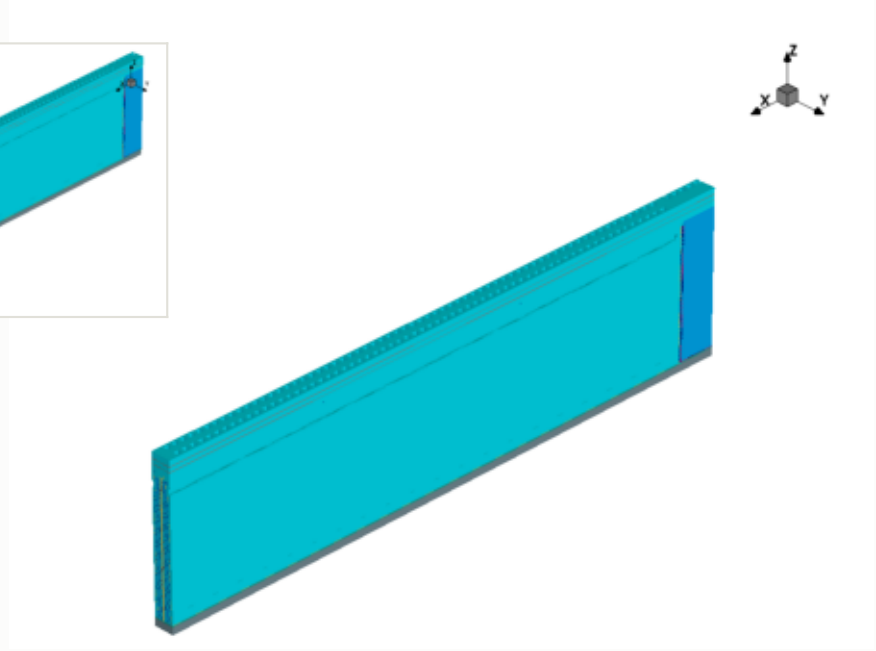
clean · Wafer_clean

0.169	778.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





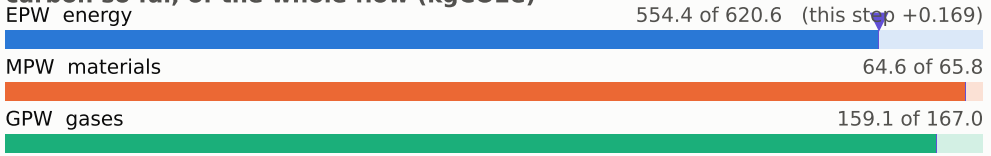
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

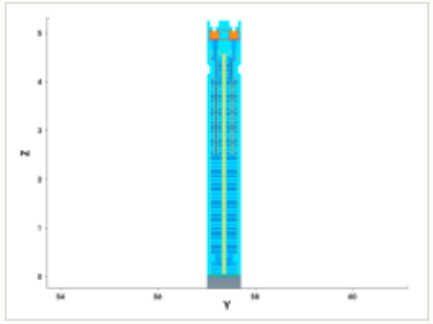
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

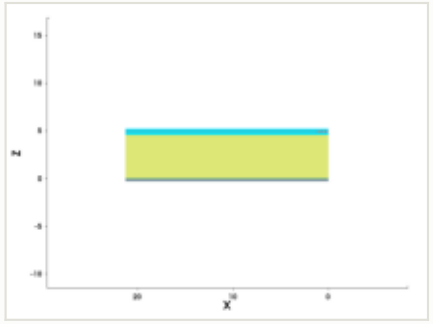


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



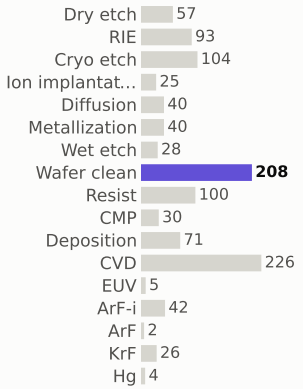
y cut at 57.351 µm



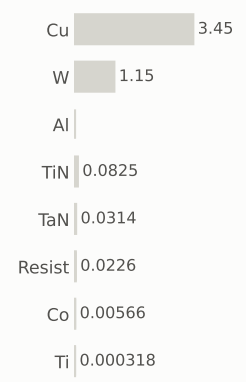
mask for this step

no mask for this step

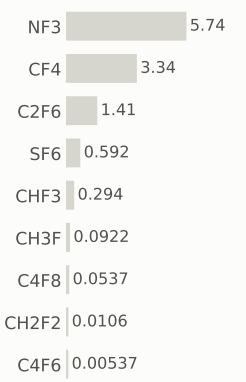
steps by process area



materials deposited (g)



gases emitted (g)



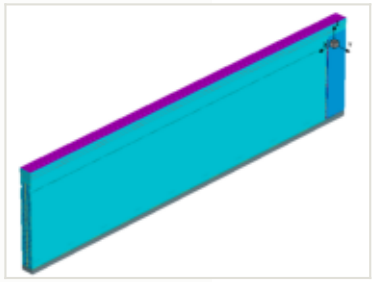
Step #12750: Metallization

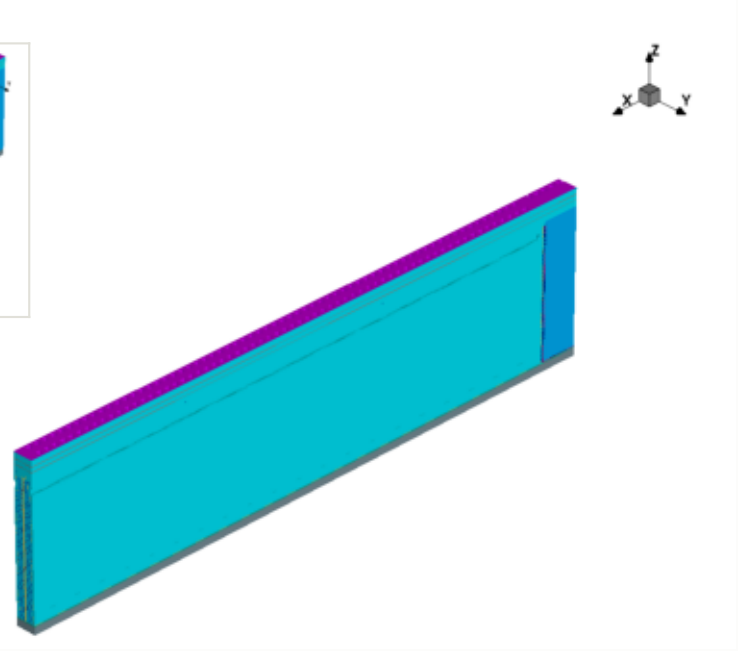
depo_isotropic · Metallization

1.2	779.3	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



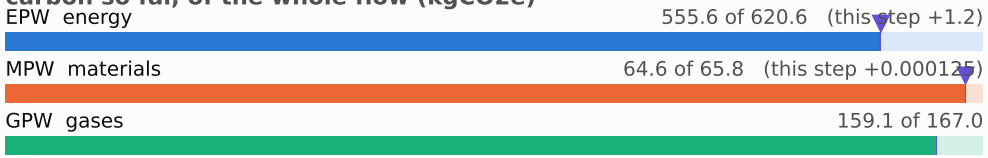


this step

process area	Metallization
material	TaN
thickness	0.005 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

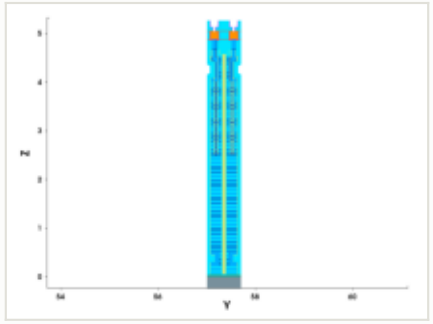
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000125 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

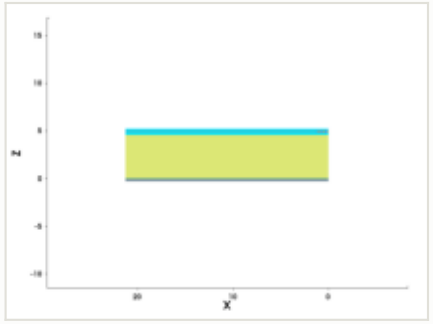


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



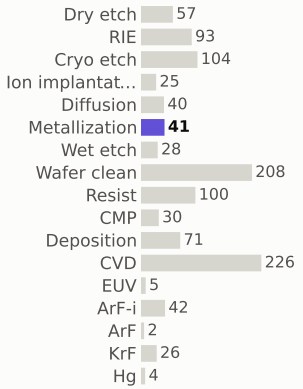
y cut at 57.351 µm



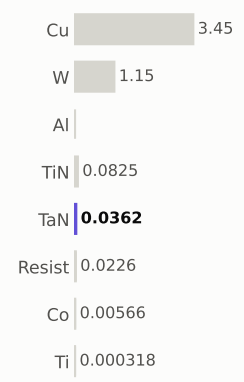
mask for this step

no mask for this step

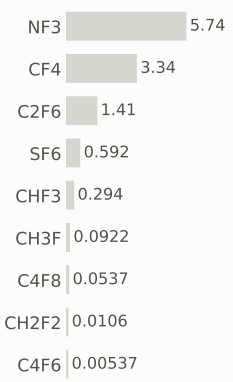
steps by process area



materials deposited (g)



gases emitted (g)



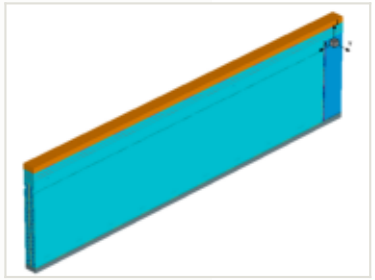
Step #12760: Metallization

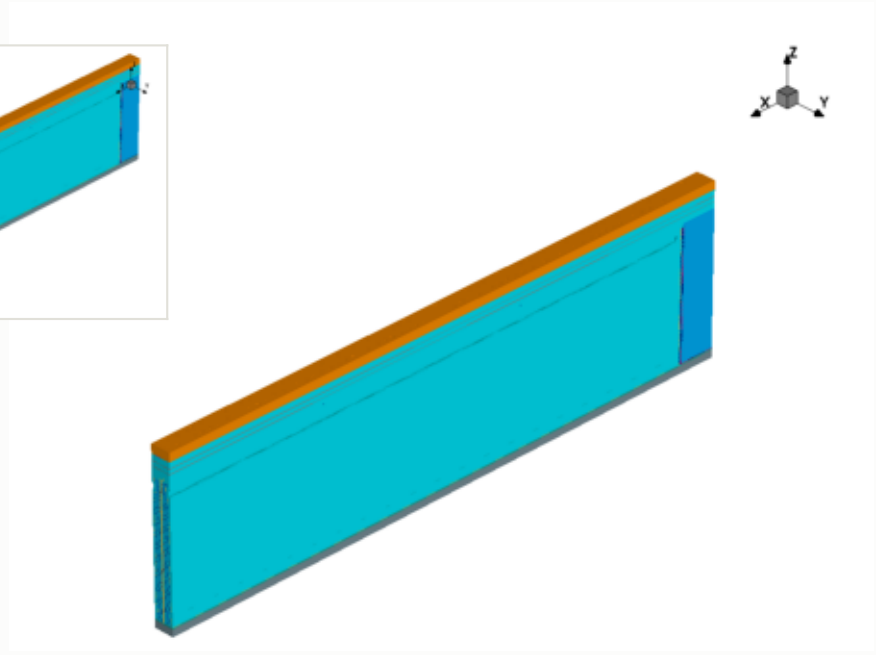
depo_isotropic · Metallization

1.21	780.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



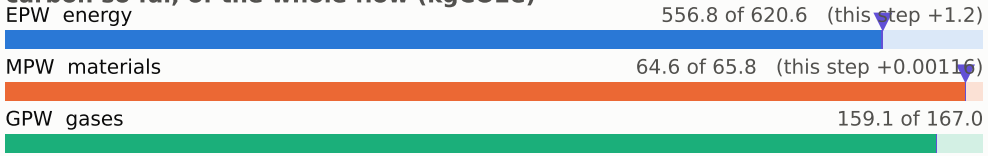


this step

process area	Metallization
material	Cu
thickness	0.3 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

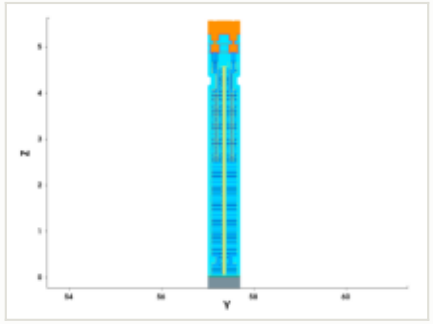
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00116 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

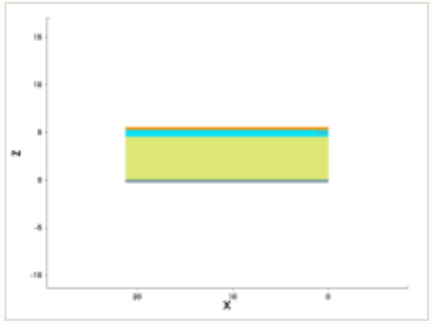


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



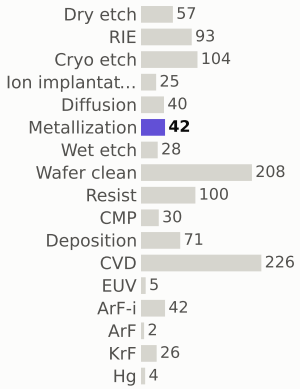
y cut at 57.351 µm



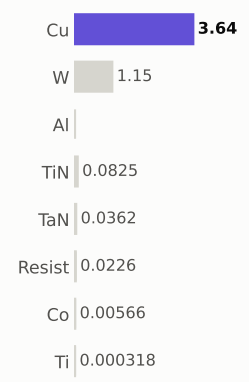
mask for this step

no mask for this step

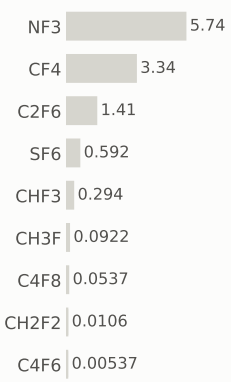
steps by process area



materials deposited (g)



gases emitted (g)



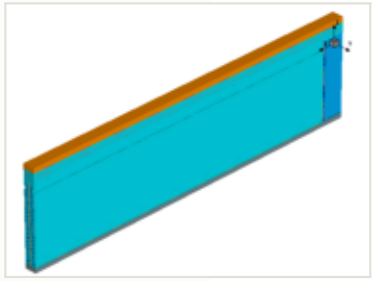
Step #12770: Diffusion

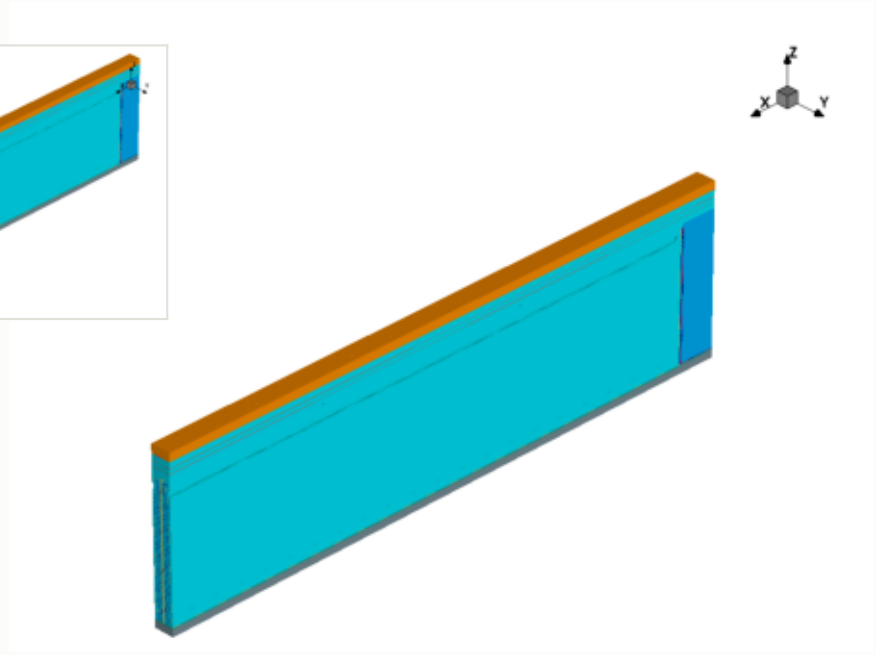
anneal · Diffusion

0.801	781.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





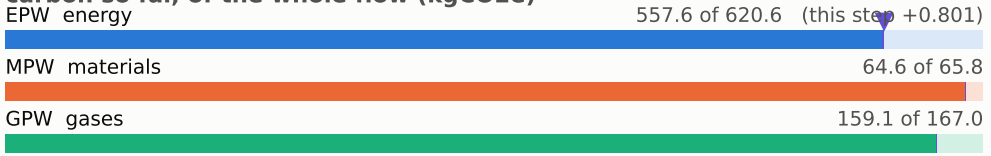
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

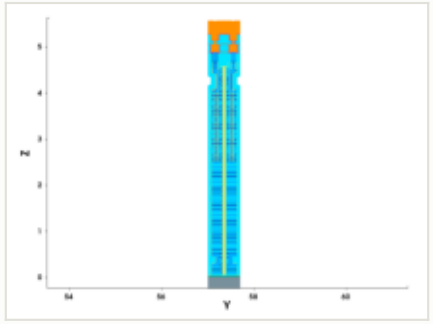
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

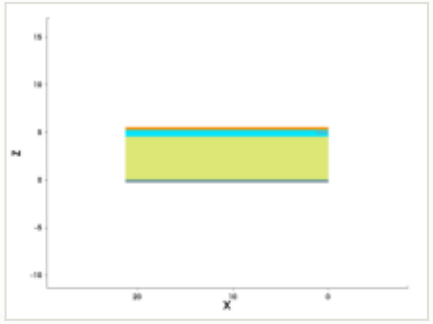


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



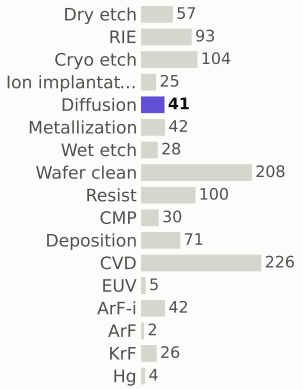
y cut at 57.351 µm



mask for this step

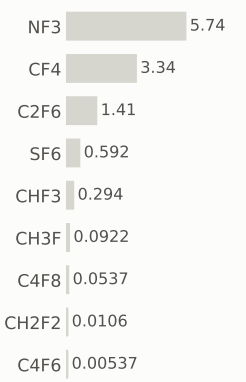
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



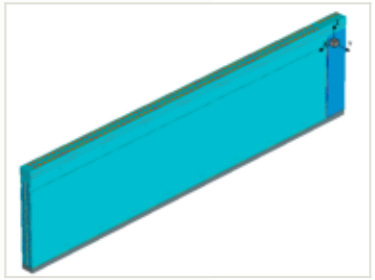
Step #12780: CMP

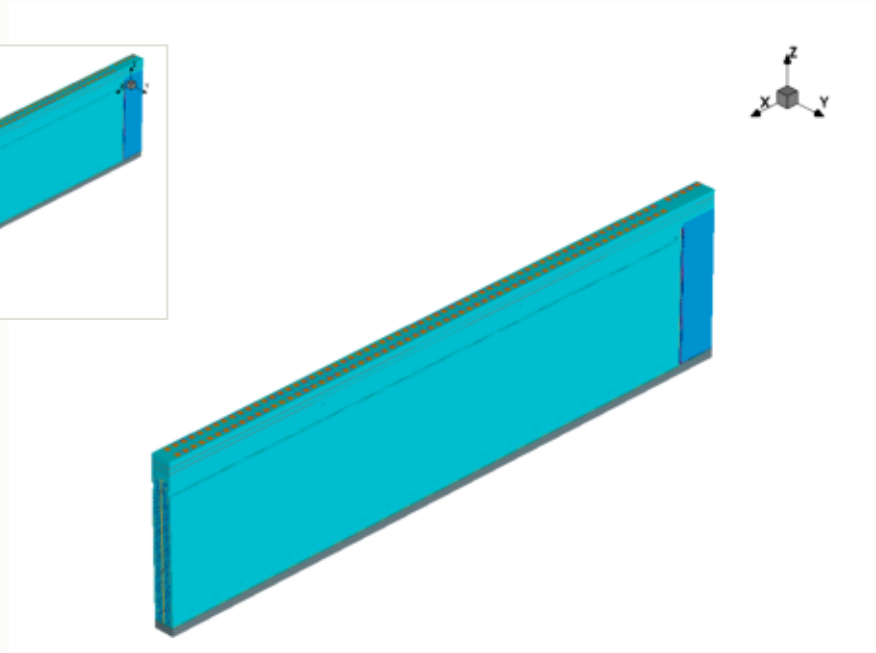
etch_cmp · CMP

0.169	781.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



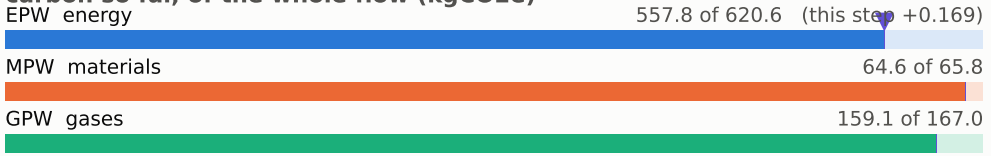


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

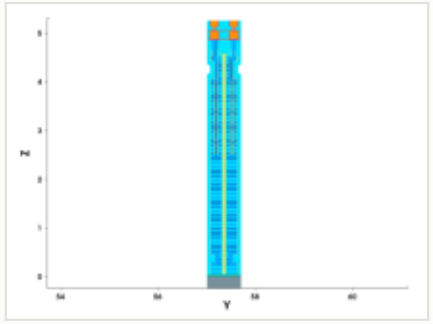
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

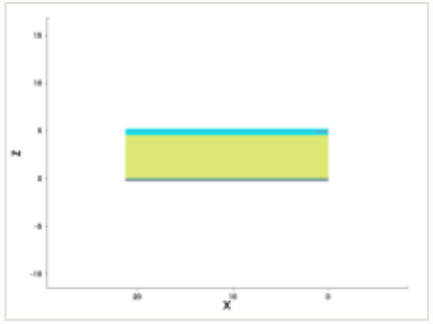


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



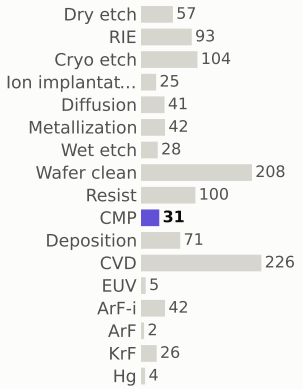
y cut at 57.351 µm



mask for this step

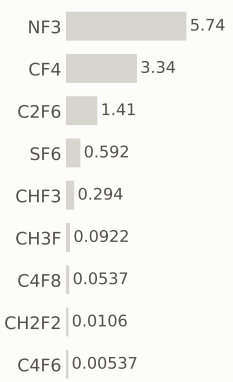
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



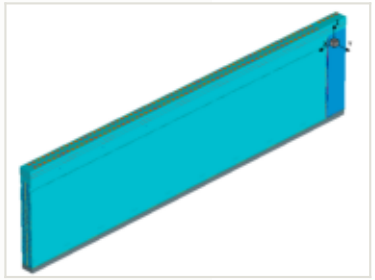
Step #12790: Wafer_clean

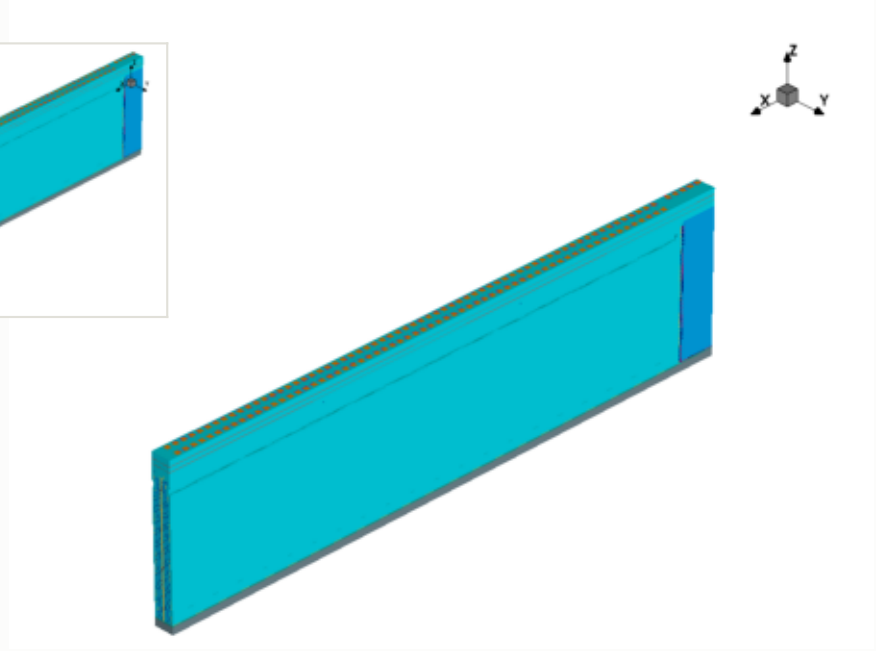
clean · Wafer_clean

0.169	781.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





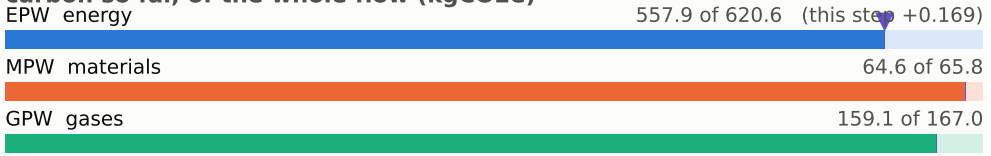
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

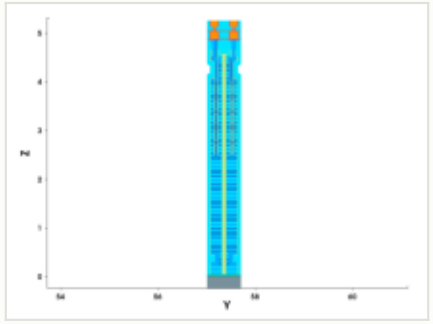
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

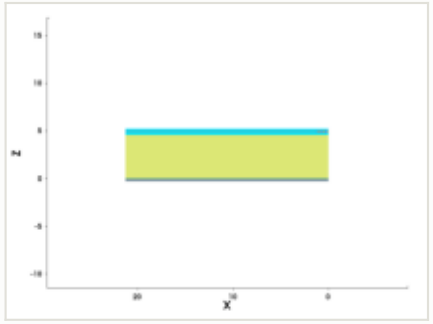


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



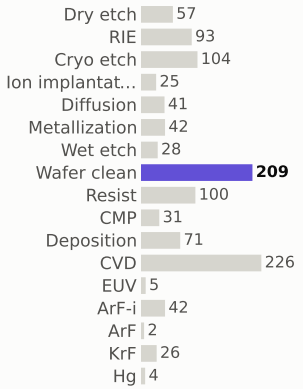
y cut at 57.351 µm



mask for this step

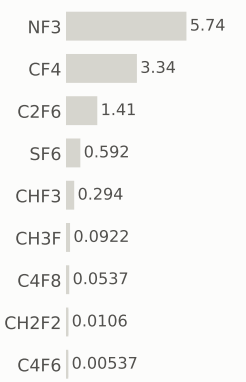
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



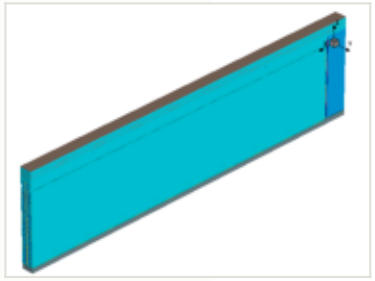
Step #12810: Deposition

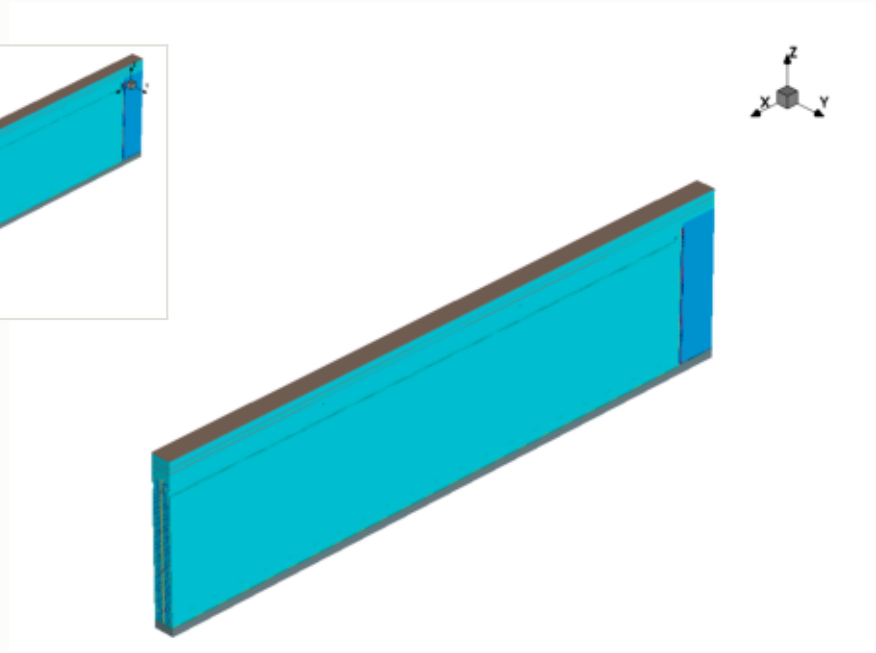
depo_isotropic · Deposition

0.656	782.3	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

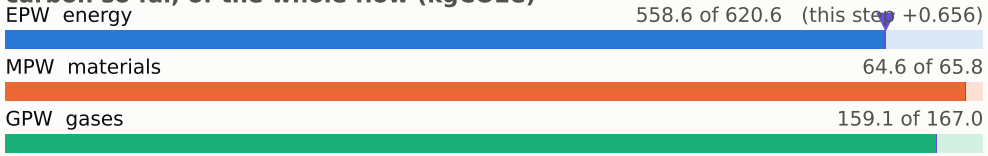




this step

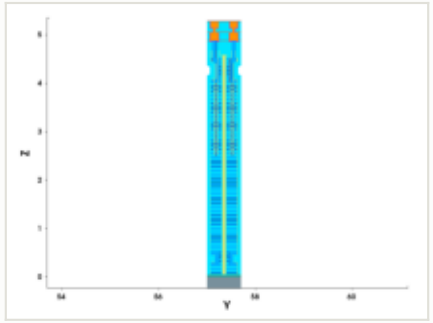
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.03 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

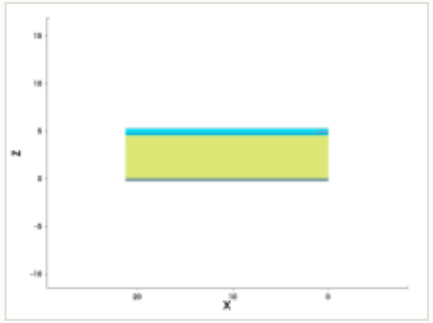


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



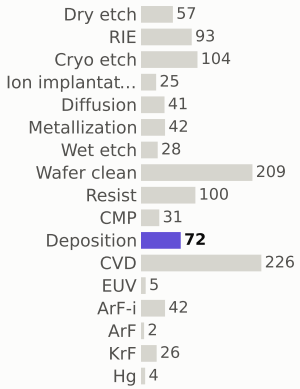
y cut at 57.351 µm



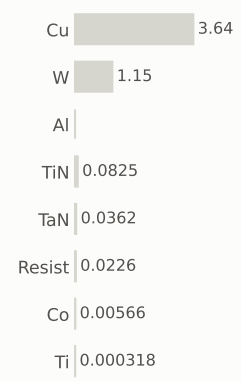
mask for this step

no mask for this step

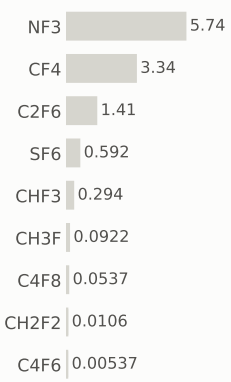
steps by process area



materials deposited (g)



gases emitted (g)



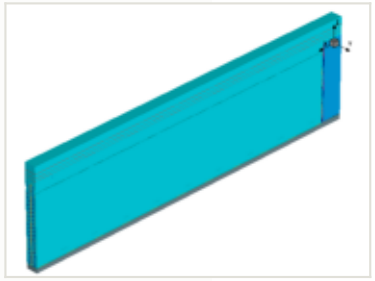
Step #12820: Deposition

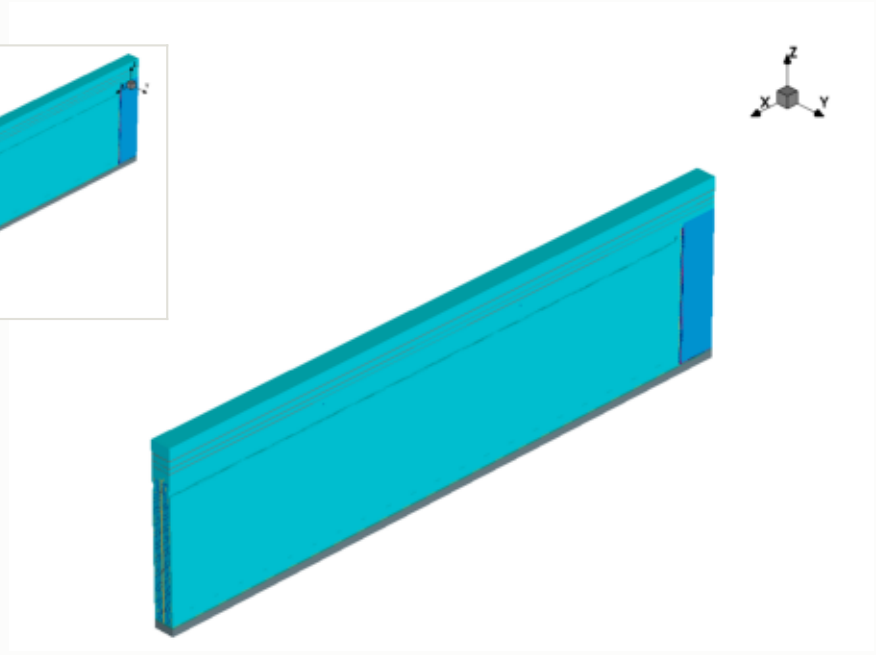
depo_isotropic · Deposition

0.656	783.0	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

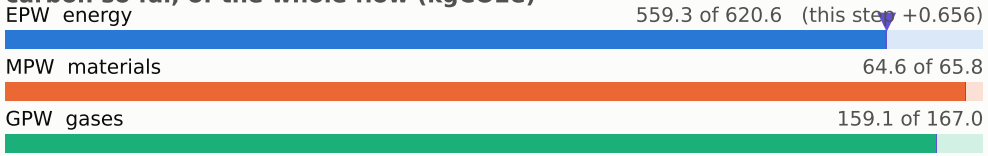




this step

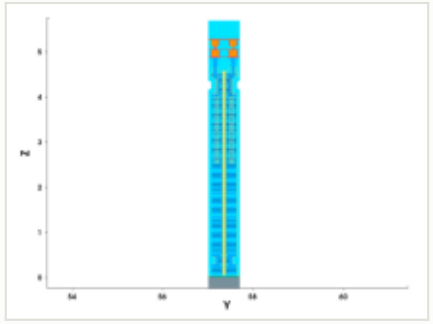
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.4 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

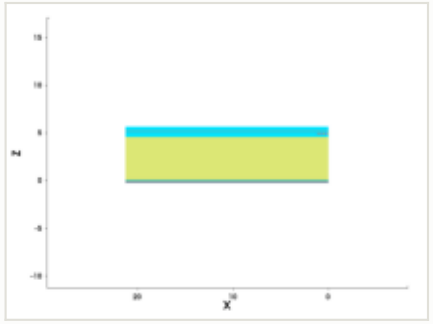


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



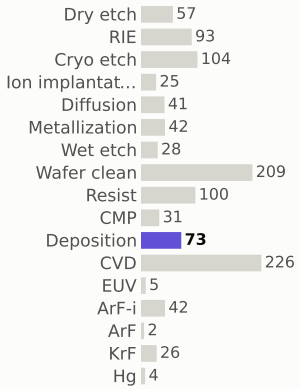
y cut at 57.351 µm



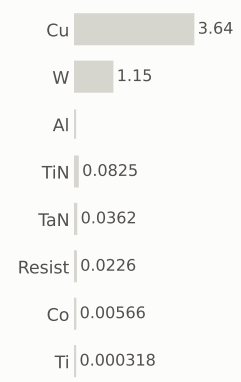
mask for this step

no mask for this step

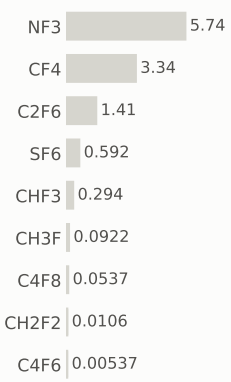
steps by process area



materials deposited (g)



gases emitted (g)



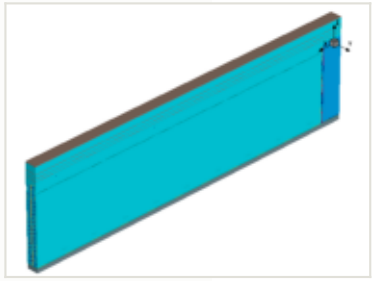
Step #12830: Deposition

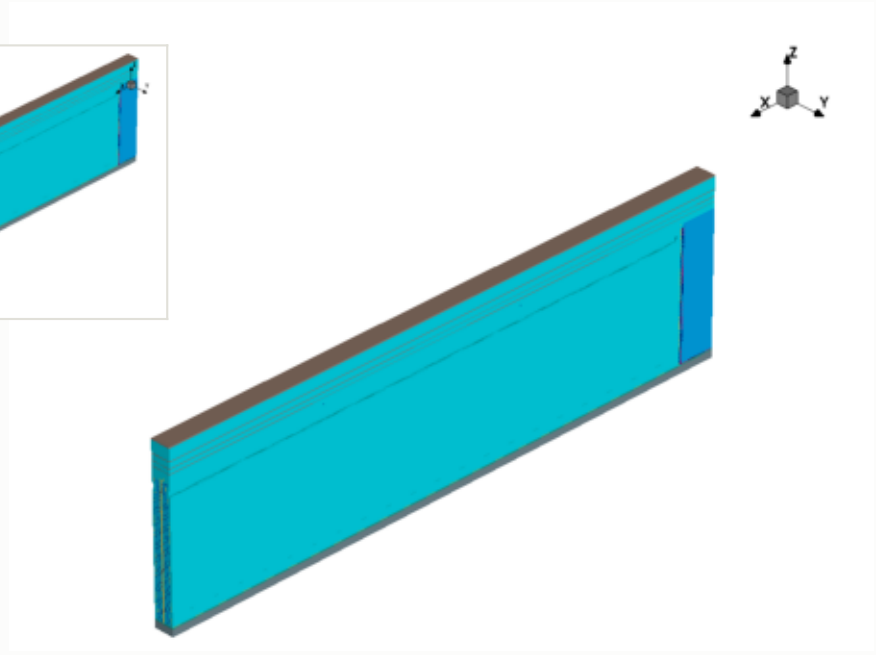
depo_isotropic · Deposition

0.656	783.6	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



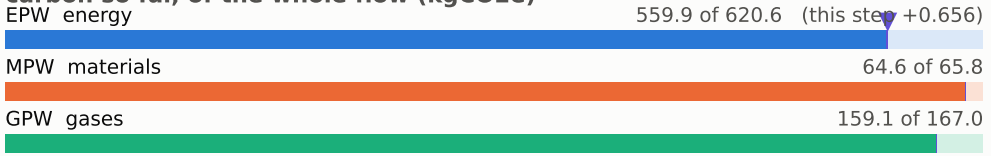


this step

process area	Deposition
material	—
thickness	0.05 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

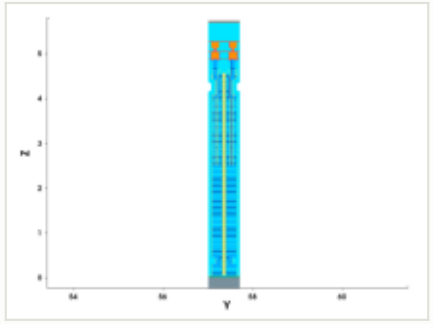
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

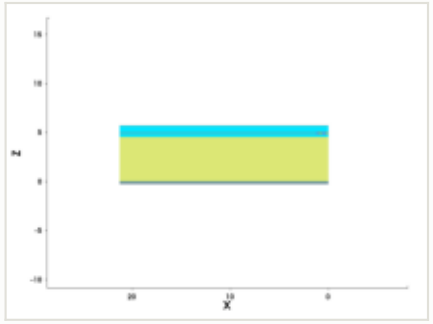


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



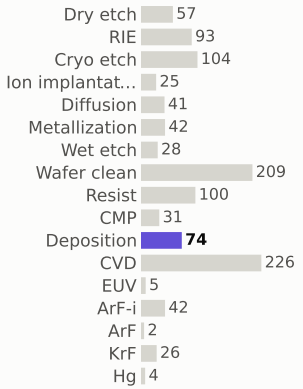
y cut at 57.351 µm



mask for this step

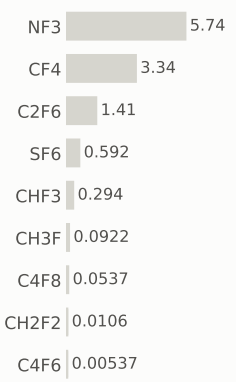
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



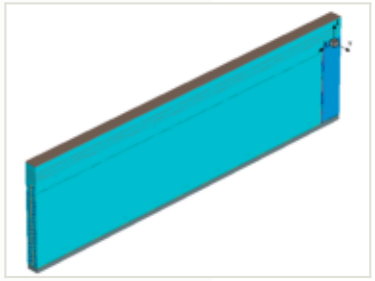
Step #12840: Wafer_clean

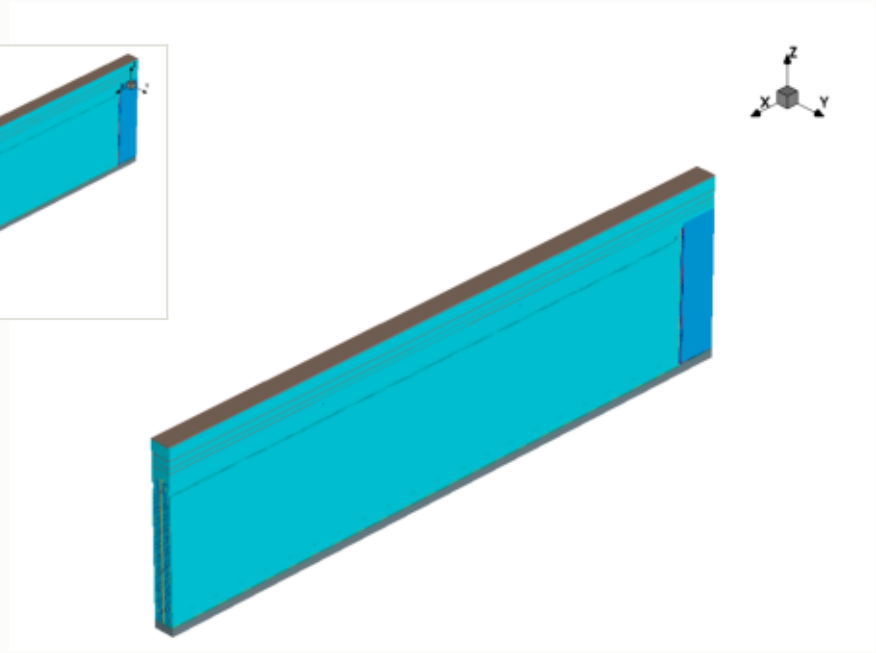
clean · Wafer_clean

0.169	783.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



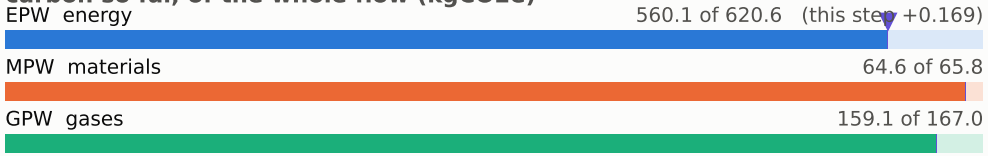


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

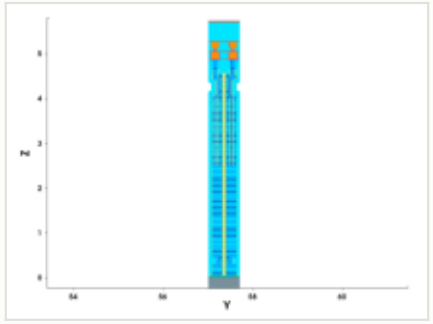
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

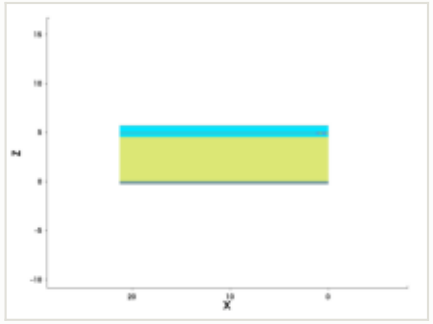


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



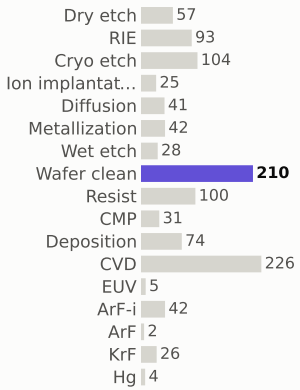
y cut at 57.351 µm



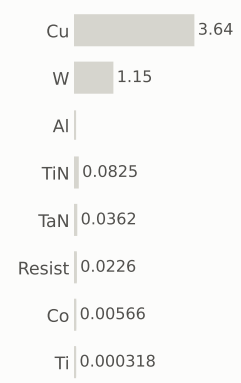
mask for this step

no mask for this step

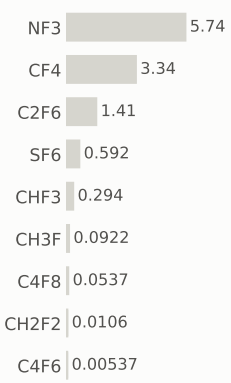
steps by process area



materials deposited (g)



gases emitted (g)



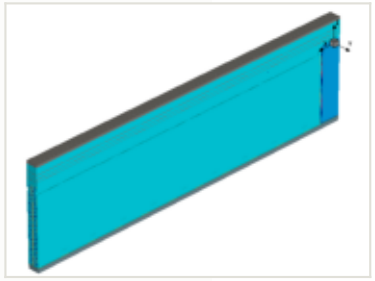
Step #12850: ArF-i

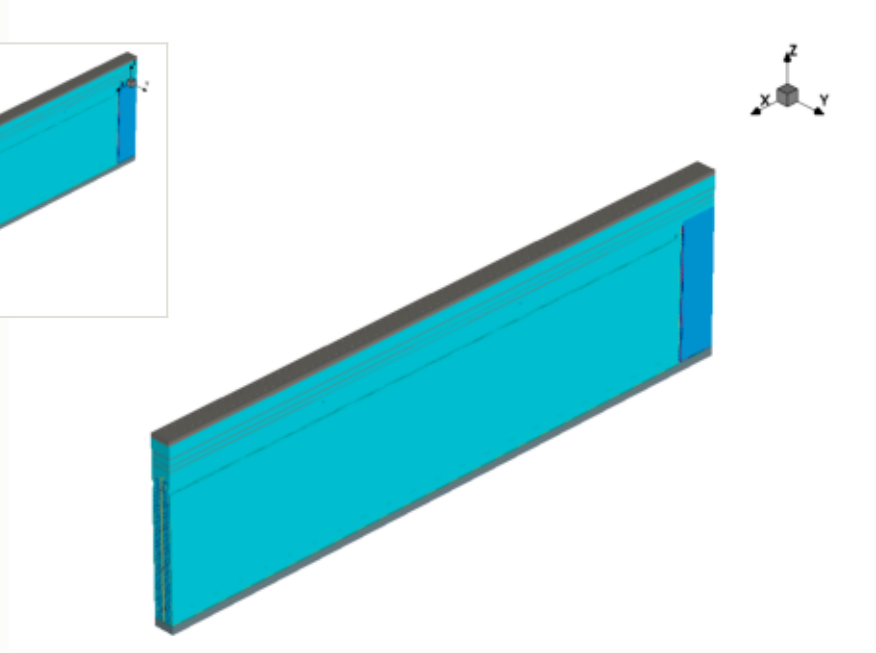
litho_patterning · ArF-i

0.217	784.0	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

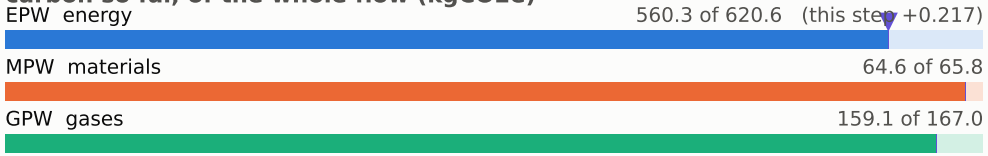




this step

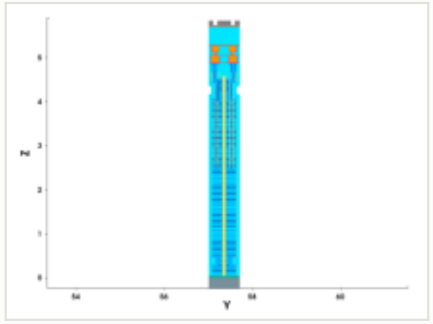
process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

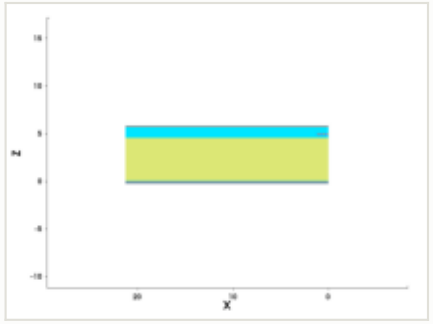


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

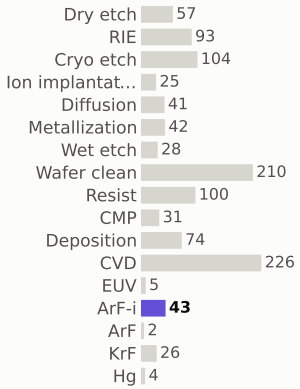


y cut at 57.351 µm



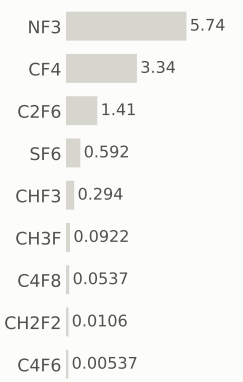
mask: layer 20:0, negative (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



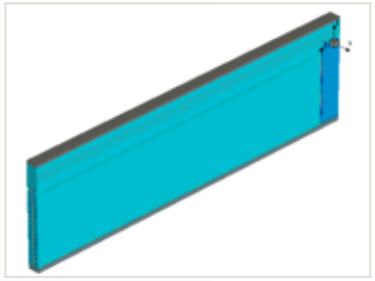
Step #12860: RIE

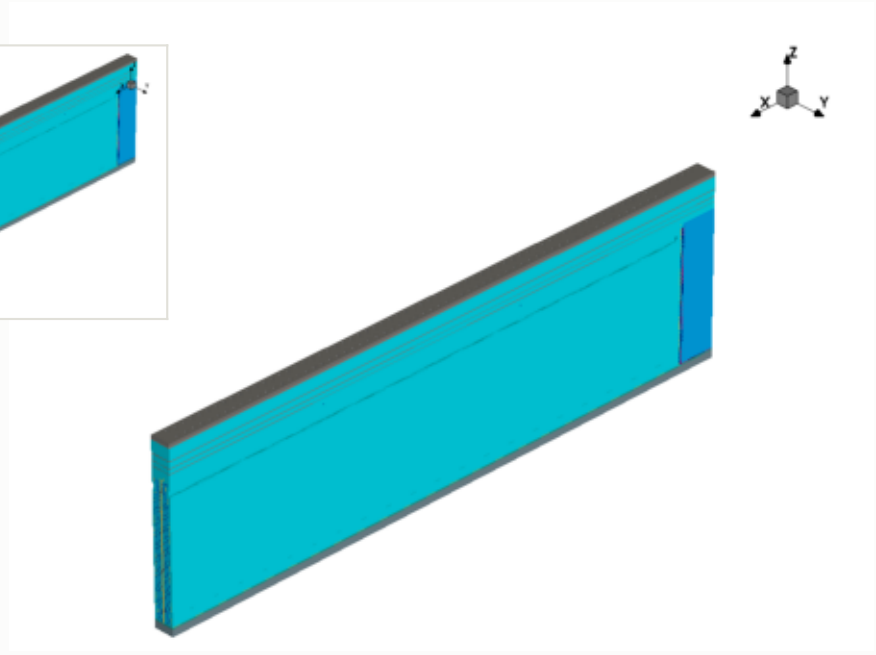
etch_anisotropic · RIE

1.04	785.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



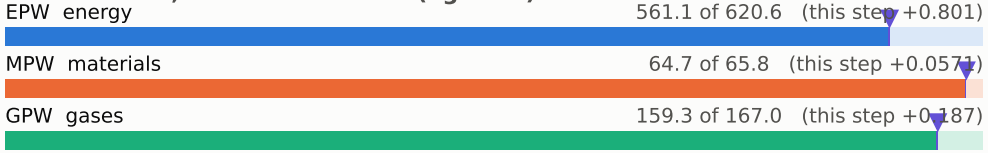


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

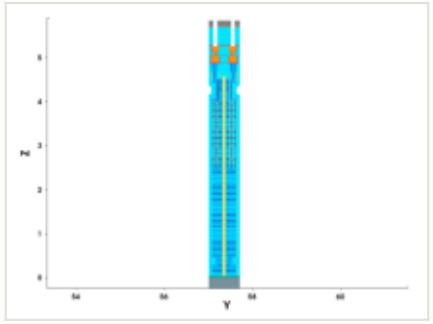
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

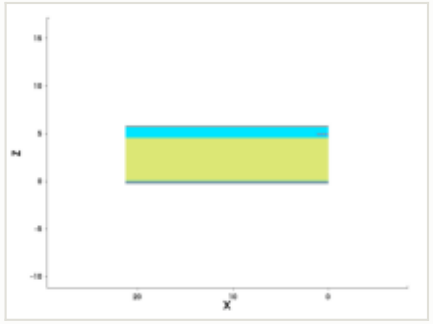


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



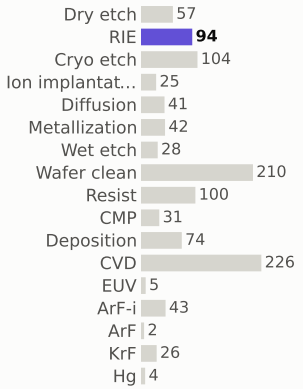
y cut at 57.351 µm



mask for this step

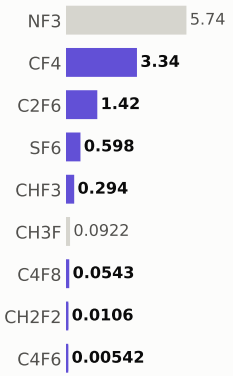
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



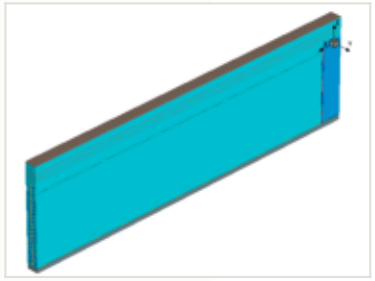
Step #12870: Dry_etch

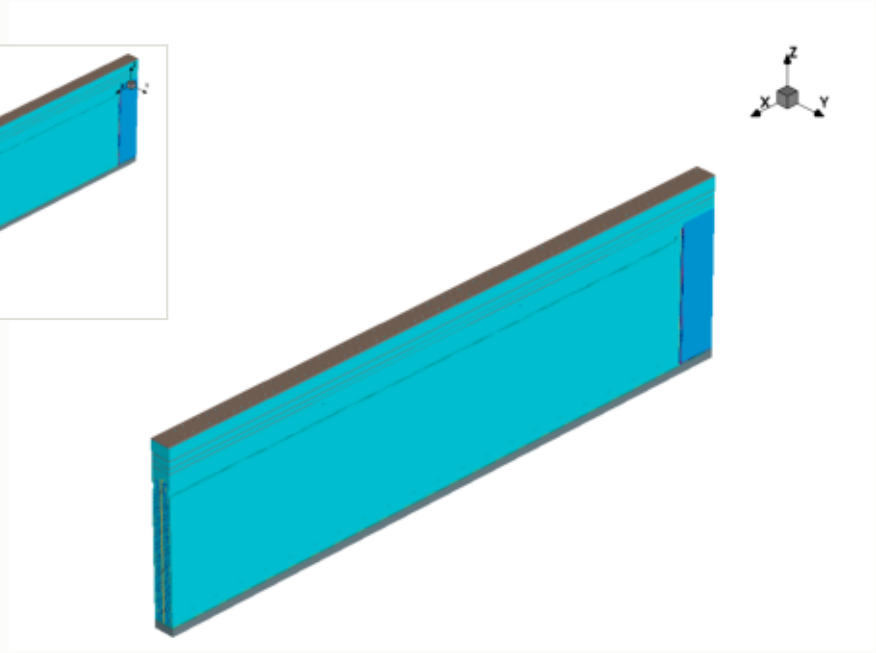
etch_strip · Dry_etch

0.801	785.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

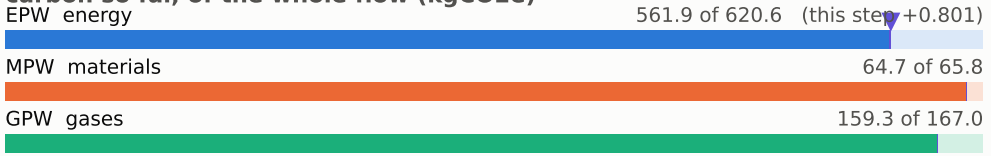




this step

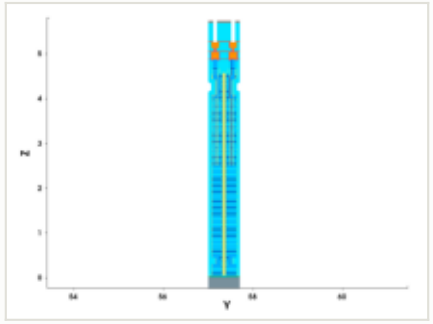
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

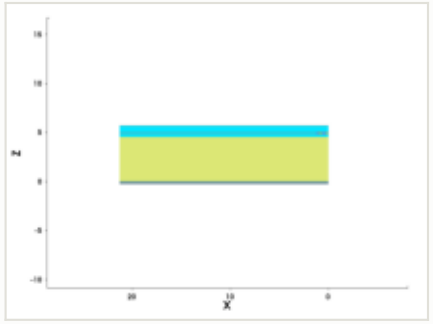


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



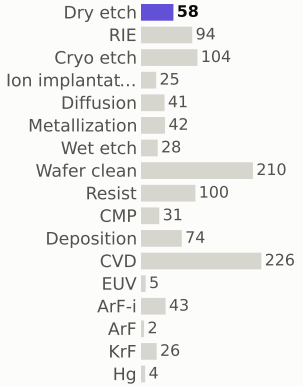
y cut at 57.351 µm



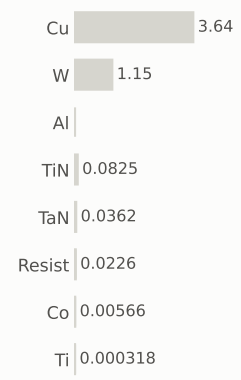
mask for this step

no mask for this step

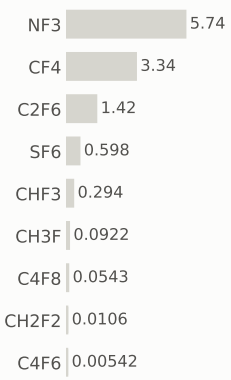
steps by process area



materials deposited (g)



gases emitted (g)



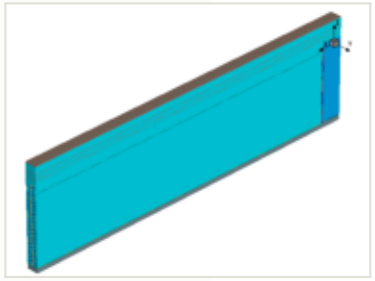
Step #12880: Wafer_clean

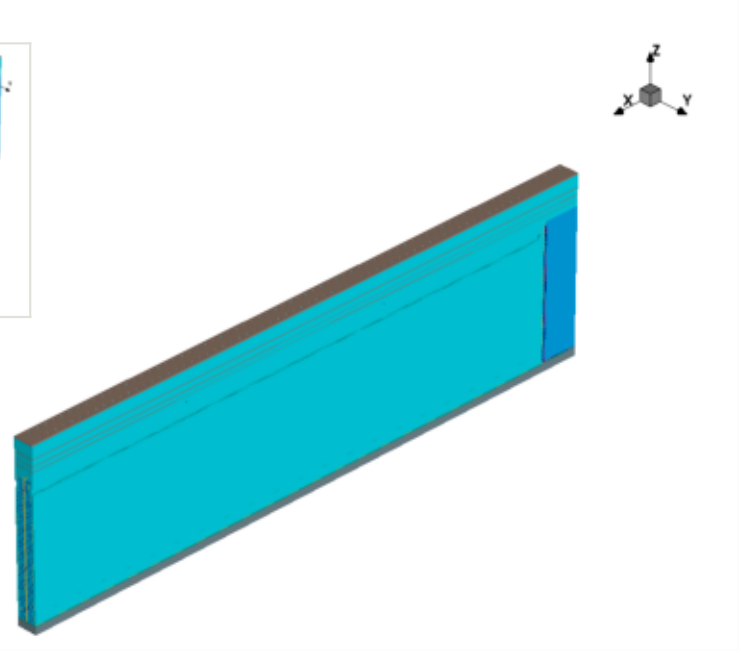
clean · Wafer_clean

0.169	786.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



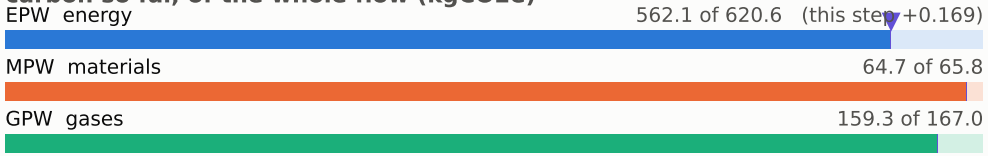


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

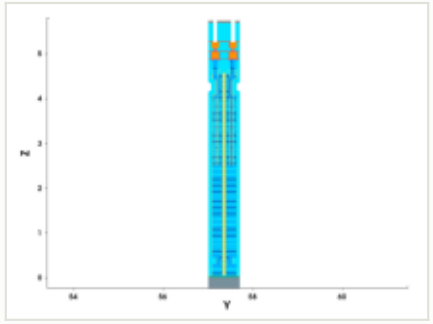
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

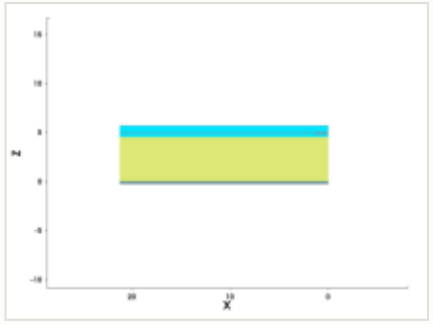


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



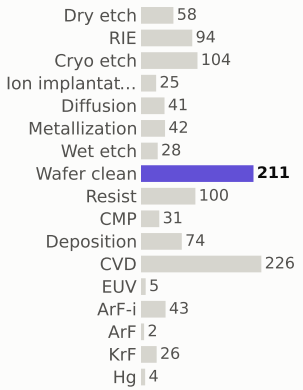
y cut at 57.351 µm



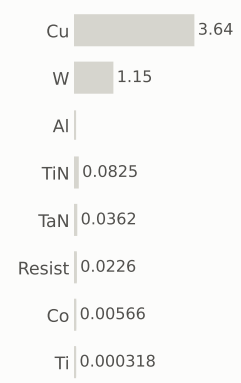
mask for this step

no mask for this step

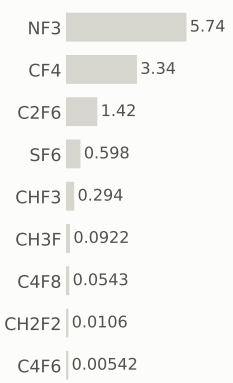
steps by process area



materials deposited (g)



gases emitted (g)



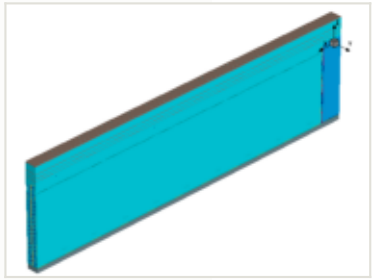
Step #12890: Wafer_clean

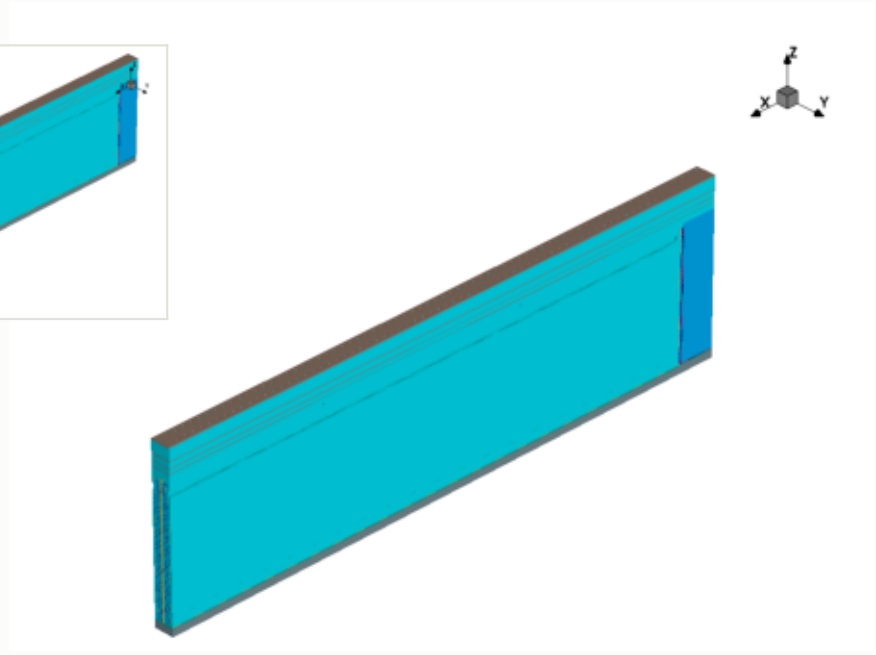
clean · Wafer_clean

0.169	786.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





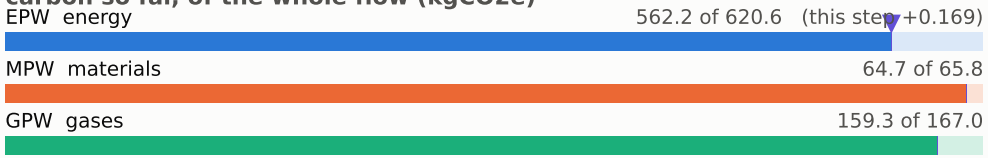
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

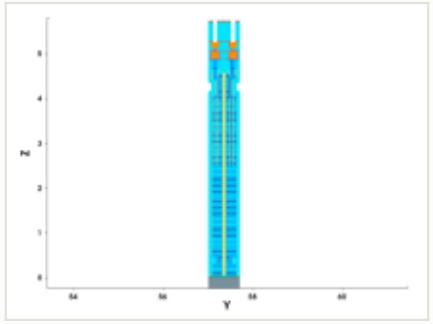
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

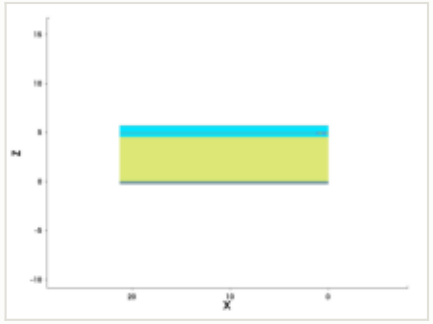


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



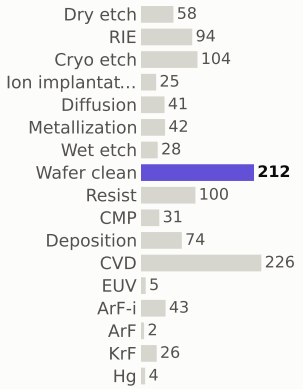
y cut at 57.351 µm



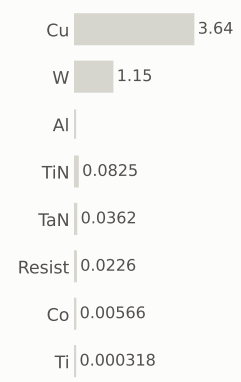
mask for this step

no mask for this step

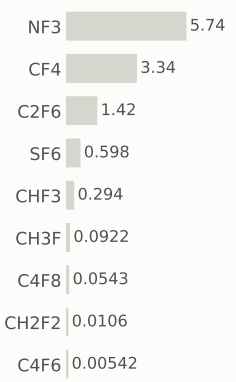
steps by process area



materials deposited (g)



gases emitted (g)



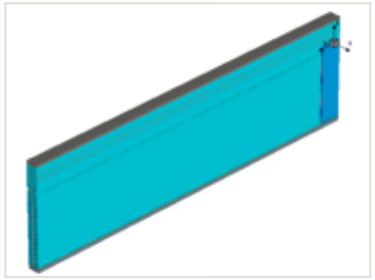
Step #12900: ArF-i

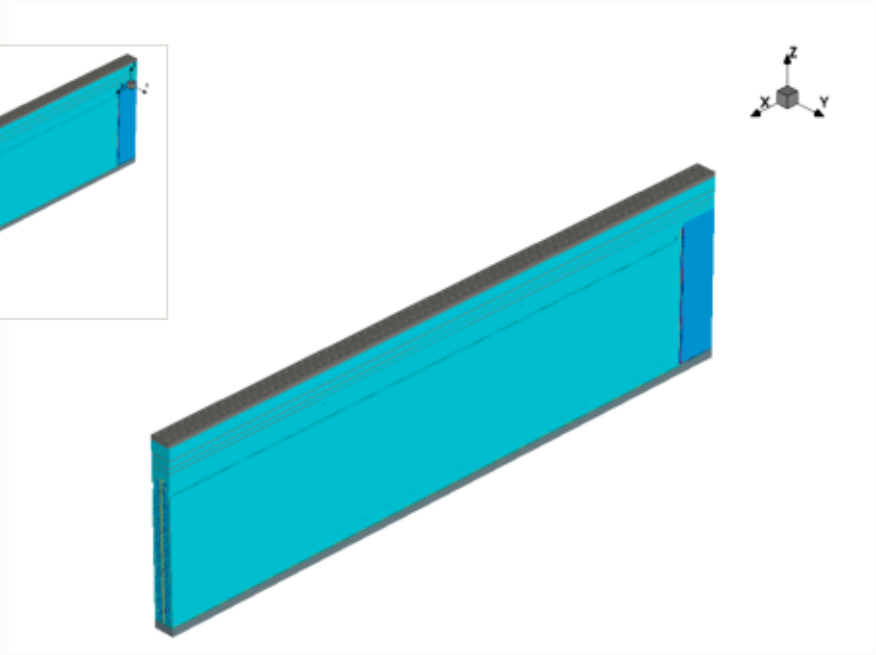
litho_patterning · ArF-i

0.217	786.4	0.025%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



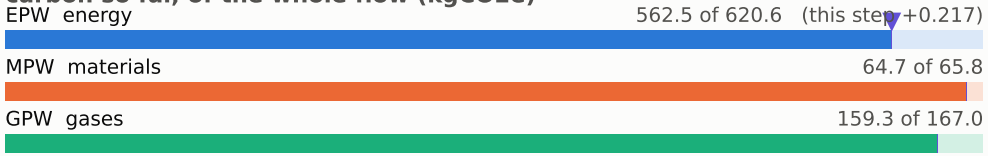


this step

process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

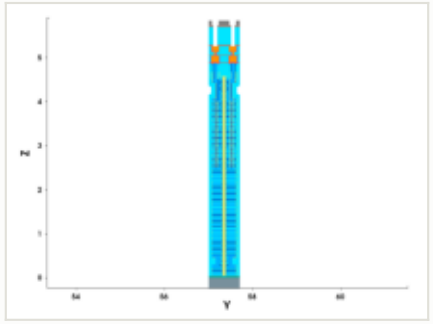
EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

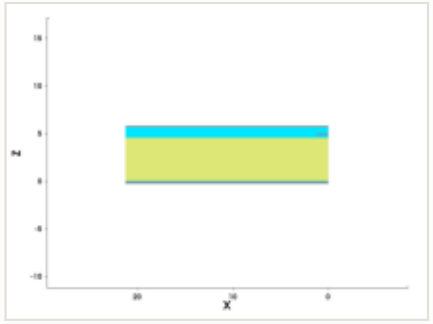


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

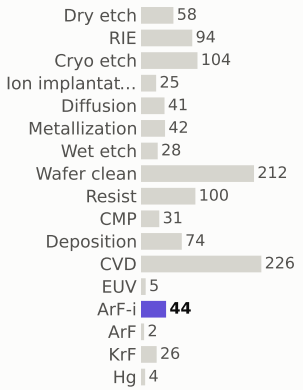


y cut at 57.351 µm

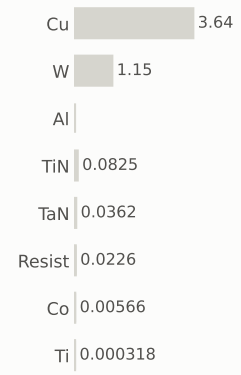


mask: layer 21:0, negative (GDSII)

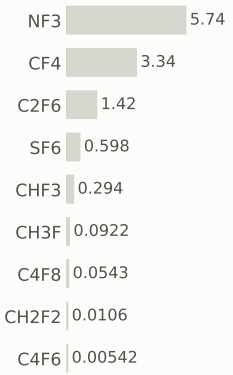
steps by process area



materials deposited (g)



gases emitted (g)



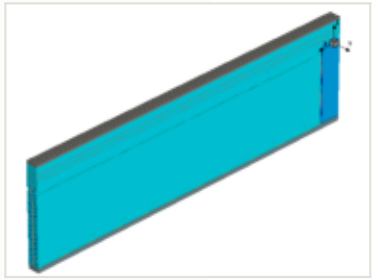
Step #12910: RIE

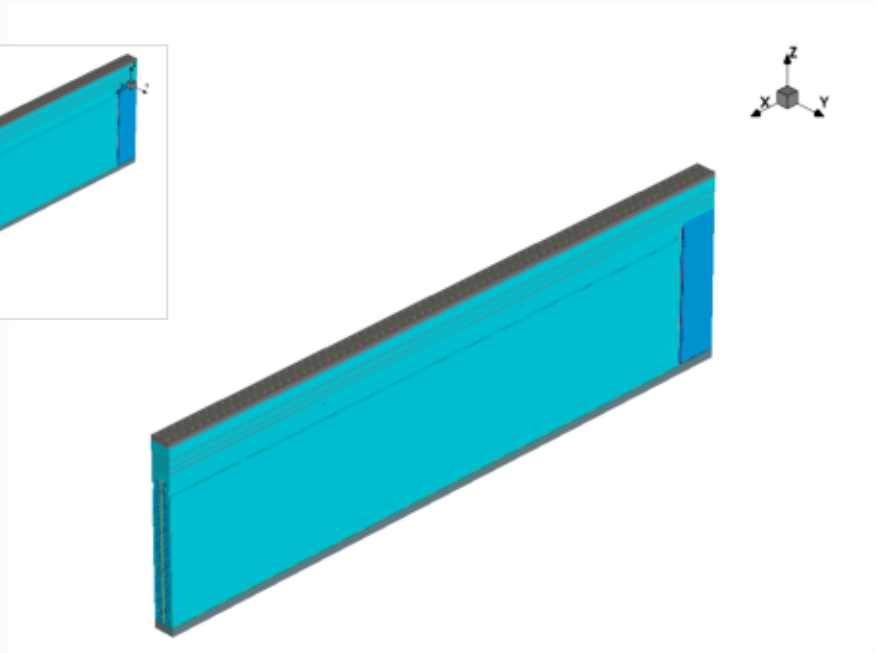
etch_anisotropic · RIE

1.04	787.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



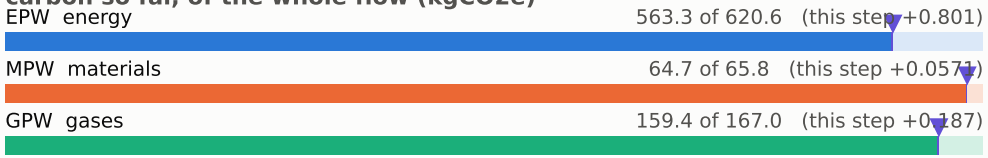


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

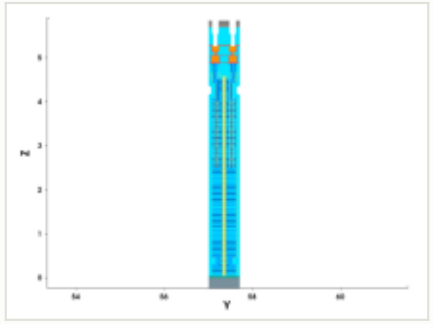
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

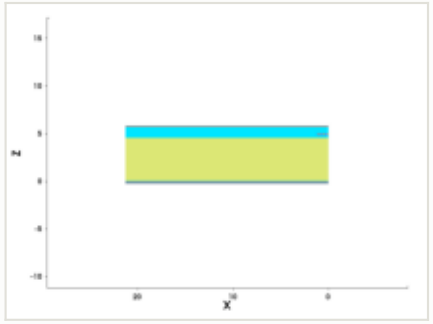


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



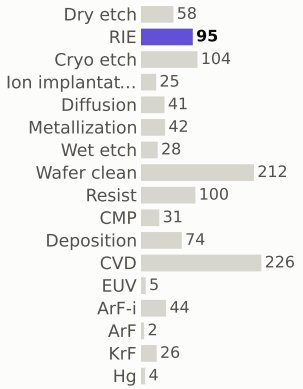
y cut at 57.351 µm



mask for this step

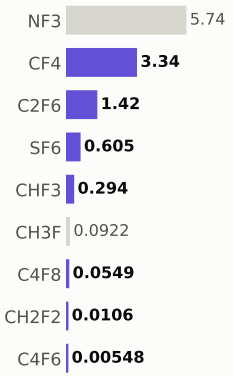
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



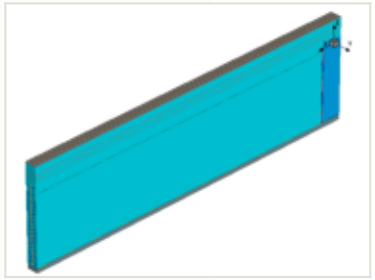
Step #12920: Dry_etch

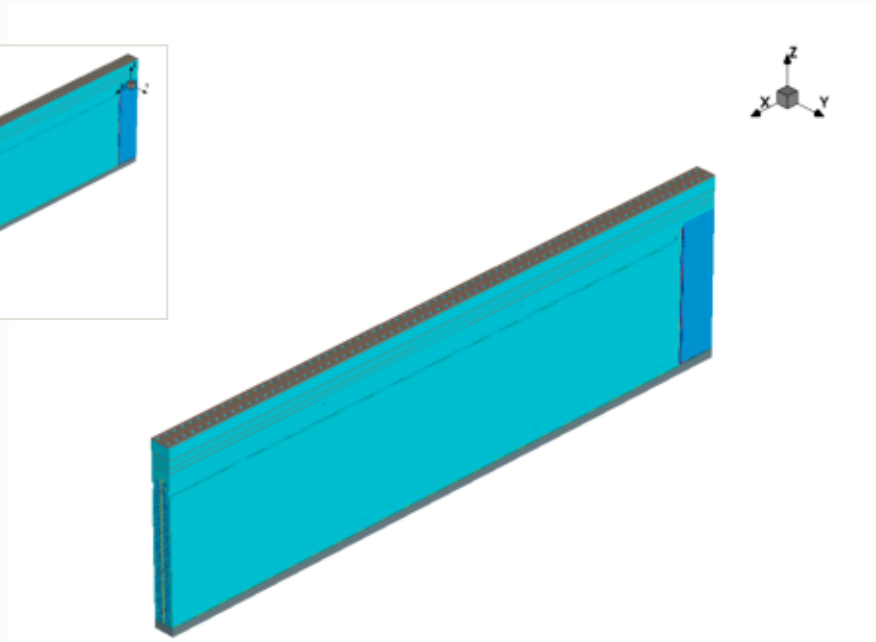
etch_strip · Dry_etch

0.801	788.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

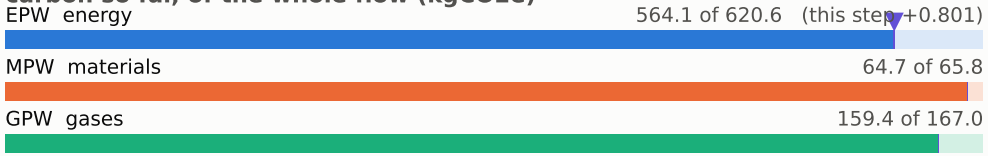




this step

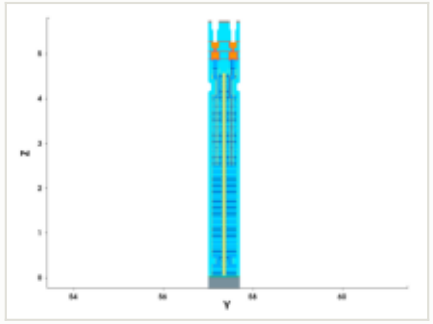
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

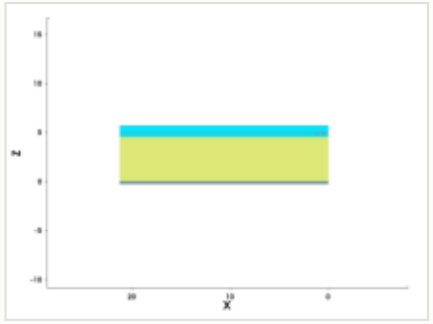


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



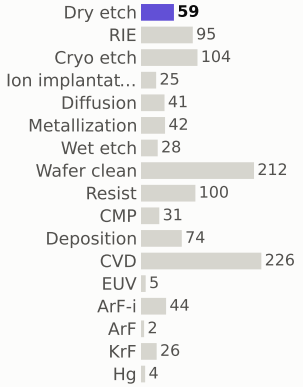
y cut at 57.351 µm



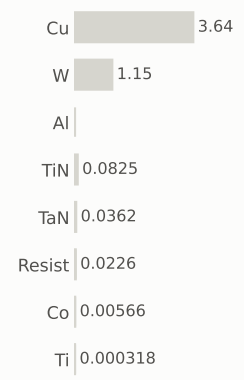
mask for this step

no mask for this step

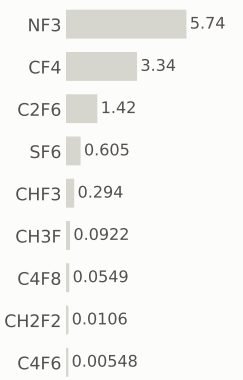
steps by process area



materials deposited (g)



gases emitted (g)



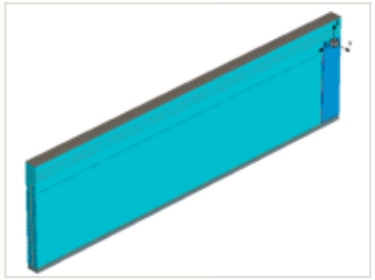
Step #12930: Wafer_clean

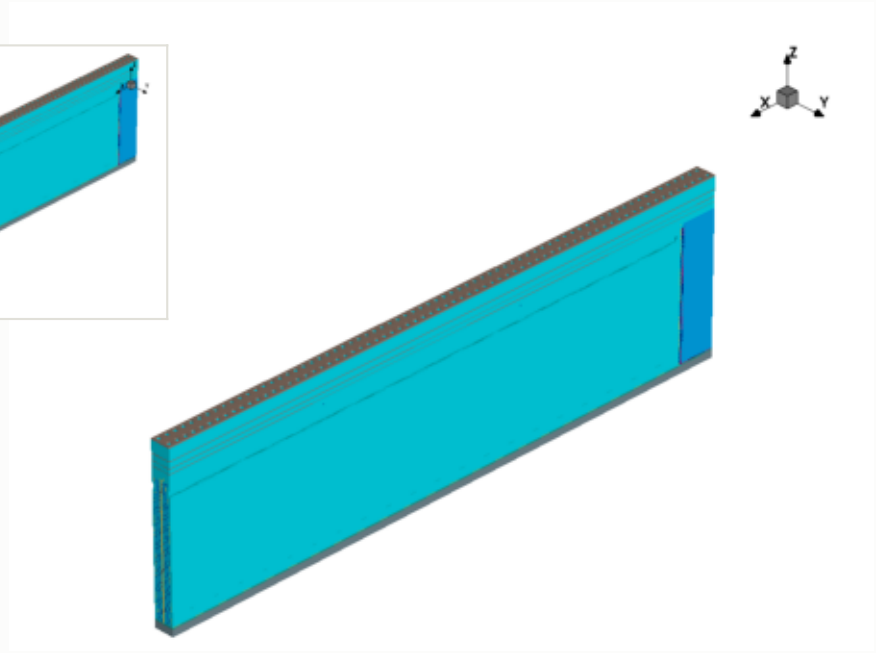
clean · Wafer_clean

0.169	788.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





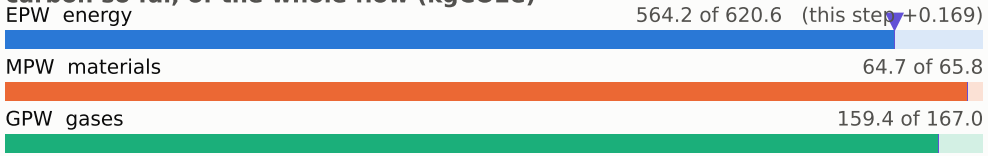
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

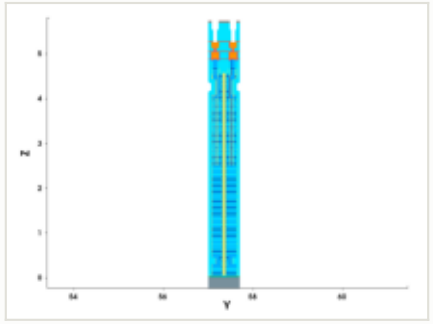
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

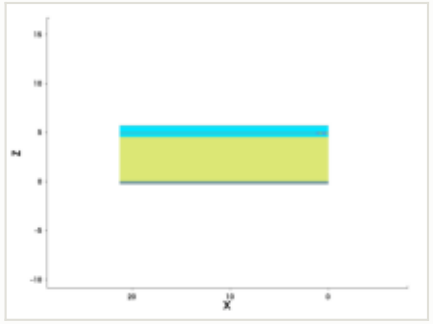


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



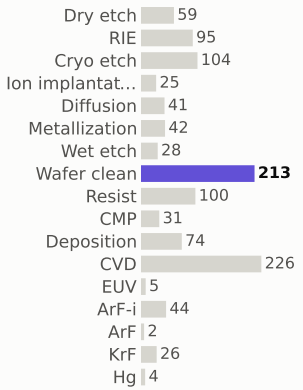
y cut at 57.351 µm



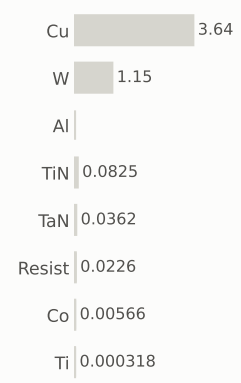
mask for this step

no mask for this step

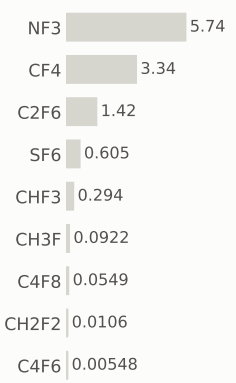
steps by process area



materials deposited (g)



gases emitted (g)



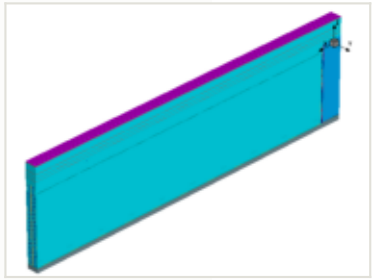
Step #12940: Metallization

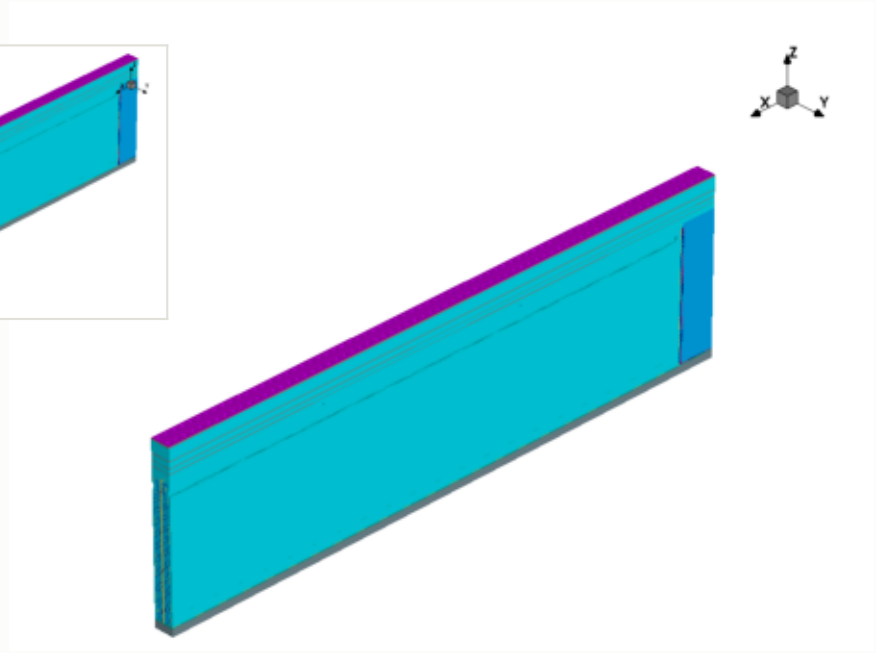
depo_isotropic · Metallization

1.2	789.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



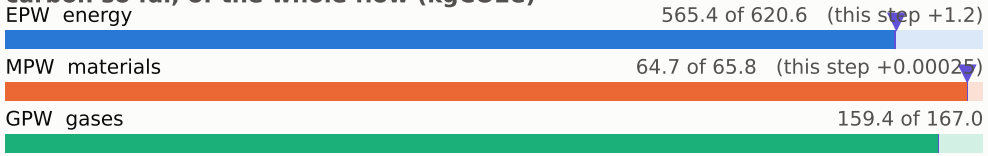


this step

process area	Metallization
material	TaN
thickness	0.01 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

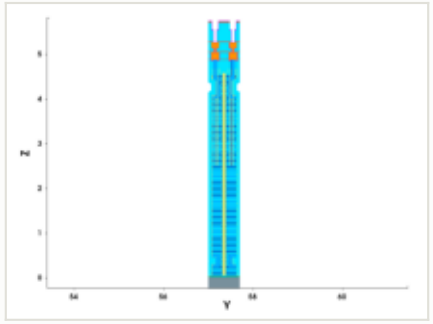
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00025 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

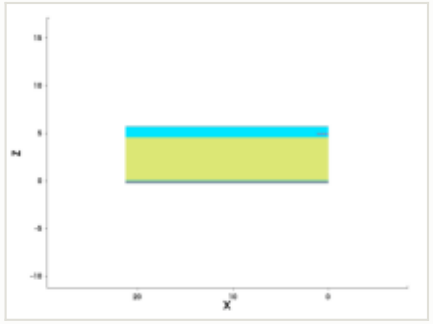


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



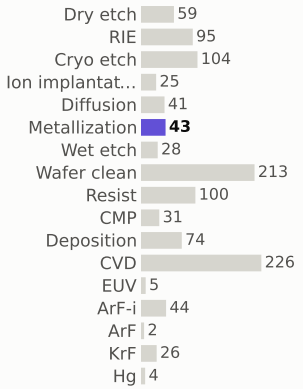
y cut at 57.351 µm



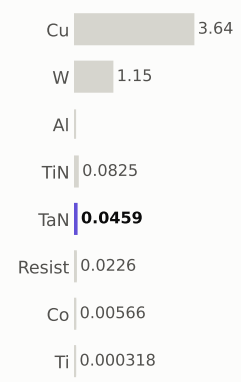
mask for this step

no mask for this step

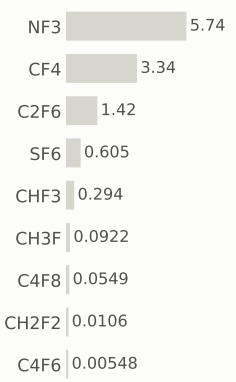
steps by process area



materials deposited (g)



gases emitted (g)



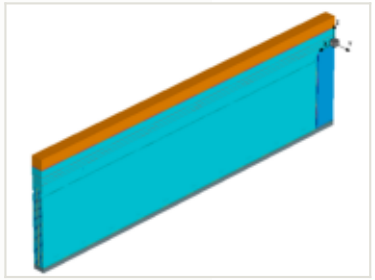
Step #12950: Metallization

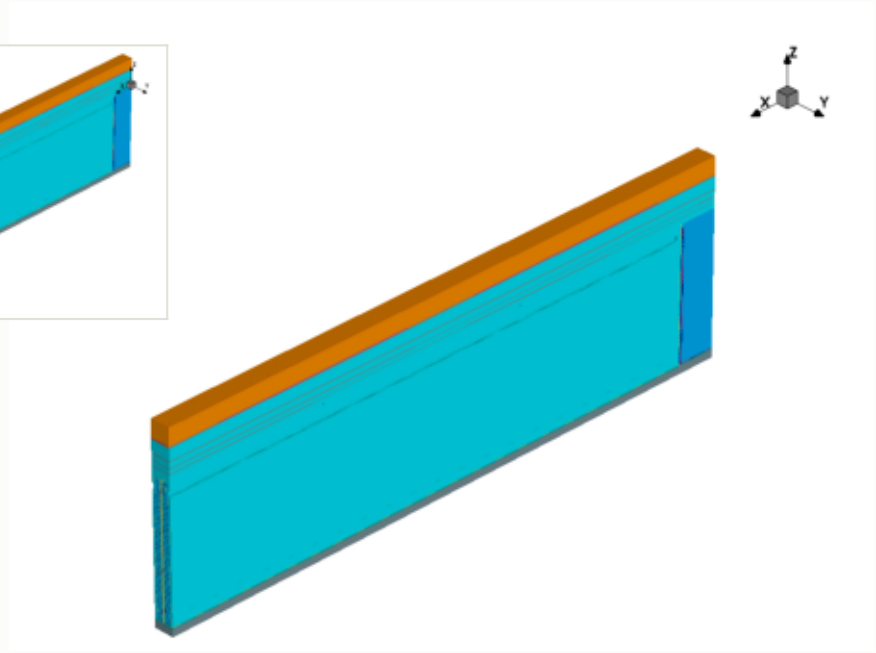
depo_isotropic · Metallization

1.21	790.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



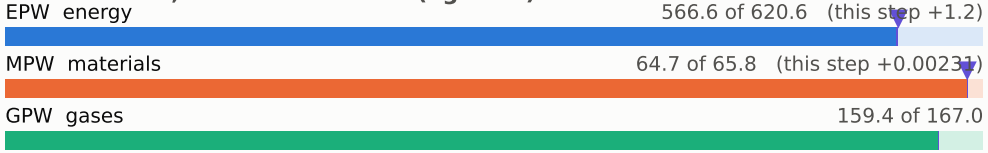


this step

process area	Metallization
material	Cu
thickness	0.6 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

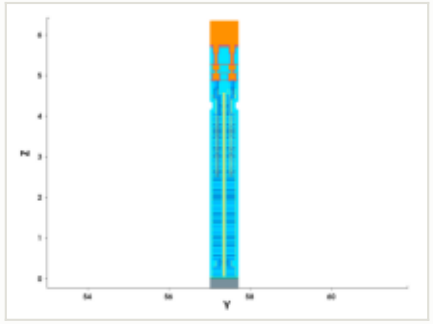
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00231 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

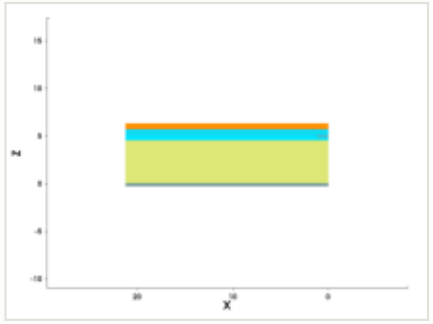


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



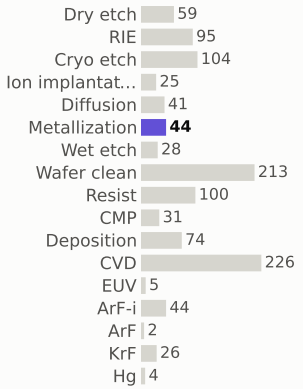
y cut at 57.351 µm



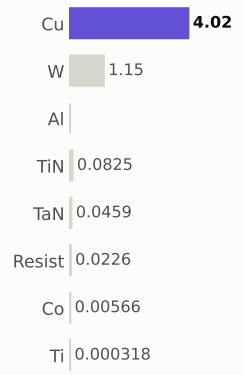
mask for this step

no mask for this step

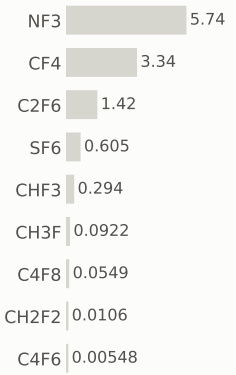
steps by process area



materials deposited (g)



gases emitted (g)



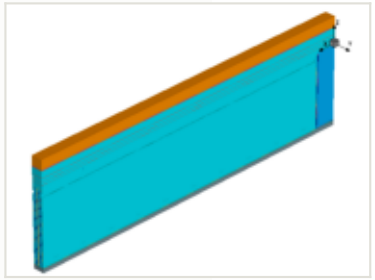
Step #12960: Diffusion

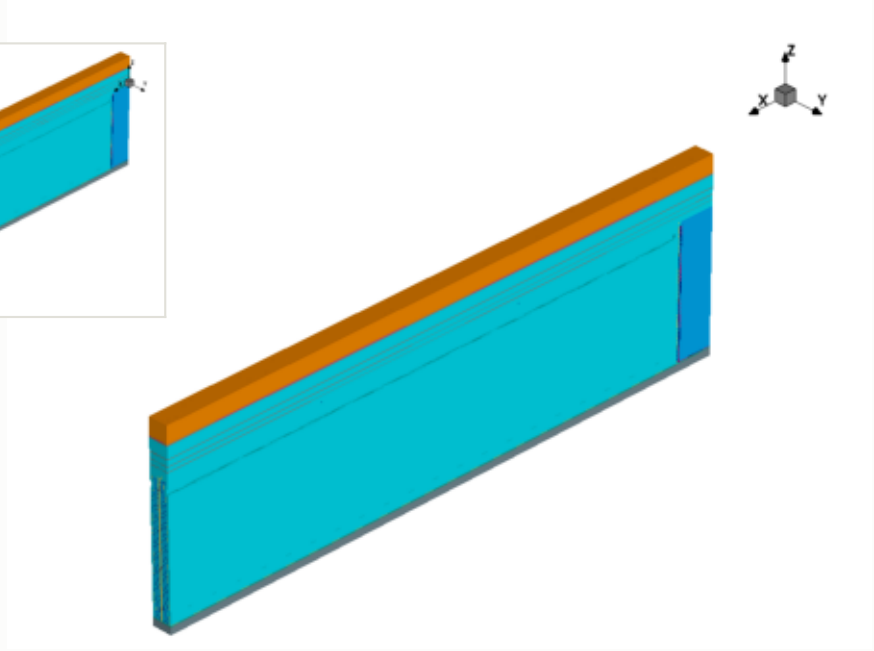
anneal · Diffusion

0.801	791.6	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





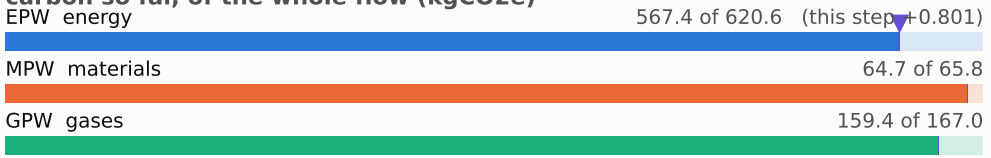
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

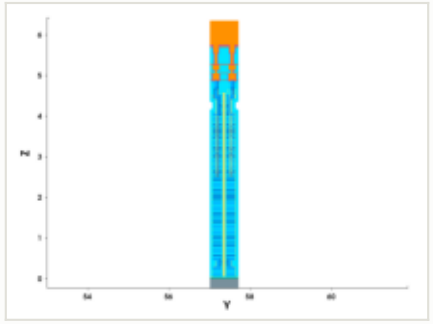
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

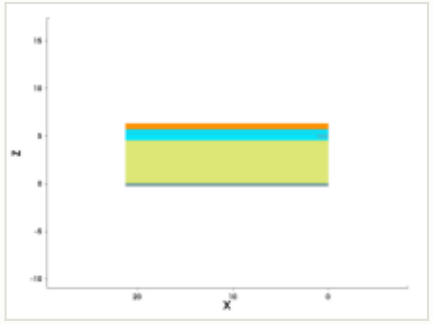


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



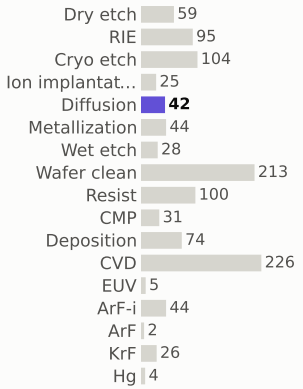
y cut at 57.351 µm



mask for this step

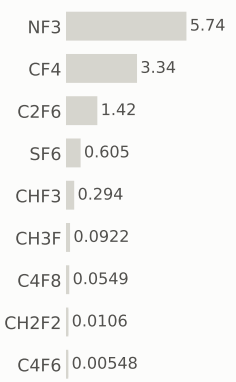
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



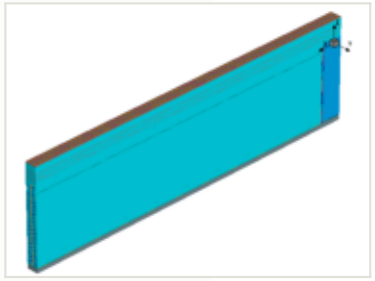
Step #12970: CMP

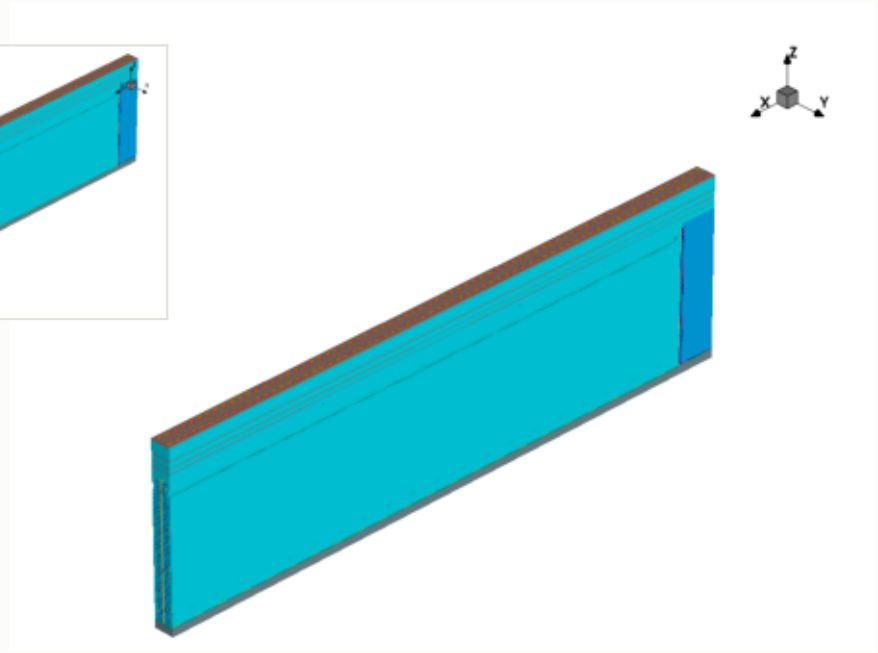
etch_cmp · CMP

0.169	791.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



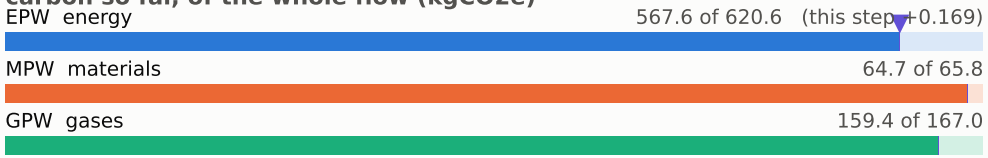


this step

process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

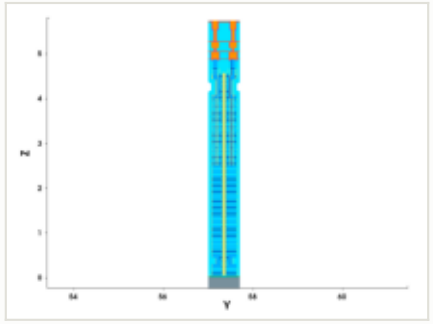
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

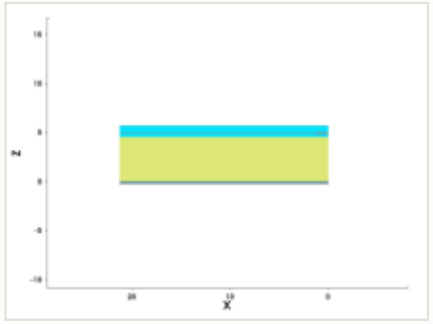


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



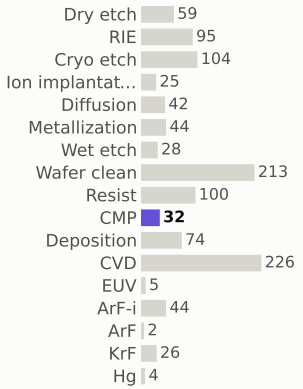
y cut at 57.351 µm



mask for this step

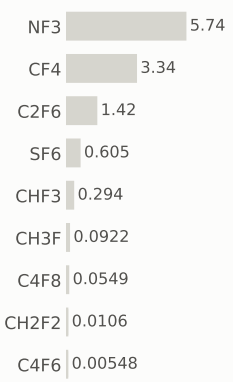
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



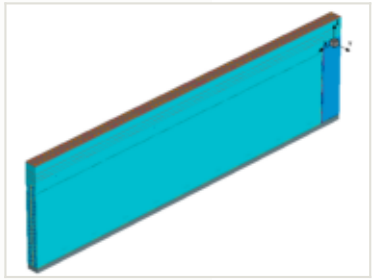
Step #12980: Wafer_clean

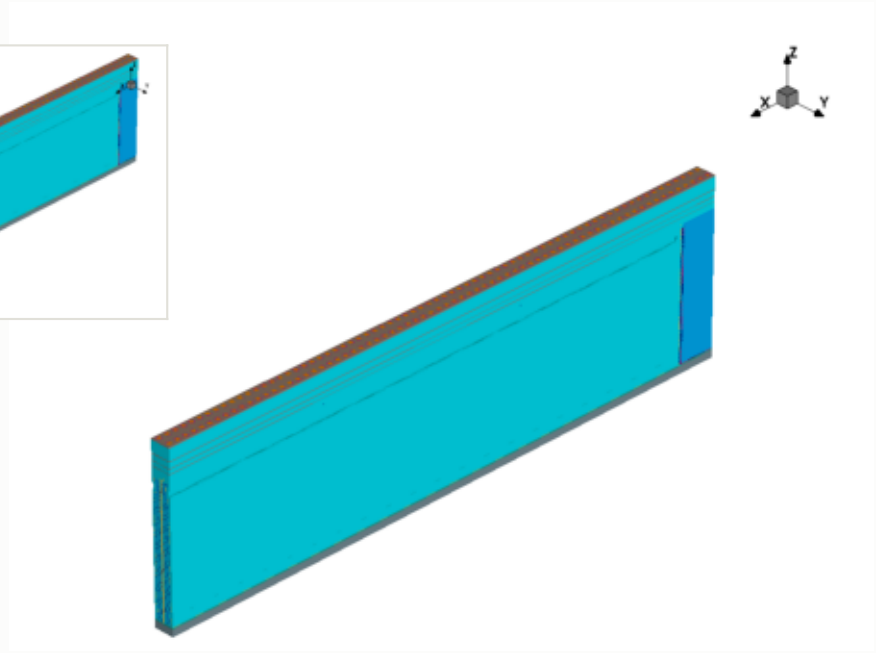
clean · Wafer_clean

0.169	792.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





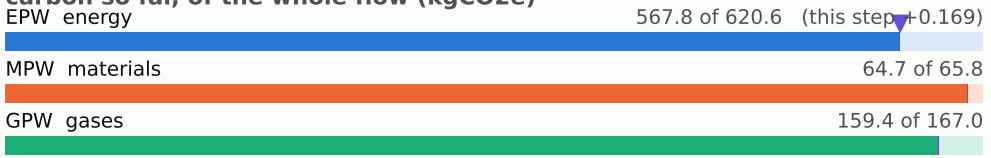
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

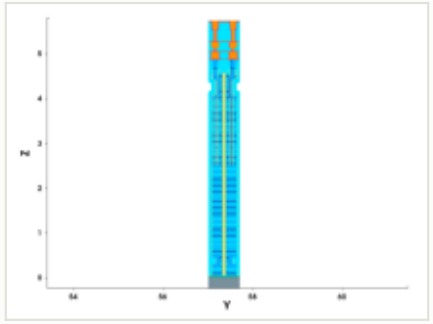
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

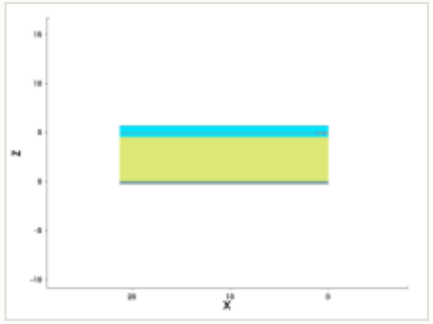


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



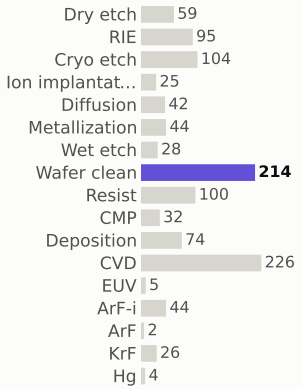
y cut at 57.351 µm



mask for this step

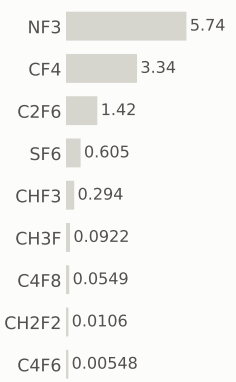
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



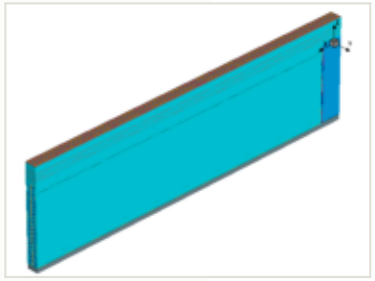
Step #12990: Dry_etch

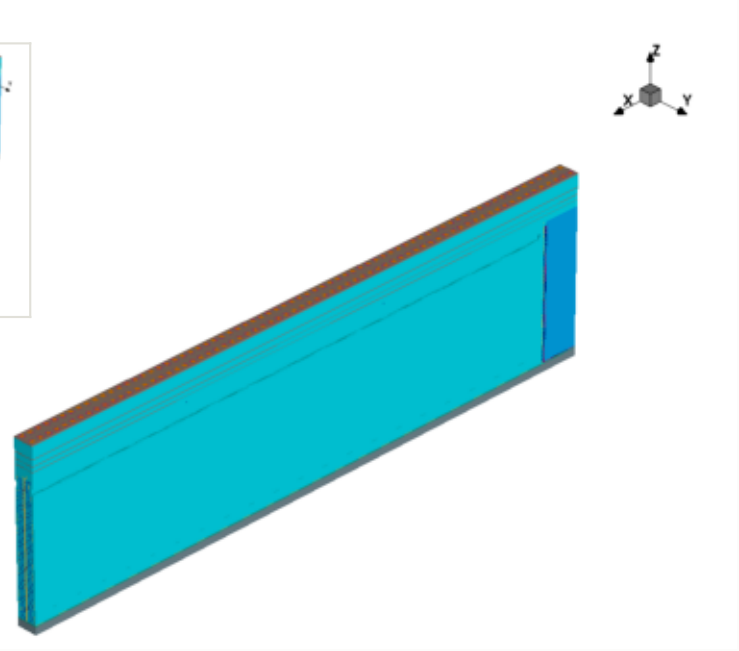
treatment · Dry_etch

0.801	792.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





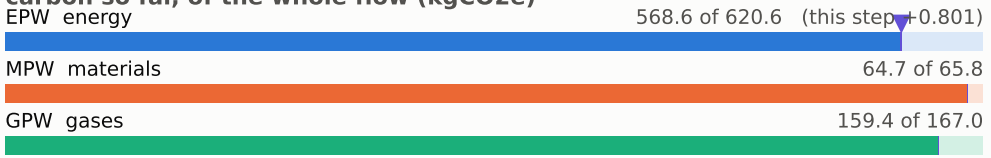
this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

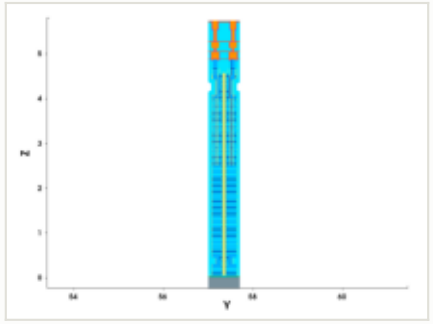
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

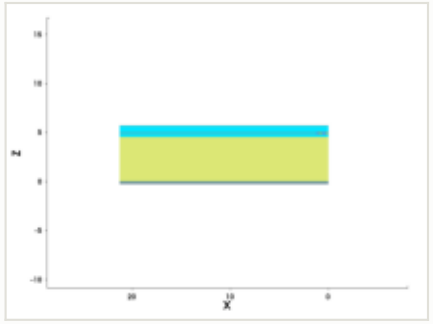


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



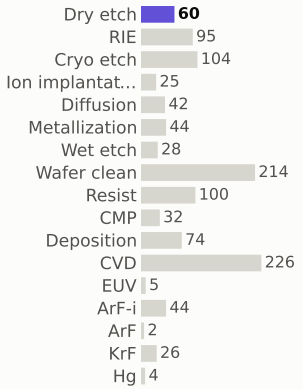
y cut at 57.351 µm



mask for this step

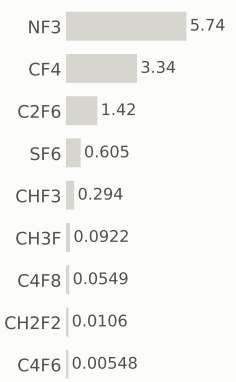
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



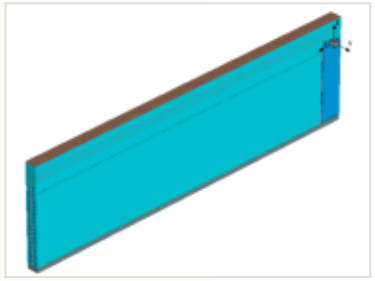
Step #13010: Wafer

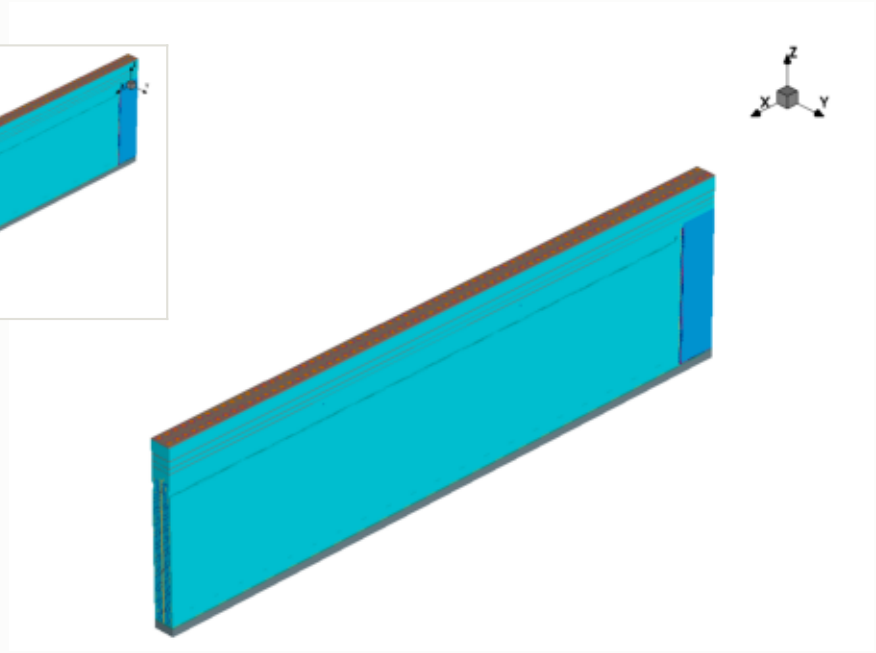
treatment · Wafer

0	792.8	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





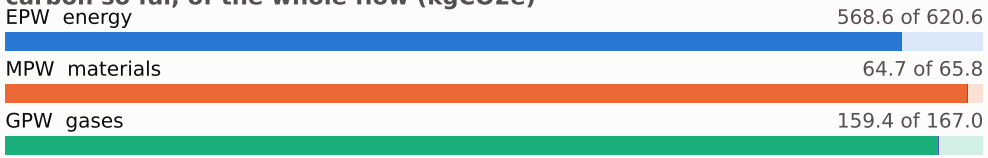
this step

process area	Wafer
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

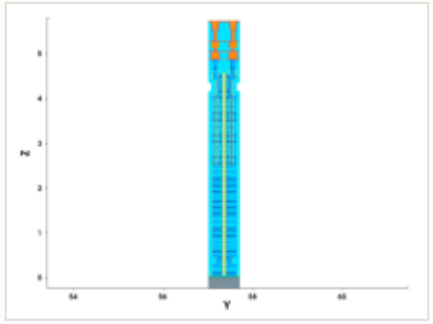
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

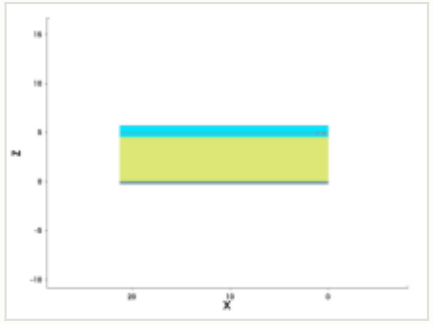


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



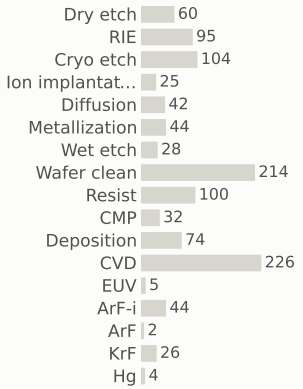
y cut at 57.351 µm



mask for this step

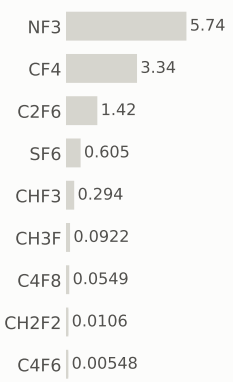
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



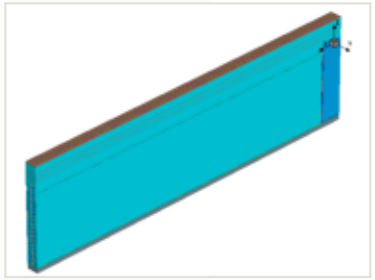
Step #13030: Wafer_clean

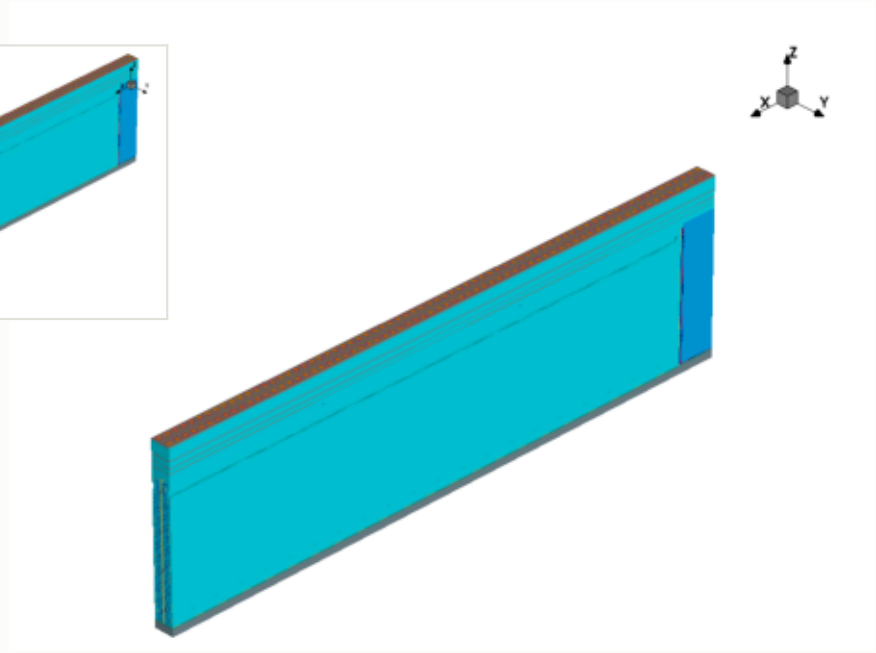
clean · Wafer_clean

0.169	792.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





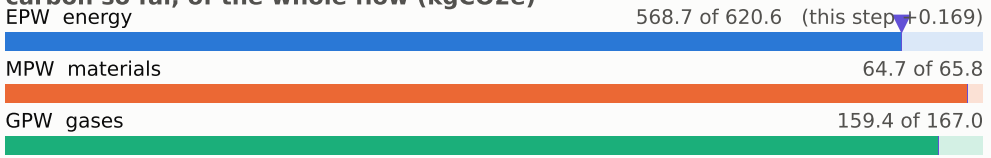
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

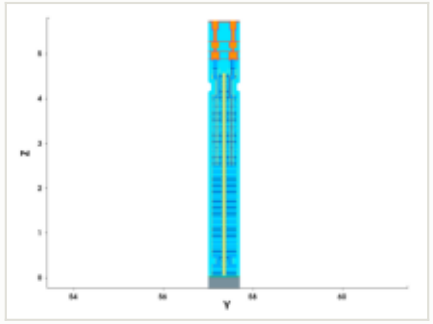
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

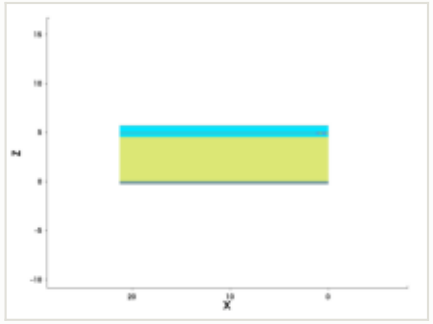


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



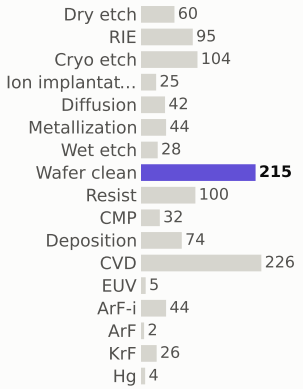
y cut at 57.351 µm



mask for this step

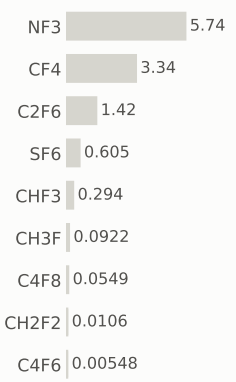
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



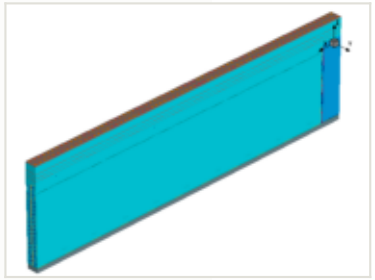
Step #13040: Diffusion

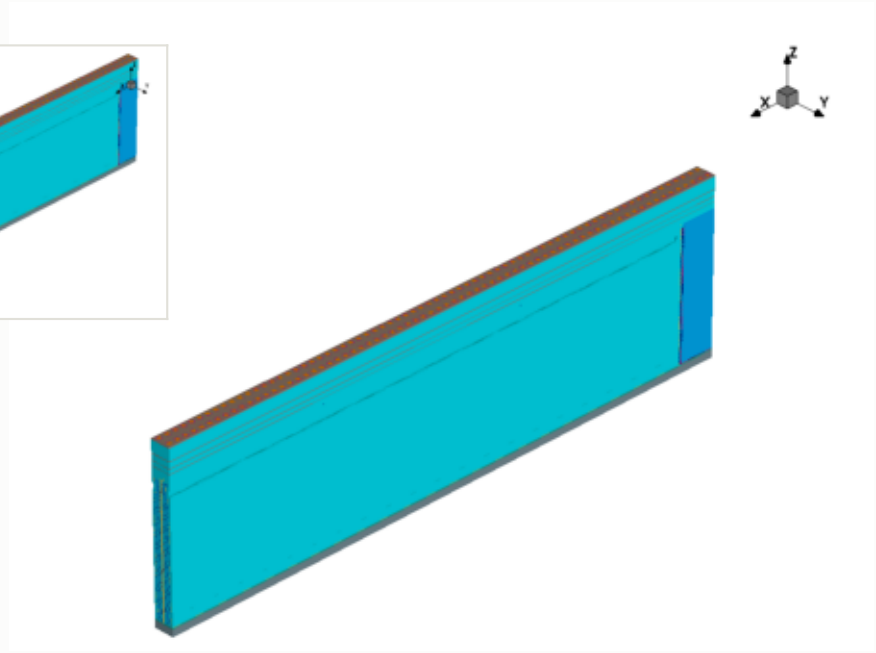
treatment · Diffusion

0.801	793.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



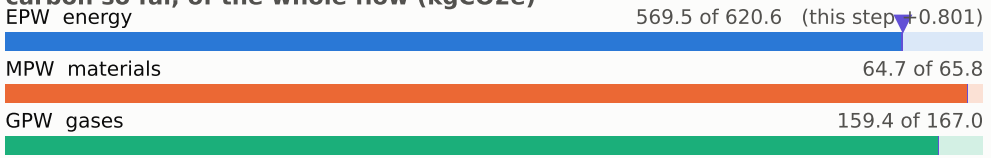


this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

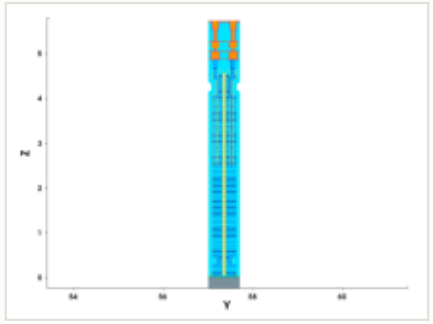
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

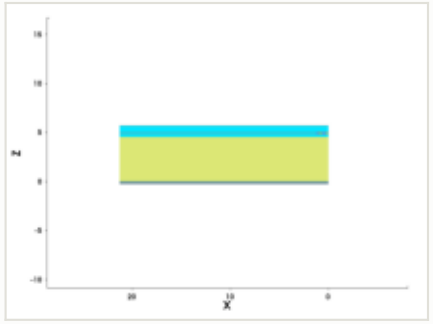


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



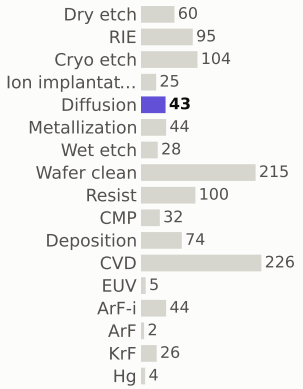
y cut at 57.351 µm



mask for this step

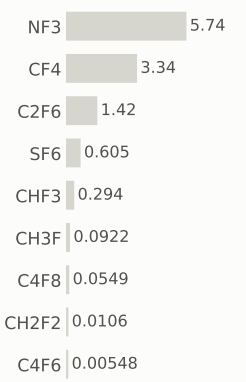
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



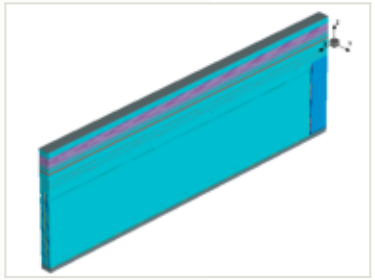
Step #13060: Dry_etch

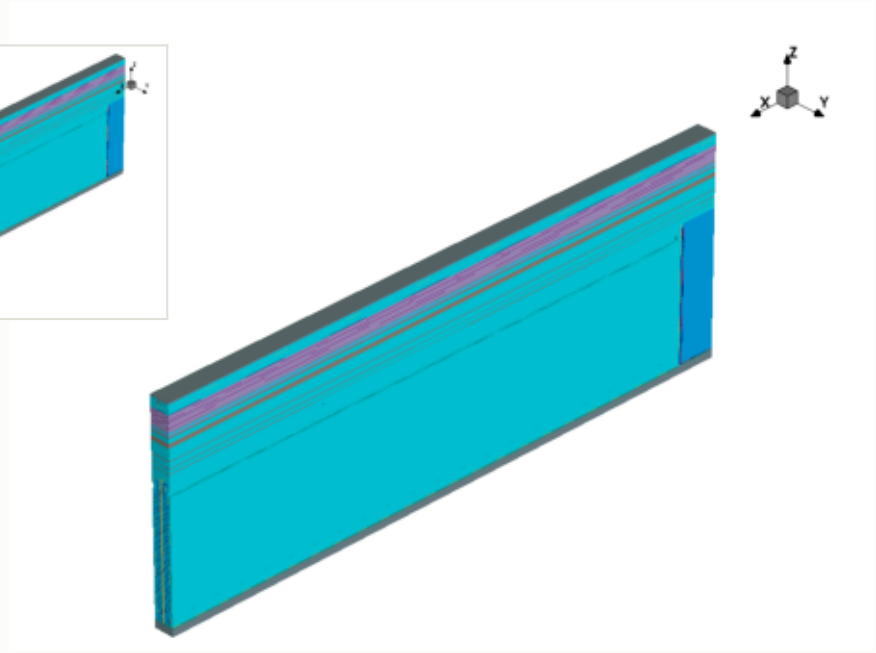
bond_structure · Dry_etch

0.801	794.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

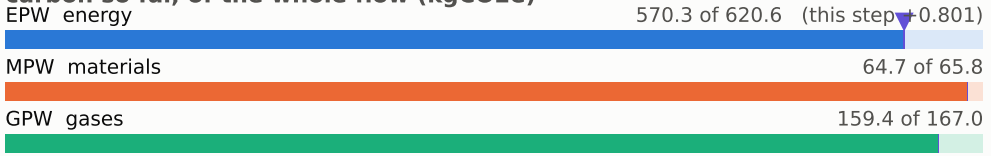




this step

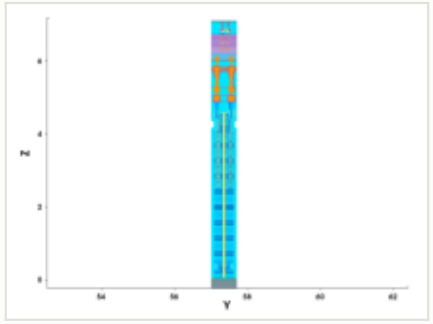
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

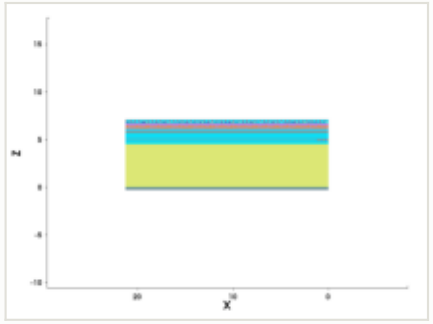


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



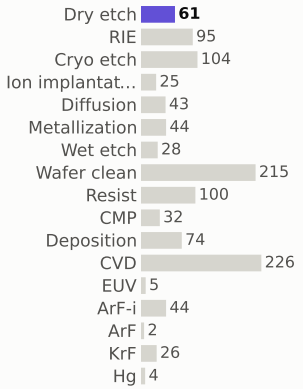
y cut at 57.351 µm



mask for this step

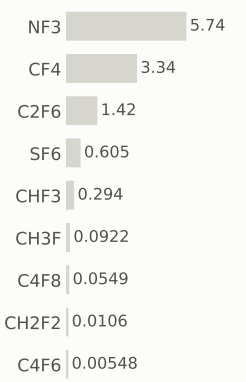
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



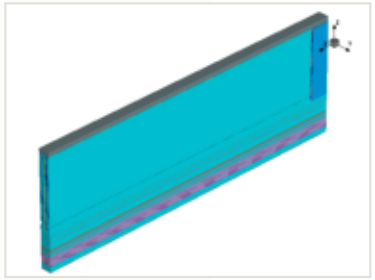
Step #13065: Wafer

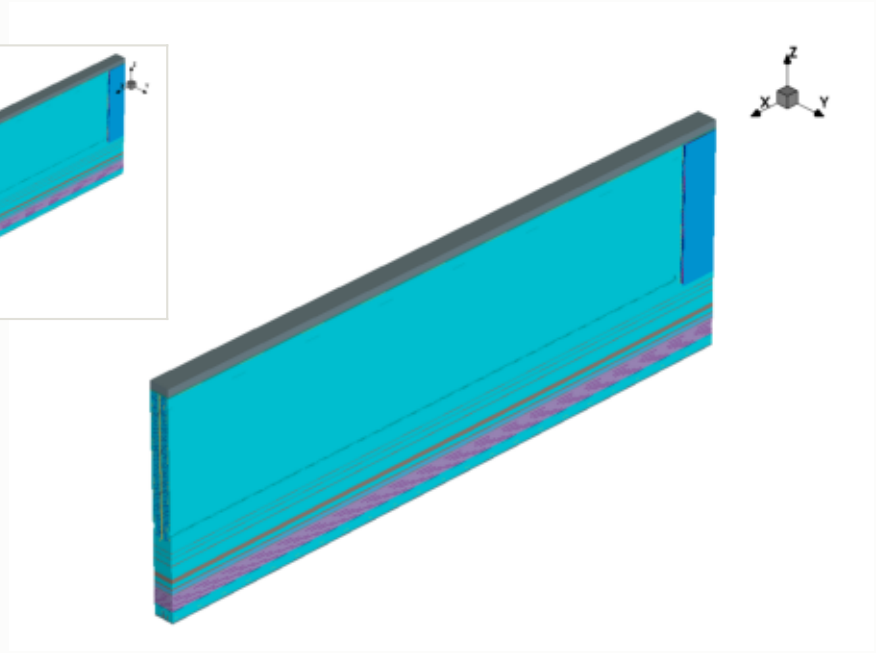
flip_structure · Wafer

0	794.5	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





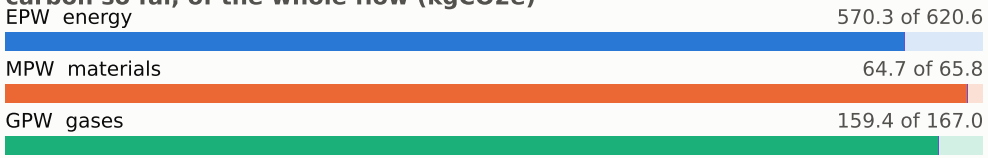
this step

process area	Wafer
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

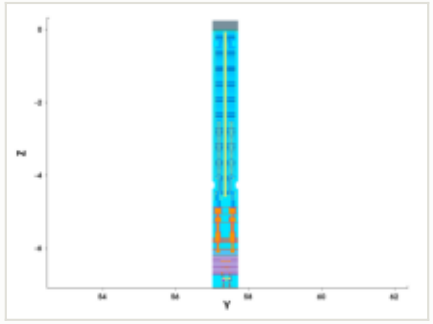
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

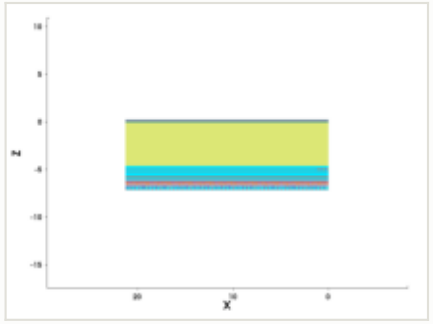


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



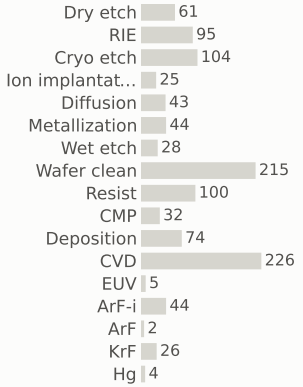
y cut at 57.351 µm



mask for this step

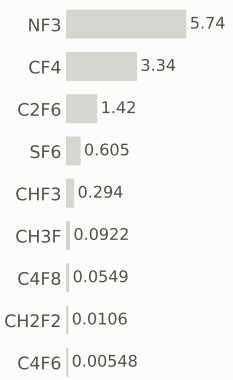
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



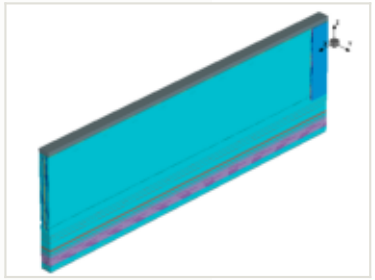
Step #13070: Dry_etch

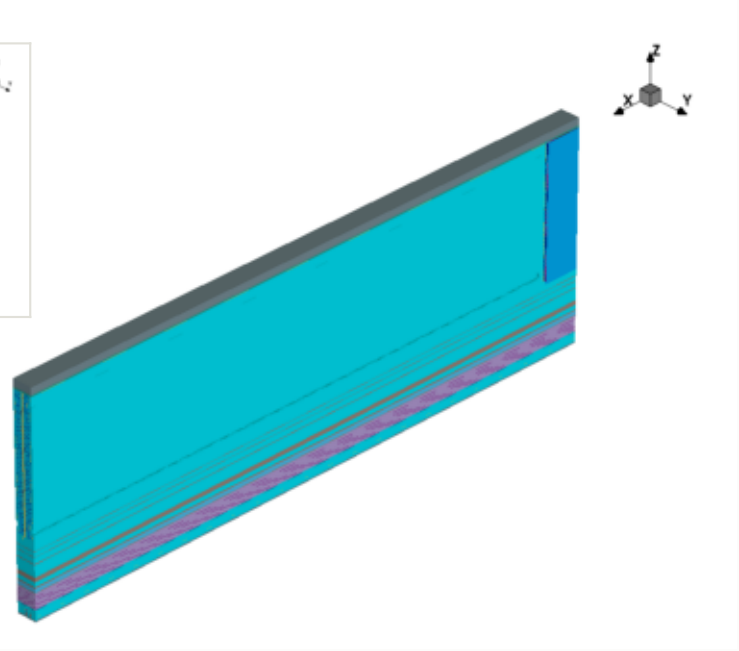
treatment · Dry_etch

0.801	795.3	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





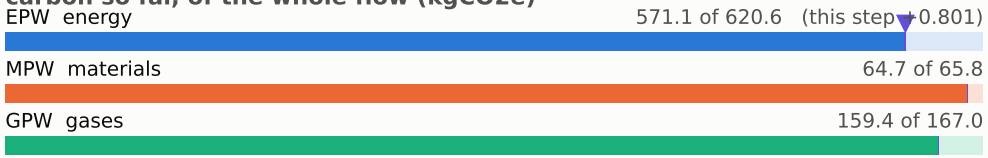
this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

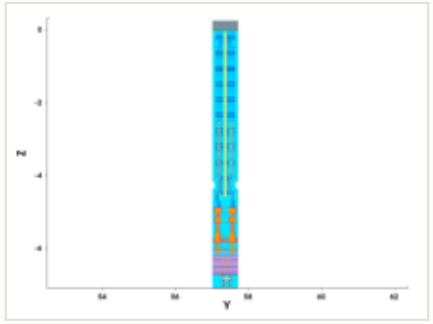
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

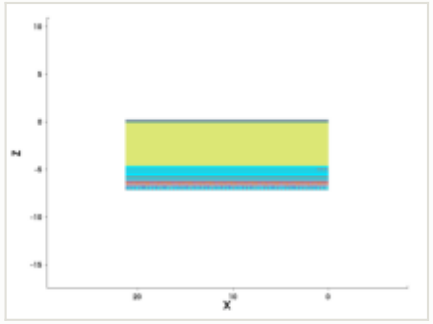


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



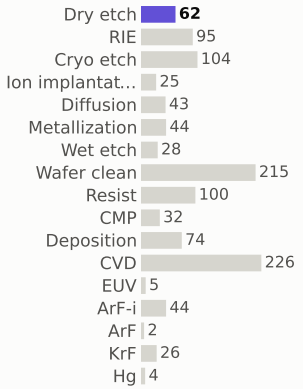
y cut at 57.351 µm



mask for this step

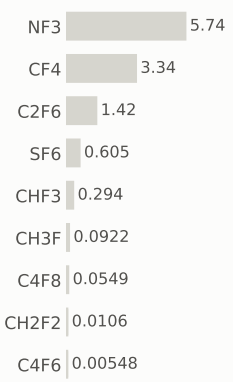
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



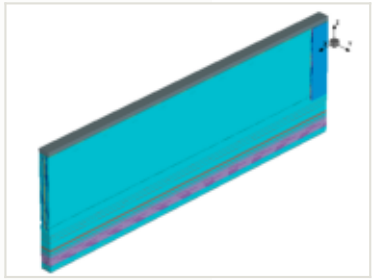
Step #13080: Diffusion

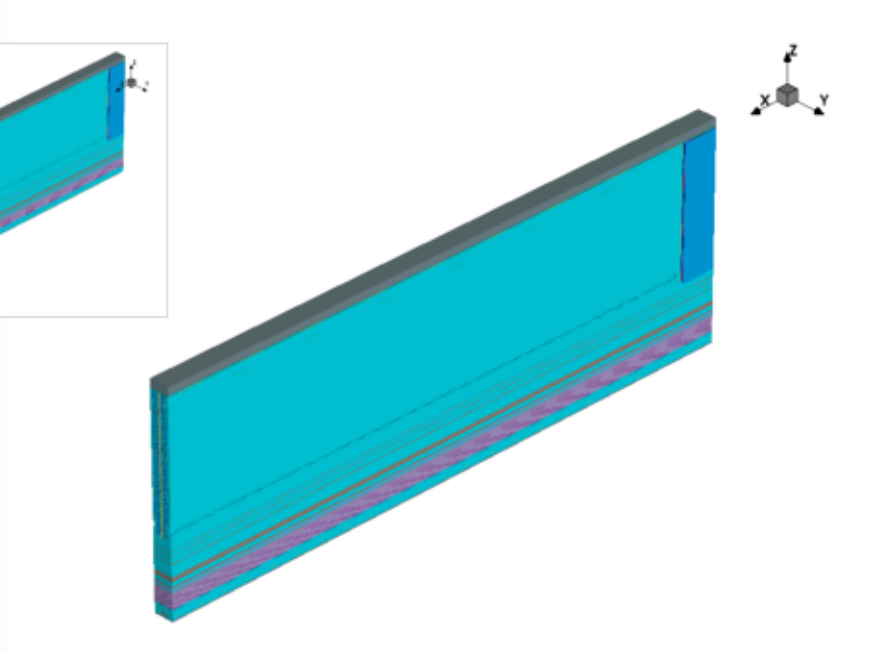
anneal · Diffusion

0.801	796.1	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



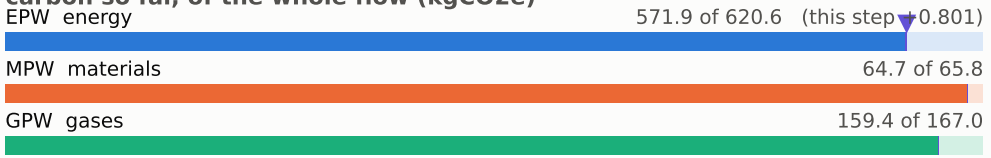


this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

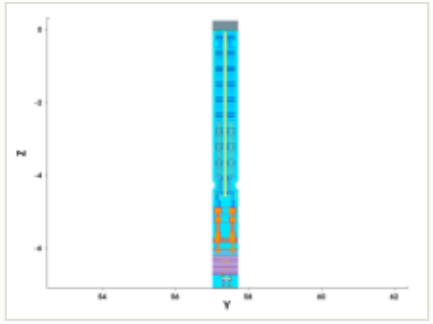
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

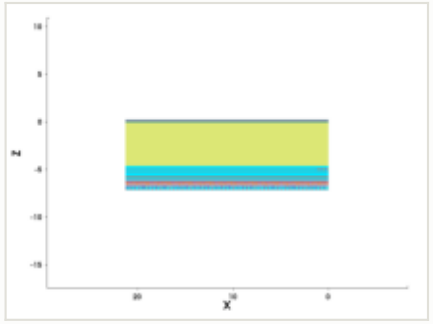


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



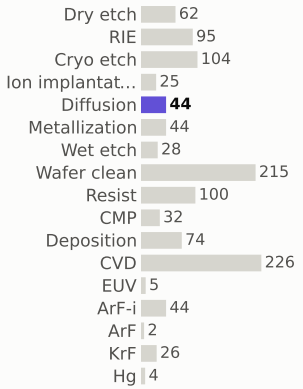
y cut at 57.351 µm



mask for this step

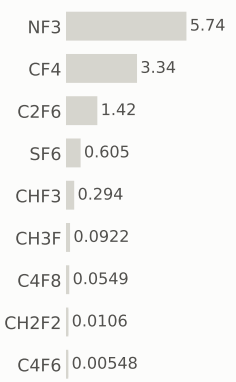
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



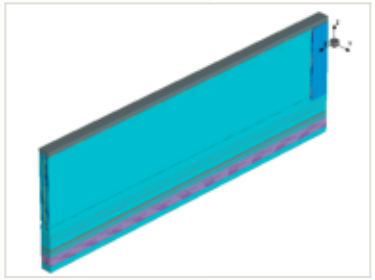
Step #13100: Dry_etch

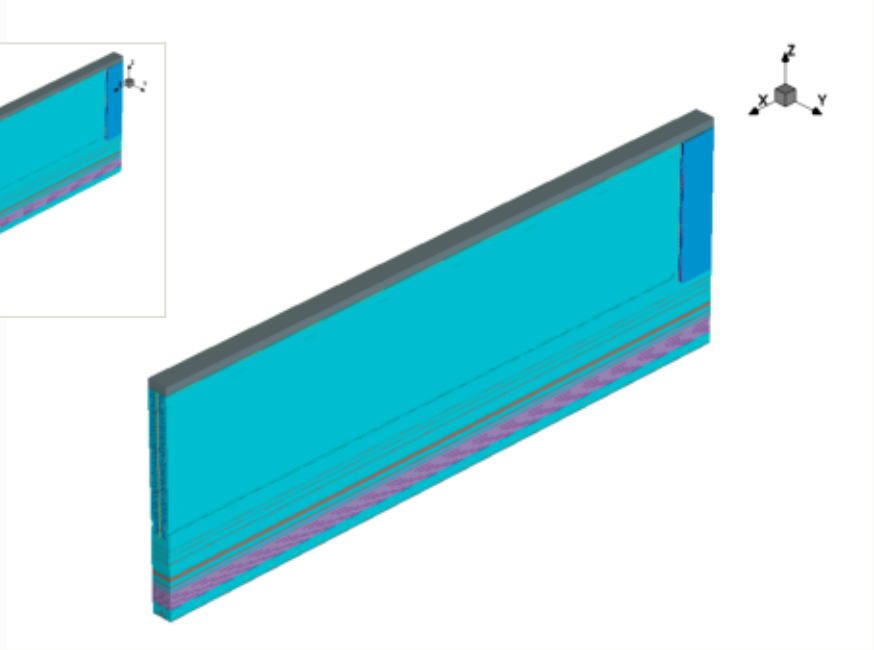
treatment · Dry_etch

0.801	796.9	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





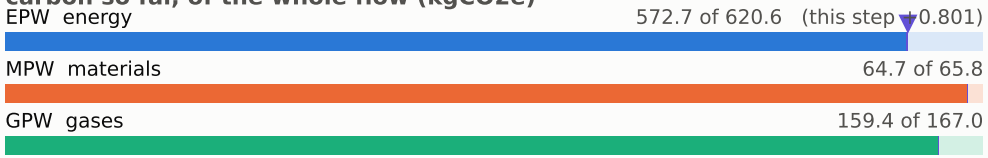
this step

process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

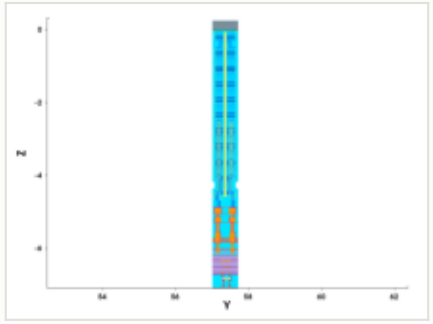
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

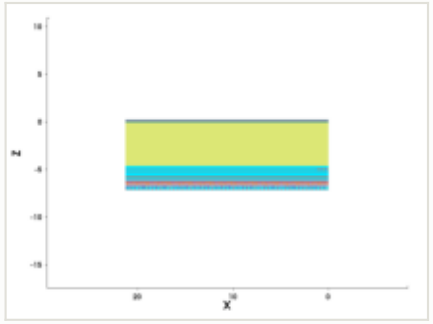


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



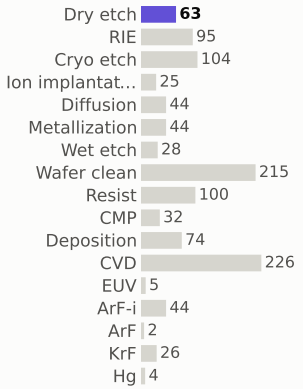
y cut at 57.351 µm



mask for this step

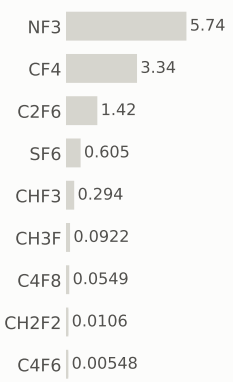
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



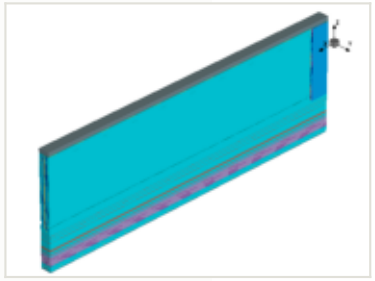
Step #13110: RIE

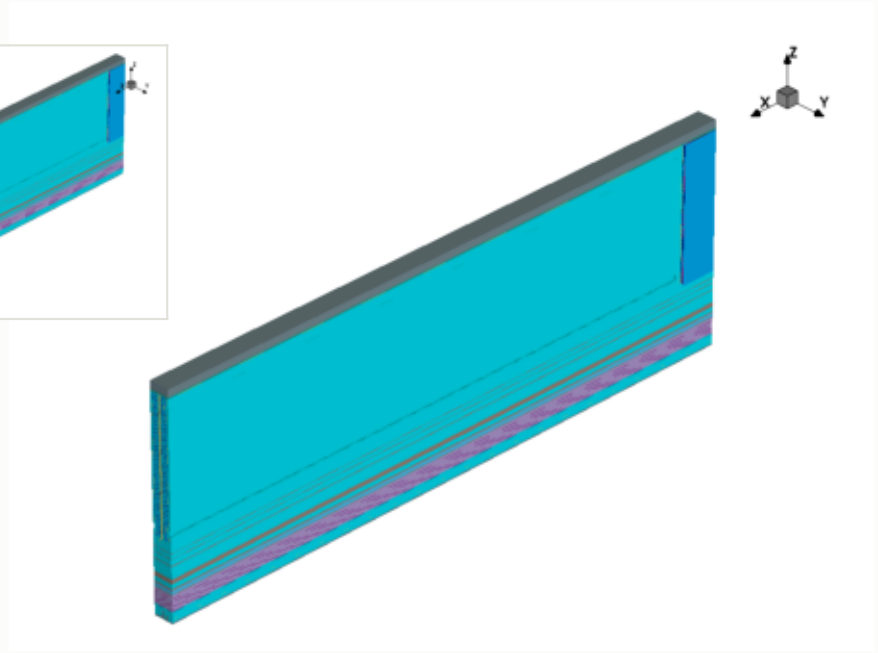
etch_noop · RIE

1.04	798.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





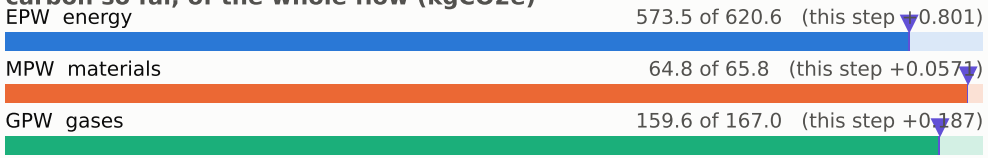
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

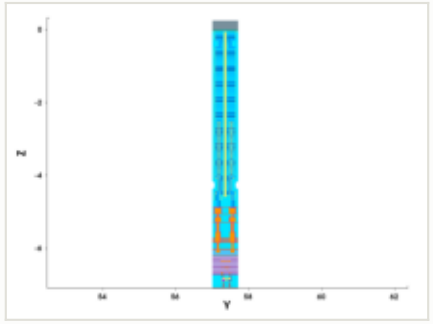
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

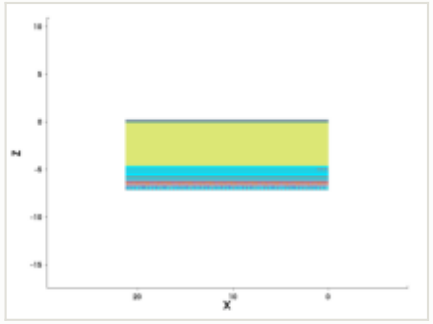


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



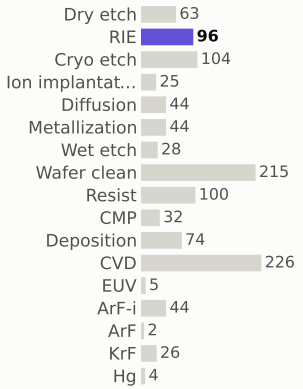
y cut at 57.351 µm



mask for this step

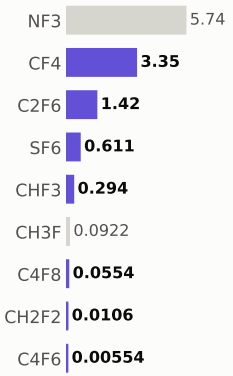
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



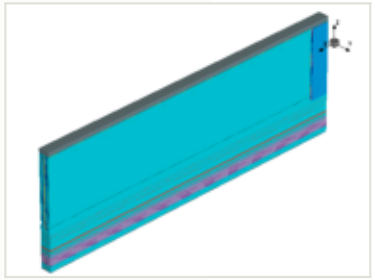
Step #13120: RIE

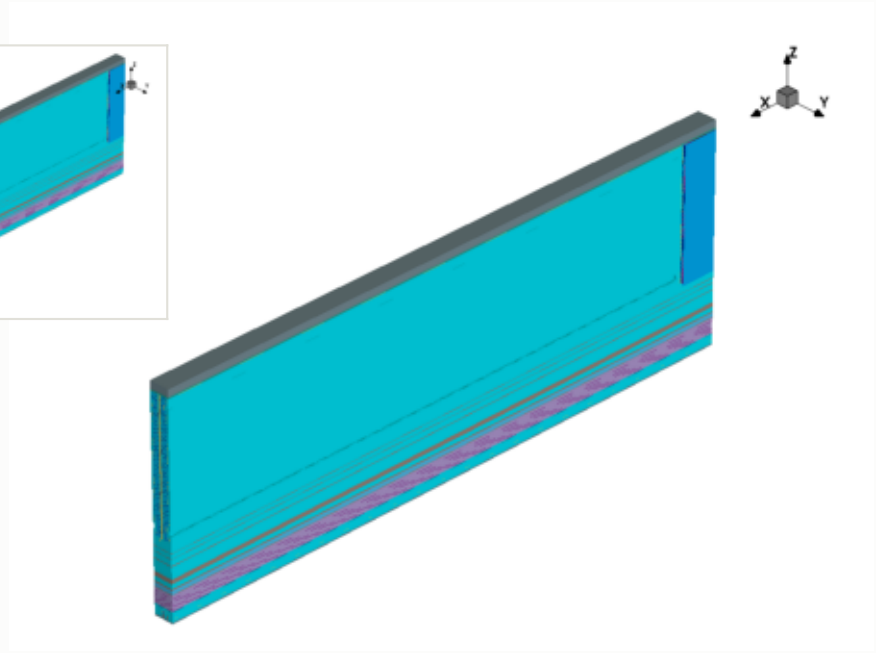
etch_noop · RIE

1.04	799.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





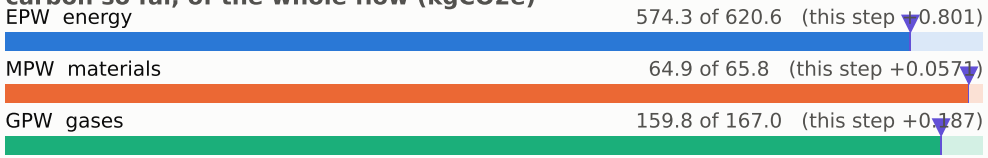
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

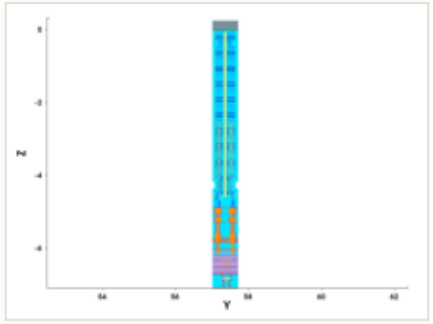
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

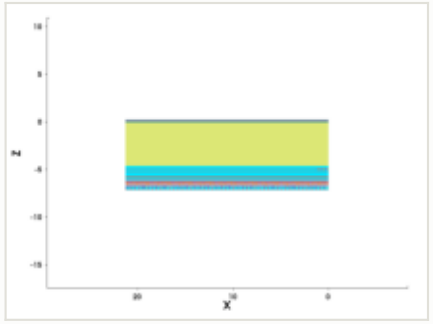


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



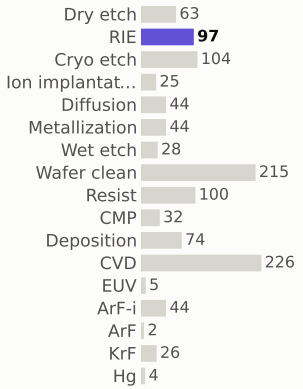
y cut at 57.351 µm



mask for this step

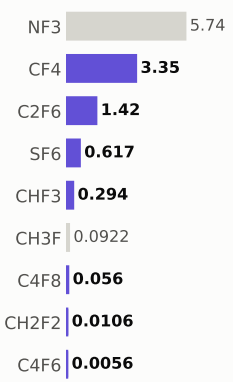
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



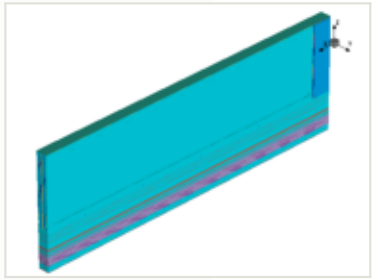
Step #13130: Wet_etch

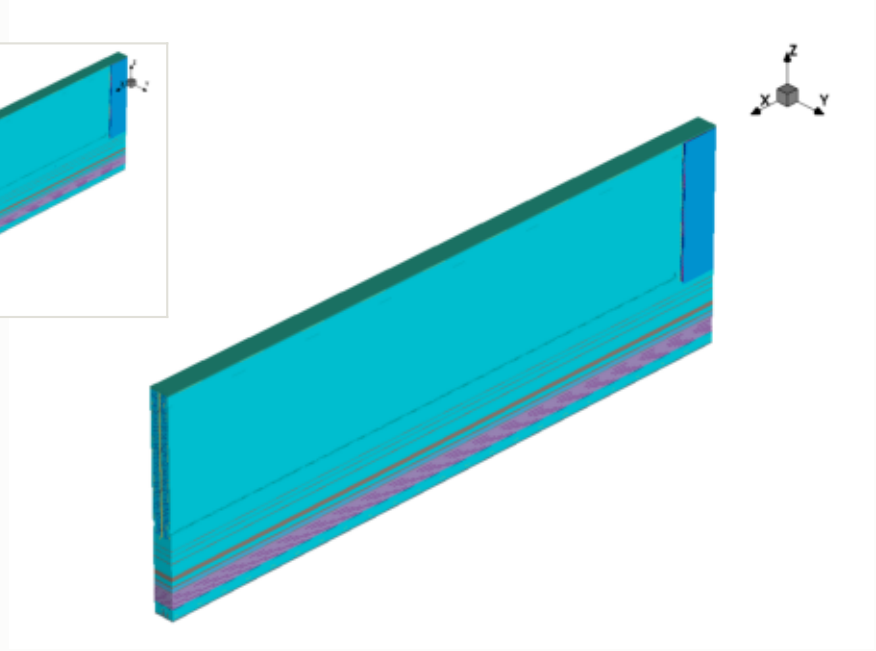
etch_isotropic · Wet_etch

0.169	799.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



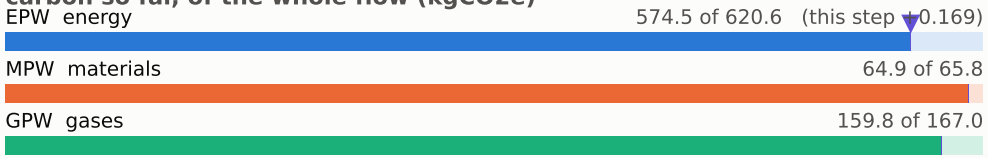


this step

process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

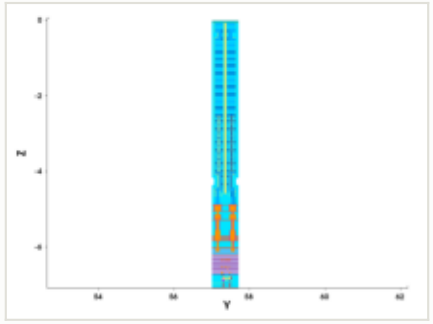
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

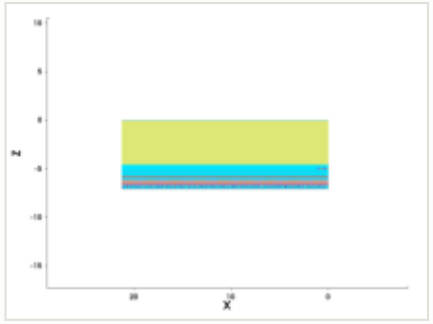


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



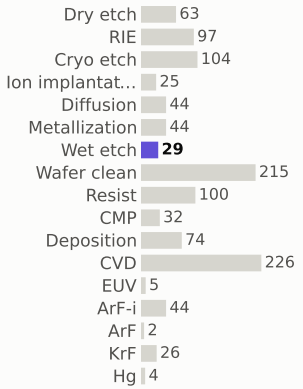
y cut at 57.351 µm



mask for this step

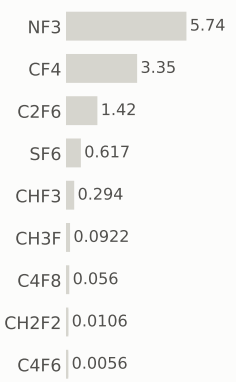
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



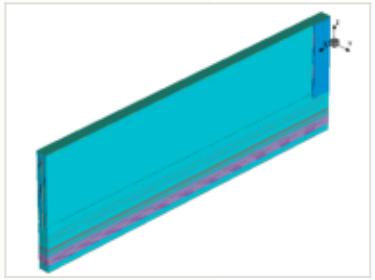
Step #13150: Wafer_clean

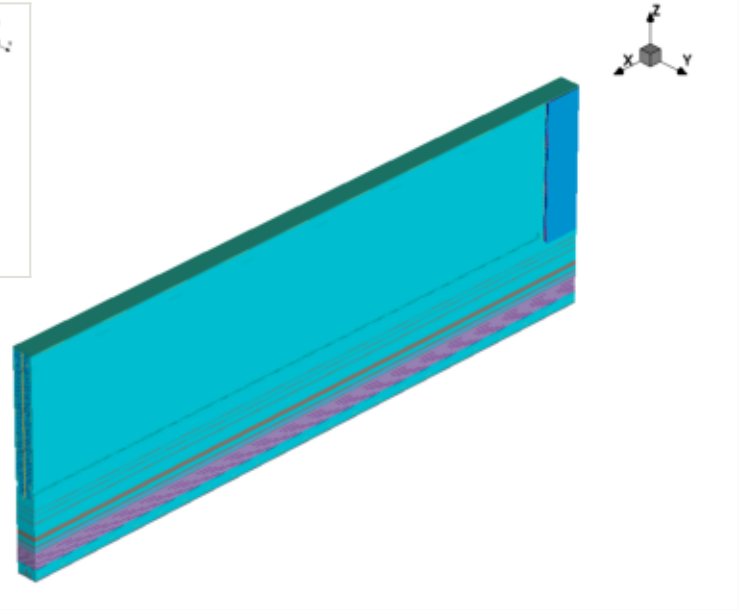
clean · Wafer_clean

0.169	799.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



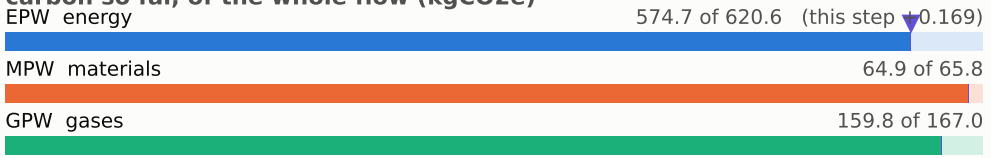


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

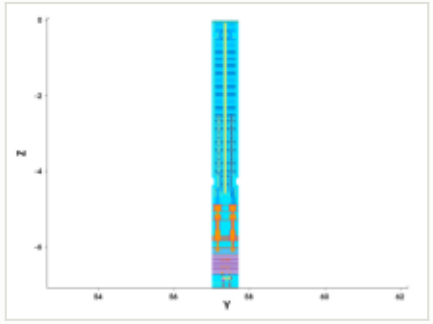
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

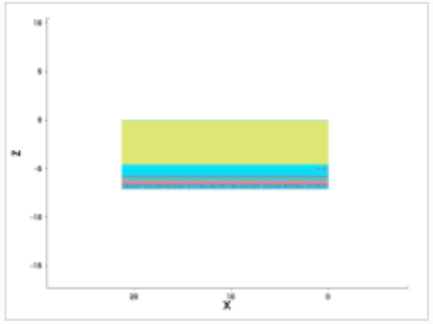


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



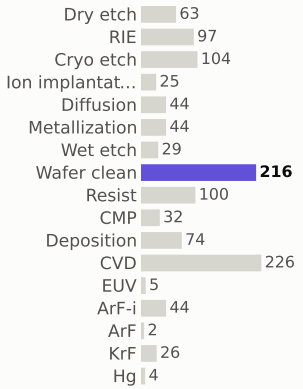
y cut at 57.351 µm



mask for this step

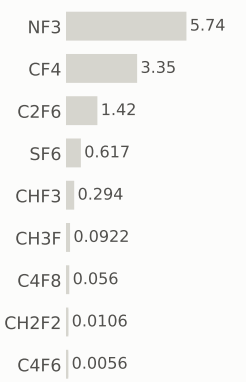
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



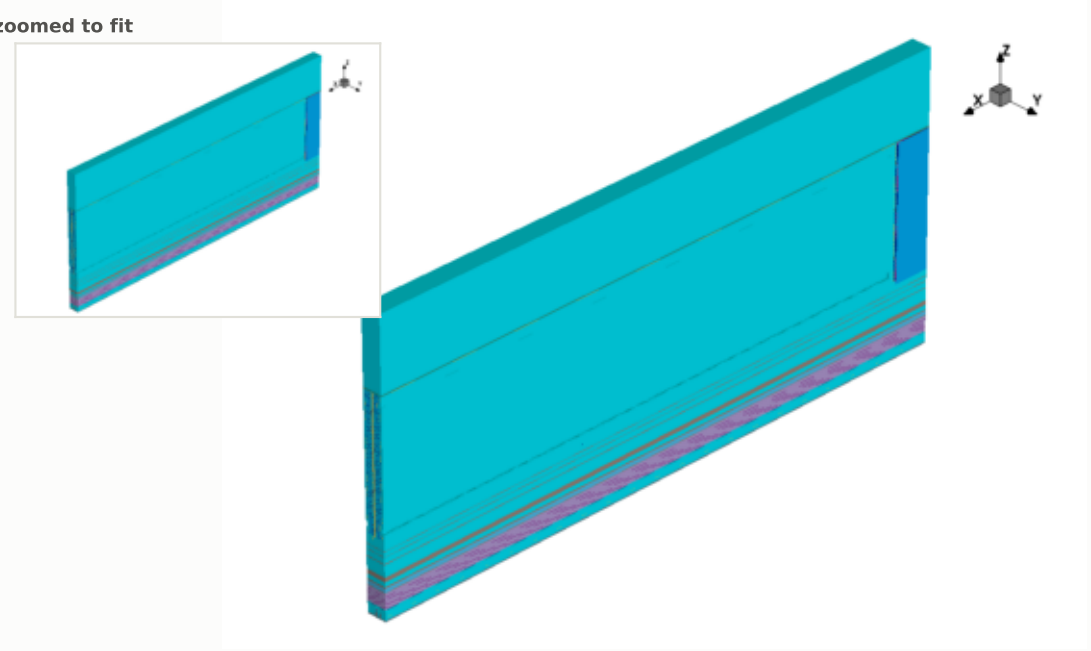
Step #13160: CVD

depo_isotropic · CVD

1.31	800.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

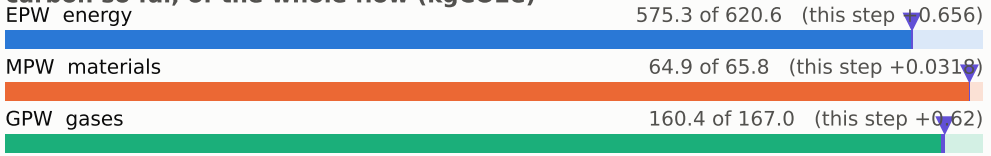
zoomed to fit



this step

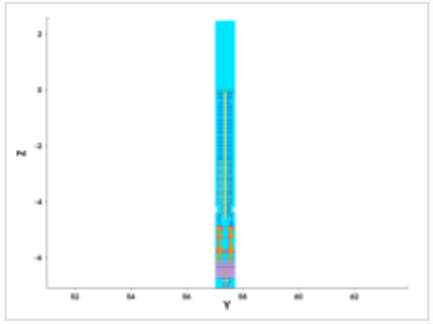
process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	2.5 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

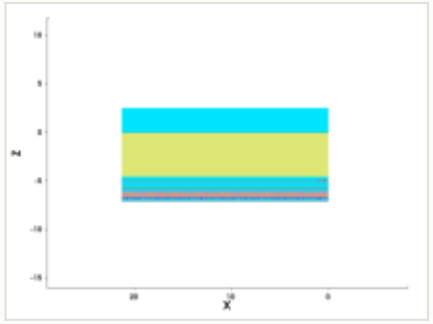


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



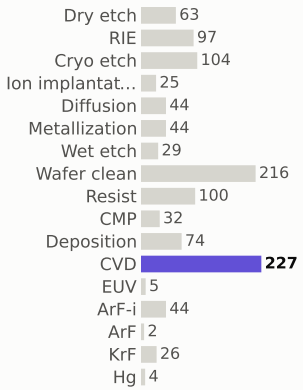
y cut at 57.351 µm



mask for this step

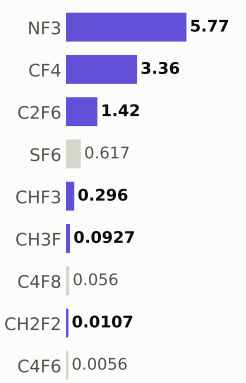
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



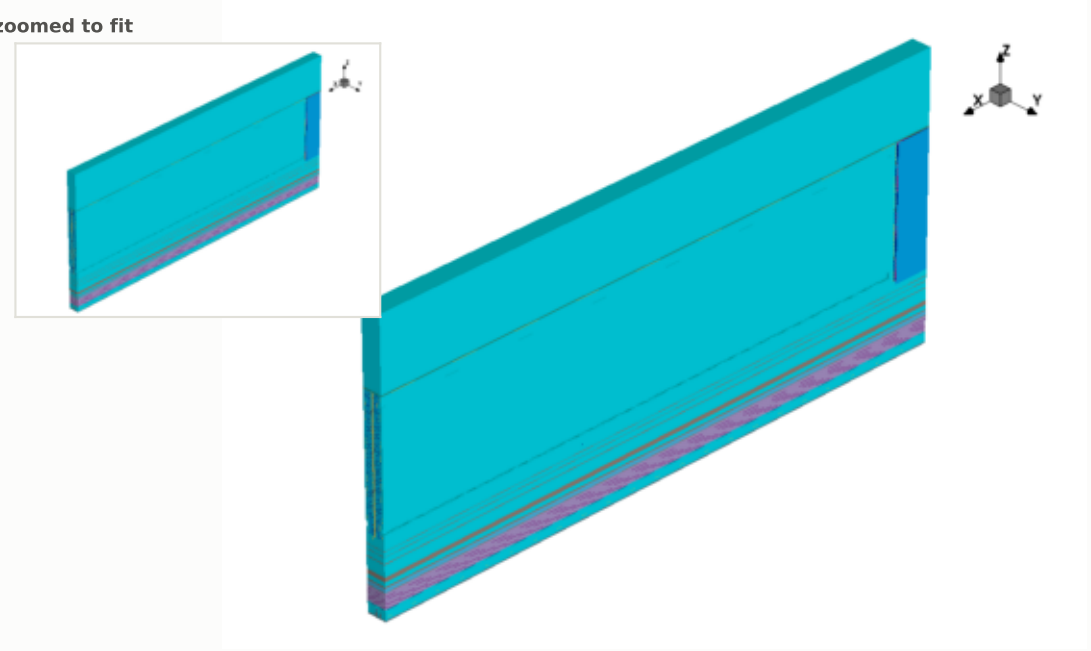
Step #13170: Diffusion

anneal · Diffusion

0.801	801.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

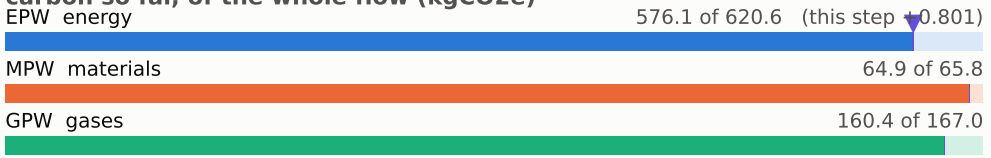


this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

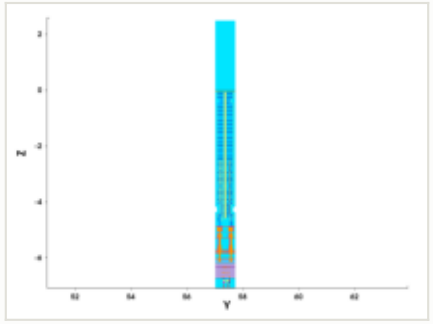
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

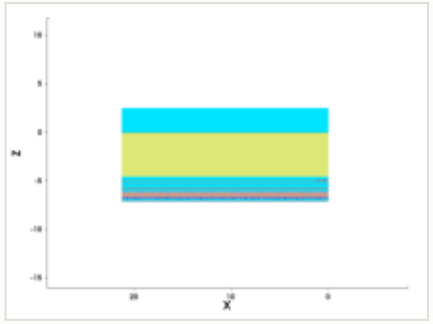


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



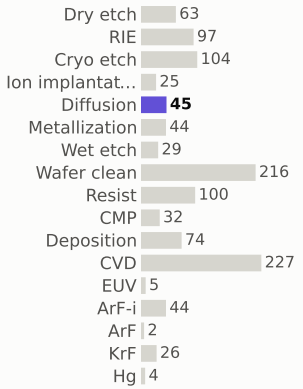
y cut at 57.351 µm



mask for this step

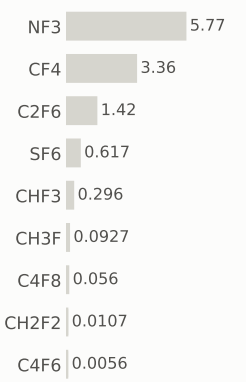
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



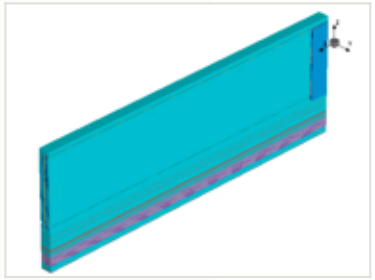
Step #13180: CMP

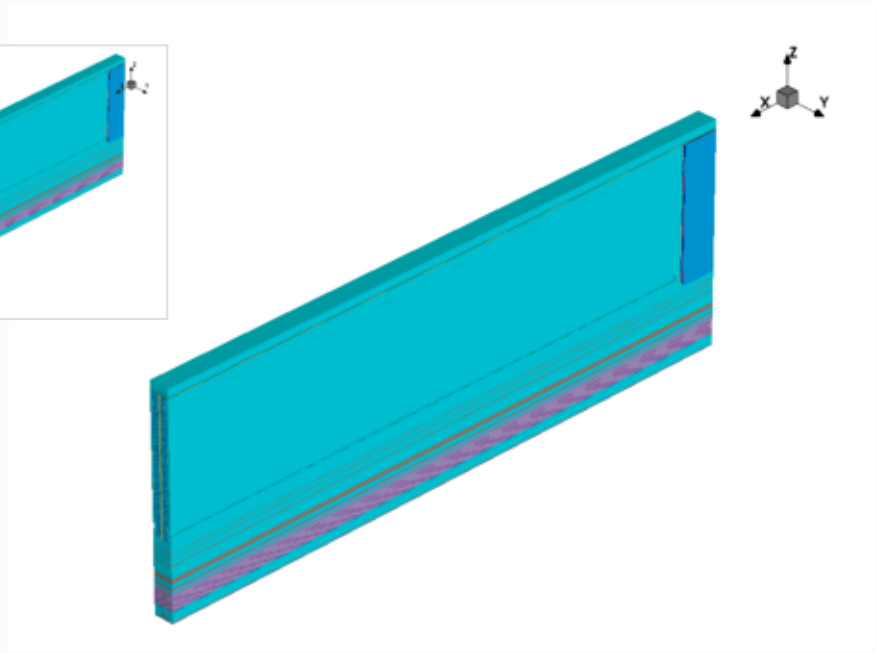
etch_cmp · CMP

0.169	801.6	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

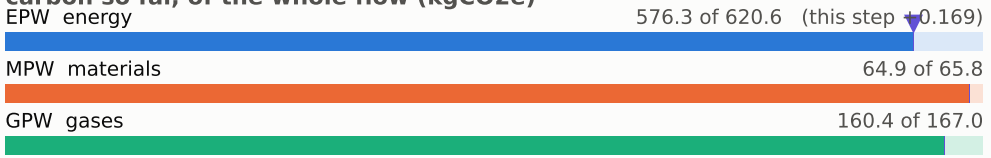




this step

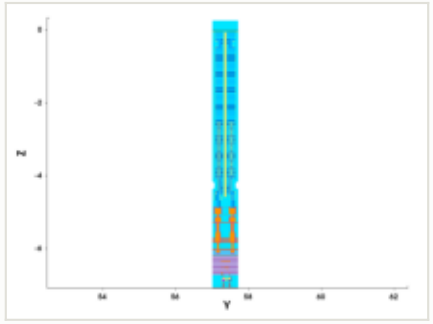
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

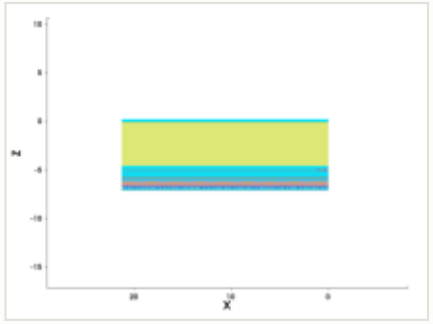


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



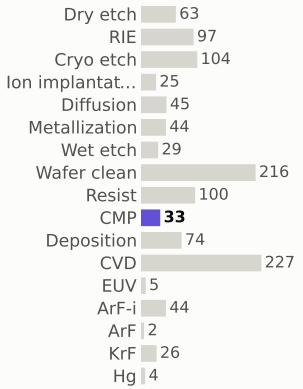
y cut at 57.351 µm



mask for this step

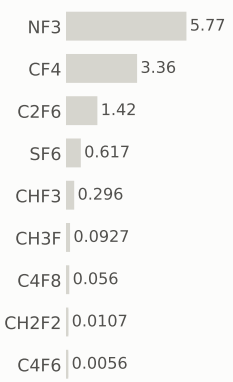
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



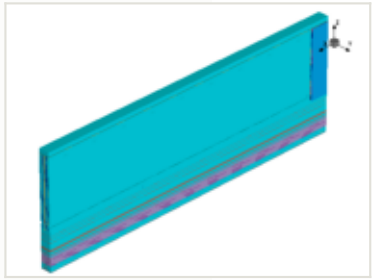
Step #13190: Wafer_clean

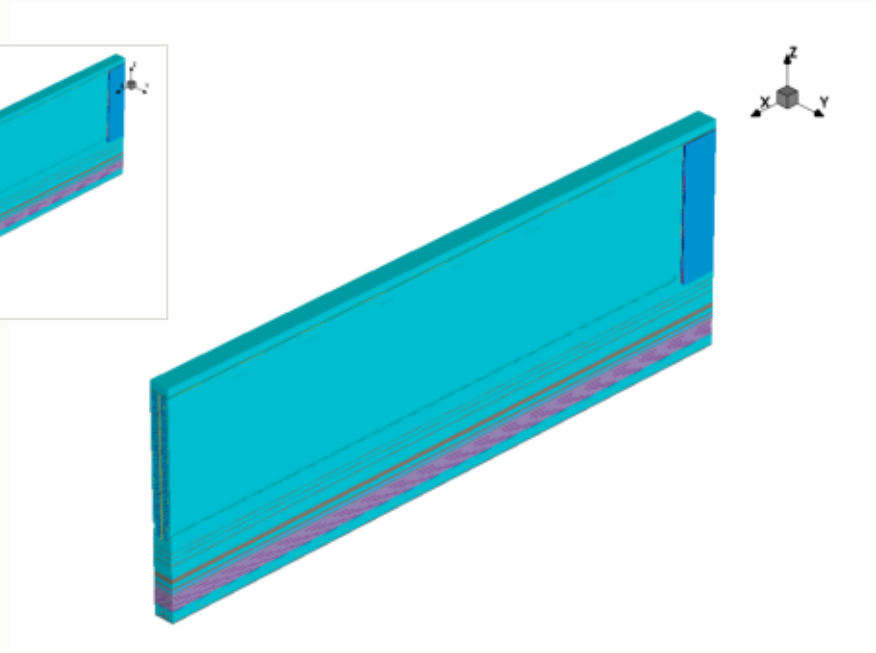
clean · Wafer_clean

0.169	801.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





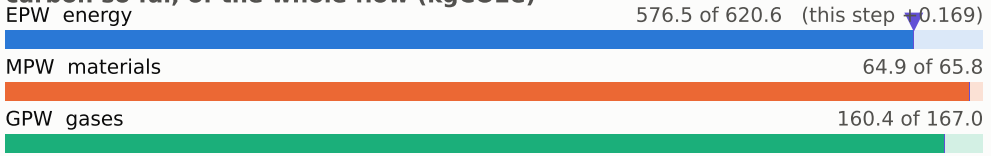
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

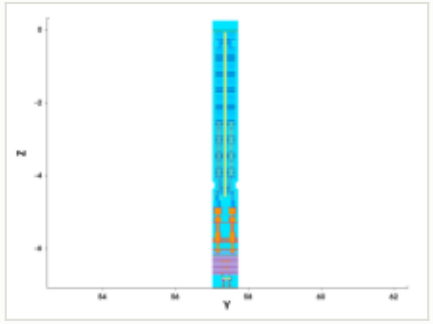
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

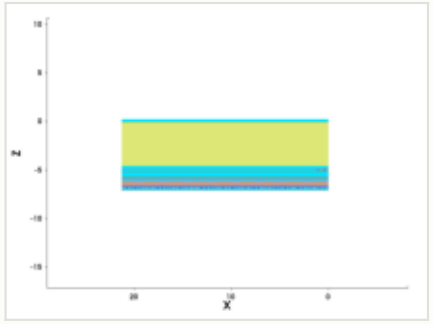


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



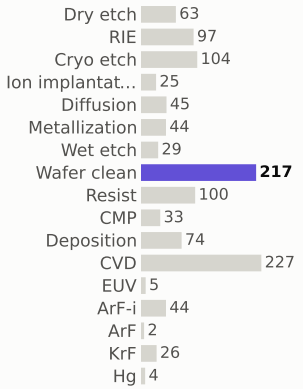
y cut at 57.351 µm



mask for this step

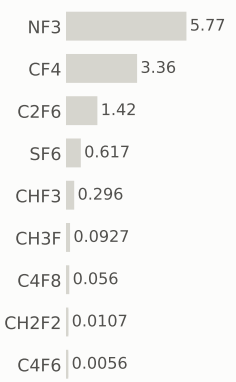
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



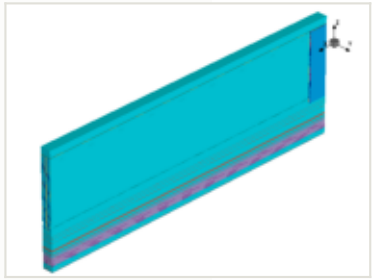
Step #13200: CVD

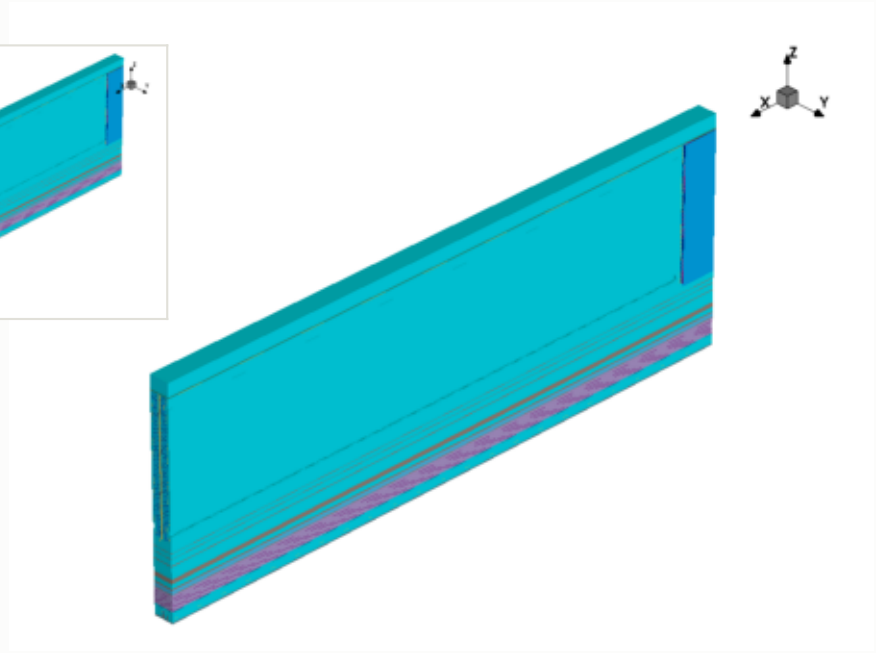
depo_isotropic · CVD

1.31	803.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



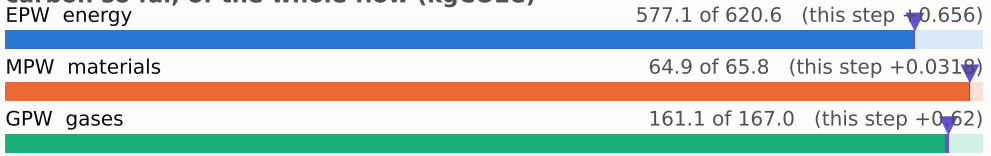


this step

process area	CVD
material	—
thickness	0.186 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

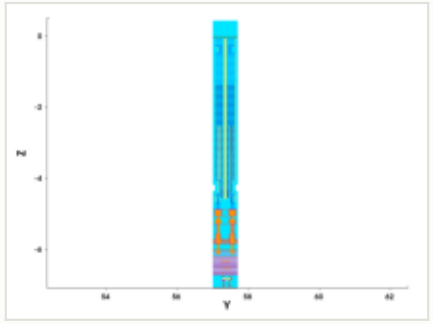
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

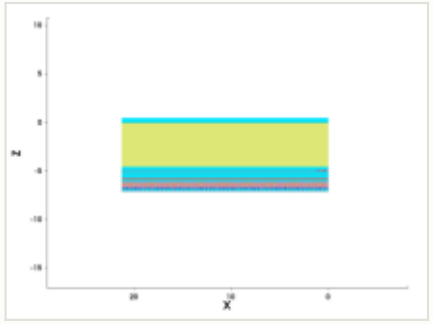


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



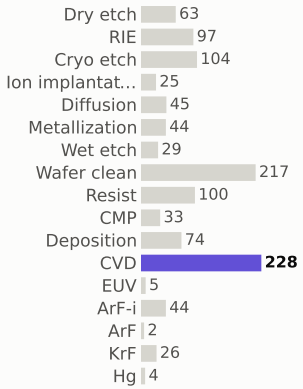
y cut at 57.351 µm



mask for this step

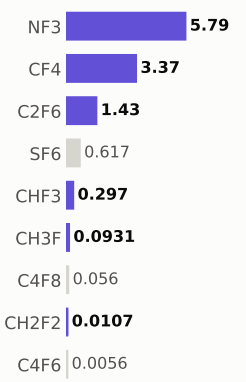
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



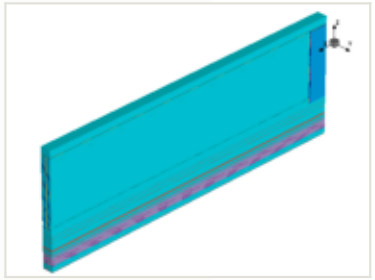
Step #13210: Wafer_clean

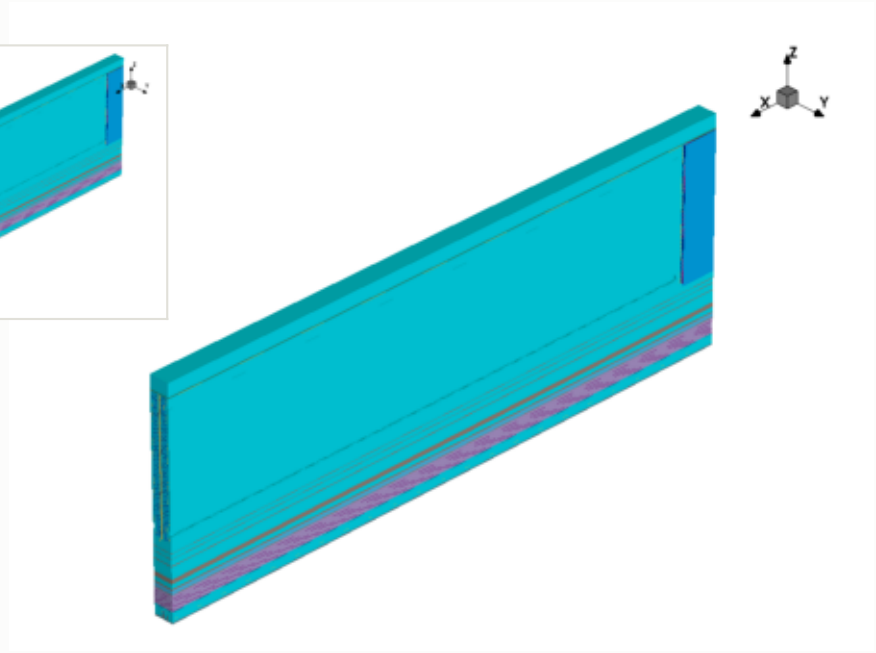
clean · Wafer_clean

0.169	803.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



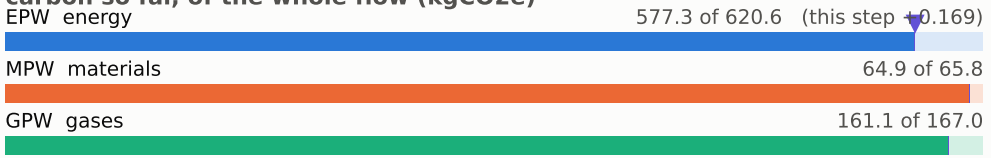


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

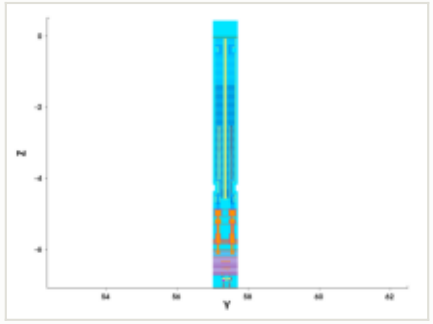
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

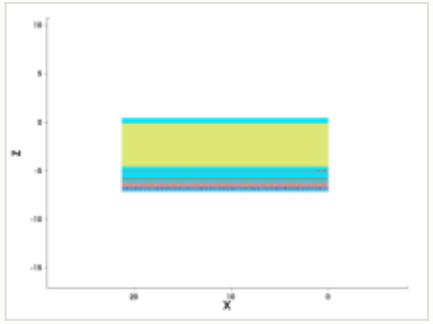


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



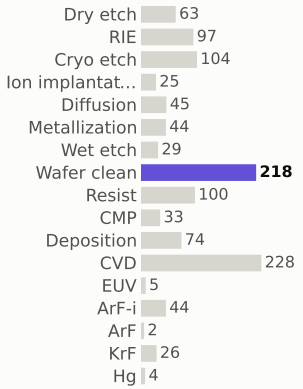
y cut at 57.351 µm



mask for this step

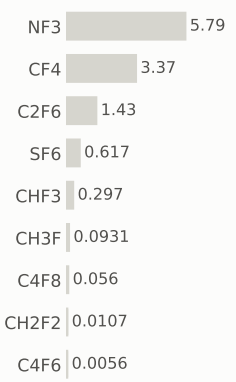
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



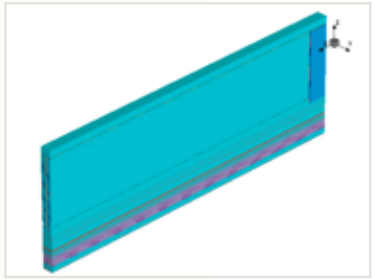
Step #13220: Resist

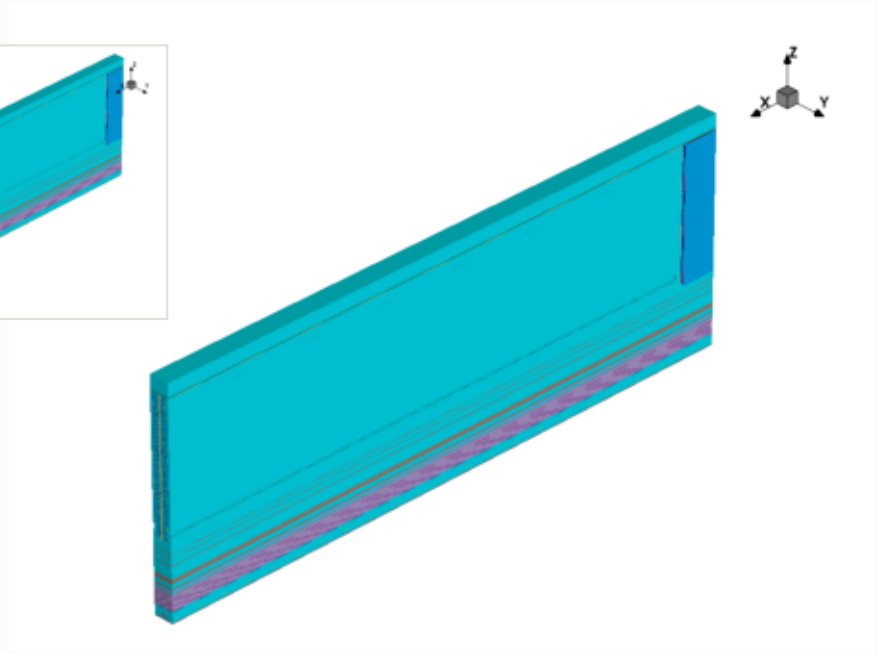
resist · Resist

0.169	803.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



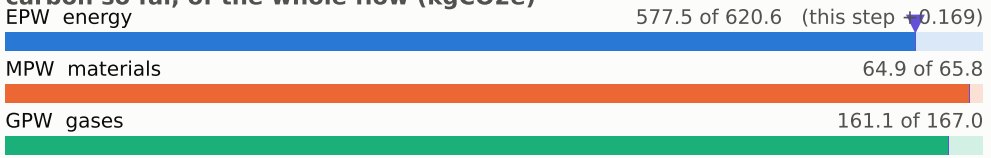


this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

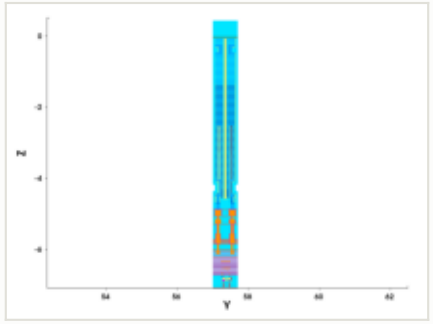
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

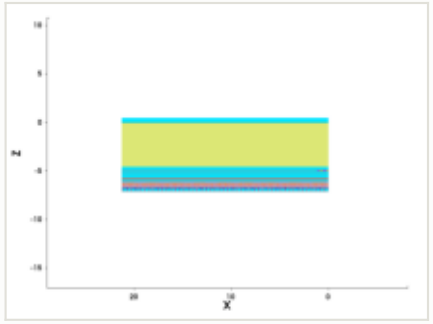


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



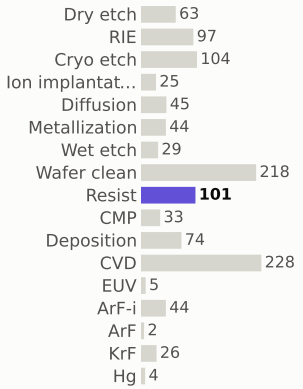
y cut at 57.351 µm



mask for this step

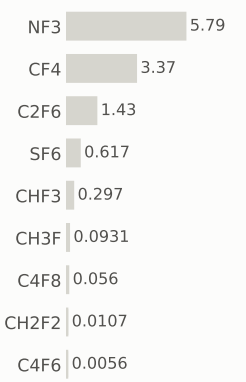
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



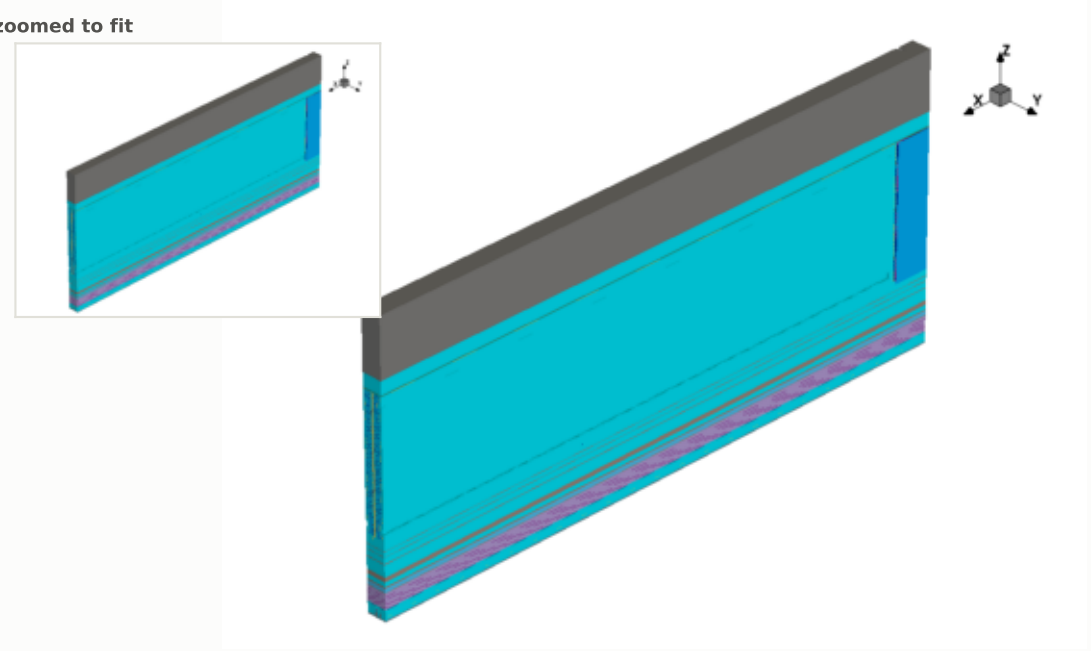
Step #13230: KrF

litho_patterning · KrF

0.199	803.7	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

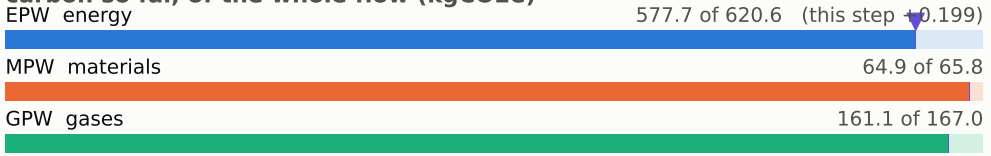
zoomed to fit



this step

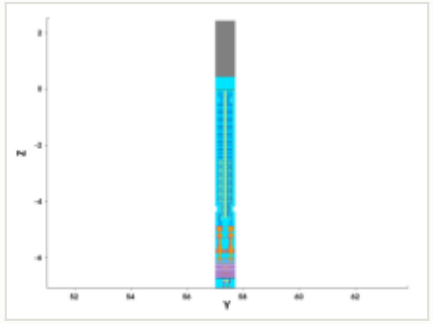
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

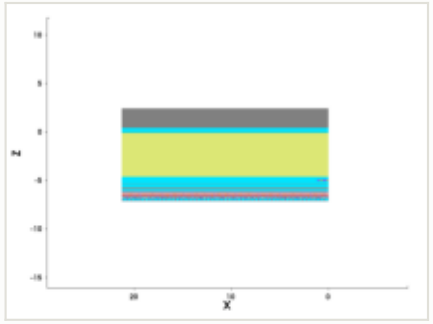


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

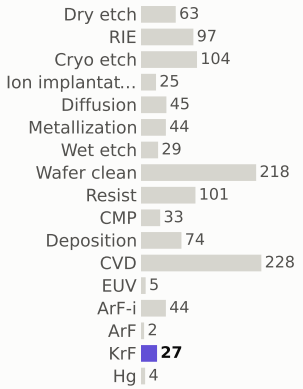


y cut at 57.351 µm



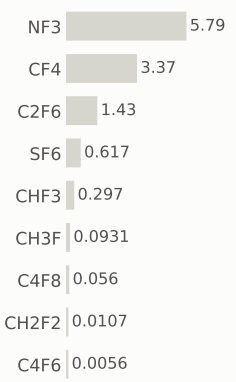
mask: layer 1410:0 (GDSII)

steps by process area



materials deposited (g)

gases emitted (g)



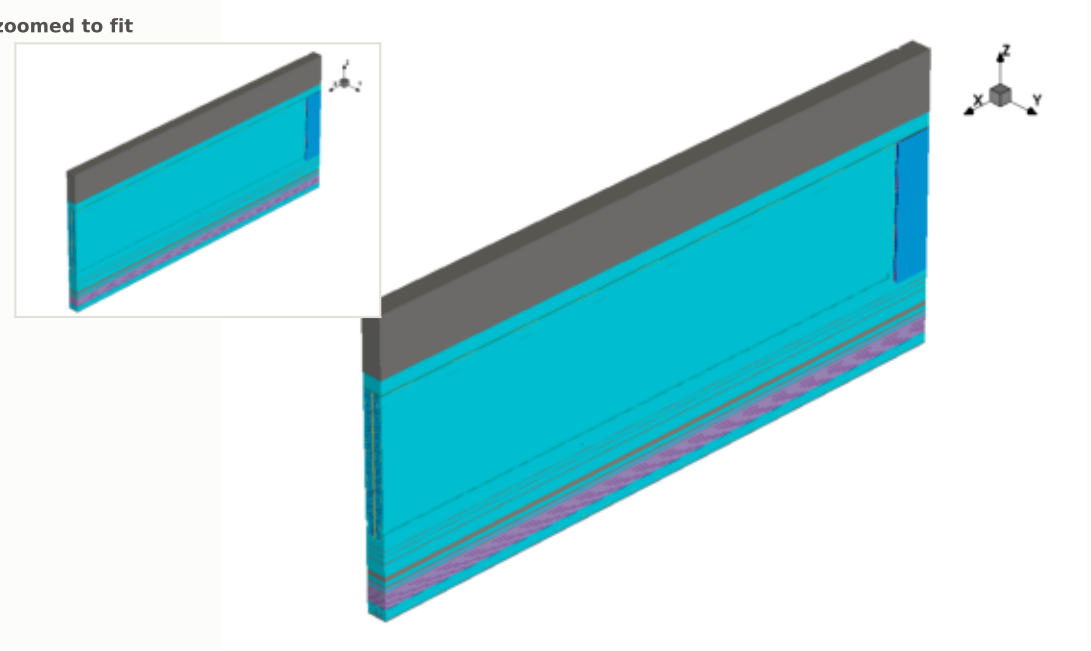
Step #13240: RIE

etch_anisotropic · RIE

1.04	804.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

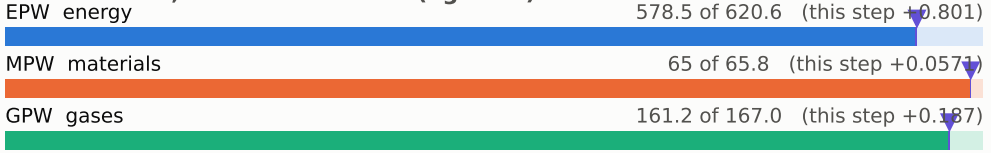


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

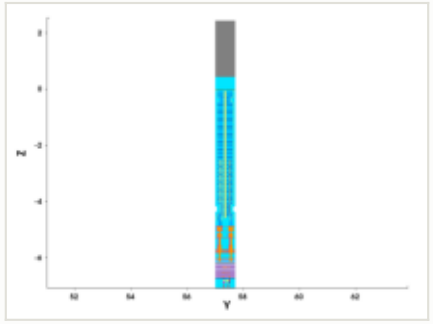
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

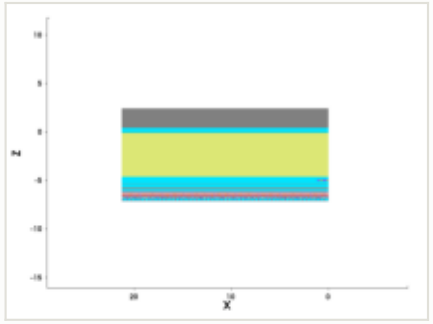


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



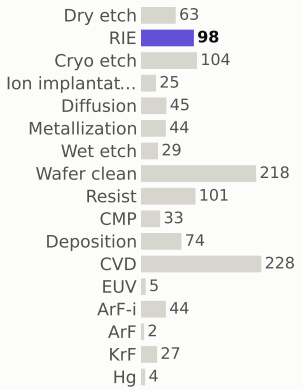
y cut at 57.351 µm



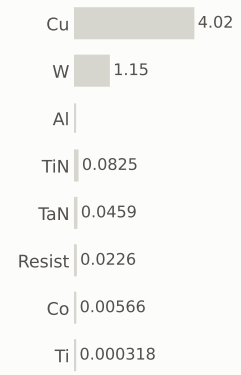
mask for this step

no mask for this step

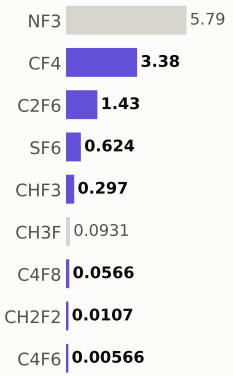
steps by process area



materials deposited (g)



gases emitted (g)



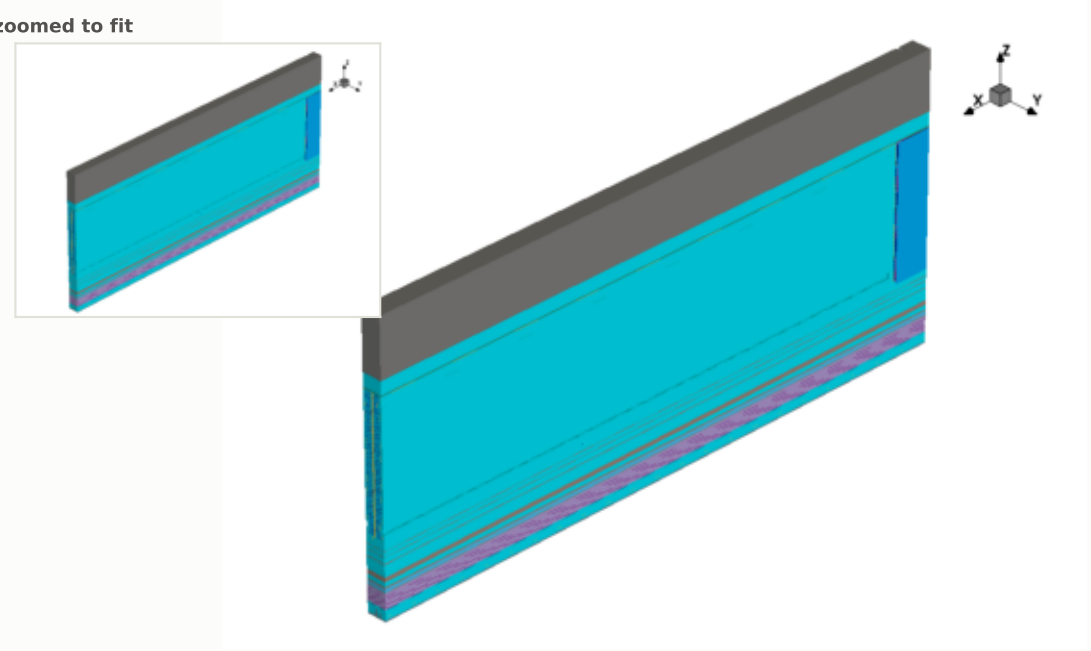
Step #13250: RIE

etch_anisotropic · RIE

1.04	805.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

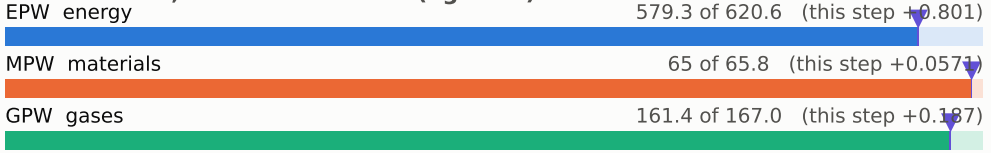


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

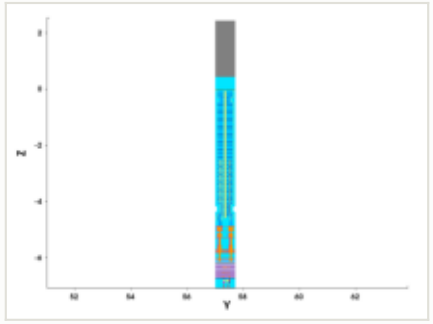
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

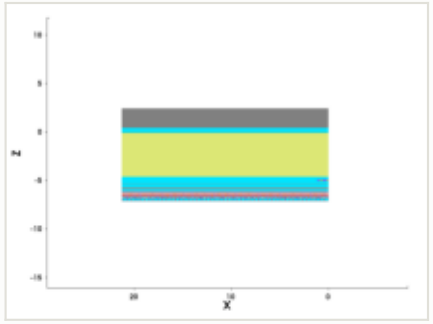


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



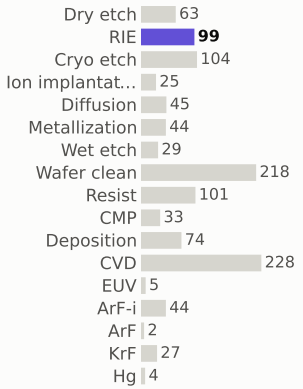
y cut at 57.351 µm



mask for this step

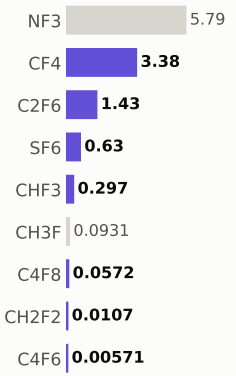
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



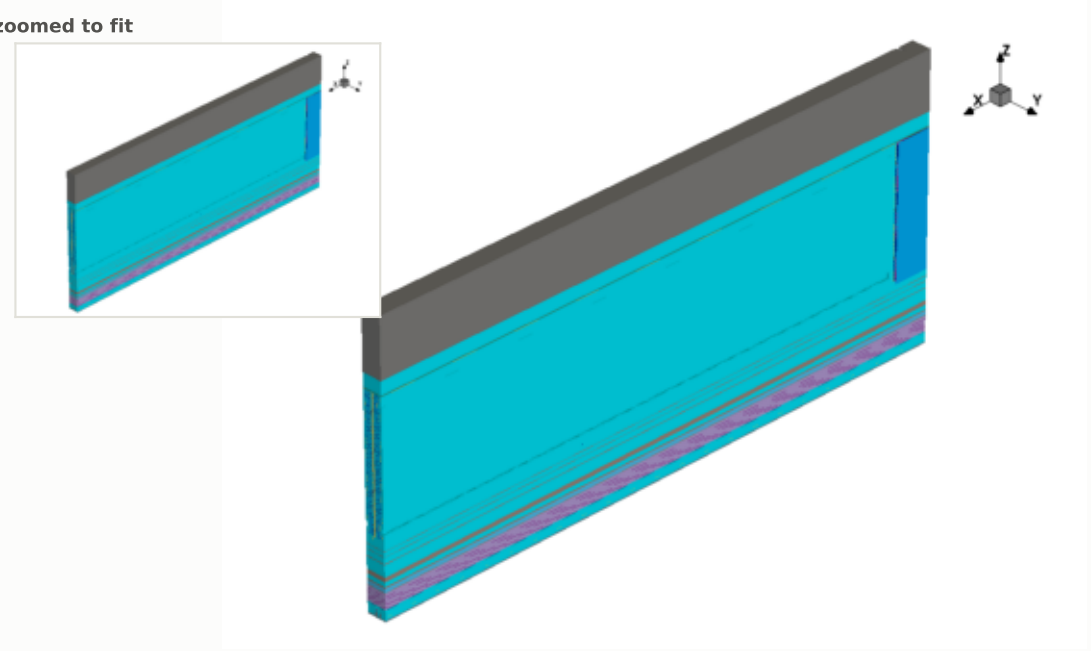
Step #13255: Spx_only

etch_anisotropic · Spx_only

0	805.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

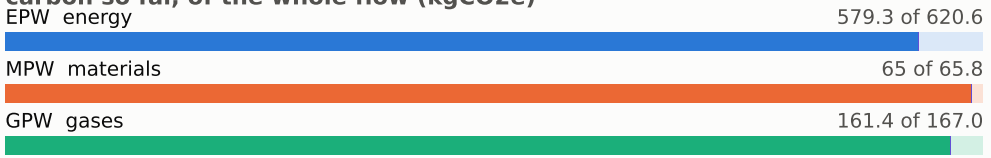


this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

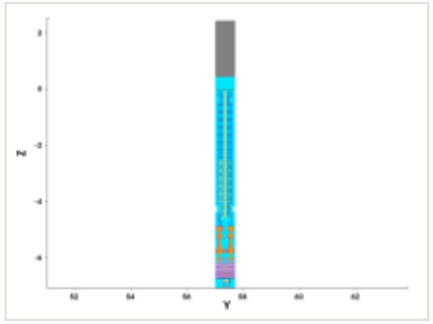
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

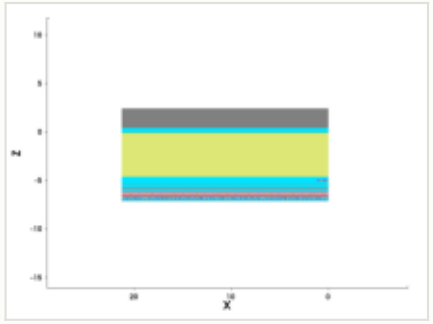


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



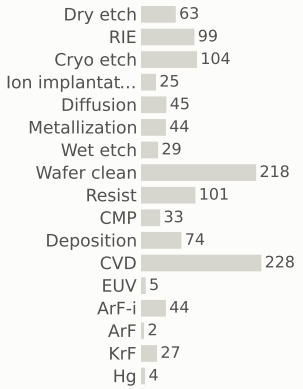
y cut at 57.351 µm



mask for this step

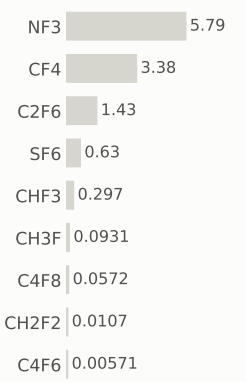
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



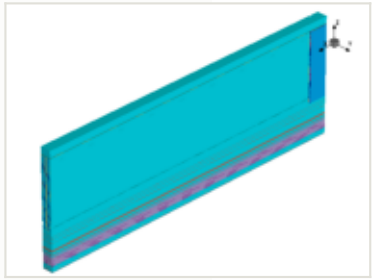
Step #13260: Dry_etch

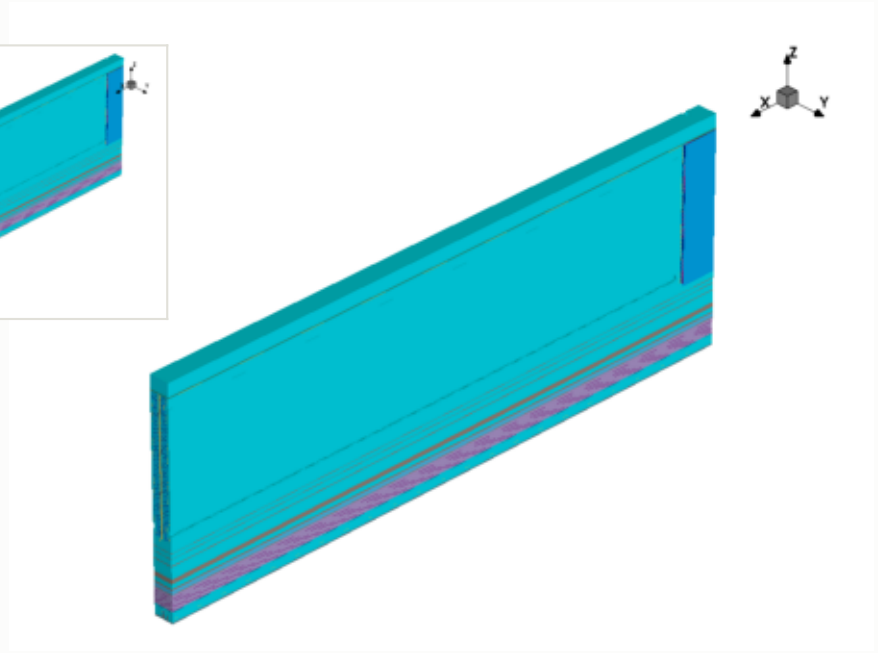
etch_strip · Dry_etch

0.801	806.5	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

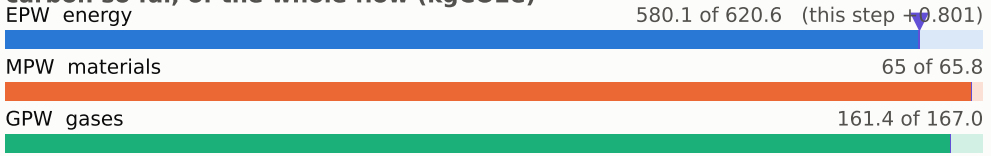




this step

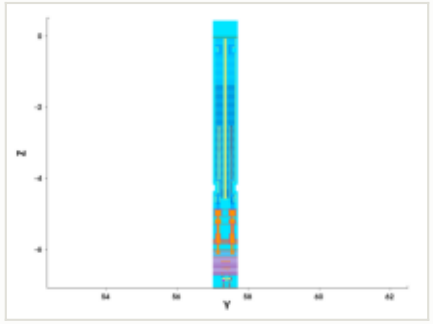
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

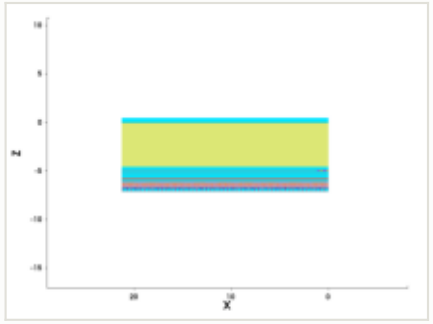


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



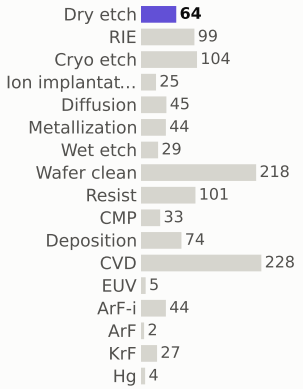
y cut at 57.351 µm



mask for this step

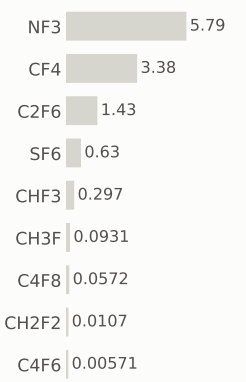
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



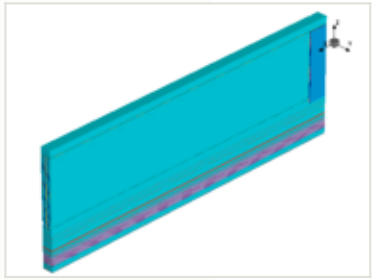
Step #13270: Wafer_clean

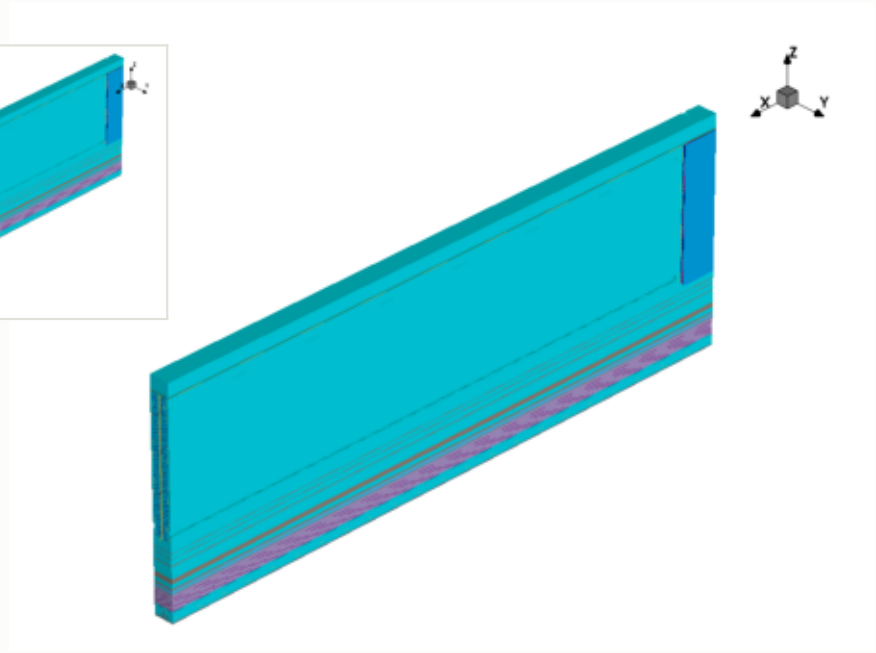
clean · Wafer_clean

0.169	806.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



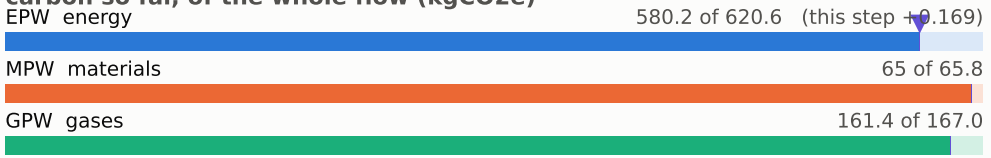


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

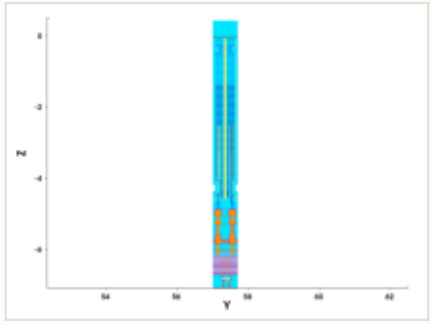
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

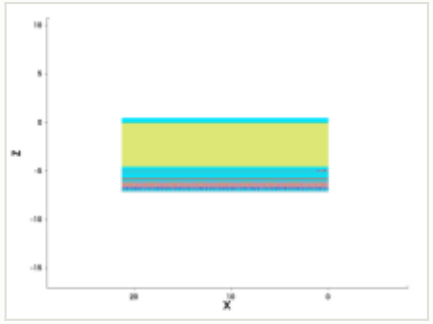


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



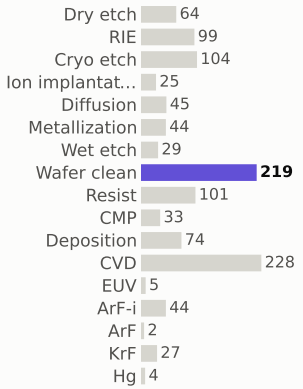
y cut at 57.351 µm



mask for this step

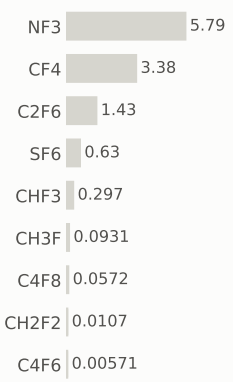
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



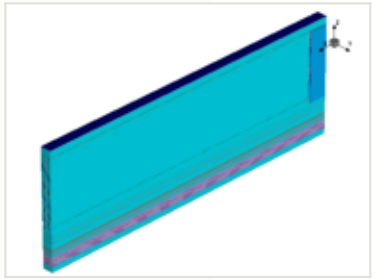
Step #13280: Deposition

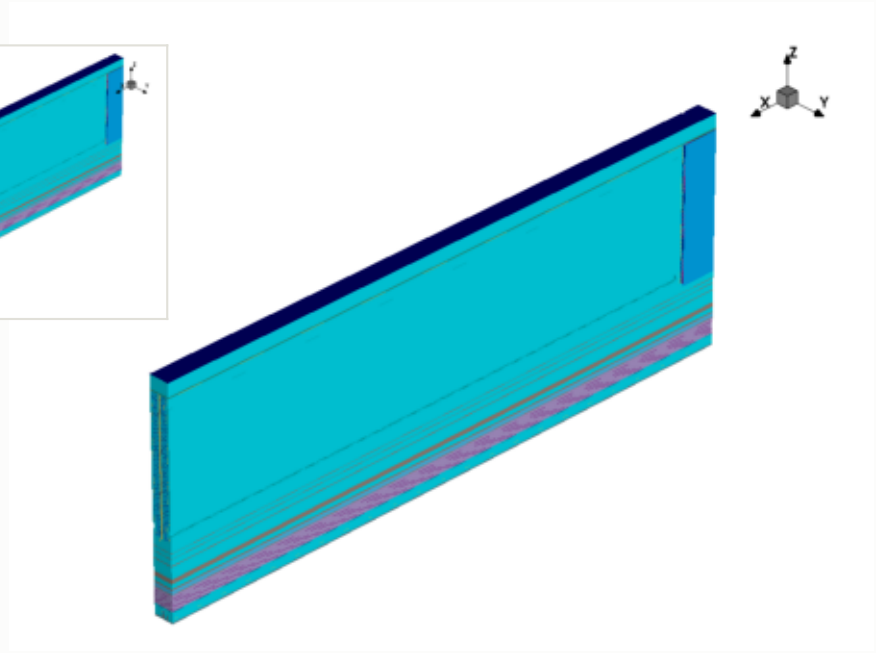
depo_isotropic · Deposition

0.656	807.4	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



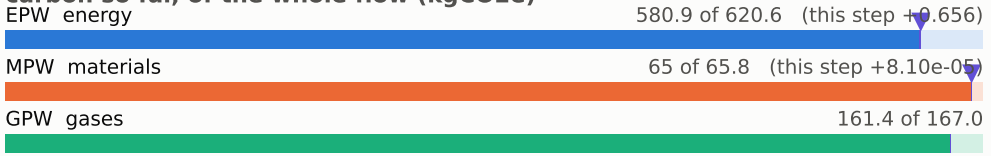


this step

process area	Deposition
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

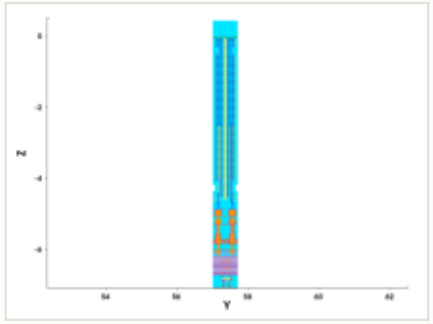
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	8.10e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

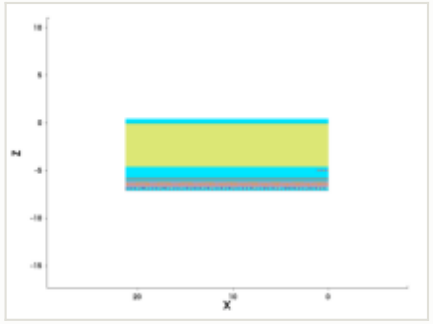


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



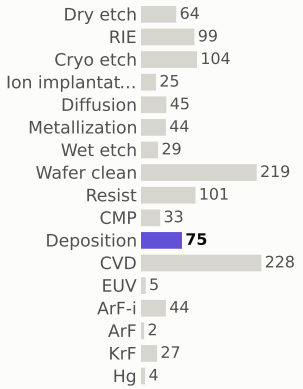
y cut at 57.351 µm



mask for this step

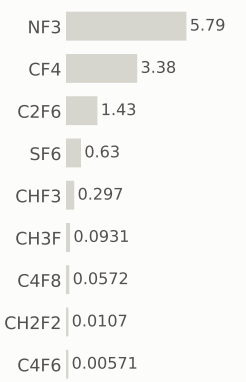
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



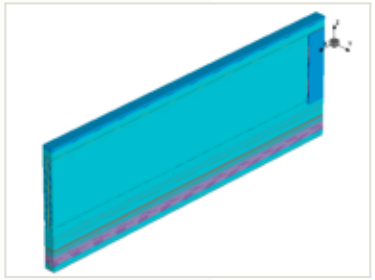
Step #13290: Metallization

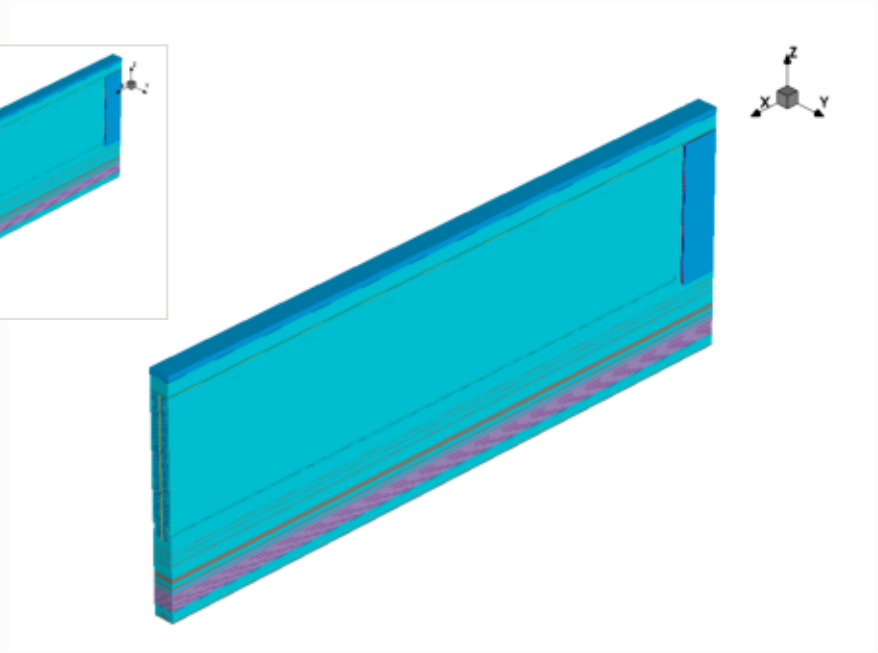
depo_isotropic · Metallization

1.22	808.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



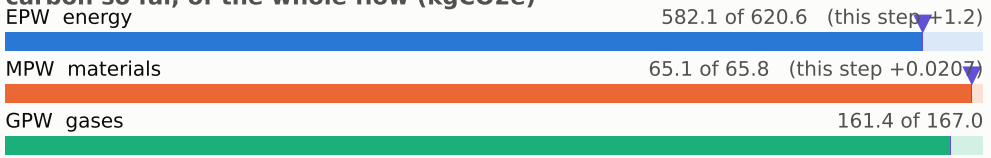


this step

process area	Metallization
material	W
thickness	0.2 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

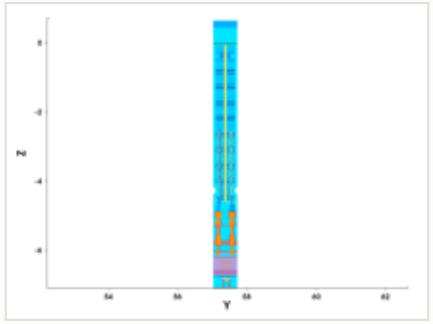
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0207 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.22 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

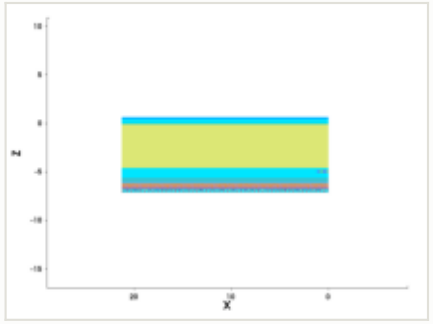


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



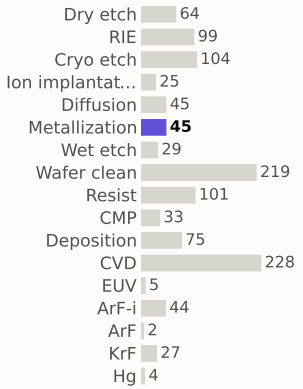
y cut at 57.351 µm



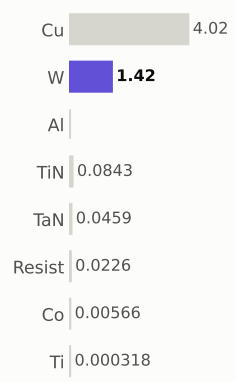
mask for this step

no mask for this step

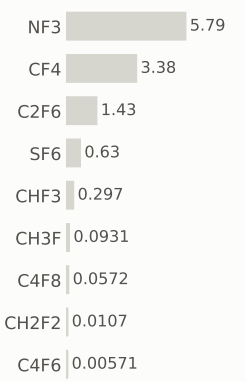
steps by process area



materials deposited (g)



gases emitted (g)



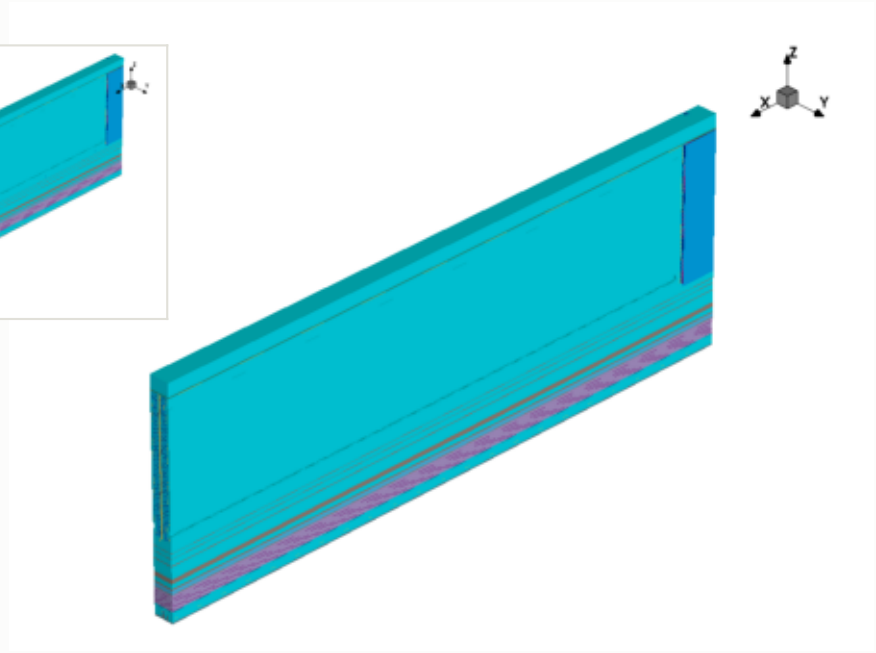
Step #13300: CMP

etch_cmp · CMP

0.169	808.8	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

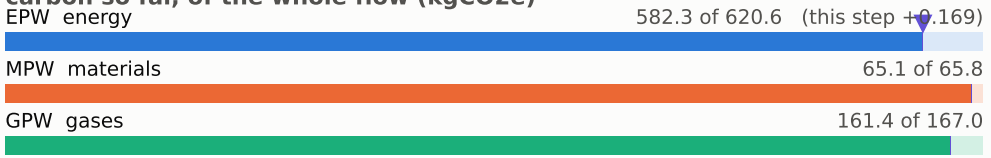
zoomed to fit



this step

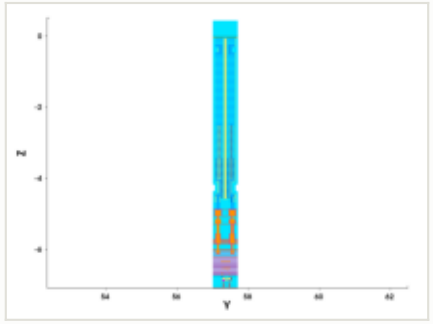
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

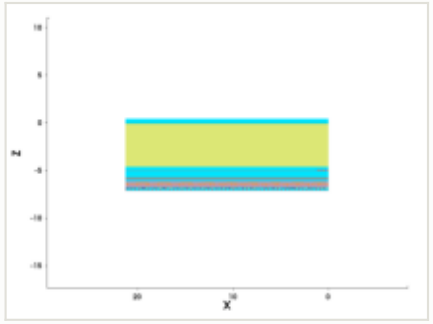


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



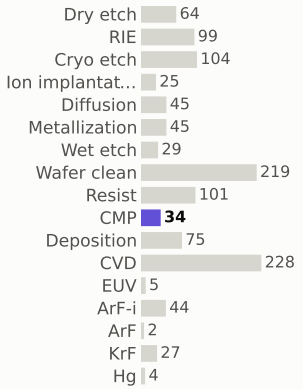
y cut at 57.351 µm



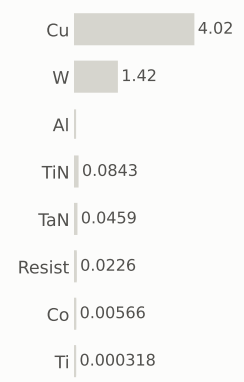
mask for this step

no mask for this step

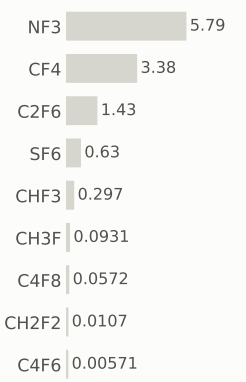
steps by process area



materials deposited (g)



gases emitted (g)



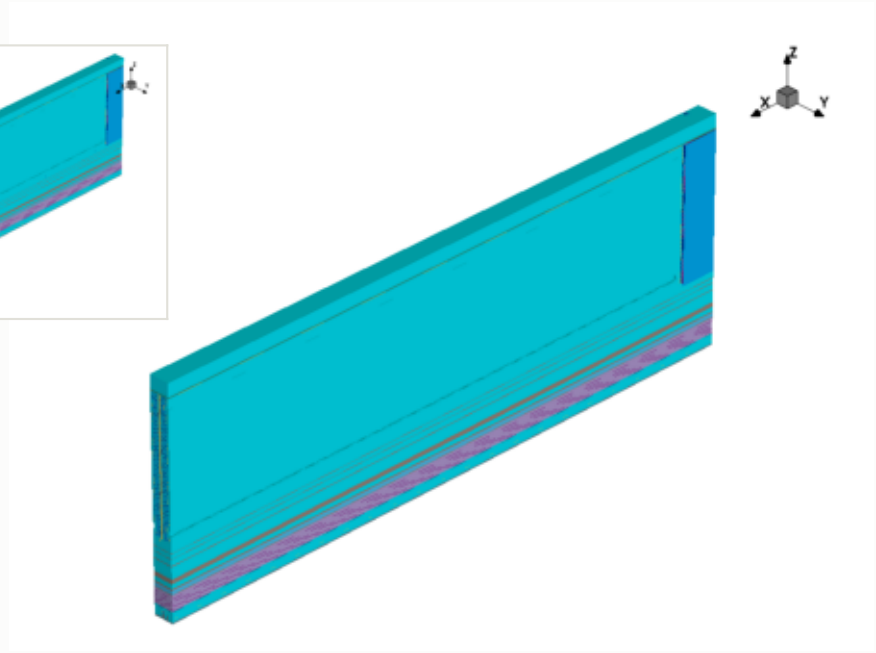
Step #13310: Wafer_clean

clean · Wafer_clean

0.169	808.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

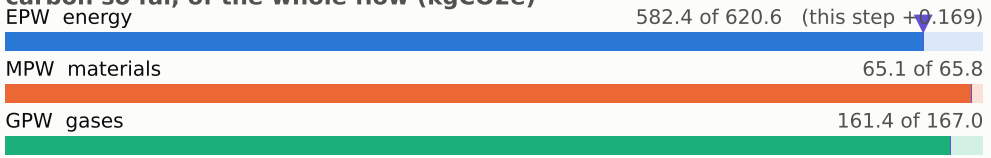


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

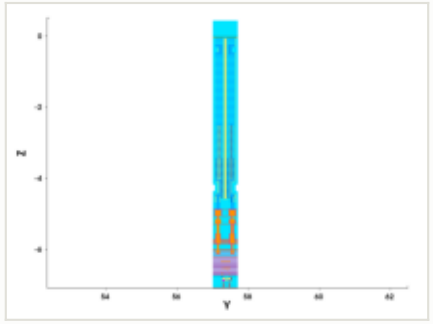
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

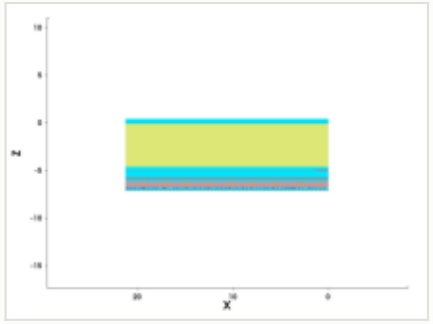


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



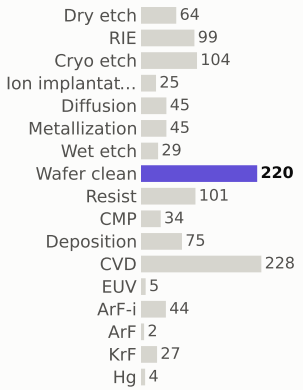
y cut at 57.351 µm



mask for this step

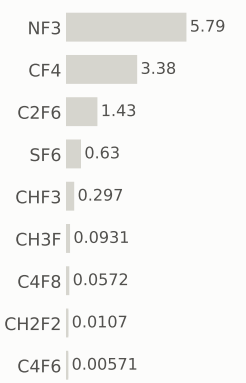
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



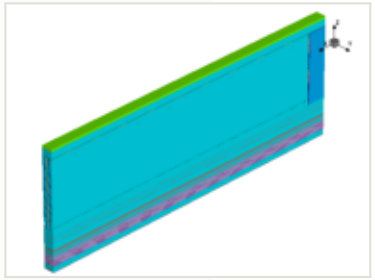
Step #13320: CVD

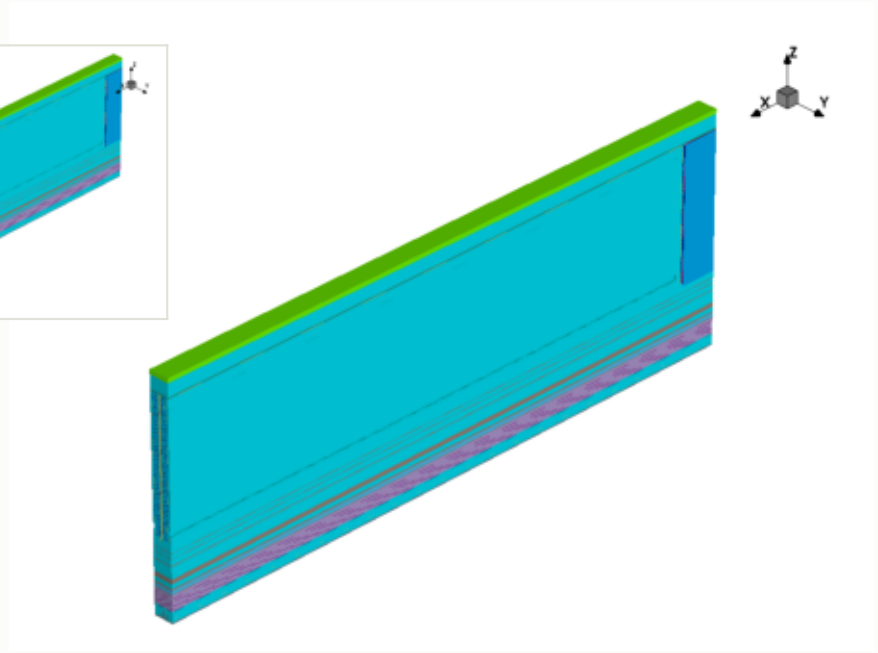
depo_isotropic · CVD

1.31	810.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

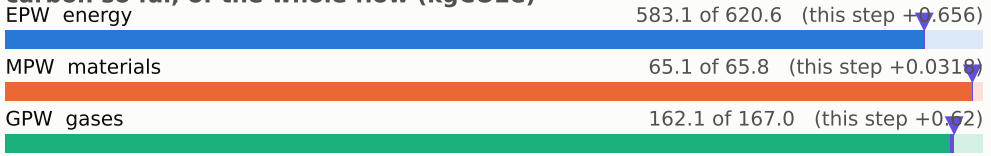




this step

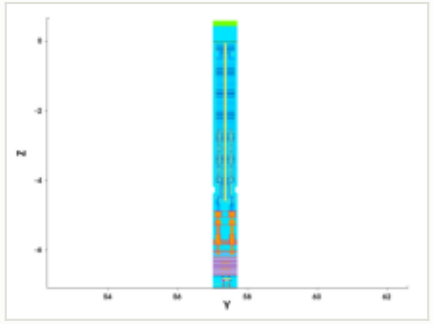
process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.15 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

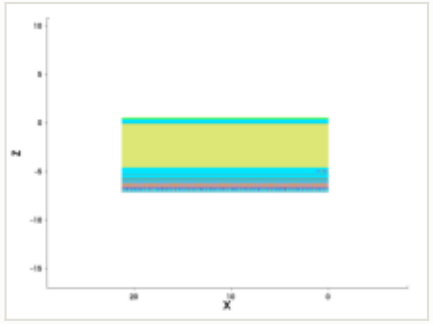


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



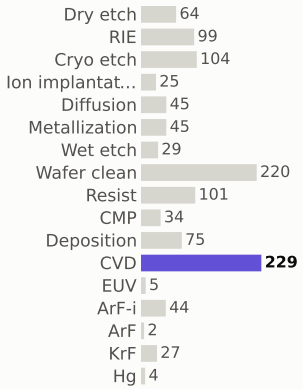
y cut at 57.351 µm



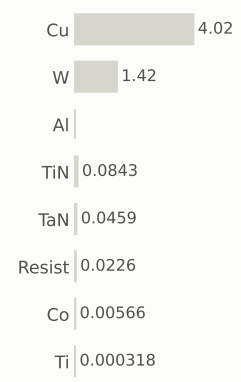
mask for this step

no mask for this step

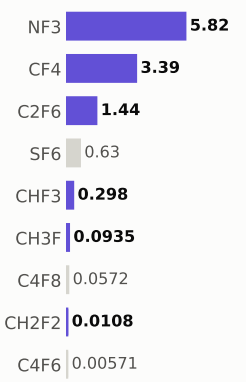
steps by process area



materials deposited (g)



gases emitted (g)



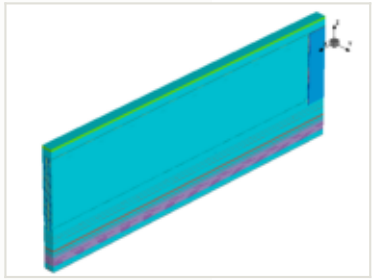
Step #13330: CVD

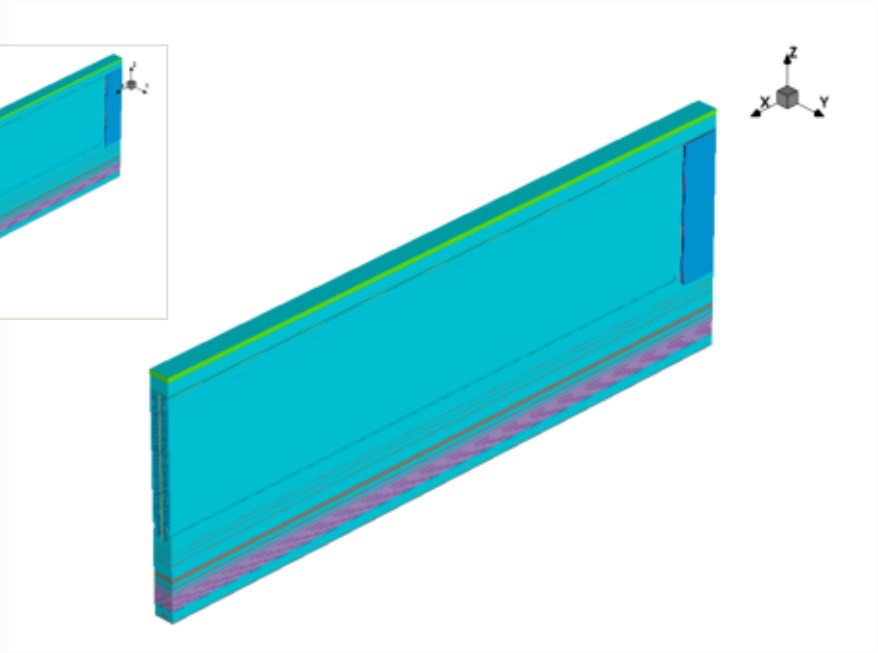
depo_isotropic · CVD

1.31	811.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

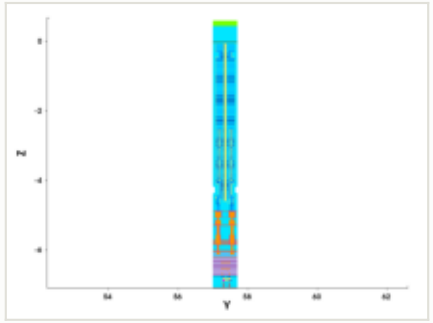
3D structure after this step

zoomed to fit

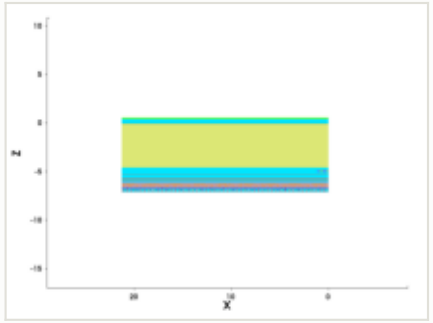




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

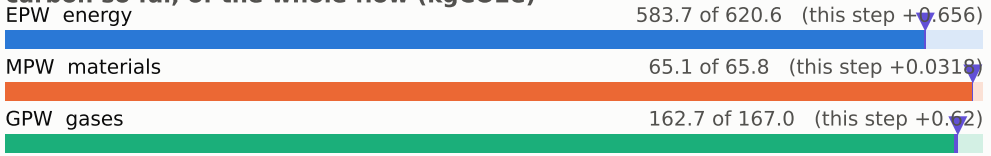
no mask for this step

this step

process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

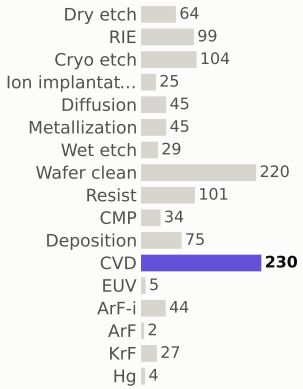
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

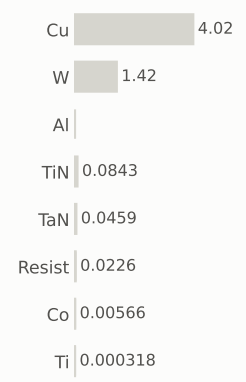


embodied carbon up to each step (kgCO2e per wafer)

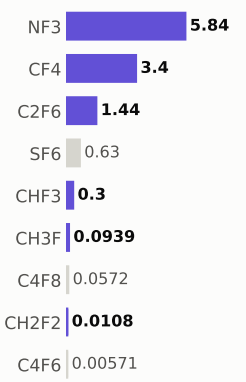
steps by process area



materials deposited (g)



gases emitted (g)



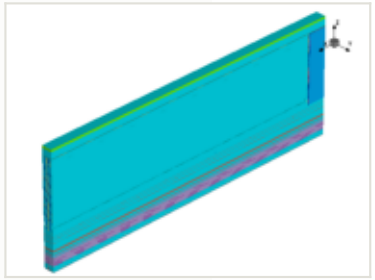
Step #13340: Wafer_clean

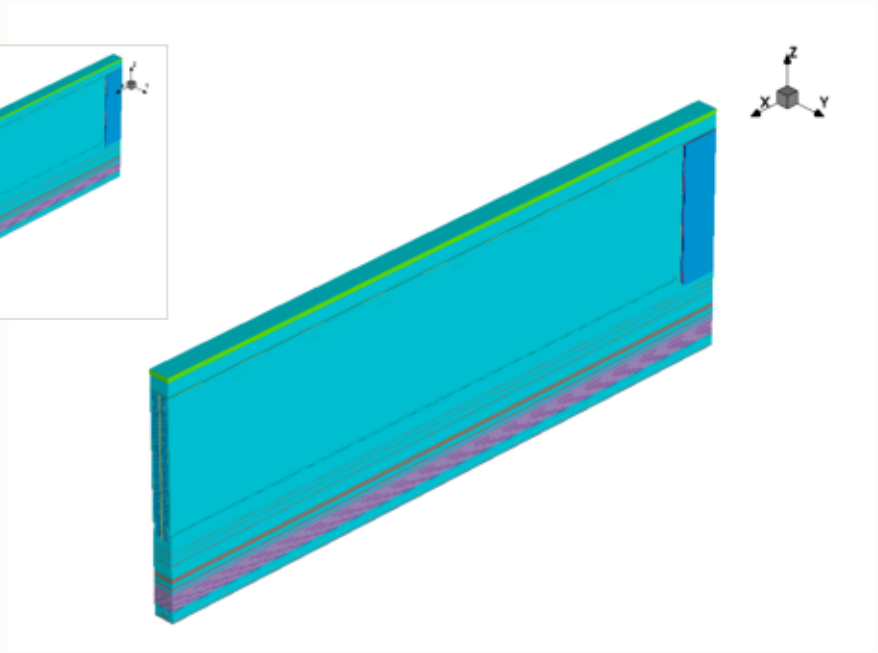
clean · Wafer_clean

0.169	811.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



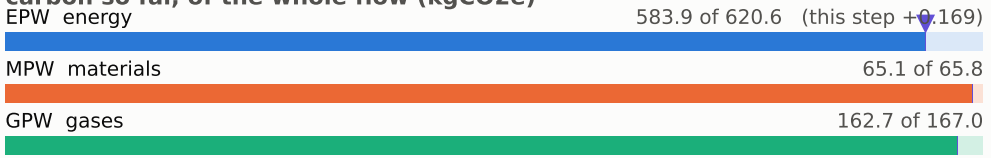


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

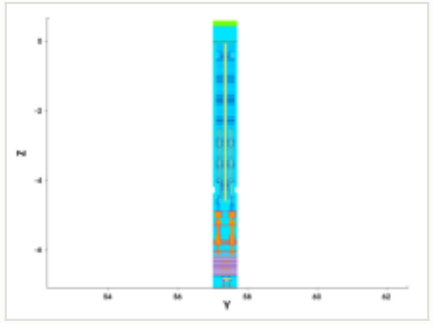
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

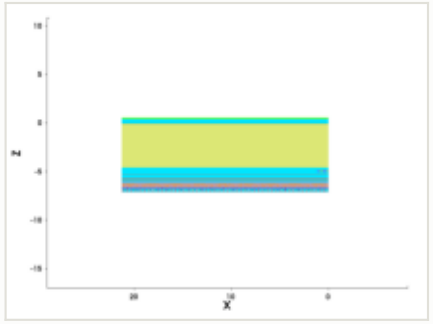


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



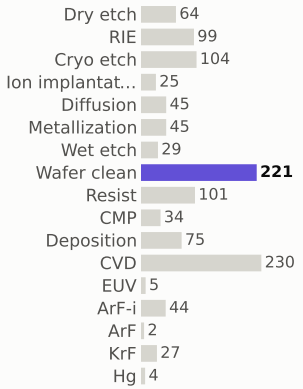
y cut at 57.351 µm



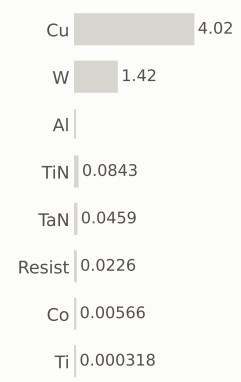
mask for this step

no mask for this step

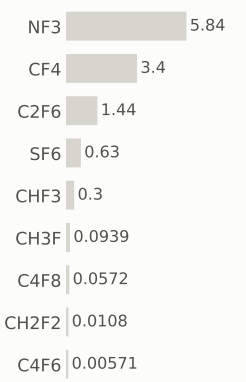
steps by process area



materials deposited (g)



gases emitted (g)



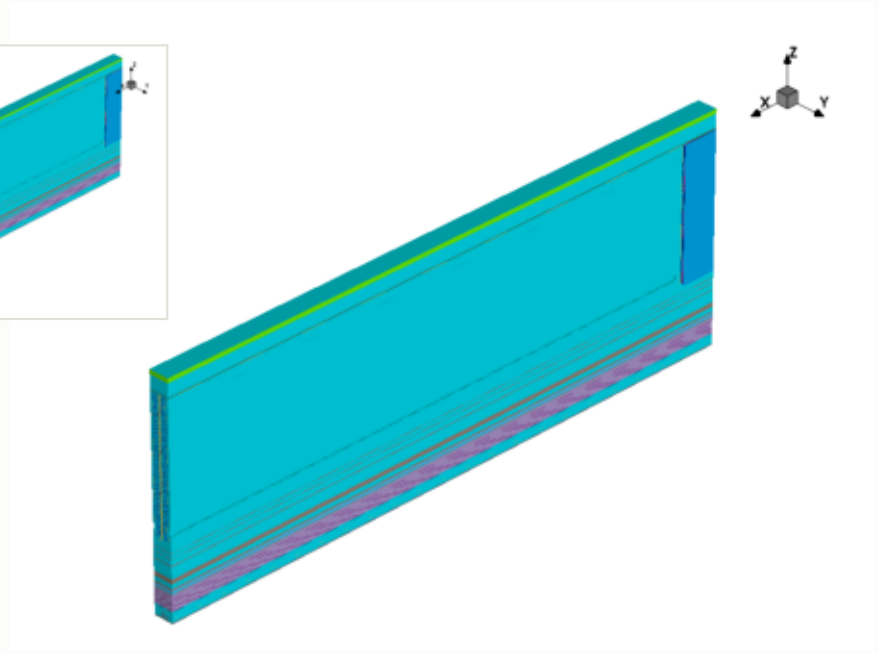
Step #13350: Resist

resist · Resist

0.169	811.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

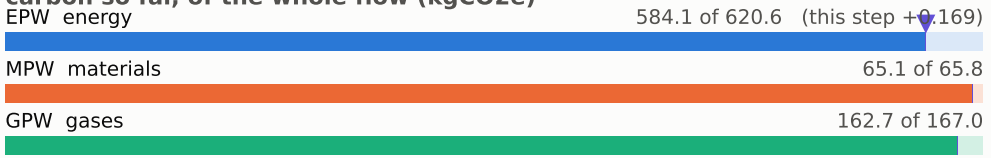


this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

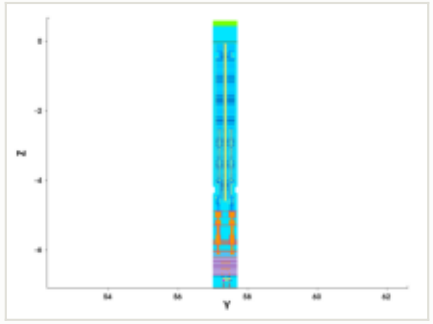
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

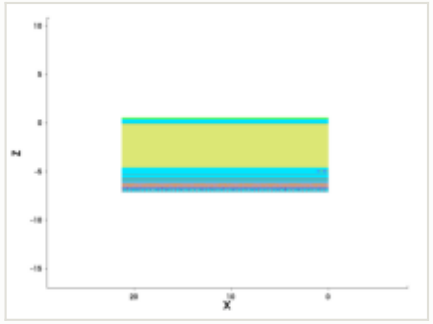


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



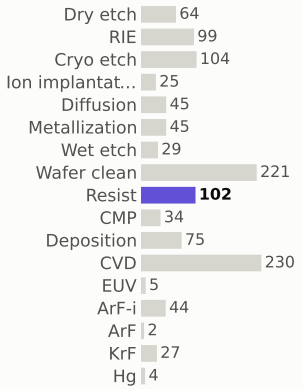
y cut at 57.351 µm



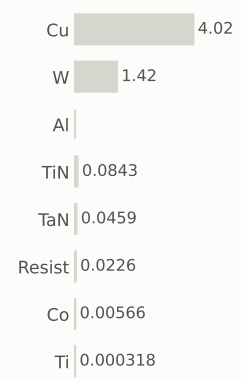
mask for this step

no mask for this step

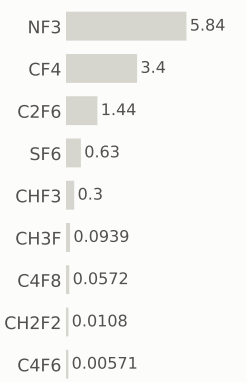
steps by process area



materials deposited (g)



gases emitted (g)



Step #13360: KrF

litho_patterning · KrF

0.199	812.1	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

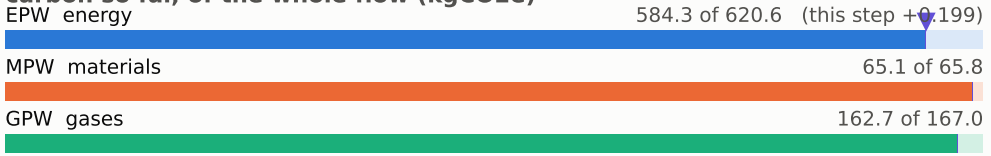
3D structure after this step

zoomed to fit

this step

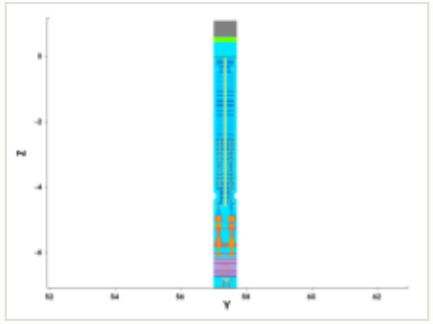
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

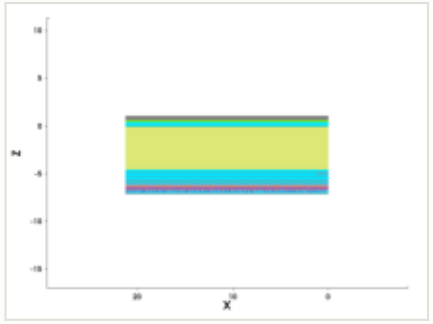


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

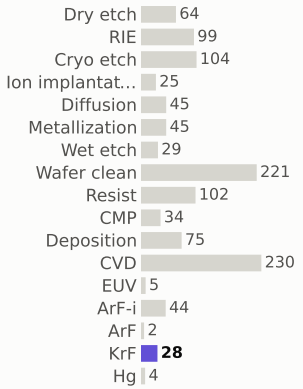


y cut at 57.351 µm

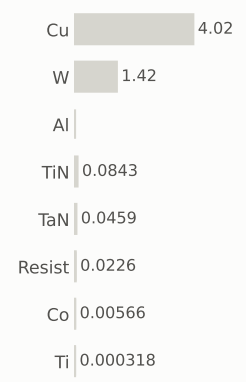


mask: layer 1420:0 (GDSII)

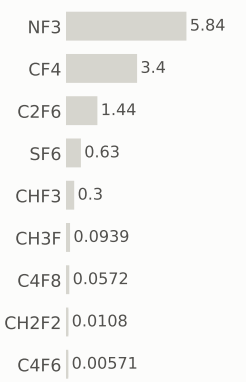
steps by process area



materials deposited (g)



gases emitted (g)



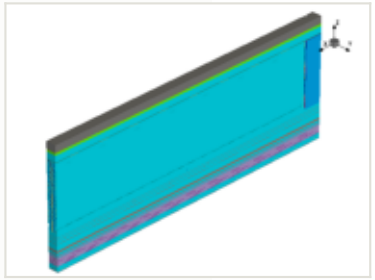
Step #13370: RIE

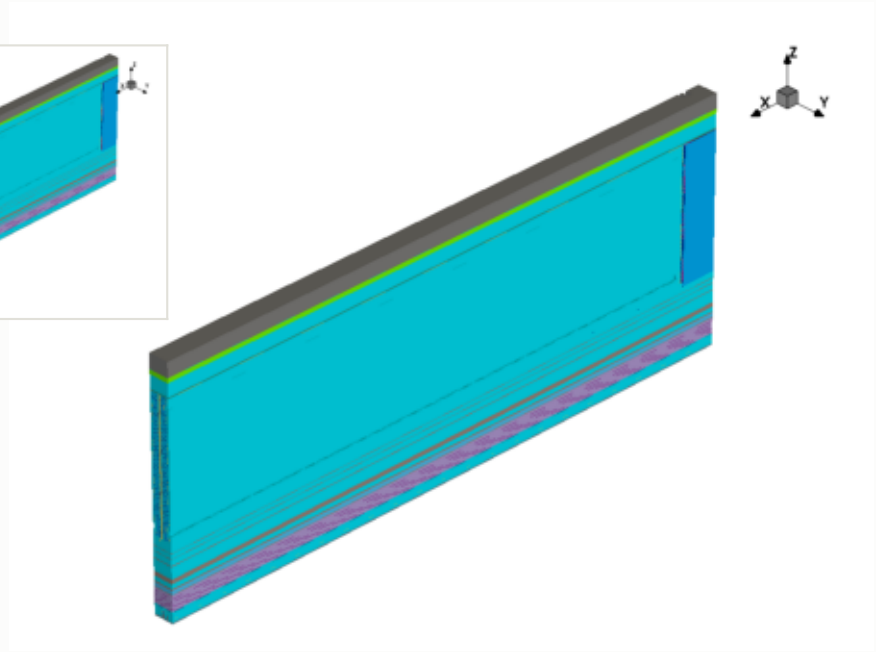
etch_anisotropic · RIE

1.04	813.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



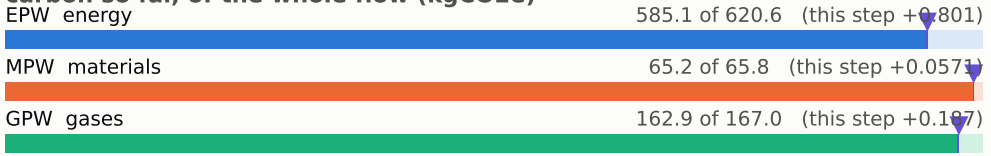


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

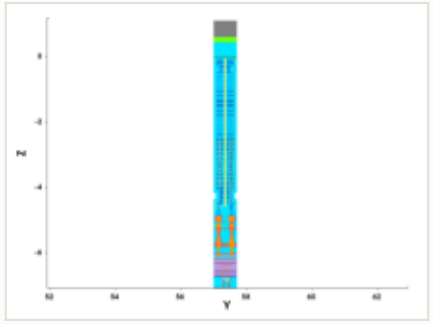
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

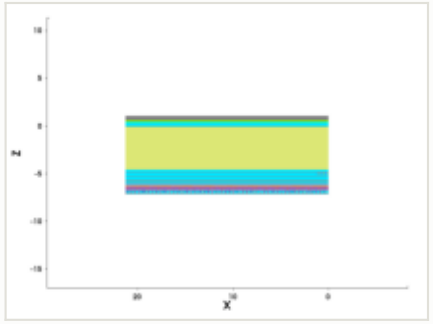


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



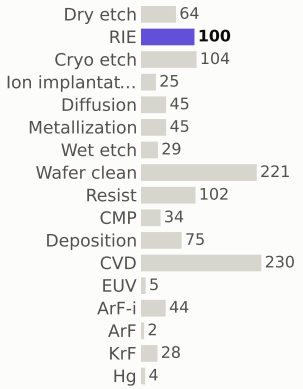
y cut at 57.351 µm



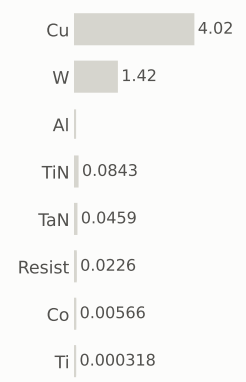
mask for this step

no mask for this step

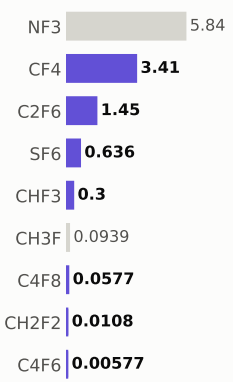
steps by process area



materials deposited (g)



gases emitted (g)



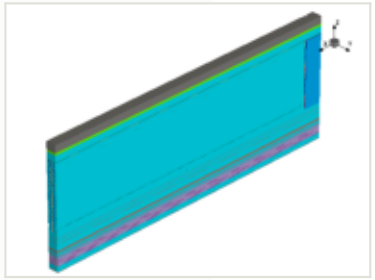
Step #13380: RIE

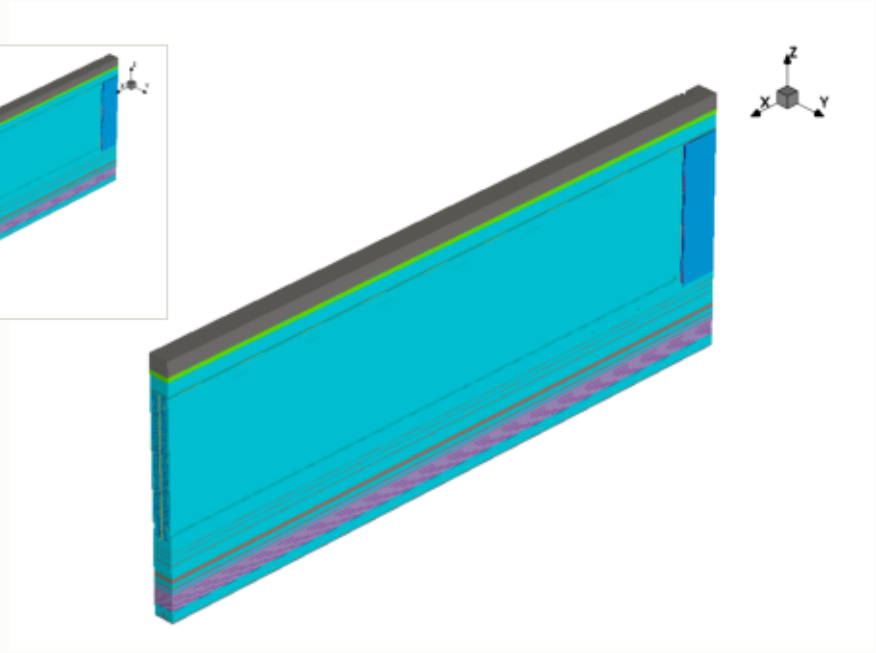
etch_anisotropic · RIE

1.04	814.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



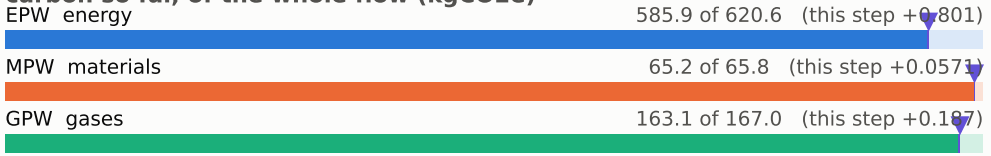


this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

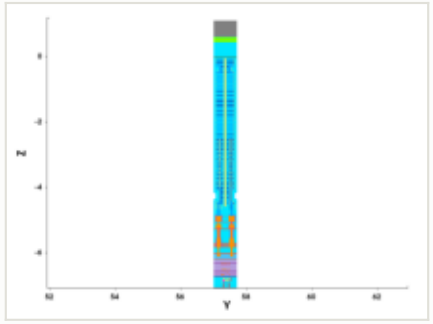
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

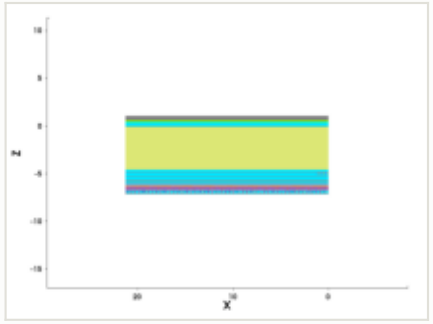


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



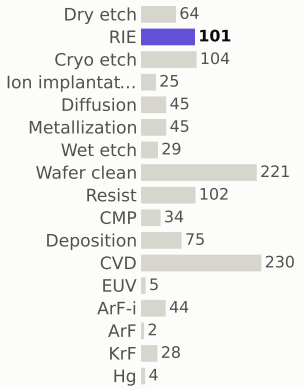
y cut at 57.351 µm



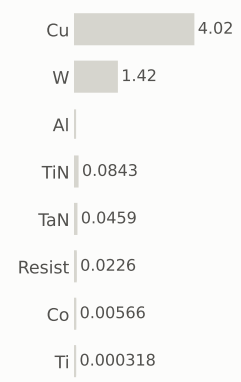
mask for this step

no mask for this step

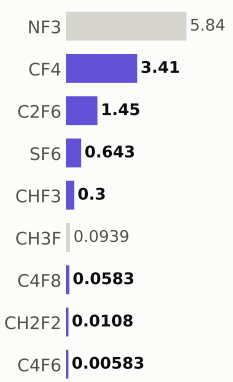
steps by process area



materials deposited (g)



gases emitted (g)



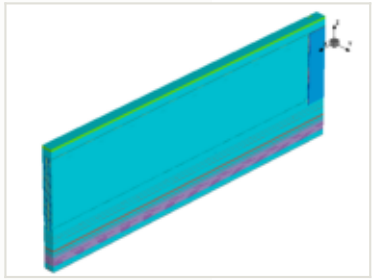
Step #13390: Dry_etch

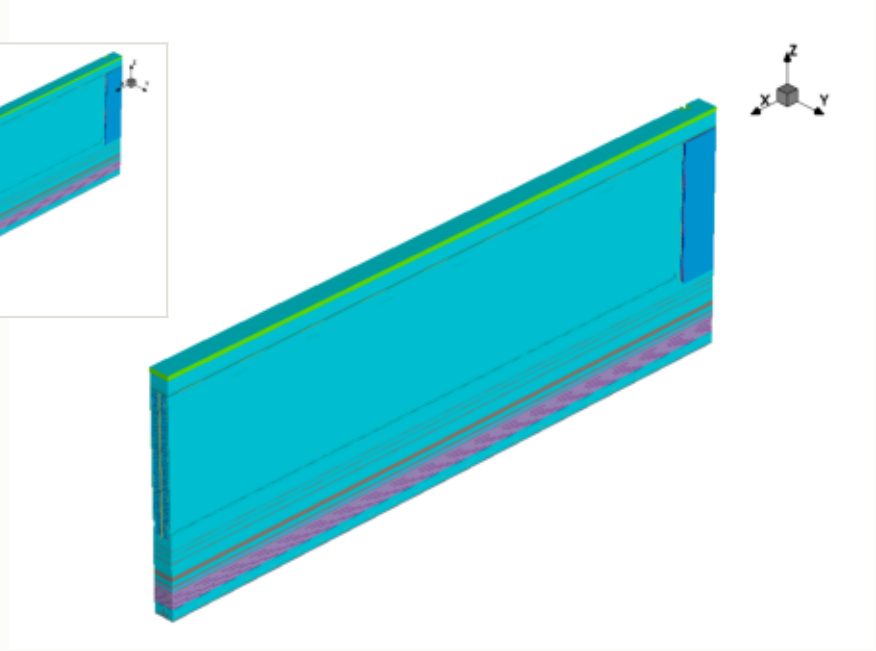
etch_strip · Dry_etch

0.801	815.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

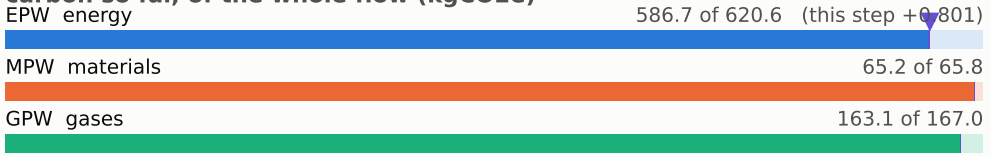




this step

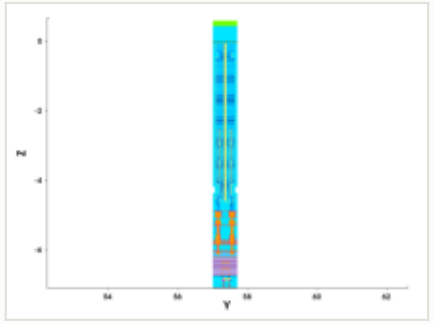
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

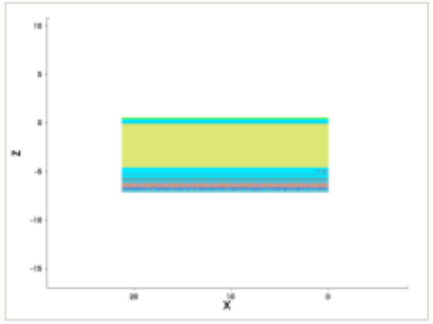


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



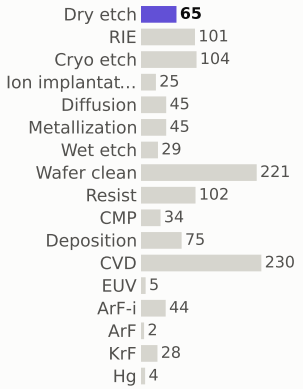
y cut at 57.351 µm



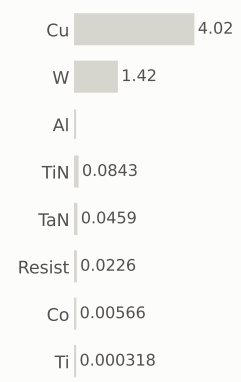
mask for this step

no mask for this step

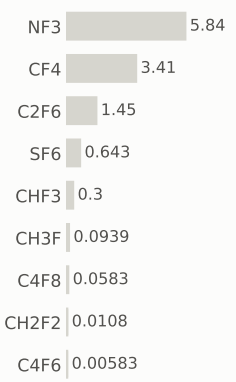
steps by process area



materials deposited (g)



gases emitted (g)



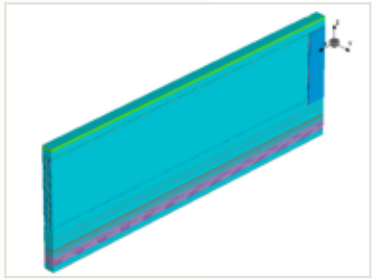
Step #13400: Wafer_clean

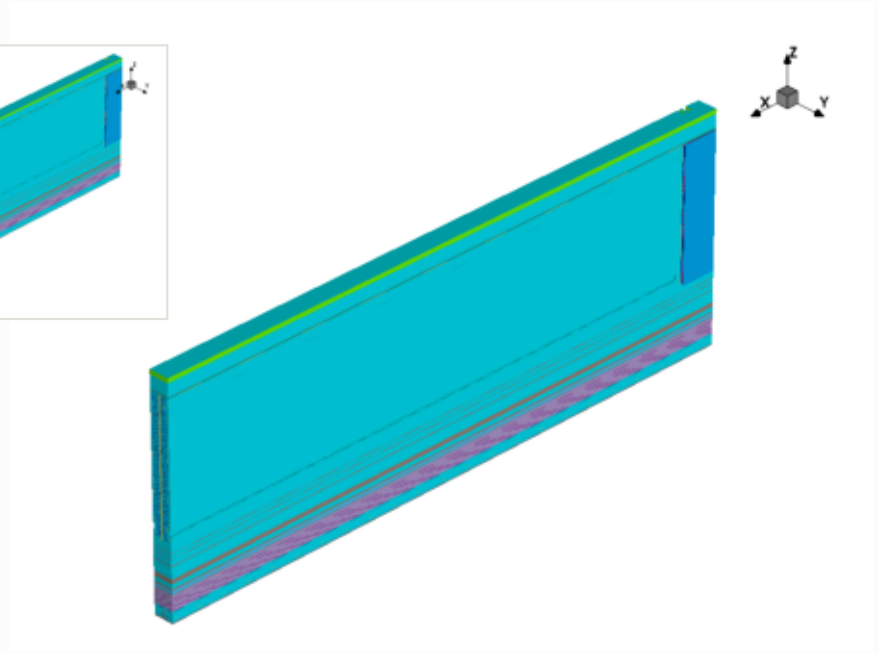
clean · Wafer_clean

0.169	815.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



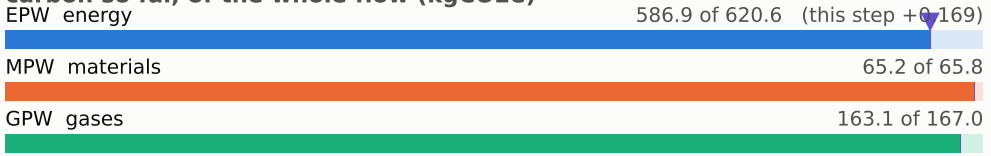


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

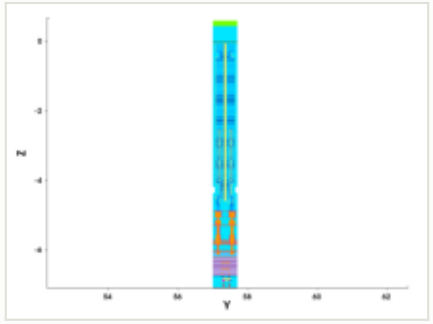
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

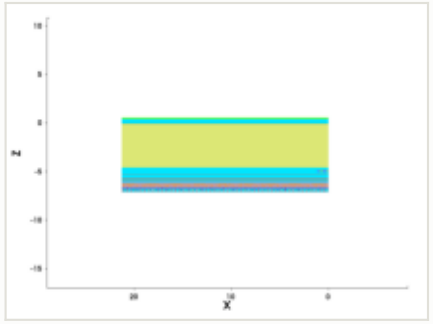


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



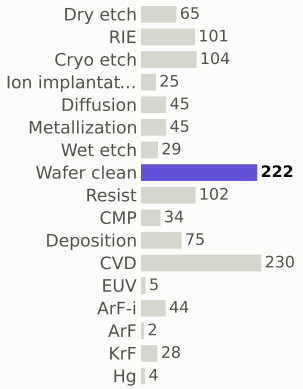
y cut at 57.351 µm



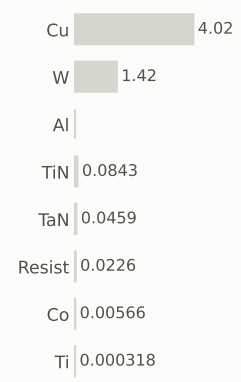
mask for this step

no mask for this step

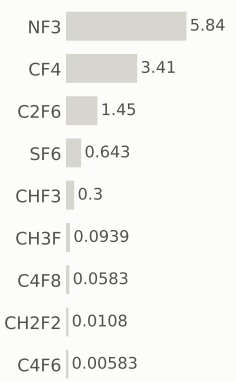
steps by process area



materials deposited (g)



gases emitted (g)



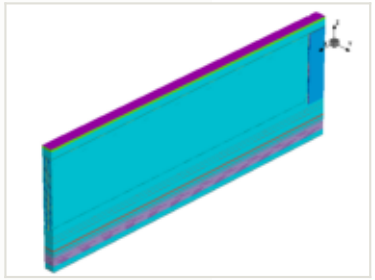
Step #13410: Deposition

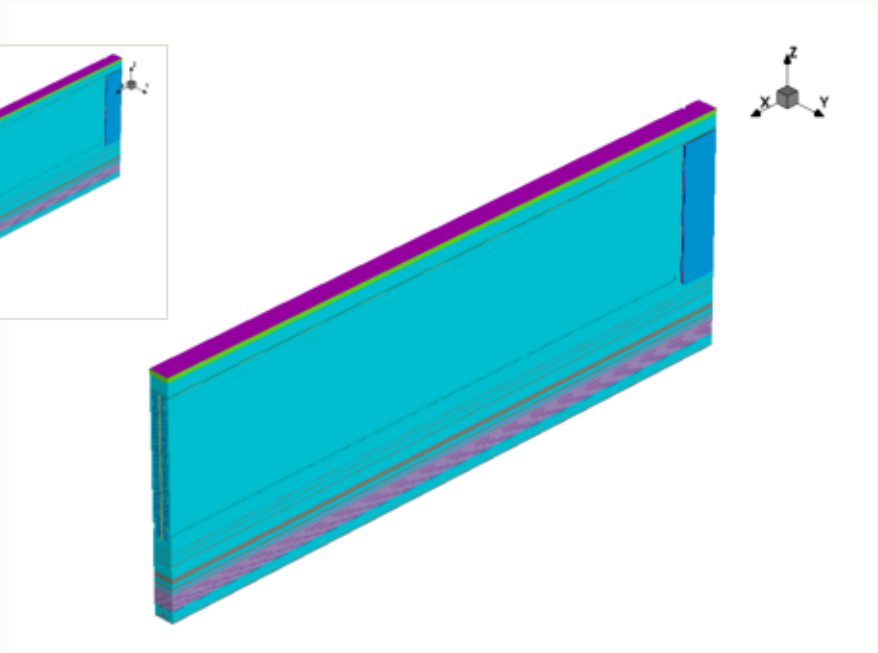
depo_isotropic · Deposition

0.657	815.8	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

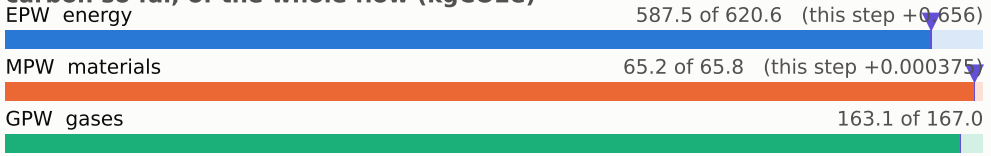




this step

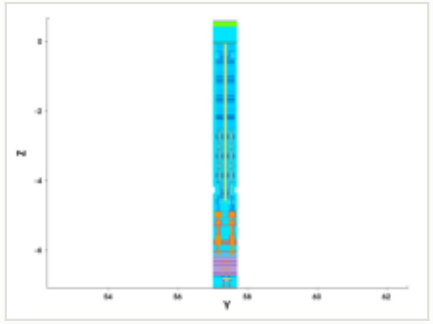
process area	Deposition	EPW	0.656 kgCO2e
material	TaN	EPW tool	0.469 kgCO2e
thickness	0.015 µm	EPW overhead	0.187 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	0.000375 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.657 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

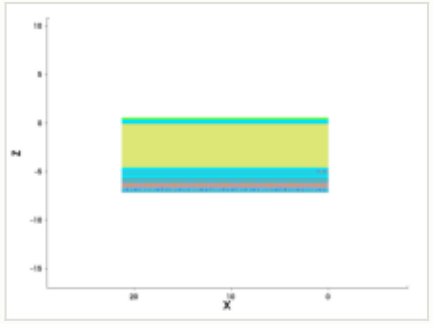


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



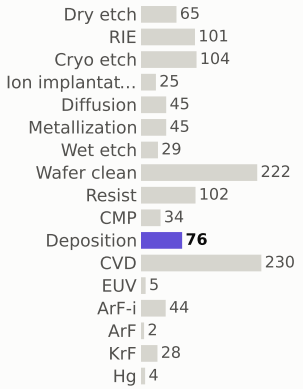
y cut at 57.351 µm



mask for this step

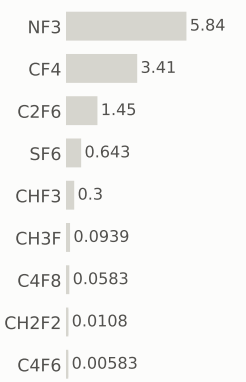
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



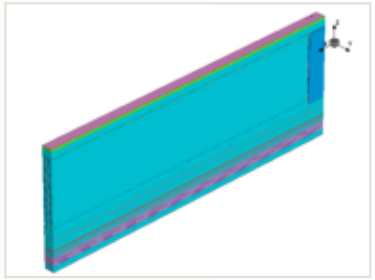
Step #13420: Metallization

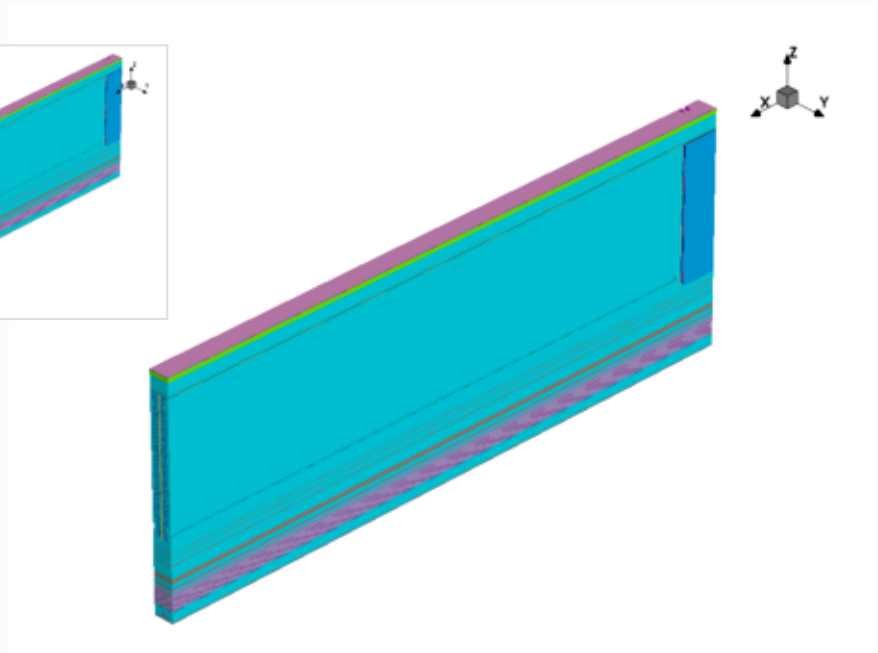
depo_isotropic · Metallization

1.2	817.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



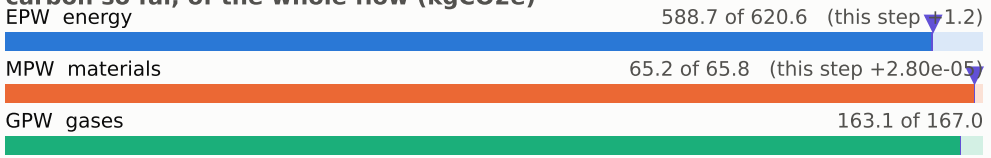


this step

process area	Metallization
material	Co
thickness	0.001 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

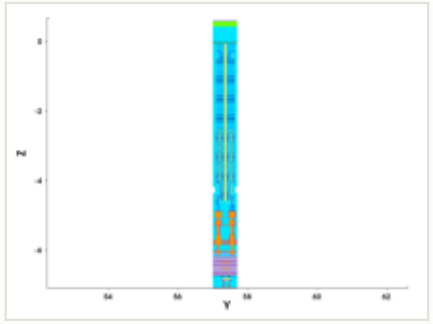
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	2.80e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

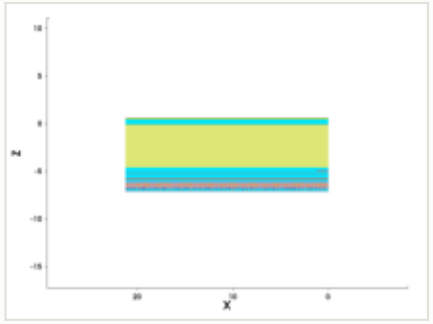


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



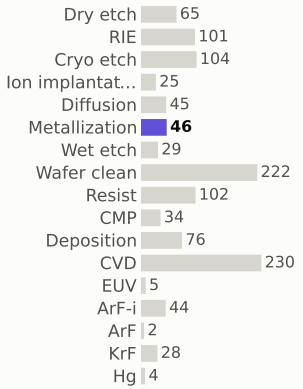
y cut at 57.351 µm



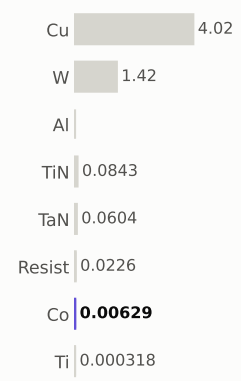
mask for this step

no mask for this step

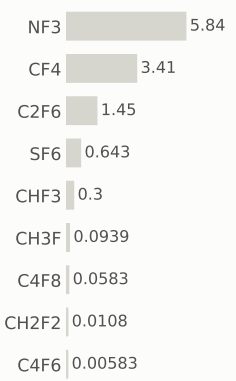
steps by process area



materials deposited (g)



gases emitted (g)



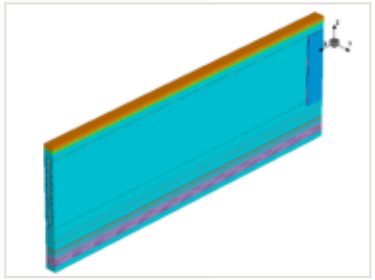
Step #13430: Metallization

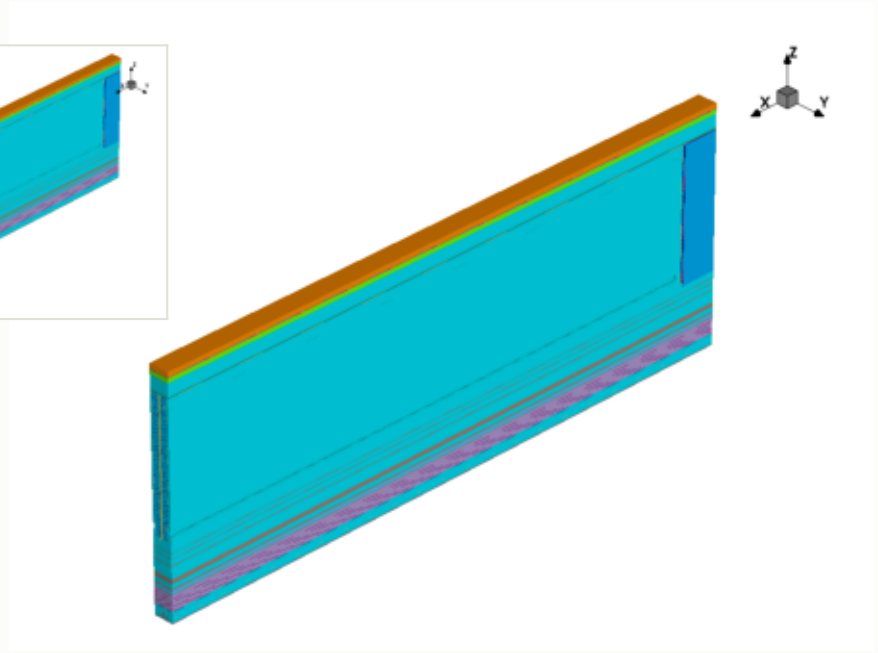
depo_isotropic · Metallization

1.2	818.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

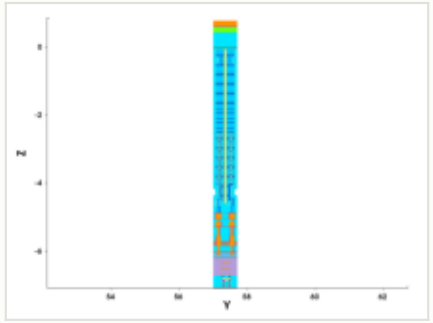
3D structure after this step

zoomed to fit

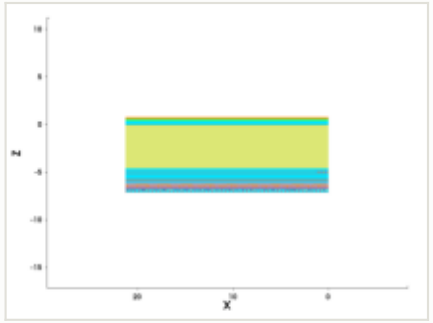




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

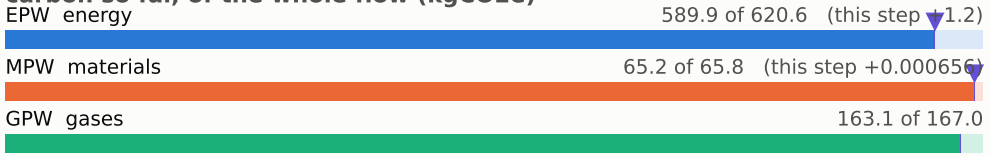
no mask for this step

this step

process area	Metallization
material	Cu
thickness	0.17 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

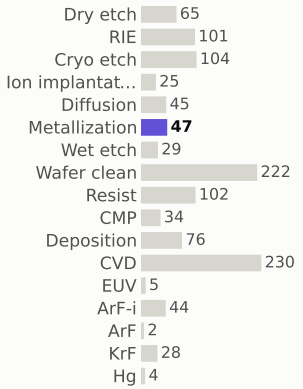
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000656 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

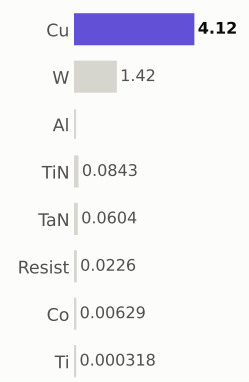


embodied carbon up to each step (kgCO2e per wafer)

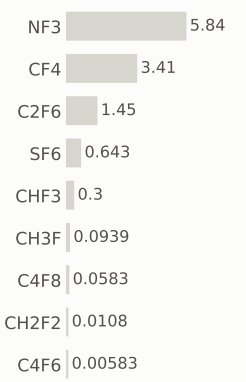
steps by process area



materials deposited (g)



gases emitted (g)



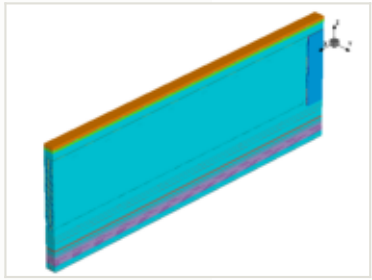
Step #13440: Diffusion

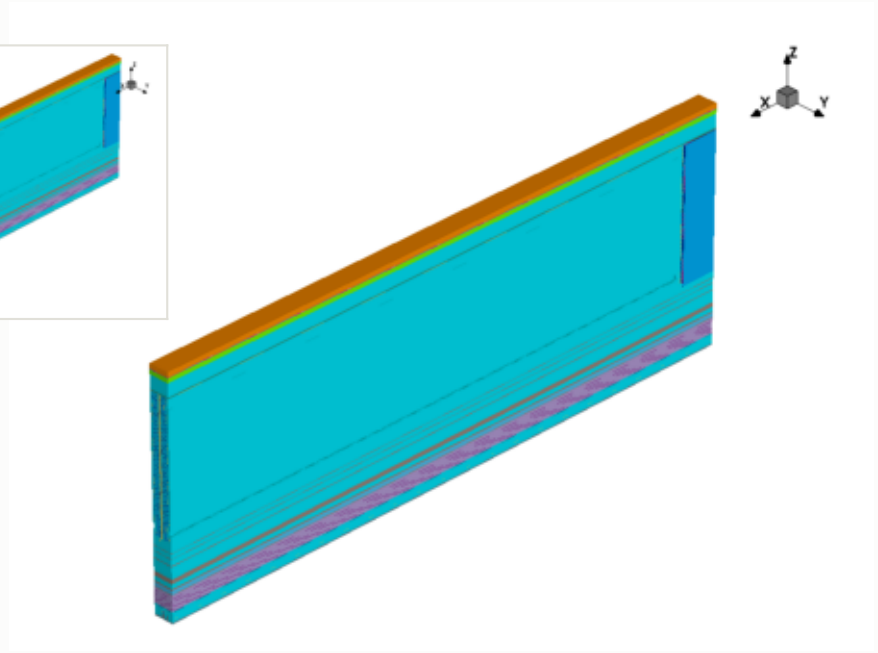
anneal · Diffusion

0.801	819.0	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

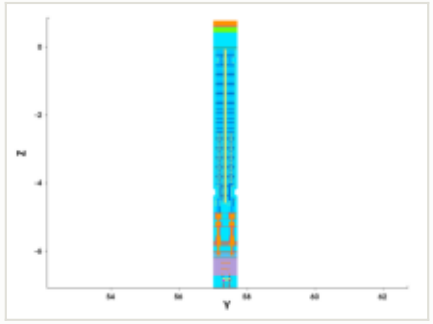
3D structure after this step

zoomed to fit

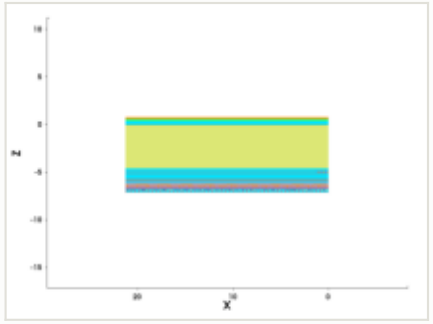




x cut at 10.650 µm



y cut at 57.351 µm



mask for this step

no mask for this step

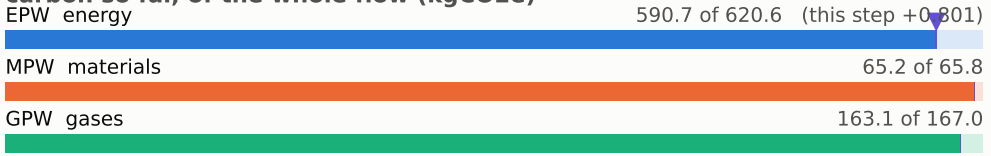
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

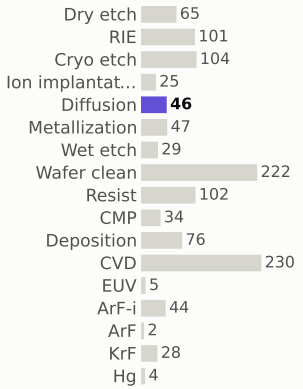
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

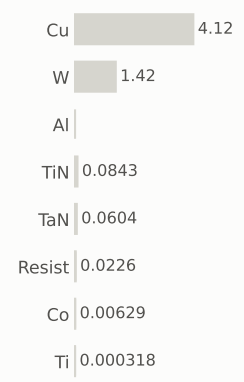


embodied carbon up to each step (kgCO2e per wafer)

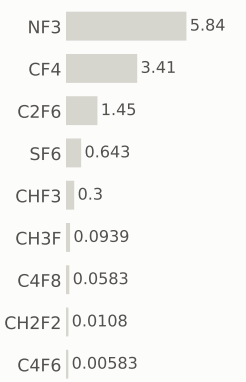
steps by process area



materials deposited (g)



gases emitted (g)



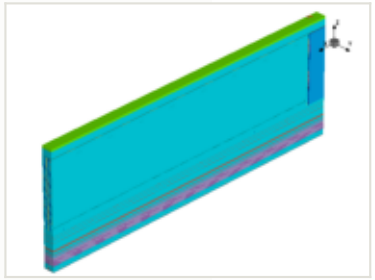
Step #13450: CMP

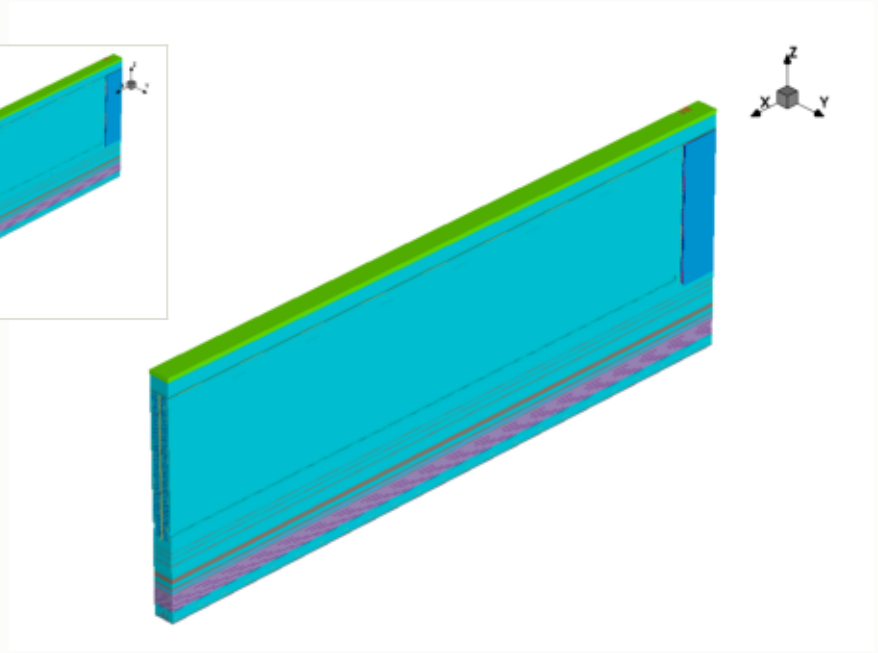
etch_cmp · CMP

0.169	819.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

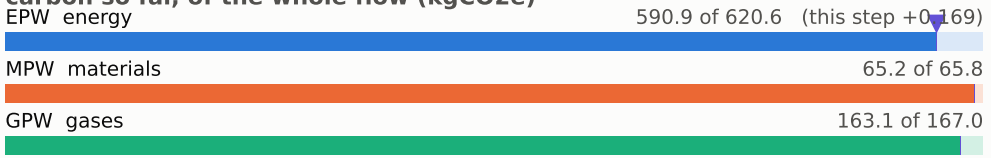




this step

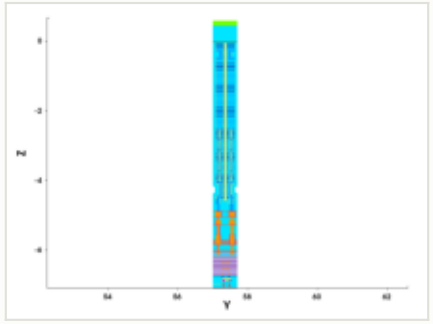
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

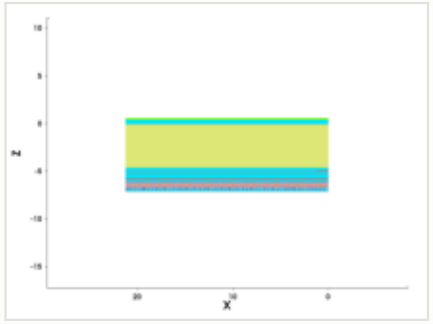


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



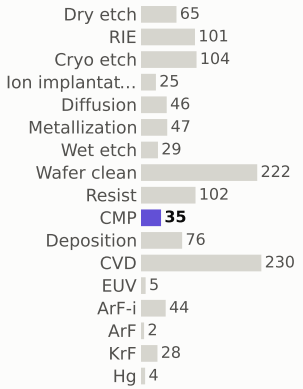
y cut at 57.351 µm



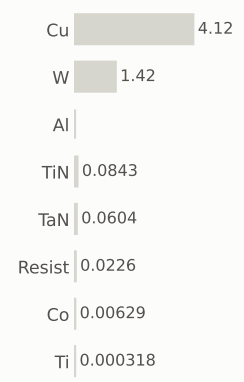
mask for this step

no mask for this step

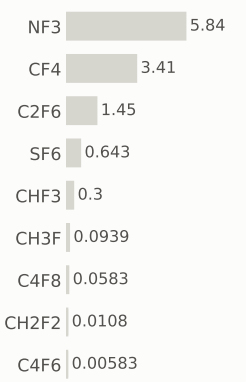
steps by process area



materials deposited (g)



gases emitted (g)



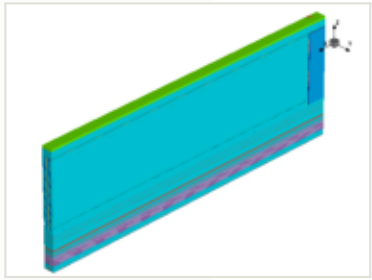
Step #13460: Wafer_clean

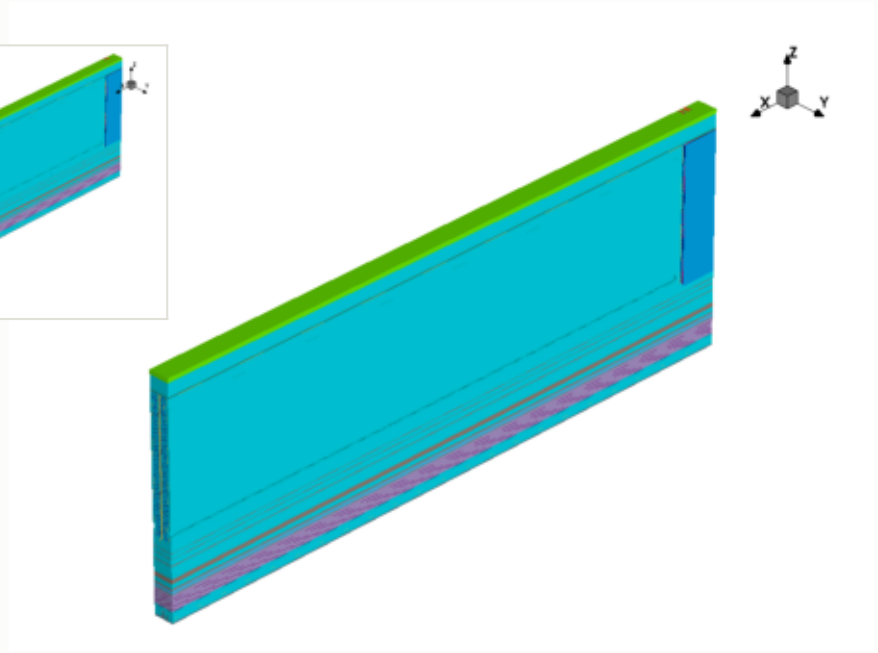
clean · Wafer_clean

0.169	819.3	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



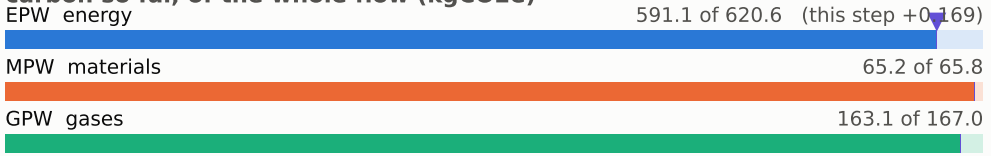


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

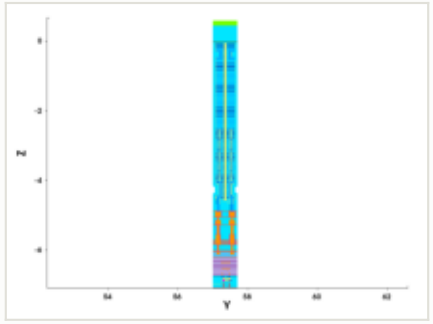
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

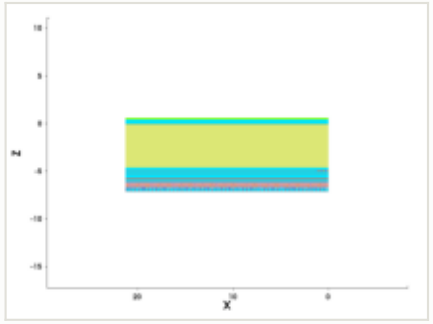


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



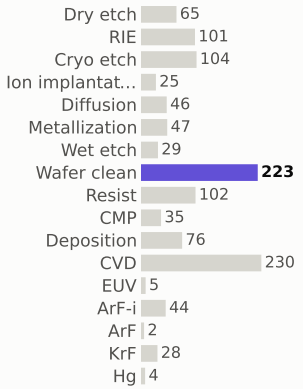
y cut at 57.351 µm



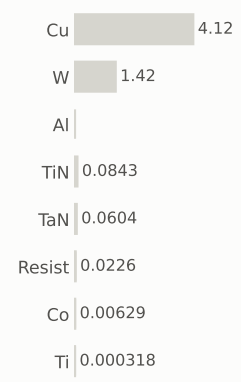
mask for this step

no mask for this step

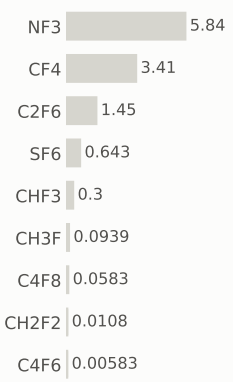
steps by process area



materials deposited (g)



gases emitted (g)



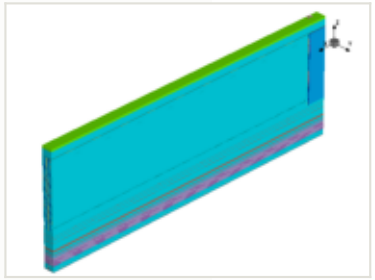
Step #13470: Metallization

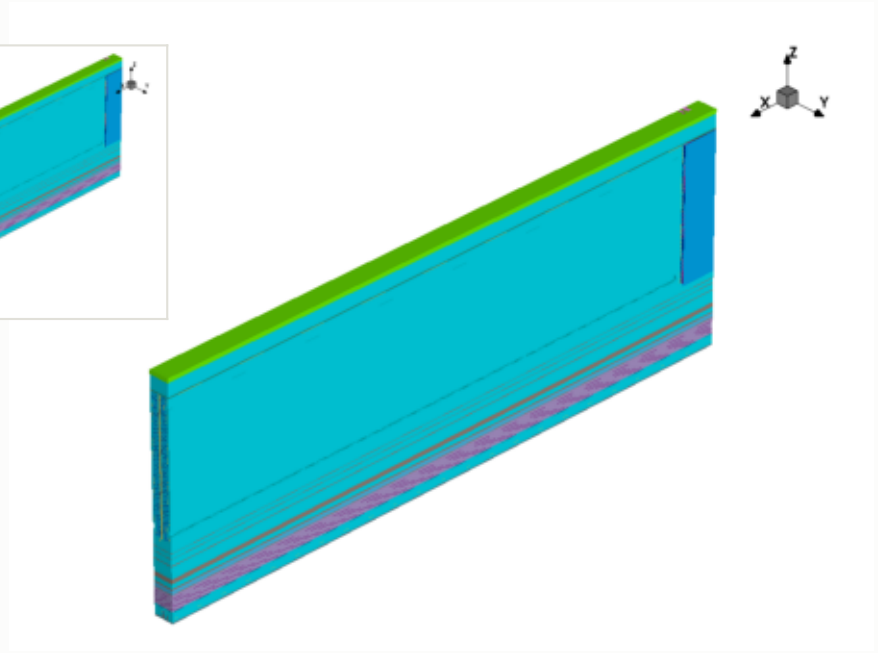
depo_isotropic · Metallization

1.2	820.5	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



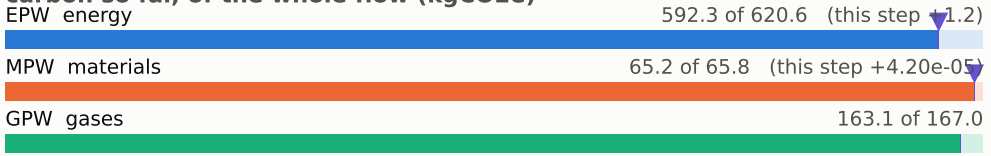


this step

process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

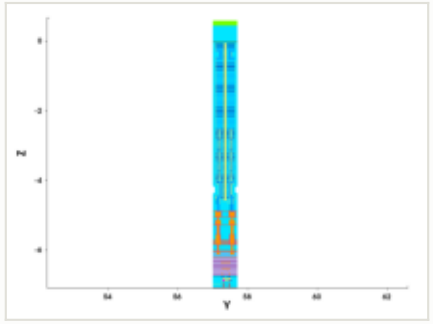
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

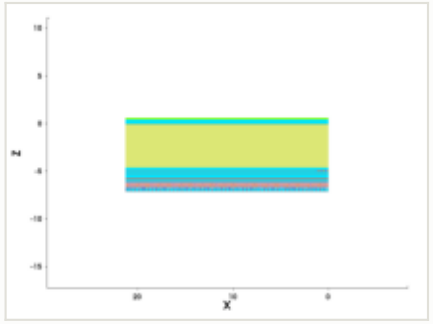


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



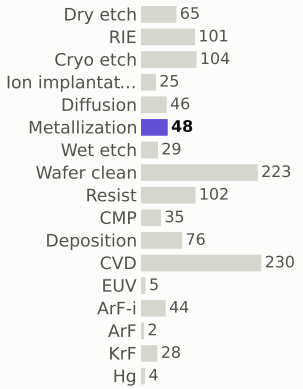
y cut at 57.351 µm



mask for this step

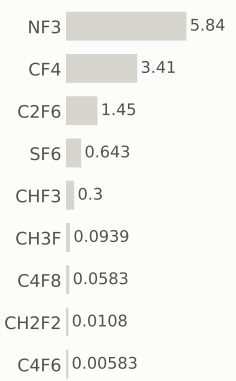
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



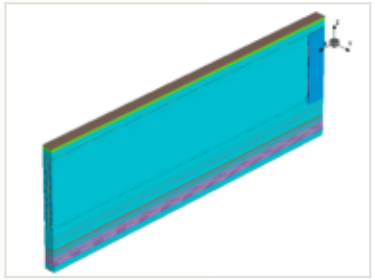
Step #13480: CVD

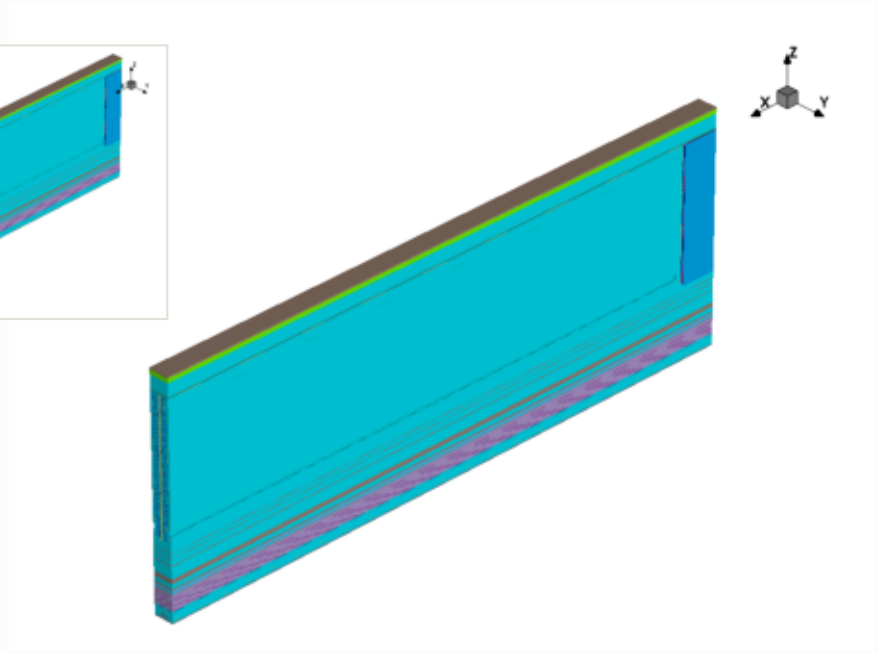
depo_isotropic · CVD

1.31	821.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



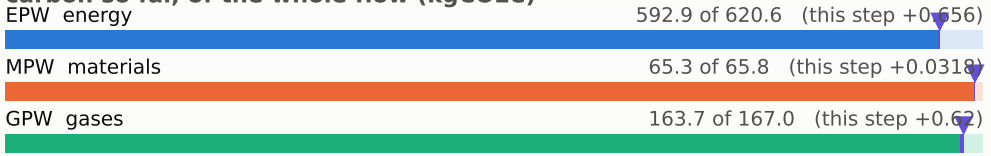


this step

process area	CVD
material	—
thickness	0.06 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

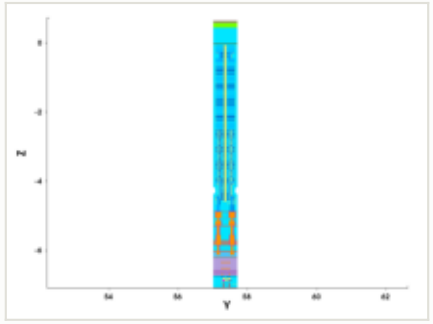
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

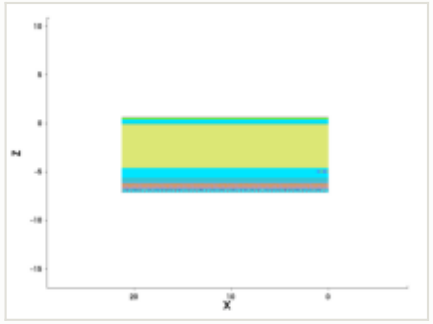


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



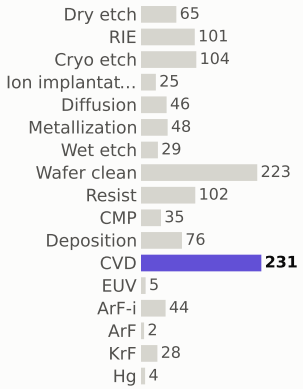
y cut at 57.351 µm



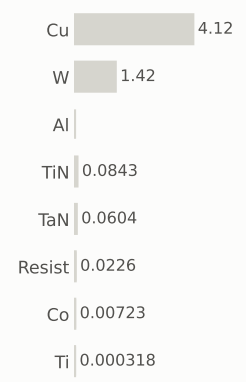
mask for this step

no mask for this step

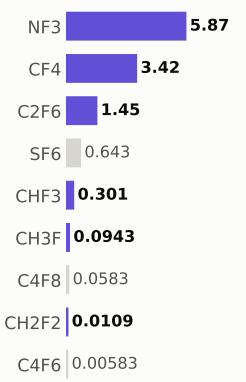
steps by process area



materials deposited (g)



gases emitted (g)



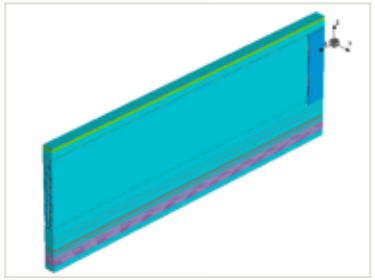
Step #13490: CVD

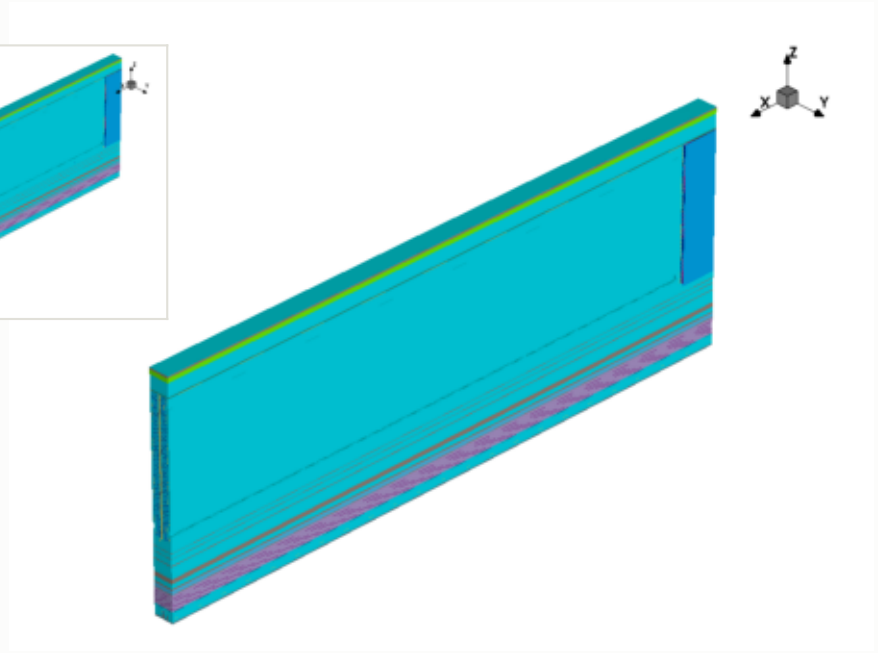
depo_isotropic · CVD

1.31	823.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



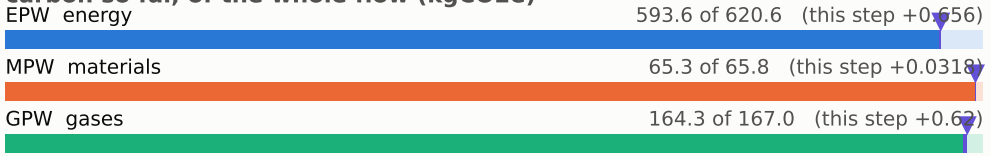


this step

process area	CVD
material	—
thickness	0.01 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

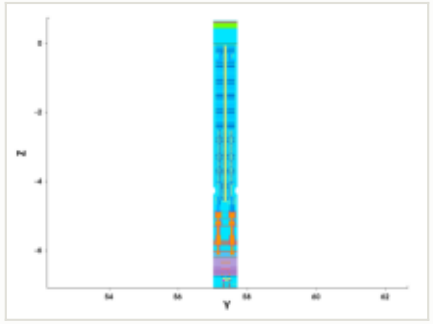
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

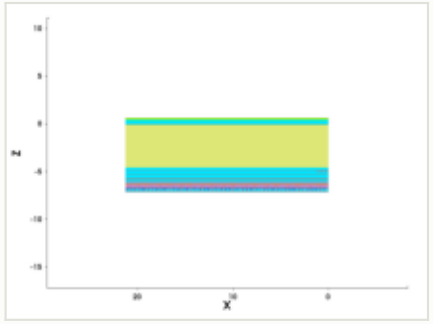


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



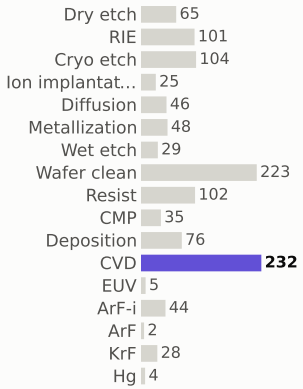
y cut at 57.351 µm



mask for this step

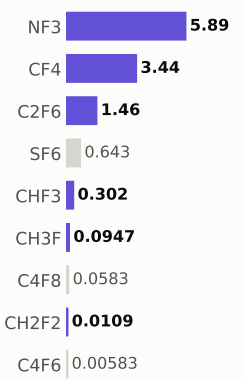
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



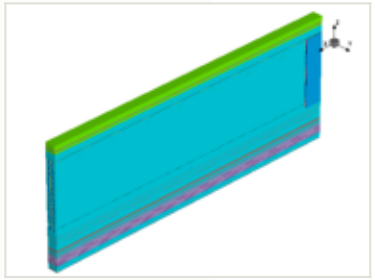
Step #13500: CVD

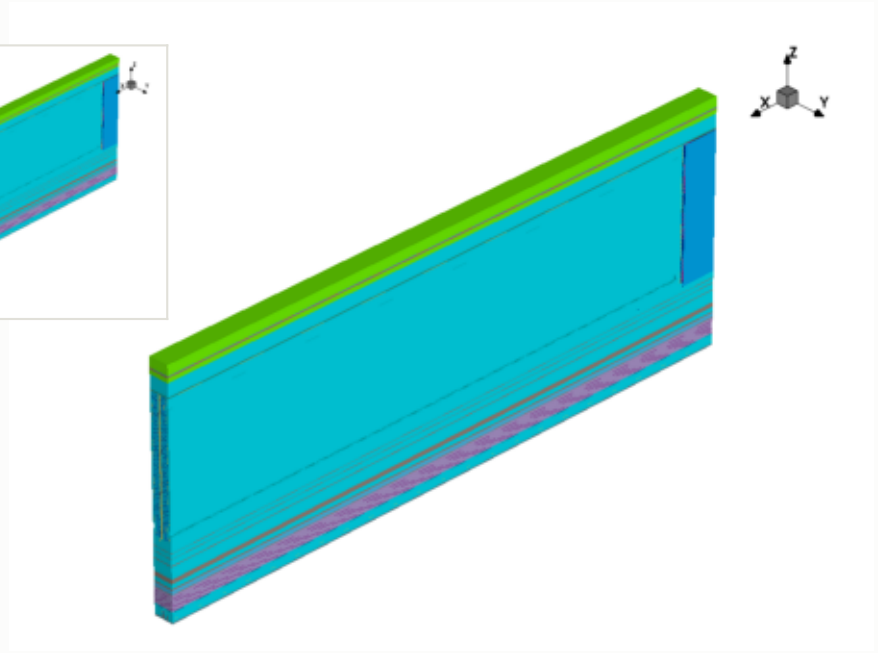
depo_isotropic · CVD

1.31	824.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



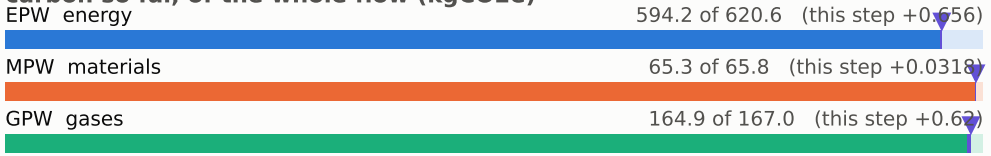


this step

process area	CVD
material	—
thickness	0.342 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

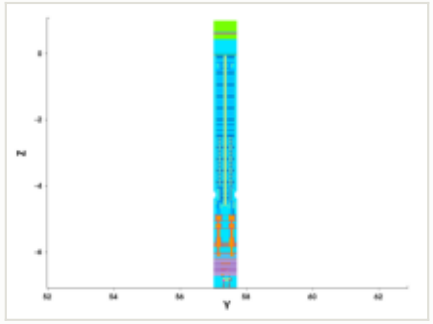
EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

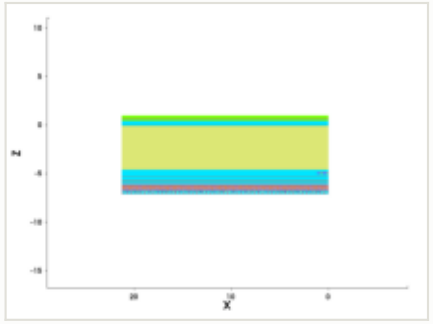


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



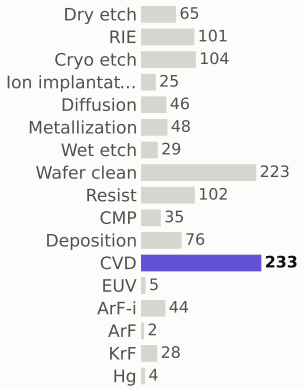
y cut at 57.351 µm



mask for this step

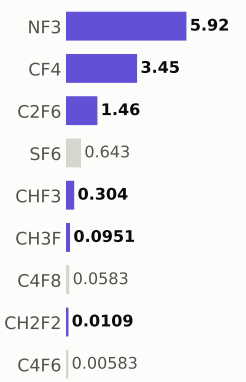
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



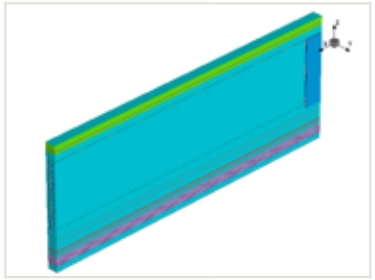
Step #13510: CVD

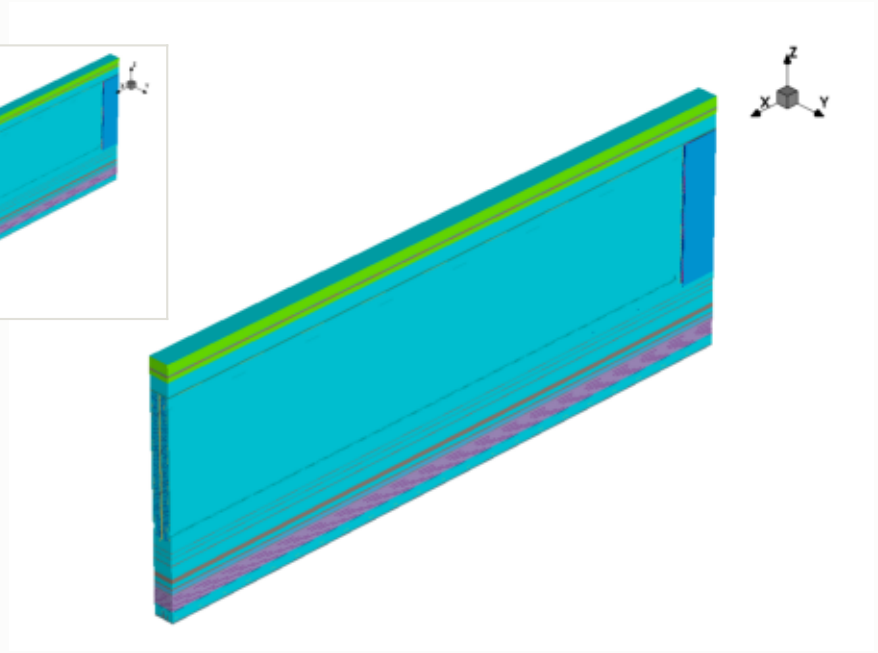
depo_isotropic · CVD

1.31	825.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



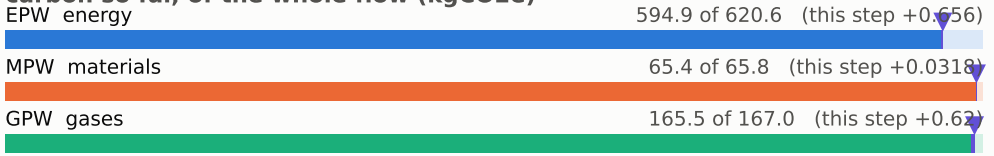


this step

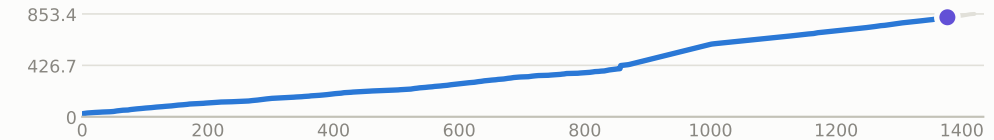
process area	CVD
material	—
thickness	0.005 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

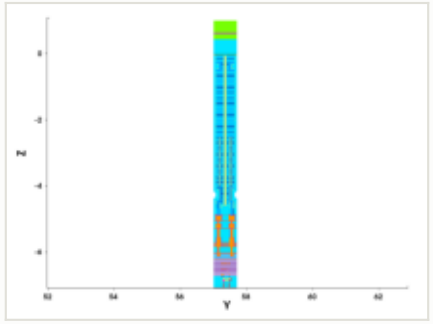
carbon so far, of the whole flow (kgCO2e)



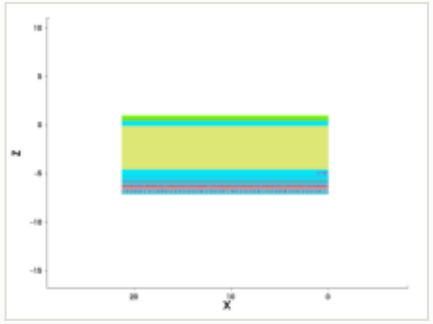
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



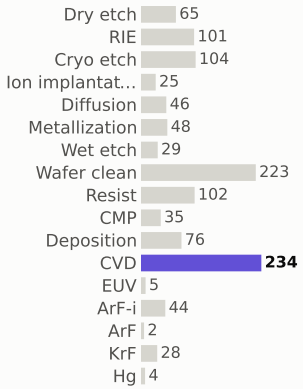
y cut at 57.351 µm



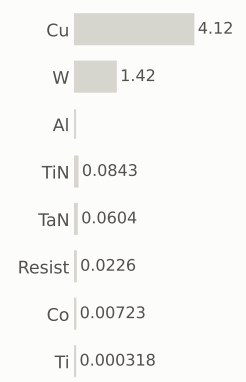
mask for this step

no mask for this step

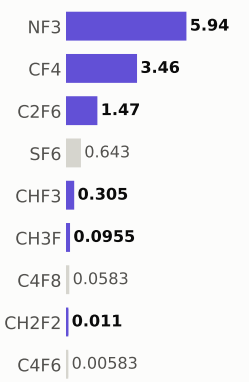
steps by process area



materials deposited (g)



gases emitted (g)



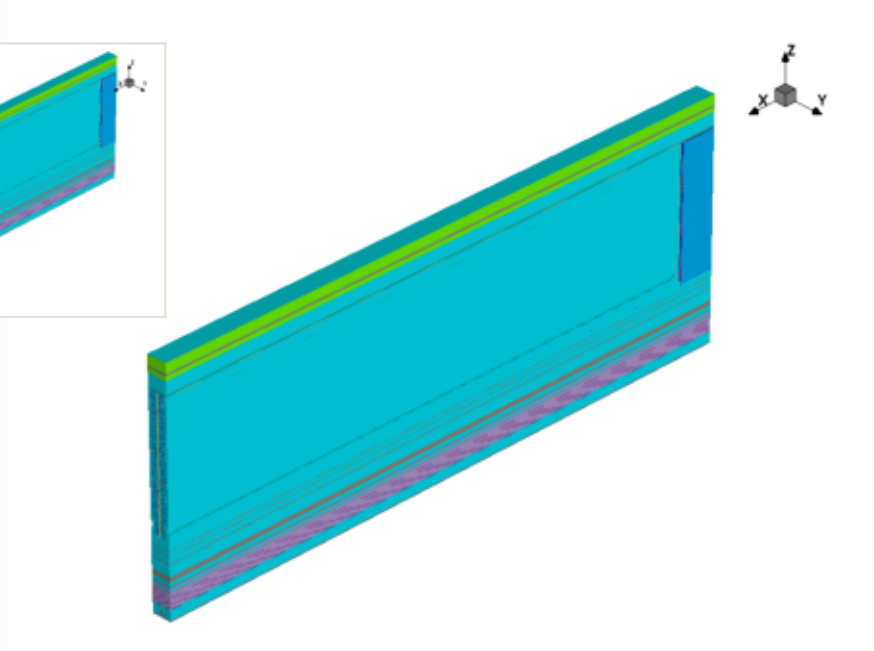
Step #13520: Wafer_clean

clean · Wafer_clean

0.169	825.9	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

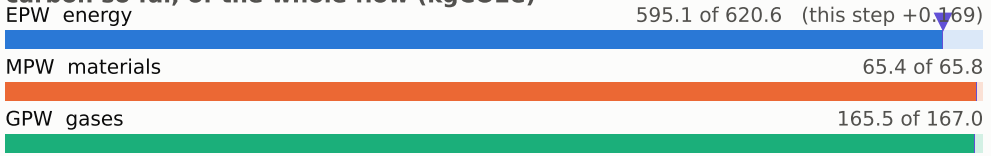


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

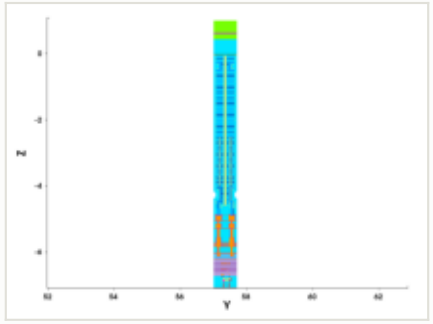
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

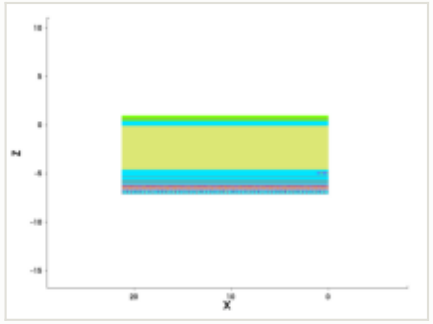


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



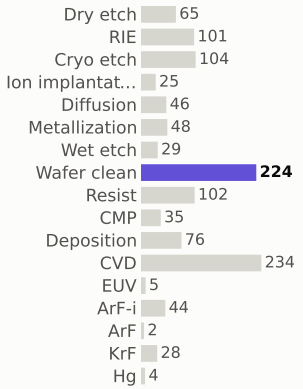
y cut at 57.351 µm



mask for this step

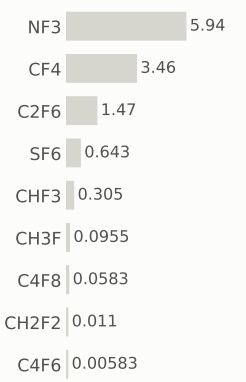
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #13530: Resist

resist · Resist

0.169	826.1	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

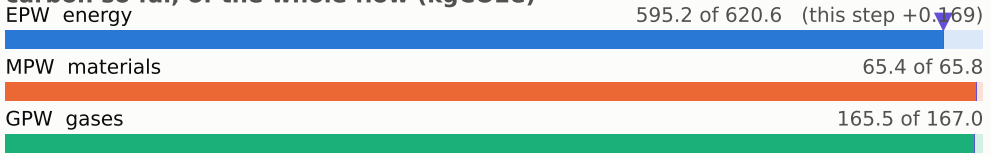
zoomed to fit

this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

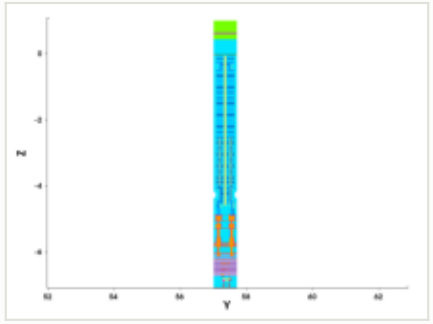
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

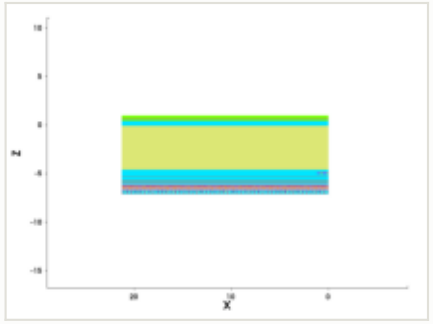


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



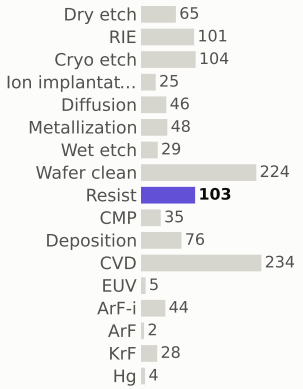
y cut at 57.351 µm



mask for this step

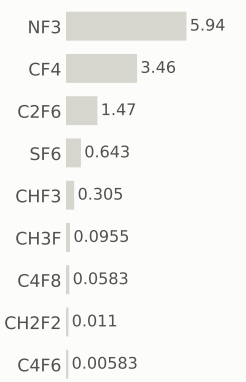
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



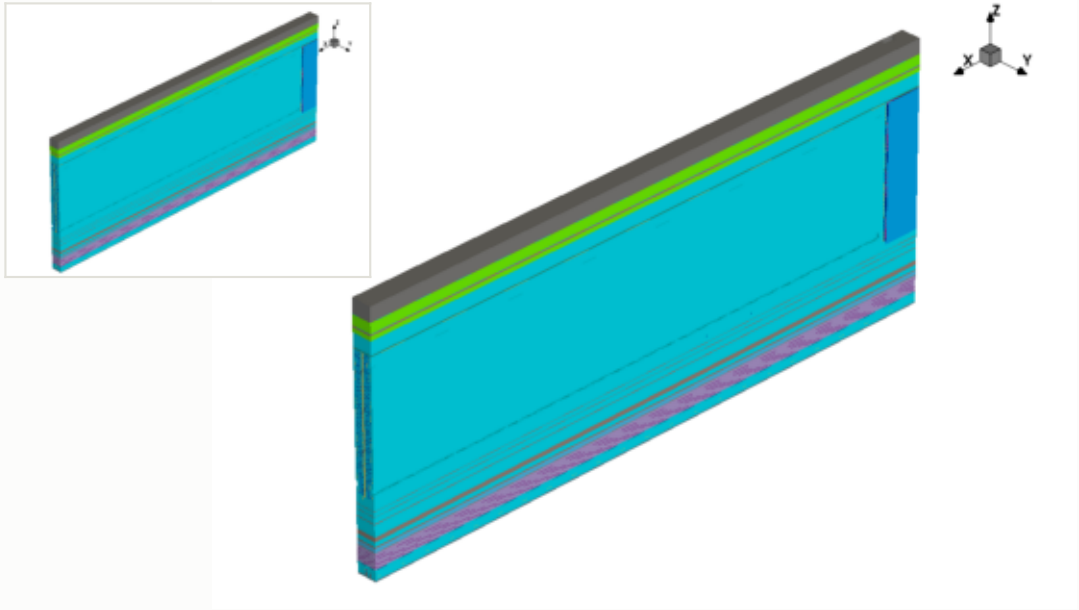
Step #13540: KrF

litho_patterning · KrF

0.199	826.3	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

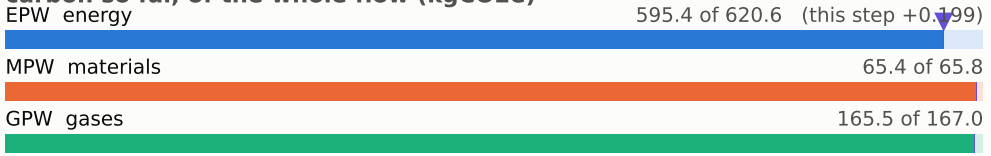
zoomed to fit



this step

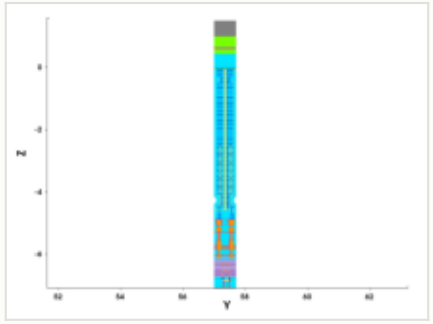
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

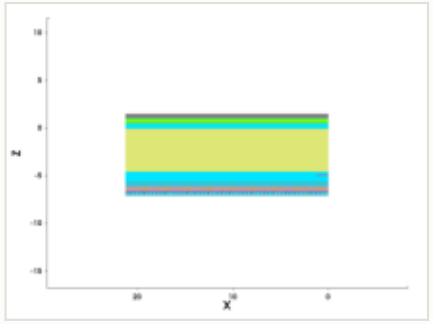


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

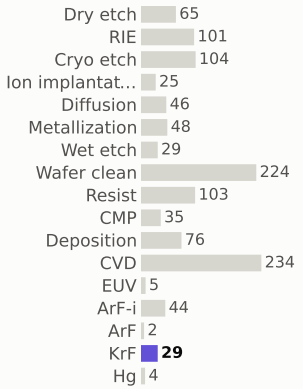


y cut at 57.351 µm

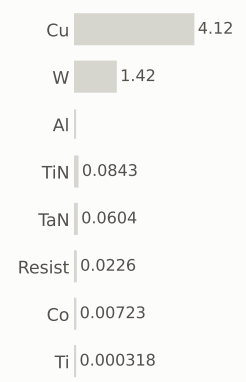


mask: layer 1430:0 (GDSII)

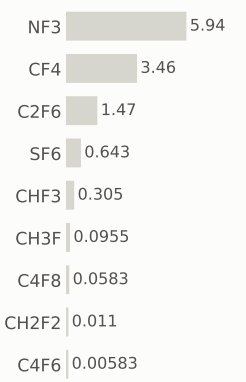
steps by process area



materials deposited (g)



gases emitted (g)



Step #13550: RIE

etch_anisotropic · RIE

1.04	827.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

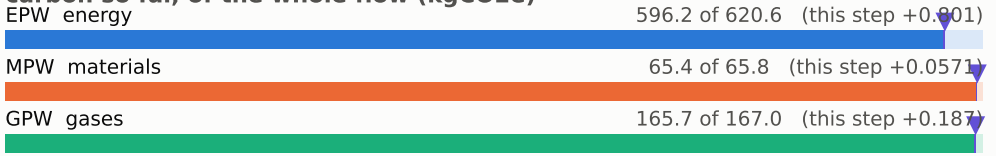
3D structure after this step

zoomed to fit

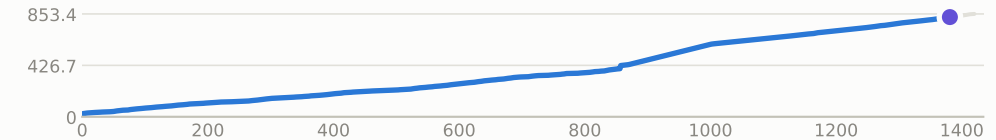
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

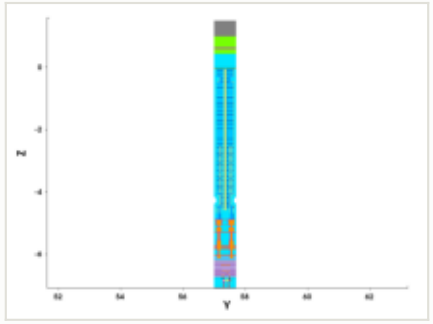
carbon so far, of the whole flow (kgCO2e)



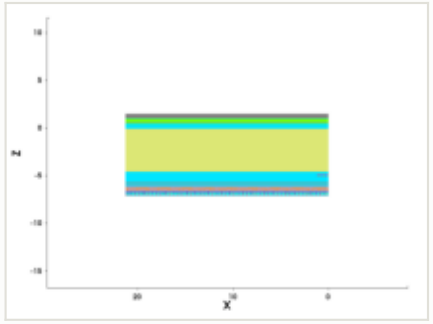
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



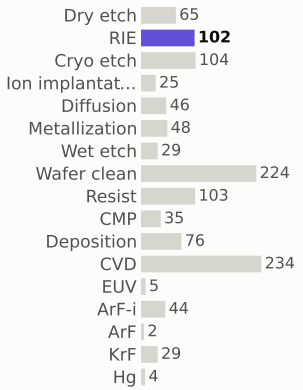
y cut at 57.351 µm



mask for this step

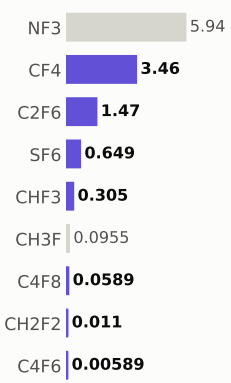
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



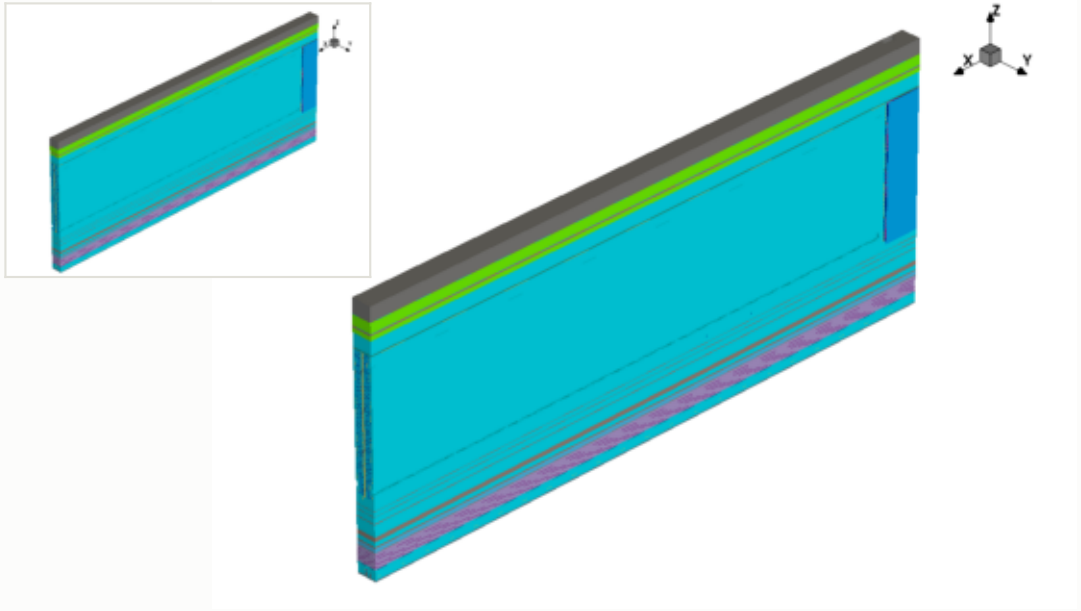
Step #13560: RIE

etch_anisotropic · RIE

1.04	828.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

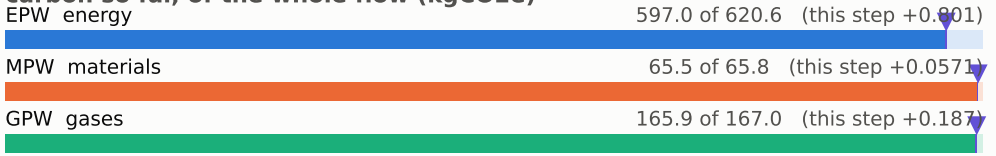


this step

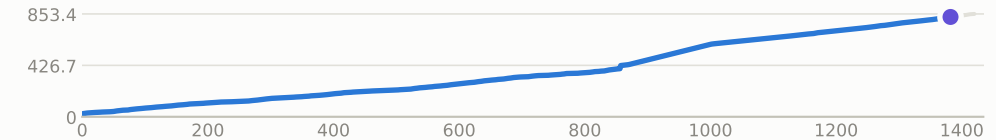
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

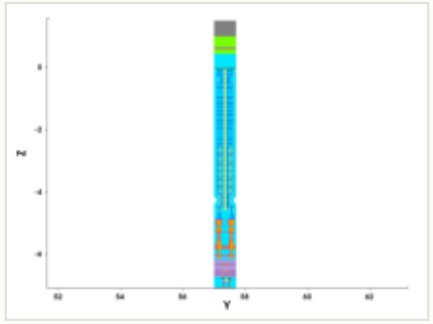
carbon so far, of the whole flow (kgCO2e)



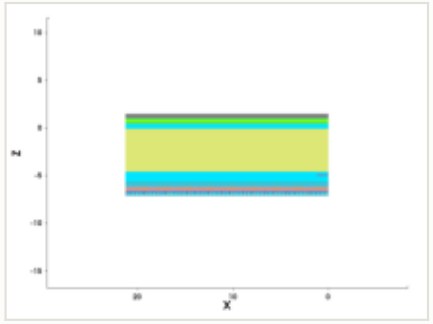
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



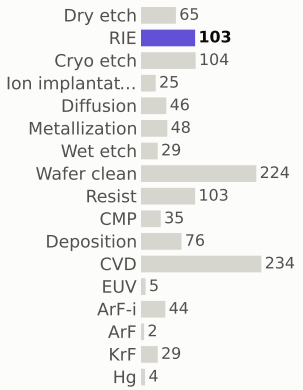
y cut at 57.351 µm



mask for this step

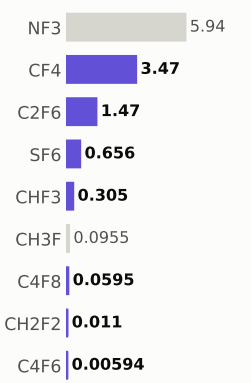
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



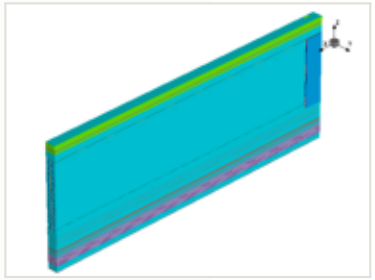
Step #13570: Dry_etch

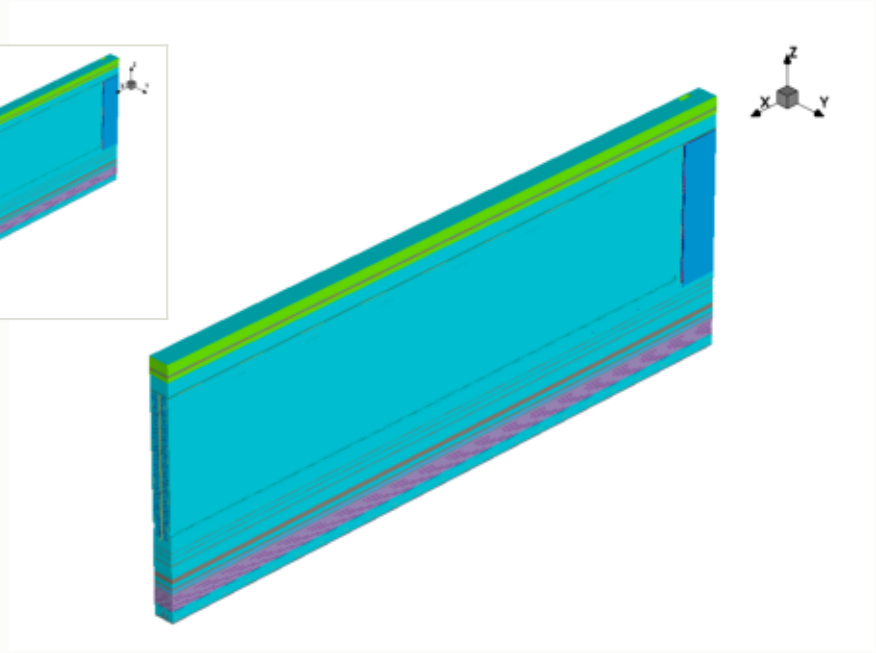
etch_strip · Dry_etch

0.801	829.2	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

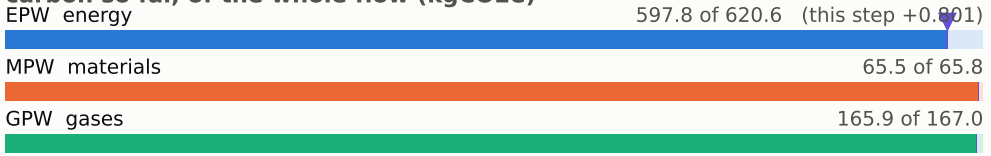




this step

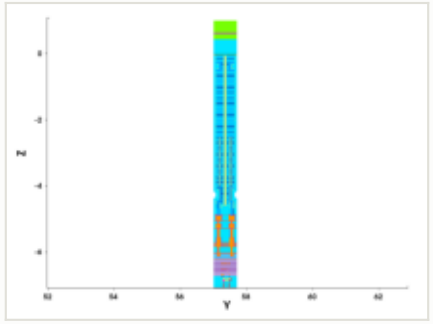
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

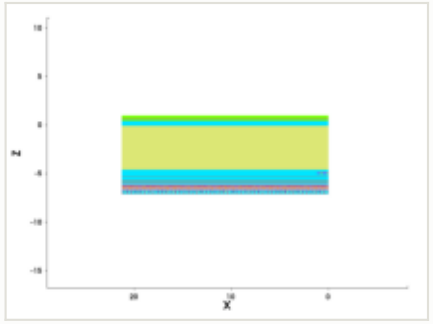


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



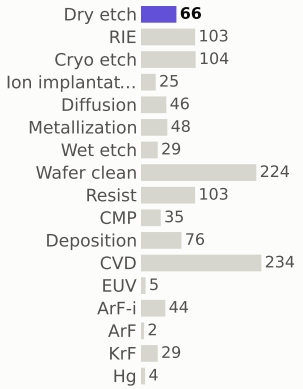
y cut at 57.351 µm



mask for this step

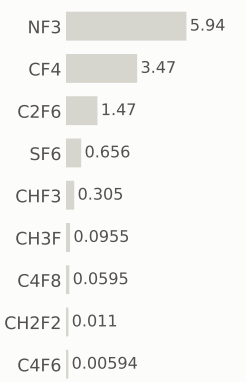
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #13580: Wafer_clean

clean · Wafer_clean

0.169	829.4	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

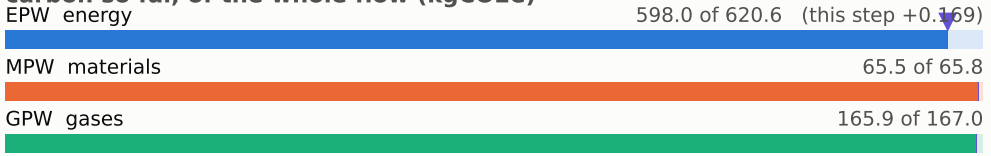
zoomed to fit

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

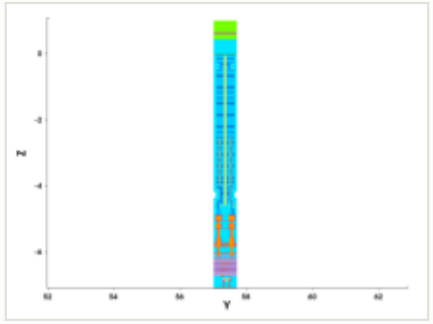
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

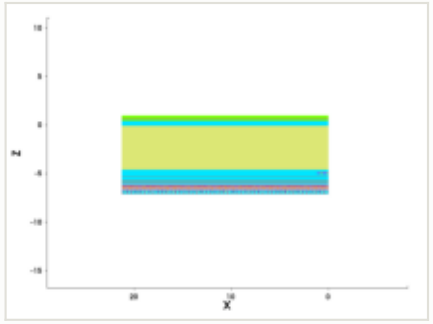


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



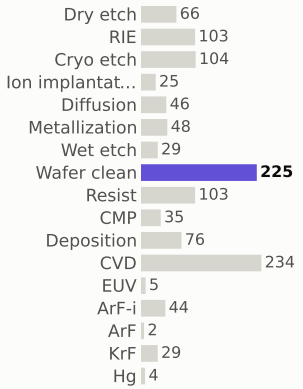
y cut at 57.351 µm



mask for this step

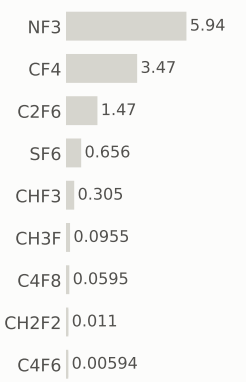
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #13590: Wafer_clean

clean · Wafer_clean

0.169	829.5	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

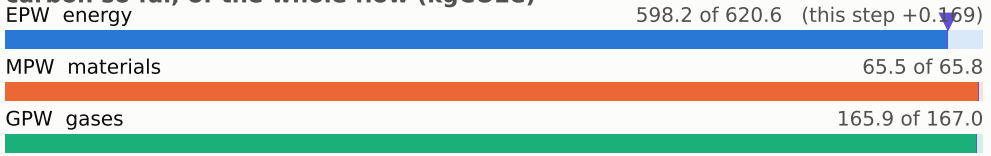
zoomed to fit

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

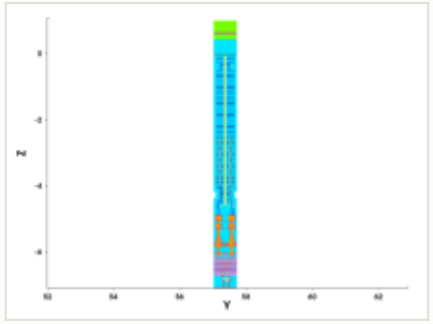
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

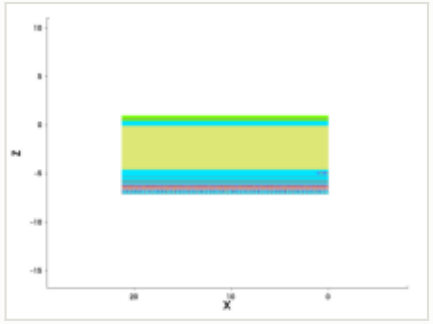


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



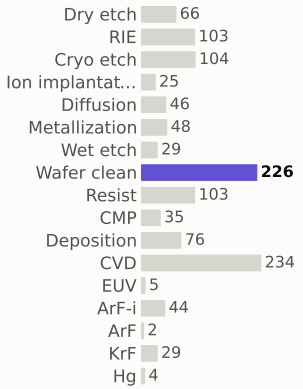
y cut at 57.351 µm



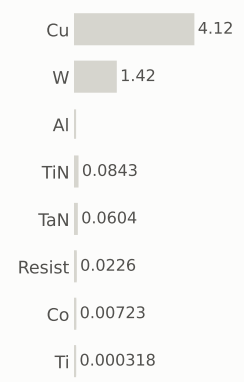
mask for this step

no mask for this step

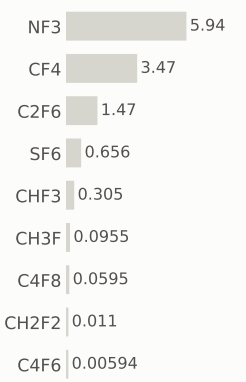
steps by process area



materials deposited (g)



gases emitted (g)



Step #13600: Resist

resist · Resist

0.169	829.7	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

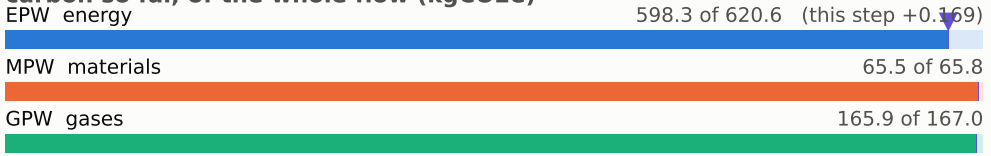
zoomed to fit

this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

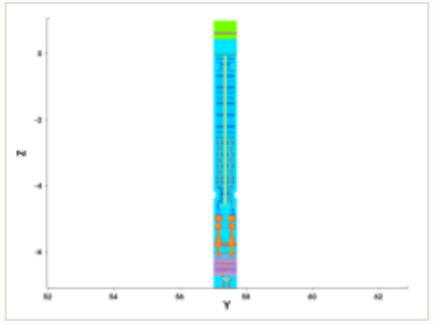
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

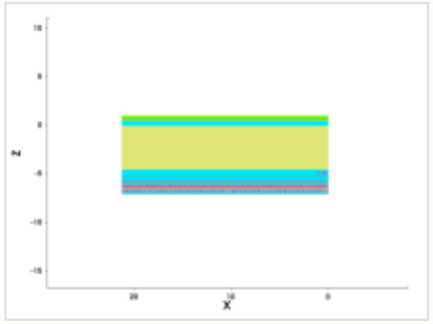


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



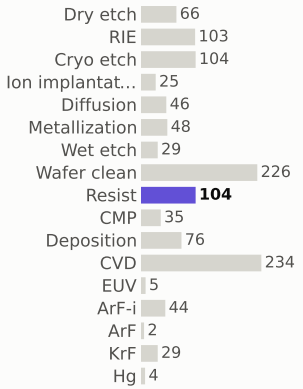
y cut at 57.351 µm



mask for this step

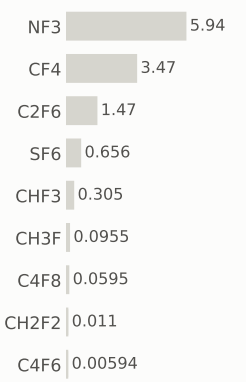
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



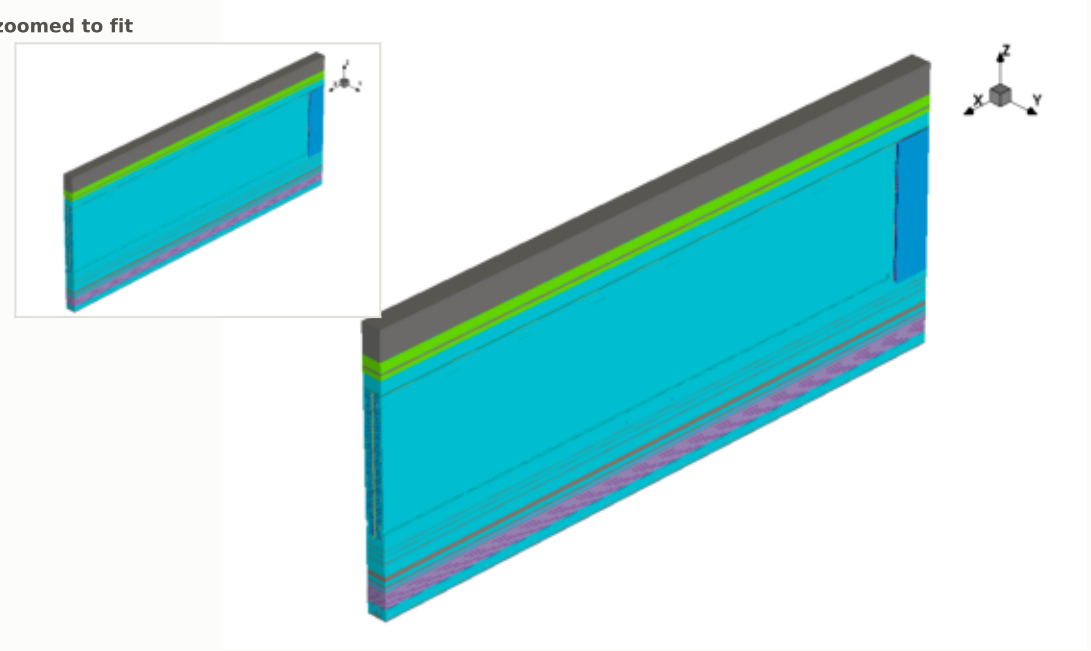
Step #13610: KrF

litho_patterning · KrF

0.199	829.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

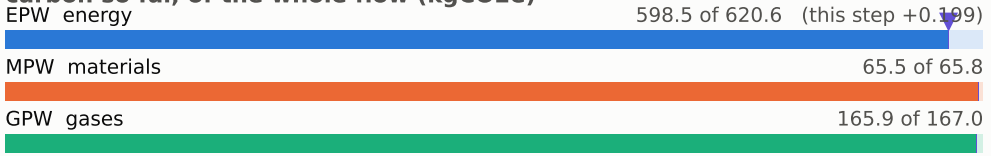
zoomed to fit



this step

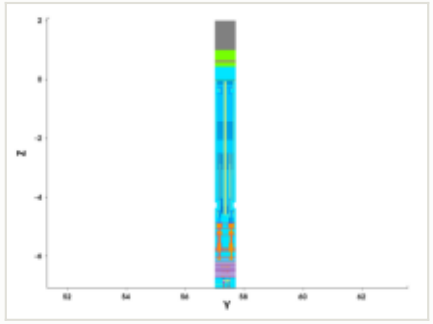
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

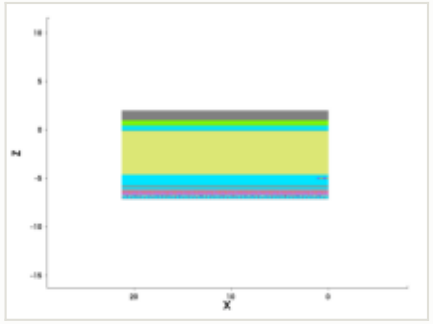


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm

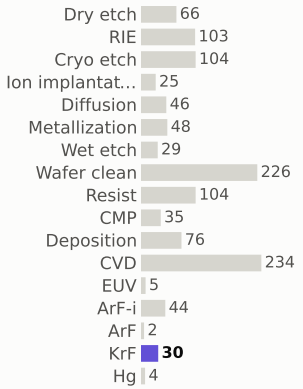


y cut at 57.351 µm

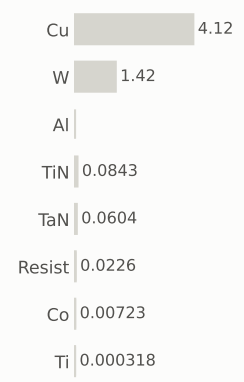


mask: layer 1440:0, positive (GDSII)

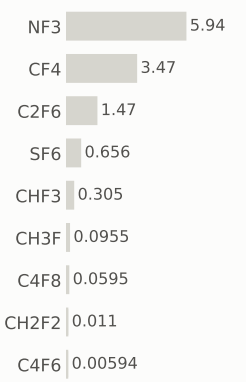
steps by process area



materials deposited (g)



gases emitted (g)



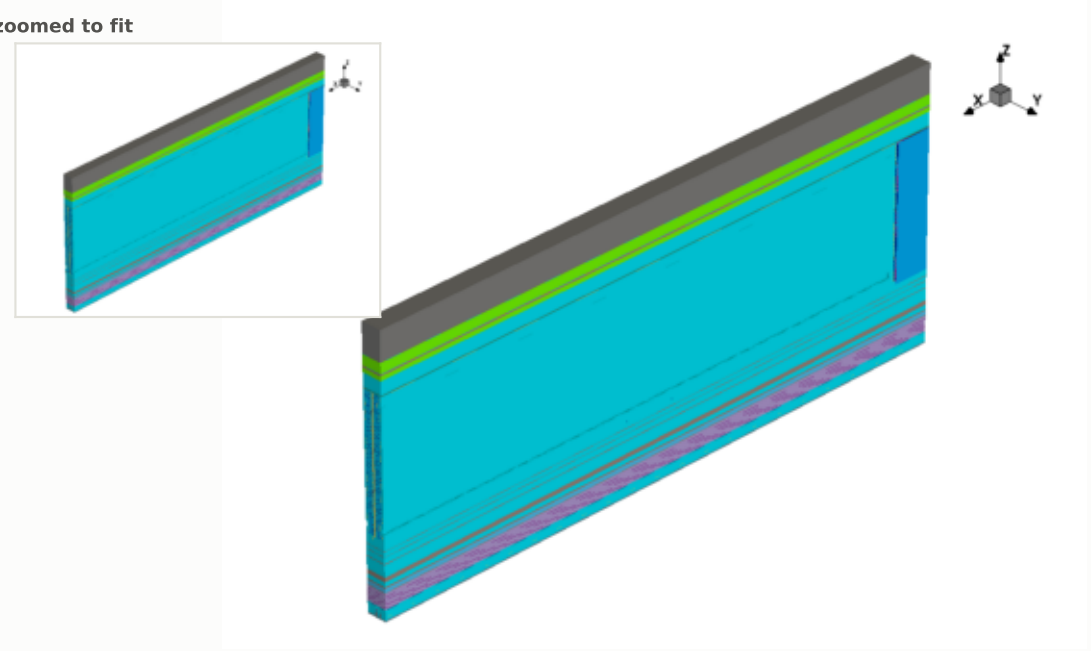
Step #13620: RIE

etch_anisotropic · RIE

1.04	831.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

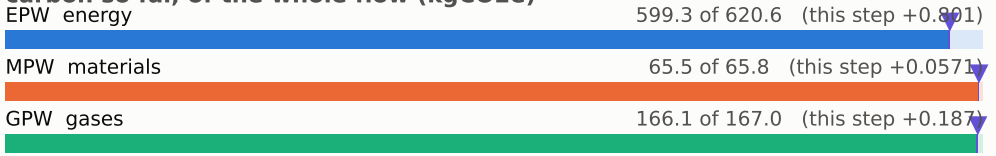
zoomed to fit



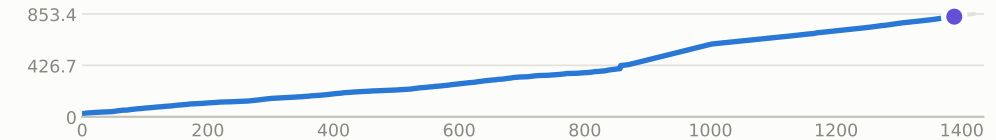
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

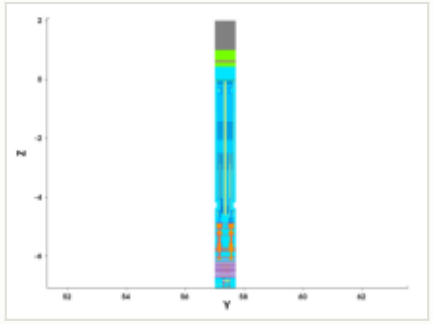
carbon so far, of the whole flow (kgCO2e)



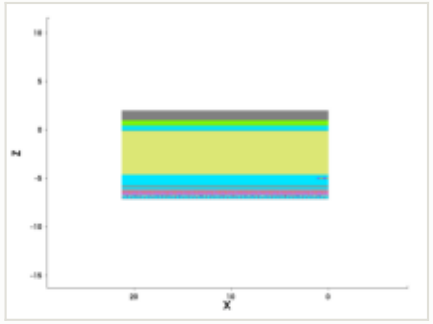
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



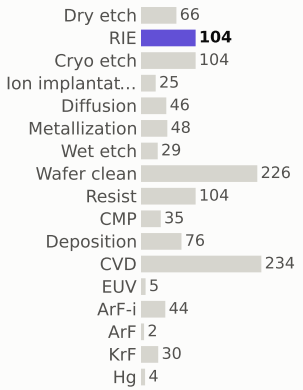
y cut at 57.351 µm



mask for this step

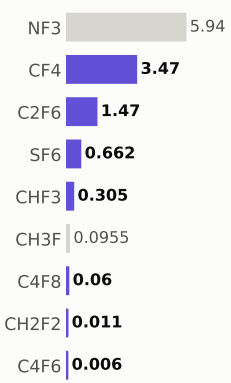
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



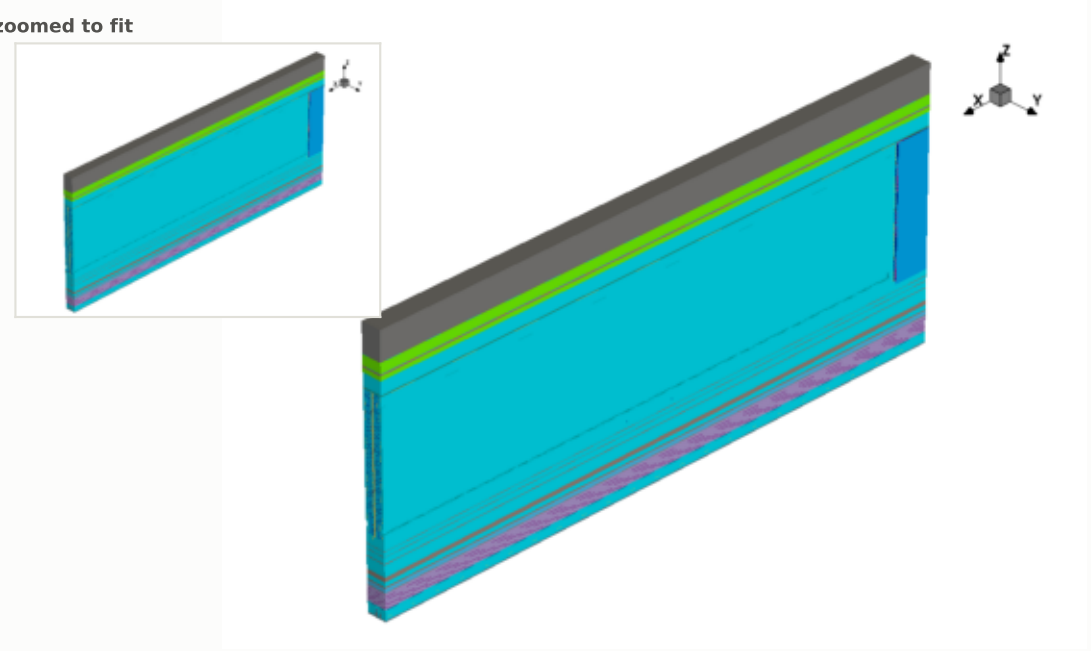
Step #13630: RIE

etch_anisotropic · RIE

1.04	832.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

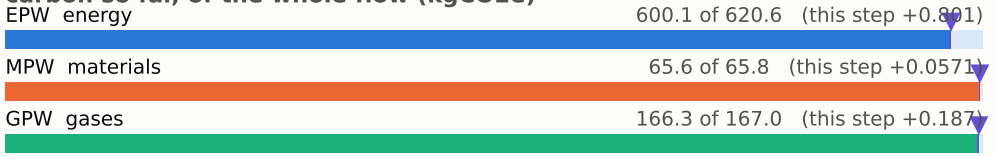
zoomed to fit



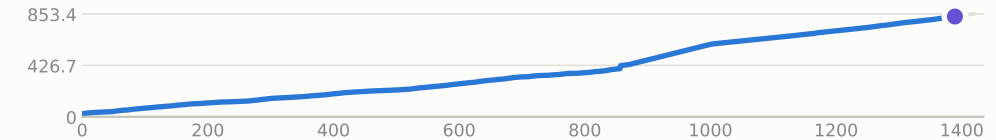
this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

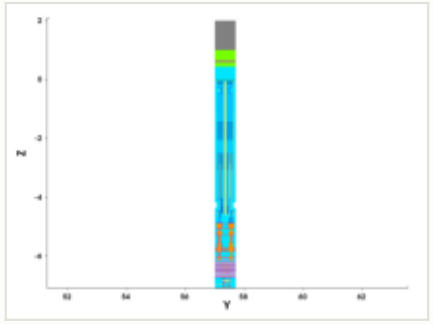
carbon so far, of the whole flow (kgCO2e)



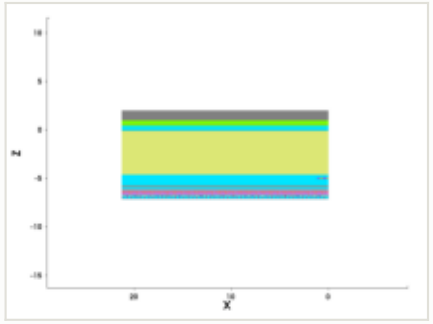
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



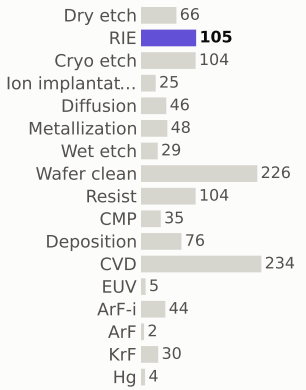
y cut at 57.351 µm



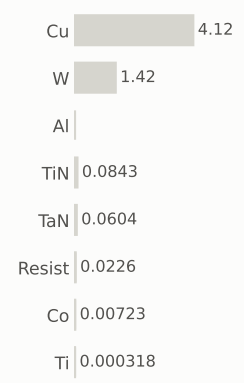
mask for this step

no mask for this step

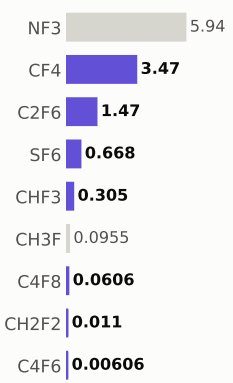
steps by process area



materials deposited (g)



gases emitted (g)



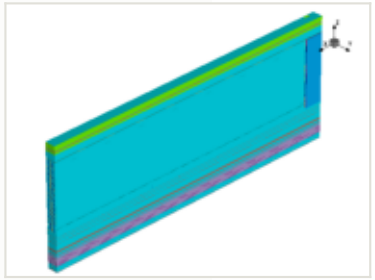
Step #13640: Dry_etch

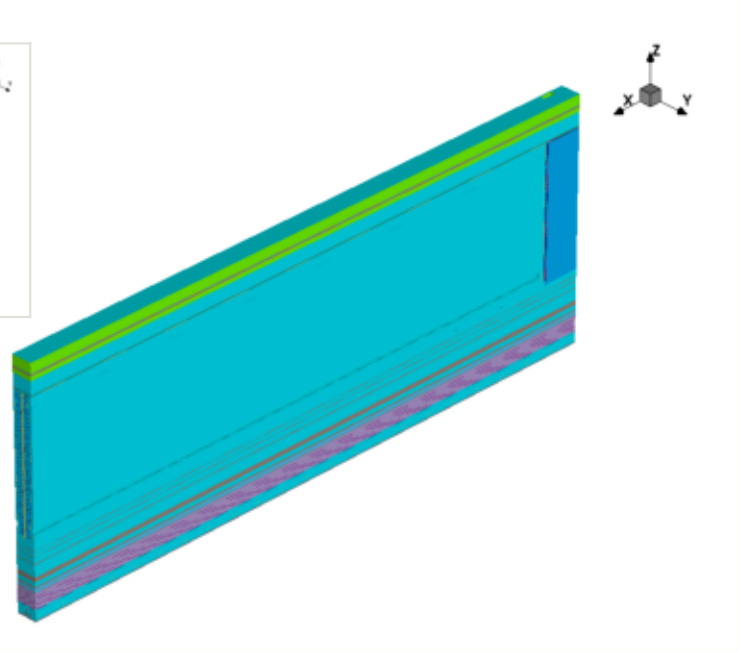
etch_strip · Dry_etch

0.801	832.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

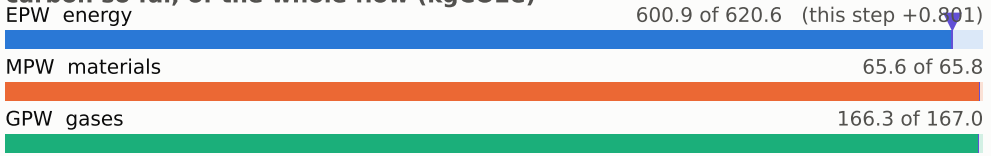




this step

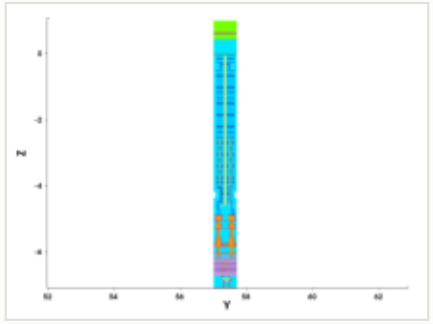
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

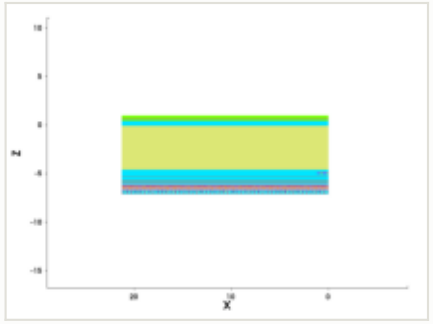


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



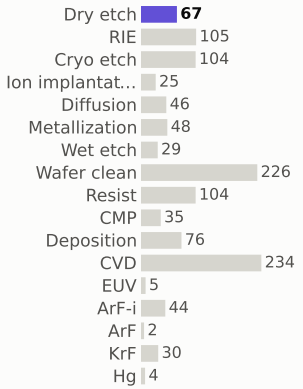
y cut at 57.351 µm



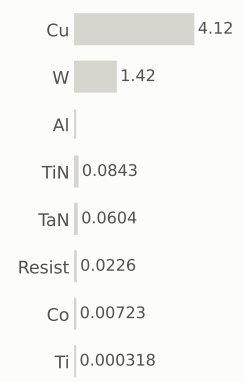
mask for this step

no mask for this step

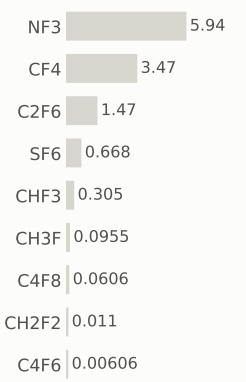
steps by process area



materials deposited (g)



gases emitted (g)



Step #13650: Wafer_clean

clean · Wafer_clean

0.169	833.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

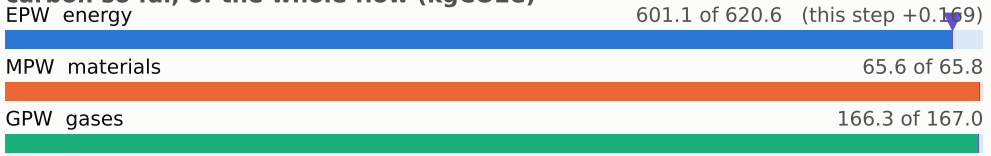
zoomed to fit

this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

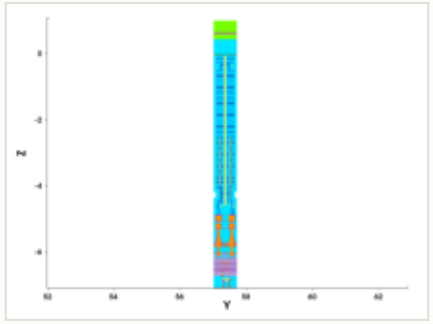
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

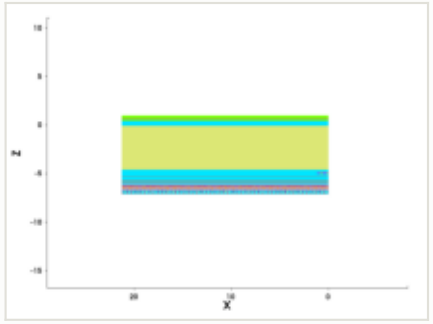


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



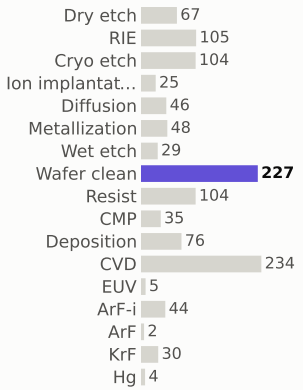
y cut at 57.351 µm



mask for this step

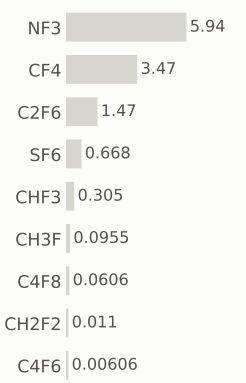
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #13660: Deposition

depo_isotropic · Deposition

0.657	833.6	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

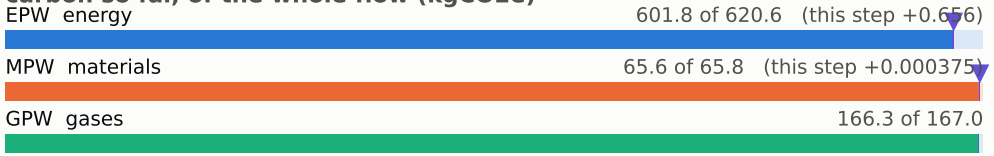
3D structure after this step

zoomed to fit

this step

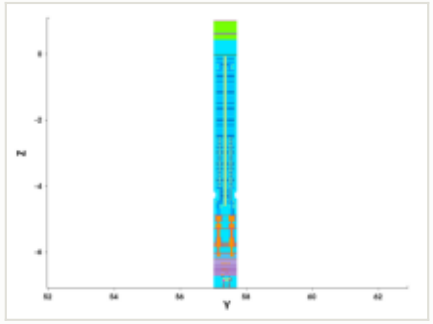
process area	Deposition	EPW	0.656 kgCO2e
material	TaN	EPW tool	0.469 kgCO2e
thickness	0.015 µm	EPW overhead	0.187 kgCO2e
material GWP	25.8159 gCO2e/g	MPW	0.000375 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.657 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

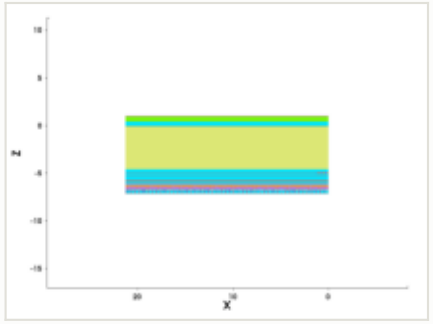


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



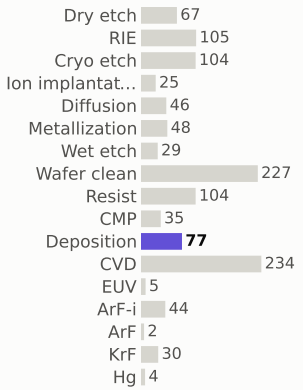
y cut at 57.351 µm



mask for this step

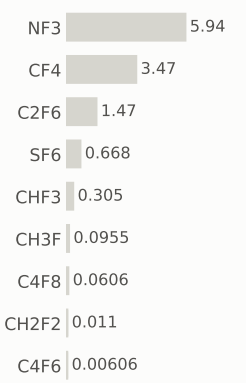
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



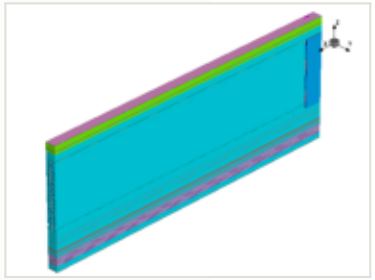
Step #13670: Metallization

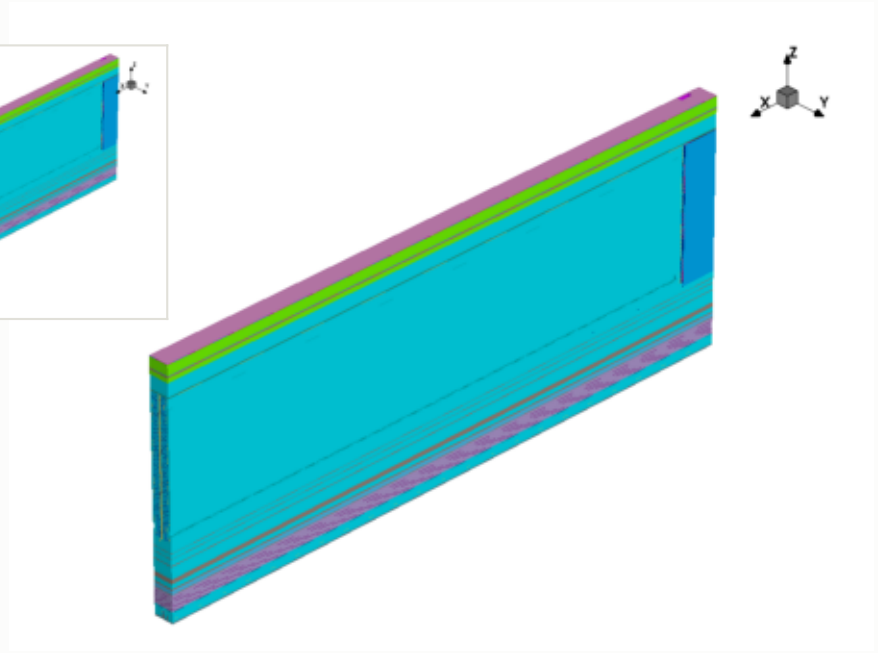
depo_isotropic · Metallization

1.2	834.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



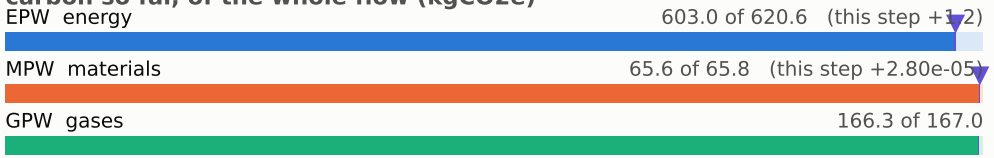


this step

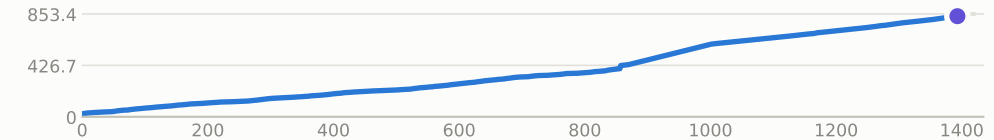
process area	Metallization
material	Co
thickness	0.001 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	2.80e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

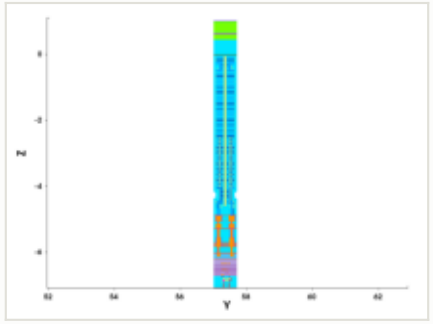
carbon so far, of the whole flow (kgCO2e)



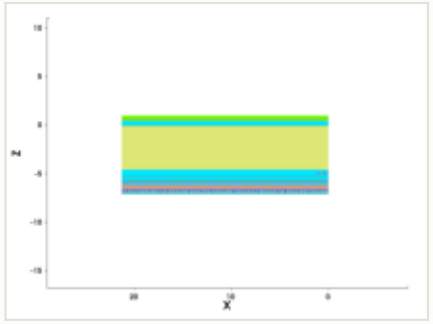
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



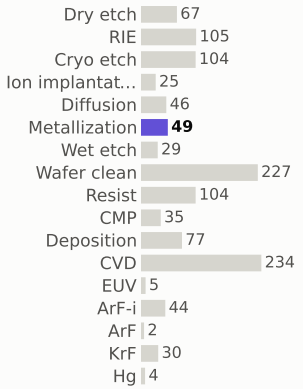
y cut at 57.351 µm



mask for this step

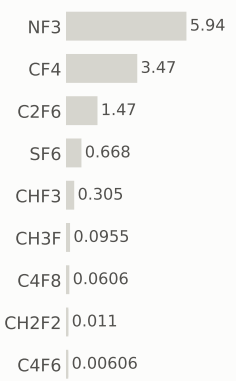
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



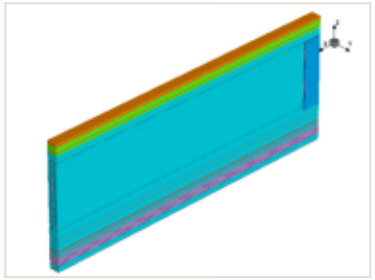
Step #13680: Metallization

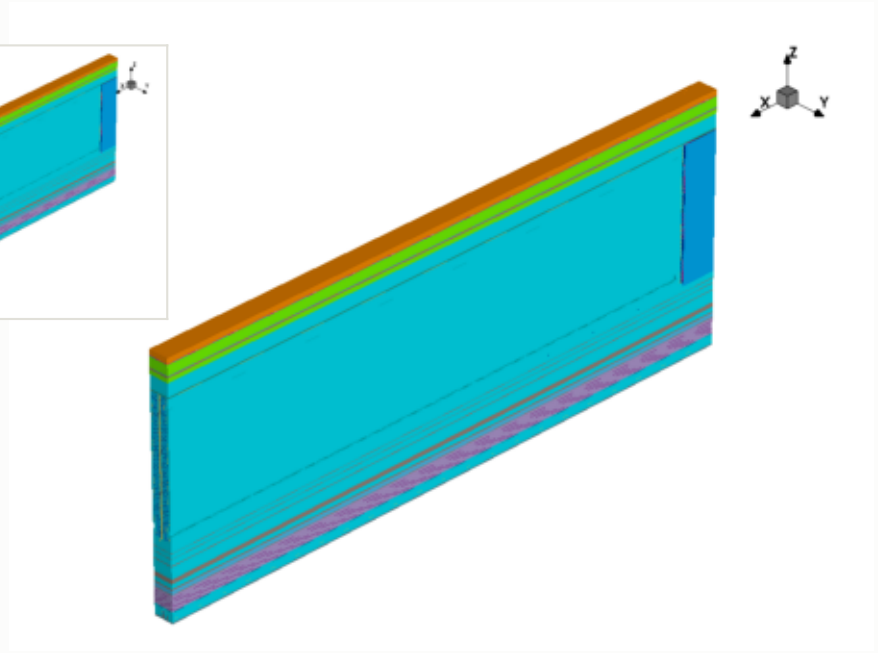
depo_isotropic · Metallization

1.2	836.0	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



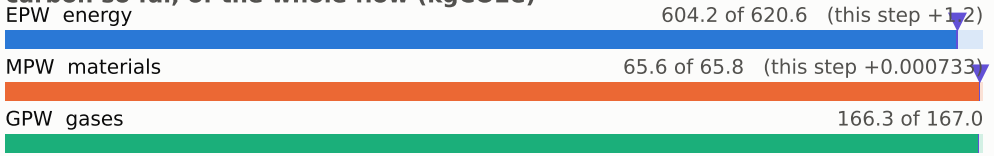


this step

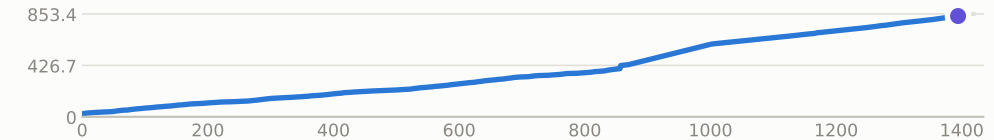
process area	Metallization
material	Cu
thickness	0.19 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000733 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

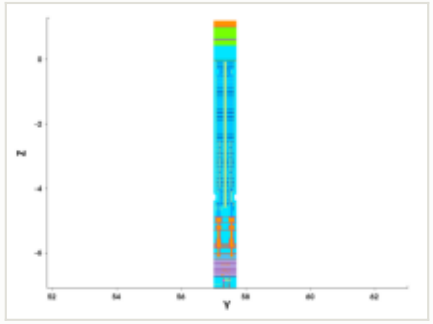
carbon so far, of the whole flow (kgCO2e)



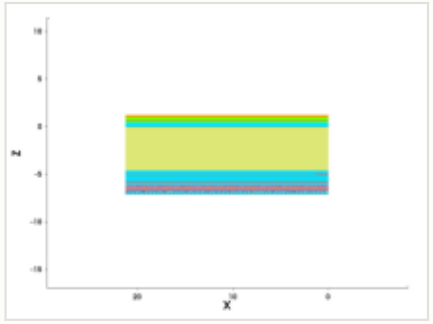
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



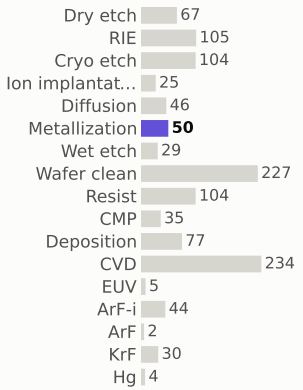
y cut at 57.351 µm



mask for this step

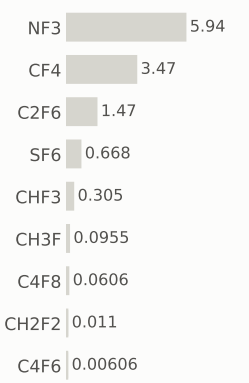
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



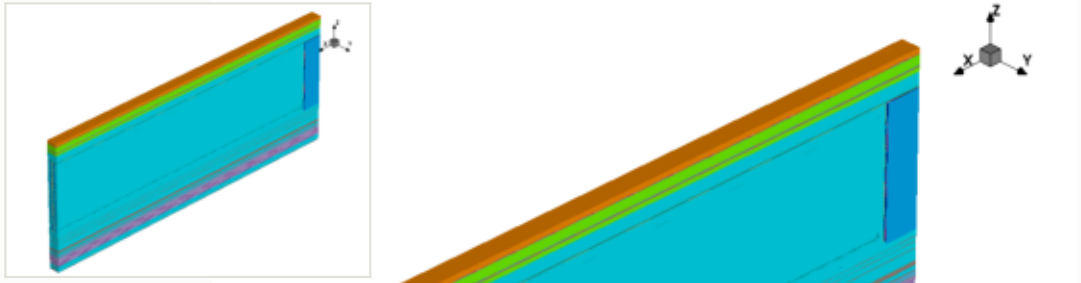
Step #13690: Diffusion

anneal · Diffusion

0.801	836.8	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

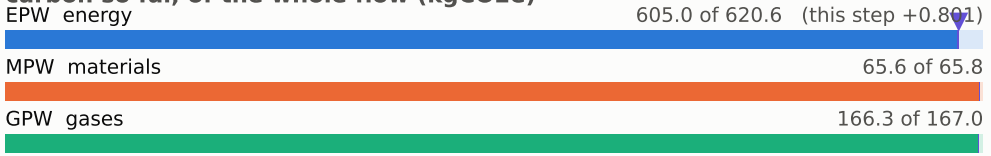


this step

process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

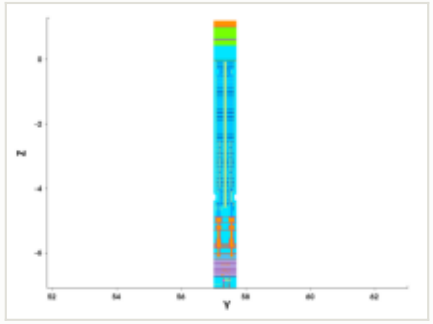
spx ran with 0 values for: TemperatureRamp

carbon so far, of the whole flow (kgCO2e)

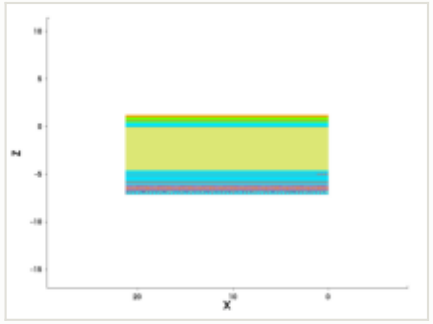


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



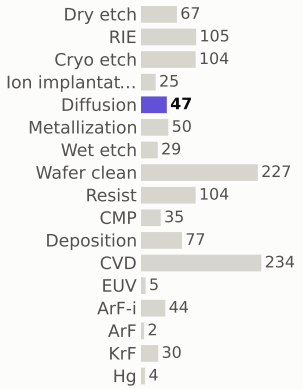
y cut at 57.351 µm



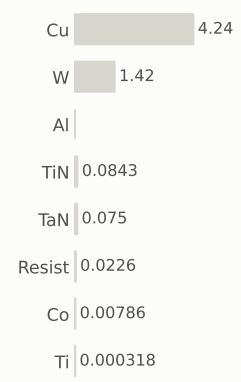
mask for this step

no mask for this step

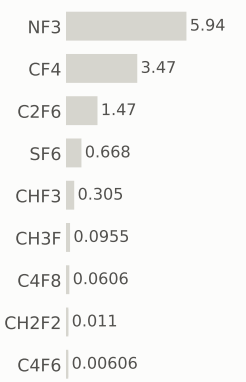
steps by process area



materials deposited (g)



gases emitted (g)



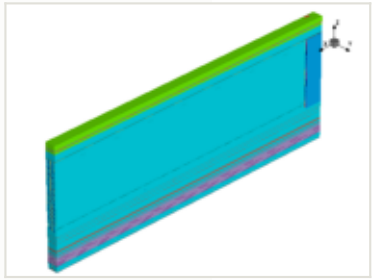
Step #13700: CMP

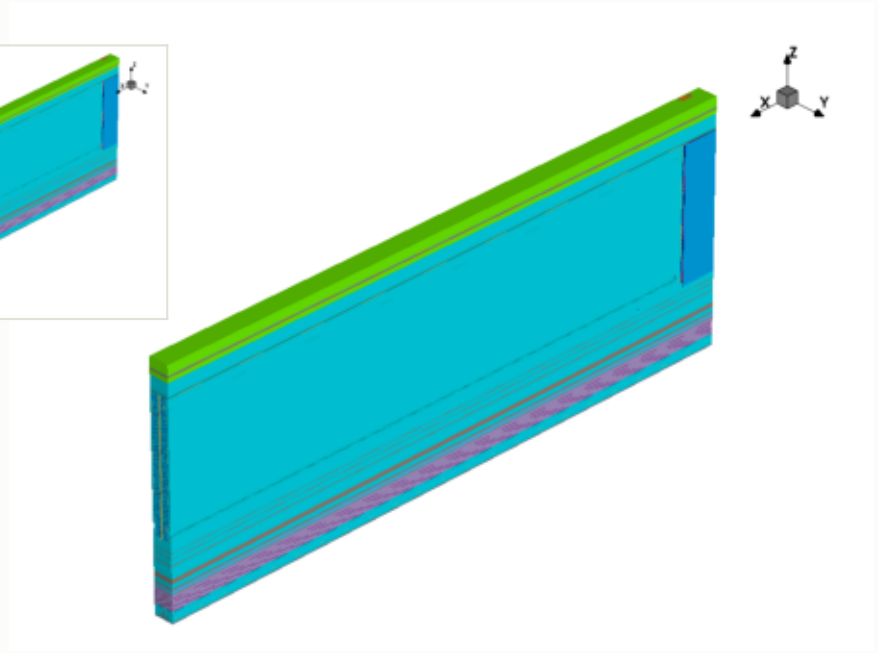
etch_cmp · CMP

0.169	837.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

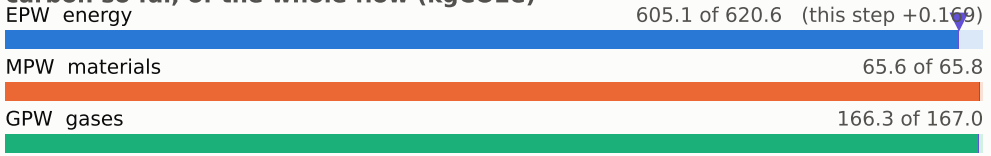




this step

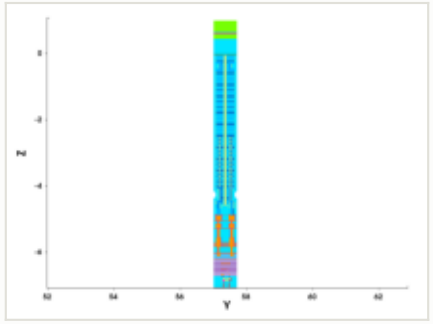
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

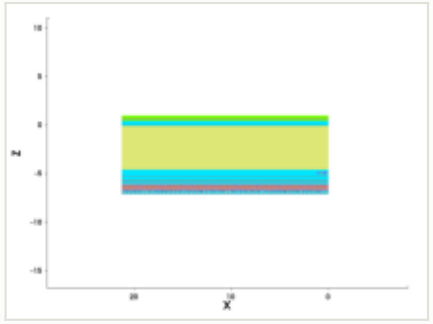


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



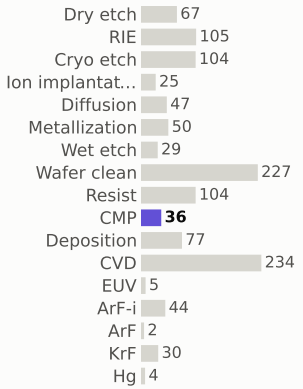
y cut at 57.351 µm



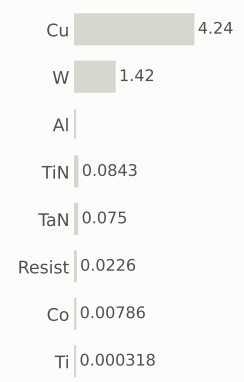
mask for this step

no mask for this step

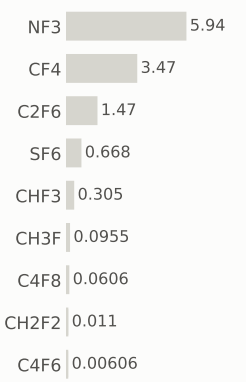
steps by process area



materials deposited (g)



gases emitted (g)



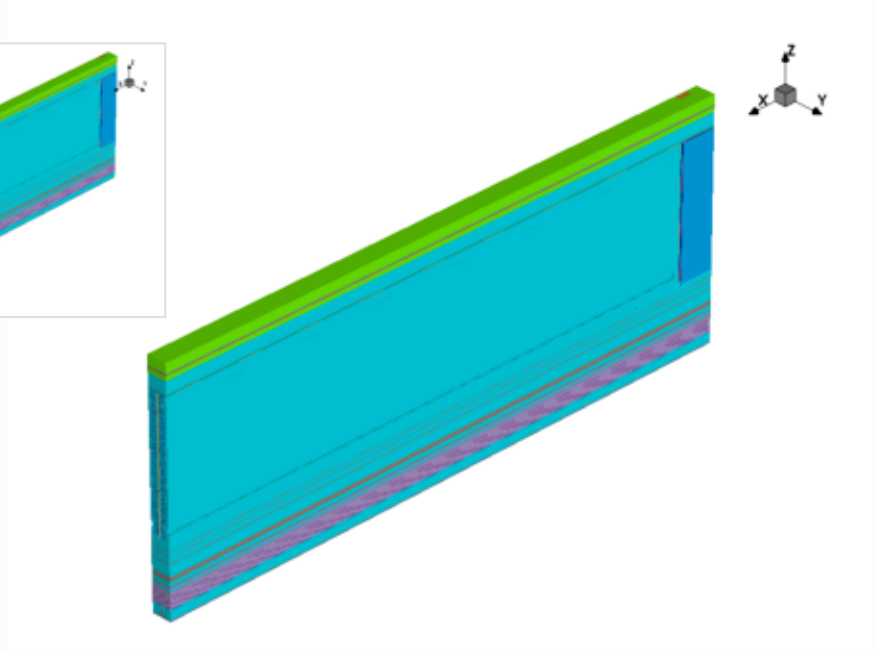
Step #13710: Wafer_clean

clean · Wafer_clean

0.169	837.2	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

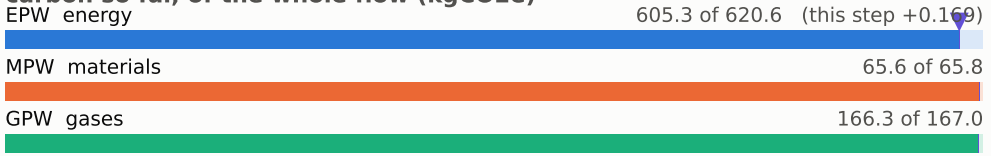


this step

process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

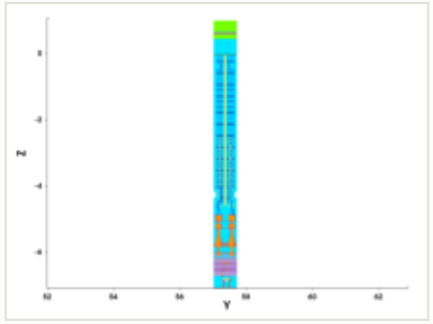
carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

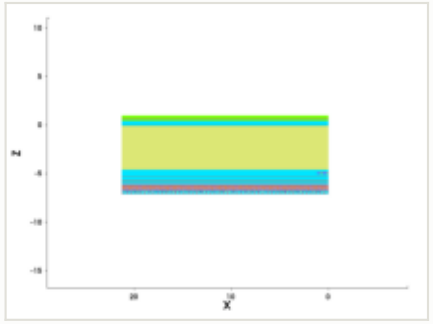


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



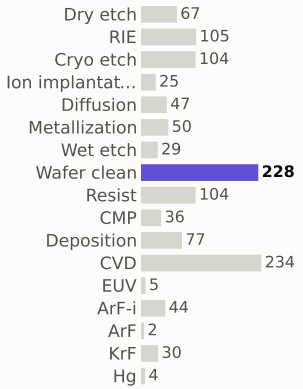
y cut at 57.351 µm



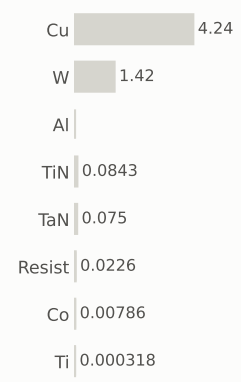
mask for this step

no mask for this step

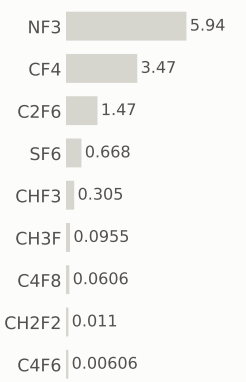
steps by process area



materials deposited (g)



gases emitted (g)



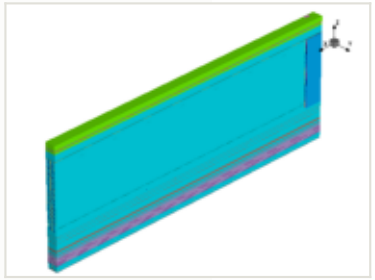
Step #13720: Metallization

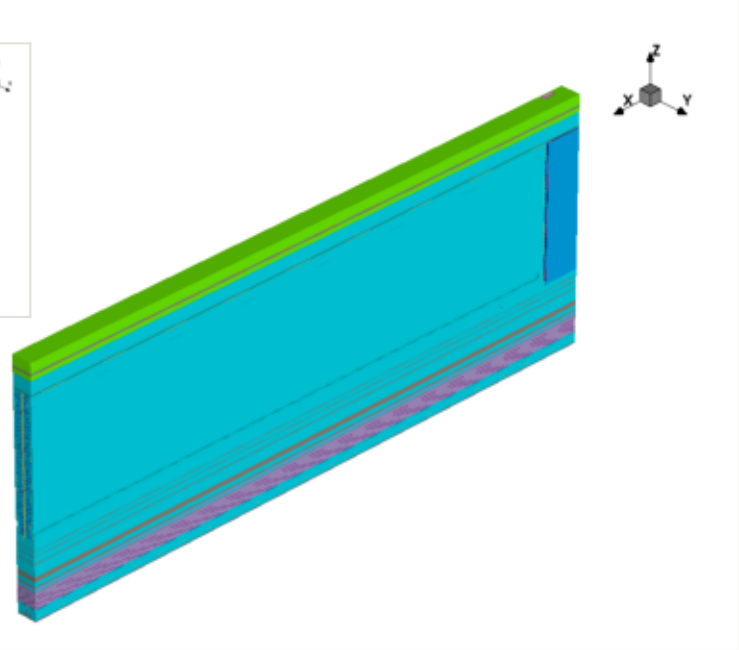
depo_isotropic · Metallization

1.2	838.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



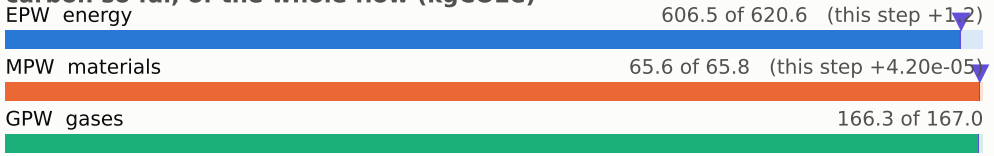


this step

process area	Metallization
material	Co
thickness	0.0015 µm
material GWP	44.5 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

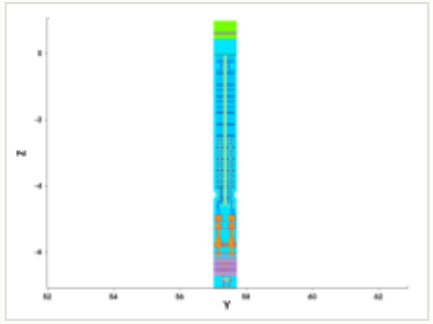
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.20e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

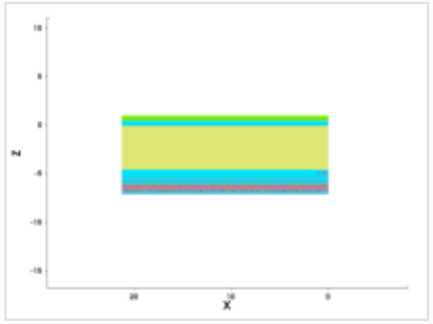


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



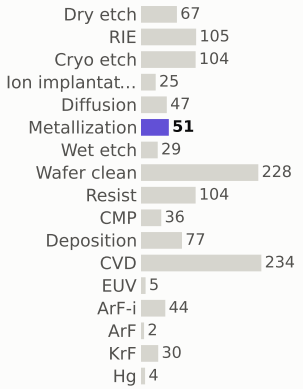
y cut at 57.351 µm



mask for this step

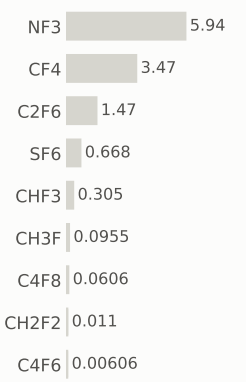
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #13740: Deposition

depo_isotropic · Deposition

0.656	839.0	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

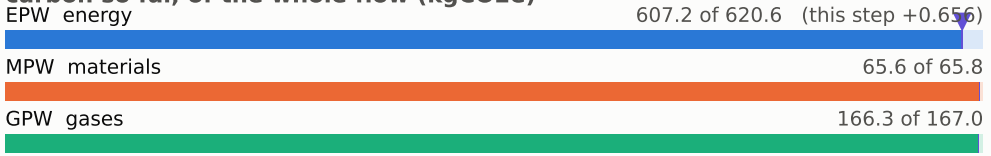
3D structure after this step

zoomed to fit

this step

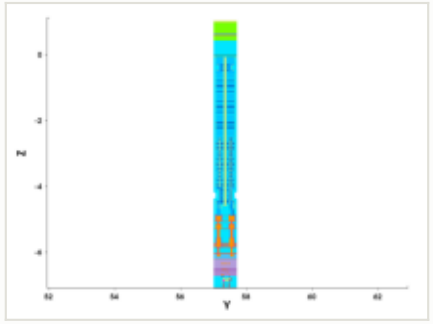
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.05 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

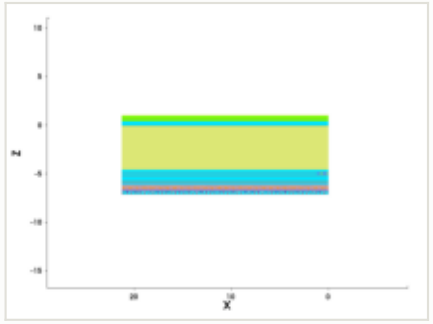


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



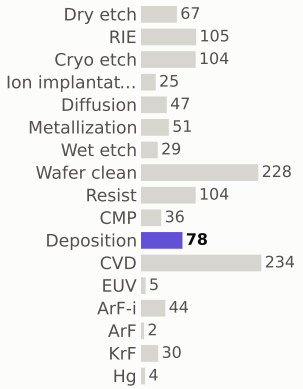
y cut at 57.351 µm



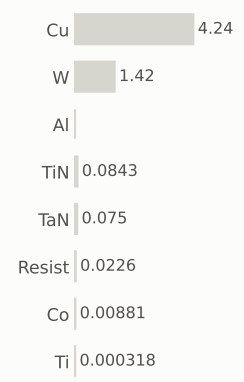
mask for this step

no mask for this step

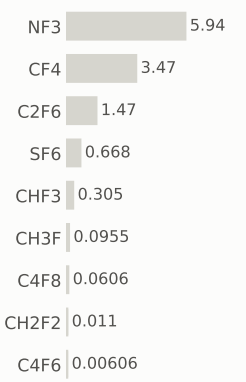
steps by process area



materials deposited (g)



gases emitted (g)



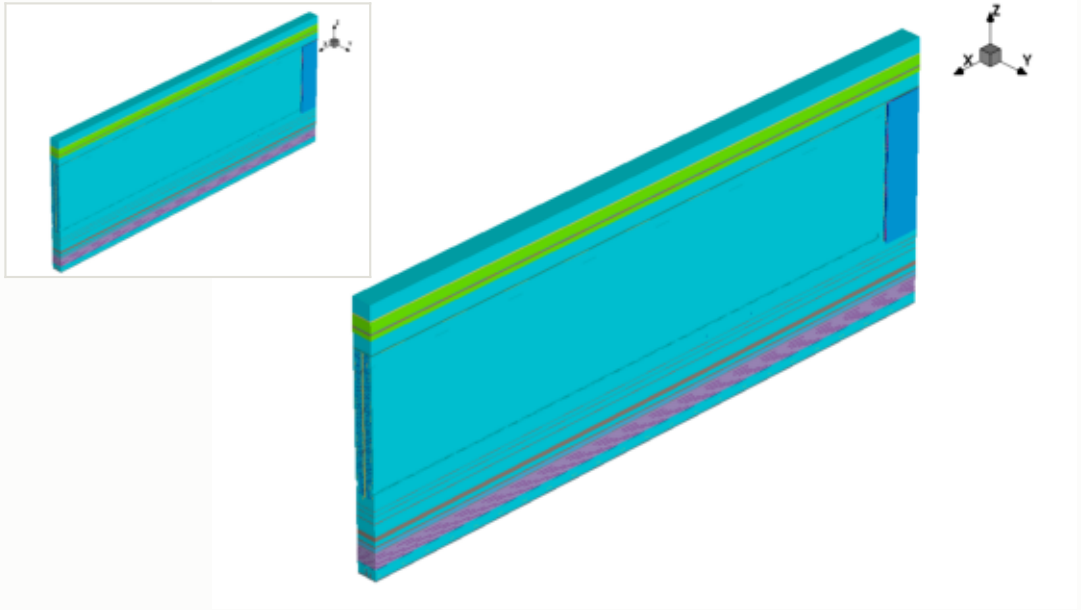
Step #13750: Deposition

depo_isotropic · Deposition

0.656	839.7	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

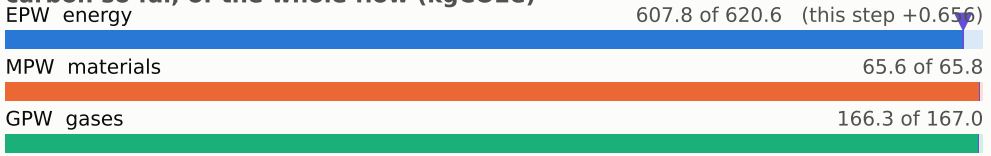
zoomed to fit



this step

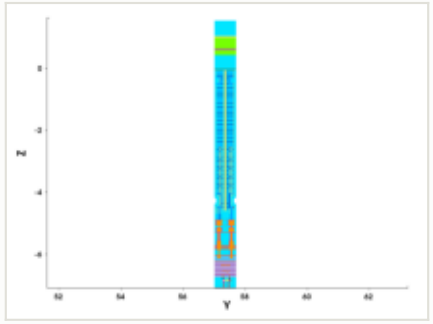
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.5 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

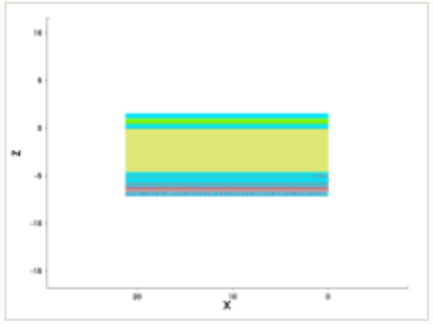


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



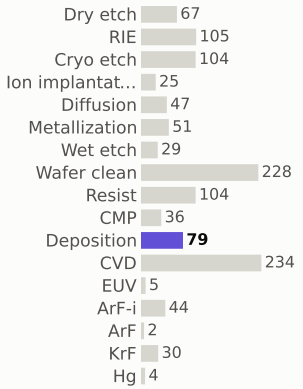
y cut at 57.351 µm



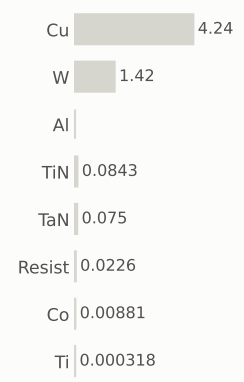
mask for this step

no mask for this step

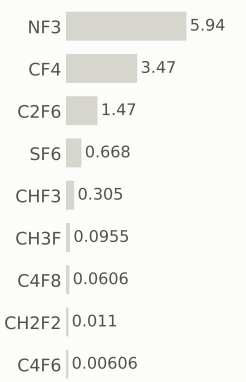
steps by process area



materials deposited (g)



gases emitted (g)



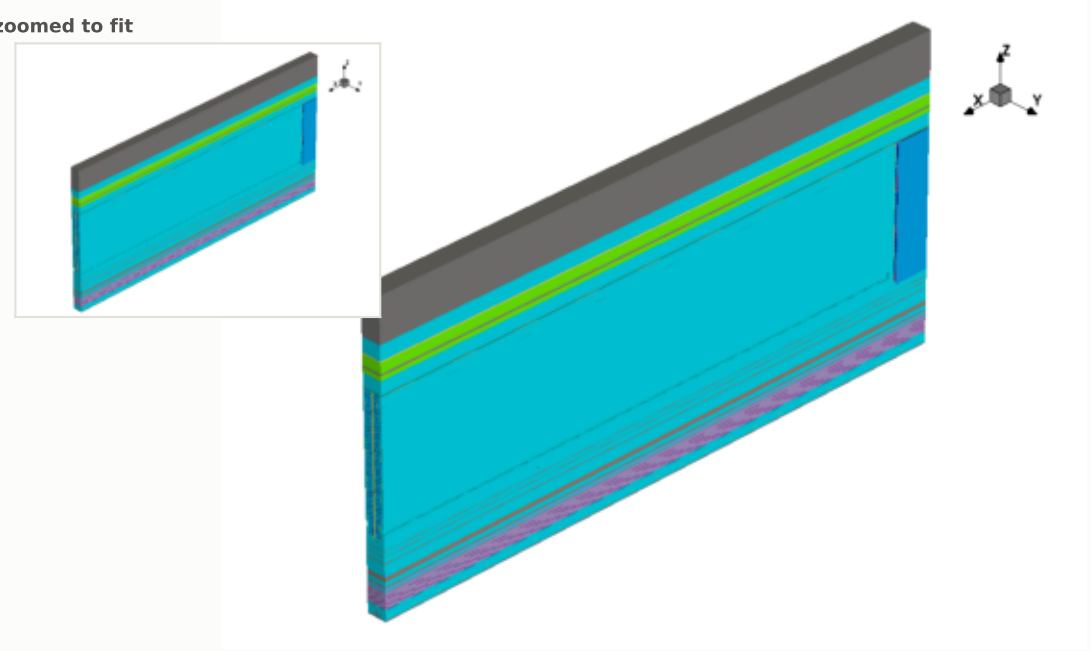
Step #13760: KrF

litho_patterning · KrF

0.199	839.9	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

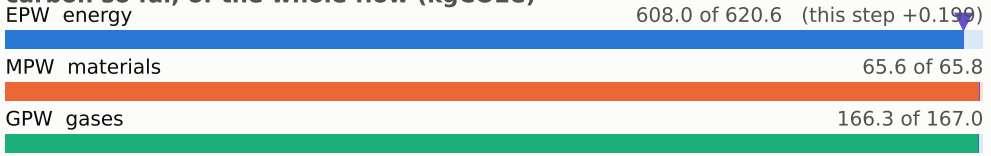
zoomed to fit



this step

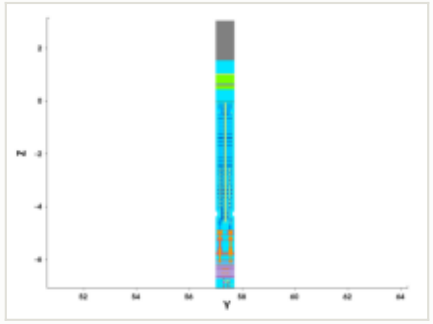
process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GPW	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

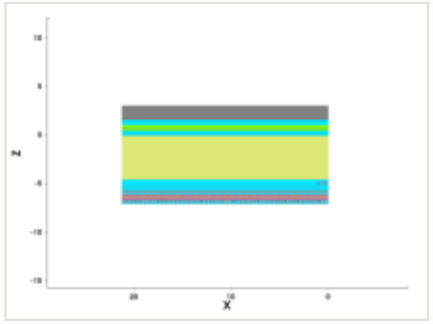


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



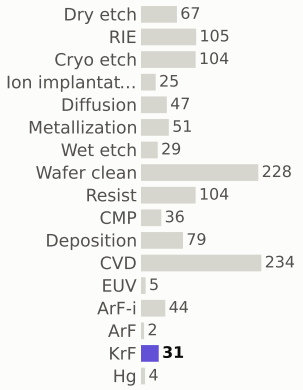
y cut at 57.351 µm



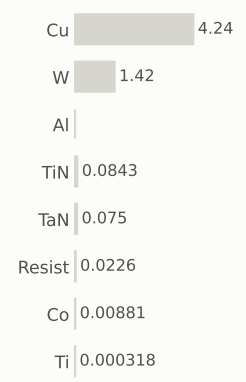
mask: layer 40:0, negative (GDSII)

nothing in the clip window

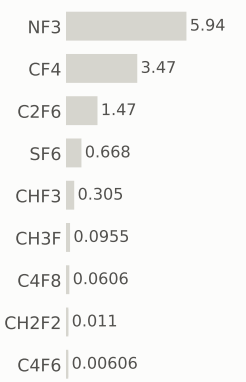
steps by process area



materials deposited (g)



gases emitted (g)



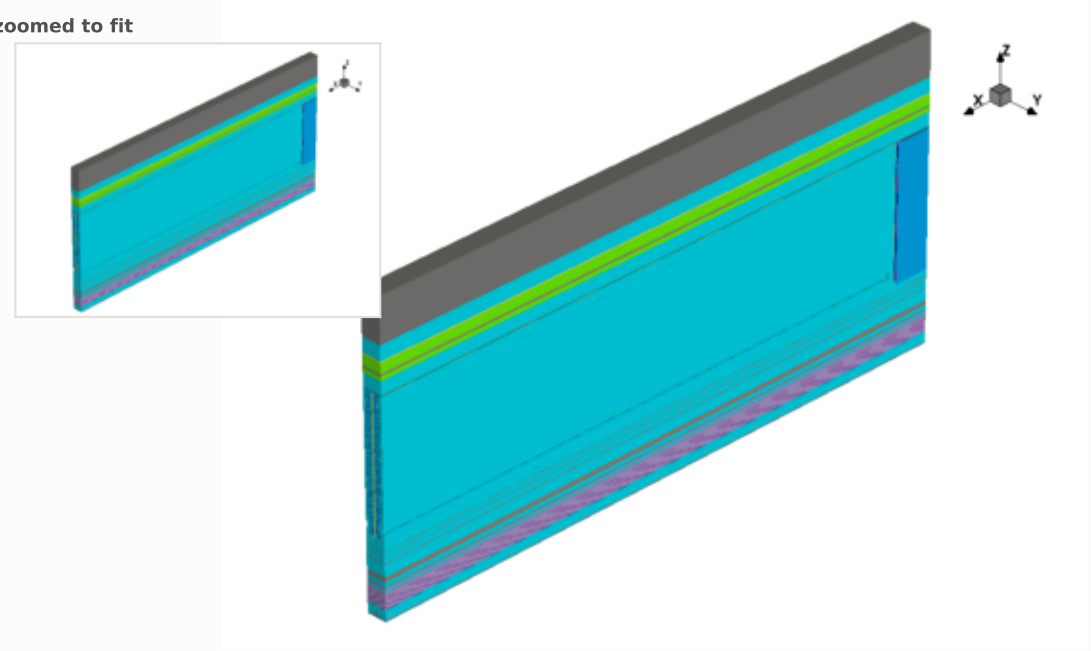
Step #13770: RIE

etch_anisotropic · RIE

1.04	840.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

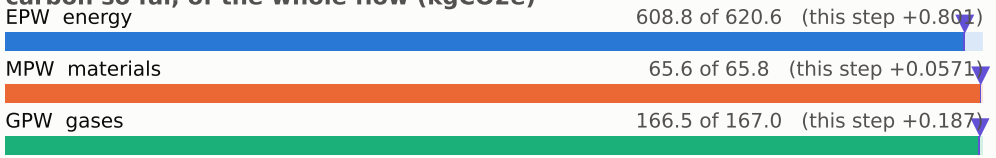


this step

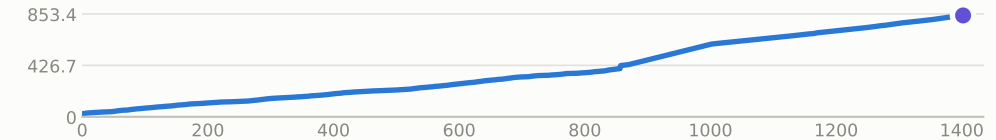
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

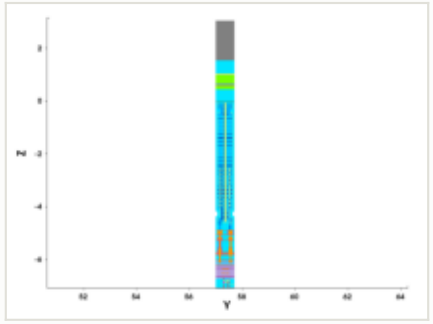
carbon so far, of the whole flow (kgCO2e)



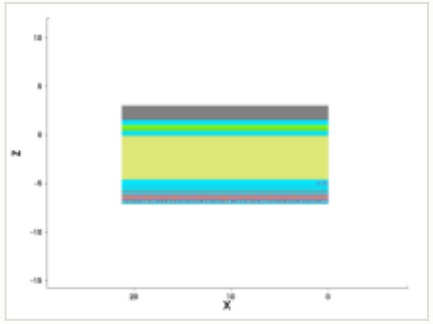
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



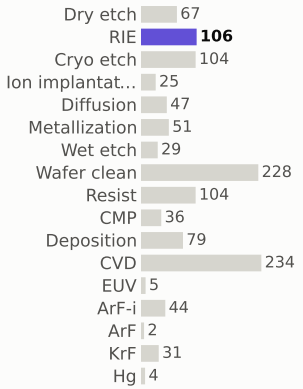
y cut at 57.351 µm



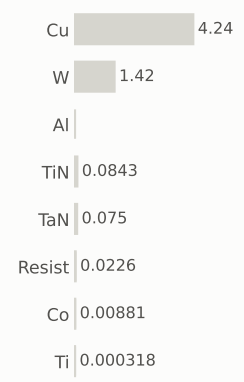
mask for this step

no mask for this step

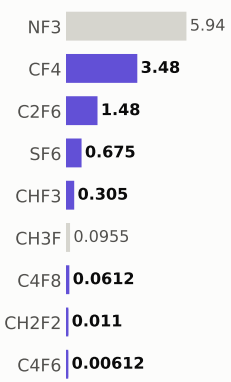
steps by process area



materials deposited (g)



gases emitted (g)



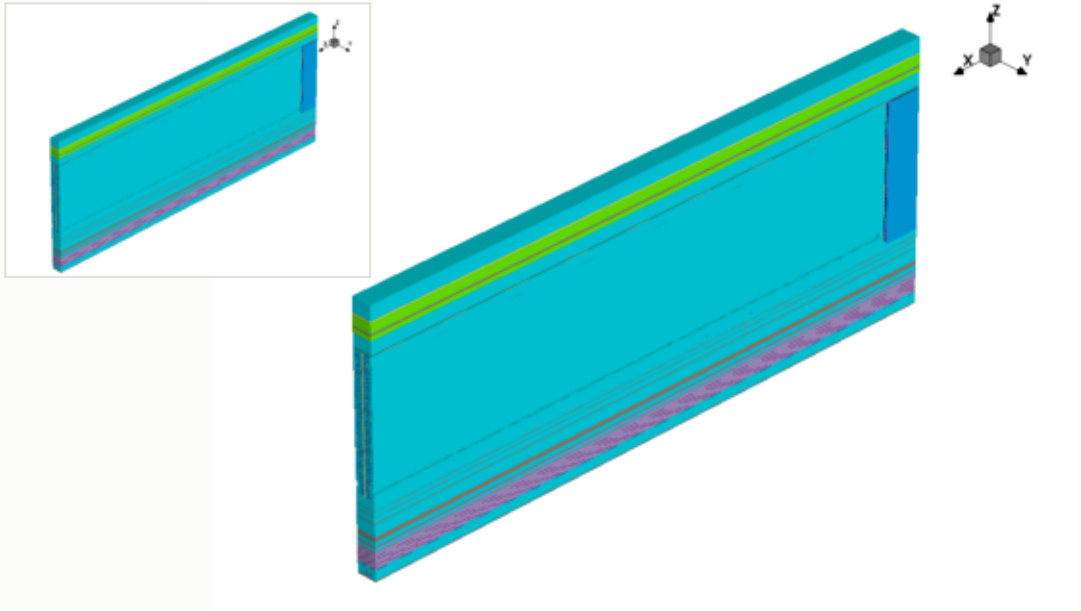
Step #13780: Dry_etch

etch_strip · Dry_etch

0.801	841.7	0.094%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

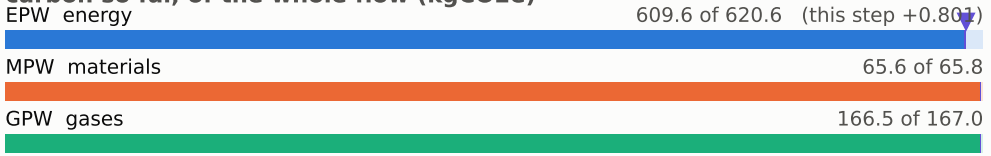
zoomed to fit



this step

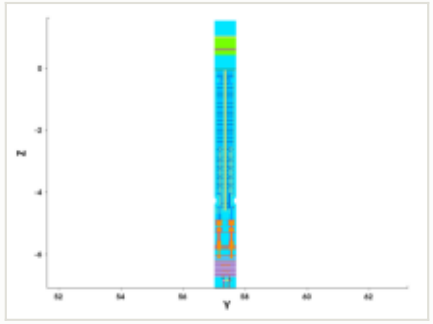
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

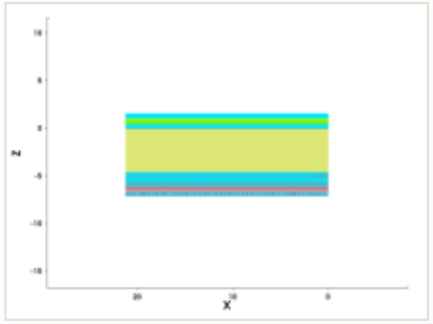


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



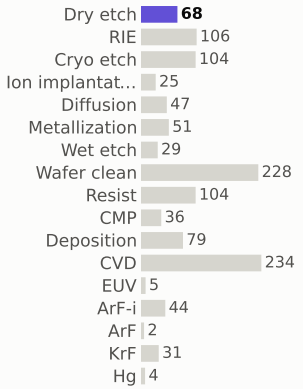
y cut at 57.351 µm



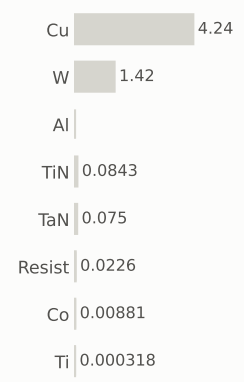
mask for this step

no mask for this step

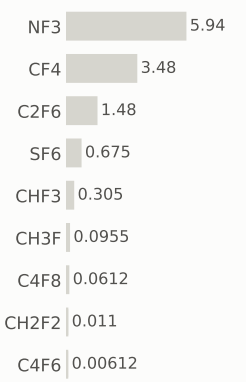
steps by process area



materials deposited (g)



gases emitted (g)



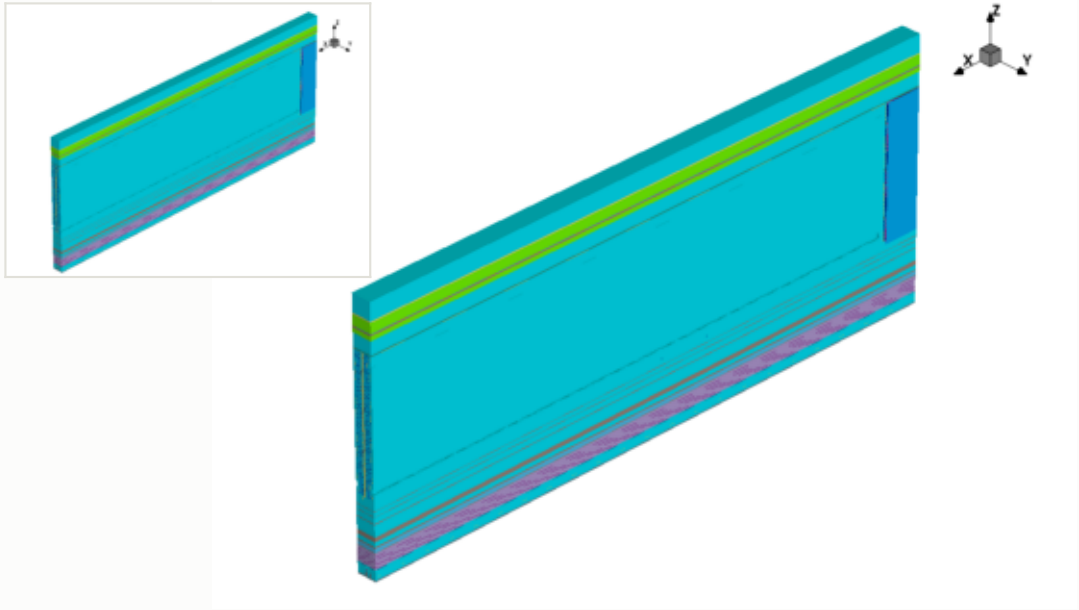
Step #13790: Deposition

depo_isotropic · Deposition

0.656	842.4	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

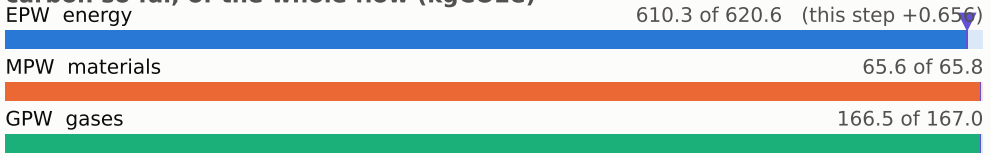
zoomed to fit



this step

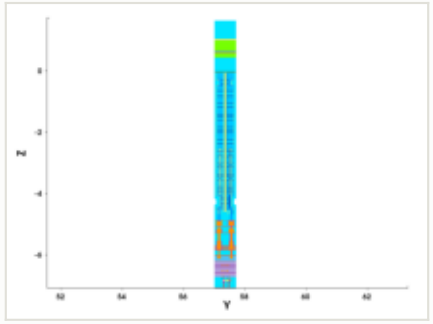
process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.1 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

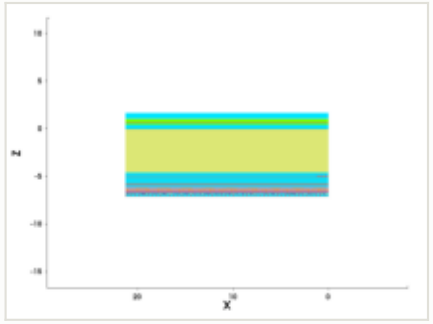


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



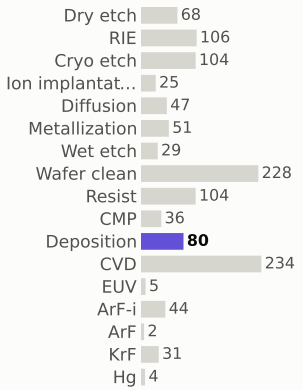
y cut at 57.351 µm



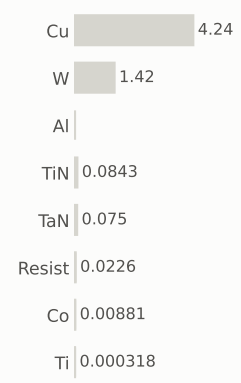
mask for this step

no mask for this step

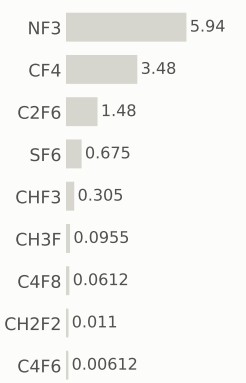
steps by process area



materials deposited (g)



gases emitted (g)



Step #13800: RIE

etch_anisotropic · RIE

1.04	843.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

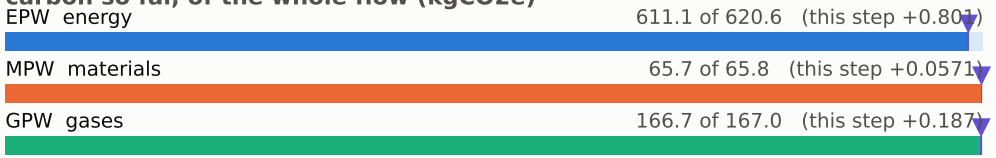
zoomed to fit

this step

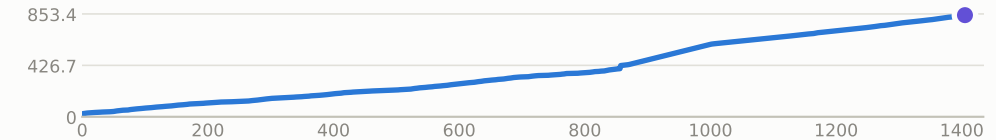
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

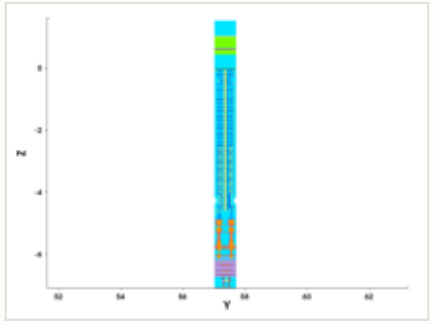
carbon so far, of the whole flow (kgCO2e)



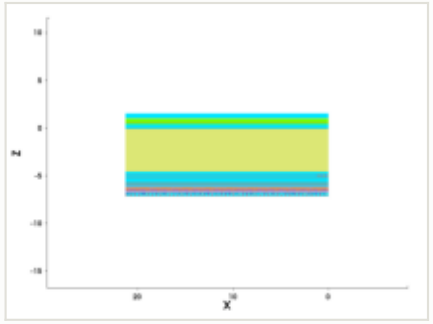
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



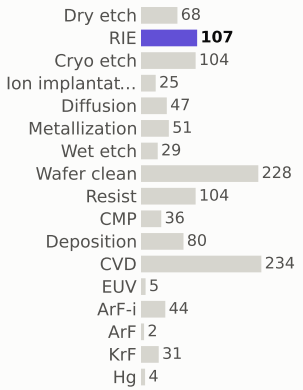
y cut at 57.351 µm



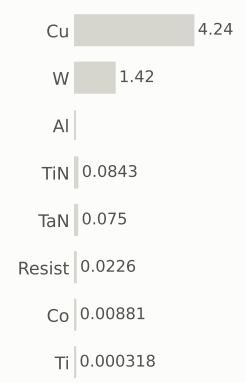
mask for this step

no mask for this step

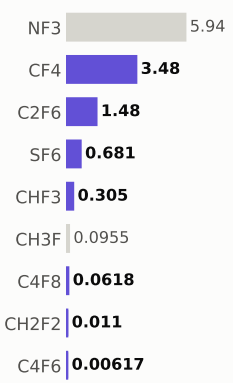
steps by process area



materials deposited (g)



gases emitted (g)



Step #13810: Metallization

depo_isotropic · Metallization

1.2	844.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

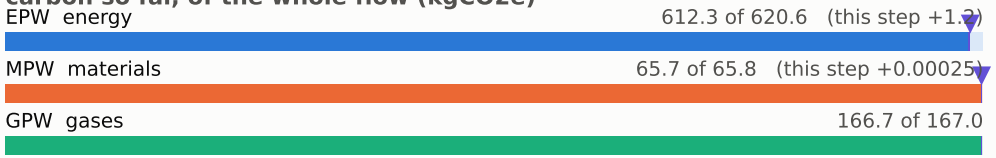
zoomed to fit

this step

process area	Metallization
material	TaN
thickness	0.01 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

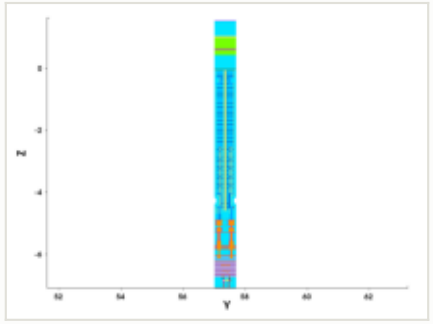
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00025 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

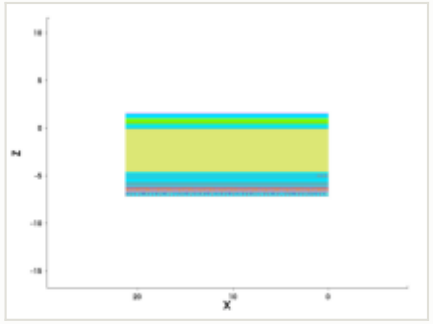


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



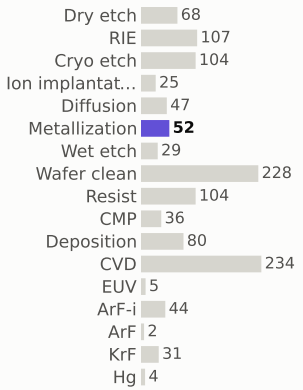
y cut at 57.351 µm



mask for this step

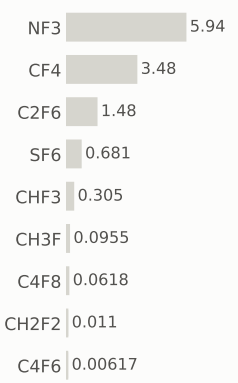
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



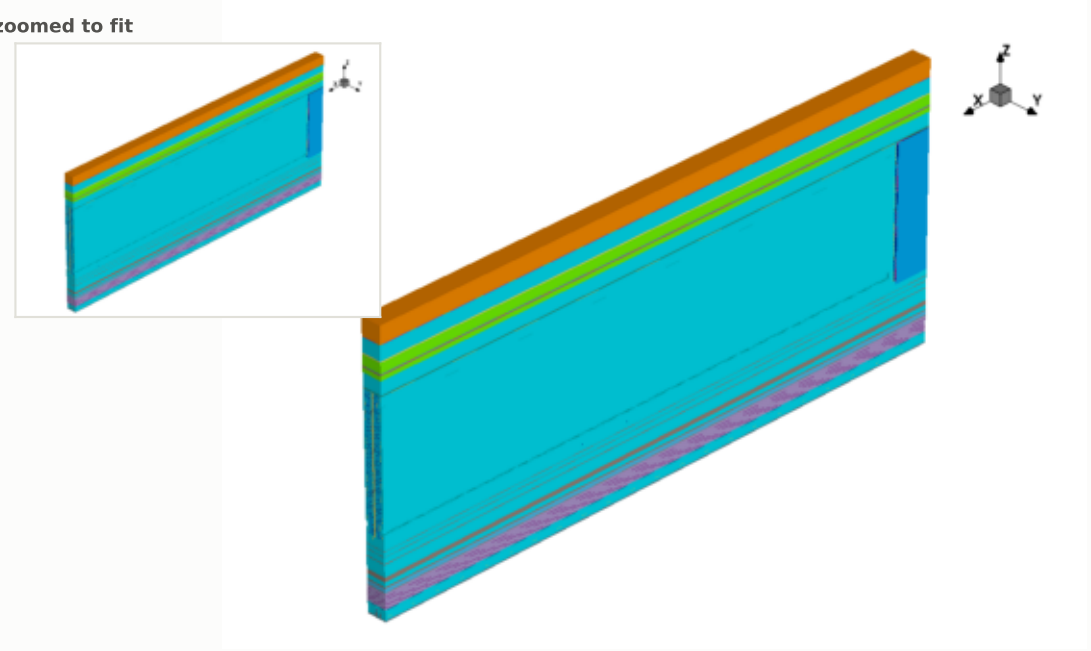
Step #13820: Metallization

depo_isotropic · Metallization

1.21	845.8	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

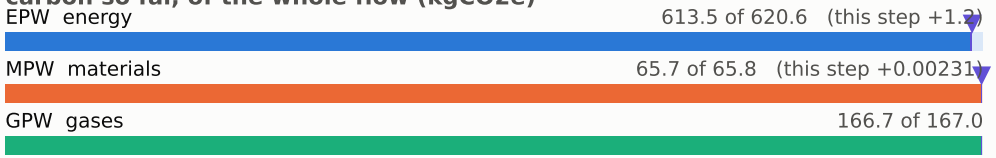
zoomed to fit



this step

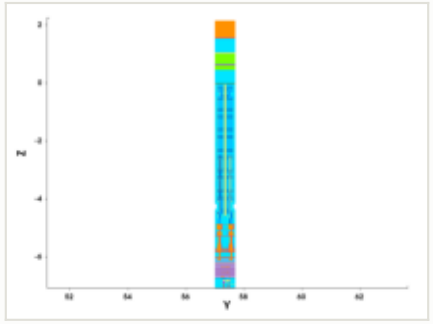
process area	Metallization	EPW	1.2 kgCO2e
material	Cu	EPW tool	0.86 kgCO2e
thickness	0.6 µm	EPW overhead	0.344 kgCO2e
material GWP	6.09011 gCO2e/g	MPW	0.00231 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

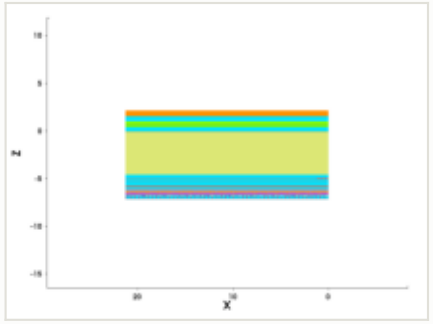


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



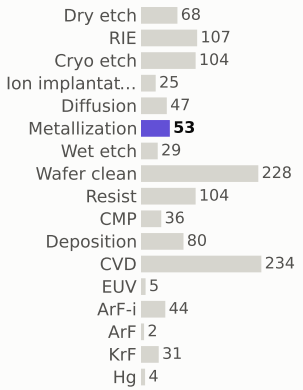
y cut at 57.351 µm



mask for this step

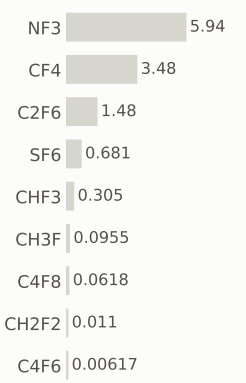
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



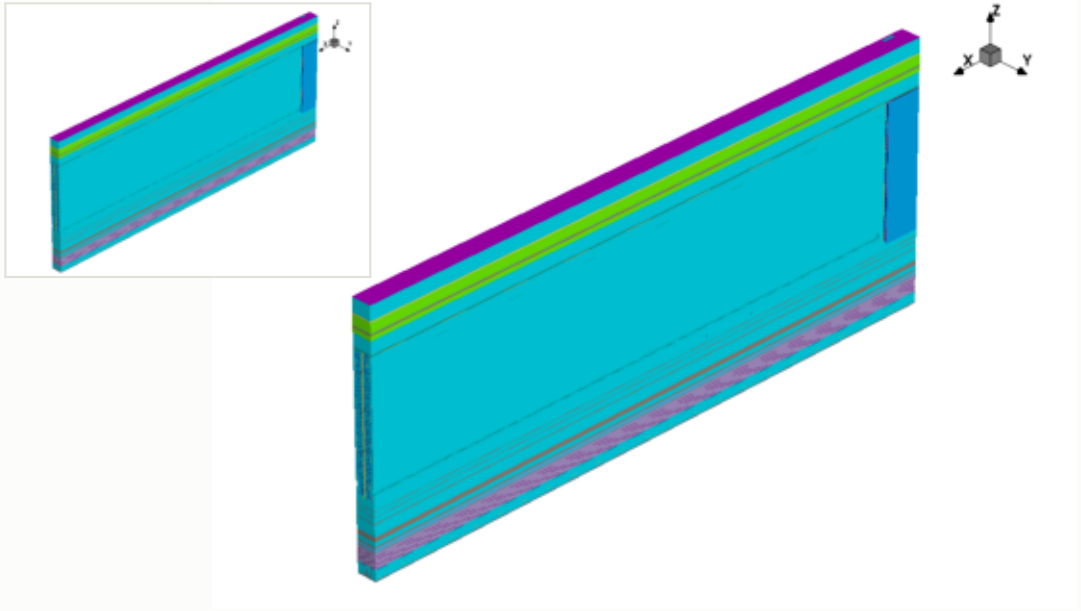
Step #13830: CMP

etch_cmp · CMP

0.169	846.0	0.02%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

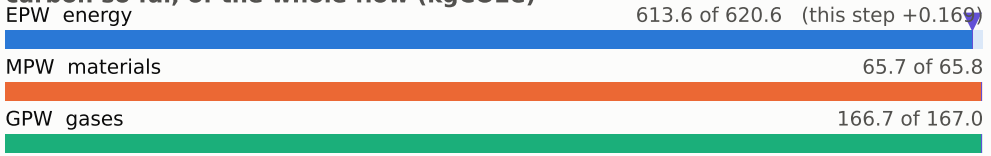
zoomed to fit



this step

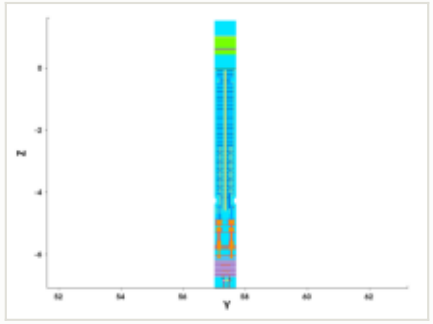
process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

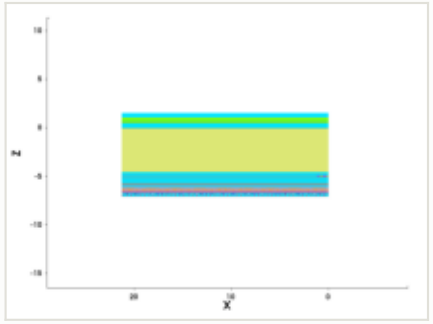


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



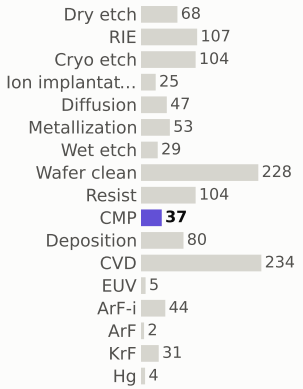
y cut at 57.351 µm



mask for this step

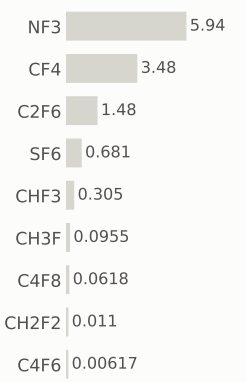
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



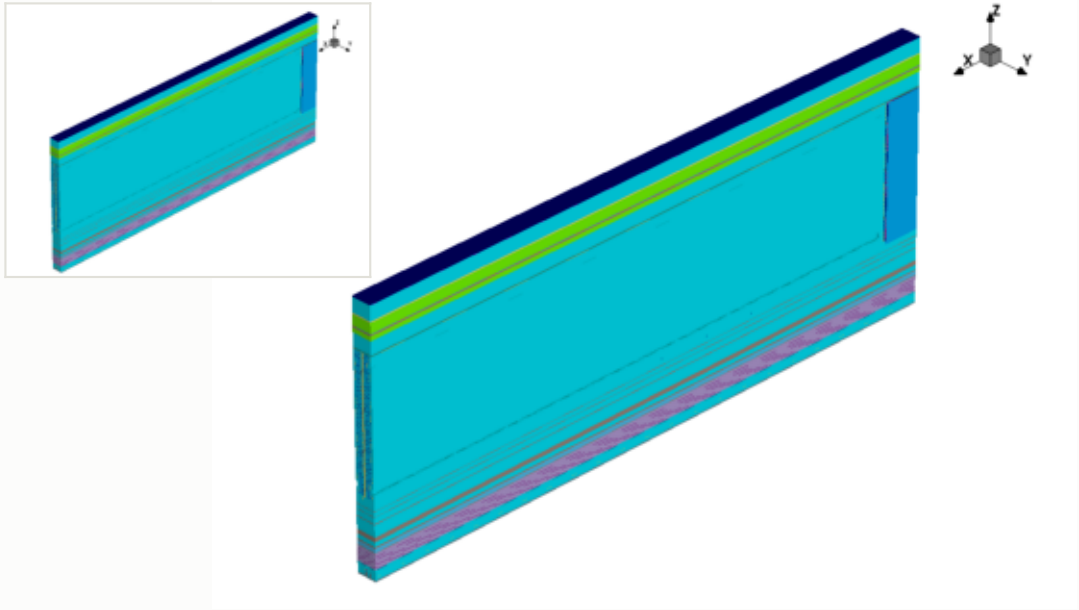
Step #13840: Metallization

depo_isotropic · Metallization

1.2	847.2	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

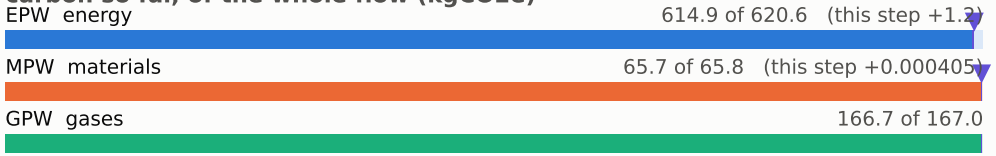
zoomed to fit



this step

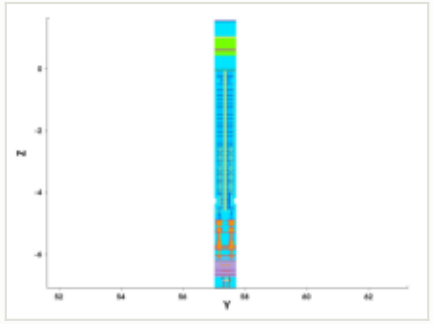
process area	Metallization	EPW	1.2 kgCO2e
material	TiN	EPW tool	0.86 kgCO2e
thickness	0.025 µm	EPW overhead	0.344 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.000405 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

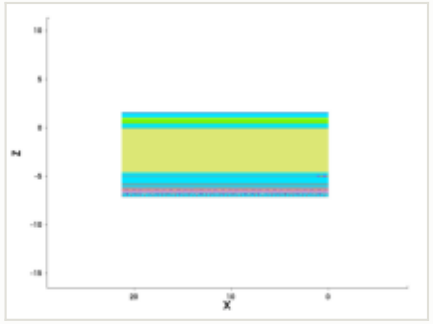


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



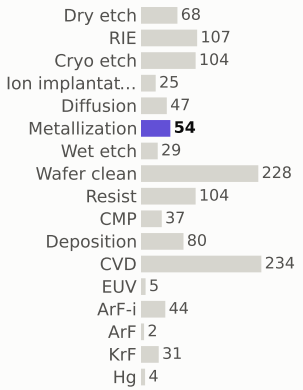
y cut at 57.351 µm



mask for this step

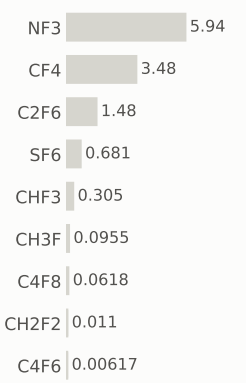
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



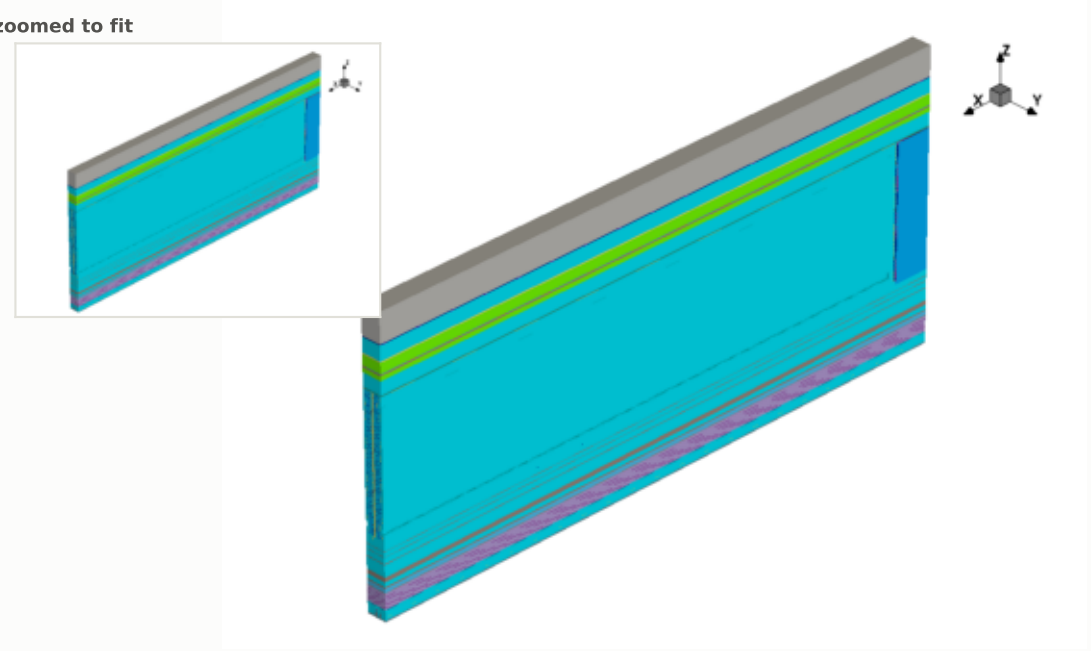
Step #13850: Metallization

depo_isotropic · Metallization

1.21	848.4	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

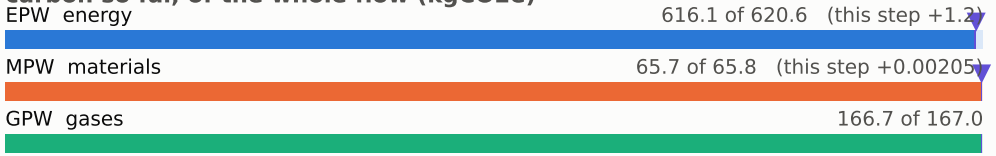


this step

process area	Metallization
material	Al
thickness	1 µm
material GWP	10.7269 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

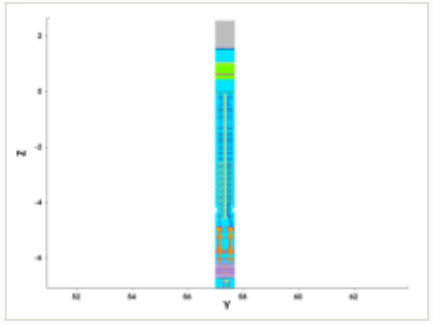
EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00205 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

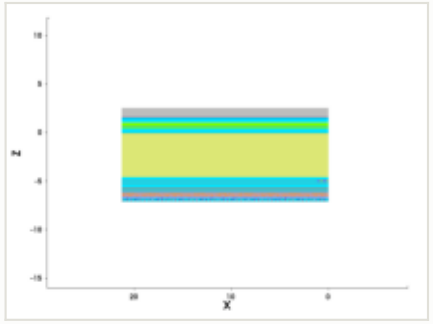


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



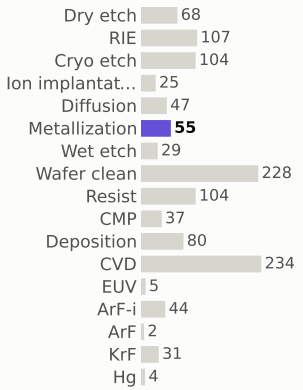
y cut at 57.351 µm



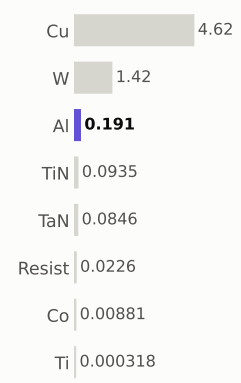
mask for this step

no mask for this step

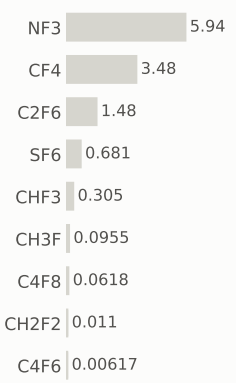
steps by process area



materials deposited (g)



gases emitted (g)



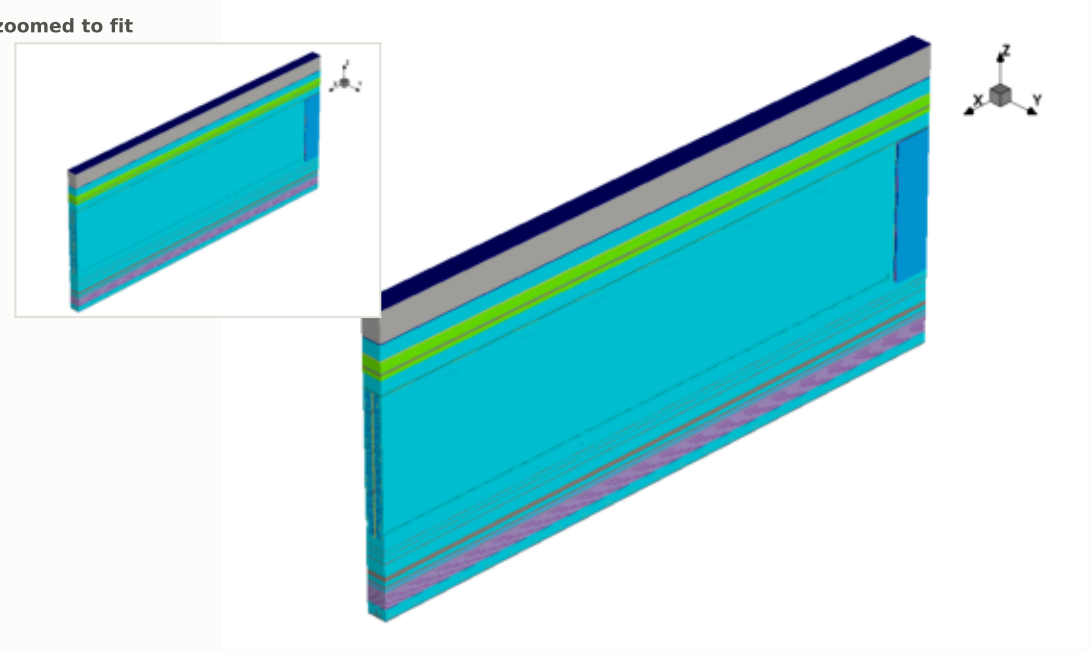
Step #13860: Metallization

depo_isotropic · Metallization

1.2	849.6	0.14%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

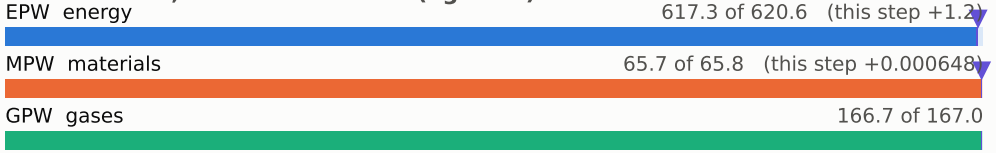


this step

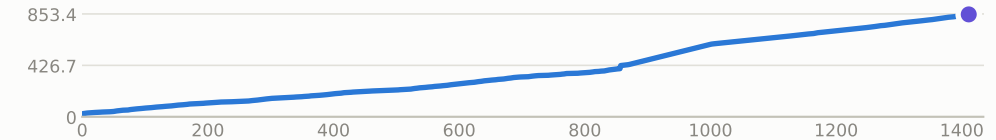
process area	Metallization
material	TiN
thickness	0.04 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000648 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

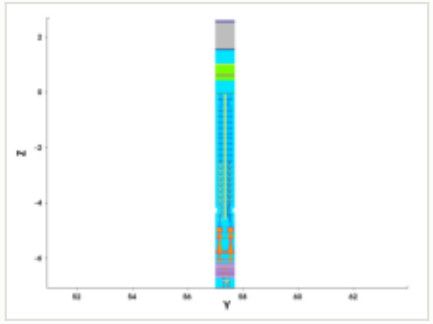
carbon so far, of the whole flow (kgCO2e)



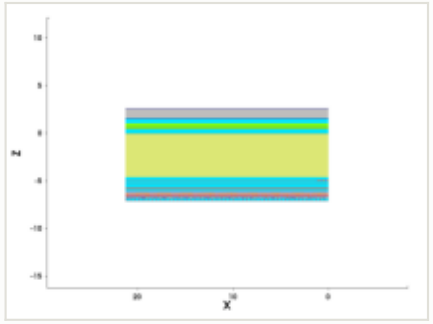
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



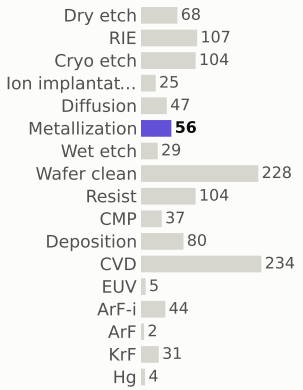
y cut at 57.351 µm



mask for this step

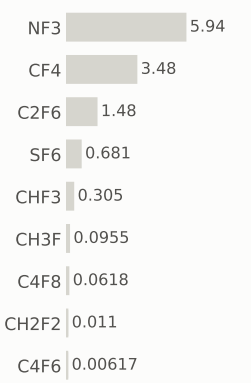
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



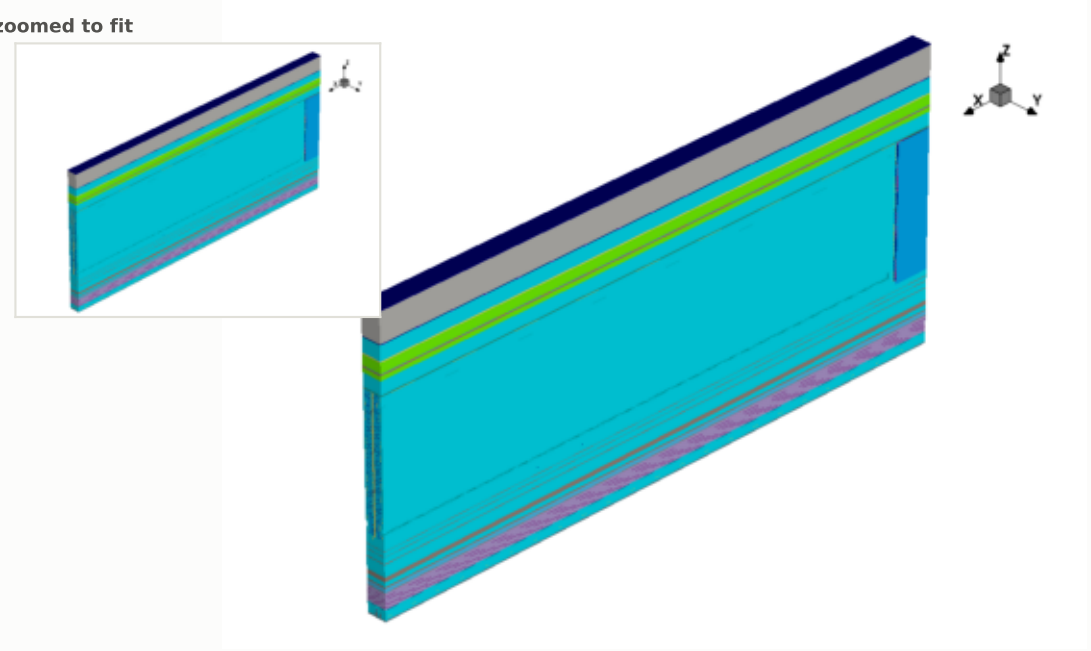
Step #13870: Hg

litho_patterning · Hg

0.199	849.8	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

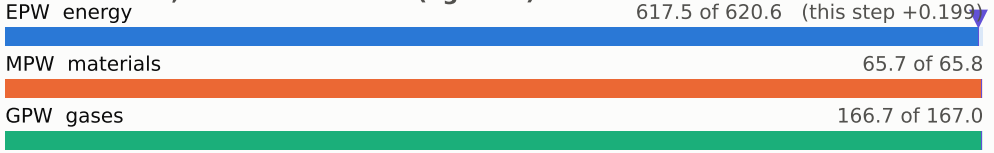
zoomed to fit



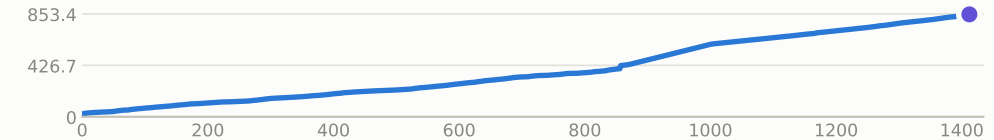
this step

process area	Hg	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

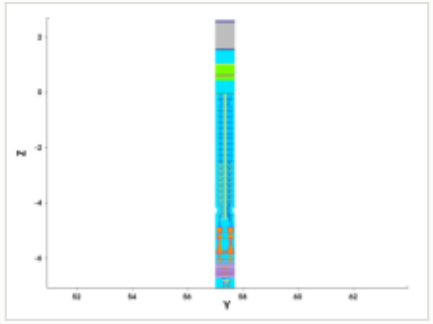
carbon so far, of the whole flow (kgCO2e)



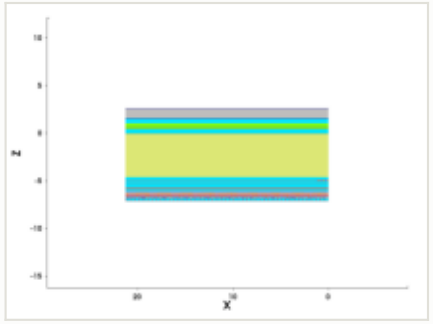
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



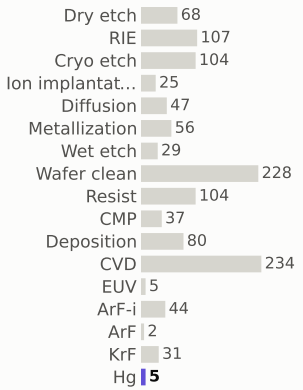
y cut at 57.351 µm



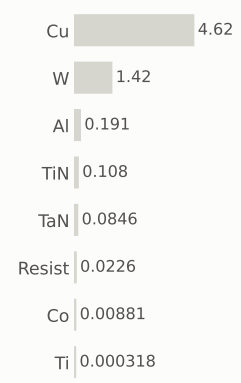
mask: layer 41:0, positive (GDSII)

nothing in the clip window

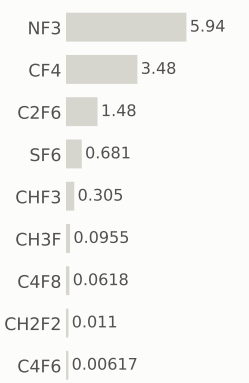
steps by process area



materials deposited (g)



gases emitted (g)



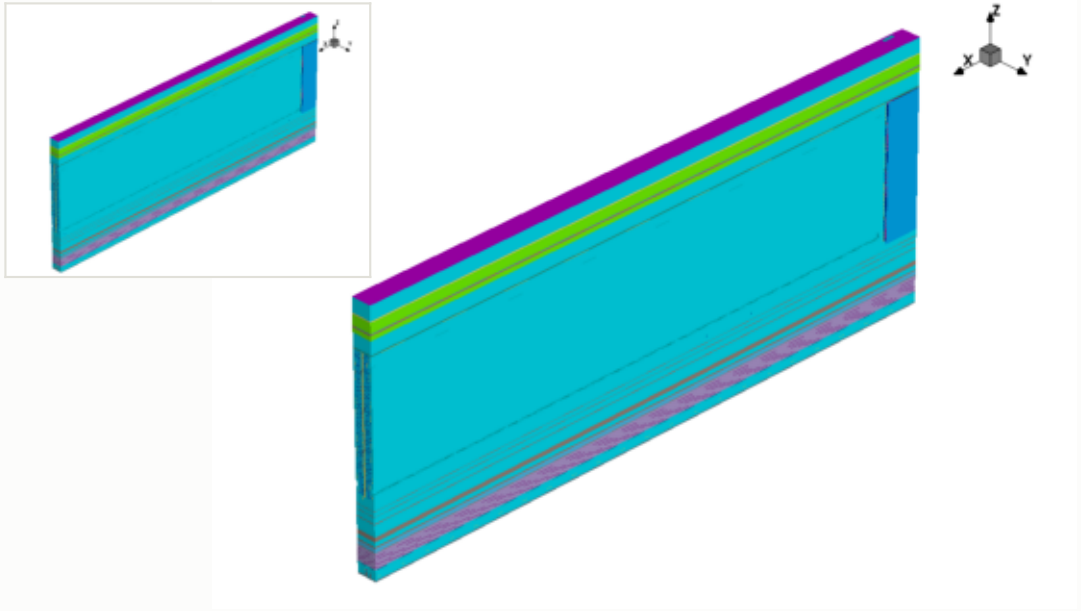
Step #13880: RIE

etch_anisotropic · RIE

1.04	850.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

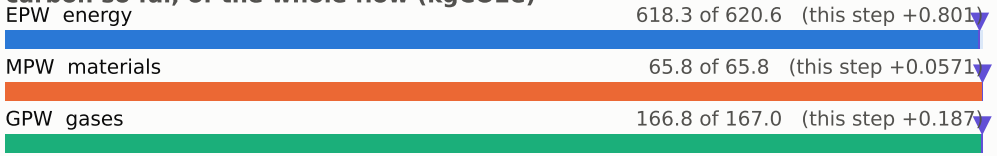


this step

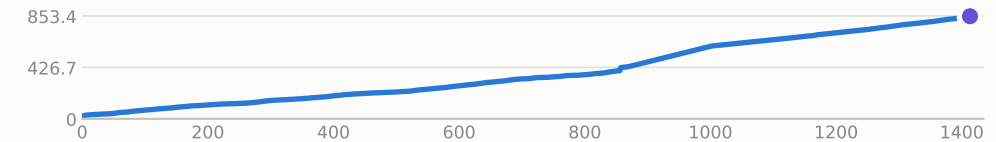
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

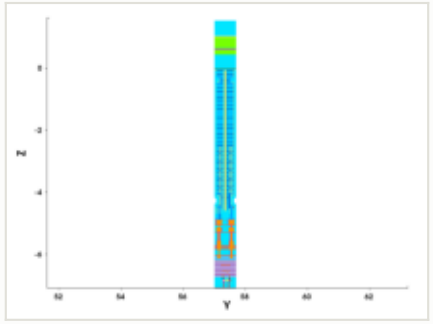
carbon so far, of the whole flow (kgCO2e)



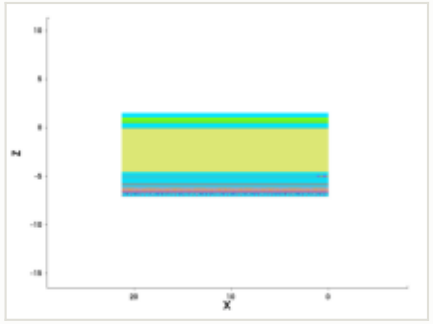
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



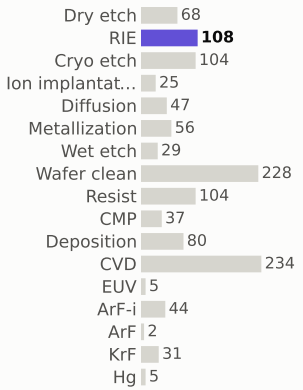
y cut at 57.351 µm



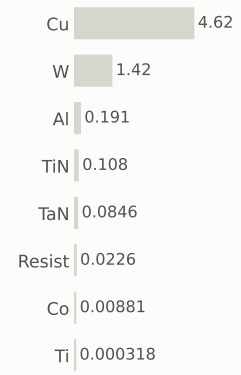
mask for this step

no mask for this step

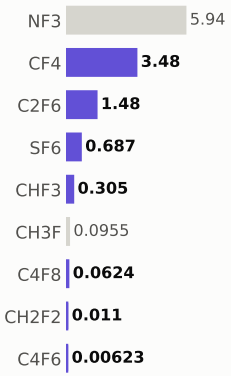
steps by process area



materials deposited (g)



gases emitted (g)



Step #13890: Spx_only

etch_strip · Spx_only

0	850.9	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

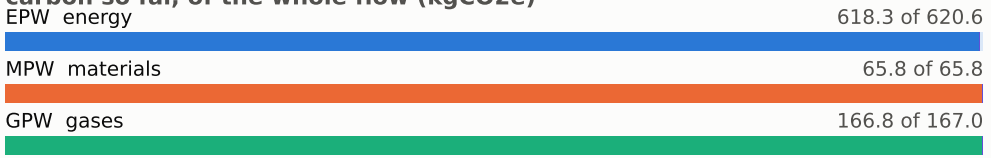
zoomed to fit

this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

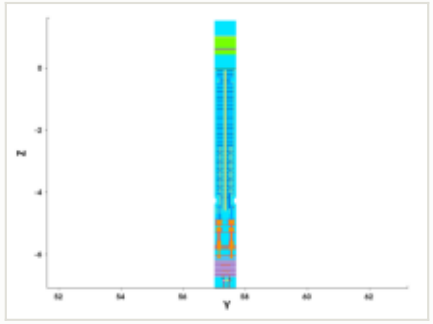
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

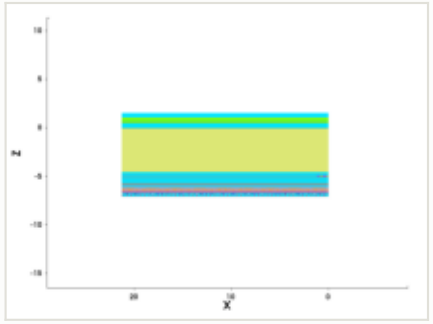


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



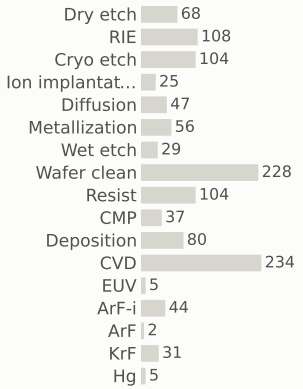
y cut at 57.351 µm



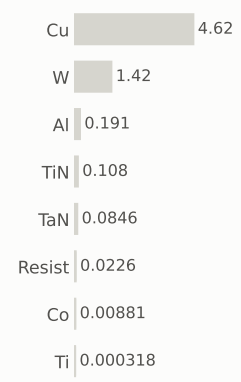
mask for this step

no mask for this step

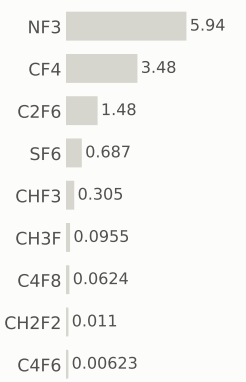
steps by process area



materials deposited (g)



gases emitted (g)



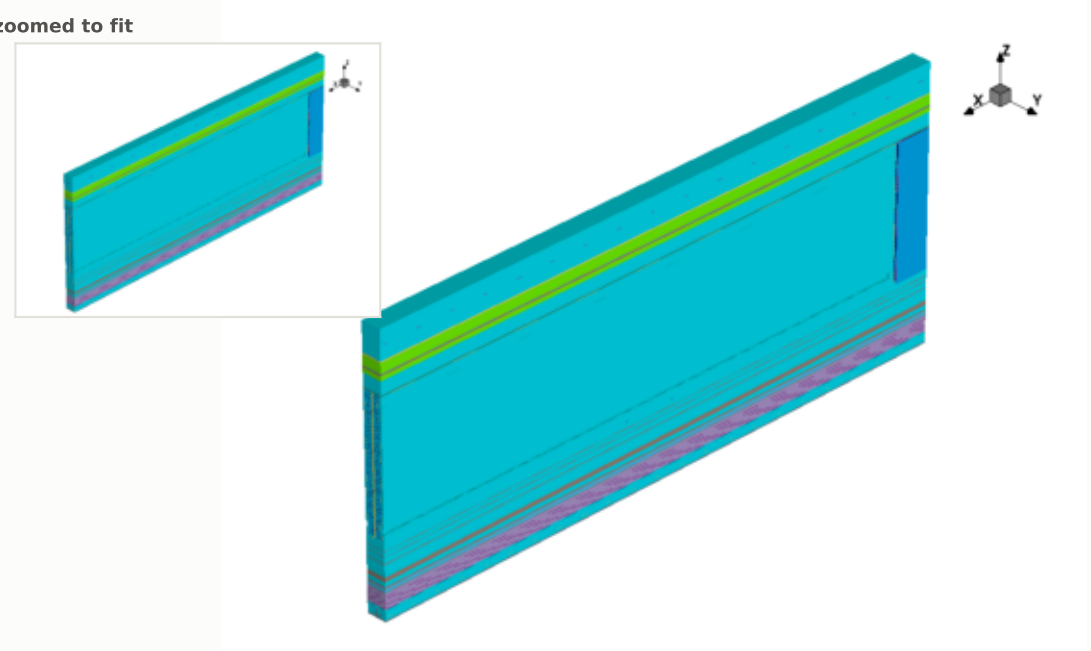
Step #13900: Deposition

depo_isotropic · Deposition

0.656	851.5	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

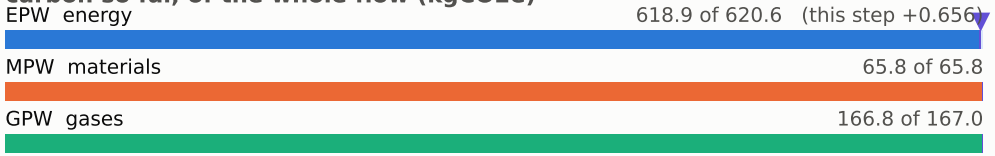
zoomed to fit



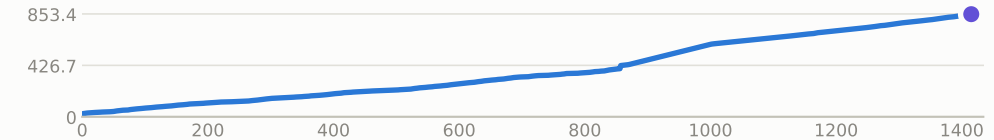
this step

process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.5 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

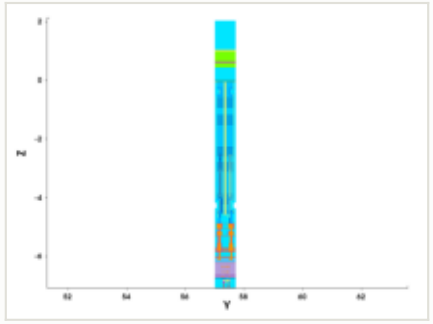
carbon so far, of the whole flow (kgCO2e)



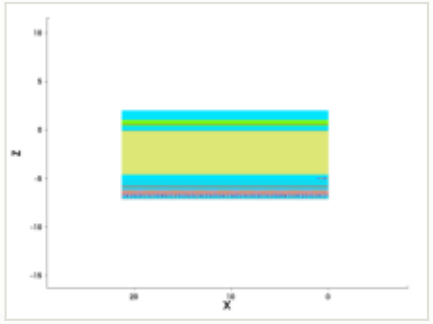
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



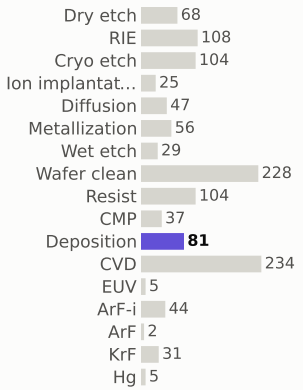
y cut at 57.351 µm



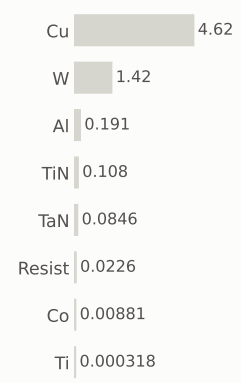
mask for this step

no mask for this step

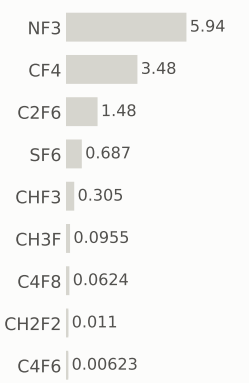
steps by process area



materials deposited (g)



gases emitted (g)



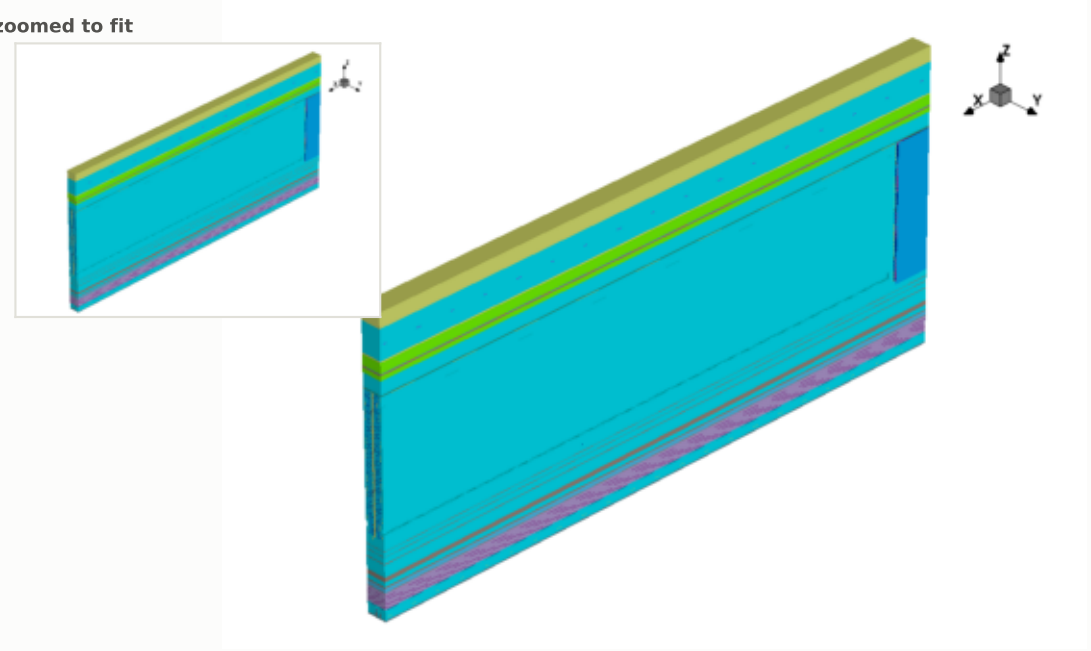
Step #13910: Deposition

depo_isotropic · Deposition

0.656	852.2	0.077%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

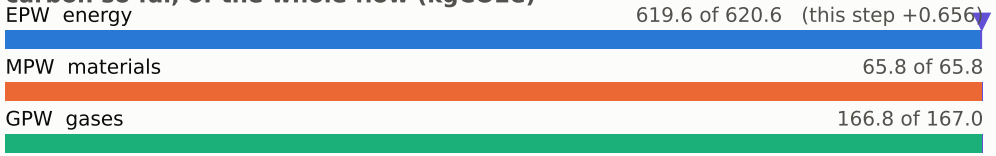
zoomed to fit



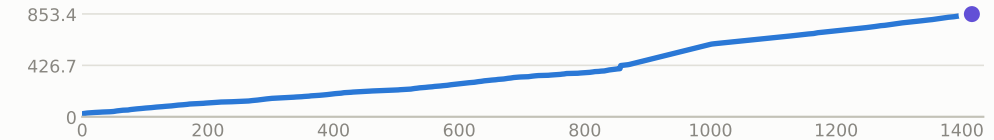
this step

process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.5 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

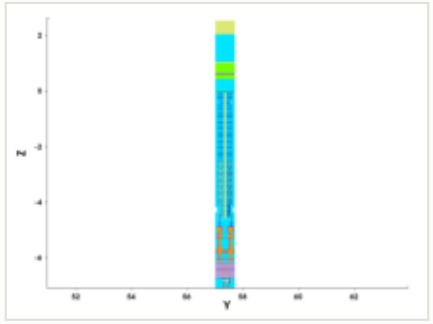
carbon so far, of the whole flow (kgCO2e)



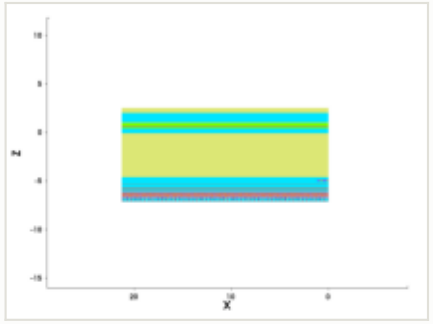
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



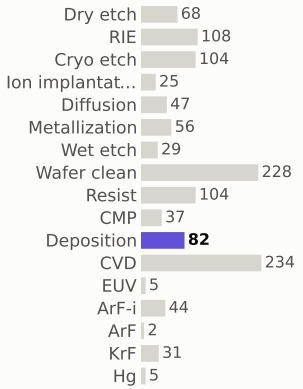
y cut at 57.351 µm



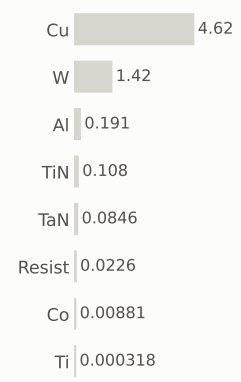
mask for this step

no mask for this step

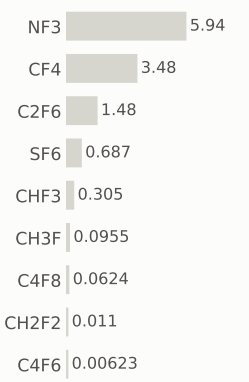
steps by process area



materials deposited (g)



gases emitted (g)



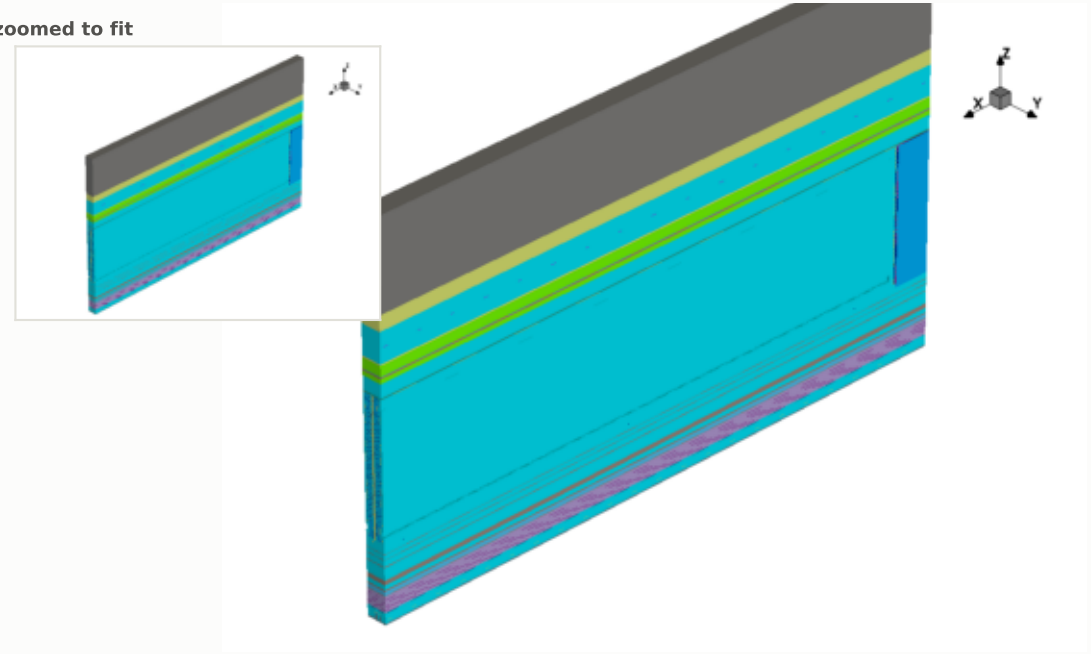
Step #13920: Hg

litho_patterning · Hg

0.199	852.4	0.023%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

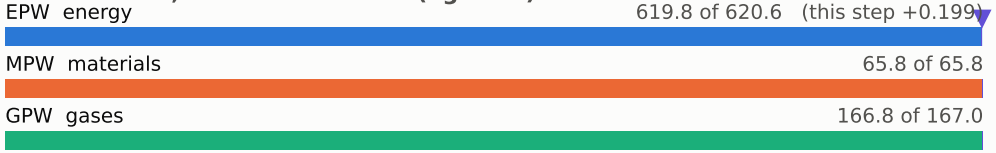
zoomed to fit



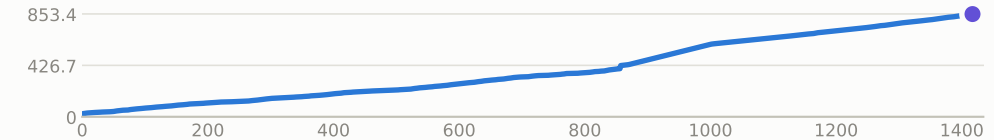
this step

process area	Hg	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

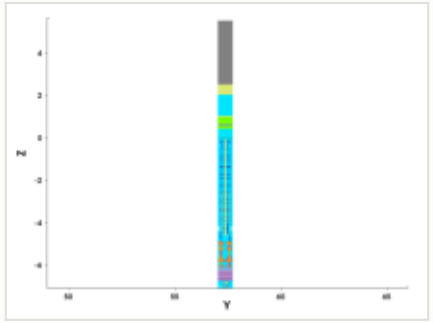
carbon so far, of the whole flow (kgCO2e)



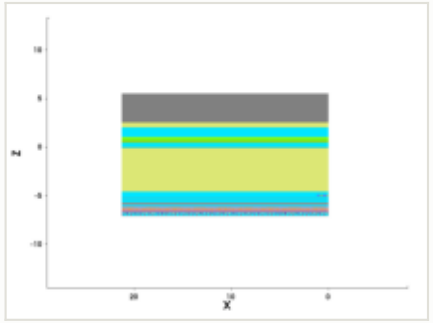
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



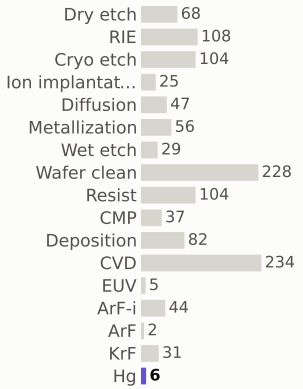
y cut at 57.351 µm



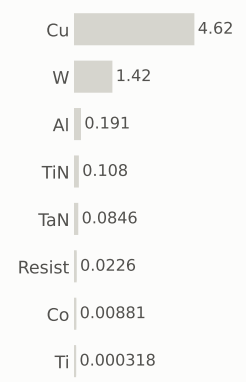
mask: layer 42:0, negative (GDSII)

nothing in the clip window

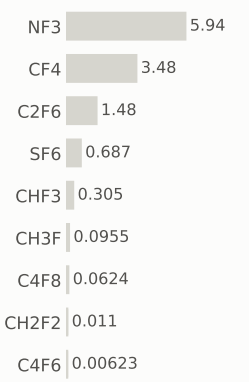
steps by process area



materials deposited (g)



gases emitted (g)



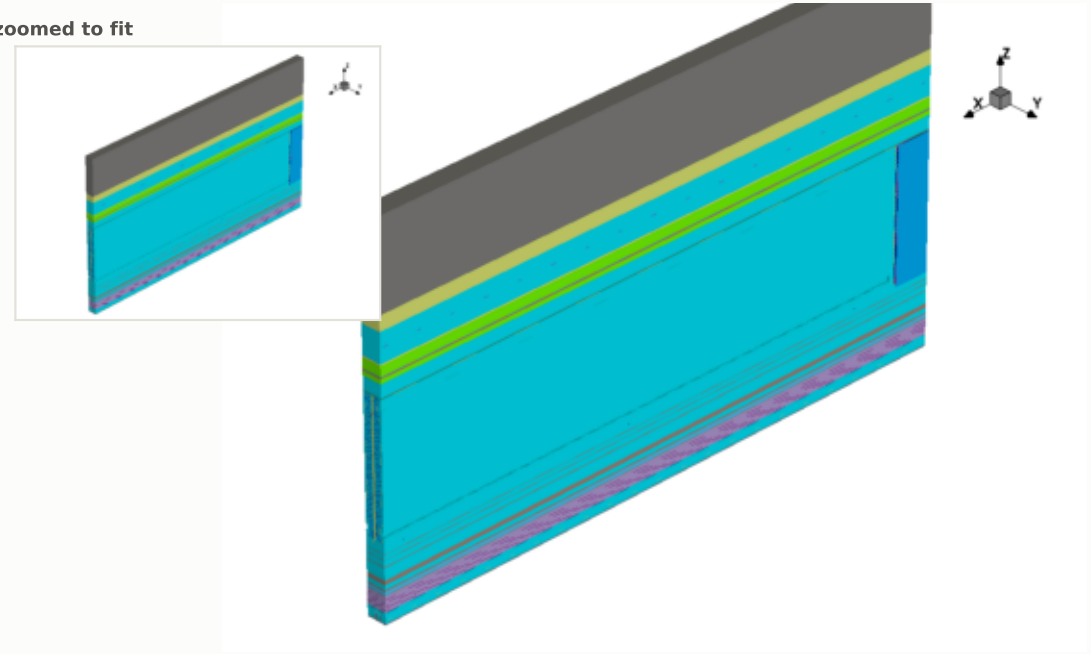
Step #13930: RIE

etch_anisotropic · RIE

1.04	853.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

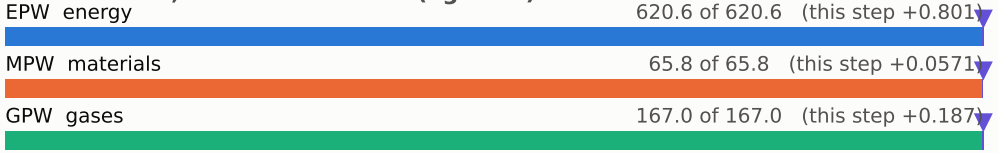


this step

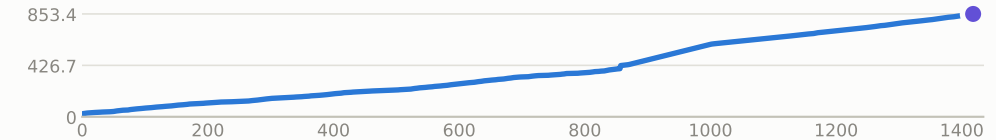
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

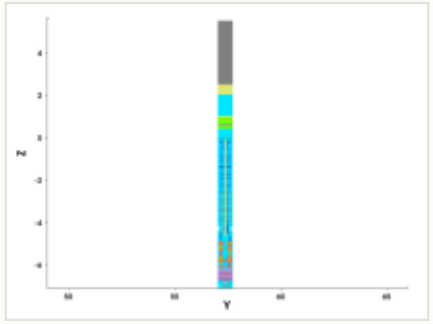
carbon so far, of the whole flow (kgCO2e)



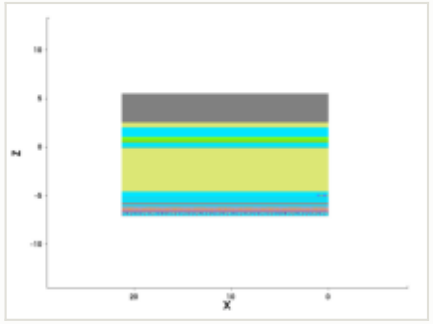
embodied carbon up to each step (kgCO2e per wafer)



x cut at 10.650 µm



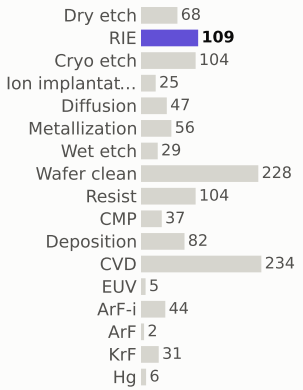
y cut at 57.351 µm



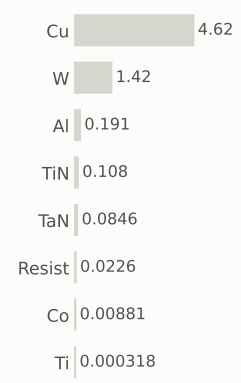
mask for this step

no mask for this step

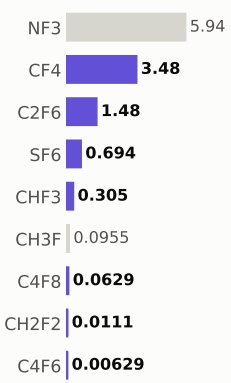
steps by process area



materials deposited (g)



gases emitted (g)



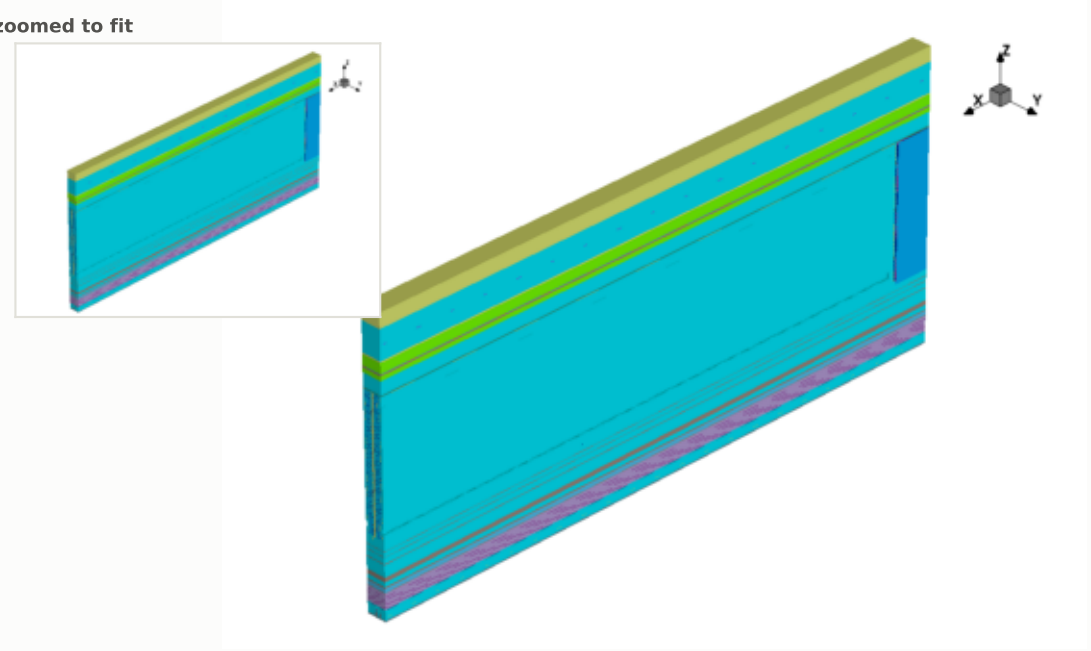
Step #13940: Spx_only

etch_strip · Spx_only

0	853.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

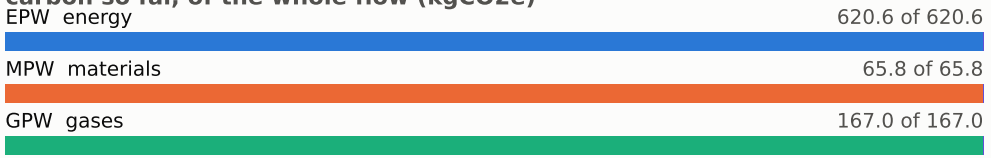


this step

process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

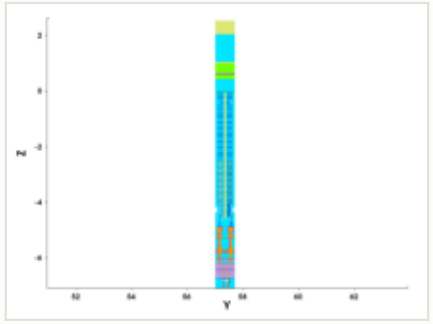
Spx_only step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

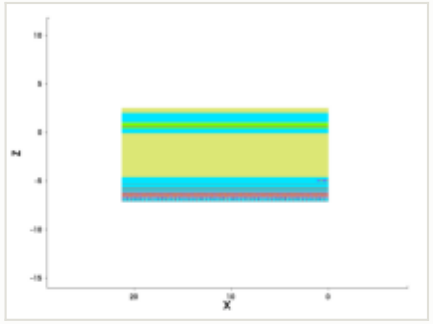


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



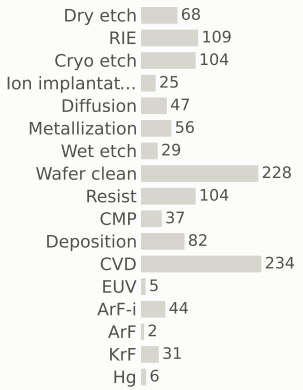
y cut at 57.351 µm



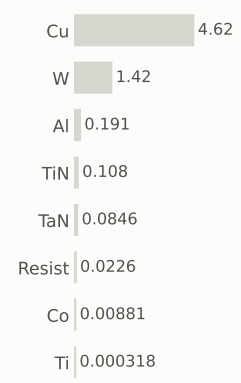
mask for this step

no mask for this step

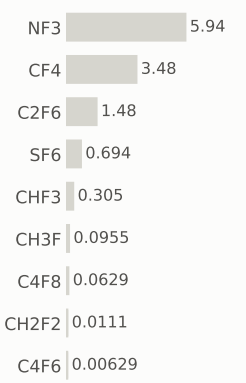
steps by process area



materials deposited (g)



gases emitted (g)



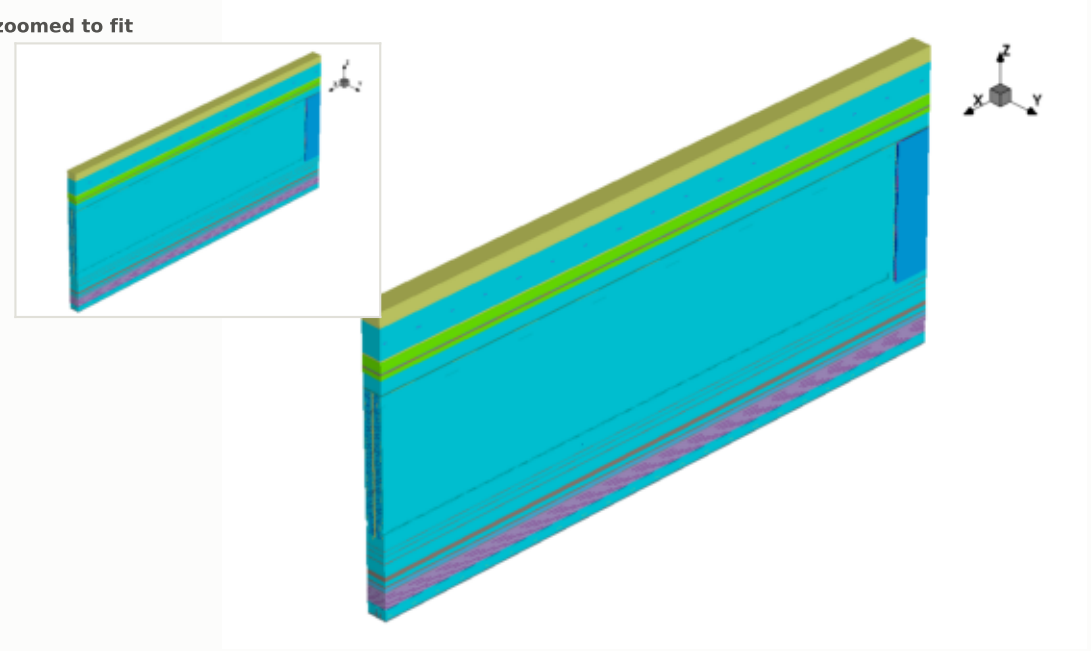
Step #13960: Wafer

treatment · Wafer

0	853.4	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

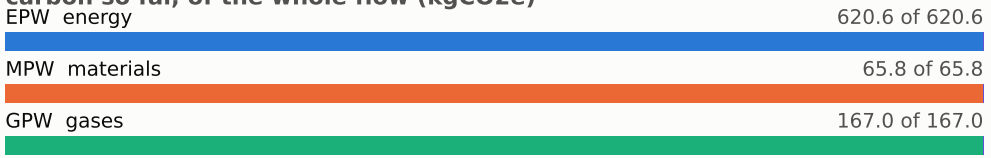


this step

process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

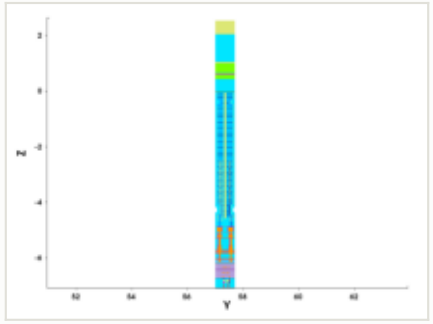
Wafer step: not counted as a process step (uncounted_process_areas)

carbon so far, of the whole flow (kgCO2e)

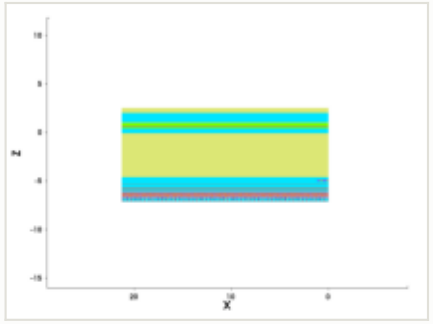


embodied carbon up to each step (kgCO2e per wafer)

x cut at 10.650 µm



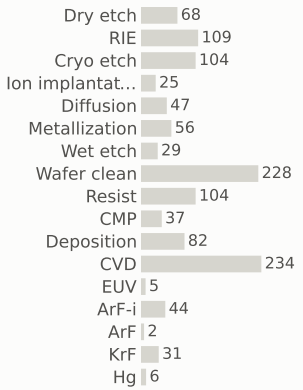
y cut at 57.351 µm



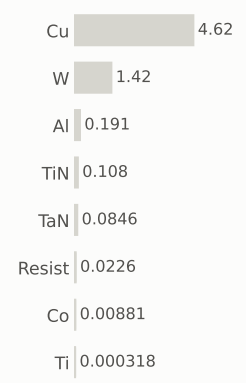
mask for this step

no mask for this step

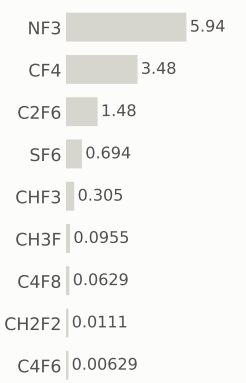
steps by process area



materials deposited (g)

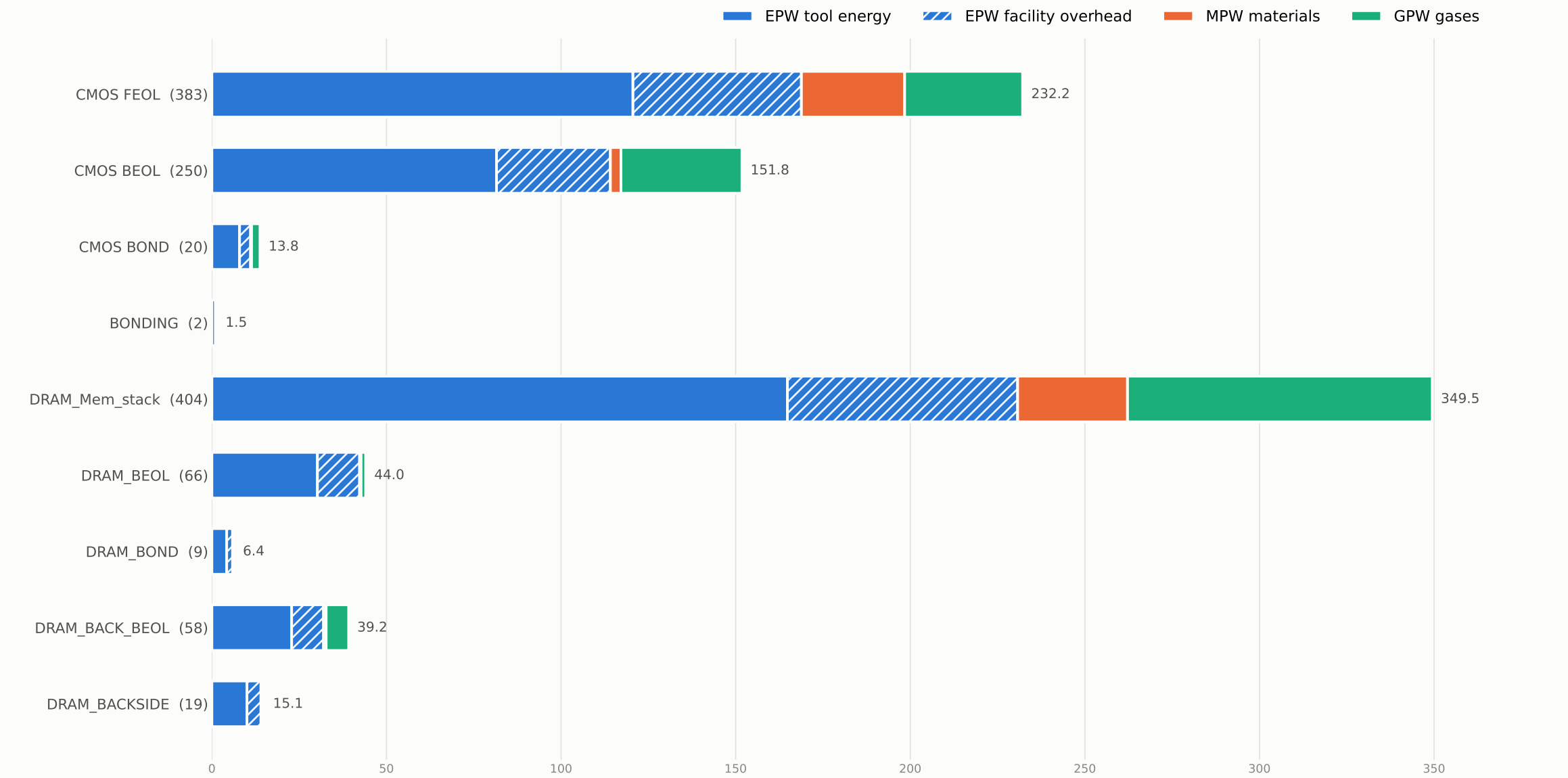


gases emitted (g)



Carbon by module

kgCO2e per wafer, in flow order (steps in brackets)



Totals by process area

kgCO2e per wafer; EPW = tool + facility overhead (overhead energy = tool energy x 0.4)

process area	steps	tool kWh	overhead kWh	EPW tool	EPW overhead	EPW	MPW	GPW	total
CVD	234	255.06	102.02	109.68	43.87	153.55	7.43	145.19	306.16
RIE	109	144.97	57.99	62.34	24.93	87.27	6.22	20.39	113.88
Metallization	56	112.00	44.80	48.16	19.26	67.42	0.13	0.00	67.56
Dry_etch	68	90.44	36.18	38.89	15.56	54.44	0.00	0.00	54.44
Deposition	82	89.38	35.75	38.43	15.37	53.81	0.00	0.00	53.81
Cryo_etch	104	86.84	34.74	37.34	14.94	52.28	0.03	1.45	53.76
Wafer_clean	228	63.84	25.54	27.45	10.98	38.43	0.00	0.00	38.43
Diffusion	47	62.51	25.00	26.88	10.75	37.63	0.00	0.00	37.63
Ion_implantation	25	33.25	13.30	14.30	5.72	20.02	0.00	0.00	20.02
Resist	104	29.12	11.65	12.52	5.01	17.53	0.00	0.00	17.53
EUV	5	16.25	6.50	6.99	2.80	9.78	0.00	0.00	9.78
ArF-i	44	15.84	6.34	6.81	2.72	9.54	0.00	0.00	9.54
CMP	37	10.36	4.14	4.45	1.78	6.24	0.00	0.00	6.24
KrF	31	10.23	4.09	4.40	1.76	6.16	0.00	0.00	6.16
Wet_etch	29	8.12	3.25	3.49	1.40	4.89	0.00	0.00	4.89
Hg	6	1.98	0.79	0.85	0.34	1.19	0.00	0.00	1.19
ArF	2	0.66	0.26	0.28	0.11	0.40	0.00	0.00	0.40
Wafer		0.00	0.00	0.00	0.00	0.00	52.00	0.00	52.00
Spx_only		0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01
all	1211	1,030.85	412.34	443.27	177.31	620.57	65.83	167.03	853.43